A History of the United States

37 chapters, each told across the same ten eras

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28 Drugs and Alcohol about 110 min 2119

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The Grid

Each row is a chapter and each column is an era. The number is the page where that era of the chapter begins. Read across a row to follow one subject through time, or down a column to see every subject in one era.

Chapter	Before 1500	The 1500s	The 1600s	1700 to 1750	1750 to 1800	1800 to 1850	1850 to 1900	1900 to 1950	1950 to 2000	2000 to today
Part 1: The Land and Its First Peoples										
1 Exploration and Discovery	17	21	31	36	39	44	64	74	82	92
2 Native Nations	99	102	108	116	120	126	134	149	158	170
3 Land and Environment	180	182	185	189	192	196	200	215	229	240
Part 2: Coming, Moving, Settling										
4 Immigration to America	256	258	260	267	271	275	283	298	310	320
5 Migration Across America	333	336	339	343	346	352	363	376	383	387
6 City Building	395	398	399	404	407	413	418	431	443	452
7 Home and Family Life	460	463	465	468	471	474	480	485	495	502
Part 3: Making and Building										
8 Technology	512	516	517	524	528	535	541	553	561	570
9 Science	581	584	585	588	591	596	604	614	633	648
10 Energy	661	663	665	669	671	677	681	691	702	711
11 Transportation	721	724	726	728	732	734	745	756	764	773
12 Landmarks	783	788	789	795	799	804	807	815	825	832
13 The Elements	844	847	848	852	855	859	863	868	886	899
Part 4: Work, Food, and Money										
14 Work and Workers	911	912	914	918	920	922	929	941	963	974
15 Food and Farming	986	989	991	995	1000	1004	1009	1015	1028	1039
16 Economy	1049	1051	1052	1057	1062	1067	1074	1084	1096	1102
17 Money	1112	1113	1115	1119	1123	1128	1136	1143	1153	1160
18 Marketplace and Consumer Life	1170	1173	1174	1178	1182	1188	1193	1204	1214	1222
19 Big Business and Its Power	1233	1234	1235	1242	1245	1250	1256	1268	1280	1291
Part 5: Power, Law, and War										
20 Government, Law, and Politics	1304	1308	1309	1317	1322	1333	1341	1356	1372	1390
21 Crime, Police, and Justice	1406	1409	1412	1422	1427	1434	1442	1456	1468	1484
22 War and the Military	1499	1502	1508	1515	1522	1534	1543	1559	1575	1585
23 America and the World	1597	1599	1600	1602	1605	1616	1623	1634	1647	1656
Part 6: Freedom and Fairness										
24 Slavery and Freedom	1666	1667	1670	1676	1684	1693	1712	1735	1741	1744
25 Rights and Movements	1755	1757	1760	1767	1770	1778	1795	1829	1871	1909
Part 7: Health and Safety										
26 Health, Disease, and Medicine	1946	1949	1954	1961	1967	1975	1983	1990	2005	2022
27 Disasters and Rescue	2032	2036	2041	2047	2054	2061	2068	2078	2093	2104
28 Drugs and Alcohol	2120	2124	2127	2133	2138	2147	2154	2165	2179	2190
Part 8: Faith, Learning, and News										
29 Religion	2203	2207	2213	2231	2242	2256	2274	2287	2304	2324
30 School and Education	2345	2348	2351	2359	2367	2375	2392	2414	2442	2474
31 News and Communication	2500	2502	2506	2512	2520	2533	2545	2559	2572	2585
Part 9: Culture and Play										
32 Art	2598	2605	2608	2613	2619	2627	2643	2660	2679	2695
33 Music	2713	2717	2719	2723	2729	2736	2746	2759	2779	2795
34 Storytelling Evolution	2814	2818	2823	2827	2833	2841	2851	2861	2877	2889
35 Styles	2908	2912	2916	2923	2928	2934	2940	2949	2959	2970
36 Sports and Play	2981	2985	2987	2995	2999	3008	3019	3028	3045	3058
37 Holidays	3075	3080	3083	3091	3094	3101	3107	3119	3125	3135

How This Book Works

This book is a history of the United States in 37 chapters, each about one subject. Every chapter is divided into the same ten eras in the same order: before 1500, the 1500s, the 1600s, 1700 to 1750, 1750 to 1800, 1800 to 1850, 1850 to 1900, 1900 to 1950, 1950 to 2000, and 2000 to today. That makes the book a grid, 37 subjects by 10 eras, and you can read down one subject or across one era.

Each chapter includes these signposts:

- **The card** at the top lists the chapter's point in one sentence, a true story, what to watch for, three questions, a section to read when you have five minutes, and the sections you can skip.
- **The first bold line of each era** states how long the era takes to read.
- **"By [year]" lines**, in the longest eras only, state where things stand partway through.
- **The end of a long era** states the one thing to remember from it, often with a few dates, and what comes next.
- **Boxes with a person's name at the top** contain many of the main stories, each built from the records.
- **The closing** at the end states the answers to the three questions, five dates to keep, one thing still not known, a place to visit and the next chapter.
- **A hard word** is explained the first time it appears in a chapter, and the glossary at the back lists them all.

Every fact in this book comes from a record someone made: a letter, a court paper, a count, a photograph, bones in the ground, or a story a nation kept. "How We Know What Happened," at the back, describes how historians check those records and what happens when two records state different things.

Before You Start

This book is long. It contains about 830,000 words, which take about 69 hours to read at 200 words a minute.

The order does not matter. Start with any chapter, or with any era inside one. A hard word is explained the first time it appears in a chapter, so if you start in the middle and meet a word you do not know, look it up in the glossary at the back.

Skipping is fine. The card at the top of each chapter lists the sections you can skip, and the chapter still makes sense without them. Read the boxes with a person's name at the top, because they contain many of the main stories.

When a page feels heavy, stop at the next bold line. The end of each long era states the one thing to remember from it. Read that line, then come back later or go to another chapter.

Four Ways to Read This Book

The times below are at 200 words a minute. That is a quick reader's pace, and many readers take longer, which is fine. Each chapter's card states that chapter's time, and the first bold line of each era states the era's.

1. Straight through

Read the 37 chapters in order, from Exploration and Discovery to Holidays. That takes about 69 hours. The shortest chapter, Home and Family Life, takes about 1 hour. The longest, Rights and Movements, takes about 4 hours 40 minutes. Most take between 1 hour 15 minutes and 2 hours 20 minutes.

2. One subject

Pick one chapter from the contents and read it start to finish. Every chapter includes all ten eras, so you follow one subject from before 1500 to today. In some chapters the first era starts thousands of years ago.

3. One era, every subject

Pick one of the ten eras and read that era in every chapter, in the order of the contents. That gives you the whole country in one stretch of time: what people were eating, building, singing and fighting over in the same years.

- Before 1500: about 2 hours 15 minutes
- The 1500s: about 2 hours 30 minutes
- The 1600s: about 5 hours
- 1700 to 1750: about 3 hours 30 minutes
- 1750 to 1800: about 5 hours 30 minutes
- 1800 to 1850: about 7 hours 30 minutes
- 1850 to 1900: about 10 hours 30 minutes
- 1900 to 1950: about 11 hours 30 minutes
- 1950 to 2000: about 10 hours 15 minutes
- 2000 to today: about 8 hours 45 minutes

4. The short road

Read "The History in Ten Paragraphs," which comes next. Then read the card at the top of each chapter and the closing at its end. The 37 cards take about 1 hour, and the 37 closings take about 1 hour 5 minutes. Reading the sections listed under "Five minutes?" on the cards adds about 1 hour 30 minutes. The whole short road takes about 3 hours 45 minutes, and it reaches every subject in the book.

The History in Ten Paragraphs

The ten paragraphs below, one for each era, state the main line of the country's history. Every fact here is told in full in one of the 37 chapters.

Before 1500

People were living in North America at least 21,000 years ago. The oldest sign found so far is a set of human footprints at White Sands, New Mexico. Some scientists questioned their age in 2021, and two other kinds of tests gave the same age in 2023. Estimates of how many people lived here before 1492 run from about 1 million to about 18 million. Cahokia, in what is now Illinois, was the largest city north of Mexico by about 1050. Norse sailors, the people often called Vikings, camped in Newfoundland around the year 1000, the first known visit from Europe. Christopher Columbus reached the Bahamas in 1492, but he never set foot in what is now the United States.

The 1500s

Spanish soldiers, and the diseases that traveled ahead of them, reached what is now the United States. Juan Ponce de León led the first European expedition to Florida that anyone wrote down, in 1513. Hernando de Soto's expedition crossed the Southeast from 1539 to 1543. Smallpox and other diseases from Europe spread along Native trade routes ahead of the newcomers and killed large shares of whole communities. Pedro Menéndez de Avilés founded St. Augustine, Florida, in 1565. In January 1599 Juan de Oñate's soldiers stormed and burned Acoma Pueblo, in New Mexico. Oñate sentenced the survivors over age 12 to twenty years of forced work, and he had one foot cut off each man over 25.

The 1600s

English colonists settled the Atlantic coast, and by the end of the century their farms ran on slavery. Colonists built Jamestown, Virginia, in 1607, and they made the colony pay by growing tobacco. The crew of an English ship, the *White Lion*, landed about 20 captive Africans at Point Comfort, Virginia, in late August 1619. The *Mayflower*'s passengers founded Plymouth in 1620. Judges in Virginia sentenced John Punch, an African who had run away, to serve for life in 1640. Far to the west, Pueblo people rose up together in 1680 and drove the Spanish out of New Mexico. By 1700 enslaved Africans outnumbered white indentured servants, people bound to work for years to pay for their crossing, on Virginia's tobacco farms.

1700 to 1750

The colonies grew on the work of people who were not free, and those people fought back. Between half and two-thirds of the white people who came from the 1630s to the Revolution came as indentured servants. About 20 enslaved men led by a man called Jemmy seized weapons at the Stono River in South Carolina in September 1739 and marched toward Spanish Florida. The rebels killed about 25 white colonists. The militia, armed colonists called up to fight, killed 35 to 50 of the rebels, and colonists set the rebels' heads on stakes along the roads. Philadelphia, the largest town, had about 13,000 people by 1750.

1750 to 1800

A new country came out of these years, and its founders wrote slavery into its constitution. British forces beat the French in a war for the continent that ended in 1763. Members of the Continental Congress adopted the Declaration of Independence on July 4, 1776. After the war for independence, American and British diplomats, the officials who spoke for their countries, agreed in 1783 that the new country's western border would be the Mississippi River. They invited no Native nation to the talks. Delegates in Philadelphia wrote the Constitution in 1787. Under it, each enslaved person counted as three fifths of a free person in deciding how many seats each state got in Congress. Census takers counted 697,681 enslaved people in 1790, in the first census, the government's count of everyone in the country.

1800 to 1850

The country grew west by purchase, by treaty and by force, and slavery grew with it. American diplomats agreed in 1803 to pay France $15 million for Louisiana, a huge stretch of land west of the Mississippi River. President Andrew Jackson signed the Indian Removal Act in 1830, and under it soldiers drove more than 16,000 Cherokee people west in 1838 and 1839. More than 4,000 of them died of cold, hunger and disease by the usual count, and one scientist who studies the size of populations puts the Cherokee losses at 8,000 or more. American armies won a war with Mexico that ended in 1848, and Mexico's leaders gave up more than 525,000 square miles, California among them. The same year, women met at Seneca Falls, New York, and signed a declaration of women's rights. Nearly 4 million people were enslaved by 1860.

1850 to 1900

Slavery ended in a war, and then the country spread west across Native land. Confederate soldiers, who fought for the Southern slave states whose leaders had broken away from the country, fired on Fort Sumter in April 1861. Settlers took up 270 million acres under the Homestead Act of 1862, much of it land that US officials had taken from Native nations. About 620,000 soldiers died in the Civil War by the usual count. Slavery ended with the 13th Amendment, a change to the Constitution, in December 1865, except as punishment for a crime. Soldiers of the 7th Cavalry killed between 250 and 300 Lakota people at Wounded Knee in December 1890. In 1896 the Supreme Court justices ruled in *Plessy v. Ferguson* that separating people by race was lawful.

1900 to 1950

Americans fought in two world wars in these fifty years and lived through hard times between them. Members of Congress declared war on Germany in April 1917. Under the 19th Amendment, from 1920, no state may deny a citizen the vote because of sex. Stock prices crashed in October 1929, and in the Great Depression of the 1930s about one worker in four had no job. Japanese pilots attacked Pearl Harbor, in Hawaii, on December 7, 1941, and more than 2,400 people were killed or missing. Army officers then removed about 120,000 people with Japanese family roots from the West Coast to camps. More than 400,000 Americans died in the war. A US bomber crew dropped an atomic bomb on Hiroshima, in Japan, on August 6, 1945, and between 90,000 and 166,000 people there died from it within four months.

1950 to 2000

Black Americans forced changes in the law, and American astronauts flew to the Moon. In 1954 the Supreme Court justices ruled that separate schools by race were unlawful. After police in Montgomery, Alabama, arrested Rosa Parks in December 1955 for refusing to give up her bus seat, most of the city's Black people stayed off its buses for a year, until federal judges ended the rule that kept Black riders apart from white riders. The Soviet Union, a large country ruled from Moscow, was the main rival of the United States for most of these years, and its engineers launched the first satellite in 1957. President Johnson signed the Civil Rights Act in 1964 and the Voting Rights Act in 1965. Neil Armstrong and Buzz Aldrin walked on the Moon on July 20, 1969. The Soviet Union broke apart in December 1991.

2000 to today

These years began with an attack and two long wars, and then came a pandemic, a disease that spread around the world. Hijackers flew airliners into the World Trade Center and the Pentagon on September 11, 2001, and killed nearly 3,000 people. American forces fought in Afghanistan for twenty years. They invaded Iraq in 2003, after American leaders said Iraq had nuclear, chemical or biological weapons, and none were found. In 2015 the Supreme Court justices ruled that same-sex couples could marry in every state. A Minneapolis police officer murdered George Floyd in May 2020. A crowd attacked the Capitol on January 6, 2021, while members of Congress counted the official votes from each state for president. By September 2026, COVID-19 had killed more than 1.2 million people in the United States.

PART 1

The Land and Its First Peoples

Chapter 1: Exploration and Discovery

CHAPTER 1 OF 37 · EXPLORATION AND DISCOVERY · 10 ERAS · ABOUT 110 MINUTES

Almost every place in this chapter was already somebody's home when the explorers arrived, and the people who lived there often showed them the way.

In the South Dakota Cultural Heritage Center there is a lead plate six inches by eight. One side has Latin words stamped into it that name King Louis XV of France. On the other side somebody scratched a date, March 30, 1743, and the names of four Frenchmen. Two of them were brothers, the sons of a fur trader, and they had just crossed the northern Great Plains with two Mandan guides. They buried the plate beside the Missouri River. It stayed in the ground for 170 years, until 1913.

This chapter describes the explorers in order, from the footprints at White Sands to the four astronauts who flew around the Moon in April 2026. For the first eight eras, watch for two things. Watch who shows the way: Hopi guides to the Grand Canyon, a Crow map to South Pass, an Ojibwe guide to the source of the Mississippi, four Inuit men at the North Pole. And watch what the explorers do to the people they meet. Some traded with them and were fed by them. Others took captives, burned towns and killed. After 1950 the only places left were the deepest ocean and space, where nobody had ever lived.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How do we know people were living here at least 21,000 years ago?
2. What did the explorers do to the people whose land they crossed?
3. Which famous "firsts" are still in doubt?

Five minutes? Read "Four captains along the coast, 1497 to 1542" in The 1500s.

Can skip: "The first recorded climb of Pikes Peak, 1820", "Alaska, 1867 to 1885", "1898: islands in two oceans", "Mapping the sea floor". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

The first people to explore this land came from Asia during the last ice age. At that time a strip of dry land joined Siberia to Alaska. Human footprints at White Sands, in New Mexico, show that people were here at least 21,000 years ago, and probably about 23,000 years ago. Over thousands of years their descendants spread across the whole continent, into its deserts, mountains, forests, coasts and plains.

Hundreds of Native nations lived here by 1500, speaking many different languages. Nobody knows how many people lived north of Mexico at that time. Estimates run from about 1 million to about 18 million.

How old are the footprints at White Sands?

The footprints at White Sands National Park, in New Mexico, are at least 21,000 years old and probably about 23,000. Scientists dated seeds of a water plant found in the layers just above and just below the prints. Tests on the seeds gave ages of about 21,000 and about 23,000 years.

For a long time scientists had thought people first arrived only about 13,500 to 16,000 years ago. When the seed dates came out in 2021, some scientists questioned them. A water plant can take in old carbon from the rock around it, and that can make its seeds test older than they are.

The second test. In 2023 a team led by Jeff Pigati, a scientist who studies rocks and the earth at the US Geological Survey, tested two other things from the same layers. They dated pollen from pine trees and other trees that grow cones. They also measured how long grains of quartz sand had been buried out of the sunlight.

Tests on the pollen gave about 21,000 to 23,000 years, and tests on the quartz gave at least 21,500 years. Every method put the footprints at about 21,000 to 23,000 years old. The park's own web page on the footprints states that people have lived in North America "for at least 23,000 years."

Trade routes and Pacific voyaging

Native people knew routes across the continent long before any European arrived. Traders carried copper from Lake Superior, shells from the Gulf of Mexico, turquoise from the Southwest and obsidian, a natural glass formed by volcanoes, from the West. Archaeologists, the scientists who dig up and study the things people left behind, have found each of these goods hundreds of miles from where it came from.

The people of Cahokia, a city on the Mississippi River near present-day St. Louis, controlled a hard stone called Mill Creek chert, which could be shaped into tools. Hoe blades made of it turn up more than 400 miles away.

Polynesian sailors, the people of the many islands of the central and southern Pacific, crossed thousands of miles of open ocean to reach the Hawaiian Islands. The archaeologist Patrick Kirch reviewed 150 years of study and put their first arrival and settlement between about the years 1000 and 1200. Older studies had put it between about the years 300 and 750. Scientists still argue about the exact century.

Leif Erikson

Who: A Norse sailor from Greenland, son of Erik the Red, who led one of the first European voyages to North America.

When and where: About the year 1000, and a Norse camp dated to 1021. L'Anse aux Meadows, Newfoundland, Canada.

A storm on the sun in the year 993 dates the Norse camp.

Norse sailors from Greenland, the people often called Vikings, reached North America about 500 years before Columbus. Two sagas, old stories written down in Iceland, tell what happened: the *Saga of the Greenlanders* and the *Saga of Erik the Red*. They say that Leif Erikson, son of Erik the Red, reached a land he called Vinland.

The sagas also describe the people the Norse met there, whom the Norse called Skraelings. According to the sagas, the two groups traded and also fought, and the fights led the Norse to leave. The sagas give the people no other name, and no record shows which Native nation they belonged to.

Finding the camp. A local man, George Decker, led the Norwegian explorer Helge Ingstad to some old mounds near the village of L'Anse aux Meadows in 1960. People there called the place the "old Indian camp." Helge Ingstad and the archaeologist Anne Stine Ingstad dug there for seven seasons, from 1961 to 1968. They found a Norse camp.

Dating it. A storm on the sun in the year 993 left a mark in the rings of trees growing on Earth. In 2021 scientists found that mark in pieces of wood at the camp that had been cut with metal blades, and they counted the rings out to the bark. The Norse cut those trees down in 1021. That is 471 years before Columbus reached the Bahamas.

Estimates of how long the Norse stayed run from a few years to about twenty. Another account says the camp was only a place to repair boats. Native people had lived on the same spot long before the Norse came.

Five earlier groups had lived there, the oldest about 6,000 years ago. A Native people that archaeologists call the Dorset lived there about 300 years before the Norse.

KNOWN:	Norse people cut trees at L'Anse aux Meadows in 1021.
NOT KNOWN:	which place in the sagas the camp was. One answer is Leif's own camp, and another is a second place the sagas call Hop. Nothing dug up at the site has settled it.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 14 MINUTES

This era starts a few years early, in 1492, with Christopher Columbus's first voyage across the Atlantic. In the century after it, European sailors mapped the Atlantic and Gulf coasts, and Spanish armies marched far inland. Every place they reached was already the home of a Native nation. Native guides showed several of them the way, including the way to the Grand Canyon. The explorers also took people captive, seized stored food and attacked towns whose people refused to let them in.

Spanish colonists built St. Augustine, in Florida, inside a Timucua town in 1565.

Christopher Columbus

Who: A sea captain from Genoa, in Italy, who sailed west for Spain looking for a sea route to Asia.

When and where: Four voyages from 1492 to 1504, to the Bahamas, the Caribbean islands and the coasts of Central and South America.

A lookout on the *Pinta* saw land at about 2 in the morning.

Christopher Columbus crossed the Atlantic four times, and he never set foot in what is now the United States. He sailed for the king and queen of Spain, and he believed that by sailing west he could reach Asia.

The lookout was Rodrigo de Triana, and the date was October 12, 1492. The land he saw was an island in the Bahamas. Its people, the Lucayan Taino, called it Guanahani. Columbus named it San Salvador. On that first voyage he took Taino people captive and carried them to Spain to show the king and queen.

The tribute. Columbus's soldiers made war on the Taino of Hispaniola, an island in the Caribbean, in 1494 and 1495. They fought until the Taino surrendered and agreed to pay a tribute, a payment that a conquered people are forced to hand over. Columbus set the rule.

In the Cibao, the part of the island with the gold mines, every Taino aged 14 or older had to bring in a hawk's bell, a small round bell, full of gold dust every six months. A priest named Bartolomé de las Casas wrote that the Spaniards cut off the hands of people who failed to pay. How often Spaniards did this is still argued over.

Sold as slaves. Columbus's men rounded up about 1,500 Taino in February 1495. They shipped about 500 of the strongest to Spain to be sold as slaves, and about 200 of them died on the voyage. Some accounts give 1,600 and 550. Within a few decades most of the Taino on Hispaniola were dead. Spanish soldiers made war on them, Spanish colonists forced them to work, and many died of disease. Counts of how many Taino lived there when Columbus arrived do not agree.

Arrested. The Spanish rulers sent an investigator, Francisco de Bobadilla, to Hispaniola in 1500. Bobadilla arrested Columbus and sent him back to Spain in chains. Bobadilla took statements from 23 witnesses and wrote them into a report. It records that Columbus punished a man who stole corn by having his ears and nose cut off. Then Columbus sold the man as a slave. The report was found again in 2005 in an archive, a building that stores old papers, in the Spanish city of Valladolid.

Columbus died in Valladolid on May 20, 1506. He never said in public that he had reached a new continent. What he believed in private is not settled. The name "America" came from another Italian sailor, Amerigo Vespucci. Vespucci wrote in 1503 that the lands across the Atlantic were a new continent. In 1507 a German mapmaker, Martin Waldseemuller, printed a world map that named the new land America.

Four captains along the coast, 1497 to 1542

Cabot, 1497. John Cabot was an Italian, born Giovanni Caboto, who sailed for King Henry VII of England. His small ship, the *Matthew*, reached the coast of North America on June 24, 1497, with a crew of 18 to 20 men. Where he landed is disputed. Some accounts say Newfoundland and others say Cape Breton Island. Cabot went ashore only once, and he went no farther inland than a crossbow could shoot. King Henry gave him 10 pounds in English money, which was about two years' pay for a laborer.

Ponce de León, 1513. Juan Ponce de León sailed from Puerto Rico in March 1513 with three ships and 65 people. Among them were two free Africans, one of them a man named Juan Garrido, and two enslaved Native people. Ponce de León named the land La Florida. His was the first European expedition to Florida that anyone wrote down. A map made in 1502, called the Cantino map, already shows a shape like Florida, so Spanish slave raiders may have reached it first without leaving a record.

On the Gulf coast the Calusa people of southwest Florida refused to trade. Calusa warriors in canoes drove his ships away. In the fights that followed, his men captured eight Calusa, four men and four women.

Ponce de León came back in 1521 to build a colony on Calusa land. Calusa warriors fought the newcomers, and they wounded him. He died of the wound in Cuba that July. A popular story says that he was searching for a Fountain of Youth. No writing from his own time supports that story, and his contract with the Spanish king says nothing about a fountain.

Verrazzano, 1524. Giovanni da Verrazzano, an Italian sailing for the king of France, followed the Atlantic coast from Florida to what is now New Brunswick, in Canada, in 1524. In New York Bay he met about 30 canoes of Lenape people. In Narragansett Bay, Wampanoag and Narragansett people visited him. Verrazzano wrote to the French king about the voyage himself.

In his letter he described what about twenty of his men did on the peninsula now called Delmarva, south of New York Bay. They found an old woman, a young woman and some children hiding in the grass. He wrote: "We took the boy from the old woman to carry back to France." His men wanted to take the young woman too, he wrote, "but it was impossible to take her to the sea because of the loud cries she uttered."

So his men kidnapped the boy, and they left the young woman behind only because she screamed. The boy was about eight years old. The letter does not give his name or his people, and no known record tells what happened to him in France.

Cartier, 1534. The Frenchman Jacques Cartier explored the St. Lawrence River, in present-day Canada, from 1534 to 1542. In the winter of 1535 to 1536, Native people at the town of Stadacona gave his men a medicine made from evergreen trees, which saved them from scurvy. These people are now called the St. Lawrence Iroquoians. Scurvy is a disease caused by a severe lack of vitamin C, which comes from fresh food. It makes people weak, damages their gums and makes them bleed under the skin.

Cartier kidnapped Donnacona, an Iroquoian chief, in 1536, along with Donnacona's two sons and seven other people. The French never brought any of them home.

Cabeza de Vaca and Estevanico

Who: Two of the four survivors of a Spanish expedition who crossed the Southwest on foot. Álvar Núñez Cabeza de Vaca was a Spanish member of the expedition. Estevanico was an enslaved African man.

When and where: 1528 to 1539. Florida, Texas, the Southwest and northern Mexico.

Four men out of an army of 600 walked from Texas to Mexico City.

Two of the four were Álvar Núñez Cabeza de Vaca and a man the Spanish called Estevanico, or Esteban. The four walked from the coast of Texas to Mexico City between 1534 and 1536.

The army belonged to Pánfilo de Narváez. It left Spain in June 1527 with about 600 men. In April 1528 Narváez landed near Tampa Bay, in Florida, with 300 men. They took over a town of the Apalachee people. Apalachee warriors attacked them there, at first 200 at once, shooting burning arrows into the houses the Spaniards had taken.

Many of the men also fell sick, and many were starving. By September only 242 of them were still alive. By then the survivors had built rafts to try to reach Mexico by sea. In November 1528 a storm washed about 80 of them onto Galveston Island, off the coast of Texas.

Estevanico was born about the year 1500 in Azemmour, in Morocco, in North Africa. Cabeza de Vaca described him as a Black man who spoke Arabic. In 1522 a Spanish nobleman, Andrés Dorantes, bought him as a slave. Dorantes and Estevanico were both on the Narváez expedition, and both lived through it.

On the Texas coast, Native people held the survivors as slaves for years. These people are now called Coahuiltecan. In 1534 the last four escaped: Cabeza de Vaca, Dorantes, Alonso del Castillo Maldonado and Estevanico. They walked west. Along the way they worked as healers among the nations they met. In July 1536 they reached Mexico City. Cabeza de Vaca published an account of the journey in 1542.

After the journey the Spanish viceroy in Mexico, Antonio de Mendoza, bought Estevanico from Dorantes or received him as a gift. A viceroy is the king's governor of a colony. Some evidence suggests that Mendoza freed him.

The last journey. Mendoza sent a friar named Marcos de Niza north in 1539 to look for seven rich cities that people called Cibola, with Estevanico as his guide. A friar is a member of a Catholic religious order. They left Culiacán, in northern Mexico, on March 7, 1539. Estevanico went ahead of the friar with Native guides from Sonora, in northern Mexico.

People of Hawikuh, a Zuni town, killed him when he got there. One Spanish account says he was "shot through with arrows." The Spanish accounts disagree about why. Francisco Vázquez de Coronado, who led a Spanish army to Hawikuh the next year, wrote in 1540 that the people of the town killed Estevanico because he had "killed and assaulted their women." Pedro de Castañeda, who marched with Coronado and wrote an account of the march in the 1560s, wrote that they killed him because he demanded turquoise and women.

Estevanico was not the first African to explore North America. Juan Garrido, a free African who sailed with Ponce de León to Florida in 1513, came before him.

Armies inland: de Soto and Coronado, 1539 to 1542

Two Spanish armies marched deep into the continent between 1539 and 1542.

Hernando de Soto landed in Florida in May 1539 with 620 men and 220 horses. For three years he marched through the Southeast, and his soldiers took the stored food of the towns they passed through. From October 1539 to March 1540 his army lived in Anhaica, the capital of the Apalachee people, where Tallahassee stands today. There his men ate the Apalachee's stored food.

Hostages. De Soto seized dozens of men, women and children to carry his army's supplies. To get his army through, he took local leaders hostage, holding them prisoner so that their people would let the army pass safely. One was the leader of the Coosa people, whom de Soto held until the army had passed the Coosa towns. At a town called Hymahi, de Soto's men took all the food. A National Park Service history states that de Soto burned at least one person from the town to death. He did it while questioning people about a town farther on, called Cofitachequi.

Mabila. De Soto's army reached Mabila on October 18, 1540. The town was most likely in the southern part of present-day Alabama, and it was surrounded by a palisade, a wall of tall upright logs. Its leader was a man the Spanish called Tascalusa. A Spanish officer, Baltasar de Gallegos, decided that a leading man of Mabila had not shown him respect. Rodrigo Ranjel, de Soto's secretary, wrote that Gallegos cut off the man's arm "with a slash."

Then a battle began. De Soto's men fought for about nine hours and set the town on fire with people still inside. One of the expedition's own record keepers wrote: "We killed them all either with fire or the sword." The counts of the dead do not agree. A National Park Service history gives about 2,000 people of Mabila killed and over 200 Spanish. Reports from the University of Alabama give thousands of people of Mabila killed, and about 20 Spanish killed and 250 wounded. The site of Mabila has never been found.

De Soto died of fever beside the Mississippi River on May 21, 1542. His men sank his body in the river. They did not want the nations they had fought to learn that he could die. The survivors reached Mexico in September 1543.

Coronado. Francisco Vázquez de Coronado left Compostela, in Mexico, in February 1540. He led about 400 European soldiers, 1,300 to 2,000 Native allies from Mexico and four friars. They were looking for cities of gold.

Close to starving, they reached the Zuni town of Hawikuh in July 1540 and demanded to be let in. The Zuni had drawn lines of sacred cornmeal on the ground and warned the Spaniards not to cross them. When the Spaniards crossed, the Zuni shot arrows. Coronado's men attacked the town, and the Zuni left it after a short fight.

Coronado sent out scouting parties from there. Pedro de Tovar's men forced their way into a Hopi town whose people had refused to let them in. Hopi guides then led García López de Cárdenas and his men to the South Rim of the Grand Canyon. They were the first Europeans to see it.

The Tiguex war. The army camped along the Rio Grande among the Tiguex pueblos in the winter of 1540 to 1541. A pueblo is a town of connected houses built of stone or dried mud. Coronado's men took over the pueblo of Coofor, near present-day Bernalillo, New Mexico, and forced its people out.

A National Park Service history states that the fighting began after "a Spanish soldier assaulted a Pueblo woman, wife to a Tiguex man." Pedro de Castañeda, who was on the march, wrote down what happened about twenty years later. A Spaniard told the woman's husband to hold his horse, went up into the house, and raped her or tried to rape her. Rape means forcing a person to have sex.

Castañeda would not name the man. A later witness, Juan de Contreras, named him in testimony as Juan de Villegas, a soldier on horseback. The husband and the town's leaders came to complain and pointed out the horse. The soldier denied it, and nobody punished him.

In the war that followed, Coronado's soldiers killed hundreds of Tiguex people and burned their villages. At a pueblo called Arenal, the defenders made the sign of peace. Two Spaniards, Pablo de Melgosa and Diego López, answered

with the same sign. The defenders put down their weapons, and the Spaniards promised not to punish them.

Arenal. Spanish soldiers then took the captives to Cárdenas's tent. Coronado had ordered Cárdenas not to take any of them alive. Cárdenas had 200 stakes set up to burn the captives alive.

When the soldiers began tying them to the stakes and burning them, about a hundred men in the tent fought back. Spanish soldiers on foot attacked the tent, and horsemen chased down the men who ran. Castañeda wrote that not one of them was left alive, except a few who hid. Those few spread the news that the Spaniards did not keep their promises of peace. Coronado's men also made Tiguex women and children slaves.

Quivira. A Native guide the Spanish called "the Turk," probably a Wichita or Pawnee man, led the army out onto the Great Plains in the spring of 1541. He told them of a rich land called Quivira. At Quivira, in present-day central Kansas, Coronado found the grass houses of Wichita towns and no gold. Before he left, Coronado ordered the Turk garroted. That means he was strangled to death with a cord or an iron collar that was tightened around his neck. The army went back to Mexico in 1542.

Spanish officials later charged Coronado and Cárdenas with crimes against Native people on the march. Coronado's friends sat on the court in Mexico that heard his case, and they cleared him. Members of the Council of the Indies, the king's council in Spain for the American colonies, found Cárdenas guilty.

St. Augustine, 1565

Pedro Menéndez de Avilés and about 800 colonists landed on the coast of Florida in September 1565. The colonists included 26 women and an unknown number of enslaved Africans. They settled at the town of a Timucua chief named Seloy. They used Seloy's council house, the building where the town's leaders met, as their first fort. That same month Menéndez's men destroyed Fort Caroline, a French settlement nearby in Timucua country. At Matanzas Inlet they killed more than a hundred Frenchmen who had surrendered.

The Timucua soon wanted the Spaniards gone from their town. Timucua warriors shot flaming arrows at the fort in 1566. Nine months after they arrived, the Spaniards moved their settlement across the bay to Anastasia Island.

BY 1600

If you remember one thing: Sailors mapped the coasts, and the two Spanish armies that marched inland took the stored food of the towns they passed and killed people who fought back and people who had surrendered.

- 1492: Columbus reaches the Bahamas. He never sees the mainland of what is now the United States.
- 1540: Hopi guides lead Cárdenas's men to the Grand Canyon. That winter Coronado's soldiers burn the Tiguex captives at Arenal.
- 1565: Spanish colonists build St. Augustine inside the Timucua town of Seloy.

Next: In 1609 a Frenchman, Samuel de Champlain, went to war beside his Wendat, Algonquin and Montagnais allies.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 7 MINUTES

European colonists built small settlements far apart from each other in the 1600s. The Spanish had held St. Augustine, in Florida, since 1565. English colonists built James Fort, at Jamestown in Virginia, in 1607, and the Spanish governor Pedro de Peralta founded Santa Fe, in New Mexico, in about 1610. French colonists settled on the St. Lawrence River, in present-day Canada. On Manhattan Island, the homeland of the Lenape people, Dutch colonists built New Amsterdam. The land between these settlements was the home of Native nations.

The explorers of this century traveled by water, in ships and in canoes. Native guides and hosts told several of them where the rivers went.

Champlain and the lake, 1609

Samuel de Champlain, a Frenchman, explored the coast of New England as far south as Cape Cod in 1605 and 1606. In 1608 he founded Quebec, on the St. Lawrence River in present-day Canada. That first winter, 16 of his 25 men died of scurvy.

Champlain went south in June 1609 with two other Frenchmen and a party of Wendat, Algonquin and Montagnais warriors. The Wendat are also called the Huron. They took him to the lake now named for him, between present-day New York and Vermont. In late July, at Ticonderoga, they met a party of Haudenosaunee men. The Haudenosaunee are also called the Iroquois. In the fight that followed, according to the historian Marcel Trudel, Champlain killed two of the Haudenosaunee men.

Henry Hudson

Who: An English sea captain hired by the Dutch East India Company to look for a sea route to Asia.

When and where: 1609 to 1611. The Hudson River in New York, then Hudson Bay in Canada.

The ship's mate wrote down every killing on the river.

Henry Hudson sailed the Dutch ship *Half Moon* up a river in what is now New York in September 1609. The Lenape called it Muhheakantuck, and the Mohicans called it Mahicannittuk. Today it is called the Hudson River. Hudson was not the first European in its harbor. Verrazzano had been there in 1524. The Lenape and Munsee people lived along the lower river, and the Mohicans lived along the upper river.

Hudson's mate, the officer just below the captain, was Robert Juet, and he kept a journal of the voyage. It records what happened on the river. Two canoes carrying 26 men attacked the ship's small boat on September 6. An arrow hit a sailor named John Colman in the throat and killed him. Two other sailors were hurt.

Three days later the crew seized two Native men who had come aboard. They dressed them in red coats to keep them on the ship. On the morning of September 15 the two men climbed out through a port, a small opening in the ship's side, and swam away. Once the ship was under sail, Juet wrote, "they called to us in scorn."

The way back. On the way back down the river, Hudson's crew killed about 10 to 12 Native men. Juet's journal records each of these killings, on October 1 and 2.

A man in a canoe climbed in through the cabin window on October 1. He took a pillow, two shirts and two belts for carrying bullets. The ship's mate shot him in the chest and killed him. Another man swam up and grabbed the side of the ship's small boat. The ship's cook cut off one of the man's hands with a sword, and the man drowned.

The next day, men in two canoes shot arrows at the ship. The crew fired their guns and killed two or three of them. About a hundred men on a point of land shot at the ship too. Juet fired a small cannon at them and killed two. He fired again at a canoe with nine or ten men in it and killed one, and the crew shot three or four more.

Then the ship anchored off the shore that Juet called "Manna-hata." Juet did not write down which nation the dead men belonged to.

The mutiny. Hudson sailed for England on his next voyage, in a ship called the *Discovery*. He searched for a northern sea passage in the great bay in Canada now named for him. The crew spent the winter in James Bay. In June 1611 most of the crew mutinied, which means they rose up against their captain. They put Hudson, his teenage son John and seven sick or loyal men into a small open boat and sailed away. No European saw any of them again.

A survivor, Abacuk Pricket, named Henry Greene and Robert Juet as the leaders. The men Pricket blamed were already dead and could not answer him, so his story cannot be checked. Only eight of the thirteen mutineers lived to reach England. English courts put some of them on trial, and none was punished.

Marquette and Jolliet

Who: Jacques Marquette, a French Jesuit priest, and Louis Jolliet, a fur trader born near Quebec. The Jesuits are a group of priests within the Catholic Church.

When and where: May to August 1673. From St. Ignace, Michigan, down the Mississippi River to the mouth of the Arkansas River.

They turned back 435 miles short of the Gulf.

Jacques Marquette and Louis Jolliet paddled down the Mississippi River in 1673 far enough to learn that it flows to the Gulf of Mexico and not to the Pacific Ocean. Native people had told them about the river first. Illinois people had described it to Marquette and invited him to visit.

The two men left St. Ignace, in present-day Michigan, on May 17, 1673. They had two canoes and five paddlers of French and Native ancestry. On the river, Michigamea warriors almost attacked them. The town's elders stopped the attack and fed the travelers. Farther south the Quapaw people hosted them for two nights near the mouth of the Arkansas River.

They turned back on July 17, about 435 miles from the Gulf. They feared the Spanish. They had also been told that the nations farther south were well armed and would attack anyone who got in the way of their trade.

On the way home they took a shorter route up the Illinois River, which their Native hosts told them about. In August they reached the mission of St. Francis Xavier, a church post where priests worked to turn Native people to the Catholic faith, at Green Bay in present-day Wisconsin.

Near Montreal, Jolliet's canoe tipped over in the Lachine Rapids, and his notes were lost. Marquette's account became the main record of the journey. Marquette died of dysentery on May 18, 1675, in present-day Michigan. Dysentery is an infection of the gut that causes severe, bloody diarrhea.

Robert de La Salle

Who: A French fur trader and explorer who claimed the whole Mississippi River valley for the king of France.

When and where: 1682 to 1687. The lower Mississippi River and the coast of Texas.

One of his own men shot him from an ambush in Texas.

Robert de La Salle was the first European known to follow the Mississippi River all the way down to the sea. In February 1682 his party came down the Illinois River to the Mississippi, built canoes there, and paddled down to the Gulf of Mexico. On the way, his men built a small fort near present-day Memphis, Tennessee.

The claim. At the mouth of the river, on April 9, 1682, La Salle claimed the whole Mississippi River basin for France. A river basin is all the land whose rain drains into one river. He named it Louisiana, after King Louis XIV. Many Native nations lived in that basin. La Salle left every one of them out of his claim. In 1803 French officials sold the claim that began that day to the United States.

The lost colony. La Salle sailed from France in July 1684 with four ships and 300 colonists to build a colony at the river's mouth. The ships sailed past the river's mouth and landed at Matagorda Bay, in Texas, on the land of the Karankawa people. Before they got there, Spanish sea raiders in the Caribbean had seized one ship with its load of supplies and tools.

In Matagorda Bay a second ship ran onto a sandbar and stuck there. The commander of the third, the escort ship, sailed it back to France. That left only *La Belle*, and it later ran aground in the bay. The colonists built their settlement on Garcitas Creek, near present-day Victoria, Texas.

One of La Salle's own men, Pierre Duhaut, killed him from an ambush on March 19, 1687, near present-day Navasota, Texas. Jean L'Archeveque had drawn La Salle into talking with him so that Duhaut could strike. Later a third member of the party shot Duhaut. In 1688 Karankawa-

speaking people killed the 20 adults still at the colony and took five children captive. The wreck of *La Belle* was found in Matagorda Bay in 1995.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

Men of the Russian navy reached the coast of Alaska in 1741, and four Frenchmen, two of them sons of a fur trader, crossed the northern Great Plains in 1742 and 1743 with Mandan guides. They may have been the first Europeans to do so. English colonists did little exploring beyond the Appalachian Mountains between 1700 and 1750. The journeys into country that no European had mapped came from outside the English colonies.

Vitus Bering

Who: A Danish-born officer in the Russian navy who led the Russian voyage that reached Alaska in 1741.

When and where: 1728 and 1741. The Bering Strait and the coast of Alaska.

In 1991 scientists dug up Bering's bones to learn how he died.

Vitus Bering led the Russian voyage that reached Alaska in 1741. On his first voyage, in August 1728, he sailed north through the narrow strip of sea between Asia and North America. It is now called the Bering Strait.

Before the second voyage, Native people in Kamchatka, in eastern Siberia, killed several Russians under Bering's command. Bering sent a group of armed men north to find the killers, with orders not to use force if they could avoid it. The armed men killed several Koryak people in the settlement of Utkolotsk and made the rest slaves.

The Russians tortured Koryak people, hurting them on purpose to make them give up the killers. The naturalist on the voyage, Georg Steller, was horrified to see it. A naturalist is a scientist who studies plants and

animals. The report of what Steller saw does not say how the Russians tortured them.

Alaska. Bering commanded the ship *St. Peter* in 1741. On July 16 his crew sighted Mount Saint Elias on the Alaska coast, and they landed briefly at or near Kayak Island. That coast was Native land. Steller described animals that European scientists had not recorded before, among them a bird now called Steller's jay and a sea animal now called Steller's sea cow.

Bering's second ship, the *St. Paul*, was under the command of Aleksei Chirikov. On July 15, 1741, Chirikov's crew sighted land in southeast Alaska, the homeland of the Tlingit people. Two days later he sent an officer, Abram Dementiev, and 10 armed men ashore in a boat to look for a safe place to anchor. After a week they had not come back. Chirikov sent another officer, Saveliev, and a few men in his second, smaller boat. They did not come back either.

Fifteen men lost. With no boats left, Chirikov could not search for them or land, and on July 27 he sailed west without them.

NOT KNOWN: what happened to the 15 men. People living on that shore may have killed them, they may have drowned in the strong tides along that coast, or, as one study argues, they went to live among the Tlingit.

Bering's death. On the way home, scurvy spread through the crew of the *St. Peter*. The ship was wrecked on an island near Kamchatka, now called Bering Island. Bering died there on December 19, 1741. About 28 of his men also died. At the time, people blamed scurvy for all of these deaths. A team of Russian and Danish scientists dug up Bering's remains in 1991 and studied his teeth and bones. They concluded that his own death was more likely caused by heart failure.

The lead plate: the La Verendrye brothers on the northern Plains, 1742 to 1743

Two French brothers and two companions buried a lead plate beside the Missouri River on March 30, 1743, at the fort of a nation the French called the Little Cherry People. The plate is six by eight inches. One side has words in Latin stamped into it that name King Louis XV, the brothers' father and the year 1741. On the other side someone scratched the date and the names of the men who placed it. The plate was found at Pierre, South Dakota, in 1913. It is kept in the South Dakota Cultural Heritage Center.

Their father, Pierre La Verendrye, was a French fur trader in what is now Canada. Two of his sons and two other Frenchmen left Fort La Reine, at present-day Portage la Prairie, Manitoba, on April 29, 1742. The son who kept the journal was known by the French title the Chevalier, which means knight. He was most likely Louis-Joseph La Verendrye. They reached the Mandan towns on the upper Missouri River on May 19 and waited there two months. On July 23 they set out west with two Mandan guides.

For months they traveled with Plains nations that the French called the Horse People, the Bow People and the Little Cherry People. Which modern nations these were is not agreed. The Bow People may have been Pawnee, and the Little Cherry People were probably Arikara.

Some of the nations they met were at war. The Horse People told the brothers that two years before, a nation the French called the Snake People had destroyed seventeen of the Horse People's villages. The Snake People had killed the men and the old women and taken the young women to sell. Which nation the Snake People were is not certain.

The brothers may have been the first Europeans to see the Rocky Mountains, probably in what is now Wyoming. Their journal is hard to follow, so nobody knows their route for certain. It lay in French archives until it was found in 1851.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 8 MINUTES

American and British diplomats signed a peace treaty in Paris on September 3, 1783. Diplomats are officials who speak for their countries, and a treaty is a written agreement between them. In the treaty the diplomats agreed that the western border of the new United States would be the Mississippi River. Native nations lived on the land between the Appalachian Mountains and that river. The diplomats did not invite them to the talks.

Before and after that treaty, colonists from Virginia and North Carolina pushed through the Appalachians into Kentucky, which the Shawnee used as a hunting ground. In the Pacific, three voyages reached places no European was known to have reached before. Spanish scouts saw San Francisco Bay in 1769. The British navy captain James Cook reached the Hawaiian Islands in 1778. In 1792 an American fur trader, Robert Gray, sailed into the mouth of the Columbia River.

Daniel Boone

Who: A hunter who marked Boone's Trace, later called the Wilderness Road, through the Cumberland Gap into Kentucky.
When and where: 1767 to 1778. The Appalachian Mountains and Kentucky.

The Shawnee took his skins and told him never to come back.

Daniel Boone marked the road that colonists used to cross the Appalachian Mountains into Kentucky. He and his brother Squire first crossed into Kentucky in 1767. In May 1769 Boone set out with five other men on a hunting and trapping trip that lasted two years. One of them, John Findley, had told him about the Cumberland Gap, a low pass through the Appalachians. Native nations had used the gap long before any colonist, and in 1750 a Virginia doctor named Thomas Walker had told colonists about it.

Kentucky was the hunting ground of the Shawnee. The Cherokee and the Haudenosaunee also claimed it. In a treaty signed at Fort Stanwix in 1768, Haudenosaunee leaders gave up their claim to Kentucky to the British. The Shawnee did not sign that treaty. In 1769 Native men chased away Joseph Martin and the men at the fort he had built near the gap for people who claimed land there.

A Shawnee party caught Boone and his brother-in-law, John Stewart, on December 22, 1769. The Shawnee took all their skins and told them to leave and never come back.

The attack of 1773. Boone led his family and about 50 other people toward Kentucky in 1773 to settle there. On October 9 a small group left the main party to get supplies. It included Boone's oldest son, James, several white men and two enslaved Black men named Charles and Adam. A band of Delaware, Shawnee and Cherokee men attacked them. The historian John Mack Faragher writes that the band had decided "to send a message of their opposition to settlement."

The attackers shot James Boone, who was 16 or 17, and Henry Russell through the hips so they could not run. The historians Michael Lofaro and Nancy O'Malley describe what the attackers did next. They tortured the two young men, slashing them with knives and tearing out their fingernails and toenails. Then they killed them. They captured Charles. Adam hid in a pile of driftwood and saw what happened. The families turned back.

Virginians fought the Shawnee in 1774 in a war called Dunmore's War. After the Virginians won a battle at Point Pleasant, Shawnee leaders gave up their claims to Kentucky. Then Richard Henderson, a judge from North Carolina, hired Boone to help start a colony in Kentucky, which Henderson called Transylvania.

The sale. Boone invited leaders of Cherokee towns to a meeting at Sycamore Shoals in March 1775, where Henderson bought the Cherokee claim to Kentucky. Three Cherokee leaders, Attakullakulla, Oconostota and Sewanooko, signed the deed, the paper that handed over the land. It is dated March 17. It covered about 20 million acres. Henderson paid in trade goods, which one estimate values at about 10,000 British pounds.

The sale broke the law. A British royal order of 1763 and the laws of Virginia and North Carolina all forbade private buyers to buy Native land. A Cherokee leader named Dragging Canoe refused to agree to the sale. He warned that Kentucky "is bloody ground, and will be dark and difficult to settle."

The road. Boone then led a work party that cut a trail through the Cumberland Gap into central Kentucky. It was later called the Wilderness Road. The work party included Sam, an enslaved Black man, and other enslaved workers. Near present-day Richmond, Kentucky, Native men attacked the camp and killed Sam and his owner. Boone built a settlement called Boonesborough on the Kentucky River.

The girls in the canoe. Three teenage girls were floating in a canoe on the Kentucky River outside Boonesborough on July 14, 1776. Two Cherokee and three Shawnee warriors, led by the Cherokee Hanging

> Maw, captured them. The girls were Boone's daughter Jemima and Elizabeth and Frances Callaway. Boone's men caught up two days later, as the warriors were building a fire for breakfast, and freed the girls. In 1778 Shawnee warriors captured Boone himself. He lived among the Shawnee for months before he escaped.

Who saw San Francisco Bay first? The Pacific, 1769 to 1792

California, 1769. Scouts under a Spanish sergeant, José Francisco Ortega, were the first Europeans known to see San Francisco Bay. A Spanish official, José de Gálvez, had sent an expedition north from San Diego in 1769 to survey California and to claim it for Spain. Native nations already lived there. The commander was Gaspar de Portolá. His party of 74 men included soldiers, an engineer, two friars and Christian Native men from Baja California. They walked past Monterey Bay without recognizing it.

In early November 1769 Ortega led a group of scouts over Montara Mountain, south of present-day San Francisco. A huge bay blocked their way north. Portolá, their commander, was not with the scouting party.

Portolá and his officers agreed to turn back on November 11. Members of the expedition reported that most of their meetings with Native people along the way were peaceful, and that people gave the Spaniards directions and food. On November 30 about a dozen Rumsen people brought the starving soldiers pinole, which is flour made from ground seeds.

Hawaii, 1778. The British navy captain James Cook reached the Hawaiian Islands on January 18, 1778. He and his crews were the first Europeans known to see them. They landed at Waimea, on the island of Kauai, and on Niihau. While Cook's men were ashore collecting supplies, one of them, John Williamson, shot and killed a Hawaiian man. Cook's crews were also the first Europeans to bring some diseases to the islands.

Cook came back in late November 1778. At Kealakekua Bay, on the island of Hawaii, about 10,000 people and 1,000 canoes met the ships, and the high chief Kalaniopuu welcomed him.

Cook's death. Hawaiians took one of the ships' boats on February 13, 1779. The next day Cook went ashore to kidnap Kalaniopuu and hold him until the boat was returned. Crowds gathered. Cook fired a warning shot, then shot one Hawaiian man dead. In the fight that followed, Hawaiians killed Cook and four marines, the soldiers who served on his ships. Cook's men killed seventeen Hawaiians. The accounts of Cook's last moments contradict each other.

The Columbia, 1792. Robert Gray was a sea captain from Rhode Island. In 1787 a group of Boston merchants sent him and another captain, John Kendrick, to the Pacific Northwest. The two were to buy sea otter furs and sell them in China. In 1790 Gray finished the first voyage by an American ship around the world.

On his first trip, in August 1788, his men fought the Tillamook people at Tillamook Bay, in Oregon. On August 16 Marcus Lopez, Gray's Black cabin boy and cook from the Cape Verde Islands, saw that a Tillamook man had taken his cutlass, a short curved sword. Lopez chased the man. Armed Tillamook men surrounded Lopez and killed him. Gray named the bay Murderer's Harbor.

Gray attacked Native villages along the coast in 1792, both before and after he found the Columbia. In April 1792 he ordered his men to destroy Opitsaht, a Nuu-chah-nulth village of about 200 houses on Vancouver Island, in present-day Canada. The village was empty at the time. One of his officers, John Boit, called it "a fine village, the Work of Ages," and wrote that Gray "had let his passions go too far."

In May he ordered an attack on another Nuu-chah-nulth village, and his men killed seven people and took their sea otter furs.

The river. Gray's crew found a safe channel over the sandbars at the mouth of a large river, where Chinook and Clatsop people lived. It was the evening of May 11, 1792. His ship, the *Columbia Rediviva*, sailed in. Gray named the river the Columbia, after his ship. He was the first navigator known to enter it who was not Native.

Gray fired on a group of Chinook people at Grays Harbor, in present-day Washington State, later in 1792, and killed 20. Back in Clayoquot Sound, near Opitsaht, a war canoe came toward his ship at night, and Gray killed or

wounded at least 25 of the men in it. In 2005 Gray's descendants formally apologized for the destruction of Opitsaht.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 27 MINUTES

The deal. American diplomats in Paris, the officials who spoke for their country, agreed to pay France $15 million for the right to take Louisiana from the Native nations who lived on most of it. The date was April 30, 1803. American officials could now take that land by a treaty, a written agreement between nations, or by war, and no European government could claim it ahead of them.

Louisiana covered about 828,000 square miles, most of the Mississippi's river basin west of the river. French officials controlled only a small part of it. Americans call the deal the Louisiana Purchase, and with it the land the United States claimed nearly doubled.

Over the next fifty years Army officers, fur trappers and a Navy squadron, a group of warships, crossed and mapped lands where Native nations already lived. Native guides showed several of them the way.

Crow people drew a map that showed trappers the way over South Pass. Two men who had run away from Spanish missions, church settlements run by Catholic priests, led a party across the Mojave Desert. An Ojibwe guide led an American to the source of the Mississippi. Some of these explorers killed the people whose land they crossed.

Meriwether Lewis and William Clark

Who: Two Army officers sent by President Thomas Jefferson to cross the Louisiana Purchase to the Pacific Ocean.

When and where: May 1804 to September 1806. From near St. Louis up the Missouri River, over the Rocky Mountains and down the Columbia River to the Pacific coast of Oregon.

Movie: *The Far Horizons* (1955). It invents a romance between Sacagawea and Clark, and a rivalry between the captains.

Every man but York came home to double pay and 320 acres.

Meriwether Lewis and William Clark led an Army expedition from near St. Louis, through the new Louisiana Purchase and beyond it, to the Pacific Ocean and back. President Thomas Jefferson gave them written orders. They were to find "the most direct & practicable water communication across this continent for the purpose of commerce." Jefferson wanted a river route that traders could use to cross the continent. Members of Congress voted $2,324 for supplies.

Jefferson also ordered them to tell the Native nations they met that their land now belonged to the United States. They were to say that the president in Washington, whom they called "their great father," now ruled over them. Lewis and Clark carried silver peace medals with Jefferson's face on them. They carried an air rifle to show off American weapons.

The doctrine of discovery. American officials used the expedition's maps, medals and flags as legal steps to claim each nation's land. They relied on a rule in American law called the doctrine of discovery. Under it, a land that no European had known before belonged to the European government whose explorers found it first, and no other European government could claim it. No one asked the Native nations who already lived there. La Salle had made the same kind of claim for France in 1682. When American diplomats bought Louisiana, they bought France's claim.

The journey. Clark and about 30 men left Camp Dubois, in Illinois, near St. Louis, on May 14, 1804. Lewis joined them at St. Charles, Missouri. In all the party had as many as 45 members, among them York, a man Clark held as a slave. Sergeant Charles Floyd died on August 20, 1804, probably of appendicitis, an infection of a small part of the gut. He was the only member of the party to die on the journey.

Near present-day Fort Pierre, South Dakota, Lakota chiefs, among them Black Buffalo, stopped the boats on September 25, 1804, and demanded gifts before they would let them pass. Lakota warriors took hold of a boat and an oar. Lewis ordered his men to show their guns, and Clark drew his sword. Black Buffalo ordered his warriors back, and nobody was killed. In his journal Clark called the Lakota the "vilest miscreants," meaning the worst villains, and in the same line he used an insulting word for Native people.

The party spent the winter of 1804 to 1805 at Fort Mandan, beside the Mandan and Hidatsa towns in present-day North Dakota. There Lewis and Clark, the expedition's two captains, hired a French-Canadian trader, Toussaint Charbonneau. They also met Sacagawea, a young Shoshone woman he held as his wife.

In the spring they went on up the Missouri to the streams where it begins. They crossed the Continental Divide at Lemhi Pass. The Continental Divide is the line of high ground that separates rivers flowing east from rivers flowing west. They crossed the Bitterroot Mountains at Lolo Pass and then went down the Clearwater, Snake and Columbia rivers by canoe. They found no water route across the continent. What they found instead was a trail that Native people used.

They first saw the Pacific on November 7, 1805. On November 24 Clark wrote down every member's choice for a place to spend the winter, including the choices of York and of Sacagawea. Accounts differ about whether this counts as a vote. The party built Fort Clatsop near present-day Astoria, Oregon. Food ran short that winter, and they had too little money to buy enough food from the Native nations nearby.

On the way home the party split up. Lewis took three men: George Drouillard and the brothers Joseph and Reubin Field. On the night of July 26, 1806, the four camped with some Blackfeet men on the Two Medicine River in present-day Montana.

The only killings. At dawn the Blackfeet men took the Americans' guns and tried to drive off their horses. Reubin Field stabbed one of the Blackfeet men to death. Lewis shot another man in the belly. The National Park Service says two or three Blackfeet were killed. Lewis wrote down no names for them. He left a peace medal around the neck of a dead man "that they might be informed who we were." It was the only fight on the expedition in which anyone was killed.

The expedition reached St. Louis on September 23, 1806. The captains had met more than 70 Native nations and drawn about 140 maps. Every man in the party except York received double pay and 320 acres of land.

Sacagawea

Who: A Lemhi Shoshone woman who interpreted for Lewis and Clark while carrying her baby son.
When and where: 1805 to 1806. From Fort Mandan in North Dakota to the Pacific and back.

Her brother turned out to be the Shoshone leader with the horses.

Sacagawea spoke for the expedition with her own Shoshone people. She was its interpreter, the person who put one language into another. She was born about 1788 near present-day Salmon, Idaho, among the Lemhi Shoshone. Most accounts say she was Shoshone. A Hidatsa account written in 2021, *Our Story of Eagle Woman*, states that she was Hidatsa by birth.

Hidatsa raiders captured her and several other children in 1800, when she was about 12. The raiders killed four Shoshone men, four women and several boys. She was held at a Hidatsa village in present-day North

Dakota. When she was about 13, Toussaint Charbonneau, a French-Canadian trader, took her as his wife. She did not agree to it. Accounts say he either bought her and another Shoshone girl, Otter Woman, from the Hidatsa, or won Sacagawea gambling. Records of the expedition describe Charbonneau as her husband and her owner.

Her son, Jean Baptiste, was born at Fort Mandan on February 11, 1805. Lewis gave her crushed rattlesnake rattle to speed the birth. She carried her son on the whole journey.

The party met a band of Shoshone in August 1805. Its leader, Cameahwait, turned out to be her brother. Sacagawea interpreted, and the Shoshone traded horses to the party for the crossing of the mountains. She also met a woman who had been taken prisoner at the same time as she was.

She was mainly the expedition's interpreter, not its guide. On the way home she did show the way in country she knew. On July 13, 1806, she told Clark to cross into the valley of the Yellowstone River through the pass now called Bozeman Pass, in Montana. Clark wrote that she "has been of great service to me as a pilot through this Country."

In the journals the men called her "the wife of our interpreter" and "the Snake woman." Snake was their name for the Shoshone. They also called her by an insulting word for a Native woman.

Her death. A clerk at a fur post on the Missouri River, John Luttig, wrote that "the wife of Charbonneau" died of a fever on December 20, 1812, at about age 25. The post was in present-day South Dakota. The next year William Clark became the legal guardian of her son and her daughter, Lizette, which meant the law made him responsible for the children. That is the strongest record, and most accounts accept it.

KNOWN:	a clerk wrote down her death in December 1812, and Clark took legal charge of her children in 1813.
DISPUTED:	Shoshone tradition holds that she lived until 1884 and died on the Wind River Reservation in Wyoming, land the US government had set aside for the Shoshone. The Hidatsa account also disputes the 1812 record.

York

Who: A Black man held as a slave by William Clark, who took him on the Lewis and Clark expedition.

When and where: 1804 to about 1815. The journey to the Pacific, then Kentucky.

He asked Clark for his freedom, and Clark had him whipped.

York crossed the continent with Lewis and Clark as a slave, and he came home still a slave. He was born into slavery in Caroline County, Virginia. His parents, Old York and Rose, were enslaved by John Clark III. In 1799 William Clark inherited York under his father's will. Before the expedition York married a woman who was enslaved by a different owner, near Louisville, Kentucky.

On the journey York hunted with a gun. Enslaved men were usually not allowed to carry one. On November 24, 1805, Clark wrote down York's choice for the winter camp alongside everyone else's.

After the expedition every other man received double pay and 320 acres of land. York received nothing. He asked Clark for his freedom because of his services on the journey. He also asked to be hired out near his wife in Louisville, or to be sold to someone there. Clark refused, and he wanted York hired out to "a Severe Master."

In 1808 Clark wrote that York had "Such a notion about freedom and his emence Services." He meant that York believed his great services on the journey had earned him his freedom. Clark did not agree. In July 1809

Clark whipped York, giving him what Clark called "a Severe trouncing," and had him put in jail.

Clark rented York out to work for other people as a wagon driver in Louisville. The last record of him in that work is from 1815. In 1811 his wife's owner planned to move to Natchez, Mississippi, far to the south, and York probably never saw her again. The National Park Service states that York was freed sometime after 1815. No paper freeing him has been found. Nobody knows for certain when or how he died.

President Bill Clinton made York an honorary sergeant in the US Army in 2001.

Zebulon Pike

Who: A US Army lieutenant sent to explore the northern and southern edges of the Louisiana Purchase.
When and where: 1805 to 1807. Minnesota, then Colorado, New Mexico and northern Mexico.

Spanish soldiers arrested him and took his papers.

Zebulon Pike led two Army expeditions to the edges of the Louisiana Purchase. His commander, General James Wilkinson, sent him out. Pike left St. Louis on August 9, 1805, to find where the Mississippi River begins. On September 21, 1805, at the mouth of the Minnesota River, he bought land from the Dakota people for a future Army fort. He named Cass Lake, in present-day Minnesota, as the river's source, but the true source is a lake farther up. An Ojibwe guide showed it to an American in 1832.

Wilkinson sent him west in 1806 to find where the Arkansas and Red rivers begin, near the Spanish lands of New Mexico and Texas. In November 1806 Pike's men tried to climb the tall peak in present-day Colorado that later took his name. They turned back in snow up to their waists. They had gone almost two days without food.

Pike's party crossed into Spanish land. No one knows whether he got lost or went there on purpose. On February 26, 1807, Spanish soldiers arrested Pike and some of his men. The soldiers took them to Santa Fe and then to Chihuahua, in Mexico. There the Spanish commander, Nemesio Salcedo, questioned Pike and took his papers. Spanish officials released Pike at the Louisiana border on July 1, 1807. Accounts still differ about whether Pike was an explorer or a spy for Wilkinson.

Pike was killed on April 27, 1813, in the War of 1812, during an American attack on the town of York, now Toronto, in Canada. By then he was a general. As British soldiers pulled back, they blew up their store of gunpowder, and the blast killed Pike and 37 other American soldiers.

John Colter

Who: A soldier of the Lewis and Clark expedition who stayed in the mountains to trap.
When and where: 1806 to 1809. Montana and Wyoming.

He ran five miles naked with the Blackfeet behind him.

John Colter walked alone through the country that is now Yellowstone National Park in the winter of 1807 to 1808. He was the first person of European descent known to see it. He had been a member of the Lewis and Clark expedition. On August 13, 1806, at the Mandan towns, the captains let him leave early to guide two trappers back up the Missouri River.

Colter joined a fur party led by the trader Manuel Lisa in 1807. In October 1807 Lisa sent him out alone to find the Crow people and ask them to trade. That winter Colter walked more than 500 miles through the country that is now Yellowstone and Grand Teton national parks. He came back in March or April 1808. Few people believed his reports of hot springs, and they called the place "Colter's Hell."

Colter fought beside Flathead and Crow men against the Blackfeet in 1808, and he was wounded in the leg. Many retellings blame Lewis's fight of 1806 for the Blackfeet's anger at Americans, and that link is disputed.

The run. He and another trapper, John Potts, were paddling a canoe on the Jefferson River, in present-day Montana, in 1808 or 1809. Several hundred Blackfeet men ordered them ashore. Colter came ashore, and the Blackfeet took his weapons and stripped him naked. Potts refused to come ashore. Blackfeet men shot and wounded him. He shot one of them, and then they killed him with arrows.

The Blackfeet told Colter to run, and they chased him. He ran about five miles to the Madison River. He killed the one man who caught up with him, using the man's own spear. Then he hid in a beaver lodge and walked for eleven days to a trading fort. All of this comes from Colter's own telling, as other people retold it later.

Colter died of jaundice, an illness that turns the skin and eyes yellow. One record gives the date as May 7, 1812, and another gives November 22, 1813.

BY 1810: Lewis and Clark were home with about 140 maps, Pike was back from Mexico, and few people believed Colter's reports of hot springs in the Yellowstone country.

The first recorded climb of Pikes Peak, 1820

Three men from an Army expedition led by Major Stephen Long reached the top of the mountain now called Pikes Peak on July 14, 1820. One of them was Edwin James, a 22-year-old naturalist, a scientist who studies plants and animals. It is the first recorded climb to the top. The Ute people call the mountain Tava-kaavi. Their tradition says that Ute ancestors were the first to climb it. Long called the high plains a "Great American Desert."

Trappers and the fur trade, 1822 to 1840

Two fur traders, William Ashley and Andrew Henry, put an advertisement in a St. Louis newspaper in 1822. They wanted one hundred "enterprising young men," meaning bold young men ready for hard work, to go up the Missouri River to its source and work there for one, two or three years. They paid $200 a year. The men who answered became known as Ashley's Hundred. Jedediah Smith, Jim Bridger and Hugh Glass were among the men who joined Ashley's company.

The Arikara fight. Ashley's boats reached the towns of the Arikara people on the Missouri River, in present-day South Dakota, in late May 1823, to buy horses. The Arikara were angry after a fight with men of another fur company, and they wanted a trading post of their own. On June 2, 1823, Arikara men attacked Ashley's party. Records give two counts of Ashley's men killed, 12 and 13.

In August an Army officer, Henry Leavenworth, led about 230 soldiers against the Arikara towns. About 750 Lakota and Yankton warriors and about 50 trappers went with him. His men fired cannon at the towns. Leavenworth and the Arikara then agreed to a peace, and the Arikara left their towns at night. Men of the Missouri Fur Company then burned one of the towns. No reliable count of the Arikara killed has survived.

The rendezvous. The trappers met every summer from 1825 to 1840 at a gathering they called the rendezvous, in the mountains of present-day Wyoming, Utah or Idaho. The word is French for a meeting. There the

trappers sold their furs and bought supplies for the next year. Native people came too, and so did the Native wives and children of many trappers.

The source of the Mississippi. An Ojibwe guide named Ozaawindib led Henry Rowe Schoolcraft to the true source of the Mississippi River in 1832. Schoolcraft was an Indian agent, a US official who dealt with Native nations, at Sault Ste. Marie, in Michigan. The source was a lake that the Ojibwe called Elk Lake. Schoolcraft renamed it Lake Itasca. He made up the name from pieces of the Latin words for "truth" and "head."

Hugh Glass

Who: A fur trapper who crawled and walked about 200 miles to safety after a grizzly bear attack.

When and where: 1823. Northwestern South Dakota.

Movie: *The Revenant* (2015), a film based on his story. An earlier film, *Man in the Wilderness* (1971), is based on it too.

He let maggots eat the dead flesh in his wounds.

Hugh Glass crawled and walked about 200 miles to safety after a grizzly bear tore the flesh from his body. The two men left to care for him had taken his gun and gone. In 1823 he joined Ashley's fur company, and he was shot in the leg in the Arikara attack of June 2. A letter he wrote to the parents of John Gardner, a trapper killed that day, still survives.

Late that summer, near the forks of the Grand River in present-day South Dakota, Glass surprised a mother grizzly with two cubs. The bear bit him and slashed him with her claws. Another trapper later said she had "torn the flesh from the lower part of the body, and from the lower limbs," and torn his neck badly. His leg was broken. The cuts on his back were so deep that his ribs showed through.

Andrew Henry, the leader of the party, asked for two men to stay with Glass until he died and then bury him. John Fitzgerald and a man written down as "Bridges" stayed. They dug a grave. Then they left, taking Glass's

rifle and knife, and told the others he was dead. Whether "Bridges" was Jim Bridger is still argued. Bridger later denied it.

Glass set his own broken leg and wrapped himself in the bear's hide. He crawled and walked about 200 miles to Fort Kiowa on the Missouri River, and part of the way he floated on a raft down the Cheyenne River. The journey took six weeks.

He left the maggots in his wounds so that they would eat the dead flesh and keep the wounds from turning to gangrene. Gangrene is the rotting of dead flesh in a living body. Glass later found "Bridges" and forgave him. Fitzgerald had joined the Army, and Glass got his rifle back.

The story was first printed in 1825, in a Philadelphia magazine called *The Port Folio*. A writer named James Hall wrote it, and other accounts dispute parts of it. No diary by Glass tells the story. It comes from other people's later retellings. In early 1833 Arikara warriors killed Glass and two other trappers, Edward Rose and Hilain Menard, on the Yellowstone River.

Jedediah Smith

Who: A fur trapper from New York who led the first party of US citizens known to reach California over land.

When and where: 1823 to 1831. The Rocky Mountains, California and Oregon.

A grizzly tore off his scalp and one ear.

Jedediah Smith led the first party of US citizens known to reach California over land. He answered Ashley's advertisement in 1822. In the autumn of 1823, in the Black Hills region, a grizzly bear broke his ribs and took his head in its mouth. Smith told another trapper, Jim Clyman, how to sew his scalp and ear back on, and Clyman did. Afterward Smith wore his hair long to hide the scar.

The map. Crow people drew a map in sand and on a buffalo hide in 1824, in the Wind River valley of present-day Wyoming, to show Smith's men the way west. With the map, Smith's men crossed South Pass, a crossing of the Rocky Mountains in present-day Wyoming. The Crow already used the pass. A trader named Robert Stuart had crossed it in 1812, but few Americans learned of his crossing. Smith's party made the pass known to Americans.

California. Smith led 15 men southwest toward California in August 1826. Mojave people near present-day Needles, California, sheltered them. Two men who had run away from the Spanish missions guided them west across the Mojave Desert. On November 27, 1826, they reached Mission San Gabriel. California then belonged to Mexico.

The Mexican governor, José María Echeandía, arrested Smith as a suspected spy and held him for about two weeks. In 1827 Smith's party became the first US citizens known to cross the Sierra Nevada mountains going east, and then the Great Basin, a huge dry region between the Sierra Nevada and the Rocky Mountains.

Smith came back in August 1827 with 18 men and two French-Canadian women. As they crossed the Colorado River, Mojave men attacked them. Trappers from Taos, in New Mexico, had fought the Mojave, and the Mojave were set on revenge. They killed ten of Smith's men, among them Silas Gobel, and took the two women captive. Smith's men shot and killed two Mojave men.

The Umpqua. In 1828 Smith led his men north from California along the Pacific coast, looking for the Columbia River and a way back to the Rocky Mountains. On July 11 an Umpqua man took an axe from their camp on the Umpqua River, in present-day Oregon. Smith and another man tied a cord around the Umpqua man's neck while the rest of the party held their guns ready.

Umpqua men attacked the camp on July 14, while Smith and two others were away. They killed 15 of Smith's men. One man, Arthur Black, escaped.

In 1831 Smith and his partners set out from St. Louis with a wagon train to trade in Santa Fe. On May 27 Smith left the wagon train on the Santa Fe Trail to look for water near the Cimarron River, in present-day Kansas. He did not come back. Weeks later his partners learned that Comanche men had killed him. His body was never found.

Jim Bridger

Who: A trapper and guide from Virginia who never learned to read or write.

When and where: 1823 to 1881. Wyoming, Utah and Montana.

He reached the Great Salt Lake in a buffalo-hide boat.

Jim Bridger was one of the first two men who were not Native known to reach the Great Salt Lake. He was born in Richmond, Virginia, in 1804. His parents died when he was 13, and he never learned to read or write. In 1823, at 19, he joined Ashley's trappers.

His boat was a bull boat, a boat made of buffalo hide, and he floated it down the Bear River to the lake in the fall of 1824. A French-Canadian trader, Etienne Provost, reached the lake at about the same time. In October 1824 Shoshone men attacked Provost's party at the Jordan River and killed eight of his men. The Shoshone, the Ute and other nations already lived around the lake.

Bridger and his partner Louis Vasquez built Fort Bridger in 1843, a trading post on the Oregon Trail in present-day Wyoming. In 1850 Bridger found a crossing of the mountains now called Bridger Pass. He later worked as a scout for the Army. By 1875 he was blind, and he died on July 17, 1881. Whether he was the "Bridges" who left Hugh Glass in 1823 is still argued. Bridger denied it.

Charles Wilkes

Who: A US Navy lieutenant who led a large exploring voyage of six ships paid for by the US government.

When and where: August 1838 to June 1842. The Pacific islands, Antarctica and the Pacific Northwest.

His men burned villages in Fiji and brought home 60,000 specimens.

Charles Wilkes commanded the United States Exploring Expedition, a Navy exploring voyage paid for by the US government. W. John Kress, a plant scientist at the Smithsonian Institution, calls it the first American exploring expedition to other parts of the world. Members of Congress approved it in May 1836. Wilkes was only a lieutenant, and he got the command in April 1838 after more senior officers turned it down.

He left Virginia on August 18, 1838, with six ships and nearly 350 men. They included naturalists, a mineralogist, who studies rocks, and a scholar of languages. One ship, the *Sea Gull*, went missing off South America in 1839.

Antarctica. Two officers on the ship *Peacock* sighted mountain peaks in the far south on January 16, 1840. The ships followed the coast. In February Wilkes wrote that they had "discovered the Antarctic Continent." His crews charted about 1,500 miles of that coast, which is now called Wilkes Land. French and British ships were exploring the same waters that year.

Fiji. Wilkes's men burned villages and killed Fijians in the Fiji Islands in 1840. On July 12 Wilkes had his men burn the villages of Solevu and Tye after people from Solevu took one of his boats. An officer, William Hudson, took a Fijian chief named Vendovi prisoner by first taking his brothers prisoner. Americans blamed Vendovi for killing American sailors in 1834.

Vendovi died in the Navy hospital in New York soon after the ships came home in June 1842. The expedition's surgeons cut off his head, and his skull became part of the first collections of the Smithsonian Institution. In December 2025 his remains were returned to his home district of Rewa, in Fiji, and buried in the chiefs' burial ground at Lomanikoro.

Off Malolo Island, later in July 1840, Lieutenant Joseph Underwood held the son of a village chief hostage while he bargained for food. The young man jumped from the boat, and the Americans fired over his head. Fighting began. Fijians killed Underwood and Wilkes's nephew, Midshipman Wilkes Henry. The Americans killed about ten Fijians that day. Accounts disagree over whether the escape or the shot started the fight.

The next day Wilkes landed about 70 men on Malolo. He ordered them to kill the men, to burn the villages and to "spare only the women and the children." His men burned the villages of Sualib and Arro and killed between 74 and 104 Fijians. In 1841 men from the *Peacock* fought about 700 islanders on Drummond's Island, in the Gilbert Islands, killed twelve and burned two villages.

Home. Wilkes's ships mapped Puget Sound, in present-day Washington State, in 1841, and his men mapped more than 800 miles of Oregon. The *Peacock* was wrecked on the sandbar at the mouth of the Columbia. The expedition came home to New York on June 10, 1842. It had sailed about 87,000 miles and lost two ships and 28 men.

It brought back more than 60,000 plant and bird specimens, which helped start the collections of the Smithsonian Institution in Washington. It also brought 254 live plants, which started what became the US Botanic Garden.

Back home, Navy officers tried Wilkes in a court-martial, a trial held by the military. They cleared him of losing the *Peacock* and of mistreating his officers. They found him guilty of illegally punishing sailors in his squadron.

His punishments had included flogging, which means beating a man's bare back with a whip. Navies of that time used a whip of nine knotted cords, called a cat-o'-nine-tails, which was made to tear the skin and cause great pain. One charge named flogging round the fleet, a sentence under which a sailor was whipped aboard each ship of the squadron in turn.

John C. Frémont

Who: An Army mapmaker, later known as "the Pathfinder," whose reports and maps guided thousands of travelers west.

When and where: 1842 to 1849. The Oregon Trail, the Great Basin, the Sierra Nevada, California and Oregon.

Kit Carson called what they did on the Sacramento River "a perfect butchery."

John C. Frémont led four expeditions to map the West, and on the third one his men killed at least 120 Wintu people in California. In 1841 he married Jessie Benton, the daughter of Senator Thomas Hart Benton of Missouri. Benton got the government to pay for Frémont's expeditions. A mountain man, a trapper who lived in the western mountains, named Kit Carson guided him. Frémont and Jessie wrote the report of the first expedition together.

The maps. Frémont led 25 men up the Platte River to South Pass in 1842. In 1843 and 1844 he went on to the Columbia River with almost 40 men and a cannon. Then he turned south and, in winter, crossed the Sierra Nevada into California. He was one of the first Americans to see Lake Tahoe. He showed that no river drains the Great Basin to the sea.

Members of the Senate and of the House each ordered 10,000 copies of his report on this trip. In 1846 members of the Senate had seven maps from his work printed. Thousands of American travelers heading west used them as a guide.

The Sacramento River, 1846. Frémont set out again in 1845 with an armed party of about 60 men. Officials of the War Department secretly told him to act as a soldier if war with Mexico began.

Some settlers claimed on April 5, 1846, that Native people were about to attack them. Frémont's party attacked a camp of Wintu people on the Sacramento River, near present-day Redding, California. Most of the people in the camp were women and children, and they were trapped against the river. One of Frémont's men, Thomas Breckenridge, wrote that "the order was given to ask no quarter and to give none." That meant no one would be spared, even people who tried to surrender.

Frémont's men fired their rifles in volleys, then charged with swords and pistols. Men on horseback chased and killed people who ran. Riflemen shot people who were swimming in the river. The men who were there gave counts of the dead from 120 to 800 or more. Breckenridge counted 120 to 150, and Thomas Martin, another of Frémont's men, counted over 175. An eyewitness, William Isaac Tustin, said at least 600 to 700 were killed on land and 200 or more in the water.

Kit Carson, who took part, later called it "a perfect butchery." None of Frémont's men was killed or wounded. Nobody was ever charged.

After that, Frémont's men shot Native people on sight. Breckenridge said the men "had orders while in camp or on the move to shoot Indians on sight." On the night of May 9, 1846, Klamath men attacked Frémont's camp and killed two or three of his party.

Three days later Kit Carson led an attack on Dokdokwas, a Klamath village at the mouth of the Williamson River, in present-day Oregon, and killed at least 14 people. The people of Dokdokwas may have had nothing to do with the first attack. Nobody was charged for these killings either. In June 1846 Frémont joined a revolt by American settlers against Mexican rule in California, the start of his part in the war with Mexico.

The San Juans. Frémont tried to find a route for a railroad through the San Juan Mountains of Colorado in the winter of 1848 to 1849. By December 12, 1848, the party needed ninety minutes to move 300 yards, and the mules had begun to die. In the end ten of the men died. Some of the starving survivors partly ate at least one of the bodies. Ute warriors later killed two of the men, Benjamin Kern and "Old Bill" Williams, when they went back for the party's gear.

Members of the new Republican Party made Frémont their first candidate for president in 1856. He died in 1890.

BY 1850

If you remember one thing: The Army officers and trappers of these fifty years crossed and mapped the West with Native guides, and some of them killed the people whose land they crossed.

In 1803: American diplomats had just bought France's claim to Louisiana, and Lewis and Clark had not yet left St. Louis.

In 1850: South Pass, the Great Salt Lake, the source of the Mississippi, the Yellowstone country and the Great Basin were known to Americans, and thousands of travelers were heading west with Frémont's maps.

Next: Jim Beckwourth, a Black trapper born into slavery, lived for years with the Crow before he found his pass through the Sierra Nevada in 1850.

1850 to 1900

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After the Civil War members of Congress paid for four survey teams, groups sent to measure and map the western lands in detail, and other Americans pushed into Alaska and the Arctic. Native nations lived on the land the surveyors mapped. One survey leader, an Army officer, had written orders to count them and to choose places for Army forts.

Secretary of State William Seward agreed in 1867 to buy Alaska from Russia, and in 1885 an Army officer crossed part of its interior.

The official in charge of the census, the government's count of everyone in the country, wrote in 1890 that there could "hardly be said to be a frontier line" left between settled and unsettled land. American explorers had already turned toward the North Pole. Two of their voyages ended with most of the men dead.

Jim Beckwourth

Who: A Black fur trapper, born into slavery, who lived with the Crow and found a low pass through the Sierra Nevada.

When and where: 1824 to 1866. The Rocky Mountains, California and Colorado.

Marysville owed him for the trail and never paid.

Jim Beckwourth found the lowest pass through the Sierra Nevada mountains in California, and it is now named for him.

He was born into slavery in Frederick County, Virginia, in 1798 or 1800. His father, Jennings Beckwith, was a white planter who held Beckwourth's mother, a Black woman, as a slave. By law he owned her, and she could not refuse him. His father freed him in papers filed in court in 1824, 1825 and 1826.

Beckwourth joined Ashley's fur company in 1824. For eight or nine years he lived with a band of Crow people, who named him "Bull's Robe." He married the daughter of a chief and rose from warrior to chief. His own book says he became the Crow's highest war chief. Other accounts suggest his fur company planned his stay so that it could win the Crow people's trade.

Beckwourth found the pass in 1850. At 5,221 feet above sea level, it is the lowest crossing of the Sierra Nevada in California. In 1851 he improved an old Native path over it into a trail that ran down to the town of Marysville. It saved gold seekers and settlers about 150 miles and kept them off steep passes such as Donner Pass.

By Beckwourth's own account, the merchants of Marysville had promised to pay him for the trail. When he came to collect in 1851, two large fires had burned the town, and its leaders were unable to pay him. In 1996 Marysville named its largest park Beckwourth Riverfront Park, to recognize the debt the town still owed him.

Beckwourth told the story of his life to a writer, Thomas Bonner, in the winter of 1854 to 1855. The book came out in 1856. Bonner never paid Beckwourth his half of the money. How much of the book is true is disputed.

Sand Creek. Colonel John Chivington hired Beckwourth as a scout in 1864. Beckwourth and about a dozen other scouts guided Chivington's soldiers to a camp of Cheyenne and Arapaho people at Sand Creek, in Colorado. The commander of a nearby Army fort had told the Cheyenne they would be safe there, and the camp was flying an American flag.

The soldiers attacked the camp on November 29, 1864. Counts of the dead run from about 70 to about 230. National Park Service counts give about 230 killed, most of them women, children and old people, and another estimate gives 70 to 163 Cheyenne. Soldiers cut scalps and other parts from the bodies of the dead.

Beckwourth spoke out in anger against the attack, and he later testified about it. Because he had guided the soldiers there, the Cheyenne refused to trade with him.

He died in a Crow village in Montana in 1866. Jim Bridger and a newspaper editor, William Byers, said the Crow poisoned him. Modern accounts name severe high blood pressure as the more likely cause.

The Great Surveys of the West, 1867 to 1879

King. Clarence King, a geologist, a scientist who studies rocks and the earth, began a survey in 1867 along the 40th parallel, from eastern California to Wyoming. A parallel is an east-west line on a map. In 1872 King's team found a field in Colorado where diamonds had supposedly been found, and showed that the diamond field was a fake.

Hayden and Yellowstone. Members of Congress gave Ferdinand Hayden $40,000 in 1871 to survey the Yellowstone country. His party of 32 included the photographer William Henry Jackson and the painter Thomas Moran. The railroad banker Jay Cooke arranged for Moran to go. Cooke wanted to make Yellowstone known to help the Northern Pacific Railroad.

Jackson took the first known photographs of Yellowstone's waterfalls. Jackson's photographs, Moran's paintings and the survey's work helped persuade members of Congress to make Yellowstone a public park. President Ulysses S. Grant signed the law on March 1, 1872.

A band of Mountain Shoshone called the Tukudika had lived in the Yellowstone country for thousands of years. In the fall of 1879 the park's superintendent, Philetus Norris, found a camp the Tukudika had recently left.

He asked the US Indian agent at Fort Washakie for help. The agent sent a party of Shoshone men, who moved the Tukudika out of their homeland to the Wind River Reservation, in Wyoming. A National Park Service history states that the Tukudika "were forcibly removed to reservations," which means they were made to go whether they wanted to or not.

Wheeler. Army Lieutenant George Wheeler mapped the land west of the 100th meridian from 1872 to 1879. A meridian is a north-south line on a map, and the 100th meridian runs through the Great Plains. Under his orders, each inch of his maps was to stand for eight miles of land.

He was also to "discover the numbers, habits, and disposition of Indians in the section" and to "select sites for future military installations." That meant counting the Native people, learning how they lived and whether they were friendly, and picking places for Army forts.

Members of Congress ended the Wheeler, King and Powell surveys in 1879 and moved their work into a new agency, the US Geological Survey. Clarence King was its first director.

Alaska, 1867 to 1885

Two cents an acre. Secretary of State William Seward and a Russian diplomat, Eduard de Stoeckl, signed a treaty at 4 in the morning on March 30, 1867. Under it, Seward agreed to pay $7.2 million for Alaska, about 2 cents an acre, for 586,412 square miles, and de Stoeckl agreed for Russia to sell it. Most reactions in the United States were positive. Some opponents called it "Seward's Folly" or "Seward's Icebox."

Seward's own figures counted about 8,000 Native people under Russian rule and about 50,000 more living beyond it. Neither Seward nor the Russian officials asked any of them.

Across Alaska on foot. Army Lieutenant Henry Allen set out in the spring of 1885 to cross Alaska with two soldiers, Sergeant Cady Robertson and Private Fred Fickett, and $2,000. They went up the Copper River. At the Native village of Taral a prospector, John Bremner, joined them. A prospector is a person who searches for gold or other minerals. Local Native people showed them how to build and use boats made of animal skins. The men ate rotten meat full of worms.

They crossed the Alaska Range by a pass that Allen named Miles Pass and went down the Tanana River and up the Koyukuk, about 1,500 miles in all. Two of the men got scurvy, a disease caused by a severe lack of vitamin C, which comes from fresh food.

John Wesley Powell

Who: A Civil War officer who had lost an arm and who led the first recorded boat journey by white men through the whole Grand Canyon.

When and where: May to August 1869. The Green and Colorado rivers in Wyoming, Utah and Arizona.

Movie: *Ten Who Dared* (1960), a Disney film about the expedition.

He ran the Grand Canyon with one arm and four wooden boats.

John Wesley Powell led the first recorded boat trip by white men through the whole Grand Canyon, in 1869. At the Battle of Shiloh, in April 1862, he was giving the order to fire when a rifle bullet hit his right arm. He lost the arm below the elbow. The raw nerve endings hurt him for the rest of his life. He went back to the Army and fought at Vicksburg.

Powell and nine other men set out from Green River, in Wyoming Territory, on May 24, 1869, in four wooden boats. At a rapid they named Disaster Falls they wrecked and lost one boat, the *No Name*. They named other places along the way, among them Flaming Gorge and Glen Canyon. One man, Frank Goodman, left the trip in July.

The three who left. Three men, Oramel Howland, his brother Seneca Howland and William Dunn, left the river on August 28. They were afraid of the rapids ahead, and they climbed out of the canyon. They were never seen again.

Shivwits Paiute men later told Powell's interpreter that they had killed the three by mistake. The Shivwits said they had taken them for prospectors who had raped and killed a Shivwits woman. Rape means forcing a person to have sex. Other accounts say Mormon settlers from Toquerville, Utah, killed them instead. Mormons are members of the Church of Jesus Christ of Latter-day Saints.

NOT KNOWN: which account is true.

> Powell and five men reached the Mormon settlement of St. Thomas, Nevada, on August 30, 1869. They had traveled almost 930 miles in 98 days. They were the first white men recorded passing through the whole canyon. Paiute people and other Native nations had lived along it long before them.
>
> Powell went down the river a second time in 1871 and 1872. From 1881 to 1894 he led the US Geological Survey.

Toward the North Pole, 1879 to 1898

The Jeannette. Twenty of the 33 men on the *Jeannette* died on the voyage, among them its captain, Navy Lieutenant George De Long. In 1879 De Long sailed the ship north through the Bering Strait. James Gordon Bennett Jr., the owner of the *New York Herald* newspaper, paid for the voyage. De Long hoped to find an open sea around the North Pole.

Ice trapped the ship for nearly two years and crushed it on June 13, 1881. The men dragged their boats across the ice and then sailed for the Lena River in Siberia. One boat, under Lieutenant Charles Chipp, was lost with all eight men in it.

De Long and the men in his boat landed in the river's delta, the low, wet land where a river splits into many channels near the sea. They starved there. His last journal entry, on October 30, 1881, lists the dead and ends "Mr Collins dying." Thirteen men came home. The voyage proved that there is no open sea at the pole.

Greely. Only six of the 25 men Army Lieutenant Adolphus Greely took to the far north came home, and Greely ordered one of the others shot. In 1881 Greely took his men to Ellesmere Island, in Canada's far north. They were to record the weather as part of the First International Polar Year, when scientists from several countries studied the far north and far south at the same time.

Two of his men, Lieutenant James Lockwood and Sergeant David Brainard, set a new record for the farthest north anyone had reached. Two dog drivers from Greenland, Jens Edward and Thorlip Frederik Christiansen, went with them. Both were Inuit, the Native people of the far north.

Supply ships failed to reach the party in 1882 and again in 1883. The men spent their last winter at Cape Sabine with about 40 days of food. Private Charles B. Henry stole food again and again.

Greely wrote an order on June 6, 1884: "Private Henry will be shot to-day." The order warned that Henry was stronger than any two of the other men. Three men, David Brainard, Francis Long and Julius Fredericks, shot him at 2 in the afternoon. On Greely's order, two of their guns held bullets and one held a blank, a cartridge with no bullet in it.

When rescue ships arrived on June 22, 1884, seven men were still alive. One died soon after, so six of the 25 lived to go home. Apart from Henry, the dead died of hunger, cold and drowning. In August 1884 doctors examined the body of one dead officer, Lieutenant Frederick Kislingbury, and found flesh cut from the bones. The survivors said they knew nothing about anyone eating the dead.

The six Inuit. Robert Peary was a Navy officer who later led expeditions toward the pole. In 1894 he reached the Cape York meteorite in northern Greenland. A meteorite is a rock that falls to Earth from space. Inuit people had relied on it to make tools. Peary took it.

In 1896 Franz Boas, a scientist at the American Museum of Natural History in New York, asked Peary to bring an Inuk, one of the Inuit, to the museum. The next year Peary deceived six Inuit into sailing with him to New York. He promised they could go home within a year with tools, weapons and gifts, and he did not keep that promise. The six arrived in September 1897. Staff at the museum kept them in its basement, and 20,000 people paid to look at them.

All six had tuberculosis, a lung disease that spreads through the air, by November 1. Four died in a New York hospital in 1898: Qisuk on February 17, Atangana and Nuktaq that spring, and a girl, Aviaq, on May 24. Qisuk's son, Minik, born about 1890, survived.

The fake burial. The museum's scientific staff wanted to keep Qisuk's body for study. Boas had the staff hold a fake burial. They filled a coffin with stones, laid a stuffed shape covered with a cloth on top, and buried it so that Minik would believe his father had been buried. Boas later said the ceremony was meant to comfort the boy.

Museum staff sent Qisuk's body to the estate of the museum's building manager, William Wallace, where Wallace had a workshop for cleaning skeletons. There the flesh was stripped from Qisuk's bones. The records do not name who did that work. Museum staff later put the skeleton on display in the museum. Wallace adopted Minik.

Of the two who lived, a young man named Uisaakassak demanded to go home to Greenland. He sailed home on the ship *Windward* in July 1898. Minik stayed in the United States. About 1906 New York newspapers reported that his father's skeleton was on display, and Minik learned it from his classmates. He died in the flu epidemic of 1918. In 1993 a Canadian writer, Kenn Harper, got the remains of Qisuk and the three others returned to Greenland, where they were buried at Qaanaaq.

1898: islands in two oceans

Eight years after the census official wrote that the frontier line was gone, President William McKinley, members of Congress and American diplomats took control of islands in the Pacific Ocean and the Caribbean Sea. Among them were the Hawaiian Islands, which James Cook had reached in 1778. Members of Congress voted to annex the Republic of Hawaii, which means to take it over as part of the United States. McKinley signed their resolution on July 7, 1898. Hawaii's former ruler, Queen Lili‘uokalani, protested to Congress.

After a war between Spain and the United States, diplomats from the two countries signed a treaty in Paris on December 10, 1898. Under the treaty, Spanish officials gave up Cuba and handed Puerto Rico, Guam and the Philippines to the United States. American officials agreed to pay Spain $20 million for the Philippines.

BY 1900

If you remember one thing: Survey teams mapped the West on land where Native nations lived. In the Arctic most of the men on two American voyages died, mainly of hunger and cold, and four of the Inuit whom Peary brought to New York died there.

One number: six. Of the 25 men Greely took to Ellesmere Island in 1881, six came home. The *Jeannette* did a little better, 13 men out of 33.

Next: Matthew Henson, a Black sailor from Maryland, worked beside Robert Peary for more than twenty years and made seven Arctic voyages with him.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 10 MINUTES

Many American explorers of the years 1900 to 1950 worked far from home. Some traveled by dog sled toward the North Pole. Pilots flew toward both poles and across the Atlantic Ocean. Teams from Yale University and from a New York museum dug in the mountains of Peru and the Gobi Desert of Mongolia.

Two of the explorers' claims are still argued over. One is whether Robert Peary's party reached the North Pole in 1909, and the other is whether Richard Byrd flew over it in 1926.

In some of the places these explorers went, local people led the way. Inuit men, women and children joined Peary's last expedition to the North Pole. The Inuit are the Native people of the far north, from Alaska to Greenland. Four Inuit men were in the small party that Peary said reached the pole, and a local man led Hiram Bingham up to Machu Picchu.

Matthew Henson (with Robert Peary)

Who: A Black sailor and explorer from Maryland who was Robert Peary's navigator and dog-sled driver on seven Arctic voyages.
When and where: 1887 to 1909. Northern Greenland and the Arctic Ocean.

Henson said he reached the spot before Peary did.

Matthew Henson went with Robert Peary on the 1909 journey that Peary claimed reached the North Pole. Members of Congress thanked Peary in 1911. Most of Henson's honors came after 1937, more than 25 years after the journey.

Henson was born in Charles County, Maryland, in 1866. His parents were sharecroppers, farmers who worked someone else's land for a share of the crop. They had been free before the Civil War. In 1867 the family left southern Maryland to escape attacks by the Ku Klux Klan, a group of white men who used violence to terrorize Black people.

At 12 Henson went to sea as a cabin boy. The ship's captain, a man named Childs, taught him to read and write. In November 1887 Henson was working in a clothing store in Washington, D.C., when he met Robert Peary, a Navy officer. Peary hired him as his helper on a trip to measure and map land in Nicaragua, where people were planning a canal. Their first Arctic trip together was in 1891 and 1892, and by 1909 they had made seven Arctic voyages.

Henson traded with the Inuit and learned their language. They called him Mahri-Pahluk. He is remembered as the only member of the expedition who was not Inuit and became skilled at driving and training dog teams the Inuit way. He was also Peary's navigator, the person who worked out where they were and which way to go.

The last expedition. Peary's ship, the *Roosevelt*, left New York in July 1908. Inuit men, women and children from Cape York, in Greenland, joined the expedition. In February 1909 the sled parties set out across the

frozen Arctic Ocean. There were 7 expedition members, 19 Inuit, 140 dogs and 28 sleds.

They carried supplies forward in relays. One party hauled loads part of the way and left them, and the next party carried them on. By the end Peary could no longer walk, and he rode on a sled. He sent Henson ahead as a scout.

Peary said that his party had reached the North Pole on April 6, 1909. The last group was six men: Peary, Henson and four Inuit men named Ootah, Egingwah, Seegloo and Ooqueah. Henson later said that he was the first of the party to reach the spot they believed was the pole.

The claim. The claim is disputed. Frederick Cook, who had been the doctor on Peary's trip of 1891 to 1892, said he had reached the pole in 1908, a year before Peary. Members of a National Geographic Society committee backed Peary. In 1989 the British explorer Wally Herbert studied Peary's records for the National Geographic Society. He concluded that the records were unreliable. They showed the party moving faster than seemed possible. Herbert found that the party may have stopped 30 to 60 miles short of the pole.

Honors, late. Members of Congress voted Peary their formal thanks in 1911, and he was promoted to rear admiral, one of the highest ranks in the Navy.

Henson published a book in 1912 called *A Negro Explorer at the North Pole*. "Negro" was the common word for a Black person at that time. He became the first African American life member of the Explorers Club, a society of explorers, in 1937, and he received a medal for the Peary expedition in 1944. Presidents Harry Truman and Dwight Eisenhower welcomed him to the White House.

He died on March 9, 1955. In 1988 he and his wife were buried again at Arlington National Cemetery. The National Geographic Society gave him its Hubbard Medal in 2000. In 2021 the International Astronomical

Union, the scientists' group that names places in space, named a crater on the Moon after him.

Hiram Bingham

Who: A Yale history teacher, not a trained archaeologist, who made the Inca site of Machu Picchu famous.
When and where: 1911 to 1915. The Andes Mountains of Peru.

Somebody had already written "A. Lizarraga 1902" on a wall.

Hiram Bingham reached Machu Picchu in 1911 and later made it famous around the world, though people had known the place long before he arrived. He taught South American history at Yale University. He was not trained as an archaeologist, a scientist who studies the past by digging up what people left behind. In 1911 he led an expedition from Yale to Peru to look for the last capital of the Inca. The Inca ruled an empire in the Andes Mountains before the Spanish arrived.

A local man named Melchor Arteaga led Bingham up to Machu Picchu on July 24, 1911. The people living in the valley next to it already knew the ruins. Explorers from the Peruvian city of Cusco are said to have reached it in 1901. One of them, Agustin Lizarraga, wrote "A. Lizarraga 1902" in charcoal on a wall there, and Bingham found the writing. A map from 1874 already shows the site. Bingham later wrote about the ruins as if he had found them himself.

Bingham guessed that Machu Picchu was the Incas' last capital, and he called it the "Lost City." The last capital was really Vilcabamba, which Bingham also visited without recognizing it.

Bingham's team dug up about 40,000 objects at Machu Picchu, among them mummies, bones and pottery. They shipped them to Yale under an order from Peru's government. Peruvian officials said the objects had only been lent. Yale and Peruvian officials agreed in 2007 on what to do with them, and Yale officials began sending the objects back in 2012.

Roy Chapman Andrews

Who: A naturalist from Wisconsin who started as a janitor at the American Museum of Natural History and later ran it.

When and where: 1922 to 1928. The Gobi Desert of Mongolia.

His team found nests full of dinosaur eggs on July 13, 1923.

From 1922 to 1928 Roy Chapman Andrews led the American Museum of Natural History's expeditions into the Gobi Desert of Mongolia. His teams went by car from Peking, in China, the city now called Beijing. He was looking for the remains of the earliest humans. His teams found none.

They found other fossils instead. Fossils are the remains of living things from long ago, turned to stone or pressed into rock. In 1922 they found the bones of a giant relative of the rhinoceros that had no horn. In 1923 they found the first nests of dinosaur eggs known to science. Scientists first thought the eggs came from a dinosaur called *Protoceratops*. In 1995 scientists showed that they belonged to a meat-eating dinosaur called *Oviraptor*.

Chinese officials seized what the expedition had found in 1928. They later gave the finds back. In 1934 Andrews became director of the museum.

Richard Byrd

Who: A US Navy pilot who led flights and expeditions to both poles.
When and where: 1926 to 1935. The Arctic and Antarctica.
Movie: *With Byrd at the South Pole* (1930), a documentary film of the trip.

His diary holds navigation readings that somebody erased.

Richard Byrd led the first flight to the South Pole, in 1929. His claim to have flown over the North Pole three years earlier is disputed. Byrd became a Navy pilot in 1918.

The North Pole claim. Byrd and the pilot Floyd Bennett flew a three-engine plane, the *Josephine Ford*, toward the North Pole on May 9, 1926. Members of Congress voted both men the Medal of Honor for the flight. Most researchers who have studied the flight believe the plane could not carry enough fuel to fly as far as Byrd said. In 1958 another pilot, Bernt Balchen, said that Bennett had told him they never reached the pole.

Byrd's diary was made public in 1996. Its erased navigation readings differ from his official report, and the records do not say who erased them. The historian Dennis Rawlins concluded that Byrd flew about 80 percent of the way and turned back because of an oil leak. Other researchers defend Byrd's claim.

The South Pole. In Antarctica, Byrd built a base camp called Little America on the Ross Ice Shelf, a huge sheet of ice floating on the sea. On November 28, 1929, Byrd, the pilot Bernt Balchen, the radio operator Harold June and the photographer Ashley McKinley took off for the South Pole. They flew there and back in 18 hours and 41 minutes. To make the plane light enough to climb onto the high ice of the Antarctic Plateau, they threw supplies out.

Alone. Byrd spent five months alone in a weather hut called Advance Base on his second expedition, in 1934. Fumes from a badly vented stove, one that did not let its smoke out properly, gave him carbon monoxide

poisoning. Carbon monoxide is a gas with no smell that stops the blood from carrying oxygen. He nearly died. Men from the base camp reached him on their third try. Byrd died in 1957.

Charles Lindbergh

Who: An airmail pilot who made the first solo nonstop flight across the Atlantic Ocean.

When and where: May 20 to 21, 1927. From Garden City, New York, to Paris, France.

He flew alone for 33 hours and 30 minutes.

Charles Lindbergh, an airmail pilot, became the first person to fly alone and without stopping across the Atlantic Ocean in May 1927. Workers at Ryan Airlines, a company in San Diego, built his single-engine plane, the *Spirit of St. Louis*.

Lindbergh took off from Roosevelt Field, in Garden City, New York, on May 20, 1927. He landed at Le Bourget airfield outside Paris the next day, after about 3,600 miles. He won the $25,000 Orteig Prize, which had been offered for the first nonstop flight between New York and Paris.

Two British fliers, John Alcock and Arthur Brown, had made the first nonstop flight across the Atlantic eight years earlier. They flew from St. John's, Newfoundland, to Clifden, Ireland, on June 14 and 15, 1919, in under sixteen hours. Lindbergh was the first to make the crossing alone.

Amelia Earhart

Who: The first woman to fly alone and nonstop across the Atlantic Ocean.

When and where: 1932, Newfoundland to Northern Ireland, and 1937, the central Pacific Ocean.

The *Itasca* heard her clearly, and she could not hear the ship.

Five years after Lindbergh, Amelia Earhart made the same kind of crossing, alone and without stopping, the first woman to do it. In 1928 she had crossed the Atlantic by air as a passenger, the first woman to do so.

She took off alone from Harbour Grace, in Newfoundland, on May 20, 1932, when she was 34. She flew through strong winds and ice, and her engine had problems. After 14 hours and 56 minutes she landed in a pasture near the city of Derry, in Northern Ireland. Members of Congress voted her the Distinguished Flying Cross, a medal for bravery or great skill in flying.

Around the world. Earhart set out in 1937 to be the first woman to fly around the world. Her route ran about 29,000 miles, near the equator. The navigator Fred Noonan flew with her. On June 29 they reached Lae, in New Guinea, with about 7,000 miles to go.

They took off for Howland Island, a tiny island in the Pacific, on July 2, 1937. A Coast Guard ship, the *Itasca*, waited there to guide them in by radio. The crew of the *Itasca* heard her clearly, but she could not hear the ship. The ship's radio direction finder could not lock onto her signal to show which way it came from.

At 7:58 in the morning she radioed that she could not hear the *Itasca*. Earhart and Noonan were never found. She was declared legally dead on January 5, 1939.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 13 MINUTES

After 1950 the places no person had ever seen were the deepest parts of the ocean and space beyond the Earth. The Soviet Union, a large country ruled from Moscow, was the main rival of the United States in those years. On October 4, 1957, Soviet engineers launched Sputnik, the first satellite to circle the Earth. A satellite is an object that travels in a path around a planet.

The next year members of Congress passed a law that set up NASA, the US space agency. Soviet and American crews then flew into space, one flight after another. Between 1969 and 1972 twelve American astronauts walked on the Moon.

Three American astronauts died in a fire on the launch pad in 1967. Seven died when the space shuttle *Challenger* broke apart in 1986. A space shuttle was a spacecraft that rode rockets up into space and landed on a runway when it came back. In the ocean, two men reached the deepest point on Earth in 1960, and scientists found living things around hot springs on the sea floor in 1977.

Don Walsh

Who: A US Navy lieutenant who, with the Swiss scientist Jacques Piccard, made the first dive to the deepest point in the ocean.

When and where: January 23, 1960. The Challenger Deep in the Mariana Trench, near Guam in the Pacific Ocean.

Below 30,000 feet a window pane cracked and the craft shook.

Navy Lieutenant Don Walsh and the Swiss ocean scientist Jacques Piccard reached the bottom of the deepest known place in the ocean on January 23, 1960. It is called the Challenger Deep, and it lies in the Mariana Trench, near the island of Guam in the Pacific. No craft of any kind, with people or without, had been there before.

They went down in the *Trieste*, a deep-diving craft called a bathyscaphe. Piccard's father, Auguste Piccard, had designed it, and it was built in Italy in 1953. The US Navy bought it. The trip down took 4 hours and 47 minutes. The pane that cracked was one of the outer window panes, and the two men went on to the bottom.

At the time they measured the depth as 37,799 feet. The figure was later corrected to 35,814 feet, which is almost seven miles. Newer measurements give between 35,797 and 36,070 feet. The two men stayed on the bottom for twenty minutes, in a cabin that was 45 degrees Fahrenheit.

Walsh and Piccard reported seeing flatfish on the bottom. Scientists now doubt it, because fish are not known to live deeper than about 26,000 to 28,000 feet. Walsh later said they could have been mistaken.

Into space, 1957 to 1972

NASA. NASA was set up on July 29, 1958, under a law called the National Aeronautics and Space Act. NASA's staff began work on October 1, 1958. Its full name is the National Aeronautics and Space Administration.

The first people in space. A Soviet cosmonaut, Yuri Gagarin, circled the Earth once on April 12, 1961, the first person to travel into space. Cosmonaut is the Russian word for an astronaut. Three weeks later, on May 5, 1961, Alan Shepard became the first American in space. His flight lasted 15 minutes. It went up into space and came straight back down without circling the Earth.

President John F. Kennedy asked Congress on May 25, 1961, to set the goal, "before this decade is out, of landing a man on the Moon and returning him safely to the Earth." The Moon program was called Apollo. At its peak 400,000 people worked on it, and it cost $25.4 billion.

John Glenn circled the Earth three times on February 20, 1962, in 4 hours and 55 minutes, in a capsule called *Friendship 7*. He was the first American to orbit the Earth, which means to travel all the way around it. Two Soviet cosmonauts had orbited before him.

Apollo 1. Three astronauts died in a fire during a practice countdown on the launch pad in Florida on January 27, 1967. They were Gus Grissom, Ed White and Roger Chaffee, the crew of Apollo 1. An electric spark set nylon material on fire inside a cabin filled with pure oxygen. The hatch opened inward, and the pressure inside the cabin held it shut. It took workers on the launch pad five minutes to open all three layers of the hatch.

The fire melted the men's suits and air hoses. Doctors who examined their bodies found that the main cause of death was carbon monoxide gas from the fire, which had stopped their hearts. All three also had third-degree burns, burns that destroy the skin all the way through. Most of the burns happened after the men had died.

The Moon. The crew of Apollo 8 became the first people to orbit the Moon in December 1968. Between December 1968 and December 1972, 24 astronauts flew around the Moon. Six Apollo crews landed on it, and twelve men walked

on it. The last landing was Apollo 17, in December 1972. No one has walked on the Moon since.

Katherine Johnson

Who: A Black mathematician from West Virginia who calculated the flight paths of the first American spaceflights.
When and where: From 1953. The Langley laboratory in Hampton, Virginia.
Movie: *Hidden Figures* (2016). The actor Taraji P. Henson plays Johnson.

Glenn refused to fly until she had checked the machines' numbers.

Katherine Johnson worked out the flight path of the first American spaceflight, in 1961, and checked the numbers for the first American flight around the Earth, in 1962. She was born in White Sulphur Springs, West Virginia, in 1918. Her county had no public high school for Black students.

Her family sent her to one on the campus of West Virginia State College, and she started there at the age of ten. Later she was one of three Black students chosen to be the first to attend the graduate school of West Virginia University.

"Colored Computers." The Langley laboratory, in Hampton, Virginia, hired her in June 1953. The laboratory belonged to NACA, the government agency that came before NASA. Her job title was "computer." At that time a computer was a person who did long calculations by hand.

Segregation means keeping people apart by race. Virginia's lawmakers had passed segregation laws, and federal workplaces had segregation rules of their own. Under them, Johnson and the other Black women computers had to work, eat and use restrooms apart from white workers. The sign on their office read "Colored Computers." "Colored" was a word for Black people used at that time.

Johnson calculated the path and the launch time for Alan Shepard's flight in 1961. For John Glenn's flight in 1962, NASA officials used electronic computers for the first time to plan the orbit. Glenn asked for Johnson by name to check the machines' numbers. He refused to fly until she had checked them. She later worked on the paths that let Apollo spacecraft meet each other in space.

President Barack Obama gave her the Presidential Medal of Freedom in 2015. Members of Congress awarded her the Congressional Gold Medal in 2019. She died on February 24, 2020, at the age of 101.

Apollo 11 crew

Who: Neil Armstrong, Buzz Aldrin and Michael Collins, the crew of the first spacecraft to land people on the Moon.

When and where: July 16 to 24, 1969. From Florida to the Moon's Sea of Tranquility and back to the Pacific Ocean.

About one in five people on Earth watched the first step live.

Neil Armstrong and Buzz Aldrin became the first people to land on the Moon on July 20, 1969. A Saturn V rocket had launched them and Michael Collins from Kennedy Space Center, in Florida, on July 16.

Armstrong and Aldrin flew the lunar module, a small landing craft named *Eagle*, down to a flat plain called the Sea of Tranquility. Collins stayed in orbit around the Moon in the command module, *Columbia*, the part of the spacecraft that would carry all three men home. About six hours after landing, Armstrong stepped out onto the Moon. Aldrin followed him 19 minutes later. An estimated 600 million people watched Armstrong's first steps live on television, and another count gives 650 million.

The two men collected 47.5 pounds of rock and soil and spent more than 21 hours on the Moon's surface. The three astronauts splashed down safely in the Pacific Ocean on July 24, 1969.

Apollo 13 crew

Who: Jim Lovell, Jack Swigert and Fred Haise, a Moon-landing crew whose spacecraft was badly damaged by an explosion.

When and where: April 11 to 17, 1970. Between the Earth and the Moon.

Movie: *Apollo 13* (1995), based on Lovell's book about the flight, *Lost Moon*.

An oxygen tank exploded two days out, and they came home anyway.

The crew of Apollo 13 survived an explosion on the way to the Moon by living in their small landing craft. They were Jim Lovell, the commander, Jack Swigert and Fred Haise. Swigert had taken the place of Ken Mattingly, who had been exposed to German measles, an illness that spreads from person to person.

The spacecraft launched on April 11, 1970. Two days later the crew stirred one of the oxygen tanks, a routine job. Damaged insulation on the wires inside the tank caught fire, and the tank exploded. The oxygen in both of the main spacecraft's tanks escaped into space.

After the explosion, mission controllers cancelled the Moon landing and worked to bring the crew home alive. The three men shut down the main spacecraft to save it for the trip back down through the atmosphere, the layer of air around the Earth.

They moved into the small lunar module, *Aquarius*, and lived in it on the way home. It was cold and wet inside, and they ran short of drinking water. The air had to be cleared of carbon dioxide, the gas people breathe out. The crew and the engineers on the ground worked out a way to make the main spacecraft's carbon dioxide filters fit the lunar module.

The crew swung around the Moon and headed home. They splashed down safely in the South Pacific on April 17, 1970. Members of the review board later blamed the way the tank had been tested before the flight, and the Teflon plastic inside it.

Sylvia Earle

Who: A marine biologist from New Jersey who set a deep-diving record and led the first all-woman team to live underwater.

When and where: 1970 to 1992. The Virgin Islands and the waters off Hawaii.

She walked the sea floor 1,250 feet down, with no line up.

Sylvia Earle holds the women's record for the deepest dive to the sea floor. She became a marine biologist, a scientist who studies life in the sea. In 1969 she asked to join the Tektite project, which let scientists live in a laboratory under the sea. She had already spent more than 1,000 hours underwater doing research. The government officials running the project turned her down, because they did not want men and women living together in the laboratory.

So in 1970 Earle led the first all-woman team of aquanauts, people who live and work underwater. They lived in a laboratory 50 feet below the surface off the Virgin Islands.

She put on a hard diving suit in 1979 that kept the water pressure away from her body. With no line connecting her to the surface, she dived to the sea floor off Oahu, Hawaii, 1,250 feet down. That women's depth record still stands. From 1990 to 1992 she was the first woman to be chief scientist of NOAA, the US government's ocean and weather agency.

Robert Ballard

Who: A Navy oceanographer and sea-floor geologist from Kansas.

When and where: 1977 to 1989. The Pacific floor near the Galápagos Islands and the North Atlantic.

First he had to look at two sunken Navy submarines in secret.

Robert Ballard helped find living things where scientists had expected none, and later he found the wreck of the *Titanic*. He became an oceanographer, a scientist who studies the ocean, and he worked at the Woods Hole Oceanographic Institution.

Ballard was a diver in 1977 on an expedition that used the submarine *Alvin* on the sea floor near the Galápagos Islands, in the Pacific. Its members found hot springs on the sea floor. Groups of living things surrounded them. They got their energy from chemicals in the water from the springs, not from sunlight. All the scientists on board were geologists, scientists who study rocks and the Earth, because they had been certain there would be no life to study.

The Titanic. Ballard asked the Navy in 1982 to pay for a search for the *Titanic*. He planned to use a new camera robot called *Argo*, which a ship could tow along the sea floor. The Navy's leaders agreed, on one condition. First he had to use *Argo* to look in secret at the wrecks of two Navy nuclear submarines, the *Thresher* and the *Scorpion*. Both had sunk in the 1960s.

After that work, in 1985, Ballard and a French team found the wreck of the *Titanic*. In June 1989 he found the wreck of the *Bismarck*, a battleship of Nazi Germany that sank in the North Atlantic in May 1941.

Sally Ride and Mae Jemison

Who: The first American woman in space and the first African American woman in space.

When and where: 1983 and 1992, aboard the space shuttles *Challenger* and *Endeavour*.

Reporters sent more than 500 requests to interview Ride.

Sally Ride became the first American woman in space in 1983, and Mae Jemison became the first African American woman in space in 1992. Ride was a physicist from California, a scientist who studies matter and energy. She joined NASA in 1978. On June 18, 1983, she flew on the space shuttle *Challenger*. She was the third woman in space. The first had been Valentina Tereshkova of the Soviet Union, in 1963. At 32 Ride was the youngest American astronaut to fly.

On the flight Ride worked the shuttle's robot arm. Two months later, on August 30, 1983, Guion Bluford became the first African American in space.

Ride later served on the Rogers Commission, which investigated the *Challenger* explosion in 1986, and on the Columbia Accident Investigation Board, which investigated the loss of the shuttle *Columbia* in 2003. She was the only person to serve on both. She died in 2012. Her obituary, the published notice of her death, named Tam O'Shaughnessy as her partner of 27 years. Ride had kept the relationship private while she was alive. She is the first astronaut known to have been LGBTQ, which stands for lesbian, gay, bisexual, transgender or queer.

Jemison. Mae Jemison was born in Decatur, Alabama, in 1956 and grew up in Chicago. She studied chemical engineering at Stanford University and then became a doctor. From 1983 to 1985 she worked as a doctor in Liberia and Sierra Leone, in West Africa. She went there with the Peace Corps, a US government program that sends volunteers to work in other countries. She joined NASA in 1987. From September 12 to 20, 1992, she flew aboard the space shuttle *Endeavour*.

The Challenger explosion, 1986

The space shuttle *Challenger* broke apart 73 seconds after it lifted off on January 28, 1986. All seven people on board died. They were the commander, Dick Scobee, the pilot, Michael Smith, and Ellison Onizuka, Judith Resnik, Ronald McNair, Gregory Jarvis and Christa McAuliffe, a schoolteacher. Many schools watched the launch live because a teacher was on board.

The shuttle broke apart because rubber rings, called O-rings, failed in a joint of one of its booster rockets. The rings sealed the joint so that hot gas could not escape. The record cold that morning had made them stiff. Test results going back to 1977 had shown that these seals were flawed. Two engineers at the Morton Thiokol company, Allan McDonald and Roger Boisjoly, had warned against launching in the cold. NASA managers ignored their warnings.

Several of the crew are thought to have survived the breakup itself. Someone switched on the emergency air supply of the pilot, Michael Smith. The crew cabin then fell and struck the ocean.

BY 2000

If you remember one thing: Between 1950 and 2000 people reached the bottom of the ocean and walked on the Moon, and ten American astronauts died on the launch pad or on the way up.

One question this era leaves open: whether people would ever go back to the Moon. Nobody has walked on it since Apollo 17 came home in December 1972. The next era states the answer.

Next: A second space shuttle, *Columbia*, was lost in 2003, seventeen years after *Challenger*.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 6 MINUTES

Since 2000, robot spacecraft steered from Earth have explored Mars and the outer solar system. Rovers, robot vehicles, drive across Mars, and probes, robot spacecraft sent to study faraway places, have flown past Pluto and out beyond the planets. People have lived on the International Space Station, a laboratory that circles the Earth, without a break since November 2, 2000.

On Earth, most of the ocean floor has still not been mapped. In April 2026 an international mapping project reported that 28.7 percent had been mapped. That same month four astronauts flew around the Moon, the first people to travel there since 1972.

The loss of Columbia, 2003

The space shuttle *Columbia* broke apart over Texas and Louisiana on February 1, 2003, as it was coming back down through the atmosphere. All seven astronauts on board died. They were the commander, Rick Husband, the pilot, William McCool, and Michael Anderson, Kalpana Chawla, David Brown, Laurel Clark and Ilan Ramon, the first astronaut from Israel.

During the launch, a piece of foam from the shuttle's big fuel tank had broken off and hit the left wing. It damaged the panels that protected the wing from the heat of coming back through the atmosphere. Foam had struck shuttles before, on 65 of the 79 earlier launches that had been photographed.

While *Columbia* was in orbit, a team of engineers asked several times for pictures of the wing, taken by military satellites or telescopes. The mission's management team said it had no need for the pictures, and Linda Ham, the mission manager, cancelled the request.

Robots to Mars and beyond

NASA's rover *Curiosity* landed in Gale crater on Mars in August 2012. In March 2013 NASA scientists reported what a rock it drilled showed: water had once filled that place, at the end of a river or in a lake, and tiny living things could have lived in it. Nothing in the rock showed that any did. *Curiosity* was still working in September 2026.

The rover *Perseverance* landed in Jezero Crater in February 2021. The mission's planners chose the crater because an ancient river once flowed into it and left a delta, a fan of mud and sand. *Perseverance* drills rock samples and stores them so that a later mission could bring them to Earth. In July 2024 it found a rock that shows a possible chemical sign of ancient life.

Perseverance carried a helicopter named *Ingenuity*, the first aircraft to fly on another planet. It was built for five flights and made 72 over nearly three years, until its rotor blades broke in January 2024.

The probe *New Horizons* left Earth in January 2006. On July 14, 2015, it became the first spacecraft to visit Pluto, passing 7,800 miles above it. In 2019 it flew past an object called Arrokoth, about 4 billion miles from the Sun.

Two older probes, *Voyager 1* and *Voyager 2*, were launched in 1977. In 2012 *Voyager 1* became the first spacecraft to reach interstellar space, the space between the stars, which begins beyond the stream of particles that blows out from the Sun. *Voyager 2* followed in 2018. *Voyager 1* is the most distant object people have ever made.

Both Voyagers get their power from plutonium, a radioactive metal that gives off heat, and their power drops a little every year. In April 2026 NASA engineers switched off one of *Voyager 1*'s instruments to save power. Two of its science instruments still work.

The James Webb Space Telescope was launched on December 25, 2021, from French Guiana, in South America. It cost just under $10 billion. It was built to see light that left the first galaxies, about 180 million years after the universe began. A galaxy is a huge group of billions of stars.

James Cameron

Who: A Canadian film director who dove alone to the deepest point in the ocean.

When and where: March 26, 2012. The Challenger Deep in the Mariana Trench, in the western Pacific Ocean.

He spent about three hours alone at the bottom of the ocean.

James Cameron, a film director from Canada, became the first person to dive alone to the Challenger Deep, the deepest point in the ocean, on March 26, 2012. He piloted a small deep-diving craft, the *Deepsea Challenger*, to the bottom alone. The craft was built in Sydney, Australia, with the National Geographic Society.

The trip down took 2 hours and 37 minutes, about twice as fast as the *Trieste* in 1960. The deepest reading on the dive was 35,787 feet. No one before him had stayed on the bottom so long. A problem with the craft's hydraulic system made it hard to use the equipment for collecting samples. The hydraulic system used fluid under pressure to move the equipment.

Cameron's was the second dive with people on board to reach the Challenger Deep, 52 years after Don Walsh and Jacques Piccard. He was the third person to go there, after Walsh and Piccard. The *Deepsea Challenger* was later given to the Woods Hole Oceanographic Institution, where Robert Ballard had worked.

Mapping the sea floor

In an international project called Seabed 2030, people from many countries are working to map the whole floor of the ocean. On April 28, 2026, staff of the International Hydrographic Organization, a group of countries that work together on sea charts, published the project's latest total. The project's leaders reported that 28.7 percent of the ocean floor had been mapped. Almost five million square kilometers, about 1.9 million square miles, had been added in the year before. In June 2025 the project had reported 27.3 percent.

Artemis II crew

Who: Reid Wiseman, Victor Glover and Christina Koch of NASA and Jeremy Hansen of Canada, the first people to travel to the Moon since 1972.

When and where: April 1 to 10, 2026. From Florida around the Moon and back to the Pacific Ocean off San Diego.

They flew 252,756 miles from Earth, farther than any people before them.

Four astronauts flew around the Moon in April 2026, the first people to go there in more than 50 years. They were the commander, Reid Wiseman, the pilot, Victor Glover, and Christina Koch of NASA, and Jeremy Hansen of the Canadian Space Agency. They launched on April 1, 2026. It was the first flight with a crew for NASA's Orion spacecraft and its SLS rocket. SLS stands for Space Launch System. The crew circled around the Moon but did not land.

According to NASA's announcement of April 10, 2026, their farthest point broke the record of 248,655 miles set by the crew of Apollo 13 in 1970. A web page of the Smithsonian's National Air and Space Museum states that Koch became the first woman to travel that far from Earth. Glover became the first Black man and Hansen the first Canadian to do so.

Koch had already spent 328 days in orbit on the International Space Station, ending in February 2020. It was the longest single spaceflight by a woman up to that time.

Some experts outside NASA objected to flying this mission without a redesigned heat shield. After Orion's test flight without a crew in November 2022, NASA engineers found that its heat shield had worn away in ways they had not expected. The heat shield protects the spacecraft from the heat of coming back through the atmosphere. NASA's leaders kept the same shield for Artemis II and changed the path the spacecraft would take on the way down.

The crew splashed down safely in the Pacific Ocean off San Diego at 5:07 p.m. Pacific time on April 10, 2026, after a trip of nearly ten days.

THE THREE QUESTIONS, ANSWERED

1. **How do we know people were living here at least 21,000 years ago?** Human footprints at White Sands, in New Mexico, lie between layers of seeds, pollen and quartz sand, and three different tests date those layers to about 21,000 to 23,000 years ago.
2. **What did the explorers do to the people whose land they crossed?** Some traded with them and were fed by them. Others took captives, from the 500 Taino Columbus's men shipped to Spain to the six Inuit Peary brought to New York, and some killed: de Soto's men at Mabila, Coronado's at Arenal, Frémont's on the Sacramento River.
3. **Which famous "firsts" are still in doubt?** Peary's North Pole in 1909 and Byrd's flight over it in 1926. Wally Herbert found Peary's records unreliable, and most researchers think Byrd's plane could not carry enough fuel. Older firsts are in doubt too: Spanish slave raiders may have reached Florida before Ponce de León, Ute tradition says Ute ancestors climbed Pikes Peak before Long's men,

and nobody knows whether the Norse camp in Newfoundland was Leif's own.

What changed: For thousands of years almost every place an explorer reached was already somebody's home, and after 1950 the only places left were the deep ocean and space, where nobody had ever lived.

FIVE DATES TO KEEP

1021: Norse sailors cut trees at L'Anse aux Meadows, in Newfoundland.

1540: Hopi guides lead Spanish soldiers to the rim of the Grand Canyon.

1806: Lewis and Clark reach St. Louis with about 140 maps.

1909: Peary says his party has reached the North Pole.

1969: Armstrong and Aldrin walk on the Moon.

> **NOT KNOWN:** Nobody knows how many people lived north of Mexico in 1500, and the estimates run from about 1 million to about 18 million.

Where to go: The La Verendrye lead plate from 1743 is kept in the South Dakota Cultural Heritage Center.

Next: Chapter 2, Native Nations, which describes what happened to the nations whose land the explorers in this chapter crossed, from the nations' side.

Chapter 2: Native Nations

CHAPTER 2 OF 37 · NATIVE NATIONS · 10 ERAS · ABOUT 100 MINUTES

Native nations were governments long before the United States existed, and through every war, treaty and law aimed at ending them, 574 of them still govern themselves today.

The Haudenosaunee (hoe-dee-no-SHOW-nee), five nations that had joined together under one council, recorded their first agreement with the Dutch on a belt of shell beads. It is called the Two Row Wampum, or Guswenta. Two purple rows of beads run along a white background. They stand for two boats traveling the same river side by side, with neither one steering the other. Haudenosaunee tradition dates the agreement to 1613.

This chapter follows the nations through ten eras. It starts before 1500, with cities and confederacies, groups of nations acting together, and ends in today's courtrooms and classrooms. Watch for the nations acting as governments. Their leaders make alliances, sign treaties, write constitutions, go to war and go to court. Against them, United States officials and soldiers take most of the nations' land, through removal at gunpoint, massacres and allotment, the cutting of a nation's shared land into private pieces. Officials also take children from their homes and send them to boarding schools. After 1950 the story turns. Nations win back schools, fishing rights and, in 2020, a reservation that five Supreme Court justices said had never ended.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did Native nations govern themselves and deal with other nations before the United States existed?

2. What did United States officials and soldiers do to take the nations' land and break the nations up?
3. How did Native nations hold on to their governments, their languages and some of their land?

Five minutes? Read "Removal and the Trail of Tears" in 1800 to 1850.

Can skip: "Comanche power" and "Treaty councils" in 1700 to 1750, and "Ironworkers from Kahnawake" in 1900 to 1950. The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

The nations of this land governed themselves, traded and made war and peace for thousands of years before any ship came from Europe. At Celilo Falls, on the Columbia River, archaeologists, the scientists who dig up and study old camps and tools, date the first signs of people to at least 10,000 years ago. Some accounts say 15,000. In those thousands of years, some peoples built cities and later left them.

When Europeans first arrived, people in what are now the United States and Canada spoke more than 300 languages. That is the count of Lyle Campbell, a scholar who studies languages, in Britannica. He sorts them into 57 families of related languages, and 20 of those families were found only in California. The nations built towns and carried goods hundreds of miles to trade. They made agreements with other nations and fought wars against them.

Nobody knows how many people lived here before 1500. Scholars have used different methods and reached different answers. Their estimates for North America north of Mexico run from about 1 million to about 18 million, and no number is settled.

Cities and kingdoms

People built Cahokia, the largest city north of Mexico, near where St. Louis, Missouri, stands today. They lived there from about 700 to about 1400. The city was biggest around 1100, when roughly 10,000 to 20,000 people lived in it. London had about as many people at the same date. Some estimates put as many as 50,000 in the country around the city.

Why they left. Cahokia's people had scattered by about 1350, and by about 1400 nobody lived there. Why they left is still argued. The causes put forward are drought, floods, and running short of the wood and food its people needed. No single cause is settled.

Ancestral Puebloan people, the ancestors of today's Pueblo peoples (*pueblo* is the Spanish word for town), built large stone buildings called great houses in Chaco Canyon, in what is now New Mexico. After long droughts, people began leaving Chaco around 1140, and by 1200 most of them had gone. They moved into the Pueblo communities of the Southwest.

The Calusa of southwest Florida built a kingdom without farming. Fish and shellfish fed about 20,000 people. Their capital, Calos, stood on mounds of shell at Mound Key. The Calusa kept live fish in pens with walls of shell, called watercourts. Scientists have dated the watercourts to between 1300 and 1400.

The Calusa also collected tribute, regular payments in goods, from other groups. Those groups lived along the coast from south of Tampa Bay down to the Florida Keys.

Traders carried turquoise, seashells, copper and obsidian, a black volcanic glass that chips to a sharp edge, hundreds of miles from nation to nation. Native trails crossed the continent, and later newcomers traveled along them.

The ancestors of the Navajo (Diné) and the Apache moved south toward the Southwest from what is now western Canada and eastern Alaska. Accounts differ about when they arrived. Some date the arrival to the 1200s, and others put it in the 1400s or as late as about 1525.

How did five warring nations become one?

Five nations that had been fighting one another, the Mohawk, the Oneida, the Onondaga, the Cayuga and the Seneca, accepted an agreement called the Great Law of Peace, or Gayanesshagowa. Under it they formed one confederacy, the Haudenosaunee, a group of nations that keep their own governments but act together on matters they share. The five nations came together under one Grand Council.

According to Haudenosaunee tradition, two men brought the five warring nations together under the Great Law. They were the Peacemaker and Hiawatha. The Haudenosaunee kept the law in memory and recited it aloud. Instead of writing it on paper, they recorded it on wampum belts, shell beads woven into patterns in which each pattern stands for the terms of an agreement.

Scholars disagree about when the Confederacy began. Barbara Mann and Jerry Fields date the founding to 1142, from an eclipse of the sun described in the founding tradition. Most other estimates place the founding between the middle 1100s and the late 1400s, and many favor the middle or late 1400s.

Writers for the Library of Congress describe the Haudenosaunee Confederacy as one of the oldest democracies, governments in which the people have a say, still existing in the world.

Few records of single persons survive from these centuries.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 10 MINUTES

Spanish armies marched inland in the 1500s, and the diseases they carried reached many Native towns before the soldiers did. These expeditions were large armed groups traveling into country new to them. The diseases spread ahead of them and behind them. A town could lose many of its people before any European arrived there, and the diseases killed large shares of whole communities. Some writers call these deaths the Great Dying.

No one can give a single death toll, because the number of people living here at the start is not known. A death toll worked out from the lowest estimate, about 1 million, and one worked out from the highest, about 18 million, differ by millions of people.

United States census takers, the officials who count the people of the country, counted about 250,000 Native people in 1900. Even the lowest estimate for 1500, about 1 million, is four times that count. Disease killed many of the missing people. Over the four centuries after 1500, soldiers, settlers and militias killed many others in wars and massacres and while forcing whole nations off their land.

The entradas

Spanish leaders called their armed expeditions into the north entradas. Hernando de Soto led one across the Southeast from 1539 to 1543. Francisco Vázquez de Coronado led another across the Southwest and the southern Plains from 1540 to 1542, with more than 300 armed Europeans and at least 1,200 Native allies.

De Soto sailed from Havana in May 1539 and landed in Florida with more than 620 men and 220 horses. For four years his men marched through the chiefdoms of the Mississippians, the town-building peoples of the Mississippi Valley and the Southeast in these centuries. Each chiefdom was a group of towns under a single ruler.

What de Soto's men did. They attacked the people of the towns on their route and forced people to work for the expedition. At the town of Hymahi, de Soto took all of the town's food. Later he burned at least one of its people to death while he questioned them about another town, Cofitachequi. His men also carried disease with them.

De Soto died of a fever in the spring of 1542, and the survivors of his expedition reached Mexico on September 10, 1543. His men were the first Europeans to see those chiefdoms, and almost the last. In the decades after they attacked the towns, forced the people to work and spread disease among them, the chiefdoms broke apart. Over the century that followed, the survivors regrouped and took in refugees, people who had fled their own towns. They formed the new nations and confederacies of the Southeast.

Spanish colonists also brought horses back to North America. Horses had lived here long before, but they had died out thousands of years earlier. Christopher Columbus carried horses on his second voyage, in 1493, and Spanish colonists brought more to the mainland through the 1500s. Coronado's expedition brought horses in 1540, and Juan de Oñate's colonists brought them to New Mexico in 1598.

Paquiquineo (Don Luís)

Who: A young man from the Chesapeake Bay country who came home with a Spanish mission in 1570 and then led an attack on it.

When and where: Left the Chesapeake in 1561. Landed back on the James River in September 1570.

He met the king of Spain, then spent nine years getting home.

Paquiquineo left the Chesapeake Bay country on a Spanish ship in 1561, as a young man. The Spanish records do not settle whether he went willingly or was taken.

He appeared before King Philip II in Spain. In Mexico City he was baptized, made a member of the Christian church in a ceremony. He was given the name of the viceroy, the man who governed the colony for the king: Don Luís de Velasco.

He came home in September 1570. He landed on the James River as guide and translator for a small Jesuit mission, a group of priests from the Jesuit order of the Catholic church, with helpers, sent to turn people to Christianity. Father Juan Bautista de Segura led this mission.

Back among his people. Paquiquineo soon left the priests. A Jesuit's letter of 1572 states that he slept in their hut no more than two nights and then went to live with his brothers, a day and a half's journey away. One of the priests wrote that the people there had lived through six years of drought and famine, a long shortage of food. The priests were trading copper and tin to other villages for corn.

Paquiquineo led an attack on the mission in February 1571, and the attackers killed the missionaries. They spared the altar boy, Alonso de Olmos, who helped the priests at church services. The Spanish records do not state why Paquiquineo attacked. The anthropologist Seth Mallios argues that the priests' trading for corn broke his people's rules of gift-giving and insulted them.

The Spanish answer. Pedro Menéndez de Avilés, the Spanish governor of Florida, sent soldiers to the James River in August 1572. They captured some of Paquiquineo's people in a fight. The Spanish found several of the captives guilty of killing the missionaries and hanged them from the yardarms, the crossbeams on the masts, of their ship. The Spanish records do not give the names of the men they hanged.

The Powhatan people returned Alonso de Olmos to the Spanish. The soldiers did not find Paquiquineo. Spanish officials sent no more missions to the Chesapeake. English colonists landed at Jamestown, in the same country, 36 years after the mission, in 1607.

NOT KNOWN: what became of Paquiquineo. European records after 1572 do not include him, and no story passed down by word of mouth that can be checked describes him.

Acoma, 1598 and 1599

Juan de Oñate led Spanish colonists north from Mexico in 1598 and founded a colony in New Mexico, the country of the Pueblo nations. Acoma Pueblo, the town of the Acoma people, stands on a mesa, a hill with a flat top and steep sides.

The paper. Before any fighting, Oñate had gathered the Acoma elders for a ceremony in which they promised to obey the king of Spain. The Spanish wrote it down as the "Act of Obedience and Vassalage by the Indians of Acoma." (A vassal is a person bound to obey a ruler.) From then on the Spanish treated the Acoma as subjects of the king of Spain, bound by Spanish law. Spanish officials later used that paper to argue that their war on Acoma was legal.

Oñate's nephew Juan de Zaldívar came to Acoma in December 1598 with about thirty soldiers. He took some of them up the mesa to demand food. In the Acoma people's own account, as reporters for the Rio Grande Sun newspaper told it in 2019, the soldiers raped an Acoma woman, forcing her into sex.

The All Pueblo Council of Governors is a council of the leaders of the Pueblo nations. In a public statement in October 2023, its leaders said that the soldiers "had demanded food and supplies, assaulted an Acoma woman, and forced allegiance to the Spanish crown," which means the soldiers made the Acoma promise loyalty to the king of Spain. Then the Acoma killed Zaldívar and eleven to fourteen of his men. Accounts differ on the number.

Oñate opened a formal inquiry, an official hearing to find out what had happened, on December 28. He questioned the survivors of Zaldívar's party. He asked the Franciscan friars with him, Catholic religious brothers of the Franciscan order, whether a war against Acoma would be just. The friars answered that he "possessed both the authority and sufficient cause," which meant he had the right to make war and a good enough reason.

The attack. Juan de Zaldívar's brother Vicente led about seventy soldiers up the mesa in January 1599. Over two or three days of fighting, his soldiers killed Acoma men, women and children. They took the town and burned much of it.

Nobody counted the dead. The only detailed written story of the fighting by someone who was there is a long poem published in 1610 by Gaspar Pérez de Villagrá, a captain who rode with the expedition. He wrote it to praise the expedition.

Estimates of the dead run from about 300 to about 1,500. The figure given most often is about 800 Acoma men, women and children, and it is the figure the leaders of the All Pueblo Council of Governors use today. The National Park Service's figure is about 1,500.

Vicente de Zaldívar's soldiers captured about 500 Acoma men, women and children. One record gives the number put on trial as 507. Oñate tried them at Santo Domingo Pueblo, though some accounts place the trial at San Juan Pueblo.

The sentences. Oñate passed sentence on February 12, 1599. He sentenced men over twenty-five to have one foot cut off and to be forced to work for the Spanish without pay for twenty years. Males from twelve to twenty-five, and females over twelve, got the same twenty years of forced work. He sentenced the old people to be handed over as slaves to the Plains Apache.

Oñate sentenced two Hopi men captured in the fighting to have their right hands cut off. Then he had them released to carry word of the punishment to their own people.

Oñate declared the Acoma children under twelve innocent. He still ordered them taken from their parents and handed to the friars. On Oñate's order, Acoma girls under twelve were sent south along the road to Mexico, to convents there, houses of nuns, women who have promised their lives to the Catholic church. Neither the National Park Service's account nor the Pueblo council's statement names the men who took the girls.

How many. The two accounts differ on the number. The National Park Service's account records about seventy girls. The Pueblo council's 2023 statement records sixty children taken to Mexico and states that none of them returned. The National Park Service's account states that almost certainly none ever came home. According to PBS, some scholars believe the children were later sold as slaves.

The feet. Oñate's sentence is in the written record. Whether anyone carried out the cutting of feet is disputed. Acoma people have passed down their own account since 1599: Oñate had the right feet of 24 captive men cut off. The New York Times reported that account in 1998. Two historians who worked for years in these records doubt that the feet were cut off. Marc Simmons first wrote that they were, and he later came to doubt it. John Kessell notes that no record describes a one-footed Acoma man.

Spanish officials charged Oñate with thirty crimes. Accounts date the charges to 1612 or 1613 and the judges' decision to 1614. The judges found him guilty of twelve, among them using too much force against Acoma. They banished him from New Mexico for life, ordering him never to return. They also barred him from Mexico City for four years (PBS gives five) and fined him 6,000 ducats, Spanish gold coins.

Oñate had rebuilt a fortune in his family's mines at Zacatecas. Rebecca Carlson, who wrote about his trial in a 2008 university paper, calls the fine unimportant to him. He spent years working to have the decision undone.

Accounts differ on whether the king ever pardoned him, which means officially forgave him. Carlson writes that he was pardoned in 1623. Other accounts say the proof of a pardon is unclear.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 12 MINUTES

Native nations still held nearly the whole continent through the 1600s. Along the eastern coast, the European settlements were a thin strip. Those settlers needed food and trade from the nations around them, and they stayed only as long as those nations let them. In New Mexico, Spanish colonists had ruled the Pueblo nations by force since 1598. Native leaders chose which newcomers to deal with, the Dutch, the French, the English or the Spanish, and used each group's rivalry with the others to bargain.

Haudenosaunee leaders made alliances with the newcomers and recorded them on wampum belts. In New England, English colonists fought two wars against coastal nations, the Pequot War of 1636 to 1638 and Metacom's war of 1675 and 1676. In the Southwest in 1680, the people of the Pueblo nations rose together and drove the Spanish off their land. They kept the Spanish out for twelve years.

Who ruled the land where Jamestown stood?

Wahunsenacawh ruled the country where English colonists founded Jamestown in 1607. He had inherited rule over six tribes in eastern Virginia and built them into a group of more than thirty tribes that paid him tribute. His country was called Tsenacommacah. The English called him "the Powhatan." His own title was mamanatowick, or paramount chief, a chief whom other chiefs answer to.

About 15,000 people lived in his country, spread across some 10,000 square miles between the James River and the Potomac. Each tribe there was a people with its own leader, called a weroance, and each weroance paid tribute to Wahunsenacawh.

Wampum diplomacy and the Covenant Chain

Haudenosaunee diplomats, the people who spoke for their nations with other nations, recorded agreements on wampum belts. Speakers kept the words of each agreement in memory, and the pattern of the belt recorded its terms.

The Two Row Wampum, Guswenta, records the Haudenosaunee account of their first agreement with the Dutch: two boats side by side on one river, neither one steering the other. A paper document dated 1613, the year Haudenosaunee tradition gives for the agreement, also exists, and accounts differ about whether it is genuine.

The Haudenosaunee alliance with the English colonies was called the Covenant Chain, from covenant, a serious promise. The oldest surviving record of the Silver Covenant Chain dates from 1677. That year Haudenosaunee leaders and the New York governor, Edmund Andros, made the alliance official at Albany.

Polishing the chain. Silver tarnishes, turning dark over time, so a silver chain has to be polished to stay bright. Haudenosaunee and English leaders "polished" the Covenant Chain again and again. They renewed the alliance at councils, with speeches and wampum.

The coastal wars

In Connecticut, English colonists and their Mohegan and Narragansett allies fought the Pequot from 1636 to 1638. The war began with the killing of two English traders. Pequot men killed one trader and his crew in 1634. In 1636 people from Block Island, who were not Pequot, killed the other. The leaders of Massachusetts Bay sent 90 soldiers in answer, and the soldiers burned the Block Island villages and then a Pequot village.

Mystic. An English-led force set fire to the walled Pequot town at Mystic before dawn on May 26, 1637. In under an hour they killed hundreds of people, most of them elders, women and children.

The English colonists and their Mohegan and Narragansett allies ended the war with the Treaty of Hartford of 1638, a treaty being a formal written agreement between nations. In it the winners declared that the Pequot nation no longer existed and made it illegal to use the name Pequot. Under the treaty, they took Pequot survivors for themselves or sold them into slavery. The Mashantucket Pequot Tribal Nation is in Connecticut today.

Sold overseas. Colonists had already begun selling captives during the war. Captain William Peirce sailed the ship *Desire* out of Boston in July 1637 with seventeen Pequot captives, fifteen of them children. Governor John Winthrop of Massachusetts wrote, "We sent fifteen of the boys and two of the women to Bermuda, by Mr. Pierce." Peirce sold them in the Caribbean.

Metacom, the Wampanoag sachem, or leader of the nation, led a coalition of nations, a group that agreed to act together, against the New England colonies. His coalition fought the colonists from 1675 to 1676, nearly forty years after the Pequot War.

The war began in June 1675, just after a hanging at Plymouth. A jury of English colonists there had found three Wampanoag men guilty of murdering John Sassamon, a Christian Native man. By report, Sassamon had warned Plymouth's governor that Metacom's men planned an attack. Plymouth officials hanged two of the three men on June 8 and shot the third a month later.

The toll. About 9,000 people died in all, and estimates put Native deaths at roughly half of all the Native people in New England. By one estimate, about two of every three of the dead were Native people, and many of them starved. For the size of the population, writers for Britannica and other histories call it the bloodiest war in American history.

The men who governed the Massachusetts colony ordered on October 13, 1675, that Native people from the praying towns, villages of Native people who had become Christians, be taken to Deer Island in Boston Harbor and kept there. The colonists feared these people would turn against the English and join Metacom. Colonists took the people of Natick to the island first.

Deer Island. The National Park Service puts the number held there at about 500 to 1,100, most of them women and children. They had too little food and shelter through a freezing winter. The two accounts of the deaths differ. The National Park Service says that very few of them survived. The Partnership of Historic Bostons, a Boston history group, says that more than half of them died over the winter.

Colonists released the survivors in May 1676. An unknown number had been smuggled off the island and sold as slaves in the West Indies or in Tangier, in North Africa. No surviving record names the men who voted for the island prison, and the National Park Service's account does not name the men who sold the captives. Every year, descendants of the nations in the war hold a gathering on Deer Island to remember them.

Inland, from the middle to the late 1600s, Haudenosaunee warriors fought the Beaver Wars against nations to the west and north. They fought for control of the fur trade, the trade in animal furs with Europeans. More than 1,000 Seneca and Mohawk warriors attacked the two main towns of the Wendat, also called the Huron, in 1649. The warriors destroyed the Wendat nation as a single people.

Some Wendat survivors fled to Quebec, and others went to the Neutral nation. Haudenosaunee warriors defeated the Neutral in 1651. They absorbed the Erie in 1657, taking the Erie survivors into the Haudenosaunee nations. By 1675,

the writers of the Canadian Encyclopedia say, they had absorbed the Susquehannock to the south and moved west into the Ohio Valley.

Metacom (King Philip)

Who: Wampanoag sachem, son of Massasoit, who led the coalition war of 1675 and 1676 against the New England colonies. The English called him King Philip.

When and where: Southern New England. Killed at Mount Hope, August 12, 1676.

"Philip's boy goes now to be sold," a minister wrote in 1677.

Metacom's father, Massasoit, had made an arrangement with the Plymouth colonists in 1621. In 1675 Metacom took a coalition of nations to war against the New England colonies. The company of the Plymouth captain Benjamin Church captured Metacom's wife and his nine-year-old son in early August 1676.

Mount Hope. A Native man named Alderman, who was serving with Church's company, shot Metacom dead at Mount Hope on August 12, 1676. Church wrote his own account of what came next. He ordered that "not one of his bones should be buried," because Metacom had left English dead unburied. Church told a Native man he called his "old Indian Executioner" to cut off Metacom's head and to quarter his body, cutting it into four pieces.

Church gave the head and one of Metacom's hands to Alderman, who showed the hand to colonists who paid him to see it. Plymouth's leaders paid Church's men thirty shillings, a sum in English money, for each enemy they had killed or captured, and they paid the same for Metacom's head. At Plymouth the head was set on a pole. Church's book does not say who put it there. By the account of a later editor of Church's book, it stayed there for more than 24 years.

The boy. The leaders of Plymouth Colony asked ministers what to do with Metacom's son. Two ministers, Samuel Arnold and John Cotton, wrote on September 7, 1676, that the children of such enemies could be put to death. Cotton wrote the line about "Philip's boy" in a letter in March 1677. The boy and his mother were sold into slavery. Cotton's letter does not say who sold them. Most accounts say the two were sent to Bermuda, and others say they were sent to the West Indies.

Colonial leaders sold hundreds of other captives into slavery to pay what the war had cost the colonies. The selling began early in the war. On August 4, 1675, the men of Plymouth's council of war ruled that 112 Native men, women and children held at Plymouth were "to be sold," and they named the colony's treasurer to sell them.

A month later the council condemned 57 more people to slavery for life. Church wrote that these people had surrendered because of promises made to them, and that about 160 in all were "sold, and transported out of the Country."

The Pueblo Revolt

The people of the Pueblo nations along the Rio Grande drove the Spanish out of New Mexico in 1680. They had lived under Spanish rule since Juan de Oñate founded the colony in 1598.

The arrests. The Spanish governor, Juan Francisco Treviño, ordered 47 Pueblo religious leaders arrested in 1675. The Spanish called their religious practice sorcery, using magic to do harm. Spanish officials sentenced four of the leaders to be hanged. Three were hanged, and the fourth killed himself. Spanish officials whipped the others in public and put them in prison. Pueblo people went to Santa Fe and demanded that the prisoners be freed, and the Spanish let them go. One of the freed leaders was Po'pay.

The people of many separate towns rose against the Spanish on the same day, August 10, 1680. They spoke Tewa, Tiwa, Towa, Tano, Keres and other languages, and the Zuni and the Hopi also promised to help. Pueblo fighters besieged Santa Fe, surrounding the town and cutting it off. They took the town and drove the Spanish out of New Mexico. In the rising they killed 401 settlers and 21 Franciscan friars. The Pueblo fighters let most of the Spanish leave, and the Spanish went south.

The Pueblos governed themselves for the next twelve years, through drought, raids and famine, a long shortage of food. Diego de Vargas led Spanish forces back into New Mexico in 1692 to retake it, and Pueblo fighters kept up armed resistance for years after that. Writers of the EBSCO Research Starters history guide call the Pueblo Revolt the most successful uprising by Native people against European rule in North America.

The horses left behind. When the Spanish went south out of New Mexico in 1680, they left their horses. An estimated 1,500 or more horses passed into Native hands at once. Writers for the Smithsonian's National Museum of the American Indian call it the largest single transfer of horses. According to the traditional account, Native people then traded horses west and north along Native trade routes. The horses reached the Ute, the Navajo, the Apache, the Comanche, the Kiowa, the Shoshone and the nations of the northern Plains.

A newer finding. A team led by William Taylor published a study in the journal *Science* in 2023, and its findings do not fit this account. Lakota, Comanche and Pawnee people were among the study's writers, and other Native partners worked on it too. The team used radiocarbon dating on the remains of horses from Wyoming, Kansas and New Mexico. Radiocarbon dating measures a form of carbon that breaks down at a steady rate, so the amount left shows how old a bone is.

The horses had been ridden and had eaten corn that people fed them. They had also worn bridles, straps around a horse's head used to steer it. They were there by the early 1600s, decades before the revolt. Native oral histories, accounts passed down by telling, had said the same thing.

KNOWN:	Plains nations had horses by the 1700s.
NOT KNOWN:	whether most of their horses came from the Spanish herds left in 1680 or had spread north decades earlier.

Po'pay (Popé)

Who: A Tewa religious leader from Ohkay Owingeh who organized the 1680 rising across dozens of towns and several languages.
When and where: New Mexico, August 1680.

Knotted cords counted down the days to the rising.

Po'pay had to get the people of dozens of towns to rise on a single day, and the people of those towns did not all speak the same language.

Pueblo accounts say that runners carried knotted cords from town to town. The knots on each cord marked the days left until the rising. Po'pay set the rising for August 11, by one account, or August 13, by another.

The runners. The Spanish captured two of the runners, Pedro Omtua and Nicolás Catua of Tesuque Pueblo, on August 9. Spanish officials tortured them, hurting them on purpose to force them to explain the

knotted cords. The Smithsonian's account of the runners does not say how they were hurt. Later the Spanish hanged them. No surviving record names the men who did it.

When Po'pay learned that the runners had been caught, he ordered the rising to begin early. It began on August 10, 1680.

A statue of Po'pay stands in the United States Capitol. It is one of New Mexico's two statues in the Capitol's Statuary Hall collection, which holds statues of two people from each state's history.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

Haudenosaunee leaders bargained with French and British officials for fifty years, because both sides wanted Native allies. Native nations also went on making their own agreements with one another.

The Haudenosaunee accepted the Tuscarora as a member nation in these years, after South Carolina forces had defeated the Tuscarora and colonists had taken their land in North Carolina. On the southern Plains, the Comanche built the strongest power in the region on horses and trade.

The printed speeches of the Onondaga speaker Canasatego, from these years, are among the best records of Native diplomacy, the work of dealing with other nations through talks and agreements, from that century.

Staying neutral

About 1,300 delegates, people sent to speak for their nations, came from more than thirty nations to meet at Montreal on August 4, 1701. There they made peace with the leaders of New France, the French colony in what is now Canada. As part of the same peace, the Haudenosaunee promised to stay neutral. That meant they would fight for neither France nor Britain.

For half a century, Haudenosaunee leaders used their neutrality to bargain with both sides.

How did the Five Nations become six?

The Five Nations of the Haudenosaunee became six after the Tuscarora lost a war, and their land, in North Carolina. The Tuscarora moved north and asked to join them.

The war. More than 400 Swiss and German settlers, led by Baron Christoph von Graffenried, founded the town of New Bern, North Carolina, on Tuscarora land in 1710. They pushed the people of the Native town of Chattoka off the spot where the Neuse and Trent rivers meet. Slave raiders were also capturing Tuscarora people to sell them as slaves. The State Library of North Carolina's history of the war does not name the raiders.

Tuscarora fighters struck back on September 22, 1711, in attacks they had planned together. The war lasted until 1713. South Carolina colonial forces, with Native allies of their own, defeated the Tuscarora. By that library's count, 1,400 Tuscarora were dead and 1,000 had been captured and enslaved.

About 1,500 Tuscarora survivors then moved north, roughly 500 miles. They asked the Five Nations of the Haudenosaunee to sponsor them, which meant to speak for them and accept them. The Oneida took them in. In 1722 the whole Tuscarora nation became the sixth nation of the Haudenosaunee Confederacy. Since then the Confederacy has had six member nations, and it is also known as the Six Nations.

Comanche power

The Comanche moved onto the southern Plains in the early 1700s and became the most powerful nation there. Comanchería, the country they controlled, later covered about 240,000 square miles. That is twenty-four times the size of Wahunsenacawh's country in Virginia.

The Comanche built their power on horses. They bred, trained and traded horses, and they controlled the horse trade from the southern Plains to the northern Plains. In the 1740s they traded bison robes and horses to French traders for guns, which were better than the weapons the Comanche could get from the Spanish. Comanche fighters also raided other peoples, making quick attacks to seize goods, horses or people. Comanche leaders made agreements with other nations as well as fighting them.

Treaty councils

At treaty councils, Native leaders and colonial officials met to bargain and make agreements. Speakers on both sides made speeches and exchanged wampum.

At Lancaster, Pennsylvania, in June and July of 1744, colonial officials paid the Haudenosaunee to give up their claims to the Shenandoah Valley.

Canasatego

Who: Onondaga leader and the Haudenosaunee's speaker at treaty councils, whose 1744 speech on union Benjamin Franklin printed.

When and where: Lancaster, Pennsylvania, June and July 1744. He gave the union speech on July 4, 1744.

On July 4, 1744, he told the colonies to unite.

Canasatego spoke for the Haudenosaunee Confederacy at treaty councils with colonial officials. He closed the council at Lancaster by advising the colonies to unite, as his own nations had done centuries earlier: "We heartily recommend Union and a good Agreement between you our Brethren… you, as well as we, will become the stronger."

Benjamin Franklin printed the minutes of the council, the written record of what was said there, including the speech.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 8 MINUTES

French and British armies fought their wars on Native land after 1750. Then the leaders, soldiers and settlers of a new country, the United States, moved onto that land and fought the nations for it.

When British forces beat the French in 1763, Native nations lost the second empire they had bargained with against Britain. Then the colonists went to war with Britain in the American Revolution. The Haudenosaunee nations took different sides in that war, and their Confederacy split for the first time in centuries. During the war, in 1778, United States officials signed their first treaty with a Native nation. In 1791 fighters from a confederacy of nations in the Ohio Country destroyed a United States army.

Pontiac and the Line

French and British commanders both needed Native allies in the French and Indian War, which lasted from 1754 to 1763. A loose confederacy of nations from the Great Lakes, the Illinois Country and the Ohio Country rose against the British in 1763. Under the Odawa leader Pontiac, their fighters took or besieged British forts across the west. Pontiac lifted his siege of Detroit in October 1763, but the fighting went on until 1766.

Fort Pitt. British officers at Fort Pitt tried to give smallpox, a disease that spreads from person to person and can kill, to the Native fighters around them in 1763. Delaware, Shawnee and Mingo fighters had besieged the fort, where Pittsburgh stands today, and smallpox had broken out among the British soldiers inside.

The blankets. The fort's commander, Captain Simeon Ecuyer, and the trader William Trent gave two Delaware men, Turtle's Heart and Mamaltee, two blankets and a handkerchief from the fort's smallpox hospital on June 24, 1763. Trent wrote in his journal, "I hope it will have the desired effect." In July

the British commander in North America, General Jeffery Amherst, asked Colonel Henry Bouquet in a letter whether smallpox could be sent among the nations at war. Bouquet answered that he would try to spread it with blankets.

Whether it worked. A month after the June meeting, Trent saw the two Delaware men again, and they showed no sign of smallpox. A man named Gershom Hicks later said that about 100 Native people died of smallpox between the spring of 1763 and April 1764. The historian Elizabeth Fenn writes that no one can know whether the attempt to spread smallpox worked.

The Line. British officials answered the war by drawing a line. They issued the Royal Proclamation, an official public announcement of an order, on October 7, 1763. In it they forbade colonists to settle west of the Appalachian Mountains, and they set aside the land beyond for the Native nations. Colonists were angry about the line and crossed it anyway. Writers for Encyclopedia Virginia call the Proclamation a foundation of later treaty law, the set of rules for agreements between governments.

Paxton. While the war went on, settlers in Pennsylvania attacked Native people who were not fighting in it. Armed settlers from Paxton Township killed the twenty Conestoga people who lived in a small town near Lancaster in December 1763. The Conestoga were unarmed. Fifty-seven men came at daybreak on December 14 and found three men, two women and a boy at home. They shot them, stabbed them and killed them with hatchets.

One of the dead was Shehaes, a very old man who had helped make a treaty with William Penn in 1701. They killed him in his bed. The attackers scalped all six, cutting away the skin and hair from the top of each head, and burned the huts.

The workhouse. To keep them safe, Lancaster's officials put the other fourteen Conestoga in the town workhouse, a building where poor people were made to work. Franklin called it "a strong Building." On December 27 about fifty men broke down the workhouse door and killed all fourteen with guns and hatchets, children a year old among them.

The Paxton men claimed the Conestoga had hidden enemy spies. The historian Michael Goode writes that the charge was made up. British soldiers stationed nearby did nothing to stop the killings, and no one was ever put on trial for them. Benjamin Franklin wrote down the names of the dead in 1764.

The Revolution splits the Haudenosaunee

Under the Great Law, the Five Nations, and then the Six, had stayed united for centuries. In the American Revolution they split. The Oneida and the Tuscarora sided with the Americans. The Mohawk, the Seneca, the Cayuga and the Onondaga fought with the British. Many of them fought under the Mohawk leader Joseph Brant, whose own name was Thayendanegea. At the Battle of Oriskany, on August 6, 1777, Haudenosaunee men fought on both sides.

George Washington ordered an expedition into Seneca and Cayuga country in 1779. In his written orders, he directed "the total destruction and devastation of their settlements." From July 31 to October 15, General John Sullivan's soldiers burned more than forty Seneca and Cayuga towns and the crops in their fields. Thousands of people fled to the British at Fort Niagara. That winter many of them starved or froze to death there.

Seneca chiefs spoke to Washington directly in 1790. "When your army entered the country of the Six Nations, we called you the Town-destroyer," they said. They told him that "to this day, when that name is heard, our women look behind them and turn pale." The name was older than the burning of their towns. Washington had been given the name Conotocaurius, Town Destroyer, in 1753, and his great-grandfather had been given it before him.

The treaty system begins

United States officials signed their country's first treaty with a Native nation at Fort Pitt on September 17, 1778. The Lenape leaders White Eyes, John Kill Buck Jr. and Pipe signed for their nation, which the treaty calls the Delaware. In the treaty, the American officials recognized Lenape sovereignty, the right of a nation to govern itself. They also asked the Lenape to let American troops cross their land safely.

A fourteenth state. The negotiators raised the idea of a Native fourteenth state, which would join the thirteen and send its own representatives to Congress. No one in Congress ever acted on that idea.

In all, United States officials and Native leaders made 374 treaties that were formally approved, or ratified. The last of them date from the 1860s. The National Archives holds all 374, and its archivists, the people who keep its records, have put every one online. Each of the 374 is an agreement between the United States and a Native nation.

The Northwest Confederacy

North of the Ohio River, the nations built a confederacy to defend their country. Its leaders included Little Turtle of the Miami, Blue Jacket of the Shawnee and Buckongahelas of the Lenape.

About a thousand confederacy fighters fought the army of General Arthur St. Clair on November 4, 1791. About 930 of his soldiers began the fight, and the confederacy's fighters destroyed his army. Counts of the soldiers they killed run from more than 630 to more than 650, and counts of the wounded from 264 to more than 270. Of every 100 soldiers who fought, about 97 were killed or wounded. No battle in the history of the United States Army has had a higher share.

Native casualties, the killed and wounded together, were roughly 100. St. Clair's defeat is still the largest defeat Native forces have ever given a United States army, larger than the Battle of the Little Bighorn, fought 85 years later. After the defeat, members of Congress held the first investigation by Congress in the country's history.

Anthony Wayne led a legion, an army unit of 3,300 men, against the confederacy in 1794. Wayne's legion defeated the confederacy at Fallen Timbers, near where Toledo, Ohio, stands today, on August 20, 1794. The confederacy's leaders had lost the war. At Greenville, on August 3, 1795, they signed a treaty with Wayne, who signed for the United States, and in it they signed away most of Ohio and part of Indiana.

Little Turtle (Mishikinaakwa)

Who: Miami war leader who co-led the confederacy that destroyed St. Clair's army in 1791, then decided the war could not be won and argued for peace.

When and where: The Wabash and Maumee country, in present-day Indiana and Ohio, in the 1780s and 1790s.

He beat two American armies, then argued for peace.

A year before St. Clair's defeat, in 1790, warriors led by Little Turtle beat another American army, under General Josiah Harmar.

After the confederacy lost at Fallen Timbers in 1794, Little Turtle decided that the war could not be won. He signed at Greenville the next year and spent his last years arguing against more war with the United States.

At his urging, members of Congress passed a law in 1802 that let the president stop traders from selling liquor to Native nations. War Department officials then banned traders from bringing alcohol onto Native land, but they did little to enforce the ban.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 10 MINUTES

East of the Mississippi River, Native leaders tried several ways to keep their nations' land. North of the Ohio River, people from many nations joined a confederacy and fought American armies. In the Southeast, Cherokee leaders wrote a constitution, a written plan for their government and its basic laws, and took their fight to court. In Florida, Seminole fighters went to war against removal.

Two Shawnee brothers, Tecumseh and Tenskwatawa, built the northern confederacy around one idea. Tecumseh said that the land belonged to all the nations together, so no single nation could sell any part of it. Soldiers under the American commander William Henry Harrison burned the confederacy's center in 1811, and American soldiers killed Tecumseh in battle in 1813.

The Cherokee changed their nation fast in the eleven years from 1821 to 1832. Sequoyah showed a way to write the Cherokee language in 1821. Cherokee leaders approved their constitution, with three branches of government, in 1827, and the nation began printing a newspaper the next year. Then, in 1832, Chief Justice John Marshall, the head of the Supreme Court of the United States, wrote the court's ruling in the Cherokee Nation's favor.

Members of Congress passed the Indian Removal Act in 1830. Under it, United States soldiers forced the nations of the Southeast out of their homes and moved them west. In the Cherokee removal alone, soldiers held about 16,000 people in stockades, pens fenced with upright logs, and more than 4,000 of them died of cold, hunger and disease. In the West, the people of those nations built their governments again.

Tecumseh and Tenskwatawa

A Shawnee man named Tenskwatawa began preaching around 1805 that Native people should return to Native ways of living. People called him the Prophet. His brother Tecumseh took the movement to other nations and asked them to join it. In 1808 the brothers built a town on the Wabash River as the center of a confederacy of many nations. The town was called Prophetstown.

William Henry Harrison, an American governor, was buying Native land through treaties. Native leaders signed the Treaty of Fort Wayne with him in 1809 and sold him land.

At Vincennes. Tecumseh answered Harrison at Vincennes in August 1810. The land, he said, "never was divided, but belongs to all for the use of each." He added, "No part has a right to sell, even to each other, much less to strangers." He called the latest sale "bad," because "it was made by a part only." By that argument, the Fort Wayne sale counted for nothing, and so did every sale made by one nation alone.

Harrison marched about a thousand men to Prophetstown on November 7, 1811, while Tecumseh was away asking more nations to join the confederacy, and fought the people there. The fight became known as the Battle of Tippecanoe. On the next day, November 8, Harrison burned the town.

Fighters from the confederacy fought on beside the British in the War of 1812. They fought the Battle of the Thames at Moraviantown, in the British colony of Upper Canada, on October 5, 1813. Their British ally, Major General Henry Procter, left the battlefield in the middle of the fight. British army officers later put Procter on trial for his failures in that campaign. American soldiers killed Tecumseh in the battle. After his death the confederacy broke apart.

Tecumseh

Who: Shawnee leader who went from nation to nation arguing that Native land belonged to all the nations together, so no single nation could sell it.

When and where: the Wabash River country and the South, 1805 to 1813. Killed in battle at the Thames, October 5, 1813.

Movie: *We Shall Remain: Tecumseh's Vision* (PBS, 2009), a documentary about him and his brother.

He carried his argument to a Creek council of 5,000 people.

Tecumseh arrived at the yearly great council of the Creek Nation at Tuckabatchee in October 1811. More than 5,000 Creek people had gathered there. He asked them to join the confederacy of the northern nations.

The Creek leader Big Warrior refused to join him, because he was afraid of losing the support of the United States.

Tecumseh was still away in the South a month later, when Harrison marched on Prophetstown.

How did the Cherokee learn to read in a few years?

A Cherokee man named Sequoyah showed a syllabary he had made in 1821, a set of 86 written characters in which each character stands for a whole syllable, not a single sound. Within a few years most Cherokee people could read and write it. Writers at the website History.com state that by the 1830s a larger share of Cherokee people could read than of the white settlers around them.

Cherokee leaders set up a capital at New Echota in 1825. Two years later they approved a written constitution, modeled on the United States Constitution, with three branches of government. Officials of the state of Georgia answered by declaring the Cherokee government abolished, ended by law.

The Cherokee began printing the *Cherokee Phoenix*, the first Native American newspaper, in 1828. They printed it in Cherokee and in English.

Sequoyah

Who: Cherokee silversmith who could not read or write any language, and made a way to write his own. He is the only person in recorded history known to have created a writing system alone, without first being able to read and write another language.

When and where: born in Taskigi, Tennessee, about 1770. First showed his syllabary in 1821. Died 1843 in Mexico.

He could not read any language, and he wrote one down.

Sequoyah could not read or write English or any other language. He worked on the problem for about twelve years before he showed his syllabary in 1821.

Seven years later, half of what the nation's newspaper, the *Cherokee Phoenix*, printed was in his characters. He died in 1843, in Mexico.

NOT KNOWN: the year he was born. The years given for it run from 1770 to 1778, and the Library of Congress lists 1770 with a question mark.

Removal and the Trail of Tears

President Andrew Jackson signed the Indian Removal Act into law on May 28, 1830. Under it, the president could give Native nations land west of the Mississippi River in exchange for their land in the East. Cherokee leaders took their fight with Georgia to the courts.

The court case. Georgia's lawmakers had made it a crime for a white person to live in the Cherokee Nation without a license from the state. Samuel Worcester, a missionary who translated parts of the Bible into Cherokee, refused to leave. Georgia guards arrested him and other missionaries in July 1831. They were tried in Georgia that September, and each was sentenced to four years in prison at hard labor, which means forced hard work. The New Georgia Encyclopedia's account of the case does not name the judge.

Chief Justice John Marshall ruled in *Worcester v. Georgia*, in 1832, that the laws of Georgia had no force inside the Cherokee Nation, which was under the protection of the United States. That ruling is still one of the foundations of tribal sovereignty law, the law about a Native nation's right to govern itself. Georgia's officials ignored it. Worcester stayed in prison until January 1833, when he accepted a pardon, an order forgiving the crime, from Georgia's governor.

The treaty. Three years after the ruling, in late 1835, a small group of Cherokee men who favored moving signed the Treaty of New Echota with United States officials. The signers gave up all of the nation's land east of the Mississippi for $5 million. Members of the Senate approved the treaty in May 1836, and United States officials treated it as an agreement by the whole nation to move west.

The roundup. President Martin Van Buren sent General Winfield Scott with about 7,000 soldiers in May 1838 to round up the Cherokee people who had not moved west. Georgia's militia, its citizen soldiers, joined them. In 1838 and 1839, soldiers came to Cherokee houses in the Southeast and took families out at gunpoint, often with no time to pack. They held about 16,000 Cherokee people in stockades in Tennessee and Alabama.

Then the soldiers forced them to go to Indian Territory, in what is now Oklahoma. More than 4,000 of them died of cold, hunger and disease, about one in every four of the people the soldiers had held.

How many died. That figure is the traditional estimate, and the counts differ. The National Park Service counts more than a thousand deaths on the trip itself, and perhaps several thousand more from what the removal did to people afterward. Russell Thornton, a scholar at the University of Minnesota, wrote in 1984 that the number of deaths "may have been twice" the traditional figure.

The Cherokee named the removal *Nunna daul Tsuny*, the Trail Where We Cried. In English it is called the Trail of Tears.

United States officials and soldiers forced all five of the large southeastern nations to move to Indian Territory: the Cherokee, the Muscogee (Creek), the Choctaw, the Chickasaw and the Seminole. They removed more than 14,000 Choctaw people from 1831 to 1833, and hundreds died. They removed more than 14,500 Creek people in 1836 and 1837. Several hundred Creek people died on the way, and about 3,200 died after they arrived, of disease, hunger and cold.

Rebuilding. In Indian Territory, the people of those nations rebuilt their governments, their courts, their newspapers and their schools.

Rebecca Neugin

Who: a Cherokee woman who was about three years old when soldiers took her family out of their home in Georgia in 1838.

When and where: Georgia to Indian Territory, 1838 and 1839. Interviewed in 1932.

Ninety-four years later, she told a historian what she remembered.

Soldiers took Rebecca Neugin's family out of their home in Georgia in 1838, when Rebecca was about three years old. The soldiers held the family in a camp in Tennessee, and in September 1838 the family set out west. Rebecca and the smaller children rode in a wagon, while her mother, her father and her older brother Dick walked the whole way.

The historian Grant Foreman interviewed her in 1932, when she was about 97 years old, and she told him what she remembered of the journey. She was one of the last people still alive who had been on it. She died on July 15, 1932.

The Seminole resistance

In Florida many Seminole people refused to leave, and United States soldiers fought them in the Second Seminole War, from 1835 to 1842. It was the longest war between United States forces and a Native nation east of the Mississippi, and the one that cost the most. National Park Service writers give its cost in a count. For every Seminole person soldiers killed, they forced four out of Florida. For every four forced out, three soldiers died and United States officials spent $32,000.

General Thomas Jesup lured the Seminole war leader Osceola to a talk on October 21, 1837, under a white flag of truce, a sign that both sides promise not to attack or seize anyone while they talk. Osceola came in under the flag, and Jesup had him seized. Many people in the United States protested the trick in public at the time. Osceola died a prisoner at Fort Moultrie, South Carolina, on January 30, 1838.

Most of the fighting stopped in 1842. A few hundred Seminole people never surrendered and stayed in the Everglades. Their descendants belong to two tribes today. Federal officials, the officials of the national government, recognized the Seminole Tribe of Florida in 1957 and the Miccosukee Tribe in 1962, accepting each tribe as a nation with its own government. Leaders of both tribes point out that the Seminole were never conquered.

Osceola

Who: Seminole war leader in the fight against removal from Florida.

When and where: Florida. Seized by General Thomas Jesup under a flag of truce, October 21, 1837. Died a prisoner at Fort Moultrie, South Carolina, January 30, 1838.

He was born Billy Powell, in a Creek town in Alabama.

Osceola was born in 1804 in the Creek town of Talisi, in what is now Alabama. His name at birth was Billy Powell.

He died a prisoner at Fort Moultrie three months after Jesup seized him. Frederick Weedon, the army doctor there, cut off Osceola's head before the burial and put it on display in his drug store.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 22 MINUTES

United States officials and soldiers forced one Native nation after another onto reservations, pieces of land set aside for each nation in a treaty or by an order, between 1850 and 1900. Soldiers and officials held the people there, and officials and settlers took the rest of each nation's country.

United States soldiers killed hundreds of Native people at a time in these years. Army officers hanged 38 Dakota men together at Mankato, Minnesota, in 1862. At Bear River in 1863 soldiers killed an estimated 250 to 500 Shoshone people, and at Sand Creek in 1864 about 230 Cheyenne and Arapaho people, most of them women, children and elders.

Cavalry, soldiers on horseback, killed about 177 Piikani people on the Marias River in 1870, by the best estimate. Soldiers at Wounded Knee in 1890 killed between 250 and 300 Lakota people.

Hunters killed off the bison herds that fed the Plains nations, and army officers encouraged the killing. Navajo leaders signed a treaty in 1868 in which United States officials agreed to let their people go home. Members of Congress stopped making treaties with Native nations in 1871. Lakota, Cheyenne and Arapaho fighters beat George Custer's soldiers at the Little Bighorn in 1876, and the next year the Nez Perce fled toward Canada with the army behind them.

An Army officer named Richard Henry Pratt opened the first federal boarding school for Native children built far from the reservations in 1879. Federal officials used Pratt's school as the model for many more. Pratt stated his aim in his own words: "Kill the Indian in him, and save the man." Federal investigators later counted at least 973 children who died at the federal boarding schools.

Members of Congress passed a law in 1887 under which federal officials cut nations' shared land into private pieces. The officials opened what was left over to white settlers.

The wars, from the nations' side

The Dakota War. In Minnesota in 1862, United States officials failed to deliver the food and the payments that they had promised the Dakota in treaties. Dakota fighters went to war, and the war lasted six weeks.

The trials. Afterward, army officers sitting as military commissions, courts run by the army, tried 392 Dakota men. Some of the trials took less than five minutes. The men had no defense lawyers and no interpreters, the people who translate between two languages. The officers on the commissions sentenced 303 of the men to death by hanging.

President Abraham Lincoln and his lawyers read the trial records. Lincoln looked first for men found guilty of rape, forcing a person into sex, and found two. Then he added men found to have taken part in killing people who were not soldiers. He cut the list of men to be executed down to 39. One of the 39 was reprieved, which means his execution was called off.

Mankato. Army officers hanged 38 Dakota men together at Mankato on December 26, 1862, while about 4,000 people watched. It is the largest mass execution, the most people put to death at one time, in United States history. Captain William Duley cut the rope. Several members of his family had been killed in a Dakota attack on the Lake Shetek settlement.

The bodies were put in a shallow grave on a sandbar. The Minnesota Historical Society's account does not say who buried them. Before morning, doctors had dug up most of the bodies to use for medical study.

Hanged by mistake. Two of the 38 men were hanged by mistake, by the Minnesota Historical Society's account. The first was Wicaŋȟ˘pi Wastedaŋpi. He went by Caske, a common name that means first-born son. Dakota families gave it to a first-born boy, so many men had it. When the name Caske was called for another man on the list, Wicaŋȟ˘pi Wastedaŋpi stepped forward. He was hanged in that man's place.

The second was Wasicuŋ, a young white man whom the Dakota had adopted as a child. The commission had found him not guilty at his trial.

Bear River. A month after the hangings, soldiers attacked a Shoshone camp far to the west. Colonel Patrick Edward Connor led them against a winter camp of Northwestern Shoshone people on the Bear River, near what is now Preston, Idaho, on January 29, 1863. Connor had announced that he would take no prisoners. His soldiers killed an estimated 250 to 500 Shoshone people. Afterward army leaders promoted Connor to brigadier general, a higher rank.

The Long Walk. Far to the south, in 1863 and 1864, United States soldiers marched about 8,500 Navajo people more than 300 miles to Bosque Redondo and held them there. This forced march is known as the Long Walk. Thousands of the Navajo died while soldiers held them at Bosque Redondo. In 1868 Navajo leaders negotiated a treaty in which United States officials agreed to let the survivors go home. The Navajo were one of the very few removed nations who got their homeland back.

Sand Creek. Colonel John Chivington led about 675 volunteer soldiers against Black Kettle's Cheyenne and Arapaho camp at Sand Creek in Colorado on November 29, 1864. The camp was flying a United States flag and a white flag.

Earlier that month Major Scott Anthony, the commander at nearby Fort Lyon, had told Black Kettle's people to camp at Sand Creek. In his report of November 16 he wrote that he told them "no war would be waged against them." That promise was to hold until his own commander decided what to do. When Chivington arrived, Anthony agreed to the attack.

The attack lasted eight hours. The soldiers killed about 230 people, most of them women, children and elders.

Captain Silas Soule, one of the officers there, refused to let his men fire. He wrote to his mother that he saw "a Massacre of three hundred Indians mostly women and children." In another letter he wrote that it was "hard to see little children on their knees have their brains beat out." Soldiers cut scalps and other body parts from the dead.

Members of a committee of Congress later wrote, using an old word for cowardly and cruel, that Chivington "deliberately planned and executed a foul and dastardly massacre." Chivington left the army and was never put on trial,

and no officer or soldier was punished.

A man shot Soule dead on a Denver street in April 1865, about two months after Soule spoke against Chivington before investigators. No one was ever punished for killing him. National Park Service staff look after the place of the massacre today as the Sand Creek Massacre National Historic Site.

Red Cloud's War. Gold was found in Montana in 1863, and the Bozeman Trail was opened as a road for white settlers to reach it. The trail ran through Lakota country, and the Lakota had never given permission for it. When soldiers built forts along it to protect the settlers, Lakota fighters went to war, from 1866 to 1868. The war is named for the Lakota leader Red Cloud.

After the war, United States officials and Lakota leaders signed the Treaty of Fort Laramie in 1868. In it, the officials agreed to close the Bozeman Trail. They also set up the Great Sioux Reservation "for the absolute and undisturbed use" of the Lakota. It covered all of what is now South Dakota west of the Missouri River, including the Black Hills.

The Marias River. Major Eugene Baker's cavalry came to a camp of Piikani (Piegan Blackfeet) people on the Marias River in Montana on January 23, 1870. His scout, Joe Kipp, told him it was the camp of Heavy Runner's band, a group of families who lived and traveled together, and that the band was at peace with the United States. By one account, Baker answered, "That makes no difference, one band or another of them, they are all Piegans and we will attack them."

By the best estimate, Baker's soldiers killed 37 men, 90 women and 50 children. The army's own count was 173 dead, and the National Parks Conservation Association's figure is more than 200. Neither Baker nor any of his men was ever punished.

Gold in the Black Hills. Members of an expedition led by George Custer announced in 1874 that there was gold in the Black Hills, inside the reservation promised to the Lakota in 1868. Federal officials ignored their own treaty. In December 1875 they ordered the winter camps that were off the reservation to come onto it by January 31, 1876. When the camps did not come in, the officials sent the army.

The Little Bighorn. George Custer led part of the 7th Cavalry, an army unit of soldiers on horseback, on June 25, 1876. He attacked a camp where Lakota, Cheyenne and Arapaho people had gathered as allies, meaning they had agreed to fight on the same side. The camp stood by the Little Bighorn River, also called the Greasy Grass. Fighters from the camp killed Custer and about 200 of the soldiers with him in about an hour. Those soldiers made up five companies, smaller units of the 7th Cavalry.

Crazy Horse. The Lakota leader Crazy Horse, who had led fighters at the Little Bighorn, surrendered on May 6, 1877, less than a year after the battle. Soldiers were arresting him at Fort Robinson on September 5, 1877. A Lakota man named Little Big Man jumped on his back and held him, and a soldier on guard stabbed him with a bayonet, a blade fixed to the end of a rifle. Crazy Horse died of the wound.

Who stabbed him. Most accounts say a soldier on guard stabbed him. The name most often given is Private William Gentles, but only one witness named him, so who the soldier was is disputed. Little Big Man later told a different story, in which Crazy Horse was cut by his own knife.

The last surrender. In the Southwest, Geronimo, of the Chiricahua Apache, surrendered on September 4, 1886. He was the last Native leader to formally surrender to the United States. United States officials then held the Chiricahua people as prisoners of war, people an army holds after capturing them in a war, for 27 years.

BY 1886: Geronimo had surrendered, the last Native leader to do so, and United States officials held most Native nations on reservations. The Black Hills, promised to the Lakota in 1868, had been taken after the gold strike.

Sitting Bull

Who: Hunkpapa Lakota leader and holy man. The camp that fought at the Greasy Grass gathered around him.

When and where: the northern Plains. Canada from 1877 to 1881. Killed by Indian police at Standing Rock, December 15, 1890.

Movie: *Sitting Bull: A Stone in My Heart* (2006), a documentary built from his own recorded words.

Buffalo Bill's show paid him fifty dollars a week.

The camp on the Greasy Grass in June 1876 formed around Sitting Bull, and Crazy Horse was one of its battle leaders.

Sitting Bull took his people north into Canada in 1877. Hunters were killing off the bison, and by 1881 there were not enough left to feed his people. He surrendered at Fort Buford in July of that year.

He signed a contract with Buffalo Bill's Wild West, a traveling show, at the Standing Rock Agency on June 6, 1885. The show paid him fifty dollars a week for a four-month season, and only he could sell his own photographs and autographs.

Indian police, Native men who worked as police for the agency that ran the reservation, came to arrest him at Standing Rock on December 15, 1890. Indian agents, the federal officials who ran the reservations, were alarmed by the Ghost Dance, a new religious dance spreading across the reservations. The policemen killed Sitting Bull.

Chief Joseph

Who: Nez Perce leader in the flight of 1877, when his people fled toward Canada with the United States Army behind them.

When and where: the Wallowa Valley of Oregon to the Bear Paw Mountains of Montana, 1877. Surrendered October 5, 1877, forty miles from Canada.

Movie: *I Will Fight No More Forever* (1975, television film).

They stopped forty miles short of Canada.

Joseph's band of the Nez Perce lived in the Wallowa Valley of northeastern Oregon. General Oliver O. Howard told the Nez Perce chiefs in 1877 that their people had 30 days to move onto the reservation in Idaho, or the army would move them by force. That year about 750 Nez Perce people fled with the army behind them. Only about 200 of them were fighters, and the rest were families.

They traveled about 1,170 miles, the National Park Service's figure for the trail. Some accounts state a longer distance, up to about 1,700 miles.

They surrendered on October 5, 1877, at the foot of the Bear Paw Mountains in Montana, forty miles short of the Canadian border. Lieutenant C. E. S. Wood recorded Joseph's words at the surrender: "From where the sun now stands I will fight no more forever."

The promise. Colonel Nelson Miles had promised Joseph that his people would be taken back to the reservation in Idaho. General William Tecumseh Sherman, the commanding general of the army, overruled the promise, cancelling it. He wanted the Nez Perce punished as an example to other nations and never allowed to go home. "I believed General Miles, or I never would have surrendered," Joseph said afterward.

United States officials sent his people to Kansas, then to Indian Territory. There, many of them were sick and short of supplies, and their number fell from 410 to 391. Two hundred sixty-eight Nez Perce people left Indian

> Territory for the Northwest in May 1885. Officials sent Joseph to the Colville reservation in Washington, and he died there, at Nespelem, in 1904. Officials never allowed him back to the Wallowa Valley.

The bison, and the end of treaties

There were roughly 30 million bison on the Plains in the early 1800s. One count in 1884 found 325, about one bison left for every 90,000 there had been. By 1900 there were fewer than a thousand.

Railroads carried hunters into the herds' country. Hunters made hide hunting into a large business, killing bison for their skins. Disease and drought killed others. Army officers encouraged the killing, because the herds were the Plains nations' food and the base of how they made a living, and the bison were part of their religion.

Members of Congress stopped making treaties with Native nations in 1871. They did it with a rider, a short passage attached to a spending bill called the Indian Appropriations Act. Treaties already made stayed in force. After 1871, United States officials dealt with Native nations through acts of Congress and orders from the president instead of treaties.

How did the nations lose 90 million acres?

Native nations lost about 90 million acres between 1887 and 1934 through allotment, the cutting of land that a whole nation owned together into private pieces. Federal officials carried it out.

Members of Congress passed the Dawes Act, also called the General Allotment Act, in 1887. Under it, federal officials divided nations' land into separate parcels, or pieces. The head of a family got 160 acres, a single adult or an orphan got 80 acres, and each other child got 40 acres. On land fit only for grazing animals, each person got twice as much. Officials called everything left over "surplus," meaning extra, and opened it to white settlers.

The land Native nations held fell from about 138 million acres in 1887 to 48 million in 1934. The lost land was about two-thirds of all the land Native nations still had in 1887. The Meriam Report, a federal study published in 1928, states the figures as 137 million and 47 million acres. The loss is about the same by either count.

Allotment went on until members of Congress ended it in 1934.

The boarding schools

Richard Henry Pratt, an Army officer, opened the Carlisle Indian Industrial School at Carlisle, Pennsylvania, on November 1, 1879. It was the first federal boarding school for Native children built far from the reservations. Federal officials used Carlisle as the model for the schools that followed.

Pratt stated the purpose himself. In 1892 he spoke to a national conference of charities in Denver. He said that "all the Indian there is in the race should be dead. Kill the Indian in him, and save the man." People often quote the line in a shorter form, "kill the Indian, save the man." Pratt's recorded words are the longer ones, printed in the conference's official report.

Members of Congress passed two laws to get the children into the schools. In 1891 they gave the Commissioner of Indian Affairs, the federal official in charge of Native affairs, the power to make and enforce rules forcing Native children to attend school. In 1893 they let federal officials hold back rations, clothing and annuity payments from parents who would not send their children. These were the food, goods and yearly payments that United States officials had promised a nation in its treaties.

Some Indian agents, the federal officials in charge of the reservations, sent reservation police to seize children. Some families sent their children willingly, hoping the schooling would help them. Many families had no real choice.

Arrival. When a child arrived, staff at school after school cut the child's hair and replaced the child's clothes with a uniform. They gave the child an English name and forbade the child to speak the language of home. Staff punished children for speaking their own languages. The federal investigation of the schools and survivors' own accounts document the punishments: beatings, solitary confinement, which means locking a child away alone, withholding food, and washing children's mouths out with soap.

Children spent half of each school day doing unpaid work that kept the school running. Pratt also ran what he called the outing system. Under it, he sent students to live and work with white families, in the summers and during school terms.

The count. Investigators at the United States Department of the Interior studied the schools and published their findings in 2022 and again on July 30, 2024. They counted 417 federal Indian boarding schools in 37 states and territories, open between 1819 and 1969. Their first report, in 2022, listed 408. They also found more than a thousand other places that had similar aims.

At least 973 children died at the federal schools. The investigators found at least 74 marked and unmarked burial sites at 65 of the schools. The report states that its count of the dead is low.

Children kept their languages by speaking them where the staff could not hear. At these schools, children made friends with children from other nations. Those friendships were one of the starting points of the Native organizing of the next century. Some students, Zitkala-Ša among them, used the reading and writing they learned at the schools to argue against the schools.

Sarah Winnemucca

Who: Northern Paiute writer, interpreter, and lecturer who argued her people's case in front of audiences across the East.

When and where: Nevada, and the lecture halls of the East, in the 1870s and 1880s.

"My people were only given one week to get ready in."

Sarah Winnemucca was working as an interpreter when she learned, just before Christmas 1878, that President Rutherford B. Hayes had ordered her people taken to the Yakama reservation under an army guard. She wrote the line about the one week. Officials were punishing the whole group because a few Paiute men had fought in the Bannock War of 1878.

United States officials forced about 550 Northern Paiute men, women and children to move 350 miles north in 1879, from the Malheur reservation in Oregon to the Yakama reservation. The march took nearly a month through snow and over two mountain ranges. The officials in charge made some of the men walk in leg irons, chains locked around the ankles.

Five children, one woman and an old man died on the road. Their bodies were left beside it because the ground was too cold to dig graves. The Oregon Historical Society's account of the march does not say who decided to leave them. The historian Sally Zanjani estimated that more than one fifth of the group died far from home after they were sent away, mostly of hunger and disease.

Schurz's promise. Winnemucca went east and gave lectures about what officials had done to her people. She met with Secretary of the Interior Carl Schurz in Washington in 1880. He gave her a written promise that the Paiute could go back to Malheur, and then he broke it. Her people never returned there. President Chester Arthur ended the Malheur reservation in 1882 and 1883, and white settlers could then claim its land.

She published *Life Among the Piutes: Their Wrongs and Claims* in 1883. It is the first known autobiography, a book a person writes about their own life, by a Native American woman. She was also the first Native woman known to hold a United States copyright, the legal right to control who copies and sells a piece of writing. In 1884 she asked Congress to give her people back their reservation.

She opened a school of her own for Paiute children near Lovelock, Nevada, in the mid-1880s. Accounts of her life give the year as 1884 or 1885. Staff at the government boarding schools forbade children to speak their own languages, but at her school Winnemucca taught in both Paiute and English. The school ran out of money and closed within a few years.

Zitkala-Ša

Who: Yankton Dakota writer, musician, and organizer, taken as a child to a boarding school in Indiana.

When and where: the Yankton reservation, South Dakota, and Washington, D.C. Born February 22, 1876. Died January 26, 1938.

An interpreter promised her all the red apples she wanted.

Zitkala-Ša wrote about how she left home in her essay "The Big Red Apples." She was in her eighth year, in March, when two white missionaries came to her village on the Yankton reservation. Her mother told her they had come "to take away Indian boys and girls to the East." A young white interpreter told the girl she could pick all the red apples she wanted and ride on "the iron horse," a train. "I had never seen a train, and he knew it," she wrote.

She begged to go. Her mother warned her, "Their words are sweet, but, my child, their deeds are bitter." Then her mother agreed. The missionaries took the girl to White's Indiana Manual Labor Institute. The essay does not name them.

The editors of the *Atlantic Monthly*, a magazine, published a set of her essays about her own life in 1900. One of them was "The School Days of an Indian Girl." In the essays she described her time at the school and criticized it. She was working at Carlisle when she published them. They appeared again in her book *American Indian Stories* in 1921.

She helped start the National Council of American Indians in 1926 and led it until she died in 1938.

The Ghost Dance and Wounded Knee

A Northern Paiute man named Wovoka, from the Walker Lake country of Nevada, had a vision on January 1, 1889. It came during a solar eclipse, when the moon blocked the sun. He taught that if people lived rightly and danced a round dance, in which people move in a circle, the dead would return and the world would be made whole.

On reservations where people were hungry, they took up the dance, which became known as the Ghost Dance. Indian agents who were frightened by it called it the start of an uprising, an armed revolt. Newspaper writers printed stories that said the same.

Indian police killed Sitting Bull at Standing Rock on December 15, 1890, while they were trying to arrest him.

Wounded Knee. Soldiers of the 7th Cavalry under Colonel James W. Forsyth were disarming Spotted Elk's band, also called Big Foot's band, at Wounded Knee Creek on December 29, 1890, taking away the people's weapons. During the disarming, a rifle belonging to Black Coyote, a deaf man, went off. By some accounts he fired first, and by others a soldier tried to grab his gun and it went off. The soldiers opened fire.

They fired carbines, which are short rifles, and four Hotchkiss guns, small cannon set on a hill above the camp. The soldiers killed between 250 and 300 Lakota people. The killing is known as the Wounded Knee Massacre.

A burial party put 146 of the dead in one grave at the site in early January 1891. They were 84 men and boys, 44 women and 18 children. Others died later of their wounds, and some bodies were never found, so no count gives the women and children among all the dead.

Twenty-five soldiers also died. Writers for Britannica and History.com say that most of those soldiers were hit by their own side's fire. Twenty soldiers of the 7th Cavalry received the Medal of Honor, the top award the United States military gives for bravery in battle, for what they did that day. Some of the medal statements praise soldiers for driving out "Sioux Indians" who were hiding in a ravine, a narrow, steep valley.

BY 1900

If you remember one thing: In these fifty years United States soldiers and officials took most of the land the nations still held, forced the people onto reservations and began taking their children to boarding schools.

One number: about 250,000. That is how many Native people census takers counted in 1900, the lowest number on record. Even the lowest estimate for 1500, about 1 million, is four times as many.

Next: In the next fifty years Native people built national organizations of their own, while the boarding schools stayed open.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 11 MINUTES

Native leaders founded national organizations of their own in these fifty years, the Society of American Indians in 1911 and the National Congress of American Indians in 1944. Federal boarding schools stayed open the whole time.

In Oklahoma in the early 1920s, killers murdered dozens of Osage people for their shares of the nation's oil money, in the years called the Reign of Terror.

Members of Congress made Native people citizens in 1924 and ended allotment, the cutting of nations' shared land into private pieces, in 1934. Native men fought for the United States in both world wars, and some of them sent messages in Choctaw and Navajo.

Organizing

A group of Native doctors, lawyers and writers met in Columbus, Ohio, on October 12, 1911, and founded the Society of American Indians. It was the first national Native rights organization that Native people built and ran themselves. Its leaders included Dr. Charles Eastman, who was Santee Dakota, and Dr. Carlos Montezuma, who was Yavapai-Apache. Zitkala-Ša edited its journal.

Many of the members had gone through the boarding schools, and some of them used that schooling to argue against federal Indian policy. The Society lasted from 1911 to 1923.

Twelve men and one woman founded the Alaska Native Brotherhood at Sitka in 1912. They wanted citizenship for Alaska Natives. The Alaska Native Sisterhood began in 1915. For the next fifty years, these two groups were the only civil rights organizations Alaska Natives had, the only groups working for their right as citizens to equal treatment under the law.

Eighty delegates from more than fifty tribes and associations met in Denver on November 16, 1944, and founded the National Congress of American Indians. Native nations still use it as one of their main ways to act together.

Ironworkers from Kahnawake

Mohawk men from Kahnawake had worked on structural iron, the metal frames that hold up bridges and tall buildings, since the 1880s. Kahnawake crews worked on the riveting gangs that put up the Empire State Building in New York in 1930 and 1931. A riveting gang joined the steel beams together with hot metal pins called rivets.

Kahnawake ironworkers kept building towers in Manhattan for generations. One of them, Harold McComber, is the subject of *High Steel*. Don Owen directed that documentary film for the National Film Board of Canada in 1965.

The Osage and the Reign of Terror

Osage people in Oklahoma earned so much from oil in the 1920s that they were among the wealthiest people in the world, counted per person. White guardians and some of the men who married Osage women stole that money, and killers murdered Osage people to get it.

Osage leaders under Chief James Bigheart won their own terms in the Osage Allotment Act, which members of Congress passed in 1906. Under the Dawes Act, federal officials had given the head of a family 160 acres, single people and children less, and opened the rest of a nation's land to white settlers. The Osage divided all of their reservation among their own people instead, 657 acres each for 2,229 people, about four times a family head's 160 acres.

Headrights. The Osage kept the minerals under the ground as shared property. Oil companies paid royalties, which are payments for the oil they took out. Each Osage member got an equal share of that money. The share was called a headright.

Members of Congress passed a guardianship law in 1921. Under it, judges in local courts appointed white guardians over the money of many Osage adults. The guardian, not the Osage owner, controlled the money. Guardians stole from the people whose money they controlled. Some of the men who married Osage women stole from their wives.

Osage headright holders died in murders and in suspicious deaths, deaths that might have been murders, between about 1920 and 1925. These years are known as the Reign of Terror. Counts of the dead disagree. Osage Nation officials count more than sixty. Writers at the Oklahoma Historical Society count about 24, and FBI writers say at least two dozen.

The investigation. Members of the Osage Tribal Council asked the federal government to send detectives in 1923. Agents of the new Bureau of Investigation, which later became the FBI, took the case. They proved that William Hale, a cattle rancher whom people called "the King of the Osage Hills," had ordered the murders of Anna Brown and members of her family so that he could inherit their oil rights, getting them after they died. Hale's nephew, Ernest Burkhart, was married to Anna's sister Mollie.

The killings. Anna Brown was shot on Hale's orders in May 1921. Two months later her mother, Lizzie Q. Kyle, died, and poison was suspected. In January 1923 Hale had Anna's cousin Henry Roan killed for insurance money. In March 1923 a bomb blew up the house of Anna's sister Rita Smith, killing Rita, her husband Bill and their servant, Nettie Brookshire.

Who went to prison. Burkhart pleaded guilty in 1926 and was sentenced to life in prison. Hale was found guilty in 1929 and sent to prison for life. Two other men, Kelsie Morrison and John Ramsey, were also sentenced to life. Hale, Ramsey and Burkhart were all later let out on parole, which means freed early under watch.

Citizenship and the end of allotment

Under the Indian Citizenship Act, which became law on June 2, 1924, every Native person born in the United States became a citizen. In some states they still could not vote, because each state's lawmakers wrote that state's voting rules. The Library of Congress records that officials in some states kept Native people from voting until 1957.

A federal survey called the Meriam Report came out in 1928. Its full title was *The Problem of Indian Administration*. It records that under allotment, tribal land had fallen from 137 million acres to 47 million. It also records the poverty and disease Native people lived with after that loss.

The Meriam Report's writers condemned the boarding schools in detail, judging them wrong and showing why. One of the things they condemned was the half-day system, under which school staff made children spend half of every school day on unpaid work that kept the schools running.

The Indian Reorganization Act became law on June 18, 1934. It was written while John Collier was Commissioner of Indian Affairs, the federal official in charge of dealings with Native nations, a job he held from 1933 to 1945. In the act, members of Congress ended allotment and extended the protections on land held in trust, land the United States holds on behalf of a Native nation or person.

Members of Congress also allowed up to $2,000,000 a year in the act for buying land for Native people. Nations could now write constitutions and set up their own governments. After 1934 federal officials began to recognize the governments that nations set up under the act. For the forty-seven years before, since 1887, federal officials had used allotment to break nations up into single landowners.

The schools stay open

Federal boarding schools kept running after 1900. The federal investigators counted the schools that ran between 1819 and 1969, and some kept going after that. Enrollment, the number of children in the schools, kept climbing for decades and reached its peak of about 60,000 children in 1973.

Collier and the officials under him moved toward day schools, where children went home at night, in the 1930s. They allowed some teaching in Native languages, and they closed some boarding schools. Federal officials still took tens of thousands of children from their homes to the schools every year.

One of those children was Chester Nez, a Navajo boy at the Fort Defiance boarding school, who later helped build the Marines' code in Navajo.

Why could no enemy read the code?

Code talkers were Native soldiers who sent military messages in their own languages, which the enemy could not understand. The first code talkers were Choctaw.

American soldiers were fighting in the Meuse-Argonne campaign, a long series of battles in France, in the autumn of 1918. German soldiers were tapping American telephone lines. They were also capturing about one in four runners, the soldiers who carried messages on foot. Choctaw soldiers of the 36th Division began passing orders in Choctaw.

They sent the first secure messages, ones the Germans could not read, on October 26, 1918. When Choctaw had no word for a piece of equipment, the men made one up. A machine gun became "little gun shoot fast."

In the Second World War, Marine Corps officers had Navajo speakers do the same work. Twenty-nine Navajo recruits arrived in San Diego on May 5, 1942, and built the code. Between 350 and 420 Navajo served as code talkers, in all six Marine divisions in the Pacific. At Iwo Jima, Major Howard Connor was the 5th Division's signal officer, in charge of its messages. He said: "Were it not for the Navajos, the Marines would never have taken Iwo Jima."

No enemy ever broke the code. It stayed classified, meaning officially secret, until 1968. The original twenty-nine received the Congressional Gold Medal, a medal that members of Congress vote to give, on July 26, 2001, and the code talkers who came after them received silver medals.

Jim Thorpe

Who: Sac and Fox athlete and Carlisle student, the first Native American to win an Olympic gold medal.

When and where: Stockholm, July 1912. Olympic officials took his medals in 1913 and named him the sole champion again in 2022.

Movie: *Jim Thorpe – All-American* (1951)

Olympic officials took his gold medals over two dollars a game.

Jim Thorpe was a citizen of the Sac and Fox Nation and a student at the Carlisle Indian Industrial School, the federal boarding school in Pennsylvania. There he was twice named a football All-American, one of the best college football players in the country.

He won the pentathlon, a contest of five track and field events, on July 7, and the decathlon, a contest of ten, on July 15, at the Stockholm Olympics in 1912. His decathlon score stood as the world record until 1927.

Olympic officials took both medals back in 1913. Thorpe had played minor-league baseball in 1909 and 1910 for a reported two dollars a game, up to $35 a week. The Olympic rules of the time allowed only amateurs, athletes who are not paid to play.

Restored. Officials of the International Olympic Committee named him co-champion in October 1982, sharing the titles with Hugo Wieslander in the decathlon and Ferdinand Bie in the pentathlon. On July 15, 2022, 110 years to the day after his decathlon win, they named him the sole champion of both events.

Ira Hayes

Who: Akimel O'odham (Pima) Marine from the Gila River community in Arizona, one of the six men who raised the second flag on Mount Suribachi.

When and where: Iwo Jima, February 23, 1945. He died in Arizona on January 24, 1955.

Movie: *The Outsider* (1961). Adam Beach also plays him in *Flags of Our Fathers* (2006).

He raised the flag on Iwo Jima and died of exposure at 32.

Ira Hayes, from the Gila River Indian Community in Arizona, was a Marine paratrooper, a soldier trained to jump from planes by parachute. He was one of the six men who raised the second American flag on Mount Suribachi, on Iwo Jima, on February 23, 1945. Americans celebrated the photograph of that flag-raising.

Hayes came home to a fame he had not asked for. He and two other men from the photograph raised a flag at a ceremony in Washington on May 9, 1945. The ceremony started a tour to sell war bonds, loans people made to the government to pay for the war. Later that month the Marines ordered Hayes back to his unit.

Americans celebrated the photograph, but Hayes got little help after the war. Police arrested him 52 times for being drunk in public. Years later President Dwight Eisenhower met him and called him a hero.

Hayes was found dead on the morning of January 24, 1955, near an empty mud-brick hut close to his home at Sacaton, in the Gila River community. He had been drinking the night before. The coroner, the official who decides how a person died, gave the cause as alcohol poisoning and exposure, dying from cold or other harsh weather without shelter. He was 32.

Chester Nez

Who: Navajo Marine, one of the twenty-nine who built the code. As a boarding-school student, he was punished for speaking the language the code was made from.

When and where: the Fort Defiance boarding school, Arizona, then the Pacific, 1942 to 1945. He died June 4, 2014.

He outlived the other twenty-eight who built the Navajo code.

When Chester Nez spoke Navajo at the Fort Defiance boarding school, staff washed his mouth out with soap.

Marine Corps recruiters signed Nez up in 1942. He was one of the twenty-nine who went to San Diego and built the code out of the language.

He wrote about what happened to him in *Code Talker*, a memoir, or book about the writer's own life, published in 2011 with Judith Schiess Avila. It is the only memoir by one of the original twenty-nine. Nez died on June 4, 2014, the last of the twenty-nine still alive.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 15 MINUTES

Native nations spent these fifty years fighting a federal plan to end them as nations, and winning back the running of their own affairs. Members of Congress declared in 1953 that the United States should end its recognition of Native nations, which means its dealings with each nation as a government. Twenty-two years later, in 1975, President Ford signed a law under which Native nations could take control of their own schools, clinics and police.

The Menominee, whose nation members of Congress had terminated, ending its recognition, organized until members of Congress restored it. Fishermen in the Northwest kept fishing until a federal judge ruled on their 1850s treaties. A group called Indians of All Tribes took Alcatraz Island and held it for nineteen months. In 1980, in the Sioux Nation's Black Hills case, the justices of the Supreme Court ruled that United States officials had broken the 1868 treaty when they took the hills.

Termination and relocation

Members of Congress passed House Concurrent Resolution 108, a formal statement of policy that they voted on, on August 1, 1953. In it they declared that the United States should end its recognition of Native nations, along with the federal services that came with it. The policy's name was termination. Under it, members of Congress and federal officials stopped treating a tribe as a nation with its own government, and the tribe lost its federal services.

Members of Congress ran about sixty termination proceedings, the formal legal steps that ended a nation's recognition, between 1953 and 1970. Those proceedings covered more than a hundred tribes and bands. More than 3 million acres of land stopped being held in trust, which meant the United States no longer held that land on the nations' behalf.

The Menominee. The Menominee of Wisconsin ran a profitable timber business. Members of Congress passed the Menominee termination act in 1954. It took effect in 1961, and the Menominee, who had done well before, became poor. Menominee people organized a movement called DRUMS, with Ada Deer among its leaders. They kept pushing until members of Congress reversed the termination. The Menominee Restoration Act became law on December 22, 1973.

Officials of the Bureau of Indian Affairs, the federal agency in charge of Indian affairs, also moved Native people off reservations and into cities in these years. They began this relocation program between about 1948 and 1952, and members of Congress backed it with the Indian Relocation Act of 1956. People who signed up got a one-way ticket and short-term help. About 30,000 people moved in the 1950s, and more than 100,000 had moved by the time the program ended in the 1970s.

Many of them lived in poverty in the cities. They also built lasting Native communities in Minneapolis, Chicago, Oakland and San Francisco, Los Angeles, and Denver. The next Native movement started in those cities.

Alcatraz and Wounded Knee, 1969 to 1973

George Mitchell, Dennis Banks, Clyde Bellecourt and about 200 other people founded the American Indian Movement, known as AIM, in Minneapolis in July 1968. Its members began with street patrols against police abuse. Many Native people had moved to Minneapolis under relocation.

Alcatraz. A group calling itself Indians of All Tribes landed on Alcatraz Island in San Francisco Bay on November 20, 1969. They occupied the island's federal prison, closed since 1963, moving in and holding it. Richard Oakes, who was Mohawk, was among the leaders. In their proclamation, a public statement, they offered to buy the island for "$24 in glass beads and red cloth." They held it for nineteen months, until June 11, 1971. After the occupation began, President Nixon ended termination as federal policy.

Blue Lake. Nixon also signed H.R. 471, a law that members of Congress had passed, on December 15, 1970. Under it, Blue Lake and 48,000 acres of sacred land went back to Taos Pueblo.

Alaska. A year after Blue Lake, members of Congress settled the land claims of Alaska Natives, ending them by agreement. The Alaska Native Claims Settlement Act became law on December 18, 1971. Under it, the claims were settled for 44 million acres and $962.5 million.

That is nearly as much land as the 48 million acres all Native nations together held in 1934. The land and money went to twelve regional corporations, later thirteen, and to more than 200 village corporations. These were businesses, and Alaska Natives became their owners, called shareholders.

Wounded Knee, 1973. About 200 Oglala Lakota and AIM members occupied the village of Wounded Knee, on the Pine Ridge Reservation in South Dakota, in 1973. They chose it because soldiers of the 7th Cavalry had killed between 250 and 300 Lakota people there in 1890. Of the 146 dead whom a burial party put in one grave there, 62 were women and children.

The occupiers held the village for seventy-one days, from February 27 to May 8, against federal marshals, officers who carry out the orders of the national courts, and agents of the FBI, the national police force that looks into federal crimes. The federal forces surrounded the village with armored vehicles.

Why they came. The occupiers were protesting against the treaties that United States officials had broken, and against Richard Wilson, the tribal chairman at Pine Ridge since 1972, who was Oglala himself.

His opponents accused him of corruption, of giving the few jobs and benefits to his family and friends, and of using an armed group of his own, the Guardians of the Oglala Nation, known as the GOONs, to attack people who opposed him. Three weeks before the occupation, members of the tribal council had brought charges against him at a hearing to remove him from office, and Wilson avoided a trial.

Both sides fired shots during the siege, the seventy-one days in which federal forces surrounded the village. Early in the siege, someone shot US Marshal Lloyd Grimm, and afterward he could not move his legs. Accounts of the siege do not say who shot him.

Frank Clearwater, an Eastern Cherokee man, was shot in the head on April 17 while he rested in a church during a gunfight with federal forces. He died on April 25. FBI officials released a report on deaths on the Pine Ridge Reservation in 2000. It found that Clearwater was killed by gunfire from federal officers, who were firing back after people shot at them.

A federal agent firing as a sniper, a hidden marksman, shot and killed Lawrence "Buddy" Lamont, an Oglala man, with a bullet through the heart on April 26.

The charges. Judge Fred Nichol threw out the charges against the AIM leaders Dennis Banks and Russell Means in 1974. He wrote that "the misconduct by the government in this case is so aggravated that a dismissal must be entered in the interests of justice." He meant that government officials had broken the rules so badly that the case had to end.

The fish-ins and the Boldt decision

Leaders of the Northwest nations signed treaties with Isaac Stevens, the governor of Washington Territory, in the 1850s. In them they gave up land and kept the right to fish.

The Treaty of Medicine Creek, signed by the Nisqually, the Puyallup and other nations on December 26, 1854, says that "the right of taking fish, at all usual and accustomed grounds and stations, is further secured to said Indians." Those words promised that they could keep fishing at all the places where they had always fished. A century later, Washington State officers were arresting their people for fishing.

Through the 1960s and into the 1970s, the Nisqually, the Puyallup and other nations held fish-ins. At a fish-in, people fished in the open, in front of the state's officers, and let themselves be arrested. They did it to force the treaties into a courtroom. Frank's Landing on the Nisqually River was the center of the fish-ins.

Judge George Boldt heard the treaty case, *United States v. Washington*, in federal court. He ruled on February 12, 1974, that the treaty rights still stood. Under them, the tribes had a right to half the harvestable salmon, the salmon that could be caught each year.

Boldt also ruled that the tribes were co-managers of the fishery, the salmon catch and the waters it comes from, so that tribal governments and the state run it together. The justices of the Supreme Court upheld the ruling in 1979, which means they let it stand.

Why do the Lakota refuse more than a billion dollars?

The Lakota have refused the money that Supreme Court justices awarded them for the Black Hills in 1980, which with the interest it has earned is now more than a billion dollars. Lakota leaders say that the Black Hills were never for sale.

United States officials guaranteed the Black Hills of South Dakota to the Lakota in the Fort Laramie treaty of 1868. Members of an Army expedition led by George Custer announced in 1874 that they had found gold there. United States officials ignored the treaty and took the hills.

The Lakota call one mountain in the Black Hills Tunkasila Sakpe, the Six Grandfathers. It is the mountain known in English as Mount Rushmore.

The justices of the Supreme Court held in *United States v. Sioux Nation*, in 1980, that United States officials had broken the treaty when they took the hills. They awarded the Lakota compensation, money paid to make up for a loss.

Self-rule, children taken and women sterilized

President Ford signed the Indian Self-Determination and Education Assistance Act in January 1975, self-determination meaning a nation deciding its own affairs. Under the act, a nation could sign a contract with the federal government and run federal programs itself, such as its own schools, clinics and police. The word Education is in the act's name because schools were at the center of it.

Children taken. While nations were taking over their own programs, workers at agencies were still taking Native children from their families, as they had for decades before 1978. Writers at the National Indian Child Welfare Association state that 25 to 35 percent of Native children were taken, between one in four and one in three. They also state that 85 percent of them were placed outside their families, even when relatives or members of their tribe could care for them.

Members of Congress passed the Indian Child Welfare Act in 1978. In it they set federal standards for those cases, meant to keep Native children with Native families. They also gave tribal courts the power to decide the cases.

Sterilized. Doctors at hospitals of the Indian Health Service, the federal agency that runs hospitals and clinics for Native people, sterilized thousands of Native women in the same years. Sterilization is an operation after which a woman can never become pregnant. The doctors at these hospitals did two kinds of operation. In a tubal ligation, a surgeon cuts or ties the two tubes that carry eggs to the womb. In a hysterectomy, the surgeon cuts out the womb.

Investigators from the General Accounting Office, a federal office, studied 4 of the agency's 12 areas. In 1976 they reported 3,406 sterilizations of Native women there from 1973 to 1976. Writers at the National Library of Medicine state that the women had not given their permission. The investigators' report does not name the doctors. No surviving record says how many of the women had each kind of operation.

Thirty-six of the women were under 21, although a court had ordered a stop to sterilizing women that young. Two years before the report, Dr. Connie Pinkerton-Uri, a Choctaw and Cherokee doctor, had found that one in four Native women had been sterilized without agreeing to it.

Gaming. Ten years after the child welfare law, members of Congress set rules for gaming run by Native nations, meaning gambling businesses such as casinos. Their Indian Gaming Regulatory Act became law on October 17, 1988. For some nations gaming became a large source of income, and the amount varies a great deal from nation to nation.

Under the act, tribal gaming falls into three classes of games, and a nation and a state can make a compact, a formal agreement about the games. In the act, members of Congress also set up the National Indian Gaming Commission to watch over tribal gaming.

President George H. W. Bush signed the Native American Graves Protection and Repatriation Act, known as NAGPRA, on November 16, 1990. Under it, Native human remains and sacred objects held in museums and federal collections belong to the nations and families they came from, and the people holding them must send them home, which the law's name calls repatriation.

Nations take the schools back

Navajo leaders founded a college of their own at Tsaile, Arizona, in 1968. It was called Navajo Community College, and today it is Diné College. It was the first college in the country controlled by a Native nation. Native nations have founded dozens of tribal colleges since.

Senators on a Senate subcommittee on Indian education, a small group of senators assigned to that one subject, published their report in November 1969. The report's title is *Indian Education: A National Tragedy, A National Challenge*. It is known as the Kennedy Report, because Senator Robert F. Kennedy led the work at first and Senator Edward Kennedy finished it.

The report states that federal policy had been "coercive assimilation," forcing a group of people to give up their own language and ways of life, with "disastrous effects." Its authors called for Native people to run Native schools.

Boarding-school enrollment peaked at about 60,000 children in 1973. After that it fell, as the leaders of Native nations took over schools under the 1975 law and other schools closed. A few boarding schools are still open, run by tribes or by the Bureau of Indian Education, the federal office in charge of Indian schools. People who had been children at the schools began publishing accounts of what had happened to them there.

Wilma Mankiller

Who: the first woman elected Principal Chief of the Cherokee Nation.

When and where: Cherokee Nation, Oklahoma. She was chief from December 14, 1985 to August 14, 1995.

Movie: *Mankiller* (2017)

Cherokee voters re-elected her in 1991 with 83 percent of the vote.

Wilma Mankiller was a child when her family moved to San Francisco under the relocation program. She joined the movement during the occupation of Alcatraz by Indians of All Tribes in 1969.

She went back to Oklahoma. There she worked on projects in which Cherokee communities built their own water lines and health clinics, with the people who lived there doing the work.

She became Principal Chief of the Cherokee Nation, the head of the nation's government, on December 14, 1985, when Ross Swimmer left the office. Cherokee voters elected her in her own right in 1987 and re-elected her in 1991. Until August 14, 1995, she led a nation of about 140,000 citizens.

She received the Presidential Medal of Freedom, an award given by the president of the United States, in 1998. Mankiller died on April 6, 2010.

Billy Frank Jr.

Who: Nisqually fisherman, a leader of the fish-ins, who later chaired the Northwest Indian Fisheries Commission.

When and where: Frank's Landing on the Nisqually River, Washington. He was first arrested at fourteen.

Officers arrested him more than fifty times for fishing the Nisqually.

Billy Frank Jr. was fourteen, in 1945, the first time officers arrested him for fishing the Nisqually River. He kept fishing through the fish-ins of the 1960s and 1970s, and he went to jail more than fifty times.

After the Boldt decision of 1974, Frank chaired the Northwest Indian Fisheries Commission.

He died in May 2014 and received the Presidential Medal of Freedom after his death, in 2015. As of 2026, Washington State officials planned to place his statue in the National Statuary Hall in the United States Capitol.

Adam Fortunate Eagle

Who: Ojibwe writer. He was a Pipestone boarding-school student who became a lead organizer of the Alcatraz occupation.

When and where: Pipestone Indian Boarding School, Minnesota, 1935 to 1945. San Francisco Bay, 1969 to 1971.

He called his boarding-school days "a little bit of heaven."

Adam Fortunate Eagle spent ten years at the Pipestone Indian Boarding School in Minnesota, from 1935 to 1945. In those years John Collier was Commissioner of Indian Affairs, and federal officials were changing some of the schools. Three decades later, Fortunate Eagle was one of the lead organizers of the occupation of Alcatraz.

He published his account of the school, *Pipestone: My Life in an Indian Boarding School*, in 2010. The line about heaven is from that book. In the same book he writes about the harm the boarding-school system did.

Other people who were children at the schools told it differently. Chester Nez was punished with soap for speaking Navajo at Fort Defiance, and Zitkala-Ša's essays of 1900 criticized her school.

BY 2000

If you remember one thing: Members of Congress set out to end the nations as governments in 1953, and by 2000 the nations were running many of their own schools, clinics and courts.

- 1973: Members of Congress restore the Menominee Nation, twelve years after its termination took effect.
- 1974: Judge George Boldt rules that the treaty tribes have a right to half the salmon that can be caught.
- 1978: Members of Congress pass the Indian Child Welfare Act and give tribal courts the power to decide Native children's cases.

Next: After 2000 the first Native cabinet secretary had her officials count the boarding-school dead, and five Supreme Court justices ruled that a reservation in Oklahoma had never ended.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 10 MINUTES

Native nations govern themselves today. There are 574 federally recognized Native nations in the United States, which means that United States officials deal with each one as a government. There are also nations recognized by their states, and nations still working to win federal recognition. More than 200 Alaska Native villages have their own corporations. Native Hawaiians have no federally recognized government of their own.

Census takers counted 9.7 million people who identified as American Indian or Alaska Native, alone or together with another group, in 2020. That was 2.9 percent of the people in the country, and almost forty times the 250,000 counted in 1900.

The Navajo Nation's land covers more than 27,000 square miles across Arizona, New Mexico and Utah. It is the largest reservation in the country. The number of Cherokee Nation citizens passed 450,000 in 2023, counting

enrolled citizens, people officially listed as members. Some counts in 2024 and 2025 put it above 475,000.

The National Museum of the American Indian opened on the National Mall in Washington on September 21, 2004. About 25,000 Native people from across North and South America walked in the opening procession.

Sovereignty in court and in office

Justices of the Supreme Court ruled on a question of Native sovereignty, a nation's right to govern itself, in 2020. When United States officials forced the Muscogee (Creek) Nation to move to Oklahoma, they promised it a reservation there. The justices decided *McGirt v. Oklahoma*, a case about whether that reservation still existed, on July 9, 2020. Five of the nine justices held that members of Congress had never ended it. Four justices disagreed.

Under the ruling, about 3.25 million acres of eastern Oklahoma, including much of Tulsa, are still reservation land. Because of that, Oklahoma's state officials can no longer put Native people on trial for major crimes there. Those cases go to federal and tribal courts instead. Justice Neil Gorsuch wrote the opinion, the written decision of the court. Its first sentence is: "On the far end of the Trail of Tears was a promise."

The Trail of Tears was the forced removal of the Cherokee and the other Southeastern nations to Indian Territory, in what is now Oklahoma, in the 1830s. In later rulings, judges applied the same reasoning to the reservations of the other nations that United States officials had forced out of the Southeast.

Seven of the nine justices upheld the Indian Child Welfare Act, which means they kept it in force, in *Haaland v. Brackeen* on June 15, 2023. The people challenging the act had claimed that it broke the Constitution. Two justices disagreed with the ruling.

Deb Haaland of Laguna Pueblo and Sharice Davids, who is Ho-Chunk, became the first Native American women elected to Congress in 2018. In 2021 Haaland became the first Native American cabinet secretary, as Secretary of

the Interior, the head of the federal department in charge of the government's dealings with Native nations.

Standing Rock, and land returned

Leaders of the Standing Rock Sioux Tribe opposed the Dakota Access Pipeline, a long pipe built to carry oil, in 2016. Its route ran under Lake Oahe, upstream of the reservation. The lake is the reservation's water supply.

People set up camps beginning in April 2016, among them one called Oceti Sakowin. Young people carried the tribe's message on foot. They ran a 500-mile relay to the office of the Army Corps of Engineers, the part of the army whose permission the pipeline needed, in Omaha, and then a 2,000-mile run to Washington, D.C. Thousands of people came to the camps. News reports at the time called the camps the largest gathering of Native nations in more than a century.

North Dakota's governor, Jack Dalrymple, declared an emergency in August 2016 and put the Morton County Sheriff's Office and other police forces under one command. On November 20, 2016, at the Backwater Bridge, officers fired tear gas, rubber bullets and bags filled with lead into a crowd of protesters. They also sprayed the protesters with a fire hose.

Sophia Wilansky. A protester named Sophia Wilansky was hit that night. In 2026 federal appeals judges wrote out her account of what happened, in a ruling on her lawsuit. In that account, three officers fired rounds at her as she ducked behind a metal sheet, and one hit her arm. She ran about 30 yards, and as she bent to pick up a piece of plywood, a round fired by Officer Jonathan Moll hit her.

The judges described the round, called an aerial warning munition, as an exploding flash-bang inside a bullet made to hurt rather than kill. "It tore the flesh off her arm, and she fell to the ground," the judges wrote. "Her left hand and forearm are permanently and severely injured." Federal judges threw out her lawsuits and those of the other injured protesters, and no officer was held responsible.

An Army official, the Assistant Secretary for Civil Works, turned down the easement, the legal permission the pipeline needed to cross under the lake, on December 4, 2016. That same week, more than 2,000 United States military veterans arrived to stand with the camps.

The oil flows. President Donald Trump ordered the Army to approve the pipeline quickly in January 2017. Army officials granted the easement two weeks later, and oil has moved through the pipeline since that June. A federal judge cancelled the easement in 2020, but the oil kept flowing for about five years without one. Army officials granted the easement again in 2026, and the tribe's leaders then asked the court to end their own appeal.

Land returned. Some land has gone back to Native nations. Federal officials took 18,800 acres out of the Flathead Reservation in Montana in 1908 for the National Bison Range, a federal wildlife refuge, land set aside to protect wild animals. Members of Congress passed a law in December 2020 that gave the range back to the Confederated Salish and Kootenai Tribes, and the tribes took over running it in 2022.

The boarding-school investigation

Secretary Haaland started an investigation of the boarding schools in June 2021. Earlier officials of her own department had run them.

The first volume of the investigators' report came out in May 2022. It identifies 408 federal schools, and marked and unmarked burial sites at 53 of them. The final volume came out on July 30, 2024. It lists 417 federal Indian boarding schools in 37 states or places that were then territories, open from 1819 to 1969. It also lists 1,025 more places with similar aims that fell outside the four rules the investigators used to count federal schools.

The report records what school staff did to the children. Staff cut the children's hair, replaced their clothes with uniforms, gave them English names and forbade them to speak the languages they had grown up speaking. The punishments the report records for speaking those languages include beatings, solitary confinement, which means being locked up alone, holding back food, and washing children's mouths out with soap.

At least 973 children died at the schools. The investigators found at least 74 burial sites at 65 schools. The report states that the real number of deaths is higher than the number the investigators could count.

President Biden apologized on behalf of the United States for the boarding-school system on October 25, 2024. He spoke at the Gila River Indian Community in Arizona, which was Ira Hayes's home community. It was the first formal presidential apology for the system. He called the system "a sin on our soul."

Coming home. Children who died at the Carlisle Indian Industrial School are buried in a cemetery at the Carlisle Barracks, an Army post. Army officials have been digging up the children's remains and returning them to their nations.

Army officials returned two children, Little Chief and Horse, to the Northern Arapaho in 2017. When they opened the grave of a third child, Little Plume, they found two other children in it. They found Little Plume and returned him

the next year. Nine Rosebud Sioux children went home to South Dakota in July 2021, and Rosebud young people escorted them.

Schools in Native languages

Boarding-school staff punished children for speaking Native languages. In the two world wars, soldiers used two of those languages, Choctaw and Navajo, to send messages the enemy could not read. Today Native nations run schools that teach children in their own languages.

Hawaii's lawmakers banned teaching in Hawaiian in a law dating from 1896. The first Pūnana Leo immersion preschool, a school where teachers give every lesson in the language itself, opened at Kekaha on Kaua'i in 1984. Children there learn arithmetic and reading in Hawaiian instead of learning Hawaiian as one subject. The Pūnana Leo schools were the first immersion preschools in the United States in an Indigenous language, a language of the people who lived in a place first.

The ban on teaching in Hawaiian was lifted in 1986, and the first public immersion classes opened in 1987. According to ʻAha Pūnana Leo, the group that started the Pūnana Leo schools, students can now be taught in Hawaiian from preschool through graduate school, the university study that comes after a first degree.

Wôpanâak, the Wampanoag language of Massachusetts, had no living speakers after the middle of the 1800s. Jessie Little Doe Baird helped found the Wôpanâak Language Reclamation Project, reclamation meaning taking something back, in 1993. She began rebuilding the language from documents written in it.

Wôpanâak is the first American language with no living speakers to come back into living use. The project opened Mukayuhsak Weekuw, the Children's House, in September 2016. It is a school in Mashpee where Wampanoag children are taught mostly in Wôpanâak.

Cherokee, Lakota, Diné and many other nations run immersion schools of their own.

Deb Haaland

Who: Laguna Pueblo. She was one of the first two Native American women elected to Congress, and the first Native American cabinet secretary.

When and where: New Mexico and Washington, D.C., from 2018.

Her grandparents were taken to boarding school as children.

Deb Haaland is a citizen of Laguna Pueblo. She describes herself as a 35th-generation New Mexican. Members of the Senate voted to approve her as Secretary of the Interior on March 15, 2021.

The line about her grandparents is from her own account, which does not name who took them. Three months after the Senate vote, she ordered officials of her department to count what earlier officials had done at the schools. They published the final count on July 30, 2024, while she was still the secretary.

Jessie Little Doe Baird

Who: Mashpee Wampanoag social worker who became a linguist and brought back a language that no one had spoken since the middle of the 1800s.

When and where: Mashpee, Massachusetts, from 1993.

Movie: *We Still Live Here: Âs Nutayuneân* (2011)

She rebuilt a language using a Bible printed in 1663.

By her own account, the work began with a dream about the language.

Among the written records of the language was the Eliot Bible, a Bible printed in Wôpanâak in 1663. Baird used that Bible and the other documents to rebuild the language.

After she helped found the project in 1993, Baird went to the Massachusetts Institute of Technology and earned a degree in linguistics, the study of how languages work. Her daughter grew up speaking Wôpanâak. In 2010 the MacArthur Foundation gave Baird a MacArthur Fellowship, a prize for people's work.

THE THREE QUESTIONS, ANSWERED

1. **How did Native nations govern themselves and deal with other nations before the United States existed?** They ran their own councils and laws, as the Haudenosaunee did under the Great Law of Peace. Their leaders made alliances and treaties and recorded them on wampum belts.
2. **What did United States officials and soldiers do to take the nations' land and break the nations up?** Soldiers forced whole nations west and onto reservations, and killed hundreds of people at a time at Sand Creek, Wounded Knee and other places. Officials cut

the nations' land from about 138 million acres to 48 million and took children to boarding schools, where at least 973 died.

3. **How did Native nations hold on to their governments, their languages and some of their land?** They went to court, from *Worcester v. Georgia* in 1832 to *McGirt v. Oklahoma* in 2020. They organized until members of Congress restored the Menominee, and they kept their languages, first in secret and now in schools of their own.

What changed: Nations that held nearly the whole continent in the 1600s had lost most of it by 1900, and after 1950 they won back the running of their own governments, schools and courts.

FIVE DATES TO KEEP

1680: Pueblo fighters drive the Spanish out of New Mexico.

1838: Soldiers force the Cherokee west on the Trail of Tears.

1887: Allotment begins under the Dawes Act.

1974: Judge George Boldt upholds the Northwest fishing treaties.

2020: Five justices rule that the Muscogee (Creek) reservation never ended.

NOT KNOWN: whether anyone carried out Oñate's 1599 sentence to cut off the Acoma men's feet, which the Acoma account says happened to 24 men.

Where to go: The National Museum of the American Indian, on the National Mall in Washington, D.C.

Next: Chapter 3, Land and Environment. It asks how Americans cleared, used up and protected the land, and who lost most at each step.

Chapter 3: Land and Environment

CHAPTER 3 OF 37 · LAND AND ENVIRONMENT · 10 ERAS · ABOUT 95 MINUTES

Americans cleared this land, used it up and then saved pieces of it, and at almost every step the people who lost most had the least say: first Native nations, later poor and Black neighborhoods.

Miantonomi, a leader of the Narragansett nation, told the Montaukett people on Long Island in 1642 what the English were doing to the land. The plains had once been full of game, he said, but the English were cutting down the grass and the trees. Their cows and horses ate the grass, their hogs spoiled the clam banks, and he warned that his listeners would all starve. An English soldier wrote his words down. The next year, English officials sentenced Miantonomi to death.

This chapter follows the land itself through ten eras, from Native nations burning forests on purpose to government fire crews learning that same kind of burning from Yurok people today. Keep an eye on two things. The first is the numbers. Beavers, salmon, forests, 30 million bison and billions of passenger pigeons fall era by era. Some, like the eastern forests and the bison, later climb back, and the passenger pigeon never does. The second is who had to move. The families driven out of the first national parks were Native. The neighborhoods beside the waste dumps of the 1900s were far more likely than others to be Black or poor.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did Native nations shape the land before Europeans came?
2. What happened to the 30 million bison and the billions of passenger pigeons?

3. Who was pushed out when parks were made, and who lived next to the waste?

Five minutes? Read "Yosemite and the Ahwahneechee" in 1850 to 1900.

Can skip: "Whaling from Nantucket", "The forest reserves", "Plastic in the rain". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 2 MINUTES

Native nations shaped the land of North America for thousands of years before any European arrived. In the eastern forests, people set fires on purpose to keep the woods open and to grow food for the animals they hunted. Farmers planted three crops together in a way that kept the soil good. In southwest Florida, the Calusa built pens in the water to hold live fish.

Animals shaped the land too. Beavers dammed streams across the continent, and bison grazed the Plains. Nobody knows how many bison lived there before 1500. The figure usually given, about 30 million, is for the early 1800s, before hunters killed off the herds.

Why Native people burned the woods

For thousands of years, Native people in what is now the eastern United States burned their forests on purpose to keep them open. They set low fires in the understory, the layer of brush and small plants under the tall trees. They burned the same ground again and again. The fires helped oaks and other trees that can live through fire. Food plants grew back after each burn, and so did food for game, the wild animals people hunted.

What the trees show. Researchers at Penn State, a university in Pennsylvania, published a study in the journal *Annals of Forest Science* in 2019. They found that which trees grew in the eastern forests depended more on Native burning than on the climate. Even small groups of people burned large areas, over and over. European colonists treated this burning as unimportant.

Three crops in one field

Native farmers grew corn, beans and squash together in the same field, three crops known as the Three Sisters. The beans put nitrogen, a natural substance that plants need to grow, into the soil. The wide squash leaves shaded the ground and held the soil in place. The tall corn stalks gave the bean vines something to climb.

Growing the three crops together kept the soil good for years. A field planted with only one crop, year after year, wears out.

Beavers, bison and fish pens

Beavers changed whole watersheds, all the land that drains into one stream or river. Their dams held back water in ponds. The ponds slowed floods, and the wet ground around them became meadows. Nobody counted the beavers, so the numbers are estimates. They run from 60 million to 400 million beavers in North America before the fur trade, the buying and selling of animal skins between European traders and Native trappers.

Bison lived on the Plains, the wide grasslands in the middle of the continent. About 30 million lived there in the early 1800s. Estimates for the years before Europeans came are less certain.

In what is now southwest Florida, the Calusa people built watercourts between 1300 and 1400. These were pens in the water that held live fish until people needed them for food.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

Two things changed the land of the Americas in the 1500s: new animals arrived from Europe, and most of the Native people who had farmed the land died. Ships brought pigs, horses, cattle and earthworms. Pigs that escaped from the Spanish explorer Hernando de Soto and his men went wild in the Southeast, and European earthworms moved into northern forests that had never had worms.

Most of the Native people died of diseases that came across the ocean with Europeans: with European settlers, and with the Africans whom Europeans had enslaved and carried over. European soldiers and settlers killed others in wars to take Native lands, and many starved. Forest grew back over the farmland of the people who died.

Pigs, horses and cattle

Pigs that escaped from Hernando de Soto's herd became the ancestors of the feral hogs of the Southeast, pigs from farm stock that live in the wild.

Thirteen pigs. De Soto landed near Tampa Bay, in what is now Florida, in May 1539, with about 600 men and a herd of pigs. The written histories of his expedition list the first herd as 13 pigs. By the time de Soto died in 1542, the herd had grown to about 700. That count leaves out the pigs the men ate or gave away and the pigs that escaped.

Horses and cattle came with the Spanish. Columbus brought horses on his second voyage, in 1493, and Spanish colonists brought horses and cattle to their colony in New Mexico in 1598.

Earthworms

Before Europeans came, the forests of northern North America had no earthworms. Glaciers, thick sheets of ice that once covered the north, had scraped the ground bare of them. For roughly the last 12,000 years, the forests that grew back after the ice stood on a thick, soft layer of fallen leaves called duff. Native wildflowers and young trees grew in it.

European earthworms crossed the ocean hidden in ballast soil, heavy dirt carried low in a ship to keep it steady. Worms also came in the soil around the roots of plants brought from Europe. Later they spread west with wagons, horses and farms. Where European worms move into a northern forest, they eat the duff that the wildflowers and young trees need.

Forest over lost farms

Most of the Native people of the Americas died after 1492. Epidemics, diseases that spread quickly to many people, killed most of them. European settlers brought the diseases, and so did enslaved Africans whom Europeans had carried across the ocean. The diseases included smallpox, measles and influenza, and Native people had never been exposed to any of them before.

European soldiers and settlers made war on Native nations to take their land, and they enslaved Native people. Famine, a lack of food so great that people starve, followed when too few people were well enough to farm.

What the deaths did to the land. A team of researchers at University College London, in England, published an estimate in 2019. They estimated that the Native population of the Americas fell from about 60 million to about 6 million within roughly a hundred years. Other estimates of the starting number differ. For North America north of Mexico alone, they run from 1 million to 18 million.

The team estimated that about 55.8 million hectares of farmland grew back into forest after its farmers died. A hectare is a square 100 meters on each side, so that is an area bigger than California.

The new trees took carbon dioxide, a gas that traps heat, out of the air. By the team's estimate, the regrowth lowered the carbon dioxide in the air by 7 to 10 parts in every million. It cooled the whole planet by about 0.15 degrees Celsius, about a quarter of a degree Fahrenheit. All of these figures are this one team's estimates.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 4 MINUTES

English colonists and the Native nations of New England had different ideas in the 1600s about what land was for. The colonists treated land as property that one owner held forever, marked off with fences and recorded in deeds, the papers that say who owns a piece of land. The Native nations used their land by season, with planting grounds, fishing places and hunting grounds, and no one person owned it forever.

The historian William Cronon describes these two ways of using land in his 1983 book *Changes in the Land*. The colonists let their cattle and hogs roam free. The animals got into Native planting fields, and colonists and Native people quarreled over it again and again, as the Narragansett leader Miantonomi told his listeners in 1642.

Trappers changed the land too. They caught so many beavers for the fur trade in these years that the beaver's numbers began a long fall.

The beaver trade

Hat makers in Europe wanted beaver fur to make felt, a thick cloth made by pressing fur together. Native trappers caught the beavers, and European traders paid for the furs with goods made in Europe. The number of beavers in the Northeast began to fall in the 1600s.

The fall went on until about 1900, when about 100,000 beavers were left in North America, most of them in Canada. Today there are roughly 15 million, against the 60 million to 400 million of the years before the fur trade. All of these numbers are estimates.

The first limits on cutting trees

A Plymouth Colony rule of 1626 forbade anyone to cut timber, wood for building, on the colony's land, or to sell it, without official permission. It was among the earliest conservation rules in the colonies, rules for protecting land, water, plants and animals so that people do not use them up.

William Penn wrote rules for settlers in Pennsylvania in 1681. For every five acres a settler cleared of trees, Penn's rules required the settler to leave one acre, a piece of land a little smaller than a football field, still covered in trees. The rules named oaks and mulberry trees, which were useful for ships and for silk. Settlers widely ignored this rule.

Miantonomi

Who: A sachem, or leader, of the Narragansett nation, who said out loud in 1642 what English settlers were doing to Native land.

When and where: A speech to the Montaukett people on Long Island in 1642, written down by the English soldier Lion Gardener. He was killed near present-day Norwich, Connecticut, in 1643.

English officials sentenced him to death and handed him to his enemy.

Miantonomi spoke to the Montaukett people on Long Island in 1642. English settlement in New England was then about one generation old, and he told them what the English were doing to the land.

Lion Gardener, an English soldier, wrote the speech down in his *Relation*, an account he set down in 1660, years later. By Gardener's record, Miantonomi said that "our fathers had plenty of deer and skins and our plains were full of game and turkeys."

Then he said that the English "with scythes cut down the grass, and with axes fell the trees" (a scythe is a long curved blade for cutting grass, and to fell a tree is to cut it down). He went on: "their cows and horses eat the grass, and their hogs spoil our clam banks, and we shall all be starved."

KNOWN:	The complaints match what colonists were doing at the time, cutting the forests and letting their cattle and hogs roam into Native fields.
NOT KNOWN:	Nobody can check Miantonomi's exact words, because an Englishman wrote them down years after the speech.

The sentence. Uncas, the sachem of the Mohegan nation, captured Miantonomi in 1643 and took him to Hartford. That September the commissioners of the United Colonies of New England met in Boston. They were English officials from the Massachusetts Bay, Plymouth,

Connecticut and New Haven colonies. They passed the case to a group of ministers, or church leaders. Then the commissioners sentenced Miantonomi to death and handed him back to Uncas to carry out the sentence.

Governor John Winthrop of Massachusetts Bay wrote in his journal that "they all agreed that he ought to be put to death." He gave their reason, spelling the name a different way: "Uncas could not be safe while Miantonomo lived."

Mohegan men took Miantonomi back to the place near present-day Norwich, Connecticut, where he had been captured. There one of Uncas's men, walking behind him, killed him with a hatchet blow to the head. Mohegan tradition and later histories name that man as Wawequa, the brother of Uncas. By the historian Neal Salisbury's account, several Englishmen went along to make sure the killing was carried out.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 3 MINUTES

Colonists of the early 1700s used land and sea as if there would always be more. Tobacco planters in Virginia, the owners of large farms called plantations, wore out the soil of a field in three or four years. Then they cleared new ground farther up the rivers and moved on.

In New England the king's officials saved one thing, the largest white pines, for the masts of the Royal Navy, Britain's navy. Whalers from the island of Nantucket were sailing far out into the ocean to hunt sperm whales by the middle of the century.

The king's pines

The king's officials kept the largest white pines in New England for the masts of the Royal Navy's ships, the tall poles that hold up the sails. The rule appeared in the Massachusetts Bay charter of 1691, the written grant from the king that set up the colony's government. It covered every white pine 24 inches across or more.

Under the White Pine Act of 1722, cutting any white pine more than 12 inches across without a license, or written permission, was against the law in New Hampshire and nearby colonies. The king's trees carried a mark called the Broad Arrow, three slashes cut with a hatchet. Royal officials enforced these laws loosely until the 1760s.

Worn-out tobacco fields

A tobacco field in colonial Virginia gave good crops for only three or four years. After that, planters cleared new ground and let the old field grow over with scrub, brush and small trees. The land around Jamestown wore out early. Planters moved up the rivers and reached the Rappahannock and the Potomac by about 1650, and they kept moving up the rivers all through the 1700s. Because new land was cheap, they kept wearing out one field after another.

Whose land it was. The new land along the Rappahannock River was the homeland of the Rappahannock Tribe. The tribe's own history states that English settlement there began illegally in the 1640s. By that history, Rappahannock leaders sold their first land to the English in 1651. They spent more than ten years in county courts trying to get paid for that sale and others, and they never received full payment.

The tribe's history goes on to say that by the late 1660s, settlers and bands of settlers who acted as their own law had forced the Rappahannock to move. The tribe moved first inland on the north side of the river, and later to its old hunting grounds on the south side. In 1683, the history states, officials of the Virginia colony forced the tribe out of their homes and moved them to a place called Portobago Indian Town.

Who did the work. Indentured servants and enslaved Africans worked the tobacco fields. An indentured servant worked for a master for a set number of years, often four to seven, to pay for the trip to America. Planters held enslaved Africans as property and forced them to work without pay for life. Under a Virginia law of 1662, a child born to an enslaved mother was enslaved for life too. Growing tobacco took a great deal of hand work, and planters wanted more and more workers to do it.

Whaling from Nantucket

Whalers from Nantucket, an island off Massachusetts, were hunting sperm whales far out in the ocean by the middle of the 1700s. Before that, New Englanders hunted right whales, another kind of whale, from the beaches. On Nantucket, people tell a story that deep-sea whaling began there in 1712, when wind blew Christopher Hussey's boat out to sea and the men aboard killed a sperm whale.

KNOWN:	Obed Macy first wrote the story down in his history of Nantucket in 1835, more than 120 years later, and historians at the Nantucket Historical Association call 1712 the traditional date.
NOT KNOWN:	No record from 1712 shows that Hussey's voyage happened the way the story tells it.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 5 MINUTES

After independence, American leaders turned the land beyond the mountains into squares that could be sold, and they took it from the nations who lived there with treaties signed under threat and then with armies. In 1785 members of Congress ordered the land north and west of the Ohio River measured into a grid. The Shawnee, the Miami, the Lenape (Delaware) and other nations lived on that land. That January, US officials had claimed most of Ohio in a treaty, a written agreement between nations, signed under threat.

Surveyors measured the land under army guard. US generals led armies against the nations, who had joined together to stop settlement, from 1790 to 1794. After General Anthony Wayne's army defeated them in 1794, leaders of eleven nations signed a treaty in 1795 in which they handed most of Ohio to the United States.

The king's claim to New England's biggest white pines ended with independence. Before that, in 1772, men in Weare, New Hampshire, beat a sheriff who came to arrest a mill owner fined for cutting them. Once the king's claim was gone, settlers crossed the Appalachian Mountains and cleared the forest for farms.

Ebenezer Mudgett

Who: A sawmill owner in Weare, New Hampshire, fined for cutting the king's white pines, who led the Pine Tree Riot.

When and where: South Weare, New Hampshire, April 13 and 14, 1772.

At dawn, 20 to 30 men with soot-blackened faces beat the sheriff.

Ebenezer Mudgett owned a sawmill in Weare, New Hampshire. He had been fined for cutting the king's white pines, the trees kept by law for the masts of the Royal Navy.

Sheriff Benjamin Whiting rode into South Weare on April 13, 1772, with a warrant to arrest Mudgett, a paper from a court that allowed the arrest. Mudgett told him he would pay in the morning.

At dawn on April 14, Mudgett came back with 20 to 30 men who had blackened their faces with soot, the black powder that smoke leaves behind. They beat the sheriff and his deputy with rods and drove them out of town. The fight became known as the Pine Tree Riot.

The trial. Eight men stood trial in the Superior Court at Amherst, New Hampshire, in September 1772, for rioting and for beating Whiting. They pleaded guilty. The four judges fined each man only 20 shillings, in British coins, and made them pay the cost of the hearing. After the riot, officials mostly stopped enforcing the law that kept the pines for the navy.

KNOWN:	Some accounts count the riot as a forerunner, an earlier event of the same kind, of the Boston Tea Party, a protest in December 1773.
NOT KNOWN:	No one has proved that the riot led to the Tea Party.

The grid, 1785

Members of the Confederation Congress, the national government of the United States before the Constitution, passed a law called the Land Ordinance on May 20, 1785. It specified how the public land would be sold. Public land was land that US officials claimed for the nation, and Native nations lived on much of it. Under the ordinance, surveyors had to survey that land, which means measure it and map it, before anyone could buy it.

The surveyors divided the land into townships, squares six miles long on each side. They cut each township into 36 sections. A section was one square mile, which is 640 acres. The sections were numbered, and members of Congress set aside section 16 in every township to support public schools.

After changes to the system in 1796, surveyors carried the grid across the continent to the Pacific Ocean. People in airplanes over the Midwest can still see its checkerboard pattern on the ground.

Whose land the grid covered

The land north and west of the Ohio River was the home of the Shawnee, Lenape, Wyandot, Miami, Ottawa, Chippewa, Potawatomi and other nations in 1785. As members of Congress saw it, the historian George W. Knepper writes, "Indian claims to that land had to be cleared." Here, cleared means ended, so that no Native nation still had a right to the land.

The treaty at Fort McIntosh. US commissioners, officials sent to do a certain job, met leaders of several nations at Fort McIntosh in January 1785, four months before the Land Ordinance passed. In the treaty they made there, the commissioners said that most of the Native people of Ohio had to stay inside a reserve, an area set aside for them, between the Cuyahoga and Maumee rivers. Under the treaty, Knepper writes, most of Ohio would be open for settlement.

The Shawnee refused to obey the treaty. Knepper writes that it was made under threat by minor chiefs who had no right to speak for their people. A year later, US commissioners forced the Shawnee into another treaty, at Fort Finney.

The first survey. Surveyors for the national government began measuring this land in eastern Ohio in September 1785, and they stopped after a few miles when they heard of an attack by Native people. They went back to work in 1786 under army guard, and they measured the land for sale while the nations who lived on it still rejected the treaty.

The war for Ohio and the Treaty of Greenville

Chapter 2 tells this war from the nations' side. The nations of the region had joined in a confederacy, a union of nations, to stop American settlement. Led by Little Turtle of the Miami and others, its warriors beat American armies in 1790 and 1791, and General Anthony Wayne's army defeated the confederacy at Fallen Timbers in 1794. Here is what the war meant for the land.

The land the nations lost. A year after Fallen Timbers, on August 3, 1795, leaders of eleven northwestern nations signed the Treaty of Greenville. In it they handed nearly 17 million acres to the United States, an area a little bigger than West Virginia. Under the treaty, the nations kept only the northwest quarter of Ohio and the lands beyond it, behind a boundary called the Greenville Treaty Line.

Clearing after independence

After independence, the British king no longer had any right to the white pines in the United States. Settlers crossed the Appalachian Mountains, some of them into the Northwest Territory, the US name for the land north and west of the Ohio River, and cut down the forest to make farms.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 5 MINUTES

New Englanders put their rivers to work and cut down most of their forests in the first half of the 1800s, and a few Americans began to write about saving wild places. Mill dams across the rivers stopped the salmon from swimming upstream to lay their eggs. By 1850 farmers had cleared most of New England's forest, partly to feed a boom in sheep.

The painter George Catlin wrote in 1832 about keeping a huge park for the bison and the Plains nations, which he called "a nation's park." He meant a park that would hold both the bison and the Native nations who lived there. In these same years John James Audubon painted the birds of North America, and Henry David Thoreau lived for two years in a cabin he built at Walden Pond, in Massachusetts.

Dams stop the fish

Atlantic salmon live in the ocean but swim up rivers to spawn, or lay their eggs. A dam across a river blocks them on the way. The first dam across the Connecticut River went up at Turners Falls, Massachusetts, in 1798, and it cut the salmon off from the water where they spawned. By the War of 1812, no salmon came into the Connecticut River at all.

The Great Stone Dam. Workers for the Essex Company, a company formed to use the river's water to power mills, built the Great Stone Dam across the Merrimack River at Lawrence, Massachusetts. They finished it in 1848. It stood 30 miles from the sea, and salmon could not get past it. By 1865 the salmon runs, the yearly trips of fish up a river, were gone from the rivers of southern New England. The runs of shad, another fish, collapsed as well.

The writers of an 1857 report in New Hampshire named three causes: the dams, fishing with no limits, and mill machinery that killed young fish.

The bare East

New Englanders had cut down most of their forest by 1850. Estimates of how much was cleared run from about 60 percent to 80 percent of the land, and some places were cleared even more. One estimate puts the forest left in 1850 as low as a quarter of New England.

A boom in Merino sheep, a kind of sheep raised for its wool, was one cause of the clearing. By 1840 New England farmers kept about 4 million sheep, more sheep than there were people, and they cleared still more forest to feed them.

Across the eastern United States, the forest reached its smallest size around 1872, roughly half the land it had covered in 1620. After that, many eastern farmers left their farms and moved west, and trees grew back on the fields they left behind.

A park for people and bison

The painter George Catlin traveled far up the Missouri River, into the West, in 1832. He feared that the bison and the Plains nations would be destroyed together. He wrote of "a magnificent park," which he called "a nation's park, containing man and beast, in all the wild and freshness of their nature's beauty." In Catlin's plan, the people who lived on the land would stay on it with the animals.

Forty years later, in 1872, Yellowstone became the first national park. By 1879 its superintendent, the official in charge, had the Tukudika people who lived there forced out and taken to a reservation, land set aside for a Native nation to live on. The next era tells how.

John James Audubon

Who: A painter and naturalist, a person who studies animals and plants in the wild. He painted *The Birds of America*, 435 life-size pictures of birds.

When and where: He lived from 1785 to 1851. His book came out from 1827 to 1838. In 1813, on the way to Louisville, Kentucky, he saw a flock of passenger pigeons that he figured at more than 1 billion birds.

Movie: *John James Audubon: Drawn from Nature* (PBS American Masters, 2007)

One flock of passenger pigeons darkened the sky for three days.

John James Audubon painted the birds of North America at their real size. His book *The Birds of America* contains 435 of these paintings. It came out from 1827 to 1838, and Audubon worked on it for about 18 years.

Audubon was riding to Louisville, Kentucky, in 1813 when a flock of passenger pigeons, wild pigeons that flew in huge flocks, passed over him. In his own account, that one flock darkened the sky for three days. He did the arithmetic himself and put it at more than 1 billion birds.

Why this story is here: About a hundred years after Audubon's count, the passenger pigeon was gone.

Henry David Thoreau

Who: A writer who also worked as a land surveyor, measuring woodlots for a living.

When and where: He lived from 1817 to 1862. He lived in a cabin at Walden Pond, near Concord, Massachusetts, from July 4, 1845, to September 6, 1847.

He lived at Walden Pond for two years, two months and two days.

Henry David Thoreau wrote about wild nature, and he earned his money measuring land. He worked as a land surveyor, measuring and mapping woodlots, pieces of land kept for growing trees.

Thoreau moved into a cabin he had built himself at Walden Pond, near Concord, Massachusetts, on July 4, 1845. He lived there until September 6, 1847. His book about those years, *Walden*, came out in 1854.

Thoreau gave a talk called "Walking" in Concord on April 23, 1851. It came out as an essay in June 1862, just after he died. In it he wrote that "in Wildness is the preservation of the world," using preservation to mean keeping something safe.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 22 MINUTES

US officials gave away the land they had taken from Native nations in these fifty years, and hunters killed nearly all the bison, the animals the Plains nations lived on. Members of Congress passed laws in 1862 for giving the land to railroad companies, to settlers called homesteaders and to the states for colleges. By one historian's count, US officials took more than 1.5 billion acres from the nations between 1776 and 1887.

On the Plains, hide hunters killed the bison for their skins, which are called hides. About 30 million bison lived there in the early 1800s, and fewer than 1,000 were left by 1900. Army generals wrote in favor of the killing, to take away the nations' food and force them onto reservations. Professional hunters killed passenger pigeons at their nesting grounds, a reported 50,000 birds a day for months at one Michigan nesting in 1878.

The first parks were made on Native land, and soldiers and officials forced out the people who lived there. Volunteer soldiers drove the Ahwahneechee people out of Yosemite Valley in 1851, and President Abraham Lincoln signed the Yosemite Grant in 1864. Yellowstone became the first national park in 1872, and its superintendent had the Tukudika people forced out in 1879. John Muir helped found the Sierra Club in 1892, in part to protect the forests and mountains of California's Sierra Nevada.

Land taken from Native nations, in acres

US officials took more than 1.5 billion acres from Native nations from 1776 to 1887, more than half the area of the United States today. They took it by treaty and by executive order, a command from the president. Historian Claudio Saunt, of the University of Georgia, made this count from maps that Charles C. Royce published in 1899, which show every piece of land US officials took from the nations by treaty or order.

Once officials had taken the land, it counted as public land, land that US officials claimed for the nation. Members of Congress then passed it on to railroads, settlers and colleges.

Railroads. President Abraham Lincoln signed the Pacific Railway Act on July 1, 1862. Under it, railroad companies received "every alternate section of public land, designated by odd numbers," meaning every other square mile. For each mile of track, a company got five of these sections on each side of the line.

The act also states: "The United States shall extinguish as rapidly as may be the Indian titles to all lands falling under the operation of this act." With those words, which use extinguish to mean end, the lawmakers told officials to end the nations' legal ownership of the land along the railroads as fast as they could. From 1850 to 1871, members of Congress granted railroad companies about 130 million acres, an area bigger than California.

Homesteads. Lincoln signed the Homestead Act on May 20, 1862. Under it, a person could claim up to 160 acres of public land and keep it after living on it or farming it for five years. Over the life of the law, US officials handed 270 million acres to private owners this way, 10 percent of the area of the United States. A National Park Service page states that much of this land "was seized from the Native Americans who previously inhabited it." Seized means taken by force.

Colleges. Lincoln signed the Morrill Act on July 2, 1862. Under it, each state received 30,000 acres of public land for each of its senators and representatives in Congress. State officials sold the land to pay for colleges

of "agriculture and the mechanic arts," which means farming and the use of machines.

A page about the act from the National Archives, the office that keeps the government's records, states that the act "resulted in the expropriation," or taking away, "of over 10 million acres from Native American tribal lands."

The reporters Robert Lee and Tristan Ahtone traced that land for the magazine *High Country News* in 2020. US officials got it through takings that the reporters call "violence-backed," meaning that violence or the threat of it forced the nations to give the land up. Officials paid the nations less than $400,000 in all. By the early 1900s, sales of the land had raised $17.7 million for 52 universities, more than 40 times what the nations were paid. In 2020 the universities still held more than 500,000 acres of it.

One example comes from Minnesota. Four bands of Dakota people signed treaties at Mendota and Traverse des Sioux in the summer of 1851. The reporters describe the conditions as "withholding of rations," or food supplies, "the threat of violence, enforced starvation." The Dakota received "less than 2.4¢ per acre."

Allotment. Chapter 2 tells how federal officials used the Dawes Act of 1887 against the nations. Here is what became of the land. Under the act, federal officials cut reservations into plots, 160 acres for the head of a family and less for others, and opened the land left over, which they called "surplus," to white settlers.

By the National Park Service's count, reservation land fell from 138 million acres in 1887 to 48 million acres in 1934. About 90 million acres, 65 percent of what the nations held, passed out of their hands.

Fire made a crime, 1850

California's state lawmakers passed a law called the *Act for the Government and Protection of Indians* on April 22, 1850. Under Section 10 of the law, it was a crime to set the prairie on fire, or to refuse to help put out a prairie fire. Section 10 states that such people "shall be subject to fine or punishment, as a Court may adjudge proper." That means a judge would choose the punishment.

Section 10 speaks of "any person or persons." The whole law was about Native people, though, and burning the land on purpose was a Native practice. Reporters at the news outlet ICT write that under California law, cultural fire, burning done by Native people to care for the land, was "explicitly banned" for more than 150 years.

The rest of the law. Under other sections of the same law, white people could take custody of Native children, meaning control over the children and where they lived, until the children were grown. Officials could also arrest Native people they called "vagrants," meaning people with no home or work. Within 24 hours, officials could hire those people out "to the best bidder" for up to four months. This was forced labor: the arrested person had to work for whoever won the bidding.

Yosemite and the Ahwahneechee

The Ahwahneechee people lived in Yosemite Valley, in California's Sierra Nevada mountains, and their headman, or leader, was Tenaya. In 1851 volunteer soldiers burned their villages and their food and drove them out of the valley.

How it started. James D. Savage was a trader who had moved onto Ahwahneechee land to mine gold and to trade. Ahwahneechee fighters attacked his trading post on the Fresno River in December 1850, took his supplies and killed two of his men. One account says Chowchilla fighters joined them.

California's governor, John McDougal, called for volunteers early in 1851 after attacks on miners and traders. About 200 men on horseback formed the Mariposa Battalion, a unit of soldiers, on January 24, 1851. Their commander was Major James D. Savage. State officials supplied their camp gear and supplies. Federal money paid for the battalion, through federal officials called US Indian commissioners.

What the battalion did. The battalion's men first entered Yosemite Valley on March 27, 1851, by the Park Service's date. A National Park Service page states that they "systematically burned villages and food supplies and forced men, women, and children away from their homes."

Lafayette H. Bunnell rode with the battalion and wrote an account of it in 1880. He writes that in Captain John Boling's company, men on foot had orders "to burn any caches they might find," the hidden stores of food. Acorns were the people's main food for the winter.

On the upper San Joaquin River, next to Yosemite, Bunnell says the huts and "adjacent supplies of acorns, were at once given to the flames." He estimated the acorns burned there at 800 to 1,000 bushels, and pine nuts too. A bushel fills a large basket, so that was 800 to 1,000 large baskets of winter food.

In Yosemite Valley, Bunnell found that "nearly all the acorns had been burnt." The battalion's men marched Tenaya and his people to a reservation on the Fresno River.

Tenaya's son. Battalion men held Tenaya's youngest son and another young man as hostages, prisoners held to force others to do something, tied to a tree. Bunnell writes that the guards let the two untie themselves "in order to fire upon them, expecting to kill them both." One guard shot Tenaya's son in the back and killed him. The other young man escaped.

Bunnell describes the shooter only as "a young man" urged on by "an old Texan," and he names neither of them. Captain Boling told the guards "that they had committed murder." Bunnell writes that "no punishment was ever inflicted upon the perpetrators," meaning the men who did it. Tenaya was killed in 1853 in a fight with Mono Lake Paiutes. Accounts of why the fight happened differ.

The park. President Lincoln signed the Yosemite Grant on June 30, 1864. With it, US lawmakers set aside Yosemite, the first land the nation's lawmakers set aside to protect and to keep for public use. California's state officials took charge of it under the grant. Yosemite became a national park on October 1, 1890.

The National Parks Conservation Association, a group that campaigns for the parks, states in an article that the 1890 park came "with authorization," or official permission, "to remove anyone making homes without permission." Ahwahneechee families came back and lived in the valley anyway, and they worked for the hotels and the park.

The count of the dead. A Park Service page states that by 1910, more than 90 percent of the Ahwahneechee who had lived in the valley were dead or missing. The page does not say how each of them died or went missing.

To Tu Ya, also called Maria Lebrado Ydrte, saw the 1851 raid as a child. She was Tenaya's granddaughter. She saw the battalion's men destroy her home in Yosemite Valley, and during the raid she saw her uncle murdered.

Because she was Tenaya's granddaughter, Tenaya's sons were her uncles, so the son the guard shot was one of them. The Park Service page does not say whether he was the uncle she saw killed, and it does not name who killed that uncle. The page describes To Tu Ya as the last survivor of the raid. She came back to Yosemite in the late 1920s and told her story.

BY 1887: Railroads, homesteaders and colleges held hundreds of millions of acres that had belonged to Native nations, and US officials had begun cutting up the reservations that were left.

Hide hunters and the railroads

Hide hunters, who sold the skins for money, killed most of the Plains bison. About 30 million bison lived on the Plains in the early 1800s. By 1900 fewer than 1,000 were left, about one for every 30,000, and one count in 1884 found only 325. The hunters called themselves "buffalo runners," because bison are also called buffalo.

The railroads. The first railroad across the country, finished in 1869, carried hunters out to the herds. Leather makers at a tannery in Pennsylvania, a place where animal skins are made into leather, found a way in 1871 to turn bison hides into leather for harnesses and machine belts. Each hide sold for $1 to $3. The Kansas Pacific and the Atchison, Topeka and Santa Fe railroads carried the hides east.

Lieutenant Colonel Richard Irving Dodge, an army officer, estimated that three railroads carried about 1.4 million hides from the southern Plains in 1872, 1873 and 1874. He had a count from one line, the Santa Fe, and guessed that the other two lines carried as many.

Dodge also figured that hunters wasted so many hides before 1874 that about three bison were killed for every hide shipped. By his estimate, that adds up to more than 3 million bison killed by white hunters in those three years, and about 1.2 million more killed by Native hunters, about 4.4 million in all. William T. Hornaday of the Smithsonian Institution reported these totals in 1889. They are Dodge's estimates, not counts.

Dodge described Kansas in the fall of 1873: "Where there were myriads of buffalo the year before, there were now myriads of carcasses. The air was foul with a sickening stench." (Myriads means huge numbers, and carcasses are the dead bodies of animals.)

Officers who wanted the bison gone

Army officers encouraged the hide hunters. Several officers also wrote that killing the bison would force the Plains nations onto reservations.

KNOWN:	No one has found a written order to kill off the bison.
DISPUTED:	Historians disagree about whether the army had a plan anyway. Robert Utley writes that the army had no official policy, but that officials cheered the killing because it would force the nations onto reservations. David Smits argues that the army followed a plan that was mostly never written down. William Dobak doubts that the army caused much of the killing at all.

What the officers wrote. General William T. Sherman wrote to General Philip H. Sheridan in May 1868 that the nations would go wherever the bison were. He suggested inviting hunters from England and America to a great buffalo hunt to "make one grand sweep of them all." In 1869 Sherman became the commanding general of the whole army.

Sheridan answered in a letter of October 15, 1868, that the best way was "to now make them poor by the destruction of their stock, and then settle them on the lands allotted to them." By "stock," Sheridan meant the nations' bison and horses. A month earlier, another officer had written in his journal that his orders were to kill all the buffalo his men found.

Not every officer agreed. Colonel William B. Hazen wrote in 1872 that killing the buffalo to take away the nations' food was a false idea, and he called the killing a waste.

Help from the forts. Officers at army posts, among them Forts Griffin, Dodge, Concho and Richardson, gave the hunters protection and supplies and helped them ship hides. In 1873 hide hunters asked Dodge, then in command at Fort Dodge, what he would do if they hunted south into Texas, on hunting grounds reserved for the Kiowa, Comanche, Southern Cheyenne and Arapaho under the 1867 Medicine Lodge Treaty. One of the hunters later recalled that Dodge told them to hunt buffalo where the buffalo were.

Colonel Nelson A. Miles wrote out his plan in December 1874, during the Red River War of 1874 and 1875: "I would leave as few Indians or Buffalo behind, and as little for them to subsist on as possible." (To subsist is to stay alive.) In a book about his own life, published in 1896, he wrote that the buffalo and the Indian had both stood in the way of what he called civilization.

Officials in Washington

Leaders of the Plains nations asked officials to stop the hunters, and the officials did not stop them. A Sioux delegation, a group sent to speak for its people, asked Columbus Delano in 1871 to keep white hunters out of Sioux country. Delano was the Secretary of the Interior, one of the president's top officials. He refused.

Colonel Edward Wynkoop had been the Cheyenne's Indian agent, the US official in charge of their reservation. In 1872 he reported words that Little Robe, a Cheyenne chief, had said to him: "What do you think my people ought to do when they see their cattle, the buffaloes, killed by your race when they are not hungry?" Only Wynkoop's report records these words.

Delano's yearly report for 1873 records that he "would not seriously regret the total disappearance of the buffalo from our western prairies, in its effect on the Indians." He saw the loss of the bison as "a means of hastening their sense of dependence upon the products of the soil and their own labors." He meant that without the bison, Native people would have to live by farming.

A law that never was. Members of both houses of Congress passed a bill to protect the bison in 1874. Under it, no one but a Native person could kill a female bison, or more males than needed for food, and the fine would be $100 for each animal. President Grant did not sign the bill. Members of Congress ended their meeting before he signed it, so the bill never became law. This is called a pocket veto.

Grant had promised a Southern Cheyenne and Arapaho delegation that hunters would be kept out of their country. Their agent, John D. Miles, wrote on September 30, 1874, that "they looked in vain." They watched for the hunters to be kept out, and the hunters kept coming.

Texas lawmakers brought in their own bill to protect the bison in 1875. The *Handbook of Texas*, published by the Texas State Historical Association, states that Sheridan protested it, "contending that peace with the Indians could be maintained only if their food supply was eliminated." The lawmakers did not pass the bill.

The speech in Texas. A speech that Sheridan is said to have given against the bill comes from one source only. The former hide hunter John R. Cook wrote about it in 1907, about 32 years later. In Cook's account, Sheridan told the lawmakers to give each hunter "a medal of bronze, with a dead buffalo on one side and a discouraged Indian on the other." Cook also has him say: "let them kill, skin, and sell until the buffaloes are exterminated," or killed off completely.

DISPUTED: No official record of Sheridan speaking to the lawmakers has turned up. Smits reports that, but he judges Cook believable. Writing in *Harper's* magazine in 2008, Christopher Ketcham called the speech "believed to be apocryphal," which means probably made up.

BY 1890: One count in 1884 had found only 325 bison left on the Plains, and no federal law protected the ones that remained.

Frank Mayer

Who: A buffalo runner, one of the hunters who killed bison for their hides, from 1872 to 1878, who later said the army helped the runners.

When and where: The Plains in the 1870s. He died in Fairplay, Colorado, in February 1954, at a reported age of 104.

He said the army gave hunters "free ammunition, all you could use."

Frank H. Mayer killed bison for their hides from 1872 to 1878. At about age 100 he told his story to a writer, Charles B. Roth. It came out in 1958 as a book, *The Buffalo Harvest*, four years after Mayer died.

The book describes the business: the rifle, the wagons, the skinners who cut the hides off the dead animals, and hides that sold for a dollar or two each.

Mayer said that "army officers in charge of plains operations encouraged the slaughter of buffalo in every possible way." They gave hunters free ammunition. "All you had to do to get it was apply at any frontier army post," he said. "I received thousands of rounds in this way." Mayer also quoted an officer he did not name: "if we kill the buffalo we conquer the Indian."

The last part of the book records Mayer's regret for what the runners did.

NOT KNOWN: Mayer told the story when he was very old, the numbers in it are his own, and some people have questioned even his age.

The pigeon nestings

Professional hunters killed passenger pigeons where the birds gathered to nest, and by the mid-1890s the wild flocks were almost gone. At their peak, an estimated 3 to 5 billion passenger pigeons lived in North America. By one estimate they made up four of every ten birds on the continent.

About 136 million pigeons nested across some 850 square miles of Wisconsin in 1871, by one estimate. Professional netters, who caught the birds in nets, and shooters worked one of the last great nesting grounds, near Petoskey, Michigan, in 1878. They killed a reported 50,000 birds a day for months. Railroads carried the dead birds to city markets. Hunters learned where each big nesting was from telegraph messages, which were sent over wires.

Yellowstone and the Tukudika

President Grant signed a law on March 1, 1872, and under it Yellowstone became the first national park. In the law, members of Congress set the park apart "for the benefit and enjoyment of the people." The Tukudika, a band of Mountain Shoshone also called the Sheep Eaters, already lived there. A Park Service page states that they had lived in that area "for thousands of years." The Shoshone, Bannock, Crow, Blackfeet and other nations also have ties to the park's land.

Chapter 1 tells this removal briefly, and here is how it happened. Philetus W. Norris, Yellowstone's superintendent from 1877 to 1882, found a Tukudika camp near Bunsen Peak in the fall of 1879 and asked the Indian agent at Fort Washakie for help. The agent, whom the writer John Clayton names as James Patten, sent a party of Shoshone, who brought the Tukudika to the Wind River Reservation. In the words of a Park Service page, the Tukudika "were forcibly removed to reservations."

NOT KNOWN: The records do not say how many Tukudika were taken or what force the Shoshone party used, and the letters between Norris and Patten have not survived.

Keeping people out. Norris met in Washington with a Crow delegation in the spring of 1880, while officials were asking the Crow to give up 1.6 million acres. That spring Norris also got a "solemn promise" from people at the Lemhi agency, the office of another Indian agent, and at the Wind River Reservation to stay out of the park.

A Park Service page states that early superintendents "actively discouraged tribes from visiting the newly established park to hunt or collect resources." In his reports, Norris called the Tukudika "feeble," or weak, "and harmless." He wrote that Native people stayed away from the geysers out of "superstitious awe," meaning fear that came from beliefs about spirits. Clayton reports that historians and Shoshone scholars treat that claim as false. With it, Norris helped support a false story that no one had lived on the park's land.

The court case. Justices of the US Supreme Court ruled on Native hunting in and around the park in 1896. Wyoming officials had arrested a Bannock man named Race Horse for hunting elk in Wyoming. He hunted under the 1868 Fort Bridger Treaty, in which US officials promised a right to hunt on "unoccupied lands."

Justice Edward Douglass White wrote the Court's decision, in *Ward v. Race Horse*. He called the treaty right "temporary and precarious," which means easily lost, and he ruled that the right had ended when Wyoming became a state. After the ruling, hunting under the treaty in and around Yellowstone was a crime under Wyoming law.

More than 120 years later, on May 20, 2019, the justices threw out the 1896 decision in a different case, *Herrera v. Wyoming*, about the Crow people's own treaty. They ruled that the Crow treaty right to hunt had not ended when Wyoming became a state.

The forest reserves

George Perkins Marsh, a scholar from Vermont who also represented the United States in other countries, published a book called *Man and Nature* in 1864. In it, Marsh argued that whole societies collapse when their people destroy the forests and wear out the soil. Writers at the Forest History Society and the National Park Service link his book to the Forest Reserve Act of 1891. Under that law, a president could set aside public forest land. Those forest reserves were the start of the national forests.

John Muir

Who: A writer born in Scotland whose magazine articles made Yosemite known across the country, and who helped found the Sierra Club.

When and where: He lived from 1838 to 1914. He came to California in 1868 and was president of the Sierra Club from 1892 to 1914.

Movie: *John Muir in the New World* (PBS American Masters, 2011)

The Sierra Club started with 182 members in San Francisco.

John Muir was born in Scotland in 1838 and came to California in 1868. He wrote magazine articles about Yosemite. Readers across the country learned about Yosemite from them, and protecting it became a national cause.

Muir and others founded the Sierra Club in San Francisco on May 28, 1892. The founders wrote down the club's aims. One was "to explore, enjoy, and render accessible the mountain regions of the Pacific Coast," which means to make them easy for people to reach. Another was to win the help of "the people and government in preserving the forests and other natural features of the Sierra Nevada."

Muir was the club's first president, and he led it until he died in 1914. In May 1903 he took President Theodore Roosevelt camping in Yosemite.

BY 1900

If you remember one thing: US officials gave away the land they had taken from Native nations, hunters killed nearly all the bison the Plains nations lived on, and soldiers and officials forced Native people out of the first parks.

- 1851: Volunteer soldiers drive the Ahwahneechee out of Yosemite Valley.
- 1862: Lincoln signs the Pacific Railway Act, the Homestead Act and the Morrill Act, the laws under which officials give away public land.
- 1872: Yellowstone becomes the first national park. In 1879 its superintendent has the Tukudika forced out.
- 1872 to 1874: Three railroads carry about 1.4 million bison hides east.

Next: After 1900, presidents put hundreds of millions of acres under federal protection, and some of that land had belonged to Native nations.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 17 MINUTES

President Theodore Roosevelt and the federal officials after him took charge of hundreds of millions of acres in the first half of the 1900s, and some of that land had belonged to Native nations. Roosevelt alone put about 230 million acres under federal protection from 1901 to 1909, in national forests, parks, bird refuges (land where wild birds are kept safe) and monuments.

Glacier National Park, created in 1910, lies mostly on land that federal officials count as sold by the Blackfeet in 1895. Before the sale, about 600 Blackfeet had starved to death in one winter. Many Blackfeet say they only rented the land out.

The last passenger pigeon died in the Cincinnati Zoo in 1914. In Yellowstone, park managers poisoned wolves until they killed the park's last pack in 1926.

On the southern Plains, farmers had plowed up the grass that held the soil, and in the 1930s that soil blew away. Forest Service leaders ordered crews in 1935 to bring every wildfire under control by 10 the next morning. Young men in the Civilian Conservation Corps planted billions of trees.

Roosevelt's acres

The 230 million acres Roosevelt protected, an area bigger than Texas, included 150 national forests, 5 national parks, the first 18 national monuments, 51 federal bird reservations and 4 national game preserves. Some counts combine the bird reservations and game preserves into one figure of 55.

A national forest is public land where federal foresters, people trained to study and manage forests, look after the trees. The U.S. Forest Service began in 1905, with Gifford Pinchot as its first chief, and its foresters ran the national forests.

The first refuge. Roosevelt named Pelican Island, in Florida, the first federal bird reservation on March 14, 1903. It became the first piece of what is now the National Wildlife Refuge System.

Roosevelt signed the Antiquities Act on June 8, 1906. Under it, a president could protect land by proclamation, an official announcement, and land protected this way is called a national monument. Devils Tower became the first one that September 24.

Whose land. Many national forests were made from the ancestral lands of Native nations, the lands their ancestors lived on. Arthur Blazer, the president of the Mescalero Apache Tribe, told a Senate committee in 2018 that the US Department of Agriculture, which runs the Forest Service, "has acknowledged that the vast majority of federal forest lands are carved out of the ancestral homelands of Indian tribes."

Forest Service staff say the same of the national forests in North Carolina, and the staff of Olympic National Forest say it of all the land in present-day Washington. A Park Service web page states that Devils Tower is sacred to dozens of Native nations, including the Arapaho, Cheyenne, Crow, Kiowa, Lakota and Shoshone. The Lakota name for it is Mato Tipila, which means Bear Lodge.

Theodore Roosevelt

Who: The president from 1901 to 1909. In those years he put about 230 million acres of land under federal protection.

When and where: The White House, 1901 to 1909. Yosemite, California, in May 1903.

Movie: *The Roosevelts: An Intimate History* (PBS, Ken Burns, 2014), a documentary about the whole Roosevelt family

Roosevelt slept under a pile of about 40 wool blankets.

President Theodore Roosevelt went camping in Yosemite with John Muir, the first president of the Sierra Club, in May 1903. Muir wanted the best of the wild land left wild. The two men spent three nights sleeping in the open.

They camped first under the Grizzly Giant, a giant sequoia tree, on May 15, and that was the night of the blankets. The next night, near Sentinel Dome, a snowstorm dropped five inches of new snow on top of about five feet of old snow. They spent the last night in a meadow near Bridalveil Fall.

Glacier and the Blackfeet

Most of Glacier National Park lies on land that federal officials say the Blackfeet sold to the United States in 1895. The deal came after a winter in which about 600 Blackfeet starved to death. Many Blackfeet say the tribe only rented the land out and never sold it.

Before the sale. US Army troops attacked Heavy Runner's camp of Piikani people, one of the Blackfeet peoples, on the Marias River in 1870, as Chapter 2 tells. By the best estimate they killed about 177 men, women and children. The army's own count was 173, and the National Parks Conservation Association's figure is more than 200.

By 1880 hide hunters had killed nearly all the bison. Members of Congress then cut the money for annuities, the food and goods that US officials had promised a tribe every year in a treaty as payment for land.

The Starvation Winter. Early storms held up supplies in the winter of 1883 and 1884. In December a wagonload of bacon reached the Blackfeet agency near Browning, the office of the US official who ran the reservation. The Montana Historical Society's textbook describes the bacon as "so old it was contaminated with maggots," the young of flies, which feed on rotting meat.

The textbook states that 600 Blackfeet died of starvation, about one in four of the tribe. Other accounts give counts from 500 to 800. The Assiniboine, another nation, lost about 300 people. That winter is known as the Starvation Winter.

No record names the person who sent the spoiled bacon. John Young had been the Blackfeet's Indian agent, the US official in charge of their reservation, since December 1876. He went to Washington in 1881 and got $15,000 put back into the reservation's budget. He protested against shrinking supplies and more cuts to the agency's budget.

No one acted on his protests, and in September 1883 he resigned. The records do not name the officials who ignored him. The new agent, R. A. Allen, did not arrive until the middle of March 1884.

The sale. The same textbook states: "After Starvation Winter, the tribes sold the only resource they had left: land." The Blackfeet sold a strip of about 800,000 acres in 1895 for $1.5 million. Prospectors, people who search for gold and other metals, hoped the strip held gold.

Many Blackfeet say the tribe offered the land as a 99-year lease, which is a rental, and never sold it. They also say that because of a surveying mistake, US officials took another 45,000 acres. Under the agreement, the Blackfeet kept the right to hunt, fish, cut wood and gather plants on the land.

Little Dog told the US commissioners, the officials sent to buy the land, that the Blackfeet "did not ask the government to come and buy their land." White Calf, a Piikani chief, said of Chief Mountain: "Chief Mountain is my head. Now my head is cut off."

The park. Prospectors found no gold. The strip became a forest reserve in 1897, under the 1891 law that let a president set aside public forest land. Members of Congress made most of the strip Glacier National Park on May 11, 1910. After that, an article by the National Parks Conservation Association states, the Blackfeet's rights on the land "essentially vanished."

After the park was created, judges ruled that the Blackfeet's right to go into the mountains in the park no longer counted. A 2019 *High Country News* report states that hunting and fishing rights ended in 1910 and that only the right to gather plants remained. By 1912 a park ranger had arrested a Blackfeet tribal member for hunting in the park. The accounts of that arrest name neither the ranger nor the hunter.

A judge of a US District Court ruled in 1932 that the Blackfeet had no right to hunt in the park. In 1973 a judge ruled that Blackfeet people could enter the park without paying the entrance fee.

Hetch Hetchy

People who wanted to protect land split into two sides over Hetch Hetchy Valley. Gifford Pinchot's side said conservation meant using the land carefully, forever. The side of John Muir, the first president of the Sierra Club, said the best of it should be left wild.

Hetch Hetchy Valley lies inside Yosemite National Park. After the earthquake and fire of 1906, San Francisco's city leaders wanted the valley for a reservoir, a lake made by a dam to store water. Pinchot backed the city. Muir and the Sierra Club fought the dam for years.

Members of Congress sided with the city and passed the Raker Act, which allowed the city to dam the valley. President Wilson signed it on December 19, 1913. Muir died a year later, on December 24, 1914. The O'Shaughnessy Dam was finished in 1923, and the valley is under water today.

Members of Congress passed the National Park Service Organic Act on August 25, 1916, partly in reaction to the Hetch Hetchy fight. With it they created a new federal office, the National Park Service, whose officials took charge of protecting the parks.

Gifford Pinchot

Who: The first American trained as a professional forester, and the first chief of the U.S. Forest Service, until President Taft fired him. Later he was twice governor of Pennsylvania.

When and where: He lived from 1865 to 1946. He ran the Forest Service from 1905 to 1910.

Movie: *Seeking the Greatest Good* (WVIA and PBS, 2013)

President Taft fired him on January 7, 1910.

Gifford Pinchot was the first American trained as a professional forester. He defined conservation as "the greatest good to the greatest number for the longest time." To Pinchot, that meant using the public lands carefully and keeping them useful forever.

Pinchot became the first chief of the Forest Service in 1905. In 1909 and 1910 he fought in public with the Secretary of the Interior over claims to federal coal lands in Alaska, and President Taft fired him for refusing to obey his bosses.

Pinchot and John Muir had been friends until they took opposite sides over Hetch Hetchy. Pinchot later served two terms as governor of Pennsylvania.

The last passenger pigeons

In about a century, the passenger pigeon went from an estimated 3 to 5 billion birds to none. After the last confirmed wild bird was shot, a few people reported seeing wild passenger pigeons until about 1902, and no one confirmed those sightings.

Press Clay Southworth

Who: A 14-year-old boy who shot the last confirmed wild passenger pigeon.
When and where: Pike County, Ohio, on March 24, 1900.

His family stuffed the bird and gave it buttons for eyes.

Press Clay Southworth, age 14, shot a passenger pigeon in Pike County, Ohio, on March 24, 1900. He asked his mother first, then used his family's 12-gauge shotgun. His bird is the last wild passenger pigeon that anyone has confirmed. Southworth told the story himself decades later.

His family had the bird mounted, which means stuffed and set up for display. It got buttons for eyes, so people called it "Buttons." The Ohio History Connection, in Columbus, keeps it today.

Martha. The last passenger pigeon of all was a bird named Martha. She died at the Cincinnati Zoo on September 1, 1914, at about 29 years old. Her body was frozen in a 300-pound block of ice and shipped to the Smithsonian Institution, which still has it. Sol Stephan, the zoo's general manager, had cared for Martha, and he saved her feathers in a cigar box.

NOT KNOWN: Nobody can say for sure where Martha came from, because Stephan's accounts changed over the years and the zoo's own records do not agree with each other.

Yellowstone's wolves

Park managers in Yellowstone killed at least 136 wolves between 1914 and 1926, and in 1926 they killed the park's last wolf pack. The National Park Service gives these numbers. The *Idaho Capital Sun* and other later accounts report that park rangers shot the last two wolf pups near Soda Butte that year.

The managers used poison, especially one called strychnine, which they put inside the bodies of dead animals for wolves to eat. Dan Stahler, a biologist, told *Smithsonian* magazine in 2026: "We're talking crates and crates of this poisoning." In 1915 the park's superintendent called wolves "a decided menace," or danger, to the animals they hunted.

The killing in Yellowstone was part of a program to wipe out predators, animals that hunt other animals for food, on public lands across the whole country. The program's targets were wolves, bears, cougars and coyotes. Rewards for dead wolves went back almost 300 years. The first recorded wolf bounty in the colonies, a reward paid for each dead animal, dates from Massachusetts in 1630.

Why did the Plains soil blow away?

Wind carried away the soil of the southern Plains in the 1930s, because farmers had plowed up the grass that held it. Wheat sold at high prices during World War I. Farmers with tractors plowed up tens of millions of acres of shortgrass, the short native grass of the southern Plains. Its roots formed sod, a thick mat of roots and earth that held the soil in place.

A drought, a long time with too little rain, began in 1931. With the sod gone, the dry soil blew away.

What blew away. By the count of the US Department of Agriculture, about 35 million acres of farmland, an area about the size of Iowa, had been ruined for growing crops by December 1934. The same count listed 100 million acres that had lost all or most of their topsoil, the top layer of soil where crops grow. Another 125 million acres were losing it fast. In 1935 alone, an estimated 850 million tons of topsoil blew off the southern Plains.

The worst single dust storm struck on April 14, 1935, a day called Black Sunday. It crossed the Plains as a black wall of dust moving at 60 miles an hour.

That spring, a soil scientist named Hugh Hammond Bennett was testifying, or giving evidence, at a hearing in Washington while dust from the Plains darkened the sky outside. Bennett said: "This, gentlemen, is what I've been talking about." Accounts of the hearing differ about the day. Some put it in late March 1935, and others in early April.

Members of Congress declared soil erosion, the wearing away of soil by wind or water, "a national menace." The Soil Conservation Service began on April 27, 1935, with Bennett as its director. The methods that followed included contour plowing, which means plowing along the curves of a slope instead of up and down it. They also included shelterbelts, rows of trees planted to slow the wind, and planting grass on the land again.

Caroline Henderson

Who: A farmer and writer who homesteaded in the Oklahoma Panhandle and stayed on her land through the dust years.

When and where: Near Eva, Oklahoma, from 1907. She lived from 1877 to 1966.

When the dust came, she kept farming and did not leave.

Caroline Henderson graduated from Mount Holyoke College. In 1907 she began homesteading near Eva, in the Oklahoma Panhandle, claiming public land and farming it for years in order to own it. When the dust storms came in the 1930s, she stayed on her land.

Henderson wrote letters about the dust years. In one, dated June 30, 1935, she described trying to rescue her home from wind-blown dust "which penetrates wherever air can go." Soon everything was covered again, she wrote, with dust that ran "from a film to actual ripples on the kitchen floor." She swept her floors with sawdust that had oil mixed in.

The *Atlantic Monthly* magazine printed her letters as "Letters from the Dust Bowl." One appeared in May 1936. That same month, Henry A. Wallace, the Secretary of Agriculture, wrote to praise her.

Across her whole life, her letters run from 1908 to 1966. The University of Oklahoma Press later collected them in a book, *Letters from the Dust Bowl*.

Every fire put out

Forest Service rules from 1935 to 1978 required crews to control every wildfire by the morning after someone spotted it. A fire called the Big Burn covered 3 million acres of the Northern Rockies in 1910 and killed at least 78 people. After it, Forest Service leaders tried to put out every fire. In 1935 they made the "10 a.m. policy," under which crews had to bring each fire under control by 10 o'clock the morning after it was spotted.

What the stopping did. For thousands of years Native people had burned the eastern forests on purpose. Researchers at Penn State call the total stopping of fires after about 1940 "an ecologically transformative event in all forests." They mean it changed how the plants and animals of every forest lived together.

The men who made it work. About 3 million young men worked in the Civilian Conservation Corps, a federal work program, from 1933 to 1942. Because the corps had so many men, Forest Service leaders could carry out the 10 a.m. policy. Estimates of the trees the corps members planted run from more than 2 billion to 3.5 billion, and the figure most often given is about 3 billion. The corps members also fought fires, worked to stop erosion and did building work in more than 800 parks.

Forest Service officials worked with the Ad Council, an advertising group, to start the Smokey Bear campaign in 1944. Its slogan, "Only YOU can prevent forest fires," dates from 1947. In 1950 a bear cub was rescued from the Capitan Gap fire in New Mexico, and the cub became the living Smokey Bear.

Aldo Leopold

Who: A forester for the U.S. Forest Service who got the first wilderness area set aside, and later wrote *A Sand County Almanac*.

When and where: He lived from 1887 to 1948. He worked in New Mexico in the 1920s and later in Wisconsin.

Movie: *Green Fire: Aldo Leopold and a Land Ethic for Our Time* (2011)

In 1924 he got the world's first official wilderness area set aside.

Aldo Leopold began as a forester for the U.S. Forest Service. In 1924 he got the Gila Wilderness, in New Mexico, set aside. It was the first area in the world officially named a wilderness area, which is land kept free of roads and buildings. Frank C. W. Pooler, the Forest Service official in charge of the region, approved Leopold's plan on June 3, 1924.

The Gila Wilderness covered about 755,000 acres. Under the plan the area had no roads, cabins or other buildings. People could still hunt and fish there, and ranchers could let their animals feed on its grass.

Whose homeland it was. The Gila had been a homeland of the Chiricahua and Warm Springs Apache, and Mogollon and Mimbres people had also lived in the region. Before Leopold came to New Mexico, US Army soldiers, federal officials and white settlers had forced many Apache people onto reservations. Army soldiers imprisoned Chiricahua and Warm Springs Apache people for defending their land.

Of the thousands of Chiricahua and Warm Springs Apache who had lived in the Gila, only a few hundred were still alive after the Army imprisoned them. US officials later sent the survivors to Fort Sill, Oklahoma, more than 500 miles away.

Later years. Leopold taught at the University of Wisconsin. He bought a worn-out farm in Wisconsin and rebuilt its soil and its woods. On April 21, 1948, he had a heart attack and died while helping to fight a grass fire on a neighbor's land.

His book *A Sand County Almanac* came out in 1949, the year after he died. In it he called for a "land ethic," an ethic being a set of ideas about right and wrong. In Leopold's land ethic, people count themselves as members of the land community and stop acting as its conquerors. The book includes his "green fire" essay, about watching a wolf die after it was shot.

BY 1950

If you remember one thing: Federal officials took charge of protecting hundreds of millions of acres, much of it Native land, while the passenger pigeon, Yellowstone's wolves and the soil of the southern Plains were lost.

In 1900: The Forest Service, the National Park Service and the first national monument did not exist yet, and a boy in Ohio shot the last confirmed wild passenger pigeon.

In 1950: Crews had orders to put out every wildfire by 10 the next morning, and a bear cub rescued from a fire in New Mexico had become the living Smokey Bear.

Next: After 1950 a chemical called DDT thinned the eggshells of bald eagles, a river in Cleveland caught fire, and families found buried chemicals seeping into their basements.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 17 MINUTES

Pollution that people could see pushed members of Congress to pass the big laws that protect the land, air and water, and then people began to count who lived next to the waste. DDT, a chemical for killing insects, left bald eagles with eggs so thin they broke. In 1969 an oil well blew out off Santa Barbara, California, and oil and trash on the Cuyahoga River in Cleveland caught fire.

Between 1970 and 1973, members of Congress passed the Clean Air Act, the Clean Water Act and the Endangered Species Act, and the Environmental Protection Agency opened.

Park Service officials forced the last Native families in Yosemite Valley out of their homes in 1969. Later in the 1970s, families at Love Canal, in Niagara Falls, New York, found chemicals seeping into their basements. They lived on and beside more than 20,000 tons of buried chemical waste.

Residents of Warren County, North Carolina, lay in the road in 1982 in front of trucks carrying poisoned soil to a landfill, a place where trash and waste are buried, in a mostly Black community. In a 1983 federal study, investigators looked at four landfills for dangerous waste in eight southeastern states. All four were in areas where Black residents were 38 to 90 percent of the people.

In a 1987 national study, Charles Lee found that race was the strongest clue to where such waste sites stood. That year Rev. Benjamin F. Chavis Jr. named the pattern "environmental racism."

DDT and the eagles

DDT is a pesticide, a chemical made to kill insects, and it was in wide use from the mid-1940s. DDT thinned the eggshells of raptors, birds that hunt other animals, until the eggs broke in the nest. By 1963 only 417 nesting pairs of bald eagles were left in the lower 48 states, meaning every state except Alaska and Hawaii.

Rachel Carson

Who: A marine biologist, a scientist who studies life in the sea, and a writer. She had worked as a writer and editor for the U.S. Fish and Wildlife Service. She wrote *Silent Spring*.

When and where: She lived from 1907 to 1964. *Silent Spring* came out in September 1962. She died in Silver Spring, Maryland.

Movie: *Rachel Carson* (PBS American Experience, 2017)

She went on television to defend her book while sick with cancer.

Rachel Carson first became famous for a book called *The Sea Around Us*. In September 1962 she published *Silent Spring*. In it she laid out what persistent pesticides, the kind that stay in soil, water and living things for a long time, were doing to birds, soil, water and people.

People in the chemical industry attacked her, and Carson answered them with evidence. She spoke on the television program *CBS Reports* on April 3, 1963. By then she was already sick with breast cancer, and she died of it on April 14, 1964. Officials of the Environmental Protection Agency banned DDT in 1972, eight years later.

Yosemite's last village

Park Service officials evicted, or forced out of their homes, the last Native families who lived in Yosemite Valley in 1969. Then they destroyed their cabins.

How the village began. Ahwahneechee and Paiute families lived in a village in the valley in the 1920s and worked in the park. The Ahwahneechee are also called the Southern Sierra Miwok. Native workers helped build the Ahwahnee Hotel, which was finished in 1927. Park Service workers built cabins at a new site, called Wahhoga, in 1933, for families that park officials had moved out of the old village.

Charles Goff Thomson was Yosemite's superintendent from 1929 to 1937. A history published by Brigham Young University states that Thomson wanted to use the new village to control the Native people in the park. The families paid rent, and they paid for their power and water.

Park Service officials made a housing rule in 1953 under which only permanent government workers could live in the village. As each home fell empty, the officials destroyed it. In 1969 they evicted the families who were still there.

How it ended. Deborah Tucker's parents lived in the village. She told the National Parks Conservation Association: "The Park Service came and burned the cabins out." Some later magazine accounts say the empty cabins were burned in a firefighting training exercise. A Park Service web page states that a Native home near Camp 4 "was razed," or torn down completely, "in the 1960s when the last Indian village was removed from Yosemite Valley."

Jay Johnson, the last Native park worker who lived in the valley, retired and left in 1996.

NOT KNOWN: No record names the official who ordered the village torn down in 1969. Erin Davenport, a Yosemite park historian, told the National Parks Conservation Association that park historians have found no record of why it was done either.

The river that burned

Oil and trash floating on the Cuyahoga River in Cleveland, Ohio, caught fire on June 22, 1969. The fire burned for about 20 to 30 minutes before it was brought under control. The river had burned at least 13 times since 1868. The 1912 fire killed five men, and the 1952 fire did $1.3 million in damage.

Whose waste it was. The river ran past steel mills and other factories. A 2019 *Smithsonian* magazine article names American Ship Building, the Sherwin-Williams Paint Company, Republic Steel and Standard Oil as Cleveland companies whose success left the river full of poisonous waste. A Cleveland State University history states that the river was "polluted from decades of industrial waste."

The 1969 fire started in oil and trash caught under two railroad bridges near the Republic Steel mill. A flare thrown from a passing train most likely lit it. Investigators from Cleveland's Bureau of Industrial Wastes looked into the fire. No record names the company whose oil burned.

Editors at *Time* magazine published a story on the burning river on August 1, 1969, and readers across the country saw it. Carl Stokes was Cleveland's mayor and the first Black mayor of a major US city. The day after the fire, he held a news conference at the river and called on the state and federal governments to act.

Santa Barbara. Earlier the same year, oil had spilled on the coast of California. Five months before the Cuyahoga fire, on January 28, 1969, an oil well blew out under Platform A. It was a Union Oil Company drilling platform about 5.5 miles southeast of Santa Barbara, California. A blowout is an uncontrolled burst of oil and gas up a well. Officials of the U.S. Geological Survey, the federal agency that approved offshore drilling, had excused Union Oil from federal safety rules when they approved its drilling permit.

Federal and state investigators found that extra steel pipe lining the well would have prevented the blowout. About 100,000 barrels of oil, more than 4 million gallons, leaked out by December 1969, and the oil blackened the beaches at Santa Barbara. By one estimate, about 9,000 birds died.

Earth Day and the new laws

An estimated 20 million Americans took part in Earth Day events on April 22, 1970, a day set aside for events about protecting the land, air and water. The 1970 census, the national government's count of everyone in the country, counted 203,392,031 people in the United States, so about one in ten Americans took part. Senator Gaylord Nelson organized Earth Day with Denis Hayes.

The new laws. The National Environmental Policy Act became law on January 1, 1970. Under it, federal officials had to study how a big federal project would affect the land, air and water before they went ahead. The Environmental Protection Agency, called the EPA, opened on December 2, 1970. Its officials enforce the federal laws against pollution.

Members of Congress passed the Clean Air Act in 1970. In 1972 EPA officials banned DDT, and members of Congress passed the Clean Water Act. The Endangered Species Act, for kinds of plants and animals in danger of dying out, became law on December 28, 1973.

After the DDT ban, bald eagles came back. They grew from 417 nesting pairs in 1963 to about 10,000 pairs by 2007. That year, federal officials took the bald eagle off the endangered list.

Love Canal

Workers for Hooker Chemical buried more than 20,000 tons of chemical waste in an old canal in Niagara Falls, New York, from the 1940s into the 1950s. Families later lived on and beside it, and the chemicals seeped into their homes.

What was buried. The canal, called Love Canal, covered 15 acres. Hooker Chemical and Plastics Corporation began using it in 1942. The sources put the waste at about 21,000 tons, and they differ slightly on the exact amount and on whether the dumping ended in 1952 or 1953. In a 1981 report, officials of New York State's health department wrote that Hooker "admits" burying 21,800 tons.

Investigators later found more than 200 chemical compounds there. The timeline lists 200 tons of dioxin, a group of chemicals that cause cancer and harm babies before they are born.

Who knew. A federal judge wrote in a 1994 decision that Hiram Young, Hooker's works manager, was responsible for the company's waste disposal. The court record does not say who first decided to use the canal as a dump.

Hooker's lawyer visited the dump in 1946 and saw children with swimsuits. He warned the company's president that Hooker was "running a real hazard," and he suggested guards or a fence. Hooker's executives talked about his ideas, but they never built the fence. City officials of Niagara Falls also used the canal as a dump for the city's own waste until 1953.

The sale for $1. Hooker sold the covered canal to the Niagara Falls Board of Education for $1 in 1953. Hooker's officials wrote into the deed "a disclaimer of responsibility for future damages due to the presence of buried chemicals," a statement that refused blame for any harm the chemicals did later. That same year, Hooker officials told Board members "that homes could not be built at the Canal site."

The school. The members of the Board of Education had a public elementary school, the 99th Street School, built on the middle third of the canal in 1954. That January their architect wrote that the builders had dug into two pits of

chemical waste, some of it in 55-gallon drums. He wrote that it was "poor policy to attempt to build over this soil."

Building of homes on 97th and 99th Streets, beside the canal, began in 1954. The Board of Education later sold part of the canal land to others. Board members talked in 1957 about a plan to sell a piece of it to developers for building homes. A Hooker lawyer told the Board on November 7, 1957, that the land was "unsuitable," or not fit, "for construction" where "basements, water lines, sewers and such underground facilities would be necessary."

Ninety-nine homes had backyards that touched the canal by the spring of 1978. No record names the companies that built the homes.

The chemicals come up. By the 1970s, chemicals were coming up into yards and basements. On October 3, 1976, chemicals from the buried waste were reported "seeping into basements of homes." In November 1976, testers found 15 chemicals, including three toxic chlorinated hydrocarbons, a kind of chemical that contains chlorine.

Families in the neighborhood counted miscarriages, the deaths of unborn babies before they can live outside the mother's body. They also counted birth defects, problems in a baby's body that are there when the baby is born, and illness. Dr. Beverly Paigen, a scientist, found a "high rate of birth defects and miscarriages among Love Canal families."

The emergency. Dr. Robert Whalen, New York State's health commissioner, the state's top health official, declared a state of emergency on August 2, 1978. That is an official order saying that people are in danger and need help fast. He closed the 99th Street School and ordered pregnant women and children under 2 moved out.

President Jimmy Carter approved emergency federal money on August 7 so that state officials could buy homes. State officials moved the first 236 families out, at a cost of $10 million. Other counts say 239 families.

The lawsuit. New York State's lawyers sued Occidental Petroleum, which owned Hooker by then, and two Hooker companies for $635 million in April 1980.

Scientists in a study paid for by the EPA reported on May 17, 1980, that they had found chromosome damage in 11 of the 36 residents tested. Chromosomes are the strands in each cell that carry a person's genes. Damage to them can raise the risk of cancer, and of birth defects in the person's children.

The second emergency. That month, residents held two EPA officials for about six hours to demand that the rest of the families be moved. Carter then declared a second emergency, which opened the way to move 710 more families. In all, about 950 families left.

Members of Congress passed a law called Superfund in December 1980. Under it, federal officials could make companies that polluted a site pay for cleaning it up. Occidental agreed in 1994 and 1995 to pay New York State and the federal government more than $225 million for the cleanup and the moves.

Lois Gibbs

Who: A 27-year-old housewife at Love Canal whose son went to the 99th Street School. She started the Love Canal Homeowners Association.

When and where: Love Canal, in Niagara Falls, New York, from 1978 to 1980.

Movie: *Lois Gibbs and the Love Canal* (CBS television movie, 1982), with Marsha Mason playing Gibbs

She was 27, and her son's school stood on buried chemicals.

Lois Gibbs was a housewife when she learned about the buried chemicals at Love Canal. Her son went to the 99th Street School, which stood on and beside the old canal. Officials brushed off her worries. In August 1978 she organized her neighbors into the Love Canal Homeowners Association.

Gibbs later founded the Center for Health, Environment & Justice. Through it, she has helped other neighborhoods fight the pollution near their homes. In 1990 she won the Goldman Environmental Prize, an award given to people who protect the environment. A CBS television movie about her aired on February 17, 1982.

Warren County

Residents of Warren County, North Carolina, lay down in the road in front of trucks in September 1982. The trucks carried soil poisoned with PCBs to a new landfill near Afton, a mostly Black community. An exhibit by the University of North Carolina Library and other histories describe the protest as the start of the environmental justice movement. People in that movement fight a pattern in which companies and government officials put polluting sites next to poor, Black, brown and Native communities.

PCBs are a group of chemicals made by people. They were used in electrical, heating and hydraulic equipment, machines worked by liquid under pressure. An EPA web page states that PCBs cause "a variety of adverse health effects," or harmful effects, "in animal and humans."

Where the soil came from. The poisoned soil came from roadsides. Robert Ward, of the Ward Transformer Company of Raleigh, hired Robert Burns and his sons to get rid of oil that contained PCBs. The Burns family ran a business called Transformer Sales, in Allegany, New York. In the summer of 1978, the Burns men sprayed about 31,000 gallons of the oil from their trucks at night, on the shoulders, or edges, of about 240 miles of roads in 14 North Carolina counties.

The site. Governor James B. Hunt Jr. announced a plan in December 1978 to bury about 60,000 tons of the soil scraped from those roadsides. He chose a landfill at Afton, in Warren County. State officials bought farmland there from a farmer who was in money trouble. Warren County had the highest share of African American residents in North Carolina, and it was one of the poorest counties in the state. The Afton community was 84 percent African American.

Most people near Afton drank water from wells. The water table there, the level underground where the soil is full of water, was only about 5 to 10 feet down. The U.S. Commission on Civil Rights later wrote that the site "was not scientifically the most suitable site, because the water table was a mere 5-10 feet below the surface."

The protest. Residents formed a group called Warren County Citizens Concerned About PCBs and sued. They lost in 1982. The trucks began to deliver the soil that September, and protesters lay in the road. Officers arrested about 500 people, and one later count says more than 500.

State officials went ahead and built the landfill. Governor Hunt promised that the soil would be cleaned of poison once a way to do it existed. The cleanup was finished by 2003 and cost $17.1 million.

Counting who lives near the waste

The authors of two studies in the 1980s counted who lived near hazardous waste sites, the places that hold waste that can harm people's health. Both found the answer tied to race.

The 1983 study. Members of Congress asked the US General Accounting Office, a federal office that investigates for Congress, to study landfills for hazardous waste in eight southeastern states. Its report came out on June 1, 1983.

The report lists four landfills in those states that took other people's waste for pay. One was the Warren County landfill. The others were in Sumter County, Alabama, Chester County, South Carolina, and Sumter County, South Carolina. All four were in areas where Black residents made up 38 to 90 percent of the population.

The 1987 study. The Commission for Racial Justice of the United Church of Christ published a national study in 1987, *Toxic Wastes and Race in the United States*. Charles Lee wrote it. He found that race was "the most potent variable in predicting where these facilities were located," more powerful than household income or the value of homes. The race of the people nearby predicted where a waste site stood better than how much money they had.

When the report came out, the commission's leader, Rev. Benjamin F. Chavis Jr., called the pattern "environmental racism." He meant "intentionally selecting communities of color for wastes disposal sites and polluting industrial facilities." Communities of color are places where most people

are not white. The authors of a follow-up study in 1994 found that people of color were "47 percent more likely to live near a hazardous waste facility than white Americans."

President Bill Clinton signed Executive Order 12898 on February 11, 1994. In it he directed the officials of federal agencies to find and deal with pollution that fell hardest on communities of color and poor communities. A 2007 report by the United Church of Christ states that the EPA "has failed to implement" the order, meaning its officials had not carried it out.

Wolves back in Yellowstone

Gray wolves brought from Canada were released in Yellowstone National Park starting in January 1995, and by the end of 1996, 31 wolves had been moved there. Park managers had killed the park's last wolf pack in 1926. For most of the 1900s, elk had eaten the park's willows and other woody plants so heavily that the plants could not grow tall. After the wolves came back, the willows began to grow.

KNOWN:	Park Service scientists wrote in 2019 that most studies agree on that growth.
DISPUTED:	Scientists still debate how much of that growth the wolves caused, by hunting the elk and changing where the elk fed.

BY 2000

If you remember one thing: Members of Congress passed the big federal laws against pollution in the 1970s, and then residents and researchers showed that waste sites stood far more often beside Black and poor neighborhoods.

One number: In 1963 there were 417 nesting pairs of bald eagles in the lower 48 states. By 2007, after the DDT ban, there were about 10,000, roughly 24 times as many.

Next: After 2000 came the warmest year ever measured, fires that burned millions of acres, and a fight in Louisiana over a chemical plant beside a school.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 15 MINUTES

The land grew hotter and drier after 2000, and fires grew bigger. The year 2024 was the warmest ever measured, both for the whole world and for the lower 48 states. Heat and drought dried forests that had filled up with fuel during about a century of stopping fires.

Fires burned about 4.2 million acres of California in 2020. Lake Mead and Lake Powell, two reservoirs on the Colorado River, fell to record lows in 2021 and 2022. Fire managers now set fires on purpose again, and Yurok people teach cultural fire to government fire crews.

Fights over pollution and protected land went on. Shell's managers agreed in 2002 to pay to move the families of Old Diamond, a Louisiana neighborhood beside a Shell chemical plant. Federal lawyers sued the owner of a plant near an elementary school in LaPlace in 2023, and they dropped the suit in 2025.

President Donald Trump cut Bears Ears National Monument, in Utah, by about 91 percent on July 13, 2026. As of September 26, 2026, the land he removed was open to mining claims, and lawsuits against the cut were waiting

in federal court.

Some animals came back. Bald eagles grew from 417 nesting pairs in 1963 to about 71,400 in the 2020s, by the newest count. The bison, fewer than 1,000 in 1900, now number more than 400,000, most of them in herds raised for sale. Wolves, wiped out of Yellowstone by 1926, numbered 2,797 in seven western states at the end of 2022.

The warmest year

Scientists at NOAA, the federal agency that tracks weather and climate, reported in January 2025 that 2024 was the warmest year on record for the planet. The world averaged 1.29 degrees Celsius, about 2.3 degrees Fahrenheit, above the average for the 20th century, the years 1901 to 2000. Scientists at NASA and other groups agreed. The lower 48 states averaged 55.5 degrees Fahrenheit in 2024, which was 3.5 degrees above their 20th-century average. Seventeen states set records.

Why are the fires bigger?

After about a century of stopping fires, the woods held heavy loads of fuel, the dead wood and brush that burns. Heat and drought dried that fuel out.

The Camp Fire. The Camp Fire of November 8, 2018, killed 85 people in California and destroyed nearly all of the town of Paradise. About 19,000 buildings burned. Those figures are the ones given by the encyclopedia *Britannica* and by a timeline from the National Institute of Standards and Technology, a federal science agency. It was the deadliest wildfire in the United States in a century.

The 2020 fires. Fires burned about 4.2 million to 4.3 million acres of California in 2020, an area bigger than Connecticut and about 4 percent of the state. By the count of CAL FIRE, California's fire agency, that was the most in modern records. One fire, called the August Complex, passed 1 million acres. It became known as the first "gigafire" in the state's modern history, meaning a fire that burned more than a million acres.

Fire managers at government agencies now set fires on purpose to burn off fuel. These planned fires are called prescribed burns, and prescribed burning is official policy again.

Margo Robbins

Who: A member of the Yurok Tribe who helped found the Cultural Fire Management Council, to bring traditional burning back to Yurok land.

When and where: The Klamath River country of northern California. The first community burn was in 2012, and the council formed in 2013.

She wanted straight hazel shoots to weave baskets for her grandchildren.

Margo Robbins, a member of the Yurok Tribe, wanted hazel for weaving baskets. When hazel is burned, it sends up new shoots that grow straight, and weavers need straight shoots. Her first goal was to weave baskets to carry her grandchildren.

Robbins and others held their first community burn on Yurok land along the Klamath River in 2012. In 2013 they formed the Cultural Fire Management Council. Cultural fire, burning done by Native people to care for the land, was the practice California's lawmakers had made a crime in 1850, in the same law that let officials hire out Native adults to the highest bidder.

After the council's burns, deer came back to the burned ground at Weitchpec. Council members now run training exchanges, called TREX, that teach cultural burning to fire crews.

The shrinking river

The Southwest had its driest 22 years in at least 1,200 years from 2000 to 2021. Scientists from UCLA and Columbia University found this by studying tree rings, which show how much a tree grew each year. They published their study in the journal *Nature Climate Change* in February 2022. They found that about 42 percent of how bad the drought was came from climate change caused by people.

About 40 million people depend on the Colorado River for water, and so does much of the country's winter vegetable crop. Lake Mead and Lake Powell, two reservoirs on the river, fell to record lows in 2021 and 2022. By the middle of 2022, Lake Mead was about a quarter full.

The cut. Officials of Arizona, California and Nevada agreed in May 2023 to cut their use of the river by at least 3 million acre-feet through the end of 2026. News reports gave that figure from the federal Bureau of Reclamation. An acre-foot is enough water to cover one acre of land one foot deep. The cut is about a tenth of the water those three states are allowed to take, and federal officials agreed to pay the states for the water they gave up.

Cancer Alley

The stretch of Louisiana between Baton Rouge and New Orleans holds a long line of chemical plants. A 2007 report by the United Church of Christ describes this strip as "so-called 'cancer alley,' the chemical manufacturing corridor between Baton Rouge and New Orleans," a corridor being a long, narrow strip of land. The towns of Norco and LaPlace are both in it.

The same report records who lived near commercial hazardous waste sites across the country in 2000, using census figures. In neighborhoods with such sites, 56 percent of the people were people of color, against 30 percent elsewhere. The poverty rate was 18 percent near the sites and 12 percent elsewhere. Forty of the 44 states with such sites showed the same gap.

Margie Richard

Who: A schoolteacher from the Old Diamond neighborhood of Norco, Louisiana, who led her neighbors' campaign to make Shell pay to move them away from its plant.

When and where: Norco, Louisiana. The campaign ended in 2002. She won the Goldman Environmental Prize in April 2004.

Her family's home stood 25 feet from a Shell plant's fence.

Four generations of Margie Richard's family lived in the Old Diamond neighborhood of Norco, Louisiana. Their home stood 25 feet from the fence of a Shell Chemical plant. Shell had bought land in Diamond and built the plant there in the early 1950s. The neighborhood sat between the chemical plant and a Shell refinery, a plant that turns crude oil into fuels.

Residents of Old Diamond kept a record of the cancers and birth defects among their neighbors.

The fires. Gas leaked from a Shell pipeline in 1973, and a spark from a lawnmower set it on fire. The fire burned Leroy Jones, age 16, to death while he was cutting grass. It also burned to death Helen Washington,

who was resting on her porch. An explosion at the Shell refinery in 1988 killed seven workers and damaged homes in Diamond.

In a 2004 interview with *Grist* magazine, Richard remembered: "The elder people were always coughing ... black soot falling on the grass, on our houses." She also said: "My sister died at a very early age from a rare bacterial disease."

The campaign. Richard led the neighborhood's campaign for about 13 years. Residents tested their own air with buckets built to catch air samples. They took their case to Shell's shareholder meetings, where the company's owners gather, and to Shell in the Netherlands.

Shell's managers agreed in 2002 to pay to move every Old Diamond family that wanted to leave. They also agreed to cut the plant's emissions, the pollution it sends into the air, by 30 percent, and to put $5 million into a community fund. Shell paid at least $80,000 for each house and $50,000 for each trailer, about $30 million in all.

Richard became the first African American to win the Goldman Environmental Prize, in April 2004.

The plant by the school

Federal lawyers sued the owner of a plant in LaPlace, Louisiana, in 2023, and they dropped the suit in 2025. The plant released a chemical likely to cause cancer into the air near an elementary school.

The plant is in St. John the Baptist Parish (a parish in Louisiana is like a county in other states). The company DuPont built the plant to make neoprene, a kind of man-made rubber. The Japanese company Denka bought it about 2015, according to AP news reporting in 2025. To make neoprene, the plant used a chemical called chloroprene, and it released chloroprene into the air.

EPA scientists found in 2010 that chloroprene is "likely to be carcinogenic to humans," meaning likely to cause cancer. They also found that children breathing it build up a lifetime risk of cancer faster than adults do.

The school. More than 300 children went to the 5th Ward Elementary School in 2023, about 450 feet from the plant, or about a football field and a half.

The EPA's recommended long-term level for chloroprene is 0.2 micrograms in each cubic meter of air, a microgram being one millionth of a gram. In their 2023 lawsuit, federal lawyers reported the averages at the five air monitors closest to the plant. The lowest was "more than four times" the recommended level, and the highest was "more than 14 times" that level.

AP reporters wrote in March 2025 that the community around the plant is about 59 percent Black. They also reported that the plant's air pollution posed the highest cancer risk from any single source to a mostly Black population in the country.

The lawsuit. Lawyers for the US Justice Department, acting for the EPA under President Biden, sued Denka under the Clean Air Act on February 28, 2023. They asked the court to force deep cuts in the plant's chloroprene pollution. A trial was set for April 2025.

Under President Trump, Justice Department lawyers dropped the suit on March 7, 2025. Their press release stated that dropping it carried out the president's order called "Ending Radical and Wasteful Government DEI

Programs and Preferencing." DEI stands for diversity, equity and inclusion.

Denka's managers announced on May 13, 2025, that the LaPlace plant would not start making chloroprene again "for an indefinite period," meaning with no end date. They named losses of more than $100 million, rising costs, falling production and problems keeping enough staff. Articles in trade magazines also listed the pollution rules as a reason. The plant was not making chloroprene as of May 2025.

Plastic in the rain

More than 1,000 tons of tiny plastic pieces fall each year on protected land in the western United States. Janice Brahney, a scientist at Utah State University, and her team found this in a study published in the journal *Science* on June 12, 2020. The pieces are called microplastic, bits of plastic too small to see easily, and they fall with rain and with dust. The team sampled 11 parks and other protected areas.

Each year's total is about as much plastic as more than 120 million plastic bottles. Most of the pieces were tiny fibers from clothing.

Comebacks

People nearly wiped out the bald eagles and the bison, and wiped out the wolves of Yellowstone and of public lands across the country. All three live in far larger numbers now. Hide hunters killed the bison, park managers and others poisoned and shot the wolves, and DDT that people sprayed thinned the eagles' eggshells.

Eagles. The U.S. Fish and Wildlife Service counted about 71,400 nesting pairs of bald eagles in the 2020s, against 417 in 1963. The figure of about 10,000 pairs for 2007 and the figure for the 2020s come from different counts. For the newer one, the service's crews surveyed nests from the air in 2018 and 2019, used bird watchers' reports for places they could not fly over, and built a computer model to estimate the total. The service's officials said the eagles had kept increasing fast since their survey of 2009.

Bison. About 30 million bison lived on the Plains in the early 1800s, and fewer than 1,000 were left by 1900. A count in 2008 found about 20,500 Plains bison in 61 herds kept for conservation. It also found about 400,000 bison in commercial herds, which are raised for sale. A U.S. Fish and Wildlife Service web page now lists about 420,000 bison in commercial herds.

Wolves. State wildlife officials estimated 2,797 wolves in seven western states at the end of 2022. Park Service biologists counted 108 wolves in nine packs in Yellowstone National Park at the end of 2024. At the end of 2025, the park's lead wolf biologist, Dan Stahler, gave a count of 84 wolves in eight packs. People are still fighting in court over whether wolves should stay on the endangered list.

Land given back

Chapter 2 tells how members of Congress gave the National Bison Range, a bison refuge on the Flathead Reservation in Montana, back to the Confederated Salish and Kootenai Tribes in 2020.

How the tribes lost it. President Theodore Roosevelt signed a law on May 23, 1908, under which federal officials could spend money to buy land on the reservation for a bison range. The tribes objected strongly. Federal officials took the land anyway and paid about $1.56 an acre.

The Fifth Amendment to the Constitution requires fair payment when the government takes private property for public use. Judges of the U.S. Court of Claims, a federal court that heard claims for money against the national government, ruled in 1971 that the price had been below the land's fair value. That made the taking a break of the Fifth Amendment. The judges awarded the tribes about $231,548.

Wahhoga. Park Service officials in Yosemite agreed in 2018 to share the care of Wahhoga, the site of the valley's last Native village, for 30 years. They share it with the American Indian Council of Mariposa County. A roundhouse, a round building for gatherings, was finished there in the winter of 2025.

Bears Ears

Three presidents have created, cut, restored and cut again Bears Ears National Monument in Utah since 2016. As of September 26, 2026, most of the land was out of the monument and open to mining claims, and no court had ruled on the latest cut.

Created and cut. President Obama created the monument on December 28, 2016, with 1.35 million acres. Five tribes had proposed it: the Hopi Tribe, the Navajo Nation, the Ute Mountain Ute Tribe, the Ute Indian Tribe and the Pueblo of Zuni.

President Donald Trump cut it by about 85 percent in December 2017, to about 201,876 acres. President Biden restored it on October 8, 2021. In June 2022, the five tribes signed an agreement with the Bureau of Land Management and the Forest Service to manage the monument together.

The 2026 cut. Trump signed a proclamation on July 13, 2026, that cut Bears Ears to 121,096 acres, a cut of about 91 percent. Reporters for the *Moab Sun News* published these figures on July 18, 2026. About 1.24 million acres left the monument. What is left is two pieces, Shash Jáa, about 106,816 acres, and Indian Creek, about 14,279 acres.

The proclamation lists minerals in the removed land: silver, copper, uranium, vanadium and zinc. The same day, Trump cut Grand Staircase-Escalante National Monument from about 1.87 million acres to about 181,500 acres.

The tribes' commission. In the July proclamation, Trump also ended the Bears Ears Commission. Elected representatives of the five tribes sat on it, and they had managed the monument with federal officials under the 2022 agreement. Reporters for the *Moab Sun News* wrote that a committee that only gives advice took its place, with tribes holding 5 of about 15 seats.

The tribes say Interior Department officials promised in writing in June 2025 to ask the commission's views before any change to the boundaries, and that no one asked it before July 13. Their statement said: "Our Tribes were not informed of or asked about this decision, and that's unacceptable."

The mining claims. Under the proclamation, the removed land opened on September 11, 2026, to new mining claims, mineral leasing, grazing and timber cutting. A mining claim is a legal claim to the right to mine minerals on public land, and mineral leasing means renting out the right to take minerals from the ground.

By noon on September 11, 16 claims had been recorded in San Juan County. By September 23 staff of a conservation group had counted 23 claims in the land taken out of Bears Ears, 20 of them filed by one mining company.

In court. On September 2, 2026, three sets of groups that had sued over Trump's 2017 cut went back to the federal court in Washington and asked it to hear the new cut as well. They argue that the Antiquities Act of 1906 lets a president create a monument and does not let a president undo one. Utah's members of Congress called the cuts "a return to an appropriate application of the Antiquities Act."

The tribes' own 2017 case, *Hopi Tribe v. Trump*, is still pending, which means undecided, in the same court. As of September 26, 2026, no court had ruled on the 2026 cuts, and no judge had ordered them stopped.

BY 2026

If you remember one thing: After 2000 the land grew hotter and drier and the fires grew bigger, bald eagles, bison and wolves came back by the thousands, and the fights went on over chemical plants beside Black neighborhoods and over Native land such as Bears Ears.

Still open: No court has yet decided whether a president can undo a national monument another president made. The tribes' 2017 case and the suits filed in September 2026 were still waiting in federal court on September 26, 2026.

THE THREE QUESTIONS, ANSWERED

1. **How did Native nations shape the land before Europeans came?** They burned the eastern forests on purpose, again and again, to keep them open and grow food for game, and they planted corn, beans and squash together to keep the soil good.
2. **What happened to the 30 million bison and the billions of passenger pigeons?** Hide hunters, carried by the railroads and helped by army officers who wanted the Plains nations' food gone, left fewer than 1,000 bison by 1900. Netters and shooters killed the pigeons at their nestings, and the last one died in a zoo in 1914.
3. **Who was pushed out when parks were made, and who lived next to the waste?** Native people were pushed out: the Ahwahneechee in Yosemite, the Tukudika in Yellowstone, the Blackfeet at Glacier. Studies from 1983 on found waste sites far more often beside Black and poor neighborhoods.

What changed: Americans cleared, sold and wore out the land that Native nations had kept with fire, then began to protect parts of it by law, and fire crews now learn the old burning from the Yurok.

FIVE DATES TO KEEP

1642: Miantonomi describes what English settlers are doing to the land.

May 20, 1785: Members of Congress pass the Land Ordinance, the start of the grid.

March 1, 1872: Yellowstone becomes the first national park.

September 1, 1914: Martha, the last passenger pigeon, dies.

April 22, 1970: About 20 million Americans take part in Earth Day.

NOT KNOWN: Nobody counted the bison on the Plains before 1500, and the usual figure of about 30 million comes from the early 1800s.

Where to go: Devils Tower National Monument, the first national monument, which the Lakota call Mato Tipila, or Bear Lodge.

Next: Chapter 4, Immigration to America. It asks why people left home to come here, and whom Americans tried to keep out.

PART 2

Coming, Moving, Settling

Chapter 4: Immigration to America

CHAPTER 4 OF 37 · IMMIGRATION TO AMERICA · 10 ERAS · ABOUT 100 MINUTES

People came here to stay from almost every country on earth, and from the 1600s on, the people who had come earlier kept deciding who could join them, by law and sometimes by mob violence.

The poet Emma Lazarus wrote a poem in 1883 to help raise money for the stone base of the Statue of Liberty. It includes the lines "Give me your tired, your poor, / Your huddled masses yearning to breathe free." Nobody read it aloud when the statue opened in 1886. In 1903 her friend Georgina Schuyler had the poem mounted on a bronze plaque inside the base. One year before Lazarus wrote those lines, President Chester A. Arthur had signed a law that shut Chinese workers out of the country.

This chapter follows the people who came to stay, from a Spanish fort in Florida in 1565 to the record number of people held in immigration detention in 2026. Watch why they came: for land and work, to worship in their own way, and to escape famine, riots against Jews, revolution and war. Watch, too, who decided whether they could stay. Lawmakers wrote the rules, from citizenship for white people only in 1790 to the Chinese Exclusion Act of 1882 and the quotas of 1924. Mobs in Philadelphia, Louisville and Rock Springs used fire and guns. The forced arrival of Africans appears here only where it meets those laws. Chapter 24, Slavery and Freedom, tells it in full.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Why did people leave home to come here?
2. Who did Americans try to keep out, and how?

3. Who could become a citizen, and how did that change?

Five minutes? Read "Angel Island" in 1900 to 1950, the Lee Puey You story after it, and the next section, "How the quota laws of 1921 and 1924 worked."

Can skip: "The Dutch and the Swedes" in the 1600s, "Jewish communities" in 1700 to 1750, "The Revolution and the Loyalists" in 1750 to 1800. The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 1 MINUTE

No one from across the ocean came to settle here for good before 1500. Hundreds of Native nations lived in the land that is now the United States, and their people spoke hundreds of different languages. Every group of newcomers who came later arrived on land where people were already living.

Nobody knows how many people lived north of Mexico in 1492. Estimates run from about 1 million to as many as 18 to 20 million. Chapter 2, Native Nations, tells who they were.

The land bridge and the Norse

The first people to reach this land came on foot from Asia, long ago. They walked across the ice-age land bridge, a stretch of dry ground that joined Asia to North America.

Chapter 1 tells the Norse voyages in full. What matters here is that the Norse did not stay. Around the year 1000, Norse sailors from Greenland, the people often called Vikings, built a camp at L'Anse aux Meadows, in Newfoundland, in what is now Canada. Wood they cut there with metal blades dates to 1021. Before Columbus, it is the only camp that anyone from across the ocean is known to have built in North America.

Native people had lived at that spot long before the Norse came. Five Native groups had lived there, the oldest about 6,000 years earlier. The Norse stayed only a short time, from a few years to about twenty by different estimates, and built no settlement that lasted.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

Europeans came to these coasts in the 1500s to conquer and explore, and almost none of them stayed. Most were soldiers, explorers, sailors and Christian missionaries, people sent to bring others into their religion. Native nations held almost all of the land for the whole century. Of three European attempts to settle, only the Spanish town of St. Augustine lasted.

Which settlement lasted?

St. Augustine, Florida, lasted. It is the oldest town that Europeans founded on the US mainland, and people have lived in it without a break since it began. The Spanish navy commander Pedro Menéndez de Avilés founded it on September 8, 1565. He named it for Saint Augustine because he had first sighted land on that saint's feast day, a day the church sets aside to honor a saint. St. Augustine is 42 years older than Jamestown and 55 years older than Plymouth.

Seloy. Menéndez landed at Seloy, a town of the Timucua people. Counts of the people who came with him differ. The National Park Service gives about 600 soldiers and settlers. The Florida Museum of Natural History gives about 800, including 26 women and an unknown number of enslaved Africans.

The newcomers used the council house of the Timucua chief, also named Seloy, as their first fort. It was the building where the town's leaders met. Timucua fighters attacked again and again to drive the Spanish away. In May 1566 Menéndez moved the settlement across the bay to Anastasia Island, and by about 1572 the Spanish had moved back to the place where the town stands today.

Fort Caroline. Two other attempts to stay failed. René de Laudonnière founded Fort Caroline, a French settlement near where Jacksonville, Florida, stands today, in 1564. The French there met Timucua people, and one of their leaders was Athore. The French settlers were Huguenots, French Protestants,

members of the Christian churches that split away from the Catholic Church led by the pope in Rome. In 1565 Menéndez destroyed Fort Caroline and killed most of the French defenders. The Spanish were Catholics, and the fight was partly over religion.

Roanoke. Nobody knows what happened to the 117 English settlers who landed on Roanoke Island in 1587. The island lies off the coast of what is now North Carolina, and it was the land of the Roanoke, Secotan and Croatoan peoples. Sir Walter Raleigh organized the colony, a settlement ruled from a faraway country, and John White led it. On August 18, 1587, a settler named Eleanor Dare gave birth there to a daughter, Virginia Dare, the first English child recorded as born in the Americas.

The settlers sent White back to England for supplies. Before he left, they agreed on a signal. If they moved, they would carve the name of their new place on a tree or a post. If they were in trouble, they would carve a cross above the name. White could not get back for three years. In 1588 Queen Elizabeth I took every English ship for her navy, to fight the Spanish Armada, a fleet of Spanish warships.

What White found. White went ashore on August 18, 1590, Virginia Dare's third birthday. He found a wall of upright logs, the houses gone, and no people. Someone had carved CRO on a tree and CROATOAN on a post, with no cross. Croatoan was a nearby island, where a man named Manteo was born, and White wrote that its people were friends of the English. A storm blew his ship out to sea before he could search there. No one found the settlers.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 9 MINUTES

Most English people who went to Virginia and Maryland in the 1600s paid for the voyage with years of their own work. People from Europe began coming to stay in these years. Some came so that they could worship in their own way. Others came to make money by farming, trading or growing tobacco.

The colonists settled along the Atlantic coast, on land where Native nations already lived. English colonists built Jamestown on Powhatan land and Plymouth on the site of a Wampanoag town. Dutch colonists built New Amsterdam on Lenape land, and Quakers under William Penn settled Pennsylvania on Lenape land too.

Jamestown and indentured servitude

Jamestown. English colonists founded Jamestown, Virginia, in 1607, and it was the first English settlement that lasted. The Virginia Company of London paid for it. Many people owned shares in the company, hoping to split a profit. The colonists built their fort inside Tsenacomoco, the land of the Powhatan, a union of Native towns and peoples under one head chief. Jamestown Island was a hunting ground of the Paspahegh, one of those peoples, and Paspahegh fighters attacked the new settlement on May 26, 1607.

Many colonists died of disease and hunger in the first years. In the fall of 1609 the Powhatan leader, also called Powhatan, cut off the food trade with the colonists and surrounded their fort. About 500 colonists were alive at the start of that winter. About 61 were alive by spring.

Tobacco. John Rolfe began growing tobacco for sale overseas in the 1610s. Once tobacco sold well, more settlers came to Virginia to grow it, and they took more Powhatan land without paying for it. On March 22, 1622, fighters from an alliance of Virginia Indian nations, led by Opechancanough, attacked the English settlements and killed as many as 347 colonists.

Most early English arrivals in the Chesapeake, the country around the Chesapeake Bay in Virginia and Maryland, came as indentured servants. An indentured servant signed a contract, called an indenture, and gave up four to seven years of work in exchange for the trip across the ocean. The contract often promised food, a place to live, and a payment at the end called freedom dues.

Servants made up about 80 to 90 percent of the European immigrants to the Chesapeake in the 1600s, eight or nine of every ten. Across all the British colonies over the whole colonial period, the share was lower: about one-half to two-thirds of white immigrants came this way.

Africans brought by force, 1619

An English privateer, a privately owned armed ship whose crew attacked the ships of enemy countries, brought more than twenty Africans to Virginia in late August 1619. The ship was the *White Lion*, and it landed them at Point Comfort, near the mouth of the James River.

Where they came from. These Africans had been seized in wars around the kingdoms of Kongo and Ndongo, in West Central Africa, in what is now Angola. In 1618 and 1619 Luis Mendes de Vasconçelos, the Portuguese governor of Angola, led two attacks on the people of Ndongo.

He fought alongside the Imbangala, African fighters paid to fight for others, and together they captured thousands of people. Their captors marched them to the Portuguese port of Luanda and loaded them onto the Portuguese ship *San Juan Bautista*, with about 350 captives bound for Mexico.

Two English privateers, the *White Lion* and the *Treasurer*, attacked that ship in the Gulf of Mexico. Their crews took 50 to 60 of the Africans between them. The *White Lion* carried its share, more than twenty people, to Virginia. They are the first Africans recorded in English Virginia, and their landing began the forced arrival of Africans in the English mainland colonies.

Richard Frethorne

Who: An English boy who came to Virginia as an indentured servant and wrote home begging for help.

When and where: Arrived in December 1622 on the *Abigail*. Martin's Hundred, Virginia. Listed among the dead there by February 1624.

"I never ate anything but peas and loblollie," he wrote home.

Richard Frethorne is one of very few ordinary people in the early colonies whose own words survive. He came to Virginia as an indentured servant, landing in December 1622 from a ship called the *Abigail*, and he worked at a place called Martin's Hundred.

In March and April 1623 he wrote letters to his parents in England. He begged them to buy out his indenture, which meant paying to free him from the rest of his years of work. If they could not do that, he asked them to send food. "I never ate anything but peas and loblollie (a thick porridge)," he wrote. In another line he wrote, "there is nothing to comfort us."

His name was on a list of the dead at Martin's Hundred by February 1624. The list does not give the cause of his death.

Plymouth and the Puritans

Farther north, the *Mayflower* carried about 102 passengers across the Atlantic in 1620. They founded Plymouth, in what is now Massachusetts, on the site of Patuxet, a Wampanoag town. A disease had killed many of Patuxet's people between 1616 and 1619. The Pilgrims among the passengers were English Separatists. They wanted to worship apart from the Church of England, the official church of their country, and before crossing the ocean they had first moved to Leiden, in Holland.

A larger group followed ten years later. In 1630 John Winthrop led a fleet of 11 ships to Massachusetts Bay with about 700 passengers. His flagship, the ship that carried the fleet's leader, was the *Arbella*. These colonists were Puritans, English Protestants who disagreed with how the Church of England was run.

Laud's courts. King Charles I made William Laud Archbishop of Canterbury, the head of the Church of England, in 1633. Laud wanted every church to follow the same rules and ceremonies. Judges in a royal court called the Star Chamber and in a church court called the High Commission took church jobs away from many Puritan ministers, the men who led church services, or banned them from preaching.

Three Puritan writers, William Prynne, Henry Burton and John Bastwick, wrote against Laud. In 1637 judges sentenced them to have their ears cut off and their cheeks branded. Branding means burning the skin with a hot iron to leave a mark that never goes away.

About 20,000 Puritans came to New England over the 1630s, in what is called the Great Migration. They were one part of a larger movement. About 80,000 people left England in that decade, and roughly 20,000 went to each of four places: Ireland, the West Indies, the Netherlands and New England. The West Indies are the islands of the Caribbean Sea.

Anne Hutchinson

Who: An English woman who followed her minister to Boston, was put on trial and banished (ordered to leave and never come back) three years later, and was killed by Siwanoy fighters in 1643.

When and where: Crossed on the *Griffin* in the fall of 1634. Boston, then Rhode Island, then land the Dutch claimed.

She asked the court why she was banished, and got no answer.

Anne Hutchinson sailed from England on the *Griffin* in the fall of 1634 with her husband, William, and their children. About 200 passengers were on board. She was following her minister, John Cotton, the preacher who led her church, to Boston.

What she taught. She held a meeting in her Boston house every week, starting in 1636, to go over the Sunday sermons. Women came at first, and then men came too. She taught that God's grace, his gift of being saved and going to heaven, did not depend on good behavior. She also said she could tell which of the colony's ministers had received that grace and which had not.

The trial. Governor John Winthrop and other colony leaders accused her of heresy, which means teaching against the church's beliefs. They counted the meetings against her too. In November 1637 the colony's lawmakers and judges, sitting as the General Court of Massachusetts, tried her at Newtown, now Cambridge, and Winthrop led the questioning.

At the end Winthrop told her, "you are banished from out of our jurisdiction (the land the court ruled) as being a woman not fit for our society." She answered, "I desire to know wherefore (why) I am banished?" Winthrop replied, "Say no more, the court knows wherefore and is satisfied."

She moved to Portsmouth, Rhode Island, in 1638. In 1642 she moved again, to land on Long Island Sound that the Dutch claimed, near what is now New York City. The Siwanoy, a Native people, lived there. The Dutch governor was Willem Kieft. On the night of February 25, 1643, Kieft's

men killed at least 80 and perhaps as many as 120 Munsee people. With those killings they started a war with the Native nations nearby. It is now called Kieft's War, and it lasted until 1645.

Her death. During that war, in August or September 1643, Siwanoy fighters attacked Anne Hutchinson's household near Pelham Bay. They killed her, her servants, and all but one of the children who were with her.

The Dutch and the Swedes

Dutch settlement began earlier than the Puritan fleet, in the 1620s, between Virginia and New England. Dutch colonists founded a colony called New Netherland. Its main town was New Amsterdam, on the island of Manhattan, in Lenapehoking, the homeland of the Lenape people. From 1638 to 1655, Swedish and Finnish settlers ran a small colony called New Sweden along the Delaware River, until the Dutch took it over in 1655.

Four English warships arrived at New Amsterdam in 1664. The Dutch governor, Peter Stuyvesant, surrendered without a fight, and the papers of surrender were signed in late August. The English renamed the colony New York, after James, the Duke of York, the younger brother of King Charles II of England.

The first Jewish refugees, 1654

Twenty-three Jewish refugees, people fleeing danger at home, landed at New Amsterdam in 1654 on a ship called the *Ste. Catherine*. They came from Recife, in Brazil. The Dutch had held that part of Brazil, and the refugees left after Portuguese forces took it back. Another Jewish man, Jacob Barsimson, had arrived earlier that year.

Governor Peter Stuyvesant tried to force the refugees out of the colony. Leaders of the Dutch West India Company, the trading company that ran New Netherland and gave the governor his orders, overruled him.

Catholics, Quakers and Huguenots

Other groups crossed the ocean for their religion too. Catholics settled in Maryland from 1634. Quakers, members of a Protestant group also called the Society of Friends, settled in Pennsylvania under William Penn from 1681. The Lenape people lived on both sides of the Delaware River there.

The Huguenots. King Louis XIV of France canceled the Edict of Nantes in 1685. King Henry IV had made that law in 1598, and under it French Protestants could worship in the towns they held. Huguenots fled France, and most went to other countries in Europe. Only about 1,500 to 2,000 had reached the American colonies by 1700. They settled in places such as New York, Massachusetts, and Charleston, South Carolina.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 6 MINUTES

Scots-Irish and German newcomers came by the thousands in these fifty years, and many other arrivals came bound to work for years or for life. The Scots-Irish came from the north of Ireland, and the Germans from the lands along the Rhine River. Small Jewish communities grew in the port towns. The colonists still lived in a thin strip along the Atlantic coast, and the land farther inland was still Native land.

Bound workers could not leave their work until their time was up, and some never could. Indentured servants had signed contracts for years of work. Merchants sold British convicts as laborers. Slave traders brought enslaved Africans by force.

Why Germans and Scots-Irish left home

German newcomers left to escape war, taxes and failed crops, and they came for cheap land. They came from southwest Germany and from the lands along the Rhine River, including the Palatinate, a region along the Rhine whose people were called Palatines. About 2,100 Palatines reached New York in 1709 and 1710.

A larger number of Germans went to Pennsylvania, mostly between about 1725 and 1775. About 12,000 German speakers had come to Pennsylvania by 1749. By the time of the Revolution, Germans made up about one-third of the people in that colony. The name Pennsylvania Dutch comes from *Deutsch*, the German word for German.

The Scots-Irish were Presbyterians, members of a Protestant church, from Ulster in the north of Ireland. Their ancestors were Scots who had settled in Ulster. Between 1710 and 1775 about 200,000 of them came to the land that later became the United States.

The Test Act. They left to escape high rents and a law called the Test Act, which Ireland's lawmakers passed in 1704. Under it, only people who took communion, a church ceremony of bread and wine, in the Church of Ireland could hold a public job. That rule shut Presbyterians out of public jobs. Marriages performed by Presbyterian ministers did not count under the law, so the law said the children of those marriages were born outside a lawful marriage.

Pennsylvania was the main place where the Scots-Irish landed. Many then moved south along the Appalachian backcountry, the land in and near the Appalachian Mountains, far from the coast. That backcountry was Native land.

John Peter Zenger

Who: A boy from the German Palatinate who grew up to be a New York printer, jailed and tried over what his newspaper printed.
When and where: Arrived in New York in June 1710. Died July 28, 1746.

His paper missed one issue in his almost nine months in jail.

John Peter Zenger reached New York in June 1710, at the age of 13. His father, Nicolaus, had died on the voyage. The family came from Rumbach, in the Palatinate, and had joined the Palatine migration in late 1709, traveling first to England. John Peter landed with his widowed mother, Johanna, and his two siblings. In 1711 he began work as an apprentice, a young worker who learns a trade from a master, to the New York printer William Bradford.

The newspaper. In 1733 Zenger became the printer of the *New-York Weekly Journal*, a newspaper started by men who opposed the governor, William Cosby. Cosby had copies of the paper burned. Two judges on Cosby's side, James De Lancey and Frederick Philipse, ordered Zenger's arrest, and on November 17, 1734, the sheriff took him to jail.

The charge. He was charged with seditious libel, the crime of printing words that blame a public official or the government. His bail, money paid to a court so that a person can leave jail until the trial, was set at 400

> pounds. Zenger did not have that much money, so he stayed in jail until his trial, almost nine months later. His wife, Anna, and his apprentices kept printing the paper, and only one issue was missed. On August 4, 1735, a jury found Zenger not guilty.

Jewish communities

Most of the Jews in the colonies were Sephardic Jews, whose families had come from Spain and Portugal. After the first group landed at New Amsterdam in 1654, Jewish communities formed at Newport, Rhode Island, and at Charleston, New York and Savannah, Georgia. On July 11, 1733, 42 Jews landed at Savannah.

Convicts sent as bound labor

The Transportation Act. British lawmakers passed the Transportation Act in 1718. Under it, merchants shipped about 50,000 convicts, people found guilty of crimes, from Britain to the American colonies. Some counts put the number as high as 60,000, and more than 20,000 went to Virginia. Officials of the British treasury, the office that handled the government's money, paid the merchants for each convict. The first merchant to get this contract was Jonathan Forward.

In the colonies the merchants sold the convicts as bound laborers, workers who had to work for a buyer for a set number of years. Most had to work seven years, and some had to work fourteen. Buyers included merchants, tradesmen, shipbuilders and ironworks owners, because a convict cost less than an indentured servant or an enslaved African. The shipping went on until 1775.

Convicts made up roughly a quarter of the people who emigrated from Britain in the 1700s, leaving their own country to live in another. Indentured servants kept coming alongside them.

Forced arrival through Charleston

The number of enslaved Africans that slave traders brought to the mainland colonies rose sharply in these years. Rice planters in the South Carolina Lowcountry, the low coastal land, bought large numbers of West and Central Africans. Some planters chose Africans who already knew how to grow rice. Charleston, South Carolina, was the main mainland port where enslaved Africans arrived.

Sullivan's Island. Many were held first at a quarantine station on Sullivan's Island, at the mouth of Charleston's harbor, where new arrivals were kept apart so that any disease they carried would not spread. They were kept in buildings called pest houses and checked for disease there. The histories of the island do not name the people who kept them. Then slave traders took them to Charleston and sold them at public auction.

The historian Peter Wood wrote in 1974 that Sullivan's Island "might well be described as the Ellis Island of black Americans." Ellis Island was the New York station where millions of European immigrants later arrived. Published estimates put at least 200,000 Africans through Charleston harbor and say that about 40 percent of African Americans today have an ancestor who came that way. Nic Butler, a historian at the Charleston County Public Library, argues that the share credited to Sullivan's Island is too high.

Across the whole slave trade, slave traders forced about 12.5 million Africans onto ships. About 10.7 million lived through the crossing, and about 1.8 million died on the slave traders' ships on the way. Fewer than 4 of every 100 people who lived through the crossing were brought to mainland North America. About 305,000 came straight from Africa, and close to 388,000 came in all, counting those brought by way of the Caribbean.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 5 MINUTES

After the Revolution, members of Congress decided who could become a citizen, a full legal member of the new country, and they limited it to free white people. Fewer people crossed the Atlantic to settle during the American Revolution, from 1775 to 1783, than before it. In the 1790s thousands of refugees arrived from a revolution in the Caribbean.

Alexander Hamilton

Who: An orphaned clerk from the Caribbean who came to New York as a teenager to go to school, and stayed.

When and where: Born on Nevis. Sailed from St. Croix in late 1772 or early 1773. King's College, New York.

Movie: *Hamilton* (2020), the filmed Broadway stage musical with its original cast, released on Disney+ on July 3, 2020. It is a dramatization, not a documentary.

A letter about a hurricane paid his way to school.

Alexander Hamilton was born on the Caribbean island of Nevis on January 11. The year is disputed: accounts of his life give either 1755 or 1757. By 1772 he was working on the nearby island of St. Croix as a clerk, a worker who keeps written records for a business.

A hurricane struck St. Croix on August 31, 1772. Hamilton wrote a letter describing the storm, dated September 6, and a newspaper called the *Royal Danish American Gazette* printed it on October 3. After reading it, men on the island raised money to send him to school in North America.

He sailed in late 1772 or early 1773. Accounts differ on the date. He studied at King's College in New York, and he stayed. Later he worked for the new national government at the Treasury, the department that handles the government's money.

The Revolution and the Loyalists

Fewer immigrants crossed the Atlantic during the Revolutionary War, from 1775 to 1783, than in the years before it. The shipping of British convicts to the colonies ended in 1775. During and after the war, tens of thousands of Loyalists, colonists who sided with the British king against the Revolution, left the country, many of them for Canada.

Under the peace of 1783, the western border of the United States was the Mississippi River. Spanish and French officials claimed the land beyond the river. Most of the land away from the Atlantic coast was still Native land.

Who could become a citizen in 1790?

Only "a free white person" of "good character" could. Members of Congress wrote those words into the Naturalization Act of 1790, the national government's first law on naturalization, the legal process by which a person born in another country becomes a citizen. To qualify, a person had to live in the United States for two years, including at least one year in the state where they applied, and swear loyalty to the Constitution. Children born abroad to citizens were included as citizens.

Under this law, an immigrant who was enslaved or was not white could not become a citizen. Members of Congress did not fully remove the limit to white people until 1952, 162 years later.

The men who wrote the Constitution in 1787 also set a date for arrivals by force. In Article I, Section 9, they barred Congress from stopping the bringing in of enslaved people before 1808. So for about twenty more years, slave traders could legally bring captives into the country.

Pierre Toussaint

Who: A man born enslaved in Saint-Domingue and brought to New York by the family that enslaved him. He became a leading hairdresser and was freed in 1807.

When and where: Brought to New York in 1787. Freed July 2, 1807. Died in New York, June 30, 1853.

He was one of New York's leading hairdressers, and still enslaved.

Pierre was born enslaved in Saint-Domingue, a French colony in the Caribbean, in about 1766, though some recent accounts give 1781. Until he was freed, he was known simply as Pierre. In 1787 Marie Elisabeth Bérard and her husband, the family that enslaved him, took him to New York. The Bérards were fleeing fighting and disorder in the colony, four years before the revolution there began.

In New York Pierre trained under a hairdresser and became one of the city's leading hairdressers. When Marie Elisabeth Bérard's husband died, Pierre supported her. He was still enslaved by her family at the time. During an outbreak of yellow fever, a deadly disease spread by mosquitoes, in New York in 1798, he nursed sick people who were kept apart from others.

Freedom. Marie Elisabeth Bérard married again, to a Mr. Nicolas, and she died in 1807. On her deathbed she made her second husband promise to free Pierre. Mr. Nicolas kept the promise, and Pierre was free on July 2, 1807. If he was born in about 1766, he was about 41, though one account says 45. As a free man he took the surname Toussaint, to honor Toussaint Louverture, a leader of the revolution in Saint-Domingue.

Toussaint used his money to care for orphans and to fund early Catholic charities in New York, groups that helped the poor. In 1811 he married Juliette Noel, an enslaved woman, after he paid for her freedom.

> **After his death.** Cardinal John O'Connor, the head of the Catholic Church in New York, had Toussaint's remains moved in 1990 to the crypt of St. Patrick's Cathedral, a burial room beneath the church. Pope John Paul II declared him Venerable in 1996, a title the pope gives to a person whom church leaders are studying as a possible saint.

Refugees from Saint-Domingue

Enslaved Africans in Saint-Domingue, a French sugar colony, rose up on August 21, 1791, against the French planters who enslaved them and against French rule. The fighting went on until January 1, 1804, when Saint-Domingue became the independent country of Haiti.

Refugees fled to cities in the United States. They included white planters and free people of color, meaning people of African or mixed descent who were not enslaved. They also included enslaved people, whom their owners brought with them. That first year a ship called the *Charming Sally* carried 15 refugees to Philadelphia. By 1794 more than 3,000 refugees had come to that city, and nearly 20,000 refugees from Saint-Domingue came to the United States across the 1790s and the 1800s.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 11 MINUTES

The first large wave of people who chose to come here arrived in these years, mostly from Germany and Ireland, and some Protestant Americans met the Irish Catholics with mobs. Germans were the largest group of newcomers from about 1840.

One kind of arrival ended early in these years. Under a national ban that began on January 1, 1808, no one could legally bring an enslaved person into the country from abroad.

Protestants, Christians whose churches had split away from the Catholic Church led by the pope, formed mobs that attacked Catholics near Boston in 1834 and burned Irish Catholic homes in Philadelphia in 1844. Then more than 1.5 million Irish people reached American ports between 1845 and 1855, during and after a famine in Ireland, a time when food ran so short that many people died of hunger. New York was the main port where they landed, and large numbers also came ashore at Boston and at New Orleans.

The end of legal arrival from Africa

President Thomas Jefferson signed a law on March 2, 1807, that banned bringing enslaved people into the United States from other countries. It is called the Act Prohibiting Importation of Slaves, importation meaning bringing people or goods in from abroad. The ban began on January 1, 1808, the earliest date the Constitution allowed.

No one could legally bring an enslaved African across the ocean after that. Smugglers kept bringing in captives against the law. From then on, the number of enslaved people in the country grew mainly through children born to enslaved mothers. Enslavers held those children as property from birth.

Germans and the Forty-Eighters

More immigrants came from Germany than from any other country from about 1840 to 1890. Many Germans left home because their crops had failed, and they came for farmland, most of all in the Midwest. Roughly three out of every five German immigrants settled there to farm. About 6 million Germans came to the United States between 1820 and World War I.

The Forty-Eighters. Revolutions broke out in Germany in 1848 and 1849. The people who fought in them wanted democratic government, in which the people choose their leaders. Soldiers from Prussia, a kingdom in what is now Germany, crushed the revolutions, and some of the revolutionaries fled abroad.

The ones who came to the United States are called the Forty-Eighters, after the year 1848. There were only about 4,000 to 10,000 of them. Many were journalists or professionals, people trained for skilled work such as law or medicine. Some of them later became leaders in their new country.

Jette Bruns

Who: A German woman who settled in Missouri with her doctor husband and wrote letters home to her brother for more than fifty years.

When and where: Born 1813 in Oelde, Westphalia. Arrived at the Westphalia Settlement, central Missouri, in 1836. Died 1899.

She wrote home to her brother for more than fifty years.

Henriette Geisberg Bruns, known as Jette, came from Oelde, a town in Westphalia, a region of Germany. In 1836 she arrived at the Westphalia Settlement in central Missouri with her husband, the doctor Bernhard Bruns. Their baby son came too, along with two of her brothers and a maid.

Bernhard became mayor of Jefferson City, the capital of Missouri, and he died in office during the Civil War. After he died, Jette had almost no money. She ran a boardinghouse, a house where people paid to sleep and eat, across from the Missouri state capitol, the building where the state's lawmakers meet.

The letters. Her letters home to her brother Heinrich survive. The University of Missouri Press printed them in 1988, along with a short account of her life that she wrote after the Civil War. Adolf E. Schroeder edited them and translated them into English. The book is called *Hold Dear, As Always: Jette, a German Immigrant Life in Letters*.

Mobs against Catholics, 1834 and 1844

Some Protestant Americans turned against the Irish newcomers and against Catholics. Before the Irish famine, Protestant mobs attacked Catholics in two cities and burned their buildings.

Charlestown, 1834. A mob of mostly Protestant workmen burned a Catholic convent school, a school that nuns ran in their own home, in Charlestown, near Boston, on the night of August 11, 1834. The nuns and 47 girl students were inside, and only six of the girls were Catholic. One Boston newspaper put the mob at 50 to 60 men, and another put it at 150 to 200. About 2,000 people gathered to watch.

The mob broke down the doors and windows, set fire to the buildings and burned them to the ground. The men also broke open the tomb where several dead nuns were buried. The records of the attack list no deaths.

Police arrested 13 men. At trial, juries found 12 of them not guilty, including their leader, John R. Buzzell. He later wrote that the evidence against him was enough "to have convicted (found guilty) twenty men, but somehow I proved an alibi (proof that he was somewhere else)." The one man found guilty was 16 years old, and Governor John Davis pardoned him, forgiving the crime and ending the punishment.

Massachusetts lawmakers at first voted down a request from Catholic church leaders to be paid for the loss. In 1846 they voted the church $10,000. Church leaders put the loss at about $100,000, ten times as much, and refused the money.

Philadelphia, 1844. In Philadelphia the fighting began over schools. Teachers in the public schools made Catholic children read the Protestant version of the Bible. Francis Patrick Kenrick, the Catholic bishop of Philadelphia, the leader of the city's Catholic churches, objected. Nativists, people who wanted power kept for native-born Protestants, then claimed that Catholics were trying to throw the Bible out of the schools.

Nativists held a rally on May 6, 1844, in Kensington, a mostly Irish Catholic neighborhood. Shooting and fighting broke out between the nativists and the Irish Catholics. An 18-year-old nativist named George Shiffler was shot dead. The records do not say who fired the shot.

Nativist crowds burned Irish Catholic homes, a fire station and a market on May 7 and 8. They also burned two Catholic churches, St. Michael's and St. Augustine's, and the school of St. Augustine's. They burned a seminary, a school that trains people for church work, run by the Sisters of Charity, a group of Catholic nuns. One count gives about 30 homes burned and about 200 people driven from their homes.

St. Philip de Neri. Nativist crowds learned in July that the brother of the priest at a third church, St. Philip de Neri, had stored guns in it to defend it. They gathered there and brought a cannon. Soldiers of the state militia, a force of citizens called up to serve as soldiers, fired on the crowd under General George Cadwalader. The crowd attacked the soldiers with bricks, stones, bottles and guns.

Accounts give different counts of the dead. For the May riots, the counts run from at least 14 dead to 20 dead. One of the Catholics killed was Joseph Rice. The Encyclopedia of Greater Philadelphia lists four soldiers and "probably a dozen rioters" dead in July. EBSCO's reference articles give figures that add up to about 34 dead and at least 126 wounded in both riots together. There is no agreed total.

Who was blamed. A grand jury, a group of citizens who decide whether there is enough evidence to put a person on trial, charged 19 of the rioters. Juries found only a handful of the 19 guilty. In June 1844 a grand jury blamed the Catholics who had been attacked. It said the May riots came from efforts "to exclude the Bible from the public schools." Catholics answered that the jury had called no Catholic to testify.

That October, Philadelphia voters elected Lewis Levin to Congress. Levin was a newspaper editor who had founded the nativist group that held the Kensington rally.

The Irish famine

A plant disease called potato blight rotted Ireland's potatoes from 1845 to 1852. The blight is a water mold, a tiny living thing that grows on the plant and makes it rot. At least two out of every three people in Ireland lived mainly on potatoes. People starved, and at least 1 million people in Ireland died. This is called the Great Famine.

Food kept leaving. Ireland had been part of the United Kingdom since 1801, ruled by the government and the lawmakers in London. Ships kept carrying food from Ireland to Britain while people starved. The historian Christine Kinealy counted almost 4,000 ships that carried butter, grain and farm animals from Ireland to British ports in 1847. Soldiers guarded some of those shipments. About 400,000 people in Ireland died that year. Accounts still differ about how much of the need that food could have met.

In an earlier food shortage, in the 1780s, British officials had banned grain from leaving Ireland, and the ban had worked. In 1846 and 1847, Prime Minister Lord John Russell, the head of the British government, and his ministers, the officials who ran its departments, refused to ban grain from leaving Ireland again. They also made Irish taxpayers pay for famine relief, the food and work given to people who were starving.

Charles Trevelyan was the senior official at the Treasury, the government's money department, who ran famine relief. In December 1846, writing to Lord Monteagle about Ireland, he said that the real problem was not the famine. He said it was "the selfish, perverse (stubbornly wrong) and turbulent (unruly) character of the people," meaning the Irish.

The quarter-acre rule. In 1847 members of the British Parliament, the lawmakers in London, added a rule to the law on help for the poor. The rule was named for one of them, William H. Gregory. Under it, a family holding more than a quarter of an acre of land could get no relief. To get food, a family had to give up its land.

Evictions. Landlords forced tenant families, families who rented the land they farmed, off their land and often pulled their houses down. This is called eviction. Police records list almost 250,000 people evicted from 1849 to 1854.

The historian James S. Donnelly Jr. judged that count too low. He also counted families forced to give up their land in other ways from 1846 to 1854, and he wrote that the total "would almost certainly exceed half a million persons." Lord Palmerston, a minister in the British government, cleared about 2,000 tenants from his land in County Sligo and paid their fares overseas. Major Denis Mahon evicted thousands of tenants in County Roscommon.

Most of the people who died in the famine died of disease, which people then called fever. Officials in Ireland did not record deaths then, so the numbers are estimates. They run from about 1 million to 1.5 million dead.

The crossing. At least 2.1 million people left Ireland from 1845 to 1855, and most of them went to the United States. People called the crowded ships of those years "coffin ships." The *Virginius* sailed to Canada in 1847 with 496 passengers, and a report in the Toronto *Globe* said that 158 of them had died, "nearly one third."

More than 1.5 million Irish people arrived at American ports between 1845 and 1855. Most of them were poor when they landed. They settled in cities of the Northeast, such as New York and Boston, and in other cities in Pennsylvania, Massachusetts and beyond.

Patrick Kennedy and Bridget Murphy

Who: Two people from County Wexford, Ireland, who married in Boston during the famine years. Patrick was the great-grandfather of President John F. Kennedy.

When and where: Patrick left Ireland in 1848 or 1849. They married in September 1849. East Boston, Massachusetts.

He made barrels in East Boston and died of cholera at 35.

Patrick Kennedy was born in 1823 at Dunganstown, in County Wexford, Ireland. He left during the Great Famine and made a new home in East Boston, Massachusetts. Accounts differ on whether he left in 1848 or in 1849. In East Boston he worked as a cooper, a person who makes barrels.

He married Bridget Murphy in September 1849. She also came from County Wexford, from a place called Owenduff. The couple had five children.

Patrick died of cholera, a disease people catch from dirty water or food, on November 22, 1858, at the age of 35. Cholera causes heavy diarrhea, and a sick person can lose so much water from the body that they die within days.

Patrick's son Patrick Joseph Kennedy, known as P.J., had a son named Joseph P. Kennedy. Joseph's son was John F. Kennedy, who later became president of the United States.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 21 MINUTES

Chinese immigrants became the first people that lawmakers barred from the country because of where they came from. Newcomers from Europe kept arriving, about 11 million through the port of New York alone between 1855 and 1890. Germans still came in the largest numbers in the 1850s. Chinese immigrants began arriving on the West Coast in 1849. Toward the end of the century, Italians, Poles, Hungarians, Greeks and Jews from the Russian Empire came in growing numbers.

Mobs in the West killed Chinese immigrants and drove whole Chinese communities out of their towns from 1871 to 1887. The Chinese Exclusion Act of 1882 was the first law in the United States to bar a group of people because of their nationality, meaning the country they came from.

Germans and the Know-Nothings

More than 950,000 Germans came to the United States in the 1850s. About 220,000 of them arrived in 1854 alone.

Members of secret societies that opposed immigrants and Catholics formed a political party in the 1850s. Its official name was the American Party. Members who were asked about the group answered, "I know nothing," and the party came to be called the Know-Nothings. The Know-Nothings feared that Catholics were a danger to the Protestant faith and to democracy. In the 1854 elections in Massachusetts, Know-Nothing candidates won all but 3 of the 400 seats in the state's lawmaking body.

Bloody Monday. Know-Nothing mobs attacked immigrants in Louisville, Kentucky, on Monday, August 6, 1855. It was election day, and people call it Bloody Monday. Irish and German immigrants made up about a third of the city. In the weeks before the vote, George D. Prentice, editor of the *Louisville Journal*, wrote newspaper articles against immigrants.

Know-Nothing gangs stopped immigrants from voting. Mobs led by party members attacked the German neighborhoods east of downtown and the Irish neighborhoods to the west. They set fires and robbed homes and businesses, and they damaged more than 100 homes and businesses.

Accounts written later say the mob burned Quinn's Row, a row of houses owned by an Irishman named Quinn, and shot people as they ran from the flames. A study published in the 2020s in the Kentucky Historical Society's journal uses court statements from the 1850s to challenge many of the stories told about the riot.

The Kentucky Historical Society lists at least 22 dead. Other counts give 19 to 22, and Catholic leaders at the time put the number at well over 100. Most of the dead were Irish and German. A grand jury charged five people, and none was found guilty. No one paid the victims for their losses. The Catholic bishop, Martin John Spalding, asked Catholics to stay calm and not to take revenge.

The party's members then split over slavery. By 1860 the party had broken apart, and many of its members had joined the new Republican Party.

Carl Schurz

Who: A German revolutionary who fled after the revolutions of 1848 and 1849 failed. In the United States he became a Union general, a senator and Secretary of the Interior.

When and where: Born March 2, 1829, in the Rhine Province of Prussia, near Cologne. Settled in Watertown, Wisconsin, in 1852. Died May 14, 1906.

He fought a revolution in Prussia and became a United States senator.

Carl Schurz was one of the Forty-Eighters. He grew up near the city of Cologne, in the Rhine Province of Prussia, a kingdom in what is now Germany. Schurz joined the revolutions of 1848 and 1849. When Prussian forces crushed them, he fled to France and then to London. In 1852 he crossed the Atlantic as a refugee and settled in Watertown, Wisconsin.

In the Civil War he served as a general in the Union army, the army of the United States government, which fought the Southern states that had broken away. From 1869 to 1875 he represented Missouri in the United States Senate, as the first senator born in Germany. President Hayes then chose him as Secretary of the Interior, one of the heads of government departments who advise the president. He held that job from 1877 to 1881.

The last known slave ship

A sailing ship called the *Clotilda* carried about 110 African captives to Mobile, Alabama, around July 8, 1860. Bringing captives into the country had been against the law for 52 years. The businessman Timothy Meaher paid for the voyage, reportedly to win a bet that he could smuggle captives in. The ship's captain, William Foster, set the *Clotilda* on fire near Twelve-Mile Island. It is the last documented slave ship to reach the United States.

Kossola. One of the captives was Oluale Kossola, born in about 1841 in what is now Benin, in West Africa. Accounts say soldiers of the kingdom of Dahomey captured him in a raid and took him to the coast. In Alabama, Meaher, his brothers and some friends enslaved the captives for the next five years, until slavery ended in 1865.

Africatown. After the end of slavery the survivors could not pay for the trip home. They built their own town near Mobile and called it Africatown. The writer Zora Neale Hurston interviewed Kossola in 1927, and her book about him, *Barracoon*, came out in 2018. The wreck of the *Clotilda* was found in the Mobile River and confirmed in 2019.

Chinese immigrants in the West

Chinese immigration to the United States began with the California Gold Rush, when crowds of people went to California to look for gold. About 325 Chinese people came in 1849. Three years later, in 1852, some 20,000 came. About 40,400 arrived in all between 1851 and 1860. Most came from Guangdong province, also called Canton, in southern China. By 1880 about 105,465 Chinese people lived in the United States.

The railroad. Chinese workers built the western part of the first railroad to cross the country. Managers of the Central Pacific Railroad had about 10,000 to 12,000 Chinese workers on the job at any one time between 1865 and 1869, about nine out of every ten of the railroad's workers. Chinese workers earned about $35 a month and paid for their own food and housing. White workers got about $42 a month plus housing.

The strike. Thousands of Chinese workers went on strike in June 1867, which means they stopped work together to press for better terms. They asked for $40 a month, a 10-hour workday instead of 11 hours, and shorter shifts in the tunnels. Charles Crocker was the railroad director in charge of building. He cut off all food and supplies to their camps and brought in the sheriff and his deputies. After about eight days the workers went back to work.

Snow, explosions and disease killed Chinese railroad workers. In the winter of 1866 and 1867, 44 storms hit the Sierra Nevada mountains. At Strong's Canyon an avalanche, a mass of snow sliding down a mountainside, carried away the camp of two work crews of Chinese workers who were building tunnels there. One worker's body was not found until spring.

Workers blasted through solid granite rock with black powder, an early kind of gunpowder. From 1867 they also used nitroglycerin, a liquid explosive that can go off from a jolt. Smallpox, a disease that passes from person to person and covers the body in sores, broke out in 1868 and 1869.

The count of the dead. The railroad's managers kept no record of how many workers died. A figure of about 1,200 dead comes from one newspaper report in 1870. A reporter for the *Sacramento Reporter* wrote that a train had

brought in "the accumulated bones of perhaps 1,200" Chinese men from along the line. Chinese aid societies paid to send such bones home to China. On the same day, the *Sacramento Union* reported about 50 dead.

KNOWN:	Chinese workers died of snow, explosions and disease, and the railroad's managers did not count them.
NOT KNOWN:	The real number. Researchers at a railroad history museum think about 100 to 150 Chinese workers died building the railroad, and other estimates run from hundreds to thousands.

The Page Act and the Chinese Exclusion Act

Members of Congress passed the Page Act in 1875, the national government's first law to limit immigration. Under it, contract laborers from China, Japan and other Asian countries could not come in. These were workers brought over without their free consent, already bound to work for someone for a set time. The law called them by an insulting name used then for Asian laborers. A Chinese worker who came freely, with no such contract, could still come in.

Chinese women. Under the Page Act, no one could bring in women for prostitution, which means selling sex for money. Immigration officers at the ports decided which women those were. In practice they treated Asian women in general as suspects and kept out most Chinese women. The stated reason, stopping prostitution, was a pretext, meaning a false excuse.

What changed in 1882. President Chester A. Arthur signed the Chinese Exclusion Act, a law to shut Chinese workers out, on May 6, 1882. Under it, no Chinese laborer could come in for 10 years, whether he had a contract or not. It was the first law in the United States to bar a group of people because of their nationality or their ethnicity, the group a person is born into, with its own language and customs. Members of Congress later renewed the law and made it wider. They did not cancel it until 1943, 61 years after Arthur signed it.

BY 1882: Chinese workers had built the western part of the first railroad across the country, and lawmakers had now shut new Chinese workers out for ten years.

Attacks on Chinese immigrants, 1871 to 1887

Before and after the Exclusion Act, mobs in the West attacked Chinese immigrants. Juries and judges punished almost none of the attackers.

Los Angeles, 1871. A gunfight broke out between two groups of Chinese residents of Los Angeles on October 24, 1871. A local rancher, Robert Thompson, fired into a Chinese store. He was hit by return fire and died about an hour later. A mob of about 500 non-Chinese men then spent about two hours pulling Chinese people out of the Chinese part of town. They shot, beat and hanged them, and they robbed Chinese homes and stores.

They killed 18 Chinese men and boys, about one in ten of the city's Chinese people. Only one of the dead is thought to have taken part in the first gunfight. The mob hanged seven men from the wooden crossbar of a porch and three more from the side of a wagon.

The dead included Chee Long Tong, a doctor who healed people with herbs, and a teenager, Ah Loo. Dr. Tong offered the mob money and the diamond ring on his finger if they would spare his life. The geographer Reece Jones writes that a man in the crowd shot him in the mouth and cut off his finger to take the ring. The mob also hanged him.

A grand jury charged 25 men. By one account eight were found guilty of manslaughter, the crime of killing someone without planning to, and by another account nine. A judge sentenced them to two to six years in San Quentin prison. About a year later the judges of the California Supreme Court threw out the guilty verdicts because of a mistake in how the charges were written. None of the men was tried again.

Rock Springs, 1885. Between 100 and 150 armed white miners and railroad workers attacked the Chinese miners they worked beside in Rock Springs, in Wyoming Territory, on September 2, 1885. A territory was land the national

government ran before it became a state. Managers of the Union Pacific Railroad had hired the Chinese miners at lower pay. Most of the attackers were immigrants themselves, from Britain, Ireland, Wales and Scandinavia.

The attackers shot Chinese miners, struck them with picks, hatchets and clubs, and burned all 79 houses in the Chinese part of town. Some Chinese people burned to death in their cellars. About 600 Chinese people fled. The Wyoming State Historical Society lists 28 Chinese killed and 15 wounded. Other counts put the dead at 40 to 50, because some who fled were never found.

President Grover Cleveland sent army troops, and the railroad's managers brought the Chinese miners back to work under army guard. Officials arrested 16 men, or 22 by some accounts. A grand jury charged none of them. It said that no one had been able to testify to "a single criminal act committed by any known white person that day."

Eureka, 1885. Other mobs drove Chinese people out of their towns. In Eureka, California, in February 1885, a stray bullet from a gunfight between two Chinese groups killed a city councilman, David Kendall. That night more than 600 white residents gathered and put up a gallows, a frame for hanging people. City leaders ordered every Chinese resident to leave within 24 hours. A committee of white residents put about 480 Chinese people, nearly all of the town's Chinese residents, on two steamships to San Francisco. No one was killed.

The Chinese families left behind their furniture, goods, shrimp boats and vegetable farms. Chinatown was torn down, and the accounts do not name who did it. A Chinese aid group, the Six Companies, sued the city in 1886, and the case was dropped. Rules keeping Chinese people out stayed in Eureka's city charter, its basic set of laws, until 1959.

Tacoma and Seattle. Mayor Jacob Weisbach of Tacoma, in Washington Territory, led a year-long campaign to drive the Chinese out. He called them "a curse." Tacoma's leaders ordered all Chinese residents to leave by November 3, 1885, and about 150 left before that date. On November 3, a mob forced about

200 more Chinese residents out of their homes in the rain and onto trains out of town. The mob then burned their houses. Officials charged 27 Tacoma men, Weisbach among them, and later dropped every charge.

Mobs in Seattle forced more than 300 Chinese residents to the docks in February 1886, and ships carried about 300 of them away. Of 13 men tried for the Seattle riot, none was found guilty.

Hells Canyon, 1887. A gang of white horse thieves shot and killed as many as 34 Chinese gold miners in May 1887. The miners were camped in Hells Canyon, on the Snake River in Oregon, and the gang took their gold dust. One of the gang, Frank Vaughan, confessed that they had agreed nearly a year earlier to murder the miners. Vaughan testified for the state and was not tried. Three of the accused men fled before trial. A jury found the three who stood trial not guilty.

A stone marker placed at the site in 2012 states, "No one was held accountable." Its words are in English, in Chinese and in Nez Perce, a Native American language.

Jews from the Russian Empire, and others from southern and eastern Europe

Italians, Poles, Hungarians and Greeks came in growing numbers late in the century. Many of them were peasants, poor farm workers, who had lost their land as farming in Europe became a business. They came to earn enough money to go home and buy a piece of land. One Slavic steelworker put the plan this way: "A good job, save money, work all time, go home, sleep, no spend." Many did go home.

Jews from the Russian Empire came too, running from riots and from laws aimed at them. They came as families and planned to stay.

Revolutionaries murdered the ruler of the Russian Empire, Tsar Alexander II, in March 1881. In the towns, many people in the crowds believed a rumor that Jews had killed him. They also believed that a special order from the government allowed them to beat Jews.

Six weeks after the murder, on April 15, 1881, a fight in a tavern in the town of Elisavetgrad, in what is now Ukraine, turned into an attack on the town's Jews. It was the first of about 250 pogroms, violent riots in which a crowd attacks Jewish people and their homes, shops and synagogues, the buildings where Jews worship. The pogroms went on until 1884, in Kiev, Warsaw and many other places. Another count gives 259 pogroms from 1881 to 1883.

Who attacked. The historian John Klier describes many of the attackers as laborers who drifted from place to place looking for work. Railroad workers, factory workers and local business rivals of Jews also took part. Peasants joined in mostly to steal. The attackers robbed and wrecked Jewish homes and shops. They beat, raped and killed Jews. Rape means forcing a person into sex.

Counts of the dead differ. One count says at least 40 Jews were killed from April to December 1881. Klier counts about 50 deaths in the whole wave. About half of those dead were Jews, and about half were rioters killed while the tsar's officials put down the riots.

For many years, accounts said that officials of the tsar's government had planned the attacks. Later, the historians Hans Rogger, I. Michael Aronson and John Klier looked for evidence of that and found none. Some local officials helped stir the attacks up. Klier writes that more than a thousand rioters were punished in some way. Another account says the tsar pardoned most of the rioters who were brought to trial. The two accounts do not agree.

The May Laws. Nikolai Ignatiev, the tsar's minister in charge of the police and of other matters inside the country, proposed rules called the May Laws in May 1882. Tsar Alexander III approved them. Ignatiev blamed the pogroms on the Jews themselves. The rules were for Jews in the Pale of Settlement, the region of the Russian Empire where Jews were allowed to live.

Under the May Laws, Jews could not move to new homes outside towns. They could not buy or rent land in the countryside, and they could not trade on Sunday mornings or on Christian holidays.

More than 2 million Jews from the Russian Empire came to the United States between 1881 and 1914. Many of them were escaping the pogroms and the May Laws. Over 1.5 million of them had arrived by 1910, most of them from Russia.

The Statue of Liberty and Emma Lazarus

The Statue of Liberty, a gift from France, was opened with a ceremony in New York Harbor on October 28, 1886. Emma Lazarus wrote the poem quoted at the start of this chapter, "The New Colossus," in 1883, to help raise money for the statue's pedestal, the stone base it stands on. She wrote it one year after lawmakers shut Chinese workers out of the country.

Lazarus was born in New York in 1849, into a Sephardic Jewish family, whose ancestors were Jews of Spain and Portugal. She had worked with Jewish refugees from Russia who arrived in the 1880s. She died in 1887. People mostly forgot her poem until 1903, when her friend Georgina Schuyler had it mounted on a bronze plaque inside the pedestal. After that, people began to link the statue with the idea of America as a refuge, a safe place, for immigrants.

Castle Garden and Ellis Island

Most immigrants who landed at New York from August 1855 to April 1890 passed through Castle Garden. It stood at Castle Clinton, in Battery Park, at the southern tip of Manhattan. It was the country's first official depot for immigrants, a building where officials received newcomers, checked them and recorded them. New York State and New York City officials ran it together. About 11 million immigrants arrived through New York in those years, and about 8 million of them passed through Castle Garden.

People had complained that officials at Castle Garden were dishonest. When it closed in 1890, officials of the national government took charge of immigration. On January 1, 1892, they opened Ellis Island, the national government's immigration station on an island in New York Harbor, in its place.

Annie Moore

Who: A 17-year-old from County Cork, Ireland, the first immigrant processed at Ellis Island. For about a century, another woman's life was told as hers.

When and where: Arrived January 1, 1892, on the steamship *Nevada*. Lived on the Lower East Side of New York. Died in 1924.

First through Ellis Island, she received a $10 gold coin.

Annie Moore sailed from Ireland on the steamship *Nevada* with her two younger brothers, Anthony, who was about 15, and Philip, who was about 12. They reached New York on January 1, 1892, the day Ellis Island opened, and officials processed her before anyone else.

The wrong Annie. For about a century, writers described the wrong woman as Annie Moore. That woman went west and died in Texas, and she had been born in Illinois. In 2006 the genealogist Megan Smolenyak, a person who traces family histories through old records, offered $1,000 to anyone who could find the real Annie. Researchers who took up the challenge found her.

The real Annie Moore never left New York. She lived in a tenement, a crowded apartment building, on Monroe Street on the Lower East Side, and she married a clerk at a fish market. She died in 1924, at the age of 50. A headstone was placed on her grave in 2008. Statues of her stand at Ellis Island and at Cobh, in Ireland.

Irving Berlin

Who: A Jewish boy from the Russian Empire who arrived at Ellis Island at five and grew up to write "White Christmas" and "God Bless America."

When and where: Born in 1888 in the Russian Empire (the place is uncertain). Arrived at Ellis Island September 14, 1893. Died September 22, 1989, at 101.

He remembered his family's house in Russia burning.

Irving Berlin was born Israel Beilin in 1888 in the Russian Empire. The place of his birth is uncertain. Recent research points to Tyumen, in Siberia, and older accounts give the region around Mogilev. His family used May 11 as his birthday, but one account gives May 23.

His memory of Russia was of his family's house burning during violence against Jews. No record says who set the fire. The family arrived at Ellis Island on September 14, 1893, and in America their surname became Baline.

He wrote his songs, among them "Easter Parade," under the name Irving Berlin. He wrote a first version of "God Bless America" in 1918, and the singer Kate Smith first performed it for the public in 1938. Berlin died on September 22, 1989, at the age of 101.

Wong Kim Ark

Who: A cook born in San Francisco to Chinese immigrant parents. Officials refused to let him back into the country, and his case went to the Supreme Court.

When and where: Born 1873, San Francisco. Refused landing August 1895. Supreme Court decision March 28, 1898.

Born in San Francisco, he was refused entry to his own country.

Wong Kim Ark was born in San Francisco in 1873. His parents were Chinese immigrants, and he worked as a cook. In August 1895 he was coming home from a visit to China on the steamship *Coptic*.

John H. Wise, the official in charge of collecting taxes on goods coming into the port of San Francisco, refused to let him land. Wise acted under the Chinese exclusion laws. He said Wong was not a citizen, because his parents had not become citizens, and he held Wong on the ship.

The court order. On October 2, 1895, the national government's court in San Francisco issued a writ of habeas corpus for him, an order that the people holding him bring him before a judge and give a lawful reason for holding him. Judge William Morrow ruled that Wong was a citizen and ordered him freed.

Henry Foote, the national government's lawyer in San Francisco, had brought the case against Wong. Lawyers for the national government took it higher. The case, *United States v. Wong Kim Ark*, went on to the Supreme Court. On March 28, 1898, the justices, the judges of the Supreme Court, decided it 6 to 2 in Wong's favor. They ruled that under the Fourteenth Amendment, an addition to the Constitution, everyone born in the United States is a citizen.

BY 1900

If you remember one thing: Lawmakers shut Chinese workers out in 1882, the first time a law barred people for the country they came from, and in 1898 the Supreme Court ruled that anyone born here is a citizen.

One number: 61 years. Members of Congress kept the Chinese Exclusion Act, renewed and widened, from 1882 until they canceled it in 1943.

Next: After 1900 more newcomers came through Ellis Island than ever before, and lawmakers answered with limits on how many could come from each country.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 17 MINUTES

The great wave of newcomers peaked in the early 1900s, and then members of Congress cut it down with quota laws, laws that fixed how many people could come from each country. Officials at Ellis Island admitted 1,004,756 people in 1907 alone. Under the 1924 law, no more than about 165,000 immigrants could come each year, and almost no one from Asia could come at all. On the West Coast, officials at Angel Island held Chinese arrivals anywhere from a few days to many months.

Officials pushed somewhere between 500,000 and 2 million people of Mexican descent out of the country from 1929 to 1936. By the two larger counts, about 6 in 10 of them were US citizens. Officials in Cuba, the United States and Canada refused to let a ship full of Jewish refugees land in 1939.

Ellis Island's busiest years and the melting pot

Ellis Island's busiest year was 1907. Officials there admitted 1,004,756 immigrants that year. On the busiest single day, April 17, 1907, 11,747 people came through. About 12 million immigrants passed through Ellis Island from 1892 to 1954. Older claims of about 17 million are too high.

The melting pot. A play by Israel Zangwill called *The Melting Pot* opened in Washington, D.C., on October 5, 1908. Theodore Roosevelt watched the first night from his box and praised it. After that the phrase "melting pot," a pot in which different metals melt together into one, spread. People used the phrase to mean that newcomers from many lands would blend together into one nation.

Frank Capra

Who: A boy from Sicily who crossed to America in the cheapest part of a ship in 1903. He grew up to win three Academy Awards for directing films.

When and where: Born May 18, 1897, in Bisacquino, Sicily. Came through Ellis Island in 1903, then went on to Los Angeles.

Movie: *Frank Capra's American Dream* (1997), a documentary made for television. Kenneth Bowser directed it, and Ron Howard hosted it.

He crossed in steerage and rode a train across the whole country.

Frank Capra was born Francesco Rosario Capra in the town of Bisacquino, in Sicily, an island that belongs to Italy. His family crossed the Atlantic on the ship *Germania*. They traveled in steerage, the cheapest space on a ship, below the decks. They passed through Ellis Island and then rode a train across the whole country to Los Angeles.

As a grown man he directed films. He won his three Academy Awards as best director in the 1930s, and he made the film *It's a Wonderful Life* in 1946.

Doukenie Bacos

Who: A Greek girl who reached Ellis Island at 16. She was traveling underage and was afraid officials would send her back.

When and where: Born December 18, 1904, to a Greek family in Thrace. Sailed from Piraeus, Greece. Reached Ellis Island January 5, 1921.

She thought of jumping into the water rather than be sent home.

Doukenie Bacos was born to a Greek family in Thrace, a region of southeastern Europe now divided among Greece, Turkey and Bulgaria. She was 16 when she sailed from Piraeus, the port city of Athens, on the ship *King Alexander* on December 22, 1920. She was traveling underage, younger than an adult, to join her uncle in America.

Three days. She reached Ellis Island on January 5, 1921, and officials held her there for three days. Her uncle was a bachelor, a man who had never married. The officials would not hand a girl her age to a bachelor alone, so he had to find a family for her to live with. She was afraid they would send her back, and she later said she had thought of jumping into the water rather than be sent home.

On the third day the officials let her go. That night her uncle came with a woman from the family he had found. "This is your aunt," he told her, and Doukenie went to live with that family.

Her full name was Doukenie Babayanie Bacos. She told her story to the Ellis Island Oral History Project, which records interviews with people who came through the island, in 1991. Her interview is number 49 in its EI series. Some records give 1919 as the year she left Greece, and National Park Service staff marked that year as a mistake.

Angel Island

Angel Island Immigration Station opened in 1910 on an island in San Francisco Bay. The officials who worked there mainly enforced the Chinese Exclusion Act, an 1882 law that barred Chinese workers from coming in, and other laws against Asian immigrants. They held many Asian arrivals for days or months. The National Park Service states that some stays lasted several years, while the longest stay in other records is 22 months.

Officials refused entry to about 18 out of every 100 people who arrived at Angel Island. At Ellis Island the figure was 1 to 3 out of every 100.

Station officials kept the detainees, the people being held, locked in crowded dormitories, large rooms full of beds. No detainee could leave without a guard. Officers read every letter the detainees sent or received. No one could visit a detainee until the officials had decided the case.

The doctors. Doctors from the Public Health Service examined the arrivals. They turned people's eyelids inside out to look for trachoma, an eye infection that can cause blindness. They took samples of blood, spit, urine or stool, which means bowel movements, and they tested Asian arrivals for worms that live inside the body, such as hookworm. When officials doubted a person's age, doctors made the person strip naked. The doctors then examined the person's hair, skin, teeth, voice and genitals, the private parts of the body.

Paper sons. Chinese Americans born in the United States were citizens, as the Supreme Court had ruled in 1898. Under the exclusion laws, most other Chinese people could get in only by showing a family tie to a citizen. Some claimed a tie that was false. They were called "paper sons" and "paper daughters," because the family tie existed only on paper.

To catch them, a group of officials called a Board of Special Inquiry questioned each Chinese applicant about home villages and relatives. The board was made up of two inspectors, a translator and a stenographer, a clerk who wrote down every word. One inspector said in 1934 that "well over ninety percent" of his time went to questioning arriving Chinese people.

A few detainees killed themselves, most of them by hanging themselves in the bathrooms. The historian Judy Yung, who gave this account in 2026, said they were "driven to suicide." She gave no names and no count.

The poems. Detainees carved poems into the wooden walls of the barracks where they were held. The poems were almost lost before a park ranger, Alexander Weiss, found them in 1970, and then they were saved.

Counts of the Chinese people who passed through the station differ. One count is about 175,000 between 1910 and 1940, and the National Park Service states 250,000. A fire burned the station's main office building on August 12, 1940, and the station closed soon after.

Lee Puey You

Who: A Chinese woman who came as a "paper daughter" in 1939. Officials held her on Angel Island for about 20 months and then sent her back to China.

When and where: Arrived at Angel Island in 1939. Deported to China in November 1940. Came back in 1947 and settled in San Francisco.

Inspectors asked her about 170 questions, then sent her back.

Lee Puey You arrived at Angel Island in 1939 as a "paper daughter," and inspectors asked her about 170 questions. Officials held her there for about 20 months, one of the longest stays known for a Chinese detainee.

In November 1940 they refused to let her in and deported her, which means they forced her to leave the country. She later said she "cried a bowlful of tears."

Members of Congress canceled the Chinese Exclusion Act in 1943. Lee Puey You came back to the United States in 1947 as a war bride, a woman engaged or married to a man who served in the war. She married and settled in San Francisco. Judy Yung later wrote about her in an article called "A Bowlful of Tears Revisited."

How the quota laws of 1921 and 1924 worked

The Emergency Quota Act of 1921 was the first law to put number limits on immigration. Under it, the yearly quota for each nationality was 3 percent of the people born in that country who lived in the United States in 1910. That count came from the census, the official count of everyone living in the country.

The 1924 law. Members of Congress set much lower limits in the Immigration Act of 1924. It is also called the Johnson-Reed Act, after the two lawmakers behind it. Representative Albert Johnson of Washington State was chairman of the House committee on immigration, and David Reed was a senator from Pennsylvania. President Calvin Coolidge signed the law on May 26, 1924.

Under the 1924 law, each country's quota was 2 percent of the people born there who lived in the United States in 1890. Lawmakers picked 1890 because it came before the large wave from southern and eastern Europe. Italy's yearly quota fell from more than 42,000 to fewer than 6,000.

Asia shut out. Lawmakers also shut out anyone "ineligible for citizenship," meaning anyone who could not become a citizen under US law. At that time the law let only white people and people of African birth or descent become citizens. Members of Congress had added people of African birth or descent in 1870, five years after the Civil War, and left Asian immigrants out. In 1922 and 1923 the justices, the judges of the Supreme Court, ruled that immigrants from Japan and from India could not count as white. Under that rule, almost no one from Asia could come in.

Lawmakers capped the total at about 165,000 people a year, down from about 355,000 under the 1921 law. About 86 of every 100 quota places went to northern and western Europe, and about 9 went to southern and eastern Europe. Under those limits, the great rush of newcomers from Europe came to an end.

Coolidge had told Congress in his first message to it, in December 1923, "America must be kept American. For this purpose, it is necessary to continue a policy of restricted immigration." The State Department's Office of the

Historian states: "The most basic purpose of the 1924 Immigration Act was to preserve the ideal of U.S. homogeneity (sameness)."

Eugenics and the men behind the 1924 law

The men who shaped the 1924 law argued for it with eugenics. Eugenics was a false belief that a nation could improve its people by controlling who had children and who came into the country. Its believers ranked groups of people as "better" or "worse" by birth.

Harry Laughlin. Harry H. Laughlin ran the Eugenics Record Office at Cold Spring Harbor, New York. He testified, meaning he spoke under oath at a hearing, to Johnson's committee three times between 1920 and 1924, and Johnson named him the committee's "expert eugenics agent."

Laughlin claimed that immigrants from southern and eastern Europe, mainly Italians and Jews, had higher rates of "feeblemindedness, insanity, criminality, and dependency." Those were old words for low intelligence, mental illness, crime and needing public charity. He said that too many of the people in mental hospitals came from southern and eastern Europe. His own figures showed high rates for German and Irish immigrants as well.

Madison Grant. Madison Grant was a New York lawyer. In 1916 he wrote a book called *The Passing of the Great Race*. In it he claimed that a "Nordic race" from northwestern Europe was better than all other people. Grant helped Johnson write the 1924 law. Adolf Hitler, who later ruled Germany, wrote to Grant and called the book his "Bible."

At home. Laughlin also turned eugenics against Americans already living here. In 1922 he wrote a model law, a sample law for lawmakers to copy, for forced sterilization.

Sterilization is an operation after which a person can never have a baby. For a woman, a doctor cut or tied the two tubes that carry eggs from the ovaries to the womb, or cut out a piece of each tube, so that no egg could reach the womb again. For a man, a doctor cut the tubes that carry sperm. The operation could not be undone.

In part because of Laughlin's work, lawmakers in dozens of states passed laws that let officials force people to be sterilized against their will. By the 1960s doctors in the United States had sterilized more than 62,000 people this way, about 60 out of every 100 of them women.

Carrie Buck. Carrie Buck grew up with foster parents in Charlottesville, Virginia. In 1923 she became pregnant, by her own account because the foster parents' nephew raped her. The foster parents took the pregnancy as proof that she was "feebleminded" and had her sent to a Virginia state colony for people called feebleminded. The colony's head, Dr. Albert Priddy, picked her for a court case that would test whether Virginia's new sterilization law was allowed.

On May 2, 1927, in the case Buck v. Bell, the justices of the Supreme Court ruled 8 to 1 that Virginia officials could sterilize her against her will. On October 19, 1927, Dr. John H. Bell cut out a piece of each of her tubes.

In Germany. In 1933 the Nazis, Hitler's party, which then ruled Germany, passed a forced sterilization law based in part on the American practice. Under it, special courts of a judge and two doctors decided each case, and an approved doctor did the operation. About 400,000 people in Nazi Germany were sterilized this way. In 1936 leaders of the University of Heidelberg, in Germany, gave Laughlin an honorary degree, a degree given as an honor instead of for study.

Mexican arrivals and the Mexican Repatriation

The Mexican Revolution was a war in Mexico that lasted from 1910 to 1920. During those years about 350,000 people fled north across the border into the United States. More kept coming in the 1920s.

The Repatriation. Officials pushed people of Mexican descent out of the United States from 1929 to 1936, in what is called the Mexican Repatriation, from a word for sending people back to their own country. By the larger counts, most of the people sent to Mexico had been born in the United States, so they were being sent out of their own country. It happened during the Great Depression, the years when many people in the United States had no work.

Secretary of Labor William N. Doak, who served under President Herbert Hoover, led a national campaign to remove Mexicans. He claimed they were taking jobs from other workers. In Los Angeles, Charles P. Visel, head of the city's unemployment relief committee, which handled public money for people without work, planned a publicity campaign.

Officials raided Mexican American neighborhoods and public places. County welfare officers, the officials who handed out relief money, told families that their relief money would stop unless they took free train tickets to Mexico. Officials of the Los Angeles County welfare office ran the trains, and the first one left on March 23, 1931.

Hospital staff carried patients on stretchers to trucks bound for the border. Families lost their property. In a 2005 California law, state lawmakers wrote that this property "often was sold by local authorities" as "payment" for the cost of removing the families.

La Placita. Immigration agents blocked the exits of La Placita, a park in Los Angeles, on February 26, 1931, while about 400 people were inside. Accounts of what happened next differ. According to the California Migration Museum, the agents arrested about a dozen people and deported a few. According to a History.com article, they rounded up more than 400 people and deported them.

Counts of the people removed run from about 500,000 to about 2 million. The California Migration Museum states about 1 million, following the historians Francisco Balderrama and Raymond Rodríguez. By that count, about 60 of every 100 were citizens born in the United States. The 2005 California law states about 2 million, and about 1.2 million of them had been born in the United States.

Officials pushed about 35,000 people out of Los Angeles, about a third of the Mexican people living there. California lawmakers apologized for the removals in that 2005 law. In February 2012 the members of the Los Angeles County Board of Supervisors, who run the county, apologized and put up a plaque, a metal sign with words on it, at LA Plaza. No apology from the federal government, the national government in Washington, is on record.

The braceros. US and Mexican officials started the Bracero Program in August 1942. Under it, Mexican farmworkers came to the United States on contracts to work for a set time and then go home. Bracero is a Spanish word for a man who works with his arms, from *brazo*, the word for arm.

The St. Louis and the first refugee law

The German ship *St. Louis* left Hamburg for Havana, Cuba, on May 13, 1939, with 937 passengers. Almost all of them were Jewish refugees. Cuba's president, Federico Laredo Brú, canceled their papers to land, and Cuban officials let only 28 passengers off at Havana. The passengers asked President Franklin D. Roosevelt for help, and he never answered them. State Department officials told them they must "await their turns on the waiting list." Canadian officials refused them too.

Why they fled. The passengers were fleeing attacks on Jews that had grown worse in 1938. The Nazi government of Germany had taken over Austria in March 1938. On the night of November 9 and 10, 1938, Nazi leaders set off attacks on Jews across Germany. That night is called Kristallnacht, the Night of Broken Glass.

Nazis burned more than 1,400 synagogues, the buildings where Jews worship, wrecked thousands of Jewish-owned shops and broke into Jewish homes. German police locked about 26,000 Jewish men in concentration camps, prison camps for people the Nazis targeted, just because they were Jewish. Afterward Nazi officials seized property that Jews owned.

The ship turned back to Europe on June 6. The passengers went to four countries: 288 to Britain, 224 to France, 214 to Belgium and 181 to the Netherlands.

The German army invaded France, Belgium and the Netherlands in May 1940. Before the invasion, 87 of the passengers in those countries had moved on to other countries. Nazis killed 254 of those who were still there in the Holocaust, their murder of Jews and other people during World War II. The other 278 survived.

The first refugee law. Members of Congress passed the Displaced Persons Act in 1948, the first refugee law in the United States. Displaced persons were Europeans who had lost their homes in the war. Under the law, 200,000

people could come at first, counted against the yearly quotas, and in 1950 lawmakers raised the total. About 350,000 to 400,000 European refugees settled in the United States between 1948 and 1952. One count gives 393,542.

BY 1950

If you remember one thing: Lawmakers cut the great wave from Europe with quota laws and shut out almost everyone from Asia, and only after World War II did they pass a law for refugees.

In 1907: officials at Ellis Island admitted 1,004,756 immigrants in a single year.

After 1924: no more than about 165,000 could come in a year, and about 86 of every 100 places went to northern and western Europe.

Next: After 1950 federal officers rounded up Mexicans in the hundreds of thousands, and lawmakers ended the quotas.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 13 MINUTES

Members of Congress threw out the country-by-country quotas in 1965, and after that most newcomers came from Asia, Latin America, Africa and the Caribbean instead of Europe. Federal officers rounded up Mexicans and Mexican Americans in 1954 and sent them to Mexico, and many of the people they sent were US citizens.

Refugees came from Cuba after 1959 and from Vietnam, Laos and Cambodia after 1975. Counts credited to the United Nations refugee agency give at least 200,000 Vietnamese who died at sea while escaping by boat, from storms, hunger, disease and attacks by pirates. About 2.7 million people who had lived in the country for years without legal permission got legal status under a 1986 law.

The braceros and the 1954 deportations

The Bracero Program lasted until 1964. Over those years, Mexican farmworkers signed about 4.5 million short-term contracts to work in the United States. The workers came for the length of a contract and were expected to go home afterward.

Federal officers carried out a mass deportation of Mexicans in 1954, in a federal operation whose official name used an insult for Mexican people. The insult was aimed at Mexicans who crossed the border without papers, and it suggested that they had swum across the Rio Grande, the river along the border between Texas and Mexico.

Who ran it. Attorney General Herbert Brownell Jr., the head of the US Justice Department under President Dwight D. Eisenhower, pushed for the operation. General Joseph M. Swing ran it. He was the commissioner, or head, of the Immigration and Naturalization Service (INS), the federal office in

charge of immigration. Swing announced the operation on June 9, 1954. INS officials planned it with officials of the Mexican government. Officers started in California and Arizona and moved into Texas in the middle of July.

The roundups. Border Patrol officers, the federal agents who guard the border, ran the roundups in a military style. Newspapers covered them heavily, and officials meant the coverage to scare people into leaving on their own. On the first day in Texas, July 15, 1954, officers caught 4,800 people. That July officers held 42,000 people in Texas, and more than 63,000 left on their own.

Officers sent people south by truck, bus, train and plane. From Port Isabel, Texas, they also put people on two ships, the *Emancipation* and the *Mercurio*, bound for Veracruz, Mexico. About a quarter of the people removed from Texas went by ship. A History.com article states that the ships were later compared to slave ships.

Seven deportees jumped from the *Mercurio*. The *Handbook of Texas* says a mutiny followed, meaning the people on board rose up against the crew. Other accounts say the seven drowned. After protests in Mexico, officials stopped sending people by ship.

The heat. The historian Mae M. Ngai reports that in July 1955 officers rounded up workers in 112-degree heat, and as a result 88 braceros, Mexican farm workers hired under contract, died of sunstroke, an illness in which the body gets so hot that it can no longer cool itself. A labor official told Ngai that more would have died without help from the Red Cross.

Ngai also quotes a Mexican labor leader. He reported that deportees were carried into Mexico on trucks "like cows" and unloaded in the desert, fifteen miles down the highway from the border. US immigration officials also left deportees in Mexicali, a Mexican border city where the summer heat averages 108 degrees. Other deportees died of disease and other causes while officers held them.

NOT KNOWN: No record gives a full count of the dead, says whether the 88 were left in the desert, names the officers who ran that roundup, or shows that any official was charged for the deaths.

The count. INS officials claimed that as many as 1,300,000 people had left, counting people who, officials said, had fled out of fear. The historian Kelly Lytle Hernández puts the true number deported at "likely closer to 300,000."

The official numbers rested on an INS report of nearly 1.1 million people caught. Katrina Burgess, a professor at Tufts University, wrote in 2024 that this figure covered a whole government year, which ended in June 1954. The operation was announced on June 9, 1954, so most of those people had been caught before it began. Many of the arrests were also the same people caught again, sometimes on the same day, Burgess wrote.

Many of the people deported were US citizens of Mexican descent. No one knows how many.

Antonio García

Who: A Mexican farmworker who joined the Bracero Program at seventeen, after having his military ID card changed to say he was eighteen.

When and where: Born June 13, 1937, in Tepehuanes, Durango, Mexico. Worked on contracts in Texas, California, Colorado and New Mexico into the early 1960s. Told his story in Chicago in 2003.

At seventeen he had his ID changed to say eighteen.

Antonio García was born in Tepehuanes, in the Mexican state of Durango. He grew up in Nuevo Ideal and left school after fourth grade. With his changed ID card, he signed up at the contracting centers in Chihuahua and in Empalme, Sonora, where Mexican workers signed contracts to work in the United States.

His first contract was for work in the cotton fields of Texas, and within a week he was seriously ill. He kept working on contracts, on and off, into the early 1960s, in California, Colorado, New Mexico and Texas.

Claudia Morales Haro recorded his life story in Chicago on March 12, 2003. The recording is item 162 in the Bracero History Archive, which contains more than 700 such interviews. His record does not say whether he settled in the United States for good.

What the 1965 law changed

With the Immigration and Nationality Act of 1965, also called the Hart-Celler Act, members of Congress ended the national-origins quotas, the rules that had limited immigration country by country since the 1920s. Under the new law, people got in mainly through family ties to people already here or through their job skills. Lawmakers also ended the bars against Asians and against southern and eastern Europeans.

For the first time, lawmakers also set a limit on the number of immigrants from the Western Hemisphere, the half of the world that includes North and South America. That limit included Mexico. Over the following decades, the main sources of immigration shifted away from Europe, toward Asia, Latin America, Africa and the Caribbean.

Cubans and the Mariel boatlift

Fidel Castro led a revolution in Cuba in 1959. Through the 1960s, many Cubans came to the United States, especially to Florida.

Mariel. Castro briefly let people leave in 1980. From April to October that year, about 125,000 Cubans crossed by boat from the port of Mariel to Florida. This was the Mariel boatlift.

About a month before the boatlift began, members of Congress passed the Refugee Act of 1980. With it, lawmakers created a permanent system for admitting refugees. They defined a refugee the same way the United Nations does: a person outside their own country who cannot go back because of a real fear of being harmed for their race, religion, nationality, group or political beliefs.

Reinaldo Arenas

Who: A Cuban novelist whom Castro's officials jailed for his writing and for being gay, and who left Cuba in the Mariel boatlift.

When and where: Born July 16, 1943, near Holguín, Cuba. Left through the port of Mariel in May 1980. Lived in New York. Died December 7, 1990.

Movie: *Before Night Falls* (2000), a drama based on his life story. Julian Schnabel directed it. Javier Bardem played Arenas, a role that earned an Oscar nomination.

He tried to escape Cuba floating on a tire inner tube.

Cuban officials persecuted Reinaldo Arenas, which means they punished him again and again, for what he wrote and for being gay, which means loving people of the same sex. In 1973, according to records at Princeton University Library, they charged him with "ideological deviation," holding ideas that Cuba's rulers banned, and with publishing his books in other countries without permission. A court convicted him on both. Another account says the court also called him "extravagant," which here means living too freely, and "immoral."

A charge he denied. The Mexican writer Rafael Lemus writes that in 1973 officials also accused Arenas and a friend of abusing two young people who were not yet adults. Arenas denied it, and none of these accounts says it was ever proved. Lemus says Arenas hid for four months to avoid arrest and was held from January 1974.

Arenas escaped from prison and tried to leave Cuba by floating away on a tire inner tube. Cuban officers caught him and locked him in El Morro, an old fort in Havana used as a prison. Officers of State Security, the political police of Fidel Castro's government, moved him to their headquarters, called Villa Marista.

Villa Marista. Lemus writes that at Villa Marista officers sent steam into the prisoners' closed cell until it became a steam room. There officers forced Arenas to sign a paper saying he was sorry for being gay and for his

books. He was released in 1976, after he signed a promise not to write again. No record names any of the officers who held him.

In May 1980 he left Cuba through the port of Mariel, one of about 125,000 Cubans in the boatlift. In the United States he lived in New York, and there he wrote the story of his own life, called *Before Night Falls*.

He became sick with AIDS, a disease caused by a virus that destroys the body's defenses against other illnesses. On December 7, 1990, at the age of 47, he killed himself. His life story came out in 1992, two years after his death, and in English in 1993.

Refugees from Southeast Asia and the boat people

North Vietnamese soldiers took Saigon, the capital of South Vietnam, in 1975, at the end of the Vietnam War. After that, refugees fled Vietnam, Laos and Cambodia. They included the Hmong, an ethnic group from the mountains of Laos. Many Vietnamese escaped in small boats, and they became known as the boat people.

Storms, hunger, disease and pirates killed many of them at sea. Counts credited to the United Nations refugee agency run from 200,000 to 400,000 dead. The journalist Barry Wain estimated 30,000 dead by 1981. No one knows the exact number.

The pirates. According to a UN refugee official in 1986 and a later study, the pirates were Thai fishermen in the Gulf of Thailand. Robert C. McCabe, a researcher at Maynooth University in Ireland, writes that some fishermen turned to piracy after big commercial fishing boats had cut the catch in their villages. No record names a single pirate.

Pirates robbed and raped refugees, murdered some and set others adrift. Rape means forcing a person into sex. In 1981 alone, the UN refugee agency counted 452 boats that reached Thailand with 15,479 refugees. Pirates had attacked 349 of those boats, on average three times each. They kidnapped 228 women, and 881 people were dead or missing.

Officials of the UN refugee agency and of the Thai government started a program against the pirates in June 1982. The governments of twelve countries paid for it, and the US government gave $1.2 million of the $3.67 million. By October 1987, 96 suspects had been arrested. No record names the officers who arrested them. Judges gave sentences of two to fifty years in prison, and one pirate was sentenced to death in December 1986.

US officials accepted 402,382 Vietnamese refugees from camps in Southeast Asia and Hong Kong from 1975 to 1997, more than any other country. A different count gives nearly 281,000 Vietnamese refugees settled in the United States from 1981 to 1990 alone. About 500,000 more Vietnamese came to the United States by 1994 through the Orderly Departure Program, a legal way to leave Vietnam without the boat trip.

Tung Trinh

Who: One of 52 survivors of a refugee boat from Vietnam that drifted at sea for 37 days in 1988.

When and where: Left Vietnam by boat in 1988. Brought ashore at Bolinao, in the Philippines. Later settled in Orange County, California.

Movie: *Bolinao 52* (2008), a documentary directed by Duc Nguyen. It won two regional Emmy awards in 2009.

More than 20 ships passed the boat, and one stopped, then left.

Tung Trinh was one of 110 people on a small boat that left Vietnam in 1988. The boat's engine stopped working, and the boat drifted at sea for 37 days.

More than 20 ships passed the boat during those days. One of them, the USS Dubuque, a ship of the United States Navy, stopped on June 10, 1988. The ship's commanding officer ordered food, water and sea charts given to the people on the boat. The sailors guessed 50 to 70 people were on the boat, not 110. Then the Dubuque left, and the boat stayed adrift.

As many as 58 of the 110 people on board died. An estimated 20 had died before the Dubuque came, and the others died after it left. During the drift, some of the people on the boat ate human flesh to stay alive, which

is called cannibalism.

The captain. Navy leaders removed the Dubuque's commanding officer from his command on August 11, 1988. At his own request, he was tried by a military court. Its officers found him guilty of breaking orders to help people in danger at sea, and he received a letter of reprimand, a written scolding.

Filipino fishermen found the boat and brought the 52 survivors to Bolinao, in the Philippines. Tung Trinh later came to the United States and settled in Orange County, California.

Sergey Brin

Who: A Jewish boy from Moscow whose family left the Soviet Union in 1979. In 1998 he started Google with Larry Page.
When and where: Born 1973 in Moscow. Arrived in the United States October 25, 1979, at six.

He left Moscow at six and later started Google with Larry Page.

Sergey Brin was born in 1973 in Moscow, the capital of the Soviet Union, a communist country where the government owned the businesses and only one party was allowed. His family was Jewish. In the Soviet Union his parents were held back in their careers and harassed because they were Jews. The accounts of his life do not name the people who did it.

The family got permission to leave in 1979. About 51,000 Soviet Jews left that year, the most in any single year. In the United States his father, Michael, taught mathematics at the University of Maryland. His mother, Eugenia, became a researcher at NASA, the US space agency. In 1998 Sergey and Larry Page started the company Google.

The 1986 law

President Ronald Reagan signed the Immigration Reform and Control Act, called IRCA, on November 6, 1986. About 2.7 million people who had lived in the country for years without legal permission got legal status under it. To qualify, they had to show they had lived in the country without a break since before 1982. Certain farmworkers could also qualify.

Under the same law, officials could for the first time punish employers who knowingly hired workers without legal permission. These punishments are called employer sanctions. After 1986 the number of people living in the country without legal permission grew again.

BY 2000

If you remember one thing: After lawmakers ended the quotas in 1965, the main sources of newcomers shifted from Europe to Asia, Latin America, Africa and the Caribbean.

The question this era leaves open: Which countries would send the most people once the quotas were gone? The 2023 count in the next era gives the answer.

Next: After 2000 the share of people born abroad climbed to its highest since 1910, and new federal agencies took over immigration.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 18 MINUTES

The share of people here who were born in another country rose almost to its highest ever, and federal officers held more people in immigration detention than ever before. About 47.8 million people living in the United States in 2023 had been born in another country, by the Census Bureau's count. That was 14.3 percent of the population, the highest share since 1910. More of them came from Mexico than from any other country.

Attorney General Jeff Sessions announced a "zero tolerance" policy in April 2018, during President Donald Trump's first term: every adult who crossed the border illegally was to be charged with a crime. Border Patrol agents then took thousands of children from their parents. The US Border Patrol counted 895 migrant deaths along the border with Mexico in the year ending September 2022, the most on record. In January 2026 immigration officers held more than 71,000 people in detention, the most ever, according to a report by two human rights groups.

Who comes now

About 11 million of the immigrants living here in 2023 were born in Mexico, 22 of every 100, Pew Research Center researchers found. Immigration from Mexico has slowed since about 2007.

India came next with 3.2 million, then China with 3 million, the Philippines with 2.1 million and Cuba with 1.7 million. Counted by region, about 52 of every 100 immigrants were born in Latin America and 27 of every 100 in Asia.

The number of immigrants from Africa south of the Sahara Desert roughly tripled from 2000 to about 2.5 million in 2024, the Migration Policy Institute found. Communities of people born in Jamaica, Haiti, the Dominican Republic and other Caribbean countries also grew.

The share of people born abroad, 14.3 percent in 2023 by the Census Bureau's count, came close to the record of 14.8 percent, set in 1890. The lowest share was 4.7 percent, in 1970.

Pew and the Department of Homeland Security both publish estimates of how many people live in the country without legal permission. Their estimates reached about 12 million in 2007, then fell to about 11 million, and rose again in the 2020s. The estimates are uncertain.

Homeland Security and DACA

Hijackers from the group al-Qaeda flew planes into the World Trade Center in New York and the Pentagon near Washington on September 11, 2001, and a fourth plane crashed in Pennsylvania. They killed 2,977 people.

A new department. After those attacks, members of Congress passed the Homeland Security Act of 2002. With it, lawmakers created the Department of Homeland Security, called DHS, which began work in 2003. They ended the old Immigration and Naturalization Service and split its work among three new agencies inside DHS.

Officers of Immigration and Customs Enforcement, called ICE, arrest and hold immigrants inside the country. Officers of Customs and Border Protection, called CBP, work at the borders, and the Border Patrol is part of CBP. Staff of the third agency, US Citizenship and Immigration Services, took over the government's other immigration work, such as handling requests to become a citizen.

DACA. Janet Napolitano, the Secretary of Homeland Security, created DACA, short for Deferred Action for Childhood Arrivals, on June 15, 2012. President Barack Obama spoke about it at the White House that day. Under DACA, some people who were brought to the country as children can apply to have their removal from the country deferred, or put off, and to work legally.

Lawyers and judges have fought over DACA in court again and again. In September 2017, in President Donald Trump's first term, Sessions advised ending it, and Elaine Duke, the acting Secretary of Homeland Security, ended it. On June 18, 2020, the justices of the Supreme Court ruled that the way officials had ended DACA broke federal law, so DACA stayed.

A federal appeals court ruled on January 17, 2025, that its work permits were unlawful and limited that ruling to Texas. About 455,000 people held DACA status on March 31, 2026, according to the National Immigration Forum, an immigrant rights group, using figures from US Citizenship and Immigration Services.

Deaths at the border

Every year hundreds of migrants die trying to cross from Mexico into the United States. People die of heat and thirst in the desert, drown in the Rio Grande, or die during or after chases by Border Patrol agents. Some die in custody, meaning while officers are holding them. Investigators inside CBP counted 171 deaths in the 2022 budget year in which CBP officers were involved "in some way," and 52 of those people died in CBP custody.

Two counts. Border Patrol officials and the staff of the International Organization for Migration (IOM), a United Nations agency, count these deaths in different ways. For 2022, Border Patrol officials counted 895 deaths along the border in the federal budget year, which runs from October to September. IOM staff counted 686 deaths and disappearances there in 2022. They call their figures minimums, because Texas county coroners, the officials who look into deaths, and Mexican search teams do not report every death to them.

Auditors at the Government Accountability Office, which checks federal agencies for Congress, found in 2022 that Border Patrol officials had left out deaths found by sheriffs, by medical examiners, the doctors who find out why a person died, and by other agencies. The auditors later reported that Border Patrol officials had made the changes they asked for. In a report of April 2025, the same auditors gave 704 deaths and more than 5,800 rescues for 2023.

Border Patrol officials' yearly counts were 281 in 2018, 300 in 2019, 254 in 2020, 568 in 2021 and 895 in 2022. The 2022 count is the highest on record, more than three times the count for 2018.

IOM staff called the US-Mexico border "the world's deadliest migration land route." Of the 686 people they counted there in 2022, 307 died crossing the Sonoran and Chihuahuan deserts, most of them from heat.

For 2025, IOM staff counted 409 migrant deaths in all of North and South America, the lowest count since 2014. They linked the drop to fewer people trying to cross the US-Mexico border and the Darién, a jungle between Colombia and Panama. In February 2026 they said the 2025 figures would not be final until the middle of 2026.

Children taken from their parents, 2018

Zero tolerance. Attorney General Jeff Sessions, the head of the Justice Department, announced a "zero tolerance" policy on April 6, 2018. He told federal prosecutors along the southwest border to charge with a crime every adult whom DHS officers sent to them for crossing the border illegally. That included parents who came with their children. Officers sent the parents to jail to face criminal charges, and Border Patrol agents took their children away from them.

The memo. Kirstjen Nielsen, the Secretary of Homeland Security, approved a plan on May 4, 2018, to send every adult caught crossing illegally to be charged with a crime, parents included. Sessions had urged her to approve it. The plan came from a memo of April 23, 2018, signed by the heads of three agencies inside DHS: Kevin K. McAleenan of CBP, Thomas Homan, the acting head of ICE, and L. Francis Cissna of US Citizenship and Immigration Services. It covered adults "initially arriving or apprehended (caught) with minors (children)."

Border Patrol leaders told their commanders along the Mexican border to begin on May 5. McAleenan left out parents with children under five who were crossing illegally for the first time. Agents asked a federal refugee office to take each child, and ICE officers held the parents. In a later court case, government lawyers wrote that there was "no prescribed specific course of action for reunification." That means no one had set a plan for putting the families back together.

Sessions said in a speech on May 7, 2018, "If you are smuggling a child, then we will prosecute you and that child will be separated from you as required by law." Officials had already separated about 280 families in 2017, under a trial program in El Paso, Texas.

What the inspector general found. Michael Horowitz, the Justice Department's inspector general, an official who checks the work of a government department, later investigated. In January 2021 Horowitz reported that Sessions and a small number of other Justice Department

officials knew, when the policy was issued in April 2018, that DHS "would change its long-standing practice" and separate families. He found that the Attorney General's office "was a driving force" in that change.

He also wrote that the department's "single-minded focus on increasing immigration prosecutions came at the expense of careful and appropriate consideration of the impact of child separations."

The count of the children. Officials deported hundreds of parents without their children before a federal judge ordered the government to put the families back together in June 2018. The counts of the children differ because each one includes a different group of children.

Taken. President Joe Biden set up the Family Reunification Task Force in February 2021. Its staff counted 3,924 children taken from their parents between January 20, 2017, and January 20, 2021.

Back together, and still apart. By February 1, 2023, 2,926 of those children had been reunited with their families. Most were reunited before the task force began, and more than 600 through its work. The other 998 were still apart, and task force staff said the true number still apart was "likely greater than 1,000."

Covered by the lawsuit. A federal judge approved a settlement, an agreement that ends a lawsuit, on December 11, 2023, in a lawsuit brought by the American Civil Liberties Union called *Ms. L. v. ICE*. It covers an estimated 4,500 to 5,000 children and their parents. By March 20, 2024, task force staff had confirmed 1,120 separated children as part of that group. No final count of children returned to their parents is on record. As of September 2026, no record shows that any official had been charged for the separations.

Arrests, detention and deportation, 2025 and 2026

Chief Justice John G. Roberts Jr. swore in Donald Trump as president for a second term on January 20, 2025, in the Rotunda, the round central hall of the US Capitol. A year later, in January 2026, ICE officers held more than 71,000 people in detention, in custody while officials decided whether to deport them. It was a record.

Human Rights Watch and Physicians for Human Rights gave that figure in a report of June 25, 2026. ABC News reported on July 21, 2026, that ICE officers held 65,765 people, citing figures ICE had posted.

Deaths in custody. The same report counted 52 people who died in ICE custody from January 20, 2025, to June 4, 2026. Thirty-nine of them died in the first year. Of those 39, seven were apparent suicides, meaning they appeared to have killed themselves, and five died of injuries. The others died of strokes, heart failure, long-term illness and failure to breathe. The yearly death rate reached about 8.4 for every 10,000 people held, which the report describes as "nearly four times" the rate under the previous president, Joe Biden.

The report lists how each of the five died of injuries. One was Silverio Villegas González, whom an ICE officer shot during an arrest near Chicago. His death is told below, with the other shootings. On September 24, 2025, a gunman fired on a van of detainees at the ICE office in Dallas, Texas. Two of the men in the van died of their wounds. Norlan Guzmán-Fuentes of El Salvador died that day, and Miguel Ángel García-Hernández of Mexico died on September 30. The gunman, Joshua Jahn, 29, shot and killed himself, NBC News reported.

José Castro-Rivera of Honduras, 24, ran into traffic in Virginia on October 23 while ICE officers chased him to arrest him. A vehicle hit and killed him, and the report does not name the driver.

Camp East Montana. Geraldo Lunas Campos, 55, from Cuba, died on January 3, 2026, inside Camp East Montana, an ICE camp at Fort Bliss in El Paso, Texas. The El Paso County medical examiner ruled his death a homicide,

the El Paso TV station KFOX14 reported. On a death record, homicide means that another person caused the death. The ruling is not a finding that anyone broke the law.

The medical examiner found that pressure on his neck and torso, the middle of his body, kept him from breathing. The human rights groups' report states that the pressure came from force that ICE staff used on him. In a later death report of their own, ICE officials said he had tried to harm himself and that staff used force to stop him. The report names none of the staff involved.

Removals. Removal is the government's word for deportation. ABC News reported that ICE had posted 356,389 removals in the 2026 budget year through about July 21, 2026. For the 2025 budget year, the immigration researcher Austin Kocher counted about 329,018 ICE removals.

DHS officials claimed on January 20, 2026, "more than 675,000 deportations" and "an estimated 2.2 million self-deportations" in the year since January 20, 2025. Self-deportation means leaving the country on one's own, without being deported by officers. Kocher notes that DHS officials have not published how they count these numbers. He found that ICE's own removals made up only about 54 percent of an earlier DHS claim of 605,000. In the year after January 20, 2025, DHS officials counted 90,084 people caught along the southwest border.

Shootings. Federal immigration officers shot and killed at least four people from January 2025 to late January 2026, PBS NewsHour reported. Three of them were Silverio Villegas González, Renée Nicole Good and Alex Pretti.

Villegas González was a citizen of Mexico. ICE officers stopped his car in Franklin Park, a suburb of Chicago, on September 12, 2025, just after he had dropped his children at school. One officer shot him in the neck, and he died.

DHS officials said that Villegas González drove his car at officers and dragged one of them, and that the officer fired "fearing for his own life." In body-camera video, an agent described his own injuries as "nothing major." Federal officials have not released the name of the officer who shot him, and federal prosecutors declined to charge that officer.

Investigators of the Illinois State Police opened their own case in spring 2026, at the request of the Franklin Park police. It was still open in September 2026, the Chicago Sun-Times and WBEZ reported.

ICE officer Jonathan Ross shot Renée Nicole Good, a US citizen, in Minneapolis on January 7, 2026. CBP officers shot Alex Pretti in Minneapolis on January 24, 2026. As of the PBS NewsHour report, no officer had been charged in either death.

Giannis Antetokounmpo

Who: A basketball player born in Greece to Nigerian parents. He grew up a citizen of no country and came to the United States when a National Basketball Association (NBA) team picked him in 2013.

When and where: Born December 6, 1994, in Athens, Greece. Came to the United States in 2013, at 18, to play for the Milwaukee Bucks.

Movie: *Rise* (2022), a Disney+ drama about the Antetokounmpo family, made with the family. It is a dramatization, not a documentary.

As boys in Greece, he and his brothers sold goods on the street.

Giannis Antetokounmpo's parents, Charles and Veronica, were immigrants from Nigeria with no legal status in Greece. Giannis was born in Athens and grew up stateless, which means no country counted him as its citizen. Under Greek law, a child does not become a citizen just by being born in Greece. In the United States, the 1898 ruling in Wong Kim Ark's case makes any child born in the country a citizen. As boys, Giannis and his brothers sold goods on the street.

Greek officials made him a citizen in 2013. A few weeks later, in June 2013, the managers of the Milwaukee Bucks picked him 15th in the NBA draft, the yearly event in which teams choose new players, and he came to the United States at 18 to play for them. He was named the league's Most Valuable Player in 2019 and in 2020. In 2021 he won the NBA championship with the Bucks and was named the most valuable player of the finals. His parents and brothers joined him in the United States.

Peter Dut and Santino Chuor, Lost Boys of Sudan

Who: Two Dinka teenagers whose parents were killed when government forces attacked their villages in Sudan's civil war. They lived for years in a refugee camp in Kenya. They were among the "Lost Boys" resettled in the United States.

When and where: From Sudan. Kakuma refugee camp, Kenya. Houston, Texas, in their first year in the United States, during the resettlement that began in 2001.

Movie: *Lost Boys of Sudan* (2003), a documentary that records the lives of these two young men. Megan Mylan and Jon Shenk made it. It was nominated for an Emmy and won the Truer Than Fiction Award at the Independent Spirit Awards.

They walked out of Sudan and spent years in a camp in Kenya.

Peter Nyarol Dut and Santino Majok Chuor were Dinka, an ethnic group of southern Sudan. Sudan's civil war was a war inside one country, between the government in the north and rebels in the south. Military forces of the central government attacked the boys' villages, destroyed them and killed their families, according to the notes for the PBS film about them. Those notes do not state which soldiers or fighters killed each boy's parents.

The militias. Researchers at Human Rights Watch report that in these years Sudan's rulers gave modern weapons to militias, armed groups outside the regular army, called the muraheleen. The militia fighters burned Dinka villages, stole cattle and food, and carried off women and children as slaves. Human Rights Watch does not say that these militias attacked the two boys' own villages.

The rebel army of the south, the Sudan People's Liberation Army, also took many of the older Lost Boys as soldiers. The two boys survived a long walk out of Sudan and then spent years in Kakuma, a refugee camp in Kenya.

Starting in 2001, about 4,000 of these young refugees from Sudan were resettled in the United States. People called them the "Lost Boys of Sudan," and Peter and Santino were among them. The documentary records their first year in the United States, which began in Houston. After a time, Peter left Houston for Kansas to go to high school.

THE THREE QUESTIONS, ANSWERED

1. **Why did people leave home to come here?** Many came for land and work, like the Germans who farmed the Midwest and the Chinese who built the western railroad, or to worship in their own way, like the Puritans. Others ran from danger: the Irish famine, the pogroms against Jews in the Russian Empire, the Nazis, and wars in Vietnam and Sudan.
2. **Who did Americans try to keep out, and how?** Lawmakers shut out Chinese workers in 1882 and almost everyone from Asia in 1924, and they cut the number of southern and eastern Europeans with quotas. Mobs burned Catholic churches in Philadelphia and drove Chinese people out of Eureka and Tacoma. Officials pushed hundreds of thousands of people of Mexican descent, many of them citizens, across the border in the 1930s and again in 1954. In 2018 agents took thousands of children from their parents at the border, and in January 2026 officers held more than 71,000 people in detention, the most ever.
3. **Who could become a citizen, and how did that change?** In 1790 only "a free white person" could become a citizen. In 1870 members of Congress added people of African birth or descent. In 1898 the Supreme Court ruled in Wong Kim Ark's case that anyone born here is a citizen, and in 1952 members of Congress removed the limit to white people.

What changed: In 1500 no one from across the ocean had come to stay, and by 2023 about 47.8 million people living in the United States had been born in another country.

FIVE DATES TO KEEP

September 8, 1565: Menéndez founds St. Augustine, the oldest European town on the US mainland.

1790: Members of Congress limit citizenship to free white people.

May 6, 1882: President Arthur signs the Chinese Exclusion Act.

May 26, 1924: President Coolidge signs the quota law.

1965: lawmakers end the national-origins quotas.

> **NOT KNOWN:** No one knows how many Chinese workers died building the railroad, and the estimates run from about 100 to the thousands.

Where to go: Ellis Island, in New York Harbor, where about 12 million immigrants came through from 1892 to 1954 and a statue of Annie Moore stands.

Next: Chapter 5, Migration Across America. It asks how people moved, and were moved, across the land once they were here.

Chapter 5: Migration Across America

CHAPTER 5 OF 37 · MIGRATION ACROSS AMERICA · 10 ERAS · ABOUT 75 MINUTES

Many of the biggest moves across this land were forced or driven. Soldiers marched whole nations west, and slave traders and enslavers forced more than a million enslaved people south. Later, about 6 million Black Southerners went north to get away from mob killings and from Jim Crow laws, the laws that kept Black and white people apart.

In about 1805, in Maryland, a slave trader from Georgia locked an iron collar around the neck of a man named Charles Ball. A padlock fastened the collar to a chain about 100 feet long. Fifty-one other people were locked to the same chain, and the trader handcuffed the men in pairs. He had just bought Ball away from his wife and children. The line of people walked south for more than a month, to Columbia, South Carolina.

This chapter follows people moving across the land that became the United States, from the first walkers out of Asia to the families who left New Orleans after Hurricane Katrina in 2005. Some went because they chose to: a church walking to the Connecticut River, wagon families on the Oregon Trail, homesteaders on the Great Plains. Others were driven out: the Acadians, the Tuscarora, the Cherokee, the Navajo, Japanese Americans in 1942. In each move, watch who decided that people would go, and whose land they went to. Before 1900 that land almost always belonged to a Native nation.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did horses reach the Native nations of the Plains?

2. How many Cherokee died when soldiers forced them west?
3. Why did about 6 million Black Americans leave the South between 1910 and 1970?

Five minutes? Read "Indian removal and the Trail of Tears" in 1800 to 1850, the Rebecca Neugin story after it, and the next section, "The overland trails."

Can skip: "Seasonal rounds and trade paths", "The cattle drives", "The frontier line disappears from the census, 1890". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 2 MINUTES

People were moving across this land for thousands of years before any European saw it. The first people came on foot from Asia during the ice age, and their descendants kept moving. Some built towns and later left them. Some nations traveled south over a long time, and many families moved with the seasons to hunt, fish and gather food.

Hundreds of Native nations lived here. There were no horses and no wagons. Trails and trade paths crossed the continent, and later newcomers followed many of them.

Cahokia and Chaco Canyon: towns people built and left

Cahokia was the largest town north of Mexico. It stood on the Mississippi River, on the Illinois side, near where the city of St. Louis is today. People lived there from about the year 700 to about 1400. Around 1100, between 10,000 and 20,000 people lived in the city, about as many as lived in London at the time. Some estimates put as many as 50,000 people in the area around it.

Most of Cahokia's people had moved away by about 1350, and by the 1400s the town was nearly empty. Nobody knows for sure why they left. The main explanations are floods, a shortage of wood and food, and drought, a long stretch with little or no rain. No single cause has been settled.

Chaco Canyon, in what is now New Mexico, was a center of the Ancestral Puebloan people, the ancestors of today's Pueblo peoples. People started to leave the canyon around 1140, after a run of bad droughts, and most had gone by the end of the 1100s. They moved out and joined the Pueblo communities of the Southwest.

The Navajo and Apache move south

The ancestors of the Navajo and the Apache came from the far north, from what is now western Canada and eastern Alaska. The Navajo call themselves the Diné. Over a long time, groups of them moved south along the edge of the Rocky Mountains and the Great Plains toward the Southwest. Their long move is called the Athabascan migration.

When they arrived in the Southwest is disputed. Many accounts place their arrival between the 1400s and about 1525. Others argue that they came earlier, in the 1200s.

Seasonal rounds and trade paths

Many peoples moved in a pattern called a seasonal round, a yearly circuit from one camp to the next. Over a year, a family moved between its fishing places, its hunting grounds and the spots where wild plants ripened.

Trade goods moved across the land as well. Traders passed turquoise, seashells, copper and obsidian, a black glass made by volcanoes, from nation to nation along trade paths. Some of these goods ended up hundreds of miles from where they were found.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 5 MINUTES

Spanish colonists brought two things north in the 1500s that later changed how people moved across this land: horses, and families who came with wagons and herds to stay. The whole land from coast to coast was still Native land. Europeans reached its edges, but no crowds of settlers moved across the interior. The first colony that came over land to stay arrived at the very end of the century, when Juan de Oñate led Spanish colonists north into what is now New Mexico in 1598.

Why there were no horses here, and how they came back

Horses lived in North America during the ice age and died out here thousands of years ago. Spanish sailors and colonists brought them back.

Christopher Columbus carried horses on his second voyage across the Atlantic, in 1493. Spanish expeditions, groups of men sent out to explore, then brought horses onto the mainland. Francisco Vázquez de Coronado's expedition had them from 1540 to 1542, and Juan de Oñate's colonists brought them to New Mexico in 1598. In the next century, horses changed how people lived on the Great Plains.

Oñate's colony on the Camino Real, 1598

Juan de Oñate and his colonists left Santa Bárbara in January 1598. It was a town in northern New Spain, the Spanish colony that included Mexico. Earlier Spanish expeditions had followed the Conchos River north. Oñate's column, the long line of people, wagons and animals, cut straight across the Chihuahuan Desert instead.

The column reached the Rio Grande in late April, at El Paso del Norte, near where El Paso, Texas, and Ciudad Juárez, Mexico, stand today. On April 30, 1598, Oñate claimed New Mexico for Spain in a ceremony called La Toma, which means "the taking." In July he set up the colony's headquarters at San Juan Pueblo, which its people call Ohkay Owingeh.

Whose land. The colonists moved onto the homeland of the Pueblo peoples. Families from Chaco Canyon had joined the Pueblo communities after the droughts of the 1100s, about 400 years earlier.

The road the colonists traveled was later called El Camino Real de Tierra Adentro, the royal road of the interior. It ran from Mexico City to northern New Mexico, and Oñate's march made it more than 600 miles longer. Its stretch inside the United States is a National Historic Trail today.

Two colonies by sea. Before Oñate, two groups of Spanish colonists had come to what is now the United States by ship. Tristán de Luna y Arellano brought about 1,500 people in 11 ships to Pensacola Bay, in what is now Florida, in 1559. They were soldiers, colonists, enslaved people and Aztec allies.

A hurricane sank many of the ships a month later, and Spanish ships took the survivors back to Mexico in 1561. Pedro Menéndez de Avilés founded St. Augustine, also in Florida, with about 600 settlers in 1565.

Juan de Oñate

Who: The Spanish leader of the first known colonists to travel over land, with families, wagons and herds, into what is now the United States, and the man who ordered the attack on Acoma Pueblo.

When and where: From Santa Bárbara in northern New Spain to San Juan Pueblo (Ohkay Owingeh), New Mexico, January to July 1598. Acoma Pueblo, December 1598 to February 1599.

He sentenced each Acoma man over 25 to lose a foot.

Oñate's column had about 400 men, about 130 of them with their families, and 83 wagons and carts. The colonists drove more than 7,000 head of livestock, or farm animals, with horses among them. The colony included soldiers, farmers, workers, servants and Franciscan friars, men who belong to a Catholic religious order. Enslaved people walked in the column too. The surviving records do not say how many.

Acoma. Oñate's nephew, Juan de Zaldívar, went to Acoma Pueblo in December 1598 with about thirty soldiers to demand food. In the Acoma people's own account, the soldiers also raped an Acoma woman, forcing her into sex. The Acoma killed Zaldívar and between eleven and fourteen of his men.

Oñate sent Zaldívar's brother, Vicente de Zaldívar, with about seventy soldiers in January 1599. Acoma stands on top of a mesa, a steep hill with a flat top. The soldiers climbed it, killed Acoma men, women and children, and burned much of the town.

Nobody counted the Acoma dead at the time. Estimates run from about 300 to about 1,500. The number given most often is about 800, which Pueblo leaders gave in a 2023 statement from the All Pueblo Council of Governors, the governors who lead the 20 Pueblo nations of New Mexico and Texas.

The sentences. The soldiers captured about 500 people. In February 1599 Oñate sentenced them in four groups. Each man over 25 was to have a foot cut off and do twenty years of forced work as a slave. Boys and men

from 12 to 25 got twenty years as slaves. Girls and women over 12 got the same twenty years. Children under 12 were to be taken from their families and given to the friars.

The Acoma's own account, passed down since 1599, gives the number of men whose feet were cut off as 24, and so do news reports. Reporters for AP and NPR, and Pueblo leaders, say it was the right foot. Two historians, Marc Simmons and John Kessell, doubt that the feet were ever cut off. Simmons had once written that they were.

A National Park Service page on the Camino Real states that about 70 Acoma girls were sent down the road to convents, homes for nuns, in Mexico. The same page states that probably none of the girls ever came home. Pueblo leaders count 60 children taken to Mexico, and they say none came home.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

Native nations of the Plains and the Rocky Mountains got horses in this century, and that changed how they moved more than anything else did. People on horseback could travel farther and faster than on foot, and they could follow the buffalo herds.

Along the Atlantic coast, English, Dutch and other colonists settled at places like Jamestown, founded in 1607, and Plymouth, founded in 1620. Almost none of them went far inland. The Appalachian Mountains were hard to cross, and many Native nations lived on the far side of them. The few colonists who did move inland went short distances. Thomas Hooker's group walked from Massachusetts to the Connecticut River in about two weeks.

How did horses spread to the Plains nations?

Two accounts explain how horses spread, and the timing is still not settled. In the older account, horses passed north through trade after the Pueblo Revolt of 1680. In the newer one, many nations already had horses by the early 1600s.

The older account. In this account, Pueblo peoples drove the Spanish out of New Mexico in 1680, and the Spanish left their horses behind. The trouble began with the Spanish governor of New Mexico, Juan Francisco Treviño. In 1675 he ordered the arrest of 47 Pueblo medicine men, who were religious leaders, on a charge of sorcery, or witchcraft. Spanish officials sentenced four of them to be hanged. They hanged three, and the fourth killed himself. Treviño's men whipped the others in public and put them in prison.

Pueblo people went to Santa Fe and demanded that the prisoners be let go, and the Spanish freed them. One of the freed men was Popé, a religious leader of the Tewa, one of the Pueblo peoples. In 1680 Popé led the Pueblo peoples of what is now New Mexico in a rising against Spanish rule. They killed about 400 Spaniards and drove the rest out of New Mexico for more than a decade.

The horses the Spanish left behind, perhaps more than 1,500, were the largest number of horses to pass into Native hands at one time. In the older account, these horses then moved north and east through trade. The Ute, Navajo and Apache got them first. Over the next few decades horses reached many other nations, as far as the northern Plains.

The newer account. It comes from a study published in the journal *Science* in 2023. The archaeologist William Taylor, a scientist who studies the past through the things people left behind, led it, working with Lakota, Comanche, Pawnee and other Native partners. The team tested horse bones from Kansas, New Mexico and Wyoming to find their age. Some dated from the 1500s and 1600s.

Marks on the teeth and bones showed that people had put bridles on these horses and fed them corn, so people were keeping and caring for them. The team found that many nations across the Plains and the Rocky Mountains had

horses by the early 1600s, decades before the Pueblo Revolt. The finding matches what Native people have long said in their nations' own histories, passed down by word of mouth.

Over the next century, horses changed how the Comanche, Lakota, Cheyenne and the other Plains nations hunted and traveled.

The first colonial moves inland

Two English ministers moved out of the colony of Massachusetts Bay in 1636 and started new towns in what are now Connecticut and Rhode Island. Thomas Hooker chose to go, and his story follows this section.

Roger Williams. The other minister did not choose. The leaders of Massachusetts ordered Roger Williams out of the colony for what he preached: he taught that the government should not force religious beliefs on people. He fled in the winter rather than be sent back to England, and Wampanoag people helped him survive.

With help from Wampanoag and Narragansett people, Williams then traveled by canoe to a freshwater spring at a place called Moshassuck. He settled there with the permission of the Narragansett and named the place Providence. In 1638 two Narragansett sachems, or leaders, named Miantonomi and Canonicus signed a deed, a paper that says who owns a piece of land, for the settlement.

Thomas Hooker

Who: A minister who led about 100 members of his church from Massachusetts to the Connecticut River, where they settled Hartford.

When and where: From Newtown (now Cambridge), Massachusetts, to Hartford, Connecticut, May and June 1636.

His church walked for two weeks, living partly on milk from 160 cows.

Thomas Hooker was the pastor of the church at Newtown, the town that is now Cambridge, Massachusetts. His church set out on foot for the Connecticut River in May 1636.

They drove 160 cattle with them and lived partly on the cows' milk. They followed Native trails, a route later called the Old Connecticut Path. John Winthrop, a leader of Massachusetts Bay, wrote about the departure in his journal on May 31, 1636: "Mr. Hooker, pastor of the church of Newtown, and most of his congregation (the people of his church), went to Connecticut." Winthrop also noted that Hooker's wife rode on a horse litter, a stretcher carried by horses.

Whose land. The place where they settled was Suckiaug, which means "black earth." It was the main village of the River Indians, and their sachem was Sequassen. In 1636 Sequassen and his people signed the land over by deed to Samuel Stone and William Goodwin, two men in Hooker's group. The land ran along the river between the towns of Wethersfield and Windsor, and six miles west of the river.

Nobody wrote the deed into the official land records at the time, and the settlers and the Native owners argued over its terms for generations. The histories of the deed do not say what Sequassen's people received for the land.

Dutch traders had built a fort there, the House of Hope, before Hooker's group came. The town the settlers built is Hartford. In 1639 the people of Hartford and the towns near it adopted a plan of government called the

> Fundamental Orders of Connecticut.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

Settler families began moving inland from the Atlantic coast in the first half of the 1700s, because land near the coast had filled up and grown expensive. They moved into the valleys and hills behind the older settlements, an area colonists called the backcountry. The main road inland was the Great Wagon Road.

Native nations were moving too. After a war with colonists in North Carolina, the Tuscarora walked north to New York. On the southern Plains, the Comanche moved in on horseback and took control, and many Apache there fled south. The country beyond the Appalachian Mountains was still Native land.

Who traveled the Great Wagon Road?

Many of the families on the Great Wagon Road were German or Scots-Irish. The road ran about 730 miles into the backcountry. It started in Philadelphia and went southwest through the Shenandoah Valley of Virginia. Then it ran down through the hill country of the Carolinas, called the Piedmont, toward Augusta, Georgia. Heavy traffic began in the 1720s.

Both groups had come to Pennsylvania. By the time of the Wagon Road, land there had grown costly, so many families moved on. The Scots-Irish were Protestants, Christians whose churches had split from the Catholic Church. Their families had moved from Scotland to the north of Ireland in the 1600s and then on to America. The Germans were called the "Pennsylvania Dutch," from *Deutsch*, the German word for German. Moravians, members of a Protestant church from central Europe, came along the road as well.

An older path. The road followed a Native path that colonists called the "Indian Road." When settlers began arriving in the Shenandoah Valley in about 1729, most of the people living near the present-day town of Winchester, Virginia, were Shawnee. Other nations used the valley too, among them the Iroquois, a league of six nations also called the Six Nations or the Haudenosaunee. The valley's north-south trail was a war path between nations to the north and the Catawba nation to the south.

The Treaty of Lancaster. Officials sent by Virginia's governor, William Gooch, met Iroquois leaders at Lancaster, Pennsylvania, in 1744. In the treaty they signed there, a written agreement between nations, Gooch's officials bought the Iroquois claim to the valley, meaning their right to the land. Colonists also got the right to use the Indian Road. Accounts of how much the officials paid do not agree.

The Native delegates at Lancaster came from five nations of the Iroquois league, the West Virginia Encyclopedia states. None came from the Shawnee, who lived in the valley. Even more settlers traveled the road after 1744.

Why did the Tuscarora walk north?

The Tuscarora nation moved about 500 miles north, from North Carolina toward Pennsylvania and New York, after colonists defeated them in war. The move took from 1713 to 1722, and they traveled in stages.

Neoheroka. The war ended at Neoheroka, a Tuscarora fort on Contentnea Creek in what is now Greene County, North Carolina. Colonel James Moore of South Carolina led the attack. He commanded militia, colonists called up to fight as soldiers, from both Carolinas, and Native warriors who fought on the colonists' side.

Moore's men laid siege to the fort, which means they surrounded it and kept attacking it, from March 1, 1713. They dug under an outer wall and set the fort on fire. It fell on March 23.

History pages from North Carolina's state government list 950 Native people killed or captured at Neoheroka. They list about 1,000 Tuscarora sold into slavery during the whole war. The colonists and allies who won the war sold these captives, and those state pages do not name who bought them. The colonists' side also forced more than 3,000 other Native people from their homes. About 200 colonists and about 1,000 Native people died in the war.

About 1,500 Tuscarora survivors made the walk north. The Oneida nation sponsored them in the Haudenosaunee league, which means the Oneida backed their joining. In 1722 the Tuscarora became its sixth nation.

How the Comanche took the southern Plains

The Comanche had moved onto the southern Plains by the early 1700s, and they became the strongest power there. The land they held became known as Comanchería, Comanche country.

Apache groups were already living on the southern Plains. The Handbook of Texas, a history reference, states that by 1700 the Comanche, Wichita and Tejas Indians were "better armed through trade with the French." Those nations took control of the southern Plains, and the Apache began moving southwest. Many Apache, most of all the Lipan and the Mescalero, "fled southward into Central Texas as well as into northern Mexico."

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 7 MINUTES

Crowds of American settlers crossed the Appalachian Mountains for the first time in these fifty years, onto land that belonged to Native nations. As many as 300,000 went through the Cumberland Gap into Kentucky and beyond between 1775 and about 1810. After the Revolutionary War ended in 1783, more settlers moved into the Ohio Valley. Members of Congress passed two laws, in 1785 and 1787, that set out how western land would be measured, sold and governed.

The Acadians. Farther north, British officers began forcing the French-speaking Acadians out of Nova Scotia, north of the British colonies, in 1755. They forced out about 10,000 to 11,500 people, and about 2,600 to 3,000 of them later made their way to Louisiana.

The Acadians forced from Nova Scotia, 1755

The Acadians were French-speaking Catholic farm families in Nova Scotia and the land around it, in what is now eastern Canada. Between 14,000 and 18,000 Acadians lived there in 1755. Nova Scotia was a British colony, and Charles Lawrence, its lieutenant governor, was one of the top British officials there.

The decision. Lawrence and his council of advisers in Halifax decided on July 28, 1755, to remove the Acadians from the colony. A navy admiral and the colony's top judge sat on the council that day. Some council members had recently come from New England and wanted the Acadians' farmland.

Lawrence had first asked Acadian delegates, men chosen to speak for their villages, to swear loyalty to Britain with no conditions. When they refused, Lawrence and his council had them jailed.

Grand-Pré. Lieutenant Colonel John Winslow commanded New England troops. On September 5, 1755, he ordered every Acadian male aged 10 and older at Grand-Pré to gather in the church there at 3 p.m., and he held them there. The order read to them said that their land, houses, cattle and other livestock now belonged to the Crown, meaning the British king. They could keep their money and household goods, and they would be removed from the colony.

Winslow wrote that the order was "very disagreeable to my nature," but that his job was "to obey such orders as I receive." Two other officers led arrests and removals in other districts.

The plan. Charles Morris, a New Englander, drew up the plan. Soldiers would surround the churches on a Sunday morning and capture as many men as they could. Then they would break the earth walls that kept the sea off the Acadians' fields, and burn their houses and crops.

New England and British soldiers did burn many Acadian houses, barns and churches that year. Around the Minas Basin, where Grand-Pré stands, Winslow's own list records 698 buildings that the soldiers under him burned by November 1755, most of them houses and barns, and one Catholic church.

The ships. When the men refused to go, the soldiers threatened their families with bayonets, blades fixed to the ends of their guns. By the fall of 1755, soldiers had put about 1,100 Acadians on ships bound for three colonies, South Carolina, Georgia and Pennsylvania, the historian James H. Marsh writes. That count includes only the people on ships for those three colonies. Winslow's own table records everyone shipped from the Minas district alone, to all places: 2,242 people, 1,510 of them by his men and 732 by a second company.

British officers and soldiers kept removing Acadians until 1764. One count gives about 10,000 Acadians forced out between 1755 and 1763, and another gives about 11,500 between 1755 and 1764. British ships carried them to the thirteen colonies, to England and to France. Many Acadians escaped into the forests, and British soldiers hunted them for the next five years. In French the forced removal is called *Le Grand Dérangement*, the great upheaval.

KNOWN:	In one group of about 3,100 Acadians removed in 1758, about 1,649 died by drowning or disease.
DISPUTED:	One count puts all the Acadian dead at close to 5,000, from disease, shipwreck or violence, and other counts put the dead at as many as half of those forced out.

How did the Acadians become Cajuns?

About 2,600 to 3,000 Acadians made their way to Louisiana between 1764 and 1785, and their descendants are the Cajuns. The word Cajun comes from the word Acadian.

Louisiana was then a Spanish colony. The largest group came in 1785, when Spanish officials paid for seven ships that carried 1,596 Acadians from Nantes, in France, to Louisiana. The first of them reached New Orleans on July 29, 1785. The Acadians settled along the bayous, slow-moving streams through low, wet land, and on the prairies west of New Orleans.

Joseph Broussard (Beausoleil)

Who: An Acadian who fought the British removal of his people, then led one of the first Acadian groups to Louisiana.

When and where: Acadia (Nova Scotia and New Brunswick), 1755 to 1762. New Orleans and the Attakapas district of Louisiana, 1765. Born about 1702, died 1765.

Freed from a British prison, he led about 200 Acadians to Louisiana.

Joseph Broussard, whom people called Beausoleil, organized armed resistance when British officers began forcing the Acadians out in 1755. He and his men raided British troops. British soldiers captured him and held him at Halifax until 1763. That year Britain and France signed the Treaty of Paris, a peace agreement that ended their war.

South. Broussard and about 600 Acadians hired a ship in late 1764 to carry them to Saint-Domingue, the island colony that is now Haiti. He and about 193 to 200 of them sailed on to Louisiana in February 1765 and reached New Orleans late that month. He was one of the first Acadians to settle there.

> Colonial officials named him a militia captain on April 8, 1765. They also made him the commander of the "Acadians of the Attakapas," the district around the present-day town of St. Martinville.
>
> A sickness spread through the new settlement that year. Most accounts say it was yellow fever, and Broussard died of it in October 1765. Officials at Parks Canada, the Canadian government office that runs historic sites, have named him a National Historic Person, a title they give to people who were important in Canada's history.

The Wilderness Road and the Cumberland Gap

The Cumberland Gap is a low notch through the mountains near the place where Kentucky, Virginia and Tennessee meet. In 1775 Daniel Boone and about 30 axmen cut a road through it called the Wilderness Road. As many as 300,000 settlers traveled that road into Kentucky and beyond between 1775 and about 1810. Kentucky filled with settlers and became a state in 1792.

The deed. Boone cut the road for Richard Henderson's Transylvania Company, a business its partners formed to buy up what they called "wild lands" and open them to settlers. At Sycamore Shoals on the Watauga River, from March 14 to 17, 1775, Henderson got a deed from Cherokee leaders, among them Attakullakulla, Oconostota and Old Tassel.

The deed covered about 20 million acres between the Cumberland and Ohio rivers, most of Kentucky and part of Tennessee. In return the Cherokee received trade goods, which one estimate values at about 10,000 British pounds.

Who objected. Dragging Canoe, Attakullakulla's son, spoke against selling land that the Cherokee nation held in common, and he rejected the sale. Other nations had claims to the land too, among them the Shawnee, who had once lived on part of it. The Tennessee Encyclopedia describes the Cherokee claim as the strongest of these.

By buying the land, Henderson broke a British rule of 1763, the Royal Proclamation, which banned private buyers from buying Native land. Officials in Virginia and North Carolina and members of the Continental Congress, the meeting of leaders from the colonies, all refused to accept Henderson's colony as lawful.

Into the Ohio Valley, and a grid laid over the land

Settlers moved in large numbers into the Ohio Valley and the Northwest Territory after the Revolutionary War ended in 1783. The Northwest Territory was the land north of the Ohio River that later became the states of Ohio, Indiana, Illinois, Michigan and Wisconsin. It was the land of the Shawnee, Miami, Delaware and other nations, and in the 1790s those nations and the settlers fought each other.

The grid. Members of Congress adopted the Land Ordinance, a law, on May 20, 1785. Under it, surveyors, workers who measure land, marked the western land into squares called townships, each six miles on a side, and divided each township into 36 sections of 640 acres. In the law, the members of Congress set aside one section in every township to pay for public schools. Because of that grid, much of the American countryside is laid out in squares today.

Two years later, on July 13, 1787, members of Congress passed the Northwest Ordinance. In it they wrote the rules for governing the territory, set out how parts of it could become new states equal to the original thirteen, and banned slavery north of the Ohio River. Five states later came from the territory.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 14 MINUTES

Two kinds of journey crossed the country in these fifty years: families who chose to go west in wagons, and whole nations and enslaved people whom soldiers and slave traders forced to move. The United States about doubled in size in 1803, and by 1850 its land reached the Pacific Ocean. Families drove wagons over the Oregon, California and Mormon trails, and about 300,000 people rushed to California for gold between 1848 and 1855.

Forced journeys. United States soldiers drove the Cherokee, Choctaw, Muscogee, Chickasaw and Seminole nations from their homelands in the Southeast. Slave traders and enslavers forced more than a million enslaved Black people to move from the Upper South to the Deep South, in chains or in ships.

The Louisiana Purchase, 1803

American officials bought about 827,000 square miles of land west of the Mississippi River from France in 1803. The price was 15 million dollars, and afterward the country was about twice as big as before. All or part of 15 later states came from this land.

Thousands of Native people, in dozens of nations, already lived on this land. None of them had a part in the sale. After the purchase, American settlers began moving onto it, into the interior of the West.

The domestic slave trade: a forced march south

Slave traders and enslavers forced more than one million enslaved people to move from the Upper South to the Deep South between about 1790 and 1860. The Upper South meant Virginia, Maryland, the Carolinas, and later Kentucky and Tennessee. The Deep South meant the cotton lands from Georgia west to Texas.

Traders and enslavers moved about 200,000 people this way every ten years from 1820 to 1860. This trade inside the country was later called the "Second Middle Passage," after the first Middle Passage, the voyage of captured Africans across the Atlantic Ocean.

Traders moved people south in two ways. They marched some on foot in coffles, lines of enslaved people chained together and marched over land. They shipped others down the coast to New Orleans, which had the largest slave market in the country.

Franklin and Armfield. The largest slave-trading company in the country was Franklin and Armfield, founded in Alexandria, Virginia, in February 1828. John Armfield bought people in Alexandria. Isaac Franklin sold them in Natchez, Mississippi, and after 1829 in New Orleans. Rice Ballard joined them in the early 1830s and ran their office in Richmond. The partners bought, moved and sold more than 8,500 people, 1,000 to 1,500 a year in their busiest years, the mid-1830s.

Near the end of each summer, the partners sent coffles "in chains over the mountains." Each coffle held from several dozen to several hundred people, and the partners timed it to reach the Deep South for the fall selling season. They also owned three ships fitted to carry enslaved people, the *Tribune*, the *Uncas* and the *Isaac Franklin*. The voyage from Virginia to Louisiana took about three weeks, and at the busiest time one of these ships left every two weeks. The last shipload left in November 1836.

Sold, or taken along. Some of these people were sold to traders. Others were forced to move when the enslavers who held them moved to new land and took them along. Historians' estimates of the traders' share run from

about 16 percent to about two-thirds. Walter Johnson puts it near the top, at about two-thirds sold and one-third taken along, and Michael Tadman gives 60 to 70 percent for the years after 1808.

Charles Ball

Who: An enslaved man who was chained into a coffle and marched from Maryland to South Carolina. He later escaped and told his life story.

When and where: Sold in Maryland about 1805. Marched to Columbia, South Carolina. His story was printed in Philadelphia in 1837.

He escaped slavery twice, hiding by day and walking at night.

The chapter opening describes how a Georgia slave trader chained Charles Ball into a coffle of 52 people in Maryland in about 1805 and marched them for more than a month to Columbia, South Carolina. At Columbia the trader sold Ball again, into Georgia.

Escape. Years later Ball escaped and walked north back to Maryland, hiding by day and moving at night. He was captured again and held as a slave. After his second escape he settled near Philadelphia.

Ball told his story to Isaac Fisher, a white abolitionist, someone who worked to end slavery. Fisher wrote it down, and it came out in 1837 as *Slavery in the United States: A Narrative of the Life and Adventures of Charles Ball*. Because Fisher did the writing, the book records Ball's story in Fisher's words. Many of the dates in Ball's life are known only roughly, even his birth, in about 1780 in Calvert County, Maryland.

Indian removal and the Trail of Tears

President Andrew Jackson signed the Indian Removal Act in 1830. He then directed the removal of the Cherokee, Muscogee (Creek), Choctaw, Chickasaw and Seminole nations from their homelands in the Southeast. Federal troops forced them to Indian Territory, which is now Oklahoma.

The Choctaw. Federal troops removed the Choctaw from 1831 to 1833, forcing the first groups out in snow and river ice. Andrew K. Frank, writing for the Oklahoma Historical Society, counts more than 14,000 Choctaw removed and says hundreds died. Other counts give about 15,000 removed and about 2,500 dead.

The Creek. After a war with the Creek, United States troops marched the remaining Creek people west in five large groups in August and September of 1836. Soldiers put chains on Creek men taken prisoner in the war and marched them to Montgomery, Alabama, and their families followed in wagons. More than 14,500 Creek were removed. Several hundred died on the way. About 3,200 more died after they arrived, from disease, hunger and exposure, the harm cold and bad weather do to the body.

The steamboat *Monmouth*, carrying about 600 Creek people, was in a collision in October 1837. About half of the Creek on board died.

The Chickasaw. Under the Treaty of Pontotoc, signed in 1832, the Chickasaw had to leave their lands in Mississippi. They moved west from 1837, and more than 500 Chickasaw died of smallpox during the removal.

The Seminole. Many Seminole in Florida fought removal in a war that lasted from 1835 to 1842. For every four Seminole sent west, US Army soldiers killed one, the National Park Service states. A few hundred Seminole stayed in the Everglades, the swamps of southern Florida, and never surrendered.

The Cherokee. The Cherokee removal of 1838 and 1839 is called the Trail of Tears. United States Army troops under General Winfield Scott began rounding up the Cherokee who had not moved west in May 1838. President

Martin Van Buren had sent Scott with about 7,000 soldiers, and state militia, part-time soldiers called up by a state, joined them, Georgia's among them. Soldiers took families from their homes and farms, often with no time to pack.

The soldiers held about 16,000 Cherokee in stockades, pens with walls of tall logs, in Tennessee and Alabama. They sent three groups west by river in June 1838. After that, Scott agreed to let John Ross, the Cherokee's Principal Chief, organize the rest. Ross organized fourteen groups, which left between August and December 1838. The walk over land was about 800 miles, and people died on the way from cold, hunger and disease.

How many died. The number of Cherokee dead is not settled. The count most often given is about 4,000. The National Park Service's history of the trail states that more than 1,000 people died on the journey itself, and "perhaps several thousand" more died later because they had been forced from their homes.

Higher estimates run from above 5,000, in an article on the History.com website, to about 8,000 or more, twice the usual figure. Russell Thornton, a scholar who studies population, gave that highest estimate in 1984. Andrew K. Frank, writing for the Oklahoma Historical Society, says as many as one in four Cherokee died.

Rebecca Neugin

Who: A Cherokee woman who, at about three years old, was forced west on the Trail of Tears. She told what she remembered when she was 97 or 98.

When and where: Soldiers took her from Georgia in 1838. Interviewed in 1932 in what is now Oklahoma.

She rode west in a wagon at three and remembered it at 97.

Soldiers rounded up Rebecca Neugin's family in Georgia in 1838. Her father was Tickaneesky, and her mother was Sallie Ketcher. The soldiers held the family in a camp in Tennessee.

The family started west in September 1838 with the group led by Mose Daniel and George Still. Rebecca and the smaller children rode in a wagon. Her mother, her father and her older brother Dick walked the whole way.

The historian Grant Foreman interviewed her in 1932, when she was 97 or 98 years old and one of the last living people who had made the journey. Her memory of it is one of the few first-hand accounts by a survivor. She died on July 15, 1932.

The overland trails

Most of the western trails started from towns on the Missouri River, mainly Independence and St. Joseph, in Missouri.

The Santa Fe Trail opened in 1821, when William Becknell led the first trading trip from Missouri to Santa Fe, which then belonged to Mexico. It was mainly a road for traders hauling freight.

The Oregon Trail ran about 2,000 to 2,170 miles, from Independence, Missouri, to the Willamette Valley in Oregon, and the trip took four to six months. About 1,000 people set out on it on May 22, 1843, with about 120 wagons and thousands of oxen and cattle, and they reached Oregon that fall. This trip was called the "Great Migration" of 1843. It is a different event from the Great Migration of Black Southerners in the 1900s, which this chapter describes later.

After it, other families knew that wagon trains could make the whole journey, and hundreds of thousands followed over the next twenty years. From the 1840s on, tens of thousands of them came to the Willamette Valley, where at least twenty Native nations lived, among them the Kalapuya, Molala and Chinook peoples.

The California Trail split off from the Oregon Trail and turned southwest toward California. Traffic on it grew fast after gold was found in 1848.

How did people die on the wagon trails?

About one in ten people who set out on the wagon trails died on the way, between 20,000 and 30,000 people. Most of the dead had cholera, a disease that causes severe diarrhea and vomiting, or typhoid, a disease that causes a high fever. Both spread through dirty water at crowded camps.

Other emigrants, people leaving one home to settle in another, died in accidents or drowned while crossing rivers. Later stories often showed Native people attacking wagon trains, but most of the emigrants who died on the trails died of disease. Families traveled in groups for safety, in covered wagons pulled by oxen.

Why the Latter-day Saints moved west

Members of the Church of Jesus Christ of Latter-day Saints, also called Mormons, moved west after armed men attacked them in Missouri and in Illinois.

Missouri. Missouri's governor, Lilburn Boggs, ordered on October 27, 1838, that the Latter-day Saints be driven out of the state. If necessary, in the order's own word, they were to be "exterminated," killed off entirely. Three days later about 200 armed Missourians led by Colonel Thomas Jennings attacked the Mormon settlement at Haun's Mill. They killed seventeen or eighteen men and boys. One of the boys, Sardius Smith, was ten years old. No record shows that anyone was ever tried for the killings.

Both sides in Missouri used force that year. The Missouri State Archives records that mobs of older Missouri settlers and armed Mormon groups roamed the countryside. Armed Mormon groups raided the towns of Gallatin and Millport in mid-October 1838 and burned property there. Over the next year, about 8,000 church members left Missouri for Illinois, many of them ragged and stripped of their property, and settled at Nauvoo.

Illinois. About 100 men from Illinois militia units broke into the jail at Carthage, Illinois, on June 27, 1844. They killed the church's founder, Joseph Smith, and his brother Hyrum. Five men were tried for the killings, and all five

were found not guilty. Vigilantes, people who use force to punish others without any legal right to, then attacked Mormon farms around Nauvoo.

Brigham Young, a leader of the church, planned to move about 15,000 people west. The first wagons crossed the Mississippi River into Iowa on February 4, 1846, and by March 1 about 500 wagons were on the move. More than 70,000 Latter-day Saints had traveled the Mormon Trail by 1869. Some of the poorer groups pulled handcarts, small two-wheeled carts pulled by hand, instead of driving wagons.

Whose land. The Salt Lake Valley lay between the lands of the Ute, to the south, and the Shoshone, to the north. A few Native people crossed the valley itself, but none had camps there. The Ute, Goshute, Shoshone, Paiute and Navajo peoples lived in what is now Utah, and their homelands had been there for thousands of years.

Brigham Young

Who: The Latter-day Saint leader who led the first company of Mormon emigrants to the Salt Lake Valley.

When and where: From Winter Quarters, in what is now Nebraska, to the Salt Lake Valley, in what is now Utah. Arrived July 1847.

Three of the men in his first company were enslaved.

The Saints called Young's first company the vanguard company, the group that goes ahead of the rest. About 148 people traveled in it in 1847, about 1,300 miles from Winter Quarters, a camp in what is now Nebraska, to the Salt Lake Valley.

Counts of who was in the company differ. One count lists 143 men, 3 women and 2 children. Three of the men were enslaved Black men: Hark Lay, Green Flake and Oscar Crosby.

The three men. William Lay and his wife, Sytha, held Hark Lay in slavery and took him along. James Madison Flake held Green Flake in slavery, and William Crosby held Oscar Crosby, according to the Century

of Black Mormons project at the University of Utah. The two enslavers sent Green Flake and Oscar Crosby ahead to build shelters and plant crops for their white families, who came later.

The three enslaved men hunted for the travelers, guarded the animals, and went into the Salt Lake Valley on July 22, 1847, ahead of most of the company. The rest reached the valley that July, and Young himself arrived on July 24.

New land from Mexico, and the Gold Rush

A journalist named John O'Sullivan wrote the phrase "Manifest Destiny" in the *Democratic Review* in 1845. It named an idea that many Americans held: that the United States was meant to spread across the continent to the Pacific Ocean. People used the idea to argue for moving west and for going to war with Mexico.

The Mexican Cession. The Mexican-American War lasted from 1846 to 1848 and ended with the Treaty of Guadalupe Hidalgo. In it, Mexican officials, after losing the war, gave up California, Nevada and Utah, most of Arizona and New Mexico, and parts of Colorado and Wyoming. The land they gave up is called the Mexican Cession. Texas had already been annexed, made part of the United States, in 1845. In 1846 officials of the United States and Britain agreed on the border of Oregon.

Gold. Gold was found at Sutter's Mill in California in early 1848, and the rush came in 1849. About 80,000 people, called forty-niners, arrived that year. From 1848 to 1855 about 300,000 people came to California, about half by sea and about half over land on the California Trail.

California's population, not counting Native people, grew from about 14,000 in 1848 to about 250,000 in 1852. San Francisco grew from about 200 people in 1846 to about 36,000 in 1852.

Whose land. The forty-niners arrived on the land of California's Native nations. About 150,000 Native people lived in California in 1848, and by 1870 only about 30,000 were left, about 120,000 fewer. The historian Benjamin

Madley documents the causes: killings, kidnapping, forced labor, disease, hunger, and newcomers driving Native people off their land.

Vigilantes and United States Army soldiers killed 1,000 or more Native people between December 1849 and May 1850. On the last day of 1854, 116 volunteer militiamen attacked the Tolowa people and killed hundreds of them. One militiaman died. Madley documents that state and federal officials took part in the killings, and that taxpayers paid for militia campaigns.

BY 1850

If you remember one thing: The country grew to the Pacific in these fifty years, and soldiers and slave traders forced hundreds of thousands of people to move across it.

In 1800: the United States ended at the Mississippi River, and the Cherokee, Choctaw, Muscogee, Chickasaw and Seminole lived in their homelands in the Southeast.

In 1850: the country reached the Pacific, soldiers had forced those five nations to Indian Territory, and wagon trains had carried families to Oregon, Utah and California.

Next: After 1850, settlers claimed free farmland on the Great Plains, and soldiers forced the nations who lived there onto reservations.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 16 MINUTES

Settlers filled the Great Plains and the far West after 1850, and United States soldiers forced the Native nations who lived there onto reservations, areas of land that federal officials set aside for them, often far from their homelands. In 1864 soldiers marched about 8,500 Navajo people more than 300 miles to one of them.

Free land and fast travel. Under a law of 1862, settlers could claim free farmland, and they filed about 4 million claims under it. After 1869 a railroad could carry them across the country in about a week. Tens of thousands of Black Southerners also moved west, to Kansas, to get away from violence and cheating in the South. By 1890 the officials who counted the country's population no longer found a clear line between settled land and thinly settled land in the West.

Still on the trail. Wagon trains kept rolling west on the Oregon Trail after 1850. One family that set out in 1853 was Amelia Stewart Knight's.

Amelia Stewart Knight

Who: A wife and mother who kept a diary, day by day, of her family's wagon journey on the Oregon Trail.

When and where: From Monroe County, Iowa, to the Willamette Valley near Milwaukie, Oregon Territory, April to September 1853.

Her diary says nothing of her pregnancy until the baby came.

Amelia Stewart Knight, her husband and their seven children left Monroe County, Iowa, for Oregon in April 1853. She was pregnant, and she kept a diary of the trip.

Her diary records rain and mud, dust, sick children and hard river crossings. It also records the daily work of handling the wagons and the cattle.

The family reached Oregon on September 17, 1853. The next day she gave birth to her eighth child, a son named Adam.

Why were the handcart companies trapped in the snow?

More than 1,000 Latter-day Saint emigrants set out late in the season in 1856 to walk to Salt Lake City pulling handcarts, small carts they pulled by hand. Church leaders urged them to go forward that late in the year, even after one of their own leaders, Levi Savage, warned that snow would catch them. Snow trapped many of them in what is now Wyoming. Counts of the dead run from about 170 to about 250.

The emigrants traveled in two companies. The Willie company, about 500 people, left Florence on August 17, and the Martin company left on August 27. Florence was in Nebraska Territory, land the United States governed that was not yet a state.

An early blizzard struck on October 19, 1856, just after the Martin company had waded across the North Platte River, near where Casper, Wyoming, is today. More than 300 rescuers came from Salt Lake with wagons, and they brought the survivors in through November.

Counts of the dead differ. Modern counts give about 67 to 77 dead in the Willie company and about 103 to 150 in the Martin company. Older accounts say about 200 to 250 people died in the stranded companies and the wagon trains traveling with them.

Levi Savage

Who: A leader in the Willie handcart company who warned the emigrants not to leave so late, then stayed with them through the blizzards. His journal records what happened.

When and where: From Florence, Nebraska Territory, to Salt Lake City, August to November 1856. Born 1820 in Ohio.

Movie: *17 Miracles* (2011), a film directed by T. C. Christensen about the Willie handcart company. It draws on Savage's journal.

He warned that leaving so late would bring snow and deaths.

Levi Savage was 36 and one of the Willie company's sub-captains, the leaders under the captain. At a camp meeting at Florence on August 13, 1856, he spoke to the roughly 500 emigrants. They still had more than 1,000 miles to go, and he urged them to wait until spring.

In his journal, Savage wrote that most of the emigrants still meant to go on if the church's leaders said go. At later meetings, church leaders urged them to go forward whatever happened. Out on the trail on September 12, the church leader Franklin D. Richards scolded Savage "very severely in open meeting for his lack of faith in God."

Savage went with the company anyway, and when the blizzards came he worked through them to keep people alive. His journal survives, and most modern accounts of the Willie company rely on it.

The Weeping Time, 1859

The domestic slave trade went on through the 1850s. Its largest single sale took place at the Ten Broeck Race Course near Savannah, Georgia, on March 2 and 3, 1859. Pierce Mease Butler sold between 429 and 436 enslaved men, women and children there to pay his gambling debts. The sale brought him $303,850. It rained hard on both days.

Joseph Bryan, a broker who arranged sales for others and the largest slave dealer in Savannah, ran the sale. For weeks he advertised it in newspapers across the South, from Richmond to New Orleans. Mortimer Thomson, a reporter for the *New York Tribune*, came to the sale dressed as a buyer and wrote a long account of it. The surviving accounts do not say where the buyers took the people they bought.

Sold in families. Butler required that people be sold in "families" as far as possible, and Bryan's advertisement promised that they would be "sold in families." Thomson told the story of a 23-year-old man named Jeffrey, who planned to marry a woman named Dorcas. Because they were not married, they could not be sold as a family. Jeffrey begged his buyer, "Please buy Dorcas," but she was being sold with her family of four and could not be bought alone, and another buyer took her. Parents were also separated from children at the sale. Black Georgians remembered the sale as "the weeping time."

Free land and the railroad

President Abraham Lincoln signed the Homestead Act on May 20, 1862. Under it, an adult who headed a household could claim up to 160 acres of public land for free, and after living on the land and farming it for five years, the settler owned it. Federal officials had taken much of that land from the Native nations who lived on it. Women could claim land. After the Civil War, freed people could claim land too, and so could immigrants who planned to become citizens.

Settlers filed about 4 million claims over the years the law was in force. Settlers and other private owners got about 270 million acres this way, about 10 percent of all the land in the United States. Many settlers moved onto the Great Plains to claim land under the law.

The railroad. The first railroad across the country was finished at Promontory Summit, Utah, on May 10, 1869. By wagon, a trip from coast to coast took four months or more, and by train it took about a week. Under the Pacific Railroad Act, which Lincoln signed in 1862, each railroad company got 6,400 acres of federal land for every mile of track it built. The railroads' owners sold that land to settlers, and their trains carried settlers and immigrants west by the trainload.

The Long Walk of the Navajo, 1864

United States soldiers forced more than 10,000 Navajo people from their homeland in what is now Arizona and New Mexico from 1863 to 1866. General James Henry Carleton ordered Colonel Christopher "Kit" Carson to make the Navajo surrender.

Burning the food. Carson's troops, guided by Ute scouts and warriors, began burning Navajo fields in August 1863. They took more than 5,000 sheep, goats and mules, and they destroyed more than 75,000 pounds of wheat and the Navajo peach orchards.

The march. Starving, 6,000 Navajo had gathered at Fort Canby and Fort Wingate by March 1864. Soldiers then marched about 8,500 people more than 300 miles to the Bosque Redondo reservation at Fort Sumner, New Mexico. The walk took from 18 to 40 days, and the soldiers gave them food planned to last 8 days. This march is known as the Long Walk.

About 200 people died on the march, in winter. Navajo accounts passed down in families say soldiers shot people who could not keep up, among them old people, sick people and pregnant women.

Soldiers held more than 8,000 Navajo and Mescalero Apache people at Bosque Redondo, and thousands more died there. On June 1, 1868, Navajo leaders signed a treaty with the United States, and under it the Navajo could go home.

On the Plains. Through the 1870s and 1880s, soldiers forced many other nations onto reservations. In the Treaty of Fort Laramie of 1868, United States officials had set aside the Great Sioux Reservation, all of present-day South Dakota west of the Missouri River, the Black Hills included, "for the absolute and undisturbed use" of the Lakota. After soldiers on an Army expedition reported gold in the Black Hills in 1874, federal officials ignored the treaty. In December 1875 they ordered the Lakota bands in their winter camps to move onto the reservation by January 31, 1876, and then they sent the Army against the bands that stayed out.

Laura Ingalls Wilder

Who: The author of the Little House books. As a child she moved with her family by covered wagon again and again.

When and where: Born February 7, 1867, near Pepin, Wisconsin. Family moves from 1869 to 1879. Her own last move, to Mansfield, Missouri, in 1894.

Movie: *Little House on the Prairie* (NBC series, 1974 to 1983). It ran for nine seasons and 205 episodes, with Michael Landon as Charles Ingalls. It is loosely based on the books.

Her family's Kansas cabin stood on land that belonged to the Osage.

Laura Ingalls Wilder's family moved six times between 1869 and 1879. In 1869 they went to Montgomery County, Kansas, near the town of Independence. Their cabin stood on the Osage Diminished Reserve, land that still belonged to the Osage Nation, and the family had no right to be there. They left in 1871 and went back to Pepin, Wisconsin.

The family moved to Walnut Grove, Minnesota, in 1874, and swarms of grasshoppers ate their wheat crops there in 1875 and 1876. Then they moved to Burr Oak, Iowa, where they helped run a hotel from 1876, and back to Walnut Grove in 1877. In 1879 they settled at De Smet, in Dakota Territory. There Laura's father, Charles Ingalls, took a job at a railroad camp and then claimed a homestead.

Her own move. Laura married Almanzo Wilder in 1885. In 1894, a year of drought, Laura, Almanzo and their daughter Rose drove a covered hack, a light wagon with a cover, about 650 miles from De Smet to Mansfield, Missouri. She kept a diary of the trip, published in 1962 as *On the Way Home*. In Mansfield she later wrote the Little House books.

What the books changed. The books came out from 1932 to 1943, and in them Wilder changed the real story. In *Little House on the Prairie*, Laura is several years older than she really was in Kansas, where she was 2 to 4 years old. The books leave out the Burr Oak years completely, and they do not say that the family was living on Osage land.

The cattle drives

Cowboys drove herds of longhorn cattle north from Texas to Kansas after the Civil War. They took the cattle to railheads, the towns at the end of a railroad line, where the cattle were loaded onto trains. Abilene was one of these towns. The railroad reached Abilene in 1867, and after that cowboys used the Chisholm Trail from Texas heavily.

About 5 million cattle or more were driven north from Texas between 1866 and 1884. Some estimates put the total for the whole era at about 6 million. About 700,000 cattle reached the railheads in Kansas in 1871 alone.

The long drives ended in the mid-1880s for several reasons. Barbed wire fences went up across the open range. Railroads reached deeper into Texas, so the cattle no longer had to walk to Kansas. Kansas lawmakers passed a quarantine law in 1885 and 1886, keeping Texas cattle out of the state to stop the spread of disease. Then the cold of the winter of 1886 and 1887 killed much of the cattle herd.

Why did the Exodusters go to Kansas?

Thousands of Black Southerners left their homes in 1879 and moved to Kansas, to get away from violence and cheating in the South. People called them Exodusters, after the Exodus, the story in the Bible of the Israelites' escape from slavery in Egypt. Many came from Mississippi, Louisiana, Texas and Tennessee. As many as 40,000 moved to Kansas, Oklahoma and Colorado between about 1879 and 1881, and about 6,000 reached Kansas in the spring of 1879 alone.

What they left. Former Confederates took away their votes, white landowners cheated them, and armed white men attacked them. Many were sharecroppers, who farmed someone else's land and paid the owner with a share of the crop.

The Confederates had been the Southerners who fought against the United States in the Civil War to keep slavery. After the presidential election of 1876, former Confederates took back control of the Southern states and took away

Black Southerners' right to vote. The historian Damani Davis calls what Black Southerners faced "relentless terror."

Henry Adams's list. In Louisiana, the Black organizer Henry Adams collected the names of nearly 700 Black people who had been attacked. For each one he listed where it happened and what was done. Most of them had been whipped, beaten, shot or killed by white men, often by armed groups. Some entries name the man. One entry says that a white man, Dr. Moore, killed a Black man named Calvin in 1871, after Calvin was accused of cursing a white woman.

Many Exodusters traveled up the Mississippi River by steamboat, and crowds of them waited on the docks at St. Louis. In 1880 a United States Senate committee of three Democrats and two Republicans looked into why they had left. The three Democrats, the majority, held that the move had been planned to settle Black voters in Northern states for political gain. A 1912 history of Kansas states that the evidence hardly supported that idea.

The two Republicans wrote their own report. It said the migrants had fled because they could no longer bear the hardship, unfair treatment and suffering that "a class of Democrats in the South" had brought on them.

The migrants saw Kansas as a safe place. John Brown, who fought to end slavery, had lived there, and before the Civil War many Kansans had wanted it to be a free state, a state without slavery.

Benjamin "Pap" Singleton

Who: A man born into slavery who organized thousands of Black Southerners to move to Kansas.

When and where: Born enslaved in Nashville, Tennessee, about 1809. Led the move to Kansas in the late 1870s. Died 1900.

He called himself the "Father of the Exodus."

Benjamin Singleton ran away from slavery several times, and in the end he escaped to the North and to Canada. He came back to Tennessee in about 1871, after the Civil War. He worked there as a carpenter, a cabinetmaker and an undertaker, a person who prepares the dead for burial.

Singleton was angry that white landowners cheated the Black sharecroppers who worked their land, and he believed Black Southerners would never get a fair chance at land in the South. He printed handbills, short printed notices, to draw Black settlers to Kansas. He formed a group called the Edgefield Real Estate and Homestead Association to move people there.

Singleton said under oath, at a hearing, that his work helped more than 7,400 people move. Between 1877 and 1879, his group steered more than 20,000 people toward Kansas. He died in 1900.

The Oklahoma Land Run, 1889

About 50,000 people raced into central Oklahoma at noon on April 22, 1889, to claim land. The land they entered was called the Unassigned Lands. It covered about 1.9 million acres in the middle of Indian Territory, and nearly 2 million acres were claimed in a single day.

Indian Territory was the land in present-day Oklahoma where federal officials had moved many Native nations. The Unassigned Lands were the last part of it that federal officials had not given to any of those nations. Federal officials had taken land in Indian Territory from the nations there.

How the land was taken. After the Civil War, federal officials pressed the Creek and Seminole nations to give up land. Dennis N. Cooley, the Commissioner of Indian Affairs, was the federal official in charge of dealings with Native nations. In 1865 he told the nations' leaders that they had lost their rights under the old treaties by joining the Confederacy.

The Creek had to sign over the western half of their land for $975,168 in treaties of 1866, and the Seminole had to sign over all of theirs for fifteen cents an acre. With these treaties the Five Tribes, the Cherokee, Choctaw, Muscogee (Creek), Chickasaw and Seminole nations, lost the western half of present-day Oklahoma.

The boomers. Groups of settlers called boomers, led by David L. Payne and William Couch, entered the Unassigned Lands against the law from 1879 to 1888. Cavalry soldiers, who fought on horseback, arrested them and took them out again and again. Representative William Springer of Illinois changed a bill in Congress in 1889 so that President Benjamin Harrison could open the land, and Harrison announced the opening on March 23, 1889.

Each settler could claim 160 acres under the Homestead Act. People who sneaked in before the start to grab land were called "Sooners." By the end of that day, Guthrie and Oklahoma City were towns of about 10,000 to 15,000 people each.

The frontier line disappears from the census, 1890

The officials who ran the United States census, the official count of the population taken every ten years, reported in 1890 that the country no longer had a clear "frontier line." The frontier line had been the edge of thinly settled land in the West, where two to six people lived on each square mile.

The historian Frederick Jackson Turner wrote an essay about that census finding in 1893, called "The Significance of the Frontier in American History." Many people later treated 1890 as the end of the years when settlers could still find open land in the West, land that had belonged to Native nations.

BY 1900

If you remember one thing: In these fifty years settlers claimed farms across the Great Plains and the far West, and soldiers forced the Native nations who lived there onto reservations.

One number: about 270 million acres. That is how much land settlers and other private owners got under the Homestead Act, about a tenth of all the land in the United States.

Next: After 1900, millions of Black Americans left the South, and Plains farm families fled the Dust Bowl.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 11 MINUTES

About 6 million Black Americans left the South between about 1910 and 1970 to get away from lynch mobs, crowds that kill a person without any trial, and from Jim Crow laws, and most of the early travelers went north by train. Plains farm families moved too. In the late 1930s more than 250,000 people left the dried-out farms of the southern Plains for California.

Forced out. United States Army officers ordered Japanese Americans out of their homes on the West Coast in 1942. Soldiers and officials forced them into camps.

The Great Migration, the first wave

The movement of Black Americans out of the rural South from about 1910 to 1970 is called the Great Migration. The first wave, from about 1910 to 1940, went mostly to cities in the North and the Midwest. Most people rode the trains north, and later travelers also went by car and by bus.

The cities. Between 1910 and 1920, Chicago's Black population grew to about two and a half times its size, and Detroit's grew to about seven times. Many newcomers also went to New York, Philadelphia and Pittsburgh.

About 90 percent of Black Americans lived in the South in 1900. By 1970 fewer than 55 percent did.

Why they left. Georgia is one example. Black Georgians had several reasons to leave. The boll weevil, a beetle that eats cotton plants, and floods ruined the cotton crop in 1915 and 1916. Lynch mobs killed more than 450 people in Georgia between 1880 and 1930, most of them Black.

Under Jim Crow laws, the Southern laws that kept Black people and white people apart, nearly all Black Georgians could not vote. About 74,000 Black Georgians left between 1910 and 1920, and about 260,000 more left in the 1920s.

The Chicago Defender. Black Southerners from across the South went north for jobs, and some people in the North urged them to come. Robert S. Abbott started a Black newspaper, the *Chicago Defender*, in 1905. In its pages, job listings and train schedules ran beside editorials, articles that give a newspaper's opinion, and the editorials urged Black Southerners to come north. The *Defender* listed May 15, 1917, as the date for a "Great Northern Drive." Some Northern employers paid migrants' train fares and then took the money back out of their wages.

White officials in the South worked to keep Black workers from leaving. Officials in Macon, Georgia, charged Northern labor recruiters, people hired to find workers, a license fee of $25,000 for permission to work there.

Chicago, 1919. Black people in the North also met violence. Eugene Williams, who was 17, and four friends drifted on a homemade raft in Lake Michigan in Chicago on July 27, 1919. They crossed an unmarked line into the water off a beach at 29th Street where only white people were allowed. George Stauber, a 24-year-old white man, threw rocks at them until Williams drowned. A white police officer, Daniel Callahan, refused to arrest Stauber and stopped a Black officer from arresting him.

Five days of attacks followed, through August 3. The white attackers included youth gangs and groups that called themselves athletic clubs. Thirty-eight people were killed, 23 Black and 15 white, and the histories of the riot do not say who killed each of them. About 537 people were injured, two-thirds of them Black. Hundreds of homes were burned or damaged. Of the 138 people charged with crimes afterward, two-thirds were Black.

Ida Mae Brandon Gladney

Who: A sharecropper's wife who left Mississippi with her family after white men nearly beat a cousin to death.

When and where: From Chickasaw County, Mississippi, by train, late October 1937. Settled in Chicago.

The turkeys turned up the next day, and no one apologized.

Ida Mae Brandon Gladney and her husband, George Gladney, were sharecroppers in Chickasaw County, Mississippi, farming another man's land for part of the crop. They grew cotton on the plantation of Edd Pearson, whom they called Mr. Edd.

In 1937 a group of white men beat a cousin of theirs, Joe Lee, nearly to death with heavy chains. The men had falsely accused him of stealing turkeys. The accounts of the beating do not name them. After the beating, George came home and told Ida Mae, "This is the last crop that we're making."

The train north. The cotton was picked by late October 1937, and George settled with Mr. Edd over what the year's work was worth. None of the family had ever been on a train or out of Mississippi. A brother-in-law drove Ida Mae and the two children, James and Velma, in his truck to the train station at Okolona. There they met George and rode north through the night.

The family settled in Chicago. One account says they stopped first in Milwaukee, where Ida Mae's sister Irene lived. In Chicago, George hauled ice up four and five flights of stairs, and Ida Mae did housework for other people and then found work as a hospital aide. In time the family bought a house. Ida Mae died in 2002. The writer Isabel Wilkerson told Ida Mae's life, from interviews with her, in the 2010 book *The Warmth of Other Suns*.

The Dust Bowl move

Drought, years with too little rain, and windstorms tore the topsoil off farms across the southern Great Plains in the 1930s. The Dust Bowl was the name for those parts of Oklahoma, Texas, Kansas, Colorado and New Mexico.

More than 250,000 people left the Dust Bowl states for California from about 1935 to 1940. Another count covers only Oklahoma but every destination: everyone who left Oklahoma for any place, minus the people who moved in. By that count Oklahoma may have lost as many as 440,000 people. Many of the travelers went west on U.S. Route 66, and in California the newcomers were called "Okies."

The "bum blockade." The Los Angeles police chief, James E. Davis, sent his officers to the state line in February 1936 to turn back poor migrants. People called it the "bum blockade."

Davis's officers, 136 of them, stood at 16 crossing points from February 3 on California's roughly 700-mile border with Arizona, Nevada and Oregon, far outside the city. Davis pointed to a 1933 state law against bringing poor people into California. His officers stopped trains, cars and people hitchhiking. They took people's fingerprints and turned back anyone they judged unable to support themselves.

The officers arrested 122 people for vagrancy, the crime of having no home or means of support, on February 8 alone. A week into the blockade, Davis announced that he would force the people arrested to work without pay in quarries, digging rock, sand and gravel out of the ground for public projects. Many of the people arrested were locked up in prisons far from the state line.

For and against. Railroad executives and some county sheriffs backed the blockade. So did the people who ran the *Los Angeles Times* and the Los Angeles Chamber of Commerce, a group of the city's business owners. California's attorney general, the state's top lawyer, Ulysses S. Webb, said Davis had "no jurisdiction beyond the city's territorial limits," meaning no power outside Los Angeles. Arizona's governor, Benjamin Moeur, called the blockade illegal and against the Constitution.

Davis called his officers home in early April 1936. He claimed they had turned back about 11,000 people. Writing in *Smithsonian* magazine in 2025, Olatunji Osho-Williams reports that the real number was probably much lower.

Florence Owens Thompson

Who: A farmworker born in what is now Oklahoma, whose parents said their families were Cherokee, who moved from crop to crop in California. She is the woman in Dorothea Lange's 1936 photograph "Migrant Mother."

When and where: Born in Indian Territory in 1903. Moved to California in the 1920s. Photographed at a pea-pickers' camp in Nipomo, California, in March 1936.

Lange took seven pictures of her in about ten minutes.

Florence Owens Thompson was born Florence Leona Christie in Indian Territory, which is now Oklahoma, in 1903. Both of her parents said their families were Cherokee.

She married Cleo Owens in 1921, and during the 1920s, years before the Dust Bowl, they moved west with several of his relatives to Oroville, California. Cleo died in 1931. In about 1933 she and her parents moved to Shafter, California, and she worked in the fields as a farm laborer, following the crops from one harvest to the next.

Nipomo. The timing chain in the family's car, an engine part the car could not run without, snapped on March 6, 1936. The family stopped at a camp of pea pickers near Nipomo, where freezing rain had destroyed the pea crop. The photographer Dorothea Lange took seven pictures of Thompson in about ten minutes. One of them became known as "Migrant Mother," and it became one of the most reproduced pictures of the 1930s.

Thompson's name was not made public until 1978, when Emmett Corrigan, a reporter for the *Modesto Bee*, found her. She died on September 16, 1983.

The forced removal of Japanese Americans, 1942

United States Army officers forced more than 110,000 Japanese Americans out of their homes on the West Coast in 1942. The United States had been at war with Japan since Japanese forces attacked Pearl Harbor, in Hawaii, on December 7, 1941. President Franklin D. Roosevelt signed Executive Order 9066, a rule issued by a president, on February 19, 1942. Under it, military commanders could keep "any or all persons" out of areas they named.

General John L. DeWitt commanded the Army on the West Coast. In March and April 1942 he issued orders aimed at "all persons of Japanese ancestry," meaning everyone whose family came from Japan. United States citizens and people who were not citizens were ordered out alike.

108 orders. DeWitt set up an Army office on March 11 to carry out the removal. Karl Bendetsen, an Army officer, commanded it, and the next day, at 34, he was made a colonel. Bendetsen's staff divided the coast into 108 areas of about 1,000 Japanese Americans each and issued a separate order for each area.

Under each order, the people in that area had to report on a set day, about a week later, and they could bring only what they could carry. Under the first order, issued on March 24, the 55 Japanese American families on Bainbridge Island, near Seattle, had six days to leave their homes. Army officers carried out the 108 orders and forced out 110,442 people.

The camps. Army engineers built 15 temporary camps, many of them at fairgrounds and racetracks. When people arrived, soldiers searched their bags. At the Tanforan camp in California, half of the people lived in horse stables, three to six people in a stall built for one horse.

After about three months, on average, officials moved them again, to 10 larger camps run by officials of a federal office called the War Relocation Authority. Staff at the National Archives count about 120,000 people removed in all, a larger total than the count for the 108 orders alone.

The end. None of the people removed was charged with a crime or tried. Officials lifted the order that kept Japanese Americans off the West Coast on January 2, 1945, and they closed Tule Lake, the last camp, on March 20, 1946.

In 1983 the members of a federal commission that studied the removal found that its causes were "race prejudice, war hysteria and a failure of political leadership." Under the Civil Liberties Act, which President Ronald Reagan signed in 1988, each surviving person who had been held received a written apology and $20,000.

George Swanson Starling

Who: A Florida fruit picker who organized the other pickers to ask for better pay, then fled north when he learned that the grove owners were plotting to hang him.

When and where: From Wildwood, Florida, to Harlem in New York City, by train, April 14, 1945.

For asking ten cents more a box, grove owners plotted to hang him.

George Swanson Starling grew up in Eustis, in Lake County, Florida. He finished first in his high school class and went to Florida Agricultural and Mechanical State College in Tallahassee. In the middle of his second year, his father told him he had had enough schooling and must go to work.

The groves. Starling went to work picking citrus fruit. The pickers climbed trees as tall as 40 feet, and some fell and broke arms or legs. Starling organized the pickers to ask for better pay. In one dispute, the packing house, where the fruit was packed for sale, would pay the pickers ten cents for each box of tangerines. The pickers asked for twenty cents. At auction in 1944, tangerines sold for an average of $4.40 a box.

The plot. The grove owners were angry. Someone overheard them planning to take Starling and two other pickers out to the cypress swamps at Blackwater Creek and hang them. Starling left within days, on April 14, 1945. A man named Roscoe Colton drove him in his truck to the train station in Wildwood.

There Starling boarded the Silver Meteor, a train to New York. He rode in the Jim Crow car, the car set aside for Black passengers, where luggage was also stored. The ride took 23 hours. His wife, Inez, stayed behind until he could send for her.

Starling settled in Harlem, where an aunt and many of his friends lived. He later got a steady job working on the same train that had carried him north. Isabel Wilkerson's book *The Warmth of Other Suns* also describes his life. The accounts of Starling's life do not name the grove owners who planned to kill him.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 5 MINUTES

Americans moved south and west after 1950, toward jobs, warmer weather and cheaper land. The population of the South grew by about 36 percent between 1970 and 1990, and the population of the West by about 51 percent. In the second wave of the Great Migration, many Black Southerners moved to cities on the West Coast. Federal officials paid for Native people to move from reservations to cities.

By the late 1990s more Black Americans were moving to the South than were leaving it.

Where the second wave of the Great Migration went

The second wave of the Great Migration lasted from about 1940 to 1970, and it was larger than the first. More of its travelers went west, to Los Angeles, Oakland, Seattle and Portland, for factory jobs, first during World War II and then after it. These travelers are counted in the same total of about 6 million people who left the South between 1910 and 1970.

Robert Joseph Pershing Foster

Who: A Black surgeon who could not work in the hospital of his own home town in Louisiana. He drove alone to California and opened his own medical practice in Los Angeles.

When and where: From Monroe, Louisiana, to Los Angeles, California, by car, April 1953.

Motel clerks in Arizona turned him away under "vacancy" signs.

Robert Joseph Pershing Foster had served as a surgeon in the Army. Back home in Monroe, Louisiana, he found that the people who ran St. Francis Hospital there would not let Black doctors do even a simple operation to take out a patient's tonsils. Isabel Wilkerson traced his trip in *The Warmth of Other Suns*.

The drive. Foster set out alone on Easter Monday, April 6, 1953, in his 1949 Buick Roadmaster, a dark red car with white-walled tires. He planned to send for his family once he was settled. Once he was long past Texas, he thought he was a safe distance from the South.

In Arizona, motel clerks turned him away, a Black man traveling alone, even under "vacancy" signs, which tell drivers that rooms are free. One clerk told him they had forgotten to turn off the sign. Others said the rooms had just been rented. He could not find a place to stay the night, so he drove on until he reached California.

> Alone in the dark desert, he asked himself what he was doing. If he turned back, Wilkerson writes, "the I-knew-its would ring in his ear," meaning the people who would say they had known he would fail.
>
> In Los Angeles, Foster opened his own medical practice. Many of his patients were migrants from the South, as he was. One of them was the musician Ray Charles.

Why did Americans move to the Sun Belt?

Many Americans moved from the older factory states of the North to the South and the West after World War II, for jobs, for warmer weather and for cheaper land. People later nicknamed the Northern factory states the "Rust Belt" and the South and the West the "Sun Belt." Air conditioning made hot places comfortable all year, which helped the Sun Belt grow.

The South grew by about 36 percent between 1970 and 1990, and the West by about 51 percent. By counts from the Census Bureau, the federal office that counts the population, the whole country grew by about 22 percent in those years, from about 203 million people to about 249 million. California passed New York as the state with the most people in 1962. It had about 17 million people then, roughly three times as many as before World War II.

The interstates. Members of Congress passed the Federal-Aid Highway Act in 1956. Under it, work began on the Interstate Highway System, a network of big highways across the country. The interstates made long moves by car easier, and they made it easier to live in a suburb, a town at the edge of a city.

Federal officials move Native families to cities

Officials of the Bureau of Indian Affairs, the federal office in charge of the government's dealings with Native nations, paid for Native people to move from reservations to cities from about 1950 into the 1970s. They started the program, called Relocation, between about 1948 and 1952, and members of Congress backed it with the Indian Relocation Act in 1956. It ran in the same years as termination, a policy that members of Congress declared on August 1, 1953, to end the government's recognition of Native nations and the services that came with it.

The officials promised people one-way tickets to a city and help for a short time after they arrived. About 30,000 Native people moved to cities in the 1950s, and more than 100,000 had moved by the time the program ended in the 1970s. They went to Minneapolis, Chicago, Oakland and San Francisco, Los Angeles, Denver and other cities. Many of them lived in poverty there. They also built new Native communities in those cities.

When did the Great Migration turn around?

A small number of Black Americans began moving back to the South in the 1970s, and the number grew through the 1990s. By the late 1990s the South gained more Black migrants than it lost to each of the other three regions of the country, the Northeast, the Midwest and the West. For 35 years before that, more Black Americans had left the South than moved to it. William H. Frey, a researcher at the Brookings Institution, a research group, calls this return the New Great Migration.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 9 MINUTES

Hurricane Katrina and the broken levees around New Orleans forced more than 1 million people from their homes on the Gulf Coast in 2005, the largest sudden move of these years. After Hurricane María hit Puerto Rico in 2017, about 130,000 people left the island for the mainland within a year.

A steadier flow. By Census Bureau estimates, the South gained about 2.7 million more people from other states than it lost to them from 2020 to 2024.

How many people moved south after 2020

Census Bureau staff count moves between states as net domestic migration: the number of people who move into a state from other states, minus the number who move out to other states. By this measure the South gained about 2.7 million people from 2020 to 2024.

Many of the people who moved between states left coastal states where homes cost a lot. California lost about 343,000 people to other states from April 2020 to July 2022, and New York lost about 300,000, by Census Bureau counts. The metro areas, cities together with their suburbs, that lost the most people included New York, Los Angeles, San Francisco and Chicago.

Working from home. Some people could work from home after 2020. Researchers at the Pew Research Center, a group that studies how Americans live and what they think, reported that working from home let some people choose where to live by the cost of a home and the space it offered, not by where their job was.

Florida slows down. Florida's gains from other states shrank after 2022. By Census Bureau counts, Florida had a net gain of about 311,000 people from other states in 2022, about 184,000 in 2023, and about 23,000 in the year from July 2024 to June 2025. Census Bureau staff released the 2025 figures in January 2026. In that same year North Carolina gained the most people from other states, about 84,000, and Texas and South Carolina each gained about 67,000.

Researchers at the National Association of Realtors, a group for people who sell houses, ranked the states from the same Census figures. In that ranking, nine of the fifteen states with the biggest gains in 2024 to 2025 were in the South. The Midwest also gained about 16,000 people from other states that year, its first such gain of the 2020s.

The New Great Migration. The movement of Black Americans to the South went on. William Frey of the Brookings Institution found that the Black population of metro Atlanta grew about five times over from 1970 to 2020, and by 2010 it had passed Chicago's to rank second in the country. Atlanta was the

most common destination of young Black adults aged 16 to 26 who moved between 2000 and 2018, and seven of their ten most common destinations were in the South.

Leaving New Orleans after Hurricane Katrina, 2005

Hurricane Katrina struck the Gulf Coast in late August 2005, and the levees around New Orleans, walls of earth or concrete built to hold back water, broke. The storm and the broken levees forced more than 1 million people from their homes in the New Orleans area and along the Gulf Coast. A month later, the people of up to 600,000 homes were still away. These counts come from The Data Center, a research group in New Orleans, and from the federal housing department.

Many never came back. New Orleans had 484,674 people at the 2000 census. By July 2006 an estimated 230,172 people lived there, fewer than half as many, The Data Center reported. In 2024 the city had about 363,000 people, about 120,000 fewer than in 2000, by an estimate reported by the news site The Center Square in 2025.

The New Orleans metro area's seven parishes, the Louisiana name for counties, had 970,849 people in July 2025, 3.6 percent fewer than in April 2020, The Data Center found.

The bridge. Police stopped some of the people who tried to walk out of the flooded city. On September 1, 2005, officers from three police forces blocked the Crescent City Connection bridge over the Mississippi River: the Gretna police, the bridge police and the Jefferson Parish sheriff's office. They turned back people on foot at gunpoint. Two paramedics from San Francisco, medical workers who ride ambulances, were in the crowd, and they reported that police "fired their weapons above their heads."

Gretna Police Chief Arthur Lawson made the decision together with officials of the sheriff's office and the bridge police. Gretna's mayor, Ronnie Harris, defended it. He said the town was not a shelter or an evacuation route and had no running water, no electricity and no food. Members of the Gretna City

Council voted to support the blockade, which lasted until Labor Day, September 5, 2005. No count survives of how many people the officers turned back.

Houston. About 250,000 evacuees, people who leave a place to escape danger, came to Houston at the peak, and about 150,000 were still there a year later. Those are Houston officials' own estimates, reported by the *Texas Tribune* in 2010. By 2010 Houston's mayor, Annise Parker, said city officials no longer counted them apart from other residents: "They are Houstonians."

Kimberly Rivers Roberts

Who: A woman of 24 from the Lower Ninth Ward of New Orleans who wanted to be a rap singer. With her husband, Scott Roberts, she filmed the Katrina flood from inside it.

When and where: New Orleans, August 2005. Then Alexandria and northern Louisiana, then Memphis, Tennessee, then home to New Orleans.

Movie: *Trouble the Water* (2008), a documentary about Kimberly and Scott Roberts, directed by Tia Lessin and Carl Deal and built largely on her own video. It won the Grand Jury Prize for documentary at the 2008 Sundance Film Festival and was nominated for an Academy Award.

She filmed the Katrina flood on a $20 camcorder.

Kimberly Rivers Roberts and her husband, Scott Roberts, were still in their New Orleans neighborhood when Katrina's flood came, because someone had recently stolen their car. "If I had wheels I'd be gone too," she told a neighbor. She began filming on August 28, 2005, two days before the storm reached the city, with a camcorder she had bought for $20.

As the water rose, the couple and their neighbors climbed into an attic. They called 911 and heard that police would not come until conditions got better. A neighbor floated people to safety on a punching bag.

The Navy base. Afterward the group tried to take shelter at a Navy base that stood nearly empty. The couple's film shows men at the gate turning them away at gunpoint, and a voice says, "Get off our property, or we're

going to start shooting." Navy officials deny that this happened. The accounts of the film do not name the men at the gate.

Kimberly's grandmother died in a hospital that was not emptied of its patients, and one of her uncles, called Uncle Ned in the film, was later found dead in his home. The accounts do not say who decided to leave the hospital's patients there, and they do not state the cause of either death.

The couple went to a Red Cross shelter in Alexandria, Louisiana, then to the trailer of another uncle, in northern Louisiana. Next they moved to Memphis, Tennessee, to stay with one of Kimberly's cousins and try to start over. They waited for a promised check from FEMA, the federal office for disaster help, but it did not arrive, and they had a hard time settling in a new city.

Home. Kimberly and Scott went back to their home in the Ninth Ward and set out to rebuild it. Scott got a job with a builder repairing houses that the flood had ruined. Kimberly started her own record label, Born Hustler Records.

Leaving Puerto Rico after Hurricane María, 2017

About 130,000 people left Puerto Rico for the mainland within a year after Hurricane María hit the island in September 2017. That was almost 4 percent of the people of Puerto Rico. The count is the Census Bureau's estimate for July 2017 to July 2018.

Other counts run higher. Researchers writing in the science journal PNAS in 2020 described estimates made from air travel records, phone records and social media data. Those estimates run from about 160,000 to 470,000, depending on which data the researchers used.

Puerto Rico is a territory of the United States, land that belongs to the country but is not a state, and Puerto Ricans are United States citizens. A move from the island to the mainland is a move within the country. About a third of the people who moved from Puerto Rico to the mainland in 2018 went to Florida, by Census Bureau figures.

THE THREE QUESTIONS, ANSWERED

1. **How did horses reach the Native nations of the Plains?** In the older account, horses spread north through trade after the Pueblo Revolt of 1680, when the Spanish left perhaps more than 1,500 of them behind. A 2023 study of horse bones found that many nations had horses by the early 1600s, decades earlier, as the nations' own histories say.
2. **How many Cherokee died when soldiers forced them west?** Nobody knows exactly. The count most often given is about 4,000, and estimates run from more than 1,000 on the journey itself to about 8,000 or more.
3. **Why did about 6 million Black Americans leave the South between 1910 and 1970?** They left lynch mobs, Jim Crow laws and cotton crops ruined by the boll weevil and floods, and they went toward jobs in Northern and Western cities. In Georgia, one

example, lynch mobs killed more than 450 people between 1880 and 1930, and nearly all Black Georgians could not vote.

What changed: Before 1900 most of the moves in this chapter went west onto Native land, many of them forced, and after 1900 millions of Black Southerners went north, until the flow began turning back toward the South in the 1970s.

FIVE DATES TO KEEP

April 30, 1598: Oñate claims New Mexico for Spain in La Toma, "the taking."

May 1838: US soldiers begin rounding up the Cherokee for the Trail of Tears.

May 20, 1862: Lincoln signs the Homestead Act.

February 19, 1942: Roosevelt signs Executive Order 9066.

August 2005: Hurricane Katrina forces more than 1 million people from their homes.

NOT KNOWN: Nobody knows for sure why most of the people of Cahokia had left their city by about 1350.

Where to go: The Trail of Tears National Historic Trail, run by the National Park Service, marks more than 2,200 miles of the removal routes across nine states.

Next: Chapter 6, City Building. It asks how Americans built the towns and cities where so many of these travelers ended up.

Chapter 6: City Building

CHAPTER 6 OF 37 · CITY BUILDING · 10 ERAS · ABOUT 90 MINUTES

American cities were built by people who decided where the streets, pipes and buildings would go, and who could live where.

About 600 men lifted a row of brick and stone buildings on Lake Street in Chicago in 1860. The row was 320 feet long, and with its sidewalks it weighed about 35,000 tons. The men turned about 6,000 lifting screws, and in five days the whole row rose 4 feet 8 inches, up to a new street that crews had built on top of the city's sewers. The shops inside stayed open the whole time, and people kept shopping while the buildings rose.

This chapter follows American cities in order, from the earth mounds of Cahokia to the empty office towers of today. As you read, watch the problems people were trying to fix: fire in wooden towns, sickness from dirty water, streets of mud, families packed into dark rooms. And watch who got to live where. Every colonial town and boomtown here went up on a Native nation's land. In the 1900s, white owners and judges kept Black families out of most neighborhoods, and federal officials refused to back home loans where Black families lived. Later, officials cleared whole neighborhoods for developers and ran freeways through others.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Why do so many American cities have straight streets that cross in a grid?
2. How did people get clean water into crowded cities?
3. Who decided which families could live in which neighborhoods?

Five minutes? Read "Central Park and Seneca Village" in 1850 to 1900.

Can skip: "Which tower would be tallest?", "Growth in the Sun Belt", "The tallest buildings now stand in other countries". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

Hundreds of years before any European came, people on this land built planned cities, deciding ahead of time where each open square, wall and mound would go. They built them between about 900 and about 1350. The largest stood beside the Mississippi River. In the Southwest, people cut and stacked stone to build towns, and their descendants still live in towns along the Rio Grande.

The people who built these places kept no written records, and no builder's, planner's or ruler's name survives. Archaeologists, the scientists who dig up and study what people left behind long ago, know them only from the mounds, walls, rooms and roads they left.

Cahokia, a city laid out on purpose

People built up the city of Cahokia quickly from about 1050, in what is now Illinois, across the Mississippi River from St. Louis, Missouri. It was the largest town north of Mexico before Europeans came. At its peak, around 1100, roughly 10,000 to 20,000 people lived in the center, about as many as lived in London at the same date. Some estimates put as many as 50,000 people in the wider settlement around the center, but those larger estimates do not agree with one another.

The people of Cahokia planned where the big pieces of their city would go. At its center they raised Monks Mound. It is the largest prehistoric structure made of earth in the Americas, built before anyone here kept written records. The mound holds about 814,000 cubic yards of earth. Workers carried every basketload of it by hand, and they raised the mound in stages.

The mound. It has four terraces, or wide steps, and its top stands about 100 feet high. Its base covers roughly 14 acres. The World Heritage listing for Cahokia, kept by UNESCO, the United Nations body for historic sites, records fourteen separate building stages in the mound.

The Grand Plaza, about 50 acres of open ground, lay at the foot of the mound. Its builders leveled it by hand and filled in the low spots, to make a flat field for ceremonies and for the game of chunkey. In chunkey, one player rolled a stone disk across the ground, and the players threw spears to see whose spear landed closest to the spot where the stone stopped.

West of Monks Mound the builders set up a circle of red cedar posts. They built five of these circles, one after another, and each new circle took the place of the last.

A calendar of posts. The posts line up with the points on the horizon where the sun rises on four days of the year. Two are the solstices, the longest day and the shortest day. The other two are the equinoxes, the days between them when day and night are the same length. So the posts worked as a calendar, and the builders set them into the plan of the city.

A palisade, a wall of upright logs about two miles long, ran around the central part of the city. People began it around 1100 and rebuilt it three times over about two hundred years. Each rebuilding took between 15,000 and 20,000 logs of oak and hickory, each roughly a foot thick and twenty feet tall.

The people of Cahokia had scattered and left the city by about 1350.

Stone towns of the Southwest

Chaco Canyon, in what is now northwestern New Mexico, was the center of the Ancestral Puebloan world. The Ancestral Puebloans were the ancestors of the people who live in the pueblos of the Southwest today. Spanish colonists later used pueblo, their word for a town, for the towns of stone and adobe, or sun-dried mud brick, that they found there.

At Chaco the people built great houses, large buildings of shaped stone. They went on building and using the great houses for roughly 280 years, from about 900 to about 1180. One of them, Pueblo Bonito, had more than 650 rooms.

The people of Chaco planned their roads too. The roads run straight for miles across the Four Corners country, where Colorado, Utah, Arizona and New Mexico meet. They average about 30 feet wide, and they connected farther-off villages to the canyon.

At Mesa Verde, in southwestern Colorado, people built rooms of sandstone blocks and mud into the hollows under overhanging cliffs. They built Cliff Palace, the largest of these cliff dwellings, between 1190 and 1280. It has 150 rooms and 23 kivas, round rooms used for ceremonies and meetings and usually sunk into the ground. More than 100 people lived there.

People began leaving Chaco in the 1140s, and by about 1180 to 1200 they had stopped building and living in the canyon. People had left Mesa Verde by the 1280s. The people who left both places gathered again in the pueblo towns along the Rio Grande, and their descendants still live in those pueblos.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

The oldest lasting European town on the US mainland began inside a Timucua town. Eight years after Spanish colonists founded it, the king of Spain set written rules for how every new Spanish town must be laid out.

St. Augustine and the Laws of the Indies

Pedro Menéndez de Avilés landed on the coast of Florida on September 8, 1565, with about 800 colonists, among them 26 women and an unknown number of enslaved Africans. He placed his town, St. Augustine, inside Seloy's town, a town of the Timucua people whose chief was named Seloy. Chief Seloy offered Menéndez his council house, and the colonists used it as their first fort. Timucua council houses were round and roofed with thatch, dried straw or reeds, and they were big enough to hold up to a thousand people.

Within months the Timucua had tired of the Spaniards living in their town, and they attacked them again and again to drive them away. In 1566, according to the Florida Museum of Natural History, the Timucua shot flaming arrows at the fort. The fort was the Timucua's own council house, and the museum's account states that this is why the attack did so much harm. On April 1, 1566, half of the fort burned, and the Spaniards lost all their supplies.

Three places in seven years. Menéndez moved the settlement across the bay to Anastasia Island in May 1566, about nine months after landing. There the Spaniards felt safer from attack, and they lived on the island for about six years. By about 1572 they had moved back to the mainland, to the site of today's St. Augustine.

Through both moves, people went on living in the town. Of all the cities Europeans founded on the mainland of the United States, St. Augustine is the oldest one that people have lived in the whole time.

King Philip II of Spain issued the Laws of the Indies in 1573. They were written orders for Spain's lands in the Americas, which the Spanish called the Indies, and they stated exactly how the men who founded Spanish towns had to lay them out.

Each town began with a central plaza, a public square of open ground with the town built around it. The plaza's sides had to face north, south, east and west, and it had to be one and a half times as long as it was wide. The streets ran out from it in a grid, straight streets crossing at right angles. The founders had to set aside certain lots, or pieces of land, for the church and for the government buildings.

St. Augustine's Plaza de la Constitucion was laid out under those orders, after the town already existed. It is the oldest public square in the country. Government buildings and the church faced onto it, and traders began holding a market there in 1598.

The 1573 orders were among the first written rules anywhere in the world that specified the same town plan for every new town, wherever Spanish officials founded one. Across what is now the Southwest, Spanish town-builders went on laying out towns in the same plaza-and-grid pattern.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 8 MINUTES

Each colonial town of the 1600s went up in the homeland of a Native nation, and some were planned while others simply grew. English colonists began Jamestown in 1607 as a fort. Spanish officials laid out Santa Fe around a plaza. Boston, founded in 1630, had no plan at all, and its streets formed around the houses and the hills.

William Penn planned the most. In 1682 he had his surveyor, the man who measured the land and marked out the streets, draw the whole of Philadelphia on paper before a single street existed on the ground. The founders of many early American towns later followed Penn's example.

Fort towns, walls, and watchmen

Three English ships sailed into the Chesapeake Bay on April 26, 1607, with 144 English men and boys. They settled on Jamestown Island, on the James River in Virginia. The island lay inside the land of the Paspahegh people, who hunted there and had a town about four miles away. The Paspahegh were one of 28 to 32 nations, each with its own chief, that paid tribute, a payment of goods, to one ruler, Powhatan.

The Paspahegh attacked the newcomers on May 26. By June 15 the colonists had finished James Fort, a three-sided fort. Its wall was a stockade, a fence of upright oak and poplar logs about 14 feet high. In the 1620s the surveyor William Claiborne laid out a settlement east of the fort called New Towne, and archaeologists have dug up the foundations of rows of houses there.

Governor Pedro de Peralta founded Santa Fe about 1610 and planned it around a central plaza, in the Spanish pattern. He and his men laid it out at the foot of the Sangre de Cristo Mountains, on the ruins of an older Pueblo town. The Tewa people call that place O'Ga P'Ogeh, which means "white shell water place." The Palace of the Governors stands on Santa Fe's plaza. People have used it as a public building without a break for longer than any other building in the United States.

English colonists founded Boston in 1630 without any plan, on a peninsula that the Massachusett people called Shawmut, a piece of land with water on almost every side. The colonists put their houses up near the waterfront, and the streets formed around the hills, brooks and marshes that were already on the ground. The biggest hill was the Trimountain, a steep hill with three tops, which is now called Beacon Hill.

The cow-path story. According to a well-known story, Boston's crooked streets follow old cow paths. The story is a legend. A writer for Boston Magazine checked it in 2018 and traced it to a line that the American writer Ralph Waldo Emerson wrote in 1860.

Boston's first houses had chimneys made of wood and roofs made of thatch, and several of them burned in the colony's first months. Thomas Dudley, one of the colony's leaders, wrote a letter to the Countess of Lincoln in March 1631. In it he wrote that he and the other leaders had ordered "that noe man shall build his chimney with wood nor cover his house with thatch." The fires, he wrote, always began "in the wooden chimneys."

The people of Boston set up a night watch that same year. The watchmen walked the streets at night to guard the town against criminals, wild animals and fire. The Boston Police Department's own history lists that watch as the start of the city's police.

Wall Street in New York City runs along the line of a wall that is long gone. The wall ran across the northern edge of New Amsterdam, the Dutch town on the southern tip of Manhattan, in the homeland of the Lenape people. It stretched from the East River to the North River, which is now called the Hudson.

Who dug it. A gang of Black men dug a trench across the island for the wall in 1653. The Dutch West India Company, the trading company whose officials ran the colony, owned them. Some of the men had gained what the Dutch called half-freedom: a half-free man could hold land of his own, but he still owed the Company work and a yearly payment, and his children stayed the Company's property.

According to a Columbia University history project, the men dug as long as daylight allowed, all the way from the East River to the Hudson. The town's records show what went up along that line: a wall of planks 9 feet high, nailed to posts, from river to river. The officials had first planned a wall of sharpened logs 12 feet long, but nobody would build it for the price they offered, so they changed the plan to planks.

Who ordered it. Petrus Stuyvesant and the town's officials ordered the wall on March 13, 1653. Stuyvesant was the director general, the man who governed the Dutch colony of New Netherland for the Company. He and the officials had heard that the English in New England were preparing to attack.

In April, Stuyvesant's council gave out the rest of the work. The townsmen, "without exception," had to dig a ditch in front of the wall from river to river. The ditch was 4 to 5 feet deep and 11 to 12 feet wide, a much bigger hole than the trench, and it was there to slow down attackers. The soldiers, the other servants of the Company and the half-free Black men, "no one excepted," had to finish building up the town's fort.

So the order sent the half-free Black men to the fort, not to the ditch. The city's archivists write that enslaved and free workers built the wall itself together. The officials put the cost at 3,166 guilders, the Dutch money of the time. Townsmen paid that money in, and the minutes of the officials' meetings, their written record, name the men who paid. Officials reported the wall finished on July 28, 1653.

The Columbia project names three of the men who dug the trench: Paulo d'Angola, Simon Congo and Anthony Portuguese. All three were half-free. They were among eleven enslaved men who had worked for the Company for eighteen or nineteen years. The eleven asked the colony's council to free them, saying the Company had promised them their freedom.

The Company's director at the time, Willem Kieft, and his council granted half-freedom to the eleven men and their wives on February 25, 1644, nine years before the wall, on conditions. Kieft and his council gave the men plots of land north of the town. But their children, "at present born or yet to be born," stayed the property of the Dutch West India Company.

Each of the eleven owed the Company a yearly payment that Dutch colonists did not owe: 30 skepels (a Dutch measure for dry goods) of grain or beans, plus one hog. And whenever the Company's officials called, each had to come back and work for the Company.

A rattle watch began walking the streets of New Amsterdam on August 12, 1658. Its members were paid night patrolmen who carried wooden rattles, noisemakers that clack loudly when spun. The rattle watch was among the first paid public police forces in America.

Philadelphia, drawn before it was built

William Penn made Thomas Holme the surveyor-general, or chief surveyor, of his colony of Pennsylvania in 1682. The land along the Delaware River was the homeland of the Lenape people. When Penn's colonists arrived, about 2,500 Lenape and about 3,000 Europeans lived on the two sides of the river. The Europeans were Dutch, Swedish and Finnish settlers, and English settlers in West New Jersey.

Penn bought land in southeastern Pennsylvania from the Lenape from 1682 to 1684, paying goods worth at least 1,200 English pounds. He landed on October 28, 1682, and Holme laid out the grid of Philadelphia that December. Holme's map, "A Portraiture of the City of Philadelphia," was printed in 1683. Penn used the map to guide the work, and to advertise the town and draw settlers to it.

William Penn

Who: The founder of Pennsylvania, a Quaker (a member of the Religious Society of Friends, a Christian group that began in England in the 1600s), who had Philadelphia planned on paper before it existed on the ground.

When and where: The grid laid out by his surveyor-general Thomas Holme in December 1682, and Holme's map published in 1683.

Movie: *Penn of Pennsylvania* (1941), released in the United States as *The Courageous Mr. Penn* (1942). The film is about Penn's life in general, not the making of the city plan.

Holme drew the streets wide partly because London had burned in 1666.

In his instructions for laying out the town, Penn asked for "a greene Country Towne which will never be burnt, and always be wholesome," meaning healthy to live in.

Holme drew a straight grid on the land between the Delaware and the Schuylkill rivers. Two wide main streets crossed it, Broad Street and High Street, which is now called Market Street. Holme drew the streets wide,

partly because of the fire that burned London in 1666. Where the two main streets crossed, he placed a large central square, with one smaller public square in each of the four quarters of the town.

All five squares are still there. The four quarter squares are called Logan, Franklin, Washington and Rittenhouse squares, and Philadelphia's City Hall stands on the center square.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 3 MINUTES

As the four biggest colonial ports grew to between 8,000 and 13,000 people each, the people in them organized to fight fire. Around 1750 about 13,000 people lived in Philadelphia, about 12,000 in Boston, about 11,000 in New York, and about 8,000 in Charleston, though some of the Charleston counts come from 1760 rather than 1750. London at the same date had over 700,000 people.

Boston was the busiest port until about 1750, when New York and Philadelphia passed it.

Who fought fires in a wooden town?

Town officers and volunteers fought the fires. In Boston, a fire burned through the town on October 2 and 3, 1711. It destroyed some of the town's main buildings and left many families without homes.

A few months later, twelve men became Boston's first Board of Fire Wards, a group of town officers for fires. The Boston Public Library's guide to the city's great fires does not state who chose them or list their duties. Charleston had "firemasters," officers chosen by a vote. At a fire, everyone had to obey a firemaster's orders as if they were the law.

People in Boston also formed mutual fire societies, groups whose members helped one another. The members of each society agreed to protect one another's houses, and they had no duty to fight a fire at the house of anyone outside the society.

Men in Philadelphia went further in 1736. They formed a company of volunteers who agreed to fight a fire at any house in the town.

Benjamin Franklin

Who: A Philadelphia printer who helped start the first volunteer fire company in the colonies that protected a whole town, not only its own members.

When and where: Philadelphia, December 7, 1736.

Every member had to bring six leather buckets to every fire.

Franklin printed a newspaper, the *Pennsylvania Gazette*. On February 4, 1735, he printed an unsigned letter in it, "On Protection of Towns from Fire," which he had written himself. In the letter he warned people about carrying live coals from room to room. His advice began with the line "an Ounce of Prevention is worth a Pound of Cure." The next year he and other men in the city set up a fire company themselves.

About twenty-five men, Franklin among them, signed the Articles of the Union Fire Company, the written rules of the group, on December 7, 1736. Each member had to bring six leather buckets and two strong linen bags to every fire. Members carried water in the buckets, and they used the bags to pack up people's belongings and carry them out of a burning house.

Every member had to come out at every fire alarm, to any house in the town, whether its owner was a member or not. Other Philadelphians soon set up more fire companies, among them the Heart-in-Hand, the Britannia and the Fellowship, and Franklin encouraged them.

Volunteer companies fought the fires in American towns for more than a hundred years. Then, on April 1, 1853, firefighters in Cincinnati, Ohio, began work as the first fire department in the United States whose firefighters were all paid. Miles Greenwood was their first chief.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 9 MINUTES

The leaders of the new country had their capital city drawn on paper before anyone built it, on ground where three Native towns had once stood. The members of Congress put the capital on the Potomac River by a law of 1790, and in 1791 President George Washington chose the exact ground. By the spring of 1791 the chosen square held two towns that shipped tobacco, and enslaved people worked the farmland around them. Pierre L'Enfant drew the whole city on paper that year.

In Philadelphia, yellow fever killed about one person in every ten in 1793. After the fever came back in 1798, workers there built the first public waterworks in the country driven by steam.

The ground the capital was built on

President George Washington signed the Residence Act, a law the members of Congress had passed, on July 16, 1790. In it they placed the capital somewhere on the Potomac River, inside a square no bigger than ten miles on each side. They left it to Washington to say exactly where, and to appoint a surveyor and three commissioners, officials put in charge of building the capital.

Washington rode out and looked at the country himself. He chose the ground at and beside Georgetown, and he told Congress his choice on January 24, 1791.

Three Native towns had stood inside that square of land. The largest was Nacotchtank, on the southeast side of the Anacostia River, and the river's name comes from the town's name. A second town, Nameroughquena, stood on the west bank of the Potomac, across from today's Theodore Roosevelt Island. The third stood on a bluff, a high, steep bank, in the northwest part of today's city, and the record does not give its name.

Captain John Smith reached that stretch of the Potomac in 1608. Within forty years of that first contact, only about a quarter as many Native people lived in the region as before 1608, according to a National Park Service history of the Native peoples of Washington. Many had died of diseases that Europeans

brought, and others had been killed in wars, though the Park Service history does not say who fought them. Others joined nations to the north, south and west.

Two towns stood inside the square by the spring of 1791, and both are still there. Virginia's colonial lawmakers voted to create the town of Alexandria in 1749. The county surveyor, John West, laid out its first sixty acres around Hugh West's tobacco warehouse.

Maryland's colonial lawmakers approved the town of Georgetown in 1751. They bought sixty acres for it from George Gordon and George Beall for 280 English pounds. The site already had a tobacco inspection house, a building where tobacco was checked before it was shipped, and Georgetown grew into a tobacco port.

The members of Congress changed the law on March 3, 1791, so that the district, the land set aside for the capital, could reach south and take in Alexandria too. Jones Point, where the survey of the district's boundary began, is at Alexandria.

The rest of the land was farms, and the main crop was tobacco. Five men owned the ground where the National Mall is now. The Histories of the National Mall project at George Mason University reports that these men enslaved people, and the enslaved men and women raised tobacco, corn, wheat and cattle on the land.

The largest of the five farms was Notley Young's tobacco plantation, a big farm worked by enslaved people. It covered 800 acres, and the Smithsonian's National Museum of African American History and Culture stands on part of it. The museum gives the number of people enslaved there as 245, from the 1790 census, the government's count of the people. The Histories of the National Mall project gives 260. Federal officials bought Young's land in 1791.

In all, nineteen men owned the land for the new city, about 6,000 acres. Washington met them at a dinner in Georgetown, and they signed an agreement on March 30, 1791.

The deal. The owners got nothing for the land that became streets. For land taken for public buildings and squares, they were paid by the acre. The rest of the land was split into building lots, and half of the lots went back to the owners. These men were later called the Original Proprietors, which means the owners of the land when the city was laid out.

Before anyone could build the city, a crew had to survey the boundary of the district, measuring the ground and marking exactly where its corners and edges lay. Someone also had to draw the plan of the city that would go inside it.

Andrew Ellicott ran the boundary survey. Benjamin Banneker, a free Black farmer from Baltimore County, Maryland, worked under him and did the survey's astronomy, the science of the stars and planets. Pierre L'Enfant, an engineer born in France, drew the city plan in 1791, and Washington dismissed him the next year.

Pierre L'Enfant

Who: The French-born engineer who designed the plan of Washington, D.C., and was dismissed before it was built.
When and where: The federal district, 1791 to February 1792.

L'Enfant would not hand over his drawing, so Washington fired him.

Pierre Charles L'Enfant drew the plan of the capital in 1791. He laid down a street grid and then drew diagonal avenues across it that meet at circles and squares.

The three commissioners Washington had appointed under the Residence Act asked for L'Enfant's original drawing, so that an engraver could cut it into a printing plate and print copies. L'Enfant refused to hand it over, and he clashed with the commissioners. In February 1792 George Washington dismissed him for insubordination, refusing to obey the people in charge of him.

Andrew Ellicott, with help from his brother Benjamin Ellicott, rebuilt the plan and changed parts of it. He straightened Massachusetts Avenue and took out some of the squares. The engravers Thackara and Vallance made an engraving of Ellicott's version in March 1792. Copies of their engraving were the first plan of the city that many people saw, and the city was built from it: L'Enfant's plan as Ellicott had changed it.

L'Enfant died poor in 1825. He was buried on a farm in Prince George's County, Maryland, that belonged to his friend William Dudley Digges.

A second burial. In 1908 the members of Congress set aside $1,000 for a new grave. On April 28, 1909, his coffin lay at the US Capitol for three hours, so that people could pay their respects. He was the first person from another country to be given that honor. That afternoon he was buried again at Arlington National Cemetery, on a hillside above the city he had planned.

Benjamin Banneker

Who: A free Black tobacco farmer and self-taught astronomer who made the observations that fixed the starting corner of the federal district.
When and where: The survey's base camp at Jones Point, Virginia, February to April 1791.

At nearly sixty, Banneker worked nights, timing the stars.

George Ellicott, Andrew Ellicott's cousin and Banneker's neighbor and friend, recommended him, and Major Andrew Ellicott hired him as scientific assistant on the survey of the ten-mile square. Banneker was nearly sixty. He farmed tobacco, and he had taught himself astronomy.

Banneker worked at night. To find a position on the earth by astronomy, an observer watches particular stars cross a particular point in the sky and records the exact moment, then calculates the position from those times. Banneker made those observations and did those calculations, and he kept the survey's regulator clock, a very exact clock, running true.

He used his observations and calculations to fix the district's south corner at Jones Point. He was paid $2 a day, an ordinary day's pay for an assistant. Andrew Ellicott was paid $5.

Banneker left the survey in the spring of 1791, before the crew had run the full forty miles of the boundary. He was in poor health. His first almanac, a book printed once a year with a calendar and tables of the sun, moon and stars, was the one for the year 1792.

According to a popular story, Banneker redrew the whole plan of Washington from memory after L'Enfant was dismissed. The story is a legend. Banneker had left the project about ten months before L'Enfant was dismissed, and Andrew Ellicott rebuilt the plan. The historian Silvio Bedini and writers for the National Park Service and the Library of Congress have worked through the records. They credit Banneker with the astronomy for the boundary survey.

Water after the fevers

Yellow fever killed about 5,000 people in Philadelphia between August 1 and November 9, 1793. About 50,000 people lived in the city, so the fever killed roughly one of every ten of them. About 20,000 other people fled the city, among them the officials of the new federal government.

Yellow fever comes from a virus that mosquitoes carry from person to person. Most people who catch it get better within a week. In about one case in seven, the illness comes back worse and the virus destroys the liver, so the skin and the whites of the eyes turn yellow. The person bleeds from the mouth, nose and eyes and vomits blood turned black by stomach acid, which people then called the black vomit. Between 30 and 60 percent of these people die.

What people blamed. Nobody then knew that mosquitoes spread the fever. Many Philadelphians thought it came from dirty water or from rotting matter in the streets, so they wanted clean water for the city. In 1797 a group of them asked the city's councils to take charge of the water supply, and the council

members formed a Watering Committee. The fever came back in 1798. The city's leaders then hired Benjamin Henry Latrobe, an English-born architect and engineer, to design a waterworks.

The waterworks. The members of the Watering Committee accepted Latrobe's design in 1799, and workers began building the Centre Square Water Works that year. Nicholas J. Roosevelt built its engines and pumps under a contract worth $33,000. The steam engine went up from February 1800.

The steam pumps started supplying the city on January 21, 1801. The engines pumped water from the Schuylkill River through a tunnel along Chestnut Street to the pump house, a white marble building on Centre Square. From there the water ran through the streets in pipes made of hollowed-out logs. Philadelphia was the first American city with a public water supply driven by steam.

What the water could not do. Clean water could not stop a disease that mosquitoes spread. The fever came back seven times in the twelve years after 1793. Each time, only the winter frost, which killed the mosquitoes, brought it to an end.

The engines broke down often and cost a great deal to keep running. Centre Square ran for only fourteen years. In 1815 the Fairmount Water Works took its place.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 7 MINUTES

Towns beside water that boats could reach grew into cities in a few years, on land that had belonged to Native nations. In New York, three commissioners laid a grid of straight streets over Manhattan's hills. After a deadly disease and a great fire, the city's leaders had clean water brought in from a river to the north.

Boomtowns

Rochester, Chicago and San Francisco were all boomtowns between 1800 and 1850, towns whose number of people grew very fast, in a few years.

The City of Rochester's own history and the Monroe County Library's Rochester Voices project both describe Rochester as America's first boomtown. The land there was Seneca land. In 1788 Seneca leaders signed a treaty, a written agreement between nations or governments, that handed about 2,600,000 acres east of the Genesee River to two land dealers, Oliver Phelps and Nathaniel Gorham.

The Seneca leaders also granted land west of the river for a mill, on the condition that a mill be built for the Seneca to use. Ebenezer Allen built a sawmill and a mill for grinding grain on that tract in 1789.

The Erie Canal ran through Rochester from 1825, and workers finished the whole canal across New York that October. The falls of the Genesee River turned the mills' wheels. Grain from the Genesee Valley came to the mills, and canal boats carried the flour away. Before the canal, shipping a ton of flour from Rochester cost $100. By canal it cost $7. By 1838 the mills at Rochester ground more flour than the mills of any other American city, and Rochester got the nickname the Flour City.

About 1,500 people lived in Rochester in 1820. Census takers, who count the country's people every ten years, counted 9,269 there in 1830. By 1834, the year Rochester became a city, about 12,630 people lived there. That was more

than eight times as many people as in 1820.

Chicago grew where the Chicago River runs into Lake Michigan, beside Fort Dearborn, an army fort built at the river's mouth in 1803. It became a town in 1833, when about 350 people lived there.

That September, in the Treaty of Chicago, Potawatomi leaders signed over about 5,000,000 acres: their last lands in Illinois and Wisconsin, and their last reservations, or lands kept for them, in Michigan. The Potawatomi leaders agreed to move west of the Mississippi River. Chicago became a city on March 4, 1837, when 4,170 people lived there.

For most of the year Chicago's streets were mud, until crews began in 1855 to raise them over sewers, underground pipes that carry waste water away.

San Francisco grew on the homeland of the Yelamu, an Ohlone people. About 200 people lived in the town in 1846, and by 1852 about 36,000 lived there. They had come in the California Gold Rush, a race to California to dig for gold.

People put the town together in a hurry. They lived in tents and in hulks, the hulls of old ships that no longer sailed. Six great fires burned the town between December 1849 and June 1851, and the fire of May 3 and 4, 1851, destroyed as much as three-quarters of it. People rebuilt after each one.

A grid laid over Manhattan's hills

New York's state lawmakers appointed three commissioners on April 3, 1807, and gave them the job of laying out the streets of Manhattan. They were Gouverneur Morris, John Rutherfurd and Simeon De Witt, and they filed their plan in March 1811.

The commissioners laid the streets out as a grid. In the remarks they filed with the plan, they gave their reason: "straight-sided and right-angled houses are the most cheap to build and the most convenient to live in." Their plan covered Manhattan from Houston Street north to 155th Street. It had twelve avenues and 155 numbered cross streets, which cut the land into rectangles.

The commissioners drew the lines straight over the island's hills without bending them. Manhattan was Lenape land, and by 1811 much of this part of it was rocky hills, swamps and marshes, with farms, houses and barns scattered across it.

The chief surveyor, John Randel Jr., was hired for the job at twenty. In 1808 he mapped the island's hills, rocks, swamps, streams and ponds, and its houses, barns, fences and fields. Then his crews set 1,549 marble markers and 98 iron bolts into the ground to fix the grid in place.

Workers then had to make the island match the drawing. To build each street, they dug away hills or filled in hollows, leveled the strip of ground and paved it. The new streets ran through the spots where about 721 of the 1,825 buildings north of Houston Street stood in 1811, so those buildings had to be moved out of the way. The work took about sixty years to reach 155th Street.

The commissioners set aside very little land for parks, and Central Park came only later, on ground where the people of Seneca Village already lived.

How New York got clean water

Cholera causes watery diarrhea so heavy that the body loses its water very fast. A person with a bad case who does not get water back into the body fast enough can die of it within hours. In 1832 cholera killed about 3,500 out of about 250,000 New Yorkers, about one person in every seventy.

New York had no clean piped water then. People drew water from wells, ponds and springs inside the city. Workers for a private firm, the Manhattan Company, sank wells and laid wooden pipes to send the water out. Carters drove barrels of country spring water around town and sold it for a penny a gallon, and that water was often dirty too. Others caught rain in tanks called cisterns.

The water from the downtown wells was dirty by the 1830s. More and more privies, outdoor toilets built over pits in the ground, stood near the wells, and liquid from old graves soaked into the ground as well. When people sick with cholera used the privies, the waste overflowed into nearby wells and cellars.

A fire started in a five-story warehouse on Merchant Street, near Wall Street in lower Manhattan, on the evening of December 16, 1835. A strong wind blew it toward the East River. It burned about seventeen blocks and destroyed between 600 and 700 buildings, and the damage came to about $20 million. Two people died in it.

No water to fight it. The firefighters could not get enough water onto the fire. There was little water to be had, and what there was froze in the wells, in the hoses and in the hydrants, the street pipes that firefighters draw water from.

The Croton Aqueduct, a channel built to carry water a long way, became the first large city water supply in the United States that ran without a break. New York's city leaders had decided by the late 1820s that they needed a cleaner and bigger supply of water. After the cholera and the fire, they had the aqueduct built. Workers began building it in 1837, with John B. Jervis as chief engineer.

The workers dammed the Croton River in Westchester County, north of the city. Then they built a covered tunnel of brick on a stone base, forty-one miles long, to carry the water down to Manhattan. Inside, the tunnel was eight and a half feet high and seven and a half feet wide. It dropped about 13 inches in every mile, so gravity carried the water along it.

Water entered the city on June 22, 1842. On July 4 the distributing reservoir, a large basin that held the water and fed it out to the city's pipes, began filling at 42nd Street and Fifth Avenue.

New Yorkers held a celebration across the whole city on October 14, 1842. A parade seven miles long marched through the streets, and the new fountain in City Hall Park threw water fifty feet into the air. People called a glass of the new water a "Croton cocktail," though it had no alcohol in it.

George Templeton Strong

Who: A young New York lawyer whose diary, kept for forty years, records the coming of Croton water.
When and where: New York City, 1842.

In 1842 Strong doubted that even Croton water could clean New York.

George Templeton Strong was twenty-two when the Croton water arrived. He kept a diary from 1835 until his death in 1875, about 2,250 pages in all. It is a first-hand record of New York in the 1800s, and two editors, Nevins and Thomas, published it in 1952.

While the water was still on its way in 1842, Strong wrote that "Croton Water is slowly flowing towards the city, which at last will stand a chance of being cleaned." The sentence goes on: "if water can clean it." He thought the city was so dirty that even water might not get it clean.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 20 MINUTES

Builders made American cities taller in these fifty years, dug sewers under them, and spread them outward along streetcar lines. In Chicago, crews raised the streets over new sewers and lifted the buildings to match. Census takers counted 1,099,850 people in Chicago in 1890, which made it the second-largest city in the country.

In New York in the same years, poor families crowded into tenements, apartment buildings put up cheaply to hold as many families as a plot of land would take.

How crews lifted Chicago out of the mud

Rain and waste had nowhere to drain off Chicago's low, almost level clay ground, so pools of dirty water stood in the streets, and for most of the year the streets were mud. Diseases spread from that standing water, among them typhoid fever, a fever caught from food or water dirtied by human waste, and dysentery, a gut infection that causes bloody diarrhea. In 1854 cholera killed six of every hundred people in the city.

The members of the Board of Sewerage Commissioners, the officials in charge of Chicago's sewers, hired Ellis S. Chesbrough in December 1855. He had been the water engineer in Boston. His plan of 1856 was the first plan for a complete sewer system covering a whole American city.

Streets on top of the sewers. Workers laid the sewer pipes along the existing streets and then buried them under several feet of new fill, earth dumped on top. That raised the level of the street by an average of four to five feet, and by as much as eight feet in places. A building left at the old level now sat below its own street, and its ground floor became a basement.

Next the buildings had to come up to the new street level. The tool for lifting them was the jackscrew, a screw inside a case that a man turned by hand with a lever. Crews set many jackscrews under a building, and the men turned them together.

The first masonry building, one built of brick or stone, went up this way in January 1858. It was a four-story brick block at Randolph and Dearborn streets that weighed about 750 tons. It rose 6 feet 2 inches on 200 jackscrews. Reporters of the time wrote that it rose "without the slightest injury" to the building. More than fifty buildings like it were raised that same year. Every figure for these raisings comes from those newspaper reports.

A group of engineers that included George Pullman raised half a city block in 1860. It was the row of brick and stone buildings on Lake Street between Clark and LaSalle streets, 320 feet long. With its sidewalks, the row weighed about 35,000 tons. About 600 men worked about 6,000 jackscrews, and the block went up 4 feet 8 inches in five days. The shops in it stayed open the whole time, and people came and went, shopped and worked in them while the buildings rose.

Up: the elevator and the steel frame

Builders could not make tall buildings without two inventions: an elevator people trusted, and a frame that could hold the building up.

The danger in an elevator was that the rope could break and drop the platform. Elisha Graves Otis made a brake from a steel wagon spring that locked into a ratchet, a bar with teeth, and held the platform if the rope broke.

The test. Otis showed his brake in 1854 at the Crystal Palace exposition in New York, a large public show of new machines and goods. He rode a platform up an open-sided shaft, and halfway up he had the rope cut with an axe. The brake caught, and the platform held.

The first passenger elevator sold for everyday use in a building went into the E. V. Haughwout Building, at 488 Broadway in New York, on March 23, 1857. A steam engine drove it. It cost $300 and rose about two-thirds of a foot per second.

Rent on the top floor. Before elevators, the higher a room was, the lower its rent, because people had to climb the stairs to reach it. Once elevators ran, floors above the fourth or fifth could rent for as much as the floors below, or more.

In a wall-bearing building, the outside walls hold up everything. The taller the building, the thicker the walls at the bottom have to be. The Monadnock Building in Chicago, finished in 1891, is 16 stories tall, the tallest building ever held up by brick walls. Its walls are six feet thick at the bottom. In a framed building, a skeleton of iron or steel columns and beams carries the weight. The walls hang on that frame as a covering.

The Home Insurance Building in Chicago is the building most often called the first skyscraper, a very tall building held up by a metal frame. William Le Baron Jenney designed it, and it was finished in 1885. It stood ten stories and 138 feet tall, and builders added two more floors in 1890. Its frame of iron and steel carried loads that walls had carried in older buildings. Workers tore it down in 1931 to make room for the Field Building.

That title is in dispute. The Home Insurance Building was not built all of steel, and it was not the first building with a metal frame. Jason Barr, who writes about the history of skyscrapers, traces the label "first skyscraper" partly to a later publicity campaign. People close to Jenney ran it, along with Chicago boosters, people who promoted their city to bring in money and business.

Other buildings have a claim to be first. One is New York's Equitable Life Building of 1870, the first office building with an elevator. A fire destroyed it in 1912 and killed six people, and the Equitable Building of 1915, about 40 stories tall, went up on the same ground. Whichever building came first, builders used the frame and the elevator to roughly triple the height of buildings within one generation.

BY 1893: Chicago's downtown stood on streets raised over its sewers, and its office buildings rose ten stories and more on frames of iron and steel.

The White City. Steel frames also held up the buildings of a world's fair, the World's Columbian Exposition, which opened in Chicago in 1893. Daniel Burnham directed the building work, and Frederick Law Olmsted drew the site plan, the layout of the grounds. Workers put up a matched set of white buildings in the style of ancient Greek and Roman buildings, with rows of columns. Electric lights lit them, and the fair got the name the White City.

The buildings were temporary, built to stand only while the fair ran. Under their fronts they were sheds on those steel frames. The fronts were made of staff, a mix of plaster, cement and plant fiber that workers poured into molds, and then painted white. More than 27 million people visited the fair in its six months. Planners in the City Beautiful movement of the 1890s and 1900s took the White City as their model. They worked to bring grand public buildings, wide avenues and parks into American cities.

High steel. Mohawk ironworkers built steel frames like these. From 1886 on, men from two Mohawk communities, Kahnawake and Akwesasne, helped build bridges and tall buildings across the Northeast. That year, at a bridge job at Kahnawake, a Mohawk town near Montreal, Mohawk men started working high steel, joining the steel frames of bridges and buildings high above the ground.

From Kahnawake and Akwesasne, the men traveled to bridge and skyscraper jobs wherever the steel work was, and they called that "booming out." An exhibition at the Smithsonian's National Museum of the American Indian, *Booming Out: Mohawk Ironworkers Build New York*, documents their work. It includes their work in the next century on the Empire State Building, the Chrysler Building, the George Washington Bridge and the World Trade Center.

The tenement

New York's state lawmakers passed a tenement law on May 14, 1867, the first broad housing law in the country. Under it, cellar apartments were banned unless the ceiling stood at least a foot above the street, and every building needed fire escapes and one toilet for every twenty people. The law also set the first rules for letting fresh air into rooms.

Officials did little to make builders obey it. To meet its rules about fresh air, builders cut windows in the walls between inside rooms, so the windows let in no outside air. The lawmakers tried again with the Tenement House Act of 1879, later known as the Old Law. Under it, every room a person lived in had to have a window that opened to outside air. Builders met that rule with the dumbbell tenement.

Builders pinched each building in at its middle, which left room for a narrow air shaft, and rooms in the middle of the building got their required window onto that shaft. Seen from above, the building was wide at both ends and thin in the middle, the shape of a dumbbell, an exercise weight with a ball at each end of a short bar.

The shaft was too narrow and too deep to bring daylight or moving air down below the top floors, so most of the rooms it was built for stayed dark and stale. The shafts filled up with garbage. The shaft also worked as a flue, a channel that draws hot air and smoke upward, as a chimney does. A fire in the shaft could climb it to the floors above.

New York's state lawmakers passed a new Tenement House Act in 1901, later known as the New Law. In it they banned the dumbbell design and required windows facing outward in every room, along with courtyards, indoor toilets and protection against fire. The rules covered old buildings as well, so owners had to change those too.

The White Wings. The reporter Jacob Riis wrote that one man's broom "saved more lives in the crowded tenements than a squad of doctors." The man was George E. Waring Jr. New York's reform mayor, William L. Strong,

appointed him street-cleaning commissioner in 1895, and Waring held the job until 1898. He put about 2,000 street sweepers in white uniforms, and people called them the White Wings. Each day they cleaned about 450 miles of street.

Waring set up the first planned systems for getting rid of the city's waste and for recycling it. He organized Juvenile Street Cleaning Leagues too, clubs for children who helped keep their streets clean.

Jacob Riis

Who: A Danish immigrant and police reporter who took up flash photography early and used it to show how the poor of a city lived.
When and where: New York's tenement districts. *How the Other Half Lives* came out in 1890.

Riis took his photographs at night, by the flash of burning magnesium.

Jacob Riis used flash photographs as evidence in an argument about how poor families in New York lived. He came to the United States from Denmark in 1870 and worked as a police reporter for the *New York Tribune* and then the *Evening Sun*. As a reporter he covered the city's poorest streets, among them Mulberry Street. The police there also ran shelters for homeless people, called police lodging houses, and Riis saw inside them and wrote about them.

Riis began using magnesium flash around 1887. Magnesium powder burns for an instant with a light bright enough to record a picture on a photographic plate, a glass sheet coated with chemicals that change where light hits them. With the flash he photographed tenements, saloons and slums at night.

Riis published an eighteen-page article in *Scribner's* magazine in 1889. The next year he published the book *How the Other Half Lives*, with seventeen photographs and a number of engravings, pictures printed from lines cut into metal or wood.

The *Dictionary of American History* states that Riis's work, along with the work of other reformers, pushed New York's lawmakers toward the tenement law of 1901, which banned the dumbbell design. Theodore Roosevelt, as New York City's police commissioner, became Riis's ally.

Bridget and Joseph Moore

Who: An Irish immigrant waiter and his wife, one tenement family the records document, whose apartment the Tenement Museum has recreated.

When and where: 97 Orchard Street, on New York's Lower East Side, 1869.

The Moores' water came from a backyard pump, beside the privies.

Bridget Moore married at seventeen, in 1865. Joseph Moore worked as a waiter, and some of the Tenement Museum's accounts record him as a bartender. In 1869 the couple moved with their daughters into 97 Orchard Street, a tenement built in 1863. They were one of the few Irish families in a building where most people were German, and they lived there about a year.

Their daughter Agnes Mary died on April 21, 1869, at five months old. The cause given in the records was marasmus, severe malnutrition, a form of starvation. The body gets too little food of every kind, so it uses up its own fat and then its own muscle, and the child wastes away.

The Tenement Museum's account describes two things linked to the deaths of babies at that time. One was swill milk, which came from cows kept in stables beside distilleries, where liquor was made. The cows ate the leftover grain mush from making the liquor. Their milk was thin and dirty, and in the 1850s New Yorkers blamed it for the deaths of many babies. The museum says Agnes Mary's marasmus came from tainted milk.

> The other was the building itself. The water at 97 Orchard Street came from a pump in the back yard, beside the privies.
>
> Of the eight children Bridget and Joseph Moore had, four lived to be adults.
>
> The building still stands, and museum staff have recreated the Moores' apartment as it was in 1869. Between 1863 and 1935, when the floors where people lived were sealed shut, an estimated 7,000 people from more than twenty countries lived at 97 Orchard Street. The Moores are one of the families among them whose names appear in the records.

Central Park and Seneca Village

The commissioners who drew the 1811 plan had left almost no land for public parks. In 1853 New York's state lawmakers set aside more than 700 acres in the middle of Manhattan for a park. Frederick Law Olmsted and Calvert Vaux won the design contest in 1858 with a plan they called Greensward, and building began that year.

People already lived on part of that land, in Seneca Village. John and Elizabeth Whitehead had divided their farmland there into about 200 lots, or pieces of land for sale, in 1825. On September 27 of that year, Andrew Williams, a twenty-five-year-old Black shoeshiner, bought the first three lots for $125. Epiphany Davis bought twelve lots for $578, and the African Methodist Episcopal Zion Church bought lots as well.

About 225 people lived in Seneca Village by 1855. Roughly two-thirds of them were African American, and Irish and German families lived there too. They had built three churches and a school.

Land and the vote. Under New York law at the time, a Black man could vote only if he owned at least $250 worth of property. In 1845 about 100 Black men in the whole city met that rule, and 10 of them lived in Seneca Village.

New York City officials took the whole park site by eminent domain, the power of a government to take private property for public use, paying the owner whether the owner wants to sell or not. Officials set a price for each piece of

land, and the owners in Seneca Village were paid. The histories of the village by the Central Park Conservancy, the group that looks after the park, do not name the commissioners who set the prices.

Many of the owners said the price of their land was too low, and they objected in court. Researchers at the Conservancy have studied the records and found that, with few exceptions, the owners got about what their land would have sold for.

The New York State Supreme Court, the state's main trial court, approved the prices in February 1856, and the land became the city's property. The officials had taken the land from its owners, including Black men whose right to vote depended on owning it.

Many people left in 1856, and others stayed on as renters of the city. In the summer of 1856, Mayor Fernando Wood sent the people of Seneca Village a final notice to leave. In October 1857, once the money to build the park was in place, city officials evicted the last of them, forcing them out of their homes under the law. In all, city officials forced about 1,600 people off the whole park site by the end of 1857.

Accounts differ on how the last people were made to leave. The historians Roy Rosenzweig and Elizabeth Blackmar studied the records from the time for their 1992 history of the park, and they found that the people left "quietly and without violence." The Conservancy's researchers say their work since the early 1990s has found no sign of a fight over the taking.

A New York Public Library research guide describes it differently. It states that the people of Seneca Village fought the police sent to put them out, and that in August 1857 the last Black landowners were "forcibly and violently scattered throughout the city." Rosenzweig and Blackmar think the stories of police clubbing people out mix up the taking of the park land with police raids on pig farms near the park's western edge two years later.

Crews tore down the buildings, and workers began building the park in 1858.

Frederick Law Olmsted

Who: The designer, with Calvert Vaux, of Central Park, whom the Olmsted Network, a group that studies his work, calls the founder of landscape architecture as a job in the United States.

When and where: New York from 1858, and parks in Brooklyn, Boston, Washington and Chicago afterward.

Movie: *Frederick Law Olmsted: Designing America* (PBS, 2014), a documentary that premiered June 20, 2014.

Editors once sent Olmsted through the slave states as a reporter.

Olmsted had been a sailor on a trading voyage to China. From 1848 to 1855 he ran a farm on Staten Island, and in 1852 he published a book called *Walks and Talks of an American Farmer in England*. That year editors at the *New York Daily Times* sent him as a reporter through the Southern states, where slavery was legal.

His travels through the South and Texas lasted until 1857, and his reports were later collected in three books. In them he argued that slavery had held the South back. By his count, the same amount of work took four times as long in Virginia as in the North, and the profits of slavery went to no more than 8,000 owners of large plantations.

He and Calvert Vaux won the Central Park contest in 1858 with the Greensward plan. They designed the land itself, its paths, lawns, trees, water and roads, which is the work called landscape architecture. Olmsted meant the park for everyone in the city, rich and poor. He and Vaux gave it meadows and woods shaped to look like open country, so that city people could get some relief from the city.

Olmsted's best-known designs after Central Park include Prospect Park in Brooklyn, the Emerald Necklace in Boston, a chain of connected parks, and the grounds of the US Capitol in Washington. In 1893 he laid out the grounds of the World's Columbian Exposition in Chicago, in Jackson Park.

Out: streetcar lines and streetcar suburbs

American cities of the 1800s were walking cities. Most people lived close enough to walk to their work and their shops. Horses pulled streetcars along rails in many cities, at 5 or 6 miles an hour.

Electric cars. Frank J. Sprague's Richmond Union Passenger Railway, an electric streetcar line in Richmond, Virginia, began running cars on a regular schedule on February 2, 1888. It started with ten cars on about twelve miles of track, and by May twenty-three cars ran on it. It was the first large electric street railway that worked well.

Electric streetcars ran at 10 to 15 miles an hour, so a ten-minute ride covered ground that took thirty minutes to walk. Within two years, about 110 electric railways using Sprague's equipment were built or under contract.

Builders put up houses along the new lines, out past the old edge of town, and many white city families moved out to them. These new neighborhoods of homes at the edge of a city, where people lived and from which they traveled in to work, were called streetcar suburbs.

A city in one day

The land run into Oklahoma, a race in which settlers rushed in at a set signal to claim pieces of land, began at noon on April 22, 1889. That morning, Guthrie was a railroad station. By nightfall about 10,000 people were camped there.

The land was part of Indian Territory, where United States officials had forced many Native nations to move from their homelands. Dennis N. Cooley was the federal Commissioner of Indian Affairs in 1865, the official in charge of the government's dealings with Native nations. He told the leaders of five of those nations that they had lost their rights to their land. His reason was that they had sided with the Confederacy, the Southern states that fought the United States in the Civil War.

In treaties signed in 1866, the Creek signed over the western half of their land for $975,168, and the Seminole signed over all of theirs for fifteen cents an acre. The treaties stated that the land was for other Native nations and for people freed from slavery.

The Unassigned Lands. The land opened in 1889, about 1.9 million acres, lay in the middle of the land the Creek and Seminole had signed over. It was called the Unassigned Lands, because federal officials had not assigned it to any nation. In early 1889, members of Congress voted to pay the Creek about $2.28 million and the Seminole about $1.91 million to drop the rule about who the land was for. Then settlers could claim it.

Congressman William Springer wrote the change to a law that let President Benjamin Harrison open the land. Harrison announced the opening on March 23, 1889.

By the morning after the run, the settlers in Guthrie had set up a temporary government, a land office where settlers filed their claims, a newspaper and three banks, all of them working out of tents. Oklahoma City filled up on the same scale on the same day.

BY 1900

If you remember one thing: In these fifty years American cities grew taller and spread farther out, and in Chicago crews raised the streets over new sewers and lifted whole buildings to match.

In 1850: Chicago's streets were mud for most of the year, buildings were held up by their own walls, and most city people walked to work.

In 1900: Chicago's streets ran on top of its sewers, frames of iron and steel held up buildings ten stories and taller, and electric streetcars carried families out to new suburbs.

Who lived where: poor families in New York crowded into tenements, city officials forced the people of Seneca Village off their land to make Central Park, and in 1889 settlers raced onto Oklahoma land that the Creek and Seminole nations had signed over in 1866.

Next: In the 1900s, officials wrote rules about what could be built in each part of a city, and white owners and judges kept Black families out of most neighborhoods.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 18 MINUTES

New York's officials wrote the first rules for a whole American city on what could be built where, and builders in New York went on to put up the tallest buildings in the world. Millions of people moved into American cities in these decades. In 1907 alone, more than a million immigrants came through Ellis Island in New York, and millions of Black Americans moved from the South to cities such as Chicago and Detroit.

White property owners and judges kept Black families out of most neighborhoods, and federal officials refused to back home loans in the neighborhoods where Black families lived. To build a secret city for the atomic bomb, Army officials took farmland in Tennessee from the families who lived on it.

Drinking water for Chicago and San Francisco

People in Chicago drank water from Lake Michigan, drawn in through intake pipes about two miles offshore. The city's sewers emptied into the Chicago River, and the river flowed into the lake. In 1885 a great storm carried the river's filth far out into the lake, and people feared it would reach the intakes. The Sanitary District of Chicago, the public body in charge of the work, was set up to solve the problem.

The Sanitary District's engineers made the river flow the other way. Workers dug a canal 28 miles long, the Chicago Sanitary and Ship Canal. Through it, the river now flowed away from the lake, toward the Des Plaines River and on down to the Mississippi, carrying the city's sewage with it.

Missouri lay downstream on the Mississippi. Sanitary District officials feared that Missouri's officials would ask a court for an injunction, an order telling someone to stop. Partly to finish before a judge could issue that order, crews blasted out the last barrier at dawn on New Year's morning, and the reversal was complete on January 2, 1900.

Missouri's state officials sued the Sanitary District that year. A court later dismissed the case because Missouri could not prove that the river's pollution came from Chicago. The sewers of St. Louis also emptied that city's waste into the Mississippi.

Stephenson Strobel, a health researcher at McMaster University in Canada, estimated in a 2025 study that the reversal cut the number of deaths in Chicago by about 4 percent in 1900. That is about 985 people who would otherwise have died that year.

An earthquake broke the main pipes from San Francisco's reservoirs on April 18, 1906. It also broke the water mains, the large pipes that carry water under the streets, in more than 300 places. When firefighters hooked their hoses to the hydrants, no water came out. The fires burned for about three days and destroyed more than four square miles of the city.

Hetch Hetchy. After the fire, city officials planned to dam the Hetch Hetchy Valley in Yosemite National Park and pipe its water to the city. Damming the valley meant flooding it, and conservationists, people who work to protect wild land, fought to stop the dam.

President Woodrow Wilson signed the Raker Act on December 19, 1913. Under it, San Francisco's city officials could build the dam, and no other American city has been allowed to build in a national park for its own use. The members of Congress who passed the Act wrote into it that nobody could make a private profit from that water and power.

Workers finished the O'Shaughnessy Dam in 1923, and by 1934 the full system was carrying water to San Francisco. City officials later sold the dam's electricity to a private company, Pacific Gas and Electric, and people fought in court for decades over whether that broke the Act.

Federal work crews. In the 1930s, federal crews laid new water and sewer pipes in cities across the country. They worked for the Works Progress Administration (WPA), a federal program that hired people to build public projects such as roads, pipes and schools, from 1935 to 1943. WPA crews laid about 24,000 miles of storm and sewer lines, which carry away rainwater and waste, and about 19,700 miles of water mains.

WPA crews built about 500 water-treatment plants, more than 4,000 new schools, hundreds of hospitals, and airports, and they built or improved about 651,000 miles of road. Counts of the WPA's work differ from one record to another, so each of these figures is close but not exact.

Zoning: rules for what may be built where

Under zoning, officials divide a city into districts and set rules for what may be built in each one. The rules state how tall a building may be, how close it may stand to the street, and what it may be used for. No American city had zoning rules that covered the whole city before 1916. That year, on July 25, New York's officials voted to adopt the first citywide zoning resolution in the country.

Popular accounts state that the Equitable Building, finished in New York in 1915, caused the resolution. It rose about 40 stories straight up from the sidewalk, with no setback, no place where the upper floors step back from the street. It cast a shadow of about seven acres on the streets around it and cut off the sun from the buildings next to it.

Jason Barr, who writes about the history of skyscrapers, states that officials had been preparing the resolution for years. In his account they would have passed it without the Equitable Building, which became the best-known example of what the rules were meant to stop.

The resolution's height rule depended on the width of the street. A building's front wall could rise straight up to a height set by multiplying the street's width by a fixed number. Above that height, the building had to step back from the edge of its lot each time it went higher. A tower could go up with no height limit if it covered no more than a quarter of the lot.

The wedding cake. Architects who followed that formula designed New York's skyscrapers of the next decades as stacked boxes, each one smaller than the one below, and people called the shape a wedding cake. The officials who wrote the resolution meant the setbacks to keep air and sunlight reaching the sidewalks.

Officials also used city laws to decide who could live where. In Louisville, Kentucky, a city law barred Black buyers from buying homes on blocks where most people were white.

William Warley led the Louisville branch of the NAACP, a group that fought for Black Americans' rights. He agreed to buy a lot in a mostly white neighborhood so that the law could be tested in court. In 1917, in *Buchanan v. Warley*, the justices of the Supreme Court all agreed that the Louisville law could not stand. After that ruling, local officials turned to zoning rules that did not name race.

Officials in the village of Euclid, Ohio, had banned two-family houses and apartment buildings from parts of the village, so families who could not afford a whole house of their own could not live there. In 1926, in the case *Village of Euclid v. Ambler Realty Co.*, the justices of the Supreme Court ruled that a local government had the power to zone. After the ruling, officials in cities across the country wrote zoning rules.

Daniel Burnham

Who: The Chicago architect who directed the building of the 1893 world's fair and then wrote the first plan for the whole of a big American city.

When and where: Chicago, from the 1893 world's fair to the *Plan of Chicago* in 1909.

Burnham spent about thirty months on Chicago's plan and charged nothing.

Daniel Burnham ran the building of the White City at the 1893 world's fair in Chicago, where the buildings were designed together as one planned city. After the fair, Burnham applied the same idea to a real city that already existed.

Burnham and Edward Bennett published the *Plan of Chicago* in 1909. The members of the Commercial Club of Chicago, a club of the city's business leaders, asked for the plan and raised about $80,000 to pay for

it. Burnham worked on it for about thirty months and charged nothing. The club spent more than $40,000 more to print 1,650 copies of the finished book.

It was the first comprehensive plan for a large American city, setting out what the whole city should become, with its streets, parks and waterfront planned together. In it, Burnham and Bennett called for the lakefront to stay open to the public, for rings of parks around the city and for wider streets. Much of Chicago's lakefront is still open to the public because of that plan.

KNOWN:	Many people know Burnham for the line "make no little plans," but people began to credit it to him only after he died in 1912.
NOT KNOWN:	No record from his lifetime has him saying or writing it.

Which tower would be tallest?

The builders of the Chrysler Building beat the builders of 40 Wall Street in 1929, in a race for the tallest building in the world. Then, in 1931, the builders of the Empire State Building went higher than both.

The hidden spire. The Chrysler Building's architect, William Van Alen, was racing the builders of 40 Wall Street. His workers put together a spire, a tall pointed top, 125 feet long. They built it in secret inside the building's fire tower, a shaft that ran up through the building, where nobody outside could see it. In October 1929 they raised the spire up through the roof and fixed it in place.

The spire brought the Chrysler Building to 1,046 feet, the official figure from the Council on Tall Buildings and Urban Habitat, though some popular accounts print 1,048. The tower at 40 Wall Street stood 927 feet tall. The Chrysler Building opened on May 27, 1930, and it was the tallest building in the world for eleven months.

The Empire State Building opened on May 1, 1931. It had 102 stories and stood 1,250 feet to the roof, taller than both. Its builders put it up in 410 days, for about $41 million. President Hoover turned on its lights by pressing a button in Washington.

The building opened during the Depression, the years of the 1930s when businesses failed and people lost their jobs. Through the Depression the building stood about half empty.

Mohawk ironworkers from Kahnawake and Akwesasne worked on the steel of both the Empire State and Chrysler buildings. From the 1920s to the 1960s, their families lived in a Brooklyn neighborhood that people called Little Caughnawaga, after an older spelling of Kahnawake.

Where Great Migration families could live

The Great Migration was the move of about 6 million Black Americans out of the South between 1910 and 1970. In its first ten years, from 1910 to 1920, the Black population of Chicago rose 148 percent. Detroit's rose 611 percent, to about seven times its 1910 size. Philadelphia's rose about 500 percent, and New York's roughly doubled.

Black families who arrived could not live wherever they chose. White property owners kept them out of neighborhoods with restrictive covenants, sections written into the deed of a house, the legal paper that records who owns it. A covenant forbade the owner to sell or rent the house to a Black buyer, and sometimes to Jewish or Asian buyers as well. Each later owner of the house had to obey it too.

Through the 1920s, white owners in Chicago and across the country signed covenants into their deeds, and judges enforced the covenants by ordering owners to obey them. Because of the covenants and the judges, Black families could rent or buy homes in only a few districts. In Chicago that was a strip of the South Side called the Black Belt, which its own residents named Bronzeville. In New York it was Harlem.

Kitchenettes. The number of people who needed rooms in those districts kept rising. Landlords in the crowded districts split their flats, or apartments, into smaller units: one apartment became several one-room units, each rented separately. Each unit had a hotplate, a small electric burner for cooking, and the families shared one bathroom. Bronzeville had more than 80,000 of these kitchenettes. For each square foot of floor, families in them paid more rent than white families paid in white neighborhoods.

The 1948 ruling. In the case *Shelley v. Kraemer*, in 1948, the justices of the Supreme Court ruled that courts could no longer enforce the covenants. The ruling did not stop owners, builders and lenders from refusing Black buyers on their own, and many kept refusing for years.

The maps with red lines

Federal officials set up an agency called the Home Owners' Loan Corporation in 1933, during the Depression, the hard times of the 1930s. Its staff worked with mortgages, loans to buy a house. They bought more than a million home loans from lenders and replaced each one with a new 15-year loan at 5 percent interest, easier terms than banks then gave. By 1936 they had swapped in new loans for about one in ten home loans outside farms.

The same staff made maps of 239 cities between 1935 and 1940, called "Residential Security" maps. They went through each city neighborhood by neighborhood and gave each one a grade for how safe they judged it was to lend money on the houses there.

Four colors. A was the best grade, shaded green on the map, followed by B in blue and C in yellow. D was the worst grade. The maps described D neighborhoods as "hazardous," and the staff shaded them red. From those red areas comes the word redlining: drawing a line around a neighborhood on a map and then refusing to lend money for houses inside it.

The appraisers, the staff members who judged each neighborhood, wrote notes that describe Black families moving in as "infiltration," a word for an enemy sneaking in. If Black families lived in a neighborhood, that all but guaranteed it a D.

What the scholars dispute. Scholars disagree about how much these particular maps changed who got loans. In a 2017 study, researchers at the Federal Reserve Bank of Chicago, a branch of the country's central bank, found lasting harm in the neighborhoods graded red.

Price Fishback, a historian of money and business, and three other scholars found in a 2021 study that the maps changed little about where the government lent. They also found that officials at the Federal Housing Administration, another federal agency, had already built their own way of redlining before these maps were drawn.

What both sides agree on. Federal Housing Administration officials insured home loans, promising to repay the lender if a buyer could not. They refused to insure loans in the neighborhoods where Black homeowners lived,

and they backed loans mostly on new houses, which were sold almost only to white buyers.

Researchers at the National Community Reinvestment Coalition compared the old maps with present-day data in a study published in 2018. They found that about 74 percent of the neighborhoods graded red are low-to-moderate income today, meaning the families there earn well below the typical family in their area. About 64 percent are majority-minority, meaning most of the people living there are not white.

Researchers at the University of Richmond scanned the maps for their Mapping Inequality project and put them online, so anyone can look up what the appraisers wrote about a street.

BY 1940: White owners and judges had kept Black families in a few crowded districts for twenty years, and federal staff had shaded nearly every neighborhood where Black families lived red on their lending maps.

Oak Ridge: a secret city on taken farmland

Brigadier General Leslie Groves, an Army officer, had just taken charge of the Manhattan Project, the Army's secret program to build an atomic bomb. On September 19, 1942, he approved taking about 59,000 acres along the Clinch River in eastern Tennessee, a site Army officials called "Site X." Officers of the US Army Corps of Engineers took the land in October 1942.

Who lived there. About 1,000 families lived on that land. Some farmed it, and others lived in its small crossroads communities: Wheat, Elza, Scarboro (also spelled Scarborough) and Robertsville, with New Bethel and New Hope also named. Many of the families had farmed the same ground for generations. Accounts of how many people lived there differ: the Tennessee Encyclopedia states roughly 4,000 people, and other accounts state more than 3,000.

To take the land, the Corps officers filed a paper called a declaration of taking. Under it, the government could take the land right away, whether or not the owners wanted to sell, and settle the price later. Army officials paid an average of $46.86 an acre.

Army officials told the residents they had to leave, sometimes only by notices nailed to their front fence posts. The Tennessee Encyclopedia's account of the removals does not name the men who put up the notices. Families who had lived there for generations got a few weeks to go, and within a year all the families were gone.

Most families accepted the price the government set, and a few sued in court over what their land had been valued at. About an eighth of Roane County's land and about a seventh of Anderson County's land went into the site.

Workers built a city for the Army on that ground from 1942. The site's official name was the Clinton Engineer Works, and the city built there is Oak Ridge. The plans were for about 13,000 people, but by 1945 about 75,000 lived there, which for a short time made it the fifth-largest city in Tennessee.

Alphabet houses. Workers built about 3,000 houses from cemesto, ready-made panels of cement and asbestos, a mineral fiber. The houses came in a few standard types named by letters, such as the A house and the B house, so people called them alphabet houses. The flat-tops, flat-roofed houses, came in pieces made ahead of time by workers for the Tennessee Valley Authority, a federal agency, and crews put them together on the site.

The city had schools, buses, a 300-bed hospital and a symphony orchestra. Fences and gates surrounded it, and people had to show a badge to get in or out.

Oak Ridge did not appear on any published map. Los Alamos in New Mexico and Hanford and Richland in Washington were built for the same program. The plants at Oak Ridge separated the uranium, a heavy metal, for the atomic bomb that American forces dropped on Hiroshima. Most of the more than 70,000 people who lived and worked there did not know what they were making until the bombs fell. They had promised not to talk about their jobs.

Colleen Black

Who: A teenager from Nashville whose family moved to a city left off every published map, and who spent the rest of her life there.

When and where: Oak Ridge, Tennessee, from 1944 until her death in March 2015.

At nineteen, she hunted leaks in pipes without knowing what they carried.

Colleen Black was born Colleen Rowan, one of more than ten children in a Nashville family. In 1944 the family moved to Oak Ridge for war work. They lived in a trailer camp. The ground was mud, with boardwalks laid across it, and people needed a pass to get through the gates.

At nineteen, Colleen worked as a leak detector at the K-25 plant. Her employer was Ford, Bacon & Davis, a company hired to do work at the plant. Her job was to find holes in the welded joints of pipes, the places where two pieces of metal had been melted together. She climbed over the pipes with a probe that sensed helium, looking for pinhole leaks.

Helium atoms are small enough to escape through a flaw too fine to see, so where the probe found helium, the weld was leaking. The plant had hundreds of miles of pipe. Even the gas's name was kept quiet: the workers were not supposed to say the word "helium." Years later she said, "I didn't know where those pipes were going, I didn't know what was going through them ... I just knew that I had to find the leak and mark it."

She learned what the plant had been separating only after American forces dropped an atomic bomb on Hiroshima, in Japan, on August 6, 1945. The bomb killed between 90,000 and 166,000 people in the city within two to four months. That count comes from the Radiation Effects Research Foundation, a research group that the United States and Japan run together and that studies the bomb's survivors. An American survey in 1946 recorded a lower count, 70,000 to 80,000. The exact number is not known.

She married a man called Blackie Black and stayed in Oak Ridge for the rest of her life. In 2005 she gave an oral history, a recorded interview about her life, to the Atomic Heritage Foundation's Voices of the Manhattan Project. The K-25 Virtual Museum includes her account, and she is featured in Denise Kiernan's 2013 book *The Girls of Atomic City*.

BY 1950

If you remember one thing: In these fifty years officials began zoning American cities, white owners and judges kept Black families out of most neighborhoods, and federal officials refused to back home loans where Black families lived.

- January 2, 1900: Chicago's river runs backward, carrying the city's sewage away from Lake Michigan, its drinking water.
- 1935 to 1940: federal staff grade the neighborhoods of 239 cities and shade the D neighborhoods red.
- October 1942: Army officials take the land of about 1,000 families in Tennessee to build Oak Ridge.

Next: After 1950, families moved out of the old city centers to new suburbs, and officials cleared whole neighborhoods for developers and ran highways through others.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 12 MINUTES

St. Louis lost about 60 percent of its people between 1950 and 2000, and Detroit lost nearly half. Families moved out of the centers of older American cities to new suburbs at their edges. Cities in the hot South and West grew fast. Phoenix, Arizona, went from about 107,000 people in 1950 to more than 1.3 million in 2000.

At the edges of metropolitan areas, cities together with their suburbs, builders put up the same kind of neighborhood again and again: the subdivision, a tract of land divided into lots for single-family houses. White families bought those houses with mortgages that federal officials insured, promising to repay the lender if a buyer could not. Federal officials refused to insure the same kind of mortgage in the neighborhoods where Black families lived.

In the old centers, officials cleared whole neighborhoods and cut freeways through others. Under the Housing Act of 1949, city officials labeled neighborhoods blighted, run-down enough to tear down, and took the property by law. They had the buildings torn down, sold the cleared land to developers, and forced at least 300,000 families out of their homes.

Federal officials paid for new freeways with money from a highway law of 1956. Highway planners ran many of those freeways through poor neighborhoods, most of them home to Black families and other people of color.

Urban renewal: clearing neighborhoods for developers

Members of Congress voted for the Housing Act of 1949. Under Title I, one section of that law, federal officials offered money to city governments for a program the law identifies as "urban redevelopment." People later called it urban renewal.

Once city officials had given an area the legal label blighted, they took the property in it by eminent domain, the government's power to take private property at a price that officials set. They had crews tear the buildings down. Then they sold the cleared land to private developers for much less than it had cost to take and clear, and federal money covered the gap. In 1954, in *Berman v. Parker*, the justices of the Supreme Court ruled that this was allowed.

Officials ran the program from 1949 to 1974, in more than 400 cities and on more than 1,200 projects, and crews tore down about 2,000 neighborhoods. Officials forced at least 300,000 families out of their homes, the lowest count the records support. Counting people instead of families, researchers estimate about 1.2 million.

Who was forced out. Black Americans were about 13 percent of the country's population in 1960, and they were at least 55 percent of the people forced out. These figures come from Brent Cebul's 2020 article in the Boston Review and from the Digital Scholarship Lab's "Renewing Inequality" project.

The writer James Baldwin said in a filmed interview in San Francisco in 1963 that urban renewal "means Negro removal," using what was then the usual word for Black Americans.

Why freeways ran through Black neighborhoods

President Dwight D. Eisenhower signed the Federal-Aid Highway Act on June 29, 1956. It provided about $25 to $26 billion for a system of 41,000 miles of freeway.

Highway planners laid many of the routes where land was cheap and where the people who lived there were least able to organize against them. Those were mostly poor neighborhoods of Black families and other people of color. Builders defended taking land in Black neighborhoods by saying it was the cheapest. It was cheap partly because of redlining: for years, lenders and federal officials had refused to lend money for homes in those neighborhoods.

Rondo, a neighborhood in St. Paul, Minnesota, is one well-documented case. About 80 percent of St. Paul's Black residents lived there. Between 1956 and 1968, crews built Interstate 94 through the neighborhood.

Counts of what Rondo lost differ. Guides from the Minnesota Historical Society state about 600 families. ReConnect Rondo and reporting in Sahan Journal state about 700 homes and 300 businesses, and they record that Rondo lost 61 percent of its residents. One in eight of St. Paul's Black residents lost a home to the building of Interstate 94.

Marvin Roger Anderson

Who: A young man whose family lost its home and its apartment buildings to a freeway, and who has spent his later years working to build a land bridge over it.

When and where: Rondo, St. Paul, Minnesota. His family learned of the freeway in 1958.

Crews loaded his father's townhomes onto flatbed trucks and moved them away.

Marvin Roger Anderson grew up in Rondo. His father built Rangh Court there, twelve two-story townhomes, in 1947 or 1948 (accounts give both years). To build them, his father got "the largest Federal Housing Authority mortgage backed loan ever given to nonwhite applicants in the state."

The family learned in 1958, the year Anderson finished Central High School, that the new Interstate 94 would run through their land. State officials condemned the buildings for the freeway, taking them by law, and called the homes "inferiorly constructed." Crews loaded the buildings onto flatbed trucks and moved them away. Four of them still stand, on other ground, on the University of Minnesota's St. Paul campus.

Anderson became the State Law Librarian of Minnesota, the librarian in charge of the state's law library. In 1983 he and Floyd Smaller founded Rondo Days to keep alive "the history, culture and memories of the Rondo that existed before the highway."

He now leads ReConnect Rondo. Its members propose to build a lid over the freeway, a land bridge of about 21 acres, with homes, businesses and community space on top. Early design studies for the land bridge list between 350 and 1,400 homes.

The *Voices of Rondo* oral-history collection records his account, and so do press interviews.

Sprawl, and the people who left the old centers

Levittown, New York, opened in 1947, and by 1951 its builders had put up more than 17,000 houses there. The builders, Levitt and Sons, mass-produced the houses, building the same house over and over. They split the work into 27 steps and trained each crew to do one step. The houses stayed in place, and the crews moved from house to house. At the busiest time, about 30 houses went up in a day.

The first houses rented for $60 a month, and a renter could buy one for $7,500 after a year. From 1949, the builders sold ranch houses for $7,990. The lease for the first houses stated that they could not be "used or occupied by any person other than members of the Caucasian race," meaning white people. In 1960 every family in Levittown was still white.

Builders at the edges of other cities copied the pattern, and the spreading of cities outward in subdivisions like these came to be called sprawl.

The older cities lost people in the same fifty years. Detroit fell from about 1.85 million people in 1950 to about 950,000 in 2000, a loss of nearly half. St. Louis fell from about 857,000 to about 348,000, a loss of roughly 60 percent, and it kept shrinking after that, to about 300,000 by 2020.

Real estate agents and speculators, people who buy property to sell it again at a profit, pushed white families to sell and move out, a practice called blockbusting. They frightened white owners into selling cheaply by telling them Black families were moving in. Then they sold the same houses to Black families for much more, often "on contract."

On contract. Under a contract sale, if a buyer missed a single payment, the seller could put the family out, and the buyer lost everything already paid. In Chicago's North Lawndale neighborhood, about 3,000 buildings were sold on contract. North Lawndale went from 87 percent white in 1950 to 91 percent Black in 1960.

At the same time, federal officials insured mortgages on the new houses at the edge, under rules they set for which loans they would insure. Under those rules, they refused to insure mortgages in the neighborhoods where Black families lived.

Jane Jacobs and Robert Moses

Who: Robert Moses, who built roads, bridges and parks across New York for decades without ever winning an election, and Jane Jacobs, a writer who helped defeat three of the projects planned for the city.

When and where: New York City, 1950s to 1968.

Movie: *Citizen Jane: Battle for the City* (2016), a documentary directed by Matt Tyrnauer, 92 minutes.

Police arrested Jane Jacobs at a public hearing in April 1968.

Robert Moses held as many as twelve appointed posts at the same time, including parks commissioner and head of the Triborough Bridge and Tunnel Authority. Officials gave him every one of those posts, and he never won an election to any of them.

From those posts he directed the building of 13 bridges, 2 tunnels, 658 playgrounds, 17 state parks, and hundreds of miles of parkways and expressways, wide roads for fast traffic. To build the Cross-Bronx Expressway, Moses's crews cut a path through crowded blocks of the Bronx and forced thousands of families out of their homes.

Jane Jacobs lived in the West Village in New York and wrote about cities. She had never been trained in a school as a city planner. In 1961 she published *The Death and Life of Great American Cities*. In it she wrote that a busy street with both homes and shops stays safe because of "eyes on the street," the people who use it and watch it.

Three fights. Jacobs helped defeat three projects. In 1958 she and other opponents stopped a road planned through Washington Square Park. In 1961 and 1962 they beat city officials' attempt to label their own West Village a slum and clear it under urban renewal.

The third was the Lower Manhattan Expressway, a road ten lanes wide, raised above the street, planned across what is now SoHo and Little Italy. Jacobs led the Joint Committee to Stop the Lower Manhattan

Expressway. Police arrested her at a public hearing in April 1968. The charges included inciting to riot, urging a crowd to violence, and they were later reduced.

Jacobs moved to Toronto in 1968. In 1969 Mayor John Lindsay declared the expressway "dead for all time," and city officials took it off the city's map that year.

Robert Caro's book about Moses, *The Power Broker*, came out in 1974 and won a Pulitzer Prize in 1975. It is the standard account of Moses's career, and in it Caro criticizes much of what Moses did. In 2007 the historians Hilary Ballon and Kenneth T. Jackson edited a book, *Robert Moses and the Modern City*. In it they give parts of his record, especially his parks and swimming pools, more credit than Caro gives them. The argument is not settled.

Carl Stokes

Who: The first Black mayor of a major American city, elected in the years when officials were clearing neighborhoods and building freeways through cities.

When and where: Cleveland, Ohio. Elected on November 7, 1967, and mayor from January 1, 1968, to 1971.

In 1969 oil and trash on Cleveland's river caught fire.

Carl Stokes was the grandson of a man who had been enslaved. His mother was a widow, and she raised him in the Outhwaite public housing in Cleveland, apartments owned by a government and rented to families with low incomes.

Cleveland's voters elected Stokes mayor on November 7, 1967. He won about 50.5 percent of the vote and defeated Seth Taft. The 1960 census had counted 876,050 people in Cleveland, the eighth-largest city in the country.

Because Cleveland was one of the ten largest cities, Stokes counts as the first Black mayor of a major American city. On the same day, voters in Gary, Indiana, a much smaller city, elected Richard Hatcher their mayor. Stokes was re-elected in 1969 and served until 1971.

He had campaigned on housing, on bringing money into neighborhoods that had gone without it, and on the city's pollution. On June 22, 1969, an oil slick and piled-up trash on the Cuyahoga River caught fire under a railroad bridge, set off by sparks from a passing train. It burned for about twenty minutes and damaged two railroad bridges. It was at least the thirteenth fire on that river since 1868, and for a century people had used the working river as a sewer.

The next day Stokes held a press conference on the riverbank and demanded that state and federal officials act. In October 1972 the members of Congress passed the Clean Water Act over President Richard Nixon's veto, his refusal to sign it. Members of Congress can pass a law over a veto when two-thirds of each house vote for it. Stokes was one of the local leaders who had pushed for the Clean Water Act.

Black mayors followed Stokes in other big cities. By 1972 voters had elected Kenneth Gibson in Newark, James McGee in Dayton and Ted Berry in Cincinnati. In 1973 they elected Tom Bradley in Los Angeles, Maynard Jackson in Atlanta and Coleman Young in Detroit.

Growth in the Sun Belt

While Detroit and St. Louis lost people, cities in the Sun Belt, the warm states of the South and the West, gained them. Phoenix, Arizona, had about 106,818 people in 1950 and 1,321,045 in 2000, and its metropolitan area went from about 323,000 people to about 3,014,000 in the same fifty years.

Phoenix grew in area too. It covered about 17 square miles in 1950, more than 475 square miles by the 2000s, and about 518 square miles now. City officials added that land by annexation, moving the city's legal boundary out to take in neighboring land.

Houston, Texas, went from 596,163 people in 1950 to 1,953,631 in 2000, and by 1990 it was the fourth-largest city in the country. Air conditioners cooled the air inside homes and offices, so people in Phoenix and Houston could live and work indoors through the hot summers. Land at the edge of these cities was cheap, and city officials kept annexing it. In the 1950s alone, Phoenix more than quadrupled in population.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 7 MINUTES

Downtowns in most big American cities gained people in the 2000s, and in the 2020s many of their office towers began to empty. In the neighborhoods that buyers and lenders have returned to, rents and prices have risen, and some of the people who lived there can no longer afford to stay. Cities run hotter than the land around them, and in Miami Beach the sea now floods some streets at high tide.

Downtown living and empty offices

Researchers at the Brookings Institution studied census counts for big American cities. Between 2000 and 2010, the downtowns grew faster than the areas around them. Of 53 downtowns they studied, 42 gained people. Young adults had already begun moving in. By 2000, people aged 25 to 34 made up nearly a quarter of downtown residents, up from 13 percent in 1970.

After the coronavirus pandemic of 2020, many office workers kept working from home for some or all of the week. Companies shrank the office space they rented, and as their leases, the contracts for renting the space, ran out, they gave up floors.

A fifth of the space. By 2024 about 20 percent of American office space stood empty, with figures for that year running from about 19.6 to 20.1 percent. That was the highest share since Moody's Analytics, a research company, began keeping track in 1979, and it came to more than 900 million square feet of empty floor. In downtown Manhattan the share was about 23 percent. Axios and ABC News reported these Moody's figures in January 2024.

Builders began turning empty office towers into apartments. In 2024 they had plans under way to make more than 55,300 apartments out of former office buildings, a record and more than four times the number in 2021. The

Washington, D.C., area led, with 5,820 apartments planned. Researchers at the firm RentCafe counted those 2024 plans, as Time reported in January 2024.

Builders had finished, or were working on, 11.8 million square feet of former office space as apartments in 2025, the highest figure on record, according to industry reports from CBRE and Brevitas.

Gentrification, and who can no longer afford to stay

In gentrification, buyers and lenders return to a neighborhood they had avoided for years. New owners fix up the houses, landlords and sellers raise rents and prices, and the people who lived there through the years without investment can no longer afford to stay.

Buyers and lenders have returned to close-in neighborhoods, the ones near downtown, that appraisers had marked red in the 1930s. The work of the National Community Reinvestment Coalition and the Mapping Inequality project documents that pattern.

Washington, D.C., is one documented case. Census figures record the city as 71 percent Black in 1970, when people called it Chocolate City. By 2011 the Black share of the population had fallen below half, to 49.2 percent, for the first time in about fifty years.

U Street. New money went mostly into Shaw, U Street and Columbia Heights, close-in neighborhoods that appraisers had redlined. Around U Street, builders put up more than 2,000 expensive condominiums (apartments that each owner buys) and apartments between 1997 and 2007. The neighborhood went from about 60 percent Black in 2000 to about 20 percent Black in 2010.

Virginia Ali

Who: A co-founder of Ben's Chili Bowl, a diner on U Street in Washington, D.C., who has helped run it since 1958 while the neighborhood around it changed.

When and where: 1213 U Street NW, Washington, D.C., from August 22, 1958.

Ben Ali wrote "Soul Brother" in soap on the diner's front window.

Virginia Rollins came from Chance, Virginia. Her husband, Mahaboob Ali, who went by Ben, had come from Trinidad. On August 22, 1958, the two of them opened a diner at 1213 U Street NW with a bank loan of $5,000.

People then called U Street "Black Broadway." Hundreds of Black-owned businesses, churches and theaters lined it and the streets nearby. At the Lincoln Theatre and the Howard Theatre, Duke Ellington, Louis Armstrong and Billie Holiday performed.

Martin Luther King Jr. was shot and killed in Memphis, Tennessee, on April 4, 1968. James Earl Ray later pleaded guilty to his murder. For the next four days, riots broke out across Washington, and the corner of 14th and U was at the center of them. Thirteen people died, and about 1,200 homes and businesses were damaged.

Open past curfew. Stokely Carmichael led the Student Nonviolent Coordinating Committee, a civil rights group. He asked the Alis to keep the Bowl open past the curfew, an order to stay off the streets at night. According to the shop's own history, they did, and they fed police officers and activists. Ben wrote "Soul Brother" in soap on the front window. Like many other Black business owners, he wrote it to mark the diner as Black-owned and keep rioters from wrecking it, and the diner came through the riots untouched.

> Drug dealing at 14th and U rose sharply in the mid-1970s. From 1986 to 1991, crews digging the Metro, Washington's subway, tore up U Street. From the 1990s, buyers and lenders returned, and the new condominiums went up around the Bowl.
>
> Ben's is still open, and Virginia Ali, in her nineties, is still one of the people who run it. When the diner turned 65 in 2023, she told a reporter: "We really mean something to this community."

Building for heat and water

Daytime air in a city is 1 to 6 degrees Fahrenheit warmer than in the areas around it, according to the Environmental Protection Agency (EPA). At night the air can be up to 22 degrees warmer, and a roof or road in the sun can be 50 to 90 degrees hotter than the air above it.

Why cities run hot. The EPA's guides state the reasons. Roads, sidewalks and buildings soak up much of the sun's heat and hold it. A city has fewer trees and plants than the country around it, so it gets less shade and less cooling from water drying off leaves. Tall buildings and narrow streets block cooling breezes. The EPA's list of ways to cool a city includes planting trees, growing plants on roofs, and building roofs and pavements that reflect sunlight.

In Miami Beach, Florida, streets flood on clear days with no storm, when the tide is at its highest. At those tides the ocean pushes the water of Biscayne Bay back up the storm drains, and bay water pours out of the drains into the lowest streets. Seawater also rises up through the soft, holey limestone under the city. In the fall of 2014, crews began raising streets, including West Avenue and Sunset Harbour, and putting in pumps to push the water back out.

Crews had put in twelve pump stations along the west side of South Beach by 2016 and 2017. The cost was first reported as about $400 million and later as about $600 million, as the project grew. During the King Tides, the name used in Miami Beach for the highest tides of the year, the raised and pumped blocks stayed mostly dry after the work.

The tallest buildings now stand in other countries

The tallest building in the world has stood in Asia since 1998. Chicago's Sears Tower, finished in 1974, was the last building in the United States to be the tallest in the world. It lost that title because of a rule about how to measure height.

In April 1996, members of the Council on Tall Buildings and Urban Habitat ruled that a spire designed as part of a building counts toward its height. Under that rule, the Petronas Towers in Kuala Lumpur, Malaysia, measured 1,483 feet to the tops of their spires, and in 1998 they became the tallest buildings in the world. The Sears Tower's roof and its highest floor that people used were both higher than those of the Petronas Towers.

The Burj Khalifa in Dubai, finished in 2010, stands 2,717 feet (828 meters). That is more than 900 feet taller than the tallest building in the United States.

Workers set the last piece of the spire on One World Trade Center, in New York, on May 10, 2013. That November, members of the Council on Tall Buildings and Urban Habitat ruled that its spire counts toward its height. The building opened in 2014 at 1,776 feet, a height its builders chose for 1776, the year of the Declaration of Independence.

One World Trade Center is the tallest building in the Western Hemisphere, the half of the world that holds North and South America. Mohawk ironworkers worked on its steel, as Mohawk ironworkers had on the towers of the 1930s and on the bridges before them.

THE THREE QUESTIONS, ANSWERED

1. **Why do so many American cities have straight streets that cross in a grid?** Town founders chose the grid on purpose, from Spain's town rules of 1573 to Penn's Philadelphia in 1682. The three men who laid a grid over Manhattan in 1811 wrote that square-cornered houses were "the most cheap to build and the most convenient to live in."

2. **How did people get clean water into crowded cities?** City leaders paid for big waterworks after epidemics and fires: steam pumps in Philadelphia in 1801, the Croton Aqueduct in New York in 1842, and a canal that turned Chicago's river around in 1900.
3. **Who decided which families could live in which neighborhoods?** White owners wrote rules called covenants into their deeds to bar Black buyers, and judges enforced them until 1948. Federal officials refused to back home loans where Black families lived, and after 1949 city officials cleared about 2,000 neighborhoods.

What changed: Builders went from carrying baskets of earth at Cahokia to raising steel towers, and owners, judges and officials wrote rules about who could live where.

Whose land: Every colonial town and boomtown in this chapter went up on a Native nation's land, from Seloy's Timucua town at St. Augustine to the Lenape homeland at Philadelphia and the Potawatomi lands at Chicago.

FIVE DATES TO KEEP

About 1100: Cahokia reaches its peak.

1573: Spain's king orders a plaza and a grid for every new Spanish town.

June 22, 1842: Croton water reaches New York City.

July 25, 1916: New York adopts the first citywide zoning rules.

1948: The Supreme Court rules that courts can no longer enforce covenants.

> **NOT KNOWN:** No builder's, planner's or ruler's name survives from Cahokia.

Where to go: The Tenement Museum at 97 Orchard Street in New York, where staff have recreated the Moores' 1869 apartment.

Next: Chapter 7, Home and Family Life. It asks who lived in these houses and apartments, and how.

Chapter 7: Home and Family Life

CHAPTER 7 OF 37 · HOME AND FAMILY LIFE · 10 ERAS · ABOUT 60 MINUTES

The American home changed shape in every era, and again and again people with power decided which families could have one, and which could stay together.

A Hidatsa woman named Buffalo Bird Woman told the story of her girlhood to a scientist, who published it in 1921. Home was her mothers' earth lodge on the Missouri River, with a floor more than forty feet across and a fire in the center. The women of the family owned it. Her father had four wives, all sisters, and she was taught to call all four her mothers. By 1921, she said, "We no longer live in an earth lodge, but in a house with chimneys."

This chapter describes American homes from the earth lodges and longhouses of the Native nations to the houses where parents worked and children went to school online in 2020. From era to era the houses change: the one-room colonial house, the log cabin, the city tenement, the sod house cut from the prairie, the bungalow and the suburban ranch house. As you read, watch four things: who lived together under one roof, who did the work of the house, whose land it stood on, and who decided which families could live there.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who lived together under one roof, from the longhouse to today?
2. Who did the work of keeping house, and did stoves, pipes and wires make it lighter?

3. How did lenders and builders keep Black families out of new neighborhoods, and what did those families do?

Five minutes? Read "New home loans, and the red lines" in 1900 to 1950, then the story of Clyde Ross in 1950 to 2000.

Can skip: "Where the log cabin came from", "Bungalows, and houses sold by catalog", "The TV in the living room". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 3 MINUTES

A Native household was built around kin, a person's relatives by birth or by marriage, and in many nations several related families lived under one roof. Each nation built the kind of house that suited its climate and the way its families got their food. Children learned to work by working beside adults, and old people lived in the same household as the young.

Who was in charge of a longhouse?

Women ran the longhouse among the Haudenosaunee, several Native nations joined together in one union. The head of each house was the clan mother. She was the eldest or most respected woman of her clan, a large group of families who come down from the same ancestor. Women owned the house's everyday goods. The women of a longhouse and their children all came down from one woman who lived long ago, and every family traced itself back through the mothers' side.

The longhouse was a long building with a frame of wooden posts, covered with bark, and it held many related families. Two families lived in each section of the house, one on each side of an aisle, and they shared one fire in the middle. By different counts, one house held 30 to 60 people, or 20 families or more.

A woman's daughters and granddaughters lived in the longhouse all their lives. When they married, their husbands came to live with them. A son moved out when he married, into his wife's longhouse, but he still belonged to his mother's.

The earth lodge

Among the Mandan, Hidatsa and Arikara nations on the upper Missouri River, women built the houses and owned them. Their house was the earth lodge, a dome made of timber and packed over with earth. An older woman directed the building, and women kept the lodge in repair.

One earth lodge held about 10 to 20 people, most often sisters and their families. Markers put up by the State Historical Society of North Dakota state that most lodges were 35 to 45 feet across, and that one lodge took about 150 trees to build. A lodge lasted about ten years, and then the women built it again. The large villages had more than 100 lodges each.

Pueblos, wigwams and tipis

Taos Pueblo is a town of many homes joined together and stacked on top of one another, the kind of Native town called a pueblo. It is the northernmost of the nineteen Pueblos of what is now New Mexico. Its people built their homes of adobe, earth mixed with water and straw. They poured the adobe into molds or shaped it into bricks and dried them in the sun.

The two main buildings, the North House and the South House, rise five stories high. By the Pueblo's own estimate, people built them between about 1000 and 1450, and families have lived in them ever since. People have lived at Taos Pueblo without a break for more than 1,000 years.

In the Eastern Woodlands, the forests of the eastern part of the continent, many families lived in wigwams. A wigwam had a frame of saplings, thin young trees, covered with bark or with woven mats. On the Plains, the grasslands in the middle of the continent, families lived in tipis covered with animal hides. A family could take a tipi down and pack it up, so the household could move with the hunt.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

Most of the Europeans who came to this land in the 1500s were soldiers, priests and sailors, so European families were few, and Native households went on living in their own towns.

The newcomers carried diseases that Native people had never had before, and those diseases made Native people sick and killed many of them. About 200,000 Timucua people lived in what is now northern Florida and southern Georgia before the Spanish came. After one outbreak of disease after another, only about 2,000 were left by the 1650s, 1 for every 100 who had lived there. No record counts how many died in the 1500s alone.

St. Augustine, built in Seloy's town

Chapter 6 tells how Pedro Menéndez de Avilés founded St. Augustine in September 1565, inside Seloy, a town of the Timucua people. The town's chief, also named Seloy, at first offered Menéndez his council house, the building where the town held its meetings, and the Spanish used it as their first fort. Within months the Timucua were attacking the newcomers to drive them away. Seloy's town covered more than 12 acres, and its remains lie under what is now the Fountain of Youth Park in St. Augustine.

Few of the newcomers came as families. Of the 800 or so colonists, 26 were women, and an unknown number were enslaved Africans, people the Spanish owned as property and forced to work without pay.

Most Spanish men in St. Augustine had no wife. Most of the men who did marry had wives who were not Spanish, and from 1565 on, many Spanish soldiers married Native women. The Timucua traced their families and their inheritance through the mother's line.

Church officials in the town wrote down the names of the church's members. Their lists are called the Cathedral Parish records, after the parish, the group of people who belong to one church. The records begin on January 24, 1594, and they are the oldest church records that survive in the United States. The

first marriage in them is dated February 2, 1594. The first baptism, the church ceremony that welcomes a person, often a baby, into the church, is dated June 25.

Teresa and Francisco Camacho

Who: One of the mixed Spanish and Native households of early St. Augustine.

When and where: St. Augustine, Florida, 1580.

A Spanish soldier, his Native wife, her sister and her son.

One household in St. Augustine in 1580 held four people: Francisco Camacho, his wife Teresa, Teresa's sister Catalina and Catalina's son Juan. Teresa was a Native woman.

Francisco had come to Florida in the 1560s from San Lúcar de Barrameda, a town near Cádiz in Spain. He was a soldier in a regiment, a unit of the army, and he also worked as a fisherman.

NOT KNOWN: Teresa's nation. The account of this household does not name it.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

At Plymouth in 1620, English colonists built their houses on the site of Patuxet, a Wampanoag village. The usual English house in Massachusetts had one room, where the family lived and worked, and many households also held servants, enslaved people and boys from other families learning a trade.

Why Patuxet's houses stood empty

An English sea captain named Thomas Hunt kidnapped 20 Native people from Patuxet, a village of the Wampanoag nation on Plymouth harbor, in 1614. He also took 7 from Nauset, another Native community, and carried them all to Spain to sell them as slaves.

Then, between 1616 and 1619, a disease killed most of the Native people on the southeast coast of what is now Massachusetts. Estimates of the deaths run from one-third of the people to as many as 90 percent.

One of the kidnapped Patuxet men was Tisquantum, also known as Squanto. He made his way home in 1619 and found the houses empty and the cornfields grown over. The English who founded Plymouth Colony arrived the next year and built their houses at Patuxet. By then he was one of the last of the Patuxet people.

NOT KNOWN: what the disease was. Researchers have proposed several deadly diseases, among them plague, yellow fever and smallpox.

One room, then two

The basic English house in Massachusetts had a single room, called the hall. Next to it stood the chimney bay, the part of the house that held the chimney, the front door and the stairs. The family cooked, kept warm, had light and did its work at the hearth, the open fireplace in the hall that gave the room its heat and light.

The next step up was the hall-and-parlor house, with a second room called the parlor. The hall was still for cooking and work. The parlor was the best room, and it usually held a bed. Colonists went on building one-room houses well into the 1700s.

Births, deaths, servants and apprentices

A New England mother had about 8 babies, on average, and about 6 of them lived to grow up. People in New England lived long lives for the time. In the Chesapeake, the colonies around Chesapeake Bay, about half of all children died before the age of 20. About 1 baby in 4 died as an infant, and about another 1 in 4 before 20. Most children in the Chesapeake lost a mother, a father or both before they reached their teens.

A household also held people who were not family. Indentured servants, who had signed a contract to work for a master for a set number of years, lived and worked in the house as members of the household. Through the 1600s, colonists owned more and more enslaved people, and they forced them to work in their households without pay.

Apprentices. At about 14, and sometimes younger, a boy could leave home for good. His parents and a master signed a written contract called an indenture, and under it the boy had to live with the master and work for him. He served until the age of 21, for as long as 7 years, and learned the master's trade.

The master took the place of a parent. In the indenture he promised to feed the boy, give him a place to sleep, train him in the trade and give him some schooling. About two-thirds of these contracts required the master to teach the

boy reading or writing. At the end, the young man got freedom dues, a payment of clothes, tools or money.

Mary Ring

Who: A widow in Plymouth Colony whose will and list of belongings record what one household owned.

When and where: Plymouth Colony, from 1629 or 1630. She died in July 1631.

Governor Bradford owed her 2 pounds, and Winslow owed her beaver.

After Mary Ring died in July 1631, her belongings were written down one by one, in a list called an inventory. It survives, and it records what one Plymouth household owned.

The inventory lists her books: a Bible, a psalm book with the songs from the Bible's Book of Psalms, and books called *A Plea for Infants* and *Garland of Virtuous Dames*. Her clothes came in violet, red, blue, grey, white and black, and in a mixed color. Her will lists green cloth "to make a coat."

The inventory also lists money owed to her. Governor William Bradford owed her 2 pounds in English money, written in the record as "Due from Gov." Edward Winslow owed her beaver fur, which the record spells the old way: "Bever due from mr Wynslow."

She wrote her own will, the paper that says who gets a person's belongings after death, and signed her name to it. The colony's court officials proved the will on October 28, 1633, more than two years after she died, which means they agreed it was real, so it could be carried out. By then both of the men she had named to oversee the will had died, and her son Andrew was still a child. Another colonist, Thomas Prence, took on the job of looking after Andrew's share.

Her husband, William Ring, died in the city of Leiden, in the Netherlands, sometime between 1620 and 1629. The Rings belonged to a church of English Separatists, people who wanted to worship apart from the Church of England. The church had moved to Leiden, and in 1620 some of its members sailed to America and founded Plymouth.

The Rings had planned to go too, but they were among 20 people who could not sail when their ship, the Speedwell, turned back. Mary Ring came to Plymouth as a widow in 1629 or 1630 with her three children, Elizabeth, Susanna and Andrew. All three married in the colony.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 2 MINUTES

Each room had its own use in the new houses that wealthy colonists built in the early 1700s, but most families still lived in one or two rooms. In every kind of house, women and children made much of what the household used, from cloth to candles, and much of that work went on at the hearth.

A room for every use: the Georgian house

Wealthy colonists built a style of house now called Georgian, with both sides built to match. Its builders took its balanced look from ancient Greek and Roman buildings. A central hall ran through the middle of the house, with rooms opening off it. The formal rooms, the parlors and the dining room, were at the front, and the rooms for work were farther back. Some Georgian houses had four chimneys, one to heat each of the four main rooms.

In a house with separate rooms, the people in a family could be apart from one another and have some privacy. Houses like this belonged to wealthy families.

What a household made at the fire

Around the open fire in the hearth, a household spun and wove cloth and made candles, soap, butter and cheese. Its members preserved food so it would keep, and the cooks worked over the same fire. Women and children did most of this work. The historian Laurel Thatcher Ulrich described it in northern New England in her 1982 book *Good Wives*.

Sarah Kemble Knight

Who: A Boston businesswoman whose journal of a ride to New York records the households where she stayed.

When and where: On the road from Boston to New Haven and New York and back, October 1704 to early 1705.

A bed "as hard as it was high," warmed with a hot stone.

Sarah Kemble Knight lived in Boston, where she ran a business and taught. In October 1704 she set out on horseback for New Haven and New York on legal business, and she kept a journal of the trip. In it she wrote about the road, the inns and the households where she stayed the night.

She stayed in New York for about two weeks and started home on December 21. Near Eastchester, New York, her party got lost in a great storm of wind and snow. One woman would hardly let them through her door, but the woman's son invited them in.

He showed Knight a bed "as hard as it was high," and he warmed it with a hot stone at her feet. The family had nothing in the house but milk. They boiled it and sweetened it with molasses for her.

Her journal stayed unprinted for about 120 years, until Theodore Dwight printed it in 1825.

Why this story is here: Few pieces of writing by American women survive from that time, and her journal is one of them.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 4 MINUTES

When men left home during the Revolutionary War, women ran the households. A household still made most of what it used, and women and children still did most of that work. On the frontier, the edge of the colonists' settlements on land where Native nations lived, families made their homes in log cabins. Families in the older settlements lived in farmhouses.

Where the log cabin came from

The log cabin was the usual first house of a frontier family by the second half of the 1700s. Settlers from Sweden and Finland had brought it to America more than a hundred years earlier. To build one, they cut notches near the ends of the logs so that the logs locked together at the corners.

These settlers came to the colony of New Sweden on the Delaware River from 1638. Among them were the Forest Finns, who were skilled at building with logs. The Nothnagle Log House in Greenwich Township, New Jersey, built in about 1638 to 1643, still stands. A common story says the first English colonists lived in log cabins, but the English at Plymouth and at Jamestown did not build them.

Whose land. New Sweden was built on Lenape land. Peter Minuit's Swedish and Dutch crew sailed up the Delaware River in the spring of 1638 and made agreements with several sachems, or leaders, of the Lenape and Susquehannock nations. With the sachems' permission, the crew built Fort Christina.

Several thousand Lenape lived in separate communities along the river, and people at the time counted them as the owners of the land beside it. The Lenape welcomed the colony because its people brought goods to trade. New Sweden lasted 17 years before the Dutch took control of it.

Abigail Adams

Who: Ran her family's farm, its money and her children's education through the Revolutionary War while her husband, John Adams, was away.

When and where: Braintree, Massachusetts, in the part that is now the city of Quincy, 1774 to 1784.

Movie: *John Adams* (2008), a miniseries made for HBO. It is about John Adams and his family, and Abigail Adams is one of its main characters.

Charles, who was 6, was given smallpox on purpose three times.

Abigail Adams ran the family farm at Braintree, Massachusetts, and managed its money from 1774, while her husband, John Adams, was away for years at a time. He served first in the Continental Congress, the meeting of leaders from the colonies, and then in Europe as a diplomat, a person who represents a country to other governments. Abigail taught the children herself, and she and John wrote each other more than 1,100 letters.

The inoculation. She took all four children to Boston in July 1776 to be inoculated, which meant giving a healthy person smallpox on purpose so that the person got a milder case. A doctor cut the skin with a small blade called a lancet. Then he put in a thread soaked in pus from a smallpox sore.

Caught the natural way, smallpox killed about 30 of every 100 people who got it. Of every 100 people inoculated, about 2 or 3 died. People who lived through smallpox usually did not catch it again, so inoculation gave them that protection at far less risk.

Abigail wrote to John from Boston on July 13, 1776: "I yesterday arrived and was with all 4 of our Little ones innoculated for the small pox." Their doctor was Dr. Bulfinch. She wrote that the children "stood the opperation Manfully," which meant they were brave about it.

Nabby, who was 10, broke out in many painful sores, while 9-year-old Johnny had a light case. Charles was 6, and he had to be inoculated three times before he caught the smallpox. Thomas, the fourth child, showed no sickness at first and had to be inoculated again.

Her years of running the farm alone ended in 1784, when she and Nabby sailed to Europe to join John.

Martha Ballard

Who: A midwife in Maine whose daily diary, kept for 27 years, records the work of an ordinary American household.

When and where: Hallowell, Maine. The diary runs from January 1785 to 1812.

Movie: *A Midwife's Tale* (1997), a documentary made for the PBS series *American Experience*. It is about Martha Ballard.

She wrote down 816 births, each one with a name.

Martha Ballard was a midwife, a woman who helps other women give birth, in Hallowell, Maine. She wrote in her diary nearly every day for 27 years, from January 1785 to 1812: about 10,000 entries over 9,965 days. The diary records 816 births that she helped with.

Around those births, the diary records the daily work of her own household: her garden, spinning, weaving, quilting and brewing, the work her daughters did, the visitors who came and the weather each day.

The historian Laurel Thatcher Ulrich, who wrote *Good Wives*, published a book about the diary, *A Midwife's Tale*, in 1990. Ulrich won the Pulitzer Prize, a yearly American prize for writing, for the book. The whole diary can be read online at DoHistory.org.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 9 MINUTES

Middle-class families in towns and cities, the families in between the rich and the poor, began in the early 1800s to treat the home as a place set apart from paid work. From the 1830s, cooks began to trade the open hearth, the fireplace where food cooked over the fire, for the iron stove. On farms, children's chores came first, and school fitted around them. In the South, enslavers could sell any member of an enslaved family at any time.

The parlor and the home apart from work

In these families the home was the mother's to run, and paid work happened somewhere else. The parlor, the front room where the family received visitors, became the room for showing off the family's best things.

The historian Aileen Kraditor gave these ideas a name in 1968: the cult of domesticity, the strongly held belief that the home was a place set apart from paid work and that it was the mother's to run. Catharine Beecher's 1841 book, *A Treatise on Domestic Economy*, a title that means a long book about running a home, described how to run a home of this kind. Readers used it as a guide.

Was the iron stove better than the fire?

For the woman who did the cooking, the stove was better in some ways and worse in others. It meant less bending and lifting, less smoke, and no ash in the food. But it took hard work to clean, and stoves were drafty and hard to control.

Iron cookstoves spread through American kitchens from the 1830s. Foundries, the workshops where iron was melted and poured into molds, made stoves of many designs. By 1850 most middle-class kitchens in the cities had a stove in place of the open hearth, and poorer families and families far from towns changed over later.

Some writers disliked the change. Henry David Thoreau complained in 1846 that the stove "concealed the fire." He wrote that cooking had become "merely a chemic process," by which he meant a kind of chemistry.

A child's day, and where school fit in

On a farm, a child's chores came first. Country schools, called district schools, usually held two terms a year. The summer term ran from about May to August, for the younger children, and women usually taught it. The winter term ran from about November to April, when older boys were not needed for work on the farm, and men were often hired to teach it.

Children missed school when the weather was bad or when there was work to do at home. The teacher often lived with the students' families, staying with one family and then the next, which people called boarding around.

The rod. At home, parents expected children to obey, and many parents hit their children with a rod, a stick or switch used for punishment. They pointed to a verse in the Bible's book of Proverbs: "He that spareth his rod hateth his son: but he that loveth him chasteneth him betimes."

The verse means that a father who does not use the rod hates his son, and a father who loves his son punishes him early. Some writers of advice books in the 1800s began to tell parents to punish children calmly and firmly instead. One of them told parents never to punish a child in anger.

Home under slavery

Enslavers could break up an enslaved family on any day. They sold enslaved people, and slave traders bought them and moved them. Lawmakers and courts in the slave states did not count the marriages of enslaved people as legal marriages, and enslavers sold husbands and wives apart.

Traders forced more than 1 million enslaved people to move from the Upper South to the Deep South, the states farther south, between about 1790 and 1860. From 1820 they moved about 200,000 people every ten years. They marched them overland in chained lines called coffles, or shipped them by sea to New Orleans, which had the largest slave market in the country.

By the count of the Equal Justice Initiative, an American organization, the traders separated nearly half of the people they moved from a husband, a wife or a parent.

Catharine Beecher

Who: Wrote a book of instructions for running an American home, and argued that keeping house was a skill to be studied and taught in schools.
When and where: Hartford, Connecticut, from 1823. Her *Treatise on Domestic Economy* came out in 1841.

One book on running a home went through about 15 editions.

Catharine Beecher's *A Treatise on Domestic Economy*, published in 1841, was the first American book to describe every part of running a home. Printers issued it in about 15 editions and put out new copies nearly every year for years.

Beecher argued that housekeeping was a profession, a skilled job that people should study and that schools should teach. She had founded the Hartford Female Seminary, a school for girls and young women in Connecticut, in 1823. In 1869 she and her sister Harriet Beecher Stowe wrote another book about the home, *The American Woman's Home*.

Buffalo Bird Woman (Waheenee)

Who: A Hidatsa woman who grew up in her mothers' earth lodge and later told the story of her life to the anthropologist Gilbert Wilson.

When and where: About 1839 to 1932. Born by the Knife River and raised at Like-a-Fishhook village on the Missouri River, in what is now North Dakota.

Her family believed that "an earth lodge was alive and had a spirit."

Buffalo Bird Woman grew up in the 1840s in an earth lodge, a large round house with a frame of timber, covered over with packed earth. Her household held her father, his four wives, her grandmother and her great-grandmother.

Late in her life she told the story of that household to Gilbert Wilson, an anthropologist, a scientist who studies how groups of people live. He interviewed her every summer from 1906 or 1907 to 1918. She spoke Hidatsa and never learned English. Wilson published her story in 1921 as *Waheenee: An Indian Girl's Story*, and her son, Edward Goodbird, drew the pictures for Wilson's books.

She was born in an earth lodge near where the Knife River meets the Missouri, about 1839.

When she was ten days old, an old man named Nothing-but-Water gave her the name Good Way. Later her father gave her the name Waheenee-wea, Buffalo Bird Woman, because she was a sickly child, and people believed a new name could move the gods to help her.

Her father, Small Ankle, was a Hidatsa chief. He had four wives, and they were sisters: her mother, Weahtee, and Red Blossom, Stalk-of-Corn and Strikes-Many Woman. "I was taught to call all these my mothers," she said. "Such was our Indian custom." Her great-grandmother White Corn and her grandmother Turtle lived with the family too.

The smallpox winter. Smallpox, a deadly disease that spreads from person to person, came up the Missouri River in June 1837, before she was born. The American Fur Company's steamboat *St. Peters* carried

passengers sick with it. Francis A. Chardon ran Fort Clark, a trading post near the Mandan and Hidatsa villages, where traders swapped goods for furs. He met the boat 30 miles downstream and heard that people on it were sick.

The boat landed at Fort Clark on June 18 or 19. In less than a day, workers from the boat and the fort unloaded goods and loaded furs, and villagers came and went from the boat.

About 2,000 Mandan people were alive in the spring of 1837. By October, smallpox had killed all but 138 of them. Buffalo Bird Woman's great-grandmother told her: "More than half of my tribe died in the smallpox winter. Of the Mandans only a few families were left alive. All the old people and little children died."

The move. When she was four, the Mandan and Hidatsa chiefs moved their people together to one new village at Like-a-Fishhook Point on the Missouri, to protect them from attacks by the Dakota and the Yanktonai. The village stood by 1845. Women and dogs carried the family goods there, some loads on their backs and some on travois, frames of two poles that the dogs dragged along the ground.

The people built about seventy summer earth lodges at the new village, most of them in the second summer. For the winter, families built small huts in the woods, and the June flood often swept them away.

In the first hard winter at the new village, when she was about six, the family camped at Many-Frogs Brook, three miles away. Smallpox broke out again with the first snow. "My brother, my mother Weahtee, and her sister Stalk-of-Corn, died," she said. Her grandmother Turtle made her a doll of deerskin with eyes of bone beads.

Inside the lodge. The women of a household owned its earth lodge. Her mothers' lodge had a floor more than forty feet across, with the fire in the center. A screen of upright split logs stood between the fire and the door

and kept out drafts and dogs. Her father's bed, of cottonwood planks and buffalo robes, stood in front of the fire, where the oldest person in the family usually slept, because it was the warmest place.

The rest of the family slept in covered beds against the back wall. The buffalo-skin bedding could not be washed, so her mothers aired it on the stage, the raised platform where they dried their corn. The family believed that "an earth lodge was alive and had a spirit."

Later years. Her first husband was Magpie, and her son Goodbird was born in 1869. Federal officials, the officials of the national government, carried out a law of 1887 called the Dawes Act in the late 1880s. They divided reservation land, the land set aside for a Native nation to live on, into allotments, plots that they assigned to one family each, and they sold the land left over.

Native nations held about 138 million acres in 1887 and about 48 million in 1934. Once the reservation was divided, the families of Like-a-Fishhook left the village to live on the plots assigned to them, and by the late 1880s the village was gone. In 1921 she said: "We no longer live in an earth lodge, but in a house with chimneys." She added that "my son's wife cooks by a stove." She still got up at daybreak to hoe her corn and sing to it. She died in 1932.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 8 MINUTES

Where an American family made its home in these years depended on how much money it had and what part of the country it was in. In New York, a working family might rent a tenement flat of small rooms, with shared toilets out in the yard. A middle-class family might have a house with a parlor for guests, a dining room for meals and a paid servant to help run it. On the plains, where few trees grew, settler families cut their houses out of the ground.

Lawmakers and officials also began to decide where children spent their days. State lawmakers required children to go to school, and federal officials took Native children away from their homes to distant schools.

The tenement flat

The tenement at 97 Orchard Street in New York, which Chapter 6 describes, was a building of small apartments, called flats, rented to working families. Families lived in small rooms, and some took in boarders, people who paid to sleep in the family's rooms. They carried water into the flat by hand and shared privies, outdoor toilets, in the yard.

Lawmakers in New York began to change what a tenement had to include. Under the Tenement House Act of 1879, builders added air shafts, narrow gaps between buildings meant to let in air and light. Governor Theodore Roosevelt appointed a commission, a group of people chosen to study the housing laws, in 1900. Lawrence Veiller, its secretary, did most of the investigating and helped write a new law.

New York's state lawmakers passed that law on April 12, 1901. It required a window in every room and a toilet in every new apartment. Owners of old tenements had to give them better light and add one toilet for every two families.

Bigger houses and paid help

Families who could pay for them built houses with a parlor for guests and a separate dining room. City homes got running water inside the house first, and farm families waited decades longer. In 1890, 24 of every 100 American households had running water, and 8 of every 100 had electricity.

Some middle-class homes also had paid help. The census of 1870, the count of the country's people taken every ten years, was the first to record the jobs of working women in full, by a set method. The census takers counted 867,354 women in domestic service, paid work in another family's home. That was about 47 of every 100 women who worked for pay, by far the most common paid job for women.

Sod houses, dugouts and the Dakota land under them

The plains had few trees, so settler families there built with sod, the top layer of prairie soil, held together by the roots of the grass. Settlers used a plow to cut about an acre of ground into strips 12 inches wide and 4 inches thick, and they cut the strips into pieces about three feet long.

Builders laid these sod bricks grass side down, turning each layer lengthwise and then crosswise, to make walls two feet thick. Most early sod houses had dirt floors. Women who lived in them complained about the dirt, the bugs, the snakes, the leaking roofs and the poor light.

Some families dug their home, called a dugout, into the side of a hill or a creek bank. A dugout took less work to build than other kinds of house. Charles and Caroline Ingalls settled in a dugout on Plum Creek, near Walnut Grove, Minnesota, in May 1874. Their crops failed three years in a row, and in 1876 they left. The hollow in the creek bank where the dugout stood is still there, on what was once the Gordon farm.

Whose land. Walnut Grove lies on land the Dakota signed away in 1851 under treaties, written agreements between nations. At Traverse des Sioux on July 23, 1851, three men bargained with the Sisseton and Wahpeton Dakota

for about 24 million acres. Luke Lea spoke for the US government. With him were Alexander Ramsey, the governor of Minnesota Territory, which was not yet a state, and Henry Sibley.

The three men agreed to pay about 7.5 cents an acre. The Dakota believed US officials would take the land by force if they refused. After the treaty was signed, the fur trader Joseph R. Brown had Dakota leaders sign another paper, and no one read it aloud to them or translated it.

Under that paper, traders could take the annuity money, the yearly payments promised to the Dakota, for debts the traders claimed the Dakota owed them. Within ten years, more than 100,000 white settlers moved onto the land.

School laws and the boarding schools

Massachusetts lawmakers passed a law in 1852 that required children aged 8 to 14 to go to school for twelve weeks a year. By 1900, 32 states had such a law, and by 1918 every state had one, Mississippi last. Under these laws, families had to send their children to school for part of the year.

Federal officials took Native children from their families to boarding schools far away. Richard Henry Pratt, an Army officer, opened the Carlisle Indian Industrial School, a boarding school for Native children, on November 1, 1879.

Forced attendance. Members of Congress passed a law in 1891 that gave the Commissioner of Indian Affairs, the federal official in charge of Native affairs, a new power. He could make and enforce rules forcing Native children to attend school. Two years later they allowed officials to hold back food rations, clothing and payments owed to parents who refused to send their children.

At the schools, staff cut the children's hair, took away their clothes and gave them English names. Staff beat children who spoke their own language, locked them up alone, held back their food or washed their mouths out with soap. Survivors and a federal investigation later reported these punishments.

Officials of the US Department of the Interior reported on these schools in 2022 and 2024. They counted 417 federal boarding schools for Native children in 37 states or territories, open between 1819 and 1969, and at least 973 children who died at those schools. Their report states that this count is low.

Nathalie Gumpertz

Who: A tenement mother of four who became a dressmaker after her husband disappeared.

When and where: 97 Orchard Street, New York. Her husband Julius disappeared in 1874. A judge declared him dead in 1883.

Three dresses a week, sewn in the front room of a tenement flat.

Nathalie and Julius Gumpertz were immigrants from East Prussia who lived at 97 Orchard Street. They were living there during the Panic of 1873, a money crash in which many people lost their jobs. One morning in 1874, Julius left for work and never came back, and Nathalie was left with four young children.

She became a dressmaker and worked in the front room of the flat, the parlor. She made about three dresses a week and earned 8 to 9 dollars.

A judge declared Julius legally dead in 1883, which meant that under the law he counted as dead. Because of that, Nathalie could claim 600 dollars that he was due from his father's estate, the money and property his father left when he died. She then moved the family to upper Manhattan.

Later, researchers who trace family histories for the Tenement Museum in New York found that Julius had gone to Cincinnati.

Rachel Calof

Who: A young woman from Russia who came to North Dakota for an arranged marriage and later wrote down what keeping a homestead house was like.

When and where: Near Devils Lake, North Dakota, 1894 to 1904. She wrote her memoir in 1936, and it was published in 1995.

In winter, in-laws and chickens shared their one 12-by-14-foot room.

Rachel Calof was 18 when she traveled from Russia to marry Abraham Calof in 1894. It was an arranged marriage, planned by other people rather than by the couple. Abraham was a homesteader, a settler who claimed land from the US government to live on and farm.

The couple lived in a shanty, a small, roughly built house, 12 feet by 14 feet with a dirt floor. In the winters, Abraham's parents and brothers, and the chickens, shared the one room with the couple to save fuel. Of all the hardships, she wrote, "the lack of privacy was the hardest to bear."

She wrote a memoir, a book about her own life, in 1936, in Yiddish, a language spoken by many Jews in Eastern Europe. It records her years on the homestead from 1894 to 1904. Indiana University Press published it in English in 1995 as *Rachel Calof's Story*.

Whose land. The land around Devils Lake was Dakota land. Under a treaty of 1867, the land south of the lake became a reservation for the Sisseton and Wahpeton Dakota and the Cut Head band of Yanktonai Dakota. The Dakota name for the lake is Mni Wakan, Spirit Water.

Settlers who were not Native illegally claimed 64,000 acres of the reservation, and the Dakota bargained with the US government and accepted 345,000 dollars for that land. Federal officials opened 100,000 more acres of the reservation to settlers in 1904, at about 4 dollars and 50 cents an acre. By 1934 the Dakota owned a little more than 50,000 acres of it.

NOT KNOWN: whether the Calofs' own claim lay on land taken from the reservation. The State Historical Society of North Dakota's history of the reservation does not say.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 12 MINUTES

Running water and electric power reached most American homes between 1900 and 1950, city homes first and many farm homes decades later. After the Great Depression, the hard times of the 1930s, federal officials made home loans easier to pay off, but they usually refused to back loans in the neighborhoods where Black families lived.

Why washday took more than a day

Without running water or electric power, washing a family's clothes by hand took more than one day. The person doing the wash first sorted the clothes and treated the stains, and then the clothes soaked in warm soapy water for one or two days. Next the washer scrubbed them on a washboard, a ridged board set in a tub of warm water and soap.

Then the washer boiled the clothes in a pot with soap and soda to kill lice, stirring them with a wooden stick called a dolly. After rinsing the clothes, the washer hung them out to dry, with the whites in the sun to bleach them. Last came the ironing, with flatirons, solid metal irons heated over the coals and held with a rag around the handle.

Pipes and wires. Water pipes and electric wires reached city homes first and farm homes decades later. In 1940, 74 of every 100 American homes had running water, and by 1950, 83 did. Officials of the Census Bureau, the federal office that counts the country's people and homes, found that more than one home in three had no flush toilet in 1940. Most of the homes without running water were in the country.

Electric power spread the same way. By 1950, 94 of every 100 households had it, but farms came last: fewer than 11 of every 100 farms had electric power in 1935, and about 9 in 10 had it by 1950.

From icebox to refrigerator. In 1930, 14 of every 100 households had a modern refrigerator, and by 1950, 80 did. A family without one used an icebox, a cabinet cooled by a block of ice, and relied on the iceman's deliveries, on milk brought each day and on canned food. The refrigerator ended the iceman's rounds and the daily trip to buy fresh food.

Cowan's argument. The historian Ruth Schwartz Cowan studied what the new machines did to a housewife's work. In her 1983 book *More Work for Mother*, she argued that washing machines, vacuum cleaners and other new conveniences mostly replaced work that servants, children and men had done before. She wrote that a housewife of 1950 kept a middle-class home clean and comfortable by herself, work that a household of 1850 had needed three or four workers to do.

At the same time, families expected clothes washed more often, houses kept cleaner and more cooking done. So, in Cowan's argument, the housewife's own hours of work did not fall. Census counts record one part of the change: the number of domestic servants fell by 32 percent between 1940 and 1950.

Mokyr's answer. Another scholar, Joel Mokyr, agreed that the hours did not fall, and in 2000 he gave a reason. In the late 1800s, scientists showed that germs cause many diseases. Doctors and nurses then told housewives that keeping germs out of the house would keep their families healthy, so, Mokyr wrote, housewives cleaned more.

Bungalows, and houses sold by catalog

The small family house of this era was the bungalow, a house of one main floor with a low roof and a front porch, and sometimes rooms tucked under the roof. In Chicago, more than 80,000 brick bungalows built between 1910 and the mid-1930s still stand.

Sears, Roebuck and Company sold whole houses by catalog from 1908 to 1940. A buyer chose a design, and the house came by railroad in boxcars, as a kit. The average kit held about 25 tons of materials and more than 30,000 parts. Many buyers put the house together themselves with relatives, friends and neighbors, and others hired local carpenters. The first Sears house catalog, in 1908, offered 44 designs priced from 360 dollars to 2,890 dollars.

NOT KNOWN: how many kit houses Sears sold. Nobody kept an official count, and the usual estimate is 70,000 to 75,000, with some estimates as high as 100,000.

The Rogarshevskys and the tailor's disease

When Abraham Rogarshevsky died of tuberculosis on July 17, 1918, his widow kept the family in its tenement home by working for the building and renting out space. The Rogarshevskys were a Jewish family from Lithuania who had lived since about 1908 at 97 Orchard Street in New York.

Tuberculosis, a disease caused by a germ, usually infects the lungs, and it spreads through the air when a sick person coughs or talks. People then called it "the tailor's disease." After Abraham died, his widow, Fanny, became the building's janitor, the worker who cleaned the building and kept it in repair. She also took in boarders, people who paid to live in the family's rooms.

New home loans, and the red lines

A home loan was hard to pay off before the 1930s. Under a mortgage, a loan to buy a house, the lender can take the house if the borrower does not repay it. The usual mortgage ran for 5 to 10 years, and each month the borrower paid only the interest, the fee for borrowing the money.

At the end, the borrower owed the whole amount at once. Buyers usually had to pay about half the price up front, as a down payment. In the Great Depression many borrowers could not pay, and lenders foreclosed, which means they took the houses back.

The 1933 law. Members of Congress passed the Home Owners' Loan Act in June 1933, and under it federal officials set up the Home Owners' Loan Corporation, or HOLC. HOLC officials replaced families' old loans with new loans that ran for 15 years at a fixed rate, which means the interest charge never changed. The family paid the same amount every month, and each payment paid off part of the loan itself.

Members of Congress set up the Federal Housing Administration, or FHA, in 1934. FHA officials insured home loans, which means they promised to repay the lender if the family could not. Those loans ran 20 to 25 years and needed only a small down payment. After the Second World War the 30-year loan became the usual kind.

Graded by color. Chapter 6 tells the story of the HOLC maps. Between 1935 and 1940, HOLC staff graded city neighborhoods with help from local bank loan officers, real estate agents and appraisers, the people whose job is to judge what a property is worth. Each neighborhood got a grade from A, "Best," down to D, "Hazardous," and the staff shaded the D neighborhoods red. Refusing loans in a neighborhood came to be called redlining, after those red areas.

FHA officials used a rating system of their own. They too rated neighborhoods where Black people lived "D," and they usually refused to insure loans there. The story of Clyde Ross, in 1950 to 2000, shows what that meant for one family.

Depression households

Families changed who lived together under one roof during the Great Depression. Couples waited to marry: there were 10.1 marriages each year for every 1,000 Americans in 1929, and 7.9 by 1932. For the first time in US history, parents had too few babies to replace their own generation.

Families doubled up in one home. Young married couples moved back in with their parents, and old parents moved in with their grown children. By 1940, 1.5 million married women lived apart from their husbands.

One room in Bronzeville. In Chicago, Black families who had moved north from the South settled in a South Side neighborhood called Bronzeville. Chapter 6 describes the restrictive covenants, written agreements not to sell or rent to Black people, that white owners signed on many other blocks. Because of those agreements, Black Chicagoans were limited mostly to the Black Belt, an area between 12th and 79th streets on the South Side that took in Bronzeville.

Landlords there split apartments into one-room units called kitchenettes and charged very high rents for them, and Bronzeville held more than 80,000 of them. A whole family lived in one room and shared a bathroom down the hall with everyone else on the floor.

Some buildings had no proper heat. Families lit kerosene lamps, and the stoves they rigged up often overheated and started fires. In one year, about 751 fires broke out in the Black Belt. The writer Richard Wright recorded this life in his 1941 book *12 Million Black Voices*.

Expert advice, and parents' letters about hitting

Many parents of the 1920s and 1930s began to take advice from experts on child development and psychology, the study of the mind and of how people behave. John B. Watson's 1928 book, *Psychological Care of Infant and Child*, sold 100,000 copies within months.

Watson told parents: "Never hug and kiss them. Never let them sit in your lap." He went on: "If you must, kiss them once on the forehead when they say goodnight." He advised them to "shake hands with them in the morning."

Letters about hitting. Parents wrote to Angelo Patri, a popular expert on raising children, for advice from 1924 to 1939. In 2004 three researchers published a study of 147 of these letters that mentioned corporal punishment, hitting a child as a punishment. Most came from middle-class parents.

The parents wrote more about how useful hitting was than about a child's right to be kept safe from harm. The reasons they gave most often were that the child had disobeyed or been rude. One letter in four described an argument about hitting with another adult close to the family.

The teenager, and the war at home

A new word for a stage of life came into use in these years: teenager. More children stayed in high school instead of going to work, and that is part of why people began to see the teenage years as a stage of their own.

Writers had printed "teen ager," as two words, as early as 1913. A 1941 article in *Popular Science* printed it with a hyphen, as "teen-agers," and this is usually named as the first time it appeared in print as one word rather than two. The word spread in the late 1940s, when advertisers used it to sell goods to young people.

The war at home. During the Second World War, individuals and families moved to cities with war factories to take well-paid jobs, and more Americans moved from one part of the country to another than in any earlier period. Richmond, California, grew from fewer than 24,000 people to more than 100,000 during the war.

Six million women worked in war-industry jobs that women had not held before. At home, federal officials limited how much of twenty kinds of goods each family could buy, and families shopped with ration books.

The Baldizzi family

Who: A Sicilian immigrant family who lived through the Great Depression in a tenement flat. Museum staff later rebuilt their apartment from the daughter's memories.

When and where: 97 Orchard Street, New York, 1928 to 1935.

Their apartment was rebuilt from what one daughter remembered.

Adolfo and Rosaria Baldizzi raised their children, Josephine and Johnny, in a flat at 97 Orchard Street from 1928 until 1935. Adolfo had come to the United States from Palermo, in Sicily, in 1923. He was a trained woodworker who made fine furniture, but during the Depression he could find only odd jobs.

Rosaria sewed the linings into coats in a garment factory. In the evenings the family sat at the kitchen table to play cards and listen to the radio. Adolfo built a corner cabinet himself.

Forced out. New York City officials required owners of old tenements to make costly repairs, such as fireproofing, in 1934 and 1935. The owner of 97 Orchard Street chose to empty the apartments rather than pay for the work. The owner evicted the families who still lived there, which means the owner forced them to leave.

Next the owner boarded up the upper windows and sealed off the floors where people had lived, and the Baldizzis had to move. The museum's accounts of the family do not name the owner.

After the Tenement Museum was founded in 1988, Josephine Baldizzi came back to 97 Orchard Street and recorded hours of memories of her childhood there. Museum staff rebuilt the family's apartment from what she remembered.

Theodore and Patricia Bladykas

Who: The first family to move into Levittown, with their 14-month-old twin daughters, Patricia Ann and Betty Ann.

When and where: October 1, 1947, 67 Bellmore Road, Levittown, Long Island, New York.

Levittown's lease allowed only white families, and servants of any race.

Theodore and Patricia Bladykas were the first family to move into a house in Levittown. They moved in on October 1, 1947, with their twin daughters, Patricia Ann and Betty Ann, who were 14 months old. A photographer from the newspaper *Newsday* took their picture that day, when the first 300 families moved into Levittown.

Chapter 6 tells how the company Levitt and Sons, run by William Levitt, built Levittown. The first houses rented for 60 dollars a month, and a renter could buy the house for 7,500 dollars after a year. From 1949 the company sold ranch houses, long, low houses of one story, for 7,990 dollars. Many of the fathers who moved in were veterans just back from the Second World War.

Clause 25. William Levitt and his company rented and sold these houses to white families only. Clause 25 of the Levittown lease, the paper that set the rules for renting, stated that no "person other than members of the Caucasian race," meaning no one who was not white, could live in the house. The one exception was for servants: a white family could employ servants who were not white, and those servants could live with the family in the house.

The justices of the US Supreme Court ruled in 1948, in the case *Shelley v. Kraemer*, that courts could not enforce rules like this. William Levitt dropped the written clause and kept refusing to sell to Black families. His sales contracts forbade buyers to resell their houses to African Americans.

He said that if a Black family bought a house in Levittown, white families would stop buying there. He told the *Saturday Evening Post* in 1954: "We can solve a housing problem, or we can solve a racial problem. But we cannot combine the two."

BY 1950

If you remember one thing: Running water, electric wires and long home loans reached most American families between 1900 and 1950, and FHA officials usually refused to back loans in the neighborhoods where Black families lived.

- In 1890, 24 of every 100 households had running water and 8 had electricity.
- In 1950, 83 of every 100 had running water and 94 had electricity.

Next: The suburbs grow after 1950, and a crowd attacks the house of the first Black family in Levittown, Pennsylvania.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 9 MINUTES

A house in the suburbs became the common picture of American family life after 1950. The share of households that owned their home rose from 43.6 of every 100 in 1940 to 61.9 in 1960. Lenders and sellers kept Black families out of many of the new suburbs, and in Levittown, Pennsylvania, a mob attacked the house of the first Black family to move in.

Inside the house, families planned their evenings around the new television set. Later in the century, more mothers took paid jobs, the divorce rate more than doubled and some parents began to teach their children at home.

Who got to move to the suburbs

Some 76 million babies were born in the United States between 1946 and 1964, the years people call the baby boom. Builders put up more towns like Levittown, including a second Levittown in Pennsylvania from 1952.

Builders, sellers and lenders went on with redlining and whites-only sales for many years after the 1948 Supreme Court ruling. Activists campaigned against these practices. In 1968 members of Congress passed the Fair Housing Act, which made it against the law to refuse to sell or rent a home to someone because of race.

The TV in the living room

Most American families bought their first television set in the 1950s. About 9 of every 100 households had a set in 1950, 64.5 had one by 1955, and 87 by 1960.

The household changes shape

The divorce rate more than doubled between 1960 and 1980, from 9.2 divorces each year for every 1,000 married women to 22.6. The rate reached its peak in 1981 and then began to fall.

More mothers took paid jobs. In 1948, about 17 of every 100 married mothers were in the labor force, which means they worked for pay or were looking for paid work. By 1985 the figure was 61 of every 100, and by 1995 about 70. With both parents at work, more families paid for day care.

Latchkey children. Some children went home after school to an empty house. People called them latchkey children, because they carried their own key to the house.

The counts of them differed widely. Census Bureau researchers found about 2.1 million children aged 5 to 13 regularly home alone in a 1987 study, while advocacy groups, which campaign for a cause, estimated 6 to 7 million. In the fall of 1995 Census Bureau workers counted 6.9 million children aged 5 to 14, or 18 of every 100, who regularly looked after themselves, on average six hours a week.

School comes home

Some parents began to teach their children at home in the 1970s and 1980s instead of sending them to school. Homeschooling had two separate starting points. John Holt wanted to change how schools taught, and he started a newsletter called *Growing Without Schooling* in August 1977. On the religious side, Raymond Moore argued for keeping young children at home in his 1981 book *Home Grown Kids*.

Taught in secret. In many states, homeschooling was against the law, or nearly impossible to do legally, into the 1980s. Most state laws of the 1970s did not say whether it was allowed, and many families taught their children in secret.

A judge in a Massachusetts court ruled in 1978 that homeschooling was legal if the parents were able to teach, covered the required subjects and showed that the children were learning. Over the next fifteen years, judges in other states ruled that homeschooling was legal, each state with its own rules.

In most homeschooling families the mother did the teaching, and many mothers stayed home to do it. Families also met in homeschool groups.

Daisy and William Myers

Who: The first Black family in Levittown, Pennsylvania: a war veteran, a teacher and their three small children, who stayed in their house while crowds attacked it.

When and where: 43 Deepgreen Lane, Levittown, Pennsylvania, from August 13, 1957, to 1961.

Movie: *Crisis in Levittown* (1957), a documentary by Lee Bobker and Lester Becker, filmed while the events were happening.

Crowds of up to 1,000 people stood outside their new house.

William and Daisy Myers and their three small children moved into 43 Deepgreen Lane in Levittown, Pennsylvania, on August 13, 1957. They were the first Black family to live in Levittown. William had served in the Army in the Second World War, and Daisy was a teacher. Lewis Wechsler, a white Jewish friend, helped them arrange to buy the house.

The crowds. Crowds of 200 to 1,000 people gathered on Deepgreen Lane for more than a week. People in the crowds threw rocks through the windows of the house and threw Molotov cocktails, bottles filled with fuel and set on fire. They burned crosses, one of them at the Wechsler house.

Members of the Ku Klux Klan, a group whose members bombed and attacked Black Americans, had used burning crosses as a threat for decades, to make the people who saw them fear they would be attacked. The account of the attacks on Deepgreen Lane does not say who in the crowds burned the crosses there.

Callers threatened the family by telephone. Milk, bread and oil stopped coming to the house, and the family's fire insurance was canceled. On the seventh night more than 500 people gathered before state troopers, the state police, broke up the crowd.

Who organized it. Members of the Levittown Betterment Committee passed around a petition, a paper that many people sign to ask for something, against letting Black families live in Levittown. The

committee's members also organized the harassment of the family, the steady stream of acts meant to frighten them. A judge ordered the Betterment group to stop.

Pennsylvania Governor George Leader sent the state police, and after about six weeks the troopers had ended the open violence. A court found two people guilty of burning the cross on the Wechsler lawn.

Neighbors, church members and Quaker families, members of the Christian church called the Society of Friends, brought the Myers family food and flowers. The family stayed in the house until 1961, when they moved to York, Pennsylvania. Daisy Myers told the story in her memoir *Sticks 'n Stones*, a book about her own life published in 2005.

NOT KNOWN: who stopped the deliveries, which company canceled the insurance, and who the two people found guilty were. The Baltimore Sun's 1997 account names none of them.

Clyde Ross

Who: A Second World War veteran and soup-factory worker who could not get a mortgage and bought his Chicago house on contract.
When and where: North Lawndale, on Chicago's West Side, from 1961.

Miss one payment, and he would lose the house and every dollar.

Clyde Ross was born in 1923 near Clarksdale, Mississippi. In the Second World War he was drafted, ordered into the armed forces, and he served on Guam. He moved to Chicago in 1947 and took a job as a taster at Campbell's Soup.

No loan. Ross tried to get an ordinary mortgage for a house in another neighborhood. A loan officer told him there was no money to lend, and the records do not name the officer or the bank. FHA officials, using a rating system of their own, rated neighborhoods where Black people lived "D" and usually refused to back loans there.

The contract. Ross bought a house in North Lawndale in 1961 for 27,500 dollars. The seller was a speculator, a person who buys property to sell it again at a profit, and had paid 12,000 dollars for the house. Ross bought on contract: the seller kept the deed, the paper that shows who owns a house, until Ross made the last payment.

If Ross missed a single payment, he would lose his 1,000-dollar down payment, every monthly payment he had made, and the house. The journalist Ta-Nehisi Coates cited one estimate that 85 of every 100 Black families who bought homes in Chicago bought them on contract.

The strike. Ross joined the Contract Buyers League, a group of Black homeowners on Chicago's South and West Sides, in 1968, and became its vice president. About 500 families in the league stopped making their payments together, as a strike against the contracts. By July 1971 the league and the sellers had worked out new terms for 155 contracts, saving each family an average of 14,000 dollars.

About 70 league families lost their homes to eviction, and the Chicago Reporter's account does not name who had each family put out. The league also took its fight to court and lost. Coates wrote that the league lost a jury trial in 1976. The Chicago Reporter says its last appeal, its last request for a higher court to change the ruling, was turned down in 1983.

Coates told Ross's story in *The Atlantic* magazine in 2014. Ross was 91, and he still lived in the house and owned it.

DISPUTED: the year and the price. Coates gives 1961 and 27,500 dollars, and a 2014 account on BillMoyers.com gives 1958 and 27,000 dollars.

David and Micki Colfax and their sons

Who: Parents who built their own house on a ranch in California and taught their four sons at home.
When and where: Near Boonville, Mendocino County, California, from 1973.

Three sons taught at home on a California ranch went to Harvard.

David and Micki Colfax taught their four sons at home on a ranch in California, where the family built its own house. Their eldest son, Grant, started public school in St. Louis, but after six weeks his parents took him out, because they found the lessons "very rote and uncreative," rote meaning learned by repeating things over and over.

The family moved to land near Boonville, in Mendocino County, California, in 1973. The land, 40 acres by the Harvard Crimson's count or 50 by the Anderson Valley Advertiser's, had no house, no electricity and no running water. Everyone, the young sons included, helped build the house, the barns and the sheds. The family raised goats, sheep, chickens and pigs and sold them to restaurants in San Francisco.

> Lessons fit around the work of the ranch. "Some days we did a lot of homework, some days we had none," Reed Colfax said later. "It depended on the weather and what had to be done on the ranch." The boys studied four or five days a week, summers included.
>
> Grant, Drew and Reed went on to Harvard University and graduated in 1987, 1990 and 1992. The youngest son, Garth, was still being taught at home in 1989, a year when about 500,000 American children were taught at home by their parents, by a count reported in *The Harvard Crimson*. David and Micki Colfax wrote a book about their method, *Homeschooling for Excellence*.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 7 MINUTES

The median new American house grew to 2,467 square feet by 2015, while the average household shrank to 2.50 people by 2025. In the crash that began in 2007, lenders took back millions of homes from families who could not pay their loans. During the pandemic of 2020, many homes became an office and a school at the same time.

More house, fewer people

The median new house built for one family, the one in the middle when all are ranked by size, grew from 983 square feet in 1950 to a peak of 2,467 square feet in 2015, by Census Bureau figures. That is about two and a half times as large. Over many of the same years, households shrank: the average household held 3.67 people in 1940 and 2.50 people in 2025.

The kinds of households changed too. In 2025, 29 of every 100 households were one person living alone, up from 20 in 1975, and married couples made up 47, down from 66. That year 58 of every 100 young adults aged 18 to 24 lived in their parents' home.

The Census Bureau counted 65 of every 100 households as owning their home in the second quarter of 2026, the months of April, May and June.

How millions of families lost their homes, 2007 to 2010

Lenders foreclosed on about 3.8 million homes between 2007 and 2010. Estimates for the whole crisis run from about 7.7 million to 10 million homes lost, because researchers counted in different ways. Federal Reserve History, a website of the country's central bank, and the Federal Reserve Bank of Chicago list these figures.

The loans. In the years before the crash, lenders made many subprime loans, which cost more and went to borrowers with weaker credit, a weaker record of paying back debts. Because the loans cost more, the monthly payments were bigger.

While house prices kept rising, a borrower who fell behind could sell the house and pay off the loan, or borrow more against the higher price. When prices stopped rising, neither way out worked. Borrowers who could not pay fell further behind, and lenders foreclosed on their homes.

Reporters at the Center for Public Integrity, a news organization, ranked Countrywide Financial the largest subprime lender of 2005 to 2007 in a 2009 report. They found that Countrywide made at least 97.2 billion dollars in high-

interest loans in those years. Angelo Mozilo founded the company and ran it for 39 years.

Foreclosure filings, the papers a lender files to start taking back a home, reached their peak in 2010, at nearly 2.9 million homes. In 2025 there were 367,460, according to ATTOM, a private company that collects property records.

More Americans came to live in households of several generations, such as grandparents, parents and children together. Researchers at the Pew Research Center, which studies American life, counted 7 of every 100 Americans in such households in 1971, and 18 of every 100, or 59.7 million people, in 2021. The number grew faster during the years of the crash than in the years before. When Pew asked people in these households why they lived that way, money was the reason they gave most often.

How many people have no home?

Each January, local volunteers count the people who are homeless on a single night, and officials of the US Department of Housing and Urban Development (HUD), the federal housing department, report the count. The volunteers counted 771,480 people in January 2024, the highest number on record and 18 percent more than a year before.

HUD's report named several likely causes: a shortage of homes people could afford, rising prices, pay that did not keep up for families with middle and low incomes, and the lasting effects of racism. Homeless people in families with children rose 39 percent that year. In 13 communities where many newly arrived immigrants had gone into shelters, family homelessness more than doubled. The next count, in January 2025, was a little lower, at 745,652 people.

2020: office and school at once

Many homes became a workplace and a classroom at once in 2020. The share of workers who worked mainly from home tripled, from 5.7 percent in 2019 to 17.9 percent in 2021, by the Census Bureau's count.

Homeschooling rose too, meaning that the family taught the children, not that teachers at a school taught them online. By the fall of 2020, 11.1 percent of households with school-age children were homeschooling, about twice the share that spring.

Many children did their schoolwork online, and some homes had no broadband, the fast home internet used for online classes. Pew researchers reported that 35 percent of households with children and incomes under 30,000 dollars a year had no broadband, against 6 percent of households with incomes over 75,000 dollars.

NOT KNOWN: how much of the rise in learning at home lasted. In a later federal survey, parents reported that about 5 of every 100 children learned at home in 2022-23, against about 4 before the pandemic, but that survey asked different questions from the one in 2020.

Pierre Solon and Katty Familia

Who: Immigrant parents, from Haiti and the Dominican Republic, who lost their Boston house to foreclosure.

When and where: Hyde Park, Boston, Massachusetts, about 2006 to 2010.

Bought for 584,000 dollars, sold at auction for 115,000.

Pierre Solon, from Haiti, and Katty Familia, from the Dominican Republic, bought a green wooden house in Boston's Hyde Park neighborhood in about 2006 for 584,000 dollars. They borrowed the whole price from Fremont Investment and Loan, a subprime lender, as two mortgages, two separate loans on the same house. Their lawyer, Nadine Cohen of Greater Boston Legal Services, said the payments on the two loans were almost 5,000 dollars a month more than the couple earned.

Each parent worked two jobs, and then each lost one. "Sometimes we don't even have food," Katty Familia said. The couple fell more than 150,000 dollars behind.

Staff at Wells Fargo, the bank that collected their payments, offered lower monthly payments. The couple turned the offer down, because the change would have raised what they owed to more than 600,000 dollars. Wells Fargo's staff then foreclosed, and late in 2009 the family moved out with their children.

The sale. Wells Fargo sold the house at auction, a sale where the highest bidder wins, in November 2010. The buyer was an investor, a buyer who hopes to sell later for more, and the price was 115,000 dollars. Two weeks later that investor sold it to another investor for 270,000 dollars.

The Massachusetts Attorney General, the state's top lawyer, sued Fremont for deceptive lending, lending by misleading borrowers. Fremont sold its loans to investors, then went bankrupt, unable to pay its debts, and closed.

The couple ran a day care center in Boston in 2010. "I worked 20 years in this country for this," Katty Familia said. The reporter Anthony Brooks told their story on the radio program *Morning Edition*, on NPR, in December 2010.

The Munoz family

Who: A Las Vegas family whose four school-age children did school online at the dining room table while both parents went out to work.
When and where: Las Vegas, Nevada, August 2020.

"It was a challenge because we couldn't log in."

Four of the five Munoz children in Las Vegas, Nevada, started the school year in August 2020 on laptops at the dining room table. Both parents worked in the food industry, away from home. While they were at work, the family's 10th-grade son helped the younger children: a fourth-grader with reading and a new middle-schooler with writing. When he had trouble, he emailed the teachers.

The father, Johnathan Munoz, recorded the family's first week as an audio diary for NPR's radio program *Weekend Edition Sunday*. On the second day he said, "it was a challenge because we couldn't log in." The middle-schooler said that one teacher kept marking her absent, and another teacher did not allow a brother or sister to be seen in the background.

THE THREE QUESTIONS, ANSWERED

1. **Who lived together under one roof, from the longhouse to today?** A longhouse held 30 to 60 relatives, and colonial households held servants, apprentices and enslaved people too. By 2025 the average household held 2.50 people.

2. **Who did the work of keeping house, and did stoves, pipes and wires make it lighter?** Women and children did most of it. In Ruth Schwartz Cowan's argument, the machines did not cut a housewife's hours, because families expected more.
3. **How did lenders and builders keep Black families out of new neighborhoods, and what did those families do?** FHA officials usually refused to back loans where Black families lived, and William Levitt sold only to white families. The Myers family stayed through weeks of attacks, and about 500 families in the league Clyde Ross helped lead stopped paying together.

What changed: Households shrank from longhouses of relatives to 2.50 people, and owning a home came to depend on long loans that lenders gave some families and refused others.

Whose land: The colonists' and settlers' houses in this chapter went up on Native land, from Plymouth, built on the Wampanoag village of Patuxet, to the Ingalls family's dugout on land the Dakota signed away in 1851.

FIVE DATES TO KEEP

September 1565: Spanish colonists settle inside Seloy's town.

1841: Catharine Beecher's book on running a home.

November 1, 1879: Richard Henry Pratt opens the Carlisle school.

October 1, 1947: the first families move into Levittown.

1968: members of Congress pass the Fair Housing Act.

NOT KNOWN: No record counts how many Timucua died of disease in the 1500s alone.

Where to go: The Tenement Museum, 97 Orchard Street, New York, where staff rebuilt the Baldizzis' apartment from Josephine's memories.

Next: Chapter 8, Technology. It asks whose names went on American inventions and who did the hard work around them.

PART 3

Making and Building

Chapter 8: Technology

CHAPTER 8 OF 37 · TECHNOLOGY · 10 ERAS · ABOUT 85 MINUTES

Many famous American inventors had help that their patents did not name, and the hardest work around the new machines often fell to people who had no say in it.

The Smithsonian's National Museum of American History holds a strip of paper tape from May 24, 1844. It records the first public message sent from Washington to Baltimore by telegraph, a machine that sent messages over a wire as pulses of electricity. The words were "What hath God wrought" (What has God done).

Samuel Morse sent the message, and the code is named for him. He also held the patents, the government grants that gave him alone the right to make and sell the telegraph for a set number of years. His partner, Alfred Vail, who built the instruments, received the message in Baltimore. In 1911 one of Vail's sons had his father's gravestone changed to read "INVENTOR OF THE TELEGRAPHIC DOT AND DASH ALPHABET."

This chapter follows American tools and machines from fish traps under a Boston street to the AI programs of the 2020s. As you read, notice whose name goes on each invention. A patent could name one man while a partner, a team of workers or an enslaved man did part of the work. In 1858 the government's chief lawyer ruled that an enslaved person's invention could not be patented.

Notice also who did the hard work: Scottish prisoners at an ironworks, enslaved people at furnaces and in cotton fields, children at spinning machines. The last era covers smartphones and AI programs, and what happened when people trusted a machine's answer.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Why did planters enslave far more people after the cotton gin?
2. Why could an enslaved inventor not get a patent?
3. What went wrong when people trusted a computer's answer?

Five minutes? Read "Who could own an invention?" in 1850 to 1900.

Can skip: "The Cambridge printing press", "Tractors replace horses", "What is inside a smoke detector?", "Batteries, printers and gene tools". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

People in North America made their tools from wood, stone, bone, clay and animal skin, and near Lake Superior from copper. With no metal machines, they built traps that caught fish with the tide, fields that grew three crops together and boats of birch bark. The names of the people who first worked out these methods are not recorded. Some of the methods are still in use, and an Ojibwe builder in Wisconsin still makes birchbark canoes.

Spear-throwers, fish traps and fields

Before the bow and arrow, hunters in western North America threw spears with an atlatl, a spear-thrower stick that holds the back end of the spear during the throw.

Hunters across the West began using the bow and arrow about 1,400 years ago, around A.D. 600 to 700. A study published in 2026 in the *Journal of Archaeological Method and Theory* gives that date. South of an east-west line called the 55th parallel, archaeologists, the scientists who dig up and study what people left behind, stop finding atlatls soon after the bow appears. North of that line, hunters used both weapons side by side for more than 1,000 years.

Fish weirs. People built fixed traps called fish weirs, fences of wooden stakes set across the water with bundles of brush woven between the stakes. Fish swam in over the mud flats with the rising tide, and the fence trapped them when the water went out. People then scooped them up with baskets and nets, or caught them by hand.

Lines of wooden stakes from fish weirs lie under Boylston Street in Boston. People built these weirs in channels that filled and emptied with the tide, between about 4,700 and 3,700 years ago. They used them for more than a thousand years, until mud filled the bay around 1550 B.C., about 3,500 years ago.

Native farmers planted corn, beans and squash together in the same field, three crops known as the Three Sisters. They also set controlled fires, started on purpose and kept in bounds, to shape the land.

Crafts, and a calendar made of posts

People made pottery from clay and wove baskets. They tanned animal hides, treating the skins so they stayed soft and did not rot, and they made canoes from the bark of birch trees.

At Cahokia, in what is now Illinois, people built the largest city north of Mexico before 1500. Between about A.D. 900 and 1100, builders there raised five circles of upright wooden posts, one after another.

Each circle had between 24 and 72 posts and measured about 240 to 475 feet (73 to 145 meters) across. The biggest was wider than a football field is long. The posts marked the places on the horizon where the sun rose at different times of the year, so the circles worked as a calendar.

Near Lake Superior, people worked copper for thousands of years. They worked it cold, shaping it without melting it.

Wayne Valliere

Who: An Ojibwe builder of birchbark canoes who has taught more than 100 students and teaches the Ojibwe language.
When and where: Lac du Flambeau, Wisconsin, in the present day.

Spruce roots are the thread that sews his canoes together.

Wayne Valliere builds birchbark canoes, the same kind of boat people built in North America before 1500. He is a member of the Lac du Flambeau Band of Lake Superior Ojibwe, in Wisconsin. A profile of him by the National Endowment for the Arts, a US government agency that gives money and honors to artists, describes the Ojibwe way of building canoes as handed down over many generations.

Valliere began learning at 16 from elders, among them Joe Chosa, Marvin DeFoe and Ojaanimigiizhig. He also studied old objects kept in museums, along with writings and photographs.

Valliere builds each canoe from birchbark, cedar, spruce roots and pitch. The hull, the body of the boat, is thick birchbark. Ribs of straight-grained cedar hold the hull in shape, and spruce roots sew the pieces of bark together. To seal the seams against water, he mixes pitch, a sticky tree sap, with oak ash and deer tallow, the fat of a deer. Each material is gathered in a different season of the year.

More than 100 students have learned from him. At least one of them now builds canoes without his help, and a second is learning as an apprentice, working beside him. Valliere also teaches the Ojibwe language at Lac du Flambeau Public School.

Valliere's National Heritage Fellowship, which he received from the National Endowment for the Arts in 2020, is the highest honor in the United States for folk and traditional arts, the crafts and music that families and communities hand down.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 1 MINUTE

European metal reached Native towns on the East Coast before most Europeans did. Native smiths, people who shape metal into tools, cut up European copper and brass, a yellow metal made by melting copper and zinc together, and made tools of their own design from it.

Copper kettles cut into arrow points

Copper and brass from Europe reached the East Coast through trade with Europeans along the shore. Both metals had been made in European furnaces from rock dug out of the ground. Native traders passed the metal from one group to the next, so it spread along the East Coast ahead of most Europeans. In 1524 Verrazzano, a European sailing along the Northeast coast, wrote that the Native people there had sheets of worked copper.

Some of the metal came as kettles, large cooking pots. Native smiths cut the kettles apart and hammered the pieces into arrow points, knives, pipe stems, bells and ornaments. They took the new metal and made it into the tools and goods they already used.

Spanish expeditions, groups of men sent out to explore, crossed parts of what is now the United States in the 1500s, through lands where Native nations lived. The men on these expeditions carried iron weapons and tools. They left behind no lasting trade in tools.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 9 MINUTES

English colonists used running water to drive machines. They dammed streams to turn waterwheels, and the wheels drove grist mills, which ground grain into flour, and sawmills, which cut logs into boards. At ironworks, places where workers made iron from rock dug out of the ground, waterwheels worked the bellows and the hammers.

The first ironworks begun in the English colonies stood on Powhatan land in Virginia. The first one that ran for years stood on Pawtucket land in Massachusetts. Its workers were skilled English ironworkers, nearby farmers, at least two Native woodcutters, and Scottish prisoners of war whom the company's agents forced to work there for seven years.

Falling Creek, 1619 to 1622

Powhatan warriors destroyed the ironworks at Falling Creek, Virginia, in 1622, three years after Virginia Company officers sent the first ironworkers there. It was the first ironworks begun in the English colonies. Colonists built it at the falls of the James River, where the river drops over rocks, near present-day Richmond. In 1619 officers of the Virginia Company, the English company that ran the colony, hired Captain Benjamin Bluett to run the ironworks and sent workers with him.

Bluett died on the voyage or soon after he arrived. Others of his men died too, of what colonists called seasoning, the sickness that killed between 50 and 90 percent of early newcomers, according to Chesterfield County's history of the site. The men who lived could not finish the furnace, the tall oven where iron is melted out of rock.

Company officers hired John Berkeley to take over in 1621. He sailed with his son Maurice and 25 other men, according to the county, and he promised iron by June 9, 1622.

Why the Powhatan attacked. Like the colonists' tobacco farms, the ironworks stood on Powhatan land. Opechancanough, a Powhatan leader, planned an attack across the colony to force the English to leave Powhatan territory, as the county's history states it. From 1618 to 1620, English tobacco planters, owners of large farms called plantations, had taken Powhatan land without paying for it. Virginia Company officials had pressed to take Powhatan children for English schooling.

The attack. Powhatan warriors attacked the ironworks on March 22, 1622. They killed 27 English people there, John Berkeley among them. Two children at Falling Creek survived. The warriors left the works in ruins. Across the colony that day, Powhatan warriors killed about 350 English colonists.

Maurice Berkeley was away from the works and lived. He tried to start the ironworks again and failed. Another Englishman tried from 1634 to 1636 and failed too.

Whether Falling Creek ever made iron is disputed. Virginia's official list of historic places states that workers produced iron there in 1620, and it calls Falling Creek the first successful ironworks in English North America. The county's history states that Bluett's men could not finish the furnace, and that in 1621 Berkeley was still promising iron for the next year. Either way, the first ironworks in the English colonies that ran for years was the one at Saugus, in Massachusetts.

Seven waterwheels at Saugus

The first ironworks in the English colonies that ran for years stood on the Saugus River in Massachusetts. By 1643 John Winthrop Jr. had gathered about two dozen investors for it, people who put in money hoping for a profit. They formed the Company of Undertakers of the Iron Works in New England, using undertakers in an old sense, people who take on a big job. The works ran from 1646 to about 1670.

Seven waterwheels drove three kinds of machine. The first was a blast furnace. Air blown into its fire made the fire hot enough to melt iron out of rock, about 3,000 degrees Fahrenheit. The air came from bellows, giant bags that squeeze air into a fire, and these bellows were 18 feet long.

Next came a forge, where 10 to 12 men hammered the iron. The last machine was a rolling and slitting mill, which pressed iron into flat sheets and cut it into thin rods.

The ironworks burned more than 3,600 cords of wood a year, each cord a stack 8 feet long, 4 feet wide and 4 feet high. Set end to end, one year's cords would have made a stack more than 5 miles long. Workers cut the wood with axes and saws made at the works.

The Saugus works closed because of debt, high costs and competition from iron brought in from overseas. The men who had lent money sued. Some of the ironworkers, such as James Leonard, went on to run forges in other places. The works have been rebuilt on the old site, which is now the Saugus Iron Works National Historic Site.

Who did the work at Saugus?

About 100 people worked at the Saugus ironworks. The company's managers brought skilled ironworkers over from the iron-making parts of England, and colonists from nearby farms did the unskilled work. Company records show that at least two Native people cut wood for the same wages as the other workers. The records do not give their names.

The workers lived in a village the company owned, called Hammersmith after the English town that many of the ironworkers came from. The works stood on the land of the Pawtucket, the Native people who lived there when the English arrived. Saugus had been a Native settlement, and the English first called it Lynn.

The Scots. Some of the workers were prisoners of war. In September 1650, Oliver Cromwell's English soldiers defeated a Scottish army at Dunbar, in Scotland, and captured Scottish soldiers. Members of England's Council of State, the leaders who ran England's government at the time, ordered 150 of the prisoners handed over to agents of two of the ironworks company's investors.

The agents shipped the prisoners across the ocean on the ship *Unity*, which left Gravesend, England, on November 11, 1650, and reached Boston in December. How many died on the voyage is not known. The company kept 62 of the prisoners for its ironworks, and about 36 of them went to work at Saugus, though a National Park Service history gives about 35. The agents sent others to mills in Maine and New Hampshire and sold the rest to farmers and merchants in New England.

The company held the Scots at Saugus to seven years of forced labor under indentures, written contracts that required a person to work for a master for a set number of years. The Scots cut wood, worked as colliers, who make charcoal, and did other labor. They received clothing, food and tobacco.

Some of the Scots are known by name. A list of 35 Scots at the works in 1653 includes James Adams, John Clark, Alexander Innes, Peter Grant, John Stewart and Thomas Tower. A November 1653 list of the property of the works

includes Alexander Innes, with a value of 10 pounds. The Scottish Prisoners of War Society prints these lists from the company's record books.

The danger. Flying sparks and pieces of hot metal were a constant danger to the men at the forge. The forge hammer weighed 500 pounds, and its noise cost many workers their hearing. The surviving records of the works do not give a count of deaths.

The Cambridge printing press

The first printing press in the English colonies, a machine that presses paper hard onto metal letters covered in ink, arrived in Cambridge, Massachusetts, in 1638. The Reverend Jose Glover, a minister, bought the press and hired a locksmith named Stephen Daye to run it. Glover died on the voyage across the ocean. His widow, Elizabeth Glover, owned the press, and Daye ran it.

Daye printed the *Bay Psalm Book* on the press in 1640. It was a book of psalms, the religious songs of the Bible, and the first book printed in what is now the United States.

Joseph Jenks

Who: A London sword maker who ran the blacksmith shop at the Saugus ironworks and asked the colony to protect his machines from copiers.

When and where: Saugus, then part of Lynn, Massachusetts. At the works from 1646. Died in 1683.

The colony gave Jenks 14 years of protection from copiers.

Joseph Jenks wanted to stop other men from copying his machines. In 1646 Massachusetts leaders gave him a patent for his water-driven mills. A National Park Service page about his shop states that US Patent Office officials credit him with the first patent for a machine in what is now the United States.

Jenks learned his trade in England. In the 1630s he worked for a sword maker whose sword mill at Hounslow ran on water power. A sword marked with Jenks's name as its maker survives in a museum in Wales.

The petition. In May 1646 Jenks asked the Massachusetts General Court, the leaders who made the colony's laws, for the only right to build water-driven mills for 14 years. The mills would make scythes, long curved blades for cutting grass. He also asked to build "a new Invented Saw Mill" and other machines for making tools with edges. His written request survives in the Massachusetts Archives, dated May 10, 1646.

In it he explained why he wanted protection. Without it, another man would see his machine and copy it, and Jenks would lose the profit. In his own spelling: "Another when he seeth it makes the like," and "soe I loose the benefit." The court's magistrates, the leaders who sat as its judges, granted his request.

The shop. The agent in charge of the ironworks let Jenks build a water-driven hammer at the tail of the blast furnace on January 20, 1647. It ran on the water that had already turned the furnace wheel. The company owned the shop, and Jenks paid rent.

There he made saw blades, felling saws for cutting down trees, and steelyards, a kind of weighing scale. He drew iron into fine wire, pulling it through small holes to make it thin. In 1652 he sent two saw blades for a water-powered sawmill that John Winthrop Jr. ran in Connecticut.

When the ironworks company went bankrupt, unable to pay its debts, Jenks bought a share of his shop in September 1656 and the rolling and slitting mill that October, with borrowed money. By October 1667 he had little money left, and he asked the court for 50 pounds to start making iron wire. He died in March 1683.

What the ground showed. An archaeologist dug up the site of Jenks's shop in 1952. He found waterwheels, the supports of a water-driven hammer, an anvil base and a stone forge.

NOT KNOWN: whether Jenks made the dies, the stamps, for the first silver coins made in New England in 1652, as a marker put up in 1898 states, because no document from the time shows that he did.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 6 MINUTES

Iron furnaces and forges went up across Pennsylvania, Maryland and Virginia, and in Maryland many of the ironworkers were enslaved people or convicts. At the Baltimore Iron Works, which opened in 1731, 42 of the first 89 workers, almost half, were enslaved. Waterwheels still drove the machines, and the furnaces still burned charcoal. In Philadelphia, the printer Benjamin Franklin designed a new kind of stove in 1741 or 1742 and refused to patent it.

How the Franklin stove saved heat

Benjamin Franklin designed a stove that sent more of a fire's heat into the room and less of it up the chimney. He made it in 1741 or 1742. The Franklin Institute, a science museum in Philadelphia, states 1742 in its history of his inventions.

Franklin's stove was a fireplace lined with metal that stood a few inches out from the chimney. At the back was a hollow iron wall, called a baffle. Air from the room flowed into the baffle, picked up the fire's heat and came back out into the room.

Behind the baffle was a U-shaped passage called an inverted siphon. The smoke had to go up over the baffle, down behind it and only then up the chimney. On that longer path, the iron took in more of the smoke's heat. The stove made less smoke than an ordinary fireplace, too.

Franklin refused to patent the stove. He gave his reason later in his autobiography, the story of his life that he wrote himself. "As we enjoy great Advantages from the Inventions of others," he wrote, "we should be glad of an Opportunity to serve others by any Invention of ours." He added, "and this we should do freely and generously." Everyone gains from other people's inventions, he meant, so people should share their own for free. He never patented any of his inventions.

Enslaved people and convicts at the Maryland furnaces

Many of the workers at the Maryland ironworks were enslaved people or convicts. In 1719, investors in Britain set up the Principio Company in Cecil County, Maryland. In 1731 Dr. Charles Carroll and four other Maryland planters, two of them also named Carroll, founded the Baltimore Iron Works on the Patapsco River. Dr. Carroll put in 2,200 acres of land and served as the first manager. The five partners supplied the money and the enslaved workers.

Whose land. Susquehannock leaders had signed away the Patapsco valley in 1652, though the Piscataway lived in the region too. Five Susquehannock war chiefs signed that treaty, a written agreement between nations, with Maryland colonial officials. In it they gave up their claim to a long stretch of the Chesapeake Bay shore that includes the valley. Emma Bilski, a history graduate student at Johns Hopkins University, points out that the land was not really the Susquehannock's to give away.

Convicts. The Baltimore Iron Works opened with 89 workers, and 42 of them were enslaved. Over the years, many other workers there were convicts, people found guilty of crimes in Britain. Under a British law of 1718, English judges sentenced many such people to be shipped to the colonies. Merchants who held contracts with the British government carried them across the ocean and sold their labor to colonists, most often for seven years. Iron makers were among the buyers.

The managers of the Baltimore Iron Works preferred enslaved workers to convicts. Many of the enslaved ironworkers were Igbo people from West Africa, who brought their own skill at working metal. By 1785 an advertisement for the works listed a workforce of "more than 200 Negroes," using a word of that time for Black people.

The work. The workers hauled charcoal, oyster shells and iron ore to the top of the furnace. They worked 12-hour shifts, day and night, six or seven days a week.

The owners lived away from the works and often quarreled, and they refused to send the food and workers they had promised. In 1775 the manager, Clement Brooke, wrote to the partners: "I cannot do my job unless you send

more slaves and more corn." The Mount Clare Museum's history of the works states that the workers and the animals were starving.

Each of the five owners took one fifth of the iron. In good years the partners earned 55 percent on their money each year. The works closed in 1799. In 1810 the 30 heirs of the original owners, the people who had inherited their shares, had the land that was left sold at auction. It came to more than 30,000 acres. The heirs had 200 enslaved people sold at the same auction.

Some workers ran away. In August 1754 the *Maryland Gazette* newspaper printed a notice for the capture of people who had escaped. Others worked slowly.

An enslaved blacksmith named Anthony and an indentured servant named Thomas Livy were jailed for robbing a store in the spring of 1771. They were caught with the stolen goods, most likely food. The Mount Clare Museum's account of the case does not say who caught them or who jailed them. The overseer, the man in charge of the workers, was William Hammond. He wrote about Anthony: "We shall feel the loss of Anthony's time very much, I cannot tell how we can carry on the Forge business without him."

Benjamin Franklin

Who: A Philadelphia printer who designed the Franklin stove and, later, the lightning rod, and refused to patent either one.

When and where: Born in Boston in 1706. Worked in Philadelphia. Died there in 1790.

Movie: *Benjamin Franklin* (2022), a two-part documentary by Ken Burns, shown on PBS. It is about Franklin's life.

A London ironmonger patented a copy of Franklin's stove.

Benjamin Franklin made his living as a printer in Philadelphia. He printed a newspaper and a yearly almanac, a book of calendars, weather guesses and sayings.

Franklin gave the model of his stove to an old friend, Robert Grace, who owned an iron furnace in Pennsylvania. Grace cast the iron plates for the stoves there, and he made money on them, because more and more people wanted one. To help sales, Franklin wrote a pamphlet in 1744 that explained how the stove worked.

Governor Thomas of Pennsylvania liked the stove so much that he offered Franklin a patent, which would have let only Franklin sell the stoves for a set number of years. Franklin turned the offer down.

In London an ironmonger, a seller of iron goods, copied much of Franklin's pamphlet. He made small changes to the stove that "rather hurt its operation," took out a patent on it in England and, Franklin wrote, "made, as I was told, a little fortune by it." Other people patented Franklin's inventions too. Franklin never challenged them, he wrote, "having no desire of profiting by patents myself, and hating disputes."

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 10 MINUTES

Machines began to do work that had taken many hands, and after one of them, the cotton gin, planters enslaved far more people. Eli Whitney built the gin in 1793. One small gin could clean about 50 pounds of cotton a day, work that took several hundred hours by hand.

Planters grew far more cotton, counted in bales, large bundles pressed tight and tied: fewer than 9,000 bales a year around 1790, more than 200,000 by 1800. To grow it, they held far more people in slavery, about 700,000 in 1790 and nearly 4 million by 1860.

Other machines came into use in the same years. Benjamin Franklin's lightning rods went up on Philadelphia roofs in the 1750s, and by 1755 the first steam engine in America was pumping water from a New Jersey mine. Samuel Slater's cotton-spinning mill began work at Pawtucket, Rhode Island, in 1790, the start of the factory cloth industry in the United States. Slater hired children to run its machines.

Franklin's lightning rod, step by step

Franklin described a way to protect buildings from what he called "electrical fire" in a letter of 1750 to Peter Collinson. After lightning tests in France in 1752, and then the kite test that Franklin described, the first lightning rods went up on buildings in Philadelphia.

Franklin printed instructions for making one in *Poor Richard's Almanack* for 1753. Readers were to take a small iron rod and push one end three or four feet into moist ground, with the other end standing six or eight feet above the top of the building. A foot of brass wire, sharpened to a fine point, went on the top end. Small staples held the rod to the house.

When lightning struck, it would hit the point and pass through the metal into the ground, "without hurting any Thing," Franklin wrote. Pointed rods soon stood on buildings throughout the colonies.

The first steam engine in America

The first steam engine in America pumped water out of a mine. It was ordered from England in 1748 for Colonel John Schuyler's copper mine, in what is now North Arlington, New Jersey. The parts arrived in 1753. An engine builder named Josiah Hornblower put it together, and it was pumping by 1755.

It was a Newcomen engine, named for Thomas Newcomen, who built the first one in England in 1712. A boiler filled a large tube, called a cylinder, with steam. Then a spray of cold water turned the steam back into water, which left the cylinder nearly empty. The weight of the air outside pushed a piston, a tight-fitting plug, down into the cylinder. The piston pulled one end of a big rocking beam, and the other end of the beam worked the pump in the mine.

The cotton gin and slavery

The cotton gin cleaned cotton so fast that planters grew far more of it, and they held far more people in slavery to grow and pick it. Upland cotton has sticky seeds caught in its fiber. Before 1793, enslaved people pulled the seeds out by hand, and one worker needed many hours to clean a single pound. Planters could grow only as much cotton as their workers could clean.

Eli Whitney's cotton gin did the cleaning by machine. It used two cylinders that combed the cotton through small holes. The seeds could not pass through the holes and stayed behind. A history of the gin by the Eli Whitney Museum states that one small gin could clean 50 pounds of cotton a day. By hand, that much cotton took several hundred hours of work.

Planters used the gin to grow far more cotton. Across the United States they grew fewer than 9,000 bales of cotton a year around 1790 and more than 200,000 bales by 1800. By 1860 the cotton crop was more than 400 times as large as in 1790.

More people enslaved. With the gin cleaning cotton so fast, planters wanted far more enslaved people to grow and pick it. The 1790 census, the government's count of everyone in the country, recorded 697,897 enslaved people in the United States. By 1860 there were nearly 4 million, more than five times as many, and about 2.5 million of them lived in the cotton-growing regions from eastern Maryland to Texas. Slaveholders spread slavery west across the South.

Enslaved people picked the cotton by hand in the summer sun, about 200 pounds each a day. Solomon Northup was a free Black man from New York. Kidnappers seized him in Washington, D.C., in 1841 and forced him into slavery. Slaveholders held him for twelve years. He picked cotton on Edwin Epps's plantation in Louisiana until he was rescued in 1853.

Northup described the rule in his book *Twelve Years a Slave*, published that year. "An ordinary day's work is two hundred pounds," he wrote. "A slave who is accustomed to picking, is punished, if he or she brings in a less quantity than

that." Each night the cotton was weighed at the gin-house, the building where the gin stood. "After weighing, follow the whippings," Northup wrote.

The punishment was the lash, one stroke of a whip on a person's bare body. Northup wrote that twenty-five lashes were "deemed a mere brush," thought of as a small punishment. Fifty was "the ordinary penalty," and one hundred was "called severe." Northup described Epps himself seizing the whip and lashing the people he enslaved.

Patsey. Epps once forced Northup to whip an enslaved woman named Patsey, and then took the whip himself. Northup wrote that her back was "covered with long welts," raised and swollen stripes, and that "the lash was wet with blood," meaning the whip itself. Epps then had her washed with salt and water.

Enslaved people ran the gins and the cotton presses too, the machines that squeezed cotton into bales. The machines had many moving parts that could catch and injure a worker. The tiny cotton fibers the gins threw into the air damaged the workers' lungs.

The people who owned the plantations kept the profit. With the gin, Southern plantation owners gained money and political power. Their cotton supplied cloth mills in the Northern states and in Britain. By 1830, cloth and other finished cotton goods were the largest group of factory-made goods the country sold to other countries.

Cotton land. Planters wanted new land for cotton. The Eli Whitney Museum's history of the gin states that this demand was "a major factor" in the removal of five Native nations from their homelands in the Southeast: the Cherokee, Chickasaw, Choctaw, Muscogee (Creek) and Seminole.

President Andrew Jackson signed the Indian Removal Act on May 28, 1830. Under it, the president could trade land west of the Mississippi River for the lands of Native nations inside the states. US officials then forced the five nations west to Indian Territory, land that is now part of Oklahoma. At least 60,000 people were forced to make the journey, and more than 8,000 of them died on the way.

US Army soldiers under General Winfield Scott began forcing the Cherokee out of their homeland in 1838. Scott was sent to another post soon after the removal began. The soldiers first held the Cherokee in crowded, dirty camps, where many died.

About 4,000 Cherokee died on the march west that the soldiers forced them to make in the fall and winter of 1838 and 1839. On the march they faced heavy rain, rivers blocked by ice and too little food. Many died of illness, and some died of disease after drinking dirty standing water. White landowners took the land the five nations had been forced to leave.

Eli Whitney

Who: The builder of the cotton gin, patented in 1794, and the subject of a famous myth about machine-made gun parts.
When and where: Georgia and New Haven, Connecticut. Born 1765, died 1825.

Workers at Whitney's gun factory never made parts that fit any gun.

Eli Whitney graduated from Yale College in 1792 and traveled south to work as a tutor, a private teacher. On the way, Catherine Littlefield Greene invited him to her plantation, Mulberry Grove, north of Savannah, Georgia. She was the widow of General Nathanael Greene, who had died in 1786 and left her dozens of enslaved people. Enslaved workers on the plantation grew mostly rice and some cotton.

There, in 1793, Whitney built the machine that pulled cotton fiber away from its seeds. He applied for a patent in 1794, and the patent was granted on March 14, 1794.

Catherine Greene's part. Whitney held the patent. In 1870 the women's rights activist Matilda Joslyn Gage wrote that Catherine Greene invented the cotton gin. The Eli Whitney Museum states that it "has been suggested that Catherine Greene also contributed ideas to the design."

NOT KNOWN: how much of the design was Catherine Greene's.

The gun parts. Whitney signed a contract in 1798 to make muskets for the federal government. He promoted the idea of interchangeable parts, gun parts made by machine so that any part would fit any gun of the same model. According to a famous story, he showed government officials such parts in 1801.

The historian Robert S. Woodbury showed in 1960 that Whitney did not start the idea, and that the credit people gave him for it rests on a myth. Gunmakers at French armories, factories that made weapons, had worked on the idea before Whitney. Workers at Whitney's own factory never made parts that truly fit any gun.

The best-documented first American use of interchangeable parts in a factory came at John H. Hall's rifle works at the Harpers Ferry armory. Hall's workers were putting rifles together from interchangeable parts by 1819.

Samuel Slater

Who: An English mill worker who rebuilt England's spinning machines from memory at Pawtucket and hired children to run them.

When and where: Trained in Derbyshire, England. The mill ran at Pawtucket, Rhode Island, from December 1790.

He crossed the ocean with England's spinning machines stored in his memory.

Samuel Slater rebuilt English cotton-spinning machines from memory in the United States. In England, Richard Arkwright had patented a spinning machine in 1769, called the water frame because a waterwheel drove it. Arkwright and his partners built a mill several floors high at Cromford, in Derbyshire, with waterwheels to drive the machines. He moved cotton spinning out of homes and into powered mills with hundreds of workers, and other mill owners copied his system.

Slater trained in a mill in Derbyshire that used Arkwright's machines. Under British law, machines for making cloth could not leave the country, and skilled cloth-mill workers could not move abroad. In 1789 Slater left England without telling officials that he was a trained mill worker.

Pawtucket. Moses Brown hired him in December 1789, and Brown's family firm, Almy & Brown, put up the money and the building. With local mechanics, people who build and repair machines, among them the Wilkinson family, Slater rebuilt two kinds of machine from memory. Carding machines combed raw cotton so the fibers lay straight, and spinning machines twisted the fibers into thread, called yarn. The machines at Pawtucket spun their first yarn in December 1790.

The children. Slater had worked in a factory for part of his own childhood. At Pawtucket he hired children to run the machines, which were built for children to run. Most of the children at Slater's mill were about 10 years old, and some were as young as six. They were paid by the hour, and they earned much less than adults.

> A Boston politician in 1801 described mills that employed "children from four to Ten years old." In the mill villages that grew up later, the child workers were 8 to 14 years old. The names of the first children who worked for Slater are not recorded.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 8 MINUTES

The men whose names went on the patents of these years had help. Samuel Morse, Charles Goodyear, the sewing machine maker Elias Howe and the reaper maker Cyrus McCormick each held a patent, a government grant of the only right to make and sell an invention for a set number of years. Morse's partner Alfred Vail built the telegraph instruments. Jo Anderson, a man enslaved by the McCormick family, worked on the first McCormick reaper.

Inventors in the United States built machines to cut grain, plow prairie soil and sew cloth between 1831 and 1846. Goodyear found a way to treat rubber with sulfur so that heat did not melt it. People in Philadelphia made a new French way of taking photographs work in 1839, within weeks of hearing about it. By 1844 Morse's telegraph line ran 40 miles, from Washington to Baltimore.

How the telegraph sent a message

The electric telegraph sent messages as pulses of electricity through a wire. At the receiving end, each pulse worked an electromagnet, a magnet that works only while electricity flows through it. The magnet moved an arm with a pen-like point, called a stylus.

A clockwork motor, which ran on gears like the inside of a wind-up clock, pulled a strip of paper tape past the stylus. For each pulse, the stylus pressed a small dent into the tape, a short one for a dot or a longer one for a dash. Every letter and number had its own pattern of dots and dashes, and the patterns are called Morse code.

Morse asked Congress for $30,000 in 1843 to build a telegraph line from Washington to Baltimore. Members of the House and then senators voted for the money, and President John Tyler signed the bill.

Ezra Cornell, a man hired to build the line, was supposed to lay the wire underground. When he ran into bad wire, he suggested stringing the wires overhead on trees and poles instead. A signal in a long wire grows weaker as it goes, and Leonard Gale showed Morse how to boost it at points along the line with a device called a relay.

The first message. Morse sent the first public message over the line from the US Capitol in Washington on May 24, 1844, and Alfred Vail received it in Baltimore. The words were "What hath God wrought," from the Bible, in the book of Numbers, chapter 23, verse 23. They mean "What has God done."

Annie G. Ellsworth chose the words. Her father was the commissioner of the Patent Office, the head of the government office that grants patents. The paper tape with the message still exists, and the Smithsonian's National Museum of American History holds it.

Samuel Morse

Who: The inventor who held the patents on the electric telegraph and sent the first public message over the Washington to Baltimore line.

When and where: The story is that the idea came to him in 1832. The first public message went from Washington to Baltimore on May 24, 1844.

Morse got the fame. Alfred Vail built the instruments.

Samuel Morse held the patents on the American electric telegraph and got the fame for it. According to a Library of Congress history, the idea of sending messages over a distance by electricity came to Morse in 1832, during a conversation on a ship coming home from Europe.

Alfred Vail built and improved the instruments in the shop at the Speedwell Ironworks. The same Library of Congress history identifies hiring Vail as one of the keys to Morse's success.

The code. Who worked out the dot and dash code is still argued. Morse held the patents, and people call it Morse code. Starting in 1887, Vail's son Stephen Vail sent letter after letter to newspaper editors claiming the credit for his father. In 1911 another son, J. Cummings Vail, had Alfred's gravestone changed to read "INVENTOR OF THE TELEGRAPHIC DOT AND DASH ALPHABET."

The writer Joseph Connor, in a 2020 article in *American History* magazine, credits Vail with the practical version, which gives the shortest codes to the letters people use most. According to Connor, Vail learned which letters were most common from printers, who kept the metal type for those letters closest at hand.

Sewing machines, rubber and the camera

Elias Howe patented a sewing machine on September 10, 1846. It sewed a lockstitch, in which two threads lock together through the cloth. Isaac Singer made a more practical machine, and it sold better than Howe's. Howe sued the makers who used his invention. In 1854 judges ruled that Howe's patent was good, and the makers had to pay him royalties, fees for using a patent. At the peak, the fees paid to Howe reached as much as $4,000 a month.

The Combination. Howe and the biggest makers, Singer, Wheeler & Wilson, and Grover & Baker, pooled their patents in 1856, so each member could use the others' patented inventions. The members of their group, called the Sewing Machine Combination, controlled the sewing machine trade until 1877.

Rubber on a hot stove. A piece of Charles Goodyear's mix of rubber and sulfur landed on a hot stove in Woburn, Massachusetts, in 1839. The rubber charred like leather instead of melting. It took Goodyear five more years to learn to make the change happen every time. He patented the process in 1844, and rubber treated this way is called vulcanized rubber.

The accounts of how the piece reached the stove differ. Mass Moments, a Massachusetts history site, states that he either dropped it or placed it there. A speech in his honor printed in 1860 states that the piece was in his hand and was thrown into the stove "by a violent gesture." The encyclopedia Britannica's account states that it flew from his hand while he was trying to sell his rubber.

The camera. François Arago announced the daguerreotype, an early way of making photographs invented by the French artist Louis Daguerre, at a meeting of the French Academy of Sciences in Paris on August 19, 1839. The news reached the United States in late August and September. Within weeks, people in Philadelphia had made the process work for themselves.

Robert Cornelius of Philadelphia made a photograph of himself in October or November of 1839. It is the oldest surviving portrait photograph made in America, and the Library of Congress holds it.

Cyrus McCormick and John Deere

Who: Two makers of farm machines: McCormick's reaper cut grain, and Deere's steel plow cut through sticky prairie soil.

When and where: The reaper at Walnut Grove and Steele's Tavern, Virginia, 1831. The plow at Grand Detour, Illinois, 1837.

An enslaved man carried the grain that tested the first reaper.

Cyrus McCormick held the patent on the McCormick reaper, a machine that cuts ripe grain, and he built the business that sold it. Other people worked on the first one.

His father, Robert McCormick, tried for several years to build a reaper at Walnut Grove, the family farm in Rockbridge County, Virginia. In 1831 he gave up. Within six weeks, Cyrus showed a working reaper by cutting oats at nearby Steele's Tavern. A blacksmith named John McCown worked on the machine. So did Jo Anderson, a man the McCormicks enslaved.

Jo Anderson's own words. Anderson was born on Robert McCormick's plantation in about 1807. In 1885 a white interviewer, William J. Hanna, wrote down Anderson's words in dialect spelling, which tries to show how a person's speech sounded. Hanna recorded Anderson as 78 years old and unable to walk.

"I was born at old Massa Rob't McCormick's house," Anderson said. "I was his slave, but when a baby was given to Massa Cyrus." Massa meant master. Robert McCormick owned Anderson from birth and gave him to his son Cyrus while Anderson was still a baby.

Anderson said he carried grain to the blacksmith shop to test the machine: "I fotched loose grain to the shop so he could see how it cut it." He also worked as "blower and striker for the blacksmith," the helper who pumps the bellows, the big bag that blows air onto the fire, and swings the heavy hammer.

Anderson gave Cyrus the credit for the reaper and said Robert "tried it and gave it up." He said Cyrus freed him before the Civil War.

The business. Another inventor, Obed Hussey, patented a reaper in December 1833. McCormick patented his own in 1834 and printed a warning against Hussey's machine in *Mechanics' Magazine*. He made his first sales in 1840 and moved to Chicago in 1847. In 1859 he and his brothers made and sold more than 5,000 reapers.

The steel plow. John Deere made plows at Grand Detour, Illinois. The sticky prairie soil there clogged iron plows. In 1837 Deere built a plow with a polished steel moldboard, the curved blade that turns the soil over. The sticky soil slid off the smooth steel, so the plow scoured, or cleaned, itself as it moved. By 1846 Deere was selling nearly 1,000 plows a year.

Whose land. Ho-Chunk people, long called the Winnebago in English, had lived in a village at the bend of the Rock River where Grand Detour stands. Settlers who came later said they could still see signs of that village.

US officials pressured the Ho-Chunk into selling more of their land in 1832. That year Ho-Chunk leaders signed a treaty with two men sent to make it for the United States, General Winfield Scott and John Reynolds, the governor of Illinois. In it the Ho-Chunk gave up their land along the Rock River and across southern Wisconsin, and kept only their land farther north. US officials agreed to pay them $10,000 a year for 27 years and hoped they would move west to Iowa.

The records do not make clear whether this treaty or an earlier one covered the exact spot where Grand Detour stands. Five years after this treaty, Deere built his steel plow there, on land where Ho-Chunk people had lived.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 16 MINUTES

Inventors after 1850 worked with patent lawyers and teams of workers, and US officials decided who could own an invention at all. In 1858 the US attorney general, the government's chief lawyer, ruled that an invention by an enslaved person could not be patented. According to the law scholar Kara Swanson, after that ruling free Black inventors could not get patents either.

Alexander Graham Bell received the first telephone patent in 1876. Thomas Edison's team at Menlo Park, New Jersey, made a long-lasting electric lamp in 1879, building on 40 years of work by other inventors.

Who could own an invention?

Only the person who made an invention could get a patent for it. In 1858 the government's chief lawyer ruled that an enslaved inventor could not get one, and that his owner could not get one either.

Ned's invention. An enslaved blacksmith named Ned, near Holmesville in Pike County, Mississippi, had invented a new farm tool by August 1857. It was a "double Cotton Scraper, and two plows" that could "Scrape both Sides of the Cotton ridge at the same time, and plough out the middles," meaning both sides of a raised row of cotton and the low strips between the rows.

Everything known about Ned comes from the letters of Oscar J. E. Stuart, a white planter and lawyer who claimed to own him. Ned's second name, if he had one, is not recorded. How his life ended is not known.

Stuart applied for a patent on Ned's invention on November 15, 1857. He swore that he owned Ned and that Ned was the inventor. Stuart threatened Ned, too. If Ned wrote to the Patent Office himself, Stuart said, he would "correct" Ned "according to our Southern usage," and to correct meant to punish. Stuart made the threat in a letter he later sent to the secretary of the interior, one of the president's top officials. He did not write in it what the punishment would be.

The patent commissioner, Joseph Holt, sent the application back. Under the law, the inventor had to swear an oath, a solemn promise before the law, that he had made the invention, and Holt decided that an enslaved man could not take the oath. Stuart complained to Jacob Thompson, the secretary of the interior. Thompson sent the question to the attorney general, Jeremiah S. Black.

The ruling. Jeremiah Black ruled in 1858 that "a machine invented by a slave, though it be new and useful, cannot, in the present state of the law, be patented." He based his ruling on the *Dred Scott* decision of 1857, in which the justices, the judges of the Supreme Court, ruled that African Americans were not citizens. Without citizenship, Jeremiah Black reasoned, an enslaved inventor could not swear the oath. The owner could not get the patent either, because only the "original and first inventor" could.

Free Black inventors had received patents before this. The law scholar Kara Swanson writes that Jeremiah Black's opinion "created a formal racial barrier" to the patent system, a bar written into the rules. After it, free Black inventors could not get patents either.

Stuart asked Congress twice for a patent and failed both times. In 1859 Senator Albert Gallatin Brown of Mississippi backed a bill that would let enslavers patent the inventions of the people they enslaved, but senators never moved the bill forward. Stuart sold the "Stuart Double Plough and Scraper" anyway, without a patent.

A propeller for boats. Benjamin Montgomery, enslaved in Mississippi by Joseph Davis, invented a better propeller for boats. In the late 1850s Joseph Davis tried to patent it through his brother, Jefferson Davis, then a US senator from Mississippi. Patent Office officials turned him down for the same reasons they had given Stuart: Joseph Davis was not the inventor, and an enslaved inventor could not swear the oath.

The Confederacy. Delegates from seven Southern states that had broken away from the United States met in Alabama in February 1861. There they set up their own government, the Confederate States of America, also called the

Confederacy, and chose Jefferson Davis as its president. On May 21, 1861, he signed a Confederate patent law under which enslavers could patent the inventions of the people they enslaved.

Benjamin Montgomery was free by June 1864, when he applied for a patent on his propeller in his own name. No patent was ever issued, and the records do not say why.

Patents again. At the end of the Civil War in 1865, members of Congress and lawmakers in the states added the 13th Amendment, a change to the Constitution, and with it ended slavery everywhere in the country, except as a punishment for a crime. Under the 13th Amendment and the amendments that followed it, Black inventors could get patents again.

Counting Black inventors. Henry E. Baker, a Black patent examiner who checked patent applications for the government, published a count of the patents Black inventors had received. He wrote it in 1913, to mark fifty years since the Emancipation Proclamation of 1863, President Abraham Lincoln's order declaring free the enslaved people in the Confederate states.

Baker called the pamphlet *The Colored Inventor*, using a word of the time for Black Americans. The publishers of *The Crisis*, the magazine of the NAACP, a group that works for the rights of Black Americans, printed it.

The Patent Office's records did not show an inventor's race. To find Black inventors, Patent Office staff wrote to thousands of patent lawyers, to newspapers and to factory owners, then checked each answer against the records. Baker counted about 800 patents granted to Black inventors. He judged that the real number was at least twice that, because many lawyers said they could not remember the names of their Black clients.

The telephone, 1876

A telephone turns the sound of a voice into electric current and back into sound. In Bell's own description from 1876, the voice makes a thin sheet, called a membrane, shake in time with the sound. The shaking makes the electric current in the wire grow stronger and weaker in the same pattern. At the other end, the changing current makes a second sheet shake, and the listener hears the voice.

Bell's lawyer filed an application for a patent called "Improvement in Telegraphy" on the morning of February 14, 1876. It was the fifth filing at the Patent Office that day. A few hours later, papers from Elisha Gray arrived as the 39th filing. Gray filed a caveat, a notice that he was working on an invention and planned to apply for a patent. Both papers described a liquid transmitter, a sending part that used a liquid. On the advice of his lawyers, Gray withdrew.

Patent Office officials issued patent number 174,465 to Bell on March 7, 1876. On March 10 Bell spoke the first sentence ever sent by telephone. He spoke to his assistant, Thomas A. Watson, who was in the next room of Bell's laboratory in Boston. Bell's notebook records the words "Mr. Watson," then "come here," then "I want to see you."

Bell filed first and held the patent. People have argued ever since about whether Bell or Gray deserved the credit.

Alexander Graham Bell

Who: A Boston inventor whose lawyer filed for the telephone patent a few hours before a rival's papers arrived.

When and where: Boston, 1876.

Bell's first working telephone used a sending part like his rival's.

Alexander Graham Bell was a professor at Boston University when he worked on the telephone. He hired Thomas A. Watson, who had worked in a Boston machine shop since 1872, as his assistant.

The telephone Bell spoke through on March 10, 1876, had a liquid transmitter like the one Elisha Gray had described in his caveat, which was supposed to stay secret. Benjamin Lathrop Brown studied whether Bell had copied Gray, and he concluded that Bell and his lawyers did not.

Brown published his study in November 2020 in an engineering journal, the *Proceedings of the IEEE*, as "The Bell Versus Gray Telephone Dispute: Resolving a 144-Year-Old Controversy." He based his finding on letters, notes kept by a man who worked with Bell, and the engineering details of the two designs. He was answering several people who had claimed that Bell saw Gray's papers on a trip to Washington and copied the idea.

Emma Nutt

Who: The first woman telephone operator.

When and where: Boston, hired September 1, 1878.

Her pay as the first woman telephone operator: $10 a month.

Emma Nutt left a job at a telegraph office on September 1, 1878, to work for the telephone dispatch company in Boston. She became the first woman to work as a telephone operator, the person who connected each caller to the person being called. Her pay was $10 a month for a 54-hour week, less than 5 cents an hour. Her sister, Stella, started work as an operator the same day.

The first operators had been teenage boys, who were rude to callers and played pranks on them. The telephone companies' managers replaced the boys with women. By the late 1880s, almost every telephone operator was a woman. It was one of the first big jobs in a new technology open to women.

Emma Nutt worked as an operator for more than 30 years. The New England Historical Society, *Boston Magazine* and *Time* give the exact length as anything from about 33 to about 37 years.

Edison's team at Menlo Park

An electric lamp makes light when an electric current heats a thin thread inside it, called a filament, until the filament glows. Inventors had worked on electric lamps for 40 years before Edison's team made its lamp in 1879. British inventors showed arc lamps, which make light from a spark between two rods, starting in 1835. Joseph Swan in England patented a lamp design before Edison did, and so did William Sawyer and Albon Man.

Edison ran a laboratory at Menlo Park, New Jersey, with a team of workers. Among them were Charles Batchelor, John Kruesi and Francis Upton. In October 1879 the team made a lamp with a filament of cotton thread that had been baked into carbon. It burned for about 14.5 hours. Edison showed the lamps to the public at Menlo Park on New Year's Eve, 1879. The team kept testing filaments and settled on one made of bamboo. It let Edison's lamps burn for up to 1,200 hours, 50 full days.

Lighting a district. Edison's team built the other parts needed to light a whole district: the dynamo, a machine that makes electricity when it is turned, the wiring, and a meter to measure how much electricity each customer used. The Pearl Street station at 257 Pearl Street in Manhattan began supplying electricity on September 4, 1882, to the first district of paying customers.

The Menlo Park team built other machines too. Two years before the lamp, in early December 1877, John Kruesi built a phonograph, a machine that records sound and plays it back, to Edison's design. The first one recorded on a sheet of tinfoil, and on December 6, 1877, Edison recorded himself saying "Mary had a little lamb."

Thomas Edison

Who: An inventor whose team at Menlo Park made a long-lasting electric lamp, the phonograph and the system that brought electric light to paying customers.

When and where: Menlo Park and West Orange, New Jersey. Born 1847, died 1931.

Movie: *Edison* (2015), a documentary in the PBS series *American Experience*. It is about Edison's life and work.

Edison's West Orange laboratory had room for 20 projects at once.

Thomas Edison worked with a team. In 1876 he moved his staff of assistants to Menlo Park, New Jersey, and set up a laboratory with all the equipment needed to work on any invention. The National Park Service calls it the first research laboratory of its kind anywhere, the model for later ones.

In November 1887 Edison opened a much bigger laboratory in West Orange, New Jersey, less than a mile from his home. It had five buildings, with machine shops, a library and separate rooms for work in physics, chemistry and metals. Because the laboratory was so large, Edison and his staff could work on as many as 10 or 20 projects at once.

Edison first showed motion pictures in 1891. Two years later he began making movies to sell, in a building on the laboratory grounds called the Black Maria. Factories to make Edison's inventions grew up around the laboratory, and at the peak, during World War I, 10,000 people worked there.

Lewis Latimer

Who: A self-taught draftsman who drew the telephone patent and patented a better way to make the carbon filaments inside electric lamps.
When and where: Boston and New York. Born 1848, died 1928.

The son of two people who escaped slavery drew the telephone patent.

Lewis Latimer's parents had freed themselves from slavery in Virginia by escaping, and he was born in Chelsea, Massachusetts, on September 4, 1848. He taught himself to be a draftsman, someone who makes exact drawings of machines and buildings. While he worked at a patent law firm, he made the patent drawings for Bell's 1876 telephone application.

The filament. Latimer patented a better way of making the carbon filaments that glow inside electric lamps. The patent, number 252,386, was issued on January 17, 1882, for a "Process of Manufacturing Carbons." His method made filaments that were more even and lasted longer.

He received his filament patent while working at the U.S. Electric Lighting Company of Hiram Maxim. In 1884 he went to work for Edison's company, where he made patent drawings and testified in court as an expert on patents. He also oversaw lamp factories and the setting up of electric lighting in New York, Philadelphia, Montreal and London.

Latimer became one of the 28 first members of the Edison Pioneers, a group of people who had worked closely with Edison before 1885, on January 24, 1918. He was its only Black member. He died in Flushing, in Queens, New York, on December 11, 1928, at the age of 80. The house in East Flushing where he lived from 1903 is now the Lewis Latimer House Museum.

Granville T. Woods

Who: A Black inventor who patented a way to send messages between moving trains and railroad stations.

When and where: Cincinnati, Ohio. Patent issued November 29, 1887.

People on Edison's side claimed his inventions twice. Woods won both hearings.

Granville T. Woods invented a way for moving trains and stations to send messages to each other. He patented it as the Synchronous Multiplex Railway Telegraph on November 29, 1887, while working in Cincinnati. With it, the people who directed train traffic, called dispatchers, could know where the trains were.

Woods's telegraph worked by induction, in which a changing electric current in one wire makes a current flow in another wire nearby, without the two touching.

Twice, people on Edison's side claimed that inventions Woods had patented were theirs. Each time Patent Office officials held a hearing, called an interference proceeding, to decide who had invented first. Woods won both times. Edison offered him a job, and Woods turned it down. Over his career Woods received dozens of patents for electrical and railroad inventions.

Nikola Tesla

Who: A Serbian immigrant inventor whose induction motor used alternating current to make a shaft turn.

When and where: New York. Patent issued May 1, 1888.

Westinghouse paid him $60,000 and promised $2.50 a horsepower.

Nikola Tesla patented a motor that ran on alternating current, or AC, electric current that switches direction back and forth many times each second.

Other motors carried current to their turning part through brushes, sliding electrical contacts. Tesla's motor worked without brushes because it used induction, the same effect Woods used for his train telegraph.

Tesla was a Serbian immigrant who came to the United States in 1884. He filed for the motor patent in October 1887. The patent, number 381,968, was issued on May 1, 1888.

The deal. Westinghouse was a company that built electric generators and motors. In July 1888 it bought a license, the right to use Tesla's patents on AC motors. Tesla received $60,000 in cash and in stock, shares in owning the company. Under the deal he was also to get $2.50 for each horsepower, the measure of a motor's power, of the motors the company sold.

At her first pay of $10 a month, Emma Nutt, the first woman telephone operator, would have had to work 500 years to earn $60,000.

The typewriter and barbed wire

Christopher Latham Sholes, Carlos Glidden and Samuel Soule patented a typewriter on June 23, 1868. E. Remington & Sons began making it on March 1, 1873. The Sholes & Glidden Type-Writer went on sale in 1874 for $125. Its keyboard began with the letters Q, W, E, R, T and Y across the top row of letters. That arrangement, called QWERTY, is still used on keyboards.

Barbed wire. Joseph Glidden, a farmer from De Kalb, Illinois, received patent number 157,124 for barbed wire on November 24, 1874. He was a different man from the Carlos Glidden of the typewriter. Joseph Glidden's barb was a short piece of wire that locked onto a wire of two strands, and he built machines to make the wire in large amounts. Nine wire-fence designs were patented, and Glidden's was the one that sold well.

With barbed wire, people could fence land where there was little wood to build fences. In the West, landowners fenced off land and water that had been open to all. Some were homesteaders, families who had claimed farms, and some were ranchers. Their fences ended the open range, the unfenced land where cattle had grazed freely.

Range wars. Other cattlemen had grazed their herds on the open range, treating it as land for everyone's use. In Texas, New Mexico, Colorado and Wyoming, they found their old grazing land fenced off. They first asked officials for the right to use it. Then they fought what the National Archives calls "fierce range wars" against the farmers who owned the fenced land.

Native peoples who moved with the seasons had always used these lands too. The fences blocked them from more of those lands, according to the National Archives, and they called barbed wire "the Devil's rope."

BY 1900

If you remember one thing: Inventing became the work of teams and patent lawyers, and the end of slavery changed who could own an invention.

- In 1858: an enslaved blacksmith like Ned could not get a patent for his own cotton scraper, and the man who claimed to own him could not get one either.
- In 1887: Granville T. Woods, a Black inventor, patented his railway telegraph in his own name.

Next: After 1900, machines moved into American factories and homes, and two bicycle builders from Dayton, Ohio, flew.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 10 MINUTES

Americans brought machines into their factories, farms and homes in these fifty years. Workers at Henry Ford's car plant in Michigan started building car frames on a moving assembly line on October 7, 1913. Electric refrigerators and radios went into American homes, tractors began to replace horses on farms, and chemists made new materials, starting in 1907 with Leo Baekeland's Bakelite, the first plastic made entirely from chemicals.

The airplane and the electronic computer both date from these years. Wilbur and Orville Wright flew a powered airplane in North Carolina on December 17, 1903. In February 1946 engineers at the University of Pennsylvania showed ENIAC, the first general-purpose electronic computer, one that could be set up to solve many kinds of problems.

Scientists at Bell Labs, the research laboratory of the telephone company AT&T, made the first working transistor in December 1947. It was a small part that could control the flow of electricity and make a weak signal stronger. Transistors later replaced vacuum tubes, glass tubes with the air taken out, like light bulbs, in machines like ENIAC.

The Wright brothers

Who: Wilbur and Orville Wright, bicycle builders from Dayton, Ohio. They invented an airplane with an engine that a pilot could steer and control.

When and where: Near Kill Devil Hills, North Carolina, December 17, 1903.

Movie: *Wright Brothers' Flying Machine* (2003), a documentary in the PBS series *NOVA*, first shown on November 11, 2003. It is about the brothers' flying machine.

The first photograph of a flight was the photographer's first photograph ever.

The Wright brothers made four flights in an airplane with an engine near Kill Devil Hills, North Carolina, on December 17, 1903. Orville flew first. He stayed in the air for 12 seconds and covered 120 feet. On the fourth flight of the day, Wilbur flew 852 feet in 59 seconds.

Five people watched. One of them, John T. Daniels of the Life-Saving Station at Kill Devil Hills, squeezed the camera's shutter bulb and took the photograph of the first flight. It was the first photograph he had ever taken.

The brothers worked out how to steer an airplane on gliders, which have no engine, in 1901 and 1902. Besides going forward, an airplane can move in three other ways, and the Wrights built a control for each.

When the wing tips move up and down, the airplane rolls. The Wrights controlled roll with cables that twisted the tips of the wings. To control pitch, the nose moving up and down, they added a movable part at the front, called an elevator, in 1901. Yaw is the nose swinging from side to side. In 1902 they added a movable upright part at the back, called a rudder, to control it.

To test the shapes of wings, they built their own wind tunnel, in which a fan blows a steady stream of air past a model. They built a light gasoline engine, too, in the workshop of their bicycle shop in Dayton.

The first death. Army officers were testing the Wright airplane at Fort Myer, Virginia, on September 17, 1908. Orville flew it with Lieutenant Thomas E. Selfridge of the Army Signal Corps as his passenger. The Signal Corps, the Army's communications branch, was then in charge of the Army's first aircraft.

One of the propellers split and hit a wire, and the plane dived into the ground. Selfridge's skull was broken. He never woke up, and he died that evening, the first person killed in an airplane crash. Orville broke a leg and several ribs and spent seven weeks in the Army hospital.

Henry Ford

Who: The carmaker whose workers at Highland Park, Michigan, built the Model T on a moving assembly line.

When and where: Highland Park, Michigan. The moving chassis line started on October 7, 1913.

Movie: *Henry Ford* (2013), a documentary in the PBS series *American Experience*, directed by Sarah Colt. It is about Ford's life and his company.

A motor and a rope pulled each car frame past 140 workers.

At Henry Ford's plant in Highland Park, Michigan, the time it took to build the frame of a car fell from more than 12 hours to about 93 minutes. The workers got that speed by building on a moving assembly line. On a moving line, a motor pulls the car or part past the workers, and each worker adds parts or does one task as it goes by.

Ford's workers tried the line first on one part. In the spring of 1913 they put together the flywheel magneto, an engine part that makes electricity, on a line, with the job split into 29 steps done by 29 men along a moving belt.

The chassis line started on October 7, 1913. Each chassis, the frame of the car with its wheels, moved along a line 150 feet long. At first the time to build one fell to under 3 hours.

Five dollars a day. Ford announced a pay of $5 for an eight-hour day in January 1914. Before that, workers had earned about $2.34 for nine hours, so the new pay was more than double. Part of his reason was to keep workers who hated the pace of the line.

Ford's workers built cars in a year when about 23,000 workers in American industry died on the job, about 63 a day. Staff at the Bureau of Labor Statistics, a federal office that gathers facts about work, made that count for 1913. It covers every kind of industrial work across the country and gives no number for Ford's plant.

Radios, refrigerators and plastic

Lee de Forest's Audion of 1906 made weak radio signals stronger. The Audion was a vacuum tube. Inside its glass were a thin wire called a filament, a metal plate, and a bent wire between them called the grid. A small electric current on the grid controlled a bigger current between the other two parts. The Audion could pick up radio signals too, and tubes like it made radios for the home practical.

Philo Farnsworth, age 21, sent the first all-electronic television picture from his lab on Green Street in San Francisco on September 7, 1927. The picture was a straight line, made with his own camera tube, called the image dissector.

The refrigerator. The General Electric company began selling the Monitor Top, an electric refrigerator for the home, the same year. It cost about $300, and by the middle of 1929 buyers had bought more than a quarter of a million of them. It was the first home electric refrigerator sold in large numbers.

The researcher Emanuela Cardia studied what American households owned, using the census, the government's count of everyone in the country, taken every ten years. In 1930, 14 percent of households had a mechanical refrigerator, one that kept itself cold with a motor instead of a block of ice. By 1940 the share was 44 percent, and by 1950 it was 80 percent, four homes in five.

Plastic. Leo Baekeland, a chemist born in Belgium who lived in the United States, made Bakelite in 1907 from two chemicals, phenol and formaldehyde. It was the first fully synthetic plastic, made entirely from chemicals rather than from anything that grew. He filed for a patent in July 1907 and received it on December 7, 1909.

Nylon, a strong thread-like material made from chemicals, came from Wallace Carothers's work at the DuPont chemical company, which he joined in 1928. Gerard Berchet made the first half ounce of nylon on February 28, 1935, and DuPont officials announced it to the public on October 27, 1938.

Tractors replace horses

Census takers counted 246,000 tractors on American farms in 1920, and about 26 million horses and mules, more than 100 for every tractor. The Fordson tractor of 1917 was cheap and made in large numbers. About three out of every four American tractors built in the 1920s were Fordsons.

The International Harvester company began selling the Farmall in 1924. It could do the jobs other tractors did, and it could also cultivate, breaking up the soil and clearing weeds between rows of growing crops. Cultivating was the last farm job that horses still did.

The Census of Agriculture of 1954 was the first to record more tractors than horses and mules on American farms: about 4.6 million tractors and about 4 million horses and mules.

The first computers and the transistor

ENIAC, the first general-purpose electronic computer, was shown to the public at the University of Pennsylvania in February 1946. John Mauchly and J. Presper Eckert designed it at the university's Moore School, and the US Army paid to build it. It was built to work out artillery firing tables, charts that told soldiers how to aim their big guns.

ENIAC had 17,468 vacuum tubes, weighed more than 27 tons, and filled 1,800 square feet. It could not store a program. For each new set of calculations, its programmers, a team of six women, moved its switches and plugged its cables into new places by hand.

The transistor. At Bell Labs, a group of scientists led by William Shockley looked for a replacement for vacuum tubes, which often failed. On December 16, 1947, John Bardeen and Walter Brattain placed two gold contacts, very close together, on a small slab of germanium, a chemical element. The electric push, or voltage, on one contact controlled the flow of electricity, or current, through the other.

The device made a signal up to 100 times stronger. It was the first transistor. Workers at Bell Labs chose the name "transistor" in a vote, and the company's officials announced it to the public on June 30, 1948. Bardeen, Brattain and Shockley shared the Nobel Prize in Physics in 1956.

The six ENIAC programmers

Who: The six women who programmed ENIAC, the first general-purpose electronic computer, and were left out of its celebration.

When and where: University of Pennsylvania, Philadelphia, 1945 and 1946.

Movie: *The Computers* (2014), a short documentary first shown at the Seattle International Film Festival in 2014. The six programmers are its subject.

Six women programmed ENIAC. None was invited to its dinner.

The six were Jean Jennings Bartik, Betty Snyder Holberton, Kathleen McNulty Mauchly Antonelli, Marlyn Wescoff Meltzer, Frances Bilas Spence and Ruth Lichterman Teitelbaum. No one invited them to the dinner held for the machine's public showing in 1946. The records do not say who made the guest list.

At first the six had only drawings of the machine's wiring to learn from. They were hired to work out the paths of artillery shells for the Army, and later they also ran calculations for the scientists at Los Alamos, the laboratory in New Mexico where the atomic bomb was built.

Their story went untold for decades, until Kathy Kleiman learned about the six while she was researching her college thesis, a long paper written at the end of college. She started the ENIAC Programmers Project to record their story. In 1997 the group Women in Technology International named all six to its Hall of Fame. Four of them were there to accept. Ruth Teitelbaum had died, and her husband, Adolph, accepted for her. Frances Spence is not named among those who were there.

Ernest Lawrence

Who: A scientist who built the cyclotron, a machine that whirls particles in a spiral and uses them to smash atoms.

When and where: Berkeley, California, from 1930 and 1931.

One of his machines had a magnet that weighed 220 tons.

Ernest O. Lawrence's first cyclotron was small enough to hold in one hand. He built it at Berkeley, California, in 1930 and 1931, out of glass, sealing wax and bronze. It sped up hydrogen ions, hydrogen atoms with an electric charge, to an energy of 80,000 electron volts, a very small unit of energy used for particles.

Each machine Lawrence built after that was bigger. The first model was followed by machines 11 inches, 27 inches and 37 inches across. The 60-inch cyclotron of 1939 had a magnet that weighed 220 tons. Inside each machine, particles whirled in a spiral and then struck atoms and broke them apart.

Scientists used cyclotrons to make elements that do not occur in nature. They made radioactive materials for use in medicine, too, materials whose atoms break down and give off energy and tiny particles. Lawrence won the Nobel Prize in 1939. One later machine at Berkeley, the 88-Inch Cyclotron, still runs.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 11 MINUTES

Computers shrank from machines that filled a room to machines that sat on a desk at home, and people linked them into networks. Jack Kilby of Texas Instruments built the first integrated circuit, a whole electronic circuit on one small chip, in 1958. The Altair 8800, a computer sold as a kit of parts, was announced in the magazine *Popular Electronics* in January 1975, and IBM began selling the IBM PC in 1981. The IBM PC became the standard office computer.

A student at the University of California, Los Angeles, or UCLA, sent the first message between two computers on ARPANET, a network that linked computers at universities, on October 29, 1969. Martin Cooper of Motorola made the first call on a handheld cell phone four years later. All 24 satellites of the Global Positioning System, or GPS, were working by 1995.

Grace Hopper

Who: A mathematician and Navy officer whose A-0 compiler let people write instructions for a computer in words closer to English.

When and where: Harvard University from 1944, then Remington Rand, the company that made the UNIVAC computer. Born in New York in 1906, died in 1992.

Movie: *The Queen of Code* (2015), a short documentary directed by Gillian Jacobs, part of the FiveThirtyEight series *Signals*. It is about Hopper's life and work.

At 79 she was the oldest officer still serving in the Navy.

Grace Hopper's A-0 compiler, built in 1951 and 1952, was a program that takes instructions written closer to human language and translates them into instructions the computer can carry out.

Hopper trained as a mathematician. During World War II she served as a lieutenant in the WAVES, the women's branch of the Navy. In July 1944 she was assigned to the Harvard Mark I computer, and she became one of its first programmers. In 1946 she wrote the first long manual on how to program a computer.

She built the A-0 compiler while working on the UNIVAC computer at the Remington Rand company. From 1955 to 1959 she worked on FLOW-MATIC, a programming language for business. Like other programming languages, it was a set of words and rules for writing instructions a computer can follow, and its words were English words. Later, the designers of COBOL, another business programming language, drew on FLOW-MATIC.

Hopper retired from the Navy in 1986 as a rear admiral, one of the Navy's highest ranks. She was 79, and at that time she was the oldest officer still serving in the Navy.

Who invented the chip?

Two engineers at two different companies, Jack Kilby and Robert Noyce, each made a chip. The encyclopedia Britannica and the German Patent and Trade Mark Office both name the two men as its inventors. A chip, or integrated circuit, puts a whole electronic circuit, the path electricity flows through and the parts along it that do jobs such as switching it on and off, onto one small piece of material.

Kilby, of Texas Instruments, showed the first integrated circuit on September 12, 1958. He made it from germanium and connected its parts with wires by hand.

Noyce, of Fairchild Semiconductor, designed a chip made of silicon, another chemical element, in 1959. On Noyce's chip the wiring lay flat and was printed onto the surface, so factories could make chips in large numbers. Every later chip was built in Noyce's flat form. Silicon Valley, the area south of San Francisco where many computer companies started, is named for this element.

Kilby received the Nobel Prize in Physics in 2000. Noyce had died in 1990. Kilby said in public that Noyce deserved to share the prize.

Martin Cooper

Who: A Motorola engineer who made the first call on a handheld cell phone.

When and where: A street in Manhattan, New York City, April 3, 1973.

The first handheld cell phone call went to his rival.

Martin Cooper, an engineer for the Motorola company, made the first call on a handheld cell phone while walking along a street in Manhattan on April 3, 1973. He called Joel Engel, his rival at Bell Labs.

The phone Motorola sold to the public came 10 years later. The DynaTAC 8000X went on sale in 1983. It cost $3,995 and weighed 28 ounces, almost 2 pounds. One charge of its battery gave about 30 minutes of talking.

The personal computer

The Altair 8800, a computer sold as a kit, a box of parts that the buyer puts together, was announced in the magazine *Popular Electronics* in January 1975. About 30 people came through a rainstorm to Gordon French's garage in Menlo Park, California, on March 5, 1975, to see an Altair. That meeting was the first of the Homebrew Computer Club, a group of people who built their own computers.

Steve Jobs, Steve Wozniak and, for a short time, Ron Wayne founded Apple Computer on April 1, 1976. The Apple II, which came out in June 1977, was a finished computer that people could plug in at home. IBM brought out the IBM PC on August 12, 1981, and its design became the standard for computers in offices.

Lee Felsenstein

Who: An engineer who ran the meetings of the Homebrew Computer Club and designed two early small computers, the Sol-20 and the Osborne 1.

When and where: Berkeley and the San Francisco Bay Area, California, from 1973. Born in Philadelphia in 1945.

He put a public computer terminal in a Berkeley record store.

Arrested at Sproul Hall. Nine years before he set up that terminal, Lee Felsenstein was a student at the University of California, Berkeley. He took part in the Free Speech Movement of 1964, a protest by Berkeley students for the right to speak and organize about politics on campus. He was at the Sproul Hall sit-in, where students sat inside a campus building and refused to leave. On December 3, 1964, hundreds of police officers went into the building and arrested nearly 800 students, Felsenstein among them.

Community Memory. Felsenstein earned a degree in electrical engineering and computer science at Berkeley in 1972. The next year he and four partners, among them Efrem Lipkin and Jude Milhon, set up a computer terminal that anyone could use, a keyboard and a printer or screen connected to a computer somewhere else. They called the project Community Memory and put the terminal in a Berkeley record store, Leopold's Records, next to the store's paper bulletin board.

People typed notes about music, politics, housing, services, women's groups and technology, and other people read them. The Computer History Museum's profile of Felsenstein describes it as one of the earliest online communities. The last version of the project closed in 1992.

Community Memory ran on a large computer called an SDS-940, which was donated for the project. It was a time-sharing computer, one that many people could use at once from different terminals. The project's

team worked with Resource One, a nonprofit group, run for a cause rather than for profit. Three students who had dropped out of computer science at Berkeley had started Resource One.

Homebrew. Felsenstein was one of the first members of the Homebrew Computer Club, and he ran its open meetings, keeping order with a blackboard pointer.

With Gordon French and Bob Marsh, at a company called Processor Technology, he designed the Sol-20, one of the first small computers sold with its own case and keyboard. More than 10,000 were sold from 1977 to 1979. He then designed the Osborne 1 for Adam Osborne's company, a portable computer with a 5-inch screen. It cost $1,795, and more than 125,000 were sold in 1982.

His motto, as the Computer History Museum records it, is "To change the rules, change the tools." In 2016 the museum's leaders gave him their Fellow Award for his influence on early personal computing.

Steve Jobs and Steve Wozniak

Who: The founders of Apple Computer: Wozniak designed its first two computers, and Jobs made the first sale.

When and where: Jobs's garage in California, from 1976.

Movie: *Steve Jobs* (2015), directed by Danny Boyle and written by Aaron Sorkin. It is a drama, not a documentary. Actors play the real people, and it is based on Walter Isaacson's biography of Jobs. *Steve Jobs: The Man in the Machine* (2015), directed by Alex Gibney, is a documentary about Jobs.

A Volkswagen and a calculator paid for the first Apple computer.

Steve Wozniak designed Apple's first two computers, and Steve Jobs found the first buyer. The two met while they were both working at the Hewlett-Packard company. In 1975 Wozniak began developing the Apple I. He later said that the Homebrew Computer Club was where the Apple I got its start.

To pay for the first working model, Jobs sold his Volkswagen and Wozniak sold his scientific calculator, for $1,300 in all. On April 1, 1976, they founded Apple Computer, with Ron Wayne as a partner for a short time. Jobs and Wozniak built the first Apple I computers in Jobs's garage.

On his first sales call, Jobs got an order for 50 of them at $500 each, $25,000 in all, from a store called the Byte Shop. The Apple I came without a case, a keyboard, a screen or a power supply. Buyers added their own.

The Apple II. Wozniak designed the Apple II in 1977. Its electronics, keyboard and power supply came together in one case, and it could show color pictures. The programming language BASIC came stored in its permanent memory, a part that keeps what is stored in it even when the power is off.

> A floppy disk drive for storing files became available for it in 1978, and in 1979 VisiCalc came out, a spreadsheet program for working with tables of numbers. With those two, the Apple II sold in large numbers. Apple's sales grew from $7.8 million in 1978 to $117 million in 1980, 15 times as much in two years.

How the internet began

The internet began with ARPANET, a computer network paid for by the Department of Defense to link computers at universities. On October 29, 1969, Charley Kline, a student programmer in Leonard Kleinrock's lab at UCLA, tried to log in to a computer at Stanford Research Institute. He typed "L" and "O," the first two letters of LOGIN, and then the system crashed. So the first message ever sent on the network was "LO." About an hour later the full login worked.

The computers on ARPANET began using TCP/IP, the internet's shared set of rules for moving data between computers, on January 1, 1983.

The web. Many people use "the web" and "the internet" to mean the same thing, but they are two different things. The internet is the network of linked computers. The World Wide Web is a system of pages linked to one another, which people read over the internet.

A British computer scientist, Tim Berners-Lee, invented the web at CERN, a European physics laboratory in Switzerland. He proposed it in 1989 and opened it to the public on August 6, 1991. Americans built the program that let ordinary people use the web easily. Marc Andreessen, Eric Bina and others at the University of Illinois released Mosaic, the first widely used graphical browser, a program that shows web pages with pictures, on April 22, 1993.

GPS. The Global Positioning System, or GPS, is a set of satellites owned by the United States. Each satellite sends out a signal that gives its position and the time, and a GPS receiver picks up these signals and uses them to work out where it is. The first GPS satellite went up on February 22, 1978, and the full set of 24 was working in 1995.

Until 2000, under a policy called Selective Availability, US government officials made the GPS signals for the public less exact on purpose, for national security. President Bill Clinton ordered the policy ended in May 2000. After that, ordinary GPS receivers could find their place to within about 10 to 20 meters (about 30 to 65 feet) or better, close enough to give driving directions turn by turn.

What is inside a smoke detector?

The ionization smoke detector, a cheap alarm made in large numbers for homes, holds about a third of a millionth of a gram of americium-241. Americium is a radioactive element, one whose atoms break down and give off energy and tiny particles, and scientists discovered it in 1944 and 1945.

The americium gives off tiny particles called alpha particles. The particles knock bits of electric charge loose from the air in a small chamber inside the detector, so a very small electric current can flow through the air. When smoke enters the chamber, it disturbs the current, and the alarm goes off.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 11 MINUTES

Nine in ten American adults carried a smartphone by 2025, a survey by the Pew Research Center found, and new programs could write answers and match faces, sometimes wrongly. Apple began selling the iPhone in 2007. In 2022 the company OpenAI released ChatGPT, a program that writes answers to typed requests, and by February 2026, 44 percent of American adults told Pew they had used it.

ChatGPT is an AI program. AI, short for artificial intelligence, is a name for computer programs that do tasks such as recognizing a face or writing a paragraph, and these programs make mistakes. In 2020 Detroit police arrested the wrong man after a face-recognition search, a computer comparison of a photo of a face with a collection of photos, returned a false match.

The smartphone

Steve Jobs of Apple announced the iPhone on January 9, 2007. It went on sale on June 29, 2007, for $499 or $599, with a phone contract, a promise to pay a phone company for service every month. On September 10, 2007, Apple officials announced that the company had sold its one millionth iPhone the day before, 74 days after the phone went on sale.

A smartphone puts a computer, a camera, a phone, a GPS receiver and a music player into one machine that fits in a pocket. Social media apps run on these phones.

Researchers at the Pew Research Center, a group that runs large surveys, have asked Americans whether they own a smartphone since 2011. In their first count, in 2011, 35 percent of adults owned one, about one in three. In a survey of 5,022 adults from February to June 2025, 91 percent did, and 98 percent owned some kind of cell phone.

Older people were less likely to own a smartphone. Among adults aged 18 to 29, 97 percent had one, against 78 percent of those 65 and older.

For some Americans, the phone is the only way to get online. In the same 2025 survey, 78 percent of adults had broadband at home, a fast internet connection that is always on. Among adults in households earning less than $30,000 a year, 34 percent reached the internet only through a smartphone. Among those earning $100,000 or more, 4 percent did.

Batteries, printers and gene tools

The battery in an electric car is a lithium-ion battery, a kind that can be recharged many times. Scientists in three countries worked on it, starting in the 1970s, when M. Stanley Whittingham began work on rechargeable lithium batteries for the company Exxon in New Jersey.

John Goodenough, an American chemist then working at Oxford in England, found the material for the battery's cathode, its positive end, in 1980: lithium cobalt oxide. Akira Yoshino in Japan built a safe early model with a carbon anode, the negative end, in 1985. The Sony company in Japan began selling the first lithium-ion batteries in 1991. Whittingham, Goodenough and Yoshino shared the Nobel Prize in Chemistry in 2019.

3D printing. A 3D printer builds a solid object out of many thin layers, one on top of another. Chuck Hull invented stereolithography, a method of 3D printing in which the machine hardens each thin layer in turn, in 1983 and 1984. He applied for a patent on August 8, 1984, and received it on March 11, 1986. In the 2010s, 3D printers spread to factories and to hobbyists' desks.

Gene tools. CRISPR is a tool for changing genes, the instructions inside living cells that shape how a living thing grows and works. Chapter 9 tells how the scientists Jennifer Doudna and Emmanuelle Charpentier made it into that tool in 2012.

Limor Fried

Who: An engineer who founded Adafruit Industries, a company that makes electronics kits and teaches people to build with them.

When and where: MIT in Cambridge, Massachusetts, then New York City, from 2005.

Adafruit's designs are open source: anyone may copy and change them.

Adafruit Industries began in 2005 in Limor Fried's dorm room at the Massachusetts Institute of Technology, or MIT, a university in Cambridge, Massachusetts. She had earned a bachelor's degree from MIT in 2003, and she finished a master's degree in engineering there in 2005. Adafruit sells electronics kits along with lessons on how to build them, and it shares its designs as open source, free for anyone to copy and change.

According to the company's website, Fried was the first woman engineer on the cover of *Wired* magazine, and women own all of the company. Adafruit's workers make its products in the company's factory in New York City. In 2024 Adafruit moved to Industry City in Brooklyn. The company's website states that it has more than 100 employees.

How AI programs work, and why they make mistakes

One kind of AI program, a neural network, learns from examples instead of following step-by-step instructions written by a person. To train a network to name what is in photographs, a computer shows it a huge number of photos whose right answers are already known.

Each time the network gives an answer, the computer compares it with the right answer and adjusts the network's settings a little, so that the next answer comes closer. After enough examples, the network can name what is in photos it has never seen, though it still makes mistakes.

A contest. Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton of the University of Toronto won ImageNet in 2012 with a neural network called AlexNet. ImageNet was a contest that scored computer programs on naming what is in photographs. Each program gave its five best guesses for each picture. AlexNet missed with all five on about 15 of every 100 pictures, and the second-place program missed on about 26 of every 100. After AlexNet's win, neural networks came into wide use, a time called the deep-learning era.

ChatGPT and programs like it produce text, pictures and computer code from requests typed in plain language. OpenAI released ChatGPT on November 30, 2022. Five days later, OpenAI's chief executive, Sam Altman, posted online that it had "crossed 1 million users." Analysts at the bank UBS estimated that by the end of January 2023, about 100 million people a month were using it.

Why they make mistakes. Staff at the National Institute of Standards and Technology, a federal agency, published a report in July 2024 on how large language models, programs that write text, work and why they make mistakes. A program of this kind predicts the next word in a sentence, over and over. That kind of guessing can produce true statements, and it can also produce false ones. The report states that the programs give false answers in a confident way, and that people who believe them can be misled.

Fake cases in court. Two lawyers in a federal court case in New York cited court decisions in 2023 that ChatGPT had made up. Lawyers cite, or point to, earlier decisions to show a judge how courts have ruled in cases like theirs.

Roberto Mata had sued the airline Avianca, saying a metal serving cart struck his knee on a flight. On March 1, 2023, his lawyers, Steven A. Schwartz and Peter LoDuca, filed a court paper that cited earlier decisions to support Mata's side. ChatGPT had made up those decisions and the quotes from them.

Schwartz later said that he asked ChatGPT whether one case was real, and it answered that the case "does indeed exist." In written orders, the judge then asked whether the cases existed, and the lawyers kept standing by them. On June 22, 2023, Judge P. Kevin Castel found that the lawyers had acted in bad faith, meaning dishonestly. He ordered the two lawyers and their law firm to pay a penalty of $5,000.

Karla Ortiz

Who: A San Francisco concept artist whose designs went into Marvel films and who told US senators how AI image programs affected artists.
When and where: San Francisco and Washington, D.C., from 2022.

"I am no longer certain of my future as an artist."

Karla Ortiz is a concept artist, an artist who draws the first designs for characters and scenes. Her designs went into Marvel films, including *Doctor Strange*, *Black Panther* and *Avengers: Infinity War*. She worked on the video game *Final Fantasy 16* and the card game *Magic: the Gathering*, too.

Ortiz first saw an AI image program in April 2022. A website showed pictures made by a machine that copied the styles of named artists, her friends among them. No one had asked them. By August and September of 2022, the programs Midjourney, Stable Diffusion and DALL-E were in wide use.

The lawsuit. Ortiz, Sarah Andersen, Kelly McKernan and other artists sued Stability AI, Midjourney and DeviantArt in federal court on January 12, 2023. They said the companies had broken copyright law, the law that protects an artist's right to decide who copies their work.

Her testimony. Ortiz gave written testimony, a statement to lawmakers, to a subcommittee of the US Senate Judiciary Committee on July 12, 2023. In it she explained how the popular image programs were built. Their makers trained them on a collection of 5.8 billion pictures with captions, "taken indiscriminately from the web," meaning without anyone choosing which ones.

People at a nonprofit group called LAION, which the company Stability AI funded, built the collection. Ortiz wrote that the collection "includes almost all of my fine art work."

She described what she had seen in her own work. She had been part of three productions where generative AI, AI that makes new pictures, text or sound, had been "utilized," or used, "at different levels." "Some of my peers have lost jobs, some have had their duties lessened (and of course, their proposed pay)," she wrote.

She also wrote, "I am no longer certain of my future as an artist." She worked with researchers at the University of Chicago on Glaze, a program meant to protect artists' pictures from having their style copied.

US District Judge William Orrick ruled on August 12, 2024, that the artists' main copyright claims could go forward toward a trial. As of October 2, 2026, the case had not gone to trial. On September 28, 2026, Judge Orrick changed the schedule so that the next major motions in the case, formal requests to the judge, are to be heard by September 29, 2027.

Arrested by a false match

Detroit police officers arrested Robert Williams, a Black man, at his home in Farmington Hills, Michigan, in January 2020, after a face-recognition search matched him by mistake. They arrested him in front of his wife and two young daughters and in view of his neighbors.

The software, which names possible matches for a face from a collection of photos, had matched a blurry picture from a store's security video to his expired driver's license photo. The video showed a theft of watches from a Shinola store in Detroit in 2018. Williams was not the man in the video.

Police used the match to build a photo lineup, a set of photos shown to someone who might pick out the thief. They showed it to a contractor hired to prevent theft at the store, who had not seen the theft and knew it only from the video.

Thirty hours. Detroit police held Williams for 30 hours. The ACLU, the civil rights group whose lawyers represented him, described the place he was held as "an overcrowded, dirty cell." Wayne County prosecutors dropped the charges, but they kept the right to bring them again.

The county's prosecutor, Kym Worthy, apologized. "This case should not have been issued based on the DPD investigation, and for that we apologize," she said, using the initials of the Detroit Police Department. His was the first publicly reported case in which police wrongly arrested someone after a false face-recognition match.

At least three Detroit arrests of this kind are publicly known, according to reporting by Michigan Public in 2024. In 2023 Detroit police wrongly arrested Porcha Woodruff, who was pregnant, for a carjacking, stealing a car from its driver.

The settlement. Williams and lawyers for the City of Detroit settled his lawsuit on June 28, 2024. Under the settlement, Detroit police may not arrest anyone based on a face-recognition result without other evidence, and they

may not build a photo lineup from a face-recognition match alone. They must train officers, including on the higher rate of wrong matches for people of color, and they must review cases back to 2017.

AI at work, by the numbers

The share of Americans who have used ChatGPT has grown in each Pew survey. Researchers at the Pew Research Center found in February 2026, in a survey of 5,119 American adults, that 44 percent had used it, up from 18 percent in 2023 and 34 percent in 2025.

About half of adults in the 2026 survey used some AI chatbot, a program that answers typed questions in sentences. Among adults with jobs, 38 percent used chatbots for tasks at work. Forty percent of adults expected AI to have a bad effect on society over the next 20 years.

Some employers say AI is the reason they are cutting jobs. Staff at Challenger, Gray & Christmas, a company that helps laid-off workers find new jobs, count the job cuts that American employers announce and the reasons the employers give.

From January through August 2026, employers named AI as the reason for 116,175 announced job cuts. That was about 22 percent of all the cuts they announced, more than one in five and more than for any other reason. These numbers are employers' own statements, not a government count of jobs lost to AI.

THE THREE QUESTIONS, ANSWERED

1. **Why did planters enslave far more people after the cotton gin?** One gin cleaned about 50 pounds of cotton a day, work that took several hundred hours by hand. Planters grew far more cotton and enslaved far more people to grow and pick it, nearly 4 million by 1860.
2. **Why could an enslaved inventor not get a patent?** Only the inventor could get a patent, and only by swearing an oath. In 1858

Attorney General Jeremiah Black ruled that an enslaved man was not a citizen and could not swear it, and that the enslaver could not take the patent instead.

3. **What went wrong when people trusted a computer's answer?** In 2023 two lawyers cited court decisions that ChatGPT had made up, and a judge ordered them and their law firm to pay a $5,000 penalty. In 2020 Detroit police held Robert Williams for 30 hours after a face-recognition search matched him by mistake.

What changed: Work done by hand moved to machines run by water, then steam and electricity, then pocket-sized computers.

FIVE DATES TO KEEP

1646: Joseph Jenks gets a patent for his water-driven mills.

1793: Eli Whitney builds the cotton gin.

May 24, 1844: Morse sends the first public telegraph message.

December 17, 1903: The Wright brothers make the first powered flights.

November 30, 2022: OpenAI releases ChatGPT.

> **NOT KNOWN:** Ned's second name, if he had one, was never recorded, and nobody knows how his life ended.

Where to go: The Saugus Iron Works National Historic Site, in Saugus, Massachusetts, where the 1640s ironworks stands rebuilt.

Next: Chapter 9, Science. It follows Americans who measured the world and the people some scientists studied without asking.

Chapter 9: Science

CHAPTER 9 OF 37 · SCIENCE · 10 ERAS · ABOUT 95 MINUTES

Science in America grew from a few people watching the sky into labs with thousands of workers, and along the way many scientists used people who never agreed to be studied.

A dead moose reached General John Sullivan in New Hampshire on April 3, 1787, badly rotted. Thomas Jefferson had asked for its skin, skeleton and horns. He wanted to show a French scientist named Buffon that American animals were not small and weak. The antlers could not be saved, so Sullivan sent another animal's antlers and wrote that they "may be fixed on at pleasure." In the book where he argued with Buffon, Jefferson also called Black people inferior to white people. He called that claim "a suspicion only."

This chapter follows American science from the sunrise posts at Cahokia to the James Webb Space Telescope. Most of it is people measuring things: a comet, the speed of light, the carbon dioxide in the air. Watch also for the people who were measured. From the 1830s to the 1970s, scientists and doctors dug skulls out of graves, had enslaved people photographed as "evidence" about race, and injected hospital patients with plutonium. The last eras show what families and Native nations did about it. Watch, too, who gets the credit: more than once it went to someone other than the person who did the work.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did American scientists learn how far away the stars are?
2. Why did scientists collect human skulls and brains, and where are those remains now?

3. Who did the work but was left out of the credit?

Five minutes? Read "How did Henrietta Leavitt measure space?" in 1900 to 1950, then "The shape of DNA, and who got the credit" in 1950 to 2000.

Can skip: "Charting the winds at sea", "The physics inside the transistor, 1947", "Heavy elements, made by many countries", "Computers that predict the shapes of proteins". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

The people of Native nations here watched the sky for many generations and learned where the sun rose at each time of year. They studied plants, fire, the weather and the tides as well. Each generation taught what it knew to the next.

They marked what they knew about the sky with posts and with carvings in rock. At Cahokia, people set up circles of wooden posts that marked where the sun rose. At Chaco Canyon, sunlight falls on two carved spirals in set places at certain times of year.

Woodhenge at Cahokia

The people of Cahokia, in what is now Illinois, tracked the sunrise with circles of upright wooden posts between about the years 900 and 1100. Chapter 8 tells how they built the circles. The place where the sun comes up moves along the horizon over a year, and the posts marked that movement. The circles are known today as Woodhenge. Nations such as the Chickasaw and the Osage descend from the people who lived at Cahokia.

The posts marked the sunrise on the solstices, the longest and the shortest days of the year. They also marked it on the equinoxes, the two days each year when day and night are about the same length.

How does the Sun Dagger at Chaco Canyon work?

Sunlight slips past three slabs of sandstone and lands on two spirals carved into the rock. The slabs lean against a cliff near the top of Fajada Butte, a steep hill with a flat top in Chaco Canyon, in what is now New Mexico. The canyon was a center for the ancestral Pueblo people from 850 to 1250. Their descendants today include the Hopi, the Pueblo peoples of New Mexico and the Navajo Nation.

The daggers. The light falls on the spirals as thin blades, called daggers, and they land in set places at the solstices and the equinoxes. Researchers in a group called the Solstice Project argue that the spirals also mark a cycle of the moon. Over 18.6 years, the points where the moon rises and sets swing out to their widest and back again. Other researchers who study ancient astronomy still debate that claim.

Worn away. Anna Sofaer, a volunteer making a record of rock art, recorded the daggers at the summer solstice of 1977. They no longer form the way she saw them. By the summer solstice of 1989, two of the three slabs had shifted. Park officials had limited visits to the butte, but visitors walking there had made the ground wear away faster.

Knowledge of living things

Native nations built up knowledge of ecology, the ways living things depend on each other and on the land, by watching closely over many generations. Scientists today call it Traditional Ecological Knowledge, or TEK for short. Part of it is knowledge of how fire changes the land and the living things on it.

The Ecological Society of America, a group of scientists who study ecology, has a section for TEK. A botanist, or plant scientist, named Robin Wall Kimmerer helped to found it.

Robin Wall Kimmerer

Who: A botanist of the Citizen Potawatomi Nation who uses Native knowledge and laboratory science together in her work.

When and where: The present day. The State University of New York's College of Environmental Science and Forestry.

Her 2013 book has been a New York Times bestseller since 2020.

The book is *Braiding Sweetgrass*, and it has been on the list since February 2020. Kimmerer is a member of the Citizen Potawatomi Nation. In her work she studies plants in two ways together. One is the knowledge that Native nations have built up and taught. The other is the science of the laboratory.

Sweetgrass at a Mohawk farm. In a study published in 2005, Kimmerer and a fellow researcher worked to bring back sweetgrass, a plant used to make baskets, at Kanatsiohareke, a Mohawk farm. Members of the Mohawk community chose the spot, because sweetgrass had grown there in the past. The researchers then tested four ways of planting it. The sweetgrass grew best beside a crop called hairy vetch.

She earned a PhD in botany, the highest degree a university gives, at the University of Wisconsin. She teaches at the College of Environmental Science and Forestry, part of the State University of New York, where she

> founded the Center for Native Peoples and the Environment. In 2022 the MacArthur Foundation named her a MacArthur Fellow.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 1 MINUTE

Europeans who crossed the ocean in the 1500s studied this land only on short visits. They collected what they found, described it in writing and published their descriptions in Europe.

Thomas Hariot's report

Thomas Hariot, an English mathematician, spent a year on Roanoke Island and wrote down the plants, the animals and the foods he found. The island lies among the shallow waters of the Outer Banks, in what is now North Carolina. Hariot went there in 1585 and 1586 with the expedition that Walter Raleigh sent.

He also wrote about the Native nations who lived there, such as the Secotan. They spoke Algonquian languages, a group of languages related to each other.

The book. Hariot published *A Briefe and True Report of the New Found Land of Virginia* in 1588. It was the first book in English about North America written by someone who had been there. A publisher named Theodor de Bry printed it again in 1590, with pictures made from drawings by John White.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 3 MINUTES

A few colonists kept careful records of the sky and the weather in the 1600s, and some of them still read the sky as a message from God. English colonists in New England founded Harvard, the first college in their colonies, in 1636, and printed their first almanac three years later. One of them, Increase Mather, measured comets but still wrote that they were warnings from God.

A college, an almanac and a telescope

Members of the Massachusetts Bay General Court, the colony's lawmakers, voted in 1636 to found a college. It was the first college in the English colonies. John Harvard died in 1638 and left the college his books and half of everything else he owned. On March 13, 1639, the same lawmakers voted to name the college Harvard, after him.

The almanac. That same year a sailor named William Pierce worked out an almanac, a yearly booklet with a calendar and tables of the sun and moon. Its title was *An Almanack for New England for the Year 1639*. Stephen Daye printed it at Cambridge, in Massachusetts, on the first printing press in the colonies. It was one of the first two things printed there.

For the next two hundred years, ordinary families learned about astronomy, the study of the sun, the moon, the planets and the stars, from almanacs.

The telescope. John Winthrop Jr., a governor of Connecticut, visited London from 1661 to 1663. While he was there, members of the Royal Society, a group of scientists in London, elected him a fellow, or full member. He was one of the first colonists to join. In about 1672 he gave Harvard his telescope, which was 3.5 feet long. It was the college's first scientific instrument.

The comet of 1680

Thomas Brattle followed the Great Comet of 1680 with a telescope from Boston. He had finished his studies at Harvard in 1676. The comet showed in the sky in November and again in December. Brattle worked out that the two sightings were one comet, which had swung around the sun in between.

A warning and a measurement. Increase Mather wrote a book about comets in 1683, called *Kometographia*. In it he listed comets as warnings sent by God. Yet he measured comets with parallax, which means looking at an object from two different places. A near object seems to shift against the stars far more than a distant one. His measurements showed that comets were distant bodies, not vapors, a kind of gas.

That same year Mather founded the Boston Philosophical Society, modeled on the Royal Society in London. People then called the study of nature natural philosophy, so a philosophical society was a group for studying nature. Mather's group met for about three years.

The English scientist Isaac Newton published his book the *Principia* in 1687. In it Newton set out his laws of motion and of gravity, and he wrote about Brattle's New England observations of the comet of 1680.

Rev. John Campanius Holm

Who: A chaplain in the colony of New Sweden who kept the first known weather record in what is now the United States.

When and where: Near what is now Wilmington, Delaware, 1644 and 1645.

Twice a day, with no instruments, he wrote down the weather.

John Campanius Holm kept the first known record of the weather in what is now the United States. He wrote it down twice a day through 1644 and 1645. He followed the same plan day after day, and he had no instruments at all.

Holm was a chaplain, a minister who serves people far from home. He served New Sweden, a Swedish colony on the Delaware River, near what is now Wilmington, Delaware.

Officials of the National Oceanic and Atmospheric Administration, the US government's weather and ocean agency, now give the John Campanius Holm Award. It goes to volunteers who take weather readings, up to 25 people each year.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 3 MINUTES

Colonists who studied nature began to share their work in these years, with each other and with scientists in London. A farmer named John Bartram and Benjamin Franklin of Philadelphia got them started. Bartram traveled through the colonies from 1730 to collect and study American plants. Franklin organized a society in Philadelphia for this kind of work in 1743. From 1747 he ran experiments with electricity and mailed the results to London. Scientists in Britain read those letters, and Franklin became the first American scientist known outside the colonies.

The American Philosophical Society, 1743

Benjamin Franklin organized the American Philosophical Society in Philadelphia in 1743. It is the oldest learned society in the United States, a group of people who meet to share and discuss what they have found out.

The botanist John Bartram first suggested the idea in 1739. Franklin started the society with a pamphlet, a short printed booklet. It was titled *A Proposal for Promoting Useful Knowledge among the British Plantations in America*, and in that title the word plantations means colonies.

The society grew out of the Junto, a club Franklin had started in 1727 with about twelve friends, most of them tradesmen. Its members met on Friday nights to argue about right and wrong, about government and about nature. Franklin meant the new society to be an American answer to the Royal Society in London.

Benjamin Franklin

Who: A Philadelphia man whose experiments with electricity made him the first American scientist known in other countries.

When and where: Philadelphia, experiments from 1747 to 1751.

Movie: *Benjamin Franklin* (2022), a two-part documentary by Ken Burns, shown on PBS. It is about Franklin's life.

Scientists still say "positive" and "negative" because Franklin chose those words.

Benjamin Franklin ran experiments with electricity in Philadelphia from 1747 to 1751. He reported his results in letters to Peter Collinson, a member of the Royal Society in London.

Franklin thought of electricity as a single fluid. He called an object positive when it held more of this fluid than normal, and negative when it held less. He argued that electric charge is never made or destroyed. It only moves from one object to another, an idea scientists call the conservation of charge. He also proposed that lightning is electricity.

His letters came out in London in April 1751 as a book, *Experiments and Observations on Electricity*. In 1753 members of the Royal Society gave him the Copley Medal, their highest honor. He was the first person from outside Britain to receive it, and the first American to be known in other countries as a scientist.

John Bartram

Who: A Quaker farmer who studied plants, built the oldest surviving botanic garden in America, and first suggested the American Philosophical Society.

When and where: His farm at Kingsessing, on the Schuylkill River, from 1728. He lived from 1699 to 1777.

He and his sons gave science more than 200 new American plants.

Scientists first learned of more than 200 American plants through John Bartram and his sons. Bartram was a farmer and a Quaker, a member of a Christian group also called the Society of Friends.

He bought a farm of 102 acres at Kingsessing, on the Schuylkill River, in 1728. There he built a botanic garden, where plants are grown so they can be studied. It is the oldest botanic garden in America that survives today.

He traveled from 1730 to 1765 to collect plants, from New York as far south as Florida. The idea that became the American Philosophical Society was his, in 1739. King George III named him the King's Botanist for North America in 1765.

Carl Linnaeus, the Swedish botanist who made the system scientists use to name living things, reportedly called Bartram "the greatest natural botanist in the world."

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 7 MINUTES

Scientists in Europe began reading and reprinting American work after 1750. Benjamin Franklin designed the 1752 test that showed lightning is electricity. Men in France ran it first, and Franklin wrote that he then ran it with a kite. In 1769 four men at Norriton timed the planet Venus as it crossed the sun, and members of the Royal Society in London reprinted their results.

Thomas Jefferson used evidence in his book *Notes on the State of Virginia*, and later a moose, to argue with a scientist in France about the size of American animals. In that same book he wrote that Black people were inferior to white people, meaning less able, in body and mind. He called the claim "a suspicion only." Benjamin Banneker, a free Black farmer who taught himself astronomy, sent Jefferson an almanac he had calculated as proof against it.

Who ran the lightning test first?

Men in France ran Franklin's lightning test before Franklin did. Franklin had published a plan for a test that would show whether lightning is electricity. Thomas-François Dalibard followed the printed plan at Marly-la-Ville on May 10, 1752, with a tall iron rod. He reported the test to the French Academy, a society of scholars, on May 13. The test showed that lightning is electricity.

The kite. Franklin wrote that he ran his own test in Philadelphia with a kite. He first described it in print on October 19, 1752, in the *Pennsylvania Gazette*, a newspaper. Nobody recorded the exact day or the exact spot. The date most often given, June 1752, comes from an account Joseph Priestley wrote in 1767.

Few writers have doubted that the kite flight took place, and no record from the day settles the question either way. One who did doubt it, Tom Tucker, argued in a 2004 book called *Bolt of Fate* that Franklin never flew the kite at all.

Timing Venus, 1769

Four men at Norriton, in Pennsylvania, timed the planet Venus as it passed across the face of the sun on June 3, 1769. An event like this is called a transit of Venus. Astronomers in many parts of the world timed the same transit. They planned to use the times to work out the distance from Earth to the sun.

Members of the American Philosophical Society chose teams to watch from three places. It was the first large, organized science project in the colonies.

Norriton. The Norriton team worked at the home of David Rittenhouse, in what is now Montgomery County, Pennsylvania. Rittenhouse built most of the instruments they used. He, William Smith, John Lukens and John Sellers made the timings. Their results filled the first volume of the society's *Transactions*, its printed record of research, which came out in 1771. Members of the Royal Society in London reprinted the results.

David Rittenhouse

Who: A Pennsylvania instrument maker and astronomer who taught himself both trades. One of the four men who timed Venus at Norriton in 1769.

When and where: Norriton, Pennsylvania, in 1769, and Philadelphia. He lived from 1732 to 1796.

He taught himself to build machines that showed the planets moving.

David Rittenhouse taught himself to make scientific instruments and to study the stars. He built orreries, machines that show the planets moving around the sun. When an orrery's gears turn, small models of the planets move along their paths.

He followed Franklin as president of the American Philosophical Society in 1791. Later he became the first director of the United States Mint, where the country's coins are made.

Jefferson, Buffon and the moose

A scientist in France named Buffon taught that animals and people grew "degenerate" in the Americas, by which he meant smaller and weaker. Thomas Jefferson argued against him with evidence in his book *Notes on the State of Virginia*. Jefferson wrote the book to answer a list of questions a French official had sent. He called each chapter a Query, an old word for a question.

A claim with no evidence. In the chapter called Query XIV, Jefferson wrote that Black people "are inferior to the whites in the endowments both of body and mind." Inferior means lower or less able, and endowments here means natural abilities. He called the claim "a suspicion only." In reason, he wrote, Black people were "much inferior, as I think one could scarcely be found capable of tracing and comprehending the investigations of Euclid," a mathematician of ancient Greece.

Jefferson himself enslaved more than 600 people during his life.

The moose. Jefferson's other evidence against Buffon was a moose. He was then the United States minister in Paris, the country's official representative to the French government. On January 7, 1786, he wrote to John Sullivan of New Hampshire, a former general and governor. He asked for "the skin, the skeleton, and the horns of the Moose."

Sullivan's party of men killed a moose in Vermont. Its body reached Sullivan on April 3, 1787, badly rotted. The skull and antlers could not be saved, so Sullivan sent the antlers of another animal. He wrote that they were "not the horns of this moose but may be fixed on at pleasure," meaning Jefferson could attach them however he liked.

The shipment reached France in about late September 1787. Around October 1, Jefferson gave Buffon the moose's bones and skin, along with the other animal's antlers. Jefferson wrote that the moose "convinced Mr. Buffon. He promised in his next volume to set these things right." Buffon died in April 1788 without printing a correction.

Benjamin Banneker

Who: A free Black tobacco farmer who taught himself mathematics and astronomy and calculated almanacs for the years 1792 through 1797.

When and where: Baltimore County, Maryland, where he was born on November 9, 1731. He died in 1806.

He sent Thomas Jefferson a handwritten almanac as proof.

Benjamin Banneker taught himself mathematics and astronomy from books and instruments that he borrowed. He was a free Black tobacco farmer in Baltimore County, Maryland. For six years of printed almanacs, 1792 through 1797, he worked out where the sun, the moon and the planets would be on each day.

He correctly predicted the solar eclipse of April 14, 1789, when the moon passed between the earth and the sun. Better-known almanac makers had calculated it differently.

The letter. On August 19, 1791, Banneker sent Jefferson his almanac in manuscript, written by hand, as proof against Jefferson's claim about Black people. In the letter that went with it, he challenged Jefferson about slavery. "However variable we may be in society or religion," he wrote, "however diversified in situation or colour, we are all of the same family."

Jefferson wrote back on August 30, 1791: "no body wishes more than I do to see such proofs as you exhibit (show), that nature has given to our black brethren (brothers), talents equal to those of the other colours of men." Jefferson sent the almanac on to a man named Condorcet in Paris.

The almanacs sell. Banneker's first almanac, for 1792, went on sale in December 1791. Goddard & Angell of Baltimore printed it, and it sold in Baltimore, Philadelphia and Alexandria. His Quaker neighbors, Joseph and George Ellicott, had already interested other Quakers in his work, and the abolition societies of Baltimore and Philadelphia, groups that worked to end slavery.

Members of the abolition societies of Pennsylvania and Maryland promoted the almanacs. Several printers published them, and they sold in the United States and in England. The 1792 almanac opened with support from James McHenry, a Maryland politician, who used the word of the time for a Black person: "I consider this Negro as fresh proof that the powers of the mind are disconnected with the colour of the skin."

Jefferson, later. Banneker died in 1806. Three years later, on October 8, 1809, Jefferson wrote to Joel Barlow: "we know he had spherical trigonometry enough to make almanacs, but not without the suspicion of aid from Ellicot, who was his neighbor & friend, & never missed an opportunity of puffing him." Spherical trigonometry is the mathematics of angles and distances on a sphere, such as the sky, and puffing meant praising someone too much. Jefferson was hinting that Ellicott had helped with the mathematics. He himself called it a suspicion.

Jefferson's fossil, 1797

Jefferson presented a paper about a fossil animal named Megalonyx, which means "giant claw," to the American Philosophical Society on March 10, 1797. The paper was printed in 1799. It was the first fossil animal from the United States to get a genus name, the name for a group of closely related kinds of animals or plants.

Caspar Wistar wrote a second paper to go with Jefferson's. With it, Wistar began the study of paleontology, ancient life through fossils, in the United States.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 10 MINUTES

Americans set up their own places for science between 1800 and 1850. Benjamin Silliman, Yale's professor of chemistry from 1802, founded the country's first lasting science journal, a magazine where scientists publish their research, in 1818. President James K. Polk signed a law in 1846 to set up the Smithsonian Institution in Washington, paid for by an Englishman who never visited the country. Its first head, Joseph Henry, set up a network of volunteers who reported the weather.

In Philadelphia in the same years, a doctor named Samuel George Morton collected hundreds of human skulls. Some of the men who supplied him robbed graves to get them. Morton measured the skulls and used the numbers to rank the races of people. The idea that skull size shows the intelligence or worth of a group of people is false.

A professor, a fallen stone and a journal

Benjamin Silliman became Yale's professor of chemistry, the study of what things are made of and how they change, in 1802. His job covered natural history too, the study of plants, animals and rocks. He gave his first lectures in 1804.

The Weston meteorite. A meteorite, a rock from space, fell at Weston, Connecticut, on December 14, 1807. Silliman and James Kingsley wrote down what was known about the fall. Then they ran chemical tests to learn what the rock was made of. It was the first meteorite fall in the Americas that scientists recorded and studied in this way.

Silliman founded the *American Journal of Science* in 1818. It was the first science journal in the country that lasted, and for decades people called it "Silliman's Journal."

Who found induction first?

Michael Faraday in London was the first to publish the discovery of electromagnetic induction, the making of an electric current with a magnet. He published it in 1831, and he keeps the credit. Joseph Henry, in Albany, New York, had found the same effect on his own but published later.

Self-induction. Henry has the credit for a related discovery, self-induction, which he published in Silliman's journal in July 1832. When the electric current in a coil of wire changes, the coil makes its own push of electricity against that change. Scientists measure the strength of self-induction in a coil with a unit called the henry, named after him.

Henry was a teacher at the Albany Academy, a school in Albany, and his best years of research there ran from 1826 to 1832. He built the strongest electromagnets of his day. An electromagnet is a magnet made by sending electricity through a coil of wire.

Samuel Morton's skulls

Samuel George Morton, a Philadelphia doctor, collected human skulls and used them to rank the races of people. He lived from 1799 to 1851 and belonged to the Academy of Natural Sciences of Philadelphia, a society of scientists. He prepared a lecture on how skulls differ in 1830 and began to collect them in the early 1830s. When he died in 1851, he owned 867 human skulls.

Where they came from. He got the skulls through 138 contacts: scientists, traders, military men and missionaries, people sent by churches to spread their religion. Army surgeons at far-off posts were his most useful suppliers. Some of his collectors robbed graves. About a dozen skulls came from a potter's field in Philadelphia, a burial ground for poor people, where poor Black Philadelphians were buried.

Morton himself got the remains, what was left of the bodies, of at least 14 Philadelphians, Black and white, while he worked as a doctor at the city's almshouse, a public home for poor people. A 2021 report for a University of Pennsylvania project does not name who dug up their bodies, and a Philadelphia news station reported that year that the Philadelphia skulls were likely bought from grave robbers.

A doctor in Havana, José Rodríguez Cisneros, dug up the graves of enslaved Africans in Cuba in 1840 and shipped about 50 of their skulls to Morton. More skulls came from battlefields and from gallows, the frames on which people were hanged. Many of the skulls were those of Native Americans.

The measuring. Morton filled each skull with peppercorns, and later with lead shot, which means tiny balls of lead. Then he poured them out and measured how much the skull had held. He believed that a bigger skull meant a more intelligent person. He ranked the races by the average size of their skulls, with white Europeans at the top.

His book *Crania Americana*, Latin for "American skulls," came out in 1839 and made him famous in other countries. The Penn Museum's own account states that Morton is sometimes called the "father of scientific racism."

Scientific racism is the use of the look of science to claim that some races are better than others.

Used to defend slavery. Two of Morton's followers, the doctor Josiah Nott and George Gliddon, used his work to defend slavery. In 1854 they published a book in Morton's honor called *Types of Mankind*. Nott argued in it that Black people were a separate species, a separate kind of living thing, naturally and forever below white people. His aim was to show that freeing enslaved people would be a disaster for both races. Nott wrote to protect his clients, who were slaveholders.

Disputed. Scholars still argue about how carefully Morton measured. Stephen Jay Gould of Harvard wrote in 1978, and again in a 1981 book, that Morton's numbers were biased, meaning his mistakes leaned toward the answer he wanted. In 2011 six scholars led by Jason E. Lewis measured 308 of the 670 skulls Morton had published, and they found his errors small and random. A 2016 paper argued against parts of their work.

KNOWN: Morton's main idea, that skull size ranks the intelligence or worth of groups of people, is false, and the argument is only about how carefully he measured.

The skulls today. The Penn Museum, at the University of Pennsylvania, holds Morton's skulls. Museum staff took them off a classroom display in 2020. On April 12, 2021, the museum's director, Christopher Woods, apologized for "the unethical possession of human remains in the Morton Collection." Unethical means wrong. He announced a plan to return the remains.

On the order of a Philadelphia court, museum staff buried the skulls of 19 Black Philadelphians from the collection in January 2024. They buried them at Eden Cemetery in Collingdale, Pennsylvania, one of the oldest Black burial grounds near Philadelphia. A group called the Black Philadelphians Descendant Community Group objected. Its members said that the Black community, not the museum, should decide what happened to the remains.

James Smithson

Who: An English chemist who left his fortune to the United States to found the Smithsonian Institution. He never set foot in the country.

When and where: Died in Genoa, Italy, on June 27, 1829. The Smithsonian was founded in Washington on August 10, 1846.

More than 100,000 gold coins crossed the ocean to found the Smithsonian.

James Smithson, an English chemist, paid for the Smithsonian Institution, even though he never set foot in the United States. He died in Genoa, Italy, on June 27, 1829.

His will left his money to his nephew, with a condition. If the nephew died without heirs, meaning children or other family to inherit the money, it was to go to the United States. The will said the money was "to found at Washington, under the name of the Smithsonian Institution, an establishment (a place set up for a purpose) for the increase & diffusion (spreading) of knowledge."

NOT KNOWN: why Smithson left his fortune to a country he had never seen. He never explained it.

The nephew died without heirs in 1835. In September 1838, more than 100,000 gold coins called sovereigns reached the mint in Philadelphia, where the country's coins are made. Workers there turned them into American coins worth more than $500,000. Members of Congress then argued for eight years about what to do with the money. President James K. Polk signed a law to set up the Smithsonian on August 10, 1846.

Joseph Henry

Who: A teacher at the Albany Academy who discovered self-induction and became the first Secretary, or head, of the Smithsonian Institution.

When and where: Albany, New York, for his research of 1826 to 1832. Princeton from 1832. Washington from 1846.

About 150 volunteers sent him the weather from where they lived.

Joseph Henry, the first Secretary, or head, of the Smithsonian Institution, drew the first weather maps of the whole country. From 1849 he ran the Smithsonian Meteorological Project, a study of the weather. About 150 volunteers wrote down the weather where they lived. They sent their reports by mail, and by telegraph at no charge. A telegraph sent messages along a wire as bursts of electricity.

A decade later, about 600 observers were sending reports. The project ran from 1849 to 1870, and it led to the National Weather Service, the weather agency of the United States government today.

How he got the job. Henry discovered self-induction while he taught at the Albany Academy, and he became a professor at Princeton in 1832. On December 3, 1846, he won seven of the twelve votes cast by the Smithsonian's Board of Regents, the group that runs it. He held the job for 32 years, until he died in 1878.

Charting the winds at sea

Matthew Fontaine Maury made charts, or maps for sailors, of the winds at sea and of the currents, the streams of moving water inside the ocean. He built them from thousands of old ships' logs. A log is the daily record a ship's crew keeps of its position, the wind and the sea.

Maury ran the Navy's Depot of Charts and Instruments, where old logs were stored, from 1842. He went through thousands of them for facts about winds and currents. In 1847 he issued the first Wind and Current Chart of the North Atlantic. He then collected logs from ship captains around the world, all kept in the same format. Captains who used his charts cut their sailing times.

He joined the Confederacy. Virginia's leaders took the state out of the United States in April 1861. Maury then resigned from the US Navy and joined the Confederacy, the group of Southern states that broke away from the United States and fought against it in the Civil War. Before the war, Maury had wanted slavery to spread into the American West and also into Cuba and Brazil.

Maria Mitchell

Who: A librarian and astronomer on Nantucket who found a comet in 1847. She was the first American and the first woman to win the King of Denmark's gold medal for finding a comet.

When and where: Nantucket, Massachusetts. She found the comet on October 1, 1847.

From a Nantucket bank roof, she found a comet nobody had charted.

Maria Mitchell was sweeping the sky with a small telescope on October 1, 1847, moving it slowly across the stars. She was on the roof of the Pacific National Bank building on Nantucket, an island off Massachusetts, where her father worked. She found a comet, a frozen body that travels around the sun, that no chart of the sky showed. It is known as "Miss Mitchell's Comet."

The King of Denmark offered a gold medal for each new comet found with a telescope. Mitchell won it, the first American and the first woman to receive it. An Italian astronomer named de Vico saw the same comet two days after she did.

Library by day. Mitchell had been the first librarian of the Nantucket Atheneum, a library on the island, since 1836. For the 20 years she held that job, she read in the library by day. At night she and her father watched the sky from the roof of the bank, where he was in charge.

She became the first woman elected to the American Academy of Arts and Sciences, a society of scholars, in 1848.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 14 MINUTES

Louis Agassiz, the most famous scientist in America, used his fame in these years to push a false claim about race. In 1850 he had seven enslaved people in South Carolina photographed, with their clothes taken off, as "evidence" that the races of people were created separately.

Officials of an Army museum in Washington asked Army doctors, from 1868, to send them the skulls of Native Americans, and scientists there measured the skulls to rank the races. Agassiz attacked Charles Darwin's book *On the Origin of Species*, and the botanist Asa Gray defended it.

Other American scientists measured light, energy and the stars with new care. Albert Michelson measured the speed of light in 1879 to within about 0.05 percent of the value used today. Josiah Willard Gibbs of Yale founded the science of energy in chemical changes with a paper published in 1876 and 1878. Williamina Fleming, first hired as a maid, sorted more than 10,000 stars at Harvard's observatory.

The argument over Darwin's book

Charles Darwin's book *On the Origin of Species* came out in London in November 1859. Darwin argued that species, the different kinds of living things, were not each created separately. He wrote that species change over time, and he called the way they change natural selection.

How it works. Some living things have a trait, a feature such as color or size, that helps them survive and have young where they live. They have more young, and the young are likely to have the trait too. In time, the trait spreads through the group.

Two sides at Harvard. Copies of the book reached Harvard quickly, and two scientists there took opposite sides. Asa Gray, a botanist, defended Darwin's theory. Gray had arranged for the book to be published in America. Louis Agassiz, who studied animals and was the most famous scientist in America, attacked it. He insisted that every species had been created separately.

The two men argued it out at the American Academy of Arts and Sciences in early 1860, after Agassiz opened his attack there on January 10. Gray kept defending Darwin in essays in the *Atlantic Monthly*, a magazine. He collected them in 1876 in a book called *Darwiniana*. In it he argued that a person could accept science and keep a religious faith at the same time.

Agassiz and the fifteen photographs

Louis Agassiz used his standing as America's most famous scientist to push a racist idea called polygenism, the false claim that the races of people were created separately. In the winter of 1850, Agassiz hired a photographer named Joseph T. Zealy to photograph enslaved people in South Carolina as "evidence" for it.

On a tour of plantations, the large farms worked by enslaved people, Agassiz looked for people who had been born in Africa. He thought that people born in Africa were racially "pure." He examined them himself. He left instructions with Robert W. Gibbes, a doctor in Columbia, South Carolina, to collect photographs as evidence. Gibbes arranged for Zealy to photograph the people Agassiz had examined.

Zealy made 15 daguerreotypes, an early kind of photograph made on a polished metal plate. The pictures show seven people: Renty Taylor, his daughter Delia, Alfred, Drana, Fassena, Jack and Jem.

Found again. Elinor Reichlin found the daguerreotypes at Harvard's Peabody Museum in 1976. They are the earliest known photographs of enslaved people in America.

The accounts of how the people were posed differ. In 1977 Reichlin wrote that each person had been posed naked. News reports in 2025 said that Renty and Delia were posed with no shirts on and photographed from several angles. Both accounts agree that the people were photographed with their clothes off, at least from the waist up. Reichlin's account does not make clear whether Gibbes or Zealy did the posing.

Who they were. B. F. Taylor enslaved Renty on his plantation, called Edgehill. Delia was Renty's daughter, born in America to parents from Africa. Jack worked in Taylor's fields as a driver, an enslaved man put in charge of other enslaved workers. F. W. Green of Columbia enslaved Jem. Reichlin's account does not name who enslaved Alfred, Drana and Fassena.

The lawsuit. Tamara Lanier, who identifies Renty as her great-great-great-grandfather, sued Harvard over the pictures in 2019. She claimed the images belonged with Renty's family, asked for money in damages, and asked Harvard to admit its part in slavery. In 2022 the judges of the Supreme Judicial Court of Massachusetts, the state's highest court, let her claim for emotional distress, harm to a person's mind and feelings, go forward.

Lanier and Harvard's lawyers settled the case on May 28, 2025. Harvard paid her an amount that was not made public, and Harvard's leaders agreed to give up the images. Lanier's lawyer said that Harvard still had not publicly accepted her family tie to Renty.

Harvard's leaders handed the pictures over in December 2025, from the Peabody Museum to the International African American Museum in Charleston, South Carolina. Harvard's statement said the move would put them "in the appropriate context and increase access to them for all Americans." Museum staff there unveiled them to the public on March 11, 2026.

Why did Army doctors send skulls to Washington?

Scientists at the US Army Medical Museum in Washington measured the skulls of Native Americans to rank the races. From the 1860s, Army doctors and other collectors sent them the skulls, and the collectors took many of them from graves and battlefields.

One man's remains. The museum got the remains of a Dakota man killed in May 1864 near an Army post in rural Minnesota. Militiamen, part-time citizen soldiers who were fighting the Native people there, shot him twice. One bullet shattered his skull. The other went through his mouth or neck.

Soldiers dragged his body to a nearby fort, where settlers beat it. Someone cut off his scalp, the skin and hair of the top of the head, and carried it away to keep. Someone buried the body in a shallow grave. A few months later, his skeleton was dug up again.

He was about 25 to 35 years old, and nobody recorded his name. The historian Samuel Redman, who told this man's story in 2016, does not name the militiamen, the settlers or the person who dug him up. Alfred Muller, an Army doctor with the title of acting assistant surgeon, examined the man's remains and sent them to the Army Medical Museum. They ended up at the Smithsonian.

The orders. Officials in the Army's Surgeon General's Office, which ran the Army's doctors, sent out a letter called Circular No. 2 in 1867. In it they asked Army medical officers to collect skulls, along with Native weapons, clothing, tools, food and medicine. From 1868, officials at the Army Medical Museum asked Army doctors in "Indian country," the lands where Native nations lived, to send them Native American skulls.

The museum's scientists did craniometry, which means measuring skulls. Craniometry was part of a false "racial science." People who did it used the measurements to sort humans into races, then ranked the races by how intelligent they believed each one was. George A. Otis, the museum's curator, the person in charge of its collection, collected and measured hundreds of the skulls.

Redman's history of these collections tells how the collectors got the skulls. They dug some out of graves, sometimes months after the burial, and emptied whole cemeteries. Others came from battlefields, from people who had just been killed there. Redman's account does not say who did the killing. Some skulls arrived at the museum without anyone asking for them.

Mulberry Creek, 1869. One of the collectors was B. E. Fryer, the Army surgeon at Fort Harker in Kansas. In late January 1869, US soldiers and Kansas settlers attacked a party of Pawnee men at a farm near Mulberry Creek, in Kansas. James Riding In, a Pawnee historian, calls it a massacre. The Pawnee said the soldiers fired first, and Army officers said the Pawnee did.

Two soldiers were wounded. The soldiers and settlers killed eight or nine Pawnee men. One of them was a man whom settlers captured and then killed, and the reason given was that he was "trying to escape." Five men escaped and hid their dead friends' bodies from the soldiers. Three of those five later died of exposure, from cold out in the open.

In all, as many as twelve Pawnee men may have died because of the attack: eight or nine at the farm, the captured man among them, and three more from the cold.

Fryer sent a civilian to cut off the dead men's heads on January 30. Over the next weeks, Fryer or men he sent searched for the hidden bodies. On March 11, 1869, Fryer shipped 26 Native skulls to Washington, and six of them were Pawnee. He wrote to Otis that two of the skulls were damaged because soldiers "shot into the bodies and heads several times after the fight."

Fryer sent Otis the remains of at least 42 people from 1868 to 1872. They were Cheyenne, Wichita, Caddo, Osage, Kansa and people of other nations. Riding In told what Fryer did in a 1992 article, quoting Fryer's letter to Otis.

Moved to the Smithsonian. The museum's collection held about 3,000 skulls by about 1900. Army officials moved the Native remains to the Smithsonian's national museum in 1898 and again in 1904. The only ones they kept back were those of interest for the study of surgery. One record counts 2,000 skulls in the transfer.

Colleges and farm research in every state

The Morrill Act became law on July 2, 1862. Under it, members of Congress gave each state 30,000 acres of public land for each of its senators and representatives. The states sold the land and used the money to start colleges that taught farming and the skills of machines and building. These are called land-grant colleges, and through them students in every state could study science.

Whose land. The land had been taken from Native nations. The reporters Robert Lee and Tristan Ahtone counted about 10.7 million acres of it. US officials had taken it from nearly 250 Native nations, bands and communities, in more than 160 land deals backed by violence. For all of it, US officials paid less than $400,000.

Under the Hatch Act of 1887, members of Congress set aside federal money for an agricultural experiment station in every state, a place where scientists run tests to improve farming. Scientists at the stations did research for the public in every state, including places far from the old college towns.

Maria Mitchell's eclipse trip, 1878

Maria Mitchell led a team of women about 2,000 miles by train to Denver in 1878 to watch a total solar eclipse, when the moon blocks all of the sun's light for a short time. The eclipse took place on July 29, 1878. Her team was made up of students and graduates of Vassar College, a college for women. The women ran the trip themselves, and reporters for newspapers across the country followed it.

Mitchell had become a professor at Vassar in 1865. She was the first American woman to be a professor of astronomy.

Josiah Willard Gibbs

Who: A Yale physicist whose long paper of 1876 and 1878 founded chemical thermodynamics, the science of energy in chemical changes.
When and where: Yale, in New Haven, Connecticut. Professor from 1871.

Yale paid him nothing for his first nine years as professor.

Josiah Willard Gibbs founded chemical thermodynamics. With its rules, chemists can work out from the energy in a chemical change whether the change will happen on its own.

He published this work in two parts, in 1876 and 1878. It came out in the *Transactions of the Connecticut Academy*, a journal that was little known, and it was very hard to follow.

A model in the mail. Scientists in Europe took notice of Gibbs's work before scientists in America did. In 1873 Gibbs had published shorter papers that described a curved surface in mathematics. In 1874 the Scottish physicist James Clerk Maxwell, a scientist who studied matter and energy, built a model of that surface with his own hands. He shaped it in clay, made copies in plaster and mailed one to Gibbs in New Haven.

No pay. Gibbs had earned the first American doctorate in engineering, at Yale in 1863. A doctorate is the highest degree a university gives. He was Yale's professor of mathematical physics from 1871, and for his first nine

years in that job he received no pay. He lived on money he had inherited. When leaders of Johns Hopkins University offered him a job in 1880, Yale's leaders began to pay him so that he would stay.

He received the Copley Medal, at that time the top science honor in the world, in 1901.

Albert Michelson

Who: An immigrant from Prussia, raised in mining towns in the West, who measured the speed of light to within about 0.05 percent of today's value.
When and where: Annapolis, Maryland, in 1879. Cleveland, Ohio, in 1887.

His speed of light was off by only 1 part in 2,000.

Albert Michelson measured the speed of light at Annapolis, in Maryland, in 1879. He got 299,864 kilometers per second, in air, and some books print his figure as 299,910. His result came within about 0.05 percent of the value scientists use today. That is an error of about 1 part in 2,000.

Michelson was born on December 19, 1852, in Strelno, a town in Prussia, a German kingdom. The town is now Strzelno, in Poland. His parents brought him to the United States in 1855, when he was 2. He grew up in mining towns in California and Nevada, and he graduated from the US Naval Academy at Annapolis.

No ether. Michelson and the chemist Edward Morley ran the test now called the Michelson-Morley experiment in Cleveland, Ohio, from April to July 1887. Scientists then thought that light traveled as waves through an unseen substance they called the ether. Michelson and Morley found no trace of the ether. A result like this, where the test finds nothing, is called a null result.

Their null result was an early clue for the theory of relativity, a new account of space, time and light that Albert Einstein put forward in 1905. Michelson became the first American to win a Nobel Prize in science,

in 1907.

Williamina Fleming

Who: An immigrant from Scotland, hired as a maid in the house of the Harvard College Observatory's director, who went on to classify more than 10,000 stars.

When and where: Harvard College Observatory, from about 1879.

She started as a maid in the observatory director's house.

Williamina Fleming, an immigrant from Scotland, took a job in about 1879 as a maid in the house of Edward Pickering. He was the director of the Harvard College Observatory, a building for studying the sky. In 1881 she moved onto the observatory's staff, and she sorted more than 10,000 stars by type for the Draper Catalogue, a list of stars published in 1890.

She was one of the observatory's "computers." At that time a computer was a person whose job was to do calculations and measurements. Fleming studied photographic plates, glass sheets that hold photographs of the night sky. On a plate from 1888 she spotted the Horsehead Nebula, a cloud of gas and dust in space.

Fleming discovered 59 nebulae and 10 novae, stars that suddenly grow much brighter for a time. She also found more than 310 variable stars, stars whose brightness changes. In 1899 she became Curator of Astronomical Photographs, the person in charge of the observatory's photographs of the sky.

The Chest of 1900. Harvard's time capsule of 1900, a box of papers sealed to be opened in the future, holds an entry by Fleming. It records that she earned $1,500 a year, while men in similar jobs earned $2,500. After talking about women's pay with Edward Pickering, the director, she wrote of him: "Does he ever think that I have a home to keep and a family to take care of as well as the men?"

BY 1900

If you remember one thing: After 1850 some American scientists measured light and the stars with new care, while Louis Agassiz and the Army's museum scientists used photographs and skulls to rank the races.

- 1850: Agassiz has seven enslaved people in South Carolina photographed as "evidence" for his false claim.
- March 11, 1869: Army surgeon B. E. Fryer ships 26 Native skulls to Washington, six of them Pawnee.
- 1879: Albert Michelson measures the speed of light to within about 0.05 percent.

Next: In the next fifty years American astronomers measured how far away the galaxies are, and the men who ran a new office in New York called their work a science of who should have children.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 25 MINUTES

American scientists measured distances across the universe and ran the first controlled nuclear chain reaction in these fifty years, and some of them used people as research material without asking them. Henrietta Leavitt found a rule for measuring space in 1912, and Edwin Hubble used it to show in 1929 that the galaxies are moving apart. Enrico Fermi's team in Chicago ran the chain reaction on December 2, 1942. Fermi, Albert Einstein and Emilio Segrè all came to American labs from Europe in the 1930s.

The staff of the Eugenics Record Office in New York collected family records from 1910 to 1939 to claim that poverty and behavior are inherited. At the Smithsonian, Aleš Hrdlička collected human brains and bones to rank the races. Doctors working for the Manhattan Project, the secret program that built the atomic bomb, injected 18 hospital patients with plutonium from 1945 to 1947 to learn where it goes in the human body.

The charge of an electron and the place of a gene

Robert Millikan measured the electric charge of a single electron in experiments with drops of oil, from 1909 to 1913. An electron is one of the tiny particles inside every atom, and it carries electricity. Millikan showed that electric charge comes in exact, equal pieces, and he won the Nobel Prize in Physics in 1923.

The Nobel Prizes were first given in 1901 under the will of Alfred Nobel, a Swedish inventor. Each year they go to people for work in physics, chemistry, medicine, writing and peace, and later also economics.

The fly room. Thomas Hunt Morgan found a fruit fly with white eyes in his lab at Columbia University in New York in 1910. People called the lab the "fly room," and after that fly Morgan and his team spent 17 years working there. They proved that genes sit in a line along the chromosomes. A gene is a piece of the instructions, passed from parents to children, that shapes a living thing, and a chromosome is a thread inside a cell that carries genes.

Morgan won the Nobel Prize in Physiology or Medicine in 1933. It was the first Nobel Prize for genetics, the study of how living things pass on their traits.

Frank Chapman and the first Christmas Bird Count

Who: An ornithologist, a scientist who studies birds, at the American Museum of Natural History. He proposed counting birds at Christmas instead of shooting them.

When and where: Across North America, from Toronto to Pacific Grove, California, on December 25, 1900.

On Christmas Day 1900, 27 people counted birds instead of shooting them.

Hunters had a Christmas custom called the "side hunt." In a side hunt, they shot at every bird and animal they could reach. Frank M. Chapman proposed that people count the birds instead.

Chapman worked at the American Museum of Natural History in New York. He had founded a bird magazine called *Bird-Lore* and was its editor.

The 27 bird-watchers ran 25 separate counts that day, from Toronto in Canada to Pacific Grove in California. Together they counted about 18,500 birds of 90 species.

The count has run every winter since 1900. It is one of the longest-running records of citizen science, research done by ordinary volunteers, in the world.

Robert Goddard

Who: A physics teacher at Clark University who built and flew the first rocket that burned liquid fuel. The New York Times mocked his rocket science in an editorial in 1920 and printed a correction in 1969.

When and where: Worcester and Auburn, Massachusetts. The first flight was on March 16, 1926.

His first liquid-fuel rocket flew 2.5 seconds and landed 184 feet away.

Robert Goddard launched the first rocket that ran on liquid fuel at Auburn, Massachusetts, on March 16, 1926. It rose 41 feet, flew for 2.5 seconds and came down in a farm field 184 feet away.

Goddard grew up in Worcester, Massachusetts, and began teaching physics at Clark University there in 1914. He later ran its physics department, and he became a full professor in 1934. In all he spent 29 years in Clark's physics department.

The editorial. The Smithsonian Institution published his paper *A Method of Reaching Extreme Altitudes*, about reaching very great heights, in January 1920. On January 13, 1920, the editors of the New York Times answered it with an editorial, an article giving the newspaper's own opinion. Its title was "A Severe Strain on Credulity." Credulity means a willingness to believe things.

The editors mocked Goddard for thinking that a rocket could push itself forward in a vacuum, a space with no air in it, like outer space. They wrote that he lacked "the knowledge ladled out daily in high schools."

The correction. The three Apollo 11 astronauts were on their way to the Moon on July 17, 1969, when the Times editors printed a correction. It read: "...it is now definitely established that a rocket can function in a vacuum as well as in an atmosphere. The Times regrets the error."

How did Henrietta Leavitt measure space?

Henrietta Swan Leavitt found that a kind of blinking star shows how bright it truly is, and that let astronomers measure how far away stars are. She was one of more than 100 women the Harvard College Observatory hired as "computers" to study photographs of the stars. She was paid 30 cents an hour, five cents more than most of the others. She had been deaf since an illness in her twenties.

The blinking stars. Leavitt studied a kind of variable star called a Cepheid, which grows brighter and dimmer over and over, in a steady rhythm. In 1912 she published a paper on 25 Cepheids in the Small Magellanic Cloud, a patch of stars in the southern sky. She showed that the time a Cepheid takes for one blink tells how bright it truly is.

Astronomers compare that true brightness with how bright the star looks from Earth. The dimmer it looks compared with its true brightness, the farther away it is. With Leavitt's rule, Cepheids became markers for measuring distance across the universe. Her rule is called the period-luminosity law: the period is the time of one blink, and luminosity means true brightness.

Leavitt died on December 12, 1921. In 1925 a Swedish mathematician who did not know she had died wrote to begin nominating her for a Nobel Prize.

Edwin Hubble

Who: An astronomer who showed that there are galaxies far outside our own and that the universe is expanding.
When and where: Mount Wilson Observatory, California, from 1923 to 1929.

On one glass plate, Hubble wrote "VAR!" beside a star in Andromeda.

Edwin Hubble found a Cepheid star on a photograph of the Andromeda "nebula," a faint cloudy patch in the sky, on the night of October 4 to 5, 1923. He was working with the Hooker telescope at Mount Wilson, which has a main mirror 100 inches across. The photograph was on a glass plate, and Hubble marked the plate "VAR!" for variable star.

Using Leavitt's law, Hubble worked out how far away the star was. His answer was about 900,000 light-years, each one the distance light travels in a year. At that distance Andromeda lay far outside the Milky Way, our own galaxy, so it had to be a separate galaxy, a huge group of billions of stars. He announced the result at a meeting of the American Astronomical Society on January 1, 1925.

The universe is expanding. In March 1929 Hubble published his finding that the farther away a galaxy is, the faster it is moving away from us. He used measurements of the galaxies' light made by Milton Humason.

Georges Lemaître, a Belgian scientist, had already worked out and published the expansion of the universe in 1927. His paper appeared in a journal that few people read. In 2018 the members of the International Astronomical Union, the world society of astronomers, renamed Hubble's law the Hubble-Lemaître law.

BY 1930: Astronomers had shown that Andromeda is a separate galaxy and that the galaxies are moving apart, and a rocket on liquid fuel had flown in a Massachusetts field. In Washington and in New York, other scientists were

collecting brains, bones and family records to rank people.

Aleš Hrdlička's brain collection

Aleš Hrdlička collected human brains and bones for the Smithsonian to try to prove that the races differ in their bodies. From 1903 into the early 1940s he ran the physical anthropology division, the study of human bodies and bones, at the Smithsonian's National Museum of Natural History in Washington. Hrdlička believed that white people were superior to other races. Scientists have since shown his theories about the races' bodies to be false.

The brain collection. Hrdlička began what he himself called the "racial brain collection" in 1903. Reporters at the Washington Post looked into his collecting and published what they found in August 2023. They reported that the museum held at least 255 brains, and that Hrdlička or people working at his direction collected most of them.

Collectors took most of the remains without the consent of the dead people or their families. Consent means agreeing to something after being told what it is. Many of these people had died in hospitals, were poor, or had no relatives nearby to claim and bury them. Collectors dug up burial grounds as well.

Kodiak Island. In the 1930s Hrdlička and a small team dug up the remains of about 1,000 people from the Uyak site on Kodiak Island, in Alaska, and shipped them to the Smithsonian. The people of Larsen Bay, who are Alutiiq, asked the Smithsonian for their dead in 1987. Museum staff sent the remains back to Larsen Bay in September 1991. It was one of the first returns of Native American remains in the country.

The Post's reporters found that in 2023 the museum held more than 30,700 human remains. More than 19,000 of them, about 62 percent, or three of every five, were collected while Hrdlička led the division.

The Eugenics Record Office

The staff of the Eugenics Record Office at Cold Spring Harbor, New York, collected family records from 1910 to 1939 and used them to argue that some people should not have children. Eugenics was the claim that a country should decide which of its people may have children, and stop the rest. The men who ran the office presented it as the science of heredity, the passing of traits from parents to children.

Who paid for it. The biologist Charles B. Davenport directed a research station at Cold Spring Harbor that belonged to the Carnegie Institution of Washington, a private group that paid for scientific research. In 1910 Davenport persuaded Mary Harriman, the widow of a railroad owner, to give $10,000. She bought an 80-acre farm next to the station and built an office on it. The Eugenics Record Office opened there in 1910, with Harry H. Laughlin in charge.

Door to door. Staff at the office ran summer courses that trained field workers, and photographs of the courses survive from 1913 to 1922. The field workers went door to door and filled out forms of questions about families. By 1924 the office's files held 750,000 index cards, along with family trees and reports.

Davenport, Laughlin and their workers traced traits through families, such as blindness and musical ability. They also traced what they called "feeblemindedness," their label for people they judged to be of low intelligence. They treated poverty and behavior as traits that parents pass to their children.

Laughlin used the office's records to push for laws to keep immigrants out and to sterilize people labeled "defective." To sterilize a person is to operate on the body so that the person can never have a child. In a woman, for example, a doctor cut out a piece of each of the two tubes that carry eggs to the womb. It cannot be undone.

The critics. Scientists who studied heredity, among them Thomas Hunt Morgan, Raymond Pearl and Herbert Jennings, pointed out flaws in eugenics research. They showed that the eugenicists' experiments were badly designed

and that the eugenicists used the new science of genes too simply. They also showed that the work was racist and unfair to poor people. In 1935 leaders of the Carnegie Institution gathered a panel of scientists to judge the office's work, and the panel found that almost none of it had any scientific worth.

Nazi Germany. In the 1930s Americans learned what Nazi Germany's leaders were doing with eugenics. Under a German law of July 14, 1933, judges and doctors sat on special courts, each made up of a judge and two doctors. By 1945 the judges and doctors on these courts had forced about 400,000 people to be sterilized.

Doctors usually cut the tubes that carry sperm in men, and tied or cut the tubes that carry eggs to the womb in women. The forced operations killed hundreds of people, most of them women.

Vannevar Bush, the president of the Carnegie Institution, cut off the office's money in 1939, and it closed in December 1939. Its records went to the University of Minnesota in 1948. Staff at Cold Spring Harbor Laboratory now publish this history on the laboratory's archive website, under the heading "Good genes, bad science."

Albert Einstein

Who: A physicist who moved to America in 1933, the year the Nazis took control of Germany. He became a US citizen in 1940.

When and where: Princeton, New Jersey, from October 1933.

In 1946 Einstein called racism "a disease of white people."

Albert Einstein arrived in the United States on October 17, 1933, the year the Nazis took control of Germany. He joined the new Institute for Advanced Study in Princeton, New Jersey, as one of its first professors.

He had done his most famous physics, the theory of relativity, in Europe between 1905 and 1915. Relativity explains how space, time and gravity are tied together.

The letter. On August 2, 1939, Einstein signed a letter to President Franklin Roosevelt. The physicist Leo Szilard wrote it, with Edward Teller and Eugene Wigner. The letter stated that Germany might build atomic bombs. Historians at the Atomic Heritage Foundation, a group that records the history of the bomb, trace the start of the Manhattan Project back to this letter.

Einstein became a citizen of the United States on October 1, 1940. He kept his Swiss citizenship.

At Lincoln. The leaders of Lincoln University, a Black university in Pennsylvania, gave Einstein an honorary degree in May 1946. An honorary degree is one a college gives to honor a person's life and work. At Lincoln he called racism "a disease of white people." He said, "I do not intend to be quiet about it."

Emilio Segrè

Who: An Italian physicist who lost his job at home in 1938 under Mussolini's laws against Jews, stayed at Berkeley, and won a Nobel Prize.

When and where: Berkeley, California, from the summer of 1938.

Italian officials took his job while he was visiting Berkeley.

Emilio Segrè was visiting Ernest Lawrence's Radiation Laboratory at the University of California, Berkeley, in the summer of 1938. That year Benito Mussolini, who ruled Italy, put antisemitic racial laws, laws hostile to Jews, into force. Italian officials used those laws to take away Segrè's job as a professor in Palermo, Italy.

Segrè stayed in America. Lawrence gave him a job as a research assistant, at low pay.

Segrè later helped discover the elements technetium and astatine, and plutonium-239, one form of plutonium. With Owen Chamberlain he discovered the antiproton. A proton is a tiny particle with a positive electric charge, found in the center of every atom, and an antiproton matches it but has a negative charge. Segrè and Chamberlain shared the Nobel Prize in Physics in 1959.

BY 1939: Einstein, Fermi and Segrè had come to American labs from Europe, the Eugenics Record Office had lost its money, and Einstein had signed a letter warning President Roosevelt that Germany might build atomic bombs.

Splitting the atom, and the big laboratories

Two chemists in Berlin, Otto Hahn and Fritz Strassmann, discovered nuclear fission in December 1938. Fission is the splitting of the center of an atom, called the nucleus, into two pieces, and the split gives off energy. In January 1939 Lise Meitner and her nephew Otto Frisch explained what the chemists' results meant. Meitner was a Jewish refugee, a person who flees their home country to be safe, living in Sweden. In 1944 the Nobel judges gave the prize for the discovery to Hahn alone and left Meitner out.

Chicago Pile-1. News of fission reached America in January 1939. On December 2, 1942, Enrico Fermi's team of 49 people ran the first controlled nuclear chain reaction. In a chain reaction, each atom that splits sets off more atoms splitting. Controlled means the team could keep it steady and stop it. The team's reactor, a machine built to run a chain reaction, was called Chicago Pile-1.

The reactor stood under the west stands of Stagg Field, an unused football field at the University of Chicago. The reaction kept itself going for 28 minutes and made about half a watt of power.

Fermi was an Italian physicist. He won the Nobel Prize in 1938 and sailed straight from the ceremony in Stockholm to New York. His wife, Laura, was Jewish, and Mussolini had just put his new laws against Jews into force.

The big labs. Chicago Pile-1 belonged to the Manhattan Project. The project's leaders set up three large laboratories: the Metallurgical Laboratory in Chicago, Clinton Laboratories at Oak Ridge in Tennessee, and Los Alamos in New Mexico. They were the start of the national laboratories, the government's big science labs.

Hiroshima and Nagasaki. American bomber crews dropped the project's two atomic bombs on the Japanese cities of Hiroshima and Nagasaki in August 1945. Researchers at the Radiation Effects Research Foundation, a research group that Japan and the United States run together, estimate that within two

to four months the bombs killed 90,000 to 166,000 people in Hiroshima and 60,000 to 80,000 in Nagasaki. The exact numbers are not known, and a US study in 1946 gave lower counts.

The lab at Berkeley. An older lab came before these, the Radiation Laboratory at the University of California, Berkeley. Ernest Lawrence built it around his cyclotrons, machines that speed tiny particles around in a circle and smash them into atoms. From 1931 the lab was housed in a converted campus building, and it became a separate laboratory on July 1, 1936. In 1939 Lawrence won the first Nobel Prize at Berkeley.

Glenn Seaborg, a chemist who found new elements with the Berkeley lab's cyclotrons, redrew the periodic table, the chart of all the chemical elements, before the war ended. He pulled a row of heavy elements out of the main table and set it as its own row at the bottom. That row is called the actinides. With his layout, chemists could predict how elements that nobody had made yet would behave.

After the war, American scientists built their labs on the Berkeley plan of big machines run by big teams.

Plutonium injected into 18 patients

Doctors working for the Manhattan Project injected 18 hospital patients with plutonium between April 1945 and July 1947. Plutonium is a radioactive metal, one that gives off radiation, which is invisible energy that can damage or kill the cells of the body. Scientists made plutonium during the war and used it in atomic bombs.

Once plutonium is in the blood, it collects in the bones, the liver and the spleen, an organ that cleans the blood. There it keeps giving off radiation that kills cells. When a person breathes it in, it scars the lungs and can cause cancer.

Why they did it. The doctors wanted to know where plutonium goes in a human body and how fast the body gets rid of it in urine. Workers on the bomb could take some in by accident, and the doctors wanted a way to judge how much a worker had taken in. Four men led the program: Wright Langham of Los Alamos, Joseph Hamilton of the University of California in San Francisco and Berkeley, Hymer Friedell of Oak Ridge, and Stafford Warren, the head of the project's medical section.

The doctors injected one patient at Oak Ridge, three in Chicago, 11 in Rochester, New York, and three at the University of California hospital in San Francisco. Doctors in the program also injected six people with uranium, five with polonium and at least one with americium, all of them radioactive metals. In all they injected about 30 civilians, people who were not in the armed forces.

Ebb Cade. Cade, a 53-year-old Black cement mixer, was hurt in a car accident at Oak Ridge on March 24, 1945. On April 10, 1945, a doctor injected him with 4.7 micrograms, or millionths of a gram, of plutonium.

The doctors later blamed each other for the injection. Dr. Joseph Howland, an Army doctor, said he gave it only after his boss, Hymer Friedell, ordered him to in writing, and that Cade never gave his consent. Friedell said he gave no

such order and that another doctor, Dwight Clark, gave the injection. A federal committee set up in 1994 to look into these experiments could not tell which account was right.

Doctors did not set Cade's broken arm and leg until April 15, five days after the injection, when they also took samples of his bone. Someone pulled out 15 of his teeth, and they were tested for plutonium. The committee's report does not name who pulled them. Its members could not find out whether the teeth were pulled mainly because they were decayed or for the test. Cade died of heart failure on April 13, 1953.

Albert Stevens. Doctors at the San Francisco hospital told Stevens, a 58-year-old house painter, that he had stomach cancer that would kill him. They injected him with plutonium on May 14, 1945, as part of Joseph Hamilton's program. The federal committee's report does not identify the doctor who gave the injection. When surgeons operated, they found a harmless ulcer, a sore in the stomach, and no cancer.

The doctors told Stevens only about "a new substance." Members of the federal committee later found that he was almost certainly never told it was plutonium. He died of heart failure in 1966, 21 years later.

Simeon Shaw. One patient came from Australia. In April 1946 officials of the Red Cross and the US Army arranged for Simeon Shaw, a 4-year-old boy with bone cancer, and his mother to fly on an Army plane to San Francisco. Within days, doctors at the university hospital injected him with a liquid that held plutonium and two other metals, yttrium and cerium, under Joseph Hamilton's direction.

His medical file holds a consent form for an operation. A witness signed it a week after the injection, stating that his mother agreed. The committee found that the form probably covered a bone test done that week, not the injection itself. Surgeons cut out a piece of his bone tumor to see how much of the metals it had taken up. He went home to Australia about a month after the injection, and nobody followed up on him. He died in January 1947.

Eda Schultz Charlton. Charlton, 49, entered Strong Memorial Hospital in Rochester on November 2, 1945. Three weeks later doctors in the hospital's research ward, run by Dr. Samuel Bassett, injected her with 4.9 micrograms of plutonium. No doctor had ever found that she was dying of any illness. She lived until 1983.

Elmer Allen. Allen was a Black railroad porter from Italy, Texas, a worker who carried passengers' bags on the trains. He was 36, and his doctors believed he had bone cancer. On July 18, 1947, doctors at the University of California hospital injected him with plutonium. The researchers never expected the injection to help him. Soon afterward, a surgeon cut off his left leg at the middle of the thigh. A test of his tumor showed a slow-growing bone cancer called chondrosarcoma.

Two doctors signed a note saying the experiment had been explained to him and that he agreed. The committee's members found it unclear whether he was told the injection was not meant to help him. According to later newspaper reporting, he died in 1991 without ever learning what had been done to him. In 1995 his daughter, Elmerine Whitfield Bell, told the committee: "We contend that my father was not an informed participant in the plutonium experiment."

Trinity and the people downwind

Manhattan Project scientists exploded the first atomic bomb at a test site called Trinity, in New Mexico, on July 16, 1945. The project's leaders decided not to move the people who lived nearby. According to the National Park Service, they worried that moving people would make them suspicious or cause panic. The nearest ranchers lived 13 miles from the site, and tens of thousands of people lived within 50 miles.

Secrecy first. General Leslie Groves, the Army officer in charge of the project, wanted the test kept secret above all. Under his rules, nobody was to be protected unless people had already been badly hurt. When the project doctor James Nolan brought him safety plans, Groves asked him, "What's the matter with you, are you a Hearst propagandist?" William Randolph Hearst owned a chain of newspapers known for stories written to shock and excite readers, and a propagandist spreads one side's story.

The scientists Joseph Hirschfelder and John Magee warned of "the definite danger of dust containing active material." Robert Oppenheimer, the director of the Los Alamos lab, approved a plan to move people only after they had taken in 60 to 100 roentgens of radiation. A roentgen is a unit for measuring radiation. Nobody was moved.

The fallout. Fallout is radioactive dust that falls back to the ground after a nuclear explosion. The fallout from Trinity spread over an area about 250 miles long and 200 miles wide. It fell heaviest on Chupadera Mesa, a flat-topped highland 30 miles away, and it fell across Lincoln, Socorro, Otero and Sierra counties. People there ate vegetables and drank water that had fallout in them.

The Ratliff family, who lived near a place called Hot Canyon, took in about 47 roentgens in two weeks. The allowed amount was 1.4, so they took in more than 30 times the limit. Nobody warned them.

After the test, Groves put out a statement to the press. It said that an ammunition dump had exploded and that nobody had been hurt.

Stafford Warren, the project's chief medical officer, reported to Groves in writing on July 21, 1945. He wrote that the dust "was potentially a very serious hazard over a band almost 30 miles wide extending almost 90 miles northeast of the site." He advised that future tests take place in an area "with a radius of at least 150 miles without population," meaning with nobody living within 150 miles.

NOT KNOWN: how many people the Trinity fallout made sick or killed, because no government official ever counted them. The National Park Service states that the exposure is "linked to cancer, stillbirth, and birth defects." A stillbirth is a baby born dead, and a birth defect is a problem in a baby's body that begins before it is born.

The physics inside the transistor, 1947

Chapter 8 tells how John Bardeen and Walter Brattain built the first transistor, a small solid part that can make an electrical signal stronger, at Bell Labs, the research laboratory of the telephone company AT&T, on December 16, 1947. The physics that led to it started with a design that failed.

William Shockley, the head of their group, designed an amplifier, a device to make a signal stronger, out of a solid crystal in April 1945. It did not work. A year later Bardeen suggested a reason: electrons sitting on the surface of the crystal might be blocking the electric push from outside, so that it could not reach into the crystal.

Bardeen and Brattain set out to study those surface electrons, and that work led them to the transistor. Bardeen explained how the electrons behave at the surface using quantum mechanics, the branch of physics that describes very tiny particles. Shockley then designed a second kind, the junction transistor, and the three men shared the Nobel Prize in Physics in 1956.

Barbara McClintock

Who: A scientist who studied the chromosomes of corn and discovered that genes can move from one place to another.

When and where: Cold Spring Harbor, New York. Her key experiments began in 1944.

In corn, she found genes that move from place to place.

Barbara McClintock discovered that some genes move from one place to another on the chromosomes, which is called transposition. She found it in maize, which is corn, in experiments she began in 1944. She called the moving pieces "controlling elements."

McClintock earned a PhD at Cornell University in 1927 and studied the chromosomes of corn under the microscope. She came to Cold Spring Harbor in 1941 or 1942. The Eugenics Record Office there had closed in December 1939.

She first described the moving genes in 1948. In 1950 she published a paper about mutable loci, places on a chromosome that can change, called *The Origin and Behavior of Mutable Loci in Maize*. She presented the work at a meeting of scientists at Cold Spring Harbor in 1951.

Slow to catch on. Other scientists respected her as an expert on chromosomes, but for decades few of them took up her idea of moving genes. In the 1960s and 1970s, molecular biologists, who study the molecules inside cells, reached the same finding.

BY 1950

If you remember one thing: Between 1900 and 1950 American scientists learned to measure the universe and to split atoms, and doctors working on the bomb injected 18 hospital patients with plutonium to see where it went in their bodies.

- In 1900: Volunteers counted birds on Christmas Day, and nobody had yet measured the distance to another galaxy.
- In 1950: Astronomers had measured the distance to Andromeda and found the galaxies moving apart, and the government ran big national laboratories that began with the atomic bomb.

Next: After 1950, a doctor cut cells from a woman's tumor without asking her, and a reporter and the families brought the hidden radiation experiments into the open.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 19 MINUTES

A doctor cut cells from Henrietta Lacks's tumor in 1951 without asking her, and for the rest of the century Americans kept learning what scientists had done to people who never agreed. The reporter Eileen Welsome made the plutonium injections public in 1993. Members of a federal committee then found nearly 4,000 radiation experiments on people, and in 1995 President Bill Clinton apologized. Under laws passed in 1989 and 1990, museum staff had to begin returning the Native American dead that scientists had collected.

In the same fifty years, James Watson and Francis Crick worked out the shape of DNA from Rosalind Franklin's X-ray pictures. By about 1970 most geologists, the scientists who study the Earth's rocks, agreed that the continents move. Charles David Keeling's measurements on Mauna Loa showed the carbon dioxide in the air rising every year.

Henrietta Lacks and the HeLa cells

A doctor at Johns Hopkins Hospital cut cells from the tumor of Henrietta Lacks in 1951 without her knowledge or consent. Nobody told her, and nobody asked her permission. Her cells became the first human cells that kept living and dividing in a lab with no end.

Henrietta Lacks was a Black woman, born on August 1, 1920. She and her husband moved to Baltimore, Maryland, in 1941, and they had five young children. In 1951 she went to Johns Hopkins Hospital because she was bleeding from her vagina.

Dr. Howard Jones was a gynecologist, a doctor who treats the parts of a woman's body that carry and deliver babies. He found a malignant, or cancerous, tumor on her cervix, the narrow lower end of the womb. Doctors treated her cancer with radium, a radioactive metal, which was the usual treatment at the time.

The cells. During the treatment she was asleep under anesthesia, a medicine that puts a patient into a deep sleep so they feel no pain. While she was asleep, Dr. Lawrence Wharton Jr. cut samples from her tumor and from healthy tissue of her cervix. They went to the laboratory of George Gey, a cell biologist at Johns Hopkins.

Most human cells died in a lab dish. Hers kept dividing, about once every 20 to 24 hours. Scientists named the cells HeLa, from the first two letters of her first and last names. Henrietta Lacks died at Johns Hopkins on October 4, 1951. She was 31.

Scientists used HeLa cells to study the polio vaccine, the shot that protects people from polio, and leukemia, a cancer of the blood. They also used them to study the AIDS virus, cancer, and the effects of zero gravity in space. Her family did not learn that her cells were being used in research until the 1970s.

The shape of DNA, and who got the credit

James Watson and Francis Crick worked out the shape of DNA in 1953, using Rosalind Franklin's X-ray pictures without her knowledge. DNA is the molecule inside cells that carries genes, and a molecule is a group of atoms joined together. Watson was a 24-year-old American, and Crick was British. They worked at Cambridge University in England. At Caltech in California, Linus Pauling and his team were racing for the same answer.

Watson and Crick published the structure in the journal *Nature* on April 25, 1953. DNA is a double helix: two long strands wound around each other in a spiral.

Franklin's picture. They depended on the work of Rosalind Franklin at King's College London. Franklin took pictures of DNA with X-rays, a kind of invisible light that passes through many things. In early 1953 Maurice Wilkins, who worked at King's College, showed Watson her picture called "Photo 51." He did it without her knowledge. Accounts differ on whether this was in late January or in February.

What the picture showed. Photo 51 showed a pattern shaped like an X. To Franklin, the X meant that the strands of DNA wind in a helix, a spiral. When Watson saw the picture, he concluded the same thing, and he also took from it measurements of the molecule. He and Crick relied on Photo 51 to work out their model of DNA's shape.

Her data reached Watson and Crick a second way, in a report for the Medical Research Council, a British government group that paid for research. Max Perutz passed it to them. Franklin's own paper came out in the same issue of *Nature*. She died in 1958, at 37. In 1962 the Nobel Prize went to Watson, Crick and Wilkins. Nobel rules do not allow a prize to go to a person who has died.

What Watson said. Watson later became the chancellor, one of the top leaders, of Cold Spring Harbor Laboratory in New York, the place where the Eugenics Record Office had run from 1910 to 1939. While he held that post, he made racist statements about the intelligence of people in Africa.

The *Sunday Times Magazine* of London quoted him in 2007. He said he was "inherently gloomy about the prospect of Africa" because "all our social policies are based on the fact that their intelligence is the same as ours." In the same sentence he went on: "where all the testing says not really."

The laboratory's trustees, the people who govern it, acted at once in 2007. They removed him from all his management jobs there and ended his post as chancellor.

In a PBS film shown on January 2, 2019, *American Masters: Decoding Watson*, he was asked whether his views had changed. He said, "No, not at all." On January 11, 2019, the laboratory's leaders called his views "unsubstantiated (not backed by proof) and reckless" and "reprehensible (deserving strong blame), unsupported by science." They took away his honorary titles.

How did geologists learn that the continents move?

Maps of the ocean floor recorded the evidence, and by about 1970 most geologists agreed that the continents move. The idea is called plate tectonics. The Earth's outer shell is broken into huge pieces, called plates, that slowly shift.

Marie Tharp and Bruce Heezen published a map of the floor of the Atlantic Ocean, the *Physiographic Diagram of the Atlantic Ocean*, in 1957. Harry Hess published a paper called "History of Ocean Basins" in 1962. In it he proposed seafloor spreading, the idea that new ocean floor forms along the undersea mountain ranges and spreads outward. Tharp and Heezen published a painted map of the ocean floor of the whole world in 1977.

Marie Tharp

Who: A geologist and mapmaker who mapped the ocean floor and found the rift valley along the Mid-Atlantic Ridge. Women were kept off the research ships, so she drew her maps on land.

When and where: Lamont Geological Observatory, Columbia University, from 1948.

Her work partner first called her discovery "girl talk."

Marie Tharp found a valley running down the middle of the Mid-Atlantic Ridge, a long chain of mountains under the Atlantic Ocean, in 1952 and 1953. It was a rift valley, a valley that forms where the Earth's crust, its rocky outer layer, is pulling apart. Her work partner, Bruce Heezen, at first dismissed her finding as "girl talk."

Drawn on land. Tharp began work at Columbia University's Lamont Geological Observatory in 1948. According to Smithsonian Magazine, Navy rules kept her off the research ships that Maurice Ewing and her other colleagues hired. She first sailed on one in 1968.

She made her maps from soundings, measurements of how deep the water is, taken on the ships. Tharp turned the depth numbers into drawings of the shape of the sea floor. The rift valley she found became part of the evidence that the sea floor spreads apart. Staff at the Library of Congress later named her one of the great mapmakers of the 20th century.

Measuring the air on Mauna Loa

Charles David Keeling of the Scripps Institution of Oceanography began measuring the carbon dioxide in the air on Mauna Loa, a volcano in Hawaii, in March 1958. Carbon dioxide is one of the gases in the air. It is a greenhouse gas, a gas that takes in heat and gives it off again. The amount in the air is rising mostly because people burn coal, oil and other fossil fuels for energy.

The Keeling Curve. His first reading was 313 parts per million, meaning 313 molecules of carbon dioxide in every million molecules of air. The average in the early part of that year was about 315. The measurements have run without a break since 1958, and the line they make on a graph is called the Keeling Curve. It rises in a saw-tooth pattern, with small ups and downs inside a steady climb.

The curve had passed 430 parts per million by 2026, more than a third above Keeling's first reading. Climate scientists use the Keeling Curve as the base measurement for their work.

What was the hiss in the Bell Labs antenna?

The hiss was the leftover glow of the Big Bang, the hot beginning of the universe. Arno Penzias and Robert Wilson picked it up in 1964 and 1965 at Bell Labs, the research laboratory of the telephone company AT&T, in Holmdel, New Jersey, with a large antenna shaped like a horn. An antenna is a device that picks up radio waves.

The hiss was made of microwaves, a kind of radio wave, and they could not get rid of it. They even cleared out pigeons that had nested in the antenna, along with their droppings.

Physicists at Princeton identified the hiss as the cosmic microwave background, the cooled light left over from the early universe. Penzias and Wilson won the Nobel Prize in 1978.

Darleane Hoffman

Who: A nuclear chemist who found plutonium in nature and whose group did chemistry on single atoms of the heaviest elements.

When and where: Los Alamos, New Mexico, and from 1984 Berkeley, California. She reported the plutonium find in 1971.

She found plutonium in rock from a California mine.

Darleane Hoffman showed in 1971 that plutonium-244, one form of plutonium, occurs in nature. Scientists had believed that all plutonium was made by people. Hoffman was a leader of a chemistry division at Los Alamos National Laboratory, and she found tiny traces of plutonium-244 in rock from the Mountain Pass mine in California.

One atom at a time. In 1984 she moved to Lawrence Berkeley National Laboratory and the University of California, Berkeley. Her group did chemistry on single atoms of the heaviest elements. The periodic table arranges the elements in patterns, and elements in the same column behave alike. Her group tested those patterns one atom at a time, to find where the heaviest elements stop following them. She died in 2025, at 98.

BY 1970: Scientists knew the shape of DNA, most geologists agreed that the continents move, and the Keeling Curve had been climbing on Mauna Loa for twelve years. The first Apollo crew had landed on the Moon in 1969.

What the Moon rocks and the space probes taught

Six Apollo crews landed on the Moon between 1969 and 1972. They brought back 382 kilograms, or 842 pounds, of Moon rock and soil, in 2,200 samples. Scientists at NASA's Johnson Space Center, part of the government's space agency, keep the samples and are still opening and studying them. Scientists used the rocks to work out the history of the Moon and the ages of surfaces across the solar system.

The Viking tests. Viking 1 and Viking 2 became the first spacecraft to land on Mars and work there, in 1976. Each carried tests for life in the Martian soil, and the results did not agree. One test, called Labeled Release, gave a result that looked like life. Another found no organic matter, the carbon-based material that living things are made of.

DISPUTED: Most scientists now think chemicals in the soil called perchlorates caused the Labeled Release result. Gilbert Levin, who led that test, argued to the end of his life that it had found life. Nobody has settled the question.

The Voyagers. The two Voyager probes were launched in 1977. Their pictures showed volcanoes on Io, a moon of Jupiter, and a cracked shell of ice on Europa, another moon of Jupiter. Other pictures showed the thick air around Titan, a moon of Saturn. The Voyager flights reached Uranus in 1986 and Neptune in 1989. Both probes are now in interstellar space, the space between the stars.

Cosmos. Carl Sagan's television series *Cosmos* began on PBS, the public television network, in the fall of 1980. In 2020 Charles Apple of *The Spokesman-Review*, a newspaper in Washington state, wrote that it was the

most-watched PBS series until Ken Burns's *The Civil War* in 1990. The news site Space Daily reported that about 500 million people around the world have seen it.

Linda Morabito

Who: A navigation engineer at NASA's Jet Propulsion Laboratory who found the first active volcano ever seen beyond Earth.

When and where: Jet Propulsion Laboratory, Pasadena, California, on March 9, 1979.

Movie: *The Farthest* (2017), a documentary about the Voyager missions, in which Morabito tells the story of her discovery.

A faint cloud at the edge of Io was an erupting volcano.

Linda Morabito found the first active volcano ever seen beyond Earth on March 9, 1979. It was on Io, a moon of Jupiter. An active volcano is one that is erupting, throwing out gas and melted rock.

Morabito was a navigation engineer for the Voyager 1 probe, a person who works out where a spacecraft is and where it is heading. She was processing a routine navigation picture of Io when she noticed a faint plume, a cloud shooting up from the surface, at the edge of the moon. It was 270 kilometers high, about 170 miles. The plume came from an erupting volcano. Morabito later wrote her own history of the discovery.

Richard Feynman

Who: A physicist who rebuilt the theory of light and matter, drew the diagrams that physicists still use, and explained hard ideas simply.

When and where: Los Alamos during World War II. Later Caltech, in Pasadena, California. He died in 1988.

Movie: *The Fantastic Mr Feynman* (2013), a BBC documentary with interviews with Feynman, his family and his colleagues.

He dropped a piece of rubber into ice water on television.

Richard Feynman sat on the commission, a group of experts, that looked into why the space shuttle Challenger broke apart in 1986. On February 11, 1986, at a hearing shown on television, he dropped a piece of O-ring rubber into ice water. An O-ring is a rubber ring that seals a joint. The cold rubber lost its spring. In about one minute he showed how launching in cold weather had caused the failure.

Challenger. The shuttle broke apart soon after launch, and all seven people aboard died. Hot gas had leaked through a joint in one of its booster rockets. The air at launch was 36 degrees Fahrenheit, 15 degrees colder than at any earlier shuttle launch. The seven who died were Francis R. Scobee, Michael J. Smith, Ellison S. Onizuka, Judith A. Resnik, Ronald E. McNair, Christa McAuliffe and Gregory B. Jarvis.

The physicist. Feynman shared the Nobel Prize in Physics in 1965 for his work on how light and electrically charged matter act on each other. Physicists everywhere still use the simple drawings he invented to show how particles act on each other, called Feynman diagrams. During World War II he worked at Los Alamos. Later he became known for explaining hard ideas plainly, in his lectures at Caltech and in his 1985 book *Surely You're Joking*. He died in 1988.

The national labs, a late Nobel and the genome

During the Cold War, the long rivalry between the United States and the Soviet Union after World War II, government money paid for big machines and big teams at the national laboratories. The main labs were Lawrence Berkeley, Lawrence Livermore, Argonne, Los Alamos and Oak Ridge. Scientists at these labs and in the Soviet Union raced each other to make new elements.

Inside the proton. Murray Gell-Mann proposed in 1964 that protons and neutrons, the particles in the center of every atom, are built from smaller particles, which he named quarks. For years nobody could find a quark.

A team from MIT and SLAC, a laboratory in California with a machine two miles long that speeds up electrons, found the first signs of quarks in 1968. They fired electrons at protons and found hard little pieces inside them. Gell-Mann won the Nobel Prize in 1969, and three leaders of the team shared one in 1990.

A late Nobel. Barbara McClintock, who had found genes that move in corn, won the Nobel Prize in Physiology or Medicine in October 1983. The prize came 39 years after she began the experiments, and she was the first woman to win that prize without sharing it.

The genome. Officials of the National Institutes of Health and the Department of Energy launched the Human Genome Project on October 1, 1990. A genome is the complete set of DNA in a living thing. The project's goal was to read all the DNA in a human genome, in order. Its planners gave it 15 years and about $3 billion.

Two laws to return the dead

Under two laws passed in 1989 and 1990, museum staff had to begin returning the Native American dead they held. Scientists had collected many of those remains, among them the skulls Army doctors sent to Washington from 1868 and the remains of about 1,000 people that Aleš Hrdlička's team dug out of burial grounds in Alaska in the 1930s. Native nations campaigned for both laws.

The National Museum of the American Indian Act became law on November 28, 1989. It requires Smithsonian staff to list the Native human remains the Smithsonian holds and to return them when asked.

The Native American Graves Protection and Repatriation Act became law on November 16, 1990. People call it NAGPRA, and repatriation means sending something back to where it came from. Under NAGPRA, staff at museums and government agencies that take federal money must list the Native American human remains and sacred objects they hold. They must return them to descendants and to Native nations.

The radiation experiments made public

The reporter Eileen Welsome made the plutonium injections public in November 1993. Her three-part series in the *Albuquerque Tribune*, called "The Plutonium Experiment," began on November 15, 1993, and it named the patients who had been injected. She won the Pulitzer Prize for National Reporting, a top award for American journalism, in 1994.

Energy Secretary Hazel O'Leary opened her department's records in 1993. President Bill Clinton set up the Advisory Committee on Human Radiation Experiments on January 15, 1994, to investigate.

What the committee found. The committee's members reported in 1995. They found "nearly 4,000 human radiation experiments sponsored by the federal government between 1944 and 1974." They also found "several hundred intentional releases of radiation into the environment for research purposes," meaning radiation let out into the air, land or water on purpose.

They found "little evidence of rules or practices of consent except in research with healthy subjects." In the 1940s and 1950s it was routine for doctors to use patients in research without the patients knowing. About the plutonium injections, the committee's members wrote: "In no case was there any expectation that these patient-subjects would benefit medically from the injections."

The Fernald School. Some of these experiments were done on children. The Fernald School was a state school in Massachusetts for children with disabilities, and researchers from MIT worked there with the school's senior staff. In 1946 they gave radioactive iron to 17 boys who lived at the school. From 1950 to 1953 they gave radioactive calcium to 57 boys, in a special breakfast.

The boys belonged to the school's "science club," whose members got extra milk and trips. One MIT researcher, Robert S. Harris, wrote that three boys who objected could be "induced to change their minds." A 1949 letter from the school's head to parents described a special diet and blood tests and did not mention radiation.

Doctors at Massachusetts General Hospital in Boston injected 11 patients with uranium from 1953 to 1957.

The prisons. Two doctors aimed X-rays at the testicles of 131 prisoners in Oregon and Washington from 1963 to 1973. The testicles are the organs that make sperm. The doctors wanted to learn how radiation affects the making of sperm, and the men could get no health benefit from it. Carl Heller did this to 67 men at the Oregon State Prison, and his former student C. Alvin Paulsen ran the study in Washington.

The men had to agree to a vasectomy at the end, an operation that cuts the tubes that carry sperm so a man can no longer father children. In Heller's study most men also had small pieces cut out of their testicles for testing five or more times, and were paid $25 each time.

Vanderbilt. Researchers at Vanderbilt University and the Tennessee state health department studied nutrition in pregnancy from 1946 to 1949. The women were poor patients at a clinic for pregnant women. Paul Hahn led the part of the study in which researchers gave about 819 of them a drink with a dose of radioactive iron at their second visit to the clinic.

Emma Craft and other women from the study sued Vanderbilt in federal court in 1994. They said the researchers called the drink a "cocktail" or a "vitamin drink" and never told them it was radioactive, and that no one gave them a chance to say no. The women had been pregnant when they drank it, and the children they had been carrying joined them in the lawsuit.

The women also said that Vanderbilt researchers ran a follow-up study in the 1960s and found "a disproportionately high incidence of cancer" in the children, meaning more cancer than expected. They said nobody told them they had been exposed or what the study found. Vanderbilt officials denied doing wrong. University officials settled the case in 1998 by paying the women $10 million.

The apology. At the White House on October 3, 1995, Clinton said that the United States "offers a sincere apology to those of our citizens who were subjected to these experiments, to their families, and to their communities."

O'Leary announced in November 1996 that the families of 12 people injected in the 1940s would get $4.8 million in all. Eleven had been injected with plutonium and one with uranium, and each family was to receive $400,000. Eleven of the 12 had died. One woman was still alive, living in upstate New York.

BY 2000

If you remember one thing: Between 1950 and 2000 Americans learned how DNA is built and that the continents move, and a reporter, the families and Native nations brought the hidden experiments and the collected dead into the open.

One number: Members of the federal committee counted nearly 4,000 radiation experiments on people paid for by the government from 1944 to 1974. That is about one new experiment every three days for thirty years.

Next: After 2000, scientists finished reading the human genome, and the Lacks family, Native nations and the Trinity downwinders won agreements, apologies and payments.

2000 to Today

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The biggest experiments after 2000 had thousands of scientists on them. About 3,000 people worked on ATLAS, one of the two experiments that found the Higgs boson in 2012. Scientists finished reading nearly all of the human genome in 2003 and read all of it in 2022, and they detected gravitational waves for the first time in 2015.

Some of the people who were used for science, and their families, have won apologies, rules and payments. The Lacks family reached an agreement in 2013 on how scientists may use the genome of Henrietta Lacks's cells. The head of the Smithsonian apologized in 2023 for how people collecting for it had taken human remains, many of them dug from graves and most taken without consent. People who lived downwind of the Trinity test became able to claim federal payments in July 2025.

The genome, finished twice

Scientists announced a working draft of the human genome in June 2000. The International Human Genome Sequencing Consortium, the teams of scientists doing the work, announced in April 2003 that the genome was essentially complete. They had read about 92 percent of it, with fewer than 400 gaps left. They timed the announcement to the 50th anniversary of the 1953 paper on the shape of DNA.

Another group of scientists, the Telomere-to-Telomere consortium, published the first complete human genome with no gaps on March 31, 2022.

The cost. The National Human Genome Research Institute, the government's genome research agency, has published what the work cost. Reading one person's genome to a high-quality draft cost about $14 million in 2006. By late 2015 it cost less than $1,500.

James Watson, who with Francis Crick had worked out the shape of DNA in 1953, died in November 2025, at 97.

The Higgs boson, and the collider that members of Congress cancelled

Scientists at CERN, the European physics laboratory in Geneva, Switzerland, announced on July 4, 2012, that they had found the Higgs boson. The Higgs boson is a particle tied to how other particles get their mass, the amount of matter in an object. Two experiments found it, called ATLAS and CMS. Both used the Large Hadron Collider, a machine that smashes tiny particles together at very high speed.

According to the US Department of Energy, about 1,700 American scientists worked on these experiments through two American labs, Fermilab and Brookhaven.

The American collider. American physicists had begun building a collider of their own, the Superconducting Super Collider, near Waxahachie, Texas. It was planned as a ring 54 miles around. Members of Congress cancelled it in October 1993, after $2 billion had been spent and more than 14 miles of tunnel had been dug. The Large Hadron Collider, the machine the Higgs experiments used, was built in Europe instead.

Monica Dunford

Who: An American physicist who worked on the ATLAS experiment at CERN. She was one of six physicists a film crew followed through the search for the Higgs boson.

When and where: CERN, in Geneva, Switzerland, up to the Higgs announcement on July 4, 2012.

Movie: *Particle Fever* (2013), a documentary directed by Mark Levinson and produced by the physicist David Kaplan. It follows six physicists, Dunford among them, from the start-up of the Large Hadron Collider to the announcement of the Higgs boson on July 4, 2012.

She was one of about 3,000 people on a single experiment.

Monica Dunford is an American experimental physicist, a scientist who runs experiments and measurements. At the time of the Higgs search she was a postdoc on the ATLAS experiment at CERN, doing research for a few years after her PhD before taking a permanent job. About 3,000 people worked on ATLAS.

First beam. Dunford tested electronics for the ATLAS detector and crawled under the floor to lay cable, as she later told the story. She was in the ATLAS control room on the day the collider first sent a beam of particles into the detector. "When beam arrived at the detector there were no lights," she said. Physicists hunched over their computers, looking for the beam's data, until someone found it and put it up on a monitor at the front of the room.

Jennifer Doudna

Who: A biochemist at the University of California, Berkeley, who with Emmanuelle Charpentier made CRISPR-Cas9 into a tool for cutting DNA at a chosen spot.

When and where: UC Berkeley. The key paper came out on June 28, 2012.

Movie: *Human Nature* (2019), a documentary about CRISPR, in which Doudna is one of the main people interviewed.

She and Charpentier made a tool that cuts DNA where scientists choose.

Jennifer Doudna and Emmanuelle Charpentier published a paper about CRISPR-Cas9 in the journal *Science* on June 28, 2012. CRISPR-Cas9 is a tool that scientists can program to cut DNA at a spot they choose. Cutting DNA at a chosen spot lets scientists change genes.

Doudna is a biochemist, a scientist who studies the chemistry inside living things, at the University of California, Berkeley. She and Charpentier won the Nobel Prize in Chemistry in 2020. It was the first science Nobel Prize shared by two women with nobody else.

Rainer Weiss

Who: An MIT physicist, who came to America as a child refugee from Nazi Germany, and who worked out the idea for the detectors that first picked up gravitational waves.

When and where: MIT, and the LIGO detectors at Hanford, Washington, and Livingston, Louisiana. The detection came on September 14, 2015.

His idea from 1972 caught the ripples from two black holes crashing.

The two LIGO detectors picked up gravitational waves for the first time on September 14, 2015. Gravitational waves are ripples in space and time. These came from two black holes that crashed into each other 1.3 billion light-years away. A black hole is a place in space where gravity is so strong that not even light can get out. Rainer Weiss had worked out the idea for the detectors in 1972.

A child refugee. Weiss was born in Berlin in 1932. His family fled the Nazis through Prague and reached New York in January 1939. He became a professor at MIT, the Massachusetts Institute of Technology.

His kind of detector is called a laser interferometer. It uses laser beams to measure very small changes in distance. The LIGO detectors were paid for by the National Science Foundation, the government agency that pays for science research. LIGO scientists announced the discovery on February 11, 2016, a century after Einstein predicted the waves. Weiss shared the Nobel Prize in Physics with Barry Barish and Kip Thorne in 2017.

The Webb telescope

The James Webb Space Telescope was launched on December 25, 2021. NASA staff released the telescope's first science images on July 11 and 12, 2022. Scientists using the telescope have found galaxies from the first few hundred million years of the universe, earlier than any galaxies found before.

Scientists using the telescope made the first clear detection, in 2022, of carbon dioxide in the air of an exoplanet, a planet that circles a star other than our Sun. The planet is called WASP-39b.

Computers that predict the shapes of proteins

Scientists at DeepMind, a British lab, showed in 2020 that their program AlphaFold2 could predict the shape of a protein well enough for practical use. A protein is a molecule that does work inside living cells, and its shape sets what it can do. AlphaFold2 is an example of artificial intelligence, or AI, computer software that learns patterns from huge amounts of data.

According to the European Molecular Biology Laboratory, free public databases held more than 200 million predicted protein shapes by 2026. Staff at the science journal *Nature* surveyed more than 1,600 researchers around the world in 2023. Reporting the results, Richard Van Noorden and Jeffrey Perkel wrote that AI tools were becoming more common in science. Many of the researchers expected AI tools to become a central part of their work soon.

The 2024 Nobel Prize in Chemistry was split in two. Half went to David Baker, an American at the University of Washington, for using computers to design new proteins. The other half went to Demis Hassabis and John Jumper of DeepMind in Britain.

Heavy elements, made by many countries

Scientists from several countries now work together to make the heaviest elements. Teams from Dubna in Russia and from Lawrence Livermore and Oak Ridge in the United States have worked on the same experiments. Scientists on joint panels of two international science groups, IUPAC and IUPAP, judge who made each new element.

The seventh row. IUPAC officials announced on December 30, 2015, that they recognized four new elements, numbers 113, 115, 117 and 118. With those four, the seventh row of the periodic table was full. A team from Dubna and Livermore got the credit for three of them. One of those, tennessine, was named for Tennessee, the home of the Oak Ridge lab, whose scientists shared the credit for it.

Scientists at Berkeley made livermorium, element 116, in 2024 by firing a beam of titanium at a target. With that result, the Berkeley team prepared to try for element 120. The physicists are looking for what they call the "island of stability," a group of superheavy elements that they predict would last longer before breaking apart.

The Lacks family and the HeLa genome

European researchers published the genome of the HeLa cells in March 2013 without the consent of Henrietta Lacks's family. Francis Collins, the director of the National Institutes of Health, met with members of the Lacks family three times between April and July 2013.

The agreement. NIH officials announced the NIH-Lacks Family Agreement on August 7, 2013. Under it, scientists must apply to NIH to use the full genome data of HeLa cells. A working group that includes members of the Lacks family reviews the requests.

The lawsuit. The Lacks estate, the property Henrietta Lacks left behind, managed for her heirs, sued Thermo Fisher Scientific in 2021. The company sold about a dozen products made with HeLa cells. The lawyers Ben Crump and Chris Seeger represented the estate. The lawsuit stated that the company "made staggering profits by using the HeLa cell line." It added that this happened "all while Ms. Lacks' Estate and family haven't seen a dime."

According to news reports in STAT and Smithsonian Magazine, the two sides announced a settlement on August 1, 2023. That day would have been Henrietta Lacks's 103rd birthday. They kept the terms of the settlement secret.

Apologies, and the dead still held

Lonnie G. Bunch III, the Secretary, or head, of the Smithsonian, apologized in August 2023 for how people collecting for the Smithsonian had taken many of its human remains. He spoke after the Washington Post published its investigation of Aleš Hrdlička's collecting. The Penn Museum's director had apologized for the Morton skulls two years earlier.

Still held. Reporters at ProPublica, a news organization, counted the Native American remains still held, using federal NAGPRA reports. As of January 6, 2025, staff at about 650 museums, universities and agencies had reported holding more than 210,000 Native American remains. They had not yet made 90,831 of them, or 42 percent, available for return. That is more than two of every five.

The Ohio History Connection had the most remains not yet made available for return, about 8,000. The Illinois State Museum had about 5,800 and Harvard University about 5,400. The Smithsonian is not on that list, because it reports under the 1989 law instead of NAGPRA.

A second count, made in September 2024 and reported by Retrospect Journal, agrees closely. It found that 126,299 of 216,804 remains had been returned, which leaves about 90,500 not returned.

The 2024 rule. A revised NAGPRA rule took effect on January 12, 2024, after Interior Department officials announced it on December 6, 2023. Under the rule, staff at museums and federal agencies must ask direct descendants, Native nations or Native Hawaiian organizations for their free, prior and informed consent. That means a yes given freely and beforehand, by people who have been told everything about the request. Staff need it before they display human remains or sacred and cultural objects, let others see them, or do research on them.

To follow the new rule, staff at the Field Museum in Chicago covered display cases in two of its halls in January 2024. A statement from the museum said that the cases held cultural items that "could be subject to" the new rules,

and that they would stay covered while staff reviewed what was known about the items.

Payment for the Trinity downwinders, 2025

Downwinders are people who lived where the wind carried fallout from nuclear bomb tests. Members of Congress passed the Radiation Exposure Compensation Act, a law to pay people for harm, in 1990. The law included people who lived downwind of the Nevada Test Site, where nuclear bombs were tested from 1951 to 1957. It did not include the Trinity downwinders in New Mexico.

The campaign. Tina Cordova of the Tularosa Basin Downwinders Consortium led the campaign to add them. In July 2025, 80 years after the Trinity test, President Donald Trump signed a budget bill that added New Mexico's downwinders to the law. The bill also added uranium workers who entered the industry after 1971.

Each person who qualifies can receive $100,000 in one payment. According to KUNM, a New Mexico public radio station, New Mexicans can apply until December 31, 2027.

THE THREE QUESTIONS, ANSWERED

1. **How did American scientists learn how far away the stars are?** Henrietta Leavitt showed in 1912 that the blink of a Cepheid star tells how bright it truly is, and so how far away it is. Edwin Hubble used her rule in 1923 to place Andromeda far outside our own galaxy.
2. **Why did scientists collect human skulls and brains, and where are those remains now?** Samuel Morton, the Army's museum scientists and Aleš Hrdlička measured them to rank the races, an idea that is false. Morton's skulls are at the Penn Museum, whose staff buried 19 of them in 2024, and the Smithsonian still held at least 255 of the brains in 2023. Many Native American remains

have been returned since the laws of 1989 and 1990, but in January 2025 museums still had 90,831 not yet made available for return.

3. **Who did the work but was left out of the credit?** Lise Meitner explained fission, and Rosalind Franklin's X-ray pictures showed the shape of DNA. The Nobel Prizes went to Otto Hahn alone in 1944 and to Watson, Crick and Wilkins in 1962.

What changed: American science grew from a few colonists watching comets to experiments with thousands of workers, and the people scientists had used without asking, and their families, began to win apologies and payments.

FIVE DATES TO KEEP

June 3, 1769: four men at Norriton time Venus crossing the sun.

1839: Samuel Morton's *Crania Americana* ranks the races by skull size.

1912: Henrietta Leavitt publishes her rule for measuring space.

April 25, 1953: Watson and Crick publish the shape of DNA.

November 16, 1990: NAGPRA becomes law.

> **NOT KNOWN:** how many people the fallout from the 1945 Trinity test made sick or killed, because no government official ever counted them.

Where to go: The International African American Museum in Charleston, South Carolina, where the 1850 photographs of Renty, Delia and five others went on public view on March 11, 2026.

Next: Chapter 10, Energy. It asks what powered American life, from muscle and wood to the atom and the sun.

Chapter 10: Energy

CHAPTER 10 OF 37 · ENERGY · 10 ERAS · ABOUT 75 MINUTES

Americans moved from burning wood to burning coal, oil and natural gas, and getting that energy cost many people their freedom, their land or their lives.

A tavern keeper named Jesse Fell burned hard coal on an iron grate in his fireplace in Wilkes-Barre, Pennsylvania, on February 11, 1808. Most people thought that coal was too difficult to burn at home, but Fell showed that it could warm a house. From the 1820s to the 1840s, people put stoves and furnaces made for that coal into their homes. Miners dug that coal in northeastern Pennsylvania, and on September 6, 1869, a fire in one of their mines, the Avondale mine near Wilkes-Barre, killed 110 people.

This chapter describes the fuels in order, from the fires Native people set on purpose before 1500 to the wind turbines and solar panels of 2025. Watch for two things. One is the main fuel. Wood came first, then coal, which passed wood around 1885, and then oil and natural gas. In 2016 natural gas passed coal as the fuel that made the most American electricity. The other is who did the work and who paid for it: the enslaved people who made charcoal for the iron furnaces, the coal miners killed underground, and the Native nations whose rivers were dammed and whose land held oil and uranium.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Did kerosene really save the whales?
2. How did electricity reach almost every American home?

3. Who paid for America's energy with their freedom, their land or their lives?

Five minutes? Read "The current war" in 1850 to 1900.

Can skip: "A stove to save wood", "Oil for a world war", "Panels on the White House roof", "New and restarted reactors". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

People here did their work with the strength of their own bodies and with fire, and wood was the fuel they burned. They burned it to heat their homes, to cook and to see after dark.

North of Mexico there were no horses, oxen or mules. People carried and dragged loads with their own muscle, and they moved heavy goods by canoe. Dogs were the only animals that hauled loads. On the Plains, dogs dragged a travois, two long poles tied together at one end to form a narrow V. The load rode on the poles, and their far ends dragged along the ground.

Why did Native people set fires on purpose?

Native people lit small fires and kept them under control to help the plants and animals they used for food, clothing and ceremonies. They also burned land to clear space for crops and travel, and they used fire to hunt game. They burned on regular cycles, and the practice is now called cultural burning.

In the eastern part of the continent, oak and chestnut trees fill many forests. According to the National Park Service, those forests grew that way because people burned them on purpose to get certain nut crops. In what is now northern California, Karuk and Yurok people burned their hazel shrubs so that the shrubs would grow new stems for weaving baskets.

After 1500. Native people went on burning land long after 1500. For most of the 1900s, though, federal officials in charge of forests tried to put out wildfires, to protect water supplies, towns and timber for sale. Under the rules those officials made, Native people could do far fewer burns.

Yurok people worked with California's state fire department on a small burn on Yurok land in 2013. In May 2014, organizers with the Cultural Fire Management Council held the first Yurok training event for this kind of burning. Writers for The Nature Conservancy reported in 2020 that Native people who burn in the old way still face punishment. Their report does not say who punishes them.

Basket stems today. Karuk and Yurok people still burn their hazel shrubs. Four researchers studied those burns and published their findings in 2021, and the study appears among the US Forest Service's research papers. Burning every three to five years gave the best stems. In the first season after a fire, the shrubs grew 13 times as many basket stems.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

Spanish explorers and soldiers brought horses here in the 1500s, and at the very end of the century Spanish colonists built water mills. A horse could drag a travois farther and faster than a dog could, and it could carry more weight. Still, most people living here did their work with muscle, fire and wood for the whole century.

The mills stood in New Mexico. A history of the state's mills, printed in a Santa Fe newspaper, calls them the first mills powered by water in what is now the United States.

Horses come ashore

Hernando de Soto landed near what is now Tampa Bay, Florida, in May 1539, with 11 ships. He brought about 250 horses, 1,000 soldiers on foot, many dogs and a herd of pigs.

Hymahi. De Soto's army reached Hymahi, a small Native village where two rivers meet, in April 1540. Writers for the National Park Service say the rivers must have been the Congaree and the Wateree, in what is now South Carolina. De Soto took all of the village's food. He burned at least one of the people who lived there to death while questioning that person about a town called Cofitachequi.

Francisco Vázquez de Coronado led a large party north from Mexico into the Southwest in 1540, and his men brought horses too. He had more than 300 armed Europeans and at least 1,200 Native people who marched on the Spanish side. Many dozens of servants and enslaved people went with them. Coronado's party went as far as what is now Kansas and returned in 1542.

The first water mills, 1598

Juan de Oñate led Spanish colonists north into what is now New Mexico in 1598. That year they founded a town called San Juan de los Caballeros on the Rio Grande, on land that belonged to the Tewa, one of the Pueblo nations. Pueblo is the Spanish word for town or village. The Spanish used it for the Native peoples of the Southwest who lived in towns built of stone and of adobe, brick made of mud and dried in the sun.

Early in 1599 the colonists moved to a new capital, San Gabriel, which they built at a pueblo town near Ohkay Owingeh, a Tewa town. Writers for the National Park Service call that site abandoned. They do not say whether Tewa people had been made to leave it.

Tub mills. Oñate's colonists built their first water mills in 1598. These were small tub mills, with the water wheel lying flat in the stream. The moving water spins the wheel, and the wheel turns the millstone directly, so the stone grinds grain into flour. The colonists hauled their millstones north on the Camino Real, the Royal Road, which ran about 1,600 miles north from Mexico City.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

Wind and falling water began turning mills in the English colonies in the 1600s. The mills ground grain and sawed logs. All other work still took muscle, and colonists burned wood in open fireplaces to heat their homes and cook.

Two other kinds of energy came into use in these years. Native people rode horses and fed them corn, and along the coast colonists began to hunt whales for oil to burn in their lamps.

The first wind and water mills

George Yeardley, the governor of the Virginia colony, had a windmill built at Flowerdew Hundred, on the James River, in 1621. It is the first windmill recorded in English America. Members of the colony's council, its group of leaders, wrote to the company in London that was in charge of the colony. Their letter records Yeardley's "good example" of building a windmill.

What the diggers found. Archaeologists, scientists who dig up the past, later dug at the spot. They found a round pit five feet across with iron hoops and brackets still in place, and they took it to be the remains of Yeardley's windmill.

Whose land, and whose work. Flowerdew Hundred was land of the Weyanock, a Native nation whose language belonged to the Algonquian family of languages. Yeardley brought 15 men from England in 1619 to grow tobacco there.

That same year, the crews of two armed ships, the *White Lion* and the *Treasurer*, attacked a Portuguese ship carrying 350 enslaved Africans and seized 50 to 60 of the captives. The *White Lion* brought its share to Point Comfort, Virginia, in late August 1619. Yeardley and a merchant named Abraham Peirsey bought them in exchange for food.

At least 11 of these Africans were taken to Flowerdew Hundred, Yeardley's land. The archaeologists' account of Flowerdew does not say who moved them there. A count of the colony's people taken in 1624 and 1625 records seven Africans at Flowerdew as laborers. It gives no names for them. The records do not say what happened to the other four.

Records from 1621 also list a gristmill, a mill that grinds grain, near Jamestown. It is the first gristmill recorded in the English colonies. Farther north, three English carpenters built a sawmill and a gristmill run by water at Newichawannock, in what is now South Berwick, Maine, in 1634. They were among the first water mills in New England.

William Chadbourne

Who: An English carpenter who helped build one of the first sawmills run by water in New England.
When and where: What is now South Berwick, Maine, from July 1634.

The mill's model arrived three years before the men who built it.

William Chadbourne, James Wall and John Goddard arrived from England in July 1634 on a ship called the *Pied Cow*. They worked under a contract with the Laconia Company of Captain John Mason. Under it, the three carpenters were to build a sawmill, to cut logs into boards, and a gristmill on the Little Newichawannock River.

The model. Thomas Eyre wrote in a letter of May 1631 that a model of a sawmill had been sent to Newichawannock on the *Pied Cow*, the same ship the carpenters took three years later. Chadbourne and his partners built the sawmill in a rocky gorge. The spot is now below the Great Works bridge on Brattle Street in South Berwick. In 1652, in a lawsuit, James Wall made a sworn statement, a promise under the law to tell the truth, about where the 1634 mills had stood.

The wheel. Writers for the Old Berwick Historical Society say the sawmill is "thought to be the first" mill in America with an overshot wheel. On an overshot wheel, water pours onto the top of the wheel, and the weight of the water turns it. People in other towns of the region say that their town had the first sawmill.

Whose river. Newichawannock is an Abenaki word, and Abenaki people lived there. Between 1616 and 1618, before the carpenters came, diseases brought by Europeans, such as measles or smallpox, killed seven out of every ten of them. The society's histories of the mills do not say that Mason's men asked the Abenaki for permission to build. A leader named Rowls survived the diseases. In 1650 he sold his village land, Quamphegan, to a man named Thomas Spencer for five pounds.

Mason died in 1638, and after that the mills fell apart. People stole their materials, and the society's timeline of the mills does not name the thieves. Later one of the mills burned.

NOT KNOWN: what became of William Chadbourne after he built the mills. The society's histories of the mills do not say.

How much wood did a family burn?

Wood was the main fuel in colonial homes, and one household burned about 15 to 40 cords of it a year. A cord is a stack of wood 4 feet high, 4 feet wide and 8 feet long. Forty cords stacked in one row 4 feet high would run 320 feet, longer than a football field from goal line to goal line.

The low figure, 15 cords, comes from one New England estimate of the 1600s. The high figures, 30 to 40 cords, are for a home that heated and cooked with open fireplaces. An open fireplace sent most of its heat up the chimney, so a family had to burn a great deal of wood to stay warm.

Horses in Native hands

Native people were riding horses, putting bridles on them and feeding them corn by the early 1600s. A team of scientists showed this in a study published in the journal *Science* in 2023. The team, which included Lakota, Comanche and Pawnee scholars, studied the bones of horses found in what are now Wyoming, Kansas and New Mexico.

The old account. For a long time, the usual account said that Native people got their horses only after the Pueblo Revolt, decades later. In August 1680, Pueblo people led by Po'pay, a Tewa religious leader, rose against the Spanish colonists in New Mexico and drove them out. Perhaps 1,500 or more horses came into Native hands at one time.

Oil from whales

English colonists on Long Island and in New England began killing whales close to shore for their oil in the 1600s. From about 1690, whale oil was one of the main fuels that Americans burned in their lamps, and it stayed one for about 150 years. By the 1840s it cost too much for most families, and they burned cheaper fuels instead.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 3 MINUTES

Iron furnaces in the colonies, tall ovens where fire melts iron out of rock, ran on charcoal, a fuel made from wood. Many of the workers who fed the fire at the Baltimore Iron Works, in Maryland, were enslaved people and convicts shipped from Britain. They hauled charcoal to the furnace around the clock.

Lamps burned whale oil, and families who could pay the most bought candles made from a waxy oil in the head of the sperm whale. Benjamin Franklin designed an iron stove in 1741 to heat a room with less wood.

Charcoal for iron

Iron furnaces burned charcoal, wood that has been heated with too little air for it to burn away. The heat drives off the water and gases in the wood. What is left is nearly pure carbon, and it burns hotter than wood. The men who made charcoal were called colliers.

The owners. Five planters started the Baltimore Iron Works on the Patapsco River in Maryland in 1731. Dr. Charles Carroll, a surgeon, gave 2,200 acres of land and ran the works at first. Two of his cousins, along with Daniel Dulany and Benjamin Tasker, put in money and enslaved workers. In the 1600s, the land around the Patapsco lay within the hunting grounds of the Susquehannock nation, whose villages stood farther north, along the Susquehanna River.

The workers. The Baltimore Iron Works opened with 89 workers, and 42 of them were enslaved. Many of its workers were enslaved people or transported convicts, people found guilty of crimes in Britain and sent to the colonies to be sold into 7 to 14 years of forced work. Judges in English courts gave that sentence. Merchants shipped the convicts across the ocean, and merchants and ship captains sold them in the colonies, some of them to iron makers.

The workers hauled cartloads of charcoal, oyster shells and iron ore, the rock that iron comes from, to the top of the furnace in 12-hour shifts. The work went on around the clock, six or seven days a week. The owners did not give the workers enough food. Some workers ran away, and others worked slowly on purpose.

The profits. In a good year the partners made back more than half of what they had put in, a profit of 55 percent. By 1785 an advertisement for the Baltimore Iron Works listed "more than 200 Negroes," the word then used for Black people. The people it listed were enslaved.

The surviving records of these early furnaces list the owners. They do not list the men who made the charcoal.

What gave the brightest light?

The brightest light in the colonies came from candles of spermaceti, a waxy oil found in the head of the sperm whale. Candle makers turned it into candles that burned brighter and cleaner than any others, and they cost the most. Writers at the New Bedford Whaling Museum call spermaceti "the most valuable product of the Yankee whaling industry," Yankee meaning someone from New England.

A stove to save wood

Benjamin Franklin designed what he called the Pennsylvania fireplace in 1741. It was a stove of cast iron, iron poured into a mold while it is melted. Franklin meant it to give more heat and less smoke from less wood than an open fireplace. Few people bought it at first. David Rittenhouse later improved the design, and after that it sold better.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 7 MINUTES

The first steam engine in America pumped water out of a copper mine in New Jersey from 1755 on. A fire under a boiler, a closed tank of water, made the steam that ran it.

The iron furnaces still burned charcoal, and they burned it by the acre. Hopewell, in Pennsylvania, used the charcoal from about an acre of forest every day it ran. At Catoctin Furnace in Maryland, enslaved Black workers and some free Black workers cut the wood, made the charcoal and hauled it to the furnace. After dark, spermaceti candles still gave the brightest light.

The first steam engine in America

Water flooded John Schuyler's copper mine at Second River, now North Arlington, New Jersey, in 1748, and the mine had to shut down. Schuyler paid an English engine maker, Jonathan Hornblower, 1,000 pounds to send him a "fire engine" and mechanics to set it up. Fire engine was the name people then used for this kind of steam engine.

The crossing. The parts arrived in New York in September 1753 on a ship called the *Irene*, after twelve weeks at sea, and Jonathan's son Josiah came with them. The parts were taken to Newark, New Jersey, and put together there. By early 1755 the engine was pumping water out of the mine.

How it worked. The engine was a kind invented in England by Thomas Newcomen. The fire under the boiler turned water into steam, and the steam flowed into a cylinder, a tall round tube. Inside the cylinder, the steam filled the space under a piston, a heavy disk that could slide up and down. Then a jet of cold water sprayed into the cylinder. The cold turned the steam back into water, which left almost nothing under the piston, so the weight of the air above pushed the piston down.

The piston was joined to one end of a beam that rocked up and down like a seesaw. The other end of the beam lifted long pump rods that hung down the mine shaft, and the pumps raised the water. Then the weight of the pump rods pulled the piston back up, and the engine started again. Engines of this kind stood about two stories high. Their power was about ten horsepower, a measure based on the work one horse can do.

Josiah Hornblower

Who: The young English engine builder who came over with the first steam engine in America and later ran it.

When and where: The Schuyler copper mine at Second River, now North Arlington, New Jersey, 1753 to 1773.

The engine house burned the same night he shut the mine.

For twenty years after it started, the engine worked only part of the time. Josiah Hornblower, who had been about 24 when he came over with it, later ran it himself. In 1761 he and a partner, John Stearndall, leased the mine from the Schuyler family. They paid one seventh of the ore they dug as rent.

The first fire. The building around the engine caught fire in March 1762 and burned to the ground. A report in the *New York Mercury* of March 22 states that the engine works had cost nearly ten thousand pounds. People at the time guessed that a careless workman had left a candle burning. The engine was working again by 1767. Richard McCormick, the historian who wrote the engine's history, does not say who repaired it or when.

Out of fuel. Hornblower told what happened next in a letter of July 1768, in which he called the engine "the Fire engine." Late in 1767, he had to stop it because he had run out of fuel. He got more, put the old pumps back in, made the repairs and started it again in April 1768.

Soon the vein, the band of ore running through the rock, dipped deep. He tried to follow it with hand pumps, but they could not keep the water out, and his partners had not sent money they owed him.

The second fire. On a Monday in July 1768, he stopped the mine and let all the workers go. At about eight o'clock that evening, the engine house caught fire. It burned to ruins in a few minutes while a crowd of

people tried to save it. Hornblower wrote that a man had gone into the engine house two hours before to light his pipe, and the man said he could not find any fire there to light it with.

A report in the *Pennsylvania Chronicle* states that the fire was "supposed to be by Design," because no one knew of any fire being used there lately. People thought someone had set it on purpose. Mining at Schuyler's stopped in 1773, and Hornblower stayed in New Jersey for the rest of his life.

KNOWN:	Half of the engine's cylinder is now in the Smithsonian Institution.
NOT KNOWN:	what became of the rest. One account says the engine was broken up around 1800, and another says it was repaired in 1793 and pumped water into the 1800s.

Charcoal by the acre

Hopewell Furnace burned the charcoal from about an acre of forest every day it ran. Mark Bird built it on French Creek, in Berks County, Pennsylvania, in 1771. When the furnace was in blast, lit and making iron, it burned up to 800 bushels of charcoal a day, a bushel being a measure for dry goods such as grain.

In a year the furnace used 5,000 to 7,000 cords of wood, from more than 200 acres of woodland. A family heating its home with open fireplaces burned 30 to 40 cords. The owners held between 4,000 and 8,000 acres of forest to keep the furnace supplied. Woodcutters, using only axes, cut an average of two cords a day each.

Whose land. The land around Hopewell had belonged to the Lenni Lenape. After the colony's founder, William Penn, left Pennsylvania in 1701, his agent James Logan handled Penn's land deals. Under Logan, the colony's land deals with the Lenni Lenape were often dishonest, and some were frauds, deals built on lies.

By about 1750, nearly all Native people had been driven out of the lower Delaware and Susquehanna river valleys, and most moved west to the Allegheny, Monongahela and Ohio river valleys. The history of Hopewell written for the National Park Service does not name the people who drove them out.

How a collier made charcoal

A collier made charcoal in a pile called a pit, built on flat, cleared ground. He drove a green pole, freshly cut and 18 feet long, into the middle of that ground and built a chimney with three corners around it. Then he stacked about 30 cords of wood around the chimney in three layers and covered the whole pile with leaves and dust.

To light the pit, he shoveled red-hot coals from his cooking fire down the chimney. The wood inside smoldered, burning slowly with very little air. The collier watched it day and night, walking on top of the pit to find soft spots where air was getting in and filling them. After 10 days to two weeks, the wood had turned into charcoal. One pit made about 1,100 bushels, enough to run the furnace for a day and a half.

Who made the charcoal?

At Catoctin Furnace, in Maryland, enslaved Black workers and free Black workers made it. Four brothers, James, Baker, Roger and Thomas Johnson, built the furnace, and it was making iron by 1776. The Johnson family enslaved more than 80 people there. One of them was a skilled man recorded only as Collier Sam, and the Catoctin Furnace Historical Society's history gives only his name and his trade.

Over the life of the furnace, at least 271 enslaved African Americans worked at Catoctin, along with an unknown number of free African Americans. They mined ore, cut wood, made charcoal, drove wagons and worked the forge, the shop where iron was heated and hammered into shape. Woodcutters cut, hauled and stacked the wood for the colliers, and other workers carried the finished charcoal to the furnace.

At Hopewell. Mark Bird enslaved 18 people in 1780: 10 men, 4 women, 3 boys and 1 girl. No one else in Berks County enslaved as many people that year. That same year Pennsylvania lawmakers passed a law to end slavery in stages, but under it mainly children born after the law passed could become free. The last record of slavery at Hopewell is from 1821, when Matthew Brooke freed one person he enslaved.

Hopewell's owners hired free Black workers only for jobs that took little training, as woodcutters, laborers and teamsters, who drove wagons. According to the National Park Service, no record from Hopewell's 112 years shows a Black worker in a skilled job or as a manager.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 5 MINUTES

Hard coal from northeastern Pennsylvania went into American stoves and furnaces in these years, and gas made from coal lit city streets. A tavern keeper showed in 1808 that the coal could heat a home, and boats, first on the rivers and then on canals, carried it to the cities. The country's first public gas street lamp was lit in Baltimore in 1817, with gas made from coal and sent through pipes.

The American whaling fleet reached its largest size in 1846, with 735 ships. Whale oil cost too much for most families, and most homes burned cheaper fuels.

Stone coal

Northeastern Pennsylvania has a hard kind of coal called anthracite, which people then called stone coal. Two brothers named Gore, who were blacksmiths, had burned it in their forges, the fires where blacksmiths heat metal, since 1769. For use at home, though, most people thought anthracite was too difficult to burn.

Fell's grate. Jesse Fell ran a tavern in Wilkes-Barre, Pennsylvania. He had believed for some time that a big enough pile of anthracite would keep burning once it was lit. He built a grate shaped like the letter L out of "Small Iron rods," to hold the coal up off the floor of the fireplace so air could flow in underneath.

Fell set his grate in the fireplace of the tavern's main room on February 11, 1808, and lit the coal. He later wrote that "on first lighting" he "found it to burn excellently well."

After Fell showed that anthracite could heat a home, the anthracite business grew. A company called Lehigh Navigation was formed in 1818, and anthracite floated down the Lehigh River to Philadelphia in boats called arks. Later,

boats on canals carried the coal into the cities. From the 1820s to the 1840s, people put stoves and furnaces made to burn anthracite into homes and other buildings.

Gaslight

Heating coal makes a gas that burns, a mix of gases that includes hydrogen and a gas called carbon monoxide. Rembrandt Peale, who ran a museum on Holliday Street in Baltimore, lit a room of his museum with gas made from coal in 1816.

The first gas company. The Gas Light Company of Baltimore received its charter, a legal paper that allowed the company to form, on June 17, 1816. It was the first gas company in the United States. Gas makers stored the gas in large tanks called gasometers and sent it through pipes. The company's first public gas street lamp, the first in the country, was lit at Market and Lemon Streets in Baltimore on February 7, 1817.

About 50 American cities and towns had gas works, plants that made gas, by 1850. Gaslight was expensive, and most Americans could not pay for it. It lit streets, shops and the homes of richer families in the larger cities.

Whale oil, and who could pay for it

The American whaling fleet counted 735 ships in 1846, the most in any year, out of about 900 whaling ships in the whole world. Other counts put the peak of the whaling business as late as 1852. New Bedford, Massachusetts, was the leading whaling port, and at its peak more than 400 whaling ships sailed from there.

Who could pay. Whale oil cost $1.30 to $2.50 a gallon, so mainly richer families burned it. Oil from sperm whales lit their homes, along with lighthouses and public halls. Most families burned cheaper fuels: tallow candles made of animal fat, lard oil made from pig fat at about 90 cents a gallon, and camphene at about 50 cents.

Camphene was one part turpentine, a liquid made from pine trees, mixed with four parts alcohol and a little camphor, added for its smell. Its fumes caught fire easily, and camphene lamps could start sudden fires.

Nelson Cole Haley

Who: A New Bedford harpooner whose book tells how a whaling crew turned a whale into lamp oil.

When and where: At sea on the whaling ship *Charles W. Morgan*, 1849 to 1853.

The crew boiled the whale's fat over a fire of whale scraps.

Nelson Cole Haley hunted whales for the oil that Americans burned in their lamps. He grew up in New Bedford, Massachusetts, and his father died when he was a boy. In 1844, when he was twelve, he ran off to sea to become a whaleman, even though his mother did not want him to go.

The chase. Haley became a harpooner on the *Charles W. Morgan* in 1849, when he was seventeen. A harpooner threw the harpoon, a heavy iron spear, into the whale from a boat that the crew rowed out from the ship. Haley called the harpoon an iron.

He later wrote about driving it into the whale's blubber, the thick layer of fat under its skin. "I raised the iron, and with all the force I was capable of, I sent it to the socket into the vast mass of blubber and meat that was now only a foot or two clear of the boat."

He drove a second harpoon in beside the first. Then the second mate, one of the ship's officers, killed the whale with a lance, a long spear with a sharp blade, and the crew rowed the dead whale back to the ship.

Whale into oil. The next morning, men stood on the dead whale and peeled off strips of blubber. The crew hoisted the strips on board and cut them into chunks they called bibles, then boiled the chunks in two big metal pots on the try-works, a brick furnace built on the deck.

The heat melted the fat into oil, which the crew poured into barrels below deck. For fuel under the pots, they burned the crisp bits of blubber left over from the boiling.

> Haley's voyage lasted four years, from 1849 to 1853. Whaling ships of that time stayed at sea three to five years on a voyage. Haley wrote a book about it, *Whale Hunt*, which the Mystic Seaport Museum first published in 1948, almost fifty years after he died.
>
> The *Charles W. Morgan* was built in New Bedford in 1841 and made 37 voyages between 1841 and 1921, with sailors from fifty countries. It is the last wooden whaling ship left in the world, and it is kept at the Mystic Seaport Museum, in Mystic, Connecticut.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 13 MINUTES

Americans began to buy two new kinds of energy in these years: kerosene for their lamps, made from oil, and electricity sent through wires. Edwin Drake's crew drilled a well in Pennsylvania that reached oil in 1859, and refiners made the oil into kerosene. Thomas Edison's workers began sending electric current to paying customers from a station in New York City in 1882.

Coal became the country's main fuel in these years. By the count of the US Energy Information Administration, coal supplied 9 percent of the energy Americans used in 1850 and 27 percent by 1870. Around 1885, Americans began to burn more energy from coal than from wood for the first time. Miners died digging it. A fire at the Avondale coal mine in Pennsylvania killed 110 people in 1869.

Drake's well, 1859

Edwin Drake's crew struck oil 69.5 feet down near Titusville, Pennsylvania, on August 27, 1859. It was the first well in the United States drilled on purpose to find oil and sell it. Writers at the American Oil & Gas Historical Society date the start of the American oil business to that day.

Drake worked for the Seneca Oil Company of New Haven, Connecticut, whose owners wanted oil to make kerosene, a cheap fuel for lamps. The driller, William Smith, known as Uncle Billy, cut through rock and shale, a soft rock made of hardened mud, at about three feet a day. A steam engine of six horsepower drove the drill.

Edwin Drake

Who: The man whose 1859 well started the American oil business, though he lost his money afterward.

When and where: Titusville, Pennsylvania, 1859.

Movie: *Born in Freedom: The Story of Colonel Drake* (1954). A 30-minute drama paid for by the oil industry's American Petroleum Institute, with Vincent Price as Drake. It first played in Titusville on March 29, 1954.

The title "Colonel" was made up to impress the town.

People called him Colonel Drake, but he never held that rank. An investor named James Townsend made up the title and addressed letters to "Colonel E. L. Drake" to impress the people of Titusville.

The drive pipe. Water from Oil Creek kept seeping into the hole, and the sides kept caving in. Drake had a pipe driven down to the rock, and then the drill worked inside the pipe, which kept the water and loose ground out.

After the well. Drake never patented the drive pipe, which means he never got a government paper giving him the only right to make and sell it. The Seneca Oil Company fired him, and he lost his money on Wall

Street, a street in New York City where people buy and sell shares in companies. When he came back to Titusville years later, he had little money, and his health was failing.

His friends raised about $4,000 for him in the late 1860s. In 1873 people of Titusville won him a pension from the state of Pennsylvania, $1,500 paid to him every year. Drake died in November 1880 in Bethlehem, Pennsylvania.

Did kerosene save the whales?

A common saying holds that kerosene saved the whales, but writers at the University of Houston's engineering college and at the website environmentalhistory.org call that claim a myth. Whaling was already shrinking before kerosene came, and in most homes kerosene replaced camphene, not whale oil.

What families burned. American families switched to kerosene for their lamps in the 1860s. Refiners made it by heating oil until parts of it boiled off as vapor, then cooling the vapor back into separate liquids. Kerosene sold for about 50 cents a gallon.

The whaling fleet had been largest in 1846, and whalers caught fewer whales through the 1850s. Whale oil cost $1.30 to $2.50 a gallon, so mainly richer families burned it. Most families burned camphene and lard oil.

The tax. During the Civil War, members of Congress passed the Internal Revenue Act to pay for the Union Army, the army of the Northern states. Under that law, alcohol was taxed at up to $2.08 a gallon from 1862 to 1864. The lawmakers meant to tax alcohol that people drank, but they did not leave camphene out of the law, and camphene was about three quarters alcohol.

Before the tax, camphene and kerosene each sold for about 50 cents a gallon. At the top rate, the tax on the alcohol in one gallon of camphene came to more than a dollar, while kerosene was taxed at 10 cents a gallon. People stopped buying camphene, and families switched to kerosene.

Pearl Street, 1882

Thomas Edison's workers started the first permanent central power station in the United States on September 4, 1882. It stood at 257 Pearl Street in Manhattan, in New York City, and it made electricity in one place for many buildings, sending it out through wires.

Coal into current. The station ran on coal. Four boilers burned coal to make steam, the steam ran engines, and the engines turned six dynamos, machines that turn motion into electric current. Wires carried the current to a district of about a quarter of a square mile around the station.

The station served 85 customers on its first day, with about 400 lamps among them. Within a year it served 513 customers with about 10,000 lamps. It ran until January 2, 1890, and in those years it went dark only once, for three hours.

Thomas Edison

Who: The inventor who built his electric system on direct current, a kind that could not travel far.

When and where: Pearl Street, Manhattan, New York City, from September 4, 1882.

When his station started, Edison stood in a banker's office.

Thomas Edison was in the office of the banker J. Pierpont Morgan, at the bank of Drexel, Morgan and Company, when the Pearl Street station began sending out current. He had built the system to run on direct current, called DC, which flows in one direction only, like the current from a battery.

The station sent DC out at a low voltage, the force that pushes electric current through a wire. Low-voltage direct current loses a lot of its energy in the wires as it travels. Because of that loss, Pearl Street could serve only the small district around the station.

The current war

Alternating current, called AC, flows back and forth, changing direction many times each second. In the United States today it changes direction 60 times a second. A machine called a transformer can easily raise or lower the voltage of AC, which could not easily be done with direct current.

George Westinghouse sold electric systems that ran on AC. Nikola Tesla received patents for motors that ran on alternating current in May 1888, and two months later Westinghouse paid for the right to use them. Chapter 8 tells how Tesla's motor worked and what Westinghouse paid for it.

The dogs. Edison and his supporters attacked AC as deadly. Harold Brown, an electricity salesman, ran tests at Edison's laboratory in West Orange, New Jersey, in 1888, killing dogs with alternating current to show that it was dangerous. The dogs were strays, bought from boys in the neighborhood for 25 cents each. Edison's supporters used the word "Westinghoused" to mean killed by electricity.

The chair. New York lawmakers replaced hanging with death by electricity in 1888, a change Governor David B. Hill had proposed. New York State officials hired Harold Brown to build the first electric chair, which kills a person by sending a strong electric current through the body. Edison secretly paid Brown to use alternating current in it. A chair that killed with AC fit Edison's campaign to convince Americans that alternating current was too dangerous to have in their homes.

The first man put to death in the electric chair was William Kemmler, of Buffalo, New York. Kemmler killed Matilda Ziegler, who lived with him as his wife, in their apartment in Buffalo, striking her more than 25 times on the head, neck and shoulders with a hatchet. At his trial in a Buffalo court in May 1889, he was found guilty of murder and sentenced to die by electricity in Auburn State prison.

His lawyer argued that death by electricity was a cruel and unusual punishment, which the New York State Constitution forbids, but judges all the way up to the US Supreme Court let the sentence stand.

The execution. Prison officials at Auburn put Kemmler in Brown's chair on August 6, 1890, and sent a current through him for 17 seconds. A doctor said he was dead, but then Kemmler groaned and began to breathe again. The officials turned the current back on, and this time it took between one and four minutes to kill him.

During the execution his fist clenched so hard that blood ran from his palm, and the back of his coat briefly caught fire. The room filled with the smell of burning flesh, and several witnesses fainted. Newspaper writers called the execution "a disgrace to civilization."

Chicago and Niagara. General Electric offered to light the World's Columbian Exposition, the world's fair in Chicago, with Edison's direct current for $554,000. Westinghouse offered to light it with alternating current for $399,000, and he got the job. In May 1893 his company's AC system lit about 100,000 electric lamps at the fair, 250 times the 400 lamps Pearl Street lit on its first day.

Westinghouse also won the job of building the three generators, machines that turn motion into current as Edison's dynamos did, for the Adams power plant at Niagara Falls. Each was rated at 5,000 horsepower. The Adams plant sent out its first alternating current, to factories in the town of Niagara Falls, on August 26, 1895. In November 1896 current from Niagara Falls lit Buffalo, New York.

George Westinghouse and Nikola Tesla

Who: The businessman who sold alternating current, and the inventor whose AC motors he bought the right to use.

When and where: New York State, 1888 to 1890.

Movie: *The Current War* (2017). A Hollywood drama, not a documentary. Michael Shannon plays Westinghouse and Nicholas Hoult plays Tesla.

Westinghouse refused to sell New York three dynamos for its electric chair.

George Westinghouse had owned the right to use Nikola Tesla's AC motors since 1888. Tesla was a Serbian immigrant who had come to the United States in 1884. Westinghouse stood to lose millions of dollars if Americans came to believe that alternating current would kill them in their homes.

New York State officials asked to buy three Westinghouse AC dynamos for the new electric chair, and Westinghouse refused to sell them. Harold Brown got AC dynamos anyway. Westinghouse then gave $100,000 toward the lawyers' fees for Kemmler's appeal to the US Supreme Court, and the appeal failed.

After Kemmler's execution in August 1890, Westinghouse said, "It has been a brutal affair. They could have done better with an ax."

Avondale, 1869: one way out

Power stations such as Pearl Street burned coal, and miners died digging it. Thirteen years before Pearl Street opened, a fire at the Avondale mine, near Wilkes-Barre, Pennsylvania, killed 110 people on September 6, 1869. The Steuben Coal Company owned the mine.

Mine operators then kept small fires burning at the bottom of their shafts, the deep holes that lead down into a mine. A fire like this, called a ventilating furnace, made a draft that moved fresh air through the mine.

One way out. Sparks from the ventilating furnace set fire to the wooden timbers in the shaft, and the flames spread up to the breaker, the wooden building where coal was broken and sorted. The breaker stood right over the shaft, and the shaft was the only way into or out of the mine. The men who ran the Steuben Coal Company had the mine and its breaker built that way.

The fire filled the mine with smoke and gas, and 108 men and boys underground died because they could not breathe. Five of the dead were boys between 12 and 17 years old. Two men who went down to rescue the miners also died.

Pennsylvania lawmakers passed a mine law in 1870. It required every mine to have more than one way out, and it forbade using a single passage both to bring fresh air in and to let bad air out.

From charcoal to coal

Hopewell Furnace in Pennsylvania burned charcoal from about 1771 until it closed in 1883. By then iron makers had built new ironworks that burned anthracite, the hard coal, and charcoal furnaces had fallen out of use. For more than a hundred years, workers at Hopewell had cut the forest around it for charcoal. Mark Bird, who built the furnace, enslaved some of those workers in its early decades, and the last record of slavery at Hopewell is from 1821.

Lafayette Houck

Who: One of the last colliers at Hopewell Furnace, who later taught the National Park Service how charcoal was made.

When and where: Hopewell Furnace, Berks County, Pennsylvania, in the later 1800s, and again in 1936.

Fifty-three years after the furnace closed, he showed how charcoal was made.

Lafayette Houck came from a family of colliers. Their work at Hopewell began in 1818, when Henry Houck worked there as a master collier, in charge of the pits and of the helpers who worked them. In winter Henry also repaired wooden wagon wheels or chopped wood to sell. In the 1830s the furnace's owners paid about 20 colliers, and each one tended eight or nine pits.

The watch. Lafayette made charcoal the way Hopewell's colliers had since the 1700s, in smoldering piles of stacked wood. A Hopewell collier lived in a rough hut near his pits from May to October, with one or two helpers, and together they tended up to eight or nine pits at once. They watched the pits day and night, from the moment a pit was lit until a teamster, a man who drove a wagon, hauled away the last load of charcoal.

Teaching the trade. Hopewell closed in 1883, and federal officials bought the land in the 1930s. In 1936 Houck and his son, William, showed workers of the Civilian Conservation Corps, a government jobs

program, how a collier lived and worked. Houck taught the National Park Service how charcoal was made, and he told them what he remembered about life at the furnace.

NOT KNOWN: his exact age, his pay, and the years he began and stopped work as a collier. One National Park Service page gives his life as 1846 to 1939, and another says he was 82 in 1936, which would put his birth around 1854.

BY 1900

If you remember one thing: Alternating current, which a transformer could raise or lower, won the current war against Edison's direct current.

- 1859: Edwin Drake's crew strikes oil near Titusville, Pennsylvania.
- 1869: A fire at the Avondale coal mine kills 110 people. The mine has only one way out.
- Around 1885: Coal passes wood as the country's main fuel.

Next: In the next fifty years, electric wires reached almost every American home, and oil shot out of a hill in Texas.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 15 MINUTES

Electric wires reached almost every American home in these fifty years. About 8 households in 100 had electricity in 1890, and 94 in 100 did by 1950. Emanuela Cardia worked out those figures in a 2008 study from the census, the government's count of the people and their homes, taken every ten years. Most farms got their wires only after 1935.

Coal still heated most homes. The 1940 census found that more than half of American homes were heated mainly with coal, and about a quarter with wood.

A single well in Texas began to pour out an estimated 100,000 barrels of oil a day in 1901. In World War II, most of the oil burned by the Allies, the countries fighting Germany, Italy and Japan, came from the United States.

Spindletop, 1901

A well on Spindletop hill, about 3 miles south of Beaumont, Texas, struck oil 1,139 feet down on January 10, 1901. Oil shot more than 100 feet into the air. The well poured out an estimated 100,000 barrels a day until workers capped it nine days later, on January 19.

Nine days. All of Texas had produced 836,000 barrels of oil in the whole year of 1900. At the estimated rate, the Spindletop well poured out about 900,000 barrels in its nine days, more than that whole year's total.

The salt dome. Anthony F. Lucas led the drilling. According to the Texas State Historical Association, he was the leading American expert on salt domes, large masses of salt buried deep underground, and Spindletop hill stood on one. Other geologists of the time doubted that oil lay in the salt domes along the coast of the Gulf of Mexico. Lucas was sure it did, and the well proved him right.

Several oil companies started at Spindletop or grew large there, among them the Texas Company, later called Texaco, and Gulf Oil.

What coal cost the miners

Coal miners died at work by the thousands in the early 1900s. Officials of the federal Mine Safety and Health Administration, part of the national government, keep a count for every year. In 1900, 1,489 coal miners died in accidents at work. The worst year was 1907, when 3,242 died. The count was still 2,063 in 1930, and it fell to 643 in 1950.

The yearly counts from 1900 through 2025 add up to about 105,000 coal miners who died in accidents. The counts do not include miners who died of disease from their work.

Monongah. Explosions in mines No. 6 and No. 8 at Monongah, West Virginia, killed 362 men by the official count on December 6, 1907. It is the worst mine disaster in American history. Later estimates run above 500, because the explosion destroyed the system that kept track of which miners were underground, so no one could make a full count.

Some of the dead were boys as young as eight. Mine companies of the time hired more workers than they listed, and their lists left out many of the children who worked underground. The dead included immigrants from Italy, Poland and Hungary, and Black workers. About 1,000 children lost a father.

What went wrong. Experts at the time did not agree on what set off the explosions. According to Clio, a history website run from Marshall University, a loaded coal car broke loose near the top of a trestle, a raised track, and ran back into mine No. 6. Some experts blamed torn electrical wires. Others blamed miners' lamps.

C. W. Watson was the president of the Fairmont Coal Company, which Clio names as the owner of the mines. He said the explosion "could only have been caused by the careless use of a mining lamp." The West Virginia Mine Wars Museum names a different owner, the Consolidation Coal Company. Neither history names anyone charged with a crime. After Monongah, members of Congress created the federal Bureau of Mines.

Ludlow. Colorado National Guard soldiers, the state's own force of part-time soldiers, killed striking coal miners and their families at Ludlow, Colorado, in 1914. The coal miners of southern Colorado had gone on strike in 1913, stopping work to demand changes. The coal they mined supplied Colorado's steel mill and its railroads. Most of them worked for the Colorado Fuel and Iron Company, whose main owner was John D. Rockefeller Jr.

The miners wanted the company's owners to obey Colorado's mining laws and to accept their union, the United Mine Workers, a group workers form so they can bargain with owners together. The strikers and their families lived in tent camps. Governor Elias M. Ammons sent the Colorado National Guard, and the mine owners agreed to pay for it.

Fighting broke out at the tent camp at Ludlow on April 20, 1914. No one knows who fired the first shot. Guardsmen burned the tents. At least 19 people were killed: one guardsman, five miners, and thirteen women and children. The women and children had hidden in a pit under a tent, and fumes from the burning filled the pit, so they died because they could not breathe.

Guardsmen killed three strike leaders who were under a flag of truce, a white flag that asks for a pause in fighting. They shot the strike leader Louis Tikas, a Greek American, three times in the back.

In the days after, miners attacked mines across the region and killed mine bosses and guards. No one knows how many people died in those days, but more died than at Ludlow. President Woodrow Wilson sent the US Army on April 28. The strike ended in December 1914, and the miners won none of the union's goals.

Samuel Insull

Who: Edison's former secretary, who tied more than 85 electric companies into one chain that collapsed in 1932.

When and where: Chicago, 1892 to 1932.

Tried three times after his companies failed, he was never found guilty.

Samuel Insull built a chain of electric companies that served millions of people, and the chain collapsed in 1932. He had worked for Thomas Edison as his private secretary. In 1892 Insull took over the Chicago Edison company, and he built it into Commonwealth Edison.

The chain. Insull tied together more than 85 companies through holding companies, companies that own shares, or pieces of ownership, of other companies. Insull's companies owned shares in one another, and they had borrowed much of the money behind them. Many small investors bought shares in his companies.

The fall. Insull's companies failed in 1932, during the Great Depression, and investors lost millions of dollars. Insull fled to Europe. In October 1932 a grand jury in Chicago, a group of citizens who decide whether there is enough proof for a trial, charged him with crimes.

He moved to Greece, and officials there refused to send him back. In 1934, after US officials pressed them, Greek officials told him to leave. Officials of Turkey arrested him, and US State Department officers brought him back to the United States to stand trial.

He was tried three times. One trial was for fraud, cheating people by lying. Another was for embezzlement, stealing money that others had trusted him to handle. The third was for breaking federal bankruptcy law, the rules for what happens when a person or a company cannot pay its debts. He was found not guilty all three times.

> Franklin Roosevelt, running for president, named Insull in a campaign speech in 1932. In 1935 members of Congress passed the Public Utility Holding Company Act, which they wrote to stop anyone from building another chain of electric companies like Insull's.

Rivers into power, and who lost their homes

Federal officials had large dams built to make electricity from 1933 to 1945. In the Tennessee Valley alone, they made an estimated 82,000 people leave their homes to do it. A hydroelectric dam holds back a river, and water rushing down through the dam turns machines that make electricity.

The TVA. President Franklin Roosevelt signed the law that created the Tennessee Valley Authority, or TVA, on May 18, 1933. The TVA was a federal corporation, a company owned by the national government. Its board and staff ran it, and TVA officials had dams built in the Tennessee Valley. Before the TVA, about 3 in 100 families in the valley had electricity. Work on the first TVA dam, Norris Dam, began in October 1933.

Who had to leave. TVA officials bought or took about 1.3 million acres from their owners from 1933 to 1945, and they made those 82,000 people leave to make room for 16 hydroelectric dams. For Norris Dam alone, TVA officials made more than 3,000 families leave.

The price. TVA officials set the price of each piece of land, and an owner who would not sell at that price lost the land anyway. TVA lawyer John Snyder put it this way: "a price is set and we buy at that price or we condemn." To condemn land is to take it by law without the owner's agreement, at a price the government sets. TVA's own economists estimated that condemning land got it for about 60 percent of what a private buyer would pay.

The TVA's board and staff made these choices together. Wayne Moore's 1995 history of the TVA land-taking does not name one official who made them.

Hoover Dam. Hoover Dam stands on the Colorado River, between Arizona and Nevada, and it was one of the big federal dams that made electricity in these years. Its first generator began sending electricity over 266 miles of wire

to Los Angeles on October 9, 1936. Some records date the start to October 26, 1936, the day that generator reached full operation. Chapter 12, Landmarks, tells how many men died building the dam.

Grand Coulee. Officials of the federal Bureau of Reclamation had Grand Coulee Dam built on the Columbia River, and it was finished in 1941. The dam stands 550 feet high, and its builders gave it no fish ladders, the rising pools that let salmon swim upstream past a dam. Salmon and steelhead, a trout that swims to the sea and back, could no longer reach the whole upper Columbia Basin, and their runs there ended.

The lake behind the dam flooded Kettle Falls, where Native people had fished for salmon for about 9,000 years. The catch there had averaged about 644,500 fish a year. Native people also lost thousands of acres of low, flat land beside the river where they grew food, and they lost their way to the places where they gathered roots and berries.

As the dam neared completion in June 1940, the Colville Confederated Tribes held a three-day Ceremony of Tears to mourn the loss of the fishery at Kettle Falls. Between 8,000 and 10,000 people came, from the Spokane, Yakima, Nez Perce, Coeur d'Alene and other nations.

Osage oil

Some of the oil Americans burned in the 1920s came from under the Osage Nation's land in Oklahoma. Under the Osage Allotment Act of 1906, the Osage kept the rights to the oil and minerals under their land, owned in common. Each Osage person with a share, called a headright, received part of the oil money. Under a law passed in 1921, local judges put many Osage adults' money under the control of white guardians, people the judges chose to handle it.

The murders. From about 1920 to 1925, husbands, guardians and others killed Osage headright holders to inherit their shares of the oil money. A cattleman named William Hale planned to have members of one Osage family killed so that his nephew, Ernest Burkhart, would inherit their headrights. It was the family of Ernest's wife, Mollie. Federal agents later proved that Hale ordered the murders.

Mollie's family. Mollie Burkhart's mother was Lizzie Q. Kyle. Her sisters were Anna Brown and Rita Smith, and Rita's husband was Bill Smith. Henry Roan was Mollie's cousin.

The killings. Kelsie Morrison shot Anna Brown in May 1921. He later admitted that he killed her at Hale's request. Henry Roan was shot and killed in January 1923. A cowboy named John Ramsey admitted that Hale had hired him to kill Roan, though Ramsey later took back his confession.

A bomb blew up Rita Smith's home on March 10, 1923. It killed Rita, Bill Smith and Nettie Brookshire, who worked for them. Ramsey confessed that he took part in those murders too, and he named Hale as the leader, along with two other men.

The sentences. Ernest Burkhart admitted his guilt in June 1926 and got life in prison for the murder of Bill Smith. Morrison got life that November for killing Anna Brown. Burkhart testified against his uncle and Ramsey. Hale got life in January 1929, and Ramsey got life that November, both for the murder of Henry Roan. Chapter 2 tells how Osage leaders brought in the federal agents who solved the case.

NOT KNOWN: whether Mollie's mother, Lizzie Q. Kyle, was poisoned when she died in July 1921, or by whom. Poison was suspected, but the Oklahoma Historical Society's and the FBI's accounts do not say.

DISPUTED: the number of dead. Writers for the Oklahoma Historical Society count about 24 violent or suspicious deaths, and the FBI's history of the case says at least two dozen. The Osage Nation counts more than 60 murders and suspicious deaths.

The night the lights came on

Fewer than 11 farms in 100 had electricity from a central power station in 1935. A Tennessee farmer, whose name was not recorded, had said in 1930 that "the greatest thing on earth is to have the love of God in your heart." The next greatest thing, he said, "is to have electricity in your house." David Nye records the words in his book *Electrifying America*. In 1984 the national association of electric cooperatives named its history of the program *The Next Greatest Thing*.

The REA. President Roosevelt created the Rural Electrification Administration, or REA, on May 11, 1935, with Executive Order 7037, a rule a president issues on his own authority, without a vote of Congress. He signed the Rural Electrification Act on May 20, 1936. Under that law, REA officials made long-term, low-cost loans to cooperatives, businesses owned by the people who use them. Farmers' cooperatives used the loans to build power lines.

Louisan Mamer, who worked for the REA teaching farm families to use electricity in their homes, toured farm towns in the 1930s and 1940s with a traveling show called the electric circus. She showed families electric appliances and how to cook with electricity.

Pedernales. In the Texas Hill Country, power first ran through the lines of the Pedernales Electric Cooperative on October 27, 1939. The cooperative had 1,800 miles of line and 422 first customers. Families coming home that night

saw their farmhouses lit for the first time. About half of American farms had electricity by 1945, about 9 in 10 by 1950, and about 97 in 100 by 1960.

Pearl Yates

Who: A country schoolteacher in western North Carolina who remembered when the lights first came on in her home.
When and where: Haywood County, North Carolina, 1939.

She and her children walked room to room, looking at the lights.

Pearl Yates taught high school English in the countryside of western North Carolina. When the electric lights came on in her home, she walked from room to room with her small children. "We'd stop and look and they were so pretty and we'd go look at another room and another room," she said. The power reached her part of Haywood County in July 1939.

Before the lines. She cooked on a wood stove, and washing clothes took a whole day, in a large pot. Officials of the power company, Carolina Power and Light, would not run lines into the countryside where the Yates family lived, so people there formed a cooperative. Many neighbors did not trust it at first, because outsiders ran it and members had to pay money up front. Organizers went from door to door and signed many of them up.

Bonnie Bishop interviewed Pearl Yates and her husband, Oral Yates, a former North Carolina state lawmaker, on May 29, 1984. The interview was part of a set recorded for the 50th anniversary of the REA, and it is kept in the Southern Oral History Program at the University of North Carolina.

Oil for a world war

Six of every seven barrels of oil the Allies used in World War II came from the United States. Writers at the American Oil & Gas Historical Society count about 6 billion of the 7 billion barrels.

Pipelines. Tanker ships carried oil from Texas along the coast, and German submarines were sinking them, so workers built two pipelines on land. The Big Inch, a pipe 24 inches wide, ran from East Texas to Illinois and on to the East, and it was finished in August 1943. Together with a second line, the Little Big Inch, it carried more than 350 million barrels of oil east before the war in Europe ended.

BY 1950

If you remember one thing: Electric wires reached almost every American home in these fifty years.

In 1890: About 8 American households in 100 had electricity.

In 1950: About 94 households in 100 had it, and about 9 farms in 10, most of them wired by farmers' own cooperatives after 1935.

Next: After 1950 Americans began splitting atoms to make electricity, and in 1973 a cutoff of foreign oil left drivers waiting in long lines for gasoline.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 12 MINUTES

Splitting atoms became a new way to make American electricity after 1951, when a test reactor in Idaho lit four light bulbs. The first full-scale American plant built only for civilian power began sending electricity into a power company's lines in Pennsylvania in 1957. A reactor at Three Mile Island, also in Pennsylvania, partly melted in 1979, and after that American power companies stopped ordering new reactors.

Americans kept burning more oil and natural gas after World War II, according to the US Energy Information Administration. Officials of the Arab members of OPEC, a group of countries that sell oil, cut off oil sales to the United States in October 1973, and drivers waited in long lines for gasoline. Members of Congress then passed new energy laws and set up a federal Department of Energy, which opened in 1977.

How does a nuclear power plant make electricity?

A nuclear power plant makes electricity from the heat of splitting atoms. Inside the reactor, the machine where the atoms are split, atoms of uranium split in two, and each split gives off a large amount of energy as heat. This splitting is called fission.

The fuel is uranium, a heavy metal dug out of the ground as ore, rock that holds a metal, and it sits in the core, the center of the reactor. The heat boils water into steam. The steam spins the blades of a turbine, a wheel with many blades, and the turbine turns a generator, which turns that motion into electric current.

Four light bulbs. A test reactor in Idaho called EBR-1 made the first usable electricity ever produced from nuclear energy on December 20, 1951. It lit four 200-watt light bulbs. The town of Arco, Idaho, about 1,200 people, ran on electricity from a nearby reactor called BORAX III for about an hour on July 17, 1955. Arco was the first town lit only by nuclear power.

Shippingport. President Eisenhower gave a speech called "Atoms for Peace" to the General Assembly of the United Nations on December 8, 1953. Work on a nuclear power plant at Shippingport, Pennsylvania, began in 1954. The plant sent its first electricity into the lines of Duquesne Light, a power company, on December 18, 1957. Five days later it reached its full power of 60,000 kilowatts, or 60 million watts. That was enough to light 300,000 bulbs like the ones EBR-1 lit.

Shippingport was the first full-scale nuclear power plant in the United States built only to make electricity for civilian use, meaning not for the military. It was not the first to make nuclear electricity. EBR-1 and smaller American reactors came before it, and so did reactors built for more than one purpose. A British plant, Calder Hall, began sending power in 1956.

Garrison Dam and the Three Affiliated Tribes

Members of Congress paid for Garrison Dam on the Missouri River in North Dakota in 1947. The dam was part of the Pick-Sloan Plan of the Army Corps of Engineers, the part of the Army that builds dams and waterways. Under that plan, the dams were to stop floods and make electricity. The Army Corps was in charge of Garrison Dam, and it was finished in 1953.

Under the lake. The dam's lake, Lake Sakakawea, flooded the farmland along the river on the Fort Berthold Reservation, the home of the Mandan, Hidatsa and Arikara, called the Three Affiliated Tribes. It covered the river bottomlands, the low, flat lands beside the river where the tribes farmed. Counts of the flooded land run from 153,000 to about 155,000 acres.

Farms, homes, schools, a hospital, businesses, churches, cemeteries, roads and whole towns went under the water, among them Elbowoods, Sanish and Van Hook. By 1954 the Elbowoods school building stood under 50 feet of water.

Who had to leave. Under the federal act for taking the tribes' land, about 300 families had to leave their homes before the lake covered them. By one count, that was more than 90 percent of the tribes' people.

How the land was taken. Federal planners had chosen the dam's plan without asking the tribes. In May 1948 tribal leaders signed a contract giving up the tribes' land for $5,105,625. Tribal chairman George Gillette was in tears as he signed. He said the contract broke the Treaty of Fort Laramie, a signed agreement the tribes had made with the United States: "The truth is, as everyone knows, our Treaty of Fort Laramie ... and our constitution are being torn to shreds by this contract."

Members of Congress passed a law in 1949 that set out how the tribes would be paid for their land, and the president signed it. In 1950 the tribes' adults voted on whether to accept the terms of that law. Of the 991 adults, 525 voted yes. Tribal members feared that if they refused, they would be paid even less later.

NOT KNOWN: whether a vote against the terms could have stopped the dam.

Black lung and the 1969 law

An explosion at the Consol No. 9 coal mine in Farmington, West Virginia, killed 78 miners on November 20, 1968. Members of Congress then passed the Federal Coal Mine Health and Safety Act of 1969, which became law on March 30, 1970. Under it, federal inspectors had to check every underground coal mine four times a year and every surface mine twice a year, and they could close a mine where they found a danger to life. Miners with black lung disease could also get federal payments.

What black lung does. Black lung is a group of lung diseases caused by breathing coal mine dust. The body's defending cells try to break down the grains of dust and cannot, so the cells burst and release chemicals that damage the lung. Over the years, scar tissue and hard lumps form in the lungs.

Miners cough up dark, inky phlegm, the thick slime that comes up from the lungs. In the worst form, large masses of scar tissue fill the lungs, and the miner gasps for every breath. According to US Department of Labor figures reported in *Smithsonian* magazine in 2017, more than 76,000 miners had died of black lung since 1968.

Officials of the Mine Safety and Health Administration count the coal miners who died in accidents at work each year. They counted 311 in 1968, the year of Farmington, and 260 in 1970, the year the law took effect. By 2000 the count had fallen to 38, and in 2014 it was 16, the lowest ever recorded.

1973: the gasoline runs short

Officials of the Arab members of OPEC announced an embargo, an order that stops trade, on oil sales to the United States in October 1973. They acted because US officials were sending military supplies to Israel in the Yom Kippur War, which began on October 6, 1973, when Egypt and Syria attacked Israel's forces. The embargo lasted until March 1974. In about six months, the price of a barrel of oil rose from about $2.90 to $11.65, about four times as much.

Long lines. Gas stations ran out of gasoline, and drivers waited in long lines. Officials in many states let drivers buy gas only on certain days, by license plate number: cars with odd numbers on odd-numbered days, and cars with even numbers on even-numbered days. Members of Congress set a national speed limit of 55 miles an hour, which became law in January 1974. A US Department of Energy report gives the average price of gasoline as 39 cents a gallon in 1973 and 53 cents in 1974.

More energy laws followed. The Energy Policy and Conservation Act became law on December 22, 1975. Under it, federal officials set up the Strategic Petroleum Reserve, a store of crude oil, the oil that comes straight out of the ground, kept for emergencies.

The law also contained the first fuel-economy standards for cars, rules for how far a car must go on a gallon of gas. New cars had to average 27.5 miles per gallon by 1985, about twice what they got then. The federal Department of Energy opened on October 1, 1977.

David Falconer

Who: A government photographer who recorded Oregon's gasoline shortage in pictures and in his own captions.

When and where: Oregon, October 1973 to May 1974.

In Portland he photographed a man in line with a gas can.

David Falconer photographed the gasoline shortage in Oregon and wrote down what he saw. He worked for Project DOCUMERICA, which officials of the Environmental Protection Agency started in 1972 to photograph environmental problems around the country. About 120 photographers took part, and the National Archives keeps Falconer's pictures and captions.

Closed pumps. At Sutherlin, on Interstate 5, he photographed closed pumps in October 1973. His caption states, "By midday gasoline was becoming unavailable along interstate highways." That month in Lincoln City, gas stations closed early and stayed closed on weekends, and "No Gas Today" signs went up.

Portland. In December 1973 he photographed a man who had run out of gas and was standing in line with a gas can. The man's name was not recorded. By February 1974, drivers there with even-numbered license plates lined up for gas on even-numbered days, Monday through Friday. Anyone could try on Saturday if gas was left, and all Portland stations closed on Sundays.

Falconer wrote that Oregon was the first state to use this odd-even plan. In May 1974, after the embargo had ended, he photographed a sign at a downtown Portland station saying its gas for the day was sold out.

What went wrong at Three Mile Island?

A stuck valve let cooling water escape from the Unit 2 reactor at the Three Mile Island power plant in Pennsylvania on March 28, 1979, and the reactor's core overheated and partly melted down. It was the worst accident in American history at a nuclear power plant that sold its electricity. In a meltdown, the reactor's fuel gets so hot that it melts.

The valve. General Public Utilities ran the reactor. A small fault in the cooling system came first. Then a relief valve, a valve that opens to let out pressure, stuck open. The operators had no gauge that showed the valve was open, and they believed the system held too much water, so they cut back the water flowing into it. The core lost its water and overheated, and at least 45 percent of it melted.

Governor Dick Thornburgh advised pregnant women and preschool children living within 5 miles to leave on March 30, and he told people within 10 miles to stay indoors. Officials gave out messages that conflicted with each other. About 140,000 people left on their own.

A small amount of radiation escaped. Radiation is energy that substances such as a reactor's fuel give off, and it can damage the cells of the body. The average dose to people within 10 miles was 0.08 millisieverts, about the same as one chest X-ray. A millisievert is a unit that measures how much radiation a body takes in. No one got more than 1 millisievert, about a third of the radiation an average American gets from nature in a year.

Scientists at the Nuclear Regulatory Commission, the federal agency that oversees nuclear plants, and more than 12 health studies since 1981, found no health effects they could detect. No deaths are linked to the accident.

Where the uranium came from

Miners dug uranium on Navajo land for the federal government and later for power plants. Federal officials did not warn them about radon, a gas from the uranium that caused lung cancer. Until 1966 the federal Atomic Energy Commission was the only buyer, and after that uranium was also sold to fuel power plants.

Miners dug nearly 30 million tons of uranium ore from Navajo land from 1944 to 1986, according to the Environmental Protection Agency. About 10,000 people worked in uranium, and about a quarter of them were Navajo.

No warning. Uranium gives off radon, a radioactive gas, which means it gives off energy that can damage the cells of the body. Navajo miners say that no one taught them about the dangers of uranium mining, and that no one gave them protective gear or fresh air in the mines.

A federal committee, the Advisory Committee on Human Radiation Experiments, reported in 1995 that "at least several hundred" uranium miners had died of lung cancer from breathing radon. Its members found that federal officials did not require the mine owners to pump fresh air into the mines. They also did not warn the miners.

Church Rock. A dam holding radioactive waste, owned by the United Nuclear Corporation, broke near Church Rock, New Mexico, on July 16, 1979, four months after Three Mile Island. About 94 million gallons of radioactive liquid and 1,100 tons of solid waste poured down the Rio Puerco, past about 1,700 Navajo people.

A 2013 study estimated that the spill released 46 curies of radioactivity, against 13 curies for Three Mile Island. Curies measure how much radioactive material was let loose. The millisieverts given for Three Mile Island measure something else: how much radiation a person's body took in. The US Geological Survey called the spill "the largest accidental radioactive release in the United States."

No official declared an emergency or ordered anyone to leave. A New Mexico state agency called the effects on people and the environment "quite limited," but its report also warned that water pumped out of the mines into the river might be a greater danger over time. Governor Bruce King turned down the Navajo Nation's request for a federal disaster declaration. Chapter 13, The Elements, tells the spill in full.

NOT KNOWN: how many of the people living along the river were harmed by the spill.

Panels on the White House roof

President Carter held a ceremony for 32 new solar panels on the roof of the West Wing of the White House on June 20, 1979. The panels used sunlight to heat water. They were in use for about seven years. In 1986, while President Reagan was in office, workers took the panels down to repair the roof, and no one put them back.

Some of the panels went to Unity College in Maine in 1991, where workers put 16 of them back up to heat water for the cafeteria. One panel went to the Smithsonian's National Museum of American History in 2009. Others are at the Carter Library and at a solar museum in Dezhou, China.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 13 MINUTES

Natural gas took coal's place as the fuel that makes the most American electricity. Drillers got more and more of that gas by fracking, which cracks rock deep underground to free the gas inside. Natural gas made more electricity than coal over a full year for the first time in 2016, according to the US Energy Information Administration, or EIA. Wind and solar power made a record share of the country's electricity in 2025, about 17 to 18 percent, by EIA figures.

Americans produced more energy than they used in 2019, for the first time since 1957. A winter storm shut down power plants across Texas in February 2021, and more than 4.5 million households lost power. A new nuclear reactor began running in Georgia in 2023, the first new American reactor to start since 2016.

What is fracking?

Fracking, short for hydraulic fracturing, is a way of getting gas and oil out of rock deep underground. Drillers pump water, sand and chemicals down a well at high pressure. The pressure cracks the rock, and the sand holds the cracks open so that gas or oil can flow out. Many of these wells turn and run sideways through the rock for thousands of feet, which is called horizontal drilling.

From Texas shale. George Mitchell's company, Mitchell Energy, began working the Barnett Shale, a layer of rock made of hardened mud, in Texas in the early 1980s. Mitchell and his company spent about twenty years trying to get gas out of it. In the late 1990s an engineer, Nick Steinsberger, used a cheaper way of fracking called slickwater fracturing, which cut the cost enough that selling the shale gas made money. In the 2000s drillers joined fracking with horizontal drilling.

American wells produced 16 times as much shale gas in 2023 as in 2007, according to the EIA. By 2018 the United States was the world's largest producer of both natural gas and crude oil.

The argument. People disagree about fracking. Supporters say it brought cheaper energy and jobs. They add that natural gas burns cleaner than coal, and that power companies have replaced coal-burning plants with gas-burning ones.

Critics worry that fracking fluid can pollute groundwater and that methane, the main gas in natural gas, leaks from wells and pipes. Earthquakes are a third worry.

Drillers pump wastewater from oil and gas wells deep underground. Scientists at the US Geological Survey wrote in 2017 that the sharp rise in Oklahoma's earthquakes is widely seen as the result of that deep pumping of waste fluids. An earthquake near Pawnee, Oklahoma, measured 5.8 in magnitude, the number that states an earthquake's size, on September 3, 2016. It was the largest ever recorded in the state.

Mine deaths, black lung and a pipeline

An explosion of gas and coal dust killed 29 miners at the Upper Big Branch mine in Montcoal, West Virginia, on April 5, 2010. It was the worst mine disaster in the United States in 40 years. Performance Coal Company, part of Massey Energy, ran the mine.

The federal Mine Safety and Health Administration's report on the explosion identifies Massey's "corporate culture," its habits and ways of working, as the root cause. Officials of the agency issued 369 citations and orders, written notices of broken safety rules, with fines of $10.8 million.

The trial. Massey's former chief executive, Don Blankenship, was convicted in December 2015 of conspiring, or planning with others, to break federal mine safety laws. It was a misdemeanor, a less serious kind of crime. Judge Irene Berger gave him the longest sentence the law allowed on April 6, 2016: one year in prison and a $250,000 fine. A year of supervised release followed, when a court officer checks on him. He was not convicted of causing the miners' deaths.

Black lung came back among coal miners in the 2000s. Researchers at NIOSH, the federal agency that studies workers' health, found 60 cases of the worst form of the disease in 20 months at one small clinic in Kentucky in 2016. Reporters at National Public Radio found 962 cases of advanced black lung in the 2010s, more than double the 441 cases NIOSH had recorded in the 40 years before. *Smithsonian Magazine* reported both counts in 2017.

Standing Rock. Pumping more oil also meant moving it. A company called Energy Transfer Partners built the Dakota Access pipeline to carry crude oil from the Bakken oil field in North Dakota to Illinois. Members of the Standing Rock Sioux Tribe opposed it in 2016 where it crosses under Lake Oahe, upstream of their reservation. The lake is the tribe's water supply, and they feared that a spill would pollute it.

Protest camps began in April 2016. Army Corps of Engineers officials denied the easement for the crossing, the permission to cross federal land under the lake, on December 4, 2016. President Trump ordered the Army Corps to

review and approve the easement quickly on January 24, 2017. Army Corps officials approved it on February 8, 2017, and oil moved through the pipeline later that year. The tribe's lawsuits continued.

Natural gas, wind and solar, 2016 to 2025

Natural gas passed coal for a full year for the first time in 2016, when it made 34 percent of American electricity, according to the EIA. By 2025 it made about 41 percent, more than twice coal's share of about 17 percent. Nuclear power made about 18 percent.

Renewables. Wind, solar, flowing water and other sources that do not run out are called renewable energy. In 2025 they made about 24 percent of American electricity: wind about 11 percent, solar about 7 and hydropower, from flowing water, about 6.

Wind and solar together made a record share. The EIA's yearly report puts it at about 17 percent, but its rounded shares for wind and solar add up to 18, so the share was about 17 to 18 percent.

Analysts at the federal Bureau of Labor Statistics expect wind turbine service technicians to be the fastest-growing job in the country from 2024 to 2034, up 49.9 percent. Solar panel installers come second, up 42.1 percent. The two jobs together are projected to add fewer than 20,000 jobs.

Electric cars. Americans also began buying cars that run partly or fully on electricity: hybrids, which have both a gasoline engine and an electric motor, and battery electric cars, which run on electricity alone. According to the EIA, about 20 percent of the new cars, SUVs and pickups sold in 2024 were hybrids or electric, and in 2025 the share rose to about 22 percent.

But 2025 was also the first year that sales of battery electric cars fell. Federal tax credits for buyers of these cars, money taken off the buyer's tax bill, ended on September 30, 2025. In each of the months left in 2025, battery electric cars made up less than 6 percent of sales.

Why did the Texas lights go out in 2021?

Texas power plants of every kind broke down in the cold during a winter storm named Uri in February 2021. To keep the state's whole power grid from failing, the people who run it cut off power to millions of customers. More than 4.5 million households, about 10 million people, lost power, some of them for days. Reports on the failure by federal energy regulators and by researchers at the University of Texas give these figures, and so do news reports.

The grid. The Electric Reliability Council of Texas, called ERCOT, runs the Texas grid, the network of power plants and wires that serves the state. Some of its gas, coal, nuclear and wind plants stopped working. ERCOT's operators cut 20,000 megawatts, a megawatt being a million watts. That was more than 300 times the 60 megawatts the Shippingport nuclear plant made in 1957, and the largest cutoff of its kind in American history.

ERCOT officials later said the grid came within 4 minutes and 37 seconds of a total collapse. Texas state health officials counted 246 deaths from the storm in 2022, but others have argued that the official count is much too low. Reporters at BuzzFeed News compared the number of Texans who died in the week of the worst blackouts with the number who would normally die. They estimated that about 700 people were killed by the storm in that week alone. Damage estimates run above $195 billion.

Cristian Pavón Pineda

Who: An 11-year-old sixth grader who died in his family's mobile home during the Texas blackout.

When and where: Conroe, Texas, February 16, 2021.

He had seen snow for the first time the day before.

Cristian Pavón Pineda was 11 years old when he died in his family's mobile home in Conroe, Texas, on February 16, 2021. The home had been without power since the Sunday before, and the temperature there fell to about 10 degrees. Al Día News reported that he had seen snow for the first time the day before he died.

The family of five slept in one room to keep warm. Cristian shared a bed with his younger brother. His aunt, Jaliza Yera, told a TV station that he went to bed in "a shirt, a sweater, two pairs of pants, socks on." His stepfather put another blanket over him. In the morning his family could not wake him.

The cause. Early reports, and his family's lawsuit, said he died of hypothermia, which means the body's temperature has fallen too low to keep a person alive. But an autopsy, an examination of a body to find the cause of death, found on February 18, 2021, that he died of carbon monoxide poisoning, Conroe police said in May 2021.

Carbon monoxide is a gas with no color or smell that comes from burning fuel. When a person breathes it in, it stops the blood from carrying oxygen. Across Texas, more than 30 people had died in the storm by February 21, Al Día News reported. Many of them died of carbon monoxide poisoning after using cars or generators to keep warm.

The lawsuit. His mother, Maria Elisa Pineda Guzman, sued ERCOT and Entergy Texas, the power company, for more than $100 million on February 22, 2021. Her lawyer was Tony Buzbee. Officials had called the

power cuts temporary rolling outages, planned cuts that move from area to area. The suit states that ERCOT and Entergy Texas did not give his family good enough information about how long the outages would last.

"Accurate information might have saved Cristian Pineda's young life," the suit states. In a statement, ERCOT officials called his death a "tragedy" and said they were "confident that our grid operators made the right choice to avoid a state-wide blackout."

NOT KNOWN: where the carbon monoxide in the home came from, and how the lawsuit ended. Published reports do not say.

New and restarted reactors

Two new nuclear reactors at the Vogtle plant in Georgia began selling power in 2023 and 2024: Unit 3 on July 31, 2023, and Unit 4 on April 29, 2024, according to the EIA. Unit 3 was the first new reactor to start in the United States since Watts Bar Unit 2 in Tennessee, owned by the TVA, began running in 2016.

Work on Watts Bar 2 had begun in 1973. Workers stopped building it in 1985 and started again in 2007, so the two Vogtle reactors were the first American reactors begun after the 1970s to start running.

Palisades. The owners of two closed plants plan to run them again. Holtec International owns the Palisades plant in Michigan, which shut down in May 2022, and it got a $1.52 billion federal loan to restart it. Its targets of 2025 and early 2026 passed without a restart.

Workers began loading fuel into the reactor in late August 2026. On August 30 a fuel assembly, a tall bundle of fuel rods, tilted while it was being loaded, and the loading was paused. Officials of the Nuclear Regulatory Commission, the federal agency that oversees nuclear plants, began a review, and their first

results showed no radiation had escaped and no workers were harmed. Holtec's officials had given no date for the restart as of late September 2026, a Michigan TV station reported.

Three Mile Island again. Three Mile Island Unit 1 is the reactor that did not melt down in 1979, and it closed in 2019. It now carries the name Christopher M. Crane Clean Energy Center. Leaders of its owner, Constellation Energy, signed a 20-year agreement to sell its power to Microsoft. Members of the Nuclear Regulatory Commission are expected to decide in May 2027 whether to give Constellation an operating license, permission to run the reactor, the news outlet WITF reported in July 2026.

What energy independence means

Americans produced more energy than they used in 2019, for the first time since 1957, according to the EIA. They produced less than 1 percent more than they used.

People call this energy independence: Americans produce more energy in total than they use. It does not mean Americans stopped buying fuel from other countries. In November 2019 Americans still bought nearly twice as much crude oil from other countries as they sold to them, by EIA figures.

Americans sold a record amount of energy to other countries in 2025, about one and a half times as much as they bought from them, the EIA reported in May 2026. Oil and products made from it were 63 percent of the energy sold abroad, and natural gas was 29 percent.

THE THREE QUESTIONS, ANSWERED

1. **Did kerosene really save the whales?** No. Whaling was shrinking before kerosene came, and in most homes kerosene replaced camphene, after a Civil War tax on alcohol raised camphene's price.
2. **How did electricity reach almost every American home?** Edison's Pearl Street station began selling current in 1882, and

alternating current, which a transformer could raise or lower, carried power from Niagara Falls to Buffalo by 1896. After 1935 federal loans paid for farmers' own power lines, and by 1950, 94 households in 100 had electricity.

3. **Who paid for America's energy with their freedom, their land or their lives?** Enslaved people fed the iron furnaces, at least 271 at Catoctin alone, and about 105,000 coal miners died in accidents from 1900 through 2025. Federal officials made 82,000 people in the Tennessee Valley leave their homes for dams, other dams flooded Kettle Falls and the Fort Berthold farmlands, and husbands, guardians and others murdered Osage people for their oil money.

What changed: Americans went from fires of wood to coal, oil, natural gas, the atom, the wind and the sun.

FIVE DATES TO KEEP

February 11, 1808: Jesse Fell burns hard coal in his tavern fireplace.

August 27, 1859: Drake's crew strikes oil in Pennsylvania.

September 4, 1882: Edison's Pearl Street station starts selling electricity.

December 6, 1907: The Monongah explosions kill at least 362 miners.

2016: Natural gas passes coal in making American electricity.

> **NOT KNOWN:** The surviving records of the early colonial iron furnaces list the owners, not the men who made the charcoal.

Where to go: The *Charles W. Morgan*, the last wooden whaling ship left in the world, is kept at the Mystic Seaport Museum in Mystic, Connecticut.

Next: Chapter 11, Transportation. It asks how Americans moved people and goods, from trails and canals to railroads, cars and planes.

Chapter 11: Transportation

CHAPTER 11 OF 37 · TRANSPORTATION · 10 ERAS · ABOUT 75 MINUTES

Each new way of moving people and goods made trips faster or cheaper, and enslaved people and immigrant workers did much of the building, on land taken from Native nations.

Eight Chinese workers laid the last stretch of track at Promontory Summit, in Utah, on May 10, 1869. It joined two railroad lines into the first one across the country, and a golden spike marked the spot. With it, a trip from coast to coast took about a week instead of about four months. Chinese workers had built the line's western part for less pay than white workers got.

Era by era, this chapter describes how people and goods moved, from foot trails and canoes before 1500 to jets, container ships and ride apps after 2000. Watch two numbers fall as the eras go by: how long a trip took and what it cost. Look for old routes under new ones, too. Many colonial roads began as Native paths, and so did part of the first federal highway. And keep an eye on who did the digging and the track-laying, whose land and homes the routes went through, and who was told where to sit.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How much faster and cheaper did travel get?
2. Who built the new routes, and whose land and homes did they take?
3. How did Black travelers fight the laws that made them sit apart?

Five minutes? Read "The Chinese workers" in 1850 to 1900.

Can skip: "Trains and Jeeps in World War II", "The minivan", "Ports, trains and bridges". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

Nobody in North America hauled loads on wheels before 1500, and nobody rode a horse. People moved themselves and their goods on foot and by water, over trails and rivers that crossed the whole continent. The horses of the Americas had died out after the ice age. North of Mexico, no large animal was left that could pull a cart.

With no animal to pull a cart, people here had no use for one. In Mesoamerica, the region that is now Mexico and Central America, people made toys with wheels, but they did not use wheels to haul loads.

Trails that became highways

Native people built and kept up long trails across the continent, and many of those trails later became colonial roads. Some of those roads became highways. Many trails had names. The Great Trail, also called the Great Path, was a network of footpaths that linked New England, the middle of the Atlantic coast and the Great Lakes. People who spoke Algonquian and Iroquoian languages, two families of related languages, used it.

The Natchez Trace. This route began as paths worn by bison walking to salt licks, places where animals lick salt from the ground, near present-day Nashville. Natchez, Choctaw and Chickasaw people followed the paths, widened them and kept them clear. They used them for trade, travel, war and talks with other nations. People have used the route for about 8,000 years.

People traded over these trails and rivers for thousands of years. As early as 6000 BC, they carried seashells and native copper, copper found pure in the ground, between the Great Lakes and the coast of the Gulf of Mexico.

Canoes, portages and the travois

On the rivers and lakes, people traveled in two kinds of canoe. A dugout was one log that builders hollowed out with fire and tools. For thousands of years dugouts were the main boats of the Southeast and of the Eastern Woodlands, the forests that covered the eastern part of the continent. Some dugouts found in Wisconsin lakes are more than 4,000 years old.

Birchbark. In the Northeast and around the Great Lakes, people built canoes from the bark of birch trees. One or two people could carry one. It floated in shallow water, and its owners repaired it with materials from the forest. Builders made them in sizes from one-person boats to big canoes that carried freight, goods being shipped.

Where one river or lake ended, travelers lifted their canoes and carried them over land to the next, a crossing called a portage. Portages and trails linked the waterways into one network across the continent.

On the Great Plains, dogs dragged loads on a travois, which Chapter 10 describes. The travois was the usual way to move goods on the Plains before horses came.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

Spanish soldiers and colonists brought horses and wagons into what is now the United States in the 1500s, and they built a port in Florida, all on land where Native nations already lived. The first horses in the Southwest came north with Coronado's soldiers in 1540. The port, St. Augustine, began inside a Timucua town, where Spanish colonists took over the chief's council house in 1565. Juan de Oñate's colonists followed in 1598 with 83 wagons and carts, bound for Pueblo land in New Mexico.

Horses, a port and 83 wagons

Horses. Christopher Columbus carried horses to the Caribbean islands on his second voyage, in 1493. In 1539 Hernando de Soto landed about 250 horses in Florida, on a march that Chapter 10 tells. The next year a Spanish commander, Francisco Vázquez de Coronado, led a long march north from the west coast of Mexico, from 1540 to 1542. His soldiers brought more than a thousand horses, the first in what is now Arizona. The march went on into New Mexico and the Great Plains.

A port. Spanish colonists built St. Augustine, in what is now Florida, in 1565, and it became a port. Spanish ships sailing between Spain and the Caribbean stopped there for supplies and for protection. People have lived there without a break ever since. It is the oldest European town and port on the mainland of the United States.

The colonists built it inside a Timucua town. Their leader, Pedro Menéndez de Avilés, landed at Seloy on September 8, 1565. Seloy was also the name of the town's chief, who answered to a greater chief. One count gives Menéndez about 1,500 soldiers and colonists, and another gives 800 colonists, including 24 women and an unknown number of enslaved Africans.

The priest with the Spanish ships wrote that the Timucua received the Spanish well and gave them their chief's large house on the riverbank. It was the chief's council house, the building where the town's leaders met. The Spanish took it over. Two of their captains had a ditch, a moat and a wall of earth built around it, and it became their first fort there. Chapter 1 tells what happened between the colonists and the Timucua after that.

Wagons. Juan de Oñate led a Spanish colony north from Santa Bárbara, in northern New Spain, the Spanish colony that included Mexico, in January 1598. About 400 men came, about 130 of them with families. They brought 83 wagons and carts and more than 7,000 farm animals. In late April the long line of wagons, people and animals reached the Rio Grande near El Paso del Norte.

The land the colony moved into was the homeland of the Pueblo peoples. Oñate claimed New Mexico for Spain on April 30, 1598, in a ceremony the Spanish called La Toma, "the taking." In July he set up the colony's headquarters, which the Spanish called San Juan de los Caballeros, at the Tewa town of Ohkay Owingeh on the upper Rio Grande. Chapter 10 tells about the colony's water mills.

The route the wagons followed became El Camino Real de Tierra Adentro, Spanish for "the royal road of the interior." It ran about 1,600 miles from Mexico City to northern New Mexico.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 3 MINUTES

Boats carried most of the goods in the English colonies of the 1600s, from canoes and small open boats called shallops to ships that sailed along the coast from one colony to another. The leaders of the Massachusetts colony licensed a ferry across Boston's harbor in 1631. On land, colonists followed paths that Native people had made long before, and a rider carrying mail began using them between New York and Boston in 1673.

The Boston Post Road, on Native paths

The first post rider, a man on horseback who carried letters, left New York for Boston in January 1673. The governor of New York, Francis Lovelace, had pushed for the route along with John Winthrop Jr. of Connecticut. The rider went by way of New Haven, Hartford, Springfield and Worcester. The trip was about 250 miles and took about two weeks. The path that he and later riders used grew into the Boston Post Road.

In time the road had two main routes between New York and Boston, an upper road inland and a lower road near the shore, and both followed Native paths.

The upper road. The first rider's route, through Hartford, Springfield and Worcester, became the upper road. Between Boston and the Connecticut River it followed the Old Connecticut Path and the Bay Path, Native trails from Massachusetts Bay to the river. Nipmuc people carried corn along them to Massachusetts Bay in the early 1630s and traded it there for copper, iron goods and wool cloth. Thomas Hooker led the members of his church along the same path in 1636 to found the town of Hartford.

The lower road. It ran near the shore, through New London and Providence. That stretch followed the Pequot Path, which Narragansett and Pequot people had traveled for hundreds of years. The path ran about 69 miles, from Providence to New London, Connecticut, and English settlers kept using it until it became part of the lower road.

What Roger Williams saw. The English colonist Roger Williams walked the Pequot Path in 1636. Along 20 miles of it he passed about a dozen Narragansett villages. He wrote: "It is admirable to see what paths their naked hardened feet have made in the wilderness in the most stony and rockie places" (by admirable he meant worth wondering at).

Edward Converse

Who: A Puritan settler and church deacon who ran the first ferry between Charlestown and Boston.
When and where: Boston harbor, licensed June 14, 1631. Lived 1589 to 1663.

The colony's rulers let him charge two pence a passenger.

Edward Converse ran a ferry across the harbor between Charlestown and Boston, and the record that licensed it still exists. The members of the Massachusetts Court of Assistants, the colony's ruling council, wrote it on June 14, 1631: "Edw: Converse hath vndertaken to sett vpp a fferry betwixte Charlton & Boston." That is how people spelled English then. In modern spelling, Converse had taken on the job of setting up a ferry between Charlestown and Boston.

The fares. The same men set his prices. A passenger crossing alone paid two pence, or two of the small English coins called pennies. When two or more crossed together, each paid one penny. Their license was a charter, a written permission from a government.

Two firsts. A history page from the West End Museum in Boston states that Converse started his ferry in 1630, the first on that crossing, and ran it for about ten years. At the time it was the only way across the river there. The history page of Boston's transit agency, the MBTA, names a different first. It states that Thomas Williams opened the first chartered transit service in the United States in 1631, with his own ferry to Charlestown.

> Converse was a Puritan, one of the English Christians who started the Massachusetts colony on land where Native people already lived. He was also a deacon, a church officer who helps the minister. The Charlestown ferry earned 50 pounds in English money in 1639. The next year the colony's leaders gave the ferry's earnings to Harvard College.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 5 MINUTES

Stage lines for passengers and a big freight wagon called the Conestoga came into use in the English colonies between 1700 and 1750. Pennsylvania wagon builders made the Conestoga to haul freight to Philadelphia, and New Jersey officials licensed one of the first stage lines in 1706. The roads were still rough, as Sarah Kemble Knight wrote in her journal after she rode from Boston to New York in 1704.

The wagon and the people it was named for

The Conestoga wagon was a big freight wagon from Lancaster County, Pennsylvania. Its name first shows up in writing on December 31, 1717. That day James Logan, who had been William Penn's secretary, wrote in his account book that he had bought a "Conestogoe Waggon."

German and Swiss wagon builders along the Conestoga River worked out the design. The wagon's bed curved down toward the middle, so the load settled toward the center, and a white cloth cover stretched over hoops above it. Drivers hitched teams of four to six horses to it and hauled loads more than 60 miles to Philadelphia. People in the East used it as their main freight wagon well into the age of railroads.

The name. The wagon took its name from the Conestoga River and from the Conestoga people who lived in Lancaster County. In December 1763 a mob of men from Paxton township, known as the Paxton Boys, killed the last of the Conestoga people. First they killed the six people at home at Conestoga Manor, a settlement under the colony's protection: three men, two women and a boy.

Lancaster officials then put the survivors in the county workhouse, a locked building where people who broke small laws were held and made to work, to protect them. On December 27 the mob broke in and killed them, 14 or 16 men, women and children by different counts. The mob killed about 20 Conestoga people in the two attacks, and no one was ever put on trial for it. Chapter 2 tells the killings in full.

Stage lines, log roads and the Great Wagon Road

A stage line carried passengers in a wagon or coach along a fixed route between towns. New Jersey officials gave Hugh Huddy a patent, a legal right, to run one between Burlington and Perth Amboy in 1706. Coaches ran between New York and Philadelphia from 1732, though not on a fixed schedule.

New lines opened from Trenton to New Brunswick in 1738 and from Bordentown to New York in 1740. By 1756 several regular coach services ran between New York and Philadelphia, and the trip took about three days.

Where a road crossed swampy ground, builders laid logs side by side across it. People called it a corduroy road, after a cloth with raised ribs.

The Great Wagon Road. It ran from Philadelphia down the Shenandoah Valley of Virginia toward Augusta, Georgia, and colonists used it heavily from the 1720s on. Accounts give its length as about 730 miles or almost 800, depending on where the count ends.

South of Lancaster and York, in Pennsylvania, the road roughly followed a route that Europeans called the Great Warriors' Path. Native people had made that path to travel through the Shenandoah Valley, and they still valued it.

The Treaty of Lancaster. Leaders of the Six Nations of the Iroquois, six Native nations joined in one league, met officials of Pennsylvania, Virginia and Maryland in the town of Lancaster in June 1744. There the two sides made a treaty, a written agreement between nations. In it the Iroquois leaders gave up their use of the path and their claims to land in the Shenandoah Valley.

The two sides later disagreed about how much land the treaty covered. Virginia's leaders claimed land as far as the Ohio River and beyond it. The Iroquois said they had given up much less.

Who built the roads. In Virginia, colonists built and repaired the roads themselves, under road laws that the colony's lawmakers passed in 1632 and 1657 and added to in 1661. Each year the members of each county court chose an overseer of the highways, a man put in charge of the roads. He was usually given all the "Labouring Male Titheables" who lived on or near the road.

Titheables, or tithables, were the people counted for a tax on each person: all free men 16 and older, and all enslaved people 16 and older, women as well as men. The road orders required the working men among them to bring their own tools, wagons and teams of animals and to work six days a year on the roads. The surviving road orders do not say how many of these men were enslaved.

Sarah Kemble Knight

Who: A Boston businesswoman who rode from Boston to New York and wrote down what the road was like.

When and where: October 1704, by road from Boston to New Haven and on to New York. Her journal was published in 1825.

She wrote down which bridges she was afraid to cross.

Sarah Kemble Knight traveled with post riders, the men who carried the mail, and hired guides from one town to the next. She crossed some rivers by canoe in the dark and others at fords, shallow places where people and horses can wade across.

Her journal is an eyewitness record of the road itself. It was published in 1825, more than a hundred years after her ride.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 4 MINUTES

Around Philadelphia, between 1750 and 1800, investors paid for a road and charged travelers a fee to use it, and an inventor ran a boat pushed by a steam engine. The investors, people in and around Philadelphia and Lancaster, put their money into the road company in 1792. Their Philadelphia and Lancaster Turnpike, finished in the mid-1790s, was the first long-distance road of its kind in the country.

John Fitch's steamboat carried paying passengers on the Delaware River on a schedule in 1790, seventeen years before Robert Fulton's steamboat ran on the Hudson.

The first turnpike

The Philadelphia and Lancaster Turnpike Company got its charter in 1792 to build a turnpike, a road whose owners charge travelers a fee, called a toll, at gates along the way. Work on the road began that year. People in and around Lancaster and Philadelphia bought 1,000 shares of the company's stock, pieces of ownership in the company, at $300 each. A Philadelphia merchant, William Bingham, led the company.

The road ran 62 miles between Philadelphia and Lancaster, mostly along the older Philadelphia-Lancaster Road, the first stretch of the Great Wagon Road. Its surface was broken stone and gravel. Some accounts say it was in use in 1794, and most say 1795.

KNOWN:	The records name the people who bought shares and the company's president.
NOT KNOWN:	who broke the stone, or how many men did the work.

The Lancaster Turnpike was the country's first long-distance engineered toll road, planned and built to a design, with a prepared surface. After it succeeded, people built many more toll roads, the historian Charlene Mires

writes. By the 1830s the country had almost 12,000 miles of turnpikes, nearly 200 times the length of the first one.

A steamboat before Fulton's

John Fitch tested a steamboat on the Delaware River on August 22, 1787, while delegates to the Constitutional Convention watched. They were the men meeting in Philadelphia that summer to write the Constitution of the United States. A steam engine on Fitch's boat worked rows of oars that pushed the boat through the water, and it moved upriver, against the current, at about three miles an hour.

Fitch's boat, the *Experiment*, carried passengers on a schedule in the summer of 1790. It was the first steamboat service anywhere in the world to carry paying passengers on a schedule. The boat carried up to about 30 passengers at a time on the Delaware, between Philadelphia and New Jersey towns such as Trenton, Burlington and Bordentown, and also to Wilmington.

John Fitch

Who: The builder of a working steamboat in 1787 and of the first steamboat service anywhere to carry paying passengers on a schedule.
When and where: The Delaware River, 1787 to 1790. Died 1798.

George Washington backed a rival's steamboat, not his.

John Fitch had little money when he decided, at about 41, that a boat driven by steam could travel upriver against the current. The country had no national patent law yet. So in 1786 he went from state to state asking lawmakers for a patent, the legal right to be the only one to build and run such boats.

Benjamin Franklin, who had his own ideas about powering boats, showed little interest. Virginia's lawmakers turned Fitch down in favor of the boat of a Virginian, James Rumsey, which George Washington backed. New Jersey's lawmakers gave Fitch the only right to run steamboats on that state's waters for 14 years.

Losing out. His 1790 passenger service failed when too few people bought tickets, the machinery broke down and the people who had put money into his boats stopped paying. In 1791 the members of the federal patent board, led by Thomas Jefferson, gave steamboat patents to both Fitch and Rumsey and would not say that Fitch had been first. Fitch sailed to France and England in 1793 to look for backers and found none.

In 1796 he went to Kentucky, where he had claimed land, and found squatters, people living on it without his permission. He killed himself in Bardstown, Kentucky, in 1798, nine years before Robert Fulton's steamboat began running on the Hudson.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 13 MINUTES

Moving people and goods got much faster and cheaper in the first half of the 1800s, on steamboats, canals and the first railroads. Robert Fulton's steamboat went from New York City to Albany in 32 hours in 1807, a trip that took a sailing boat about four days. After the Erie Canal opened in 1825, hauling a ton of goods from Buffalo to New York City cost under $10, against about $100 by wagon.

Who did the work. American-born laborers and Irish immigrants dug the Erie Canal. In the South, enslaved Black men dug most of the Dismal Swamp Canal, and enslaved people built most of the railroads. Owners rented the men out to the builders and kept their pay, and the men running Southern railroad companies also bought enslaved people outright.

Steamboats on the rivers

Robert Fulton's North River Steamboat left New York City on August 17, 1807, and reached Albany, 150 miles up the Hudson River, 32 hours later. Sailing boats called sloops took about four days to make the same trip. Fulton's boat began carrying passengers on a regular schedule right away. People later called it the *Clermont*.

Many later accounts say that people who doubted the boat called it "Fulton's Folly" (a folly is a foolish project). Few records from 1807 show anyone using that name. It comes from stories told afterward.

The western rivers. Within a few years, steamboats were running on the Ohio and Mississippi Rivers. Twenty-one steamboats arrived at New Orleans in 1814, and 2,770 arrived in 1846, more than 130 times as many. A steamboat covered about 50 miles a day in the 1810s and about 100 by the 1820s.

Much of what the boats carried was cotton, grown and picked by enslaved people whom their enslavers forced to work. Farms in the state of Mississippi grew 20 million pounds of cotton in 1820 and 535 million pounds in 1860. Before the Civil War, more than half of all the cotton grown in the United States passed through New Orleans.

Boilers. Steamboat boilers, the tanks where water was heated into steam under pressure, exploded again and again on the Mississippi and the rivers that flow into it in these years. The explosions killed thousands of people.

Robert Fulton

Who: The man who made the steamboat a business that lasted.
When and where: The Hudson River in New York, from August 17, 1807.

His second boat steamed 1,800 miles from Pittsburgh to New Orleans.

Robert Fulton's partner, Robert Livingston, paid for the building of his boats. Livingston held a monopoly on steamboats in New York, which meant that by law no one else could run them there.

Fulton and Livingston paid for a second boat, the *New Orleans*, built at Pittsburgh. It left Pittsburgh on October 20, 1811, traveled more than 1,800 miles down the Ohio and Mississippi Rivers, and reached New Orleans on January 10, 1812. It was the first steamboat on the western rivers.

The National Road

The National Road was the first highway in the United States built entirely with money from the federal government, the national government in Washington. Members of Congress approved it in 1806, while Thomas Jefferson was president. Work began at Cumberland, Maryland, in 1811. The road reached Wheeling, on the Ohio River, in 1818, and by the 1830s it went on to Vandalia, Illinois, where the federal money stopped.

An older path. West of Cumberland, the road followed a path that a Delaware chief named Nemacolin and a colonist named Thomas Cresap marked and cleared in 1749 or 1750. It ran between the Potomac and Monongahela Rivers, from the site of Cumberland to the mouth of Redstone Creek, where Brownsville, Pennsylvania, now stands.

George Washington used the path on his first trip to the west, and General Edward Braddock's army used it in the French and Indian War. People called it Braddock's Road until the National Road was built on the same route.

Who built it. Irish laborers built sections of the National Road for six dollars a month.

How much did the Erie Canal save?

Hauling a ton of goods from Buffalo to New York City cost about $100 by wagon. On the Erie Canal it cost under $10, and some accounts put it at about $6 a ton.

The Erie Canal was a waterway that people dug so boats could travel where no river ran. Workers dug it from 1817 to 1825, and it stretched 363 miles, from Albany on the Hudson River to Buffalo on Lake Erie. While it was being dug, people who mocked it called it "Clinton's Ditch," after DeWitt Clinton, who pushed for it.

The canal opened on October 26, 1825, and cost about $7.1 million. The tolls that boats paid to use it covered the cost of building it in about nine years.

The hoggees. Boys drove the mules that pulled the canal boats, and people called them hoggees. Many were about 10 years old. A hoggee walked 18 to 20 miles a day, six hours on and six hours off, for about $8 to $10 a month. Many hoggees were cheated out of their pay. Columns on the canal boys in the Lockport *Union-Sun & Journal*, a newspaper in a canal town, tell of the cheating but do not name the men who did it.

Who dug the canals, and whose land they crossed

American-born workers and Irish immigrants dug the Erie Canal. Between 2,000 and 3,000 laborers were working on it in 1818, with a horse for every second man. Ads in Ireland that called for workers promised wages of 8, 10 or 12 dollars a month, and meals came on top of that pay: meat three times a day, and liquor.

The pay. Many men were paid by the day instead, 75 cents to $1.50 a day, and less as more workers arrived. They got no pay on rainy days, but they still had to pay for their lodging. Some contractors, the men who ran sections of the work, started fights among the workers to get out of paying them.

At Rochester, a contractor named William Brittin brought about 30 convicts, people serving prison sentences, from Auburn State Prison to work. Many escaped, and Irish workers took their places.

Malaria. The canal crews reached the Montezuma Marshes, wet ground full of mosquitoes, in 1819. The mosquitoes spread malaria, a disease that brings high fevers and chills again and again. The canal commissioners, the state officials in charge of the canal, reported that about 1,000 workers fell sick between the middle of July and October 1, 1819. Many died, and no one kept a count of the dead.

Whose land. The Erie Canal ran through the homelands of the Oneida, Onondaga, Cayuga and Seneca nations. The Oneida Indian Nation's history of the canal states that the Oneida lost the land for it through a series of 27 treaties with the State of New York, and it describes all 27 as illegal. New York State officials took parts of the Seneca Oil Spring Reservation, land set aside for the Seneca, from 1858 to 1871 for a canal reservoir, a lake made to store water.

In the South. Owners rented out enslaved men to dig canals and kept the men's pay. Enslaved men dug most of the Dismal Swamp Canal, which runs 22 miles from Deep Creek, Virginia, to South Mills, North Carolina, from 1793 to 1805.

Before the canal, George Washington and his brother John Augustine Washington ran a business called the Dismal Swamp Company. Through it they bought 54 enslaved men, women and children to dig ditches and cut timber in the swamp.

DeWitt Clinton

Who: The New York governor who pushed the Erie Canal through while people mocked it.
When and where: New York State, 1811 to 1825.

He predicted the Iroquois would be gone from New York.

DeWitt Clinton pushed for the Erie Canal while critics called it "Clinton's Ditch" and "Clinton's Folly." In 1811 he was the mayor of New York City and a new canal commissioner, one of the state officials planning the canal. Its planned route ran through the land of the Iroquois, six Native nations joined in one league and also called the Six Nations.

The speech. That year Clinton spoke to the New-York Historical Society. He predicted that "before the passing away of the present generation, not a single Iroquois will be seen in this state." He was wrong. More people of the Six Nations live in New York now than lived there at the end of the American Revolution.

The wedding of the waters. As governor, Clinton rode on the *Seneca Chief*, the first boat to travel the whole length of the canal, in 1825. At the opening he poured water from Lake Erie into New York harbor, and people called it the "wedding of the waters."

James Garfield, canal boy

Who: A 16-year-old who worked on an Ohio canal boat. He later became president of the United States.

When and where: The Ohio and Erie Canal, 1848, on the boat *Evening Star*.

At 16 he ran away to drive a canal boat's mules.

James Garfield was 16 when he ran away in 1848 to work on a canal boat called the *Evening Star*. The only job open was driving the mules that pulled the boat. The boat carried freight, goods being shipped, on the Ohio and Erie Canal, and for about six weeks he helped haul coal, copper ore and wood.

Later he caught malaria and went home. His six weeks on the canal made a strong picture for his campaign for president in 1880. Writers of biographies for that campaign told stories of him falling into the canal, fourteen times by their count. That number comes from those later stories, not from the record of his time on the boat.

Moses Grandy

Who: An enslaved man who ran boats through the Dismal Swamp canals, bought his own freedom three times, and described the men who dug the canals.

When and where: Camden County, North Carolina, and the canals to Norfolk, Virginia, from the early 1800s. His book about his life was published in London in 1843.

He paid for his own freedom three times.

Moses Grandy was born into slavery in Camden County, North Carolina, around 1786. His owners hired him out each year to other men. He worked canal boats that carried shingles, boards and other freight through the Dismal Swamp canals to Norfolk.

Three prices. The first time, Grandy paid $600 to his owner, who kept the money and sold him to a man named Trewitt.

The second time, Grandy paid Trewitt $600, money he earned over two and a half years. But Trewitt had pledged Grandy as security for a loan, which meant the lender could take Grandy if the loan was not paid back. When Trewitt's business failed, the lender took Grandy and sold him to Enoch Sawyer, who put him to work in the fields.

The third time, a captain from Deep Creek, Edward Minner, lent Grandy $650, and Sawyer set him free for that price. Grandy paid Minner back over three years.

His wife. While Grandy worked on the canal, his wife's owner sold her. A group of enslaved people passed along the road beside the canal, and she called out to him. The man who had bought her, Mr. Rogerson, pulled out a pistol and said he would shoot Grandy if he came near the wagon. Grandy wrote: "I have never seen or heard of her from that day to this."

The diggers. Grandy also described the enslaved men who dug the canals. They worked "up to the middle or much deeper in mud and water, cutting away roots and baling out mud" (baling out means scooping out).

They slept in open huts, still covered in mud, with no bedding. Once a month their owners came to collect the men's pay.

An overseer named Wiley M'Pherson, the man in charge of the workers, gave every man the same amount of digging each day. When men had not finished the day before, M'Pherson tied them up and flogged them. Flogging means whipping a person's bare back until the skin splits.

Then he had "pork or beef brine," the salt water used to keep meat, put "on their bleeding backs, to increase the pain." He left men tied all day with their feet barely touching the ground, while flies and mosquitoes settled on the cuts.

Grandy wrote: "I have seen him flog slaves with his own hands, till their entrails were visible" (entrails are the organs inside the body). He went on: "I have seen the sufferers dead when they were taken down. He never was called to account in any way for it."

After freedom. Grandy later married again. He lived in Boston, sawed wood, worked in coal yards, loaded and unloaded ships, and went on more voyages. He saved enough to buy his second wife out of slavery in Virginia, and she joined him in Boston.

He raised $450 to buy his son's freedom. His daughters in New Orleans paid for their own freedom, $1,200 for each of them. His book was sold to raise money for his relatives who were still enslaved.

The first railroads

The Baltimore and Ohio Railroad was the first railroad in the United States to carry both passengers and freight on a schedule for anyone who paid. Its builders laid the first stone on July 4, 1828. Its first 13 miles of track, from Baltimore to Ellicott's Mills, opened in 1830.

A legend. In a well-known story, Peter Cooper's steam engine, the *Tom Thumb*, raced a horse pulling a railroad car on August 28, 1830, and lost when a belt on the engine slipped. The story comes from a talk that John H. B. Latrobe gave in 1868, 38 years later. No record from 1830 shows that the race took place, and the Baltimore and Ohio Railroad Museum's page on the *Tom Thumb* describes the race as a legend.

New York State's first steam passenger train ran on August 9, 1831, behind a locomotive, or train engine, named the *DeWitt Clinton*. It went 16 miles from Albany to Schenectady in about an hour, on the Mohawk and Hudson line.

In the city. Abraham Brower ran a horse-drawn coach called an omnibus on the streets of New York City from 1827. The *John Mason*, a horse-drawn car that rolled on rails, began carrying passengers in lower Manhattan for the New York and Harlem Railroad on November 14, 1832. It was the first street railway in the country.

Who built the Southern railroads. In the South, enslaved people built most of the railroads. The men running Southern railroad companies bought hundreds of enslaved men between the ages of 16 and 35 as early as 1841, and in the 1850s they bought or rented enslaved workers in groups of hundreds.

The historian Theodore Kornweibel Jr. found enslaved workers on 85 of the 113 railroads in the Confederate states, the Southern states that later fought the United States in the Civil War. Southern railroad companies owned about 15,000 enslaved people in 1860, and women and children were among them.

The president of the Mississippi Central Railroad told the people who owned shares in the company in 1855 that "the slave is preferable to free labor, and far better adapted to the construction of railways in the south." By free labor he meant workers who were not enslaved.

BY 1850

If you remember one thing: Steamboats, canals and the first railroads made moving people and goods far faster and cheaper, and in the South enslaved people did much of the digging and track-laying.

- 1807: Fulton's steamboat reaches Albany in 32 hours. Sloops took about four days.
- 1825: The Erie Canal opens. Hauling a ton from Buffalo to New York City drops from about $100 by wagon to under $10.
- 1841: The men running Southern railroad companies are buying enslaved men by the hundreds.

Next: After 1850 railroad builders pushed track across the whole country, through the homelands of many Native nations.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 14 MINUTES

Railroad builders laid track across the whole country in the second half of the 1800s. The country had about 9,000 miles of railroad track in 1850 and more than 190,000 by 1900, more than twenty times as much. Counts for 1900 run from about 193,000 to 215,000 miles, depending on whether side tracks are counted.

Coast to coast. The first railroad across the country was finished in 1869. Central Pacific managers hired about 10,000 to 12,000 Chinese workers to build its western part. The tracks crossed the lands of the Cheyenne, Lakota, Arapaho, Shoshone and Paiute nations.

In the cities, streetcars ran on rails laid in the streets, and by the end of the century the companies that ran them were switching from horse-drawn cars to electric ones. Many Americans, women among them, were riding bicycles.

The railroad across the continent

Two companies built the first railroad across the country. Union Pacific crews built west from Omaha, Nebraska, while Central Pacific crews built east from Sacramento, California. The two lines met at Promontory Summit, Utah, on May 10, 1869.

Land and loans. Members of Congress voted to give the two companies land and loans, and President Abraham Lincoln signed their Pacific Railroad Act on July 1, 1862. Under it, each company got 6,400 acres of public land, ten square miles, for every mile of track it built. Each also got up to $48,000 in government bonds, a kind of loan, for every mile. The top rate was for miles built through mountains.

The act also stated: "The United States shall extinguish as rapidly as may be the Indian titles to all lands falling under the operation of this act" (to extinguish a title means to end someone's ownership of land). With those words, the lawmakers directed federal officials to end Native nations' ownership of the land along the railroad as fast as they could.

Members of Congress voted to give railroads about 130 million acres of federal land from 1850 to 1871, and some counts run as high as 175 million acres. State lawmakers gave railroads about 51 million acres more.

Whose land the railroad crossed

The companies' builders and US officials took land from many Native nations while the two lines were built, and they drove people from their homes. Among those nations were the Cheyenne, Lakota (also called Sioux), Arapaho, Shoshone and Paiute. Historians at the Utah Division of State History count 15 Native homelands that the track crossed.

The bison. The Lakota, Arapaho and Cheyenne hunted bison, also called buffalo, for food, shelter and trade. As the railroad crews pushed west, soldiers, railroad workers and travelers hunted the bison too. National Park Service writers state that the number of bison fell from tens of millions in the early 1800s until they had almost died out.

Sand Creek. The railroad crews worked in the middle of a war between the US Army and the Cheyenne, Arapaho and Lakota nations. US soldiers on horseback led by Colonel John Chivington attacked a camp of Cheyenne and Arapaho people at Sand Creek, in Colorado, on November 29, 1864. The camp flew a US flag and a white flag.

The soldiers killed about 230 people, most of them women and children, and the people they killed were not armed. Britannica and the National Park Service give that count, and PBS writers give 150. Chapter 2 tells Sand Creek in full.

Striking back. Cheyenne, Arapaho and Lakota fighters struck back for Sand Creek. They burned the town of Julesburg, Colorado, on January 7, 1865, and cut the telegraph line, the wire that carried messages by electric signals.

Guards for the crews. Six days later, on January 13, 1865, four army units of Pawnee soldiers, called companies, joined the US Army under Major Frank J. North. Union Pacific managers and US officials hired hundreds of Pawnee, who guarded the work crews alongside the Army. Some Paiute people worked for the Central Pacific.

The fighting reached the tracks. At Plum Creek, Nebraska, on August 27, 1867, Cheyenne men pulled up railroad track, wrecked a work train and killed all but one of its crew.

The Chinese workers

Central Pacific managers hired about 10,000 to 12,000 Chinese workers from 1865 to 1869. Most came from Guangdong, a province in southern China. The Chinese workers were paid about $35 a month, and they had to buy their own food and housing. White workers got about $42 a month, plus housing.

The dead. The Central Pacific's managers kept no record of the Chinese workers who died, so no one knows how many there were. Avalanches, masses of snow sliding down the mountains, killed workers in the winter of 1866 to 1867, when 44 storms hit. In one, snow carried away the camp at Strong's Canyon.

Blasts of black powder and nitroglycerin, explosives used to cut through rock, killed others. Smallpox, a disease that spreads from person to person, killed more in 1868 and 1869.

Guesses of the number run from about 50 into the thousands. On June 30, 1870, a writer at the *Sacramento Union* reported "about fifty." The same day, a writer at the *Sacramento Reporter* told of "the accumulated bones of perhaps 1,200," about 20,000 pounds of bones being shipped home to China. Researchers on the online forum of the Central Pacific Railroad Photographic History Museum estimate 100 to 150 dead, and other estimates run to hundreds or thousands.

The strike. Chinese workers in the Sierra Nevada mountains went on strike on June 25, 1867, which means they stopped work to demand better terms. Estimates of how many struck run from 2,000 to 5,000. They asked for $40 a month instead of $35, for workdays of no more than ten hours, and for shorter shifts in the cramped tunnels. It was the largest strike by workers in the country up to that time.

The construction boss, Charles Crocker, and his foreman, James Harvey Strobridge, who ran the work crews, cut off food and supplies to the Chinese camps. After a week with little food, Crocker told the strikers that men who went back at once would only be fined. Men who stayed out would lose all their pay for June.

Most men, weak from lack of food, went back. The men who refused faced a group of armed white men, and work started again. Crocker gave the strikers no raise, though pay for experienced men later rose a little, according to the Library of Congress.

Ten miles in a day. Crews laid 10 miles and 56 feet of track in one day on April 28, 1869. Eight Irish workers carried the rails, and several thousand Chinese workers did the rest of the work.

The last track. Twelve days later, on May 10, 1869, a different crew of eight, all of them Chinese, laid the last track at Promontory Summit, joining the two lines, and a golden spike marked the place. Three of those eight, Ging Cui, Wong Fook and Lee Shao, rode on a float at the celebration of the railroad's 50th anniversary on May 10, 1919.

Hung Wah

Who: The leading Chinese headman on the Central Pacific.
When and where: The Central Pacific Railroad, from January 1864.

The payroll names him, and not most of his workers.

Hung Wah is one of the few Chinese railroad workers whose names appear in the records. The company's payroll, its list of the people it paid, usually named only headmen, the men who led the Chinese work gangs. The names of most of the thousands of Chinese laborers are not in the records.

Hung Wah first appears on the Central Pacific's payroll in January 1864. With another headman, Ah Toy, he supervised the railroad's first Chinese crew, about 23 men. He became the leading Chinese headman on the line, over hundreds of Chinese workers and at times thousands. He handled the workers' pay and their living arrangements, and he dealt with the company for them.

Cornelius Vanderbilt

Who: A ferryman who moved his money from steamboats into railroads and left the largest American fortune of his time.

When and where: New York, 1810 to 1877.

The ferryman of 1810 died in 1877 with about $100 million.

Cornelius Vanderbilt took control of the New York Central Railroad, in the East, in 1867, the same year Chinese crews were building the Central Pacific through the Sierra Nevada. He had owned steamboats, and early in the 1860s he moved his money into railroads. He joined the New York Central with his Hudson River Railroad on November 1, 1869, and in 1871 he opened Grand Central Depot, a train station in Manhattan.

The ferryman. Vanderbilt had started at 16, in 1810, carrying freight and passengers between Staten Island and Manhattan in a periauger, a shallow sailboat with two masts. Other boat captains called him "the Commodore," the title of a navy officer who commands a group of ships, at first as a joke.

Two stories. Accounts of how he got the boat differ. One says he borrowed $100 from his mother and named the boat *Swiftsure*. The first account of his life, printed in 1853, says the boat was his father's and he kept half the profit. The website Biography.com describes the story of the young ferryman as a legend.

When he died in 1877, he left the largest American fortune of his time.

Mark Twain, steamboat pilot

Who: Samuel Clemens, who steered steamboats on the Mississippi River before he wrote books as Mark Twain.

When and where: The Mississippi River. He got his pilot's license on April 9, 1859.

Movie: *Mark Twain* (2001), a documentary by Ken Burns about his whole life. PBS showed it in January 2002.

"Mark twain" was a river call meaning twelve feet of water.

Steamboats still ran on the Mississippi River while the railroads spread. Samuel Clemens was a licensed steamboat pilot, the person who steers the boat, on the Mississippi. He got his license on April 9, 1859, when he was 23. He piloted boats until 1861, when the Civil War began and traffic on the river stopped.

He took his pen name from a call used on the river. A leadsman, the crew member who measured the depth of the water, called the depth out to the pilot. "Mark twain" meant two fathoms, and a fathom is six feet, so the call meant twelve feet of water. In 1883 he published *Life on the Mississippi*, a book about his years as a pilot.

Clocks, sleeping cars and danger on the railroads

Time zones. Railroad managers in the United States and Canada began using standard time zones, regions where all clocks show the same time, on November 18, 1883. Before then, each railroad company ran its trains by its own clocks. William F. Allen, the editor of the *Traveler's Official Railway Guide*, a book of train schedules, drew up the plan.

On the day of the switch, railroad workers reset every railroad clock to the new time, even clocks that had already passed noon. In those places noon came twice, so people called it "the day of two noons." Members of Congress did not vote time zones into law until the Standard Time Act of March 19, 1918.

Sleeping cars. George Pullman built the *Pioneer*, his first sleeping car built for luxury, in 1865. It had upper beds that folded away and seats that pulled out. His company, the Pullman Palace Car Company, began in 1867.

Danger. About 8.5 of every 1,000 American trainmen, the workers who ran the trains, were killed on the job in 1889, twice the rate in Britain.

In the same year, 1.73 of every 1,000 trainmen died while coupling, the job of joining railroad cars together.

With the old link-and-pin coupler, a brakeman, the crew member who set the brakes and joined the cars, had to stand between two moving cars to join them. Mangled hands were known as "the badge of the experienced worker."

Safer equipment. Safer couplers and brakes existed for years before lawmakers required railroad owners to use them. Eli Janney got a patent, the legal right to be the only one to make and sell an invention, for an automatic coupler in 1873. It joined cars without a worker standing between them. George Westinghouse's air brake existed from 1869. A pump on the locomotive pushed air through hoses to every car, and when the engineer opened a valve, the air set the brakes on all the cars at once.

Members of Congress passed the Safety Appliance Act in 1893. Under it, railroad owners had to put automatic couplers and air brakes on trains that crossed state lines, starting in 1900. Accidents while coupling cars fell from

about 11,000 in 1892 to about 2,000 in 1902, while the number of railroad workers grew.

Who decided where Black passengers sat?

Train crews and lawmakers decided where Black passengers could sit. Ida B. Wells had a first-class ticket on a train in Tennessee in 1884, but the train crew ordered her into the car for Black passengers. She refused, and they forced her off the train. She sued the railroad and won $500 in a Tennessee court. The judges of the Tennessee Supreme Court threw out that decision in 1887.

Plessy. Louisiana lawmakers passed the Separate Car Act on June 10, 1890. Under it, railroads had to provide "equal but separate" cars for Black and white passengers. A passenger who broke the law could be fined $25 or jailed for 20 days.

Homer Plessy, a shoemaker in New Orleans, offered to test the law. To test it, he would break it on purpose so that judges would have to decide whether the law was allowed. He bought a ticket on the East Louisiana Railroad on June 7, 1892, and sat in the car for white passengers. The railroad's officials agreed to the test, because separate cars cost the company money. A private detective hired by the group planning the test, the Citizens' Committee, arrested him.

Judge John H. Ferguson ruled against Plessy. On May 18, 1896, the justices of the US Supreme Court ruled against him too, 7 to 1, in *Plessy v. Ferguson*. With that ruling, the justices allowed states to keep separating Black and white passengers by law. Nearly sixty years later, in 1954, the justices of the Supreme Court threw out that ruling's reasoning in *Brown v. Board of Education*.

Electric streetcars and bicycles

Electric streetcars replaced horse-drawn ones in American cities between 1888 and 1902. Electric streetcars had run earlier in East Cleveland in 1884, in Baltimore in 1885 and in Montgomery, Alabama, in 1886. The Dictionary of American History describes the system that Frank Sprague built in Richmond, Virginia, in 1888 as the first successful one.

The owners of about 200 streetcar systems had switched to electricity or planned to by 1891. Street railway track grew from about 8,000 miles in 1890 to 22,500 miles in 1902. In those same years, the share of streetcar service pulled by animals fell from 75 percent to 1 percent. The last horse-drawn streetcar in paid service in New York City ran on July 26, 1917.

Bicycles. Cycling became a craze in the 1890s, something that very many people took up at the same time. Riders used the safety bicycle, which had two wheels of the same size, a chain drive and air-filled tires. Women rode as well as men.

Susan B. Anthony told the reporter Nellie Bly in the *New York World* on February 2, 1896, that bicycling "has done more to emancipate women than anything else in the world" (to emancipate is to set free). She said: "I stand and rejoice every time I see a woman ride by on a wheel."

Frank Sprague

Who: The engineer whose streetcar system in Richmond showed that electric streetcars could work in a city.

When and where: Richmond, Virginia. Regular service began February 2, 1888.

Ten electric cars began running in Richmond on February 2, 1888.

Frank Sprague built the Richmond Union Passenger Railway, which began regular electric streetcar service that day with 10 cars. By May 1888 it had about 12 miles of track and 23 cars.

Within about two years, about 110 electric railways had been built or ordered using Sprague's equipment. Members of the Institute of Electrical and Electronics Engineers, a society of engineers, have named his Richmond railway a milestone in the history of engineering. Chapter 6 tells how builders put up new neighborhoods along the electric lines.

BY 1900

If you remember one thing: Railroad builders laid track from coast to coast, across the homelands of many Native nations, and Chinese workers built the western part of the first line.

In 1850: the country had about 9,000 miles of railroad track, and a trip from one coast to the other took about four months.

In 1900: it had more than 190,000 miles of track, and since 1869 the trip had taken about a week.

Next: After 1900 Ford's workers built cars by the millions, and airplanes began carrying mail and passengers.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 10 MINUTES

Orville and Wilbur Wright flew a powered airplane in 1903, and Ford's Model T car went on sale five years later, for $825. Workers at Ford built 15 million Model Ts by 1927. The first airline to fly on a schedule began in 1914, across Tampa Bay in Florida, and by 1927 airplanes were carrying mail and paying passengers on schedules.

Trains carried about 43.7 million members of the armed forces in World War II, from 1941 to 1945. In Virginia, bus drivers had the power of police officers to make Black passengers sit apart from white passengers and to arrest riders who refused.

The Model T and the new roads

The Ford Model T went on sale on October 1, 1908, at $825, and its price fell to $260 by 1925. From 1913 Ford's workers built it on a moving assembly line, and the time it took to build a car's frame fell from more than 12 hours to about 93 minutes. Chapter 8 tells how the line worked. The 15 millionth Model T left the factory on May 26, 1927, the last day Ford's workers made the car.

The Lincoln Highway. A group of private backers, among them Carl Fisher and Henry Joy, laid out the Lincoln Highway in 1913. It was the first named road for cars across the whole country, from Times Square in New York City to San Francisco. After 1926 highway officials made parts of it into sections of numbered routes.

Route 66. State and federal highway officials gave the main highways of the United States numbers on November 11, 1926, and Route 66 got its number that day. It ran 2,448 miles, from Chicago to Santa Monica, California, and people called it the "Main Street of America." It was fully paved only by the late 1930s.

Crashes. Car crashes killed about 4,100 to 4,200 people in the United States in 1913. In 1937 they killed about 37,800 to 39,600, about nine times as many, which was about 30 of every 100,000 Americans. The counts differ because

the National Safety Council, a private safety group, and the federal officials who kept the national tables counted deaths in different ways.

The Wright brothers

Who: Orville and Wilbur Wright, the brothers who flew their powered airplane four times in one day.
When and where: Near Kill Devil Hills, North Carolina, December 17, 1903.

The first flight lasted 12 seconds and covered 120 feet.

Orville and Wilbur Wright made four flights in a powered airplane near Kill Devil Hills, North Carolina, on December 17, 1903. Orville flew first. Wilbur made the fourth and last flight of the day, 852 feet in 59 seconds. Five people watched, and one of them, John T. Daniels, took the photograph. He belonged to a lifesaving station, a crew that rescued people from wrecked ships.

About ten years after those flights, an airplane in Florida was carrying paying passengers on a schedule. Chapter 8 tells how the brothers built their airplane.

How the airlines began

The St. Petersburg-Tampa Airboat Line in Florida was the first airline in the world to fly on a schedule. Its first flight left on January 1, 1914. The pilot, Tony Jannus, flew a Benoist flying boat, a plane that takes off from and lands on water. The trip across Tampa Bay took about 23 minutes, low over the water.

Abram C. Pheil, a former mayor of St. Petersburg, paid $400 at an auction for the first round-trip ticket. The airline ran for about four months.

Airmail. Army pilots began flying mail on a regular route on May 15, 1918, 218 miles from Washington to Philadelphia to New York. Post Office pilots took over in August 1918. Members of Congress passed the Kelly Act in 1925, and under it private companies flew the airmail on contract. The airline companies began with those mail contracts.

After Lindbergh. Charles Lindbergh, an airmail pilot, flew the *Spirit of St. Louis* alone and without stopping from New York to Paris on May 20 and 21, 1927, about 3,600 miles in 33 hours and 30 minutes. Chapter 1 tells the flight in full. Airline passengers in the United States rose from 5,782 in 1926 to 173,405 in 1928, about 30 times as many. Applications for pilot licenses tripled in 1927.

The DC-3. The Douglas DC-3 airliner first flew passengers in 1936. It carried 21 or more passengers at up to 207 miles an hour. In the words of van der Linden, a curator, or collection keeper, at the Smithsonian's National Air and Space Museum, the DC-3 is "widely considered to be the first airliner capable of making money just by carrying passengers." By 1939 DC-3s carried about 90 percent of the world's airline travel.

Jack Knight

Who: An airmail pilot who flew through the night and a snowstorm to finish a test that kept the airmail going.

When and where: North Platte, Nebraska, to Chicago, the night of February 22 to 23, 1921.

The next pilot would not fly in the storm, so Knight did.

Post Office officials ran a test in February 1921 to show that airplanes could carry mail across the country faster than trains. Pilots flew the mail in a relay, day and night, each pilot handing it to the next. Jack Knight flew his part, from North Platte, Nebraska, to Omaha.

At Omaha the next pilot refused to take off in the storm. Knight flew the mail on to Chicago himself, at night, through snow and fog, over ground he had never flown before. The mail crossed the country, 2,629 miles, in 33 hours and 20 minutes. After the test, members of Congress voted money to keep the airmail flying.

The sleeping-car porters

Pullman Company managers hired Black men as the porters who served passengers in the company's sleeping cars. After the Civil War, George Pullman looked for men who had been enslaved to hire as his porters. By the early 1900s the Pullman Company was the largest single employer of Black men in the United States.

400 hours. Porters often had to work at least 400 hours a month to earn a full month's pay. A porter who worked every day of a 30-day month would still need more than 13 hours a day to reach 400. Their pay was among the lowest of all the workers on the train. Company managers built tips from passengers into the porters' pay, which saved the company money. To earn those tips, porters had to do whatever passengers asked.

"George." Passengers called porters "George," whatever their real names were, after George Pullman. Calling an enslaved person by the owner's name was a practice from slavery. Some passengers called the porters "boy."

The porters formed a union, a group of workers who bargain together with their employer. The union's leaders won a contract in 1937 that limited the porters' work to 240 hours a month.

The porters of Miles of Smiles

Who: C. L. Dellums, E. D. Nixon and six other retired Pullman porters, with Rosina Tucker, a porter's wife, who described the porters' work in their own words.

When and where: Pullman sleeping cars across the country, from the 1910s to the 1940s. Filmed in 1982.

Movie: *Miles of Smiles, Years of Struggle* (1982), a documentary by Jack Santino and Paul Wagner, made from the porters' own accounts.

On the job, the porters had to smile for the passengers.

Eight retired Pullman porters told two filmmakers in 1982 what their job had been like. Rosina Tucker, a porter's wife who worked to build the porters' union, tells the story in the film. The porters describe the smile Pullman managers required them to wear and the nights they worked without sleep.

Two of the men in the film are C. L. Dellums and E. D. Nixon. Pullman managers fired Dellums in October 1927 for organizing porters into a union. Nixon later called the meeting that organized the Montgomery bus boycott. From December 1955 to December 1956, most of the 40,000 Black people of Montgomery, Alabama, stayed off the city's buses, until federal judges ended bus segregation there. Chapter 25 tells the boycott in full.

Trains and Jeeps in World War II

Trains carried about 43.7 million members of the armed forces from December 1941 to August 1945, more than 97 percent of all the troops moved in organized groups. About 100,000 of them rode troop trains on an average day, and half of all Pullman sleeping cars were in use carrying troops.

The Jeep. About 640,000 Jeeps, small army cars, were built in the war. About 363,000 of them were the Willys MB, and about 280,000 were the Ford GPW.

Drivers with the power to arrest

Under Virginia law, buses had to seat Black and white passengers apart, and bus drivers were "special policemen." A driver could decide a passenger's race, order the passenger to another seat, throw the passenger off the bus and make an arrest. Refusing a driver was a crime with a fine of $5 to $25. Greyhound's own rules stated that the company had "full control and discretion as to the seating of passengers" (discretion means the power to decide).

Riders thrown off. Bessie Nelson and Mamie Kinchlow were thrown off a Greyhound bus in Alexandria, Virginia, in August 1932 and jailed for more than a week. Mildred Turpin and Caroline Johnson, clerks for the Navy Department, were jailed at Clarendon, Virginia, on January 16, 1943. Tonito Holmes was beaten, removed from a Greyhound bus and fined $14.25 in Johnson City, Tennessee, on February 19, 1945. The Encyclopedia Virginia account of these cases does not name the drivers and officers who threw the riders off, jailed them or beat them.

Irene Morgan

Who: A Baltimore war worker who refused to give up her bus seat in Virginia and won her case in the US Supreme Court.

When and where: A Greyhound bus near Saluda, Virginia, July 16, 1944. The Supreme Court decided her case on June 3, 1946.

She tore up the sheriff's warrant and threw it out the window.

Irene Morgan, of Baltimore, was 27 and built B-26 bombers in a wartime factory. She had been visiting her mother in Gloucester County, Virginia, and on July 16, 1944, she took a Greyhound bus home to her husband and to see a doctor. Near Saluda, two white passengers got on the crowded bus. The driver told Morgan and the woman next to her, a Black woman holding a baby, to move to the back seat.

"I refused to move," Morgan said years later. The driver drove the bus to the jail in Saluda, where the sheriff showed her a warrant, a paper that allows an arrest. She "just took it and tore it up and just threw it out the window." "So then he put his hands on me," she said, and she kicked him. The sheriff and a deputy forced her off the bus.

The courts. Her mother paid $500 to get her out of jail. Morgan paid a $100 fine for resisting arrest on October 18, 1944. She would not admit breaking the seating law. A judge found her guilty of it anyway, and she refused to pay the $10 fine. The Encyclopedia Virginia account of her case does not name the judge.

Her lawyer was Spottswood W. Robinson III, and the lawyers Thurgood Marshall and William H. Hastie of the NAACP, a civil rights group, joined him. The judges of the Virginia Supreme Court of Appeals ruled against her on June 6, 1945.

The justices of the US Supreme Court ruled for her on June 3, 1946. Justice Stanley Reed wrote that a passenger crossing several states could meet a different seating rule in each one, and under Virginia's law might

have to change seats again and again inside Virginia. Seating on trips across state lines, he wrote, needed "a single, uniform rule," so Virginia's law could not apply to those buses.

The ruling covered only trips between states. Reporters at the *Afro-American* newspaper wrote that bus companies' seating rules did not change right away.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 11 MINUTES

Three new ways of moving people and goods over long distances began between 1956 and 1958: Interstate highways, shipping containers and American jet airliners across the Atlantic. President Dwight Eisenhower signed the law for the Interstates, the big highways that join the states, in 1956, the same year a ship called the *Ideal X* carried 58 shipping containers from New Jersey to Texas.

Pan Am's jets crossed the Atlantic about twice as fast as the older planes. Private railroads were losing money on passenger trains, and in 1971 a new company, Amtrak, took over most of them.

Homes taken. Highway officials forced the people of more than 475,000 households, more than a million people, out of their homes to make room for roads paid for by the national government. That number is the US Department of Transportation's estimate, a figure worked out rather than an exact count.

The Interstate highways

President Eisenhower signed the Federal-Aid Highway Act on June 29, 1956. The law allowed for 41,000 miles of new highways, called the Interstate Highway System. It was the largest public works program, a set of projects that the government pays for and the public uses, in the history of the United States up to then. The system is about 48,900 miles long today, and in 1990 members of Congress added Eisenhower's name to it.

The new Interstates went around Route 66. Officials of AASHTO, the association of state highway officials, took Route 66 off the list of numbered US highways on June 27, 1985.

Dwight Eisenhower

Who: The president who signed the Interstate highway law, 37 years after he crossed the country in an Army motor convoy.

When and where: The 1919 convoy from Washington, D.C., to San Francisco. The highway law, June 29, 1956.

In 1919 an Army convoy needed 62 days to cross the country.

The Army's first convoy of motor vehicles, a group of 81 vehicles traveling together, crossed the country in 1919. It went 3,251 miles, from Washington, D.C., to San Francisco. Its vehicles sank in mud and broke down, and bridges failed under them.

Dwight Eisenhower was a young Army officer, a lieutenant colonel, and he volunteered to go. He wrote a report on the trip. At the end of World War II, in 1945, he saw the autobahns, Germany's highways. After those two experiences, he argued for a system of highways across the United States, and as president he signed the law for it on June 29, 1956.

The homes the highways took

The people who planned the Interstates ran many of them through city neighborhoods, and they took the homes that stood in the way. By the US Department of Transportation's estimate, highway officials took the homes of more than 475,000 households, more than a million people, for federal roads.

The planning scholar Alan Altshuler estimated in those years that the new system would push out up to a million people, "a disproportionate number of them Black," meaning more than their share. The law professor Deborah N. Archer says the neighborhoods chosen "were largely Black and poor."

Montgomery. Interstate 85 ran south of Montgomery, Alabama, in the plans drawn up in 1955. Sam Engelhardt became head of the Alabama State Highway Department in 1959. His business cards read "I Stand for White Supremacy and Segregation," a stand for white rule and for keeping Black and white people apart by law.

Under Engelhardt, state highway officials chose a final route through Montgomery's only middle-class Black neighborhood. Ralph Abernathy, a minister and civil rights leader, lived there, and Alabama State University stood next to it. An engineering firm had suggested a route through Oak Park, which city officials had closed rather than let Black people use it.

Highway officials took about 356 homes in the Centennial Hill, Bel Air and Bottoms neighborhoods and paid owners an average of $3,300 a house. Abernathy sent a telegram, a short message sent over electric wires, to President John F. Kennedy in 1961. He wrote that the route "will destroy one of the best negro neighborhoods in the south" ("negro" was the common word for Black people in his time). Rebecca Retzlaff, a researcher at Auburn University, told this history in a 2019 study.

Rondo. Highway officials built Interstate 94 through Rondo, a Black neighborhood in St. Paul, Minnesota, from 1956 to 1968. Counts of what was lost differ: about 600 families by one count, about 700 homes and 300 businesses by another. Chapter 6 tells Rondo's story in full.

Jet airliners, a crash and cheaper fares

The Grand Canyon crash. Two airliners, TWA Flight 2 and United Air Lines Flight 718, left Los Angeles three minutes apart on June 30, 1956, and flew into each other over the Grand Canyon in Arizona. The crash killed all 128 people on the two planes, the deadliest airplane crash in the world up to that time. It was a major reason the Federal Aviation Agency was set up in 1958.

The jet. A Pan Am Boeing 707 jet flew from New York to Paris with 111 passengers and 12 crew members on October 26, 1958, in about 8 hours. It was the first jet service across the Atlantic by an American airline. The 707 flew about twice as fast as the older propeller airliners and carried about twice as many people.

Fares. Until 1978 the members of a federal board, the Civil Aeronautics Board, or CAB, decided which airlines could start or stop flying and what they could charge. President Jimmy Carter signed the Airline Deregulation Act on October 24, 1978, and under that law the board lost its control of fares and routes. Under the CAB Sunset Act, a part of that law, lawmakers set dates for the board to lose its powers one by one, and the board shut down on December 31, 1984.

Average fares fell about 40 to 45 percent in the decades after 1978, counted in dollars adjusted for rising prices, by government and industry figures. Two economists, experts on money, prices and trade, studied the change for a research group, the Brookings Institution, and found that fares ended up lower than they would have been under the old rules.

Barbara "Dusty" Roads

Who: An American Airlines stewardess who fought the airline's rule that fired women at age 32 or when they married.

When and where: American Airlines, from 1950. She lived from 1928 to 2023.

Movie: *Fly With Me* (2024), a PBS *American Experience* documentary about flight attendants, directed by Sarah Colt and Helen Dobrowski. Roads appears in it.

American's managers fired stewardesses at 32 or when they married.

Barbara Roads, called Dusty, joined American Airlines in 1950 as a stewardess, a woman who served passengers on an airplane. The job is now called flight attendant. In a 1953 contract, American's managers put in a rule that any stewardess hired after November 1953 would be fired at age 32. A stewardess who married was fired too.

In a radio interview, Roads gave her view of the reason: "It was cheaper to have someone that'll never get married, never have babies, never get pensions, never get the highest rate of pay."

Fighting the rule. Roads lobbied members of Congress, trying to talk them into acting, on her own flights to Washington, D.C. In 1963 she held a news conference that made the papers. At news conferences she also asked reporters to pick the 32-year-old stewardesses out of a lineup. Representative Martha Griffiths helped her.

The stewardesses came close to a strike, a refusal to work until their demands were met, in August 1968. By Roads's account, American's managers dropped the age rule the day after someone asked Griffiths to press the Equal Employment Opportunity Commission, the federal office that handles complaints of unfair treatment at work.

Roads helped start the Association of Professional Flight Attendants, a union, in 1977. She worked for American Airlines for 44 years.

The Freedom Rides

Black and white riders traveled together by bus into the South in May 1961, and people called them Freedom Riders. They were testing a court order that banned separating Black and white travelers in bus stations used for trips between states. Two groups left Washington, D.C., on May 4 on a Greyhound bus and a Trailways bus, headed for New Orleans. Among the Black riders was John Lewis, who later led the Student Nonviolent Coordinating Committee, a civil rights group.

Anniston. A white mob with clubs and bats attacked the Greyhound at Anniston, Alabama, on May 14. They broke its windows and slashed its tires, then chased the bus until the flat tires stopped it several miles out of town. One of the attackers threw a firebomb, a bomb that starts a fire, through a window. The riders and other passengers barely got out of the flames.

A state officer riding the bus in plain clothes held the attackers back with his gun. The mob beat the riders while the bus burned, until police arrived. Later that day white men got on the Trailways bus at Anniston and beat the riders on it. Their beating left one rider, Walter Bergman, a white man, with permanent brain damage.

Birmingham. The same day, a crowd met the Trailways bus in Birmingham with pipes, chains and baseball bats. Many in the crowd belonged to the Ku Klux Klan, a group that used violence against Black people and their supporters. Eugene "Bull" Connor, the city official in charge of the police, had told Klan leaders that the police would hold back from the bus station for 15 minutes, and he encouraged the Klan to attack the riders.

With no police there, Klansmen beat the riders in the waiting room and in a dark hallway. They beat James Peck, a white rider, until he was unconscious, and he needed more than 50 stitches. Walter Bergman was among the riders they beat, for the second time that day.

Montgomery. A white crowd in Montgomery attacked a new group of riders arriving by Greyhound on May 20. They beat John Seigenthaler, a Justice Department official, until he was unconscious. Alabama's public safety director

had arranged protection for the buses up to Montgomery, but Montgomery police did not take it over, and the director went into the fighting himself. Governor John Patterson blamed the riders for the violence.

The ban. The rides went on, and over the summer of 1961 the number of Freedom Riders grew to more than 400. Justice Department officials asked the members of the Interstate Commerce Commission, the federal officials in charge of trade and travel between states, to act. The commissioners banned keeping Black and white passengers apart in bus and train stations. They issued the ban on September 22, 1961, and it took effect on November 1, 1961. People in many Southern towns kept ignoring it.

Amtrak takes over the passenger trains

Private railroad companies were losing money on their passenger trains. Members of Congress passed the Rail Passenger Service Act in 1970 and created a new company called Amtrak. Amtrak began running most of the passenger trains between American cities on May 1, 1971, after the private railroads gave them up.

How much did a box save?

Loading a ton of cargo onto a ship by hand cost about $5.86 in 1956. Loading it in containers cost about 16 cents, less than a thirtieth as much, by the count of the writer Marc Levinson in his book *The Box*.

Before containers, dockworkers loaded cargo onto ships by hand. A ship called the *Ideal X* left Port Newark, New Jersey, on April 26, 1956, carrying 58 metal boxes, each 35 feet long, the size of a truck trailer. These boxes are called shipping containers. The ship reached Houston, Texas, five days later.

Members of an international standards group set standard sizes for containers in the late 1960s, and the same boxes came into use around the world.

Malcom McLean

Who: A North Carolina trucking owner who sold his trucking company to start shipping truck-sized boxes by sea.
When and where: Port Newark, New Jersey, April 26, 1956.

His trucking company began with one used truck in 1934.

Malcom McLean built that one truck into a trucking company in North Carolina. Then he sold the trucking company and put the money into carrying containers by sea, and in 1956 he sent the *Ideal X* to sea with them.

The deadliest year on the roads

Car crashes killed 54,589 people in the United States in 1972, the most in any single year before or since. By 1975 the yearly count had fallen to 44,525.

Federal safety officials have given part of the credit to a lower speed limit. During the oil shortage of 1973 and 1974, which Chapter 10 tells, members of Congress set a national speed limit of 55 miles an hour to save fuel, and by March 1974 every state had lowered its limit to match. A federal task force later estimated that the 55-mile limit saved 20,000 to 30,000 lives from 1974 to 1978.

The minivan

Workers in Windsor, Ontario, in Canada, built the first Dodge Caravan and Plymouth Voyager minivans on November 2, 1983. The minivans went on sale as 1984 models.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 12 MINUTES

With a phone in hand, Americans after 2000 could find their way and hire a ride, and some of them bought electric cars. About 8 of every 100 new cars sold in 2024 ran on batteries alone, according to Kelley Blue Book. About 7 of every 100 bridges in the country were rated in poor condition in federal records in 2025, by the count of a road builders' group.

Deaths. Federal safety officials counted 43,230 deaths in car crashes in 2021, and their early estimate for 2025, published in April 2026, is about 36,640. A container ship that had lost power hit a bridge in Baltimore in 2024, and the bridge's fall killed six road workers. An Army helicopter and a passenger jet collided near Washington, D.C., in 2025, and the crash killed all 67 people aboard.

Maps and rides on a phone

GPS, the Global Positioning System, uses signals from satellites to show a GPS device where it is. Until 2000 a setting called Selective Availability made GPS less exact for ordinary users. Federal officials switched the setting off a few minutes after midnight at the end of May 1, 2000, and the error of an ordinary GPS device shrank from about 100 meters to about 20, according to the government's GPS website.

Ride apps. People also began hiring rides through apps on their phones. Uber began in March 2009 as a company called UberCab and started its app service in San Francisco on July 5, 2010. In October 2010, officials of San Francisco's city transit agency and of California's Public Utilities Commission ordered UberCab to stop operating. Its owners changed the company's name to Uber that month, so that it would not look like a taxi business. Lyft started its service in June 2012.

In a ride-hailing app, a rider asks for a ride, and a driver using a personal car takes the job. The companies' managers called their drivers independent contractors, people in business for themselves rather than employees. One driver, Barbara Ann Berwick, asked the state of California to decide whether she was Uber's employee.

Barbara Ann Berwick

Who: A San Francisco Uber driver who asked the state of California to decide whether she was Uber's employee, and won.

When and where: San Francisco, July to September 2014. The decision came on June 3, 2015.

A hearing officer ruled that Uber owed her 56 cents a mile.

Barbara Ann Berwick drove for Uber in San Francisco, in her own car, for about eight weeks in the summer of 2014, and she quit on September 18. Uber's managers called her an independent contractor, and the company did not pay her back for the costs of her car. She took a claim to the California Labor Commissioner, the state office that handles pay disputes.

The hearing. Berwick argued her case herself on March 10, 2015, at a hearing, a meeting where each side tells its story to an official who decides. Uber sent two lawyers and a product manager, Brian Tolkin.

The hearing officer, Stephanie Barrett, listed the rules Uber's managers set for its drivers. The car could be no more than 10 years old, counted by its model year, and the company had to approve it. Drivers needed $1,000,000 in insurance. They had to keep a rating from riders of 4.6 stars or higher, or the company turned off their app. The agreement said riders did not tip. Barrett wrote that Uber was "involved in every aspect of the operation."

The ruling. Barrett ruled on June 3, 2015, that Berwick was Uber's employee, and she ordered Uber to pay Berwick $4,152.20. That was 56 cents a mile for 6,468 miles, plus $256 in bridge tolls and $274.12 in

> interest, extra money owed for the time she had waited to be paid. Barrett turned down Berwick's claim for unpaid wages, because Berwick agreed that Uber had paid her.
>
> Uber's lawyers asked a court in San Francisco to undo the decision. In a statement, Uber's spokespeople said the ruling applied to Berwick alone, and Uber's managers kept treating the company's other drivers as contractors.

The fight over drivers' pay

What was at stake. If app drivers were employees, the companies would have to pay back the costs of their cars, as the hearing officer ordered Uber to do for Berwick. As contractors, drivers paid those costs themselves.

California voters decided the question for app drivers in their state in November 2020. They passed Proposition 22, a law put to the voters directly, 58.6 percent to 41.4 percent. Under it, app drivers stayed contractors, but the companies had to pay them at least 120 percent of the minimum wage for the time they spent working on a ride request, plus 30 cents a mile.

The money. Uber, Lyft, DoorDash, Instacart and Postmates paid for the campaign for the law. It raised $205.7 million, the most for any California ballot question up to then, by the count of Ballotpedia, which tracks state ballot questions.

In court. Four drivers and a union, the Service Employees International Union, went to court on January 12, 2021, to overturn the law. The justices of the California Supreme Court ruled on July 25, 2024, that the law could stand.

Massachusetts. In June 2024 Uber and Lyft settled a lawsuit brought by Massachusetts, agreeing to pay $175 million and to pay drivers there at least $32.50 an hour for time spent driving to a rider or carrying one, while the drivers stayed contractors.

Bigger vehicles and electric ones

Americans bought more light trucks, meaning pickups, sport utility vehicles (SUVs) and minivans, than cars for the first time in 2001. Light trucks made up 50.8 percent of new vehicle sales that year, against 49.2 percent for cars, according to trade magazines that follow the car business. The gap grew wider in the years after.

Tesla. The company Tesla delivered its first Model S electric cars to buyers in Fremont, California, on June 22, 2012. It is named after the inventor Nikola Tesla, who died in 1943 and had no connection to it.

Americans bought about 1.3 million battery electric cars, which run on electricity alone, in 2024 and almost as many in 2025, by the counts of Kelley Blue Book and Cox Automotive, car-market research companies, published in January 2026. They made up 8.1 percent of new car sales in 2024 and 7.8 percent in 2025. Chapter 10 gives larger shares that also count hybrids, which have both a gasoline engine and an electric motor. Tesla sold nearly half of all the battery electric cars bought in 2025.

The tax credit. Members of Congress passed a budget law in July 2025, and President Donald Trump signed it on July 4. Under it, the federal tax credits for buying electric cars, money taken off what a buyer owes in taxes, ended for cars bought after September 30, 2025. In the last three months of 2025, Americans bought 234,000 electric cars, 46 percent fewer than in the three months before.

Scooters. A company called Bird put electric scooters for rent on the streets of Santa Monica, California, in September 2017. Another company, Lime, followed in 2018, and city officials began trial programs for scooters.

Ports, trains and bridges

Ports. Ships carrying containers backed up outside American ports in 2021. Staff at the Marine Exchange of Southern California counted a record 109 container ships waiting off the ports of Los Angeles and Long Beach on January 9, 2022. By October 2022 the line was down to about 4 ships.

A fast train. California voters approved $9.95 billion in bonds, a kind of loan, for high-speed trains from San Francisco to Anaheim in November 2008. The state's high-speed rail officials then put the cost of the whole system at $33.6 billion. Workers broke ground in Fresno on January 6, 2015, and the first section planned is 171 miles, from Merced to Bakersfield.

US Transportation Secretary Sean Duffy canceled about $4 billion in federal grants, money the national government had promised, for the California line in July 2025. He said the state's rail agency had "no viable plan," meaning no plan that could work. California officials sued, then dropped the case in December 2025.

The project now gets $1 billion a year through 2045 from cap-and-trade, a state program that makes polluters pay for permits to pollute. Associated Press reporters wrote in December 2025 that cost estimates are now above $100 billion, about three times the $33.6 billion estimate. No passenger had ridden a California high-speed train as of 2026.

Bridges. Researchers at ARTBA, a group for companies that build roads, counted the federal bridge records for 2025. The United States had 624,167 bridges, and 41,677 of them were rated in poor condition, about 7 of every 100. Drivers crossed those poor bridges 163 million times a day.

Deaths on the roads and in the air

Car crashes killed 43,230 people in the United States in 2021, the most in recent years, and the count then fell four years in a row. In April 2026 officials at the National Highway Traffic Safety Administration published an early estimate of 36,640 deaths in 2025. That is 1.10 deaths for every 100 million miles driven, and the only lower rate on record was 1.08, in 2014.

The Key Bridge. The container ship *Dali* was leaving Baltimore for Sri Lanka at about 1:30 in the morning on March 26, 2024, when it lost power and steering and hit a support of the Francis Scott Key Bridge. The bridge fell into the Patapsco River. The fall killed six men who were filling potholes on the bridge on an overnight shift.

Maryland Governor Wes Moore named them in a statement: Dorlian Ronial Castillo Cabrera, Carlos Daniel Hernandez Estrella, Alejandro Hernandez Fuentes, Miguel Angel Luna Gonzalez, Jose Mynor Lopez and Maynor Yassir Suazo Sandoval. They were immigrants from Mexico and Central America.

The cause. Members of the National Transportation Safety Board, the federal board that investigates crashes, reported the cause in November 2025. A label band had been put on a wire the wrong way. The band kept the wire from going all the way into its connector, so the connection came loose and cut the ship's power.

The board's press release does not say who put the band on or when. The board's members sent safety advice to several groups, among them the ship's builder, HD Hyundai Heavy Industries of South Korea.

The charges. Federal prosecutors, the government lawyers who bring criminal cases, charged the two Synergy Marine companies that ran the ship with crimes on May 12, 2026. They also charged the *Dali*'s technical superintendent, Radhakrishnan Karthik Nair, the manager in charge of the ship's machinery and upkeep. The charges include causing the deaths of the six workers.

A charge is an accusation, not proof. Under the law, the companies and Nair are innocent unless a judge or jury finds them guilty. The same day, Attorney General Anthony Brown, Maryland's top lawyer, announced a $2.25 billion settlement, a deal that ends a legal claim, with the ship's owner, Grace Ocean, and with Synergy.

Near Washington. An Army Black Hawk helicopter collided with an American Eagle passenger jet that was landing at Reagan National Airport, near Washington, D.C., on January 29, 2025. The crash killed all 67 people on the two aircraft, 64 on the jet and three soldiers on the helicopter. It was the deadliest crash of an American passenger airline in more than 20 years.

The head of the National Transportation Safety Board, Jennifer Homendy, said on January 27, 2026: "This was preventable. This was 100% preventable." CNN reported that the board's members blamed Federal Aviation Administration officials for helicopter routes that ran close to the path planes took down to the runway, and for failing to act on years of reports of near misses.

THE THREE QUESTIONS, ANSWERED

1. **How much faster and cheaper did travel get?** On the Erie Canal a ton of goods went from Buffalo to New York City for under $10 instead of about $100 by wagon, and on the first railroad across the country a coast-to-coast trip took a week instead of four months. By 1956, loading a ton of cargo onto a ship cost about 16 cents in containers, against about $5.86 by hand.
2. **Who built the new routes, and whose land and homes did they take?** Irish and American-born laborers dug the Erie Canal, enslaved people dug the Dismal Swamp Canal and built most of the Southern railroads, and Chinese workers built the western part of the first railroad across the country. Builders and US officials took Native nations' land, and highway officials building federal

roads took the homes of more than 475,000 households, many in Black neighborhoods.

3. **How did Black travelers fight the laws that made them sit apart?** They refused to move, went to court and stayed off the buses: Ida B. Wells sued a railroad in 1884, Homer Plessy tested a Louisiana law and lost in 1896, Irene Morgan won in the Supreme Court in 1946, and Black riders in Montgomery, Alabama, stayed off the city buses for more than a year in 1955 and 1956. After white mobs and Klansmen beat the Freedom Riders in Alabama in 1961, federal officials banned separating passengers in bus and train stations.

What changed: People here went from moving on foot, by canoe and with dogs before 1500 to railroads, highways, jets and container ships today, each faster or cheaper than what came before.

FIVE DATES TO KEEP

January 1673: the first post rider leaves New York for Boston.

October 26, 1825: the Erie Canal opens.

May 10, 1869: Chinese workers lay the last track of the first railroad across the country.

May 18, 1896: the Supreme Court rules against Homer Plessy.

June 29, 1956: Eisenhower signs the Interstate highway law.

NOT KNOWN: Nobody knows how many Chinese workers died building the Central Pacific, because its managers kept no record of their deaths.

Where to go: Golden Spike National Historical Park, at Promontory Summit in Utah, marks where the last track was laid, and from May to October you can see working copies of the engines Jupiter and No. 119 there.

Next: Chapter 12, Landmarks. It asks who built the country's best-known structures and monuments, and what people have argued about them.

Chapter 12: Landmarks

CHAPTER 12 OF 37 · LANDMARKS · 10 ERAS · ABOUT 75 MINUTES

Many of the people who built America's landmarks were enslaved, forced to work or killed on the job. Americans still argue about whose land the landmarks stand on and which people they honor.

In Emancipation Hall, in the Capitol Visitor Center in Washington, there is a block of sandstone from Aquia Creek, Virginia. It was part of the Capitol's East Front, built from 1824 to 1826. Members of Congress unveiled it as a marker on February 28, 2012. Its plaque states that the stone "was quarried," or cut from the ground, "by laborers, including enslaved African Americans." In the 1790s the officials in charge of building the Capitol rented enslaved men from their owners by the year, and the owners collected the pay.

This chapter describes landmarks from earthworks more than 3,000 years old to memorials opened in 2024. Watch for two things as you read. One is who did the building: Native men drafted to cut stone for a Spanish fort, enslaved men rented out by their owners, immigrants killed on the Brooklyn Bridge. The other is what people argued about afterward. A golf club used an ancient earthwork in Ohio as a golf course from 1910 to 2024. Workers carved Mount Rushmore on land that members of Congress had taken from the Sioux. Since 2015, officials and work crews have taken down, moved or renamed hundreds of Confederate memorials, and in 2025 federal officials began putting some back.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who built the White House and the Capitol, and who was paid for their work?
2. How many workers died building Hoover Dam?
3. What has happened to the Confederate monuments since 2015?

Five minutes? Read "Enslaved men who built the capital" in 1750 to 1800.

Can skip: "The Woolworth Building", "The largest private house", "The Space Needle". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 6 MINUTES

Native people built the oldest landmarks in the United States, hundreds and even thousands of years before any European arrived. They piled up mounds of earth, and one of them covers more ground than the Great Pyramid in Egypt. In the Southwest they built stone houses with hundreds of rooms and set whole villages under cliffs.

No one wrote down the names of the people who built these places. In the Southwest, the builders' descendants, among them the Pueblo peoples, still live in the same region.

Monks Mound and the mound builders

Monks Mound, at Cahokia in what is now Illinois, is the largest structure made of earth in North and South America. People piled up its earth in stages, mostly between about the years 1050 and 1200. It stands about 100 feet tall. Its base covers about 14 acres, and the base of the Great Pyramid of Giza, in Egypt, covers about 13. The pyramid is far taller, so the mound is bigger only in the ground it covers.

Poverty Point, in Louisiana, is much older. People built it between about 1700 BC and 1100 BC, which is about 3,700 to 3,100 years ago. The builders lived by hunting, fishing and gathering wild food. They heaped up earth into five mounds and six curved ridges, set one inside another around a wide open space in the middle. It is the largest earthwork of its age north of Mexico.

A world list. Poverty Point is on the World Heritage List, which includes places chosen because they matter to all people. The World Heritage Committee of UNESCO, a part of the United Nations, keeps the list. Its members added Poverty Point on June 22, 2014, at a meeting in Doha, Qatar, as the 1,001st place on the list and the 22nd in the United States.

Serpent Mound, in Ohio, is a snake shaped out of earth, about a quarter of a mile long. It is the largest snake-shaped mound in the world. Nothing found inside it shows who built it.

Two dates. Archaeologists, scientists who study the past from the things people left behind, have tested the mound with radiocarbon dating, a way to measure the age of anything that was once alive. Tests done in 1991 gave about the year 1070, the time of a people archaeologists call the Fort Ancient culture. Tests done in 2014 gave about 300 BC, the time of a people they call the Adena culture. Later work has found evidence for both dates.

NOT KNOWN: which people built Serpent Mound. Archaeologists now think it possible that people built it in one period and rebuilt it in another.

Stone buildings of the Southwest

In Chaco Canyon, in what is now New Mexico, people built large stone buildings called great houses between about 850 and 1150. The largest, Pueblo Bonito, rose four stories high. It held about 600 to 650 rooms, or more. The counts differ because some of them include a guess at how many rooms the upper floors held, and some do not.

At Mesa Verde, in what is now Colorado, people built Cliff Palace between about 1190 and 1280. They set it under an overhang, a place where the rock of the cliff sticks out above like a roof. Cliff Palace has about 150 rooms and 23 kivas, rooms used for religious ceremonies. It is the largest cliff dwelling in North America.

The National Park Service lists 26 tribes with a special tie to Mesa Verde. Twenty-one of them are Pueblo nations, among them the Hopi Tribe and the Pueblos of Zuni, Taos and Acoma in New Mexico. The other five, the Navajo Nation, three Ute tribes and the Jicarilla Apache Nation, also have a tie to the place.

Earthworks under a golf course

Native people in southern Ohio built long walls of earth in the shapes of circles and eight-sided figures about 2,000 years ago. Archaeologists call these people the Hopewell culture. Hopewell is a name archaeologists gave them, not the name of any nation.

Accounts differ on how long the building went on, from about 400 years to about 700, in the centuries between about 200 BC and the year 500.

The Octagon Earthworks, in Newark, Ohio, is a large circle of earth joined to an eight-sided wall. Together the two walls surround nearly 70 acres. The walls line up with the points where the moon rises, in a pattern that repeats every 18.6 years.

The golf course. The Moundbuilders Country Club began leasing, or renting, the Octagon in 1910 and used it as a golf course. Since 1933 the land has belonged to the Ohio Historical Society, now called the Ohio History Connection, a group set up under state law and not run to make money. For decades its leaders kept leasing the site to the club.

Members of UNESCO's World Heritage Committee voted on September 19, 2023, to add eight Hopewell earthwork sites to the World Heritage List. The Octagon was one of them.

Glenna Wallace

Who: The first woman to lead the Eastern Shawnee Tribe of Oklahoma, chief from 2006 to 2026, who fought for years to get a golf course off the Octagon Earthworks.

When and where: The Octagon Earthworks, Newark, Ohio, from the late 2000s to 2025.

A golfer shouted at her, "Out of the way! You don't belong here!"

The golfer. Wallace first visited the Octagon Earthworks in the late 2000s, in 2007 by the account in the Smithsonian's *American Indian* magazine and in 2009 by other accounts. That was the visit when the golfer shouted at her. She decided to get the golf course off the earthworks and to win World Heritage status for them.

Forced out of Ohio. Wallace's own people had once lived in western Ohio, near Lewistown. Their band of Shawnee and Seneca, joined as one, gave up that land in a treaty, a written agreement between nations, signed on July 20, 1831. James B. Gardiner, a special official, signed it for the United States.

Under the treaty, US officials forced the people out of Ohio in 1832. They walked about 800 miles to a reservation, land set aside for a Native nation, in what is now Ottawa County, Oklahoma. More than 15 percent of them died on the way, according to the Ohio History Connection. No record of the march names the officials who led it. Wallace was born in that county in about 1939.

Her own words. At Ohio Statehood Day in 2019, Wallace asked, "Why had I never read anything about Newark Earthwork Mounds?" Then she answered her own question: "Because you here in Ohio didn't even realize the treasures that you had."

The court case. Leaders of the Ohio History Connection went to court to end the club's lease. They used eminent domain, a legal power to take property for public use as long as the owner is paid. On December 7, 2022, the judges of the Ohio Supreme Court ruled 5 to 1 that they could do it.

Wallace gave the main speech at the 2023 meeting in Riyadh, Saudi Arabia, where members of UNESCO's World Heritage Committee voted to list the earthworks. In August 2024 leaders of the club and of the Ohio History Connection agreed to end the court case, and they did not make the terms public. "At last," Wallace said. "So many people have worked so long for this to become reality."

The club ran the golf course until the end of 2024. The Octagon opened to the public on January 1, 2025.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 1 MINUTE

Nothing that Europeans built here in the 1500s still stands. The Spanish built their first forts in Florida of wood and earth. Those forts burned many times, and the Spanish rebuilt them again and again.

Spanish priests ran missions, settlements where Native people lived and were taught the Catholic religion. The mission buildings of this century, in Florida and in the Southwest, survive only as remains in the ground that archaeologists dig up. Native mounds and stone buildings of earlier centuries, such as Monks Mound and Cliff Palace, were standing then, and they still stand today.

Where did St. Augustine begin?

St. Augustine, Florida, began in a Native town. Pedro Menéndez de Avilés landed at Seloy, a town of the Timucua people, on September 8, 1565. The Spanish made their first settlement in Seloy's village. People have lived in St. Augustine without a break ever since. Of all the cities that Europeans founded on the mainland of the United States, it is the oldest that people never left.

Francis Drake and his men burned the town in 1586. The only landmark left from this century is the place where the Spanish settled. Nothing they built there in the 1500s survives above ground. The stone fort that still stands in St. Augustine, about a mile south of Seloy, was begun in 1672.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 9 MINUTES

Spanish officials forced Native men from their towns to cut the stone for a fort at St. Augustine, in Florida. The fort still stands, and it is among the oldest buildings that Europeans put up in the United States. So are two churches of this century: a chapel in Santa Fe, New Mexico, made of adobe, bricks of mud and straw dried in the sun, and a wooden church building in Massachusetts called a meetinghouse.

A fort built of shell stone

Spanish officials ordered a stone fort, the Castillo de San Marcos, built at St. Augustine after pirates attacked the town. About a hundred English pirates came ashore at midnight on May 28, 1668, and looted St. Augustine. They killed more than 50 Spaniards, and they said they would come back and take the town's fort.

The order. Queen Mariana of Spain then named Manuel de Cendoya governor of Florida. She sent him to the viceroy of New Spain, the king's highest official in Mexico, to collect 12,000 pesos, which were Spanish coins, to begin a fort of stone.

Workers broke ground on October 2, 1672, and Cendoya laid the first foundation stone that November. The last stones were set in 1695 or 1696, by different accounts. By 1696 the fort had cost close to 100,000 pesos. It is the oldest masonry fort, a fort built of stone or brick, in the continental United States.

Stone made of shells. The fort is made of coquina, a soft stone made of packed seashells. Workers cut the coquina from quarries, pits where stone is cut out of the ground, on Anastasia Island, across the bay from the town. They cut grooves in the rock with picks and axes, then broke the blocks loose with bars and wedges.

The largest blocks were 2 feet thick and 4 feet long. Albert Manucy, who wrote a history of the fort for the National Park Service in 1942, says six strong men "could hardly lift" one of them. Oxen dragged the blocks to a dock, and rafts carried them across the bay.

The stones are held together with mortar, a paste that hardens. Workers made lime, a white powder, for the mortar by burning oyster shells in kilns, or very hot ovens. By the spring of 1672 they had stored about 7,000 bushels of lime, measures of about 8 gallons each. The walls they built are 14 feet thick at the base and about 9 feet thick at the top.

Two sieges. English soldiers attacked the fort twice and failed both times. Governor James Moore of Carolina came with an army in November 1702, with Colonel Robert Daniel at his side. Almost 1,500 people crowded into the fort for safety. The siege, an attack in which an army surrounds a place and tries to force the people inside to give up, lasted about two months.

The English cannonballs did not shatter the walls. The shell stone crushed inward where each ball hit, so the balls sank into the wall or bounced off. When a Spanish force arrived to help, the English set fire to their own ships and retreated. As they left, Daniel's troops and their Native allies burned the rest of St. Augustine to the ground.

The fort and the people inside it survived. General James Oglethorpe of Georgia besieged the fort for 27 days in 1740, and he failed too. The fort is now the Castillo de San Marcos National Monument.

Who cut the stone?

Spanish officials forced Native men to work on the Castillo and in the king's corn fields near the town. At times up to 300 Native people were working for the Spanish king in St. Augustine, counting the men in the fields as well as the men on the fort. The men came from three nations: the Guale of what is now Georgia, the Timucua of eastern Florida and the Apalache of western Florida.

Officials drafted the men in their home towns, which means they ordered them to come and work. Some of the men had to walk about 200 miles to reach St. Augustine. Manucy wrote that "many of them served unwillingly." A drafted man had two choices. He could bring his family with him, or he could leave his wife and children in the village to feed themselves.

Past their terms. Each group of drafted men was supposed to work for a set time and then go home, but officials often held the men long past it. They drafted even some chiefs, and one chief was forced to work on the fort for more than three years without once going home. Some governors used Native men as their own servants. Years before the work on the Castillo began, Spanish officials had sentenced the Apalache to forced labor on the forts as a punishment for rebelling.

The pay. The Native laborers were paid 1 real, a Spanish silver coin, a day, plus a daily share of corn. Manucy counted the real as worth 12 and a half cents, and a park handbook written in 1993 puts it at about 20 cents. Spanish master masons on the same job earned 20 reales a day.

The fort alone needed about 150 laborers: 50 to quarry and haul the stone, 50 to gather oyster shells and help at the lime kilns, and 50 to dig trenches, carry sand and mix mortar. Officials found it hard to get even 100. In the first winter a sickness the Spanish called El Contagio, "the contagion," spread among the Native laborers, and the crew shrank to almost no one. Manucy's history does not count how many men died.

No names. The payrolls, the lists of workers and their pay, counted the Guale, Timucua and Apalache men who cut the stone. Manucy's history and the 1993 park handbook were both written from those payrolls. They give the men's nations and numbers, but not their names.

Convicts and enslaved men. Spanish officials also put convicts on the fort, men sentenced to forced labor there for a set number of years. One Spaniard caught smuggling English goods got 6 years of labor on the fort, and his term would double if he ran away. Enslaved Black people owned by the Spanish king worked there too, and in 1687 officials added eighteen of them to the crew.

That same year, eleven people escaped slavery in Carolina and reached St. Augustine. They were eight men, two women and a nursing baby. Governor Diego de Quiroga y Losada gave them shelter and had them baptized in the Catholic Church, a ceremony with water that makes a person a member of the church.

The governor hired the men as paid workers, and six of them worked on the Castillo. In 1693 King Charles II of Spain ordered that enslaved people who escaped from the English colonies and became Catholic would be free.

John Collins

Who: An English mason the Spanish held prisoner, who made the lime and ran the quarry for the Castillo de San Marcos.

When and where: St. Augustine, Florida, from 1670 for 19 years or more.

He made more lime in a week than Spanish workmen in two.

John Collins was an English mason, a builder in stone. In 1670 a ship sailing for the new English colony at Charleston, in Carolina, landed by accident at the Spanish mission of Santa Catalina, near the Savannah River. The Spanish there captured two Englishmen from the ship.

A small boat came from Charleston to get them back, and Collins and another man carried a message ashore. The Spanish held them too and sent all four men to St. Augustine as prisoners.

Juan Calens. Governor Manuel de Cendoya put the prisoners to work on the fort, and the Spanish payroll lists Collins as Juan Calens. According to Manucy's history, Collins could burn more oyster shells into lime in a week than Spanish workmen could in two. The lime went into the mortar that held the coquina blocks together. As a prisoner he was paid 8 reales a day, when a master workman was owed 20.

Collins rose from master of the lime kilns to master of the quarry. Later he ran the dugout boats, the food stores and the convict workers. At last he held the job of pilot, guiding ships between St. Augustine and Charleston. Officials of the Spanish king recognized his work after he had spent 19 years or more in the town.

NOT KNOWN: Manucy's history and the 1993 park handbook do not say when he was born or died, or whether he ever went home.

The oldest churches

San Miguel Chapel in Santa Fe, New Mexico, is often called the oldest church in the country. Its site and its foundations are that old, but most of the adobe walls standing today date from a rebuilding in 1710. The chapel was built around 1620 as a church for the soldiers, servants and laborers of the town's Spanish officials, and it was in use before 1626.

The Pueblo Revolt. The Pueblo people of northern New Mexico rose up against Spanish rule in 1680. Spanish officials had been collecting tribute, payments of goods, from the Pueblos, and they had used their power to keep the Pueblo people from practicing their own religion. A religious leader named Popé planned the uprising from Taos. The Pueblo people drove the Spanish out of the region, and the Spanish stayed out until 1692, when Diego de Vargas led them back.

The chapel was partly destroyed in the revolt. The histories of the chapel written for the National Park Service and the Historic Santa Fe Foundation do not say who damaged it. In 1710 Governor José Chacón had it rebuilt on the old foundations.

The Old Ship Meetinghouse in Hingham, Massachusetts, was built in 1681. The Puritans, a group of English Protestants who settled in New England, met in buildings like this one for church. It is the only Puritan meetinghouse from the 1600s still standing in the country, and the oldest church building in the United States that has been in use without a break.

The name. Its roof rests on a frame of heavy timbers, left open to view from inside. To the people who worshiped there, the frame looked like the hull, or body, of a ship turned upside down. That is how the building got its name.

The two claims. San Miguel's foundations are about 60 years older than the Old Ship Meetinghouse, but San Miguel's standing walls are about 30 years younger.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 5 MINUTES

Colonists in Boston and Philadelphia put up public buildings that are still landmarks today. The first lighthouse in what became the United States was lit at the mouth of Boston Harbor in 1716, and Boston's Old North Church went up in 1723. In Philadelphia, builders began a statehouse for Pennsylvania's lawmakers in 1732. It is now called Independence Hall.

In Texas, Spanish priests ran a mission at the place later known as the Alamo. The first stone of its stone church was laid in 1744.

The first lighthouse

Boston Light stands on Little Brewster Island, at the entrance to Boston Harbor. It was first lit on September 16, 1716, the first lighthouse in what became the United States. British forces blew up the tower in 1776. It was rebuilt in 1783, and the 1783 tower is the one that stands today.

George Worthylake

Who: The first keeper of Boston Light, who drowned with his family two years into the job.

When and where: Little Brewster Island, Boston Harbor, 1716 to 1718.

The keeper, his wife and their daughter drowned coming home from church.

George Worthylake became the first keeper of Boston Light in 1716. He was paid £50 a year, 50 British pounds, to keep the light burning from sundown to sunrise. Worthylake lived on the island with his wife Ann and his daughter Ruth. He earned extra money guiding ships in and out of the harbor, and he kept a flock of sheep on Great Brewster Island, next door.

The canoe. The family had gone into Boston for a church sermon on November 3, 1718. Just after noon, on the way back to the island, their canoe tipped over. Worthylake, Ann and Ruth all drowned. Shadwell, a man Worthylake enslaved, drowned with them, and so did George Cutler, a servant, and John Edge, a friend of the family. Within weeks, in a second accident, two of the men hired to keep the light until a new keeper was chosen drowned near the island too.

Franklin's song. Benjamin Franklin was 12 years old and working for his older brother, a printer in Boston. His brother had him write a ballad about the drowning, a song that tells the story, called "The Lighthouse Tragedy." Once it was printed, Franklin's brother sent him around the town to sell copies. Franklin later called his own verses "wretched stuff," and no copy survives.

A steeple and a statehouse

Old North Church in Boston, also called Christ Church, was built in 1723. It is the oldest church building still standing in Boston. Its steeple, the tall pointed tower on top, was finished in 1740 and is 191 feet high.

Two lanterns. On April 18, 1775, the church's sexton, or caretaker, Robert Newman, and Captain John Pulling Jr. hung two lanterns in the steeple. The lanterns were a signal, worked out by Paul Revere, that British troops were moving by water.

Work on the Pennsylvania State House in Philadelphia began in 1732 and ended in 1753, 21 years later. Andrew Hamilton designed it with Edmund Woolley, a master carpenter. Workers added its tower from 1750 to 1753, and in 1753 they raised a bell into it. Words on the bell read "Proclaim liberty throughout all the land." The bell later became known as the Liberty Bell.

The State House is now called Independence Hall. Leaders of the colonies debated and signed the Declaration of Independence inside it in 1776.

The mission church later called the Alamo

A Spanish priest, Father Antonio de San Buenaventura y Olivares, founded Mission San Antonio de Valero in San Antonio, Texas, on May 1, 1718. Soldiers sent by Governor Martín de Alarcón came with him.

The people who lived at the mission came from more than a hundred Native groups, among them the Payaya, the Pamaya, the Xarame, the Apache and the Karankawa. Diseases brought by Europeans had already killed many Native people in the region. Many came to the San Antonio missions for protection from their enemies and for a steadier supply of food.

Kept by force. As a rule, the Spanish forced no one to come in. Once people had joined a Texas mission, though, priests and soldiers used force to keep them. Soldiers went out to bring back people, and whole groups, who left. Smallpox and measles spread through the mission in 1739. In 1745, 311 people lived there.

The cornerstone of a stone church, the first stone set at a corner of the new building, was laid on May 8, 1744. That church fell down in the 1750s. Work on the church that stands today began in 1758. The missionaries never finished it, and the bell towers and the dome they planned were never built. The records do not say who laid the church's stones.

Spanish officials closed the mission in 1793 and divided its lands among the Native people and others who still lived there.

The Alamo Company. From 1803 a company of Spanish soldiers lived in the old mission. They had been stationed before at Alamo de Parras, a town in what is now Mexico, so people called them the Alamo Company, and the mission took its name from them.

Texans were fighting to break away from Mexico in 1836, in a war called the Texas Revolution. That year Mexican soldiers attacked the old mission and killed the Texan soldiers defending it, 187 men by the National Park Service's count.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 6 MINUTES

Enslaved people worked at every stage of putting up the first buildings of the new national capital. Leaders of the United States began building the capital beside the Potomac River in the 1790s, and its two main buildings were the President's House and the Capitol. The officials in charge rented enslaved men from their owners, and the owners collected the pay. Enslaved workers also built two houses of the time that are landmarks today, Mount Vernon and Monticello.

Raising the capital's buildings

The land chosen for the capital had been home to the Nacotchtank people. The English captain John Smith traveled up the Potomac River in 1608 and found their villages there. European settlers brought diseases that killed many of the Nacotchtank. In colonial times most of their groups joined other Native peoples, and some moved away.

James Hoban, an architect born in Ireland, designed the President's House, later called the White House. Its cornerstone was laid on October 13, 1792. George Washington laid the cornerstone of the Capitol, the building where Congress meets, on September 18, 1793. Neither building was done by 1800. When John Adams moved into the President's House in November 1800, it was still unfinished, and the Capitol had only one wing.

Enslaved men who built the capital

Enslaved men quarried sandstone for the President's House and the Capitol. They also cut timber, made bricks, sawed boards, did carpentry and hauled loads. The three commissioners, the officials in charge of the new District of Columbia, bought a sandstone quarry on Government Island, in Aquia Creek, in Stafford County, Virginia, for $6,000 on December 2, 1791. In the fall of 1792 they allowed the man running the quarry to rent forty enslaved men to work there, beside a number of free workers.

Rented by the year. William C. Allen, a historian in the office of the Architect of the Capitol, studied the Capitol's payrolls for Congress in 2005. His report states that the commissioners rented enslaved men from their owners by the year. They called this "Negro hire," using a word of that time for Black people. The Capitol's accounts list 385 payments for it between February 11, 1795, and May 17, 1801.

Who got the money. The money went to the owners. At first an owner got $60 a year for each man, when a white laborer with no special skills earned $70 a year. The commissioners raised the owners' rate to $70 in 1797. When a man missed work, the commissioners cut his owner's pay.

Two of the commissioners, Gustavus Scott and William Thornton, rented out men they held in slavery to the project, and Thornton also designed the Capitol. James Hoban was paid for enslaved men too. So were the commissioners' treasurer, the man who oversaw the Capitol's laborers and a land developer. Many Maryland slaveholders rented out men as well, and several of them were women, such as Ann Digges and five women of the Brent family.

The commissioners told the overseers to keep the rented men at work "from sunrise to sunset," six days a week. They housed the men in huts and gave them rations, a set share of food and drink, of salt meat and whiskey. Allen found that the overseers probably kept the enslaved sawyers, men who sawed logs into boards, at work seven days a week. For work at night, on Sundays and on holidays, the men themselves got small extra wages.

First names only. Only a few of the men's names were written down, and only their first names. The payment records list Dick and Tom, rented from Ann Digges in 1795, and Nace, Gerrard, David, Charles, Harry, Gabe and Sil, rented from the Brent women in April 1795. Allen wrote that "virtually nothing is known of their lives." A man named Jacob ran away from the work in 1793, and his owner offered a $6 reward for him.

KNOWN:	The historian Bob Arnebeck counts at least 200 known enslaved laborers on the Capitol and the President's House together.
NOT KNOWN:	the full number. Richard Baker, the Senate Historian, and Kenneth Kato of the House history office, who introduced Allen's report, wrote that no one will ever know how many enslaved people helped to build the Capitol.

Tom, Peter, Ben, Harry, and Daniel

Who: Five enslaved carpenters who helped build the White House, known only by the first names on one month's payroll.
When and where: The President's House building site, Washington, May 1795.

Peter and Ben each worked 28 days and got none of the pay.

Tom, Peter, Ben, Harry and Daniel were enslaved carpenters who worked on the President's House in May 1795. The carpenters' payroll for that month lists them by their first names only. James Hoban, the architect of the building, held four of them in slavery, and his assistant, Pierce Purcell, held the fifth.

The payroll records how many days the men worked that month. Peter and Ben each worked 28 days, Daniel worked 25 and Harry worked 9. Hoban signed for their wages. The pay went to the men's owners and not to the men.

NOT KNOWN: which of the five men Purcell held. The White House Historical Association's own accounts differ.

Who built Mount Vernon and Monticello?

Enslaved workers built and enlarged Mount Vernon, George Washington's home in Virginia, and Monticello, Thomas Jefferson's home in Virginia.

Washington had the roof of his one-story house at Mount Vernon raised in the late 1750s to make a full second floor. The people he enslaved did most of the work, while he was away at war. He began enlarging the house again in 1774, and its last new room was finished in about 1787.

Jefferson began building Monticello in 1769 and left it unfinished in 1784. From 1796 to about 1809 he redesigned it and had it rebuilt, and enslaved workers did much of the building and made much of its fine woodwork. One of them was John Hemmings, a joiner, or maker of fine woodwork.

Thomas Jefferson

Who: The owner of Monticello, who designed his own house and worked on a design for Virginia's capitol that copied a Roman temple.

When and where: Monticello, Virginia, 1769 to about 1809. The design for the Virginia State Capitol in Richmond, 1785.

Virginia's capitol was modeled on an ancient Roman temple in France.

Thomas Jefferson designed his own house at Monticello, and with a French draftsman, a man who drew building plans, he made the design for the Virginia State Capitol.

The temple. Virginia's Directors of Public Buildings asked Jefferson for a design for a new state capitol in Richmond in 1785. He was then the United States minister to France, the man who represented the country to the French government. The French draftsman, Charles-Louis Clérisseau, persuaded him to model the outside of the building on the Maison Carrée, an ancient Roman temple in the city of Nîmes, in France.

Clérisseau drew the final plans. A plaster model reached Richmond late in 1786 to guide the workmen, and Samuel Dobie built the capitol. A history of the building by the Virginia General Assembly, the state's lawmakers, states that the capitol "introduced the Classical Revival temple style to the modern world," a style that copies the temples of ancient Greece and Rome. The same history states that the capitol "has inspired countless public buildings for over 200 years."

NOT KNOWN: who Dobie's workmen were, or whether any of them were enslaved. The General Assembly's history names only Dobie.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 4 MINUTES

Work on two tall stone monuments, one to the Battle of Bunker Hill and one to George Washington, stopped when the people raising them ran out of money. They were among the first monuments Americans built to the Revolution and to Washington. Workers finished the Bunker Hill Monument near Boston in 1842. The Washington Monument in the capital stood at about 150 feet when work stopped in 1854, and the stump stayed unfinished for 25 years.

Enslaved craftsmen at Monticello, in Virginia, went on rebuilding Thomas Jefferson's house until about 1809. One of them, the joiner John Hemmings, made much of its fine woodwork.

Two obelisks that ran out of money

Both monuments are obelisks, tall stone pillars with four sides that narrow toward a pointed top. The Bunker Hill Monument marks the Battle of Bunker Hill, fought on June 17, 1775, and it stands on Breed's Hill in Charlestown, Massachusetts. A group of leading citizens of the Boston area formed the Bunker Hill Monument Association in 1823 to build it.

The cornerstone. Daniel Webster and the Marquis de Lafayette, a French general, led the ceremony that laid its cornerstone on June 17, 1825, 50 years to the day after the battle. About 40 men who had fought in the battle came, along with about 190 other men who had fought in the Revolutionary War.

The association bought a granite quarry in Quincy, Massachusetts. Horses pulled the granite blocks three miles to the Neponset River on a railway. A National Park Service article describes it as the first horse-drawn railway built to carry goods as a business. Ships then carried the blocks to Charlestown.

Out of money. The association's money ran out in December 1828, after 19 months of work, and the work stopped. To pay off its debts, its leaders sold 10 acres of the old battlefield in 1839. Sarah Josepha Hale organized a Ladies' Fair at Quincy Market in Boston in 1840, and the money taken in at the fair came to more than $30,000, about a quarter of the monument's cost.

Workers finished the 221-foot granite obelisk in July 1842. It was dedicated, or opened with a public ceremony, on June 17, 1843, with President John Tyler there.

The Washington Monument. Work on a monument to Washington began in the capital on July 4, 1848. About 20,000 people watched members of the Washington National Monument Society lay the cornerstone. By 1854 the society's money was gone, and work stopped at about 150 feet.

The Pope's stone. In March 1854, members of the Know-Nothing movement, a political movement against Catholics, stole a block of marble meant for the monument. Pope Pius IX, the head of the Catholic Church, had given it. By the usual account, the thieves threw the stone into the Potomac River, and it was never found. Know-Nothings took control of the monument society from 1855 to 1858. The stump of the monument stood unfinished until 1879.

The Castle. One building in the capital in these years was paid for out of one man's gift. James Smithson, an English chemist who never set foot in the United States, left his fortune to the country in his will, to found the Smithsonian Institution.

Workers put up the Smithsonian Institution Building, of red sandstone, from 1847 to 1855, and the money for the work came from the interest that Smithson's gift had earned. The architect James Renwick Jr. designed it, and its cornerstone was laid on May 1, 1847. People call it the Castle.

John Hemmings

Who: An enslaved joiner who made the fine woodwork of Monticello's rebuilt rooms, and one of only five people Jefferson freed in his will.

When and where: Monticello and Poplar Forest, Virginia. Born 1776, died 1833.

Of more than 600 people Jefferson enslaved, his will freed five.

Thomas Jefferson held John Hemmings in slavery at Monticello. From the age of 17, Hemmings learned the joiner's trade from white joiners whom Jefferson hired.

When Jefferson rebuilt Monticello, Hemmings made the fine woodwork of the remodeled rooms and much of the furniture. He also led the woodwork at Poplar Forest, Jefferson's other house in Virginia.

Freed by the will. Jefferson enslaved more than 600 people during his life, and in his will he freed five of them. Hemmings was one of the five, and he became free in 1826. Jefferson also left him the tools of the joiner's shop.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 10 MINUTES

The Washington Monument, finished in 1884, was the tallest structure in the world at 555 feet. In New York, workers hung the Brooklyn Bridge from cables, and at least 21 of them died. Other workers raised a bronze statue onto the new dome of the Capitol and set up the Statue of Liberty in New York Harbor. As St. Louis grew, between 1850 and 1904, people dug away almost all of the Native mounds that had stood there for centuries.

The Statue of Freedom

Workers set the last section of a bronze statue called Freedom on top of the new dome of the Capitol on December 2, 1863. Clark Mills, a sculptor who had taught himself, cast the statue from 1860 to 1862 in his foundry, a workshop where metal is melted and poured into molds. One of the men who cast it was Philip Reid, whom Mills held in slavery.

Philip Reid

Who: An enslaved foundry worker who helped cast the Statue of Freedom for the dome of the Capitol, and who was free by the day it was raised.

When and where: Clark Mills' foundry, Washington, 1860 to 1863.

Reid kept Sunday's pay, and his owner took the other six days.

Philip Reid worked out how to take apart the plaster model of the Statue of Freedom so it could be cast in bronze. He was born in slavery in Charleston, South Carolina, in about 1820, and Clark Mills bought him there. Mills said he paid $1,200 for Reid "because of his evident talent for the business."

The puzzle. An Italian sculptor on the job refused to show how to take the model apart unless he got a raise in pay. Reid worked out how to separate the sections himself, using a pulley and tackle, a set of ropes and wheels for lifting heavy loads.

The pay. Reid earned $1.25 a day, but he kept only Sunday's pay. Mills collected the pay for the other six days of each week.

Free. President Abraham Lincoln signed the District of Columbia Compensated Emancipation Act on April 16, 1862. It was a law for setting free, or emancipating, the enslaved people of the capital, and Reid became free that day.

Under the law, owners who had stayed loyal to the Union, the side of the United States government in the Civil War, could be paid, or compensated, up to $300 for each person. The government of the District of Columbia counts 3,100 people freed under it, and the US Senate's history office counts 2,989.

When the last section of Freedom was set on the dome in December 1863, Reid had been free for more than a year and a half. He later signed his name Philip Reed.

The Brooklyn Bridge and the bends

The Brooklyn Bridge, in New York, opened on May 24, 1883. It is a suspension bridge, which means its roadway hangs from cables strung between tall towers. At least 21 workers died building it, and most of the dead were immigrants.

How many died. The counts differ. The historian David McCullough counted 21 deaths. C. C. Martin, an assistant engineer on the bridge, told the *Brooklyn Eagle* newspaper that 27 men died in accidents. E. F. Farrington, the head mechanic on the job, said the number could be as high as 40.

The towers stand on caissons, huge boxes, open underneath, that were sunk to the bottom of the river. Men worked inside them, digging out the riverbed in air pumped in under pressure to keep the water out.

The bends. When men came up out of that air too quickly, bubbles of gas formed in their blood. The bubbles caused great pain in the joints, and they could leave a man paralyzed, unable to move parts of his body, or kill him. This illness is called caisson disease, or the bends.

Dr. Andrew Smith, the bridge company's doctor, gave it the name caisson disease. Between January 25 and May 31, 1872, he treated 110 cases among the roughly 600 men who worked in the air under pressure. Three of them died of it.

Caisson disease killed John Myers on April 22, 1872, Patrick McKay eight days later, and Daniel Reardon within the month. After those deaths, Washington Roebling, the chief engineer, stopped digging the New York caisson any deeper.

Other workers died too. John French and Thomas Douglas died before the caisson deaths, and Neil Mullen died in 1877. In 1878 a strand of cable snapped at the Manhattan end and threw Thomas Blake and Harry Supple, two men from Newfoundland, to their deaths. Caisson disease also crippled many workers who lived.

John, Washington, and Emily Roebling

Who: The man who designed the Brooklyn Bridge, the son who took charge of building it, and the son's wife, who carried his instructions to the engineers for 11 years.

When and where: New York, 1869 to 1883.

Movie: *Brooklyn Bridge* (1981), Ken Burns' first documentary, narrated by David McCullough. It was nominated for the Academy Award for Best Documentary Feature. It is about the bridge and the Roeblings.

Emily Roebling rode across first, holding a live rooster.

John A. Roebling designed the Brooklyn Bridge, but he died before building began. In June 1869 he was measuring the site for a tower when a ferry crushed his foot against the posts that held up the dock.

Doctors told him the foot should be cut off. He refused and allowed only a water treatment, with cold water dripped steadily onto the wound. His crushed toes were cut off, and tetanus, an infection from a wound that makes the muscles tighten and lock, set in. He died on July 22, 1869.

The son. His son, Washington Roebling, took over as chief engineer. He went down into the caissons with the workers and got caisson disease. He collapsed in 1872 and was left partly paralyzed. For the next 11 years, until the bridge opened in 1883, he directed the work from a room in Brooklyn Heights, watching the bridge through a telescope.

The son's wife. Emily Warren Roebling carried her husband's instructions to the engineers for 11 years. She studied cable construction, the forces on a bridge and the mathematics of bridge building, so she could pass on his orders correctly and answer the engineers' questions. When the bridge's trustees, the people chosen to look after the project, tried to remove her husband, she defended him before them.

Some writers today call her the bridge's "effective chief engineer," meaning she did much of that job in practice. Washington kept the title and made the engineering decisions from his sickroom. Emily was the person the engineers dealt with every day.

> When the bridge opened in 1883, Emily Roebling was the first person to cross it by carriage. She carried a live rooster as a sign of victory.

The Washington Monument, finished

Members of Congress voted in 1876 to have the federal government take over the unfinished Washington Monument. Army engineers led by Lieutenant Colonel Thomas Lincoln Casey started work again in 1879, and the workers finished the monument on December 6, 1884.

An aluminum tip. On its point the workers placed a cast aluminum tip that weighed 100 ounces. Aluminum then cost about a dollar an ounce, roughly a day's wage for a laborer, and the whole tip cost $225.

The monument was dedicated on February 21, 1885. At 555 feet, it was the tallest structure in the world until the Eiffel Tower was built in Paris in 1889.

A line in the stone. About 150 feet up, the color of the marble changes. The stone laid after 1879 came from different beds of rock in the quarries than the stone laid from 1848 to 1854. The line shows where the work stopped for 25 years.

Coins for Liberty's pedestal

People in France gave the money for the Statue of Liberty, and Americans had to raise the money for its pedestal, the tall stone base it stands on. Édouard de Laboulaye proposed the statue in 1865. The sculptor Frédéric Auguste Bartholdi designed it.

The statue's finished arm and torch were shown at the Centennial Exhibition, a big fair in Philadelphia in 1876 that marked 100 years since 1776. From 1877 to 1882 the arm stood in Madison Square Park in New York to draw gifts of money for the pedestal.

Two pennies thick. The engineer Gustave Eiffel joined the work in 1879 and designed the iron frame inside the statue. The statue's copper skin hangs on that frame, and it is 3/32 of an inch thick. In the words of the National Park Service, that is "about the thickness of two pennies placed together."

The newspaper campaign. The architect Richard Morris Hunt designed the pedestal, which was built from 1884 to 1886. Money for it came in slowly until 1885. That year Joseph Pulitzer, who published the *New York World* newspaper, ran a campaign in its pages. More than $100,000 came from about 120,000 donors. Most gave less than a dollar, and many of the gifts were coins from schoolchildren. Every donor's name was printed in the paper.

President Grover Cleveland led the dedication ceremony on October 28, 1886. The statue is 151 feet tall, and with its pedestal it stands about 305 feet high.

Frédéric Auguste Bartholdi

Who: The French sculptor who designed the Statue of Liberty and spent more than 15 years on it.

When and where: Paris, 1870 to 1885. New York Harbor, dedicated October 28, 1886.

The statue crossed the ocean in 350 pieces, packed in 214 crates.

Frédéric Auguste Bartholdi worked on the Statue of Liberty from 1870 to 1886. He traveled to the United States in 1871, and in a little under three months he chose Bedloe's Island in New York Harbor as the site and met many of the leading people in America. He and Laboulaye set up committees in both countries to raise money.

Hammered copper. The statue's skin was made of thin copper sheets shaped by hammering, a method called repoussé. The arm and torch were finished in 1876, and the head and shoulders in 1878. That year the head and shoulders were shown at the Paris Universal Exposition, a world's fair. The whole statue was put together in Paris between 1881 and 1884.

Then it was taken apart again. The French ship Isère carried it across the Atlantic Ocean in 350 pieces packed in 214 crates, and it arrived on June 17, 1885. Bartholdi came back to New York with a group sent from France for the dedication in 1886.

Mounds dug away for fill

St. Louis, Missouri, was once called the Mound City. It was built among about 40 mounds, hills of earth that Native people built between about the years 600 and 1300. As the city grew, people dug away most of them. One group of 25 mounds stood along the north riverfront, and by the 1850s all of them were gone except part of the largest, Big Mound.

Sold for fill. Big Mound stood about 30 to 34 feet high. Workers dug it away completely in 1869, and its earth was sold to the North Missouri Railroad to build up the beds for its tracks. The workers who removed it found 32 burials, the graves of people, in two chambers inside it.

No surviving record says what became of the bones, and no archaeologist is known to have recorded the digging. The history of the mounds written for the Landmarks Association of St. Louis does not name the men who dug the mound out or the person who sold its earth.

Before the World's Fair of 1904 in Forest Park, two more groups of mounds along the River des Peres were destroyed, 16 mounds in all. That same history does not name who destroyed them.

One mound is left in St. Louis, Sugarloaf Mound, on the south side of the city. In 1928 someone built a house on top of it. The story of how the Osage Nation got it back is in 2000 to Today.

The largest private house

George W. Vanderbilt had Biltmore House built in Asheville, North Carolina, between 1889 and 1895. It has 250 rooms and covers about 179,000 square feet, and it is still the largest privately owned house in the United States. Vanderbilt opened it to his family on Christmas Eve, 1895.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 14 MINUTES

Workers died building three of the giant landmarks that went up between 1913 and 1941. Five died on the Empire State Building and 11 on the Golden Gate Bridge. At Hoover Dam the official count is 96, and it leaves out 42 more deaths that were listed as pneumonia.

The Lincoln Memorial and Mount Rushmore went up in the same years. Mount Rushmore was carved into a mountain in the Black Hills of South Dakota, on land that members of Congress took from the Sioux in 1877, breaking a treaty, a written agreement between nations.

The Woolworth Building

Frank W. Woolworth paid the whole $13.5 million cost of the Woolworth Building in New York himself, in cash. Workers finished it in 1913. At 792 feet, it was then the tallest building in the world. Woolworth had made his fortune from stores that sold goods for five and ten cents.

A roped-off section at the Lincoln Memorial

At the opening of the Lincoln Memorial in Washington in 1922, a white Marine ordered Black guests into a roped-off section, apart from white guests. William Howard Taft, who led the group in charge of the memorial, told the only Black speaker to cut his speech.

The building. Work on the memorial to Abraham Lincoln began on February 12, 1914, Lincoln's 105th birthday. Joseph Blackburn, a member of the Lincoln Memorial Commission, the group in charge of the building, broke the ground. He had been an officer in the Confederate army, the army of the Southern states that fought to keep slavery in the Civil War. The architect Henry Bacon designed the building, and the sculptor Daniel Chester French made the seated statue of Lincoln inside.

About 50,000 people came to the dedication on May 30, 1922. The crowd was segregated, which means Black people and white people were kept apart. According to an account in Smithsonian magazine, the Marine was rough and rude as he sent the Black guests to their section. Twenty-one of them left in protest.

The telegram. Robert Russa Moton was the only Black speaker. A week before, on May 23, Taft sent him a telegram, a message sent by wire. Taft was the Chief Justice, the head of the US Supreme Court, and also the president of the Lincoln Memorial Commission. He wrote, "Shall have to ask you to cut five hundred words."

Taft added, "I am sure you wish to avoid any insinuation (hint) of attempt to make the occasion one for propaganda (messages meant to push people to one side)." Moton made the cuts. One passage he removed said that the work Lincoln died for was "still unfinished."

A concert on the steps. Marian Anderson, a famous African American singer, sang from the memorial's steps in 1939, after the Daughters of the American Revolution refused to let her sing in Constitution Hall. The group owned the hall, in Washington, and its rule allowed only white performers there. Eleanor Roosevelt, the president's wife, quit the group in protest.

Harold Ickes, the Secretary of the Interior, got President Franklin D. Roosevelt's approval for a concert at the memorial. On Easter Sunday, April 9, 1939, Anderson sang from its steps to about 75,000 people of all races.

Whose land is Mount Rushmore on?

Mount Rushmore is carved on land that US officials promised to the Sioux in a treaty and that members of Congress later took. The Lakota, one of the Sioux peoples, call the mountain Tunkasila Sakpe, the Six Grandfathers. It stands in the Black Hills of South Dakota.

The treaty. In the Fort Laramie Treaty of 1868, US officials set aside a large reservation for the Sioux, and the Black Hills were part of it. The treaty states that the land was "for the absolute and undisturbed use and occupation" of the Sioux, theirs alone to live on and use. It also states that no part of it could be given up unless three-fourths of the adult Sioux men signed.

Lieutenant Colonel George Armstrong Custer led close to 1,000 soldiers and teamsters, men who drove wagons, into the Black Hills on July 2, 1874. Lieutenant General Philip Sheridan had sent them, and by the middle of August they had found gold.

The gold rush. Hundreds of miners rushed into the hills once the news spread. At first soldiers tried to keep them out of the land the treaty had set apart for the Sioux. In late 1875 President Ulysses S. Grant ordered the Army to leave the miners alone.

The food law. Members of Congress then set out to get the hills for the United States. In August 1876 they passed a law saying no more money would be spent on food for the Sioux until they gave up the Black Hills. Under it, officials would give the Sioux no more food rations, and the Sioux then depended on those rations to eat.

A group of officials led by George Manypenny brought the Sioux a paper in September 1876. It was already written, and it stated that the Sioux gave up the Black Hills. Only about one in ten of the adult Sioux men signed it, far fewer than the three-fourths the treaty required. Members of Congress made that paper a law anyway on February 28, 1877, and with it they took the Black Hills.

The judges of the US Supreme Court, who are called justices, decided the case *United States v. Sioux Nation* in 1980. Justice Harry Blackmun wrote for the court that the taking had broken the treaty. The justices agreed with an earlier award of $17.1 million for the land, plus interest, extra money of 5 percent for every year since 1877. The total came to about $106 million.

The Sioux nations have never taken the money. They say the Black Hills were never for sale. The money sits in a federal account held by the US Interior Department, and officials keep its balance secret. South Dakota Public Broadcasting put it at about $2 billion in 2022. Frank Star Comes Out, president of the Oglala Sioux Tribe, has said, "We will never sell out our holy lands, the Black Hills, to the United States for monetary compensation (payment in money)."

The carving. Workers carved the mountain from October 4, 1927, to October 31, 1941. According to the National Park Service, nearly 400 men and women worked there, among them drillers and blacksmiths. They did about 90 percent of the carving with dynamite and removed about 450,000 tons of rock in all.

Then drillers made rows of holes close together, a method called honeycombing, and smoothed the faces with hand tools. Workers were lowered down the 500-foot face on steel cables, sitting in bosun chairs, seats hung from a cable. No worker died in the 14 years of carving. Today the memorial's staff consult with 21 Native nations connected to the site.

Gutzon Borglum

Who: The sculptor who designed Mount Rushmore, after starting a Confederate carving in Georgia that the Ku Klux Klan helped pay for.

When and where: Stone Mountain, Georgia, from 1923. The Black Hills of South Dakota, 1927 to 1941.

Klan money helped pay for Borglum's carving at Stone Mountain.

Before he carved Mount Rushmore, Gutzon Borglum went to meetings of the Ku Klux Klan and took a deep part in Klan politics, according to reports in the Washington Post and Smithsonian magazine. No record proves he was a member.

The Klan. Confederate veterans had started the first Klan in Pulaski, Tennessee, in December 1865. Its members whipped, burned the homes of and murdered freed Black people and the people who helped them. Federal officials put that Klan down in the early 1870s.

William J. Simmons and 16 other men climbed Stone Mountain in Georgia on Thanksgiving night in 1915. They burned a cross there and announced that the Klan was born again. Members of the new Klan preached white supremacy, the belief that white people should rule over everyone else. They were also against Catholics and Jews, and they wanted fewer immigrants let into the country.

At Stone Mountain, Borglum designed a carving of three Confederate leaders on horseback, Robert E. Lee, Stonewall Jackson and Jefferson Davis. Klan money helped pay for it. Work began in 1923, and Borglum unveiled the head of Lee in 1924.

Borglum quarreled with the businessmen who ran the Stone Mountain project, and they fired him. He then destroyed his models to protect his design, which made the directors angry. A warrant, a court paper ordering his arrest, was issued, and he fled Georgia.

Borglum began Mount Rushmore in 1927 and directed the carving until he died on March 6, 1941. His son, Lincoln Borglum, finished the work that year.

Nick Clifford

Who: The youngest worker on Mount Rushmore, and the last of its carvers to die.

When and where: Mount Rushmore, South Dakota, 1938 to 1940. Died in 2019, at 98.

The Borglums hired the 17-year-old partly to pitch for their baseball team.

Donald "Nick" Clifford was 17 when he was hired at Mount Rushmore in 1938, the youngest worker on the job. The Borglums wanted him for the crew's baseball team, where he pitched.

Clifford earned 55 cents an hour. He ran the winch, a machine that winds a cable, and the cable carried men up the face of the mountain. He also drilled holes for the dynamite. He worked there until 1940.

When Clifford died on November 23, 2019, at the age of 98, he was the last of the carvers.

The Empire State Building and the Mohawk ironworkers

Five workers died building the Empire State Building in New York, by the official count. A story that hundreds died is false.

Builders took about 410 days to put up the building, from the first steel to the opening on May 1, 1931. Up to 3,400 workers were on the job each day, and during the fastest stretch the steel frame rose about four and a half floors a week. The photographer Lewis Hine took pictures of the men working on the steel for his book *Men at Work*.

The building has 102 stories and stands 1,250 feet tall, or 1,454 feet to the tip of today's antenna. It was the tallest building in the world until 1970, when the steel frame of the World Trade Center's North Tower rose higher.

High steel. Some of the riveting crews were ironworkers from Kahnawake, a community of the Mohawk Nation in Canada. Ironworkers put up the steel frames of buildings and bridges, and riveters joined the steel beams with rivets, hot metal pins that lock the beams together. Kahnawake men had done this work since the 1880s.

The Quebec Bridge. Kahnawake ironworkers had died on an earlier job. On August 29, 1907, the south half of the Quebec Bridge fell while it was being built across the St. Lawrence River in Canada. James H. Marsh, writing in the Canadian Encyclopedia, gives 75 workers killed, and Eve Cable, writing in *The Eastern Door*, the Kahnawake newspaper, gives 76. Thirty-three of the dead were Kahnawake men.

A Royal Commission, a group set up to find the cause, blamed two engineers. Theodore Cooper, an expert brought in to oversee the design, had made mistakes in it and had lengthened the bridge to save money. John Deans, the chief engineer, kept the crews working after Cooper warned of danger.

The women's rule. The women of Kahnawake made sure after 1907 that the men of one family did not all work on the same job at the same time. *The Eastern Door* gives the rule this way: "If a lady had seven sons, they're not all

going to the same job." Crosses made from the bridge's steel stand in the village cemetery. In 2007 a monument went up with a picture of each man who was killed and a 40-foot steel model of the bridge.

The Golden Gate Bridge

Eleven workers died building the Golden Gate Bridge in San Francisco. One died in October 1936, and ten died on February 17, 1937, when a falling work platform tore through the safety net under the bridge. The bridge opened in May 1937, and on May 27, the first day, about 200,000 people walked across it.

Who designed it. Joseph Strauss was the chief engineer. Charles Ellis, the engineer who worked out how the bridge's structure would hold its weight, did the detailed design and thousands of calculations from 1929 to 1931. Strauss fired Ellis before building began, and Ellis got no public credit for decades. In a 2007 report, officials of the Golden Gate Bridge District gave Ellis the main credit for the design.

Joseph Strauss

Who: The chief engineer of the Golden Gate Bridge, who hung a safety net under the bridge while it was built.

When and where: The Golden Gate Bridge, which opened in May 1937.

A $130,000 net under the bridge caught 19 falling men.

Joseph Strauss required hard hats on the Golden Gate Bridge job. He also spent about $130,000 on a safety net that hung under the bridge while it was built. It was the first large safety net used on an American bridge job.

The net caught 19 men who fell. They called themselves the Halfway to Hell Club.

Al Zampa

Who: An ironworker who fell from the Golden Gate Bridge and lived, and who later had a bridge named for him.

When and where: The Golden Gate Bridge, October 19, 1936. From Crockett, California.

Zampa fell into the net where it sagged down onto the rocks.

Alfred Zampa was an ironworker from Crockett, California. On October 19, 1936, he slipped on wet steel and fell into the safety net. He was near the shore, where the net sagged down onto the rocks, and he broke four bones in his back.

Zampa spent twelve weeks in the hospital and two years in a back brace. Then he went back to ironwork. He was one of the first members of the Halfway to Hell Club and later its unofficial spokesman, the man who spoke for the club without holding any title.

California lawmakers named the new bridge over the Carquinez Strait the Alfred Zampa Memorial Bridge in 2003.

How many men died building Hoover Dam?

By the official count, 96 men died building Hoover Dam. The count comes from the Bureau of Reclamation, the federal office that runs the dam. It includes deaths from falls, rock slides, machinery and drowning, and it leaves out deaths from heat, heart trouble and pneumonia, a sickness of the lungs. The exact number of deaths is not known.

The 42. The main builder was a company called Six Companies Inc. The records list 42 more workers who died of pneumonia. Workers said these men really died of carbon monoxide poisoning. They said officials of Six Companies listed the deaths as pneumonia so the company would not have to pay compensation claims, the payments owed when a worker is hurt or killed on the job.

Carbon monoxide is a gas with no color or smell that comes from burning fuel. When a person breathes it in, it stops the blood from carrying oxygen. The workers blamed the gasoline-powered vehicles running inside the diversion tunnels, which were dug to carry the river around the building site. The tunnels often reached 140 degrees Fahrenheit and filled with thick exhaust. In nearby Boulder City, no one who did not work on the dam was recorded as dying of pneumonia in those years.

Father and son. The first deaths on the official list came in 1922, during the surveys for the dam. J. G. Tierney drowned on December 20 of that year, and Harold Connelly also died that year. The last death on the list came 13 years to the day after Tierney's. On December 20, 1935, his son, Patrick Tierney, an electrician's helper, fell from one of the dam's intake towers, the towers that take in water.

People tell a story that men are buried inside the dam's concrete. It is false. Workers poured the dam as many separate blocks, in shallow layers, with crews standing over each pour, so a body could not have been left inside. President Franklin D. Roosevelt dedicated the dam on September 30, 1935, and its generators began making electricity in 1936.

BY 1950

If you remember one thing: Workers died building three of the giant landmarks of these years, and Mount Rushmore was carved on land that members of Congress took from the Sioux.

96 and 42: Hoover Dam's official count of the dead is 96. The records list 42 more workers as dying of pneumonia, deaths that workers blamed on carbon monoxide in the tunnels.

Next: After 1950, two men climbed the unfinished Gateway Arch in St. Louis to protest who was kept off the job.

1950 to 2000

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Americans in these years fought over who was allowed to build a landmark and over what a memorial to the dead should look like. In St. Louis, Justice Department lawyers later charged in court that leaders and members of the building trades unions kept Black workers out of the unions, and so off the job building the Gateway Arch. Two men climbed the unfinished arch in protest.

In Washington, critics attacked the design of the Vietnam Veterans Memorial, a black stone wall set into the ground. The names of more than 58,000 Americans who died in the Vietnam War are cut into it, and it was dedicated in 1982. Builders also put up two towers in New York, one of them for a few years the tallest building in the world, and workers restored older landmarks, including the Statue of Liberty and Ellis Island.

The Space Needle

Builders put up the Space Needle in Seattle in about 400 days for a world's fair called the Century 21 Exposition. It opened on April 21, 1962, the first day of the fair. The tower is 605 feet tall, with a restaurant on top that turns around.

The Gateway Arch and the men kept off the job

The Gateway Arch in St. Louis is 630 feet tall and 630 feet wide, and its outside is stainless steel. The architect Eero Saarinen won the contest to design it, and the winner was announced in February 1948. Saarinen died in 1961, before building began. Workers built the arch from February 12, 1963, to October 28, 1965. Insurers had predicted 13 deaths, and no worker died.

Black workers were kept out of the building jobs on the arch, a federal monument. In St. Louis, the building trades unions, groups that carpenters, plumbers, ironworkers and other builders join to bargain together for jobs and pay, had almost all white members. Justice Department lawyers later charged in court that the leaders and members of these unions kept Black men out of the unions and so out of the jobs.

The climb. Percy Green and Richard Daly climbed a 125-foot construction ladder on the unfinished north leg of the arch on July 14, 1964. The two men were activists, people who campaign for a cause, and they climbed to protest. An article by the Gateway Arch Park Foundation states that they stayed up for 6 hours, and a St. Louis Public Radio report states that Green was up there for more than four hours. Police arrested both men.

Green later helped start ACTION, a St. Louis group that worked for fair hiring. ACTION stood for the Action Committee to Improve Opportunities for Negroes, a word of that time for Black people.

Two Black plumbers, Elijah Smith and Oliver Parker, were hired to work on bathrooms in the arch's visitor center. White union workers walked off the job on January 7, 1966, to protest the hiring of the two men.

The lawsuit. Lawyers in the US Justice Department sued the St. Louis Building and Construction Trades Council and five of its unions about a month later, on February 4, 1966. They charged the unions under the Civil Rights Act of 1964, which made it against the law for unions and employers to treat workers unfairly because of their race.

Officials of several of the unions settled the case. They signed an agreement to end "all practices that were discriminatory," meaning every rule and habit that kept Black workers out, and to judge everyone who applied by the same written standards. Two unions, the sheet metal workers and the electricians, fought the case in court. A judge ruled for them in 1968. In 1969 appeals judges overturned that ruling, found that the two unions' rules still kept Black workers out, and ordered the rules changed.

Harold McComber

Who: A Mohawk ironworker from Kahnawake, in Canada, who left home to work on the steel frames of Manhattan skyscrapers.

When and where: Manhattan and Kahnawake, filmed in 1965.

Movie: *High Steel* (1965), a documentary by the National Film Board of Canada, directed by Don Owen. It is about McComber's working life.

McComber's sons grew up without him while he built New York skyscrapers.

Harold McComber was a Mohawk ironworker from Kahnawake, in Quebec. Like many Kahnawake men, he left home to work on the steel frames of skyscrapers in Manhattan.

The director Don Owen built his 13-minute documentary *High Steel* around McComber's working life, and the National Film Board of Canada produced it in 1965. The film shows McComber and the men he worked with on the steel high above New York City. It also shows the quieter life of their community at home. To film the men at work, the camera crew reached the site by crossing on a ladder from the building next door.

In the film, McComber is sorry that his sons have had to grow up without their father. The film also tells how Kahnawake men first earned their name for steelwork, on a railway bridge built in the late 1800s.

The World Trade Center

The architect Minoru Yamasaki designed the two towers of the World Trade Center in New York, each 110 stories tall. The ribbon-cutting was on April 4, 1973. The North Tower stood 1,368 feet. Its steel frame had passed the height of the Empire State Building in 1970, which made it the tallest building in the world. It kept that title until May 1973, about a month after the ribbon-cutting, when the steel frame of the Sears Tower in Chicago rose higher.

Philippe Petit

Who: A French high-wire walker who walked on a cable between the two towers of the World Trade Center.

When and where: New York, August 7, 1974.

Movie: *Man on Wire* (2008), directed by James Marsh. It won the Academy Award for Best Documentary Feature and is about the walk.

Petit crossed between the towers eight times in 45 minutes.

Philippe Petit walked on a wire strung between the tops of the Twin Towers, about 1,350 feet above the ground, on August 7, 1974. He and his friends had smuggled a cable up to the roofs and strung it from one tower to the other. Petit crossed eight times in 45 minutes.

Afterward police arrested him. The charges against him were dropped. In return, he put on a show for children in Central Park.

The Vietnam Veterans Memorial and its critics

Two long walls of polished black granite, set into the ground of the National Mall in Washington, form the Vietnam Veterans Memorial. The names of more than 58,000 Americans who died in the Vietnam War are cut into the stone, in the order they died. It was dedicated on November 13, 1982, and it became the most visited memorial in Washington.

The attacks. Critics attacked the design before it was built. One veteran called it "a black gash (a deep cut) of shame." Another critic said that burying the dead "in a black stone sarcophagus," a stone coffin, "is like spitting on their graves." Opponents of the design included the radio host Patrick Buchanan and Congressman Henry Hyde. They campaigned to make the wall white and to add an eight-foot statue of soldiers.

The members of the US Commission of Fine Arts, a federal board that reviews designs in Washington, kept the wall. They also added a statue of three soldiers by Frederick Hart nearby, in 1984. Hart's design had won third place in the contest for the memorial.

Maya Lin

Who: The 21-year-old college student whose design won the contest for the Vietnam Veterans Memorial.

When and where: Yale University and Washington, 1981 to 1982.

Movie: *Maya Lin: A Strong Clear Vision* (1994). It won the Academy Award for Best Documentary Feature and is about Lin.

Entry number 1026 beat 1,420 others.

Maya Lin was a 21-year-old architecture student at Yale University when she won the contest to design the Vietnam Veterans Memorial. She had first drawn the design for a class. She was born in Ohio, where her parents were professors. They had escaped from mainland China as young adults.

The contest. It was the biggest design competition held in the United States up to then. The judges looked at the entries without knowing who had made them, and they chose entry number 1026 over 1,420 others. In May 1981 officials came to Lin's college room to tell her she had won the $20,000 first prize.

The critics. Lin argued against the campaign to make her wall white and add a statue, and she said Congressman Henry Hyde was "drawing mustaches on other people's portraits." Some of the wall's critics also used her age and her Chinese family background as reasons to change the design or drop it. The speakers at the dedication in 1982 did not mention her.

Restoring the old landmarks

Workers restored the Statue of Liberty for its 100th birthday from 1984 to 1986. They took down its original 1886 torch in 1984, and it is now in the statue's museum. In its place they put a copy made of copper and covered in gold leaf, thin sheets of gold. The statue was dedicated again on July 4, 1986.

Ellis Island, in New York Harbor, was where federal officials checked arriving immigrants. On May 11, 1965, President Lyndon B. Johnson signed Proclamation 3656, a formal order from the president, under which Ellis Island became part of the Statue of Liberty National Monument. Ellis Island's main building had opened in 1900. Workers restored it, and in 1990 it opened as a museum of immigration.

On the West Coast, from 1910, US officials held immigrants in barracks on Angel Island, in San Francisco Bay. Chinese immigrants held there had carved more than 200 poems into the barracks walls, and in 1970 the poems were found again. After that, people campaigned to save the barracks. In 1997 Angel Island Immigration Station became a National Historic Landmark.

2000 to Today

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Memorials to people who were killed or enslaved went up after 2000: to the victims of the September 11 attacks, to the Black Americans murdered by mobs without a trial, and to the people held in slavery. In the same years, officials and work crews took down, moved or renamed hundreds of memorials to the Confederacy, the Southern states that fought to keep slavery in the Civil War.

These memorials include statues and place names. Researchers at the Southern Poverty Law Center, a group that campaigns for civil rights, counted 415 of them removed, moved or renamed between 2015 and April 2025, and 2,086 still standing. In 2025 federal officials began putting some of the removed monuments back.

Ground Zero rebuilt

The National September 11 Memorial in New York stands where the Twin Towers stood. On September 11, 2001, hijackers took over four airliners. They crashed two of them into the North Tower and the South Tower, and fires from the crashes made both towers collapse. They crashed a third into the Pentagon, the headquarters of the US military, near Washington, and the fourth, Flight 93, also crashed.

The dead. The hijackers killed 2,977 people, not counting the 19 hijackers themselves, according to the 9/11 Memorial and Museum. Of those, 2,753 were killed in New York, 184 at the Pentagon and 40 on Flight 93.

The memorial opened on September 11, 2011, ten years after the attacks. It has two pools set into the footprints of the two towers. Each pool covers nearly an acre. Water falls 30 feet into each pool and then drops another 20 feet. Around the edges, bronze panels carry the names of 2,983 people. They are the people killed on September 11, 2001, and the six people killed in an earlier bombing of the towers in 1993.

The design, called *Reflecting Absence*, won a contest in 2004 that drew 5,201 entries from 63 countries. The museum under the memorial opened on May 21, 2014. On November 3, 2014, One World Trade Center opened nearby. It stands 1,776 feet to the tip of its spire. The height was chosen to match 1776, the year of the Declaration of Independence.

Michael Arad

Who: The architect, then 34 and working for New York City, whose design for the National September 11 Memorial was chosen from 5,201 entries.

When and where: New York. He won the design contest in 2004, and the memorial opened in 2011.

His first sketches showed two empty spaces in the Hudson River.

Michael Arad was 34 years old and working as an architect for a New York City government office when he entered the contest for the memorial. In his first sketches, he pictured two empty spaces cut into the surface of the Hudson River. His finished design put two pools where the towers had stood.

Arad worked with the landscape architect Peter Walker, who plans outdoor spaces such as parks. The contest judges chose their entry from 5,201 in 2004.

New memorials in Washington

New memorials to Black Americans went up in Washington after 2000. The Martin Luther King, Jr. Memorial was dedicated on October 16, 2011, the first memorial to an African American in the area of the National Mall. At its center is a 30-foot carving called the Stone of Hope.

The Capitol's builders. Members of Congress unveiled a marker to the enslaved workers who helped build the Capitol on February 28, 2012, in Emancipation Hall, the great hall of the Capitol Visitor Center. The marker is a block of Aquia Creek sandstone taken from the Capitol's East Front, which was built from 1824 to 1826. Its plaque states that the stone "was quarried by laborers, including enslaved African Americans, and commemorates (honors) their important role in building the Capitol."

The National Museum of African American History and Culture opened on September 24, 2016, beside the Washington Monument. It is part of the Smithsonian. The architect David Adjaye led its design. He based the building's outside, a bronze-colored shape in three levels called the corona, on the art of the Yoruba, an African people. The pattern of its metal screen is copied from ironwork made by enslaved and free Black craftsmen in the South.

Memorials to lynching and slavery

The National Memorial for Peace and Justice, in Montgomery, Alabama, is a memorial to Black people who were lynched, killed by a mob without a trial. The lawyer Bryan Stevenson founded the Equal Justice Initiative (EJI), a legal group that does not make a profit. EJI built the memorial and opened it on April 26, 2018, with a museum called the Legacy Museum.

805 blocks. The memorial holds 805 steel blocks, each six feet tall, hanging from above. Each one stands for a county in the United States where EJI researchers found a record of a lynching. In their study *Lynching in America*, EJI researchers counted almost 4,400 lynchings of Black people from 1877 to 1950.

A copy of each county's block stands at the site, and EJI offers each copy to groups in that county to set up at home. A team called MASS Design Group designed the memorial, and it cost about $20 million, paid by private foundations.

122,000 names. EJI opened the Freedom Monument Sculpture Park nearby in March 2024, on 17 acres with nearly 50 sculptures. Its National Monument to Freedom lists more than 122,000 last names taken from the 1870 census, the government's count of everyone in the country that year. That census was the first time people who had been enslaved could claim a last name that the government would recognize. In it, nearly five million Black people took these names as their own.

The monument stands 43 feet tall and 155 feet long, and EJI dedicated it on Juneteenth, June 19, 2024. Juneteenth marks June 19, 1865, the day Major General Gordon Granger of the US Army reached Galveston, Texas, with the news that enslaved African Americans were free.

Confederate monuments coming down and going back up

Who put them up. Members of the United Daughters of the Confederacy had put up hundreds of Confederate statues at courthouses and other public places since the Civil War, researchers at the Southern Poverty Law Center (SPLC) reported in 2018. The researchers found two main periods of building.

The first began around 1900, while lawmakers in Southern states were passing Jim Crow laws, which they wrote to force Black and white people apart and to take away Black people's rights. The second ran from the early 1950s through the 1960s, when segregationists, people who wanted to keep the races apart, fought the civil rights movement.

Hundreds of these memorials came down or were renamed after people were killed in three places: Charleston, Charlottesville and Minneapolis.

Charleston. A 21-year-old white man named Dylann Roof sat in on a Bible study at the Emanuel African Methodist Episcopal Church in Charleston, South Carolina, on June 17, 2015. After about an hour he opened fire and killed nine Black worshippers. Roof had posted pictures of himself with Confederate flags on his website, and he had written racist views there. A federal jury found Roof guilty of all 33 federal charges against him on December 15, 2016, and the federal court sentenced him to death on January 10, 2017.

The Confederate battle flag had flown at the South Carolina State House, in Columbia, since 1961. After the murders, state lawmakers voted to take it down, and Governor Nikki Haley signed the law on July 9, 2015. The next day an honor guard of the state Highway Patrol lowered the flag, and it went to the Confederate Relic Room at the state museum.

Charlottesville. Paul Goodloe McIntire paid for a statue of the Confederate general Robert E. Lee in Charlottesville, Virginia, and it was dedicated in 1924. The members of the city council voted 3 to 2 on February 6, 2017, to remove it. City workers took the statue down on July 10, 2021, and city officials gave it to

the Jefferson School African American Heritage Center. In October 2023 the staff of the center's Swords into Plowshares project had the statue melted down at a foundry.

The rally. White supremacists organized a rally called Unite the Right around the statue for August 2017, six months after the council's vote. On August 12 James Alex Fields Jr. drove his car into a crowd of people protesting against the rally. He killed Heather Heyer, a 32-year-old paralegal, a worker who helps lawyers, and injured many others. Counts of the injured run from 19 to 29.

A Virginia jury found Fields guilty in December 2018 of first-degree murder, the most serious kind of murder, done on purpose. In 2019 he pleaded guilty to 29 federal hate crimes, crimes against people chosen because of their race, religion or a similar trait, and a federal judge sentenced him to life in prison.

Minneapolis. Minneapolis police officer Derek Chauvin murdered George Floyd on May 25, 2020, by kneeling on his neck, back and shoulder for about ten minutes. A jury convicted Chauvin of murder in 2021. Protests followed Floyd's murder, and SPLC researchers counted at least 168 Confederate symbols removed in 2020 alone.

Virginia Governor Ralph Northam ordered the removal of Richmond's Lee statue, a 12-ton monument put up in 1890, in June 2020. The judges of the Supreme Court of Virginia then refused to block the removal in two lawsuits. A crew led by Devon Henry, who owned the construction company doing the work, took the statue down on September 8, 2021, ABC News reporters wrote.

SPLC researchers counted 2,086 Confederate memorials still standing in public places in April 2025. In their count, memorials include place names and other symbols as well as monuments such as statues, and 685 of the 2,086 were monuments. Rivka Maizlish of the SPLC called the memorials "the results of a mass propaganda campaign," meaning a huge effort to push one side of the story.

Going back up. Federal officials began putting some monuments back in 2025. Protesters had pulled down a statue of the Confederate general Albert Pike in Washington in June 2020 and set it on fire. National Park Service

workers put it back up in October 2025. They cited a historic preservation law and an executive order, a written command from the president, that President Donald Trump had signed on restoring removed monuments.

Members of the United Daughters of the Confederacy put up the Confederate Memorial at Arlington National Cemetery in 1914. The sculptor, Moses Ezekiel, had fought for the Confederacy. The memorial shows "an enslaved man following his owner to war." It also shows an enslaved woman as a "mammy," holding a white officer's child.

The "mammy" was a made-up picture of an enslaved Black woman who cared for a white family's children and seemed happy in slavery. The picture hid the truth that enslavers separated these women from their own children. A commission set up by Congress found that the memorial showed "highly sanitized depictions of slavery," pictures of slavery with the cruelty cleaned out.

On that commission's advice, the memorial was taken down in December 2023. *Stars and Stripes*, the newspaper that reported it, does not say who did the work. In August 2025 Defense Secretary Pete Hegseth announced that it would return in 2027, and *Stars and Stripes* reported that the repairs would cost about $10 million.

Who took down the slavery exhibit in Philadelphia?

National Park Service workers took down an exhibit about slavery in Philadelphia in 2026. A federal judge ordered it put back, but appeals judges later let officials of the Trump administration replace it with panels of their own. The President's House site marks where George Washington lived while he was president. Its exhibit panels described the lives of nine people Washington held in slavery there.

The court case. The workers removed the panels in January 2026, following President Trump's executive order called "Restoring Truth and Sanity to American History." Lawyers for the City of Philadelphia sued. On February 16, 2026, US District Judge Cynthia Rufe ordered the panels restored while the case went on, and workers began putting them back three days later.

Lawyers for the federal government appealed, which means they asked a higher court to change the ruling. On June 18, 2026, judges of the Third Circuit Court of Appeals ruled that the city had no right to decide what the exhibit showed. On July 15 officials of the Trump administration replaced the old panels with their own.

The new panels still describe the nine enslaved people, but they leave out some of the old panels' detail, such as a map of the routes of the slave trade and a timeline of slavery. As of September 2026, the administration's panels were in place, and the city's case was still going on in the lower court.

Old landmarks, new names and owners

Sugarloaf Mound, on the south side of St. Louis, is the only one left of the roughly 40 mounds that Native people built in the city. People dug away all the others as the city grew, Big Mound and its 32 burials among them, as 1850 to 1900 describes. Someone built a house on top of Sugarloaf Mound in 1928.

Back to the Osage. The Osage Nation bought that house and about a third of the mound in 2009, and had the house torn down in 2017. In 2025, working with an arts group called Counterpublic, the Osage Nation got the last two pieces of the property transferred to it. The Osage Nation's announcement describes Sugarloaf Mound as a sacred Osage site.

Afterward Andrea Hunter, the Osage Nation's Historic Preservation Officer, said, "It has been a long 17-year journey, and I am thrilled to finally see the entire mound come under Osage Nation control." She said her office must now "remove all the structures from the properties and stabilize (make firm) the mound." Osage leaders plan to build a center there to teach visitors about the mound.

New names. Workers finished restoring the barracks of the Angel Island Immigration Station in California in 2009. On February 22, 2018, President Donald Trump signed a law under which the Jefferson National Expansion Memorial in St. Louis, the grounds of the Gateway Arch, became Gateway Arch National Park. It is the smallest national park in the country.

THE THREE QUESTIONS, ANSWERED

1. **Who built the White House and the Capitol, and who was paid for their work?** Enslaved men did much of the work, from quarrying the stone to sawing boards and hauling loads. The officials in charge rented them from their owners by the year, and the owners collected the pay.
2. **How many workers died building Hoover Dam?** By the official count, 96. The records list 42 more as dying of pneumonia,

deaths that workers blamed on carbon monoxide in the tunnels, so the exact number is not known.

3. **What has happened to the Confederate monuments since 2015?** By April 2025 officials and work crews had taken down, moved or renamed 415 Confederate memorials, from statues to place names. Hundreds of them came down or were renamed after Dylann Roof killed nine Black worshippers in Charleston in 2015, James Alex Fields Jr. killed Heather Heyer in Charlottesville in 2017 and the police officer Derek Chauvin murdered George Floyd in Minneapolis in 2020. In 2025 federal officials began putting some back.

What changed: No one wrote down who built the oldest landmarks here, and since 2000 Americans have built memorials to the enslaved and the murdered and fought over monuments to the Confederacy.

FIVE DATES TO KEEP

October 2, 1672: Workers break ground for the Castillo de San Marcos.

September 18, 1793: George Washington lays the cornerstone of the Capitol.

February 28, 1877: Members of Congress take the Black Hills from the Sioux.

May 24, 1883: The Brooklyn Bridge opens, after at least 21 workers died building it.

June 17, 2015: Dylann Roof kills nine Black worshippers in Charleston.

NOT KNOWN: Nobody knows how many enslaved people helped to build the Capitol, and two congressional historians wrote that no one ever will.

Where to go: The Octagon Earthworks in Newark, Ohio, a golf course from 1910 to 2024, opened to the public on January 1, 2025.

Next: Chapter 13, The Elements. It describes the copper, iron, gold and other elements that people here dug up, used and discovered.

Chapter 13: The Elements

CHAPTER 13 OF 37 · THE ELEMENTS · 10 ERAS · ABOUT 85 MINUTES

From the first ironworks to the uranium mines, the people whose land held the metal often lost that land. Many of the people who dug and handled it paid with their freedom or their health.

A lump of pure copper that weighs 3,708 pounds has been at the Smithsonian, in Washington, since 1860. It once lay in the Ontonagon River country near Lake Superior. The Ojibwe had known it for a long time, held it sacred and made offerings at it. In 1843 a Detroit merchant hauled it away, and a federal agent made him hand it over to the government. The Keweenaw Bay Indian Community asked for it back in 1991, and Smithsonian officials refused in 2000.

This chapter follows the elements people here dug up, shaped, named and made, from copper hammered 9,500 years ago to the hunt for element 120. The gold, copper and uranium in this chapter lay under Cherokee, Ojibwe, Sioux and Navajo land, so watch what happened to the people who lived there. Watch, too, who did the work with the metal: Scottish prisoners, enslaved ironworkers, hatters, dial painters, men making a lead additive for gasoline, and uranium miners. Nobody in the United States discovered a new element until 1940. Between 1940 and 1974, teams in American labs made or found at least twelve.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who lost their land when Americans found gold and copper on it?
2. What did the people in charge say about the radium, lead and uranium that poisoned workers?

3. When did someone in the United States first discover a new element?

Five minutes? Read "Lead in gasoline and paint" in 1900 to 1950.

Can skip: "Meanwhile in Europe", "Lead at Galena", "Elements at home". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

Native miners near Lake Superior were digging and shaping copper at least 9,500 years ago. A 2021 study places them among the world's first coppersmiths, people who make things out of copper. All the land that later became the United States was Native land before 1500. No miner or metalworker from these thousands of years is known by name.

Copper is one of the chemical elements. Everything is made of atoms, tiny pieces far too small to see, and an element is a substance made of only one kind of atom. Copper, iron, gold and silver are all elements.

Numbers and rows. Each element has a number. It counts the protons, tiny particles in the center of each atom, so copper is element 29 because every copper atom has 29 protons. Chemists arrange the elements in order of their numbers in a chart called the periodic table.

The numbers rise from left to right along each row, and the elements in each column often act alike. The seventh row ends with element 118. Heavier elements, such as element 120, which scientists are trying to make, would go in an eighth row.

Nobody anywhere on Earth knew what an element was before 1500. People here mined copper, gathered salt and colored earth, and chose hard stone for tools. In some regions they also worked iron that had fallen from space. They turned these materials into tools, beads, ornaments and paint.

Copper from Lake Superior

In the Lake Superior country, in what are now Michigan, Wisconsin and Minnesota, veins of pure copper metal, called native copper, run through the rock. Native miners dug it out and hammered it into spear points, knives, fishhooks and ornaments. They also made awls, thin pointed tools for punching holes.

How old the work is. David Pompeani, a scientist who studies rocks and the Earth, led a team that dated the copper work. They used 53 radiocarbon dates, ages measured from things that were once alive, such as wood or bone. The team published its results in 2021 in *Radiocarbon*, a magazine for scientists' studies. The copper work began at least 9,500 years ago, about 3,500 years earlier than scientists had thought. It was busiest between about 7,000 and 5,000 years ago.

The oldest known copper work in the Middle East is a pendant, an ornament that hangs from a cord, about 8,700 years old. So the Lake Superior work is as old as the earliest copper work known in the Middle East, or older. Writers at the journal *Science* reported the study under the headline "Ancient Native Americans were among the world's first coppersmiths."

Traders carried Lake Superior copper hundreds of miles along Native trade routes. Along the same routes they carried shells, blue-green turquoise stone, and obsidian, a natural glass that forms from volcanic lava.

Iron from space

The only iron metal people here could work before 1500 came from meteorites, pieces of rock or metal from space that have fallen to Earth. Nobody here yet got iron out of rock by smelting, which means heating rock until the metal inside it melts out.

The beads from Havana. Archaeologists, the scientists who dig up and study what people left behind long ago, found 22 iron beads in 1945 in burial mounds near Havana, Illinois. Native people of the Hopewell way of life, which lasted in the eastern woodlands from about 2,000 to 1,600 years ago, had built the mounds over the graves of their dead. They made the beads about 2,000 years ago.

Timothy McCoy of the Smithsonian and four other scientists studied the beads and published their findings in 2017. The iron came from the Anoka meteorite, an iron meteorite from Minnesota. So people had carried or traded the metal several hundred miles before anyone made it into beads. The beadmakers shaped the iron by hammering it, the same way people worked copper.

Gold, silver, ochre, salt and stone

In some regions, metalworkers made ornaments of gold and silver. People also gathered salt and traded it, and they made tools of flint and chert, two hard stones.

A compound. Red and yellow ochre, a colored earth, went into paint. Ochre gets its color from iron oxide, a compound. A compound is two or more elements joined together, and iron oxide is iron joined with oxygen.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 1 MINUTE

Spanish explorers crossed the Southeast and the Southwest in the 1500s on long journeys called expeditions, looking for gold and silver. They found no cities of gold in the lands that later became the United States. The land was still Native land from coast to coast, and people here worked the same copper, gold and meteorite iron as before. The Spaniards brought some metal of their own into their few outposts, the small forts and settlements they built.

Spanish gold hunters, 1539 to 1543

Hernando de Soto led a Spanish expedition across the Southeast from 1539 to 1543. Francisco Vázquez de Coronado led another across the Southwest from 1540 to 1542. Both men were following stories of cities full of gold, and neither found one. In the same century Spaniards did find large amounts of silver, but in Mexico and South America, not in the future United States.

No science of elements yet. Nobody here or anywhere else had a science of elements. In Europe, the modern idea of an element had not been written down. People there who studied what things are made of still worked by alchemy, the older practice that came before chemistry. Alchemists tried to turn ordinary metals, such as lead, into gold by heating them and trying to make them purer. They kept most of their methods secret.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

The first ironworks English colonists built here stood on the land of the Powhatan, a union of Native towns in eastern Virginia, and Powhatan fighters destroyed it in 1622. At the second, at Saugus in Massachusetts, workers melted iron out of ore, rock that contains metal, from 1647 until about 1670. The company's men forced Scottish prisoners of war to work there. People in Europe, Asia and Africa had been smelting iron for thousands of years already.

Falling Creek, 1619 to 1622

English colonists began building an ironworks on Falling Creek, near present-day Richmond, in 1619. They built it for the Virginia Company, the English company that King James I had allowed to start a colony in Virginia in 1606. By March 1622 the furnace was ready to make iron, or close to it.

Opechancanough was the head chief of the Powhatan. The English colony stood on Powhatan land, and English colonists had been moving farther up the James River and settling more of it. From 1618, Virginia Company officials gave land to new colonists, who planted it in tobacco. By 1620 colonists were taking the land they wanted from the Powhatan without asking or paying, according to the historian Betty Wood.

Company officials, George Thorpe most of all, also pressed the Powhatan to send their children to live among the English. The English wanted to school the children and make them Christian. Opechancanough planned attacks on many parts of the colony to drive the English out for good.

The attack. Powhatan fighters destroyed the ironworks at Falling Creek on March 22, 1622, and killed 27 people there. The colonists never managed to rebuild the works.

Hammersmith at Saugus

Richard Leader began building an ironworks on the Saugus River near Boston in 1646. He worked for the Company of Undertakers of the Iron Works in New England, and in the 1600s an undertaker meant a person who took on a business project. People called the works Hammersmith. Its furnace was running by the next year.

From ore to iron. Hammersmith was the first ironworks in North America where every stage of the work happened on one site. The ore was bog iron, which workers collected from swamps and ponds. A blast furnace, a tall furnace with air blown into its fire, got hot enough to melt the iron out of the ore. The iron that came out of it was called pig iron.

At the forge, a workshop where workers heat iron and hammer it, they turned pig iron into wrought iron, a tougher iron that bends instead of breaking. In the rolling and slitting mill, rollers pressed the iron flat and blades cut it into strips. The strips were the raw material for nails, tools and horseshoes. Seven waterwheels powered the machines.

The works ran until about 1668 to 1670. The site is now Saugus Iron Works National Historic Site. Chapter 8 tells more about both ironworks and the people who built and ran them.

Scottish prisoners at Saugus

Scottish prisoners of war did forced labor at Saugus. English soldiers under Oliver Cromwell took these Scottish soldiers prisoner at the Battle of Dunbar in 1650 and the Battle of Worcester in 1651. About 400 prisoners from the two battles reached New England in all, according to the National Park Service. The prisoners had no choice about going or about the work.

The march to Durham. Cromwell's soldiers won the Battle of Dunbar in September 1650. They marched about 3,900 of the fittest prisoners south to the English city of Durham. Their English captors gave them no food and no shelter on the march, and killed some of them for refusing to keep walking. At Durham the English locked the prisoners in the city's cathedral, a large church that was not in use at the time.

A group of English leaders called the Council of State decided to send most of the prisoners to the colonies. In November 1650 they ordered 150 of the Dunbar prisoners handed over to agents of John Becx and Joshua Foote, two of the main investors in the Company of Undertakers. The prisoners sailed from England on a ship called the *Unity* and reached Boston in December 1650. The records do not show how many died on the voyage.

The men running the company sent 36 of these prisoners to work at Saugus, according to the historian Marsha Hamilton. They put others to work at a company warehouse in Boston and sold the rest to farmers, merchants and other mills.

Bound to work. The company's men forced many of the Scots to work five to seven years at Saugus under indentures, written agreements that bound a person to work for a master for a set number of years. These men had no say in theirs. They had to work to pay back the cost of their voyage, food, shelter, care and clothing. National Park Service staff give the term at Saugus as seven years.

Most of the Scots at Saugus cut wood, worked as colliers, the men who made charcoal, or did heavy work that needed no special skill. Other prisoners did hard and dangerous forced labor elsewhere in Massachusetts and in New

Hampshire and Maine, for the farmers, merchants and mill owners who bought them.

Meanwhile in Europe

Chemists in Europe, people who study what substances are made of and how they change, began to separate their work from alchemy in the 1600s. Robert Boyle's book *The Sceptical Chymist* ("chymist" is an old spelling of chemist), published in 1661, is one sign of that split. Every element discovered in this century was found in Europe. Phosphorus, for one, was found in the city of Hamburg in 1669.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 3 MINUTES

At many of the iron furnaces colonists built between 1700 and 1750, the owners forced enslaved people to do much of the work. Colonists had built ironworks in New England, Pennsylvania, Maryland, Virginia and New Jersey. By about 1750 they were shipping raw iron, iron not yet made into finished goods, from their own furnaces to Britain. Nobody discovered an element anywhere in the Americas in these years, and element science still happened only in Europe.

Enslaved and indentured ironworkers

British investors set up the Principio Company in Maryland in 1719. Its furnace began with workers who were mostly indentured servants from Britain. By the middle of the 1700s, more of the workers there were enslaved African Americans, many of them in skilled jobs.

The owners fed their enslaved and indentured workers mostly corn and salt pork. Some of these workers tried to escape. Historians at the University of Maryland and the Maryland Historical Trust, who surveyed the state's old industries, wrote that escape attempts were "not uncommon."

The Baltimore Iron Works. Five rich Maryland planters, owners of large farms called plantations, founded the Baltimore Iron Works in 1731. They were Dr. Charles Carroll, his cousins Charles and Daniel Carroll, Daniel Dulany and Benjamin Tasker. The partners put up money and supplied enslaved people, whom they held as property and forced to work without pay. The works opened with 89 workers, and 42 of them were enslaved, according to the staff of the Mount Clare Museum House in Baltimore.

Workers hauled cartloads of charcoal, oyster shells and iron ore to the top of the furnace in 12-hour shifts. The furnace ran day and night, six or seven days a week. Many of the workers were enslaved people. Others were convicts, people found guilty of crimes in Britain and shipped across the ocean as punishment. The owners did not give the workers enough food, according to the museum. Some enslaved and convict workers ran away, and others worked slowly or carelessly on purpose.

The Iron Act of 1750

Members of Parliament, Great Britain's lawmakers, passed the Iron Act in 1750. They wrote it to keep colonists making pig iron and bar iron, iron hammered into bars, and sending it to Britain. The act also stated that colonists could not build new mills to make finished ironware, iron goods ready for use. The members of Parliament wanted that business kept in Britain.

Colonists ignored the ban. After 1750 they built more than sixty new ironworks, about twenty of them in Pennsylvania. By the American Revolution, colonists made more iron than people in England and Wales did. By a common ranking, the colonies were third in the world for iron.

At home. Colonists used other elements every day. Lead went into gunshot and into pewter, a gray metal made mostly of tin and used for plates, bowls, mugs and spoons. Kettles and buttons were made of copper or of brass, a metal made partly of copper. Silver and gold arrived in the colonies as Spanish coins.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 5 MINUTES

Chemists in Europe turned chemistry into a modern science in the second half of the 1700s. Among them were Carl Wilhelm Scheele in Sweden and Antoine Lavoisier in France, who worked out what an element is and found new ones. Nobody in the United States discovered an element in these years. Americans put elements into their coins and dug ores out of the ground.

Joseph Priestley, one of those chemists, collected oxygen in England in 1774 and moved to Pennsylvania twenty years later. Meanwhile owners of iron furnaces in Maryland held hundreds of Black people in slavery. And in North Carolina in 1799, a 12-year-old boy found a gold nugget in a creek on his father's farm.

Three elements in the new coins

Lawmakers in Congress passed the Coinage Act, and it became law on April 2, 1792. Under it, government officials set up the United States Mint, where workers make the country's coins. The act also stated that the coins would be made of three elements: gold for coins called eagles, silver for dollars and smaller coins, and copper for cents.

Enslaved ironworkers in Maryland

After the Revolution, Maryland officials took a one-fifth share of the Baltimore Iron Works from Daniel Dulany, who had sided with Britain in the war as a Loyalist. They offered the share for sale in 1785 in an advertisement in the *Maryland Gazette*, a newspaper. The advertisement listed more than 200 enslaved Black people at the works, according to the Mount Clare Museum House. Their owners made them work without pay.

Catoctin Furnace. Four brothers, Thomas, James, Baker and Roger Johnson, built Catoctin Furnace in Maryland, and it was making iron by 1776. Thomas Johnson later became the first governor of the state of Maryland. In 1803 Baker Johnson became the only owner.

One Johnson owner by himself enslaved more than 80 people at the furnace, according to the Catoctin Furnace Historical Society, which names him only as "Johnson." The society names some of the enslaved men, among them Collier Sam, Waggoner's Henry and Harvey. A collier made charcoal, and a waggoner drove wagons.

Counting all the years the furnace ran on enslaved labor, at least 271 enslaved people and an unknown number of free African Americans worked at Catoctin, according to a 2023 study in the journal *Science*. They mined ore, made charcoal, worked the forge, filled the furnace, drove wagons and cut wood. Others worked in the owners' houses and fields. The furnace had a cemetery for its African American workers, with burials from 1774. Between about 1825 and 1850, the owners switched to paid workers, most of them white.

The cemetery. The same study traced the DNA, the material in the body's cells that parents pass down to their children, of 27 people buried in the Catoctin cemetery. Scientists from Harvard Medical School, the Smithsonian and the company 23andMe planned and led the work, and they published it on August 4, 2023. They linked the 27 people to nearly 42,000 living relatives.

Joseph Priestley

Who: The English chemist credited with discovering oxygen, who spent his last ten years as an immigrant in Pennsylvania.

When and where: Isolated oxygen on August 1, 1774, in England. Moved to America in 1794. Lived in Northumberland, Pennsylvania, until his death on February 6, 1804.

Priestley pointed sunlight through a lens and caught a new gas.

He did it in England on August 1, 1774. With a burning lens, a glass lens that bends sunlight to one hot point, he heated mercuric oxide, a compound of the element mercury and oxygen. He isolated a gas from it, which means he collected the gas apart from everything else. The gas was later named oxygen.

Carl Wilhelm Scheele, in Sweden, had found the same gas earlier but published his work later. The French chemist Antoine Lavoisier gave oxygen its name. Priestley had already invented carbonated water, water with bubbles of gas in it, in Leeds in 1767. In 1770 he gave rubber its English name, because he found it was good for rubbing out pencil marks.

A mob, then Pennsylvania. Priestley was also a minister, a leader of a church, and he preached outside the official Church of England. Many people in Birmingham, England, turned against him for his views on religion, and in 1791 a mob there burned his home and his laboratory. In 1794 he left England for Pennsylvania and settled in Northumberland, where his house was finished in 1798.

There he built what was then the best chemistry laboratory in the United States and identified the gas carbon monoxide. He turned down a job as a professor at the University of Pennsylvania. Priestley died in Northumberland in 1804. American chemists met at his house in 1874, one hundred years after he isolated oxygen, and that meeting was a step toward the founding of the American Chemical Society in 1876. The house is now a National Historic Landmark.

Conrad Reed

Who: A 12-year-old farm boy who found the first gold in the United States that an expert confirmed was real.

When and where: 1799, Little Meadow Creek, Cabarrus County, North Carolina.

The Reeds used a 17-pound gold nugget to hold a door open.

Conrad Reed pulled a heavy yellow rock out of Little Meadow Creek in 1799. The creek ran through the farm of his father, John Reed, in Cabarrus County, North Carolina. The rock was a gold nugget, a lump of metal found in the ground, and it weighed 17 pounds.

Nobody in the family knew the rock was gold. They used it as a doorstop for about three years. In 1802 a jeweler identified it as gold and bought it from John Reed for $3.50, about a week's wages. It was worth about $3,600 at the time.

Conrad's find was the first gold found in the United States that an expert checked and showed was real. After it, people in North Carolina began the first gold mining in the United States.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 5 MINUTES

Thousands of miners rushed onto Native land to dig gold in Georgia from 1829 and copper in Michigan from 1843. The Georgia gold lay under Cherokee land, and the Michigan copper lay under Ojibwe land. Georgia lawmakers gave the Cherokee gold country away by lottery, a drawing of names by chance. In 1842 a United States official signed a treaty, a written agreement between nations, under which Ojibwe bands, groups within the Ojibwe nation, gave up the copper country.

Nobody in the United States discovered a new element in these years. In Europe, chemists such as Humphry Davy and Jöns Jacob Berzelius found one element after another.

Gold on Cherokee land

North Carolina came first. After Conrad Reed's find in 1799, more people dug for gold on the Reed farm and along the creeks near it, and that mining grew from 1803. Miners in North Carolina dug more gold than miners in any other state until the California gold rush of 1849.

The first printed report of gold in Georgia appeared in the *Georgia Journal*, a newspaper, on August 1, 1829. The gold was in the Cherokee Nation, the homeland of the Cherokee people. By the spring of 1830, about 4,000 miners had crowded onto Cherokee land to dig. The gold town of Dahlonega takes its name from the Cherokee word for "golden."

The Gold Lottery. Members of the Georgia General Assembly, the state's lawmakers, passed a law on December 24, 1831. Under it they divided "the original Cherokee Indian territory" into lots and gave it away by lottery. In the Gold Lottery, winners received lots of 40 acres in the gold country for a fee of $10 a lot. The drawings ran from October 22, 1832, to May 1, 1833.

The Cherokee who lived on the land could not enter, because the lawmakers barred anyone who lived in the Cherokee territory. The lots went to white Georgians. Men who were United States citizens, widows and the families of orphans could enter if they had lived in Georgia for three years and had not mined gold in the Cherokee territory since June 1, 1830.

Forced out. President Martin Van Buren sent General Winfield Scott in 1838 to force the remaining Cherokee off their land and west of the Mississippi River. Scott commanded about 7,000 soldiers and members of state militias, part-time armies run by the states. His troops forced the Cherokee into stockades, pens fenced with tall wooden posts.

A missionary wrote in June 1838 that the Cherokee "have been dragged from their houses." In Georgia, he wrote, "multitudes were allowed no time to take any thing with them except the clothes they had on."

Scott's soldiers forced more than 16,000 Cherokee to leave for the west in 1838 and 1839, on the journey called the Trail of Tears. Over 4,000 died of cold, hunger and disease along the way, by the usual count. Russell Thornton, a scientist who studies the size of populations, puts the Cherokee losses at about 8,000 or more. Chapter 2, Native Nations, tells the Trail of Tears in full.

Copper on Ojibwe land

Douglass Houghton was Michigan's state geologist, a scientist the state paid to study its rocks. In 1840 he published a report on the copper of the state's Upper Peninsula. The Keweenaw Peninsula there has veins, long thin bands running through the rock, of pure native copper. Native miners had worked those same veins for thousands of years.

The land belonged to Ojibwe bands. Robert Stuart, an official the United States sent to make an agreement, signed the Treaty of La Pointe with Ojibwe bands on October 4, 1842. Under the treaty, the bands gave up land along the south shore of Lake Superior, including the Keweenaw copper country.

The treaty states that the Ojibwe would receive $12,500 a year in coins and $10,500 a year in goods for 25 years. It also states that $75,000 would go toward debts the Ojibwe owed to traders. The Ojibwe kept the right to hunt on the land "until required to remove by the President of the United States." That meant they could hunt there only until the President ordered them to leave.

The treaty never uses the word copper, but people came to call it the Copper Treaty. The Bad River Band of Ojibwe and the Lac Courte Oreilles Ojibwa College both say that United States officials wanted the copper.

The copper rush. United States officials opened an office at Copper Harbor in 1843 to handle land with metal in it. From that year on, thousands of prospectors, people who search the ground for metal, poured into the Upper Peninsula in the nation's first great copper rush. The Keweenaw stayed the country's biggest source of copper into the 1880s.

The Ontonagon Boulder

Julius Eldred, a merchant from Detroit, made a deal in the summer of 1841 for the sacred Ontonagon Boulder, the 3,708-pound lump of pure copper. He made it with Okandikan, a leader of the Ontonagon band of Ojibwe. Eldred agreed to pay $45 right away and $105 more if he managed to move the boulder. He hauled it away in 1843.

Walter Cunningham was the United States government's agent in charge of the lands with metal in them. Before Eldred could leave Copper Harbor, Cunningham made him hand the boulder over to the government in Washington. Later, members of Congress paid Eldred money to make up for his loss.

The boulder sat in a yard of the War Department, the part of the government that ran the army, until 1860. Then it was moved to the Smithsonian, the national museum in Washington, D.C., and put on display.

The Keweenaw Bay Indian Community, made up of the L'Anse and Ontonagon bands of Lake Superior Ojibwe, formally asked the Smithsonian to return the boulder in 1991. Smithsonian officials refused in 2000. They said the boulder did not fit the meaning of a sacred object under the law on returning Native objects, and it stayed at the Smithsonian.

Lead at Galena

Miners also dug lead in these years, in Missouri and in the Galena district of Illinois and Wisconsin. The town of Galena, Illinois, is named for galena, a lead ore.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 7 MINUTES

Mercury from the gold mines got into California's rivers and fish, and mercury in the hat shops of Danbury, Connecticut, poisoned the hatters who breathed it. California's gold miners released about 13 million pounds of it at their mines and into rivers. Miners also took gold from the Black Hills, land that belonged to the Sioux, and in 1877 members of Congress voted to take the hills from them.

Aluminum, rare at the start of these years, was cheap by the end of them. In 1886 Charles Martin Hall, a 22-year-old in Oberlin, Ohio, found a cheap way to make it. By the middle of the 1890s its price had fallen from about $12 a pound to less than $1. Scientists abroad already knew aluminum as an element. What Hall found was a process, a series of steps for making the metal.

Mercury in the rivers and the hat shops

Mercury is an element, a metal that stays liquid at room temperature, also called quicksilver. Gold miners used it to catch the fine bits of gold in the gravel they dug, and they lost mercury into the rivers.

How much mercury. California's gold miners used about 26 million pounds of mercury, and about half of it, some 13 million pounds, escaped into the ground and the rivers. Scientists at the United States Geological Survey, the government office for studying the country's rocks and water, made the count.

Placer miners, who wash gold out of sand and gravel in streams, lost about 10 million pounds of mercury, 80 to 90 percent of it in the Sierra Nevada mountains. Miners at hardrock mines, which are dug into solid rock, lost about 3 million pounds more.

Still in the fish. The Geological Survey's scientists list three ways this mercury reaches people: in fish they eat, in mud they handle and in the air they breathe. In the air it is a vapor, a gas rising off a liquid or a solid. The form that builds up most easily in living bodies, fish included, is called methylmercury. It is also the form most poisonous to people.

California officials had warned people about eating fish from about 20 lakes, rivers and other waters by 2005, because of mercury. Fish in the Bear and Yuba river areas carry enough mercury to put people's health at risk, the scientists found. They give no count of people poisoned by the gold miners' mercury.

The Danbury shakes. People in Danbury, Connecticut, had made hats since 1780. To make felt, a thick cloth of pressed fur, they soaked fur in a liquid that contained mercury nitrate, a compound of mercury, and the fur matted together into felt. This step was called carroting. As they worked, the hatters breathed in the mercury as vapor and as dust.

Doctors first described the hatters' mercury poisoning in medical writing in 1860, and people called it the Danbury shakes. The poisoned hatters had tremors, shaking they could not stop. Their skin itched, burned or peeled. Some lost their hair and teeth. They sweated, swelled up, felt dizzy and got

confused. The mercury slurred their speech and made them stumble, and people mistook the shaking hatters for drunks. In bad cases, hatters had hallucinations: they saw or heard things that were not there.

The members of Connecticut's State Board of Health watched the mercury's effects in Danbury's hat factories in the 1880s and 1890s. They did not act, because the carroting was not harming people outside the factories. The historian Shirley Wajda, who wrote the history of the Danbury shakes for Connecticut Humanities, names no factory owner and gives no count of hatters poisoned before 1900.

Silver, copper and a table from Russia

Miners found the Comstock Lode, a long, rich band of ore inside the rock, at Virginia City, Nevada, in 1859. It was the richest find of silver ore ever made in North America. Its ore held gold as well as silver, and miners rushed there too.

Dmitri Mendeleev, in Russia, published his periodic table of the elements in 1869: a chart that arranges the elements in order, with elements that act alike grouped together. Mendeleev used his table to predict elements that nobody had found yet. Every element that Americans discovered later went into it.

Americans strung copper wire through their cities in these years for electric lights and the telephone. The copper came from Michigan, and later from Butte, Montana.

Gold from the Black Hills

The Black Hills, in what is now South Dakota, belonged to the Sioux under the Fort Laramie Treaty of 1868. Lieutenant Colonel George A. Custer led an Army expedition into the hills in 1874, with miners along, and the men confirmed that there was gold. Miners poured in, and in late 1875 President Ulysses S. Grant ordered the Army to stop keeping them out.

Members of Congress then voted to cut off the Sioux's food until they gave up the hills. In 1877 they took the hills by law, though only about one in ten adult Sioux men had signed the paper giving them up. Chapter 12, Landmarks, tells that story in full. It also tells about the money the justices of the Supreme Court awarded the Sioux in 1980, about $106 million, which the Sioux have refused.

An aluminum tip for the Washington Monument

Chapter 12, Landmarks, tells how workers finished the Washington Monument in 1884. Its tip was a pyramid of aluminum, a metal that was rare then. Aluminum cost about $1 an ounce, the same as silver, though some price records from those years give a little less.

Colonel Thomas Lincoln Casey of the Army Corps of Engineers was the engineer in charge of finishing the monument. He talked over the metal for the tip with William Frishmuth, the owner of a Philadelphia foundry, a factory where workers melt metal and pour it into molds. Frishmuth chose aluminum, and Casey agreed. Aluminum matched the color of the monument's granite and would not stain. It also polished well and could be engraved.

Frishmuth cast a pyramid that weighed 100 ounces, the largest piece of aluminum anyone had yet cast. After some bargaining over his bill, he was paid $225.

Before the tip went up, Frishmuth showed it at the Tiffany's store in New York without permission. The tip sat on the floor, so visitors could step over the top of the Washington Monument. Workers set it in place on December 6, 1884.

Charles Martin Hall and Julia Brainerd Hall

Who: A 22-year-old Ohio college graduate who invented the cheap way to make aluminum, and his older sister, a chemist whose notes helped win the fight over his patent.

When and where: February 23, 1886, in the family woodshed in Oberlin, Ohio.

Charles Hall made aluminum in a woodshed, with batteries he built himself.

Hall was 22 and had just finished at Oberlin College. In the family woodshed he melted a substance called cryolite and dissolved aluminum oxide, a compound of aluminum and oxygen, in it. On February 23, 1886, he ran electricity from batteries he had built himself through the melted mix, and beads of aluminum metal came out.

His older sister, Julia Brainerd Hall, had studied chemistry at Oberlin College. She helped with the experiments, watched them and wrote them down. Later Hall had to fight for his patent, the government paper that gave him alone the right to make and sell his invention for a set number of years. Julia's notes helped him win.

In France, Paul Héroult found the same process in the same year, working on his own. Both men were born in 1863, and both died in 1914. The method is called the Hall-Héroult process, and aluminum makers around the world still use it.

Cheap aluminum. Aluminum cost about $12 a pound in the early 1880s. In 1888 the investors who backed Hall formed the Pittsburgh Reduction Company to make aluminum his way. After aluminum makers began to use his process, the price fell to about $2 a pound, and by 1893 to about 78 cents. The company's owners renamed it the Aluminum Company of America, or Alcoa, in 1907. Aluminum became cheap enough for kitchen foil and pots and pans.

Helium, almost found in America

William Hillebrand, a chemist at the United States Geological Survey, got a gas out of uranium ore around 1890. The gas did not join with other substances. Hillebrand recorded it as nitrogen and missed the chance to find helium.

William Ramsay in London followed up on Hillebrand's paper in 1895. He got a gas out of cleveite, a uranium mineral, and examined it with a spectroscope, an instrument that splits light into its colors. He saw a yellow line that belongs to helium. Scientists had seen helium's line in the light of the sun since 1868, but nobody had found helium on Earth before. The credit for finding it on Earth goes to Ramsay.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 25 MINUTES

Radium, lead, mercury, arsenic and uranium poisoned many of the Americans who worked with them in the first half of the 1900s. Young women who painted watch dials swallowed radium because their supervisors told them to shape their brushes with their lips, and many of them died of radium poisoning. At least 15 workers died of lead poisoning in the 1920s at the plants that made a lead additive for gasoline.

Arsenic from a copper smelter in Montana killed farm animals. From 1941 Navajo and other Native miners dug uranium, and Navajo miners later reported that no one had told them of its dangers.

At the same time, scientists began making new elements, substances of only one kind of atom, with a machine at Berkeley, California. The first, in 1937, was identified in Italy, in material that had come out of that machine. By the end of 1949, teams working in the United States had made or found seven more: astatine, neptunium, plutonium, curium, americium, promethium and berkelium.

The gas well that put out the fire

Helium is an element, a gas that does not burn. A drilling company struck a roaring gas well on William Greenwell's farm near Dexter, Kansas, in May 1903. The people of Dexter planned a celebration. They pushed a burning bale of hay toward the gas to light a giant flame, and the gas put the fire out. Local people made fun of it as "wind gas."

Erasmus Haworth was a Kansas geologist, a scientist who studies rocks. He carried samples of the gas to the University of Kansas. There the chemists Hamilton P. Cady and David F. McFarland tested it. On December 7, 1905, they showed that the gas was 1.84 percent helium. Scientists had thought helium was rare on Earth, and Cady and McFarland found large amounts of it in the natural gas under the Great Plains.

Helium for airships. Airships flew in World War I, and helium became valuable to armies because it does not burn. Members of Congress passed the Helium Act in 1925. Under it, federal workers, people who worked for the national government, began producing helium and storing it underground in the Bush Dome of the Cliffside gas field, near Amarillo, Texas. That store became the Federal Helium Reserve, and for decades it held nearly all of the world's supply of helium.

Under a 1927 law, no one could sell American helium to other countries. For that reason, the German airship *Hindenburg* flew on hydrogen, a gas that burns. On May 6, 1937, the *Hindenburg* caught fire as it came in to land at Lakehurst, New Jersey. Of the 97 people on board, 35 died, and so did a man working on the ground.

After the fire, President Franklin D. Roosevelt considered selling helium to Germany. Then in 1938 Hitler took over Austria and made it part of Germany, and Harold Ickes, the Secretary of the Interior, refused to sell. Ickes headed the federal department in charge of public lands.

Radium on the watch dials

Marie and Pierre Curie found the element radium in France in 1898. Radium is radioactive: its atoms break apart on their own and give off radiation, energy that shoots out of an atom as it breaks apart. Some radiation travels as tiny, fast particles that can damage the living cells of a body. The kind radium gives off is called alpha radiation.

Workers at American companies mixed radium into paint in the early 1900s so that watch numbers glowed in the dark. Other sellers put radium into tonics, drinks they claimed would make people healthy.

The companies. Two doctors, Sabin Albin von Sochocky and George S. Willis, founded the Radium Luminous Materials Corporation in New Jersey in 1914. After World War I the company was reorganized and renamed the U.S. Radium Corporation, with a plant in Orange, New Jersey. At its biggest, the company employed as many as 250 young women, some only 15 years old.

Arthur Roeder, who had been the company's treasurer, in charge of its money, replaced von Sochocky as president in January 1921. Howard Barker was the vice president and chief physicist. In Ottawa, Illinois, the Radium Dial Company ran another dial-painting plant. It belonged to the Standard Chemical Company of Pittsburgh, and Joseph Kelly was its president.

KNOWN:	More than 3,000 girls and young women painted radium dials across the country in the 1910s and 1920s, by the count of the National Institute of Standards and Technology, a federal science laboratory.
NOT KNOWN:	The full count, since Northern Public Radio gives about 4,000 and neither count comes with a list of the painters.

The Radium Girls

Who: Young women, some of them teenagers, who painted glowing radium numbers on watch and instrument dials and were poisoned by the paint.

When and where: The U.S. Radium Corporation in Orange, New Jersey, from 1917, and the Radium Dial Company in Ottawa, Illinois, into the 1930s.

The method had a name: "lip, dip, paint."

Young women at the U.S. Radium Corporation plant in Orange, New Jersey, painted watch numbers with radium paint from 1917. To shape each brush to a fine tip, a painter drew it between her lips, then dipped it in the paint and painted: lip, dip, paint. Drawing the brush through the lips was called lip-pointing. Each time a painter lip-pointed, she swallowed a little radium. The women went home glowing, and some people called them ghost girls.

Who told them to. Their supervisors told them to lip-point and said the paint was safe. Von Sochocky, one of the company's founders, grew worried about lip-pointing and told the painters to stop, but the supervisors contradicted him, according to Northern Public Radio. Arthur Roeder was the company's president from January 1921, and he still signed papers for the company in 1926. Lip-pointing there went on until after 1926. No surviving record names the New Jersey supervisors.

What the radium did. Radium belongs to the same chemical family as calcium, the mineral in bones and teeth. So the painters' bodies took the swallowed radium into their bones and teeth, the way they take in calcium, and there it kept giving off radiation from inside the body. The radiation damaged the bone marrow, the soft center of the bones that makes blood cells, until the marrow stopped making them. Doctors call this aplastic anemia.

The jawbone died and broke down, which doctors call necrosis of the jaw, and some women got bone cancer. A history published by the National Institute of Standards and Technology states that the radiation "had

eaten away their bones from the inside out." Some women's spines collapsed, and they had to wear braces. Their jawbones shattered under light pressure. When doctors examined dial painters' bodies after death, they found 99 percent of the radioactivity in the skeleton.

The first deaths. Grace Fryer began painting dials at the Orange plant on April 10, 1917, and worked there until 1920. Another painter there, Amelia Maggia, called Mollie, painted dials at the company until she died in 1922. In 1924 a dentist found damage to Grace Fryer's jawbone of a kind he had never seen.

In June 1925 a chemist at the company died of aplastic anemia, and a dial painter died 13 days later. The medical examiner of Essex County, New Jersey, the doctor who looks into sudden and suspicious deaths there, examined the chemist's body. He gave the jaw disease the name "radium jaw."

The lawsuit. Grace Fryer and four other dial painters sued the company in May 1927. Their lawyer was Raymond Berry. Mollie Maggia's body was dug up on October 25, 1927. It held 48.4 micrograms of radium, millionths of a gram, which is 500 times the amount later set as the limit for a human body.

Company officials first offered each woman $10,000, with her medical and legal costs taken out of it. The women refused. In June 1928 they accepted $10,000 each, their medical bills and a pension of $600 a year, money paid every year for the rest of their lives. A committee of three doctors had to approve the bills. Some accounts give the amount each woman got as $15,000.

A federal judge arranged the deal on June 4, and the agreement was signed on June 8. The company's officials admitted no blame, and Berry took 10 percent. Von Sochocky, one of the two doctors who founded the company, died of radium poisoning on November 14, 1928. Grace Fryer died in 1933.

Ottawa, Illinois. The Radium Dial Company advertised for painters in Ottawa in September 1922. The advertisement named a Miss Murray as the person young women should apply to. Catherine Wolfe, who later married and took the name Donohue, described how Miss Murray taught them to point the brushes between their teeth. Supervisors at Radium Dial told the women the paint was safe, and even healthy, and kept them lip-pointing.

Radium Dial's managers hired doctors to examine the painters from 1925. The doctors told one painter that she was very healthy. Nobody told her that she had tested positive for radiation, her niece said later.

Catherine Wolfe Donohue painted dials there and began to feel sick in 1925. Radium Dial's managers fired her in 1931. In her own words, "my limping was causing much talk." She lost half her body weight, and parts of her jaw fell out.

Joseph Kelly lost his post as Radium Dial's president in 1934. He opened a new dial company, Luminous Processes, a few blocks away and hired many of the same workers. That plant ran until 1976, when officials of the Nuclear Regulatory Commission, a federal agency, found radiation there 1,666 times the allowed level.

The hearing. A lawyer took Catherine Donohue's case for no pay, before the Illinois Industrial Commission, a state board that decided claims by workers hurt on the job. At a hearing on February 10, 1938, she brought a box of pieces of bone that she said had come from her jaw. When the judge asked her doctor about her chances of living, she collapsed, and several men carried her from the room. She weighed 71 pounds.

The hearing moved to her home. On April 5, 1938, the hearing officer ruled that her work at Radium Dial had made her sick and awarded her $5,661, which included a pension of $277 a year for life. On July 6 the members of the commission threw out the company's appeal and added $730.

> Catherine Donohue died on July 27, 1938, at the age of 35. According to NPR Illinois, she weighed less than 60 pounds. Her lawyer said hers was the eleventh death among women who had done that work at the plant, and Northern Public Radio later counted fifteen Ottawa workers dead of radium. Radium Dial's lawyers appealed all the way to the United States Supreme Court and lost, and the case ended in October 1938. Because Radium Dial had left Illinois, the Ottawa women together collected only $10,000.

Radium sold as medicine

William J. A. Bailey owned Bailey Radium Laboratories in East Orange, New Jersey. He sold Radithor, which was radium dissolved in water. Bailey had falsely claimed degrees from Harvard and from Vienna, and in 1927 he had pleaded guilty to practicing medicine without the right to do it. His company sold more than 400,000 bottles of Radithor at $1 each.

Eben Byers was a rich Pittsburgh businessman. He hurt his arm in 1927 at a party on a train after the Harvard-Yale football game, and he began drinking Radithor to heal it. He stopped in 1930, when his teeth began falling out. Accounts of how much he drank differ. The Museum of Radiation and Radioactivity at Oak Ridge Associated Universities says he averaged three bottles a day for about two years from December 1927. Other accounts give about 1,400 bottles in all.

The radium destroyed his bones. A government investigator reported that his whole upper jaw except two front teeth, and most of his lower jaw, had been removed. The rest of his bones were slowly breaking down, and holes were forming in his skull.

Byers gave testimony to the Federal Trade Commission, the federal office whose officials act against companies that cheat or mislead buyers. On December 19, 1931, the commission's officials ordered Bailey to stop claiming that Radithor was harmless or that it healed people. Byers died months later, in 1932, at the age of 51.

BY 1938: Doctors had named radium jaw, a federal judge had arranged the New Jersey painters' settlement, and Illinois officials had ruled that Catherine Donohue's work at Radium Dial made her sick.

Lead in gasoline and paint

Lead is a soft, heavy metal and an element, and it poisons people who swallow it or breathe it in. Scientists at the National Institute for Occupational Safety and Health (NIOSH), the federal office that studies dangers at work, list what large doses of lead do. Their list includes paralysis (being unable to move part of the body), madness, gout (painful, swollen joints), colic (sharp pains in the belly) and death.

The additive. Thomas Midgley Jr., a researcher at General Motors (GM), found that tetraethyl lead, a liquid made of lead joined with other chemicals, stopped engine "knock," a pinging sound in gasoline engines. In August 1924 GM and Standard Oil of New Jersey set up the Ethyl Gasoline Corporation to sell it, and the DuPont company made it.

The dead. Workers who made tetraethyl lead died of lead poisoning at three plants. Standard Oil ran one, the Bayway refinery in Elizabeth, New Jersey, a plant that turned oil into gasoline and other products. Seven men died there in October 1924, and 33 went to the hospital, according to Bill Kovarik, who counted the dead in a 2005 article in a journal about workers' health.

Ten more workers died at a DuPont plant in Deepwater, New Jersey, between September 1923 and the fall of 1924. At GM's laboratory in Dayton, Ohio, at least two died, and 40 went to the hospital.

KNOWN:	At least 15 workers died from making tetraethyl lead in the 1920s, the count NIOSH scientists gave in 2021.
DISPUTED:	Kovarik's plant-by-plant numbers add up to at least 19, while other accounts give five dead at Bayway, not seven, and 35 in the hospital.

How the work poisoned them. The tanks that held the chemicals were not closed off from the workers, who had to open them between steps, Kovarik writes. Workers squeezed half-melted lead through metal grates with shovels and with their boots.

At Bayway, poisoned men had to be held down and tied into straitjackets, coats that pin a person's arms to the body. Kovarik's account does not say who restrained them. Their bodies were black and blue from muscle spasms they could not stop. They saw things that were not there, such as flying insects, and some could suddenly turn violent or want to kill themselves. Workers called tetraethyl lead "loony gas."

What the bosses said. Frank Mann, the Bayway plant supervisor in 1924, said, "These men probably went insane because they worked too hard." A month before the deaths, in September 1924, engineers from DuPont had visited Bayway and called its safety measures "grossly inadequate."

Joseph G. Leslie, a Bayway worker poisoned there, was kept in a mental hospital for 40 years. Only his wife, Gertrude, and later his son knew he was alive, and everyone else believed he had died. Kovarik does not say who sent Leslie to the hospital or who spread the story that he was dead. He names no manager of the Deepwater plant or the Dayton laboratory apart from Midgley and Charles Kettering, who ran GM's research.

Midgley held a press conference on October 30, 1924. He poured tetraethyl lead on his hands and breathed its vapor for about 60 seconds, to show it was safe. A few months earlier, lead had made him seriously ill. At the same event he acknowledged the two deaths in Dayton and said more than 50 workers were "under observation" for lead poisoning.

The meeting in Washington. Hugh Cumming, the Surgeon General, ran the Public Health Service, the federal office in charge of the nation's health. On May 20, 1925, he brought more than 100 people from industry and from public health to a meeting in Washington.

Frank Howard of Standard Oil said the industry could not give up "what has come to the industry like a gift from heaven." Yandell Henderson, who studied how the body works at Yale, called tetraethyl lead "one of the most dangerous

things in the country today." Alice Hamilton of Harvard told Kettering there were "thousands of things better than lead."

Members of a Surgeon General's committee reported in 1926 that they found "no good grounds" for a ban, and asked for more study. The Ethyl company paid Robert Kehoe of the University of Cincinnati to study lead. Kehoe set out the industry's "safe level" theory, the claim that lead below a certain amount did no harm, and the companies held to that claim for decades.

Lead paint. Painters also used paint that contained lead. Between 1925 and 1927, more painters died of lead poisoning than workers in all other jobs combined, NIOSH states.

Delegates of the International Labour Organization, a world body that sets rules for workplaces, met in Geneva in November 1921. They agreed on a treaty to ban lead paint for the insides of buildings. The ban took effect in 1923 in the countries that accepted it. The United States did not belong to the organization until 1934, and it was still outside the ban in 2017.

Mercury in Danbury's hat shops

Danbury's hatters kept breathing mercury at work after 1900, because they still used it to turn fur into felt. Connecticut lawmakers added mercury poisoning to the state's workmen's compensation law in 1913. Under a law of that kind, employers pay workers who are hurt or made sick by their jobs.

Alice Hamilton of Harvard examined 100 Danbury hatters in 1922 and found 43 with mercury poisoning. Some shook so hard from tremors that they could not feed themselves.

Who pushed for a fix. The hatters in the union, a group of workers who join together to bargain with their employers, kept demanding investigations. Grace Mailhouse Burnham, who had set up the Workers' Health Bureau, found a way to make felt without mercury in the Soviet Union. She could not get the hat manufacturers to agree to use it.

Government researchers made a full study only in 1937. They found many hatters poisoned by mercury. They also found that drinking and smoking had not caused the hatters' symptoms, as some manufacturers and insurance

companies had claimed in court. The historian Shirley Wajda's account of the Danbury shakes names no factory owner.

The end of mercury felt. Officials of the Public Health Service then worked out an agreement with the hatters' unions and the hat manufacturers in the 26 states that made hats. Under it, hat makers stopped using mercury and used hydrogen peroxide in its place.

Connecticut Governor Robert A. Hurley announced his state's ban on December 1, 1941. He said "it was possible to manufacture good fur felt hats without making physical wrecks out of hundreds of workers." Mercury is still in Danbury's soil and in the mud of the Still River.

Arsenic smoke in Montana

Arsenic is an element and a strong poison. The Anaconda Copper Mining Company started its Washoe smelter, a plant where workers melt ore, rock that holds metal, to get the metal out, at Anaconda, Montana, in 1902. It stood about 25 miles from the copper mines of Butte. When workers melted copper ore there, arsenic went up the smokestack with the smoke, and farmers nearby reported livestock dying in the smelter's first year.

Farmers in the Deer Lodge Valley formed an association in 1904 to sue the company. Fred J. Bliss, whose 320-acre farm lay about five miles from the smelter, filed the case, *Bliss v. Anaconda Copper Mining Co.*, in 1905. In January 1909 a federal judge, William Hunt, agreed that arsenic from the smelter had poisoned livestock and the hay and grass they ate, but he refused to close the smelter.

The judge's reason. Closing the smelter, Hunt wrote, would throw thousands of the company's workers out of their jobs and do harm that could never be undone to the cities of Anaconda and Butte. Bliss could ask a court for money for his losses instead. The smelter's superintendent, a Mr. Mathewson, told the court that no known way of catching the arsenic would work at a smelter that size, and in April 1909 Hunt dismissed the case.

Buying the smoke rights. The company's officers then bought out many of the farmers. From the rest they bought "smoke easements," legal rights to send the company's smoke over a farm. Staff of the Environmental Protection Agency (EPA), the federal office whose staff work to protect the country's air, water and land, wrote a history of the site. Neither that history nor the 1909 rulings name the officers who chose the buyouts, and neither counts the people harmed by the arsenic.

Uranium from Navajo land

Uranium is a heavy metal element whose atoms break apart slowly and give off radiation. From 1941, Native American miners, among them Navajo, Pueblo and Hopi people, worked in mines in the Southwest that dug uranium and vanadium, another metal element. The Vanadium Corporation of America began mining uranium in secret on Navajo land in 1943.

Miners working for mining companies dug nearly 30 million tons of uranium ore out of Navajo land from 1944 to 1986, EPA staff report. The companies mined under leases with the Navajo Nation, agreements to use land for a set time in return for payment.

BY 1943: Hat makers had stopped using mercury, Anaconda's owners had bought the right to send smoke over the farms that were left, and a company was secretly mining uranium on Navajo land.

A false first, and the machine at Berkeley

While those poisonings went on, physicists, scientists who study matter and energy, were hunting for new elements. Fred Allison, a physicist at Alabama Polytechnic, now Auburn University, announced two new elements in 1931. He called element 85 "alabamine" and element 87 "virginium," and said he had found them with a machine in his lab.

Scientists at other labs tried his method and could not get the same results. They showed that the signals Allison saw came from his equipment, not from any element, and scientists stopped accepting his claims. A Berkeley team made the real element 85 in 1940.

The machine at Berkeley. Ernest Lawrence built his first cyclotron, a machine that whirls tiny charged particles in a spiral up to high speed, at Berkeley, California, in 1930 and 1931. When the fast particles hit a target, they can turn atoms of one element into atoms of another. With his machines, Berkeley's Radiation Laboratory became the place most of the new elements made in the United States came from.

Ernest Lawrence

Who: The physicist who invented the cyclotron, the machine behind most of the elements discovered in the United States.

When and where: Berkeley, California, from the 1930s to the 1950s. He won the Nobel Prize in Physics in 1939 and died in 1958.

Lawrence's first cyclotron was small enough to hold in one hand.

Lawrence made his first cyclotron out of glass, sealing wax and bronze, and it was small enough to hold in one hand. He and his teams then built larger and larger ones, and the 60-inch cyclotron of 1939 had a magnet that weighed 220 tons. The Radiation Laboratory at Berkeley became an independent laboratory in 1936. Lawrence won the Nobel Prize in Physics, a world prize given each year, in 1939.

His machines made the material for several new elements. His 37-inch cyclotron supplied the metal sheet that held the first element made by people, and teams used his 60-inch cyclotron to make astatine and plutonium in 1940. Lawrence died in 1958. Element 103, lawrencium, carries his name.

Elements made by people

Mendeleev's periodic table of 1869, the chart of the elements in order, had a gap at element 43, below manganese. Early in 1937, Lawrence mailed a thin sheet of a metal called molybdenum to Emilio Segrè in Palermo, Sicily. The sheet had been part of the 37-inch cyclotron.

The first element made by people. Segrè and Carlo Perrier, a scientist who studies minerals, took a few billionths of a gram of element 43 out of the sheet. They named it technetium, from a Greek word meaning "artificial." Technetium was the first element made by people. Italian scientists did the chemistry, and the material came from an American machine.

The first elements found in America. Teams in American labs made or found seven new elements in the 1940s. Most of their names came from planets, places and people.

Segrè and two other Berkeley scientists made element 85 in 1940 by striking the element bismuth with alpha particles in the 60-inch cyclotron. It was the element Allison had called "alabamine." In 1947 its makers named it astatine, from a Greek word for "unstable."

Edwin McMillan and a colleague found element 93 the same year, in uranium that had been struck in the cyclotron. It was the first element heavier than uranium. Uranium was named for the planet Uranus, so they named element 93 neptunium, for Neptune, the next planet out.

Plutonium. Glenn Seaborg's team made element 94 in the 60-inch cyclotron on December 14, 1940. On the night of February 23 and 24, 1941, they proved by chemistry that it was a new element, in Room 307 of Gilman Hall at Berkeley. The room is now a National Historic Landmark.

They named the element plutonium, for the planet Pluto. Seaborg chose the symbol Pu instead of "Pl" as a joke, because "pee-yoo" is what people say about a bad smell, and the members of the naming committee accepted it. Government officials kept the work secret from the start. During World War II, American scientists used plutonium and uranium to build atomic bombs.

96 before 95. Seaborg's team made element 96 in the summer of 1944 and named it curium, for Marie and Pierre Curie. They identified element 95 late that year and named it americium, for the Americas.

An element's number counts the protons in its atoms, not the order in which it was found. These two were so hard to separate from each other that the team nicknamed them "pandemonium" and "delirium," words for wild uproar and madness. Much of this work took place at the Metallurgical Laboratory, a secret wartime lab in Chicago.

Told on a quiz show. Seaborg was a guest on a children's radio show called *Quiz Kids* on November 11, 1945. A boy asked him whether any new elements had turned up during the war. Seaborg said yes, elements 95 and 96. He told the boy five days before he planned to announce them to the American Chemical Society, a national group of chemists.

Charles D. Coryell and two other chemists found element 61 in 1945 at Clinton Laboratories in Oak Ridge, Tennessee, now Oak Ridge National Laboratory, in the leftovers of uranium atoms that had been split. Grace Mary Coryell, Charles Coryell's wife, suggested the name promethium, for Prometheus, who stole fire from the gods in Greek stories. The team announced the element in 1947. Promethium is still the only element ever discovered in Tennessee.

Seaborg's team made element 97 at Berkeley in December 1949 by striking americium with alpha particles in the 60-inch cyclotron. They named it berkelium, for Berkeley.

Emilio Segrè

Who: An Italian physicist who stayed in the United States after Benito Mussolini and his officials barred Jews like him from university jobs in Italy, and who helped find two elements.

When and where: Palermo, Sicily, and Berkeley, California, from 1937 into the 1940s. He lived in the United States from 1938 and died in 1989.

Mussolini's officials took Jews' university jobs while Segrè was visiting Berkeley.

Emilio Segrè was Jewish, and he was in Berkeley for the summer when he learned the news. Benito Mussolini and his officials in Italy had issued new racial laws against Jews. Under them, Italian officials took university jobs away from Jews. Segrè stayed in the United States and became an American citizen.

He had already helped identify technetium in Palermo. At Berkeley, Segrè helped make astatine in 1940, and with Glenn Seaborg he showed in 1941 that atoms of plutonium-239 could be split.

He won the Nobel Prize in Physics in 1959 for his work on the antiproton, a tiny particle with the opposite electric charge of the proton, a particle in the center of every atom.

Glenn Seaborg

Who: The chemist whose teams found plutonium and nine more elements, and the first person to see an element named for him while he was alive.

When and where: Berkeley, California, and wartime Chicago, from 1940 to 1974. He died in 1999.

A letter addressed only in element names would reach him, Seaborg joked.

The address was Seaborgium, Lawrencium, Berkelium, Californium, Americium. That stood for Seaborg, at the Lawrence laboratory, in the city of Berkeley, in California, in America. Element 106 was named seaborgium after him in 1997.

Glenn T. Seaborg led the team that made plutonium in December 1940 and proved it was a new element in February 1941. In all, he helped discover ten elements, numbers 94 through 102 and number 106. One of the ten, nobelium, was later credited to scientists at Dubna in the Soviet Union.

Seaborg shared the 1951 Nobel Prize in Chemistry with Edwin McMillan for their work on the elements heavier than uranium. From 1961 to 1971 he was chairman of the Atomic Energy Commission, the federal agency in charge of atomic energy.

BY 1950

If you remember one thing: The elements Americans used poisoned many of the people who handled them, and in 1940 American labs began finding new elements of their own.

- June 8, 1928: The New Jersey dial painters sign their settlement with the U.S. Radium Corporation.
- 1940: Teams at Berkeley make astatine and neptunium, the first elements discovered in the United States, and then plutonium.
- December 1, 1941: Connecticut's governor announces the state's ban on mercury in hat making.

Next: After 1950, Berkeley and a lab in the Soviet Union argued over who had made the heaviest elements first, and uranium mining on Navajo land reached its peak.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 17 MINUTES

Scientists at Berkeley and at Dubna, a lab in the Soviet Union, argued for decades over who had made the newest elements first. Teams at Berkeley and other American labs made or claimed elements 98 through 106 between 1950 and 1974, and Dubna's scientists claimed several of the same ones. In 1992 and 1993 the members of an international panel gave some credit to Berkeley, some to Dubna, and split the rest.

Uranium mining on Navajo land reached its peak in the same years. Miners breathed radioactive gas underground, and most small mines had no fresh air blown in until the 1960s or later. Government researchers and mine owners did not tell the miners the danger, and hundreds of miners died of lung disease and cancer. Other scientists proved that lead was harming children's brains, and between 1971 and 1996 federal officials limited and then banned lead in house paint and in gasoline for cars.

Elements from Berkeley and from a bomb test

Kenneth Street Jr., Stanley G. Thompson, Albert Ghiorso and Glenn Seaborg made element 98 at Berkeley early in 1950 by striking curium with helium particles. They announced it on March 17, 1950, and named it californium, for the state and its university.

Two elements from a bomb. Elements 99 and 100 turned up in the dust of Ivy Mike, the test of a hydrogen bomb, a bomb far more powerful than the first atomic bombs. The test took place at Enewetak Atoll, a ring of small coral islands in the Pacific Ocean, on November 1, 1952. The blast wiped out one of the atoll's islands, Elugelab, turning it to vapor.

Drone planes with no one aboard flew through the mushroom cloud, carrying filter papers to catch the dust. Workers then shipped tons of coral from the blast site to Berkeley. In the dust and coral, Ghiorso and teams from Berkeley, Argonne and Los Alamos found two new elements. In the blast, uranium atoms had soaked up huge numbers of neutrons, tiny particles in the center of an atom.

Government officials classified the discoveries, keeping them secret. The scientists published them on August 1, 1955, with the names einsteinium and fermium. By then both men they honored had died, Enrico Fermi in November 1954 and Einstein in April 1955. Neither man was told that an element would carry his name.

Seventeen atoms, one at a time

A Berkeley team made element 101 in 1955 from about a billion atoms of einsteinium, all the einsteinium that existed. The team was Ghiorso, Bernard G. Harvey, Gregory R. Choppin, Thompson and Seaborg. They struck the einsteinium with particles, and the new atoms flew off onto a gold foil, a way of catching atoms that Ghiorso had worked out. The team dissolved the foil and rushed it through the chemistry.

They made 17 atoms of element 101 in all. It was the first element identified one atom at a time. In the scientists' own telling, Ghiorso wired the detector to the building's fire alarm, so the bell rang for each new atom. He forgot to disconnect it, and the alarm later emptied the building late at night.

The team named element 101 mendelevium, for Dmitri Mendeleev, the Russian who made the periodic table. That was during the Cold War, the long contest between the United States and the Soviet Union, and Seaborg asked the United States government for approval before he used a Russian's name. A French chemist later told Ghiorso that the name did more for American and Soviet relations than most diplomacy, the work of officials who handle dealings between countries.

The fight over elements 102 to 106

Berkeley and Dubna each claimed to have made elements 102 through 106 first. Each side gave the elements its own names and argued for them in science journals for decades. In 1992 and 1993 a panel of chemists and physicists called the Transfermium Working Group, named for the elements after fermium, element 100, split the credit between the two labs, and the names were settled in 1997.

Element 102. Scientists in three countries worked on element 102. A team in Stockholm, Sweden, claimed it in 1957 and named it nobelium, for Alfred Nobel, then took back its claim. Berkeley scientists claimed the element in 1958, and scientists at Dubna did experiments on it in 1966. The panel's members credited Dubna. The International Union of Pure and Applied Chemistry (IUPAC), the world organization of chemists whose officials approve element names, kept the Swedish name nobelium anyway, because scientists had used it for more than 30 years.

The panel's members split the credit for elements 103, 104 and 105 between Berkeley and Dubna. They credited Berkeley alone with element 106. Element 103 kept the name lawrencium, for Ernest Lawrence.

Seaborgium. IUPAC's officials proposed a rule in 1994 against naming elements for living people. Under that rule, they could not have approved the Berkeley name for element 106, seaborgium. In 1997 they set the proposal aside and approved the final names: rutherfordium for element 104, after the scientist Ernest Rutherford, dubnium for element 105, after the town of Dubna, and seaborgium for element 106.

Glenn Seaborg was the first person to see an element named for him while he was alive. He kept a periodic table with seaborgium on it until he died in 1999.

Albert Ghiorso

Who: The builder of instruments who helped discover more elements than anyone else in history, without ever earning a doctorate.

When and where: Berkeley, California, from 1944 to 1974, with work into the 1990s. He died in 2010.

He helped discover 12 elements and never earned a doctorate.

Albert Ghiorso helped discover americium through seaborgium, numbers 95 to 106, without a doctorate, the highest university degree. Guinness World Records lists his 12 as the most elements discovered or co-discovered by one person.

Ghiorso built the tools the element hunters needed, among them the detectors. He worked out the method of catching new atoms on a foil, which his team used to find mendelevium in 1955. And he ran the HILAC, a machine at Berkeley that sped up particles to make new elements.

One story his colleagues told about him has him driving fresh, radioactive samples across the lab in a Volkswagen car.

Darleane Hoffman

Who: A nuclear chemist who proved that plutonium, an element people had made in labs, also occurs in nature.

When and where: Los Alamos, New Mexico, from 1953, and Berkeley, California, from 1984. She died in 2025, at the age of 98.

She found plutonium, an element made in labs, in ancient California rock.

Darleane C. Hoffman was a nuclear chemist, a chemist who studies atoms that break apart and give off radiation. She began work at Los Alamos in 1953. In 1971 she found traces of plutonium-244 in ancient rock from the Mountain Pass mine in California. Until then plutonium had been known only as an element made in labs, and her find showed that tiny amounts of it also occur naturally on Earth.

Hoffman showed that atoms of fermium-257 could split in two on their own. She was the first woman to lead a science division at Los Alamos. From 1984 she led work at Berkeley on the chemistry of the heaviest elements, studying them one atom at a time.

She won the National Medal of Science in 1997. In 2000 she won the American Chemical Society's Priestley Medal, named for Joseph Priestley, as only the second woman ever to receive it.

Elements at home

Many home smoke detectors hold a speck of americium-241, about a third of a millionth of a gram. The americium gives off alpha particles inside a small chamber. The particles let a tiny electric current run through the air in the chamber. Smoke that drifts in disturbs the current, and the alarm goes off.

Silicon is an element that was discovered in Sweden in 1824. Computer chips are made of it, and Silicon Valley takes its name from the element in the chips made there.

The federal helium program was $1.4 billion in debt by 1996. That year members of Congress passed the Helium Privatization Act and ordered the Federal Helium Reserve sold.

Uranium miners on Navajo land

While the element hunters argued over credit, uranium miners on Navajo land were breathing radioactive gas. Their ore went to one buyer. EPA staff state that the Atomic Energy Commission (AEC), the federal agency in charge of atomic energy, was the only buyer until 1966 and kept buying ore until 1970. Doug Brugge and Rob Goble, writing in the *American Journal of Public Health* in 2002, say the government was the only buyer from World War II until 1971.

About 10,000 people worked in uranium mining in all, and about a quarter of them were Navajo. The mining peaked in the middle of the 1950s, with about 750 mines working. The companies named in the records include the Vanadium Corporation of America, Kerr-McGee, the United Nuclear Corporation and Hydro Resources Inc.

What radon does. When uranium atoms break apart, they turn into radon, a radioactive gas. Radon atoms then break apart into radioactive specks called radon daughters, and miners breathed them in. Brugge and Goble explain that the daughters stay in the lungs next to delicate cells and hit those cells with strong doses of radiation. The harm named in the records is lung cancer and lung fibrosis, scarring that makes the lungs stiff, so that breathing gets harder.

Navajo miners report that no one told them about the dangers of uranium mining. They say they got no protective equipment and no ventilation, the fans and shafts that blow fresh air into a mine and push radon out. Scientists already knew about the danger from uranium mines in Europe. Brugge and Goble found that few protections were in place for American miners before 1962, and they came slowly after that.

A warning turned down. Members of a federal panel, the Advisory Committee on Human Radiation Experiments, traced who made these choices in their 1995 report. In 1948 Merril Eisenbud, a safety expert in the AEC's New York office, urged AEC officials to write safety rules into the agency's contracts to buy ore. Officials of two AEC divisions, Biology and Medicine and Raw Materials, rejected his advice. They said that under the law the AEC had no power over safety in the mines.

The study that did not warn them. Public Health Service workers led by Duncan Holaday and Victor Archer followed the miners from 1950 to measure their cancer risk. Accounts give the study group two sizes. Linda M. Richards, writing in 2013 for *Distillations*, the magazine of the Science History Institute, counts 3,415 miners, 779 of them Native American. The 1995 federal committee counted about 4,100. Brugge and Goble found that the researchers did not tell the miners about the risks they were studying.

At first the researchers left the minority miners out of the results, saying they wanted "a homogeneous population," a group of people who are all alike. Duncan Holaday later testified in court that the researchers made a spoken agreement with the mine owners not to tell the miners what they found. They made that deal so the owners would let them into the mines.

Staff of the AEC and of the Public Health Service met on January 25, 1951. The staff at that meeting believed the radon in the mines was at levels that would cause cancer. They agreed that ventilation could lower it, but they called it "unrealistic" to set a limit the mine owners had to meet.

Hueper's speech. Wilhelm Hueper, a cancer researcher at the National Cancer Institute, warned about radon. His figures showed lung cancer in 40 to 75 percent of the radioactive ore miners in Europe. In 1952 Shields Warren of

the AEC asked the institute's director to stop Hueper from giving those figures in a speech to the Colorado Medical Society. The director ordered Hueper to cut them, and Hueper did not give the speech.

Warren then asked that Hueper be fired. Hueper kept his job, but Victor Archer later said Hueper was forbidden to travel west of the Mississippi River for his research. The 1995 committee report that gives Archer's account does not say who gave that order.

Public Health Service officials urged the mining companies to ventilate. Some large mines got ventilation in the 1950s and 1960s, but most small mines got none until the 1960s or later. Secretary of Labor Willard Wirtz set the first federal radon limit for these mines in 1967. The committee's members found that the mine owners, the state governments and the federal government all had a duty to ventilate the mines after 1951.

In court. A federal judge found in 1984, in a case called *Begay v. United States*, that Public Health Service officials, working under the Surgeon General, decided not to warn the miners because they feared many would quit. The judge still ruled that the miners could not sue the United States government. The law that lets people sue the government for harm does not cover choices that officials make as a matter of policy, and the judge found that the choice not to warn the miners was one of those.

In the ruling, the judge wrote that the miners' situation "cries for redress," meaning it called out to be put right, and called it a "tragedy of the nuclear age." Appeals judges upheld the ruling in 1985.

A study from 2000, cited by Brugge and Goble, found 94 deaths from lung cancer among Navajo people from 1969 to 1993, and 63 of them had been uranium miners. Richards reported that by the early 1980s "hundreds of Indian miners had died from lung diseases and cancers."

KNOWN:	The 1995 federal committee found that 410 of the 4,100 miners in the Public Health Service study had died of lung cancer by 1990, more than five times the 75 deaths expected.
NOT KNOWN:	How many Navajo miners died in all, since none of these accounts gives a full count.

The law of 1990. Members of Congress passed the Radiation Exposure Compensation Act on October 5, 1990, and President George H. W. Bush signed it on October 15. Under it, a uranium worker made sick by radiation on the job could claim one payment of $100,000, if the worker met the law's rules.

Harris Charley was a Navajo uranium miner. His records show he breathed 1,192 "working level months" of radon, a unit that measures how much radon a miner breathed over time. He died of lung fibrosis in 1986.

His son, Perry H. Charley, became an environmental scientist and worked with Stewart Udall, a former Secretary of the Interior, to investigate the miners' illnesses. In 1996 he helped start the Uranium Education Program at Diné College, which carries the Navajo people's own name for themselves. Navajo organizers, among them Harry Tome, Perry Charley and Phil Harrison, pushed for help for the miners from the 1960s on.

The spill at Church Rock

An earthen dam owned by the United Nuclear Corporation broke near Church Rock, New Mexico, on July 16, 1979. The dam held waste from the company's uranium mill, a plant that takes uranium out of crushed rock. The waste was tailings, the ground-up rock left behind after the uranium is taken out.

The waste ran at least 80 miles down the Rio Puerco, a river that flows past the homes of about 1,700 Navajo people. Accounts give slightly different amounts. Richards gives 1,000 tons of radioactive tailings and 93 million gallons of acid, radioactive water. New Mexico's Environmental Improvement Division, the state office that checked for pollution, gives 1,100 tons and 94 million gallons.

Measured in curies, a unit of radioactivity, the spill released 46. The accident at the Three Mile Island nuclear power plant, four months earlier, released 13. The U.S. Geological Survey called Church Rock "the largest accidental radioactive release in the United States."

The authors of the Environmental Improvement Division's own report saw it differently. They concluded that the spill's effects "on people and the environment were quite limited." They wrote that the water pumped out of the mines into the Puerco River might be a greater long-term danger.

What officials did. Within a few hours, officials of the Environmental Improvement Division began testing the river. They ordered United Nuclear to stop sending waste to the broken pond, and the company stopped milling for about three months. No official declared an emergency or ordered anyone to leave.

New Mexico's governor, Bruce King, turned down the Navajo Nation's request to have the area declared a federal disaster area, according to news accounts. The state's report on the spill names no United Nuclear official as responsible for the dam.

Lead: the proof and the bans

Thousands of American children went to the hospital each year in the 1960s with lead encephalopathy, swelling and damage of the brain, and about one in four of them died. A 2007 review in the journal *Environmental Health Perspectives* states those numbers.

Where the lead came from. Before the 1960s, the poisoned children usually lived in old, poorly kept slum housing, David Rosner and Gerald Markowitz wrote in 2012. The children had swallowed flakes of lead paint and went into convulsions, fits of violent shaking they could not control. Rosner and Markowitz call lead paint in older buildings the major source of lead around children.

Lead harms a child's growing brain. It lowers intelligence and causes trouble with reading, paying attention and behavior, and the 2007 review states that no safe level of lead is known. The review names the main ones to blame: the paint and pigment industry, the oil industry and a few scientists paid by industry.

The proof. Clair Patterson, a Caltech scientist who studied the chemistry of rocks and the Earth, measured lead in the air at about 1,000 times its natural level. In a 1965 paper he called the "safe level" idea of Robert Kehoe, the scientist the Ethyl company had paid to study lead, "an ill-defined opinion unsupported by any evidence."

Herbert Needleman, a children's doctor in Pittsburgh, measured the lead in the baby teeth of about 2,500 schoolchildren in the 1970s. Children with more lead in their teeth did worse in school. Scientists paid by the lead industry challenged his work, and later reviews backed his findings, according to Bill Kovarik, who wrote the 2005 article about the tetraethyl lead deaths.

The bans. Members of Congress passed the Lead-Based Paint Poisoning Prevention Act in 1971. Under it, homes built or repaired with federal money could have only limited amounts of lead paint, and in 1972 the limit was set at 0.5 percent of the paint. Officials of the U.S. Consumer Product Safety

Commission banned lead paint in all homes in 1978, and from then on house paint could hold no more than 0.06 percent lead. Homes built before 1978 are still likely to have some lead paint.

EPA officials began work on a rule to cut the lead in gasoline in 1972 and 1973. Gas stations began selling unleaded gasoline, and car makers built cars for it, in 1975 and 1976. The Clean Air Act's ban on leaded gasoline for cars and trucks took effect on January 1, 1996. On January 29, EPA Administrator Carol M. Browner announced the last step in 25 years of cutting lead out of gasoline. She said the amount of lead in children's blood had fallen 70 percent.

Poisoned ground put on the cleanup list

EPA officials put both the Anaconda smelter site in Montana and the Church Rock mill on the National Priorities List in 1983. The list includes the country's most dangerous waste sites, chosen for cleanup under the federal Superfund program.

Smelting at Anaconda had stopped in 1980. EPA staff count nearly 100 years of smelting at the whole site, which left arsenic, cadmium, copper, lead and zinc over more than 200 square miles. EPA staff list Atlantic Richfield, the company that bought the smelter in 1977, as responsible for the cleanup.

BY 2000

If you remember one thing: Between 1950 and 2000, researchers proved what radon did to uranium miners and what lead did to children, and federal officials banned leaded gasoline and began paying sick uranium workers.

One number: 70 percent. That is how far the amount of lead in children's blood had fallen by January 1996, after 25 years of cutting lead out of gasoline, by the count Carol Browner announced.

Next: After 2000, Russian and American scientists finished the seventh row of the periodic table together, and lead from old pipes got into the tap water of Flint, Michigan.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 11 MINUTES

The heaviest elements yet came from Russian and American scientists working together. Joint teams made elements 114 through 118 between 1998 and 2010. The atoms formed at Dubna, in Russia, and scientists from American labs brought detectors and, for element 117, the only target material in the world that would work. In 2015 and 2016 IUPAC's officials recognized the last four elements of the periodic table's seventh row and gave them names.

The poisons of earlier years were still in the ground and the pipes. As of June 2026, EPA staff count 523 abandoned uranium mines on the Navajo Nation. Managers appointed by Michigan's governor switched the water supply of Flint, Michigan, in 2014, and into 2015 the water carried lead from old pipes to the city's people.

Finishing the seventh row

Scientists at the Joint Institute for Nuclear Research in Dubna, Russia, worked with scientists from Lawrence Livermore National Laboratory in California. For elements 115 and 117, scientists from Oak Ridge National Laboratory and Vanderbilt University joined them. The first atom of element 114 formed at Dubna in December 1998, and element 116 followed in 2000.

IUPAC's officials named element 114 flerovium in May 2012, for Russia's Flerov Laboratory. They named element 116 livermorium, for the Livermore laboratory and the city of Livermore, California.

The row filled. IUPAC's officials recognized elements 113, 115, 117 and 118 on December 30, 2015, which filled the seventh row of the periodic table, and they announced the names on November 28, 2016. Scientists at RIKEN, a research institute in Japan, made element 113, the first element discovered in Asia, and it became nihonium. Element 115 was named moscovium, for the region around Moscow, and element 117 tennessine, for Tennessee.

Element 118 became oganesson, for Yuri Oganessian, the Dubna physicist who led the hunt for these heavy elements. It was only the second element named for a living person, after seaborgium. His colleagues asked him to leave the

conference call before they voted on the name. IUPAC's officials credited these elements to the joint teams, made up of scientists from several countries who worked together.

Tennessine and the berkelium target

Joe Hamilton of Vanderbilt University started the partnership on element 117 when he introduced Jim Roberto of Oak Ridge to Yuri Oganessian. To make element 117, the Dubna team needed a target made of berkelium-249, and the only place that could make it was a research reactor at Oak Ridge National Laboratory. The berkelium spent 250 days in the reactor. Oak Ridge chemists then spent 90 days working on it and ended up with 22 milligrams, thousandths of a gram.

Five trips across the Atlantic. The berkelium was shipped in June 2009. Russian customs officials, who check what comes into a country at its border, twice rejected the paperwork, so the package crossed the Atlantic Ocean five times. At Dubna, scientists fired a beam of calcium-48 atoms at the target for six months and made six atoms of element 117. The team published the discovery in April 2010.

Tennessee became the second American state named on the periodic table, after California. The atoms of tennessine formed in Russia. Promethium, found at Oak Ridge in 1945, is still the only element discovered in Tennessee.

Clarice Phelps

Who: A nuclear chemist at Oak Ridge National Laboratory and a U.S. Navy veteran. Her laboratory's staff profile describes her as the first Black woman involved in the discovery of an element.

When and where: Oak Ridge National Laboratory, Tennessee, from 2009. Her berkelium work began in 2010, and the laboratory recognized her in 2019.

She helped purify berkelium for the experiments that confirmed element 117.

Clarice Phelps served in the U.S. Navy and graduated from Tennessee State University. She joined Oak Ridge National Laboratory in 2009, and her first job there was as a nuclear operations technician.

She worked in the laboratory's Nuclear Security and Isotope Technology Division from 2010. There she was part of a team that helped purify berkelium-249, removing everything else from it. Scientists used that berkelium to confirm the discovery of element 117. The berkelium for the first run at Dubna had shipped in June 2009, before she began this work.

A 2019 news release from Oak Ridge National Laboratory described her as "the first African American woman to be involved with the discovery of an element, tennessine." That first is her laboratory's claim. IUPAC's officials credited Oak Ridge as a laboratory, not its scientists one by one. The same year, she was named to the Periodic Table of Younger Chemists, a project of IUPAC and the International Younger Chemists Network.

Toward element 120, and the end of federal helium

Scientists in the Heavy Element Group at Berkeley Lab made two atoms of livermorium in a new way in 2024. They fired a beam of titanium-50 at plutonium-244 in the lab's 88-Inch Cyclotron, and Berkeley Lab's news center reported the result on July 23, 2024. The test showed that the method could work for element 120.

Element 120 would start the eighth row of the periodic table, and it would be the heaviest atom ever made. The Berkeley team began runs to make it in 2025, firing titanium-50 at californium-249. As of August 2026, no one had announced a discovery.

The end of federal helium. Federal officials sold the Federal Helium System, which included the reserve, the Cliffside gas field and a pipeline 423 miles long, to Messer, a company that sells industrial gases. The sale was completed in June 2024. Messer's accepted bids, opened in January 2024, totaled $423.35 million. Officials of the Bureau of Land Management, a federal agency, stated in 2024 that the sale in the end sent about $460 million to the U.S. Treasury, the government's money office.

Helium is still scarce. Hospitals need it for MRI machines, which take pictures of the inside of the body, and rocket makers and computer chip makers use it too.

Uranium on Navajo land today

Elsie Mae Begay, a Navajo woman, lived in Monument Valley below the abandoned Skyline uranium mine. Two of her sons who had lived with her died, one of lung cancer and the other of a tumor. Linda Richards, who reported her story in 2013, calls them rare cancers. Richards does not say whether doctors linked the sons' cancers to the mine.

Begay's family hogan, a traditional Navajo house, gave off radiation up to 100 times the acceptable level, the Associated Press reported. When her children were young, they had played in piles of uranium tailings. Richards writes that people testing Navajo homes found dangerous radioactivity where families had built with mine debris or mixed it into concrete floors. EPA officials treated the Begay hogan as nuclear waste, and in 2001 men in radiation suits took it apart and hauled it away.

The Navajo ban. Members of the Navajo Nation Council passed the Diné Natural Resources Protection Act in 2005. Under it, uranium mining is banned on Navajo land.

How many mines. EPA staff count 523 abandoned uranium mines on the Navajo Nation as of June 2026. They have money to study and clean up 230 of them, fewer than half, and they list 46 as priority mines. Richards, counting abandoned mine sites rather than mines, found at least 1,032 on Navajo land in 2013, and 38 water sources with more radioactive elements than health limits allow.

EPA officials had reached agreements and settlements over the mines worth more than $1.7 billion by June 2026. A settlement is an agreement to pay money to end a legal fight.

About 3,975 uranium workers had applied for payments under the Radiation Exposure Compensation Act by March 2001, according to Brugge and Goble. The law ran out for a time. Members of Congress renewed it as part of the One Big Beautiful Bill Act, which became law on July 4, 2025. According to the

Department of Justice in September 2026, workers must file claims by December 31, 2027, and a uranium worker who qualifies receives $100,000, paid once.

Lead in the water of Flint

Flint, Michigan, is a majority-Black city. Emergency managers switched its water supply in April 2014. Michigan Governor Rick Snyder had appointed them to run the city in place of its elected leaders. Flint had been getting Detroit's water, drawn from Lake Huron, and the new supply came from the Flint River.

Who made the switch. Governor Snyder later appointed the Flint Water Advisory Task Force to find out what went wrong. Its 2016 report states that Flint's emergency manager Ed Kurtz approved using the Flint River, and that Darnell Earley, the emergency manager in April 2014, oversaw the switch. A timeline published by the Centers for Disease Control and Prevention (CDC), the federal health agency, states that the river water had no corrosion control, a treatment that keeps water from eating into pipes.

Detroit's water carried a chemical called orthophosphate, which protects pipes from corrosion. The Flint River water had none, and it was high in chloride, a salt. The river water ate into the city's old pipes.

Flint's service lines carry water from the main line under the street into each house. Between 10 and 80 percent of them were made of lead, according to an estimate published in 2016. Lead was in the water for about 18 months, reporters for NPR, a national radio network, said in 2023.

The children's blood. Mona Hanna-Attisha, a children's doctor at Hurley Medical Center in Flint, and her colleagues compared blood tests of 1,473 children in Flint with 2,202 outside the city, in a study published in 2016. Among Flint children under 5, the share with high lead in their blood about doubled, from 2.4 percent in 2013 to 4.9 percent in 2015. In the parts of the city with the most lead in the water, it rose from 4.0 to 10.6 percent.

Children absorb 40 to 50 percent of the lead they swallow in water, and adults absorb 3 to 10 percent. Babies who drank formula mixed with tap water were at the highest risk.

The warnings. Miguel Del Toral, an EPA expert on lead in water, warned officials of Michigan's Department of Environmental Quality (MDEQ) on February 27, 2015. He wrote that MDEQ's instructions for collecting water samples made the lead readings come out low, and that a federal rule, the Lead and Copper Rule, required Flint to have corrosion control. The same day, Stephen Busch of MDEQ wrote back that Flint had working corrosion control. The task force found that this was wrong.

Del Toral asked another MDEQ official on April 25, 2015, how Flint could meet the rules with no corrosion control. That official traded e-mails about his question with Busch and Mike Prysby of MDEQ. In July 2015 MDEQ's spokesman, Brad Wurfel, wrote that Flint's residents "do not need to worry about lead in the water supply."

Hanna-Attisha and her colleagues at Hurley Medical Center told pregnant women and children to stop drinking the water in September 2015. Flint was connected again to a regional water system in 2015. Snyder asked for federal help on January 14, 2016, President Barack Obama declared a federal emergency in Flint two days later, and EPA officials issued their own emergency order on January 21.

Legionnaires' disease. Legionnaires' disease, a lung infection and a kind of pneumonia, is caused by bacteria that grow in poorly treated water. Genesee County, where Flint is, had 90 cases from June 2014 to October 2015. NPR reporters linked the outbreak in 2018 to low chlorine, the chemical that kills germs in drinking water, in Flint's water.

KNOWN:	State health officials counted 10 dead in one report and 12 in another.
DISPUTED:	The official count, because reporters for the PBS program *Frontline* found more deaths from pneumonia that it missed.

The charges. None of the officials charged over Flint's water was convicted of a felony, one of the most serious kinds of crime. Michigan's attorney general, the state's top lawyer, charged 15 officials in 2016 and 2017, Busch, Prysby and Earley among them. Seven of them took plea deals for lesser crimes called misdemeanors. In June 2019 the prosecutors who had taken over the cases dropped the charges against the other eight, saying the cases were only partly built.

The same prosecutors charged Governor Snyder and eight other officials in January 2021, Earley among them. Snyder faced two counts of willful neglect of duty, knowingly failing to do his job. The state's health director and its chief medical executive each faced nine counts of involuntary manslaughter, causing a person's death through carelessness without meaning to kill.

How the cases ended. A one-judge grand jury, a single judge who decides alone whether to charge someone with a crime, had brought the 2021 charges. The judges of the Michigan Supreme Court ruled in 2022 that this broke the state constitution. The 2021 cases ended on October 31, 2023, with no one found guilty.

Flint's people sued. U.S. District Judge Judith Levy approved a settlement of $626.25 million on November 10, 2021, for more than 90,000 Flint residents and businesses.

THE THREE QUESTIONS, ANSWERED

1. **Who lost their land when Americans found gold and copper on it?** The Cherokee lost theirs after gold turned up in Georgia in 1829, when Georgia lawmakers gave it away by lottery and soldiers forced them west in 1838. The Ojibwe bands of Lake Superior gave up the copper country in the Treaty of La Pointe in 1842, and members of Congress took the Black Hills from the Sioux in 1877, after Custer's men found gold there.
2. **What did the people in charge say about the radium, lead and uranium that poisoned workers?** Supervisors told the dial

painters the paint was safe, the Bayway supervisor said his poisoned men had worked too hard, and Thomas Midgley breathed the vapor of tetraethyl lead in front of reporters to show it was safe. Public Health Service researchers agreed with mine owners not to tell the uranium miners what they found.

3. **When did someone in the United States first discover a new element?** In 1940, at Berkeley, California, when teams using Ernest Lawrence's cyclotron made astatine and neptunium, and then plutonium. Every element before that had been found abroad, and Fred Allison's 1931 claims were false.

What changed: Nobody anywhere knew what an element was in 1500, and by 2010 scientists from American labs had helped make element 118, the heaviest yet.

FIVE DATES TO KEEP

1799: Conrad Reed finds a 17-pound gold nugget in North Carolina.

October 4, 1842: Ojibwe bands give up the Keweenaw copper country in the Treaty of La Pointe.

June 8, 1928: The New Jersey dial painters sign their settlement with the U.S. Radium Corporation.

December 14, 1940: Glenn Seaborg's team makes plutonium at Berkeley.

April 2014: Emergency managers switch Flint's water to the Flint River.

> **NOT KNOWN:** How many Navajo uranium miners died of their work in all, because each count that exists covers only some of them.

Where to go: Saugus Iron Works National Historic Site, on the Saugus River near Boston, is the place where Hammersmith's furnace, forge and mill worked from the 1640s until about 1670.

Next: Chapter 14, Work and Workers. It asks what the daily job felt like and how working people organized, and Grace Fryer's fight comes back there from the workers' side.

PART 4

Work, Food, and Money

Chapter 14: Work and Workers

CHAPTER 14 OF 37 · WORK AND WORKERS · 10 ERAS · ABOUT 85 MINUTES

For most of American history, employers and masters decided how long people worked, how much they earned and how safe the job was, and workers won shorter days and higher pay mainly by acting together.

In October 1836 the girls in one room of a Lowell mill, in Massachusetts, could not decide whether to turn out, their word for walking off the job together. The managers had just taken more money out of their pay for room and board. Harriet Hanson was 11 and had worked in the mill since she was 10. She later remembered telling them: "I don't care what you do, I am going to turn out, whether anyone else does or not." She walked out, and they followed her.

This chapter follows the people who did the work, era by era, from the farmers and hunters of the Eastern Woodlands before 1500 to the warehouse and delivery workers of today. Watch what happens when workers act together, the way Harriet's room did. Some won higher pay or a shorter day. Others were beaten, jailed or shot by police, hired guards and soldiers. Watch the deaths at work too. Some of the laws that made jobs safer came only after workers died, in a burning mine or behind a locked factory door.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Which workers were not free to leave their work?
2. What happened when workers went on strike?
3. What made work less deadly?

Five minutes? Read "The shirtwaist makers and the Triangle fire" in 1900 to 1950.

Can skip: "The telephone operators", "The Mohawk ironworkers", "The pandemic, 2020". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 2 MINUTES

The people of North America worked for their own households and towns before 1500, and they planned the work around the seasons. They farmed, hunted, fished and built canoes. Nobody worked for wages, pay from an employer, and no one hired another person, set their hours or fired them. Work for wages began here only after people from Europe arrived.

Who did which work?

In the Eastern Woodlands, the forests of eastern North America, women and men did different work. Women farmed corn, beans and squash and gathered wild foods, and men hunted and fished. On the biggest jobs they worked together in larger groups: at planting time and at harvest, for building canoes, and for maple sugaring, boiling down the sap of maple trees to make sugar.

Where farmers grew enough food for whole towns, some people could spend their time on one kind of work. They became traders, stoneworkers or religious leaders.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

The first work that newcomers organized in this land was forced work. Spanish settlers in Florida and the Southwest made Native people work for them, with little pay or none. Spanish officials ran that forced work under two systems, the encomienda and the repartimiento.

The encomienda and the repartimiento

Under the encomienda, a system set up by the king of Spain, Spanish settlers could force Native people to work for them and to hand over tribute, payments of goods or work that rulers force people to give. In return the settlers were supposed to give the Native people "protection and religious instruction." Settlers used the encomienda from Florida across the Southwest.

The repartimiento. Spanish officials in the colonies added a second system in the middle of the 1500s. Under it, the officials sent Native workers in groups, by turns, to work on Spanish projects, and they were supposed to pay them. In New Mexico, Spanish governors used the repartimiento to farm and to trade. They also forced enslaved people to do the work, though Spanish law forbade it.

Acoma, 1599: forced work as a sentence

Juan de Oñate, the leader of the Spanish colony in New Mexico, sentenced the survivors of Acoma Pueblo to twenty years of forced labor in February 1599. That meant work they could not refuse and could not leave. Acoma Pueblo is the Acoma people's town, with their fields around it.

What came before. Spanish soldiers led by Oñate's nephew, Juan de Zaldívar, came to Acoma in December 1598 to demand food. In the Acoma people's own account, the soldiers raped an Acoma woman: they forced her into sex. The Acoma killed Zaldívar and eleven to fourteen of his men, and accounts differ on the number.

The attack. Oñate sent about seventy soldiers back to Acoma in January 1599, led by Vicente de Zaldívar. Over two or three days they killed Acoma men, women and children and burned much of the town. Nobody counted the dead, and estimates run from about 300 to about 1,500.

The sentences. Oñate gave the men over 25 two punishments: one foot cut off, and twenty years of forced labor in Spanish households, with no right to leave. Boys and men from 12 to 25 got the twenty years of forced labor only. So did girls and women over 12.

The feet. The sentences are in the written record, but whether anyone carried out the cutting of feet is disputed. In the Acoma people's own account, Oñate had the right feet of 24 captive men cut off. Two historians who studied the Spanish records for years, Marc Simmons and John Kessell, came to doubt it, and Kessell points out that no record describes an Acoma man with one foot.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

Three of every four newcomers around Chesapeake Bay between 1630 and 1680 were servants bound to a master for years, and they could not quit. Most got no wages. Over the whole colonial period, between one-half and two-thirds of the Europeans who came to the English colonies came this way.

Masters worked them in the tobacco fields, and a sickness in their first summer killed more than half of them. Virginia judges had runaways whipped and added years to their time. In the same century, English colonists began to hold Africans in slavery for life.

Bound for years: indentured servants

An indentured servant was bound by a written contract, called an indenture, to work for a master for a set number of years. Most terms ran four to seven years, and the servant could not leave. Most servants got no wages. A skilled servant named Buckland was paid 20 English pounds a year on top of his food, drink, washing and a place to sleep, and his contract was unusual.

How many. Counts differ. Most estimates run from one-half to two-thirds of all European immigrants to the colonies, and some go as high as three-quarters. Around Chesapeake Bay, about 50,000 servants arrived between 1630 and 1680, three of every four newcomers. In the 1630s there were about six men for every woman among them.

Servants there worked tobacco "from dawn until dusk, six days a week through the growing season." Many fell sick in their first summer. People called this sickness the seasoning, and it killed more than half of the new arrivals. One writer of the time said immigrants died "like cats and dogs."

Women servants. Virginia's lawmakers punished a woman servant who had a baby without being married. She had an extra year added to her term, or she had to pay her master 1,000 pounds of tobacco.

At the end of the term, a master owed the servant "freedom dues," goods to start a free life with. In Virginia a man got ten bushels of corn (a bushel is about eight gallons), thirty shillings in English coins and a musket, a long gun. A woman got fifteen bushels of corn and forty shillings.

Masters also took children and teenagers as apprentices, who worked for them to learn a trade such as printing or shoemaking. At home, families did their own spinning, weaving, candle-making and brewing, the making of beer.

Martin's Hundred, on the land of Tsenacomoco

Why this place. Richard Frethorne, an indentured servant whose letters home survive, worked at Martin's Hundred. He arrived in December 1622, nine months after a Powhatan attack there.

Martin's Hundred was an English plantation on the James River in Virginia, and its main settlement was Wolstenholme Towne. The plantation stood inside Tsenacomoco, the land of the Powhatan chiefdom, a union of many Native nations under one high chief.

Tsenacomoco ran north from the James River to the Potomac River, and from Chesapeake Bay west to the places where the rivers drop in falls from higher ground. About 15,000 people lived there.

The attack of 1622. The Powhatan leader Opechancanough led an attack on most of the English settlements on March 22, 1622. Colony officials recorded 347 colonists killed. They got no report from some settlements, so the real number was probably higher. The Powhatan fighters killed at least 58 colonists at Martin's Hundred. Its main settlement, Wolstenholme Towne, lost more people than any other settlement.

Richard Frethorne

Who: An indentured servant in Virginia, probably a young teenager, whose letters home to his parents survive.

When and where: Martin's Hundred, Virginia, from December 1622 until his death, recorded by February 1624.

"When we are sick there is nothing to comfort us," he wrote home.

Richard Frethorne reached Martin's Hundred in December 1622 on a ship called the *Abigail*, nine months after the Powhatan attack. He had come as an indentured servant, and he was probably a young teenager.

His letters. He wrote to his parents in England in March and April 1623. He begged them to buy out his indenture, or else to send food. Since he left the ship, he wrote, he had eaten nothing but peas and loblollie, a thin porridge boiled in water.

The muster list. A muster list made by February 1624, a count of the living and the dead at each settlement, records Frethorne, spelled "Frethram," among the dead at Martin's Hundred.

Runaways, and a sentence for life

Servants who ran away and were caught faced the whip and more years of work. The governor of Virginia and his council sat together as the colony's highest court, called the General Court.

Three runaways. These judges heard the case of three runaway servants of Hugh Gwyn, a planter, or owner of a plantation, on July 9, 1640. They ordered two of them, a Dutchman called Victor and a Scotsman named James Gregory, whipped thirty times each. Both men also had to serve extra years.

The third runaway was John Punch, a Black man. For the same escape, the same judges ordered him to serve his master "for the time of his natural Life," which meant for the rest of his life.

Branding. Thirteen days later, on July 22, the same judges ordered other runaways whipped thirty times. They also ordered three of them, Christopher Miller, Peter Wilcocke and a Black man named Emanuel, "to be Burnt in the cheek with the letter R." This punishment is called branding. To brand a person was to press a red-hot iron into the skin, and the burn left the letter on the cheek as a scar for life.

The court records name the letter R and the cheek, but they do not describe the iron used, and they do not name the men who did the whipping and the branding. The judges also ordered Miller and Emanuel to work in shackles, iron rings locked around the legs, for a year or more.

The 1643 law. Virginia's lawmakers wrote two punishments for runaways into law in March 1643. A runaway had to serve double the time he had been gone. A servant who ran away a second time was to be "branded in the cheek with the letter R." The law did not state who was to do the branding.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 2 MINUTES

A boy who learned a trade between 1700 and 1750 climbed three ranks: apprentice, journeyman and master. A journeyman, the middle rank, was a skilled worker paid wages by the day. Sailors and dock workers were the other large group who worked for wages, and officers of the British navy could seize them and force them to serve at sea.

Around Chesapeake Bay and in the Lowcountry along the southern coast, colonists held more and more people in slavery. Slave traders brought about 8,600 enslaved Africans to Virginia between 1701 and 1710, about 6,200 between 1711 and 1720, and about 13,000 between 1721 and 1730. Chapter 24, Slavery and Freedom, tells the work and the lives of enslaved people in full.

The ladder of a trade

A trade such as printing, shoemaking, carpentry or blacksmithing had those three ranks. A child or teenager started as an apprentice, bound to a master to learn the work. When the apprentice's time was up, he became a journeyman and worked for wages. A journeyman who set up a shop of his own became a master.

In the towns, enslaved men also worked at skilled trades. Owners "hired out" some of these men to other employers, who paid wages for the work, and the owners kept the wages.

Benjamin Franklin

Who: A runaway apprentice, bound at 12 to work in his brother James's print shop.

When and where: Boston, New York and Philadelphia, 1718 to 1723.

Movie: *Benjamin Franklin* (2022), a two-part documentary by Ken Burns, shown on PBS in April 2022. It is about Franklin's life.

At 17, with about four years left to serve, Franklin ran away.

Benjamin Franklin was 12 in 1718 when he was bound as an apprentice to his brother James, a printer in Boston. Under his indenture he had to work in James's shop until he was 21, and only in the last year would he get a journeyman's wages.

The beatings. Franklin later wrote that his brother beat him. He also admitted that he himself may have been "too saucy and provoking," which meant rude enough to make James angry. He ran away in September 1723, first to New York and then to Philadelphia. He reached Philadelphia on October 6, 1723, with almost no money.

Sailors, press gangs and the Knowles Riot

Sailors and dock workers earned wages, and officers of Britain's Royal Navy could impress them: seize them and force them to serve on a navy ship. The groups of men sent to grab them were called press gangs.

The Knowles Riot. Press gangs working for Commodore Charles Knowles, a navy officer who commanded a group of ships, seized 46 men in Boston in November 1747. Among them were boatmen, sailors and even carpenters. Beginning on November 17, a crowd made up mostly of working people rioted in Boston for three days. They took some of Knowles's officers as hostages and forced Knowles to free the seized men in exchange.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 2 MINUTES

Hired workers in Philadelphia began to act together against their employers in the late 1700s. In 1786 journeymen printers there went on strike, stopping work together until their employers met their demands, and they won pay of at least $6 a week. In 1794 journeymen shoemakers in the same city formed a trade union, a group of workers in one trade who join together to bargain, or make a deal, with their employers over pay and hours. Their union lasted.

The printers' strike of 1786

The journeymen printers of Philadelphia struck in 1786 against a cut in their wages. A US Labor Department list of important events in labor history records the start of the strike as June 2, 1786. The printers won, and their employers agreed to pay at least $6 a week.

The printers also set up a fund of money to support one another during the strike, the first strike fund on record. The same Labor Department list calls theirs "the earliest authenticated strike of workers in the United States in a single trade" (authenticated means proven by records).

The bakers of 1741. The labor historian Selig Perlman, writing in 1922, noted that the usual list of American strikes began with New York's bakers, who stopped work in 1741. He found that those bakers were master bakers, the shop owners, protesting a city rule that set the price of bread. By his account, the 1786 printers' walkout was the earliest real strike of workers against their employers known in America.

The shoemakers' union of 1794

Shoemakers in those days were called cordwainers. Journeymen shoemakers in Philadelphia formed the Federal Society of Journeymen Cordwainers in May 1794, the first trade union in the United States that lasted. Its members called a "turn-out," their word for a strike, to hold a standard wage, the same pay for the same work across the trade.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 9 MINUTES

Factory owners in the first half of the 1800s ran their workers' days by bells. In the mills of Lowell, Massachusetts, young women worked about 12 hours a day. Children tended machines in the mills, and Irish immigrants dug canals and railroads. Through these years workers fought for a working day of ten hours.

Forming a union, a group of workers who join together to bargain with their employers, could be a crime. A Philadelphia jury found shoemakers guilty in 1806 for forming one to raise their pay. By 1842 a Massachusetts judge had ruled that a union could be lawful.

Unions on trial, 1806

The union shoemakers of Philadelphia, the cordwainers, held a turn-out, their word for a strike, in November 1805. Four months later, in March 1806, lawyers led by Jared Ingersoll and Joseph Hopkinson put eight of them on trial. Judge Moses Levy heard the case in the Mayor's Court of Philadelphia, a city court.

The verdict. The jury found the eight men guilty of conspiracy, a secret plan by two or more people to break the law. In the jury's words, their crime was "a conspiracy to raise their wages." Each man had to pay a fine of $8 and the costs of the trial. After this verdict, prosecutors could take workers to court as criminals for forming a union to raise their pay.

Children, diggers and charcoal burners

Children worked in the mills, and that was legal. They tended the machines and did the doffing: they took the full bobbins, the spools that held the spun thread, off a spinning machine and put empty ones on.

Gangs of laborers dug the new canals and graded the new railroad lines, which means they leveled the ground for the tracks. Many of these men were immigrants from Ireland.

Making iron took many workers too, and much of the work was cutting wood and turning it into fuel. At Hopewell Furnace, a place in Pennsylvania where workers made iron, the furnace burned charcoal, wood burned slowly with little air until only a black fuel is left. Hopewell paid more than 200 workers from 1835 to 1837, and more than 100 of them were woodcutters.

Most woodcutters worked part of the year, cutting and splitting hardwood through the winter with only axes. A woodcutter cut about two cords a day and was paid by the cord, a stack of cut wood of a set size.

About 20 colliers, the men who made the charcoal, worked there in the 1830s. A collier tended slowly burning mounds of wood day and night, for up to two weeks at a time, and lived in a hut beside them. He was paid by the bushel of charcoal, about eight gallons.

Who killed the men at Duffy's Cut?

The records do not say. All 57 Irish workers at Duffy's Cut, in Pennsylvania, died in 1832, some of them of cholera, and their remains show that someone killed others.

The 57 men arrived on the ship *John Stamp* in late June 1832. They came from Donegal, Tyrone and Derry in Ireland, and their average age was 22. A contractor named Philip Duffy hired them to dig a cut for the Philadelphia and Columbia Railroad, a channel through high ground so the tracks could run level. They also built up a bank of earth for the tracks. The site was at mile 59 of the line, in Chester County.

The cholera. By early August cholera had reached their camp. Cholera is an infection of the gut, spread by water or food that carries traces of a sick person's waste. In bad cases it causes watery diarrhea and vomiting, and the body loses water so fast that a person can go into shock and die. Local people stayed away from the camp.

The skulls. A team of researchers dug up some of the men's remains from 2009 to 2012. Some of the skulls had been struck by blows at about the time of death, and one man had a bullet in his skull, fired at close range. None of the remains showed wounds from trying to defend themselves. Someone beat several of the men and shot one.

No record shows that anyone was charged. To explain the killings, the historian who led the dig, William E. Watson, points to fear of cholera and perhaps to nativism, hatred of immigrants.

Lowell and the mill girls

The owners of the mills built the mill city of Lowell, Massachusetts, at Pawtucket Falls on the Merrimack River. Most of the mill workers were young New England women, aged 15 to 25, known as the mill girls. Under the companies' rules they had to work at least a year, live in a company boardinghouse and go to church. On average a mill girl stayed about three years.

The Lowell mills belonged to Francis Cabot Lowell and other Boston merchants, called the Boston Associates. Their mills spun cotton that enslaved people had grown in the South. The mills also made a cheap cloth that the owners sold to plantations to clothe enslaved people.

The working day. A report of the Massachusetts lawmakers in 1845 recorded the hours. From May through August, work began at 5 in the morning and ended at 7 at night, with half an hour for breakfast and 45 minutes for the midday meal. From September through April, work began "as soon as they can see" and ran until 7:30 at night. Across the year the mills ran an average of twelve hours and ten minutes a day, and bells rang the start and the end of work.

Whose land it was. The mill owners built on land where the Pennacook people had lived until a war in the 1670s. The falls were a Pennacook fishing and gathering place, and in the 1600s two Pennacook towns stood where Lowell stands now. Their leaders were the sachems, or chiefs, Passaconaway and his son Wannalancet.

King Philip's War. Native nations led by Metacom, a Wampanoag leader the English called King Philip, fought the English colonies of New England in 1675 and 1676. About 9,000 people died in the war. Wannalancet kept peace with the English.

Cocheco, 1676. Wannalancet led about 400 people from the nations allied with him to Cocheco, in New Hampshire, to make peace with an English captain named Richard Waldron. Waldron broke the peace and took them all prisoner. He later let Wannalancet and about half of the group go.

Waldron sent the others to Boston, where the governor and council of Massachusetts decided what to do with them. Officials in Boston hanged seven or eight of them. About 200 were enslaved for life and shipped to Barbados, an island in the Caribbean Sea, and the records do not name the men who sold them.

Wannalancet and his people moved north to St. Francis in Canada. By the end of the war they no longer lived in the lower Merrimack valley.

The Lowell turn-outs of 1834 and 1836

The mill managers announced wage cuts of 12 to 25 percent in February 1834. About 800 women turned out, and the strike failed.

The board increase. The women who ran the boardinghouses told the mill managers in October 1836 that they could not make ends meet. The managers then took more money out of each woman's weekly pay for room and board, which was the same as a cut in pay. About 1,500 women struck and formed the Factory Girls' Association.

They were better organized than in 1834, and this time they won part of what they asked for. Managers at the Boott and Merrimack mills took back the whole increase. At the Hamilton mill, managers took it back only for workers paid by the day.

Harriet Hanson Robinson

Who: A child worker in a Lowell mill who led the girls of her room out on strike at age 11, and later wrote a book about the mill girls.

When and where: Lowell, Massachusetts, 1835 to 1848. Born February 8, 1825. Died 1911.

She was 11 when she led her room out on strike.

Harriet Hanson had started work in a Lowell mill at 10, as a doffer. On the day of the turn-out in October 1836, the girls in her room could not decide whether to leave. She later remembered saying, "I don't care what you do, I am going to turn out, whether anyone else does or not." She walked out, and the others followed her.

She stayed at the mills until 1848. In 1898 she published a book about her years there, *Loom and Spindle, or Life Among the Early Mill Girls*, and in later life she worked for women's right to vote.

The ten-hour day

Workers in Philadelphia, in Lowell and on federal building jobs wanted a working day of ten hours. Coal heavers, the men who moved coal on and off boats on the docks of the Schuylkill River in Philadelphia, stopped work in 1835, and other workers joined them.

The first general strike. By June 10, workers in more than forty trades, or kinds of skilled work, had struck, nearly 20,000 people in all. It was the first general strike in the United States, with workers in many different trades stopping work at the same time. Philadelphia's city officials agreed to the ten-hour day and to raises.

President Martin Van Buren ordered a ten-hour day on March 31, 1840, for laborers on building projects of the United States government. His order covered only workers paid by the federal government, not anyone who worked for a private employer.

Commonwealth v. Hunt, 1842

The judges of the highest court in Massachusetts, the Supreme Judicial Court, heard a case called Commonwealth v. Hunt in 1842. Chief Justice Lemuel Shaw ruled that in Massachusetts a trade union was not a criminal conspiracy just for being a union. Workers there could lawfully form a union, and they could lawfully strike. Thirty-six years earlier, a Philadelphia jury had found union shoemakers guilty of a crime for doing the same thing.

Sarah Bagley and the ten-hour petition

Lowell mill workers formed the Lowell Female Labor Reform Association in the winter of 1844 and 1845, with Sarah Bagley as its president. The women sent petitions, written requests signed by many people, to the Massachusetts lawmakers, asking for a ten-hour day. After the petitions came in, the lawmakers looked into conditions in the mills.

The hearing. Bagley spoke to a committee of the Massachusetts House in 1845 and told them that she had "worked in the Lowell Mills eight years and a half." The chairman, William Schouler, and all the other members agreed "that legislation is not necessary at the present time," meaning that no new law was needed. They turned down the ten-hour petition. Bagley said afterward that the committee had "misrepresented" what she told them.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 15 MINUTES

Police, private guards, state militia (the part-time soldiers of each state) and federal soldiers broke many strikes between 1850 and 1900, and they shot and killed strikers. The workers had formed the first national unions, to fight for higher pay and an eight-hour day.

The work itself was dangerous, in mines, mills, rail yards and sweatshops, crowded workshops where people sewed clothing for long hours and low pay. Children worked beside adults. In the cotton mills of the South, they worked 12-hour days, six days a week.

The Chinese railroad strike, 1867

Thousands of Chinese workers built the Central Pacific Railroad through the Sierra Nevada mountains of California in the 1860s. White workers on the line earned $40 a month, and the Chinese workers earned $35. Several thousand of them went on strike on June 25, 1867, and estimates of their number run from 2,000 to 5,000.

They asked for the same pay as white workers, shorter hours and an end to whipping. The records do not name the men who whipped them. The company's managers cut off the food supply to their camps. After eight days the strike ended with no promises from the company, though pay later rose a little. It was the largest strike by workers in the country up to that time.

Why 110 people died at Avondale, 1869

A fire killed 110 people at the Avondale coal mine, in Plymouth Township near Wilkes-Barre, Pennsylvania, on September 6, 1869. Among the dead were five boys and two men who tried to rescue the workers.

One way out. Above the mine stood the breaker, a tall wooden building where coal was broken up and sorted. It stood over the mine's only shaft, the one way in and out. Sparks from the furnace that pushed air through the mine set the breaker on fire, and the men and boys below had no other way out. Pennsylvania's lawmakers passed a mine safety law in 1870 that required more than one exit from every mine.

The Knights of Labor

A tailor named Uriah Smith Stephens founded the Knights of Labor in Philadelphia on December 9, 1869, as a secret society. Terence V. Powderly later led it and opened it up to the public. The Knights had fewer than 10,000 members in 1879 and about 700,000 by 1886.

The Knights let women join, and in some places Black workers too. Besides higher wages, they called for an eight-hour day, equal pay for equal work, and an end to child labor and to the work of prisoners leased to companies. They also wanted the public, not private owners, to own utilities, the companies that supplied things like gas and water.

Servants, children and sweatshops

Census takers count everyone in the country for the United States government every ten years, and in 1870 they counted women's work carefully for the first time. They found 867,354 domestic servants, people paid to cook, clean and care for other families' homes, about half of all women who worked for pay. Domestic service stayed the largest job for women in every census from 1870 to 1940.

Children at work. At least 18 percent of all children aged 10 to 15 had jobs from 1890 to 1910, nearly 1 in 5. In Philadelphia, the pay children brought home made up 28 to 46 percent of their family's income, depending on the family's ethnic group.

In the South, cotton mill owners hired children in growing numbers. Of the nearly 100,000 textile workers in the South in 1900, 25,000 were children under 16. By 1904 there were 50,000 children in the mills, and 20,000 of them were under 12. They worked 12-hour days, six days a week. Mill owners said the fingers of adult farm workers were too "knotted and gnarled" for mill work.

In the hard-coal fields of Pennsylvania, breaker boys sat on benches in the breakers and picked slate, a kind of rock, out of the coal. In the cities, people sewed clothing in sweatshops set up in lofts, the open upper floors of buildings, and in tenements, cheap apartment buildings packed with families.

The Great Railroad Strike, 1877

John W. Garrett, the president of the Baltimore and Ohio Railroad, cut his workers' wages by 10 percent in July 1877, and it was not the first cut. Workers at Martinsburg, West Virginia, walked off the job. The strike spread through West Virginia, Maryland, Pennsylvania, New York, Missouri and Illinois.

Governors called up their state militias, and President Rutherford B. Hayes sent federal troops. He issued the orders for West Virginia on July 18, Maryland on July 21 and Pennsylvania on July 23.

Pittsburgh. Militiamen brought in from Philadelphia opened fire on a crowd in Pittsburgh on July 21. They killed more than twenty people, among them a woman and three small children. The official death count for Pittsburgh was 26, and the real number was likely higher.

Reading. Two days later, in Reading, Pennsylvania, a crowd threw bricks and stones down on militiamen marching through a deep cut for the railroad. Some of the soldiers fired into the crowd and killed at least ten people. By September 1877, state and federal troops had broken the strike, and the workers won nothing at the time.

The telephone operators

A new job came with the telephone. Calls went through a switchboard, where an operator connected each caller to the person being called, and the first operators were teenage boys. Emma Nutt became the first woman operator on September 1, 1878, at a telephone company in Boston, and her sister Stella started the same day. Emma Nutt earned $10 a month for a 54-hour week. The companies' managers replaced the boys with women, and by the late 1880s nearly all operators were women.

The eight-hour day and Haymarket, 1886

Workers across the country went on strike for an eight-hour day on May 1, 1886, about 35,000 of them in Chicago. Two days later, Chicago police fired on strikers at the McCormick Harvesting Machine Company plant and killed at least two of them.

The bomb. Workers held a meeting in Chicago's Haymarket Square on May 4 to protest the killings. Inspector John Bonfield marched on the meeting with 176 police officers and ordered the crowd to break up. Then someone threw a dynamite bomb at the police, and the police drew their guns and fired wildly. No one has ever proved who threw the bomb.

The dead. Counts differ. A Library of Congress history page lists seven police officers and at least four people in the crowd killed. A Chicago History Museum page lists eight police officers dead and sixty wounded, and an unknown number in the crowd killed or wounded.

The trial. A Chicago jury convicted eight anarchists, people who believe there should be no government, of conspiracy. None was shown to have thrown the bomb. Judge Joseph E. Gary sentenced seven of them to death and the eighth, Oscar Neebe, to fifteen years in prison.

Officials at the Cook County Jail hanged four of the men on November 11, 1887: Albert Parsons, August Spies, George Engel and Adolph Fischer. A fifth, Louis Lingg, killed himself in his cell with a small bomb that someone

smuggled in to him. The last two, Samuel Fielden and Michael Schwab, had asked for mercy, and the governor of Illinois at the time changed their sentences to life in prison.

The pardon. A later governor of Illinois, John Peter Altgeld, pardoned Fielden, Schwab and Neebe, the three men still in prison, in June 1893. The pardon freed them from the rest of their punishment. Altgeld found that the trial had been unfair, and he wrote that Judge Gary ran it with "malicious ferocity," meaning cruel, spiteful anger.

Altgeld's reasons. A court officer named Ryce had the job of calling in the men who would sit on the jury. The state's attorney, the lawyer who brought the case, had suggested Ryce for that job. Altgeld wrote that Ryce called in jurors who were already against the men on trial.

Altgeld also wrote that "much of the evidence given at the trial was a pure fabrication," which means it was made up. Some leading police officials, he wrote, had threatened men with torture and had offered money and jobs to anyone who would swear to what the police wanted.

The Knights. After Haymarket, most members left the Knights of Labor. By 1890 the Knights had about 100,000 members, down from about 700,000 in 1886.

The American Federation of Labor

Leaders of craft unions, each made up of the skilled workers of one trade such as cigar makers or carpenters, met in Columbus, Ohio, on December 8, 1886. There they founded the American Federation of Labor, the AFL. Its union leaders bargained with employers for higher wages and shorter hours, a way of running unions called "pure and simple" unionism.

Samuel Gompers

Who: A cigar maker who became the first president of the American Federation of Labor.

When and where: Born in London, January 27, 1850. Worked in New York. President of the AFL from 1886 to 1924, except for 1895.

A cigar roller from London led the AFL for almost forty years.

Samuel Gompers made cigars from boyhood, and in New York he rolled them in tenement shops. He became the first president of the American Federation of Labor and served from 1886 to 1924, except for one year, 1895.

Gompers built the AFL around wages and hours. Its unions had about 140,000 members in 1886 and nearly 3 million by 1924, about twenty times as many.

Why brakemen lost their fingers

A brakeman on a railroad joined, or coupled, the cars of a train by hand. He stood between two cars as they moved together and guided an iron link into a socket. Brakemen's hands were crushed, and missing fingers were called "the badge of the experienced worker." Thousands of brakemen died.

A safer way. Eli Janney got a patent for an automatic coupler in 1873, the right to be the only one to make and sell it for a set time. The Westinghouse air brake, a brake worked by air pressure, had existed since 1869. Railroad owners did not put either one on most of their cars.

Members of Congress voted through the Safety Appliance Act in 1893, and it required both on trains by 1900. Coupling accidents fell from about 11,000 in 1892 to about 2,000 in 1902, while the number of railroad workers grew.

Homestead, 1892

The lockout. The steelworkers' union contract at Carnegie Steel's works in Homestead, Pennsylvania, was running out in the summer of 1892. Henry Clay Frick, the chairman of Carnegie Steel, cut pay for hundreds of the workers and refused to bargain with the union. On June 29 he shut the plant and locked out all 3,800 workers, keeping them out of their jobs. The workers went on strike, and armed strikers surrounded the plant and closed off the town.

On July 6, 300 agents of the Pinkerton detective agency, a private company that Frick had hired, tried to land from barges and take the plant back. Strikers and Pinkertons fought a gun battle, and seven workers and three Pinkerton agents were killed. Some accounts count more.

Eight thousand state militiamen, sent by the governor of Pennsylvania, arrived at Homestead on July 12. By November the strike had failed, and the steelworkers at Homestead no longer had a union.

Pullman, 1894

The Pullman Palace Car Company's workers made and repaired railroad cars near Chicago. They lived in Pullman, Illinois, a company town, where the company owned the houses and its managers collected the rent. The company's managers cut wages by about 25 to 35 percent in 1893 and 1894, and they did not cut the rents.

The workers belonged to the American Railway Union, founded in June 1893. It was the first industrial union, taking in every kind of railroad worker in one union instead of one craft at a time.

The boycott. The union's members held a convention, a big meeting of delegates sent by its branches, in June 1894, and the delegates voted for a national boycott of Pullman cars. In the boycott, railroad workers across the country refused to handle trains that carried Pullman cars. While it spread, on June 28, 1894, President Grover Cleveland signed a law making Labor Day a federal holiday.

The court order. Attorney General Richard Olney, the federal government's top lawyer, took the union to court. Federal judges William Woods and Peter Grosscup issued an injunction, a court order to stop doing something, on July 2, 1894. They ordered Eugene V. Debs, the union's founder, and its other officers to stop anything that got in the way of the mail or of trade between the states.

The troops. The next day, over the protest of Governor Altgeld, President Cleveland ordered federal troops into Chicago. More than 16,000 troops were called up across the country, and the soldiers arrested boycotters and rioters and sometimes shot them.

About 30 people were killed across the country, and some counts reach 34. Accounts of the strike do not say who killed each of them. Federal judges and soldiers ended the boycott.

Jennie Curtis

Who: A seamstress in the Pullman shops whose speech helped start the Pullman boycott.

When and where: Pullman, Illinois, 1893 and 1894. Born 1871. Died 1917.

Her pay fell from as much as $2.25 a day to 80 cents.

Jennie Curtis spoke to the American Railway Union's convention in June 1894, and her speech helped persuade the delegates to vote for the Pullman boycott.

She had worked for five years as a seamstress in the sewing room of Pullman's repair shops. She told how the women there made "all the carpets, and all the silk, satin, plush, and velvet drapings for the dining cars," and they sewed the cloth covers for the seats.

Two cuts in one week. Before the depression of 1893, a time when many businesses failed and many people lost their jobs, she earned $1.50 to $2.25 a day. Pullman's managers cut her pay twice in one week in November 1893, and it fell to about 80 cents a day. Many of the girls she worked with averaged 40 to 50 cents.

Curtis was president of Local 269, the union's branch for young women workers, which had 125 members. Later in 1894 she told the United States Strike Commission, a panel that looked into the strike, about wages, rents and abusive supervisors. She said her father had worked for Pullman for 13 years. When he died, he owed back rent, and Pullman's managers collected it.

Eugene V. Debs

Who: A locomotive fireman who founded the American Railway Union and went to jail for leading the Pullman boycott.

When and where: Terre Haute, Indiana, and Chicago. Founded the union in June 1893.

Movie: *American Socialist: The Life and Times of Eugene Victor Debs* (2017), a documentary directed by Yale Strom. It is about Debs's life.

He ran for president from a prison cell, as Convict 2253.

As a young man in Terre Haute, Indiana, Eugene V. Debs was a locomotive fireman, the worker who shoveled coal into the engine's fire. He became an officer of the Brotherhood of Locomotive Firemen, a union for that one craft.

He founded the American Railway Union in 1893, for all railroad workers. After the Pullman boycott, judges found him guilty of contempt of court, disobeying a court's order, and he served six months in the jail at Woodstock, Illinois, in 1895.

Convict 2253. Debs later ran for president five times as a Socialist, one of those who wanted the public, not private owners, to own the big industries. He gave a speech against the war in 1918, and he was sentenced to ten years in prison for it.

He ran for the last time in 1920 from his prison cell, as Convict 2253, and got between 900,000 and 1 million votes. President Warren Harding shortened his sentence, and Debs went free at Christmas 1921.

Lattimer, 1897

Nearly 400 immigrant coal miners marched from Harwood to Lattimer, Pennsylvania, on September 10, 1897, to support a strike, and they carried no weapons. Luzerne County Sheriff James Martin had deputized nearly 90 men, which means he had given them the power to act as officers of the law. The deputies fired on the marchers. They killed 19 men, who were Polish, Slovak and Lithuanian, and wounded 38. Many were shot in the back.

Martin and his deputies were tried for murder from February 1 to March 2, 1898, and the jury found every one of them not guilty. By 1900 the United Mine Workers, the coal miners' union, was the fastest-growing union in the country.

Convict leasing

State officials in the South leased, or rented out, prisoners to mines, railroads and planters after the Civil War, and the people who leased them forced them to work. This was called convict leasing. Most of the prisoners were Black men, and in Alabama more than 90 percent of the state prisoners leased out were Black.

The Black Codes. Sheriffs arrested many of them for small offenses under the Black Codes, laws that Southern lawmakers passed from November 1865 to make Black people prove they had work. In Alabama, contractors whipped prisoners who tried to escape or did not work hard enough. Every year, 4 or 5 of every 100 leased state prisoners in Alabama died.

BY 1900

If you remember one thing: Workers built the first national unions in these fifty years, and police, deputies, militiamen and federal soldiers broke many of their strikes by force, firing on strikers and crowds from Pittsburgh to Lattimer.

- 1869: A fire at the Avondale mine kills 110 people, because the mine had only one way out.
- 1886: Workers strike for an eight-hour day, and after the Haymarket bomb officials hang four men who were never shown to have thrown it.
- 1894: Federal judges and soldiers end the Pullman boycott, and about 30 people are killed.

Next: After 1900 children still worked in the mines and mills, and a fire in a New York garment factory killed 146 workers.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 25 MINUTES

Members of Congress passed the first federal laws for a lowest wage, a shorter work week and the right to form a union in the 1930s, after three decades of child labor and deaths on the job. Under the Wagner Act of 1935, forming a union became a legal right. A law of 1938 set a minimum wage of 25 cents an hour and a work week of 44 hours, falling to 40 by 1940, and it banned most work by children under 16.

Children still worked in mills and mines in the first years of the century. In 1900, 1,489 coal miners were killed at work, and in 1907, 3,242. A fire killed 146 workers in a New York garment factory in 1911.

At Ludlow, Colorado, in 1914, National Guard soldiers burned a camp of striking miners' tents, and the fire killed 13 women and children hiding in a pit under one tent. In Chicago in 1937, two years after the union law, police shot and killed ten people who were marching to support a steel strike.

The hard-coal strike, 1902

In Pennsylvania, 147,000 hard-coal miners and breaker boys, the boys who picked waste rock out of the coal, went on strike in May 1902. They stayed out for about five months, until October. President Theodore Roosevelt stepped in as a mediator, a go-between who helps both sides reach a deal. It was the first time a president had done this in a strike. Earlier presidents had sent troops to break strikes, as in 1877 and 1894.

The commission. Roosevelt set up a commission, a group of people chosen to look into the strike. Its members heard 558 witnesses, and some of them were breaker boys younger than nine. The commission's members ruled in March 1903 that the miners should get a 10 percent raise and a nine-hour day.

Mother Jones

Who: Mary Harris Jones, a union organizer for coal miners whom a prosecutor called "the most dangerous woman in America."

When and where: Baptized in Cork, Ireland. Organizing from the 1870s, and for the United Mine Workers from the 1890s. Died 1930.

She led striking mill children on a march toward the president's home.

The children's march. Children who worked in textile mills in Philadelphia were on strike in July 1903. Mary Harris Jones, known as Mother Jones, led them on the March of the Mill Children, from Philadelphia toward the home of President Theodore Roosevelt at Oyster Bay, New York. She wanted people across the country to look at child labor.

How she became an organizer. She had worked as a dressmaker. Her husband and all four of her children died of the disease yellow fever in Memphis, Tennessee, in 1867, and she lost her home and her shop in the great fire in Chicago in 1871. After that she became a full-time organizer, and from the 1890s she worked above all for the United Mine Workers. In 1902 a prosecutor in West Virginia called her "the most dangerous woman in America."

> **Her age.** She celebrated her "100th" birthday on May 1, 1930, and died later that year. The record of her baptism in Cork gives August 1, 1837, which would have made her 92.

How many miners died?

The worst year on record was 1907, when 3,242 coal miners were killed at work out of 680,492, about 1 miner in every 210. In 1900 the number killed had been 1,489.

Monongah. Explosions in mines No. 6 and No. 8 at Monongah, West Virginia, killed hundreds of men and boys on December 6, 1907, and some of the boys were as young as eight. The official count was 362 dead. The West Virginia Mine Wars Museum and later estimates put the dead at more than 500, and about 1,000 children lost their fathers.

Accounts differ on which company ran the mines, the Fairmont Coal Company or the Consolidation Coal Company. The Fairmont company's president, C. W. Watson, blamed "the careless use of a mining lamp." No record shows that anyone was charged.

Lewis Hine photographs the children

Lewis Hine left his job as a teacher in 1908 to take photographs of working children. He worked for the National Child Labor Committee, a group trying to end child labor. From 1908 to 1917 he made thousands of pictures of children at work in mills, mines, food-canning plants and on city streets.

Addie Card

Who: A child spinner in a Vermont cotton mill whose photograph by Lewis Hine appeared on a United States postage stamp in 1998.
When and where: North Pownal, Vermont, 1910. Died 1993.

Hine wrote her name down wrong, as "Addie Laird."

Lewis Hine photographed Addie Card at her spinning machine in the cotton mill at North Pownal, Vermont, in 1910. Hine wrote that she was 12, but the girls at the mill told him she was 10. He called her an "anemic little spinner," anemic meaning pale and weak because the blood is not carrying enough of what the body needs. The picture appeared on a United States postage stamp in 1998.

Her real name. A researcher named Joe Manning, working with the novelist Elizabeth Winthrop, found her real name and traced her life in 2005 and 2006. She married Edward Hatch in 1915, at 17. They split in 1925, and he won custody of their daughter. She married again, to Ernest Lavigne, and got another job in a mill in Cohoes, New York. In her last years she lived in a housing project in Cohoes, and she died in 1993.

The shirtwaist makers and the Triangle fire

Many young immigrant women in New York City sewed shirtwaists, women's blouses, in garment shops. The shirtwaist makers held a meeting at Cooper Union, a hall in New York, on November 22, 1909.

The Uprising of the 20,000. Clara Lemlich, a shirtwaist maker, was still bruised from a beating by men hired by her company. She stood up and spoke in Yiddish, the language of many Jewish immigrants. "I have no further patience for talk," she said. "I move we go on strike." About 20,000 garment workers struck for eleven weeks.

The fire. A fire broke out on the top floors of the Triangle Waist Company's building near Washington Square on March 25, 1911, and it killed 146 workers. Most of them were young immigrant women, and some were only 14. On the ninth floor, the door to one stairway was locked, and no record proves who locked it. Dozens of workers jumped from the building.

The trial. The owners of the company, Max Blanck and Isaac Harris, were tried for manslaughter, the crime of killing someone without meaning to, through a wrongful act or through carelessness. The jury found them not guilty on December 27, 1911, because the prosecutors could not prove that the owners knew the door was locked. New York's lawmakers then set up the Factory Investigating Commission, and they rewrote the state's factory safety laws.

Rose Rosenfeld Freedman

Who: A worker who survived the Triangle fire and spent the rest of her life telling what happened there.

When and where: New York City, March 25, 1911. Died February 15, 2001, at 107, the last survivor of the fire.

For ninety years she told the truth about the locked doors.

Rose Rosenfeld was two days short of her 18th birthday when the Triangle fire broke out. She got out by following the company's executives up the stairs to the roof of the building, on the tenth floor, and from the roof rescuers pulled the survivors across.

She later said that the owners offered her money to keep quiet about the locked doors, and that she turned it down. She died in Beverly Hills, California, in 2001, at 107, the last survivor of the Triangle fire.

Frances Perkins

Who: A woman who watched the Triangle fire, later became the first woman in a president's Cabinet, and pushed for the law that set the country's first federal minimum wage.

When and where: New York, 1911. Secretary of Labor in Washington, 1933 to 1945.

Movie: *Summoned: Frances Perkins and the General Welfare* (2020), a documentary by Mick Caouette, shown on PBS. It is about Perkins's life.

Frances Perkins watched the Triangle workers jump from the building.

She was having tea near Washington Square on March 25, 1911, when the fire broke out, and she ran to the building. She later said that the New Deal, the set of programs President Franklin Roosevelt started in the 1930s, began on that day.

In the Cabinet. After the fire, Perkins worked as an investigator for New York's Factory Investigating Commission. President Roosevelt made her Secretary of Labor in 1933. She was the first woman in a US Cabinet, the group of top advisers who run the government's departments, and she held the job until 1945. She pushed for the Fair Labor Standards Act and for Social Security, the government's program of payments to people who are old or cannot work.

Lawrence, 1912

Massachusetts lawmakers cut the legal work week from 56 hours to 54 in 1912, and mill owners in Lawrence, Massachusetts, then cut their workers' pay. The workers struck on January 12, 1912. Most of them worked for the American Woolen Company. More than half of the mill workers were women and children, and they spoke more than a dozen languages. Leaders of a union called the Industrial Workers of the World sent organizers, who ran committees that gave out food and help.

The children. The strikers began sending their children to other cities in February, to live with relatives or with strangers who could feed them and keep them safe during the strike. On February 24, police at the Lawrence railroad station clubbed women and children who were trying to send children away, and they jailed 30 women. The owners agreed on March 12 to a 15 percent raise, double pay for overtime (extra hours of work) and no punishment for the strikers.

The assembly line

Managers at Henry Ford's plant at Highland Park, Michigan, started a moving line to build car frames, called chassis, on October 7, 1913. The work moved past 140 assemblers along 150 feet of line, and each man did the same few motions over and over, at the speed of the line. Building a chassis had taken more than 12 hours. On the line it took about 93 minutes.

The pace. Many men could not stand the pace, and they quit. Worker turnover at the plant reached 370 percent in 1913, which means that for every job, Ford's managers had to hire nearly four men in a year. Ford announced a pay of $5 for an eight-hour day on January 5, 1914. Before that, pay had been about $2.34 for nine hours.

The $5 day came with conditions. Ford's managers called part of the money "profit-sharing," and they paid that part only to men who passed inspections of their homes. Company investigators, from an office that the managers called the Sociological Department, checked whether a man drank, whether he saved money, how clean his home was and whether he took in boarders, people who paid to live in a room of his home. The managers left out workers under 22 and, at first, most women.

Ludlow, 1914

Coal miners in Colorado went on strike in September 1913. They wanted the state's mining laws enforced, and they wanted the mine owners to recognize their union, the United Mine Workers, which means to accept it and bargain with it. Most of them worked for the Colorado Fuel and Iron Company, whose main owner was John D. Rockefeller Jr. The strikers and their families lived in tent colonies.

The Death Special. Men in an armored car belonging to the company, called the "Death Special," fired on the tent colony at Forbes in October 1913 and killed one person. Governor Elias M. Ammons sent in the Colorado National Guard, the state's soldiers, and the mine owners paid for it.

The fire at Ludlow. Fighting broke out at the tent colony at Ludlow on April 20, 1914, and nobody knows who fired the first shot. Guardsmen burned the tents. At least 19 people were killed: one guardsman, five miners, and thirteen women and children who had hidden in a pit dug under a tent. The women and children suffocated, which means they died because they could not get air. Guardsmen shot the strike leader, Louis Tikas, three times in the back.

Miners then attacked mines across the region and killed supervisors and guards, and nobody knows exactly how many. President Woodrow Wilson sent in the US Army on April 28. The strike ended in December 1914, and the miners won none of their union's goals.

BY 1914: Henry Ford paid $5 for an eight-hour day, and New York had rewritten its factory laws after the Triangle fire, but no federal law yet limited children's work or set a lowest wage.

The first child-labor law, and the justices who threw it out

Members of Congress passed the Keating-Owen Act in 1916, the first federal law against child labor. Under it, a company could not ship goods to another state if it had used child workers to make them. It covered children under 14 in mills, food-canning plants and factories, and children under 16 in mines. Children aged 14 to 16 could not work more than 8 hours a day or 6 days a week, or at night.

Hammer v. Dagenhart. Roland Dagenhart and his sons Reuben, 14, and John, 12, worked in a cotton mill in Charlotte, North Carolina. Under the new law John would have to stop working, and Reuben would lose hours from his 60-hour week. A committee of Southern cotton mill owners recruited Dagenhart to sue and paid his legal costs. He argued that the law took away the money his sons earned.

The justices of the Supreme Court ruled 5 votes to 4 on June 3, 1918, in the case Hammer v. Dagenhart, that the law broke the Constitution and could not be used. The five justices in the majority said that Congress's power over trade between the states did not cover how goods were made, and Justice Oliver Wendell Holmes disagreed. Members of Congress passed a second child-labor law in 1919, and the justices threw that one out too.

Grace Fryer

Who: A painter of glowing watch dials who was poisoned by radium at work and sued the company while she was dying.

When and where: Orange, New Jersey. Worked from 1917 to 1920. Sued in 1927. Settled in June 1928.

Movie: *Radium Girls* (2018), a drama inspired by the dial painters' case. Its main characters are two invented sisters, not Grace Fryer.

"Our instructors told us to point [our brushes] with our lips."

Grace Fryer painted watch dials with glowing paint at the United States Radium Corporation in Orange, New Jersey, from 1917 to 1920. The paint glowed because it contained radium, which gives off radiation, energy that damages the body from the inside. Supervisors taught the women to shape the tip of the brush with their lips, so the radium paint went into their mouths, and they told the women the paint was safe.

What the radium did. Radium is in the same chemical family as calcium, so the body takes it into the bones and teeth, and almost all of it stays in the skeleton. Scientists who studied the dial painters found that it stopped the body from making enough new blood cells, killed the bone of the jaw so that the jawbone broke down, and caused bone cancer.

A dentist named Theodor Blum examined Grace Fryer in 1924 and found that her jawbone was wasting away in a way he had never seen before. As the dial painters fell sick and died, company officials played down and disputed the poisoning.

The lawsuit. A lawyer named Raymond Berry sued the company in May 1927 for five of the women: Grace Fryer, Katherine Schaub, Quinta Maggia Macdonald, Albina Maggia Larice and Edna Hussman. Reporters called them "the five women doomed to die," and some of them were too sick to travel to court.

> The company's officials agreed in June 1928 to pay the women to end the lawsuit, in a deal called a settlement. Each woman received $10,000 (some accounts say $15,000), $600 a year for as long as she lived and payment for her medical bills, which a committee of three doctors had to approve. The company's officials never admitted that the company was to blame.

Catherine Wolfe Donohue's case, 1938

A supervisor named Miss Murray also taught the dial painters at the Radium Dial Company plant in Ottawa, Illinois, to point their brushes with their lips. Catherine Wolfe Donohue was one of them. "We would first dip the brush into water, then into the powder, and then point the ends of the bristles between our teeth," she said.

The box of bones. She came to a hearing on February 10, 1938, with a box of pieces of bone that she said had come from her jaw. When the judge asked her doctor about her chances, she collapsed, and several men carried her from the room. She weighed 71 pounds.

The ruling. The hearing officer, the official who ran the hearing, ruled on April 5, 1938, that her illness came from her work at the plant. He awarded her $5,661, paid as $277 a year for life. The company's lawyers appealed, asking for a new ruling. The members of the Illinois Industrial Commission, a state board that decided workers' claims, turned down the appeal on July 6, 1938, and added $730.

Donohue died later that month, at 35. The last appeal in her case ended in October 1938, after her death.

How many men died digging the Hawks Nest tunnel?

Nobody knows exactly. Almost 5,000 men dug the Hawks Nest tunnel near Gauley Bridge, West Virginia, from 1930 to 1932, most of them Black men from the South, and counts of the dead run from 476 to close to 1,000.

Who had it dug. Managers of the Union Carbide company had the tunnel dug to carry water to a power plant, which was built to supply power to Union Carbide's alloy works, where metals were mixed. They hired the Rinehart and Dennis company of Charlottesville, Virginia, to do the digging.

The dust. The tunnel went through sandstone made mostly of silica, the hard mineral that sand is made of, and a third of it was dug wider so that the silica could be mined. The men worked with no dust control and nothing to protect their breathing, in poor air underground.

Breathing silica dust causes silicosis. The dust forms scar tissue in the lungs, and the scarred lungs cannot take in enough oxygen. Silicosis cannot be cured, and in bad cases it kills.

The counts. Members of a US House of Representatives committee held hearings in 1936 and put the deaths at 476. The scientist Martin Cherniack studied the records in 1986 and found as many as 764. Scientists at the National Institute for Occupational Safety and Health, the federal agency that studies workers' health, said close to 1,000 in 2002.

The workers and their families filed 538 lawsuits. They ended in a settlement of $200,000 in all, and single payments ran from $30 to $1,600.

Building the skyscrapers, the bridge and the dam

Up to 3,400 workers a day built the Empire State Building in New York, and by the official count five of them died. Lewis Hine photographed the ironworkers high on the open steel frame in 1931. Mohawk ironworkers from Kahnawake, a Mohawk community just south of Montreal in Canada, worked in the building crews.

The Golden Gate Bridge. The chief engineer, Joseph Strauss, required workers on the bridge in San Francisco to wear hard hats, and he spent about $130,000 on a safety net under it. The net caught 19 men who fell, and they called themselves the "Halfway to Hell Club." One of them, the ironworker Al Zampa, fell on October 19, 1936, broke four bones in his back and went back to ironwork.

Eleven men died building the bridge. Ten of them died on February 17, 1937, when a falling platform tore through the net.

Hoover Dam. The federal Bureau of Reclamation's official count is 96 deaths on the job. It leaves out 42 deaths that were recorded as pneumonia, a disease of the lungs. The true number of deaths at the dam is not known.

Managers at Six Companies, the company that hired the dam workers, had crews drive gasoline trucks inside the tunnels the men dug at the dam site. The trucks' exhaust filled the tunnels with carbon monoxide, a poison gas from engines. When tunnel workers fell very ill or died, company doctors said they had a form of pneumonia. A history of the dam kept by the University of Nevada, Las Vegas library describes this, and so does a book of dam workers' memories by the historians Andrew Dunar and Dennis McBride.

The lawsuit. Six workers sued Six Companies in 1933. They said the gas had poisoned them and left them with lasting illness, and each asked for $75,000 plus lost pay. Company officials hired a private detective to look into the men's pasts. More workers came forward and joined the case, and in the end the officials settled with 50 workers and paid them an amount they kept secret.

The right to organize and the sit-down strikes

Many businesses closed and millions of people lost their jobs during the Great Depression of the 1930s. At its worst, in 1933, about 12.8 million people, almost one worker in four, had no job. Federal officials hired people who were out of work through programs such as the Works Progress Administration and the Civilian Conservation Corps. Under the Wagner Act, which members of Congress passed in 1935, forming a union became a legal right for workers.

The CIO. That same year John L. Lewis and other union leaders formed a new group inside the American Federation of Labor, the AFL, a large group of unions of skilled workers, to organize whole factories. The AFL's leaders threw them out, and the group became the Congress of Industrial Organizations, the CIO. Its unions took in every worker in a mass-production plant, such as a car factory or a steel mill.

Flint. Autoworkers in Flint, Michigan, sat down inside General Motors' Fisher Body plants on December 30, 1936. In a sit-down strike, workers stop working but stay inside the plant, so the company cannot bring in other workers. They stayed for 44 days, until February 11, 1937.

The Women's Emergency Brigade, led by 23-year-old Genora Johnson, guarded the plants. When police fired tear gas at the men inside one of General Motors' plants, the women broke its windows. "We used our clubs to smash the windows out so the men inside could get some air," Johnson remembered. General Motors' managers agreed to recognize the United Auto Workers.

The Memorial Day Massacre, 1937

Several hundred people marched toward the Republic Steel plant on Sunday, May 30, 1937. The plant stood on Chicago's far Southeast Side, and the marchers were supporting the steelworkers' strike there. They were white, Black and Latino men, women and children, and about 200 of them were women. There, 264 Chicago police met them.

The police threw tear gas, fired their guns and beat the marchers with clubs. They killed ten people and shot 40, and more than 90 people were wounded in all. The Paramount company filmed it, and its officials at first held back the newsreel.

What the senators found. Senators on a committee led by Senator Robert La Follette Jr. looked into the killings. They found that the police had used "excessive force," far more force than was needed, and had made "baseless allegations," claims with nothing true behind them. The senators said the police "did not carry out an honest investigation." No record names any police officer who was ever punished.

BY 1937: Workers had a legal right to form unions, and the Flint sit-down had made General Motors bargain with the autoworkers, but no federal law yet set a lowest wage or kept children out of factories.

A. Philip Randolph

Who: A union leader who organized the Pullman porters and won the first contract between a major American company and a union led by Black workers.

When and where: Founded the Brotherhood of Sleeping Car Porters in August 1925. Contract signed August 25, 1937. Washington, 1941.

Movie: *10,000 Black Men Named George* (2002), a television drama with Andre Braugher as Randolph. Also *A. Philip Randolph: For Jobs and Freedom* (1996), a documentary shown on PBS.

Passengers called the porters "George," whatever their real names were.

A. Philip Randolph founded the Brotherhood of Sleeping Car Porters in August 1925. Porters worked on the Pullman Company's sleeping cars on trains, and all of them were Black men. Pullman's managers required them to serve passengers with a smile.

The name. After the Civil War, George Pullman, whose company ran the sleeping cars, set out to hire men who had once been enslaved to work as his porters. Passengers called the porters "George," after him. The Chicago History Museum says this carried on a practice from slavery, when an enslaved person was called by the owner's name.

The contract. The Pullman Company's officials signed a contract with the union on August 25, 1937, twelve years after it began. It was the first contract between a major American company and a union led by Black workers. The porters' working month had run to about 400 hours, and under the contract it fell sharply. The porters also got $1.25 million in raises.

Order 8802. Randolph threatened in 1941 to lead a march on Washington. On June 25, 1941, President Franklin Roosevelt signed Executive Order 8802, an order from the president. It stated there could be "no discrimination in the employment of workers in defense industries or government because of race, creed, color, or national origin."

> Under the order, defense companies, which made weapons and supplies for the military, and government offices could not refuse to hire people because of their race, religion or the country their families came from. Randolph directed the March on Washington for Jobs and Freedom in 1963.

A lowest wage for the whole country, 1938

President Roosevelt signed the Fair Labor Standards Act on June 25, 1938, and it took effect on October 24. The law set the first federal minimum wage at 25 cents an hour, rising to 40 cents by 1945, and a work week of 44 hours, falling to 40 hours by October 1940. It also banned most work by children under 16 in the jobs it covered, but it did not cover farm work.

The justices of the Supreme Court ruled the other way on child labor in 1941. They decided that Congress did have the power to limit child labor, under the 1938 law.

War work and the braceros

US officials started the bracero program, named for the Spanish word for a man who works with his arms, in 1942 to bring Mexican men to work on American farms and railroads during the war. Over 22 years, more than 4.5 million contracts were signed.

The missing pay. American employers held back 10 percent of every bracero's pay from 1942 to 1946, and many braceros said the holdbacks went on after that. The money went through American banks to banks in Mexico, for a savings fund the braceros were to receive when they went home. Most braceros never got that money, and the records of where it went do not add up. Mexican officials agreed in 2009 to pay former braceros or their families 38,000 pesos each.

Women in the war plants. Women took jobs in war plants in large numbers during World War II. They made up 27 percent of the workforce in 1940 and nearly 37 percent by 1945, when about 6.5 million more women were working. Before the war about 1 percent of aircraft workers were women. In 1943 more than 310,000 women built aircraft, about 65 percent of all the workers in that industry.

Naomi Parker Fraley

Who: A war worker in the machine shop of the Naval Air Station at Alameda, California, and the woman in a 1942 photograph linked to the "We Can Do It!" poster.

When and where: Alameda, California, 1942. Died January 20, 2018, at 96.

She drilled, patched airplane wings and riveted in a Navy machine shop.

Naomi Parker went to work at the Naval Air Station in Alameda, California, during World War II. She was assigned to its machine shop, where her jobs included drilling, patching airplane wings and riveting. In March 1942 a photographer took her picture at work on a turret lathe, a metal-cutting machine for making many copies of the same part.

The photograph and the poster. People later linked her photograph to a poster of a woman worker with the words "We Can Do It!", which became famous in the 1980s. For decades the woman in the photograph was said to be someone else. Then a scholar named James J. Kimble found the original print, and its caption named Naomi Parker. Nobody has proved that the poster's artist used this photograph.

The strike wave of 1945 and 1946

When the war ended, workers in steel, coal, railroads and car plants struck in greater numbers than ever before. There were 4,750 strikes involving 3.47 million workers in 1945. The Bureau of Labor Statistics, the federal office that counts jobs and workers, recorded 4,985 strikes involving 4.6 million workers in 1946, and no other year in its records has more.

BY 1950

If you remember one thing: In these fifty years workers won federal laws for a lowest wage, a 40-hour week, an end to most child labor and the right to form unions.

In 1900: Children worked in mills and mines, no federal law set a lowest wage, and 1,489 coal miners were killed at work.

In 1950: Federal law set a minimum wage and a 40-hour week, banned most work by children under 16, and made forming a union a legal right.

Next: After 1950 the share of workers in unions reached its peak and then fell, while new groups of workers organized.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 15 MINUTES

Union membership reached its peak in 1954, when about one American worker in three belonged to a union, and after that the share fell for the rest of the century. By 1956 more Americans worked in offices, stores and schools than with their hands in factories, mines and building trades. Factory jobs reached their highest number in June 1979 and then fell, and by 2000 about 13 percent of workers belonged to a union.

New groups of workers organized in these years: farmworkers, city garbage collectors, flight attendants and office clerks. The share of women with paid jobs rose from 34 percent in 1950 to 60 percent in 2000.

Unions at their peak, and the move to the office

In the peak year, 1954, 34.8 percent of Americans who worked for wages or salaries belonged to a union. Leaders of the AFL and the CIO joined their groups into one, the AFL-CIO, on December 5, 1955. It had about 15 million members, and George Meany was its first president.

Offices and stores. For the first time, by 1956, more Americans worked in white-collar jobs, in offices, stores and schools, than in blue-collar jobs, working with their hands in factories, mines and building trades.

Women went to work for pay in growing numbers. In 1950, 34 percent of women were in the labor force, which means they had a paid job or were looking for one, and by 2000 the figure was 60 percent. Over the same years, women's share of all workers rose from 30 percent to almost 47 percent.

The braceros at work

The bracero program went on after the war. Men who wanted contracts went through reception centers at the border, run by officials of the federal agency that handled immigration. One was the Rio Vista farm in El Paso, Texas, and men who went through it remembered "poor food, humiliating medical examinations, and the fumigation."

The spraying. Officials had the men strip off their clothes, line up and be sprayed with DDT, a white powder that kills insects. The officials believed that Mexican workers carried insects and germs.

Growers could hire braceros right at the border from 1948, which let them skip the cost of moving the men and avoid government checks. Braceros worked six or seven days a week on farms and railroads in 26 states. Some were paid little and lived in poor housing, and supervisors cursed at some of them and attacked some of them physically. They worked with pesticides, chemicals that kill insects and weeds, and researchers later found these chemicals very harmful to farmworkers.

The end of the program. Members of Congress agreed in late 1963 to keep the program going for only one more year. They did not renew it again, so the law behind it, Public Law 78, ran out on December 31, 1964, and the program ended. Labor unions had pushed to end it. By then growers were also using more farm machines, and the public had learned about the braceros' working conditions.

The Mohawk ironworkers

Ironworkers from the Mohawk communities of Kahnawake, just south of Montreal, and Akwesasne, which lies across the border of the United States and Canada on the St. Lawrence River, worked high steel from the 1920s onward. That means they built the steel frames of bridges and skyscrapers, walking on narrow beams far above the ground. They helped build the Empire State Building, the Chrysler Building and Rockefeller Center in New York, and later the World Trade Center. The 9/11 Memorial and Museum has an exhibit about them called "Skywalkers."

Cesar Chavez and Dolores Huerta

Who: Organizers who built the farmworkers' union, together with the Filipino American grape pickers led by Larry Itliong, who walked out first.

When and where: Delano, California, 1965 to 1975.

Movie: *Cesar Chavez* (2014), a drama directed by Diego Luna, with Michael Peña as Chavez. *Dolores* (2017), a documentary about Dolores Huerta directed by Peter Bratt.

About 1,500 grape pickers, most of them Filipino American, walked out first.

Cesar Chavez and Dolores Huerta led farmworkers to win the first union contracts in the California grape fields. The pickers who walked out of the vineyards around Delano, California, on September 8, 1965, belonged to the Agricultural Workers Organizing Committee, or AWOC, a union led by Larry Itliong. Eight days later, on September 16, members of Chavez and Huerta's National Farm Workers Association voted to join the strike.

The march. Strikers marched about 300 miles from Delano to Sacramento, the state capital, from March 17 to April 11, 1966, and about 10,000 supporters were with them when they arrived. The two unions joined together and became the union now called the United Farm

Workers. Chavez went without food for 25 days in February and March of 1968 as a call for nonviolence, and Senator Robert Kennedy was with him when he ended the fast on March 10.

The boycott. Across the country, people stopped buying California table grapes, in what was called the grape boycott. Twenty-six grape growers in Delano signed the first union contracts on July 29, 1970. Dolores Huerta negotiated the contracts, and she made up the union's slogan, "Sí, se puede," Spanish for "Yes, it can be done."

The federal labor law of 1935 left out farmworkers. California's lawmakers passed the Agricultural Labor Relations Act in 1975, the first law in the United States to give farmworkers the right to collective bargaining, bargaining as a union with their employers.

The accounts of 2026. The New York Times reported in March 2026 that Chavez had sexually abused two girls in the 1970s. The two women, Debra Rojas and Ana Murguia, are daughters of union organizers, and they said the abuse began when they were 12 and 13. Rojas said Chavez raped her in a motel room when she was 15, forcing her into sex.

Dolores Huerta then said publicly that Chavez had pressured her into sex once and had raped her in a car in 1966, and that she became pregnant both times. She said she had kept the secret for 60 years because she believed the truth would hurt the farmworker movement.

Chavez died in 1993, so no court has ever heard these accusations. Leaders of the United Farm Workers canceled the union's events for Cesar Chavez Day. On March 26, 2026, California's lawmakers voted to rename the state holiday Farmworkers Day, and Governor Gavin Newsom signed the change into law.

Pesticides in the fields

Farmworkers who worked in fields sprayed with pesticides had high rates of birth defects, cancer and burns. Cesar Chavez and Dolores Huerta campaigned to ban DDT on grape and lettuce ranches. In the 1980s, children in several towns of California's San Joaquin Valley had cancer at rates "800% above the norm."

Fired for marrying or turning 32

Managers at US airlines fired stewardesses who married or who turned 32, and at some airlines 35, into the late 1960s. They also weighed the women and checked that they wore girdles, tight undergarments.

Dusty Roads. Barbara "Dusty" Roads flew for American Airlines from 1950. She pressed members of Congress to change the rules and helped found the Association of Professional Flight Attendants, a union. The Civil Rights Act of 1964 banned treating workers unfairly in hiring and firing because they were women or men. Flight attendants brought cases under that law, and in those years airline managers dropped the age and marriage rules.

Memphis, 1968

Two Memphis garbage collectors, Echol Cole, 36, and Robert Walker, 30, took shelter from the rain on February 1, 1968, inside the compactor of their garbage truck, the part that crushes the garbage. The truck was not working properly, and the compactor crushed both men to death.

Eleven days later, 1,300 Black sanitation workers went on strike. They wanted the city's leaders to recognize their union, which means to accept it and bargain with it, and they wanted the work made safe and pay that was enough to live on. Many of them were paid so little that they were on welfare, government help for people without enough money, and hundreds relied on food stamps to feed their families. The strikers carried signs that read "I AM A MAN."

The mayor. The members of the City Council voted to recognize the union on February 22, and Mayor Henry Loeb rejected the vote. He said only the mayor could recognize a union, and he would not. He also kept the worn-out trucks on the streets and would not pay overtime for late-night shifts. The next day, police sprayed Mace, a stinging chemical, and fired tear gas at marchers walking to City Hall.

Violence broke out during a march on March 28, and some people looted downtown shops. Members of a young Black Power group called the Invaders were accused of starting it. Police followed the marchers back to Clayborn Temple, a church, fired tear gas inside and clubbed people lying on the floor.

Larry Payne. A Memphis police officer shot and killed Larry Payne, a 16-year-old student. The officer said Payne came at him with a knife. Most people who saw it said Payne came out of a basement with his hands up, and none of them saw a knife. The officer was never charged or disciplined.

Dr. Martin Luther King Jr. came to Memphis to support the strikers. James Earl Ray shot and killed him on the balcony of the Lorraine Motel on the evening of April 4, 1968. City negotiators agreed on April 16 that the City Council would recognize the union and raise wages. The union was AFSCME Local 1733, part of a national union of city and state workers.

Coal dust and a new safety law

An explosion in the Consol No. 9 mine at Farmington, West Virginia, killed 78 miners on November 20, 1968. Members of Congress then passed the Federal Coal Mine Health and Safety Act of 1969. Under it, federal inspectors had to inspect every underground coal mine four times a year, and they could shut a mine down if it put lives in danger. The law also provided federal payments for miners with black lung.

Black lung is a disease caused by breathing coal dust, which scars the lungs. In the worst form, masses of scar tissue leave the miner "gasping for every breath." More than 76,000 miners have died of black lung since 1968.

Fewer deaths. The number of coal miners killed at work fell from 311 in 1968 to 133 in 1980 and 38 in 2000. President Richard Nixon signed the Occupational Safety and Health Act, a federal law on safety at work, on December 29, 1970. That year about 14,000 workers were killed on the job, about 38 a day.

Office workers organize

Two office workers in Boston, Karen Nussbaum and Ellen Cassedy, founded 9to5 in 1973. Its members were women who did clerical work, such as typing, filing and answering phones. They organized over pay, promotion and harassment on the job, treatment meant to upset, shame or frighten a worker. In an early lawsuit, they won $1.5 million in back pay, the pay the women were owed, from publishing companies in Boston.

At the checkout. With barcode scanners, cashiers no longer had to key in each price by hand. The scanners also recorded, hour by hour, what sold, and managers later used those records to set workers' schedules. The first sale rung up by scanning a barcode came at a Marsh supermarket in Troy, Ohio, at 8:01 in the morning on June 26, 1974, when the cashier Sharon Buchanan scanned a ten-pack of Juicy Fruit gum.

The mills close

The managers of Youngstown Sheet and Tube suddenly shut the company's Campbell Works steel mill in Youngstown, Ohio, on September 19, 1977, and 5,000 workers lost their jobs. People in Youngstown remember that day as "Black Monday." Factory jobs in the United States reached their highest number, 19.6 million, in June 1979, and after that the number fell for forty years.

The share of wage and salary workers in unions was 20.1 percent in 1983 and about 13.4 percent by 2000.

Frank Lumpkin

Who: A Black steelworker who lost his job, his last paycheck and his pension when his Chicago steel mill shut down, and who fought for 17 years to win the money back.

When and where: Wisconsin Steel, South Deering, Chicago, from the 1950s. Born 1916. Died March 1, 2010.

In April 1980, the bank said their last paychecks were worthless.

Frank Lumpkin spent 17 years fighting to win back the pay and pensions, money paid to workers after they retire, that he and more than 3,000 other steelworkers lost when their mill closed in 1980.

A working life. Lumpkin was born in Georgia in 1916 to a family of sharecroppers, who farm land that belongs to someone else and pay the owner with part of the crop. At 6 he hauled stone for an uncle, and at 15 he left school to work on citrus plantations. He also boxed for money as "KO" Lumpkin.

He moved north to Buffalo, New York, in 1940, worked in construction, in an aircraft factory and at Bethlehem Steel, and served in the Merchant Marine, the crews of American cargo ships. In the 1950s he went to work at Wisconsin Steel in the South Deering neighborhood of Chicago, and he stayed about 30 years.

How the mill closed. Wisconsin Steel ran out of money in 1980. Its main buyer, International Harvester, had stopped buying its steel, and Harvester's managers then demanded the money they were owed.

Harvester had owned the mill from 1902 until 1977. That year its managers sold the mill to Envirodyne, a company with no experience in steel, and lent Envirodyne the money to buy it. When Harvester's own workers went on strike in late 1979, Harvester's managers bought none of the mill's steel while the strike lasted. That steel had been about 40 percent of what the mill made.

In March 1980 Harvester's managers moved to collect the money Envirodyne still owed for the mill. They also told Chase Manhattan Bank, which had lent Envirodyne money, that they would no longer back the bank's emergency loans to the mill. The *International Directory of Company Histories*, a reference book about companies, describes these steps.

The mill closed without warning in late March 1980, while Envirodyne owned it, and the accounts do not name the person who gave the order. The workers lost their jobs, their health benefits, their back pay and their pensions. In April, officers of Chase Manhattan Bank told them that their final paychecks were worthless.

Save Our Jobs. Lumpkin and his wife, Bea Lumpkin, started the Wisconsin Steel Save Our Jobs Committee. Its members were Black, Mexican American and other steelworkers and their families, and they helped families with food, medical care and utility bills. First the members tried to get the mill reopened. Then they fought to recover the lost pay and benefits, and they sued International Harvester and Envirodyne.

Six hundred former workers had died in the seven years since the closing, the committee counted in 1987. The lawyer Thomas Geoghegan won a settlement of $14.5 million from International Harvester in 1988. Over 17

> years the committee won about $17 million in all, about a third of the $40 million the workers were owed. Lumpkin also ran to become an Illinois state lawmaker and lost.

The air traffic controllers, 1981

About 13,000 federal air traffic controllers, the workers who guide airplanes in and out of airports, went on strike on August 3, 1981. Federal law did not allow federal workers to strike. President Ronald Reagan gave them 48 hours to return to work, and on August 5 he fired 11,345 of them. Federal officials also stopped accepting their union as the workers' union. The historian Joseph McCartin, who wrote a book about the strike, argues that the firings showed private employers that they could fight strikes and beat their unions.

El Monte, 1995

State and federal agents and El Monte police raided a fenced apartment complex in El Monte, near Los Angeles, on August 2, 1995. Inside they found 72 Thai garment workers who had been forced to sew clothes and were not allowed to leave, and they freed them. Suni Manasurangkun and her sons ran the operation. The recruiters who brought the workers from Thailand took their passports when they arrived, and the workers owed about $5,000 for the trip.

Behind the fence. A high fence topped with razor wire surrounded the buildings, and an armed guard sat in the courtyard. The workers sewed clothes 84 hours a week, from early morning to midnight, for about $1.60 an hour. One worker, Rotchana Cheunchujit, was paid five to seven cents a garment. The owners made the workers buy food at their own store at high prices, and some workers had not been outside the fence for as long as seven years.

One woman escaped through a second-story window. A letter in pencil with a hand-drawn map reached the authorities, and the raid followed. The clothes carried "Made in the USA" labels and were sold at chain stores including Miller's Outpost, Mervyn's and Montgomery Ward.

The guilty pleas. Suni Manasurangkun, three of her sons and three other people pleaded guilty to federal crimes in February 1996, which means they told the court they had broken federal law. Two of her sons fled and were never caught. Officials of the companies that had sold the clothes agreed in 1997 to pay the workers the wages they were owed, to end lawsuits against them. One account puts the total at more than $4.5 million, and another at $4 million.

After the case, and partly because of it, members of Congress passed a federal law in 2000 against human trafficking, the buying, selling and moving of people to force them to work.

BY 2000

If you remember one thing: The share of workers in unions peaked in 1954 and fell for the rest of the century, while farmworkers, garbage collectors, flight attendants and office clerks organized.

One number: 13.4 percent. That was the share of wage and salary workers in unions in 2000. In 1954 it had been 34.8 percent, more than two and a half times as high.

Next: After 2000 some workers found jobs through apps on their phones, and workers at Starbucks and Amazon voted in their first unions.

2000 to Today

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Millions of Americans now work for themselves and sell their work to companies, outside most labor laws, including the minimum wage law. About 11.9 million of them were independent contractors in their main job in July 2023, and some find their work through apps on their phones. One worker in ten belonged to a union in 2025, according to the federal Bureau of Labor Statistics. Workers at Starbucks coffee shops and at an Amazon warehouse voted in their first unions in 2021 and 2022.

Workers are still killed on the job. In 2024, 5,070 workers were killed at work, about 14 a day.

Unions and pay today

Union members were 10.0 percent of wage and salary workers in 2025, 14.7 million people, according to the Bureau of Labor Statistics. They were 32.9 percent of government workers and 5.9 percent of workers at private companies. The share had been 20.1 percent in 1983 and about one worker in three in 1954.

The federal minimum wage has been $7.25 an hour since July 24, 2009, and as of September 2026 members of Congress have not raised it. The first one, in 1938, was 25 cents.

Working through an app

Some independent contractors find work through apps that send them rides to drive or food to deliver, and the laws on the minimum wage and overtime pay do not cover them. Counting only people for whom it was their main job, independent contractors were 7.4 percent of all workers in 2023.

Gig work. Researchers at the Pew Research Center, a research group, counted more widely: anyone who had ever earned money through an online gig platform, an app or website that hands out short jobs one at a time. In their survey of August 2021, 16 percent of American adults had done so at least once, and 9 percent had done so in the past year.

Proposition 22. The managers of the app companies treat their drivers as independent contractors. California voters approved Proposition 22, a question put on the ballot, on November 3, 2020, with 58.63 percent of the vote. Under it, the app companies could keep treating drivers as independent contractors, but they had to pay drivers at least 120 percent of the minimum wage for the time they spent on a ride or a delivery, plus pay for each mile.

Raef Lawson

Who: A food-delivery driver for Grubhub whose court case asked whether an app driver is an employee.

When and where: Los Angeles, 2015 and 2016. A federal judge ruled on his case in 2018 and 2023, and approved a settlement in 2026.

After years in court, a judge awarded him $65.11.

Raef Lawson was an actor, writer and director in Los Angeles who was trying to make a career. He signed up in August 2015 to deliver food for Grubhub, a company whose app takes food orders from restaurants.

He picked blocks of two to five hours around mealtimes, week by week, and Grubhub paid him for the blocks he worked, 105 hours in his busiest month. It paid nothing toward his gas or his phone, so those costs came out of his own pocket. In 2016 Grubhub's managers classed all 4,000 of its California delivery drivers as independent contractors.

Let go. Grubhub's managers ended the company's agreement with him on February 15, 2016, saying he had not worked many of the blocks he signed up for. The judge at his trial also found that he sometimes "gamed" the app, which means he tricked it: he logged on late, and he switched his phone to airplane mode so no new orders came.

The case. Lawson sued Grubhub for the minimum wage, for overtime pay, which is extra pay for hours past 40 in a week, and for his costs, saying that he had been its employee. Judge Jacqueline Scott Corley ruled in 2018 that he had been an independent contractor. Three months later, the justices of California's Supreme Court set a new rule: a company must prove that a worker is not an employee.

A federal appeals court, a higher court that checks the rulings of lower courts, sent Lawson's case back to Judge Corley on September 20, 2021. She ruled on March 30, 2023, that under the new test Lawson had been

an employee. She wrote that "food delivery is not outside the usual course of" Grubhub's business, and she awarded him $65.11 in unpaid minimum wages for four months of work.

The settlement. Grubhub's officials agreed in 2026 to pay $24.75 million to California drivers to end their claims, and the judge approved the deal on July 30, 2026. About 60,000 drivers are in the settlement, and each gets at least $25. Judge Corley called it "my oldest case."

The pandemic, 2020

About 48.7 million employed people, 35.4 percent of them, worked from home in May 2020 because of the COVID-19 pandemic, according to the Bureau of Labor Statistics. By April 2022 that share had fallen to 7.7 percent, and many office workers kept working from home for part of the week. Workers in grocery stores, warehouses, delivery and health care, called essential workers, kept going to work in person.

Unions at Starbucks and Amazon

Baristas, the workers who make the coffee drinks, at a Starbucks on Elmwood Avenue in Buffalo, New York, voted 19 to 8 to form a union on December 9, 2021. The Washington Post and ABC News reported that it was the first union at one of the roughly 9,000 US stores that Starbucks owned itself. Workers at hundreds of other stores followed.

Workers at Amazon's JFK8 warehouse on Staten Island, New York, voted 2,654 to 2,131 for the Amazon Labor Union. Officials of the National Labor Relations Board, the federal agency that runs union elections, announced the result on April 1, 2022. It was the first union at an Amazon warehouse in the United States.

Chris Smalls

Who: An Amazon warehouse worker who was fired and then organized the first union at an Amazon warehouse in the United States.

When and where: The JFK8 warehouse, Staten Island, New York, 2020 to 2022.

Movie: *Union* (2024), a documentary by Brett Story and Stephen Maing about the union campaign at JFK8. It won a Special Jury Award at the Sundance Film Festival in 2024.

He built a union from a bus stop across the road.

Chris Smalls worked as a process assistant at Amazon's JFK8 warehouse on Staten Island. He led a walkout over safety from COVID-19 in March 2020, with workers leaving their jobs together in protest, and Amazon's managers fired him.

Smalls and other workers then started the Amazon Labor Union. At a bus stop across the road from the warehouse they set up a table with food and talked with workers as they came and went, thousands of conversations over many months. The workers won the vote in April 2022.

Where the new unions stand in 2026

Baristas at 600 Starbucks stores, about 12,000 workers, had joined the union Starbucks Workers United by June 2025, according to the union. They began a national strike on November 13, 2025, and it grew to about 4,500 workers in more than 130 cities. The union's leaders called it the longest strike in the company's history. As of September 2026, no Starbucks store's workers have voted to approve a first contract.

Members of the National Labor Relations Board ordered Amazon on April 2, 2026, to begin talks with the union at JFK8 on a contract for pay and working conditions. They found that Amazon had "illegally and willfully" ignored the union, according to a Teamsters union press release. As of September 2026, no contract has been reported.

Injuries in the warehouses

Staff members working for Senator Bernie Sanders, who led a US Senate committee on labor, published a report on Amazon warehouses in December 2024. It was a report by his own staff, not by the whole committee, and they interviewed 135 workers for it.

What they found. In each of the past seven years, the staff found, Amazon workers had been nearly twice as likely to be injured as other warehouse workers.

In 10- to 12-hour shifts, workers repeat the same movements "hundreds and thousands of times," the report states. A computer tracks how fast each worker goes, and workers who fall behind get penalties from the computer system automatically, which can end in firing.

Children in the slaughterhouses

Managers at Packers Sanitation Services, a cleaning company based in Wisconsin, put 102 children aged 13 to 17 to work on night shifts, officials of the US Department of Labor reported in February 2023. The children worked at 13 meatpacking plants in 8 states, where animals are killed and cut up for meat, according to reporting by KCUR, a public radio station in Kansas City.

They cleaned back saws, brisket saws and head splitters, machines that cut up animals, using chemicals strong enough to burn skin, and at least three of them were injured. The JBS plant in Grand Island, Nebraska, had 27 of the children, and the Cargill plant in Dodge City, Kansas, had 26.

The fine. Packers Sanitation Services paid $1.5 million, the most the law allowed. Jessica Looman of the Labor Department's Wage and Hour Division called the violations "systemic," which means built into the way the company worked, and "a corporate-wide failure."

On farms, federal law still allows children to work at younger ages than in other jobs. Outside school hours, a child under 12 may work on a farm owned or run by a parent, or, with a parent's permission, on a small farm that does not have to pay the minimum wage. Children of 12 and 13 may work with a parent's permission or on a farm where a parent works, and any child 14 or older may do farm jobs outside school hours.

Heat, floods and mines

Sixty-eight crop workers died of heat from 1992 to 2006, according to a 2008 report by the Centers for Disease Control and Prevention. Their death rate from heat was 0.39 for every 100,000 workers, against 0.02 for all civilian workers, nearly 20 times as high. A 56-year-old tobacco worker in North Carolina died in July 2005 with a body temperature of 108 degrees. He was in the United States on a temporary farmworker visa, and the report does not name him.

Across all jobs, 345 workers died of heat from 2011 to 2019, 43 of them in 2019, according to the Bureau of Labor Statistics. Officials of the Occupational Safety and Health Administration, the federal job-safety agency, proposed a rule on August 30, 2024, to protect workers from heat, a requirement that employers would have to obey by law. The officials held public hearings from June 16 to July 2, 2025. As of September 2026 there is no final federal rule on heat at work.

Erwin, Tennessee. Floods from Hurricane Helene reached the Impact Plastics plant in Erwin, Tennessee, on September 27, 2024, and five employees and a contractor who cleaned the offices drowned. One worker who survived and the families of some who died said no one told the workers they could leave until the flood had covered the road and the plant had lost power.

Inspectors from Tennessee's job-safety agency found that the company had "exercised reasonable diligence," which means it had taken reasonable care. They gave the company no citations, the official notices for breaking safety rules.

The local chief prosecutor, District Attorney General Steven Finney, looked into whether the workers had been told they could not leave, or that they would be fired if they left. He found "no evidence" of either and closed the case with no charges in July 2025. The family of Johnny Peterson, one of the men who died, filed a lawsuit.

Upper Big Branch. An explosion of gas and coal dust killed 29 miners at the Upper Big Branch mine in West Virginia on April 5, 2010. The mine belonged to Massey Energy. Its chief executive, Don Blankenship, was found guilty in December 2015 of conspiring, plotting with others, to break federal mine safety laws. Judge Irene Berger gave him the longest sentence the law allowed on April 6, 2016, one year in prison and a fine of $250,000. He was not convicted of causing the deaths.

The count. Injuries at work killed 5,070 workers in 2024, according to the Bureau of Labor Statistics, one worker every 104 minutes, about 14 a day. Road crashes and other transportation accidents killed 1,937 of them. In 1970 about 38 workers a day had been killed on the job.

Wage theft and the tomato fields

Some employers do not pay workers all the money the law says they have earned, and this is called wage theft. Researchers at the Economic Policy Institute, a research group that sides with labor unions, studied the ten states with the most people in 2017. They found that 2.4 million workers there were paid less than the minimum wage, and together they lost $8 billion a year. A worker who worked all year lost about $3,300, nearly a quarter of what they had earned.

The Fair Food Program. Tomato pickers in Immokalee, Florida, formed the Coalition of Immokalee Workers in 1993, and the coalition's Fair Food Program has run since 2011. Big buyers such as McDonald's and Walmart pay a little extra for each pound of tomatoes and buy only from growers who follow a code of conduct written by the workers. The code sets a lowest pay, requires shade and water and bans harassment. Checkers visit the farms, and workers can call a complaint line.

More and more farmworkers come to the United States on temporary H-2A visas, which allow them to do farm work for a season. About 48,000 H-2A jobs were approved in fiscal year 2005, the government's budget year, and nearly 400,000 in fiscal year 2025, more than eight times as many. More than 90 percent of the visas go to Mexican workers.

THE THREE QUESTIONS, ANSWERED

1. **Which workers were not free to leave their work?** Native people forced to work under Spanish rule, indentured servants bound for years, enslaved people, prisoners leased to Southern mines and planters, and the Thai garment workers held behind a fence in El Monte until 1995. Virginia judges sentenced John Punch in 1640 to serve for the rest of his life.
2. **What happened when workers went on strike?** Some won, like the Philadelphia printers in 1786 and the Flint autoworkers in 1937. Many times police, hired guards and soldiers broke the strike, and at Lattimer in 1897, Ludlow in 1914 and Chicago in 1937 they killed strikers, their families and people marching with them.
3. **What made work less deadly?** Laws written after workers died: more than one way out of every Pennsylvania mine after the Avondale fire, safer couplers after thousands of brakemen died, new factory laws in New York after the Triangle fire, and the 1969 coal mine law after Farmington. Deaths on the job fell from about 38 a day in 1970 to about 14 a day in 2024.

What changed: Work moved from households, forced labor and bound servants to jobs for wages, and wage workers won a shorter week, a lowest wage and safety laws, though their share in unions has fallen since 1954 to one in ten.

FIVE DATES TO KEEP

July 9, 1640: Virginia judges sentence John Punch to serve his master for life.

June 2, 1786: Philadelphia's printers begin the earliest proven strike of workers in one trade in the United States.

March 25, 1911: The Triangle fire kills 146 workers in a New York factory where a stairway door was locked.

June 25, 1938: President Roosevelt signs the law that sets the first federal minimum wage, 25 cents an hour.

April 1, 2022: Workers at an Amazon warehouse on Staten Island win the first union at an Amazon warehouse in the country.

NOT KNOWN: No one has ever proved who threw the bomb at Haymarket Square in 1886.

Where to go: Hopewell Furnace National Historic Site, in Pennsylvania, is the National Park Service site of the ironworks where more than 100 woodcutters and about 20 colliers worked in the 1830s.

Next: Chapter 15, Food and Farming. It asks what Americans ate and how they grew it, and the braceros and farmworkers of this chapter appear there too.

Chapter 15: Food and Farming

CHAPTER 15 OF 37 · FOOD AND FARMING · 10 ERAS · ABOUT 75 MINUTES

American food began with crops that Native farmers bred, and since then much of it has been grown on taken land by people who were enslaved or paid little.

On Cape Cod in November 1620, men from the *Mayflower* found baskets of corn buried in the ground. Nauset farmers had buried the corn to plant the next spring. The Englishmen dug it up and named the spot Cornhill. They wrote that they would "satisfy them for their corn," meaning they would pay the owners later. It was the first corn the Plymouth colonists ate. Native farmers had spent thousands of years turning a wild grass into that corn.

This chapter follows American food from those fields to the drive-in and the lab. Watch two things as you go. The first is whose land and whose work the food came from: Native farmers, enslaved Africans in the rice and cotton fields, families who farmed for a share of the crop, Mexican farmworkers, and today's crop pickers. The second is how far the food travels. When the country began, about nine of every ten working people were farmers. Then came tin cans, pond ice, cold railroad cars, frozen dinners and fast food. In 2022 the country had fewer than 2 million farms for the first time since before the Civil War.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did Native farmers turn a wild grass into corn?
2. Who did the hardest farm work, and what did they get for it?
3. How did Americans come to eat food grown far away by people they never met?

Five minutes? Read "Who got the cotton money?" in 1900 to 1950.

Can skip: "Farming began in the East too", "Why turn corn into whiskey?", "The potato blight", "Organic farming rules become law". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 5 MINUTES

Native farmers turned a wild grass into corn, and by 1500 corn fed whole towns. Farming here was already thousands of years old. People in each region also ate what grew or lived best around them: bison on the Plains, salmon in the Northwest, wild rice around the Great Lakes.

Europeans later carried corn, beans and squash from these fields across the ocean.

How corn was made

Native farmers made corn out of a wild grass called teosinte. A teosinte seed head holds only a few hard kernels. Farmers began changing it about 9,000 years ago in the Balsas River valley of southern Mexico. Each year they took seed only from the plants they chose and planted it again, so the crop changed a little at a time. After thousands of years of this breeding, the grass had become corn.

Scientists have confirmed the date and the place in two ways. Ancient corn dug from the ground shows it, and so do the genes of corn and teosinte, the tiny instructions inside every living thing that parents pass on to their young.

Corn is so changed from the wild grass that it cannot reseed itself. It grows only where people plant it. Native farmers grew it a little farther north at a time, over thousands of years, until it reached what is now the United States.

The Three Sisters

Native farmers planted corn, beans and squash together in the same field, and each crop helped the other two grow. The three crops are called the Three Sisters.

The corn grew tall, and the bean vines climbed the corn stalks. Bean roots hold tiny living things called bacteria. These bacteria take nitrogen, a gas in the air, and put it into the soil. Plants need nitrogen to grow, so the beans fed the soil that all three plants shared. The squash spread wide leaves over the ground, and their shade kept water in the soil and blocked the sunlight that weeds needed.

Corn and beans eaten in the same meal also feed people better than either food does alone. Together they give the body a fuller mix of protein, the part of food the body uses to build muscle.

Farming began in the East too

Native people in the eastern part of North America started farming on their own, long before corn reached them. They took wild plants and bred them into crops. About 5,000 years ago they had tamed a kind of squash.

They also grew six plants for their seeds: sunflower, goosefoot, sumpweed, maygrass, little barley and erect knotweed. Sunflower and sumpweed seeds are rich in oil. The other four have starchy seeds. Maygrass and little barley are grasses whose grain can be ground into flour, and people cooked and ate goosefoot's small seeds.

Scientists call this group of crops the Eastern Agricultural Complex. Eastern North America is one of the places in the world where people started farming on their own, without learning it from anyone else.

Foods by region

Cahokia was a Native town near where St. Louis is today. Around the year 1100, between about 10,000 and 20,000 people lived there, and corn farming fed them.

People in other regions lived mainly on other foods. On the Great Plains, hunters killed bison. By a widely used estimate, about 30 million bison still lived there in the early 1800s. Along the Northwest coast and its rivers, salmon was the main food. People on the coasts gathered shellfish. In the Northeast, people collected sap from maple trees.

Around the Great Lakes, people harvested manoomin, a wild rice that grows in shallow water, from canoes. In Ojibwe, the word means "the good berry." The Ojibwe are one of the Anishinaabe peoples, a family of related nations that also includes the Odawa and the Potawatomi. In an Anishinaabe prophecy, a teaching about what was to come, the people were led to the place where they would find "the food that grows on water." Manoomin is still a central food of Ojibwe life.

No farmer from before 1500 is known by name. Because corn cannot reseed itself, every generation of Native farmers had to save the seed and plant it again. Seed keepers in Native nations still grow these crops.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

European ships began carrying crops and farm animals across the Atlantic in both directions in the 1500s. Corn, potatoes and turkeys went east to Europe, and wheat, cattle and pigs came west. Africans, most of them carried across in chains by slave traders, brought crops of their own. The historian Alfred W. Crosby named this movement the Columbian Exchange, in his 1972 book of that name. Native people here still grew and ate their own foods, and in Florida a Spanish army lived on the corn it took from them.

What crossed the ocean

Europeans carried American crops and animals home to Europe. These included corn, potatoes, tomatoes, beans, squash, peanuts, peppers, vanilla, turkeys and cacao, the bean that chocolate is made from. Corn and potatoes later became main foods across Europe, Africa and Asia.

Europeans brought wheat, rice, sugarcane, citrus fruit (the family of fruit that includes oranges and lemons), and honeybees to the Americas on their ships. They also brought cattle, pigs, sheep, goats and chickens. Columbus brought horses on his second voyage, in 1493. Spanish colonists founded St. Augustine, in Florida, in 1565, and began a colony in New Mexico in 1598. In both places they planted wheat and citrus and raised cattle.

The word exchange can sound like a friendly trade, but Europeans took American crops home from lands they were invading. Much of what arrived was grown to feed the colonists. Before long, planters were forcing enslaved people to grow sugarcane on plantations.

Enslaved Africans brought their own crops and the skill to grow them, among them okra, watermelon, yams, sorghum, black-eyed peas and African rice.

De Soto's army and Apalachee corn

The Spanish soldier Hernando de Soto landed near Tampa Bay, in Florida, in May 1539, with an army of more than 600 people. He carried no food supplies except a small herd of pigs. Men who marched with him wrote accounts of the trip, and those accounts say the herd began with about 13 pigs. By 1542 it had grown to about 700. The army rarely killed the pigs for food, and pigs that escaped ran wild across the Southeast.

De Soto's army moved from town to town and took the corn that Native villagers had stored, and the soldiers ate it. Without that corn, the villagers were often at risk of starving. De Soto also seized dozens of Native men, women and children. His soldiers forced them to carry the army's equipment and supplies and to do the work of the camp.

The captives. Rodrigo Ranjel, de Soto's secretary, kept a diary of the march. An account written from it states that the Spaniards took the men "to have more slaves and servants." Some of the captives died and others ran away, so the soldiers seized more. The same account records that the soldiers wanted the women "for their lewdness and lust," which means they wanted them for sex. The women were prisoners and could not refuse. The soldiers raped them, forcing them to have sex.

De Soto's army seized Anhaica, the main town of the Apalachee, in October 1539. Tallahassee, Florida, stands there today. The army lived in the town until March 1540 and took its stored food, which was enough to feed 600 men and 220 horses for five months.

Archaeologists, scientists who study the past by digging up what people left behind, have found burned corncobs at the site. The Apalachee fought back with quick raids and ambushes, and they went after the Spaniards' horses.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 6 MINUTES

Native corn kept the first English colonists at Jamestown and Plymouth alive in the 1600s. The colonists bought some of that corn and grew some of it with Native teaching, and they took some by force. In Virginia, planters then grew tobacco in place of food.

Jamestown: hunger, and corn the English took by force

The colonists at Jamestown, in Virginia, planned to live partly on food they traded from the Powhatan people and partly on supply ships from England. In the autumn of 1609, Powhatan leaders stopped trading food with them, and Powhatan warriors surrounded the fort. The winter of 1609 to 1610 is called the Starving Time. About 500 colonists were alive when it began. By spring, about 61 were still alive.

The colonists ate their horses, dogs and rats, and they ate leather. Some ate the bodies of people who had died. Archaeologists found the bones of a girl whom researchers call Jane. The bones show that colonists ate her body after she died. Scientists at the Smithsonian announced the finding in 2013.

Percy's raid. English soldiers also took corn by force and destroyed Native fields. On August 10, 1610, George Percy led English soldiers against the Paspahegh town near Jamestown. They burned the houses, cut down the cornfields and took corn. Percy wrote that they killed 15 or 16 people. Other accounts put the number of villagers killed as high as 75.

The soldiers captured the wife and two children of Wowinchopunck, the Paspahegh weroance, or leader. When Percy hesitated, his men complained. Percy then agreed to kill the children. His soldiers threw the children overboard from a boat and shot them in the head in the water.

That night at Jamestown, Captain Davis told Percy that Lord De La Warr, the colony's governor, wanted the children's mother burned alive. Captain Davis and two soldiers took her into the woods and killed her with a sword. Percy wrote that he did not believe the order really came from the governor.

Ten years of raids. Powhatan warriors attacked the English settlements in March 1622. After that, colonial soldiers marched out almost every summer for ten years to attack Powhatan towns and cornfields. At a peace meeting in 1623, colonists poisoned the Powhatans' wine, shot them, and then destroyed their cornfields.

Why Virginia planters grew tobacco instead of food

Tobacco sold for so much money that many Virginia planters grew it in place of food crops. John Rolfe grew the colony's first successful tobacco crop in 1612, from a sweet kind of tobacco.

Members of Virginia's assembly, the colony's lawmaking body, passed laws again and again to make planters grow food. Under a tobacco law of 1629 and 1630, a family could grow 2,000 tobacco plants for each family member and no more. The lawmakers set the limit partly so that families would still plant food crops.

Plymouth: corn dug up, then corn planted

The first corn the Plymouth colonists ate was corn they took. In November 1620, men from the *Mayflower* went exploring on Cape Cod. There they found baskets of corn that the local people had buried in the ground. The owners were saving it as seed for the next spring.

The Englishmen dug up about ten bushels of the corn (a bushel holds about 8 gallons) and a bag of beans. They wrote that they planned to "satisfy them for their corn," which meant to pay the owners later. They named the place Cornhill.

The same group of men dug up graves nearby. They opened bundles that held the bones of a man and a small child. They wrote that they "brought sundry of the prettiest things away," meaning they took several of the objects buried with the dead.

The owners of the corn were Nauset people. In the summer of 1621, the colonists met one of them and promised to pay him. They offered to bring the owners as much corn as they had taken. Later in 1621 the colonist Edward

Winslow wrote that the owners had been given "full content," meaning they were fully satisfied. His letter does not say what they were given.

The colonists built their town on the land of Patuxet, a Wampanoag town. An epidemic, a disease that spreads quickly through many people, had killed most of the people of Patuxet between 1616 and 1619. One reason the colonists gave for settling there was the "good corn-ground ready to our hands." The Patuxet people had cleared those fields.

The colonists planted about 20 acres of corn in the spring of 1621. The corn did well, and their first harvest, that autumn, fed them through their second winter.

Squanto (Tisquantum)

Who: A Patuxet man who taught the Plymouth colonists to plant corn, after an English captain had kidnapped him.

When and where: Patuxet, where the colonists built Plymouth, in what is now Massachusetts. Kidnapped 1614, home 1619, taught the colonists in the spring of 1621, died 1622.

Movie: *Squanto: A Warrior's Tale* (1994), a feature film that mixes his real story with invented events.

He taught the colonists to bury fish with their corn.

Squanto, also called Tisquantum, showed the Plymouth colonists how to plant corn in the spring of 1621. He also interpreted between the colonists and the Wampanoag, which means he put each side's words into the other's language.

An English captain named Thomas Hunt kidnapped Squanto in 1614 and took him to Spain to be sold as a slave. Squanto reached England and learned English there. In 1619 he came home with an English expedition and found his town of Patuxet empty. An epidemic had killed most of its people.

The colonist Edward Winslow described the planting in a letter of December 1621. Following "the manner of the Indians," he wrote, "we manured our ground with herrings, or rather shads," two kinds of fish. The colonists buried the fish in the soil with the corn seed to feed the crops, and Squanto taught them to do it.

Scholars have argued for decades about where Squanto learned the method. In 1975 Lynn Ceci, an anthropologist (a scientist who studies how people live), argued in the journal *Science* that burying fish with corn was a European method. She found little evidence that many Native farmers used fish this way, and she argued that Squanto probably learned it during his years among Europeans. Other scholars defended the older account, the one Winslow gave.

KNOWN:	Squanto showed the colonists how to bury fish with their corn.
NOT KNOWN:	Where he learned the method, from Native farmers or among Europeans, is still not settled.

What the colonists ate

The English colonists learned corn dishes from Native cooks, and Americans went on eating them for centuries. One was hominy, which is dried corn cooked in water. For samp, cooks pounded corn into coarse meal and boiled it into a mush, and the name comes from the Narragansett word *nasàump*. Johnnycakes were flat cakes baked from cornmeal. Along with their corn, the colonists ate beans, and once their hogs had multiplied, they ate pork and drank cider made from apples.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 6 MINUTES

Planters in the Carolina Lowcountry, the flat, wet land along the coast of South Carolina, became some of the richest people in the colonies by forcing enslaved Africans to grow rice. The enslaved Africans built the rice fields and did the work in them. Planters paid more for captives from West Africa who already knew how to grow rice.

Farther north, farmers in Pennsylvania, New York and New Jersey grew more wheat than they could eat and shipped flour to other places. New Hampshire's town histories say that settlers there planted the first potato patches in North America that lasted.

Carolina rice and African skill

English settlers first brought a small amount of rice to Charles Town, now Charleston, in 1670. No record shows that anything grew from it. By July 1699, samples of Carolina rice were on sale in London to test whether buyers wanted it. The first crops grew on dry, high ground. By the 1720s, planters had moved their rice into freshwater swamps inland, and they had enslaved workers build dams to hold the water in ponds.

The rice boom. Carolina planters shipped out about 269,000 pounds of rice a year from 1698 to 1702. Forty years later, from 1738 to 1742, they shipped out more than 30 million pounds a year, more than a hundred times as much. One kind of Carolina rice later became known as Carolina Gold.

Planters put out advertisements asking for captives from the part of West Africa known as the Rice Coast. That coast included Senegambia, the land along the Senegal and Gambia rivers, and Sierra Leone. Planters paid more for people from there because those people knew how to grow rice.

African methods. Records from Carolina fields show African methods at work. Enslaved workers built earth banks around fields beside the tidal rivers so they could flood and drain them. They sowed seed by pressing it into the mud with the heel and toe. They separated the grain from its husks by tossing it in coiled baskets called fanner baskets.

What is certain, and what is argued. It is certain that planters paid more for captives from the Rice Coast and that enslaved Africans used African methods in the Carolina fields. Scholars argue about how much the whole rice business owed to that African knowledge. In 2001 the scholar Judith Carney argued that African knowledge was its base, and in 2005 three historians answered that she had made the African share too large.

Most scholars now hold a middle view: planters built the rice business with their money and by force, and they also depended on African skill. Records from the colony also show that in its early years, more captives came from Angola than from any other place, and Angola is not on the Rice Coast.

Whose land the rice grew on

The Kiawah, Etiwan and Stono people lived along the Ashley, Cooper and Stono rivers. They were still there in 1670, when English colonists settled among them at Charleston Harbor.

English colonists captured Kussoe people who fought back in 1671, and Stono people in 1674, and sold them into slavery in the Caribbean. The Kiawah left the Ashley River in the early 1680s and moved to the island now named for them. By then no nation of the Lowcountry had more than fifty bowmen, and the Stono had sixteen hunters. In the spring of 1684, leaders of eight nations signed away their land claims to the English who ran the colony.

English colonists in Carolina made their biggest profits before 1720 from two kinds of goods they shipped out. One was ship supplies, such as tar made from pine trees. The other was Native people, whom they sold as slaves. In 1770 a colonist named William Bull wrote of the Native nations along the coast that "there remains now, except a few Catawbas, nothing of them but their names, within three hundred miles of our sea coast."

The rice workers, unnamed in the records

Planters' records list the enslaved rice workers by first name, age and price. They say almost nothing else about them. No single rice grower from these years can be followed through life in the records.

Planters gave each enslaved worker a set task every day. The usual task was to work a quarter of an acre of land. A worker who finished the task had the rest of the day. Workers labored in direct sun, standing in water or mud, among mosquitoes, horseflies, snakes and alligators.

Enslaved Africans dug the banks and canals of the rice fields. Nic Butler, a historian at the Charleston County Public Library, has written that the earth they moved across the Lowcountry is about as much as all the great pyramids of Egypt put together.

More enslaved people died in the Lowcountry than were born there until the 1760s. Many of the dead were children. Planters kept buying new captives to replace them, and by 1708 more Black people than white people lived in South Carolina.

The slave ships. Slave traders rarely landed more than 300 enslaved Africans a year at Charles Town in 1710. By 1720 they landed more than 1,000 a year, and by 1770 more than 3,000. More than four of every ten Africans brought to North America in the slave trade came in through Charleston Harbor.

Eliza Lucas Pinckney

Who: A planter's daughter who ran her family's South Carolina plantations as a teenager and made the colony's first successful indigo dye.

When and where: Wappoo plantation, near Charleston, South Carolina. Indigo experiments from 1739, first successful dye in 1744.

She was about 16 when she took charge of three plantations.

Eliza Lucas produced South Carolina's first successful dye from indigo, a plant that colors cloth blue, in 1744. Indigo is a farm crop but not a food.

Her father, George Lucas, went back to the island of Antigua, in the Caribbean, in 1739 to serve as its lieutenant governor, the official just below the governor. Eliza was about 16. She took charge of Wappoo plantation and of the 20 enslaved people who worked it. She also oversaw the family's two other plantations, which were much larger.

That same year she began experimenting with indigo seed. Turning the plant into dye was skilled work done by experts. She first worked with an indigo maker from the island of Montserrat. Her best results came with an indigo maker of African descent whom her father hired from the French islands of the Caribbean.

She used her 1744 crop to make seed and gave the seed to other planters. Planters in South Carolina shipped out about 5,000 pounds of indigo in the shipping year 1745 to 1746. By 1748 they shipped out 130,000 pounds. She later married and took the name Pinckney.

Wheat, potatoes and everyday meals

Pennsylvania, New York and New Jersey were called the breadbasket colonies. Farmers there grew more wheat than their families needed. Merchants shipped the extra wheat as flour out of the ports of Philadelphia and New York.

Settlers known as the Scots-Irish, families of Scottish background who had lived in the north of Ireland, planted potatoes at Londonderry, New Hampshire, in 1719. New Hampshire's histories count their patches as the first lasting potato fields in North America. The potato first grew as a crop in the Andes mountains of South America. People carried it to Europe, and settlers later brought it back across the ocean to North America.

Most colonial families ate corn, pork and bread, and drank cider. Rich families also bought tea, sugar and wine from overseas, and they ate bread made of wheat.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 5 MINUTES

About nine of every ten working people in the United States were farmers when the country began, by a widely used estimate from the US Department of Agriculture. The first census, a count of the people, was taken in 1790. It counted about 4 million people.

Thomas Jefferson wrote that farmers were God's chosen people, while people he enslaved farmed his land. In 1796 Amelia Simmons became the first American to write a cookbook for American kitchens, and some of its recipes used American cornmeal.

Matthew Patten

Who: A New Hampshire farmer, carpenter and judge whose diary records what his family grew, caught and ate.

When and where: Bedford, New Hampshire, on the Merrimack River. His surviving diary runs from 1754 to 1788.

In ten days of fishing, he sent 354 fish home.

Matthew Patten kept a diary that shows how one farm family in New Hampshire fed itself. He headed each little booklet of it "A Day Book continued." His entries for the year 1767 record the family's food through the seasons.

Patten was a joiner, a carpenter who does the finer woodwork in a house. He also farmed, worked as a judge, and surveyed land, which means he measured it and marked its borders.

What the family ate. The Pattens grew corn, rye, wheat, peas, potatoes, barley and turnips. They butchered their own hogs, calves and sheep, and they caught shad and salmon in the Merrimack River. They bought tea, sugar, molasses and rum. The family paid and was paid in bushels of grain as often as in money.

Winter. Neighbors paid Patten in corn for surveying their land and for writing deeds, the papers that prove who owns a piece of land. In January he took corn and rye to a neighbor's mill to be ground into meal and flour, and on January 30 he butchered the family's hogs.

Spring. The family's potatoes had spent the winter buried in a pit. Patten opened it on March 4 and wrote that they were "sound and good." In April and May the family sowed peas and planted potatoes and corn, and "the boys" spread manure from the barn on the field behind it. They also sowed hemp and flax, plants whose fibers are made into rope and cloth.

Fishing. In May Patten helped haul a net for shad, a fish the family took from the Merrimack each spring. From May 28 to June 6 he fished at Amoskeag Falls and sent 354 fish home.

The name Amoskeag comes from a Pennacook word that means roughly "good fishing place." Long before Bedford was a town, Pennacook people strung large nets across the river at the falls and caught sturgeon, salmon and a fish called the alewife. Bedford lies in Pennacook country.

Summer. In June and July Patten caught salmon and split them with two neighbors. In August he reaped his rye, and later that month he mowed his meadow for hay.

Patten died on August 27, 1795, in one of the town's fields. He had carried dinner out to the men who were mowing.

Jefferson's farmers and Monticello's rations

Thomas Jefferson believed that family farms made free citizens. He wrote about the idea in his book *Notes on the State of Virginia*, first printed in 1785: "Those who labour in the earth are the chosen people of God, if ever he had a chosen people." For about a hundred years afterward, American leaders wrote land laws meant to put land in the hands of family farmers.

People Jefferson enslaved did the farm work at Monticello, his plantation in Virginia. He enslaved more than 600 people during his adult life. In the years Jefferson had tobacco grown at Monticello, enslaved children picked worms off the tobacco plants.

The rations. Each week, Jefferson gave each enslaved worker at Monticello a set amount of food, called a ration: a peck of cornmeal (about 8 quarts), half a pound of pork or pickled beef, and four salted fish. Nursing mothers received extra, and workers got whiskey at harvest. Researchers who studied the Monticello records found that these rations were not enough food for a person to live on. So in their little free time, enslaved people hunted, fished, trapped, kept gardens and raised animals.

Enslaved families also sold chickens and vegetables from their own gardens and flocks to the Jefferson family. Jefferson's granddaughter Anne Cary Randolph wrote down these purchases in the household accounts. Most of the

sellers she recorded were enslaved people who worked in the house, many of them members of the Hemings family.

Why turn corn into whiskey?

Farmers in the backcountry, the land far from the coast beyond the older farms and towns, could not easily haul wagons of corn over the long roads to buyers. So they turned their extra corn into whiskey, which they could carry to market in jugs.

Amelia Simmons

Who: The author of *American Cookery*, the first cookbook written by an American for American kitchens.
When and where: Hartford, Connecticut, 1796.

On her title page she called herself "an American orphan."

Amelia Simmons wrote *American Cookery*, the first cookbook written by an American for American kitchens. It was printed in Hartford, Connecticut, in 1796. Beyond those words on her title page, nothing about her life is known.

Her book contains the first known printed recipes that used American cornmeal. Among them were johnnycake, "Indian slapjack" and Indian pudding. Colonists called corn "Indian corn" and cornmeal "Indian meal," so the word Indian in these names meant that the dishes were made with cornmeal.

It also contains the first printed recipe that used pearlash, a salt made from wood ashes, to make dough rise. Bakers used pearlash before baking powder existed. The book includes some of the earliest printed uses of the words "cookey" (cookie) and "slaw" (a salad of shredded cabbage), which came into American English from Dutch settlers in America.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 7 MINUTES

Sealed in tin cans or kept cold with ice cut from frozen ponds, food could now be sold far from the farm where it grew, to people the grower never met. In the Deep South, planters, the owners of large farms worked by enslaved people, spread cotton fields over land that federal troops had forced Native nations to leave. On the prairie of the Midwest, farmers used a new steel plow to turn thick grassland into fields of wheat and corn.

Food in cans, and ice from ponds

A Frenchman named Nicolas Appert proved in 1809 that food sealed in a container and then heated would keep without spoiling. The method is called canning.

Americans soon took it up. William Underwood opened a cannery in Boston in 1820 or 1821. In New York, Thomas Kensett canned seafood, fruit and vegetables, and in 1825 he received an American patent for tin cans, a government paper that gave him the sole right to make and sell them for a set number of years.

Other merchants sold ice. Workers cut blocks of ice from frozen ponds in New England in winter, including Walden Pond. Merchants shipped the ice to cities. People used it to keep food, fresh fish and drinks cold. Families kept ice in a box called an icebox, which became a common piece of kitchen furniture.

Frederic Tudor

Who: A Boston merchant, called the Ice King, who built a business shipping New England pond ice to warm places around the world.

When and where: Boston. First cargo of ice to Martinique in 1806. Ice to India in 1833. Died 1864.

His ship carried 180 tons of pond ice to India.

Frederic Tudor built a business selling New England pond ice in hot countries far away. His first cargo of ice left Boston in 1806, bound for the Caribbean island of Martinique. A Boston newspaper, the *Boston Gazette*, printed: "No joke. A vessel has cleared at the Custom House for Martinique with a cargo of ice." The ship had permission from the port's officials to sail, and the paper hoped the trip would "not prove a slippery speculation," a speculation being a risky business deal.

Much of the ice melted during the three-week voyage. Tudor sold what was left. He lost about $4,500 on the trip, though some accounts say $4,000. He kept at the business for years, and in 1812 and 1813 he spent time in debtors' prison, a jail for people who could not pay what they owed.

His ship the *Tuscany* sailed from Boston for Calcutta, in India, on May 12, 1833. It carried 180 tons of ice cut the winter before. The ship reached India in September, about four months later. About 100 tons of the ice were still frozen, and about 80 tons had melted on the way.

Cotton fields on Native farms

Across the Deep South, planters spread cotton so widely that they pushed food crops to the edges of their land. Many planters bought corn and pork from Midwestern farms instead of growing enough food on their own plantations.

Who did the work. Enslaved people did the work in the cotton fields. The planters held them as property and forced them to work without pay. They worked in groups called gangs, from sunup to sundown, six days a week.

Solomon Northup, a free Black man who was kidnapped and sold into slavery, was held in Louisiana from 1841 to 1853, nearly ten of those years on Edwin Epps's cotton plantation. He wrote that "an ordinary day's work is two hundred pounds." Each night the cotton was weighed, and pickers who brought in less were whipped, usually fifty strokes. Epps did some of the whipping himself.

Whose land it was. The new cotton fields were the farms and hunting grounds of the Cherokee, Muscogee (Creek), Choctaw, Chickasaw and Seminole nations. In 1830 President Andrew Jackson signed the Indian Removal Act. Under it, federal troops forced these nations off their land and moved them west. Chapter 2, Native Nations, tells the removals in full.

The Trail of Tears. The Cherokee were one of those nations, and their removal of 1838 and 1839 is called the Trail of Tears. Soldiers under General Winfield Scott forced more than 16,000 Cherokee from their homes and held them in crowded, dirty camps, with too little food and water and no shelter from the weather.

Measles, whooping cough and dysentery, a disease of the gut that causes bloody diarrhea, spread through the camps. More than 4,000 Cherokee died of disease, hunger and cold in the camps and on the march west, and one scholar has estimated that twice as many died.

Breaking the prairie

Farmers moving onto the prairie of Illinois found thick, sticky sod, the top layer of soil, held together by the tangled roots of grass. Prairie sod stuck to iron plows and stopped them.

John Deere, a blacksmith in Grand Detour, Illinois, made a plow of polished steel in 1837. The sod slid off its smooth blade instead of sticking, so the plow could cut through the prairie. By 1846 Deere was selling nearly a thousand plows a year.

Cyrus McCormick had shown his reaper, a machine that cuts grain, in Virginia in 1831, six years before Deere made his plow. Before it, farmers cut wheat by hand, with tools such as the sickle, a curved blade. A reaper cut grain many times faster than a person with hand tools, so a family could plant far more wheat. Within one generation, farmers had turned the prairie into fields of wheat and corn.

Deere's plows broke land that Native nations had lived on. Early settlers at Grand Detour saw signs of a Ho-Chunk village inside the bend of the river.

War on the Mississippi. US soldiers and militia, citizens called up to fight, went to war in 1832 against Sauk and Meskwaki people led by Black Hawk. On August 1 and 2, at Bad Axe on the Mississippi River, Black Hawk's band of perhaps 500 men, women and children tried to cross the river. Soldiers on the army steamboat *Warrior* fired on them.

About 1,300 soldiers and militia under General Henry Atkinson attacked the people still on the east bank. The soldiers shot some of the Sauk and Meskwaki before they could reach the far shore, and others drowned trying to get across. Thomas Ford, a governor of Illinois, later estimated that 150 were killed and as many drowned. Others have disputed his count.

The treaties. With two treaties, written agreements between nations, US officials got Native land in northern Illinois and southern Wisconsin in 1832 and 1833, a few years before Deere made his plow. Farmers moved onto that land and plowed it.

Ho-Chunk leaders signed the first treaty with General Winfield Scott and the governor of Illinois in September 1832. They gave up their lands in southern Wisconsin and northern Illinois in return for yearly payments. At a treaty meeting in Chicago in 1833, Potawatomi leaders sold the last of their land in northern Illinois and all of their land in southeastern Wisconsin. In return they got land farther west, in Iowa, and they had to leave Wisconsin by 1838.

The potato blight

A potato disease called late blight broke out near the ports of Philadelphia and New York in 1843. It is caused by a water mold, a tiny living thing that destroys potato plants. The blight spread through potato fields in the eastern United States. American farmers lost potato crops, but they had other food to eat.

The blight reached Europe by 1845. In Ireland, poor families depended on potatoes and had no other food, and the blight destroyed their crop. About 1 million people died in Ireland's Great Famine, a time of so little food that people starved, from 1845 to 1852.

1850 to 1900

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Families claimed millions of acres of free farmland under the Homestead Act, land that US officials and soldiers had taken from Native nations. By 1900, growing, packing and shipping food had become a big business, from Texas ranches to Chicago meatpacking plants to the cities of the East. Cowhands drove millions of cattle north from Texas, and meatpackers in Chicago shipped their beef east in cold railroad cars.

Farmers formed groups of their own, starting in 1867, to buy supplies together and to speak up against the rates railroad companies charged them.

Homestead farms on taken land

The Homestead Act became law on May 20, 1862. Under it, a person could claim 160 acres of public land for free. To keep the land, the person had to live on it and farm it for five years. Women could claim land. After 1868 Black Americans could claim land too. Most Native people could not.

Over the years the law was in force, people filed about 4 million claims. About 1.6 million of them finished the five years and received a land patent, the government paper that made the land theirs. Together they received about 270 million acres, close to one tenth of all the land in the United States.

All of this land had belonged to Native nations. US officials and soldiers had taken it by treaty, by purchase and by war.

The Dawes Act. Members of Congress passed the Dawes Act in 1887. Under it, federal officials broke up land that each Native nation held together into separate plots for single families. Officials then sold the land they called surplus, meaning left over, and much of it went to white farmers. Native nations held about 138 million acres in 1887 and about 48 million by 1934. Through the plots and the sales, federal officials had taken about 90 million acres from them, two of every three acres they had held.

Oklahoma. Federal officials opened land in Oklahoma to settlers in 1889, two years after the Dawes Act. US officials had taken that land from Native nations, and settlers raced in to claim it.

Uriah and Mattie Oblinger

Who: A homesteading family whose letters describe the food and work of a Nebraska farm claim.

When and where: Fillmore County, Nebraska. Uriah filed his claim in 1872. Mattie and their baby, Ella, arrived in May 1873. The family letters run from 1862 to 1911.

Uriah built the family's house out of blocks of prairie sod.

The letters of Uriah and Mattie Oblinger describe what one homestead family in Nebraska grew and ate in the 1870s. The Library of Congress and the Nebraska State Historical Society keep the family's letters, and the Library of Congress collection holds 318 of them, written from 1862 to 1911.

Uriah went ahead from Indiana to Nebraska in the fall of 1872. He filed a claim about eleven miles west of the town of Geneva, in Fillmore County, broke the prairie sod and built a house out of blocks of sod. On December 1 he wrote about the other young families starting out around him: "We will all be poor here together."

The Otoe-Missouria and the Pawnee had signed over most of their land in this part of Nebraska in treaties of 1854 and 1857. Homesteaders began filing claims in the county in 1866.

Mattie and their baby daughter, Ella, joined Uriah in May 1873. On June 16 Mattie wrote that Uriah had planted sod corn, corn set straight into newly plowed prairie sod. She spelled words her own way: "Uriah has 23 acres of sod corn planted it looks real well."

An Iowa settler, L. S. Coffin, described in 1902 how sod corn was planted. A man struck an axe deep into the sod, and a boy or girl dropped three or four kernels into the slot and stamped it shut with one foot.

> Mattie's garden had "Squashes and Cucumbers & Mellons & Beans comeing on," and tomato plants too. She bought milk and butter from a neighbor, Mrs. Furgison, and paid twelve cents a pound for the butter. "I like the sod house the best," she wrote, and "every lick we strike is for our selves and not half for some one else."
>
> Over the next years the family lived through blizzards, drought, swarms of grasshoppers and low prices for their crops. In February 1880 Mattie died giving birth, and her newborn son died with her. She was 36. She left three daughters, Ella, Stella and Maggie, and Uriah moved the family to Minnesota that year.

Sharecropping: growing a crop someone else sold

After the Civil War, most freedpeople, the men, women and children who had been freed from slavery, had no land of their own. Many became sharecroppers, families who farmed someone else's land for a share of the crop. A landowner gave a sharecropping family food, seed, tools and a cabin on credit, which meant the family would pay for them later. This was called the furnish. The family usually worked about twenty acres.

At harvest the landowner sold the crop and took out the cost of the furnish plus interest, an extra charge for paying later. Then the landowner paid the family its share, usually half of the crop's value or less.

The debt. The landowner held a lien on the crop, a legal claim to it until the debt was paid, and kept the account books. Most sharecropping families stayed in debt to the landowner year after year. They grew food and cotton but did not control the money from selling it.

The bison killed, the cattle driven north

About 30 million bison lived on the Great Plains in the early 1800s, and by 1900 fewer than 1,000 were left. Hunters killed them to sell their hides, and the herds also shrank from railroad building, disease and drought. US Army commanders encouraged the killing, because the herds were the food supply of the Plains nations.

The people of the Lakota leader Sitting Bull were starving by 1881, because the bison herds were nearly gone. They surrendered at Fort Buford in July 1881.

Ranchers put cattle on the grasslands where the bison had lived. Between 1866 and 1884, cowhands drove about 5.2 million Texas cattle north to railroad towns. In 1867 Joseph McCoy built a market for cattle at Abilene, Kansas, where the railroad line reached, and herds came up to it along the Chisholm Trail. In Texas, as many as one cowhand in four was Black. Many others were Mexican or Mexican American.

Fences. The long drives ended when barbed wire fences and quarantine lines closed the open range, the unfenced grassland where cattle could graze and travel freely. Joseph Glidden received a patent for barbed wire on November 24, 1874. Under a Kansas law of 1885 and 1886, passed to keep out a cattle disease called Texas fever, Texas cattle could not cross the quarantine line into Kansas. Herds could no longer use the trails that crossed such lines.

Charles Goodnight

Who: A Texas cattleman who invented the chuckwagon, the kitchen on wheels that fed cowhands on cattle drives.

When and where: Texas, 1866.

His chuckwagon's pantry box dropped flat to make a table.

Charles Goodnight invented the chuckwagon in 1866. A cattle drive crossed open grassland for months, and its crew had to eat every day. Goodnight took a leftover Army wagon and built a hinged pantry box onto it. Part of the box dropped down flat to make a table where the cook could work.

The chuckwagon carried the food for the trail, such as beans, bacon, biscuits and coffee. On a drive, the cook ranked second only to the trail boss, the man in charge of the whole drive. Some characters in the novel *Lonesome Dove* are partly based on Goodnight.

Beef from the Plains to a city plate

The Union Stock Yards opened in Chicago on December 25, 1865. Their pens, where cattle, hogs and sheep were held until they were sold, became the largest market for farm animals in the world.

Gustavus Swift was a meatpacker, a businessman who ran a plant where animals were killed and cut up into meat for sale. He shipped beef east from Chicago already butchered, instead of shipping live animals.

An engineer named Andrew Chase designed a railroad car for Swift in 1878 that was cooled with ice, and beef packed in it stayed cold all the way to the cities of the East. A steer, one of the male cattle raised for beef, could now be raised in Texas, killed in Chicago and eaten in New York.

Farmers join together

Oliver Hudson Kelley, a Minnesota farmer and a clerk in the federal Department of Agriculture, founded the Patrons of Husbandry in 1867 (husbandry is an old word for farming). People knew it better as the Grange. It was the first large farmers' group with members across the whole country. Grange members met in their own halls and bought supplies together to get lower prices, and they spoke up together against the rates the railroad companies charged to carry their crops.

Colleges for farming. Members of Congress had already passed the Morrill Act in 1862. Under it, federal officials gave each state 30,000 acres of public land for each of the state's members of Congress, land that had been taken from Native nations. State officials sold the land and used the money to start colleges that taught farming and the skills of working with machines. These schools are called land-grant colleges.

Experiment stations. Under the Hatch Act of 1887, each state's land-grant college received $15,000 a year at first to run a farm experiment station. The act was named for William Hatch, a member of the US House of Representatives from Missouri who led the House committee on farming. At these stations, which today run in all 50 states, scientists studied soil, seed and farm animals and sent their results to farmers.

1900 to 1950

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Federal inspectors went to work inside the meatpacking plants in 1906, and in the years that followed federal officials took on many new jobs in how Americans got their food. In the 1930s, officials paid farmers to grow less, bought up extra food and gave it to hungry families. During World War II they limited how much sugar, meat and coffee each person could buy. In 1946 President Harry Truman signed a law that made school lunch a permanent federal program.

On the farms, tractors began to take the place of horses, and new kinds of seed corn and chicken farming began.

What was in the meat: the 1906 food laws

President Theodore Roosevelt signed two food laws on the same day, June 30, 1906. Under the Pure Food and Drug Act, no one could sell food or medicine from one state to another if it was mislabeled or adulterated, meaning made impure or mixed with something that should not be in it.

Scientists at the Bureau of Chemistry enforced this law at first, and since 1930 the Food and Drug Administration, or FDA, has done that work.

The second law was the Meat Inspection Act. Under it, federal inspectors worked inside the packing plants and checked the meat. Members of Congress passed both laws within a few months after a novel about Chicago's meatpacking plants came out.

Upton Sinclair

Who: A novelist whose book about the Chicago meatpacking plants came out months before the 1906 food-safety laws.

When and where: Chicago, 1904. His novel *The Jungle* came out as a book in February 1906.

He wrote about the workers, and readers noticed the meat.

Upton Sinclair wrote *The Jungle* to show how immigrant workers in Chicago's meatpacking plants lived. In 1904 he spent about seven weeks working undercover in the plants. His novel was printed in parts in 1905 and came out as a book in February 1906.

Readers fixed instead on what Sinclair described going into the meat. In the novel, spoiled ham and moldy sausage were dosed with chemicals and ground up again, along with meat that had fallen on the dirty floor. The packers put out poisoned bread for the rats in the storerooms, and then "rats, bread, and meat would go into the hoppers together." The hoppers fed the machines that ground the sausage.

Within months, Roosevelt signed the two food laws of June 30, 1906. Later that year Sinclair wrote about how readers had taken his book: "I aimed at the public's heart, and by accident I hit it in the stomach."

George Washington Carver

Who: A scientist who taught poor Southern farmers how to rebuild worn-out cotton land and grow their own food.

When and where: Born enslaved near Diamond, Missouri, about 1864. Taught at Tuskegee Institute, Alabama, from 1896 until his death in 1943.

Movie: *George Washington Carver: An Uncommon Life* (Iowa PBS, 2018), a documentary.

His most famous booklet listed 105 ways to prepare peanuts.

George Washington Carver taught Southern farmers to plant crops that rebuilt their soil. Moses and Susan Carver enslaved his mother, Mary, and so they claimed him as their property too.

When he was a few months old, a band of men kidnapped him and his mother from the farm to sell them. A neighbor got the baby back. His mother was not brought back. Moses and Susan Carver raised him and his older brother, Jim.

He was the first Black student at Iowa State, and later he taught there. He took charge of farming studies at Tuskegee Institute in Alabama in October 1896 and stayed until he died in 1943.

Year after year of cotton had worn out the soil on many Southern farms. Carver taught crop rotation, which means changing the crop planted in a field from one year to the next. A farmer planted peanuts, cowpeas (a kind of bean) or sweet potatoes, which built the soil back up, and then went back to cotton, which wore it down.

Carver wrote 44 practical bulletins, small printed booklets, and gave them to farmers for free. He also sent out the Jesup wagon, a wagon pulled by a mule that was used to show farmers new ways of farming. His best-known bulletin came out in 1916 and was reprinted for decades. It was called *How to Grow the Peanut and 105 Ways of Preparing It for Human Consumption*. In his laboratory he made hundreds of test products from peanuts and sweet potatoes.

> Many people believe that Carver invented peanut butter. He did not. The Aztecs of Mexico made peanut paste centuries earlier. Marcellus Gilmore Edson received a patent, the sole right to make and sell an invention, for a peanut paste in Canada in 1884, and John Harvey Kellogg received one in 1895.

Pellagra: a disease from a corn diet

About 3 million Americans got pellagra, a disease caused by a diet missing a vitamin called niacin, between 1900 and 1940. About 100,000 of them died of it. Those hit hardest were poor farmers, sharecroppers (families who farmed someone else's land for a share of the crop) and cotton-mill workers in the South. They lived mostly on cornmeal, molasses and dried pork, with few fresh vegetables and little meat or milk.

Many mill owners paid workers in scrip, a kind of paper money that could be spent only at the company's own store. Those stores sold mostly cornmeal, dried meat and molasses.

A person with pellagra got skin rashes, diarrhea and dementia, which means confusion and loss of memory. The disease could end in death.

The prison test. Most scientists of the time thought pellagra was an infection, spread by germs. Joseph Goldberger, a doctor with the US Public Health Service, thought the diet caused it, and he tested the diet on prisoners in 1915, with the agreement of Mississippi's governor, Earl Brewer.

At the Rankin State Prison Farm, Goldberger fed eleven volunteer prisoners the corn-based meals poor Southerners ate, starting on April 19. They got biscuits, mush, grits, gravy, cornbread, sweet potatoes, cane syrup and coffee. In return, the men were promised pardons, meaning they would be let out of prison.

About half of the eleven men got pellagra, five or six depending on the account, and the others showed some signs of it. The men had skin rashes and stomach illness, and the men who got pellagra recovered. Goldberger had shown that he could cause pellagra by keeping fresh food from people.

In 1916 Goldberger injected himself, his wife and other volunteers with blood from pellagra patients, to show that the disease did not spread from person to person.

A scientist in Wisconsin found in 1937 that the missing nutrient was niacin, also called vitamin B3. Federal and state laws passed during World War II required that niacin be added to bread.

Sugar and pineapple from Hawaii

Sugar plantations made up most of the business of the Hawaiian Islands. The planters brought in contract workers from overseas, bound to work for a set number of years, many of them from Japan and, after 1908, from the Philippines. Around 1899 a Japanese contract worker earned about $13 a month, and his food and the cost of his trip both ways came out of that pay.

James Dole started the Hawaiian Pineapple Company in 1901, and workers at his company canned pineapple in huge amounts for kitchens on the mainland. US officials had taken over the islands three years before, in 1898.

The takeover. Hawaii was a kingdom with its own queen until 1893. On January 17 of that year, a group of mostly American businessmen moved to overthrow Queen Liliuokalani. Some of the group's members were sons of partners in the companies that controlled more than half of Hawaii's sugar crop, and the group pushed for the United States to take over the islands.

John L. Stevens, the top US official in Hawaii, had 162 US sailors and Marines brought ashore from the warship USS *Boston*, and the queen gave up her power under protest.

Native Hawaiians signed 21,269 names on petitions in 1897, asking that Hawaii not be joined to the United States. On July 7, 1898, President William McKinley signed the Newlands Resolution, a measure to annex Hawaii, which means to take it over and make it part of the country.

Tractors, hybrid corn and chickens

Henry Ford's company began selling a cheap tractor called the Fordson in 1917. Even so, American farms in 1920 had 246,000 tractors and about 26 million horses and mules, and animals still did most of the pulling. The International Harvester company brought out the Farmall in 1924. It could also cultivate row crops, meaning it could work the soil between rows of growing plants, the last big farm job that still needed horses.

A farm horse ate crops too. It took about five acres to grow the feed for one horse. When a farmer sold a horse or stopped using it, the farmer could plant those acres with food for people.

Henry A. Wallace founded Pioneer Hi-Bred in 1926. It was the first company that sold hybrid seed corn, seed that comes from crossing two different parent plants.

BY 1930: Federal inspectors were working inside the meatpacking plants, tractors were starting to replace farm horses, and farmers could buy hybrid seed corn.

Cecile Steele

Who: A Delaware farmer who began the business of raising chickens for meat after a hatchery sent her 500 chicks by mistake.

When and where: Ocean View, Delaware, 1923. She lived from 1900 to 1940.

She ordered 50 chicks, and the hatchery sent her 500.

Cecile Long Steele had ordered the chicks in 1923 for her farm in Ocean View, Delaware. She raised all 500 indoors and sold them at 16 weeks old for 62 cents a pound.

She ordered 1,000 chicks on purpose in 1924, and by 1926 she was raising 10,000. By 1928 about 500 farms on the Delmarva Peninsula, the land where Delaware, Maryland and Virginia meet, were raising broilers, young chickens raised for their meat.

Chicken later became the meat Americans eat most, by weight. By the count of the National Chicken Council, a group that speaks for chicken companies, each American ate about 32 pounds of chicken in 1965. Its figures for 2025 put each American at about 103 pounds of chicken, 59 pounds of beef and 49 pounds of pork.

Dust, pigs and food stamps in the 1930s

Farmers on the southern Great Plains had plowed up the grassland to plant wheat during World War I, when wheat sold for high prices. A drought began in 1931, and the wheat died. The bare, dry soil blew away in dust storms. The storm of April 14, 1935, is called Black Sunday. Crops failed year after year. Farm families lived on vegetables from their gardens, canned food, food handed out by federal officials, and jackrabbits.

No one kept an official count of the people the dust killed. In Kansas alone, dozens of people died of dust pneumonia in 1935. Dust pneumonia happened when dust settled deep in a person's lungs. It stopped the cilia, the tiny hairs that sweep the airways clean, from moving. The person had fever, chest pain, coughing and trouble breathing.

Paid to grow less. Under the Agricultural Adjustment Act of May 1933, federal officials paid farmers to grow less, so that crop prices would rise. That year farmers plowed under about 10 million acres of cotton that was already growing, turning the young plants back into the ground. Federal officials bought about 6 million pigs and had them killed. At the same time, hungry people across the country stood in breadlines, long lines for free food.

Judges threw out the farm law in 1936, and members of Congress wrote a new one in 1938. Federal payments to farmers have continued ever since.

Many Americans were angry that federal officials were having food destroyed while people went hungry. Federal officials set up the Federal Surplus Relief Corporation to give extra farm food to families on relief, meaning families getting government help.

Food stamps. The first Food Stamp Program began in Rochester, New York, on May 16, 1939, and Mabel McFiggin was the first buyer. People bought orange stamps that paid for food. With them came free blue stamps, which could buy only foods that federal officials listed as surplus, meaning extra.

Who got the cotton money?

The landowners got most of it. Under the cotton contracts of 1934 and 1935, federal officials paid landowners to take land out of cotton. The sharecroppers who farmed that land, families who farmed someone else's land for a share of the crop, were supposed to get a part of the payment, but under the contracts their share was only about 11 percent.

A landowner could keep the whole payment by evicting the sharecroppers, meaning forcing them off the land, and hiring day laborers, workers paid one day at a time, instead. When planters cut their cotton land by 40 or 50 percent, they cut the number of families working it by the same share.

The lawyers. Some lawyers at the Agricultural Adjustment Administration, the federal office that ran the farm payment program, tried to stop the evictions. Led by Jerome Frank, they decided that landowners had to keep the same families on their land. In February 1935 the office's head, Chester Davis, had Secretary of Agriculture Henry Wallace fire them.

The union. Seven Black men and eleven white men met at the Sunnyside School in Poinsett County, Arkansas, in July 1934 to fight the evictions. They started a union, a group of workers who join together to demand better pay and treatment, and called it the Southern Tenant Farmers' Union (a tenant farmer works land that someone else owns).

Planters and lawmen answered with force. Sheriff J. D. Dubard of Poinsett County arrested a white organizer, Ward Rodgers, at the end of a union meeting in Marked Tree in January 1935, and Rodgers was held in jail there.

About forty armed planters broke up a union crowd at Birdsong on March 15, 1935, along with lawmen and the planters' riding bosses, the men on horseback who watched over the field workers. Later that month a mob attacked the home of the union's lawyer, Cornelius Tyree Carpenter.

At a union meeting in Gilmore, four armed white men broke in. They ordered the Black members to leave and told them they would be lynched, meaning killed by a mob, if they were caught at another union meeting. Many members were beaten and threatened with guns. The Encyclopedia of Arkansas account of the union does not name the men who did it.

Ration books and victory gardens

Federal officials rationed food during World War II, which means they limited how much of it each person could buy. Every person got a ration book of stamps, even babies. Sugar was the first food rationed, in May 1942. Coffee followed that November. In March 1943 meat, cheese, fats and canned foods were added. Sugar was also the last food to come off the list, in June 1947.

Federal officials allowed each person 26 pounds of sugar a year, about 8 ounces a week. In 1945 they cut the amount to as little as 4.5 ounces a week. Every month each person got 64 red points for meat and fats and 48 blue points for processed foods, such as canned goods. A shopper had to hand over the right points, along with money, to buy these foods.

Victory gardens. Americans planted about 20 million victory gardens, home vegetable gardens grown to help feed the country during the war. By 1943 and 1944, these gardens grew about 40 percent of the country's fresh vegetables, roughly 9 to 10 million tons. Accounts differ on which of the two years was the peak.

Frozen food was already on sale by the war years. Clarence Birdseye's quick-frozen foods first went on sale on March 6, 1930, in 18 stores in Springfield, Massachusetts.

Adelaide Wisdom Benjamin

Who: A girl whose family grew a victory garden and shopped with ration stamps during World War II.

When and where: New Orleans, Louisiana, and El Paso, Texas, 1941 to 1945. She was born in New Orleans in 1932.

She put 10 cents of her 25-cent allowance into war stamps.

Adelaide Wisdom grew up in New Orleans. During the war her father tried to join the Army and the Navy but did not meet their requirements. The Marines accepted him. After boot camp he was posted to El Paso, Texas, as a Marine recruiter, and the family moved with him.

At the store, the family could buy sugar, butter and some vegetables only with the right ration stamps. Beef and lamb were hard to find. With so little to serve, the family could not have guests over for meals.

The family grew a victory garden in New Orleans and in El Paso. "We grew mirlitons, and string beans and tomatoes in our backyard," she said in a 2013 oral history for the National WWII Museum (a mirliton is a pale green squash). The crops grew well in New Orleans but not in El Paso. She collected scrap with her brother, such as old coat hangers and bicycle tires, and she put 10 cents of her 25-cent weekly allowance into war savings stamps.

When Germany surrendered in 1945, she bought a chocolate soda, a comic book and bubble gum. She later became a lawyer and practiced law with her husband, Eddie Benjamin, for 55 years.

Farms taken from Japanese Americans

President Franklin Roosevelt signed Executive Order 9066, an order from the president that has the force of law, on February 19, 1942. In March and April, General John L. DeWitt ordered about 120,000 people whose families came from Japan forced out of their homes on the West Coast. Under his orders, US officials sent both citizens and noncitizens away and locked them up in prison camps behind barbed wire.

Many of these families were farmers. By 1940 Japanese American farmers and their children grew almost 40 percent of California's vegetables. When they were forced out, many had to leave their orchards and fields with crops ready to harvest.

Farms lost. The Farm Security Administration was the federal office put in charge of their farms. Its officials handed more than 1,000 of the farms to new farmers in March 1942 alone.

Farming in the camps. The prisoners farmed inside the camps too. At Tule Lake, California, they harvested almost 30 crops, among them potatoes, rutabagas and daikon radishes. Most of the field workers there were paid $12 a month, a quarter of what farmworkers outside earned, and camp officials sold more than 2 million pounds of what they grew.

Braceros in the fields

US officials started the bracero program in 1942 to bring Mexican workers, called braceros, into the country for farm and railroad work. Over 22 years, workers signed more than 4.5 million contracts. Braceros picked much of the country's fruit and vegetables until the program ended in 1964.

US employers also held back 10 percent of each bracero's pay, and many braceros never got that money back. Chapter 14, Work and Workers, tells the story of the missing pay.

Why school lunch became a law

Many young men called up to serve in World War II failed their physical exams because of malnutrition, not having had enough of the right food. General Lewis Hershey ran the draft, the system that called men into the military. After the war he told members of Congress that malnutrition put the country's safety at risk.

Towns and charities had already run school meal programs for decades, and in the 1930s federal officials gave these programs extra food they had bought from farmers. On June 4, 1946, President Harry Truman signed the National School Lunch Act. The law states that Congress passed it "as a measure of national security," a way to keep the country safe, and to protect the health of the nation's children. Under it, federal money for school lunches became permanent.

BY 1950

If you remember one thing: Between 1900 and 1950, federal officials began inspecting Americans' meat, paying farmers to grow less, rationing food and paying for school lunches.

- June 30, 1906: Theodore Roosevelt signs the Pure Food and Drug Act and the Meat Inspection Act.
- May 16, 1939: The first food stamps go on sale in Rochester, New York.
- June 4, 1946: Harry Truman signs the National School Lunch Act.

Next: After 1950 farms grew fewer and bigger, and Americans started buying frozen TV dinners and fast-food hamburgers.

1950 to 2000

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Fewer, bigger farms grew more food on each acre after 1950. Farmers used tractors, hybrid seed, fertilizer and pesticides. Swanson sold 25 million frozen TV dinners in 1955, and fast-food stands served hamburgers in seconds. In the 1980s the owners of about 300,000 farms could not pay their loans, after land prices fell and interest rates rose. Meanwhile some farmers and cooks began to grow food by organic methods, without man-made chemical fertilizers and pesticides, and to serve food from nearby farms.

Tractors outnumber horses, and farms grow bigger

The 1954 Census of Agriculture, the federal government's count of the country's farms, was the first in which American farms had more tractors than horses and mules. Farms had about 4.6 million tractors that year and about 4 million horses and mules. In 1920 there had been about 25 to 26 million horses and mules.

The number of farms fell too. In 1935 the country had 6.8 million farms, the most it ever had. After that the number kept dropping, toward 2 million. The farms that were left were bigger. Many of them grew only a few kinds of crops and relied on chemical fertilizers and pesticides.

Nearly all the corn in the country grew from hybrid seed by the 1950s. For about a hundred years before that, farmers had grown roughly 25 to 30 bushels of corn on an acre, a bushel being about 8 gallons. By the 1970s farmers were growing more than 100 bushels an acre, more than three times as much.

Fast food and the TV dinner

Richard and Maurice McDonald opened a drive-in restaurant in San Bernardino, California, in 1940. In 1948 they rebuilt it around what they called the Speedee Service System, with a menu cut down to 15-cent hamburgers, french fries and milkshakes. The carhops, the servers who had carried food out to people's cars, lost their jobs. Workers made the food on a kind of assembly line, where each worker did one job over and over, and customers got their food in seconds.

The Swanson company put a frozen turkey dinner in an aluminum tray on sale in 1953 and called it a TV dinner. It cost about 98 cents. The company sold more than 10 million in 1954 and 25 million the next year. According to company stories, Swanson had too many turkeys left over after Thanksgiving.

Whose idea it was. Who thought of the TV dinner is disputed. For decades a salesman named Gerry Thomas said the idea was his. The company's owners were the brothers Gilbert and Clarke Swanson. Their relatives, and a Swanson scientist named Betty Cronin, said the brothers came up with it at the company and that Thomas took the credit. Frozen meals had been sold before either claim: in 1945 the Maxson company made frozen "Strato-Plates" to serve on airplanes.

Ray Kroc

Who: A milkshake-machine salesman who became the McDonald brothers' franchise agent and then bought them out.

When and where: Visited San Bernardino, California, in 1954. Opened his first McDonald's in Des Plaines, Illinois, on April 15, 1955. Bought out the brothers in 1961.

Movie: *The Founder* (2016), a drama based on his story and the brothers'.

He bought out the McDonald brothers for $2.7 million in 1961.

Ray Kroc sold milkshake machines. In 1954 he went to see the McDonald brothers' stand in San Bernardino. He became the brothers' agent for franchises, deals that let other people open stores under the brothers' name and system in return for fees.

Kroc opened his own McDonald's in Des Plaines, Illinois, on April 15, 1955. In 1961 he bought the brothers out for $2.7 million. The brothers said Kroc had made a handshake promise to keep paying them royalties, a share of the money from every sale, and never kept it. Kroc also opened a McDonald's near their original stand, in competition with it.

Norman Borlaug

Who: An Iowa-born plant scientist whose wheat helped poor countries grow far more grain, in what was called the Green Revolution.

When and where: Mexico from 1944. India and Pakistan in the 1960s. Nobel Peace Prize in 1970.

Movie: *The Man Who Tried to Feed the World* (PBS American Experience, 2020), a documentary.

His short-stemmed wheat came from crossing thousands of kinds.

Norman Borlaug bred wheat that grew more grain on the same land. Starting in 1944, he worked in Mexico for a program run by the Rockefeller Foundation, a charity started by the Rockefeller family. He crossed thousands of kinds of wheat. The result was a semi-dwarf wheat, with short stems, that resisted plant diseases. By 1956 Mexican farmers grew all the wheat the country needed.

Experts in the 1960s predicted famines, times when there is so little food that people starve, in India and Pakistan. Farmers there planted Borlaug's wheat, and the famines the experts predicted did not happen. Borlaug won the Nobel Peace Prize in 1970, the only farm scientist to have won it.

Many writers credit his work with saving as many as a billion lives, a common estimate rather than a count. The rise in grain harvests in poor countries that followed is called the Green Revolution.

Farmers grew his wheat together with chemical fertilizer and irrigation, water brought to the fields by pipes and ditches, so they came to depend on chemicals they had to buy. Many farmers also stopped planting old kinds of crops, so fewer kinds were grown.

Pesticides in the fields

Farmers sprayed their crops with pesticides, chemicals that kill insects and weeds. One of them, DDT, which kills insects, was in wide use from the mid-1940s, and it made the shells of eagles' eggs thin. Rachel Carson's book *Silent Spring*, published in September 1962, described what these long-lasting pesticides did to birds, soil, water and people. By 1963 only 417 pairs of bald eagles were nesting in the lower 48 states. Officials at the Environmental Protection Agency banned DDT in 1972.

Farmworkers in steady contact with pesticides had high rates of burns, cancer and children born with birth defects. In the 1980s, children in several towns of California's San Joaquin Valley had cancer at rates "800% above the norm," according to the Walter P. Reuther Library at Wayne State University.

The farmworkers' union. Cesar Chavez and Dolores Huerta campaigned to ban DDT on grape and lettuce ranches. With the United Farm Workers, they organized the people who picked the crops. They are known for the grape strike in Delano, California, when grape pickers refused to work until growers met their demands.

In March 2026 the New York Times reported that two women said Chavez had sexually abused them as girls in the 1970s, and Dolores Huerta then said that he had raped her, forcing her into sex. Chavez died in 1993, so no court has heard these accusations. Chapter 14, Work and Workers, tells the strike and these accounts in full.

Black farmers and their land

Black farmers ran more than 900,000 farms in 1920. Between 1910 and 1997, the amount of farmland Black Americans owned fell by nearly 90 percent.

Loans held back. Researchers at the Union of Concerned Scientists, a science group, named two of the causes in a 2023 report. The first was that officials of the US Department of Agriculture, or USDA, had stopped Black farmers from getting the loans and grants that would have kept them on their land. Lloyd Wright, who once ran the department's civil rights office, called it "the last plantation," after the big farms where enslaved people had been forced to work.

Heirs' property. The second cause was heirs' property. When a landowner died without a will, a paper that says who gets a person's things after the person dies, the land passed to the landowner's heirs, the family members who inherit it. All the heirs owned the land together. Each owned a share of the whole farm, not a piece of land of their own.

Any one of the heirs could ask a court to force a sale of the whole farm. A land buyer could buy one heir's share and then ask a court to sell the farm at auction, and farms sold that way went for far less than they were worth. USDA officials have called heirs' property "the leading cause of Black involuntary land loss," meaning land lost when the owners did not choose to sell.

What it cost. In a study published in 2022, five economists put the value of the land Black farmers lost from 1920 to 1997 at about $326 billion in today's dollars. By 2007 Black farmers ran 32,938 farms, about 1.5 percent of the country's farms.

The court case. Black farmers took the USDA to court in a case called *Pigford v. Glickman*. They showed that from 1983 to 1997, USDA officials treated Black farmers worse than white farmers because of their race, and that the officials failed to look into the farmers' complaints.

Members of local county committees made the decisions on USDA farm loans. Black farmers said committee members turned down their loans, or made them wait longer for approval than white farmers. The Congressional Research

Service report on the case does not name the committee members.

Judge Paul L. Friedman approved a settlement, an agreement that ends a lawsuit, on April 14, 1999. Under it, 15,645 farmers had their claims approved, out of 22,721 who qualified. Those approved under the part of the settlement called Track A each received $50,000, and USDA officials canceled what they owed on their USDA loans. The payments and debt relief came to about $1.06 billion. A second settlement, of $1.25 billion, was announced on February 18, 2010, for farmers who had missed the first deadline.

Organic farming rules become law

Organic farmers grow food without man-made chemical fertilizers and pesticides, and they build up their soil with compost, rotted plant and animal matter. J.I. Rodale began a magazine called *Organic Farming and Gardening* in Emmaus, Pennsylvania, in 1942 and spent decades urging farmers and gardeners to farm this way.

Members of Congress passed the Organic Foods Production Act in 1990. It required national rules for what could be sold as organic. The USDA organic seal, a label for food that meets those rules, came into use on October 21, 2002.

Alice Waters

Who: A cook who built her restaurant around food from nearby farms and started a school garden where children grow and cook their own food.

When and where: Chez Panisse restaurant, Berkeley, California, opened in 1971. The Edible Schoolyard, Berkeley, started in 1995.

Movie: *Alice Waters and Her Delicious Revolution* (PBS American Masters, 2003), a documentary.

In her school garden, students grow food, cook it and eat it.

Alice Waters opened a restaurant called Chez Panisse in Berkeley, California, in 1971. She cooked with food that was fresh, grown nearby, in season and organically grown. Chez Panisse was one of the first restaurants of the style later called California cuisine, cooking built on fresh food from local farms. Waters helped start the farm-to-table idea, which means buying food straight from local farms.

Waters started the Edible Schoolyard at Martin Luther King Jr. Middle School in Berkeley in 1995. Students there grow a garden, cook what they grow and eat it.

The farm crisis of the 1980s

American farmers sold large amounts of grain to other countries in the 1970s. They planted "fencerow to fencerow," every acre up to the fences. Expecting the good years to last, many farmers borrowed from banks to buy more land and machines.

Then, in the early 1980s, interest on loans, the extra money a borrower pays back on top of a loan, rose high. At the same time, the price of farmland fell. In parts of the Midwest, farmland lost as much as 60 percent of its value between 1981 and the mid-1980s. By 1984 American farmers owed about $215 billion, twice what they had owed in 1978, and many could not earn enough to keep up with their loan payments.

The owners of an estimated 300,000 farms defaulted during the 1980s, meaning they could not pay back their loans. About 300 banks in farm country failed, running out of money and closing.

Farm Aid. Willie Nelson, John Mellencamp and Neil Young held a concert called Farm Aid in Champaign, Illinois, on September 22, 1985. About 80,000 people came, and more than $7 million was raised for farm families. Farm Aid concerts still take place.

Russel and Mary Jane Jordan

Who: An Iowa farm couple who sold their machinery, livestock and household goods at auction to pay their bank, while their daughter filmed it.

When and where: Iowa. The family had owned its farm for 125 years. Their bank was in Atlantic, Iowa.

Movie: *Troublesome Creek: A Midwestern* (1995), a documentary about the family. It won the Grand Jury Prize at the Sundance Film Festival and was nominated for an Academy Award.

To pay the bank, they sold everything but the land.

Russel and Mary Jane Jordan owned a farm in Iowa that had been in their family for 125 years. A company called Norwest, which owned banks in several states, bought their bank in Atlantic, Iowa, and a man from the company's headquarters, Tim Wolf, took over their loans.

The debt. In the spring of 1990 the bank's officers called in an old debt of $70,000, which meant they demanded that it be paid. That same spring the Jordans had taken out their yearly loan of $150,000 to pay for the season's farming. To keep the bank from taking the farm, they decided to pay off both, about $220,000 in all.

The auction. The Jordans sold all their machinery, their livestock and even their household goods at an auction. They kept only the land. The sale raised about $300,000, more than enough to pay what they owed. Russ and Mary Jane moved to a rented house in Atlantic. Their son Jim took over farming the home place with his own equipment. Later the crop land was rented out to neighbors.

Their daughter, Jeanne Jordan, and the filmmaker Steven Ascher made the film about the family.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 11 MINUTES

Americans have argued since 2000 over what goes into their food, who picks it and who can get enough of it. Meanwhile the number of farms kept falling. The 2022 Census of Agriculture, the USDA's count of the country's farms, found 1.9 million farms in the United States, 7 percent fewer than in 2017. It was the first count below 2 million since before the Civil War. In 1935 there had been 6.8 million.

Weight, hunger and food stamps

Researchers at the Centers for Disease Control and Prevention, or CDC, measure obesity, a body weight high enough to raise the risk of illness. By their count, about three American adults in every ten had obesity in 1999 and 2000, and about four in every ten in the years 2021 to 2023.

Doctors link diet to heart disease and to type 2 diabetes, an illness in which the body cannot keep the sugar in the blood at a healthy level.

Some Americans live far from a place to buy fresh food. USDA officials map low-income areas that are far from a supermarket, and people call these areas food deserts. In a city, far means more than 1 mile away. In the country, it means more than 10 miles. An area counts when at least 500 people, or a third of the people who live there, are that far away.

By that measure, about 19 million people, about 6 percent of the population, lived in food deserts in 2019.

USDA economists had counted households that did not always have enough food since 1995. In September 2025, USDA officials under Agriculture Secretary Brooke Rollins announced that they would end the count. They called it "redundant, costly [and] politicized," meaning a repeat of other work, too expensive and slanted by politics. The last report came out on December 30, 2025. It reported that in 2024 about one household in seven did not always have enough food: nearly 48 million people, including more than 14 million children.

President Donald Trump signed the One Big Beautiful Bill Act on July 4, 2025. Under it, federal spending on SNAP, the Supplemental Nutrition Assistance Program once called food stamps, goes down. Economists at the Congressional Budget Office, which works out costs for Congress, figured that the cut will come to $186.7 billion over ten years.

Farmers' markets and the 2020 pandemic

Farmers' markets, where farmers sell their own crops straight to shoppers, grew in number. USDA officials counted 1,755 of them in 1994 and 8,771 in 2019, five times as many. The number grew about 7 percent a year until about 2011, and then it stopped growing much.

The meat plants. Other food reaches shoppers from people they never see, such as the workers in plants that process meat and poultry, meaning chickens and turkeys. The disease COVID-19 spread through those plants in 2020.

Officials in 23 states reported 16,233 cases of COVID-19 among workers in 239 plants in April and May 2020, and 86 of those workers died. This count comes from a CDC report of July 2020. Of the sick workers whose race was recorded, 87 percent were members of racial or ethnic minority groups.

Changed seeds and new kinds of meat

Most of the country's corn, soybeans and cotton grows from genetically engineered seed. Scientists made it by changing the seed's DNA, the chemical instructions inside every cell, in a laboratory, so that the plants survive sprays that kill weeds. In 2025 farmers planted it on about 92 percent of the country's corn acres, 96 percent of its soybean acres and 93 percent of its cotton acres, according to USDA economists.

Industry histories name the Flavr Savr tomato, sold in 1994, as the first engineered food sold in stores. Scientists of the National Academies of Sciences, a group of leading American scientists, reviewed about 1,000 studies in 2016. They wrote that they found no convincing proof that eating these foods harms people's health. Many Americans still argued against them. Under federal rules in force since January 2022, foods that contain changed DNA that tests can detect must carry a "bioengineered" label.

New kinds of meat. The Impossible Burger, made from plants, came out in 2016. Beyond Meat also makes meat from plants. The public could buy its shares, small parts of the company, for the first time in 2019, and their price rose more on the first day than that of any other company's new shares that year.

USDA officials gave two companies, UPSIDE Foods and GOOD Meat, the first approval in the United States to sell chicken grown from animal cells instead of from a live bird in June 2023. People call it lab-grown meat.

On vertical farms, workers grow vegetables indoors on shelves stacked high. Investors put money into these companies, and the business grew fast. AeroFarms, an early company of this kind, filed for bankruptcy, a court process for a business that cannot pay its debts, in June 2023. The company later reorganized and stayed in business.

Who picks the food?

Most hired crop workers in the United States were born in other countries, USDA economists report. In recent years about 15 percent of crop-farm jobs have gone to workers on H-2A visas, papers that let a farmworker from another country come for one season of farm work. More than 90 percent of H-2A workers come from Mexico.

The number of H-2A jobs grew more than eight times over from 2005 to 2025. Chapter 14, Work and Workers, gives the numbers.

Heat and children. Crop workers in the fields died of heat at nearly 20 times the rate of all American workers from 1992 to 2006, a CDC count found. Federal law also lets children work on farms at younger ages than in other jobs. Chapter 14 tells the death of one of those crop workers, a 56-year-old tobacco picker in North Carolina, and sets out the rules for children on farms.

Lucas Benitez

Who: A tomato picker who helped start the Coalition of Immokalee Workers and its Fair Food Program.

When and where: Born in Guerrero, Mexico. Moved to Immokalee, Florida, at 17. The coalition began in 1993, and the Fair Food Program began in 2011.

Movie: *Food Chains* (2014), a documentary about the coalition. Benitez appears in it.

In one case, he helped free about 700 workers held in slavery.

Lucas Benitez came to Immokalee, Florida, from the state of Guerrero in Mexico when he was 17. He picked tomatoes in the fields there. In 1993 he helped start the Coalition of Immokalee Workers, a group of farmworkers.

The orange groves. In 2000 Benitez looked into slavery in the orange and grapefruit groves around Lake Placid, Florida, north of Immokalee. Two brothers, Ramiro and Juan Ramos, supplied as many as 700 pickers

to the growers there. They made the workers pick fruit ten hours a day, six days a week, and threatened them at gunpoint with torture and death if they tried to escape.

The brothers charged each worker $1,000 for the trip to Florida. Each week they took most of the workers' pay, saying it went to that debt and to food and rent, but no worker was ever set free. In April 2001 four of the workers escaped with the coalition's help. A federal court convicted the brothers in 2002, and Ramiro Ramos was later sentenced to 15 years in prison. WGCU, a public radio station in Florida, credits Benitez with helping to free about 700 workers in that one case.

The Fair Food Program. The coalition and Florida's tomato growers began the Fair Food Program in 2011. Big buyers such as McDonald's and Walmart pay about a penny more for each pound of tomatoes. The program's managers suspend any farm where workers are forced to work, children are put to work as a regular practice, or workers suffer sexual assault, any sexual touching or attack against a person's will, including rape. Chapter 14 tells the rest of how the program works.

Benitez received the University of Michigan's Wallenberg Medal, an award for people who defend human rights, in October 2023.

Black farmers in the new century

The 2022 Census of Agriculture counted 46,738 Black farmers on 32,653 farms, according to a USDA report of June 2024. They made up 1.4 percent of all the country's farmers and worked 5.3 million acres. The number of farms run by Black farmers was 8 percent lower than in 2017. In 1920 Black farmers had run more than 900,000.

USDA officials paid about $2 billion on July 31, 2024, to more than 43,000 farmers, ranchers and forest owners who had faced discrimination in USDA farm loans, meaning they had been treated worse because of who they were, such as their race. The NAACP Legal Defense Fund reported the payments that day. The money came from a section of the Inflation Reduction Act, a law that members of Congress passed in 2022.

The first farmers, still farming

People in Native nations are working to control their own food, which is called food sovereignty. Tribes run bison herds, seed banks that save the seeds of old crops, and programs that bring back traditional foods. Native chefs and seed keepers are rebuilding the ways their nations cooked and ate. Among the Ojibwe of the Great Lakes, manoomin, or wild rice, is harvested by canoe and is still a central food.

Nephi Craig

Who: A White Mountain Apache and Diné (Navajo) chef who opened a café built around Apache foods.

When and where: Café Gozhóó, Whiteriver, on the White Mountain Apache Reservation in Arizona, opened in 2021.

Movie: *Gather* (2020), a documentary about Native food sovereignty that features him.

He cooks Western Apache foods like parched corn and acorn powder.

Nephi Craig is a chef who cooks the foods of the Western Apache. He opened Café Gozhóó, a café built around Apache foods, in Whiteriver, Arizona, on the White Mountain Apache Reservation, in 2021. Apaches who are recovering from addiction to alcohol or drugs work in its kitchen through a job-training program run by the tribe's Rainbow Treatment Center.

The food. Craig's Western Apache seed mix has pine nuts, amaranth (a plant whose tiny seeds are eaten like grain), pumpkin seeds and parched corn. To parch corn, he swirls white corn kernels in a hot pan until they toast. He has also served grilled squash with Apache acorn powder, made from ground acorns.

Craig said in 2026 that he had been sober himself for 15 years. That year he published a book about his life and his cooking, *Our Knives Will Save Us*. He also founded the Native American Culinary Association, a group for Native cooks.

The 2020 documentary *Gather* follows him and others who are bringing back Native foods. Among them are Twila Cassadore, a San Carlos Apache woman who gathers wild foods, and Elsie DuBray, a student scientist from the Cheyenne River Sioux.

THE THREE QUESTIONS, ANSWERED

1. **How did Native farmers turn a wild grass into corn?** For thousands of years, starting in southern Mexico about 9,000 years ago, they saved seed only from the teosinte plants they chose, until the grass became corn.
2. **Who did the hardest farm work, and what did they get for it?** Enslaved people worked the rice and cotton fields for no pay, and sharecroppers got half the crop's value or less and stayed in debt. From 1992 to 2006, crop workers died of heat at nearly 20 times the rate of all American workers.
3. **How did Americans come to eat food grown far away by people they never met?** With canning, pond ice and railroad cars cooled with ice, food could be sold far from the farm, and the farms grew fewer, down to 1.9 million in 2022.

What changed: Growing food went from the work of nearly every family to the business of fewer, bigger farms, while the hardest field work stayed with people who were enslaved, in debt or far from home.

FIVE DATES TO KEEP

About 9,000 years ago: farmers in southern Mexico begin breeding corn.

November 1620: the *Mayflower* men dig up Nauset seed corn.

May 20, 1862: the Homestead Act becomes law.

June 30, 1906: Roosevelt signs the pure food and meat inspection laws.

April 14, 1999: a judge approves the *Pigford* settlement.

NOT KNOWN: No farmer from before 1500 is known by name.

Where to go: You can read 318 of the Oblinger family's letters online at the Library of Congress, in its Prairie Settlement collection.

Next: Chapter 16, Economy. It follows how the country has made its living, through booms and panics.

Chapter 16: Economy

CHAPTER 16 OF 37 · ECONOMY · 10 ERAS · ABOUT 80 MINUTES

Much of this country's wealth came from work that people were forced to do for little or no pay, and in crash after crash, the poorest families lost the most.

The census of 1860 lists Mayer Lehman, a merchant in Montgomery, Alabama, as the owner of seven enslaved people. He and his brothers, Henry and Emanuel, took cotton grown by enslaved people as payment for the goods they sold, and then they began trading cotton. Their company, Lehman Brothers, failed on September 15, 2008, owing about 619 billion dollars. It was the largest bankruptcy, the court case of a company that cannot pay its debts, in the country's history.

This chapter follows how people here made a living, from Native trade before 1500 to the jobs of 2026. They made it first mostly by farming, then more and more in mines, mills and factories, and by 2026 mostly in services: stores, offices, hospitals and schools. Two things come back in era after era. One is who did the work and who kept the money: servants, enslaved people, convicts and sharecroppers did much of it for little or nothing. The other is the crash: the panics of 1819, 1837, 1857, 1873 and 1893, the Great Depression of the 1930s and the crash of 2008. In each one, watch who lost a job, a farm or a home.

Four words for hard times. In a panic, fear spreads, many people rush to take their money out of the banks at once, and businesses close. A crash is a sudden fall in prices, of stocks or of homes. A depression is the long stretch of hard times that can follow, lasting years. A recession is a shorter one, measured in months.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did slavery make money for people far from any plantation?
2. What happened to ordinary families when a boom turned into a crash?
3. When did the national government start helping people who lost their jobs?

Five minutes? Read "The Panic of 1837" in 1800 to 1850.

Can skip: "Shell, copper and obsidian", "Parliament's molasses and iron laws", "Where the name Gilded Age comes from", "Why farmers joined the Grange". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 2 MINUTES

Native people here traded without money. They had economies of their own, ways of making, sharing and trading the things they needed. Nobody had coins or wages, and nothing had a set market price. They traded by barter, swapping one good for another, and by giving gifts between people and between nations. Stone tools, shells and copper traveled hundreds of miles this way. No record names any one person who took part in this trade.

How far did Cahokia's hoe blades travel?

The people of Cahokia controlled the making and trading of stone blades for hoes, farm tools for breaking up soil. Archaeologists, scientists who study the past from the things people left behind, have found these blades more than 400 miles away, from Louisiana to Minnesota and in the Arkansas River valley.

The city. Cahokia stood near what is now St. Louis. It was the largest city north of Mexico, and it was at its biggest from about 1050 to 1150. Its farmers grew more corn than they ate, and that surplus, the extra corn, fed specialists: people who spent most of their time on one kind of work, such as making tools, instead of growing food.

The stone. Workers chipped the blades from Mill Creek chert, a tough stone found only in southern Illinois. They dug it out of quarries, open pits for stone, and some of the pits were 10 to 20 feet deep.

Shell, copper and obsidian

People carried the shells of whelks, large sea snails, north from the coast of the Gulf of Mexico. Traders moved copper from the land around Lake Superior. At Cahokia, workers made shell beads in large numbers, and families on small farms made them too.

In the West, people traded obsidian, a natural glass that forms in volcanoes, across their own trade networks. Archaeologists have traced pieces found in western Canada back to Obsidian Cliff, in what is now Wyoming, and Bear Gulch, in what is now Idaho. Some pieces had traveled as far as 750 miles from where the stone formed.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

Explorers found no gold or silver in the land that became the United States, though the kings and queens of Spain, France and England paid for their voyages in the 1500s to find it. The Spanish silver came from mines in Mexico and South America. Fishing was the first steady European business in these waters. Some ship crews traded with Native people on the coast, and some of that trade was for furs. Almost no Europeans settled here yet, so there was no colonial economy.

Fish, furs and a rope from a cliff

European fishing crews worked the cod waters off the northeast coast all through the 1500s and dried their catch on shore.

The rope. Abenaki people on the coast of what is now Maine traded with the crew of Giovanni da Verrazzano's ship in 1524 from the top of a cliff. They would not let the crew come ashore, so they lowered baskets of goods to the ship on a rope.

Crews of earlier European ships had already taken Native people from the Maine coast to sell as slaves. That is why the Abenaki kept this crew off the land. Verrazzano called the place "the land of bad people." The records do not say whether the baskets held furs.

Furs on sticks. Mi'kmaq traders held up beaver furs on sticks to the crew of Jacques Cartier at Chaleur Bay, north of Maine in what is now Canada, in July 1534. They traded the furs for knives and other iron goods. Cartier wrote the trade down in his own account of the voyage, one of the earliest records of furs traded between Europeans and Native people on the Atlantic coast of North America. The fur trade reached the land that became the United States in the next century.

St. Augustine. The Spanish town of St. Augustine in Florida began in 1565 as a military post, and the Spanish king's government paid its costs. Unlike the English colonies that came later, it was not set up to make a profit.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 7 MINUTES

Tobacco became Virginia's main business in the 1600s. The planters who owned the fields changed who did the work. At first the field workers were indentured servants from England, people who had signed contracts to work for a master for a set number of years. Near 1700 the planters switched to buying enslaved Africans, whom they could hold for life. Farther north, Dutch traders shipped beaver skins to Europe, and lawmakers in London wrote rules that required colonists to ship their main crops only to England or its colonies.

Tobacco in the Chesapeake

The first colonists at Jamestown hoped to find gold. John Smith wrote that there was "no talk, no hope, no work but dig gold, wash gold, refine gold, load gold." They found none. Colony officials tried glassmaking and other small businesses, and none of them made money.

Tobacco. Then Virginia's planters began to grow tobacco as a cash crop, a crop grown to sell rather than to eat. They shipped about 2,300 pounds of tobacco in 1615 and 1616, about 20,000 pounds by 1617, and about 1.5 million pounds by 1629 and 1630.

The trade kept growing. Planters in Virginia and Maryland, the two colonies around Chesapeake Bay called the Chesapeake colonies, together sent about 22 million pounds of tobacco a year to England by 1700. They also used tobacco as money.

John Rolfe

Who: A Jamestown colonist whose tobacco shipments to London began Virginia's trade in its main crop.

When and where: Jamestown, Virginia, 1612 to 1614.

Rolfe grew tobacco from West Indies seed and shipped it to London.

John Rolfe began Virginia's export trade in tobacco, the business of sending it away to be sold in another place. He lived at Jamestown, the English colony in Virginia. By July 1612 he was growing tobacco from seed brought in from the West Indies, the islands of the Caribbean Sea. He sent his first shipments to London in 1614.

Who worked the tobacco fields?

Indentured servants did the tobacco work for most of the 1600s. Each servant's contract was called an indenture. In return for the servant's years of work, the master paid for the trip across the ocean and gave food, clothing and a place to sleep.

The terms. Adults usually served four to seven years, and children sometimes served much longer. At the end of the term, the master owed the servant "freedom dues." Depending on the contract and the colony, these could be land, tools, clothes or farm animals.

How many. Of about 120,000 English people who moved to Virginia and Maryland in the 1600s, about 90,000 came as servants, three out of every four. These are historians' estimates from ship records and court records.

The headright. Planters gained from each servant in two ways. Under Virginia's headright rule, whoever paid for a person's passage, the cost of the ocean trip, received 50 acres of land for that person. So a planter who paid a servant's passage got the servant's years of work and 50 acres as well. The large tobacco planters built their estates, their large holdings of land, this way.

What servants went through

Officers of the Virginia Company could hang, burn to death or shoot runaway servants they caught, under the company's first form of contract, used from 1609 to 1619. The Virginia Company of London, a group of investors in England, started Jamestown in 1607 and ran the colony until 1624. That year the king of England took the colony away from the company and put it under his own control.

Whippings and added years. Masters could whip their servants. When a servant ran away, judges in the colony's courts added years to the servant's term. They also added years when a woman servant had a child outside marriage. Under a Virginia law of March 1662, she served two more years or paid her master 2,000 pounds of tobacco.

When the master was the father. Some of the fathers were the masters themselves. Virginia's lawmakers wrote a second law in December 1662 about a woman servant "gott with child by her master," that is, made pregnant by him. When her years ran out, officers of the local church, called churchwardens, sold her to work two more years "for the good of the parish," the church district where she lived. Under this law the churchwardens sold the woman, not the master who was the father.

Some masters also forced women servants into sex. Julie Richter, a historian of colonial Virginia, writes that some masters "physically and sexually abused their servants." She gives no count of how many.

The rules for masters. A Virginia law required masters to give servants "Competent Dyet, Clothing & Lodging," which meant enough food, clothes and shelter. The same law required masters not to "exceed the Bounds of moderation in correcting them," which meant not to go too far in punishing them.

Many servants died of disease before their contracts ended. Most of those who lived to the end of their terms stayed poor.

From servants to enslaved Africans

About 300 Africans lived in Virginia by 1650, next to about 4,000 white servants. In the 1660s, members of Virginia's assembly, the colony's lawmakers, passed the colony's first laws that treated Africans as slaves for life.

The first law. Under a law of March 1661, an English servant who ran away together with Africans had to serve extra time for the Africans' missing time as well as his own. Judges punished runaway servants by adding years to their terms. But the law's words said the Africans were "incapable of makeing satisfaction by addition of time." Judges could add no years to their time, because they were already held for life.

Servants and enslaved Africans fought on the same side in Bacon's Rebellion of 1676 and 1677, an armed uprising in Virginia. Nathaniel Bacon led colonists who wanted to kill or drive out every Native person in Virginia. When the governor, William Berkeley, tried to stop him, Bacon's men turned on the governor and burned Jamestown in September 1676.

The rebellion showed that poor white and Black people could unite, and the planters who ruled Virginia feared it would happen again. After it, the planters passed new laws. Historians at the College of Charleston describe the purpose of those laws as "to harden Virginia's racial caste system," a system that sorts people at birth into ranks they can never leave.

The slave code. Members of Virginia's assembly gathered the colony's slave laws into one slave code in 1705. Under it, enslaved people were property, and being baptized, going through the church ceremony that makes a person a Christian, did not free anyone. A master who killed an enslaved person while "correcting" them faced no punishment.

The planters' choice. Planters chose to buy enslaved Africans instead of servants near 1700. Fewer English servants were coming by then, because life in England had improved. In the same years the Atlantic slave trade grew, as ship crews carried captive Africans across the ocean to be sold.

The economist David Galenson, a scholar who studies how people make and spend money, explains the choice in money terms. A servant worked for a master for about five years. A planter who bought an enslaved person held that

person, and that person's children, for life. So the planter got a whole lifetime of unpaid work, and the work of the children too, instead of five years.

Beaver skins, and rules from London

One ship of the Dutch West India Company, the *Arms of Amsterdam*, carried 7,246 beaver skins to Amsterdam in a single cargo in 1626, along with the skins of otters, minks, lynxes and muskrats. A letter known as the Schaghen letter lists the cargo. People in Europe bought hats made of felt from beaver fur, and those buyers were the reason traders wanted the skins.

English leaders followed an idea called mercantilism, the belief that colonies existed to make the home country richer. Colonists were to send raw goods, such as tobacco, to England and buy back finished goods, things already made in England.

The Navigation Acts. Members of Parliament, England's lawmakers, wrote this idea into laws called the Navigation Acts. Under the act of 1651, colonists had to ship their trade goods only in English or colonial ships. The act of 1660 listed "enumerated goods," a phrase that means listed goods, and colonists could ship these only to England or to other English colonies. The list included tobacco, sugar, cotton, ginger and indigo, a plant used to make blue dye.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 6 MINUTES

Much of the colonies' wealth, most of all in the South, came from the work of enslaved people, who were paid nothing. Free colonists were doing well. By 1774 they earned more on average than the people of England, the economic historians Peter Lindert and Jeffrey Williamson found. Among free colonists only, income was spread more evenly than in any other place the two could measure at that time. In the rice country of South Carolina, though, the wealth from rice stayed with a small group of white planters.

Convicts for sale. From 1718, British judges sent about 50,000 convicts across the ocean to be sold as laborers.

What the colonies sold

Tobacco came from the Chesapeake colonies, wheat and flour from the middle colonies, such as Pennsylvania, New York and New Jersey, and fish, rum and ships from New England. In the Lowcountry of South Carolina, the low land along the coast, enslaved workers grew rice and indigo and prepared them for sale.

Timber cost little in the colonies, so colonial shipbuilders could build ships for less than builders in England. By the Revolution, about one-third of the British merchant fleet, all the ships that carried Britain's goods for trade, had been built in America.

Rice, and the people who knew how to grow it

Enslaved Africans built and worked the rice fields of the South Carolina Lowcountry. Rice was the Lowcountry's main cash crop from the early 1700s. Enslaved Africans were a majority of the people in South Carolina from about 1708, and by 1740 they were about two out of every three.

Nic Butler of the Charleston County Public Library writes that enslaved people "moved earth on a massive scale" to build the rice fields. By the 1720s, Carolina planters were asking slave traders for Africans from the region called the Rice Coast, because people from there already knew how to grow rice. The wealth from rice, Butler writes, "remained concentrated in the hands of a small minority of white planters."

The children. Historians at the College of Charleston record that enslaved children in the Lowcountry died at especially high rates. Their account gives no count of those children's deaths in the early 1700s. The number of enslaved people there did not grow through births until the 1760s.

Eliza Lucas Pinckney

Who: A young woman who ran her father's three South Carolina plantations from the age of 16 and directed the colony's first successful indigo dye crop.

When and where: South Carolina, 1744 to the 1750s.

Her first crop, in 1744, made 17 pounds of blue dye.

Eliza Lucas directed the first successful crop of indigo dye in South Carolina in 1744 and shared her seed with neighboring planters. From the age of 16 she had run her father's three plantations. She later married and became Eliza Lucas Pinckney.

What followed. South Carolina's indigo exports grew from about 5,000 pounds in 1745 and 1746 to about 130,000 pounds by 1748. By the middle of the 1750s, planters there shipped more than a million pounds a year. Indigo became the colony's second biggest export after rice, and by the Revolution it made up more than a third of the value of the colony's exports.

Whose work. Enslaved workers on her father's plantations did the growing. Her father sent skilled dye-makers from the Caribbean, and enslaved workers did the processing, the work of turning the plant into dye. She relied on their knowledge and their labor.

Convicts sold at the dock, 1718 to 1775

British judges sent about 50,000 people convicted of crimes to the North American colonies between 1718 and 1775, and merchants sold them there as laborers. The convicts were about one in four of all the people who moved from Britain to colonial America in those years.

The law. Members of Parliament passed the Transportation Act in 1718. Transportation here meant sending convicts away across the ocean as a punishment. Under the act, a judge could send a person convicted of a felony, a serious crime, to the colonies instead of hanging them: for seven years for one of the less serious felonies, and for fourteen years for worse crimes.

Sold at the dock. Most convicts went to Maryland and Virginia. There, merchants sold them at the dock as bound laborers, people who had to work for whoever bought them until their term ended. In the records of four Maryland counties, about 82 of every 100 convicts were men and 18 were women. On two ships of 1721 and 1724, most convicts were between 20 and 29 years old, and about half were laborers with no trade.

The money. British government officials paid a London merchant named Jonathan Forward £3, three British pounds, for each convict he carried, and later £5. A Maryland firm, Stevenson, Randolph and Cheston, sold convicts in Maryland from 1767 to 1775. The firm's own sales books record that most men sold for £11 to £14 and most women for £9 to £10. The planters who bought them got years of work without paying wages.

Convicts died on the crossing. On 38 voyages from London between 1718 and 1736, the ships carried 3,599 convicts, and 385 of them died, about 11 of every 100. The historian A. Roger Ekirch points out that the records of some of these voyages list no deaths at all. If those blanks are gaps in the clerks' records, the true count was about 15 of every 100. On 12 voyages from Bristol in the 1770s, 23 of 990 convicts died.

Rhode Island's slave ships

Merchants in Rhode Island sent out at least 1,000 slaving voyages in the century before 1807. That year members of Congress made a law that banned bringing captive Africans into the United States, starting on January 1, 1808. The Rhode Island ships carried more than 100,000 Africans into slavery.

About 60 of every 100 slaving voyages that left North America sailed from Rhode Island, and in some years more than 90 of every 100 did. These numbers come from *Slavery and Justice*, a 2006 report from Brown University.

The loop. The merchants who sent out these ships made their money by buying and selling human beings. Their crews sailed a loop around the Atlantic Ocean. Colonists made rum from Caribbean molasses, a thick, dark syrup left over when sugar is made.

Captains carried the rum and other goods to West Africa and traded them for captive people. Crews then carried the captives across the ocean, on the crossing called the Middle Passage, to be sold in the Caribbean and on the mainland. The ships brought sugar and molasses back north.

Parliament's molasses and iron laws

Members of Parliament passed the Molasses Act in 1733. Under it, colonists had to pay a duty, a tax on goods brought in, on molasses from colonies that did not belong to Britain. The duty was sixpence, six British pennies, for each gallon. Many colonists dodged it by smuggling, bringing the molasses in secretly.

Under the Iron Act of 1750, raw iron from the colonies could enter Britain with no duty. The act forbade colonists to build new steel furnaces, rolling and slitting mills or plating forges, the works that turned raw iron into finished goods. Colonists built them anyway.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 6 MINUTES

Americans paid for the Revolution with paper money and debt. In 1790 the new national government and the states still owed about 79 million dollars for the war, and Alexander Hamilton wrote the plan to pay those debts in full. Members of Congress taxed whiskey to help pay them, and in 1794 farmers in western Pennsylvania rose up against that tax. Both before and after the war, Rhode Island merchants sent ships to Africa to buy captive people.

The Browns and the slave ship Sally

Four merchant brothers of Providence, Rhode Island, sent their ship *Sally* to West Africa to buy people in 1764. They were Nicholas, John, Joseph and Moses Brown. The ship carried 17,274 gallons of rum, along with chains, guns and shackles, metal rings locked around a person's wrists or ankles. By that year Rhode Island had about 30 rum distilleries, places where rum was made, and 22 of them were in Newport. The distillers made the rum from molasses shipped from the Caribbean.

The voyage. The captain, Esek Hopkins, bought 196 Africans, and at least 109 of them died, by the count in Brown University's report. Hopkins kept the ship on the African coast for more than nine months while he bought them. The captives died from disease and starvation, by killing themselves, and in an uprising.

Nineteen died before the ship left Africa. They included several children and a woman who hanged herself below deck. On the day the ship sailed, another woman was "left for dead," in the words of the report, which does not say who left her or what became of her.

Sixty-eight people died during the crossing. Four of them, a woman and three children, died in the first week at sea. Twenty more died soon after the ship reached the West Indies, and one more died on the way to the Brown brothers in Providence.

The uprising. When the captives rose up, Hopkins's crew fired on them. Hopkins wrote in his account book: "Slaves Rose on us was obliged fire on them and Destroyed 8 and Several more wounded badly." By "Destroyed 8" he meant that his crew killed eight people. The survivors were so sick that buyers paid only £5 for some of them.

Brown's next ship. John Brown kept trading in people. In 1795 the crew of his ship *Hope* took 229 Africans from the Gold Coast of Africa, and 198 of them lived to be sold in Cuba. Under a law of 1794, Americans could not own, equip, put money into or work on ships that carried captive people to other countries.

The trial. Members of the Providence Abolition Society, a group working to end slavery, took Brown to a United States court for breaking that law. On August 5, 1797, he became the first American tried under it. The jury found him not guilty and ordered the society's members who had charged him to pay his court costs. He did not get the *Hope* back.

Paying for the war with paper money

Members of the Continental Congress had almost no power to tax. From 1775 they paid for the war by printing paper money called Continental dollars, and the paper dollars quickly lost their value. By the end of 1777 it took about 4 paper dollars to buy what 1 silver dollar bought, and by 1781 it took about 100. The paper money was close to worthless. The saying "not worth a Continental," meaning worth nothing at all, comes from this money.

Alexander Hamilton

Who: The author of the plan to pay the Revolution's debts in full.
When and where: The new United States government, 1790 and 1791.
Movie: Alexander Hamilton (American Experience, 2007), a two-hour documentary shown on PBS, with Brían F. O'Byrne as Hamilton.

After a dinner deal, members of Congress voted his debt plan through.

Alexander Hamilton planned how the new United States would pay the debts of the Revolution. He set out the plan in his First Report on Public Credit in January 1790. He counted the national debt at about 54 million dollars, and the states owed about 25 million dollars more for the war.

Paying in full. Hamilton proposed that the national government pay every debt in full and take over the states' war debts, a step called assumption. The people who were owed money would then look to the new government to be paid, and they would want it to last so that they got their money. Paying in full would also build the government's credit, the trust lenders have that a borrower will pay back what it borrows.

The dinner. Assumption did not have enough votes to pass in the House of Representatives in June 1790. Thomas Jefferson invited Hamilton and James Madison, a member of the House from Virginia, to dinner, and the three men made a deal.

The deal. Madison agreed to stop blocking assumption and to win over enough Southern members to pass it. In return, Hamilton agreed to find Northern votes for putting the country's permanent capital on the Potomac River, where Washington, D.C., stands today. Members of Congress passed both: first the capital on the Potomac, then assumption.

Factories. Hamilton's Report on Manufactures is dated December 5, 1791. In it he laid out a plan to grow American manufacturing, the making of goods in workshops and mills, with tariffs (taxes on goods brought in from other countries) and subsidies (money paid to help a business grow). Members of Congress never brought the plan to a vote.

Hamilton also helped found the Society for Establishing Useful Manufactures. Through it, he and the other founders started a factory town at the falls of the Passaic River, which is Paterson, New Jersey.

Why farmers fought the whiskey tax

Farmers in western Pennsylvania sent their corn to market as whiskey, so a tax on whiskey fell on the crop they sold. Members of Congress passed that tax in 1791 to help pay the war debt. It was an excise, a tax on goods made and sold inside the country, of 7 cents a gallon on distilled spirits such as whiskey.

Tar and feathers. A gang in disguise caught a tax collector named Robert Johnson in September 1791 and tarred and feathered him: they coated him with hot tar and stuck feathers to it.

Bower Hill. About 600 armed men attacked the house of John Neville, the tax inspector, at a place called Bower Hill in July 1794. When their leader, James McFarlane, stepped into the open, someone inside the house shot and killed him. The accounts do not say who fired. The attackers then set the house on fire. This uprising is called the Whiskey Rebellion. President George Washington called up about 13,000 militia, soldiers drawn from the states, and the rebels broke up without a battle.

The Dreadful Night. Troops under General Henry Lee broke into houses on the night of November 13, 1794, pulled people from their beds and arrested about 150 of them. They held the prisoners in animal pens and basements. For years afterward, people in the area called it "the Dreadful Night."

The troops took 20 of the prisoners to Philadelphia for trial. Two men, Philip Wigle and John Mitchell, were found guilty of treason, the crime of making war against their own country. Washington pardoned them both, setting aside their punishment.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 11 MINUTES

In 1800 most American families grew and made most of what they used. By 1850 far more of them grew crops and made goods to sell, and bought what they needed, a change called the market revolution. The biggest thing Americans sold to other countries was cotton grown by enslaved people, and by 1860 enslavers valued the people they held at 3 billion dollars or more.

Two panics. In the panics of 1819 and 1837, many people rushed to get their money out at once, and businesses closed. About 50,000 people in three big cities had no steady work in 1819, and by one estimate a third of all workers had no jobs by the fall of 1837.

From household to market

Farm families and workshops could sell goods far from home once moving them got cheaper. Before the Erie Canal opened in 1825, it cost about 100 dollars to haul a ton of goods by road from Buffalo to New York City. By canal boat the same ton cost about 10 dollars, and the trip took about ten days instead of twenty.

The owners of the new cotton mills in New England hired young women from farm families and paid them wages in cash. These women were among the first large groups of Americans who worked for cash wages.

Copper and gold. A rush for copper began on the Keweenaw Peninsula of Michigan, a point of land with water on three sides, in 1843. It began after Douglass Houghton, a scientist who studied rocks for the state as its geologist, reported on the copper there. By 1849 miners there dug 96 of every 100 pounds of copper mined in the United States.

Gold was found at Sutter's Mill in California early in 1848. About 80,000 people went to dig for it in 1849, and about 300,000 people came to California between 1848 and 1855. The merchants who sold the miners their supplies made steadier money than most of the miners did.

King Cotton

Cotton was the most valuable American export, and enslaved people did the work of growing it. They planted it, picked it and got it ready for sale. The crop was known as King Cotton.

More than half. Cotton made up 51.6 percent of the value of everything the United States sold to other countries in 1840, and it stayed above half almost every year until the Civil War. About three of every four pounds grown here were sent abroad. From 1820 to 1860, about 80 of every 100 pounds of cotton used in the world came from the United States.

The cotton crop more than doubled in six years. Enslaved people grew about 300 million pounds of cotton in 1831 and more than 600 million pounds in 1837.

Forced south. To grow it, enslavers forced people out of the older slave states and into the new cotton states. They moved about 155,000 enslaved people this way in the 1820s and at least 250,000 in the 1830s. Between about 1790 and 1860, enslavers moved more than 1 million people from the Upper South, the older slave states farther north such as Virginia, to the Deep South, the cotton states farther south.

People counted as property worth billions

Enslavers counted the people they held as capital, money or property that an owner uses to make more money. Census takers, the government workers who count every person in the country, counted 3,953,760 enslaved people in the United States in 1860. Enslavers valued them at 3 billion dollars or more. That was the price enslavers put on human beings as property, and it was more than all the money invested in American railroads and factories put together.

Enslavers used enslaved people as capital in two ways. The people grew the cotton and were paid nothing for it. Enslavers could also borrow money by promising the people themselves to a lender, and if the loan was not repaid, the lender could take the people and sell them.

Banks and bond sellers

Planters in Louisiana and Mississippi set up their own banks in the 1830s. The banks lent planters money on mortgages, loans in which the borrower promises property to the lender, and these mortgages covered land and enslaved people. The banks then sold bonds, paper promises to repay a loan with interest, to investors in Europe and in the Northern states. The mortgages on land and people backed those bonds.

Far from any plantation. Investors in Europe and the North who bought these bonds made money from loans backed by enslaved people. Bankers in London, Amsterdam and New York sold the bonds to them. Baring Brothers of London and its partners in New York sold 7 million dollars in bonds for the Union Bank of Louisiana in 1832. Hope and Company of Amsterdam sold 12 million dollars in bonds for the Citizens' Bank of Louisiana in 1833.

A bank's own report. Two Louisiana banks, the Citizens Bank and the Canal Bank, lent planters money and took about 13,000 enslaved people as collateral, property a borrower promises to hand over if the loan is not repaid. When planters failed to pay, the two banks came to own about 1,250 of those people.

JPMorgan Chase, a bank that grew in part out of those two, made these figures public in a report in January 2005. JPMorgan Chase's leaders apologized and set up a 5-million-dollar scholarship fund for students in Louisiana. They made the report because, under a 2003 Chicago law, companies that did business with the city had to search their records for ties to slavery.

Insurers, cotton traders and Northern mills

Northern insurance companies sold enslavers policies, written deals, on the lives of enslaved people. The enslaver paid the company a fee, and when an enslaved person died, the company paid the enslaver.

Two insurers. New York Life, then called Nautilus Insurance, insured the lives of 520 people from 1846 to 1848. Those people appear to have been enslaved, and the company paid out money for 15 of their deaths. In March 2000, leaders of the insurer Aetna apologized for policies the company sold on enslaved people's lives in the 1850s. They said Aetna would pay no reparations, payments made to make up for a wrong.

The Lehmans. The Lehman brothers of Montgomery, Alabama, whose story opens this chapter, made their start as merchants who took slave-grown cotton as payment and then traded it.

The Lowell mills. Northern mill owners made money from slave-grown cotton too. Francis Cabot Lowell, Nathan Appleton, Patrick Tracy Jackson and the other Boston merchants who built the Lowell mills were called the Boston Associates. They founded the Boston Manufacturing Company in 1813, and their mills spun cotton grown by enslaved people in the South.

The mills also made a cheap, rough cloth known then as "negro cloth," from the word white Americans of that time used for Black people. The mill owners sold the cloth to planters, who used it to clothe the people they enslaved. By 1845, 31 cloth companies tied to the Boston Associates, in Massachusetts, New Hampshire and southern Maine, made one-fifth of all the cotton and wool cloth in the United States.

Senator Charles Sumner of Massachusetts named the two sides by their tools: the lash, a whip, and the loom, a machine that weaves thread into cloth. He called the tie between planters and mill owners an "unholy union ... between the lords of the lash and the lords of the loom."

The Panic of 1819

The Panic of 1819 began the first major depression that the American public lived through, a long stretch of hard times, from about 1819 to 1822, when businesses closed and many people lost their work.

What caused it. The prices that buyers paid for cotton fell sharply. Officers of the Second Bank of the United States called in loans, asking borrowers to pay back money right away. Under an 1817 order from Congress, buyers of land from the national government had to pay in gold or silver coin. Owners closed American factories because buyers chose cheaper goods from Britain.

No work. During the panic, employers let workers go, and judges jailed people who could not pay their debts. In New York, Philadelphia and Baltimore, about 50,000 people had no work or only odd jobs. One observer from abroad estimated that half a million people across the country had no work. Nobody made a real count.

Debtors' prison. In those years, a person who was owed money could sue, and a judge could order the borrower arrested and locked in a debtors' prison, with no set date for release. In Philadelphia, judges sent 1,808 people to debtors' prison this way. In Boston, judges sent 3,500.

Land lost its value. Farmland in Pennsylvania fell from 150 dollars an acre in 1815 to 35 dollars in 1819, and property in Richmond, Virginia, lost half its value.

The poor. In New York City, a city of about 120,000 people, the number of paupers, people too poor to feed themselves without help, rose from 8,000 in 1819 to 13,000 in 1820. Mathew Carey, a publisher and economist in Philadelphia, estimated that 3 million people, about one-third of the nation, were hurt by the panic.

Small cotton planters went bankrupt in 1819, and again in 1837: they could not pay their debts, so their lenders could take what they owned. Larger planters bought up the small planters' land and the enslaved people those planters had held.

The Panic of 1837

A depression began with the Panic of 1837 and lasted until about 1843. In the years before it, buyers speculated in western land and in cotton, buying in the hope of selling later at a higher price. Then British lenders made loans harder to get, and the price of cotton fell, by half in New Orleans.

The banks stop paying. The bankers of New York City stopped paying out gold and silver on May 10, 1837. People who brought in the banks' paper money could no longer trade it for coins.

Of the country's 850 banks, 343 closed, about 4 of every 10, and 62 more partly failed, by the count in a history collection edited by Hubert Howe Bancroft in 1902. A college textbook gives a smaller count, 200 banks closed. The governments of seven states and one territory did not pay back their loans.

How many lost work. By one widely repeated estimate, a third of all workers had no jobs by the fall of 1837, and within two years employers cut the pay of people who kept their jobs by 30 to 50 percent. Other estimates put New York's unemployment, the share of people who want work and cannot find it, at about 25 or 30 percent in 1837 and 1838.

Nobody counted the jobless at the time, so these are later estimates, and the teaching guides that repeat them do not say who made them.

The flour riot. Food had already grown costly before the banks stopped paying. A barrel of flour in New York rose from 7 dollars to 12 between 1836 and 1837. About 5,000 New Yorkers met in City Hall Park on February 13, 1837, to protest the price of food and fuel.

Part of the crowd broke into the store of the flour merchants Eli Hart and Company. They took or destroyed 500 to 600 barrels of flour and 1,000 bushels of wheat, about 60 pounds of wheat to each bushel.

Who paid. Mississippi planters owed 33 million dollars to the banks of New Orleans, and their 1837 cotton crop sold for only 10 million dollars. What happened next depended on whom a debtor owed.

Some planters had borrowed from banks in Mississippi and Florida whose bonds those states had backed. A bond is a paper promise to repay borrowed money with interest, and a state that backed a bank's bonds promised to pay the buyers itself if the bank could not. Lawmakers in both states later refused to pay those bonds. So the investors in Europe and the North who had bought them lost their money, and the planters who owed those banks kept their land and the people they enslaved.

Families sold apart. Debtors who owed other creditors, the people they owed money to, could lose the people they enslaved. When such a debtor could not pay, a judge could order his property taken, and the creditors seized the people he enslaved and sold them. Two Louisiana banks came to own about 1,250 enslaved people when planters failed to repay their loans. Many debtors fled to Texas at night and took the people they enslaved with them.

A single creditor's claim against a debtor who failed to pay could force the sale of a whole enslaved community. The creditors sold its people to different buyers and scattered them, and sales like these split families. By the count of the Equal Justice Initiative, a group that studies the history of slavery, enslavers and traders separated nearly half of all the people they moved inside the country from a husband, a wife or a parent.

Philip Hone

Who: A New York merchant whose diary recorded the Panic of 1837 as it happened.

When and where: New York City. He kept the diary from 1828 to 1851.

Hone wrote down the crash of 1837 and lost money in it.

Philip Hone sold goods at auction in New York City, and he had served as the city's mayor in 1826 and 1827. During the Panic of 1837 he wrote in his diary how business in the city stopped: "No goods are selling, no business stirring, no boxes encumber (crowd or block) the sidewalks of Pearl Street."

Hone lost part of his own fortune in the crisis years. His diary was printed in 1889, after his death, as *The Diary of Philip Hone*.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 14 MINUTES

American mills made 11.4 million tons of steel in 1900, more than Britain and Germany together. By then the United States was an industrial country, one where much of the work is done in mines, mills and factories. Railroad track grew from about 9,000 miles in 1850 to 193,346 miles by 1900. The era is called the Gilded Age.

In the South. After slavery ended, Southern landowners farmed with sharecroppers, families who worked a landowner's land for a share of the crop and stayed in debt to him. Officials in Southern states leased prisoners, renting them out to private companies, and in Alabama 90 or more of every 100 prisoners were Black.

Hard times. About one in four workers in New York City lost their jobs after the Panic of 1873, by estimates of the time. For the Panic of 1893, estimates for the whole country run from about 17 to 25 percent of workers.

Where the name Gilded Age comes from

The name Gilded Age comes from an 1873 novel by Mark Twain and Charles Dudley Warner, *The Gilded Age: A Tale of Today*, in which the two writers made fun of their own time. Something gilded is covered with a thin layer of gold, so the name means a thin shine of gold over serious problems underneath.

Steel, rails, cattle and metal

Steel rails got cheap in these years. According to histories of the steel industry, a ton of steel rails cost about 140 dollars in 1872, about 35 dollars in 1882 and about 20 dollars by 1900. American mills made more steel than Britain's by about 1889 or 1890.

Cattle. Cattle ranchers in Texas drove herds north to the railroads in Kansas, where a Texas longhorn worth up to about 5 dollars at home could sell for up to about 40 dollars. From 1866 to 1884, cowboys drove more than 5 million cattle north.

Ranchers also kept cattle on the open range, land with no fences, where the animals grazed wherever they wandered. Four things ended the open-range cattle business. Barbed wire fences went up across the range, and more railroads were built. Officials put a quarantine on cattle in 1885 and 1886, keeping animals apart from others to stop the spread of a disease. The last was the winter of 1886 to 1887.

Homesteads. Members of Congress passed the Homestead Act in 1862. Under it, a settler could claim up to 160 acres of land and own it after living on it and farming it for five years. Settlers filed about 4 million claims, and about 270 million acres, about a tenth of the country's land, became private farms. US officials had taken that land from Native nations.

Silver, copper and aluminum. The Comstock Lode, a long vein of silver running through rock near Virginia City, Nevada, was the richest body of silver in North America. It was found in 1859. Copper from Michigan, and later from Butte, Montana, became the wire for electric lights and telephones.

Charles Martin Hall, a 22-year-old college graduate, made aluminum by a new method in his family's woodshed in Oberlin, Ohio, on February 23, 1886, with help from his sister Julia. The Pittsburgh Reduction Company began making the metal by his method in 1888, and a new industry began. A pound of aluminum cost about 12 dollars in the early 1880s and less than 1 dollar by the middle of the 1890s.

The panics of 1857 and 1873

The Ohio Life Insurance and Trust Company failed on August 24, 1857. Its manager in New York had stolen from it, and its officers had made bad loans. News of the failure went out by telegraph, and people in many places rushed to take their money out of banks, a rush called a bank run. A short but deep depression followed.

The Panic of 1873 began on September 18, when Jay Cooke and Company failed. It was an investment bank, a firm that sold bonds and shares to raise money for businesses such as railroads, and Jay Cooke and his partners had put too much money into bonds of the Northern Pacific Railroad.

The exchange closes. The New York Stock Exchange, where people buy and sell shares of companies, closed for ten days. It had never closed like this before. About 18,000 businesses failed. Economists at the National Bureau of Economic Research set the dates when hard times begin and end, and by their count this depression lasted 65 months, until March 1879. It was the longest stretch of hard times on their record until the 1930s.

New York's jobless. About one of every four workers in New York City had lost their jobs by the winter of 1873 to 1874, according to estimates drawn from newspapers of the time. The historian Howard Zinn wrote that more than 90,000 homeless workers, almost half of them women, slept in New York police stations.

Tompkins Square. About 7,000 people with no work met in Tompkins Square Park in New York City on January 13, 1874. They wanted Mayor William F. Havemeyer to create public jobs. Soon after 10 in the morning, New York police, some of them on horseback, charged into the crowd. The police commissioner, the official in charge of the city's police, was Abram Duryée.

The police beat people with clubs, and men, women and children ran. The accounts give no count of the people who were hurt. Men from a German workers' club fought back, and someone hit one policeman on the head with a hammer. The police drove the crowd out of the square and arrested 46 people.

Why farmers joined the Grange

Farmers were angry about what railroads charged to ship their grain and what grain elevators, the tall buildings where grain is stored before it is shipped, charged to hold it. In 1867 Oliver Hudson Kelley founded the Patrons of Husbandry, known as the Grange (husbandry is an old word for farming). By the middle of the 1870s it had about 800,000 members.

Grange members pushed state lawmakers to pass the Granger laws, which set the highest rates that railroads and grain elevators could charge. In *Munn v. Illinois* in 1877, the justices of the Supreme Court ruled that those laws were allowed under the Constitution.

The Panic of 1893 and Coxey's Army

About 500 banks and 15,000 businesses failed in the Panic of 1893, and more than 150 railroad companies later went bankrupt. Nobody counted the people out of work at the time. Later estimates for the worst of it run from about 17 to 25 percent of workers. The economic historian Stanley Lebergott estimated that unemployment rose from about 3 percent in 1893 to 18.4 percent in 1894.

The march. A businessman named Jacob Coxey led a march of men without work from Massillon, Ohio, to Washington, D.C., in 1894. About 100 men left on March 25, and about 500 arrived on May 1, while a crowd of about 30,000 watched them. It was the first large protest march on Washington.

The marchers, called Coxey's Army, asked the national government for a 500-million-dollar program of public works. Under it, federal officials would have hired people without jobs to build public projects. When Coxey tried to speak at the Capitol, police arrested him for walking on its grass, and he served 20 days in jail.

Sharecropping and debt

Most freedpeople in the South, the people freed from slavery, had no land after the Civil War. Many families went to work as sharecroppers.

The furnish. A landowner gave a family a plot of land, usually about twenty acres. He also "furnished" the family with food, seed, tools and a cabin on credit, to be paid for later, and this was called the furnish. On the furnish, landowners charged interest, the fee a borrower pays for borrowing, far above a fair rate.

At harvest the landowner sold the crop and took out the cost of the furnish and the interest. Then he gave the family its share, usually half of what was left or less.

Landlords held a lien on the crop, a legal claim on it until the debt was paid, and they kept the account books. So most sharecroppers were still in debt when the landlord settled the yearly accounts, year after year. A landlord could put a sharecropping family off the land whenever he chose.

Sharecropping lasted into the 1930s. Census takers counted 1,831,470 tenant farmers in the South in 1930, farmers who worked land they did not own, about half of all farmers there. About 700,000 of them were share tenants, who owned mules or tools and kept two-thirds to three-quarters of the crop.

Most Black farmers in the country owned no land by 1935: 77 of every 100. Nearly half of white farmers owned none. In 1937 members of the President's Committee on Farm Tenancy counted about 775,000 sharecroppers in the South, almost equal numbers of white and Black.

Debt peonage and Alonzo Bailey

Members of Congress banned peonage, forcing a person to work to pay off a debt, in the Peonage Abolition Act of 1867. Lawmakers in Southern states then wrote laws that had the same effect. Peonage was most common in the cotton belt, the part of the South where cotton was grown, and in the turpentine camps of southern Mississippi, Alabama, Georgia and the Florida panhandle, where workers collected sap from pine trees to make turpentine, a liquid used in paint.

Alonzo Bailey, a Black farm laborer in Montgomery County, Alabama, signed a work contract on December 26, 1907, with the manager of the Riverside Company, a company that ran a farm. Bailey received 15 dollars before he started work, to be paid back out of wages of 12 dollars a month. He quit after about a month.

The charge. Under Alabama law, a worker who quit without paying back money he got this way was a criminal. An Alabama court found Bailey guilty, fined him 30 dollars and sentenced him to 136 days of hard labor, heavy work done as a punishment.

The appeal. A Montgomery lawyer, Edward S. Watts, and a city judge, William H. Thomas, helped Bailey take his case to the United States Supreme Court, and Booker T. Washington helped in secret. On January 3, 1911, in *Bailey v. Alabama*, the justices ruled that the Alabama law broke the Thirteenth Amendment, the part of the Constitution that bans slavery. Other states' peonage laws stayed in force until the justices struck down Georgia's and Florida's in the 1940s.

Convict leasing in Alabama

The Thirteenth Amendment bans slavery "except as a punishment for crime." Under that exception, state and county officials in the South rented out prisoners to private companies, which was called convict leasing. The companies paid the government a fee for each prisoner, and the companies' bosses made the prisoners work in mines, on farms and in lumberyards.

Officials in Alabama leased prisoners from 1846 to July 1, 1928. After a money crisis in the state government in 1875, the warden John G. Bass, the official in charge of the state's prisoners, leased them to coal mines, farms and lumberyards for monthly fees.

Three coal companies. From 1883, under the warden John Hollis Bankhead, officials leased most state prisoners to three coal companies: the Pratt Coal and Iron Company, the Tennessee Coal, Iron and Railroad Company, called TCI, and the Sloss Iron and Steel Company. In 1888 TCI signed a ten-year contract for every state prisoner able to work, paying the state 9 to 18.50 dollars a month for each prisoner.

More than 95 of every 100 county prisoners in Alabama were Black, and so were 90 of every 100 state prisoners. The companies' bosses whipped prisoners to make them work. Coal company officials wrote false reports of "bad conduct" so that the prisoners would have to serve longer sentences.

What the state made. Alabama's state officials collected money from leasing prisoners. Fees for leased prisoners were about 10 of every 100 dollars the state took in during 1883, and in 1898 they were nearly 73 of every 100 dollars.

Deaths. Leased prisoners died at about ten times the rate of prisoners in states that did not lease them, according to Digital History, a history website of the University of Houston. By the same source, 25 of every 100 Black leased prisoners in Alabama died in 1873. At Sloss's Coalburg mine, 9 of every 100 prisoners died each year, Dr. Thomas D. Parke, the health officer of Jefferson County, reported in 1895.

Alabama's mines held 1,710 state prisoners and 745 county prisoners in 1896. In the two years before, 233 state prisoners and 181 county prisoners had died.

James Knox. A prisoner named James Knox, from Mobile, was killed at Sloss's Flat Top mine in 1924. Men there whipped him with a steel wire and lowered him into a vat of boiling water. The published accounts of his death do not name the men who did it.

His death was first reported as a suicide by poison, and it led to an investigation. An autopsy, an examination of a dead body by doctors, found that his heart had stopped and that the poison had been put into his body after he died. Investigators found that the whipping and the boiling water caused the injuries that killed him.

Sloss's managers kept county prisoners working in the company's mines until 1928. That year Alabama's officials ended convict leasing. Alabama was the last state to end it, about five years after Florida.

Breaking strikes. TCI's managers also used leased prisoners in Tennessee, where officials leased prisoners from 1871 to 1896. A TCI official told the *New York Times* that "one of the chief reasons" the company took up convict leasing "was the chance it offered for overcoming strikes," the times when workers stop working until their employer agrees to better pay or conditions.

The official said free miners would not want to strike once they saw the company had plenty of prisoners to put to work. The account of the interview does not give his name. TCI's managers used prisoners to keep unions, groups of workers who join together to bargain with their employer, out of the mines.

Andrew Carnegie

Who: A Scottish immigrant who started work as a mill boy, built a steel company, sold it for 480 million dollars and gave away more than 350 million dollars.

When and where: Scotland and Pennsylvania, 1835 to 1919. He built steel mills from the 1870s and sold his company in 1901.

Movie: The Richest Man in the World: Andrew Carnegie (American Experience, 1997), a documentary about him shown on PBS.

His first job, at about 13, paid 1.20 dollars a week.

Andrew Carnegie's family moved from Dunfermline, Scotland, to Allegheny, Pennsylvania, in 1848. At about 13 he went to work in a cotton mill as a bobbin boy, changing the spools that held the thread. He worked as a telegraph messenger and then as a manager for the Pennsylvania Railroad. From the 1870s he built steel mills, among them the Edgar Thomson works.

Selling and giving. In 1889 Carnegie wrote an essay called "The Gospel of Wealth." In 1901 he sold Carnegie Steel to a group led by J. P. Morgan for 480 million dollars. He gave away more than 350 million dollars, including money for about 2,500 public libraries. Counts of the libraries vary, and some go above 2,800 around the world.

Homestead. Workers at Carnegie Steel's works in Homestead, Pennsylvania, went on strike in 1892. Henry Clay Frick, one of the men running Carnegie's company, hired 300 armed agents of the Pinkerton detective agency, a private company that hired out guards and detectives, to take back the plant. On July 6 the Pinkertons came up the river on barges, and they and the strikers fought a gun battle.

Seven workers and three Pinkertons were killed, and some accounts count more. No record names the men who fired the shots that killed them. Governor Robert E. Pattison of Pennsylvania sent about 8,000

state soldiers. By November the strike had failed, and Frick and the company's other managers had broken the steelworkers' union at Homestead.

John D. Rockefeller

Who: The founder of Standard Oil, which by 1880 controlled about 90 percent of oil refining in the United States.

When and where: Cleveland, Ohio, from 1870. He lived from 1839 to 1937.

Movie: The Rockefellers (American Experience, 2000), a documentary shown on PBS about four generations of his family, not about him alone.

Lamp oil dropped from about 26 cents a gallon to 7 cents.

John D. Rockefeller built the company that controlled most of America's oil refining, the work of turning crude oil from the ground into fuels people can use. He founded the Standard Oil Company in Cleveland in 1870, and by 1880 his company controlled about 90 percent of the oil refining in the United States.

Cheap light. Kerosene was the fuel that ordinary families burned in their lamps. A gallon of it cost about 26 cents in 1870, about 9 cents by 1880 and about 7 cents by 1890, so lamp light became cheap for ordinary homes.

Rockefeller became the richest man in the country. Later he gave large sums of money to universities and to medical research.

BY 1900

If you remember one thing: After slavery ended, Southern landowners and state officials used debt and prison leasing to keep many Black workers laboring for little or nothing.

One number: 73. In 1898 nearly 73 of every 100 dollars that Alabama's state government took in came from leasing prisoners, up from about 10 in 1883.

Next: After 1900 Henry Ford roughly doubled his workers' pay, and after the crash of 1929 hungry families stood in breadlines.

1900 to 1950

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Stock prices, the prices of shares in companies, crashed in October 1929, and by 1933 about one of every four workers had no job. In Chicago and Pittsburgh, more than half of Black workers had no job in 1932, by later estimates. In 1935 members of Congress and President Franklin D. Roosevelt set up old-age pensions and unemployment insurance in the Social Security Act. The pensions were regular payments to people too old to work, and the insurance paid workers who had lost their jobs.

Farm and household workers. The law did not include farmworkers and domestic workers, people who cleaned, cooked and cared for children in other people's homes. Most Black workers did those two jobs, so most of them could get neither payment. Two accounts disagree about who left these workers out. One points to Southern members of Congress, and the other to the Treasury Secretary, who asked for the change for reasons that had to do with collecting the tax.

War work. Unemployment fell to 1.2 percent in 1944, when factories were making goods for the Second World War.

Henry Ford

Who: The carmaker who roughly doubled his workers' pay in 1914 and cut the price of his Model T car from 825 dollars to 260 dollars.

When and where: Ford's Highland Park car factory. He announced the five-dollar day on January 5, 1914.

Movie: Henry Ford (American Experience, 2013), a documentary about him shown on PBS.

Ford's managers hired almost four new workers a year for every job.

Henry Ford roughly doubled his workers' pay in 1914. He did it to stop workers from quitting, and workers who earned more could also buy the cars they built.

Why they quit. Ford's managers put in an assembly line, which moves each car past a row of workers who each do one step over and over. After that the work never let up and was dull. Turnover, the count of workers who quit and had to be replaced, reached 370 percent at the Highland Park factory in 1913.

The five-dollar day. Ford announced five dollars for an eight-hour day on January 5, 1914, and the new rate began on January 12. Before, a Ford worker earned about 2.34 dollars for a nine-hour day.

The raise came with conditions. Ford paid the extra money as profit-sharing, a share of the company's profits, and a worker qualified only after six months. Staff of the company's Sociological Department inspected workers' homes before granting the full rate. They checked whether workers saved their money and whether they drank alcohol, and they checked for what they called "proper" living. Ford left out men under 22, and at first most women.

After the raise, fewer workers quit. The price of the Model T kept falling, from 825 dollars in 1908 to 260 dollars by 1925.

> Ford fought unions, the groups workers form to bargain with their employer, for decades. He owned a newspaper, the *Dearborn Independent*, and through the 1920s he published antisemitic articles in it, articles hateful toward Jewish people. Ford later apologized in public for the articles.

The crash and the Great Depression

The Dow Jones average is a number built from the stock prices of a group of large companies. It climbed through most of the 1920s and reached 381 on September 3, 1929. Prices fell hard on Black Thursday, October 24, 1929, and again on Black Tuesday, October 29. By July 8, 1932, the Dow stood at 41, about 89 percent below its peak.

The hard times of the 1930s are called the Great Depression. By 1933 about 25 percent of workers, one in four, had no job, and about 9,000 of the country's 25,000 banks had failed and closed. Government officials and scholars made the jobless estimate later, because nobody counted the jobless every month at the time.

Output and prices. GDP, the total value of all the goods and services made in the country in a year, fell from about 104 billion dollars in 1929 to about 56 billion dollars in 1933. Part of that drop came only from lower prices: from October 1929 to April 1933, the prices of the things people bought every day fell by about 27 percent.

Economists take the change in prices out of GDP to see how much less the country actually made. Counted that way, as real GDP, output fell by about 30 percent.

Hungry people stood in breadlines, lines for free food. Some lived in shantytowns, groups of shacks, which people called Hoovervilles.

The Bonus Army

Veterans of the First World War camped in Washington, D.C., in the summer of 1932 to ask for a bonus they were owed for their war service. The bonus was not due until 1945, and the veterans asked to be paid early in cash. Estimates of their number run from about 10,000 to more than 20,000, and many had their families with them. Members of the Senate voted the payment down on June 17.

The shooting. Police trying to clear veterans out of some buildings in the city on July 28 shot and killed two veterans, William Hushka and Eric Carlson. Hushka had come from Lithuania. He sold his butcher shop to join the army in the First World War and became a citizen while he served. He was buried at Arlington National Cemetery less than a week after he was shot.

The Army. President Hoover then ordered the Army to clear the camps. General Douglas MacArthur sent in infantry, soldiers on foot, and cavalry, soldiers on horseback, with tanks and tear gas. The gas stings the eyes, nose and throat, so people cough and their eyes stream with tears. MacArthur's soldiers burned the main camp.

Members of Congress later voted to pay the bonus over the veto of President Franklin D. Roosevelt, who had refused to sign the bill. The payments began in June 1936.

Benjamin Roth

Who: A lawyer in Youngstown, Ohio, who kept a diary of the Great Depression for more than ten years.

When and where: Youngstown, Ohio. He began the diary in 1931.

People sold their frozen bank savings for 50 cents on the dollar.

Benjamin Roth's diary records the banks that closed in Youngstown, the stocks that became worthless, prices, and what lawyers and workers earned. When the banks closed, people could not get their savings out. Even after the banks reopened, savers could take out only a small part. Roth wrote down how people with cash bought up these frozen bank accounts for 50 cents on the dollar, half of what was in them.

When cash ran out, he wrote, people turned to barter, trading goods instead of paying money, and to scrip, paper that stood in for money.

The umbrella. In 1931 Roth wrote that "banks are absolutely terrible in their insistence on payments of notes and mortgages," meaning that bankers demanded that borrowers pay back their loans. "It is the old story of lending you an umbrella when the sun is shining and then demanding it back when it rains."

His son, Daniel B. Roth, and James Ledbetter edited the diary in 2009, and the publisher PublicAffairs printed it as a book, *The Great Depression: A Diary*.

Last hired, first fired

Black workers lost their jobs first and at higher rates than white workers. Employers hired Black workers last and fired them first, a pattern called "last hired, first fired." By 1932 nearly half of all Black Americans had no work, according to the National Museum of African American History and Culture.

More than half of Black workers in Chicago and Pittsburgh had no job in 1932, History.com reports, drawing on the Library of Congress. About a quarter of white workers in Northern cities had none. In Atlanta in 1934, nearly 70 percent of Black workers had no job. These numbers are all later estimates, because nobody kept an official count by race.

Forced to Mexico. Government officials and private employers forced people whose families came from Mexico to leave for Mexico, starting in 1929. Federal officials and the officials of local welfare agencies blamed Mexican immigrants for taking scarce jobs and for costing public money when they got relief, the aid given to people without work.

California's lawmakers found in 2005 that about 400,000 American citizens and legal residents had been forced to go from California alone. They estimated that about two million people were forced out across the country, and that about 1.2 million of them had been born in the United States. Local officials sold some families' property to pay for sending them away.

The New Deal

The federal government, the national government in Washington, paid no unemployment insurance and no old-age pensions before 1933. Members of Congress set up both in the Social Security Act of 1935. President Roosevelt and his officials started many programs from 1933 on, including the CCC, the WPA and Social Security, and together they are called the New Deal.

Jobs. President Roosevelt set up the Civilian Conservation Corps, or CCC, a federal jobs program for young men, on March 31, 1933. About 3 million young men worked in it over the years it ran. They earned 30 dollars a month and sent about 25 dollars of it home.

The Works Progress Administration, or WPA, ran from 1935 to 1943. About 8.5 million people worked for it. WPA workers built about 650,000 miles of roads, 125,000 public buildings and 75,000 bridges.

The first check. President Roosevelt signed the Social Security Act on August 14, 1935. The first monthly old-age check went to Ida May Fuller of Ludlow, Vermont, on January 31, 1940. It was for 22.54 dollars, and she had paid a total of 24.75 dollars in Social Security taxes.

Frances Perkins

Who: The Secretary of Labor who chaired the committee that planned Social Security, and the first woman in a United States Cabinet.

When and where: Secretary of Labor from 1933 to 1945. She lived from 1880 to 1965.

Movie: Summoned: Frances Perkins and the General Welfare (2020), a documentary about her.

Her committee wrote the plan for Social Security in under seven months.

As chair of the Committee on Economic Security, Frances Perkins led the planning of Social Security. She was Secretary of Labor from 1933 to 1945 and the first woman in a United States Cabinet, the group of the president's top officials.

The committee. President Roosevelt set up the committee on June 29, 1934, and its members planned old-age pensions and unemployment insurance. They sent their report to Roosevelt on January 15, 1935, and he signed the Social Security Act on August 14, 1935.

According to the historian Larry DeWitt, Perkins wanted farmworkers and domestic workers covered by the new law, and so did Harry Hopkins, the official in charge of aid to the jobless.

Who was left out of Social Security

Of every 100 Black workers in 1930, 65 were farmworkers or domestic workers, the two kinds of work the Social Security Act of 1935 did not include. So about two-thirds of Black workers could get neither an old-age pension nor unemployment insurance, and in the South the share was higher. The president of the NAACP, a civil rights group, gave these figures in 2020.

What Houston said. Charles Hamilton Houston spoke for the NAACP at the hearings on the bill in 1935. He used the word Negro, the common word for Black people at that time. The bill, he said, would leave out "Negro sharecroppers and Negro cash tenants, who are just about at the bottom of the economic scale." Sharecroppers paid for the land they farmed with part of the crop, and cash tenants paid in money.

Houston called the law "a sieve with the holes just big enough for the majority of Negroes to fall through," comparing it to a strainer full of holes.

Who made the choice. Two accounts disagree about who left these workers out. Jon Dubin, in a 2024 article in a Stanford law journal, points to Southern members of Congress. One of them, Representative Howard Smith of Virginia, argued for keeping Black sharecroppers and domestic workers dependent on their employers.

Dubin also points to Senator Harry Byrd of Virginia, whose words show what Southern members feared from the bill. Byrd opposed a rule that payments be high enough for "decency and health." He said that rule "might serve as an entering wedge," a first step that opens the way to more, "for federal interference with the handling of the Negro question." Byrd did not want federal officials to have a say in how matters touching Black people were handled.

The executive director of Perkins's committee reported that Byrd's position was "supported by nearly all of the southern members of both committees," the two committees in Congress that worked on the bill.

The other account. Larry DeWitt, a historian at the Social Security Administration, gave a different account in 2010. He found that Treasury Secretary Henry Morgenthau asked for the changes to the bill that left these workers out, for reasons that had to do with collecting the tax. DeWitt wrote that charges of racial bias, unfair treatment because of race, in the founding of Social Security did "not hold up under detailed scrutiny," meaning close study.

The changes. Members of Congress changed the law in 1950 to cover farm and domestic workers who worked regularly for one employer, and in 1954 they covered the rest. Representative Smith opposed these changes too. Of eight House members who voted against them, six were Southern Democrats. In the Senate, Byrd opposed them, and so did a senator from Mississippi.

Farm payments and evicted sharecroppers

Federal farm payments in the 1930s went to landowners, and many sharecroppers lost their places on the land. The landowners are also called landlords, or planters.

Paid to grow less. Members of Congress passed the Agricultural Adjustment Act in May 1933. Under it, officials of the Agricultural Adjustment Administration, or AAA, ran a farm program under Agriculture Secretary Henry Wallace, and they paid farmers to grow less. Their aim was to raise farm prices: with less cotton and pork for sale, buyers would have to pay more for them.

That year farmers plowed under about 10 million acres of growing cotton, turning the plants back into the soil. AAA officials bought about 6 million pigs and had them killed.

Who got the checks. Under the AAA's cotton contracts of 1934 and 1935, officials paid landowners to take land out of cotton. Agriculture Secretary Henry Wallace "rented" about three-eighths of the country's cotton land in this way. Sharecroppers got only about 11 percent of the payments.

A landlord could keep the whole check by evicting the sharecroppers, putting them off the land and out of their homes, and hiring day laborers, who were paid by the day. When planters cut their cotton land by 40 or 50 percent, they

cut the number of their tenants, the families who farmed their land, by the same share. More than two-thirds of Black sharecroppers made no profit or went into debt, according to the National Museum of African American History and Culture.

The union. Seven Black men and eleven white men founded the Southern Tenant Farmers' Union in Arkansas in July 1934 to protest the evictions. Planters and the men who worked for them attacked the union.

Lawmen arrested and jailed a white organizer. Four armed white men told Black members they would be lynched, killed by a mob, if they were caught at another union meeting. Many members were beaten and threatened with guns, and the accounts do not name the men who beat them. A mob attacked the home of the union's lawyer. Chapter 15 tells the union's story in full.

The purge. A group of AAA lawyers, led by Jerome Frank, ruled that landlords had to keep the same tenants. AAA administrator Chester Davis had Wallace fire those lawyers in February 1935, a firing called the AAA purge.

The justices of the Supreme Court ruled the Agricultural Adjustment Act unconstitutional, against the Constitution, in 1936. Members of Congress wrote a new version in 1938, and federal officials have paid farmers ever since.

War work ends mass unemployment

Factory workers made war goods during the Second World War, and unemployment fell from 14.6 percent in 1939 and 1940 to 1.2 percent in 1944. The figure for 1939 and 1940 was estimated later. The official yearly records begin in 1944, and 1.2 percent is the lowest rate in them.

Workers in American factories built 299,230 military aircraft from January 1940 to August 1945. Real GDP rose about 72 percent from 1940 to 1945. By 1945 the United States made about half of all the manufactured goods in the world.

BY 1950

If you remember one thing: During the Great Depression, members of Congress set up the first national payments for people out of work or too old to work.

In 1929: The federal government paid no unemployment insurance and no old-age pensions.

In 1950: Old-age checks had gone out since 1940, and members of Congress changed the law to cover farm and domestic workers who worked regularly for one employer.

Next: In 1956 a ship carried 58 shipping containers from Port Newark to Houston, and in 1985 three musicians played the first Farm Aid concert to raise money for farm families.

1950 to 2000

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The income of the typical American family roughly doubled between 1947 and 1973, counted with the rise in prices taken out. Families who were poor, in the middle and rich all gained. In the 1970s prices and unemployment rose at the same time, and after June 1979, when factory jobs peaked at 19.6 million, their number fell.

Mills and farms. About 50,000 people in the Mahoning Valley around Youngstown, Ohio, lost jobs in steel and related work within about five years of 1977. The owners of about 300,000 farms could not repay their loans in the 1980s. By the 2000s, more than 80 of every 100 jobs outside farming were in services, in places such as stores, offices, hospitals and schools. The country went 120 months in a row without a downturn from March 1991 to March 2001.

The postwar boom

Pay rose for families at every level of income for about twenty-five years after the Second World War. Between 1947 and 1973 the median family income, the income of the family in the middle, roughly doubled, counted with the rise in prices taken out. Half of all families earned more than the median family, and half earned less. A general rise in prices like that is called inflation, and it means each dollar buys less than before.

The economists Claudia Goldin and Robert Margo studied the gap between high wages and low wages. They found that the gap narrowed sharply in the 1940s and stayed narrow into the 1970s, and they named this the Great Compression, from compress, to squeeze together.

The G.I. Bill. Under the G.I. Bill, veterans of the Second World War could use federal money for school or job training, and federal officials backed their home loans, promising to repay the lender if a borrower could not. By July 1956, when the first version ended, 7.8 million of the 16 million veterans had used its money for education or job training. From 1944 to 1952, officials of the VA, the federal agency for veterans, backed about 2.4 million home loans.

A duty to keep people working. Members of Congress made "maximum employment, production, and purchasing power" (how much people can buy with their money) a stated duty of the federal government in the Employment Act of 1946. Under the same act they created the Council of Economic Advisers, a group of economists who advise the president.

The ideas behind the act came from the British economist John Maynard Keynes, who argued for government spending against slumps, the times when business falls off. Members of Congress took a promise of full employment out of the bill before they passed it.

Shipping boxes. Malcom McLean's ship *Ideal X* carried 58 large shipping boxes, called containers, from Port Newark to Houston on April 26, 1956. By the economist Marc Levinson's calculation, loading loose cargo onto a ship by hand cost about 5.86 dollars a ton in 1956, and loading it in containers cost about 16 cents a ton.

What was stagflation?

Stagflation was the name for a mix of inflation and a stagnant economy, one that is not growing. In the 1970s prices and unemployment rose together, which most economists had thought could not happen.

The oil embargo. Leaders of Arab oil-selling countries stopped the sale of their oil to the United States in October 1973, a ban on trade called an embargo. The price of oil roughly quadrupled in six months. Consumer prices, the prices of the things people buy for daily life, rose about 11 percent in 1974, and unemployment reached 9.0 percent in May 1975.

Inflation passed 13 percent in 1979 and 1980, and in 1980 prices rose 13.5 percent. Unemployment reached 10.8 percent in November and December 1982, the highest rate since the Great Depression.

Factory jobs fall

A shrinking share of American workers had jobs in manufacturing, making goods in factories, after 1950. In 1955, 30.6 of every 100 jobs outside farming were in manufacturing, by 2005 about 11 of every 100 were, and by the 2020s about 8. The number of factory jobs peaked at 19.6 million in June 1979. By the 2000s, more than 80 of every 100 jobs outside farming were in service industries.

Black Monday. The managers of Youngstown Sheet and Tube announced on September 19, 1977, that they would close the company's Campbell Works steel mill in Youngstown, Ohio. About 5,000 workers lost their jobs at once, and the day became known as Black Monday. Within about five years, about 50,000 people in the Mahoning Valley around Youngstown lost jobs in steel and related work. These numbers come from later news reports.

Gerald Dickey

Who: A Youngstown steelworker who proposed that the workers and the town buy their closed mill and reopen it.

When and where: Youngstown, Ohio, 1977 to 1979.

The mill stayed closed after federal officials said no in 1979.

After the Campbell Works closed, Gerald Dickey proposed that the steelworkers and the people of Youngstown buy the mill and reopen it. To buy it, they needed about 100 million dollars in federal loan guarantees, the government's promise to repay the loans if the borrowers could not. From 1976 to 1979 Dickey was the recording secretary, the keeper of the written record of meetings, of Local 1462, a branch of the United Steelworkers union.

The coalition. Church leaders led the Ecumenical Coalition of the Mahoning Valley, a group that brought together people of different churches, and its members took up Dickey's plan.

The answer. Officials in President Carter's government first signaled that they supported the guarantees. In 1979 officials of the Economic Development Administration, a federal agency, turned the guarantees down, and the mill stayed closed. Accounts differ a little on when officials first signaled support, and all of them agree on the refusal and the closing.

Dickey and the coalition spread the idea of workers owning their own workplaces among unions and their members.

The farm crisis of the 1980s

The owners of about 300,000 farms defaulted in the 1980s, which means they could not repay their loans. About 300 farm banks failed.

Borrowing. Farmers sold more crops abroad in the 1970s, planted more of their land and borrowed at low rates of interest, the extra money a borrower pays a lender for a loan. They borrowed against the value of their land.

Higher rates. Then officials of the Federal Reserve, the country's central bank, which sets interest rates for other banks, raised interest rates to bring down inflation, which passed 13 percent in 1979 and 1980. Under their chairman, Paul Volcker, the rate they set for banks passed 20 percent in 1981. Every new loan now cost farmers more than the cheap loans of the 1970s.

The grain embargo. After Soviet troops invaded Afghanistan, President Jimmy Carter stopped many sales of American farm goods to the Soviet Union, the main rival of the United States. With his order of January 4, 1980, he canceled contracts for about 17 million metric tons of wheat, corn and soybeans (a metric ton is about 2,205 pounds). Farmers lost those sales abroad, and exports fell.

With loans costing more and fewer crops sold abroad, the land that farmers had borrowed against lost value. In parts of the Midwest, farmland lost up to 60 percent of its value from 1981 to 1986. Farm debt reached about 215 billion dollars by 1984, double the debt of 1978.

Farm Aid. Willie Nelson, John Mellencamp and Neil Young played the first Farm Aid concert at Memorial Stadium in Champaign, Illinois, on September 22, 1985. About 80,000 people came, and the three musicians raised more than 7 million dollars for farm families.

The 1980s and 1990s

Income tax rates on the highest incomes fell in the 1980s. Under the Economic Recovery Tax Act of 1981, President Reagan's tax cut, the top income-tax rate, the tax on each dollar of income above a high level, fell from 70 percent to 50 percent. Under the Tax Reform Act of 1986, it fell to 28 percent.

Inflation fell from 13.5 percent in 1980 to between 3 and 4 percent by the middle of the 1980s. The national debt, the total that the federal government owes on money it has borrowed, roughly tripled during the 1980s.

The long boom. The country went 120 months in a row without a downturn, from March 1991 to March 2001. A stretch of growth like this is called an expansion, and it was the longest on record until then, according to the National Bureau of Economic Research, the group of economists that dates the country's booms and downturns.

Unemployment fell to 3.9 percent in the fall of 2000. In the government's budget years of 1998 through 2001, federal tax money coming in was more than federal spending, which is called a budget surplus.

The NASDAQ index, a single number built from the stock prices of many companies, among them many technology companies, rose about 600 percent from 1995 to March 2000.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 8 MINUTES

Three recessions have come since 2000, stretches of months when output falls and people lose their jobs. The longest ran from December 2007 to June 2009, after home prices crashed. In that crash lenders foreclosed on homes, taking them because the borrowers could not keep up the loan payments. They foreclosed on Black and Latino families at higher rates than on white families.

Where things stand. The unemployment rate was 4.1 percent in August 2026, and about 86 of every 100 jobs outside farming were in services. From April to June 2026, the richest tenth of households held about 69 of every 100 dollars of household wealth, and the poorer half held about 2.

The dot-com crash

Technology stock prices crashed from 2000 to 2002. The NASDAQ index peaked at 5,048 on March 10, 2000, and by October 2002 it had fallen about 78 percent. The boom and the crash are called the dot-com bubble, after companies that did business on the internet: prices climbed far above what the companies were worth and then fell. A recession followed. It lasted eight months, from March to November 2001.

The crash of 2008

The Great Recession, from December 2007 to June 2009, began with a crash in home prices. Home prices rose for years before the crash, and lenders made home loans, called mortgages, that many borrowers could not repay. Bankers bundled many of these mortgages together into investments and sold the investments as safe. When home prices fell, the investments failed, and some of the biggest financial firms began to fail too.

Lehman Brothers. The financial firm Lehman Brothers failed on September 15, 2008, owing about 619 billion dollars. It was the largest bankruptcy, the court process for a person or company that cannot pay its debts, in the country's history. It was the same firm that the Lehman brothers began in Alabama, trading cotton grown by enslaved people.

The rescue. Members of Congress passed a rescue plan for financial firms in October 2008, the Troubled Asset Relief Program, or TARP, worth up to 700 billion dollars. Treasury officials paid out about 443 billion dollars under it, and the firms that got the money paid most of it back.

Employers cut about 8.7 million jobs in the Great Recession, and unemployment peaked at 10.0 percent in October 2009. Lenders foreclosed on about 3.8 million homes from 2007 to 2010, and they kept taking homes for years after. Counting those later years, estimates for the whole crisis run from 7.7 million to 10 million homes.

Who lost homes and wealth

Black and Latino homeowners lost their homes at higher rates than white homeowners, and they lost more of their wealth. Latino and Hispanic both name Americans whose families came from Spanish-speaking countries, and African American is another name for Black Americans.

The foreclosure counts. The Center for Responsible Lending, a research group that campaigns on lending, studied the people who took out home loans from 2005 to 2008. By 2010, lenders had foreclosed on about 8 percent of the African American borrowers in that group and 8 percent of the Latino borrowers, against 4.5 percent of the non-Hispanic white borrowers. Because there were more white borrowers, white families still lost more than half of all the homes that lenders took.

Countrywide. Lawyers at the Justice Department said that from 2004 to 2008, Countrywide Financial, a mortgage lender, charged at least 200,000 qualified Black and Latino borrowers higher fees and rates or steered them into subprime loans, loans with higher interest rates meant for borrowers that lenders think are more likely to fall behind.

Bank of America had bought Countrywide, and in December 2011 its officers agreed to pay 335 million dollars in a settlement, an agreement to pay money to end a legal case. It was the largest settlement up to that time in a case about lenders treating borrowers unfairly because of their race.

Wells Fargo. The Justice Department's lawyers said Wells Fargo charged Black and Latino borrowers higher rates and fees, and steered some into subprime loans when they qualified for regular ones. The lawyers said at least 34,000 borrowers were treated this way from 2004 to 2009.

Officers of the bank agreed on July 12, 2012, to pay at least 175 million dollars. Of that, 125 million dollars went to borrowers, and 50 million dollars went to help with down payments, the part of a home's price that the buyer pays at the start.

Assistant Attorney General Tom Perez said that a Black borrower in the Chicago area with one kind of Wells Fargo loan paid on average 2,937 dollars more than a similar white borrower. Wells Fargo's officers denied that the bank discriminated, treating people unfairly because of their race, and said they settled to avoid a fight in court.

What families lost. Researchers at the Pew Research Center reported in 2011 on wealth, or net worth, what a household owns minus what it owes. From 2005 to 2009, with the rise in prices taken out, the wealth of the typical Hispanic household fell 66 percent. For the typical Black household it fell 53 percent, and for the typical white household 16 percent.

The typical white household had 113,149 dollars in wealth in 2009. That was 20 times as much as the typical Black household had, and 18 times as much as the typical Hispanic household had. These were the largest gaps since government officials began collecting the numbers in 1984.

About 35 of every 100 Black households and 31 of every 100 Hispanic households owned nothing or owed more than they owned, against 15 of every 100 white households.

Ed Neufeldt

Who: A worker who built cabinets for recreational vehicles in Elkhart, Indiana, lost his job in 2008 and introduced President Obama in 2009.
When and where: Elkhart, Indiana, from 2008 into the middle of the 2010s.

He built RV cabinets for 32 years, then lost his job.

Ed Neufeldt worked at Monaco Coach, a plant in Elkhart, Indiana, until the fall of 2008. The plant built RVs, recreational vehicles that are homes on wheels, and Elkhart was known as the "RV capital of the world." As the RV plants shut, unemployment in Elkhart's county went from about 5 percent to nearly 20 percent in about a year.

The president's visit. Neufeldt introduced President Obama at a town hall meeting in Elkhart on February 9, 2009. Obama was on his first trip to promote the stimulus bill, a plan to spend federal money to bring back jobs.

Reporters from ProPublica, NPR and NBC News followed Neufeldt for years. He worked three low-wage jobs, and later he got a factory job again. NPR reporters visited him again in 2015 and 2016.

The long recovery and the shift to services

The recovery after 2009 was slow at first, with years of high unemployment. It then became the longest expansion in the country's records, which go back to 1854. It lasted 128 months, from June 2009 to February 2020, and the National Bureau of Economic Research announced these dates on June 8, 2020.

Gig work. Researchers at the Pew Research Center found that by August 2021, 16 percent of American adults had earned money at some time through an online gig platform, an app or a website that connects workers with short, one-time jobs.

Where the jobs are. The country had 159.1 million jobs outside farming in August 2026. About 12.6 million of them, about 8 percent, were in manufacturing, and about 137.5 million, about 86 percent, were in services, counting government jobs. The Bureau of Labor Statistics, the government office that counts jobs and prices, gave these numbers. They are early figures, and the bureau's staff can still change them.

2020 and the price rises after

The shortest recession on record lasted two months, from February to April 2020, as the COVID-19 pandemic, a disease that spread around the world, began. Unemployment reached 14.8 percent in April 2020.

The CARES Act. Members of Congress passed the CARES Act, a relief law of about 2 trillion dollars, on March 27, 2020. It was the largest relief law up to that time. GDP fell at a yearly rate of 31.4 percent from April to June 2020, which means output would have fallen that far in a full year at that speed. From July to September, output rose at the fastest rate on record.

Prices jump. Consumer prices rose 9.1 percent in the 12 months to June 2022, according to the Bureau of Labor Statistics, the largest rise since November 1981. By the middle of 2023, prices were rising near 3 percent a year.

Economists disagree about the causes. Some blame breakdowns during the pandemic in supply chains, the chains of factories, ships and trucks that bring goods to stores. Others point to federal relief spending, or to higher energy prices after Russian forces invaded Ukraine.

The economy in 2026

Who holds the wealth. In 2026 the richest tenth of American households held most of the country's wealth, and the poorer half held almost none of it. Economists at the Federal Reserve, the country's central bank, keep the count. Of every 100 dollars of household wealth from April to June 2026, the richest 10 percent of households held about 69 dollars, the next 40 percent held about 29, and the poorer half held about 2.

The same economists also divide the wealth by race. Households they count as white held about 82 of every 100 dollars, Black households about 3, and Hispanic households about 2.

Jobs and prices. The unemployment rate was 4.1 percent in August 2026, about 7.0 million people. It was 6.0 percent for Black workers, 4.8 percent for Hispanic workers, 3.7 percent for white workers and 3.2 percent for Asian workers. Consumer prices rose 3.4 percent in the 12 months to August 2026. As of September 2026, the National Bureau of Economic Research had announced no recession since the two months in 2020.

THE THREE QUESTIONS, ANSWERED

1. **How did slavery make money for people far from any plantation?** Bankers in London, Amsterdam and New York sold bonds backed by mortgages on enslaved people, Northern companies insured their lives, and Boston mill owners spun the cotton that enslaved people grew. Rhode Island merchants sent out ships that carried more than 100,000 Africans into slavery.
2. **What happened to ordinary families when a boom turned into a crash?** Judges jailed debtors in 1819, creditors sold enslaved families apart in 1837, one worker in four had no job by 1933, and lenders took 7.7 million to 10 million homes in the crisis that began in 2007. In the 1930s and after 2007, Black families lost jobs and homes at higher rates than white families.

3. **When did the national government start helping people who lost their jobs?** In the 1930s. Before 1933 the federal government paid no unemployment insurance or old-age pensions, and members of Congress set up both in the Social Security Act of 1935, though farm and domestic workers had to wait until 1950 and 1954.

What changed: People here went from trading by barter, with no money at all, to working for wages, and by 2026 about 86 of every 100 jobs outside farming were in services.

FIVE DATES TO KEEP

1614: John Rolfe sends his first tobacco to London.

1860: Census takers count 3,953,760 enslaved people.

October 29, 1929: Stock prices fall hard on Black Tuesday.

August 14, 1935: President Roosevelt signs the Social Security Act.

September 15, 2008: Lehman Brothers fails.

> NOT KNOWN: Nobody kept a regular count of the jobless in 1837, in 1893 or in the early 1930s, so every figure for them is a later estimate.

Where to go: Lowell National Historical Park, in Lowell, Massachusetts, tells the story of the Lowell mills and the slave-grown cotton they spun.

Next: Chapter 17, Money. It follows coins, paper dollars, banks and the Federal Reserve.

Chapter 17: Money

CHAPTER 17 OF 37 · MONEY · 10 ERAS · ABOUT 70 MINUTES

Most American money since 1690 has been a written promise to pay, and when the people who made the promise could not keep it, the people holding the money took the loss.

Soldiers from Massachusetts came home from a failed attack on Quebec in 1690, and the colony's treasury held too few coins to pay them. In December the colony's lawmakers began voting to pay them in paper. Each bill stated that the treasurer would accept it "in all Public payments" and that it could later be traded for coins. No government in Europe or its colonies had printed paper money before. People took the bills and passed them from hand to hand.

This chapter follows money from the shells and copper traded before 1500 to bitcoin. As you read, watch what stood behind a dollar, from Spanish silver to nothing but the law. Watch, too, who took the loss when a promise to pay was broken: whoever held Continental bills in 1781, savers at the banks that failed from 1837 to 1933, and about 31,000 depositors of the Freedman's Bank who never got any money back. In Louisiana in 1837, enslaved people pledged for loans were sold to pay the debts.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. What has stood behind the dollar, and what stands behind it now?
2. Why did people rush to pull their money out of banks, and who lost when a bank failed?
3. How did enslavers and bankers use enslaved people to raise money?

Five minutes? Read "1930 to 1933: the runs, the holiday and deposit insurance" in 1900 to 1950. The Lydia Lobsiger story follows it.

Can skip: "The ATM", "War bonds", "Payday loans". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 1 MINUTE

Wampum, beads made from shell, later passed as money between colonists and Native traders. Before 1500, though, nobody in this land used money of any kind: there were no coins, paper bills, banks or prices. People got what they needed by barter, trading one thing straight for another, and by gifts, and Chapter 16 tells how they carried shells, copper and obsidian, a volcanic glass, hundreds of miles.

Was wampum money?

Wampum was not money before Europeans came. People used it to keep records, to make agreements between nations, in ceremonies and as gifts.

The white beads came from the shells of whelks, large sea snails. The purple-black beads came from the shells of quahogs, hard-shelled clams of the Atlantic coast. People strung the beads on cords or wove them into belts.

Among the Haudenosaunee, a league of Native nations in what is now New York State, the beads of a belt recorded spoken agreements and histories. People presented the belts at meetings of leaders, called councils, and read them there.

Wampum first passed as money in the 1600s, when colonists and Native traders used it in their dealings with each other.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

The American dollar was later based on a Spanish silver coin, and Spanish officials forced Native men to dig the silver for it. The mines lay far to the south, in lands the king of Spain ruled. Workers at Spanish mints, workshops where coins are made, struck the silver into a coin called the piece of eight. Almost none of those coins reached the land that became the United States in the 1500s. This land still had no mint of its own.

The piece of eight

The piece of eight was a silver coin worth 8 of the Spanish money units called reales. In English it was also called the Spanish dollar. Workers at Spanish mints in the Americas made it from silver dug at Potosí, in what is now Bolivia, and at Zacatecas, in Mexico. Silver was found at Potosí in 1545.

Native men of the Andes, the high mountains of South America, did the digging at Potosí. Francisco de Toledo, the viceroy who governed Peru for the king of Spain, began forcing thousands of them to work in the mines in 1573. They worked in turns, under a system called the mita. Toledo took the name and the idea of turns of work from an older system used by the Inca.

Toledo's rules applied to Native men aged 18 to 50 in the highland provinces between Potosí and Cuzco. Each year about one man in seven from those villages had to go to the mines, more than 11,000 men a year.

The men worked in shifts, one week in the mines and then two weeks of rest. Many brought their families and carried most of their own food. They were paid a little, and most of that pay went to the tax they owed the colonial government. The work was dangerous and unhealthy. By sending the men away from their home villages, Toledo broke up life there too.

Across the Andes, disease killed far more Native people than work in the mines did. No reliable count shows how many of the forced workers died from the work at Potosí.

Traders carried the piece of eight from South America to Europe, and across the Pacific Ocean to Asia. Writers on the history of money, among them Alejandra Irigoin and Bridget Millmore, call it the first money used around the whole world.

Almost none of that silver reached this land. Some did reach St. Augustine, the Spanish military post in Florida that Chapter 16 describes. Pedro Menéndez de Avilés had founded it in September 1565 at Seloy, a Timucua village. Its soldiers were paid from Spanish colonial funds, in the coin of a foreign empire.

Later, Thomas Jefferson and members of Congress based the United States dollar on the piece of eight. Until February 21, 1857, Spanish coins were legal tender in the United States: the law said people had to accept them to pay a debt. The piece of eight was divided into eight parts called bits, so two bits made a quarter of a dollar. Americans still call a quarter "two bits."

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 4 MINUTES

English colonists never had enough coins, so they paid each other in goods and then began to make money of their own. Little English coin reached the colonies. Colonists spent what they did get on goods shipped in from overseas, and those coins went back across the ocean. To trade with each other, they paid in tobacco, corn, beaver skins and wampum. Massachusetts leaders opened a mint without England's permission. In 1690 they began printing paper money.

Money made of goods

Colonists called payment in goods "country pay." Colonial lawmakers set how much each kind of goods was worth when it was used to pay a debt. Money of this kind is called commodity money, because the money is a useful thing in itself.

English money was counted in pence, shillings and pounds. Twelve pence made a shilling, and twenty shillings made a pound.

Virginia lawmakers set a value on tobacco so it could be used as money from the 1620s, and in 1642 they made it legal tender. People in Virginia, Maryland and North Carolina paid with tobacco through much of the colonial period. In the northern colonies, beaver pelts and corn passed as money too.

New England colonists used wampum for small payments, though the nations that made it had not used it as money before colonists came. Members of the Massachusetts General Court, the colony's lawmakers, accepted wampum for small sums from 1637, made it legal tender for small debts in 1650 and canceled that law in 1661.

The pine-tree shilling, 1652

Members of the Massachusetts General Court voted in 1652 to set up a mint in Boston. They did not ask anyone in England first. By English law, only the king could make coins. In 1652 England had no king.

Why England had no king. Supporters of the king had fought a civil war against supporters of Parliament, England's lawmakers. Afterward the judges of a court set up by Parliament found King Charles I guilty of making war on Parliament and on the people of England. Their sentence called him a "traitor" and a "murderer." Fifty-nine of the judges, Oliver Cromwell among them, signed the order for his death. On that order the king was beheaded in London on January 30, 1649.

Workers at the Boston mint struck silver coins worth threepence, sixpence and a shilling. One was the pine-tree shilling, named for the tree stamped on it. They struck pine-tree shillings from about 1667 to 1682, but nearly all of the colony's silver coins carry the date 1652.

A date as a cover. Coin experts who write in the magazine CoinWeek and in older books on early American coins explain the 1652 date this way. Charles II became king of England in 1660. A coin dated 1652 could pass as a leftover from the years without a king. A coin with a later date would show that the men of Massachusetts were still making coins without the king's permission.

The mint closed in 1682. In 1684 officials acting for King Charles II canceled the Massachusetts charter, the legal paper that gave the colony the right to exist and stated what it could do. The illegal mint was one of their complaints against the colony.

John Hull

Who: A Boston silversmith who ran the colony's mint against English law and struck the pine-tree shilling.

When and where: Boston, Massachusetts. Mintmaster from 1652 until the mint closed in 1682. Lived 1624 to 1683.

He kept a share of every batch of coins he struck.

John Hull ran the Massachusetts mint from 1652 until it closed in 1682, without permission from England. He was a blacksmith's son from Market Harborough, England, and came to Boston with his parents in 1635, when he was 11. He trained as a silversmith, a craftsman who makes things out of silver.

Members of the Massachusetts General Court named him mintmaster, the man in charge of the mint, in 1652, when he was 27. His appointment was dated May 27. Robert Sanderson worked with him as his partner. Under his contract he was paid in coins from every batch, and that pay made him one of the richest men in New England. He later became the colony's treasurer, the official in charge of its money.

A story says Hull gave his daughter Hannah her weight in pine-tree shillings as her dowry, the money or goods a bride's family gives when she marries. The story is a legend. The writer Nathaniel Hawthorne retold it in his book *Grandfather's Chair* and added details of his own.

The first government paper money, 1690

The soldiers in the chapter's opening were paid in paper bills called bills of credit, and the first ones, voted from December 1690, totaled 40,000 pounds. The soldiers had come home from a failed attack on the French city of Quebec, in Canada, during King William's War. According to accounts from the time, the unpaid troops were close to mutiny, an uprising against their officers.

These bills were the first paper money printed by any government in Europe or the lands Europeans had settled. China had used paper money centuries earlier.

The paper spreads. By 1714 Massachusetts lawmakers had put out 240,000 pounds in paper money. By 1718 lawmakers in the other New England colonies were printing their own.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

Lawmakers in colony after colony printed paper money in the early 1700s, until every colony had its own. The bills held their value only where the lawmakers printed limited amounts.

Benjamin Franklin argued for paper money in Pennsylvania and then printed it. In South Carolina, which had no banks, enslavers borrowed money by pledging land and the people they enslaved. If a loan was not paid back, the lender could take the land or the enslaved people who had been pledged.

Paper that held its value and paper that lost it

Economists at the Federal Reserve Bank of Cleveland, who study how money and trade work, compared the paper money of the colonies. In Pennsylvania, New York and New Jersey, lawmakers printed limited amounts, and the bills held their value. In New England and the Carolinas, lawmakers printed large amounts. Prices counted in paper money rose there, and each bill was worth less silver than before.

South Carolina's paper. Lawmakers there began printing paper money in 1703, when 1.5 South Carolina pounds were worth one English pound. By the late 1720s it took about 7.

In Virginia, people began to pass paper receipts for tobacco as money. Virginia lawmakers set up public warehouses for tobacco, with government inspectors, in the 1720s and 1730s. A planter, a farmer who grew tobacco to sell, brought his tobacco to a warehouse. An inspector checked it, and it stayed in the warehouse. The inspector then wrote a receipt, called a tobacco note, for the inspected tobacco. People passed tobacco notes from hand to hand as money.

Benjamin Franklin

Who: A Pennsylvania printer who argued for paper money, then printed it with leaf patterns that were almost impossible for counterfeiters, people who make fake money, to copy.

When and where: Pennsylvania, from 1729.

Movie: *Benjamin Franklin* (2022), a documentary by Ken Burns, shown on PBS. It is about his whole life, not only his work on money.

He printed the veins of real leaves on paper money.

Benjamin Franklin printed paper money for three colonies and made it hard to fake. In 1729, when he was 23, he published a pamphlet, a short printed booklet, arguing that Pennsylvania needed more paper money. Its title was *A Modest Enquiry into the Nature and Necessity of a Paper-Currency*, a modest look at what paper money is and why it is needed.

Partly because of the pamphlet, Franklin won the contract to print Pennsylvania's bills in his shop. His firm printed money for Pennsylvania, New Jersey and Delaware.

To stop counterfeiters, Franklin printed the shapes of real leaves on the bills. Every leaf has its own pattern of veins. A counterfeiter could almost never carve that pattern into a printing plate by hand. The method is called nature printing. Franklin almost certainly worked it out from leaf prints he made with his friend Joseph Breintnall. The printing plates were probably cast in metal from plaster molds of real leaves.

Scientists at the University of Notre Dame looked closely at 600 notes printed by Franklin's firm. In their 2023 study they found the leaf prints, tiny shiny flakes of a mineral called mica in the paper, and colored fibers mixed into it. Printers kept putting leaf prints like Franklin's on colonial money into the Revolution.

Colonial bills carried the words "To Counterfeit is Death." Under colonial laws, judges could sentence a counterfeiter to death.

Paper money lent against land and people

South Carolina's lawmakers lent some of the colony's paper money out on land. In 1712 they set up a public land bank, through which officials lent paper money to people who pledged their land as security, something a borrower promises to hand over if a loan is not paid back.

People also borrowed from one another through private agreements. One kind was a bond, a written promise to pay back a loan. Another was a mortgage, a loan in which the borrower pledges property as security, so the lender can take that property if the borrower does not pay.

A mortgage could be on land, or on movable property, called chattel. In these loan papers, enslavers and lenders listed enslaved people as chattel. So when an enslaver did not pay, the lender could take the people he had pledged.

Nic Butler of the Charleston County Public Library described in 2024 how these loans worked, including loans from slave brokers, dealers in enslaved people. "A slave owner could mortgage their human chattel to a slave broker to secure credit used to purchase more slaves," he wrote. The enslaver pledged the people he enslaved to get a loan and used the loan to buy more people. Then he paid off the debt with what the people's forced work earned.

Hundreds of colonial land mortgages survive in the Charleston County Register of Deeds, the county office that keeps records of land. Thousands of early chattel mortgages survive in the South Carolina Department of Archives and History.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 6 MINUTES

Paper money was one of the colonists' quarrels with Britain, and then it was how they paid for the Revolution. Members of the British Parliament banned new paper money as legal tender in New England in 1751 and in all thirteen colonies in 1764. Colonists added the ban to their complaints against Britain. During the war, members of the Continental Congress paid with paper bills that were nearly worthless by 1781.

After the war, Thomas Jefferson, Alexander Hamilton and members of Congress planned a new money system. Members of Congress chose a dollar divided into tenths and hundredths in 1785, set up a national bank in 1791 and created a mint in 1792.

The Currency Acts

Members of Parliament, Britain's lawmakers, passed the first Currency Act in 1751 because New England's paper money had lost so much of its value. Under the act, Massachusetts Bay, Rhode Island, Connecticut and New Hampshire could no longer print new paper bills as legal tender.

A second Currency Act followed thirteen years later. On April 19, 1764, members of Parliament extended the ban to all thirteen colonies. Colonists were already short of coins. Without new legal-tender paper, they had even less money to pay debts and buy goods. Colonists listed the Currency Acts among their complaints against Britain.

Not worth a Continental

People working for the British side printed fake Continental bills on purpose, so that the real bills would lose their value faster. Continental bills were the paper money that members of the Continental Congress, the men from the thirteen colonies who led the Revolution, printed from June 1775 to pay for the war. In all, they put out $241,552,780 in Continental bills, and Chapter 16 tells how fast the bills lost their value.

Made in New York. Most of the British faking was done in New York City, which the British army held for most of the war. Someone placed an advertisement in the *New York Gazette* of April 14, 1777, asking people going into the other colonies to take fake bills with them.

The general's letter. General Henry Clinton, the British commander, wrote about the plan in January 1780. His letter went to Lord George Germain, the official in London in charge of the American colonies. Stuart Hatfield, writing about the British faking in the *Journal of the American Revolution* in 2015, calls that letter the clearest sign that British leaders approved the plan.

People had stopped using the paper by 1781. In North Carolina it took 210 paper dollars to buy what one silver dollar bought.

Continental bills with about $226 million printed on them ended up worthless. Counted in silver, the loss was about $40 million. Whoever was holding the bills as they lost value lost that money, and most of them were ordinary people. The saying "not worth a Continental" comes from these bills, and by the late 1800s people in Britain were using it too.

Counterfeiting was a hanging crime

Officials in New York hanged Owen Sullivan, whose name was also spelled Syllavan, in 1756 for faking that colony's paper money. Colonial bills were printed with the warning "To Counterfeit is Death," and New York officials carried it out.

Sullivan was born in Ireland in about 1720. As an engraver, he cut designs into metal plates for printing, and as an armorer he had made and repaired weapons for colonial soldiers. He ran gangs of counterfeiters who passed fake colonial bills from New England to New York.

Before that, officials in Rhode Island had punished Sullivan for faking that colony's bills. On October 9, 1752, the *Boston Evening Post* printed a report from Providence, Rhode Island. The report states that he had "lately had both his Ears crop'd, and been branded on both his Cheeks with the Letter C."

Cropping meant cutting off part or all of each ear. Branding meant burning the letter C, for counterfeiter, into the skin of each cheek with a hot iron. The C marked him as a counterfeiter to anyone who saw his face. The report does not name the officials who carried out the punishment.

Sullivan went on counterfeiting. At a session of the Supreme Court for the City and County of New York, he was found guilty of faking and passing New York's bills of credit from 1737. He was hanged on Monday, May 10, 1756. A pamphlet about his life, which says it was "taken from his own mouth," was printed the same year.

Why a dollar has a hundred cents

A dollar has a hundred cents because the men who planned American money chose to count it by tens. In notes he wrote in 1784, Thomas Jefferson proposed the Spanish dollar as the basic unit, divided into tenths and hundredths. The English system of pounds, shillings and pence did not divide by tens.

Members of Congress voted on July 6, 1785, "that the money unit of the United States of America shall be one dollar." They also voted that the coins should "increase in decimal ratio," counting up by tens. With that vote, the United States became the first nation whose coins were fully based on tens.

Under the Coinage Act of April 2, 1792, members of Congress set up the United States Mint in Philadelphia. The Mint's building was the first building the national government put up under the Constitution. The act defined the dollar as the unit of money. Under it, the decimal system of dollars, dimes, cents and mills became law (a mill is one tenth of a cent).

The act listed coins in three metals: gold coins, including a ten-dollar coin called the eagle, silver coins from the dollar down to the half-dime, and copper coins, the cent and the half-cent.

Alexander Hamilton

Who: The first Secretary of the Treasury, who planned the national bank and the United States Mint.

When and where: Secretary of the Treasury, 1789 to 1795. Lived 1757 to 1804.

Movie: *Alexander Hamilton* (2007), a documentary in the PBS series *American Experience*.

Jefferson and Madison said the Constitution did not allow his bank.

Alexander Hamilton planned the national bank and the Mint in reports he wrote for Congress. President George Washington chose him on September 11, 1789, to be the first Secretary of the Treasury, the head of the government department in charge of money.

Hamilton sent Congress his *Report on a National Bank* in December 1790. He proposed a bank that would hold the government's money and put out banknotes, paper money printed by a bank, that people could trust.

Lawmakers in Congress then passed a bill to charter the First Bank of the United States, in Philadelphia, for twenty years. Thomas Jefferson and James Madison argued that the Constitution did not allow Congress to create a bank. In the House, Madison asked whether the power to start a bank was "among the powers" listed for Congress in the Constitution.

President Washington asked Hamilton to answer the objections, so that he would have "the arguments for and against the measure." Then Washington signed the bill, on February 25, 1791. Twenty years later, in 1811, a new charter lost by one vote in each house of Congress.

Hamilton had already sent Congress his *Report on the Establishment of a Mint*, in January 1791. In it he defined the dollar in both silver and gold, with 15 ounces of silver worth 1 ounce of gold. Members of Congress built the Coinage Act of 1792 on his plan and set up the Mint under it.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 11 MINUTES

Americans in these years paid with paper money printed by hundreds of different banks. Some of that paper was lent out against enslaved people, and some against land that had belonged to Native nations.

The country had a national bank twice before 1850, and both banks lost their federal charters, the government papers that allowed them to exist. The First Bank's charter ran out in 1811. President Andrew Jackson vetoed, or refused to sign, a new charter for the Second Bank in 1832, and its federal charter ran out in 1836.

With no national bank and no national paper money from 1836 until 1862, people used gold and silver coins and the notes of state banks. A bank's notes were worth less the farther from that bank a person tried to spend them.

Banknotes lent against enslaved people

Planters in Louisiana, the enslavers who ran large farms, had set up banks by the 1830s that lent out their own paper money against mortgages on enslaved people. In such a loan the enslaver pledged the people he enslaved, and the bank could take them if he did not pay. The paper notes people carried in their pockets in Louisiana had been lent out against enslaved human beings.

At the Consolidated Association of the Planters of Louisiana, an enslaver pledged "slaves and cultivated land" to the bank in a mortgage. He could then borrow up to half of what the bank's officers judged those people and that land to be worth. He received the loan in the bank's own paper notes.

To raise money to lend, the banks put out bonds. Each bond was a paper promise to repay a loan with interest, an extra payment for the use of the money. Louisiana lawmakers backed $2.5 million of the Consolidated Association's bonds, at 5 percent interest, which meant the state promised to pay the bonds if the bank did not.

Baring Brothers, bankers in London, paid the money up front and sold the bonds to investors in Europe. Chapter 16 lists the other Louisiana banks that raised money this way and the firms that sold their bonds.

A typical bond was for $1,000. Edward E. Baptist wrote in 2010 that this was "roughly the price of a field hand," an enslaved man who worked in the fields. An investor in Europe could own a share of a debt backed by thousands of enslaved people without owning any one person.

Before the banks. The merchant Vincent Nolte lent $48,000 to the Louisiana enslaver Alonzo Walsh in 1824. Walsh pledged "from 90 to a 100 head of first rate slaves," using head, the word for counting cattle. The account of the loan gives no names for these people. At New Orleans auction prices they were worth up to $80,000.

The crash. Many enslavers had placed several mortgages on the same person. Then cotton prices fell, from 18 cents a pound in 1834 to under 10 cents by late 1836, and by March 1837 ten large New Orleans cotton firms had failed.

When enslavers could not pay, the people they had mortgaged were sold in forced sales to pay their debts. The accounts of the crash do not name who ran each sale. Buyers spent what little cash they had at the first sales, and then the prices of land and of enslaved people fell sharply.

Chapter 16 tells the rest of the crash. It explains why some enslavers kept the people they held, and it gives a 2005 bank report's count of the enslaved people that two Louisiana banks took as security.

The Bank War

The Bank War was the fight over whether the country should keep a national bank, a bank chartered by the national government to hold the government's money. Members of Congress chartered the Second Bank of the United States in 1816, for twenty years, after the money troubles of the War of 1812. Its main office was in Philadelphia.

Nicholas Biddle became president of the Bank in 1823. Under him the Bank held the government's deposits, the money the government kept in the Bank's care, and moved money across the country. Its officers also collected the notes of state banks and presented them to those banks for payment in gold and silver coins. A state bank had to hold enough coins to pay, so its owners could not print as many notes as they wanted.

Andrew Jackson and his supporters argued that the Bank was a private company with public power, with owners in other countries and no control by voters. They said its officers gave favors to rich people with connections. Biddle and the Bank's defenders argued that with the Bank, the country's money was worth the same everywhere and kept its value. They said that without it, nothing would stop state bankers from printing too much paper.

Henry Clay was running against Jackson for president in 1832. That July, four years before the old charter ran out, Clay led the push for a new one, and members of Congress passed it. Jackson vetoed the charter on July 10, 1832.

The veto. In his veto message Jackson called the Bank "unauthorized by the Constitution" and "dangerous to the liberties of the people." He said it weakened the rights of the states. That fall Jackson beat Clay and won a second term.

The deposits. Jackson ordered the government's money moved out of the Bank in 1833 and into state banks he chose, which his opponents called "pet banks." Two Treasury secretaries would not move the money, and Jackson replaced them. Roger Taney, the third, moved it. Biddle struck back in 1833 and 1834 by cutting the Bank's lending to force a new charter. Borrowing became hard, and Biddle lost supporters.

Members of the Senate censured Jackson on March 28, 1834, voting to blame him officially for what he had done. The Bank's federal charter ran out in 1836, and the Bank, which went on as a Pennsylvania state bank, failed for good in 1841.

Andrew Jackson

Who: The president who vetoed the Second Bank's new charter and moved the government's money out of it.

When and where: The United States. Veto in 1832, removal of the deposits in 1833.

Movie: *Andrew Jackson: Good, Evil and the Presidency* (2007), a two-hour PBS documentary narrated by Martin Sheen. It covers his whole presidency, not only the Bank War.

He sold 68,000 acres for paper notes that turned out worthless.

Andrew Jackson distrusted banks and paper money after he lost money on a land deal. In August 1795, at 28, he sold 68,000 acres of land to David Allison. The accounts of the sale do not say where the land lay or which Native nation had lived on it. Jackson was selling for himself, for John Overton and for others.

Allison paid with promissory notes, written promises to pay money later, and Jackson used the notes to buy supplies for a trading post. In 1797 Allison could not pay his debts, and his notes were worth nothing. The people holding the notes came to Jackson for payment instead. It took him about 15 years to pay his way out.

Allison's creditors, the people he owed, had him put in debtors' prison in Philadelphia, a jail for people who could not pay what they owed. He died there on September 28, 1798.

A 2001 article in *American Heritage* magazine describes what Jackson took from all this. By the end, the article states, he had settled ideas about debt, paper money and speculation, which is buying something only to sell it later at a higher price. "He hated them all."

State banknotes and counterfeit detectors

Each bank that state lawmakers chartered printed its own notes, in its own designs. A note was worth only as much as people trusted the bank that printed it. Far from that bank, people would take its notes only for less than the amount printed on them, and the difference was called a discount.

Store owners checked notes against a printed guide before they took them. The guides were called bank note reporters and counterfeit detectors, and publishers put them out weekly or monthly from about 1826 to 1866. Each issue listed the discount on each bank's notes and described the fake notes going around. Shopkeepers kept them by the cash drawer.

Some banks were called wildcat banks. They were set up under state laws, called free-banking laws, with few rules for opening a bank. The story went that some stood so far out in the woods that only wildcats lived nearby, so nobody could get there to trade the notes for coins.

Economists today, among them Gerald Dwyer and the team of Rolnick and Weber, have found that most free banks were not frauds. The people holding their notes did lose money, but less than the legend says.

Whose land the Michigan banks lent against

Michigan's wildcat bankers lent money in the late 1830s against land in the southeast of the state, around Ann Arbor. Until 1807 that land belonged to the Potawatomi, Ojibwe, Odawa and Wyandot nations.

In the Treaty of Detroit, a written agreement between nations, leaders of those four nations signed the land over to the United States on November 17, 1807. William Hull, the governor of Michigan Territory, signed for the United States. The land included the place that later became Ann Arbor.

How much land, for how much. United States officials paid $57,717.32 for the land, according to a history page of the Ann Arbor District Library, which puts the land's estimated value at $6,400,000. That price was less than 1 dollar for every 100 dollars the land was worth. The library counts the land at about 5.6 million acres, and the treaty page of the Nottawaseppi Huron Band of the Potawatomi counts it at about 8 million.

Alpheus Felch

Who: A Michigan bank commissioner, later the state's governor and a US senator, who wrote down the frauds he found in 1838.

When and where: Michigan, 1838 and 1839. Lived 1804 to 1896.

Under a top layer of coins, the boxes held nails and glass.

Alpheus Felch was one of three officials who inspected Michigan's banks in 1838, and he wrote down the frauds he found. A lawyer in Monroe, Michigan, he had been one of four state lawmakers who voted against Michigan's free-banking law.

Michigan lawmakers passed that law on March 15, 1837. Before it, Michigan had 9 banks. Within months there were 18, and then 40. Three bank commissioners were appointed on December 30, 1837, to inspect them, and Felch served as one of them in 1838.

Felch described the inspections in 1880. Some banks kept "grossly insufficient books," records far too poor to show what the bank held. One bank had nothing of value at all. At another, bankers showed the commissioners boxes of coin. Under a top layer of coins, the boxes were filled with nails and glass.

At three banks in a row, the bankers showed the commissioners the same coins as their own. Local accounts say the bankers sent word ahead of the commissioners' route and carried the coins from bank to bank.

The commissioners could close a bank by court order. They could also hand it to a receiver, an official who takes over a failed bank and pays its debts from what is left. After the commissioners' reports, many banks closed and criminal cases followed in the courts.

Michigan lawmakers suspended the free-banking law in 1838. By September 1839 only 9 banks were left. In 1845 the judges of the Michigan Supreme Court ruled that the law broke Michigan's constitution, the state's basic set of laws.

How much people lost. People holding the notes of Michigan's failed banks lost an estimated 60 percent of the amount printed on them, according to the Ann Arbor District Library. A note printed for 10 dollars ended up worth about 4.

Economists' estimates of the total lost run from about $350,000, for the "perhaps a dozen" true wildcat banks alone, to about $1 million, for all of Michigan's failed free banks. Some of those economists argue that the judges' 1845 ruling against the law itself made many notes worthless.

The Panic of 1837: the runs

Chapter 16 describes what the Panic of 1837 did to jobs and food prices. At the banks, it began with a rush for coins.

In a money panic, fear spreads and many people try to get their money out of banks at once. At a single bank, that rush is called a run: people crowd in to trade the bank's notes or their deposits for coins. A bank keeps only part of its money in coin, so it cannot pay everyone at the same time.

The officers of New York City's banks stopped paying out gold and silver for their notes and deposits on May 10, 1837. Bank officers across the country did the same. This was called suspending specie payments, specie meaning coins of gold and silver. People who lined up at the banks got paper promises or nothing.

KNOWN:	Hundreds of the country's about 850 banks closed in the hard times that lasted to about 1843, about 4 in 10 by the count most often given.
NOT KNOWN:	No surviving record counts the depositors who lost their savings.

1850 to 1900

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Members of Congress gave the country one national paper money during the Civil War. For the rest of the century, Americans argued over what should stand behind it: gold alone, or gold and silver together.

Before the war, more than 10,000 different kinds of banknotes were in use by 1860. Federal officials estimate that from one-third to nearly one-half of that paper money was fake. Members of Congress created the national paper money in 1862 and then taxed state banknotes until bankers stopped printing them.

Greenbacks

The new national paper money was printed with green ink on the back, so people called the notes greenbacks. Members of Congress passed the Legal Tender Act on February 25, 1862, to help pay for the Civil War. Under the act, they approved $150 million in paper money called United States Notes, and later acts raised the total to $450 million.

No gold behind them. Nobody could trade these notes at the Treasury for gold or silver. They were money because members of Congress made them legal tender, which means the law said people had to accept them to pay debts.

Salmon P. Chase, the Treasury Secretary, managed the new paper money. He put his own portrait on the first $1 United States Note, in 1862.

Members of Congress created national banks, which got their charters from the national government, in the National Banking Acts of February 25, 1863, and June 3, 1864. These banks put out national bank notes of one design, backed by government bonds.

The tax. In 1865 members of Congress put a tax of 10 percent on state banknotes. Bankers stopped printing their own notes. The state notes still in use fell from $143 million in 1865 to $4 million by 1867. By then, people everywhere in the country used the same national paper money.

In God We Trust. The words "In God We Trust" first appeared on American coins in these same years. In November 1861 a Pennsylvania minister, the Reverend M. R. Watkinson, wrote to Chase asking that God be named on the coins. His letter survives in Treasury records. Chase told officials at the United States Mint to prepare a motto.

A two-cent coin was the first United States coin to carry the words IN GOD WE TRUST. Members of Congress approved it in the Coinage Act of April 22, 1864. Chapter 29, Religion, tells how the words became the national motto in 1956.

Counterfeiters and the Secret Service

Fake bills made up somewhere between a third and a half of the paper money in use in the 1860s, but nobody made a full count at the time. History pages of the US Marshals Service and the Secret Service give about one-third. A 2025 release from the Department of Homeland Security gives nearly one-half.

Treasury officials formed the Secret Service Division on July 5, 1865, as part of the Treasury Department, to catch counterfeiters. Its first chief was William P. Wood, and Treasury Secretary Hugh McCulloch swore him in.

Guarding presidents was not part of the job at first. On September 6, 1901, Leon F. Czolgosz shot President William McKinley at a public event in Buffalo, New York. McKinley died on September 14. Afterward, members of Congress asked the Secret Service to protect presidents. In 1902 its agents took on full-time protection of the president.

Emanuel Ninger

Who: A sign painter who drew fake money by hand and was caught when his ink ran on a wet bar top.

When and where: New Jersey and Manhattan, 1883 to 1896. Lived 1845 to 1924.

He drew each fake bill by hand, and one took weeks.

Emanuel Ninger drew fake banknotes by hand for thirteen years, until his ink smudged on a wet bar. He was a sign painter, born in Germany, who lived in New Jersey. He drew United States banknotes, mostly $50 and $100 bills, with a pen, a pencil and a brush made of camel's hair.

To make a bill, Ninger laid a real note against glass and traced over it. He used colored inks for the parts printed in color. One bill took him weeks. From 1883 to 1896 he spent his fakes in Manhattan.

Ninger laid one of his bills on a wet saloon bar on March 28, 1896, and the ink smudged. A man working for the bartender followed him and had him arrested. The tellings of the story differ a little on how the arrest happened, but all of them include the smudged ink on the wet bar.

Ninger pleaded guilty in May 1896, and a judge sentenced him to six years in prison. Newspaper writers and experts at the time marveled that each fake had cost him more work than honest sign painting would have paid.

The Freedman's Savings Bank, 1865 to 1874

The Freedman's Savings Bank held the savings of people freed from slavery. It failed in 1874, and about 31,000 of its depositors never got any of their money back.

President Abraham Lincoln signed the bank's charter into law on March 3, 1865. The full name was the Freedman's Savings and Trust Company. By 1871 it had about 37 branches in 17 states and Washington, D.C. Most accounts were small, from about $5 to $50. Depositors put money in and took it out over the years, and all their deposits together added up to more than $57 million.

At first the bank could hold only safe government bonds. In 1870 members of Congress changed its charter to let it make loans on land and buildings, and other loans.

Henry Cooke. Henry D. Cooke sat on the bank's board and also served his family's banking business. The Office of the Comptroller of the Currency, the government office that oversees national banks, published a history of the bank. It states that Cooke used the Freedman's Bank's money and property "to benefit the latter," meaning his family's business.

The headquarters. The bank's headquarters, on Lafayette Square across from the White House, cost more than $200,000 to build and furnish. The Comptroller's history notes that the bank spent that money when it already had problems.

Charles Meigs, an examiner who inspected banks for the Comptroller's office, reported in February 1873 and again in 1874 that the bank was in growing danger of failing. In between, in the Panic of 1873, many of the Freedman's Bank's loans went bad.

Frederick Douglass became the bank's president in March 1874. He put $10,000 of his own money into the bank to restore trust in it. Then he saw that it could not be saved. The bank closed on June 29, 1874.

When the bank closed, it owed about 61,000 depositors nearly $3 million, by the count in the Comptroller's history. Only about half of them ever collected anything.

Who got paid. Between 1875 and 1883 the commissioners who closed out the bank's business made five rounds of payments. Depositors who turned in their passbooks, the small books that recorded their savings, got back 62 cents for every dollar they were owed. About 31,000 depositors never turned in their passbooks and got nothing, and the records do not say why.

The bank's signature books and deposit records from 29 branches survive at the National Archives. They list depositors' names, ages, jobs and family members. Black families today use them to trace their history.

Bank runs of 1873 and 1893

There was no deposit insurance in the 1800s, no promise backed by the government to repay people's savings if their bank failed. Every depositor in a failed bank waited for whatever the receiver, the official who took over the failed bank, could collect from the bank's leftover property.

At least 100 banks failed in the Panic of 1873, in Washington, D.C., Pennsylvania, New York, Virginia, Georgia, Indiana, Illinois and Ohio. On September 20, 1873, the New York Stock Exchange, the market in New York City where people buy and sell shares of companies, closed for ten days. It was the first time it had ever closed.

More than 440 banks suspended in the Panic of 1893, from June to mid-August, by the count of writers for the Federal Reserve's history website. To suspend meant to shut the doors or stop paying out cash, and some of those banks later reopened. Chapter 16 gives the count of banks and businesses that failed.

No surviving record counts how many depositors lost savings in 1873 or 1893, or how much they lost.

Gold against silver

Americans argued over whether gold alone or gold and silver should stand behind the dollar. Farmers and others who owed money were on one side, and bankers on the other.

With the dollar tied to gold, the supply of money grew slowly. Prices fell through most of the years from 1865 to 1896. A farmer's crop brought fewer dollars each year, but his mortgage payment, the money he owed each year on a loan against his farm, stayed the same.

Supporters of "free silver" wanted the government to coin all the silver brought to it, at 16 ounces of silver to 1 ounce of gold. That would have put more money in use and raised prices, which would have made debts easier to pay. Debtors and many people in the states that mined silver wanted silver. Bankers and bondholders, people who had lent money by buying bonds, wanted gold alone.

Members of Congress dropped the standard silver dollar from the list of coins in the Coinage Act of February 12, 1873. After that, in practice, the dollar was tied to gold alone and not to silver, just as miners in the West were digging more and more silver. People on the silver side called the act "the Crime of '73."

Two later laws required the government to buy some silver. The Bland-Allison Act of 1878 required the Treasury to buy $2 million to $4 million of silver each month and coin it into silver dollars. Lawmakers passed it over President Hayes's veto.

Under the Sherman Silver Purchase Act of 1890, the purchases rose to 4.5 million ounces a month. Treasury officials paid for the silver in notes that could be traded for gold. As holders traded the notes in, the Treasury's gold fell, from $190 million in 1890 to about $100 million in 1893. In 1893, during the panic, members of Congress canceled the Sherman Act because President Cleveland insisted on it.

William Jennings Bryan

Who: A 36-year-old former congressman from Nebraska who became the Democratic Party's candidate for president with a speech for silver.
When and where: The Democratic convention in Chicago, July 9, 1896.

"You shall not crucify mankind upon a cross of gold."

William Jennings Bryan became the Democratic Party's candidate for president in 1896 with a speech for silver. On July 9, 1896, at the Democratic convention in Chicago, he gave the last speech in the debate over the party's platform, its list of positions.

He ended the speech with words aimed at the supporters of gold. "You shall not press down upon the brow of labor this crown of thorns," he said. Then he said, "you shall not crucify mankind upon a cross of gold," crucify meaning to put someone to death by nailing them to a cross.

Both pictures come from the Bible's account of the death of Jesus. Bryan was comparing that suffering to the gold standard, a dollar backed by gold alone. Under it prices fell, and farmers and working people found their debts harder to pay each year.

After the speech, the delegates, the party members sent to the convention, chose him as their candidate for president on the fifth round of voting. Republicans wrote support for the gold standard into their platform. Their candidate was William McKinley. Bryan's votes came mostly from the South, the rural Midwest and the Rocky Mountain states. McKinley was strongest in the cities and in the Northeast, the Upper Midwest and the Pacific Coast.

McKinley won the election, with about 51 percent of the votes from the people to Bryan's 47 percent. Gold stayed behind the dollar. Bryan ran against McKinley again in 1900 and lost again.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 11 MINUTES

The Federal Reserve and deposit insurance, the two protections Americans still have for their savings, both began in these fifty years. Members of Congress and two presidents set them up.

The Federal Reserve, a bank for the nation's banks, came in 1913, six years after the Panic of 1907. Deposit insurance, a government promise to pay savers back if their bank fails, became law in 1933 and began on January 1, 1934, after thousands of banks had failed. Lawmakers meant both to stop bank runs, when many people rush to take their money out of a bank at once.

What stood behind the dollar changed too. In the Gold Standard Act of March 14, 1900, members of Congress made gold the only metal behind the dollar, at $20.67 an ounce. After that, silver was no longer a standard in law. In 1933 President Franklin Roosevelt ordered Americans to turn in their gold coins. From 1934 on, a dollar in an American's hand was paper that could not be traded for gold.

The Panic of 1907

Depositors rushed to pull their money out of New York's trust companies in October 1907. A trust company was a kind of bank that took people's deposits and managed their money. The country had no central bank, a bank that holds other banks' cash and lends to them in a crisis, so a private banker, J.P. Morgan, organized the rescue.

The trouble began with a failed attempt to corner the stock of United Copper, buying up so much of it that the buyers could set its price. When the attempt failed, the banks of the men behind it were ruined, and depositors grew afraid of other banks linked to those men.

So many depositors demanded their money from the Knickerbocker Trust Company on October 22, 1907, that its officers stopped paying out money. It was one of the largest trust companies in New York. Depositors then ran on the Trust Company of America and others, standing in lines around the block. Some hired people to hold their place in line. Charles T. Barney, the Knickerbocker's former president, shot himself on November 14, 1907.

J.P. Morgan

Who: A New York banker who ran the rescue in the Panic of 1907 from his own library.

When and where: New York City, October and November 1907.

Locked in his library, the bankers pledged their own $25 million.

J.P. Morgan was 70 years old when he took charge of stopping the Panic of 1907. He worked from his private library at 36th Street and Madison Avenue in New York City.

Under Morgan's rescue, teams of examiners went through the books of the trust companies and decided which ones were sound enough to save. Officials of the Treasury, the government department in charge of its money, deposited $25 million of the government's money with Morgan's group of banks. Morgan also got the presidents of banks and trust companies to promise money to shared rescue funds.

The locked room. Morgan gathered the presidents of the trust companies in his library on the night of November 2 to 3, 1907, and had the doors locked. He kept them in the room until they agreed to put up $25 million of their own, separate from the Treasury's $25 million, to save the weakest trusts. They signed the deal near 5 in the morning.

In the account told by the Morgan Library & Museum and others, Morgan kept the key in his pocket. Memoirs differ on who held the key and on the exact hour, but not on the locked doors. The museum's exhibition on the panic describes the meeting.

People at the time drew a lesson from the panic: the country's money system had depended on one 70-year-old private banker. In the Aldrich-Vreeland Act of 1908, members of Congress created the National Monetary Commission to design a better system.

Jekyll Island and the Federal Reserve, 1913

The Federal Reserve, the central bank of the United States, began with a secret meeting in November 1910. Six men traveled to the Jekyll Island Club, on an island off Georgia, and to keep the trip secret they traveled as a duck-hunting party. They were Senator Nelson Aldrich, his secretary, a Treasury official and three bankers, among them Paul Warburg. On the island they wrote a plan for a central bank.

The men who were there did not admit to the meeting in public until the 1930s. The Federal Reserve's own published history describes the trip. Members of Congress never passed their plan, called the Aldrich Plan. Three years later they used much of its design in the law they did pass.

President Wilson signed the Federal Reserve Act on December 23, 1913. The act provided for twelve Reserve Banks in different parts of the country, run under a board in Washington. The Reserve Banks opened on November 16, 1914.

The Federal Reserve is a bank for banks. It holds their reserves, the money a bank keeps on hand instead of lending it out, and its officers move money between banks. They can lend to a sound bank that is facing a run, so the bank can pay its depositors and stay open.

Black-owned banks, and Greenwood's losses

Black Americans started their own banks in the early 1900s, among them Maggie Lena Walker in Richmond and Jesse Binga in Chicago. Black depositors and business owners later lost money when banks failed and when a white mob burned Tulsa's Black district in 1921.

Maggie Lena Walker opened the St. Luke Penny Savings Bank in Richmond, Virginia, in November 1903, and became its first president. She was the first Black woman to found a bank in the United States. On the first day, people deposited more than $8,000, in amounts from 31 cents to over $100.

During the Great Depression of the 1930s, Walker merged her bank with other Black-owned banks into the Consolidated Bank and Trust Company. That bank lasted into the 2000s as the longest continuously run Black-owned bank in the country. Walker lived from 1864 to 1934.

Jesse Binga opened a private bank in Chicago in 1908, mainly for Black customers, because many bank managers turned them away. In 1921 he opened the Binga State Bank. Binga lived at 5922 South Park Avenue, in a white neighborhood. His home and his offices were bombed at least five times. The records do not say who set the bombs.

Illinois state officials closed the Binga State Bank in 1930, when it could not meet a requirement to hold $400,000 in cash. Accounts date the closing to June or to July 31, 1930. Binga lost his own fortune. Many depositors lost their savings, and the records do not count them.

A jury found Binga guilty in 1933 of embezzlement, taking money a person was trusted to look after. A judge sentenced him to ten years in prison. Governor Dwight H. Green of Illinois pardoned him in 1941, an official act that forgave the crime and ended the punishment.

Greenwood, 1921. A white mob attacked Greenwood, the Black district of Tulsa, Oklahoma, where about 10,000 people lived, on May 31 and June 1, 1921. The mob included men whom the Tulsa police had deputized, or sworn in as temporary officers, and National Guard units took part in the attack. The attackers destroyed more than 1,000 homes and businesses.

Estimates of the number of people killed run from 50 to 300. The count is uncertain because many of the dead were never identified and lie in unmarked graves. Most Black Tulsans lost their homes, and thousands lived in tents through the winter of 1921 to 1922.

No white person went to prison for the murders or the burning. An all-white grand jury, a group of citizens who decide whether to charge people with crimes, blamed Black Tulsans.

Greenwood's residents filed more than $1.8 million in damage claims with the city of Tulsa between June 1921 and June 1922. All but one of the claims were turned down. A 2021 Brookings study of the claims does not name the officials who turned them down. The one claim paid went to a white shop owner, for guns taken from his store.

Officers of the insurance companies refused to pay for the burned property. They called the attack a "riot," and their policies did not cover riot damage. No rebuilding money came from the city, the state or the federal government.

Greenwood's residents rebuilt the district. Members of the National Negro Business League, whose name used the common word of the time for Black people, held their national meeting in Greenwood in 1925. By the 1940s Greenwood had 242 Black-owned businesses.

1930 to 1933: the runs, the holiday and deposit insurance

More than one American bank in three failed from 1930 to 1933, and their depositors lost about $1.34 billion. Federal deposit insurance did not begin until 1934, so the losses fell on the depositors.

The count. Banks had failed at a rate of more than 600 a year even in the 1920s, mostly small banks in farm country. The count grew after 1929. The records of the Federal Deposit Insurance Corporation, or FDIC, the government agency that insures bank deposits, list 9,096 banks that suspended from 1930 to 1933, meaning they closed their doors or stopped paying depositors. The country had about 25,000 banks. By one estimate, about nine million savings accounts had been lost by March 1933.

The Bank of United States in New York City failed on December 11, 1930, the largest bank failure in the country up to then. It had about 400,000 depositors. Many were immigrants who worked in the clothing factories, and small savers in the Bronx and Brooklyn.

A history published by the FDIC in 1998 states that officials of the Federal Reserve, the bank for the nation's banks, "did little to ease the liquidity problems of banks," meaning the banks' lack of cash on hand to pay depositors. Most of the banks that failed in 1930 had not joined the Federal Reserve system, and its officials felt no responsibility for them.

The governor of Michigan closed every bank in the state on February 14, 1933, for eight days. That left 900,000 depositors unable to reach their money. Runs, when many people rush to take their money out of a bank at once, then spread from state to state.

Franklin Roosevelt became president in March 1933. On March 6, two days after he took office, he closed every bank in the country, and this was called the bank holiday. On March 9 members of Congress passed the Emergency Banking Act.

The fireside chat. On the night of Sunday, March 12, Roosevelt gave the first of his radio talks called fireside chats, and about 60 million people listened. He explained in simple terms how banks work, and he said money was safer in a

reopened bank than under the mattress. Banks reopened from March 13 to 15, after federal officials reviewed them. When they opened, people put in more money than they took out.

Deposit insurance. Members of Congress created the Federal Deposit Insurance Corporation in the Banking Act of June 16, 1933. Its insurance began on January 1, 1934, and covered each depositor up to $2,500. In July 1934 the limit rose to $5,000. If an insured bank failed, its depositors would be paid back from the insurance fund, up to the limit. With that promise, depositors no longer had a reason to rush to the bank to get their money out first.

The gold. In the same months, Roosevelt and members of Congress took gold out of Americans' hands. Roosevelt signed Executive Order 6102, a written order from the president, on April 5, 1933. It required Americans to turn in their gold coins, gold bars and gold certificates, paper notes that could be traded for gold, to the government at $20.67 an ounce. Americans could keep small keepsakes and collections.

Members of Congress moved all the gold the country used as money to the Treasury in the Gold Reserve Act of January 30, 1934. They set a new price of $35 an ounce. From then on, Americans could not trade their dollars for gold. Foreign governments still could.

Lydia Lobsiger

Who: The first depositor in the United States to be paid back by federal deposit insurance.

When and where: East Peoria, Illinois, 1934. Her grave record gives her life as 1870 to 1956.

A check for $1,250, dated July 3, 1934.

Lydia Lobsiger received the first federal deposit insurance check ever paid. She was about 64 years old, and the $1,250 was her life savings, by the FDIC's own history.

Her bank was the Fond du Lac State Bank of East Peoria, Illinois, and it failed in the spring of 1934. It was the first bank with FDIC insurance to fail, a few months after the insurance began.

When Lobsiger's bank failed, reporters at newspapers across the country wrote that a depositor with a small account had not lost her savings.

War bonds

About 85 million Americans, roughly half the country, lent money to the government during World War II by buying war bonds. Together they paid in about $185.7 billion.

One kind was the Series E bond. A buyer paid $18.75 for it and got back $25 ten years later. Children bought war stamps for 10 cents or 25 cents each and saved them toward a bond.

Officials of the Treasury and the Office of Education ran a program called Schools at War from 1942 to 1945. Students bought stamps and bonds toward goals set at their schools. About 200,000 schools and 30 million students took part, and they raised more than $2 billion.

Their money paid the price of at least 90,000 jeeps and several thousand airplanes for the armed forces. Eleanor Roosevelt spoke in support of the program on the radio on September 25, 1942.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 9 MINUTES

President Nixon cut the dollar's last tie to gold on August 15, 1971, when he ended the right of foreign governments to trade dollars for gold. Since then no metal has stood behind the dollar. In the same years Americans began to pay with plastic cards and take cash from machines.

Prices rose fast for about seventeen years, from about 1965 to 1982, and a dollar saved in 1965 bought about a third as much in 1982. Paul Volcker, head of the Federal Reserve, stopped that rise in the early 1980s by letting interest rates rise past 20 percent.

Plastic: the charge card and the credit card

The first card that people could use to pay at many different businesses was the Diners Club card, in 1950. Frank McNamara started the company with Ralph Schneider and other partners. The cards were made of cardboard, and at first a couple dozen restaurants in New York took them. The card was first used at Major's Cabin Grill in February 1950.

The Diners Club card was a charge card: the holder had to pay the whole bill each month.

The forgotten wallet. A story often told says that McNamara had the idea in 1949 after he forgot his wallet at dinner. Matty Simmons, the company's first publicity man, was there, and he said later that the story was made up for publicity. According to Simmons, McNamara simply had the idea and worked on it.

Officers of Bank of America mailed about 60,000 ready-to-use cards, called BankAmericards, to people in Fresno, California, in September 1958. None of those people had asked for a card. The BankAmericard was a true credit card: the holder could pay part of the bill and owe the rest later.

About 22 percent of the accounts fell behind on payments, against the 4 percent the bank's officers had planned for. There was fraud too, cheating to get money or goods. The bank lost millions of dollars in the card's first years.

Members of Congress banned mailing credit cards to people who had not asked for them in 1970. Officers of Bank of America licensed the card to other banks across the country, and in 1976 it was renamed Visa.

A lender who turned down a woman for credit because she was a woman broke no federal law until 1974. A blog post by the Consumer Financial Protection Bureau, a federal agency, states that "discrimination against women applying for credit was common."

Lenders who made mortgages, or home loans, "often discounted a married woman's income, especially if she was of childbearing age," counting her pay as worth less when they decided whether to lend to her. Lenders were more likely to turn down single women.

The law. Members of Congress passed the Equal Credit Opportunity Act in October 1974. Under the act, it became illegal for lenders to treat people differently because of their sex or whether they were married. In March 1976 members of Congress added race, color, religion, national origin, age, and receiving public assistance, money from a government aid program.

The ATM

The first automated teller machine in the United States, a machine that pays out cash without a bank clerk, opened on September 2, 1969. It stood at a Chemical Bank branch in Rockville Centre, New York. Docutel, a company in Dallas, built the machine, called the Docuteller, from an idea by Don Wetzel. The bank's advertisement stated: "On September 2, our banks will open at 9:00 and never close again."

August 15, 1971: the last tie to gold

Foreign governments could still trade dollars for gold after 1934, at $35 an ounce, though Americans could not. This deal was part of an agreement that the governments of many countries made after World War II, called the Bretton Woods system.

Far more dollars were held outside the country by 1971 than the United States had gold to pay for them. Foreign governments were trading in dollars for American gold, and the country's gold was running down. It was a run on the nation's gold.

President Nixon spent a weekend at Camp David with his advisers. They included Arthur Burns, the chairman of the Federal Reserve, John Connally, the Treasury Secretary, and Paul Volcker, a Treasury official. On Sunday night, August 15, 1971, Nixon announced that Treasury officials would no longer trade gold for the dollars foreign governments turned in. He called the change temporary. No president has reversed it.

Since that night the dollar has been fiat money, money backed by no metal and worth what it can buy. People accept it because the law makes it legal tender and because they trust the government that prints it. The greenbacks of 1862 worked the same way.

The Great Inflation

Prices in the United States rose fast year after year from about 1965 to 1982. Economists call it the Great Inflation, because inflation is the word for prices rising so that each dollar buys less.

Prices had risen only 1 or 2 percent a year in the early 1960s. Food prices rose more than 20 percent over the year to late 1973, and prices overall rose about 11 percent in 1974.

WIN buttons. President Ford started a campaign called "Whip Inflation Now" in 1974, with buttons printed with the word WIN. Prices went on rising. At the peak, in 1979 and 1980, they rose more than 1 percent a month, and for all of 1980 they rose 13.5 percent.

A dollar saved in 1965 bought about a third as much in 1982, so savings put aside in 1965 had lost about two-thirds of their buying power. Borrowing became costly. In October 1981 the average interest rate on a 30-year home mortgage was 18.63 percent. That is the highest rate ever recorded in the Freddie Mac survey of home loan rates.

Bruce Bent and Henry Brown

Who: Two men who set up the first money market fund, a way for small savers to earn more than the legal limit on bank interest.

When and where: The United States, 1969 to 1973. Bent was born in 1937. Brown lived from 1926 to 2008.

By federal rule, a bank could pay savers no more than 5 percent.

Bruce Bent and Henry Brown set up a fund that let ordinary savers earn the higher interest that only large investors could get before.

Federal Reserve officials put a rule called Regulation Q in place in September 1933. It forbade banks to pay more than a set rate of interest on a savings account. In July 1973 they raised the limit on an ordinary savings account from 4.5 percent to 5 percent. Most banks paid exactly the limit. Prices rose about 11 percent in 1974, so money in a savings account at 5 percent lost about 6 percent of its buying power that year.

Large investors could earn more by buying short-term debt from the government and from large companies, such as Treasury bills, short loans to the government. Only large investors could buy that debt directly.

Bent and Brown met while working at TIAA-CREF, a company that manages retirement money. In August 1969 Bent proposed a fund that would pool the money of many small savers and buy that debt for them.

They set up the Reserve Fund in 1970. A saver needed $1,000 to start, and each share cost $100. Sources differ on whether the fund opened to savers in 1971 or 1972.

Bent and Brown had no sales staff, so they phoned financial advisers and handed out brochures. The reporter Robert D. Hershey Jr. wrote an article about the fund in the *New York Times* on January 7, 1973. By the end of that year the fund held $100 million.

Federal Reserve History lists 1 money market fund in 1972, 36 in 1975, 90 in 1980 and 649 in 1990. The limits of Regulation Q were phased out by 1986 under a law members of Congress passed in 1980. The Reserve Fund later grew into the Reserve Primary Fund, which held about $62 billion in 2008.

Paul Volcker

Who: The chairman of the Federal Reserve who brought down the Great Inflation with interest rates above 20 percent, at the cost of a recession with 10.8 percent unemployment.
When and where: Washington, D.C., 1979 to 1983.

Interest rates passed 20 percent, and farmers drove tractors to Washington.

Paul Volcker stopped the Great Inflation by letting interest rates rise past 20 percent. He was six feet seven inches tall and known for smoking cheap cigars. President Jimmy Carter appointed him chairman of the Federal Reserve in August 1979.

Volcker announced a new policy on Saturday night, October 6, 1979. He and the other Federal Reserve officials would control how much money was in the economy. They would let interest rates rise as high as that took them. The federal funds rate, the rate banks charge one another for overnight loans, passed 20 percent in 1981.

In the recession of 1981 and 1982, a time when business shrank and people lost jobs, unemployment reached 10.8 percent, the highest since the 1930s. Farmers drove tractors to the Federal Reserve's building in protest. Home builders in debt mailed sawed-off pieces of two-by-four lumber to the Federal Reserve.

Inflation fell from 13.5 percent in 1980 to 3.2 percent in 1983.

The savings and loan crisis

Federal officials closed 747 failed savings and loans from 1989 to 1995. These were businesses, also called thrifts, that took in people's savings and made long-term home loans at fixed interest rates. Kenneth Robinson, writing for the Federal Reserve's history website, estimates that taxpayers paid as much as $124 billion of the cost.

When interest rates climbed in the late 1970s and early 1980s, the thrifts lost money in two ways. Under Regulation Q, federal officials kept the interest thrifts could pay on savings far below what savers could earn elsewhere, so savers took their money out. And the thrifts' old home loans kept paying the low fixed rates of years before, while the rates the thrifts had to pay to attract savers' money rose sharply.

Riskier loans. In the Garn-St Germain Act of 1982, members of Congress let thrifts make new kinds of loans beyond home mortgages. Many of those loans carried more risk. By 1983 closing the thrifts that could not pay their debts would have cost about $25 billion. Their insurance fund, the Federal Savings and Loan Insurance Corporation, or FSLIC, held only $6 billion.

Members of Congress closed the FSLIC in a 1989 law and moved thrift deposit insurance to the FDIC. They also created the Resolution Trust Corporation to shut down failed thrifts. By the end of 1995 its officials had closed the 747 savings and loans, which together held more than $407 billion in loans and other property. Savers with insured accounts were paid in full.

Banks failed in the same years. In 1984 federal officials rescued Continental Illinois National Bank, which held $45 billion in loans and other property, instead of letting it fail. FDIC records list 748 failed insured banks from 1985 through 1988.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 10 MINUTES

Most Americans pay with cards today, and the runs came back where deposit insurance did not reach. Cards made up 65 percent of payments in 2024, according to the Federal Reserve's yearly payment diary. Cash was used for 14 percent of payments in 2025.

Savers ran from money market funds in 2008, and depositors ran from Silicon Valley Bank and other banks in 2023. Both times, federal officials promised to cover money that no insurance protected. Since 2009 people have also used digital coins such as bitcoin, which no government and no bank puts out.

2008: the run comes back

The run of September 2008 hit money market funds, the kind of fund Bruce Bent and Henry Brown had started. In such a fund, the managers pool many people's money to buy short-term loans to the government and to large companies. Many people treated these funds like bank accounts. A share was meant to stay worth exactly $1. The funds had no deposit insurance.

The large financial firm Lehman Brothers failed on September 15, 2008. The next day the Reserve Primary Fund "broke the buck": its shares fell below $1, to 97 cents. The fund held $62.5 billion, and $785 million of it was in Lehman's debt.

According to a study in the *Journal of Financial Crises*, published by Yale University, investors pulled more than $40 billion out of the fund in two days. On September 17 alone they pulled about $144.5 billion out of money market funds across the country.

Treasury officials announced a guarantee for money market funds on September 19, 2008. President George W. Bush approved it, and Treasury Secretary Henry Paulson put up to $50 billion of the Treasury's Exchange Stabilization Fund behind it. The guarantee worked like deposit insurance, set up within a week, and the run stopped.

Banks failed too. According to a history the FDIC published in 2017, 489 insured banks failed from 2008 through 2013. Their failures cost the deposit insurance fund about $73 billion.

IndyMac failed on July 11, 2008, with losses of about $12 billion, the most costly failure in the FDIC's history. Washington Mutual, which held $307 billion in loans and other property, failed on September 25, 2008, the largest bank failure in the country's history. Members of Congress raised the deposit insurance limit from $100,000 to $250,000 on October 3, 2008, and made it permanent in 2010.

The trial. Officials of the Securities and Exchange Commission later charged Bruce Bent and his son with fraud over the Reserve Primary Fund's collapse. In 2012 a jury cleared both of fraud and found the son careless, which the law calls negligence, on one charge.

Money issued by nobody

Bitcoin is a digital coin that no government and no bank puts out. It runs on a public record of who owns which coins, kept by many computers that check one another's work.

A person or group using the name Satoshi Nakamoto published a paper describing Bitcoin on October 31, 2008. On January 3, 2009, Nakamoto started the system by creating its first block of records.

The code of that first block includes a headline from that day's *Times* of London: "Chancellor on brink of second bailout for banks." The headline was about the British government's top money official, the Chancellor, and a bailout, government money given to businesses in trouble. No one has shown who Nakamoto is.

People hold bitcoin as an investment, hoping its price will rise, and others use it to pay for things. Economists disagree about whether it is money, property owned for its value, or a bet on a price.

Laszlo Hanyecz

Who: A programmer who made the first recorded purchase with bitcoin: two pizzas for 10,000 coins.

When and where: Florida, May 22, 2010.

Two pizzas for 10,000 bitcoins, later worth about $1.1 billion.

Laszlo Hanyecz paid 10,000 bitcoins for two pizzas on May 22, 2010. It was the first recorded time anyone traded bitcoin for a real thing.

Hanyecz was a programmer in Florida who had worked on Bitcoin's code. He wrote the first version of the Bitcoin program for Apple's Mac computers. He was the first to use a computer's graphics card to create new bitcoins, a job called mining.

In May 2010 he posted an offer on the Bitcoin forum, a website where users wrote to one another. He would give 10,000 bitcoins to anyone who got two large pizzas delivered to him. A forum user named jercos took the deal and ordered the pizzas, which cost jercos about $25. At the time of the offer, the 10,000 bitcoins were worth about $41, according to the bitcoin.it wiki.

The same wiki estimates that at the high point of May 2025, those 10,000 coins would have been worth about $1.1 billion. In a 2019 interview, reported by the news site CoinDesk, Hanyecz said, "A trade happens because both parties think they're getting a good deal." May 22 is now marked each year as Bitcoin Pizza Day.

Payday loans

A payday loan is a small loan meant to be paid back from the borrower's next paycheck. Researchers at the Consumer Financial Protection Bureau released a study of these loans on April 24, 2013. They looked at a $350 payday loan with the typical fee of $15 for every $100 borrowed. Its annual percentage rate, the cost over a full year as a share of the amount borrowed, came to 391 percent.

Storefront payday lenders charged fees of $10 to $20 for every $100 borrowed, the study found. Nearly half of payday borrowers took out more than 10 loans in a year, and 14 percent took out 20 or more. The typical borrower was in debt to a payday lender 199 days of the year, more than half of it.

Inflation returns, 2021 to 2026

Prices rose fast again from 2021 to 2023, and Federal Reserve officials raised interest rates faster than at any time since Volcker. Prices rose 9.1 percent in the 12 months to June 2022, according to the Bureau of Labor Statistics, the largest rise since November 1981. Energy prices rose 41.6 percent that year and food prices 10.4 percent.

Federal Reserve officials raised their main interest rate from near zero in March 2022 to more than 5 percent in 2023. By the middle of 2023, prices were rising at close to 3 percent a year.

According to a report by the Congressional Research Service, which does research for members of Congress, economists still disagree about the causes. The causes they name include broken supply chains (the network of factories, ships and trucks that moves goods), federal spending on pandemic relief and higher energy prices.

Prices rose 3.4 percent in the 12 months to August 2026, the Bureau of Labor Statistics reported. Energy prices rose 16.3 percent, and food prices 2.7 percent.

FTX and stablecoins

FTX was a company where people bought, sold and kept cryptocurrency, digital money such as bitcoin. It went bankrupt in November 2022: it could not pay what it owed and went to court to settle its debts. Its founder was Sam Bankman-Fried. According to the *Washington Post*, he had used customers' money to buy luxury real estate, to make investments and to give political donations.

A federal jury found him guilty in November 2023 of wire fraud, cheating people by phone, email or the internet. The jurors also found him guilty of two conspiracies, plans made with other people to commit a crime. One was to commit fraud, and the other was to commit money laundering, hiding where money from a crime came from. Judge Lewis Kaplan sentenced him on March 28, 2024, to 25 years in prison and ordered him to give up more than $11 billion.

The people running the bankruptcy paid customers back what their accounts were worth on the day FTX filed for bankruptcy in November 2022. On that day one bitcoin sold for $16,080. By May 2024 it sold for close to $62,675, according to CBS News. Customers who had held bitcoin did not get back that later value.

Another payment went out to customers on July 31, 2026. An FTX press release in 2026 states that, after it, customer claims had received 105 percent of their value in all, and small claims 120 percent. Payments above 100 percent were interest added to the claims. FTX's chief executive, John Ray, described the plan in 2024 as "100% of bankruptcy claim amounts plus interest."

President Trump signed the GENIUS Act on July 18, 2025. The law contains rules for stablecoins, digital coins whose makers promise to trade each one for a fixed amount, such as one dollar. Under the law, a company that puts out stablecoins must hold at least one dollar of approved reserves, such as cash, insured bank deposits or Treasury bills, for every coin. The law states that stablecoins are not federally insured, according to a 2025 report by the Congressional Research Service.

Like the old state banknotes. In the 1800s, before national paper money, banks chartered by the states printed their own notes. Each note was only that bank's promise to pay in coin. A stablecoin works the same way: a private promise to pay, backed by what the company holds, with no federal insurance behind it.

2023: the run on Silicon Valley Bank

Federal officials protected every depositor at two failed banks in March 2023, including money above the insurance limit. Depositors took $42 billion out of Silicon Valley Bank in California in a single day, March 9, 2023. About 88 percent of the bank's deposits were above the $250,000 insurance limit. If the bank failed, the owners of that money stood to lose it.

Officials of California's Department of Financial Protection and Innovation closed the bank on March 10. On Sunday, March 12, officials of New York's Department of Financial Services closed Signature Bank, where about 90 percent of the deposits were uninsured.

The rescue. Treasury Secretary Janet Yellen, after consulting the president, used an emergency power that same Sunday and protected every depositor at both banks, insured or not. FDIC Chairman Martin Gruenberg told Congress on March 27, 2023, that FDIC officials estimated the cost at about $20 billion for Silicon Valley Bank and $2.5 billion for Signature Bank. They planned to collect that money from other banks through special fees, not from taxpayers.

California officials closed First Republic Bank on May 1, 2023, and JPMorgan Chase took over all its deposits.

Cash fades, and the penny ends

Cash was the third most used way to pay in the United States in 2025, after credit cards and debit cards. Federal Reserve researchers track this with the Diary of Consumer Payment Choice, in which people record the payments they make.

Cash was the most used way to pay in 2016, at about a third of all payments. By 2024 cash was used for 14 percent of payments, credit cards for 35 percent and debit cards for 30 percent. Adults aged 18 to 24 paid by phone 45 percent of the time that year.

People made about 47 payments a month in 2025, according to the diary's report released in May 2026. Cash was 14 percent of them, about 6 a month. It was the sixth year in a row that cash came third.

Who still uses cash. People aged 55 and older paid cash about 10 times a month, against about twice for people aged 18 to 24. In the countryside the count was about 9 a month, and in cities and suburbs about 6. Ninety percent of people said they planned to keep using cash.

Each penny cost the US Mint 3.69 cents to make and ship in the 2024 budget year. The Mint's losses on pennies came to about $85 million that year. On February 9, 2025, President Trump said he had told the Treasury Secretary to stop making new pennies. A PBS and Associated Press report at the time stated that it was unclear whether he had that power, because members of Congress set the rules for coins.

The last pennies. The last pennies made for everyday use were struck at a ceremony at the Philadelphia Mint on November 12, 2025. The first federal cents had been made 232 years earlier, in 1793. The expected savings were about $56 million a year. Pennies already made are still legal tender.

Members of the House of Representatives passed the Common Cents Act on September 14, 2026, and senators passed it on September 28. Under the bill, penny production would end in law. When a customer paid cash and exact

change was not available, the total would round up or down to the nearest five cents. Payments by card or check would not round. As of October 2, 2026, the bill was waiting for President Trump's signature.

THE THREE QUESTIONS, ANSWERED

1. **What has stood behind the dollar, and what stands behind it now?** Silver and gold stood behind it, then gold alone. Since August 15, 1971, only the law and people's trust in the government that prints it have stood behind it.
2. **Why did people rush to pull their money out of banks, and who lost when a bank failed?** A bank keeps only part of its money in cash, so the first in line got paid. Until deposit insurance began in 1934, the rest took the loss, about $1.34 billion from 1930 to 1933.
3. **How did enslavers and bankers use enslaved people to raise money?** Enslavers pledged them for loans, and in the 1830s Louisiana banks lent out their own paper money against them. When the loans went unpaid, the people were sold.

What changed: People here went from living without money to paying with cards, and the dollar went from silver and gold to paper backed only by the law.

FIVE DATES TO KEEP

December 1690: Massachusetts lawmakers vote for paper money.

July 6, 1785: members of Congress vote for a dollar counted in tens.

December 23, 1913: President Wilson signs the Federal Reserve Act.

January 1, 1934: federal deposit insurance begins.

August 15, 1971: President Nixon ends the dollar's last tie to gold.

NOT KNOWN: No surviving record counts the depositors who lost savings when banks failed in 1837, 1873 and 1893.

Where to go: The National Archives keeps the Freedman's Savings Bank's signature books, which list depositors' names, ages and jobs.

Next: Chapter 18, Marketplace and Consumer Life. It asks how Americans bought things, from the peddler at the door to the online store.

Chapter 18: Marketplace and Consumer Life

CHAPTER 18 OF 37 · MARKETPLACE AND CONSUMER LIFE · 10 ERAS · ABOUT 75 MINUTES

Americans have taken goods home and paid later since the 1600s, and new ways of selling, from the peddler's trunk to the online store, kept changing who could buy and what they paid.

In Springfield, Massachusetts, the merchant John Pynchon kept a page in his account book for John Stewart, a Scottish soldier captured in battle in 1650 and "sold for eight years service." On December 29, 1653, Pynchon wrote that Stewart owed him 30 pounds in English money "for releasing him from my Service." Stewart had bought his freedom on credit. For 36 years his page filled with iron, boots, pepper and sugar. He paid in shell beads, blacksmith's work, land and a day of haymaking, almost never in coins.

This chapter describes buying and selling from the trading place on the Columbia River, long before there were shops, to buying online. The peddler at the door, the department store with a price tag on every item, the mail-order catalog, the supermarket, the mall and the credit card come in between. Watch how people paid: in shells and crops, in work, on the shopkeeper's book, a little each month, and at last with plastic. Watch, too, who was turned away and who was sold. Printers ran advertisements offering enslaved people for sale in the same newspapers as everything else, and in the South the people who ran Woolworth's lunch counters served white customers only.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did colonists pay for goods when they had almost no coins?
2. Where were people bought and sold like goods?
3. How much of what Americans buy today is bought online?

Five minutes? Read "The ledger" and the story of John Stewart, in the 1600s.

Can skip: "Barnum's American Museum", "The franchise", "Buying secondhand clothes". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 3 MINUTES

People in this land traded copper, seashells and stone blades across hundreds of miles long before there were any stores. They traded one thing for another, and goods passed from hand to hand. At a few great trading places, people from many nations gathered to fish and to trade.

Where was the biggest market in the Northwest?

The biggest trading place in the Northwest was a ten-mile stretch of the Columbia River. It ran from Celilo Falls down to a place called The Dalles, on the border of what are now Oregon and Washington. People fished and lived there. Archaeologists, scientists who study the past from what people left behind, date the first human use of the site to at least 10,000 years ago. Some newspaper and museum writers give 15,000 years.

Traders came from as far north as what is now British Columbia and as far south as California. Some accounts, cited by the Oregon Historical Society, state that traders came from as far east as the Great Lakes.

Northern traders brought coiled baskets, skins and berries. Traders from the south carried obsidian, a black volcanic glass that makes sharp blades, along with bows and captives, people seized by force and traded away with the other goods. The accounts of the trade do not identify who seized them. From the east came buffalo hides and camas, a root that people dug up and ate. From the coast came whale oil, dentalium shells and wapato, another root people ate.

Dentalium shells. These are the shells of a small sea animal. Each one is shaped like a tiny elephant tusk and can grow a few inches long. Most of them came from the sea off Vancouver Island, and people passed them from hand to hand over a wide area. So many people accepted them that traders used them to measure what other goods were worth.

Horses came later, after 1500. Spanish colonists had brought horses to the Americas. In the early 1700s, Native traders passed horses north to this part of the river.

Much later, in October 1805, two American explorers, Meriwether Lewis and William Clark, reached this stretch of river. President Thomas Jefferson had sent them west to look for a water route to the Pacific Ocean. In his journal, Clark called the place a "mart," an old word for a market: the "Great Mart of all this Country." People had been trading there long before his visit.

Under the lake. The trading place is now under water. Workers for the US Army Corps of Engineers, the part of the army that builds dams and other large works, built The Dalles Dam eight miles downstream. At 10 o'clock on the morning of March 10, 1957, the dam's floodgates were closed.

Four and a half hours later, Celilo Falls lay under the lake behind the dam. The villages of Wyam, on the Oregon side, and S'kin, on the Washington side, went under too. US government officials later paid the Warm Springs, Yakama, Umatilla and Nez Perce nations $26.8 million for the fishing sites they lost.

Goods that traveled without a store

Archaeologists have found copper from Lake Superior, shells from the Gulf of Mexico, obsidian from the West and turquoise from the Southwest hundreds of miles from where each was dug up or gathered. People carried and traded these goods over those distances, one exchange at a time.

Cahokia was a city near what is now St. Louis. It was at its largest from about 1050 to 1150. Around 1100, about 10,000 to 20,000 people lived there. People at Cahokia controlled the trade in hoe blades made from Mill Creek chert, a hard stone that holds a sharp edge. Archaeologists have found those blades more than 400 miles away. People that far from Cahokia got the blades by trading for them.

No trader from before 1500 is known by name.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

Native traders chose which European goods to take, and copper and brass kettles were among the things they wanted most. They traded for them with the crews of European whaling and trading ships off what is now eastern Canada. There were still no stores. Archaeologists can see the traders' choices in the kettles they cut up and made into new things, and in the metal they passed far inland, as far as the Great Lakes.

What Native customers chose

Native buyers in the Northeast and around the Great Lakes traded for copper and brass kettles in the late 1500s. Basque, French and Dutch traders carried the kettles across the ocean because those buyers asked for them. Crews from the Basque Country, a mountain region on the border of Spain and France, crossed the ocean to hunt whales. They traded with the Beothuk of Newfoundland, the Innu of Labrador and the Mi'kmaq.

Many of these kettles were not used for cooking. Archaeologists dig them up cut into flat sheets. Native craftspeople turned the sheets into beads, tubes, arrow points and pendants, which are ornaments that hang from a cord. For a long time, people in these nations had made the same shapes from copper found as pure metal in the ground. Buyers like these used a kettle as a supply of metal.

Pieces of Basque copper kettles from the late 1500s and early 1600s turn up from the Canadian Maritimes, on the Atlantic coast, to the lower Great Lakes. Native traders passed the metal along among themselves, far inland. Basque kettles and axes reached the lands of the Huron and their neighbors in what is now Ontario.

Native traders chose carefully what they took in trade, and they kept doing so as the trade grew. In the fur trade of later years, they were skilled bargainers who asked for useful tools and for fine goods alike, such as iron knives, wool

cloth and glass beads.

Good guns. One fur-trading company, the Hudson's Bay Company, began in 1670. Its Native customers demanded good guns, so the men running the company sent gunsmiths to its trading posts to check the guns and repair them.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

Colonial shoppers in the 1600s took their goods home first and paid later, mostly in crops, other goods and work. In Springfield, Massachusetts, John Pynchon wrote down in his account books what each customer owed and let the debts run for months. In the same century, public markets opened in the colonial towns, and colonists built trading posts to trade with Native people.

Market days

A market day was a set day when farmers brought meat, milk, butter, cheese, fruit and vegetables into town to sell. Town people could not travel out to the farms for these foods, so the farmers came to them.

Governor John Winthrop of Massachusetts ordered a market for Boston in 1634. It is the first public market of the European kind recorded by officials in the English colonies. Hartford, in Connecticut, had a market in 1643. New York had one by 1686, and Philadelphia had one in 1693.

Trading posts

A trading post was a building where colonists traded goods with Native people and with other traders. In 1627, the colonists of Plymouth Colony built one at Aptucxet, on the Manomet River on Cape Cod. It was about 20 miles from Plymouth. The post stood on land of the Wampanoag people, who used the place as summer fishing grounds. The name Aptucxet meant "little trap in the river."

At Aptucxet the Plymouth colonists traded with the Wampanoag for corn and beans, and with Dutch traders. That same year, Isaack de Rasieres, the secretary of the Dutch colony, brought wampum, small beads made from shells and strung together, into Plymouth's trade.

The ledger

A colonial shopkeeper kept two books. The first, the journal, listed every sale and every purchase in the order it happened. The second, the ledger, gave each customer a page of their own. The customer's purchases went on the left side of the page and the customer's payments on the right.

Customers bought on credit, taking the goods home and paying for them later. A shopkeeper usually let a customer's debt run for up to a year. Customers paid in crops, in other goods, in work and sometimes in coins.

John Pynchon kept a shop in Springfield, Massachusetts. The town and the shop stood on land on both sides of the Connecticut River that his father, William Pynchon, bought from the Agawam people on July 15, 1636. William Pynchon paid for the land in goods, with no coins: 18 fathoms of wampum, strings of beads six feet long, and 18 coats, 18 hatchets, 18 hoes and 18 knives. The Agawam kept their planted fields and the right to fish, hunt deer and gather nuts there.

Seven of John Pynchon's account books survive, with entries from 1651 to about 1702. Almost every adult in Springfield bought goods at his shop. The editors who printed the books found only a few payments in coins. Books like these record what ordinary colonists bought, item by item.

John Stewart

Who: A Scottish soldier, captured in war and sold into forced labor, who became the blacksmith of Springfield and bought on credit at John Pynchon's shop for more than thirty years.
When and where: Springfield, Massachusetts, 1653 to 1689.

A prisoner of war who owed 30 pounds for his freedom.

Soldiers of Oliver Cromwell's army captured John Stewart, a Scottish soldier, at the Battle of Dunbar, in Scotland, in 1650. The men on the council that ruled England ordered the governor at Newcastle to hand over 150 of the Dunbar prisoners to the agents of two men who had put money into an ironworks in Massachusetts. In his own recorded words, Stewart was "sold for eight years service." He was forced to work for whoever owned his service until the eight years ran out.

Stewart and the other prisoners crossed the ocean on a ship called the *Unity*. He reached New England in December 1650 and worked at the ironworks at Lynn, in Massachusetts. Then the merchant John Pynchon bought his service and took him to Springfield.

Buying his freedom. Pynchon wrote in his account book on December 29, 1653, that Stewart owed him 30 pounds, the main English unit of money, "for releasing him from my Service." While Pynchon waited for the money, Stewart agreed to pay an extra 12 shillings, a smaller unit, each year for three years. He paid that extra in blacksmith's work, "in mending or making of Iron worke."

Stewart became the Springfield blacksmith, and his page in Pynchon's ledger lists what he bought, item by item. For his forge, the fire where a blacksmith heats iron to shape it, he bought 241 pounds of iron by weight in one purchase, then more bars of iron every few weeks, bars of steel and a coal basket. Pynchon charged him for using Pynchon's tools at the shop and for an old bellows, the air pump that blows on a forge fire.

He bought a sword and belt, a grindstone, a pair of boots, wool cloth, silk and buttons, a bushel of apples, soap, a comb, pepper and ginger. Almost none of it was paid in coins. By December 23, 1654, he had paid in two lots of wampum and in smith's work, including work on Pynchon's mill, and neighbors had paid wheat to Pynchon for him. He still owed 13 pounds 6 shillings. Later he paid in land and in a day of haymaking.

The editors of the account books explain that he was fined in 1660 for striking a man named William Morgan and breaking his nose, and in 1662 for playing cards at his house.

Ten years on. By 1664 he was still buying on the book: sugar again and again, salt, gunpowder, scissors, pins, a knife for "the Boy," and iron for the horseshoes of the troopers, soldiers on horseback. On February 20, 1666, he owed 11 pounds 17 shillings and 1 penny.

His wife. By then Stewart was married. The books charge him for church seats for himself and his wife, but they never give her name. His page also records a charge of 1 pound, a fine for the two of them, for "unseasonable night meetings" (meetings at an improper hour of the night).

Pynchon's books list his purchases until 1689. They do not say what became of his wife. Stewart died at Springfield on April 21, 1691. The Scottish Prisoners of War Society, a group that traces the Dunbar prisoners, records that he left "no family, so far as is known."

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 5 MINUTES

Goods made in Britain became cheap enough in the 1700s for colonial families to buy them in large amounts. Shops in the port towns sold these goods, and in Philadelphia Benjamin Franklin made advertisements pay for much of his newspaper. Colonists bought and sold enslaved people in the same towns. In New York, city officials set up a market for those sales, and printers, Franklin among them, ran advertisements offering people for sale.

The consumer revolution

Colonists began buying many of the clothes, tools and kitchen things they had once made for themselves, mostly from Britain. Historians call this big change in buying by consumers, the people who buy goods to use them, the "consumer revolution."

In Britain, makers found better ways to make goods. Merchants got better at shipping goods across the ocean and at selling them on credit. Prices fell. At the same time, colonial families earned more. Goods that only rich families had owned became ordinary.

Families who drank tea bought a whole set of goods to go with it: a teapot, cups, sugar and a table to set them on. Buying goods like these became a sign that a family was respectable.

A market where people were sold

In New York, colonists bought and sold enslaved people at a market created by the men who ran the city, the aldermen of the Common Council. They set it up by law on November 30, 1711, at the bottom end of Wall Street, near Pearl Street and Water Street.

Slaveholders sold enslaved Africans at this market and rented them out to others as workers. They sold and rented out Native people held in slavery there too. Under the law, owners who rented out enslaved people as workers by the day had to do it at this market. Buyers inspected the people for sale there, looking them over before buying. The accounts of the market do not describe how the inspections were done.

The building. The market was a wooden building with a roof and open sides. Slaveholders used it for 51 years, until 1762.

New York had about 5,000 people in 1700, and at least 750 of them, about one person in seven, were enslaved. Jim O'Grady, a reporter for the New York radio station WNYC, wrote in 2015 that thousands of people were sold at the market: men, women, shiploads of children and captured Native people. The accounts of the market do not identify the men who bought and sold people there.

A sign now marks the place. Jumaane Williams, a member of the New York City Council, asked for it in 2015, and in June of that year Mayor Bill de Blasio officially opened it at Wall Street and Water Street.

Newspaper advertising begins to pay

Benjamin Franklin bought a newspaper in Philadelphia, the *Pennsylvania Gazette*, from a printer named Samuel Keimer on October 2, 1729. It had about 90 subscribers, the people who paid to get every copy. Under Franklin, the money that sellers paid to print their advertisements became one of the main ways the paper made money.

Franklin changed the way the advertisements looked. He put more empty space around them and used letters of different sizes. He made the *Gazette* four pages instead of two, partly to make room for more advertisements. By 1748 it had more than 1,500 subscribers, and it was the most widely read newspaper in the colonies.

The partnership. Franklin stepped back from printing in 1748, when he was 42. He made a partnership with a printer named David Hall, who ran the shop while Franklin kept a share of the profits. Much of the money that let Franklin stop working at his trade so early came from the paper's advertising.

Benjamin Franklin

Who: The Philadelphia printer who ran advertisements offering enslaved people for sale in his newspaper, and took the fees.

When and where: Philadelphia, from October 2, 1729.

Movie: *Benjamin Franklin* (2022), a documentary by Ken Burns, shown on PBS. It covers Franklin's whole life.

At least 277 of his paper's advertisements listed enslaved people for sale.

Benjamin Franklin published the *Gazette* for 37 years, counting the years when he was David Hall's partner in the paper. In those 37 years, the *Gazette* contained at least 277 advertisements offering at least 308 enslaved people for sale. Franklin ran these notices and took the fees for them. He also acted as the go-between for the buyers and the sellers.

The runaways. Slaveholders paid Franklin to print notices about enslaved people who had run away from them. He printed notices about runaway servants too: workers bound to a master for a set number of years, who had left before their years were up.

Shops and shopkeepers in the port towns

Shopkeepers in the port towns ordered cloth and other goods from London and sold them. Some sold haberdashery, small sewing goods such as thread, ribbons and buttons. Colonists who bought on credit could end up in debt to a local shopkeeper or to a merchant in London.

Women ran some of these shops. Elizabeth Murray came to Boston from Scotland in 1749, when she was 23. She opened a shop that sold cloth and other goods from Britain. Her brother's business friends helped her get started. She traveled to London to choose the goods for her shop, and she earned enough to support herself. Later she helped other women open shops.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 7 MINUTES

Many colonists refused to buy British goods from 1767 to 1770, as a protest against British taxes. Merchants promised not to bring the goods in, and families made their own cloth instead of buying it. Members of merchants' committees warned the shopkeepers who kept selling British goods. In Boston, Ebenezer Richardson fired from his house during a protest at one of those shops, next door, and killed a boy named Christopher Seider. Away from the ports, a single storekeeper sold a whole farming neighborhood its goods on credit until the next harvest.

The boycott of British goods

Members of Parliament, the lawmakers in London, passed a tax law called the Townshend Revenue Act in 1767. Under it, customs officials collected duties, taxes on goods brought into the colonies from Britain. Colonists who opposed the duties answered with a boycott, refusing to buy British goods as a protest.

The seniors at Harvard College in Massachusetts voted in the autumn of 1767 not to drink tea or wear clothes from Britain at their graduation, as a protest against the Townshend duties. At the next year's graduation, the seniors' theses, a printed sheet of the ideas they were ready to argue for in public at the ceremony, were on paper made in New England.

The agreement. About 60 Boston merchants and traders signed the Boston Non-Importation Agreement on August 1, 1768. They promised not to bring in most British goods from January 1, 1769, to January 1, 1770. Within two weeks, all but 16 of the town's merchants had joined.

The value of British goods brought into Boston fell by about half, from about 420,000 pounds in 1768 to about 208,000 pounds in 1769. In 1770, members of Parliament took back most of the Townshend taxes. The agreement in Boston ended that year.

Spinning instead of buying

Families who did not buy British cloth had to make their own, called homespun. To make it, women first spun wool or flax, a plant whose stems give the threads for linen, into yarn on a spinning wheel. Then the yarn was woven into cloth.

Women in New England held at least 46 spinning meetings between 1768 and 1770, where they spun together in public. In all, 1,644 women took part. Ministers held 30 of the meetings, and 94 out of every 100 spinners came to those 30.

One day in Brookfield. Women in Brookfield, Massachusetts, met at the minister's house from 5 o'clock in the morning one day in August 1769. The young woman who did best on the linen wheel spun 70 knots, measured lengths of yarn. One married woman did her morning work at home and made cheese first. Then she rode more than two miles carrying her wheel, spun from 9 in the morning to 7 at night, and finished 53 knots. After that she went home to do the milking.

Members of a society in Hartford, Connecticut, offered a prize of 12 pounds to whoever made the most yards of woolen cloth in a year. Printers ran newspaper stories about the meetings. Only six of those stories called the spinners "Daughters of Liberty," a name for women on the side of the protest.

Ame and Elizabeth Cuming

Who: Two sisters who kept a shop in Boston, refused to join the boycott and were named in the newspaper and the town record for it.

When and where: Boston, 1765 to 1776, then Halifax, Nova Scotia.

A shipment worth 300 pounds, and a warning of "consequences."

Ame and Elizabeth Cuming kept a shop in Boston and would not stop selling British goods when the town's merchants told them to. The sisters were in their middle thirties and not married, and by 1765 they earned their own living. They sold fashionable goods from London, taught embroidery, sewing pictures and patterns with colored thread, and took in young women who paid to live with them. Elizabeth Murray had helped them start their business.

The warning. A shipment of silk and haberdashery worth 300 pounds reached them from London in late 1769. In the middle of November, men from the merchants' committee came to the shop and warned the sisters of "consequences" if they sold the goods. The sisters refused. They said they had never signed any agreement to stop bringing in goods. They called themselves "two industrious Girls" (two hard-working girls) who were trying "in an honest way" to earn their bread.

A notice in the *Boston Gazette* described them as "enemies to their country" who "preferred their own little private Advantage to the Welfare of America." On March 19, 1770, the men at Boston's town meeting voted to write the sisters' names into the town record. The wording of the vote states that this was so that "POSTERITY" (the people who live later) "may know" they had "opposed their Country."

The sisters left Boston in March 1776 with the Loyalists, the colonists who sided with Britain in the Revolutionary War, and went to Halifax, Nova Scotia. By 1780 they were "more prosperous than ever they were in Boston."

A killing at a shop door, 1770

A Boston man named Ebenezer Richardson shot and killed a boy, Christopher Seider, during a protest at a shop door in February 1770. Theophilus Lillie kept the shop, in Boston's North End, and he went on selling British goods during the boycott.

A carved head on a long pole went up in front of Lillie's shop door on February 22, 1770. According to some accounts, it was a sign that read "Importer." Thomas Hutchinson, the acting governor of Massachusetts, wrote about that day in a letter. Hutchinson was running the colony for the king while the governor was away in England. A crowd of boys and young men gathered and let no one go into the shop.

Richardson lived next door to Lillie and worked as a customs informer. Customs officials collected the taxes on goods brought in by ship, and they paid informers like him to report smugglers. When Richardson tried to pull the pole down, the crowd threw stones and sticks at him, chased him into his house and broke his windows. From inside, he fired a musket at the boys. The shot hit Christopher Seider in the chest, and he died soon after.

Hutchinson wrote that Richardson "killed one & wounded another." The second boy was hit in the thigh. Christopher was 10 and a half years old in Hutchinson's letters, and 11 in most other accounts.

The funeral. Christopher Seider's funeral was on February 26, 1770. Hutchinson wrote that people said the procession, the line of people walking behind the coffin, "reached from the Town House to Liberty Tree," two places in Boston. The modern editors of his letters call it the largest procession ever seen in Boston, as people reported it at the time.

Richardson was tried for murder on April 20, 1770, and the next day the jury found him guilty. The judges believed he was innocent, and they put off his sentence. Richardson stayed in jail for almost two years. On March 10, 1772, the news came that the king had pardoned him, and he went free.

The country store and the crop ledger

Away from the port towns, one country storekeeper sold goods to a whole farming neighborhood. Customers paid in crops, in work and in promises to pay later. The storekeeper wrote each family's debt in a ledger. A farm family bought goods all through the year and paid after the harvest. When a harvest failed, the family could not pay, and the debt carried into the next year.

A shop near the city. Not every store worked this way. Robert Townsend kept a shop in Oyster Bay, on Long Island, close to New York City, and kept account books there and in New York. His surviving books record sales from 1773 to 1785. Staff at the East Hampton Library have copied them out and published them.

One book records sales from April 1, 1773, to November 15, 1779. In 1773 it lists tea at 4 shillings 8 pence a pound, stockings, calico, a cotton cloth, linen cloth, gloves, packs of pins, molasses, a dark sweet syrup, and large barrels of spirits, a strong liquor.

Townsend sold on credit too, and he let a customer's debt run for months, but usually not for more than a year. Unlike the farm families above, most of his customers paid in cash. Library staff counted the payments in his book from 1773 to 1779. In their count, 73 out of every 100 were cash, 22 were farm goods and 5 were notes, written promises to pay. Some customers paid with their work, as a washerwoman, a doctor, a barber or a tailor.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 6 MINUTES

Peddlers carried goods from house to house in the first half of the 1800s, selling to farm families in the country. In the cities, P.T. Barnum sold tickets to his shows with posters and newspaper advertisements, and he lied in them. His first show was an enslaved woman, Joice Heth, whom he advertised as 161 years old. A. T. Stewart opened a store in New York in 1846 with several floors of goods and a set price on each one.

The peddler's trade

A peddler traveled a route through the countryside and sold goods at people's doors. Some went on foot. Others drove a wagon. They carried tinware, clocks, pins, needles, combs, books and patent medicines, bottled remedies sold under a brand name. The best known were the tin peddlers from Connecticut, who sold tinware: pots, pans, cups and other things made of thin sheets of metal coated with tin.

James Guild

Who: A Vermont farm boy who set out to peddle with a trunk of goods and wrote down what happened to him.

When and where: Tunbridge, Vermont, and the roads beyond it, 1818 to 1824.

Three days on the road, and 30 cents' worth of goods sold.

James Guild grew up in Tunbridge, Vermont, bound out from the age of 9 to work on a farm, which meant he had to work for a farmer until he turned 21. His journal does not say who bound him out. He turned 21 on July 9, 1818. He wrote that his "disposition," his nature, "would not allow me to work on a farm."

Guild held a note, a written promise that someone would pay him money. He could not turn the note into cash, so he traded it for "a trunk of goods." The first entry in his journal is headed "Tunbridge Oct, 5th 1818." Tunbridge is a town about thirty miles from Montpelier. He felt ashamed of the work and wondered why he "should stoop so low as to follow so mean a calling" (so low a trade).

At the houses he visited, he asked, in his own spelling, "do you wish to by some harecombs, needles, buttons, buttonmolds, sewing silk." Customers who wanted to buy "would want to banter" (haggle over the price) "until they could get it for nothing."

On the road. At first he sold very little. On the road toward the Dog River, he wrote, "I think I traveled 3 days and selling but 30 cent worth of goods." At poor houses that wanted "a needle or a few pins," he gave them away, and he believed he gave away more than he sold.

He killed a partridge with a stone and traded it for "a dish of bread and milk." A flood had carried off a bridge, so he crossed the river on a log and fell in with his trunk.

Over the next six years, Guild went from peddling to mending pots and pans as a tinker, then to teaching school, then to painting portraits. In the end he went to London. In 1937 an editor at the Vermont Historical Society printed his journal as *From Tunbridge, Vermont, to London, England*.

P.T. Barnum

Who: A showman who sold tickets through advertising, and whose first paid show was an enslaved woman he put on show under a false claim.
When and where: New York City and towns across the Northeast, from 1835.
Movie: *The Circus* (2018), a two-part documentary in the *American Experience* series on PBS. It is about the American circus and tells a large part of Barnum's story.

He advertised an elderly blind woman as 161 years old.

Phineas Taylor Barnum started his career in show business by showing an enslaved woman named Joice Heth to paying crowds. In 1835 a man named R. W. Lindsay of Kentucky signed a contract selling Barnum possession of Heth, the right to hold her and show her for ten months. Heth was Black, elderly and blind. Barnum advertised her as George Washington's nurse and said she was 161 years old. He lied about her age.

On show. For seven months, Barnum and his partner, Levi Lyman, showed Heth to paying crowds in cities and towns across the Northeast. They put her on show in taverns and inns, places where people drank, ate and stayed the night, and in museums, railway houses and concert halls. To bring in the crowds, they used posters and advertisements in the *New York Sun*.

When fewer people came, Barnum wrote a letter to a Boston newspaper without signing his name. In it he claimed that Heth was a machine made of whalebone and old leather, and not a person. People came back to see for themselves.

After her death. Joice Heth died in February 1836. Barnum then sold tickets to a dissection of her body, the cutting open of a dead body to look at its insides. On February 25, 1836, at the City Saloon in New York, 1,500 people paid 50 cents each to watch.

Dr. David L. Rogers, a surgeon, cut open her body while doctors and medical students watched from the crowd. Rogers examined her heart, her lungs, her brain and the places where the bones of her skull join. He decided that she had been no more than 75 years old, or at the most 80.

Barnum called himself the "Prince of Humbug," from humbug, a trick or a false claim made to fool people. He wrote a book called *The Humbugs of the World*, which came out in the middle of the 1860s. In it he wrote that humbug did not include crimes or swindles, tricks to cheat people out of their money.

The Greatest Showman, a musical movie about Barnum, came out in 2017. Its story is mostly made up, and Joice Heth does not appear in it.

Barnum's American Museum

Barnum bought John Scudder's American Museum, at Broadway and Ann Street in New York, in 1841. A day's ticket cost 25 cents.

Barnum showed the Feejee Mermaid there in 1842. His advertisements showed pictures of a beautiful creature, half woman and half fish. The object in the museum was the head and upper body of a monkey sewn to the tail of a fish. Large crowds came to see it in New York for about a month.

The Marble Palace

Alexander Turney Stewart opened a dry goods store, selling cloth, clothing and sewing supplies, at 280 Broadway in New York on September 21, 1846. The building was faced with Tuckahoe marble, and it became known as the Marble Palace.

Stewart sold goods to shoppers on several floors, and that was new. Before, shopkeepers sold to shoppers only on the ground floor and used the upper floors for wholesale, selling goods in bulk to other sellers, or as living space. Stewart set one standard price on his goods, so that customers did not haggle, and he held sales to bring in women customers.

Many writers call the Marble Palace the first American department store, a large store with many kinds of goods, each kind in its own section, called a department. Other writers name Macy's or Wanamaker's as the first, because definitions of a department store differ. The building still stands and is now called the Sun Building.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 13 MINUTES

City shoppers got the department store between 1850 and 1900, and farm families got the mail-order catalog. Store owners such as John Wanamaker made price tags and money-back promises common. Edward Clark, a partner in the Singer sewing machine business, sold the machines across the whole country on monthly payments. Sellers used advertisements to sell medicines that harmed people. In 1859 a slave dealer advertised the largest recorded sale of people in the United States.

Paying a little each month

Edward Clark began selling Singer sewing machines on installments in 1856, letting buyers take a machine home and pay for it in small amounts over time. Isaac Singer's company made the machines, and Clark, a lawyer, was Singer's business partner.

The trade-in. Clark announced in February 1856 that customers could trade in any old sewing machine, even a worthless one, for $50 off a new Singer. In September he began the installment sales. Furniture makers in New York and clock makers in New England already sold goods this way. But no business selling across the whole country had done it before, Peter Lyon wrote in 1958, in a history of Singer's machine for the magazine *American Heritage*.

Paying more. A writer for the magazine *Scientific American* thought buyers would pay twice as much in order to pay a little at a time. He said that a woman "would rather pay $100 for a machine in monthly installments" than pay $50 all at once. A Singer machine that sold for $110 could cost as little as $23 to make.

The department store

Rowland Hussey Macy opened a small dry goods store at Sixth Avenue and 14th Street in New York on October 28, 1858. It grew into Macy's department store. On the first day he took in $11.06, by most accounts. In his first year he took in about $85,000.

John Wanamaker opened a store called the Grand Depot in Philadelphia in 1876. It stood at Thirteenth Street and Market Street, in an old freight station of the Pennsylvania Railroad, a building where trains loaded and unloaded goods.

Wanamaker gets the credit for making three practices common: one price for every customer, a printed price tag on the goods and a promise of money back on anything returned. Before stores like his, customers bargained over prices, as James Guild's customers did when he peddled in Vermont. Wanamaker was not the first to use all three. A. T. Stewart had set fixed prices in 1846, and histories of the Wanamaker stores describe Wanamaker as "one of the first" to use the other two.

With a money-back promise, a customer who bought something poor could return it and get the money back. Montgomery Ward, a mail-order seller, made the same promise to customers who bought goods without seeing them first.

John Wanamaker

Who: The Philadelphia store owner who bought the first full-page newspaper advertisement for an American department store, and who later ran the national mail service.

When and where: Philadelphia, from 1861, and the Grand Depot from 1876.

A whole newspaper page for one advertisement, in 1879.

John Wanamaker bought a whole page of a newspaper in 1879 and filled it with one advertisement. It was the first full-page newspaper advertisement bought by an American department store. His first store, which he opened in Philadelphia in 1861 with his brother-in-law Nathan Brown, sold men's clothes.

Postmaster General. From 1889 to 1893, Wanamaker was the US Postmaster General, the official in charge of the national mail service. He pushed for rural free delivery, a service in which postal workers carried mail to farm families' homes. Later, mail carriers on those rural routes brought catalogs from mail-order companies, his competitors, to the same farms.

People sold at auction: Savannah, 1859

Slave dealers advertised sales of people in the newspapers, the same way sellers advertised goods. Pierce Mease Butler held people in slavery on Georgia plantation land. On March 2 and 3, 1859, he sold between 429 and 436 of them, men, women and children, to pay his gambling debts. The sale took place at the Ten Broeck Race Course, a racetrack near Savannah, Georgia. It was the largest single sale of enslaved people recorded in the United States.

The advertising. Joseph Bryan, a slave dealer in Savannah, ran the sale. He advertised it for weeks in the Savannah newspapers and in papers in seven other Southern cities, from Richmond to New Orleans. The sale brought Butler $303,850. It rained hard on both days.

Mortimer Thomson, a reporter for the *New York Tribune*, went to the sale pretending to be a buyer. He wrote a long account of what he saw, and the *Tribune* printed it under the title "What Became of the Slaves on a Georgia Plantation?"

Sold in families. Butler had ordered that the people be sold in families as far as possible, and Bryan's advertisement promised it. But a couple counted as a family only if they were married. Thomson told of a young man named Jeffrey, 23, who planned to marry a woman named Dorcas. Dorcas was to be sold in a family of four, and no buyer could buy her alone.

The man who bought Jeffrey thought about buying Dorcas too, until he learned that she could not be bought without the rest of her family. Jeffrey begged him, "Please buy Dorcas, mas'r" (master). Another buyer bought her. At this sale, parents were sold away from their children, and couples who planned to marry were sold away from each other. Black Georgians remembered the two days as "the weeping time."

Chain stores and the five-cent store

George Gilman started a business in New York selling tea and coffee in 1859. By most accounts, George Huntington Hartford was his partner from the start.

The A&P. The owners named the business the Great Atlantic and Pacific Tea Company in 1869, to mark the finishing of the first railroad across the country, and people called it the A&P. It grew into a chain store, one of many stores that belong to the same owners and sell the same kinds of goods. The A&P had 70 stores by 1878 and nearly 200 by 1900.

Frank Winfield Woolworth opened a store in Utica, New York, on February 22, 1879. Everything in it cost five cents. The store did well for about two weeks and then failed. On June 21, 1879, he opened another five-cent store in Lancaster, Pennsylvania, and that one succeeded.

Advertising agencies and patent medicines

Francis Wayland Ayer was 21 when he started an advertising agency in Philadelphia in 1869. He planned advertisements for sellers and bought space for them in newspapers and magazines. His father, Nathan Wheeler Ayer, gave him $250, and Francis named the business after him: N. W. Ayer & Son. It is usually called the first advertising agency in the country.

Ayer showed his customers what each newspaper charged for space. That way the sellers knew he was working for them and not for the newspapers.

Lydia E. Pinkham's Vegetable Compound, a patent medicine, went on sale in the middle of the 1870s. Its advertisements stated that it cured almost any "female complaint," meaning a health problem of women. Smithsonian museum records and other histories state that the mixture was 18 to 20 percent alcohol. Its label stated that the alcohol was there to dissolve and preserve the other ingredients.

Pinkham's sons put her picture in the newspaper advertisements. She asked customers to write to her personally, and she used their letters to sell more of the compound.

A syrup for babies. Mrs. Winslow's Soothing Syrup was older. Jeremiah Curtis and his partner, Benjamin A. Perkins, began selling it in 1849, to mothers to give to babies who were teething, growing their first teeth. The syrup held alcohol and morphine, about 65 milligrams of morphine in each ounce. Morphine is a strong painkilling drug. It slows a person's breathing, and too much can kill, especially a child.

One teaspoon of the syrup held enough morphine to kill an average child. The makers advertised it on recipe cards, in newspapers, on calendars and on trade cards, small printed cards given out to customers.

Members of Congress passed a food and drug law in 1906. Under it, makers had to list the drugs in their medicines on the label. Some time after that law, the makers of Mrs. Winslow's syrup took out the morphine and dropped the word "soothing" from its name.

The American Medical Association, the national organization of doctors, printed a book in 1911 called *Nostrums and Quackery*, which means fake cures and the selling of fake medicine. The book listed Mrs. Winslow's syrup under the heading "Baby Killers." No one counted how many children the morphine in the syrup killed.

Mail order reaches the farm

Aaron Montgomery Ward mailed his first catalog, a printed list of goods that customers could order by mail, from Chicago on August 18, 1872. It was one sheet of paper. It listed 163 items, by the Chicago History Museum's count.

Leaders of the Illinois Grange, an organization of farmers, made Ward their buyer late in 1872. His catalog grew to 32 pages in 1874 and to 152 pages in 1876, with about 3,000 items.

Richard Warren Sears began selling watches in 1886, while he was the railroad worker in charge of a station in Minnesota. He moved his watch business to Chicago, where he worked with a watchmaker, Alvah Roebuck. In 1893 their company became Sears, Roebuck and Company, and that year its catalog ran to 196 pages.

Mail to the farm. In 1890 nearly 41 million Americans, about 65 out of every 100, lived in the countryside. Many city people had their mail brought to their homes for free from 1863, but farm families had to pick up their mail at the post office. A Postal Service history describes farmers who had to unhitch the horse from the plow and go five or six miles into town to get it.

Postal workers began bringing mail to farm families' homes in West Virginia on October 1, 1896, on three routes that started from Charles Town, Halltown and Uvilla. This service was called rural free delivery. Within a year, 44 routes were running in 29 states.

Aaron Montgomery Ward

Who: A traveling salesman who worked out that a farm family could buy from a printed list instead of from a country store.

When and where: Chicago, from August 18, 1872.

A newspaper called his company a "swindling firm," then took it back.

Aaron Montgomery Ward built his mail-order business on a complaint he heard from farmers about their country stores. He had worked as a clerk at the Chicago store that later became Marshall Field's, and then as a traveling salesman for a St. Louis company that sold goods in bulk to stores.

On the road, Ward heard the farmers' main complaint. Country general stores charged high prices and had few goods to choose from. The farmers blamed the middlemen, the sellers between the maker and the buyer, for taking too much profit. Ward's plan was to buy straight from the factories, pay cash and sell straight to customers by mail.

The Chicago Fire of 1871 destroyed Ward's savings. In 1872 he started over with $1,600 of his own and $800 from his brother-in-law, George R. Thorne.

"Grangers, Beware!" Editors at the *Chicago Tribune* printed an article with that heading on November 8, 1873. It described Montgomery Ward & Co. as a "swindling firm" that hid behind a post-office box. Ward sued the *Tribune* for libel, printing something false that harms a person's name, and asked for $20,000.

The editors printed a notice taking back the charge on December 24, 1873. In it they called Ward's company "a bona fide firm" (a real one), "composed of respectable persons, and doing a perfectly legitimate business."

Ward promised that customers could send goods back for their money. By 1875 his catalog included the words "satisfaction guaranteed or your money back." When he died in 1913, his company sold $40 million worth of goods a year and had 6,000 workers.

Richard Sears

Who: A railroad station agent who started by selling watches and went on to build the Sears catalog.

When and where: Minnesota in 1886, then Chicago.

Movie: *Mr. Sears' Catalogue* (1989), a documentary in the *American Experience* series on PBS, about Richard Sears and his catalog.

A box of watches a jeweler refused, sold for $14 each.

Richard Sears was 22 when 1886 began. He was the station agent for the Minneapolis and St. Paul Railroad at North Redwood, Minnesota, next to the town of Redwood Falls.

That year a jeweler refused a shipment of watches that came to Sears's station. Sears got permission to sell the watches himself, and he sold them to other station agents along the line for $14 each. Then he quit the railroad and started the R. W. Sears Watch Company in Minneapolis. He moved the business to Chicago, where he hired a watchmaker, Alvah C. Roebuck, to repair watches that customers sent back.

The big book. Sears wrote the words in the catalog himself. At its biggest, the Sears catalog ran to about 1,500 pages and listed about 100,000 items. The documentary *Mr. Sears' Catalogue* states that it went to about 20 million people at its height.

Sears resigned as president of the company in 1908 or 1909. He died on September 28, 1914.

Persis Foster Eames Albee

Who: The first woman to sell door to door for the company that later became Avon.

When and where: Winchester, New Hampshire, and across the Northeast, from 1886.

At 50, she took perfume from door to door.

Persis Foster Eames Albee was 50 years old in 1886 when David McConnell hired her to sell perfume door to door. McConnell had sold books from house to house. He gave away perfume as a free gift with the books, and he found that customers liked the perfume more than the books. He started a perfume company, the California Perfume Company. It later became Avon.

Albee, who lived in Winchester, New Hampshire, became the company's first General Traveling Agent. Her product was the Little Dot Perfume Set, which held five scents: violet, white rose, hyacinth, lily of the valley and heliotrope, a small purple flower. She sold it from house to house in Cheshire County, New Hampshire, and then across the Northeast, and she found other women to sell for the company and trained them.

Most of what is known about Albee comes from Avon's own history of itself.

Selling without a vote. Albee could sell across the Northeast and train other women to sell, but she could not vote. When she started selling, women could vote in only three western territories, lands that were not yet states: Wyoming, Utah and Washington. In 1887, lawmakers in Congress took the vote away from women in Utah, and judges on Washington's highest court struck down that territory's law.

A change called the Nineteenth Amendment became part of the Constitution in 1920, 34 years after Albee started selling. It states that no state and no federal official may keep a citizen from voting because of whether the citizen is a man or a woman.

BY 1900

If you remember one thing: Between 1850 and 1900, shoppers began paying by the month, buying at marked prices and ordering from catalogs, while dealers in the South still sold people like goods.

- 1856: Edward Clark starts selling Singer sewing machines on monthly payments.
- 1859: Pierce Butler sells between 429 and 436 people at a racetrack near Savannah, the largest recorded sale of people in the United States.
- 1872: Montgomery Ward mails his first catalog, one sheet of paper.
- 1876: John Wanamaker opens the Grand Depot in Philadelphia. He gets the credit for making price tags and money-back promises common.

Next: After 1900, one grocery chain grew to about 15,000 stores, and shoppers began taking goods off the shelves themselves.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 14 MINUTES

Chain stores spread across the country, and the biggest grocery chain, the A&P, grew from about 200 stores in 1900 to about 15,000 by 1930. Grocers began letting customers take goods off the shelves themselves, and families bought cars and radios on monthly payments. Sellers began paying radio and television stations to air their advertisements. Coal companies paid miners in their own money, which only the company store accepted at full value. During the Second World War, shoppers needed ration stamps as well as money to buy many foods.

The chain store at full size

The A&P had 1,600 stores by 1915. By 1930 it had about 15,000 stores and sold $2.9 billion worth of goods a year. That made it the world's largest retailer, a business that sells goods to the people who use them. The A&P was the largest grocery seller in the United States from 1915 to 1975.

The lunch counter. Woolworth's stores had lunch counters, where customers sat and ate. In the South, the people who ran these counters served white customers only, and the rule was still in force in 1960. On February 1 of that year, four Black students from North Carolina A&T, a college in Greensboro, North Carolina, sat down at the Woolworth counter there and asked to be served. They were Ezell Blair Jr., Franklin McCain, Joseph McNeil and David Richmond. The counter staff refused to serve them.

Hundreds of people joined their protest over the next six months. On July 25, 1960, the staff of the Greensboro counter began serving Black customers. By the account of Geneva Tisdale, who worked in the store's kitchen, the manager of the lunch counter picked her and two other Black kitchen workers to be the first. He told them to get dressed, walk around the store like customers and then sit at the counter. Tisdale ordered an egg-salad sandwich. Chapter 25, Rights Movements, tells the sit-ins in full.

F. W. Woolworth

Who: The storekeeper who built a national chain of five-cent and ten-cent stores and paid cash for the tallest building in the world.

When and where: Utica and Lancaster, 1879, and New York, 1911 to 1913.

$13.5 million for the world's tallest building, paid in cash.

Frank Winfield Woolworth turned his five-cent stores of 1879 into the largest store company in the world. On November 12, 1911, he signed agreements to join his chain with four rival chains, the Knox, Kirby, Charlton and C. S. Woolworth chains.

The new F. W. Woolworth Company started with 596 stores and $65 million in capital, the money put into a business to run it. The company was formally set up in 1912, so some histories list that year as its start.

The building. The Woolworth Building in New York was finished in 1913. At 792 feet, it was the tallest building in the world at the time. It cost $13.5 million, and Woolworth paid for it himself, in cash.

Who first let shoppers serve themselves?

Clarence Saunders started self-service shopping when he opened the first Piggly Wiggly store at 79 Jefferson Street in Memphis, Tennessee, on September 6, 1916, by most accounts. His customers picked goods off the shelves themselves instead of asking a clerk to fetch them.

The turnstile. Customers came in through a turnstile, a gate with turning arms that lets one person through at a time. Then they had to follow a single fenced path past every shelf in the store before they reached the place where they paid.

Saunders asked the government for a patent, a paper that lets only the inventor use an invention for a number of years. He applied for one on his "Self-Serving Store" on October 21, 1916, and received Patent 1,242,872 on October 9, 1917.

Michael J. Cullen opened King Kullen in Jamaica, in Queens, New York, on August 4, 1930. He had worked for the Kroger grocery company and had sent its managers a plan for a supermarket, which they ignored.

By the account of the Smithsonian, the national museum in Washington, King Kullen was the first supermarket. It met the museum's five tests: separate departments, self-service, low prices, many stores under one name and buying goods in huge amounts. Cullen's slogan, a short phrase repeated in advertising, was "Pile it high. Sell it low."

Clarence Saunders

Who: The Memphis grocer who started self-service shopping and then lost his company fighting traders on Wall Street.

When and where: Memphis, Tennessee, 1916 to 1923, and later stores. He died in 1953.

Orders for 196,000 of the 200,000 shares, and he still lost.

Clarence Saunders built Piggly Wiggly into a chain of 1,268 stores and then lost it in a fight over its stock. By 1923 the stores sold $100 million worth of groceries a year, and Piggly Wiggly was the third-largest grocery business in the country. Saunders had put the company's stock up for sale on the New York Stock Exchange, a market where people buy and sell shares, small pieces of ownership in a company.

Writers on the history of money and business describe the fight this way. In early 1923, some Piggly Wiggly stores in New York that other owners ran under the Piggly Wiggly name failed. Traders then began selling Piggly Wiggly stock short. Selling short is a bet that a stock's price will fall. The trader sells shares he has borrowed, and later he must buy the same number of shares to give back. If the price has fallen by then, he buys them for less than he sold them for.

Saunders fought back. He borrowed about $10 million from bankers in the South, added his own money and kept buying shares. In the end he held orders for 196,000 of the 200,000 shares that existed.

Why he still lost. Many of the shares Saunders bought came from the short sellers, who now had to hand them over. Saunders held almost every share, so if they had to buy from him, he could name the price.

Then the men who ran the Stock Exchange stopped trading in the stock. They gave the short sellers five days to hand over the shares, instead of the usual one day. In those five days, shares poured in from distant places, and the short sellers used them to pay what they owed.

> Saunders still owed the bankers, and a payment of $2.5 million on his loan was due on September 1, 1923. He did not have the money. In August he gave up the company and handed what he owned to the people he owed. He was bankrupt, which means he could not pay his debts.
>
> He started again with a new chain called the "Clarence Saunders, Sole Owner of My Name" stores. They did well until the 1930s, when they failed during the Great Depression, the years of hard times that began in 1929. Later he built the Keedoozle, a grocery store that worked like a vending machine. It never made a profit. Saunders died on October 15, 1953.

The company store

Many coal companies in Appalachia owned the store in their mining towns and paid their miners mostly in scrip, the company's own money, made as metal coins or paper. A miner could draw scrip against pay he had already earned. The company store accepted it at full value, but a miner could not turn it in for cash.

Scrip at other stores. Other merchants nearby took scrip only at a discount of 10 to 30 percent. A miner who spent $1 of scrip at another store got only 70 to 90 cents' worth of goods.

Researchers for the US Coal Commission, a government group that studied the coal industry, checked prices in coal towns in 1922. Food cost 4.2 percent more at company stores than at independent stores in southern West Virginia, and 7 percent more in Alabama. Later, Price Fishback, an economist who studies how money, work and business fit together, looked at these numbers. He found that at most companies a miner's store debt ran no longer than two weeks.

Two views. A page by the West Virginia Mine Wars Museum describes scrip differently. It describes the system as debt bondage, being forced to keep working to pay off a debt. In its words, scrip "created a kind of debt bondage

that kept coal miners and their families tied to their employer for almost everything they needed to survive." Fishback found small price gaps and short debts.

West Virginia lawmakers passed a law in 1891 under which companies could give out scrip only as pay for work already done or as an advance on wages. In 1925 they made it against the law to pass scrip from one person to another. Scrip disappeared when the coal industry collapsed in the 1950s.

The catalog under Jim Crow

Jim Crow is the name for the laws and customs in the South under which white lawmakers and officials kept Black people apart from white people in schools, railroad cars, waiting rooms, courtrooms and cemeteries, and at water fountains. White mobs enforced it with murder. Between 1877 and 1950, white mobs lynched more than 4,400 Black people, killing them without any trial, by the count of the Equal Justice Initiative.

Hyman's argument. Louis Hyman, a historian at Cornell University, argued in 2018 that mail-order catalogs helped Black farm families in the South get around Jim Crow at the store. NPR, the *Washington Post* and America's Black Holocaust Museum reported his argument.

According to Hyman, many Black farm families had to buy on credit at a white-owned country store, often owned by the landlord whose land they farmed. The storekeeper set the prices and made Black customers wait until white customers were served. A catalog order cost the same national price for everyone, and no clerk decided who was served first. Instead of the landlord's credit, the family could use the mail-order company's.

Hyman says that some Southern storekeepers held bonfires to burn catalogs, and told white customers that Sears was owned by Black people to keep them from buying there. These are Hyman's claims.

Buying on time

Americans began buying cars, radios, furniture and household machines on installments in the 1920s, paying a little at a time, as buyers of sewing machines had since the 1850s. By the end of the 1920s, about 60 percent of cars and about 80 percent of radios were bought on installments, according to the reference site Encyclopedia.com. The number of cars registered in the United States grew from about 9 million in 1920 to nearly 27 million by 1929, nearly three times as many.

Members of the National Association of Finance Companies, a group of lending companies, agreed on standard terms for car buyers in 1924. A buyer paid one-third of the price at the start for a new car and two-fifths for a used car. The buyer then had no more than twelve months to pay the rest.

Advertising on radio and television

The first paid radio advertisement went on the air on August 28, 1922, on the station WEAF in New York. The Queensboro Corporation paid $50 for about ten minutes of air time to promote the Hawthorne Court apartments in Jackson Heights, Queens.

No television station could sell advertising until officials of the Federal Communications Commission, the government office that licenses stations, gave licenses to 10 stations on May 2, 1941. Under those licenses, the stations could sell advertising from July 1, 1941. Two years earlier, people at NBC had shown test advertisements for three companies.

Nine dollars for ten seconds. The first commercial under the new licenses ran on WNBT in New York on July 1, 1941, before a Brooklyn Dodgers baseball game against the Philadelphia Phillies. For about ten seconds it showed the face of a Bulova watch over a map of the United States, while a voice said, "America runs on Bulova time." It cost Bulova $9: $4 for the air time and $5 in station charges. There were about 4,000 television sets in the New York area at the time.

Mary Frances Gerety wrote advertisements at N. W. Ayer & Son, the advertising agency. For the diamond company De Beers, she wrote the slogan "A Diamond Is Forever" in the late 1940s. In 1999 the editors of the magazine *Ad Age* named it the most memorable slogan of the 1900s.

Shopping under rationing

Officials of the US government's Office of Price Administration rationed many goods during the Second World War, limiting how much of each one a person could buy. Sugar, meat, cheese, fats, shoes, gasoline and processed foods, foods that were canned, dried or frozen, were among them. At the start, each person could buy 26 pounds of sugar a year, about 8 ounces a week. Every person in the country, babies included, got a ration book full of stamps.

War Ration Book Two came into use on March 1, 1943. Its blue stamps bought processed foods, and its red stamps bought meat, cheese, fats and oils. Each stamp was worth a number of points, such as 8, 5, 2 or 1, and every food had a price in points as well as in money. In a cartoon the US government made to teach the system, a shopper's string beans cost 3 blue points.

Each person got 48 blue points and 64 red points a month, so a family of four had 192 blue points a month for processed food. Every cut of almost every kind of meat had its own point value.

At the store a shopper paid with stamps and with money together. The price could not go above the ceiling price, the highest price government officials allowed. Point values changed often. Newspaper editors printed them, radio announcers read them out and members of local ration boards posted them. By the end of the war, more than 100,000 volunteers were running the program through about 5,600 local boards.

Emilia Ruth Palmquist Lundberg

Who: A widowed mother of six in Jersey City who shopped for her family with ration books during the Second World War.

When and where: Jersey City, New Jersey, about 1942 to 1943.

Six children, $100 a month and a ration book for each.

Emilia Lundberg, who was born in Göteborg, Sweden, in 1894, fed six children on a small, fixed income, and during the war she had to count ration stamps as well. In 1930 her husband was killed in an explosion at the dry-cleaning plant where he was in charge. As her son Donald remembered it, his father left her with six children, no relatives to turn to and no money.

She raised the children on workmen's compensation, money paid to a worker's family after the worker is hurt or killed on the job, of $100 a month. The rent was $35 a month. Her oldest daughter, 16, moved to night school and took a job keeping accounts for $12 a week. Donald packed and delivered groceries at a store after school and on Saturdays for $5 a week.

At the store. Rationing changed little at the family's table. Before the war, they ate a roast on Sunday after church and potatoes and macaroni the rest of the week. Under rationing, they still bought meat only for Sunday and ate macaroni, spaghetti and mashed potatoes the rest of the week.

Everyone in the family had a ration book with stamps for meat and other goods, and they took the books to the store and bought what the stamps allowed. The ration was counted for each person, so a large family got more in all than a small one. Shoes were rationed too. Each child had one good pair for school and one pair of sneakers for play.

Emilia had a 1933 Plymouth car. Donald remembered the family's gasoline ration as about a gallon and a half, but he did not say whether that was for a week or a month. When they told the ration board they

were driving to visit his older brother, a soldier at Camp Kilmer in New Jersey, they got a coupon for about two more gallons. Donald described all of this in an interview for the Rutgers Oral History Archives in 2008, about 65 years later.

BY 1950

If you remember one thing: Between 1900 and 1950, shoppers began picking goods off the shelves themselves in ever-bigger chain stores, and they bought cars and radios on monthly payments.

One number: 15,000. The A&P had nearly 200 stores in 1900 and about 15,000 in 1930, about 75 times as many.

Next: After 1950, shoppers drove to the first enclosed malls and paid with plastic cards.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 10 MINUTES

Americans began shopping at enclosed malls and discount stores after 1950, and paying with plastic cards. The first fully enclosed shopping mall in the United States opened in Minnesota in 1956. Kmart, Target and Wal-Mart all opened their first stores in 1962. In 1974 a cashier in Troy, Ohio, made the first sale with a scanner that read a printed code on a package. By 2000, people could shop at home on television shopping channels, and from 1995 on a computer.

The enclosed mall

Southdale Center opened in Edina, Minnesota, on October 8, 1956. It was the first fully enclosed shopping mall in the United States with its own heating and cooling. It had 72 stores on two levels around a brightly lit central court. Its two biggest stores were the department stores Dayton's and Donaldson's.

The Dayton family owned department stores in Minneapolis. In 1952 they hired the architect Victor Gruen and his firm, Victor Gruen & Associates, to design the mall. Building began in 1954. Southdale cost $20 million.

Victor Gruen

Who: The architect who designed the first enclosed American mall and later rejected what developers built from his idea.

When and where: Vienna, then New York and Minnesota. Southdale opened October 8, 1956.

Movie: *The Gruen Effect: Victor Gruen and the Shopping Mall* (2009), a documentary by Anette Baldauf and Katharina Weingartner, about Gruen and the malls he started.

He arrived in New York with eight dollars and no English.

Victor Gruen designed Southdale as the center of a new neighborhood, and developers copied only the mall. Gruen was a Jewish architect from Vienna, in Austria.

Leaving Vienna. In March 1938 the leaders of Nazi Germany, the Germany ruled by Adolf Hitler's Nazi Party, took over Austria and made it part of Germany, a takeover called the Anschluss. In the weeks after, Austrian Nazis and others beat, attacked and humiliated Jews. In Vienna, Nazis forced Jews to scrub the city's streets while crowds jeered at them. Soon Nazi officials put Austria's Jews under the same laws that German officials already used to take away Jewish rights in Germany.

Gruen fled Vienna that year. He reached New York on the ship *Statendam* with an architect's degree, eight dollars and no English. One of his early jobs was a shop front on Fifth Avenue in New York, for Ludwig Lederer's leather-goods shop. Gruen designed its entrance as an arcade, a covered walkway, meant as a "customer trap" to draw people inside.

For Southdale, the Dayton company bought about 500 acres of land in Edina. Gruen's plan for the land around the mall had apartment buildings, houses, schools, a medical center, a park and a lake. He said he wanted "the Minneapolis downtown you would get if you started over and corrected all the mistakes." The medical center opened in 1965. The Daytons sold the rest of the land for housing.

> In a speech in London in 1978, Gruen rejected the malls that American developers had built. He said he refused to pay alimony, the money a person pays a former husband or wife after a divorce, for those developments, and he used a rude word to describe them. He went back to Vienna and objected to a shopping center built there.

Plastic at the counter

With a credit card, a shopper could take goods home from any store that took the card and pay the bank later. Chapter 17, Money, describes how the cards began: the cardboard Diners Club card of 1950, which holders had to pay off in full every month, and the BankAmericards that officers of Bank of America mailed to people in Fresno, California, in September 1958.

At the counter. The Fresno cards came ready to use, though nobody had asked for them. Each of the 60,000 or so cardholders got $300 to $500 of credit at once, good in any store that took the card, and each store that took it paid the bank 6 percent of every sale. The bank's officers later licensed the card across the country, and in 1976 it was renamed Visa. In 1970 members of Congress banned mailing credit cards to people who had not asked for them.

Women and credit. Before 1974, lenders often turned women down for credit because they were women, or because they were married or single. Mortgage lenders, who lend money to buy houses, often counted only part of a married woman's pay, especially if she was young enough to have children. Lawmakers in Congress passed the Equal Credit Opportunity Act in October 1974. Under it, lenders could no longer treat people differently because of their sex or whether they were married.

In March 1976 the lawmakers added more reasons lenders could not use: race, color, religion, national origin, which means the country a person or their family came from, age, and getting money from government aid programs.

Four discount chains in one year

Four discount chains, selling goods at prices below what other stores charged, began in 1962. The first Kmart store opened in Garden City, Michigan, on March 1, 1962, and reports stated that 4,000 customers waited in line on the first day. Target's first store followed on May 1, in Roseville, Minnesota. On July 2, Sam Walton opened the first Wal-Mart in Rogers, Arkansas. Woolco's first store opened that year too.

Kmart grew out of the S. S. Kresge chain of five-and-dime stores, which sold small goods for 5 and 10 cents, and Woolco out of F. W. Woolworth's company. Target grew out of the Dayton Company's department stores, run by the same Dayton family that had hired Victor Gruen to design Southdale.

Sam Walton

Who: The Arkansas storekeeper who put discount stores in small towns and built the largest retailer in the United States.

When and where: Rogers, Arkansas, from July 2, 1962.

Movie: *Wal-Mart: The High Cost of Low Price* (2005), a documentary by Robert Greenwald. It argues against the company and is not a life story of Walton.

He put discount stores in towns too small for the big chains.

Sam Walton built Walmart, spelled Wal-Mart at first, by opening discount stores in small towns. The heads of the large chains had judged those towns too small to bother with. Before 1962 Walton ran Ben Franklin variety stores in Arkansas, which sold many kinds of small, cheap goods. He bought goods in huge amounts and priced them below the stores nearby. Walmart became the largest retailer in the United States and the largest private employer, which means it had more workers than any other company.

Economists have measured what happened to shoppers and to towns when Walmart stores opened. Two economists, one at MIT and one at the US Department of Agriculture, reported in 2007 that shoppers gained

a great deal from Walmart's low prices, most of all households with lower incomes.

Jobs and sales. Three other economists reported in 2008 that when a Walmart opened, the county around it lost about 150 retail jobs, or 2.7 percent. By their count, each Walmart worker replaced about 1.4 workers at other stores.

Small towns. Kenneth Stone, an economist at Iowa State University, has studied Walmart in Iowa since 1988. In a 2012 study, he and a co-author followed 28 Iowa towns of 3,000 to 20,000 people for 15 years after each got a Walmart. Total retail sales in those towns rose, but most of the gain went to Walmart, and other stores in town that sold the same things lost sales.

None of the study's Walmart towns had fewer than 3,000 people. In Iowa's smallest towns, under 2,500 people, retail sales fell 30 percent, about $1.5 billion, in the years after Walmart came to the state. In those years, shopping in Iowa moved to the bigger cities and to the Walmart towns, where people from nearby towns now came to shop.

The franchise

Roy Allen bought a recipe for root beer in 1919. In 1924, by most accounts, he began selling A&W franchises: the right for other people to run a business under the A&W name, following his rules. Writers on the history of franchising often call A&W the first successful food franchise. J. Willard Marriott was one of its early franchise owners.

The Howard Johnson's restaurant business sold its first franchise to Reginald Sprague in 1935. By the end of 1936 there were 41 Howard Johnson's restaurants. By 1939 there were 107, with about $10.5 million in sales.

The barcode

The Universal Product Code, or UPC, is the barcode printed on packages in stores, read by a scanner at the checkout.

The people who planned the first scan chose a small pack of gum on purpose. They wanted to show that the code could be printed clearly on a package that size. The Smithsonian's National Museum of American History keeps one of the first ten scanners, a Spectra-Physics Model A.

Sharon Buchanan

Who: The supermarket cashier who made the first sale ever scanned with a barcode.

When and where: Marsh Supermarket, Troy, Ohio, 8:01 a.m., June 26, 1974.

A ten-pack of Juicy Fruit gum, 67 cents, at 8:01 in the morning.

Sharon Buchanan was the cashier at the first sale ever made with a barcode scanner. At 8:01 in the morning on June 26, 1974, she was working at the Marsh Supermarket in Troy, Ohio. The customer was Clyde Dawson, who headed research and development, the work of trying out new ideas, at Marsh. He bought a ten-pack of Wrigley's Juicy Fruit gum. Buchanan passed it over the scanner. The price was 67 cents.

Stamps, television selling and the end of the big catalog

Green Stamps. About 80 percent of American households collected S&H Green Stamps in the 1960s and 1970s. These trading stamps were small stamps that shoppers got with purchases at grocery stores, gas stations and other stores. Shoppers pasted them into books and traded full books for goods at a Green Stamps store or from a catalog. In the 1960s more S&H stamps were printed than US postage stamps, and about 35 million S&H catalogs went out each year.

Shopping on television. The Home Shopping Network began with a radio station. In 1977 a company that advertised on Lowell "Bud" Paxson's radio station paid him in electric can openers instead of money. The station's host, Bob Circosta, sold them on the air for $9.95 each. In 1982 Paxson and Roy Speer started the Home Shopping Club on cable television in Pinellas County, Florida, and it went national on July 1, 1985.

Joe Segel founded QVC, another shopping channel, in 1986. It first went on the air that November, and the first thing it sold was a radio made to be used in the shower.

The last Sears catalog. Sears officials announced on January 25, 1993, that they were closing the company's catalog business, cutting 50,000 jobs and closing 113 stores. The last general catalog went out that spring. The catalog had run for about a hundred years.

Buying on a computer begins

Jeff Bezos filed papers to start a company called Cadabra, Inc., on July 5, 1994, working from a rented house with a garage in Bellevue, Washington. The company became Amazon. Amazon.com opened for orders on July 16, 1995. The first book it sold was *Fluid Concepts and Creative Analogies*, by Douglas Hofstadter. In its first week it took in about $12,000 in book orders.

Pierre Omidyar started a website called AuctionWeb, where people sold goods at auction to whoever offered the most money. He started it over Labor Day weekend, on September 3, 1995. It later became eBay. The first thing sold on it went for $14.83. By eBay's account, it was a broken laser pointer.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 12 MINUTES

About one dollar in six of all American retail spending went to online sales from April to June 2026, by the count of the US Census Bureau. Since 2005, members of Amazon Prime have paid a fee for two-day shipping. Drivers carry the orders to people's doors, including on days of summer heat. Sellers now buy the right to show advertisements to one person at a time, chosen from records of what that person has done online.

How much shopping is online?

Online sales were 17.1 percent of all retail sales in the United States from April through June 2026, by the count that workers at the US Census Bureau publish every three months. In all of 2025 they were 16.4 percent. The April-to-June figure came to $340.2 billion, 12.2 percent more than in the same months of 2025. The bureau's staff released those numbers on August 18, 2026. By those numbers, more than four dollars in five were still spent in stores.

Researchers at the Federal Reserve, which acts as a bank for other banks, ask people to keep diaries of how they pay. In 2024, credit cards were the most-used way to pay in the United States, used for 35 percent of all payments. Cash was used for 14 percent.

Jeff Bezos

Who: The founder of Amazon, which began as an online bookstore and grew into a store selling books, music, movies and many other kinds of goods.

When and where: Bellevue and Seattle, Washington, from July 1994.

Movie: *Amazon Empire: The Rise and Reign of Jeff Bezos* (2020), a documentary by James Jacoby and Anya Bourg in the PBS series *Frontline*. It is an investigation that argues a case against Amazon's power.

Web use was growing 2,300 percent a year, so he quit.

Jeff Bezos was a vice president at D. E. Shaw, an investment company in New York, when he read in 1994 that use of the web had been growing by 2,300 percent a year that spring. He quit his job to sell goods online.

He and his wife, MacKenzie Bezos, moved from Manhattan to the Seattle area. They chose it because software engineers lived and worked there, at Microsoft and the University of Washington, and because of its taxes. Bezos studied which product would sell best online, and he chose books. Music and DVDs were the next goods Amazon added.

Prime. When Amazon Prime began on February 2, 2005, members paid $79 a year and got two-day shipping on their orders at no extra charge, according to Amazon's announcement. By 2021, more than 200 million people around the world were members. In 2026 Prime in the United States cost $139 for members who paid once a year, or $14.99 a month for members who paid monthly, about $180 over a year. Those prices had not changed since February 2022, according to the shopping news site dealnews.

The workers. Senate committee staff, drivers and union representatives have complained about how Amazon treats the people who pack and deliver its orders. A Senate committee's staff counted more injuries in Amazon's warehouses than in other warehouses. In 2023 Amazon ended its contract with one delivery company soon after the company's drivers

> joined a union, a group that workers form to bargain together with their employer. Amazon's representatives said the contract ended for other reasons.

Malls empty out

As shoppers moved online or to newer outdoor shopping centers, the anchor department stores of many malls closed. Anchor stores are the big stores that draw shoppers to the smaller shops around them, and the smaller shops that had depended on the department stores' crowds lost customers. Mall owners could not find new stores to fill the empty space. Owners and builders have turned some empty malls into apartments, clinics, colleges and warehouses.

How empty. Nobody keeps an official government count of malls, and the private counts disagree, because people define a mall in different ways. In 2017, Green Street Advisors, a company that grades malls for investors, sorted them by how well they were doing. The more than 300 malls it gave its lowest grades were only about 72 percent full, while the best malls were about 95 percent full. In the weakest malls, more than a quarter of the space stood empty.

Across the whole country, researchers at Coresight Research, a company that studies stores, counted about 3,000 more stores closing than opening in 2025.

Advertising aimed at one person

When a person visits a website, the company that runs the site can save small files and codes on the person's device that record what the person looked at. Companies called data brokers collect records like these and combine them with records from outside the internet, such as store purchases, warranty cards and property records. Then they sell what they have gathered, including the chance to show an advertisement to a chosen group of people.

The staff of the Federal Trade Commission, the government office that protects shoppers, published a report in May 2014 called *Data Brokers: A Call for Transparency and Accountability*. The commission's staff found that data brokers held "billions of data elements," meaning billions of separate facts, about nearly every American shopper. They gathered most of it without people knowing.

The report describes how brokers attached offline records to the files in a person's web browser, so the person could be followed from site to site and shown advertisements chosen for them. The commissioners asked Congress to pass a law under which people could see the information held about them.

Website and app owners use money from this advertising to keep their services free to use, and the advertisements can show a person something they want to buy. But companies the person has never heard of buy and sell records of what the person reads and buys. As of 2026, members of Congress had passed no general federal privacy law. By June 29, 2026, lawmakers in 24 states had passed their own, according to the law firm Husch Blackwell.

Selling to children inside a game

Fortnite is a video game made by Epic Games that came out in 2017. It is free to play. Epic sells V-Bucks, game money that players buy with real money to get costumes and dance moves for their characters. By 2022, the game had more than 400 million players around the world, according to the Federal Trade Commission.

The $520 million. Officials of the Federal Trade Commission announced on December 19, 2022, that Epic had agreed to pay $520 million. Of that, $275 million was a penalty under the Children's Online Privacy Protection Act, a law that requires a company to get a parent's permission before it collects personal information from a child under 13. The officials said Epic had collected personal information from children under 13 who played Fortnite without telling their parents or getting their permission.

The other $245 million went back to customers as refunds. The officials said Epic had used "dark patterns," confusing buttons and screens that led players into charges they did not mean to make. They said children could buy V-Bucks just by pressing buttons, with no parent or card owner agreeing, and that the game could charge a player who was only waking it from sleep mode.

The officials cited "hundreds of millions of dollars in unauthorized charges," meaning charges that players had not agreed to. Lina Khan was the head of the commission at the time.

Payday loans

A payday loan is a small loan from a storefront lender that the borrower is supposed to pay back on the next payday. Researchers at the Consumer Financial Protection Bureau, a government office that watches over lenders, published a study of these loans in April 2013. They found that a borrower who took the typical $350 loan, with a fee of $15 for each $100, had to find more than $400 in 14 days.

Nearly half of payday borrowers took out more than 10 loans in a year, and the typical borrower was in debt 199 days of the year, more than half of it. Chapter 17, Money, describes the rest of the study, including how a fee like that comes to 391 percent a year.

Buying secondhand clothes

The staff of ThredUp, a company that sells used clothes online, publish a yearly report on the market for secondhand clothes. Because ThredUp sells used clothes, its report is a company's estimate, not a government count.

By ThredUp's 2025 report, the market for used clothes in the United States grew 14 percent in 2024, and online sales of used clothes grew 23 percent. In 2024, 58 percent of shoppers bought something secondhand, up from 52 percent in 2023. The firm GlobalData did the research for ThredUp's 2026 report, which states that in 2025 used clothing sales grew almost four times as fast as all clothing sales. GlobalData's researchers forecast that the US secondhand market will reach $78.8 billion by 2030.

The last step: heat in the delivery truck

The drivers who make fast delivery possible keep delivering in summer heat. Injury records kept by OSHA, the government's workplace safety office, list at least 40 UPS drivers taken to the hospital since 2015 for heat illness, which happens when the body gets too hot and cannot cool itself down. Danielle Kaye of NPR gave that count in July 2023.

José Cruz Rodriguez was 23. He died of heat illness during a UPS delivery shift in Waco, Texas, in August 2021, according to OSHA. His family sued UPS for causing his death, and the case was settled.

Esteban Chavez Jr., a UPS driver for four years, died after he was found unconscious in his truck. It was June 25, 2022, the day after his 24th birthday, and he was on his route in Pasadena, California. The temperature that day reached the upper 90s. UPS trucks had no air conditioning then. His family said at least 20 minutes went by before anyone noticed him and called for help.

The medical examiner, the doctor whose job is to find out why a person died, found that his heart stopped working. His family believes the heat caused it.

UPS managers and leaders of the Teamsters union, which represents UPS drivers, agreed in June 2023 that every small delivery truck UPS bought after January 1, 2024, would have air conditioning, heat shields and fans.

Inside the warehouses. Warehouse workers pack the orders before drivers carry them. In December 2024, Senator Bernie Sanders released a report by staff of the Senate committee he led, staff members who work for the party that holds the most seats in the Senate. They found that in 2023 Amazon recorded more than 30 percent more injuries at its warehouses than the average for the warehouse industry. For seven years in a row, Amazon warehouse workers were nearly twice as likely to be hurt as workers at other warehouses.

The report states that Amazon had plans to lower injuries and did not use them, because the plans would slow down work.

Renica Turner

Who: A driver who delivered Amazon packages in the California desert for a small delivery company, and who joined a union to push for protection from the heat.

When and where: Palmdale and Victorville, California, about 2022 to 2023.

110 degrees, a broken air conditioner and 300 packages to go.

Renica Turner delivered Amazon packages for a company called Battle Tested Strategies. It was one of about 3,000 small companies under contract to deliver Amazon's packages. Its vans were leased from Amazon, and its drivers worked out of an Amazon building in Palmdale, California.

The hot day. In a July 2023 story by the NPR reporter Danielle Kaye, Turner described a day in Victorville, California, when the temperature reached 110 degrees. The van's air conditioning and fans were not working, she said. When she rolled the windows down, hot air came in. Her body began to tingle, and she thought she might pass out.

She told the Amazon dispatcher, the worker who sends drivers out and keeps track of them, and the dispatcher gave her a 20-minute break. In her words, "They never sent no one out to help me with the rest of the route." The route was 300 packages, and she was expected to deliver 25 every hour. She finished it herself, feeling woozy and numb.

Amazon's representatives told NPR that its vans had working air conditioning and that any van without it was taken off the road at once. The delivery companies were in charge of fixing the vans, they said, and Amazon had changed some routes the year before so drivers could take more breaks to cool down. The owner of Battle Tested Strategies, Johnathon Ervin, said Amazon regularly failed to fix broken air conditioning, sometimes for weeks or months.

The union. Turner and 83 other drivers at the company joined the Teamsters union in April 2023 and bargained a contract with Battle Tested Strategies. NPR reported that they joined the union in large part

to push for protection from the heat.

Amazon ended its contract with the company in June 2023. Teamsters representatives and the company's owner said Amazon was punishing the drivers for joining a union. Amazon's representatives said Amazon ended the contract over safety violations and missed insurance payments by the company. The drivers went on strike, stopping work as a protest.

What the drivers got. Lawyers for the National Labor Relations Board, the government office that enforces labor law, brought a case against Amazon. They argued that Amazon was the drivers' employer too, even though another company had hired them. In April 2026 they settled the case with Amazon, agreeing on terms that ended it. Turner and the other drivers, about 84 of them, were each to get two weeks of pay, about $250,000 in all.

Under the settlement, Amazon did not admit that it had been the drivers' employer. A judge approved the settlement in May 2026, and the Teamsters asked to appeal. The news stories about the case do not say what Turner did after the strike.

THE THREE QUESTIONS, ANSWERED

1. **How did colonists pay for goods when they had almost no coins?** They took goods home on credit and paid later in crops, other goods, work and wampum, while the shopkeeper wrote each debt in a ledger.
2. **Where were people bought and sold like goods?** At the market New York's aldermen set up on Wall Street in 1711, in newspaper advertisements such as the 277 or more in Franklin's *Pennsylvania Gazette*, and at a racetrack near Savannah, where Pierce Butler sold between 429 and 436 people in 1859.
3. **How much of what Americans buy today is bought online?** About one dollar in six: online sales were 17.1 percent of all retail

sales from April to June 2026, by the US Census Bureau's count.

What changed: Before 1500, goods passed hand to hand at trading places like the one on the Columbia River, and by 2026 one dollar in six of American retail spending went to online orders carried to the door.

FIVE DATES TO KEEP

1653: John Stewart owes John Pynchon 30 pounds for his freedom.

1711: New York's aldermen set up a market for selling enslaved people.

1872: Montgomery Ward mails his first catalog.

1916: The first Piggly Wiggly lets shoppers serve themselves.

1995: Amazon.com opens for orders.

NOT KNOWN: No one counted how many children the morphine in Mrs. Winslow's Soothing Syrup killed.

Where to go: A sign at Wall Street and Water Street, in New York City, marks where the city's market for enslaved people stood from 1711 to 1762.

Next: Chapter 19, Big Business and Its Power. It describes how a handful of companies grew huge, and what Americans did about their power.

Chapter 19: Big Business and Its Power

CHAPTER 19 OF 37 · BIG BUSINESS AND ITS POWER · 10 ERAS · ABOUT 95 MINUTES

From the Virginia Company of 1606 to the trillion-dollar companies of 2026, the biggest businesses here kept getting bigger, and at every step people fought over how much power their owners should have.

Members of a protest group called the Sons of Liberty smashed open 340 chests of tea with axes on December 16, 1773, and dumped the tea into Boston Harbor. The tea belonged to a company, the East India Company of London. British lawmakers had just let the company's agents sell its tea straight to the colonies, more cheaply than colonial merchants could. In February 1774 the company's directors sent a bill for the tea to the British minister in charge of the colonies. It came to 9,659 pounds, 6 shillings and 4 pence.

This chapter describes the biggest companies of each era, from the London investors behind Jamestown to the technology companies in court in 2026. It follows where each company's power came from. At first it came from a government's written permission. Later it came from secret deals with railroads and from buying up rivals until a few owners ran whole industries. People pushed back. Kings and lawmakers took that permission away, workers went on strike, the reporter Ida Tarbell dug through the records, and presidents and judges split companies apart. The owners of some companies made money from shipping enslaved people. Others kept selling leaded gasoline after workers who made its additive died, and cigarettes after studies linked smoking to lung cancer.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did a few companies come to control whole industries?
2. What did owners do when their workers went on strike?
3. Did breaking up a giant company ever work?

Five minutes? Read "Homestead, 1892" in 1850 to 1900.

Can skip: "The Hudson's Bay Company, 1670 to 1825", "Who ran the corporations? 1932", "What was a conglomerate?". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 1 MINUTE

No company existed in this land before 1500, no business owned by a group of people who put money into it and shared what it earned. Starting a corporation, a company that goes on after its first owners are gone, took a written grant from a government.

Native nations traded goods over hundreds of miles, and Chapter 18, Marketplace and Consumer Life, tells that trade. The traders belonged to many different nations, and none of them ran the trade as a company whose owners each held a part of it.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 1 MINUTE

Merchants in London started forming joint-stock companies in the 1500s, and later companies of that kind paid for England's first lasting colonies on this coast. Kings and queens paid for the European voyages that reached this coast during that century. No company with a charter, the written grant from a government that creates a company and lists its rights, traded here in those years.

What was a joint-stock company?

A joint-stock company had many owners at once. Each investor bought shares, parts of the ownership, and got a part of the profit that matched the number of shares he owned. The money the investors paid in was called stock.

The explorer Sebastian Cabot and a group of London merchants formed the Muscovy Company in 1555. It held a monopoly on English trade with Russia, the right to be the only English company trading there. Before it, investors had taken their money back after each voyage. The Muscovy Company was the first English joint-stock company that kept its investors' money in the business from one voyage to the next.

Sir Edward Osborne founded the Levant Company on September 11, 1592, by joining two older companies, the Venice Company and the Turkey Company. Its merchants traded in the eastern Mediterranean Sea, not in North America.

A charter from the queen. A nobleman and 218 other men asked Queen Elizabeth I for a charter for a new company. She gave one to the East India Company of London on December 31, 1600.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 10 MINUTES

The first lasting English colonies here were started by companies, and the investors who owned them hoped the colonies would earn them a profit. A king signed each company's charter, investors bought its shares, and the company's officers sent the settlers. The kings who signed the charters gave the companies land where Native nations already lived. Two of those companies lost their charters before the century ended. King James I took the Virginia Company's in 1624, and English judges cancelled the Massachusetts Bay Company's in 1684.

Companies that traded people. The owners of other companies made money from enslaved people. The Dutch West India Company owned the enslaved men who dug the trench for a wall at New Amsterdam. The ships of the Royal African Company, chartered in 1672, carried more enslaved Africans to the Americas than those of any other company or group in the history of the slave trade, the historian William Pettigrew found.

The Virginia Company, a colony run as a business

King James I of England gave the Virginia Company of London a charter in 1606 to start a colony in North America. The company's settlers founded Jamestown on May 13, 1607. They built it inside Tsenacomoco, the land of the Powhatan chiefdom, a group of towns under one leading chief.

The shares. A share in the company cost 12 pounds and 10 shillings, at 20 shillings to the pound, and it lasted seven years. By one estimate, the first sale of shares raised as much as 10,000 pounds. In 1616 the company paid its shareholders in land alone.

Who ran it. Under a second charter, dated May 23, 1609, the king gave up some of his power over the company, and from then on the investors elected a treasurer to run it. A third charter, dated March 12, 1612, added the island of Bermuda to the company's lands and allowed the company to run lotteries.

The lotteries raised 7,000 pounds of the company's 9,831 pounds in cash in the year before May 1620. Members of the House of Commons, one of the two groups of lawmakers in England's Parliament, attacked the lotteries in a debate in February 1621. King James I banned them soon after.

The first Africans in Virginia. The first Africans known to reach English Virginia came in 1619, while the company ran the colony. Earlier that summer the crews of two privateers, ships whose captains had a government's permission to attack and rob enemy ships, had attacked a Portuguese slave ship, the *São João Bautista*, off the coast of Campeche, in what is now Mexico. They seized 50 to 60 Africans from it.

One of the two, the *White Lion*, had sailed from the Netherlands, and its captain, John Colyn Jope, carried Dutch papers giving him that permission. The *White Lion* landed "20. and odd" Africans at Point Comfort in Virginia in late August 1619, in the words of the colonist John Rolfe. Governor George Yeardley and the merchant Abraham Peirsey bought them and paid for them with food.

The other ship, the English *Treasurer*, arrived three or four days later with more of the stolen people. Its owners were Robert Rich, Lord Warwick, a nobleman, and Samuel Argall, Virginia's deputy governor.

The deaths. Of the 4,270 people counted in Virginia from 1619 to 1622, only 1,240 were alive at the end. About 700 settlers lived in the colony in 1619. Over the next three years the company's officers shipped 3,570 more men, women and children there.

English settlers had been taking Powhatan land to grow tobacco. An alliance of Virginia Indians led by Opechancanough, the leading chief of the Powhatan, attacked the English settlements on March 22, 1622, and killed as many as 347 colonists. Besides those 347, 2,683 people were gone. One investor asked what had happened to them. Most of them had died of disease, Encyclopedia Virginia's history of the company states.

The end of the company. King James I's advisers on the Privy Council, his inner council, named a group of officials in May 1623 to look into the company's business. On May 24, 1624, the king took away the company's charter. From then on Virginia was a royal colony, run by officials the king chose.

Sir Edwin Sandys

Who: The man who ran the Virginia Company while its officers shipped thousands of settlers to a colony where most of them died.

When and where: London, 1619 to 1624.

Every person shipped to Virginia earned someone 50 acres.

The investors of the Virginia Company elected Sir Edwin Sandys their treasurer, the man who ran the company, on April 28, 1619. In the next three years most of the people counted in the company's colony died or were killed.

To draw settlers, Sandys and the company's officers used a plan called the headright. They promised 50 acres of land for each person whose trip to Virginia someone paid for. The payer got the land, whether he paid for his own trip or for another person's. The settler then paid the company a quitrent, a small yearly rent on the land, of one shilling a year. The land handed out this way was Powhatan land.

Who could have stopped it. The historian Edmund S. Morgan studied these years in a book published in 1975. He and later scholars have shown that a responsible king would have stopped the company from recklessly shipping his people to their deaths, Encyclopedia Virginia's history of the company states. King James I did not stop the shipments while they went on, from 1619 to 1622. He took the company's charter away in 1624.

The Dutch West India Company

Members of the States General, the lawmaking assembly of the Netherlands, gave the Dutch West India Company a charter on June 3, 1621. For 24 years no one else from the Netherlands could sail to or trade with the Americas. The same rule held for the coast of Africa from the Tropic of Cancer, a line on maps north of the equator, to the Cape of Good Hope, near Africa's southern tip.

Fort Orange and Manhattan. Men working for the company built a trading post called Fort Orange in 1624, where Albany, New York, stands today. It was the first lasting Dutch settlement in New Netherland, the colony along the Hudson River. The upper Hudson was the country of the Mohican people.

The company's settlers then moved onto Manhattan Island, the home of the Lenape people. Company officials said they bought Manhattan in 1626 for trade goods worth 60 guilders in Dutch coins. A guide to Lenape history from the Pratt Institute Libraries says the Dutch likely misunderstood what the Lenape meant by that exchange.

People the company owned. The company owned people as well as ships and forts. Enslaved men owned by the company dug the trench for a wall at New Amsterdam, the town on Manhattan, in 1653. Three of them are known by name: Paulo d'Angola, Simon Congo and Anthony Portuguese.

The company's director and council had given these three men, eight other men and their wives what the officials called "half-freedom" in 1644, after 18 or 19 years of forced work. The men could own land, but their children stayed the company's property. The men still owed the company a yearly payment of grain or beans and a hog, and they had to work for the company whenever its officials called them.

The Massachusetts Bay Company

Investors in the Massachusetts Bay Company started the colony of Massachusetts, and the company's leaders governed it under the company's charter. King Charles I signed that charter in March 1629. In late August 1629 twelve men, among them John Winthrop, signed a paper called the Cambridge Agreement. They promised to sail to New England if the company's charter and its government moved there with them.

Boston. Winthrop arrived at the Shawmut Peninsula in 1630 with about 1,000 Puritan settlers, English Protestants who wanted to "purify" the king's official church, the Church of England. Winthrop renamed the peninsula Boston.

The land around the peninsula belonged to the Massachusett people, whose leading chief was Chickataubut. About 100,000 people had lived in the region, according to the West End Museum in Boston, and an epidemic, a disease that spread quickly through many people, had killed about 75,000 of them from 1616 to 1618. In 1633 Chickataubut agreed to a treaty with Winthrop, a written agreement that gave up his rights to the peninsula. He died of smallpox that same year.

The charter cancelled. English judges cancelled the company's charter on October 23, 1684, for three reasons. Colonists had traded with other countries against English trade laws. The colony's leaders had made their own coins, without the king's picture on them. And the colony's lawmakers had passed laws against members of the Church of England and against Quakers, a Christian group, also called the Society of Friends, that began in England in the 1600s.

The Hudson's Bay Company, 1670 to 1825

King Charles II gave a charter to the Hudson's Bay Company on May 2, 1670. Prince Rupert, the king's cousin, had persuaded the king and a group of merchants and noblemen to back the company. Under the charter, the company alone could trade in all the lands whose rivers flow into Hudson Bay, in what is now Canada. In the charter those lands are called Rupert's Land.

The Hudson's Bay Company was an English company. Much later, in 1821, it took over the trading posts of its rival, the North West Company, in the Pacific Northwest. The company's governor, George Simpson, and the trader Dr. John McLoughlin opened Fort Vancouver on the Columbia River, in what is now Washington state, on March 19, 1825. Chinookan people had lived along that river for thousands of years. They called the place where the fort stood Skatcutxat, which means "Mud Turtles."

The Royal African Company

The Royal African Company's investors made money from shipping enslaved Africans to the Americas. King Charles II gave the company a charter in 1672, and his brother James, who held the title Duke of York, led it. James later became King James II. The company held the only English right to trade with West Africa, a trade in gold, silver and enslaved people.

Ships that the company put money into carried 186,748 enslaved people on 652 voyages to English colonies in the Americas between 1672 and 1731. Of those people, 16,077 died on the way. The count comes from the Trans-Atlantic Slave Trade Database, a record of known slave voyages. From 1680 to 1686 the company's ships carried about 5,000 people a year, most of them to English colonies in the Caribbean and to Virginia.

The brands. Thousands of the people on the company's ships arrived with the company's initials branded on their chests. To brand a person means to press a red-hot metal stamp against the skin. The heat burns the letters into the flesh and leaves a raised scar for life. Many of the brands read DoY, for

Duke of York, the company's leader. Others read RAC, for Royal African Company. The accounts of the branding name the company but not the men who pressed the hot stamps into people's skin.

The historian William Pettigrew wrote in his 2013 book *Freedom's Debt* that the company "shipped more enslaved African women, men and children to the Americas than any other single institution (company or other group) during the entire period of the transatlantic (across the Atlantic) slave trade." He also wrote that its investors knew what the company did and meant to make money from it.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

The historian Robert E. Wright counts only 8 business corporations chartered in the colonies before independence, so almost no business in the colonies from 1700 to 1750 was a corporation. He writes that before the Revolution, American business corporations were "few, small, and largely inconsequential," which means mostly unimportant. Under British rules, Wright writes, getting a charter cost a lot, and its owners got little in return. So most colonists in business found other ways to organize.

Partners, and the trade in people

Many colonial businesses were partnerships, in which two or more men put their money together and shared the profits. Under the law of the time, each partner was liable for all of the firm's debts, which meant he could be made to pay them. If the firm failed, the people it owed could make any one partner pay all of it from his own property.

The trade opened. The Royal African Company's monopoly ended in practice in 1689, after King James II lost his throne, and the company's own leaders admitted it that year. Under a law that members of the English Parliament passed in 1698, any English merchant could carry enslaved people from Africa. He had to pay the company a fee of 10 percent on the goods he shipped out of Africa. The Royal African Company itself stopped trading in enslaved people in 1731.

Boston's trade in people. Boston merchants bought and sold people. The *Boston Gazette*, a newspaper, ran about 2,300 advertisements for about 2,000 enslaved people from 1719 to 1781. Charles Apthorp, a Boston merchant born in Britain, ran a firm that brought in enslaved people and sold them along with other goods. By 1746 he was one of the richest men in Boston.

Apthorp placed advertisements in the *Boston Gazette* in the 1730s and 1740s for "a parcel of likely negros just imported," a group of newly arrived Black people he called healthy, good-looking and fit for work. In 1742 he offered a reward of five pounds for the return of "a negro man named Jack about 35 years old," who had escaped from his enslaver, Stephen Eastwick. Apthorp's business partner was Thomas Hancock, one of the richest merchants in New England.

Thomas Hancock

Who: The Boston merchant whose firm rented out the ships that carried the Acadians away from their homeland in 1755.

When and where: Boston, from about 1724 until his death in 1764.

His firm's ships, rebuilt like slave ships, carried away Acadian families.

Thomas Hancock built one of the largest fortunes in New England, and part of his firm's business was carrying people whom British officials had forced from their homes. He started out at 13 or 14 as a bookseller's apprentice in Boston, learning the trade.

What he traded. Hancock's ships took codfish, whale oil, logwood (a wood that gives a dye) and potash (a salt made from wood ashes) to other countries. With different partners he made paper. He sold rum and molasses to the crews of the fishing fleet off Newfoundland, and he controlled a group of fishing boats of his own.

Smuggling. Under English trade laws, colonial merchants could buy and sell only where the laws allowed. Hancock's ships carried tea and paper from Amsterdam to the West Indies anyway, and when his ships brought French molasses into Boston, he hid it.

The Acadians. Hancock's partner Charles Apthorp was a paymaster and agent for the British army and navy, the man who paid out their money, and he supplied goods and money to British forces in Nova Scotia and Boston. In the autumn of 1755 British officials forced the Acadians, French-speaking Catholic farm families, out of Nova Scotia and took their houses and farms. The officials who gave the orders were Governor Charles Lawrence of Nova Scotia and Governor William Shirley of Massachusetts.

Lawrence and Shirley hired about 24 sailing ships from the firm of Apthorp and Hancock to carry the Acadians away. The partners charged 40 to 48 pounds a month for each ship, and the governors also paid a

fixed fee for each person carried. In Boston the ships were rebuilt inside to hold two people for every ton of cargo the ship could carry, and some held 300 to 500 people. The historian Mark Peterson writes that they were "fitted out in the manner of slave ships."

The ships carried the Acadians to British colonies along the Atlantic coast, among them Maryland and South Carolina. The ships were in poor condition. On board, Acadians went without enough food, caught diseases and died. The ships carried no passenger lists, so no record shows which Acadians died at sea or how many.

Hancock made his nephew, John Hancock, his equal partner in 1763. Thomas Hancock died of a stroke on August 1, 1764, as he entered the council chamber in Boston. He left property and money worth about 70,000 pounds.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 6 MINUTES

Colonists in Boston destroyed tea in 1773 that belonged to the East India Company of London. British lawmakers had just given that company the right to sell its tea straight to the colonies. After independence, American lawmakers began chartering companies of their own.

Charters at home. Most of the new charters went to banks and to insurance companies, which took a regular fee from people and paid them back after a loss such as a fire. Others went to companies that piped water to towns or built canals, bridges and toll roads, roads that people paid a fee to use.

Robert E. Wright counts 319 business corporations chartered in the United States before 1801. Only 8 of them came from the colonial years and 21 from 1776 to 1789. The other 290 were chartered from 1790 to 1800.

Whose tea went into Boston Harbor?

The tea belonged to the East India Company of London, the company chartered in 1600. The company held about 17 million pounds of tea, by weight, that it could not sell. Members of the British Parliament passed the Tea Act on May 10, 1773. Under it, the company could ship tea straight to the American colonies, and its own agents could sell the tea there at a lower price.

The Act contained no new tax. The tax on tea that members of Parliament had set in 1767 still had to be paid on the company's tea. But the company's agents could sell tea more cheaply than colonial merchants could, so the merchants who sold tea in the colonies lost business.

The three ships. Ships carrying the company's tea reached Boston in late 1773. Thomas Hutchinson, the royal governor of Massachusetts, refused to let the three ships, the *Beaver*, the *Dartmouth* and the *Eleanor*, leave the harbor until they had unloaded.

On December 16, 1773, members of the Sons of Liberty, a protest group, boarded the ships. Some of them were dressed as Mohawk men. They dumped all of the tea into the harbor, more than 92,000 pounds of it, about 46 tons. The company's directors later asked the British government to pay them back for it.

Americans start chartering companies

Robert Morris planned the first commercial bank in the United States, the Bank of North America, a bank that kept people's money safe and lent money to merchants and businesses. Members of the Continental Congress, the leaders from the 13 states who ran the country during the Revolution, approved his plan on May 26, 1781. They granted the bank its charter on December 31, 1781, and it opened on Chestnut Street in Philadelphia on January 7, 1782.

Morris started the bank with a shipment of French silver worth $450,000 that had been sent to Congress, and the bank's officers then lent the money back to Congress. Ninety-nine Philadelphians and members of Congress bought shares.

A fight over the charter. Pennsylvania's lawmakers gave the bank a second charter of their own in 1782. In March 1785 they received petitions, written requests signed by many people, asking them to cancel it. The bank's critics said its officers favored their friends, charged too much interest on loans and had too much power over trade and politics. They pointed to the 14 percent dividends, shares of the profits, that the bank paid its shareholders.

The lawmaker William Findley called the bank "a monopoly." That September the lawmakers voted to cancel the charter. In 1787 they gave the bank a narrower charter that lasted 14 years. Under it the bank's capital, the money its owners put in, could be no more than $2 million, and the bank could own no property beyond what its business needed.

Toll bridges. Only 7 of the 319 corporations that Wright counts before 1801 were for making goods. Lawmakers in the Northeast chartered 59 private companies to build toll bridges from 1786 to 1798. The first was the Charles River Bridge in Boston, and in its first six years its investors earned an average of 10.5 percent a year on their money.

What a charter cost. Getting a charter cost money. Wright found that people who wanted one paid for signatures and for help getting a bill passed by the lawmakers. Some paid lobbyists, people hired to persuade lawmakers, and

some paid bribes.

How much an owner could lose. In a corporation, each owner risks the money he paid for his shares. Under a rule called limited liability, owners cannot also be made to pay the company's debts from their own property. The historians Oscar and Mary Handlin found only 16 early charters that spelled out that promise. Wright argues that owners' risk was limited anyway, because a company could not charge its shareholders more than the price of their shares.

Robert Morris

Who: The merchant who ran the country's finances near the end of the Revolution, then went to a debtors' prison owing nearly $3 million.
When and where: Philadelphia, 1781 to 1801. Born 1734, died 1806.

When he left prison, he owed $2,948,711.11.

As Superintendent of Finance, the official in charge of the Continental Congress's money, Robert Morris wrote the charter of the Bank of North America. He used the bank to keep the value of the country's money steady and to keep the national government from running out of money.

After the war, Morris bought land with borrowed money. Starting in 1793, he and his partners, among them John Nicholson, bought thousands of building lots in the new District of Columbia. They also bought millions of acres in Pennsylvania, New York, Kentucky, Virginia, Georgia and the Carolinas.

Seneca land. Morris's land in western New York was Seneca land. His son Thomas Morris and his agent Charles Williamson met Seneca leaders at Geneseo, New York, from August 20 to September 16, 1797. Nearly 3,000 people came from the Seneca and the other nations of the Haudenosaunee, the union of six Native nations to which the Seneca belong.

The Seneca leaders at Geneseo included Cornplanter, Red Jacket and Handsome Lake, along with the clan mothers, the women leaders who choose their clan's chiefs and can remove a chief who acts badly. At this meeting the Seneca leaders signed the Treaty of Big Tree. They gave up almost all of their remaining land in New York, nearly 3.5 million acres, and kept 12 small pieces.

Morris paid $100,000. Federal officials held the money and put it into shares of the Bank of the United States, which paid the Seneca up to $6,000 a year. Morris asked that the $100,000 go back to his own heirs, the family members who would inherit from him, if the Seneca nation should "become extinct," meaning if every Seneca died. The heads of the federal Treasury and State departments refused. Most of the land went on to the Holland Land Company.

Prison. Morris and Nicholson together owed their creditors, the people they had borrowed from, about $12 million by 1797. Officers arrested Morris at his home in February 1798 and took him to the Prune Street debtors' apartment, next to the Walnut Street Prison in Philadelphia. He was held there about three and a half years because he could not pay his debts.

Members of Congress passed a bankruptcy law in 1800. Under it, a court could divide what a person owned among the people he owed and free him from the rest of the debt. Accounts differ about whether Morris's time in prison helped get that law passed. He was released under it in 1801. The three officials who handled his case found that he himself owed $2,948,711.11.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 9 MINUTES

State lawmakers chartered thousands of corporations, companies the law treats as one body, in the first half of the 1800s, and the fights over them reached the Supreme Court and the president. At the start of the century the lawmakers had to pass a special law for each charter. The justices of the Supreme Court ruled in 1819 that a state's lawmakers could not change a charter they had already granted. By 1837 a Connecticut law let people start a corporation for any lawful business by following set steps.

The mills and the bank. Merchants in Boston used corporations to build cotton mills, and by 1850 their group controlled a fifth of the country's cotton cloth making. In the 1830s President Andrew Jackson fought the Second Bank of the United States, one of the largest companies in the country, and its federal charter ran out in 1836.

From special charter to general law

A group that wanted a corporation before 1811 had to ask lawmakers to pass a special law just for them. New York's lawmakers changed that on March 22, 1811, when they passed the country's first general incorporation law, a law that let any group form a corporation by following set steps, without a special vote.

Under the New York law, only makers of goods could form such a company: makers of goods of wool, cotton or linen, and of iron, steel, copper, lead and similar metals. Any five or more people could form one. Its charter lasted 20 years, and it could raise no more than $100,000.

An emergency law. The historian Seavoy called the law "essentially an emergency measure" in a 1972 article in the *Business History Review*. Under two federal laws that banned trade, the Embargo of 1807 and the Non-Intercourse Act of 1809, cloth from Europe had stopped coming in. New York's lawmakers wanted thread made at home so that families could weave their own cloth.

Connecticut's lawmakers went further in 1837. Their new law let people incorporate, or form a corporation, for "any other lawful business." It was the first such law that did not list which trades were allowed or set a time limit on the corporation's life. Banks still needed a special act.

Who was shut out. Lawmakers in some Southern states shut particular groups out of their general incorporation laws, most often people who were not white. In those states, the economic historian Eric Hilt found, a Black American could not form a corporation under the general law.

How many. State lawmakers created 22,419 business corporations by special laws, one charter at a time, from 1790 to 1860, by the count of the economic historians Richard Sylla and Robert E. Wright. That is about 70 times the 319 corporations chartered before 1801. Together their charters allowed their owners to raise at least $4.5 billion. Several thousand more corporations formed under general laws.

Could lawmakers take back a charter? The Dartmouth case, 1819

The justices of the Supreme Court ruled in 1819 that a charter is a contract, and that a state's lawmakers cannot change it at will. The case was about a college's charter, but business corporations could rely on the same rule to protect their own.

The case. King George III had granted Dartmouth College its charter in 1769. The college's trustees, the board that ran it, removed its president, John Wheelock, in 1815. Wheelock then persuaded the governor of New Hampshire to change the charter, and in 1816 the state's lawmakers rewrote it. Under the new charter the governor chose trustees and a state board could overrule them. The lawmakers also renamed the college Dartmouth University.

The old trustees sued William Woodward, who served as secretary under the new plan. Daniel Webster argued the trustees' case before the Supreme Court. "It is, sir, as I have said, a small college, and yet there are those who love it!" he said. Chief Justice John Marshall wrote the Court's decision and ruled that the 1769 charter was a contract. The Constitution states that no state may pass a law "impairing," or weakening, "the obligation of contracts," and New Hampshire's lawmakers had broken that rule.

The way around it. After the ruling, state lawmakers could not take back a charter that earlier lawmakers had granted. Justice Joseph Story, in his own written opinion in the case, pointed to a way around this. Lawmakers could write into a charter, at the start, a power "to alter, modify, or repeal" it, which means to change it or cancel it. State lawmakers began writing such lines into new charters.

Dartmouth College's own history of the case describes it as a basis for the Constitution's protection of corporations. Supreme Court justices still cite it, as they did in *Citizens United v. FEC*, a 2010 case about money spent on elections, and in *Burwell v. Hobby Lobby*.

The Boston Associates and the mills

A group of Boston merchants, known today as the Boston Associates, built cotton mills with far more money behind them than their rivals had. By 1850, the National Park Service counts, they controlled a fifth of the country's cotton cloth making.

Francis Cabot Lowell, Patrick Tracy Jackson and Nathan Appleton organized the Boston Manufacturing Company in 1813. It started with $400,000 in capital, the money its owners put in, more than ten times the money behind the average cotton mill in Rhode Island. By 1820 the owners had put in $200,000 more.

Waltham. They built a mill at Waltham, on the Charles River, in 1814. It was the first mill in the world to spin cotton into thread and weave it into cloth under one roof, so every step from raw cotton to finished cloth happened in one building.

Within six years the mill ran 5,376 spindles, the rods that twist cotton fibers into thread, and 175 power looms, weaving machines driven by water power. A typical Rhode Island mill ran fewer than 1,000 spindles. The Waltham mill used 450,000 pounds of cotton a year, and by 1820 it used all the power the Charles River could give.

Lowell. The same investors then built mills at Pawtucket Falls, on the Merrimack River. The falls were a fishing place of the Pennacook people, and in the 1600s two Pennacook towns stood there, under the leaders Passaconaway and his son Wannalancet.

What happened to the Pennacook. Native nations fought the English colonies of New England in 1675 and 1676, and Wannalancet kept peace with the English. In 1676 he led about 400 people from the nations allied with him to make peace with an English captain, Richard Waldron. Waldron took them all prisoner.

Officials in Boston hanged seven or eight of the prisoners, and about 200 were enslaved for life and shipped to Barbados. Wannalancet, whom Waldron let go, moved north to Canada with his people. Chapter 14, Work and Workers, tells this in full.

The mill village became a town named Lowell on March 1, 1826. It had 2,500 people that year and 6,477 in 1830.

Whose cotton. Enslaved people in the South grew the cotton the mills spun. The mills also made a cheap cloth, which people at the time called "negro cloth," that planters bought to clothe the people they enslaved. Senator Charles Sumner of Massachusetts called the tie between the mill owners and the slaveholders an "unholy union" between the lords of the loom and the lords of the lash, or whip.

Nicholas Biddle

Who: The head of the Second Bank of the United States, one of the largest companies in the country, who cut off its loans in his fight with President Andrew Jackson.

When and where: Philadelphia, from 1822 or 1823 to 1839.

He cut off loans, hoping people would blame Jackson.

Nicholas Biddle became president of the Second Bank of the United States in 1822 or 1823. The Federal Reserve's own history of the bank describes it as "one of the largest U.S. corporations." Its charter, which President James Madison signed into law on April 10, 1816, ran for 20 years, and the bank opened in Philadelphia in January 1817.

The bank had $35 million in capital, three and a half times the $10 million of the first national bank. The federal government owned a fifth of its stock, and the president chose 5 of its 25 directors, the board that ran it. By the end of the 1820s the bank had at least 25 branches and did about $70 million of business a year.

The Federal Reserve's history states that Biddle "was not above allowing the Bank to make loans to his friends while denying loans to those less friendly."

The fight with Jackson. Andrew Jackson, president from 1829, called the bank unconstitutional, or against the Constitution, and corrupt. Both houses of Congress voted to renew its charter early, in 1832, 28 to 20 in the Senate and 107 to 85 in the House, and Jackson vetoed the bill, refusing to sign it. Chapter 17 tells that fight in full.

After Jackson won a second term, he ordered the government's money moved out of the bank and into banks chartered by the states, and by late 1833 most of it was gone.

Biddle's answer. Biddle cut the bank's lending and called in loans, telling borrowers to pay back what they owed. He believed that a shortage of credit, money to borrow, might turn people against Jackson. In April

1834 a majority of the House of Representatives voted against renewing the charter.

The federal charter ran out in 1836, and Pennsylvania's lawmakers chartered the bank again as a state bank. In 1839 the bank's officers suspended payment: they stopped handing out coins to people who brought in its paper notes. Biddle resigned that year.

What the stockholders found. The bank's stockholders, the people who owned its shares, chose a committee in 1841 to look into its books. The committee members reported that Biddle had used the bank's money to buy cotton, tobacco and other crops and ship them to Europe. The bank closed for good on April 4, 1841.

Charged with a crime. A grand jury, a group of citizens who decide whether there is enough proof for a trial, charged Biddle and two other bank officers with a crime on December 10, 1841. The charge stated that the three men had cheated the stockholders out of $400,000 in 1836 and had tried to cover it up with false records in 1841. Biddle was arrested. Judges later dropped the charges.

Biddle paid the bank's creditors, the people it owed money, with what was left of his own fortune. He faced lawsuits until he died in 1844. In one of them he was sued for nearly $25 million.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 18 MINUTES

A few companies came to control whole industries between 1850 and 1900. One railroad, the Pennsylvania, had more capital in 1880 than the federal government took in that whole year. John D. Rockefeller's Standard Oil refined 90 to 95 percent of the country's oil by 1880, and by 1895 one company controlled about 98 percent of the country's sugar refining.

Rules and soldiers. Members of Congress wrote the first federal rules for big business in these years, and federal lawyers used one of them against striking railroad workers. When workers struck, owners and officials answered with armed guards, court orders and soldiers.

How big was the biggest railroad?

The largest corporation in the country in 1880 was a railroad, the Pennsylvania. That year it had 30,000 workers and $400 million in capital, and it carried 10 percent of all the freight in the United States and 20 percent of the passengers. Federal officials collected about $334 million in all in the year that ended on June 30, 1880, less than the railroad's capital. By 1882 the Pennsylvania was the largest corporation in the world.

Free land. Federal and state officials also gave the railroads land grants, gifts of land from a government. Federal officials granted about 130 million acres to railroad companies from 1850 to 1871, and state lawmakers granted about 51 million acres more. The Illinois Central got the first federal grant in 1851, under a law passed the year before: nearly 2.6 million of the law's 3.75 million acres.

Whose land. The granted land came from land that federal officials had taken, or were taking, from Native nations. The Pacific Railroad Act of July 1, 1862, states: "The United States shall extinguish as rapidly as may be the

Indian titles to all lands falling under the operation of this act." In that law, members of Congress told federal officials to end the Native nations' rights to the land along the railroad as fast as they could.

John D. Rockefeller

Who: The Cleveland businessman who built Standard Oil and set up the first large trust.

When and where: Cleveland and New York, 1870 to 1911.

Movie: *The Men Who Built America* (2012), a History Channel series about Rockefeller, Carnegie, J. P. Morgan and others. It is a docudrama: actors play the real people in scripted scenes.

"Take Standard Oil Stock, and your family will never know want."

John D. Rockefeller used secret deals with the railroads to take over nearly all of the country's oil refining, the work of cleaning crude oil from the ground and separating it into products that people can use.

Rockefeller formed the Standard Oil Company in Ohio on January 10, 1870, with his brother William Rockefeller, Henry M. Flagler and two other partners. The company had $1,000,000 in capital. Its refineries in Cleveland could handle about 1,500 barrels of crude oil a day, out of about 10,000 to 12,000 barrels a day for all the refineries in Cleveland together.

The secret deal. Thomas A. Scott of the Pennsylvania Railroad set up the South Improvement Company late in 1871, a secret deal between a few refiners and the railroads. Rockefeller and his partners controlled 900 of its 2,000 shares. Under the deal, the railroads would charge $2.56 a barrel to carry crude oil to New York, and members would get back $1.06 on every barrel they shipped. That payment was called a rebate.

Members would also get $1.06 in cash on every barrel that their rivals shipped, a payment called a drawback. A rival refiner would pay the full price, and part of what he paid would go to his own competitors.

Oil producers in western Pennsylvania learned of the plan and fought it. They organized, raided the shipments of the scheme's members and asked Pennsylvania's lawmakers and Congress for help. Pennsylvania officials suspended the South Improvement Company's charter on April 2, 1872, and the company never shipped any oil.

The Cleveland buyout. While the scheme stood, in early 1872, Rockefeller bought out at least 21 of the 26 refiners in Cleveland within three months. One refiner later testified that "our establishment was worth $150,000, and we sold it for about $65,000," about forty or forty-five cents for each dollar his business was worth. Rockefeller told the sellers, "Take Standard Oil Stock, and your family will never know want," meaning they would never be poor.

In those months Standard Oil went from refining about 1,500 barrels a day to 10,000. Ida Tarbell, a reporter who later wrote a history of Standard Oil, found that it then held more than a fifth of the country's refining capacity. Rockefeller kept buying out rivals and making deals with railroads for lower rates. By 1880 Standard Oil refined 90 to 95 percent of all the oil produced in the United States.

What a trust was

A trust was a way to run many companies as one business, with one small group in charge, and the Standard Oil Trust was the first large one. More than 40 people who owned shares in Standard Oil signed the Standard Oil Trust Agreement on January 2, 1882.

They handed over their shares in about 40 companies to nine trustees, people who hold and manage property for others. Rockefeller was one of the nine, and the nine trustees themselves owned most of the stock. The trust was valued at about $70 million. In time the trustees ran some 40 corporations, 14 of them fully owned.

Owners in other industries copied the plan, and laws against such combinations of companies are still called antitrust laws.

The Ohio ruling. The justices of the Ohio Supreme Court ruled in 1892 that the Standard Oil Trust was a monopoly, a business with no real rivals, and that it broke Ohio law. Rockefeller moved the property to companies in other states. Ida Tarbell later wrote of the trust, "You could argue its existence from its effects, but you could not prove it." She meant that people could see what the trust did, but they could not prove it existed.

Tobacco and sugar. James B. Duke had gone into the cigarette business in 1881. He leased new machines that rolled cigarettes, advertised heavily and bought up smaller rivals. On January 31, 1890, Duke and the owners of four other firms joined their businesses into one, the American Tobacco Company, the largest tobacco business in the world. Duke's own firm was W. Duke, Sons and Company of Durham, North Carolina. In sugar, the American Sugar Refining Company controlled about 98 percent of the country's sugar refining by 1895.

Franklin Tarbell

Who: An oil producer who fought Standard Oil from 1872 into the 1890s, and the father of the reporter Ida Tarbell.

When and where: Oil Creek and Titusville, Pennsylvania, from 1859 until his death in 1905.

He and scores of other producers pledged not to sell to Standard Oil.

Franklin Tarbell fought Standard Oil for twenty years, and in 1893 he had to mortgage his family's home to pay a ruined partner's debts. His daughter, the reporter Ida Tarbell, told his story in her memoir, published in 1939.

Tanks. Tarbell was on his way home to Iowa in 1859 when he heard about the oil well at Titusville, Pennsylvania. He guessed that oil men would need tanks to hold their oil, and he believed he could build a wooden one that held 500 barrels or more. He was told, "Show me a model that won't leak, and I'll give you an order." He made the model, got the order and built the first large round wooden tank on Oil Creek. For ten years he built tanks.

He began to produce oil himself in 1867, on farms near Rouseville, and in 1870 he moved his family into a new house he had built in Titusville. Ida was thirteen.

The pledge. The oil producers learned of the South Improvement scheme in 1872. Ida remembered her father coming home one night with a grim look on his face. He told the family that he and scores of other producers had signed a pledge not to sell oil to Standard Oil, the one company that had gained from the scheme. Ida, fifteen that year, later wrote that in 1872 "there was born in me a hatred of privilege, privilege of any sort."

Twenty years. The fight between the independent producers and Standard Oil went on for twenty years. The producers waited for slow decisions from the Interstate Commerce Commission, the federal board

set up in 1887 to watch over the railroads, and some of their businesses failed while they waited.

One of Franklin Tarbell's business partners lost everything in this fight and shot himself between 1891 and 1893, while Ida was living in Paris. Ida's memoir does not state the man's name. The partner left Franklin Tarbell owing money on notes, written promises to repay loans. To pay them, during the money panic of 1893, he mortgaged the family's home: he borrowed money and pledged the house, which the lender could take if the loan was not repaid.

His warning. Ida Tarbell later spent most of five years investigating Standard Oil. Her father warned her that Rockefeller and Standard Oil could crush her, just as, in his view, they had crushed her home town of Titusville, a 2012 account in *Smithsonian* magazine says. Franklin Tarbell died in Titusville on March 1, 1905.

Andrew Carnegie

Who: The Pittsburgh steelmaker whose company owned every step of making steel, and whose manager broke the union at Homestead.
When and where: Pittsburgh, until he sold his company in 1901.
Movie: *Andrew Carnegie: The Richest Man in the World*, an American Experience documentary shown on PBS on January 20, 1997.

One company owned the coke, the ore, the ships and the railroad.

Andrew Carnegie cut his costs and raised his profits by owning every step of making steel. That way of running a business is called vertical integration: one company owns each stage of making a product, from the raw material to the finished goods.

Coke and ore. Carnegie already owned 73.5 percent of Henry Clay Frick's coke company by 1888. The company made coke, a fuel baked from coal that steel makers burn to melt iron ore. Carnegie leased newly

found iron ore deposits in the Mesabi Range of Minnesota in the 1890s. The name Mesabi comes from the Ojibwe name for the range, Misaabe Wajiw, which means Big Man Mountain.

Ojibwe land. The Mesabi Range was Ojibwe country. Ojibwe bands gave up northeastern Minnesota to the United States in a treaty of 1854. Federal officials wanted a vein of copper there, on the north shore of Lake Superior. In a treaty of 1855 Ojibwe bands gave up most of north-central Minnesota, where federal officials wanted land for logging and mining.

Ships and a railroad. Carnegie bought ships on the Great Lakes to carry his ore, and he bought a railroad and extended it to link his steel works to Lake Erie. He no longer had to buy coke, ore or shipping from other companies, and that cut his costs. Carnegie Steel was the largest steel maker in the world by the end of the 1890s, and it made a quarter of all the steel produced in the United States. In 1900 it earned $40 million in profits.

Homestead. Carnegie was in Scotland in the summer of 1892. In the words of the encyclopedia Britannica, he gave his general manager, Henry Clay Frick, "carte blanche," or full freedom, "to break the union" at the Homestead works.

BY 1892: Standard Oil had refined 90 to 95 percent of the country's oil as early as 1880. The American Tobacco Company was the largest tobacco business in the world, and Andrew Carnegie owned most of Henry Clay Frick's coke company.

Homestead, 1892

Henry Clay Frick broke the steelworkers' union at the Homestead works near Pittsburgh in 1892 with a fence, a lockout and about 300 armed Pinkerton men. Frick was the general manager of Carnegie Steel. That year he became its chairman, and he owned 11 percent of it. The union was the Amalgamated Association of Iron and Steel Workers, and its contract ran out on July 1, 1892.

The lockout. Frick cut wages and had a fence built around the plant, topped with barbed wire, which the workers called "Fort Frick." In late June he locked the workers out, closing the plant to them. On July 2 he fired all 3,800 workers.

The Pinkertons. Frick signed a contract with William and Robert Pinkerton, who ran the Pinkerton National Detective Agency, for 300 armed men at five dollars a day for each man. Agency officials gathered 316 men in New York and Chicago. Some were Pinkerton regulars. Others had no jobs, some were criminals on the run, and a few were college students earning money for the summer. Their crates held 250 Winchester rifles and 300 pistols.

Early on July 6, 1892, tugboats towed two barges carrying the men up the Monongahela River to the plant. Dawn was breaking when the barges reached the riverbank, where about 10,000 people were waiting, by the count of the historian Leon Wolff. When the Pinkerton men tried to come ashore, shooting broke out on both sides. No one knows who fired the first shot.

The surrender. The gun battle went on all day. Strikers threw dynamite at the barges and sent a burning raft toward them. By late afternoon the Pinkerton men had agreed to surrender, and the crowd beat them as they walked up to the town.

The dead. Counts of the dead differ. The encyclopedia Britannica lists at least 7 workers and 3 Pinkerton men killed in the battle and the days after it. Wolff counts about 9 strikers and 7 Pinkerton men dead from bullets, beatings, drowning and suicide.

Wolff writes that about 40 strikers and 20 Pinkerton men were shot, and he names some of the dead. A Pinkerton guard named Klein died on the gangplank, shot through the head. Joseph Sotak went to help a wounded worker, Martin Murray, and was shot in the mouth and killed. A worker named Streigle was shot through the throat and died at once. Neither Wolff nor Britannica names the men who fired the shots that killed them.

The soldiers. After the battle, Frick asked Governor Robert Pattison of Pennsylvania for help. Pattison sent about 8,500 soldiers of the state National Guard, and they took over the plant on July 12. By July 15 Carnegie Steel's managers had the plant running again with replacement workers.

Many of the replacement workers were Black men from the South, and the union did not let Black workers join it. In November 1892 about 2,000 white workers attacked Black workers and their families, and gunfire badly wounded several people. The accounts do not say who fired. The union gave up on November 21. By then, Wolff counts, 35 men had died from the battle and the violence that followed.

After the strike. Officials charged scores of union leaders and workers with crimes, and almost all of them were found not guilty. Carnegie Steel's managers cut wages in all of Carnegie's mills by about half. A heater's helper, one of the workers who heated the steel, was paid by the ton of steel: he earned 4.85 cents a ton in February 1892 and 2.22 cents a ton in February 1894. The managers enforced the twelve-hour workday.

Employers blacklisted the members of the union's advisory committee for life, putting them on a list of workers the employers agreed never to hire. Members of Congress investigated the use of private armed guards at Homestead, and Frick testified before them.

John W. Holway

Who: A 23-year-old medical student hired as a Pinkerton guard for the Homestead job.

When and where: Chicago to Homestead, Pennsylvania, July 1892.

Movie: *The River Ran Red* (1993), a documentary about the Homestead lockout, directed by Steffi Domike and Nicole Fauteux. It tells the story from the workers' side.

The agency promised "the element of danger" would be "entirely lacking."

John W. Holway was hired to guard the Homestead works, and a crowd there kicked him and beat him with stones. He was studying to be a doctor and was one of the college students who signed up with the Pinkerton agency for summer work at five dollars a day.

The train. The man in charge of the agency's Chicago office would not tell the men where they were going. "The element of danger which is usually found in such expeditions will be here entirely lacking," he told them.

After dark, agency men put Holway and the other Chicago men on a train at the Lake Shore depot. Pinkerton detectives stood guard to stop anyone from getting off, even at the stops in Toledo and Cleveland. The thought struck Holway, Wolff writes, that he was "a sort of prisoner," and he had a feeling that there would be shooting.

The barges. Holway was on the barges at Homestead on July 6. According to Wolff, he does not seem to have fired a shot all day. On the walk up to the town after the surrender, three strikers knocked Holway down, and one of them said, "You have killed two men this morning. I saw you!" As they pushed him up the hill, a stone hit him in the head.

Holway told the congressional committee under oath what happened next. He ran about half a mile, but he was weak and had had nothing to eat or drink, and his legs gave out. A man grabbed him by the back of his

> coat. Then, in Holway's words, "about 20 or 30 men came up and kicked me and pounded me with stones."
>
> "I thought I was about going," he said, meaning about to die, "and commenced to scream." Two or three strikers with rifles then ran up and kept the crowd off him.

The first federal rules, and how they were used

Members of Congress passed the first federal laws to control big business in 1887 and 1890. In the 1890s the justices of the Supreme Court let the sugar trust stand, and federal officials used the second law to break a railroad strike.

The railroads. The Interstate Commerce Act, which set rules for interstate commerce, or trade between the states, became law on February 4, 1887. Under the Act, the five members of a new board, the Interstate Commerce Commission, watched over the railroads. The railroads became the first industry under federal regulation, rules made and enforced by the national government, and the commission was the first federal board of its kind.

The Sherman Act. The Sherman Antitrust Act, named for Senator John Sherman of Ohio, became law on July 2, 1890. Under it, every contract or combination "in restraint of trade" became illegal, meaning any deal that blocks free buying and selling between the states. Trying to monopolize, or to become the only seller of something, also became a crime.

A person found guilty could be fined $5,000 and sent to jail for a year, and people harmed could sue for three times their losses. The law did not define the words "trust," "combination," "conspiracy" or "monopoly."

The sugar case. The justices of the Supreme Court ruled on the sugar trust in 1895, in *United States v. E. C. Knight Co*. The American Sugar Refining Company controlled about 98 percent of all sugar refining in the United States, and the justices ruled that it had not broken the law. They reasoned that making sugar was manufacturing, and that manufacturing was not trade.

Pullman. The Pullman Palace Car Company ran sleeping cars, railroad cars with beds, on trains across the country. Its managers laid off more than 2,000 workers and cut wages by 25 percent in 1893, during a depression, a long time when many businesses fail and many people lose their jobs. The company owned the town where its workers lived, and its managers did not lower the rents. Company president George M. Pullman refused to meet the workers' representatives.

Railroad workers in 27 states, between 125,000 and 250,000 of them, joined a boycott of Pullman cars in 1894, refusing to handle trains that carried them. That year federal officials used the Sherman Act against the workers. A special lawyer working for the US attorney general claimed in court that the strike broke the Sherman Act.

The court order. On July 2, 1894, two federal judges granted the government's lawyer an injunction, a court order that tells people to stop doing something and lets judges jail anyone who breaks it. President Grover Cleveland used the injunction as his reason to send federal troops. The soldiers arrested boycotters and rioters and sometimes shot them, according to the Federal Judicial Center's history of the case.

About 30 people were killed across the country during the strike, and the histories of it do not say who killed each of them.

BY 1900

If you remember one thing: Between 1850 and 1900 a few owners took control of whole industries, and when workers struck, a governor and a president sent soldiers against them.

- 1882: Standard Oil's owners sign the first large trust agreement.
- 1890: Under the new Sherman Antitrust Act, deals "in restraint of trade" become illegal.
- 1892: Frick's Pinkerton men and the Homestead steelworkers fight a gun battle.

Next: After 1900 bankers joined hundreds of firms into giant corporations, and two presidents had their lawyers sue to break up the biggest.

1900 to 1950

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The biggest companies the country had yet seen took shape in these years, and presidents and judges began to break some of them apart. Between 1895 and 1904 owners and bankers joined more than 1,800 manufacturing firms into 157 large corporations. The largest, US Steel, issued more than a billion dollars in stock, shares sold to investors, in 1901.

Breakups and new rules. Presidents Theodore Roosevelt and William Howard Taft had their lawyers sue the biggest combinations. In 1911 the justices of the Supreme Court ruled that Standard Oil and the American Tobacco Company had broken the law, and under court orders both were split apart. Members of Congress passed two more antitrust laws in 1914. In 1934, after the stock market crash of 1929, they passed a law that required companies selling stock to report their finances to the public.

Soldiers of the Colorado National Guard, whose costs the mine owners paid, burned a strikers' tent camp at Ludlow in 1914, and Chicago police killed 10 people during a steel strike in 1937. During the Second World War, federal

officials gave about two-thirds of the value of their main war contracts to 100 companies.

The great merger movement and US Steel

The 157 large combinations of 1895 to 1904 were mergers, in which two or more companies join into one, and more than 60 of them formed in 1899 alone.

Naomi Lamoreaux names the causes in her book *The Great Merger Movement in American Business*. Factories cost a great deal to build, and industries grew fast after 1887. After the depression of 1893 began, rival owners cut their prices against each other in price wars. The owners had made agreements, called pools, to hold prices up, but the price wars went on anyway.

Lamoreaux writes that "there was nothing natural or inevitable," nothing certain to happen, about these combinations. Most of them used more money and work to make the same goods than the new rivals that soon appeared, and in her words they "quickly lost their positions of market dominance," their lead in their markets. The firms that built barriers against rivals, such as US Steel through its control of iron ore, kept their lead longer.

US Steel. The banker J. P. Morgan put US Steel together as a corporation in New Jersey in February 1901. He bought Carnegie Steel for Andrew Carnegie's asking price of $480 million. Other companies went in with it, among them makers of steel, pipe and bridges and the Lake Superior Consolidated Iron Mines, whose ore came from Ojibwe land in Minnesota. Charles M. Schwab was president from 1901 to 1903, and Elbert H. Gary was chairman from 1901 to 1927.

A billion dollars in stock. US Steel was the first company in the world with more than a billion dollars in stock. It issued stock with a stated value of about $1.4 billion.

Government investigators at the federal Bureau of Corporations put the real value of what the company owned at about $682 million, about half the stock issued. So about half of the stock was watered stock, stock issued for more than the property behind it.

Two-thirds of the steel. US Steel made about two-thirds of all the steel in the United States at the start. In 1902 the company employed 168,000 people. By 1911 its share had fallen to less than half.

J.P. Morgan

Who: The Wall Street banker who joined rival companies into giant ones, among them US Steel in 1901.

When and where: New York, 1901 to 1904.

Carnegie named $480 million. Morgan said yes at once.

J. P. Morgan, a banker on Wall Street in New York City, led a wave of mergers around 1900, joining rival companies into single giant ones.

The steel deal. Charles M. Schwab, then president of Carnegie Steel, gave a speech at the University Club in December 1900, and the speech led Morgan to the steel deal. Carnegie named a price of $480 million for his company, and Morgan accepted it at once. Morgan then formed US Steel in 1901 and sat on its finance committee.

Northern Securities. Morgan and the railroad owner James J. Hill also formed the Northern Securities Company in 1901. It was a holding company, a company that owns controlling shares in other companies and does little else. Through it they controlled three railroads: the Great Northern, the Northern Pacific, and the Chicago, Burlington and Quincy. President Theodore Roosevelt's Justice Department lawyers sued to break it up in 1902.

Ida Tarbell

Who: The reporter whose history of Standard Oil, pieced together from court records and sworn testimony, showed how Rockefeller took over the oil business.

When and where: New York and the Pennsylvania oil region, 1900 to 1904.

Rockefeller called her "That poisonous woman."

Ida Tarbell spent most of five years, by her own count, tracing from public records how John D. Rockefeller built Standard Oil. She began in 1900. Her father was Franklin Tarbell, one of the oil producers who had signed the 1872 pledge not to sell oil to Standard Oil.

The records. Tarbell read hundreds of thousands of documents from across the country, among them court records and depositions, sworn testimony taken outside a courtroom and written down. She interviewed employees and lawyers. The writer Mark Twain arranged for her to meet Henry H. Rogers, an officer of Standard Oil, and she interviewed him too.

Her *History of the Standard Oil Company* came out in 19 parts in *McClure's* magazine from November 1902 to October 1904, and as a book in two volumes in 1904.

The widow's plant. In one part, Tarbell told the story of a Cleveland widow, Mrs. Backus, who sold her late husband's plant for making oil for machines to Rockefeller in 1878. Tarbell wrote that he paid her next to nothing. Rockefeller's biographer Ron Chernow tells it differently: the plant was out of date, the widow asked for as much as $200,000, and Rockefeller paid $79,000, including $10,000 added on top. The two accounts of that sale do not agree.

Chernow does call Rockefeller's buyouts of his Cleveland rivals in 1872 "one of the most shameful episodes in his career." Rockefeller called Tarbell "That poisonous woman." He told his advisers, "Not a word about that misguided woman."

What came of it. Her series made many Americans furious with Standard Oil. Smithsonian magazine writes that this public fury is credited with the company's breakup. The breakup came after the justices of the Supreme Court ruled against Standard Oil in 1911.

Theodore Roosevelt

Who: The president known as the trust-buster, whose lawyers won the Northern Securities case.

When and where: Washington, 1901 to 1909.

Movie: *TR: The Story of Theodore Roosevelt*, an American Experience documentary first shown on PBS in October 1996.

His lawyers brought 44 antitrust suits. Taft's brought more.

Theodore Roosevelt is known as the trust-buster, but the president after him had his lawyers bring more antitrust suits. Roosevelt's Justice Department lawyers sued the Northern Securities Company, J. P. Morgan and James J. Hill's railroad holding company, in 1902 under the Sherman Act, the federal law of 1890 against deals that block trade.

The ruling. The justices of the Supreme Court ruled 5 to 4 for the government on March 14, 1904, and upheld the order to break the company up. Justice John M. Harlan wrote for the majority. Justice Oliver Wendell Holmes disagreed, writing, "Great cases like hard cases make bad law." He meant that a case the whole country is excited about stirs up feelings that can bend the judges' thinking.

Who brought more. Roosevelt's lawyers brought 44 antitrust suits in seven and a half years. William Howard Taft, the next president, had his lawyers bring more in four years, at least 75. Taft's lawyers sued US Steel in 1911, and Roosevelt then accused Taft of being unable to tell the good trusts from the bad ones.

Standard Oil and American Tobacco are broken up, 1911

The justices of the Supreme Court ruled against Standard Oil and the American Tobacco Company in May 1911, and both companies were split apart. On May 15 they ruled that Standard Oil of New Jersey had broken the Sherman Act. Chief Justice Edward D. White wrote the court's decision. He wrote that the Sherman Act banned only "unreasonable" limits on trade, a test called the rule of reason. The justices gave the company's owners six months to break it up, and Standard Oil was split into 34 separate companies.

Richer after the breakup. Rockefeller was worth about $300 million in May 1911 and more than $900 million by the end of 1913, three times as much. According to Chernow, Roosevelt said that Wall Street's prayer was now "Oh Merciful Providence, give us another dissolution," asking God for another breakup. In 1916 newspaper headlines called Rockefeller the world's first billionaire. Companies that grew out of Standard Oil later became Exxon, Mobil and Chevron.

Tobacco. The justices ruled against the American Tobacco Company on May 29, 1911. They found that James B. Duke, Benjamin N. Duke, George W. Watts, R. J. Reynolds and others had broken the Sherman Act "by seeking to monopolize practically all sectors of the tobacco industry." Federal judges in New York divided the company's property in November 1911 among a new American Tobacco Company, Liggett and Myers, P. Lorillard, R. J. Reynolds and others. James B. Duke's tie to American Tobacco ended that year.

Two new laws, 1914

Members of Congress passed two laws in 1914 that added to the Sherman Act. President Woodrow Wilson signed the first, the Federal Trade Commission Act, on September 26, 1914. Under it, "unfair methods of competition" became illegal, and the members of a new Federal Trade Commission had the power to make companies obey that rule. The commission opened on March 16, 1915.

The Clayton Act. The second law named business practices the Sherman Act had left vague. Wilson signed it, the Clayton Antitrust Act, on October 15, 1914. It banned price discrimination, charging different buyers different prices for the same goods. It also banned deals that tied a buyer to one seller, mergers that cut down competition, and the same people sitting on the boards of competing companies.

The Clayton Act also stated "that the labor of a human being is not a commodity or article of commerce," not a thing to be bought and sold. Under that section, federal lawyers and judges could no longer treat unions as illegal combinations, as they had used the Sherman Act against the railroad workers of the Pullman boycott in 1894.

Ludlow, 1914

Soldiers of the Colorado National Guard, the state's part-time army, burned the tent camp of striking coal miners at Ludlow, Colorado, on April 20, 1914. The mine owners were paying the Guard's costs. At least 19 people were killed, 13 of them women and children. Most of the strikers worked for the Colorado Fuel and Iron Company, whose main stockholder, the person who owned the most shares, was John D. Rockefeller Jr.

The strike. The miners struck in September 1913. In October men in a company armored car known as the "Death Special" fired on their tent camp at Forbes and killed one person. Governor Elias M. Ammons sent the Colorado National Guard.

The fire. Fighting broke out at the tent camp at Ludlow on April 20, 1914. Nobody knows who fired the first shot. Guardsmen set the tents on fire. Thirteen women and children had hidden in a pit dug under a tent, and they suffocated there, dying for lack of air. One guardsman and five miners were also killed in the fighting. Guardsmen shot the strike leader Louis Tikas three times in the back.

President Wilson sent the US Army on April 28. The strike ended in December 1914, and the union won none of its goals.

Leaded gasoline, 1924 to 1926

The leaders of three of the largest companies in the country kept selling a lead fuel additive in the 1920s after workers who made it died. The additive was tetraethyl lead, a liquid made with lead. Researchers at General Motors added it to gasoline to stop engine knock, a pinging sound in the engine. General Motors and Standard Oil of New Jersey formed the Ethyl Gasoline Corporation in August 1924 to sell it, and DuPont made it. Chapter 13 tells the whole story of the additive.

The dead. Workers died at plants and a laboratory of all three companies. Seven men died and 33 went to the hospital after working at Standard Oil's Bayway refinery in Elizabeth, New Jersey, in October 1924, by the count of Bill Kovarik, who wrote a history of leaded gasoline in 2005. Other accounts count five dead. Kovarik counts ten more dead at a DuPont plant and at least two dead at General Motors' laboratory in Dayton, Ohio.

Poisoned workers at Bayway were "black and blue from uncontrolled muscle spasms." Some had to be tied into straitjackets, coats that hold a person's arms tight against the body.

"A gift from heaven." The Surgeon General, the head of the federal public health service, called a meeting about the additive on May 20, 1925. Frank Howard of Standard Oil said there that the industry could not give up "what has come to the industry like a gift from heaven." The companies kept selling it. Robert Kehoe, a scientist paid by the Ethyl company, then set out a theory of a "safe level" of lead that the companies' leaders relied on for decades.

Who ran the corporations? 1932

Hired managers, not owners, ran the 200 largest corporations outside banking, two writers showed in 1932, and those 200 held about half of the country's corporate wealth. Adolf Berle and Gardiner Means counted them in their book *The Modern Corporation and Private Property*. By 1930 the 200 held 49.2 percent of the country's corporate wealth. The scholar Gerald F. Davis later worked out that at the same pace they would have held all of it by 1959.

Berle and Means also found that the ownership of these companies was split among many scattered shareholders, so the managers made the decisions.

The Securities and Exchange Commission, 1934

After the stock market crash of 1929, members of Congress passed a law that required companies selling stock to report their finances to the public. President Franklin D. Roosevelt signed it, the Securities Exchange Act, on June 6, 1934. Under it, the members of a new Securities and Exchange Commission, or SEC, watched over the stock markets. The yearly and quarterly reports that such companies still file with the SEC began with this law.

The fox and the hen house. Roosevelt made Joseph P. Kennedy the first chairman of the SEC in the summer of 1934. Kennedy had made money in the stock market using the same practices the new law banned, and he had given, lent and raised money for Roosevelt's 1932 campaign.

Newspaper writers said the choice was "like putting the fox in the hen house." In private, Roosevelt answered, "set a thief to catch a thief." Both sayings meant the same thing: Roosevelt had put a man who had used those practices in charge of stopping them. Kennedy resigned in September 1935.

Spies, guns and company police, 1936 to 1940

Senators on a committee led by Senator Robert M. La Follette Jr. of Wisconsin held hearings from 1936 to 1940 on how employers fought unions. Witnesses told the senators that company managers had planted spies inside unions and hired their own police forces. The managers had also stored up weapons, among them submachine guns, rifles and tear gas.

Memorial Day, 1937. Several hundred men, women and children marched toward the Republic Steel plant in Chicago on Memorial Day, May 30, 1937, to support striking steelworkers. Republic Steel's chairman was Tom M. Girdler.

At the plant, 264 Chicago police threw tear gas at the marchers, fired their guns and beat them with clubs. The police killed 10 people and shot 40, and more than 90 people were hurt in all. The senators on the La Follette committee found that the police had used "excessive force," far more force than was needed.

Hawaii's Big Five

Five companies in Hawaii, known as the Big Five, controlled about 94 percent of the islands' sugar crop by 1920, by the count of the historian Carol MacLennan. Two of them were Castle and Cooke and Alexander and Baldwin.

The same people sat on the boards of several of the firms, and together they controlled shipping, banking and stores in the islands. Four of the firms together owned nearly three-quarters of the Matson shipping line, which carried 80 percent of the cargo between the mainland and Hawaii.

The overthrow. Hawaii was a kingdom with its own rulers until 1893. On January 17 of that year, a group of mostly American businessmen calling themselves the Committee of Safety moved against Queen Lili‘uokalani. John L. Stevens, the United States minister to Hawaii, was the American government's representative there. He had 162 US sailors and Marines brought ashore from a warship.

The queen's soldiers and police did not fight. She gave up her power under protest, to the United States rather than to the committee. Chapter 23, America and the World, tells the overthrow in full.

Some children of Big Five partners were members of the Committee of Safety, which then pressed for the United States to annex the islands, or take them over. The Big Five's plantations stood on the land of Native Hawaiians, the Kanaka Maoli. From 1893 into the 1950s, the owners of the Big Five dominated Hawaii's economy and government.

War contracts, 1940 to 1944

A small number of large companies received most of the government's war business during the Second World War. Robert Higgs, writing for the Independent Institute in 1995, gives the figures for prime contracts, contracts made directly between the government and a company, from 1940 to 1944. The 100 companies with the most contract money received about two-thirds of the value of those contracts, the top 10 received about 30 percent, and General Motors alone received nearly 8 percent.

Federal officials also paid for new factories and rented them to private companies. Just 26 companies had the use of half the value of all those government-paid factories as of June 30, 1944.

BY 1950

If you remember one thing: Judges split up Standard Oil and American Tobacco in 1911, but the biggest corporations kept growing through the next forty years.

One number: 49.2 percent. By 1930 the 200 largest corporations outside banking held that share of the country's corporate wealth, and at the same pace, Gerald F. Davis worked out, they would have held all of it by 1959.

Next: After 1950 the largest companies had hundreds of thousands of workers each, and their leaders organized to win influence in Washington.

1950 to 2000

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Companies grew so large in these years that General Motors had 618,365 workers in the United States in 1979, and the Bell telephone system had about a million in the early 1980s. From the 1970s, business leaders organized to win influence in Washington. By 1982 more than 2,400 firms paid lobbyists registered there to persuade lawmakers.

Products and disease. The leaders of some companies kept selling products that studies linked to disease and death. Tobacco companies paid for advertisements in 1954 in which their leaders disputed studies linking smoking to lung cancer, and in 1994 heads of tobacco companies told a House hearing that they did not believe nicotine is addictive.

Breakups. AT&T's leaders agreed with federal lawyers to split up the Bell System, and the split took effect in 1984. Federal lawyers won a ruling in 1999 that Microsoft held a monopoly, and in 2001 an appeals court agreed that Microsoft had broken the law to keep it.

General Motors

General Motors was the country's largest car maker in the 1950s, and in 1953 it sold more to the armed forces than any other company. In 1954 it sold about half of all new cars in the country, 54 percent by the Associated Press's count.

A billion dollars. In 1955 GM became the first company to earn more than a billion dollars in profit in one year. In the first three quarters of that year alone it made $912,887,537 in profit. *TIME* magazine wrote, "Never before in history has one corporation made so much in so short a time." GM's workforce in the United States peaked at 618,365 in 1979, which made it the largest private employer in the country, according to an Associated Press timeline.

"Good for General Motors." President Dwight D. Eisenhower chose GM's president, Charles E. Wilson, as Secretary of Defense in January 1953. Wilson gave up a salary of $600,000 a year for one of $22,000. He owned 40,000 shares of GM stock, and GM sold more to the armed forces than any other company. As Secretary of Defense he could make decisions that helped or hurt GM, so at his hearing senators pressed him on that conflict.

Senator Robert C. Hendrickson asked whether Wilson could make a decision that went strongly against GM's interest. Wilson answered, "Yes, sir, I could." But he said he could not imagine one, "because, for years, I thought what was good for our country was good for General Motors, and vice versa," meaning the other way around. People soon repeated his words as "What is good for General Motors is good for the country."

Ralph Nader. Ralph Nader criticized the safety of GM's Corvair car in a 1965 book called *Unsafe at Any Speed*. GM managers hired private detectives to look into Nader's private life, and in 1966 GM's president, James M. Roche, apologized to a Senate committee. That same year members of Congress passed the National Traffic and Motor Vehicle Safety Act, the first federal rules for how cars must be built.

Nader sued GM for invading his privacy and settled for $425,000, which he used to start groups that work for consumers. GM stopped making the Corvair in 1969.

Eisenhower's warning, 1961

President Dwight D. Eisenhower warned the country about the power of the arms makers and the armed forces together in his farewell speech, on January 17, 1961. "In the councils of government," he said, "we must guard against the acquisition of unwarranted influence, whether sought or unsought, by the military-industrial complex."

By the military-industrial complex he meant the ties between the armed forces and the companies that made their weapons. He meant that the people making the government's decisions had to keep the arms makers and the armed forces from gaining power they had no good reason to have, whether they went looking for it or not. The weapons makers were the same kinds of large companies that had received most of the war contracts of 1940 to 1944.

What was a conglomerate?

A conglomerate was one company that owned many businesses that had nothing to do with each other. Harold Geneen ran the company ITT from 1959 to 1977, and in those years ITT bought more than 250 companies in more than 80 countries. They included the Sheraton hotel chain, Avis car rental, the Hartford insurance company and Continental Baking, the maker of Wonder Bread.

ITT's yearly sales grew from about $700 million to $17 billion, and it became the eleventh-largest corporation in the country. Geneen believed that good managers could make a success of any business.

ABC and Chile. ITT's leaders tried to buy the ABC television network for $700 million in 1966, and federal antitrust officials stopped the deal. In Chile, officials under President Salvador Allende took over ITT's company there. ITT's leaders later admitted that the company had given the CIA, the American spy agency, $350,000 for "political" purposes in Chile.

Business organizes in Washington

Heads of large companies built up their own political power in Washington in the 1970s.

The Powell memo. The lawyer Lewis F. Powell Jr. sent a private memo to Eugene B. Sydnor Jr. of the US Chamber of Commerce, a national group of businesses, on August 23, 1971. Its title was "Attack On American Free Enterprise System." Powell wrote that critics of business now came from "the college campus, the pulpit (preachers in churches), the media" and from politicians. He urged business leaders to keep "constant surveillance," or a close watch, on textbooks and television programs.

President Richard Nixon nominated Powell to the Supreme Court on October 21, 1971, less than two months later. The newspaper columnist Jack Anderson made the memo public in September 1972. That year the Chamber's leaders set up a 40-person working group on the memo, and in late 1973 they approved plans for business to act in education, in communications, in politics and in the courts.

The Business Roundtable. Three groups of company heads joined to form the Business Roundtable on October 13, 1972. Its founders included John D. Harper of Alcoa and Roger Blough, the retired head of US Steel. They set out to fight the growing power of unions, especially construction unions.

Offices and lobbyists. Company heads opened offices in Washington, and by 1978 more than 500 corporations had one. They also hired lobbyists to persuade lawmakers. Firms with registered lobbyists went from 175 in 1971 to 2,445 in 1982, about 14 times as many.

Money for candidates. Companies also set up political action committees, or PACs, groups that collect money and give it to candidates. Officials of the Federal Election Commission counted 89 company PACs at the end of 1974 and 1,206 at the end of 1980, more than 13 times as many.

Rules removed. In the same years, members of Congress voted to take federal rules off three industries, and President Jimmy Carter signed each law. The airline law of 1978 began ending federal officials' control over which

routes airlines flew and what they charged, and the federal board that had set them closed in 1984.

The banking law of 1980 ended, over six years, the limits on the interest banks could pay savers. The trucking law of 1980 let trucking companies set many of their own prices and made it easier for new companies to start. Each law took power over prices from federal officials and gave it to the companies.

What did the companies say when their products hurt people?

Heads of tobacco companies publicly disputed the evidence that smoking causes disease and that nicotine is addictive for forty years, from 1954 to 1994. The major American tobacco companies ran an advertisement called "A Frank Statement to Cigarette Smokers" in more than 400 newspapers on January 4, 1954, and it reached about 43 million people.

In it, the companies' leaders disputed new scientific studies that linked smoking cigarettes to lung cancer. The public relations firm Hill and Knowlton, a company paid to shape what the public thinks of its clients, designed it.

The 1963 memo. Addison Yeaman, the chief lawyer of the Brown and Williamson tobacco company, wrote a memo on July 17, 1963. To protect its earnings, he wrote, the industry must "either a) disprove the theory of causal relationship or b) discover the carcinogen or carcinogens...and demonstrate our ability to remove or neutralize them." He meant that the companies had to either show that smoking does not cause cancer, or find the substances in smoke that do cause it and show they could take them out or make them harmless.

The 1994 hearing. Representative Ron Wyden asked seven tobacco company heads whether nicotine is addictive at a hearing in the House of Representatives on April 14, 1994. William Campbell of Philip Morris answered, "I believe nicotine is not addictive, yes." James Johnston of R. J. Reynolds said cigarettes "clearly do not meet the classic definition of addiction." Joseph Taddeo of US Tobacco said, "I don't believe that nicotine or our products are addictive."

The settlement. Four cigarette makers, Philip Morris, R. J. Reynolds, Brown and Williamson and Lorillard, signed an agreement on November 23, 1998, with the attorneys general, the chief lawyers, of 46 states. The attorneys general had sued to get back what their states spent treating sick smokers. The companies' leaders agreed to pay at least $206 billion over the first 25 years and to limit their advertising.

Asbestos. Asbestos is a mineral that separates into thin, strong fibers, and builders used it to fireproof and insulate buildings. When people breathe asbestos fibers in, the fibers can stay in the lungs for years. They scar the lungs and can cause asbestosis, a lung disease that brings shortness of breath, coughing and lasting lung damage. They can also cause mesothelioma, a cancer of the thin linings of the chest and belly. The illness can take 10 to 40 years or more to show.

Johns-Manville was once the largest maker of asbestos products in the country. Andrew Hansen began working at a shipyard in 1940, tearing old insulation out of ships and putting new insulation in. Most of it was made by Johns-Manville and contained asbestos. In 1978, at 65, he was found to have asbestosis and mesothelioma, and he died 13 months later. His widow sued the company.

What the company knew. Dr. Kenneth Smith, Johns-Manville's medical director until 1966, gave sworn testimony in the case. He said that in the 1940s he had told the company's leaders about the danger asbestos posed to insulation workers and had urged warnings on its insulation products, and that they did not act on his advice.

Johns-Manville's lawyers answered that the company began putting warnings on its products in 1964, as soon as science proved a link between its products and asbestosis. The jury found for Hansen's widow, and in 1984 a federal appeals court upheld that finding against the company.

By 1981, while her case was still in the courts, Johns-Manville faced 9,300 lawsuits brought by 12,800 people who sued over illness from asbestos. Its leaders filed for bankruptcy, a court process for a company that cannot pay its debts, in August 1982. Under the filing, all the injury lawsuits against the

company were put on hold. By the end of 1988, more than 12,600 claims, almost $500 million in all, had been paid from a fund set up in the bankruptcy to pay for injuries.

Bhopal. More than 40 tons of a poison gas called methyl isocyanate leaked from a pesticide plant in Bhopal, India, a factory that made poisons to kill insects, on December 3, 1984. Union Carbide India Limited, part of the American company Union Carbide, ran the plant.

How it happened. Union Carbide's leaders had planned to close the plant, which was losing money. Instead they cut many jobs to run it for less, among them maintenance engineers. In June 1984 the unit that kept the gas cold was shut down, to cut the plant's use of electricity by 18 percent. Weeks before the leak, the machine meant to clean escaping gas was turned off too: on October 23 by the date in a report by France's environment ministry, or three weeks before by a 2005 medical review.

On December 2, 1984, the plant's senior managers ordered some pipes cleaned with water. A broken valve let about a ton of that water into a tank holding 40 tons of the gas, and on December 3 the gas escaped into the air. Neither report names the people who turned off the gas-cleaning machine or shut down the cooling unit.

What the gas did. The gas damaged people's eyes, causing redness and sores, and it filled their lungs with fluid. It killed about 2,000 people in the first hours, according to Britain's Health and Safety Executive, a government safety agency. The 2005 medical review, by Edward Broughton, gives about 3,800 dead at once, and 15,000 to 20,000 early deaths reported over the next twenty years.

At first Union Carbide's leaders tried to keep the company free of legal blame. Later they said the company accepted "moral responsibility," and the company paid $470 million to the government of India.

The Bell System is split, 1984

AT&T's Bell System held about $150 billion in property and employed about a million people in the early 1980s. Justice Department lawyers had sued AT&T in 1974. In January 1982 AT&T's leaders and the Justice Department's lawyers agreed that AT&T would give up its local telephone companies.

In return, a 1956 court order that had kept AT&T out of the computer business was lifted, and most newspapers reported the deal as a win for AT&T. Judge Harold H. Greene approved the deal with changes, and the case was closed on August 24, 1982.

The Baby Bells. The local companies became seven regional companies, called the Baby Bells, on January 1, 1984. AT&T kept its long-distance service, its research center, Bell Labs, and its equipment maker, Western Electric. It kept $34 billion of the $149.5 billion in property it had held the day before, less than a quarter, and it had 636,000 fewer workers.

Bill Gates

Who: The head of Microsoft when federal and state lawyers sued it for keeping an illegal monopoly on the software that runs personal computers.

When and where: Redmond, Washington, and Washington, D.C., 1998 to 2001.

Movie: *Inside Bill's Brain: Decoding Bill Gates* (2019), a Netflix series directed by Davis Guggenheim.

A judge ordered his company split in two.

Bill Gates founded Microsoft and ran it as its chief executive until January 2000, so he was in charge when a federal judge found that it held an illegal monopoly. Microsoft made Windows, the operating system, or main program, that ran most personal computers and the other programs on them.

The case. Lawyers for the Justice Department, the attorneys general of twenty states and the District of Columbia sued Microsoft on May 18, 1998. The main question was whether Microsoft could bundle its web browser, Internet Explorer, the program people use to look at websites, with Windows. To bundle it meant to build it into Windows, which made it harder for rival browsers such as Netscape Navigator to win users.

On tape. Gates answered the government lawyers' questions in a deposition that was filmed on videotape. A person who was there called him "evasive and nonresponsive," dodging the questions. According to *Businessweek* magazine, the government lawyers used snippets of emails to show that many of his denials were false.

The rulings. Judge Thomas Penfield Jackson found on November 5, 1999, that Microsoft held a monopoly. On June 7, 2000, he ordered the company split into two, one for Windows and one for its other software. On June 28, 2001, seven judges of the federal appeals court agreed that Microsoft had illegally kept its monopoly, but they threw out the breakup order, partly because Jackson had talked about the case to reporters while it was still going on.

No breakup. Attorney General John Ashcroft announced on September 6, 2001, that Justice Department lawyers would no longer seek a breakup. The government's lawyers settled with Microsoft on November 1, 2001. Microsoft's leaders agreed to share some of the company's technical information with other companies and accepted a panel of three investigators with full access to its systems.

Steve Young

Who: A Vlasic pickle executive who described what it was like to sell to Walmart.

When and where: The late 1990s to January 2001. His account was published in December 2003.

Movie: *Is Wal-Mart Good for America?* (FRONTLINE, PBS, 2004), a documentary about Walmart.

Walmart sold a gallon of Vlasic pickles for $2.97.

Steve Young watched Walmart's buyers push his company's pickles to a price that left almost no profit. He was vice president of grocery marketing for pickles at Vlasic. The reporter Charles Fishman told his story in *Fast Company* magazine in December 2003.

The jar. A Walmart buyer saw Vlasic's gallon jar of pickles in the late 1990s and began talking about it with Pat Hunn, the head of Vlasic's Walmart sales team. The jar went into every Walmart, about 3,000 stores, at $2.97. At that price Vlasic and Walmart each made a penny or two a jar, if that. Young said each store sold about 80 jars a week, which came to 240,000 gallons of pickles a week.

Walmart was 30 percent of Vlasic's business. Young says shoppers who used to buy Vlasic's pickle spears and chips at supermarkets bought the cheap gallon jar instead, and Vlasic's profits from pickles fell by 25 percent or more.

The threat. Young says Vlasic's managers asked to raise the price. He remembers the Walmart buyers' answer: "If you do that, all the other products of yours we buy, we'll stop buying." He called it "a clear threat." Hunn remembers the buyers saying, "We want the $2.97 gallon of pickles. If you don't do it, we'll see if someone else might."

Later, Walmart's buyers let Vlasic shrink the jar to just over half a gallon, at $2.79. Vlasic filed for bankruptcy in January 2001, though everyone involved agrees that the gallon jar was not a main cause.

Why this story is here: A buyer as large as Walmart could name its price, even a price that left the seller almost nothing.

BY 2000

If you remember one thing: Between 1950 and 2000 the biggest companies had hundreds of thousands of workers, and from the 1970s their leaders built up their own power in Washington.

One question this era leaves open: The Bell System was split up in 1984 under a deal a judge approved, but in 2001 appeals judges threw out the order to split Microsoft in two. Would judges order a giant company split up again? The next era answers it.

Next: After 2000 three American companies came to be worth more than $4 trillion each, and government lawyers took four of the largest technology companies to court.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 13 MINUTES

Three American companies, Nvidia, Apple and Alphabet, the company that owns Google, were each worth more than $4 trillion in September 2026. In the same years government lawyers took four of the largest technology companies to court over monopoly power. Walmart employed about 1.6 million people in the United States in 2026, by its own count, about 2.6 times General Motors' American workforce at its peak in 1979.

Fraud and a rescue. A federal jury convicted Enron's former chief executives of fraud in 2006. Purdue Pharma, the maker of the pain pill OxyContin, pleaded guilty to federal crimes in 2007 and again in 2020. Lehman Brothers failed in 2008, and members of Congress set aside $700 billion that year to rescue banks and other companies.

The four cases. The four technology companies taken to court were Google, Meta, Amazon and Apple. By September 2026 judges had ruled twice that Google broke the law and once for Meta, and the Amazon and Apple cases had not come to trial.

How big are the biggest companies?

Nvidia became the first company worth $5 trillion on October 29, 2025, according to the business news network CNBC. A company's worth, its market value, is the price of one share times the number of shares, and it changes every day the stock market is open. The investment website The Motley Fool listed the three most valuable companies in the world on September 7, 2026: Nvidia at about $5.6 trillion, Apple at about $4.7 trillion and Alphabet at about $4.1 trillion.

Workers. Walmart reported $713.2 billion in revenue, the money it took in, for the year that ended on January 31, 2026. Its 2026 annual report records about 2.1 million workers around the world, about 1.6 million of them in the United States, and about 92 percent of its American workers were paid by the hour. At that time it had more than 10,900 stores in 19 countries.

Amazon's yearly report to the government for 2025 states that it employed about 1,576,000 people at the end of that year. It also used independent contractors, workers paid for each job who are not employees, and temporary workers, and neither group is counted in that number.

Fraud and harm at the top

Judges and juries found that leaders of several large companies of these years had deceived the public, and Purdue Pharma's leaders admitted it in guilty pleas.

Enron. Enron, an energy company in Houston, Texas, filed for bankruptcy on December 2, 2001. Its workers had much of their retirement savings in Enron stock. At the end of 2000, 62 percent of the money in the company's retirement plan was in its own shares, according to the Congressional Research Service, which does research for Congress.

The shares fell from more than $80 each in January 2001 to less than 70 cents in January 2002, less than a hundredth of their old price. A lawsuit claimed that Enron had blocked its workers from moving their retirement money out of Enron stock while the price fell.

A federal jury in Houston convicted Enron's former chief executives, Kenneth L. Lay and Jeffrey K. Skilling, on May 25, 2006. The jury found them guilty of conspiracy, planning a crime together with others, and of fraud, lying or cheating to get money. According to the Justice Department, the two men had lied to investors and others to make Enron look healthy, as if it were growing steadily and had no large debts or losses. Their lies pushed up the price of Enron's stock.

Tobacco in court. Federal Judge Gladys Kessler ruled on August 17, 2006, that the major tobacco companies had broken the federal racketeering law, the law against running a business through a pattern of fraud. She found that the companies had "defrauded the American people by lying for decades about the health risks of smoking and their marketing to children."

She wrote that they had "marketed and sold their lethal products with zeal, with deception," meaning with great eagerness and with lies, and that they had cared only about their profits, "without regard for the human tragedy or social costs that success exacted," the price other people paid for it. She ordered them to publish statements about the harm of smoking and about how addictive nicotine is.

Purdue Pharma. Purdue Pharma made OxyContin, a pain pill and an opioid, a drug from the same family of chemicals as heroin. The first wave of opioid overdose deaths, from 1999, came from prescription opioids, after doctors wrote more opioid prescriptions in the 1990s, according to the federal Centers for Disease Control and Prevention. CDC researchers count about 806,000 Americans who died of opioid overdoses from 1999 to 2023, from all kinds of opioids.

Guilty pleas. Purdue pleaded guilty in May 2007 to misleading the public about the risk of becoming addicted to OxyContin, and it agreed to pay $600 million. Its president, Michael Friedman, its top lawyer, Howard R. Udell, and its former chief medical officer, Paul D. Goldenheim, also pleaded guilty. They admitted to misbranding, selling a drug with false or misleading claims about it, and together they paid $34.5 million.

Purdue pleaded guilty again on November 24, 2020, to three federal felonies, which are serious crimes. No member of the Sackler family, which owned Purdue, was charged with a crime.

Attorneys general of 55 states and territories joined a $7.4 billion settlement with Purdue and the Sacklers on June 16, 2025, according to the New York attorney general's office. Federal Judge Madeline Cox Arleo sentenced Purdue to pay $5.5 billion on April 28, 2026, as CNBC and Reuters reported. She said she could not send Purdue's executives or owners to prison, because Justice Department lawyers had charged only the company.

More than 40 people spoke at the sentencing. Judge Arleo said, "Your government failed you." New York's attorney general announced on May 1, 2026, that Purdue had shut down. A new company, Knoa Pharma, took over, with no role for the Sacklers.

The 2008 rescue

The investment bank Lehman Brothers failed in 2008, and members of Congress voted public money to rescue banks and other companies.

Lehman Brothers. Lehman Brothers filed the largest bankruptcy in the country's history on September 15, 2008. It had about $639 billion in property and $619 billion in debts, according to the reference publisher EBSCO. Its chief executive was Richard S. Fuld Jr.

Investigators later looked into whether Lehman's executives, Fuld among them, had moved money and raised their own pay just before the bankruptcy. EBSCO's entry states that Fuld "walked away with $480 million" but does not say over what years he received it.

Repo 105. Anton R. Valukas, an investigator named by the bankruptcy court, reported in March 2010 on deals that Lehman's managers called Repo 105. They had used the deals to make the firm's books look stronger at the end of each quarter, the three-month periods for which companies report their results. Officials of the Securities and Exchange Commission reported in 2011 that it would be hard to prove that Lehman had broken the law. No Lehman executive was charged with a crime.

The rescue. Members of Congress passed the Emergency Economic Stabilization Act on October 3, 2008. It allowed up to $700 billion to rescue banks and other companies. Federal officials paid out $443.5 billion in all, and most of it was paid back. Auditors at the Government Accountability Office, which checks federal spending for Congress, put the final cost at $31.1 billion in December 2023.

Some parts of the rescue made money, and others lost it. The money put into banks under the Capital Purchase Program earned a profit of $16.3 billion. The rescue of the insurance company AIG, the loans to car makers and the housing programs all cost money, and the housing programs cost the most. The car loans began with $13.4 billion that President George W. Bush announced in December 2008.

Antitrust returns

Government lawyers brought monopoly cases against four of the largest technology companies in the 2020s. By September 2026 they had won two cases against Google and lost one against Meta, and two more were waiting for trial. Neither judge in the Google cases ordered Google to sell off part of its business.

Google search. Federal Judge Amit P. Mehta ruled on August 5, 2024, that Google had illegally kept monopolies in general internet search and in search advertising. On September 2, 2025, he refused to make Google sell its Chrome browser or its Android phone software. Then, on December 5, 2025, he ordered Google to share its search index, the huge list of web pages a search engine keeps, and some user data with rival companies. A committee of five technical experts was to oversee it.

Google's lawyers appealed. In late July 2026 Justice Department lawyers filed a brief, a written argument, with the appeals court, according to the news service Courthouse News. They asked the court to keep the ruling and to ban Google's large payments to companies such as Apple. Under those deals, Google is the search engine that the companies' products use unless the user changes it.

Google advertising. In a second case, federal Judge Leonie Brinkema ruled in April 2025 that Google had illegally monopolized the tools that websites use to sell advertising space. Her opinion on what Google must do was made public on September 16, 2026. She refused to make Google sell its advertising exchange, the online market where websites sell ad space to advertisers, and ordered new rules for how the tools work.

The National Law Review reported the opinion in an article by a writer for Chamber of Progress, a group funded by the technology industry.

Meta. The Federal Trade Commission's case against Meta was about its purchases of Instagram in 2012 and WhatsApp in 2014. The trial began on April 14, 2025, and Judge James Boasberg ruled for Meta on November 18,

2025. He found that Meta did not hold a monopoly, because TikTok and YouTube competed with it. FTC lawyers appealed on January 20, 2026, and in August 2026 the appeals judges had not yet set a date to hear the case.

Amazon and Apple. FTC lawyers and 17 state attorneys general sued Amazon on September 26, 2023, for keeping an illegal monopoly. Judge John H. Chun set the trial for March 29, 2027, and he will decide the case himself, without a jury. Justice Department lawyers and 16 state attorneys general sued Apple on March 21, 2024, for monopolizing the market for smartphones. On June 30, 2025, Judge Julien Neals refused Apple's request to throw the case out.

Lina Khan

Who: A law student whose 2017 article argued that the people who enforce antitrust law were missing Amazon's kind of power, and who later led the Federal Trade Commission.

When and where: New Haven, Connecticut, and Washington, D.C., 2017 to 2025.

At 32 she became the youngest head of the FTC.

Lina Khan argued that the judges and officials who enforce antitrust law had stopped looking at the kind of power Amazon held, and she later ran the agency that sued Amazon. She published her argument in the *Yale Law Journal* in 2017, while she was a student at Yale Law School, under the title "Amazon's Antitrust Paradox," a paradox being something that seems to contradict itself.

Her argument. She wrote that those judges and officials had come to judge competition mainly by whether prices went up. In her view, that test missed the power Amazon's leaders built by accepting losses while the company spread into selling, delivery, publishing and cloud computing, renting out computer storage and computing power over the internet. They could use the company's place in one business to gain in another.

The diapers. Her article gave an example. In 2009 Amazon's leaders wanted to buy Quidsi, the company that ran the website Diapers.com, and Quidsi's owners said no. Amazon then cut its prices for diapers and other baby goods by up to 30 percent. Quidsi's executives worked out that Amazon was on track to lose $100 million in three months on diapers alone. Amazon bought Quidsi the next year, and by February 2012 it had cut the discounts it gave parents from 30 percent to 20 percent, she wrote.

At the FTC. Members of the Senate confirmed her to the Federal Trade Commission by a vote of 69 to 28 on June 15, 2021. President Joe Biden chose her as the commission's chair, and she was sworn in the same day. At 32 she was the youngest chair in the agency's history, according to CNBC. Under her, FTC lawyers sued Amazon in 2023 and carried on the case against Meta.

Her critics. Robert D. Atkinson and Michael R. Ward of the Information Technology and Innovation Foundation, a research group funded partly by technology companies, argued in a 2023 report that her 2017 article misread how online markets work. Khan served as chair until January 20, 2025.

Corporate money in politics

Corporations and unions have been able to spend without limit on elections since 2010, as long as they do not work directly with a candidate. The justices of the Supreme Court ruled 5 to 4 on January 21, 2010, in *Citizens United v. Federal Election Commission*, that "the Government may not suppress political speech based on the speaker's corporate identity." That meant the government could not silence speech about politics just because a corporation was doing the speaking.

The case. The case began with a film that a non-profit corporation had made to attack Hillary Clinton. Federal law had barred the group from advertising it close to the election.

Super PACs. After the ruling, committees called super PACs could raise and spend unlimited sums. Researchers at the Brennan Center for Justice, a law and policy institute, count about $6.4 billion that super PACs spent on federal elections from 2010 to 2022. In the 2024 election they spent at least $2.7 billion. That year, spending by groups that do not have to name their donors passed $1 billion, spending people call dark money.

Lobbying. Companies and other groups also pay lobbyists. Organizations spent $5.08 billion on lobbying the federal government in 2025, the first year above $5 billion, according to OpenSecrets, a group that tracks money in politics. That year 15,768 organizations reported lobbying, and the US Chamber of Commerce spent the most, $72.1 million.

Working for a company without being employed by it

Some companies now pay people for work without hiring them as employees. Ride and delivery companies call their drivers independent contractors. The federal law on the minimum wage and overtime pay is the Fair Labor Standards Act, and its rules apply to employees but not to independent contractors, according to the Department of Labor.

How many. The count of Americans who do this kind of work depends on the question asked. Researchers at the Bureau of Labor Statistics, the federal agency that counts jobs, found 1.6 million people in May 2017 who found work through an app or a website, 1.0 percent of all workers. Researchers at the Pew Research Center found in 2021 that 16 percent of American adults had ever earned money through an online job app, and 9 percent had done so in the past year.

THE THREE QUESTIONS, ANSWERED

1. **How did a few companies come to control whole industries?** Their owners made secret deals with railroads, bought out rivals and joined companies into trusts and mergers. By 1880 Standard Oil refined 90 to 95 percent of the country's oil.
2. **What did owners do when their workers went on strike?** They locked workers out and hired armed guards, and officials sent soldiers and court orders. At Homestead in 1892 Frick's Pinkerton men fought the steelworkers, and at Ludlow in 1914 soldiers paid by the mine owners burned the strikers' tent camp.
3. **Did breaking up a giant company ever work?** Not clearly. Judges split Standard Oil into 34 companies in 1911, yet Rockefeller was three times as rich by 1913. AT&T gave up its local telephone companies in 1984 in a deal most newspapers called a win for AT&T, the order to split Microsoft was thrown out in 2001, and judges in 2025 and 2026 would not make Google sell off parts of its business.

What changed: In 1500 no company existed in this land, and by 2026 three American companies were each worth more than $4 trillion.

FIVE DATES TO KEEP

May 13, 1607: The Virginia Company's settlers found Jamestown.

December 16, 1773: Colonists dump the East India Company's tea into Boston Harbor.

July 2, 1890: The Sherman Antitrust Act becomes law.

May 15, 1911: The justices of the Supreme Court rule against Standard Oil.

January 21, 2010: The justices decide *Citizens United*, on corporate spending in elections.

NOT KNOWN: No record shows which Acadians died at sea on the ships Thomas Hancock's firm rented out in 1755, or how many.

Where to go: Lowell National Historical Park stands in Lowell, Massachusetts, the mill town the Boston Associates built.

Next: Chapter 20, Government, Law, and Politics. It asks who wrote the rules of American government, and who got to vote.

PART 5

Power, Law, and War

Chapter 20: Government, Law, and Politics

CHAPTER 20 OF 37 · GOVERNMENT, LAW, AND POLITICS · 10 ERAS · ABOUT 140 MINUTES

The rules of American government were written by people, and in every era people with power rewrote them, most often to decide who could vote and who could not.

One copy of a report survives in the National Archives of the United Kingdom. John Pory wrote it in 1619. It records six days, July 30 to August 4, when twenty-two elected men sat in the church at Jamestown, Virginia, the only building on the island big enough to hold them all. They were the first elected lawmakers in English North America. They made rules about tobacco, gambling and land. No record says who voted for them.

This chapter follows the rules from that church to the Capitol on January 6, 2021. Watch three things as you read. Who may vote: the answer changes in nearly every era, and the people who changed it usually wrote down why. What judges may do: in 1803 the justices of the Supreme Court first cancelled a law, and they have been cancelling and allowing laws ever since. What happens to people in power who break the law: a governor in 1614, a cabinet secretary in 1929, a president in 1974.

The chapter also follows the laws about schools, which say where children must go and what they may be taught. From 1790 on, it tracks the laws on who may come to live in the country and who may become a citizen.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who decided who could vote, and how did the answer change?

2. When did judges first cancel a law, and what have they done with that power since?
3. What happens when people in power break the law?

Five minutes? Read "The Zenger trial, 1735" in 1700 to 1750, then the John Peter Zenger and Andrew Hamilton stories after it.

Can skip: "Land for colleges, 1862", "The food laws of 1906", "New kinds of schools, 1991 to 1996", "Immigration law after 2001". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 6 MINUTES

Long before Europeans arrived, the Native nations here governed themselves, and they did it in different ways. Most of this era is about the Haudenosaunee, five nations in the Northeast that joined in a league and lived under one law, because their own full telling of that law survives. Two short spans at the end cover the Pueblo towns along the Rio Grande, where each town ran its own affairs, and Cahokia, a city of thousands on the Mississippi River led by chiefs.

How five nations at war made one law

The Haudenosaunee are a league of Native nations whose homelands lie mainly in what is now New York State and the Great Lakes region. Their name means "people who build a house." Five nations joined first: the Mohawk, the Oneida, the Onondaga, the Cayuga and the Seneca. The Tuscarora joined in 1722, which made six. Each nation stayed a separate nation inside the league.

The law that joined them is called the Great Law of Peace. In the Haudenosaunee account, the five nations were once at war with each other. A man called the Peacemaker wanted to spread peace among them. He came to the house of an Onondaga leader named Hayo'wetha, often written Hiawatha. When the nations made peace, they pulled up a white pine tree and threw their weapons into the hole. Then they planted the tree again on top of the weapons and named it the Tree of Peace.

The Haudenosaunee kept the Great Law in memory and passed it on by speaking it. A full telling takes several days. They also wove belts of purple and white shell beads, called wampum. The patterns in the belts recorded the laws of the league, the stories told at ceremonies, and important dealings with other nations. The belts were presented at meetings of the Grand Council, the league's lawmaking body. The Haudenosaunee used wampum to keep these records, and never as money.

KNOWN:	The league was in place before Europeans arrived.
DISPUTED:	When it began. The dates people give run from about 1142 to the 1400s or 1500s.

The Grand Council and the clan mothers

The league's lawmaking body is the Grand Council, and it has fifty seats. Each seat belongs to a leader called a *hoyaneh*. The Onondaga hold fourteen seats, the Cayuga ten, the Mohawk nine, the Oneida nine and the Seneca eight. Council members are not paid, and they serve for life.

The Haudenosaunee count the Onondaga Nation as the league's capital. They say the council fire burns there, as the Peacemaker promised, and they call the Onondaga the "Keepers of the Central Fire." The Mohawk, in the east, are the "Keepers of the Eastern Door," and the Seneca, in the west, are the "Keepers of the Western Door."

How a law passes. Before the council members pass a law, every member must agree to it. This way of deciding is called consensus. Under it, no law passes while even one member is against it.

Women hold power in this government through the clan mothers. A clan is a large group of related families. The members of a clan choose a clan mother together, and she serves for life. The clan mothers choose the council members. A clan mother can also remove any *hoyaneh* who does not serve the good of his nation.

Did the Constitution borrow from this law? In 1988 members of both houses of Congress voted for a resolution, a written statement that lawmakers vote on, which said yes. Representative Morris Udall put it forward. Its title states that it was meant "to acknowledge the contribution of the Iroquois Confederacy of Nations to the development of the U.S. Constitution." Iroquois is the more common English name for the Haudenosaunee, and a confederacy is a league of nations or states. The Constitution is the written plan for how the national government of the United States works.

Scholars disagree about how much the men who wrote the Constitution took from the Great Law. Bruce Johansen and Donald Grinde argue that the founders admired Native governments and borrowed ideas from them. Elisabeth Tooker, an anthropologist, a scientist who studies how peoples

live, argued the other side. She wrote that the founders took their models from European ideas and that the evidence of borrowing from the league is very thin.

The Pueblo towns

Groups of Spanish explorers began coming up the Rio Grande in 1540. They found farming peoples living in settled towns, which the Spanish called pueblos, the Spanish word for towns. The people of each town governed themselves. Each town had its own leaders, customs and language. No one leader or council ruled all the pueblos.

At Zuni, a council of priests held power over the town. One group of them, the bow priests, handled dealings with outsiders.

Cahokia, a city led by chiefs

Cahokia stood near the Mississippi River, east of where St. Louis is now. The city covered more than 4,000 acres, and its largest years came between about 1050 and 1150. Archaeologists, the scientists who dig up what people left behind, disagree widely about how many people lived there then. Their estimates run from 10,000 to 20,000, and some go higher.

Cahokia was a chiefdom, a society led by chiefs and by officials of high rank. The people built large mounds of earth with flat tops. Buildings for religious leaders and for other officials stood on top of the mounds. Other mounds held the graves of people of high rank.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

The first governors of Florida and of New Mexico served the king of Spain. Pedro Menéndez de Avilés held his office in Florida by a contract with the king, who lived across the ocean in Spain. Juan de Oñate held his office in New Mexico by a contract with the viceroy, the official who ruled in the king's name from Mexico City. Menéndez used a Timucua council house as his first fort. Oñate made a Tewa town his first capital.

A governor for Florida, 1565

Pedro Menéndez de Avilés founded St. Augustine, in Florida, on September 8, 1565. King Philip II of Spain had ordered the voyage. Menéndez held the title of *adelantado*, a Spanish office that a man received by signing a contract with the king. He soon became Florida's first governor. St. Augustine is the oldest European settlement in the mainland United States where people have lived without a break since its founding.

Menéndez made his first settlement in the town led by Seloy, a chief of the Timucua people. About 800 colonists came with him, among them 26 women and an unknown number of Africans whom the Spanish held as slaves. Timucua council houses were usually round, built of thatch, and big enough to hold up to a thousand people. Seloy offered Menéndez his council house to use as a fort, and the Spanish used it as their first fort in Florida.

Officials of the king of Spain set rules in 1573 for laying out new towns. The rules belong to a body of royal orders known as the Laws of the Indies. They required a town's streets to form a grid around a central square, or plaza. The main government buildings and the church were to face the plaza. St. Augustine's grid of streets was laid out in 1603, thirty-eight years after the town was founded.

A governor for New Mexico, 1598

Juan de Oñate made a contract with the viceroy in Mexico City. He promised to settle New Mexico and govern it. In return he got loans, supplies and soldiers. He left the town of Santa Bárbara early in 1598 with 600 Spanish settlers and Native allies from Tlaxcala, in Mexico.

On the Rio Grande he set up his first capital at Ohkay Owingeh, a town of Tewa-speaking people. The Spanish named the town San Juan de los Caballeros in 1598. A year later Oñate moved his capital across the river to a place called San Gabriel. The people of Ohkay Owingeh still live in their town.

Oñate went to the pueblos one by one in the autumn of 1598. Through translators, he made the people of each pueblo swear loyalty to the king of Spain. He also placed a missionary, a priest sent to bring people into his religion, in each one. Oñate governed New Mexico until 1608. The next era tells what the king's judges later found he had done there.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 10 MINUTES

Spanish judges in Mexico City put a former governor of New Mexico on trial in 1614. The English built Jamestown, in Virginia, in 1607, and twelve years later twenty-two elected men met there as the first elected assembly in English North America. The next year, forty-one men aboard the *Mayflower* signed an agreement to make laws for themselves and obey them.

The colonists' own lawmakers decided who could vote, and they changed the rules many times. In 1662 Virginia's lawmakers passed a law that made every child born to an enslaved woman a slave as well. In Massachusetts, lawmakers made towns keep a school.

What the judges found Oñate guilty of, 1614

Royal judges in Mexico City tried Juan de Oñate, the former governor of New Mexico, in 1614. Officials of the king of Spain had brought 30 charges against him. The judges found him guilty of 12 and not guilty of 18. They ordered him out of New Mexico for life and barred him from Mexico City for four years, or five, since accounts differ. They also fined him 6,000 ducats in the money of Castile, in Spain.

The charges. One of the twelve was using too much force against Acoma Pueblo. He was found guilty of it because of the sentences he gave the Acoma people after the fighting there in 1599. He was also found guilty of hanging two Native men without just cause, of the murders of two of his own captains, of ordering his captains to kill two soldiers who had run away from the army, and of lying to the viceroys and the king about how rich the land was.

What happened at Acoma. Oñate's nephew Juan de Zaldívar came to Acoma in December 1598 with about thirty soldiers. He took some of them up the mesa, the flat-topped hill where the town stands, to demand food. In the Acoma people's own account, the soldiers raped an Acoma woman. Rape means forcing a person into sex. The Acoma killed Zaldívar and eleven to fourteen of his men. Accounts differ on the number.

Oñate sent Juan's brother, Vicente de Zaldívar, to Acoma in January 1599 with about seventy soldiers. Over two or three days of fighting, those soldiers killed Acoma men, women and children, and they burned much of the town. No one counted the dead at the time.

KNOWN:	The soldiers killed hundreds of people. The Pueblo nations' own leaders use the figure of about 800.
DISPUTED:	The count. Estimates run from about 300 to about 1,500, and PBS reports that about 200 people of Acoma survived out of nearly 2,000. Those figures come from different sources and do not fit together.

The sentences. Afterward Oñate sentenced the Acoma men over twenty-five to have one foot cut off, and to be forced to work as slaves of the Spanish for twenty years. Younger men and boys from twelve up, and women and girls over twelve, got the same twenty years of forced work. He handed the old people to the Plains Apache as slaves.

The children. He took the children under twelve from their families and gave them to the friars, the Catholic religious brothers with his colony. On Oñate's order, about 70 of the girls were sent to convents in Mexico, houses where nuns live. The National Park Service's account states that almost certainly none of the girls ever came home.

The Hopi men. Oñate also sentenced two Hopi men captured in the fighting to have their right hands cut off. Then he had them released to carry word of the punishment to their own people.

The feet. The sentences are in the written record. Whether the feet were cut off is disputed. In the Acoma people's own account, Oñate had the right feet of 24 captive men cut off. The historians Marc Simmons and John Kessell doubt that it was done, and Kessell points out that no record describes a one-footed Acoma man.

The first General Assembly, 1619

The leaders of the Virginia Company of London, the business that ran the colony, put out a paper called the Great Charter in 1618. A charter is a written grant of rights and powers. Sir Thomas Smythe and Sir Edwin Sandys wrote it. It allowed the governor to "summon a General Assembly to legislate as appropriate." That meant the governor could call a meeting of men to make laws.

In the summer of 1619 Governor George Yeardley used that power. The colony had eleven settlements, and two elected men came from each one to speak and vote for it. They were called burgesses. A burgess was an elected member of the assembly. The twenty-two burgesses met in the Jamestown church with Yeardley and his council, the six men who advised him, from July 30 to August 4, 1619.

In six days the members set rules for how they would do business and protected colonists' claims to their land. They also made the Church of England the colony's official church. They passed laws about tobacco, gambling, behavior and dealings with Native nations.

Whose land it was. Jamestown stood on land that belonged to the Paspahegh. They were one of twenty-eight to thirty-two groups, each with its own chief, that paid tribute, goods given to a more powerful ruler, to Powhatan. He was the chief over all the other chiefs of Tsenacomoco. The anthropologist Helen C. Rountree estimates that about 15,000 people lived in Tsenacomoco, across about 6,000 square miles. The English built Jamestown on Paspahegh land in 1607 and stayed there without asking the Paspahegh's permission.

The assembly outlives the company. The king's officials took away the Virginia Company's charter in 1624. Virginia became a royal colony in 1625, with a governor and council chosen by the king. No one in England had given the assembly permission to go on meeting, but the burgesses kept meeting.

Whether the assembly had any legal right to meet stayed unclear until 1627. That year the king asked the General Assembly to help make rules for the tobacco trade. By asking, he treated the assembly as a lawful body, though no written order said it could meet.

John Pory

Who: The colony's secretary, who served as Speaker of the first elected assembly in English North America and wrote the record of what it did.
When and where: Jamestown, Virginia, July 30 to August 4, 1619.

Pory, the colony's secretary, ran the 1619 assembly and wrote its record.

John Pory wrote the record of the first meeting of Virginia's General Assembly. He called his report "The Proceedings of the First General Assembly, July 30, 1619." One copy of it survives, in the National Archives of the United Kingdom. Any copies kept in Virginia, Brent Tarter wrote on the Library of Virginia's blog in 2019, would have been destroyed in one of the British raids on Richmond during the American Revolution, along with most of the colony's records.

Pory knew how a lawmaking body ran its business. He had sat in the English Parliament, the group of lawmakers who made England's laws. The leaders of the Virginia Company made him secretary of the colony on December 2, 1618. He sailed from England in January 1619 with the governor, George Yeardley, and reached Jamestown on April 18.

That summer Yeardley made him Speaker of the assembly, the officer who runs its meetings. Pory was not one of the elected burgesses. He ran the meetings because the governor appointed him. He left Virginia in the early summer of 1622, after Christopher Davidson was made secretary in his place.

Why forty-one men signed a paper off Cape Cod, 1620

The *Mayflower* anchored off Cape Cod, in what is now Massachusetts, in November 1620. The passengers had permission, called a patent, to settle near the Hudson River. That land lay inside the Virginia Company's territory. Off Cape Cod they were outside it, so no one had given them the right to govern themselves there.

Forty-one men signed an agreement on November 11, 1620, to keep order until a new patent came. It is known as the Mayflower Compact. The signers agreed to "covenant & combine our selves together into a Civill body politick." That meant they promised to form a government together. They also promised to make and obey "just & equall lawes." The women and children did not sign. Most of the servants did not sign either. The original paper has been lost.

The passengers founded Plymouth Colony. They settled at Patuxet, where a Wampanoag community had lived until "a sickness had killed most of them." Plymouth remained a separate colony from Massachusetts Bay until 1691.

Who could vote

The colonies' lawmakers decided who could vote. In Virginia they went back and forth for more than fifty years over one question: whether every free man could vote, or only a man who owned land or headed a household. The rules came from an old English idea that only people who owned property should have a say in running a place. No record says who voted for the first burgesses in 1619.

The members of the General Assembly let all freemen vote in 1646. A freeman was a man who was neither enslaved nor bound as a servant. They allowed only heads of households to vote in 1655, and they opened the vote to all freemen again in 1656. In 1670 they limited the vote to men who owned land or headed a household.

That limit lasted, except for one year. In 1676, during a revolt called Bacon's Rebellion, all freemen could vote again. The Assembly's members put the limit back in 1677, after the revolt fell apart. In 1684 men who rented land for their whole lives could vote as well.

The members passed a law in 1699 that barred women and anyone under twenty-one from voting. Women voting went against English custom, and no one knows whether any Virginia woman had tried to vote before then.

Candidates in Virginia gave voters food and drink before and after an election. People called it treating. Virginia's lawmakers limited treating during the week of an election, so candidates had other people do the treating for them.

Massachusetts: church members only. In Massachusetts Bay, the officers of the colony's company sat together as its lawmaking group, called the General Court. In May 1631 they ruled that only church members could become freemen, the men who could vote. Women could not vote at all.

To join a church, a person had to stand before the church members, tell how they had come to their faith, and be accepted. Most men in the colony could not pass that test. Under a new charter from the king in 1691, men who owned property could vote whether or not they belonged to a church.

A Virginia law on the children of enslaved women, 1662

The members of Virginia's General Assembly passed a law about the children of enslaved women in December 1662. It states that "all children borne in this country shalbe held bond or free only according to the condition of the mother." Bond meant held in slavery. Under English law, whether a child was born free or bound depended on the father. Under the new Virginia law, a child born to an enslaved woman was enslaved, whoever the father was. Enslavers gained every such child as a worker they owned.

A town made to keep a school

The lawmakers of Massachusetts Bay put the duty of teaching children on each household in 1642. Parents and masters had to try to teach their children and apprentices, young people bound to a master to learn a trade, to "read the English tongue," or have someone else teach them. They also had to teach them the colony's capital laws, the laws whose penalty was death.

If a household failed, the town's selectmen, its elected leaders, could fine it. They could also take the children away and place them with another master.

In 1647 the lawmakers passed a second law. Its opening words state that "that old deluder, Satan" wants to keep people from knowing the Scriptures, the Bible. A deluder is someone who tricks people. Once a town had fifty households, the people of the town had to appoint someone to teach reading and writing. The parents or the town paid that teacher. A town of one hundred families also had to set up a grammar school to prepare boys for the university.

Under the 1647 law, the penalty fell on the town rather than on the parents. If the people of a town went more than a year without doing what the law required, the town had to pay five pounds to the nearest school until they did it.

1700 to 1750

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A New York jury set a printer free in 1735 though the governor wanted him in prison. In the same years, the men elected to the colonial assemblies held the power to propose every law and to decide how public money was spent, which gave them control over the pay of the governors the king appointed. The men of Virginia's assembly used their power another way. They took the vote away from every free Black man, every free man of mixed Black and white family, and every Native man in the colony.

Why the governors could not control their own pay

After 1680 the members of the colonial assemblies had the power to propose all of a colony's laws. In a royal colony, the king chose the governor and his council, and the colonists elected the assembly. By 1763 nine of the thirteen colonies that later rebelled had royal governors.

The members of each colonial assembly decided how public money was spent, and they checked the spending. That gave them control over the salaries of the governor and of the colony's judges.

Officials in London tried to make the assemblies set fixed yearly salaries. They also wanted a fixed civil list, a set list of paid government posts. That list would have given each governor jobs to hand out to his supporters. The assembly members turned down every one of these attempts.

Virginia's burgesses take away the vote, 1723

The burgesses of Virginia's General Assembly passed a law in May 1723 about enslaved people accused of crimes punished by death. It also set punishments for plots and revolts, and rules to control Black and Native people, enslaved or free.

One clause of it states that "no free negro, mullatto, or indian whatsoever, shall hereafter have any vote at the election of burgesses, or any other election whatsoever." Negro was a word used then for a Black person. Mulatto was a word used then for a person with both Black and white parents or ancestors. With that clause, the burgesses took the vote from every free Black man, every free man of mixed Black and white family, and every Native man in the colony.

A lawyer in London objected. Richard West advised the Board of Trade, the group of officials in London who oversaw the colonies. On January 10, 1724, he wrote: "I cannot see why one Freeman should be used worse than another meerly upon account of his complexion." Complexion meant skin color.

William Gooch, Virginia's lieutenant governor, an officer of the king one rank below the governor, defended the law. On May 18, 1736, he wrote that free Black people had been suspected of helping a planned revolt of enslaved people before the 1723 session. He said the law was meant "to make the free-Negros sensible that a distinction ought to be made between their offspring and the Descendants of an Englishman." Sensible here meant aware. The ban was still in force when Gooch defended it in 1736.

That same year, 1736, Virginia's lawmakers set a land test for voters. A man had to have owned 100 acres of unfarmed land, or 25 acres with a house on it, for at least a year.

The Zenger trial, 1735

A jury in New York refused in 1735 to find the printer John Peter Zenger guilty of seditious libel. Libel means printed words that harm someone's good name. Seditious libel was the crime of publishing written blame of any public official, of the law, or of anything the law set up. Under that law, it made no difference whether the printed words were true.

The judge told the jury to decide only one question: whether Zenger had printed the newspaper. The jury decided more than that and found him not guilty.

What the verdict did not change. The jurors set Zenger free, but the law stayed the same. The case set no rule that later judges had to follow. In New York, the truth of the printed words did not become a defense, a reason that can keep a person from being found guilty, in a libel case until April 6, 1805, seventy years later.

That day the state's lawmakers passed a law allowing truth as a defense when the words were printed "with good motive and for justifiable ends." In 1821 New York's leaders wrote the same rule into the state's second constitution.

John Peter Zenger

Who: A New York printer jailed for more than eight months for printing attacks on the governor, then found not guilty by a jury.

When and where: New York City, November 1734 to August 1735.

Eight months in jail, and the paper missed only one issue.

John Peter Zenger sat in jail from November 17, 1734, to August 5, 1735, for printing a newspaper that attacked the governor of New York. The trouble began in 1733. The chief judge, Lewis Morris, wrote a dissent, a judge's written disagreement with the other judges, against Governor William Cosby in a court case. Cosby removed Morris from office. Morris's allies answered by starting a newspaper, the *New-York Weekly Journal*, and Zenger printed it. The first issue came out on November 5, 1733.

Until then the only paper in the colony had been the *New York Gazette*, which printed the governor's side. In the *Journal*, writers accused Cosby's government of tyranny, meaning cruel and unfair rule, with articles, jokes and mocking verses, issue after issue. Cosby offered a reward of fifty pounds to anyone who found out who wrote them. He also ordered copies of the paper burned in public by the hangman.

In January 1734 Chief Justice James De Lancey asked a grand jury to charge Zenger with a crime. A grand jury is a group of citizens who decide whether there is enough proof to charge a person with a crime. The grand jury refused. In October a second grand jury refused too. Its members said no one could tell who had written the pieces.

On November 17, 1734, the sheriff arrested Zenger and took him to the Old City Jail. De Lancey and another judge named Philipse, both allies of Cosby, had signed the order for his arrest. His bail, the money he had to pay to get out of jail until his trial, was £400. That was more money than he had, so he stayed in jail. His wife Anna and his apprentices kept printing the paper. Only one issue was missed.

On April 16, 1735, his lawyers, James Alexander and William Smith, argued that the court itself had not been set up lawfully. De Lancey struck both of their names off the list of lawyers allowed to practice. Zenger was left with no lawyer.

The trial took place on August 4, 1735, on the second floor of City Hall. Andrew Hamilton, a lawyer from Philadelphia, defended Zenger. Hamilton tried to show that what Zenger printed was true. De Lancey stopped him: "The law is clear that you cannot justify a libel." Hamilton then spoke to the jury. He told them they had "the right beyond all dispute to determine both the law and the fact." The jury found Zenger not guilty. He was let out of jail the next day.

Andrew Hamilton

Who: The Philadelphia lawyer who defended Zenger without pay and told the jury it could judge the law as well as the facts.
When and where: New York City, August 4, 1735. Philadelphia before and after.

He took the case for nothing and told the jury to judge the law.

Andrew Hamilton defended John Peter Zenger for free. He was a lawyer in Philadelphia, and the members of the Pennsylvania Assembly elected him their Speaker. He traveled to New York for the trial and worked on the case with another lawyer, John Chambers.

The judge would not let Hamilton prove that Zenger's words were true. So Hamilton built his defense on the power of the jury. He told the jurors that the case was "not of small or private concern." He called it "the cause of liberty." He argued that the law gave people "a right to liberty of both exposing and opposing arbitrary power," meaning power used without limits, "by speaking and writing truth." The jury found Zenger not guilty.

The phrase "Philadelphia lawyer" entered American speech because of this case. Dictionaries define it as a very clever lawyer, skilled at the fine points of the law.

> Back in Pennsylvania, Hamilton led the planning of the Pennsylvania State House and pushed to get it finished. The building is now called Independence Hall.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 15 MINUTES

Fifty-six men signed the Declaration of Independence, and in the next fifteen years the leaders of the new country wrote down how it would be governed. They wrote a plan of government, a constitution to replace that plan, and a list of rights added to the constitution. The men who wrote the Constitution put clauses in it that protected slavery. They set no national rule on who could vote, so each state's lawmakers decided.

In the 1790s the country's leaders split into the first two political parties. In 1798 President John Adams signed a law that made it a crime to publish false and scandalous writing against the government. Federal judges sent people who criticized him to jail under it.

The Declaration of Independence, 1776

The delegates to the Continental Congress voted for independence from Britain on July 2, 1776. The Continental Congress was the meeting of delegates from the thirteen colonies. Two days later, on July 4, they approved the Declaration of Independence, which gave their reasons.

Five men made up the committee that wrote the first version: Thomas Jefferson, John Adams, Benjamin Franklin, Robert R. Livingston and Roger Sherman. Jefferson did most of the writing. The delegates began signing the finished copy on August 2, 1776. In the end 56 men signed it. Livingston, who helped write it, never signed.

Who could vote in the new states

Leaders in many of the new states wrote state constitutions, each with its own rules for who could vote. The constitutions written in ten states between 1776 and 1790 stated that voters had to be "male" or "freemen." The election laws of five states contained no rule about sex.

New Jersey's constitution of 1776 gave the vote to "all inhabitants of this colony of full age, who are worth fifty pounds." That meant every adult living there who owned fifty pounds' worth of property could vote. It said nothing about sex or race. So unmarried women and free Black people who owned enough property could vote. Fifty pounds was about the price of three horses or eight cows in 1806.

Married women found it much harder to meet the property test, because under the law of the time a wife's property was usually in her husband's hands.

New Jersey's lawmakers passed an election law in 1790 that said a voter must vote in the town where "he or she" lived. In 1797 they passed a law that did the same for the whole state. It also dropped the words "clear estate," an old legal phrase meaning owning property outright, from the property rule. What wives and widows held was often not theirs outright, so with those words gone, it was easier for them to qualify.

Then in 1798 the lawmakers passed a law that let only white male taxpayers vote for town officers. Property owners of any race or sex still voted for county, state and national offices. The next era tells how that ended.

The Articles of Confederation

Before the Constitution, the country's plan of government was the Articles of Confederation. The members of Congress approved it on November 15, 1777. It took effect on March 1, 1781, when Maryland's leaders approved it, the last of the thirteen states to do so.

Under the Articles, each state had one vote in Congress, whatever its size. Congress had no power to tax. It could not make rules for trade between the states. Any change to the Articles needed every state to agree.

The Constitutional Convention, 1787

Delegates from the states met in Philadelphia in 1787 to write a new plan of government. The meeting was called for May 14, but too few delegates had arrived to begin until May 25. About 55 delegates took part. No delegates came from Rhode Island. The delegates worked until September 17, 1787, and 39 of them signed the finished Constitution.

The delegates settled how to share seats in Congress with a deal called the Connecticut Compromise. Every state got two senators, and the number of seats each state got in the House of Representatives depended on its population.

Three-fifths. The delegates also had to decide whether enslaved people counted as part of a state's population. They agreed to count each enslaved person as three-fifths of a free person when working out a state's seats in the House. This rule is in Article I, Section 2. Under it, the states with the most enslaved people got extra seats in the House.

Article II states that each state has as many electors, the people who pick the president, as it has senators and representatives. So those states also got extra votes for president. Under the law, the enslaved people being counted were property, with no rights at all, so none of them could vote.

Slavery and voting in the Constitution

Two other clauses of the Constitution dealt with slavery. Article I, Section 9 stated that Congress could not ban "the Migration or Importation of such Persons as any of the States now existing shall think proper to admit" before 1808. That meant members of Congress could not stop the shipping of enslaved Africans into the country for twenty years. Article V stated that no amendment could change that clause before 1808.

Article IV, Section 2 stated that a person "held to Service or Labour in one State" who escaped into another state "shall be delivered up on Claim of the Party to whom such Service or Labour may be due." That meant a person who escaped slavery into a free state still had to be handed back to the enslaver.

The Constitution set no national rule for who could vote. Article I, Section 2 stated that the voters for the House in each state must meet the same rules as the voters for the largest house of that state's own lawmaking body. So each state's lawmakers decided who could vote.

Ratification

The Constitution would take effect once nine states ratified it. To ratify means to approve formally. Under Article VII, a meeting of delegates in each state, called a convention, voted on it. From October 1787 to May 1788, Alexander Hamilton, James Madison and John Jay wrote 85 essays to persuade New Yorkers to vote for it. They signed them with a made-up name, Publius. The essays came out mainly in two New York newspapers, and they are known as *The Federalist*.

On June 21, 1788, New Hampshire's delegates made it the ninth state to ratify. The new government began on March 4, 1789. Delegates in North Carolina and Rhode Island approved the Constitution only after the first president took office. Rhode Island's lawmakers turned down eleven attempts to ratify it between September 1787 and January 1790. When Rhode Island's delegates finally voted on May 29, 1790, they approved it 34 to 32.

George Washington

Who: The first president, elected twice with every electoral vote, who chose to leave office after two terms.

When and where: Federal Hall, New York, April 30, 1789. Philadelphia, September 19, 1796.

Washington stopped after two terms, a limit presidents kept for about 150 years.

George Washington is the only person ever elected president with every electoral vote. It happened twice, in 1789 and in 1792. The electoral votes are the votes cast by electors, the people each state chooses to pick the president. The votes for the first election were counted on April 6, 1789, more than a month late. Members of the new Congress could not reach New York in time because of a cold, snowy winter. Washington had all 69 votes.

On April 30, 1789, Washington swore the oath of office, the promise each new president makes, on the second-floor balcony of Federal Hall in New York. Robert R. Livingston, a New York judge whose title was chancellor, read him the oath while a crowd watched from the street. Then Washington went inside and read his first speech as president, called the first inaugural address, to both houses of Congress.

Nothing in the Constitution limited how many times a president could be elected. In 1796 Washington decided to stop after two terms. He announced it in a written message to the country, printed on September 19, 1796, in a Philadelphia newspaper. It is called his Farewell Address, though he never read it aloud.

James Madison had written a first version in 1792. Alexander Hamilton did most of the writing in 1796, and Washington made changes to it himself. In it he told the country of "the resolution I have formed, to decline being considered among the number of those out of whom a choice is to be made." That meant he would not run again.

In the same address he warned "against the baneful effects of the spirit of party." Baneful means harmful. He also wrote that it was "our true policy to steer clear of permanent alliances with any portion of the foreign world."

For about 150 years the two-term limit stayed a habit, not a law. Franklin Roosevelt broke it when he won a third term in 1940 and a fourth in 1944. The Twenty-second Amendment of 1951 made the limit law. Era 9 tells how.

The people he owned. Washington was an enslaver. He owned 123 people at his plantation, Mount Vernon. He also managed 153 more who belonged to the estate of his wife Martha's first husband, the property that man had left when he died. Under the law, Washington could not sell or free those 153.

In his will, dated July 9, 1799, he freed his valet, or personal servant, William Lee at once. He ordered the other 122 people he owned to be freed after Martha died. Martha Washington freed them on January 1, 1801. The other 153 people went back to her first husband's estate. Martha's grandchildren, who inherited that estate, divided them among themselves, and in dividing them they split up families.

The custom of reading the Farewell Address aloud in Congress began on February 22, 1862, during the Civil War, after a thousand people in Philadelphia asked Congress to mark Washington's birthday. Since 1896 a senator has read it aloud in the Senate every year. It runs 7,641 words.

James Madison

Who: A delegate to the Constitutional Convention who first argued against a bill of rights, then wrote one and steered it through Congress.
When and where: Philadelphia and New York, 1787 to 1791. Born March 16, 1751. Died June 28, 1836.

Madison wrote the Bill of Rights after arguing that none was needed.

James Madison introduced the amendments that became the Bill of Rights. An amendment is a change or addition to the Constitution. Madison had been the loudest voice against adding a list of rights. He had argued that the Constitution already limited the government's powers. Then delegates in several states agreed to ratify only if the first Congress would consider the amendments they proposed. After that, Madison changed his position.

On June 8, 1789, Madison stood up in the House of Representatives and proposed his list. He told the House, "I think we should obtain the confidence of our fellow citizens, in proportion as we fortify the rights of the people against the encroachments of the government." To fortify means to make stronger. Encroachments means steps that go beyond what is allowed.

Madison wanted each change written into the body of the Constitution, in the place where it belonged. In August 1789 the House members followed Roger Sherman's plan instead. They voted to add the changes at the end, as separate amendments. That is why the Bill of Rights sits at the back of the Constitution.

The House members passed 17 amendments. The senators cut and combined them into 12. Members of Congress proposed those 12 amendments on September 25, 1789, and President Washington sent

them to the states on October 2. By December 15, 1791, lawmakers in three-quarters of the states had ratified ten of them. Those ten are the Bill of Rights.

At the Constitutional Convention, Madison had backed the rule counting each enslaved person as three-fifths of a free person. In essay No. 54 of *The Federalist*, he defended it. He said enslaved people had a "mixed status as property and persons." He said many times that slavery went against the principles of a republic, a country governed by elected representatives. He never put forward a workable plan to end it.

Madison enslaved people all through these years. About 100 enslaved people worked at his plantation, Montpelier, at any one time, and nearly 300 over the life of the plantation. When he was short of money, he sold sixteen of them to his cousin William Taylor, who owned a sugar plantation and a cotton plantation in Louisiana. He freed no one in his will. He asked his wife, Dolley, not to sell the people he left her without their consent unless they misbehaved. She sold some of them and kept others enslaved.

Who could become a citizen, 1790

The first federal law on who could become a citizen allowed only a "free white person." Members of Congress passed it in 1790 and called it the Naturalization Act. Naturalization is the legal process by which a person born in another country becomes a citizen.

The person had to be of good character, had to have lived in the United States for two years and in the state for one, and had to swear an oath to support the Constitution. Part of that racial limit stayed in federal law until 1952.

The Fugitive Slave Act, 1793

In 1793 the members of Congress passed a law about "persons escaping from the service of their masters." President George Washington signed it on February 12, 1793. Under the law, an enslaver, "his agent or attorney" could "seize or arrest" a person in any state. He could take that person before a single federal judge, or a county, city or town magistrate. A magistrate is a local judge. If the enslaver swore to his claim, spoken or in writing, the judge could give him a certificate to take the person away.

The law provided no jury for the seized person. It stated nothing about letting that person give evidence. Anyone who blocked the seizure, rescued the person, or hid the person after being told of the claim had to pay the enslaver $500.

The first two parties

In the 1790s the country's leaders split into the first two political parties. A political party is a group of people who work together to win elections and run the government their way.

Alexander Hamilton and John Adams led the Federalists. They wanted a strong national government and an economy built on trade, manufacturing and banking, and they leaned toward Britain. Thomas Jefferson and James Madison led the Democratic-Republicans. They wanted more of the government's power left with the states, opposed a national bank, supported farming and leaned toward France. Washington warned against parties in his Farewell Address of 1796.

The Sedition Act, 1798

In 1798 members of Congress passed four laws, and President John Adams signed them. Three of them dealt with people from other countries. The fourth, signed on July 14, 1798, was the Sedition Act. Sedition means words or acts that stir people up against the government. Under it, a person committed a crime by writing or publishing "any false, scandalous and malicious writing" against the government. The punishment was a fine of up to $2,000 and up to two years in prison. The law ran out on March 3, 1801.

The only newspaper editors put on trial under the Sedition Act worked for Democratic-Republican papers.

DISPUTED: The number found guilty. Historians have long counted 17 people charged and 10 found guilty, and a later study found 16 more people charged. The National Constitution Center counts more than two dozen found guilty, most of them publishers who opposed the government.

One of the people found guilty was Matthew Lyon, a Democratic-Republican congressman from Vermont who had been born in Ireland. In a letter, he wrote that he saw the public good "swallowed up in a continual grasp for power, in an unbounded thirst for ridiculous pomp, foolish adulation, or

selfish avarice." Pomp means showy display. Adulation means too much praise. Avarice means greed. He had also read aloud at campaign meetings a letter by the poet Joel Barlow that blamed Adams and the Senate for trouble with France.

A federal grand jury in Vermont charged Lyon on October 5, 1798. A deputy marshal, a federal law officer, arrested him the next day. On October 9 he stood trial at Rutland before Justice William Paterson of the Supreme Court. Lyon defended himself.

Paterson told the jury that its job had "nothing whatever to do with the constitutionality or unconstitutionality of the sedition law." Constitutionality means whether a law is allowed by the Constitution. The jury found Lyon guilty within an hour. Paterson sentenced him to four months in prison, a $1,000 fine and $60.96 in court costs.

Lyon was held in the jail at Vergennes. While he was there, the voters of his district elected him to Congress again. He was released on February 9, 1799.

Justice Samuel Chase sentenced another writer, James Callender, to nine months in prison and a $400 fine. After Jefferson became president, he pardoned Callender. To pardon means to officially forgive a crime. In 1804 the members of the House voted to impeach Chase, partly for how he ran Callender's trial. To impeach means to charge an official with wrongdoing in office, so that the Senate puts him on trial. The senators found him not guilty.

BY 1800

If you remember one thing: Between 1776 and 1791 the country's leaders wrote down how it would be governed, protected slavery in the plan, and left the vote to each state.

- 1787: Thirty-nine delegates signed the Constitution in Philadelphia.
- 1791: Ten amendments, the Bill of Rights, became part of it.
- 1798: Under the Sedition Act, a jury found a congressman guilty of writing against the president, and his voters re-elected him from jail.

Next: In 1803 the justices of the Supreme Court claimed a power the Constitution does not name: cancelling a law.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 11 MINUTES

The Supreme Court's justices struck down part of a law passed by Congress for the first time in 1803. From then on, American judges claimed the power to cancel laws that break the Constitution. In the same years, lawmakers in several states dropped the rule that a white man had to own property to vote. Lawmakers in some of those states took the vote away from Black men.

The first national party convention met in 1831. President Andrew Jackson handed out government jobs to his supporters. He also signed a law that let him make treaties to push Native nations out of the East and send them west. By the end of his presidency, his officials had removed about 50,000 Native people from their homelands in the East.

John Marshall

Who: The Chief Justice who wrote that judges may refuse to follow a law that breaks the Constitution.

When and where: Washington, D.C., 1801 to 1835. *Marbury v. Madison* decided in February 1803.

A judge's job, he wrote, is "to say what the law is."

John Marshall wrote the ruling in which the Supreme Court's justices first claimed the power to strike down a law passed by Congress. He was the Chief Justice, the head of the Supreme Court. President John Adams named Marshall for that post on January 20, 1801, and Marshall took his seat on February 4.

He held the seat for 34 years, until he died in Philadelphia on July 6, 1835. In that time he heard more than a thousand cases and wrote more than five hundred opinions. An opinion is a judge's written explanation of a ruling.

In 1801, in his last days in office, Adams appointed William Marbury a justice of the peace, a local judge, for Washington, D.C. The paper that gave Marbury the job, called a commission, was never delivered to him. The new Secretary of State, James Madison, refused to hand it over. The Secretary of State is the president's chief adviser on dealings with other countries. Marbury asked the Supreme Court for an order forcing Madison to deliver it.

In February 1803 Marshall and the other justices decided the case, called *Marbury v. Madison*. They said they could not give Marbury the order he asked for. A section of the Judiciary Act of 1789 said the Supreme Court could give such orders. The justices decided that the section went beyond what the Constitution allowed, so they refused to follow it. It was the first time the justices had declared a law passed by Congress and signed by the president unconstitutional, meaning that it broke the Constitution.

> Marshall wrote: "It is emphatically the province and duty of the judicial department to say what the law is." The judicial department means the courts. Province here means proper job. A law that goes against the Constitution is void, which means it has no force, he wrote. And when the two conflict in a case, judges must follow the Constitution. This power is called judicial review. The justices did not declare another act of Congress unconstitutional until the *Dred Scott* case of 1857.

The Louisiana Purchase, 1803

The deal. Two American officials, Robert Livingston and James Monroe, signed a treaty, a written agreement between countries, with France on April 30, 1803. Under it the United States bought 828,000 square miles of land west of the Mississippi River. The price was $15 million, about 4 cents an acre. President Jefferson had sent Monroe to buy only New Orleans and the land to its east called West Florida, and to spend no more than $10 million.

The question it raised. President Thomas Jefferson usually read the Constitution strictly, which means he held that the government could do only what the document said it could. He wondered whether the Constitution allowed the government to take on new land at all.

In a letter in 1803 to Wilson Cary Nicholas, Jefferson wrote: "Our peculiar security is in possession of a written Constitution. Let us not make it a blank paper by construction." Peculiar here meant special. By construction he meant by stretching the meaning of its words. He wrote that he would rather ask the nation for more power than take it. He went ahead with the purchase anyway.

French officials sold France's claim to the land. Thousands of Native people, in dozens of nations, lived on it. They were not part of the sale. Article VI of the treaty states that the United States would carry out the treaties Spain had made with "the tribes and nations of Indians" there.

Ending the import of enslaved people, 1808

Under the Constitution, the members of Congress could not ban the shipping of enslaved Africans into the country before 1808. On March 2, 1807, President Jefferson signed a law that banned it from the first day allowed. The ban took effect on January 1, 1808. From then on, bringing enslaved people across the ocean into the United States was against federal law.

Who gained the vote and who lost it

Between 1807 and 1838, the lawmakers of several states rewrote their rules on voting. They let white men without property vote. In the same laws and constitutions, they shut out Black men. New Jersey's lawmakers also ended voting by women.

New Jersey. New Jersey's lawmakers passed an election law in 1807 that let all white male taxpayers vote and shut out everyone else. Women and free Black people in New Jersey who met the property test had voted since 1776. The new law ended that. The law's words gave its aim as a "uniform practice" across the state, meaning the same rules everywhere.

New York. The delegates who rewrote New York's constitution in 1821 dropped the property test for white men. In the same document, they required a Black man to own property worth $250, free of debt, before he could vote. In 1825 only 298 of the nearly 6,000 free Black adults in New York met that test.

North Carolina and Pennsylvania. North Carolina's free Black men lost the vote in 1835. The delegates to the state's constitutional convention, a meeting of men chosen to rewrite the state constitution, took it away from them. Pennsylvania's convention delegates did the same in January 1838. They added the word "white" before "freemen," the men allowed to vote, in the section on who could vote.

Robert Purvis wrote a protest called the *Appeal of Forty Thousand Citizens, Threatened with Disfranchisement*. To disfranchise means to take away the vote. He read it aloud on March 14, 1838, at the First African Presbyterian Church in Philadelphia. Pennsylvania's voters approved the new constitution on October 9, 1838, by 113,971 votes to 112,759. Black men in Pennsylvania

could not vote again until the Fifteenth Amendment in 1870. That change to the Constitution barred the states from denying a man the vote because of his race.

The first national party convention, 1831

Men who opposed the Freemasons, a secret society, formed the Anti-Masonic Party. In September 1831 the party's members met in Baltimore to choose their candidate for president. It was the first national nominating convention held by any party. A nominating convention is a meeting where a party's members choose its candidates.

The Anti-Masons were also the first party to give voters a platform, a written list of what the party stood for. The delegates chose William Wirt. In the 1832 election he won only Vermont and its seven electoral votes.

Andrew Jackson

Who: The president who signed the law to remove Native nations from the East, used the veto to fight the national bank, and gave government jobs to his supporters.
When and where: Washington, D.C., 1829 to 1837.

He signed the removal law, and his officials forced about 50,000 people west.

Andrew Jackson was the seventh president, from 1829 to 1837. On May 28, 1830, he signed the Indian Removal Act. The act allowed the president to make treaties with Native nations living east of the Mississippi River. Under those treaties the nations had to give up their eastern land and move west. The senators passed it 28 to 19. The members of the House passed it by five votes, 102 to 97, on May 26, 1830, by the count of the House's own historians.

What the law did. By the end of Jackson's presidency, his officials had made nearly 70 removal treaties. Federal officials removed about 50,000 Native people from the East to Indian Territory, land west of the Mississippi River that the government set aside for them. The Choctaw,

starting in 1831, were the first of the southeastern nations to be forced out. The forced removal of the Muscogee, or Creek, began in 1836. Forcing the Seminole from their land took the government almost twenty years and $15 million.

White settlers got about 25 million acres of the nations' eastern land, which federal officials opened to them.

The Trail of Tears. In 1838, a year after Jackson left office, United States Army troops and state militia, part-time soldiers called up by a state, forced more than 16,000 Cherokee people out of their homelands. The soldiers held the Cherokee in camps and then sent them west. The journey is called the Trail of Tears. The National Archives puts the Cherokee dead at about 4,000. The National Park Service counts more than a thousand who died on the way and perhaps several thousand more who died from its effects.

The ruling Georgia ignored. Georgia's lawmakers had also passed a law about white people living on Cherokee land. It required any white person living in Cherokee territory after March 1, 1831, to get a license from Georgia's governor and to swear an oath to support Georgia's constitution and laws. Samuel Worcester, a Christian missionary, a man sent to spread his religion, lived on Cherokee land without that license. He was found guilty in a Georgia court and sentenced to four years of hard labor, forced heavy work, in the state prison.

In 1832 the justices of the Supreme Court decided *Worcester v. Georgia*. Chief Justice John Marshall wrote that Georgia's law was "void, and the judgment a nullity," meaning it had no force. Georgia's officials did not carry out the ruling. In April 1832 Jackson wrote to John Coffee that the Court's decision "has fell still born." Stillborn means born dead.

Many schoolbooks quote Jackson saying, "John Marshall has made his decision, now let him enforce it." No record from his own time shows that he said it. The sentence first appears in a history book by Horace Greeley printed in 1864, about twenty years after Jackson died.

The veto. On July 10, 1832, Jackson vetoed a bill to renew the charter of the Second Bank of the United States, a national bank. The charter was the paper that gave the Bank the right to operate. A veto is a president's refusal to sign a bill into law. His veto message called the Bank "dangerous to the liberties of the people."

He was the first president to make a veto the center of a major fight over the economy. He made the veto a part of his campaign in 1832 and won a second term.

Jackson then moved the government's money out of the Bank. On March 28, 1834, the senators voted to censure him for it. To censure is to scold someone officially. It is the only time the Senate has censured a president. In January 1837, after Jackson's party won a majority, the senators voted to expunge the censure, which means to strike it from the record.

Jackson also filled government jobs with his own supporters. He called this rotation in office. He argued that government work could be made "so plain and simple" that the jobs could pass around among men who deserved them. Senator William L. Marcy of New York gave the practice its nickname when he said "to the victor belong the spoils of the enemy." Spoils are the goods an army takes from the side it beats. The practice became known as the spoils system.

The fight over the school tax, 1834 and 1835

In 1834 Pennsylvania's lawmakers passed the Free Public School Act, a law for a statewide system of common schools paid for by taxes. A common school was a free public school open to the children of the town.

Many voters did not want to pay the tax. In 1835 petitions, written requests signed by thousands of them, reached the state's lawmakers, asking them to repeal, or cancel, the law. No one in the state Senate spoke against repeal, and few members of the House did.

Thaddeus Stevens, a member of the Pennsylvania House, spoke against repeal on April 11, 1835. He said: "Many complain of the school tax, not so much on account of its amount, as because it is for the benefit of others and not themselves." He pointed out that a man "cheerfully pays the tax which is necessary to support and punish convicts." The same man, Stevens said, "loudly complains of that which goes to prevent his fellow being from becoming a criminal."

After his speech, the lawmakers dropped the repeal, and the schools stayed. Stevens later called the speech the "crowning utility" of his life, the most useful thing he ever did.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 21 MINUTES

Black men in Louisiana registered to vote in 1897 numbered 130,344. In 1900 they numbered 5,320. This era tells how both numbers came to be.

Members of Congress passed three amendments to the Constitution between 1865 and 1870, and lawmakers in enough states approved them. Under them, slavery ended, people born in the country became citizens, and no state could deny a citizen the vote because of race. The justices later ruled that the citizenship rule did not cover Native people born into their own nations. In the 1870s and 1880s Black men in many Southern states voted and held office.

Starting in 1890, the delegates who rewrote Southern state constitutions added rules meant to stop Black men from voting. The justices of the Supreme Court ruled that those rules were allowed. They also ruled that laws requiring Black and white people to be kept apart were allowed. In the same years, lawmakers in Washington passed the first laws to shut out immigrants by nationality, and to fill some government jobs by exam. Lawmakers in more and more states required children to go to school.

Laws that made children go to school

In 1852 the lawmakers of Massachusetts passed the first state law that required children to go to school. Children from 8 to 14 had to attend. The school committee in each town or city had to look into every case of a child who did not. A family that broke the law could be fined up to $20. By 1900, thirty-two states had a law like it. Mississippi's lawmakers were the last to pass one, in 1918. These laws worked alongside laws against child labor.

Dred Scott v. Sandford, 1857

Dred Scott, an enslaved man, went to court to win his freedom. He argued that he had become free by living in a territory where slavery was banned. The justices of the Supreme Court decided his case, *Dred Scott v. Sandford*, on March 6, 1857, by a vote of 7 to 2.

Chief Justice Roger B. Taney wrote the ruling. He wrote that a Black person whose ancestors were brought to the country and sold as slaves was not a citizen under the Constitution. So Scott had no right to sue in a federal court. Taney also wrote that Black people "had no rights which the white man was bound to respect."

Taney's ruling also said that Congress had no power to ban slavery in the federal territories, the lands that were not yet states. With that, the justices struck down the Missouri Compromise, a law of 1820 that had banned slavery in the northern part of the Louisiana Purchase lands. That made *Dred Scott* only the second time the justices had struck down a law passed by Congress. The first was *Marbury v. Madison* in 1803.

Secession, 1860

The delegates of a South Carolina convention signed an ordinance of secession, a law declaring that their state was leaving the United States, in Charleston on December 20, 1860. Four days later the delegates approved a statement of their reasons. In it they wrote of "an increasing hostility on the part of the non-slaveholding States to the institution of slavery." They also wrote that the Northern states had joined to elect a man "whose opinions and purposes are hostile to slavery." The man they meant was Abraham Lincoln.

Abraham Lincoln

Who: The president who suspended a basic legal right during the Civil War, was told by the Chief Justice that he had no power to do it, and kept doing it.

When and where: Washington and Baltimore, April to July 1861.

Movie: *Lincoln* (2012), a film about the fight in January 1865 to get the House of Representatives to pass the Thirteenth Amendment. The film is about the amendment, not about habeas corpus.

Lincoln kept holding people without a judge, against the Chief Justice's ruling.

In the first months of the Civil War, President Abraham Lincoln suspended the writ of habeas corpus. A writ is a written order from a court. Habeas corpus is a person's right to be brought before a judge, who decides whether the government may go on holding that person. When it is suspended, the government can hold a person without taking them before a judge at all. The Constitution allows the writ to be suspended "in Cases of Rebellion or Invasion." It does not say who may suspend it.

Lincoln wrote to General Winfield Scott on April 25, 1861. Governor Hicks of Maryland had called the state's lawmakers to a special meeting, and they looked likely to vote to leave the United States. Lincoln told Scott to stop any effort to arm Maryland's people against the federal

government. He approved "the most prompt, and efficient means." Those means included "the bombardment of their cities" and, "in the extremest necessity," suspending the writ.

On April 27 he gave Scott the power to suspend it anywhere along the routes the army's troops used between Philadelphia and Washington. No federal official had suspended the writ since the War of 1812.

The arrest. At two in the morning on May 25, 1861, federal soldiers entered the country house of John Merryman, a planter, or owner of a large farm, in Baltimore County, and woke him from his bed. They took him to Fort McHenry. General George Cadwalader held him there. The charge was that Merryman was an officer in a company of men who supported secession and had federal weapons.

On May 26 Chief Justice Roger Taney, who also served as a judge in Maryland, ordered Cadwalader to bring Merryman to court. Cadwalader refused in a letter. He wrote that Lincoln had given officers the power to suspend the writ. A deputy marshal, an officer of the federal courts, carried Taney's next order to the gate of the fort, and a guard would not let him in.

On May 28, with a crowd gathered outside, Taney ruled in Baltimore that the president had no power to suspend the writ, under the Constitution or any law. He said Merryman must be freed at once. Taney issued no order to make that happen. He was 84 years old. He was also the judge who had written *Dred Scott*, so almost no one at the time saw him as neutral.

On July 4, 1861, Lincoln explained himself to Congress. He said the power to suspend the writ had "purposely been exercised but very sparingly." Then he asked, "are all the laws, but one, to go unexecuted, and the government itself go to pieces, lest that one be violated?" Unexecuted means not carried out, and lest means for fear that. He argued that the men who wrote the Constitution could not have meant that "the danger should run its course, until Congress could be called together."

Lincoln's officials went on arresting people and suspending the writ without a vote of Congress. On September 24, 1862, Lincoln suspended it across the whole country for anyone who resisted the draft, the law that made men serve in the army. In March 1863 the members of Congress passed a law that let the president suspend the writ for the rest of the war. With that law, members of Congress settled the dispute. No court settled it.

The amendment. Nearly four years after Merryman's arrest, Lincoln put his name on a change to the Constitution. On January 31, 1865, the members of the House passed that change, the Thirteenth Amendment, which ended slavery, by 119 votes to 56. Lincoln signed the resolution, the paper sending the amendment to the states, the next day, though a president does not have to sign an amendment. Enough states had ratified it, or approved it, by December 6, 1865.

Land for colleges, 1862

Senator Justin Morrill of Vermont put forward the bill that became the Morrill Act on July 2, 1862. Under it, each state received 30,000 acres of federal land for each of its senators and representatives in Congress. The land was to pay for colleges that taught "agriculture and the mechanic arts," meaning farming and the skills of building and machines. More than 10 million acres of the land given out under the act had been taken from Native nations by United States officials, through treaties, agreements and seizure.

Colleges paid for by these land gifts are called land-grant colleges. A second Morrill Act, in 1890, set a rule for each state. The state had to set up separate land-grant colleges for Black students, or show that its colleges did not turn students away because of race.

The Fourteenth and Fifteenth Amendments

Members of Congress approved the Fourteenth Amendment on June 13, 1866, and it became part of the Constitution on July 9, 1868. It states that "all persons born or naturalized in the United States" are citizens. Naturalized means made a citizen by a legal process. It also states that no state may "deny to any person within its jurisdiction the equal protection of the laws." Within its jurisdiction means inside the area where its laws apply. Equal protection means a state has to apply its laws the same way to everyone.

Who it left out. John Elk, a Native man who lived in Omaha, Nebraska, tried to register to vote in 1880. The registrar refused him because he was Native, and Elk sued. The justices ruled against him 7 to 2 in *Elk v. Wilkins* in 1884. They held that Native people born into their own nations were no more citizens by birth than the children of foreign diplomats, officials who represent other countries. Under a law of 1924, which Era 8 describes, American Indians became citizens.

Members of Congress passed the Fifteenth Amendment on February 26, 1869, and it was ratified on February 3, 1870. It states that the right of citizens to vote "shall not be denied or abridged by the United States or by any State on account of race, color, or previous condition of servitude." Abridged means cut back. Previous condition of servitude means having once been enslaved. The amendment names race, color and past slavery. It says nothing about sex, so states could still keep women from voting.

New constitutions in the South

Reconstruction was the rebuilding of the Southern states' governments after the Civil War. Members of Congress passed laws for it called the Reconstruction Acts. Under those acts, a convention met in Charleston on January 14, 1868, to write a new constitution for South Carolina. Black men in South Carolina had voted for the first time in the election of its delegates. Three out of every five delegates were Black.

Article X of the constitution they wrote required a system of free public schools, "free and open to all the children and youths of the State, without regard to race or color." Schools were to be open at least six months a year. Once there were enough schools, children would have to attend for at least twenty-four months. For decades, state officials did not carry out Article X.

The Tweed Ring and Tammany Hall

In the 1860s and 1870s, William M. Tweed held several offices in New York City and New York State at once. He was a state senator from 1867 to 1871 and chaired the state Senate's finance committee. From 1868 he was the Grand Sachem, or head, of Tammany Hall. Tammany Hall was a political machine of the Democratic Party. A political machine is a party organization that wins votes by handing out jobs and favors. Through Tammany Hall, Tweed had great power in the Democratic Party.

How the ring stole. Tweed and the men around him, called the Tweed Ring, stole money from the city in several ways. They sent the city bills for more than the work cost. Through printing and supply companies they controlled, Tweed's men billed the city too much for supplies. They ran up the cost of public buildings far past their budgets.

A new county courthouse was expected to cost half a million dollars. It ended up costing the city's taxpayers about $13 million. Modern estimates of what the ring stole in all run from $30 million to $200 million, by the count of the reference guide EBSCO Research Starters.

How it ended. The cartoonist Thomas Nast attacked Tweed in the magazine *Harper's Weekly* from 1869. In July 1871 editors at *The New York Times* printed evidence from inside the city's accounts, after people who knew about the thefts passed it to the paper.

In late 1873 a court found Tweed guilty and sentenced him to twelve years in prison and a fine of $12,750. He appealed, which means he asked a higher court to look at the case again, and the higher court cut his sentence to one year. He died of pneumonia in the Ludlow Street jail on April 12, 1878.

Minor v. Happersett: citizens without the vote

In October 1872 Virginia Minor, a citizen of Missouri, tried to register to vote in St. Louis. The registrar, the official who signed up voters, was Reese Happersett. He refused her because she was not a "male citizen." Minor sued. She argued that the Fourteenth Amendment made voting a right of every citizen.

The justices of the Supreme Court ruled against her, and none of them disagreed. They decided the case in their session that began in October 1874. The Chief Justice read the ruling. It states that "the Constitution of the United States does not confer the right of suffrage upon any one." Suffrage means the right to vote. Under the ruling, each state could keep the vote for men only.

The first laws against immigrants by nationality

The Page Act of 1875 was the first federal law to limit immigration. It barred Chinese laborers brought over under contract. In practice, officials used it to keep out most Chinese women too.

On May 6, 1882, President Chester A. Arthur signed the Chinese Exclusion Act. Under it, no Chinese laborer could come into the country for ten years. This was the first federal law to shut out a whole group of people because of their nationality. Members of Congress later renewed the act and widened it, and they did not repeal it until 1943.

Federal officials also took over the work of admitting immigrants. Officials of New York State had run the landing station at Castle Garden from 1855 to 1890. In 1892 the federal station at Ellis Island opened.

The election of 1876 and the end of Reconstruction

The presidential election of 1876 was between Rutherford B. Hayes, a Republican, and Samuel J. Tilden, a Democrat. The electoral votes of Florida, Louisiana, South Carolina and Oregon were in dispute. On January 18, 1877, Senator George Edmunds of Vermont proposed an Electoral Commission to settle the count. It had fifteen members: five representatives, five senators and five Supreme Court justices. Members of Congress passed the bill.

The commission's members voted 8 to 7, along party lines, to give every disputed electoral vote to Hayes. Hayes became president. On April 24, 1877, on his orders, federal troops left the Louisiana state house. It was the last state house in the South that federal troops had guarded.

The Equal Justice Initiative, in its calendar *A History of Racial Injustice*, marks that withdrawal as the end of Reconstruction. Its calendar says the withdrawal opened the way for white supremacist rule to come back in the South with nothing to hold it back. White supremacy is the belief that white people should rule over everyone else.

Joseph Hayne Rainey

Who: A man born to enslaved parents, forced to work on Confederate defenses, who became the first Black member of the US House and the first to preside over it.

When and where: Georgetown, South Carolina, and Washington, D.C. Born June 21, 1832. Sworn in December 12, 1870. Died August 2, 1887.

Rainey became the first Black member of the House in 1870.

Joseph Hayne Rainey was the first African American to serve in the House of Representatives. He was born in Georgetown, South Carolina, to enslaved parents. His father, Edward L. Rainey, worked as a barber and used his earnings to buy the family's freedom in the early 1840s.

The Confederacy was the group of Southern states that broke away from the United States and fought it in the Civil War. During the war, the Confederate army forced Rainey to work on the defenses of Charleston. He also worked as a steward, a ship's servant, on a Confederate ship that slipped through the blockade, the line of United States warships that tried to keep ships out of Southern ports.

In 1862 he and his wife escaped to Bermuda. There he ran a barbershop and she ran a dress shop. They came home in 1866.

Rainey won election to a full term in Congress on October 19, 1870, with 63 percent of the vote. On November 8 he won a special election for the rest of the current term, with 86 percent. He was sworn in on December 12, 1870.

The Klan law. On April 1, 1871, he spoke for the Ku Klux Klan Act. The Ku Klux Klan was a vigilante group, people who act outside the law, that attacked Black Americans and their white allies in the South. Klansmen whipped men, women and children, and they burned Black schools and churches. In July 1870 Klansmen killed six people, among them a schoolteacher, at Cross Plains, Alabama. Members of Congress gathered testimony that Klansmen in Alabama alone committed more than 100 murders between 1868 and 1871.

Under the Ku Klux Klan Act, it became a federal crime to deny anyone the rights the Constitution protects. President Ulysses S. Grant signed it on April 20, 1871. Rainey said the Constitution owed "protection to the humblest citizen, without regard to rank."

Rainey also supported the Civil Rights Act of 1875. He pressed for equal schooling, and he said "mixed schools," schools for Black and white children together, would "annihilate" prejudice. Annihilate means wipe out.

On April 29, 1874, he led a meeting of the House from the Speaker's chair, the seat of the House's leader. He was the first African American to do so.

On January 25, 1877, Rainey spoke against the Electoral Commission. He warned it could lead to "the election of a President by one man rather than by a majority of the people of the United States." Members of Congress passed the bill anyway.

Rainey lost his seat in 1878, by 62 percent to 38, and left the House on March 3, 1879, after more than eight years. No Black member of Congress served longer than he had until the 1950s. He was the first of fourteen Black representatives who served during Reconstruction. After Reconstruction ended, the number of Black members of Congress fell to zero and stayed near zero for decades.

The Pendleton Act, 1883

Since Andrew Jackson's time, presidents had handed out government jobs to their supporters. Then Charles Guiteau, a man who had wanted a government job and had not gotten one, killed President James A. Garfield. Chester A. Arthur became president after Garfield. Arthur had come up through New York machine politics.

On January 16, 1883, Arthur signed the Pendleton Act. Under it, some federal jobs went to people who passed a competitive exam, not to people who had worked for the winning party. At first the exam rules covered only about 10 percent of the government's 132,000 workers. The other 90 percent were not covered.

Taking away the vote in Mississippi and Louisiana

The Fifteenth Amendment barred states from denying the vote because of race. In 1890 the delegates who wrote Mississippi's new constitution set out to stop Black men from voting without naming race. Senator J. Z. George led the convention. Its committee on voting included its one Black delegate, Isaiah Montgomery. The committee members agreed that their task was to take away Black men's votes.

The rules. The delegates added a poll tax, a tax of $2 a year that a man had to pay before he could vote. They required two years of living in the state and one in the county. A voter had to swear he had not committed certain crimes, crimes the committee believed Black men were more likely to commit.

State officials printed the ballots, and only a man who could read could use them. A man who could not read could be read a section of the constitution. If he could explain it, he could vote. The registrar alone decided whether he had explained it well enough. The delegates put the new constitution into effect themselves, without a vote of the people.

What the justices said. In 1898 the justices of the Supreme Court heard *Williams v. Mississippi*. Williams was a Black man found guilty of murder. Jurors in Mississippi were chosen from the lists of voters. He argued that the

voting rules in the 1890 constitution kept Black men off those lists, and so off his jury.

On April 25, 1898, the justices ruled against him, and no justice disagreed. Justice Joseph McKenna wrote that the rules "do not on their face discriminate between the races, and it has not been shown that their actual administration was evil." He went on: "only that evil was possible under them." McKenna meant that the rules did not name any race, and no one had proved that officials used them unfairly, only that they could.

How long they lasted. Mississippi's poll tax lasted until 1964 in federal elections and until 1966 in state elections. Its reading test lasted until the Voting Rights Act of 1965. Era 9 describes how each one ended.

Louisiana. Louisiana's delegates wrote a new constitution in 1898 and also put it into effect without a vote of the people. Their rules for voters were a poll tax of $1 a year, with receipts for two past years, a reading test given by parish registrars, a test of owning $300 in property, and a grandfather clause. The registrars were the officials who signed up voters in each parish, Louisiana's name for a county. The reading test had no fixed standard.

Under the grandfather clause, a man who failed the tests could still register if his father or grandfather had voted in 1867, a year when formerly enslaved men could not vote in Louisiana. Thomas Semmes chaired the convention's committee on the courts. He said: "We met here to establish the supremacy of the white race, and the white race constitutes the Democratic Party of this State."

The count. By the count of the Louisiana encyclopedia *64 Parishes*, 130,344 Black voters were registered in 1897. In 1898 there were 12,902. In 1900 there were 5,320, and in 1922 there were 598.

Plessy v. Ferguson, 1896

A Louisiana law required railroads to give Black and white passengers separate cars, or cars split by a partition. A person who broke the law could be fined $25 or jailed for 20 days. On May 18, 1896, the justices of the Supreme Court ruled 7 to 1 in *Plessy v. Ferguson* that the law was allowed.

Justice Henry Billings Brown wrote the ruling. He called it a "fallacy," a mistake in reasoning, to believe that "the enforced separation of the two races stamps the colored race with a badge of inferiority." A badge of inferiority means a mark that one group is lower than another.

Justice John Marshall Harlan was the only justice who dissented, or disagreed. He wrote: "Our constitution is color-blind, and neither knows nor tolerates classes among citizens."

Born here, a citizen: Wong Kim Ark, 1898

Wong Kim Ark was born in San Francisco in 1873 to parents who had come from China. In August 1895 he came home from a trip to China on the ship *Coptic*. Officials refused to let him land, under the Chinese exclusion laws. On October 2, 1895, a writ of habeas corpus was filed on his behalf.

The justices of the Supreme Court ruled 6 to 2 on March 28, 1898, that he was a citizen under the Fourteenth Amendment, because he was born in the United States, whatever his parents' nationality.

Cumming v. Richmond County, 1899

In June 1897 the members of the school board of Richmond County, Georgia, closed Ware High School, the county's high school for Black students. The board members said they could not afford a high school for 60 Black students while 300 Black children had no primary schooling. They kept paying for the high schools for white students.

J. W. Cumming, James S. Harper and John C. Ladeveze, Black taxpayers, sued. They asked the court to stop the board from spending public money on the white high schools until there was an equal school for their children. They also asked it to stop taxing them for schools their children could not attend.

The justices of the Supreme Court turned them down on December 18, 1899. The justices found no violation of the Fourteenth Amendment and treated the closing as a question of money, not of race.

Justice John Marshall Harlan wrote the ruling. Three years earlier he had been the one dissenter in *Plessy*. Here he wrote that "if it appeared that the board's refusal to maintain such a school was in fact an abuse of its discretion and in hostility to the colored population because of their race, different questions might have arisen." Discretion means the power to decide.

BY 1900

If you remember one thing: The three amendments of 1865 to 1870 made Black men citizens and voters, and by 1900 the delegates of the Southern states and the justices of the Supreme Court had taken most of that vote back.

In 1870: The Fifteenth Amendment barred every state from denying a man the vote because of race, and Joseph Rainey of South Carolina sat in the House.

In 1900: Louisiana had 5,320 Black voters on its rolls, down from 130,344 three years earlier, and the justices had allowed Mississippi's voting rules and Louisiana's separate railroad cars.

Next: Between 1913 and 1920, four amendments changed who paid, who chose the senators, who could drink and who could vote.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 23 MINUTES

Four amendments, changes to the Constitution, went in between 1913 and 1920. Under them, Congress could tax incomes, voters chose their senators, alcohol was banned, and no state could deny women the vote. A fifth amendment ended the ban on alcohol in 1933.

President Franklin Roosevelt was elected four times. The justices of the Supreme Court struck down parts of his New Deal, the set of laws he backed in the 1930s, and then he tried to add more justices to the Court. The justices ruled that a Virginia law was allowed, and under it doctors sterilized about 8,300 people, operating on them so that they could never have children. The justices also held that the Army's orders forcing people of Japanese descent from their homes during World War II were justified. About 122,000 people were forced into camps under those orders.

In big cities, the leaders of party organizations called political machines still traded jobs and favors for votes. One of them explained how, and this era opens with him.

George Washington Plunkitt

Who: A New York machine politician who explained to a reporter, from a shoeshine stand at the New York County Court House, how he made his money and kept his votes.

When and where: The Fifteenth Assembly District, New York City. His talks were printed in 1905. He died in 1924.

"I seen my opportunities and I took 'em."

George Washington Plunkitt was a district leader for Tammany Hall, the Democratic Party's political machine in New York City. A district leader was the party's boss in one part of the city. Plunkitt led Tammany's work in the Fifteenth Assembly District. He also served as a state senator and as a member of the state assembly, one of the two groups of lawmakers in New York State. He grew up poor. By the time he died in 1924, he was rich.

A newspaperman named William L. Riordon wrote down Plunkitt's talks and published them in 1905 as *Plunkitt of Tammany Hall*. The title page states that Plunkitt gave the talks "from His Rostrum." A rostrum is a speaker's stand, and his was the shoeshine stand at the New York County Court House. The book is not a word-for-word record. Riordon gathered and shaped what Plunkitt said.

Plunkitt split graft into two kinds. Graft is money a person gets by using a public office. Dishonest graft, he said, meant blackmailing gamblers and squeezing money out of illegal businesses. To blackmail people is to make them pay money by threatening to tell their secrets.

Honest graft meant learning where city officials planned to build something, buying the land first and then selling it to the city at a profit. "I ain't no looter," he said. "The looter hogs it. I never hogged." His

> summary of himself is cut into his gravestone: "I seen my opportunities and I took 'em."
>
> He also described the everyday work of a machine. When a building burned at night, Plunkitt and his captains, the party workers under him, came to the fire and found the families shelter and clothes on the spot. He knew employers across the city and got people jobs. His men found families in trouble before the charity groups did. He said he did it because those people remembered him on election day.

The food laws of 1906

President Theodore Roosevelt signed two laws on the same day, June 30, 1906. The Pure Food and Drug Act banned selling food, drink or drugs across state lines if they were mislabeled or made impure. Harvey Wiley and the officials of his Bureau of Chemistry enforced it. In 1930 that office was renamed the Food and Drug Administration. The Meat Inspection Act required federal inspectors in meatpacking plants, set rules for cleanliness and banned false labels. The laws followed Upton Sinclair's book *The Jungle*, published in February 1906, and years of work by Wiley's team.

Two amendments in 1913

The income tax. Members of Congress put a 2 percent tax on incomes over $4,000 into a tariff bill in 1894. A tariff is a tax on goods brought in from other countries. The Supreme Court's justices struck the income tax down, 5 to 4.

Members of Congress passed the Sixteenth Amendment on July 2, 1909. According to the National Archives, the conservatives who proposed it expected it never to be ratified. An amendment is ratified, or approved, when lawmakers or special meetings in three-quarters of the states have voted for it. The Sixteenth Amendment was ratified on February 3, 1913. Under it, Congress may tax incomes "without apportionment among the several States." That meant the money did not have to be divided among the states by population.

Senators chosen by voters. The Seventeenth Amendment was ratified on April 8, 1913, when Connecticut's lawmakers voted for it and completed the three-quarters needed. Before it, the lawmakers of each state chose that state's senators. After it, the voters chose them. Augustus Bacon of Georgia was the first senator elected by the voters under the new amendment, on July 15, 1913.

The grandfather clause struck down, 1915

Under a 1910 amendment to Oklahoma's constitution, Black men had to pass a reading test to vote, and white men were excused from it. The amendment excused anyone who could vote "on January 1st, 1866, or at any time prior thereto." It also excused any "lineal descendant" of such a person, meaning a child or grandchild. Almost no Black man could vote before 1866, so nearly all white men were excused and Black men had to take the test. A rule like this is called a grandfather clause, because it excused a man whose grandfather could vote.

Federal prosecutors, the government lawyers who bring criminal charges for the United States, charged two Oklahoma election officers, Frank Guinn and J. J. Beal, with keeping Black citizens from voting under this rule. The justices of the Supreme Court ruled against the 1866 clause in *Guinn v. United States* on June 21, 1915.

They held that it was void, meaning it had no force, because it broke the Fifteenth Amendment. That amendment bars the United States and every state from denying a citizen the vote "on account of race, color, or previous condition of servitude." Chief Justice Edward White wrote the opinion.

Prohibition and repeal

Members of Congress passed the Eighteenth Amendment on December 18, 1917, and lawmakers in the states ratified it on January 16, 1919. It banned "the manufacture, sale, or transportation of intoxicating liquors," meaning alcoholic drinks. The ban began one year after ratification, in January 1920.

Fourteen years later, members of Congress voted to end the ban. They passed the Twenty-first Amendment on February 20, 1933. Its first section reads: "The eighteenth article of amendment to the Constitution of the United States is hereby repealed." To repeal a law is to cancel it. Members of Congress sent this amendment to special conventions, meetings held in each state to decide on it, and not to the state lawmakers. It was ratified on December 5, 1933. No other amendment repeals another, and no other was sent to state conventions.

The Nineteenth Amendment, 1920

Members of the House passed the Nineteenth Amendment on May 21, 1919, and senators passed it on June 4, 1919. On August 18, 1920, Tennessee's lawmakers ratified it. Their vote made Tennessee the thirty-sixth state to ratify, which was enough. Secretary of State Bainbridge Colby certified it, or officially declared it part of the Constitution, on August 26, 1920.

Its words are: "The right of citizens of the United States to vote shall not be denied or abridged by the United States or by any State on account of sex." Abridged means cut down. After it, no state could keep the vote for men only, as the justices had allowed in *Minor v. Happersett*.

BY 1920: Four amendments in seven years had given Congress the income tax, given voters the choice of their senators, banned alcohol, and barred every state from keeping the vote from women. The rules that kept Black men from voting in the South still stood, except the grandfather clause.

Quotas, and citizenship for Native people, 1921 and 1924

In 1921 members of Congress passed the Emergency Quota Act, the first law with a set limit on how many immigrants could come. A quota is a fixed share. Under the act, the yearly limit for each nationality was 3 percent of the people born in that country who were living in the United States in 1910.

In 1924 members of Congress passed the Immigration Act, which set a lower limit. It was 2 percent of the people born in each country who were living in the United States in 1890, as the census of that year counted them. A census is a count of all the people in the country. Under the 1924 act, fewer immigrants could come from Southern and Eastern Europe than before, and nearly all immigrants from Asia were barred. The total was capped at about 165,000 a year.

Citizens, but not voters. President Calvin Coolidge signed the Indian Citizenship Act on June 2, 1924. Under it, American Indians became citizens. Since 1884, under the justices' ruling in *Elk v. Wilkins*, Native people born into their own nations had not been citizens by birth. Even as citizens, they were not on the lists of voters in every state, because each state's lawmakers wrote their own rules about who could vote.

An Arizona law barred people "under guardianship" from voting. In 1928 the judges of the Arizona Supreme Court held that Native people on reservations were wards of the federal government, people under its guardianship. So, the judges held, the law kept them from voting. New Mexico's constitution shut out "Indians not taxed."

In 1948 judges in both states ended these bans. On July 15 the judges of the Arizona Supreme Court overturned their own 1928 ruling in *Harrison v. Laveen*. Justice Levi Udall wrote that ruling. In August a federal court in New Mexico ruled in *Trujillo v. Garley* that the "Indians not taxed" rule broke the Constitution. In Utah, Native people could not vote until 1957, later than in any other state.

The anti-lynching bill, 1922

Lynching is murder by a mob, carried out without any trial. Members of Congress introduced bills to make lynching a federal crime, one the national government could charge. George Henry White, then the only Black member of Congress, introduced one on January 20, 1900. Leonidas Dyer, a Republican representative from Missouri, introduced another on April 8, 1918. Members of the House passed Dyer's bill on January 26, 1922. Senators opposed to it blocked it in the Senate, and it never became law. No federal anti-lynching law passed in this era.

Teapot Dome

In the Teapot Dome case, a member of the president's cabinet leased Navy oil land to two oil men who gave him money. The cabinet is the group of officials who head the big government departments. Albert B. Fall was Secretary of the Interior under President Warren G. Harding. Fall persuaded Harding to move the Navy's oil reserves, land kept for the Navy's future oil, to the Interior Department, and the Secretary of the Navy carried out the transfer.

The leases and the money. Fall then leased, or rented out, the Teapot Dome reserve in Wyoming to Harry F. Sinclair's Mammoth Oil company. He leased the Elk Hills reserve in California to Edward Doheny's Pan-American Petroleum. Doheny gave Fall a $100,000 "loan" with no interest. Sinclair gave him about $300,000 in cash and government bonds, papers the government promises to pay back with interest.

The investigation. In October 1923 the senators on the Public Lands Committee opened an investigation. Senator Thomas J. Walsh of Montana led it for about two years, and he found Doheny's $100,000. In 1927 the justices of the Supreme Court ruled that the oil men had gotten the leases corruptly, meaning by dishonest deals.

On October 25, 1929, a jury found Fall guilty of taking a bribe from Doheny. A bribe is money given to an official in return for a favor. A judge sentenced Fall to one year in prison and a $100,000 fine. He was the first person found guilty of a crime he committed while serving in a president's cabinet.

Doheny was found not guilty of paying the same bribe. Sinclair went to jail for several months, for contempt of Congress, the crime of refusing to obey Congress or one of its committees, and for having the jury in his case followed.

School law: who decides, and who pays

Who decides where a child goes to school. Oregon's Compulsory Education Act of 1922 required parents to send children aged 8 to 16 to the public school in their district. Compulsory means required. The Society of the Sisters of the Holy Names of Jesus and Mary, which ran Catholic schools, sued Oregon's governor, Pierce, over the law. A second case was brought for Hill Military Academy.

The justices of the Supreme Court struck the law down in *Pierce v. Society of Sisters* on June 1, 1925. Justice James McReynolds wrote that it interfered with the liberty of parents to direct their children's education. He wrote: "The child is not the mere creature of the state." He meant that a child does not belong to the government, and that the people who raise a child have the right to guide the child's future.

Under that ruling, state lawmakers may require a child to be educated, but they may not require every child to attend a government school. Private schools and church schools are legal because of it. Homeschooling depends on the same ruling, but it became legal only later, state by state, in the 1980s and early 1990s.

Who decides what a teacher may teach. Tennessee's governor, Austin Peay, signed the Butler Act on March 21, 1925. It made it unlawful to teach in the state's public schools "any theory that denies the story of Divine Creation of man as taught in the Bible, and to teach instead that man has descended from a lower order of animals." That meant a teacher could not teach evolution, the scientific idea that living things, people included, developed over a very long time from earlier forms.

Civic leaders in Dayton, Tennessee, heard that the American Civil Liberties Union would pay for a court challenge to the law. They set up a test case, a trial arranged on purpose so that a court would rule on the law. They asked John T.

Scopes, a football coach and science teacher, whether biology could be taught without evolution. He said it could not, because the state's own approved textbook included it.

His trial opened in Dayton on July 10, 1925, before Judge John T. Raulston. On July 21 the jury found him guilty, and he was fined $100, the smallest fine the law allowed.

In January 1927 the judges of the Tennessee Supreme Court threw out his conviction. The state constitution required a jury, not a judge, to set any fine over $50, and Raulston had set the fine himself. So the judges never ruled on the Butler Act itself. It stayed in force until 1967.

Who pays. Between 1920 and 1950, more and more of the money for public schools came from the states, and less from local school districts. In the school year 1919-20, local money made up 83.2 percent of the money for public schools. By 1949-50 the local share had fallen to 57.3 percent. Nearly all the rest came from the state, and less than 3 percent from the federal government. Small districts were merged, and their number fell by almost a third between 1937-38 and 1949-50.

Ending segregated schools. Governor Earl Warren of California signed the Anderson Bill in June 1947. With it, California's lawmakers and governor became the first in the country to repeal their own state's law on segregated schools. Segregated schools kept children of different races apart.

Buck v. Bell, 1927

Virginia's lawmakers passed a law in 1924 that allowed state officials to sterilize people they judged "feeble-minded" and likely to have "socially inadequate offspring." Sterilization is an operation that makes a person unable ever to have children. "Feeble-minded" was a label officials put on people they judged to be slow to learn. Offspring means children. Aubrey Strode wrote the law and then defended it in court.

Carrie Buck was born on July 2, 1906, in Charlottesville. She was raised by foster parents, the Dobbs family. In 1923 she became pregnant. By her own account, as Encyclopedia Virginia reports it, Clarence Garland, the foster parents' nephew, raped her. Rape means forcing a person into sex. Her foster parents believed the pregnancy showed promiscuity, meaning sex with many partners, and so showed that she was "feebleminded." On January 23, 1924, they asked a court to have her committed to a state institution, meaning sent there to live by a court's order.

The ruling. Her case went to a trial court under Judge Bennett T. Gordon on November 18, 1924, then to Virginia's Supreme Court of Appeals on November 12, 1925, and then to the Supreme Court. On May 2, 1927, the justices ruled 8 to 1 that Virginia's law was allowed. Justice Oliver Wendell Holmes wrote the opinion. It includes the sentence "Three generations of imbeciles are enough." Imbecile was another insulting label for a person judged slow to learn. Justice Pierce Butler dissented, meaning he disagreed with the ruling, without writing an opinion.

The operation. She had been sent to the Virginia State Colony for Epileptics and Feebleminded, a state home for people who had seizures or were judged slow to learn. Dr. Albert Priddy ran it. He picked her for the operation, and on September 10, 1924, the colony's board approved it. Carrie Buck did not agree to it. Near the end of her life she said she "kicked against it," and she told the historian Paul Lombardo, "They done me wrong. They done us all wrong."

On October 19, 1927, Dr. John H. Bell performed the operation on Carrie Buck. It is called a salpingectomy. He cut her fallopian tubes, the tubes that carry eggs to the womb, so that she could never become pregnant again.

The 8,300. Between 1927 and 1972, doctors in Virginia sterilized about 8,300 people under the law. Encyclopedia Virginia names the two main operations: a vasectomy for men, which cuts the tubes that carry sperm, and a salpingectomy for women. It does not say which operation was done on each person. Virginia's lawmakers repealed the law in 1974. On May 2, 2002, Governor Mark Warner apologized for the state's program. No later group of justices has overruled *Buck v. Bell*, or cancelled it with a new ruling.

BY 1927: The justices had struck down the grandfather clause and allowed the sterilization law. Tennessee's teachers still could not teach evolution, and most of the money for schools was still local.

Franklin D. Roosevelt

Who: The president who lost key cases in the Supreme Court, tried to add justices to it, and was elected four times.

When and where: Washington, D.C., 1933 to 1945. His plan for the courts went to Congress on February 5, 1937.

After losing in the Supreme Court, Roosevelt tried to add up to six justices.

Franklin D. Roosevelt tried to change the size of the Supreme Court after its justices struck down one of his laws. On May 27, 1935, in *Schechter Poultry Corp. v. United States*, the justices struck down part of the National Industrial Recovery Act. That part let the president approve "codes of fair competition," which were rules for whole industries.

The justices held that members of Congress had given the president powers that Congress had no right to give away. They also held that the poultry business in the case was not trade between the states, so Congress could not make rules for it.

On February 5, 1937, Roosevelt sent Congress a plan to reorganize the federal courts. Under the plan, the president could name one new justice to the Supreme Court for each justice over the age of 70, up to six more. The Court could have grown from nine justices to fifteen.

On March 29, 1937, in *West Coast Hotel Co. v. Parrish*, the justices ruled that Washington State's minimum wage for women and minors was allowed. A minimum wage is the lowest pay the law allows, and minors are young people who are not yet adults. Soon after, they ruled that other New Deal laws were allowed too.

Did the plan change the justices' minds? Many people at the time believed it had. As the historian William Leuchtenburg describes it, they believed the justices, and Justice Owen Roberts above all, had "crossed over." The legal historians G. Edward White and Barry Cushman argue against that. Court papers show that Roberts had already voted for the Washington law at the justices' private meeting on December 19, 1936, weeks before Roosevelt sent his plan to Congress.

On June 14, 1937, the members of the Senate Judiciary Committee told the Senate to reject the plan. They called it "needless, futile, and utterly dangerous." On July 22, 1937, the senators rejected it, 70 to 20. According to the Federal Judicial Center, Roosevelt lost a great deal of political support for proposing it.

Four terms. The senators turned down his court plan, but the voters elected him twice more. George Washington had served two terms, and presidents after him had kept that custom. Roosevelt was elected four times, in 1932, 1936, 1940 and 1944.

An order on jobs. In his third term, Roosevelt barred discrimination in defense work and government jobs with an order of his own. He signed Executive Order 8802 on June 25, 1941. An executive order is a written order from the president to the officials under him.

The order states: "There shall be no discrimination in the employment of workers in defense industries and in Government, because of race, creed, color, or national origin." Discrimination means treating people worse because of who they are. A creed is a religion. With the order, Roosevelt created the Committee on Fair Employment Practice. After the war, members of Congress refused to make that committee permanent.

The camps. Eight months after the jobs order, Roosevelt signed a second one, Executive Order 9066, on February 19, 1942. It let the Secretary of War and military commanders mark off military areas and shut "any or all persons" out of them. Lieutenant General John L. DeWitt, commander of the Army's Western Defense Command, used it to force people of Japanese descent from their homes. He issued Public Proclamation No. 4, an official public order, on March 29, 1942.

Over six months, under his orders, about 122,000 men, women and children were forced into assembly centers, which were short-term holding places. From there they were forced into ten camps in remote parts of six western states and Arkansas. Nearly 70,000 of them were American citizens, by the count of the National Archives. None was charged with a crime, and none had any way to appeal.

The courts and the camps

A federal court in San Francisco found Fred Korematsu guilty of breaking the military's exclusion orders, the orders that shut people of Japanese descent out of the military areas. It gave him five years of probation, a time when a person found guilty lives under a court's watch instead of in prison. On December 18, 1944, in *Korematsu v. United States*, the justices of the Supreme Court let his conviction stand, 6 to 3. They held that forcing citizens out of their homes in wartime was justified.

On the same day, in *Ex parte Endo*, all the justices agreed that federal officials could not hold a citizen whom they admitted was loyal. According to the National Archives, that ruling "paved the way for the end of mass confinement."

In the 1980s federal judges threw out the convictions of Korematsu and others. Judge Marilyn Hall Patel vacated, or cancelled, Korematsu's conviction on November 10, 1983. In the Civil Liberties Act of 1988, members of Congress apologized for the imprisonment and voted to pay $20,000 to each person who had been held.

Poll taxes and the white primary

The poll tax. Under Georgia law, a voter owed a poll tax of $1 a year, and officials let no one vote who had not paid it. People under 21 or over 60, blind people, and women who did not register to vote did not have to pay. A poll tax kept anyone who could not pay it from voting. In 1890 the delegates who wrote Mississippi's constitution had added one as part of their plan to stop Black men from voting.

Nolan Breedlove sued Georgia's tax collector, T. Earl Suttles. All the justices of the Supreme Court ruled that the tax was allowed, in *Breedlove v. Suttles* on December 6, 1937. Justice Pierce Butler wrote that "the equal protection clause does not require absolute equality." The equal protection clause is the part of the Fourteenth Amendment that requires states to treat people equally under the law.

The white primary. In Texas, the Democratic Party was the only party that won elections, so its primary was the election that counted. A primary is the election in which a party chooses its candidates. On May 24, 1932, the leaders of the Texas Democratic Party voted that only "white citizens" could be members. The election judges of precinct 48 in Harris County refused Lonnie Smith, a Black voter, a ballot in the primary. A precinct is one small voting area. Smith sued one of those judges, a man named Allwright, and the others.

In *Smith v. Allwright*, decided on April 3, 1944, the justices held that the party's leaders were acting for the state when they ran the primary. Keeping Black voters out of it therefore broke the Fifteenth Amendment. Justice Stanley Reed wrote the opinion. Justice Owen Roberts dissented.

1948: the armed forces, refugees and conventions on television

On July 26, 1948, President Harry Truman signed Executive Order 9981. It states: "There shall be equality of treatment and opportunity for all persons in the armed forces without regard to race, color, religion or national origin." By about 1953 Black and white troops served together in nearly all the armed services.

The Displaced Persons Act of 1948 was the first American law for refugees, people forced from their home countries. It first allowed 200,000 Europeans who had lost their homes in the war to come to the United States. Members of Congress raised the number later. Between 1948 and 1952 about 350,000 to 400,000 people came in under it.

Both major parties held their national conventions in Philadelphia in 1948, and television networks broadcast them. A national convention is a large meeting of a party. Delegates, the people each state's party sends, choose the party's candidate for president there. They also write its platform, the list of positions it stands for. The National Constitution Center puts the television audience at as many as 10 million viewers on the East Coast.

On July 12, 1948, after a speech by Hubert Humphrey, the Democratic delegates added a civil rights plank to their platform. A plank is one position in a platform. Civil rights are the rights every citizen has to equal treatment under the law. Two Southern delegations walked out on live television.

On July 17, Southern delegates met in Birmingham, Alabama, and formed the States' Rights Democratic Party, called the Dixiecrats. They chose Strom Thurmond, the governor of South Carolina, as their candidate. Their platform stated, "We stand for the segregation of the races." Segregation means keeping people of different races apart. Thurmond won four states and 39 electoral votes.

BY 1950

If you remember one thing: In this era the justices allowed a state to sterilize people who had not agreed to it and allowed the Army to force 122,000 people into camps, and later officials apologized for both.

About 122,000 men, women and children of Japanese descent were forced into camps in 1942, nearly 70,000 of them American citizens, more than half. None was charged with a crime. In 1988 members of Congress apologized and voted to pay each person who had been held $20,000. Virginia's governor apologized for the sterilizations in 2002. No later justices have overruled *Buck v. Bell*.

Next: In 1954 the same Court ruled that segregated schools broke the Constitution, and the fight moved to how fast anything had to change.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 25 MINUTES

A senator with a brain tumor, unable to speak, pointed to his eye to vote "aye" on June 10, 1964, and the Senate cut off debate on a civil rights bill for the first time. That law banned segregation in public places. Under a second law, in 1965, officials in some states and counties had to ask federal officials before changing any voting rule. Between 1954 and 1977 the justices of the Supreme Court decided cases on schools, on how votes and voting districts were counted, and on how police question people they arrest.

Five amendments were added to the Constitution between 1951 and 1971. In 1974 President Richard Nixon resigned before members of the House could vote to impeach him. To impeach a president is to charge him formally with wrongdoing, which leads to a trial in the Senate. In 1998 the members of the House impeached President Bill Clinton, and the senators found him not guilty.

Five amendments, 1951 to 1971

Each of the five amendments added between 1951 and 1971 governs who holds office or who votes.

Since the Twenty-second Amendment was ratified on February 27, 1951, "No person shall be elected to the office of the President more than twice." Before it, Washington's two terms had been a custom, and Franklin Roosevelt had broken it. Members of Congress proposed the amendment in 1947, and Minnesota's lawmakers made it the thirty-sixth state to approve it.

The District of Columbia, the capital city, got electors in presidential elections through the Twenty-third Amendment, ratified on March 29, 1961. The District got no senators and no voting member of the House.

The Twenty-fourth Amendment, ratified on January 23, 1964, bans poll taxes only in elections for president, vice president, senator and representative. A state could still charge a poll tax in its own elections.

Then, on March 24, 1966, in *Harper v. Virginia State Board of Elections*, the justices of the Supreme Court struck down state poll taxes too. Virginia's constitution had set a poll tax of up to $1.50 a year. Justice William O. Douglas wrote that making "the affluence of the voter or payment of any fee an electoral standard" breaks the Equal Protection Clause. Affluence means wealth.

The Twenty-fifth Amendment was ratified on February 10, 1967. It states what happens when a president dies, resigns or is removed. It also sets out a way to fill an empty vice presidency: the president names someone, and the members of both houses of Congress vote to confirm, or approve, the choice.

Eighteen-year-olds got the vote under the Twenty-sixth Amendment. It was ratified on July 1, 1971, faster than any other amendment.

Brown v. Board of Education and all deliberate speed

On May 17, 1954, in *Brown v. Board of Education*, the justices of the Supreme Court held that segregated public schools broke the Constitution. The ruling did not state how or when to end segregation.

A year later, on May 31, 1955, they decided a second case, often called *Brown* II. Chief Justice Earl Warren wrote for a unanimous Court, meaning every justice agreed. The justices sent the cases back to federal district judges, the judges of the lowest federal courts. They told school boards to make "a prompt and reasonable start toward full compliance." Compliance means obeying the ruling.

They ordered the judges to admit the children who had sued to public schools "on a racially nondiscriminatory basis with all deliberate speed." A racially nondiscriminatory basis meant that no child could be turned away because of race. Deliberate means careful and unhurried, so under those words school boards could move slowly. The order set no date. School board members who moved slowly were not clearly breaking it.

Thurgood Marshall

Who: The lawyer who won 29 of the 32 cases he argued in the Supreme Court and later became its first Black justice.

When and where: Baltimore, New York and Washington, D.C. Born 1908. Died 1993.

The law school that turned him away had to admit Black students three years later.

Thurgood Marshall wanted to study law at the University of Maryland, and officials at the law school turned him away because he was Black. He went to Howard University's law school instead and graduated first in his class in 1933. One of his first cases, *Murray v. Pearson*, ended with Maryland's judges ordering that same law school to admit Black students. The Maryland Court of Appeals ruled in the case on January 15, 1936.

He went on to hold four jobs in the American legal system. From 1940 to 1961 he ran the Legal Defense and Educational Fund of the NAACP, a law office whose lawyers took civil rights cases to court. The NAACP is the National Association for the Advancement of Colored People. Colored was a word used then for Black people.

After that he was a federal appeals judge, then Solicitor General, the officer who argues the government's own cases in the Supreme Court, and then a justice of the Supreme Court. He was the first Black justice.

Before he became a justice, he argued 32 cases in the Supreme Court and won 29.

DISPUTED: When he argued them. The NAACP's own history puts all 32 in his years at the Fund. The *New York Times* put 18 of them in his years as Solicitor General, and the National Park Service counts 19 cases in that job, 14 of them won.

His NAACP cases included *Smith v. Allwright* in 1944, *Morgan v. Virginia* in 1946 on segregated buses between states, *Sweatt v. Painter* in 1950 on the University of Texas law school, and *Brown v. Board of Education* in 1954.

On October 5, 1961, President John F. Kennedy made Marshall a judge of the United States Court of Appeals for the Second Circuit, a court that hears appeals from trial courts. He did it by recess appointment, which means a president fills the seat while the Senate is away. A recess appointment ends unless the senators confirm it. Southern senators opposed Marshall, and the senators did not confirm him until September 11, 1962, eleven months later.

He left that court on August 23, 1965, to become Solicitor General of the United States. President Lyndon Johnson nominated him, or named him, to the Supreme Court on June 13, 1967. The senators confirmed him on August 30, 1967, and he took the oath as a justice on October 2. He stopped full-time work on the Court on October 1, 1991, and died on January 24, 1993.

After his death, Chief Justice William Rehnquist said most justices are remembered for their work on the Court. "Justice Marshall, however, is unique because his contributions to constitutional law before becoming a member of the Court were so significant."

Prayer and the Bible in public schools, 1962 and 1963

The members of the New York State Board of Regents, the officials who oversaw the state's schools, wrote a prayer for public school children to say each morning. It read: "Almighty God, we acknowledge our dependence upon Thee, and we beg Thy blessings upon us, our parents, our teachers and our Country." No child had to say it.

On June 25, 1962, in *Engel v. Vitale*, the justices of the Supreme Court held the prayer unconstitutional anyway, 6 to 1. Unconstitutional means against the Constitution. The justices ruled against the prayer because government officials had written it.

The prayer was short and a child could stay silent, and the justices held that neither fact changed the ruling. Justice Hugo Black wrote that the First Amendment's ban on establishing a religion, meaning setting up an official religion, "must at least mean that in this country it is no part of the business of government to compose official prayers for any group."

The justices decided two more cases together in *Abington School District v. Schempp* on June 17, 1963. In Pennsylvania's public schools, the school day opened with Bible readings. In the Baltimore city schools, children said the Lord's Prayer together. The justices struck down both. Justice Tom Clark wrote the opinion, and Justice Potter Stewart was the only one to dissent.

One person, one vote

Under Georgia's county unit system for statewide elections, the candidate who won a county got all of that county's unit votes. So a vote in a small rural county counted many times as much as a vote in a city. The justices struck the system down, 8 to 1, in *Gray v. Sanders* on March 18, 1963. Justice William O. Douglas wrote: "The conception of political equality from the Declaration of Independence, to Lincoln's Gettysburg Address, to the Fifteenth, Seventeenth, and Nineteenth Amendments can mean only one thing." He ended the sentence with the words "one person, one vote."

On June 15, 1964, in *Reynolds v. Sims*, the justices applied the same rule to the lawmaking bodies of the states. Alabama's lawmakers had not redrawn the lines of the state's legislative districts, the areas that elect the state's lawmakers, since 1901. They still used the 1900 census. By 1960, 25.1 percent of Alabamians could elect a majority of the state senate. In the worst case, one senate district had about 41 times as many people as another.

Chief Justice Earl Warren wrote the opinion. Under it, districts had to hold about equal numbers of people, and lawmakers in almost every state had to redraw their districts.

Lyndon B. Johnson

Who: A Senate leader who kept count of how each senator would vote, then a president who used the same skill on civil rights laws.
When and where: Texas and Washington, D.C., 1948 to 1969.

Johnson's 87-vote win in 1948 earned him the nickname "Landslide Lyndon."

Lyndon Johnson reached the Senate in 1948 by 87 votes, out of nearly a million cast. That was his final margin over Coke Stevenson in the Texas Democratic runoff, a second election between the top two candidates in the party's primary. A landslide is a win by a huge margin. For the rest of his life people called him "Landslide Lyndon."

Precinct 13. On the night of the runoff, Johnson seemed to have lost. Six days after the polls closed, an election judge named Luis Salas reported 202 more votes from Precinct 13 in Jim Wells County. Of them, 200 were for Johnson and 2 for Stevenson. He had changed the total on the tally sheet from 765 to 965. The added names were in alphabetical order, in the same ink and the same handwriting. Some of the people listed said they had not voted that day.

The members of the state party's executive committee voted 29 to 28 to declare Johnson the winner. A federal judge, T. Whitfield Davidson, blocked that result. Then Supreme Court Justice Hugo Black, acting alone, ruled for Johnson. He ruled aloud on September 28, 1948, and in writing the next day. Johnson's name went on the ballot.

In 1977 Salas told a reporter, James Mangan of the Associated Press, that the local political boss, George B. Parr, had ordered about 200 votes added. No court ever ruled that the votes were forged, and Johnson was never charged.

The vote counter. Johnson became his party's whip in the Senate in 1951. The whip counts votes and keeps party members in line. In 1953 he became its minority leader, the leader of the smaller party, and from 1955

to 1961 he was majority leader, the leader of the larger party. At 45, still in his first term, he was the youngest party leader the Senate had ever had.

The Senate's historians describe how he worked on other senators. He stood close to a senator's face and pulled "clippings, memos, statistics" out of his pockets. They list the tones he used: begging, accusing, coaxing, high spirits, scorn, tears, complaint and "the hint of threat."

In 1957 he pushed through the first civil rights law since Reconstruction, the years after the Civil War. He wanted it passed, but he feared a long filibuster would split his party. A filibuster is a way of blocking a bill by talking on and on so that no vote can happen. To get the bill through, he took out its main enforcement sections. Members of Congress passed it without them. According to the Senate's own history, very little changed because of it.

As president, Johnson signed the Civil Rights Act on July 2, 1964. On April 11, 1965, he signed the Elementary and Secondary Education Act, and he said, "no law I have signed or will ever sign means more to the future of America." On August 6, 1965, he signed the Voting Rights Act. It included the kind of enforcement tools that the 1957 law had been passed without. Federal examiners could register voters themselves, and some states had to ask federal officials before changing any voting rule.

How the civil rights laws passed

The filibuster. Southern senators led by Richard Russell of Georgia tried to block the Civil Rights Act of 1964 with a filibuster. They held the Senate floor, meaning they kept talking so that no vote could happen, for 60 working days, including seven Saturdays. On June 10, 1964, the senators voted 71 to 29 for cloture, the motion that ends debate. It was the first time senators had ever cut off debate on a civil rights bill.

Senator Clair Engle of California had a brain tumor and could not speak. He raised his weakened arm and pointed to his eye to vote aye, which means yes. The senators passed the bill 73 to 27 on June 19.

What the 1964 act says. The act has ten parts, called titles. Title II bans segregation in places that serve the public. Its words are: "All persons shall be entitled to the full and equal enjoyment of the goods, services, facilities, and privileges ... of any place of public accommodation ... without discrimination or segregation." Title VI bans discrimination in any program paid for with federal money.

Later, federal officials used Title VI to cut off money to school districts that stayed segregated. Title VII bans discrimination in jobs. In Title VII, members of Congress also created the Equal Employment Opportunity Commission.

What the 1965 act says. Section 4 of the Voting Rights Act of 1965 contains a formula for which places were covered. A state or county was covered if it had used a literacy test, a reading test for voters, or a similar device on November 1, 1964. It also had to have fewer than half of its voting-age people registered or voting in the 1964 presidential election. With the act, members of Congress outlawed literacy tests in the covered places.

Under Section 5, officials in every covered place had to get permission before changing any voting rule at all. The permission had to come from the Attorney General, the head of the Justice Department, or from a federal court in Washington. By the end of 1965, a quarter of a million new Black voters had been registered, one-third of them by federal examiners, according to the National Archives.

BY 1965: The justices had ruled that segregated schools, official school prayers and unequal voting districts all broke the Constitution. Members of Congress had passed the two civil rights laws, and federal examiners were registering Black voters in the South.

Federal money for schools, 1965

Johnson signed the Elementary and Secondary Education Act on April 11, 1965, outside the former Junction Elementary School near Stonewall, Texas. Elementary and secondary schools are the schools for children from first grade through high school. Junction Elementary was the one-room school where he had started at the age of four. His first teacher, Kate Deadrich Loney, sat beside him.

Under Title I of the act, federal officials send money to school districts according to how many children in them come from families with low incomes. The money comes with conditions the districts have to meet. In the 2015 budget year, about $14.3 billion went out under Title I, according to the National Center for Education Statistics in 2019.

Immigration law after 1965

Members of Congress passed the Immigration and Nationality Act in 1965. Under it, the national quotas set in the 1920s ended, and officials admitted immigrants on the basis of family ties and skills. The act also set the first cap on immigration from the Americas.

The Refugee Act of 1980 set out a permanent system for admitting refugees. The Immigration Reform and Control Act became law on November 6, 1986. Under it, about 2.7 million people who had lived in the country for years without permission got legal status, meaning the right to stay. The act also set penalties for employers who hired workers without permission to work.

Miranda v. Arizona, 1966

On March 13, 1963, police arrested Ernesto Miranda at his home in Phoenix, Arizona. They questioned him in a police interrogation room, a room for questioning suspects. After about two hours he signed a written confession. At his trial the officers admitted that no one had told him he could have a lawyer with him. An Arizona court found him guilty of kidnapping and rape.

The justices of the Supreme Court set a rule for police in *Miranda v. Arizona* on June 13, 1966. Chief Justice Earl Warren wrote the opinion. Before questioning a person in custody, meaning held by police and not free to leave, police must give four warnings. The person has the right to remain silent. Anything the person says can be used against them in court. The person has the right to a lawyer. If they cannot pay for one, a lawyer will be appointed, or provided free. If the person asks for those rights, the questioning must stop.

School law in the 1970s

The Amish fathers. Wisconsin law required children to attend school until age 16. Three fathers did not send their children to high school after the eighth grade. Jonas Yoder and Wallace Miller belonged to the Old Order Amish church, and Adin Yutzy to the Conservative Amish Mennonite church. Their children were Frieda Yoder and Barbara Miller, both 15, and Vernon Yutzy, 14. Each father was fined $5.

The justices of the Supreme Court held 6 to 1 in *Wisconsin v. Yoder*, on May 15, 1972, that Wisconsin officials could not make the fathers send their children to high school. Chief Justice Warren Burger wrote the opinion. The justices found that the families' way of life was a religious practice, lived every day and centuries old. They held that Wisconsin's officials had not shown that two more years of school mattered enough to override that practice, which the First Amendment protects.

Title IX. The Education Amendments of 1972 became law on June 23, 1972. Title IX of that law states: "No person in the United States shall, on the basis of sex, be excluded from participation in, be denied the benefits of, or be subjected to discrimination under any education program or activity receiving Federal financial assistance." That means a school or college that gets federal money may not treat a student worse because the student is a girl or a boy.

Is education a right? Texas schools were paid for partly with local property taxes. A property tax is a tax on the value of land and buildings. In the school year 1967-68, two school districts in San Antonio spent different amounts on each pupil. Edgewood, whose students were mostly Mexican American, spent $356 on each pupil. Alamo Heights spent $594.

In *San Antonio Independent School District v. Rodriguez*, on March 21, 1973, the justices decided 5 to 4 that education is not a fundamental right. A fundamental right is a right the Constitution itself protects. They found no right to education written in the Constitution or clearly implied by it. Justice Lewis Powell wrote the opinion.

Texas officials only had to show that their way of paying for schools was rational, meaning it had a sensible reason. Keeping schools under local control counted as a sensible reason. After 1973, people who sued over unequal school money had to go to state courts, under state constitutions.

Children with disabilities. President Gerald Ford signed the Education for All Handicapped Children Act on November 29, 1975. Handicapped is an older word for disabled. Under the act, every child with a disability had a right to a free, appropriate public education.

The Department of Education's history of the law states that in 1970, schools in the United States educated only one in five children with disabilities. It also states that members of Congress were concerned about "the more than 1 million children with disabilities excluded entirely from the education system."

The history also records that under the laws of many states, certain children were not allowed in school. Those laws applied to children who were deaf or blind. They also applied to children with an intellectual disability, a condition that makes learning and thinking much harder, and to children who were emotionally disturbed, meaning they had serious trouble controlling their feelings and behavior.

The paddle. In October 1970, James Ingraham was a student at Drew Junior High School in Dade County, Florida. Because he was slow to follow a teacher's instructions, he was held over a table in the principal's office and hit more than 20 times with a wooden paddle. School policy allowed one to five blows, called licks, on the buttocks with a flat wooden paddle less than two feet long, three to four inches wide and about half an inch thick.

The beating raised a hematoma, a pocket of blood under the skin. He needed a doctor, and he missed school for several days.

His family sued three school officials: the principal, Willie J. Wright, an assistant principal, Lemmie Deliford, and an assistant to the principal, Solomon Barnes. The Supreme Court's opinion does not say who held him or who struck him. Another student, Roosevelt Andrews, was paddled several times for small offenses. Once he was hit on the arms and lost the full use of an arm for a week. The opinion does not say who struck him either.

The justices decided *Ingraham v. Wright* 5 to 4 on April 19, 1977. They held that the Eighth Amendment's ban on "cruel and unusual punishments" does not cover school discipline. They wrote that it "was designed to protect those convicted of crimes."

They also held that school staff do not have to give a child a hearing, a chance to tell his side, before hitting him, because the child can sue afterward if the beating was too severe. The Constitution, as the justices read it, did not forbid paddling in schools. So paddling ended only in states where lawmakers voted to end it, one state at a time.

BY 1977: The justices had let Amish children leave school after eighth grade, had refused to call education a right, and had let school staff go on paddling. Members of Congress had opened schools to every child with a disability and barred schools that take federal money from treating girls worse than boys.

Congress and the president: war powers and Watergate

War powers. Members of Congress passed the War Powers Resolution in 1973, a law that limits how long a president may keep troops fighting without the approval of Congress. President Nixon vetoed it, meaning he rejected it, on October 24, 1973. On November 7, 1973, the representatives voted 284 to 135 and the senators voted 75 to 18 to override the veto, which made it law without his signature.

Under it, when a president sends troops into fighting without a declaration of war, he must report to Congress in writing within 48 hours. Within 60 days he must end the fighting unless members of Congress declare war, vote to approve it or give more time.

The break-in. Early on June 17, 1972, five men broke into the offices of the Democratic National Committee at the Watergate building in Washington, D.C. Police arrested them with cash and equipment for spying on the offices. Investigators later tied them to Nixon's re-election campaign. Officials in Nixon's White House at first denied any link. The break-in and the scandal that followed are called Watergate. A scandal is wrongdoing that becomes publicly known.

The tapes. On February 7, 1973, the senators voted to create a committee to investigate, led by Sam Ervin of North Carolina. A White House aide, Alexander Butterfield, told the committee that the president kept a tape recorder in rooms of the White House that switched on at the sound of a voice. In October 1973 Nixon ordered Attorney General Elliott Richardson to fire the special prosecutor, Archibald Cox, a lawyer named to investigate the case, who had demanded the tapes. Richardson and his deputy resigned in protest over Nixon's order.

In *United States v. Nixon*, decided on July 24, 1974, the justices of the Supreme Court ruled 8 to 0 that Nixon had to hand the tapes over to the prosecutor. Justice William Rehnquist took no part. Chief Justice Warren Burger wrote that the president's right to keep his talks private "must yield to the demonstrated, specific need for evidence in a pending criminal trial."

The charges and the resignation. In late July the members of the House Judiciary Committee approved three articles of impeachment, or formal charges. They approved obstruction of justice on July 27, abuse of power on July 29 and contempt of Congress on July 30. Obstruction of justice means trying to block an investigation. Abuse of power means using the powers of an office for wrong purposes.

On the morning of August 9, 1974, Nixon signed a one-sentence letter of resignation. He addressed it to the Secretary of State, because a law passed in 1792 states that a president's resignation is filed there. It took effect at 11:35 a.m., when Secretary of State Henry Kissinger initialed it, or wrote his initials on it. The members of the full House never voted on the articles.

Gerald Ford took the oath of office at noon. He said: "My fellow Americans, our long national nightmare is over." He went on to say that the Constitution worked. He also said: "I am acutely aware that you have not elected me as your President by your ballots."

Vice President Spiro Agnew had resigned in October 1973. Nixon had named Ford to replace him under the Twenty-fifth Amendment, and the members of both houses had confirmed him. As president, Ford named Nelson Rockefeller

as his vice president the same way. For about two and a half years, the country had a president and a vice president whom no voter had chosen for either office.

New kinds of schools, 1991 to 1996

In 1991 Minnesota's lawmakers passed the first state law allowing charter schools. Under it, a local school district could grant a group a charter, a written agreement, to run a school. The law's text calls them "outcome-based schools," and it allowed only eight in the whole state. State Senator Ember Reichgott Junge was its chief author. The first one, City Academy in St. Paul, opened in 1992.

Homeschooling became legal across the country state by state during the 1980s and early 1990s, through new laws and court rulings. Michigan was the last state, and it took two steps there, in 1993 and 1995.

On May 25, 1993, in *People v. DeJonge*, the justices of the Michigan Supreme Court struck down the state's rule that parents who taught their children at home must use certified teachers, meaning teachers licensed by the state. The ruling covered only families whose religious beliefs forbade certified teachers.

On the same day, in *People v. Bennett*, the same justices ruled that the rule still applied to a family with no religious objection. Michigan's lawmakers wrote a home-education exemption into the state's school attendance law in 1995. An exemption is a rule that lets some people off a requirement. It took effect on July 1, 1996.

An impeachment at the end of the century

The members of the House of Representatives impeached President Bill Clinton on December 19, 1998. They voted 228 to 206 to charge him with perjury, which means lying under oath, before a grand jury, a group of citizens who hear evidence in secret and decide whether someone should be charged with a crime. They voted 221 to 212 to charge him with obstruction of justice in the matter of Monica Lewinsky. They voted down two other charges.

The Constitution requires the votes of two-thirds of the senators, 67 votes, to find a president guilty and remove him. On February 12, 1999, the senators voted 45 guilty and 55 not guilty on the perjury charge. On the obstruction charge the vote was 50 to 50. Clinton stayed in office.

BY 2000

If you remember one thing: In this era members of Congress and the justices, between them, ended the poll taxes and literacy tests that officials in many Southern states used to keep Black Americans from voting, and a president resigned rather than face a vote on three charges.

Who counts the votes when the count itself is disputed? In 1948 a party committee and a single justice settled Lyndon Johnson's 87-vote win. The next era opens with that question in Florida in 2000, and the answer this time came from the Supreme Court.

Next: Since 2000, judges have ruled on recounts, on money in elections and on the shape of voting districts, and on January 6, 2021, a crowd attacked the Capitol while the electoral votes were being counted.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 21 MINUTES

Since 2000, judges have ruled on how votes are counted, how voting districts are drawn and whether corporations may spend money on elections. In 2000 the justices of the Supreme Court stopped a recount, a second count of the votes, in Florida during the presidential election. On January 6, 2021, while members of Congress counted the electoral votes, a crowd attacked the Capitol, the building where Congress meets. Judges also ruled on public money for religious schools and on how children are assigned to schools.

Bush v. Gore, 2000

The presidential election of 2000, between George W. Bush and Al Gore, came down to Florida. In Florida's official count, Bush led by 537 votes out of nearly 6 million. The justices of the Florida Supreme Court ordered recounts by hand of the ballots that counting machines had read as blank. They also ordered votes found in two counties' earlier recounts added to the official count.

The justices of the United States Supreme Court decided *Bush v. Gore* on December 12, 2000, in an unsigned opinion, one that no single justice signed as its author. Seven justices saw a problem under the Constitution. Different counties used different rules to decide what a ballot meant, so two identical ballots could be counted differently depending on where they were cast. Five justices voted to stop the recount instead of sending it back to be fixed, because there was no time left.

Justices John Paul Stevens, David Souter, Ruth Bader Ginsburg and Stephen Breyer dissented. The opinion states that the ruling was "limited to the present circumstances." The recount ended. Bush won Florida's 25 electoral votes and became president, with 271 electoral votes to Gore's 266. Gore had received more votes across the country than Bush.

Money and maps

Money. The justices decided *Citizens United v. Federal Election Commission* on January 21, 2010, 5 to 4. Citizens United, a nonprofit corporation, had made a film attacking Hillary Clinton. A nonprofit corporation is a group organized like a company that does not pay profits to owners. Its leaders wanted to advertise the film during a period before an election when federal law barred corporations from spending money on elections.

Justice Anthony Kennedy wrote for the majority that "the Government may not suppress political speech based on the speaker's corporate identity." Suppress means stop. That means government officials may not ban a group's political speech just because the group is a corporation. The justices overruled a 1990 decision, *Austin v. Michigan Chamber of Commerce*, and part of another, *McConnell v. Federal Election Commission*.

Maps. The justices decided *Rucho v. Common Cause*, also 5 to 4, on June 27, 2019. Gerrymandering is drawing the lines of voting districts so that one side wins more seats. Two maps were before the Court: North Carolina's map for Congress drawn by Republicans in 2016, and Maryland's drawn by Democrats in 2011.

Chief Justice John Roberts wrote that "partisan gerrymandering claims present political questions beyond the reach of the federal courts." Partisan means favoring one political party. Justice Elena Kagan wrote in dissent: "For the first time ever, this Court refuses to remedy a constitutional violation because it thinks the task beyond judicial capabilities." After 2019, a voter who believed a map was drawn to favor one party could not take that complaint to a federal court. Judges in state courts and members of Congress could still act on it.

Asking permission. Under Section 5 of the Voting Rights Act, officials in the places the act covered had to get federal permission before changing any voting rule. On June 25, 2013, in *Shelby County v. Holder*, the justices struck down the formula in Section 4(b) that set which places were covered. The ruling states that the formula "can no longer be used" to make a place ask permission. So the places it had covered no longer had to ask.

The justices left Section 5 itself in place and wrote that "Congress may draft another formula based on current conditions." As of September 2026, members of Congress had passed no new formula.

Schools and the courts

Tests and money. The No Child Left Behind Act is named for 2001, but it became law on January 8, 2002. Members of Congress wrote it as a new version of the Elementary and Secondary Education Act of 1965, and they kept that act's method: federal money for schools, with conditions attached. The Congressional Research Service, the office that does research for members of Congress, states that with No Child Left Behind, members of Congress began to focus on accountability for test results. Accountability here means judging schools by how their students scored on tests.

On December 10, 2015, members of Congress replaced it with the Every Student Succeeds Act. State officials still had to set standards, lists of what students should learn, and test students every year in reading, mathematics and science. Under the new law, state officials had more say over what to do about the results. In the 2023 budget year, federal spending on all the programs under the 1965 act came to $29.0 billion, according to a Congressional Research Service report of February 2024.

Public money for religious schools. Four times between 2002 and 2022, the justices of the Supreme Court ruled on public money for religious schools. On June 27, 2002, in *Zelman v. Simmons-Harris*, the justices ruled 5 to 4 that Ohio's scholarships for students in Cleveland were allowed. The scholarships paid up to $2,250 a year for families with low incomes, and up to $1,875 for other families.

According to the Court's 2002 opinion, 82 percent of the private schools taking part were religious, and 96 percent of the children using the scholarships went to a religious school. The justices held that such a program is allowed if it treats religion neutrally, favoring no religion and no lack of one. The money also had to reach a religious school only because a family chose that school.

In the next three rulings the justices went further. They held that when state officials pay for private schools or programs, they may not leave out religious ones. Missouri officials had refused a church preschool a grant, a gift of public money, for resurfacing its playground just because it was a church, though the grant was open to others. The justices ruled against them in *Trinity Lutheran Church of Columbia v. Comer* on June 26, 2017.

Three years later, five justices barred Montana officials from keeping scholarship money from religious schools, in *Espinoza v. Montana Department of Revenue*, decided 5 to 4 on June 30, 2020. Maine officials paid tuition, the cost of attending a school, for children from towns with no high school, but only at nonreligious schools. In *Carson v. Makin*, decided 6 to 3 on June 21, 2022, the justices held that Maine officials could not require that.

Race and school assignment. On June 28, 2007, in *Parents Involved in Community Schools v. Seattle School District No. 1*, the justices struck down two plans that used a child's race to decide which school the child attended. The vote was 5 to 4. Officials of Seattle's school district had used race to break ties, meaning to choose between students who ranked the same for a high school place. School officials in Jefferson County, Kentucky, had used it in placing children in elementary schools.

Chief Justice Roberts wrote: "The way to stop discrimination on the basis of race is to stop discriminating on the basis of race." Only four justices joined that sentence. Justice Kennedy cast the fifth vote and wrote separately. He wrote that school officials may still work toward a mix of students.

He listed ways to do it. Officials could choose where to build new schools, draw attendance zones and pay for special programs. They could also recruit students and teachers, and track enrollment and results by race. Attendance zones are the areas whose children go to each school. After 2007, school officials could still try to mix their schools, but they could not assign a single child to a school because of that child's race.

Counting the votes, January 6, 2021

Every four years, the members of the House and Senate meet together in the House chamber to count the electoral votes for president. The vice president presides, or runs the meeting. Tellers, the members chosen to count, open each state's certificate and read out its votes. A certificate is an official paper that states a fact, here the state's electoral votes. Until 2022, one member of the House and one senator together could object in writing to a state's votes. The members of the two houses then met separately, debated and voted.

Joe Biden won the 2020 election, with 306 electoral votes to Donald Trump's 232, by the count of the National Archives.

In some states, groups of Republicans signed papers stating that they were their state's electors, and they sent the papers to the Senate and the National Archives. Federal law requires each state's governor or other top official to sign a certificate naming the state's electors. No such official had signed one for these groups, so the papers were left out of the count, as a Congressional Research Service report of December 2024 states.

The attack. A crowd attacked the Capitol while the count was under way on January 6, 2021. The Government Accountability Office is a federal agency that checks the government's work. Its 2023 report states that over about seven hours, members of the crowd assaulted, or attacked, at least 174 police officers. Of them, 114 were from the United States Capitol Police and 60 from the police department of Washington, D.C.

The same office's 2022 reports record about $1.5 million in damage. They state that the plans Capitol Police officials made for the day "did not reflect the potential for extreme violence aimed at the Capitol." Those officials had information that protesters might be armed and were planning to target Congress.

That day, members of Congress took up two objections. According to the Congressional Research Service report, senators rejected the objection to Arizona's votes 93 to 6, and representatives rejected it 303 to 121. The

objection to Pennsylvania's votes failed 92 to 7 among senators and 282 to 138 among representatives. The members of Congress then finished the count and declared the result.

The new rules. Afterward, members of Congress changed the rules. The Electoral Count Reform and Presidential Transition Improvement Act became law on December 29, 2022. Under it, an objection must have the support of one-fifth of the members of each house, about 87 representatives and 20 senators. Only the state's executive, usually the governor, may certify the state's electors. Electors must be chosen under state laws passed before election day.

The President of the Senate is the vice president, and the act states: "The President of the Senate shall have no power to solely determine, accept, reject, or otherwise adjudicate or resolve disputes over the proper certificate of ascertainment of appointment of electors, the validity of electors, or the votes of electors." That means the vice president cannot decide alone which electors or which votes count.

The pardons. On January 20, 2025, the day he took office again, President Donald Trump signed Proclamation 10887, a formal order from the president. With it, he cut the prison sentences of fourteen named members of two groups, the Oath Keepers and the Proud Boys, to the time they had already served. Among them were Stewart Rhodes and Kelly Meggs.

A pardon is a president's order that forgives a federal crime, so the person is no longer punished for it. He gave "a full, complete and unconditional pardon" to everyone else found guilty of crimes connected with the events at the Capitol on January 6, 2021. He also ordered everyone still in prison for them released.

Donald Trump's two impeachment trials

Senators can remove a president from office only if two-thirds of them, 67 votes, find him guilty. Senators have never found a president guilty.

The members of the House impeached President Donald Trump on December 18, 2019, on two charges: abuse of power and obstruction of Congress, meaning blocking Congress's investigation. On February 5, 2020, the senators voted 48 guilty and 52 not guilty on the abuse of power charge, and 47 to 53 on the obstruction charge.

The members of the House impeached him a second time on January 13, 2021. The charge was incitement of insurrection, meaning stirring people up to rise against the government. On February 13, 2021, the senators voted 57 guilty and 43 not guilty, ten votes short of conviction. Seven Republican senators voted guilty: Richard Burr, Bill Cassidy, Susan Collins, Lisa Murkowski, Mitt Romney, Ben Sasse and Pat Toomey.

Wandrea ArShaye "Shaye" Moss

Who: A Fulton County election worker who was publicly and falsely accused of cheating in the 2020 count by a lawyer for the Trump campaign and by Rudolph Giuliani, and who told a committee of Congress under oath what followed.

When and where: Fulton County, Georgia, 2017 to 2022. Sworn testimony in Washington, D.C., June 21, 2022.

The object on the video was a ginger mint.

Shaye Moss worked in the Department of Registration and Elections in Fulton County, Georgia, from 2017 until 2022. She handled applications from voters and requests for absentee ballots, which are ballots mailed to voters who vote from home. She also helped count the votes. Her mother, Ruby Freeman, worked as a temporary election worker in the same county for the 2020 election.

On June 21, 2022, Moss testified under oath to the House committee that investigated January 6. She said: "I have always been told by my grandmother how important it is to vote and how people before me, a lot of people, older people in my family did not have that right."

Older voters were her favorite part of the job: "the older voters like to call, they like to talk to you, they like to get my card, they like to know that every election I am here." She once drove to a hospital to bring a woman her absentee ballot application.

The count. About five million Georgians voted in 2020. From November 11 to 19, 2020, election workers counted every ballot by hand, in 41,881 batches. In no county did the margin, the gap between the two candidates' totals, change by more than 0.73 percent. Secretary of State Brad Raffensperger certified the result on November 20, and Governor Brian Kemp certified it too. After a machine recount, requested on behalf of the Trump campaign, state officials certified the result again on December 7, 2020. Joe Biden won Georgia and its 16 electoral votes.

On December 3, 2020, a lawyer working for the Trump campaign played clips of a security video from the State Farm Arena at a hearing of the Georgia Senate. The lawyer said election workers there had pulled 18,000 hidden ballots from "suitcases of ballots under a table, under a tablecloth." The two women in the clips were Freeman and Moss.

The women's lawsuit, as the federal court described it in 2022, states what happened next. Georgia's state election officials and agents of the Georgia Bureau of Investigation watched the whole security tape, which ran for hours, and interviewed everyone who had been there. They found no evidence for any of the claims. Early on December 4, the state's Voting Implementation Manager posted that the video "Shows normal ballot processing." The containers held ballots that had been processed earlier that night, and no observer had been told to leave.

On December 10, 2020, Rudolph Giuliani, a lawyer for President Trump, told a hearing of the Georgia House that the two women were "quite obviously surreptitiously passing around USB ports as if they are vials of heroin or cocaine." Surreptitiously means secretly. USB ports are the sockets that computer memory sticks plug into. He said their homes and workplaces should have been searched. Moss later testified that the object her mother handed her in the video was "a ginger mint."

On January 2, 2021, Trump telephoned Raffensperger and named Ruby Freeman again and again. He called her "a vote scammer, a professional vote scammer and hustler." Raffensperger answered on the call that Giuliani "or his people, they sliced and diced that video and took it out of context."

What followed. Moss testified that Trump supporters sent her messages. She described them as "a lot of threats, wishing death upon me, telling me that, you know, I will be in jail with my mother, and saying things like, 'Be glad it's 2020 and not 1920.'" Many were racist, she said. The historian Tera W. Hunter wrote that the "1920" message, and messages telling the two women they should hang, pointed back to the history of lynching, murder by a mob without a trial.

Strangers went to her grandmother's house and pushed their way in. They said they had come to make a citizen's arrest, an arrest by someone who is not a police officer, and needed to find Moss and her mother. Agents of the FBI, the Federal Bureau of Investigation, a national police agency, advised Freeman to leave her home around the week of January 6, 2021, and she stayed away about two months. "I've lost my name," Freeman said, "and I've lost my reputation."

Moss said she stopped giving out her business cards and stopped going to the grocery store. Asked how many of the permanent election workers and supervisors in that video were still doing the job, Moss answered: "There is no permanent election worker or supervisor in that video that is still there." She had left too.

The lawsuit. In December 2021 Freeman and Moss sued Giuliani for defamation in federal court in Washington, D.C. Defamation is harming a person's good name with false statements. On August 30, 2023, Chief Judge Beryl Howell ruled against Giuliani on all their claims, as a penalty for refusing to hand over evidence. On December 15, 2023, a jury awarded the two women about $148 million, a figure reported by PBS. The judge later reduced the amount to about $146 million, according to NBC News.

On April 15, 2024, Judge Howell refused Giuliani's request to throw out or change the verdict, the jury's decision. In January 2025 a federal judge in New York, Lewis Liman, found Giuliani in contempt of court, the offense of disobeying a court's orders, for failing to give information about his property. A court filing showed that the judgment had been satisfied, meaning its terms had been met, on February 21, 2025.

In November 2025 President Trump pardoned Giuliani and others who had worked to overturn the 2020 result. A presidential pardon covers only federal crimes, and none of the people named had been charged with a federal crime over the 2020 election. A pardon does not cancel a civil judgment, a court's order in a lawsuit to pay money, such as the award to Freeman and Moss. That award had already been paid.

Who does not get a vote

In the 2020 census, Census Bureau workers counted about 690,000 people living in the District of Columbia, more than in Wyoming or Vermont. The voters of each of those states elect two senators and a voting representative. The District has no voting member of the House and no senators. It has three electors for president because of the Twenty-third Amendment.

The same census found about 3.3 million people in Puerto Rico, more than lived in 21 of the 50 states. Its residents are United States citizens. While they live there, they cannot vote for president. Their Resident Commissioner, their one member in the House, cannot vote on the House floor.

Just under 339,000 people lived in four smaller island areas: Guam, the United States Virgin Islands, American Samoa and the Northern Mariana Islands. None of the four has a voting member of Congress or electors for president. People born in American Samoa are nationals, who belong to the United States without being full citizens.

In all, about 4.3 million people lived in 2020 in places with no voting member of Congress. The Census Bureau states the rule itself: "The populations of the District of Columbia and Puerto Rico are excluded from the apportionment population because they do not have voting seats in Congress." The apportionment population is the count used to divide the seats in the House among the states.

Felony convictions. In 2024, about 4 million Americans could not vote because of a felony conviction, 1.7 percent of adults of voting age. A felony is a serious crime. The count is an estimate by the Sentencing Project, not a government count. In 2016 the number had been 5.9 million. Among Black Americans of voting age, 1 in 22, or 4.5 percent, could not vote for this reason. That was more than three times the rate for everyone else, 1.3 percent.

About 72 percent of the people counted were not in prison. They were on probation, which is supervision in place of prison, or on parole, which is early release from prison under supervision, or they had finished their sentences. State lawmakers and state constitutions set these rules. Under the laws of Maine and Vermont, people in prison can vote.

A senator convicted, 2024

On September 22, 2023, federal prosecutors in Manhattan charged Senator Bob Menendez of New Jersey, his wife, Nadine Menendez, and others with bribery. According to news reports of the case in 2023, agents had found $480,000 in cash and more than $100,000 in gold bars in the Menendez home. On July 16, 2024, a jury found him guilty on all counts, or charges, including acting as a foreign agent of Egypt, meaning acting on behalf of Egypt's government.

He resigned from the Senate in August 2024. On January 29, 2025, a judge sentenced him to 11 years in prison. He reported to prison in June 2025.

Immigration law after 2001

In the Homeland Security Act of 2002, members of Congress created the Department of Homeland Security. Its officials began work in 2003. Under the act, the old Immigration and Naturalization Service was split into three agencies: U.S. Citizenship and Immigration Services (USCIS), Immigration and Customs Enforcement (ICE) and Customs and Border Protection (CBP).

On June 15, 2012, officials of the Department of Homeland Security began a policy called Deferred Action for Childhood Arrivals, or DACA. Under it, certain people brought to the country as children could get protection from being deported, or sent out of the country, and permission to work. Lawsuits over DACA went on for years.

BY TODAY

If you remember one thing: Since 2000 the fights over the rules have been about counting: who counts the votes, whose money counts, and who gets no vote to count.

- 2000: Five justices stopped the Florida recount, and George W. Bush became president with fewer votes across the country than Al Gore.
- 2021: A crowd attacked the Capitol during the electoral count, and members of Congress then finished the count and declared the result.
- 2020: About 4.3 million people lived in places with no voting member of Congress, and in 2024 about 4 million adults could not vote because of a felony conviction.

Next: The chapter closing is below. Chapter 21, Crime, Police, and Justice, follows.

THE THREE QUESTIONS, ANSWERED

1. **Who decided who could vote, and how did the answer change?** Each colony's and then each state's lawmakers decided, and the men who wrote the Constitution left it that way. Virginia's lawmakers took the vote from free Black men in 1723, New Jersey's lawmakers took it from women and free Black men in 1807, and Southern delegates took it from Black men again after 1890. Members of Congress took most of that power from the states with four amendments (1870, 1920, 1964, 1971), which state lawmakers approved, and with the Voting Rights Act of 1965.
2. **When did judges first cancel a law, and what have they done with that power since?** In 1803, in *Marbury v. Madison*. The justices have used it to allow forced sterilization, separate railroad cars and the wartime camps, and to strike down school segregation, official prayers, poll taxes and unequal districts.

3. **What happens when people in power break the law?** A jury, a court or the Senate decides. In 1614 royal judges found Juan de Oñate, a former governor of New Mexico, guilty of 12 charges and ordered him out of New Mexico for life. A jury found a cabinet secretary guilty of taking a bribe in 1929, and a judge sentenced him to a year in prison. A president resigned in 1974 rather than face a vote. Bill Clinton and Donald Trump were impeached, Trump twice, and the senators found neither guilty. A jury found a senator guilty of bribery in 2024, and he went to prison.

What changed: The rules began as whatever the lawmakers of each colony chose. Over ten eras they became a written plan that judges enforce, and the vote, which began with men who owned property, reached women in 1920 and eighteen-year-olds in 1971.

FIVE DATES TO KEEP

1619: Twenty-two elected burgesses met at Jamestown, the first elected assembly in English North America.

1803: The justices of the Supreme Court first cancelled a law passed by Congress.

1870: The Fifteenth Amendment barred every state from denying a man the vote because of race.

1920: The Nineteenth Amendment barred every state from denying a woman the vote.

1974: Richard Nixon resigned the presidency rather than face a House vote on three charges.

NOT KNOWN: Whether the feet of the Acoma men were cut off in 1599: the sentence is in the written record, the Acoma account says it was done, and two historians doubt it.

Where to go: Independence Hall in Philadelphia, where the delegates signed the Constitution on September 17, 1787, is open to visitors as part of a national park.

Next: Chapter 21, Crime, Police, and Justice. It asks who enforced these laws, and what happened to the people they caught.

Chapter 21: Crime, Police, and Justice

CHAPTER 21 OF 37 · CRIME, POLICE, AND JUSTICE · 10 ERAS · ABOUT 140 MINUTES

Americans stopped whipping and branding people in public and started locking them in prisons, and from the first colonial courts on, whether a person got a fair trial depended on who that person was.

Clarence Earl Gideon sat in the Florida state prison at Raiford in 1961, with a five-year sentence for a break-in at a pool hall. He was 50 and had no money for a lawyer. Florida gave free lawyers only to people who could be sentenced to death, so the judge had refused him one. Gideon wrote to the US Supreme Court in pencil. In 1963 all nine justices agreed that a state must give a lawyer to a poor person charged with a serious crime. At his second trial, the jury found him not guilty after an hour.

This chapter follows crime and punishment in order, from the clan laws of the Cherokee to the police killings and trials of the 2020s. In the 1600s judges had people whipped, branded and hanged in front of their neighbors. By 2009 about 1.6 million people were locked inside American prisons. The people who kept order changed too: settlers took turns on a night watch, then cities hired police, then the national government hired agents. Thousands of people got no trial at all. Mobs burned, hanged and shot them, and for most of those killings nobody was ever punished.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did punishment move from the whipping post to the prison cell?
2. Who kept order before there were police?

3. What happened to the people in mobs who killed without a trial?

Five minutes? Read "From street work to the cell, Philadelphia" in 1750 to 1800, then "Francis McIntosh, St. Louis, 1836" in 1800 to 1850.

Can skip: "Counterfeiting was a hanging crime", "A cabinet member convicted, 1929", "Leaders in court", "Whose courts: McGirt, 2020". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

When one person killed another, the families of the two people settled it between them, and a life was owed for a life. No nation north of Mexico kept its laws in writing. Among the Cherokee, the dead person's clan had the duty to take a life from the killer's side. In accounts of other nations, the killer's relatives had the duty to give up a life in exchange. Some families took a payment instead: furs, other goods, or wampum, strings or belts of beads made from shells.

KNOWN:	The rules, which the nations remembered and later wrote down. Some come from the nations' own spoken traditions, and some from accounts by Native and European writers after about 1600.
NOT KNOWN:	The name of any person tried or punished before 1500. No written record names one.

The Cherokee law of blood

Among the Cherokee, a killing was settled between two clans. A clan is a large family group. The Cherokee had seven clans, among them the Wolf, the Deer and the Paint, and a child belonged to the mother's clan.

What the nation's newspaper said. The editors of the *Cherokee Phoenix*, the Cherokee Nation's own newspaper, printed an article on the old rules in 1829. The article states that the Cherokee "as a nation, had nothing to do with murder." The nation's leaders did not punish a killing. The clans did. The clan of the dead person had the duty to take revenge by killing the killer.

If the killer ran away, his brother or his nearest relative "was liable to suffer in his stead," in the article's words. The clan could kill the brother or the relative in the killer's place. Every killing counted as murder, so a person who killed someone by accident could be killed in return. A man who killed his own brother answered to no law and no clan.

The same article states that about twenty years before 1829, the Cherokee clans met in a council, a large meeting to decide the nation's business, and ended these rules. The Cherokee Nation dates that council to 1810. From then on, murder was a crime against the Cherokee government.

Words of comfort instead of revenge

The Haudenosaunee are five nations in what is now New York: the Mohawk, Oneida, Onondaga, Cayuga and Seneca. Their own tradition says the five nations were once caught in feuds, long runs of killings in which each side took revenge for the last death.

In that tradition, a man named Hiawatha, whose daughters had been murdered, strung shells into wampum. He used the strings to speak words of comfort instead of words of revenge. These words of condolence, comfort for people in grief, became central to the Great Law of Peace, and the five nations joined together under it.

The Haudenosaunee told the Great Law aloud for generations. Two written versions existed by 1910 at Six Nations of the Grand River, a Haudenosaunee community in Ontario. Seth Newhouse, a Mohawk, wrote one, and the chiefs of that community wrote the other. Arthur C. Parker printed them in English in 1916, for the New York State Museum.

What the written law did to a murderer. In that version, a chief who committed murder lost his title, and the other chiefs removed him. A war chief then ordered him to leave the land of the five nations and never come back.

An outsider adopted into one of the nations got a warning from a war chief for a first wrong. After a second wrong, the written law says, the person was to be sent out of the land of the five nations for good.

Forgiveness at the Green Corn

Once a year, the people of each Muscogee town forgave every crime except murder. The Muscogee, also called the Creek, and other nations of the Southeast did this at the Green Corn Ceremony, held each midsummer when the first corn ripened. Its Muscogee name, the busk, comes from their word for "to fast." People fasted, and they lit the sacred fire again in the town square.

The trader James Adair wrote about the ceremony in the middle 1700s. So did the naturalist William Bartram, who studied plants and animals, and Benjamin Hawkins, a government agent, an official sent to deal with a Native nation. The most detailed descriptions were written in the late 1800s and early 1900s.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 4 MINUTES

The only European courts in this land sat at a few Spanish outposts, and there the same officials governed the settlers, commanded the soldiers and judged the cases. Spanish settlers founded St. Augustine, in Florida, in 1565. Juan de Oñate led a Spanish colony into New Mexico in 1598. Everywhere else, the Native nations kept their own laws. The two men who led those colonies, Pedro Menéndez de Avilés and Oñate, each decided what happened to hundreds of prisoners.

In a Spanish town, an official called an alcalde acted as the judge. His power covered smaller crimes and quarrels, but in practice he heard most of the cases in his area. A person who lost could appeal, asking a higher judge or the governor himself to change the decision.

Killed without a trial, 1565

The Spanish commander Pedro Menéndez de Avilés had French Protestants killed after they surrendered to him, in September and October of 1565. Protestants are Christians outside the Catholic Church. The men had no trial.

The count. The National Park Service's handbook for Fort Matanzas lists 111 men killed the first time and 134 the second, about 245 in all, with 16 spared each time. Chuck Meide, speaking on the PBS program *Secrets of the Dead*, gives wider ranges for the two killings, from 110 to 200 and from 70 to 150.

The Acoma trial, 1599

Juan de Oñate, the leader of the Spanish colony in New Mexico, put about 500 Acoma people on trial in February 1599. He had ordered the attack on their town a month before, and he sat as their judge.

The trial followed a fight at Acoma Pueblo, the Acoma people's town. Spanish soldiers under Juan de Zaldívar came to Acoma in December 1598 and demanded food. Acoma people killed Zaldívar and eleven or twelve of his soldiers. The Acoma people's own account says the soldiers had also raped an Acoma woman, which means they forced her into sex.

Oñate opened an inquiry on December 28, an official hearing to find out what had happened. He asked the Franciscan friars with the colony, Catholic religious brothers, whether a war on Acoma would be just, meaning right and fair. They said yes. In January 1599 Oñate sent Juan de Zaldívar's brother, Vicente de Zaldívar, with about 70 soldiers against Acoma. His soldiers broke into the town, killed Acoma men, women and children, and burned it.

The count of the dead. The counts differ. New Mexico's state historian gives about 600. The Pueblo leaders of the All Pueblo Council of Governors give 800, and a National Park Service page gives about 1,500.

The trial of the captives began on February 9, 1599, and lasted three days. Oñate named a defender for the prisoners, and the defender asked for mercy. Acoma witnesses said many of their people had been working in their fields

during the first fight. They said that "some wanted to surrender but some refused, and so they fought."

The sentences. Oñate gave his sentence on February 12. He sentenced no one to death. Every man over 25 was to have one foot cut off and to be made a slave for twenty years, forced to work for Spanish colonists with no pay and no right to leave. Boys and young men from 12 to 25, and women and girls over 12, were also to be made slaves for twenty years.

Oñate sentenced the old people to be handed over as slaves to the Plains Apache. Two Hopi men, from a different people, had been captured in the fighting. Oñate sentenced each of them to have his right hand cut off and to be sent home to carry word of the punishment to his own people.

The children. Oñate declared the Acoma children under 12 innocent. He still ordered them taken from their families and handed to the friars. A National Park Service page states that about 70 girls were sent to convents in Mexico, houses where nuns live, and that in all probability none of them ever came home. The PBS program *POV* states that some scholars believe the children were later sold as slaves. The Pueblo leaders of the All Pueblo Council of Governors give the number of children taken as 60.

The feet. The Spanish record states the order, the ages of the prisoners, the twenty-year terms and the taking of the children, and on those points the accounts agree. It states Oñate's order to cut off a foot. It does not say how the cutting was to be done, or whether it was done to each man.

The Acoma people's own history says that 24 men had a foot cut off, and reporters for the Associated Press and NPR give the same number. The historian Marc Simmons first wrote that the feet were cut off, and he later came to doubt it. The historian John L. Kessell doubts it too. He wrote that "the historical record makes no mention of a one-footed Acoma slave."

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 16 MINUTES

Judges in the English colonies punished crime on the body and in public. They ordered whippings, brandings and cut-off ears, and they sent people to sit in the stocks where their neighbors could see them. Ordinary settlers in Massachusetts Bay took turns walking a watch to keep order at night. A jail held people waiting for a trial or a punishment, people who had done small wrongs, and people who owed money. Locking people in a prison for years as the punishment itself did not begin until the 1790s.

Colonial judges in Massachusetts Bay and Plymouth also tried Native people who lived near the colonists. Virginia's lawmakers passed a law in 1669 under which a master who beat an enslaved person to death, after the person resisted him, was not charged with a crime. Judges in Massachusetts had 19 people hanged for witchcraft in 1692.

Jamestown under martial law, 1610 to 1619

The leaders of the Jamestown colony in Virginia ruled it under martial law, rule by army officers instead of by ordinary courts, from 1610 to 1619. Jamestown stood in Tsenacomoco, the land of the Powhatan people.

Two of the colony's leaders issued the first orders in May and June of 1610, and Sir Thomas Dale added military rules on June 22, 1611. William Strachey printed the whole set in 1612 as the *Lawes Divine, Morall and Martiall*. Rule under these laws ended when Sir George Yeardley arrived in Virginia on April 18, 1619.

The death crimes. The historian Brent Tarter lists the punishments in the *Encyclopedia Virginia*. The punishment was death for murder, for robbery and for blasphemy, insulting God. It was also death to speak treason, words against the king's rule, to speak against the company that ran the colony, or to trade with Indians without permission.

The printed laws of 1612 also specify death for any man who would "ravish or force any woman, maid or Indian," which means rape. They specify death for "the horrible, and detestable sins of Sodomie" too, and they do not say which

acts that word covered.

The whipping crimes. The punishment was a whipping for gambling, for missing church, for sex between two unmarried people, for killing farm animals without need, and for stealing crops or farm tools.

The sources disagree about adultery, sex between a married person and someone outside the marriage. The printed laws themselves specify death for it, and Tarter lists it among the whipping crimes.

What George Percy wrote. The colonist George Percy wrote an account of the colony's early years in 1625. He wrote about the building of a fort at Henricus in 1611, and about colonists there who ran away to live with the Native people nearby. When they were caught, Dale had them put to death: "Some he apointed to be hanged some burned some to be broken upon wheles others to be Staked and some to be shott to deathe." Percy wrote that Dale did these things "To terrefy the reste," meaning to frighten the other colonists.

Breaking on the wheel was a way of putting a person to death in Europe. The executioner broke the person's arms and legs with a heavy bar or club and tied the body to a large wooden wheel. Percy does not describe how it was done at Jamestown, and he does not explain what "Staked" meant.

Percy also wrote that Dale had men who robbed the colony's storehouse tied to trees and left there until they starved to death. Percy wrote his account partly to defend his own record in Virginia.

Oñate convicted, 1614

Juan de Oñate was the leader who had sentenced the Acoma people in 1599. Spanish officials later put him on trial in Mexico City for what he had done in New Mexico. They charged him with 30 crimes in 1612 or 1613. They found him guilty of 12 in 1614. Among them were using too much force against the Acoma and hanging two Indians without cause.

The officials banished Oñate from New Mexico for life, ordering him to leave and never return. They also barred him from Mexico City for four or five years and fined him 6,000 ducats, which were Spanish coins. Whether he was later pardoned, officially forgiven and freed from the punishment, is not settled. No record shows that any Acoma person was asked to testify.

The watch and the whipping post in Massachusetts

The court records of the Massachusetts Bay Colony from the early 1630s show who kept order and how the colony's judges punished people. The colonists had settled among the Massachusett people and the Pawtucket, who belonged to a larger group of nations led by Massachusett sachems, or leaders.

From the colony's first year, its judges tried Native people who lived near the colonists. They fined the Massachusett sachem Chickatabut one beaver skin in 1631 for shooting a pig that belonged to Sir Richard Saltonstall. Governor John Winthrop wrote in his journal that Chickatabut paid it right away.

The watch. The settlers themselves kept watch at night. On April 12, 1631, the judges ordered a watch of four men every night at Dorchester and four more at Watertown, "to begin att sunsett." Anyone who fired a gun after the watch was set had to pay a fine of 40 shillings, which was English money, or be whipped. In July the judges ordered a watch of six men and an officer every night at Boston.

The watchmen were settlers taking turns, and some of them did not take their turn. The judges fined Daniel Abbott "for refuseing to watch" and Richard Norman for "negligence in watching."

The whipping post. Whipping meant being struck on the bare back with a whip, often at a post in a public place. Bartholomew Hill got a whipping for stealing a loaf of bread from John Hoskins, and John Baker got one for shooting at birds on the Sabbath, the day of rest and worship.

John Goulworth confessed to a felony, a serious crime, and the judges had him whipped and then set in the stocks. The stocks were a wooden board with holes that locked a person's ankles while the person sat in a public place. For helping in a crime, John Boggust and John Pickryn had to sit in the stocks at Salem for four hours.

The pillory was a wooden frame on a post, set on a platform in a public place. It locked a person's neck and wrists while the person stood. Britannica's account for students states that it was meant to expose people to "public scorn and ridicule," and to cause "bodily discomfort," meaning the body ached from standing locked in place.

The bilboes were iron rings locked around the ankles and joined to a bar, so the person could not walk away. Edward Seale, punished for drunkenness, sat in the bilboes until the court day ended, "& then to bee severely whiped."

The branding iron. Richard Hopkins sold guns, powder and shot to Indians. The court sentenced him in September 1632 to be whipped and "branded with a hott iron on one of his cheekes." Branding meant pressing a red-hot iron into the skin. It burned a mark into the flesh that stayed for life. Robert Scarlett, "a knowen theife," got a whipping in 1635 and the letter T, for thief, branded on his forehead. Then his master, Benjamin Felton, had to send him out of the colony.

Officers of the colony also cut off ears. Three Quakers, members of a Christian group also called the Society of Friends, were held in the Boston prison in September 1658: Christopher Holder, John Copeland and John Rous. Their sentence stated that each man's right ear was to be "cut off by the Hangman."

On September 16 the deputy of the marshal, the colony's officer who carried out the court's orders, came to the prison with a group of men and shut the door. They cut off the right ear of each of the three men. George Bishop, a Quaker, recorded it in his book *New England Judged* in 1661.

NOT KNOWN: How many strokes any of these whippings had, which of the deputy's men cut off the Quakers' ears, and with what tool.

Philip Ratcliffe

Who: A servant whose ears were cut off for speaking against the Massachusetts government and the church at Salem.

When and where: Boston, June 1631.

Whipped, ears cut off, fined 40 pounds and banished, for words.

The judges of the Massachusetts Bay court at Boston sentenced Philip Ratcliffe on June 14, 1631. Their order states that he "shalbe whipped, haue his eares cutt of, fyned 40£, & banished." In today's spelling, he was to be whipped, have his ears cut off, pay a fine of 40 pounds, and leave the colony for good. Governor John Winthrop's journal calls him a servant of a Mr. Cradock.

His crime was "vttering mallitious & scandulous speeches against the govmt & the church of Salem." He had said hateful and shameful things about the colony's government and about the church at Salem. The record says what he said was "proved vpon oath," which means people swore in court that he said it.

Winthrop wrote in his journal that the sentence "was presently executed," which in his day meant it was carried out right away. Neither the court record nor Winthrop says who cut off Ratcliffe's ears, or how. The same judges that day ordered Philip Swaddon whipped for running away from his master, Robert Seely, to go to Virginia.

When killing an enslaved person was no crime, 1669

The members of Virginia's assembly, the colony's lawmakers, passed "An act about the casuall killing of slaves" in October 1669, while Sir William Berkeley was governor. An enslaved person was a person owned by another person as property and forced to work for no pay.

Under the act, if an enslaved person resisted and died under "correction" by the master, or by someone the master sent, the death "shall not be accompted felony." Correction meant a beating. Under this law, a master who beat an enslaved person to death, after the person resisted him, was not charged with any crime.

The lawmakers gave their reason in the act. They wrote that no one would kill on purpose when the enslaved person was his own "estate," meaning his property. The act is printed in William Waller Hening's collection of Virginia's old laws, the *Statutes at Large*.

A murder trial at Plymouth, 1675

The judges of Plymouth Colony tried three Wampanoag men in 1675 for killing another Native man, John Sassamon. Plymouth stood in the country of the Wampanoag. Sassamon, a Christian, had reportedly warned Plymouth's governor that men of the Wampanoag leader Metacom, whom the English called Philip, were planning an attack. Sassamon died at Assawompset Pond on January 29, 1675.

In March the judges ordered one of the three, a man named Tobias, to answer questions about the death. Three men put up their land, worth 100 pounds, as a promise that Tobias would come back to court in June: Tobias himself, a sachem named Tuspaquin, and Tuspaquin's son William.

In June the judges tried Tobias, Tobias's son Wampapaquan and a third man, Mattashunannamo, for Sassamon's murder. The charge stated that they had struck him or twisted his neck until he died, then pushed his body through a hole in the ice of the pond. The three men said they were innocent.

The jury was twelve English colonists. The judges added six Native men to advise the jury. The six had not taken the jurors' oath, but the record states that they "fully concurred," meaning they agreed with the verdict. The jury found all three men guilty, and the judges sentenced them to hang.

Plymouth officials hanged Tobias and Mattashunannamo on June 8, 1675. The judges put off Wampapaquan's hanging for a month. Within that month he was shot to death, and the colony's court record does not say who shot him. Later that June, Metacom's men and the English colonists went to war, starting at Swansea. The English called it King Philip's War.

Salem, 1692

English settlers built Salem on Naumkeag, a home of the Pawtucket people. A plague brought by English traders killed many of the Pawtucket between 1616 and 1619, the City of Salem's tourism office states.

Colonial judges in Massachusetts tried people for witchcraft, using the Devil's power to hurt others, in 1692 and 1693. Accusers named more than 200 people, Smithsonian magazine reports. The judges had 19 of them hanged, and Sheriff George Corwin carried out the hangings. Corwin also had one man pressed to death under stones. At least five more accused people died in jail.

The first accusers were three girls in Salem Village: Betty Parris, who was 9, Abigail Williams, 11, and Ann Putnam Jr., about 12. They said three women were hurting them by witchcraft: Tituba, a woman enslaved in the house of the minister Samuel Parris, and Sarah Good and Sarah Osborn.

The magistrates, or local judges, John Hathorne and Jonathan Corwin questioned the three women on March 1, 1692. Tituba denied it. The magistrates kept at her with questions again and again, and then she confessed. Members of the Putnam family later made accusations in dozens of cases.

The special court. Governor William Phips set up a special court to try the accused on May 27, 1692. It was called a Court of Oyer and Terminer, words that mean "to hear and to decide," and William Stoughton was its chief judge.

The judges accepted spectral evidence: an accuser's word that the spirit of the accused person had come to her in a dream or a vision and hurt her. Only the accuser could say what she had seen. Stoughton was sure such evidence could be used. The minister Increase Mather wrote: "It were better that ten suspected witches should escape than one innocent person be condemned."

Pressed to death. Giles Corey refused to plead guilty or not guilty. Accounts give his age as 71 or 81. Under English law, a court could force a plea by pressing. The prisoner lay on the ground under a wooden board, and officers piled heavier and heavier stones on it until the prisoner answered or died.

Sheriff Corwin made Corey strip naked and lie face up on the ground. Then Corwin placed large rocks on the board one at a time. The pressing went on for two days, and Corey died on the third, September 19, 1692. Because he was never convicted, his children could keep his land under the law. The Massachusetts Historical Society gives that as his reason for refusing to plead.

A child in the jail. Sarah Good's daughter Dorothy was a small child when the magistrates had her jailed with her mother in March 1692. The warrant, the written order for her arrest, calls her four years old. Her father later said four or five, and one archive says six.

Her father, William Good, wrote in 1710 that she was kept in prison seven or eight months and "chain'd in the dungeon." He wrote that she was "so hardly used and terrifyed," meaning treated so harshly and frightened so badly, and that she had been a heavy cost to care for ever since. Sarah Good gave birth in jail, and her baby died there. Sarah Good was hanged on July 19, 1692.

Phips shut the special court in October 1692, after accusers named his own wife. The judges of the new court that replaced it would not allow spectral evidence, and they condemned 3 of 56 defendants. Phips had pardoned everyone still held by May 1693. The governor and lawmakers of Massachusetts passed an act in October 1711 that cancelled the convictions of many of the condemned, and they granted 600 pounds to the heirs. The state's lawmakers made a formal apology in 1957.

Rebecca Nurse

Who: A member of the Salem church whom a jury found not guilty of witchcraft, and who was then convicted and hanged.

When and where: Salem, Massachusetts, March to July 1692.

A jury found her not guilty, and the judges sent it back.

Edward and Jonathan Putnam filed a complaint against Rebecca Nurse on March 23, 1692. They said she had hurt Ann Putnam Sr., Ann Putnam Jr. and Abigail Williams by witchcraft. Marshal George Herrick arrested her the next day, on a warrant signed by the magistrates John Hathorne and Jonathan Corwin. She had been sick for eight or nine days and could not leave the house.

At her questioning on March 24 she said, "I am as clear as the child unborn." When she moved her hands, the record states, the accusers fell into fits. Thirty-nine of her neighbors signed a statement that they had never had "Any: cause or grounds to suspect her."

Women searched her body for a mark of the Devil. The most experienced of them found nothing that could not come from a natural cause. On June 28 Nurse asked the court in a petition, a written request, for other women to examine her again, two of them midwives, women who help other women give birth. She signed the petition with her mark, because she could not write her name.

At her trial the jury found her not guilty. The judges objected. They pointed to something she had said about two other accused women, Goodwife Hobbs and her daughter: "they used to come among us." Goodwife was the title for a married woman, like "Mrs." today.

The jury went out again. Back in court Nurse did not answer, and one juror, Thomas Fisk, took her silence as evidence against her. The jury then found her guilty. Afterward she explained that she had meant the two women were fellow prisoners. She wrote that she was "something hard of hearing, and full of grief," and that no one had told her how the court took her words.

Governor Phips pardoned her. Several men from Salem then persuaded him to take the pardon back. On July 3 the minister Nicholas Noyes led the members of the Salem church in putting her out of the church, "by unanimous vote," which is called excommunication. Sheriff George Corwin had her hanged on July 19, 1692. Her house in Danvers still stands. Her case file, number 94, is in the University of Virginia's *Salem Witch Trials Documentary Archive*.

BY 1700

If you remember one thing: Colonial judges punished crime by hurting people where their neighbors could watch: they had people whipped, branded and hanged, and had their ears cut off.

- 1610: Jamestown's leaders put the colony under martial law, with death for murder, robbery and speaking against the company.
- 1669: Virginia's lawmakers wrote that a master who beat a resisting enslaved person to death had committed no felony.
- 1692: Massachusetts judges had 19 people hanged for witchcraft, and the sheriff pressed Giles Corey to death.

Next: In the next fifty years the colonies set up separate courts and patrols for enslaved people, and British judges shipped tens of thousands of convicts across the ocean.

1700 to 1750

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An enslaved person accused of a crime faced a different kind of trial from a free white person. Under a Virginia law of 1723, a panel of local justices tried an enslaved person accused of a crime that carried death, with no jury. A white defendant in New York could turn away up to twenty jurors without giving a reason, and two enslaved men, Quack and Cuffee, went to trial there in 1741 with no lawyer. White men in South Carolina rode on slave patrols from 1704 to watch and control enslaved people.

Free colonists in Virginia took their quarrels over land and debts to a county court of local justices, which met about once a month. A lender could have a person who could not pay a debt locked in jail. British judges also sent about 52,200 convicts across the ocean to work in the colonies.

The county court and the debtors' jail

Local courts sat in the Virginia settlements by 1622, made up of local judges called commissioners of the peace, later called justices of the peace. The justices usually met once a month. The members of Virginia's assembly named these courts county courts in 1642, and by the end of the colonial years Virginia had 61 counties.

The justices gave out licenses to marry and to keep taverns, recorded deeds and wills, and settled quarrels over land and over debts. A deed is a paper showing who owns land. Emily Salmon, writing in the *Encyclopedia Virginia*, calls serving as a justice "the logical stepping stone" to a seat in the House of Burgesses, Virginia's assembly.

The debtors' jail. A jail in these years held people waiting for a trial or a punishment, and people who owed money. From the first colonies until the middle 1800s, a lender could have a person who could not pay a debt locked in jail. Under English law, which the colonists brought with them, the prisoner even paid the cost of being kept there.

Robert C. Voigt, writing in the *Encyclopedia of North Carolina*, explains that jailing a debtor gave the lender "a forceful means" of collecting the money. A person in jail could not work to earn it.

Convicts shipped from Britain

British judges sent about 52,200 convicts by ship to the American colonies between 1700 and 1775, a punishment called transportation. More than 20,000 of the convicts went to Virginia, and about 80 percent of all of them went to Maryland or Virginia. Emily Jones Salmon gives these numbers in the *Encyclopedia Virginia*.

Members of Parliament, Britain's lawmaking body, passed the Transportation Act in 1718. Under it British officials paid merchants to carry convicts across the ocean, and Jonathan Forward was the first merchant to hold the contract. The convicts were sold as soon as they landed, and Salmon does not name the sellers. Most of them were forced to work for their buyers for seven to fourteen years, mostly in tobacco fields. Others went to tradesmen, shipbuilders and ironworks.

Virginia's lawmakers tried to stop the shipments. They passed a law in 1722 meant to make shipping convicts too expensive. Forward complained, and the members of the Board of Trade, officials in London who watched over the colonies, overturned the law. Benjamin Franklin wrote against the shipments in 1751. The last convict ship reached the James River in Virginia in 1776.

Slave patrols and courts without juries

White South Carolinians set up official slave patrols in 1704, copying the island colony of Barbados. A slave patrol was a group of white men who rode out to watch and control enslaved people. The patrollers were the same men who served in the militia, the colony's part-time citizen army. At first wealthy men, called gentlemen, rode alongside poorer white farmers. The patrollers were paid for the work from 1734 to 1737.

Enslaved people at Stono, in South Carolina, rebelled against the people who enslaved them in 1739. After that, the members of the South Carolina assembly ordered "constant, regular patrols" in the Negro Act of 1740. Negro was an old word for a Black person. In Charleston, the town's night watch patrolled the streets.

What a patrol did. A patrol's route ran 10 to 15 miles. In later years a typical patrol was a few men on horseback. They searched the cabins where enslaved people lived, broke up their gatherings and guarded the roads. The *South Carolina Encyclopedia* states that this patrol system lasted until the end of the Civil War.

Courts with no jury. A Virginia law of 1723 states how enslaved people were to be tried for crimes that carried death. Justices named for each case tried the accused person at the county courthouse. They could convict on a confession or on the oath of one witness. All of this was done "without the solemnity of a jury," which means with no jury at all. If they found the person guilty, they ordered the execution. The owner of the accused person could come and speak in the person's defense.

The justices could also convict on the word of Black or Native witnesses if it "shall seem convincing" to them. The law calls these witnesses "Negroes, Mulattos, or Indians," old words for Black people, for people with one Black and one white parent or ancestor, and for Native people.

The same law states a punishment for any Black or Native witness found to have lied. The person's ear was to be nailed to the pillory, and the person was to stand there for an hour. Then the ear was to be cut off. An hour later the

other ear was to be nailed and cut off the same way.

Then the person was to get 39 lashes of the whip, "well laid on, on his or her bare back, at the common whipping-post." The justices had this warning read aloud to every such witness.

NOT KNOWN: Who was to carry out this punishment, and how often it was done.

The New York trials of 1741

New York judges tried about 200 people in 1741 for a plot to burn the city, after a string of fires that began in March. Daniel Horsmanden was one of the three judges. Officials burned 13 Black men at the stake and hanged 17 more. They also hanged four white people, two men and two women. John and Sarah Hughson and Peggy Kerry were hanged on June 12.

Burning at the stake meant chaining a person to an upright post with wood piled around it. Then the wood was set on fire, so the person burned to death.

The witnesses. The first witness before the grand jury, a group of citizens who decide whether a person should be charged with a crime, was Mary Burton. She was a 16-year-old servant of the tavern keeper John Hughson. At first she refused to talk. She began to talk as she was being taken to the dungeon at City Hall. Enslaved witnesses named Sandy and Fortune testified after being promised their lives. The grand jury told Sandy that enslaved people who confessed and named others "were certainly pardoned, and shipped off."

Judge Horsmanden questioned suspects himself before he sent them to the grand jury. White defendants had more protection under New York law: each could turn away up to twenty jurors without giving a reason. Quack and Cuffee, two enslaved men, had no lawyer.

Douglas Linder, who wrote an account of the trials for the Famous Trials project at the University of Missouri-Kansas City, writes of "grave questions about the contours of the suspected conspiracy." He means serious doubts about what the supposed plot really was and who was in it.

Quack and Cuffee

Who: Two enslaved men in New York City who were burned at the stake in the trials of 1741.

When and where: New York City, tried and burned in May 1741.

At the stake, each man was told the other had confessed.

Quack was enslaved by a painter named Roosevelt, and Cuffee by Adolph Philipse. On May 12 the undersheriff, the sheriff's deputy, put Cuffee in a cell with a white prisoner, Arthur Price. He gave the two men "a tankard of punch now and then," a large cup of a mixed drink, so that Price could draw Cuffee into talking. Price then swore that Cuffee had spoken about the fires.

Attorney General Richard Bradley, the colony's chief lawyer, led the case against the two men. His side called eleven witnesses. Quack and Cuffee had to defend themselves. They called ten witnesses, and most of them said the two men had been somewhere else when the fires started. The men said they were innocent. The jury, Horsmanden wrote, was "soon returned" and found them guilty. Judge Horsmanden sentenced them to be "chained to a stake, and burnt to death" on Saturday, May 30, 1741.

The sheriff brought them to the stake at about three o'clock that afternoon, with piles of wood around them. Horsmanden wrote that the crowd "were very impatient" for the fire. The deputy secretary, a Mr. Moore, and Quack's owner, Mr. Roosevelt, told each man that the other had already confessed. Both men then confessed and named other people.

Moore asked the sheriff to wait while someone asked the governor for a reprieve, an order to put off the burning. The sheriff said he could not take the men back through the crowd without a strong guard. Horsmanden called the governor's answer "conditional and discretionary," meaning it came with conditions and left room for choice. He wrote that "the execution proceeded." He does not name the men who lit the fire.

> The names they gave led to more trials. Horsmanden told the story in his own book, *A Journal of the Proceedings*, in 1744.

1750 to 1800

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Pennsylvania's lawmakers began to replace the whip and the gallows with the prison cell. Before then, a convict in Pennsylvania could be fined, made to pay back what he took, whipped, branded or hanged. From 1786 he could also be put to hard labor in the streets. The lawmakers turned the Walnut Street Jail in Philadelphia into the country's first state prison in 1790. After that, a judge could send a convict there for years, to work and to spend part of the time alone in a cell.

The rights of accused people were written down too. Lawmakers in enough states approved the Bill of Rights, the first ten additions to the Constitution, in 1791, and four of them list rights for accused people. Not everyone got a trial. Armed settlers in Pennsylvania killed twenty Conestoga people in 1763, and no one was ever charged.

Counterfeiting was a hanging crime

Colonial paper money carried the warning "To Counterfeit is Death." Counterfeiting means making fake money. Owen Sullivan was an engraver, a person who cuts designs into metal for printing, and he was born in Ireland. He ran gangs that passed fake bills from New England to New York.

The *Boston Evening Post* printed a report from Providence, Rhode Island, on October 9, 1752. It said that Sullivan "has lately had both his Ears crop'd, and been branded on both his Cheeks with the Letter C." Cropping meant cutting off part of the ear, and the C was for counterfeiting. He had been punished for faking Rhode Island's paper money. The newspaper does not name who cut and branded him.

New York judges later convicted him of faking and passing New York's bills, and he was hanged on May 10, 1756.

The Conestoga killings, 1763

Armed white settlers from Paxton Township and nearby places in Pennsylvania killed all twenty people of the Conestoga town in December 1763. No one was ever charged with the killings. The Conestoga were descendants of the Susquehannock people. Their town stood near Lancaster, on land that the Penn family had set aside for them. The land was supposed to be under the Penn family's protection.

Native nations to the west were at war with the British in 1763, in what is called Pontiac's War. The killers claimed that the unarmed Conestoga were hiding spies for the enemy. The historian Michael Goode calls that charge "a fabrication," which means a lie. Earlier in the war, Lieutenant Governor James Hamilton had offered a reward of 25 pounds for each Native scalp. A scalp is the skin and hair cut away from the top of a person's head.

The first attack. Benjamin Franklin wrote about the killings in a pamphlet in 1764. He counted twenty people in the town: seven men, five women and eight children. At daybreak on December 14, 1763, fifty-seven men rode in with

guns, short swords and hatchets. They found six people at home, three men, two women and a boy. The men shot them, stabbed them and killed them with hatchets.

Shehaes was a very old man who had been at a treaty meeting with William Penn in 1701. He was in his bed, and they cut him to pieces there. The attackers scalped all six. Then they burned the houses.

The workhouse. Lancaster's magistrates put the other fourteen Conestoga people in Lancaster's workhouse, a building where poor people were made to work, to keep them safe. On December 27 about fifty of the Paxton men broke open the workhouse door. Franklin wrote that the families knelt, with the children holding on to their parents, and said they were innocent.

The men killed all fourteen with guns and hatchets, among them children about a year old and babies still nursing. Will Soc was shot at the door. A company of British soldiers stationed nearby did nothing to stop the attack.

Franklin named some of the dead. They included Shehaes, his daughter Peggy, and her husband John Smith, a Cayuga, with their child of about three. They also included Betty and her son Peter, and Sally, whose Native name was Wyanjoy.

Governor John Penn offered a reward for the killers' names. In January 1764, as many as 200 of the Paxton men marched toward Philadelphia. They said they were protesting the government's failure to protect them, and they wanted to reach the Native refugees on Province Island, near the city. They stopped at Germantown when they heard that the city's militia had gathered. Franklin worked out a compromise: the marchers agreed to break up, and their leader, Matthew Smith, was allowed into the city to publish their complaints.

No one ever came forward with the killers' names. A modern introduction to Franklin's pamphlet, published by Miami University, states that no one was ever charged and tried.

Three trials in Boston, 1770

Judges in Boston heard three murder trials in 1770. One was of a customs informer who had shot a boy. The other two were of a British captain and of eight British soldiers.

Ebenezer Richardson fired a gun from his house into a crowd of boys on February 22, 1770, and killed Christopher Seider, who was 10 or 11. Richardson was a customs informer, a man who told on smugglers to the customs officials, the officers who collected taxes on goods brought in by ship. A jury found Richardson guilty of murder on April 21. The judges thought he was innocent, and they put off sentencing him.

The soldiers. British soldiers fired into a crowd on March 5, 1770, and killed five people. Crispus Attucks, Samuel Gray and James Caldwell died that night. Samuel Maverick, 17, died the next morning, and Patrick Carr died nearly two weeks later. Captain Thomas Preston and eight soldiers were arrested and jailed.

No lawyer in Boston would defend them until John Adams agreed, with Josiah Quincy and Robert Auchmuty. Adams later wrote that in a free country, a lawyer "ought to be the very last thing that [an] accused Person should want." In his day, want meant lack: an accused person must never be left without a lawyer.

A jury found Preston not guilty on October 30. The soldiers' trial began on November 27. After two and a half hours, the jury found six soldiers not guilty. It found Hugh Montgomery and Mathew Kilroy guilty of manslaughter, killing someone without planning to, and manslaughter could be punished by death. The two men used an old English rule called benefit of clergy, which let a first-time offender who could read escape a death sentence once.

Instead of being hanged, each man was branded on the hand, at the base of the right thumb. On December 14, 1770, Sheriff Stephen Greenleaf pressed a red-hot iron onto each man's thumb in the courtroom, and it burned a mark into the skin that stayed for life. John Adams remembered in 1822 that when the sheriff came near, the two soldiers "burst into tears." The *Boston Gazette* reported that after the branding they were "discharged," which means let go.

From street work to the cell, Philadelphia

Pennsylvania's lawmakers tried two new punishments in place of hanging between 1786 and 1790. The historian Louis Masur found that in the ten years before 1786, Pennsylvania officials had hanged about 40 people for robbery or burglary.

The wheelbarrow men. The lawmakers passed a law in 1786 under which judges sentenced convicts to "continued hard labour publicly and disgracefully imposed." People called the prisoners "wheelbarrow men." Ann Warder wrote in her diary on March 30, 1787, that the convicts cleaned the streets with "an iron collar around their neck and waist to which a long chain is fashioned and at the end a heavy ball."

Their heads were shaved, and a guard went with each gang. Warder wrote that at first the prisoners "preferred death" to being shown in public this way. Thirty wheelbarrow men escaped in one night in October 1788. The lawmakers ended the law by 1790.

The reformers. Thirty-seven Philadelphians formed the Philadelphia Society for Alleviating the Miseries of Public Prisons in 1787, a group to ease the suffering in the prisons. Among them were the doctor Benjamin Rush, Benjamin Franklin and Bishop William White. They drew on the ideas of John Howard, an English prison reformer.

In the city's old jail, men, women and children were held together, whatever their crime. Annie Anderson, writing in the *Encyclopedia of Greater Philadelphia*, states that the sheriff there forced money out of prisoners, sold them liquor from a bar and held back their food.

The cells. Pennsylvania's lawmakers passed a new law in April 1790. Under it the Walnut Street Jail became the Walnut Street Prison, a state prison for convicts from all of Pennsylvania. The lawmakers ordered a separate building of cells, called a penitentiary house, for "the more hardened and atrocious offenders," the convicts who had done the worst crimes. The word penitentiary comes from penitence, being sorry for wrongdoing.

Each cell was six feet wide, eight feet long and nine feet high, and thirty-six of them were built. Judges could sentence a prisoner to time alone in one of these cells, which is called solitary confinement.

The law professor David Shapiro studied the prison's records for the *Harvard Law Review* in 2019. He found that judges sent 748 people to the prison between 1795 and 1800, and sentenced 29 of them to time in solitary confinement. Margaret Price, convicted of counterfeiting in 1795, got four years with four months in solitary.

Prisoners worked in a nail factory inside the prison. Keepers locked a prisoner who refused to work in a cell. Caleb Lownes wrote in 1799 that such a prisoner stayed there "some weeks, without labour, bed, or furniture, of any kind, except a vessel to hold his drink, and another his mush, and a blanket."

Rights for the accused, 1791

Lawmakers in enough states approved the first ten amendments to the Constitution, called the Bill of Rights, on December 15, 1791. An amendment is an addition or change to the Constitution. Four of them list rights of people accused of crimes:

- The Fourth Amendment requires officers to have a warrant, a judge's written order, to search a home. A judge may give one only when there is good reason to believe a crime took place.
- The Fifth forbids forcing anyone to be a witness against himself. That means no one can be forced to confess, or to say anything that could help prove him guilty.
- The Sixth requires, for an accused person, "a speedy and public trial, by an impartial jury," a jury whose members favor neither side, and the help of a lawyer.
- The Eighth forbids "cruel and unusual punishments" and bail set too high. Bail is money paid to the court so an accused person can go free until the trial.

At first these rules applied only to the federal government and its courts. State courts did not have to follow them.

Patrick Lyon

Who: A Philadelphia blacksmith held three months in the Walnut Street Prison for a bank robbery he did not commit.

When and where: Philadelphia, 1798 to January 1799.

He walked 150 miles to clear his name and was locked up.

Patrick Lyon, a Philadelphia blacksmith born in Scotland, did a rush job for the Bank of Pennsylvania in 1798. He changed the locks on two of its iron vault doors. It was his last job before he left the city. He told the bank's carpenter that the doors were not right for a bank and the locks were not safe. A stranger came to his shop with the carpenter and spent a long time looking at the doors, keyholes and locks.

Then yellow fever spread through Philadelphia, and about 1,300 people died of it. Lyon left the city by boat for Lewes, Delaware, with his 19-year-old apprentice, a young man learning his trade. The apprentice died of the fever there.

On the night of August 31, while Lyon was away, someone took $162,821 from the bank's vault without breaking in. Lyon heard that he was the main suspect, and he walked back to Philadelphia to clear his name. "I had come 150 miles to surrender myself," he wrote.

High Constable John Haines, the chief of the city's constables, wanted the $2,000 reward. He and the bank's officers questioned Lyon. The bank's officers were sure that the man who changed the locks had made an extra key. A magistrate sent Lyon to the Walnut Street Prison and set his bail at $150,000.

Lyon called the prison "cold, damp, unwholesome and solitary." His cell was 12 feet by 4 feet. "I read until I was tired and walked until I was weary," he wrote. He caught yellow fever there and lived.

The robbers were Isaac Davis, a carpenter, and the bank's porter, Thomas Cunningham, who died of the fever. Davis was the stranger who had studied the locks. He confessed and gave back the money, and the

governor promised him a pardon. Davis seems never to have spent a day in prison. The bank's officers still said Lyon had helped by making a key. He stayed in prison until January 1799, when a grand jury refused to charge him.

Lyon told what had happened to him in a book, his 1799 *Narrative*. He sued the bank's officers and the constable for keeping him in prison without cause. A jury awarded him $12,000 in 1805. The men he sued appealed, and in 1807 they paid him $9,000 to settle the case. The painter John Neagle later painted him as *Pat Lyon at the Forge*.

1800 to 1850

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In the new prisons of New York and Pennsylvania, keepers locked convicts, people found guilty of crimes, alone in cells: at night in New York, and all day and night in Pennsylvania. Both kinds of prison kept prisoners silent. Officials in Boston and New York hired full-time police. Pennsylvania's lawmakers moved hangings inside jail walls in 1834, and Michigan's lawmakers voted to end the death penalty twelve years later.

In the South, white men still policed enslaved people. South Carolina's slave patrols, small groups of white men on horseback, searched the places where enslaved people lived and broke up their gatherings. Virginia's local judges, called justices, tried enslaved people with no jury. White mobs also killed people with no trial at all. A white mob in St. Louis burned Francis McIntosh to death in 1836, and no one was charged.

The Cherokee Nation's police, 1808

Under the old Cherokee law of blood, a dead person's clan, not the nation, punished a killing. Leaders of the Cherokee Nation began to change that in 1808, when they set up their own police. Two years later the clans gave up revenge.

The Cherokee lived then in their own country, in parts of what are now Georgia, Tennessee, Alabama and North Carolina. The leaders on the Cherokee national council, who made the nation's laws, passed a written law at Brooms Town on September 11, 1808. Under it they set up "regulating parties," companies of men who made sure people obeyed those laws. The principal chief, or head leader, Black Fox signed it with Path Killer, Toochalar, and Charles Hicks, the council's secretary.

The regulators. Each company had six men: a captain, a lieutenant who was second in command, and four privates. The law states the pay: fifty dollars for the captain, forty for the lieutenant and thirty for each private. The regulators went after horse thieves and robbers. They also protected what children got from their dead parents and what widows owned.

A man found guilty of stealing a horse got "one hundred stripes on the bare back," one hundred blows of a whip. Blows like these cut the skin and leave welts, raised stripes of swollen skin, and scars.

If a regulator killed a man who fought back with a weapon, "the blood of him or them shall not be required." That meant the dead man's clan, his mother's large family group, could not kill in return.

The end of clan revenge. The seven clans met in council at Oostanallah on April 10, 1810, and passed an "act of oblivion for all lives." Oblivion means forgetting: the clans forgave every life that one clan still owed another. From then on, a person who killed someone "without malice intended," meaning by accident, would not be held guilty. A man who killed his brother in anger would be guilty of murder.

With this act the clans ended the law of blood. The editors of the *Cherokee Phoenix*, the nation's own newspaper, printed both laws on March 13, 1828.

Two kinds of prison: Auburn and Eastern State

Prison keepers in New York and Pennsylvania ran two different kinds of prison in the 1820s. In both, they kept prisoners from talking to each other.

Auburn. At Auburn Prison in New York, prisoners worked together in groups during the day and slept alone in cells at night. They were not allowed to speak at any time. Elam Lynds became the head keeper in 1821. He had prisoners flogged for breaking even small rules. Flogging means beating a person's bare back with a whip, which cuts the skin and leaves welts and scars.

Lynds and John D. Cray made prisoners march in step with their arms locked together, a way of marching called the lockstep. Lynds dressed them in striped uniforms.

New York's officials used the same plan at Sing Sing, their third state prison. Officials in Connecticut, Massachusetts, Maryland and Washington, D.C., had taken up the Auburn plan by 1829. Officials in most other states chose it later.

Eastern State. Pennsylvania's officials opened Eastern State Penitentiary, a prison in Philadelphia, in October 1829. The architect John Haviland laid out seven blocks of cells joined to a center hall. He said he designed it for "watching, convenience, economy and ventilation." By economy he meant saving money, and by ventilation, fresh air.

Each prisoner lived and worked alone in a cell about 8 feet wide and 12 feet long, with its own small exercise yard. A Bible was the only thing he was allowed to keep. Prisoners made shoes and wove cloth in their cells. Whenever a prisoner left his cell, a guard put a hood over his head so that no other prisoner could see his face.

The iron gag. Guards punished prisoners who broke the rules. They gave them cold showers, and one account states that guards apparently did this once outdoors in winter. They also used the iron gag. The guards clamped an iron piece over the prisoner's tongue, like the bit in a horse's mouth, and chained it to his hands, which were held behind his back. With his hands chained to it, he could not take it out.

Prison officials put a prisoner named Mathias Maccumsey in the iron gag in 1833 as a punishment for talking, and he died. The prison doctor gave the cause as apoplexy, an old word for a sudden stroke. The historian Jennifer Lawrence Janofsky, writing in the *Encyclopedia of Greater Philadelphia*, states that the prison's leaders tried to cover up the death.

A committee of Pennsylvania lawmakers investigated charges in 1834 and 1835 that the prison's officials had been cruel to prisoners. The lawmakers did not find the warden, Samuel Wood, guilty of cruel and unusual punishment.

The English writer Charles Dickens visited Eastern State in 1842. He wrote that he believed the system "in its effects, to be cruel and wrong." He also wrote: "I hold the slow and daily tampering with the mysteries of the brain to be immeasurably worse than any torture of the body." He meant that keeping a person alone and silent for years did more harm to the mind than any torture does to the body.

Pennsylvania's prison officials gave up keeping prisoners alone in 1913. Eastern State was built for 250 prisoners, and by the middle 1920s it held about 1,700.

Charles Williams

Who: The first prisoner at Eastern State Penitentiary.

When and where: Philadelphia, October 1829 to October 1831.

Prisoner Number 1 got two years alone for stealing a watch.

Charles Williams was an 18-year-old Black farmer. He had been found guilty of breaking into a house and stealing a silver watch, a gold seal and a gold key, worth $25 in all. A seal was a small stamp, often hung on a watch chain, for pressing a mark into wax. His sentence was two years alone and in silence, with work.

Guards led him into the new prison in October 1829 with a hood over his head. From then on he was Prisoner Number 1, and Jane Brox, who tells his story in the *Public Domain Review*, writes that he was not called Charles again while he was there.

His cell was twelve feet by eight. *Smithsonian* magazine reports that prisoners there spoke to the guards through a small feeding hole and worked alone in their cells. He was let out in October 1831, two years to the day after he went in.

After Nat Turner's revolt, 1831

Nat Turner, an enslaved man, led a revolt in Southampton County, Virginia, in August 1831. Turner and the rebels who joined him killed 55 white men, women and children, by the count of the historian Patrick Breen. A Library of Virginia article gives the number as 57.

White people from Southampton and beyond killed Black people with no trial in the days after the revolt. Breen writes that they killed about three dozen in Southampton County. He also writes that armed units formed "to put down the revolt and in many cases get revenge." A Library of Virginia article states that they killed hundreds, with no exact count.

The county's justices tried the accused rebels in a court of oyer and terminer. In Virginia that meant local justices judged enslaved people with no jury. Breen counts thirty enslaved people and one free Black man sentenced to death. Nineteen were put to death on the justices' orders. Governor John Floyd changed the sentences of the other twelve so they would not die.

Turner hid for two months. He gave himself up on October 30, 1831. The justices tried him on November 5 and ordered him "hanged by the neck until he be dead" on November 11. He was hanged that day, as the justices had ordered. In the same court record, the justices set his value as property at $375. Virginia officials paid that sum to his owner.

Hangings move inside the jail

Pennsylvania's lawmakers became the first in any state to ban public executions, hangings in front of crowds, in 1834. The Death Penalty Information Center, a group that collects facts about the death penalty, records this. After that, officials hanged prisoners who had been sentenced to death inside the county prisons.

Michigan's lawmakers voted to end the death penalty on May 18, 1846. An article by the news outlet Michigan Public states that Michigan was the first English-speaking government in the world to do so.

Francis McIntosh, St. Louis, 1836

A white mob in St. Louis, Missouri, burned Francis McIntosh alive on April 28, 1836, and no one was ever charged. McIntosh was a free Black man, 26 years old, from Pittsburgh. He worked on the steamboat *Flora* as a cook and as a porter, who carried passengers' bags.

That day Deputy Constable William Mull, a local law officer, arrested McIntosh for getting in the way of officers who were arresting two sailors. Deputy Sheriff George Hammond, a county law officer, helped take him to jail. McIntosh was told he would get five years in prison. He stabbed both officers with a knife. Hammond bled to death. Mull was badly wounded, and accounts differ on whether he died later.

A crowd of several hundred white people gathered, and a city alderman, a member of the city council, was among them. Men from the crowd broke into the jail and took McIntosh out. They chained him to a locust tree near Seventh and Chestnut Streets, then piled wood around his knees and set it on fire. The crowd cheered as he burned, according to a history written at the University of Missouri-St. Louis. He died within about twenty minutes. By one report, he asked to be shot.

What the judge told the jurors. Judge Luke Lawless then spoke to the grand jury. He told the jurors to charge any person they could identify. But he said the killing might be the act of a whole crowd moved by one feeling, and if

it was, it was beyond the reach of the law. He also blamed the writings of abolitionists, people who worked to end slavery. The jurors, led by their foreman, the head juror, John O'Fallon, charged no one.

Elijah Lovejoy, the editor of the *St. Louis Observer*, wrote against the killing and against the judge. A mob of people who were against ending slavery killed Lovejoy in Alton, Illinois, in November 1837.

A young lawyer named Abraham Lincoln spoke about the McIntosh killing in Springfield, Illinois, in January 1838. He said McIntosh had been "chained to a tree, and actually burned to death." He added that it was "all within a single hour from the time he had been a freeman, attending to his own business, and at peace with the world." Lincoln's words left out the stabbing and Hammond's death.

Police take over from the night watch

The people of Boston had kept a town watch, men who walked the streets at night to keep order, since 1631.

Boston. A small day police force began work in Boston in 1838, with officers on duty full time beside the night watch. The crime historian Gary Potter calls this the first organized police force paid for with public money and with officers on duty full time, a 2017 *TIME* article reports. Boston's city officials replaced the watch and the day police with one Police Department of 250 officers in 1854.

New York. About 320,000 people lived in New York before 1845. A night watch and 182 other officers guarded them: 100 city marshals, 31 constables and 51 city police officers, all kinds of local law officers. New York's state lawmakers passed a Municipal Police Act, a law for a city police force, on May 7, 1844. Under it, city officials were to end the night watch.

The mayor and the city council argued over who would choose the officers. Mayor William Havemeyer and the council finally adopted the law for the city on May 23, 1845. Accounts differ on the size of the force: the website

CultureNow states that the 1844 law allowed 1,200 officers, and another count gives 800. After a long argument, the officers first wore a navy blue uniform in 1853.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 21 MINUTES

Mobs killed people with no trial in the South, in the West and in New Orleans, and the men in those mobs were almost never punished. After the Civil War, white mobs in the South killed Black people. A killing like this, by a mob, with no trial, is called a lynching. Mobs in the West killed Chinese immigrants, and a mob in New Orleans killed eleven Italian American men.

The count. The Equal Justice Initiative, a group in Montgomery, Alabama, founded by the lawyer Bryan Stevenson, published a report called *Lynching in America* in 2017. It records 4,075 Black people lynched in twelve Southern states between 1877 and 1950, more than one a week for 73 years. Counting lynchings in other states too, its total is more than 4,400.

The report states that the people who took part "were celebrated and acted with impunity," meaning with no fear of punishment, and that state and federal officials mostly let the killings happen.

Private detectives such as Allan Pinkerton's men worked for railroads and factory owners. Officials in Alabama and other states rented prisoners to mining companies and farm owners, and in Alabama more than 95 percent of the county prisoners and 90 percent of the state prisoners were Black. New York state senators on a committee that met in 1894 and 1895 estimated that New York City's police officers took more than $10 million a year in payments from people breaking the law.

Vigilance committees in San Francisco, 1851 and 1856

A vigilance committee was a group of private citizens who arrested, tried and punished people themselves, outside the courts. About 700 San Francisco men formed one on June 9, 1851. The next day they held their own trial and hanged John Jenkins, a man from Sydney, Australia, for stealing a safe. At that time California law allowed a death sentence for that kind of theft.

That year the committee's members hanged four men. They held and questioned people at their headquarters without due process, the rules a court must follow to give an accused person a fair trial. Of the others they held, they whipped one, forced 28 out of California, handed 15 to the police and courts, and let 41 go.

San Francisco men formed the committee again on May 14, 1856, after James Casey shot the newspaper editor James King of William. William Tell Coleman led it, and its leaders claimed 6,000 members. The committee's members hanged four more men that year, among them Casey and Charles Cora. The boxer James "Yankee" Sullivan killed himself after being held in one of the committee's cells. The members broke up the committee on August 11, 1856. Members of the two committees hanged eight people in all.

The Dakota trials, 1862

US Army officers tried 392 Dakota men after the war between the United States and the Dakota in Minnesota in 1862, and hanged 38 of them together at Mankato. The Minnesota River valley was the Dakota homeland.

Colonel Henry Sibley set up a military commission, a court of army officers, to try Dakota prisoners on September 28, 1862. The officers held the trials at the Lower Sioux Agency, also called the Lower Agency. US officials set it up in 1853 to run the new Dakota reservation, and it had a school, shops and homes for the government workers. The war began there in August 1862.

Five-minute trials. The trials took place in François LaBathe's summer kitchen, one of the only buildings there still standing. Some trials lasted no longer than five minutes. The officers sentenced 303 men to death and 16 to prison.

The law professor Carol Chomsky of the University of Minnesota wrote that "the evidence was sparse," meaning thin, and that the officers were "biased," meaning they had taken a side before they heard the cases. She wrote that the men on trial had no lawyers, and that the trials were held in a language they did not know.

President Abraham Lincoln and his lawyers read all 303 trial records. Lincoln wrote to the Senate that too much mercy might encourage another war, and that too much harshness would be "real cruelty."

He looked first for men found guilty of rape, which means forcing a person into sex, and he found two. Next he added men found to have taken part in "massacres," the killing of many civilians, people who were not soldiers. Men found only to have fought in battles stayed off his list. He approved 39 hangings, and one of those men was later reprieved, which means spared from hanging.

Army officers hanged 38 Dakota men together at Mankato on December 26, 1862. About 4,000 people watched, and 2,000 soldiers stood guard. An article by the Death Penalty Information Center states that it was the largest mass execution in United States history. Two of the 38 were not men Lincoln had

approved for hanging: the officers hanged them by mistake. One was Caske, whose Dakota name was Wicaŋhˇpi Wastedaŋpi. The other was Wasicuŋ, a white man the Dakota had adopted, whom the officers had found not guilty.

The bodies were cut down and put together in one shallow grave on a sandbar between Mankato's main street and the Minnesota River. Before morning, doctors had dug up most of them to use as cadavers, dead bodies that doctors and medical students cut open to study the human body.

A year later, the army's chief lawyer found that Sibley had never had the power to set up the trials. By then the 38 men had already been hanged.

Police against the draft riots, 1863

Mobs in New York City attacked the police, burned buildings and killed Black New Yorkers from July 13 to 16, 1863. The rioters were fighting the new Civil War draft, a law that required men to serve in the army when called. The state militia, the state's part-time soldiers, had been sent to Gettysburg, so at first the Metropolitan Police were the only force in the city.

On the first day, rioters beat Police Superintendent John Kennedy, the head of the city's police officers, until he was nearly unconscious and cut his hand with a knife. Police Commissioners Thomas Acton and John Bergen, two of the officials who ran the police, took command. Their officers kept the rioting out of Lower Manhattan, below Union Square. Rioters burned two police station houses. They also burned the Colored Orphan Asylum, a home for Black children, and the more than 200 children inside all got out.

Rioters beat, tortured and killed Black New Yorkers. Village Preservation, a New York history group, records that William Jones, a Black cart driver, had gone out to buy a loaf of bread. A mob chased him on Clarkson Street, beat him, hanged him and set him on fire. Another Black man was attacked by a crowd of 400, hanged from a tree and set on fire. Hundreds of Black New Yorkers left the city, among them the doctor James McCune Smith and his family.

Secretary of War Edwin Stanton sent five regiments, units of soldiers, from Gettysburg to stop the riot, and it ended on July 16. By the official count, 119 or 120 people were killed in the riots. A New-York Historical Society article records the official count of 119 and states that it may be too low.

Mobs that killed Chinese immigrants

White mobs in the West killed Chinese immigrants. After the killings at Los Angeles in 1871 and at Rock Springs in 1885, every man who was charged or arrested went free. The justices of the California Supreme Court had ruled in 1854 that Chinese people could not testify against white people in court.

Los Angeles, 1871. A white man named Robert Thompson was killed in Los Angeles on October 24, 1871, during a gunfight between two Chinese groups. A crowd of about 500 people, one tenth of the city, then attacked the Chinese part of town. They killed 18 Chinese men and boys, by the count of the Los Angeles Public Library, or 19 in the list of names the historian Scott Zesch published in 2008. Among the dead was a doctor, Gene Tong.

The mob hanged seven of them from the crossbar of Goller's wagon shop and three from a freight wagon. The next morning seventeen bodies lay in the jail yard.

Grand jurors in Los Angeles brought 25 charges of murder. Only ten men were tried, and seven or eight of them were found guilty of manslaughter. Judges later threw out the verdicts on a technical point of law. District Attorney Cameron Thom chose not to try the men again, and they went free in 1873.

Rock Springs, 1885. Rock Springs was a town in Wyoming, which was then a territory, land under the federal government that was not yet a state. Between 100 and 150 armed white miners, railroad workers and townspeople attacked the Chinese miners there on September 2, 1885. They killed 28, wounded 15, and robbed and burned all 79 homes in the town's Chinatown. Federal troops arrived nearly a week after the killings and stayed for 13 years.

Sixteen white miners were arrested and let out on bail, money paid as a promise to come back for trial. The historian Tom Rea, writing for WyoHistory.org, a Wyoming history website, states that "though the killing

had been done in daylight, in front of other people, no one could be found who would swear to having seen any crimes." No charges were filed. Chinese diplomats, officials who spoke for China's government, listed nearly $150,000 in losses. Under pressure from the president, members of Congress agreed to pay.

BY 1885: San Francisco's committees had hanged eight men, army officers had hanged 38 Dakota men at Mankato, and every man charged or arrested for the killings of Chinese immigrants at Los Angeles and Rock Springs had gone free.

Prisoners rented out: convict leasing

State and county officials in Alabama and other states rented prisoners to private companies after the Civil War. This was called convict leasing. The companies' bosses made the prisoners work in mines, on farms and on railroads. The companies' owners paid the state or the county, not the prisoners.

The historian Mary Ellen Curtin gives the Alabama numbers in the *Encyclopedia of Alabama*. Officials there leased prisoners from 1875 to 1928. More than 95 percent of the county prisoners and 90 percent of the state prisoners there were Black. Most county prisoners had been found guilty of misdemeanors, small crimes such as being drunk in public. They had to work longer to pay off their court costs.

The whip and the deaths. The companies' bosses whipped prisoners who tried to escape or talked back, and they whipped others to make them work harder. Each year, four or five of every hundred state prisoners died. Of the 745 county prisoners in the mines in 1896, 181 had died in the two years before. At the Coalburg mine, 90 of every 1,000 prisoners died.

Allan Pinkerton

Who: A Scottish barrel maker who started a private detective agency in Chicago.

When and where: Chicago and across the country, 1850 to 1884.

His sign was an open eye over the words "We never sleep."

Allan Pinkerton's detectives were paid by the companies that hired them. They were not police officers. Pinkerton was a cooper, a barrel maker, from Glasgow, Scotland, and in 1843 he settled in Dundee, Illinois. His house there was a hiding place for people escaping slavery. While cutting wood one day he found a gang of counterfeiters, people who made fake money, and told the sheriff, who arrested them.

Pinkerton and the lawyer Edward Rucker started a detective agency in Chicago in 1850, which became the Pinkerton National Detective Agency. The men who ran express companies, which carried money and packages by train, hired his detectives to catch criminals, and so did the men who ran the railroads.

His men kept photographs of known criminals, and some of them joined gangs in disguise. Pinkerton guarded Abraham Lincoln on his way to Washington in February 1861, against what Pinkerton said was a plan to kill him in Baltimore.

Pinkerton's detectives and local helpers raided the Missouri farm of the family of the outlaw Jesse James, a robber on the run from the law, in January 1875. They threw a firebomb into the house. It exploded and killed Jesse's eight-year-old half-brother, Archie. It wounded his mother, Zerelda Samuel. Jesse was not there. Afterward, many people took the side of the James family and blamed the agency for the raid.

Factory owners also hired Pinkerton men after the Civil War as armed guards against workers on strike, workers who had stopped work together to press their employers for changes. Pinkerton died in Chicago in 1884.

The Haymarket trial, 1886

Someone threw a bomb at police at a workers' meeting in Haymarket Square in Chicago on May 4, 1886. Seven officers were killed and sixty were hurt. The bomb thrower was never identified.

Prosecutors charged eight anarchists, people who wanted to do away with all government. They tried the eight as accessories, people who help a crime happen without doing the act. The prosecutors said the men's speeches had led to the bombing. A jury found all eight guilty in three hours. The judge sentenced seven to hang and Oscar Neebe to 15 years in prison.

Four of the men, Albert Parsons, August Spies, George Engel and Adolph Fischer, were hanged on November 11, 1887, as the judge had sentenced them. Louis Lingg had already killed himself in his cell with an explosive someone had smuggled in. Governor Richard Oglesby changed the sentences of two men to life in prison.

Governor John Peter Altgeld pardoned the three men still in prison, Samuel Fielden, Oscar Neebe and Michael Schwab, in June 1893. The pardon freed them and cancelled their punishment. Altgeld wrote that their trial had been unfair.

The first electric chair, 1890

William Kemmler became the first person put to death in an electric chair, on August 6, 1890, at Auburn Prison in New York. He had killed Matilda Ziegler in Buffalo in 1889 with a hatchet, or with an axe by one account. Electrocution means killing a person by sending a strong electric current through the body. The current stops the heart and burns the skin where the metal touches it.

A Buffalo dentist, Dr. Albert Southwick, had first suggested it in 1881. He argued that it was kinder than hanging, in which a person sentenced to death could hang for up to 30 minutes before dying. Kemmler's lawyers argued that the electric chair was a cruel and unusual punishment, which the Constitution forbids. The justices of the US Supreme Court turned them down.

Kemmler's chair had two metal plates covered with damp sponges, one on his head and one on his back. Warden Charles Durston, the head of the prison, gave the order, with 17 witnesses in the room. The first shock lasted 17 seconds. It was about 700 volts by one account, or 1,000 by another. A volt is a measure of how strongly electricity pushes through a wire or a body.

The first shock did not kill him. The witnesses smelled burning cloth and burning flesh. A second shock of about 1,030 volts went on for about two minutes, until smoke rose from his head, and Kemmler died. Doctors who examined the body found that the plate on his back had burned through to his spine, the HISTORY.com account states.

The chair ran on alternating current, a kind of electricity that flows back and forth many times a second. Harold Brown, whom New York officials hired to build the chair, used it. *Smithsonian* magazine reports that Thomas Edison, whose company sold the other kind, direct current, was secretly paying Brown to use alternating current. George Westinghouse's company built electric systems that used alternating current. After Kemmler's death Westinghouse called it "a brutal affair" and said: "They would have done better with an axe."

Eleven men killed in a New Orleans prison, 1891

A mob in New Orleans broke into the city's prison on March 14, 1891, and killed eleven Italian American men. Five months earlier, the police chief, David Hennessy, had been shot. Nineteen Italian Americans were charged with his murder, among them a 14-year-old boy. Nine were tried. The jury found six not guilty, and the trial of the other three ended in a mistrial, with no verdict. A *Smithsonian* magazine article states that "there was no solid evidence against them."

A newspaper notice stated that citizens should "come prepared for action." A crowd of thousands gathered at the Henry Clay statue. The lawyer William Parkerson told them to "set aside the verdict of that infamous jury" (infamous means known for something bad).

Who led the mob. Parkerson and three other city leaders led the killers: the lawyer Walter Denegre, the politician and businessman James Houston, and John Wickliffe, the editor of the *New Delta* newspaper. John M. Parker, who later became governor of Louisiana, and Walter Flower, who later became mayor of New Orleans, were in the mob.

The prison warden, Lemuel Davis, let the 19 prisoners hide. The mob found eleven of them. Men dragged Emmanuele Polizzi outside, hanged him from a lamppost and shot him. He was mentally ill. They hanged Antonio Bagnetto, a peddler who sold fruit, from a tree and shot him. They shot or clubbed to death nine others inside the prison.

The eleven were Antonio Bagnetto, James Caruso, Loreto Comitis, Rocco Geraci, Joseph Macheca, Antonio Marchesi, Pietro Monasterio, Emmanuele Polizzi, Frank Romero, Antonio Scaffidi and Charles Traina. Three of them had been found not guilty, three had had no verdict, and five had never been tried.

No one charged. Grand jurors, who met under Judge Robert Marr, a friend of several men in the mob, said they could not identify the killers. They called the killers "several thousands of the first, best, and even the most law-abiding, of the citizens." They charged no one.

Italian officials called home their ambassador, their government's chief representative in the United States. President Benjamin Harrison agreed to pay the families $25,000 to make up for the killings, and federal officials paid $2,211.90 to each family. Mobs lynched at least eight more Italian Americans in Louisiana in the 1890s. No one was charged and tried in court for those killings either.

Henry Smith

Who: A Black man burned to death by a white mob in front of thousands of people, without a trial.
When and where: Paris, Texas, February 1, 1893.

Thousands came by train to watch a man burned to death.

A white mob in Paris, Texas, tortured Henry Smith with hot irons and burned him to death on February 1, 1893, in front of thousands of people. He never had a trial, and no one was ever arrested for killing him.

The accusation. Myrtle Vance, a girl of three or four, was found dead in the woods near Paris in January 1893. Newspaper reports said that she had also been raped, which means forced into sex. The *New York Sun* called Smith her "ravisher," an old word for a man who rapes someone.

Her father, Henry Vance, was a Paris police officer. Several days before, he had arrested Henry Smith for being drunk and disorderly and had beaten him. Smith was accused of killing the girl. The two main accounts of what followed, by the journalist Ida B. Wells and the historian Terry Anne Scott, do not name the person who first accused him. Smith ran away to Arkansas.

Posses of Black and white men went after him. A posse is a group of men gathered to chase someone. They rode the Texas & Pacific and the St. Louis & San Francisco railways for free. Near Hope, Arkansas, a Black constable named Robinson, a local law officer, arrested Smith. Texas Governor James Hogg telegraphed Sheriff D. S. Hammond to hold Smith

for a trial. Hogg also ordered that Smith not be taken to Paris. The sheriff and the county attorney answered that they were "helpless" against "an enraged public."

Men brought Smith to Paris by train. The mayor closed the schools. Trains brought people from Dallas, Fort Worth, Texarkana, Fort Smith in Arkansas, and other towns. The *New York Sun* stated that the crowd numbered 10,000, and other accounts state from 5,000 to 15,000. Men paraded Smith through town on a carnival float to a prairie near the railroad depot. There they tied him on a wooden platform ten feet high.

The torture. Henry Vance and men of his family burned Smith with red-hot irons for fifty minutes, starting at his feet. The *Sun* identifies them as the girl's father, her brother and two uncles. Another account identifies the father, his 15-year-old son and his brother-in-law. They burned out Smith's eyes and pushed hot irons down his throat.

Then men piled wood and cottonseed hulls, the dry outer shells of cotton seeds, around the platform, poured oil on it and set it on fire. The *New York Sun* reported the torture and the burning on February 2, 1893, and Wells reprinted the report in her book *The Red Record* in 1895. The *Sun* states that people carried off pieces of the burned wood.

Scott, writing on the website BlackPast, adds that Smith rolled out of the fire and fell ten feet to the ground. People in the crowd threw a rope around him and pulled him back into the fire. She writes that men cut his shirt from his body piece by piece and threw the pieces to the crowd.

What the witnesses said. The *Sun* stated that Smith had confessed at Hope. People who had seen the girl's body told Wells they saw no sign of the rape the newspapers reported. They said they saw only a small scrape and a bruise, mostly on her neck.

The Reverend King, a Black minister in Paris, said Smith was mentally ill. King said he had tried two years before to get him into an asylum, a hospital for mentally ill people. Men drove King out of Paris for speaking

against the lynching. Wells wrote that he was "ridden out of Paris on a rail," carried out of town on a wooden pole.

The day after Smith was killed, Governor Hogg telegraphed the sheriff of Lamar County and asked for the killers to be arrested. A few days later a mob hanged Smith's stepson, William Butler, at Hickory Creek, five miles from Paris, and shot his body. Scott writes that no one was ever punished for killing William Butler either.

Police who took payments: New York, 1894

Senator Clarence Lexow and other New York state senators, sitting as a committee, investigated the New York City police in 1894 and 1895. They found that officers took payments from people who broke the law, in return for letting them go on breaking it. Their report ran to 10,576 pages. In it, they estimated that officers extorted more than $10 million a year, forcing people by threats to pay.

The owners of brothels, houses where people paid for sex, paid police captains $30,000 a year. Owners of gambling houses paid $15 to $300 a month to stay open. Officers also paid bribes, secret payments for a favor, to be promoted. Captain Timothy Creedon told the committee he had paid $15,000 to become a captain, although he had scored 97.82 on the promotion test.

An Encyclopedia.com history of the hearings states that the chief of police, Thomas Byrnes, was said to be worth $350,000. The senators also found officers taking part in bribery, frightening voters, cheating in elections and beating people.

Roosevelt's changes. Mayor William Strong put Theodore Roosevelt on the city's Board of Police Commissioners in 1895. Roosevelt brought in civil service exams for new officers, tests that a person had to pass to get a government job, and raised the physical standards. He walked the streets at midnight to check that officers were on duty, and he started police patrols on bicycles. He left in 1897 to become assistant secretary of the navy.

A court for children, Chicago, 1899

Illinois lawmakers voted to set up a court for children in Chicago in the spring of 1899. A 2019 article by the Chicago radio station WBEZ states that it was the first juvenile court in the world: a separate court for children accused of crimes, run to help the child instead of only punishing.

Before then, police arrested children, put them in handcuffs and held them overnight in jails and police stations with adult prisoners. In 1882 the Cook County jail in Chicago held more than 250 children aged 14 and under, and at least 20 of them were younger than 11. A 10-year-old boy who stole a pair of shoes waited almost two weeks in the county jail for a grand jury to hear his case.

Julia Lathrop worked at Hull House, a house on Chicago's Near West Side where Jane Addams and other women lived among poor immigrant families and helped them. Lathrop visited Illinois jails and wrote down what she found. Lucy Flower proposed a separate court for children. In 1907 a building for the children's court, with rooms where children were held, opened at Halsted and Ewing Streets.

BY 1900

If you remember one thing: White mobs killed Black, Chinese and Italian American people with no trial, and the courts punished almost none of the killers.

In 1850: A death sentence in New York was carried out with a rope, and a child arrested in Chicago could be held in jail with adult prisoners.

In 1900: New York's prisons used the electric chair, and Chicago had a court just for children, which WBEZ calls the first in the world.

Next: After 1900 the national government hired its own agents, and mobs went on taking Black prisoners out of jails.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 19 MINUTES

The national government built a police force of its own, while some officers in the South whipped Black suspects or left Black prisoners to mobs. The US attorney general put 34 federal agents, agents of the national government, to work in 1908. By the 1930s the staff of their Bureau of Investigation kept the nation's fingerprint files and worked in a crime laboratory. Federal agents hunted gangsters such as John Dillinger, and they raided the gangs that sold alcohol during Prohibition, the years when making and selling it was against the law.

Mobs still took Black prisoners out of jails and killed them. A deputy sheriff and other white men in Mississippi whipped three Black men until they confessed to a murder. The sheriff in Chattanooga sent his guards away in 1906 on the night a mob came for Ed Johnson. Three times in the 1930s, the justices of the US Supreme Court threw out verdicts because of how Black defendants had been treated.

White men, many of them made deputies by the police and given a police officer's powers, burned Greenwood, the Black district of Tulsa, in 1921 and killed as many as 300 people. No white person was sent to prison for it.

Ed Johnson

Who: A Black man wrongly convicted, or found guilty, of rape and sentenced to death, then lynched after the justices of the US Supreme Court ordered that he be kept safe.

When and where: Chattanooga, Tennessee, January to March 1906.

The justices ordered him protected, and the sheriff sent his guards away.

A jury found Ed Johnson guilty of a rape he did not commit. Rape means forcing a person into sex. The justices of the US Supreme Court stopped his hanging. A mob then lynched him on a bridge in Chattanooga, which means the men in it killed him with no trial.

A man attacked Nevada Taylor, a young white woman, from behind in January 1906, as she walked home from a streetcar stop. He choked her with a leather strap. A day or two later, officers arrested Ed Johnson, a Black man who worked as a porter at the Last Chance Saloon.

A witness, Will Hixson, may have given the police Johnson's name to collect a $375 reward. At the trial, thirteen witnesses swore that Johnson was at the saloon at the time of the attack. An all-white, all-male jury found him guilty of rape in February, and the judge sentenced him to hang. The court had chosen his lawyers, and they dropped his appeal. An appeal is a request for a higher court to look at the case again.

Two Black lawyers, Noah Parden and Styles Hutchins, took up the appeal. They asked a federal court for habeas corpus, an order to review whether a prisoner is held lawfully, and the federal judges turned them down.

Parden went to Washington and met Justice John Marshall Harlan of the US Supreme Court on March 17. Within two days the justices issued a stay, an order that stopped the hanging until they could hear the case, and ordered Sheriff Joseph Shipp to keep Johnson safe.

The night of March 19. The order appeared in Chattanooga's evening newspapers that day. That evening Shipp sent away the usual guards and left only the night jailer, Jeremiah Gibson. Between nine o'clock and midnight, a mob broke into the jail and took Johnson to the county bridge over the Tennessee River. Shipp came to the jail during the break-in and made no effort to stop the mob or to send for help.

The men put a rope around his neck, threw it over a beam and pulled him up. The rope broke or slipped, and he fell. They pulled him up a second time and shot him. One account says a deputy sheriff fired five more shots into his head. Johnson's gravestone gives his last words: "God Bless you all. I AM A Innocent Man."

A trial in the Supreme Court. For the first and only time, the justices of the US Supreme Court held a criminal trial themselves. The charge was contempt of court, the crime of defying a court's order. Twenty-five men were tried. The justices found six guilty in May 1909: Shipp, Luther Williams, Nick Nolan, Jeremiah Gibson, Henry Padgett and William Mayes.

Shipp, Williams and Nolan got 90 days in jail, and the other three got 60 days. Shipp was welcomed home as a hero. The case is *United States v. Shipp* (1909).

Judge Doug Meyer of the Hamilton County Criminal Court set aside Johnson's conviction in February 2000. He ruled that the all-white jury and the trial judge's refusal to move the trial out of Chattanooga had made it unfair. A memorial to Johnson was dedicated in Chattanooga on September 19, 2021.

A federal police force, 1908 to 1934

Attorney General Charles Bonaparte ordered the lawyers of the Department of Justice on July 26, 1908, to send most of their investigations to his chief examiner, Stanley W. Finch. Finch handed each one to one of 34 agents. Nine of the agents came from the Secret Service, and Bonaparte hired 25 himself. The group was later called the Bureau of Investigation. Today it is the Federal Bureau of Investigation, or FBI.

Attorney General Harlan Fiske Stone made J. Edgar Hoover, then 29, the Bureau's director in 1924. Hoover fired agents he called "political hacks," men who had their jobs as favors from politicians. He required new agents to pass background checks and physical tests, and from January 1928 he trained them. The Bureau had 441 agents in 1924 and 339 five years later.

Fingerprints. Hoover set up a division to hold the country's fingerprint files on July 1, 1924. A fingerprint is the pattern of ridges on a fingertip, and it is different for every person. A print left on a glass or a gun can show who touched it. Police anywhere could send in prints and ask whether they matched a known criminal.

Bureau officials began keeping the national crime counts in 1930. The FBI's own history gives November 24, 1932, as the official birthday of its crime laboratory, which agent Charles Appel did most of the work to build.

Gangsters. Gunmen killed four lawmen, one of them a Bureau agent, at the train station in Kansas City in June 1933. The FBI's history names a gangster known as "Pretty Boy" Floyd as one of the gunmen. Agents and local police killed Floyd in a shootout on an Ohio farm in October 1934.

John Dillinger was a different gangster, and his gang robbed banks across the middle of the country. Agents led by Melvin Purvis found Dillinger in Chicago in July 1934 and shot him dead as he left a movie theater. In these years members of Congress also gave the Bureau's agents the power to carry guns and make arrests.

Mobs and the courts, 1908 and 1930

White mobs attacked Black residents of Springfield, Illinois, for nearly three days in August 1908. They lynched two men, Scott Burton and William Donegan, and destroyed dozens of homes. The mobs drove nearly 2,000 Black residents out of the city for good. Officials called out the state militia, the state's part-time soldiers. About 150 people were arrested, and few of them were convicted, a National Park Service article states.

A white mob of 5,000 to 10,000 people broke into the Grant County Jail in Marion, Indiana, on August 7, 1930. They beat and hanged two Black men, Thomas Shipp and Abram Smith, both 19. A third prisoner, James Cameron, 16, survived. He was later convicted as an accessory, a person who helps with a crime, and served four years. He was pardoned, officially forgiven, in 1993.

The NAACP, a group that worked for the rights of Black Americans, named 27 suspects in the mob. All-white juries found the leaders who were tried not guilty.

Tulsa, 1921: no one sent to prison

As many as 10,000 white Tulsans attacked Greenwood, the Black district of Tulsa, Oklahoma, on the night of May 31, 1921, and the police helped organize them. Greenwood stood on Muscogee (Creek) land. The attack lasted into the afternoon of June 1. The attackers burned 35 city blocks and killed Black residents.

The count of the dead. Estimates run from 50 to 300. Lawyers at the US Department of Justice wrote in 2025 that the massacre "killed as many as 300 Tulsans, perhaps even more." Some bodies were thrown in the Arkansas River, and some were buried in unmarked mass graves.

The attack began with an arrest. Tulsa police arrested Dick Rowland, a 19-year-old Black man, after a white man accused him of attacking Sarah Page, the white woman who ran an elevator he used. Police Chief John Gustafson later told a federal agent that Page had no scratches or bruises. She said

Rowland had not said anything bad to her, and she would not press charges. A white crowd gathered at the courthouse. Armed Black men from Greenwood also came to the courthouse.

The deputies. Tulsa police then made hundreds of white men special deputies, men given a police officer's powers for a time. The Justice Department's 2025 review states that many of them had been calling for a lynching and had been drinking. Police officers helped organize these deputies into the forces that attacked Greenwood.

At Gustafson's later trial, a witness named John Oliphant said the police "were the chief fellows setting fires." The 2025 review reports that Oliphant named two men as the chief arsonists, the leaders in setting the fires: a police officer named Brown, and a man called "Cowboy Long."

Dr. A. C. Jackson, a Black surgeon, walked out of his house with his hands up to surrender. A young white man shot him, and Jackson died of his wounds. No one was ever charged with killing him.

The camps. Governor James Robertson declared martial law, rule by the military instead of by ordinary officials, at 11:30 in the morning on June 1. Police, special deputies and National Guard soldiers arrested Black Tulsans whether or not anyone suspected them of a crime. They held between 4,000 and 6,000 people under armed guard, at Convention Hall, at McNulty Park, a baseball park, and at the fairgrounds. Guard officers arrested some white residents for stealing, but they held no white residents in the camps.

The grand jury. A state grand jury, a group of citizens who decide whether to charge people with crimes, heard almost 200 witnesses. Its members charged about 70 men, many of them Greenwood's Black leaders, with starting a "riot." The *Tulsa Tribune* ran the headline "Roundup of 64 Indicted Blacks Is On: No Warrants Issued for Whites" on June 17. Indicted means charged with a crime. The grand jurors' report, printed on June 26, identified the armed Black men who had gone to the courthouse as a "direct cause."

Most of the cases were later thrown out, including the one against Dick Rowland. Some white men were accused of looting, stealing during a riot, and none of them went to prison. No white person was ever sent to prison for the

killings or the burning.

The grand jurors also charged Police Chief Gustafson with neglect of duty, which means failing to do his job, and with dishonest dealings from before the massacre. He was convicted and removed from his job. Less than a year before the massacre, a mob had taken a white prisoner, Roy Belton, from the same jail and lynched him, and Gustafson had ordered his police not to step in.

Members of a state commission published a report on the massacre in February 2001 that recommended paying the survivors. Lawyers in the Civil Rights Division of the US Department of Justice published their own review, *Review and Evaluation: Tulsa Race Massacre*, in January 2025. The review states that police and white residents worked together in the attack.

The review also states that if today's civil rights laws had been in force in 1921, the attackers could have been tried and convicted. The youngest of them would now be more than 115 years old, so no one can be tried.

The Osage murders, 1921 to 1929

Men murdered members of the Osage Nation in Oklahoma in the early 1920s to take their oil money. The Osage earned more than $30 million in 1923 from oil under their land, the Osage reservation, land set aside for the nation. Osage people held headrights, shares of that money, and when an owner died, a person who was not Osage could get the share.

The guardians. Under a 1921 law, local judges appointed white guardians to control the money of many Osage adults, and the judges handed many of these jobs to their friends. Guardians stole from the Osage people whose money they controlled, and some guardians and husbands killed Osage people to inherit their headrights.

Mollie Burkhart's family. The cattleman William Hale, known as the "King of the Osage Hills," set out to get the headrights of Mollie Burkhart's family. Mollie was married to his nephew, Ernest Burkhart. Hale's plan needed four women of the family to die in a set order: first Mollie's sister Anna, then their

mother, then two more of Mollie's sisters. After those four deaths, all their headrights would pass to Ernest, and through his nephew Hale could take control of them. Federal agents later proved that Hale ordered the murders.

Members of Mollie's family died one after another:

- Anna Brown, Mollie's sister, was shot on May 22, 1921.
- Lizzie Q. Kyle, their mother, died in July 1921, and people suspected poison.
- Henry Roan, their cousin, was shot in January 1923 on Hale's orders, for his insurance money.
- Rita Smith, another sister, died on March 10, 1923, when men blew up her house. The bomb also killed her husband, Bill Smith, and Nettie Brookshire, who worked in their house.

Mollie herself was probably being poisoned.

The count. Counts of the dead differ. The Oklahoma Historical Society and the FBI each give about two dozen, and the Osage Nation counts more than 60. People who told others what they suspected got death threats, and some were killed. Someone threw a lawyer, W. W. Vaughn, from a train.

Members of the Osage Tribal Council asked federal officials to send detectives in April 1923, and Bureau of Investigation officials sent agents. Tom White led four agents who worked undercover, in secret and under false jobs, as a cattle buyer, an insurance salesman, an oil prospector and an herb doctor.

Ernest Burkhart pleaded guilty in June 1926 and got a life sentence. A jury convicted Hale in January 1929, and he was sent to prison for life. Kelsie Morrison and John Ramsey also got life sentences. Hale, Ramsey and Burkhart were later let out on parole, let out early and watched by officials. Governor Henry Bellmon pardoned Burkhart in 1965.

A cabinet member convicted, 1929

Albert Fall was secretary of the interior, a member of the president's cabinet in charge of federal lands. He leased two oil fields that the government kept for the navy to two oil men. The fields were Teapot Dome in Wyoming and Elk Hills in California. Edward Doheny gave Fall a $100,000 "loan" with no interest, and Harry Sinclair gave him about $300,000 in bonds and cash.

A jury found Fall guilty on October 25, 1929, of taking a bribe, a secret payment for a favor, from Doheny. He was sentenced to one year in prison and a $100,000 fine. He was the first person convicted of a crime he committed as a cabinet member. A jury found Doheny not guilty of paying the bribe. Sinclair was jailed for contempt of Congress, which means refusing to obey Congress, and for trying to influence a jury. His term was six months by one account and nine by another.

The end of convict leasing in Alabama, 1928

Alabama's officials rented county prisoners, more than 95 percent of them Black, to coal companies until 1928. This was called convict leasing. The companies' bosses forced the prisoners to work in the mines and whipped them for talking back, for trying to escape and to make them work harder. An explosion at the Banner mine killed 123 Black county prisoners on April 8, 1911.

James Knox. James Knox, a prisoner from Mobile, was killed at the Flat Top mine in 1924. His death was first reported as a suicide by swallowing poison. An autopsy, a doctor's examination of the body after death, showed that his heart had failed and that someone had injected the poison into his body after he died.

The City of Birmingham's history of the case states that his fatal injuries came from being whipped with a steel wire and held in hot water. The historian Mary Ellen Curtin, writing in the *Encyclopedia of Alabama*, describes him as a prisoner tortured by being lowered into "a vat of boiling water." Neither

account names the men who whipped him, put him in the water or injected the poison. Alabama's officials ended convict leasing on June 30, 1928, after his death, the last in any state to do so.

Eliot Ness and Al Capone

Who: The Chicago gang boss who sold illegal beer and liquor during Prohibition, and the federal agent whose team raided his breweries.

When and where: Chicago, 1925 to 1932.

Movie: *The Untouchables* (1987), a film directed by Brian De Palma. It is a drama, not a documentary, and it changes many of the facts.

The gang boss went to prison for not paying his taxes.

Making and selling alcohol was against the law from 1920 to 1933, under the Eighteenth Amendment to the Constitution, and the ban was called Prohibition. Al Capone became the boss of a Chicago gang in 1925, after its leader, Johnny Torrio, was wounded and quit. His men made and sold beer and liquor.

Federal officials hired Eliot Ness, who was 26, to run the Prohibition agents in Chicago in 1929. Ness's team grew from six agents to about ten. From March 1931 they raided Capone's stills, where liquor was made, and his breweries. Gangsters offered them bribes, and the agents turned them down. A Chicago newspaper writer, Charles Schwarz, called them "untouchables."

Capone was arrested several times. Police in Philadelphia arrested him for carrying a hidden gun on May 17, 1929, and he served nine months. A federal judge sentenced him to six months for contempt of court in February 1931.

The tax case. Agents of the Treasury Department built the case that sent him away for years. A jury found him guilty of tax evasion, not paying taxes that are owed, on October 18, 1931. On November 24 the judge sentenced him to eleven years in federal prison and fined him $50,000. He also had to pay $215,000 in back taxes.

> Ness was one of the agents who took Capone to the train for the federal prison in Atlanta on May 3, 1932. It is the only time the two men are known to have met. Capone later served part of his sentence at Alcatraz, a federal prison. He was released on November 16, 1939, with brain damage caused by syphilis, a disease passed on through sex. Ness died of a heart attack in 1957, and his book *The Untouchables* came out months after his death.

Beaten into confessing: the third degree

The "third degree" meant police using cruelty to force a prisoner to confess. President Herbert Hoover set up a national commission to study crime and law enforcement in May 1929. The commission's 1931 report on police in 15 cities states that "physical brutality, protracted questioning, threats, prolonged illegal detention" were "widespread." Protracted questioning means questioning that goes on for hours or days, and prolonged illegal detention means holding a person for a long time when the law does not allow it.

Brown v. Mississippi. A man named Raymond Stewart died in Mississippi on March 30, 1934, and three Black men were later charged with his murder. That night Deputy Sheriff Dial and a group of white men seized Arthur Ellington, a Black man. They hanged him by a rope from the limb of a tree, let him down, and hanged him again. When he still said he was innocent, they tied him to the tree and whipped him. Days later the deputy arrested him again and whipped him until he agreed to confess.

On April 1 the same deputy, the jailer and other white men made Ed Brown and Henry Shields take off their clothes. They bent the two men over chairs and cut their backs open with a leather strap that had buckles on it. They kept whipping until the men confessed in the words the deputy demanded. The three men were charged with Stewart's murder on April 4, tried the next day and sentenced to death.

In court the deputy admitted the whippings. Speaking of Ellington's whipping, he said it was "Not too much for a negro." He added, "not as much as I would have done if it were left to me." Negro was a word used then for a Black person.

All the justices of the US Supreme Court agreed in 1936, in *Brown v. Mississippi*, to throw out the convictions. Chief Justice Charles Evans Hughes wrote: "The rack and torture chamber may not be substituted for the witness stand." The rack was an old device for torturing a person by stretching the body.

The Scottsboro cases, 1931 to 1937

Nine Black teenagers and young men, aged 13 to 20, were riding a freight train in Alabama on March 25, 1931. When the train stopped near the town of Paint Rock, local police and a mob seized them. Two white women on the train, Victoria Price and Ruby Bates, accused them of rape. The nine were Olen Montgomery, Clarence Norris, Haywood Patterson, Ozie Powell, Willie Roberson, Charlie Weems, Eugene Williams, and the brothers Andy and Roy Wright.

Juries sentenced eight of them to death. At the trial of Roy Wright, who was 13, the jurors could not agree between death and life in prison, and the judge declared a mistrial, which means the trial ended with no verdict. In 1932, in *Powell v. Alabama*, the justices of the US Supreme Court ordered new trials because the young men had not had real lawyers.

Ruby Bates later said that Price had made up the story to avoid being charged with vagrancy, having no home or job. Bates then testified for the defense. Judge James Horton set aside a guilty verdict against Patterson. The justices threw out the convictions again in 1935, in *Norris v. Alabama*, because officials had kept Black people off the county's jury lists.

Seven more years of trials followed. The men were let out of prison between 1943 and 1976. Governor Robert Bentley of Alabama signed a law in 2013 that led to pardons for all nine.

BY 1950

If you remember one thing: The national government now had detectives of its own, and the justices of the Supreme Court began throwing out verdicts won by torture or by trials without real lawyers.

One number: The Department of Justice started with 34 agents in 1908. The Bureau had 441 by 1924, about thirteen times as many.

Next: After 1950 the number of people in American prisons grew more than eight times over, and the justices wrote new rules for the police station.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 24 MINUTES

The number of people in American prisons grew more than eight times over in these fifty years, by the count of the Bureau of Justice Statistics, part of the US Department of Justice. In 2000, about 46 of every 100 sentenced prisoners were Black. From the 1970s on, lawmakers in Congress and the states wrote long prison terms that judges had to give, many of them for drug crimes.

New rules. The justices of the US Supreme Court set new rules for police and courts between 1961 and 1968. Police could not use evidence from an illegal search. State courts had to give a lawyer to a poor person charged with a serious crime. Police had to warn people of their rights before questioning them.

The justices struck down the death-penalty laws then in use in 1972 and allowed new ones in 1976. Some trials still ended with no one convicted, among them the 1955 murder trial for the killing of Emmett Till and the 1992 state trial of the officers who beat Rodney King.

Emmett Till

Who: A 14-year-old Black boy from Chicago, kidnapped and murdered by two white men whom an all-white jury then found not guilty.

When and where: Money and Sumner, Mississippi, August and September 1955.

Movie: *Till* (2022), a feature film about his mother, Mamie Till-Mobley, and her fight after his death.

An all-white jury set his killers free after 67 minutes.

Emmett Till was 14 and lived in Chicago. He was visiting relatives near Money, Mississippi, in August 1955. On August 24 he went into Bryant's Grocery and Meat Market, owned by a white couple, Roy and Carolyn Bryant. As Carolyn Bryant left, he whistled at her. Lawyers at the US Department of Justice tell these facts in their 2021 notice closing the case.

Between 2 and 3:30 in the morning of August 28, Roy Bryant and his half-brother, J. W. Milam, came to the house of Till's great-uncle, Mose Wright, who was 64. Milam carried a pistol and a flashlight. The two men took the boy away by force, with at least one other person.

Three days later his body was found in the Tallahatchie River. His killers had beaten him on his back and hips and shot him above the right ear, and one eye had been forced from its socket. The men had fastened a cotton gin fan around his neck with barbed wire to sink the body. The fan weighed 70 to 75 pounds.

He was identified by his father's silver ring. His mother, Mamie Till Bradley, later Mamie Till-Mobley, held an open-casket funeral in Chicago so that people could see what had been done to him.

The trial. Bryant and Milam were tried for murder in the Tallahatchie County Courthouse in Sumner in September 1955. Women and Black people were kept off the jury, and the Justice Department's 2021 notice does not name the officials who kept them off. Mose Wright stood up in court and pointed to Milam as one of the men.

The all-white, all-male jury found both men not guilty on September 23, after 67 minutes. One juror said, "If we hadn't stopped to drink pop, it wouldn't have taken that long." In November a grand jury refused to charge them with kidnapping, although they had admitted taking the boy.

Bryant and Milam told the writer William Bradford Huie, for *Look* magazine, in January 1956 that they had kidnapped and murdered Till. The magazine paid them between $3,600 and $4,000. They could not be tried again for the murder, because of the rule called double jeopardy: a person who has been found not guilty cannot be tried a second time for the same crime.

FBI agents reopened the case in 2004. A Mississippi grand jury did not charge Carolyn Bryant in February 2007. Justice Department lawyers reviewed the case again after 2017 and closed it in December 2021, stating that they could not charge anyone with a federal crime. President Joe Biden made places connected to Emmett Till and his mother a national monument on July 25, 2023.

Clarence Earl Gideon

Who: A poor man in Florida who was refused a lawyer, wrote to the US Supreme Court from prison in pencil, and won the right to a lawyer for every person charged with a serious crime.

When and where: Panama City, Florida, 1961 to 1963.

Movie: *Gideon's Trumpet* (1980), a television film with Henry Fonda as Gideon, based on Anthony Lewis's 1964 book.

He asked the Supreme Court for a lawyer, in pencil, from prison.

Someone broke into the Bay Harbor Pool Room in Panama City, Florida, on June 3, 1961. The thief took coins from the jukebox and the cigarette machine, and beer, wine and soda. A 22-year-old man named Henry Cook said he had seen Clarence Earl Gideon leave with his pockets full of coins.

Gideon, who was 50, had no money for a lawyer. Under Florida's law, only people who could be sentenced to death got free lawyers, so Judge Robert McCrary Jr. refused to give him one. Gideon defended himself in court. The jury convicted him on August 4, 1961, and the judge sentenced him to five years in prison.

From the state prison at Raiford, Gideon wrote to the US Supreme Court in pencil, asking the justices to hear his case. The justices gave him a lawyer, Abe Fortas. All nine justices agreed on March 18, 1963, in a ruling written by Justice Hugo Black, that a state must give a lawyer to a person charged with a felony, a serious crime, who cannot pay.

At Gideon's second trial, on August 5, 1963, a local lawyer, W. Fred Turner, showed that Cook had lied at the first trial about his own criminal record. The jury found Gideon not guilty after an hour. He died in 1972.

Rules for the police station, 1961 to 1968

The justices of the Supreme Court wrote several more rules for police and courts in these years.

Mapp. Cleveland police forced their way into the home of Dollree Mapp on May 23, 1957. They waved a paper they called a search warrant, a judge's written order allowing a search, and no such paper was ever shown in court. They searched the house and found books they called obscene, meaning too sexual for the law to allow. She was sentenced to one to seven years.

Six of the nine justices ruled on June 19, 1961, in *Mapp v. Ohio*, that evidence police get from an illegal search cannot be used in a state trial.

Miranda. Phoenix police officers Carroll Cooley and Wilfred Young arrested Ernesto Miranda on March 13, 1963, and questioned him for two hours about the kidnapping and rape of an 18-year-old woman. He confessed. No one had told him he could stay silent or have a lawyer.

Five of the nine justices ruled on June 13, 1966, in *Miranda v. Arizona*, that police must tell a person they are holding about these rights before questioning. Chief Justice Earl Warren wrote the ruling. He based it on the Fifth Amendment of 1791, which forbids forcing anyone to be a witness against himself.

Afterward, officers across the country began reading people their rights aloud from "Miranda warning" cards. Arizona prosecutors, the government lawyers who bring the charges in court, tried Miranda again without the confession, and a jury convicted him again on March 1, 1967.

Gault. The sheriff of Gila County, Arizona, took Gerald Gault, 15, into custody in June 1964 without telling his parents. A neighbor, Ora Cook, had complained that he made an obscene phone call. Judge Robert McGhee sent him to the State Industrial School until he turned 21, about six years. An adult could have been fined for the same act. Eight of the nine justices ruled in 1967 that children in juvenile court have the right to hear the charges, to have a lawyer, to question witnesses and to stay silent.

Terry. Cleveland detective Martin McFadden stopped John W. Terry on October 31, 1963, patted the outside of his clothes and found a gun. Eight of the nine justices ruled on June 10, 1968, in *Terry v. Ohio*, that police may stop a person they reasonably suspect is armed and about to commit a crime. They may pat the outside of the person's clothes for weapons, even without grounds to arrest, which is called a stop and frisk.

BY 1968: A poor person charged with a serious crime got a free lawyer, police had to warn a suspect of the right to stay silent, and evidence from an illegal search could not be used in a state trial.

The men who killed civil rights workers, in court

Members of the Ku Klux Klan and some Southern officers killed people who worked for Black Americans' rights in the 1960s. The Ku Klux Klan is a secret white group that used terror against Black people and their supporters. In the six cases below, nobody was convicted of murder at the time.

Jackson, Mississippi, 1963. Byron De La Beckwith shot Medgar Evers, who led the NAACP's work in Mississippi, in the driveway of his home in Jackson in June 1963. Beckwith was tried twice in 1964, and both juries could not agree. A jury convicted him in 1994, 31 years after the killing.

Birmingham, 1963. A bomb exploded under the steps of the 16th Street Baptist Church in Birmingham, Alabama, on September 15, 1963. It killed four girls: Addie Mae Collins, 14, Denise McNair, 11, Carole Robertson, 14, and Cynthia Wesley, 14. Klan members had set it.

FBI agents named Robert Chambliss as the main suspect. FBI Director J. Edgar Hoover closed the case in 1968 without charging anyone and ordered the evidence sealed. Chambliss was convicted in 1977. Thomas Blanton was convicted in 2001 and Bobby Frank Cherry in 2002, with the help of secret FBI recordings made in 1964. Another suspect died before he could be tried.

Neshoba County, 1964. Local police in Neshoba County, Mississippi, jailed three civil rights workers for "speeding" on June 21, 1964. They were James Chaney, 21, a Black college student from Mississippi, and Andrew Goodman,

20, and Michael Schwerner, 24, white men from New York. After dark the police let them go "into the hands of the Ku Klux Klan." The three were stopped again, forced into a sheriff's car, driven to a country road and shot dead.

A federal jury convicted seven of the 18 men charged, among them a deputy sheriff and the head of a Mississippi Klan group. The verdict came on October 20, 1967, and the sentences ran from three to ten years. Edgar Ray Killen, a Klan leader and former preacher who had organized the killings, went free because one juror would not convict a preacher. A jury in Mississippi convicted Killen of manslaughter, killing a person without planning to murder, on June 21, 2005, and he died in prison in 2018.

Marion, Alabama, 1965. Alabama State Trooper James Bonard Fowler shot Jimmie Lee Jackson, 26, in the stomach in a cafe in Marion in February 1965, and Jackson died eight days later. Fowler pleaded guilty in 2010 to misdemeanor manslaughter, a lesser charge for killing a person. He was sentenced to six months and served five.

Selma, 1965. Four men in Selma attacked James Reeb, a minister from Boston, in March 1965. One of them struck him in the head with a club, and Reeb died two days later. The prosecutor let a man whose brother was a suspect sit on the jury, and the jury found all three men tried not guilty.

Highway 80, Alabama, 1965. Three Klan members shot and killed Viola Liuzzo, a mother of five from Detroit, on March 25, 1965, as she drove on Highway 80. A paid FBI informer rode in their car. An informer is a person who secretly passes information to the police or the FBI. State juries did not convict the three men. Federal juries convicted all three of violating her civil rights, and each got 10 years in prison.

In these six cases, every conviction came from a federal jury or came 14 to 45 years after the killing. No one was ever convicted of killing James Reeb.

Attica, September 1971

About 1,281 of the roughly 2,200 men held at Attica, a New York state prison, took over the prison on September 9, 1971. They held 42 guards and other workers as hostages. In July the prisoners had made demands about food, mail, medical care and parole. During the takeover someone beat a guard, William Quinn, and he died. Governor Nelson Rockefeller refused to come to the prison.

Retaking the prison. Officers dropped tear gas into the prison yard at 9:46 in the morning on September 13. State troopers, deputies and guards fired into the smoke, some with shotguns loaded with buckshot. They killed at least 39 people: 29 prisoners and 10 of the hostages. Counting the whole four days, 43 people died.

Rockefeller and other officials first said the prisoners had cut the hostages' throats. The medical examiners, the doctors who examine bodies after a death, found that the hostages had died of gunshot wounds from the officers' guns. Afterward, prison officers tortured prisoners who had survived. Pages of an official report released in 2015 describe torture, burning and sexual abuse.

Who was charged. Prosecutors charged 62 prisoners with 1,289 crimes. They charged one state trooper, for reckless endangerment, putting people in danger through carelessness. A jury found the prisoner John Hill guilty of murdering William Quinn on April 5, 1975, and found another prisoner, Charley Joe Pernasilice, guilty of assault in Quinn's death.

Governor Hugh Carey pardoned the prisoners who had pleaded guilty in December 1976. Hill received clemency, a governor's mercy that cut his punishment short, about a year after his trial. New York State officials paid $8 million in settlements to prisoners and their families in 2000 and $12 million to the families of the prison workers in 2005.

Executions stop and start again, 1972 to 1999

Five of the nine justices struck down the death-penalty laws then in use on June 29, 1972, in *Furman v. Georgia*. More than 600 people were on death row, waiting to be executed. Justice Potter Stewart wrote that the death sentences were "cruel and unusual in the same way that being struck by lightning is cruel and unusual." Being struck by lightning happens to very few people, by chance.

Lawmakers in 35 states and in Congress wrote new death-penalty laws in the next four years. The justices allowed them on July 2, 1976, in *Gregg v. Georgia*.

Gary Gilmore murdered Max Jensen, 25, a gas station worker, on July 19, 1976, and Bennie Bushnell, 26, a motel manager, the next night, in Utah. He asked to be executed. At 8:07 in the morning on January 17, 1977, a firing squad shot him at the Utah State Prison. A firing squad is a group of shooters who fire rifles at the prisoner's heart. There were five shooters, and prison officials said one of them had a blank. Gilmore was the first person executed after the 1976 ruling.

Texas officials carried out the first execution by lethal injection on December 7, 1982. Lethal injection means killing a person with drugs put into a vein through a tube in the arm. The Death Penalty Information Center, a group that collects facts about the death penalty, describes the usual three-drug method. The first drug puts the person to sleep, the second stops the muscles from moving, which stops the breathing, and the third stops the heart. Some states later used a single large dose of a drug such as pentobarbital.

New death sentences reached their highest number, 325, in 1986. Executions rose to 98 in 1999, the most in any year since 1976.

Wounded Knee, 1973: the marshals and the trial

Members of the American Indian Movement took over the village of Wounded Knee, South Dakota, on February 27, 1973, and held it for 71 days. The movement had begun in Minneapolis in 1968, when its members patrolled the streets to watch for police abuse of Native people.

About 200 Oglala Lakota and movement members held the village. They were protesting against the head of the tribal government on the Pine Ridge Reservation and against the treaties the United States had broken. They chose Wounded Knee because soldiers of the 7th Cavalry had killed between 250 and 300 Lakota there in 1890. US marshals, FBI agents and other federal officers surrounded the village with armored vehicles and automatic weapons.

The shootings. Someone opened fire on a group of US marshals on the night of March 16. A bullet hit one of them, Lloyd Grimm, and left him paralyzed from the waist down. The Associated Press reported in 2000 that no suspects were ever found.

A bullet hit Frank Clearwater, an Eastern Cherokee man inside the village, in the head on April 17, and he died on April 25. The FBI released a report on deaths on the Pine Ridge Reservation, where Wounded Knee lies, in 2000. It found that Clearwater was killed by gunfire from federal officers, who were firing back after people shot at them.

A federal sniper shot and killed Lawrence "Buddy" Lamont, an Oglala man, on April 26. The FBI report gives the date as April 27 and says he died in a gunfight with federal officers at a roadblock. A US attorney, a federal prosecutor, reviewed his killing and charged no one.

Federal prosecutors charged the movement's leaders, Dennis Banks and Russell Means, for their part in the takeover. After their trial had run for eight and a half months, US District Judge Fred Nichol threw out the charges in 1974. He wrote that "the misconduct by the government in this case is so aggravated that a dismissal must be entered in the interests of justice." Aggravated here means serious.

Drug laws and long sentences, 1973 to 1994

Lawmakers wrote the long sentences that filled the prisons. Some of those sentences were mandatory minimums. A mandatory minimum is the shortest sentence a law allows for a crime, and a judge cannot give less.

New York's lawmakers passed Governor Nelson Rockefeller's drug laws in 1973. Selling two ounces or having four ounces of heroin, cocaine or marijuana brought 15 years to life in prison.

The basketball player Len Bias died from cocaine on June 19, 1986. House Speaker Tip O'Neill had Democratic leaders write a new federal drug law, and the Republicans who then ran the Senate passed their own. President Ronald Reagan signed the law on October 27, 1986.

Crack and powder. Under that law, 5 grams of crack cocaine brought at least five years in federal prison, the same sentence as 500 grams of powder cocaine. Crack is a form of cocaine that is smoked. A person with crack faced the same sentence as a person with 100 times as much powder.

Staff of the US Sentencing Commission reported that in 1992, about 91 of every 100 people sentenced in federal court for crack were Black, and about 3 were white. In 2004, of the people who entered treatment for crack, 53 of every 100 were Black and 38 were white.

Members of Congress passed the largest crime law in the country's history in 1994. Representative Jack Brooks of Texas sponsored it in the House. Senator Joe Biden of Delaware wrote the Senate version, and President Bill Clinton signed it. The law provided money for 100,000 new police officers and $9.7 billion for prisons. Under it, 60 more federal crimes could be punished by death, and people in prison could no longer get federal college grants.

BY 1994: Five grams of crack brought the same five-year federal sentence as 500 grams of powder cocaine, and members of Congress had voted money for 100,000 new police officers and $9.7 billion for prisons.

Kemba Smith

Who: A college student sentenced to 24 and a half years in federal prison for a crack cocaine ring run by a man who beat her, though she never sold or used drugs.

When and where: Virginia, and federal prison, 1994 to 2000.

Movie: *Kemba* (2024), a film about her life, shown on BET+.

A first-time offender, she got 24 and a half years, no parole.

Kemba Smith grew up near Richmond, Virginia, and went to Hampton University. There she met Peter Hall, who ran a crack cocaine ring. The NAACP Legal Defense Fund, a group of civil rights lawyers, states that she was a victim of his violence at home.

She turned herself in to federal agents on September 1, 1994. She pleaded guilty to conspiracy to sell cocaine, to lying to federal agents and to conspiracy to launder money. Conspiracy means agreeing with others to commit a crime, and to launder money means to hide where money from crime came from.

The NAACP Legal Defense Fund states that she never sold or used drugs, and that federal prosecutors still charged her with conspiracy to sell cocaine. Federal judge Richard B. Kellam sentenced her to 24 years and 6 months in federal prison, with no parole.

She was pregnant during her case. She said of her son, "I had two days with him before giving him to my parents."

President Bill Clinton cut her sentence short on December 22, 2000, after about six and a half years in prison. The Clinton Library describes her as "a first time non-violent drug offender." President Joe Biden pardoned her on January 19, 2025.

How many people were in prison

Officials of the Bureau of Justice Statistics have counted prisoners since 1925. These are their counts of people in state and federal prisons:

- 1950: about 166,000
- 1972: about 196,000
- 1990: about 771,000
- 2000: about 1.38 million

From 1950 to 1972 the count rose by about 30,000. Then it began to climb in 1974, and by 2000 it was about seven times the count of 1972. The share of Americans in prison grew more than four times over: about 1 of every 1,000 Americans in 1950, and almost 5 of every 1,000 in 2000.

In 2000, almost 10 of every 100 Black men aged 25 to 29 were in prison, against about 1 of every 100 white men that age.

Killed for his shoes, 1989

James David Martin, 17, killed Michael Eugene Thomas, 15, for his basketball shoes in 1989. Martin pleaded guilty to a lesser charge than murder and got seven years in prison. After his release he killed two more people.

Thomas was a ninth grader at Meade Senior High School in Anne Arundel County, Maryland. He had bought the shoes, Air Jordans, for $115.50. On May 2, 1989, Martin led him into the woods near the school and strangled him to death. Then Martin took the shoes and ran. The *New York Daily News*, the *New York Times*, the *New York Post* and *Sports Illustrated* reported that Martin raped the boy before he killed him.

Seven years. The school stood on federal land, so Martin was tried in federal court. He pleaded guilty to voluntary manslaughter, a charge for killing that is less serious than murder, and he was sentenced to seven years in prison. In May 1990 *Sports Illustrated* ran a cover story by Rick Telander called "Your Sneakers or Your Life."

Two more killings. Martin was released in 1996 and moved to New York City. In the Bronx on February 23, 1998, he beat 14-year-old Marleny Cruz and strangled her to death. In 2005 he murdered his wife, Cicela Santiago, in Allentown, Pennsylvania, and he was sentenced there to 22 to 40 years in prison.

Bronx detectives matched DNA from under Cruz's fingernails to Martin in 2011. He pleaded guilty to manslaughter in her killing in 2016 and got 20 more years. The Bronx district attorney said the plea was for "raping and strangling" her.

Rodney King, 1991 and 1992

Los Angeles police officers Laurence Powell, Timothy Wind and Theodore Briseno and Sergeant Stacey Koon shocked Rodney King with a Taser, a weapon that fires an electric shock, early on March 3, 1991, after a car chase. They hit him with their batons while other officers watched. Police Chief Daryl Gates said the officers struck him "between fifty-three and fifty-six times." King had a broken bone in his face, a broken right ankle, and bruises and cuts. George Holliday filmed the beating from his apartment.

The trial was moved to Simi Valley, in Ventura County. The jury included ten white jurors. The jurors found all four officers not guilty of assault on April 29, 1992, and they could not agree on one charge against Powell. Rioting spread across Los Angeles from April 29 to May 3. One count gives 63 dead and another gives 55. Police or National Guard soldiers shot 10 or 11 of them. More than 12,000 people were arrested.

A federal jury convicted Koon and Powell of violating King's civil rights in 1993, and each got 30 months in prison. The jury found Wind and Briseno not guilty. King won $3.8 million from the city of Los Angeles.

Two deaths at police hands in New York, 1983 and 1999

Michael Stewart, 1983. New York City Transit Police officers arrested Michael Stewart, 25, a Black graffiti artist, at the First Avenue subway station on September 15, 1983, for writing graffiti. In their custody he was beaten unconscious. He was hogtied, his ankles bound and his hands tied to his feet with an elastic strap. The account of his death does not name which officers struck him.

He reached Bellevue Hospital in a coma and died on September 28, after 13 days. The cause of death was listed as cardiac arrest, which means the heart stopped. The city's medical examiner, Elliot Gross, gave three different findings, among them an injury to the neck and blows to the body. Doctors hired by Stewart's family found that he had been strangled.

Six officers were tried for criminally negligent homicide, causing a death through carelessness, and for assault and perjury, lying under oath. An all-white jury found them not guilty on November 24, 1985. A lawsuit against eleven officers and the transit authority ended with a payment of $1.7 million.

Amadou Diallo, 1999. Four New York City police officers in plain clothes, Sean Carroll, Richard Murphy, Edward McMellon and Kenneth Boss, fired 41 shots at Amadou Diallo outside his building in the Bronx on February 4, 1999. Diallo, 23, was an immigrant from Guinea. Nineteen of the shots hit him and killed him. He had no gun, and he had taken out his wallet.

The officers were charged with second-degree murder, and the trial was moved to Albany. A jury found all four not guilty on February 25, 2000. City officials paid his family $3 million in 2004.

The wrong man: DNA and the Central Park case

DNA is the chemical code inside the cells of a living body, and it differs from person to person. Cells left at a crime can be tested to show whether they came from a certain person.

Kirk Bloodsworth. A Maryland jury convicted Kirk Bloodsworth in 1985 of the 1984 rape and murder of Dawn Hamilton, a nine-year-old girl in Rosedale, Maryland. The jury convicted him on the word of five witnesses, and he was sentenced to death.

A DNA test in 1993 showed that the attacker's cells, saved in a paper bag in the judge's chambers, were not his. He was freed after more than nine years in prison, and Governor William Donald Schaefer pardoned him. He was the first person on death row in the United States cleared by DNA. In 2003 the cells were matched, through a DNA database, to Kimberly Shay Ruffner, who had been held in a cell one floor below Bloodsworth. Ruffner pleaded guilty in 2004.

The lawyers Barry Scheck and Peter Neufeld had founded the Innocence Project, a group that uses DNA tests to free people who were wrongly convicted, at Cardozo Law School in New York in 1992, a year before Bloodsworth was cleared.

Central Park, 1989. A man attacked and raped Trisha Meili on April 19, 1989, while she was running in Central Park in New York City. Police questioned five teenagers for at least seven hours before recording their statements on video. They were Antron McCray, 15, Kevin Richardson, 14, Yusef Salaam, 15, Raymond Santana, 14, and Korey Wise, 16.

Four of them confessed on video, then took it back within weeks, saying the police had lied to them and pressured them. The DNA found at the scene matched none of the five. Donald Trump paid for full-page ads in four New York newspapers on May 1, 1989, calling for the death penalty. The ads named no defendant. Juries convicted all five teenagers in 1990.

BY 2000

If you remember one thing: The justices gave accused people new protections in the 1960s, and from the 1970s lawmakers wrote the long sentences that filled the prisons.

In 1960: Police did not have to warn a suspect that he could stay silent or have a lawyer.

In 2000: Police had to give that warning before questioning, and prisons held about seven times as many people as in 1972.

Next: In 2002 a judge cleared the five teenagers from Central Park, and after 2009 the prison count began to fall for the first time since 1972.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 18 MINUTES

American prisons held about 1.62 million people at the end of 2009, the most ever counted. Then the count fell for the first time since 1972, and by the end of 2023 it was about 1.25 million. The gap between Black and white Americans stayed wide: Black Americans were in prison at almost five times the rate of white Americans in 2023.

Judges and governors cleared thousands of people of crimes they had not committed. Police killed more than a thousand people a year in 2024 and 2025, by one group's count. Some officers were charged and convicted, and others were never charged, or juries found them not guilty. Murders jumped in 2020, then fell, and the 2025 murder rate tied the lowest the FBI has recorded.

Fewer prisoners, a gap that shrank but stayed wide

The prison count reached 1,617,970 at the end of 2009, the highest in the Bureau of Justice Statistics count, which goes back to 1925. It fell in 2010 for the first time since 1972.

Shorter drug sentences. Lawmakers also cut some of the long drug sentences. New York's lawmakers ended the Rockefeller minimum sentences in April 2009. Senator Dick Durbin of Illinois wrote the Fair Sentencing Act, and President Barack Obama signed it on August 3, 2010. Under the new law, it took 28 grams of crack, not 5, to bring the five-year minimum, so the gap between crack and powder cocaine sentences went from 100 to 1 down to 18 to 1.

Under the First Step Act, which President Donald Trump signed on December 21, 2018, that change reached people already in prison. In its first year, more than 2,000 federal prisoners got shorter sentences under the crack change, and more than 3,000 were let out under changed rules for time off for good behavior.

The gap. About 1,254,000 people were in state and federal prisons at the end of 2023, and about 657,500 more were in local jails in the middle of 2024, by the bureau's count. For every 100,000 Black Americans, 929 were serving prison sentences in 2023. For every 100,000 white Americans, 190 were. Ten years earlier, in 2013, the two rates had been 1,348 and 237.

A conviction for a felony can also take away a person's right to vote. About 4 million Americans could not vote in 2024 because of a felony conviction, including 1 in every 22 Black adults, by the count of the Sentencing Project, a group that studies prisons and sentences.

The Exonerated Five

Who: Five New York City teenagers convicted in 1990 of a rape in Central Park, cleared in 2002 when another man confessed and his DNA matched.

When and where: New York City, 1989 to 2014.

Movie: *When They See Us* (2019), a Netflix drama series directed by Ava DuVernay. *The Central Park Five* (2012) is a documentary by Ken Burns, Sarah Burns and David McMahon.

Another man confessed in 2002, and his DNA matched.

Antron McCray, Kevin Richardson, Yusef Salaam, Raymond Santana and Korey Wise were 14 to 16 years old when juries convicted them in 1990 of the attack on Trisha Meili in Central Park. Another man had committed the rape alone.

Judge Thomas Galligan sentenced the four younger boys to 5 to 10 years in custody for young people. Wise, who was 16, was tried as an adult. His jury found him not guilty of rape and attempted murder and guilty of sexual abuse, assault and riot. He was sentenced to 5 to 15 years in an adult prison. Santana was released in 1995, McCray in 1996, and Salaam and Richardson in 1997.

A prisoner named Matias Reyes met Wise at Auburn prison in New York in 2001. Reyes was serving a life sentence for other rapes. He told officials in 2002 that he had raped Trisha Meili and that he had acted alone. A DNA test matched him. Wise was released in August 2002, after about 13 years locked up.

Prosecutors in District Attorney Robert Morgenthau's office looked into the case again and asked the court to throw out the convictions. Justice Charles Tejada vacated, or cancelled, all five convictions on December 19, 2002.

A panel chosen by Police Commissioner Raymond Kelly accepted the DNA result in January 2003 but said the five had "most likely" also taken part in the attack. Justice Tejada and the district attorney did not accept

that view. Reyes was never charged for the attack on Meili, because the five-year time limit for charging the crime had passed.

New York City officials paid the five $41 million in 2014 to settle their lawsuit. Four of them got about $7.1 million each. Wise got $12.2 million, because he had served six more years than the others.

Proving convictions wrong

An exoneration means a judge or a governor has cleared a person of a crime after a conviction, because new evidence shows the person did not do it. The National Registry of Exonerations counted 3,767 exonerations in the United States from 1989 through the end of 2025. The Innocence Project states that DNA tests helped in more than 600 of them.

Ninety-seven people were exonerated in 2025 alone. On average each of them had lost 14.2 years in prison, and 59 of the 97 were Black. In at least 70 of the cases, police, prosecutors or other officials had broken the rules, which the Registry calls official misconduct. Payments to people who were cleared have added up to more than $5.5 billion since 1989. The National Registry of Exonerations gave these figures in its 2026 report.

George Stinney Jr. A judge threw out the conviction of George Stinney Jr., a 14-year-old Black boy, in 2014, seventy years after South Carolina officials executed him. Two white girls, Betty June Binnicker, 11, and Mary Emma Thames, 8, were killed in Alcolu, South Carolina, on March 22, 1944. Stinney was accused. He was tried, convicted and sentenced to death in one day in April 1944, and the all-white jury took less than ten minutes.

Governor Olin Johnston refused to spare him, and South Carolina officials put him to death in the electric chair on June 16, 1944. Judge Carmen Mullen threw out his conviction on December 16, 2014. She ruled that he had not had a real defense and that his confession had probably been forced. No court has named anyone else as the killer.

The death penalty, 2000 to 2026

Executions in the United States fell from 85 in 2000 to 11 in 2021, then rose to 47 in 2025. Officials had carried out 27 executions in 2026 by September 18. Judges and juries handed down 23 new death sentences in 2025, and 2,024 people were on death row on October 1, 2025. Twenty-three states have no death penalty. The Death Penalty Information Center gave these figures in September 2026.

Officials in the United States have carried out 1,681 executions since 1977. They killed 1,490 of those people by lethal injection. They killed the rest by electrocution, in gas chambers, sealed rooms filled with poison gas, by firing squad or by hanging.

The victims. Of the victims in the crimes behind these executions, 76 of every 100 were white and 14 were Black. Of all murder victims in the United States from 1985 to 2023, about 49 of every 100 were white and 47 were Black, by FBI counts. A killing was far more likely to end in an execution when the person killed was white.

The justices of the Supreme Court ruled in 2002 that states may not execute people with intellectual disabilities, a lasting limit on a person's ability to learn and reason. They ruled in 2005 that states may not execute people for crimes committed before age 18.

Cleared from death row. The Death Penalty Information Center counts more than 200 people sentenced to death since 1973 who were later cleared and freed.

The group's staff also list 21 people executed since 1976 who, they say, may have been innocent. Among them are Cameron Todd Willingham, executed in Texas in 2004, and Troy Davis, executed in Georgia in 2011. No court or governor has declared any of the 21 innocent. The group notes that courts do not usually hear a claim of innocence once the person is dead.

The Danziger Bridge, New Orleans, 2005

New Orleans police officers shot six unarmed Black people on the Danziger Bridge on September 4, 2005, in the days after Hurricane Katrina. They killed James Brissette, 17, and Ronald Madison, 40, who was shot in the back. Among the officers were Kenneth Bowen, Robert Gisevius, Anthony Villavaso and Robert Faulcon. The officers then worked to cover up what they had done.

Officers were convicted in 2011. Federal Judge Kurt Engelhardt threw out those convictions in September 2013 and ordered a new trial, pointing to what he called "highly unusual, extensive and truly bizarre actions" by the prosecutors. Five former officers pleaded guilty on April 20, 2016, and they were sentenced to 3 to 12 years in prison.

In a separate case, Roland Bourgeois Jr., 55, of the Algiers Point neighborhood, was sentenced in 2019 to 10 years in prison. He had shot three Black men who were trying to flee the area.

Police killings and what happened in court

No government agency keeps a full count of the people police kill. Campaign Zero, a group that campaigns for a cause, counted at least 1,383 people killed by police in 2024 and 1,314 in 2025. By its count, police killed Black Americans at 2.6 times the rate of white Americans in 2025. In 2020, 62 of every 100 local police departments used body cameras, cameras worn on an officer's uniform, the Bureau of Justice Statistics reported in 2023.

Officer Darren Wilson shot and killed Michael Brown, 18, in Ferguson, Missouri, on August 9, 2014. A St. Louis County grand jury did not charge Wilson. Federal prosecutors did not charge him in 2015, and in 2020 the county prosecutor, Wesley Bell, reviewed the case and did not charge him either.

The Ferguson report. Lawyers of the US Department of Justice reported on the Ferguson police on March 4, 2015. They found a "pattern or practice of unlawful conduct" that broke the First, Fourth and Fourteenth Amendments,

which protect free speech, protect people from unfair searches and arrests, and promise equal treatment under the law. Police there had broken the law again and again, in ways that went against rights the Constitution protects.

The city's finance director had written to the police chief in 2010 that "unless ticket writing ramps up significantly," it would be hard to raise more money the next year. Of the city court's 2013 cases in which a judge issued an arrest warrant, an order to arrest someone, 92 of every 100 were against Black people.

The report describes a 32-year-old Black man who sat in his parked car in a park in the summer of 2012. An officer blocked in his car and demanded his Social Security number. The officer accused him, with no cause, of being a pedophile, a person who wants to sexually abuse children. When the man spoke of his rights, the officer arrested him at gunpoint on eight city charges.

One charge was for giving his name as "Mike" instead of "Michael." Another was for not wearing a seat belt while sitting in a parked car. The man told investigators he lost his job doing work for the federal government because of the charges.

Officer Daniel Pantaleo killed Eric Garner, 43, on Staten Island, New York, on July 17, 2014. Pantaleo held him around the neck in a chokehold, which police rules banned. Garner said "I can't breathe" 11 times. The medical examiner ruled his death a homicide, a killing by another person. A grand jury did not charge Pantaleo. In July 2019 Attorney General William Barr decided against federal charges, and in August 2019 Police Commissioner James O'Neill fired Pantaleo.

Tamir Rice, 12, was holding a toy gun at a park in Cleveland on November 22, 2014. The person who called 911 said twice that the gun was "probably fake" and the person "probably a juvenile," meaning a young person. The dispatcher, the worker who sent the call to the officers, did not pass that on. Officer Timothy Loehmann shot Tamir almost as soon as he arrived and killed him. A grand jury did not charge Loehmann. Loehmann was fired in 2017 for lying on his job application.

Breonna Taylor. Officers in Louisville, Kentucky, forced open Breonna Taylor's door on March 13, 2020, with a warrant to search for drugs. Her boyfriend, Kenneth Walker, fired once and hit officer Jonathan Mattingly. Officers fired 32 shots, and Myles Cosgrove fired the shot that killed Taylor, who was 26. A grand jury did not charge Mattingly or Cosgrove.

A former detective, Kelly Goodlett, pleaded guilty in 2022 to helping put false facts on the warrant. A federal jury convicted Brett Hankison in November 2024 of using excessive force, more force than the law allows. His shots hit no one. He was sentenced to 33 months in July 2025. Louisville's city officials paid $12 million.

George Floyd. Officer Derek Chauvin knelt on George Floyd's neck in Minneapolis for more than nine minutes on May 25, 2020, and killed him. Floyd was 46. A jury convicted Chauvin of second-degree murder, third-degree murder and second-degree manslaughter, different levels of charges for causing a death, on April 20, 2021. Judge Peter Cahill sentenced him to 22 and a half years on June 25, 2021. The state's sentencing rules set 12 and a half years, and the judge gave more after finding "particular cruelty."

Chauvin also pleaded guilty to a federal crime, and Judge Paul Magnuson sentenced him to 21 years for it in July 2022, served at the same time as the state sentence. The three other officers got federal sentences: Thomas Lane 2 and a half years, J. Alexander Kueng 3 years and Tou Thao 3 and a half years. Minneapolis city officials paid Floyd's family $27 million.

Tyre Nichols. Five police officers in Memphis, Tennessee, pulled Tyre Nichols, 29, from his car on January 7, 2023. They were Tadarrius Bean, Demetrius Haley, Emmitt Martin III, Desmond Mills Jr. and Justin Smith. They sprayed him with pepper spray and shocked him with a Taser. When they caught him near his mother's house, they punched him, kicked him and beat him with a baton. He died on January 10 of injuries to his head.

Mills and Martin pleaded guilty. A federal jury convicted Haley on all counts in October 2024, and Bean and Smith only of trying to influence witnesses. A state jury found Bean, Haley and Smith not guilty of all state charges,

including murder, in May 2025. Judge Sheryl Lipman ordered a new federal trial for the three in August 2025, because the first judge may have been biased. No new verdict had been reported by September 2026.

Who was charged after school shootings

After some shootings at schools, prosecutors charged officers who did not act, or a parent who supplied the gun.

Parkland. Nikolas Cruz, 19, a former student, shot and killed 14 students and three staff members at Marjory Stoneman Douglas High School in Parkland, Florida, on February 14, 2018, and wounded 17 others. The report of a Florida state commission states that Deputy Scot Peterson "fled to a position of personal safety while Cruz shot and killed" students and staff. A jury found Peterson not guilty of felony child neglect and other charges on June 29, 2023.

Uvalde. A gunman killed 19 children and two staff members at Robb Elementary School in Uvalde, Texas, on May 24, 2022. Officers killed the gunman 77 minutes after the first officers entered the school. A Department of Justice review states that the school district police chief, Pete Arredondo, told officers at several points to wait. A grand jury charged Arredondo and a former school officer, Adrian Gonzales, with child abandonment and endangerment. A jury found Gonzales not guilty of all 29 counts on January 21, 2026.

Winder, Georgia. A gunman killed two 14-year-old students and two teachers at Apalachee High School in Winder, Georgia, on September 4, 2024. The gunman, Colt Gray, pleaded guilty to all 55 counts, and he was sentenced to life in prison without parole on July 27, 2026. His father, Colin Gray, had given him the AR-15-style rifle as a Christmas present. A jury convicted Colin Gray of two counts of second-degree murder and other crimes on March 3, 2026, and he was sentenced to 15 years on July 30, 2026.

Crime goes up, then down

Murders in the United States rose from 16,669 in 2019 to 21,570 in 2020, up 29 percent. The Pew Research Center, using FBI data, called it the largest one-year rise in more than a century. The murder rate in 2020 was 7.8 for every 100,000 people, still lower than the rate of 10 in 1991.

The violent crime rate dropped 9.3 percent from 2024 to 2025, from 362.9 to 327.6 for every 100,000 people. FBI officials called it the largest one-year drop since the bureau began making national estimates in 1936. Murders fell about 18 percent, robberies 18.5 percent and property crimes 12.4 percent. The 2025 murder rate, 4.1 for every 100,000 people, tied 1955 and 1956 as the lowest the FBI has recorded. FBI officials released these figures in August 2026.

Leaders in court

A jury in Manhattan convicted former President Donald Trump of 34 felonies on May 30, 2024. The charges were for falsifying business records to hide a payment of hush money, money paid to keep someone quiet, to Stormy Daniels. He was the first US president convicted of a felony. Judge Juan Merchan sentenced him on January 10, 2025, to an unconditional discharge, which means no jail, no fine and no probation. His appeal had not been decided by September 2026.

A federal jury convicted Senator Bob Menendez of New Jersey on July 16, 2024, of taking bribes, secret payments for favors, and of acting as an agent of Egypt's government, which means working for it in secret. Agents had found $480,000 in cash and more than $100,000 in gold bars in his home. He resigned in August 2024, and on January 29, 2025, a judge sentenced him to 11 years in prison.

President Trump pardoned, on January 20, 2025, everyone convicted over the attack on the US Capitol on January 6, 2021, with one exception. The Government Accountability Office found that about 140 police officers were attacked that day. For fourteen members of two groups, the Oath Keepers and the Proud Boys, he cut their sentences to the time they had already served.

Whose courts: McGirt, 2020

Five of the nine Supreme Court justices ruled on July 9, 2020, in *McGirt v. Oklahoma*, that the Muscogee (Creek) reservation had never been ended. Members of Congress had set that land aside for the nation in the 1800s. Justice Neil Gorsuch wrote the opinion.

US officials forced the Muscogee (Creek), the Cherokee and three other nations from their homelands in the Southeast to Indian Territory, which is now Oklahoma, in the 1830s. Those forced removals are called the Trail of Tears. Gorsuch's opinion begins: "On the far end of the Trail of Tears was a promise." Under the ruling, Oklahoma's prosecutors could not charge Native people for major crimes there, and federal and tribal courts took those cases.

Jimcy McGirt, a Seminole citizen, had been convicted in a state court of sexually abusing his wife's granddaughter. A federal jury convicted him after the ruling, in November 2020.

The judges of Oklahoma's highest criminal court then applied the same reasoning to the lands of other nations. As a result, almost all of eastern Oklahoma, including much of Tulsa, counts as Indian country, the legal name for Native land, in criminal cases. The justices ruled in 2022, in *Oklahoma v. Castro-Huerta*, that state prosecutors may also charge people who are not Native for crimes against Native victims there.

Tulsa's survivors, 2020 to 2026

No white person was ever sent to prison for the 1921 Tulsa massacre. Survivors of the massacre sued the city, the county and state bodies in 2020. Tulsa County Judge Caroline Wall dismissed the case in July 2023, and the justices of the Oklahoma Supreme Court upheld her ruling in June 2024.

Viola Ford Fletcher was 7 in 1921. She spoke to Congress on May 19, 2021, and died on November 24, 2025, at 111. Lessie Benningfield Randle, born on November 10, 1914, was the last known living survivor in May 2026.

Tulsa's mayor, Monroe Nichols, announced the Greenwood Trust on June 1, 2025, a private fund with a goal of $105 million by June 1, 2026. The plan listed $24 million for housing, $60 million for buildings and for clearing blight, which means run-down property, and $21 million for land, scholarships and grants to small businesses. It included no direct payments to the survivors. No total raised had been published by September 2026.

BY TODAY

If you remember one thing: The prison count began to fall after 2009, but Black Americans were still in prison at almost five times the rate of white Americans in 2023.

- 2009: State and federal prisons held 1,617,970 people, the most ever counted.
- 2021: A jury convicted Officer Derek Chauvin of murdering George Floyd.
- 2025: Ninety-seven people were cleared of crimes they had not committed, and the murder rate tied the lowest the FBI has recorded.

Next: The chapter closing answers the three questions from the opening.

THE THREE QUESTIONS, ANSWERED

1. **How did punishment move from the whipping post to the prison cell?** Colonial judges had people whipped, branded and hanged in public, and had their ears cut off. Pennsylvania's lawmakers turned the Walnut Street Jail into the country's first state prison in 1790, and by 2009 American prisons held about 1.62 million people.
2. **Who kept order before there were police?** Among the Native nations, families and clans settled killings, and in the colonies settlers took turns on a night watch while white men in the South rode on slave patrols. Full-time police began work in Boston in 1838 and in New York in 1845, and the US attorney general put 34 federal agents to work in 1908.
3. **What happened to the people in mobs who killed without a trial?** In almost every case the chapter describes, nobody punished them. No one was charged for killing twenty Conestoga people in 1763, for burning Francis McIntosh in 1836 or for burning Henry Smith in 1893, and no white person went to prison for the attack on Greenwood, the Black district of Tulsa, in 1921.

What changed: The chapter begins with families settling a killing life for a life and ends with about 1.25 million people in state and federal prisons. From the first colonial courts on, whether a person got a fair trial depended on who that person was.

FIVE DATES TO KEEP

1631: Massachusetts judges order settlers to keep a night watch.

1790: The Walnut Street Jail becomes the first state prison.

1893: A white mob burns Henry Smith to death in Paris, Texas.

1963: The justices rule that a poor person charged with a serious crime must get a lawyer.

2009: The prison count reaches its highest point, 1,617,970.

NOT KNOWN: Nobody knows how many people the attackers killed in Tulsa in 1921, and the estimates run from 50 to 300.

Where to go: Eastern State Penitentiary in Philadelphia, where Charles Williams became Prisoner Number 1 in 1829, is now a historic site.

Next: Chapter 22, War and the Military. It follows the wars the country fought, the draft, and the soldiers and veterans.

Chapter 22: War and the Military

CHAPTER 22 OF 37 · WAR AND THE MILITARY · 10 ERAS · ABOUT 130 MINUTES

Until World War II, sickness killed more American soldiers than weapons did. In almost every war, at home and abroad, soldiers also killed people who could not fight back.

A small photograph on glass, of three young children, was the only thing Sergeant Amos Humiston had on him when he was killed at Gettysburg on July 1, 1863. The 1850 to 1900 era tells how that picture reached his wife. In his war, about five soldiers died of disease for every three who were killed in battle like him.

This chapter follows, in order, the wars fought on this land and then the wars American soldiers fought overseas, from a town on the Missouri River destroyed about 1325 to the war with Iran that began in 2026. As you read, keep three things in view: what killed the soldiers, who was made to go, and what happened to soldiers who killed people who could not fight back. For most of this history, the flu killed more soldiers than bullets did, and so did typhoid and dysentery, infections spread by dirty water and food. In the Civil War a drafted man, one ordered into the army by law, could pay $300 and go home. The men who burned towns and shot prisoners were rarely punished, though a few people stood in their way, like the helicopter pilot Hugh Thompson at My Lai.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. What killed most American soldiers before World War II?
2. Who was made to fight, and who could get out of it?

3. Were soldiers who killed people who could not fight back ever punished?

Five minutes? Read "The Mexican War, 1846 to 1848" in 1800 to 1850 and "Who had to fight in the Civil War?" in 1850 to 1900.

Can skip: "Louisbourg, 1745", "The forts that were destroyed, 1568 and 1586", "The Bonus Army, 1932". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 5 MINUTES

People fought wars in North America long before Europeans came, and everything known about those wars comes from what the fighters left in the ground. No one north of Mexico wrote them down. Archaeologists, scientists who dig up and study what people left behind, have found skeletons with wounds, houses burned where they stood, and ditches and walls built around towns.

From that evidence they can tell that a town was attacked, about when it happened, and what a weapon did to a bone. No fighter's name survives from these wars, and no word that any of them said, including their reasons for fighting.

On the Great Plains. Signs of violence there go back to about 250 B.C. Fighting grew more common after about the year 1250. From then on attackers often burned villages, and skeletons regularly show cut marks from scalping and from bodies being cut apart. Scalping means cutting the skin and hair off the top of a person's head.

Walls, ditches and stone weapons

Farming villages along the Missouri River, in what are now North and South Dakota, often had a ditch dug around them and a palisade, a fence of tall logs set upright in the ground. Some of these villages held up to 1,000 people.

Cahokia. The largest town north of Mexico stood near present-day St. Louis. Between about 1050 and 1150, perhaps 15,000 to 20,000 people lived in its center. Its people began a wall of upright logs about two miles long around the middle of the town about 1100. They rebuilt it three times over about 200 years, so four walls stood there one after another.

Staff at Cahokia Mounds State Historic Site estimate about 20,000 logs for each wall. Towers called bastions stuck out from the wall at even distances. Archers could stand in them and shoot arrows at attackers.

Archaeologists disagree about what the Cahokia wall was for. Some point out that bastions are built for fighting, and that some stretches of the wall went up in a hurry and cut right through neighborhoods where people lived. Others, in lesson pages from the University of Illinois, think it may have marked off the town's powerful families from ordinary townspeople, or insiders from outsiders.

What the bones show. Wounds show which weapons people used. At Norris Farms, in central Illinois, archaeologists studied a cemetery of 264 people from a village of about the year 1300. At least 43 of them, about 16 in every 100, had died by violence. Counting only the adults, at least one death in three came from fighting.

Arrows had hit some of them. Others had been struck from the front, the side and the back with heavy weapons such as stone axes. The bones do not show who the attackers were.

How many died at Crow Creek?

At Crow Creek, on the Missouri River in South Dakota, archaeologists found the bones of at least 486 men, women and children killed in a single attack. The town there belonged to ancestors of the Arikara Nation. Archaeologists from the University of South Dakota dug into the ditch around the town in 1978. They dated the bones by measuring the carbon left in them, a method called radiocarbon dating, and found that the attack happened about 1325.

Nearly the whole town. Douglas Bamforth, an archaeologist at the University of Colorado, calls the 486 dead nearly the whole town. P. Willey, the scientist who studied the bones, thinks they were about 60 in every 100 of the people who lived there. Bamforth thinks the killings came from a fight over food. The likely reason, he says, was too many people in the area for the food they could get, at a time when the weather was getting worse.

Many of the people at Crow Creek had been scalped, had their heads cut off, or had their bodies cut apart. Scalping leaves cut marks on the skull, and cracked skulls were common.

The bones also show that these people were badly underfed before they were killed. Animals had chewed many of the bones, so the bodies lay in the open for some time before anyone buried them in the ditch.

The bones do not show who the attackers were. Fewer young women lie in the grave than a whole town would have had. Willey thinks the most likely reason is that the attackers carried young women away as captives.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 9 MINUTES

Small Spanish armies marched into the homelands of Native nations in the 1500s, and the reports their leaders wrote are the first written record of war in what is now the United States. The soldiers brought horses and trained attack dogs, and they fought with swords and lances. Their reports tell only the Spanish side.

Forts and prisoners. Native people destroyed every fort one Spanish captain built inland, all of them by 1568. Spanish soldiers and French colonists also fought each other on the coast of Florida. There the Spanish commander had his men kill about 245 French prisoners who had already surrendered.

De Soto's march, 1539 to 1543

Hernando de Soto landed near Tampa Bay, in Florida, in May 1539, with about 600 men, 220 horses, packs of dogs and a herd of pigs. For the next three years his army crossed the Southeast. De Soto took hostages to force the towns to let his army pass. He held the leader of the Coosa towns hostage to get through those towns, and his men ate the food the Coosa had stored.

The National Park Service's page on de Soto states what his men did to towns that did not cooperate. They threw people to the dogs, setting trained attack dogs on captives to bite and tear them, often to death. They burned people alive. They raped Native people, which means they forced them to have sex. They cut off people's noses and hands, and they enslaved men and women to carry the army's loads.

The end of the march. De Soto died of a fever on May 21, 1542. His men sank his body in the Mississippi River and hid his death, so the nations they had been fighting would not learn that he could die. In June 1543 the rest of the army went down the Mississippi in seven boats. Only 311 of the Spanish reached the Gulf of Mexico. They took enslaved Native people with them, and no one recorded how many.

How the battle at Mabila began

De Soto's army reached Mabila, a town with a palisade around it, in October 1540, a year and a half before de Soto died. The battle there began on October 18, when a Spanish officer decided that a Mabila man had not shown him enough respect. The argument ended with the officer cutting off the man's arm. An article at HistoryNet, a history magazine website, names the officer as Captain Baltasar de Gallegos. Mabila's leader was a man the Spanish called Tascalusa.

The battle lasted about nine hours. At the end de Soto's soldiers burned the town while people were still inside it. The army's own record-keeper wrote, "We killed them all either with fire or the sword."

The counts of the dead do not agree. The National Park Service's page lists more than 200 Spanish dead and about 2,000 people of Mabila. A University of Alabama news report states about 20 Spanish killed and 250 wounded, with thousands of Mabila dead. Searchers from the University of Alabama have looked for the town, and no one has ever found where Mabila stood.

The Tiguex towns, 1540 to 1541

Far to the west, in the winter of 1540 to 1541, the Spanish commander Coronado and his army fought the Tiguex towns on the Rio Grande in New Mexico. The Tiguex were Pueblo people, who lived in towns of houses several stories tall. They spoke the Tiwa language and lived in twelve or more towns near present-day Albuquerque. Tiwa is still spoken today at the pueblos of Isleta, Sandia, Taos and Picuris.

Corn and cloaks. Coronado's soldiers took over the town of Coofor and made its people leave. The army's horses and mules ate the corn stalks the Tiguex needed to burn for heat in winter. Then Coronado demanded about 300 or more pieces of cloth from the towns, and his men did not give the towns time to decide. People had to take off their own cloaks and hand them over. Soldiers took better cloaks off any Tiguex man they saw wearing one.

A complaint to Coronado. Then a Spanish soldier went into a Tiguex house and raped the wife of the man who lived there, or tried to. Pedro de Castañeda, a soldier on the expedition who later wrote its history, says so. The husband and the town's leaders complained to Coronado.

The husband could not pick out the soldier in a line of men, but he knew the soldier's horse. The owner of the horse denied it, and the husband got no justice. Castañeda knew the soldier's name and chose not to write it down.

The next day Tiguex men killed a Native man who guarded the army's horses and drove off many of the horses. Coronado sent Don García López de Cárdenas to surround the town where the soldier had raped the woman, or tried to. After a day and a night, Cárdenas's men and their Native allies from Mexico smoked the defenders out of the cellars.

The defenders made the sign of the cross to ask for peace. Two Spanish soldiers on the roof made the sign back. The men put down their weapons, and the Spanish promised them they would not be punished.

The stakes. Coronado had ordered Cárdenas not to take anyone alive and to make an example of them to frighten the other towns. Cárdenas ordered 200 stakes set up to burn the prisoners alive. Castañeda wrote that when the

prisoners saw the Spanish tying men to the stakes and starting to burn them, about 100 men in a tent fought back. Spanish foot soldiers and horsemen killed them.

Castañeda gives no exact count of the people burned. A history from the New Mexico Office of the State Historian names the town as Arenal, across the river from Coofor, and states that the Spanish killed dozens of captives there. At his trial, Cárdenas told a different story. He said some soldiers had speared men in a tent without his order.

The fighting went on for about three months. The Tiguex shut themselves inside their houses and shot arrows from the roofs and through slots in the walls. The Spanish charged on horseback and set fires to force people out. By the count on a National Park Service page, the Spanish killed hundreds of Tiguex, and the Tiguex killed a smaller number of Spanish and Mexican soldiers. The Spanish burned villages and enslaved Tiguex women and children.

Cárdenas on trial. Spanish officials later put Cárdenas on trial. A royal prosecutor, the lawyer who brought the case for the king, charged him with killings, with rapes committed by his men, with setting dogs on Native people, and with taking food and clothing. The judges of the Council of the Indies, the king's council for Spain's colonies, found him guilty in 1549 and said so again in 1551.

They first ordered him to serve 30 months as a soldier at Oran, in North Africa, and to pay 800 ducats, which were Spanish coins. He asked for a lighter punishment and got one: 12 months as a soldier in Spain, a fine of 200 ducats, and 10 years banned from the Americas.

Why the inlet is called Matanzas

An inlet south of St. Augustine, in Florida, has been called Matanzas, the Spanish word for "slaughters," since 1565. There Spanish soldiers killed about 245 French prisoners who had surrendered.

Fort Caroline. About 200 French colonists built Fort Caroline near present-day Jacksonville in 1564. They were Huguenots, French members of the Protestant churches, which had broken away from the Catholic Church, the church led by the pope in Rome. The land belonged to the Timucua, who had lived in northern Florida for at least 1,000 years. King Philip II of Spain claimed Florida, and he was Catholic.

The Spanish founded St. Augustine, farther south, on September 8, 1565. Twelve days later their commander, Pedro Menéndez de Avilés, marched 500 soldiers overland and attacked Fort Caroline at dawn. His men killed most of the settlers they found. In his own letter to the king, Menéndez counted 132 men at the fort whose throats his soldiers cut, and 10 more killed the next day. He spared about 50 women and children and sent them by ship to Havana, in Cuba.

At the inlet. A hurricane had already wrecked the French ships. About 127 French survivors came ashore at the inlet with no food and no weapons, and they surrendered to Menéndez. He told them to give up their Protestant faith. They refused, and his soldiers killed 111 of them.

Menéndez spared 16 men. A few said they were Catholic. Some were sailors from Brittany who had been forced onto the ships. Four were craftsmen the Spanish settlers at St. Augustine needed. Menéndez wrote to King Philip on October 15, 1565, "I had their hands tied behind them and put them to the knife."

A second group of French survivors came to the same inlet with their commander, Jean Ribault. They surrendered and refused the same demand, and Menéndez's men killed 134 of them. In his letter Menéndez gave his reason: the French "spread the Lutheran sect," meaning Protestant teaching, in land the king of Spain claimed.

The Florida Museum of Natural History dates the first killing to September 29, 1565. It dates the second to October 11, and the National Park Service dates it to October 12.

The forts that were destroyed, 1568 and 1586

Captain Juan Pardo built a chain of small Spanish forts inland between 1566 and 1568. He built Fort San Juan in January 1567 at the Native town of Joara, in the upper Catawba Valley of North Carolina, and thirty soldiers held it. In all, Pardo left about 120 men in forts across the Carolinas and east Tennessee.

The soldiers demanded more food than the towns around them could give, and they attacked the people they lived among. By May 1568, Native people had attacked every one of the forts, destroyed them all and killed nearly every soldier. Archaeologists have found Fort San Juan. It is the Berry site in Burke County, North Carolina.

Drake at St. Augustine. English sailors under Francis Drake burned St. Augustine in 1586. Drake was a privateer, a sea captain allowed by his government to attack Spanish ships and towns. He arrived with 21 ships between May 28 and 30, 1586, and the town had about 150 soldiers. His men burned the town and its wooden fort and took about 2,000 pounds in money from the treasury.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 10 MINUTES

English colonists fought their first wars on this coast against the nations whose land they had settled on: the Powhatan in Virginia, the Pequot in Connecticut, and Metacom's Wampanoag and their allies in New England. In every one of these wars, Native allies fought beside the colonists. Mohegan and Narragansett fighters, for example, joined the English against the Pequot in 1637.

The militia. The English colonies had no army. They had a militia instead, made up of ordinary men, mostly farmers, who left their work to fight when their leaders called them and went home when the fighting ended.

The lawmakers of Massachusetts Bay, meeting as the General Court, ordered every healthy man between 16 and 60 to join the militia on December 13, 1636. Ministers and judges did not have to. The lawmakers organized the men into three regiments, large units of soldiers. Leaders of today's Army National Guard date its beginning to that order.

Colonial soldiers struck at towns and food. They attacked the Pequot town at Mystic before dawn in 1637. In Virginia, from 1622 on, they burned Powhatan corn in the fields and seized the corn people had stored.

The Powhatan wars in Virginia, 1622 to 1646

The English colony in Virginia stood on the land of the Powhatan, a group of nations led by one head chief. In 1622 the Powhatan leader Opechancanough judged that the English would keep taking more Powhatan land unless they were stopped.

The attack of 1622. Opechancanough sent fighters to the English plantations, or large farms, along the James River in one planned attack on March 22, 1622. The men came to houses and fields as if to trade or share a meal, then attacked. They killed as many as 347 colonists, out of about 1,200 in the colony.

A Powhatan man who worked for a settler named Richard Pace warned him, and Pace warned Jamestown in time. The English accounts of the warning do not give the Powhatan man's name.

The poisoned toast. Opechancanough agreed to meet English leaders on the Pamunkey River in May 1623 to make peace. Captain William Tucker led the English party of about 12 men. The English offered poisoned drinks to toast the agreement. When the Powhatan leaders fell sick, the English shot them, and some of the English cut off scalps.

The English accounts of the dead do not agree. A settler named Robert Bennett wrote to his brother on June 9, 1623, that "some tooe (two) hundred" were thought to have been poisoned, and that the English killed about 50 more. Governor Francis Wyatt and his council answered in 1624 that no one had been poisoned at a feast and that only nine were shot. Their own report says the men took poison along, to be used if nothing else worked. Neither account names the poison.

Burning the corn. The English also went after the Powhatan food supply, from 1622 on. In their own words, they waited "till their corne was ripe," then raided through the fall and winter to "surprize their corne," meaning to seize it. They agreed to pauses in the fighting so both sides could plant, then

attacked again after the harvest. In July 1624, English soldiers burned fields with enough corn, by their own count, to feed 4,000 men for a year. The war lasted until 1632.

Opechancanough attacked again on April 18, 1644, and killed about 400 colonists. Virginia's governor, Sir William Berkeley, led soldiers who captured him in 1646 and took him to Jamestown. Within two weeks, one of his English guards shot him dead in his cell. Encyclopedia Virginia's history of Opechancanough does not name the guard.

The Pequot War, 1636 to 1638

The Pequot War grew out of two killings of English traders. Pequot men killed the trader John Stone and his crew on the Connecticut River in 1634. Manisses men from Block Island, a different nation from the Pequot, killed the trader John Oldham in July 1636.

Endecott's raid. John Endecott of Massachusetts Bay led 90 soldiers to Block Island in August 1636 and burned the villages there. Then his men sailed to Pequot land and burned a Pequot village too. Pequot fighters attacked Wethersfield, Connecticut, on April 23, 1637, killed nine men and women, and took two girls captive.

Mystic. Captain John Mason of Connecticut and Captain John Underhill of Massachusetts Bay reached the Pequot town at Mystic before dawn on May 26, 1637. Mohegan and Narragansett fighters came with them. The English set fire to the palisade and the houses inside it, and they shot the people who tried to get out.

Historians at the Mashantucket Pequot Museum count more than 400 Pequot killed in one hour, and other counts go as high as 700. Most of the Pequot fighting men were away with their sachem, or leader, Sassacus, so most of the dead were old men, women and children. The English lost two dead and twenty wounded.

The English fought the surviving Pequot in a swamp near present-day Fairfield, Connecticut, in July 1637, for about a day and a night. Sassacus escaped west, and Mohawk fighters killed him later that month. The Pequot

museum's history of the war states that some captive Pequot women were killed after the English argued over who would get them, and it does not name who killed them. The English sold other captives into slavery in New England and the Caribbean.

Captives for sale. John Winthrop, the governor of Massachusetts Bay, wrote in his journal on July 13, 1637, that the colony's leaders had sent 15 Pequot boys and 2 women away as slaves. A ship captain, William Peirce, carried them. Massachusetts leaders held a day of thanksgiving on October 12, 1637, for defeating the Pequot and bringing the soldiers home safe. In February 1638 Peirce came back on the ship *Desire* with cotton, tobacco and enslaved Africans.

The Treaty of Hartford. Leaders of the Connecticut colony at Hartford and Mohegan and Narragansett leaders signed the Treaty of Hartford, a written agreement, in 1638. Under it they divided the surviving Pequot between the Mohegan and the Narragansett. The signers barred the Pequot from their homeland and forbade them ever to call themselves Pequot again.

Pequot survivors held together. By 1651, Robin Cassacinamon had led some of them back to 500 acres of their land. In 1666, leaders of the Connecticut colony set aside a reservation, land kept for a Native nation to live on, for the Pequot at Mashantucket. The Mashantucket Pequot Tribal Nation is in Connecticut today.

King Philip's War, 1675 to 1676

King Philip's War was fought in New England in 1675 and 1676. The English called the Wampanoag sachem Metacom by the name King Philip.

A National Park Service article calls it one of the deadliest wars in American history, when the number killed is compared with how many people lived in New England. One estimate, published by HistoryNet, counts about 9,000 people killed out of about 80,000 in New England, about one person in nine. About two out of every three of the dead were Native people, and many of them died of starvation.

A death and a trial. John Sassamon, a Christian Native man, was found dead under the ice of a pond in January 1675. Days before, by the accounts of the time, he had warned Plymouth's governor, Josiah Winslow, that Metacom's men were planning an attack.

English judges at Plymouth tried three Wampanoag men for his murder on June 1, 1675, before an all-English jury. Six Native men sat with the jury to advise it. Plymouth's officials hanged two of them, Tobias and Mattashunannamo, on June 8. They held back the third, Wampapaquan, for a month and then shot him to death. The fighting began that June at the English town of Swansea.

The swamp fort. When the war began, Metacom sent the old men, women and children of the Wampanoag across Narragansett Bay to live with the Narragansett. The Narragansett stayed out of the fighting. In October 1675 their leaders promised Massachusetts Bay in a treaty that they would not join Metacom. The leaders in Boston then demanded that the Narragansett hand over the Wampanoag who had fled to them, as hostages. The Narragansett refused.

Leaders of the English colonies feared the two nations would join together. About 1,000 English soldiers attacked a fortified Narragansett winter village in a frozen swamp in Rhode Island on December 19, 1675. About 1,000

Narragansett men, women and children were living there. Governor Winslow of Plymouth and two majors, from Massachusetts and Connecticut, led the soldiers of their colonies.

The soldiers took the fort and burned every house in it. The National Park Service counts 300 Narragansett men and almost 400 women and children killed. James Oliver, an English officer, wrote that the English lost about 68 dead and 150 wounded.

After the attack, the Narragansett joined Metacom, and Native fighters struck back across New England. By one count, they attacked 52 of the 90 English towns and burned 17 of them to the ground. Narragansett fighters came to Providence in late March 1676. Roger Williams of Providence tried to bargain with them, and they turned him down and burned the town.

A Native man fighting with the colonists shot Metacom dead at Mount Hope on August 12, 1676. The colonists put his head on display at Plymouth. They sold his wife and his nine-year-old son into slavery.

What happened on Deer Island?

Massachusetts Bay's leaders held Christian Native people, who were allies of the English, on Deer Island in Boston Harbor through the winter of 1675, and many of them starved. The leaders moved them there that autumn because they did not trust them. The first were Nipmuc people from the town that is now South Natick. Between 500 and 1,100 people were held there, most of them women and children, with little shelter and too little food.

The colonists let the survivors go in May 1676. The two counts of the dead do not agree. Mass Humanities, a history group, states that about half of some 900 people held there died of hunger and cold. A National Park Service page states that very few survived, so by its account most of them died.

Some people were smuggled off the island and sold as slaves in the West Indies, the islands of the Caribbean, or in Tangier, in North Africa. The National Park Service page gives no number and does not name the men who sold them. Massachusetts colonists also put about 45 Native people to death in public on Boston Common.

Sold for life. Governor Josiah Winslow of Plymouth and Governor John Leverett of Massachusetts Bay signed papers that allowed Native men, women and children to be sold into slavery for life. A colonial record notes one sale: "that day wee sold Capt Davenport 47 Indians young & old for 80 pd," meaning 80 pounds for 47 people. Ships carried captives to Barbados, Bermuda, Jamaica, the Azores, Spain, and Tangier.

Descendants of captives sent from Rhode Island to Bermuda have since reconnected with Native nations in New England and Canada.

The Beaver Wars and the Pueblo Revolt

Other wars of the 1600s had no English colonists in them. Chapter 2, Native Nations, tells these two from the side of the nations who fought them. In the Beaver Wars, fighters from the Haudenosaunee nations, also called the Iroquois, raided villages of the Wendat, also called the Huron, for years, to control the trade in furs. French and Dutch traders armed their Native trading partners with guns, and the guns made the raids deadlier.

The raids broke the Wendat apart between 1642 and 1649. Haudenosaunee fighters destroyed the Wendat homeland, Wendake, in 1649 and 1650. They killed many Wendat and adopted many others into Haudenosaunee nations. About 500 Wendat moved east to the Quebec City region.

The Pueblo Revolt. People from the Pueblo towns of the Rio Grande rose together against Spanish rule in New Mexico on August 10, 1680. They surrounded Santa Fe, took it, and drove the Spanish out. They killed 401 settlers and 21 Franciscan friars, who were Catholic priests, and let most other Spanish people leave. The Pueblos governed themselves for 12 years. Diego de Vargas began the Spanish return in 1692, and armed resistance went on for years after.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 9 MINUTES

Both wars that reached North America in these years began in Europe, one over who would become king of Spain and one over who would rule Austria. Colonial militia and Native fighters did most of the fighting here, and the Native fighters had their own reasons. Here the two wars are called Queen Anne's War, from 1702 to 1713, and King George's War, from 1744 to 1748.

The Haudenosaunee stay out. The leaders of the Haudenosaunee Confederacy, an alliance of Native nations with the biggest fighting force in the northeast, chose to stay out of these wars.

At Montreal on August 4, 1701, about 1,300 representatives from more than 30 Native nations made peace with New France, the French colony in Canada. There the Haudenosaunee leaders promised to take neither side between France and Britain. They worked to keep that promise for about fifty years, and French and British commanders both had to plan around it.

The winter raid. In the north, the common act of war was the winter raid. Fighters came at a sleeping town before dawn, killed some of the people and took others captive. Then they marched the captives hundreds of miles to Canada on foot. A captive might be adopted into a Native nation, sold back to family for money, or killed on the trail for not keeping up.

Deerfield, February 29, 1704

About 250 fighters attacked Deerfield, a town in western Massachusetts, before dawn on February 29, 1704. Their commander was Jean-Baptiste Hertel de Rouville, a French lieutenant, or junior officer, born in Canada. He led 47 French Canadian soldiers and about 200 fighters from the Abenaki, the Wendat and the Kanien'kehaka, as the Mohawk people call themselves. The historians of Deerfield's museum write that these nations fought to keep English settlers out of their homelands.

About 300 people lived in Deerfield, and more than half were killed or taken captive. Deerfield's history museum, the Pocumtuck Valley Memorial Association, which keeps the town's old records, lists 48 killed and 111 captured. A Library of Congress history page lists 57 killed and 112 captured.

In the house of the town's minister, John Williams, the attackers killed his six-week-old daughter Jerusha and his six-year-old son John. They also killed Parthena, a Black woman Williams had bought and held as a slave. Her husband Frank, also enslaved by Williams, was taken captive.

The march north. The attackers drove the captives across the frozen Deerfield River and marched them 300 miles to New France in the middle of winter. Each captive belonged to the man who had taken them, and those men killed captives who could not keep up. Williams wrote that on one day they killed two women "so faint they could not travel."

Mary Brooks, a young woman, told Williams that her falls on the ice had caused a miscarriage, which means she had lost the baby she was carrying. She said she was too weak to walk far and knew they would kill her. They killed her that day. Williams did not record the names of the men who killed them. Of the more than 100 captives, only about half ever came back to New England.

John Williams

Who: Deerfield's minister, captured with five of his children in the 1704 raid. He came home and wrote the best-known captivity story of colonial America.

When and where: Born in Roxbury, Massachusetts, December 16, 1664. Minister at Deerfield from 1686. Captured February 29, 1704. Back in New England late in 1706. Died 1729.

The raiders took him, his wife and five of their children.

Soon after the march began, at the Green River in what is now Greenfield, Massachusetts, his wife, Eunice Mather Williams, fell while crossing and went under the icy water. She walked on a short way. At the foot of a mountain, the man who had taken her captive killed her with one blow of a hatchet. Williams did not record his name.

Williams and four of his children came home. His daughter Eunice, who was seven when the raiders took her, did not. He returned late in 1706 and preached about his captivity in Boston on December 5 that year, and in 1707 he went back to his church in Deerfield.

His book about the raid and the march, *The Redeemed Captive Returning to Zion*, was printed in seven editions before 1795.

Eunice Kanenstenhawi Williams

Who: John Williams's daughter, taken from Deerfield at seven. She became a member of the Kanien'kehaka nation at Kahnawake and chose to stay there all her life.

When and where: Born in Deerfield, September 17, 1696. Taken February 29, 1704. Died at Kahnawake, near Montreal, in 1785, aged 89.

Carried off at seven, she refused every offer to bring her home.

Eunice Williams was seven when the attackers carried her out of Deerfield. At Kahnawake, a Kanien'kehaka town near Montreal, a family adopted her in a formal ceremony. She was first named A'ongote, "she was taken and placed." Later she was named Kanenstenhawi, "she brings in the corn," and was accepted as a full member of the nation. She was also baptized a Catholic, in the Christian ceremony with water that makes a person a member of a church, and given the name Marguerite.

Her father and English officials tried again and again to bring her back. She refused. At 16 she married François-Xavier Arosen, a man of her town. They had at least three children: two daughters, Catherine and Marie, born in 1736 and 1739, and a son, John. The Pocumtuck Valley Memorial Association's history says John seems to have died at Lake George, in New York.

She visited her Williams brothers and sisters in New England four times, in 1740, 1741, 1743 and 1761. At her brother Stephen's house in Longmeadow, Massachusetts, she slept outside in the orchard instead of inside the English house. Each time, she went home to Kahnawake. New Englanders called her the "unredeemed captive," the one they never bought back.

Why the Tuscarora went to war, 1711 to 1713

The Tuscarora of North Carolina had three complaints, as the historians Elizabeth A. Fenn, Peter H. Wood and others name them in the account NCpedia publishes. They objected to the way white traders dealt with them and to the enslaving of Native people. Most of all, they objected to settlers taking their land. English settlers had founded the town of New Bern near the Tuscarora in 1710.

The Tuscarora began a series of planned attacks on the settlers on September 22, 1711. Soldiers from South Carolina, fighting with Native allies, defeated them by March 1713. By then about 1,400 Tuscarora were dead, and the winners had captured and enslaved about 1,000.

About 1,500 surviving Tuscarora walked north. In 1722 they became the sixth nation of the Haudenosaunee Confederacy.

The Yamasee War, 1715 to 1717

The Yamasee War in South Carolina began over debt and the slave trade in Native people. Yamasee towns stood along the Savannah River and around Port Royal. Traders from Charleston let the Yamasee take goods and pay later, hoping to take Yamasee land when the debts could not be paid. By 1715, paying off the debt would have taken at least two years of work from every Yamasee man.

The traders also enslaved Native people and whipped them, and settlers were moving onto Yamasee land. The Yamasee themselves had raided Apalachee towns in Florida and sold the people they captured to Carolina merchants as slaves.

Pocotaligo. Thomas Nairne, the colony's agent for dealing with Native nations, met Yamasee leaders at Pocotaligo Town on April 14, 1715, with two other colonists, William Bray and Samuel Warner. They hoped to stop the fighting before it started. The next morning, Good Friday, Yamasee men killed Bray and Warner. They held Nairne for several days and tortured him to death by sticking burning splinters of wood into his skin.

Across the South, Creek, Choctaw, Apalachee and other men killed the traders who worked among them. By June 1715, about nine in every ten of those traders were dead.

The first Yamasee attacks, on plantations near Port Royal, killed about 100 colonists. Over the whole war, about 400 settlers were killed, and colonists had to leave half the farmland in the colony. The South Carolina Encyclopedia and NCpedia give no count of the Yamasee dead or the other Native dead.

Craven's war. South Carolina's governor, Charles Craven, called up every white man and armed some enslaved Black men as well. About 300 families who owned plantations escaped by ship. Craven's forces beat the Yamasee at Port Royal and at the Salkehatchie River and drove them south of the Savannah River. By the spring of 1716, most Yamasee had moved to Spanish Florida.

Carolina officials talked the Lower Cherokee into fighting the Creek in 1716. That began a war between the Cherokee and the Creek that lasted about forty years. Carolina officials ended their own war by 1718.

Louisbourg, 1745

King George's War began in this part of the world in 1744, after a group of English scalp hunters, men who hunted Native people for their scalps, killed or wounded several Penobscot people. In 1745 New England soldiers laid siege to Louisbourg, a French fortress on Cape Breton Island, in what is now Canada, surrounding it to cut it off until it gave up.

Massachusetts leaders raised an army of their own colonists to take the fortress. The soldiers sailed from Boston on March 24, 1745, and nearly 2,000 of them landed near Louisbourg on April 30. Colonel William Pepperrell commanded them. Commodore Peter Warren, a British navy commander, waited offshore with a group of warships. Up to about 4,300 New Englanders served there over the whole siege, but only about half that many at any one time.

The acting French governor, Louis Du Pont Duchambon, asked on June 15 what conditions he would get if he surrendered. He and his soldiers surrendered the fortress on June 17, 1745.

The winter after. Far more New Englanders died holding Louisbourg than died taking it. During the siege, French fire killed about 100, and 30 died of illness. Then the soldiers spent the winter in wrecked, dirty buildings, and sickness spread. The counts of the dead do not agree. A HistoryNet account counts 561 burials between late November and mid-February. The Dictionary of Canadian Biography gives 1,200 to 2,000 deaths over the winter and spring.

British officials gave Louisbourg back to France in 1748 in the Treaty of Aix-la-Chapelle, the agreement that ended the war. In return, French officials gave back Madras, a city in India they had taken from Britain.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 17 MINUTES

Soldiers from the colonies, and then from the new United States, were at war for most of the forty years after 1754. In the Revolution, sickness killed more of them than enemy weapons did.

The French and Indian War began that year. In the treaty that ended it, in 1763, French officials signed over Canada and all French land east of the Mississippi River to Britain. The Revolution lasted from 1775 to 1783 and ended with the thirteen colonies independent. From 1785 to 1795, US soldiers fought a group of Native nations north of the Ohio River.

Counting the dead. Historians at the American Battlefield Trust estimate that about 6,800 Americans were killed in battle in the Revolution. At least 17,000 more died of disease, about two and a half for every one killed in battle, and 8,000 to 12,000 of those were prisoners of war. The Defense Department's official list shows 4,435 American battle deaths, and its own note says its records from before World War I are often incomplete.

The French and Indian War, 1754 to 1763

British and colonial soldiers fought French soldiers and the Native nations allied with France in the French and Indian War. French soldiers forced a young Virginia colonel, George Washington, to surrender at Fort Necessity, in western Pennsylvania, in 1754.

Braddock's defeat. The next summer, the British general Edward Braddock marched about 1,450 men toward Fort Duquesne, the French fort where Pittsburgh stands now. About 900 French and Canadian soldiers and Ottawa, Shawnee, Lenape and Mingo fighters caught his column, a long line of marching soldiers, near the Monongahela River on July 9, 1755.

The Native fighters shot from the woods on both sides of the road. They and the French killed 456 of Braddock's men and wounded 422. The French side lost 23 killed. Braddock was badly wounded, and he died on the night of July 13.

Fort William Henry. The French general Montcalm surrounded the British at Fort William Henry, on Lake George in New York, with about 8,000 men, nearly 2,000 of them Native fighters. The British commander, Lieutenant Colonel George Munro, surrendered on August 9, 1757. Under the terms, his men could march away with their muskets but no ammunition.

The next day, Native fighters allied with Montcalm killed and scalped sick and wounded men left in the fort. Then they attacked the column of British soldiers, and the women and children with it, on the road.

The counts of the dead run from several dozen up to about 185 men, women and children. Early reports said 1,500, but a HistoryNet account finds it unlikely that more than 200 were killed. Montcalm got 500 captives released, and the Native fighters kept about 200 more.

British soldiers took Quebec on September 13, 1759, after a battle outside the city that lasted about twenty minutes. The British general James Wolfe was shot and died on the field. Montcalm was hit by a musket ball below the ribs and died the next morning.

Did the blankets from Fort Pitt spread smallpox?

Officers at Fort Pitt, the British fort at present-day Pittsburgh, gave two Lenape men blankets from the fort's smallpox hospital in June 1763, meaning to give them the disease. No one can tell whether the blankets caused the smallpox that spread among the Ohio nations that year and the next. Smallpox spreads from person to person, mostly through the air people breathe, and it killed about one in three people who caught it. It was already in the area that spring.

The siege. In 1763, the year the war with France ended, Native nations of the Ohio country went to war against the British in what is called Pontiac's War. Shawnee, Lenape, Mingo and Seneca fighters surrounded Fort Pitt on June 22, 1763. Guyasuta was one of their leaders. Captain Simeon Ecuyer commanded the fort, with the trader and militia captain William Trent.

That spring the fort's officers had set up a hospital for soldiers sick with smallpox. On June 24, two Lenape men, one named Turtleheart, came to the fort to warn the soldiers to leave. Trent wrote in his journal, "we gave them two Blankets and an Handkerchief out of the Small Pox Hospital. I hope it will have the desired effect."

Amherst's letters. Two weeks later, the British commander in North America, General Jeffery Amherst, suggested the same thing in a letter to Colonel Henry Bouquet. "Could it not be contrived (arranged) to Send the Small Pox among those Disaffected (unfriendly) Tribes of Indians?" he wrote on July 8. Bouquet answered that he would try to spread it "by means of Blankets."

On July 16 Amherst told him to try it, and to try "Every other method that can serve to Extirpate (wipe out completely) this Execreble (hateful) Race." Amherst also told Bouquet to take no Native prisoners.

The Revolution, from Lexington to Yorktown

The Revolution was fought between British soldiers and colonists who wanted independence. The shooting started at Lexington, Massachusetts, at about 5 in the morning on April 19, 1775. British soldiers killed eight militiamen on the town green and wounded ten. The fighting went on all day along 16 miles of road back to Boston, and by night 73 British soldiers and 49 militiamen were dead.

Bunker Hill. British soldiers attacked the colonists' earth walls on Breed's Hill on June 17, 1775, in the battle named for nearby Bunker Hill. They charged three times in two hours. On the third charge they climbed the walls and fought with bayonets, the long blades fixed to the ends of their muskets. The colonists had run out of ammunition and fell back. British casualties, meaning the killed and wounded together, came to 1,054, and the colonists lost about 450.

That June, the delegates, or representatives, from the thirteen colonies who met as the Continental Congress chose Washington to command a new Continental Army. About 231,000 men served in it during the war, but never more than 48,000 at one time. About 7,000 of them were Black.

Who had to serve. Under the rules of the time, the delegates in Congress had no power to draft anyone, which means to order a person into the army by law. So in 1778 they asked each state to draft men from its militia for one year in the Continental Army. Officials in some states drafted more men than others, and the army stayed short of men. A drafted man could pay someone else to go in his place.

Lawmakers in the colonies and the states wrote laws that kept enslaved men out of the militia. That ban covered the militia, not the Continental Army, where about 7,000 Black men served.

Pennsylvania lawmakers made Quakers, whose religion forbids fighting, pay a fine if they would not join the volunteer militia companies called Associations. The fine matched the value of the time a man would have spent drilling,

practicing marching and handling weapons with his company. Officials took property from Quakers who refused to pay.

Saratoga and Yorktown. The British general John Burgoyne surrendered his army of 5,895 British and German soldiers at Saratoga, New York, on October 17, 1777. When the news reached France that December, the French foreign minister decided to make an alliance with the United States.

Four years later, Washington and the French general Rochambeau led about 17,000 American and French soldiers to Yorktown, Virginia. They surrounded the British general Charles Cornwallis there, while a French fleet held the sea behind him. About 8,000 British soldiers surrendered on October 19, 1781. It was the last big land battle of the war. When the news reached London, the British prime minister, Lord North, said, "Oh God. It is all over." The peace talks that followed ended with the Treaty of Paris in 1783.

Smallpox and Valley Forge

Washington ordered his whole army inoculated against smallpox on February 6, 1777. To inoculate a soldier, a doctor cut open a sore on a person sick with smallpox, then pushed the same blade under the healthy soldier's skin. The soldier then had a milder case of smallpox and could never catch it again.

Valley Forge. More than 12,000 soldiers, with about 400 women and children, marched into winter camp at Valley Forge, Pennsylvania, on December 19, 1777. They built about 1,500 log huts, each 14 feet by 16 feet. Four days later, Washington wrote to Congress that 2,898 of his men were "unfit for duty, because they are barefoot and otherwise naked."

Nearly 2,000 people in the camp had died by June 1778. Most died of the flu, of typhus, a fever spread by body lice, or of typhoid or dysentery, which spread through dirty water and food. Dysentery is an infection of the gut that causes bloody diarrhea, and a person can lose so much water that they die. Two out of three of those deaths came in March, April and May.

The prison ships of New York

The British held New York City from 1776 to the end of the war. Their officers kept captured American soldiers and sailors in prisons in the city and on ships in the harbor.

The British admiral in command at New York was Richard Howe. The ship called the *Jersey* held about 1,200 prisoners. The British officers who ran the ships gave the prisoners too little food and left the sick without care. By the National Park Service's account, smallpox, hunger and that neglect killed thousands.

One survivor wrote that the ship's biscuits were eaten through by weevils, which are small beetles. The bread was sour and often moldy, and the meat was rotten with age. Every morning, he wrote, a boat from each of the hospital ships carried the dead ashore, where the bodies "were all tumbled together into a hole dug for the purpose."

The Sugar House and the Provost. On land, the British held more than 4,500 Americans in the Sugar House, a sugar warehouse five stories tall. Another prison, the Provost, was run by Captain William Cunningham. His prisoners were packed so tightly that at night they could turn over only if they all turned at once.

Leaders of the Congress asked Washington to protest to Howe in January 1777. Washington wrote to Howe about prisoners the British had sent back from New York. He said their "miserable emaciated Countenances," meaning faces so thin from hunger that the bones showed, proved what they said.

The counts of the dead do not agree. A National Park Service page gives more than 11,500 dead on the ships and in the prisons around New York. Historians at the American Battlefield Trust estimate 8,000 to 12,000 deaths among all American prisoners in the whole war.

Gnadenhutten, March 8, 1782

Gnadenhutten was a town of Lenape and Mohican Christians in the Tuscarawas Valley, in what is now Ohio. They belonged to the Moravian church, and they refused to fight for either side in the Revolution. In February 1782, more than 150 of them went back to their old towns to gather the corn they had left in the fields.

About 160 Pennsylvania militiamen marched there under David Williamson, the officer they had chosen to command them. They blamed the Moravian Lenape for raids on Pennsylvania settlements, which the Lenape denied. The militiamen promised to take the people to Fort Pitt for safety and collected their weapons. Then they held a vote on whether to kill them. Only 16 or 18 of the militiamen voted against it.

The killings. The prisoners spent the night praying and singing hymns, with the men in one building and the women and children in another. On the morning of March 8, 1782, the militiamen tied them up. Taking turns, they struck each person on the head with a heavy hammer called a mallet, and then they scalped them.

They killed 96 people. Eric Sterner, writing in the *Journal of the American Revolution*, counts 62 adults and 34 children. Another count gives 28 men, 29 women and 39 children.

A militiaman named Obadiah Holmes Jr., who opposed the killing, said another militiaman, Nathan Rollins, killed 19 people himself. Two boys lived, and one of them survived being scalped. The militiamen burned the buildings with the bodies inside, burned the other Moravian towns, and carried off goods that needed 80 horses to move.

People talked of bringing the killers to court, but no one charged any of the militiamen with a crime. Later the Moravian missionary John Heckewelder, a man sent to spread his religion, gathered the bones and buried them at the town.

George Washington

Who: A Virginia officer in the French and Indian War who went on to command the Continental Army for the whole of the Revolution.

When and where: Fort Necessity, 1754. The Monongahela, July 9, 1755. Commander in chief from June 1775. Resigned at Annapolis, Maryland, December 23, 1783.

Movie: *Washington* (History channel, 2020), a documentary series in three parts, with acted scenes.

Four bullets went through his coat at the Monongahela.

As a young officer, George Washington was on the losing side twice in two years. At Fort Necessity in 1754 he surrendered to the French, and he promised that he and his men would go back to Virginia and stay out of the region for a year.

A year later he rode with Braddock as a volunteer aide, a helper to the general. After the battle at the Monongahela he wrote to his brother, "I had 4 Bullets through my Coat, and two Horses shot under" him.

The delegates of the Continental Congress chose him to command their army in June 1775, and he held the command until the war ended. On December 23, 1783, he handed his commission, the paper that made him commander, back to Congress at Annapolis. The next day he left for his home at Mount Vernon.

Joseph Plumb Martin

Who: A Connecticut teenager who joined the army at 15, served as a private and then a sergeant, and later wrote down what it was like.

When and where: Born in Becket, Massachusetts, November 21, 1760. Joined the army in June 1776 and was still serving at Yorktown in 1781. Died in Prospect, Maine, May 2, 1850, aged 89.

He made his shoes from raw cowhide on the march to Valley Forge.

Joseph Plumb Martin was 15 when he joined the Connecticut troops in June 1776. He fought on Long Island that August, and later at White Plains, Germantown and Monmouth. At Valley Forge he went out with groups of soldiers searching the countryside for food for the army.

On the march into the camp he made himself shoes from a piece of raw cowhide. His other choice, he wrote, was to go barefoot, as hundreds of other men did, "till they might be tracked by their blood upon the rough frozen ground." A National Park Service article points out that his words do not make clear whether he saw the bleeding feet himself or repeated a story people told after the war.

At Yorktown he was a sergeant in the sappers and miners, soldiers who dug the trenches that let the American and French armies surround Cornwallis. He took part in the attack on Redoubt No. 10, a small British fort of earth.

After the war he farmed in Prospect, Maine, married Lucy Clewley, and raised five children. He lost a fight over his land with Henry Knox, a former general. By 1818, when officials approved him for a pension, a payment of $96 a year for his service, he owned almost nothing. In 1830, at 70, he published his memoir, the story of his own life. It was printed again in 1962 as *Private Yankee Doodle*.

Deborah Sampson

Who: A Massachusetts woman who joined the army as a man named Robert Shurtleff, fought in the light infantry, was wounded, and later won a soldier's pension.

When and where: Born in Plympton, Massachusetts, December 17, 1760. Enlisted May 1782. Discharged October 1783. Died 1827.

She enlisted as Robert Shurtleff and served 17 months as a man.

At 10, Deborah Sampson was bound out as an indentured servant to a farmer's family in Middleborough, which meant she owed the family a set number of years of unpaid work. In May 1782, at 21, she joined the 4th Massachusetts Regiment as a man, under the name Robert Shurtleff. She served 17 months near West Point, New York, in a light infantry company, a unit of lightly armed foot soldiers.

She was wounded in combat. The Paul Revere House museum says it was in a fight with a raiding party of Loyalists, colonists who fought for the British king. The National Women's History Museum describes a sword cut to her forehead and a musket ball in her left thigh. The historian Alfred F. Young found accounts that put the wound in her shoulder or her chest.

The musket ball story. The story that she cut a musket ball out of her own thigh, so no army surgeon would find out she was a woman, comes from a book Herman Mann printed about her in 1797. The New England Historical Society calls Mann's book "a poor history," because he added a great deal that he imagined.

KNOWN:	Sampson was wounded in combat.
NOT KNOWN:	how, where on her body, or when. The accounts differ on all three.

A doctor treating her for a fever in Philadelphia found out she was a woman. General Henry Knox gave her an honorable discharge in October 1783, a record that she had served well. In 1785 she married Benjamin Gannett of Sharon, Massachusetts, and they had three children.

Paul Revere visited her at Sharon in 1804 and wrote to her congressman to support her claim for a soldier's pension. She went on the federal pension list in 1805. In 1806 she wrote to Revere asking to borrow ten dollars. She died in 1827. Ten years later, members of Congress voted to pay her husband, who had outlived her, for her service. He died before the money reached him.

St. Clair and Fallen Timbers, 1791 to 1794

Several Native nations north of the Ohio River joined in a confederacy, an alliance of nations, and fought US soldiers from 1785 to 1795. General Arthur St. Clair led about 2,000 men north from the Ohio River in 1791. They were a mix of militia, volunteers signed up for six months, and regular soldiers.

On the way, volunteers whose six months were up went home. About 60 militiamen ran off. St. Clair sent 300 regular soldiers back to guard his food supplies, and about 1,400 men were left on November 3.

The Wabash. The next day, on the Wabash River, more than 1,000 fighters under the Miami leader Little Turtle attacked St. Clair's camp. More than 900 of the army's soldiers, and the women and children traveling with them, were killed or wounded. That count comes from a US Army history article, and other counts split the dead and wounded differently. Most were left where they fell.

Members of Congress held their first special investigation ever into what went wrong, and they cleared St. Clair. President Washington then put General Anthony Wayne in charge of a new force called the Legion of the United States. Its soldiers drilled over and over until they could march and fight together.

Wayne's soldiers beat the Native confederacy at Fallen Timbers, in what is now Ohio, on August 20, 1794, in less than an hour and a half. US Army historians trace today's regular army back to the Legion.

BY 1800

If you remember one thing: In the Revolution, disease killed about two and a half American soldiers for every one killed in battle, and thousands of those who died of it were prisoners.

- 1763: Officers at Fort Pitt hand two Lenape men blankets from the smallpox hospital.
- 1777 to 1778: Nearly 2,000 people die of disease in the winter camp at Valley Forge.
- March 8, 1782: Pennsylvania militiamen kill 96 Christian Lenape and Mohican people at Gnadenhutten, and no one is charged.

Next: The new country's army fought Britain once more, then Mexico, and Native nations again and again on their own land.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 13 MINUTES

US soldiers went to war with Britain and with Mexico between 1800 and 1850, and they fought Native nations again and again on the nations' own land, from the Creek in Alabama to the Sauk in Illinois and the Seminole in Florida.

The army's officers now trained at a school of their own. President Thomas Jefferson signed the law that set up the US Military Academy at West Point, New York, on March 16, 1802. US soldiers fought the British from 1812 to 1815 and Mexico from 1846 to 1848, and they marched into Mexico City in September 1847.

Disease again. In the war with Mexico and in the Second Seminole War, disease killed most of the soldiers who died. For the war with Mexico, Defense Department records list 1,733 American soldiers killed in battle and 11,550 dead of other causes. About seven of every eight American dead died of something other than enemy fire.

Why Madison asked for war in 1812

President James Madison asked Congress for war with Britain on June 1, 1812, and listed three complaints. The first was impressment: British navy officers stopped American ships, took off sailors they said were British, and forced them to serve on British warships. The second was British orders that blocked trade by countries outside the war, including the United States. The third was that British officials encouraged Native nations to attack American settlers in the West.

The members of the House voted for war 79 to 49, and the senators voted 19 to 13, the closest votes on any declaration of war in US history. Madison signed the declaration on June 18, 1812.

Washington burns. British soldiers beat the American defenders at Bladensburg, Maryland, on August 24, 1814, and marched into Washington. Their commanders, Major General Robert Ross and Rear Admiral George Cockburn, had their men set fire to the Capitol, the President's House and the Treasury. Madison and many people in the city had already fled.

Fort McHenry. Three weeks later, British warships fired between 1,500 and 1,800 shells at Fort McHenry, which stood between the British navy and the city of Baltimore. The shelling lasted 25 hours, starting at 6 in the morning on September 13. The shells killed four men in the fort and wounded 24. Major George Armistead and his men held the fort, and the British ships sailed away.

Francis Scott Key watched the attack from an American ship in the harbor. He had gone to the British fleet to get a prisoner released, and the British would not let him go back to Baltimore until the battle was over. He wrote the verses that became "The Star-Spangled Banner."

New Orleans. American and British diplomats, officials sent to make the peace, signed a treaty to end the war at Ghent on December 24, 1814. The news did not reach New Orleans until March.

Before it came, on January 8, 1815, British soldiers under Edward Pakenham attacked General Andrew Jackson's lines outside the city. Jackson's force included frontiersmen from Tennessee and Kentucky, free Black soldiers,

Choctaw fighters, and Louisiana militia, part-time citizen soldiers. Jean Lafitte's privateers, ship crews allowed by a government to attack enemy ships, fought with them too.

Of 3,000 British soldiers in the main attack, about 2,000 were killed, wounded or captured in less than 30 minutes. Pakenham rode forward to rally them, was hit by American fire, and died of his wounds. Counts of the British dead run from 285 to about 700. Jackson's side lost 13 killed.

For the whole war, Defense Department records list 2,260 Americans killed in battle and 4,505 wounded. They list no count of soldiers who died of disease.

The Creek War, 1813 to 1814

While US soldiers fought the British, a second war began in the South, on the land of the Creek, or Muskogee, people, in what is now Alabama and Georgia. It started as a war among the Creek themselves, between Creek towns friendly to the United States and a group from the Upper Towns called the Red Sticks.

Mississippi militia ambushed Red Sticks at Burnt Corn Creek on July 27, 1813, as the Red Sticks came home with ammunition from the Spanish at Pensacola. A month later, on August 30, Red Sticks attacked Fort Mims, a stockade, or fort with a wall of upright logs, north of Mobile. They killed 250 Creek and American settlers there. Tennessee's governor then sent Andrew Jackson with Tennessee militia into Creek country.

Horseshoe Bend. Jackson's army of about 3,300 men attacked the Red Sticks' walled town of Tohopeka, at Horseshoe Bend on the Tallapoosa River, on March 27, 1814. The army included Tennessee militia, US soldiers, and Cherokee and Lower Creek allies. About 1,000 Red Sticks under Chief Menawa held the town.

Jackson's army killed more than 800 of them. Soldiers counted 557 bodies on the battlefield, and soldiers led by a man named Coffee shot an estimated 300 more in the river as they tried to swim away. Jackson's side lost 49 killed and 154 wounded. Cherokee and Lower Creek fighters took about 350 Upper Creek women and children prisoner.

Menawa was wounded seven times. He lay among the dead until dark and escaped in a canoe. Later in 1814, in the Treaty of Fort Jackson, US officials made Creek leaders give up land. An Encyclopedia of Alabama article states more than 21 million acres, and a National Park Service page states 23 million.

The Arikara towns and the Bad Axe, 1823 and 1832

The Arikara, whose towns stood on the Missouri River in present-day South Dakota, fought US soldiers in 1823. Arikara men attacked a party of fur traders led by William Ashley below their towns on June 2, 1823, and killed twelve or thirteen of Ashley's men. Colonel Henry Leavenworth came in August with about 230 soldiers of the 6th Infantry, an army unit of foot soldiers. Fur trappers and about 750 Lakota and Yankton allies came with him.

They attacked the towns with cannon. After a peace was signed, the Arikara left their towns at night. As the soldiers marched away on August 15, two men of the Missouri Fur Company set the empty towns on fire. No reliable count of the Arikara dead survives.

Black Hawk. The Sauk leader Black Hawk led about 1,000 Sauk, Meskwaki and Kickapoo people back across the Mississippi River into Illinois in April 1832. US officials said the Sauk and Meskwaki had signed the land away in an 1804 treaty, and Black Hawk rejected that treaty. About 7,000 US soldiers and militiamen went out against his band. Abraham Lincoln served as a militiaman in this war.

The Bad Axe. By August the band, perhaps 500 men, women and children, was trying to cross back over the Mississippi near the Bad Axe River, in present-day Wisconsin. Soldiers aboard an army steamboat, the *Warrior*, under Captain Throckmorton, fired on them on August 1. The next day about 1,300 soldiers and militiamen under Brigadier General Henry Atkinson attacked the people still on the east bank.

A Wisconsin Historical Society marker put up in 1955 states that the people were "driven into the water by their pursuers." There, it states, "warriors, old people, women and children" were "shot down or drowned as they tried to escape." Illinois governor Thomas Ford later estimated 150 killed, as many drowned, and 50 taken prisoner.

About 70 people reached the far bank, where Dakota fighters, allies of the United States, captured or killed them. Neither the marker nor Ford's account names the men who fired on the people in the water.

The Second Seminole War, 1835 to 1842

US soldiers fought for seven years to force the Seminole out of Florida and west to Indian Territory, now Oklahoma. The Seminole had taken in people who escaped slavery in Georgia, South Carolina and Florida, and that turned slaveholders and army officers against them.

Dade's column. About 180 Seminole fighters led by Micanopy, Alligator and Jumper attacked Major Francis Dade's column of soldiers on December 28, 1835, on the road from Fort Brooke to Fort King. They killed almost all the soldiers in the column. A Florida Department of State history page states that one soldier survived, and other accounts count two or three.

Osceola. General Thomas Jesup ordered soldiers to seize the Seminole leader Osceola near St. Augustine in October 1837, while Osceola was meeting them under a white flag of truce, the sign that both sides agree not to fight while they talk. A National Park Service history describes a "public outcry" against Jesup's "dishonorable ploy," meaning a trick that broke the rules of honor. Soldiers then took Osceola and 237 of his people to Fort Moultrie in South Carolina. He died there as a prisoner on January 30, 1838.

More than 1,500 US soldiers had died in the war by 1842, most of them of disease, and US officials had spent more than $20 million on it. The National Park Service gives no count of the Seminole dead for the whole war. Its page gives one rate instead: for every four Seminole people the soldiers forced out of Florida, they killed one.

The Mexican War, 1846 to 1848

Texas became part of the United States in 1845, and Mexican officials claimed the strip of land between the Nueces River and the Rio Grande. President James K. Polk sent soldiers under General Zachary Taylor into that strip in 1846. An article in the Colorado Encyclopedia states that Polk "goaded" Mexico into war this way, meaning he pushed Mexico into starting the fighting.

Mexican soldiers under Brigadier General Anastasio Torrejón surrounded a US patrol of cavalry, soldiers on horseback, under Captain Seth Thornton on April 25, 1846. The patrol surrendered. Mexican officials said Thornton's men were the invaders.

Polk told Congress that Mexico had "shed American blood upon the American soil." House members voted for war 174 to 14 and senators 40 to 2, and Polk signed the declaration on May 13, 1846. Ulysses S. Grant, a young officer who fought in the war, later wrote that he regarded it "as one of the most unjust ever waged by a stronger against a weaker nation."

Veracruz. US soldiers won at Monterrey in September 1846 and at Buena Vista in February 1847. General Winfield Scott's guns fired more than 6,700 shells into the Mexican port city of Veracruz over four days in March 1847, and the city's defenders surrendered on March 29. One count, in a HistoryNet article, states about 100 Mexican civilians, 80 Mexican soldiers and 13 Americans dead there. Other counts run higher. Scott's soldiers then fought their way inland and took Mexico City in September 1847.

Civilians. American soldiers also killed Mexican civilians. Early in 1847, near Saltillo, Mexican ranchers dragged an Arkansas private, Samuel Colquitt, to death. Arkansas volunteers from Captain C. C. Danley's company and another company then found the civilians they believed were guilty and shot between 17 and 30 of them.

General Taylor was furious. He threatened to send the two companies back to the Rio Grande to do hard labor, and he did not, because the Mexican general Santa Anna's army arrived. No one counted all the Mexican civilians killed in

the war.

The surgeons. Army surgeons had to cut off an arm or leg shattered in battle, an operation called amputation. In the spring of 1847 they used anesthetics in battle for the first time, drugs that put the patient to sleep so the patient feels no pain. Most of the men who died, though, died of dysentery, the gut infection that causes bloody diarrhea, or of yellow fever and malaria, fevers spread by mosquitoes.

The treaty. Mexican officials signed the Treaty of Guadalupe Hidalgo on February 2, 1848. Under it they gave up 55 percent of Mexico's land, more than 525,000 square miles, to the United States, and US officials agreed to pay the Mexican government $15 million. That land became California, Nevada, Utah, Arizona, New Mexico and much of Colorado. In the 1840s, Comanche, Apache and Ute people controlled much of Mexico's northern lands, including part of this land.

About 78,700 Americans served in the war and 13,283 died, about one in six. By an estimate quoted on Smarthistory, an education website, about 25,000 Mexican soldiers died.

John Riley

Who: An Irish-born US Army sergeant who crossed to the Mexican side before the war began and led a unit of deserters, the San Patricios, against the US Army.

When and where: Company K, 5th US Infantry. Crossed the Rio Grande in the spring of 1846. Captured at Churubusco, near Mexico City, August 20, 1847.

Movie: *One Man's Hero* (1999), a feature film about him, with Tom Berenger as Riley. It is a drama, not a documentary.

Army officers hanged fifty of his men and branded his face.

John Riley had trained West Point cadets to fire cannon. In the spring of 1846, before members of Congress declared war, he crossed the Rio Grande and joined the Mexican army, which made him a deserter, a soldier who leaves his army without permission. He organized other deserters into the San Patricio Battalion, a unit of a few hundred soldiers named in Spanish for Saint Patrick. Many of the men were Irish immigrants, and many were Catholic.

A National Park Service page lists their reasons. American officers treated Catholic soldiers with "anti-Catholic bigotry and brutal discipline," bigotry meaning hatred of people for who they are or what they believe. A Smithsonian article states that officers quite often forced Catholic soldiers to attend Protestant services.

Some men were against the war. Mexican officials offered land to foreigners who joined them. Some San Patricios were escaped slaves and free Black men, who saw Mexico, where slavery was against the law, as freer than the United States.

The San Patricios fought at Monterrey, Buena Vista, Cerro Gordo and Churubusco. At Churubusco on August 20, 1847, US soldiers captured between 75 and 80 of them. Army officers tried 72 as deserters in two mass courts-martial, or military trials, held 3 and 6 days after the capture. The men had no lawyers, and no one wrote down what was said.

The hangings. Army officers hanged fifty of them, at San Angel on September 10 and at Mixcoac on September 13, 1847. On September 13, Colonel William Harney hanged 30 of them within sight of Chapultepec Castle as the US flag went up over it. He left their bodies on the gallows to rot.

The whip and the iron. Riley had deserted before the war was declared, so the court could not sentence him to death. Instead the officers of the court ordered him whipped with fifty lashes, blows from a whip on his bare back that cut the skin. They also ordered him branded. A metal letter D, for deserter, was heated red-hot and pressed into the skin of his face near the eye, and it burned a scar in the shape of the letter that he carried for life.

Some accounts say it was done to both cheeks. One says the first D went on upside down and a second D was burned on the right way up. The Texas State Historical Association and *American Heritage* accounts do not name the men who held the whip and the iron.

Riley stayed in the Mexican army after the war. Records of where he went stop a few years later, and the date of his death is not known.

BY 1850

If you remember one thing: The army fought its way to Mexico City and the country took more than half of Mexico's land, and disease killed most of the Americans who died on the way.

One number: 7 of every 8. That many of the American soldiers who died in the war with Mexico died of something other than enemy fire: 11,550 of disease and other causes, against 1,733 killed in battle.

Next: More American soldiers died in the Civil War than in any other war the country has fought, and sickness killed more of them than bullets did.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 23 MINUTES

More American soldiers died in the Civil War, from 1861 to 1865, than in any other war the United States has fought. The Union was the United States government and the states that stayed loyal to it. The Confederacy was made up of Southern slave states whose leaders broke away.

How many died. Between 650,000 and 850,000 soldiers died in the Civil War. The historian J. David Hacker worked out that range in 2011 from census records, the government's counts of everyone in the country, from 1850 to 1880. His best estimate, about 750,000, is nearly twice the 405,399 Americans who died in World War II.

The long-used count was lower, about 620,000. The armies' own records count fewer still, because they are incomplete. The Union officer in charge of the draft, for example, counted only 133,821 Confederate dead in his final report.

What killed them. Disease killed more soldiers than bullets did. Union records list 140,414 Union soldiers killed in battle and 224,097 dead of disease and other causes. That is close to the rate an American Battlefield Trust article gives for the war: five deaths from disease for every three in battle.

Measles spread through the new army camps in 1861. In August that year Lee wrote, "The soldiers everywhere are sick." Army doctors recorded 79,462 cases of typhoid, an infection from bacteria in food or water that contains human waste. It causes high fever and stomach pain, and it can kill.

US soldiers fought the nations of the Plains and the West from the end of the Civil War until 1890. American sailors beat a Spanish fleet in Manila Bay, in the Philippines, in 1898, and American soldiers fought Spain in Cuba. A new war began in the Philippines in February 1899, between US soldiers and Filipino fighters led by Emilio Aguinaldo.

The Civil War battles, 1861 to 1865

Confederate gunners opened fire on Fort Sumter, in the harbor at Charleston, South Carolina, at 4:30 in the morning on April 12, 1861. Over 34 hours they fired more than 4,000 shots and shells. Major Robert Anderson and the Union soldiers who held the fort surrendered it the next afternoon.

Antietam. The two armies had about 23,000 casualties, soldiers killed, wounded or missing, in twelve hours of fighting at Antietam Creek in Maryland, on September 17, 1862. More soldiers were lost at Antietam than on any other single day of the war.

Gettysburg. The Union army under George Meade and the Confederate army under Robert E. Lee had 51,112 casualties between them at Gettysburg, Pennsylvania, from July 1 to 3, 1863. About 7,000 of those men were killed. On the last day, Lee sent about 12,500 men to attack the Union soldiers on Cemetery Ridge, in the attack called Pickett's Charge. Close to 60 percent of them were killed, wounded or captured.

Ulysses S. Grant's army surrounded the river town of Vicksburg, Mississippi, for 46 days, in a siege that cut the town off until it gave up. On July 4, 1863, about 29,000 Confederate soldiers at Vicksburg marched out, stacked their rifles and surrendered to him.

Who had to fight in the Civil War?

Officials on both sides drafted soldiers when too few men volunteered, and both drafts let richer men out. Members of the Confederate Congress passed their draft law first, on April 16, 1862. Under it, officials could draft any white man aged 18 to 35. On October 11, 1862, they passed the Twenty-Slave Law, under which any man who held twenty or more enslaved people did not have to serve. Drafted soldiers called it "a rich man's war."

The Union draft. Members of the US Congress passed the Enrollment Act on March 3, 1863. Army officers called provost marshals ran the draft. They drafted men in each district where too few had volunteered to fill the number of soldiers the army asked for. A drafted man could hire a substitute to go in his place, or pay a $300 fee and go home. In 1863, $300 could take the average person a whole year to earn.

Substitutes were often 18- or 19-year-olds too young to be drafted, or immigrants who were not yet citizens. People called it "a rich man's war and a poor man's fight." More than 20 percent of the men drafted did not report. Some fled west and some hid.

The New York riots. The draft began in New York City on Saturday, July 11, 1863. A rich man could pay his way out, and many New Yorkers were angry that poor men had to go instead. Many poor white workers, most of them Irish, also feared that if slavery ended, Black people would move to New York and take their jobs.

From Monday, July 13, to Thursday, July 16, mobs looted stores, burned a draft office, and attacked police and soldiers. Their main target was the city's Black people. The rioters lynched at least a dozen Black men, which means they killed them without any trial.

On Clarkson Street on July 13, a gang led by an Irish bricklayer, John Nicholson, beat a Black man named William Jones, who drove a cart for a living, hanged him and set his body on fire. The mobs burned homes and businesses. Rioters set fire to the Colored Orphan Asylum ("Colored" was a

word used then for Black people), a home for Black children without parents, on Fifth Avenue with more than 200 children inside. All of the children got out.

The official count of the dead was 119, and the New-York Historical Society notes that many historians think more died. Union soldiers came from Gettysburg to stop the riots. Afterward many Black families moved to Brooklyn and New Jersey, and the number of Black people living in New York City fell by 20 percent.

Why did surgeons cut off so many arms and legs?

Soldiers on both sides fired the minié ball, a soft lead bullet shaped like a cone, and it often flattened when it hit a body. It splintered bone, tore muscle, and drove dirt and bits of cloth into the wound. A surgeon could not mend an arm or leg smashed that way, so he cut it off. Three of every four operations in the war were amputations, close to 60,000 in all.

The operation. The surgeon first put the soldier to sleep with chloroform or ether. He tied a tight band called a tourniquet around the arm or leg to slow the bleeding, then cut through the skin with a knife called a scalpel and through the muscle with a long knife. Then he sawed through the bone. Last, he tied off the blood vessels and bandaged the stump.

Soldiers whose limbs were cut off within two days of the wound lived more often than those operated on later. Infection, loss of blood and poor food killed many of the men operated on later.

Amos Humiston

Who: A sergeant in the 154th New York Infantry, killed at Gettysburg. He was identified only because of a photograph of his three children found in his hand.

When and where: Born in Owego, New York, April 26, 1830. Killed at Gettysburg, Pennsylvania, July 1, 1863. Identified in November 1863.

A photograph of three children was all he had on him.

Amos Humiston had made harnesses for horses and had sailed on a whaling ship from 1850 to 1854. He married Philinda Smith on July 4, 1854, and they had three children, Franklin, Alice and Frederick. He joined the army in July 1862 and was a sergeant by January 1863.

He was killed on the first day at Gettysburg. The records do not say what wound killed him. His body was found near the corner of Stratton and York Streets in the town. He had nothing on him to show who he was except a small photograph on glass, called an ambrotype, of three young children. He was buried as an unknown soldier.

"Whose Father Was He?" A doctor named John Francis Bourns had hundreds of copies of the photograph made. On October 19, 1863, editors at the *Philadelphia Inquirer* printed the story under that headline. In November, Philinda Humiston saw a copy and knew her children. Without the photograph, his grave in the Soldiers' National Cemetery would read "Unknown," like the graves of 979 other soldiers there.

In October 1866, his three children were among the first thirty children at a home for soldiers' orphans that opened in Gettysburg. A monument to Humiston went up on North Stratton Street in 1993. It is the only monument on the battlefield to a single enlisted soldier, meaning a soldier who was not an officer.

The prison camps

Why the camps filled. Union and Confederate commanders traded captured soldiers under an agreement made in July 1862. In December 1862, Confederate President Jefferson Davis announced that captured Black soldiers and their white officers would not be traded. President Abraham Lincoln stopped the trades in the summer of 1863 because the Confederates would not include Black soldiers.

By August 1863, large trades had ended, and the prison camps filled up. By the end of the war, about 400,000 men had been held in more than 150 prisons.

Andersonville. Confederate officers opened a prison camp at Andersonville, Georgia, early in 1864. It was a walled pen of 26.5 acres, built for 10,000 men, and in August 1864 it held more than 33,000. One stream was the only water for most of them. About 19 feet inside the wall ran a line called the deadline, and guards had orders to shoot any prisoner who crossed it.

More than 45,000 Union soldiers were held there in 14 months. Almost 13,000 of them, more than one in four, died there under the Confederate officers who ran the camp, of disease, hunger, filth, crowding, and exposure to heat and cold.

A court of army officers found the camp's commander, Captain Henry Wirz, guilty of war crimes, acts that break the laws of war. The court sentenced him to death, and Union army officers hanged him in Washington on November 10, 1865. He was the only person put to death for war crimes in the Civil War.

The Union camps. Between 26,000 and 31,000 Confederate soldiers died in Union prison camps. At Elmira, New York, the barracks had no protection from the winter cold. Close to 3,000 Confederate prisoners died there, about one in four of the men held. An American Battlefield Trust article blames Colonel William Hoffman, the Union officer in charge of all prisoners.

At Camp Douglas, in Chicago, people from the US Sanitary Commission, a group that checked on the health of soldiers and their camps, reported standing water, filthy toilet trenches and "rotten bones." The camp's commanders cut the prisoners' food to make money for themselves. At Point Lookout, Maryland, more than 4,000 prisoners died.

Fort Pillow, April 12, 1864

Black men who joined the Union army served in regiments, large army units, called the US Colored Troops. The Confederate general Nathan Bedford Forrest attacked Fort Pillow in Tennessee with about 1,500 men on April 12, 1864. The fort held 262 Black artillery soldiers, who fired the big guns, and 295 white Tennessee cavalrymen. Forrest's men took the fort and kept shooting soldiers who had surrendered. A witness told congressional investigators, "They killed all the men after they surrendered, until orders were given to stop."

Private George Shaw, a Black soldier, testified that a Confederate shot him after he surrendered, told him "You are fighting against your master," and threw him into the river. Shaw lived. He saw Confederate soldiers shoot three teenage boys who had escaped slavery in the forehead while the boys begged for their lives.

Forrest's men killed 182 of the Black soldiers that day or wounded them so badly that they died later, about 70 percent of them. A sign in the museum at the fort lists all 182 by name. About 20 percent of the white soldiers were killed. Forrest's men also killed some Black civilians, and no one knows how many. After Fort Pillow, Black soldiers went into battle shouting "Remember Fort Pillow."

Christian Fleetwood

Who: A Black man born free in Baltimore, who carried his regiment's flag through the attack at New Market Heights and received the Medal of Honor.

When and where: Born in Baltimore, Maryland, July 21, 1840. Enlisted in the 4th US Colored Infantry, August 1863. New Market Heights, outside Richmond, Virginia, September 29, 1864. Died September 28, 1914.

Two flag bearers fell, and he picked up the flag.

Christian Fleetwood's parents were free. He graduated from the Ashmun Institute in Oxford, Pennsylvania, in 1860, and helped start the *Lyceum Observer*, an early Black newspaper in the South. When he enlisted in August 1863, his officers made him sergeant major, the regiment's top sergeant, because he was so well educated.

His regiment charged at New Market Heights on September 29, 1864. Two soldiers carrying the flag were shot down. Fleetwood picked it up and carried it forward under fire. In his diary that day he wrote that the regiment "got used up" and that he "saved colors," meaning the regiment's flag.

He was one of 14 soldiers of the Colored Troops who received the Medal of Honor, an award for bravery in battle, for that day. Every officer in his regiment asked Secretary of War Edwin Stanton to make Fleetwood an officer. Stanton refused.

After the war Fleetwood worked as a clerk in the War Department in Washington. He also organized a battalion of the District of Columbia National Guard and served as its major.

Ulysses S. Grant and Robert E. Lee

Who: The two commanding generals at the end of the Civil War. Lee led the Confederate Army of Northern Virginia from June 1862. Grant led all the Union armies from March 1864.

When and where: Lee was born at Stratford Hall, Virginia, on January 19, 1807, and died in Lexington, Virginia, on October 12, 1870. Grant was born in Point Pleasant, Ohio, on April 27, 1822, and died on July 23, 1885. They met at Appomattox Court House, Virginia, on April 9, 1865.

Movie: *Grant* (History channel, 2020), a documentary in three parts about Grant, with acted scenes.

Wesley Norris said Lee had salt water poured into his cuts.

Both men trained at West Point, where Lee finished second in the class of 1829 and Grant 21st of 39 in 1843, and both fought in the Mexican War. In 1854 Grant quit the army so he would not be thrown out, after heavy drinking at remote army posts in the West.

Wesley Norris. From 1857 Lee was in charge of 189 enslaved people on the estates of his dead father-in-law. Three of them ran away in 1859. They almost reached the Pennsylvania line, but at Westminster, Maryland, they were caught and jailed, and Lee was told. No surviving record names the men who caught them.

One of the three, Wesley Norris, gave his account in 1866. A county constable named Dick Williams, he said, whipped him and his cousin fifty times each and whipped his sister twenty times while Lee watched. Then, Norris said, Lee ordered the overseer to wash their backs with brine, very salty water, poured into the cuts.

Harpers Ferry. In October 1859, the year of the escape, Lee led the 90 US Marines who ended John Brown's raid at Harpers Ferry, Virginia. Brown and 18 of his men had seized the US Armory and Arsenal there, where the government made and stored guns, to fight a war against slavery.

Lee resigned from the US Army on April 20, 1861, after Virginia's leaders voted to leave the Union. On December 29, 1862, he signed the paper freeing the people enslaved on his father-in-law's estates, because the will required it. At Gettysburg on July 3, 1863, he ordered Pickett's Charge.

Grant took Fort Donelson in February 1862, after telling the fort's commander he would accept "no terms except an unconditional and immediate surrender." He took Vicksburg on July 4, 1863. In March 1864 he became a lieutenant general and the commander of every Union army.

Appomattox. Lee surrendered to Grant in Wilmer McLean's parlor at Appomattox Court House on April 9, 1865. Under Grant's terms, Lee's soldiers could go home once they signed a promise not to fight the United States again. The officers kept their side arms, the small weapons they carried, and their own horses and belongings. About 28,000 Confederate soldiers signed. Grant finished writing his memoirs a few days before he died of cancer in 1885.

BY 1865: Lee had surrendered at Appomattox, and somewhere between 650,000 and 850,000 soldiers of the two armies were dead.

Bear River, Sand Creek and the Marias, 1863 to 1870

During and after the Civil War, US soldiers attacked Native camps in the West where families were living through the winter.

Bear River. Colonel Patrick Edward Connor led the 3rd California Volunteer Infantry against a Northwestern Shoshone winter camp on the Bear River, near present-day Preston, Idaho, on January 29, 1863. He had announced that he did not mean to take prisoners. The soldiers attacked in the early morning and began firing without a word.

People jumped into the freezing river. A woman named Anzee Chee hid under the riverbank with other women, and she drowned her own crying baby so the soldiers would not find them. Yeager Timbimboo, a 12-year-old boy, lived by lying still among the dead.

Soldiers killed an estimated 250 to 500 Shoshone, and Connor was promoted to brigadier general for the attack. An article in Smithsonian magazine states that the attack "is often described as" the deadliest killing of Native people by US soldiers, with a death toll believed to be higher than at Sand Creek or Wounded Knee.

Sand Creek. Colonel John Chivington led about 675 Colorado volunteer soldiers against the Cheyenne and Arapaho camp of Black Kettle at Sand Creek, in Colorado, on November 29, 1864. Black Kettle's camp flew a US flag and a white flag. The soldiers killed about 230 people by the National Park Service's count, most of them women, children and old people. Soldiers cut off scalps and other parts of the bodies.

Captain Silas Soule and Lieutenant Joseph Cramer would not let their men fire. Soule put the number of dead at three hundred. He also wrote that it was "hard to see little children on their knees have their brains beat out by men professing to be civilized."

After Soule and Cramer sent their letters, army officers held an investigation and members of Congress held two more. Members of Congress's Joint Committee on the Conduct of the War wrote that Chivington "deliberately

planned and executed a foul and dastardly (cowardly) massacre," a killing of many people who could not defend themselves.

Chivington left the army and was never tried, and no officer or soldier was punished. About two months after Soule testified against Chivington, on April 23, 1865, someone shot Soule dead on a street in Denver. No one was ever brought to justice for his murder.

The Marias. General Philip Sheridan sent Major Eugene Baker of the 2nd Cavalry against Mountain Chief's band of Piikani, or Piegan Blackfeet, in Montana in January 1870. "Tell Baker to strike them hard," Sheridan ordered.

At daybreak on January 23, Baker found a camp on the Marias River. His scout, Joe Kipp, told him it was the peaceful band of Heavy Runner. By one account, Baker answered, "That makes no difference, one band or another of them. They are all Piegans and we will attack them." He ordered a sergeant to shoot Kipp if Kipp warned the camp.

Heavy Runner came out of his lodge waving his paper of safe passage, and the soldiers shot him first. They knocked down lodges, the band's tent homes, with people inside them, set them on fire, and burned the band's winter food.

Army officers counted 173 dead. A later estimate, reported by HISTORY, is that the soldiers killed 37 men, 90 women and 50 children, which makes 177. Other counts run to more than 200. Baker took about 140 women and children prisoner. When he found smallpox among them, he left them with no food or shelter. Neither Baker nor his men were ever punished.

The army in the West, 1866 to 1890

From 1866 to 1890, US soldiers fought the nations of the Plains and the West. US officials wanted the nations on reservations, land set aside for them, and soldiers attacked the bands that would not go.

The Buffalo Soldiers. Members of Congress created six regiments of Black soldiers in 1866, later merged into four: the 9th and 10th Cavalry and the 24th and 25th Infantry. They became known as the Buffalo Soldiers, and army commanders posted them west of the Mississippi River. Counts of how many of them received the Medal of Honor in the wars against the Plains nations run from 17 to 23.

Little Bighorn. An army expedition under George Armstrong Custer reported gold in the Black Hills in 1874. An 1868 treaty had promised the hills to the Lakota, and US officials ignored it. In December 1875 they ordered the winter camps off the reservation to come onto it by January 31, 1876, and then sent the army after the camps that stayed out.

Lakota and Cheyenne fighters killed Custer, the commander of the 7th Cavalry, beside the Little Bighorn River in Montana on June 25 and 26, 1876. They killed every man in the five companies, or groups of soldiers, under his own command. In all, 268 US soldiers and scouts died in the two days.

The Nez Perce. General Oliver O. Howard ordered Chief Joseph's band of Nez Perce to leave their homeland in the Wallowa country of northeast Oregon in 1877 and move to the reservation. Fighting began that June. About 750 Nez Perce people, about 200 of them fighters, fled from US soldiers from June to October. They covered about 1,170 miles and fought the soldiers again and again as they went. They surrendered at Bear Paw, in Montana, on October 5, 1877, 40 miles from Canada.

Wounded Knee. Soldiers of the 7th Cavalry under Colonel James Forsyth surrounded a Lakota camp at Wounded Knee Creek in South Dakota on December 29, 1890, and began taking away the Lakota's guns. The soldiers had four Hotchkiss guns, small cannon on wheels that fired exploding shells one after another, on a bluff above the camp.

A rifle belonging to Black Coyote, a deaf man, went off. Some accounts say he fired it, and others say it went off when a soldier grabbed it. Soldiers of the 7th Cavalry then killed between 250 and 300 Lakota, and almost half of the dead were women and children.

At least 25 soldiers died, and an article in Smithsonian magazine states that other soldiers' bullets likely hit many of them. Twenty soldiers of the 7th Cavalry received the Medal of Honor for that day.

Cathay Williams

Who: A woman born into slavery who enlisted in the army as William Cathay and served two years with a Black infantry regiment in the West.

When and where: Born in Independence, Missouri. Enlisted at St. Louis, November 1866. 38th US Infantry, in Kansas and New Mexico. Discharged October 14, 1868. Applied for a pension in 1891 and was refused.

Her papers said William Cathay, age 22, 5 feet 9 inches.

Cathay Williams was born enslaved. During the Civil War, Union soldiers of the 8th Indiana Infantry freed her and took her as "contraband," the army's word for enslaved people who came under Union control. She worked for the regiment, washing clothes and cooking, and followed it to the battle of Pea Ridge in Arkansas and to the Shenandoah Valley in Virginia.

She enlisted at St. Louis as William Cathay in November 1866. She was sent to the 38th Infantry, one of the new Black regiments. She caught smallpox soon after she enlisted and recovered. Over two years she marched about 1,000 miles with her company, from Fort Riley in Kansas to Fort Bayard in New Mexico.

An army doctor at her post found out she was a woman in 1868. She was discharged on October 14 with a certificate of disability, a paper saying she was not fit to serve. In 1876 she told a St. Louis newspaper, "I played sick... The post surgeon found out I was a woman."

> She asked for a pension, the regular payment a former soldier can get, as a disabled veteran in 1891. She had spent eighteen months in a hospital in Trinidad, Colorado, and had lost some of her toes. Officials at the Pension Bureau turned her down, even though her discharge papers recorded illness. The date of her death is not known.

1898, Cuba and the Philippines

From 1898 on, American soldiers fought most of their wars overseas. The US Navy ship *Maine* blew up in the harbor at Havana, Cuba, on February 15, 1898, and 266 of the 354 men aboard died. People still argue over the cause. Later in 1898, a court of navy officers blamed a mine. In 1976, a study led by Admiral Hyman Rickover found that a fire in a coal bin most likely set off the explosives stored next to it.

San Juan Hill. The Black soldiers of the 10th Cavalry fought at Las Guasimas, Cuba, on June 24, 1898, and again in the attack on San Juan and Kettle Hills on July 1. About one in five of them became casualties in that attack. The fighting with Spain stopped on August 13, 1898.

Typhoid killed more American soldiers in 1898 than Spanish soldiers did. Defense Department records list 385 Americans killed in battle in the war with Spain and 2,061 dead of other causes, most of them of typhoid. In the army's camps in the United States there were 20,738 cases of typhoid and 1,590 deaths.

The Philippines. Private William Walter Grayson of the 1st Nebraska Volunteer Infantry was standing guard in Manila on February 4, 1899. He fired the shot counted as the first of a new war, between US soldiers and Aguinaldo's fighters. US soldiers captured Aguinaldo in 1901, and President Theodore Roosevelt declared the war over on July 4, 1902, though fighting went on in places.

Historians at the State Department count more than 4,200 American and more than 20,000 Filipino fighters dead. They count as many as 200,000 Filipino civilians dead of violence, hunger and disease. Other estimates of the

civilian dead run much higher, up to 1,000,000, and historians disagree about them. The State Department's history does not say how many of the civilians American soldiers killed.

BY 1900

If you remember one thing: More American soldiers died in the Civil War than in any other war, and after it US soldiers spent twenty-five years fighting the nations of the West.

In 1861: Americans were fighting each other in a war at home in which between 650,000 and 850,000 soldiers died, most of them of disease.

In 1900: American soldiers were fighting across the Pacific, in the Philippines, where as many as 200,000 Filipino civilians died by the State Department's count, and more by other counts.

Next: In the next fifty years more than 20 million Americans served in two world wars, and most of them were drafted.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 24 MINUTES

Most of the Americans who fought in the two world wars were drafted. World War II was the first American war in which more of them died in battle than of disease, accidents and infections.

Who they fought. Members of Congress declared war on Germany in April 1917. German submarines had sunk American ships, and German officials had tried to draw Mexico into an alliance against the United States. American soldiers then fought on the side of the Allies, the countries already at war with Germany, among them Britain. In World War II the United States fought Japan, Germany and Italy, after Japanese pilots attacked Pearl Harbor in December 1941 and Germany and Italy declared war four days later.

The two wars. About 4.7 million Americans served in World War I, from 1917 to 1918, and 116,516 died, more of them of disease and accidents than in battle. More than 16 million served in World War II and 405,399 died, in counts that run from 1941 to 1946. These are the Defense Department's counts.

The draft. In World War I, 2.8 million men were drafted, which means they were ordered into the armed forces by law, and 2 million volunteered. In World War II about 10 million of the 16 million who served were drafted.

Military leaders kept Black service members in separate units until after President Harry Truman ordered equal treatment in 1948, and nearly every unit was mixed by about 1953. In 1945, American airmen dropped the first two atomic bombs used in war, on two cities in Japan. An atomic bomb gets its huge power from the tiny centers of atoms. One bomb could kill tens of thousands of people, and unlike other bombs it also gave off radiation, invisible energy that sickened and killed people for weeks afterward.

What was the water cure?

The water cure was a torture American soldiers used on Filipino prisoners during the war in the Philippines. Soldiers held a man down and poured water into his mouth and nose until his stomach swelled. Then they pressed or stood on his stomach to force the water out, and started again.

Igbaras. Soldiers did this to Tobeniano Ealdama, the mayor of the town of Igbaras, on the island of Panay, on November 27, 1900. Captain Edwin Glenn, an army lawyer, oversaw it. Ealdama said later, "My stomach and throat pained me, and also the nose where they passed the salt water through." That night Glenn ordered the town's 400 to 500 houses burned.

A sergeant who had been there, Charles Riley, told a Senate committee about it on April 14, 1902. Army officers tried Glenn and found him guilty. His punishment was a $50 fine and one month in which he was not allowed to do his job. He retired in 1919 as a brigadier general.

Samar. Filipino fighters killed 48 of the 74 men in an American company at Balangiga, on the island of Samar, in late 1901. Brigadier General Jacob Smith then ordered Marine Major Littleton Waller to turn the island into a "howling wilderness." Smith told him, "The more you kill and burn, the better it will please me." Smith said he thought Filipino boys of ten were old enough to fight.

Waller had eleven Filipino guides and carriers shot without a trial. At his court-martial, a trial run by military officers, the officers voted 11 to 2 to find him not guilty. At Smith's own court-martial in 1902, officers reprimanded him, which means they scolded him officially. President Theodore Roosevelt then ordered Smith to retire from the army at once. The historian Paul Kramer writes, "The scale of abuses in the Philippines remains unknowable."

The draft of 1917 and the men who refused

Members of Congress passed the Selective Service Act, the law for the World War I draft, on May 18, 1917. Men had to register on three days in 1917 and 1918, and by the end about 24 million men had filled out draft cards, about 23 percent of everyone living in the country in 1918.

Some drafted men refused to fight because their religion forbade it, and they were called conscientious objectors.

The Hutterite brothers. Three brothers, David, Joseph and Michael Hofer, and Joseph's brother-in-law Jacob Wipf were drafted in 1918. They were Hutterites, members of a Christian community in South Dakota that does not fight in wars. They refused to put on uniforms or obey military orders. Officers at a court-martial at Camp Lewis sentenced them to 20 years in prison.

Alcatraz. Soldiers took the four men, chained in pairs, to the military prison on Alcatraz Island on July 25, 1918. Guards put them in cold, wet cells in the basement. At first they got half a glass of water a day and no food.

Guards chained them standing to the cell bars with their hands crossed and pulled up high, so only their toes touched the floor. This was called high cuffing, and it strained their arms and shoulders until they swelled. After five days their arms were too swollen to put on their jackets. Duane Stoltzfus, a historian at Goshen College, writes that guards were reported to have beaten them with knotted whips as well.

Guards moved the four in chains to Fort Leavenworth, in Kansas, on November 19, 1918, after the fighting in Europe had stopped. Joseph Hofer died there on November 29, and Michael Hofer on December 2. Officials had Joseph's body dressed in the army uniform he had refused to wear.

The prison doctor listed pneumonia, a sickness that fills the lungs with fluid, during the flu outbreak in the prison. The Hutterite church's record states that they died "as a result of cruel mistreatment by the United States military." On December 6, 1918, Secretary of War Newton Baker ordered an end to chaining prisoners to cell bars.

Houston, 1917

Army leaders sent 654 Black soldiers of the 3rd Battalion, 24th Infantry, to guard the building of Camp Logan outside Houston, Texas, in July 1917. People in Houston insulted them with racial slurs every day.

Houston police arrested a Black soldier on August 23, 1917, after he questioned the arrest of a Black woman. When Corporal Charles Baltimore asked about it, police beat him, shot at him and held him. A false report that he had been killed spread through the camp.

About 150 soldiers took rifles and marched into the city. About 20 people died. Townspeople, police officers and National Guardsmen were among them, and so were soldiers of the 24th whom other soldiers of the 24th shot by accident. The Department of Veterans Affairs history of that night does not say who shot each of the others.

The trials. Army officers charged 118 soldiers with disobeying orders, mutiny (refusing orders as a group), trying to kill and murder. The officers found only eight not guilty. At sunrise on December 11, 1917, they hanged the first 13 men sentenced to death. They did it in secret, within a day of the sentence, and did not tell the War Department.

After protests, War Department officials ordered that every death sentence be reviewed first. At later trials, officers sentenced 16 more men to death, and President Woodrow Wilson changed ten of those sentences to prison terms. In all, army officers hanged 19 soldiers. Most of the others got life in prison. Only Black soldiers were sentenced to death.

The Secretary of the Army canceled all the convictions in 2023 and changed the men's records to honorable discharges, which show that a soldier left the army in good standing. The Army's reviewers found that the soldiers "were wrongly treated because of their race and were not given fair trials."

The Meuse-Argonne, the flu and the Choctaw code talkers, 1918

More than a million American soldiers fought in the Meuse-Argonne, in northern France, from September 26 to November 11, 1918. By the National Archives' count, 26,277 of them were killed and 95,768 were wounded. An Air Force Times report on the battle's 100th anniversary called it "America's deadliest battle ever."

The flu. Influenza, the flu, spread through the army camps and troopships in 1918. Most of the soldiers it killed in army hospitals died of the pneumonia that followed it, and some died within hours. A sailor on one troopship to France remembered 15 burials at sea in one day.

Eric Durr, writing on the Army's website, counts about 45,000 soldiers dead of flu and pneumonia by the end of 1918. That is more than the 26,277 killed in the Meuse-Argonne. A narrower count, of soldiers admitted to army hospitals for flu, is 24,664 deaths.

The Choctaw code talkers. German soldiers were listening in on American telephone lines. On October 26, 1918, officers of the 142nd Infantry had Choctaw soldiers send orders over the phone in the Choctaw language, and the Germans could not understand them. The soldiers made up code words. Artillery, the big guns, was "big gun," and a machine gun was "little gun shoot fast."

Nineteen Choctaw soldiers are honored today as these code talkers. Their work was kept quiet until the late 1980s. Some of them had signed their enlistment cards as "natural born citizens." In 1901 members of Congress had changed the law to make the Choctaw citizens of the United States, along with four other nations in Indian Territory, now Oklahoma: the Cherokee, Chickasaw, Seminole and Creek. Many other Native people did not become citizens until a law of 1924.

Henry Gunther

Who: A drafted bank clerk from Baltimore, whom General John Pershing's army listed as the last American soldier killed in World War I.

When and where: Born in Baltimore, Maryland, in June 1895. Company A, 313th Infantry, 79th Division. Killed near Chaumont-devant-Damvillers, France, at 10:59 a.m. on November 11, 1918.

He charged a German machine gun one minute before the fighting stopped.

Henry Gunther kept the accounts at the National Bank of Baltimore when he was drafted in September 1917, and he was engaged to be married. By the time his regiment reached France in July 1918 he was a supply sergeant. Then army censors, who read soldiers' mail, opened a letter he had written to a friend. He complained about life in the trenches and told his friend not to join. He lost his rank and became a private again.

German and Allied leaders signed the armistice, the agreement to stop fighting, early on November 11, 1918. It would take effect at 11 a.m., and a runner brought word of it to Gunther's unit. At 10:59 Gunther charged a German machine gun with his bayonet locked onto his rifle. The German soldiers fired, and a bullet hit him in the left temple, the side of his head. He died there. The records do not say why he charged.

Army officials later gave him back his sergeant's rank and awarded him the Distinguished Service Cross, an army medal for bravery in battle. His body came home in 1923, and he is buried in Most Holy Redeemer Cemetery in Baltimore.

BY 1919: The fighting in Europe was over, and 116,516 of the Americans who served had died, about 45,000 of them of flu and pneumonia.

The Bonus Army, 1932

Members of Congress voted World War I veterans a bonus, an extra payment for their war service, in 1924, but set the payment date for 1945. In 1932, during the Great Depression, thousands of veterans came to Washington and camped there to ask for the money early. Counts of the veterans run from about 10,000 to more than 20,000. Counting their wives and children, a count in George Mason University's *Histories of the National Mall* reaches about 43,000 people.

Police clearing the camps on July 28, 1932, shot two veterans, William Hushka and Eric Carlson, and both died. Hushka was a butcher from Lithuania who had sold his shop to join the army and became a citizen while he served. He was buried at Arlington National Cemetery.

Tanks and tear gas. The same day, President Herbert Hoover ordered the army to clear the camps. General Douglas MacArthur, the army's chief of staff, sent soldiers with six tanks, and they drove the veterans out with tear gas and bayonets.

A National Park Service history states that the soldiers used torches to set the camps on fire, and accounts differ on who set the fires in the main camp, at Anacostia. Members of Congress voted to pay the bonus in 1936, over President Franklin Roosevelt's veto, his refusal to sign the law.

The draft of 1940 and the objectors

President Franklin Roosevelt signed the Selective Training and Service Act on September 16, 1940, the first draft in the country's history in a time of peace. Every man aged 21 to 45 had to register. By 1945, 50 million men aged 18 to 45 had registered and 10 million had been drafted.

About 43,000 men registered as conscientious objectors. To be accepted as one, a man usually had to belong to a church that taught peace, such as the Quakers, the Mennonites or the Church of the Brethren. Of the 43,000, about 25,000 served in the armed forces without weapons, most of them as medics, and about 12,000 did unpaid work in Civilian Public Service camps. About 6,000 refused to cooperate with the draft at all and went to prison, and more than 4,400 of those were Jehovah's Witnesses.

World War II, 1941 to 1945

Japanese Navy pilots attacked the US Navy base at Pearl Harbor, Hawaii, on December 7, 1941. They killed 2,403 Americans, counting the missing: 2,008 sailors, 109 Marines, 218 soldiers and 68 civilians. A Japanese bomb of 1,760 pounds hit the forward deck of the USS *Arizona* and set off an explosion. Of her crew, 1,177 died. More than 900 of them are still inside the sunken ship.

A mess attendant. Doris Miller was a mess attendant, a sailor who cooked and served food, on the USS *West Virginia* that morning. It was one of the few jobs Navy leaders let Black sailors do. Miller carried his dying captain. Then he fired a machine gun he had never been trained to use. Admiral Chester Nimitz gave him the Navy Cross, a navy medal for bravery in battle, in May 1942.

The crew of a Japanese submarine torpedoed his ship, the *Liscome Bay*, on November 24, 1943. The torpedo set off the bombs stored below deck, and the ship sank in about 23 minutes. About three of every four men aboard died, Miller among them. His body was never found.

Bataan. Major General Edward King Jr. surrendered about 75,000 American and Filipino soldiers on the Bataan Peninsula, in the Philippines, on April 9, 1942. Japanese soldiers marched them about 65 miles. The guards starved and beat the prisoners and stabbed with bayonets men who were too weak to walk.

By a Department of Veterans Affairs estimate, Japanese soldiers killed 650 American and 16,500 Filipino soldiers during and after the march, and other counts run higher. After the war, an American military court tried the Japanese commander, Lieutenant General Homma Masaharu. The court sentenced him to death, and American soldiers shot him on April 3, 1946.

D-Day and Iwo Jima. Soldiers of the Allies, the countries fighting Germany, landed in Normandy in France on June 6, 1944, called D-Day. Researchers at the National D-Day Memorial Foundation have found the names of 4,436 Allied soldiers, sailors and airmen killed that day. Of them, 2,519 were American, and the rest came from Britain, Canada and Norway.

On the Pacific island of Iwo Jima, from February 19 to March 26, 1945, 6,821 Americans were killed and 19,217 were wounded. Of about 21,000 Japanese defenders, 216 were captured. The rest were killed.

More than 350,000 American women served in uniform, in new women's branches of the Army, Navy and Marines.

Soldiers in separate units

Black soldiers, sailors and airmen served in separate units. Black pilots trained at Tuskegee, and the 332nd Fighter Group, made up of these pilots, guarded American bombers flying over Europe.

The bomber story. Reporters began to write in March 1945 that the group never lost a bomber. That story is false. Daniel Haulman, an Air Force historian, counted the bombers lost: enemy planes shot down 27 bombers that the 332nd was guarding. The other fighter groups in the same air force lost 46 each, on average.

Port Chicago. Two ships being loaded with ammunition blew up at the Port Chicago Navy base in California on July 17, 1944. The explosion killed 320 men, and 202 of them were Black sailors. Navy officers had given the dangerous loading work to Black sailors without enough training or equipment.

Afterward, 258 sailors refused to go back to loading under the same conditions. Navy officers convicted fifty of them of mutiny. Thurgood Marshall, a lawyer for the NAACP, a group that works for Black Americans' rights, worked to get a higher court to look at their case again. In 2024, Navy Secretary Carlos Del Toro cleared the names of all the men who were punished.

Japanese Americans. After Japan's attack, Army leaders would not let Japanese Americans join the army. President Franklin Roosevelt signed Executive Order 9066, an order from the president that officials had to follow, on February 19, 1942.

Under it, federal officials forced more than 120,000 people of Japanese descent from their homes and locked them in camps behind fences and guard towers. Two thirds of them were American citizens. None of them was charged with a crime.

Army leaders dropped the ban on joining in 1943. From the camps, 1,208 people volunteered, fewer than 6 of every 100 who could. In Hawaii, where officials did not lock up Japanese Americans in large numbers, more than 10,000 volunteered. About 14,000 men served in the 442nd Regimental Combat Team, an army unit of Japanese American soldiers. They received 9,486 Purple Hearts, the medal for being wounded or killed in battle, and 21 Medals of Honor, the country's highest award for bravery in battle.

More than 375 Navajo Marines sent messages in a code built from the Navajo language. In it, a fighter plane was a "hummingbird" and a submarine was an "iron fish." Japanese code breakers never broke it. At Iwo Jima six Navajo code talkers sent more than 800 messages without an error. Military officials kept the code secret until 1968.

Truman's order. President Truman signed Executive Order 9981 on July 26, 1948. It required "equality of treatment and opportunity for all persons in the armed services without regard to race, color, religion or national origin." According to a National Park Service history, the labor leader A. Philip

Randolph, who had threatened a protest march, pushed Truman to sign it. Military leaders were slow to obey it, and nearly every unit was mixed by about 1953.

Benjamin O. Davis Jr.

Who: Commander of the Black fighter pilots trained at Tuskegee, and the first Black general in the Air Force.

When and where: Born in Washington, DC, December 18, 1912. West Point, 1932 to 1936. Commanded the 99th Fighter Squadron and the 332nd Fighter Group in Italy. Died July 4, 2002.

For four years his West Point classmates spoke to him only on duty.

Benjamin O. Davis Jr.'s father, Benjamin O. Davis Sr., was one of only two Black officers in the Army. The son entered West Point in 1932. For four years his white classmates spoke to him only when their duties required it. He never had a roommate, and he ate his meals in silence. He finished 35th of 276 in the class of 1936.

The Army had no Black flying units, so Army leaders made him an infantry officer, an officer of foot soldiers. When Army leaders opened pilot training for Black men at Tuskegee, he trained there and earned his pilot's wings in March 1942. He led the 99th Fighter Squadron and then the 332nd Fighter Group in combat.

Years later, Davis told the historian Alan Gropman that he himself doubted the claim that the group never lost a bomber. In 1998 President Bill Clinton made him a four-star general.

Daniel Inouye

Who: A Japanese American soldier from Hawaii who lost his right arm in Italy, and later a US senator.

When and where: Born in Honolulu, September 7, 1924. 442nd Regimental Combat Team. Wounded near San Terenzo, Italy, April 21, 1945. Died December 17, 2012.

With his left hand he pulled a live grenade from his right.

Daniel Inouye wanted to become a surgeon. When he first tried to join the army, he was turned away under the rule that barred Japanese Americans. He enlisted in 1943, after Army leaders dropped the ban.

He led an attack up a ridge in Italy against three German machine guns on April 21, 1945. A bullet hit him in the stomach, and he kept going. Then, in his own words, a German "rifle grenade smashed into my right elbow and exploded. I looked at my dangling arm and saw my grenade still clenched in a fist that suddenly didn't belong to me anymore."

He pulled the live grenade out of his right hand with his left and threw it at the last machine gun. Then he was shot in the leg and passed out.

Surgeons at a field hospital cut off his right arm on May 1. Doctors had already given him so much morphine, a painkiller, that they could give him nothing to put him to sleep for the operation. He was given other people's blood 17 times. He left the army in 1947 as a captain and represented Hawaii in the US Senate for 50 years. On June 21, 2000, President Clinton gave him the Medal of Honor, with 19 other Japanese American veterans.

Chester Nez

Who: A Navajo Marine, one of the first 29 code talkers, who fought on four Pacific islands.

When and where: Born in Chi Chil Tah, New Mexico, 1921. Guadalcanal, Bougainville, Guam and Peleliu, 1942 to 1944. Died in Albuquerque, June 4, 2014.

Punished at school for speaking Navajo, he built a code in it.

At his boarding school at Fort Defiance, school staff punished students, often by hitting them, for speaking Navajo. Marine recruiters came to the reservation in April 1942, and Chester Nez joined. He and 28 other Navajo Marines built the code at Camp Elliott in California.

He went into combat on Guadalcanal in November 1942, then on Bougainville. On Guam, in July and August 1944, he was wounded in the foot. He fought again on Peleliu in September 1944.

At home after the war he had nightmares and memories of combat that kept coming back, what doctors now call post-traumatic stress disorder. A Navajo healing ceremony for warriors, the Enemy Way, helped him. He finished high school, studied art at the University of Kansas on the G.I. Bill, a law under which federal officials paid for veterans' schooling, and worked 25 years for the Veterans Administration. He was the last of the first 29 when he died at 93.

BY MARCH 1945: Allied soldiers had been fighting in France since D-Day, and Marines were fighting on Iwo Jima.

Tokyo, March 1945

Major General Curtis LeMay commanded the American B-29 bombers attacking Japan, and in early 1945 he changed their method. Instead of dropping ordinary bombs from high up in daylight, the planes flew low at night and dropped bombs made to start fires. Many were filled with napalm, gasoline thickened into a jelly, which sticks to whatever it lands on, including skin, and keeps burning.

B-29 crews in 279 planes dropped more than 1,665 tons of these bombs on Tokyo's neighborhoods of wooden houses on the night of March 9 to 10, 1945. The fires joined into one huge fire, called a firestorm, that used up the oxygen in the air. People suffocated or burned to death. Sixteen square miles of the city burned, and more than a million people lost their homes.

Historians at the National WWII Museum estimate 110,000 dead, and other estimates start at 80,000. That is close to the 116,516 Americans who died in all of World War I. Most of the dead were civilians. LeMay and his commander, General Hap Arnold, called the raid a success. American bomber crews firebombed many more Japanese cities after Tokyo.

The atomic bombs, August 1945

Scientists and engineers of the Manhattan Project, the secret program to build the bomb, built the first atomic bombs at a cost of about $1.9 billion.

The order. General Thomas Handy, the acting Army chief of staff, gave the order on July 25, 1945, at the direction of the Secretary of War, Henry Stimson, and the Army chief of staff, George Marshall. He told General Carl Spaatz to have the 509th Composite Group, a bomber unit, drop the first bomb on a Japanese city as soon as the weather allowed. President Harry Truman approved.

The crew of the bomber *Enola Gay*, piloted by Colonel Paul Tibbets, dropped a bomb made with uranium, a heavy metal, on Hiroshima at 8:16 a.m. on August 6, 1945. On August 9, Major Charles Sweeney's crew dropped a bomb made with plutonium, a different heavy metal, on Nagasaki. Japan's leaders surrendered on August 14.

Scientists at the Radiation Effects Research Foundation, run by Japan and the United States together, estimate that 90,000 to 166,000 people in Hiroshima and 60,000 to 80,000 in Nagasaki died within four months. The exact numbers are not known. A US study in 1946, the Strategic Bombing Survey, gave lower counts: 70,000 to 80,000 dead in Hiroshima and more than 35,000 in Nagasaki.

Most of the dead were killed by burns from the flash of heat and by injuries from the blast. The same US study found that 15 to 20 percent died of radiation sickness.

Radiation sickness. The bombs gave off invisible energy called radiation, which killed the cells that make blood and line the gut. People with radiation sickness ran fevers within two days. After about two weeks their hair fell out, and they bled from the gums and nose and under the skin. Deaths from radiation began about a week after the bombs and were highest three to four weeks later.

Coming home: the G.I. Bill and Isaac Woodard

President Franklin Roosevelt signed the G.I. Bill on June 22, 1944. Under it, federal officials paid for veterans' schooling and promised banks that the Veterans Administration would repay a veteran's home loan if the veteran could not. By 1956, 7.8 million of the 16 million World War II veterans, about one in two, had used it for school or job training. From 1944 to 1952, Veterans Administration officials made that promise on nearly 2.4 million home loans.

Who got the loans. Local white bank officers and real estate agents decided who got the loans. Reporters at *Ebony* magazine found in 1947 that of 3,229 of these home loans in 13 Mississippi cities, only two went to Black veterans.

Isaac Woodard. Police in South Carolina beat a Black veteran blind on the day he left the Army. Sergeant Isaac Woodard, of Winnsboro, South Carolina, left the Army on February 12, 1946, after four years of service. On the bus home from Camp Gordon, Georgia, he argued with the driver after asking for a rest stop. At Batesburg, South Carolina, the driver had police take him off the bus.

Woodard said that Police Chief Lynwood Shull and Officer Elliot Long beat him with blackjacks, which are short clubs, and tried to gouge out his eyes. Shull later said, "I hit him across the front of the head." A local judge fined Woodard $50. Doctors at the veterans' hospital in Aiken found bleeding in both eyeballs. He was blind for the rest of his life.

Federal officials charged Shull with violating Woodard's civil rights. In November 1946, an all-white jury found Shull not guilty after thirty minutes. Chapter 25, Rights and Movements, tells how his case reached President Truman.

BY 1950

If you remember one thing: World War II was the first American war in which more soldiers died in battle than of disease, and most of the men who fought it had been drafted.

Left open: The draft of 1940 was the first ever in a time of peace. Would young men still be drafted after the world wars were over? The next era answers it.

Next: American forces went to war in Korea in 1950 and in Vietnam in 1965.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 15 MINUTES

American forces fought two long wars in Asia between 1950 and 2000, in Korea and in Vietnam. Drafted men fought in Vietnam until the draft ended in 1973, and after that everyone who joined the armed forces had volunteered.

The dead. By the Defense Department's counts, 36,574 Americans died in Korea, from 1950 to 1953, and 58,220 died in Vietnam. Many more people died in those wars: at least 2.5 million in the Korean War, most of them Koreans, and by Vietnam's own count about 3 million fighters and civilians in the Vietnam War.

About 697,000 American troops went to war against Iraq in 1991. The fighting lasted from January 17 to February 28, and 383 Americans died.

Korea, 1950 to 1953

North Korean troops invaded South Korea on June 25, 1950, and American troops went to fight for the South. About 120,000 Chinese soldiers surrounded about 30,000 American and allied troops at the Chosin Reservoir in North Korea in November 1950. The temperature fell to 20 degrees below zero. The fighting at Chosin ended on December 13. Of about 8,000 Marines in the fighting, 836 were killed, and the survivors called themselves "The Frozen Chosen."

No Gun Ri. American soldiers of the 7th Cavalry Regiment fired on South Korean refugees, people fleeing from danger, in the last week of July 1950. The refugees had gathered at and under a railroad bridge near the village of No Gun Ri. Survivors say about 400 people died, and Korean officials have confirmed at least 163 dead or missing.

Army officers who reviewed the shooting said in January 2001 that they could not count the dead and found no order to shoot. That month President Bill Clinton said that "things were done which should not have been done." He expressed regret and did not apologize.

The historian Sahr Conway-Lanz found a letter in 2006 from the US ambassador, John Muccio, dated July 25, 1950. It said that refugees who kept coming toward American lines after warning shots "will be shot." The Army's report did not include the letter.

By the Korean War Legacy Foundation's count, 38 percent of the Americans taken prisoner in Korea died in captivity, more than one in three. The fighting stopped with an armistice signed on July 27, 1953. At least 2.5 million people died in the war, most of them Korean soldiers and civilians. No single official count exists.

The hydrogen bomb tests, 1952 and 1954

Radioactive dust from American bomb tests after World War II sickened people who were not at war: Marshall Islanders, American service members and Japanese fishermen. American scientists and military officers tested nuclear weapons, the atomic bombs and the stronger bombs that came after them, in the Marshall Islands, in the Pacific, on the land of the Marshallese people.

The first hydrogen bomb. American scientists set off the first hydrogen bomb, a far more powerful kind of nuclear weapon, on Elugelab Island in Enewetak Atoll, a ring of small coral islands around a lagoon, on November 1, 1952. It was more than 450 times as powerful as the bomb dropped on Nagasaki. It destroyed the island and left an underwater crater 6,240 feet wide.

Castle Bravo. The test called Castle Bravo, at Bikini Atoll on March 1, 1954, was 1,000 times as powerful as the bomb dropped on Hiroshima, and two and a half times as powerful as its planners expected. A cloud 100 miles wide dropped fallout on Marshall Islanders, American service members, and the crew of a Japanese fishing boat. Fallout is radioactive dust and ash that falls back to the ground after a nuclear blast. It burns the skin it lands on and damages the body from inside when it is breathed in or swallowed.

The military had made no plans to move the people living nearby. On Rongelap, about 110 miles away, the fallout looked like snow, and children played in it. About three days after the blast, American crews took more than 230 people off Rongelap, Rongerik and Utirik, among them 28 American service members. The fallout left Rongelap unsafe to live on.

Twenty-three Japanese fishermen became sick, and one died. The physicist Theodore Taylor said later, "obviously we didn't irradiate (expose to radiation) the natives on purpose. Obviously, we didn't know what the hell we were doing."

What followed. American officials made the exposed Marshall Islanders the subjects of a secret medical study called Project 4.1. They told the people working on it not to discuss its purpose, "due to possible adverse public reaction," meaning for fear of how the public would react. The United States later gave the exposed people medical care. A screening from 1993 to 1997 found 57 cases of thyroid cancer, a cancer of a gland in the neck, among Marshallese people born before 1954.

The war in Vietnam, and who had to go

Marines landed at Da Nang, South Vietnam, on March 8, 1965. By April 1969, about 543,000 American troops were in Vietnam. The deadliest year was 1968, when 16,899 Americans died there. The last American combat troops left on March 29, 1973. The war ended on April 30, 1975, when North Vietnamese troops and their allies in the South took Saigon, the capital of South Vietnam.

Who had to go. Counts of the men drafted during the Vietnam years run from about 1.8 million to 2.2 million. Full-time college students could get a student deferment, which put off their draft call while they stayed in school. Until 1969, local draft boards, the officials who chose which men in an area were drafted, called the oldest eligible men first.

The lottery. Selective Service officials, who ran the draft for the whole country, held a lottery instead on December 1, 1969. They put 366 birthdays in blue plastic capsules and drew them one at a time. The first date drawn was September 14, so men born on that day were called first. For 1970, the draft reached number 195.

Many men refused to go. Officials accused 209,517 men of breaking the draft law, and fewer than 9,000 were convicted. Some men fled to Canada. In 1972 more men were recognized as conscientious objectors than were drafted. The last draft call came on December 7, 1972, and the law allowing the draft ran out on June 30, 1973. Since July 1980, young men have had to register for the draft again, but no one has been drafted.

The pardon. Jimmy Carter pardoned, or officially forgave, the men who had broken the draft law between 1964 and 1973 on January 21, 1977, his first full day as president. His pardon left out crimes that involved force or violence,

and it did not cover men who had deserted the armed forces after joining.

Ron Kovic

Who: A Marine from Long Island, paralyzed by a bullet in Vietnam, who came home and spoke against the war.

When and where: Born July 4, 1946. Grew up in Massapequa, New York. Wounded in Vietnam on January 20, 1968.

Movie: *Born on the Fourth of July* (1989), directed by Oliver Stone and based on Kovic's memoir. Kovic helped write the screenplay.

The second bullet went through his shoulder and into his spine.

Ron Kovic joined the Marines and then volunteered for a second tour in Vietnam, a second stretch of service there. On January 20, 1968, he was shot in the right foot. Then a second bullet went through his right shoulder. It collapsed one lung and cut into his spinal cord, the bundle of nerves inside the backbone that carries messages between the brain and the body.

From the middle of his chest down, he could no longer move. In the hospital at Da Nang, doctors told him he would never walk again.

He spent months at the Veterans Administration hospital in the Bronx, New York. *New Mobility* magazine describes the hospital as short of staff, with out-of-date equipment and conditions that were sometimes abusive. Patients called it "the Bronx Zoo."

He gave his first speech against the war in 1969. In 1972, at a protest at President Richard Nixon's campaign office in Los Angeles, he was thrown out of his wheelchair and arrested. The *New Mobility* account does not name who threw him. His book *Born on the Fourth of July* came out in 1976.

Muhammad Ali

Who: The heavyweight boxing champion, who refused to be drafted and won his case in the Supreme Court.

When and where: Refused induction in Houston, Texas, April 28, 1967. Supreme Court decision, June 28, 1971.

When an officer called his name, he would not step forward.

Muhammad Ali was a minister of the Nation of Islam, a Muslim religious group. He asked his draft board to list him as a conscientious objector. The board said no, and so did the appeal board.

He went to the army's induction center in Houston, where new soldiers were sworn in, on April 28, 1967. When the officer called him, he stayed where he was. The officer warned him that refusing was a crime that could bring five years in prison and a $10,000 fine. Boxing officials took away his heavyweight title that day.

He was found guilty in a federal court of refusing the draft on June 20, 1967, and sentenced to five years in prison and a $10,000 fine. His lawyers appealed, and he did not box again for more than three years. On June 28, 1971, all eight justices who heard the case overturned his conviction. They ruled that the appeal board had never given a reason for turning down his claim.

My Lai, March 16, 1968

American soldiers of Charlie Company, 1st Battalion, 20th Infantry, under Captain Ernest Medina, killed the unarmed people of My Lai, a village in South Vietnam, on March 16, 1968. The counts of the dead differ. The Army's own investigation, led by Lieutenant General William Peers, found at least 175 killed and possibly more than 400. Another Army count gives 347. A memorial in the village lists 504 names, from babies of one year old to a person of 82.

What the soldiers did. The Peers report describes it. Soldiers of the 1st Platoon herded 60 to 70 villagers into an irrigation ditch and shot them. In another part of the village, soldiers of the 2nd Platoon made 10 to 20 women and children squat in a circle, fired grenades into them, and then shot the wounded with rifles.

Soldiers raped women and girls, forcing them into sex by violence or threat. The investigators found that one platoon leader watched several rapes that day and did nothing to stop them. Soldiers also burned homes, killed farm animals and destroyed crops.

The cover-up. Officers at every level of the division took steps that kept the killings secret, and the story did not become public until late 1969. The Peers investigation named 30 men who knew about the killings. Fourteen of them were charged with crimes. Only one was convicted.

Calley. Lieutenant William Calley was tried at a court-martial. On March 29, 1971, a panel of six army officers found him guilty of murdering 22 people. Two days later, the same officers sentenced him to life in prison. Three days after the guilty verdict, President Richard Nixon ordered him held under house arrest, kept under guard in his own quarters.

Lieutenant General Albert Connor, the commander who reviewed the case, cut the life sentence to 20 years in August 1971. Secretary of the Army Howard Callaway cut it to 10 in April 1974. Calley was freed early on parole, under rules he had to follow, in November 1974. No one was convicted of rape.

Hugh Thompson Jr.

Who: An Army helicopter pilot who landed between American soldiers and Vietnamese villagers at My Lai to stop the killing.

When and where: Born in Atlanta, Georgia, April 15, 1943. My Lai, South Vietnam, March 16, 1968. Died in Pineville, Louisiana, January 6, 2006.

He set his helicopter down between the soldiers and the villagers.

Hugh Thompson Jr. flew a small scout helicopter with two crewmen, Glenn Andreotta and Lawrence Colburn. Over My Lai that morning, he dropped green smoke next to wounded villagers as a signal that they needed help. When he came back, they were dead. He watched Captain Medina shoot a wounded young woman, and when he saw an irrigation ditch full of bodies he radioed, "There's something wrong here."

He landed, and Lieutenant Calley told him to mind his own business. Then Thompson saw soldiers chasing about ten villagers, some of them children, toward a shelter. He set his helicopter down between the soldiers and the villagers. He told Colburn and Andreotta to fire their machine guns at the soldiers if the soldiers shot at him or the villagers.

He got 11 people out of the shelter, and two other helicopter pilots flew them to safety. Andreotta climbed into a ditch of about 100 bodies and pulled out a living boy, Do Ba. Thompson flew the boy to a hospital and reported the killings. Medina then told his men to stop.

Andreotta was killed in combat three weeks later. In 1969 Mendel Rivers, the congressman who led the House Armed Services Committee, said Thompson was the one soldier at My Lai who should be punished, for threatening American soldiers. Thompson got death threats.

In March 1998, Army leaders gave all three men the Soldier's Medal, Andreotta after his death. It is the Army's highest award for bravery outside combat with an enemy. That year Thompson and Colburn went back to My Lai and met survivors.

What the Vietnam War cost

Vietnamese officials published their first count of the war's dead on April 3, 1995. They said 1.1 million fighters of North Vietnam and of the Viet Cong, the communist fighters in South Vietnam who fought on the North's side, had died over 21 years, and nearly 2 million civilians in the North and the South. Between 200,000 and 250,000 South Vietnamese soldiers died, by the US military's estimate.

Operation Homecoming. American aircrews flew 591 American prisoners of war home from North Vietnam from February to April 1973. Many had been held in Hoa Lo prison in Hanoi, which the prisoners called the "Hanoi Hilton." The historian Andrew Lipps, quoted in a Defense Department news story, writes of the "mental and physical torture" they went through.

Agent Orange. American forces sprayed plant-killing chemicals over South Vietnam. One of them, Agent Orange, was tainted with a poison called dioxin. Today officials of the Department of Veterans Affairs treat a list of illnesses in veterans who were exposed to Agent Orange as caused by it, so the veterans do not have to prove it. The list includes several cancers, such as bladder cancer, lung cancer and prostate cancer, and also Parkinson's disease, type 2 diabetes and heart disease.

The Vietnam Veterans Memorial in Washington, designed by Maya Lin, opened in 1982. It lists 58,281 names. The count is higher than the Defense Department's 58,220 because the two lists count some deaths differently.

Women and gay service members

Under a law signed on June 12, 1948, women could serve as permanent members of the armed forces. Under the same law, women could make up no more than 2 percent of the forces and could not serve in combat units. Members of Congress removed the cap in 1967. In July 1976, 119 women entered the Military Academy at West Point, and 62 of them graduated in 1980.

Under a 1993 law called "Don't Ask, Don't Tell," gay, lesbian and bisexual service members could serve only if they kept it secret. Military officials put more than 13,000 of them out of the armed forces under it before it ended in 2011.

The Gulf War, 1991

About 100,000 Iraqi soldiers invaded Kuwait on August 2, 1990. American and allied pilots began bombing Iraq on January 17, 1991. The ground attack began on February 24 and lasted 100 hours, and President George H. W. Bush declared a cease-fire on February 28. About 697,000 American troops went to the region, and 383 died, 148 of them in battle.

No official US count of the Iraqi dead exists. In 2003, Carl Conetta of the Project on Defense Alternatives, a research group, estimated that American and allied forces killed 20,000 to 26,000 Iraqi soldiers and about 3,500 Iraqi civilians.

Gulf War illness. At least one in four of the American troops came home sick. In 2008 scientists on a committee advising the Department of Veterans Affairs found that 175,000 to 210,000 Gulf War veterans had an illness. Its signs were memory problems, headaches, pain all over the body, and stomach trouble. Doctors had found no treatment that worked, and few of the sick veterans had gotten better.

The committee's scientists linked the illness to pills the soldiers took to protect them from nerve gas, and to pesticides, the chemicals used to kill insects. Other scientists, on panels of the Institute of Medicine, weighed the evidence differently.

BY 2000

If you remember one thing: Drafted men fought in Vietnam, and after the draft ended in 1973 everyone who served had volunteered.

- July 1950: Soldiers of the 7th Cavalry fire on refugees under a railroad bridge at No Gun Ri.
- March 16, 1968: American soldiers kill the people of My Lai. Only Lieutenant William Calley is convicted, and he goes free on parole in 1974.
- June 30, 1973: The law allowing the draft runs out.

Next: Every American who fought in the wars after September 11, 2001, had volunteered.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 13 MINUTES

Every American who fought in the wars after the attacks of September 11, 2001, had volunteered. American forces fought in Afghanistan for almost 20 years and in Iraq for almost 9.

Who served. Researchers at the Pew Research Center found in 2011 that in the ten years after 9/11 about one American in 200 was on active duty, serving full-time in the military, at any one time. At the height of World War II it had been nearly one in 11.

The dead. A 2020 report to Congress, using Defense Department records up to July 2020, lists more than 2,400 American service members dead in the Afghanistan war and about 4,500 in the Iraq war. More than 53,000 were wounded in action in the two wars together.

Far more people in those countries died. Researchers in the Costs of War project at Brown University study the wars after 9/11 in Iraq, Afghanistan, Syria, Yemen and Pakistan. They estimate that more than 940,000 people in those countries were killed in the fighting itself from 2001 to 2023, and more

than 432,000 of them were civilians. They estimate that 3.6 to 3.8 million more died because of what the fighting destroyed: the ways people earned a living, hospitals and health care, roads and water systems, and the land.

Counts of the Iraqi civilian dead differ widely. A 2006 survey published in the medical journal *The Lancet* estimated about 600,000 violent deaths, and US and Iraqi officials disputed it. Chapter 23, America and the World, explains why the counts are so far apart.

Afghanistan and Iraq, 2001 to 2021

American bombing of Afghanistan began on October 7, 2001, less than a month after the September 11 attacks. The invasion of Iraq began on March 20, 2003. The last line of American military vehicles, carrying about 500 troops, left Iraq for Kuwait on December 18, 2011.

Leaving Afghanistan. American officials signed a deal with the Taliban in February 2020 and agreed to take American troops out of Afghanistan. The Taliban were the armed group American forces had fought there since 2001. President Joe Biden announced in April 2021 that the last of them would be out by September. Taliban fighters took Kabul, Afghanistan's capital, on August 15, 2021, and several thousand more American troops came to the airport to fly American embassy staff and some Afghans out.

Abbey Gate. A bomber from ISIS-K, a terror group in Afghanistan that was also fighting the Taliban, set off explosives he was wearing in a crowd at Abbey Gate, an entrance to the Kabul airport, on August 26. He killed 13 American service members: 11 Marines, a Navy medic and an Army soldier. Another 45 were wounded. US Justice Department officials said in 2025 that about 160 Afghan civilians were killed, and other reports say about 170.

The drone strike. Three days after the Abbey Gate bombing, on August 29, 2021, American forces fired a missile from a drone, an aircraft with no pilot aboard, at a car in Kabul. Officers believed it carried an ISIS-K bomb. The driver was Zemari Ahmadi, a 43-year-old aid worker, and the car was at his family's home. The missile killed ten civilians, seven of them children. The next night, at 11:59 p.m. on August 30, the last American plane left Kabul.

General Mark Milley first called it a "righteous strike." On September 17, 2021, General Frank McKenzie, the head of the military's Central Command, called it "a tragic mistake." On December 13, 2021, Defense Secretary Lloyd Austin decided that no one would be punished.

Abu Ghraib, 2003 and 2004

American military police guarding prisoners at Abu Ghraib prison near Baghdad committed what an Army report calls "sadistic (taking pleasure in hurting people), blatant (done openly), and wanton (done for no reason) criminal abuses" of them, from October to December 2003. Army Major General Antonio Taguba investigated and wrote that report in 2004.

What the guards did. The report lists guards punching, slapping and kicking prisoners, jumping on their bare feet, keeping them naked for days, and forcing them into sexual poses to photograph them. Guards poured the liquid from broken glow sticks on prisoners.

Guards used dogs without muzzles to frighten prisoners, and in at least one case a dog bit a prisoner and badly hurt him. Guards also threatened prisoners with a loaded pistol, and attached wires to one prisoner to make him think he would be tortured with electricity.

The CBS program *60 Minutes II* broadcast the guards' photographs on April 28, 2004. Army officers at courts-martial convicted Specialist Charles Graner, who was sentenced to 10 years in prison, Private First Class Lynndie England, sentenced to 3 years, and other low-ranking soldiers.

Haditha and Kandahar, 2005 and 2012

A roadside bomb killed Lance Corporal Miguel Terrazas, a 20-year-old Marine, in the Iraqi town of Haditha on November 19, 2005. Marines of Staff Sergeant Frank Wuterich's squad then shot five unarmed Iraqi men who had been ordered out of a car. Wuterich told his men to "shoot first and ask questions later." They broke into nearby houses with grenades and rifle fire. In about 45 minutes they killed 24 Iraqi men, women and children, among them an old man in a wheelchair. No weapons were found.

Military prosecutors charged eight Marines. They dropped the charges against six, and one was found not guilty at a court-martial. Wuterich pleaded guilty to one charge of neglecting his duty in January 2012. The judge sentenced him to 90 days, a pay cut and demotion to private, but under his plea deal, an agreement to admit a crime in return for a lighter punishment, he served no time in prison.

Kandahar. Army Staff Sergeant Robert Bales left his base in Kandahar, Afghanistan, on the night of March 11, 2012, and went into two villages, Alkozai and Najiban. He killed 16 unarmed Afghan civilians and wounded six more. Seventeen of the 22 people he shot were women or children, and eleven of the dead came from one family. Bales pleaded guilty, and on August 23, 2013, he was sentenced to life in prison.

Pat Tillman

Who: A professional football player who left the Arizona Cardinals to join the Army Rangers, and was killed by soldiers of his own platoon.

When and where: Enlisted 2002. 75th Ranger Regiment. Killed in eastern Afghanistan, April 22, 2004. A committee of the House of Representatives held a hearing on his death on April 24, 2007.

He called out "Cease fire" and his own name until he stopped.

Pat Tillman walked away from a $3.6 million contract with the Arizona Cardinals after September 11, 2001. In 2002 he and his brother Kevin joined the Army together. Pat became a Ranger, a soldier of a specially trained Army unit, in the 75th Ranger Regiment, and served tours in Iraq and Afghanistan.

Soldiers in a truck from Pat's own platoon fired on the hillside where his group was on April 22, 2004, after what Kevin Tillman called "a small harassing ambush." The soldier beside him said later that they yelled "Cease fire. Friendlies." Tillman threw a smoke grenade, and the shooting stopped.

When the shooting started again, Tillman was hit. He called out "Cease fire" and his own name over and over until he stopped. Three bullets struck his forehead, so close together that Army medical examiners judged he was shot with an M-16 rifle from about 10 yards. An Afghan soldier with them was killed too.

The first account. His family was told that enemy fighters had killed him. In the words of Congressman Henry Waxman, who led the hearing, "the military told the world" the same thing. Army commanders awarded him the Silver Star, a medal for bravery in battle, with a written account of a battle with the enemy.

Kevin Tillman told Congress that Pat's uniform, equipment and notebook were destroyed, and his testimony does not name who destroyed them. On May 29, 2004, five weeks after the shooting, an Army general

> announced that Tillman had "probably" been killed by his own side. Kevin Tillman called the Army's first account "utter fiction."

Wounds that do not show, and veterans at home

Military doctors diagnosed 535,544 service members with a traumatic brain injury, damage to the brain from a blow or a blast, from 2000 through 2025, according to a Defense Department count published in June 2026. About 83 percent were mild, most of them concussions, injuries in which a blow shakes the brain, and many happened in training or daily life, not in combat.

Veteran suicides. A total of 6,398 veterans killed themselves in 2023, an average of 17.5 a day, according to a report the Department of Veterans Affairs released in early 2026. That is nearly as many as the about 6,900 service members who died in the Afghanistan and Iraq wars together. Sixty-one percent of them had not received care from the department in the last year of their lives. The country had about 18 million veterans in 2023, about 6 of every 100 adults, the Pew Research Center found.

Tammy Duckworth

Who: An Army helicopter pilot who lost both legs when her Black Hawk helicopter was hit in Iraq, and later a US senator.

When and where: Born in Bangkok, Thailand. Illinois Army National Guard. Wounded in Iraq, November 12, 2004.

"The explosion vaporized my right leg," she wrote.

Tammy Duckworth became an Army officer in 1992 and learned to fly helicopters. In 2004 she went to Iraq as a Black Hawk pilot with the Illinois Army National Guard, a part-time state army force that can be sent to war.

A rocket-propelled grenade, a small rocket fired from a tube carried on a fighter's shoulder, hit the front of her helicopter on November 12, 2004. In her memoir she wrote, "The explosion vaporized my right leg. It blew my left leg up into the bottom of the instrument panel." It also tore through her right arm.

Dan Milberg, the other pilot aboard, landed the damaged helicopter. Matt Backues, another member of the crew, dragged her, unconscious, across the desert. She lost both legs and part of the use of her right arm, and she recovered at Walter Reed Army Medical Center.

She was an assistant secretary of Veterans Affairs from 2009 to 2011. Illinois voters elected her to the US House in 2012 and to the Senate in 2016. In 2018 she became the first senator to give birth while in office.

Who may serve

"Don't Ask, Don't Tell" ended on September 20, 2011, and gay, lesbian and bisexual service members could serve openly. Defense Secretary Ashton Carter opened every combat job to women on December 3, 2015. About 220,000 jobs, roughly one in ten, had been closed to them.

Transgender service members. The justices of the Supreme Court let President Donald Trump's ban on transgender service members take effect on May 6, 2025, while lawsuits went on. On June 1, 2026, judges of a federal appeals court in Washington ruled 2 to 1 that the ban likely broke the Constitution, and they blocked the discharge of the service members who had sued.

A few weeks later a judge of a lower court extended that protection to all transgender troops then serving, about 4,240 people by the count of the Gender Justice League, a group that works for transgender people's rights. New people were still not allowed to join. On August 28, 2026, Justice Department lawyers asked the Supreme Court to lift the lower court's order.

The Wounded Knee medals. Twenty soldiers of the 7th Cavalry received the Medal of Honor for Wounded Knee in 1890, where soldiers of the regiment killed between 250 and 300 Lakota people, almost half of them women and children. Defense Secretary Lloyd Austin ordered a review of those medals in 2024. In September 2025, Defense Secretary Pete Hegseth announced that the soldiers would keep them. A bill called the Remove the Stain Act, which would take the medals back, had not passed as of September 2026.

The department's name. President Trump signed an order on September 5, 2025, letting officials call the Defense Department the "Department of War." Members of Congress set its legal name. Members of the House voted 216 to 212 on July 22, 2026, to change the legal name, but as of late September 2026 the change had not become law.

2025 and 2026: boats, Venezuela and Iran

US military crews began firing on boats in the Caribbean Sea and the eastern Pacific on September 2, 2025. President Trump's officials said the boats were carrying drugs. By September 19, 2026, the Associated Press counted at least 231 people killed in 69 strikes. In at least one case, the Associated Press reported, military officials gave no evidence that the boat carried drugs.

Lawyers for the American Civil Liberties Union are suing for the families of two men from Trinidad, Chad Joseph and Rishi Samaroo, who were killed in a strike on October 14, 2025. The group says they were fishermen and farm workers.

Maduro. Soldiers of the Army's Delta Force and officers of the CIA, the American spy agency, captured Venezuela's president, Nicolás Maduro, and his wife, Cilia Flores, in the capital, Caracas, on January 3, 2026. They flew the two to a Navy ship, the USS *Iwo Jima*, and then took them to New York. The next day federal officials made public the charges against them, crimes tied to smuggling drugs. On January 5, in a federal court in New York, both said they were not guilty.

By the Council on Foreign Relations' account, American forces killed 32 Cuban officers in the attack. Members of Congress had not voted to approve the raid beforehand.

Iran. American and Israeli forces began bombing Iran on February 28, 2026, and Iranian forces fired back at American troops. By the Pentagon's public count on September 22, 2026, 19 American service members had died and more than 820 had been wounded. Five officials told the *Washington Post* that at least 22 service members had died in the Middle East since the war began, not all of them in the fighting. Chapter 23 gives a count of the people killed in Iran. The war was still going on in late September 2026.

THE THREE QUESTIONS, ANSWERED

1. **What killed most American soldiers before World War II?**
   Disease. In the war with Mexico seven of every eight American dead

died of something other than enemy fire, and in World War I flu and pneumonia killed more soldiers than the war's deadliest battle did.

2. **Who was made to fight, and who could get out of it?** Drafts filled the armies of the Civil War, both world wars and Vietnam. In 1863 a drafted man could pay $300 and go home, and in the Vietnam years college students could put off the draft. Since 1973 everyone who serves has volunteered.
3. **Were soldiers who killed people who could not fight back ever punished?** Rarely. No one was charged for killing 96 Christian Lenape and Mohican people at Gnadenhutten in 1782, and no one was tried for Sand Creek. Lieutenant William Calley, the only man convicted for My Lai, went free on parole in 1974.

What changed: The fighters changed from militia farmers, who went home when the fighting ended, to millions of drafted men in the world wars, to a small army of volunteers, about one American in 200.

FIVE DATES TO KEEP

May 26, 1637: English soldiers burn the Pequot town at Mystic.

April 19, 1775: The shooting starts at Lexington.

July 1 to 3, 1863: The battle of Gettysburg.

August 6, 1945: An American bomber crew drops an atomic bomb on Hiroshima.

June 30, 1973: The law allowing the draft runs out.

> **NOT KNOWN:** Nobody knows how many Filipino civilians died in the war of 1899 to 1902, where the counts run from 200,000 to 1,000,000, or how many of them American soldiers killed.

Where to go: The Amos Humiston monument stands on North Stratton Street in Gettysburg, Pennsylvania, the only one on the battlefield to a single enlisted soldier.

Next: Chapter 23, America and the World. It asks how the country dealt with other nations, from treaties and alliances to the islands it came to rule.

Chapter 23: America and the World

CHAPTER 23 OF 37 · AMERICA AND THE WORLD · 10 ERAS · ABOUT 90 MINUTES

From 1494 to today, people far away decided who would rule this land and its islands, and the people living there almost never got to choose.

Native Hawaiians put 21,269 signatures on a petition against letting the United States take their islands. They signed at meetings on all five main islands between September 11 and October 2, 1897. The signers were more than half of the Native Hawaiians counted that year. Four men carried the petition to Washington, and the treaty to take Hawaii failed. In July 1898 members of Congress took the islands anyway, by a method that did not need the votes the Hawaiians had blocked.

This chapter follows America's dealings with other countries, from a line drawn across the Atlantic in 1494 to the five territories of today. For the first four eras there was no United States, and rulers in Europe traded this land among themselves. In 1775 the new country started out weak. Its leaders needed France's help, and for years they could not free American sailors enslaved in Algiers. Then watch American leaders become the ones deciding for others, from the land taken from Mexico to the elected governments they helped overthrow after 1950. In every era, watch who was in the room when a deal was signed, and who was left out.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who got to choose when this land and its islands changed hands?
2. How did a weak new country come to rule islands and overthrow governments?

3. Are the people of the territories American citizens?

Five minutes? Read the Hawaii sections of 1850 to 1900, from "Hawaii: a kingdom overthrown, 1887 to 1893" through "21,269 names against annexation, 1897".

Can skip: "Trade laws made in London", "A committee for secret letters, 1775 to 1789", "Perry's warships in Japan, 1853 and 1854", "Trade and the climate agreement". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 2 MINUTES

Native nations on this land made alliances, promises to stand together against their enemies, long before any European arrived. Their leaders met in councils and gave gifts to seal an agreement. No Native diplomat, a person sent to speak and bargain for a nation, is known by name from the years before the Europeans came. There was no United States yet. American leaders declared independence in 1776, 276 years after 1500.

Far out in the Pacific Ocean, Polynesian voyagers, people of the many islands of the central and southern Pacific, sailed to the islands now called Hawaii and settled them. Then, in 1494, the rulers of Spain and Portugal signed a treaty, a written agreement between countries, that split the lands across the Atlantic between their two kingdoms.

When did people first reach Hawaii?

Polynesian voyagers were the first people to live on the islands of Hawaii. Scientists using revised radiocarbon dating, a way of measuring the age of old wood, bone and charcoal, now put their landing at about the years 1000 to 1200.

Older estimates were much earlier. For a long time most archaeologists, scientists who study the past through the things people left behind, put the first landing between about the year 300 and the year 750. Some newer studies put it a little later, between about 1200 and 1290.

Only one early home site in the islands has been dated with confidence, the Bellows Beach site at Waimānalo, on the island of Oʻahu. The argument over the first date is still open.

A line drawn at Tordesillas, 1494

The rulers of Spain and Portugal divided the lands across the Atlantic between them in a treaty signed in the Spanish town of Tordesillas on June 7, 1494. King Ferdinand II of Aragon and Queen Isabella I of Castile signed for Spain. King John II signed for Portugal.

The treaty described a line running from the North Pole to the South Pole, 370 leagues, or about 1,185 miles, west of the Cape Verde Islands in the Atlantic Ocean. The king of Portugal could claim new lands east of the line. The king and queen of Spain could claim new lands west of it.

Millions of people lived in the Americas, on the land these three rulers had divided. The rulers had never seen it. None of the people who lived there was at Tordesillas, and the rulers did not ask them or tell them.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 1 MINUTE

Kings in Spain, Portugal, France and England claimed land on this side of the ocean from their capitals in Europe. French Protestants, Christians who had broken away from the Catholic Church, built a fort in Florida, on land the king of Spain claimed. Spanish soldiers settled the matter in 1565 by killing hundreds of French soldiers and sailors on the Florida coast. Most of the dead had already surrendered or been captured.

Fort Caroline and Matanzas, 1565

Fort Caroline, the French fort, stood in the homeland of peoples who spoke the Timucua language. No French or Spanish leader asked them about it.

The king's contract. King Philip II of Spain made a contract in 1565 with Pedro Menéndez de Avilés, a Spanish commander. Under it, Menéndez was to settle Florida and drive the French out of their fort. Menéndez arrived in August. In his own words, he came "to plant the Gospel in these parts," which meant spreading his Catholic faith there.

The killings. Menéndez's soldiers took Fort Caroline on September 20, 1565. By his own count they cut the throats of 132 men there and killed 10 more the next day, and he sent about 50 women and children by ship to Havana. Timucua people told him where shipwrecked French sailors had come ashore. At that inlet, a narrow arm of the sea south of St. Augustine, his soldiers killed about 245 Frenchmen who had surrendered, among them the French commander, Jean Ribault. Chapter 22, War and the Military, tells the killings in full.

Menéndez wrote to King Philip that punishing the French this way was "serving God Our Lord, and Your Majesty," because they were Protestants. The inlet is still called Matanzas, the Spanish word for slaughters.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 3 MINUTES

New Netherland, the Dutch colony on Lenape land, changed hands three times in ten years. English forces took it in 1664, Dutch forces took it back in 1673, and in 1674 Dutch diplomats handed it to England again.

The colonies on this coast belonged to England, the Dutch Republic, France and Spain. Officials in London, Amsterdam, Paris and Madrid set each colony's borders, chose its wars and decided who it could trade with. Colonists had no ambassadors, officials sent to live in another country and speak for their own, and they could not sign treaties in their own names.

Trade laws made in London

Members of Parliament, England's lawmakers in London, decided what English colonists could buy and sell. They did it through a series of laws called the Navigation Acts.

They aimed the first act, in 1651, mainly at Dutch merchants. It required all trade between England and its colonies to travel in English or colonial ships, and it was one of the causes of a war at sea that English and Dutch sailors began fighting in 1652. The act of 1660 listed goods that colonists could ship only to England or to another English colony: sugar, tobacco, cotton, wool, ginger and indigo, a plant used to make blue dye.

Under the Staple Act of 1663, European goods bound for the colonies had to land in England first. Customs officers there, who collected the taxes on goods coming in, checked them, and then merchants shipped the goods on. Under these laws, merchants in Boston could not legally buy rope from Dutch traders or sell tobacco to French ones.

Dutch diplomats gave up New Netherland for a nutmeg island

The Dutch colony of New Netherland stood on the homeland of the Lenape people, which they call Lenapehoking. It covered what is now New York City, parts of Long Island and the Hudson Valley, New Jersey, northeastern Delaware and eastern Pennsylvania.

Mannahatta. Dutch colonists said they had bought the island of Mannahatta in 1626 for trade goods worth 60 guilders, the Dutch money of the time. The Dutch most likely misread what the Lenape meant by the exchange. Diseases that the colonists brought cut the number of Lenape people by 85 percent by 1700.

An English fleet arrived at New Amsterdam, the colony's main town, in 1664, and the Dutch surrendered the colony without a fight. Accounts give different days for the surrender because the English and the Dutch used different calendars at the time.

The trade. English and Dutch diplomats ended their war with the Treaty of Breda, signed on July 21, 1667. No Lenape leader took part. The Dutch had done better in the fighting, but the Dutch men who bargained at Breda let England keep New Netherland. In return the Dutch took Suriname and Run, an island in what is now Indonesia where nutmeg grew. Suriname had sugar plantations, large farms where the planters forced enslaved people to do the work.

Dutch forces sailed in during a third war, in August 1673, took the colony back and renamed its main town New Orange. Under the Treaty of Westminster, signed on February 19, 1674, Dutch diplomats handed the colony back to England, and it stayed English. By then the English called it New York.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

New England soldiers captured a French fortress in 1745, and three years later British diplomats gave it back without asking them. Colonists lived under the rule of the king of Britain, France or Spain. When their king went to war, they were at war too, and they had no say in the peace terms.

Native nations made treaties with English and French officials in their own names. In the summer of 1701 the Haudenosaunee, a league of five Native nations in what is now New York, made two agreements about two weeks apart, one with the English and one with the French.

Two agreements in the summer of 1701

At Albany. Haudenosaunee representatives met the acting English governor of New York at Albany on July 19, 1701. The English called them the Five Nations. They made a deed, a signed paper that hands over land or rights to land, to King William of England. It was about hunting lands that ran west from what is now western New York toward what is now Chicago.

People still argue over what the Haudenosaunee kept and what they gave up in that deed. Its English text states that they handed the land to the king, and that he would protect their "free hunting" there forever. Haudenosaunee groups read it as protecting their right to hunt there.

The raccoon case. In Canada, government lawyers argued against that reading in a court case. In 1987 Ontario officials charged Jesse Ireland and David Jamieson, two men of the Oneida, one of the five Haudenosaunee nations, with hunting raccoons without a license in southern Ontario, inside the area the deed describes.

The government lawyers said the hunting promise counted only if the English and the Haudenosaunee won the land back from the French and their Native allies, and that never happened. In 1990 an Ontario judge rejected that argument and ruled that the two men had a right under the deed to hunt there.

At Montréal. About two weeks after the Albany meeting, on August 4, 1701, the governor of New France, the French colony in North America, signed a peace at Montréal with about 1,300 representatives of 39 Native nations. It is called the Great Peace of Montréal, and with it the French and the Native nations ended decades of war.

French officials meant the peace to keep the Five Nations neutral, taking neither side, if the French and the English colonists went to war. The Haudenosaunee kept out of the next war between France and Britain.

Kondiaronk

Who: A Huron-Wendat chief who argued for the Great Peace of Montréal while he was dying of a fever. The French called him "the Rat."
When and where: Montréal, August 1701. Born around 1649.

He spoke for peace on August 1 and died within hours.

Kondiaronk was a chief of the Huron-Wendat people. The French nicknamed him "the Rat" for the tricks he used when he bargained.

A disease was spreading through Montréal while the great council met, and Kondiaronk caught it. On August 1, 1701, weak with fever, he spoke to the council and pushed the leaders toward agreement. Afterward he was carried to the Hôtel-Dieu, a hospital in the town, and he died a few hours later. Accounts give his death as August 1 or August 2.

The governor of Montréal, French officers, war chiefs and priests walked in the procession at his large funeral on August 3, and he was buried at the town's Notre-Dame church. The Great Peace of Montréal was signed the next day, August 4, 1701.

What British diplomats won at Utrecht, 1713

In the Treaty of Utrecht, British diplomats won a contract to ship kidnapped African people across the ocean and sell them as slaves. The treaty was signed on April 11, 1713, at Utrecht in the Netherlands. With it the diplomats ended a war in Europe that the colonists called Queen Anne's War.

Under the contract, called the asiento, British traders had the only right for thirty years to carry enslaved Black people into Spain's colonies in the Americas and sell them there. The diplomats wrote the asiento into the peace as one of Britain's prizes, beside land.

The land came from France. French diplomats signed over three things to Britain: all the land whose rivers drain into Hudson Bay, France's claims to Newfoundland, and Acadia, a French colony on the Atlantic coast of what is now Canada. A New England force had taken Acadia's capital, Port-Royal, in 1710.

Louisbourg, taken in 1745 and given back in 1748

New England soldiers under Colonel William Pepperrell sailed from Boston on March 24, 1745, to take Louisbourg, a French fortress on Cape Breton Island. With British warships offshore, they laid siege to it, surrounding it and cutting it off, and the French surrendered on June 17, 1745, after 49 days.

The cost. Far more of the New Englanders died holding the fortress than taking it. In June 1746 Pepperrell estimated that about 1,200 of his men had died, most of them of disease in the fortress that winter. Chapter 22, War and the Military, tells the siege and that winter in full.

Given back. British diplomats gave Louisbourg back to France in 1748, in the Treaty of Aix-la-Chapelle. In exchange French diplomats gave back Madras, a city in India that French forces had taken from Britain. British officials did not ask the colonists, and New Englanders were furious.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 16 MINUTES

In the middle of their war for independence, American leaders looked in secret for anyone abroad who would help them fight Britain. They started in November 1775. French and American diplomats signed an alliance in 1778, and in 1783 American diplomats in Paris signed the peace that ended the war.

Weak after the war. The new country had almost no navy and little money. Rulers in North Africa seized American ships, and their officials enslaved the crews. When the French and the British went to war with each other in the 1790s, American leaders tried to keep out of it. President Washington accepted a treaty that favored Britain because he judged the country too weak to fight the British. From 1798 to 1800 American sailors fought an undeclared war at sea against France.

A committee for secret letters, 1775 to 1789

Members of the Second Continental Congress, the meeting of delegates each colony chose to speak for it, set up the Committee of Secret Correspondence on November 29, 1775. The committee's job was to make secret contact with people in other countries who might help. Its first five members included Benjamin Franklin and John Jay.

The delegates renamed it the Committee for Foreign Affairs on April 17, 1777. Later they set up a Department of Foreign Affairs. Accounts disagree on whether that happened on January 10, 1780, or January 10, 1781. On August 10, 1781, they chose Robert R. Livingston of New York as the first Secretary for Foreign Affairs.

Under the new Constitution, members of Congress set up the department again in a law of July 27, 1789. In a second law, on September 15, 1789, they renamed it the Department of State, because they had given it record-keeping work at home on top of foreign affairs. It still has that name.

Which country recognized the United States first?

Sultan Mohammed III, the ruler of Morocco, was the first ruler of any country to recognize the United States, or accept officially that it existed, in Encyclopaedia Britannica's account. On December 20, 1777, he declared that ships flying the American flag could enter Moroccan ports freely. That was about six weeks before French officials signed their treaties with the Americans.

His declaration named ships of several other nations too. It was part of his plan to trade in peace with Christian countries, and he did not aim it at the Americans alone.

A treaty with Morocco followed nine years later. The American agent Thomas Barclay signed it at Marrakesh on June 28, 1786. Thomas Jefferson and John Adams added their signatures in Paris and London, and delegates to Congress ratified it, giving it final approval, in July 1787.

It was the first treaty American officials made with an African country or a Muslim one. Officials of the two countries rewrote parts of it in 1836 (some accounts say 1837), and it is still in force. No other treaty relationship in United States history has lasted as long without a break.

What it took to win over France, 1778

France's foreign minister, the Comte de Vergennes, a French nobleman, agreed to an alliance only after the Americans won a major battle. Benjamin Franklin sailed to France in late 1776. His ship reached Auray, in Brittany, on December 3 or 4, 1776, and he was in Paris by December 21. He, Silas Deane and Arthur Lee were the American commissioners, the men sent to bargain for the United States.

Saratoga. A British army surrendered to American forces after the Battle of Saratoga, in New York. The State Department's own historians write that Vergennes "finally decided in favor of an alliance" when that news reached him in December 1777.

Two treaties. American and French diplomats signed two treaties on February 6, 1778. In the Treaty of Amity and Commerce, a treaty of friendship and trade, the French formally recognized American independence. In the Treaty of Alliance, the two sides promised that neither would make peace with Britain without the other. A secret section stated that other countries in Europe could join.

Spain's leaders took their country into the war against Britain on June 21, 1779, as France's ally. They did not sign a treaty with the United States, and they did not recognize American independence during the war.

Benjamin Franklin

Who: The American who spoke for his country in France for more than eight years, and one of the three Americans who made the peace with Britain.

When and where: Paris, December 1776 to May 1785.

Movie: *Franklin* (2024), an eight-part drama series on Apple TV+, with Michael Douglas as Franklin. It is about his time in France and is based on Stacy Schiff's book *A Great Improvisation*. It is a drama, not a documentary.

In Paris, Franklin wore a fur cap instead of a wig.

His biographer Walter Isaacson identifies the fur as marten. Franklin wore plain clothes too, and the French took the cap as a sign that he was a plain, rugged American. Augustin de Saint-Aubin made an engraving of Franklin in the cap in 1777, and the print is in the Philadelphia Museum of Art.

Franklin represented the United States in France from December 1776 to May 1785, first as one of the commissioners and then as minister. Delegates to Congress named him Minister Plenipotentiary to France in 1778, a title meaning he had full power to act for his country there. He presented his credentials, the letters that proved his position, to the French government the next spring.

The delegates also named him minister to Sweden, in 1782. Franklin never went to Sweden, but he worked out a treaty with it from Paris and signed it in April 1783. In the same years he was one of the three Americans who made the peace with Britain. His mission in France ended in May 1785.

What British diplomats signed away in 1783

In the Treaty of Paris, British diplomats accepted that the United States was independent. They also handed the United States Britain's claim to land that Native nations lived on.

Benjamin Franklin, John Adams and John Jay bargained for the United States, and they signed the Treaty of Paris on September 3, 1783.

Under the treaty, the British accepted a western border for the United States at the Mississippi River. Americans could fish off Newfoundland, and British lenders could still collect the debts Americans had owed them since before the war. The American side promised Loyalists, colonists who had sided with Britain, payment for property that Americans had taken from them. British troops had to leave the thirteen states. Franklin asked the British to hand over Canada as well, and they refused.

The treaty took effect in 1784, after delegates to the Confederation Congress, meeting at Annapolis, Maryland, and British officials had both approved it.

Not invited. No Native nation was invited to Paris, though the land inside that western border was their home. Among them were the Haudenosaunee in what is now New York and Pennsylvania. The Tuscarora had joined their league in about 1722, and after that it was called the Six Nations: the Mohawk, Oneida, Onondaga, Cayuga, Seneca and Tuscarora.

The British king had set aside much of that land for Native nations in 1763, in a royal order called the Royal Proclamation. In 1783 British diplomats handed it to the United States, and they ignored many earlier British treaties with Native nations.

At Fort Stanwix in 1784, American officials backed by a strong force of soldiers treated the Six Nations as nations they had beaten in war, and they took huge parts of Six Nations land.

John Adams

Who: The first American minister to Great Britain, sent to the king he had rebelled against.

When and where: London. Presented to King George III on June 1, 1785.

In 1785 Adams met the king who had called the colonies rebels.

Adams met the king as the representative of the United States on June 1, 1785. Nearly ten years earlier, in August 1775, the same king had declared the colonies to be in open rebellion against him, and the next year Adams signed the Declaration of Independence.

Two days after the meeting, on June 3, 1785, Adams wrote to Thomas Jefferson that the king had received him with more kindness than he had expected. The Library of Congress holds that letter.

BY 1786: British diplomats had accepted American independence, and John Adams was the first American minister in London. The alliance with France, signed in 1778, was still in force.

Americans enslaved in Algiers, 1785 to 1796

Officials of the ruler of Algiers held American sailors as slaves from 1785 to 1796, and American leaders could not free them. Algiers, Tunis, Tripoli and Morocco, called the Barbary states, were states on the coast of North Africa. Many of their rulers sent armed ships to seize the ships of weaker countries, to make those countries pay them for safe passage, and they held the captured sailors for ransom.

Moroccan ships seized an American merchant ship in 1784, and a treaty with Morocco followed in 1786. In 1785 Muhammad, the Dey, or ruler, of Algiers, declared war on the United States, and his ships captured several American ships.

A Boston ship. An Algerian warship took the Boston ship *Maria* about three miles off Cape St. Vincent, in Portugal, on July 25, 1785. The *Maria* reached Algiers on August 4, and the Dey's officials enslaved its crew. More American crews followed.

American leaders had almost no navy and very little money. They could not free the sailors by force, and for years they could not pay what the Dey asked for them. In 1795 they sent Joel Barlow, Joseph Donaldson and Richard O'Brien to North Africa, and the three men made treaties with Algiers, Tunis and Tripoli. The treaty with Algiers was signed on September 5, 1795, and the American ratification is dated March 7, 1796.

The price. The treaty freed 83 American sailors, by the count of the State Department's historians. Counts of the captives differ with the year, from 83 freed in 1796 to 115 held in 1793.

In all, the treaty cost the United States about $1 million, by the Treasury's estimate. The money for the first payment reached Algiers late, and the Dey demanded more, which roughly doubled the price. Other accounts give a smaller sum paid at once, plus supplies for the Dey's ships every year.

The Dey's officials released the surviving sailors on July 8, 1796. Six captives had died shortly before. Several others carried lasting injuries from their years of captivity.

James Leander Cathcart

Who: An American sailor enslaved in Algiers for eleven years, who rose to be the Dey's chief Christian clerk and put the 1795 treaty into English.

When and where: Born June 1, 1767, in County Westmeath, Ireland. Captured July 25, 1785, off Cape St. Vincent, Portugal. Held in Algiers until May 1796. Died October 6, 1843.

Captured at 18, Cathcart spent eleven years as a slave in Algiers.

He had come to America from Ireland as a boy. During the Revolution the British had already captured him once and held him on prison ships at New York, until he escaped.

The capture. Cathcart was one of six crewmen on the Boston ship *Maria* when an Algerian warship with 14 guns took it on July 25, 1785. He later explained why Algerian warships were out in the Atlantic that summer. Spain's war with Algiers had kept them out until the two made peace in 1785.

Then Charles Logie, the British consul in Algiers, spoke to the Dey's government. A consul is an official who looks after his country's trade and citizens in another country. Logie told them that American ships were no longer under the British king's protection, so the Algerians could take them.

The Algerian crew drove the six Americans into a boat and took their hats, handkerchiefs and shoes. On the voyage the captors locked all 42 of their prisoners every night in the sail room in the ship's hold. The drinking water was so foul, Cathcart wrote, that "we were, literally, obliged to strain through our teeth." The *Maria* reached Algiers on August 4. On August 12 the *Dauphin* of Philadelphia arrived with 15 more American captives, which made 21 in all.

Dey Muhammad chose some of the captives to work in his palace and garden. His officials sent others to the slave prisons and to hard labor at the harbor. A man at hard labor got one suit of clothes, worth

about a dollar and a half, and two rough blankets meant to last his whole captivity.

The beating. About four months after he arrived, officials who ran the Dey's household ordered Cathcart beaten with the bastinado, blows with sticks on the soles of the bare feet. Four men threw him down and tied his arms and legs. In the usual method, Cathcart wrote, the man lay face down with his ankles tied to a pole that two men held up, and a guard sat on his shoulders. Cathcart took 28 blows and lost four toenails.

He wrote that all fourteen slaves in the garden got the same beating, "old men of sixty and children of ten years of age" among them.

The clerk. Cathcart rose step by step. He served coffee to the minister of the harbor, then became clerk of the harbor, then clerk of a slave prison, and at last chief Christian clerk to the Dey. Foreign ambassadors who could not get a meeting with the Dey went through him. He helped Joseph Donaldson, who bargained for the Americans, and he translated the Turkish text of the 1795 treaty into English.

By Cathcart's own list, 9 of the 21 Americans taken in 1785 died in Algiers, most of them of plague, a disease that spread quickly and killed many people. The other 12 went home at different times. Family or friends bought some of them free before 1796, and the rest were freed in 1796. Cathcart left Algiers on May 8, 1796, carrying dispatches, official letters for the American government.

In January 1799 he sailed from Philadelphia as the American consul for Tunis and Tripoli. His daughter, J. B. Newkirk, published his account, *The Captives: Eleven Years a Prisoner in Algiers*, in 1899.

Staying out of Europe's war, 1793 to 1796

President George Washington decided in 1793 to keep the United States out of the new war between France and Britain.

The French minister. France's new minister, Edmond-Charles Genêt, landed at Charleston, South Carolina, on April 8, 1793. Two weeks later, on April 22, Washington issued the Proclamation of Neutrality, which stated that the United States would take neither side.

Genêt hired American privateers, privately owned armed ships with a government's permission to attack enemy ships, to go after British ships. When Washington would not do as Genêt wished, Genêt threatened to go over the President's head to the American people. Washington demanded that the French government replace him.

The treaty with Britain. John Jay signed a treaty with Britain on November 19, 1794, and senators approved it on June 24, 1795, by 20 votes to 10. Under it, British troops were to leave the forts in the Northwest that British diplomats had promised to give up in 1783. British officials agreed to give American traders terms as good as those of any other country, and they put heavy limits on trade with the British islands in the Caribbean.

The two sides sent several quarrels to arbitration, a way of settling a dispute by letting a neutral panel decide: the border between Maine and Canada, the old debts, and British seizures of American ships. British officials kept the right to seize American cargo bound for France, with payment, and to take French goods off American ships without paying.

The State Department's historians sum it up this way: the treaty "addressed few U.S. interests, and ultimately granted Britain additional rights." It was very unpopular with Americans. Washington signed it anyway, because he judged that the country was too weak to fight Britain.

The treaty with Spain. Thomas Pinckney signed a treaty with Spain on October 27, 1795, after bargaining with a Spanish minister, Manuel de Godoy. Senators approved it on March 7, 1796. Under it, the border with Spanish West

Florida was set at the 31st parallel, an east-west line on the map, and Americans got free use of the Mississippi River and of the port of New Orleans without paying tax.

That mattered to farmers in Kentucky and Tennessee. They could not haul their crops over the mountains to sell them, so they floated the crops down the rivers to New Orleans, which Spanish officials controlled. Under Pinckney's Treaty they could sell them there.

The Farewell Address. Washington's Farewell Address was printed in the *American Daily Advertiser*, a Philadelphia newspaper, on September 19, 1796. Alexander Hamilton had helped him write it. In it Washington said he would not run for a third term, and he argued that American leaders should not make permanent alliances with any foreign country.

The XYZ Affair and a war at sea, 1798 to 1800

American navy crews fought French ships in the Caribbean from 1798 to 1800, in a war that neither government declared. It followed a failed attempt at talks. President John Adams had sent three commissioners to France: Elbridge Gerry, Charles Cotesworth Pinckney and John Marshall.

French agents told the three Americans that talks could not begin until the United States met their demands. The agents asked for a low-interest loan to France, and they wanted the United States government itself to pay what American merchants said France owed them. On top of that, they wanted "a substantial bribe" for the French foreign minister, Talleyrand.

W, X, Y and Z. Adams made the Americans' reports public, with the four agents' names replaced by the letters W, X, Y and Z. The agents were Nicholas Hubbard, Jean Hottinguer, Pierre Bellamy and Lucien Hauteval.

The fighting at sea that followed is called the Quasi-War, from a word meaning "almost." The State Department's historians call it "a limited, undeclared war." American and French diplomats ended it with an agreement called the Convention of 1800, also called the Treaty of Mortefontaine, and in the same agreement they ended the alliance of 1778 between France and the United States. Senators did not ratify it until December 18, 1801.

BY 1800

If you remember one thing: For its first twenty-five years the United States was too weak to free its sailors by force or to fight Britain, so its leaders made treaties and paid ransom instead.

- In 1775: The colonies were at war with Britain, and their leaders were looking in secret for help abroad.
- In 1800: American diplomats had signed treaties with France, Britain, Spain, Morocco and Algiers, and American navy crews had just fought French ships at sea.

Next: In the next fifty years American leaders doubled the size of the country with one purchase and pushed its border to the Pacific.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 10 MINUTES

With one purchase in 1803, American diplomats roughly doubled the size of the United States, and by 1848 the country's border reached the Pacific Ocean. American leaders bought land from France, took Florida from Spain, split the Oregon country with Britain and took the Southwest from Mexico in a war. Native nations lived across all that land, and none of them signed the sales and treaties.

American sailors fought two wars against rulers on the coast of North Africa in the same years. In 1823 President James Monroe told Europe's rulers to start no new colonies in the Americas.

Tribute and the Barbary wars, 1801 to 1815

American leaders had been paying tribute, protection money, to the rulers of the Barbary states, the four states on the coast of North Africa: Algiers, Tunis, Tripoli and Morocco. American officials paid the rulers so that their ships would not seize American ships and hold the crews for ransom.

When American officials would not raise the payments, the pasha, or ruler, of Tripoli declared war on the United States on May 14, 1801. The fighting ended with a treaty signed on June 10, 1805. Under it, American officials paid no more yearly tribute to Tripoli, but they did pay a ransom to free the crew of the *Philadelphia*, an American warship that Tripoli's forces had captured. A second Barbary war followed in 1815.

The Louisiana Purchase, 1803

American envoys, men sent to bargain for their government, signed a treaty on April 30, 1803, to buy about 828,000 square miles of land west of the Mississippi from France. The price was $15 million, and the purchase roughly doubled the size of the United States.

The envoys had been sent to buy only the port of New Orleans, and French officials offered them all of Louisiana instead.

Why Napoleon sold. The people enslaved in the French colony of Saint-Domingue, now called Haiti, had risen against French rule. Napoleon, the ruler of France, had sent an army to put them down. The attempt failed, and he lost the army he sent. With a war against Britain coming, Napoleon sold Louisiana to raise money and to keep the British from taking it.

A written promise. Article III of the treaty states that the people living in the territory would be "incorporated in the Union of the United States," made part of the country. It also states that they would be admitted to "all the rights, advantages and immunities of citizens of the United States" (immunities are the protections the law gives every citizen).

So in 1803, the land the United States bought came with a written promise that its people would become full citizens. A century later, judges on the Supreme Court ruled that no such promise covered Puerto Rico, Guam or the Philippines.

Native nations lived across those 828,000 square miles, but they were not part of the sale and no one asked them. The promise of citizenship in Article III did not include them.

How Florida changed hands, 1818 and 1819

General Andrew Jackson led American troops into Spanish Florida in 1818. They attacked the Seminole people and people who had escaped slavery, whom the American officials who approved the raid saw as a threat to Georgia. Jackson's troops seized the Spanish forts at Pensacola and St. Marks, and Jackson had two British citizens put to death, on charges of stirring up the Seminoles and the escaped slaves.

President James Monroe and his advisers seriously thought about speaking out against Jackson. John Quincy Adams defended him. As Secretary of State, Adams ran the country's dealings with other countries, and he used Jackson's raid to put a choice to Spain: control the people of East Florida, or hand it over to the United States.

The treaty. Adams and the Spanish minister, Luis de Onís, signed the Adams-Onís Treaty in 1819. Under it, Spain's leaders handed East Florida to the United States and gave up their claim to West Florida. Spanish officials received no payment. Instead, American officials agreed to pay American citizens, up to $5 million in all, the money those citizens said Spain owed them, so Spain would not have to pay it.

The treaty also fixed the western edge of the Louisiana Purchase. Spain's leaders gave up their claims to the Pacific Northwest, and American officials accepted Spanish rule over Texas. Leaders in both countries ratified the treaty in 1821.

John Quincy Adams

Who: The Secretary of State who got Florida from Spain and wrote the warning now called the Monroe Doctrine.

When and where: Washington, D.C., 1817 to 1825.

Adams talked President Monroe out of sharing his warning with Britain.

Monroe had thought of making his 1823 warning to Europe's rulers jointly with Britain. Adams talked him out of it, so Monroe made it alone. The State Department's historians credit Adams with writing the statement.

Adams was Secretary of State from 1817 to 1825, under President James Monroe. He had already used Jackson's raid to press Spain to give up Florida, and he negotiated the 1819 treaty with Spain's minister, Luis de Onís, which carries both their names.

The Monroe Doctrine, 1823

President James Monroe told Europe's rulers to start no new colonies in North or South America in his yearly message to Congress, on December 2, 1823. In his words, the American continents "are henceforth," meaning from now on, "not to be considered as subjects for future colonization by any European powers."

Who enforced it. The United States Navy was too small to enforce the warning in 1823. British leaders wanted other European powers kept out of Latin America too. Spanish officials had blocked British merchants from trading there under Spain's colonial trade rules, and British leaders wanted the new Latin American countries open to British trade. British warships kept those powers' fleets away.

Texas, Oregon and the land taken from Mexico, 1845 to 1848

American leaders took Texas and the Southwest in the 1840s, and they split the Oregon country with Britain.

Texas. Texans had won a war of independence from Mexico in 1836 and set up their own republic. President John Tyler made a treaty with the Republic of Texas in April 1844 to bring it into the United States, but senators voted the treaty down that June. A treaty needs the votes of two thirds of the senators.

So members of the House and the Senate passed a joint resolution instead, on March 1, 1845, to annex Texas, which means to take it and make it part of the country. A joint resolution needs only more than half of the votes in each house. Texas joined the United States on December 29. In 1898 members of Congress took Hawaii the same way.

Oregon. The Oregon country stretched from the Pacific coast to the Rocky Mountains, covering what is now Oregon, Washington and most of the Canadian province of British Columbia. American and British officials set the border there at the 49th parallel, an east-west line on the map, on June 15, 1846. The United States got the part south of that line, and the two sides settled it by treaty, without a war.

Mexico. The land in the Southwest came from a war with Mexico, fought from 1846 to 1848. American and Mexican officials signed the Treaty of Guadalupe Hidalgo on February 2, 1848, at Guadalupe Hidalgo, a city north of Mexico City where Mexico's leaders had fled. Mexican officials, whose army had lost the war, signed over more than 525,000 square miles, and American officials agreed to pay $15,000,000. The land became all or part of California, Nevada, Utah, Arizona, New Mexico, Colorado and Wyoming.

Between about 80,000 and 100,000 Mexican citizens lived on that land, depending on the count, and without moving they found themselves living in the United States. Senators changed the treaty's promises to them before they approved it on March 10, 1848.

Rights. The treaty promised that Mexicans who became American citizens would keep "the free enjoyment of their liberty and property" and the free practice of their religion. Senators rewrote that part of the treaty. In the version that took effect, members of Congress would decide the "proper time" for these new Americans to get all the rights of citizens.

Land titles. Senators cut out completely the part of the treaty that kept Spanish and Mexican land grants valid under American law. Land grants were legal titles to land given by Spanish or Mexican officials. American officials then wrote a side agreement, the Protocol of Querétaro, telling Mexican officials that the cut had not canceled those titles. American judges did not treat the side agreement as equal to the treaty.

Under a new law of 1851, Mexican landowners in California had to prove their deeds, the papers showing they owned their land, before officials of the United States government. The cases took an average of 17 years. Many families sold most of their land during those years to pay English-speaking lawyers and land taxes. The Peralta family of San José and the East Bay kept only 700 of their 49,000 acres.

English-speaking American squatters, newcomers who settled on land that was not theirs, took more. In some cases, as with the Suñol and Berryessa families, squatters lynched the Mexican landowners, killing them without any trial, and

stayed on the land. Many Mexican American families lost their land or fell into debt. The historians William Carrigan and Clive Webb counted at least 597 Mexicans whom mobs in the United States lynched between 1848 and 1928.

The miners' tax. California's new lawmakers passed the Foreign Miners' Tax in 1850, $20 a month for each miner born outside the country. They aimed it at Chinese and Latino miners, and officials collected it mostly from them. Thousands of foreign-born miners left the mining towns, and lawmakers ended the tax the next year.

Village land. In northern New Mexico and southern Colorado, some land grants belonged to whole villages together. The families of a village shared its common land, including forest on the nearby mountains. In the late 1800s American officials and judges broke up these shared grants, and the villages lost most of their common land.

A special Court of Private Land Claims judged these grants from 1891. Its judges, and United States land officials before them, often recognized a grant only up to the foot of the mountains, leaving out the forests.

The judges of the Supreme Court decided a case called *United States v. Sandoval* in 1897. They agreed with government lawyers that a village's common lands belonged to the United States government, and seven grants lost between 95 and 99 percent of their land under that ruling. Other villages lost their common land in lawsuits that split it into private pieces, or because they could not pay the taxes on it.

Another part of the treaty describes the Native nations who lived on the land as "savage tribes," an insulting name. It states that "a great part" of the land was "now occupied" by them, and that from then on they would "be under the exclusive control of the Government of the United States." The article also states that the United States government would "forcibly restrain" their raids into Mexico. Among the nations on that land were the Tohono O'odham. No Native nation signed the treaty.

Nicholas Trist

Who: The American diplomat who made peace with Mexico after President James K. Polk ordered him home.

When and where: Mexico, 1847 and 1848.

Polk fired the diplomat who made his peace with Mexico.

President James K. Polk sent Trist to Mexico to bargain for an end to the war, then recalled him in October 1847. Trist ignored the order and kept bargaining with no authority to do so. In February 1848 he signed the Treaty of Guadalupe Hidalgo.

Polk sent the treaty to the Senate anyway, because under it he got what he wanted, including California. Then he fired Trist, and he refused to pay Trist's expenses or his salary for his last months of work.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 14 MINUTES

Americans claimed their first land overseas in 1856, small islands covered in seabird droppings, which make crops grow. American officials bought Alaska from Russia in 1867, and in 1898 they took Hawaii and, after a war with Spain, Puerto Rico, Guam and the Philippines. No one asked the people who lived in these places. They became territories, lands that belonged to the United States but were not states.

How it grew. In the 1790s the United States could not free its sailors from Algiers. By 1900 it reached from the Atlantic to the Pacific, its railroad track had grown from about 9,000 miles in 1850 to 193,346 miles, and its mills made 11.4 million tons of steel a year, more than Britain and Germany together.

Hawaii. A group of mostly American businessmen overthrew Hawaii's queen in 1893 with the help of American Marines. More than half of the Native Hawaiian people signed a petition against annexation, and members of

Congress annexed the islands anyway.

Perry's warships in Japan, 1853 and 1854

Commodore Matthew Perry opened two Japanese ports to American ships by bringing warships to Japan. Perry, the navy officer in command of the group, sailed into Edo Bay, beside the city of Edo, now called Tokyo, in July 1853, and he came back in 1854. Japanese and American officials signed the Convention of Kanagawa, a written agreement that opened the two ports, on March 31, 1854. The Japanese officials signed it under the threat of Perry's warships.

Islands claimed for bird droppings, 1856

Americans first claimed land overseas for the United States in 1856, forty-two years before the war with Spain. They wanted guano, the droppings of seabirds, because it makes crops grow.

Members of Congress passed the Guano Islands Act on August 18, 1856. Under it, an American citizen who found an island with guano on it could take the island for the United States, unless another country already claimed it. Americans claimed about a hundred islands under this law.

The judges of the Supreme Court ruled that the law was valid in 1890, in a case called *Jones v. United States*. Lawmakers have never ended it, and it is still part of United States law.

Who was left out of the Alaska sale?

American officials bought Alaska from Russia in 1867, and the diplomats who wrote the treaty left Alaska's Native peoples out of American citizenship by name. American and Russian diplomats signed the treaty on March 30, 1867, for a price of $7,200,000. Russian officials formally handed over the territory at a ceremony on Castle Hill in Sitka on October 18, 1867, and then the United States flag went up.

Tlingit land. Russian officials sold land that they did not really control. Alaska Native nations lived on it, and they had no part in the sale. Castle Hill itself was Tlingit land. Alexander Baranov, the first Russian governor, had attacked the hill in 1804 to take it from the Kiksadi clan of the Tlingit people, and after a six-day battle the Russians seized the Kiksadi fort and burned it. In 1867 a Tlingit village still stood beside the Russian town of Sitka.

Article III of the treaty states who would become an American citizen. People living in Alaska who stayed would be admitted to "the enjoyment of all the rights, advantages, and immunities of citizens of the United States." The article then leaves out "uncivilized native tribes," an insulting name for Alaska's Native peoples. They would instead be "subject to such laws and regulations as the United States may from time to time adopt."

Alaska had no civil government, one run by elected or appointed officials outside the armed forces, until 1884. For 17 years American officers ruled it, first from the Army, then from the Treasury Department, then from the Navy. The historian Rebecca Poulson, writing for the Alaska Historical Society, says that American officials considered Native people "savages," an insulting word, and did not allow them to be citizens.

Kake. Under Russian rule, officials of the Russian company that ran Alaska had paid compensation, a payment to the dead person's family, when a Tlingit person was killed. Under Tlingit law, a death was settled with a payment, not with punishment. American officers refused to pay. In early 1869 an army guard at Sitka killed two men from the village of Kake, deaths Poulson calls accidental. Army officers at Sitka refused to pay compensation for them, and relatives of the dead men then killed two white men.

General Jefferson C. Davis ordered the American warship USS *Saginaw* to attack Kake. It was winter. The crew fired exploding shells from the ship's guns at the Kake villages and burned 28 of the villages' 29 clan houses, the houses of the Kake clans, which are large family groups. They also burned the villagers' canoes, totem poles and stored food.

NOT KNOWN: No one has worked out how many people died after the attack, reporters at KTOO, a public radio station in Juneau, wrote in 2024. Kake oral history, the accounts the Tlingit of Kake passed down by word of mouth, records many deaths, especially among elders and children.

Hawaii: a kingdom overthrown, 1887 to 1893

A group of mostly American businessmen overthrew Queen Liliʻuokalani of Hawaii in January 1893, and American Marines and sailors helped them. Two of their leaders had already used armed men to force a Hawaiian king to do what they wanted.

The Bayonet Constitution. A group of lawyers and sugar planters who were haole, the Hawaiian word for white people who are not Native Hawaiian, forced King Kalākaua to sign a new constitution on June 30, 1887. An armed militia, a force of armed citizens, backed them, and their leaders were two lawyers, Sanford B. Dole and Lorrin Thurston. The document became known as the Bayonet Constitution, after the blade fixed to the end of a rifle.

The overthrow. Early in 1893 members of the Annexation Club, Thurston among them, contacted John L. Stevens, the United States Minister to Hawaii and the top American diplomat there. They wanted to know whether the Americans would back them. The group, calling itself the Committee of Safety, began to move against the queen on January 15, 1893.

Stevens had 162 American sailors and Marines landed from the warship USS *Boston* on January 16. The queen's government had several hundred soldiers and police, and they did not fight. Dole became president of the new provisional government, a government set up for a short time until a lasting one is formed.

Queen Liliʻuokalani gave up her throne under protest. She addressed her protest to the United States, not to the men who had taken power. She wrote that she yielded "to the superior force of the United States of America," whose minister, John L. Stevens, "has caused United States troops to be landed at Honolulu."

Two American reports that disagree, 1893 and 1894

Two American investigations of the overthrow reached opposite findings. President Grover Cleveland sent James H. Blount to Hawaii to find out what had happened, and Blount reported that Stevens had landed the Marines on a false excuse to help the plotters.

What Cleveland said. Cleveland sent Congress a message about it on December 18, 1893. He said that the show of armed force, the "military demonstration on the soil of Honolulu," was "of itself an act of war." He also said that "the provisional government owes its existence to an armed invasion by the United States."

Albert S. Willis, the new American minister in Hawaii, apologized for what Stevens had done, which he called "the reprehensible conduct of the American minister," reprehensible meaning deserving blame. He also apologized for "the unauthorized presence on land of a military force."

What the Senate said. Cleveland could not get the queen put back on her throne. Members of a Senate committee made their own investigation in 1894, called the Morgan Report, and they cleared everyone involved. On July 4, 1894, the men who had taken power declared the Republic of Hawaii.

Queen Lili‘uokalani

Who: The last queen of Hawaii, overthrown with American help and then tried in her own palace. She wrote the song "Aloha ‘Oe."

When and where: Honolulu. Overthrown January 17, 1893. Held prisoner in ‘Iolani Palace in 1895.

The queen was put on trial in her own throne room.

The rising of 1895. Two years after the overthrow, Hawaiians loyal to the queen, among them Robert Wilcox and Prince Kūhiō, tried to restore her by force in January 1895. Within 36 hours officials of the Republic of Hawaii had caught the rebels and jailed them. Then they arrested the queen, and on January 24, 1895, they forced her to sign a paper giving up her throne for good.

The trial. They tried her before a military court in the throne room of ‘Iolani Palace, where she had ruled. The charge was misprision of treason, knowing about a plot against the government and not reporting it. The officers who sat on the court found her guilty, fined her $5,000, and sentenced her to five years in prison at hard labor, forced heavy work.

The Republic's officials later reduced the sentence. For nearly eight months they held her prisoner in an upstairs bedroom of the palace, and they allowed her no visitors except one woman companion. She spent her days praying, reading, writing music, crocheting and quilting.

21,269 names against annexation, 1897

Native Hawaiians defeated a treaty to annex their islands by signing their names against it. Two Hawaiian groups held meetings on the five main islands between September 11 and October 2, 1897: Hui Aloha ʻĀina, which had separate branches for men and for women, and Hui Kālaiʻāina. They gathered 21,269 signatures of Native Hawaiians against annexation. That year about 39,000 Native Hawaiian and part-Hawaiian people were counted, so more than half of them signed.

Four men carried the petition to Washington: James Kaulia, David Kalauokalani, John Richardson and William Auld. They arrived on December 6 and presented it to the Senate on December 9, 1897. A treaty needs the votes of two-thirds of the senators. When the four men left Washington on February 27, 1898, only 46 senators were still willing to vote for the annexation treaty, short of the two-thirds it needed. The treaty failed.

Taken anyway. Supporters of annexation in Congress then used a method that did not need the Senate votes the Hawaiians had blocked: a joint resolution, which passes when more than half of the House and more than half of the Senate vote for it. Members of the House voted for theirs, the Newlands Resolution, on June 15, 1898, and senators on July 6. President William McKinley signed it on July 7, 1898, and under it American officials took over Hawaii.

1898: Puerto Rico, Guam and the Philippines

After a war with Spain in 1898, American officials ruled Puerto Rico, Guam and the Philippines. The American battleship USS *Maine* blew up in the harbor of Havana, in Cuba, on February 15, 1898, and in April members of Congress declared war on Spain.

Guam and Puerto Rico. Juan Marina, the Spanish governor of Guam, surrendered the island to officers of the American warship USS *Charleston* on June 21, 1898. American troops landed in Puerto Rico at Guánica on July 25.

Spanish and American diplomats signed the Treaty of Paris on December 10, 1898. Spanish diplomats gave up Cuba and signed Puerto Rico and Guam over to the United States. They sold the Philippines to the United States for $20 million.

No one from Puerto Rico, Guam or the Philippines took part in the talks in Paris. Several million people in those places got new rulers without being asked. Among them on Guam were the CHamoru, the island's Native people. In the Philippines, Filipinos had already declared their own independence.

Why Americans and Filipinos went to war, 1899

War between American troops and the Philippine Republic began on February 4, 1899, because American officials had decided to keep the islands. Filipinos had been fighting Spain before the Americans arrived. On June 12, 1898, they declared their independence. American officials kept the islands instead of recognizing the new republic, and the two sides went to war.

The dead. The war lasted until 1902. About 4,200 Americans and about 20,000 Filipino soldiers were killed.

> **DISPUTED:** Counts of the Filipino civilians, people who were not soldiers, who died run from about 200,000 to as many as 1,000,000. The State Department's historians give the lowest count: as many as 200,000 dead "from violence, famine, and disease." Other estimates run much higher, and historians disagree about them.

Much of the dying came from hunger and from a cholera epidemic in 1902, an outbreak of a gut disease spread by dirty water. The epidemic was worse because the fighting had broken up daily life in the islands, and it killed more than 200,000 people in all.

What American troops did. The State Department's historians also record what American troops did. In their words, "U.S. forces at times burned villages, implemented civilian reconcentration policies, and employed torture on suspected guerrillas." Reconcentration meant forcing civilians out of their homes into camps or zones that soldiers guarded, and the guerrillas were fighters in small bands who hid among the people and attacked by surprise.

One kind of torture American soldiers used was the water cure, in which they forced water into a prisoner until his stomach swelled and then pressed it out of him. Chapter 22, War and the Military, tells where they used it and on whom.

Emilio Aguinaldo

Who: The Filipino leader who declared the Philippines independent from Spain and then led the war against the United States.

When and where: Born March 22 or 23, 1869, near Cavite, on the island of Luzon. President of the Philippine Republic from January 1899. Captured at Palanan on March 23, 1901. Died February 6, 1964.

American officers captured him by pretending to be prisoners.

Aguinaldo was the mayor of his town, Cavite Viejo, in 1896, and a local leader of the Katipunan, a secret society fighting Spanish rule.

Two deals. Aguinaldo made a deal with the Spanish governor in December 1897: he would leave the islands in return for money and a promise of reforms. From Hong Kong and Singapore he then made plans with American consuls, officials who looked after American trade and citizens there, and with officers of Commodore George Dewey's fleet. He would come back and help the Americans fight Spain. He landed in the Philippines again on May 19, 1898.

Filipinos declared independence on June 12, 1898, with Aguinaldo at the head of their government. Six months later Spanish diplomats sold the islands to the United States in the Treaty of Paris. Filipinos proclaimed a constitution on January 23, 1899, and elected Aguinaldo president of the Philippine Republic. The fighting with American troops began on February 4, and from November 1899 his forces fought as guerrillas.

The trick. General Frederick Funston captured him at Palanan, in northern Luzon, on March 23, 1901. Macabebe scouts, Filipinos who served the United States Army, dressed as rebel soldiers, and Funston and a few other American officers went along, pretending to be their prisoners. Fake letters sent ahead to Aguinaldo said that more of his own soldiers were coming and were bringing American prisoners to him. At Palanan the scouts opened fire on his guards, killed two and seized him.

> Aguinaldo swore loyalty to the United States. On April 19, 1901, he called on his fighters to lay down their weapons and accept American rule. He lived until 1964 and saw the Philippines become independent in 1946.

Samoa divided, 1899

Diplomats divided the Samoan islands between the United States and Germany on December 2, 1899, in an agreement called the Tripartite Convention, from a word meaning three-sided. The three sides that signed it were the United States, Germany and Great Britain.

By signing it, the German and British ambassadors gave up their countries' claims to Tutuila and every other Samoan island east of a north-south map line, 171 degrees west. John Hay, the American secretary of state, gave up American claims to Upolu, Savaiʻi and the other islands west of that line, leaving them to Germany.

The chiefs' papers. The chiefs of the island of Tutuila signed a paper handing their island to the United States on April 17, 1900. Tui Manuʻa Elisala, the ruler of the Manuʻa islands, signed his islands over on July 16, 1904.

BY 1900

If you remember one thing: In the second half of the 1800s American leaders took lands across two oceans, and most of the people who lived in them were never asked.

One question this era leaves open: Were the people of Puerto Rico, Guam and the Philippines now American citizens, with all the rights in the Constitution? The next era answers it.

Next: After 1900 American officials ruled the new islands, and the judges of the Supreme Court decided how much of the Constitution reached the people there.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 17 MINUTES

Puerto Rico "belongs to, but is not a part of, the United States," the judges of the Supreme Court ruled in 1901. In a set of cases between 1901 and 1922 they held the same for the other islands taken in 1898, and the people living there did not get the whole Constitution. American troops finished a war in the Philippines and occupied Haiti, the Dominican Republic and Nicaragua.

Workers, most of them from islands in the Caribbean, built the Panama Canal for the United States, on a strip of land that American officials held forever under a treaty. Americans fought in two world wars. After the first, senators refused to join the League of Nations that President Woodrow Wilson had proposed, and after the second, American leaders helped found the United Nations and the NATO military alliance.

Finishing the war in the Philippines, 1900 to 1902

American soldiers in the Philippines tortured Filipino prisoners and killed civilians. Chapter 22, War and the Military, tells what they did at Igbaras and on Samar in full. Here is what happened in Washington, and to the officers who gave the orders.

Igbaras and Samar. At Igbaras, on November 27, 1900, Captain Edwin F. Glenn, an Army lawyer, and a doctor working for the Army directed the water cure of the town's mayor, Tobeniano Ealdama, until Ealdama said he was a captain among the Filipino fighters. That night Glenn had the town burned. On Samar, townspeople and Filipino fighters at Balangiga killed about 48 American soldiers on September 28, 1901, the Army's worst loss in a single fight since the Little Bighorn. General Jacob H. Smith then ordered, "I want no prisoners. I wish you to kill and burn."

DISPUTED: Counts of the people American troops killed on Samar are far apart. Most estimates fall between 2,000 and 2,500, and a British researcher, Bob Couttie, who studied it for ten years, put it at about 2,500. An American soldier who was there said 39, and other counts reach 5,000. An old figure of 50,000 is now traced to misread documents.

What happened to the officers

The torture came out in the Senate. Senator George Frisbie Hoar asked for an investigation, and Senator Henry Cabot Lodge kept it inside his own Committee on the Philippines. On April 14, 1902, Charles S. Riley, a former sergeant, told the committee what he had seen at Igbaras.

Glenn. Secretary of War Elihu Root ordered Glenn court-martialed, tried in a military court. Army officials moved the trial from San Francisco to Catbalogan, in the Philippines, far from the witnesses. The officers on the court found Glenn guilty, suspended him for one month and fined him fifty dollars. Ealdama, the mayor he had tortured, was then serving a prison sentence as a "war traitor."

Waller and Smith. On Samar, Marine Major Littleton Waller had eleven Filipino guides and carriers shot without a trial. He told his court-martial that he had followed Smith's orders, and the officers on the court found him not guilty.

Officers at another court-martial found Smith guilty of "conduct to the prejudice of good order and military discipline," behaving in a way that harmed order and discipline in the armed forces. They asked only that he be formally scolded. President Theodore Roosevelt went further and forced Smith to retire, two years before he was due to leave the Army. Smith was not sent to prison.

Roosevelt's letter. In a private letter, Roosevelt called the water cure "the old Filipino method of mild torture" and wrote that "nobody was seriously damaged." In the same letter he also wrote, "torture is not a thing that we can tolerate."

Roosevelt declared the war over on July 4, 1902. American troops kept fighting in the Moro provinces of the southern Philippines until 1913.

The Insular Cases, 1901 to 1922

The justices, the judges of the Supreme Court, decided between 1901 and 1922 that the islands taken in 1898 belonged to the United States but were not fully part of it. Their decisions are called the Insular Cases, from a word meaning having to do with islands.

The justices decided *Downes v. Bidwell* on May 27, 1901, holding that Puerto Rico "belongs to, but is not a part of, the United States." From these cases came the idea of the "unincorporated territory," a place the United States owns and rules, where the people do not get the whole Constitution unless Congress chooses to give it to them.

A trial by jury. In *Balzac v. Porto Rico* in 1922, the justices held that Puerto Ricans had no constitutional right to a trial by jury at home. By then Puerto Ricans were American citizens.

The justices who decided these cases believed the people of the islands were not fit for full constitutional government. In 2022 Justice Neil Gorsuch wrote that the Insular Cases "have no foundation in the Constitution and rest instead on racial stereotypes," and he added, "They deserve no place in our law."

The justices of the Supreme Court have never overturned them. In 2026 they are still the law for the people of Puerto Rico, Guam, the U.S. Virgin Islands, American Samoa and the Northern Mariana Islands.

Who became a citizen, and when

Members of Congress decided, island by island, whether the people of each territory would be American citizens. The answers came at different times. People born in American Samoa are still not citizens at birth.

Puerto Rico. Members of Congress set up a government for Puerto Rico under the Foraker Act, signed on April 12, 1900. The law identified the island's people as "citizens of Porto Rico," under the protection of the United States but not citizens of it. President Wilson signed the Jones-Shafroth Act on March 2, 1917, and under it Puerto Ricans became American citizens. Their citizenship came from a law passed by Congress, not from the Constitution, and it did not come with a vote for president.

One month later Americans entered the First World War. Two months after that, Wilson signed the law for the draft, which required young men to serve in the military. Accounts disagree on how many Puerto Ricans were drafted: the figure usually given is about 18,000, and History.com gives 20,000.

Hawaii and the Virgin Islands. Hawaii's territorial government was set up under the Hawaiian Organic Act, signed on April 30, 1900. Under the same law, the people who had been citizens of the Republic of Hawaii in August 1898 became American citizens. Danish officials agreed in a treaty signed on August 4, 1916, to sell the Danish West Indies to the United States for $25 million in gold. The islands changed hands on March 31, 1917, and became the U.S. Virgin Islands, but members of Congress did not make their people American citizens until 1927.

Guam. Officers of the United States Navy governed Guam from 1899. Its people did not become citizens until President Harry Truman signed the Organic Act of Guam on August 1, 1950, fifty-two years after Navy officers took the island. Under the act, Guam got a government run by civilians and stayed an unincorporated territory.

American Samoa. Members of Congress did not approve the Samoan chiefs' deeds of 1900 and 1904 until 1929, and Navy officers governed American Samoa until 1951. Under federal law its people are still American nationals,

people who can live in the United States and carry an American passport but are not citizens. American Samoa is the only territory with people living on it where they do not become citizens at birth.

The Philippines. Under laws members of Congress passed in 1902 and 1916, American officials ran a colonial government in the Philippines, with a promise of independence someday. Under the Tydings-McDuffie Act, signed on March 24, 1934, the Philippines would become independent on July 4, 1946, after ten years of self-rule. In the same act, members of Congress put Filipinos in a different group under immigration law, so far fewer could move to the United States. The Philippines became independent on July 4, 1946.

Filipinos served in the American armed forces in the Second World War, at the call of the United States. Members of Congress then passed the Rescission Act in 1946, which stated that their service "shall not be deemed," or counted, "to have been active military, naval, or air service" for most veterans' benefits. Under it, they could not get benefits that other veterans of the same war received.

The Frenchman who signed for Panama, 1903 to 1914

American officials backed Panama's break from Colombia in 1903. President Theodore Roosevelt sent American warships to Panama, and Panama's leaders declared independence on November 3. They at once named Philippe-Jean Bunau-Varilla, a French engineer who had worked on France's failed attempt to dig a canal there, to bargain for them in Washington.

Bunau-Varilla signed the Hay-Bunau-Varilla Treaty for Panama on November 18, 1903, and no Panamanian signed it. Under the treaty American officials got a strip of land 10 miles wide across Panama, forever. Panama got $10,000,000 at once and $250,000 a year, starting nine years later.

The cost in lives. The canal opened on August 15, 1914. The official count of the dead during the American building of the canal is 5,609, out of about 56,000 workers, about one worker in ten. Most of the workers came from Barbados, Jamaica and other islands of the Caribbean. The historian Caroline Lieffers, writing for The Conversation, says many historians think the real number was several times higher.

Marines in Haiti, the Dominican Republic and Nicaragua

American presidents sent the Marines to occupy three nearby countries, holding them with troops and controlling their governments, for years at a time.

Haiti. American Marines took $500,000 from Haiti's national bank in December 1914 and moved it to New York. Haitian President Jean Vilbrun Guillaume Sam was killed in July 1915, and the State Department's historians do not name who killed him. President Woodrow Wilson then sent the Marines into Haiti, and they stayed for nineteen years, until 1934.

Under a treaty of 1915, American officials took full control of Haiti's money and set up a Haitian police force, the Gendarmerie, commanded by Marines. They forced the election of Philippe Sudré Dartiguenave as Haiti's president in August 1915.

American officials wanted a new Haitian constitution that let foreigners own land in Haiti. When Haiti's lawmakers resisted it, American officials forced President Dartiguenave to send the lawmakers home, and they did not meet again until 1929. The new constitution, written under American direction, went to a public vote on June 12, 1918. The official count recorded it as approved by 99 percent of the votes, with fewer than 5 percent of Haitians voting.

Marines and the Gendarmerie brought back the corvée, a system of forced labor, and made Haitians build roads without pay.

The killing. Haitian fighters called cacos rose against the occupation. The Marines and the Gendarmerie killed more than 2,000 cacos, by estimates in a report by the Congressional Research Service, and American forces lost 10 killed and 26 wounded. The report gives no count of other Haitians killed.

General George Barnett, a former head of the Marine Corps, spoke in October 1920 to a navy court of inquiry, a panel of officers looking into what Marines had done in Haiti. He said the Marines had recorded 2,250 Haitians killed from 1915 to 1920, 1,861 of them in 1919.

Marines fired on about 1,500 protesters at Les Cayes on December 6, 1929, and killed at least 12 Haitians. President Franklin D. Roosevelt pulled the Marines out in 1934.

Charlemagne Péralte. Péralte, born in 1885 or 1886 by different accounts, led the armed resistance from 1917, and Marine officers ordered their men to "get Charlemagne." Sergeant Herman H. Hanneken went into Péralte's camp in disguise and shot him dead near Sainte-Suzanne on November 1, 1919. The Marines photographed the body and passed the picture around.

The Dominican Republic. Marines landed in the Dominican Republic in May 1916, and American officers declared a military government that November. From 1917 to 1921 they fought guerrillas in the east, called gavilleros. The Library of Congress's country study describes the Americans' methods as "often brutal" but gives no count of the Dominicans killed. Dominicans took back control when President Horacio Vásquez took office on July 13, 1924.

Nicaragua. American officials encouraged a revolt against President José Santos Zelaya in 1909, and Marines landed at Bluefields. Marines came again in 1912 to support President Adolfo Díaz, and a guard of 100 Marines stayed at the American embassy until 1925. About 2,000 more Marines arrived in 1927.

Augusto César Sandino fought them for as long as they stayed. The Marines trained a Nicaraguan National Guard and left in January 1933, with Anastasio Somoza García in command of the Guard.

Somoza and other high officers agreed in 1934 to kill Sandino. On February 23, 1934, Sandino came to what looked like a peace meeting, and National Guardsmen seized him and murdered him. Somoza became president on January 1, 1937, and ruled for about twenty years.

NOT KNOWN: No reliable count exists of the Nicaraguans killed in the fighting with the Marines.

Woodrow Wilson

Who: The president who proposed the League of Nations and sent the Marines into Haiti.

When and where: Washington, D.C., 1915 to 1920.

Senators voted twice against the League of Nations Wilson proposed.

Wilson was president when Americans entered the First World War on April 6, 1917. He set out his terms for peace, called the Fourteen Points, on January 8, 1918. One of them was a League of Nations, a group of countries that would settle quarrels by talking instead of fighting. Senators voted against the peace treaty and the League in November 1919 and again in March 1920, and the United States never joined it.

Wilson also sent the Marines into Haiti in 1915, and they stayed until 1934. In 1917 he signed both the law that made Puerto Ricans American citizens and the draft law that followed it.

BY 1935: The Marines had left Nicaragua and Haiti, and members of Congress had promised the Philippines independence on July 4, 1946. Puerto Ricans had been American citizens since 1917, but they could not vote for president.

Puerto Rico under an appointed governor, 1937

Police in Puerto Rico shot marchers at Ponce in 1937, under Governor Blanton Winship, and the same year officials began using public money to sterilize Puerto Rican women, which means operating on them so they could never become pregnant again. Winship was an American general whom the President of the United States appointed to govern the island. Puerto Ricans did not elect him.

Ponce. Members of the Nationalist Party marched in Ponce on Palm Sunday, March 21, 1937. They marked the end of slavery in Puerto Rico and protested the jailing of their leader, Pedro Albizu Campos. The Insular Police, who were under Winship's direct military command, opened fire on the marchers.

Accounts give the dead as between 19 and 21 and the wounded as between about 150 and 235. A human rights commission that looked into it found that police had shot all the dead. The killings are known as the Ponce Massacre.

Law 116. Law 116 took effect the same year, under Winship. Under it, officials set up a Eugenics Board, and public money paid for sterilizations. The board took its name from eugenics, a set of beliefs, now rejected, that some groups of people should be kept from having children.

In Puerto Rico the usual method of sterilization was tubal ligation, which Puerto Ricans called *la operación*. In it a doctor cut or tied the two tubes that carry eggs from the ovaries to the womb. Some women instead had a hysterectomy, in which a surgeon removed the womb.

Forced and pressured. Members of the Eugenics Board forced 97 sterilizations from 1937 until the law was repealed in 1960. Many thousands more women had the operation under pressure. Some women were told, falsely, that the operation could be undone. Others were told that they would not be hired unless they were sterilized, and employers did favor hiring sterilized women. The accounts do not name who told them these things. Hospital staff turned away some pregnant women unless they agreed to be sterilized after giving birth.

Clarence Gamble, an heir to the Procter and Gamble fortune who believed in eugenics, set up clinics on the island. By 1968 about one Puerto Rican woman in three of childbearing age had been sterilized, and a 1982 study found 41 percent of married women in Puerto Rico sterilized.

The record on consent is mixed. Some women chose the operation because it was the only birth control offered to them. The doctor Helen Rodríguez-Trías said, "Women make choices based on alternatives, and there haven't been many alternatives in Puerto Rico."

Pedro Albizu Campos

Who: A United States Army officer in the First World War who became the leader of the movement for Puerto Rican independence.
When and where: Born in Ponce, Puerto Rico, on September 12, 1891.

Albizu Campos could be drafted but could not vote for president.

Under the 1917 law he was an American citizen, so officials could draft him into the army, but no one living in Puerto Rico could vote for president. He was also the first Puerto Rican to graduate from Harvard Law School.

In the First World War he served in the United States Army and rose to second lieutenant. Army officers assigned him to the 375th Infantry Regiment, which was all Black, because Army leaders kept Black and white soldiers in separate units, a practice called segregation. The segregation he met there shaped his politics.

He became president of the Puerto Rican Nationalist Party, whose members wanted independence from the United States, in 1930. He was later convicted of sedition, the crime of urging people to rise against the government, and sent to prison. Nationalists marched in Ponce in 1937 partly to protest his imprisonment, and police shot them.

The ship American officials turned away, 1939

American officials refused to let a ship full of Jewish refugees land in 1939. The *St. Louis* sailed from Hamburg, in Germany, on May 13, 1939, with 937 passengers, almost all of them Jews escaping Nazi Germany.

Cuban officials let 28 of them land, and American and Canadian officials refused the rest. The ship carried them back to Europe, where officials in Great Britain let in 288 of them, in the Netherlands 181, in Belgium 214 and in France 224.

Nazis later killed 254 of the passengers in the Holocaust, their mass murder of the Jews of Europe, according to the United States Holocaust Memorial Museum.

Guam under Japanese occupation, 1941 to 1944

The people of Guam lived through two and a half years of Japanese occupation as people of the United States who were not American citizens. Japanese forces invaded Guam on December 8, 1941, hours after the attack on Pearl Harbor. Japanese soldiers beat CHamoru people, forced them to work without pay and put some of them to death.

Japanese officers had put every able-bodied man into a labor battalion, working without pay, by the spring of 1944. In the last weeks, Japanese troops force-marched the people to concentration camps, where soldiers held civilians under guard. The largest camp was at Manenggon. American forces landed on July 21, 1944, and people on Guam mark that day as Liberation Day.

The United Nations, the Marshall Plan and NATO, 1945 to 1949

After the Second World War, American leaders helped build the kind of world organization that senators had refused in 1920. Delegates from many countries signed the United Nations Charter in San Francisco on June 26, 1945, and it took effect on October 24, 1945.

The Marshall Plan and NATO. Members of Congress passed the law for the Marshall Plan in March 1948, under which American officials sent money to rebuild Western Europe after the war. The money eventually rose to more than $12 billion. Diplomats of twelve countries signed the treaty that created NATO, the North Atlantic Treaty Organization, a military alliance, on April 4, 1949. The State Department's historians call it "the first peacetime military alliance the United States entered into outside of the Western Hemisphere."

BY 1950

If you remember one thing: In these fifty years American officials ruled islands whose people did not get the whole Constitution, and American troops held three nearby countries for years at a time.

- 1915: President Wilson sends the Marines into Haiti. They stay nineteen years.
- 1917: Puerto Ricans become American citizens by a law of Congress, without a vote for president.
- 1950: The people of Guam become citizens, fifty-two years after Navy officers took the island.

Next: Two territories became states after 1950, and American presidents sent aid, weapons and troops abroad with one rival in mind, the Soviet Union.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 12 MINUTES

Three elected governments, in Iran, Guatemala and Chile, were overthrown with secret American help in these years. For forty years after 1950, American presidents chose where aid, weapons and troops went with one rival in mind, the Soviet Union. That rivalry is called the Cold War, and American troops fought in Korea and Vietnam as part of it.

The territories. Alaska and Hawaii became states, and Puerto Rico got a new name but stayed a territory. On Pacific islands that American officials governed for the United Nations, with a duty to look after the people, they set off nuclear bombs.

What was the Cold War?

The Cold War was a long rivalry between the United States and its allies on one side and the Soviet Union and its allies on the other. The Soviet Union was a large country ruled from Moscow, in Russia, from 1922 to 1991. It was a communist country, which meant that one party ruled it and the government owned the businesses. From 1947 on, American leaders' plan, which they called containment, was to block the spread of Soviet power wherever they saw a risk of it growing.

The end. The Berlin Wall, which had divided the city of Berlin, was opened on November 9, 1989. The Soviet Union broke apart on December 26, 1991, and the Cold War was over.

The Cuban Missile Crisis, October 1962

American and Soviet leaders came closer to nuclear war in October 1962 than at any other time. Soviet leader Nikita Khrushchev and Cuban leader Fidel Castro had secretly agreed in July 1962 to put Soviet nuclear missiles in Cuba. Before that, a force backed by the United States had tried and failed to overthrow Castro in the Bay of Pigs invasion.

The pictures. An American U-2 spy plane took pictures of missile sites being built in Cuba on October 14, 1962. On October 22 President John F. Kennedy ordered American warships to stop ships carrying weapons to Cuba. He called it a quarantine, and he told the country about it on television. American forces went on alert for war, and on October 27 a U-2 was shot down over Cuba.

A secret promise. Kennedy then offered to promise not to attack Cuba if the missiles were removed. His brother Robert Kennedy secretly told the Soviet ambassador, Anatoly Dobrynin, that American missiles in Turkey would also be taken out, though not as part of any public deal. Khrushchev announced on October 28 that the Soviet missiles would leave Cuba, and the American Jupiter missiles left Turkey in April 1963.

The Peace Corps, 1961

President Kennedy created the Peace Corps, a program for American volunteers to serve in other countries, by an order he signed on March 1, 1961. Sargent Shriver was its first director, and the first volunteers went to Ghana, in Africa, that same year.

Lillian Carter

Who: A nurse from Plains, Georgia, who joined the Peace Corps at 68. Mother of President Jimmy Carter.

When and where: India, 1966 to 1968.

Movie: *Miss Lillian: More Than a President's Mother* (2021), a documentary film about her directed by Vivian Winther.

She joined the Peace Corps at 68 and turned 70 in India.

Lillian Carter, a nurse, worked in India for the Peace Corps from 1966 to 1968. She told the recruiter she wanted to go "where it's warm, people have dark skins and need a nurse's service."

She was sent to the Godrej Colony, about 30 miles from Mumbai. After she came home, her son Jimmy Carter became president. She died in 1983, and in 1986 Peace Corps officials created the Lillian Carter Award for volunteers who serve at the age of 50 or older.

Governments overthrown with American help

American officials secretly helped overthrow elected governments in Iran, Guatemala and Chile, in coups, sudden takeovers of a government by a small group, often the army.

Iran, 1953. Prime Minister Mohammad Mosaddegh had taken control of Iran's oil industry away from British companies. In June 1953 President Dwight Eisenhower approved a secret American and British plan to overthrow him. Officers of the Central Intelligence Agency, or CIA, the American government's spy service, carried it out. The CIA's director, Allen Dulles, oversaw the planning, and Kermit Roosevelt, a CIA officer, ran the plan in Iran. CIA officers paid journalists, religious leaders and politicians to attack Mosaddegh as corrupt and power-hungry.

General Fazlollah Zahedi replaced Mosaddegh on August 19, 1953, and the Shah, Iran's king, took back power. The Shah had Mosaddegh convicted of treason. Mosaddegh served three years in prison and lived the rest of his life under house arrest, and officials of the Shah's government put to death one of his advisers and more than a dozen military officers and student leaders on his side. In 2013 CIA officials released the agency's own history, which states that the agency "planned and helped implement the coup."

Guatemala, 1954. President Jacobo Árbenz began a land reform, a plan to share out farmland, with a law called Decree 900 on June 27, 1952. Under it his officials took land from owners and gave about 1.5 million acres to about 100,000 families, and the owners were paid more than $8 million in all. The United Fruit Company was one of the landowners. Árbenz's officials offered United Fruit $2.99 an acre for its land, and State Department officials demanded $75 an acre.

Eisenhower approved a CIA operation to overthrow Árbenz in August 1953. CIA officers ran what they called "psychological warfare and political action" against Árbenz's government, using fear, rumors and false stories to turn people against their leaders. The officers also drew up lists of Guatemalan officials to be killed, and one list had 58 names.

A force under Carlos Castillo Armas invaded from Honduras on June 18, 1954, after planes had harassed the capital, and on June 27 Árbenz resigned. Castillo Armas became president, and his officials gave the land back to the companies and broke up the labor unions. Human rights groups estimate that between 1954 and 1990, the forces of Guatemala's military governments murdered more than 100,000 civilians.

Chile, 1970 to 1973. Salvador Allende was elected president of Chile in September 1970. On September 15, President Richard Nixon met CIA Director Richard Helms and ordered him to stop Allende. Helms wrote down Nixon's words in his notes, among them "make the economy scream" and "$10,000,000 available, more if necessary."

CIA officers then ran a plan they called Track II, whose stated aim was "a firm and continuing policy that Allende be overthrown by a coup." They gave weapons to men who planned to kidnap the head of Chile's army, General René Schneider. A different group of plotters killed Schneider on October 22, 1970.

General Augusto Pinochet led the army coup that overthrew Allende on September 11, 1973. An American military attaché, an officer posted to the American embassy, called the coup "close to perfect." Pinochet ruled Chile until 1990.

Chile's official commissions count 3,216 people whom Pinochet's forces killed or "disappeared" between 1973 and 1990. The disappeared were people his forces took who were never seen again. The commissions also count 38,254 people who survived political imprisonment, torture or both.

Two new states and a ballot with two answers, 1959

Alaska became the 49th state on January 3, 1959, and Hawaii became the 50th on August 21, 1959. Hawaii's voters could only say Yes or No to becoming a state right away. Their ballot of June 27, 1959, asked: "Shall Hawaii immediately be admitted into the Union as a State?"

Not on the ballot. Independence was not one of the choices. Members of the United Nations General Assembly had passed Resolution 742 in 1953, which stated that votes like this one should offer a free choice among several futures, including independence. A large majority of Hawaii's voters chose statehood. Hawaii and Alaska were then taken off the United Nations list of territories that did not govern themselves.

Puerto Rico: a new name, and armed revolt, 1950 to 1954

Puerto Ricans got the right to write their own constitution in 1950, but the island stayed a territory under the Insular Cases. Members of Congress passed Public Law 600 that year, and Puerto Rico's voters approved the law in 1951 and the constitution in 1952.

A new name. Governor Luis Muñoz Marín proclaimed the Commonwealth, the *Estado Libre Asociado*, on July 25, 1952, the anniversary of the American landing at Guánica in 1898. Under that new name, the island was still a territory of the United States. Members of the United Nations General Assembly voted on November 27, 1953, to take Puerto Rico off the list of territories that did not govern themselves, and American officials then stopped sending the United Nations reports about the island.

The revolt of 1950. Some Puerto Ricans fought the new arrangement with guns. Nationalists rose up in several towns on October 30, 1950, and *Time* magazine reported that about 70 of them took Jayuya and killed four policemen there. Soldiers and pilots of the Puerto Rico National Guard, under Major General Luis R. Esteves, attacked the towns of Jayuya and Utuado with P-47 Thunderbolt warplanes, artillery, mortars and grenades. Pilots in American military aircraft fired on and bombed towns full of American citizens.

Governor Muñoz Marín counted 31 Puerto Ricans dead across the island in the revolt, *Time* reported. *Time*'s report does not state how many people the Guard killed in Jayuya and Utuado.

Blair House. Two Nationalists, Griselio Torresola and Oscar Collazo, attacked Blair House in Washington, where President Harry Truman was living, on November 1, 1950. Torresola shot a White House guard, Leslie Coffelt, three times. Coffelt was dying, but he got up and shot Torresola dead, and then Coffelt died of his wounds. Collazo was sentenced to death, and in 1952 Truman changed the sentence to life in prison.

The House, 1954. Four Puerto Rican Nationalists went into the visitors' gallery of the House of Representatives on March 1, 1954: Lolita Lebrón, Rafael Cancel Miranda, Andrés Figueroa Cordero and Irvin Flores Rodríguez. They unrolled a Puerto Rican flag and fired down onto the floor, wounding five members of the House, and all five recovered. All four Nationalists were convicted and sent to prison.

The Pacific: a trust, nuclear tests and one vote to join

After the Second World War, members of the United Nations placed more than 2,000 Pacific islands under American rule: the Marianas, the Carolines and the Marshalls. American officials governed them for the United Nations from 1947 to 1986, with a duty to look after the people. The arrangement was called the Trust Territory of the Pacific Islands.

American officials set off 67 nuclear tests between 1946 and 1958 at Bikini and Enewetak, two atolls, rings of small coral islands around a lagoon, in the Marshall Islands. They moved the people off those atolls before the tests.

Castle Bravo. The largest test was Castle Bravo, on March 1, 1954. It was two and a half times as powerful as its planners expected, and its fallout, the radioactive dust from the blast, came down on islands where people lived. All 64 people on Rongelap, about 110 miles away, took in radiation from it. American officials moved the people of Rongelap and Utirik away about three days after the blast, and Rongelap could no longer be lived on. Chapter 22, War and the Military, tells what the fallout did to them and to a Japanese fishing crew.

The study. Officials of the Atomic Energy Commission, the government agency in charge of nuclear weapons, then ran a study of the exposed people, called Project 4.1. The people in the study lived on islands that the United States governed for the United Nations, with a duty to look after them. Radioactive iodine from fallout collects in the thyroid, a gland in the neck, and in later years thyroid disease, including thyroid cancer, was the most common lasting harm among the people exposed.

One vote to join. The people of the Northern Mariana Islands voted to join the United States. Of today's five territories with people living in them, it is the only one whose people voted to come in. A Covenant, a formal agreement, between the islands and the United States was signed on February 15, 1975, and 78.8 percent of the people who voted approved it on June 17, 1975. The Commonwealth of the Northern Mariana Islands began in 1978, and its people became American citizens in 1986.

The Marshall Islands and the Federated States of Micronesia became independent countries in 1986, tied to the United States by agreements called Compacts of Free Association. Palau's compact took effect in 1994.

The capital, the canal and an apology

The capital. Washington, D.C. is not a state either. Until 1961 the people living there could not vote for president at all, just as the people of the territories still cannot. That year lawmakers in enough states ratified, or approved, the Twenty-third Amendment to the Constitution, and it took effect on March 29, 1961.

The city got back a delegate in the House in 1971, one who can speak there but cannot vote on bills. Under the Home Rule Act of 1973, its people elected their own mayor from 1975. Members of Congress passed an amendment in 1978 to give the city full seats in Congress, but lawmakers in too few states approved it, and it ran out of time in 1985.

The canal. American and Panamanian leaders signed the Torrijos-Carter Treaties on September 7, 1977. Under them, American officials handed the Panama Canal to Panama on December 31, 1999.

An apology. Members of Congress passed, and President Bill Clinton signed, the Apology Resolution, Public Law 103-150, in 1993. It is about the overthrow of the Kingdom of Hawaii a hundred years earlier. On January 17, 1893, a group of mostly American businessmen had overthrown Queen Lili‘uokalani, and John L. Stevens, the top American diplomat in Hawaii, had landed 162 American sailors and Marines to back them.

The resolution states that Stevens "conspired," or plotted in secret, with "a small group of non-Hawaiian residents of the Kingdom of Hawaii, including citizens of the United States." It names the men who took power as a Committee of Safety that "represented the American and European sugar planters, descendents of missionaries, and financiers."

The resolution states that the overthrow was illegal. In it, members of Congress apologized to Native Hawaiians "on behalf of the people of the United States" for the overthrow, carried out "with the participation of agents and citizens of the United States." It also states that the Native Hawaiian

people never directly gave up their claims to sovereignty, the right to govern themselves. Its last section states that it does not settle any claims against the United States.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 8 MINUTES

More than 3.6 million people live in the five American territories with people on them, and none of them can vote for president while living at home. They still live under the Insular Cases. Since 2001 American leaders have sent troops to wars in Afghanistan, Iraq and, in 2026, Iran, and they have argued over trade and over the Paris agreement on climate.

September 11, 2001, and the wars after it

Nineteen hijackers attacked the United States on September 11, 2001, and killed 2,977 people, not counting the hijackers. The next day, representatives of NATO's member countries called on Article 5 of their treaty, the promise that an attack on one member is an attack on all of them, for the first and only time.

American forces went into Afghanistan on October 7, 2001, and the last American troops left on August 30, 2021. American forces invaded Iraq on March 20, 2003. American leaders said Iraq had weapons of mass destruction, meaning nuclear, chemical or biological weapons. None were found.

The dead. Chapter 22, War and the Military, gives the counts of the people killed in these wars.

DISPUTED: For Iraq, counts built from deaths recorded one by one come out much lower than estimates built from surveys of households. Workers at Iraq Body Count, a group that records deaths one at a time, had counted more than 187,000 civilians killed from 2003 to 2024.

A survey published in the medical journal *The Lancet* in 2006 estimated about 600,000 violent deaths in Iraq in just the first three years of the war, and American and Iraqi officials disputed it at the time. A survey by the World Health Organization and Iraq's health ministry estimated about 151,000 violent deaths over the same three years.

Guantánamo. American officials began holding prisoners at the navy base at Guantánamo Bay, in Cuba, on January 11, 2002. Most of them were never tried in court. According to the journalist Andy Worthington, who tracks the prison, 779 men had been held there by January 2025, and 15 were still held.

Trade and the climate agreement

Members of the World Trade Organization, a group of countries that agree on rules for trade between them, approved China's entry on November 10, 2001, and China joined at the end of that year. A new agreement among the United States, Mexico and Canada took effect on July 1, 2020, according to the Office of the United States Trade Representative. It replaced the North American Free Trade Agreement, called NAFTA.

The United States has joined and left the Paris climate agreement twice. The governments that join it promise to cut the gases that are warming the earth. American presidents took the country out of it in November 2020 and brought it back in February 2021. President Donald Trump signed an order to leave again on January 20, 2025, and researchers at the Congressional Research Service, which studies issues for Congress, report that the United States was out of the agreement as of January 27, 2026.

Strikes at sea, Venezuela and Iran, 2025 and 2026

Strikes at sea. From September 2, 2025, American forces fired on boats in the Caribbean Sea and the eastern Pacific Ocean that President Trump's officials said were carrying drugs. By September 19, 2026, they had killed at least 231 people, the Associated Press reported. Chapter 22, War and the Military, tells the strikes in full, with the families who went to court over two of the dead.

Venezuela. American troops captured Venezuela's president, Nicolás Maduro, and his wife in Caracas, Venezuela's capital, on January 3, 2026, and took them to New York to face charges tied to drugs. According to the Brookings Institution, a research group, the raid went ahead without approval from Congress beforehand.

Iran. American and Israeli forces began a war with Iran by bombing it on February 28, 2026. Workers at the Human Rights Activists News Agency, a rights group that tracks Iran, counted 3,636 people killed in Iran by April 7, 2026, and 1,701 of them were civilians. Chapter 22 gives the count of American service members killed.

The territories today

The 2020 census, the government's count of everyone who lives in the country and its territories, recorded more than 3.6 million people in the five American territories with people on them. Census Bureau workers counted 3,285,874 people in Puerto Rico, 11.8 percent fewer than in 2010.

They counted 153,836 in Guam, about 87,146 in the U.S. Virgin Islands, 49,710 in American Samoa and 47,329 in the Northern Mariana Islands. Every one of the five lost people between 2010 and 2020.

Citizens and nationals. People born in Puerto Rico, Guam, the U.S. Virgin Islands and the Northern Mariana Islands are American citizens. People born in American Samoa are American nationals. Their passports say so, and to vote they must go through naturalization, the process a person from another country uses to become a citizen.

No one living in a territory can vote for president. Voters in each territory elect one delegate to the House of Representatives, who can speak and work on committees but cannot vote on whether a bill passes. Puerto Rico's delegate is called the Resident Commissioner.

La junta. Puerto Rico's government owed more than $72 billion by 2016, and more than $55 billion in pensions, the money its officials had promised to pay its workers after they retired. Puerto Rico's leaders had no legal way to change the terms of the debt. Members of Congress passed a law called PROMESA on June 30, 2016. Under it, a board of seven members, appointed in Washington, has power over Puerto Rico's budget, and Puerto Ricans call it *la junta*.

The people whose budget the board controls did not elect its members. By the board's own account, its members have changed the terms of about 80 percent of the debt, cutting it from more than $70 billion to about $37 billion.

President Trump fired several members of the board on August 5, 2025, and CBS News named five of them. Under PROMESA a president may remove a member "only for cause," which means for a real reason such as wrongdoing.

Three of the fired members, Arthur J. González, Andrew G. Biggs and Betty A. Rosa, went to court. Judge María Antongiorgi-Jordán of the federal court in Puerto Rico put them back on the board on October 3, 2025. In 2026 she made that order permanent and called the firings "plainly contrary to law," according to the Democracy Defenders Fund, a group that campaigns on political causes.

The status votes. Puerto Ricans have voted on their island's status several times, and members of Congress have not acted on any of the results. Statehood won 52.52 percent in 2020. On November 5, 2024, statehood won 58.61 percent, free association 29.57 percent and independence 11.82 percent, according to the island's certified results. Free association would make the island a separate country tied to the United States by a formal agreement, as the Marshall Islands are.

Some voters stayed away from these votes in protest. Members of Congress do not have to follow the results, and only Congress can change Puerto Rico's status.

Washington, D.C. The capital had about 690,000 people in the 2020 census, more than the state of Wyoming, which had 576,851, or Vermont. It has one delegate in the House who cannot vote on bills, and no senators. Members of the House of Representatives voted 216 to 208 to make it a state on April 22, 2021, but senators did not take up the bill.

John Fitisemanu

Who: A man born in American Samoa and living in Utah who sued to be counted an American citizen from birth.

When and where: Utah. His case was decided by a federal appeals court in 2021.

Fitisemanu sued for citizenship, and his home territory's leaders opposed him.

Because he was born in American Samoa, Fitisemanu is an American national, not a citizen, and a national must become a citizen before he can vote. He sued the United States to be recognized as a citizen by birth, and a federal judge in Utah agreed with him.

The appeal. In 2021 judges of a higher federal court, the Court of Appeals for the Tenth Circuit, overturned that decision, holding that the Insular Cases were still the law. On October 17, 2022, the justices of the Supreme Court refused to take his case.

American Samoa's own elected leaders opposed his lawsuit. They argued that citizenship ordered by a court could put at legal risk *fa'a Sāmoa*, the Samoan way of life, which includes land held in common by families and the matai system of chiefs. Fitisemanu wants to be a citizen. The leaders of his home territory did not want citizenship granted to Samoans that way.

José Luis Vaello Madero

Who: A man the United States government sued for $28,081 after he moved to Puerto Rico.

When and where: New York, then Puerto Rico. His case was decided by the Supreme Court on April 21, 2022.

Moving to Puerto Rico ended his right to a federal payment.

While Vaello Madero lived in New York, he lawfully received Supplemental Security Income, a federal payment for people with very low incomes who are elderly, blind or disabled. Members of Congress have not made that benefit available to people living in Puerto Rico. After he moved, the payments kept coming, and he did not know he was no longer allowed to get them. Lawyers for the United States government sued him to get back $28,081.

At the Supreme Court. His case reached the justices, and on April 21, 2022, they ruled 8 to 1 that the Constitution does not require Congress to pay that benefit to people in Puerto Rico. Justice Neil Gorsuch agreed with the result, but he also wrote a separate statement of his own, and in it he called for the Insular Cases to be overturned.

THE THREE QUESTIONS, ANSWERED

1. **Who got to choose when this land and its islands changed hands?** Almost never the people who lived there: no Native nation was at Paris in 1783, and no one from Puerto Rico, Guam or the Philippines was at Paris in 1898. Of today's five territories, only the Northern Mariana Islands voted to join, and Hawaii's voters chose statehood in 1959 from a ballot that left out independence.
2. **How did a weak new country come to rule islands and overthrow governments?** First it grew: by 1848 it reached the Pacific, and by 1900 its mills made more steel than Britain's and

Germany's together. Then its leaders used force. Marines helped overthrow Hawaii's queen in 1893, American leaders took Puerto Rico, Guam and the Philippines after a war with Spain in 1898, and after 1950 CIA officers secretly helped overthrow governments in Iran, Guatemala and Chile.

3. **Are the people of the territories American citizens?** Most are, but people born in American Samoa are nationals, not citizens, and no one living in a territory can vote for president.

What changed: The chapter begins with rulers dividing land they had never seen and ends with a country that began as colonies holding five island territories whose people cannot vote for president.

FIVE DATES TO KEEP

June 7, 1494: The rulers of Spain and Portugal split the lands across the Atlantic.

September 3, 1783: British diplomats accept American independence in Paris.

February 2, 1848: Mexican officials sign over the Southwest.

December 10, 1898: Spanish diplomats sign over Puerto Rico, Guam and the Philippines.

May 27, 1901: The justices rule that Puerto Rico belongs to the United States without being part of it.

NOT KNOWN: No one has worked out how many people died at Kake after the crew of the USS *Saginaw* burned the villages in 1869.

Where to go: ʻIolani Palace, in Honolulu, where Queen Liliʻuokalani was tried and held prisoner in 1895.

Next: Chapter 24, Slavery and Freedom. It asks how slavery was built here one law at a time, and what happened to the freedom enslaved people won in 1865.

PART 6

Freedom and Fairness

Chapter 24: Slavery and Freedom

CHAPTER 24 OF 37 · SLAVERY AND FREEDOM · 10 ERAS · ABOUT 110 MINUTES

Judges, lawmakers and traders built slavery here on purpose, one law and one sale at a time. Enslaved people fought it until it ended in 1865, then fought again when white lawmakers and mobs took much of their new freedom away.

In October 1865 a freed man named Thornton Copeland placed a few lines in the *Colored Tennessean*, a Black newspaper in Nashville. "INFORMATION is wanted of my mother," they began. Her name was Betty. He had last seen her in Fauquier County, Virginia, in 1844, the year he was sold in Richmond to a man named Samuel Copeland. His advertisement is now one of more than 5,000 like it that anyone can read online. No record shows whether he ever found her.

This chapter follows slavery from the captives of Native wars before 1500 to today's arguments over repair, meaning payments or other steps to make up for what slavery did. Watch for two things as you read.

First, watch who made each rule: the Virginia judges who sentenced John Punch to serve for life in 1640, the lawmakers who made a child's status follow the mother's in 1662, the traders who sold a million people south. Second, watch what enslaved people did back. They ran away, sued and learned to read in secret, and some rose up with weapons.

After 1877 the chapter follows the forced labor that replaced slavery and the vote taken from Black men. It also describes the court ruling and new laws of 1954 to 1965, under which federal officials began to enforce Black Americans' rights. Chapter 25 describes the campaigns people organized for those rights.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did a servant in Virginia become a slave for life, and a child a slave from birth?
2. How did enslaved people fight back?
3. What happened to the freedom people won in 1865?

Five minutes? Read "How did a servant become a slave for life?" in The 1600s, the Robert Smalls story in 1850 to 1900, and "The promises broken" in the same era.

Can skip: "The ships from Rhode Island", "Enslaved builders of the capital", "Slavery exhibits in the national parks, 2025 and 2026". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 1 MINUTE

Native nations took captives in war long before Europeans arrived, but in most of them a captive could become a member of a family.

Holding a person as property for life, and passing that status on to the person's children, is called chattel slavery, from chattel, an old word for property. Europeans later wrote chattel slavery into the laws of the colonies they set up here.

Captives who could become family

Warriors in some Native nations took captives in raids on their enemies. Among the Haudenosaunee, the nations also called the Iroquois, a family that had lost someone in war could adopt a captive to take the dead person's place. People call this custom the mourning war, and the records describe it more fully than any other kind of captive-taking. In most of these nations a captive could stop being a captive, and a captive's children did not automatically become captives too.

On the Northwest Coast, among the Tlingit, the Haida and the nations near them, slaves were property. Their owners traded them, and a slave's children were born slaves. Leland Donald studied this system in a 1997 book and found that it was slavery in the full sense of the word.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 3 MINUTES

The first people Europeans enslaved in what is now the United States were Native people. Spanish raiders took them captive on the coasts of Florida and the Carolinas from the early 1500s. Enslaved Africans came with the Spanish too. About 100 of them rebelled at a Spanish colony on the Atlantic coast in October 1526 and escaped. It is the first known revolt of enslaved Africans in what became the United States.

Spanish slavery and the first revolt

The encomienda. In their colonies, Spanish colonists ran a system called the encomienda. Under it they forced Native people to work for them and to hand over goods. King Charles I of Spain issued the New Laws on November 20, 1542. They said that no one could enslave Native people, and that an encomienda ended when its holder died. Spanish colonists opposed the New Laws and kept forcing Native people to work.

The first revolt. Lucas Vázquez de Ayllón founded a colony called San Miguel de Gualdape in 1526, on the coast of what is now Georgia or South Carolina. Nobody knows the exact spot. Between 600 and 700 colonists came with him, and about 100 of them were enslaved Africans. Ayllón died, and the colonists fought among themselves.

In October 1526 the enslaved Africans rose up and escaped. Where they went is not certain. The accounts say they most likely found shelter with Native people nearby, probably the Guale. Within months the surviving colonists gave up the colony.

St. Augustine. Pedro Menéndez de Avilés and about 800 Spanish colonists founded St. Augustine, in Florida, on September 8, 1565. They moved into the Timucua town of a chief named Seloy and made his council house their first fort. Africans were among St. Augustine's first residents, some free and some enslaved. They worked as craftsmen, farmers and house servants.

Spanish slave law came from an old code called the *Siete Partidas*. It listed rights for enslaved people that English colonial law later left out. Under it, owners were not to split up families. Enslaved people could join the Catholic Church and marry in it, and the code listed ways for them to become free.

Estevanico

Who: An enslaved man from Morocco who walked across the south of the continent after a shipwreck, then was killed while guiding a Spanish expedition into Zuni country.

When and where: Shipwrecked on the Texas coast in 1528. Reached Mexico City in 1536. Killed at the Zuni town of Hawikuh in 1539.

Shipwrecked in 1528, he walked to Mexico City as a healer.

Estevanico was born in about 1500 in Azemmour, in Morocco. In 1522 a Spanish nobleman named Andrés Dorantes de Carranza bought him. Six years later both men were on a Spanish expedition that was wrecked on the coast of Texas. Native people on that coast held the survivors, Estevanico among them, as slaves for several years.

In 1534 four of the survivors escaped: Estevanico, Dorantes, Álvar Núñez Cabeza de Vaca and Alonso del Castillo Maldonado. They walked west, working as healers along the way, and reached Mexico City in July 1536.

The Spanish official who ruled Mexico, Viceroy Antonio de Mendoza, then got Estevanico from Dorantes, either by buying him or as a gift. Some evidence suggests Mendoza freed him. On March 7, 1539, Estevanico set out as the guide for Marcos de Niza, a friar, a member of a Catholic religious order, who was heading north. Estevanico went ahead of the friar with Native men from Sonora.

The people of Hawikuh, a Zuni town, killed him there in 1539. The Spanish accounts give three different reasons. The commander Coronado wrote in 1540 that they killed him because he had "killed and assaulted their women." Pedro de Castañeda wrote that he had demanded turquoise and women. Hernando de Alarcón reported that he had boasted that armed men were coming behind him.

KNOWN:	The people of Hawikuh killed Estevanico in 1539.
NOT KNOWN:	why they did it. Zuni tradition does not agree with the Spanish accounts.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 9 MINUTES

English colonists made slavery part of their law step by step in the 1600s, one court case and one new law at a time. When about twenty Africans reached Virginia in 1619, Virginia's law had no category for a person held as property for life. Judges and elected lawmakers in Virginia, Massachusetts and Carolina created that category between 1640 and 1705. In 1662 Virginia's lawmakers ruled that a child born to an enslaved mother was enslaved for life, whoever the father was.

Ship crews fed the system from across the ocean. Over the whole Atlantic slave trade they loaded about 12.5 million captive Africans onto ships bound for the Americas. About 1.8 million of them died on the way, about one person in every seven.

Who landed at Point Comfort in 1619?

An English ship called the *White Lion* landed at Point Comfort, in Virginia, in late August 1619. It carried "20. and odd Negroes," in the words of the colonist John Rolfe, using Negro, the word English writers of the time used for Black Africans. The *White Lion* was a privateer, a privately owned armed ship that attacked the ships of enemy countries. Its crew had taken these Africans from a Portuguese slave ship.

The Africans came from the Ndongo and Kongo region of West Central Africa. The exact day they landed is disputed. The National Park Service says late August. Some writers give August 20, and the Hampton historian Beth Austin argues for about August 25.

Whose land it was. Point Comfort belonged to the Kecoughtan, one of the Powhatan nations, who lived where the city of Hampton stands today. Their language belonged to the Algonquian family, a large group of related Native languages. English colonists built a fort beside their main village in 1609.

Soon after Lord De La Warr arrived, on July 9, 1610, the colonists seized the village. They drew the people out with a man playing a tambourine and then attacked them. The records do not say how many Kecoughtan they killed. The

survivors fled to other towns of the Powhatan nations.

KNOWN:	In 1619 Virginia's law did not yet have a category for slaves.
NOT KNOWN:	whether the colonists held the Africans of 1619 as servants for a set number of years or as slaves for life. The records that survive do not settle it.

The Middle Passage

The Middle Passage was the crossing of the Atlantic Ocean that ship crews forced captive Africans to make. Over the whole history of the trade, crews loaded about 12.5 million Africans onto slave ships. About 10.7 million lived through the crossing, and about 1.8 million died at sea. These figures come from the SlaveVoyages database of slave-ship voyages.

The average voyage took just over two months, and the trip from Upper Guinea, in West Africa, about three weeks. Crews chained the captives below the deck, in spaces too low for a person to stand up. They held the women apart from the men and kept the men in chains for long periods so they could not revolt.

Crew members often raped the women and children, which means they forced them into sex. About one captive in four was a child.

People died of disease and of thirst, and some killed themselves. The crew threw the bodies of the dead into the sea. Historians at the College of Charleston report that 13 to 19 percent of the people on board died during a crossing.

Ship crews landed most of the captives who survived in the Caribbean and in Brazil. Fewer than 4 percent came to mainland North America. Crews brought about 305,000 people there straight from Africa. Counting those who came by way of the Caribbean, the total is close to 388,000.

One survivor's account

Olaudah Equiano wrote that slave traders carried him across the ocean on a slave ship when he was a child. He published his account in London in 1789, in a book called *The Interesting Narrative of the Life of Olaudah Equiano*. Two other records list his birthplace as Carolina. If they are right, he never made the crossing he describes.

In his account, the crew put Equiano below deck, and he was too sick to eat. Two white crewmen tied him across the windlass, the winch that pulled up the anchor chain, and flogged him for refusing food. Flogging means that a person is tied down and struck over and over on the bare body with a whip. Each stroke cuts the skin, and deep or repeated flogging leaves raised scars for life. Equiano wrote that captives were "hourly whipped for not eating."

He wrote that the hold was "so crowded that each had scarcely room to turn himself." The heat and the foul air made people sick, "of which many died." Children fell into the tubs the captives used as toilets and nearly smothered.

One day two men who were chained together broke through the netting around the deck and jumped into the sea. A third man followed them. The crew lowered a boat. The two chained men drowned. The crew pulled the third man out "and afterwards flogged him unmercifully."

The ship landed at Barbados, an island in the Caribbean, and from there traders sold him on to Virginia.

KNOWN:	Equiano's own book says he was born in Africa.
NOT KNOWN:	whether that is true. In 2005 the historian Vincent Carretta reported the two records, a 1759 baptism record in London and a 1773 Royal Navy crew list. The book and the records cannot both be right.

John Punch

Who: The first African known to have been sentenced by a Virginia court to serve for life.

When and where: Virginia. Sentenced on July 9, 1640.

Three servants ran away together. Only the African got life.

The three men were servants of a man named Hugh Gwyn. The court record calls them "Victor, a dutchman, the other a Scotchman called James Gregory, the third being a negro named John Punch." They ran away together toward Maryland and were caught.

The judges of the General Court of Virginia heard the case on July 9, 1640. They ordered that "the said three servants shall receive the punishment of whipping and to have thirty stripes apiece," thirty strokes of the whip each. Victor and James Gregory then had to finish their time with their master, serve him one more year, and serve the colony for three years after that.

For John Punch the judges wrote a different sentence. He "shall serve his said master or his assigns for the time of his natural Life here or elsewhere." His assigns meant anyone the master sold or handed him to. Punch was the only African among the three.

It is the first known sentence of lifetime slavery in Virginia.

How did a servant become a slave for life?

Judges and lawmakers made a separate rule for Africans, one case and one law at a time between 1640 and 1705. Under it, Africans could be held for life, and so could their children. In the early years of Virginia, many workers were indentured servants, bound by a contract to work for a master for a set number of years and then set free.

Massachusetts, 1641. Lawmakers in New England wrote slavery into law too. The lawmakers of the Massachusetts Bay colony, called the General Court, adopted a code of laws named the Body of Liberties on December 10, 1641. It allowed slavery for "lawful captives taken in just wars, and such strangers as willingly sell themselves or are sold to us."

That meant people captured in a war the colonists called fair could be held as slaves. So could outsiders who sold themselves or were sold to the colonists. The Massachusetts Bay lawmakers were the first in any English colony on the mainland to put slavery into a written law.

John Casor. Anthony Johnson, an African, had arrived in Virginia in about 1621, gained his freedom and become a landowner on the Eastern Shore. He claimed a Black man named John Casor as his servant for life. Casor said the time on his contract had run out and he was free, and he went to work for Johnson's white neighbor, Robert Parker. In 1654 Johnson took Parker to court in Northampton County to get Casor back.

The county judges ruled for Johnson on March 8, 1655. They ordered Casor back to Johnson as his servant for life, which made Casor Johnson's slave, and they made Parker pay the costs of the case. It was the first court ruling in the English mainland colonies that a person who had committed no crime could be held for life.

The child follows the mother. Virginia's House of Burgesses, the colony's elected assembly, passed a law about children in 1662. Under the old rule in English law, a child's place in the law, free or bound, came from the father.

Virginia's lawmakers broke with that on purpose. A child born to an enslaved mother was now enslaved for life, whoever the father was. Lawyers called the rule by a Latin phrase, *partus sequitur ventrem*, "the child follows the womb."

Baptism. Some colonists were unsure whether baptism, the ceremony that made a person a Christian, set an enslaved person free. Virginia's lawmakers, meeting as the General Assembly in September 1667, passed a law that said it did not. It stated that "the conferring of baptisme doth not alter the condition of the person as to his bondage or ffreedome."

The lawmakers gave their reason in the law itself. With that doubt gone, they wrote, masters would be more willing to let enslaved people be baptized.

The slave codes

Virginia's lawmakers gathered the colony's earlier slave laws into one slave code in 1705: a set of laws listing what enslaved people could not do, how they could be punished and what their owners could do to them. Under it, enslaved people counted as real estate, the same kind of property as land. The code said that baptism did not free anyone. A master who killed an enslaved person while "correcting" him, which meant punishing him, faced no penalty.

Carolina's codes. Carolina's lawmakers, who met as the colony's assembly, passed their first slave law in 1690. The Lords Proprietors, the English noblemen who owned the colony, struck that law down. The assembly's members passed a new code in 1696. They based both laws on the slave code that lawmakers in Barbados had written in 1661.

The 1690 Carolina law already allowed branding, nose-slitting and castration as punishments. Branding meant pressing a metal iron, heated red-hot, into the person's skin. The iron burned a mark, often a letter, that stayed on the body as a scar for life. Nose-slitting meant cutting the person's nose open.

Under the Barbados code, an enslaved person who hit a Christian a second time was to be whipped hard, have the nose slit and be burned on the face with a hot iron. Castration meant cutting off a man's testicles, which left him unable to father children.

These codes of the 1600s set rules for enslaved people's bodies and for who could be held for life. They said nothing about reading or writing. South Carolina's lawmakers passed the first law against teaching enslaved people to write in 1740.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 10 MINUTES

Enslaved people rose up in the North and in the South between 1700 and 1750, and after each revolt colonial lawmakers wrote harsher laws. Their slave codes ordered whipping, branding and cutting off ears. South Carolina planters grew rice to sell overseas, a crop that enslaved West Africans already knew how to grow. In 1739 about 20 enslaved men near Charleston took up arms and marched toward Spanish Florida, where the king of Spain had promised freedom to people who escaped from the English colonies.

English colonists enslaved Native people as well as Africans. Slavery was a Northern business too: in New York City in 1712, about one person in five was enslaved.

How many Native people did Carolina's traders enslave?

English traders in Carolina enslaved between 24,000 and 51,000 Native people of the Southeast from 1670 to 1715. That range comes from the historian Alan Gallay. The traders shipped many of the captives to the Caribbean. In those years they shipped more enslaved Native people out of Charleston than slave ships brought enslaved Africans in.

The ships from Rhode Island

Merchants in the northern colonies sent out slave ships too. Rhode Island merchants paid for about 1,000 slaving voyages, and about half of all the slave ships that left mainland North America sailed from Rhode Island ports. The historian Jay Coughtry counted 934 of these voyages, which carried about 106,000 Africans. In the 1700s, Rhode Island merchants controlled between 60 and 90 percent of the American trade in African captives.

The DeWolf family of Bristol, Rhode Island, sent out about twice as many slaving voyages as the families closest behind them. Bringing captives into the United States from abroad became illegal in 1808. George DeWolf kept sending out slave ships anyway, against the law, until 1825.

New York, 1711 and 1712

New York City had about 5,000 people in 1700, and at least 750 of them were enslaved. The city's aldermen, the men who governed the city, set up a market at the foot of Wall Street on November 30, 1711. There, enslaved people were sold, rented out and inspected by buyers. The market stood from 1711 to 1762. The accounts of the market do not name the men who bought and sold people there.

The fire on Maiden Lane. More than 20 enslaved people in New York City set fire to a building on Maiden Lane on April 6, 1712. Accounts describe most of them as Coromantee, Akan people from the Gold Coast of West Africa, which is Ghana today. When white people came to fight the fire, the rebels attacked them. They killed 8 or 9 white people and wounded 6 or 7. Six of the rebels killed themselves before they could be captured.

Governor Robert Hunter called out the militia, the colony's armed citizens. Officials arrested more than 70 Black people. They put 27 on trial, found 21 guilty and put all 21 to death.

One account says 20 were burned alive and one was broken on the wheel. Breaking on the wheel was a public execution: the person was tied to a large wheel, and the executioner broke the person's arms and legs with an iron bar and left the person to die. Other accounts say only that the punishments ran from burning alive to breaking on the wheel.

Afterward, the members of the colony's assembly made it illegal for enslaved people to gather in groups of more than three or to own guns.

Rice and the Lowcountry

The Lowcountry is the coast of South Carolina around Charleston. Rice planters there relied on the skill and the labor of West Africans from rice-growing parts of Africa, and they paid more for captives from those coasts. Rice shipped out of Carolina rose from about 268,602 pounds a year around 1700 to more than 30 million pounds a year around 1740, more than a hundred times as much. Charleston became the largest port on the mainland where slave ships landed their captives.

Sullivan's Island. Ships arriving at Charleston stopped first at Sullivan's Island, in the harbor. The island had a quarantine station, called the pest house, where captives who might carry disease were held apart from everyone else.

Many accounts say about 40 percent of all the enslaved Africans brought to mainland North America passed through the island itself. The Charleston historian Nic Butler and others say that figure is too high, because many ships landed elsewhere in the harbor. Their doubt is about the island's share only.

Whose land it was. The rice plantations stand on Native land. When English colonists settled Charleston in 1670, about fifteen Native groups lived in the Lowcountry, among them the Stono, the Kiawah, the Etiwan, the Edisto and the Combahee. By the middle 1700s these groups were nearly gone. Many of their people had died of disease or been killed in wars, and English colonists had taken their land.

The runaway law. South Carolina's assembly passed a law in June 1712 that listed punishments for any enslaved person who stayed away for twenty days or more. For a first escape, it ordered a public whipping of up to forty lashes, forty strokes of the whip. For a second escape, it ordered the person "branded with the letter R, on the right cheek." For a third, it ordered forty lashes and one ear cut off.

A man caught a fourth time was to be castrated. A woman caught a fourth time was to be whipped, branded with an R on the left cheek, and have her left ear cut off. The 1712 law does not name the people who carried out

these punishments.

For a fifth escape, the law let two justices of the peace, who were local judges, and three landowners decide. They could order the person killed. Or they could order the person's Achilles tendon, the cord just above the heel, cut through, so that the person could never run again. From 1723 on, South Carolina's lawmakers dropped these lists from their codes and left punishment to the enslaver.

A free town to run to

Florida belonged to Spain. In 1693 the king of Spain promised freedom to enslaved people who escaped from the English colonies and reached Florida, if they became Catholics. Men also had to serve in the militia.

Florida's governor, Manuel Montiano, set up a town for these free people in 1738, about two miles north of St. Augustine. Its Spanish name was Gracia Real de Santa Teresa de Mose, and people call it Fort Mose. It was the first free Black town in what is now the United States to be recognized by law. About 100 people lived there, in more than 20 households.

British forces surrounded St. Augustine and attacked it in 1740. Fort Mose was destroyed in the fighting, which people remember as the battle of Bloody Mose. The people rebuilt it in 1752. In 1763, when the Spanish left Florida, the people of Mose left with them for Cuba. The site is now Fort Mose Historic State Park.

Francisco Menéndez

Who: A man from West Africa who escaped slavery in Carolina and became captain of the free Black militia at Fort Mose.

When and where: Born in the Gambia region of Africa. Enslaved in Carolina. Led the militia at Fort Mose, Florida, from 1738.

He escaped slavery in Carolina and became a captain in Florida.

Francisco Menéndez was a Mandinga man, born in the Gambia region of West Africa. Slave traders carried him to Carolina, where he was enslaved.

He escaped south to Spanish Florida. There he was baptized as a Catholic and took the name Francisco Menéndez. From 1738 he led the militia at Mose, the armed men who defended the town of people who had escaped from the English colonies.

Spanish colonial records document his life. Of all the people who lived at Mose, he is the one the records describe most fully.

The Stono Rebellion, 1739

About 20 enslaved men rose up at the Stono River, south of Charleston, on September 9, 1739. It was the largest uprising of enslaved people in the mainland colonies before the American Revolution. A report written in 1739 describes them as Angolan and says "one who was called Jemmy was their Captain."

The rebels seized weapons and marched south toward Spanish Florida, and others joined them along the way. The same report says they were "calling out Liberty" and marched "with Colours displayed, and two Drums beating." The rebels killed about 25 white colonists.

Lieutenant Governor William Bull met the rebels on the road that day. He rode off to warn the planters and raised the militia, which attacked the rebels the same day. Bull also hired Chickasaw and Catawba men, and some enslaved men, to hunt down those who got away. About a week later, the militia fought a group of rebels again, about 30 miles from the first fight.

The count. The militia killed between 35 and 50 rebels in the first fight. A historical marker says at least 54. The 1739 report gives its own count: "about 40 Negroes and 20 whites were killed."

What the colonists did to the captured. The report says the militia "shot on the Spot" rebels they took in the field. Colonists cut off the heads of rebels and set them on stakes along the main roads as a warning to others. The colonists put most of the captured rebels to death. One account says the militia took 43 rebels who had no weapons and hanged them. The colonists sold the survivors to the West Indies, the islands of the Caribbean.

What the Negro Act took away

South Carolina's assembly passed the Negro Act on May 10, 1740, in answer to Stono. Under it, enslaved people could not gather in groups, grow their own food, earn money or move about freely. Teaching an enslaved person to write became a crime. That was the first law in what became the United States against teaching enslaved people. It named writing, not reading.

Under the Negro Act, a white person who murdered an enslaved person was guilty only of a misdemeanor, a minor crime punished by a fine. Any white person, not only an owner, could arrest and punish a Black person who broke the act. A court could put an enslaved person to death for plotting a revolt, running away, setting fires, or teaching others about poisonous plants.

New York, 1741

New York officials put about 200 people on trial in 1741 for a plot to burn the city. Their main witness told of the plot only after a judge had threatened her and promised her a reward.

A string of fires began in New York City in March 1741, the first at Fort George, the governor's home. City officials claimed that enslaved people and poor white people had plotted to burn the city. Three judges of the colony's Supreme Court heard the cases, among them Daniel Horsmanden. The trials began on May 11, 1741.

The witness. Mary Burton was a young servant bound to a tavern owner named John Hughson. When officials threatened her with jail, she testified about a robbery. Judge Horsmanden threatened her again and promised her an end to her contract and a reward in money. After that, she began to describe a plot she had never mentioned before. The New York courts' historical society says the sentences rested on doubtful testimony.

The count. The same society counts 30 people sentenced to death and 70 sentenced to be sold into slavery in the Caribbean. The historian Jill Lepore counts 34 put to death between May 11 and August 29, 1741.

By Lepore's count, officials burned 13 Black men at the stake and hanged 17 Black men, two white men and two white women. John Hughson, his wife Sarah and their daughter Sarah were among the white people put to death. In 1744 Horsmanden published his record of the trials.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 10 MINUTES

The men who led the Revolution wrote in 1776 "that all men are created equal," and enslaved people used words like those to claim their freedom. Elizabeth Freeman won hers in a Massachusetts court in 1781 with the words of the new state constitution. About 20,000 Black people joined or fled to the British during the war, after Virginia's royal governor promised freedom to enslaved men who fought for Britain. In 1796 Ona Judge walked out of President George Washington's household and never went back.

Slavery stayed legal in most of the new country. The first census, a count of every person in the United States, found 697,681 people held in slavery in 1790. Lawmakers and judges in Vermont and six northern states began to end slavery between 1777 and 1804, but most of their laws freed only children born after a set date, and only once those children grew up. The men who wrote the Constitution in 1787 protected slavery in three of its clauses, or parts, without once using the word slave.

Phillis Wheatley

Who: A girl kidnapped from West Africa who became the first African American to publish a book of poetry.

When and where: Landed in Boston in 1761. Her book was published in London on September 1, 1773. Freed by November 1773. Died in 1784.

Her first name came from the slave ship that carried her.

Slave traders kidnapped her from West Africa when she was about seven years old. In 1761 she landed in Boston on a slave ship called the *Phillis*. John Wheatley, a Boston merchant, and his family bought her. Her first name came from the ship, and her last name from the family who bought her.

She never went to school. John Wheatley wrote that she learned "by only what she was taught in the Family." Within sixteen months of landing, he wrote, she had learned English well enough to read the hardest parts of the Bible.

Twelve years after she landed, she was a published poet. Her book, *Poems on Various Subjects, Religious and Moral*, was published in London on September 1, 1773. It was the first book of poetry by an African American. The Wheatleys freed her by November 1773, shortly after it appeared.

She married John Peters in 1778. She died poor in 1784, at about 31 years old.

Two promises of freedom in the Revolution

Both sides in the Revolution promised freedom to some of the enslaved men who would fight for them. Neither side kept that promise to all of them.

Before any promise. A Black man died in the quarrel with Britain years before either side promised anything. British soldiers fired on a crowd of colonists in King Street, in Boston, on March 5, 1770. People call it the Boston Massacre. The first person killed was Crispus Attucks, a sailor with African and Wampanoag ancestors. Most accounts say he had escaped from slavery in Framingham, Massachusetts, around 1750. Accounts differ about whether he was legally free by 1770 or still an escaped slave.

Lord Dunmore's promise. Virginia's royal governor, Lord Dunmore, made a promise on November 7, 1775, to the men enslaved by Virginians who were fighting the British. Any of them who joined the British army would be free. About 300 men joined his Ethiopian Regiment within a month, and it grew to about 800.

The Patriot promise. The Patriots, the colonists fighting the British, made promises too. Rhode Island's lawmakers voted on February 14, 1778, that any enslaved man in the state could join the army for the rest of the war. He would be set free as soon as he passed the army's inspection, and the lawmakers promised to pay his owner up to 120 pounds.

The count. Over the whole war, by the estimates of the Gilder Lehrman Institute and the American Battlefield Trust, about 20,000 Black people joined or fled to the British. About 9,000 served the Patriots, and about 5,000 of those fought as soldiers.

Which states ended slavery first?

Vermont's constitution of 1777 was the first constitution to ban slavery. It banned holding adults in slavery. Vermont then governed itself as its own country, and it joined the United States in 1791.

Pennsylvania, 1780. Pennsylvania's lawmakers passed the Act for the Gradual Abolition of Slavery on March 1, 1780, a law to end slavery step by step over many years. No elected lawmakers anywhere in the world had passed a law to end slavery before. No enslaved person went free on the day it passed. Children born after it were to go free when they turned 28.

In Massachusetts, juries and judges ended slavery in court cases decided between 1781 and 1783. Lawmakers in Connecticut and Rhode Island passed gradual laws in 1784. New York's lawmakers passed one in 1799, and the last enslaved people in New York were freed in 1827. New Jersey's lawmakers passed theirs last, in 1804. Under these gradual laws, people in the North stayed enslaved for decades.

The members of Congress passed the Northwest Ordinance on July 13, 1787, a law for the lands north of the Ohio River. It banned slavery there.

Elizabeth Freeman ("Mum Bett")

Who: An enslaved woman who sued for her freedom under the new Massachusetts constitution and won.

When and where: Sheffield and Great Barrington, Massachusetts. Jury verdict in August 1781. Died in 1829.

Her lawyer argued that seven words in a constitution set her free.

Before she was Elizabeth Freeman, she was known as Bett, or Mum Bett. She was enslaved in the household of Colonel John Ashley in Sheffield, Massachusetts.

The new Massachusetts constitution of 1780 states that "all men are born free and equal." Bett and an enslaved man named Brom sued Ashley for their freedom. Their lawyer, Theodore Sedgwick, argued that those words made slavery illegal.

A jury in Great Barrington set them free in August 1781. The jury also ordered Ashley to pay them 30 shillings for the harm done to them. The case was called *Brom and Bett v. Ashley*, Brom and Bett against Ashley. The juries and judges in that case and in the cases of an enslaved man named Quock Walker, decided from 1781 to 1783, ended slavery in Massachusetts.

Bett took the name Elizabeth Freeman. For the rest of her life she worked for pay as a housekeeper and a nurse. She died in 1829.

Sally Hemings and Thomas Jefferson

Thomas Jefferson was the main author of the Declaration of Independence. Over his lifetime he owned more than 600 enslaved people. One of them was Sally Hemings. Jefferson owned her and used her for sex from the time she was a girl, and she had his children.

Her family. Sally Hemings was born enslaved in 1773. Her mother, Elizabeth Hemings, was enslaved. Her father was most likely John Wayles, the man who owned her mother. By law Wayles owned Elizabeth Hemings, and she had no legal right to refuse him sex. Wayles was also the father of Jefferson's wife, Martha. That made Sally Hemings the half-sister of Jefferson's wife.

Paris, 1787. Jefferson was the United States minister to France in 1787, the man who spoke for his country in Paris. That year Sally Hemings went to Paris with his daughter Polly. Sally was 14 and enslaved by Jefferson. He was 44. By law he owned her, and she could not refuse him. He had the legal power to sell her or her family.

Her son Madison Hemings said in 1873 that in Paris she "became Mr. Jefferson's concubine," using an old word for a woman a man kept for sex without marrying her. Madison Hemings said she was pregnant when they came back to Virginia.

Under French law she could have claimed her freedom in Paris. Madison Hemings said Jefferson promised that her children would be freed at age 21 if she went back with him. She had children from 1790 to 1808. Four lived to be adults: Beverly, Harriet, Madison and Eston.

The DNA. In 1998 scientists compared DNA and found that a man of the Jefferson family was the father of Eston Hemings. That test could not tell Thomas Jefferson apart from other men of his family, such as a brother or a nephew. In January 2000 the Thomas Jefferson Foundation, which runs Jefferson's home at Monticello, concluded that Jefferson "was probably the father" of her children. By 2018 the foundation's historians treated this as a fact.

In September 2026 a study paid for by the Smithsonian tested DNA from a lock of Thomas Jefferson's own hair, cut on the day he died. Richard Kurin of the Smithsonian said some Hemings descendants "had more Thomas Jefferson DNA than the known Jefferson descendants." The scientists concluded that Jefferson himself fathered her children.

Historians disagree about the right word for what Jefferson did. Jessica Marie Johnson, a historian of slavery at Johns Hopkins University, wrote in 2026 that Jefferson was a rapist. In her 2008 history of the Hemings family, the historian Annette Gordon-Reed argued that Jefferson and Sally Hemings loved each other. Another historian, Eric Foner, answered in a review that neither of them left any record of their feelings.

Jefferson died in 1826. Neither he nor anyone after him ever legally freed Sally Hemings. His daughter Martha Jefferson Randolph let her live in Charlottesville with her sons. There she lived as a free woman, though no paper made her free by law, and the census of 1833 listed her as free. She died in 1835.

The Constitution's protections for slavery, 1787

The men who wrote the Constitution in 1787 put three protections for slavery into it. None of the three clauses uses the word slave.

Three-fifths. Under the three-fifths clause, each enslaved person counted as three-fifths of a person. The number of seats a state got in the House of Representatives depended on that count. Enslaved people could not vote, but their owners' states got extra seats for them. The slave states held 47 of the 105 seats in the House in 1793. Counting only their free people, they would have had 33.

Runaways. The fugitive clause said that a person "held to Service or Labour" who escaped to another state had to be "delivered up," handed back to the person who claimed that person's work.

The trade. The slave-trade clause barred Congress from ending the import of enslaved people before 1808.

Enslaved builders of the capital

Enslaved people worked at every stage of building the White House and the Capitol in Washington. The cornerstone of the White House was laid on October 13, 1792, and that of the Capitol on September 18, 1793.

Enslaved men cut sandstone at the quarries on Aquia Creek, the open pits where the stone was dug. They also cut timber, made bricks, sawed lumber and did carpentry. The officials in charge of the building hired them from their owners, and the owners kept the pay.

The payroll of the President's House for May 1795 names five enslaved carpenters: Tom, Peter, Ben, Harry and Daniel. The architect James Hoban owned four of them, and his assistant Pierce Purcell owned one. The payroll records the days four of the men worked that month: Peter 28, Ben 28, Daniel 25 and Harry 9. Hoban signed for their wages.

Ona Judge

Who: An enslaved woman who escaped from President George Washington's household and was never caught.

When and where: Walked out of the President's House in Philadelphia on May 21, 1796. Lived free in New Hampshire. Died in 1848.

She walked out while the Washingtons were eating dinner.

Ona Judge was enslaved as Martha Washington's personal servant. On May 21, 1796, while the Washingtons sat at dinner in the President's House in Philadelphia, she left. She boarded a sloop, a small sailing ship, called the *Nancy*, and sailed to Portsmouth, New Hampshire.

Three days later a runaway advertisement ran in the *Philadelphia Gazette*, offering ten dollars for her capture. George Washington tried for years to get her back. She refused an offer to return. She married and lived in New Hampshire until she died in 1848.

In the 1840s she told her own story to newspaper interviewers, including one from *The Granite Freeman* in 1845. She told them that she was poor and that she was free.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 26 MINUTES

Cotton planters spread slavery west and south between 1800 and 1850. The number of people held in slavery kept rising, even after a federal law against bringing captives from abroad took effect on January 1, 1808. Most of the growth came from children born into slavery. Census takers counted 697,681 enslaved people in 1790 and 3,953,760 in 1860, almost six times as many.

Slave traders moved more than a million people from the older slave states of the Upper South, such as Virginia, to the new cotton states of the Deep South and sold them to planters there. Traders and buyers split families by selling husbands, wives and children to different owners.

Enslaved people resisted every day, and some rose up in armed revolt. Thousands escaped to the free states and to Canada. In most of the slave states, lawmakers made it a crime to teach an enslaved person to read.

Cotton

Eli Whitney built a cotton gin, a machine that pulls the seeds out of cotton, in 1793. With the gin, planters could make money growing upland cotton, the kind that grows away from the coast. Enslavers took enslaved people west to clear and plant new cotton lands. Cotton growers in the United States produced fewer than 9,000 bales, big pressed bundles, a year around 1790. By 1800 they produced more than 200,000 bales, and by 1860 about 4 million, more than 400 times the crop of 1790.

The number of slave states grew from six in 1790 to fifteen in 1860. By 1860 enslaved people on Southern plantations grew about 75 percent of the world's cotton. Cotton made up more than half of everything Americans sold to other countries, measured in dollars.

People as money. By 1860 the people who held four million people in slavery valued them at $3 billion or more. That was more than all the money invested in the country's railroads and factories put together. Enslavers bought, sold and priced people as property. They also mortgaged them: they borrowed money and promised to hand the enslaved people over to the lender if they could not pay it back.

Whose land it was. The cotton land of these years was Native land. Choctaw leaders signed the Treaty of Dancing Rabbit Creek, a written agreement with the United States, on September 27, 1830. In it they signed away more than 10 million acres in Mississippi and Alabama. It was the first treaty signed after members of Congress passed the Indian Removal Act of 1830. The Chickasaw signed away their land in the Treaty of Pontotoc Creek in 1832.

Under the Removal Act, United States officials then forced both nations west to Indian Territory, today's Oklahoma. They moved the Choctaw in groups in 1831, 1832 and 1833. By the Choctaw Nation's own count, more than a quarter of the Choctaw people removed in those years died.

Enslavers then brought in enslaved people to clear and plant the land the Choctaw and Chickasaw had lost. Mississippi's cotton crop grew from about 20 million pounds in 1820 to 193.2 million pounds in 1839. The number of

enslaved people in Mississippi grew from 65,659 in 1830 to 195,211 in 1840.

Sold south

Slave traders moved more than one million people from the Upper South to the cotton lands of the Deep South between about 1790 and 1860. From 1820 on, they moved about 200,000 people every ten years. They marched people overland in coffles, lines of people chained together. They also shipped people along the coast to New Orleans, which had the largest slave market in the country.

Franklin and Armfield. The biggest trading firm belonged to Isaac Franklin and John Armfield. From 1828 to 1836 they ran it from 1315 Duke Street in Alexandria, Virginia. Their compound took up half a city block, with separate pens for men and for women. At their busiest they moved between 1,000 and 1,500 people a year, and about 8,500 in the nine years of the partnership. They marched people to Natchez, in Mississippi, and shipped others to New Orleans on their own ships.

Franklin and Armfield also sold enslaved women and teenage girls to men who bought them in order to rape them, to force them into sex. This was known as the "fancy" trade, and the traders charged much higher prices for these women and girls. The historian Joshua Rothman read the partners' own letters. He found that Franklin, Armfield and their partner Rice Ballard also raped some of these women themselves, and joked about it in their letters.

Families. Traders and buyers broke families apart as a matter of routine. The Equal Justice Initiative, a group that studies racial injustice, has added up the effects. Its researchers found that traders separated nearly half of the people they moved from a husband, a wife or a parent. About one enslaved marriage in three ended because an owner sold the husband or the wife away. Traders sold children away from their mothers.

Work, food and home

Enslaved people worked from sunup to sundown, six days a week. On cotton and sugar plantations, an overseer, a man the owner paid to run the work, drove them in gangs, row by row across the fields. This was called the gang system.

On the rice plantations of the Lowcountry, the coast around Charleston, the overseer gave each person a set task for the day. When it was done, the rest of the day was the person's own. This was called the task system.

Two hundred pounds. Solomon Northup, a free New Yorker whom two men drugged and sold into slavery in 1841, picked cotton in Louisiana in the 1840s. His own story comes later in this era. In his book *Twelve Years a Slave* (1853), he wrote: "An ordinary day's work is two hundred pounds."

Every night each person's cotton was weighed at the gin-house, the building where the cotton gin stood. When a person's cotton came up short, the owner or the overseer whipped that person.

On the plantation where Northup was held, the owner, Edwin Epps, did the whipping himself in the yard. Northup wrote that "Twenty-five are deemed a mere brush," fifty lashes were "the ordinary penalty," and "one hundred is called severe."

Food and clothes. Frederick Douglass lived as an enslaved child on a plantation in Maryland. In his *Narrative* (1845), he recorded the food allowance there for one month: eight pounds of pork, or the same amount of fish, and one bushel, about a big basketful, of cornmeal.

The enslaver gave out clothing once a year. It was so thin that small children too young for field work often had only a long shirt to wear. Enslaved people added to their food with gardens, fishing and trapping where their owners allowed it.

Formerly enslaved people interviewed in the 1930s remembered cabins of one room, with dirt floors and beds made of boards. Under the law, their marriages did not count. An owner could sell a husband, a wife or a child on any day.

Punishment

Owners and overseers flogged enslaved people for picking too little cotton, for arriving late and for running away. A flogging meant being tied and whipped again and again on the bare body. Each stroke cut the skin, and deep or repeated whipping left raised scars for life. The whip named most often in Southern sources is the cowskin, also called the cowhide, a whip made of dried, twisted cow's hide.

Frederick Douglass described a flogging he saw as a child. His enslaver, Aaron Anthony, stripped Douglass's Aunt Hester from her neck to her waist. He tied her crossed hands to a hook in a ceiling beam, so that she stood on her toes. Then he whipped her with "the heavy cowskin" until her blood dripped to the floor.

The enslavers' own words. Theodore Weld, an abolitionist, one of the people working to end slavery, published *American Slavery As It Is* in 1839. It collected reports from eyewitnesses and advertisements from Southern newspapers. One witness saw an overseer whip a man named Joe "with a heavy cow-hide full six feet long" for arriving five minutes late.

The enslavers wrote the advertisements themselves. Micajah Ricks of Nash County, North Carolina, placed one in the Raleigh *Standard* on July 18, 1838. It began: "Ranaway, a negro woman and two children," using negro, an old word for a Black person. The ad went on: "a few days before she went off, I burnt her with a hot iron, on the left side of her face, I tried to make the letter M." The advertisement does not give the woman's name.

In another advertisement that Weld printed, Robert Nicoll of Mobile, Alabama, offered ten dollars for "my woman Siby, very much scarred about the neck and ears by whipping."

Sexual violence

By law, an enslaved woman or girl had no right to refuse the man who owned her, and no right to defend herself from him. Harriet Jacobs later wrote a book about what one of those men tried to do to her.

Jacobs was born enslaved in Edenton, North Carolina, in 1813. In 1825 she was left in a will to a three-year-old girl, Mary Matilda Norcom. That put her under the control of the girl's father, Dr. James Norcom. From her early teens, Norcom tried again and again to rape her. He held her as property, and for years she could not get away from him.

To protect herself she began a relationship with a white lawyer, Samuel Tredwell Sawyer, and had two children with him, Joseph and Louisa.

Seven years in a crawl space. Jacobs ran away in 1835. She hid in a crawl space above a storeroom in her grandmother's house for seven years. The space was so small she could barely sit up. In 1842 she escaped north by boat. Ten years later the wife of her employer, Cornelia Grinnell Willis, bought her freedom.

Jacobs told her story in *Incidents in the Life of a Slave Girl*, dated 1861 and published under the name "Linda Brent." Historian Jean Fagan Yellin later proved from Jacobs's letters that she wrote it herself. She died in Washington, D.C., in 1897.

Churches and hush harbors

Richard Allen was born enslaved. In Philadelphia he founded the African Methodist Episcopal Church, known as the AME Church, which was formally organized in 1816. It was the first Black church denomination, a group of churches that share the same beliefs and leaders, in the country.

In the slave states, enslaved people also worshipped out of their owners' sight. They met in brush arbors, shelters built of branches, which they called "hush harbors." They sang spirituals, religious songs that expressed their faith and also had hidden messages in them.

BY 1830: Cotton planters were moving onto Choctaw land in Mississippi, Franklin and Armfield were marching people south from Alexandria, and Georgia's lawmakers had just passed a new law against teaching enslaved people.

Reading made a crime

The first laws against teaching enslaved people named writing, not reading. South Carolina's lawmakers banned teaching enslaved people to write in 1740, and Georgia's followed in 1755.

Later laws banned teaching reading too. Virginia's lawmakers banned enslaved people in 1819 from any school "for teaching them reading or writing." Between 1829 and 1834 lawmakers in Georgia, Louisiana, North Carolina, Virginia, Alabama and South Carolina passed new laws or made their old ones harsher. North Carolina's law of 1830 forbade any free person to "teach or attempt to teach any slave within this State to read or write." Most of the slave states had such a law, but not all of them.

The punishments. The punishments depended on who broke the law. A white person who taught was fined. In North Carolina the fine was up to $200. The laws of Georgia and other states added jail. For Black teachers and learners, the laws ordered whipping. North Carolina's law set the punishment at 39 lashes, and in Virginia a person caught at a school could get up to 20 lashes. Several of these laws covered free Black people as well as the enslaved.

Enslavers had several fears. With writing, an enslaved person could forge, or fake, a pass, a note from an owner that allowed an enslaved person to travel. Readers could get the newspapers of the abolitionists, and they could read the Bible without an owner choosing which parts they heard. After Nat Turner, an enslaved preacher, led a revolt in Virginia in 1831, enslavers feared organized revolt most of all.

Enslaved people learned to read anyway. Estimates reported by Encyclopedia Virginia and History.com say about 5 to 10 percent of enslaved people could read by 1860. Some estimates for the last years of slavery run as high as 20 percent.

Frederick Douglass

Who: A man born enslaved who learned to read in secret against the orders of the man who held him, escaped, and became the most heard abolitionist of his century.

When and where: Born in Talbot County, Maryland, in February 1818. Escaped on September 3, 1838. Died in 1895.

Movie: *Becoming Frederick Douglass* (2022), a documentary by Stanley Nelson, shown on PBS. It is about Douglass's life.

He traded bread with hungry white boys for reading lessons.

As a boy Frederick Douglass lived in Baltimore, in the household of Hugh and Sophia Auld. Sophia Auld began to teach him the alphabet. Hugh Auld stopped her. Teaching a slave to read was against the law, he told her, and reading "would forever unfit him to be a slave."

Douglass saw in those words how white men kept Black people enslaved: by keeping them from learning. "From that moment," he wrote in his *Narrative* (1845), "I understood the pathway from slavery to freedom." In a later book, in 1855, he called Auld's words "the first decidedly anti-slavery lecture" he ever heard.

He kept learning in secret, paying hungry white boys with bread for lessons. At about age twelve he bought a book of speeches called *The Columbian Orator*.

When Douglass was a teenager, the man who owned him, Thomas Auld, took him away from Baltimore to St. Michael's, Maryland. Thomas Auld hired him out for a year to Edward Covey, a farmer known for "breaking young slaves." Covey whipped him until his back bled. After six months Douglass fought back. They fought for nearly two hours, and Covey never whipped him again.

Thomas Auld later sent him back to Baltimore to learn a trade in the shipyards. Douglass earned $1.50 a day, and every Saturday night he had to hand "every cent" of it to Hugh Auld. He escaped on September 3, 1838. He rode a train north through Baltimore, dressed in a sailor's clothes and carrying a sailor's papers that he had borrowed. Anna Murray, a free Black woman, helped him, and he married her.

He published his *Narrative of the Life of Frederick Douglass* in 1845, and two years later he started a newspaper, the *North Star*. For the rest of the century he spoke and wrote against slavery. More people heard him speak than any other abolitionist, and he was photographed more than any other abolitionist.

No record gives the day he was born, in February 1818. He chose February 14 as his birthday. He died in 1895.

Milla Granson (Lilly Ann Granderson)

Who: An enslaved woman who ran a secret night school in Natchez, Mississippi, and by her own count taught hundreds of people to read and write.

When and where: Born in Petersburg, Virginia, about 1816. Enslaved in Kentucky, then near Natchez, Mississippi. Died in 1889.

Her school opened at midnight and closed at two in the morning.

Milla Granson ran a school at night in Natchez, Mississippi, while she was enslaved. Her pupils were enslaved people who came after a full day's work. In March 1864 she told her story to Laura Haviland, a teacher who belonged to the Quakers, a Christian group. Haviland printed it in a book in 1881, and it is the only record of the school.

Granson was enslaved in Kentucky, where the children of her enslaver taught her to read and write. When that enslaver died, she was sold south to an owner near Natchez and put to work in the fields. She could not keep up with the field hands, and the overseer whipped her. She told Haviland: "sometimes I thought I would die from such cruel whippings upon my bared body." The record does not give the overseer's name.

The school opened "at eleven or twelve o'clock at night, and closed at two o'clock A. M." Every door and window was shut so no one would find out. Her pupils came into a back alley, "each with a bundle of pitch-pine splinters for lights."

She taught twelve at a time. When those twelve could read and write, she sent them off and took twelve more, "until she had graduated hundreds." Some of her pupils wrote their own passes and set out for Canada. She never heard from them again, and she believed they got away.

After slavery ended, she pushed for the founding of Natchez Seminary, the school that became Jackson State University. She married Oliver Granderson. She died in 1889 and is buried in the Natchez City Cemetery.

Resistance and revolt

Enslaved people resisted slavery in small ways every day. They worked slowly, broke tools, pretended to be sick and took food. Most of all, they ran away.

Southern newspapers were full of advertisements for runaways, placed by enslavers who wanted them back. Some of the ads warn that the person "can read," or "can write and may forge a pass." Researchers at Cornell University, in a project called Freedom on the Move, are putting thousands of these ads online.

Gabriel, 1800. An enslaved blacksmith named Gabriel planned to lead a march on Richmond, Virginia, in 1800. On August 30, the day it was to begin, two enslaved men, Pharoah and Tom, who belonged to members of the Sheppard family, told Mosby Sheppard about the plan. Sheppard told Governor James Monroe. A storm that day scattered those who had gathered.

Virginia judges, sitting in special courts with no juries, found 26 men guilty, Gabriel among them. On the judges' orders, all 26 were hanged. Encyclopedia Virginia's account does not name the men who carried out the hangings.

The German Coast, 1811. Charles Deslondes led an uprising in January 1811 on the German Coast of the Territory of Orleans, upriver from New Orleans. Between 64 and 125 enslaved people began it, and accounts put the number who joined at 200 to 500. They marched about twenty miles down the river road toward New Orleans and killed two white men, Gilbert Andry and François Trépagnier.

Governor William C. C. Claiborne, General Wade Hampton and a militia, a band of armed local men, put the uprising down. Hampton led U.S. Army troops and Navy sailors. Manuel Andry and Charles Perret led the militia, which was made up of planters. On January 10, in about half an hour of fighting, they killed 40 to 45 rebels. They killed others without any trial as they caught them.

Over the next two weeks, planters sat as judges in special courts called tribunals. Planters and U.S. officials put 44 more captured rebels to death. About 95 rebels were killed in all. The militia cut off the heads of the dead and

set nearly 100 of them on poles along the levee, the river wall, from New Orleans upriver. Deslondes was captured on January 11 and given no trial. Militiamen cut off his hands, shot him several times and burned him alive.

Charleston, 1822. White officials in Charleston, South Carolina, accused Denmark Vesey, a free Black carpenter, of planning a revolt to free thousands of enslaved people. No revolt ever took place, and no white person was harmed. Officials in Charleston put Vesey and 34 other men to death.

The officials claimed the city's African Methodist Episcopal church had helped plan the revolt. They ordered its members to scatter and destroyed the church. The Equal Justice Initiative says they burned it to the ground, and other accounts say it was torn down. Black Charlestonians rebuilt their church. In 1834 Charleston's white city leaders outlawed independent Black churches, the historian Manisha Sinha writes. The accounts do not name the officials who passed that ban. The congregation later became the church known today as Mother Emanuel.

Nat Turner

Who: An enslaved preacher who led the deadliest slave revolt in United States history.

When and where: Southampton County, Virginia, August 21 to 23, 1831. Hanged on November 11, 1831.

Movie: *The Birth of a Nation* (2016), a feature film about Turner's revolt. Its makers took the title on purpose from a 1915 film by D. W. Griffith that inspired a new Ku Klux Klan.

After his three-day revolt, white militias killed more than 100 Black people.

Nat Turner was an enslaved preacher in Southampton County, Virginia. In jail after the revolt, he told a lawyer, Thomas R. Gray, that "the Spirit" had appeared to him in 1828. It told him that "the first should be last and the last should be first," and that signs in the sky would show him when to begin. When the sun went dark in an eclipse in February 1831, he took it as the sign to "slay my enemies with their own weapons."

Turner and a few other enslaved men began on the night of August 21, 1831. They killed the man who held him, Joseph Travis, and Travis's family as they slept. Then they went on from farm to farm, killing white men, women and children.

More men joined them until they were fifty or sixty, on horseback and armed. They headed for Jerusalem, the town where the county court met. About three miles from the town, armed white men fired on them, and the rebels scattered. By August 23 the rebels had killed at least 55 white people.

Militias and bands of armed white men answered by killing more than 100 Black people. Many of them had nothing to do with the revolt. Some estimates put the number killed at 200.

Turner hid until he was caught on October 30. Judges in Southampton County put about 30 people on trial, and Virginia officials put to death 19 of those found guilty. They hanged Turner on November 11, 1831. Gray published Turner's account that year as *The Confessions of Nat Turner*.

Lawmakers in Virginia and other slave states answered the revolt with new laws against teaching enslaved people to read and against their gathering in groups.

Sengbe Pieh (Joseph Cinqué)

Who: A Mende man who led the captives of the ship *Amistad* in taking control of it, and who won freedom for them in the United States Supreme Court.

When and where: From the Mende country, in the region of Sierra Leone in West Africa. Took the ship on July 1, 1839. Freed by the Supreme Court in 1841. Home in 1842.

Movie: *Amistad* (1997), a feature film about the case.

Fifty-three captives took over the ship that was carrying them.

Spanish slave traders had carried Sengbe Pieh and the others, most of them Mende people from the region of Sierra Leone, to Cuba. There the 53 were bought for a Cuban plantation and put aboard a sailing ship, the *Amistad*. On July 1, 1839, led by Pieh, they rose up and took control of the ship. Pieh is also known as Joseph Cinqué.

In late August 1839, the U.S. Navy ship *Washington*, commanded by Lieutenant Thomas R. Gedney, seized the *Amistad* off Long Island. Officials took the Africans to New London, Connecticut, and jailed them. Some died at sea or in prison while they waited for trial.

Their lawyers took the case all the way to the Supreme Court, and former president John Quincy Adams argued part of it for them. On March 9, 1841, the justices set them free. In 1842, 35 of the survivors sailed home to Sierra Leone. They had never been enslaved on American soil.

Solomon Northup

Who: A free man from New York who was kidnapped and held in slavery in Louisiana for twelve years, and who named the people who did it.

When and where: Kidnapped in Washington, D.C., in 1841. Freed on January 4, 1853. Where and when he died is not known.

Movie: *12 Years a Slave* (2013), directed by Steve McQueen. It won the Academy Award for Best Picture.

He woke up "alone, in utter darkness, and in chains."

Solomon Northup was born free in Minerva, New York, in 1808, and worked as a farmer and a fiddler. In 1841 two white men who called themselves Merrill Brown and Abram Hamilton offered him work playing music for a circus. They took him to Washington, D.C. There they gave him a drink that left him "insensible," or unconscious.

He woke in a slave pen, a building where traders locked up enslaved people until they sold them. It was Williams's slave pen, called the Yellow House, near the U.S. Capitol. A slave trader named James H. Burch held him there. When Northup said he was a free man, Burch beat him with a paddle, "a piece of hard-wood board, eighteen or twenty inches long," and with a cat-o'-nine-tails, a whip of rope with many strands, each with a knot tied at the end.

Burch told Northup that if he ever again said he was free, what came next would be worse. "He swore that he would either conquer or kill me."

Twelve years. In New Orleans, Burch's partner, a trader named Theophilus Freeman, told Northup that his name was now Platt. Freeman sold him under that name to William Ford, a preacher who owned a lumber mill. In the winter of 1842 Ford sold him to a carpenter, John Tibeats. Then Edwin Epps held him for almost ten years on a cotton plantation on Bayou Boeuf, in Louisiana.

The letter. A Canadian carpenter named Samuel Bass mailed letters for Northup to people he knew in Saratoga Springs, New York. Two storekeepers there sent one on to Northup's wife, Anne. She went to

Henry B. Northup, a lawyer who had been Solomon's friend since childhood, and whose family had once enslaved Solomon's father, Mintus, and freed him in a will. With help from the governor of New York, Washington Hunt, Henry B. Northup won Solomon's freedom on January 4, 1853.

No one punished. That year Northup published *Twelve Years a Slave*, which sold about 30,000 copies. Burch was tried in Washington in 1853 and found not guilty, because the law did not allow Northup, a Black man, to testify against a white man. In May 1857 the New York case against the kidnappers ended without a trial, in a dispute over which court had the power to hear it. No one was ever convicted.

NOT KNOWN: where and when Northup died. After about 1857 he disappears from the records.

BY 1840: Lawmakers in Virginia and other slave states had answered Nat Turner's revolt with new laws against reading and gathering, the *Amistad* captives were waiting in Connecticut for the Supreme Court, and thousands of enslaved people were escaping north anyway.

The Underground Railroad and the abolitionists

The Underground Railroad was a network of people who helped enslaved people escape to the free states and to Canada, with guides and safe houses. Free Black communities and their churches did much of the work. By an estimate the National Park Service uses, it helped about 100,000 people escape over several decades. Many more tried to escape and were caught.

In the free states, Black people started their own churches, newspapers and groups. Samuel Cornish and John Russwurm published the first issue of *Freedom's Journal* in New York on March 16, 1827, the first newspaper in the United States owned by Black people. Free Black communities in the North ran vigilance committees, groups that hid people who had escaped and helped them move on.

Sojourner Truth

Who: A woman who walked away from slavery in New York, then sued to get back her son, who had been sold south against the law, and won.
When and where: Ulster County, New York. Walked to freedom in 1826. Won her son's return in 1828. Died in 1883.

Her five-year-old son was sold to Alabama. She sued and won.

She was born enslaved in about 1797 in Ulster County, New York, and her name then was Isabella Baumfree. In 1826 she walked away from her enslaver, carrying her baby daughter. One year later, on July 4, 1827, New York's law ending slavery took effect.

Her son Peter was five years old. A little before she walked away, her enslaver, John Dumont, sold the boy for $20 to a doctor, who sold him on to his brother, Solomon Gedney. Solomon Gedney sent Peter to his sister's husband, a planter in Alabama. Selling a child out of the state that way was against New York law. She did not find out for months.

She sued, and in 1828 she won his return. The Historical Society of the New York Courts says she was the first Black woman to win a case of this kind against a white man in an American court.

She took the name Sojourner Truth in 1843. Her life story, the *Narrative of Sojourner Truth*, was published in 1850.

Harriet Tubman

Who: A woman who escaped slavery in Maryland and went back about 13 times to lead about 70 people out.

When and where: Born in Dorchester County, on Maryland's Eastern Shore, in March 1822. Escaped in September 1849. Died in 1913.

Movie: *Harriet* (2019), a feature film about her life, with Cynthia Erivo as Tubman.

At 13 she was hit in the head with a two-pound weight.

She was born in March 1822 in Dorchester County, on Maryland's Eastern Shore, and her name then was Araminta Ross. In 1835, when she was 13, she was at a store when an enslaved person tried to escape. An overseer threw a two-pound weight. It hit her in the back of the head and broke her skull. For the rest of her life she had pain and "sleeping spells," times when she fell asleep suddenly. The accounts of her injury do not name the overseer.

She escaped to Philadelphia, Pennsylvania, in September 1849. Between 1850 and 1860 she went back into Maryland about 13 times and brought out about 70 people, most of them family and friends, among them her parents, Rit and Ben.

An old story says she made 19 trips and freed 300 people. The National Park Service calls that story a legend. The numbers 13 and 70 come from her own statements and the records.

During the Civil War she worked as a scout for the Union army, the army of the United States. She died of pneumonia in 1913, at about 91 years old, in a home for elderly Black people.

BY 1850

If you remember one thing: Cotton planters spread slavery onto land the Choctaw and Chickasaw had been forced to leave, traders sold about a million people south to work it, and enslaved people answered by running, learning to read in secret and rising up.

In 1830: Mississippi had 65,659 enslaved people.

In 1840: it had 195,211, three times as many, on the land the Choctaw and Chickasaw had lost.

Next: In 1854 federal troops marched Anthony Burns, a man who had escaped from slavery, through Boston to a ship that took him back to Virginia.

1850 to 1900

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Slavery ended in this era, and the people it had held spent the rest of the century fighting to keep the freedom they had won. In 1850 the members of Congress passed a law that forced people in the free states to help catch those who escaped.

Fifteen years later, after the Civil War, the 13th Amendment, a change added to the Constitution, forbade slavery, and about four million people became free. Many of them had already freed themselves during the war by escaping to the Union army, the army of the United States.

Freed people searched for family members who had been sold away. They married in law, chose their own names, built churches and schools, voted and served in Congress. White men answered with new laws that forced Black people to work, with whippings and burnings, and with massacres, the killing of many people who could not defend themselves. President Rutherford B. Hayes ordered the last federal troops out of the Southern statehouses in April 1877. That ended the period of rebuilding called Reconstruction.

The Fugitive Slave Act, 1850

The Fugitive Slave Act, a law about fugitives, people who had escaped, became law on September 18, 1850. Under it, federal officials called commissioners, who worked for the national government, decided whether a captured person was a runaway slave. There was no jury. The captured person was not allowed to testify, to speak as a witness in court.

A commissioner was paid $10 for ruling that a person was a slave and $5 for ruling that the person was free. Officials could force bystanders to help catch people. The law's rules held in the free states too, and free Black people in the North fled to Canada.

Anthony Burns. Anthony Burns escaped from slavery in Virginia by stowing away on a ship. Charles F. Suttle, the man who held him in slavery, came to Boston with William Brent to claim him. On May 24, 1854, the two men had Burns arrested there. After a hearing that lasted a week, federal troops marched him through the city on June 2, 1854, to a ship that took him back to Virginia. He passed crowds draped in black, the color of mourning.

Celia, 1855

A Missouri judge refused in 1855 to tell a jury that an enslaved woman had the right to fight off a man who was raping her. Her name was Celia. Robert Newsom, a farmer in Callaway County, Missouri, bought her in 1850, when she was 14 years old. From 1850 to 1855 he raped her again and again. She had two children, and at least one was his. In 1855 she was pregnant again.

Newsom told Celia on the night of June 23, 1855, that he would come to her cabin. She had put a large stick in the corner. When he came, she struck him on the head with it twice, and the second blow killed him. She burned his body in her fireplace.

The trial. Missouri's law made it a crime to force "any woman" into sex against her will, and it allowed a woman to use force to resist. Celia's trial began in October 1855. Her lawyer, John Jameson, asked Judge William Hall to tell the jury that Celia could be found not guilty if she had been resisting rape. Hall refused. That left the jury no way to treat the killing as self-defense.

A jury of twelve white men found her guilty of murder. Her lawyers appealed, asking a higher court to look at the case again. The judges of the Missouri Supreme Court turned the appeal down on December 14, 1855. Celia was hanged in Fulton, Missouri, on December 21, 1855. The accounts of her death do not name the officials who hanged her.

Missouri's governor pardoned her in 2024, at the request of her descendants.

Dred and Harriet Scott

Who: An enslaved couple who sued for their freedom and lost in the United States Supreme Court, whose justices ruled that no Black American could be a citizen.

When and where: Suit filed in St. Louis, Missouri, in 1846. Supreme Court decision on March 6, 1857. Freed in May 1857.

A chief justice wrote that Black Americans could never be citizens.

Dred Scott was born enslaved in Virginia in about 1800. Peter and Elizabeth Taylor Blow held him. The Blows moved to St. Louis in 1830 and sold him to Dr. John Emerson, an army surgeon.

Emerson took Scott to Fort Armstrong, in Illinois, in 1833, and then to Fort Snelling, in Wisconsin Territory, in 1836. Slavery was banned in both places. At Fort Snelling, Dred Scott married Harriet Robinson, who was also enslaved. They had two daughters, Eliza and Lizzie. Eliza was born on a steamboat in free waters. After Emerson died in 1843, his widow, Irene Emerson, claimed the family as her property.

Eleven years in court. The Scotts sued for their freedom in St. Louis on April 6, 1846. They argued that living where slavery was banned had made them free. A jury ruled in their favor on January 12, 1850. The judges of the Missouri Supreme Court reversed that ruling on March 22, 1852, and Irene Emerson held the Scotts as her property again.

The Scotts and their lawyers took the case to the United States Supreme Court. On March 6, 1857, the justices ruled against them, 7 votes to 2. Chief Justice Roger Taney wrote that Black Americans were not and could never be citizens, and that Black people "had no rights which the white man was bound to respect." The justices also ruled that Congress could not ban slavery in the territories of the West.

Freedom. Irene Emerson's second husband, Calvin Chaffee, handed ownership of the family to Taylor Blow, who freed them on May 26, 1857. Dred Scott worked as a porter, a man who carried guests' bags, at Barnum's Hotel in St. Louis, and he helped Harriet with her laundry work. He died of tuberculosis, a disease of the lungs, on September 17, 1858.

Sold and smuggled, 1858 to 1860

Pierce Mease Butler sold between 429 and 436 men, women and children at a racetrack near Savannah, Georgia, on March 2 and 3, 1859, to pay his gambling debts. The National Park Service describes it as the largest single sale of people in the United States. The slave dealer Joseph Bryan ran it and advertised it for weeks in newspapers from Charleston to Richmond. The sale brought in $303,850.

A reporter from the *New York Tribune*, Mortimer Thomson, went disguised as a buyer. He wrote a long article about the sale for the *Tribune*, "What Became of the Slaves on a Georgia Plantation?" Black Georgians remembered those two days as "the weeping time."

The *Wanderer*. Bringing captives from Africa had been against the law since 1808, but some men kept doing it. In 1858 Charles A. L. Lamar led a group of investors who paid for a ship called the *Wanderer* to bring captives from Africa. Its captain, William R. Corrie, bought 487 Africans at the Congo River. About 409 of them were still alive when the ship landed at Jekyll Island, Georgia, on November 29, 1858. The others died on the voyage. Six trials followed, and none of them ended in a conviction.

The *Clotilda*. Timothy Meaher, a businessman and shipbuilder in Mobile, Alabama, bet in 1860 that he could sneak a ship full of captives into Mobile Bay. Captain William Foster sailed the ship *Clotilda* to West Africa. He bought 125 people and carried 110 of them across the ocean. On July 8, 1860, the ship entered the Gulf of Mexico. Later that month Foster anchored it near Twelve-Mile Island and set it on fire to destroy the evidence. The only punishment anyone received was a $1,000 fine charged to Foster.

Oluale Kossola (Cudjo Lewis)

Who: A man carried to Alabama on the *Clotilda*, the last known slave ship to reach the United States, who helped found the town of Africatown after he was freed.

When and where: Born about 1841 in what is now Benin, in West Africa. Landed near Mobile, Alabama, in July 1860. Freed in 1865. Died in 1935.

Movie: *Descendant* (2022), a documentary by Margaret Brown about the *Clotilda*, Africatown and the families of the survivors.

The crossing took 45 days. He was one of 110 aboard.

Oluale Kossola was born in about 1841 in what is now the country of Benin, among the Yoruba Isha people. The army of Ghezo, the King of Dahomey, attacked his town in April 1860. Kossola was captured and held in a barracoon, a pen for captives, at the port of Ouidah. There he was sold to William Foster, the *Clotilda*'s captain. The crossing to Alabama took 45 days, and Kossola suffered from terrible thirst.

In Mobile, James Meaher, Timothy Meaher's brother, enslaved him. Kossola worked on a steamship and lived with others from the *Clotilda* under Meaher's house, which stood high off the ground. He was freed in 1865. He took the name Cudjo Lewis and married Abile, a woman who had also been on the *Clotilda*.

Africatown. The people from the *Clotilda* could not afford to sail home. Timothy Meaher was responsible for what had been done to them, so they decided to ask him to give them land, free. They chose Kossola to speak for them. Meaher refused. So they bought land from him and from others, and about 30 of them founded Africatown, north of Mobile.

Kossola and Abile had six children, and all six died young. In 1902 he was hurt in a train accident and sued the railroad company. He became the caretaker of the Africatown church.

The writer Zora Neale Hurston interviewed him in 1927. Her book about his life, *Barracoon*, was not published until 2018. Kossola died on July 26, 1935, at about 94. In May 2019 officials of the Alabama Historical

> Commission announced that the burned wreck of the *Clotilda* had been identified in the Mobile River.

BY 1861: The justices of the Supreme Court had ruled that no Black American could be a citizen, the *Clotilda* had landed its captives near Mobile, and Southern states had broken away from the United States.

Freedom taken during the war

Three enslaved men, Frank Baker, Shepard Mallory and James Townsend, rowed to Fort Monroe, a Union fort at Old Point Comfort in Virginia, on May 23 and 24, 1861. It was the same ground where the *White Lion* had landed Africans in 1619. The men were about to be taken south to build batteries, gun positions, for the Confederate army, the army of the Southern states that had broken away.

General Benjamin Butler refused to send them back. He called them "contraband of war," meaning property of the enemy that an army may seize. Word spread, and enslaved people walked, rowed and rode to Union lines across the South. By the end of the war, about half a million people had freed themselves by reaching the Union army. The people who escaped to Fort Monroe called it "Freedom's Fortress."

The camps. Union army officials, and later officials of the Freedmen's Bureau, the federal office set up to help freed people, ran camps for people who had escaped slavery. These were called contraband camps, and they were crowded and dirty.

More than 60,000 freed people died of smallpox between 1862 and 1868, according to the historian Jim Downs. Smallpox is a disease caused by a virus that spreads from person to person. It brings a fever and a rash that spreads over the skin and turns to scabs. About 3 in every 10 people who caught it died, and many who lived had scars for life, most of all on their faces.

Doctors knew how to prevent it. They could vaccinate people, giving them a shot that protects against the disease, and they could keep the sick away from the well, which is called quarantine. The quarantine tents for freed people were

made of old, ripped army cloth.

Philip Reid. Philip Reid was an enslaved craftsman at the foundry of Clark Mills in Washington, D.C., a workshop where metal is melted and cast. Mills owned him. Mills's workers were casting the Statue of Freedom for the top of the Capitol dome. The paid foreman demanded more money before he would take apart the plaster model, and Reid worked out how to separate it with a pulley and tackle. He was paid $1.25 a day, but he kept only the pay for Sundays. Mills took the rest.

Members of Congress passed a law that freed the enslaved people of Washington, D.C., on April 16, 1862, and Reid was one of them. The government of the District of Columbia counts 3,100 people freed there, and the US Senate's history office counts 2,989. Under the law, owners loyal to the Union could be paid up to $300 for each person freed. The statue was set on the dome on December 2, 1863. In his later life Reid signed his name Reed.

Robert Smalls

Who: An enslaved harbor pilot who sailed himself, his family and his crew out of slavery on a Confederate ship, and later served five terms in Congress.

When and where: Charleston, South Carolina, before dawn on May 13, 1862. Later Beaufort, South Carolina. Died in 1915.

He wore the captain's straw hat and sailed past Fort Sumter.

Robert Smalls was 23 years old and enslaved. He worked as the pilot, the man who steered the ship, on a Confederate transport ship called the *Planter*, in Charleston harbor.

Before dawn on May 13, 1862, Smalls took the *Planter* out with an enslaved crew and their families aboard. He wore the captain's hat and gave the right signals as they passed Fort Sumter. Out at sea he surrendered the ship to the Union navy ships that were blockading, or closing off, the harbor.

> Smalls went on to become a ship's captain in United States service. After the war he served as one of South Carolina's state lawmakers. He was elected to the United States House of Representatives five times. He died in 1915.

Who did the Emancipation Proclamation free?

President Abraham Lincoln's Emancipation Proclamation declared free the enslaved people in the states that were fighting against the United States. Emancipation means setting people free from slavery. Lincoln announced it on September 22, 1862, and it took effect on January 1, 1863. Wherever the Union army advanced after that, its soldiers freed the enslaved people they reached. In the proclamation, Lincoln also opened the army to Black men.

Lincoln left out the slave states that had stayed loyal to the Union: Delaware, Maryland, Kentucky and Missouri. He also left out Tennessee and the parts of Virginia and Louisiana that the Union army already held. Enslaved people in those places had to wait for their state's lawmakers to end slavery, or for the 13th Amendment.

Black soldiers and the scourged back

About 180,000 Black men served in the Union army in the regiments called the U.S. Colored Troops. By the end of the war, about one Union soldier in ten was Black. About 19,000 more Black men served in the navy. At the Battle of New Market Heights in Virginia, on September 29, 1864, fourteen Black soldiers earned the Medal of Honor, the highest award for bravery in battle.

Peter. A man named Peter escaped in March 1863 from the 3,000-acre plantation of Captain John Lyons in St. Landry Parish, Louisiana, and reached Union troops at Baton Rouge. Army doctors examined him before he joined the army. His back was covered in thick raised scars, called keloid scars, which grow over a healed wound.

Peter said the overseer, Artayou Carrier, had whipped him in October 1862. He said he lay in bed for two months, sore "from the whipping and salt brine" that was poured on the wounds. The accounts do not say who poured it.

The photographers William D. McPherson and J. Oliver photographed his back about April 2, 1863. The picture became known as "The Scourged Back," scourged meaning whipped. *Harper's Weekly* printed pictures made from the photograph on July 4, 1863, and called the man "Gordon." Historians now think the magazine's staff put "Gordon" together from pictures of three different men, and that the man in this photograph was Peter. No record shows what became of Peter after he joined the Union army.

William Harvey Carney

Who: A man born enslaved in Virginia who carried the flag at Fort Wagner with the 54th Massachusetts, a Black regiment. His actions there were the earliest act of bravery by a Black American to be recognized with the Medal of Honor.

When and where: Born in Norfolk, Virginia, on February 29, 1840. Fort Wagner, South Carolina, July 18, 1863. Died on December 9, 1908.

"The dear old flag never touched the ground!"

William Harvey Carney was born enslaved in Norfolk, Virginia, on February 29, 1840. Accounts differ about how he became free. Some say he escaped on the Underground Railroad. Others say his father escaped and then bought his freedom. The family settled in New Bedford, Massachusetts.

Carney joined Company C of the 54th Massachusetts, a regiment of Black soldiers, in February 1863, and he became a sergeant in March. On July 18, 1863, the 54th attacked Fort Wagner, in South Carolina. The soldier carrying the flag was shot.

Carney took the flag and went forward with several serious wounds. He planted it on the wall of the fort and then carried it back to the Union lines. The National Park Service quotes his words afterward: "Boys, I did but my duty," and then, "the dear old flag never touched the ground!" The attack appears in the film *Glory* (1989), which is about the 54th Massachusetts.

He received the Medal of Honor in May 1900. The National Park Service says his actions were "the earliest act of African American bravery to be recognized with the medal." He was one of 22 African Americans to receive it for service in the Civil War. He died at Boston City Hospital on December 9, 1908.

Combahee Ferry, Juneteenth and the 13th Amendment

Harriet Tubman guided Union gunboats and soldiers of the 2nd South Carolina Volunteers up the Combahee River in South Carolina on the night of June 2, 1863. More than 700 enslaved people escaped to freedom on the boats that night.

The 13th Amendment. The members of Congress passed the 13th Amendment on January 31, 1865. It became part of the Constitution on December 6, 1865, when lawmakers in enough states had approved it. Its text forbids slavery everywhere in the United States, "except as a punishment for crime whereof the party shall have been duly convicted." That exception still allowed forced labor for people convicted of crimes.

Juneteenth. Major General Gordon Granger arrived in Galveston, Texas, on June 19, 1865, with about 2,000 federal soldiers. He issued General Order No. 3. It said that "all slaves are free," with "absolute equality of personal rights." More than 250,000 people were still enslaved in Texas, two and a half years after the Emancipation Proclamation, because the Union army had not been there to enforce it. June 19 became a day of celebration, called Juneteenth.

The Five Tribes. Slaveholders in the Choctaw and Chickasaw nations, the nations that United States officials forced west to Indian Territory in the 1830s, had also held Black people in slavery. So had slaveholders in the Cherokee, Muscogee (Creek) and Seminole nations. Together these five nations were known as the Five Tribes. At emancipation as many as 2,500 people were enslaved in the Cherokee Nation.

The leaders of each of these nations signed treaties with the United States in 1866. In them the leaders agreed to free the people their nations had enslaved and to make them citizens of the nation. In Article 9 of the Cherokee treaty, Cherokee leaders promised the freed people, called Freedmen, "all the rights of native Cherokees."

BY 1866: Slavery was illegal everywhere in the United States, about four million people were free, and Union soldiers had announced freedom as far away as Galveston, Texas.

Freedom, day to day

For decades after slavery ended, freed people searched for family members who had been sold away. They placed advertisements in Black newspapers, under headings such as "Information Wanted," asking for news of mothers, children, husbands and wives. The searches ran into the 1900s. Few ended in a reunion, and when one did, people wrote about it with joy.

Researchers at Villanova University and Mother Bethel AME Church in Philadelphia, in a project called "Last Seen: Finding Family After Slavery," have put more than 5,000 of these ads online.

Marriages and names. Enslaved people's marriages had never counted in law. After the war, officers of the Freedmen's Bureau recorded thousands of marriages. Couples who had been together for decades stood up to be married in law. The Bureau's marriage registers survive at the National Archives.

Freed people chose and registered their own last names. Some kept a former owner's name, so that family members who had been sold away could find them. Others chose new names. Freeman was one.

Churches. Congregations that had met in hush harbors built their own churches. Within ten years of the war, freed people had joined or founded AME and AME Zion churches across the South. They used their churches as schools, as meeting halls and as places to organize for elections.

The bank. Members of Congress set up the Freedman's Savings Bank by law to hold the savings of freed people, and President Lincoln signed the law on March 3, 1865. The bank's managers ran it badly and lost money on risky deals. It failed in June 1874, holding nearly $3 million saved by about 61,000 people.

The officials in charge of closing the bank, called commissioners, paid back 62 percent of what was owed between 1875 and 1883. Only about 30,000 depositors turned in their bank books to collect, and about 31,000 never got anything back.

Thornton Copeland

Who: A freed man who placed a newspaper advertisement in 1865 searching for his mother, from whom he had been sold away 21 years earlier.

When and where: The *Colored Tennessean*, a newspaper in Nashville, Tennessee, October 1865. He had been sold in Richmond, Virginia, in 1844.

He had not seen his mother in 21 years.

Thornton Copeland was sold away from his mother in 1844. The buyer was Samuel Copeland, in Richmond, Virginia. The ad does not say who sold him. Before that, a man named Robert Rogers had held him.

In October 1865, after slavery ended, he placed an advertisement in the *Colored Tennessean*, a Black newspaper in Nashville, to look for her. Her name was Betty. His advertisement reads: "INFORMATION is wanted of my mother, whom I left in Fauquier county, Va., in 1844, and I was sold in Richmond, Va., to Saml. Copeland."

His advertisement is item 918 in the "Last Seen" collection. No record shows whether he ever found his mother.

Schools, the moment freedom came

People who had been forbidden to read started schools as soon as the Union lines reached them. They opened schools in churches, in army camps and in buildings the Union army had taken over. Grandparents and grandchildren learned in the same rooms. Grown men and women came to night classes after a full day's work.

Who paid. Freed people paid for schools themselves wherever they could. They paid school fees out of their first wages, and they gave land and their own labor to build schoolhouses. Northern aid societies helped, the American Missionary Association most of all. Thousands of teachers came south, many of them women, and a growing number of them Black.

Freedmen's Bureau officials had counted more than 2,600 day and night schools with about 150,000 students by 1870. Counts that include every kind of school are higher.

New colleges opened for freed people. Fisk, in Nashville, opened in January 1866. Members of Congress passed a law that set up Howard University in 1867. By 1872 there were about 25 colleges for freed people.

Delegates in the Southern states wrote new state constitutions in 1868 and 1869. Their conventions were the first public bodies in United States history with many Black members. The delegates created the South's first statewide systems of free public schools, open to white children too.

Susie King Taylor

Who: A girl who learned to read in secret schools in Savannah, escaped at 13, and became the first Black teacher of freed children in a school working openly in Georgia.

When and where: Born in Liberty County, Georgia, on August 6, 1848. Taught on St. Simons Island, Georgia, in 1862. Died in 1912.

At 14 she taught 40 children by day and adults at night.

As a child in Savannah, Georgia, Susie King Taylor went to two secret schools run by Black women. Two white young people also taught her, breaking the law to do it.

In April 1862, at 13, she escaped with others to St. Simons Island, which the Union army held. She turned 14 that August. There she became the first Black teacher of freed students in a school for freed people working openly in Georgia. She taught about 40 children by day and grown people at night, until the Union army moved everyone off the island in October 1862.

She married Edward King, a Black noncommissioned officer, a soldier who had risen through the ranks to lead other soldiers. She served with the 33rd U.S. Colored Troops as a nurse and laundress, and she taught soldiers to read. In 1902 she published a book about those years, *Reminiscences of My Life in Camp*. She died in 1912.

Memphis and New Orleans, 1866

White mobs in Memphis, Tennessee, attacked the city's Black neighborhoods in early May 1866. The mobs included city policemen, firemen and some businessmen. The attacks began after a clash between the police and Black Union soldiers who had just left the army. General George Stoneman had taken the soldiers' guns away. The mobs killed 46 Black people. Two white people also died.

Members of Congress formed a committee to look into the killings. Its members counted 75 people injured, 100 robbed and five women raped. The mobs burned 91 homes, 4 churches and 8 schools, every Black church and school in the city. No one was ever punished for the killings, as far as the records show.

Frances Thompson. Frances Thompson testified to the committee. She said that on May 1, 1866, seven white men broke into her home and raped her and her roommate, 16-year-old Lucy Smith. Accounts of her testimony differ on whether two or three of the men were police officers. The men also stole about $300.

Memphis police arrested Thompson in July 1876 for wearing women's clothing. Dr. Joseph Nuttall, who worked for the city, and three other doctors examined her body against her will and declared that she was a man. She could not pay the fine, and she was sentenced to 100 days on the chain gang, a crew of prisoners chained together and forced to work, cleaning streets.

Her opponents used the arrest to attack her testimony about the rapes. She died of dysentery, a disease of the gut, in the city hospital on November 1, 1876.

New Orleans. The men who wrote Louisiana's constitution of 1864 ended slavery in the state but did not give Black men the vote. In 1866 Republicans who wanted Black men to vote called these delegates back together to change the constitution. Mayor John T. Monroe declared the meeting unlawful and asked the army for help arresting the delegates.

The delegates met anyway, at the Mechanics' Institute in New Orleans, on July 30, 1866. About 200 unarmed Black men, most of them Union veterans, marched in support.

Sheriff Harry T. Hays had given police powers to many men, a lot of them former Confederate soldiers. These police attacked, with white mobs and firemen. By the National Park Service's count, they killed 34 Black people and wounded 119, and they killed 3 white delegates and wounded 17. The official report records higher totals: 38 killed and 146 wounded in all. General Philip Sheridan called it "an absolute massacre by the police."

BY 1868: Under the 14th Amendment, Black Americans born in the United States were citizens, and white mobs and police had killed dozens of Black people in Memphis and New Orleans.

Black political power

The 15th Amendment became part of the Constitution on February 3, 1870. It states that the right to vote "shall not be denied or abridged by the United States or by any State on account of race, color, or previous condition of servitude," where abridged means cut down and servitude means slavery.

In the ten years that followed, 16 Black men served in Congress, 14 in the House and 2 in the Senate. All of them were Republicans from the South. Joseph Rainey of South Carolina became the first Black member of the House of Representatives on December 12, 1870. Robert Smalls served five terms in the House.

Across the South, about 2,000 Black men held public office during Reconstruction, some elected and some appointed. The historian Eric Foner has found more than 1,500 of them by name and listed them in his book *Freedom's Lawmakers*.

Hiram Revels and Blanche K. Bruce

Who: The first two Black United States senators. Revels was the first to serve. Bruce, born enslaved, was the first to serve a full six-year term.

When and where: Both represented Mississippi. Revels served from 1870 to 1871. Bruce served from 1875 to 1881.

Senators argued he had not been a citizen long enough.

Hiram Revels was born free in Fayetteville, North Carolina. He was a minister in the African Methodist Episcopal Church and a teacher. During the Civil War he served as a chaplain, a minister to soldiers, for Black regiments. In 1869 he won a seat in the Mississippi state senate.

Mississippi's state lawmakers voted 85 to 15 on January 20, 1870, to send him to the United States Senate. Democrats in the Senate argued against seating him. Some said he had not been a citizen until the 14th Amendment was adopted in 1868, so he had not been a citizen long enough to serve. The senators voted 48 to 8 to seat him, and on February 25, 1870, he became the first Black United States senator.

Revels served until March 3, 1871. That year he became the first president of Alcorn University. He died in 1901.

Blanche K. Bruce. Bruce was born enslaved near Farmville, Virginia, and Pettus Perkinson held him in slavery. When the Civil War began, Bruce escaped to Lawrence, Kansas, where he taught school. After the war he moved to Mississippi, where he was a sheriff and a tax collector in Bolivar County and the county's superintendent of education.

He was elected to the Senate on February 4, 1874. He served from March 4, 1875, to March 3, 1881, the first Black senator to serve a full term. Later he served as Register of the Treasury, an office in the federal government. He died in Washington, D.C., in 1898.

The promises broken

Within a year of the war's end, President Andrew Johnson and Southern lawmakers had taken back much of what freed people had been promised. Johnson had become president in April 1865, when John Wilkes Booth shot and killed President Lincoln.

Forty acres. Union general William T. Sherman issued Special Field Orders No. 15 on January 16, 1865. In it, Sherman set aside about 400,000 acres of coastal land, from Charleston to northern Florida, for freed families in plots of forty acres. About 40,000 people settled on the land. In the fall of 1865, President Andrew Johnson reversed the order and gave the land back to its former owners. The phrase "forty acres and a mule" comes from this broken promise.

The Black Codes. Mississippi's lawmakers passed the first Black Codes in November 1865, and lawmakers in other Southern states followed. Under the new vagrancy laws, which made it a crime to have no job or home, officials could arrest a Black person who could not prove he had a job. Under apprenticeship laws, officials could force Black children to work for white employers. Other laws required licenses to work in certain jobs. Officials hired out the people they arrested under these laws to work for others.

Citizens. The members of Congress answered with the Civil Rights Act of 1866. President Johnson refused to sign it, and they made it law anyway on April 9, 1866. Under it, people born in the United States were citizens, except Native people who paid no taxes. Black citizens had the same right as white citizens to make contracts, go to court and own property.

The members of Congress also approved the 14th Amendment, which became part of the Constitution on July 9, 1868. It states that people born in the United States are citizens, and that no state may deny any person "the equal protection of the laws," the same protection the laws give everyone else.

Sharecropping. Without land of their own, many freed families became sharecroppers. A sharecropper farmed a landlord's fields in exchange for a share of the crop. The landlord lent the family tools, seed and food, charged at

the plantation store's prices, and took the cost out of the family's share. The debt carried over from year to year, and families could not leave the land until it was paid.

Convict leasing

The 13th Amendment allowed forced labor as a punishment for crime, and Southern officials used that exception. Sheriffs arrested Black men, often for vagrancy or small crimes under the new codes, and county judges sentenced them. State officials then leased the prisoners, or rented them out, to planters, mine owners and railroads. The prisoners worked without pay. This was called convict leasing.

Alabama. Alabama officials leased the state's prisoners to companies from 1875 to 1928. State officials signed the leases, and company managers ran the prison camps. Officials sent most of the prisoners to coal mines run by companies such as the Tennessee Coal, Iron and Railroad Company, Sloss Iron and Steel, and Pratt Coal and Iron.

More than 95 percent of the county prisoners and 90 percent of the state prisoners were Black. In one count from the 1890s, 181 of 745 county prisoners in the mines died within two years, about one in four. County prisoners also had to work off their court costs, which made their sentences longer. In 1912 Alabama's state government took in more than $1 million from prison mining.

Terror

Confederate veterans founded the Ku Klux Klan in Pulaski, Tennessee, in December 1865. Klansmen, armed and in disguise, whipped freed people and the white people who worked with them, burned their homes and murdered them. They attacked mostly at night.

Schools and teachers. Klansmen went after schools and teachers. In late 1869, at Greensboro, Georgia, they drove away a teacher named R. H. Gladding. Abram Colby, a Black man, had just been elected to Georgia's state lawmaking body, and he supported the school. Klansmen took him out of his house and beat him nearly to death.

In Alabama, Klansmen burned many Black schools and churches. On July 11, 1870, at Cross Plains, Alabama, they killed a teacher of freed people, William Luke, a preacher from Ontario in Canada. Accounts say six or eight people were killed there.

Under the Enforcement Acts of 1870 and 1871, federal officials broke up the Klan for a time.

Colfax, 1873. White men carried out a massacre at Colfax, Louisiana, on Easter Sunday, April 13, 1873. The historian Eric Foner calls it the bloodiest single act of racial violence in Reconstruction. After a disputed election, Black militiamen and Republicans held the courthouse of Grant Parish, a parish being what other states call a county.

Christopher Columbus Nash, who claimed to be the sheriff, led more than 150 armed white men against them. Other accounts say about 300. The attackers fired a small cannon, set the courthouse on fire and shot the men who came out. That night some of the attackers killed their prisoners, most of them with shots to the head. Estimates of the number of Black men killed run from about 60 to 153. Two white men died.

A jury found three of the attackers guilty of conspiracy, planning a crime together. On March 27, 1876, in the case *United States v. Cruikshank*, the justices of the Supreme Court threw out the convictions. They ruled that the

rights at stake protected people only against actions by governments, not against private men. No one was ever punished for the killings.

Louisiana officials put up a marker in 1950 calling it a riot in which "three white men and 150 Negroes were slain." It was taken down in 2021.

The end of Reconstruction, 1877

The presidential election of 1876 ended in dispute. Five senators, five representatives and five Supreme Court justices decided it. They voted 8 to 7, Republicans against Democrats, to give the Republican, Rutherford B. Hayes, every disputed vote.

Democrats in Congress then stopped blocking the count. In return, Republicans agreed to end the army's control over the South. No written copy of this deal has ever been found. It is called the Compromise of 1877.

President Hayes then ordered the last federal troops out of the Southern statehouses. The last ones left the Louisiana statehouse in New Orleans on April 24, 1877. The 13th, 14th and 15th Amendments stayed in the Constitution, but federal officials stopped enforcing them in the South.

Plessy. The justices of the Supreme Court ruled 7 to 1 in *Plessy v. Ferguson*, on May 18, 1896, that states could keep Black and white people separate by law, under the rule known as "separate but equal." Justice Harlan was the only one to disagree. With that ruling, the justices upheld the segregation laws, laws separating people by race, that Southern lawmakers had already passed.

BY 1900

If you remember one thing: Hundreds of thousands of enslaved people freed themselves during the war, the 13th Amendment ended slavery in 1865, and within twelve years white lawmakers, mobs and judges had taken back much of what freedom had promised.

- January 1, 1863: Lincoln's Emancipation Proclamation takes effect.
- December 6, 1865: The 13th Amendment becomes part of the Constitution.
- April 24, 1877: The last federal troops leave the Louisiana statehouse.

Next: Sheriffs, judges and mine owners kept Black people working without pay well into the 1900s, and white mobs lynched thousands of Black people, killing them with no trial.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 8 MINUTES

Southern lawmakers, sheriffs, judges and company managers spent the first half of the 1900s running a system that took slavery's place. Lawmakers passed laws that kept Black and white people apart, and they took the vote from most Black men. Sheriffs arrested Black men for small crimes, judges sentenced them, and state officials rented them to mine companies, whose managers worked them without pay. White mobs lynched Black people by the thousands. Researchers at the Equal Justice Initiative have counted more than 4,400 such killings between 1877 and 1950.

The people who had been born into slavery were old by the 1930s. Writers paid by the federal government wrote down the life stories of more than 2,300 of them between 1936 and 1938.

Segregation and the stolen vote

Jim Crow. The justices' 1896 ruling in *Plessy v. Ferguson*, told at the end of the last era, let states keep Black and white people apart by law. Lawmakers across the South then passed laws that kept them apart in schools, railroad cars, waiting rooms, courtrooms and cemeteries, and at water fountains. This segregation, the separating of people by race, was written into what people called the Jim Crow laws.

The stolen vote. Southern lawmakers also took the vote from most Black men. The men who wrote Mississippi's constitution of 1890, the state's set of basic laws, set the pattern with three rules for voters. One was a poll tax, a fee a man had to pay before he could vote. Another was a literacy test, a reading test.

The third was the understanding clause, and it let one local officer decide who passed. Under it, a man had to be able to read any part of the state constitution. A man who could not read could still sign up to vote if the officer judged that he understood a part read aloud to him, or gave "a reasonable interpretation" of it.

Henry Williams. In Mississippi only qualified voters could sit on a jury, and officials made their jury lists from the books of registered voters. So the same rules that kept Black men from voting kept them off juries. An all-white jury found Henry Williams, a Black man, guilty of murder, and the judge sentenced him to be hanged.

Williams's lawyer took the case to the Supreme Court. In 1898 he told the justices that the officer who signed up voters was "the sole judge" of each test, so the officer could turn away any man he chose. The justices ruled against Williams.

Much later, in a 1965 case about the same rules, lawyers for the United States government told the Supreme Court that Black men had once been more than half of Mississippi's qualified voters. By 1899 they were about 9 percent.

The grandfather clause. The men who wrote Louisiana's new constitution added a grandfather clause in 1898. It let a man vote if his father or grandfather had voted before January 1, 1867. At that time most Black

Southerners had been enslaved, so the clause let white men who could not pass the tests vote anyway.

Louisiana had about 130,000 registered Black voters in 1896. By 1900 there were 5,320. By 1904, by the count of Equal Justice Initiative researchers, there were about 1,000.

Lynching

Researchers at the Equal Justice Initiative have documented more than 4,400 lynchings, killings by a mob with no trial, of Black people carried out as racial terror between 1877 and 1950. Other counts, which use different years and different rules for what counts, give different totals. Mobs hanged people and shot them.

Ed Johnson. An all-white jury in Chattanooga, Tennessee, found a Black man named Ed Johnson guilty of rape in 1906, though about a dozen witnesses said he was somewhere else. His lawyers asked the Supreme Court to look at the case again, and the justices stopped his execution while they did.

That night the sheriff, Joseph F. Shipp, took away the jail's usual guard and left only the night jailer. A mob broke into the jail, took Johnson to a bridge over the Tennessee River, hanged him from a beam and shot him. In 2000 a Tennessee judge threw out the guilty verdict.

White mobs used lynching to frighten Black people into staying in low-paid work and into obeying white Southerners. Black Southerners left the South to get away from the terror, in a movement known as the Great Migration.

Forced labor after slavery

Sheriffs, judges, state officials and company managers kept Black people working without pay well into the 1900s, through convict leasing and debt peonage. Under convict leasing, state officials rented prisoners out to companies, and company managers worked them without wages. Debt peonage meant forcing a person to work to pay off a debt.

Douglas Blackmon told this history in his book *Slavery by Another Name*, which won the Pulitzer Prize, one of the top American prizes for writing. In 2012 PBS showed a film made from the book.

The mines. Alabama's convict leasing, told in the last era, went on until 1928. In 1911, 123 Black county prisoners whom officials had leased to the Banner Mine were killed in an explosion there.

A prisoner named James Knox, from Mobile, died at the Flat Top mine in 1924. The first report of his death stated that he had killed himself by swallowing poison. An autopsy, an examination of the body after death, showed something else, according to a City of Birmingham article. He had died of heart failure, and the poison had been put into his body with a needle after he was dead.

The same article states that the injuries that killed him came from being whipped with a steel wire and lowered into hot water. The historian Mary Ellen Curtin writes that he was tortured by being lowered into a vat, a large tub, of boiling water.

Both accounts state that Knox was tortured to death, and they differ only on whether the water was hot or boiling. Neither account names who whipped him and put him in the water, who put in the poison, or who first called his death a suicide.

Alabama officials ended convict leasing on June 30, 1928. Alabama was the last state to end it.

Alonzo Bailey. Alonzo Bailey, a farm worker in Alabama, signed on to work for the Riverside Company for $12 a month and took $15 in advance, paid before the work was done. After about a month he left. Under Alabama law,

leaving a job after taking an advance was a crime, and the law barred him from telling the court why he had left. He was fined $30 and sentenced to 136 days of hard labor.

Booker T. Washington secretly helped pay for Bailey's lawyers. On January 3, 1911, in *Bailey v. Alabama*, the justices of the Supreme Court ruled 7 to 2 that the law broke the 13th Amendment, the part of the Constitution, added in 1865, that forbids slavery.

Lawmakers in other Southern states kept laws like it for decades. The justices struck down Georgia's law in 1942 and Florida's in 1944.

The last voices

Federal officials running the Federal Writers' Project, a government program that gave jobs to writers, sent interviewers to talk with people who had been born into slavery. Between 1936 and 1938 the interviewers worked in 17 states, wrote down more than 2,300 life stories and took 500 photographs. The Library of Congress keeps them in a collection called "Born in Slavery."

The Library of Congress also keeps sound recordings of people who were born into slavery, made between 1932 and 1975. The collection is called "Voices Remembering Slavery."

Fountain Hughes

Who: A man born into slavery in Virginia, one of the few people born enslaved whose own voice survives on a recording.

When and where: Born in Charlottesville, Virginia. Recorded by the Library of Congress in Baltimore, Maryland, on June 11, 1949.

"They'd sell us like they sell horses and cows and hogs."

Hermond Norwood of the Library of Congress interviewed Fountain Hughes in Baltimore on June 11, 1949, and the recording still exists. Hughes began: "My name is Fountain Hughes. I was born in Charlottesville, Virginia."

His grandfather, Wormley Hughes, was the enslaved head gardener at Monticello, Thomas Jefferson's plantation. Fountain Hughes said his father "got killed in the Army," and that left the children "to live on whatever people choose to give us."

He was never sold, he said, because "I was too young to sell." But he described how others were sold: "They'd sell us like they sell horses and cows and hogs and all like that. Have an auction bench, and they'd put you on up on the bench and bid on you."

He could not go anywhere without his enslaver's permission: "I couldn't go from here across the street, or I couldn't go through nobody's house without I have a note or something from my master." He said of being enslaved: "You're nothing but a dog. You're not a thing but a dog!"

NOT KNOWN: the year he was born. He said he was 101 years old, which would mean about 1848. Some reference books give 1854, and other accounts argue for a later year.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 4 MINUTES

Federal judges and lawmakers began to enforce the promises of the Civil War amendments in these fifty years, almost a hundred years after they were written.

The 14th and 15th Amendments, described in the 1850 to 1900 era, are changes added to the Constitution in 1868 and 1870. The 14th states that no state may deny any person "the equal protection of the laws," the same protection the laws give everyone else. Under the 15th, no one may be kept from voting because of race. For almost a hundred years, federal officials did not enforce them in the South.

The justices of the Supreme Court ruled in 1954 that public schools kept separate by race were against the Constitution. Members of Congress passed two laws in 1964 and 1965 that gave federal officials the power to enforce the amendments. After Alex Haley's book *Roots* came out in 1976 and ABC showed it on television in 1977, many Black families began tracing their ancestors back to slavery.

The amendments enforced

The justices of the Supreme Court all agreed in *Brown v. Board of Education*, on May 17, 1954, that "separate educational facilities are inherently unequal," unequal by their very nature. With that ruling, the justices threw out the "separate but equal" rule of *Plessy v. Ferguson* for public schools.

Two laws. The Civil Rights Act became law on July 2, 1964, and the Voting Rights Act on August 6, 1965. Under the Civil Rights Act, hotels, restaurants and other places open to the public could no longer turn people away or keep them apart because of race, and employers could no longer refuse people jobs because of race. Under the Voting Rights Act, no state or county could use any rule to keep a citizen from voting because of race.

With these laws, the members of Congress who voted for them gave federal officials the power to enforce the 14th and 15th Amendments. They came about a hundred years after the 13th Amendment forbade slavery.

Mississippi, 1995. Mississippi's state lawmakers voted in 1995 to approve the 13th Amendment, 130 years after it became part of the Constitution. No one filed the paperwork that would have made their vote official.

Roots and the search for family

An estimated 130 million people watched the *Roots* miniseries, a story told on television over several nights, when ABC showed it in January 1977. It was the largest audience a television series had ever had.

Alex Haley

Who: A writer whose book *Roots* told his family's story back to an enslaved African ancestor, and whose work led many Black Americans to search for their own families.

When and where: Born in Ithaca, New York, in 1921. *Roots* published in 1976. Died in Tennessee in 1992.

Movie: *Roots* (1977), a television miniseries on ABC based on Haley's book. It tells his family's story as drama. It is not a documentary about Haley.

A family story said an ancestor insisted "his real name was Kin-te."

Alex Haley spent part of his childhood in Henning, Tennessee, his mother's hometown. There his aunts and grandmothers told the family's stories. One was about an African ancestor named Toby, who "insisted that his real name was Kin-te."

Haley served twenty years in the United States Coast Guard, and he became a writer while he was in it. Later he traveled to Gambia, in West Africa. There a griot, a keeper of history who tells it from memory, told him the story of "his ancestor Kunta Kinte, captured by slave traders."

Roots came out in 1976 and won a special Pulitzer Prize in 1977. The television miniseries ran on eight nights in a row in January 1977 and won 145 awards, including nine Emmys, the top awards for television.

The challenges. In 1978 Haley agreed to pay the writer Harold Courlander $650,000 to settle a lawsuit claiming that Haley had copied passages from Courlander's novel *The African*. Professional genealogists,

people who trace family histories from records, have challenged links in the chain of records behind Haley's family story. Haley died in a Tennessee hospital in February 1992, of problems caused by diabetes.

NOT KNOWN: whether the records support every link Haley drew between his family and Kunta Kinte.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 10 MINUTES

Americans since 2000 have built museums about slavery, opened its records to anyone who wants to search them, and argued over whether to repair what it did. The Smithsonian's National Museum of African American History and Culture opened in Washington in 2016. In a project begun in 2015, volunteers made about 1.8 million names from the records of the Freedmen's Bureau, the federal office set up after the Civil War to help freed people, searchable online.

Lawmakers and city leaders have argued over payments or other repairs to the descendants of enslaved people. Federal officials removed or changed exhibits about slavery at some national parks in 2025 and 2026, and judges were still deciding in September 2026 whether that was lawful.

Museums, a holiday and a late vote

The Smithsonian's National Museum of African American History and Culture opened in Washington, D.C., on September 24, 2016. Its history galleries begin with the slave trade. Among the things on display is a cabin where enslaved people lived. The International African American Museum opened in Charleston, South Carolina, on June 27, 2023. It stands at Gadsden's Wharf, one of the main places where slave ships landed their captives.

Members of Congress passed the Juneteenth National Independence Day Act, and President Joe Biden signed it on June 17, 2021. Under it, June 19 is a federal holiday. It marks the day in 1865 when Union soldiers announced freedom in Galveston, Texas.

Two Mississippi residents found in 2012 that the paperwork for the state's 1995 vote approving the 13th Amendment had never been filed. It was officially recorded on February 7, 2013.

Celia's pardon. Celia was the enslaved girl Robert Newsom bought at 14 in 1850 and raped again and again until 1855, when she killed him. She was hanged that December, and the accounts do not name the officials who hanged her. On December 20, 2024, at the request of her descendants, Missouri's governor, Mike Parson, pardoned her, an official act that forgave her for the crime.

The records opened

For millions of Black Americans, the records of the Freedmen's Bureau are the first papers in which their ancestors appear by name, according to FamilySearch and the Smithsonian. The records include marriage registers, work contracts and school lists. Volunteers for FamilySearch, a genealogy group, typed up about 1.8 million names from them so that people could search them, in a project begun in 2015. The names can be searched at discoverfreedmen.org.

The GU272. The Maryland Jesuits, priests of a Catholic religious order, sold 272 enslaved people to plantations in Louisiana in 1838. They sold them in part to keep Georgetown College, now Georgetown University, from running out of money. It is one of the best-documented mass sales of people in American history. The people sold are now called the GU272.

Georgetown's leaders announced in 2016 that descendants of the GU272 would get an advantage when they applied to the university, and they renamed buildings. Genealogists have traced thousands of descendants, according to the Georgetown Memory Project in 2021.

The Jesuits and leaders of the descendants set up the Descendants Truth and Reconciliation Foundation in 2021, reconciliation meaning making peace after a wrong. The Jesuits promised to give $100 million to it. The Boston Globe reported that it is the largest pledge of its kind by a religious group in the United States.

The Cherokee Freedmen. The Cherokee treaty of 1866, told in the 1850 to 1900 era, promised the people Cherokee slaveholders had held in slavery, called Freedmen, "all the rights of native Cherokees."

In 2017, in the case *Cherokee Nation v. Nash*, a federal judge ruled that the treaty still bound the Cherokee Nation. Descendants of Cherokee Freedmen had a right to full citizenship in the nation. Cherokee leaders accepted the ruling. In 2021 the justices of the Cherokee Nation's own Supreme Court struck the words "by blood" out of its rules on citizenship.

Mélisande Short-Colomb

Who: A descendant of two people sold by the Maryland Jesuits in 1838, who enrolled at Georgetown University at the age of 63.

When and where: From New Orleans. Entered Georgetown University in Washington, D.C., in the fall of 2017.

She enrolled at Georgetown at 63, 179 years after the sale.

Mélisande Short-Colomb was a retired chef from New Orleans. She descends from Abraham Mahoney and Mary Ellen Queen, two of the 272 people the Jesuits of Maryland sold in 1838 to keep the college going.

She grew up hearing family stories from her grandmother and her great-aunts. In the summer of 2016 she learned that Mahoney and Queen had been part of the 1838 sale. In April 2017 Georgetown's leaders held a Liturgy of Remembrance, a religious service honoring the people who were sold. After that, she applied to the university, and she started there that fall.

At Georgetown she wrote a play, *Here I Am*, about her ancestors and the university.

Repair debated

Reparations are payments or other actions meant to make up for a wrong. Members of Congress have argued since 1989 over whether the United States should pay reparations for slavery.

H.R. 40. Representative John Conyers first introduced a bill called H.R. 40 in 1989. It would set up a commission, a group of people chosen to study slavery and its effects and to recommend ways to repair them, including a formal apology and payments. The number 40 comes from the forty-acre plots that General William T. Sherman set aside for freed families in January 1865, land President Andrew Johnson took back that fall.

The members of the House Judiciary Committee voted to move the bill forward for the first time in 2021. Representative Ayanna Pressley of Massachusetts introduced it again on January 3, 2025. As of September 2026, neither the full House nor the full Senate had voted to pass it, according to the records of Congress.

Evanston. The members of the city council of Evanston, Illinois, voted in 2019 to create the first reparations program paid for by a city. Council member Robin Rue Simmons led the effort. City officials set aside $10 million and in March 2021 approved its first grants of $25,000 for housing. The program is for Black residents who lived in Evanston between 1919 and 1969.

In those years Evanston's city officials used city laws to separate where Black and white people could live. The members of the city council passed Evanston's first zoning law, a law that sets what may be built in each part of a city, in January 1921, and Mayor Harry P. Pearsons signed it.

The law marked almost every block where Black people lived outside the west side of town for stores and businesses. As those blocks were turned over to businesses, the Black families living there were pushed out to the west side, the historian Andrew Wiese writes.

By August 2026, according to the Evanston RoundTable newspaper, 137 people had received a total of $3.47 million. A conservative group called Judicial Watch sued the city in May 2024, arguing that paying people based on

race breaks the 14th Amendment. On June 16, 2026, lawyers for the U.S. Department of Justice asked to join the lawsuit against the city. The case was still open in September 2026.

California and Maryland. In California, a state task force, a group the state set up to study reparations, delivered its final report on June 29, 2023. According to the state's Department of Justice, it made more than 115 recommendations, including a formal apology. Governor Gavin Newsom signed the state's apology in September 2024.

Newsom signed a law on October 10, 2025, creating a state Bureau for Descendants of American Slavery. The bureau is to check who qualifies, handle claims and recommend forms of repair. The law provides for no payments.

In Maryland, Governor Wes Moore vetoed a bill for a state reparations commission in May 2025, which means he refused to sign it into law. On December 16, 2025, Maryland's state lawmakers passed it over his veto, 31 to 14 in the Senate and 93 to 35 in the House. The 23-member commission is to study state and local policies from 1877 to 1965.

Slavery exhibits in the national parks, 2025 and 2026

President Donald Trump signed an executive order, a written command from the president to the people who run the federal government, on March 27, 2025. It was called "Restoring Truth and Sanity to American History." Interior Secretary Doug Burgum, who is in charge of the national parks, then ordered a review of how the signs and exhibits in the parks present the history of the United States.

Peter's photograph. Reporters for the Washington Post and the New York Times wrote in September 2025 that officials had ordered a copy of an 1863 photograph removed from an exhibit at Fort Pulaski National Monument, near Savannah, Georgia. The photograph shows the back of Peter, a man who escaped slavery in Louisiana in 1863, covered in scars from whipping. On September 16, 2025, Elizabeth Peace, who spoke for the Interior Department, told The Hill newspaper that park sites "were not asked to remove the photo." The two accounts differ.

The President's House. Officials removed the exhibits about slavery at the President's House site in Philadelphia in January 2026. That was the house where President George Washington lived, and where Ona Judge, an enslaved woman, walked out to freedom in 1796. On February 16, 2026, Judge Cynthia M. Rufe ordered the exhibits put back. Government lawyers appealed, and the judges of an appeals court, a higher court that hears appeals, later ruled that the administration had the power to replace that exhibit.

The national case. In a separate case, Judge Angel Kelley of the federal court in Massachusetts ruled on June 12, 2026, that the order could not be carried out at any of the park system's 433 sites. She ordered the removed exhibits put back by July 3. On July 2, 2026, the judges of the First Circuit Court of Appeals paused her order, so removals could go on while the case continued.

By the government's own count, in court papers filed in June 2026, exhibits had been removed or changed at 37 park sites. The case was still undecided in September 2026.

Joycelyn Davis

Who: A descendant of Charlie and Maggie Lewis. Charlie Lewis was one of the Africans smuggled into Mobile, Alabama, as captives on the *Clotilda* in 1860.

When and where: Africatown, in Mobile, Alabama, where she has lived all her life.

Movie: *Descendant* (2022), a documentary by Margaret Brown about the *Clotilda*, Africatown and the survivors' families. Joycelyn Davis appears in it.

"Our community is suffering, and you're asking about a ship?"

Joycelyn Davis descends from people who survived the last known slave ship to reach the United States, and she organizes gatherings in their honor in the town they founded. Her ancestors Charlie and Maggie Lewis were among the 110 Africans whom Captain William Foster carried from West Africa and smuggled into Mobile on the *Clotilda* in 1860, more than fifty years after bringing captives into the country became illegal.

She has lived in Africatown all her life. As a child she visited her great-grandmother in Lewis Quarters, one of the first pieces of land the *Clotilda* survivors bought, named for Charlie and Maggie Lewis. She went to the school and the church that her ancestors founded. In 2017 she sued the International Paper Company over pollution from its old mill near Mobile. She is a cancer survivor.

The ship. She said those words early in the search for the wreck of the *Clotilda*. After the wreck was confirmed in 2019, she came to see the ship as a way to bring money and attention to Africatown. That year she organized a reunion of *Clotilda* descendants. Of the crowd that came, she told National Geographic: "I never in a million years thought it would be that big."

A yearly festival she organizes is called Spirit of Our Ancestors. The 2021 festival, the third, honored the women who survived the *Clotilda*. She appears in *Descendant*, the documentary by Margaret Brown. It won a

special jury award at the Sundance Film Festival in January 2022 and was released in theaters and on Netflix on October 21, 2022.

THE THREE QUESTIONS, ANSWERED

1. **How did a servant in Virginia become a slave for life, and a child a slave from birth?** Virginia's judges and lawmakers did it one ruling at a time. Judges sentenced John Punch to serve for life in 1640, and in 1662 lawmakers ruled that a child born to an enslaved mother was enslaved for life.
2. **How did enslaved people fight back?** They worked slowly, ran away, sued, learned to read in secret and rose up with weapons. During the Civil War about half a million freed themselves by reaching the Union army.
3. **What happened to the freedom people won in 1865?** Freed people voted, built schools and sent 16 men to Congress. But within twelve years Southern lawmakers passed Black Codes, Klansmen whipped and murdered freed people, and federal officials stopped enforcing their rights in the South until members of Congress passed the laws of 1964 and 1965.

What changed: Slavery grew from a few court rulings in Virginia into a system that held four million people, and the freedom that ended it in 1865 took another hundred years to enforce.

FIVE DATES TO KEEP

July 9, 1640: John Punch is sentenced to serve for life.

1662: Virginia's law makes a child follow the mother into slavery.

January 1, 1808: Bringing captives from abroad becomes illegal.

December 6, 1865: The 13th Amendment ends slavery.

July 2, 1964: The Civil Rights Act becomes law.

NOT KNOWN: whether Virginia's colonists held the Africans of 1619 as servants for a set number of years or as slaves for life.

Where to go: The Smithsonian's National Museum of African American History and Culture, in Washington, D.C., whose history galleries begin with the slave trade.

Next: Chapter 25, Rights and Movements. It follows the campaigns for the rights the Constitution promised, from 1877 on.

Chapter 25: Rights and Movements

CHAPTER 25 OF 37 · RIGHTS AND MOVEMENTS · 10 ERAS · ABOUT 280 MINUTES

Most of the rights Americans have today were won by people who were first refused them and kept asking for years. Organizing often cost them their jobs, their freedom or their lives. Some of those rights were later taken away again.

Four men at Germantown, in Pennsylvania, signed a paper against owning enslaved people on April 18, 1688, and sent it to their Quaker church. The members of four church meetings read it, one after another, and passed it on without doing anything. Then the Quakers forgot it. People working to end slavery found it again in 1844, and it went missing a second time. In March 2005 it turned up in a vault in a Philadelphia meetinghouse. Haverford College, in Pennsylvania, keeps it now.

This chapter follows people who were told no: women, Black Americans, Native nations, Chinese and Mexican Americans, Japanese Americans, gay and transgender Americans, and disabled Americans. It goes era by era, from the clan mothers of the Haudenosaunee to the courtrooms of 2026. Watch how the methods grow. The first tools are a petition and a printed booklet. Later come the planned arrest, the boycott, the march, the sit-in, and the lawsuit brought on purpose to test a law. Watch, too, what each step costs, and how often a right that was won had to be won a second time.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did people without power get the people in power to listen?
2. What did organizing cost the people who did it?

3. Can a right that has been won be taken away again?

Five minutes? Read "Selma and the Voting Rights Act, 1965" in 1950 to 2000.

Can skip: "Two men climb the Gateway Arch, 1964", "Suffragists remember the clan mothers, 1891 and 1893", "Equal pay for the women's soccer team, 2016 to 2022". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 3 MINUTES

No record shows a rights movement in this land before 1500, but in at least one Native nation women held real power in government. A rights movement is a group of people who organize together to change a rule that denies them something.

About six hundred Native nations lived north of the Rio Grande, the river along part of today's border with Mexico, in 1492. Each one governed itself in its own way. The clearest record of women's power comes from the Haudenosaunee, a union of five nations in what is now New York, where women chose the chiefs and could remove them.

The clan mothers of the Haudenosaunee

The Haudenosaunee Confederacy has five founding nations: the Seneca, the Cayuga, the Onondaga, the Oneida and the Mohawk. The Onondaga Nation's own account places the founding on the shores of Onondaga Lake, in what is now central New York. The Onondaga still describe how their women choose leaders.

Each clan, a group of related families, has a clan mother. She chooses the man who speaks for her clan in council, the meeting where the chiefs talk and decide together. His title is Hoyane, which means chief. In the Onondaga account, a figure called the Peacemaker placed deer antlers on the headdress of every Hoyane to show that the man was a chief.

Taking the antlers. The clan mother can also start the removal of a chief. If he stops speaking for his clan, she warns him that he is not following the ways of the Great Peace, the laws the five nations agreed to live by when they joined together. If he still does not change, the clan removes him in council and takes his antlers off. The Onondaga say he has been "de-horned."

Haudenosaunee women held other powers too. An article by the National Park Service, the federal office that runs the national parks and historic sites, states that the women grew the food and shared it out, and through that work "controlled the economy" of their nations. They also held "the final authority over land transfers and decisions about engaging in war." They had the last word on giving land to anyone else and on going to war.

When a husband and wife separated, the wife took with her all the property she had brought into the home. The children went with her too.

When the union began. No one can prove when the five nations agreed to the Great Law of Peace. The Haudenosaunee keep their own account in spoken history and in wampum, belts and strings of shell beads that record agreements. By that account the date falls between about the year 1000 and 1400. Scholars working from written records mostly date it around 1450, the encyclopedia Britannica reports, and some date it to the late 1100s.

What the records can and cannot show

Almost everything written down about how these nations governed themselves was written after Europeans arrived, by Europeans. The nations kept their own records in spoken accounts and in wampum, cut from the white and purple parts of clam shells and woven into strings and belts.

The Onondaga Nation describes how a belt works. A speaker puts the words of an agreement into the wampum while the belt is woven. Every speaker after that uses the belt to remember the agreement and what has happened since. The Onondaga have kept the Confederacy's wampum since its founding.

NOT KNOWN: No record from before 1500 in this land names a single person who claimed a right that others denied.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 5 MINUTES

Under Spanish law in the 1500s, settlers forced Native people to work for them. Nobody here organized to change that law. The loudest objection came from a Spanish priest, Bartolomé de Las Casas, who argued in Spain that Native people had rights.

People also fought back. Enslaved Africans at San Miguel de Gualdape, a Spanish settlement on the coast of what is now Georgia or the Carolinas, rose up and escaped in October 1526. It is the first rebellion of enslaved Africans on record in what became the United States.

What was the encomienda?

The encomienda was a grant from the Spanish king that gave a settler the labor of a named group of Native people. It was one of the Spanish systems for forcing Native people to work.

On paper, the settler had to protect the people and teach them Christianity. In practice, he forced them to work for him and took the food and goods they produced. The historians who wrote the open textbook *The American Yawp* call it "an exploitive feudal arrangement in which Spain tied Indigenous laborers to vast estates." It used Native workers for the settlers' gain and bound them to huge farms, the way farm workers in old Europe were bound to a lord's land.

The same historians write that the settlers who held these grants "brutalized their laborers," beating and hurting the people who worked for them. Las Casas reached the Caribbean in 1502. He later wrote that he saw Spaniards "cut off the Hands, Noses, and Ears, both of Indians and Indianesses," meaning Native men and women. They did it, he wrote, "for no other reason, but only to gratify their bloody mindedness," for the pleasure of hurting people.

Hispaniola. On the Caribbean island of Hispaniola, Spaniards enslaved the Taíno, the Native people of the islands, and made them dig for gold. When the gold ran out, the Spaniards forced them to work on the encomienda farms. The Native population of Hispaniola fell from 60,000 in 1508 to fewer than 20,000 ten years later. Fewer than one person in three was left.

The authors of *The American Yawp* write of the Spaniards' "casual violence and dehumanizing exploitation," treating people as if they were not human. Within a few generations, the same authors write, the whole island had been emptied of its Native people.

King Charles I of Spain ended the encomienda in 1542 with a set of rules called the New Laws. Spanish officials replaced it with another forced-labor system, the repartimiento, and Spanish settlers went on forcing Native people to work for them.

Bartolomé de Las Casas

Las Casas was a priest with the Spanish soldiers who killed the Taíno gathered in a Cuban town in 1513. He spent the rest of his life arguing for Native people. At first he also asked for enslaved Africans to take their place.

He went to Cuba in 1513 as a chaplain, a priest who travels with soldiers, with the Spanish forces who were taking the island from the Taíno. At the town of Caonao, the Taíno had made cassava bread and fish for the soldiers under Captain Narváez. About two thousand people sat in the town square, and more than five hundred were in a large house nearby.

In his *History of the Indies*, Las Casas wrote what happened next. One Spaniard drew his sword, and then the whole hundred soldiers drew theirs. They began "to rip open the bellies, to cut and kill" the men, women, children and old people, who were all sitting down. They went into the large house and killed the people there too. Some escaped by climbing to the top of the house. Las Casas was in the town, but he had stepped away to another square shortly before the killing began.

Giving up his grant. For serving with the soldiers, Las Casas received land and an encomienda. In 1514 he told the people of his church that the whole encomienda system was wrong. He freed the Native people he held and began arguing for them to the Spanish officials on the island.

The request he later regretted. Two years later, in 1516, Las Casas asked the Spanish king to bring enslaved Africans to the Indies, Spain's islands and lands across the Atlantic, to take the place of Native workers. He believed Africans were better suited to hard physical labor. Slave traders had already been carrying enslaved Africans across the Atlantic for about sixteen years.

King Charles I of Spain, later Emperor Charles V, gave a man named Lorenzo de Gorrevod a charter, a written permission, to ship enslaved Africans across the Atlantic. The date was August 18, 1518.

Las Casas also wrote a short book about what Spanish settlers did to Native people, *A Brief Account of the Devastation of the Indies*. Its Spanish title is *Brevísima relación de la destrucción de las Indias*. Soon after he wrote it, King Charles issued the New Laws of 1542.

The debate at Valladolid. Charles V called Las Casas and a scholar named Juan Ginés de Sepúlveda to the town of Valladolid, in Spain, in 1550. The question was whether the Spanish had the right to make war on Native people and rule them by force. Sepúlveda argued that they did. Las Casas argued that they did not.

The two men spoke in front of a panel of churchmen and royal officials, among them the church scholars Domingo Soto and Melchor Cano. On the first day Sepúlveda spoke for three hours. The panel held its final meeting in 1551. Andrea Daher, a scholar writing for an encyclopedia of Sorbonne University in Paris, records no ruling by the panel.

Later in life, Las Casas changed his mind about enslaving Africans. In his *History of the Indies* he wrote at length against the African slave trade and the slave markets of Spain and Portugal. He did not publish that book while he was alive, so his change of mind was not public while the trade grew.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 10 MINUTES

In the English colonies of the 1600s, the men in charge punished people who spoke against them. The records name the people they punished. English colonists also brought English law with them, and under one of its rules a married woman did not count in law as a person separate from her husband. Nobody organized against that rule in this century.

The colony's leaders banished Anne Hutchinson from Massachusetts in November 1637 for what she taught in her own house. Massachusetts officials hanged Mary Dyer on Boston Common in 1660 for coming back to the colony as a Quaker, a member of a new religion that the colony's leaders had outlawed. In 1688 four men in Pennsylvania signed a protest against slavery and sent it to their church.

Coverture: what marriage did to a woman in law

The rule was called coverture. In law, a married woman was "covered" by her husband. The legal name for her was *feme covert*, a covered woman, and she had no legal standing of her own.

She did not control property, even property she had brought into the marriage. Her husband had full rights to any money or wages she earned. She could not sign a contract, a written deal that a court can make people keep. Nobody could take her to court, and she could not bring a case to court in her own name.

An adult woman who had never married, or whose husband had died, was a *feme sole*, a woman on her own. She could buy and sell property, sign contracts and go to court in her own name. Under English law, a woman lost those rights on the day she married.

Daily life. Wives ran businesses beside their husbands, and for husbands who were away. Some married women traded in their own names under a legal exception, and the people of their town held everyone to the deal. Under the law, whatever a wife held, she held because her husband allowed it.

Anne Hutchinson

Who: A Boston woman whom the colony's leaders tried and banished for what she taught at religious meetings in her own house.

When and where: Boston and Newtown, Massachusetts Bay Colony. Meetings from 1636, trial in November 1637, put out of her church in March 1638.

"You have power over my body," she told the men judging her.

Anne Hutchinson came to Boston, an English town built on Massachusett land, with her husband, William, and their children in 1634. From 1636 she held a weekly meeting in her house to go over the Sunday sermons. At first only women came. Then men came too, including some of the most important men in the colony.

Like the colony's leaders, she was a Puritan, one of the English Protestants who had come to Massachusetts to worship God in their own way. Puritans used the word grace for God's gift of being saved, of going to heaven after death. Hutchinson taught that God saved people by grace alone, and not because they behaved well. She also said she could tell which ministers and which neighbors had received it.

The trial. In November 1637 the colony's lawmakers, called the General Court, tried her for heresy, teaching against the church's beliefs. They sat as judges at Newtown, the town now called Cambridge, and the trial lasted two days. Governor John Winthrop told her she was "called here as one of those that have troubled the peace of the commonwealth," meaning the colony. He called her meetings "a thing not tolerable nor comely in the sight of God nor fitting for your sex."

She answered his Bible verses with Bible verses of her own. At the end she said God had spoken to her directly, by "an immediate revelation," and that God would curse the Puritans and their families for generations if they harmed her. The judges called that blasphemy, speaking against God. She told them, "You have power over my body, but the Lord Jesus hath power over my body and soul."

The trial record states the sentence: "Mrs. Hutchinson, the sentence of the court you hear is that you are banished from out of our jurisdiction as being a woman not fit for our society." A jurisdiction is the area a government rules, and banished meant forced out of it and forbidden to return. She asked, "I desire to know wherefore," or why, "I am banished?"

The second trial. Colony officials held her through the winter. In March 1638 the leaders of the Boston church tried her again and put her out of the church. Their charge said she had "rather been a Husband than a Wife and a preacher than a Hearer." It went on, "and a Magistrate than a Subject." A magistrate was an official with the power to judge cases.

That spring the family moved to Aquidneck Island, in what became Rhode Island, on land the minister Roger Williams helped them buy from the Narragansett. One of the first written rules of their new town, Portsmouth, said no person should be punished or "called into question on matter of religion … so long as he keeps the peace."

Her death. After her husband died, she moved in 1642 to Dutch-held land on Long Island Sound, the country of the Siwanoy, a Native people. The next winter the Dutch governor, Willem Kieft, and his council approved attacks by Dutch soldiers and armed settlers on the Native people of the region. On the night of February 25, 1643, those men murdered at least eighty Munsee people, and perhaps as many as 120.

The war that followed lasted until 1645. In August or September 1643, during that war, Siwanoy warriors killed Anne Hutchinson, her servants and all but one of the children who were with her. One daughter survived.

Massachusetts leaders outlaw Quakers, then hang four of them

Massachusetts officials hanged a woman named Mary Dyer in 1660 because she kept coming back to the colony as a Quaker. The Quakers followed George Fox, an Englishman who taught that men and women were equal in God's eyes. Quakers refused to take their hats off to people of higher rank, because they believed all people were equal. They also refused to swear oaths, solemn promises made in God's name, to any government. Massachusetts Governor John Endecott called them a "pernicious sect," a harmful religious group.

The laws. The men of the General Court passed a series of laws against Quakers. In late 1656 and 1657 they outlawed "the cursed sect of heretics commonly called Quakers." They fined ship captains who brought Quakers to the colony and people who owned Quaker books.

In October 1657 they set punishments for Quakers who came back after being sent away. A man who came back once was to have one ear cut off with a blade, and a man who came back a second time was to lose the other ear. For a third return, the law stated, "they shall have their tongues bored through with a hot iron," a hole burned through the tongue with a heated iron rod. For women who came back, the law set whipping on the bare back.

Three men's ears. The colony's leaders used that law in 1658 against a Quaker named Christopher Holder and two other Quaker men. One of the two was John Rous. Governor Endecott sent Rous to prison, and on September 7, 1658, Rous was sentenced to have his right ear cut off. A British reference book, the *Dictionary of National Biography*, states that the cutting was done inside the prison, although the law called for it to be done in public.

A chronology by the law professor Douglas Linder gives a different date for the cutting of the three men's ears, June 3, 1658. Linder adds that the men were also whipped, and then beaten twice a week for nine weeks. Neither account names the judges or the men who did the cutting, the whipping and the beating.

On October 19, 1658, the men of the General Court voted 12 to 11 to banish Quakers "upon pain of death." Under that law, a Quaker who came back after being sent away could be hanged.

Mary Dyer. Dyer had lived in Boston in the 1630s and was a close friend of Anne Hutchinson. When church leaders put Hutchinson out in March 1638, Dyer walked out of the church beside her. Around 1650 she met George Fox in England, and she came back to Boston as a Quaker in 1657. Officials jailed her as soon as she landed. She spent two and a half months in a dark cell with its windows boarded up.

Her husband got her out by promising she would not preach, and the family moved to Rhode Island. She kept going back to Massachusetts to stand by Quakers who were in prison there.

Governor Endecott sentenced Mary Dyer and two Quaker men to death on October 20, 1659. The men were William Robinson, a merchant from London, and Marmaduke Stephenson, a farm worker from Yorkshire. Dyer's son, William Dyer, asked the governor to spare her.

On October 26 Endecott signed an order giving her 48 hours to leave the colony. First, the order said, she must be taken to the gallows, the wooden frame used for hangings, and stand there with a rope around her neck while the two men were hanged. Officials hanged Robinson and Stephenson in front of her on October 27, 1659. Then they took her back to prison and sent her out of the colony.

She came back to Massachusetts in 1660. Her husband asked for her life, and she refused to say she was sorry. On June 1 officials hanged Mary Dyer near the Frog Pond on Boston Common, on the sentence of Governor Endecott and the General Court. About 200 soldiers stood around the gallows with guns, swords, long spears called pikes, and flags.

The count. Officials hanged a fourth Quaker in Boston in 1661, William Leddra, a cloth maker and Quaker minister from Cornwall, in England. An English Quaker named Edward Burrough counted what had been done to Quakers in Massachusetts between 1656 and 1660. His count lists 64 jailings

and 25 banishments. King Charles II of England ordered the punishments stopped on September 9, 1661, and said accused Quakers must be sent to England instead. By 1677, Quakers could meet openly in Massachusetts.

Four men sign a protest against slavery, 1688

Four men at Germantown, in Pennsylvania, signed a paper against owning enslaved people on April 18, 1688. Their names were Garret Hendericks, Francis Daniel Pastorius, Derick op de Graeff and Abraham op de Graeff. They sent it up through the meetings of their Quaker church. A National Park Service page calls it the first protest against African slavery made by a religious body in the English colonies.

Germantown stood on the land of the Lenape, the Native people of the Philadelphia region. Pastorius had bought the land in 1683 from William Penn, a Quaker to whom King Charles II had given a charter for Pennsylvania in 1681. The surviving records of that sale do not name the Lenape families who lived on it.

The protest rested on the Golden Rule, the teaching that people should treat others as they want to be treated. In the paper's own spelling, the rule was "that we shall doe to all men licke as we will be done ourselves."

Up through the meetings. A Quaker meeting is a group of Quakers who worship and make decisions together. A monthly meeting is a local group, a quarterly meeting joins several local groups, and a yearly meeting joins the Quakers of a whole region. The members of four meetings, one after another, read the paper and passed it on without acting. The first three were Germantown Monthly Meeting, Dublin Monthly Meeting and Philadelphia Quarterly Meeting.

The last was Burlington Yearly Meeting, in New Jersey, and its members voted to take no stand either way. Nothing changed because of the protest.

Lost and found. The Quakers put the paper aside and forgot it. Abolitionists, people working to end slavery, found it again in 1844.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 5 MINUTES

Single men, each writing on his own, printed the arguments against slavery in these fifty years. No organized group yet worked for anyone's rights, and the Quakers began to argue about slavery among themselves. Quaker leaders punished the members who raised the question first. By 1776 the Quakers of Philadelphia had changed their own rules anyway and banned slaveholding among their members. The era's last section runs past 1750, to 1778, to finish that story.

A Boston merchant writes a pamphlet against slavery, 1700

Samuel Sewall, a Boston merchant and magistrate, wrote a three-page pamphlet called *The Selling of Joseph* in 1700. A pamphlet is a short printed booklet. A Massachusetts Historical Society page calls it the first antislavery pamphlet published in America. Bartholomew Green and John Allen printed it in Boston.

Sewall argued that taking people and selling them is the crime the Bible calls man stealing, and he pointed to the verse Exodus 21:16. Then he went through the usual arguments in favor of slavery and answered each one.

Adam's case. Sewall wrote it because of a case in Boston. That year a petition, a written request signed by many people, went around Boston asking for the freedom of an enslaved man named Adam and his wife. The petition said they were "unjustly held in Bondage" by a merchant named John Saffin. Saffin had promised Adam his freedom after seven years of good behavior, and when the seven years were up, he would not free him.

Saffin printed an answer to Sewall in 1701, defending slavery and his own treatment of Adam. Adam went to court, and in 1703 he won his freedom. Sewall answered Saffin again in 1705 by reprinting an English attack on the slave trade.

Quakers who spoke first, and what their church did to them

Among the Quakers, an argument about slavery kept going in these years, and their own church leaders spoke out against the first Quakers who pressed it. Ralph Sandiford published *A Brief Examination of the Practice of the Times* in Philadelphia in 1729. It argued against owning slaves, and Quaker leaders spoke out against it in public.

Benjamin Lay in Barbados. Benjamin Lay was born in Essex, in England, in 1682. From 1718 he and his wife spent about eighteen months in Barbados, an island in the Caribbean where enslaved Africans already outnumbered the white settlers by 1660, about 27,000 to 26,000. He later wrote that there he "saw and heard of such very great Barbarity," or cruelty, "used toward the Slaves, Night and Day."

A man named Richard Parrot whipped the people he enslaved "very severely" every Monday morning. One of them, a barrel maker, told Lay he would not be whipped on Monday morning, and he killed himself on a Sunday. Hungry enslaved workers came into the Lays' shop to buy, to beg, and sometimes to steal food. Lay wrote that when he caught them he sometimes whipped them, "but I have been sorry for it many times."

Near Philadelphia. Lay then settled near Philadelphia. He was just over 4 feet tall, with a hunched back and thin arms and legs. For a time he lived in a cave with a library of about 200 books. His wife, Sarah Lay, was also small and had a curved spine. She was a recorded Quaker minister, a speaker her meeting officially recognized.

Lay published a book against slaveholding in 1737 with a long title: *All Slave-Keepers That Keep the Innocent in Bondage, Apostates*. An apostate is a person who has abandoned their faith. Benjamin Franklin printed it and left his own name off it.

Quaker leaders spoke out against the book, and the Quakers of Philadelphia Yearly Meeting disowned Lay. Disowning was a formal statement that a member no longer shared an essential part of the faith. The member's name came off the list, and he could no longer hold certain offices, though he could still come to worship. Sarah Lay stayed a member all her life.

The sword and the book. At the yearly meeting's gathering in Burlington, New Jersey, in 1738, Lay stood up in front of the Quakers there. His words began, "Oh, all you negro masters, who are contentedly holding your fellow-creatures in a state of slavery during life," using the word of the time for Black people. He threw off his plain Quaker cloak. Underneath he wore a soldier's uniform, in front of people who were committed to peace.

Holding people as slaves, he said, was as right, or "as justifiable," as "if you should thrust a sword through their hearts, as I do through this." Then he drove a sword into a hollowed-out book. Inside it he had hidden an animal bladder filled with pokeberry juice, and the red juice sprayed the people near him.

The Quakers carried him out and laid him on the doorstep. He lay there for the rest of the day, so everyone leaving had to step over him. Lay died in 1759.

The Quaker meetings change their rules, 1754 to 1778

Quakers changed their rules about slavery step by step after 1750. The Quakers of Philadelphia Yearly Meeting published a paper in 1754 that called slavery a sin, and sent it to other meetings. Four years later the members of the yearly meeting spoke out against buying and selling people, but still allowed members to keep the people they already held.

The ban. Philadelphia Yearly Meeting's members banned slaveholding among themselves in 1776, 88 years after the four men at Germantown signed their protest. A Quaker who held people in slavery could now be disowned, and Philadelphia Quakers disowned the first members for it in 1778.

Quakers in other regions made the same change in those years. The members of New England Yearly Meeting made slaveholding a reason to disown a member in 1773. New York Quakers spoke out against slaveholding in 1774, and North Carolina Quakers made it a reason to disown a member in 1776.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 12 MINUTES

People left out of the Revolution's promises answered with the Revolution's own words. The delegates to the Continental Congress voted on July 4, 1776, to adopt the Declaration of Independence, which states that "all men are created equal." The men who wrote the new state constitutions of 1776 kept women from voting, except in New Jersey.

Black taxpayers in Massachusetts asked why they paid taxes without a vote. In New Jersey, some women and some free Black men voted for thirty-one years, until the state's lawmakers took the vote away in 1807. In Philadelphia, a group of men formed a society in 1775 to help free people who had been seized and held as slaves against the law.

Dinah Nevil and the first antislavery society, 1775

The Philadelphia antislavery society of 1775 started because one woman said she was free. Her name was Dinah Nevil, and the records call her an Indian woman, meaning a Native American. A Virginia man named Benjamin Bannarman bought Nevil and her four children from Nathaniel Lowry of New Jersey in 1773. When they were delivered in Philadelphia, Nevil said in public that she and her children were "free people."

The mayor of Philadelphia sent Nevil and her children to the city's Work House, a building where the city held people, to wait for a hearing. A group of Quakers led by Israel Pemberton brought a lawsuit for her. The case ran for two years. At the end, the judges ruled that Dinah Nevil and her children were slaves. The society's records do not name the judges.

The society. After that ruling and others like it, a group of men met at the Rising Sun Tavern in Philadelphia on April 14, 1775. They formed the Society for the Relief of Free Negroes Unlawfully Held in Bondage. Sixteen of the 24 first members were Quakers. Thomas Harrison was the secretary, and Anthony Benezet was a member.

A Historical Society of Pennsylvania page calls it the first abolition society in the western world. Its first work was helping free people of color, people who were not white, who had been seized and held as slaves. The members stopped meeting in November 1775. Most of them were Quakers, who refused to fight in the war with Britain, and they believed that refusal would turn people against the society.

How Nevil got free. Dinah Nevil never won her case. Thomas Harrison, who had been caring for her and her children, used his own money in 1781 to buy the family from the man who claimed them. Then he freed them. Judges had kept rejecting her claim, and buying her was the only way left. Eight years had passed since she first said she was free.

The society grows. The society began meeting again on April 23, 1787, and let in members who were not Quakers. Its members elected Benjamin Franklin president. Later members included Benjamin Rush, Thomas Paine and John Jay. People in other states set up societies like it. New York had one by 1785, two years before the Philadelphia society met again. In 1787 Richard Allen, Absalom Jones and other Black Philadelphians formed their own group, the Free African Society.

The Philadelphia members got a charter in 1789, an official paper from the Pennsylvania government that recognized the society. They also took a longer name: the Pennsylvania Society for Promoting the Abolition of Slavery, for the Relief of Free Negroes Unlawfully Held in Bondage, and for Improving the Condition of the African Race. By 1790 Delaware, Maryland and Connecticut had societies too.

Abigail Adams

Who: A Massachusetts woman who asked her husband, while he helped lead a new country, to "Remember the Ladies" in its laws.

When and where: Braintree, Massachusetts. Letters of March 31, April 14 and May 7, 1776.

Movie: John Adams (2008 HBO miniseries). The series is about John Adams and his family. Abigail is a lead character, not the subject.

"Remember the Ladies," she wrote. "I cannot but laugh," he answered.

John Adams had been away at the Continental Congress in Philadelphia since 1774, and that summer he helped draft the Declaration of Independence. Abigail Adams stayed at home in Braintree, the town now called Quincy, Massachusetts, and ran the family farm and its money. The couple wrote to each other often. The Massachusetts Historical Society keeps their letters.

On March 31, 1776, she wrote, in her own spelling, "I long to hear that you have declared an independency." She went on: "and by the way in the new Code of Laws which I suppose it will be necessary for you to make I desire you would Remember the Ladies, and be more generous and favourable to them than your ancestors."

"Do not put such unlimited power into the hands of the Husbands," she wrote. "Remember all Men would be tyrants if they could." Tyrants are rulers who use their power cruelly. "If perticuliar care and attention is not paid to the Laidies we are determined to foment a Rebelion, and will not hold ourselves bound by any Laws in which we have no voice, or Representation." To foment a rebellion is to stir one up.

What she was asking about. In 1776 coverture still governed marriage in the colonies, and a husband controlled everything his wife owned and earned. The rule was centuries old. An English lawyer, William Blackstone, had written it down a few years earlier in his *Commentaries on the Laws of England*, published from 1765 to 1769.

"By marriage, the husband and wife are one person in law," Blackstone wrote, "that is, the very being or legal existence of the woman is suspended during the marriage." While she was married, a wife did not exist in law as her own person.

The colonists' own words. In her March letter, Abigail Adams warned that women would not obey laws in which they had "no voice, or Representation." She was using the colonists' own complaint against Britain. That July the delegates accused the British king of "imposing Taxes on us without our Consent." They called "the right of Representation in the Legislature," a say in the body that makes the laws, "a right inestimable to them," too precious to measure.

His answer. John Adams wrote back on April 14, 1776: "As to your extraordinary Code of Laws, I cannot but laugh." He wrote that the war had "loosened the bands of Government every where," He listed who had grown bold: "That Children and Apprentices were disobedient," "that schools and Colledges were grown turbulent," meaning rowdy, and "that Indians slighted their Guardians and Negroes grew insolent to their Masters."

Slighted means treated without respect, and insolent means rude to someone in charge. By "Negroes" and "their Masters" he meant enslaved Black people and the men who enslaved them. He would not give up the men's power, he wrote, because that "would compleatly subject Us to the Despotism of the Peticoat." Despotism is rule by a tyrant, and a petticoat was a woman's underskirt. He meant rule by women.

Her reply. She wrote back on May 7, 1776: "I can not say that I think you very generous to the Ladies, for whilst you are proclaiming peace and good will to Men, Emancipating all Nations, you insist upon retaining an absolute power over Wives," emancipating meaning setting free. She added that "Arbitary power is like most other things which are very hard, very liable to be broken," arbitrary power being power used however the ruler likes, with no rules to limit it.

> No lawmakers in any state changed a law for women because of these letters, which were private. Her grandson Charles Francis Adams published a book of her letters, *Letters of Mrs. Adams, the Wife of John Adams*, in 1840, more than twenty years after she died.

Some women and free Black men vote in New Jersey, 1776 to 1807

Unmarried women and free Black men who owned enough property could vote in New Jersey from 1776 to 1807. The state's constitution of 1776 gave the vote to "all inhabitants of this colony, of full age, who are worth fifty pounds ... and have resided within the county ... for twelve months." It did not say male, and it did not say white. The other states' constitutions of 1776 limited the vote to a "freeman," a "man" or a white man, and every one of them left women out.

Who could pass the test. Because of coverture, only unmarried women and widows could meet New Jersey's property test. A married woman could not own fifty pounds of anything, because in law everything she had belonged to her husband. Members of the state assembly said so in 1800: "Our Constitution gives this right to maids or widows black or white," maids meaning unmarried women. Some losing candidates later complained that married women had voted anyway.

The names on the lists. The state's election laws of 1790 and 1797 described the voter as "he or she." Researchers at the Museum of the American Revolution have found 163 women's names on nine lists of voters from 1800 to 1807. That count is the lowest possible number, because only nine lists survive.

Sarah Eoff, Margaret McDonald and Eleanor Boylan are on the 1800 list for Bedminster Township. Eleanor Boylan was a widow of about 51 when she voted. Four women of one family, Rebecca, Ann, Catherine and Sarah VanDike, voted together in October 1801. The 1801 list for Montgomery Township holds at least 46 women and at least four free Black men, among them Caesar Trent of Princeton.

The vote taken away. The men who made New Jersey's laws took the vote away in November 1807. Earlier that year, in February, Essex County held a vote on whether to build a new courthouse in Newark or in Elizabeth. People

on both sides broke the rules and brought in voters who had no right to vote. In Elizabeth the votes counted came to 279 percent of the people allowed to vote, nearly three votes for every voter.

The new law limited the vote to "free, white, male citizens." It took the vote from women, from free Black men and from recent immigrants. In its opening paragraph the lawmakers wrote of "doubts" about letting "aliens, persons of color, or negroes" vote.

Aliens meant people from other countries who were not citizens. An exhibit by the Museum of the American Revolution says the lawmakers blamed women, people of color and newcomers for all the cheating, which the museum calls scapegoating, putting the blame on people who were not at fault.

Why they did it. Writers for the National Park Service, the New Jersey Historical Commission and the Museum of the American Revolution all say the lawmakers acted for party reasons, not out of a debate about women's rights. The museum's historians also link the law to the cheating in the courthouse vote. The two parties of the time were the Federalists and the Democratic-Republicans.

The three disagree about which party women's votes had helped. A National Park Service page says women often voted for the Federalists, and that the law helped the rival Democratic-Republicans before the 1808 presidential election. A New Jersey Historical Commission teaching sheet says Federalists had come to believe that women's votes helped the Democratic-Republicans.

No vote, no taxes: Paul and John Cuffe, 1780

Seven free Black men of Dartmouth, Massachusetts, told the state's lawmakers in February 1780 that they should not be taxed if they could not vote. They were John Cuffe, Adventur Child, Paul Cuffe, Samuel Gray, Pero Howland, Pero Russell and Pero Coggeshall. John and Paul Cuffe were brothers. Gray, Howland and Russell signed with a mark instead of a written name.

The petition said that "by reason of long bondage and hard slavery, we have been deprived of enjoying the profits of our labor or the advantage of inheriting estates from our parents." They were taxed all the same, although "we are not allowed the privilege of freemen of the State, having no vote or influence in the election of those that tax us." It also said that "many of our colour (as is well known) have cheerfully entered the field of battle in the defence of the common cause," meaning they had fought in the Revolution.

The new Massachusetts constitution of March 1780 contained nothing that answered their complaint. The Cuffe brothers wrote to the county court and to the town's leaders, calling themselves both "Indian men" and "free Negroes." They refused to pay their taxes, and Dartmouth's town officials put them in jail for it.

Prince Hall asks for schools, 1787

Prince Hall and other Black Bostonians asked the Massachusetts lawmakers for schools for their children in 1787. No law barred Black children from Boston's free schools, but the National Park Service records that Black students there "found themselves at the mercy of hostile teachers and fellow students." The petition said they "receive no benefit from the free schools in the town of Boston, which we think is a great grievance," a wrong worth complaining about, "as by woful experience we now feel the want of a common education."

The lawmakers turned the petition down. They turned down a second petition in 1796. In 1798 the families opened their own school, the African School. It met in the home of Primus Hall, Prince Hall's son.

Judith Sargent Murray argues that women's minds are equal, 1790

Judith Sargent Murray published an essay called "On the Equality of the Sexes" in 1790. She was born on May 1, 1751, in Gloucester, Massachusetts. The essay ran in the *Massachusetts Magazine* under the pen name "Constantia," a made-up name for signing her writing.

She argued that men and women have equal minds, and that women fell behind only because girls did not get the education that boys got. One line of the essay asks "in what the minds of females are so notoriously deficient, or unequal." Unlike Abigail Adams's private letters, the essay was printed where anyone could read it.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 25 MINUTES

Women who organized against slavery in these fifty years learned to gather signatures, print their arguments and speak to crowds, and then some of them turned the same methods to their own rights. Ministers, lawmakers and mobs attacked them for it. Black families in Boston campaigned to get their children into the city's schools.

Teachers in Hartford, Connecticut, began teaching deaf children together in sign language in 1817. At Seneca Falls, New York, in July 1848, more than 300 people met. One hundred of them signed a declaration that listed the wrongs men had done to women, and the people at the meeting voted to demand the vote.

Deaf children taught together, 1817

The American School for the Deaf opened at Hartford, Connecticut, on April 15, 1817, in a building called Bennett's City Hotel. Seven pupils came full time in the first classes. Teachers taught them in sign language, using a method from France.

Thomas Hopkins Gallaudet founded the school with Laurent Clerc, a deaf French teacher whom Gallaudet brought back from Paris. A Hartford doctor, Mason Fitch Cogswell, and ten other Hartford men backed them. Cogswell's deaf daughter, Alice Cogswell, was one of the first pupils.

Public money. Connecticut's lawmakers voted the school $5,000 in October 1816, before it opened. An essay by John Crowley for the Disability History Museum calls that grant the first public money for educating Americans with disabilities. In 1819 members of Congress voted the school 23,000 acres of public land in Alabama, after a motion by Henry Clay.

It was land the United States had taken from Native nations, and the surviving accounts do not say which nation's land it had been. The school's leaders sold the land to build up an endowment, a sum of money kept and invested so that

its earnings pay for the school's work.

Girls ask for the same lessons as boys

Some of the first organized demands for women in these years were about schooling. Girls' schools usually taught manners and fine arts, and a few women opened schools that taught girls the subjects boys learned.

Emma Willard wrote *A Plan for Improving Female Education* in 1819 and addressed it to New York's lawmakers. She argued that young women needed the same subjects young men were taught. With the support of Governor DeWitt Clinton, she opened the Troy Female Seminary in Troy, New York, in 1821. A seminary was a school for older girls or young women.

Catharine Beecher and her sister Mary opened the Hartford Female Seminary in Connecticut on May 20, 1823. Their pupils studied geography, world history, Latin, rhetoric (the art of speaking and writing well), philosophy and the natural sciences. Within three years the school had 100 full-time pupils.

Mount Holyoke. Mary Lyon began raising money for a school for women in 1834. Gifts ran from six cents to $1,000. Women's sewing circles sent quilts and bedding, and the school's trustees, the people in charge of its money and buildings, gave stoves and furniture. She opened Mount Holyoke Female Seminary at South Hadley, Massachusetts, in 1837. Tuition was $60 a year.

A history page by Mount Holyoke College says Lyon "endured ridicule from those who felt her ambitious undertaking would be 'wasted' on women." It adds that "The idea of a college for women was horrifying to many people in the 1830s."

Oberlin. At Oberlin, in Ohio, women won the chance to earn a college degree in two steps. Oberlin stands on land that the college's own art museum calls "unceded ancestral lands" of the Erie, Wyandotte, Mingo and other Native nations, unceded meaning never given up by treaty or agreement.

Women studied beside men at Oberlin from its opening in 1833, but in a Ladies Course that ended with a diploma, not a degree. Four women entered the course that led to a bachelor's degree in 1837: Mary Kellogg, Mary Caroline

Rudd, Mary Hosford and Elizabeth Prall. Rudd, Hosford and Prall graduated in 1841. Kellogg ran out of money, left, and came back later.

Oberlin was the first college to give bachelor's degrees to women in classes with men. It also became, in 1835, the first college with a rule of admitting Black students.

Boston's Black families and the city's schools

Boston's Black families opened the African School in Primus Hall's house in 1798. They moved it into the basement of the African Meeting House in 1808, and in the mid-1830s it moved into the Abiel Smith School. The Smith School became the segregated school where most of Boston's Black pupils were taught, kept apart from white pupils by race. The members of Boston's Primary School Committee kept separate schools for Black and white children.

The boycott. William Cooper Nell and others organized a boycott of the Smith School in the late 1840s, which means families kept their children out of it in protest. They also gathered signatures against it.

Sarah Roberts. A father, Benjamin Roberts, asked the Primary School Committee four times to let his five-year-old daughter, Sarah, into the school nearest their home. To reach the Smith School she had to walk past five schools for white children. The committee's members refused each time, and in 1848 he sued the city of Boston. The Roberts family lost the case, and the committee's members kept the schools separate.

Women petition Congress, 1829 to 1830

A petition is a written request to people in power, signed by the people who are asking. Women had no vote, and petitions were the way they first spoke to Congress about national questions. The House of Representatives' own historian says women first petitioned on political issues in 1829 and 1830.

The question was Indian removal, the plan to force the Cherokee and other Native nations off their land. Catharine Beecher wrote a *Circular Addressed to Benevolent Ladies of the United States* on December 31, 1829. "It may be, that female petitioners can lawfully be heard, even by the highest rulers of our land," she wrote. More than 1,500 women signed petitions to Congress against removal. The women of Steubenville, Ohio, called themselves "the feeblest of the feeble" in theirs, on February 15, 1830.

Members of Congress mocked the women in public. Senator Thomas Hart Benton of Missouri told "these ladies" to "sit down on the way side, and wait for the coming of the conquerors." In 1830 a majority in Congress voted for the Indian Removal Act.

Maria W. Stewart speaks in Boston, 1832

Maria W. Stewart, a free Black woman, spoke at Franklin Hall in Boston in September 1832, to an audience that included men. A National Park Service page calls it one of the first recorded times an American woman of any race spoke in public.

Stewart was born in 1803 in Hartford, Connecticut. In her own words she "was left an orphan at five years of age" and "was bound out in a clergyman's family," sent to work as a servant in a minister's house. In Boston she and her husband, James W. Stewart, lived among the city's free Black families. William Lloyd Garrison printed her essay "Religion and the Pure Principles of Morality" in 1831 in his antislavery newspaper, *The Liberator*.

Her words. "How long shall the fair daughters of Africa be compelled to bury their minds and talents beneath a load of iron pots and kettles?" she asked in her speeches. She urged her listeners, "Sue for your rights and privileges."

Audiences turned against her. She scolded Black men in public for drinking too much, and some people objected that it was improper for a woman to speak in front of men. She gave a farewell speech on September 21, 1833. She asked, "What if I am a woman," and said it was "no use for me as an individual, to try to make myself useful among my color in this city."

She moved to New York in 1834 and taught school. She later worked as matron, the woman in charge of housekeeping and care, at the Freedmen's Hospital in Washington, where she died on December 17, 1879.

Prudence Crandall and Sarah Harris

Who: A young Black woman who asked to be taught, and the white teacher who was arrested, tried and attacked for teaching her and other Black girls.
When and where: Canterbury, Connecticut, 1832 to 1834.

"I want to get a little more learning," Sarah Harris told her teacher.

Sarah Harris had finished primary school, and she wanted enough schooling to teach Black children in the town of Norwich. Her father handed out *The Liberator*, Garrison's antislavery newspaper. In 1832 she asked Prudence Crandall to let her into Crandall's school for girls in Canterbury, Connecticut. "I want to get a little more learning, if possible enough to teach colored children," she said, using the word of the time for Black people. "If you think it will be the means of injuring you, I will not insist on the favor."

Crandall admitted her. The white families took their daughters out of the school.

A school for Black girls. In the spring of 1833 Crandall went to Boston to see Garrison. Then she traveled around New England asking Black families to send their daughters. Her advertisements said the school would reopen on April 1, 1833, for "young ladies and little misses of color." About 24 pupils came, and most of them boarded in the house.

The man who led the fight against the school was Andrew T. Judson, a lawyer and politician who lived next door. "We mean there shall not be such a school set up anywhere in our State," he told the minister Samuel J. May. He also said, "They are an inferior race of beings, and never can or ought to be recognized as the equals of whites."

The Black Law. The lawmakers of the Connecticut General Assembly passed what people called the Black Law on May 24, 1833. Under it, a school in the state that took Black pupils from outside Connecticut

without the town's permission was breaking the law. They wrote it to close Crandall's school.

Connecticut officials arrested Crandall on June 27, 1833, and she spent a night in jail. Judson led the case against her. At her first trial the jury could not agree. At the second, in October 1833, Judge David Daggett told the jury that Black Americans were not citizens, so the protections her lawyers pointed to did not cover her pupils. The jury found her guilty.

The judges of the Connecticut Supreme Court threw out the guilty verdict, the jury's decision, on a technical point on July 26, 1834. They did not rule on the law itself, so the Black Law still stood.

The town against the pupils. Canterbury's town officials went after the pupils too. Ann Eliza Hammond was a 16-year-old pupil from Providence, Rhode Island, who arrived in April 1833. Town officials used the state's vagrancy law against her, a law against people with no home or work in the town. Its punishment was whipping "on the naked body not exceeding ten stripes," up to ten lashes of a whip on bare skin. Crandall paid so that no one whipped her.

Shopkeepers would not sell to the school, and people threw stones and eggs at the pupils. Someone hung a dead cat at the school door. In January 1834 a fire set by an unknown person damaged the building.

The last night. A mob of armed men broke the windows of the house on the night of September 9, 1834, and went through it with weapons while Crandall and her pupils were inside. A National Park Service page states that the mob broke more than ninety windows. An account in the Online Review of Rhode Island History says five main windows on the first floor. No one was ever charged, and no record names the attackers. Crandall closed the school the next morning.

Crandall married a Baptist minister, Calvin Philleo, and moved west to Illinois and then Kansas. Connecticut's lawmakers voted her a pension of $400 a year in 1886, and she died in Kansas in 1890, aged 87. Sarah Harris became an abolitionist and a conductor on the Underground

> Railroad, helping people escape from slavery. In 1877 she traveled to Kansas to visit Crandall, and the two women wrote to each other about their old pupils.

BY 1834: Women had petitioned Congress for the first time, a free Black woman had spoken to men and women in Boston, and a mob had closed a school for Black girls in Connecticut.

Mobs attack the organizers, 1835 to 1838

White mobs in Boston, Alton and Philadelphia attacked people who organized against slavery between 1835 and 1838. The mobs broke up meetings, destroyed printing presses, burned a hall and killed a newspaper editor. No one was convicted for any of the three attacks.

Boston. A mob of about 5,000 men surrounded a meeting of the Boston Female Anti-Slavery Society at its room on Washington Street on October 21, 1835. Maria Weston Chapman, her sisters Caroline, Anne and Deborah Weston, and eight other women had founded the society in 1834. Deborah Weston called it "the day when 5,000 men mobbed 45 women." A report in the *Hampshire Gazette* of October 28, 1835, described the crowd as citizens "calling themselves 'respectable.'"

When Mayor Theodore Lyman asked the women to leave, Maria Weston Chapman answered, "If this is the last bulwark of freedom, we may as well die here as anywhere." A bulwark is a strong wall that protects. The women walked out through the hissing crowd and finished their meeting at the Chapman house.

The mob caught William Lloyd Garrison and dragged him through the streets at the end of a rope. Mayor Lyman and other men took him to the Leverett Street Jail and kept him there overnight so the mob could not reach him. No record names a man in the mob or shows that anyone was charged.

Alton. A mob in Alton, Illinois, shot and killed Elijah Lovejoy on November 7, 1837. He was a Presbyterian minister who printed an antislavery newspaper. After threats to hurt him, he had moved from St. Louis to Alton, 25 miles away,

for safety. Mobs in Alton smashed three of his printing presses.

That night Lovejoy and a group of men guarded a fourth press in a warehouse. About 20 or 30 men of the town surrounded the building. Men in the armed mob shot Lovejoy five times as he stood in an open doorway. He died within the hour, two days before his 35th birthday.

Judge William Martin and a jury in Alton's city court tried the men who had defended the press, for riot, in January 1838. A riot is a violent disturbance by a crowd. The jury found the warehouse owner, Winthrop Gilman, not guilty in about fifteen minutes, and the prosecutor dropped the charges against the other eleven defenders.

Then eleven men of the mob, among them John Solomon, Levi Palmer and James M. Rock, were tried for riot for breaking in and destroying the press. The jury found them not guilty on January 20, 1838. No one was charged with killing Lovejoy, and the court record does not name who fired the shots.

Philadelphia. A mob in Philadelphia burned down Pennsylvania Hall, a new meeting hall where antislavery groups met, on May 17, 1838. The Pennsylvania Hall Association had raised about $40,000 to build it, and the Philadelphia Female Anti-Slavery Society raised much of that money. The hall opened on May 14, and that week the Anti-Slavery Convention of American Women met in it. A convention is a large organized meeting.

Three days after the hall opened, the mob set it on fire, using its own gas lights. Between 10,000 and 15,000 people watched, and 200 to 300 of them took part in the attack.

Mayor John Swift made what one history of the city calls "half-hearted efforts" to clear the crowd. Sheriff John G. Watmough arrested about a dozen men that night, and the crowd took them away from him. In all, officials arrested dozens of men and investigated five, and no court found any of them guilty. The same history records the rioters only as "dock workers and white rioters," and names none of them. The judges of the Pennsylvania Supreme Court ordered the county to pay $27,942.27 for the damage in 1847.

Sarah and Angelina Grimké

Who: Two sisters from a South Carolina slaveholding family who spoke against slavery to crowds of men and women, and were condemned from the pulpit for speaking at all.

When and where: Charleston, then Philadelphia and New England. Their speaking tour ran in 1837 and 1838.

In 23 weeks they spoke to more than 40,000 people.

Sarah and Angelina Grimké grew up in Charleston, South Carolina. Their father, John Faucheraud Grimké, was a judge and cotton planter who held people in slavery. Both sisters became Quakers, and Angelina left Charleston for Philadelphia in November 1829. In 1835 the sisters joined the American Anti-Slavery Society, and they became its first women agents, or organizers.

That July a mob in Charleston broke into the post office and burned antislavery papers. Angelina wrote a letter to William Lloyd Garrison about it, and he printed it in *The Liberator*. After that the sisters were not welcome in South Carolina.

The tour. At Poughkeepsie, New York, they spoke to a crowd of men and women together. In 23 weeks across 1837 and 1838 they spoke at 88 or more meetings in 67 towns, to more than 40,000 people.

The ministers' letter. In July 1837 the Congregational ministers of Massachusetts, leaders of a Protestant church, issued a Pastoral Letter to their churches, and ministers read it aloud from their pulpits. The letter states that women like the sisters put "the female character" in danger of "wide spread and permanent injury." It criticizes "females" who take "an obtrusive and ostentatious part in measures of reform," pushing forward to be seen. The ministers were attacking the sisters for being women who spoke in public.

Sarah Grimké answered the same month. The first of her "Letters on the Equality of the Sexes" ran in the *New England Spectator*. At a convention of antislavery women in 1837, Angelina proposed that "the right of

petition is natural and inalienable," a right that nobody can take away.

The gag rule. Since 1836 the House of Representatives had a rule that set aside every antislavery petition without reading it, and people called it the gag rule. Women of Brookline, Massachusetts, petitioned the House to end it on February 14, 1838. They called it "an assumption of authority, at once dangerous and destructive to the fundamental principles of republican government, to the rights of minorities, to the sovereignty of the People."

They meant that the rule went against the idea that the people hold the final power in their government, and against the rights of the smaller side in a vote. The first two names on the petition were Sarah M. Grimké and Angelina E. Grimké. Members of the House voted to renew the gag rule each time a new Congress met, every two years, until 1844.

At the State House. On February 21, 1838, Angelina Grimké gave a committee of the Massachusetts lawmakers antislavery petitions signed by about 20,000 Massachusetts women. She spoke to the committee in front of an overflowing crowd. The House of Representatives' historian calls her "the first woman in the United States to speak before a legislative body," a group of lawmakers.

"I stand before you as a southerner, exiled from the land of my birth by the sound of the lash and the piteous cry of the slave," she told them, exiled meaning driven from home and piteous meaning full of pain. She also said, "We are citizens of this republic and as such our honor, happiness, and well-being are bound up in its politics, government, and laws."

Both sisters lived to see the end of slavery in the United States. Sarah died in 1873 and Angelina in 1879, before American women won the vote.

Dorothea Dix

Who: A Massachusetts woman who visited every jail and poorhouse in her state, wrote down how the people who ran them held people with mental illness, and handed the record to the state's lawmakers.
When and where: Massachusetts, 1841 to 1843. Born 1802, died 1887.

At Medford she found one person in a stall for seventeen years.

Dorothea Dix went to teach a Sunday school class at the jail at East Cambridge, Massachusetts, on March 28, 1841. She found women with mental illness, called *insane* at the time, locked in with convicted criminals in "a bare, foul-smelling, unheated jail room." The word *insane* then meant a person whose mind worked in ways that their family or their town could not manage. The jail officials denied her account. Then they improved the women's conditions.

Every jail in the state. Between 1841 and 1843 Dix visited every jail, poorhouse and house of correction in Massachusetts. A poorhouse was a building where a town kept people who had no money. A house of correction was a kind of jail for people found guilty of small crimes. She wrote down what she saw in each one.

In January 1843 she gave the Massachusetts lawmakers a memorial, a formal written statement, of what she had found. "I come as the advocate of the helpless, forgotten, insane men and women held in cages, closets, cellars, stalls, pens!" she wrote. "Chained, naked, beaten with rods, and lashed into obedience!" A stall and a pen are enclosures built for animals, a rod is a stick used for beating, and lashed means whipped.

Her notes named places. At Lincoln she found a woman in a cage. At Medford she found one person "chained, and one in a close stall for seventeen years."

Printers reprinted the memorial as a pamphlet, and newspapers printed parts of it. The lawmakers voted money to make the state hospital at Worcester bigger. The United States had 13 hospitals for people with

> mental illness in 1843. By 1880 it had 123, and Dix had helped found 32 of them.

BY 1843: Mobs had killed an antislavery editor and burned a hall, women speakers had answered the ministers who told them to keep quiet, and a woman had put the treatment of people with mental illness before a state's lawmakers.

Coverture begins to break, 1837 to 1848

Coverture, the old English rule that a married woman had no legal existence apart from her husband, was still in force in 1848. The first law in any state that let a married woman own property in her own name passed in Mississippi in 1839. It came out of a court case about a Chickasaw woman, and the property in that case was an enslaved man.

Betsy Love's case. Betsy Love was Chickasaw, through her mother, Sally Colbert. She had inherited an enslaved man named Toney from her father, Thomas Love, and had given him to her daughter, Susan. Betsy Love's husband owed money to a man named John Fisher, and Fisher went to court to have Toney sold at auction to pay the husband's debt.

The judges of the Mississippi Supreme Court ruled in 1837 that Betsy Love was the legal owner, and that her husband had no rights over her property. They gave three reasons. Chickasaw families trace their family lines and pass on property through the mother. Since 1830, contracts made under Chickasaw law, including marriage contracts, had counted as binding in Mississippi. The third, in the words of the National Park Service's account, was that "Property in the Chickasaw Nation was owned by the women of the tribe."

The new laws. Mississippi's lawmakers passed the Married Women's Property Act in 1839. A married woman could now own property she got by gift, will, purchase or inheritance, as long as it did not come from her husband.

New York's lawmakers passed their own Married Women's Property Act in 1848, months before the Seneca Falls convention. The property a woman owned when she married now stayed hers. Her husband could not sell it, and

the people he owed money to could not take it to pay his debts. Lawmakers in other states copied New York's law. One part of coverture, the part about property a wife brought into a marriage, no longer applied in New York. The rest of coverture stayed the law.

Sojourner Truth takes her name, 1843

The woman born Isabella Bomfree in Ulster County, New York, in 1797 changed her name in 1843. She said the Spirit had called her to preach the truth, and she named herself Sojourner Truth. A sojourner is a traveler who stays in a place for a while. She traveled through Long Island, Connecticut and Massachusetts, speaking and singing.

She joined the Northampton Association of Education and Industry in Massachusetts in 1844. Abolitionists had founded it as a community. Its members supported women's rights and religious tolerance and opposed war, and the community was a stop on the Underground Railroad. It broke up in 1850, and her formal lecture tour began the next year.

Elizabeth Cady Stanton and Lucretia Mott

Who: Two women who were refused seats as delegates at an antislavery meeting in London because they were women, and who called the first women's rights convention eight years later.

When and where: London, 1840. Waterloo and Seneca Falls, New York, July 1848.

A New York publisher printed their declaration to make fun of it.

Elizabeth Cady Stanton and Lucretia Mott met in London in 1840, at the World Anti-Slavery Convention. Mott was a Quaker who had helped found the Philadelphia Female Anti-Slavery Society in 1833. American antislavery groups had elected women as delegates, and in the words of a National Park Service account, the women "were refused admission because of their sex." That account does not name the men who refused them.

After an argument, the men at the convention decided that the women "would sit at the rear of the hall, but not participate." Stanton was in London on her honeymoon. She found Mott in the women's section, and the two agreed to hold a women's rights convention when they got home.

Seneca Falls. Stanton moved to Seneca Falls in 1847. The town is on Cayuga land. The Cayuga Nation kept a 64,000-acre reservation there by a 1789 treaty, a written agreement, with New York. New York officials bought almost all of it in 1795 and the last three square miles in 1807. A federal judge in western New York ruled in 2017 that Congress had never approved those purchases, as a law of 1790 required.

Stanton later wrote about what pushed her to act: "The general discontent I felt with women's portion as wife, mother, housekeeper ... impressed me with a strange feeling that some active measures should be taken."

Tea at Waterloo. Five women met for tea at Waterloo, New York, on July 9, 1848, and decided to hold the convention. They were Stanton, Mott, Mary Ann M'Clintock, Martha Coffin Wright and Jane Hunt. They put a notice in the local paper calling people to "a Convention to discuss the social, civil and religious rights and condition of woman."

The convention met on July 19 and 20, 1848, at the Wesleyan Chapel in Seneca Falls, three miles east of Waterloo. More than 300 people came. The first day was meant for women only, and men took part on the second. Lucretia Mott's husband, James Mott, chaired the convention, which means he ran the meeting. Mary M'Clintock was the secretary.

The declaration. Stanton wrote a Declaration of Sentiments with the M'Clintock family, modeled on the Declaration of Independence. It listed, one by one, the wrongs men had done to women. "He has never permitted her to exercise her inalienable right to the elective franchise," meaning the right to vote. "He has taken from her all right in property, even to the wages she earns." That was coverture. "He has denied her the facilities for obtaining a thorough education," it went on, "all colleges being closed against her."

The vote on the vote. The people at the convention voted on a set of resolutions, written statements of what they wanted. Eleven passed with no one against them. The resolution demanding the vote met opposition and passed only narrowly. Frederick Douglass stood with Stanton and argued for it. He had escaped from slavery in Maryland ten years before and now published an antislavery newspaper, the *North Star*.

One hundred people signed the Declaration of Sentiments, 68 women and 32 men. A Library of Congress exhibition states that seventy-two years of organized campaigning for women's votes followed that resolution, up to the 19th Amendment in 1920.

The signers had written that they expected "misconception, misrepresentation, and ridicule," and it came. James Gordon Bennett, publisher of the *New York Herald*, printed the whole declaration in

his paper to make fun of it. Some signers later took their names off after people criticized them harshly for signing. The records do not say how many.

BY 1850

If you remember one thing: Women who learned to organize against slavery used the same petitions, speeches and meetings to demand rights of their own, ending with a vote at Seneca Falls to demand the vote.

- February 21, 1838: Angelina Grimké speaks to a committee of Massachusetts lawmakers, with petitions from about 20,000 women.
- 1839: Mississippi's lawmakers pass the first law in any state that lets a married woman own property in her own name.
- July 19 and 20, 1848: More than 300 people meet at Seneca Falls, and 100 sign the Declaration of Sentiments.

Next: After 1850, people who were denied their rights began to plan court cases on purpose.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 52 MINUTES

Americans who were denied rights in the second half of the 1800s learned to plan court cases on purpose. They chose a law to challenge, raised money, hired a lawyer and found a person willing to be arrested or refused. Susan B. Anthony voted in 1872, the Tape family sued a school board in 1884, and Black men on a committee in New Orleans arranged an arrest on a train in 1892.

The judge at Anthony's trial told the jury to find her guilty. The Tapes won in court, and school officials still kept their daughter out of the white school. Homer Plessy, the man the committee chose, lost in the Supreme Court.

Around the cases. After the Civil War, members of Congress and state lawmakers added three amendments, or changes, to the Constitution, and in 1869 the leaders of the women's movement split over whether Black men or women should get the vote first. Women first voted under territorial law in Wyoming and Utah.

Members of Congress voted for laws that shut Chinese immigrants out, and Chinese Americans fought those laws in court. In Memphis, the newspaper editor Ida B. Wells began counting the people killed by lynch mobs, crowds that killed people outside the law, with no trial.

Sojourner Truth

Who: A woman born into slavery in New York who spoke in public for more than thirty years for the rights of Black people and of women.

When and where: Akron, Ohio, May 1851. Washington, D.C., 1864. Died at Battle Creek, Michigan, November 26, 1883.

The famous "Ain't I a woman?" version was written by someone else.

Sojourner Truth published *The Narrative of Sojourner Truth: a Northern Slave* in 1850 and bought a house in Northampton, Massachusetts. She paid for part of it by selling photographs of herself. On them she printed the words "I sell the shadow to support the substance."

Akron, 1851. She spoke at the Ohio Women's Rights Convention in the Old Stone Church at Akron, Ohio, on May 28, 1851. She had not been invited. (One account gives the date as May 29. The convention ran for two days.)

Marius Robinson, one of the convention's secretaries, printed a report of her speech in the *Anti-Slavery Bugle* on June 21, 1851, about three weeks later. Truth was staying with Robinson and his wife, Emily, at the time, and a National Park Service page says it is believed he went over the text with her.

His version begins, "May I say a few words? I want to say a few words about this matter. I am a woman's rights. I have as much muscle as any man, and can do as much work as any man." She went on, "I have plowed and reaped and husked and chopped and mowed, and can any man do more than that?"

Later she said, "You need not be afraid to give us our rights for fear we will take too much, for we can't take more than our pint will hold." She ended, "But man is in a tight place, the poor slave is on him, woman is

coming on him, and he is surely between a hawk and a buzzard." Robinson wrote that it was "impossible to transfer it to paper, or convey any adequate idea of the effect it produced upon the audience."

The other version. Frances Dana Gage, who had run the convention, wrote the famous version, with the question "And ain't I a woman?" repeated through it. Gage published it in 1863, twelve years after the speech, while she was in South Carolina, in land held by the Union Army during the war. She did not check it with Truth, and she called it "but a faint sketch of her speech."

Gage's version gives Truth a Southern way of speaking. Truth grew up in Ulster County, New York, nearly 100 miles north of New York City. She spoke only Dutch until she was nine, and she was never enslaved in the South. Gage's version also has her say she bore thirteen children. She had five. A National Women's History Museum exhibit says Gage "fabricated" her version, which means she made it up. Robinson's version is the one written down that summer.

Fighting the rules. After 1857 Truth lived in Michigan. During the Civil War she recruited Black men for the Union Army, among them her grandson James Caldwell, who joined the 54th Massachusetts regiment in April 1863. That year officials in Indiana arrested her under a state law that barred Black people from entering the state, and held her for ten days.

She worked in Washington, D.C., in 1864 for the National Freedmen's Relief Association, which helped people who had just been freed from slavery. She rode the city's streetcars to force the companies to carry Black passengers. When a streetcar conductor tried to stop her from riding by force, she had him arrested and won her case. On October 29, 1864, she met President Abraham Lincoln at the White House.

From 1870 she campaigned for the federal government to give land in the West to people freed from slavery, and the campaign failed. She went to the polls in Battle Creek, Michigan, to vote in 1872, and election officials turned her away. That same year, Susan B. Anthony voted in Rochester,

> New York, and was arrested for it. Sojourner Truth died at Battle Creek on November 26, 1883. Her exact age is not known: counting from 1797, she was about 86, and she herself believed she was 105.

Massachusetts lawmakers outlaw segregated schools, 1855

After the Roberts family lost in court, William Cooper Nell and other Black Bostonians kept up their boycott of the segregated Smith School and their petition drives. Prince Hall and other Black Bostonians had first asked the state's lawmakers for schools for their children in 1787. In 1855, 68 years after that first petition, Massachusetts lawmakers voted to outlaw segregated public schools. A National Park Service page records that race-based segregation continued in Boston's public schools for decades afterward, even though state and federal laws forbade it.

A wife's wages, New York, 1860

A married woman in New York did not own the money she earned until 1860. Under coverture, her wages belonged to her husband. Women's rights campaigners carried petitions to every county in the state. They held conventions in 40 of its 60 counties and gave talks in 150 towns. Frances D. Gage, Antoinette Brown Blackwell and Susan B. Anthony were among the speakers.

The bill. Anthony gave a draft bill, a proposed law, to a state senator, Andrew J. Colvin, and he brought it to the state's lawmakers. State senators passed it on February 28, 1860, and members of the assembly on March 20, and Governor Edwin D. Morgan signed it.

One newspaper of the time printed the date of passage as March 30. The law states that "the earnings of any married woman, from her trade, business, labor or services, shall be her sole and separate property, and may be used or invested by her in her own name."

Three amendments, and the violence around them

Members of Congress and state lawmakers added three amendments to the Constitution after the Civil War. An amendment becomes part of the Constitution when lawmakers in enough states vote to approve it, which is called ratifying it. The 13th, ratified on December 6, 1865, bans slavery.

The 14th, part of the Constitution from July 9, 1868, states that "All persons born or naturalized in the United States" are citizens, naturalized meaning made a citizen after coming from another country.

It also says no state may "deny to any person within its jurisdiction the equal protection of the laws," so a state must apply its laws the same way to everyone. The 15th, ratified on February 3, 1870, states that the right to vote cannot be denied "on account of race, color, or previous condition of servitude," servitude meaning slavery.

The violence. In these same years white men attacked Black Americans who were living as free people. In May 1866, mobs that included Memphis police officers, firemen and businessmen killed 46 Black residents of the city. By the count of a committee of Congress, the mobs also burned 91 homes, 4 churches and 8 schools and raped five women. Rape means forcing a person into sex.

More than 150 armed white men led by Christopher Columbus Nash killed between 62 and 153 Black men at Colfax, Louisiana, on April 13, 1873.

The soldiers leave. After the war, federal soldiers stayed in the South, and some of them guarded its state houses. These were the years called Reconstruction, when members of Congress wrote Black Americans' rights into the three amendments and into new federal laws. President Rutherford B. Hayes ordered the soldiers out of the Louisiana state house in April 1877, the last state house in the South that they guarded. Reconstruction ended when they left. Chapter 24, Slavery and Freedom, tells that story in full.

A law struck down. Members of Congress voted for a Civil Rights Act in 1875. Under it, no one could lawfully deny anyone "the full and equal enjoyment of any of the accommodations, advantages, facilities, and privileges of inns, public conveyances on land or water, theaters and other places of public amusement." Public conveyances are trains, boats and coaches for hire.

The justices of the Supreme Court struck the law down in 1883. They held that the 14th Amendment did not let Congress stop private individuals from treating people unfairly because of race.

Why the women's movement split in 1869

The people who had worked together against slavery split in 1869 over who should get the vote next. The 15th Amendment, which members of Congress passed on February 26, 1869, bars denying the vote because of race and says nothing about sex. The 14th Amendment had already become the first place the Constitution used the word "male." Under its second section, the number of seats a state got in Congress depended on the number of its "male inhabitants" who could vote.

The meeting. The break came on May 12, 1869, at the third annual meeting of the American Equal Rights Association. People working to end slavery and women's rights campaigners had formed that group together in 1866. At the 1866 meeting, the Black campaigner Frances Ellen Watkins Harper had told the white women there, "You white women speak here of rights. I speak of wrongs."

At the 1869 meeting, Frederick Douglass argued that Black men needed the vote first. "I do not see how any one can pretend that there is the same urgency in giving the ballot to woman as to the negro," he said, using the word of the time for Black people. "With us, the question is a matter of life and death, at least, in fifteen States of the Union." He described Black families "in danger of having their homes burnt down over their heads," and children "not allowed to enter schools."

Susan B. Anthony answered, "If you will not give the whole loaf of suffrage to the entire people, give it to the most intelligent first," suffrage meaning the right to vote. Elizabeth Cady Stanton said, "If intelligence, justice, and morality are to have precedence in the government, let the question of women brought up first and that of the negro last."

Stanton's words. A National Park Service page records that Stanton used racist language in this fight. She called Black Americans "ignorant of the laws and customs" of the country. Stanton and Anthony argued that educated white

women should vote before Black men, and their words angered Black campaigners, among them Douglass and Harper.

Two groups. Stanton and Anthony started the National Woman Suffrage Association in 1869, which worked for a change to the Constitution. Lucy Stone, Henry Blackwell and Julia Ward Howe started the American Woman Suffrage Association, which worked state by state and supported the 15th Amendment. The two groups stayed apart until 1890, when they joined as the National American Woman Suffrage Association. Two daughters of the founders, Alice Stone Blackwell and Harriot Stanton Blatch, helped bring them together.

BY 1870: Three new amendments had ended slavery, made people born in the country citizens and barred the states from denying the vote by race. White mobs were killing Black Americans who lived as free people, and the women's movement had split in two.

Women vote first in the West, 1869 and 1870

The lawmakers of Wyoming Territory passed the first law in United States history that openly gave women the vote, and they had mixed reasons for it. William H. Bright, a Democrat from South Pass City, brought the bill to the territory's first group of lawmakers. Governor John Allen Campbell approved it on December 10, 1869. Women could now vote and hold office. A National Park Service page records that some of the lawmakers also wanted to draw more women to a territory that had very few.

Shoshone country. South Pass City was a gold-mining camp that grew up after gold was found in 1867. It stood in country that United States officials had recognized as the Eastern Shoshone's in an 1863 treaty, a homeland of about 44 million acres. White prospectors, men searching for gold, had been slipping into that country since the 1850s, in spite of the treaties. In the 1868 treaty of Fort Bridger, United States officials left the Shoshone only a reservation of about 3.2 million acres on the Wind River.

Utah. The lawmakers of Utah Territory passed a women's vote law two months later, and the acting governor signed it on February 12, 1870. Two days after that, a Salt Lake City schoolteacher named Seraph Young voted in the city election. She was the first woman in the United States to vote under a law that gave women the vote on the same terms as men.

The first offices. Esther Hobart Morris became justice of the peace at South Pass City, a local judge for small cases, on February 17, 1870. She was the first woman in the country to hold that office. That spring, women in Laramie were called to serve on juries, the first women in the country to do so. Later, the jury rule was read to cover only "male citizens over 21 years of age."

The Wyoming State Museum, which records this, does not say who read it that way. Under that reading, no women sat on Wyoming juries until 1949. Louisa Swain of Laramie voted on the morning of September 6, 1870.

Taken and given back. Members of Congress voted to take Utah women's vote away in the Edmunds-Tucker Act of 1887, as part of a federal campaign against the Mormon church. Utah women got the vote back in the state constitution when Utah became a state on January 4, 1896. When Wyoming became a state in 1890, its women kept the vote, so Wyoming was the first state where women could vote.

Susan B. Anthony

Who: One of fifteen women who voted in Rochester, New York, in 1872, and the only one put on trial for it.

When and where: Rochester, New York, November 5, 1872. Trial at Canandaigua, New York, June 17 to 19, 1873.

"I shall never pay a dollar of your unjust penalty."

The leaders of the National Woman Suffrage Association decided in 1871 that women should try to register to vote, be refused, and then sue in federal court. Their argument was that the 14th Amendment made voting one of the rights of every citizen. Anthony went to register in Rochester, and the officials let her. On election day she went back, and the officials let her vote.

Election day. Anthony and fourteen other women voted in the national election in the first district of Rochester's Eighth Ward on November 5, 1872. A poll watcher for one party, Sylvester Lewis, challenged her. The three election inspectors did what New York law required. They put her under oath and asked whether she was a citizen, whether she lived in the district and whether anyone had paid her. Her answers satisfied them, and they put her ballots in the boxes.

The arrest. Nine days later, a federal official, Commissioner William Storrs, issued warrants, papers ordering an arrest, for all fifteen women. A deputy marshal, a federal officer who makes arrests, came to Anthony's house on November 18. She told the story many times. "'What for?' I asked. 'To arrest you,' he said. 'Is that the way you arrest men?' 'No.' Then I demanded that I should be arrested properly." He took her downtown on the streetcar, and the government paid her fare.

All fifteen women were arrested, and only Anthony was tried. The charge stated that she had voted "without having a lawful right to vote," being "then and there a person of the female sex." At a hearing on November 29, 1872, the assistant prosecutor, John Pound, asked the election

inspectors, "Was Miss Anthony dressed in the apparel of a woman and had she the appearance of a woman?" Her lawyer objected. The inspectors answering had been arrested themselves for letting the women vote.

Talking to the jurors' towns. Federal courts at that time did not let a person on trial for a crime speak to the jury. So in March and April 1873 Anthony spoke to the people the jurors would come from. She gave a lecture, "Is It a Crime for a U.S. Citizen to Vote?", in twenty-nine villages and towns of Monroe County. The federal prosecutor, Richard Crowley, had the trial moved to Ontario County, and Anthony scheduled a speech in every village there too.

The ordered verdict. The trial opened at Canandaigua on June 17, 1873, before a jury of twelve men. The judge was Ward Hunt, a justice of the Supreme Court who was also hearing cases in that part of New York. When the prosecution finished, Hunt read out an opinion he had already written. He said voting was not one of the rights the 14th Amendment protected, and that Anthony had known New York let only men vote.

Then he told the jury to bring in a verdict of guilty. Her lawyer, Henry Selden, asked for each juror to be asked one by one for his own verdict. The clerk refused. The jury never talked the case over.

Selden asked for a new trial on June 19, because Hunt had taken the decision away from the jury, and Hunt refused. Then the judge asked the question a court asks before it sets a punishment. Three reports of what Anthony said next survive, and none comes from the court's own record. The words below are from the version she published herself.

What she said. "Has the prisoner anything to say why sentence shall not be pronounced?" Hunt asked. Anthony answered, "Yes, your honor, I have many things to say, for in your ordered verdict of guilty, you have trampled under foot every vital principle of our government." She went on, "Robbed of the fundamental privilege of citizenship, I am degraded from the status of a citizen to that of a subject."

"The Court cannot allow the prisoner to go on," Hunt said. She kept going: "since the day of my arrest last November, this is the first time that either myself or any person of my disfranchised class has been allowed a word of defense before judge or jury," disfranchised meaning denied the vote. "The prisoner must sit down," said Hunt. "The Court cannot allow it."

She said that every man who had judged her was her "political superior," whether "native or foreign born, white or black, rich or poor, educated or ignorant, awake or asleep, sober or drunk." Hunt said she had been tried "according to the established forms of law." She replied, "Yes, your honor, but by forms of law all made by men, interpreted by men, administered by men, in favor of men, and against women."

The fine. Hunt fined her $100 and the costs of the trial. "May it please your honor, I shall never pay a dollar of your unjust penalty," she said. Hunt answered, "Madam, the Court will not order you committed until the fine is paid." If he had jailed her for not paying, she could have appealed to the Supreme Court.

The deputy marshal sent to collect reported in July 1873 that he could "find no goods or chattles land or tenements" to take, no belongings and no buildings. Federal officials dropped the matter, and she never paid.

The three election inspectors were tried after her, found guilty and fined $25 each. They refused to pay too, and the marshal put them in jail on February 25, 1874. Anthony sent telegrams to Representative Benjamin Butler and Senator Aaron Sargent. President Ulysses S. Grant pardoned the inspectors on March 3, 1874, canceling their punishment. The same day, the voters of the Eighth Ward elected all three to their posts again.

The Comstock Act, 1873

Members of Congress voted on March 3, 1873, for "An Act for the Suppression of, trade in, and circulation of obscene literature and articles of immoral use," known as the Comstock Act. Sending certain things by mail became a federal crime. They included anatomy books, some novels, photographs of boxing matches and information about contraception, ways to prevent pregnancy. Under the act, it was a crime to mail a woman the information she needed about her own body.

One man's word. Obscene means indecent or shocking, but the act never said what counted as obscene. Anthony Comstock had pushed members of Congress to pass it, and he was appointed a special agent of the Post Office. In practice, he decided what the word meant. In court he often refused to let jurors see the item he had seized, saying it was too vile to show them. He claimed that over his life he seized 150 tons of books, made 4,000 arrests and drove 15 people to suicide.

A Harvard doctor says school will harm girls, 1873

Edward H. Clarke, a professor at Harvard Medical School, published a book in 1873 that said studying would damage girls' bodies. The book was *Sex in Education*, with the subtitle *A Fair Chance for the Girls*. Clarke argued that a girl's body spends the teenage years building the organs used to carry a pregnancy, and that studying the way boys studied would take the energy that work needed.

"Unless the work is accomplished at that period, unless the reproductive mechanism is built and put in good working order at that time, it will never be perfectly accomplished afterwards," he wrote. He warned that the result would be "monstrous brains and puny bodies."

The harms he listed. A Smith College history page lists the harms Clarke said would follow. They were atrophy of the uterus and ovaries, meaning those organs shrinking and stopping working, and masculinization, a girl's body taking on a man's traits. They also included sterility, being unable to have children, and insanity and death.

He said women should study no more than four hours a day, and not at all during their monthly bleeding. He described a patient he called Miss A: "She graduated at nineteen, the first scholar, and an invalid," a person too sick to live a normal life.

Leaders of the new women's colleges answered him. Smith College opened in 1875, and its first president, Laurenus Clark Seelye, said, "It is to preserve her womanliness that this College has been founded." Smith's leaders built supervised houses for its students in answer to worries like Clarke's.

Mary Putnam Jacobi's evidence. Mary Putnam Jacobi answered with evidence. She had earned a pharmacy degree, a degree in preparing medicines, in 1863, the first woman in the country to do so. She earned a medical degree in 1864 and a second medical degree in Paris in 1871.

Jacobi sent out a thousand printed lists of questions through women's rights groups and collected answers from hundreds of women, rich and poor, in towns and on farms. Her essay *The Question of Rest During Menstruation*

won Harvard's Boylston Prize in 1876, and she was the first woman to win it. She found no evidence that monthly bleeding made women unfit to study or work.

Supreme Court justices rule that the Constitution gives no one the vote

Virginia Minor, a woman in St. Louis, Missouri, used the same argument Anthony had used. The registrar, Reese Happersett, refused to register her to vote, and she sued him, arguing that she was a citizen and that citizens have the right to vote.

All the justices of the Supreme Court ruled against her in *Minor v. Happersett*, decided in the court term that began in October 1874. They held that "the Constitution of the United States does not confer the right of suffrage upon any one," and that a state could give the vote to men only.

Women were citizens, but in the justices' reading, being a citizen did not include a right to vote. After that ruling, the only way left to win the vote everywhere at once was a new amendment to the Constitution.

BY 1875: Women voted in Wyoming and Utah, Anthony had been found guilty for voting in New York, and the Supreme Court's justices had ruled that the Constitution gave no one the vote.

Getting into college, and into court as a lawyer

Women won their first places in colleges, in medicine and in law in these years. A woman who got in could still be kept from staying, or from working.

Colleges. Vassar College, a college for women at Poughkeepsie, New York, opened on September 26, 1865, with 353 students. Smith, Wellesley and Bryn Mawr opened within the next twenty years. Two teachers from New England started Spelman in the basement of a Baptist church in Atlanta, Georgia, on April 11, 1881. Its first class was eleven students: ten women, some of them formerly enslaved, and one girl.

Universities. Some universities that had taken only men began to take women. The first woman to enter Cornell, Jennie Spencer of Cortland, came in 1870. She had won a state scholarship, and she left because there was nowhere near the campus for a woman to live. Women entered Cornell regularly from 1872.

Doctors and lawyers. Rebecca Lee Crumpler received the degree "Doctress of Medicine" from the New England Female Medical College on March 1, 1864. She was the first African American woman in the United States to earn a medical degree. After the war she moved to Richmond, Virginia, and worked with the Freedmen's Bureau, the federal office that helped people freed from slavery, treating patients who had no other doctor.

Arabella Mansfield passed the Iowa bar examination on June 15, 1869, and became the first woman admitted to a state bar, the body of lawyers allowed to practice in a court. Iowa's law described "any white male citizen" as eligible, and she argued that nothing in it forbade a woman.

Charlotte E. Ray graduated from Howard University School of Law in 1872 and was admitted to the District of Columbia bar, the first Black woman lawyer in the country. A page by the District of Columbia attorney general's office says she later had to close her law office because, as a woman of color, she could not get enough clients.

Myra Bradwell. Other women passed the law examination and were still refused a license. Myra Bradwell passed the Illinois bar examination in 1869, and the judges of the Illinois Supreme Court refused her a license because she was married.

She took her case to the Supreme Court of the United States and lost, eight to one, in *Bradwell v. The State*, in the term that began in October 1872. Justice Samuel F. Miller wrote for the Court that each state decides who may practice law, and that the 14th Amendment does not protect that right.

Justice Joseph Bradley wrote separately, "The paramount destiny and mission of woman are to fulfil the noble and benign offices of wife and mother. This is the law of the Creator," paramount meaning most important and benign offices meaning kind duties. He meant that a woman's place was in the home. He also wrote, "A married woman is incapable, without her husband's consent, of making contracts which shall be binding on her or him." That sentence states the rule of coverture.

Belva Lockwood. The justices of the Supreme Court refused to let Belva Lockwood argue cases before them in 1876. They said "none but men are permitted to practice before [us] as attorneys and counselors." Lockwood lobbied Congress, pressing its members in person, until they voted for "An Act to relieve certain legal disabilities of women." On March 3, 1879, she became the first woman admitted to the bar of the Supreme Court.

Laws that shut Chinese immigrants out, 1875 and 1882

While Americans argued about who should be added to the voters, members of Congress voted for laws to keep Chinese immigrants out. President Ulysses S. Grant signed the Page Act on March 3, 1875. A National Park Service page calls it the first federal law to restrict immigration. Agents at the ports could stop and question "any subject of China, Japan, or any Oriental country," using an old word for Asian, to find out whether the person had come "without their free and voluntary consent."

Who the agents stopped. Under the act, no one could bring in women "for the purposes of prostitution," which means to be paid for sex, and officials decided which women those were. The officers applied it to Asian women in general, and in practice almost no Chinese women could enter the country.

The Exclusion Act. President Chester A. Arthur signed the Chinese Exclusion Act on May 6, 1882. Chinese laborers could not enter the country for ten years. Section 14 states that "no State court or court of the United States shall admit Chinese to citizenship." Members of Congress extended the law in 1892, made it permanent in 1902 and repealed it, or canceled it, in 1943. Britannica calls it "the first and only major federal legislation to explicitly suspend immigration for a specific nationality."

Frances Thompson arrested for living as a woman, 1876

Memphis police arrested a Black woman named Frances Thompson for wearing women's clothing on July 10, 1876. Writers today, among them reporters at CNN, describe her as a transgender woman, a woman whose body was counted as male at birth. She told police and reporters she had lived her whole life as a woman.

Ten years earlier, in 1866, Thompson had told a committee of Congress that seven white men had raped her and another woman, Lucy Smith, during the Memphis massacre.

What the city did to her. After the 1876 arrest, four doctors, among them Dr. Joseph Nuttall, a doctor working for the city, examined her body against her will at the central police station on Adams Street and declared her a man. Modern reports on her case by MLK50 and CNN do not say how the doctors examined her. The city's recorder, an official who judged small cases, fined her $50.

She could not pay, so the recorder sentenced her to 100 days on the city chain gang, a group of prisoners chained together and made to work outdoors. She cleaned the streets for 50 cents a day toward her fine. Crowds gathered to stare at her. Officials made her pose for photographs in men's and women's clothing, and newspapers printed them. After harassment from other prisoners and the public, city officials took her off the chain gang and held her at the police station for the rest of her sentence.

Memphis newspapers printed the doctors' finding, and writers for several of the papers used it to try to show that her 1866 testimony about the rape was false. Her neighbors later found her sick and took her to the city hospital. She died there of dysentery, a disease of the gut, on November 1, 1876.

Sarah Winnemucca

Who: A Northern Paiute woman who worked as an interpreter for the United States government and the Army, saw what United States officials and soldiers did to her people, and spent the rest of her life saying so in public.

When and where: Nevada and Oregon, 1844 to 1891. The removal to the Yakama Reservation, 1878 to 1879.

"My people were only given one week to get ready in."

Sarah Winnemucca's Paiute name was Thocmetony, which means Shell Flower. She was the daughter of Chief Winnemucca and the granddaughter of Chief Truckee, and both men had guided white travelers across Paiute country. A National Park Service page lists what she saw growing up: "the murder of two uncles, the attempted rape of her sister, the Pyramid Lake War and Bannock War, and the burning of Paiute food supplies."

Working for the government. She spoke several languages. In the 1870s she worked as an interpreter for the Indian Office and as a go-between for the Paiute and the United States Army. Part of that work was persuading Paiute bands to move onto reservations. During the Bannock War of 1878, she found her father's band, who were being held by Bannock fighters, and led them out to the Army's protection. She traveled 230 miles in three days.

The removal. Army soldiers then removed Paiute people who had not fought in the war at all. The House of Representatives' historian writes that the Paiute were punished for the part a small number of them had taken in the fighting. Winnemucca wrote that "just a little before Christmas" in 1878 she learned that President Rutherford B. Hayes had ordered it. "My people were only given one week to get ready in," she wrote.

Soldiers took them off the Malheur Reservation, which lay inside their own homeland, and sent them 350 miles north to the Yakama Reservation, the Yakama Nation's land. About 550 Paiute men, women and children made the journey. It took nearly a month, through snow and over two mountain ranges. The women and children rode in wagons.

Some of the men walked in leg irons, metal rings locked around the ankles and chained together. The records do not say which soldiers or officials put the irons on them. Five children, one woman and an old man died on the road. Their bodies were left beside it, because the ground was too frozen to dig graves.

At Yakama. The historian Sally Zanjani estimated that more than a fifth of the group died at Yakama afterward, most of them from hunger and disease. W. V. Rinehart, the government agent at Malheur, wrote a report in August 1879. In it he complained about what the removal had cost. He also wrote that officials had sent the Paiute off without enough supplies, even though supplies were on the way.

The campaign. After the removal, Winnemucca stopped working as an interpreter and became a campaigner. In the early 1880s she gave more than 300 speeches. She met President Hayes and the Secretary of the Interior, Carl Schurz, in Washington in 1880. Hayes did not change his decision. Schurz gave her a written promise that her people could go back to Malheur, and then he did not keep it. President Chester Arthur opened the Malheur land to white settlers in 1882 and 1883.

She petitioned Congress in 1884 to give Malheur back to her people. She asked the government to "sweep away the [Indian] agency system" and "give us homes to live in, for God's sake and for humanity's sake." She published *Life Among the Piutes: Their Wrongs and Claims* in 1883, with help from the reformers Elizabeth Peabody and Mary Peabody Mann. A page by the Oregon Historical Society calls it the first autobiography published by a Native American woman.

Her school. She also started a school for Paiute children and taught them in Paiute and in English. It closed after three years, when the money ran out. In the same years, United States officials were opening boarding schools, where Native children lived far from their families. An Army officer, Richard Henry Pratt, opened the first of them, Carlisle, in Pennsylvania in 1879.

Staff at these schools cut the children's hair, gave them English names and forbade their languages, and punished children who spoke them with beatings, solitary confinement and withheld food. A federal count published in 2024 found that at least 973 children died at the schools.

Sarah Winnemucca died in 1891. Her statue stands in the United States Capitol for the state of Nevada. It shows her at 35, holding up a shell flower in her right hand, with a book at her side.

BY 1879: Women were entering colleges and courtrooms, members of Congress had passed the first federal law to keep immigrants out, and soldiers had marched the Paiute away from their reservation in winter.

Standing Bear goes to court, 1879

A Ponca leader named Standing Bear went to federal court in 1879 and won a ruling that a Native American is a person under the law. Then he toured the country arguing his case in public.

The Ponca land. The Ponca lived at the mouth of the Niobrara River in Nebraska. They had never fought the United States and had never sold that land. But in 1868 United States officials made a treaty with the Sioux that created the Great Sioux Reservation, and the Ponca land lay inside its lines.

Members of Congress set aside $25,000 in 1876 to move the Ponca to Indian Territory "if they consented to go." Indian Territory was the land where United States officials sent Native nations they forced from their homes.

Officials took eight Ponca chiefs south to choose new land. The chiefs turned down every place they saw and asked to go home. When officials refused, the chiefs walked more than 500 miles back to Nebraska in winter, in 40 days, "with only a few dollars and one blanket each."

The forced move. Federal officials ordered the move anyway, "using Army troops if necessary." An Indian Office inspector, E. C. Kemble, led the first 170 people south on April 30, 1877. He ordered the agent, E. A. Howard, to move the rest of the Ponca. They left on May 16, 1877, with 25 soldiers guarding them, and left behind their homes, crops and burial grounds. Standing Bear's daughter died on the way.

In July they crossed into Indian Territory onto Quapaw land. Later they chose 101,894 acres that had been the Cherokee Nation's land. A third of the Ponca died after they arrived.

Carrying his son home. Standing Bear's son, Bear Shield, died of a fever at sixteen in late December 1878. In early January 1879 Standing Bear set out on foot for Nebraska with a small group of his people, mostly women and children, carrying his son's bones to bury at home. The walk took about two months. Soldiers arrested the group and held them at Fort Omaha, an Army post, under General George Crook.

The court. Two Omaha lawyers, John L. Webster and Andrew J. Poppleton, took his case. They asked for a writ of habeas corpus, an old legal rule under which a person being held can make the government show its legal reason for holding him.

Judge Elmer S. Dundy of the United States District Court in Omaha heard the case on May 1 and 2, 1879. Standing Bear told the court, "That hand is not the color of yours. But if I pierce it, I shall feel pain ... I am a man. The same god made us both."

Dundy ruled on May 12, 1879, that "an Indian is a person within the meaning of the laws of the United States," and ordered the Ponca released. From June 1879 Standing Bear toured the country. He spoke in Chicago, Pittsburgh, Philadelphia, Boston, New York, Washington and Baltimore. The journalist Thomas Tibbles went with him, and so did Susette La Flesche of the Omaha Nation, called Bright Eyes, who interpreted, and her brother Frank.

What the ruling did not do. The ruling covered Standing Bear and the people held with him. Native people did not become citizens because of it. John Elk, a Native man who had left his nation and lived in Omaha, tried to register to vote in April 1880. Charles Wilkins, the registrar of his ward, refused him because he was Native.

The justices of the Supreme Court ruled against Elk, seven to two, in *Elk v. Wilkins* on November 3, 1884. Justice Horace Gray wrote that a Native person born into a Native nation did not become a citizen by being born in the United States. Justices John Marshall Harlan and William B. Woods dissented, which means they disagreed with the ruling.

Reformers who asked for allotment

Standing Bear's speeches reached white reformers in the East. They organized, lobbied and petitioned, and what they won was a law under which Native nations lost about two-thirds of the land they still held.

Helen Hunt Jackson heard Standing Bear speak in Boston in 1879. Two years later she published *A Century of Dishonor*, a book about how the United States government had treated Native nations. She sent a copy to every member of Congress with a note: "Look upon your hands: they are stained with the blood of your relations."

The reform groups. Two Philadelphia women, Mary Lucinda Bonney and Amelia Stone Quinton, founded the Women's National Indian Association in 1879. A group of wealthy Philadelphia men founded the Indian Rights Association on December 15, 1882, after Herbert Welsh and Henry S. Pancoast came back from a visit to the Sioux in Dakota Territory. They said their aim was to secure for Native people "the political and civil rights already guaranteed to them by treaty and statutes."

Its members believed that Native people would survive only by becoming Protestant Christians and living like white Americans. From 1883 a Quaker hotel owner, Albert K. Smiley, gathered these reformers every year at his resort at Lake Mohonk, New York. Mark S. Joy, writing about the conferences in 2019, names no Native people taking part. The reformers asked members of Congress to break reservation land into plots owned by single families, which was called allotment.

The Dawes Act. Members of Congress did it in the Dawes Act, signed on February 8, 1887, and named for Senator Henry Dawes of Massachusetts. The head of each family got 160 acres, and each person who took a plot became a citizen. Federal officials could buy the land that was "not allotted" from the Native nations and sell it to settlers.

Under the Dawes Act and the laws that followed it, the land held by Native nations went from about 138 million acres in 1887 to about 48 million by 1934. That is about 90 million acres gone, two acres out of every three. Another

count gives about 137 million and 47 million.

Mary and Joseph Tape

Who: Two Chinese American parents who took San Francisco's school leaders to court because they would not let their eight-year-old daughter into the school near their home. They won, and school officials still kept her out.

When and where: San Francisco, 1884 and 1885.

"Is it a disgrace to be Born a Chinese?" Mary Tape wrote.

San Francisco's schools had been separated by race from the start. The first public schools, opened in 1851, were for white children only. A school for Black children opened in 1854, and one for Chinese children in 1859. The city's school officials closed the Chinese school in 1870, which left Chinese American children with no public school at all.

The Ward case. A Black mother, Harriet A. Ward, had taken the same question to court in January 1874 for her daughter, Mary Frances, who was eleven. Noah F. Flood, the principal of the Broadway Grammar School, had refused to take her.

California law said, "The education of children of African descent, and Indian children, shall be provided for in separate schools." The judges of the California Supreme Court ruled against the Wards. A Black child could be kept out of a white school, they held, wherever a separate school existed, and San Francisco had one.

The family. Mary Tape was born near Shanghai, China, in 1857 and came to the United States alone in 1868, when she was eleven. Women of the Ladies' Protection and Relief Society took her in. She learned English there and took the name Mary McGladery, after one of the women who cared for her.

Joseph Tape, born Jeu Dip, arrived in 1864. He worked as a servant, ran a milk delivery business, and worked as a broker, a go-between in business deals, and as an interpreter for the Imperial Consulate of China. They had

four children, Mamie, Frank, Emily and Gertrude, and lived in the Cow Hollow part of San Francisco.

The lawsuit. They took Mamie to Spring Valley Primary School, the all-white school near their home, in 1884. The principal, Jennie Hurley, refused her because her family came from China. Joseph Tape sued the San Francisco Board of Education, the school superintendent Andrew Jackson Moulder, and Hurley herself.

The judge of the Superior Court of California ruled for the Tapes on January 9, 1885. He held that keeping out "children of Chinese parents" broke California law and the 14th Amendment's promise of equal protection. The members of the Board of Education appealed, and the judges of the California Supreme Court agreed with the lower court.

Shut out again. Moulder sent a telegram to the state's lawmakers as soon as the decision came down. The lawmakers passed a law that let school officials set up "separate schools for children of Mongolian or Chinese descent," using a word white Americans then used for people from East Asia. On April 8, 1885, officials at Spring Valley refused Mamie again. This time they said she had no vaccination certificate and the classes were full.

Her letter. That day Mary Tape wrote to the Board of Education, in English, her second language. The *Daily Alta California* printed her letter, with her own spelling, on its front page on April 16, 1885.

"I see that you are going to make all sorts of excuses to keep my child out of the Public schools. Dear sirs, Will you please to tell me! Is it a disgrace to be Born a Chinese? Didn't God make us all!!! What right have you to bar my children out of the school because she is a chinese Decend."

"It seems no matter how a Chinese may live and dress so long as you know they Chinese. Then they are hated as one. There is not any right or justice for them. ... Mamie Tape will never attend any of the Chinese schools of your making! Never!!! I will let the world see sir What justice

there is When it is govern by the Race prejudice men! ... I guess she is more of a American then a good many of you that is going to prevent her being Educated." She signed it "Mrs. M. Tape."

Mamie and her brother Frank became the first two pupils at the new Chinese Primary School in Chinatown on April 13, 1885. A history page by the San Francisco Unified School District states that the district's officials reopened the separate Chinese school "as a way to avoid integrating white schools," to keep Chinese children out of white children's schools.

BY 1887: A federal judge had ruled that a Native American is a person under the law, reformers had won the Dawes Act, and the Tapes had won their case and still lost their daughter's place in the school.

Chinese Americans take the law to court, 1886 to 1898

Chinese Americans used the 14th Amendment to challenge laws aimed at them, and sometimes they won. The members of San Francisco's Board of Supervisors, the officials who made the city's laws, voted in 1885 to require a permit to run a laundry in a wooden building. About 200 Chinese laundry owners applied, and the supervisors refused every one. About 80 owners who were not Chinese applied, and all but one got a permit.

Yick Wo. Officials arrested more than 150 Chinese men for working without a permit. One of them was Yick Wo, who had run a laundry at 349 Third Street for 22 years, and the sheriff, Peter Hopkins, held him in jail. All the justices of the Supreme Court ruled for Yick Wo in *Yick Wo v. Hopkins* on May 10, 1886. Justice Stanley Matthews wrote that a law "applied and administered by public authority with an evil eye and an unequal hand" denied people the equal protection of the laws.

The Geary Act. Members of Congress voted for the Geary Act in 1892. It kept Chinese exclusion going, and every Chinese laborer had to register and carry a certificate of residence at all times. A man caught without one could be

sentenced to up to a year of hard labor and then deported, sent out of the country by force.

To prove he had lived in the country, he needed a white witness. The act, as the Supreme Court quoted it, asked for "at least one credible white witness." A KQED news report says two.

The leaders of the Chinese Six Companies, a Chinese American association, told people not to register. They collected a dollar from each Chinese resident to pay lawyers and put out leaflets that said, "The Geary Act is an unjust law." About 100,000 Chinese people lived in the United States, and only about 13,000 registered in the first period. A 2026 KQED news report, drawing on the historians Kelly Lytle Hernández, Alexander Jin and Russell Jeung, calls the refusal the first mass resistance to deportation in the United States.

The test case. Three men became the test case: Fong Yue Ting, Wong Quan and Lee Joe. Lee Joe had lived in the country since about 1874. He had tried to get a certificate in April 1893 and was refused, because his only witnesses were Chinese. The justices of the Supreme Court upheld the law, five to three, in *Fong Yue Ting v. United States* on May 15, 1893. Justice David Brewer wrote in dissent that deportation meant "removal from home, from family, from business, from property."

Wong Kim Ark. Wong Kim Ark was born at 751 Sacramento Street in San Francisco in 1873, to Chinese parents. When he came home from a visit to China in August 1895, officials at the port of San Francisco refused to let him back in.

The justices of the Supreme Court ruled six to two on March 28, 1898, that he was a citizen under the 14th Amendment, because he was born in the United States. Justice Horace Gray wrote for the Court. Chief Justice Melville Fuller dissented, and Justice John Marshall Harlan joined him.

Black men in New Orleans plan an arrest, 1891 to 1896

A group of Black men in New Orleans formed a committee in 1891 to fight a new law in court, and they planned the arrest that started the case. Louisiana's lawmakers had passed Act 111, the Separate Car Act, on June 10, 1890.

Railroads had to give "equal but separate" seats to Black and white passengers. A passenger who sat in the wrong car could be fined $25 or jailed for 20 days. Black New Orleanians protested, and 16 of the state's lawmakers at the time were Black. The majority passed it anyway.

The committee. French-speaking men of African descent in New Orleans formed the Comité des Citoyens, the Citizens' Committee, in September 1891. Its full name was the Citizens' Committee to Test the Constitutionality of the Separate Car Law, which means whether the law fit the Constitution. The members raised money and hired a lawyer, Albion W. Tourgée.

The arrest. Homer Plessy volunteered to be arrested. He was born Homère Patris Plessy in New Orleans on March 17, 1863, to free parents of color, and he worked as a shoemaker. On June 7, 1892, he bought a ticket to Covington, Louisiana, on the East Louisiana Railroad and sat in the car for white passengers. Railroad officials were in on the plan, because the company did not want to pay the cost of running extra cars.

As planned, a private detective hired for the job arrested him, so that he would be charged with exactly the offense the committee wanted to fight. Judge John H. Ferguson of the Orleans Parish criminal court ruled against Plessy. The case is called Plessy v. Ferguson after the two men.

The ruling. The justices of the Supreme Court ruled against Plessy, seven to one, on May 18, 1896. Under their ruling, a state could make Black and white passengers ride in separate cars, as long as the cars were equal.

Justice Henry Billings Brown, who wrote the decision, said such laws did not mark either race as lower than the other. If Black Americans felt marked as lower, he wrote, that was "solely because the colored race chooses to put that construction upon it," because they chose to read the law that way.

Justice John Marshall Harlan disagreed, alone. "In view of the constitution, in the eye of the law, there is in this country no superior, dominant, ruling class of citizens," he wrote. "Our constitution is color-blind, and neither knows nor tolerates classes among citizens." The ruling stood for nearly sixty years, until *Brown v. Board of Education* in 1954. Homer Plessy died in New Orleans in 1925.

Ida B. Wells

Who: A Memphis newspaper editor who counted the people killed by lynch mobs, published the count, showed that the reason given for the killings was false, and was driven out of the South for it.

When and where: Born at Holly Springs, Mississippi, July 16, 1862. Memphis, Tennessee, 1884 to 1892.

She counted 728 lynchings in eight years and printed the number.

Ida B. Wells was born into slavery and freed in 1865. In 1884 she bought a first-class railroad ticket, and the train crew ordered her into the car for Black passengers. She refused, and they removed her from the train by force. She sued the railroad and won $500 in the circuit court, and the judges of the Tennessee Supreme Court overturned the award.

She taught in a segregated public school in Memphis. In 1891 she wrote a newspaper article showing that the board of education gave the Black schools less money, and she lost her teaching job for it. Neither the National Park Service nor the Mississippi Department of Archives and History, in their accounts of her life, names the official who fired her.

The People's Grocery. Her friend Thomas Moss was part owner of the People's Grocery, a store run by Black owners in a mostly Black part of Memphis. It was taking business from a white-owned store nearby. William Barrett, one of the rival owners, got a judge to declare the People's Grocery a public nuisance, a place that harms the people around it, and to make him a deputy.

Barrett and a group of deputies in ordinary clothes marched on the store. Shots were fired, and three deputies were wounded. "Thirty-one men were arrested and thrown in jail as 'conspirators,'" Wells wrote. Moss and two others from the store were among them.

The lynching. In the early hours of March 9, 1892, a mob took Thomas Moss, Calvin McDowell and a third man out of their cells in the Memphis jail. The men of the mob took them to the rail yards and shot them to death. Calvin McDowell was shot with a shotgun at close range. The third man was shot in the neck, and records give his name two ways, as Will Stewart and as Henry Steward.

Moss's last words were reported as "Tell my people to go west. There is no justice for them here." Black Memphis "urged and waited for the authorities to act in the matter and bring the lynchers to justice. No attempt was made to do so," Wells wrote. No one was ever arrested for the killings.

Black Memphis answers. Black Memphians organized in answer. "The black men left the city by thousands," Wells wrote. The people who stayed stopped riding the streetcars. The streetcar company's superintendent, manager and treasurer came to the office of Wells's newspaper, the *Free Speech*, and asked its editors to tell people to ride again.

The editorial. Wells printed an editorial in the *Free Speech* on May 21, 1892: "Nobody in this section of the country believes the old thread-bare lie that Negro men rape white women." The excuse given in public for lynching was that Black men had raped white women. Wells wrote that white men were killing Black men over relationships that white women had chosen.

She described what happened next in a pamphlet later that year. After editorials in the Memphis daily papers, men of the city held a meeting at the Cotton Exchange Building on May 24, 1892. "A committee was sent for the editors of the Free Speech," she wrote, "and the only reason the

open threats of lynching that were not carried out was because they could not be found." Her business partner, J. L. Fleming, "had to leave town to escape the mob, and was afterwards ordered not to return."

Wells was on vacation in New York. Letters and telegrams warned her "that bodily harm awaited my return." Then "Creditors took possession of the office and sold the outfit, and the Free Speech was as if it had never been." She did not go back. Her return home, she wrote, was "forbidden under penalty of death."

The count. Her pamphlet *Southern Horrors: Lynch Law in All Its Phases* came out in 1892. "The dark and bloody record of the South shows 728 Afro-Americans lynched during the past eight years," she wrote. "Only one-third of the 728 victims to mobs have been charged with rape, to say nothing of those of that one-third who were innocent." Two out of three had never been accused of the crime that lynching was said to punish. Using the *Chicago Tribune*'s yearly lists, she later counted 241 people lynched in 1892.

She wrote *The Reason Why the Colored American Is Not in the World's Columbian Exposition* for the world's fair in Chicago in 1893, printed in three languages. In 1894 she toured Britain speaking about lynching, and people there formed a London Anti-Lynching Committee. She married Ferdinand Barnett in 1895, and they had four children. The same year she published *A Red Record*, 100 pages of lynching figures and history.

Frances Willard. One white reformer fought her in public. Frances Willard was president of the Woman's Christian Temperance Union, a women's group that campaigned against alcohol. Willard had told a newspaper, the *New York Voice*, in 1890 that "The colored race multiplies like the locusts of Egypt," and that "The safety of woman, of childhood, the home, is menaced," or threatened, "in a thousand localities."

Wells printed those words in Britain in 1894. At her group's convention in Cleveland on November 5, 1894, Willard said Wells had "put an imputation upon half the white race in this country that is unjust." An imputation is an accusation. Wells answered her in *A Red Record*.

Suffragists remember the clan mothers, 1891 and 1893

A National Park Service page records that the power of Haudenosaunee women inspired leaders of the women's rights movement. Elizabeth Cady Stanton told the National Council of Women in 1891 that Haudenosaunee women "did not hesitate, when occasion required, 'to knock off the horns,' as it was technically called, from the head of a chief." She meant the clan mothers' power to remove a chief.

Matilda Joslyn Gage wrote in the *New York Evening Post* that among the Haudenosaunee, "In the home the wife was absolute." The Wolf Clan of the Mohawk Nation adopted Gage as an honorary member in 1893 and gave her the name Ka-ron-ien-ha-wi, "She who holds the sky."

A city government overthrown, Wilmington, 1898

Armed white men overthrew the elected government of Wilmington, North Carolina, on November 10, 1898. Black and white Republicans held the city's offices, and Alex Manly edited the *Daily Record*, the city's Black newspaper. Manly had printed an editorial that challenged white beliefs about relationships between Black and white people, and Democratic Party leaders used it to stir up white voters. Before the election two days earlier, the Red Shirts, a group of Democratic Party supporters, had frightened Black people and forced white voters to vote for Democrats.

The attack. Alfred Moore Waddell, a former Confederate officer and congressman, led the attack. About 500 white men gathered at the armory, a building where weapons were stored, and the crowd grew to perhaps 2,000. They broke into the *Daily Record* office and set the building on fire. Violence broke out across the town.

KNOWN:	The white men murdered at least 14 men, and every one of the dead who was reported was Black. The coroner, the county official who looks into deaths, counted 14.
DISPUTED:	The state's commission report of 2006 gives as many as sixty. The records do not give an exact count.

The new government. That afternoon Waddell's men forced the Republican mayor, Silas P. Wright, the aldermen, the members of the city council, and other city officers to resign. The new aldermen made Waddell mayor. The new city leaders banished Black leaders and white Republicans, forcing them to leave the city for good. Manly escaped.

Governor Daniel Russell did not ask for federal help, and President William McKinley sent no troops. Lawyers in the United States Attorney General's office investigated and closed the case in 1900 without charging anyone. The records do not name the men who fired.

BY 1900

If you remember one thing: People who were denied their rights learned to bring test cases on purpose, and judges ruled against many of them, from Anthony's vote to Plessy's train seat.

One number: 728. That is how many Black Americans Ida B. Wells counted as lynched in the eight years before 1892, about 91 a year, or nearly two every week.

Next: In the new century, organizers took their cause to the White House gates and to the courts, one law school at a time.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 64 MINUTES

The people who organized for their rights between 1900 and 1950 worked out the methods that organizers in the 1960s later used. Alice Paul put women on a picket line at the White House gates in 1917. Disabled New Yorkers sat down in a city relief office in 1935 and would not leave.

Charles Hamilton Houston planned a chain of court cases, one law school at a time. A. Philip Randolph planned marches on Washington in 1941 and 1948, and both times a president signed an order, first against unfair hiring by race in defense factories, then against unequal treatment in the armed forces.

What they were up against. Women won the vote in 1920, and white officials in the South kept many Black women from voting with the same rules and violence they used against Black men. White mobs attacked Black Americans in Springfield in 1908 and in 25 cities in 1919. White Tulsans burned the Black district of Greenwood in 1921 and killed as many as 300 people.

Officials in 30 states had at least 60,000 people operated on so that they could never have children. After President Franklin D. Roosevelt signed an order in 1942, Army officers forced about 120,000 people whose families came from Japan out of their homes and into camps, without charging them with any crime. Those who disobeyed, such as Fred Korematsu, were charged with breaking the orders themselves. The justices of the Supreme Court ruled in 1927 that the operations were allowed, and in 1944 that the removal was allowed.

Two sections run past 1950 to finish their stories: the Tulsa survivors' fight to be paid goes on to 2026, and Native Americans' fight for the vote goes on to 1957.

A mob in Lincoln's town, and the NAACP, 1908 and 1909

A group of Black and white reformers founded the National Association for the Advancement of Colored People, the NAACP, after a white mob attacked Black residents of Springfield, Illinois, in August 1908. Colored was a common word for Black people at the time. Springfield was Abraham Lincoln's home town.

Springfield. Reporters for Springfield's newspapers wrote on August 14, 1908, that a Black man had tried to attack a white woman in her home the night before. Then they wrote about a second attack. A white mob went looking for the men accused, and when the men in it could not find them, they turned on the city's Black neighborhoods.

The violence lasted nearly three days. The mob lynched two men, Scott Burton and William Donegan, killing them without any trial. Neither man had anything to do with any crime. The rioters destroyed dozens of homes. Part-time soldiers of the state militia, the state's own army, stopped the violence. Officials arrested about 150 people, and few of them were ever found guilty of anything. Nearly 2,000 Black residents left Springfield and never came back.

The founding. The NAACP's founders started it on February 12, 1909, a hundred years after Lincoln was born. Its white founders included Mary White Ovington, Oswald Garrison Villard, William English Walling and Dr. Henry Moscowitz. About 60 people signed the call for its first conference, and seven of them were Black. They included the scholar W. E. B. Du Bois, Mary Church Terrell and Ida B. Wells-Barnett. The founders said their aim was to "secure for all people the rights guaranteed in the 13th, 14th, and 15th Amendments."

Women march in Washington, 1913

Women marched down Pennsylvania Avenue in Washington to demand the vote on March 3, 1913, the day before Woodrow Wilson became president. Alice Paul and Lucy Burns organized the march. They ran the Congressional Committee of the National American Woman Suffrage Association. They chose the same route the president's parade would take the next day, and at least 250,000 people came to watch.

The crowd. The crowd pushed off the sidewalks and blocked the street. Men shoved the marchers, spat on them and shouted insults and crude requests for sex at them. The police did little to stop it. United States Army troops arrived about an hour later and cleared the street. Afterward, members of Congress investigated why the police had done so little.

Ida B. Wells-Barnett steps in. More than forty Black women marched, in their state groups or with women of their professions. Ida B. Wells-Barnett came with the Illinois group. As the march was forming, the parade's national organizers told the Illinois women that their group was to be "entirely white" and that Black women were to march at the tail end of the parade. The leader of the Illinois group went along with it.

Wells-Barnett refused and left. She waited at the side of the avenue until the Illinois women came past. Then she stepped into the line between two white marchers, Belle Squire and Virginia Brooks. The same year, Black women in Chicago formed the Alpha Suffrage Club, the first Black women's suffrage club in Illinois. Ida B. Wells-Barnett died in Chicago on March 25, 1931.

The Silent Sentinels and the Occoquan Workhouse, 1917

Women stood at the gates of the White House six days a week from January 10, 1917, holding banners that asked President Wilson to support their vote. A National Park Service article states that Alice Paul invented picketing the White House, standing outside it with signs as a public protest. One banner read "Mr President What Will You Do For Woman Suffrage?" Another read "How Long, Mr President, Must Women Wait For Liberty?" The women became known as the Silent Sentinels.

The arrests. Police began arresting the women in 1917 on a charge of "obstructing traffic," blocking the road. Over the whole campaign, judges found more than 150 women guilty of it. In November 1917 alone, 41 women were arrested, among them Lucy Burns, Dorothy Day, Dora Lewis and Mary Nolan. Their sentences ran from six days to six months.

The Night of Terror. Many of them were sent to the Occoquan Workhouse, a prison in Virginia. On the night of November 14, 1917, on the orders of the superintendent, W. H. Whittaker, the guards attacked them. The guards threw the women around, hung some of them by their arms and refused them food and medical care. Lucy Burns kept a daily log on scraps of paper and got it smuggled out of the workhouse.

Her entry for that night reads: "Seized by guards from behind, flung off my feet, and shot out of the room. All of us were seized by men guards and dragged to cells in men's part. Dorothy Day was roughly used-back twisted. Mrs. Mary A. Nolan (73-year-old picket from Jacksonville, Florida) flung into cell."

The guards handcuffed Burns all night, and for part of the time they chained her hands to the bars of her cell. By her own account, they did it because she had asked the other women how they were. That night is known as the Night of Terror.

Fed by force. Many of the prisoners went on hunger strike, refusing to eat as a protest. The prison doctor, Dr. Gannon, answered by feeding them by force. Forcible feeding meant holding a woman down and pushing a tube through

her nose or mouth, down her throat and into her stomach. Then the doctor poured liquid food through the tube. Burns wrote in her log what was done to her:

"I was held down by five people at legs, arms, and head. I refused to open mouth. Gannon pushed tube up left nostril. I turned and twisted my head all I could, but he managed to push it up. It hurts nose and throat very much and makes nose bleed freely. Tube drawn out covered with blood. Operation leaves one very sick. Food dumped directly into stomach feels like a ball of lead. Left nostril, throat and muscles of neck very sore all night."

Dora Lewis was held down by five people while Gannon forced the tube through her lips. "I didn't know where to breathe from and everything turned black when the fluid began pouring in," she wrote. Alice Paul and Rose Winslow were fed this way three times a day for three weeks, until they were let out. Winslow got a note out of the prison: "We think of the coming feeding all day. It is horrible."

Alice Paul

Who: The organizer who put the first picket line outside the White House, and who served six prison terms for women's right to vote.

When and where: Born at Mount Laurel, New Jersey, January 11, 1885. Washington, D.C., 1913 to 1920.

A nurse shone a light on her face every hour all night.

Alice Paul grew up in a Quaker family and earned a doctorate, the highest university degree, in economics from the University of Pennsylvania in 1912. She also studied in London and Birmingham. In England she joined Emmeline Pankhurst's campaign for the vote and was arrested for it more than once.

Back home, she became chairman of the National American Woman Suffrage Association's Congressional Committee in 1913. She then founded the Congressional Union and, in 1916, the National Woman's

Party. She organized the 1913 march in Washington, and she started the picket line at the White House gates on January 10, 1917.

Seven months. In October 1917 she was arrested for picketing and sentenced to seven months for obstructing traffic. On her way to serve it, she told reporters, "We are being imprisoned, not because we obstructed traffic, but because we pointed out to the President the fact that he was obstructing the cause of democracy at home, while Americans were fighting for it abroad." The United States was then fighting in the First World War.

The psychopathic ward. In the District of Columbia jail she went on hunger strike. Commissioner Gardner, one of the officials who governed the District of Columbia, told Dr. Gannon, "Go ahead, take her and feed her." She was carried on a stretcher to the jail's psychopathic ward, the part of the hospital for people thought to be insane. Gannon ordered one of her windows nailed shut and her door replaced with a barred cell door. A prisoner on a ladder nailed heavy boards across the other window from outside.

A nurse was ordered to check on her every hour through the night by turning an electric light onto her face. Under the rules of the time, the signatures of two doctors were enough to send a person to St. Elizabeth's, the government asylum, a hospital where such people were locked up. "No trial, no counsel, no protest from the outside world!" she wrote later, counsel meaning a lawyer.

The doctor who said no. The psychiatrist sent to examine her, Dr. William A. White, ran St. Elizabeth's, and he did not find her insane. A psychiatrist is a doctor who treats illnesses of the mind. "I felt myself in the presence of an unusually gifted personality," he wrote, and he praised the "most admirable, coherent, logical and forceful way" in which she argued her case. A lawyer, Dudley Field Malone, forced his way in and got her out of the ward.

She was released after five weeks of her seven-month sentence. She served six prison terms for the vote in all, three in England and three in the United States.

After the vote. She earned a law degree in 1922 and wrote the Equal Rights Amendment, which said that men and women must have equal rights under the law.

She had it introduced at Seneca Falls on July 20, 1923, where it was called the Lucretia Mott Amendment. In 1945 she worked on the words about women in the United Nations Charter, and later on the part about sex in the Civil Rights Act of 1964. The amendment she wrote was still not part of the Constitution when she died at Moorestown, New Jersey, on July 9, 1977, aged 92.

Margaret Sanger and the Comstock laws, 1914 to 1936

Margaret Sanger went to jail for giving women information about birth control, ways to keep from becoming pregnant. The Comstock Act of 1873 made it a crime to send that information by mail. Sanger was charged under the Comstock laws in 1914 for publishing *The Woman Rebel*, a paper about birth control, and she left for England.

The clinic. She opened a birth control clinic in Brooklyn, New York, on October 16, 1916. Police arrested her about a week later, and she spent 30 days in jail. Judges ruled in 1918 that doctors could give women birth control for medical reasons. Sanger founded the American Birth Control League in 1921.

The judges of a federal court ruled in 1936 that doctors could give their patients birth control. A National Women's History Museum biography states that Sanger also supported parts of the eugenics movement, whose members wanted to stop people they judged unfit from having children.

BY 1918: The NAACP had been working for nine years, and women who picketed the White House had been jailed and fed by force.

Attacks in 25 cities, and a court case out of Elaine, 1919 to 1925

White mobs attacked Black Americans in 25 major cities in 1919, and the NAACP's lawyers turned one of those attacks into a Supreme Court ruling. The year's violence is called the Red Summer. Writers at the Equal Justice Initiative, a group that records racial violence, state that white mobs and police especially targeted Black veterans coming home from the First World War.

Chicago. A 17-year-old Black boy named Eugene Williams was swimming off a raft near the 29th Street beach in Chicago on July 27, 1919. A 24-year-old white man, George Stauber, threw stones at the Black boys, and Eugene Williams drowned. A police officer, Daniel Callahan, refused to arrest Stauber.

White youth gangs and neighborhood "athletic clubs" then attacked Black neighborhoods, and the fighting went on until August 3. Thirty-eight people were killed, 23 Black and 15 white. About 537 were hurt, and two out of three of them were Black. Two out of three of the 138 people charged with crimes afterward were Black too.

Elaine. In Elaine, Arkansas, Black sharecroppers had joined a union. A sharecropper farmed land owned by someone else and paid for it with a share of the crop, and a union is a group of workers who bargain together. This one, the Progressive Farmers and Household Union of America, was led by Robert Lee Hill. Its members wanted a better price for their cotton from the white landowners.

About 100 of them met in a church at Hoop Spur on the night of September 30, 1919. Men in a car outside and the union's armed guards exchanged shots. A white security officer, W. A. Adkins, was killed, and a deputy sheriff, Charles Pratt, was wounded.

The killing. Between 500 and 1,000 armed white men came to Elaine from the surrounding Arkansas counties and from across the river in Mississippi. A white witness, H. F. Smiddy, swore in 1921 that "several hundred of them" began "to hunt negroes" and shoot them as they found them. Governor Charles Hillman Brough sent more than 500 soldiers from Camp Pike, and they arrived on October 2. Some accounts say the soldiers also killed Black people at random.

Four brothers, David, Gibson, Lewis and Leroy Johnston, were killed on October 2 on their way home from a hunting trip. The Encyclopedia of Arkansas does not say who killed them. Five white people were killed.

DISPUTED: The count of Black dead. The Encyclopedia of Arkansas gives a count that "ranged into the hundreds," and a National Archives page gives more than 100. The Army commander, Colonel Isaac Jenks, reported that his soldiers had killed two.

The trials. Phillips County officials charged 122 Black people with crimes. By November 5, 1919, twelve men had been tried in Phillips County and sentenced to death. One of their white defense lawyers admitted he "had not interviewed any witnesses." Walter White of the NAACP went to Arkansas to investigate, and a Black lawyer from Little Rock, Scipio Africanus Jones, defended the twelve.

The justices of the Supreme Court ruled on February 19, 1923, in *Moore v. Dempsey*, that the trials had been unfair. Justice Oliver Wendell Holmes wrote that one trial "lasted about three-quarters of an hour and in less than five minutes the jury brought in a verdict of guilty of murder in the first degree." An angry crowd had filled the courthouse and the streets around it.

Sworn statements in the case said Black witnesses had been "whipped, placed in the electric chair and strangled by something put in their noses to make them testify." Governor Thomas McRae freed the last of the twelve in January 1925.

Anti-lynching bills in Congress, 1900 to 1924

NAACP leaders pushed Congress for a federal law against lynching, and in the whole of this era members of Congress never passed one. Under such a law, federal judges could have tried the members of a lynch mob when state officials would not.

Representative George Henry White of North Carolina brought the first federal anti-lynching bill to Congress on January 20, 1900. He was the last Black member of Congress of his generation. Members of Congress did not pass his bill.

The Dyer bill. Representative Leonidas Dyer of Missouri introduced a bill on April 8, 1918, "to protect citizens of the United States against lynching in default of protection by the States," wherever the states failed to protect them. Members of a lynch mob would have been tried in federal court for murder, and the county where a lynching happened would have been fined.

Any sheriff or jailer who handed a prisoner to a mob, or failed to stop one, could have been jailed and fined up to $5,000. People who had been part of a mob could not sit on the jury.

NAACP leaders published a report in April 1919, *Thirty Years of Lynching in the United States, 1889-1918*. It recorded 3,224 people killed by lynch mobs in those thirty years, about two every week. Of them, 2,522 were Black, and 50 of the Black victims were women.

They sent James Weldon Johnson to Washington to press members of Congress in person. "I tramped the corridors of the Capitol and the two office buildings so constantly that toward the end, I could, I think, have been able to find my way about blindfolded," he wrote.

The vote that never came. Members of the House of Representatives passed the bill on January 26, 1922, by 230 votes to 119, with several hundred Black visitors crowded into the public gallery. NAACP organizers held a silent march in Washington for it on June 14, 1922, with more than 5,000 Black

marchers. Senators never voted on it. Southern Democratic senators filibustered it, which means they kept talking so that no vote could be held. They did the same in 1923 and 1924, and Republican senators let the bill drop.

The 19th Amendment, 1920, and the women it did not reach

The 19th Amendment to the Constitution was ratified in August 1920. After it, no official of the United States or of any state could refuse citizens the vote because of their sex. Its first section is one sentence: "The right of citizens of the United States to vote shall not be denied or abridged by the United States or by any State on account of sex," abridged meaning cut down.

Forty-two years. The same words had been in front of Congress for 42 years. Senator Aaron Sargent of California introduced them on January 10, 1878. Senators voted them down in 1887, 1914, 1918 and February 1919. Susan B. Anthony died in Rochester, New York, on March 13, 1906, and suffragists named the measure the Anthony Amendment in her memory.

Representatives passed it on May 21, 1919, and senators passed it on June 4, 1919, by 56 votes to 25. Then lawmakers in three-fourths of the states had to ratify it. Tennessee's lawmakers made it the 36th state to do so on August 18, 1920, the last one needed. Secretary of State Bainbridge Colby signed the paper declaring it part of the Constitution on August 26, 1920, 72 years after the meeting at Seneca Falls.

Who still could not vote. State officials could no longer refuse a woman the vote because she was a woman, but they could still refuse her for other reasons. In the South, white officials already kept Black men from voting, by rules and by violence.

A National Park Service article states that Black women in the South "encountered the very same disfranchisement strategies and anti-Black violence that led to the disfranchisement of Black men." A National Archives page states that many women "remained unable to vote long into the 20th century because of discriminatory state voting laws," laws unfair to people because of who they were.

BY 1920: Women had won the vote in the Constitution, white officials in the South still kept many Black women and men from using it, and senators had blocked every federal bill against lynching.

Tulsa, June 1, 1921: white Tulsans burn Greenwood

Thousands of armed white Tulsans went into Greenwood, the Black district of Tulsa, Oklahoma, on June 1, 1921, and burned it to the ground. They killed as many as 300 people. The attackers threw bodies into the Arkansas River, carried them off on railroad flatcars and buried them in unmarked graves, so the true count may never be known. Red Cross workers counted about 700 people hurt.

DISPUTED: The number of dead. The historian Scott Ellsworth writes that "credible estimates of deaths range from fifty to three hundred." In 2025 lawyers at the United States Department of Justice wrote that the attack "killed as many as 300 Tulsans, perhaps even more."

Greenwood. About 10,000 Black people lived in Greenwood. It had two newspapers, more than a dozen churches, a library, two "fine hotels," four drug stores and many grocery stores. Loula Williams owned the Dreamland Theater, and J. B. Stradford owned a hotel of 54 rooms. Doctors, lawyers and teachers lived on Detroit Avenue, and families owned pianos and good furniture.

Black Tulsans spent their money in Greenwood because most white business owners in Tulsa would not serve them or treated them with contempt. Under city laws, Black families could not live in white blocks unless they worked there as servants. White officials controlled the city's tax money and spent none of it in Greenwood on running water or on carrying away sewage and garbage, and they paved fewer of its streets. All of Tulsa, Greenwood included, stood on land in the Muscogee (Creek) Nation's reservation.

How it began. Police arrested Dick Rowland, a 19-year-old Black man, on May 31. A white woman, Sarah Page, ran the elevator he used, and a white man had accused him of attacking her. She did not press charges. That evening a white crowd gathered at the courthouse to lynch Rowland, and armed Black men went there to stop it. Shooting broke out.

Tulsa police then deputized hundreds of white men, swearing them in as temporary police officers. Many of those men had been calling for a lynching and had been drinking.

The night. White men set the first fires in Greenwood at about 1 in the morning. When firemen came, armed white men threatened them and told them to let the Black homes burn. A little after 5 in the morning, a whistle blew. White men who had been organized into companies during the night gathered at three railroad stations and went into Greenwood.

How many. Counts of the white men who went in run from 8,000 to 10,000, and the state commission of 2001 gave about 10,000. The head of Oklahoma's National Guard, the state's own army of part-time soldiers, gave the *New York Times* a figure of 25,000. The Justice Department's review says many take that to be a misprint for 2,500.

House by house. The attackers went from house to house. They took what they wanted, piled bedding and furniture in the middle of a room, poured kerosene over it and lit it. They backed trucks up to houses and loaded "everything moveable and of value." White women came with shopping bags and took clothes, silverware and jewelry. The attackers opened safes and destroyed people's legal papers before burning the buildings.

They burned more than 1,200 houses and 35 city blocks. Among the buildings they burned were a dozen churches, five hotels, 31 restaurants, four drug stores, eight doctors' offices, more than two dozen grocery stores and the library.

The killing. They killed people who ran and people who gave up. The Justice Department's review describes a grandfather shot in front of his family in a wagon, and an old man who could not walk, shot in his home when he said he

could not "march." Four white men tied a blind man who had lost both legs to the back of a car and dragged him through the streets until he died.

An old couple on Greenwood Avenue were shot in the back of the head while they prayed. A man was shot as he came out of a burning building and thrown back into the fire. Survivors could not name the men who did these things. The Red Cross's relief director in Tulsa, Maurice Willows, who was white, wrote: "This was NOT a RIOT."

Fighting back. Some Black men fought back. A group fired from the bell tower of the new Mount Zion Baptist Church until National Guard soldiers turned a machine gun on it. J. B. Stradford shot from the second floor of his hotel. He gave up when the attackers promised not to burn it, and then they burned it.

The Guard. Three Guard units were in Tulsa, and before the governor took control the guardsmen took orders from Police Chief John Gustafson. By their own reports, they helped the police organize white men into companies and went into Greenwood to round up Black residents. The Justice Department's review states that the guardsmen's main work was capturing Black men. That left Black homes empty, and other white men looted and burned them.

Some guardsmen tried to hold back white attackers. Governor James Robertson declared martial law, which put the Guard in charge of the city, at 11:30 that morning.

The airplanes. Airplanes flew over Greenwood during the attack. The Justice Department's review calls it "undisputed" that the men in them watched where Black residents ran and told the police. Many survivors said men in the planes also dropped firebombs or shot at people fleeing. The lawyer B. C. Franklin, who saw it, described turpentine bombs. The review calls that part "hotly disputed."

The camps, the passes and the grand jury, June 1921

Police, deputies and guardsmen arrested Black people all over Tulsa after the burning, whether or not they were suspected of anything. The Justice Department's review states that the arrests were not based on "probable cause or even reasonable suspicion," the legal words for a good reason to believe a person has done something wrong. Guardsmen also arrested Black servants in the servants' rooms of white homes.

They held the people they arrested under armed guard in three places: Convention Hall, McNulty Park, which was a baseball park, and the stalls of the fairgrounds. At the fullest, between 4,000 and 6,000 people were held, some for a few days and some for up to two weeks. Guardsmen arrested some white men for looting, but they held no white person in the camps.

The cards. To leave a camp, a Black person needed an identification card, and a white person had to vouch for it. People who worked in white homes and businesses got green cards that did not run out. People without jobs got red cards that were good for one day.

Anyone found without a card was arrested again. By June 7, officials had given out 7,500 of these cards, and the rule lasted at least a month. People forced to live under it said they were treated "as though they [were] dogs."

Forced labor. The day after the burning, the Guard commander ordered "all able bodied [N]egro men" in the camps to "render such service and perform such labor as is required." The men cleared the ruins and buried the dead. Red Cross officials paid up to 25 cents an hour, and a meal cost 20 cents.

For several weeks officials did not allow Black Tulsans to own or buy guns. Most of Greenwood's people had no home left. Many lived in Red Cross tents for more than a year, and many of the tents had no floor.

The grand jury. A state grand jury, a group of citizens who decide whether there is enough evidence to charge someone with a crime, heard almost 200 witnesses. It charged about 70 men, many of them Black leaders of Greenwood, with starting the "riot." The *Tulsa Daily World* printed its report

on June 26, 1921, under the headline "Grand Jury Blames Negroes for Inciting Race Rioting: Whites Clearly Exonerated," exonerated meaning cleared of blame.

Most of the cases were later dropped, including the case against Dick Rowland. Some white men were charged with looting. No white man went to prison for the killings or the burning. The grand jury also charged Police Chief Gustafson with neglect of duty, failing to do the job he was paid to do, and with dishonest dealings from before the massacre. He was found guilty in court and removed from his post.

Building Greenwood again, against the city

Greenwood's people rebuilt their district with no rebuilding money from the city, the state or the federal government, after winning a court fight against city leaders who tried to stop them. Tulsa's city commissioners passed Fire Ordinance No. 2156, a city law, on June 7, 1921. It put most of Greenwood inside the city's fire limits, an area with strict building rules meant to stop fires spreading. Every new building there had to be "of concrete, brick, or steel and ... at least two stories high."

The first newspaper story about the rule said the land would "never again be a [N]egro quarter but will become a wholesale and industrial center." The Justice Department's review stated in 2025 that "those who drafted the ordinance planned to drive Black people out of Greenwood."

The lawyers in a tent. B. C. Franklin, a Black lawyer, set up an office in a tent on Archer Street with two partners, I. H. Spears and T. O. Chappelle. Before they could get to court, police had arrested many people for trying to rebuild. The lawyers won a court order that stopped the rule for a time, and then, in September 1921, an order that stopped it for good.

The claims. Greenwood's people filed more than $1.8 million in damage claims with the city between June 1921 and June 1922. City officials turned down all but one, and the one they paid went to a white shop owner for guns taken from his store. Franklin's firm filed about $4 million in claims against the city and against insurance companies.

Officials of the insurance companies refused to pay, because their policies did not cover damage done in a "riot." The justices of the Oklahoma Supreme Court ruled in 1926 that no one could sue the insurance companies over it. Judges threw out the lawsuits against the city, Mayor T. D. Evans, the police and the airplane company.

Greenwood again. The National Negro Business League held its national meeting in Greenwood by 1925, and by the 1940s Greenwood had 242 Black-owned businesses. From the 1950s, city officials took land along its edges for highways. The Inner Dispersal Loop and the Cherokee Expressway, finished in the 1970s, run along Greenwood's south and east sides.

The survivors ask to be paid, 1997 to 2026

Greenwood's survivors began asking to be paid for what was taken from them in June 1921, and in more than a hundred years neither the city nor the state paid them. Oklahoma's lawmakers set up an 11-member commission to study the massacre in 1997. Its members handed in their report on February 28, 2001. They asked the state to pay the survivors and their descendants directly, and to fund scholarships, business growth in the area and a memorial.

The lawmakers passed the 1921 Tulsa Race Riot Reconciliation Act on June 1, 2001. They voted no money for the survivors in it.

The lawsuits. Survivors and descendants sued the state and the city in 2003. Federal judges threw the case out in 2004, ruling that they had waited too long to sue. The last three known survivors sued again in 2020: Viola Fletcher, her brother Hughes Van Ellis and Lessie Benningfield Randle. They sued the City of Tulsa, the Tulsa Regional Chamber, the county's commissioners, Sheriff Vic Regalado and the Oklahoma Military Department. On May 19, 2021, the three spoke to a committee of Congress.

A president's visit. President Joe Biden met the three survivors and spoke at the Greenwood Cultural Center on June 1, 2021. He was the first president in office to visit Greenwood about the massacre. "It wasn't a riot, it was a massacre," he said. He did not call for the survivors to be paid.

The end of the case. The justices of the Oklahoma Supreme Court ended the survivors' lawsuit on June 12, 2024, by eight votes to one. They wrote that the harm the massacre left behind "can only be resolved by policymakers, not the courts." Hughes Van Ellis had died on October 9, 2023, aged 102. Viola Fletcher died on November 24, 2025, aged 111. Lessie Benningfield Randle, born on November 10, 1914, was the last known survivor in 2026, and neither the city nor the state had ever paid her.

The graves. Tulsa's mayor, G. T. Bynum, started a search for the massacre's unmarked graves in 2018. By September 2026 the searchers had dug up the remains of 55 people in Oaklawn Cemetery, and they have named two victims of the massacre. C. L. Daniel was identified by DNA in July 2024. James Goings was identified from Veterans Administration records in June 2025, and his grave has not been found. Both men had served in the Army in the First World War.

Viola Fletcher

Who: A seven-year-old girl whose family fled Greenwood in 1921, who never went to school past the fourth grade, and who at 107 asked Congress for justice.
When and where: Tulsa, Oklahoma, 1921. Washington, D.C., May 19, 2021.

"I am 107 years old and have never seen justice."

Viola Fletcher was born Viola Ford on May 10, 1914, the daughter of Lucinda Ellis and John Wesley Ford of Tulsa. She grew up in Greenwood with her parents and five brothers and sisters. "The neighborhood I fell asleep in that night was rich," she told Congress in 2021, "not just in terms of wealth, but in culture, community, and heritage." "My family had a beautiful home. We had great neighbors and I had friends to play with. I felt safe."

The night they left. On the night of May 31, 1921, her family woke her and told her they had to leave. "I will never forget the violence of the white mob when we left our house," she said. "I still hear airplanes flying overhead. I hear the screams. I live through the Massacre every day."

"When my family was forced to leave Tulsa, I lost my chance at an education," she said. "I never finished school past the fourth grade." During the Second World War she worked in the shipyards in California. "For most of my life, I was a domestic worker serving white families," she said, cleaning and cooking in other people's homes.

Before Congress. At 106, in 2020, she joined her brother and Lessie Benningfield Randle in suing the City of Tulsa and others. She spoke to the House of Representatives' subcommittee on civil rights on May 19, 2021. "Today, I am in Washington, D.C. for the first time in my life. I am here seeking justice," she told them. "I am 107 years old and have never seen justice." She asked them to "open the courtroom doors to us."

She and her grandson, Ike Howard, published a book about her life in 2023, *Don't Let Them Bury My Story*. The justices of the Oklahoma Supreme Court ended the survivors' lawsuit in June 2024. Viola Fletcher died on November 24, 2025, aged 111, and she never saw the survivors paid.

Grace Fryer

Who: A watch-dial painter who sued the company whose paint had poisoned her, and who had to fight a state time limit that ran out before she knew she was ill.

When and where: Orange, New Jersey, 1917 to 1928.

Movie: Radium Girls (2018)

The painters shaped each radium brush between their lips.

Grace Fryer painted numbers on watch dials at the U.S. Radium Corporation's plant in Orange, New Jersey, from 1917 to 1920. The paint glowed in the dark because it had radium in it. Radium is a radioactive metal, which gives off radiation, energy that can damage the body's cells.

To keep a fine point on the brush, the painters put the tip between their lips. The physicist Robley Evans wrote down why: a cloth or their fingers "were too harsh, but by wiping the brush clean between their lips the proper erasing point could be obtained."

What radium did. The body takes up radium as if it were calcium and builds it into the bones and teeth. Once there, it stays and keeps giving off radiation from the inside. When doctors examined the bodies of dial painters after death, they found 99 percent of the radiation in the skeleton. Radium in the bones stopped the marrow from making blood properly, killed the jaw bone so that it broke apart, and caused bone cancers.

Fryer left the plant in 1920. In 1924 she went to a dentist, Theodor Blum, who found that her jaw bone was wasting away. He had never seen anything like it. Doctors came to call it radium jaw.

The time limit. A lawyer, Raymond Berry, filed suit against the company in May 1927 for Fryer and four women she had worked with: Katherine Schaub, Quinta Maggia Macdonald, Albina Maggia Larice and Edna Hussman.

The company's lawyers argued that New Jersey's statute of limitations, the time limit after which a person can no longer sue, meant the company could not be blamed. New Jersey's limit for injuries at work was two years. Radium poisoning took longer than two years to show itself, and Fryer had left the plant seven years before.

The settlement. A federal judge, William Clark, brought the two sides to a settlement outside court on June 4, 1928, and they signed it on June 8. Each woman got $10,000, though some sources give as much as $15,000. Each also got $600 a year for as long as she lived, and her doctors' bills paid. The settlement papers contain no admission that the company was to blame.

Ottawa, Illinois. Ten years later, women who had painted dials for the Radium Dial Company in Ottawa, Illinois, fought the same fight. One of them, Catherine Wolfe Donohue, collapsed at a hearing on February 10, 1938, and was carried out of the room.

The officer hearing her case ruled on April 5, 1938, that her work had made her ill and awarded her $5,661, plus $277 a year for life. On July 6 the members of the Illinois Industrial Commission, the state board that heard workers' claims, threw out the company's appeal and added $730.

She died on July 27, 1938, aged 35. The last ruling in her case came that October, after her death, and her family received a small payment.

BY 1925: White Tulsans had burned Greenwood, its people were rebuilding it without help from the city, and a dentist had found a watch-dial painter's jaw bone wasting away from radium.

Native Americans become citizens and fight for the vote, 1911 to 1957

Native Americans organized their own national rights groups in these years, and it took them 33 years after they became citizens to win the vote in every state. Native people founded the Society of American Indians at Columbus, Ohio, on October 12, 1911, the first national rights organization run by Native people.

Charles Eastman and Carlos Montezuma were among its leaders, and Zitkala-Ša edited its journal. She founded the National Council of American Indians in 1926. Eighty delegates from more than 50 tribes met in Denver on November 16, 1944, and founded the National Congress of American Indians.

Citizens without the vote. Members of Congress passed the Indian Citizenship Act on June 2, 1924, and President Calvin Coolidge signed it. American Indians became citizens. That did not put them on the voter rolls, the lists of people allowed to vote, because officials in each state set their own rules about who could vote.

Arizona and New Mexico. Under Arizona's law, no one could vote who was an "idiot, insane person, person non compos mentis, or under guardianship." Non compos mentis is Latin for "not of sound mind," and a person under guardianship has been placed in someone else's care by law.

The justices of the Arizona Supreme Court ruled in 1928 that Native people living on reservations were under the guardianship of the federal government, and so could not vote. New Mexico officials used the state constitution's phrase "Indians not taxed." People on reservations paid no state taxes, they argued, and so could not vote.

1948. Judges threw out both rules in 1948. The justices of the Arizona Supreme Court threw out their own court's 1928 ruling in *Harrison v. Laveen* on July 15, 1948. Justice Levi Udall wrote that a Native person who met the other requirements could not be kept off the rolls for living on a reservation.

Three federal judges in New Mexico ruled in August 1948, in *Trujillo v. Garley*, that the "Indians not taxed" rule broke the Constitution. Utah's officials were the last in any state to let Native people vote, in 1957.

A society for gay rights in Chicago, 1924

Henry Gerber founded the Society for Human Rights in Chicago on December 10, 1924. A National Park Service article calls it the first organization in the United States set up only to work for the civil rights of gay people. A gay person is a man who loves men or a woman who loves women.

Gerber was born Josef Heinrich Dittmar in Bavaria, in Germany, on June 29, 1892. He came through Ellis Island in October 1913, served in the United States Army and worked for the Post Office. He got a charter for the society from the State of Illinois, a paper that made it a legal group, by describing its members as people "abused and hindered in the legal pursuit of happiness." Its members printed two issues of a newsletter, *Friendship and Freedom*. No copy survives.

The arrests. Chicago police arrested Gerber and other members without a warrant, a court paper that allows an arrest. They took his typewriter, his writings and his diaries, and they gave back only the typewriter. Post Office officials fired him "for conduct unbecoming a postal employee." The society ended in 1925. Gerber died in 1972. His house in Chicago became a National Historic Landmark in June 2015.

Eugenics: operating on people so they could not have children

Eugenics was the belief that a country could improve its people by deciding who was fit to have children and stopping the rest. Francis Galton, an English scientist who studied numbers about people, made up the word in 1883. A fact sheet from the National Human Genome Research Institute calls eugenics "the scientifically erroneous and immoral theory of 'racial improvement' and 'planned breeding,'" erroneous meaning wrong. The idea became popular in the early 1900s.

Sterilization. American eugenicists pushed for laws under which officials could force people to be sterilized. To sterilize a person is to operate on them so that they can never have a child, and the operation cannot be undone. When Dr. John H. Bell operated on Carrie Buck in Virginia in 1927, he cut out a piece of each of the two tubes that carry eggs to her womb.

The model law. Harry H. Laughlin of the Eugenics Record Office wrote a model sterilization law for lawmakers in each state to copy. A page from the United States Holocaust Memorial Museum states that "in part due to Laughlin's efforts, dozens of American states enacted laws that forced certain people to be sterilized against their will." The same page states that "more than 62,000 sterilizations had been performed in the United States by the 1960s," and that about six in ten of the people operated on were women.

The National Human Genome Research Institute's fact sheet counts at least 60,000 people sterilized under the laws of 30 states by the 1970s. The people targeted included Latinos, Native Americans, African Americans, poor white people and disabled people.

Germany. Germany's Nazi rulers passed the Law for the Prevention of Offspring with Hereditary Diseases in 1933, a law about children and illnesses passed from parents to children. The Holocaust Museum's page states that the law was "based in part on the practice of forced sterilization in the US." Doctors in Nazi Germany sterilized about 400,000 people under it.

Leaders of the University of Heidelberg in Germany gave Harry Laughlin an honorary degree in 1936, a degree given as an honor without study. He wrote back on May 28, 1936, that "To me this honor will be doubly valued because it will come from a nation which for many centuries nurtured the human seed-stock which later founded my own country." Seed-stock means the animals or plants that farmers keep for breeding. Laughlin used the word for people.

Carrie Buck

Who: A young Virginia woman chosen as a test case by the man who wanted to sterilize her, and defended by a lawyer who wanted the same thing.

When and where: Virginia, 1924 to 1927. Died at Waynesboro, Virginia, January 28, 1983.

Her own lawyer wanted her sterilized too.

Carrie Buck's case was arranged so that Virginia's officials would win it. Virginia's lawmakers passed the state's sterilization law on March 20, 1924, and a lawyer, Aubrey Strode, wrote it. Albert Priddy ran the Virginia State Colony for Epileptics and Feeble-Minded, a place where Virginia's officials held people they had ordered locked up. Epileptics are people with epilepsy, an illness that causes seizures. Feeble-minded was a label then given to people who scored low on intelligence tests.

How she came to the Colony. Carrie Buck was born in Charlottesville on July 2, 1906, and raised by foster parents. Encyclopedia Virginia states that she was "allegedly raped" by their nephew and became pregnant. Rape means forcing a person to have sex. Her foster parents said the pregnancy showed that she was feeble-minded, and on January 23, 1924, they asked a court in Charlottesville to have her sent to the Colony.

Priddy wanted to know whether the new law would hold up in court. So he picked one of the people held there, Carrie Buck, and had a case brought over her. Priddy died on January 13, 1925, and Dr. John H. Bell took his place at the Colony and in the case. That is why it is called *Buck v. Bell*.

Two lawyers on one side. Strode, who had written the law, argued for the Colony in court. Carrie Buck's lawyer was Irving P. Whitehead, who had sat on the Colony's own board and supported sterilization. The historian Paul Lombardo found that Whitehead did not seriously question any of the witnesses against her and did not call one witness for her. Both lawyers wanted the judges to rule that the law was allowed.

A eugenics researcher, Arthur Estabrook, interviewed the family for one day. Then he testified that Carrie, her mother Emma and her baby daughter Vivian were "a defective strain." Officials had committed Emma Buck to the Colony in April 1920, ordering her held there. Vivian was born on March 28, 1924. A social worker said of the baby that she had "a look about it that is not quite normal."

Three courts. Judge Bennett T. Gordon of the trial court ruled for the Colony on November 18, 1924. The judges of the Virginia Supreme Court of Appeals agreed with him on November 12, 1925. The justices of the Supreme Court of the United States heard the case on April 22, 1927, and ruled for the Colony, eight to one, on May 2, 1927.

The operation. Dr. Bell performed a salpingectomy on Carrie Buck on October 19, 1927. He cut out a section of each of her two tubes that carry an egg from an ovary to the womb. After that, no egg could reach her womb, so she could never become pregnant. The operation could not be undone. The staff released her from the Colony the next month.

Officials had brought her sister, Doris Buck, to the Colony in June 1926. Colony officials signed an order to sterilize Doris in December 1927, and doctors operated on her in January 1928.

Afterward. Virginia doctors sterilized about 8,300 people under the law between 1927 and 1972. Virginia's lawmakers ended the law in 1974. Lawyers for the American Civil Liberties Union sued for the survivors in 1980, and the two sides settled the case out of court. On May 2, 2002, 75 years to the day after the ruling, Governor Mark Warner apologized for what he called "a shameful effort in which state government never should have been involved."

> Carrie Buck married twice. She died in a nursing home in Waynesboro, Virginia, on January 28, 1983. The justices of the Supreme Court have never overturned *Buck v. Bell*, which means they have never ruled that it was wrong.

BY 1929: Native Americans were citizens who still could not vote in Arizona or New Mexico, and the Supreme Court's justices had allowed Virginia doctors to sterilize Carrie Buck.

Citizens sent to Mexico, 1929 to 1936

Federal, state and county officials sent or forced hundreds of thousands of people whose families came from Mexico out of the United States during the Great Depression, the years of hard times after 1929. Many of them were American citizens. Counts run from about 500,000 to about 2 million. In a 2005 law, California's lawmakers found that about 1.2 million of the 2 million people they counted had been born in the United States, which made them citizens.

Los Angeles. William N. Doak, President Herbert Hoover's Secretary of Labor, led a national campaign to remove Mexicans, saying they were taking jobs. In Los Angeles, Charles P. Visel, head of the city's unemployment relief committee, planned the publicity. Immigration agents closed the exits of La Placita park in Los Angeles on February 26, 1931, with about 400 people inside.

Los Angeles County welfare officers told families their relief money, the public money that poor families lived on during the Depression, would stop unless they took free train tickets to Mexico. The first county train left on March 23, 1931. Hospital staff carried patients on stretchers to trucks bound for the border. Local officials often sold the property that families left behind.

The apologies. California's governor signed a law on October 7, 2005, in which the state's lawmakers apologized "for the fundamental violations of their basic civil liberties and constitutional rights committed during the period of illegal deportation and coerced emigration." That means people were sent out

of the country against the law, or forced to leave it. The Los Angeles County supervisors, who run the county, apologized in 2012. Representative Hilda Solis called on the federal government to apologize in 2006. No president and no Congress has apologized.

Two California school cases, 1931 and 1946

Mexican American parents in California won two school desegregation cases, fifteen years apart, and the first, in 1931, was the first school desegregation case won in the United States, desegregation meaning an end to keeping children of different races apart.

Lemon Grove. The principal of Lemon Grove Grammar School, near San Diego, Jerome T. Green, stood in the doorway on January 5, 1931. He let in every child except the Mexican American ones and told them to go to a separate two-room building. Members of the school board had planned this the summer before. The school had 169 pupils: 87 counted as European American, about 75 Mexican American and 7 Japanese American.

The parents formed a committee, the Comité de Vecinos de Lemon Grove, the Lemon Grove Neighbors Committee. They kept their children out of the separate building and hired two lawyers, Fred Noon and A. C. Brinkley, and they sued in the name of one of the pupils, Roberto Alvarez.

Judge Claude Chambers of the San Diego County Superior Court ruled on March 30, 1931. He ordered the district to admit the children "on a basis of equality with other children of said school district, and without separation or segregation in a separate school because of race or nationality." Putting "all the Mexicans in one group can only be done by infringing," or breaking, "the laws of the State of California," he said. Some accounts date the ruling March 31.

Mendez. Fifteen years later, five families in Orange County, among them the Mendez family, sued four school districts there. Their lawyer had won a case in 1944 against segregated pools and parks in San Bernardino.

Judge Paul J. McCormick ruled for the families on February 18, 1946. "A paramount requisite in the American system of public education is social equality," he wrote. He meant that treating all children as equals came first in

American public schools.

When the districts' officials appealed, asking a higher court to change the ruling, other groups joined the fight. NAACP lawyers, among them Thurgood Marshall, filed a friend-of-the-court brief, an argument from people who are not part of a case. Four other groups filed briefs too, among them a Japanese American group and a Latin American group.

The judges of the federal appeals court ruled for the families on April 14, 1947. Governor Earl Warren signed the Anderson Bill in June 1947, and with it California's lawmakers took segregated schools out of the state's school laws, the first lawmakers in any state to do so. Warren became Chief Justice of the United States in 1953.

The Munemitsu farm. While the Mendez family fought school segregation, they were farming land whose Japanese American owners federal officials had locked up. The farm was 40 acres in Westminster that belonged to Seima Munemitsu and his family. Federal officials held the Munemitsus at the Poston camp in Arizona from May 1942 to September 1945.

Could a school turn a disabled child away?

Yes. School officials could turn a disabled child away in these years, because no law required them to teach one. The rule came from a Massachusetts case decided on January 4, 1893.

Members of the Cambridge school committee had put a boy named John A. Watson out of the public schools in 1885. Their record states that he was excluded "because he was too weak-minded to derive profit from instruction," and that he was "troublesome to other children, making uncouth noises, pinching others," uncouth meaning rude and rough. The judges of the Massachusetts Supreme Judicial Court ruled that the committee's decision was final and that no court could review it. That rule held through the whole of this era.

Parents organize. Parents in Cleveland, Ohio, founded the Cuyahoga County Council for the Retarded Child in 1933. "Retarded" was the word then used for children with intellectual disabilities. The county's own history describes it as a group led by parents and "founded to assist children of the region who had been excluded from their public schools."

A sit-in for jobs, New York, 1935

Six disabled New Yorkers sat down in the city's Home Relief Bureau in May 1935 and refused to leave until they were given a fair chance at government jobs. During the Depression, federal officials gave out jobs to people who were out of work through the Works Progress Administration, the WPA. Workers at the city's Home Relief Bureau stamped disabled people's applications "PH," for "physically handicapped." Officials treated anyone with the stamp as unable to work, and gave them no WPA jobs.

The league. About 300 disabled people formed the League of the Physically Handicapped. Most of them lived with the effects of polio, a disease that can paralyze muscles, or of cerebral palsy, a condition from birth that affects movement. After the bureau's director refused to see them, six of them began the sit-in while others picketed outside. Later they sat in at WPA headquarters too.

One member said afterward that "everybody was getting jobs." The militant members, the same member said, "just refused to accept the fact that we were the only people who were looked upon as not worthy, not capable of work," militant meaning ready to fight hard for a cause.

Officials had made about 1,500 jobs for disabled people in New York City by 1936, and the league had ended by 1938. Lawmakers in Congress passed the Social Security Act in August 1935. It defined disability as "inability to engage in substantial gainful work," so a person who received disability insurance could not easily take a job.

BY 1935: Officials had forced hundreds of thousands of people of Mexican descent, many of them citizens, out of the country, and disabled New Yorkers had sat in to demand a fair chance at jobs.

Charles Hamilton Houston's plan: law schools first, 1935 to 1949

Charles Hamilton Houston planned the NAACP's attack on segregated schools as a series of court cases, starting with law schools. He was born in Washington, D.C., on September 3, 1895, and he was the first Black editor of the *Harvard Law Review*. He became dean, the head, of Howard University's law school in 1929, where he trained Black lawyers to fight segregation in court. One of his students was Thurgood Marshall.

The plan. Houston became the NAACP's first special counsel, its chief lawyer, in the 1930s. His plan, in his words, was to "start with the law schools and higher education, and integrate schools from the top down." To integrate schools is to let Black and white students learn in the same schools.

Donald Murray. The first case was Donald Gaines Murray's. He applied to the University of Maryland School of Law on January 24, 1935, and was refused because he was Black. Houston, Marshall and a Baltimore lawyer, Nicholas Gosnell, took his case to the Baltimore City Court.

Judge Eugene O'Dunne ordered the university's president, Raymond A. Pearson, to admit him. The judges of the Maryland Court of Appeals agreed on January 15, 1936, and Murray graduated in 1938. The second case, Lloyd Gaines's in Missouri, went to the Supreme Court in 1938.

Ada Lois Sipuel. Ada Lois Sipuel was born in Chickasha, Oklahoma, on February 8, 1924. She applied to the University of Oklahoma College of Law on January 14, 1946. The university's president, George Lynn Cross, put in writing that there was no academic reason to reject her. He refused her because state law forbade teaching Black and white students in the same room.

Thurgood Marshall took her case, and the justices of the Supreme Court decided it on January 12, 1948. Oklahoma's leaders answered by setting up a separate law school for Black students in five days, in rooms at the State Capitol. She was admitted to the University of Oklahoma on June 18, 1949, three years and five months after she applied, and graduated in August 1952. Houston died on April 22, 1950.

Lloyd Gaines

Who: A history graduate who sued to study law in his own state, won the NAACP's first school case at the Supreme Court, and then disappeared.
When and where: Missouri, 1935 to 1938. Last seen in Chicago, March 1939.

He won his case and then disappeared.

Lloyd Gaines graduated with honors from Lincoln University, Missouri's college for Black students, with a degree in history, in August 1935. His father, Henry Richard Gaines, was a tenant farmer, a farmer who rents the land he works, and had been a teacher. Lloyd was one of eleven children.

Sent out of state. He applied to the University of Missouri School of Law. The registrar, the official in charge of signing up students, was S. W. Canada, and he sent Gaines to the president of Lincoln University. Lincoln's president pointed him to a Missouri law that paid a Black student's fees to study law in a neighboring state. Missouri's officials offered to pay for him to attend law school in Kansas, Nebraska, Iowa or Illinois rather than let him into its own.

He refused the offer and sued. Charles Hamilton Houston of the NAACP and a St. Louis lawyer, Sidney Redmond, filed his case in the circuit court, a state court in Missouri, on January 24, 1936. The circuit judge ruled against him in July 1936, and the judges of the Missouri Supreme Court ruled against him on December 9, 1937.

The ruling. The justices of the Supreme Court of the United States heard the case on November 9, 1938, and decided it on December 12, 1938, six to two, for Gaines. "The State was bound to furnish him within its borders facilities for legal education substantially equal to those which the State there afforded for persons of the white race," Chief Justice Charles Evans Hughes wrote.

A state had to give a Black student a law school as good as the white one, inside the state itself. Justice James McReynolds dissented, and Justice Pierce Butler joined him.

Gone. Missouri's lawmakers did not let Gaines into the university. They created a separate law school for Black students at Lincoln University in 1939. Gaines never attended either school. He was last seen in Chicago in March 1939, at a fraternity house, and he left a bag of his belongings behind. No one formally reported him missing until October 1939.

KNOWN:	FBI agents reopened the case in 2007 as part of their work on unsolved civil rights crimes.
NOT KNOWN:	What happened to Lloyd Gaines. He has never been found.

Marian Anderson sings at the Lincoln Memorial, 1939

Leaders of the Daughters of the American Revolution, a women's group, barred the Black singer Marian Anderson from singing in their hall in Washington, Constitution Hall, in 1939. Under the hall's rules, only white performers could appear there. Interior Secretary Harold Ickes arranged for her to sing on the steps of the Lincoln Memorial instead.

About 75,000 people came on Easter Sunday, April 9, 1939. Ickes introduced her with the words "Genius draws no color line." She opened with "My Country 'Tis of Thee." At the memorial's opening ceremony in 1922, the organizers had seated the crowd by race, and they had checked the speech of the one Black speaker, Robert Russa Moton, in advance.

A. Philip Randolph

Who: A union leader who twice got a president to sign an order against discrimination by threatening to bring a march to Washington.

When and where: Washington, D.C., 1941 and 1948.

"Ten, twenty, fifty thousand Negroes on the White House lawn."

A. Philip Randolph founded the Brotherhood of Sleeping Car Porters in 1925, a union of the Black Pullman porters who worked on railroad sleeping cars. In 1935 the Brotherhood became the first African American union admitted to the American Federation of Labor, a national group of unions.

The first march. Americans were building weapons for the Second World War in 1941, and defense factories were hiring. Randolph and Bayard Rustin began organizing a march on Washington against discrimination in the defense industries, which means treating people worse because of their race. Randolph and other Black leaders met Eleanor Roosevelt and members of the president's cabinet. Randolph told them he would bring "ten, twenty, fifty thousand Negroes on the White House lawn," using the word of the time for Black people.

The order. President Roosevelt signed Executive Order 8802 on June 25, 1941. An executive order is a written order from the president to the people who work for the federal government. "There shall be no discrimination in the employment of workers in defense industries and in Government, because of race, creed, color, or national origin," it said, creed meaning religion.

With the order, Roosevelt set up a Committee on Fair Employment Practice to take complaints and look into them. Federal agencies, companies with defense contracts and the unions working on them all had to follow the order, and Randolph called off the march. A National

Archives page states that most historians judge that the committee did little, especially in the South. After the war, members of Congress refused to make it permanent.

The second march. Randolph planned another march in 1948, this time against segregation in the armed forces, where officers still kept Black and white troops apart. President Truman signed Executive Order 9981 on July 26, 1948: "There shall be equality of treatment and opportunity for all persons in the armed forces without regard to race, color, religion or national origin." Military leaders resisted the order. By about 1953, the end of the Korean War, Black and white troops served together in nearly all of the armed services.

Randolph was the chief organizer of the March on Washington in 1963. He received the Presidential Medal of Freedom in 1964 and died at his home in New York City on May 16, 1979, aged 90.

BY 1941: NAACP lawyers had won their first school case at the Supreme Court, and A. Philip Randolph's threat of a march had won a president's order against unfair hiring in defense factories.

About 120,000 people held without charges, 1942 to 1945

Army officers, acting on President Roosevelt's order, forced about 120,000 people of Japanese ancestry out of their homes on the west coast in 1942, and federal officials held them in camps. People of Japanese ancestry are people whose families came from Japan. Officials locked them up without charging them with any crime or giving them a trial. Fred Korematsu and others who disobeyed the orders were charged with breaking them. Japan had attacked Pearl Harbor, Hawaii, on December 7, 1941.

The order. President Franklin D. Roosevelt signed Executive Order 9066 on February 19, 1942. Army commanders could now mark off "military areas," as large as they decided, and order "any or all persons" out of them. Lieutenant General John L. DeWitt, who commanded the Army on the west coast, used it against people of Japanese ancestry. He set curfews, orders to stay indoors during set hours, that applied only to Japanese Americans, and he gave families as little as 48 hours' notice to leave their homes.

Members of Congress passed a law on March 21, 1942, making it a crime to disobey his orders, punishable by up to a year in prison and a $5,000 fine. A page from the federal courts counts more than 120,000 people removed, two out of three of them American citizens. A National Archives page counts about 122,000, of whom nearly 70,000 were citizens.

Topaz. Officials held them first in temporary centers such as the Tanforan racetrack near San Francisco, and then in ten camps built inland, in Wyoming, California, Utah, Arizona, Colorado, Idaho and Arkansas. One of the ten was Topaz, the Central Utah Relocation Center, 15 miles northwest of Delta, Utah. It opened on September 11, 1942, and closed on October 31, 1945. More than 11,000 people passed through it, most of them from around San Francisco, and at its fullest more than 8,000 lived there at once.

The camp had 42 blocks, each with 12 barracks and 250 to 300 people. The barracks were built of pine boards covered with tarpaper. The furniture handed out was pot-bellied stoves, army cots, blankets and mattress covers.

The people of each block shared one dining hall and one building of washrooms, toilets and laundry. A fence with seven watchtowers surrounded the camp.

James Wakasa. James Hatsuki Wakasa, a 63-year-old man born in Japan, was walking his dog inside the fence at Topaz on April 11, 1943. Private First Class Gerald Philpott, the military policeman in Guard Tower 8, shot him dead. Philpott said Wakasa had been too close to the fence and had not turned back when warned. Witnesses believed Wakasa was hard of hearing.

Investigators found that his body lay several feet inside the fence and that he had been facing the guard when he was shot. Army officers tried Philpott at a court-martial, a military trial, for manslaughter, killing a person without planning to, and found him not guilty.

Four cases at the Supreme Court, and the one that won, 1943 and 1944

Four Japanese Americans took the government's orders to the Supreme Court. Three of them lost, and one, Mitsuye Endo, won.

Breaking the curfew on purpose. Gordon Hirabayashi and Minoru Yasui each broke the curfew on purpose so that they could challenge it in court. Yasui, a lawyer in Portland, Oregon, walked the streets after curfew on March 28, 1942, asked police officers to arrest him, and then turned himself in at the police station.

Hirabayashi, a student at the University of Washington, stayed out past curfew, refused to report for removal, and went to the FBI himself on May 16, 1942. The justices ruled in both cases in 1943 that the curfew was allowed, and they put off the question of whether the removal itself was lawful.

Korematsu's case. They answered that question in Fred Korematsu's case on December 18, 1944, and they ruled by six votes to three that the removal was allowed. Justice Hugo Black wrote for the majority that "all legal restrictions which curtail the civil rights of a single racial group are immediately suspect" and needed "the most rigid scrutiny," the closest possible look. Then he allowed this one.

Justice Frank Murphy dissented. He called the removal "the legalization of racism." He compared it to the "abhorrent and despicable treatment of minority groups by the dictatorial tyrannies which this nation is now pledged to destroy," meaning the countries the United States was fighting, which were ruled by dictators. Justice Robert Jackson also dissented. Korematsu, he wrote, had been found guilty for "being present in the state whereof he is a citizen."

Mitsuye Endo. On the same day, the justices ruled in Mitsuye Endo's case, and the government lost. Endo was born in Sacramento, California, on May 10, 1920, and worked as a clerk for the state of California. A lawyer, James Purcell, chose her to bring the case because a judge would find it hard to call her disloyal. She worked for the state, she was a Christian, she spoke only English, she had never been to Japan, and her brother was in the United States Army.

Purcell filed a petition for habeas corpus for her in the federal court in San Francisco on July 12, 1942. That petition asks a judge to order that a person held prisoner be brought to court, and freed if no one can show a lawful reason to hold her.

Staying in the camps. While the case went on, officials of the War Relocation Authority, the agency that ran the camps, offered to let her out if she dropped it. She refused and stayed in the camps, because leaving would have ended the case for everyone else. Officials moved her from the Sacramento assembly center to Tule Lake and then to Topaz.

"Mitsuye Endo should be given her liberty," Justice William O. Douglas wrote for the Court. The government, he wrote, "has no authority to subject citizens who are concededly loyal to its leave procedure." He meant that officials could not keep holding citizens whom they themselves agreed were loyal.

The end of the camps. Army commanders had issued Public Proclamation No. 21 the day before the rulings, December 17, 1944. It ended the exclusion, the order keeping people of Japanese ancestry away from the west coast, on January 2, 1945. The last camp, Tule Lake, closed on March 20, 1946. Endo

moved to Chicago in May 1945 and worked as a secretary for the Mayor's Committee on Race Relations. She married Kenneth Tsutsumi, raised three children and died in Chicago on April 14, 2006.

Fred Korematsu

Who: A welder who refused to leave his home when Japanese Americans were ordered out, lost at the Supreme Court, and had his conviction thrown out in 1983.
When and where: Oakland, California, 1942. San Francisco, 1983.

"If anything, I should be pardoning the government."

Fred Korematsu was born in Oakland, California, on January 30, 1919, and named Toyosaburo. A teacher suggested he call himself Fred. His parents, Kakusaburo and Kotsui, ran the Stonehurst Flower Nursery in East Oakland.

Staying behind. When the orders came in 1942, his family went to the camps and he did not. He was 23. He had a girlfriend, Ida Boitano, and a job as a welder. He got identification papers in a false name, Clyde Sarah, and kept working. Police stopped him and his girlfriend on a street in San Leandro, California, on May 30, 1942, and arrested him. The charge was breaking Civilian Exclusion Order No. 34, the Army's order for people of Japanese ancestry to leave his area.

The case. Ernest Besig, a lawyer with the American Civil Liberties Union, visited him in the Army stockade, a military jail, at the Presidio in San Francisco and asked to take his case to the higher courts. Korematsu agreed. A federal judge found him guilty and gave him five years' probation, which meant he stayed free under a court's watch. Then Army officers sent him to the Tanforan racetrack on June 28, 1942, and to the Topaz camp with his family that September.

The justices of the Supreme Court heard his case on October 11, 1944, and ruled against him on December 18, 1944. After he left the camp, he moved to Detroit and married Kathryn Pearson. They came back to

California and raised two children.

The hidden reports. In the early 1980s, a team of lawyers who worked without pay found that the government's own lawyers had hidden evidence from the Supreme Court. Intelligence reports had told them that Japanese Americans were no military threat, and they had kept those reports out of the case. The team reopened his case.

Judge Marilyn Hall Patel ruled on November 10, 1983, in the same San Francisco courthouse where he had been found guilty, 39 years after the Supreme Court's ruling. She wrote that "the conviction that was handed down in this court and affirmed by the Supreme Court in *Korematsu v. United States* is ... vacated," or cancelled, "and the underlying indictment," the criminal charge against him, "dismissed."

About a pardon, a president's official forgiveness of a crime, he said, "I don't want a pardon. If anything, I should be pardoning the government." He received the Presidential Medal of Freedom in 1998 and died at Larkspur, California, on March 30, 2005, aged 86. In California, January 30 is Fred Korematsu Day.

Elizabeth Peratrovich and the Alaska law, 1945

Alaska's lawmakers approved the Alaska Anti-Discrimination Act on February 16, 1945, after the Tlingit leader Elizabeth Peratrovich spoke for it before them. Alaska was then a territory, land ruled by the United States that was not yet a state. The act was the first law against discrimination in the United States or its territories. Signs such as "No Natives Allowed" hung in Alaska, and the lawmakers wrote the act against them.

Peratrovich belonged to the Lukaax.ádi clan of the Tlingit. For half a century the Alaska Native Brotherhood, founded at Sitka in 1912, and the Alaska Native Sisterhood, founded in 1915, were the only civil rights groups for Alaska Natives. In Alaska, February 16 is Elizabeth Peratrovich Day every year.

A soldier blinded on his way home, and the president's committee, 1946

Sergeant Isaac Woodard was discharged from the Army after four years on February 12, 1946, and took a bus home to South Carolina. After he argued with the bus driver, police at Batesburg, South Carolina, took him off the bus. Police Chief Lynwood Shull and Officer Elliot Long beat him with blackjacks, short heavy clubs. Woodard said they tried to gouge out his eyes, to dig them out. A local judge fined Woodard himself $50.

Blind. At the veterans' hospital in Aiken, doctors found bleeding inside both of his eyeballs and a torn cornea, the clear front of the eye, in his right eye. He was blind for the rest of his life. Justice Department lawyers charged Shull with breaking Woodard's civil rights. In November 1946 an all-white federal jury took thirty minutes to find Shull not guilty.

The president. Walter White of the NAACP led a group to see President Harry Truman on September 19, 1946, and told him what had been done to Woodard. "My God! I had no idea that it was as terrible as that! We've got to do something!" Truman said. He signed Executive Order 9808 on December 5, 1946, which set up the President's Committee on Civil Rights. The committee's report, *To Secure These Rights*, came out in 1947.

BY 1950

If you remember one thing: Organizers in these fifty years invented the picket line, the sit-in, the threatened march and the planned chain of court cases, while mobs and officials burned, jailed, sterilized and removed the people they targeted.

In 1900: Representative George Henry White brought the first federal bill against lynching to Congress, and it failed.

In 1950: Women had the vote under the Constitution, Native Americans could vote in every state but Utah, and NAACP lawyers had won their first law school cases. Congress had still passed no law against lynching.

Next: After 1950, the court cases reached the public schools, and the marches filled the streets.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 56 MINUTES

Five groups of Americans ran organized campaigns after 1950: Black Americans, women, Mexican Americans, gay Americans and disabled Americans. Each group won changes in the country's laws. The justices of the Supreme Court ruled in 1954 that segregated public schools broke the Constitution, and in 1956 they struck down segregation on Montgomery's buses. Members of Congress then passed new laws.

In the Civil Rights Act of 1964 they made segregation in businesses and public places illegal. Later they banned literacy tests for voting in 1965, sex discrimination in schools that took federal money in 1972, and refusing a disabled person a job or a way into a public building in 1990. A literacy test was a reading and writing test that officials used to keep Black citizens from voting.

The cost. Each of those changes followed years of organizing, and the organizers paid for it. People who signed lawsuits lost their jobs, police beat marchers with nightsticks, and firemen turned high-pressure hoses on children. Members of the Ku Klux Klan, a secret white group that used terror against Black Americans, killed four girls with a bomb in a Birmingham church in 1963.

The next year, white men in Mississippi killed three civil rights workers, and a Neshoba County deputy sheriff was among the men later found guilty over it. Civil rights are the rights every citizen has to equal treatment under the law and to vote.

An Alabama state trooper shot Jimmie Lee Jackson in 1965. Doctors operated on Black women such as Fannie Lou Hamer, and on poor girls, without asking them, so that they could never have children. Federal officials fired thousands of gay people from government jobs.

The law school cases, 1950

NAACP lawyers following the plan Charles Hamilton Houston made in the 1930s won two more cases in June 1950, six weeks after he died. The justices of the Supreme Court decided both cases on June 5, 1950, and all of them agreed on both.

Heman Sweatt. Heman Marion Sweatt had applied to the University of Texas Law School in February 1946, and university officials turned him away because he was Black. Texas officials then opened a separate law school for Black students in Austin in February 1947. Its four teachers were part-timers borrowed from the university, and it had no full-time librarian. About 10,000 books were on order, and few had arrived. No body of lawyers had approved it as a real law school.

Chief Justice Fred Vinson wrote that the education it offered was not "substantially equal" to what Sweatt would get at the university. "Equal protection of the laws is not achieved through indiscriminate imposition of inequalities," he wrote. A state could not call it equal treatment to give Black students something worse. The justices ordered Texas officials to admit Sweatt.

George McLaurin. George W. McLaurin already had a master's degree and wanted a doctorate in education. University of Oklahoma officials let him in and then made rules for him alone, and the justices' written opinion lists them.

He "was required to sit apart at a designated desk in an anteroom adjoining the classroom," a desk set aside for him in a small room next to the classroom. He had to sit at an assigned desk on the library's upper floor, "but not to use the desks in the regular reading room."

In the cafeteria he had "to sit at a designated table and to eat at a different time from the other students." Before the case was argued, university officials eased the rules a little. McLaurin could sit in a marked row in the classroom and eat at the usual hours, still at a table of his own. The justices struck down the whole arrangement.

What signing cost in Clarendon County, 1949 and 1950

In Clarendon County, South Carolina, 107 Black parents and children signed a petition against the county's schools. For it, employers fired them, merchants refused to sell to them and lenders turned them down for loans.

One bus. School District 22 ran more than 30 school buses for white children and none for Black children, and the parents began by asking for one bus. The court record describes the all-white Summerton High School as "modern, safe, sanitary, well equipped, lighted and healthy." It describes the Black schools as "inadequate…unhealthy…old and overcrowded and in a dilapidated condition," falling apart.

The Reverend Joseph A. DeLaine, a minister and teacher, brought the NAACP together with the Pearson brothers to start the case. Six white men later attacked DeLaine. He got away by pretending he had a gun in his pocket and jumping into the passing truck of another man who had signed.

The signers. The petition went round for signatures on November 11, 1949, and the parents' lawyers filed the lawsuit on May 16, 1950. It is named for Harry Briggs, whose name was first on the list. A National Park Service page

records what happened to the signers. They were "fired from their jobs, denied service when trying to purchase supplies for their farms, or being turned down for loans on their properties."

Barbara Rose Johns

Who: A sixteen-year-old who led her whole school out on strike, three years before *Brown*.
When and where: Farmville, Prince Edward County, Virginia, April 23, 1951.

A fake phone call got the principal out of the building.

Barbara Rose Johns was a student at Robert Russa Moton High School in Farmville, the high school for Black students in Prince Edward County, Virginia. The building had been made for 180 students, and it held between 360 and 450.

Classes met in the auditorium, in school buses and in three wooden sheds wrapped in tar paper, which people compared to chicken coops. The school had no cafeteria, no science laboratory and no gym. One day a half-empty school bus for white children passed her without stopping, and she decided to act.

The strike. A student named John Watson telephoned the principal's office on April 23, 1951, and disguised his voice. He said Moton students were making trouble at the bus station downtown. The principal, M. Boyd Jones, left to see. While he was gone, the strike Johns led began, and the students stayed out of class for two weeks.

The lawyers' condition. She wrote to two NAACP lawyers in Richmond, Oliver W. Hill and Spottswood William Robinson III. They agreed to take the case on one condition: the parents had to sue to end segregation itself, rather than ask for a better Black school. The Reverend L. Francis Griffin called the parents to a meeting at the First Baptist Church in Farmville on May 3, 1951, to hear the offer, and they agreed.

> On May 23, 1951, 74 parents filed the lawsuit, *Davis v. County School Board of Prince Edward County*, for 118 Moton students. It became one of the five cases the justices of the Supreme Court decided as *Brown v. Board of Education*.
>
> Her parents sent her out of the county. She finished her last year of high school in Montgomery, Alabama, living with her uncle, the Reverend Vernon Johns, and his wife, Altona. Her family's house in Prince Edward County later burned down, and the family believed someone had set the fire. Barbara Rose Johns died of bone cancer in Philadelphia on September 25, 1991, aged 56.

Brown v. Board of Education, May 17, 1954

All nine justices of the Supreme Court ruled on May 17, 1954, in *Brown v. Board of Education of Topeka*, that separating children by race in public schools broke the 14th Amendment and its promise of "the equal protection of the laws." Chief Justice Earl Warren wrote, "Separate educational facilities are inherently," or by their very nature, "unequal."

With that ruling, the justices rejected the reasoning of *Plessy v. Ferguson* for public schools, 58 years after that case. In *Plessy*, in 1896, the justices had allowed a Louisiana law that kept Black and white train passengers apart in "equal but separate" cars.

Brown was five lawsuits that the justices bundled together. They were *Brown v. Board of Education* from Topeka, Kansas, *Briggs v. Elliott* from Clarendon County, South Carolina, *Davis v. County School Board* from Prince Edward County, Virginia, *Belton v. Gebhart* from Delaware, and *Bolling v. Sharpe* from Washington, D.C.

Rosa Parks and Martin Luther King Jr.

Who: The NAACP branch secretary whose arrest set off the Montgomery bus boycott, and the 26-year-old minister the boycotters chose to lead it.
When and where: Montgomery, Alabama, December 1955 to December 1956.

The Women's Political Council printed 50,000 leaflets after her arrest.

Rosa Parks had been the secretary of the Montgomery NAACP for twelve years on the day she was arrested for keeping her bus seat. In the early 1940s she started the branch's Youth Council. As secretary of the NAACP's Alabama State Conference, she traveled the state interviewing victims of discrimination and witnesses to lynchings. She had broken the bus seating rules many times before, and a driver had once put her off a bus for it.

Claudette Colvin. Nine months before Parks's arrest, on March 2, 1955, a fifteen-year-old student named Claudette Colvin refused to give up her seat on a Montgomery bus when white passengers got on. She told police she had a right under the Constitution to sit anywhere on the bus, as white passengers did. The officers swore at her, hit her with a nightstick and dragged her off the bus.

E. D. Nixon, a Montgomery organizer, and others decided not to build a campaign around her case. Nixon thought she was too young. When she became pregnant later that year, he looked for someone who fit the strict customs of the time. Colvin believed the choice also had to do with her family's standing outside Montgomery's Black middle class.

The boycott. Police arrested Rosa Parks on December 1, 1955, for refusing to give up her bus seat. The Women's Political Council, a group of Black professional women formed in 1949, had already been pressing white city officials about the buses. Jo Ann Robinson of the council had

proposed a boycott, a refusal to use or buy something as a protest, more than a year before. After Parks's arrest, council members printed 50,000 leaflets calling for a one-day boycott on Monday, December 5.

After that Monday, Black riders stayed off the buses. The same day, Montgomery's Black ministers and organizers formed the Montgomery Improvement Association and elected Martin Luther King Jr. as its president. He was 26, the pastor of Dexter Avenue Baptist Church, and still finishing his doctorate at Boston University. About 40,000 Black people lived in Montgomery, and for a year most of them walked to work or school, shared cars, took taxis or hitchhiked.

The bombs. City leaders had nearly 100 of the boycott's leaders charged with conspiracy, the crime of planning together to break the law. Someone bombed King's house on January 30, 1956, while his wife, Coretta Scott King, their seven-week-old daughter, Yolanda, and a neighbor were inside. No one was hurt, and no one was ever charged with a crime for the bombing. On February 1 a bomb went off in E. D. Nixon's yard.

Four other women. Four other women brought the lawsuit that ended bus segregation in Montgomery. The lawyer Fred Gray filed *Browder v. Gayle* on February 1, 1956, for Aurelia Browder, Susie McDonald, Mary Louise Smith and Claudette Colvin, then sixteen. Federal judges in Montgomery struck down Alabama's bus segregation laws in June 1956, and in November the justices of the Supreme Court agreed. Black riders went back on the buses on December 21, 1956.

Colvin later worked as a nurse's aide in New York for fifty years. A judge erased her 1955 arrest record on December 16, 2022, and she died on January 13, 2026, aged 86. Rosa Parks lost her job as a tailor and received death threats. She and Raymond Parks moved to Detroit in 1957. She worked in the office of Representative John Conyers from 1965 to 1988, and she died in Detroit on October 24, 2005.

Memphis. King led the movement for twelve more years and won the Nobel Peace Prize in 1964, at 35. He went to Memphis, Tennessee, twice in 1968 to support 1,300 Black sanitation workers on strike. On February 1, 1968, two of them, Echol Cole and Robert Walker, had been crushed to death in the compactor of their garbage truck. The strikers carried signs that read "I AM A MAN."

King was shot in Memphis on April 4, 1968, and died at 7:05 that evening at St. Joseph's Hospital. He was 39. After his death, city officials recognized the workers' union.

DISPUTED: Who was behind the killing. James Earl Ray pleaded guilty to King's murder in March 1969, and a judge sentenced him to 99 years. For the rest of his life Ray tried to take back his plea, until he died in April 1998. In 1999 the King family brought a civil case, a lawsuit rather than a criminal trial, against a Memphis man named Jowers.

The jury found that Jowers and "others, including government agencies" had plotted the killing. In 2000 Justice Department investigators reported that they found those claims not believable.

BY 1956: The Supreme Court's justices had ruled segregated public schools unconstitutional, and Black riders in Montgomery had stayed off the buses for more than a year until the justices struck down bus segregation there too.

After Brown: Little Rock, Prince Edward County and New Orleans, 1957 to 1964

After *Brown*, Black families still had to force their way into white schools one district at a time, and white officials fought them with new city rules and by shutting down public schools.

Little Rock. Nine Black students entered Central High School in Little Rock, Arkansas, in 1957. Daisy Bates organized them. She was president of the Arkansas State Conference of the NAACP, and with her husband, L. C. Bates, she published a newspaper, the *Arkansas State Press*. The nine students met at her house every school day and left for Central High from there.

A rock came through her living room window in August 1957 with a note tied to it: "STONE THIS TIME. DYNAMITE NEXT." Twice, people burned crosses in her front yard, and someone fired shots into the house from a passing car. The National Park Service's page on the Bates home does not name the people who did it.

The membership lists. Members of the city councils of Little Rock and North Little Rock then passed rules that made every organization hand its list of members to the city clerk, to be made public. Daisy Bates kept the records of the Little Rock branch of the NAACP. She refused to hand over the list, and she was found guilty in court for refusing.

The justices of the Supreme Court threw out that verdict on February 23, 1960. Justice Potter Stewart wrote that keeping a group's members private "may in many circumstances be indispensable to preservation of freedom of association, particularly where a group espouses dissident beliefs." People have a right to join a group without officials forcing the group to name its members. Dissident beliefs go against those of the people in power.

Prince Edward County. Prince Edward County, Virginia, was home to the Moton students' case. Members of the county's board of supervisors voted on June 26, 1959, to stop paying for public schools, and the county had no public

schools until September 8, 1964. White children went to a private school, Prince Edward Academy. About 2,000 Black children went without formal schooling for five years.

Attorney General Robert F. Kennedy said in 1963 that "the only places on earth not to provide free public education are Communist China, North Vietnam, Sarawak, Singapore, British Honduras." Then he added, "and Prince Edward County, Virginia."

The justices of the Supreme Court decided a case brought by Black schoolchildren of the county, the first of them named Cocheyse J. Griffin, on May 25, 1964. Closing the county's schools, while every other county in Virginia kept its schools open, denied the children "the equal protection of the laws," they ruled. A federal judge could make the supervisors raise taxes to reopen the schools.

New Orleans. A federal judge, J. Skelly Wright, ordered New Orleans school officials to start letting Black children into white schools in the spring of 1960. Four Black six-year-old girls went into two white schools on November 14, 1960. Ruby Bridges went alone into William Frantz Elementary. Leona Tate, Tessie Prevost and Gail Etienne went together into McDonogh 19, in the Lower Ninth Ward, with federal marshals, officers who work for the federal courts, walking them in.

White parents took their children out. By the end of the month, fewer than ten white children attended the two schools combined. From January to May 1961, Ruby Bridges was the only pupil at William Frantz.

Levittown, 1957: a family moves in

Levittown, Pennsylvania, was a town of houses built and sold by one builder, William Levitt. Daisy and William Myers became the first Black family to live there in August 1957, and white mobs gathered outside their home for nearly three months. William Myers was an Army veteran of the Second World War, and Daisy Myers was a teacher. They had three young children.

The lease. In the first Levittown, in New York, the standard lease had barred anyone "other than members of the Caucasian race," meaning anyone who was not white, from living in a house. The justices of the Supreme Court ruled in *Shelley v. Kraemer* in 1948 that no court could enforce clauses like that, so no judge could order anyone to obey them. Levitt dropped the written clause and kept selling only to white buyers.

The mobs. When the Myers family moved in, mobs gathered outside the house. People threw rocks through the windows, burned crosses and made death threats by telephone. Delivery men stopped bringing milk, bread and heating oil, and state police guarded the house. A documentary, *Crisis in Levittown*, was filmed while it was happening, in 1957. Daisy Myers later wrote a book about it, *Sticks 'n Stones*, published in 2005.

Sit-ins and Freedom Rides, 1960 and 1961

Black and white students used their own bodies to test segregation in 1960 and 1961, first at lunch counters and then on buses.

Greensboro. Four freshmen from the Agricultural and Technical College of North Carolina walked into the Woolworth store in Greensboro on February 1, 1960. Franklin McCain, Joseph McNeil, David Richmond and Ezell Blair Jr. sat at the lunch counter, which was for white customers only, ordered coffee and waited. The waitress and the manager ignored them. A police officer walked up and down behind them. Some white customers shouted at them, and others encouraged them.

The next day 19 students came. The day after that 85 came, from several colleges. Before the week was out there were 400, taking turns so they could still go to class. Students started sit-ins in seven other North Carolina cities and in Hampton, Virginia, and Nashville, Tennessee.

Woolworth's managers opened the Greensboro counter to Black customers on July 25, 1960. By that summer, restaurants and lunch counters in 33 Southern cities served Black customers, and a year later those in 126 cities did.

The Freedom Riders. The justices of the Supreme Court struck down Virginia laws that required segregation in travel between states in *Boynton v. Virginia* in 1960. Leaders of the Congress of Racial Equality, known as CORE, organized riders to find out whether the people running bus stations in the South would obey.

The first thirteen Freedom Riders, Black and white, left Washington, D.C., on May 4, 1961, in two groups, one on a Greyhound bus and one on a Trailways bus. A National Park Service page gives the number as eleven.

Anniston. A white mob that included members of the Ku Klux Klan attacked the Greyhound bus at Anniston, Alabama, on May 14, 1961. They slashed its tires, followed it out of town and firebombed it. The riders were trapped on the burning bus. They got out after an explosion, and then the mob beat them. The Reverend Fred Shuttlesworth organized a line of cars and got them to Birmingham within a few hours.

A photographer, Joe Postiglione, took pictures of the bus as it burned. Charles Person, one of the Freedom Riders, later wrote that the photograph appeared "on front pages of newspapers across the country and the world." Accounts name different men as the mob's leaders. No record shows that anyone went to prison for burning the bus.

The same day, young white men climbed onto the second bus, the Trailways bus, at Anniston and beat the riders until they moved into segregated seats. One of the riders they beat was Walter Bergman, a white man. The Encyclopedia of Alabama states that he suffered permanent brain damage.

Birmingham. When that bus reached Birmingham, a crowd armed with pipes, chains and baseball bats met it, and many of the men in it belonged to the Klan. Bull Connor, Birmingham's public safety commissioner, had told Klan leaders that his police would hold back from the station for 15 minutes, and he encouraged the Klan to attack the riders.

With no police there, Klansmen beat the riders in the waiting room and in a dark hallway. They beat James Peck, a white rider, until he was unconscious, with heavy wounds to his face and head. Staff at a hospital for white people only refused to treat him, and doctors at Jefferson Hillman Hospital closed his wounds with 53 stitches. In the historian Raymond Arsenault's account, Bergman was beaten for the second time that day. Over the summer of 1961 the number of Freedom Riders grew to more than 400.

Fannie Lou Hamer

Who: A plantation timekeeper in Mississippi who tried to register to vote, lost her job the same day, was beaten in a jail for her organizing, and told the country about it on national television.

When and where: Ruleville, Sunflower County, Mississippi, 1962 and 1963. Atlantic City, New Jersey, August 22, 1964.

"I question America," she told the country on television.

Fannie Lou Hamer was the youngest of the twenty children of Ella and James Lee Townsend. She began working in the cotton fields at six, weeding first and then picking. In 1944 she married Perry Hamer, whom people called Pap. For eighteen years she worked on the W. D. Marlow plantation outside Ruleville, keeping the time records for the other workers.

The operation. She went into a hospital in 1961, and a doctor performed a hysterectomy on her without asking her or telling her. A hysterectomy is an operation that cuts out the womb, the organ in which a baby grows. After it, a woman can never become pregnant, and it cannot be undone.

Hamer called operations like hers a "Mississippi appendectomy," after the ordinary operation that takes out the appendix, a small part of the gut, and her name for them came into use. The Mississippi Department of Archives and History's account of her life does not name the surgeon.

Registering. She and seventeen others went 26 miles to the Sunflower County courthouse in Indianola to register to vote on August 31, 1962. Police let only two of them in to take the literacy test. Registrars, the officials who signed people up to vote, used that test to turn Black voters away.

That evening the plantation owner told her, "If you don't go down and withdraw your registration, you will have to leave." She answered, "I didn't try to register for you. I tried to register for myself." She lost her job

that day. On September 10 someone fired sixteen bullets into the Tucker family's house in Ruleville, and other houses were shot at the same night.

Winona. She was one of ten people coming home by Continental Trailways bus from a voter registration workshop on June 9, 1963. At the bus station in Winona, Mississippi, some of the group tried to use the restaurant and the washroom. Winona's police chief, Thomas Herrod, ordered them arrested.

In the jail, state highway patrolmen made two Black prisoners, Roosevelt Knox and Sol Poe, beat her with a blackjack. The patrolmen ordered her to lie face down on a bunk. The first prisoner beat her until he was exhausted. Then the patrolman gave the blackjack to the second prisoner and made the first one sit on her feet so she could not move them.

"I began to scream and one white man got up and began to beat me in my head and tell me to hush," she said. The officers also had June Johnson, who was fifteen, and Annell Ponder beaten. Hamer lost the sight of one eye and had kidney damage for the rest of her life. "After I got out of jail, half dead, I found that Medgar Evers had been shot down in his own yard," she said.

Federal prosecutors tried five white Winona and Montgomery County officers, among them Chief Herrod, in December 1963 for violating the prisoners' civil rights. An all-white, all-male jury found them not guilty on every count.

Atlantic City. Mississippi's delegation to the 1964 Democratic National Convention, the group of people the state party sent to vote at the national party's meeting, was all white. Hamer and others founded the Mississippi Freedom Democratic Party, and in August 1964 they went to the convention in Atlantic City to ask to be seated in its place. She was the group's vice chair.

On August 22 she told the convention's credentials committee, the group that decided which delegates could take their seats, what had been done to her. Television carried it across the country. A National Archives

page records that her testimony "was interrupted by President Lyndon Johnson, in an effort to suppress the truth and stabilize the Democratic Party."

She ended, "All of this is on account of we want to register, to become first-class citizens. And if the Freedom Democratic Party is not seated now, I question America. Is this America, the land of the free and the home of the brave, where we have to sleep with our telephones off the hooks because our lives be threatened daily, because we want to live as decent human beings, in America?"

She started the Freedom Farm in 1969 to grow food and give families in Sunflower County a living. She used to say, "Nobody's free until everybody's free." She died on March 14, 1977, at 59, of heart failure brought on by cancer, diabetes and high blood pressure. Writers at the Equal Justice Initiative state that the kidney damage from the beating helped cause her death.

BY 1962: Students had sat in at lunch counters across the South, Klansmen had burned a Freedom Riders' bus and beaten the riders, and Fannie Lou Hamer had lost her job for trying to register to vote.

Birmingham, 1963: the children and the church

Birmingham, Alabama, police arrested hundreds of Black children for marching against segregation in May 1963, and firemen turned hoses on them. That September, members of the Ku Klux Klan killed four girls with a bomb at the church the marches had started from.

The children's march. Students aged eight to eighteen gathered at the 16th Street Baptist Church on May 2, 1963, and marched downtown to ask the mayor to end segregation. On the orders of Bull Connor, the public safety commissioner, police arrested them. National Park Service pages give two counts for that day. One says 600 children were arrested, some as young as six, and another says 1,200. By the next day at least 1,000 children were in jail, and over the following days Connor's police jailed more than 3,000 demonstrators.

Hoses and dogs. Connor ordered fire hoses and police dogs turned on the marchers on May 3. A National Park Service page describes it: "Firemen blasted the protestors, including children, with high-pressure fire hoses, and police used their nightsticks indiscriminately," without caring whom they hit. "Police K-9 units loosed their dogs into crowds of peaceful demonstrators, pinning down one woman and severely wounding three teenagers." These were the hoses firemen used on burning buildings, and the water came out hard enough to knock people off their feet.

The bomb. A bomb went off under the steps of the 16th Street Baptist Church just before 11 in the morning on Sunday, September 15, 1963. It killed four girls: Addie Mae Collins, Carole Robertson and Cynthia Wesley, who were fourteen, and Denise McNair, who was eleven.

A few minutes before, they had been together in the basement women's room in their Sunday clothes, ready to sing and to show people to their seats at the adult service, talking about the new school year. Addie Mae's sister, Sarah, survived and lost her right eye.

The bombers. The bombers were members of the Ku Klux Klan. FBI agents named Robert Chambliss as their main suspect. FBI Director J. Edgar Hoover closed the investigation in 1968 without charging anyone and ordered the evidence sealed. Investigators reopened the case after Hoover died in 1972.

Juries later found three of the bombers guilty: Chambliss on November 14, 1977, fourteen years after the bombing, Thomas Blanton on May 1, 2001, and Bobby Frank Cherry in 2002, 39 years after it. Chambliss died in prison in 1985. Blanton was given four life sentences and died in prison in 2020. A fourth Klansman, Herman Cash, died in 1994 without being tried.

Organizers killed in Mississippi, 1963 and 1964

White men in Mississippi killed civil rights organizers in 1963 and 1964, and for decades no jury in the state found anyone guilty of murder for it.

Medgar Evers. Medgar Evers was born at Decatur, Mississippi, on July 2, 1925, and fought in Germany and France in the Second World War. From 1954 he was the NAACP's field secretary for the state of Mississippi. He led marches, prayer vigils, boycotts and drives to register voters, and he worked to get James Meredith admitted to the University of Mississippi in 1962.

Byron De La Beckwith, a member of the White Citizens' Council, shot Evers in the back in the driveway of his home in Jackson in the early hours of June 12, 1963. The Citizens' Council was a group founded in Mississippi in 1954, after *Brown*, to keep segregation in place. Evers's wife, Myrlie, and their children had waited up for him and were watching television. When they heard the shot, the children did what they had practiced: they dropped to the floor and crawled to the bathroom.

Myrlie found her husband collapsed and bleeding, still holding the T-shirts he had carried in from the car. He died at University Hospital from his wounds and loss of blood. Beckwith was tried twice in 1964, and both juries failed to agree on whether he was guilty. A jury found him guilty on February 5, 1994, 31 years and three trials after the killing, and the judge sentenced him to life in prison. He died on January 21, 2001, aged 80.

Freedom Summer. Volunteers came to Mississippi in the summer of 1964 to register Black voters and run summer schools of their own, called Freedom Schools. It was called Freedom Summer. On June 21, 1964, three of the workers went to look at a Black church that had been burned after a civil rights meeting was held there.

James Chaney, 21, was a Black college student from Mississippi. Andrew Goodman, 20, and Michael Schwerner, 24, were white New Yorkers working with CORE. Local police arrested the three for "speeding" and held them in jail. The National Archives records that they were "released after dark into the hands of the Ku Klux Klan."

Searchers found their bodies near Old Jolly Farm in Neshoba County on August 4, 1964. Goodman and Schwerner had each been shot once in the head. Chaney had two broken arms, a broken jaw, a crushed right shoulder and injuries to his groin.

The trials. Twenty-one men were arrested. A federal jury found seven of the eighteen men on trial guilty of violating the three workers' civil rights on October 20, 1967. They included Cecil Price, the Neshoba County deputy sheriff, and Samuel Bowers, the imperial wizard, or top leader, of the White Knights of the Ku Klux Klan of Mississippi. Their sentences ran from three to ten years.

One juror would not vote to find Edgar Ray Killen guilty. Killen was a former preacher and Klan leader who had organized the killings. A Mississippi jury found Killen guilty of manslaughter on June 21, 2005, 41 years to the day after the killings. He was sentenced to 60 years and died in the state prison at Parchman on January 11, 2018.

The March on Washington, August 28, 1963

About 250,000 people came to Washington on August 28, 1963, for the March on Washington for Jobs and Freedom. A National Park Service page counts about 190,000 of them as Black and about 60,000 as white. A. Philip Randolph, who had threatened a march on Washington in 1941 and called it off, led this one. Bayard Rustin organized it in less than three months.

All men. All ten of the march's sponsors were men. Besides Randolph, they included Martin Luther King Jr., John Lewis of the Student Nonviolent Coordinating Committee, the heads of three other civil rights groups and four other men, among them the union leader Walter Reuther. Dorothy Height of the National Council of Negro Women helped plan the march, and the Park Service page states that she "operated in the background of this male dominated, leadership group." No woman gave a formal speech.

The day. The marchers walked the mile-long National Mall to the Lincoln Memorial, where civil rights and religious leaders spoke for three hours. The sponsors issued ten demands, among them fair wages, fair hiring and desegregated schools. The Park Service page records that John Lewis's speech was "toned down...at the request of white liberals and moderate black allies," people on his own side who asked him to say it more softly. What he said included, "we do not want our freedom gradually but we want to be free now."

King's speech. The National Archives calls King's speech, known as "I Have a Dream," the high point of the march. He told the crowd that he dreamed his four little children would one day live in a nation where they "will not be judged by the color of their skin but by the content of their character."

The law that followed. In 1964, Southern senators kept talking for 60 working days so that no vote could be held on a civil rights bill. On June 10, enough senators voted to end the talking, the first time that had happened on a civil rights bill, and nine days later they passed it, 73 to 27. President Lyndon Johnson signed the Civil Rights Act on July 2, 1964.

Under the act, segregation became illegal in businesses such as theaters, restaurants and hotels, and in public places such as swimming pools, libraries and public schools. Job discrimination became illegal too. A Government Publishing Office page states that this act and the Voting Rights Act of 1965 "reflected the demands of the march."

Bayard Rustin

Who: The organizer who planned the March on Washington in under three months, and who was found guilty in a California court in 1953 for having sex with men.

When and where: Born at West Chester, Pennsylvania, March 17, 1912. Los Angeles, 1953. Washington, D.C., 1963.

California's governor pardoned him 33 years after he died.

Bayard Rustin was raised a Quaker in a family already doing civil rights work, and he was a pacifist all his life, a person who refuses to take part in war or violence. In 1942, in California, he visited Japanese Americans held at the Manzanar camp and reported on the conditions there.

The first Freedom Ride. He and George Houser organized the Journey of Reconciliation in 1947, a bus trip to test a Supreme Court ruling against racial discrimination in travel between states. A National Park Service page calls it the first of the Freedom Rides. Police arrested Rustin for breaking a state law on segregated seating, and he served 22 days on a chain gang.

Los Angeles, 1953. He was found guilty of vagrancy in the Los Angeles County Superior Court on January 22, 1953. Vagrancy is a crime that means wandering with no job or home. What he had done was have sex with other men who agreed to it. The judge sentenced him to 60 days in jail and ordered him to register as a sex offender, which put his name on California's list of people found guilty of sex crimes.

California's governor wrote in 2020 that "Mr. Rustin was sentenced pursuant to a charge commonly used to punish gay men for engaging in consensual adult sexual conduct," sex that both adults agreed to. The same document says his conviction caused "an injustice that was compounded by his political opponents' use of the record of this case to try to undermine him, his associates, and the civil rights movement."

The march. He taught Martin Luther King Jr. how to practice nonviolent resistance, and he worked for years with A. Philip Randolph. Randolph made him deputy director of the March on Washington, and Rustin planned the whole of it.

He died in New York City on August 24, 1987, of a burst appendix, aged 75. President Barack Obama awarded him the Presidential Medal of Freedom on August 8, 2013, and called him "an unyielding activist for civil rights, dignity, and equality for all." Governor Gavin Newsom of California pardoned him on February 4, 2020, an official forgiveness of the crime. In the same order, Newsom told his office to find and pardon other people who had been arrested and charged for sex between adults who agreed to it.

Two men climb the Gateway Arch, 1964

Percy Green and Richard Daly climbed 125 feet up the unfinished north leg of the Gateway Arch in St. Louis, Missouri, on July 14, 1964. They were protesting the fact that Black workers were kept out of the jobs building it. Green belonged to CORE and later founded a group called ACTION.

BY 1964: About 250,000 people had marched on Washington, Klansmen had killed four girls in a Birmingham church and three civil rights workers in Mississippi, and Fannie Lou Hamer had told her story on national television.

Selma and the Voting Rights Act, 1965

Marchers in Selma, Alabama, campaigned for the right to vote in 1965, and state troopers, other officers and Klansmen attacked them. The 15th Amendment had barred states from denying the vote because of race since 1870, and yet in much of the South officials still used literacy tests to keep Black citizens from registering.

Jimmie Lee Jackson. State Trooper James Bonard Fowler shot Jimmie Lee Jackson in the stomach inside a cafe in Marion, Alabama, during a march for voting rights on February 18, 1965. Jackson was 26 and had been protecting his mother and grandfather. He died of an infection from the wound on February 26. Fowler was charged with murder in May 2007. In 2010 he pleaded guilty to misdemeanor manslaughter, a minor charge, and served five months of a six-month sentence.

Bloody Sunday. About 600 people, most of them Black, walked out of Brown Chapel in Selma on Sunday, March 7, 1965, to march to Montgomery, the state capital. At the Edmund Pettus Bridge, Major John Cloud of the Alabama State Troopers gave them two minutes to turn back. Then state troopers and other officers attacked them "with nightsticks and teargas," in the National Park Service's words. At least 50 marchers needed hospital care. The day is called Bloody Sunday.

James Reeb. About 2,000 people marched again on March 9, led by King. The marchers turned back at the bridge, and the day is called Turnaround Tuesday. That evening four men attacked three ministers outside a restaurant in Selma. One of the attackers hit the Reverend James Reeb of Boston in the head with a club, and Reeb died two days later.

Elmer L. Cook, William Stanley Hoggle and Namon O'Neil Hoggle were tried for his murder, and the jury found all three not guilty on December 10, 1965. The prosecutor had allowed a man whose brother was a suspect in the attack to sit on the jury.

Viola Liuzzo. The march set out again on March 21, 1965, with more than 8,000 people, and by the time it reached the capitol in Montgomery on March 25 the crowd had grown to nearly 25,000. That night, Viola Liuzzo was driving on Highway 80 between Selma and Montgomery. She was a 39-year-old white mother of five from Detroit who had come to help. In Lowndes County, four members of the Ku Klux Klan fired into her car and killed her. Her passenger, Leroy Moton, a 19-year-old Black volunteer, survived.

The four men were Collie Leroy Wilkins Jr., William Orville Eaton, Eugene Thomas and Gary Thomas Rowe. Rowe was a paid FBI informant, a person who secretly passed information to the FBI, and prosecutors promised not to charge him. Alabama juries did not find the other three guilty. A federal jury found them guilty of violating her civil rights, and the judge sentenced each to ten years.

FBI Director J. Edgar Hoover called Liuzzo an "outside agitator" and spread rumors about her to the press. Her family sued the FBI in 1979, and a federal judge dismissed the suit in 1983.

The Voting Rights Act. President Lyndon Johnson signed the Voting Rights Act on August 6, 1965. It banned literacy tests, and federal officials sent examiners into the places that had used them, with the power to sign up voters themselves. States and counties on the act's list had to get approval from federal officials in Washington before changing any voting rule.

By the end of 1965, a quarter of a million new Black voters had registered, a third of them through the federal examiners. By the end of 1966, only four of the thirteen Southern states still had fewer than half of their Black citizens registered.

The Chicano movement, 1954 to 1970

Mexican Americans ran their own rights campaigns from the 1950s. Many called themselves Chicanos, and their campaigns the Chicano movement.

Hernandez v. Texas. They won a case at the Supreme Court in 1954. A jury in Jackson County, Texas, found a Mexican American man, Pete Hernandez, guilty of murder, and he was sentenced to life in prison. About 14 of every 100 people in the county had Mexican or Latin American last names. In 25 years, no one with such a name had served on a jury commission, the board that chose jurors. More than 6,000 jurors had been called in that time, and none had such a name.

On the courthouse grounds, one of the two men's toilets was marked "Colored Men" and "Hombres Aqui," which means "Men Here." All nine justices ruled on May 3, 1954, in *Hernandez v. Texas*, that shutting Mexican Americans out of juries broke the 14th Amendment and its promise of equal protection. The ruling came two weeks before *Brown*.

The walkouts. About 15,000 students walked out of seven schools in East Los Angeles from March 1 to 8, 1968. About 60 of every 100 Mexican American students there left high school without finishing. Staff forbade students to speak Spanish and steered them into trade classes instead of classes that prepared them for college. The students asked for teaching in both languages, changes to what they were taught, and Mexican American principals.

Members of the Board of Education turned them down, saying the changes would cost too much. Police arrested thirteen organizers on March 31 on charges of felony conspiracy, planning together to commit a serious crime. One of them, the teacher Sal Castro, was fired. He got his job back after months of sit-ins at the board's offices, and on October 2 police arrested 35 people there.

The Moratorium. Between 20,000 and 30,000 people marched through East Los Angeles in the National Chicano Moratorium on August 29, 1970. They were protesting how many Mexican Americans were being drafted, wounded and killed in the Vietnam War. About 1,500 police officers came to Laguna Park, fired tear gas into the crowd and beat protesters. Three or four people were killed that day, and 150 were arrested.

Los Angeles County Sheriff's Deputy Thomas Wilson fired a tear gas projectile into the Silver Dollar Bar on Whittier Boulevard. It hit Rubén Salazar, a reporter for the *Los Angeles Times*, in the head and killed him. The coroner, the county official who decides how a person died, ruled his death a homicide, a killing by another person. District Attorney Evelle Younger decided not to charge Wilson. City officials later renamed Laguna Park for Salazar.

Two more dead. Two young members of the Brown Berets, a Chicano youth group, were also killed that day. Police said that Angel Gilberto Díaz drove through a police barricade and that officers shot him. Raúl Ruiz, a reporter who was at the march, said Díaz was sitting in the passenger seat when he was shot in the back of the head. Neither account names the officers who shot him.

Witnesses said Lyn Ward died when a tear gas canister exploded, and police said a bomb planted by protesters killed him.

DISPUTED: The art historian Ruben C. Cordova counts three dead that day, because the fourth man often named, Gustav Montag, was killed at a later protest, on January 31, 1971, when a pellet from a police shotgun bounced and struck him in the heart. PBS SoCal counts four dead on August 29.

The farmworkers. In the same years, farmworkers organized under Larry Itliong, Cesar Chavez and Dolores Huerta. About 1,500 mostly Filipino American grape workers led by Itliong walked out at Delano, California, on September 8, 1965. Eight days later, members of the union of Chavez and Huerta, the National Farm Workers Association, voted to join them.

What the strike won. The two unions joined together and became the union now called the United Farm Workers. People across the country stopped buying California table grapes to back the strikers. On July 29, 1970, 26 grape growers in Delano signed the farmworkers' first union contracts. Chapter 14, Work and Workers, tells the strike in full.

What was reported about Chavez. The *New York Times* reported in March 2026 that Chavez had sexually abused two girls in the 1970s. The two women, Debra Rojas and Ana Murguia, are daughters of union organizers. They said the abuse began when they were 12 and 13, and Rojas said Chavez raped her in a motel room when she was 15.

Dolores Huerta then said publicly that Chavez had pressured her into sex once and had raped her in a car in 1966, and that she became pregnant both times. She said she had kept the secret for 60 years because she believed the truth would hurt the farmworker movement.

Chavez died in 1993, so no court has ever heard these accusations. The United Farm Workers canceled its events for Cesar Chavez Day. California's lawmakers voted on March 26, 2026, to rename the state holiday Farmworkers Day, and Governor Gavin Newsom signed the change into law.

The women's movement and the Equal Rights Amendment, 1966 to 1982

Women organized in the late 1960s for equal treatment at work and under the law, and they won some laws and lost a constitutional amendment by three states.

NOW. Federal law had banned sex discrimination at work since 1964. At a government conference on women in Washington on June 30, 1966, some delegates wanted the conference to demand that a federal office, the Equal Employment Opportunity Commission, enforce that ban. Conference officials ruled the demand out of order, against the meeting's rules, saying a government commission could not act against another government office.

That night 28 people met in the hotel room of Betty Friedan, whose 1963 book *The Feminine Mystique* had been a best-seller, and founded the National Organization for Women, known as NOW. They set it up "to take action to bring women into full participation in the mainstream of American society NOW."

The amendment. Alice Paul wrote the first version of the Equal Rights Amendment, and it was first brought before Congress in 1923. The first section of the version members of Congress voted on in the 1970s reads: "Equality of rights under the law shall not be denied or abridged by the United States or by any State on account of sex." Representative Martha Griffiths of Michigan pushed it through the House in 1971. Senators passed it on March 22, 1972, by 84 votes to 8.

Three-fourths of the states, 38 of them, had to ratify it within seven years. Hawaii ratified first, and 35 states had ratified by 1977.

STOP ERA. Phyllis Schlafly led the campaign against it, called STOP ERA, from 1972. She argued that under the amendment women could be drafted into the military. She also argued that widows would lose Social Security money paid to them as wives, and that mothers would lose their edge in court fights over who keeps the children. Lawmakers in Nebraska, Tennessee, Idaho, Kentucky and South Dakota voted to take back their states' approval.

Members of Congress extended the deadline to 1982, and no more states ratified before it ran out. When the deadline passed, the amendment was three states short of the 38 it needed.

Credit. Members of Congress passed the Equal Credit Opportunity Act in October 1974. No lender could now legally refuse a woman credit, a loan or the right to buy now and pay later, because of her sex or because she was married or single.

What did Title IX change?

Title IX barred schools and colleges that take federal money from shutting anyone out of a program because of sex, and school sports teams are among those programs. It became law on June 23, 1972. Its main sentence reads: "No person in the United States shall, on the basis of sex, be excluded from participation in, be denied the benefits of, or be subjected to discrimination under any education program or activity receiving Federal financial assistance."

Who passed it. Representative Patsy Takemoto Mink of Hawaii, Representative Edith Green of Oregon and Senator Birch Bayh of Indiana worked together to pass it. Green, who had taught school for 11 years, led the hearings in the House in 1970. After Mink died on September 28, 2002, members of Congress renamed the law the Patsy T. Mink Equal Opportunity in Education Act.

The numbers. Fewer than 300,000 girls played high school sports before Title IX, and more than 3 million did by 2012, more than ten times as many. That year the White House reported that American high schools still offered girls 1.3 million fewer chances to play sports than boys. In the 2013-14 school year, the Government Accountability Office found that girls were 49 of every 100 students at public high schools that offered sports, and 43 of every 100 players.

Gay Americans organize, 1953 to 1979

Gay Americans organized openly from the 1950s against being fired, arrested and photographed for who they were.

Fired. President Dwight Eisenhower signed Executive Order 10450 on April 27, 1953. The order listed "sexual perversion," the government's term then for being gay, among the reasons to remove a person from a federal job. The historian David K. Johnson estimates that "somewhere between 5,000 and 10,000 people lost their jobs" under it. Frank Kameny, who worked at the Army Map Service, was fired in 1957 and fought his firing as a civil rights case.

Protesters picketed the White House against the firings on May 21, 1965. The members of the Civil Service Commission, which set the rules for federal jobs, announced in 1975 that federal workers would no longer be fired for being gay.

Stonewall. State rules in New York in the 1960s forbade bars to serve alcohol to gay people. Police raided gay bars, arrested the customers and photographed them. In the early hours of June 28, 1969, police led by Deputy Inspector Seymour Pine raided the Stonewall Inn, a gay bar in Greenwich Village. This time the crowd outside fought back. People threw coins, bottles, bricks, garbage cans and burning trash.

The police shut themselves inside the bar until the Tactical Patrol Force, a police unit trained to break up riots, arrived. The protests went on for six nights, and police arrested 13 people the first night. A National Park Service page counts fifty to sixty gay groups in the country before Stonewall and at least 1,500 a year later.

Thousands of people marched from Greenwich Village to Central Park in the Christopher Street Liberation Day March on June 28, 1970, the first anniversary of the raid. Craig Rodwell led the organizing.

Harvey Milk. Harvey Milk was one of the first openly gay people elected to public office in the United States, as a member of San Francisco's Board of Supervisors. Dan White, a former supervisor, shot Mayor George Moscone and then shot Milk five times at City Hall on November 27, 1978. A jury found

White guilty of voluntary manslaughter, a lesser crime than murder, in May 1979, and he was sentenced to seven years and eight months. He served about five. Crowds rioted at City Hall after the verdict.

Native activists hold their ground, 1969 to 1974

Native activists occupied land, and some broke fishing laws on purpose to be arrested. They held Alcatraz Island in San Francisco Bay from November 20, 1969, to June 11, 1971. In 1973 they held the village of Wounded Knee, on the Pine Ridge Reservation in South Dakota, the Oglala Lakota's land, from February 27 to May 8.

Fishing. Native fishermen in Washington State fished where their treaties said they could, and got arrested for it on purpose. Billy Frank Jr. of the Nisqually Nation was arrested more than 50 times, starting at age fourteen, at Frank's Landing.

Judge Boldt ruled on February 12, 1974, in *United States v. Washington*, that the tribes were entitled to half of the salmon that could be caught. The justices of the Supreme Court agreed with the ruling in 1979. President Barack Obama awarded Frank the Presidential Medal of Freedom in 2015, after his death.

BY 1974: Mexican American students had walked out of their schools, women had won Title IX, gay New Yorkers had fought back at the Stonewall Inn, and Native activists had held Alcatraz and Wounded Knee.

Done to children without asking them, 1965 to 1973

Between 1965 and 1973, surgeons sterilized two poor girls in Alabama after nurses told their mother they would get shots, and staff at a New York state school beat and sexually abused disabled children.

The Relf sisters. Nurses in Montgomery, Alabama, told Minnie Relf in June 1973 that her daughters would get birth control "shots." The nurses worked at the family planning clinic of Montgomery Community Action, which was paid for with federal money.

Minnie Relf could not read or write, and she signed the form with an "X." Surgeons then sterilized the two girls, who were 12 and 14, by tubal ligation. In that operation a surgeon cuts or ties the two tubes that carry eggs to the womb, so that the girls could never become pregnant.

Their father, Lonnie Relf, saw the surgical scars when they came home from three days in the hospital. The records do not name the surgeons. Lawyers for the Southern Poverty Law Center sued for the family. The judge in the case found that an estimated 100,000 to 150,000 poor people were being sterilized each year under programs paid for with federal money, and barred federal money for sterilizations done against people's will.

Willowbrook. Willowbrook State School, on Staten Island in New York City, was a place run by the state for children with intellectual disabilities. It was built for 4,000 people and held 6,000 by 1965. Senator Robert F. Kennedy toured it that year and said the residents were "living in filth and dirt, their clothing in rags." The television reporter Geraldo Rivera showed crowded wards, filthy bathrooms, and staff beating and sexually abusing residents in 1972.

Parents of 5,000 residents sued the State of New York on March 17, 1972. New York officials agreed to a court order in 1975, and the last children left Willowbrook on September 17, 1987.

Parents sue to get their children into school, 1971 to 1975

Parents of disabled children sued in 1971 and 1972 to get their children into public schools, and they won.

The Pennsylvania law. Pennsylvania's school law let a school board "refuse to accept or retain beginners who have not attained," or reached, "a mental age of five years." A child's mental age was a test score given as the age of an ordinary child who would score the same.

The law also allowed the "temporary or permanent exclusion from the public school of children who are found to be uneducable and untrainable," judged unable to be taught. A child who scored under 30 on an intelligence test was called uneducable, and officials handed the child to the state welfare department for placement and perhaps training, instead of school.

The lawsuit. The parents' organization was the Pennsylvania Association for Retarded Children, using the word of the time, in laws and in the case, for people with intellectual disabilities. The association's lawyer, Thomas K. Gilhool of Philadelphia, sued the state for fourteen children on January 7, 1971. The association's leaders estimated that about 100,000 children in Pennsylvania were being denied a free education.

The two sides had reached an agreement by October 1971, and three federal judges issued orders on October 8, 1971, and May 5, 1972. Pennsylvania officials agreed to give every such child from 6 to 21 a free public education. Schools had to offer a hearing before labeling a child retarded, and had to test each labeled child again at least every two years. A federal judge ruled the same way for seven children in Washington, D.C., in *Mills v. Board of Education*, on August 1, 1972.

A national law. Parents were bringing cases like these in twenty states. The federal Bureau of Education for the Handicapped estimated that in 1975 about a million disabled children were still shut out of school completely. At least 300,000 more were getting schooling that did not meet their needs. President Gerald Ford signed the Education for All Handicapped Children Act on November 29, 1975. It promised a free public education suited to each disabled child, in every state.

Section 504, Gallaudet and the Americans with Disabilities Act, 1973 to 1990

Disabled Americans occupied federal offices in 1977 and forced officials to enforce a law that had been passed four years before. Section 504 of the Rehabilitation Act of 1973 was the first federal law protecting the civil rights of disabled people. It says that no qualified person with a disability "shall, solely by reason of his or her disability, be excluded from the participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving Federal financial assistance."

No one who ran a school, hospital or office paid with federal money could legally turn a person away only because of a disability. But federal officials could not enforce the law until someone signed the rules for carrying it out, and for four years no one did. Joseph A. Califano Jr., the Secretary of Health, Education, and Welfare, refused to sign them after leaders of schools and hospitals objected to what the rules would cost.

The sit-in. Disabled people demonstrated at federal offices in eleven cities on April 5, 1977: San Francisco, Washington, Seattle, Los Angeles, Denver, Chicago, Dallas, Atlanta, New York, Boston and Philadelphia. In San Francisco, about 100 of them occupied the department's offices at 50 UN Plaza and stayed. National Park Service pages give the length of the sit-in as 26 days in one account and 24 days in another. The sit-in worked: Califano signed the rules.

Who fed them. Members of the Black Panther Party, a Black political group, brought in one hot meal a day and food for two more. Leaders of a union, the International Association of Machinists, paid bills and paid to fly a group to Washington. The Brick Hut, a business owned and run together by a group of lesbian women, women who love women, and Glide Memorial Church also helped. Judy Heumann, Kitty Cone and Dennis Billups led the occupation from inside.

Judy Heumann

Who: A woman refused a place at school as a child and a teaching license as an adult, who sued, won, and then helped lead the 1977 sit-in that won the Section 504 rules.

When and where: New York City, 1949 to 1970. San Francisco, April 1977.

"I was willing to make a fuss about it."

Judy Heumann got polio at eighteen months old, which paralyzed her, and she used a wheelchair from early childhood. New York City school officials refused to take her. Her mother, Ilse, spent years pressing them. Judy got in at about fourth grade, into a separate program for disabled students called Health Conservation 21.

The license. She finished college and applied for a license to teach. New York Board of Education officials turned her down in 1970 because she used a wheelchair. She asked the American Civil Liberties Union for help, and its lawyers refused, saying that turning someone down for a medical reason was not discrimination.

That year the *New York Times* put her story on its front page, and a lawyer, Roy Lucas, offered to represent her. She sued the Board of Education at 22, before Judge Constance Baker Motley. Board officials settled before trial and gave her the license. She and her supporters then started a group called Disabled in Action.

50 UN Plaza. In April 1977 she was one of the three leaders inside the occupied federal offices at 50 UN Plaza in San Francisco. During the sit-in she and Dennis Billups flew to Washington.

She spent the rest of her life on the same work. From 1993 to 2001 she was the federal official in charge of special education and rehabilitation. She advised the World Bank on disability from 2001 to 2005, and from 2010 to 2017 she was the federal government's special adviser on international disability rights. She died in Washington on March 4, 2023.

> "Some people say that what I did changed the world," she once said. "But really, I simply refused to accept what I was told about who I could be. And I was willing to make a fuss about it."

Deaf President Now, 1988, and the Capitol Crawl

Gallaudet University, in Washington, D.C., is the country's university for deaf students. In March 1988 members of its board of trustees, the group that runs the university, chose a hearing candidate, Elisabeth Zinser, as president over deaf candidates. Students marched, stopped going to classes and let the air out of the tires of school buses in front of the university's gates.

Four demands. Zinser had to go and a deaf president be chosen. The board's chair, Jane Spilman, had to resign. At least 51 of every 100 board members had to be deaf. No one who protested could be punished. After eight days the trustees gave in. I. King Jordan became Gallaudet's first deaf president. "A Deaf person can do anything a hearing person can, except hear," he said.

The Capitol Crawl. While members of Congress were considering the Americans with Disabilities Act, disabled protesters left their wheelchairs, walkers and crutches at the bottom of the steps of the Capitol and climbed the steps without them. The protest is called the Capitol Crawl. President George H. W. Bush signed the Americans with Disabilities Act on July 26, 1990. Employers, state and city governments, businesses open to the public and public bus and train systems all had to give disabled people an equal chance at their jobs, services and goods.

Busing, Detroit and Boston, 1974

When Black parents in the North sued over segregated schools in the 1970s, they won in Boston and lost at the Supreme Court over Detroit's suburbs.

Detroit. The justices ruled five to four against busing children across school district lines on July 25, 1974, in *Milliken v. Bradley*. A judge could not send city children to suburban schools, or suburban children to city schools, unless someone proved that one district had caused segregation in the other. A judge's plan that would have covered Detroit and 53 suburban districts could not go ahead. Justice Thurgood Marshall, who had argued *Brown* before the Supreme Court as the NAACP's lawyer, was one of the four who dissented.

Boston. Tallulah Morgan, a Black mother of three, and thirteen other families sued the Boston School Committee, the city's elected school board, in 1972. Judge W. Arthur Garrity ruled on June 21, 1974, that "the entire school system of Boston was unconstitutionally segregated," kept apart by race in a way the Constitution forbids. School officials, he found, "had knowingly carried out a systematic program of segregation affecting all of the city's students, teachers and school facilities."

Busing, carrying children by bus to schools outside their neighborhoods, began on September 9, 1974, and paired South Boston with Roxbury. Protesters threw rocks and bottles at the buses, broke the windows and shouted racial insults. Of the 800 Black students assigned to South Boston High that first day, 56 came. Garrity handed the job of watching over the schools to state officials in 1982 and issued his final order in 1985.

Warren County residents lie down in the road, 1982

Residents of mostly Black Warren County, North Carolina, lay down in front of dump trucks in September 1982 to stop a landfill for poisoned soil. Someone had illegally dumped chemicals called PCBs along North Carolina's roadsides in 1978, poisoning the soil. State officials chose Afton, in mostly Black, rural Warren County, as the place to bury it.

After the courts failed them, residents and civil rights leaders, among them the Reverend Benjamin Chavis of the United Church of Christ's Commission for Racial Justice, blocked the trucks. Police arrested about 500 people over six weeks, and state officials had the landfill built anyway. The protest is widely named as the start of the environmental justice movement, the campaign against placing pollution where poor and nonwhite people live. Members of the church's commission published a report, *Toxic Wastes and Race*, in 1987.

An apology and $20,000, 1980 to 1990

Japanese Americans who had been held in the camps of the Second World War campaigned for an apology and payment, and in 1988 they won both. The Japanese American Citizens League and the National Council for Japanese American Redress led the campaign.

The commission. President Jimmy Carter signed a law creating a commission to study the camps on July 31, 1980. Its members held 20 days of hearings in 11 places in 1981 and heard more than 750 witnesses. Their report, in February 1983, found that the causes were "race prejudice, war hysteria and a failure of political leadership." A federal judge also threw out the finding that Fred Korematsu had been guilty in 1983.

The law. Senators Daniel Inouye and Spark Matsunaga and Representatives Norman Mineta and Robert Matsui led the bill through Congress. President Ronald Reagan signed the Civil Liberties Act on August 10, 1988. Each person still living who had been held received an apology and $20,000. The first letters and checks went out on October 1, 1990.

BY 2000

If you remember one thing: Marchers, students, parents and disabled protesters won court rulings against segregated schools and buses, and new laws against segregation in public places, literacy tests, sex discrimination in schools and shutting disabled people out. Many of them were beaten, jailed or killed for it.

Left open: Would the justices of the Supreme Court keep the Voting Rights Act as members of Congress wrote it in 1965?

Next: After 2000, judges and lawmakers widened some of these rights and narrowed others.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 50 MINUTES

Judges and lawmakers since 2000 have widened some of the rights earlier movements won and taken back others. Same-sex couples, two men or two women, won the right to marry in every state in 2015, twelve years after police in some states could still arrest two men or two women for having sex at home. Members of Congress passed the first federal law against lynching in 2022, after about 200 earlier bills had failed.

Rights narrowed. The justices of the Supreme Court cut back the Voting Rights Act of 1965 in 2013 and again in 2026. They overruled the 1973 ruling that protected abortion in 2022, declaring it wrong and ending it, and a year later they stopped colleges from considering an applicant's race.

Their rulings of 2025 and 2026 allowed states to ban some medical care for transgender young people and to keep transgender girls off girls' school teams. A transgender person is someone whose own sense of being male or female, called gender identity, differs from the sex recorded at birth.

New tools. Organizers used phones and social media alongside the old tools of meetings, lawsuits and marches. A seventeen-year-old recorded a Minneapolis police officer kneeling on George Floyd in 2020, and after people saw her video they protested in cities around the world. Two words posted on Twitter in October 2017, "me too," were used more than 19 million times within a year.

Police officers and federal agents also killed people in these years, among them George Floyd in 2020, and Renée Good and Alex Pretti, two American citizens shot by federal immigration agents in Minneapolis in 2026. As of September 2026 no one had been charged with killing Good or Pretti, and the fights over voting maps, DACA and the Equal Rights Amendment were still going on.

The justices strike down the Voting Rights Act's list, 2013

The justices of the Supreme Court struck down the list of places that had to get federal permission before changing a voting rule on June 25, 2013. That permission is called preclearance. Since 1965, officials in the listed states and counties had needed it from federal officials in Washington.

The list. Nine states were on the list in full. Alabama, Georgia, Louisiana, Mississippi, South Carolina and Virginia were covered from November 1, 1964. Alaska, Arizona and Texas were covered from November 1, 1972. Named counties in California, Florida, New York, North Carolina and South Dakota were covered too, and so were two townships in Michigan.

Shelby County v. Holder. The justices split five to four. Chief Justice John Roberts wrote for the majority that "the Act imposes current burdens and must be justified by current needs." The majority struck down Section 4(b), the formula that decided which places went on the list, and left the preclearance rule itself standing with no list of places it applied to.

"We issue no holding on §5 itself, only on the coverage formula," Roberts wrote, and "Congress may draft another formula based on current conditions." Justice Ruth Bader Ginsburg wrote the dissent, joined by Justices Stephen Breyer, Sonia Sotomayor and Elena Kagan. A Justice Department web page

states the result. The places that had been on the list "no longer need to seek preclearance for the new voting changes," unless a separate court order covers them.

No new list. Members of Congress tried to write a new list in the John R. Lewis Voting Rights Advancement Act, named for John Lewis of the Student Nonviolent Coordinating Committee, one of the speakers at the 1963 March on Washington. Under one version, a state with 15 or more voting rights violations in the previous 25 years would be covered for 10 years.

Senators voted 49 to 51 on ending debate over the bill on January 19, 2022. Ending debate needed 60 votes, so the bill never came to a final vote. Later that night senators voted 52 to 48 to keep the 60-vote rule in place. By September 2026 members of Congress had passed no new list.

Proving intent: Louisiana v. Callais, 2026

Anyone suing under Section 2 of the Voting Rights Act must now show that state officials meant to discriminate by race, under a ruling of April 29, 2026. Section 2 lets people go to court over a voting rule or a map of election districts that gives a racial group less chance to elect the candidates it wants. Members of Congress had rewritten Section 2 in 1982 so that a person suing would not have to prove what the officials intended.

The Louisiana map. Louisiana's lawmakers drew a map of six congressional districts with one majority-Black district after the 2020 census. A federal judge held that the map likely broke Section 2. The lawmakers then drew a second majority-Black district. Other voters sued, saying the new map was a racial gerrymander, a set of districts drawn by race in a way the Constitution forbids.

The new rule. Justice Samuel Alito wrote for six justices that the Voting Rights Act had not required the second district, so the map broke the Constitution. He set a new rule for every future case. Section 2 "imposes liability only when the evidence supports a strong inference that the State intentionally drew its districts to afford minority voters less opportunity because of their race." A person suing must now show that officials meant to treat voters unfairly because of race.

The Congressional Research Service, which writes explanations of the law for members of Congress, put it this way on May 14, 2026. A person suing "now must show that a state drew a redistricting map based on racial, and not political, considerations." That person must also "present evidence of current-day, intentional voting discrimination that is based on race."

The dissent. Justice Elena Kagan dissented, joined by Justices Sonia Sotomayor and Ketanji Brown Jackson. To show how lawmakers can draw districts so that some people's votes count for less, she asked readers to imagine a round district that was 90 percent Black, in a state that was otherwise 90 percent white.

Lawmakers could slice that district "into six pie pieces," she wrote, and spread its Black voters across six mostly white districts. In that imagined state, "Election after election, Black citizens' votes are, by every practical measure, wasted." "'Cracked' is the usual term," she said, and the majority had made "a nullity of Section 2," a rule with no force left in it.

The new map. Louisiana's Republican lawmakers passed a new map in May 2026. It removed one of the state's two majority-Black districts and was expected to elect five Republicans and one Democrat. About a third of Louisiana's people are Black. State Senator Jay Morris wrote the bill. "We focused on the Democrat numbers, not the racial numbers when drawing," said State Representative Beau Beaullieu, a Republican. State Representative Kyle Green Jr., a Democrat, said the map took Black Louisianians "from 33% of the population to 16% of the representation."

Louisiana officials put off the House primary elections set for May 16, 2026. To use the new map, they threw out about 40,000 votes that people had already cast.

Other states. The justices let Alabama keep using a 2023 congressional map with one majority-Black district on June 2, 2026. They stopped a lower court's order against it, saying Alabama was "likely to succeed on the merits as to both claims," likely to win.

The Congressional Research Service reported in May 2026 that "some state legislatures are considering or have already enacted modifications to their redistricting maps that eliminate majority-minority districts for the upcoming 2026 congressional elections." Lawmakers in some states were drawing new maps without districts where most voters were Black or Latino. In September 2026 members of Congress had passed no new rule for Section 2.

Separate schools, counted in 2021 and 2022

School officials have been forbidden by law since 1954 to send children to separate schools because of their race. In the 2020s, federal and university researchers still found many schools where three-quarters or more of the students were of one race.

The federal count. The federal Government Accountability Office, which checks on government work for Congress, published a report on June 16, 2022, about the school year 2020-21. Its researchers found that about 18.5 million children, more than a third of American public school students, "attended schools where 75 percent or more students were of a single race or ethnicity," ethnicity meaning a group's shared background, such as Hispanic. Fourteen percent of students attended schools where 90 percent or more were of one race or ethnicity.

The longer count. Researchers at the Civil Rights Project at UCLA, a university research center, count a longer stretch using the federal government's school data. In a report of April 2024, Gary Orfield and Ryan Pfleger gave the share of public schools that were 90 to 100 percent non-white. It was 7.4 percent in 1988 and 19.8 percent in 2021, or about one school in five. In those schools in 2021, 78 of every 100 students were poor.

The same report follows Black students in the South. The share who attended a school where most students were white reached a peak of about 43 percent in the mid-1980s, and it was about 16 percent in 2021. The authors tie the turn to a 1991 Supreme Court ruling, *Board of Education of Oklahoma City v. Dowell*, which let judges end court orders to desegregate. After it, the share fell for nearly 30 years in a row.

Race in college admissions, 2014 to 2023

The justices of the Supreme Court ruled in 2023 that colleges may not give applicants a boost because of their race. In Justice Sonia Sotomayor's count, the justices had let colleges consider race "in a limited way" for 45 years before that, to gain the benefits of racial diversity. The practice is often called affirmative action.

The lawsuits. Students for Fair Admissions, a nonprofit group founded in 2014, sued Harvard College and the University of North Carolina in November 2014. Its first board had three members, Edward Blum, Abigail Fisher and Richard Fisher. At Harvard, readers and committees considered race at several steps of the process, and for a significant share of the Black and Hispanic students admitted, race tipped the decision. At the University of North Carolina, readers could give an applicant a "plus" because of race.

The ruling. Chief Justice John Roberts wrote for six justices on June 29, 2023, that both programs broke the 14th Amendment's promise of equal protection. "The guarantee of equal protection cannot mean one thing when applied to one individual and something else when applied to a person of another color," he wrote. A college may still read an applicant's own account of "how race affected the applicant's life," he added, if it is tied to the applicant's character or abilities. The justices left open the question for the military academies.

The dissent. Sotomayor dissented, joined by Justice Elena Kagan, and by Justice Ketanji Brown Jackson in the North Carolina case. Jackson took no part in the Harvard case. "The Court long ago concluded that this guarantee can be enforced through race-conscious means in a society that is not, and has never been, colorblind," Sotomayor wrote, colorblind meaning taking no notice of race at all. "Today, this Court stands in the way and rolls back decades of precedent," the Court's own earlier rulings, "and momentous progress."

From arrest to marriage, 2003 to 2015

Two men or two women in some states could still be arrested in 2003 for having sex at home. Twelve years later the justices of the Supreme Court ruled that same-sex couples could marry in every state.

Lawrence v. Texas. The justices voted six to three on June 26, 2003, to strike down a Texas law that made it a crime for two people of the same sex to have sex in private. Justice Anthony Kennedy wrote the opinion. The justices overruled a 1986 ruling that had allowed such a law: "*Bowers* was not correct when it was decided, is not correct today, and is hereby overruled."

Massachusetts. The Supreme Judicial Court of Massachusetts, the state's highest court, ruled on November 18, 2003, that the state constitution guaranteed same-sex couples the right to marry. The judges held back their own order for 180 days "to permit the Legislature to take such action" as it chose, the Legislature meaning the state's lawmakers. Massachusetts became the first state where same-sex couples could marry on May 17, 2004.

Edith Windsor. Edith Windsor and Thea Spyer married in Ontario, Canada, in 2007 and lived in New York. Spyer died in February 2009 and left everything she owned to Windsor. The federal Defense of Marriage Act defined marriage as one man and one woman for more than 1,000 federal laws and rules.

So federal officials counted Windsor as unmarried, and she paid $363,053 in estate taxes, a tax on what a person leaves at death. She sued for the money back, and the justices struck down that part of the law, five to four, in *United States v. Windsor* on June 26, 2013.

Obergefell. By 2015 voters and lawmakers had changed their marriage laws to include same-sex couples in eleven states and the District of Columbia, by the justices' count. The justices ruled five to four on June 26, 2015, in *Obergefell v. Hodges*, that same-sex couples may marry in every state. Kennedy wrote the opinion. Chief Justice John Roberts and Justices Antonin Scalia, Clarence Thomas and Samuel Alito each wrote a dissent.

Within moments of the ruling, LGBT people gathered in Christopher Park in New York. LGBT is short for lesbian, gay, bisexual and transgender. President Barack Obama made that park, a triangle of about 0.12 acres, the Stonewall National Monument on June 24, 2016, in memory of the 1969 uprising at the Stonewall Inn.

Jim Obergefell

Who: A widower who sued Ohio to have his name put on his husband's death certificate, and won the case that made same-sex marriage legal in every state.
When and where: Ohio, and Baltimore, Maryland, 2011 to 2015.

The couple married inside a medical plane on a Baltimore runway.

Jim Obergefell sued the state of Ohio over a death certificate. He wanted to be named on it as the husband of John Arthur, and Ohio law would not allow it.

In the justices' words. The justices set out his story in their written opinion, where "petitioner" and "plaintiff" both mean a person who brought the case. "Petitioner James Obergefell, a plaintiff in the Ohio case, met John Arthur over two decades ago. They fell in love and started a life together, establishing a lasting, committed relation. In 2011, however, Arthur was diagnosed with amyotrophic lateral sclerosis, or ALS."

ALS destroys the nerve cells that carry orders from the brain to the muscles. The muscles stop getting those orders, waste away and stop working. It takes the hands and legs first, then swallowing, then breathing. The Court called the disease "progressive, with no known cure."

The wedding. The two men decided to marry before Arthur died. Ohio did not allow same-sex couples to marry, and Maryland did. The Court went on: "It was difficult for Arthur to move, and so the couple were wed

inside a medical transport plane as it remained on the tarmac in Baltimore. Three months later, Arthur died. Ohio law does not permit Obergefell to be listed as the surviving spouse on Arthur's death certificate." The wedding was on July 11, 2013, and Arthur died on October 22, 2013.

The case. Obergefell sued. His case was joined with cases from Michigan, Kentucky and Tennessee, brought together by 14 same-sex couples and two men whose partners had died. The justices decided them together on June 26, 2015. The last lines of Justice Kennedy's opinion read: "They ask for equal dignity in the eyes of the law. The Constitution grants them that right."

BY 2015: The justices had struck down the Voting Rights Act's list of places that needed federal permission, and same-sex couples could marry in every state.

The Respect for Marriage Act, 2022, and Kim Davis

Members of Congress wrote the protection for same-sex marriages into federal law in 2022, and the Respect for Marriage Act became law on December 13, 2022. It does not require any state to issue marriage licenses. It forbids any state official to refuse to recognize a marriage from another state because of the couple's sex, race, ethnicity or national origin. The Attorney General, or the person harmed, may sue anyone who breaks it.

The votes. Senators passed it on November 29, 2022, by 61 votes to 36, with twelve Republican senators voting yes. The House passed it on December 8, 2022, by 258 votes to 169, with 39 Republicans voting yes. The law opens with a sentence from Kennedy's *Obergefell* opinion: "No union is more profound than marriage, for it embodies the highest ideals of love, fidelity, devotion, sacrifice, and family."

Kim Davis. Kim Davis was a county clerk in Kentucky. After the 2015 ruling she refused, on religious grounds, to issue a marriage license to a gay couple, David Ermold and David Moore, and she spent six days in jail.

A jury later ordered her to pay the couple $100,000 for the harm done to them and $260,000 for their lawyers' fees. In 2025 she asked the justices of the Supreme Court to overturn *Obergefell*, the first such request since 2015. The justices refused to hear it on November 10, 2025.

Black Lives Matter and Ferguson, 2012 to 2016

Black Lives Matter began in 2013, after a Florida jury found the man who killed Trayvon Martin not guilty.

Trayvon Martin. George Zimmerman, a neighborhood-watch volunteer, shot and killed Martin, a 17-year-old Black teenager who was walking back from a convenience store, in Sanford, Florida, on February 26, 2012. Martin was unarmed. The Sanford police chief said on March 12, 2012, that no charge could be filed. After demonstrations across the country, Governor Rick Scott appointed a special prosecutor, a lawyer chosen to handle this one case, in April 2012.

The special prosecutor charged Zimmerman with second-degree murder, a murder that was not planned ahead. The jury found him not guilty on July 13, 2013. Justice Department officials announced in 2015 that they would not bring federal hate-crime charges.

After the verdict, three Black organizers, Alicia Garza, Patrisse Cullors and Opal Tometi, started a project to build a movement and called it #BlackLivesMatter. The mark # turns a phrase into a hashtag, a label that people can search for on social media.

Michael Brown. Officer Darren Wilson of the Ferguson Police Department in Missouri shot and killed Michael Brown, an unarmed 18-year-old, at about noon on August 9, 2014. Justice Department officials announced on March 4, 2015, that the evidence did not support charging Wilson with a federal crime. Their report states that Wilson did not willfully, knowingly and on purpose, violate Brown's right to be free from unreasonable force.

The Ferguson report. The same day, lawyers in the Justice Department's Civil Rights Division published a report on the whole Ferguson police department. Its investigators had interviewed half the department's officers and read more than 35,000 pages of police records, and spent about 100 days of work in the town.

They found "a pattern or practice of unlawful conduct within the Ferguson Police Department that violates the First, Fourth, and Fourteenth Amendments to the United States Constitution." Ferguson's police broke the Constitution again and again, as a regular way of working.

Who the police stopped. The police department's own numbers for 2012 to 2014 showed who the police stopped. Black people were 67 percent of Ferguson's population. They were 85 percent of the drivers stopped, 90 percent of the people given tickets and 93 percent of the people arrested. Police searched Black drivers more than twice as often as white drivers, and found something illegal on them 26 percent less often.

The investigators found that "nearly 90% of documented force used by FPD officers was used against African Americans." In every dog-bite case where the race was recorded, the person bitten was Black. The report states that "Ferguson's law enforcement practices are shaped by the City's focus on revenue rather than by public safety needs." The city's finance director wrote to the police chief in March 2010 that "unless ticket writing ramps up significantly before the end of the year, it will be hard to significantly raise collections next year."

Recording the police. The investigators also found that officers punished people for recording them and for protesting. One officer saw a couple filming him. He arrested the father, then pulled the mother over and arrested her too, saying "no more mercy, since she wanted to videotape," and "nobody videotapes me." He took the phone from their daughter. When the parents got out of jail on bond, money paid so a person can wait for trial at home, the video had been deleted.

A federal judge ordered Ferguson officers in November 2014 to stop interfering with people "lawfully and peacefully" recording the police. On February 9, 2015, six months after Michael Brown's death, people protested outside the police station. Officers pushed a man who was recording the arrests to the ground and handcuffed him. The report states that his video shows he "was neither interfering nor resisting." Officers also arrested a protester in a wheelchair who was streaming the protest live. The report names none of these people.

Colin Kaepernick. Some athletes joined the protests against police killings. Colin Kaepernick, the quarterback of the San Francisco 49ers football team, sat and then knelt during the national anthem in the 2016 preseason, "in protest of police brutality and racial inequality." He chose to leave his contract in March 2017, and no team signed him. He accused the owners of the National Football League in October 2017 of secretly agreeing to keep him out. He settled with the league on February 15, 2019, and the terms were kept secret.

Minneapolis, May 25, 2020

A Minneapolis police officer, Derek Chauvin, knelt on George Floyd for about ten minutes on May 25, 2020, while Floyd lay handcuffed and face down in the street. Floyd died, and a jury later found Chauvin guilty of murdering him. Chauvin later admitted these facts in writing, in a plea agreement he signed in federal court on December 15, 2021, a statement in which a person accused of a crime admits to it under a deal with prosecutors.

What the officers did. Chauvin and his partner, Officer Tou Thao, answered a call about a counterfeit bill, a fake banknote. Officers J. Alexander Kueng and Thomas Lane were already there. They tried to put Floyd in a squad car. Then Chauvin, Kueng and Lane moved Floyd, who was handcuffed, out of the car and face down on the street.

He "remained restrained, prone and handcuffed on the ground for approximately ten minutes," prone meaning lying face down. Chauvin "held his left knee on Mr. Floyd's neck, back, and shoulder area and his right knee on Mr. Floyd's left arm and upper back" the whole time.

Floyd stopped resisting soon after the officers first held him down, and Chauvin went on kneeling on him. He admitted that he knew he had no legal reason to continue, "because he was aware that Mr. Floyd not only stopped resisting, but also stopped talking, stopped moving, stopped breathing, and lost consciousness and a pulse." By his own account he "heard Mr. Floyd repeatedly explain that he could not breathe, was in pain, and wanted help."

Bystanders asked him again and again to check for a pulse and to get off Floyd. Kueng checked for a pulse at least twice and said twice that he could not find one. Chauvin admitted he did not see Thao or Kueng do or say anything to get him off Floyd. Lane twice asked whether Floyd should be rolled onto his side.

Darnella Frazier. Darnella Frazier was seventeen. She was walking with her nine-year-old cousin, and she recorded what the officers did on her phone. The Pulitzer Prize Board gave her a special citation in 2021. News reports of the announcement quote it: "for courageously recording the murder of George Floyd, a video that spurred protests against police brutality around the world." She testified at Chauvin's state trial.

The verdicts. A Hennepin County jury found Chauvin guilty on April 20, 2021, of two kinds of murder, second-degree and third-degree, and of second-degree manslaughter. Judge Peter Cahill sentenced him to 22 and a half years. In federal court Judge Paul Magnuson sentenced him to 21 years, to be served at the same time. Thao, Kueng and Lane got federal sentences of between two and a half and three and a half years.

The boy in 2017. In the same plea agreement, Chauvin admitted to an attack four years earlier. He answered a call on September 4, 2017, from a mother who said her 14-year-old son had hit her. Chauvin held the boy by the throat and hit him in the head with a flashlight, cutting him near the left ear.

Then he knelt on the boy's neck, shoulders and upper back "for between fifteen and sixteen minutes" while the boy lay face down, handcuffed and not resisting, and "at times, cried from pain." Chauvin left all of it out of his report that night. The plea agreement does not name the boy.

The protests. People protested in cities and towns across the country that summer. Researchers at the Pew Research Center, an organization that runs surveys, asked 9,654 American adults about the protests between June 4 and 10, 2020. Six of every 100 said they had gone to a rally or protest on these issues in the past month. Two-thirds said they supported the Black Lives Matter movement, and 38 of every 100 said they strongly supported it.

The Crowd Counting Consortium, a project run by researchers at Harvard and the University of Connecticut, counts protests. Its data supported a *New York Times* analysis that the 2020 protests were the largest and broadest in American history up to then. The same data showed the protests were "overwhelmingly peaceful, especially relative to the heavy-handed police response," in the words of Harvard's Ash Center.

What Congress and the Justice Department did. Members of the House of Representatives passed the George Floyd Justice in Policing Act on March 3, 2021, by 220 votes to 212, and senators never voted on it. Under President Joe Biden, Justice Department lawyers sued Minneapolis and Louisville to force changes in how their police worked, through consent decrees, agreements that a judge enforces.

On May 21, 2025, under President Donald Trump, Harmeet Dhillon, the head of the Justice Department's Civil Rights Division, dropped both lawsuits. She said she was ending "the Biden Civil Rights Division's failed experiment of handcuffing local leaders and police departments with factually unjustified consent decrees."

BY 2021: Protests after police killings had filled streets across the country, and a jury had found Derek Chauvin guilty of murdering George Floyd.

The Emmett Till Antilynching Act, 2022

The Emmett Till Antilynching Act became law on March 29, 2022, the first federal law against lynching. Members of the House passed it on February 28, 2022, and senators passed it on March 7.

About 200 bills. Members of Congress had introduced about 200 anti-lynching bills before it, and none had become law. The House Judiciary Committee's own 2022 report on the bill gives two different starting dates.

One section says the first bill came from Congressman George Henry White in 1900, when he was the only Black member of Congress. Another says the first came from Congressman Leonidas Dyer of Missouri in 1918. Both sections agree that the House passed three anti-lynching bills between 1920 and 1940, and that none of them became law.

What the law covers. The law is named for Emmett Till, whom the report describes as "a 14-year-old African American youth from Chicago who was lynched in 1955 while visiting an uncle in Mississippi." Under federal law, a hate crime is hurting a person on purpose because of their race, color, religion or national origin, and in some cases their gender, sexual orientation, gender identity or disability.

The new law covers a person who conspires, or plans with others, to commit a hate crime. When the crime kills someone or badly injures them, that person can be sent to prison for up to 30 years.

Me too, 2017 to 2026

The actress Alyssa Milano wrote on Twitter on October 15, 2017, "If you've been sexually harassed or assaulted write 'me too' as a reply to this tweet," harassment meaning unwanted sexual words or touching and assault meaning a sexual attack.

The count. Researchers at the Pew Research Center later counted public English-language posts on Twitter. They found the hashtag #MeToo used more than 19 million times between that day and September 30, 2018, an average of 55,319 times a day. The researchers counted posts on one website in one language, so the figure is a count of posts and not of people.

Harvey Weinstein. Milano posted the words during the reporting about the film producer Harvey Weinstein. *The New York Times* and *The New Yorker* reported in October 2017 that more than a dozen women accused him of sexual harassment, assault or rape. He denied "any nonconsensual sex," sex the other person did not agree to. He was dismissed from his company that month.

A New York jury found him guilty of rape and of a criminal sexual act in 2020, and he was sentenced to 23 years. Under New York law, a criminal sexual act means forcing a person into sex acts such as oral sex. Judges of the New York Court of Appeals, the state's highest court, threw out those verdicts in 2024 because of "egregious errors," very serious mistakes, by the trial judge. A Los Angeles jury found him guilty of rape and two other charges in 2022.

At a new trial in New York, on June 11, 2025, a jury found him guilty of forcing oral sex on Miriam Haley, a former production assistant, in his Manhattan apartment in 2006. Justice Curtis Farber sentenced him to 15 years in prison on September 23, 2026. "You took what you wanted by force, and you have never accepted responsibility," Farber told him.

The words "me too" were already in use. A youth worker in Selma, Alabama, named Tarana Burke had been using them with girls and young women for years before 2017.

Tarana Burke

Who: A youth organizer who used the words "me too" with survivors of sexual violence for years before the words became a hashtag.

When and where: The Bronx, New York, and Selma, Alabama, 1973 to 2018.

She was saying "me too" to girls years before the hashtag.

Tarana Burke is herself a survivor of sexual abuse. She grew up in the Bronx, in New York City, and at fourteen she joined a group called the 21st Century Youth Leadership Movement. After Alabama State University and Auburn University, she moved to Selma, Alabama, and kept working for the same group there.

The girl at the camp. A National Women's History Museum biography dates the start of her work to 1996, when a girl at a youth camp told Burke that she had been sexually abused. The same biography states that years later Burke "began using the phrase 'me too' to promote the idea of 'empowerment through empathy.'" Burke's own account says her work turned toward survivors of sexual violence while she lived in Selma.

Just Be. She started an organization for Black girls called Just Be.

DISPUTED: When Just Be and her use of the name "Me Too" began. Her own website says she created Just Be in 2005, and the museum's biography gives 2007. A Colorado State University page states that she made a "Me Too" page on MySpace, a social network of that time, in 2006.

2017. The words spread across Twitter in October 2017, during the Harvey Weinstein reporting, and Burke became a leader of the movement around the world. *Time* magazine named "the Silence Breakers," Burke among them, its Person of the Year for 2017. In 2018 she started an organization called "me too." International.

Equal pay for the women's soccer team, 2016 to 2022

Five players on the United States women's national soccer team, Hope Solo, Carli Lloyd, Alex Morgan, Megan Rapinoe and Becky Sauerbrunn, filed a complaint in April 2016 that they were paid less because they were women. They filed it with the Equal Employment Opportunity Commission, the federal office that enforces the law against job discrimination.

The players and US Soccer's leaders settled the case on February 22, 2022, for $24 million, and $22 million of it went to the players. Under new contracts signed on May 18, 2022, the women's and men's teams get equal pay and bonuses through 2028.

The Women's March, 2017, and No Kings, 2025

Women marched in cities across the country on January 21, 2017, the day after Donald Trump became president. Teresa Shook, a grandmother in Hawaii, had proposed a march on Washington in a Facebook post the day after the November 2016 election. Between 3.2 million and 5.3 million people marched in the United States, about 500,000 of them in Washington, by the count in Encyclopaedia Britannica.

Erica Chenoweth of Harvard and Jeremy Pressman of the University of Connecticut founded the Crowd Counting Consortium to count this march. Their data showed it was the largest one-day demonstration in American history up to that time. Britannica records that the "No Kings" protests of June 14, 2025, in which about five million people marched, passed that record.

The end of Roe v. Wade, 2022

The justices of the Supreme Court ended the right to an abortion in 2022, when they overruled an earlier decision of their own court. An abortion is a medical procedure that ends a pregnancy. In *Roe v. Wade*, in 1973, the justices had decided that the Constitution protected that right, and the ruling stood for 49 years.

Dobbs. Mississippi's lawmakers banned abortion after fifteen weeks of pregnancy, except in a medical emergency or when the fetus, the medical word for the unborn offspring growing in the womb after the first weeks of pregnancy, had a serious defect.

Justice Samuel Alito wrote for five justices on June 24, 2022, in *Dobbs v. Jackson Women's Health Organization*: "We therefore hold that the Constitution does not confer a right to abortion. Roe and Casey must be overruled, and the authority to regulate abortion must be returned to the people and their elected representatives."

Casey was a 1992 ruling, *Planned Parenthood v. Casey*, that had kept *Roe* in place. Chief Justice John Roberts agreed that the Mississippi law could stand but did not join in overruling *Roe*. Justices Stephen Breyer, Sonia Sotomayor and Elena Kagan dissented together.

The states. After *Dobbs*, lawmakers in each state set their own rules. KFF, a health research group, counted them on August 10, 2026. Thirteen states banned abortion in almost every case. Seven more banned it after six to twelve weeks of pregnancy. Nine states and the District of Columbia set no limit by stage of pregnancy.

The Equal Rights Amendment, 2017 to 2026

Lawmakers in three more states ratified the Equal Rights Amendment after 2000, and in September 2026 it was still not part of the Constitution. The amendment reads: "equality of rights under the law shall not be denied or abridged by the United States or any State on account of sex." When the 1982 deadline ran out, 35 states had ratified it, and 38 were needed.

Three more states. Nevada's lawmakers ratified it in 2017, treating the deadline as one they could ignore. Illinois followed in 2018 and Virginia in 2020. The three states asked the Archivist of the United States, the federal official who certifies and publishes amendments, to add it, and the Archivist did not. Officials of the three states sued in January 2020. A federal judge threw out the suit, and the judges of a federal appeals court agreed in February 2023.

The deadline. Lawyers in the Justice Department's Office of Legal Counsel, which writes legal opinions for the department, wrote in 2020 and 2022 that the deadline "is valid and enforceable." In December 2024 the Archivist, Colleen Shogan, and her deputy, William J. Bosanko, wrote that the amendment "cannot be certified as part of the Constitution due to established legal, judicial, and procedural decisions." Earlier court rulings and rules, they meant, stopped them from declaring it part of the Constitution.

President Joe Biden said on January 17, 2025, three days before he left office: "I affirm what I believe and what three-fourths of the states have ratified: the 28th Amendment is the law of the land." His statement ordered no one to do anything, and Shogan did not publish the amendment.

Hair, 2018 to 2026

Lawmakers in about 30 states passed laws between 2019 and 2026 against punishing people at work or at school for their natural hair, or for styles such as braids, locs and twists. The count of about 30 states comes from the CROWN Coalition, the group of organizations that campaigned for the laws. Dove, a soap and beauty brand, founded it with Color of Change, the National Urban League and the Western Center on Law and Poverty.

Andrew Johnson. The referee Alan Maloney told Andrew Johnson, a junior wrestler at Buena Regional High School, to cut his dreadlocks or forfeit, lose the match without wrestling, at a home match in Buena, New Jersey, on December 19, 2018. Maloney gave him 90 seconds. The team's athletic trainer cut Johnson's hair beside the mat. The New Jersey Division on Civil Rights and the state's school sports association agreed on September 18, 2019, that Maloney would be suspended for two seasons.

The CROWN Act. California's CROWN Act, for Creating a Respectful and Open World for Natural Hair, was written by State Senator Holly J. Mitchell. California's governor signed it on July 3, 2019, the first state law of its kind. It added hair texture and styles such as braids, locs and twists to the state's law against discrimination at work and in schools.

Governor Phil Murphy signed New Jersey's version on December 19, 2019, a year to the day after Johnson's match. Members of the House passed a federal CROWN Act on September 21, 2020, and again on March 18, 2022. Senators never voted on it.

BY 2022: Members of Congress had passed the first federal law against lynching, and the justices had ended the right to an abortion that the Court had protected since 1973.

Disability: narrowed by judges, restored by Congress, dropped in 2026

Disabled Americans fought after 1990 over who the Americans with Disabilities Act covered and whether they could live outside institutions. On a single day in 1999 the justices of the Supreme Court narrowed who counted as disabled and ruled that keeping disabled people in institutions without need was discrimination. Members of Congress restored the wider meaning of disability in 2008. A federal judge struck down the federal rule that carried out the 1999 ruling on institutions in 2026.

Who counted as disabled. The first fight was over who counted as disabled. Twin sisters applied to fly for United Air Lines. Without glasses their eyesight was 20/200 or worse, and with glasses it was 20/20 or better. Perfect ordinary eyesight is called 20/20, and a bigger second number means worse sight. United's hiring rule required 20/100 or better without glasses and turned the sisters down.

The justices ruled on June 22, 1999, in *Sutton v. United Air Lines*, that whether a person is disabled must be judged with glasses or other aids in use. So the airline could refuse the sisters because of their eyes, and they could not sue, because with their glasses on they did not count as disabled.

A second ruling in 2002 narrowed the definition again. Members of Congress later wrote that lower courts had then "incorrectly found in individual cases that people with a range of substantially limiting impairments are not people with disabilities."

The 2008 law. President George W. Bush signed a law to reverse those rulings on September 25, 2008. Its title reads: "To restore the intent and protections of the Americans with Disabilities Act of 1990." In its text, members of Congress named the rulings and said they had "narrowed the broad scope of protection intended to be afforded by the ADA." An illness that comes and goes now counts as a disability.

Whether a person is disabled is decided without counting the help of aids. The aids include medicine, artificial limbs, hearing aids, cochlear implants (devices placed in the ear that help deaf people hear), wheelchairs and other aids for

getting around, or oxygen equipment. Ordinary glasses and contact lenses are the one kind of aid the law leaves off that list.

Where disabled people live. The second fight was over where disabled people live. Lois Curtis and Elaine Wilson were two Georgia women with mental illness and developmental disabilities. They agreed to go into the psychiatric unit, the ward for mental illness, of Georgia Regional Hospital, a state hospital. Mental health workers said each of them was ready to move to a program in the community, and yet each of them remained locked in the state hospital for several years more.

They sued under the ADA. The justices ruled on June 22, 1999, in *Olmstead v. L.C.*, that keeping a disabled person in an institution when the person could live in the community is discrimination. They wrote that "confinement in an institution severely diminishes the everyday life activities of individuals, including family relations, social contacts, work options, economic independence, educational advancement, and cultural enrichment."

Colorado. Justice Department lawyers went on enforcing that ruling. They wrote to Governor Jared Polis on March 3, 2022, that Colorado was breaking the ADA by keeping people with physical disabilities in nursing homes "unnecessarily." More than 15,000 Coloradans received care in the state's 200-plus nursing facilities in a year, and "some individuals have remained in nursing facilities for many years with no movement toward discharge."

The rule struck down. Justice Department lawyers changed sides in 2026. Officials of seventeen states, Texas first among them, sued over the federal rules for Section 504 in September 2024. By 2026 five states were left in the case: Texas, Alaska, Louisiana, Montana and Florida. They asked the judge to strike the rule that disabled people get services in "the most integrated setting," at home or in their community rather than in an institution.

The Justice Department's Office of Legal Counsel wrote on June 18, 2026, that federal law "does not include an integration mandate." In the lawyers' reading, the law does not require officials to offer disabled people services at home or in their community instead of in an institution. Justice Department lawyers

joined the five states on August 31, 2026, in asking the judge to strike the rule. The judge struck it for the whole country on September 23, 2026. The Supreme Court's 1999 ruling and the laws Congress passed still stand.

Today. Census Bureau officials reported on July 22, 2026, that 45.8 million Americans, 13.7 percent, had a disability in 2024, about one person in seven. That figure counts the "civilian noninstitutionalized population," everyone except people who live in institutions. The people the *Olmstead* fight is about are left out of it.

Alice Wong

Who: A disabled organizer who collected disabled people's own accounts of their lives, and who served on the federal council that advises the President on disability.
When and where: San Francisco, California, 2002 to 2025.

She went to the White House by robot.

Alice Wong founded the Disability Visibility Project in 2014 to collect disabled people's accounts of their own lives, in their own words. She used a wheelchair and breathed through a tracheostomy, a hole cut through the front of the neck into the windpipe, with a short tube held in it.

Her tube was connected to a ventilator, a machine that pushed air through the tube into her lungs so that she kept breathing. Neither the University of California, San Francisco, where she studied and worked, nor the National Council on Disability names the condition behind it.

At the university. As a student there she founded the university's Disability Interest Group in 2002. She earned a master's degree in medical sociology, the study of health and illness in society, there in 2004. She worked at three of its research centers: the National Center for Personal Assistance, the Disability Statistics Center and the Community Living Policy Center.

The robot. President Barack Obama appointed her to the National Council on Disability, a federal body whose members advise the President and Congress, on January 25, 2013, and she served until 2015. The White House held a reception in the Blue Room on July 20, 2015, for the 25th anniversary of the Americans with Disabilities Act. Wong could not travel to it, so she took part through a robot, a screen on wheels that carried her face and voice into the room.

The White House photographer Pete Souza took a picture of it. Its caption reads: "President Barack Obama greets Alice Wong, Disability Visibility Project Founder and Project Coordinator via robot."

She edited *Disability Visibility: First-Person Stories from the Twenty-First Century* (2020), a collection of essays by disabled writers, and wrote her own life story, *Year of the Tiger: An Activist's Life* (2022). "Disability is part of the greater continuum of diversity, and we need to allow for variation and not get so focused on what is the norm," she said. She died on November 14, 2025, aged 51.

Transgender Americans and the law, 2020 to 2026

Transgender Americans won one Supreme Court case over jobs between 2020 and 2026 and lost cases over medical care and school sports. The justices also let a military ban and a passport rule take effect while the lawsuits over them went on.

Bostock. The justices ruled six to three on June 15, 2020, in *Bostock v. Clayton County*, that "an employer who fires an individual merely for being gay or transgender violates Title VII." Title VII is the part of the Civil Rights Act of 1964 that bans job discrimination because of sex. Justice Neil Gorsuch wrote the opinion. Three fired workers had brought the cases.

Gerald Bostock had worked for Clayton County, Georgia, for ten years and was fired soon after he joined a gay softball league. Donald Zarda, a skydiving instructor in New York, was fired days after he told his employer he was gay. Aimee Stephens worked at a funeral home in Michigan. She had been hired presenting as a man, and six years later she told her employer she meant to "live and work full-time as a woman." Her employer fired her, saying the arrangement "is not going to work out."

Skrmetti. Tennessee's lawmakers passed a law that stops doctors from giving puberty blockers or hormones to anyone under 18 for two purposes. One is to help the young person live as a gender other than their sex at birth. The other is to treat distress about the mismatch between the two.

A puberty blocker is a drug that pauses the hormones that start puberty, so the body's changes do not begin while the drug is taken. Hormone treatment gives a person the hormone of the sex they identify as, which changes the body in that direction.

Doctors in Tennessee may still give the same drugs to young people for other reasons, such as puberty that starts very early. The justices ruled that the law could stand on June 18, 2025, in *United States v. Skrmetti*. Chief Justice John Roberts wrote the Court's opinion. Justice Sonia Sotomayor dissented, joined by Justice Ketanji Brown Jackson and in part by Justice Elena Kagan, and Kagan also wrote her own dissent.

School sports. Idaho's lawmakers passed the Fairness in Women's Sports Act in 2020, and West Virginia's passed the Save Women's Sports Act in 2021. Both laws say that girls' and women's teams are for students the laws define as female by biology.

Justice Brett Kavanaugh wrote for six justices on June 30, 2026, that Title IX "allows schools to provide separate women's and men's sports teams defined by biological sex." The two laws did not break the promise of equal protection, he wrote. The majority gave "the inherent physical differences between the sexes" as a reason, along with the safety and fairness of the competition. It also wrote that "27 States and various sports-governing bodies have all drawn the same line."

B.P.J. The West Virginia student is named in the record only as B.P.J., and the case was brought by her mother, Heather Jackson. Justice Sotomayor described her in dissent as "a transgender girl who wants to live her life consistent with her gender identity." When B.P.J. was 11, Sotomayor wrote, "she sought the opportunity to do what she and so many other children love to do: play sports." She wanted to run on the cross-country and track teams.

Sotomayor wrote that "B. P. J. is the only transgender girl publicly identified in the State who has sought to play sports with other girls" in the five years the ban had been in place. She agreed that the Title IX claim failed. On the equal protection question, she wrote that the trial court should have been allowed to find the facts first.

Executive orders. President Donald Trump acted on his first day back in office. He signed an executive order on January 20, 2025, titled "Defending Women From Gender Ideology Extremism and Restoring Biological Truth to the Federal Government." On January 27 he signed an order on the military, and on February 5, 2025, he signed one titled "Keeping Men Out of Women's Sports." National Park Service officials took the words "transgender" and "queer" off the Stonewall National Monument's website in February 2025.

The military ban. Defense Department officials issued a military ban on February 26, 2025. It disqualifies "anyone who either has gender dysphoria or has undergone medical interventions to treat gender dysphoria." Gender

dysphoria is the medical term for the distress a person feels when the sex recorded at their birth does not match their own sense of their gender.

Commander Emily Shilling, a Navy pilot for nearly twenty years, led a lawsuit by transgender service members, and District Judge Benjamin Settle blocked the ban. The justices let the ban take effect while the case went on, on May 6, 2025. Justices Sotomayor, Kagan and Jackson would have kept it blocked.

Passports. The justices let the government's new passport rule take effect on November 6, 2025. Every new passport now shows the holder's sex at birth. "Displaying passport holders' sex at birth no more offends equal protection principles than displaying their country of birth," the order reads. Justice Jackson dissented, joined by Sotomayor and Kagan.

Chiles v. Salazar. The justices ruled for Kaley Chiles, a licensed counselor in Colorado who uses only talk therapy, on March 31, 2026. A 2019 Colorado law barred licensed counselors from "conversion therapy" with young people, meaning any practice that tries to change a person's sexual orientation or gender identity. Justice Gorsuch wrote for eight justices that the law, as used against her talk therapy, regulated her speech based on her point of view. The case went back to the lower courts. Justice Jackson dissented alone.

BY SEPTEMBER 2026: The justices had let states ban some medical care for transgender young people and keep transgender girls off girls' teams, and a federal judge had struck down the rule that disabled people get services at home or in their community.

Standing Rock: the right to protest, 2016 to 2026

Members of the Standing Rock Sioux Tribe and their supporters protested the Dakota Access oil pipeline in 2016, and police officers answered with tear gas, rubber bullets and a fire hose. The route chosen by Energy Transfer, the company building the pipeline, ran under Lake Oahe, less than a mile north of the Standing Rock Reservation. In treaties signed in the 1800s, officials of the United States agreed that the land the route crosses belonged to the Sioux Nation, and the Sioux never gave that land up.

The police command. North Dakota's governor, Jack Dalrymple, declared an emergency on August 19, 2016, when about 1,500 people were at the protest and 28 had been arrested. He organized a joint police command. Three police forces took part: the Morton County Sheriff's Office under Sheriff Kyle Kirchmeier, the Stutsman County Sheriff's Office under Sheriff Chad Kaiser, and the Mandan police under Chief Jason Ziegler. Each chief left decisions about using force to his individual officers.

The bridge. Officers had blocked the Highway 1806 bridge with two dump trucks chained to concrete barriers and barbed wire. On the night of November 20, 2016, protesters used a semi-trailer truck to drag one dump truck away, and the crowd grew to several hundred. The officers fired tear gas canisters, rubber bullets and bags filled with lead into the crowd. They sprayed protesters with a fire hose from a Mandan fire truck. A lawsuit for the protesters says more than 200 people were hurt.

Sophia Wilansky. Sophia Wilansky was hit that night. In 2026 the judges of a federal appeals court set out her account, which they accepted as true for the appeal. Officers Dvorak, Arndt and Hanson fired rounds at her as she ducked behind a metal sheet near the remaining truck, and one hit her arm.

She ran. About 30 yards away, as she bent to pick up a piece of plywood, Officer Jonathan Moll fired a round called an aerial warning munition. "It tore the flesh off her arm, and she fell to the ground," the judges wrote.

A footnote in their ruling describes that round as "a flashbang embedded in a less-lethal bullet," a bullet made to hurt rather than kill. A flash-bang is an explosive device that makes a loud noise and a bright flash of light, and aerial ones are made to be fired above a crowd. Protesters carried her away. Her account says officers laughed and congratulated Moll, and the officers dispute that. "Her left hand and forearm are permanently and severely injured," the judges wrote.

No one held responsible. No officer or police agency was held responsible for the injuries that night. Judges of the federal appeals court ruled against the injured protesters in 2023, and again on June 24, 2026, in Wilansky's case. They said that in 2016 the law had not yet made it clear that firing at a crowd to drive it off was a "seizure" under the Constitution, a way of stopping or holding a person by force.

That mattered because of a rule called qualified immunity. It shields government officials from being sued unless the right they broke was already clearly established, meaning already made clear in the law.

Police made hundreds of arrests at the protests. A Morton County judge, Allan Schmalenberger, threw out many trespass cases, charges of going onto private land without permission, because no one had told the protesters to leave.

The pipeline. Federal officials changed their decision more than once. The Army's Assistant Secretary for Civil Works refused permission to cross under the lake on December 4, 2016. President Donald Trump ordered the Army on January 24, 2017, to approve the pipeline "in an expedited manner," quickly. Army officials gave permission on February 8, 2017, and oil has flowed since June 2017.

A federal judge, James Boasberg, ruled in 2020 that Army officials had broken the law by approving the crossing without a full study of its effects. The pipeline kept running for about five years while the Army Corps of Engineers did the study. Corps officials finished it in December 2025, and in May 2026 Brigadier General William C. Hannan signed a decision to grant permission again.

Tribal leaders asked the judges of a federal appeals court to drop the tribe's own appeal in June 2026, and the judges did. "We will continue to fight for our homelands and the promises the United States made to our Tribe in Treaty," Standing Rock Chairman Steve Sitting Bear said in December 2025.

Greenpeace. Energy Transfer's lawyers sued the environmental group Greenpeace over the protests. A jury in Morton County ordered Greenpeace to pay about $667 million on March 19, 2025, and Judge James Gion cut the amount to $345 million. Greenpeace officials said the group had six employees at the camps and called the lawsuit an attempt to punish the group for supporting the protests. "The oil is still flowing. The pipeline hasn't been shut down," Nola Taken Alive, a member of the Standing Rock tribal council, said in September 2026.

The DREAM Act and DACA, 2001 to 2026, and two killings in Minneapolis

Young people who had been brought into the United States as children, without legal permission to live there, organized from 2010 for a way to become citizens, and members of Congress never passed one. Senator Richard Durbin of Illinois and Representative Howard Berman of California first introduced the DREAM Act in 2001. It would have let these young people become citizens after two years of college or military service.

The walk. Four students from Miami Dade College, Gaby Pacheco, Carlos Roa, Felipe Matos and Juan Rodriguez, set out on January 1, 2010, to walk 1,500 miles from Miami to Washington. Three of them were in the country without legal permission. They said so in public all along the way, which put them at risk of arrest and deportation. They gathered 30,000 signatures.

A Secret Service agent refused to take their petition on April 28, 2010, and White House officials refused them a meeting with President Barack Obama. They walked the last four miles to the White House on May 1, 2010. The DREAM Act came up in the Senate on December 18, 2010. Fifty-five senators voted to end debate and 41 voted against. Sixty votes were needed, so the bill failed.

DACA. Janet Napolitano, the Secretary of Homeland Security, announced a program called Deferred Action for Childhood Arrivals, or DACA, on June 15, 2012. It made no one a citizen. People who had arrived as children could apply if they met rules about age, schooling and criminal records. Those accepted could ask the government to put off deporting them for two years at a time, and could get permission to work. About 700,000 people signed up, by the Supreme Court's count.

Attorney General Jeff Sessions wrote in September 2017 that the program should end, and the acting Secretary of Homeland Security, Elaine Duke, ended it. DACA holders, universities and states sued. The justices of the Supreme Court ruled five to four on June 18, 2020, that federal officials had ended the program unlawfully, so DACA stayed.

Texas. Officials of Texas and other states also sued. A federal appeals court ruled in January 2025 that DACA's work permits were unlawful in Texas. By July 2026 the judge in Texas, Andrew Hanen, had not issued the order that would carry that out.

The numbers. Agents of ICE, a federal immigration agency, arrested 261 DACA holders and deported 86 of them in the first year of President Donald Trump's second term, by the count of FWD.us, an immigration advocacy group. About 455,000 people held DACA on March 31, 2026, by government figures reported by the National Immigration Forum, another immigration advocacy group. Federal immigration officials were renewing permits and not approving new ones.

Minneapolis, 2026. Federal officials ran a large immigration operation in Minneapolis in early 2026, called Operation Metro Surge. During it, federal immigration agents shot and killed two American citizens, Renée Macklin Good on January 7 and Alex Pretti on January 24. The *Star Tribune* named ICE officer Jonathan Ross as the man who shot Good. ProPublica named Border Patrol agent Jesus Ochoa and Customs and Border Protection officer Raymundo Gutierrez as the agents who shot Pretti.

Federal officials said Good had used her car as a weapon, and local officials said video contradicted that. Federal officials called Pretti a domestic terrorist, and NPR reported that video contradicted their account of his death.

At the Pretti shooting, federal officials physically blocked state investigators from the scene. Hennepin County Attorney Mary Moriarty and Minnesota Attorney General Keith Ellison sued for the evidence in March 2026 and received it in June and July. Acting Attorney General Todd Blanche said there would be no civil rights investigation into Good's killing. As of September 2026 no one had been charged in either death.

THE THREE QUESTIONS, ANSWERED

1. **How did people without power get the people in power to listen?** They signed petitions, printed pamphlets and spoke to crowds, and later they planned test cases, boycotted, marched, sat in and filmed the police, from the Germantown protest of 1688 to Darnella Frazier's video in 2020.
2. **What did organizing cost the people who did it?** Their jobs, their homes and their freedom, and sometimes their bodies and their lives. A prison doctor fed Lucy Burns by force through a tube, state highway patrolmen had Fannie Lou Hamer beaten in a Mississippi jail, Byron De La Beckwith shot Medgar Evers in his driveway, and Klansmen killed four girls with a bomb in a Birmingham church.
3. **Can a right that has been won be taken away again?** Yes. New Jersey's lawmakers took the vote from women and free Black men in 1807, members of Congress took Utah women's vote in 1887, and the justices of the Supreme Court cut back the Voting Rights Act in 2013 and 2026 and ended the right to an abortion in 2022.

What changed: In 1688 four men's protest against slavery went nowhere, and by 2026 the laws that once shut women, Black Americans, gay Americans and disabled Americans out of voting, schools, marriage

and public buildings had been struck down or rewritten, while the fights over several of them went on.

FIVE DATES TO KEEP

April 18, 1688: Four men at Germantown sign a protest against slavery.

July 19 and 20, 1848: The Seneca Falls convention votes to demand the vote for women.

August 26, 1920: The 19th Amendment becomes part of the Constitution.

May 17, 1954: The justices rule in *Brown v. Board of Education*.

August 6, 1965: President Lyndon Johnson signs the Voting Rights Act.

NOT KNOWN: How many people white Tulsans killed in Greenwood in 1921, because the attackers hid the bodies, and credible estimates run from fifty to three hundred.

Where to go: Haverford College, in Pennsylvania, keeps the 1688 Germantown protest, the paper this chapter began with.

Next: Chapter 26, Health, Disease, and Medicine. It follows illness and healing in America: epidemics, doctors, hospitals and the fight over public health.

PART 7

Health and Safety

Chapter 26: Health, Disease, and Medicine

CHAPTER 26 OF 37 · HEALTH, DISEASE, AND MEDICINE · 10 ERAS · ABOUT 115 MINUTES

For most of American history nobody knew what caused disease, and after doctors learned how to stop it, the people with the least power were often the last to be protected and the first to be tested on.

Cotton Mather, a Boston minister, wrote to scientists in London in 1716 about a scar on the arm of Onesimus, an African man Mather enslaved. Onesimus said it came from an operation that gave him a small case of smallpox, a disease of fever and blisters that often killed. The small case, he said, would keep him from ever catching it again. When Boston doctors tried the method in 1721, about 2 in 100 of the people they treated died, against 14 or 15 in 100 of those who caught smallpox the ordinary way.

This chapter follows sickness and healing through ten eras, from what ancient bones show to COVID-19. In the first eras, look for the new diseases that came after 1492 and emptied whole Native towns. Then for a long stretch doctors guessed at what caused disease and bled and purged their patients, until scientists showed in the late 1800s and in 1900 that germs and mosquitoes spread it. Through all of it, notice who paid for the new knowledge: enslaved women operated on with nothing to stop the pain, Black men in Alabama kept from treatment for 40 years, and children fed a virus on purpose.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Why did new diseases kill so many Native people after 1492?
2. How did Americans learn to stop smallpox, yellow fever and polio?

3. Who was used in medical experiments without agreeing, and what changed because of it?

Five minutes? Read "The Tuskegee study, 1932 to 1972" in 1900 to 1950, then "The Tuskegee study ends, 1972 to 1997" in 1950 to 2000.

Can skip: "Midwives and surgeons in New Amsterdam", "Medical care for sailors, 1798", "New operations and new pictures of the body, 1954 to 1963". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

Almost everything known about sickness in this land before 1500 comes from bones and teeth, because nobody here wrote about illness then. They show that people had diseases, but not the ones that arrived after 1492 and killed great numbers of them. Smallpox, measles, influenza, yellow fever and cholera had never reached the Americas. Neither had mumps, chickenpox, malaria, diphtheria, typhus, leprosy or bubonic plague. Every one of these twelve diseases is an infection caused by a germ, a living thing too small to see.

What do the bones show?

Some diseases leave marks on bone, so archaeologists, scientists who learn about the past from what people left behind, can find them in skeletons hundreds of years old. Two researchers, Debra Martin and Alan Goodman, wrote in 2002 that Native people before 1492 "were most definitely not living in a pristine, disease-free environment," pristine meaning untouched and pure.

Tuberculosis was here. It is caused by bacteria, one kind of germ, and it most often infects the lungs, though it can spread to the bones, including the spine. Skeletons with its damage appear after about the year 900. Most of them come from inland places in the middle of the continent and in the Southwest.

Bones also show diseases in the same family as syphilis, a disease that is passed during sex. Two of them, yaws and bejel, spread mainly by touching skin and mouths, not by sex. Bejel is also called non-venereal syphilis, which means syphilis that is not passed by sex.

A team of scientists studied the DNA of these bacteria, taken from five people buried in the Americas long ago. DNA is the set of chemical instructions inside every living cell. In 2025 the team reported that the bacteria behind syphilis and its whole family most likely began in the Americas. Their results fit that idea for every sample of these bacteria that scientists have studied, ancient or modern.

Martin and Goodman list other illnesses that were present: rabies, whooping cough, polio, hepatitis (an infection of the liver), herpes (a virus that causes sores), tularemia (a fever people catch from animals), and stomach infections from tiny living things called giardia and amoebas. They add that almost all of these were probably rare in any one group of people.

Farming and weak blood. People who took up farming became sicker in some ways. At Dickson Mounds in Illinois, people moved over many years from hunting and gathering to farming. Researchers studied the skeletons of

people buried there, among them farmers who lived in the 1100s. Those farmers came to depend more and more on maize, which is corn, and the body cannot get much iron from maize.

Among the Dickson Mounds people, signs of anemia, having too few red blood cells, rose sharply. Anemia leaves a person weak and tired. In a skull it shows as spongy patches full of tiny holes on the top of the head and in the eye sockets. Those patches form where the marrow, the soft center of the bone, swelled to make more blood. Bone infections among the same people became nearly four times as severe, a team of researchers found in 1977.

Why the crowd diseases were missing

The diseases that later struck this land in epidemics can keep spreading only where great crowds of people live. An epidemic is a disease that spreads to many people in one place at the same time.

A person sick with measles can pass it on for less than two weeks. Then the sick person either dies or becomes immune, which means the body can fight off that disease and will not catch it again. So measles keeps spreading only where new children who have never had it are always being born.

One scientist worked out in 1957 how big a population must be for measles to stay in a place for good. His answer was 200,000 to 300,000 people, with about 7,000 of them still able to catch it. Another put the number at 500,000 in 1975.

In small towns spread far apart, a disease like that infects everyone who can catch it. Then it dies out, because nobody is left to infect. The diseases that last in small towns are slow ones, like yaws and pinta, another skin disease in the syphilis family, the scholar Linda Newson explained in 1993.

When the crowd diseases arrived after 1492, nobody here had been exposed to them and nobody was immune. The historian Alfred Crosby gave what followed a name in 1976: "virgin soil epidemics." In a virgin soil epidemic, everyone in a town can be infected at the same time.

Newson also reported research on genes, the instructions inside every cell that pass from parents to children. It suggested that among Native Americans, the body's defenses against germs varied less from person to person, which may have made whole towns easier to infect. Newson called that a partial explanation.

NOT KNOWN: how many people lived north of Mexico in 1492. Scholars' estimates run from about 1 million, made in 1949, to about 18 million, made in 1983. Others fall between, such as about 1.85 million (made in 1976) and about 3.8 million (made in 1992). How many people died after 1492 depends on which starting number is right.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 6 MINUTES

Three Europeans saw Native people in this land begin dying of diseases they had never had before, and each wrote down what he saw. None of the three could name the disease. One was a Spanish man stranded on the Texas coast in 1528. Another was a Portuguese man in Hernando de Soto's army in 1540. The third was an English scientist on the Carolina coast in 1585 and 1586, and only his account points to who carried the disease.

The Texas coast, 1528 and 1529

About 80 Spanish men came ashore in two boats on or near Galveston Island, on the Texas coast, in November 1528. They were survivors of a Spanish expedition, a group sent out to explore and conquer, led by a commander named Narváez. One of them, Cabeza de Vaca, later wrote an account of what happened. He wrote that the people of the island spoke two languages, and he called them the Capoque and the Han.

The islanders watched most of the Spaniards die. Only 15 of the 80 stayed alive. Five Spaniards camped on the coast grew so desperate that they ate the bodies of their own dead as each one died. Their names were Sierra, Diego López, Corral, Palacios and Gonçalo Ruiz. Nobody ate the body of the last one to die. Cabeza de Vaca wrote that the islanders were in an uproar when they learned of it.

Then a disease of the bowels, the long tube in the belly that food passes through after the stomach, killed half of the island's people, Cabeza de Vaca wrote. He did not name the disease. The islanders believed the Spaniards had caused the deaths, and they planned to kill the survivors. The Native man who had charge of Cabeza de Vaca talked them out of it. He told them that if the Spaniards had such power, they would have saved their own men, whom the islanders had watched die.

Healing for food. The islanders then told the Spaniards to heal the sick, and they gave them no food until they did. Cabeza de Vaca described how the island's own healers worked. A healer made small cuts in the skin over the place that hurt and sucked at the cuts. Healers also burned the skin at the painful spot on purpose, a treatment called cautery, and then blew on it.

Cabeza de Vaca tried the burning on himself and wrote that it helped him. The Spaniards blessed the sick, breathed on them, and said two Christian prayers, the Lord's Prayer and the Hail Mary.

Years later, on a long walk west, people brought Cabeza de Vaca a man with an arrowhead lodged in his chest above the heart. The year and the place are not certain, because accounts of the route differ. Cabeza de Vaca cut open the

man's chest with his knife and pulled the arrowhead out. He closed the cut with two stitches, using a deer bone as a needle, and stopped the bleeding with hair from an animal skin. He wrote that the next day he cut the stitches and the man was well.

Empty towns at Cofitachequi, 1540

The army of the Spanish commander Hernando de Soto reached Cofitachequi in the spring of 1540. It was a town on a river in what is now South Carolina or Georgia, and accounts place it in different spots.

A Portuguese member of the army, known only as the Gentleman of Elvas, wrote that there were large empty towns nearby, grown over with grass. People at Cofitachequi told the Spaniards that a disease had struck the land two years before. The people of those towns had moved away to other towns. The Gentleman of Elvas did not name the disease, and he did not say how many people died.

That disease arrived before de Soto's army did. No record shows where it came from. Accounts differ about whether de Soto's own army carried new diseases inland. The records do show that de Soto's men killed and enslaved Native people and took their food.

Roanoke, 1585 and 1586

English colonists sent by Sir Walter Raleigh lived on Roanoke Island, in what is now North Carolina, in 1585 and 1586. The island lay in the land of the Carolina Algonquian peoples. The colonists visited Algonquian towns, and within a few days after they left a town, people there began to die quickly.

The scientist Thomas Harriot, who was with the colonists, wrote down the deaths. In some towns about 20 people died, in some 40, in some 60, and in one town 120. Harriot wrote that, as far as he could learn, the deaths happened only in towns the English had visited. The oldest people told him nothing like it had ever happened before. During the same months, he wrote, no English colonist died or became seriously sick.

Harriot did not name the disease. Peter Mires, writing in 1994 in the journal *Historical Archaeology*, compared Harriot's report with John Smith's later accounts. Smith had heard about an epidemic among the Accomac, a people the Roanoke colonists visited in the winter of 1585 to 1586. Mires concluded that the English colonists carried the disease, and that it was probably influenza. He called influenza probable, not certain.

Thomas Harriot

Who: An English mathematician and scientist who wrote down, town by town, an epidemic among the Carolina Algonquians. His name is also spelled Hariot.

When and where: Roanoke Island and the Carolina Algonquian towns, 1585 and 1586. His report was printed in London in 1588.

The Algonquians said invisible bullets were killing them.

Thomas Harriot recorded what the Carolina Algonquians believed was killing them in 1585 and 1586. Some believed the English God was killing people for the English. Others believed that more English people were still to come, to kill them and take their places. These coming English were already in the air, invisible and without bodies, and they were shooting "invisible bullets" into people at the English colonists' request.

Algonquian healers sucked what they called "strings of blood" out of the sick. They said these were the strings tied to the invisible bullets.

Harriot had taught how to find the way across the open ocean at Sir Walter Raleigh's house in London. He sailed with the colony that reached Roanoke Island on June 26, 1585, and there he and the artist John White mapped the area.

Wingina was the wiroans, or chief, of the people the English lived among. Twice he was so sick that he nearly died, and he sent for the English to pray for him. Wingina and others also asked the English to pray that their enemies would die of the same disease. Harriot wrote that he told them such prayers were ungodly. When their enemies did die, they came to thank the English anyway.

Harriot explained the deaths as "the special work of God" for the English. He also described how the Algonquians used a plant they called uppowoc, which is tobacco, as a medicine. They breathed in its smoke through clay pipes, and Harriot believed it kept them healthy.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 10 MINUTES

Epidemics emptied Native towns along the Atlantic coast in the 1600s, sometimes years before colonists settled beside them. English colonists also died in large numbers in their own first years, from disease and hunger. The people who treated the sick in the colonies included a weaver with no known medical training and a minister.

In New Amsterdam, officials of a Dutch trading company paid the town's midwives. At the end of the century yellow fever reached Boston, after the virus that causes it and the mosquito that carries it had crossed the Atlantic on the ships of the slave trade.

Jamestown, 1607

A group of 104 English men and boys built a fort at Jamestown, in Virginia, in 1607. The fort stood in the land of the Paspahegh, one of the nations led by the Powhatan paramount chief, a leader over the chiefs of many nations. By the end of September 1607, half of the 104 colonists were dead, most of them from sickness.

George Percy, one of the colonists, wrote down each death with its date and cause. John Asbie died on August 6 of the "bloudie Flixe," Percy's spelling of bloody flux. Flux was the old word for diarrhea, so a bloody flux was diarrhea with blood in it. Percy also listed "Swellings" and "Burning Fevers," and deaths from wounds in fighting.

Percy blamed hunger most of all. Each day, five men shared one small can of barley boiled in water. They drank from the river. At high tide its water was salty, and at low tide it was "full of slime and filth." Many nights three or four men died.

Then Native people of the area, whom Percy called "our mortall enemies," brought bread, corn, fish and meat to the fort in great plenty. Percy wrote that "divers Kings," meaning several Native leaders, brought more. He did not name them.

The river and the drought. The geographer Carville Earle pointed to the river in 1979. At Jamestown the river water was part salt and part fresh, so the colonists' waste stayed in it instead of washing away. Tree rings show a drought that lasted seven years, from 1606 to 1612. The historian Dennis Blanton wrote that colonists who were hungry and thirsty from the drought were the most likely to fall sick.

Only about 60 of about 240 colonists in the fort lived through the winter of 1609 to 1610. That winter is called the Starving Time.

An epidemic on the New England coast, 1616 to 1619

A disease killed Native people along about 200 miles of the coast of southern New England, as far as Cape Cod, from 1616 to 1619. As far as the records show, the European fishermen and traders nearby did not catch it.

No count of the dead exists. Estimates of the share who died rest on weak records, and they run from one third to nine tenths of the people in the area. Before the epidemic, the Wampanoag Nation had about 21,000 to 24,000 people, and the Patuxet town had about 2,000. The colonist Daniel Gookin later wrote down what Native people told him about the sick. "The bodies all over were exceedingly yellow," they said, both before the people died and after. Gookin's account dates from 1674.

No one has settled what the disease was. The ideas put forward include yellow fever, smallpox, plague, typhus and several other diseases. Two researchers, John Marr and John Cathey, proposed leptospirosis in 2010. Leptospirosis is an infection by bacteria that rats carry in their urine. Marr and Cathey suggested that rats from European ships spread it into the water and soil, and they called their idea a proposal for others to test, not an answer.

When the Plymouth colonists arrived in 1620, the Patuxet town was empty. The colonists built Plymouth on its site. Governor William Bradford wrote that so many had died that the living could not bury the dead. He wrote that in many places in the area, skulls and bones still lay above the ground where houses had stood.

Squanto, also called Tisquantum, was one of the few Patuxet survivors. Bradford wrote that Squanto told other Native people the English kept the plague buried in the ground. Squanto said the English could send it among them whenever they wished.

Plymouth's first winter, 1620 and 1621

In two or three months of their first winter, about half of the roughly 100 Mayflower passengers at Plymouth died. Sometimes two or three died in one day. Bradford blamed the cold, the lack of houses, and scurvy and other diseases from the long voyage. Scurvy is a disease caused by too little vitamin C, which comes from fresh fruit and vegetables. The gums swell and bleed, the teeth loosen, and the person grows too weak to work.

At the worst, only six or seven people were well enough to care for everyone else. They cut wood, made fires, cooked, made beds and washed the sick people's clothes. William Brewster and Myles Standish were two of them.

Samuel Fuller

Who: A wool weaver with no known medical training who came on the Mayflower and became the doctor for Plymouth and its neighbors.
When and where: Plymouth, Salem and Charlestown, in Massachusetts, 1620 to 1633.

A weaver of woolen cloth became the doctor for three towns.

Samuel Fuller treated the sick of Plymouth and the towns near it for about thirteen years, although no record shows he ever trained as a doctor. Before the voyage he lived in Leiden, in the Netherlands, where he wove woolen cloth. He came to Plymouth on the Mayflower in 1620. His brother Edward and Edward's wife died in Plymouth's first winter.

Settlers at Salem fell sick in 1629 with scurvy and with a fever that had spread on their ships. Their leader, John Endicott, asked Governor Bradford to send Fuller. Endicott had heard that Fuller had cured people "by letting blood and other means."

> Letting blood, or bloodletting, meant cutting open a vein, most often in the arm, and letting blood run out into a bowl. Doctors of the time believed it balanced the fluids of the body. It does not cure scurvy or fever, and it left sick people weaker. In 1630 Fuller let blood from about 20 newly arrived colonists at Charlestown.
>
> A fever killed more than 20 men, women and children at Plymouth in 1633, and many Native people nearby. Fuller treated the sick. Then he died of the same fever, sometime between August and late September 1633. The colonist Nathaniel Morton later wrote that Fuller "was their surgeon and physician, and did much good in his place."

Smallpox on the Connecticut River, 1634

About 1,000 Native people closed themselves inside a fort on the Connecticut River in the winter of 1633 to 1634, upriver from Plymouth's trading house. A disease struck them, and more than 950 of them died, Bradford wrote. Many of the bodies lay rotting above the ground, unburied. Bradford did not name the nation or the disease.

That spring, smallpox struck the Native people living near the trading house itself. Smallpox is caused by a virus, a germ even smaller than bacteria, and it spread from person to person through close contact, face to face. It caused a fever and a rash, and the rash turned into blisters and then scabs. In general, about 3 of every 10 people who caught smallpox died, though the share in a single epidemic could be higher or lower. Many survivors had scars for life, most of all on their faces.

Bradford described the sick lying on hard mats. The blisters broke open and ran, and the skin stuck to the mats. When a sick person turned over, the skin of a whole side came off at once. So many were sick that no one could make a fire, fetch water or bury the dead. Some crawled on their hands and knees to get water and died on the way.

The English at the trading house brought the sick wood, water and food, and buried the dead. None of the English there caught the disease. Very few of the Native people survived. Their chief sachem, or leader, died, and so did almost

all of his friends and relatives. Bradford did not write down his name. No record shows who brought the disease to the river.

Midwives and surgeons in New Amsterdam

The officials who ran New Amsterdam paid some of the people who cared for its sick, and in 1658 they set up a place for sick patients. They were officials of the Dutch West India Company, a trading company, and they governed the whole colony of New Netherland. New Amsterdam stood on Manhattan, in the homeland of the Munsee Lenape.

Company officials appointed and paid the town's midwives, women who help other women give birth. The midwife Lysbert Dircksen lived in a house built for her with public money in 1638. The officials also made rules for the town's surgeons, such as Hans Kierstede, who arrived in 1638 and worked there for nearly 30 years. Under rules of 1652 and 1657, barbers from the ships needed permission to treat wounds on shore, and surgeons had to report how each wounded person had been hurt.

The Company surgeon Jacob Hendrichsen Varvanger asked in 1658 for a place to care for sick patients. Members of the colony's council agreed. They made Hilletje Wilbruch the matron, the woman in charge, and paid her 100 florins, which were Dutch coins, a year. Richard Amerman, who wrote about medicine in New Netherland, called it the first hospital.

Tryntje Jonas

Who: A midwife from Amsterdam who served as an official midwife of New Amsterdam.

When and where: New Amsterdam, now New York City, by 1644.

The mother named one man as the father, then another.

Tryntje Jonas was the official midwife of New Amsterdam by 1644. Officials of the Dutch West India Company appointed her and paid her, and she had her own house on Pearl Street. She came from Amsterdam. Her daughter Anneke Jans arrived in New Netherland in 1630.

Tryntje Jonas swore to the colony's secretary on July 7, 1644, what a mother had said about her baby's father. She had been called to help a woman named Hillegont Joris give birth. Hillegont Joris first named "Jan, the pilot" as the father of the child. Afterward she named Laurens Cornelisz. The secretary copied the statement into the colony's record book, where it survives. The record does not say why the midwife was asked to report this.

After Tryntje Jonas died, her daughter Anneke Jans collected the pay that the Company still owed her mother for her work. The date of her death is not known.

Smallpox advice and yellow fever, 1677 to 1693

After smallpox spread through Boston in 1676 and 1677, the minister Thomas Thacher of the Old South Church wrote a guide for families. It was printed in 1677 or 1678 on a single large sheet called a broadside. Librarians at Harvard University's medical library call it the first medical paper published in the American colonies.

Thacher told families not to put a smallpox patient under heavy bedclothes, in a hot room, or give strong hot drinks. He told them to give weak beer, a thin porridge of cornmeal and water, and boiled apples. He warned that bloody or black urine, and diarrhea, were danger signs.

Yellow fever. Yellow fever came from Africa. It is a disease caused by a virus, and a mosquito called Aedes aegypti carries the virus from one person to another when it bites. The virus and the mosquito both crossed the Atlantic aboard the ships of the slave trade, a team of scientists reported in 2007. Nobody knew about the mosquito's part until the early 1900s.

Yellow fever struck Boston in 1693. The makers of the PBS history series *American Experience* call Boston the first British colony in North America to have a yellow fever epidemic. That June a fleet of English warships arrived in Boston, and the first Boston death from the fever came on June 25. No surviving count gives the number of dead. Charleston and Philadelphia had yellow fever soon after.

Smallpox along the slave-raiding paths, 1696 to 1700

Smallpox broke out in Virginia in 1696 and spread south and west along the trading paths. It traveled with Native slave raiders and the captives they took for the colonial slave trade. By 1699 it had reached the Tunica people, in what is now Louisiana. It came back in waves until 1715.

By one estimate, some Native nations lost more than 60 of every 100 people. A South Carolina colonist wrote that smallpox had killed so many Native people that there was "little reason to believe they will be capable of doing any harm to us for several years to come."

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 7 MINUTES

Doctors in Boston and Charleston began protecting people against smallpox in these fifty years by inoculation: giving a healthy person a small case of smallpox on purpose, so that the person would not catch it again. Epidemics still killed thousands of people. Diphtheria killed about 5,000 people in New England, most of them children, and smallpox killed about half of the Cherokee Nation.

Smallpox and inoculation in Boston, 1721

The ship *Seahorse* sailed into Boston from the island of Barbados on April 22, 1721, with sailors sick with smallpox. Boston had about 10,600 people. Close to 5,800 of them caught smallpox that year, and 844 died.

The minister Cotton Mather led the Boston ministers who backed inoculation. He had first heard of it from Onesimus, a man he enslaved. To inoculate someone, a doctor took pus from a smallpox sore and dried it. Then the doctor rubbed or scratched the pus into a cut in a healthy person's skin.

Almost all of Boston's doctors feared that inoculation would spread the disease. Zabdiel Boylston was the one Boston doctor who tried it, starting in June 1721. By the end of the epidemic, about 287 people had been inoculated, most of them by Boylston, and 6 of them died. That is about 2 in every 100. Among people who caught smallpox the ordinary way, about 14 or 15 in every 100 died.

The fight. The doctor William Douglass led the people who opposed inoculation. He called it a "poisonous humor," humor being an old word for a fluid of the body. James Franklin printed attacks on inoculation in his newspaper, the *New-England Courant*. The town's selectmen, its elected officials, ordered Boylston to stop. He started again two days later.

Someone threw a bomb through a window of Mather's house on November 14, 1721. The note that came with it ended with the words "I will inoculate you with this, with a pox on you!" The records do not name who threw it.

Onesimus

Who: An enslaved African man who described inoculation to Cotton Mather years before Boston first tried it.

When and where: Boston, enslaved by Cotton Mather from 1706 to 1716.

He showed his enslaver the scar on his arm.

Onesimus told Cotton Mather how people in Africa protected themselves against smallpox. Mather later used that knowledge to back inoculation during Boston's smallpox epidemic of 1721.

Members of Mather's North Church gave Onesimus to Mather as a gift in December 1706. Another account gives the year as 1707. Mather named him Onesimus, after an enslaved man in the Bible. The name he was born with is not recorded. He was probably Coromantee, one of the Akan peoples of the Gold Coast, which is now Ghana.

Mather wrote to the Royal Society in London, a group of leading scientists, in 1716. He wrote that he had asked Onesimus whether he had ever had smallpox. Onesimus answered "both, Yes, and No." He had "undergone an Operation, which had given him something of the Small-Pox, and would forever preserve him from it." He showed Mather the scar on his arm.

Mather wrote down the method in Onesimus's words: people "take Juice of Small-Pox" and "Cut the Skin, and put in a drop." Mather wrote that he heard this from Onesimus months before he read of the same method in reports from Europe.

Onesimus married and had two children. Both died before they were ten. His son Onesimulus died in 1714. His daughter Katy died of tuberculosis.

Onesimus paid money in 1716 toward buying another enslaved man, Obadiah, to take his place. Mather then freed him, but with conditions. Onesimus still had to work for Mather's household, shoveling snow and

chopping wood, whenever Mather called for him. He also had to repay five pounds that Mather said he had stolen. Few records of his life after 1716 survive.

Zabdiel Boylston

Who: The one Boston doctor who tried inoculation in 1721, when almost all the others feared it.

When and where: Boston, 1721 and 1722. Born 1679, died 1766.

His first three patients were his son and two enslaved people.

Zabdiel Boylston inoculated three people on June 26, 1721. One was his son Thomas, who was about six. The other two were people Boylston enslaved: Jack, a man of 36, and Jack's son Jackey, who was two and a half. By law Jack and Jackey were Boylston's property, so they could not refuse. The accounts of that day do not say whether Jack agreed.

Boylston wrote down how each one's illness went. Thomas and Jack each had about "an hundred" blisters full of pus. On the ninth day Boylston gave them a medicine to make them vomit. All three recovered.

He went on to inoculate between 242 and 248 of the about 287 people inoculated in the epidemic, by different counts. Six of them died. When the selectmen ordered him to stop, he started again two days later. For a time he stayed on his farm, because he could not leave it without risk of being attacked.

Later he sailed to London, where the scientists of the Royal Society made him a member. In 1726 he published a book about his inoculations, *An Historical Account of the Small-Pox Inoculated in New England*.

Elizabeth Phillips

Who: A licensed midwife from London who delivered babies in Boston for more than forty years.

When and where: Boston and Charlestown, Massachusetts, 1719 to 1761.

Her gravestone says she delivered more than 3,000 babies.

The gravestone stands in the Phipps Street Burying Ground in Charlestown, Massachusetts. It says that Elizabeth Phillips "brought into this world above 3000 Children." She delivered them in and around Boston.

She was born in Westminster, in London, in about 1685. The Bishop of London gave her a license to work as a midwife in 1718. She came to Boston the next year and worked there until she died in 1761.

The throat distemper, 1735 to 1740

A disease began in Kingston, New Hampshire, in May 1735, and spread along the coast into Maine and Massachusetts. People called it the throat distemper, distemper being an old word for a sickness. Doctors today identify it as diphtheria, an infection by bacteria.

The bacteria killed the living tissue of the throat and nose. The dead tissue formed a thick coating across the throat, until the child could no longer breathe.

About 5,000 people in New England died of it by 1740, and more than three quarters of them were children. That count comes from Ernest Caulfield's 1939 book on the epidemic. In the town of Hampton Falls, New Hampshire, 210 people had died by July 1736, and 160 of them were under ten. The *Boston Gazette* newspaper printed treatments: bloodletting, borax (a white powder made from a mineral) or honey on the throat, and drinks made from herbs.

In Ipswich, Massachusetts, Mark and Hephzibah Howe lost eight children in November 1736. They were Abijah (1), Mark (2), Aaron (5), Mary (7), Lucy (9), Moses (11), Love (12) and Hannah (13).

Smallpox among the Cherokee and Catawba, 1738 and 1739

Smallpox struck the Cherokee towns in the southern Appalachian Mountains in 1738 and 1739. Between 7,000 and 10,000 Cherokee people died, about half of the nation. About half of the Catawba Nation died too.

How the disease reached them is disputed. The trader James Adair wrote in 1775 that it came into Charleston on slave ships from West Africa. Historians writing for NCpedia, a North Carolina encyclopedia, give two other routes. In one, traders brought it. In the other, Cherokee warriors carried it home after fighting beside the British against the Spanish in Florida.

Cherokee healers treated the sick in sweat houses and then plunged them into icy streams. The NCpedia historians write that this only added to the number who died. Adair wrote that some survivors whom the disease had scarred killed themselves.

In Charleston, during the same epidemic, the doctor James Kilpatrick inoculated 800 to 1,000 people. Eight of them died.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 10 MINUTES

Philadelphia doctors opened a hospital for the sick poor in 1752 and a medical school in 1765, while epidemics went on killing people in the colonies and the Native nations. Smallpox spread through the colonies and the Native nations during two wars, and in 1763 the British commander at Fort Pitt and a trader tried to spread it on purpose.

Yellow fever killed about 4,000 to 5,000 people in Philadelphia in 1793, about one person in ten. Under a 1798 law that members of Congress passed and President John Adams signed, federal officials took money from sailors' pay for their medical care.

Caesar

Who: An enslaved healer whose cure for poison led South Carolina's lawmakers to pay his enslaver for his freedom.

When and where: St. Paul's Parish, South Carolina, near the Edisto River. Freed in 1750. Died in 1754.

Lawmakers paid 500 pounds for one man's cure for poison.

South Carolina's lawmakers bought Caesar's freedom in 1750 in exchange for his cure for poison. Caesar was enslaved by John Norman of Beach Hill, near the Edisto River. He was known for curing people of poisoning and of rattlesnake bite.

White South Carolinians feared that the enslaved people who cooked their food would poison them. In 1740 the colony's lawmakers had made poisoning a crime punished by death. In November 1749 the members of the Commons House, the colony's lawmaking body, voted to pay Norman 500 pounds for Caesar's freedom. They made the payment final on May 31, 1750. The lawmakers also promised Caesar 100 pounds a year for life, and he received his first payment in 1751.

> The *South Carolina Gazette* newspaper printed Caesar's cure in May 1750. It was made by boiling plantain roots, horehound and goldenrod with rum, sugar, tobacco and lye made from wood ash.
>
> The same lawmakers passed a law in 1751 aimed at enslaved healers, "commonly called doctors." Under it, enslaved healers could not give anyone medicine without permission from white people.
>
> Caesar died in early 1754, at about 77 years old. He left an estate worth 62 pounds, and it included an enslaved woman. The record of his estate does not give her name or say how he came to own her. His wife Lilly and his daughter Hannah were never freed by the lawmakers. They were still enslaved by John Norman.

Pennsylvania Hospital and the first medical school

Dr. Thomas Bond and Benjamin Franklin founded Pennsylvania Hospital in Philadelphia on May 11, 1751. They founded it to care for "the poor, sick and insane." Franklin's petition, a written request, described people "deprived of their rational Faculties," meaning they had lost their reason. It stated that they could be "contained and confined" there.

The first patients came on February 6, 1752, to a house on Market Street that the hospital used until its own building was ready. Writers for the Mütter Museum, a museum of medical history in Philadelphia, give the year as 1753. The permanent building opened in 1756. Its basement held rooms for patients with mental illness. The first floor was for men, and the second floor was for women.

Older places in the colonies also housed people in need. Philadelphia's almshouse, a home for the poor, opened in 1732. A New York workhouse of 1736, a place where poor people lived and were put to work, later became Bellevue Hospital. The place for the sick that Dutch officials set up in New Amsterdam in 1658 was older than both.

Dr. John Morgan founded a medical school at the College of Philadelphia, now the University of Pennsylvania, in 1765. The museum's writers call it the first medical college in the colonies. Students learned at the bedsides of the hospital's patients.

Smallpox as a weapon at Fort Pitt, 1763

Two British men at Fort Pitt gave two Delaware men blankets from a smallpox hospital in 1763, hoping to spread the disease. Fort Pitt was a British fort where Pittsburgh stands now. That year Delaware, or Lenape, and Shawnee and Mingo warriors surrounded the fort, during the war led by the Ottawa leader Pontiac. Smallpox had broken out among the soldiers inside.

The blankets. Captain Simeon Ecuyer, the fort's commander, and the trader William Trent met two Delaware men, Turtle's Heart and Mamaltee, on June 24, 1763. Most accounts give that date, though one copy of Trent's journal dates the entry May 24. Ecuyer and Trent gave them two blankets and a handkerchief taken from the fort's smallpox hospital. Trent wrote in his journal, "I hope it will have the desired effect." Ecuyer signed the bill that charged the army for the blankets.

The letters. In July, General Jeffery Amherst, the British commander in North America, wrote to Colonel Henry Bouquet. He asked: "Could it not be contrived to send the Small Pox among those disaffected tribes of Indians?" By contrived he meant arranged, and by disaffected, unfriendly. Bouquet answered that he would "try to inocculate the Indians by means of Blankets."

Amherst also wrote of trying "to extirpate this Execrable Race," which means to wipe out completely a people he called hateful. Ecuyer and Trent had given away the blankets in June, before these letters, and on their own.

KNOWN:	Smallpox spread among the Native nations, and a man named Gershom Hicks reported in 1764 that about 100 people died of it between the spring of 1763 and April 1764.
NOT KNOWN:	whether the blankets caused any of it. The historian Elizabeth Fenn writes that no one can know whether the attempt worked, and a month after the meeting Turtle's Heart and Mamaltee showed no sign of smallpox.

Smallpox in the Revolution, 1775 to 1782

Lord Dunmore, Virginia's royal governor, promised freedom in 1775 to people enslaved by the rebels if they joined his army. In the spring and summer of 1776, smallpox and other fevers spread through his camps. An estimated 500 Black soldiers of his Ethiopian Regiment died on Gwynn's Island, in Virginia, along with about 150 white soldiers.

General George Washington ordered the Continental Army to be inoculated on February 6, 1777. He wrote that if smallpox spread through the army, "we should have more to dread from it, than from the Sword of the Enemy." Dr. William Shippen ran the inoculations in secret, in winter, at Morristown, New Jersey, and in Philadelphia, so the British would not learn of them.

What started the Doctors' Riot of 1788?

Medical students in New York were digging up the dead. To learn how the body is built, students cut open dead bodies and studied the parts, which is called dissection. They dug up bodies from the Negroes Burial Ground, where Black New Yorkers buried their dead ("Negro" was a word for Black people used at the time). They also dug up bodies from the paupers' graveyard, where the poor were buried.

Free and enslaved Black New Yorkers asked the city's Common Council, the group that ran the city, to stop it in February 1788. Their petition described "young gentlemen in this city who call themselves students of the physic," meaning students of medicine, who were digging up "bodies of our deceased friends and relatives." The council members took no action to stop the digging.

The riot. Boys outside New York Hospital saw a human arm from the dissecting room in April 1788. Crowds attacked the hospital. The mayor had a doctor and medical students taken to the city jail to protect them, and about 5,000 people then gathered at the jail.

Militiamen, ordinary citizens called up to serve as soldiers, had orders not to fire. A rock hit the politician John Jay in the head, a brick hit General Baron von Steuben, and the militiamen opened fire. The accounts of the riot do not name an officer who gave the order. As many as 20 people were killed, among them at least 3 rioters and 3 militiamen.

New York lawmakers banned grave robbing in 1789. The same law let judges order that the bodies of executed murderers, arsonists and burglars be dissected.

Yellow fever in Philadelphia, 1793

Yellow fever killed 4,041 people in Philadelphia from August to November 1793, by the count of Mathew Carey, who wrote a book about the epidemic. Other counts run to about 5,000, which was about one person in every ten in the city. About 20,000 people left the city. Nobody in 1793 knew that mosquitoes carried the disease.

Dr. Benjamin Rush treated patients by bleeding them heavily and giving them purges, medicines that emptied the bowels. Rush's purges were made with mercury, a metal. Losing that much blood could send a weak patient into shock, which happens when too little blood is left in the body. A person in shock becomes confused, the heart beats fast, and the blood pressure drops. A French doctor, Jean Devèze, used milder treatment at the Bush Hill hospital.

Mayor Matthew Clarkson issued a public order about the epidemic on September 10, 1793. From September 14, a committee of 26 citizens worked with him to run the city's response.

Absalom Jones and Richard Allen, two Black Philadelphians who led volunteers in nursing the sick, printed an account of the epidemic in 1794. They wrote that some prisoners were let out of jail after the elders of the African church asked, on the condition that they nurse the sick at Bush Hill. Two thirds of those prisoners were Black. Jones and Allen's account does not name the officials who let them out.

Benjamin Rush

Who: A Philadelphia doctor who treated patients in the 1793 yellow fever epidemic, and a signer of the Declaration of Independence.

When and where: Philadelphia, 1793. He worked at Pennsylvania Hospital from 1783 to 1813.

He told Philadelphians that Black people could not catch yellow fever.

Benjamin Rush printed that belief in the newspapers. In fact Black Philadelphians caught the fever and died of it while they nursed the sick.

Rush bled his patients heavily during the 1793 epidemic and gave them purges made with mercury. Several doctors died, and most of the rest were worn out by sickness or work. Then Rush asked Absalom Jones and Richard Allen, who both knew how to bleed a patient, to care for the sick. He told them where to get medicine, how to give it, and at what stage of the fever to bleed. When they could not tell what to do, he came himself or sent his student Edward Fisher.

At Pennsylvania Hospital, Rush moved patients with mental illness out of the basement and onto hospital floors with windows.

Absalom Jones and Richard Allen

Who: Two Black Philadelphians who led Black volunteers to nurse the sick and bury the dead.

When and where: Philadelphia, September to November 1793. Their account was printed in 1794.

They buried several hundred people and never asked for pay.

Absalom Jones and Richard Allen led members of the Free African Society, a group of Black Philadelphians, in nursing the sick and burying the dead in 1793. In early September a notice in the newspapers asked "the people of colour" to help the sick. It stated that they could not catch the fever. Jones, Allen and a few others met to talk over how to act. The next day they asked the mayor how they could help.

They carried away and buried the dead, several hundred poor people and strangers in all. Five men they hired helped them, and two of those men were Allen's brothers. Rush taught them when to bleed a patient, and they bled more than 800 people.

The fever killed Black Philadelphians too. In 1792, 67 Black people were buried in the city. In 1793 the number was 305. When some Black nurses fell sick, the families they worked for put them out of their houses. Jones and Allen knew of one who died in a stable.

Mathew Carey wrote that Black nurses had overcharged the sick and stolen from them. In 1794 Jones and Allen printed their own account to answer him. For burying the dead, they wrote, they had "never received, nor never asked any compensation," meaning any pay. They also named Mary Scott, a Black nurse who refused to take more than her wage from a Mrs. Mason. Mrs. Mason then gave her six pounds a year for life.

Medical care for sailors, 1798

President John Adams signed the Act for the Relief of Sick and Disabled Seamen on July 16, 1798, after members of Congress passed it. Under it, federal officials took 20 cents a month out of every merchant sailor's pay to pay for sailors' medical care. For a time, the first marine hospital, a hospital for sailors, stood on Castle Island in Boston Harbor. Today's United States Public Health Service began as this program.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 12 MINUTES

A safer protection against smallpox reached the United States in 1800, and in 1846 surgeons learned to operate on patients who felt no pain, but neither one reached everybody. The new protection, vaccination, scratched matter from cowpox, a mild disease of cows, into the skin. A federal official left the Mandan people out of a vaccination program, and in 1837 smallpox killed at least nine in ten of them on the upper Missouri River.

A Boston dentist showed surgeons in 1846 that a drug called ether, breathed as a vapor, could keep a patient from feeling an operation. From 1845 to 1849 an Alabama doctor, J. Marion Sims, operated again and again on enslaved women to work out a new surgery, and he gave them nothing to stop the pain during the operations.

Cholera, a disease spread in dirty water, killed thousands of people in the growing cities in 1832 and 1849. In 1849 Elizabeth Blackwell became the first woman in the United States to earn a medical degree.

Vaccination arrives, 1800 and 1802

To vaccinate a person, a doctor scratched fluid from a cowpox sore into the skin, and it protected the person against smallpox. In the older method, inoculation, doctors put pus from a real smallpox sore into a cut in the skin, and some inoculated people died of smallpox.

Dr. Benjamin Waterhouse of Cambridge, Massachusetts, vaccinated his young son Daniel Oliver and a servant, Samuel Carter, on July 8, 1800. He used cowpox matter on threads sent from England. It is the first recorded vaccination in the United States. Doctors in other places claimed earlier ones, but Waterhouse kept official records. He then vaccinated three more of his children and a second servant, Kesiah Flag.

Waterhouse sent vaccine to other doctors and to President Thomas Jefferson. Jefferson vaccinated his family, the enslaved people at his plantation, Monticello, and his neighbors. The enslaved people were Jefferson's property by law, so they could not refuse. The accounts of these vaccinations do not say whether they agreed.

The test. Waterhouse and six other doctors vaccinated nineteen boys on August 16, 1802. That fall the boys, whom the record calls volunteers, were exposed to smallpox to test the vaccine. The Harvard medical library's account of the test does not say who exposed them, whose sons they were, or whether their parents agreed. The members of Boston's Board of Health then reported that cowpox was "a complete preventive against all the effects of the Small-pox." That meant it kept a person from getting smallpox at all.

Smallpox on the upper Missouri, 1832 and 1837

President Andrew Jackson signed a law on May 5, 1832, that set aside $12,000 to vaccinate Native people. Members of Congress had passed it. Secretary of War Lewis Cass designed the program and ran it through federal Indian agents, the government officials sent to deal with each Native nation. By February 1833, more than 17,000 Native people had been vaccinated. Some Army doctors refused to take part because the pay was too low.

Cass left the Mandan, Hidatsa and Arikara nations on the upper Missouri River out of the program. The accounts of the program do not give his reason.

The steamboat. The American Fur Company's steamboat *St. Peters* stopped at Fort Clark, beside the Mandan villages in what is now North Dakota, on June 18, 1837. A deckhand had shown signs of smallpox weeks earlier, and three Arikara women who had caught it had boarded to go home to their village. The fort's trader, Francis Chardon, learned from the captain that people on the boat had smallpox. The crew unloaded goods and loaded furs in less than a day, and the boat went on upriver.

The historian R. G. Robertson blames the captain, Bernard Pratte Jr., who would not keep the sick apart from other people. Robertson calls that "criminal negligence," a failure to take care so serious that it should count as a crime.

Chardon wrote on July 14 that a young Mandan had died that day of smallpox. Later he wrote that the Mandan "die so fast that it is impossible" to keep count. About 2,000 Mandan lived in the villages in the spring. By October only 23 to 138 were alive, and the counts differ.

The Arikara lost between half and two thirds of their people, by different counts, and the Hidatsa lost many people too. Along the Missouri River more than 17,000 people died. About two thirds of the Blackfoot died, about half of the Assiniboine, about one third of the Crow and about one quarter of the Pawnee.

The Mandan leader Four Bears caught smallpox. Before he died he accused white people of ingratitude, of giving no thanks in return for kindness. As Chardon recorded his words: "I have loved the Whites ... and how have they

repaid it! With ingratitude!"

Cholera in the river cities, 1832 and 1849

Cholera is an infection of the gut. People catch it from food or water that carries its germ. It causes diarrhea that can start two hours after a person swallows the germ, and the body loses so much water and salt that the person can die. Cholera first reached the United States in 1832. It killed about 3,500 of New York City's roughly 250,000 people that year, about 1 New Yorker in 70.

Two steamboats from New Orleans reached St. Louis, on the Mississippi River, on January 2, 1849, with 66 passengers sick with cholera. In the spring more boats and gold seekers headed for California came through, and deaths rose. St. Louis had grown from about 5,000 people in 1830 to 77,000 by 1850, and it had no sewers. About three of every four people drank from shallow wells, often dug near outhouses. Pigs rooted through the garbage in the streets.

Doctors did not know what caused cholera. They blamed night air, bad food and alcohol, and told people to stop eating vegetables and then meat. At night they lit fires to clean the air. Some bled their patients, which left the sick person weaker.

Cholera killed 65 people in St. Louis on June 13, 1849, and 85 on June 21. In early July more than 100 a day died.

The cleanup. The city's lawmakers gave a Committee of Public Health $50,000 on June 27. The committee members stopped steamboats at a quarantine station, a place where people who may carry a disease are kept apart from others, on an island in the river. The city engineer drained standing water, and city officials banned hogs from the city. By late July, deaths had dropped to about 10 a day.

Reports from 1849 give 4,557 cholera deaths in St. Louis that year. Many deaths were never reported, so the real number was higher. About 7 of every 10 of the dead were Irish and German immigrants. Two researchers identified the germ that causes cholera in 1854 and showed that it spreads in dirty water.

Ether, 1842 and 1846

Before anesthesia, the use of a drug so that a patient does not feel pain during an operation, surgeons worked as fast as they could. A long operation could send a patient into shock.

Dr. Crawford Long of Jefferson, Georgia, cut a tumor, a lump of tissue that grows where it should not, from the neck of James Venable on March 30, 1842, while Venable breathed ether. Long had noticed that people who breathed ether for fun at parties felt little pain when they got hurt. He did not publish what he had done until 1849.

The dentist William T. G. Morton gave ether to Edward Gilbert Abbott, a young house painter, at Massachusetts General Hospital in Boston on October 16, 1846. Abbott breathed the vapor through an inhaler, a breathing device, that Morton had made. Dr. John Collins Warren cut a tumor from Abbott's neck. Abbott said afterward that he had not been fully asleep but "had felt no sensation of pain."

J. Marion Sims's operations on enslaved women, 1845 to 1849

Dr. J. Marion Sims of Montgomery, Alabama, operated again and again on enslaved Black women and girls from 1845 to 1849 to work out a new surgery. He gave them nothing to stop the pain.

The women had an injury called a vesicovaginal fistula. It is a hole torn through the wall between the bladder, where urine is held, and the vagina, the passage a baby is born through. A long, blocked childbirth crushes the tissue and makes the hole. Afterward urine runs out of the woman's body all the time. Her skin becomes sore and infected, and she smells of urine. Some women also had a hole into the bowel, so stool leaked too.

The women's enslavers sent them to Sims. The enslavers paid for their clothes and taxes, and Sims fed and housed them in a small hospital behind his office.

Seven of the women and girls are known by name. Sims named three of them, Anarcha, Betsey and Lucy. Another doctor, Nathan Bozeman, later named four more: Ann McRee, 16, Lavinia Boudurant, 13, Delia, 23, and Julia McDuffie,

20. How many there were in all is not settled. One count gives twelve. A museum of the history of anesthesia says that Sims left six or seven women unnamed.

Sims wrote that he promised to operate only with "the full consent of the patients." An enslaved woman had no power to refuse.

No drug for the pain. Ether was shown in Boston in 1846, partway through these years. Sims still used nothing to stop the pain during any of the operations. He wrote that the operations were "not painful enough to justify the trouble, and risk attending their administration." He meant the trouble and risk of giving a drug against the pain. Many white doctors of the time wrongly believed that Black people felt less pain. After the operations Sims often gave the women opium, a drug that dulls pain and brings sleep.

For each operation Sims had the woman placed naked on her knees and elbows so that he could see inside her. Sims operated on Lucy, who was about 18, for about an hour while about twelve doctors watched. Sims left a sponge inside her bladder to draw off urine. She nearly died of an infection that spread through her blood. Sims wrote: "Lucy's agony was extreme."

Sims closed the hole in Anarcha in 1849, on about the thirtieth operation. He sewed it shut with silver wire, which caused fewer infections than silk thread. A sponge or a tube drained urine from her bladder while it healed. He published the method in 1852.

The babies. Sims also operated on enslaved babies sick with neonatal tetanus, an infection in newborn babies that locks the jaw and other muscles in spasms. He believed the bones of the baby's skull had been pushed in on the brain. The historian Stephen Kenny later studied Sims's own reports of these cases.

Sims kept the baby son of an enslaved woman named Ann at his home in 1847. He cut through the baby's scalp with the point of a knife and pried a skull bone outward. Over the next two months he went on puncturing the same baby's scalp from time to time. Now he used the point of a crooked awl, a pointed tool for punching holes, to pry the skull bones outward. Kenny does not say whether Ann's baby lived.

Sims blamed the baby's illness on Ann. He said she had carried her son with his head pressed against her arm. He also blamed an enslaved nurse for not following his orders.

Sims used the awl in the same way on the baby of another enslaved woman, Frances, prying up the skull bone on one side and then the other. Her baby died twelve hours later, and Sims then cut open the body.

In his reports, every baby he operated on or cut open after death was enslaved. When a white baby, the daughter of Wyatt Peters, had the same disease, Sims gave no medicine and had her laid on her side on a pillow. Sims's two articles on the disease report 27 cases, 17 of them from his own notes. Kenny's study of those articles does not give a total of how many enslaved babies Sims operated on or how many died.

New York City officials voted in April 2018 to take down Sims's statue in Central Park.

Anarcha

Who: An enslaved young woman on whom J. Marion Sims operated about thirty times, with nothing to stop the pain, to work out his fistula surgery.
When and where: Montgomery, Alabama, 1845 to 1849.

Sims operated on her about thirty times before his surgery worked.

Anarcha's childbirth had been long and blocked. It tore two holes, one from her vagina into her bladder and one into her rectum, the end of the bowel. Urine and stool leaked from her body all the time. The accounts of her age differ. Some give 17 when she gave birth. One estimate says she was 13 when Sims first treated her.

Her operations went on for about four years, until the one in 1849 closed the hole into her bladder. The records do not show what happened to Anarcha, Betsey and Lucy after Sims finished with them.

Elizabeth Blackwell

Who: The first woman to earn a medical degree in the United States.

When and where: Geneva, New York, 1847 to 1849, then New York City. Born in Bristol, England, in 1821. Died in Hastings, England, in 1910.

About 29 medical schools turned her down before one said yes.

Elizabeth Blackwell graduated at the head of her class from Geneva Medical College in New York on January 23, 1849. She was the first woman in the United States to earn a medical degree.

The yes came in 1847, when the men studying at Geneva Medical College voted to let her in. Two accounts explain the vote. In one, the students thought the question was a joke by their teachers. In the other, they thought it would be funny to have a woman in the class.

Later in 1849 she worked at La Maternité, a hospital in Paris. She caught an eye infection from a baby she was treating and lost the sight of that eye, so she had to give up her plan to become a surgeon.

With her sister, Dr. Emily Blackwell, and Dr. Marie Zakrzewska, she opened the New York Infirmary for Indigent (poor) Women and Children in 1857. In 1868 she added a medical college for women beside it.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 9 MINUTES

Disease killed more Civil War soldiers than fighting did, and in peacetime too Americans died of sicknesses spread by dirty water, filthy streets and unwashed hands. Smallpox killed more than 60,000 people who had just been freed from slavery between 1862 and 1868. Officers of New York's new health board ordered filth cleared from the streets in 1866, and cholera killed a smaller share of the city than in earlier years. Yellow fever killed about 20,000 people along the Mississippi River in 1878.

Most American doctors did not yet kill germs on their hands and tools. When President James Garfield was shot in 1881, his doctors probed his bullet wound with unwashed fingers. He died weeks later, and the doctors who examined his body found it full of infection. By 1900 women were training as nurses and doctors. Among them were the first Black woman to earn a nursing license and the first Native American woman to earn a medical degree.

Disease in the Civil War armies, 1861 to 1865

Between about 620,000 and 850,000 soldiers died in the Civil War, and the count is disputed. For every three soldiers killed in battle, about five died of disease.

The diseases that killed them included diarrhea and dysentery, which are infections of the gut. Others were typhoid fever, an infection spread by food and water that has touched human waste, and pneumonia, an infection of the lungs. Measles, malaria (a fever spread by mosquitoes) and scurvy killed soldiers too.

Soldiers dug latrine trenches, the ditches they used as toilets, too shallow, and they did not cover the waste with dirt. Waste could then wash into the water the soldiers drank. One army surgeon found a camp's latrine uphill of "the stream, from which all water in the camp is obtained."

Clara Barton

Who: A clerk at the US Patent Office who took supplies to Civil War battlefields and later founded the American Red Cross.

When and where: Antietam, Maryland, September 17, 1862. Born in North Oxford, Massachusetts, on December 25, 1821.

A bullet passed through her sleeve and killed the soldier she helped.

Clara Barton brought a wagon of bandages and supplies to the battlefield at Antietam, Maryland, on September 17, 1862. She had spent a year collecting them. At a farmhouse she held wounded men's heads, gave them water and cooked for them. The lanterns she brought let surgeons keep operating after dark.

She was giving a soldier a drink when the bullet went through her sleeve. She wrote that it cut "through his chest from shoulder to shoulder." A surgeon there, Dr. James Dunn, called her "the angel of the battlefield."

Before the war Barton had been a teacher and then a clerk. She and others formed the American Association of the Red Cross, the group now called the American Red Cross, in Washington, D.C., on May 21, 1881. She was its president until 1904, when she resigned after growing criticism.

Hannah Ropes

Who: The head nurse of a Civil War hospital in Washington, who reported a thief on the staff to the Secretary of War and died of typhoid caught at work.

When and where: Union Hotel Hospital, Georgetown, Washington, D.C., 1862 and 1863. Born in 1809 in New Gloucester, in what is now Maine.

She took her complaint about a thief to the Secretary of War.

Hannah Ropes ran the nurses at a Civil War hospital in Washington until typhoid killed her in January 1863. Her son Edward was a Union soldier. In 1862 she volunteered as a nurse. She became matron, the woman in charge of the nurses, at the Union Hotel Hospital. It was an old hotel turned into a hospital for 300 to 400 patients.

The writer Louisa May Alcott nursed under her from December 1862. "A more perfect pestilence-box than this house I never saw," Alcott wrote in her diary, a pestilence being a deadly disease that spreads.

Ropes found that the hospital steward, the man in charge of supplies, was stealing supplies meant for the wounded. The chief surgeon ignored her complaints. She took them to the Secretary of War, Edwin Stanton, and the steward and the surgeon were both arrested. The accounts of her life do not say who arrested them.

She caught typhoid in late December 1862 and died on January 20, 1863. Alcott fell sick in the same outbreak and went home on January 21, 1863. Ropes's diary and letters were printed as a book in 1980.

Smallpox among freed people, 1862 to 1868

Smallpox killed more than 60,000 people who had just been freed from slavery from 1862 to 1868, the historian Jim Downs counts. Women and children died at higher rates than men.

During the war, people who escaped slavery came to camps the Union army set up for them. The army called them contraband camps, contraband meaning goods taken from an enemy in war. The camps were crowded and dirty, and many people in them lacked the basic things they needed to stay alive. Smallpox broke out in Washington, D.C., in 1862 and spread to nearby Alexandria, Virginia.

Doctors already knew how to stop smallpox. They could vaccinate people and keep the sick apart from the healthy. Union army officers and officials of the Freedmen's Bureau, a federal office set up to help people freed from slavery, ran the camps. They put sick freed people in tents made of old, ripped army tarps, with blankets full of insects. Sick white soldiers went to army hospitals. The accounts of the epidemic do not name the men in charge.

Harriet Jacobs, who had escaped slavery herself, came to Alexandria to open a school and worked against the epidemic.

Health boards and yellow fever, 1866 to 1879

New York State lawmakers passed a law in February 1866, and Governor Reuben Fenton signed it, giving New York City a Metropolitan Board of Health. The doctor Stephen Smith had worked to get it passed. By April its officers had given 7,000 orders to clear away piled horse manure, dead animals and heaps of garbage. Crews spread chloride and lime, chemicals that kill germs, where cholera struck. Cholera killed 1,137 New Yorkers in 1866. That was a smaller share of the city than in 1832 or 1849.

Yellow fever spread up the Mississippi River from New Orleans in 1878. It began after the ship *Emily B. Souder* arrived in New Orleans in late May, and crew members fell ill and died. The first official yellow fever death in the city that year was a four-year-old girl, in July. Along the river about 120,000 people caught the fever and about 20,000 died.

In Memphis, Tennessee, about 25,000 to 27,000 of the city's 47,000 people fled. More than 5,000 people died in the city. Annie Cook ran a brothel, a house where men paid women for sex. She turned it into a hospital, nursed the sick, and died of the fever.

Members of Congress set up a National Board of Health in 1879. Nobody proved that a mosquito spreads yellow fever until 1900.

Germs and the death of President Garfield, 1881

The English surgeon Joseph Lister urged doctors to kill germs on their hands and tools, to stop wounds from getting infected. Most American doctors thought this was too much trouble.

Charles Guiteau shot President James Garfield at a railroad station in Washington, D.C., on July 2, 1881. Doctors pushed their unwashed fingers and unwashed tools into the wound to search for the bullet. Dr. Willard Bliss, who took charge, probed the wound several times a day and never cleaned his hands first. One doctor punctured Garfield's liver while probing. The accounts of Garfield's care do not name him.

Garfield died on September 19, 1881. Doctors who examined his body found pneumonia in both lungs and his body full of pus. A medical paper of 2013 argues instead that a burst artery near his spleen killed him. An artery is a large tube that carries blood from the heart, and the spleen is an organ beside the stomach. Candice Millard, who wrote a 2011 book about Garfield's death, said in an interview that Garfield "would have been better off if they had just left him alone."

Women become nurses and doctors

Mary Eliza Mahoney had worked at the New England Hospital for Women and Children in Boston for 15 years before she began its nursing course. The hospital had only women doctors, and the course took 16 months. Of 42 students who began in 1878, four finished in 1879, and Mahoney was one of them.

She became the first Black woman in the United States to earn a professional nursing license. In 1908 she helped found a group for Black nurses, the National Association of Colored Graduate Nurses ("Colored" was a word then used for Black people).

Susan La Flesche Picotte

Who: An Omaha woman who became the first Native American woman to earn a medical degree, and went home to be the doctor for her own nation.

When and where: The Omaha Reservation, Nebraska. Born 1865, graduated 1889, died 1915.

As a child she watched a white doctor refuse to treat a dying woman.

The woman was a sick Native woman, and she died because the local white doctor would not treat her. Susan La Flesche had been born in 1865 on the Omaha Reservation in northeastern Nebraska. Her father was Chief Joseph La Flesche, and her mother was Mary, also called One Woman.

She studied at the Woman's Medical College of Pennsylvania. She finished its three-year course in two years and graduated first in her class. In 1889 she became the first Native American woman in the United States to earn a medical degree.

> Back on the reservation she cared for more than 1,300 people spread over 450 square miles. In 1913 she opened a hospital at Walthill, Nebraska, on the reservation. She died in 1915.

Typhoid in the army camps, 1898

During the war with Spain in 1898, soldiers in army camps in the United States caught typhoid. There were 20,738 cases and 1,590 deaths in the camps. Typhoid caused 87 of every 100 deaths from disease in those camps. Major Walter Reed and two other army doctors, Victor Vaughan and Edward Shakespeare, studied the camps. They found that the disease spread mostly from man to man, and some of it by flies.

1900 to 1950

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The first American patient saved by penicillin, a drug made from a mold that kills bacteria, was Anne Miller. She got her first dose in 1942. In the same years, government doctors kept that same drug from Black men in Alabama who had syphilis. Army doctors proved in 1900 that a mosquito spreads yellow fever.

An influenza epidemic, a flu that spread to huge numbers of people at once, killed about 675,000 people in the United States in 1918. A medicine mixed with a poison killed more than 100 people in 1937, and in 1938 a new law required drug makers to prove a drug safe before selling it.

Doctors of the United States Public Health Service began their study of 399 Black men in Alabama who had syphilis in 1932. For 40 years the doctors kept the men from being treated, while telling them they were being treated. Public Health Service doctors also exposed 1,308 people in Guatemala to diseases passed during sex, on purpose, from 1946 to 1948.

Proving the mosquito, 1900 and 1901

Four army doctors near Havana, Cuba, proved in 1900 that mosquitoes pass yellow fever, a disease caused by a virus, from sick people to healthy ones. Major Walter Reed led the four, and the others were James Carroll, Jesse Lazear and Aristides Agramonte.

Lazear raised mosquitoes and let them feed on yellow fever patients. A mosquito that had fed on a patient at least 12 days before passed the disease on when it bit someone else. Carroll and a soldier who volunteered caught yellow fever from these bites in late August. Lazear was probably bitten by an infected mosquito himself, and he died of yellow fever on September 25, 1900.

Two buildings. The doctors tested two buildings in November 1900. In one, volunteers slept in bedding soiled by yellow fever patients, and they stayed well. In the other, a clean building, infected mosquitoes bit volunteers, and several caught yellow fever. All of them recovered. The volunteers signed written contracts that named the danger, and they were paid. After workers killed the mosquitoes in Havana, yellow fever there nearly stopped.

Clara Maass

Who: A New Jersey nurse who died after letting infected mosquitoes bite her in a yellow fever experiment.

When and where: Havana, Cuba, 1901. Born in East Orange, New Jersey, in 1876.

She let infected mosquitoes bite her twice.

Clara Maass died of yellow fever at 25, after she let infected mosquitoes bite her in Havana in 1901. She came from a poor German immigrant family in East Orange, New Jersey. She started training as a nurse at 17, at the Newark German Hospital, and she was its head nurse by 21. She nursed soldiers in the 1898 war with Spain and then in the Philippines.

In Havana volunteers were paid $100, and $200 if they caught the fever. Maass let infected mosquitoes bite her in March 1901. She had a mild case and got well. In August 1901 she volunteered again, caught yellow fever, and died on August 24.

Soon after her death, the yellow fever commission was ordered to break up. The museum account of her life does not name who gave the order.

Mary Mallon, held on an island

Mary Mallon was a cook in New York who carried the germs of typhoid, a fever spread through food and water, in her body without being sick herself. Health officials counted 47 people infected through her cooking, and 3 of them died.

Officials of the New York City Board of Health held her on North Brother Island, in the East River, from 1907. In a letter in June 1909, she wrote that they kept her "a prisoner without being sick nor needing medical treatment." She also wrote, "I have been in fact a peep show for everybody."

The officials let her go in February 1910, after she signed a promise not to work as a cook again. In October 1914 she took a job as a cook at the Sloane Maternity Hospital, under the name Mary Brown. Early in 1915 typhoid broke out there, and 25 people caught it and 2 died. Health officials found her in March 1915 and sent her back to the island. She stayed there until she died on November 11, 1938, and she became known as "Typhoid Mary."

Fewer medical schools, 1910

Abraham Flexner published a survey of the country's 148 medical schools in 1910. Leaders of the Carnegie Foundation had hired him to do it. He called several schools "beyond repair." In the years after, the number of medical schools fell to 66.

Writers at JSTOR Daily, a history magazine, say that members of the American Medical Association, most of them white men, used the report to close schools and protect their own jobs and earnings. Seven medical schools had trained Black doctors. Only two were left: Howard University in Washington, D.C., and Meharry Medical College in Nashville, Tennessee.

Polio in New York, 1916

Polio is a disease caused by a virus that can attack the nerves in the spine. It can leave a person unable to move the arms, the legs or the muscles used for breathing, a loss of movement called paralysis.

In the summer of 1916 polio spread through New York City. More than 9 of every 10 of the city's cases were in children five or younger. One count gives more than 27,000 cases and more than 6,000 deaths across the whole country. Accounts of New York City alone differ, from about 2,000 deaths to about 6,000.

Dr. Haven Emerson was the city's health commissioner. Every day the names and addresses of the sick appeared in the newspapers. Signs went up on their houses, and the accounts of the epidemic do not say who put them up.

Shut in. A team of Health Department workers led by Dr. Simon Blatteis kept the families shut inside. Under Health Department rules, a family that could not give the sick child a separate room, a separate toilet and a nurse had to let the department take the child to a hospital, free of charge. Police stood at the railroads, roads and rivers to turn back anyone trying to leave the city.

Writers in the *New York Times* blamed Italian immigrants for the disease. Italian immigrants were shunned, cursed and beaten up, and the accounts of the epidemic do not name who beat them.

The 1918 influenza

Influenza, or flu, is a disease caused by a virus that infects the nose, throat and lungs. In 1918 a new kind of influenza spread around the world. About 500 million people caught it, and at least 50 million died. About 675,000 of the dead were in the United States. Many of them were healthy adults between 20 and 40.

There was no vaccine against it, no medicine that trains the body to fight off a germ before the germ can make a person sick. There was no antibiotic either, a medicine that kills bacteria. Many people died of pneumonia, with their lungs full of fluid.

The parade. In Philadelphia, Dr. Wilmer Krusen, the city's director of public health, let a parade to sell war bonds go ahead on September 28, 1918. People bought the bonds to lend money to the government for the war. About 200,000 people crowded onto Broad Street. Krusen said in public that the flu was not a threat, although army camps nearby were full of it.

Within three days every bed in the city's 31 hospitals was full. City officials closed schools, churches and theaters on October 3. More than 12,000 people in Philadelphia died within weeks.

The flu reached Brevig Mission, an Alaska Native village, in 1918. Within five days, 72 of the 80 villagers were dead. Miners from the town of Nome thawed the frozen ground with steam and dug one grave for all of them.

Josie Mabel Brown

Who: A Navy nurse from Missouri who worked the influenza wards at Great Lakes, Illinois, and described them in an interview when she was about 100.

When and where: Great Lakes Naval Training Station, Illinois, fall 1918 to 1919. Born on a Missouri farm in 1886.

The ambulance brought four sick men and took away four dead.

Josie Mabel Brown nursed sailors dying of influenza at the Great Lakes Naval Training Station in Illinois in 1918. At the worst of the epidemic, 6,000 of the 173,000 men there were in the hospitals. She had finished nurse training in 1917 and been called into the Navy, and in the fall of 1918 she was ordered to Great Lakes.

On her first ward, one man lay dying on the bed and another lay on the floor. A third waited on a stretcher for the bed. The nurses wrapped each dead man in a sheet and tied a shipping tag to one big toe. The ambulance, she said, "would bring us four live ones and take out four dead ones."

The nurses had no time to take temperatures. Patients had nosebleeds so strong that "the blood would just shoot across the room," she said. In some men, air leaked from torn lungs into the body, and she could see bubbles under the skin of their arms.

She worked 16 hours a day until just before Christmas 1918. In March 1919 she caught a fever of 104 or 105 herself, and she recovered. She told her story to her niece in 1986 and died at 104.

BY 1920: Army doctors had proved that mosquitoes carry yellow fever, and influenza had just killed about 675,000 Americans. Doctors still had no vaccine against the flu and no drug that killed bacteria.

Radium in the body, 1917 to 1932

Radium is a metal that gives off radiation, a stream of energy that damages living cells. Radium belongs to the same family of elements as calcium, so growing bones take it up as if it were calcium. There it keeps giving off radiation from the inside.

Women painted watch dials with glowing radium paint at the U.S. Radium factory in Orange, New Jersey, from 1917. They shaped their brushes into a point with their lips. The radium destroyed their bone marrow, the soft tissue inside bones that makes blood, so their bodies could not make enough blood. Their jaw bones died and broke apart. Some grew cancers of the bone. In 1924 a dentist, Theodor Blum, examined the dial painter Grace Fryer. Her jaw bone was wasting away, and he called the damage "radium jaw."

Chapter 13, The Elements, tells the dial painters' story and their lawsuit in full. Five of them, Grace Fryer among them, signed an agreement with the company on June 8, 1928. Each woman received $10,000, though some accounts give $15,000, plus $600 a year for as long as she lived, and her medical costs. The company's officers never admitted that the company was to blame.

A radium medicine. William Bailey sold radium dissolved in water as a medicine called Radithor. He falsely claimed to have medical training and paid doctors to tell their patients to take it. Eben Byers, a Pittsburgh businessman, drank about 1,400 bottles from 1927 to 1930. A government investigator found in 1931 that the bone in Byers's body was falling apart and holes were forming in his skull. He died on March 31, 1932, of a cancer of the jaw bone caused by the radium.

Native health, 1928

A team of investigators led by Lewis Meriam reported in 1928 on the lives of Native people on reservations and in government boarding schools. They wrote that "the health of the Indians as compared with that of the general population is bad." They wrote that the share of Native people who died each year was high, and so was the share of babies who died.

The investigators found tuberculosis, the lung disease that spreads by coughing, "extremely prevalent," which means very common. They found trachoma, an eye disease that "produces blindness," to be "a major problem." In the government boarding schools the children's food was "deficient in quantity, quality, and variety": there was not enough of it, and it was poor and the same every day. "The medical service rendered the boarding school children is not up to a reasonable standard," they wrote, meaning the care doctors gave the children was not good enough.

The Tuskegee study, 1932 to 1972

Doctors of the United States Public Health Service watched hundreds of Black men in Alabama live with syphilis for 40 years, from 1932 to 1972, and they kept the men from being treated. They wanted to see what the disease did to a body when no one treated it.

What syphilis does. Syphilis is a disease caused by bacteria. It is passed during sex, or from a mother to her unborn baby. Left untreated, it goes through stages. At first the person gets a painless sore where the germ entered. Weeks later the person gets a rash, a fever and a sore throat.

Then the germ can hide in the body for years with no signs. After 10 to 30 years it can damage the heart and the large blood vessels, the brain and the nerves. It can cause blindness, and it can cause dementia, which is the loss of memory and reason. It can kill, and a baby born with syphilis can die.

Who ran it. Dr. Taliaferro Clark began the study in 1932. He was head of the Public Health Service's office for diseases passed by sex. The study took place in Macon County, Alabama, near the town of Tuskegee. Dr. Raymond

Vonderlehr ran it there and later took Clark's place. Most of the men were poor farmers. Many were sharecroppers, who farmed someone else's land for a share of the crop.

The doctors chose 399 men who had syphilis and 201 men who did not, to compare them. They planned no treatment at all for the infected men. The study was planned to last six months, and it ran for 40 years.

The lie. The doctors told the men they had "bad blood" and that they were being treated for it. They did not tell the men that the disease was syphilis, and they never asked whether the men agreed to be studied. The men got free exams, hot meals on exam days and rides. From 1935 they were also promised about $50 toward their burials, if they let doctors cut open their bodies after death to study them.

Eunice Rivers, a Black nurse, kept in touch with the men for all 40 years. She gave them aspirin and a "spring tonic" that did nothing against syphilis. Dr. Oliver Wenger, who ran the clinic in the region, wrote, "we have no further interest in these patients until they die."

The exams also included spinal taps. In a spinal tap a doctor pushes a needle between the bones of the lower back and draws out fluid from around the spinal cord. The doctors wanted to see whether syphilis had reached the brain and nerves. A spinal tap often causes a lot of pain and other harm. The doctors' letter to the men called it their "Last Chance For Special Free Treatment."

Keeping the cure away. Keeping treatment away from the men was part of the study. Its doctors gave local doctors lists of the men's names and asked them not to treat them. In 1941 army officials drafted some of the men and told them to start treatment. Public Health Service officers gave the draft board, the group that chose men for the army, a list of 256 names to leave untreated, and the board members did as they asked.

Accounts differ on when penicillin, the drug made from a mold, became the usual cure for syphilis: 1943, by 1947, or the early 1950s. The study doctors never offered it to the men. Vonderlehr later wrote, "I hope that the availability of antibiotics has not interfered too much with this project."

What their own reports show. Reports that the study doctors printed show what the disease did to the men. In 1936 the doctors reported that the infected men had much more disease of the heart and blood vessels than the men without syphilis. Only 16 of every 100 infected men showed no sign of illness, against 61 of every 100 of the uninfected men.

About ten years later, nearly twice as many infected men as uninfected men had died. The doctors worked out that the disease had cut the men's lives by about a fifth.

Their 1955 report covered the infected men whose bodies they had cut open after death. More than 30 of every 100 had died because syphilis destroyed their heart and blood vessels or their brain and spinal cord.

A reporter made the study public in 1972. By then, by one count, at least 28 of the men had died of syphilis itself and 100 more of problems it caused. Forty of their wives had caught it. Nineteen of their children were born with it.

Charles Pollard

Who: An Alabama farmer who was kept in the Tuskegee study for 40 years without being told he had syphilis, and who then brought the lawsuit for the men.
When and where: Near Tuskegee, Macon County, Alabama, 1932 to 1972.

At 66, he learned from the news that he had syphilis.

That was in July 1972, and Charles Pollard had had syphilis for 40 years. He farmed 66 acres of land he had inherited, about three miles from Tuskegee. His father was the son of two enslaved people.

Men from the study told Pollard in 1932 that he could get a free physical exam at a one-room school the next afternoon. "So I went on over and they told me I had bad blood," he said. "And that's what they've been telling me ever since." For 40 years doctors came around "from time to time and check me over."

> Pollard went to Fred Gray, a lawyer in Tuskegee. Their lawsuit, *Pollard v. United States*, ended in a settlement of $10 million. In 1997 Pollard was one of the survivors at the White House when President Bill Clinton apologized to them.

A poison sold as medicine, and a new law, 1937 and 1938

A poison in a new medicine killed more than 100 people in 1937, and the next year a new law required drug makers to prove a drug safe before selling it. The S. E. Massengill Company of Bristol, Tennessee, sold a liquid medicine called Elixir Sulfanilamide for strep throat. Its chemist, Harold Watkins, dissolved the drug in diethylene glycol, a chemical used in antifreeze and a poison. No law required the company to test the medicine for safety before selling it.

The medicine killed more than 100 people in 15 states in September and October 1937, many of them children. It shut down their kidneys. They stopped passing urine and had strong stomach pain, vomiting and seizures before they died.

A mother wrote to President Franklin Roosevelt about her daughter Joan: "we can see her little body tossing to and fro and hear that little voice screaming with pain." The owner, Dr. Samuel Massengill, said, "I do not feel that there was any responsibility on our part." Watkins killed himself.

President Roosevelt signed the Federal Food, Drug, and Cosmetic Act on June 25, 1938, after members of Congress passed it. Under it, a company had to prove to officials of the Food and Drug Administration, the FDA, that a new drug was safe before it could sell the drug.

Penicillin, 1942

Penicillin is a drug made from a mold. It kills many kinds of bacteria, the germs behind infections such as blood infections and syphilis. It was the first antibiotic made in large amounts.

Anne Miller of New Haven, Connecticut, had an infection of the blood in March 1942, after a miscarriage, the loss of a baby before it can live outside the womb. For weeks her fever stayed at 103 degrees or higher. Her doctors got 5.5 grams of penicillin, half of all the penicillin in the United States at the time. She got her first dose on March 14. By the next morning her temperature was back to normal. She was the first patient in the United States whose life was saved by penicillin.

Exposed on purpose in Guatemala, 1946 to 1948

Doctors of the United States Public Health Service, led by Dr. John Cutler, gave people in Guatemala syphilis, gonorrhea and chancroid on purpose, or tried to, from 1946 to 1948. All three are diseases passed during sex. The doctors infected people without asking whether they agreed.

The doctors exposed 1,308 people to these diseases: prisoners, soldiers and patients in the national mental hospital. They ranged from 10 to 72 years old. The records show some kind of treatment for 678 of them.

How they did it. The doctors used women who were paid for sex to pass on the diseases. They infected most of these women first, then sent them to the prisoners. They also rubbed pus from sick men painfully onto people's genitals and into the opening where urine leaves the body. They injected germs into people's spinal fluid.

For tests, the doctors also took blood or spinal fluid from 5,128 people, among them children in orphanages and schools and people with leprosy, a disease caused by bacteria that damages the skin and nerves.

The money came from the Public Health Service, the National Institutes of Health and the Pan American Sanitary Bureau. Guatemalan officials supplied staff and buildings. Cutler's records note 83 deaths while the experiments went

on. A presidential commission studied the experiments in 2011 and found it unclear how many of those deaths the experiments caused. The doctors never published their results. The public learned of the experiments in 2010.

Federal money for hospitals, 1946

The Hospital Survey and Construction Act became law on August 13, 1946. People call it the Hill-Burton Act, after the two senators who put it forward, Lister Hill of Alabama and Harold Burton of Ohio. Over the years, federal lawmakers set aside $3.7 billion under it to build hospitals, and states and towns added $9.1 billion.

Under the law, the people who ran hospitals built with the money could keep Black and white patients apart, as long as the separate care was called equal. Accounts differ on who asked for that rule. Judges struck it down in 1963.

Pellagra, silica dust and forced sterilization

A disease of diet. Pellagra killed about 100,000 Americans between 1900 and 1940, out of about 3 million who had it, about 1 in 30. Pellagra is a disease caused by too little niacin, a vitamin, in the diet. Most of the sick were poor farmers, sharecroppers and cotton-mill workers in the South who lived on little but cornmeal, molasses and dried pork. Scientists identified niacin in 1937.

Silica dust. Almost 5,000 workers dug the Hawks Nest tunnel in West Virginia from 1930 to 1932, and Chapter 14, Work and Workers, tells its story in full. It was the Union Carbide company's tunnel, and the contractor was Rinehart and Dennis. Most of the workers were Black men who had come from other parts of the South. The rock was full of silica, the mineral in sand.

The men drilled through it with no masks and no way to control the dust. The West Virginia Encyclopedia's account of the tunnel does not name the company men who made that choice. Breathing silica's fine dust scars the lungs until a person cannot breathe, a disease called silicosis. Death counts run from 476 to close to 1,000.

Forced sterilization in Puerto Rico. Eugenics was the false idea that a country could improve its people by stopping some people from having children. In Puerto Rico, under a law that took effect in 1937, members of a government Eugenics Board ordered 97 women sterilized by force. Sterilized means that a person can never become pregnant. Doctors did it by cutting or tying the tubes that carry eggs to the womb, or by removing the womb.

Many thousands of other women were pressured into the same operation. Clarence Gamble, an heir to the Procter and Gamble soap fortune, set up birth control clinics in Puerto Rico with money from the Puerto Rico Reconstruction Administration, a federal office. Gamble made sterilization a rule in his clinics. A woman with three living children could be seen there only if she agreed to be sterilized, historians on the history podcast DIG report.

Some women were told the operation could be undone. Factory owners hired sterilized women first, and some women were told they would not get a job unless they were sterilized. Some employers opened clinics inside their factories where workers could be sterilized. Pregnant women were sometimes turned away from hospitals unless they agreed to be sterilized right after giving birth. The historians' accounts do not name the factories or the hospitals.

BY 1950

If you remember one thing: Doctors learned how to stop yellow fever and got penicillin to cure infections, and in the same years government doctors kept that cure from the Tuskegee men and infected people in Guatemala on purpose.

- 1900: Army doctors near Havana prove that mosquitoes spread yellow fever.
- 1932: Public Health Service doctors begin the Tuskegee study of 399 Black men with syphilis.
- 1942: Penicillin saves Anne Miller's life in New Haven.

Next: After 1950 a vaccine ended polio in the United States, and the Tuskegee study went on until a reporter made it public.

1950 to 2000

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A vaccine ended polio in the United States in these fifty years, and new federal insurance paid for the care of older and poorer Americans, while reporters, judges and investigators exposed what doctors had done to people who never agreed. Doctors reported more than 21,000 Americans paralyzed by polio in 1952. A vaccine tested on about 1.8 million children was announced in 1955, and the last polio case caught inside the United States came in 1979.

President Lyndon Johnson signed Medicare, health insurance for people 65 and older, and Medicaid, for people with little money, in 1965. A reporter exposed the Tuskegee study in 1972, and a federal health official announced its end that November.

In the 1970s doctors in federally paid clinics sterilized poor Black women and girls, and doctors in the Indian Health Service sterilized Native women. Many of these women and girls never gave permission. Doctors reported the first cases of the disease later named AIDS in 1981. By the end of 2000, 448,060 Americans had died of it.

Henrietta Lacks

Who: A Black woman whose cancer cells doctors took without asking her. Her cells were the first human cells to keep growing in a laboratory.

When and where: Johns Hopkins Hospital, Baltimore, 1951. Born Loretta Pleasant in Roanoke, Virginia, on August 1, 1920.

Movie: *The Immortal Life of Henrietta Lacks* (2017), a drama made for HBO about her and her daughter Deborah.

Doctors cut two samples from her cervix without asking her.

Doctors at Johns Hopkins Hospital in Baltimore cut cells from Henrietta Lacks's body in 1951 without her permission or knowledge. Her cells were the first human cells to keep growing and dividing in a laboratory. At that time no rule required doctors to ask.

She grew up in Clover, Virginia, in a log cabin that had been slave quarters on a plantation. Her white great-grandfather and great-uncle had owned it. In January 1951 she went to Johns Hopkins with a lump in her womb. Doctors found cancer of the cervix, the opening of the womb. They treated her with radium placed inside her body, and then with X-rays. During her treatments, doctors cut the two samples.

Dr. George Gey grew the cells. They kept dividing, and they were named HeLa, from the first two letters of her first and last names. By 1954 Jonas Salk's team used HeLa cells in its polio vaccine work.

Another doctor, Chester Southam, injected HeLa cells into cancer patients, prisoners and healthy people to see whether cancer could be passed on. He did not tell the people he injected that the cells were cancer cells, and prisoners were hardly free to refuse.

The members of New York's Board of Regents, the state board that licenses doctors, later found Southam guilty of fraud, deceit and unprofessional conduct. Those are legal words for cheating, lying and breaking the rules of his profession. They put him on probation, a trial period under watch, for a year.

> She died on October 4, 1951, at 31, after the cancer spread through her body. Her family did not learn about the cells until 1975. In 2021 her family sued a company, Thermo Fisher Scientific, for making money from HeLa cells without her permission, and the two sides settled in 2023.

The polio vaccine, 1952 to 1979

Doctors in the United States reported more than 21,000 people paralyzed by polio in 1952. Fewer than 1 in 100 children who catch polio are paralyzed. When polio paralyzes the muscles used for breathing, the person needs a machine to breathe. The iron lung was a metal tube about seven feet long, and the patient lay inside it. Bellows, a pump with folding sides, changed the air pressure inside the tube to pull air into the lungs.

Jonas Salk of the University of Pittsburgh made a vaccine from polio virus that had been killed. The money for his work came from the National Foundation for Infantile Paralysis, a charity that later became the March of Dimes.

The test. Dr. Thomas Francis Jr. of the University of Michigan ran the test in 1954. About 1.8 million children in the United States, Canada and Finland took part, counting children who got no shot and were only watched. From April 26 to July 10, 1954, more than 443,000 children got at least one shot of vaccine. More than 210,000 got a placebo, a shot with no medicine in it. Neither the child nor the nurse knew which one it was.

The children were called Polio Pioneers, and their parents signed permission forms. The results were announced at the University of Michigan on April 12, 1955: the vaccine was 80 to 90 percent effective. It was approved for use the same year.

The bad batch. That April, Cutter Laboratories in Berkeley, California, made 120,000 doses that still held live virus. About 40,000 children who got it had a mild form of polio. Fifty-six were paralyzed, and five died.

The children passed the virus to others, and 113 more people were paralyzed and 5 more died. Cutter's managers took the vaccine off the market on April 27, 1955. At a hearing that June, members of Congress said the staff of the

federal laboratory that was supposed to check the vaccine were mostly to blame, because they had not checked it closely.

Albert Sabin's vaccine, which people swallowed instead of getting a shot, was approved for use from 1961 to 1963. The last case of polio caught inside the United States was in 1979. Between 15 and 40 years after they got sick, 25 to 40 of every 100 people paralyzed by polio as children get new muscle pain and new weakness, which is called post-polio syndrome.

Jonas Salk

Who: The doctor who made the first polio vaccine and did not patent it.
When and where: University of Pittsburgh, 1947 to 1955. Born in New York City in 1914.

"Could you patent the sun?" he asked.

Jonas Salk made the polio vaccine that was approved for use in 1955. He never took out a patent on it, the legal right to be the only one allowed to make and sell an invention.

He started work at the University of Pittsburgh School of Medicine in 1947. In 1952 he made a vaccine from polio virus that had been killed, so that it could not cause the disease.

Thomas Francis Jr. ran the 1954 test. Salk had first learned about viruses in Francis's laboratory in 1941. At first Salk did not want some children to get a dummy shot, because the dummy shot could not protect them against polio. He gave way, because only that kind of test could prove the vaccine worked.

The television reporter Edward R. Murrow asked Salk who owned the patent, and the question about the sun was Salk's answer. People at the charity that paid for the work and at the university had looked into a patent. They judged that the method was probably not new enough to patent. Salk died in La Jolla, California, in 1995.

Paul Alexander

Who: A Dallas boy paralyzed by polio at six who lived in an iron lung for more than 70 years and became a lawyer.

When and where: Dallas, Texas, 1952 to 2024. Born in Dallas on January 30, 1946.

He lived inside an iron lung for more than 70 years.

Polio paralyzed Paul Alexander in 1952, during an outbreak in Dallas, when he was six. Within days he could not move below his neck or breathe on his own. Doctors saved him with a tracheotomy, an operation that cuts an opening in the front of the neck into the windpipe so that air can get in. Then they put him in an iron lung.

He learned to breathe outside the machine for hours at a time, starting in 1954, by swallowing air into his lungs. Every night he went back into it.

He held a stick in his mouth to turn pages, to paint, and later to type. He studied at Southern Methodist University and the University of Texas, earned a law degree in 1984, and worked as a lawyer in Dallas. In 2020 he published a book about his life, *Three Minutes for a Dog*.

"I've never thought of myself as a cripple," he said in 2017, using an old, unkind word for a person who cannot use part of the body. "I'm Paul Alexander, human being." Guinness World Records named him the longest-surviving iron lung patient. He died in Dallas on March 11, 2024, at 78.

The Indian Health Service, 1954 to 1976

President Dwight Eisenhower signed the Indian Health Transfer Act in 1954. Under it, Public Health Service officials took over health care for Native people from the Bureau of Indian Affairs, the federal office that ran the government's dealings with Native nations. The new office was called the Indian Health Service.

That same year Dr. Thomas Parran reported to the Interior Department on the health of Alaska Natives. He wrote that they suffered "sickness, crippling conditions and premature deaths to a degree exceeded in very few parts of the world," meaning that people in very few places anywhere were sicker or died younger. Health care for Alaska Natives moved to the Indian Health Service in 1955, and its staff led the work against tuberculosis.

Under a 1959 law, Public Health Service workers could bring clean water and ways to get rid of sewage to reservations, and tribes took part in the decisions. In 1976 President Gerald Ford signed the Indian Health Care Improvement Act. Under it, Medicare and Medicaid could pay the Indian Health Service for the care its staff gave.

BY 1955: Doctors had a polio vaccine that worked, and Public Health Service officials ran health care for Native people through the new Indian Health Service.

New operations and new pictures of the body, 1954 to 1963

Dr. Joseph Murray moved a healthy kidney from Ronald Herrick into his identical twin brother, Richard, at a Boston hospital on December 23, 1954. Richard was dying of kidney disease. The body's defenses attack an organ from almost anyone else, which is called rejection. Because the twins were identical, Richard's body accepted the kidney. He lived eight more years, married and had two children.

Scientists at Brookhaven National Laboratory in New York built the first machine to make technetium-99m, a radioactive chemical, in 1958. A doctor injects a tiny amount into a patient. It gives off rays that a camera outside the body picks up, to make pictures of the bones, the heart and other organs. Doctors in the United States first reported scans with it in 1963. Doctors around the world now use it in tens of millions of scans a year.

Fallout on Rongelap, 1954

United States scientists set off a hydrogen bomb called Castle Bravo at Bikini Atoll in the Marshall Islands on March 1, 1954. It was nearly three times as strong as they expected. Fallout, the radioactive dust and ash that falls after a nuclear blast, came down on Rongelap, an island of the Marshallese people about 110 miles away. It looked like snow, and children played in it.

Twenty-three Japanese fishermen on the boat *Lucky Dragon* got sick, and one died. More than 230 people were moved off Rongelap, Rongerik and Utirik. The Rongelap and Utirik people were moved about 72 hours after the blast. The accounts of the evacuation do not name who moved them or who decided when.

Radioactive iodine from the fallout collected in their thyroid glands, which sit in the neck. In the years after, doctors found more thyroid lumps and thyroid cancers among them. Staff of the Atomic Energy Commission, the federal office in charge of nuclear weapons and nuclear power, studied the exposed people. Doctors working for the United States government later gave them follow-up medical care.

Radiation tested on people, 1944 to 1974

President Bill Clinton appointed a committee in 1994 to investigate radiation experiments paid for by the federal government. He acted after reports that doctors had injected plutonium, a radioactive metal, into hospital patients who did not know it. In its 1995 report the committee counted nearly 4,000 such experiments from 1944 to 1974. Researchers paid by the federal government had also run experiments on children and on prisoners.

The committee wrote that in the 1940s and 1950s it was common for doctors to use patients in research "without their awareness or consent." In some studies on children, the radioactive doses raised the children's lifetime risk of thyroid cancer. The committee also found studies in which patients died soon after receiving radiation doses as large as those given to treat disease.

Miners who dug uranium underground breathed radon, a radioactive gas that comes off uranium ore. At least several hundred of them died of lung cancer. The committee found that federal officials neither required the mines to be ventilated, which means fresh air pumped in, nor properly warned the miners. Under a 1990 law, the Radiation Exposure Compensation Act, some of the miners get payments from the government.

Thalidomide, 1960 to 1962

Dr. Frances Kelsey, a new FDA reviewer, refused to approve thalidomide because the company's evidence was weak. The drug company Richardson-Merrell had asked the FDA for permission to sell it in 1960. It was a sleeping pill that doctors also gave pregnant women for morning sickness, and it was already sold in more than 20 other countries. For more than a year, under steady pressure from the company, Kelsey kept asking for more.

Doctors in Europe linked thalidomide in November 1961 to babies born with arms and legs that were very short or missing. Around the world more than 10,000 babies were born harmed by it, and about 40 of every 100 died at birth or soon after.

Richardson-Merrell staff had already given more than 2.5 million tablets to more than 1,000 American doctors to "test." Nearly 20,000 patients took them, and several hundred of those patients were pregnant. At least 17 American babies were born with thalidomide damage.

Company officials withdrew the application in March 1962. Members of Congress passed the Kefauver-Harris Amendments with no votes against in October 1962. Drug makers then had to prove that a drug worked, report its harms to the FDA, and get people's permission before testing drugs on them.

Smoking and asbestos, 1954 to 1998

Surgeon General Luther Terry, the country's top public health doctor, released a report on smoking on January 11, 1964. Ten experts had written it after reading more than 7,000 articles. They found that smokers died at a rate 70 percent higher than non-smokers. The average smoker was 9 to 10 times as likely to get lung cancer, and a heavy smoker at least 20 times.

Smoking was the main cause of chronic bronchitis, a long-lasting swelling of the air passages, the experts found, and it was linked to emphysema and to heart disease.

Ten years earlier, in January 1954, the big tobacco companies had paid for an advertisement in more than 400 newspapers that disputed the studies linking cigarettes to lung cancer. In 1957 Surgeon General Leroy Burney said the evidence showed that smoking caused lung cancer.

Lung cancer grows in the cells that line the air passages of the lungs. It causes a cough that does not go away, trouble breathing, and coughing up blood. Smoking causes about 9 of every 10 lung cancers in men and 8 of every 10 in women. In emphysema, a lung disease, the thin walls between the lungs' tiny air sacs break down, so the person cannot get enough air.

Under a 1965 law, every pack of cigarettes had to carry a health warning. From 1970 cigarette ads were banned from television and radio. On April 14, 1994, the heads of seven tobacco companies told Congress that nicotine is not addictive. In a 1998 settlement, a deal to end lawsuits, the tobacco companies agreed to pay 46 states at least $206 billion over 25 years.

Fibers in the lungs. Asbestos is a group of minerals that split into thin fibers. Workers used it to insulate buildings and ships. Its fibers, breathed in, stay in the lungs and scar them. They cause lung cancer and mesothelioma, a cancer of the thin lining of the chest and belly. The sickness can start 10 to 40 years or more after the person breathed the dust.

Shipyard workers, insulation workers, miners and firefighters were among those who breathed it. Johns-Manville, the biggest maker of asbestos, went bankrupt in 1982 while facing thousands of lawsuits.

BY 1965: Surgeon General Luther Terry's report had found smokers 9 to 10 times as likely to get lung cancer, and members of Congress had made drug makers prove that their drugs worked. The Tuskegee study was still running.

Medicare and Medicaid, 1965 and 1966

Two thirds of Americans over 65 had incomes below $1,000 a year in 1950, and only 1 of every 8 had health insurance, a plan that pays doctor and hospital bills.

President Lyndon Johnson signed Medicare and Medicaid into law at the Truman Library in Independence, Missouri, on July 30, 1965. Former President Harry Truman was with him. Truman had asked Congress for national health insurance 20 years earlier. Medicare is health insurance for people 65 and older. Medicaid is health insurance for people with little money. Nearly 20 million people signed up for Medicare in its first three years.

Separate wards. Many hospitals still kept Black and white patients apart. In the South, the people who ran hospitals mostly put Black patients in separate basement wards or separate buildings, or turned them away. In many northern cities, hospital care was just as segregated through unwritten rules.

Judges of a federal appeals court ruled in 1963, in the case *Simkins v. Moses H. Cone Memorial Hospital*, that hospitals built with Hill-Burton money could not keep patients apart by race. The justices of the Supreme Court refused to take up the case in 1964.

The inspectors. The Civil Rights Act of 1964 barred federal money for any place that treated people differently by race, and Medicare payments to hospitals were set to begin on July 1, 1966. Before then, almost 1,000 federal workers volunteered to inspect about 6,000 hospitals. They checked beds, waiting rooms, lunch rooms and hiring.

Inspectors drove at high speed to get away from local groups of the Ku Klux Klan, a secret white group that used terror against Black people and their supporters. Some inspectors were jailed on false charges. Bullets were fired at their rental cars, and crosses were burned on the lawns of at least two of them. Several hospital workers who helped the inspectors had their homes bombed,

and one may have been murdered, the historian David Barton Smith wrote. Smith's account does not name who jailed the inspectors, who fired, who burned the crosses or who set the bombs.

By July 1, hospitals holding 97 of every 100 beds for short-term care in the country had met the rule.

Hepatitis experiments at Willowbrook, 1956 to 1971

Willowbrook State School on Staten Island, New York, was a state home for children with intellectual disabilities. It was built for 4,000 children and held 6,000 by 1965.

Dr. Saul Krugman of New York University and Dr. Robert McCollum of Yale ran experiments with hepatitis, a disease that inflames the liver, on children at Willowbrook from 1956 to 1971. In one study they fed 60 healthy children live hepatitis virus taken from other people's stool, their solid waste. The children's skin and eyes turned yellow, their livers swelled, and they vomited and would not eat. Every one of them became ill, some badly.

Parents signed permission letters, often because signing got their child a place in the crowded school. Critics said the letters played down that the children would be infected on purpose. At times the only beds open were in the study ward.

The television reporter Geraldo Rivera filmed the school's crowded wards and filthy bathrooms in 1972, and staff who beat and sexually abused the residents. That March, parents of 5,000 residents sued the State of New York. The last children left Willowbrook in 1987.

The Tuskegee study ends, 1972 to 1997

The Tuskegee study ended in November 1972, four months after a reporter made it public. Peter Buxtun, a Public Health Service worker in San Francisco, had written in 1966 to the head of the service's office for diseases passed by sex, saying the study was wrong.

Officials of the CDC, the federal disease agency, ran the study by then. They decided to keep it going until every man had died. They also decided that doctors should then cut open each man's body and examine it. Members of local chapters of two doctors' groups, the National Medical Association and the American Medical Association, backed that decision. At a review in 1969 CDC officials decided again to keep it going.

The story breaks. In the early 1970s Buxtun went to the press. Jean Heller, a reporter for the Associated Press, wrote the story. It ran in the *Washington Star* on July 25, 1972, and on the front page of the *New York Times* the next day.

At the start of that year, 74 of the untreated men were still alive. In October a government panel advised stopping the study. In November 1972 Dr. Merlin DuVal, the top health official in the federal health department, announced the end of the study.

Senator Edward Kennedy held hearings in 1973, and federal officials began paying for the survivors' medical care. Their wives, widows and children were added in 1975. The men's lawsuit ended in a $10 million settlement out of court in 1974. Some accounts date it 1973.

The new rule. Members of Congress passed the National Research Act in 1974. Since then, a study on people that uses federal money must be approved by a review board. Every person in it must first be told what will be done and what the risks are, and must agree, which is called informed consent.

President Bill Clinton apologized at the White House on May 16, 1997, to five of the survivors, Charles "Charlie" Pollard, Herman Shaw, Carter Howard, Fred Simmons and Frederick Moss. "What the United States government did was shameful, and I am sorry," he said.

Sterilized without consent, the 1970s

To sterilize a person is to operate so that the person can never have a baby. In an operation called a tubal ligation, a surgeon cuts or ties the two tubes that carry eggs from the ovaries to the womb. In a hysterectomy, a surgeon cuts out the womb.

A federal judge ruled in 1974 in the case of Minnie Lee and Mary Alice Relf, two Black sisters in Alabama who had been sterilized as children. The judge found that doctors in programs paid for by the federal government were sterilizing an estimated 100,000 to 150,000 poor people a year. Some doctors told women on welfare, government money for poor families, that they would lose it unless they agreed. The judge barred federal money for sterilizing people against their will, and barred the threats to cut off welfare.

The Indian Health Service. Investigators of the General Accounting Office, which checks how federal money is spent, reported on the Indian Health Service in 1976. In four of its 12 regions, its doctors had sterilized 3,406 Native women from 1973 to 1976. Writers for the National Library of Medicine and the National Archives describe the operations as done without the women's permission. Thirty-six of the women were under 21, although a federal judge had ordered that women that young not be sterilized.

The investigators' report does not name the doctors. Two years before it, in 1974, Dr. Connie Pinkerton-Uri, who is Choctaw and Cherokee, had found in her own study that 1 in 4 Native women had been sterilized without consent.

One woman's case. Norma Jean Serena, who was Shawnee and Creek, gave birth in Armstrong County, Pennsylvania, on August 16, 1970. Two days later Drs. Donald Carter and James Houston cut out her fallopian tubes, the tubes that carry eggs to the womb. She said no doctor ever told her. A welfare caseworker, Jean Burgess, had advised the operation. Burgess later said, "This was a burden on society. I felt that Mrs. Serena should have no further children."

Burgess and two other county child welfare workers, Patricia Walker and Natalie Leezer, had also taken three of Serena's children from her in 1970. Serena sued in 1974. The jurors found that the welfare workers had given a

false picture of the children's health and of how long the children would stay in foster homes. They awarded her $17,000 for the taking of her children.

On the operation, the same jurors found against her. They decided that she had agreed to it, though she said no doctor ever told her about it.

In Puerto Rico, about 1 in 3 women of child-bearing age had been sterilized by 1968, and 37.4 of every 100 by 1976, historians on the history podcast DIG report. Some chose it because it was the only birth control offered to them. Others were told it could be undone, or were pushed into it by their employers.

Minnie Lee and Mary Alice Relf

Who: Two Black sisters in Alabama who were surgically sterilized as children through a federally paid clinic, after their mother was told they would get birth control shots.
When and where: Montgomery, Alabama, June 1973.

Their mother was told the girls would get birth control shots.

Surgeons in Montgomery, Alabama, sterilized Minnie Lee and Mary Alice Relf in June 1973. The girls were 12 and 14, and accounts differ on which sister was older. The surgeons tied or cut the tubes that carry eggs to the womb, so the girls could never have children. No record names the surgeons.

Nurses from a family planning clinic had told their mother, Minnie Relf, about the shots. Montgomery Community Action ran the clinic with federal money. She could not read or write, and she signed the form with an X. Both girls had intellectual disabilities. The family lived in public housing that the same agency had found for them in 1971. In all, 11 children from the same clinic's program were sterilized, and 10 of them were Black.

Their father, Lonnie Relf, saw the scars when the girls came home from the hospital. The same day a nurse asked their sister Katie, who was 17, to agree to be sterilized. Katie locked herself in her room.

> Lawyers from the Southern Poverty Law Center sued for the family. The federal officials they sued included Caspar Weinberger, the Secretary of Health, Education and Welfare, and Alvin Arnett of the Office of Economic Opportunity. The judge barred federal money for sterilizing people against their will, and barred doctors from threatening to cut off welfare money.

AIDS, 1981 to 2000

CDC doctors reported a rare kind of pneumonia in five young gay men in Los Angeles on June 5, 1981. Two of them had died. The disease was later named AIDS. It is caused by HIV, a virus that destroys the white blood cells that fight infection. As those cells die, other infections and cancers kill the person. HIV passes through blood, semen and other body fluids, not through touch.

More Americans were diagnosed with AIDS in 1985 than in all earlier years together. By early 1986 about half of the adults ever diagnosed had died, and patients died about 15 months after they were diagnosed. People with hemophilia, whose blood does not clot, took a clotting medicine made from blood given by many donors. Of those treated with it between 1979 and 1984, nearly 9 in 10 caught HIV, hepatitis C or both.

ACT UP. The writer Larry Kramer started ACT UP, the AIDS Coalition to Unleash Power, in New York in March 1987. That month FDA officials approved AZT, the first drug for AIDS. On October 11, 1988, more than 1,000 ACT UP members shut down FDA offices in Maryland, and police made 176 arrests. Eight days later FDA officials announced rules to approve drugs faster.

AIDS was the leading cause of death for Americans aged 25 to 44 in 1994. From 1996, doctors gave several drugs together, and the next year AIDS deaths fell by 47 percent. By the end of 2000, 774,467 Americans had been reported with AIDS, and 448,060 of them had died.

Ryan White

Who: An Indiana teenager who caught HIV from his hemophilia medicine, was kept out of school, and then spoke across the country about AIDS.

When and where: Kokomo and Cicero, Indiana, 1984 to 1990. Born in Kokomo on December 6, 1971.

Movie: *The Ryan White Story* (1989), a film made for ABC television. Ryan White had a small part in it.

When he came back to school, 151 of 360 students stayed home.

Ryan White was 13 when doctors told him, on December 17, 1984, that he had AIDS. They gave him six months to live. He had hemophilia and took a clotting medicine made from blood given by many people. Some of it carried HIV.

James O. Smith, the head of his school district, refused in 1985 to let him back into Western Middle School. In all, 117 parents and 50 teachers signed a petition, a written request, to keep him out. Health officials of the state and the CDC said he was no danger to anyone. When he went back for one day in February 1986, 151 of the school's 360 students stayed home.

After he returned, school staff made him use throwaway forks and spoons and a separate bathroom. Someone fired a bullet through his family's living room window while no one was home.

The family moved to Cicero, Indiana. On his first day at Hamilton Heights High School, the principal, Tony Cook, and students who had learned about AIDS shook his hand. In 1988 he spoke to a presidential commission on AIDS. He died on April 8, 1990, at 18. That August, members of Congress passed the Ryan White CARE Act. Under it, federal money pays for care for people with HIV.

BY 2000

If you remember one thing: A vaccine ended polio in the United States and Medicare paid for older Americans' care, while reporters and judges exposed experiments and operations done to people who never agreed.

In 1952: Polio paralyzed more than 21,000 Americans, and the Tuskegee men were being kept from penicillin.

In 2000: No one had caught polio inside the United States since 1979, Medicare paid for the care of people 65 and older, and a federally paid study on people needed each person's informed consent.

Next: After 2000 doctors learned to edit genes, and COVID-19 killed more than a million Americans.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 9 MINUTES

COVID-19 and drug overdoses cut American lives short in these years, even as doctors learned to edit the genes in a patient's own cells to treat sickle cell disease. COVID-19 killed 1,245,791 people in the United States from 2020 to September 19, 2026, by the early count of the CDC, the federal disease agency. Drug overdoses killed more than 100,000 Americans a year in 2022 and 2023.

Life expectancy, the average age a newborn would reach if that year's death rates held, fell from 78.8 years in 2019 to 76.4 in 2021. It was back to 79.0 in 2024, federal health officials who count births and deaths reported in January 2026. President Barack Obama signed the Affordable Care Act in 2010. In 2025, 26.7 million people had no health insurance all year, the Census Bureau reported in September 2026.

Reading and editing genes, 2003 to 2023

A genome is the full set of DNA instructions in a person's cells. Scientists of the Human Genome Project, begun in 1990, finished the first reading of the human genome in April 2003.

Sickle cell disease bends red blood cells into a curved shape like a sickle, a farm tool with a curved blade. The bent cells slow the flow of blood in small blood vessels and carry less oxygen. That causes attacks of severe pain, and it can disable or kill a person. About 100,000 people in the United States have it, the Food and Drug Administration reported in 2023. It is most common in African Americans.

FDA officials approved Casgevy on December 8, 2023. It was the first treatment made with the gene-editing tool called CRISPR that the FDA has approved. Doctors take some of the patient's blood-making cells and edit their DNA. Then the patient gets high doses of chemotherapy, strong drugs that kill the cells left in the bone marrow. Doctors put the edited cells back in a single drip into a vein.

The Affordable Care Act and the uninsured, 2010 to 2026

The Census Bureau counted 49.9 million people in the United States with no health insurance in 2010. That was 16.3 percent of everyone, about 1 person in 6.

President Obama signed the Patient Protection and Affordable Care Act on March 23, 2010. Members of the House had passed it 219 to 212. Most of its parts started in January 2014. Lawmakers in many states opened Medicaid to more adults with low incomes, and people could buy insurance on new insurance marketplaces, websites where people compare plans and sign up. By 2016, about 20 to 24 million more people had insurance, and the share without it had fallen by about half.

The Census Bureau reported on September 15, 2026, that 26.7 million people, or 7.9 percent, had no insurance for the whole of 2025. That is about 1 person in 13. Census Bureau staff changed their survey questions in 2013, so the 2010 and 2025 counts were not measured in exactly the same way.

New Medicaid rules. President Donald Trump signed a budget law with new rules for keeping Medicaid on July 4, 2025. From 2027, adults who got Medicaid when their state widened it under the Affordable Care Act must show 80 hours a month of work or similar activity to keep it. They can instead show a reason they are excused. In July 2025 staff of the Congressional Budget Office, which estimates the cost of laws for Congress, estimated that 10 million more people would have no insurance in 2034 because of the law.

Extra federal help with the price of marketplace insurance ended after 2025. In 2026, sign-ups fell by more than a million, to 23.1 million, the health research group KFF reported. The average monthly payment rose from $113 to $178.

Rural hospitals have closed too. Since 2005, 197 rural hospitals have closed or stopped keeping patients overnight, by the December 2025 count of the Sheps Center at the University of North Carolina.

COVID-19, 2020 to 2026

COVID-19 is an illness caused by a coronavirus. It spreads when an infected person breathes out droplets and tiny particles that carry the virus. It causes fever, cough, shortness of breath and loss of taste or smell. In severe cases the person has trouble breathing and pain or pressure in the chest, becomes confused, and the lips or skin turn pale, gray or blue.

The first case in the United States was confirmed in a laboratory on January 20, 2020. On March 11, 2020, officials of the World Health Organization declared a pandemic, a disease spreading around the world. FDA officials allowed the first vaccine on December 11, 2020. Recorded deaths in the United States passed 500,000 on February 21, 2021, and 1 million on May 12, 2022. The federal health emergency ended in May 2023.

The CDC's early count, as of September 24, 2026, is 1,245,791 COVID-19 deaths from 2020 to September 19, 2026. The year with the most deaths was 2021, with 463,267. In 2025 there were 20,682. These early counts rise as late death records come in.

For American Indian and Alaska Native people, life expectancy fell from 71.8 years in 2019 to 65.2 in 2021. That was a drop of 6.6 years, federal health officials who count births and deaths reported in 2022, and it brought them to the same life expectancy that all Americans had in 1944. COVID-19, accidents and liver disease caused most of that drop.

Sandra Lindsay

Who: A nurse who led the intensive care nurses at a hospital in Queens through the pandemic, and was the first American outside a vaccine trial to get a COVID-19 vaccine.

When and where: Long Island Jewish Medical Center, Queens, New York, 2020 and 2021. She came from Jamaica.

She got the first COVID-19 vaccine given outside a trial.

Sandra Lindsay got that shot on Monday, December 14, 2020, the first given in the United States outside a trial. She directed the nurses who cared for the sickest COVID-19 patients at Long Island Jewish Medical Center in Queens, New York.

She had questions about the vaccine at first. She did her own research and listened to experts before she decided. "We needed a more powerful tool," she said.

During the pandemic her aunt died, and she could not be with her mother. Her grandson Avery was born too soon, in March 2020, and spent about four and a half months in a hospital unit for newborns. Lindsay could not sleep or focus, and she began to meditate.

President Joe Biden gave her the Outstanding Citizen By Choice award, which honors immigrants who became citizens, on July 2, 2021. Her hospital scrubs and her ID badge went to the Smithsonian's National Museum of American History.

Opioids and overdose deaths, 1999 to 2026

Opioids include strong painkillers such as oxycodone and fentanyl, and the illegal drug heroin. Too much of one slows a person's breathing, and the person can die. Fentanyl is 50 to 100 times stronger than morphine. A drug called naloxone reverses an overdose if someone gives it quickly.

The CDC describes three waves of deaths. The first came from pain pills, after doctors in the 1990s began writing more prescriptions, or written orders, for opioids. The second came from heroin, from 2010. The third came from fentanyl made illegally, from 2013. From 1999 to 2023, about 806,000 Americans died of opioid overdoses, the CDC counted in 2026.

Guilty pleas. Purdue Pharma, the maker of the pain pill OxyContin, pleaded guilty in 2007 to misleading the public about how easily people became addicted to it. It paid $600 million. Three of its leaders, Michael Friedman, Howard Udell and Paul Goldenheim, pleaded guilty too. Purdue pleaded guilty to federal crimes again in 2020.

Fifty-five attorneys general, the top lawyers of state and territory governments, joined a $7.4 billion settlement with Purdue and the Sackler family, who owned it, on June 16, 2025. In the settlement, Purdue and the Sacklers agreed to pay money to end lawsuits against them.

FDA officials approved a naloxone nasal spray for sale in stores without a doctor's order on March 29, 2023. Overdose deaths from all drugs peaked at about 112,000 in the 12 months ending August 2023. The CDC's early count for the 12 months ending April 2026 is about 67,000. About 42,000 of those deaths involved opioids.

Smoking, measles and vaccines, 2006 to 2026

US District Judge Gladys Kessler ruled on August 17, 2006, that the big tobacco companies had cheated the public. In her words, they had "defrauded the American people by lying for decades about the health risks of smoking and their marketing to children." Smoking and breathing other people's smoke still kill more than 480,000 Americans a year, by the CDC's count last updated in 2024. That is nearly 1 of every 5 deaths in the country.

Measles is a disease caused by a virus. It spreads through the air when an infected person coughs or sneezes, and a person can catch it in a room up to 2 hours after the sick person has left, CDC doctors write. It starts with a high fever, a cough and red, watery eyes, and then a rash spreads down from the face.

As many as 1 of every 20 children with measles gets pneumonia, the most common way measles kills young children. About 1 child in 1,000 gets a swelling of the brain.

Measles had stopped spreading nonstop inside the United States for at least a year by 2000, which health officials call elimination. In 2025 there were 2,289 confirmed cases and 3 deaths, by the CDC's count. In an early count of 2,144 of those cases, 93 of every 100 were in people who had not been vaccinated. In 2026, by September 24, the CDC had counted 3,659 cases and 1 death.

The vaccine panel. Robert F. Kennedy Jr., the Secretary of Health and Human Services, removed all 17 members of the panel of experts that advises the CDC on vaccines on June 9, 2025. He wrote that the panel had become "little more than a rubber stamp for any vaccine," meaning a panel that approved every vaccine it was shown. Leaders of the American Medical Association said the move "upends a transparent process that has saved countless lives," meaning it threw out an open way of deciding about vaccines.

The last of the Tuskegee men

The last man from the Tuskegee study died in January 2004. The last widow in the benefit program died in January 2009. CDC officials report that 10 of the men's children receive medical care through the program.

THE THREE QUESTIONS, ANSWERED

1. **Why did new diseases kill so many Native people after 1492?** The crowd diseases, such as smallpox and measles, had never reached the Americas, so nobody here was immune and a whole town could fall sick at once. Smallpox killed about half of the Cherokee in 1738 and 1739.
2. **How did Americans learn to stop smallpox, yellow fever and polio?** Boston doctors tried inoculation in 1721, using a method Onesimus described, and safer vaccination followed in 1800. Army doctors proved in 1900 that mosquitoes spread yellow fever, and after the Salk and Sabin vaccines the last polio case caught inside the country came in 1979.
3. **Who was used in medical experiments without agreeing, and what changed because of it?** Enslaved women operated on by J. Marion Sims, the Black men of the Tuskegee study, prisoners and patients in Guatemala, and children at Willowbrook. After the Tuskegee study became public, members of Congress passed the National Research Act in 1974, and researchers in a federally paid study on people must now get each person's informed consent.

What changed: At the start of this chapter nobody knew what caused disease, and by its end doctors could prevent polio and measles with vaccines and edit the genes in a patient's own cells.

FIVE DATES TO KEEP

1721: Boston doctors inoculate against smallpox.

1900: Army doctors near Havana prove that mosquitoes spread yellow fever.

1932: Public Health Service doctors begin the Tuskegee study.

1955: The polio vaccine is shown to work and approved for use.

2020: COVID-19 reaches the United States.

NOT KNOWN: Nobody knows how many people lived north of Mexico in 1492, so nobody knows how many died of the new diseases.

Where to go: Elizabeth Phillips's gravestone, in the Phipps Street Burying Ground in Charlestown, Massachusetts, says she "brought into this world above 3000 Children."

Next: Chapter 27, Disasters and Rescue. It follows fires, floods, storms and other sudden disasters, and what people changed after them.

Chapter 27: Disasters and Rescue

CHAPTER 27 OF 37 · DISASTERS AND RESCUE · 10 ERAS · ABOUT 130 MINUTES

Fires, floods, storms and earthquakes strike suddenly, but people's choices decided how many died and who got help, and most safety rules came only after the deaths.

The South Fork Dam stood 69 feet high, 14 miles up the valley from Johnstown, Pennsylvania. It belonged to a club of rich Pittsburgh men. Their workers lowered its top by as much as 3 feet for a carriage road, and screened the channel that let extra water out, to keep the club's fish in. On May 31, 1889, after the heaviest rain ever recorded there, the dam broke and the flood killed 2,208 people. In court, two club members who were lawyers called it an "act of God," and the survivors lost.

This chapter follows sudden disasters in order, from a volcano in Oregon about 7,700 years ago to a flood in the Texas Hill Country in 2025. Each era has three things to look for. The first is the choice someone made before the disaster: a town built of wood, a locked door, a warning nobody passed on. The second is who got help afterward, and who was made to do the work or left to wait. The third is the rule that came after, from Boston's first fire wardens in 1711 to the smell put into natural gas in 1937.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. What rules did Americans make after disasters, and did they work?
2. Was anyone punished when people's choices killed others?
3. Did everyone get the same help after a disaster?

Five minutes? Read "The Triangle fire, March 25, 1911" in 1900 to 1950.

Can skip: "Fire insurance, 1752", "Hurricanes and the war on the Gulf Coast, 1779 and 1780", "Black Sunday, April 14, 1935". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 5 MINUTES

Sandals woven from sagebrush bark lay under the ash of an Oregon volcano for thousands of years, and they show that people lived through this land's disasters long before anyone wrote one down. Those people belonged to Native nations. When ash buried their fields or the rain stopped for decades, many of them moved away. What is known comes from layers of ash, from the growth rings of old trees, and from the accounts the nations told to each new generation.

Two volcanoes

Mount Mazama is a volcano in the Cascade Range of southern Oregon, about 60 miles north of the California line. About 7,700 years ago it erupted, and its top fell in. The eruption lasted only a few days. Scientists estimate that it was more than forty times as powerful as the eruption of Mount St. Helens in 1980. Its ash fell as far away as what is now central Alberta, in Canada.

Over several hundred years, rain and snow filled much of the hole that the collapse left behind. The water became Crater Lake. It is 1,943 feet deep, the deepest lake in the United States. In the Klamath language its name is *giiwas*.

The sandals. People had lived in the region for about 1,500 years or more before the eruption. In 1938 Luther Cressman, an archaeologist who studied the past from the things people left behind, dug in Fort Rock Cave, in Lake County, Oregon. Under a layer of ash from Mount Mazama, he found dozens of sandals woven from sagebrush bark.

Radiocarbon dating, a test that measures how much of one kind of carbon is left in something that was once alive, later showed that the sandals were more than 9,000 years old. Sandals of the same kind from half a dozen sites date from between about 10,500 and 9,100 years ago. The families who wore them lived there long before the mountain erupted.

The Klamath Tribes say that they have lived in the Klamath Basin for 13,000 years. They tell the eruption as a battle between Llao, chief of the Below World, and Skell, chief of the Above World. In their account, Llao attacked the people with fire after a chief's daughter refused him. Skell defended the people, and two holy men gave up their lives. Skell drove Llao back into Mount Mazama, the mountain fell in, and water filled the basin.

Sunset Crater. A second volcano, Sunset Crater, near what is now Flagstaff, Arizona, erupted around the year 1085. Scientists fixed the date from chemical changes in the growth rings of trees that were alive then. Ash fell on land 12 to 15 miles away and buried people's homes. Crops cannot grow in ash deeper than about 6 to 8 inches.

The Sinagua farmers who lived closest to the volcano were ancestors of today's Pueblo peoples. National Park Service scientists think they had to leave their fields because of the eruption and move farther away. Where the ash lay thin, it worked as a mulch, a covering that keeps soil from drying out, and crops there grew better. No record counts any dead. The people of the area told stories of the eruption, and those stories are still told today.

How do tree rings show a drought?

A tree adds one ring of new wood each year, and in a year of drought, with too little rain for crops, the ring is thin. Scientists read those rings in very old trees and wooden beams to find out which years were dry.

The rings show that the Colorado Plateau, in the dry Southwest, had too little rain from about 1130 to 1180. From 1140 to 1162 there were 23 dry years in a row. Susan Ryan, a researcher at the Crow Canyon Archaeological Center in Colorado, calls it the greatest drought recorded for North America. The Colorado River carried less water in the middle 1100s than at any other time in the past 1,200 years.

At Chaco Canyon, in present-day New Mexico, builders had stopped putting up the large stone buildings called great houses by about 1130. In the decades after that, the people of the Chaco towns moved away.

The Great Drought. A second drought, called the Great Drought, lasted from 1276 to 1299. Ryan found it less severe than the drought of the 1100s. By about 1300 the Pueblo people of the Mesa Verde region, in present-day southwestern Colorado, had moved south. They went to the Rio Grande valley, to Zuni and Acoma, and to the Hopi mesas, where rain came more reliably.

Archaeologists at Crow Canyon count the drought as one cause of that move among several. Other causes on their list are changes in how the communities got along, how they were ruled and how they made a living. In their view, the drought probably caused food shortages in a region crowded with people, and the hard times that followed may have led to quarrels and fighting.

Attackers killed at least 41 people at Castle Rock Pueblo and at least 8 at Sand Canyon Pueblo around 1280. Both places are in the Mesa Verde region. Kristin Kuckelman and two other scientists studied the skeletons. They found that someone had cut the bodies apart and broken, crushed and burned the bones, during or after the fighting. They also found signs that people probably ate human flesh there. No evidence shows who the attackers were, or who cut up the dead.

A fault under the northwest coast

The outer layer of the earth is broken into huge slabs called plates. Off the coast of present-day Washington, Oregon and northern California, one plate slides under another. The line where they meet is a fault, a crack in the earth's crust where the rock on each side can move. Scientists call this one the Cascadia Subduction Zone, from subduction, one plate sliding down under another.

When the rock along a fault moves suddenly, the ground shakes. Scientists have found signs of 19 great earthquakes on this fault in the last 10,000 years. The most recent one struck in January 1700.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 7 MINUTES

A hurricane in Pensacola Bay sank six Spanish ships in 1559 with the food for 1,500 colonists still on board. Spanish and English colonists of the 1500s wrote down the disasters they lived through, and this storm is one of them.

Two kinds of disaster hit the colonists hardest. Hurricanes, huge storms that form over warm ocean water with winds that spin in a circle, sank their ships and flooded their towns. Droughts left the colonists and their Native neighbors short of corn, and hungry colonists went looking for food in Native towns.

A hurricane wrecks a colony, 1559

A hurricane in September 1559 sank most of the ships of the largest Spanish colony yet sent to the Southeast. Nearly all of the colony's food went down with them. The colonists had landed on August 15, 1559, at Pensacola Bay in present-day Florida. The Native name of the bay was Ochuse. They came from Veracruz, in Mexico, on 11 ships, towing a 12th, smaller ship behind them.

About 1,500 people made the voyage. They included an army of 500 soldiers and about 200 Aztec warriors and craftsmen from Mexico City. Servants, enslaved Africans, and some wives and children came too. For five weeks they unloaded people and tools and began building a town on high, flat ground above the bay. They left the food for all 1,500 people on board the ships, because they thought it was safer there than on land.

The storm. One of the 11 ships, a galleon, had already sailed back to Mexico in late August or early September. On the night of September 19 to 20, 1559, a hurricane came into the bay. It sank six ships and pushed a seventh a short way onto the land. Three of the 11 ships came through the storm, and the small towed ship later carried supplies on relief voyages. The people who drowned were mostly the sailors and passengers still on board. No record gives their number.

After the storm the colonists did not have enough to eat. One ship came from Havana, in Cuba, with supplies soon after the hurricane. Relief ships from Mexico came after that, one every five to seven months. The colonists sent about 200 men inland to look for towns with food to spare. The men found a large town called Nanipacana, on the Alabama River in present-day central Alabama. Its people left before the Spaniards arrived. As they went, they destroyed the food they had stored.

Many colonists grew sick and died from lack of food. Native people attacked and killed others, and the records do not state which Native people they were. The colonists shipped many survivors back to Mexico. By September 1560, 362 people were left at the bay, and another 200 were far inland.

A new commander, Ángel de Villafañe, arrived in April 1561. He took most of the roughly 150 settlers still at the bay to Havana. Two small ships of his fleet later sank in another storm off Cape Hatteras, with everyone aboard. The last Spanish soldiers left the bay in August 1561. In February 1562 the viceroy, the Spanish king's governor in Mexico, gave up the colony.

Tristán de Luna y Arellano

Who: A Spanish noble and soldier who led the colony at Pensacola Bay when the hurricane of 1559 destroyed its ships and food.

When and where: Pensacola Bay, Florida, and the Alabama River, 1559 to 1561.

Luna ordered his hungry soldiers to march, and every one refused.

Tristán de Luna y Arellano led the colony at Pensacola Bay. It was the largest Spanish attempt up to that time to settle the Southeast, and the one with the most money behind it. Years before, Luna had marched with the Spanish explorer Coronado.

Looking for food. From February to June 1560, Luna moved most of his people inland to Nanipacana, the town the first search party had found. In April he sent a second group of 200 men, under Mateo del Sauz, farther north to look for food. They went to Coosa, a chiefdom, a group of towns under one head chief, in present-day northwestern Georgia. They stayed there until November.

The refusal. Luna ordered 100 soldiers and 50 servants to march to Coosa in late summer 1560, on short rations, meaning with less food than they needed. All of his soldiers refused, and the march never took place.

When Ángel de Villafañe arrived at the bay with a small fleet in early April 1561, he dismissed Luna and ordered him back to Spain. Luna died in New Spain, the Spanish colony in Mexico, on September 16, 1573.

The droughts behind the Lost Colony

Tree rings from old bald cypress trees on the border of Virginia and North Carolina show two long droughts in the 1500s. A team of scientists led by David Stahle published the record in 1998. The first drought lasted from 1562 to 1571 and was worst from 1565 to 1569.

A Spanish priest in the Chesapeake Bay region, Father Juan Batista de Segura, wrote about it in September 1570. He wrote that the region had gone six years with too little corn and wild fruit, with hunger, death and dry soil.

The driest years. From 1587 to 1589 came the driest three years in the whole 800-year record. English colonists on Roanoke Island, in present-day North Carolina, disappeared during those years. They are known as the Lost Colony. The Stahle team counts the drought as one of several causes of the colony's end. An earlier English colony on Roanoke had come in wetter years, 1585 and 1586. Only 4 of its 108 people died.

The Croatan people were allies of the English colonists, and in 1587 their crops were poor. That same year the people of the town of Dasemunkepeuc killed George Howe, an adviser to the English governor, John White. They spoke an Algonquian language, one of a large family of Native languages along the coast.

They killed Howe in revenge. Men under Ralph Lane, leader of the earlier colony, had attacked and killed their leader, Wingina, in June 1586. After Howe's death the people of Dasemunkepeuc left their town.

The wrong people. John White led his soldiers in an attack on Dasemunkepeuc before daylight on August 8, 1587. The people they found there were Croatan, who had come to the empty town looking for food. White's men killed one of them and wounded many before they saw their mistake. The Stahle team links the killing to the drought, because hunger had brought the Croatan to the empty town.

St. Augustine burns and floods

St. Augustine, in present-day Florida, was burned, raided or flooded again and again in its first 35 years. Spanish soldiers and colonists founded it in 1565 at Seloy, a town of the Timucua people. The Timucua were tired of the Spaniards living in their town. In 1566 Timucua warriors shot flaming arrows at the fort. On April 1, 1566, the fort was half burned, and all its supplies were lost. The records do not state whether the arrows started that fire.

Thirty-five hard years. The colonists moved the town across the bay to Anastasia Island that year. In 1570 soldiers there rose against their officers and destroyed the fort. The sea was washing away the rebuilt fort by 1572. Native people attacked the town in 1577 and burned many houses. Then, in 1586, the English privateer Francis Drake, a ship captain who raided enemy ships and towns with his own government's permission, burned the whole town and its fort with his men.

A fire swept through the town in March 1599. It was built of wood, with roofs of thatch, meaning dried plants. In September of the same year a hurricane flooded the buildings that the townspeople had not yet repaired.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 8 MINUTES

Of the 104 English colonists who landed at Jamestown in 1607, only 38 were alive a year later. The English colonies of the 1600s were that small, and one drought or one storm could kill a large share of the people in them.

Colonists built their towns mostly of wood, so a fire in one house could spread to the next. After two big fires, Boston's leaders bought a fire engine from England and ordered new buildings made of brick, with roofs of slate or tile. Governors and ordinary colonists wrote down what they lived through, in letters and journals full of detail.

Jamestown: drought and a hurricane

English colonists landed at Jamestown, in present-day Virginia, in April 1607. They built their fort inside Tsenacomoco, the land of the Powhatan people. Tsenacomoco was a paramount chiefdom, a group of many towns and chiefs under one head chief. Its head chief was Powhatan.

Chapter 26, Health, Disease, and Medicine, tells how salty river water, sickness and hunger killed most of the first colonists. Tree rings show that the colonists arrived during the driest seven years in 770 years, from 1606 to 1612. The drought hit the Powhatan towns as well as the English.

The relief fleet. The leaders of the Virginia Company of London, the business that ran the colony, sent nine ships and about 600 people to Jamestown on June 2, 1609, with farm animals and enough food to last a year. It was the largest English fleet yet to sail west.

A hurricane struck the fleet on July 24, 1609, about seven days from land. It drove the lead ship, the *Sea Venture*, away from the others. The ship began to leak so badly that one passenger wrote, "we almost drowned within."

The *Sea Venture* reached Bermuda, an island in the Atlantic. All 150 people on board got ashore, among them the colony's new governor, Sir Thomas Gates. Before the ship sank, the crew saved many of the supplies. The castaways

stayed on Bermuda for ten months and built two small ships, the *Patience* and the *Deliverance*. The playwright William Shakespeare drew partly on their story for his play *The Tempest*.

The other eight ships reached Jamestown in late August 1609. Many passengers were sick, and the crews had thrown many of their supplies overboard in the storm.

Why Jamestown starved, 1609 to 1610

Most of the people at Jamestown starved to death in the winter of 1609 to 1610, the winter called the Starving Time. The colonists had not farmed, hunted or fished to feed themselves. They lived on supplies from England and on food they bargained for from the Powhatan people, or often stole from them. In the drought the Powhatan towns had little food to spare.

The killing began on both sides in the autumn of 1609. The colonist Francis West cut off the heads of two warriors of the Patawomeck people while he was trading with them. In November 1609 warriors ambushed a party of colonists led by John Ratcliffe at Orapax. They tortured Ratcliffe to death. That same month Powhatan ordered his people to surround Jamestown and cut off its food.

How many died. Counts of the colonists differ. The historian Martha McCartney counts 240 people at Jamestown at the start of the winter. Other counts record about 500 in the colony. By May 1610 about 60 were alive. Some survivors dug up and ate the bodies of dead colonists and of Native people the colonists had killed.

The *Sea Venture* castaways reached Jamestown in their two small ships on May 24, 1610. Gates decided to give up the colony and sail home with the survivors. On the river they met Thomas West, Lord De La Warr, an English nobleman. He was bringing a year's supply of food, and everyone went back to Jamestown. Between 1607 and 1625, the company's leaders sent about 6,000 people to Jamestown. About 4,800 of them died.

The Great Colonial Hurricane, August 1635

A hurricane struck New England on August 15, 1635. The English then used an older calendar that ran ten days behind the one used today. By today's calendar the date was August 25. William Bradford, the longtime governor of Plymouth Colony, wrote that no one living there, "either English or Indeans, ever saw" a storm like it (*Indeans* is his spelling of Indians).

South of Plymouth the sea rose more than 20 feet above its normal height. Bradford wrote that many Native people climbed into trees to escape the water. No record gives the number of Native people who drowned. The wind blew down houses and, in Bradford's words, "many hundered thowsands of trees." The worst of the storm lasted five or six hours.

The same storm wrecked the *Angel Gabriel*, a ship full of English settlers, at Pemaquid, in present-day Maine. Three or four passengers and one sailor died. Off Cape Ann, in Massachusetts, it wrecked a small boat carrying two families.

Anthony Thacher

Who: An English colonist who survived the wreck of a small boat in the hurricane of 1635 and described it in a letter to his brother Peter.

When and where: Off Cape Ann, Massachusetts Bay Colony, August 11 to 15, 1635.

Of 23 people on the boat, only he and his wife lived.

Anthony Thacher wrote down what happened in a letter to his brother Peter, a few days after the wreck. On August 11, 1635, Thacher and his wife, Elizabeth, and their family left Ipswich, in Massachusetts, for the town of Marblehead. With them went his cousin John Avery, a minister, and Avery's family. The people of Marblehead had asked Avery to be their minister, and they sent a pinnace, a small sailing ship, to carry the two families.

Thacher's letter lists 23 people on board: 11 in Avery's family, 7 in his own, a man named William Elliot, and four sailors.

On the rock. At about 10 on the night of August 14, the wind split the pinnace's old sails. Before daylight the hurricane struck. The anchor dragged along the bottom, and the storm drove the boat onto the rocks of a small island. A wave threw Thacher onto a rock with his oldest daughter, Mary. His cousin Avery and Avery's oldest son joined them there.

The next wave broke the boat to pieces. It carried his wife toward the shore on a piece of the deck. All the others still on board drowned. Thacher's letter names four of his children who died. They were his oldest daughter Mary, his son William, his daughter Edith, and baby Peter, whom Edith was holding in her arms.

Another wave swept the four on the rock into the sea: Thacher, Mary, Avery and Avery's son. Thacher was in the water for at least a quarter of an hour before he crawled ashore.

On the island. He and his wife found a knapsack with a flint, a piece of steel for striking sparks from it, and dry gunpowder. They also found a drowned goat, two cheeses, some butter, a hat and his son William's coat. He struck a fire and they cooked some of the goat. They had only salty water to drink. He found the body of Avery's oldest daughter on the shore. She is buried on the island.

On the following Monday a passing boat picked them up. Thacher named the island "Thatchers Woe" and the rock "Avery's Fall." On September 3, 1635, the colony's lawmakers, meeting as the Massachusetts General Court, voted him 40 marks, an English unit of money, from the treasury toward his losses. In March 1637 they gave him the island. It is still called Thacher Island.

An earthquake, 1638

An earthquake shook New England on about June 1, 1638, by the old calendar. Bradford wrote that people heard it before they felt it. He compared the sound to faraway thunder. Dishes rattled and fell from shelves. People outdoors could not stand without holding on to posts. Native people felt it inland, and ships along the coast shook.

How strong. Scientists at Boston College's Weston Observatory estimate the 1638 earthquake at about magnitude 6.5, on the scale scientists use for an earthquake's strength. Each step up the scale, such as from 6 to 7, means the ground moves about ten times as far, and the quake lets out about 32 times as much energy.

The scientists think it was centered in central New Hampshire, but the few reports of the time make that uncertain. People felt the aftershocks, smaller quakes that follow a large one, for 20 days. No deaths were recorded.

Bradford wrote that the shaking looked like a sign that God was displeased. He added that for years afterward the summers were not warm enough for the corn to ripen well. He left the cause, in his words, "to naturallists to judge," meaning people who study nature.

Boston burns, and the town buys a fire engine

A fire in Boston on January 14, 1653, destroyed many buildings and killed three children. After it, town officials hired the owner of a water engine, a pump that sprays water, to bring it to fires. They bought six long ladders, four strong iron hooks and fire buckets. They required every property owner to keep a ladder and a swab pole, a long pole with a mop on the end for putting out sparks on roofs.

A fire in Boston's North End destroyed 45 buildings on November 27, 1676. Among them were several warehouses, the North Meeting House, a church, and the home of its minister, Increase Mather.

The first paid firefighters. Town officials brought a fire engine from England in 1678 and hired men to run it. A Boston Public Library guide calls them the first paid fire department in what is now the United States. Officials also made a law that roofs be slate or tile and walls be brick. Large fires burned parts of Boston again in 1679, 1682 and 1691.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 9 MINUTES

Every member of Philadelphia's Union Fire Company kept six leather buckets ready for the next fire. In these fifty years townspeople began to organize against the fires that kept burning their wooden towns. In 1722 about two-thirds of Boston's nearly 3,000 houses were built of wood. Boston got its first fire wardens after a fire in 1711, and neighbors in Boston and Philadelphia formed fire companies.

When a fire came, the help and the blame did not fall evenly. After Charleston burned in 1740, most of the money from London went to rich merchants. After a string of fires in New York in 1741, officials blamed enslaved Black people and poor white people and put at least 30 people to death. On the coasts, an earthquake in 1700 sent a tsunami across the Pacific to Japan, and a hurricane in 1715 wrecked a Spanish treasure fleet off Florida.

How Japan's records dated the Cascadia earthquake, 1700

The Cascadia fault broke along about 620 miles of the Pacific coast at about 9 at night on January 26, 1700. The break ran from Vancouver Island, in present-day Canada, to northern California. Scientists at the Pacific Northwest Seismic Network call it one of the largest earthquakes known anywhere in the world. They put its size at about magnitude 9.

The coast dropped when the fault broke. Salt water flooded forests of cedar trees that had grown above the reach of the tides. The salt water killed the trees. Their dead trunks, called ghost forests, still stand along rivers and bays in Washington and Oregon.

The waves in Japan. The earthquake also set off a tsunami, a series of huge waves that can cross a whole ocean. The waves reached Japan the next day. At Miyako the water destroyed 13 houses, and a fire afterward burned 20 more. Near Nakaminato the waves drove a ship carrying rice onto rocks, and two sailors died.

The Japanese records describe the waves but no shaking. In 1996 scientists used those Japanese records to work out the day and the hour of the earthquake. Tree rings from the ghost forests confirmed the year.

No one counted the dead on the American coast. The Native nations there had no written records then. In their accounts, whole villages were lost. The Makah, in present-day Washington, tell of a great earthquake at night. The Quileute tell of a flood that carried canoes far inland. Archaeologists have found Native villages on the coasts of Washington and Oregon that sank, flooded and were left empty after 1700.

The raised water. Annie Miner Peterson, a Coos woman from Oregon, told what her grandfather had seen. Melville Jacobs, a scholar who was writing down Coos stories, recorded her words in 1933. Her grandfather had seen an old woman who survived the "raised water."

When the old woman was a girl, the water had left her hanging high in a tree. She tied her pack line, the rope she used to carry loads, to a branch to climb down, and she fell. The fall broke her back, and she had a humped back for the

rest of her life. Scientists who studied the account date the flood she described to between 1690 and 1805.

A hurricane sinks the Spanish treasure fleet, 1715

A hurricane drove eleven Spanish treasure ships onto the reefs of Florida's east coast early on July 31, 1715. At least 1,000 people died. The fleet of twelve ships had left Havana, in Cuba, a few days before, bound for Spain. The 1715 Fleet Society gives the day it sailed as July 24, and a NOAA writer gives July 27.

The eleven Spanish ships carried silver and gold from Spain's colonies in the Americas. They were wrecked along the coast from present-day Fort Pierce to Wabasso, north of Vero Beach. The twelfth ship, the French *Grifon*, escaped, because its captain had sailed farther from shore.

The count. About 2,500 people were on board the fleet. Scientists at NOAA, the US government's ocean and weather agency, count about 1,000 dead. The 1715 Fleet Society, a group that studies the wrecks, says well over 1,000. Spanish records from Cuba state about 1,500. The fleet's commander, Juan Esteban de Ubilla, was among the dead.

The survivors camped on the beach in the land of the Ais people. The Ais lived along the coast from Cape Canaveral to the Indian River. On August 6, Admiral Francisco Salmón sent the pilot Nicolás de India and 18 men in a small boat to Havana for help. They reached it 11 days later.

Ships from Havana and from St. Augustine brought emergency supplies, officials, soldiers and tools for salvage, which means bringing up goods from a wreck. By late October 1715 the salvagers had raised more than 5 million pieces of eight, the silver coins of Spain.

A man the records call Captain Barrett came for the silver too. Ned Grant was an enslaved man, and his enslaver hired him out to Barrett. Barrett took Grant to the wrecks to work as a diver.

The pirates. The pirates Henry Jennings and John Wills attacked the Spanish salvage camp in the winter of 1715 to 1716. The 1715 Fleet Society dates the raid to early January 1716 and says the pirates took about 120,000 pieces

of eight. Other accounts put the raid in December 1715, name the pirate Charles Vane beside Jennings, and give the haul as about 87,500 pounds in English money.

Boston's first fire wardens, 1711

Boston's fire of 1711 began in a building of rented rooms off Cornhill on the evening of October 2. A Scottish woman named Mary Morse was working there with oakum, old rope fiber picked apart and used to seal the seams of ships. A flame nearby set it on fire.

By 2 in the morning the fire had burned about 100 buildings. Among them were the Town House, where the colony's government met, and the town's first meeting house, a building used as a church. The fire left 110 families without homes.

Four sailors climbed the meeting house steeple to save its bell. The roof fell in, and they died. Other people died when they went too close to the fire. Still others died in gunpowder blasts, because people fighting the fire blew up houses in its path so that it would have nothing to burn. The full count of the dead is not known.

After the fire. Members of Boston's churches raised 700 pounds, in English money, for the families who lost their homes. A few months after the fire, 12 men became Boston's first board of fire wardens. Each was in charge of the firefighting tools in one part of the town. Bostonians rebuilt the Town House and the meeting house in brick. The new Town House is known today as the Old State House.

Charleston: a hurricane in 1700 and a fire in 1740

A hurricane struck Charleston, South Carolina, on September 3, 1700. It knocked down thousands of trees and dozens of buildings. The Scottish ship *Rising Sun* sank at anchor. It carried people from a failed Scottish colony at Darien, in Panama. About 100 people on board drowned, including the captain.

A fire broke out in Charleston at 2 in the afternoon on November 18, 1740. A strong wind from the northwest carried burning pieces across the town. In less than a day the fire destroyed more than 300 houses and shops and several docks. The officers and crews of British navy ships in the harbor pulled down and blew up houses to stop it. The records that survive do not say whether anyone died.

Lieutenant Governor William Bull ordered every townsman to fight the fire, "by themselves and slaves." Townsmen were to send the people they enslaved to fight it as well. He called out the militia, the colony's armed citizens, to stop people from stealing from the ruins.

Who got the money. Most of the help from London after the fire went to the rich. Officials in London promised 20,000 pounds, in the king's name, for the people "most affected." They paid it out by the size of each person's losses, so most of it went to wealthy merchants, whose losses were large. Poorer people depended on gifts and loans from the vestry, the governing board of St. Philip's Church.

Mary Bedon, an innkeeper with three children, and Catharine Joor, a shopkeeper, lost their businesses in the fire and opened again in rented houses. By February 1741 the two women needed aid from the St. Philip's vestry to get through the winter.

New York, 1741: fires and blame

Thirteen fires broke out in lower Manhattan in March and April 1741. The first, in March, burned the governor's house and its chapel inside Fort George. Accounts give its date as March 8 or March 18. New York officials claimed that enslaved Black people and poor white people had plotted together to burn the city. A witness named Mary Burton told the court that the fires were such a plot.

Three judges, James De Lancey, Daniel Horsmanden and Frederick Philipse, tried about 200 people. Officials burned 13 Black men alive at the stake, which means they tied each man to a post and set a fire around him. They hanged 17 Black men and 4 white people, two men and two women. Counts of the people put to death range from 30 to 34. Officials shipped 70 to 84 more people to the Caribbean islands to be sold into slavery there.

Historians disagree about whether any plot existed. Judge Horsmanden had offered Mary Burton, a young servant bound to a tavern keeper, an end to her years of service and a reward in money. Other enslaved witnesses spoke after they were promised their lives. Two of the men burned at the stake, Quack and Cuffee, had no lawyers at their trial. Chapter 24, Slavery and Freedom, tells the trials in full.

An earthquake in New England, 1727

An earthquake shook New England at about 10:40 on the night of October 29, 1727. By today's calendar the date was November 10. Scientists at the Weston Observatory estimate it at about magnitude 5.6. At Newbury, Massachusetts, it knocked down many stone walls and the tops of almost all the chimneys.

Some solid ground turned to mud, and some marshes dried up. People felt it from Maine to the Delaware River, and smaller shocks went on for months. The records that survive show no one badly hurt.

What the ministers said. In Boston the minister Cotton Mather heard "a horrid rumbling like the noise of many coaches." Ministers in New England preached that God had sent the earthquake to show his anger. Writers blamed it on sins such as swearing and failing to keep the Sabbath, the day of rest. Towns such as Medford, Massachusetts, held a day of fasting, when people went without food and prayed.

John Barnard, a minister at Marblehead, came to a different view. He decided that the surface of the earth had moved because of shocks deep underground.

Who came when a house caught fire?

In Boston, groups of neighbors formed mutual fire societies. Members promised to help save one another's houses, and only those. On December 7, 1736, Benjamin Franklin and about 25 other men in Philadelphia signed the articles, or rules, of the Union Fire Company. Every Union member kept six leather buckets for water and two linen bags for carrying goods out of burning houses. They promised to come to every fire in the town, not only to the houses of members.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 8 MINUTES

Massachusetts lawmakers voted 3,000 pounds in 1760 for the Boston families whose homes had burned, and lawmakers in Pennsylvania and New York sent money too. Help after a disaster was starting to come from public money that lawmakers voted, and not only from churches and neighbors.

Philadelphians started a company to insure houses against fire in 1752. After big fires, officials in Boston and New Orleans ordered new buildings made of brick. During the Revolutionary War, hurricanes sank Spanish warships and troop ships on the Gulf Coast, and a fire burned a large part of New York while British troops held it.

Charleston under water, 1752

A hurricane pushed the sea more than 10 feet above the normal high-water mark at Charleston, South Carolina, on the morning of September 15, 1752. The water covered the whole town. People fled to the upper floors of their houses. Before 11 o'clock nearly every ship in the harbor had been thrown onto land. Only one, the British warship *Hornet*, rode out the storm.

About 500 buildings were destroyed. No one kept an exact count of the dead. Many people drowned, and others died when houses fell apart around them. A second storm on September 30 ruined crops and killed farm animals. Officials then limited how much corn, peas and rice could be shipped out of the colony, to keep food prices down.

Fire insurance, 1752

Benjamin Franklin organized the Philadelphia Contributionship, a fire insurance company, in Philadelphia in 1752. Under fire insurance, many owners each pay a small amount into a shared fund, and the company that keeps the fund pays any owner whose house burns. The Contributionship's inspectors looked at each house before insuring it. They set the price by how likely the house was to burn. Insured houses carried a sign showing four hands.

Owners who insured with the Contributionship had to remove the trees from their land, because trees got in the way of firefighters. A rival company, the Mutual Assurance Company, let its customers keep their trees. People called it the Green Tree.

The Cape Ann earthquake, 1755

An earthquake centered in the ocean about 25 miles off Cape Ann, Massachusetts, shook New England at about 4:30 in the morning on November 18, 1755. Scientists estimate it at about magnitude 6.2. In Boston the shaking lasted more than a minute. It threw down about 100 chimneys and damaged as many as 1,500 more. Some church steeples were left leaning. People felt it from Nova Scotia, in present-day Canada, to South Carolina. No deaths were recorded in the Boston area.

The argument. After the quake, a Boston minister and a Harvard professor argued in public over what had caused it. Seventeen days before the Cape Ann quake, a large earthquake had struck Lisbon, in Portugal. New England ministers held prayer services, and officials named days of fasting. Writers in New England published at least 27 sermons (talks given in church), poems and accounts about earthquakes after the one at Cape Ann.

The Boston minister Thomas Prince wrote that Boston had more lightning rods, metal rods on buildings that carry lightning into the ground, than any other New England town. He argued that the rods had drawn electricity into the ground and made the shaking worse. John Winthrop, a professor at Harvard College, answered him in lectures. He said that heat and gases deep in the earth had caused the earthquake.

Boston burns, and other colonies send help, 1760

A fire started in a house on Cornhill in Boston at about 2 in the morning on March 20, 1760. No one ever found the cause. The fire burned 349 buildings, 174 of them homes. It left 220 families, more than 1,000 people, without homes. No one was killed. A Boston Public Library guide describes it as the worst fire in the American colonies up to that time.

Lawmakers in Massachusetts voted 3,000 pounds for the families who lost their homes. Lawmakers in Pennsylvania and New York voted to send money too. People in Nova Scotia raised money for Boston, and people as far away as London sent gifts. Afterward Massachusetts lawmakers passed a law that any building more than 7 feet high had to be made of brick or slate.

Fires in wartime: New York, 1776, and Charleston, 1778

British troops under General William Howe took New York City in September 1776, during the Revolutionary War. On the night of September 20, a fire broke out near the southern tip of Manhattan. Accounts differ on the exact spot. John Joseph Henry, an American prisoner on a British ship, said it began in the Fighting Cocks Tavern, near Whitehall Slip.

Dry weather and strong winds spread the fire north and west. By morning it had burned between 400 and 1,000 buildings. That was between one tenth and one quarter of the city. Trinity Church burned.

Killed without a trial. British commanders blamed rebel arsonists, people who set fires on purpose. British soldiers and sailors killed a few men they suspected of setting the fire. They killed them on the spot, without a trial. The records do not give the men's names or an exact number. British officers questioned more than 200 suspects and charged none of them.

Historians still disagree about how the fire started. The historian Barnet Schecter found that no accusation of arson holds up. The historian Benjamin Carp argued in 2023 that rebels most likely set the fire on purpose. Gene Procknow, who reviewed Carp's book, wrote that Carp's evidence is not strong enough to prove it for certain.

Charleston, 1778. In Charleston, people found a fire in a bakery at 4 in the morning on January 15, 1778. By noon it had destroyed more than 250 houses and killed six people, both Black and white. People who had lost their homes got beds and food in the city's public buildings, paid for with public money. The next day South Carolina's lawmakers, meeting as the General Assembly, voted 20,000 pounds to help them.

Hurricanes and the war on the Gulf Coast, 1779 and 1780

Spain's leaders joined the war against Britain on June 21, 1779. That year Bernardo de Gálvez, the Spanish governor of Louisiana, was getting ships ready to attack the British fort at Baton Rouge. A hurricane destroyed the ships in less than three hours. Gálvez had to put off his attack for ten days.

Another hurricane struck a Spanish fleet of 64 ships under Admiral José Solano on October 20, 1780. The fleet was carrying Gálvez and 4,000 soldiers from Havana to attack the British town of Pensacola, in Florida. An unknown number of the ships sank. A NOAA writer counts half of the soldiers, about 2,000, as lost. The NOAA account does not say how many of them drowned. The attack on Pensacola did not go ahead that year.

A different storm that same month crossed the islands of the Caribbean. It is called the Great Hurricane of 1780. About 22,000 people died in the Lesser Antilles, the chain of small islands at the eastern edge of the Caribbean, from October 10 to 16. The storm also struck Puerto Rico. Scientists at NOAA count 1780 as the deadliest Atlantic hurricane season on record.

Why New Orleans was rebuilt in brick

New Orleans was a Spanish town in 1788. On March 21 of that year, a fire started at about 1:30 in the afternoon in the house of Vicente José Núñez, the army treasurer. That day was Good Friday, the Christian holy day before Easter. Church bells were the town's fire alarm, but the priests refused to let the bells ring on Good Friday. The fire burned 856 of the town's 1,100 buildings. Governor Esteban Miró set up tents for the people who lost their homes.

Another fire burned 212 more buildings on December 8, 1794. Spanish officials then required new buildings to be built with thick brick walls. Builders rebuilt the town in brick, with courtyards and iron balconies.

Charleston, 1796

A fire started in Lodge Alley in Charleston on the afternoon of June 13, 1796, and burned until nearly dawn. It burned the buildings on more than 250 lots and left at least 300 families without homes. A committee of church leaders and town leaders handed out the money that other townspeople gave, according to what each family needed. The fire reached the roof of St. Philip's Church. An enslaved man named Will climbed up and saved it.

Will

Who: An enslaved man in Charleston who put out the fire on the roof of St. Philip's Church, and whom the church's leaders then bought and freed.

When and where: Charleston, South Carolina, June to September 1796.

The church's leaders bought him for $707.14, then set him free.

Will put out the fire on the bell tower of St. Philip's Church during the Charleston fire of June 13 to 14, 1796. He climbed to the very top of the tower and tore off the burning wooden shingles of its roof. In 1854 the writer Charles Fraser wrote that he did it "at the risk of his life."

Will was enslaved by Charles Lining. Lining was a member of the vestry, the board that ran St. Philip's Church. On August 14, 1796, two months after the fire, the vestry's members voted to reward Will by making him free.

The next month they bought Will from Charles Lining for $707.14. Then they freed him with a deed of manumission, a legal paper that freed an enslaved person.

The vestry's records include only his first name. A later history of the church describes the man who saved it as a Black boatman.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 11 MINUTES

All four boilers of the steamboat *Moselle* burst at once near Cincinnati in April 1838, and about 150 people died. Between 1800 and 1850, steam engines brought a new kind of sudden death to the country's rivers and lakes. Boilers exploded, and the steamboat *Erie* burned on Lake Erie. In 1838 members of Congress passed the first federal law, a law for the whole country, to inspect steamboats, but no federal officer enforced it.

Earthquakes, a city fire and a tornado killed hundreds more people. After the New Madrid earthquakes of 1811 and 1812, members of Congress voted to give land to people whose farms the quakes had ruined. Most Missourians came to believe that speculators who moved fast got most of the benefit.

The New Madrid earthquakes, 1811 and 1812

Three strong earthquakes shook the middle Mississippi Valley in the winter of 1811 to 1812. They are named for New Madrid, Missouri, the largest town on the Mississippi River between St. Louis and Natchez. The first struck northeast Arkansas at about 2:15 in the morning on December 16, 1811. The others came on January 23 and February 7, 1812.

Experts at the U.S. Geological Survey estimate these quakes at about magnitude 7.3 to 7.5, on the scale scientists use for an earthquake's strength. The seismologist Susan Hough, a scientist who studies earthquakes, has argued that they were closer to magnitude 7.0.

The shaking threw down log cabins and chimneys as far away as Cincinnati and St. Louis. The first quake woke people in New York City, Washington, D.C., and Charleston, South Carolina. The Encyclopedia of Arkansas states that the shaking rang church bells in Boston. No story about it appeared in Boston's newspapers of the time. Susan Hough thinks the story began with bells that rang in Charleston, South Carolina, and that someone mixed up Charleston with Charlestown, a part of Boston.

The river. On the Mississippi River, high banks and whole islands fell into the water. Huge waves sank boats or threw them onto the shore. The river ran backward. Most accounts say it did so for a few minutes, and others say for up to three days.

Land sank near the St. Francis River in Arkansas. In the northwest corner of Tennessee the ground sank and formed Reelfoot Lake, about 18 miles long and 5 miles wide. Aftershocks, the smaller quakes that follow a large one, went on into 1816.

Reports of the time counted fewer than 100 deaths on land. No one counted the dead on the river or in the Native towns of the region. The historian William E. Foley writes that recent scholars believe the real number of deaths was far higher. Groups of Cherokee people who had already moved west from their homelands farmed along the St. Francis River. The earthquakes ruined their land. They left their farms and cattle and moved west again, where the Osage people living there resisted them.

Who got the land. Missouri was then a territory, land that belonged to the country but was not yet a state. In 1814 its governor, William Clark, and two other leaders asked Congress for help. In February 1815 members of Congress passed a law for people whose land the quakes had ruined. Each owner could claim the same amount of public land somewhere else, up to 640 acres. It was one of the country's earliest laws to help people after a disaster.

Speculators, people who buy things cheaply to sell them later for a profit, learned of the law first. In the spring of 1815 they hurried to New Madrid. They bought the ruined land at low prices from owners who had not yet heard of the law. Most Missourians came to believe that the speculators got most of the benefit. The records do not identify them by name.

Fire in the cold: New York, 1835

A gas pipe burst in New York's business district on the night of December 16, 1835. A coal stove lit the gas. The temperature that night fell as low as 17 degrees below zero. The rivers were frozen. Firefighters cut holes in the ice to reach water, and the water froze again in their hoses. They had fought two large fires two days before, and the city's water tanks were nearly empty.

The fire burned 17 blocks and between 530 and 700 buildings in New York's business district. Two people died. At about 3 in the morning, U.S. Marines and sailors brought gunpowder from the Brooklyn Navy Yard. They blew up buildings in the fire's path so that it had nothing left to burn. The damage came to about 20 million dollars. Twenty-three of New York's 26 insurance companies went out of business. The people who investigated the fire blamed no one.

Why steamboat boilers exploded, 1838

A steamboat's boiler is a large metal tank where a fire turns water into steam to drive the engine. A boiler can burst if the steam pressure grows too high. It can also burst if the water inside runs low and someone then lets cold water in on the hot metal.

All four boilers of the steamboat *Moselle* burst at once on April 25, 1838, as it pulled away from a landing just east of Cincinnati. The captain, Isaac Perin, had kept the boilers at full steam during the stop, to impress people watching from shore. Many witnesses said he was racing another boat, the *Ben Franklin*. The blast threw Perin onto the riverbank and killed him.

A committee of Cincinnati citizens met the next day. Its members counted at least 280 passengers on board and estimated about 150 dead. Later accounts give 160.

The *Pulaski*. The steamship *Pulaski* ran between Savannah, Georgia, and Baltimore, Maryland. On June 14, 1838, it left Charleston, South Carolina, for Baltimore. At about 11 that night, about 40 miles off the coast of North Carolina, its starboard boiler, the one on the ship's right side, exploded. The ship broke in two and sank in about 45 minutes.

About 128 people died, and 59 were saved. The records of the numbers do not agree. One record lists 168 people on board, 37 crew and 131 passengers. Yet 128 dead and 59 saved add up to 187.

Investigators at the end of 1838 found that the engineers had run the boilers wrongly. An account written with the help of two survivors identifies the cause. The second engineer let the water boil away in the boiler and then let a full load of cold water in on the hot copper. The account does not include his name.

Families on the *Pulaski* brought enslaved people with them. The passenger list does not name them. It puts the word "servant" after the family's name. The list also names three Black women, Jenny, Priscilla and Sallie Middleton, and does not say whether they were enslaved or free. The records do not say how many enslaved people died.

Rebecca Lamar

Who: A survivor of the *Pulaski* explosion who lived on a floating piece of the wreck from Thursday night until the next Tuesday, and later wrote down what happened.
When and where: Off the coast of North Carolina, June 14 to 19, 1838.

She tore her dress into squares to cool the children's heads.

Rebecca Lamar survived on a floating piece of the *Pulaski* from Thursday night, June 14, until the next Tuesday, and later wrote down what happened there. She was traveling north with her brother, the Savannah banker Gazaway Bugg Lamar. With them were his wife, Jane, and their six children. Another niece of hers was also on board. Her brother owned shares in the ship.

The explosion woke her in her cabin. Her bed had jammed the door, so she began to climb out through the small window above it. Her brother put her and a niece into a small boat. A moment later something struck her, and she was in the sea. She caught hold of a piece of wood and pulled

herself onto a floating section of the ship. Her niece Martha and her nephew Charles reached her there. A stranger brought her nephew Thomas to her.

On the wreck. The people on that section of the wreck had almost no food or water. They saved the wine and peppermint for the children. Rebecca Lamar tore strips from her dress into squares. She wet them and laid them on the children's heads during the day, to keep the children from getting thirsty. On the second day six men who could swim took the best small boat to row for help. Her brother was one of them. They reached the beach 40 miles above Wilmington, North Carolina.

People on the wreck died one by one. A man named Mr. Brown drank sea water. In a storm a woman named Mrs. Woart died, and a wave carried off her body and her husband, the Reverend Woart. Two men became delirious, which means confused and out of their minds, and walked into the sea.

A man named Mr. Smith set off on a raft he had built and was never heard from again. On the Monday night her nephew Thomas died in her lap.

The third search. On the following Tuesday the schooner *Henry Cameron*, a sailing ship with two masts or more, under Captain Eli Davis, found them. The *Pulaski*'s sailing master, the officer in charge of sailing the ship, Captain Pearson, was already on board, rescued from another piece of the wreck. That group numbered 23 people in all. He asked Davis three times to search for more survivors. The third search found Rebecca Lamar's group.

Her brother's wife, five of their six children and a niece died. Her brother and his son Charles, age 14, lived. Charles later led the investors who paid for the ship *Wanderer* to smuggle captive Africans into Georgia in 1858. Chapter 24, Slavery and Freedom, tells that voyage.

> Rebecca Lamar later married Hugh McLeod. She wrote her account of the wreck without planning to publish it. The *Georgia Historical Quarterly* later printed it.

A safety law no one enforced, 1838

Members of Congress passed a law "to provide better security of the lives of passengers" on steamboats on July 7, 1838. It was the first federal law that required steamboats to be inspected. Under it, federal judges named engineers as inspectors. The inspectors checked each boat and gave the owner two certificates, signed papers the owner needed to get a license to run the boat. Owners paid 5 dollars for each inspection, repeated every 6 or 12 months.

No one to enforce it. No federal officer enforced the law. Only a person who sued the owner in court could make an owner obey it. After 1838 steamboat disasters came more often and killed more people.

A tornado at Natchez, 1840

A tornado, a spinning column of wind that reaches down from a storm cloud to the ground, struck the Mississippi River at Natchez, Mississippi, at about 1 in the afternoon on May 7, 1840. It struck Vidalia, Louisiana, across the river, too. At the Natchez landing it threw 116 of the 120 flatboats into the river. Flatboats were large, flat-bottomed boats that carried farm goods and people downriver, and their crews and passengers drowned.

At least 317 people died, 269 of them on the river and 48 on land. The courthouse at Vidalia fell and killed Judge Keeton. Many bodies were never identified. By that count it is the second deadliest tornado in the country's history.

Left out of the count. The count of 317 leaves out the enslaved people killed on plantations in Louisiana. P. M. Lapice reported that on his plantation below Vidalia the tornado blew down all the houses of the enslaved workers. Four or five people were killed outright. About twenty were badly hurt, and some of them had wounds so bad that they were dying.

The *Mississippi Free Trader*, a Natchez newspaper, printed reports of hundreds of enslaved people killed on plantations up to 20 miles away, but no one made an exact count.

After the storm Stephen Duncan, a planter, the owner of a large farm worked by enslaved people, paid to turn the Tremont House hotel into a hospital. Doctors and several thousand dollars came from New Orleans. Other slaveholders in the area sent large groups of enslaved people to clear the ruins and pull out the bodies.

The steamboat Erie burns, 1841

The steamboat *Erie* carried passengers between Buffalo, New York, and Chicago. On the evening of August 9, 1841, it left Buffalo with about 340 people on board. The boat had just been painted and varnished.

A group of six painters, hired by William G. Miller of Buffalo, planned to ride only to the first stop, in Pennsylvania. They had left jugs of paint, varnish and turpentine, a paint thinner that burns easily, on the deck, near the boiler's vent pipe. That spot on the deck had caught fire three times before.

At about 8 p.m., off Silver Creek, New York, on Lake Erie, the turpentine exploded. Fire spread over the freshly painted boat within seconds. Two of the lifeboats turned over. The third carried the captain, Thomas Jefferson Titus, and four others. At 10 p.m. the crew of another steamboat, the *DeWitt Clinton*, arrived and pulled survivors from the lake. Fewer than 100 people were saved. The first counts put the dead at about 170. Later counts found as many as 254. All six painters died.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 17 MINUTES

More than 2,000 Union soldiers were crowded onto the steamboat *Sultana*, built to carry 376, in April 1865. Its sinking, the Peshtigo fire and the Johnstown Flood each killed more than 1,000 people. They are still the country's deadliest shipwreck, deadliest wildfire and deadliest dam failure. After the *Sultana* and Johnstown, survivors and investigators blamed named men, and no court punished any of them.

In the same years Americans built groups whose job was to warn people of danger or to help them afterward. Under a law of 1870, Army soldiers took on the job of warning people about storms. Clara Barton founded the American Red Cross in 1881.

What the Steamboat Act of 1852 changed

Steamboat explosions and fires went on after the law of 1838. From 1847 to 1852, boiler explosions caused a long run of disasters. On August 30, 1852, members of Congress passed a stronger law, the Steamboat Act. Under it, officials of the Treasury Department enforced the rules through supervising inspectors, who were in charge of the local inspectors. Under the old law, only a lawsuit could make an owner obey.

Inspectors had to test every boiler by filling it with water under pressure, to see whether it leaked or burst before it was used with steam. Owners had to fit each boiler with a safety valve, a part that lets steam out before the pressure gets too high. Pilots and engineers had to earn a license from the inspectors. The law left out freight boats, ferries and tugboats. A law of 1871 covered all steam vessels and their crews.

The Sultana, 1865: 2,000 men on a boat built for 376

The worst shipwreck in the country's history happened on the Mississippi River in April 1865, just as the Civil War was ending. The steamboat *Sultana* was built to carry 376 passengers. At Vicksburg, Mississippi, Union Army officers crowded more than 2,000 Union soldiers onto it. Most of the soldiers had just been freed from Confederate prison camps at Cahaba, Alabama, and Andersonville, Georgia.

The deal. Army officials paid boat owners for each soldier they carried home. They paid 2 dollars and 75 cents for each enlisted soldier, meaning one who was not an officer, and 8 dollars for each officer. Captain Reuben Hatch was the Army's chief quartermaster, or supply officer, at Vicksburg. He offered the *Sultana*'s captain, J. Cass Mason, a full load of about 1,000 soldiers. In return Hatch wanted a kickback, a secret payment for the favor.

One of the boat's boilers was leaking. Mason had it patched instead of making the longer repairs a full fix would take. Captain George Williams, the officer in charge of loading, put every man at the Vicksburg parole camp, where freed prisoners waited to go home, on board. He believed there were fewer than 1,500.

The explosion. The boilers exploded at about 2 in the morning on April 27, 1865, about seven miles north of Memphis, Tennessee. The boat caught fire. Investigators blamed low water in the boilers while the overloaded boat leaned to one side, along with the patched boiler. Men who were thrown into the river floated past Memphis in the dark.

Crews of other steamboats and of U.S. Navy ships pulled survivors from the water. On the Arkansas shore, the Fogelman family tied logs into a raft and took 25 soldiers off the burning boat. Franklin Hardin Barton, a Confederate soldier of the 23rd Arkansas Cavalry, was among the rescuers. The *Sultana* burned and sank at about 7 a.m.

Almost everyone on board was a Union soldier. Counts of the dead range from about 1,100 to about 1,800. General William Hoffman investigated for the Army and counted 1,238 in May 1865. Officials of the Customs Service, who

kept records of ships, counted 1,547. The researcher Jerry Potter estimates about 1,800. About 200 of the men pulled out alive later died of their burns. Mason and most of the boat's officers died.

No one punished. Captain Frederic Speed of the Army was tried and found guilty of overcrowding the boat. The Army's top lawyer, the Judge Advocate General, overturned the verdict. He found that Speed had spent the whole day at the camp and had not put a single soldier on board. Hatch quit the Army and refused three orders to come and testify. No one was punished.

Fire on October 8, 1871: Chicago and Peshtigo

On the evening of October 8, 1871, two fires began that together killed well over a thousand people. The smaller one burned the city of Chicago. The larger one burned the forests and towns of northeastern Wisconsin and Michigan's Upper Peninsula.

The Chicago fire started at about 8:30 p.m. in or near a small barn that belonged to Catherine O'Leary's family. More than two-thirds of the buildings in Chicago were made entirely of wood, and most had roofs of tar or wooden shingles, which burn easily. Only 1 inch of rain had fallen since early July. Strong winds from the southwest carried burning pieces toward the center of the city.

The fire burned until October 10. It destroyed about 17,500 buildings over more than 3 square miles and left 90,000 to 100,000 people without homes. Searchers found 120 bodies. The county coroner, the official who looks into deaths, said that an exact count was impossible. The death toll may have been as high as 300.

The cow story. A newspaper reporter, Michael Ahern, wrote that Catherine O'Leary's cow and a lantern had started the fire. She was a poor Irish Catholic immigrant, which made her an easy person to blame. In 1893 Ahern admitted that he had made the story up. In 1997 the members of the Chicago City Council voted to clear her name. The historian Richard Bales has suggested that a man named Daniel Sullivan started the fire.

Mayor Roswell B. Mason put the city under martial law, rule by the army, and placed General Philip Sheridan in charge from October 11 to 24. He also gave a private charity, the Chicago Relief and Aid Society, control of the relief. Donors in New York City gave $450,000 and donors in St. Louis $300,000. Money came from London too.

Peshtigo. The fire in Wisconsin began with people who were clearing forest for farms and railroads. They set small fires to burn away the brush. On October 8 a cold front, the edge of a large mass of cold air, moved east and made the wind stronger, and the wind joined several of those fires into one.

It burned about 1.2 million acres. It destroyed Peshtigo, Wisconsin, a town of about 1,700 people, and 16 other towns. People jumped into the Peshtigo River, into wells and into other water nearby to escape the flames. Some drowned, and others died of cold in the water.

The fire burned the town's records, so no one knows exactly how many people died. Estimates run from 1,200 to 2,400. In 1873 a report to Wisconsin's lawmakers listed 1,182 names of people who died or went missing. More than 350 bodies were buried in one grave, because no one was left alive who could name them. It is the deadliest wildfire in the country's recorded history.

Peshtigo stood on forest land that had belonged to the Menominee Nation. Henry Dodge, the governor of Wisconsin Territory, signed a treaty, a written agreement between nations, with the Menominee in 1836, at a place called The Cedars, on the Fox River. Under the Treaty of the Cedars, the Menominee handed over about 4 million acres to the United States, and United States officials agreed to pay $700,000. Loggers and white settlers then moved onto the land.

The Red Cross, 1881, and an earthquake in Charleston, 1886

Clara Barton and a group of supporters founded the American Red Cross in Washington, D.C., on May 21, 1881. On September 5, 1881, forest fires burned more than a million acres in the Thumb, the part of Michigan shaped like a thumb. The fires killed 282 people and left thousands without homes. Red Cross workers sent money, clothes and furniture to about 14,000 people in need there. It was the Red Cross's first disaster relief.

An earthquake of about magnitude 7 shook Charleston, South Carolina, at about 9:50 on the night of August 31, 1886. It killed 60 people and damaged about 2,000 buildings. Most of the damage, 81 percent of it, was to brick buildings. Wooden buildings were damaged much less than brick ones.

Storm warnings and the blizzards of 1888

President Ulysses S. Grant signed a law on February 9, 1870, that started a national weather service inside the Army's Signal Service, the branch that handled signals and telegraph messages. The law required the Secretary of War to have weather readings taken and to send notice of coming storms, by telegraph and signals, to the northern lakes and the seacoast.

The Schoolchildren's Blizzard. A blizzard, a snowstorm with very strong wind and bitter cold, struck the northern Great Plains without warning on January 12, 1888. The morning had been warm, near 50 degrees. In Huron, Dakota Territory, many children left for school without winter coats. About 235 people died in Dakota Territory, Nebraska, Minnesota and Iowa, by one count. Other estimates range from 250 to 500, and some go as high as 1,000. People call it the Schoolchildren's Blizzard.

Near Ord, Nebraska, the wind tore the door and the roof off the schoolhouse of Minnie Freeman, a teacher of 19. The schoolhouse was made of sod, blocks of earth held together by grass roots. Freeman tied her students together with twine and carried the youngest in her arms. She led them to a farmhouse about half a mile away, and all of them lived.

In Plainview, Nebraska, the teacher Loie May Royce tried to lead three students to her home, less than 90 yards from the school. The wind-driven snow blinded them, and they lost their way. Searchers found them the next morning. The three children were dead. Royce's arms and legs were frostbitten, which means the cold had frozen and damaged the flesh.

In Turner County, in present-day South Dakota, five boys were found dead four days after the storm. They were John Albrecht, Peter Graber, and Henry, John and Elias Kaufman.

A new Weather Bureau. Another blizzard struck the Northeast in March 1888 and killed about 400 people. Forecasters of the Signal Service had predicted brisk winds and rain, then fair weather. After the two storms, members of Congress passed a law on October 1, 1890. Under it, weather forecasting moved from the Army to a new Weather Bureau in the Department of Agriculture. The bureau's forecasters were civilians, people who were not in the army.

Who weakened the Johnstown dam?

The dam that broke above Johnstown, Pennsylvania, belonged to a private club of wealthy Pittsburgh men. The changes they made had weakened it. The South Fork Dam was 69 feet high and held back a lake 14 miles up the valley from Johnstown. Henry Clay Frick led a group of Pittsburgh businessmen who bought the lake for the South Fork Fishing and Hunting Club. The club had more than fifty members, most of them steel, coal and railroad owners.

Workers the club hired lowered the top of the dam by as much as 3 feet to make room for a road for carriages. They put a screen across the spillway, the channel that lets extra water out, to keep the club's fish in the lake. Someone had removed the pipes at the bottom of the dam, which could let water out, before the club bought it, and club members never replaced them. The club's members had no way to lower the lake in an emergency.

May 31, 1889. The heaviest rain ever recorded in that part of the country fell on western Pennsylvania on May 30 and 31. Army weather officers estimated 6 to 10 inches in 24 hours. The dam broke in the afternoon of May 31, 1889.

Earlier that day the club's president, Elias Unger, had twice sent a man named Parke on horseback to a telegraph office to warn Johnstown. The people who received the warnings did not pass them on to the town's leaders, because there had been many false alarms before.

The water rushed down the valley. The railroad engineer John Hess drove his engine backward toward East Conemaugh with the whistle blowing the whole way. His warning saved many people. At the Stone Bridge in Johnstown, the wreckage piled up against the bridge and caught fire. At least 80 people died there, and the fire burned for three days.

The dead. The flood killed 2,208 people. The first count was 2,209, until Leroy Temple, counted among the dead, turned up alive eleven years later. Ninety-nine whole families died. National Park Service staff count 396 children under the age of 10 among the dead. Of all the dead, 777 were never identified.

Who was left. The flood left 124 women and 198 men widowed. It left 98 children orphaned, and by the Park Service's count 568 children lost one or both parents.

Who paid. The American Society of Civil Engineers, a group of engineers, released a report in 1891. It stated that the dam would have failed even if the club had kept it as first built. The group's president in 1890, William Shinn, had been a business partner of Andrew Carnegie. The report was kept from the public for two years. The authors of a study published in 2016 found that the club's changes cut the amount of water the dam could safely let out by half.

Survivors sued the club. Two of its members, the lawyers Philander Knox and James Hay Reed, defended it in court. They argued that the flood was an "act of God," a natural event that no person could be blamed for. The survivors lost, and no one ever paid them for their losses. Frick and about half of the club's members gave money to help the survivors. Andrew Carnegie built the town a new library.

Victor Heiser

Who: A 16-year-old who rode the Johnstown Flood on a barn roof, lost both his parents, and later became a doctor.

When and where: Johnstown, Pennsylvania, May 31, 1889.

From a barn roof he watched his house crushed at 4:20 p.m.

Victor Heiser was 16 when the Johnstown Flood destroyed his home and killed his parents. As the river rose that afternoon, his father sent him to the barn to untie the horses. He heard what he called a "dreadful roar." He looked back and saw his parents at a window of the house. His father waved at him to climb onto the barn roof.

From the roof he saw his house "crushed like an eggshell." He noted the time: 4:20 p.m. The barn floated away with him on its roof. When it was about to hit a neighbor's house, he jumped onto that house. It began to fall in, and he fell back onto the barn roof. The water carried him down to the Stone Bridge.

He spent the night in a house with 19 other people. The next day he went looking for his parents. He tried to help people trapped in the burning wreckage at the bridge. For two weeks he searched among the bodies of the dead. He found the body of his mother, Mathilde Heiser. He was never able to identify his father, George.

Heiser later studied at Jefferson Medical College and became a doctor. He worked in many countries against diseases such as leprosy, cholera and malaria. In 1936 he told his story in a book, *An American Doctor's Odyssey*.

The Sea Islands hurricane, 1893

A hurricane came ashore near Savannah, Georgia, on August 27, 1893, and drove the sea over the low Sea Islands of South Carolina. Between 1,000 and 2,000 people died, most of them drowned. Most people on the islands were Black. On St. Helena Island about 6,000 Black people lived, and most of them owned their land. Many of their houses stood no more than 2 feet above the ground.

Who got the food. Red Cross workers arrived on October 1, more than a month after the storm, and ran a relief effort for 10 months. By the Red Cross's own count, Clara Barton's workers helped about 30,000 people left homeless by the storm, most of them Black. The historian Caroline Grego wrote about the relief work in 2019. She found that South Carolina politicians and newspaper writers pushed Red Cross leaders to send supplies to white farmers instead of to Black islanders. The Democratic Party leaders who ran the state government worked against the efforts of Black islanders to earn a steady living again.

In an online exhibit for the College of Charleston, Grego names some of the people involved. In March 1894 Eugene Gregorie, a white farmer at Pocotaligo, asked the politician William Elliott to tell Clara Barton that the free food the Red Cross handed out was hurting the white planters. Gregorie wrote that Black workers were refusing to work in the planters' fields while the Red Cross fed them, and he wanted the food stopped before planting time in May.

Black farmworkers on the rice plantations nearby sent seven petitions, written requests signed by many people, to Barton that spring. They wrote that they wanted to work, but the planters had little work to give and paid about 40 cents a day.

The *Columbia Daily Register* printed an article titled "Cursing the Red Cross!" in May 1894. It said that outside the towns of Beaufort and Port Royal, not a dozen white men in Beaufort County had a good word for the Red Cross. Governor Benjamin Tillman backed a collection of money for white farmers at Bluffton, after newspapers claimed that the Red Cross was discriminating against white sufferers.

Clara Barton

Who: The founder of the American Red Cross, who led its relief work after fires, floods and hurricanes.

When and where: Michigan, 1881. Johnstown, Pennsylvania, 1889. The South Carolina Sea Islands, 1893. Galveston, Texas, 1900.

Her house was built from the lumber of Johnstown's Red Cross buildings.

Chapter 26, Health, Disease, and Medicine, tells how Clara Barton carried supplies to Civil War battlefields and founded the American Red Cross. She led its relief work after disasters for more than 20 years.

She reached Johnstown on June 5, 1889, five days after the flood. She was 67. About 50 Red Cross workers came with her from Washington, D.C., and they stayed five months. Her workers built "Red Cross hotels" where people who had lost their homes could stay. Most of the people who rented rooms in them were merchants and businessmen. Red Cross workers gave out supplies worth $211,000 and helped about 25,000 people.

A separate Red Cross group from Philadelphia ran a hospital in Johnstown. The historian David McCullough wrote that Barton "spoke her mind once or twice" to the head of that group. Within a few days neither group would have anything to do with the other.

After the Sea Islands hurricane of 1893 she led the Red Cross relief on the islands. Her last relief work came after the Galveston hurricane of September 1900.

Her house in Glen Echo, Maryland, was built partly from lumber of the Red Cross buildings at Johnstown. She lived there from 1897 to 1912.

BY 1900

If you remember one thing: The deadliest shipwreck, wildfire and dam failure in the country's history all came in these fifty years, and no one was punished for the *Sultana* or for Johnstown.

- 1865: More than 2,000 freed Union prisoners crowd onto the *Sultana*, and between 1,100 and 1,800 of them die.
- 1871: The Peshtigo fire kills between 1,200 and 2,400 people on the same night that Chicago burns.
- 1889: The South Fork Dam breaks, and 2,208 people die in Johnstown.

Next: In September 1900 a hurricane flooded Galveston, Texas, and killed more people than any other natural disaster in the country's history.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 25 MINUTES

The Galveston hurricane of September 1900 killed between 6,000 and 12,000 people, more than any other natural disaster in the country's history. In the fifty years after it, many more people died because of choices other people made.

Witnesses said the owners of a New York blouse factory kept an exit door locked. Near Los Angeles the city's chief engineer and his assistant judged a leaking dam safe, and in Texas members of a school board piped waste gas into a school to save money. After the Triangle fire, the *Titanic*, the New London school explosion and the Cocoanut Grove fire, lawmakers wrote new safety rules for factories, ships, gas lines and nightclubs.

In three disasters, the people in charge treated Black survivors and Black victims worse than white ones. In Galveston, members of an armed militia forced Black workers at gunpoint to bury the dead. In the 1927 Mississippi flood, the local leaders who ran many relief camps gave supplies to white

people first and forced Black people to work. After the 1928 hurricane in Florida, officials saved the few coffins for white victims, and the bodies of Black victims were burned in piles or thrown into mass graves.

Galveston, September 8, 1900

A hurricane flooded Galveston, Texas, on September 8, 1900, and killed more people than any other natural disaster in the country's history. Galveston was a port city of about 38,000 people on an island in the Gulf of Mexico. At about 6:30 that evening, a storm wave put the city under as much as 15 feet of water. The highest water mark was 15.7 feet. The storm destroyed a third of the city, including between 2,636 and about 3,600 homes by different counts.

Between 6,000 and 8,000 people died in the city. Estimates for the whole island and the nearby coast run from 10,000 to 12,000. The *Galveston News* printed a list of 4,263 dead in early October, months before searchers found the last body.

Searchers found about 70 bodies a day during the first month. The last body was found on February 10, 1901. Galveston's leaders put the city under martial law, which means soldiers or armed guards take over from the police and courts. A volunteer militia, a group of armed citizens, patrolled the streets against looters, people who steal from damaged buildings. Some accounts say that men accused of looting were shot, but the records do not give their names or a count.

Forced at gunpoint. Too few volunteers came forward to handle the dead, so armed men rounded up other men at gunpoint or bayonet point to do it. The writer Elizabeth Blasius found that members of the volunteer militia did this, and that many of the men they forced were Black workers. About one fifth of the people of Galveston were Black. Even before the storm, the city's segregation rules, which kept people apart by race, kept Black Galvestonians from drinking at the public fountains or going to the public schools.

At first, workers loaded bodies onto barges, tied heavy rocks to them and dropped them into the Gulf. Days later, some of the bodies floated back to shore. After that, workers burned the dead in large fires. The fires burned into November.

Clara Barton, 78, arrived with Red Cross workers on September 17, nine days after the storm. Mayor Walter C. Jones led a Central Relief Committee that included the banker John Sealy. The committee's members spent its money on 483 new houses and on partial help toward 1,114 more.

The seawall. Afterward, workers built a seawall that stood 17 feet above the low-tide line. Its first section ran three and a half miles. Workers then raised the ground under the city by as much as 17 feet. They lifted about 2,000 buildings on jacks first. St. Patrick's Church, which weighed 3,000 tons, went up five feet.

Isaac Cline

Who: The chief of the U.S. Weather Bureau office in Galveston, whose pregnant wife died in the 1900 hurricane.
When and where: Galveston, Texas, September 8, 1900.

He had called a hurricane harming Galveston "a crazy idea."

Isaac Cline took charge of the Galveston weather office in 1889, when the Army's Signal Service still ran the country's weather service. The office passed to the new Weather Bureau in 1890, and his brother Joseph joined him there in 1892. Isaac wrote an article for the *Galveston Daily News* in 1891, nine years before the storm. In it he called the thought of a hurricane seriously harming Galveston "a crazy idea." Many people in the city had asked for a seawall. His article was one reason no seawall was built before the storm.

The flags. On September 8, 1900, water began flooding the low parts of the city. Around noon, Cline raised the hurricane warning flags without waiting for approval from the Weather Bureau's main office in Washington. Weather Bureau rules required that approval.

Cline later wrote that he drove a horse-drawn cart along the beach and warned people to leave. No eyewitness ever reported seeing him do it. The writer Erik Larson argued that Cline warned no one before his office sent out its official warning.

> Cline's wife, Cora, was pregnant with their fourth child, and she died in the storm. Cline nearly drowned. He saved his youngest daughter, Esther, who was 6. Joseph saved the two older girls, Allie May, 12, and Rosemary, 11.
>
> Cline moved to New Orleans in 1901. Between 1900 and 1924 he charted the paths of 16 storms.

San Francisco, 1906: the fire after the earthquake

Fire did most of the damage after the San Francisco earthquake of April 18, 1906. The earthquake struck at 5:12 in the morning. It had a magnitude, the number scientists use for an earthquake's strength, of about 7.9, and it shook the city for about 42 seconds. It broke gas pipes, and within three days more than 30 fires had spread across the city. Between 80 and 95 percent of the damage came from the fires.

Firefighters tried to blow up buildings with dynamite to make empty gaps the fire could not cross. They had no training with dynamite, and some of their blasts started new fires. The city's fire chief, Dennis T. Sullivan, had been hurt in the earthquake itself, when a building next door fell onto the fire station where he lived. He died four days later.

General Frederick Funston sent in Army soldiers. Some building owners set fire to their own damaged buildings, because their insurance paid for fire damage and not for earthquake damage.

More than 80 percent of the city was destroyed. Between 227,000 and 300,000 people lost their homes, out of about 410,000. The first counts of the dead ran from 375 to more than 500. Those counts left out hundreds of deaths in Chinatown that no one recorded. In 2005 the members of the Board of Supervisors, San Francisco's city lawmakers, voted to accept more than 3,000 as the official count. They voted for a written statement by the novelist James Dalessandro and the city historian Gladys Hansen.

On the day of the earthquake, Mayor Eugene Schmitz announced that he had given soldiers and police permission "to kill any and all persons found engaged in Looting or in the Commission of Any Other Crime." The records do not say how many people they killed. People also accused soldiers themselves of looting.

Chinatown. About 15,000 Chinese people lived in Chinatown, and they lost nearly everything. Even before the earthquake, some city officials had wanted to move the Chinese out of Chinatown to Hunters Point. After it, they tried to use the ruins to take Chinatown's land for business buildings.

The Chinese were the only group that officials kept apart in the camps for people who had lost their homes. Army officers moved the Chinese survivors to camps farther and farther from Chinatown. Most Chinese survivors left for Oakland, and only about 400 stayed in the city. The officials' plan to move Chinatown failed. Chinese residents rebuilt Chinatown.

Hugh Kwong Liang

Who: A 15-year-old in San Francisco's Chinatown who lived through the 1906 earthquake and fire.
When and where: San Francisco, April 1906.

He was 15, and the building across the street fell down.

Hugh Kwong Liang was 15 when the earthquake woke him in Chinatown. He is one of the few Chinese survivors whose own account of the earthquake has lasted to today. He remembered seeing "pieces of plaster falling down like water." He ran out into Washington Street, and the building across from his home fell down.

As the fire spread, city officials who were directing the people fleeing told him and his family to go to the open ground of the Presidio. The Presidio was an Army post by the Golden Gate, the entrance to San Francisco Bay. He remembered the moment he left: "I turned away from my dear old Chinatown for the last time."

> At the Presidio, Army officers kept the Chinese survivors apart from everyone else. They moved them from camp to camp. The last camp stood in a cold, windy corner of the Presidio near Fort Point.

The Triangle fire, March 25, 1911

A fire at the Triangle Waist Company in New York City killed 146 workers on March 25, 1911. Witnesses said the owners kept one of its exit doors locked. The company made women's blouses on the top three floors of the ten-story Asch Building. Most of its workers were young immigrant women, and many were teenagers. The youngest who died, Kate Leone and Rosaria Maltese, were 14.

The fire started on the eighth floor just before the 4:45 p.m. quitting time. Workers ran down the Greene Street stairs, crowded into the elevators, and climbed onto the one fire escape, which collapsed.

The locked door. On the ninth floor, the door to the Washington Place stairs would not open. Witnesses told the court that the owners, Max Blanck and Isaac Harris, kept it locked so workers could not leave with stolen blouses. Harris later admitted that all the blouses ever stolen were worth less than 25 dollars.

The firefighters' ladders reached only to the seventh floor. Workers trapped on the ninth floor could not get past the fire, and many jumped from the windows. A crowd in the street watched 62 people jump or fall to their deaths. Joseph Flecher, who worked in the company office, watched from the roof and saw his workers "hit the sidewalk spread out and still."

On the tenth floor, workers climbed to the roof. A law professor at New York University, Frank Sommer, and his students found ladders that painters had left and set them across to the school's roof. Workers climbed over them to safety.

The trial. Blanck and Harris were put on trial for manslaughter, the crime of killing a person without meaning to, through carelessness. Judge Crain told the jury it had to find that the owners knew the door was locked. On December

27, 1911, after less than two hours, the jury found them not guilty. One juror, Victor Steinman, said he believed the door was locked, "but we couldn't find them guilty unless we believed they knew the door was locked."

The families in 23 lawsuits against the building's owner settled on March 11, 1914, which means they agreed to take money and end the lawsuits. They got an average of 75 dollars for each life.

New York lawmakers set up a nine-member Factory Investigating Commission. From 1911 to 1914 they passed 36 new laws to change the state's rules for workplaces. A state law already in force at the time of the fire had required factory doors to open outward where possible and to stay unlocked during working hours.

Kate Alterman

Who: A worker who escaped the Triangle fire and told the jury how another worker, Margaret Schwartz, died at the locked door.
When and where: New York City, March 25, 1911, and the trial in December 1911.

With her coat turned inside out, she ran through the flames.

Kate Alterman saw another worker, Margaret Schwartz, catch fire at the locked Washington Place door, and she told the jury what she saw. Margaret died in the fire. Kate worked on the ninth floor. She had lived in Philadelphia for nine years, and she had been in New York and worked at the Triangle factory for four months.

When the fire reached the ninth floor, the two women ran to the Washington Place door. The manager's brother, whose last name was Bernstein, tried it and could not open it. Margaret tried next. Then Kate pulled the handle "in and out, all ways," and it still would not open.

Kate saw Margaret kneel down by the door. She saw the hem of Margaret's dress and the ends of her loose hair start to burn. Margaret screamed, "Open the door! Fire! I am lost, there is fire!"

> Kate turned her coat inside out and put it over her head with the fur against her face. She covered her head with a bunch of dresses. Then she ran out through the flames. "The whole door was a flame right to the roof," she said.
>
> **In court.** At the trial the prosecutor, Charles Bostwick, used her account to show that the locked door caused Margaret Schwartz's death. The owners' lawyer, Max Steuer, made her tell the story again and again. She used almost the same words each time. In this way Steuer made her story seem less believable without ever saying she was lying. She said she told it the same way "because he asked me the very same story over and over."

BY THE END OF 1911: Thousands had died in Galveston and San Francisco, a jury had found the Triangle owners not guilty, and New York lawmakers were writing new rules for the state's factories.

Ships: the Titanic, 1912, and the Eastland, 1915

The *Titanic* carried lifeboats for only some of the people on board, and United States senators blamed British officials for that. The *Titanic* was a British passenger ship. It hit an iceberg in the North Atlantic and sank on April 15, 1912. More than 1,500 passengers and crew died. The ship *Carpathia* picked up about 710 survivors.

The ship had too few lifeboats for everyone, but it met the rules of the British Board of Trade, the government office in charge of British ships. Senator William Alden Smith of Michigan led a Senate investigation. It began in New York on April 19, four days after the sinking. The senators found that Captain Edward Smith's "indifference to danger" was one of the direct causes of the deaths. They blamed the Board of Trade's officials for the missing lifeboats.

The senators called for ships to slow down near ice, to carry enough lifeboats for all on board, and to keep a radio operator on duty day and night. In March 1915, after the *Titanic* sinking, members of Congress passed the Seamen's Act, pushed by Senator Robert La Follette of Wisconsin.

The *Eastland*. The Great Lakes passenger ship *Eastland* rolled over on July 24, 1915, while it was tied to a dock in the Chicago River. It was taking workers of the Western Electric Company and their families to a company picnic in Indiana, and it had already filled to its limit of 2,500 passengers.

The ship had tipped badly before, including in 1906. Under the new law, its owners had added a full set of lifeboats so it could carry more people. That added weight may have made the ship more dangerous. At 7:28 a.m. it rolled onto its side and settled on the river bottom, only 20 feet down. In all, 844 people died. Many of the dead were immigrants, and 220 of them were Czech.

A deckhand, Peter Boyle, drowned while saving people. A nurse, Helen Repa, led much of the rescue work on shore. An Illinois grand jury, a group of citizens who decide whether there is enough evidence for a trial, charged the president of the steamship company and three other company officers with manslaughter. It charged the captain and the engineer with criminal carelessness.

No trial. The men were outside Illinois, and a federal court refused to send them there for trial. The court found "barely a scintilla of proof," meaning almost no evidence, because the four company officers had not been on the ship. None of the men was ever tried in Illinois.

Who was helped first in the 1927 flood?

About 637,000 people had to leave their homes in the Mississippi River flood of 1927. In the relief camps, local leaders gave supplies to white people first and forced Black people to work. In the spring the river broke through its levees, walls of earth built along a riverbank to hold the water in, in at least 145 places. Water covered 27,000 square miles, as deep as 30 feet. About 500 people died.

The dynamite. A group of New Orleans bankers arranged to set off about 30 tons of dynamite on the levee at Caernarvon, Louisiana, below their city, on April 29. They wanted to let the river out there to keep it from badly damaging New Orleans. The water flooded St. Bernard Parish and part of Plaquemines Parish, as Louisiana calls its counties. Blowing up the levee turned out to be unnecessary. The New Orleans businessmen never paid the people whose homes and land they flooded.

In the camps. About 325,000 people lived in relief camps, and 69 percent of them were Black. Near Greenville, Mississippi, more than 13,000 people escaped to the top of a levee that had held. Many of them were stranded there for days without food or clean water.

Local leaders from the countryside ran many of the camps. They gave supplies and boats to white people first, and Black people got what was left. A Black person got supplies only by giving the name of a white employer. Black people in the camps were forced to work and were not allowed to leave. The records do not name the camp leaders who gave these orders.

A white police officer ordered a Black man at gunpoint to unload a relief boat. The man refused, and the officer shot and killed him. The records do not give the name of the man who was killed.

Near Helena, Arkansas, a plantation overseer, the man who bossed the workers, tried to force a Black man, Owen Flemming, to rescue the plantation owner's mules. Flemming killed the overseer. Then he was lynched, which means he was killed by a mob without a trial. The records do not name the men who killed him.

Herbert Hoover, the Secretary of Commerce, led the national government's relief work. A group of Black leaders headed by Robert Russa Moton studied the camps and reported what was happening in them. Hoover asked that their reports be kept out of the newspapers, and they were. After the flood, members of Congress passed the Flood Control Act of 1928.

Two disasters of 1928: the St. Francis Dam and Lake Okeechobee

The St. Francis Dam in northern Los Angeles County broke just before midnight on March 12, 1928. That same day its builder had looked at a new leak and judged it safe.

William Mulholland was the chief engineer of the city's Bureau of Water Works and Supply. His workers had built the dam from 1924 to 1926. Leaks and cracks showed up in 1927 and 1928. That morning the dam keeper, Tony Harnischfeger, found a new leak. Mulholland and his assistant, Harvey Van Norman, looked at it and decided it was not dangerous.

The dam gave way at about 11:57 that night. Harnischfeger and his family were probably among the first people killed. The flood ran 54 miles to the Pacific Ocean and killed at least 431 people. Among them were 84 workers of the Southern California Edison company at a construction camp beside the river.

At the investigation afterward, the *Express* newspaper reported that Mulholland said, "the only ones I envy are those who are dead." The *Los Angeles Times* printed his words as "on an occasion like this I envy the dead."

The jury's finding. A coroner's jury, a group of citizens who decide how people died, ruled on the collapse on April 12, 1928. The jurors found an error in engineering judgment about the rock under the dam, and they put that blame on Mulholland's bureau and its chief engineer. They found a second error, in the rules meant to keep the public safe. They put that blame on the people the chief engineer answered to: the city's water and power commissioners, the lawmakers of the city and the state, and the public.

The jurors wrote that a great dam should never be left "to the sole judgment of one man, no matter how eminent," meaning no matter how famous and respected. They found no crime and recommended that no one be charged.

Mulholland's career as chief engineer ended after the dam broke.

Lake Okeechobee. Six months later, early on September 17, 1928, a hurricane came ashore near West Palm Beach, Florida. It had already killed 312 people in Puerto Rico. The storm pushed the water of Lake Okeechobee over the low dike, a bank of earth built to keep water back, at the lake's south end. Floods more than 20 feet deep covered the farm towns of Belle Glade, Canal Point, Chosen, Pahokee and South Bay.

At least 2,500 people died in the United States. For many years the official count was the Red Cross estimate of 1,836. About three out of four of the dead were migrant farm workers, people who moved from farm to farm to pick crops. Most of them were Black.

Black workers did most of the cleanup after the storm. Officials saved the few coffins they had for the bodies of white people. White victims were buried with a service, in a shared grave at Woodlawn Cemetery in West Palm Beach. The bodies of Black victims were burned in piles or thrown into mass graves. One mass grave at Port Mayaca holds 1,600 bodies. The records do not name the officials who made these decisions.

Robert Hazard of West Palm Beach started a group to win recognition for the Black victims buried in the city's mass grave. The city bought the grave site back in 2000. In 2003 a state historical marker was put up there.

BY 1929: The Mississippi flood had driven about 637,000 people from their homes, the St. Francis Dam and the Okeechobee hurricane had killed thousands more in 1928, and members of Congress had passed the Flood Control Act of 1928.

Black Sunday, April 14, 1935

A wall of blowing dirt swept across the southern Great Plains on April 14, 1935, with winds of up to 60 miles an hour. It reached Beaver, Oklahoma, at about 4 p.m. and Boise City, Oklahoma, at about 5:15 p.m. For a short time the sky was so black that people could not see their own hands in front of their faces. The day became known as Black Sunday.

An Associated Press reporter, Robert Geiger, and a photographer, Harry Eisenhard, were caught in the storm six miles from Boise City. The next day Geiger wrote a news story that began, "Residents of the southwestern dust bowl marked up another black duster today." A duster was a dust storm. His story is generally accepted as the first use of the words "dust bowl."

Why natural gas smells: New London, 1937

A gas explosion destroyed a school in New London, Texas, on March 18, 1937, and killed about 298 people, most of them children. Members of the school board had piped in the gas to save money. Early that year, board members canceled the school's contract for natural gas. They had plumbers connect the school to a pipe of leftover waste gas from a nearby company, the Parade Gasoline Company.

Natural gas that has not been treated has no smell and no color. Gas leaked from the new connection and built up in the closed crawlspace, the low space under the floor, which ran the whole 253-foot length of the building's front. Students had been complaining of headaches for some time, but few people paid attention.

About 500 students and 40 teachers were in the building that afternoon. At 3:17 p.m. Lemmie R. Butler, who taught shop, turned on an electric sander. A spark from its switch probably set off the gas. The *Handbook of Texas*, published by the Texas State Historical Association, records about 298 dead. Other counts run from 294 to more than 400, and the exact number is not known.

More than 70 lawsuits were filed. Judge Robert T. Brown dismissed the ones that came to trial, for lack of evidence, and no one was ever found responsible. The school superintendent, W. C. Shaw, whose son was killed, resigned. Texas lawmakers gave the state's Railroad Commission the power to require a chemical with a strong smell in natural gas, so that people can smell a leak. Within weeks of the explosion, that chemical was being mixed into the gas.

The Hindenburg, 1937, and the Cocoanut Grove, 1942

The German passenger airship *Hindenburg* caught fire on May 6, 1937, as it came in to land at the Naval Air Station in Lakehurst, New Jersey. An airship is a giant aircraft held up by a gas lighter than air, with engines to push it and a cabin for passengers. The *Hindenburg* was filled with hydrogen, a gas that burns very easily. Of the 97 people on board, 35 died. A man working on the ground, Allen Hagaman, died too.

Cameramen from four newsreel companies, which made short news films shown in movie theaters, were filming the landing and filmed the fire. None of them caught the moment the fire started. The radio reporter Herbert Morrison recorded what he saw as it happened. What caused the first spark has never been settled. Passenger travel by giant airship ended after the fire.

The Cocoanut Grove. Fire spread through the Cocoanut Grove nightclub in Boston at about 10:15 p.m. on November 28, 1942, and killed 492 people. Many of them were trapped behind doors that were locked, hidden or blocked. More than 1,000 people were inside a club that was allowed to hold 460. The owner, Barnett Welansky, had locked some exits, hidden others behind curtains, and bricked one up.

The main entrance was a single revolving door. Bodies piled up on both sides of it and jammed it until it broke. Other doors opened inward, so people could not open them against the crush of the crowd. Fire officials later told the court that if the doors had swung outward, at least 300 lives could have been saved. How the fire started is not known.

Welansky was found guilty of manslaughter and spent nearly four years in prison. Lawmakers then passed new laws that banned decorations that burn easily. The laws required emergency exits to stay unlocked, and a revolving

door could no longer be a building's only exit.

Explosions at the docks: Port Chicago, 1944, and Texas City, 1947

Two ships being loaded with ammunition blew up at the Port Chicago Naval Magazine in California, a Navy base for storing explosives, on July 17, 1944. The blast killed 320 men, and 202 of them were Black sailors. Navy officers had given Black sailors the job of loading the ships without enough training or equipment.

Texas City. A fire broke out in the cargo of the French ship *Grandcamp* at the docks of Texas City, Texas, on the morning of April 16, 1947. The ship carried about 2,300 tons of ammonium nitrate, a chemical used as fertilizer that can explode.

The ship's captain ordered his crew not to use water on the fire, so that the cargo would not be ruined. The crew closed the hatches and pumped in steam. The unusual smoke drew a crowd of about 500 people to the shore to watch, many of them children.

At 9:12 a.m. the ship exploded. Flying pieces of metal set fire to oil refineries and chemical tanks along the waterfront. The Monsanto chemical plant beside the ship took the worst of the blast, and seven broken oil tanks at the Republic Refinery caught fire. Another ship, the *High Flyer*, which carried 961 tons of ammonium nitrate, blew up at 1:10 the next morning.

At least 581 people died. The first explosion killed 26 of the town's volunteer firefighters, all but one member of the fire department, and destroyed all of the city's firefighting equipment. The anchor monument in Texas City lists 576 people known to have died. Sixty-three of them were buried without names.

More than 8,000 people sued the United States government. They used a 1946 law that let people sue the government for harm caused by the carelessness of its workers. In 1953 the justices of the Supreme Court ruled against them, 4 to 3. In 1955 members of Congress voted to pay the claims anyway. The government paid 1,394 awards, worth nearly 17 million dollars in all.

BY 1950

If you remember one thing: People's choices turned fires, floods and gas leaks into mass deaths in these fifty years, and of all the owners and officials blamed, one went to prison: Barnett Welansky, who locked the Cocoanut Grove's exits.

In 1900: Galveston had no seawall, and natural gas had no smell to warn anyone of a leak.

In 1950: Galveston stood behind a seawall 17 feet high, Texas officials could require a strong smell in natural gas, and a revolving door could no longer be a building's only exit.

Next: After 1950 forecasters began giving hurricanes names, and a president put most of the national government's disaster work under one office.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 17 MINUTES

Hurricane Audrey killed at least 416 people on the Louisiana coast in June 1957. Warnings got faster after 1950, and help got organized. Weather Bureau forecasters began giving hurricanes human names in 1953. In 1979 President Jimmy Carter put much of the national government's disaster work under one new office, the Federal Emergency Management Agency, called FEMA.

Choices made by owners and engineers still killed hundreds of people. A coal company's waste dam in West Virginia broke in 1972 and killed 125. Two hotel walkways in Kansas City fell in 1981 and killed 114, after an engineer approved a design change without checking the math. News reporters broadcast a rescue from a frozen river in Washington, D.C., live as it happened. After Hurricane Andrew in 1992, Florida got a single building code, a set of rules for how strong buildings must be.

Hurricanes with names: Audrey, 1957, and Camille, 1969

By the time Hurricane Audrey struck in 1957, hurricanes had names. In 1953 forecasters of the U.S. Weather Bureau began giving human names to storms. In 1956 Weather Bureau officials opened a hurricane center in Miami.

Forecasters warned the whole Louisiana coast at 10 a.m. on June 26, 1957, that a hurricane was coming. About 75,000 people left. Audrey came ashore the next day near Cameron, about 24 hours after the warning. The storm had sped up and arrived sooner than forecasters or residents expected.

Its storm surge, the wall of seawater that a hurricane's winds push onto land, rose as high as 12 feet. The water did damage as far as 25 miles inland. At least 416 people died, most of them in Cameron Parish. The final count may never be known.

Camille. Hurricane Camille came ashore near Bay St. Louis, Mississippi, late on August 17, 1969, with winds of 175 miles an hour. The water rose 24.6 feet at Pass Christian, Mississippi. Camille killed 172 people in Mississippi and 9 in Louisiana. As the storm moved north, its rain caused floods in Virginia that killed more than 153 people there. At least 343 people died in all.

The Alaska earthquake, March 27, 1964

The strongest earthquake ever recorded in North America struck southern Alaska at 5:36 p.m. on March 27, 1964. Most of the people it killed drowned in the giant waves it caused. The earthquake had a magnitude of 9.2 and lasted four minutes and 38 seconds.

The shaking and underwater landslides sent tsunamis, giant waves set off by an earthquake or a landslide under the sea, into the coast. Scientists of the U.S. Geological Survey count 131 dead: 122 killed by the tsunamis and 9 by the shaking. Other counts reach 139. At Valdez, 32 people died when the town's harbor and docks collapsed. Tsunamis also reached Oregon and California. In Crescent City, California, one killed 10 people.

Chenega. Chenega was a village of Chugach Alutiiq people, also called Sugpiaq, on an island in Prince William Sound. Their people had lived there since before Russians came to Alaska. About four minutes after the shaking began, a wave 27 feet high struck the village. It knocked down every building except the school and one house.

Between 23 and 26 of the roughly 70 people who lived there died, by different counts. Scientists of the Geological Survey found that Chenega lost a larger share of its people than any other town. The survivors outran the wave and climbed to high ground.

Rescuers took the survivors to Cordova. Officials of the federal Bureau of Indian Affairs later moved them for good to the village of Tatitlek. In 1971 members of Congress passed the Alaska Native Claims Settlement Act. Under it, Alaska Natives received 44 million acres and $962.5 million, held by companies that Alaska Natives own. Former Chenega residents formed one of them, the Chenega Corporation, and won the right to choose 76,093 acres of land. People began living in a new village, Chenega Bay, in 1984.

President Lyndon Johnson declared all of Alaska a disaster area the day after the earthquake. Army engineers of the U.S. Army Corps of Engineers led the work of rebuilding roads, clearing wreckage and laying out new town sites, at a

cost of 110 million dollars. The people of Valdez moved their town to higher ground four miles to the west. In 1967 federal scientists of the Coast and Geodetic Survey opened a tsunami warning center at Palmer, Alaska.

Floods of 1972: Buffalo Creek and Rapid City

The Pittston Company's coal waste dam above Buffalo Creek, West Virginia, broke at about 8 a.m. on February 26, 1972, and the flood killed 125 people. The Buffalo Mining Company, part of Pittston, owned and ran three dams made of coal waste on the creek's Middle Fork.

The third dam sat on a layer of soft coal mud instead of solid rock. Neither Pittston nor the mine's earlier owner had sent the dams' plans to the state for approval. Four days before the flood, a federal mine inspector called the dam "satisfactory."

The dam gave way and let out 132 million gallons of water, coal waste and mud. The flood ran down the narrow valley through 16 coal towns. It killed whole families, injured about 1,000 people, and destroyed 507 houses and 44 mobile homes. About 4,000 of the 5,000 people who lived along the creek lost their homes.

"An act of God." Pittston officials blamed three days of rain and called the flood "an act of God," meaning a natural event that no person could have prevented. Members of the governor's commission that investigated wrote that "the Pittston Company, through its officials, has shown flagrant disregard for the safety of residents of Buffalo Creek," meaning that its officials had openly ignored the residents' safety.

Members of a separate commission formed by citizens said the company was guilty of murdering at least 124 people. The records show no Pittston official charged with a crime.

Pittston paid 13.5 million dollars in 1974 to settle a lawsuit by survivors and family members. One count gives 645 of them, and another gives 625. That came to about 13,000 dollars each after the lawyers were paid. Governor

Moore settled the state's own lawsuit in 1977 for 1 million dollars. Members of Congress and West Virginia lawmakers passed new laws on how such dams must be built and kept up.

Rapid City. Less than four months later, on June 9 and 10, 1972, about 15 inches of rain fell on part of the Black Hills of South Dakota. Rapid Creek flooded Rapid City, where people had built houses and businesses on the creek's flood plain, the low land beside a river that the river covers when it floods. The flood killed 238 people. Afterward, new zoning rules set what could be built where, and most of the flood plain was rebuilt as parks.

FEMA and Three Mile Island, 1979

President Jimmy Carter put much of the national government's disaster work under one new office in 1979, the Federal Emergency Management Agency, or FEMA. Before then, officials of more than 100 separate federal offices could each claim a part in the response to a single disaster. On April 1, 1979, Carter signed two executive orders, written orders from the president, that started FEMA. In July he signed another order that moved disaster relief into it.

Members of Congress passed the Stafford Act in 1988. It set the rules for when a president may declare a disaster, so that federal money and help can be sent.

Three Mile Island. Four days before FEMA started, the worst accident in the history of the country's nuclear power plants began at Three Mile Island, near Harrisburg, Pennsylvania. A nuclear reactor makes heat from uranium fuel in its core, and water must keep flowing around the core to cool it.

At 4 a.m. on March 28, 1979, a small fault in the cooling system of the Unit 2 reactor set off a chain of failures. A relief valve stuck open and let cooling water escape. The operators had no instrument that showed them the valve was open. They thought the system held too much water, so they cut the flow of new water. The core overheated, and at least 45 percent of it melted.

The operators at the plant, officials of Metropolitan Edison, the power company that ran it, and state and federal officials gave the public unclear and conflicting reports. Station manager Gary Miller declared an emergency at the plant. Federal nuclear officials had no power to order people to leave.

On March 30, Governor Richard Thornburgh advised pregnant women and small children within five miles to leave. He told people within ten miles to stay indoors. About 140,000 people left on their own. In at least 12 studies since 1981, scientists have found no deaths and no health harm that can be measured from the radiation that escaped. A commission of 12 members, led by John G. Kemeny, reported on the accident on October 31, 1979.

Mount St. Helens, 1980, and the Hyatt Regency walkways, 1981

Mount St. Helens, a volcano in Washington State, blew out its north side at 8:32 a.m. on May 18, 1980, and killed about 57 people. An earthquake loosened the side of the mountain, and the whole weakened north face slid away. Hot gas and shattered rock blasted north toward Spirit Lake. The blast turned hundreds of square miles into wasteland. Most of the people who died were suffocated, and a few died of burns.

Among the dead were Harry R. Truman, who owned a lodge at Spirit Lake, the photographers Reid Blackburn and Robert Landsburg, and David Johnston, a volcano scientist of the U.S. Geological Survey. Johnston was on duty at a lookout post about six miles north of the mountain.

The walkways. Two walkways fell onto a crowd in the lobby of the Hyatt Regency Hotel in Kansas City, Missouri, on the evening of Friday, July 17, 1981. Engineering mistakes caused the fall. About 1,600 people had come to a dance in the lobby. The two walkways hung over the lobby. They killed 114 people and injured 216.

Engineers at Jack D. Gillum and Associates, the firm of the engineer Jack D. Gillum, designed the walkways. Their first design was already too weak. It could hold only 60 percent of the weight that Kansas City's building code required.

Then the makers of the steel parts, at a company called Havens Steel, proposed a change in how the walkways hung. The engineer Daniel M. Duncan accepted the change in a phone call without doing the math to check it. The change doubled the load on the walkway beams on the fourth floor. Investigators found that the walkways would have failed under one third of the weight they held that night.

Members of a Missouri state board found the engineers to blame for gross negligence, which means a serious failure to take the care their job required. They also found them guilty of misconduct and unprofessional conduct, meaning they had broken the rules of their profession. Gillum's company lost its engineering licenses in Missouri, Kansas and Texas. The engineers were cleared of the crimes they were first charged with, and no one was convicted of a crime.

Air Florida Flight 90, January 13, 1982

Air Florida Flight 90 took off from Washington National Airport on the snowy afternoon of January 13, 1982, with snow and ice on its wings. It crashed into the Potomac River. The captain, Larry Wheaton, 34, and the first officer, Roger Pettit, 31, had not turned on the engines' ice protection. The jet hit the 14th Street Bridge, struck seven cars with people in them, and fell through the ice of the river. Of the 79 people on the plane, 74 died. Four drivers on the bridge died too.

In the water. Six people held on to the wreck in water of about 34 degrees. The rescuers who reached the riverbank had no equipment that could reach them. Roger Olian, the boss of the sheet-metal workers at St. Elizabeths Hospital, was the first to jump into the water to try. A U.S. Park Police helicopter arrived at about 4:20 p.m. and pulled most of the survivors out.

One passenger, Arland D. Williams Jr., 46, was a bank examiner, a person who checks that banks follow the rules. He was strapped into the wreckage and could not free himself. Twice he passed the rescue line to others. When the helicopter crew came back for him, the wreckage had rolled and he was under the water. The coroner, the official who finds out how people died, found that he was the only passenger who drowned. The bridge span was later renamed for him.

The Carnegie Hero Fund, which gives medals for bravery, says that its staff could not firmly establish who the sixth person in the water was. The coroner and the renamed bridge name Williams.

A passenger, Priscilla Tirado, was badly hurt and blinded by jet fuel, and she was too weak to grab the line when the helicopter came back to her. Lenny Skutnik, an assistant at the Congressional Budget Office, took off his coat and boots, swam out in short sleeves and pulled her to shore. "Somebody had to go into the water," he told *Time* magazine. News reporters recorded the rescue and broadcast it live.

The cause. Investigators of the National Transportation Safety Board, the federal office that looks into crashes, found that the pilots caused the crash. The investigators blamed the pilots' failure to use the engines' ice protection and the pilots' decision to take off with snow or ice on the wings. The captain did not listen when the first officer told him the engine readings looked wrong. He did not stop the takeoff while there was still room on the runway. Both pilots died in the crash.

Melvin Windsor

Who: A U.S. Park Police officer who stood on a helicopter's skid over the frozen Potomac River to pull a survivor out.
When and where: Washington, D.C., January 13, 1982.

Patricia Felch had been in the frozen river for 30 minutes.

Melvin E. "Gene" Windsor, 41, pulled the last survivor of Air Florida Flight 90 out of the Potomac. He did it standing on one of the helicopter's skids, the long metal bars it lands on, with no safety line. He had joined the Park Police in 1971, and by 1982 he was a rescue technician.

He flew with the chief pilot, Donald W. Usher, 31. Earlier that day Usher had decided that no routine flights would go up. At about 4:09 p.m. a radio call came in. They took off at 4:16 and reached the river at about 4:20. Usher steered by watching the highway below through the windows in the helicopter's floor.

The people holding on to the wreck were in water up to their shoulders and necks. Windsor dropped life jackets and lines to them while Usher flew so low that the skids went under the water.

> **The last one out.** The last survivor out of the water was Patricia Felch. She had been in the river for 30 minutes. Windsor unbuckled his seat belt and stepped out onto the right skid, which was under the water. He grabbed her with one arm and held the back of her head with the other.
>
> Of the passenger who twice passed the line to others, Windsor told *Time*: "In a mass casualty, you'll find people like him, but I've never seen one with that commitment." A mass casualty is an event in which many people are hurt or killed at once. Windsor, Usher, Skutnik and Olian each received the Carnegie Medal for bravery. Windsor served 22 years with the Park Police.

Hurricane Hugo and the Loma Prieta earthquake, 1989

Hurricane Hugo hit two U.S. territories before it reached the mainland in September 1989. On St. Croix in the U.S. Virgin Islands, it damaged more than 90 percent of the buildings and killed three people. In Puerto Rico it killed eight people and left nearly 28,000 without homes. On September 22 it came ashore on Sullivan's Island, South Carolina. The storm surge reached 20.2 feet near McClellanville, and 13 people died in South Carolina.

Loma Prieta. An earthquake of magnitude 6.9 struck about 10 miles northeast of Santa Cruz, California, at 5:04 p.m. on October 17, 1989, and killed 63 people. A World Series baseball game was about to begin at Candlestick Park in San Francisco.

In Oakland, the upper deck of the Cypress Street Viaduct fell onto the lower deck, and 42 people were crushed to death in their cars. The viaduct was a two-level section of the Nimitz Freeway, a highway raised above the ground. A section of the San Francisco-Oakland Bay Bridge also fell, and a 23-year-old woman, Anamafi Moala, died there.

Why help was slow after Hurricane Andrew, 1992

Hurricane Andrew struck Florida on August 24, 1992, and destroyed more than 63,500 houses. Help from the national government was slow to come. Andrew came ashore at Homestead with winds as high as 165 miles an hour. It damaged more than 124,000 other houses and wrecked 99 percent of the mobile homes in Homestead. It killed 44 people in Florida. Fifteen died in the storm itself and 29 from its effects afterward. In all, the storm left 65 people dead.

"Where in the hell is the cavalry?" Three days after the storm, Dade County's emergency management director, Kate Hale, went on national television. "Where in the hell is the cavalry on this one?" she asked. "They keep saying we're going to get supplies. For God's sake, where are they?" The cavalry were soldiers on horseback, and she meant the rescuers and supplies that had not come.

Many people criticized FEMA for the slow start. Some critics blamed Florida's governor, Lawton Chiles, instead. FEMA officials believed they could not act until he sent a formal request for federal help. By that account, he waited five days to send it.

Engineers who surveyed the damaged roofs found that concrete roof tiles had been glued to felt paper and shingles stapled on the wrong way. A single building code for the whole state of Florida was set up in 1998, and it took effect in 2002.

Northridge, 1994, and the Chicago heat wave, 1995

The Northridge earthquake, magnitude 6.7, struck Los Angeles at 4:30 a.m. on January 17, 1994. The official count of the dead is 57, and other counts run from 60 to 72. Sixteen people died when the Northridge Meadows apartment building collapsed. Buildings with parking on the bottom floor held up poorly in the shaking.

The heat wave. A heat wave in Chicago in mid-July 1995 killed 739 more people in one week than usually die in a week. Most of the dead were poor, old people. Many had no air conditioning, or had it and could not afford to run it. Many kept their windows shut and would not sleep outside, because they were afraid of crime. Black residents were more likely to die than white residents.

City officials did not issue a heat emergency warning until the last day of the heat wave. The records do not say who made that choice. Few people used the city's five cooling centers.

The morgue run by the medical examiner, Edmund Donoghue, a doctor who examines the dead to find out how they died, could hold 300 bodies, and it filled. Refrigerated trucks were parked outside to hold the rest.

Donoghue said later that his office was not allowed to hold press conferences, and that Mayor Richard M. Daley was not told until the weekend. A later study counted 692 deaths above normal over the whole summer, from June 21 to August 10. That is fewer than the worst week's 739, because some of the people the heat killed would have died later that summer anyway, the study found. In August, 41 victims whose bodies no one claimed were buried together in one grave in Homewood, Illinois.

BY 2000

If you remember one thing: Warnings and federal help got faster after 1950, but the choices of owners and engineers still killed hundreds of people, and no one in these fifty years was convicted of a crime for those deaths.

More than 100: the number of federal offices that could each claim a part in the response to a single disaster before 1979. That year President Carter put most of that work under one office, FEMA.

Next: After 2000, people waited days for help in New Orleans and months for electricity in Puerto Rico, and fires started by power lines burned whole towns.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 20 MINUTES

Ethel Freeman, 91, died in her wheelchair outside the New Orleans Convention Center in September 2005, waiting for a bus that police had said would come.

In the disasters after 2000, many of the dead were old or sick, like her, and many died in the days and months after the storm or the fire, waiting for help, for electricity or for a warning. Hurricane Katrina killed 1,392 people in 2005, by the National Hurricane Center's count as corrected in 2023. Most of them died in Louisiana, many of them in New Orleans, where floodwalls broke. Investigators found that the U.S. Army Corps of Engineers was responsible for those floodwalls.

After Hurricane Maria struck Puerto Rico in 2017, officials of the island's government counted 64 dead. In 2018 they raised the official count to 2,975, after a study of deaths in the months after the storm. Power lines owned by utility companies started deadly fires in California in 2018 and 2025 and in Hawaii in 2023. In late September 2026, members of Congress were still debating whether to keep FEMA.

September 11, 2001, and the illnesses afterward

Three hundred forty-three members of the New York City Fire Department died in the attacks on the World Trade Center on September 11, 2001, and by 2026 more of its members had died from breathing the dust of the ruins.

That morning 19 hijackers from the group al-Qaeda took over four passenger jets by force. They flew two into the Twin Towers of the World Trade Center in New York City and one into the Pentagon, near Washington, D.C. On the fourth plane, United Airlines Flight 93, passengers and crew fought back, and it crashed into a field near Shanksville, Pennsylvania.

The attacks killed 2,977 people, not counting the 19 attackers. Seventy-one police and other law officers also died in New York.

The dust. For months afterward, firefighters, police officers, construction workers and volunteers dug through the ruins. They breathed dust of crushed concrete, glass fibers, asbestos, lead and mercury. Asbestos is a mineral whose tiny fibers lodge in the lungs and can cause cancer years later. Fires in the ruins burned for three months.

Many of the workers later got cancer or lung disease. In September 2026 leaders of the Fire Department reported that 453 more of its members had died of illnesses tied to the dust. That is more than the 343 it lost on the day of the attacks.

In December 2010 members of Congress passed the James Zadroga 9/11 Health and Compensation Act. President Barack Obama signed it on January 2, 2011. It set up the World Trade Center Health Program, which gives medical checkups and treatment to responders and survivors with illnesses tied to the attacks. Members of Congress renewed it in December 2015.

Hurricane Katrina, August 29, 2005

Most of the people Hurricane Katrina killed died in Louisiana, where New Orleans flooded after its floodwalls broke. On August 28, 2005, Mayor Ray Nagin ordered the first mandatory evacuation in the city's history, meaning everyone was ordered to leave. Governor Kathleen Blanco opened all lanes of the highways to traffic leaving the city, and about one million people left. Between 100,000 and 150,000 people stayed in the city. City officials opened the Superdome stadium as a "refuge of last resort," a place to go when there was nowhere else.

Katrina came ashore on August 29. The levees and floodwalls broke in more than 50 places, and water covered 80 percent of the city. In the Lower Ninth Ward, a neighborhood where most homeowners were Black, the water was more than 10 feet deep.

Who built the floodwalls. Members of Congress had put the Corps of Engineers in charge of the city's flood protection in a law of 1965. Investigators found that the Corps was responsible for the floodwalls that broke, and that the flood protection system the national government built had engineering flaws. Judges of a federal appeals court later ruled that, under a law of 1928, the Corps could not be made to pay for the damage.

Scientists at the National Hurricane Center lowered their count of the dead from 1,833 to 1,392 in 2023, after new studies. Other studies found that most of the dead in New Orleans were old people who lived near the broken levees, in the Lower Ninth Ward and Lakeview neighborhoods.

Waiting. Coast Guard crews rescued more than 33,500 of the roughly 60,000 people stranded in the city. People waited for days at the Convention Center, which had no food, water or medical help. Early news reports of mass crime at the Superdome later turned out to be false or exaggerated.

Federal officials did not begin to act in earnest to help the people at the Convention Center until September 2. Michael Brown, the head of FEMA, lost charge of the relief work to Vice Admiral Thad Allen and resigned on September 12.

Guns on the bridges. Some police officers turned their guns on survivors. On September 1, police from the city of Gretna and from the Crescent City Connection bridge police, with deputies of the Jefferson Parish sheriff, blocked the bridge over the Mississippi River. They turned back people walking out of New Orleans at gunpoint. Gretna's police chief, Arthur Lawson, made that decision with the sheriff's office and the bridge police.

On September 4, New Orleans police officers shot six unarmed Black people on the Danziger Bridge. They killed James Brissette, 17, and Ronald Madison, 40, a man with a mental disability. Madison was shot seven times, five times in the back. Among the officers were Kenneth Bowen, Robert Gisevius, Anthony Villavaso and Robert Faulcon. Police officers then tried to cover up the killings. In 2016 five former officers pleaded guilty and were sentenced to between 3 and 12 years in prison.

Memorial Medical Center. Patients, visitors and workers were trapped in Memorial Medical Center, a New Orleans hospital, during the flood. After the storm, 45 bodies were found there. In July 2006, the office of Charles Foti, Louisiana's attorney general and top lawyer, finished an investigation of nearly a year. A judge then ordered the arrest of a doctor, Anna Pou, and two nurses, Lori Budo and Cheri Landry, for murder, which Louisiana law called second degree murder.

They were accused of giving deadly doses of morphine and midazolam to four patients: Emmett Everett Sr., 61, Hollis Alford, 66, Ireatha Watson, 89, and Rose Savoie, 90. Morphine is a strong painkiller, and midazolam is a drug that makes people calm and sleepy. In large doses, both slow and then stop a person's breathing. In 2007 a grand jury refused to charge any of the three. The state later paid Pou's legal bills of more than 450,000 dollars.

Herbert Freeman Jr.

Who: A New Orleans man who took his 91-year-old mother, Ethel Freeman, out of their flooded home by boat, then stayed beside her body outside the Convention Center after she died waiting for help.
When and where: New Orleans, August 31 to early September 2005.

He borrowed a friend's boat to get his mother out.

Herbert Freeman Jr. rescued his mother from their flooded house, and she died the next day outside the New Orleans Convention Center, where police had sent them to wait for a bus. Ethel Mayo Freeman was 91. She had a pacemaker for her heart and a feeding tube, and she used a wheelchair. She was often sick, and she decided to stay home when Katrina came.

The floodwater filled her house several feet deep. On August 31, Herbert, 58, borrowed a friend's boat. He put his mother in her wheelchair, lifted her into the boat and took her to dry ground. New Orleans police officers there sent them to the Convention Center. Herbert told officers his mother needed a doctor. They told him a bus would come.

At the Convention Center. The Convention Center had no food, no water, no medical help and no way out. "She was calling out for a doctor or a nurse, but there was nothing there," Herbert said. Ethel Freeman died at the Convention Center on September 1. Her body stayed outside for days, slumped in the chair and covered with a poncho. Pictures of it were shown on national television.

Herbert watched over his mother's body for nearly four days, until he was ordered onto a bus. The NBC News report of his story does not say who gave the order. Before he left, he put a note with his name and phone number in her pocket. It took him seven weeks to find her body, at a morgue in St. Gabriel, Louisiana. Her funeral was held in New Orleans in November 2005.

> He and the families of two other people who died sued the national government. In 2009 federal appeals judges ruled that, under the Stafford Act, the government could not be sued over decisions like these in a disaster, and they threw out the case.

How many died after Hurricane Maria, 2017?

Hurricane Maria left all 3.4 million people in Puerto Rico without electricity on September 20, 2017, and thousands died in the months that followed. Puerto Rico is a U.S. territory, and its people are U.S. citizens. Maria struck two weeks after Hurricane Irma crossed the region. It caused the worst blackout in U.S. history. Some customers did not get their power back until August 2018, 11 months later.

After Irma, FEMA officials had shipped many of the supplies they kept in Puerto Rico to the U.S. Virgin Islands. After Maria, they admitted to failures in moving supplies, too few staff and poor communication.

The mayor of San Juan, Carmen Yulín Cruz, spoke of the national government's relief work on September 29, 2017: "We are dying and you are killing us with the inefficiency." On October 3, President Donald Trump compared the count of dead at that time, 16, with "a real catastrophe like Katrina."

How many died. Officials of Governor Ricardo Rosselló's government counted 64 dead for months. Researchers found that many more people than usual died in the months after the storm. People who needed dialysis, a treatment that cleans the blood of people whose kidneys have failed, could not reach emergency care for long stretches of time.

Researchers at George Washington University estimated between 2,658 and 3,290 extra deaths from September 2017 to February 2018. On August 28, 2018, Rosselló's officials accepted their study and raised the official count to 2,975. A Harvard University study in 2018 estimated 4,645 extra deaths from September to December 2017, with a very wide range, from 793 to 8,498.

Trump wrote on Twitter on September 13, 2018, that "3000 people did not die" and blamed Democrats for the number. The official count of 2,975, based on the George Washington University study, has stood since August 28, 2018.

Jesse Vazquez

Who: A retired veteran in Bayamón, Puerto Rico, who lived for months without electricity after Hurricane Maria with his 89-year-old mother, Mercedes Mercado.

When and where: Calle Alameda, Bayamón, Puerto Rico, September to December 2017.

His children flew in from New York with a generator.

Jesse Vazquez and his mother, Mercedes Mercado, went 96 days without electricity after Hurricane Maria. Maria cut the power to their house on Calle Alameda in Bayamón on September 20, 2017. For weeks the family ran electric fans and charged their phones from the battery of Jesse's car.

A small generator cost about 600 dollars on sale, more than Jesse could afford. In October his son, also named Jesse, and his daughter, Megan, flew in from New York to bring him a generator. "It's basically cheaper to buy a flight, get a generator and bring it down here," his son said. Running it five hours a day took about 80 dollars of gasoline a month.

Exhaust from a neighbor's generator blew into the family's living room. Megan said they needed a detector for carbon monoxide, a poison gas in engine exhaust that has no smell.

By Christmas the house had been without power for 96 days. Jesse and his mother used the generator to run the refrigerator, and fans at night so they could sleep. A FEMA inspector told Jesse he could not get help to fix his damaged roof. "It's kind of humiliating," he said. With his son's help he bought four solar panels. He and his mother decided to stay in Puerto Rico.

BY 2018: Puerto Rico's officials had raised the count of Maria's dead from 64 to 2,975, and federal judges had thrown out the lawsuit that Herbert Freeman Jr. and two other Katrina families brought against the national government.

Floods: Harvey, 2017, Helene, 2024, and the Texas Hill Country, 2025

Hurricane Harvey struck Texas in late August 2017 and dropped more rain than any other hurricane or tropical storm in U.S. records. Scientists at the National Hurricane Center recorded 60.58 inches near Nederland, Texas. Rescuers made more than 17,000 rescues. Their report records at least 68 people killed by the storm's direct effects in Texas and about 35 more by causes such as electric shock, car crashes and being cut off from medical care.

Hurricane Helene came ashore in Florida, near Perry, late on September 26, 2024. The next day its rain fell on North Carolina and Tennessee. Busick, North Carolina, got 30.78 inches of rain, and the rain set off more than 2,000 landslides. Scientists at the National Hurricane Center count at least 250 dead in the United States, 107 of them in North Carolina.

The plastics factory. In Erwin, Tennessee, six people died at the Impact Plastics factory as the Nolichucky River rose on September 27. The flood swept 11 people away, and five were rescued. A worker who survived and the families of the dead spoke out. They said the workers were not told they could leave until floodwater had covered the road to the plant and the power had gone out. Five employees and a contractor who cleaned the offices were killed.

Inspectors of Tennessee's workplace safety agency found in April 2025 that the company had "exercised reasonable diligence," meaning it had taken reasonable care, and that the deaths were not work-related. They issued no citations, which are official notices that a company broke a safety rule. District Attorney General Steven Finney filed no charges against the company's president, Gerald O'Connor, in July 2025, after a review by the Tennessee Bureau of Investigation.

The investigators said they found no evidence that workers were told they could not leave or that they would be fired for leaving, and that the workers had a little more than an hour to get out. The family of Johnny Peterson, one

of the dead, sued the company.

Camp Mystic. As much as 20 inches of rain fell on the Texas Hill Country early on July 4, 2025, and the Guadalupe River rose about 26 feet in 45 minutes. The flood killed 136 people, 117 of them in Kerr County, by counts reported in July 2026. At Camp Mystic, a Christian summer camp for girls, 25 campers, two counselors, Chloe Childress and Katherine Ferruzzo, and the camp's owner and director, Dick Eastland, died.

The first flash flood warning for Kerr County went out at 1:14 a.m. At 4:03 a.m. a flash flood emergency, a more urgent alert, told people in the towns of Hunt and Ingram to "seek higher ground now!"

Kerr County had no warning system built just for floods. In 2016 the county sheriff had asked for one, with sirens. County Commissioner H. A. "Buster" Baldwin said, "I think this whole thing is a little extravagant for Kerr County, with sirens and such."

After appeals from the camp, FEMA officials took 15 of its buildings on the Guadalupe River off the federal map of land likely to flood in 2013. They took 15 buildings at a newer camp site off the map in 2019 and 2020, the Associated Press reported.

Texas lawmakers passed new camp safety laws in September 2025, at an extra meeting called a special session, among them the Heaven's 27 Camp Safety Act. On June 18, 2026, members of the Texas Senate and House investigating committees released a report. The lawmakers found that the camp had no written emergency plans, which state rules required. They also found that the camp's leaders did not move the campers out in time even though they had chances to.

By July 2026 Kerr County had six new sirens. In the week before July 18, when floods came to the Hill Country again, officials sounded three of them to warn people, an official of the Upper Guadalupe River Authority said.

Fires started by power lines: Paradise, 2018, Lahaina, 2023, and Los Angeles, 2025

A single metal hook broke on a power line tower owned by Pacific Gas and Electric, called PG&E, on the morning of November 8, 2018. The tower was part of a line built in the early 1900s, and it was poorly maintained. The line started a fire. The Camp Fire burned through the town of Paradise, California. It killed 85 people and destroyed more than 18,000 buildings. On June 16, 2020, the company pleaded guilty in court to 84 counts of involuntary manslaughter, the crime of killing people through carelessness.

Lahaina. A wildfire destroyed the town of Lahaina on the island of Maui, Hawaii, on August 8, 2023, and killed 102 people. That is the most people killed in a U.S. wildfire since 1918. High winds knocked down power lines of Hawaiian Electric, and the lines started a fire that morning. Maui firefighters declared that fire under control and left. Less than an hour later, embers from it started the fire again.

Firefighters lost water pressure in the hydrants. Many people died in their cars. Some jumped into the ocean to escape the flames.

The fire destroyed a town that had been the capital of the Kingdom of Hawaii for 35 years. It burned the homes of many Native Hawaiian families who could afford to live in Lahaina only because they had inherited houses that were already paid for.

Maui has more than 80 outdoor warning sirens, and none of them sounded. Herman Andaya, the head of the Maui County Emergency Management Agency, said the sirens would have sent people toward the mountains and into the fire. He resigned the next day.

In an October 2024 report, federal fire investigators and the Maui Fire Department called the fire accidental, meaning no one set it on purpose. They found that the power lines brought down by the wind had started it, but they did not hold any person or company responsible.

Under a settlement of the survivors' lawsuits, Hawaiian Electric, Maui County and others agreed to pay 4 billion dollars. As of August 2026, none of that money had been paid, because of a court fight over the lawyers' fees, the

Honolulu Star-Advertiser reported.

Los Angeles. Fires pushed by winds of up to 100 miles an hour burned parts of Los Angeles in January 2025. They killed at least 31 people and destroyed more than 18,000 homes and other buildings. The Eaton Fire in Altadena killed 19 people. In August 2026 Los Angeles County and state fire investigators reported that it started from electric sparks on an unused Southern California Edison tower.

Seventeen of the 19 died in west Altadena, a neighborhood with a long history as a Black community. Reviewers from the firm Citygate Associates found in May 2026 that county officials' orders to leave reached west Altadena hours after the fire turned toward it. The records do not name the officials who made those decisions. The dead had an average age of 77. California's attorney general, Rob Bonta, opened an investigation into whether race, age or disability affected how officials warned people.

The Palisades Fire killed 12 people. In October 2025 Jonathan Rinderknecht, 29, was arrested and charged with starting it. Prosecutors say he set a small fire on January 1, and that strong winds rekindled it, or started it burning again, on January 7 as the Palisades Fire. Firefighters had thought the January 1 fire was out. A Los Angeles Fire Department battalion commander ordered crews to leave that site, although they reported that the ground was still burning slowly.

His first trial ended in June 2026 when the jury could not agree: ten jurors voted not guilty and two voted guilty. Prosecutors charged him again with arson, the crime of setting a fire on purpose, and he pleaded not guilty. As of October 26, 2026, his second trial had not begun, and he had not been convicted of any crime.

Cold, a collapse and FEMA's future: 2021 to 2026

A winter storm struck Texas in February 2021, and more than 4.5 million homes and businesses lost power, some of them for days. Officials of the Texas Department of State Health Services counted 246 deaths from the storm. Almost two-thirds of the dead died of hypothermia, which means the body gets so cold that it stops working. Some estimates run as high as 702 deaths. Cristian Pavón Pineda, 11, died of carbon monoxide poisoning in his family's mobile home in Conroe, Texas. The home had no power.

Surfside. Part of Champlain Towers South, a 12-story apartment building in Surfside, Florida, fell down at 1:22 a.m. on June 24, 2021, and killed 98 people. For years, water had seeped into the concrete and steel supports of the parking garage under the pool deck, and the steel had rusted.

Engineers of the firm Morabito Consultants reported the damage in 2018. They warned that if the waterproofing was not replaced soon, the damage to the concrete would spread fast. Each family owned its own apartment, and the owners together had a board. On April 9, 2021, Jean Wodnicki, the president of that board, wrote to the residents that "the concrete deterioration is accelerating," meaning the concrete was breaking down faster. A 15-million-dollar repair had been approved, but the main work had not started.

What followed. In June 2022 Judge Michael Hanzman approved a settlement of the lawsuits. The families of the dead and the people who were hurt shared nearly a billion dollars, paid by 20 defendants, the people and companies they had sued, among them insurance and engineering companies. None of the defendants admitted doing anything wrong. The apartment owners split about 96 million dollars from the sale of the land.

Florida lawmakers passed a law in May 2022. Under it, apartment buildings owned by their residents, three stories or taller, must have a safety inspection of their structure when they turn 30, or 25 if they stand within three miles of the coast, and every 10 years after. The owners must also set aside money for the repairs that a study says the building will need.

In June 2026 engineers of the National Institute of Standards and Technology, a federal science office, reported that the collapse began in early June 2021, when two connections between garage columns and the pool deck broke. They found that the original design strayed widely from the building rules of its time, and that the builders did not follow the drawings. Changes to the pool deck and years of rust weakened the building further.

FEMA's future. President Trump called for getting rid of FEMA. On February 17, 2025, officials of the Trump administration fired more than 200 FEMA workers. On March 24, 2025, Homeland Security Secretary Kristi Noem said she would move to eliminate FEMA. The acting head of FEMA, Cameron Hamilton, told members of Congress: "I do not believe it is in the best interest of the American people to eliminate the Federal Emergency Management Agency." He was removed from his job the next day.

More than 180 current and former FEMA workers signed a public statement of protest in August 2025 that they called the "Katrina Declaration." On May 7, 2026, a review council the president had set up approved a report with ten recommendations. On September 15, 2026, members of a House committee voted for one bill to change FEMA, the Post-Disaster Protection Act. In late September 2026, FEMA still existed, and members of Congress were still debating its future.

BY TODAY

If you remember one thing: Since 2000 many of the dead were old or sick, and many died waiting: for a bus at the Convention Center, for electricity in Puerto Rico, for an order to leave west Altadena.

Still open: As of August 2026 none of Lahaina's 4 billion dollars had been paid, and in late September 2026 members of Congress were still debating whether to keep FEMA.

Next: The chapter closing is below. Chapter 28, Drugs and Alcohol, follows.

THE THREE QUESTIONS, ANSWERED

1. **What rules did Americans make after disasters, and did they work?** Some worked, some failed and some are still being tested. Lawmakers and officials wrote new rules after many disasters, from Boston's fire wardens in 1711 to one building code for Florida after Hurricane Andrew. The rule to put a smell in natural gas after the New London school explosion was in use within weeks, but the 1838 steamboat law had no one to enforce it. In late September 2026 members of Congress were still debating FEMA's future.
2. **Was anyone punished when people's choices killed others?** Rarely. No one was punished for the *Sultana*, the Johnstown club won in court, a jury found the Triangle owners not guilty, and no Pittston official was charged after Buffalo Creek. Barnett Welansky went to prison for the Cocoanut Grove fire, and PG&E pleaded guilty to 84 counts of involuntary manslaughter after the Camp Fire.
3. **Did everyone get the same help after a disaster?** No. Officials paid out London's money for Charleston in 1740 by the size of each loss, so most of it went to rich merchants. Armed men forced Black workers to handle Galveston's dead in 1900, and camp leaders forced Black people to work in the 1927 flood. In 2005 police turned back people walking out of New Orleans at gunpoint, and in 2025 orders to leave reached west Altadena hours late.

What changed: Over ten eras, help after a disaster moved from neighbors and churches to public money and a national office, and warnings moved from church bells to the telegraph and sirens. After 2000, FEMA's help was slow at Katrina, its officials admitted failures after Maria, and none of Maui's sirens sounded in 2023.

FIVE DATES TO KEEP

January 26, 1700: The Cascadia fault breaks, and its tsunami reaches Japan the next day.

May 31, 1889: The South Fork Dam breaks, and 2,208 people die in Johnstown.

September 8, 1900: A hurricane floods Galveston, the deadliest natural disaster in the country's history.

March 25, 1911: The Triangle fire kills 146 workers in New York.

August 29, 2005: Hurricane Katrina comes ashore, and New Orleans floods when its floodwalls break.

NOT KNOWN: No one knows how many people died in the Galveston hurricane of 1900, and the estimates run from 6,000 to 12,000.

Where to go: Clara Barton's house in Glen Echo, Maryland, built partly from the lumber of the Red Cross buildings at Johnstown, is now the Clara Barton National Historic Site.

Next: Chapter 28, Drugs and Alcohol. It follows the drinks, tobacco and drugs Americans have used, and the laws they made to control them.

Chapter 28: Drugs and Alcohol

CHAPTER 28 OF 37 · DRUGS AND ALCOHOL · 10 ERAS · ABOUT 110 MINUTES

People in this land have used tobacco for at least 12,000 years, and when Americans turned against a drug, the punishment fell hardest on the people who had the least power.

Judges in the Massachusetts Bay Colony ordered a colonist named Robert Coles in 1634 to wear a letter D for a year. The D was made of red cloth, set on a white background, and it hung from his neck over his clothes whenever he was among other people. It was his fourth punishment for drunkenness in three years. Governor John Winthrop wrote in his journal that Coles, "having been oft (often) punished for drunkenness, was now ordered to wear a red D about his neck for a year."

This chapter follows tobacco, alcohol and other drugs through all ten eras, from a fire pit 12,300 years old to the fentanyl deaths of the 2020s. Again and again a drug is sold freely, then feared, then taxed or banned, and sometimes allowed again.

Each time, look at who gets punished. Coles was a free colonist with the right to vote. The laws that came later fell on Native people, Chinese immigrants, Black Americans and poor drinkers. No member of the Sackler family, which owned the company that made the painkiller OxyContin, has been charged with a crime. The businessman whose ginger medicine left tens of thousands of people unable to use their hands and feet never served a day in jail. Look, too, for the people who quit and then helped others quit.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How long have people here used tobacco, and what did they use it for?
2. What happened when the whole country banned alcohol from 1920 to 1933?
3. Who has been punished for drugs, and who has not?

Five minutes? Read "Poisoned alcohol and jake leg" in 1900 to 1950.

Can skip: "Drinks for voters" and "The tavern as the town's meeting place" in 1700 to 1750, and "Chewing tobacco" in 1800 to 1850. The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 6 MINUTES

People in this land used plants that change how the body and the mind feel for thousands of years before Europeans came. The oldest one found so far is tobacco. In the records written later, many of these uses were part of prayer and ceremony. Native people offered tobacco in prayer, men in the Southeast drank Black Drink to make themselves pure, and the O'odham drank cactus wine at a rain ceremony. The Apache drank their corn beer mostly at everyday social times.

Beer and wine, but no liquor. Before Europeans came, people made beer and wine mainly in the Southwest, and people elsewhere used alcohol only here and there. Nobody in this land made distilled liquor, such as rum or whiskey. To distill a drink, people heat it and catch the steam, which cools into a liquor stronger than beer or wine.

How old is tobacco use here?

People in North America were using tobacco about 12,300 years ago. Scientists found burned tobacco in a hearth, an old fire pit, at the Wishbone site, in the desert of western North America. The find is 9,000 years older than any earlier evidence of tobacco use, and it was first published online in October 2021. How the people at that fire used the tobacco is not known.

A pipe found in central Washington State is about 1,430 years old. Chemists found nicotine, the drug in tobacco, inside it. The pipe also held traces of a wild tobacco that grows here, mixed with a plant called smooth sumac.

Tobacco as an offering. The "Keep It Sacred" program of the Inter-Tribal Council of Michigan describes tobacco used as an offering, which it calls traditional tobacco. People grow or gather it for ceremony or for medicine. It is the plant known by its Latin name, *Nicotiana rustica*. People offer it to the Creator, or to another person, place or being, by burning it in a fire or smoking it in a pipe.

In the program's words, the smoke carries "thoughts and prayers to the spirit world." The smoke is generally not breathed into the lungs. The program's writers state that traditional tobacco used this way is not linked to addiction, a need for a drug so strong that stopping is very hard, or to harm to health. They set it apart from commercial tobacco, the kind sold in stores. The ways people used tobacco differed from one nation to another.

Black Drink, cacao, peyote and datura

Black Drink was a tea made from the leaves of yaupon holly or dahoon holly, two kinds of holly bush. It contains caffeine, the drug in coffee that keeps people awake. Chemists found traces of it inside beaker cups from Cahokia, in what is now Illinois, dated from 1050 to 1250. These hollies grow far to the south of Cahokia, so people carried the leaves north. Records from later centuries describe men in the Southeast drinking Black Drink in a ceremony to make themselves pure.

Chocolate from the south. Cacao is the bean that chocolate is made from, and it grows in Mexico and Central America. Jars at Pueblo Bonito, in Chaco Canyon in New Mexico, held traces of a cacao drink. They date from about 1000 to 1125, and the beans came north by trade. It is the first known use of cacao drinks north of the Mexican border. Drinks made from cacao or holly turn up at 18 sites in the Southwest and northern Mexico, from about 750 to 1400.

Peyote. Peyote is a small cactus whose top holds mescaline, a drug that causes visions. Two dried peyote tops kept at the Witte Museum in San Antonio are about 5,700 years old. They were probably found in Shumla Cave, on the Rio Grande in Texas. When chemists tested them, they still held mescaline. The chemists wrote that Native North Americans knew what peyote does as long as 5,700 years ago.

Datura. Datura is a flowering plant that holds two drugs, scopolamine and atropine. They cause hallucinations, which means seeing and hearing things that are not there. In Pinwheel Cave in California, people chewed wads of datura and pressed them into the painted ceiling. Records from later times show Native Californians using datura to enter a trance, a state like a dream while awake. People used the cave from before 1500 into the years after Europeans arrived, so the wads have no exact date.

Who made alcohol before 1500?

Native nations in the Southwest made beer and wine before Europeans came, and other nations used alcohol only here and there. Patrick J. Abbott, a doctor, found this in a 1996 study of how Native nations used alcohol before Europeans came. Most Native people first met alcohol when Europeans brought it in the 1500s and 1600s.

Cactus wine. The Akimel O'odham and the Tohono O'odham made wine from a cactus. Older books call them the Pima and the Papago. Each late summer the women gathered the red fruit of the saguaro cactus. They made most of it into jam and the last of the harvest into wine. People drank the wine at the rain ceremony held at their New Year, late in the summer. When Abbott wrote, many Tohono O'odham villages still held that ceremony.

Corn beer. The Apache made a corn beer called tiswin, which they learned to make from peoples of northern Mexico. They drank it mostly in everyday dealings with one another, and only sometimes in ceremony. The Apache reached the Southwest sometime between 1000 and 1550. For that reason, nobody knows whether they made tiswin here before 1500.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 4 MINUTES

Europeans learned to use tobacco from the peoples of the Americas in the 1500s and carried the habit home. A French official sent tobacco seeds to the queen of France in 1560, and English colonists at Roanoke smoked it the Algonquian way in 1585. Spanish colonists brought wine with them to Florida.

Most Native people in this land first met alcohol in the 1500s and 1600s, when Europeans arrived. A regular trade in alcohol with Native nations began in the 1600s.

Why nicotine is named after a Frenchman

Jean Nicot was the French ambassador to Portugal, the official who spoke for the king of France there. In 1560 he sent tobacco seeds to Queen Catherine de Médicis of France, with a letter about the plant's uses as a medicine. The next year he gave her tobacco leaves and showed her how to grind them into a powder to sniff for headaches. The people around the French king and queen took up the powder and sniffed it too.

The word nicotine comes from Nicot's name. So does *Nicotiana*, the name the Swedish scientist Carl Linnaeus gave the tobacco plants in 1753.

Tobacco reached England in 1565, perhaps brought from Florida by Sir John Hawkins. By 1588 English men and women of high rank were using it, and so were some well-trained doctors.

Thomas Hariot

Who: An English mathematician who smoked tobacco the Algonquian way at Roanoke and wrote down how the Algonquian people used it.

When and where: Roanoke Island, in what is now North Carolina, 1585 to 1586. London until his death in 1621.

He smoked the Algonquian plant uppowoc and died of nose cancer.

Thomas Hariot studied mathematics and the stars. In 1585 he sailed to Roanoke Island with a group of English ships and men sent to start a colony. Most of the colonists were soldiers. Hariot and the artist John White were the best known of the few who were not.

Whose island it was. Roanoke Island was the land of the Roanoke, Secotan and Croatoan peoples. Their leading werowance, or chief, was Wingina, and he let the English build a fort on the north end of the island. Hariot learned the Algonquian language from two Native men of the region, Manteo and Wanchese.

The plant. The people called it uppowoc. They planted it in plots of its own, dried the leaves, ground them to powder and drew the smoke through clay pipes. They believed the smoke opened the passages of the body and kept them healthy. As an offering, they threw the powder into sacred fires, and onto the water in storms and after escaping a danger.

Hariot wrote that the English "vsed to suck it after their maner" on the island. In today's words, they smoked it the way the Algonquian people did, and they went on smoking it after they came home.

The killing of Wingina. Wingina had let the English build their fort in 1585, and the next year they killed him. On June 1, 1586, men under the English commander Ralph Lane attacked his village. They shot Wingina and chased him down, and an Irishman named Edward Nugent cut off his head. A week later the English sailed home, and Hariot went with them.

> Hariot described uppowoc in a book printed in 1588, *A Briefe and True Report of the New Found Land of Virginia*. In it he also wrote that near the end of the colony's year, some of the English had been "too fierce, in slaying some of the people." In the same sentence he wrote that the people had "iustly deserued" (justly deserved) it. He did not name the dead.
>
> Hariot died in London on July 2, 1621, of cancer of the nose. The writers of Encyclopedia Virginia state that the cancer was "probably caused by heavy tobacco use."

Spanish wine in Florida

Pedro Menéndez de Avilés and his Spanish colonists founded St. Augustine, in Florida, on September 8, 1565. They built it at Seloy, a town of the Timucua people.

Wine and olive oil came by ship in big Spanish storage jars called olive jars. Historians at the Florida Museum of Natural History write that the two were perhaps even more important to the Spanish than wheat and beef. They also write that the Spanish soon turned to the Timucua for food, cooking pots and wives. Their account does not say whether the Timucua women chose these marriages.

Alcohol as a gift. The historian Peter C. Mancall writes that some Europeans surely offered alcohol to Native people when they first met, though no record from this century names a case. No record from the 1500s describes the Spanish trading wine to Native nations as a regular business.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 9 MINUTES

Drinking came over on the first English ships, and colony leaders spent the 1600s trying to keep it in bounds. The colonists' first drinks were cider and beer, and they believed alcohol kept the body warm and strong. Their leaders licensed taverns and punished drunks with fines, iron shackles, the stocks and public shame. In Virginia, planters grew tobacco to sell in England, and the smoking habit spread.

Liquor for furs. English traders sold rum and French traders sold brandy to Native nations in exchange for furs. Colony leaders passed laws against these sales, and traders kept breaking them.

Beer, cider and bad water

The first drinks the colonists made were cider and beer. They brewed beer from corn, wheat, oats, persimmons and green cornstalks. They tried to grow grapes for wine, but the vines they brought from Europe died of American pests and diseases.

Colonists believed that alcohol warmed the body, helped digestion and made people strong. The land here had plenty of clean water, but some colonists drank from dirty sources and got sick. At Jamestown, George Percy wrote that the settlers drank river water that was salty at high tide and "full of slime and filth" at low tide. He called it "the destruction of many of our men." In Europe, where rivers were often dirty, people drank alcohol in place of water, and the colonists kept that habit.

Officials of the Virginia Company of London complained about it. In a letter to Governor Francis Wyatt in 1622, they wrote that the colonists' drinking was harming the colony.

Bolts, fines and the stocks

Virginia's first lawmakers passed a law against drunkenness in 1619, at the colony's first General Assembly. The first time a man was found drunk, the minister scolded him in private. The second time, the minister scolded him in public. For a third offense he had to lie for 12 hours in bolts, iron shackles, in the house of an officer called the provost marshal, and pay the marshal's fees.

If he still kept drinking, the governor and his council, the men who advised him, could give him whatever harsh punishment they chose.

Massachusetts. The lawmakers of Massachusetts passed a set of drinking laws between 1633 and 1647. Nobody could run a tavern without a license from the county court. A tavern keeper could not serve one person more than half a pint of wine at a time. Nobody could sit drinking for more than half an hour, or stay after nine at night.

The fine. A person found drunk in Massachusetts paid 10 shillings, half a pound in English money. The law defined drunk as unable to think clearly, which showed in the way the person talked or moved. A person who could not pay was to be jailed, or set for up to three hours in the stocks, a wooden frame that locked a person's ankles while the person sat in a public place.

Prices. The lawmakers also set what an innkeeper could charge. In Boston in 1634 a meal cost sixpence and a quart of beer a penny, with a fine of 10 shillings for charging more. A traveller named John Josselyn reported that a constable, a town officer who kept order, would stop a stranger in a Boston tavern from ordering more drink than the constable thought right.

Samuel Cole

Who: The keeper of the earliest tavern in Boston known to have had a license. He is not Robert Coles, whom Massachusetts judges punished for drunkenness.

When and where: Boston, in the Massachusetts Bay Colony, 1634 to 1638.

About twenty Narragansett men ate their peace-treaty meal at his inn.

Samuel Adams Drake, a historian of Boston's taverns, dates the opening of Samuel Cole's tavern to March 4, 1634. A tavern was then called an ordinary.

Cole was a comfit-maker, a maker of sugar-coated sweets. He came to Boston in 1630, in the fleet of ships that brought Governor John Winthrop. His ordinary stood on Washington Street, and the house had a brew-house, a building for making beer.

The Narragansett visit. The English and the Narragansett people made a peace agreement in 1636, and Governor Henry Vane hosted Narragansett leaders in Boston. The Narragansett leader Miantonomi and two sons of the leader Canonicus ate at the governor's house. About twenty other Narragansett men ate their meal at Cole's inn.

The colony's judges let Cole sell his house as an inn in 1638, and he sold it to Robert Sedgwick. Cole died in 1666.

Robert Coles

Who: A colonist whom judges in Massachusetts punished four times for drunkenness, the last time by making him wear a red letter D. He is not Samuel Cole, the Boston tavern keeper.

When and where: Roxbury and Agawam, now Ipswich, in the Massachusetts Bay Colony, 1631 to 1634.

A paper on his back said DRUNKARD in large letters.

The first fine. Robert Coles came to Massachusetts in 1630 with Governor John Winthrop's fleet. In 1631 the colony's judges fined him five marks, about 3 pounds, 6 shillings and 8 pence, for being drunk on the ship *Friendship* and at Winnisimmet, now Chelsea. The *Friendship* was carrying a spiced wine made from honey.

The paper. At Agawam in 1633 he was charged with drunkenness for the third time. He had been drunk, had encouraged John Shatswell's wife to drink, and had urged her to be unfaithful to her husband. The judges fined him 10 pounds and made him stand in public with a paper on his back. On it the word drunkard was written in large letters.

The red D. He was drunk again at Roxbury in 1634. This time the judges ordered him to wear the red D for a year, and they took away his right to vote.

Afterward. Coles got back his right to vote about two months later, on May 14, 1634. The records show no later charge of drunkenness against him, though the Roxbury church later put him out.

By 1637 he had moved to Providence, a town built on land that Roger Williams got in 1636 from the Narragansett leaders Canonicus and Miantonomi. Coles was one of the 13 men who first owned land there, and in 1640 he helped write the Providence Combination, a written plan for governing the town. He died in Warwick, Rhode Island, in 1655.

Liquor for furs

Henry Hudson used liquor to test a group of Mahican leaders in September 1609. His ship lay near what is now Albany, in Mahican land. Hudson and his mate took the leaders into the ship's cabin and gave them "so much Wine and Aqua vitae," a strong liquor like brandy, "that they were all merrie."

Why Hudson did it. Robert Juet, who kept a journal on the ship, wrote that Hudson and the mate wanted to find out whether the leaders were planning "treacherie," a plan to betray them. In the end one of the men got drunk. Juet wrote that this was strange to the Mahican, "for they could not tell how to take it." The next day the people came back with tobacco and beads for Hudson, and a platter of venison.

The regular trade. The historian Peter C. Mancall dates the start of a regular trade in alcohol to the middle of the 1600s. From 1650 on, he writes, liquor was a common item in the fur trade. English traders sold rum to Native nations, and French traders sold brandy, in exchange for furs. Mancall writes that young men who controlled the furs from a season's hunt traded them for liquor, and more families became poor.

What the liquor did. Native and colonial witnesses both wrote down what happened when Native people drank the trade liquor. Drinkers fought each other and members of their own families. Some fell into fires or off cliffs, some drowned, and some killed other people. No one counted the deaths.

The bans. Colony leaders passed laws against the trade. In New Netherland, the Dutch colony that is now New York, Director General Willem Kieft, the colony's governor, and his council banned selling liquor to Native people on June 18, 1643. Their order said that "large quantities of strong liquors are daily sold to the Indians." The fine was 25 guilders, a guilder being a Dutch coin.

They passed the ban again in 1645. The next Director General, Petrus Stuyvesant, and his council passed it a third time in 1647, and raised the fine to 500 guilders plus the cost of any damage. The laws of the colonies name the officials who banned the trade. They rarely name the traders who broke the bans.

A false belief. Mancall writes that Native people organized against the trade themselves. Some colonists came to believe that Native people were born more likely to become alcoholics, people who cannot stop drinking. Mancall states that no evidence supports that belief.

Tobacco and rum

King James I of England attacked smoking in 1604, in a pamphlet called *A Counterblaste to Tobacco*. He called it "lothsome (disgusting) to the eye, hatefull to the Nose, harmefull to the braine, dangerous to the Lungs." In the same pamphlet he insulted Native people. He mocked the English for copying "the barbarous and beastly maners of the wilde, godlesse, and slauish Indians."

A milder leaf. The tobacco that the Powhatan people grew tasted dark and bitter to the English. John Rolfe, a colonist at Jamestown who smoked heavily, got seeds of a milder Spanish tobacco from Trinidad and Caracas. He was growing it by July 1612. The colonists shipped 20,000 pounds of it to England in 1617 and grew 60,000 pounds in 1622. After 1622 English planters took over land that the Powhatan had already cleared, to grow more tobacco.

Smoking rules in Massachusetts. Massachusetts officials told constables in 1633 to report "tobacco takers" as idle people. By 1638 the lawmakers had canceled their earlier laws against tobacco, and that year they wrote that since then smoking was "more abused than before." So they banned smoking near any house, barn, corn or haystack, because a pipe that close might set them on fire.

They also banned smoking outdoors in the streets and fields, except on a journey or at a meal, and in the public rooms of an inn. They passed these rules again in 1647, and that version put the distance from buildings and crops at about 330 feet.

Rum. Rum is a strong liquor made from molasses, the thick syrup left over when sugar is made from sugarcane. The first known name for rum comes from the island of Barbados, in about 1651. A planter there called it "Kill-Divill" and "a hott hellish and terrible liquor." In the 1600s New England merchants began shipping fish and lumber to the islands of the West Indies. They traded them for molasses and made the molasses into rum at home.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 7 MINUTES

Colonists drank more rum than any other strong drink between 1700 and 1750, and it was cheap. A gallon of rum made in the colonies cost 1 shilling 8 pence in Philadelphia in 1740. People drank in taverns, and they also met there to do the town's business. Colony leaders controlled drinking with tavern licenses and with rules for elections.

Georgia's ban. A Yamacraw leader named Tomochichi asked Georgia's leaders to stop the sale of rum, and they banned strong liquor for seven years, from 1735 to 1742. Nobody counted how much colonists drank in these years. The first estimates come from the 1770s.

Rum, molasses and tobacco

Colonial distillers made rum from molasses shipped in from the sugar plantations of the Caribbean. Some of the molasses came in legally, and some was smuggled, which means merchants brought it in secretly without paying the tax on it. In Philadelphia in 1740, a gallon of colonial rum cost 1 shilling 8 pence. A gallon of rum from the Caribbean cost 2 shillings 5 pence.

The Molasses Act. At the urging of large sugar planters on the British islands in the Caribbean, the members of the British Parliament, the lawmakers of Britain, passed the Molasses Act in 1733. Under it, importers owed sixpence a gallon on molasses from the French islands. Colonial merchants kept smuggling French molasses anyway, much of it from Saint Domingue, now Haiti. The tax should have brought in 25,000 pounds in British money each year. British officials collected 259 pounds in 1735.

Rum for people. Rum was part of a trade in people. New England ships carried rum to Africa, enslaved Africans to the West Indies, and molasses back to New England. Distillers made an extra-strong rum for this trade, so that ships could carry more of it in less space. When the ships reached Africa, the captains added water to the rum. They traded it for people to sell into slavery.

Tobacco and slavery. Enslaved Africans outnumbered white indentured servants, workers bound to a master for a set number of years, on Virginia's tobacco farms by 1700. Virginia planters grew 29 million pounds of tobacco in 1709, the most the colony ever grew.

Drinks for voters

Only white men aged 21 or older who owned land could vote in colonial Virginia. Enslaved people, free Black people, Native people, servants, white men with no land and all women could not. Each voter said his choice out loud in front of the crowd. Voters expected the men running for office to give them liquor and food.

The dry weeks. Virginia's lawmakers limited these gifts, called treats, in a law of 1705. Candidates could give treats before the sheriff read out the notice of an election, and again after the voting. The weeks in between were supposed to be "dry," with no treats. Some candidates got around the rule. They stayed away and had a friend, a servant or even a wife hand out the drinks.

The tavern as the town's meeting place

Boston men used taverns as meeting places for a lodge, for the colony's lawmakers, and for signing up sailors in wartime. Francis Holmes kept the Bunch of Grapes on King Street, now State Street, from 1712. One traveller called it the best punch house in Boston. Punch was a mixed drink made with liquor.

The first grand lodge. Henry Price, a Boston tailor, organized a grand lodge of Freemasons at the Bunch of Grapes on July 30, 1733. The Freemasons were a society of men who met in lodges, and a grand lodge was a head lodge over the others.

Vardy's. Luke Vardy kept the Royal Exchange tavern. When Boston's town-house burned in the winter of 1747, the colony's lawmakers finished their meetings at Vardy's.

A duel. Two young men quarrelled at Vardy's in July 1728, over their cards or their wine. They fought with swords on Boston Common, and one of them killed the other, a man named Woodbridge. The killer's friends put him on a ship that sailed at once. The historian Samuel Adams Drake, who told the story in 1886, does not give the killer's name.

Signing up sailors. During a war between Britain and France in 1745, notices posted in Boston told sailors who wanted to join a privateer to report to the Widow Gray's Crown Tavern. A privateer was a privately owned ship allowed to attack enemy ships. Women kept several of Boston's taverns. Drake's list includes the Widow Gray, the Widow Anna Swords, a Mrs. Wardwell and a Mrs. Glover.

Georgia bans rum, 1735 to 1742

James Oglethorpe and the first colonists of Georgia settled on Yamacraw Bluff, where Savannah stands today, in February 1733. Tomochichi, the leader of a Yamacraw town of about 200 Creek and Yamasee people, gave them the land on the bluff.

Tomochichi's request. Tomochichi saw more of his people drinking heavily. In 1734 he asked Oglethorpe and the Trustees, a board of men who met in London and ran the colony from there, to ban the sale of rum.

The Trustees banned rum and brandy in Georgia in 1735, and King George II and his council approved the law. It called these liquors "hurtful and pernicious (harmful) to Man's Body." The Trustees named an officer to enforce the ban and set fines that grew larger with each offense. Taverns could still sell beer and wine with a license.

The Malcontents. Colonists soon accused the officials who ran the ban of being dishonest, and they protested the ban. A group known as the Malcontents, a word for people who are unhappy and complain, argued that the ban hurt trade. Georgia merchants, they said, needed rum to trade for timber and other goods, which they then shipped to the British islands in the Caribbean. The Trustees ended the ban on September 29, 1742, and let rum in again for use in the lumber trade. Oglethorpe disagreed with their decision.

Traders from South Carolina had kept carrying rum up the Savannah River all along, against the Trustees' order. In a letter of 1737, Oglethorpe wrote that the Creek had learned drunkenness "from our Traders."

Dr. Alexander Hamilton

Who: A Scottish-born doctor from Annapolis, Maryland, who kept a diary of the taverns and drinking clubs he saw on a long ride north. He is not the Alexander Hamilton who later ran the Treasury.

When and where: From Maryland to York, Maine, and back, May 30 to September 27, 1744.

He drank three "bumpers" with the Hungarian Club and left at ten.

Dr. Alexander Hamilton had moved to Annapolis in 1738 to work as a doctor, and in 1743 he was very sick. Mainly to get his health back, he set out on horseback on May 30, 1744. He rode 1,624 miles, from Maryland to Maine and back, and reached home on September 27.

Crooked riders. On May 31, at Tradaway's inn in Maryland, he met "a drunken Club" leaving. Most of the men had climbed onto their horses and sat on them crooked. At an inn near Philadelphia on June 5, the landlady did all the work because the landlord was drunk. Hamilton wrote that the liquor had taken away "the use of his tongue." The landlord sat without moving in a corner, smoking his pipe.

The Hungarian Club. On June 15 he ate supper in New York with the Hungarian Club at Todd's tavern. After supper the men drank "bumpers," glasses filled to the brim and drunk off at once, at each round. Hamilton did not want to drink and drank only three. He left at ten, "pretty well flushed (red in the face) with my three bumpers."

Some of the men, he wrote, thought nothing was finer than being able "to pour down seas of liquor, and remain unconquered, while others sank under the table." He wrote that some members were "very fond of making a stranger drunk."

The doctors' club. In Boston he met with a club of doctors at the Sun Tavern. "We drank punch, smoaked tobacco, and talked of sundry physical matters," he wrote, meaning various matters of medicine. In

> 1745 he started a club of his own in Annapolis, the Tuesday Club. He died on May 11, 1756.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 13 MINUTES

Rum made fortunes in the colonies before the Revolution, and whiskey took its place after. By 1770 the colonies had more than 140 rum distilleries, making about 4.8 million gallons a year. Ship captains from Rhode Island carried rum to Africa and traded it for African people to sell into slavery. After the war, Americans switched to whiskey made from their own corn and rye.

Who pushed back. Native leaders asked colonists from the 1750s on to stop selling them liquor. The Philadelphia doctor Benjamin Rush argued in 1785 that heavy drinking was a disease. When members of Congress taxed whiskey in 1791, armed men in western Pennsylvania attacked the tax collectors, and President George Washington called up nearly 13,000 militiamen, part-time citizen soldiers, against them. In 1799 Handsome Lake, a Seneca religious leader, stopped drinking and preached against it.

How much did the colonists drink?

Nobody measured how much colonists drank before the 1770s, and the estimates made since then each measure something different. The writer Ed Crews, in the *Colonial Williamsburg Journal*, reports an estimate that the average grown man in the 1770s may have drunk as much as three pints a week. The writer Ian Williams gives 3.7 gallons of rum a year for each colonist by the time of the Revolution. The historian Peter C. Mancall cites an estimate of perhaps seven shots of liquor a day by 1770.

The first government count. Government figures for 1790 showed that the average American over the age of 15 drank 34 gallons of beer and cider, 5 gallons of liquor and 1 gallon of wine in a year.

Drink as medicine. People believed alcohol was good for the body. They took whiskey for stomach pain and for a sore throat, and hot brandy punch for cholera, a disease that causes severe diarrhea. Pregnant women and women in labor drank a shot of liquor to ease the pain. Benjamin Franklin collected more than 200 words that meant drunk.

Rum and the slave trade

Rhode Island had about 30 rum distilleries by 1764, 22 of them in the town of Newport. Rhode Island merchants traded the rum on the coast of West Africa for African people to enslave. In the hundred years before 1807, Rhode Islanders made at least 1,000 voyages to buy and carry enslaved people. They carried more than 100,000 Africans into slavery.

The *Sally*. Chapter 16 tells the voyage of the slave ship *Sally* stage by stage. Here is the rum's part in it. In 1764 the brothers Nicholas, John, Joseph and Moses Brown of Providence sent the ship to West Africa with 17,274 gallons of rum to trade, along with chains, shackles and guns. Its captain, Esek Hopkins, used the rum to buy 196 people.

At least 109 of the 196 died, by the count in Brown University's report on the voyage. Several were children, and one woman hanged herself below deck. When the captives rose up against the crew at sea, Hopkins's men fired on them. The ship's account book records that the crew "Destroyed 8," which means they killed eight people. After that, the account book says, "Some Drowned themselves Some Starved and others Sickened & Dyed."

The Sugar Act. Rum was made from molasses. Since the Molasses Act of 1733, colonial merchants had smuggled molasses from the French islands to avoid its tax of sixpence a gallon. In 1764, the year the *Sally* sailed, members of the British Parliament cut the tax on foreign molasses to threepence a gallon, in a law called the Sugar Act. They also pushed the British navy to stop the smuggling, and navy officers who caught smugglers could seize half of the smuggled goods they found.

From rum to whiskey. During the Revolution, British warships stopped much of the shipping, and molasses became costly and hard to get. Scotch-Irish farmers in Pennsylvania, Maryland, western Virginia and western North Carolina made whiskey from rye and corn instead. Before 1800, distillers in Kentucky were shipping whiskey down the rivers to New Orleans.

Native leaders against the liquor trade

Hagler, a leader of the Catawba people in the hill country of the Carolinas, spoke to officials sent by North Carolina in 1754. He told them, "You Rot Your grain in Tubs, out of which you take and make Strong Spirits." Hagler and other Native leaders asked the colonists to stop the liquor trade, because it brought violence and despair.

What the British official wanted. Sir William Johnson, the British superintendent of Indian affairs for the northern colonies, handled dealings with Native nations, and he wanted the trade to go on. In 1764 he wrote to his bosses in London, the Lords of Trade, that rum could cause problems. He also wrote that "the Trade will never be so extensive" without rum. After the Revolution, traders carried liquor farther west, and they added water to it so they had more to sell.

Handsome Lake. United States troops burned the villages of the Haudenosaunee, also called the Iroquois, in New York in 1779. After the war, at Fort Stanwix in 1784, American negotiators backed by soldiers took huge sections of land from the Six Nations of the Haudenosaunee. By 1799 the Haudenosaunee had lost most of their land in New York and Pennsylvania, and many of them drank heavily.

One was Handsome Lake, a Seneca man from the Genesee River. In 1799, sick after many years of heavy drinking, he had visions. Soon after, he stopped drinking and began preaching against drunkenness. His teaching became the Longhouse Religion, which Haudenosaunee people still practice.

Benjamin Rush

Who: A Philadelphia doctor who signed the Declaration of Independence and wrote that heavy drinking was a disease.

When and where: Philadelphia, 1785 to 1790.

His chart put water at the top and the gallows at the bottom.

Before Benjamin Rush, most Americans thought a drunkard simply lacked the will to stop. Rush argued that heavy drinking was an addiction.

The book. In 1785 he published a book about ardent spirits, meaning strong liquor such as rum, gin and whiskey. Its title was *An Inquiry into the Effects of Ardent Spirits Upon the Human Body and Mind*, and some records date the first printing to 1784.

Rush wrote that a person begins drinking by free choice, and the habit then becomes a need. In his words: "The use of strong drink is at first the effect of free agency. From habit it takes place from necessity." He said the only cure was to stop drinking completely. His advice to heavy drinkers was "Taste not, handle not."

The thermometer. An edition of 1790 added a chart that Rush called "A Moral and Physical Thermometer." Water and milk stood at the top. Lower down, punch led to "idleness, sickness and debt." At the bottom, gin, brandy and rum taken every day led to "murder, madness" and the gallows.

Few people changed their drinking at the time. The historian Mark Lender says that "the conception we have of addiction today can generally be traced back to Benjamin Rush."

Why farmers went to war over a whiskey tax

Farmers in western Pennsylvania turned their grain into whiskey and used the whiskey as money. From 1791 some of them fought a new federal tax on whiskey, and in 1794 hundreds of them took up guns against it. The fight is called the Whiskey Rebellion.

Why whiskey. The roads over the mountains to the east were bad, and grain spoiled on the way. Whiskey kept, and it was easier to carry. Coins were scarce on the frontier, so people paid for goods and paid their workers in whiskey.

Whose land it was. The whiskey farmers worked land that the Shawnee, Delaware and Mingo peoples had used. Representatives of the Six Nations of the Haudenosaunee had signed that land over to Britain in 1768, at Fort Stanwix in New York. That was an earlier treaty than the one made at the same fort in 1784. The Shawnee and Delaware, who lived on much of the land, were at the meeting but did not sign.

The tax. Alexander Hamilton, the Treasury Secretary in charge of the country's money, asked for a tax on whiskey made in the United States in January 1791, to pay the debt from the Revolutionary War. Members of Congress voted it into law. The law set the tax at 7 cents a gallon, but big distillers paid less than that and small ones paid more.

A big distiller could instead pay a flat fee for each still, the equipment used to distill liquor, and that came to as little as 6 cents a gallon. A farmer with a small still paid about 9 cents a gallon. The law required payment in coin, which the farmers rarely had.

Two men tarred and feathered. A gang of armed men in disguise caught Robert Johnson, a new tax collector, at Pigeon Creek in Washington County, Pennsylvania, on September 6, 1791. Some records give the date as September 11. They tarred and feathered him, which meant pouring hot tar on his skin and sticking feathers to the tar. They cut off his hair and took his horse, so he had to walk a long way. Hamilton's 1794 report on the attacks names none of the men.

Officials sent a man named John Connor to serve court papers on Johnson's attackers. Men seized Connor, whipped him, and tarred and feathered him too. They took his money and his horse, blindfolded him and tied him up in the woods, where he stayed for five hours. No surviving record names who did it.

The fight at Bower Hill. John Neville, the regional tax inspector, went with a United States marshal to serve court papers on July 15, 1794. A marshal was an officer who carried out the orders of the national courts. At a farm about 10 miles south of Pittsburgh, men fired warning shots at them.

The next day, at least 30 men from the Mingo Creek militia surrounded Neville's house, called Bower Hill. Neville shot and killed one of them, Oliver Miller. Neville had the people he enslaved help him defend the house. They could not refuse.

The men came back on July 17, this time between 400 and 600 of them, led by Major James McFarlane, who had fought in the Revolution. By then 10 United States soldiers from Pittsburgh had come to defend the house, and Neville had left to hide nearby.

After about an hour of shooting, McFarlane called a stop. When he stepped into the open, someone in the house shot and killed him. The records do not say who fired. The attackers burned Neville's house and the quarters where the enslaved people lived. The soldiers surrendered, and the attackers sent them away.

KNOWN:	The dead at Bower Hill included McFarlane and one or two other militiamen.
NOT KNOWN:	How many men died there in all, and whether one of the soldiers later died of his wounds.

The army. Washington called up 12,950 militiamen from New Jersey, Maryland, Virginia and Pennsylvania. He went with them as far as Bedford, Pennsylvania, and Fort Cumberland, Maryland, then went back to Philadelphia. He left Henry Lee, the governor of Virginia, in command, and Hamilton went along as an adviser. The rebels scattered without a battle. About 12 soldiers died of illness or accidents.

The Dreadful Night. Soldiers under Lee broke into houses on the night of November 13, 1794, and seized people, rebels and witnesses alike. The soldiers drove the prisoners barefoot, in their nightclothes, over muddy roads and through streams, and locked them in animal pens and in cellars with no floors. The soldiers ruined the health of some of the prisoners, and at least one prisoner died. In all, the soldiers arrested about 150 men during the march west.

The trials. Soldiers marched twenty of the prisoners to Philadelphia and paraded them down Broad Street on Christmas Day 1794, as trophies of the march. Ten were tried for treason, the crime of making war against one's own country, and two were found guilty. Members of Congress voted to end the whiskey tax in 1802.

Philip Vigol

Who: A man from western Pennsylvania, one of the first two Americans found guilty of treason against the United States, for attacking whiskey-tax officers.

When and where: Westmoreland and Fayette counties, Pennsylvania, 1794. Philadelphia, 1795.

His party carried guns, beat drums and burned a tax officer's house.

The court report spells Philip Vigol's name this way. Other records spell it Wigle, Weigel or Wigal.

Reigan. During the rebellion, Vigol went with an armed party, carrying guns and beating drums, to the house of Reigan, a tax officer in Westmoreland County. They forced Reigan to hand over his official papers and made him swear never to collect the tax again.

Wells. The same party then rode to the house of Wells, the tax officer in Fayette County, and swore to destroy Wells and his house. Wells had run and hidden. The men ransacked his house, which means they tore

through it and wrecked it. Then they burned it, with everything inside, including the tax records. Later they found Wells, locked him up and made him swear to quit his job.

The trial. A jury in the United States court in Philadelphia found Vigol guilty of treason in 1795, and the judges sentenced him to death by hanging. At the trial, Justice William Paterson told the jury that fear excused a crime only when a person's life was in real danger at that moment. John Mitchell, the other man found guilty, was also sentenced to hang.

The pardon. President Washington pardoned Vigol and Mitchell in November 1795, which means he freed them from their punishment. The records differ on the day, November 2 or November 14. In his yearly message to Congress, Washington wrote, "The misled have abandoned their errors."

George Washington's rum and whiskey

George Washington used alcohol to win votes and to keep soldiers working, and he handed it out to the people he enslaved on the days they worked. At the end of his life he owned one of the largest whiskey distilleries in the country.

Votes. Washington ran for the Virginia House of Burgesses, the colony's elected lawmaking body, in Frederick County in 1758. He gave the voters 28 gallons of rum, 50 gallons of rum punch and 82 gallons of wine, beer and cider. He won the seat.

Soldiers and the enslaved. As commander of the Continental Army, he wrote in June 1779 that he wanted "a pretty good stock of rum" at the forts for soldiers on work duty.

At his plantation, Mount Vernon, he gave the people he enslaved an allowance of liquor for each day's work. They could not refuse the work. Once he accused his overseer, James Butler, the man who ran their work, of giving the enslaved people 36 pints of rum. Butler answered that he had given "no more than a pint of rum & a pound of Meat to each man."

The distillery. His farm manager, James Anderson, advised him to build a whiskey distillery at Mount Vernon in 1797. It made 600 gallons that first year. In 1799, the year Washington died, it made almost 11,000 gallons, worth $7,500, when the average distillery in Virginia made about 650 gallons. Anderson's son, John, ran it with an assistant distiller and six enslaved workers. Their names were Hanson, Peter, Nat, Daniel, James and Timothy.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 9 MINUTES

Americans of the early 1800s drank almost three times as much alcohol as Americans do now, and the first big campaign against drinking began. The average American of drinking age drank 7.1 gallons of pure alcohol in 1830, against 2.48 gallons in 2023. People drank at breakfast, in the fields and at work.

Ministers and reformers began asking people in 1826 to sign a pledge, a written promise not to drink, and more than a million signed. By the 1840s and 1850s drinking had fallen to about 3.5 gallons, half the 1830 figure.

Other drugs. Anyone could buy laudanum, a painkiller made of opium and alcohol, without a doctor's order, and doctors gave it to adults and to children. Among Native nations, leaders such as Little Turtle of the Miami asked members of Congress to stop the traders who sold liquor to their people.

A dram with breakfast

Men, women and even children in the early 1800s drank "a healthful dram" with breakfast. A dram was a small drink of liquor. Farmers took cider, beer or whiskey into the fields. Employers stopped work in the middle of the morning so their workers could drink. When people got sick, a doctor's first medicine was often a drink of alcohol.

Counting the alcohol. Beer, cider, whiskey and rum hold different amounts of alcohol, so the usual way to compare them is to count only the pure alcohol inside the drinks. By the count of the historian W. J. Rorabaugh, Americans of drinking age drank an average of 5.8 gallons of pure alcohol each in 1790 and 7.1 gallons in 1830. In 2023 Americans aged 14 and older drank an average of 2.48 gallons each, by the count of the National Institute on Alcohol Abuse and Alcoholism.

Drinking fell to about 3.5 gallons of pure alcohol a person each year between 1840 and 1860. Those were the years when the temperance societies were most active.

Temperance and the pledge

People in Connecticut, Virginia and New York had formed temperance societies, groups that urged drinking little or no alcohol, by 1808. Lyman Beecher, a minister in Connecticut, began giving talks against all drinking of liquor in 1825.

The American Temperance Society. Reformers in Boston founded the American Temperance Society on February 13, 1826. Its members signed a pledge not to drink liquor, such as whiskey and rum. The pledge still allowed beer, wine and alcohol taken as medicine. Within five years the society had 2,220 local groups and 170,000 members. By the late 1830s it had more than 8,000 groups and more than 1,250,000 members.

No alcohol at all. The reformers split in the late 1830s. Many moved to total abstinence, which means no alcohol at all, not even beer or wine. Some stopped trying to talk people out of drinking and began asking lawmakers to ban the sale of liquor.

The first laws. Massachusetts lawmakers banned selling hard liquor, such as whiskey and rum, in amounts of less than 15 gallons in 1838. Only people with money could buy 15 gallons at once, so under this law only the rich could buy hard liquor. The lawmakers canceled the law within two years. In the 1840s, lawmakers in many states let voters in each town or county decide on liquor licenses, the papers from the government that allowed a person to sell liquor.

The Navy's ration. Sailors in the Navy got a daily ration of liquor, a set share handed out each day. Almost 400 people signed a petition, a paper people sign to ask for something, to members of Congress in 1843, asking them to end it. The petition was more than 10 feet long.

The Washingtonians, 1840

Six drinkers met at Chase's Tavern on Liberty Street in Baltimore on April 2, 1840, and began keeping each other sober. They were William K. Mitchell, John F. Hoss, David Anderson, George Steers, James McCurley and Archibald Campbell. They called their group the Washington Temperance Society.

Telling stories. The members stayed sober by telling each other the stories of their own drinking. Then they went looking for other drinkers and told them the same stories. Estimates of how many people joined run from tens of thousands up to 600,000. Within a few years the group faded away.

Lincoln's speech. Abraham Lincoln, then 33 years old, spoke to the Washingtonians of Springfield, Illinois, on February 22, 1842. He said that people had blamed and scolded drinkers, and the men who sold them liquor, too much, and that this was both unwise and unfair. "If you would win a man to your cause," he said, "first convince him that you are his sincere friend." He also said that "a drop of honey catches more flies than a gallon of gall," a bitter liquid.

John B. Gough

Who: An English-born bookbinder, a worker who sews and covers books, whose drinking cost him his work. His wife and baby died while he was drunk. He signed the temperance pledge and spent his life asking others to sign it.

When and where: New York, and Worcester, Massachusetts, 1829 to 1845.

Sent out to buy two quarts of rum, he drank them himself.

Drinking. John B. Gough was born in Sandgate, England, in 1817, and came to America at the age of 12, landing in New York in August 1829. He learned bookbinding in New York City. After his mother died, he began drinking heavily. He lost one job after another and sang comic songs in theaters.

Delirium tremens. Once, when he stopped drinking all at once, he had delirium tremens, an illness of heavy drinkers who suddenly stop. The body shakes and the mind is confused, and the person sees things that are not there. The illness can kill. In *An Autobiography*, printed in 1845, Gough wrote: "Hideous faces appeared on the walls, and on the ceiling, and on the floors." He saw "millions of monstrous spiders" around him.

His wife. When his wife fell sick, the women nursing her sent him to buy two quarts of rum. He drank it himself. Ten days later his wife and their baby both died. Gough wrote that the whole town learned "that my wife and child were lying dead, and that I was drunk!"

The pledge. In Worcester, Massachusetts, in late October 1842, a stranger touched his shoulder and asked, "Why do you not sign the pledge?" The man was Joel Stratton, a waiter at the town's temperance hotel. The next evening Gough told his story to a meeting at the Worcester town hall. Then he signed the pledge.

About five months later he drank again, and then he signed the pledge a second time, in public. Between May 1843 and January 1845, he wrote, he traveled more than 12,000 miles and gave 605 talks. At those talks,

> 31,760 people signed the pledge.

Whiskey and enslaved people at Christmas

Frederick Douglass, who was enslaved in Maryland in the 1830s, wrote in his book of 1845 that slaveholders used whiskey to control the people they enslaved. Enslaved people had the days between Christmas and New Year's off work. Slaveholders wanted them to spend those days drunk.

"It was deemed (thought to be) a disgrace not to get drunk at Christmas," Douglass wrote. Slaveholders made "bets on their slaves, as to who can drink the most whisky without getting drunk." In this way, Douglass wrote, they got "whole multitudes to drink to excess," which means far too much.

Why they did it. Douglass wrote that the slaveholders did this to keep enslaved people from rising up in an armed revolt. He wrote that he and the others felt "that we had almost as well be slaves to man as to rum."

Liquor and Native nations

Little Turtle, also called Michikinikwa, was a leader of the Miami people. He urged members of Congress to stop traders from selling liquor to Native nations. Congress members gave the president the power to stop those sales in 1802. War Department officials then made rules that banned traders from bringing liquor onto Native land, and did little to enforce them.

The traders. Traders kept selling because they made more money on liquor than on other goods. They also watered it down. Members of Congress made it a crime in 1834, with a fine of up to $300, to bring liquor onto Native land.

Native leaders. Native leaders worked against drinking among their own people. The Shawnee religious leader Tenskwatawa and a Delaware woman named Beate told their followers to give up alcohol. Pawnee leaders limited drinking in their towns.

Leaders of the Mohican-Stockbridge people wrote to Congress in 1827. They had been living on part of the Oneida Reservation in New York. They wrote that one reason they had agreed to move to what is now Wisconsin was to "flee

from" liquor in their community. The land in Wisconsin was the homeland of the Menominee people.

Opium sold as medicine

Opium is a drug made from the opium poppy. It stops pain and brings sleep, and a person who takes it every day comes to need it. Laudanum was opium dissolved in alcohol. Until the early 1900s, anyone could buy laudanum without a doctor's order.

For colds and for babies. Doctors gave laudanum for colds, for heart disease and for pain, to adults and to children. Nurses fed it to babies with a spoon. Patients having surgery in the late 1700s and early 1800s got laudanum and alcohol first. Then the surgeon's helpers held them down while the surgeon cut.

Morphine. A German druggist named Friedrich Sertürner took a drug called morphine out of opium in 1804. In 1817 he tested morphine on himself and three boys, and all four nearly died. A German pharmacy that later became the drug company Merck began making morphine for sale in 1827.

Chewing tobacco

The English writer Charles Dickens visited the United States in 1842. In his book *American Notes*, he complained about how common it was in Washington to chew tobacco and spit out the juice. In the courts of law, he wrote, "the judge has his spittoon, the crier his, the witness his, and the prisoner his." A spittoon is a bowl for spitting into, and the crier was the officer who called the court to order.

In hospitals, Dickens saw signs asking medical students to spit their tobacco juice into boxes and not onto the stairs. On a steamboat, he watched two young men chew and spit on the deck for a quarter of an hour before breakfast.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 16 MINUTES

Lawmakers in thirteen states banned liquor between 1851 and 1855, while drugstores went on selling opium, morphine and cocaine openly in medicine bottles. Maine's lawmakers were the first of the thirteen, in 1851. In Portland, Maine, in 1855, soldiers fired on a crowd at the mayor's order and killed a young sailor. In the Civil War, surgeons gave wounded soldiers morphine and opium, and thousands of veterans came home addicted.

Medicine and the law. Their medicines held opium, morphine, cocaine (a drug made from coca leaves) and alcohol, and the labels did not have to say so. City, state and federal lawmakers passed laws against opium. A federal law of 1887 named Chinese people outright, and accounts differ about whom San Francisco's opium law of 1875 was aimed at.

Women against the saloon. Women marched into saloons to pray, and in 1874 they founded the Woman's Christian Temperance Union. In 1880 Kansas voters approved a ban on liquor in their state constitution, the state's highest set of laws.

The Maine Law and the Portland Rum Riot

Lawmakers in Maine banned the sale of all alcoholic drinks on June 2, 1851. The law still allowed alcohol for medicine and for use in factories. Neal Dow, the mayor of Portland, had helped write it. By 1855, lawmakers in twelve more states had passed bans of their own. Judges threw out some of these laws, and lawmakers passed them again.

Dow and the Irish. For years Dow had ordered police raids on saloons in Portland. A saloon was a bar, mostly for men, that sold drinks by the glass. Irish immigrants ran many of the saloons Dow raided, and he gave speeches about "rum-swilling foreigners." About one in nine of Portland's 21,000 people was an Irish immigrant in 1850.

The liquor at City Hall. Dow was mayor again in 1855. He had city officials buy $1,600 worth of liquor for medicine and for use in workshops and factories, and he stored it at City Hall. The law required a city committee to make such a purchase, and no committee had been named.

Under the Maine Law, any three voters could ask a judge for a search warrant, a paper from a judge that allows a search. Three men got a warrant to search City Hall on June 2, 1855. The police would not let them in. By evening a crowd of between 1,000 and 3,000 people, most of them Irish, stood outside. Some threw stones at the door.

The shooting. Dow called out the militia, a force of part-time soldiers, and ordered them to fire. John Robbins, a 22-year-old sailor from Deer Isle, Maine, broke a hole in the door of the room where the liquor was kept. The soldiers shot him dead. He had planned to marry the next day. The soldiers kept firing as the crowd ran and wounded seven more people. The New England Historical Society's account of the riot does not name the soldiers who fired or their officers.

Dow was tried for buying the liquor improperly and found not guilty. Voters turned him out of office, and Maine lawmakers ended the ban in 1856.

How the Civil War made thousands of addicts

Soldiers in the Civil War were given whiskey as a "stimulant," a drink to give them energy. Members of Congress put a tax on liquor and tobacco in 1862 to help pay for the war. Navy officials ended the sailors' daily ration of liquor on September 1, 1862.

The needle. Doctors had begun using a new tool just before the war, between 1853 and 1855. The hypodermic syringe is a hollow needle for pushing a drug under the skin, and with it doctors gave more morphine. In the war, surgeons gave morphine shots and opium pills to soldiers with gunshot wounds and to soldiers whose arms or legs they had cut off. Diarrhea spread through the army camps, and doctors gave opium for that too.

Withdrawal. Morphine and opium stop pain, and a person who keeps taking them comes to need them. When the drug stops, the body goes into withdrawal, the sickness that comes when the body loses a drug it depends on. The person sweats, aches and cannot sleep. Later come stomach cramps, diarrhea, sick stomach and vomiting.

"Opium slavery." Thousands of veterans came home addicted, and they called it "opium slavery." Addicted veterans found little sympathy. Many of them could not get medical care, and many died of accidental overdoses. An overdose is too much of a drug at once, enough to make a person very sick or kill them.

NOT KNOWN:	how many veterans were addicted. The historian Jonathan S. Jones, who read their letters and records, says there were thousands.
DISPUTED:	a claim that 400,000 veterans came home with a "soldier's disease" of morphine addiction. The earliest known use of the phrase "soldier's disease" is from 1915, fifty years after the war.

John Tackett Goolrick

Who: A soldier for the Confederacy, the Southern states that broke away to keep slavery, in a unit that fired cannons, whose family's letters record his 30 years of morphine addiction.

When and where: Petersburg and Richmond, Virginia, 1864. Fredericksburg, Virginia, 1865 to 1925.

His family locked him in a farmhouse room for weeks.

The wound. John Tackett Goolrick served in the Fredericksburg Artillery in Robert E. Lee's army, and he strongly supported the Confederacy's fight to keep slavery. In August 1864, at the siege of Petersburg, an attack in which an army surrounded the city to force it to give up, a bullet hit him in the left leg.

He was taken to Chimborazo, a hospital in Richmond. Jones, who read the family's letters, writes that surgeons there apparently gave him morphine for the pain. At Chimborazo, or in the years after, he became addicted. He stayed in the army until Lee surrendered at Appomattox in April 1865.

The breakdown. After the war he worked as a lawyer in Fredericksburg. His wife was Frances Bernard Goolrick. He took larger and larger doses of morphine. In 1896 he broke down and could no longer work, and the family ran short of money.

Frances told him that she would divorce him unless he gave up morphine completely. On February 5, 1896, she wrote to her brother, "I can bear neither for myself or the children, this life any longer."

The locked room. The family could have sent him to a state hospital for the mentally ill, which took addicted veterans. His brother wrote that sending him there would be "commented upon and asked about" in the newspapers. So the family took him to his brother's farmhouse outside Fredericksburg instead. They locked him in a room, and nurses they paid

> kept him there for several weeks while he went through withdrawal. His sister-in-law Nora wrote on March 10, 1896, "my heart aches to look at him."
>
> Goolrick got well and went home. The family's letters suggest that he went back to morphine several times before he died in 1925.

What was in the medicine bottle?

Many medicines sold in stores held opium, morphine, cocaine or alcohol, and some held mercury and lead, which are poisons. These were patent medicines, sold under a brand name. Their makers did not have to tell buyers what was in them.

A syrup for babies. Jeremiah Curtis and Benjamin A. Perkins began selling Mrs. Winslow's Soothing Syrup in 1849, as a medicine for babies who were teething. Each ounce held about 65 milligrams of morphine. One teaspoon held enough morphine to kill a young child. Writers for the American Medical Association, a group of doctors, called it a "baby killer" in 1911. No record gives a count of the children who died after taking it.

A medicine for women. Lydia E. Pinkham's Vegetable Compound was sold to women from the 1870s. It was 18 to 20 percent alcohol. The label said the alcohol was there only to dissolve the other ingredients and to keep the medicine from spoiling. Critics pointed out that women who had signed temperance pledges took it.

Who was addicted. The historian David T. Courtwright writes that people addicted to opium drugs in these years were most often middle-class or rich women who had started taking morphine for an illness. Surveys in the late 1800s found that two out of every three people addicted to laudanum or morphine were women. There was also a "hard core" of opium smokers, mainly Chinese workers and white criminals.

Courtwright writes that, for the size of the population, addiction to these drugs was at least as common in the late 1800s as in 1992. Most of the people who took these medicines left no record of what happened to them.

Cocaine. Cocaine came to America as a medicine in 1884, and doctors used it to numb the eye for surgery. Soon there were reports of doctors and druggists addicted to it. Oregon lawmakers banned cocaine in 1887, the first in any state to do so. Lawmakers in Montana, Colorado, Illinois and Massachusetts passed bans by 1898.

John Stith Pemberton

Who: A Georgia druggist, addicted to morphine after a war wound, who invented Coca-Cola.

When and where: Columbus and Atlanta, Georgia, 1865 to 1888.

The first Coca-Cola had cocaine in it and cost five cents.

The wound. John Stith Pemberton had a medical degree and ran a drugstore in Columbus, Georgia. He fought for the Confederacy at the Battle of Columbus in April 1865 and took a sword wound to the chest. He took morphine for the pain and became addicted.

The drink. From 1866 he tried to make a painkiller that had no morphine in it. In 1885 he registered Pemberton's French Wine Coca, a wine mixed with coca, at his drugstore in Columbus. A ban on liquor took effect in Atlanta and Fulton County in 1886, so Pemberton made a version of his drink without wine and called it Coca-Cola. Jacob's Pharmacy in Atlanta first sold it on May 8, 1886, for five cents a glass. Pemberton claimed it cured headaches and morphine addiction.

The makers of Coca-Cola stopped using fresh coca leaves in 1903. They took out the last traces of cocaine by 1929.

The end. Pemberton was still addicted and poor when he sold his rights to the drink in 1888. He died of stomach cancer on August 16, 1888. His son Charles was addicted to opium. In June 1894 Charles was found unconscious with opium beside him, and he died ten days later, at the age of 40.

Opium laws and Chinese immigrants

Chinese immigrants came to California during the Gold Rush, and some of them brought the habit of smoking opium. Opium dens opened in San Francisco's Chinatown in the 1850s. An opium den was a room where people paid to lie down and smoke opium.

San Francisco, 1875. The members of San Francisco's Board of Supervisors, the city's lawmakers, made it a crime in 1875 to keep or visit an opium den. Reporters for the *San Francisco Chronicle* wrote that the supervisors acted after they learned that Chinese owners were running dens for white customers. The police chief, Phillip Crowley, said his officers would "keep them from opening places where whites might resort to smoke." His officers often left alone the dens that sold only to Chinese people.

Why they passed it. Writers disagree about the reason for the 1875 law. George Fisher, of Stanford Law School, argues that the supervisors wanted to protect white opium smokers. Writers for the Drug Policy Alliance, a group that works to change drug laws, point out that the dens the law closed were mainly in Chinese immigrant neighborhoods.

A false story. Irish members of the Church of the Transfiguration, a Catholic church in New York, and its assistant pastor, a priest who helped lead the church, spread a false story in 1883. They said that Chinese men were luring girls as young as 11 and 12 into opium dens. Writers for the *New York Times* called the story "preposterous," which means too silly to believe. About a week later the story was shown to be a hoax, a lie made up on purpose.

Laws that named race. Later laws named race and nationality outright. Idaho lawmakers in 1887 set punishments for "every white person" who kept or used an opium den. Fisher calls the early drug laws "laws about whites." In his view, their makers cared only about what drugs did to white people.

In the same year as Idaho's law, members of Congress named Chinese people in a law of their own. They banned any "subject of the Emperor of China" from bringing opium into the United States. By 1896, lawmakers in 22 states had banned opium dens.

Praying in the saloons: the WCTU

Eliza Jane Thompson led groups of women into the saloons of Hillsboro, Ohio, after she heard the lecturer Diocletian Lewis speak against drink on December 23, 1873. The women sang hymns and prayed until the owners closed. Women in 22 other states copied them. In Anoka, Minnesota, a saloon keeper and his wife poured cold water from an upstairs window onto the praying women.

The WCTU. Women met in Cleveland in November 1874 and founded the Woman's Christian Temperance Union, or WCTU. Its members worked to ban the making and selling of liquor. Frances Willard led the WCTU from 1879 until she died in 1898. She argued that women needed the vote to protect their homes from drunk men who beat women.

Black women in the WCTU. WCTU leaders set up a Department of Work Among Colored People in 1880. "Colored" was a word for Black people at the time. From 1883 Frances Watkins Harper, a Black poet and public speaker, ran it in the North.

Willard let the southern branches follow "states rights" on race, the idea that people in each state should settle such questions for themselves. In the 1880s Willard and the other national leaders let state and local branches form separate groups for Black women and white women. They did it to win over white women in the South, and Willard said women of both races wanted it. In practice, white members could use the separate groups to keep Black women out of their branches.

Harper called the split a short-term measure. She said it should last only until white members in the Deep South gave up "the spirit of caste," their belief that one race ranked above another.

Ida B. Wells and Willard. On speaking tours in Britain in 1893 and 1894, the journalist Ida B. Wells said that Willard had stayed silent about lynching, the killing of people by mobs without any trial. Wells said that the only public words Willard had spoken on the subject excused lynching. When people in Britain asked for proof, Wells sent them a copy of an interview Willard had given a New York newspaper in 1890, after a trip to the South.

In it, Willard said that "the colored race multiplies like the locusts of Egypt," comparing Black people to the swarms of insects sent on Egypt in the Bible. She said that "the grog shop (saloon) is its center of power" and that "the safety of woman, of childhood, of the home, is menaced (threatened)."

Defenders of lynching claimed that mobs killed Black men for raping white women, and Wells said Willard's words had been used to support that claim. Willard denied that she had stayed silent. She later spoke against lynching.

The Anti-Saloon League. Howard Hyde Russell, a minister, founded the Anti-Saloon League in Oberlin, Ohio, in 1893. Its leaders judged lawmakers only by how they voted on liquor, even if the lawmakers drank themselves. The league became a national group in 1895.

Kansas goes dry, 1880

Kansas voters were the first in any state to put a ban on liquor into their state constitution. They elected John St. John, who wanted a ban, as governor in 1878. Kansas lawmakers wrote a ban on making and selling liquor into a change to the constitution, and voters approved it in November 1880.

The ban began on May 1, 1881. Two years later Topeka, the state capital, still had 43 illegal bars. Their owners stayed open by paying a fine of $100 each month. The ban stayed in the Kansas constitution until 1948.

Liquor laws and Native nations

Members of Congress changed the 1834 law against bringing liquor onto Native land in 1854. The change stated that Native people themselves could not be charged under it. In 1895 they made it a crime to sell or give alcohol to anyone in Indian Territory, now Oklahoma, punished by a fine and one month to five years in prison.

Allotments. Members of Congress set a punishment of at least 60 days in prison, with no upper limit, in 1897 for giving alcohol to Native people who held allotments under the Dawes Act. Under that law of 1887, federal officials broke up land that a Native nation had held together into plots for single families, called allotments, and sold off the land that was left over. Native nations held about 138 million acres in 1887 and about 48 million by 1934.

Who was in prison. By 1900, about one in every 20 people in federal prisons had been found guilty under these liquor laws. The count does not say how many of them were Native people. Native people could not be charged with bringing liquor onto Native land. Under the law of 1895, though, anyone who sold or gave alcohol in Indian Territory could be charged, Native people included.

Native leaders' own rules. Native religious leaders made their own rules about drink and drugs. The Paiute religious leader Wovoka told followers of the Ghost Dance, a religious movement he led, not to drink. On the Comanche reservation in Oklahoma, Quanah Parker led ceremonies in which people took peyote, the small cactus that causes visions, as a sacred act of worship. He argued for the right to use peyote in this way.

A gold cure and a cigarette machine

Dr. Leslie Keeley opened a clinic for heavy drinkers in Dwight, Illinois, in 1879. He gave them shots of his medicine four times a day, and other patients got it by mail. He called it the "Gold Cure" and said the shots held "bichloride of gold," a chemical made with gold. By 1900 more than 300,000 people had taken it. Other doctors criticized it.

The cigarette machine. A skilled worker rolling by hand could make about four cigarettes a minute. In 1880 James Albert Bonsack of Virginia built a machine that rolled about 200 a minute. James Buchanan Duke signed a deal with Bonsack to use the machines, in 1884 or 1885 by different accounts. By 1890 Duke sold 40 percent of the cigarettes in the country. That year he joined his four biggest rivals with his own business to form the American Tobacco Company.

BY 1900

If you remember one thing: By 1900 states were banning liquor and cities were banning opium dens, while anyone could still buy opium, morphine and cocaine in a medicine bottle that did not have to say so.

- 1851: Maine lawmakers ban the sale of all alcoholic drinks. Four years later soldiers in Portland shoot dead a sailor, John Robbins, at the mayor's order.
- 1874: Women in Cleveland found the Woman's Christian Temperance Union.
- 1887: Members of Congress ban any "subject of the Emperor of China" from bringing opium into the country.

Next: In 1906 drug makers had to start listing what was in the bottle, and in 1920 a ban on alcohol covered the whole country.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 20 MINUTES

Drugs and alcohol came under federal control in the first half of the 1900s, first through labels and then through bans. From 1906, drug makers had to list opium, morphine and alcohol on their labels. From 1915, everyone who sold opium or cocaine had to register and keep records, and Treasury officials, who worked in the department that handles the government's money and taxes, took doctors to court for giving addicts their drug. In 1937 members of Congress taxed marijuana.

Prohibition. Under federal law, making, selling and carrying alcoholic drinks was banned anywhere in the United States from 1920 to 1933. This ban is called Prohibition. People kept drinking in hidden bars and made liquor at home, and federal agents made more than half a million arrests a year. Treasury officials had more poison added to factory alcohol. Bootleggers, people who made or sold liquor against the law, stole that alcohol and sold it as drink, and the poison killed thousands of drinkers.

Cigarettes. Americans smoked an average of 54 cigarettes a year each in 1900, and by 1963 the figure was 4,345.

Carry Nation

Who: A Kansas woman who smashed saloons to force officials to enforce the state's ban on liquor.

When and where: Kiowa, Wichita and Topeka, Kansas, 1900 to 1901, and Kansas City, Missouri, 1901. Arrested more than 30 times through 1910.

She walked into Dobson's Saloon carrying rocks.

Kansas voters had put a ban on liquor into their state constitution in 1880, and it took effect in 1881, but saloons stayed open.

Her name. She was born Caroline Amelia Moore in Garrard County, Kentucky, in 1846. Official records spell her name Carrie. Later she spelled it Carry, and she said her name meant "Carry A Nation for Prohibition."

Two husbands. In 1867 she married Charles Gloyd, a doctor who drank heavily. They separated before their daughter, Charlien, was born in 1868, and Gloyd died of alcoholism, the illness of being addicted to alcohol, in 1869. In 1874 she married David A. Nation, a lawyer and minister. In Medicine Lodge, Kansas, she started a branch of the Woman's Christian Temperance Union and pressed officials to enforce the ban. She kept at it for about five years.

Kiowa. On June 5, 1900, she wrote later, she heard a voice telling her, "GO TO KIOWA." Two days later she walked into Dobson's Saloon in Kiowa, Kansas, carrying rocks. She said, "Men, I have come to save you from a drunkard's fate," and smashed the saloon's stock of liquor. She smashed two more saloons in Kiowa.

The hatchet. She wrecked a saloon in Wichita in January 1901 and spent two weeks in jail. Her husband joked that she should use a hatchet next time. "That is the most sensible thing you have said since I married you," she answered. In February 1901, in Topeka, she met the governor and spoke to Kansas lawmakers.

> She was arrested more than 30 times between 1900 and 1910. She paid her fines with money from her speeches and by selling small pins shaped like hatchets. In 1901 she opened a home in Kansas City, Missouri, for the wives and children of men who drank. Kansas officials did not really enforce the ban until 1907. She died on June 9, 1911.

Why the label had to say morphine

The reporter Samuel Hopkins Adams wrote 11 articles about patent medicines, the brand-name medicines whose makers did not have to say what was in them, for *Collier's* magazine in 1905. He wrote that Americans would spend about $75 million that year on medicines that held large amounts of alcohol and opium.

By official tests, Hostetter's Bitters was 44 percent alcohol. One bottle of a medicine called Peruna, he wrote, held as much alcohol as five bottles of beer. He called the opium syrups sold for babies medicines "which stunt or kill helpless infants."

The 1906 law. President Theodore Roosevelt signed the Pure Food and Drug Act on June 30, 1906. Under it, drug makers had to print on the label whether a medicine held any of 10 drugs. They included alcohol, morphine, opium and cannabis, the plant marijuana comes from. The law did not ban any of these drugs. Chemists led by Harvey W. Wiley, at the Department of Agriculture, enforced it.

What changed. After the law passed, the makers of Mrs. Winslow's Soothing Syrup took out the morphine. The historian David T. Courtwright writes that after 1900 fewer people became addicted through medicine. Doctors had grown more careful, and after the 1906 law and the articles of reporters like Adams, more people were wary of dosing themselves.

The Harrison Act, 1914: doctors and addicts

Members of Congress passed the Harrison Narcotics Act in 1914. Under it, every seller of opium and cocaine had to register with the government and keep records.

The claims. In the years before the law passed, some officials and newspaper writers blamed drug use on Black and Chinese people. A citizens' panel called the Committee on the Acquirement of the Drug Habit wrote in 1903 that if a Chinese man "cannot get along without his dope we can get along without him." In place of "a Chinese man," the panel used an insulting name for Chinese people.

Hamilton Wright. Theodore Roosevelt named Dr. Hamilton Wright the first United States opium commissioner, the official in charge of the country's work against opium, in 1908. Wright told members of Congress that cocaine was often "the direct incentive to the crime of rape of white women by Negroes." He was claiming that cocaine drove Black men (Negroes, in the word of the time) to rape white women, that is, to force them to have sex.

A *New York Times* headline of 1914 read "Negro Cocaine 'Fiends' Are a New Southern Menace."

The dispute. Historians disagree about how much these claims shaped the drug laws. David T. Courtwright writes that cocaine became linked in the public mind with crime sprees, especially by Black men. George Fisher of Stanford Law School points out that the first state cocaine bans, from 1887 to 1898, came before these newspaper stories. He argues that racial hatred came later, in the way police and other officers enforced the laws.

The law. Representative Francis Burton Harrison of New York proposed the law, and President Woodrow Wilson signed it on December 17, 1914. It took effect on March 1, 1915. In the law, narcotics meant drugs made from opium or from coca, the plant that cocaine comes from. Doctors and druggists who handed out these drugs had to register too, and every seller had to pay a small tax.

Doctors in court. Treasury officials read the law to mean that a doctor could not give an addict regular doses of a drug, and they took doctors who did to court. The justices of the Supreme Court agreed with them on March 3, 1919, in the case of *Webb v. United States*. A doctor could not write a prescription, a doctor's written order for a drug, for an addict just to keep the addict comfortable. Dr. Webb had written more than 80 morphine prescriptions a week, at 50 cents each.

The clinics. City officials then opened 35 public clinics that sold morphine cheaply to addicts who registered. Treasury officials closed every one of them. The last, in Shreveport, Louisiana, closed on February 10, 1923. After that, an addict could get drugs only from dealers, or go to prison.

Heroin. Heroin is a drug made from morphine. Young men in the poor immigrant neighborhoods of cities began sniffing small packets of heroin powder in the 1910s and early 1920s. President Calvin Coolidge signed a law on June 7, 1924, that banned bringing opium into the country to make heroin. Representative Stephen G. Porter of Pennsylvania had introduced it.

The narcotic farm. Officials of the Public Health Service, the federal health agency, opened a hospital for addicts in Lexington, Kentucky, in 1935. It was called a "narcotic farm," and prisoners could be sent there. One doctor remembered it as "more like a prison than a hospital and more like a hospital than a prison." Courtwright writes that several studies found that 90 percent or more of the people released from Lexington soon went back to drugs.

Cigarettes become an everyday habit

Americans smoked an average of 54 cigarettes a year each in 1900, by the count of the Centers for Disease Control and Prevention. By 1963 the figure was 4,345. The justices of the Supreme Court ordered James B. Duke's American Tobacco Company broken into four smaller companies in 1911.

"Tobacco as much as bullets." General John Pershing led the American army sent to Europe in World War I. Asked what America needed to win the war, he answered, "Tobacco as much as bullets." People in community groups, businesses and relief groups such as the Red Cross, the Salvation Army and the Young Men's Christian Association set up "tobacco funds" to send cigarettes to soldiers. Editors at the *New York Sun* asked readers to give as a way of "proving your patriotism."

The first cancer studies. The scientist Ernst Wynder and others began studies in the 1940s that linked smoking cigarettes to lung cancer.

Prohibition: the law and the people who wrote it

Wayne Wheeler of the Anti-Saloon League played a major part in winning a national ban on liquor. Wheeler and other prohibitionists gained support from the fight to win the vote for women. Frances Willard of the WCTU had joined the two causes in 1879, when she asked for votes for women as a way to protect their homes from drink.

The war. In World War I, League leaders also used Americans' anger at Germany to win support. German Americans had been a major force against a ban, and after members of Congress declared war on Germany in April 1917, their protests were ignored.

The amendment. Members of Congress approved the Eighteenth Amendment to the Constitution on December 18, 1917. An amendment is a change added to the Constitution. Lawmakers in the 36th state voted for it on January 16, 1919, which made it part of the Constitution. Only the lawmakers of Connecticut and Rhode Island never voted for it.

What it banned. The amendment banned making, selling and carrying alcoholic drinks. It did not ban owning them or drinking them. Wheeler helped draft the Volstead Act, the law that set out the rules for the ban. It is named for Andrew Volstead of Minnesota, who led the House committee in charge of laws about courts and crimes. President Wilson refused to sign it and sent it back, and members of Congress passed it anyway on October 28, 1919. The ban began on January 17, 1920.

The gaps. A family could make up to 200 gallons of wine and cider a year at home. Doctors could prescribe whiskey as medicine, and they wrote about 11 million prescriptions for it a year through the 1920s. Officials caught 1,000 doctors and 400 druggists selling signed prescription forms to bootleggers in 1931. Twelve of the doctors and 13 of the druggists were charged with crimes.

Speakeasies. People drank in speakeasies, hidden bars that sold liquor against the law. By the end of the 1920s New York City alone had about 30,000 of them, by the count of the writer Deborah Blum. Officials in some states, among them Maryland and New York, refused to enforce the federal ban.

Izzy Einstein

Who: A Jewish immigrant and former postal clerk who became a federal Prohibition agent. He arrested people while wearing disguises.

When and where: New York City, 1920 to 1925.

Movie: *Izzy and Moe* (1985), a television film directed by Jackie Cooper, with Jackie Gleason as Izzy Einstein and Art Carney as Moe Smith. It is about the two agents.

One Sunday he and his partner made 71 raids in 12 hours.

Eight languages. Isidor "Izzy" Einstein was born in 1880 to a Jewish family in Austria-Hungary. He came to the United States around 1901, speaking Yiddish, Hungarian, Polish and German, and some Russian, French, Spanish and Italian. He worked as a salesman and then as a postal clerk, and he struggled to support his family on a clerk's pay.

More than 100 disguises. With no police training, he joined the new federal Prohibition service in 1920. He brought in his friend Moe Smith, who owned a cigar store. The two men said they used more than 100 disguises, among them a streetcar conductor, a gravedigger, a fisherman, an iceman and an opera singer. They usually carried no guns. Sometimes Einstein told a doorman that he was a Prohibition agent. The doorman laughed and let him in.

The count. One Sunday, with reporters following them, they made 71 raids in a little more than 12 hours. Together they made 4,932 arrests by one count, and more than 5,000 by the count of the National Archives. Of the people they arrested, 95 percent were found guilty. Thousands of bartenders, bootleggers and speakeasy owners went to jail because of their work.

Fired. Their boss, General Lincoln C. Andrews, fired them along with 36 other men in November 1925, when he reorganized the New York office. Writers at *Time* magazine suggested that their fame in the newspapers

had annoyed their bosses. Both men went on to sell insurance. In 1932 Einstein published a book, *Prohibition Agent No. 1*. He died on February 17, 1938.

Prohibition agents: who enforced the ban

The federal government had 1,450 front-line Prohibition agents in 1930. From 1921 to 1930 the agents made more than half a million arrests a year and seized more than 45,000 cars, by the count of the historian Richard F. Hamm.

Political jobs. Politicians handed out many of the agents' jobs. Judge William S. Kenyon said in 1927 that "three-fourths of the 2,500 dry agents are ward heelers and sycophants named by the politicians." He meant that most agents were political helpers and flatterers. When agents had to pass a civil service test, the kind of test used to give out many government jobs, 60 percent failed it.

Fired for bribes. In the six years from 1920, 752 Prohibition officials were fired, most for drunkenness or for taking bribes. Mabel Walker Willebrandt was the Assistant Attorney General, a top lawyer in the Justice Department, in charge of Prohibition cases from 1921 to 1929. She said the agents were "as devoid of honesty and integrity" as the lawbreakers, meaning they were just as dishonest.

The dead. By their own count, Prohibition agents killed 89 people. The Association Against the Prohibition Amendment, a group working to end the ban, counted about 1,000 people killed in enforcing it. Near the Capitol in Washington in 1924, an agent shooting at a bootlegger's car hit Senator Frank L. Greene of Vermont in the head. Greene never got back the full use of one arm.

The writer Daniel Okrent describes the first agent to kill a suspected bootlegger. That agent had joined under a false name. He had killed a man at the age of 14, and he got his badge while he was still locked up in Dannemora State Prison.

Poisoned alcohol and jake leg

Prohibition, the national ban on alcohol, lasted from 1920 to 1933. In 1926 and 1927, Treasury Department officials under President Calvin Coolidge ordered more poison put into factory alcohol, and the poison killed thousands of people who drank it.

Factory alcohol, used for fuel and in factories, had chemicals added to it so that no one could drink it. This is called denatured alcohol. Bootleggers, people who made or sold liquor against the law, stole it, paid chemists to take the chemicals out and sold it as liquor.

More poison. The officials then ordered makers to add far more poison. By 1927 the mixes included kerosene, gasoline, benzene and mercury salts, and up to 10 percent methyl alcohol, also called wood alcohol, which blinds and kills people who drink it. *Time* magazine reported in January 1927 that three ordinary drinks of the new mix could cause blindness. The records do not name the official who signed the order.

Christmas 1926. More than 60 people had come to Bellevue Hospital in New York sick from poisoned liquor by dusk on Christmas Day 1926, and eight of them had died. In the next two days, 23 more people died in the city.

KNOWN:	the poisoned alcohol killed thousands of Americans.
DISPUTED:	how many. Deborah Blum gives 400 dead in New York City in 1926 and 700 in 1927, and another count gives 585 in the city in 1926. For the whole country, one estimate gives more than 5,000 dead, and Blum gives an estimate of at least 10,000 by 1933.

What officials said. Charles Norris was New York City's chief medical examiner, the doctor in charge of finding out why people in the city died. He said, "The government knows it is not stopping drinking by putting poison in alcohol." The government, he said, "must be charged with the moral responsibility for the deaths that poisoned liquor causes, although it cannot be held legally responsible."

Seymour Lowman, the Treasury official in charge of Prohibition from 1927 as Assistant Secretary of the Treasury, said drinkers were "dying off fast from poison 'hooch.'" If the result was a sober America, he said, "a good job will have been done." Wayne Wheeler said that a person who drank this alcohol "is a deliberate suicide." Senator Edward I. Edwards of New Jersey called the program "legalized murder."

Jake leg. Tens of thousands of people were poisoned in 1930 after they drank Jamaica ginger, a medicine that was 70 to 80 percent alcohol. Under Prohibition one form of it could still be bought without a doctor's order, so people drank it to get alcohol.

The poison came from Harry Gross, the president of Hub Products Corporation in Boston, who wanted a cheaper ingredient for his Jamaica ginger. He chose a chemical called TOCP, used in making lacquer, a shiny coating for wood and metal. People at the Celluloid Corporation, which made TOCP, told him it was not poisonous.

TOCP destroys nerve cells in the spinal cord, the bundle of nerves inside the backbone that carries signals from the brain to the body. People who drank Gross's Jamaica ginger lost the use of their hands and feet. They walked with their toes flopping down. People called the illness "jake leg" and the walk the "jake walk." For most of them the damage never healed.

NOT KNOWN: the exact number harmed. The figure most often given is between 30,000 and 50,000. Most were poor, and many were immigrants.

Gross's sentence. A judge fined Gross and his partner Max Reisman $1,000 each. The judge also gave each of them a two-year jail sentence that they did not have to serve. Blues singers recorded songs about jake leg in 1930.

Alcoholics Anonymous, 1935

Two heavy drinkers named Bill Wilson and Dr. Bob Smith met in Akron, Ohio, in 1935, through a Christian group called the Oxford Group. Together they started Alcoholics Anonymous, or AA (anonymous means without names), where drinkers help each other stop drinking. At meetings, members share their own stories of drinking and of getting sober.

AA members published the group's Twelve Steps, a list of steps for staying sober, in 1939, in a book that members call the Big Book. Wilson and Smith had never heard of the Washingtonians, the Baltimore drinkers of 1840 who also helped each other stay sober by telling their stories.

Did drinking fall under Prohibition?

Drinking fell under Prohibition, and then much of it came back. The economists Miron and Zwiebel found in a 1991 study that drinking fell at first to about 30 percent of what it had been before the ban. Over the next several years it rose again, to about 60 to 70 percent. Deaths from cirrhosis, damage to the liver from heavy drinking, fell by nearly two-thirds. Blum writes that insurance companies charted a rise in alcoholism of more than 300 percent in the 1920s.

KNOWN:	drinking fell sharply when the ban began.
NOT KNOWN:	how much drinking changed over the whole ban, because the counts point different ways.

Pauline Sabin. Pauline Sabin, a leader of the Republican Party in New York, first backed the ban. "I felt I should approve of it because it would help my two sons," she said. She changed her mind. By 1929 she said that "children are growing up with a total lack of respect for the Constitution and for the law." In May 1929, in Chicago, she founded the Women's Organization for National Prohibition Reform to end the ban.

Repeal. President Franklin Roosevelt signed a law on March 22, 1933, that allowed beer with 3.2 percent alcohol by weight. Delegates in Utah made it the 36th state to approve the Twenty-first Amendment, which ended national Prohibition, on December 5, 1933. After that, lawmakers in each state and town made their own rules about when, where and to whom alcohol could be sold. Liquor stayed banned in Kansas until 1948.

Marijuana and the 1937 tax

Marijuana is made from the dried leaves and flowers of the cannabis plant. Harry J. Anslinger ran the Federal Bureau of Narcotics, the federal drug police, from 1930 to 1962. "The best cure for addiction? Never let it happen," he said.

Anslinger's stories. His articles against marijuana in the 1930s often told stories that blamed the drug for crimes by Black and Mexican people. One story in his files began, "Two Negroes took a girl fourteen years old..." The historian Courtwright writes that Anslinger played on Americans' fear of immigrants and of other races. Anslinger told the story of Victor Licata, who killed his family with an axe in Florida in 1933, as a marijuana crime. Doctors had found that Licata had a severe mental illness when he was young.

The tax. Anslinger drafted the Marihuana Tax Act, and doctors of the American Medical Association objected to it. President Franklin Roosevelt signed it on August 2, 1937, and it took effect on October 1. Selling marijuana without a federal tax stamp, which cost $1 an ounce, became a crime.

The first arrests. Federal narcotics agents and Denver police arrested Samuel R. Caldwell, a 57-year-old laborer, for selling marijuana on October 2, 1937. They arrested Moses Baca, 26, for having it. The tax stamps had not yet been printed. Judge J. Foster Symes sentenced Caldwell to four years of hard labor at Leavenworth prison and a $1,000 fine, and Baca to 18 months. "Under its influence men become beasts," Symes said of marijuana. Caldwell was released in 1940 and died in 1941.

A study. Mayor Fiorello La Guardia of New York named a committee to study marijuana in 1938. In their report of 1944, the committee's members disputed Anslinger's claims.

Liquor laws and Native nations

Federal bans on liquor on Native land stayed in force through all of these years. Members of Congress made it a crime in 1910 to bring alcohol onto the land of the Pueblo nations of New Mexico, as a condition of New Mexico becoming a state.

That same year Attorney General George Wickersham, the nation's top lawyer, asked for lighter punishments for selling, giving or bringing liquor to Native people. He wanted these to be lesser crimes, which his lawyers could charge without a grand jury, a group of citizens who decide whether there is enough proof for a trial. The punishments stayed as they were.

The Native American Church was organized in Oklahoma in 1918. Its members take peyote as a sacred act of worship.

BY 1950

If you remember one thing: In these fifty years federal lawmakers took charge of drugs and alcohol, with labels in 1906, a drug law in 1914, a national ban on alcohol from 1920 to 1933 and a tax on marijuana in 1937.

One number: Americans smoked 54 cigarettes a year each in 1900 and 4,345 in 1963, about 80 times as many.

Next: After 1950 scientists showed that cigarettes cause cancer, and a president declared a war on drugs.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 15 MINUTES

Americans learned in these fifty years that smoking kills, and the federal fight against drugs grew from $81 million a year in 1969 to $18.4 billion in 2000. About 42 of every 100 American adults smoked cigarettes in 1965, and by 1997 the number was about 25. In 1996 a fired tobacco scientist told a national television audience that his company changed its tobacco to strengthen the effect of nicotine. Drinking rose until 1981 and then fell.

The war on drugs. President Richard Nixon called drug abuse "public enemy number one" on June 17, 1971, and reporters named his program the war on drugs. Candy Lightner started Mothers Against Drunk Driving in 1980, after a drunk driver killed her daughter, and by 1988 lawmakers in every state had set the drinking age at 21.

Under a 1986 law passed by members of Congress and signed by President Ronald Reagan, 5 grams of crack, a kind of cocaine that can be smoked, carried the same prison term as 500 grams of powder cocaine. A company owned by the Sackler family began selling a strong painkiller called OxyContin in 1996.

How much people drank and smoked

Americans aged 15 and over drank an average of 2.04 gallons of pure alcohol each in 1950. From 1970 the count covers people aged 14 and over, and by that count the figure rose to 2.76 gallons in 1981, the highest since Prohibition. By 1995 it had fallen to 2.15 gallons.

Scientists at the National Institute on Alcohol Abuse and Alcoholism work these figures out from sales of alcohol, not from asking people what they drank. Between 1974 and 1981, 68 to 71 of every 100 adults told the Gallup company, which asks adults questions about what they think and do, that they drank.

Cigarettes. Cigarette smoking peaked in 1963, at 4,345 cigarettes a year for each person. By 1998 the figure was 2,261. In the late 1990s the Centers for Disease Control and Prevention (CDC) counted about 430,000 deaths a year from tobacco, one of every five deaths in the country.

The Surgeon General's report. The studies that Ernst Wynder and others began in the 1940s went on into the 1950s. In 1964 a committee of experts working for the Surgeon General, the nation's top public health doctor, concluded that smoking causes lung cancer and cancer of the voice box. Nixon signed a law on April 1, 1970, that banned cigarette ads on radio and television from January 2, 1971.

Billie Holiday and the narcotics agents

The singer Billie Holiday was arrested for drugs three times. Federal narcotics agents made the last two arrests, the last one in her hospital bed.

New York, 1947. Holiday was arrested on May 16, 1947, for having narcotics in her New York apartment. In court she said, "It was called 'The United States of America versus Billie Holiday'. And that's just the way it felt." She pleaded guilty, and a judge sent her to Alderson Federal Prison Camp in West Virginia.

After she was found guilty, she lost the police license she needed to sing in New York City clubs that sold alcohol. The federal agent George Hunter White arrested her again in a San Francisco hotel on January 22, 1949.

The hospital bed. Holiday was in Metropolitan Hospital in New York in 1959, being treated for liver and heart disease. On May 31, Harry Anslinger ordered agents of his Federal Bureau of Narcotics to arrest her there for having drugs. The agents handcuffed her to her hospital bed. A police guard stayed at her room until a judge ordered the handcuffs taken off.

She died in that room on July 17, 1959, at the age of 44. The cause was heart failure and fluid in her lungs, which came from cirrhosis, scarring of the liver. She had 70 cents in the bank.

DISPUTED: why the agents went after her. The writer Johann Hari has argued that Anslinger's agents targeted Holiday from 1939 because of her song "Strange Fruit," which is about lynching, killing by a mob. The jazz writer Lewis Porter disputes this. He points out that the claim first appears in Hari's own book, in 2015.

Native nations and the end of the federal ban, 1953

Members of Congress changed the federal liquor laws for Native people on August 15, 1953. Since 1834 it had been a federal crime to bring liquor onto Native land. Under later laws it was also a crime to sell or give alcohol to many Native people even off that land. After the 1953 change, Native people could buy alcohol off their land like anyone else.

The law's title states its purpose: "to eliminate certain discriminatory legislation against Indians," meaning laws that treated Native people worse than others. The federal ban on liquor on Native land had lasted 20 years longer than national Prohibition.

Each nation's choice. On a nation's own land, a sale of alcohol now had to follow both state law and a law passed by that nation's government, approved by the Secretary of the Interior, the head of the federal department that handles dealings with Native nations.

Members of Congress gave each nation the choice of keeping its ban, and some nations kept it. The Oglala Sioux Tribe, on the Pine Ridge Reservation in South Dakota, was one, according to the National Alcohol Beverage Control Association, a group of state alcohol officials.

LSD and marijuana

LSD is a drug that makes people see and hear things that are not there, and in the early 1960s it was legal. Timothy Leary, a teacher at Harvard University who studied how the mind works, ran experiments from 1960 in which he gave people psilocybin, a drug from certain mushrooms that causes visions. He also used LSD.

Harvard officials stopped paying him on April 30, 1963, and Harvard's president, Nathan Pusey, said he had "failed to keep his classroom appointments." Nixon later called Leary "the most dangerous man in America."

Marijuana. A man named Lowell Eggemeier lit a marijuana cigarette in San Francisco's Hall of Justice in 1964 and asked to be arrested. The writer Martin A. Lee dates the start of the modern campaign to make marijuana legal to that act. Oregon lawmakers made having a small amount of marijuana punishable by a fine instead of jail in 1973. In November 1996, California voters approved Proposition 215, 56 percent to 44 percent, and California became the first state to allow marijuana as a medicine.

Nixon declares the war on drugs, 1970 and 1971

Nixon signed a law on October 27, 1970, that sorted drugs into five lists, called schedules. The lawmakers put marijuana and heroin on Schedule I, the list for drugs with a high risk of abuse and no accepted use as medicine. Under the same law, having a small amount of a drug became a lesser crime, with at most one year in jail for a first offense. Selling drugs carried harsher punishments than before.

Heroin in Vietnam. Two members of Congress, Morgan Murphy and Robert Steele, came back from the war in Vietnam in May 1971. They reported that "as many as 10 to 15 percent of our servicemen are addicted to heroin."

"Public enemy number one." Nixon spoke at the White House on June 17, 1971. "America's public enemy number one in the United States is drug abuse," he said. He asked for $155 million in new money, for a total of "over $350 million" a year to fight drugs. When he took office in 1969, the federal drug budget had been $81 million. While Nixon was president, about 70 of every 100 federal dollars spent against drugs went to treatment and prevention.

Arrests. Agents of Nixon's new Office of Drug Abuse Law Enforcement made 6,000 drug arrests in 18 months, from 1972 to 1973, and most of the people arrested were Black. In 1973 Governor Nelson Rockefeller of New York signed laws that set 15 years to life in prison for selling two ounces, or holding four ounces, of heroin, cocaine or marijuana.

The Ehrlichman quote. The journalist Dan Baum published notes in 2016 of a 1994 interview with John Ehrlichman, one of Nixon's top advisers. In Baum's notes, Ehrlichman said that "by getting the public to associate the hippies with marijuana and blacks with heroin," and then making both a crime and punishing them heavily ("criminalizing both heavily"), "we could disrupt those communities." Ehrlichman had died in 1999. His children and other officials who had worked for Nixon disputed the quote.

Betty Ford

Who: The wife of President Gerald Ford. She told the country she was addicted to pills and alcohol, and then she founded a treatment center.
When and where: Long Beach and Rancho Mirage, California, 1978 to 1982.

At times she took as many as 20 pills a day.

The pills. A pinched nerve in Betty Ford's neck put her in the hospital for two weeks in 1964. Doctors gave her drugs for the pain, including strong painkillers and Valium, a drug that calms people. At times she took as many as 20 pills a day. She also drank. Her father and her brother Bob had both been addicted to alcohol.

The intervention. Her family held an intervention, a meeting to ask her to go into treatment, on April 1, 1978, after the Fords had left the White House. On April 11 she checked into the Naval hospital in Long Beach, California. She had arthritis, a disease that makes the joints swell and hurt, and a family spokesman read her statement: "I have found I am not only addicted to the medication I have been taking for my arthritis, but also to alcohol."

She later wrote, "I liked alcohol, it made me feel warm. And I loved pills. They took away my tension and my pain."

The center. She and Leonard Firestone, a former ambassador, founded the Betty Ford Center in Rancho Mirage, California, in 1982, for people recovering from addiction to alcohol and other drugs. It opened on October 4, 1982. She died on July 8, 2011.

Why the drinking age is 21

Clarence Busch was drunk when he hit and killed 13-year-old Cari Lightner with his car on May 3, 1980. She was walking to a church festival in Fair Oaks, California. He drove away and left her body there. He had a long record of arrests for drunk driving.

The sentence. A judge ruled that Busch's alcoholism had made it harder for him to control his decisions. The judge gave him two years in prison for involuntary manslaughter, the crime of killing someone without meaning to. He entered prison in December 1980 and was paroled nine months later, which means he was let out early under watch.

He went back to prison in 1982 for breaking the rules of his parole, and he was freed again in February 1983. Police arrested him again in April 1985, drunk and driving without a license, after a crash.

MADD. Cari's mother, Candy Lightner, founded Mothers Against Drunk Driving, or MADD, in 1980. By MADD's count, nearly 25,000 people were killed in crashes involving alcohol in 1980. That was half of all traffic deaths.

The drinking age. Lawmakers in 29 states had lowered the drinking age to 18, 19 or 20 between 1970 and 1975, after the voting age was lowered to 18. Senator Frank Lautenberg of New Jersey wrote a law to raise the drinking age again, and Reagan signed it on July 17, 1984. Under it, a state that let people under 21 buy alcohol lost part of its federal money for highways. By 1988 every state had set the age at 21.

DISPUTED: how many lives the higher drinking age saved. A review for the Institute of Medicine, a national group of health experts, found strong evidence that it worked. The economists Jeffrey Miron and Elina Tetelbaum found in 2009 that the effect faded after a year or two.

Crack cocaine and the 1986 law

Nancy Reagan, the president's wife, began a campaign in the early 1980s, with help from an advertising company, in which she told young people to "Just Say No" to drugs. Later studies found little or no change in how many young people used drugs after the campaign. The FBI's budget for fighting drugs rose from $8 million in 1980 to $95 million in 1984.

Crack. Crack is cocaine that dealers cooked with water and baking soda and dried into small rocks that can be smoked. Smoked crack reaches the brain in 6 to 8 seconds, and the high lasts 5 to 15 minutes. Around 1984 a dose cost as little as $2.50 in some cities. Hospital emergencies from cocaine more than doubled in one year, from 26,300 in 1985 to 55,200 in 1986.

The 1986 law. The basketball player Len Bias died from cocaine on June 19, 1986. Afterward, Speaker of the House Tip O'Neill had Democratic leaders write a new drug law, and Reagan signed it on October 27, 1986. In it, members of Congress added $1.7 billion to the fight against drugs and set 29 new minimum prison terms that judges had to give. The law set five years in federal prison for 5 grams of crack, the same as for 500 grams of powder cocaine.

Who went to prison. Most of the people sentenced for crack were Black. In 1992, 91 of every 100 people sentenced in federal court for crack were Black and about 3 were white, by the count of the US Sentencing Commission.

The people treated for crack were a more mixed group. Of those who entered treatment for crack in 2004, 53 of every 100 were Black and 38 were white. The historian Elizabeth Hinton writes that Reagan "led Congress in criminalizing drug users, especially African American drug users."

The bag on television. President George H. W. Bush held up a bag of crack on television on September 5, 1989. He said it had been "seized a few days ago in a park across the street from the White House." It came out later that agents of the Drug Enforcement Administration had been asked to make an arrest there, and that they had to lure the seller to the park to do it. So the sale across from the White House was one the agents had set up.

The cost. By 2000 the federal government spent $18.4 billion a year against drugs, more than 200 times the $81 million of 1969. Nearly half went to law enforcement, the police and agents who catch lawbreakers, and one-sixth went to treatment.

Jeffrey Wigand

Who: A scientist who ran research at a tobacco company and then told the public that the company changed its tobacco to strengthen the effect of nicotine.

When and where: Brown & Williamson Tobacco, Louisville, Kentucky, 1989 to 1996.

Movie: *The Insider* (1999), directed by Michael Mann, with Russell Crowe as Jeffrey Wigand. It is about Wigand.

He said his company added ammonia to boost nicotine's effect.

Fired. Jeffrey Wigand trained as a biochemist, a scientist who studies the chemistry of living things. He became vice president for research at Brown & Williamson Tobacco in Louisville, Kentucky, in January 1989. Brown & Williamson fired him on March 24, 1993. That September the company sued him and cut off his health insurance. In November 1993 he signed an agreement not to talk about his work.

Testimony. Under a judge's order, Wigand gave sworn testimony in Pascagoula, Mississippi, in late November 1995, for the state's lawsuit against the tobacco companies. The company's lawyers got a Kentucky court order against him and sued him for "theft, fraud, breach of contract," which means they said he stole, cheated and broke his agreement with the company. Lawyers for the CBS television network stopped the news program *60 Minutes* from showing an interview with him that November.

On the air. *60 Minutes* showed the interview on February 4, 1996. Wigand said the company added chemicals such as ammonia to its tobacco to increase the effect of nicotine. He said he had been harassed and had received death threats from people who did not give their names.

> The company hired investigators, who wrote a 500-page report attacking him and gave it to reporters. The reporters checked it and found that many of its charges were unproven or small.
>
> After he was fired, Wigand taught science and Japanese at a high school in Louisville. He was named Kentucky's Teacher of the Year for 1996.

OxyContin goes on sale, 1996

Opioids are strong painkillers made from the opium poppy, or from chemicals that act the same way in the body. Too much of an opioid slows a person's breathing until it stops.

The Sacklers. Three doctors, the brothers Arthur, Mortimer and Raymond Sackler, bought a drug company called Purdue Frederick in 1952. Their company, later named Purdue Pharma, began selling OxyContin in 1996. OxyContin is a pill of oxycodone, an opioid, made to release the drug slowly. Purdue sold it to be taken every 12 hours, but most patients found that it wore off after eight hours or less.

The approval. Dr. Curtis Wright, of the Food and Drug Administration (FDA), reviewed OxyContin before it went on sale. He approved a label that said the slow release "is believed to reduce the abuse liability of a drug," meaning the risk that people would misuse it. A year later Wright left the FDA and went to work for Purdue.

The sales push. Purdue's managers trained its salespeople to tell doctors that the risk of addiction was "less than one percent." More than 5,000 doctors, pharmacists and nurses went to Purdue meetings with all their costs paid. Richard Sackler, Raymond's son, ran the company's sales and marketing, and he became its president in 1999.

Prescriptions of OxyContin for pain not caused by cancer rose from about 670,000 in 1997 to about 6.2 million in 2002. From 1995 to 2001, Purdue took in $2.8 billion from sales of OxyContin. David Kessler, who led the FDA when the pill was approved, later called the approval "one of the worst medical mistakes."

BY 2000

If you remember one thing: Federal spending against drugs grew from $81 million a year in 1969 to $18.4 billion in 2000.

- 1964: A committee of experts working for the Surgeon General reports that smoking causes lung cancer.
- 1986: A new law gives a person caught with crack the same prison term as one caught with 100 times as much powder cocaine.
- 1996: The Sackler family's company starts selling the painkiller OxyContin.

Next: After 2000, deaths from opioids, starting with pills such as OxyContin, rose for more than twenty years.

2000 to Today

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Opioids, first pain pills such as OxyContin, then heroin, then fentanyl, killed about 806,000 Americans from 1999 to 2023, by the count of the Centers for Disease Control and Prevention (CDC). An overdose is too much of a drug at once, enough to make a person very sick or kill them.

Most of the deaths from 2013 on came from fentanyl made outside the law. Overdoses of all drugs killed 107,941 people in 2022, the peak year in the CDC's final figures. By the CDC's early count, the yearly total had fallen to about 67,000 in the 12 months ending in April 2026.

The rest of the story. A federal judge sentenced Purdue Pharma, the maker of OxyContin, in 2026, and the company shut down. No member of the Sackler family, who owned it, has been charged with a crime. Over the same years, voters and lawmakers in 24 states made marijuana legal for adults. In July 2026, 54 of every 100 adults told Gallup pollsters that they drink, the lowest share since Gallup began asking in 1939. About 9 of every 100 adults smoked cigarettes in 2025, by the CDC's count.

Overdose deaths, 1999 to 2026

CDC scientists describe the opioid deaths as coming in three waves. The first wave began in 1999, after doctors wrote more and more prescriptions for opioid pills in the 1990s. Deaths from heroin rose from 2010. From 2013, deaths rose from opioids made in labs, mostly fentanyl made and sold outside the law.

The death rate. About 6 of every 100,000 Americans died of a drug overdose in 1999, by the CDC's count. The rate rose to about 33 of every 100,000 in 2022 and fell to about 23 in 2024.

The counts. Overdose deaths from all drugs rose to a peak in 2022 and then fell. CDC scientists give two kinds of count. The final count for a year is published about a year later. The early count is published within months, and the scientists warn that it is often incomplete.

By the final count, overdoses killed 107,941 Americans in 2022, about 300 a day, and opioids were involved in about three of every four of those deaths. The final count for 2024 was 79,384. The early count for the 12 months ending in April 2026 is 66,937.

Who dies. Some groups die at far higher rates than others. American Indian and Alaska Native people died of overdoses at a rate of 51.6 of every 100,000 in 2024, by the CDC's count. The rate was 33.8 for Black Americans and 24.7 for white Americans. By age, people aged 35 to 44 died at the highest rate, 44.2 of every 100,000.

Fentanyl

Fentanyl is an opioid made in a lab. By the CDC's count, it is 50 to 100 times stronger than morphine. Too much of it slows a person's breathing until the breathing stops. A drug called naloxone can undo the overdose and save the person if someone gives it fast enough. Officials of the Food and Drug Administration (FDA) approved a naloxone spray for the nose that anyone can buy without a doctor's order on March 29, 2023.

The rise and fall. Deaths from fentanyl and similar lab-made opioids rose from 1 of every 100,000 Americans in 2013 to 22.7 in 2022, by the CDC's count. In 2024 the rate fell to 14.3.

Who makes it. Most of the illegal fentanyl sold in the United States has come from Mexico since about 2019. A 2025 report by the Congressional Research Service, which studies problems for members of Congress, names the makers. Mexican crime groups called cartels, especially the Sinaloa Cartel and the Jalisco New Generation Cartel, make it outside the law. Companies in China are the largest source of the chemicals they need. The cartels' members use machines to press fentanyl into fake pills made to look like real medicine.

Why no Sackler went to prison

No member of the Sackler family has gone to prison. When Judge Madeline Cox Arleo sentenced Purdue Pharma in 2026, she said she could not send its leaders or owners to prison because Justice Department lawyers had charged only the company.

The first guilty plea. Purdue Pharma pleaded guilty in May 2007 to misleading the public about OxyContin. The company admitted that it had sold OxyContin as less addictive and less likely to be abused than other painkillers. It paid $600 million. Three of its leaders also pleaded guilty: its president, Michael Friedman, its top lawyer, Howard Udell, and its former chief medical officer, Paul Goldenheim. Together they paid $34.5 million. In 2009 Purdue changed OxyContin into a form that was harder to crush and snort or inject.

Bankruptcy. Purdue filed for bankruptcy, a court process for a business that cannot pay its debts, on September 15, 2019. Purdue hired a firm called AlixPartners to check the company's money records. In December 2019 the firm reported that members of the Sackler family had taken $10.7 billion out of the company after lawsuits and investigations began.

The second guilty plea. Purdue pleaded guilty to three federal crimes on November 24, 2020. One was a plot to cheat the United States. The other two were plots to pay doctors and others to push its pills. In that case, no company leader and no Sackler was charged.

The settlement. Five justices of the Supreme Court threw out a bankruptcy plan on June 27, 2024, because it would have protected the Sacklers from lawsuits. A year later, on June 16, 2025, 55 attorneys general, the top lawyers of states, agreed to a $7.4 billion settlement with Purdue and the Sacklers, a deal that ended lawsuits in return for money.

The sentence. Judge Arleo, in Newark, New Jersey, sentenced Purdue to $5.5 billion in fines and penalties for its 2020 guilty plea on April 28, 2026. For nearly seven hours she heard from people harmed by the crisis. More than 40 spoke in court, and more than 200 sent letters.

Ed Bisch's son Eddie died of an overdose in 2001. "Punishment by a fine means 'legal for a price,'" Bisch said. Alexis Pleus said her son Jeff had first been prescribed OxyContin for a football injury, before he died of an overdose. "We still deserve justice, and this isn't it," she said. "Your government failed you," Arleo told the families.

Who gets the money. Most of the $5.5 billion in fines and penalties will never be paid. Federal officials collect $225 million of it. Purdue's other money goes to the states and towns it owes, as part of the $7.4 billion settlement. Purdue and the Sacklers pay the settlement over 15 years, to fund treatment, prevention and recovery. It includes $865 million for people harmed one by one. Many victims said they could not qualify because they could not find their old prescription records.

Purdue shuts down. New York Attorney General Letitia James announced three days after the sentence, on May 1, 2026, that Purdue had shut down. A new company, Knoa Pharma, owned by a charity, took over. As part of the $7.4 billion, the Sacklers must pay $2.9 billion by 2029, and none of them may sell opioids in the United States. The Sackler family has always said it did nothing wrong.

Other companies. The three largest companies that ship drugs to pharmacies, McKesson, Cardinal Health and AmerisourceBergen, agreed in 2021 to pay up to $21 billion over 18 years. The drug maker Johnson & Johnson agreed to pay up to $5 billion. In 2022 the drug makers Teva and Allergan and the pharmacy chains CVS, Walgreens and Walmart agreed to pay up to about $18.5 billion more among them. Under the settlements, states and towns must spend at least 85 of every 100 dollars they get on the overdose crisis.

Nan Goldin

Who: A photographer who became addicted to OxyContin after surgery, overdosed on fentanyl, went through treatment, and led protests against the Sackler family.

When and where: Berlin and New York, the 2010s to 2021.

Movie: *All the Beauty and the Bloodshed* (2022), a documentary by Laura Poitras about Goldin's life and her protests.

She threw OxyContin bottles into a museum's temple pool.

Addicted. Nan Goldin is a photographer known for her book of photographs *The Ballad of Sexual Dependency* (1986). A doctor in Berlin prescribed OxyContin for her after surgery on her wrist. "Though I took it as directed I got addicted overnight," she wrote in the magazine *Artforum* in January 2018. "I went from three pills a day, as prescribed, to eighteen."

The overdose. When her doctors stopped prescribing it, she bought pills from a dealer in New York. "When I ran out of money for Oxy I copped dope," she wrote, which means she bought illegal drugs from dealers. "I ended up snorting fentanyl and I overdosed." She waited a year before she went into treatment, because she was afraid of withdrawal, the sickness that comes when the body loses a drug it depends on. Then she spent two and a half months in treatment.

The protest. Goldin started a group called P.A.I.N., which stands for Prescription Addiction Intervention Now, in 2018. On March 10, 2018, she and other members protested in the Sackler Wing of the Metropolitan Museum of Art in New York. They threw pill bottles labeled OxyContin into the pool around the Temple of Dendur, an Egyptian temple about 2,000 years old that stands in the wing. Then they lay on the floor and chanted, "Sacklers lie, people die."

The names come down. Officials at the National Portrait Gallery in London turned down a £1 million gift from the Sacklers in 2019, after Goldin threatened to pull her show. Leaders of the Tate galleries in

> London said they would take no more Sackler gifts. Officials of the Metropolitan Museum and members of the Sackler families announced on December 9, 2021, that seven of the museum's spaces would no longer carry the Sackler name.

Marijuana becomes legal in 24 states

Voters in Colorado and Washington made marijuana legal for adults on November 6, 2012, the first two states to do so. Washington's law took effect first, on December 6, 2012. In 2018 Vermont's lawmakers became the first to make it legal without a vote of the public.

The count. By the count of the National Conference of State Legislatures, a group that serves state lawmakers, 24 states and the District of Columbia allow adults to use marijuana. The group's page was last updated in September 2026.

The 24 states are Alaska, Arizona, California, Colorado, Connecticut, Delaware, Illinois, Maine, Maryland, Massachusetts, Michigan, Minnesota, Missouri, Montana, Nevada, New Jersey, New Mexico, New York, Ohio, Oregon, Rhode Island, Vermont, Virginia and Washington. By the same count, 41 states allow marijuana as medicine. In November 2024, voters in Florida, North Dakota and South Dakota voted against making it legal.

Federal law. Under federal law, marijuana was still on Schedule I, the list for drugs with a high risk of abuse and no accepted use as medicine. President Donald Trump ordered officials on December 18, 2025, to speed up moving it to Schedule III, a list of drugs that doctors may order as medicine.

The Acting Attorney General moved marijuana sold under state medical licenses to Schedule III on April 28, 2026. All other marijuana stayed on Schedule I. In June 2026 officials of the Drug Enforcement Administration began hearings on moving the rest. As of late September 2026, they had not decided.

Crack sentences. Punishments for crack also changed. President Barack Obama signed the Fair Sentencing Act, written by Senator Dick Durbin, in 2010. Under it, the difference between crack and powder cocaine sentences dropped from 100 to 1 to 18 to 1. Under the First Step Act, which President Trump signed in 2018, the change also covered people already in prison. In its first year, more than 2,000 federal prisoners had their sentences shortened.

Vaping and smoking

An e-cigarette heats a liquid, usually one with nicotine in it, into a mist that the user breathes in. Using one is called vaping. About 27.5 of every 100 high school students vaped in 2019, by the count of a national survey released by the CDC.

The lung injuries. A lung injury from vaping put 2,807 people in the hospital or killed them in 2019 and early 2020, by the CDC's count. Of them, 68 died. CDC scientists found that vitamin E acetate, a chemical added to illegal marijuana vaping liquids, was strongly linked to the injuries. They linked most cases to those liquids, which most of the sick people had bought from friends, family or dealers. The records do not name who made them.

Age 21. President Trump signed a law on December 20, 2019, that raised the age to buy any tobacco product, including e-cigarettes, from 18 to 21. By 2024 the share of high school students who vaped had fallen to 7.8 of every 100, by the CDC's count. The newest survey, released by the FDA in June 2026, counted middle school and high school students together: 5.2 of every 100 of them vaped in 2025.

Cigarettes. About 9.1 of every 100 adults smoked cigarettes in 2025, by the CDC's count, against 42.4 in 1965. Smoking and secondhand smoke, the smoke other people breathe in, still kill more than 480,000 Americans a year, by the CDC's newest count, from 2024.

Drinking today

Americans aged 14 and over drank an average of 2.48 gallons of pure alcohol each in 2023, by the count of the National Institute on Alcohol Abuse and Alcoholism. That is below the peak of 2.76 gallons in 1981. Americans now drink more liquor and less beer than in 2000. Pure alcohol from beer fell from 1.23 gallons a person in 2000 to 0.99 in 2023, and pure alcohol from liquor rose from 0.64 gallons to 1.08.

Fewer drinkers. Gallup pollsters asked adults about drinking in July 2026, and 54 of every 100 said they drink, the lowest share since Gallup began asking in 1939. In 2023 the figure was 62. In the same 2026 poll, 51 of every 100 adults said that one or two drinks a day is bad for health.

The harm. The CDC counts about 178,000 deaths a year from drinking too much, from data for 2020 and 2021. On average, each of those people lost 24 years of life. Surgeon General Vivek Murthy reported on January 3, 2025, that alcohol causes about 100,000 cases of cancer and 20,000 cancer deaths a year in the United States. He asked Congress to put a cancer warning on alcohol labels.

Federal health officials dropped their old advice in January 2026, which had been to drink no more than one drink a day for women and two for men. The new advice says only, "Consume less alcohol for better health."

Young people and AA. About 41 of every 100 high school seniors said in 2025 that they had drunk alcohol in the past year, by the count of the Monitoring the Future survey. Alcoholics Anonymous estimates that it has more than two million members in more than 123,000 groups around the world.

THE THREE QUESTIONS, ANSWERED

1. **How long have people here used tobacco, and what did they use it for?** For at least 12,300 years, since the fire at the Wishbone site. Native people offered it in prayer, burning it or

smoking it in pipes, and the Algonquian people at Roanoke smoked it to keep the body healthy.

2. **What happened when the whole country banned alcohol from 1920 to 1933?** Drinking fell at first to about 30 percent of what it had been, then rose to about 60 to 70 percent. Treasury officials had more poison added to factory alcohol, and the poison killed thousands of drinkers before delegates in Utah ended the ban on December 5, 1933.
3. **Who has been punished for drugs, and who has not?** Robert Coles with his red D, the laborer Samuel Caldwell with four years of hard labor, Billie Holiday handcuffed to her hospital bed, and the people sentenced for crack, 91 of every 100 of them Black in 1992, were all punished. Harry Gross never served his jail sentence, and no member of the Sackler family has been charged with a crime.

What changed: Tobacco went from an offering in prayer to a habit that kills more than 480,000 Americans a year, and opium, cocaine and liquor went from open sale in the 1800s to bans in the 1900s, with liquor made legal again in 1933.

FIVE DATES TO KEEP

1634: Massachusetts judges order Robert Coles to wear a red D for a year.

February 13, 1826: Reformers in Boston found the American Temperance Society.

January 17, 1920: National Prohibition begins.

June 17, 1971: President Nixon calls drug abuse "public enemy number one."

1996: Purdue Pharma begins selling OxyContin.

NOT KNOWN: The number of people killed by poisoned alcohol during Prohibition is not known, and the estimates run from more than 5,000 to at least 10,000.

Where to go: The Temple of Dendur stands in the Metropolitan Museum of Art in New York, in the wing where Nan Goldin's group threw pill bottles into the pool in 2018 and that no longer carries the Sackler name.

Next: Chapter 29, Religion. It follows the many faiths of this land and the fight for the freedom to worship.

PART 8

Faith, Learning, and News

Chapter 29: Religion

CHAPTER 29 OF 37 · RELIGION · 10 ERAS · ABOUT 200 MINUTES

Freedom of worship in the United States grew one fight at a time, and most of those fights were started by people who were being punished for their faith.

A statue of a Pueblo man named Po'pay stands in the US Capitol in Washington. New Mexico gave it to the National Statuary Hall Collection in 2005. The sculptor, Cliff Fragua of Jemez Pueblo, carved scars on its back. In 1675 Spanish officials had Po'pay whipped in public for taking part in Pueblo ceremonies. Five years later he led the rising that drove the Spanish out of New Mexico for twelve years.

This chapter follows faith in America from the sun posts at Cahokia to the shootings of worshippers at prayer in this century. First, watch who was punished for the way they worshipped, and who did the punishing: Quakers hanged in Boston, Catholic churches burned in Philadelphia, Native dancers whose food was cut off, Black churches bombed. Then watch the people who pushed back, from Baptists who organized against the church tax to families who took their cases to the Supreme Court. Revivals, meetings meant to stir people to a stronger faith, run through the chapter too. By the early 1800s the Presbyterians, Baptists and Methodists who backed them were the largest Protestant churches in the country.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Why did colonists who crossed the ocean to worship freely punish other people for their faith?
2. How did Americans stop paying taxes to support churches?

3. How long did US officials treat Native ceremonies as crimes?

Five minutes? Read the Mary Dyer story in The 1600s, "How Virginia ended its church tax" in 1750 to 1800, and "Hiram Price's rules against ceremonies, 1883" in 1850 to 1900.

Can skip: "Three more colleges with church backing", "Laws against teaching evolution, 1921 to 1929", "Worship on a screen", "Teaching religion at home". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 5 MINUTES

Hundreds of Native nations lived on this land before 1500, and each one had its own religious life. So there were hundreds of Native religions, and many are still practiced today. The nations shared no holy book and no single founder. Each held its ceremonies in a set place, at a set season, with its own community.

No written account of these religions survives from before the 1500s, because the written record of this land begins with the Europeans who came then. What is known comes from two places. One is what people built and buried. The other is the traditions that the nations kept and still keep.

What the ground can show, and what it cannot

Archaeologists, scientists who dig up and study old buildings, graves and objects, can measure which way a doorway faces. They can check whether the sun rises in line with it on a certain day. A wall lined up with the midwinter sunrise shows that the builders lined it up on purpose. It does not show what the builders believed.

The nations themselves hold spoken histories that go back before 1500, and the people of each nation decide how much of that to make public. Written descriptions of Native religions begin after 1500. Europeans wrote them from the outside, and most of those writers were hostile to the religions they described. Many were missionaries, people sent to turn others to their own religion.

Chaco Canyon

Chaco Canyon, in what is now northwestern New Mexico, holds large stone buildings of many stories. People there began building them in the middle and late 800s, and today they are called great houses. Three of the first were Pueblo Bonito, Una Vida and Peñasco Blanco. Building went on in the canyon for more than 300 years.

The builders lined many great houses up with the sun, the moon, and north, south, east and west. They also built roads out from the canyon. The roads joined it to more than 150 other great houses across the region.

Pueblo people are the descendants of the people of Chaco. They say Chaco was a gathering place where many peoples and clans, groups of families who count themselves as related, came together to share their ceremonies and knowledge. Many Native people of the Southwest also say that Chaco is an important stop on their clans' sacred migration paths. Those are the routes their peoples traveled as they moved from one home to another.

Cahokia and its circle of posts

Cahokia stood in what is now Illinois, a few miles east of the Mississippi River and of present-day St. Louis, Missouri. People lived there mainly between about 800 and 1400. It was the largest town in North America north of Mexico before Europeans came. National Park Service figures put the number of people living in Cahokia itself at 10,000 to 20,000.

The people of Cahokia built about 120 mounds of earth. Some held public buildings on top, and others held graves. The biggest is now called Monks Mound, and no earth structure that Native people built north of Mexico is larger. It went up in fourteen stages. It rises in four steps, called terraces, to about 100 feet.

Woodhenge. Near Monks Mound stood a circle of upright wooden posts, called Woodhenge today. Posts in the circle marked where the sun came up on the longest day of the year and on the shortest. Others marked the two equinoxes, the days in spring and fall when day and night are the same length. The post for the equinox sunrise lines up with the front of Monks Mound.

Bill Iseminger, an archaeologist at the Cahokia site, gives these figures for the circle. The people rebuilt it about five times. It measured from about 240 to 480 feet across. The first circle had 24 posts, and each new circle had twelve more than the one before.

NOT KNOWN: what the people said or believed as they watched the sun rise over the posts. No record of it survives.

The earthworks of Ohio

People in the woodlands of eastern North America built long walls of earth between about the years 1 and 400. They laid the walls out in geometric shapes and built mounds of many shapes beside them. Many of these earthworks are in Ohio. In 2023 they were added to UNESCO's list of World Heritage Sites as the Hopewell Ceremonial Earthworks, ceremonial meaning built for religious gatherings and ceremonies. UNESCO is the part of the United Nations that keeps that list of important places.

Objects found at the earthworks came from far away. There is mica, a shiny rock that splits into thin sheets. There are shark's teeth, sea shells, copper, and obsidian, a black glass made by volcanoes. The builders had contacts with peoples from the Atlantic coast to the Rocky Mountains, and these things reached Ohio through them.

Whose name? Archaeologists call the builders the Hopewell culture, but that is not the name of any Native nation. Mordecai C. Hopewell owned a farm near Chillicothe, Ohio, where some of the mounds stood. Warren K. Moorehead dug into twenty-eight mounds on that farm, and the objects he took out went on show at the world's fair in Chicago in 1893. After that, archaeologists used the farmer's name for the builders. The National Park Service's account of the earthworks states that different Native groups built them.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 8 MINUTES

Spanish priests, with Spanish soldiers at their side, set out in the 1500s to turn the peoples of this land into Catholics, Christians who follow the pope in Rome. The priests belonged to orders, groups of Catholic men who promised to live by one set of rules. The Jesuits were one order. The Franciscans were another, and their priests and brothers are called friars.

Franciscan friars built missions, churches they ran inside Native towns, among the Timucua and Guale peoples of what is now Florida and Georgia. From 1598 they built missions among the Pueblo nations of what is now New Mexico as well. In each mission town the friars baptized the people, washing them with water in the Christian ceremony that makes a person a member of the church. They also taught them Catholic beliefs.

Missions and government. Spanish officials used the missions to bring the Native towns under the Spanish government. The officials demanded that the people promise loyalty to that government, and they ordered each mission town to send men to work for the Spanish. Nearly all of the written record of the missions in this century comes from the missionaries themselves, so it records their view.

The missions in Florida and Georgia

Luis Cáncer de Barbastro, a Dominican friar from Spain, argued that friars should bring Native people to Christianity without soldiers. He had come to the Caribbean in 1518. King Charles V of Spain gave him a written order to try it in Florida, and he sailed from Havana, Cuba, in early June 1549. On June 26 he went ashore near Tampa Bay to greet a group of Tocobaga people, and they killed him. The Tocobaga left no account of why.

The Jesuits tried missionary work in Florida from 1568 to 1571 and left without success. Franciscan friars came next. After a new group of them arrived from Spain in 1587, they began building missions up the coast into what is now Georgia.

San Pedro de Mocama, on the coast of what is now Georgia, was the first successful mission there. By the end of 1595 the friars had missions in at least five main towns of the Guale people, who spoke a Muskogee language.

A mission town with a friar living in it was called a *doctrina*. There the friar baptized people and led Mass, the main Catholic church service. Friars also taught the catechism, a set of questions and answers about Christian beliefs that people learn by heart. The friar Francisco Pareja had a catechism in Spanish and in the Timucua language printed in Mexico City in 1612.

Workers for St. Augustine. Spanish officials used the mission towns to get workers as well. Under a system called the *repartimiento*, each mission town had to send a set number of unmarried men to St. Augustine every year to work for the Spanish.

Killed for refusing to change faith, 1565

Spanish soldiers killed French Protestants, Christians whose churches had broken away from the pope, in Florida in 1565 because of their religion. It was the first killing over religion between Europeans in what became the United States.

The first group. A hurricane wrecked the French ships that year, and about 127 French Protestants came ashore with no food and no weapons. They surrendered to Pedro Menéndez de Avilés, the Spanish commander. Menéndez told them to give up their Protestant faith, and they refused. His soldiers killed 111 of them. Menéndez spared 16 men: a few who said they were Catholic, some sailors forced onto the ships, and four craftsmen St. Augustine needed.

Of the others, Menéndez wrote to the king of Spain on October 15, 1565: "I had their hands tied behind them and put them to the knife." That meant his men killed them with knives.

The second group. About two weeks after the first killing, a second group of Frenchmen under Jean Ribault surrendered. They refused the same demand, and Menéndez's soldiers killed 134 of them. The two killings are dated September 29 and October 11 or 12, 1565.

Guale, 1597

Guale people killed five Franciscan friars and destroyed their missions in the autumn of 1597. One of the five was Blas Rodríguez, the friar at the town of Tupiqui. A sixth friar was held captive and freed in 1598.

NOT KNOWN: which Guale people killed the friars, and why.

The historians J. Michael Francis and Kathleen Kole translated the earliest Spanish records of the uprising and published them in 2011. They wrote that those two questions cannot "be answered with any certainty." Older retellings name a young leader and give a reason. In the earliest records, only two of the seven witnesses named that young man at all, and neither said he alone ordered the killings.

The governor's revenge. Gonzalo Méndez de Canzo, the Spanish governor of Florida, led soldiers into Guale country himself. On October 27 people at the town of Ospo shot arrows at his men. Over the next ten days the governor and his officer Alonso Díaz burned Ospo and four more Guale towns, with their houses and storehouses, some of them full of corn. The governor could not capture anyone, so he ordered his men to destroy the corn fields and other crops as well. The records describe the burning and give no count of Guale people killed.

Lucas. The governor took seven young Guale captives in July 1598. Francis and Kole describe them as boys. One was Lucas, the son of don Felipe, the chief of Tupiqui. Spanish officials questioned Lucas a second time on July 27, now under the threat of torture, and he confessed that he had been there when Blas Rodríguez was killed. They sentenced him to death on July 28. He was hanged at St. Augustine on July 29, 1598, and the records do not name the man who hanged him.

Lucas was the only Guale person the Spanish officially charged and punished for the uprising. Spanish officials divided the other six boys among themselves as personal servants. The boys were captives and had to do forced labor for the men who held them. The Spanish left the northern missions empty until 1604.

New Mexico, 1598

Juan de Oñate claimed for Spain the land the Spanish called New Mexico in early May 1598, at a crossing of the Rio Grande near what is now El Paso. It was the homeland of the Pueblo nations, and Franciscan friars traveled with him.

That autumn Oñate went to every pueblo, as the Spanish called each Pueblo town. Through translators he demanded that the people of each town promise loyalty to the Spanish government. At those meetings he gave each town a missionary, and the Franciscan missions in New Mexico began that way.

Acoma. The friars also gave their approval for a war on Acoma Pueblo. In December 1598 Oñate's nephew, Juan de Zaldívar, went up to Acoma Pueblo with soldiers to demand food. In the Acoma people's own account, the soldiers raped an Acoma woman, forcing her into sex. The Acoma killed Zaldívar and eleven to fourteen of his men.

What the friars said. Oñate then asked the Franciscan friars whether a war against Acoma would be just. The friars answered that he "possessed both the authority and sufficient cause." That meant the friars said he had the right to make war and a good enough reason.

The killing. Oñate's soldiers attacked in January 1599. They killed Acoma men, women and children and burned much of the town. No one counted the dead, and estimates run from about 300 to about 1,500.

The sentences. Oñate sentenced the men over twenty-five to have one foot cut off. He sentenced them and every other captive over twelve to twenty years of forced labor, and he handed the old people to the Plains Apache as slaves. Whether the feet were really cut off is disputed.

The children. On Oñate's orders, Acoma children under twelve were taken from their parents and given to the friars. About seventy of the girls were sent south to convents in Mexico, houses of nuns, women who have promised their lives to the Catholic church. The records do not name the men who took them. Almost certainly none of the girls ever came home.

Doña María Meléndez

Who: The Timucua chief of Nombre de Dios, the town beside St. Augustine, and a Catholic.

When and where: Spanish Florida, the late 1500s and early 1600s.

The friars' success in her town was "largely owing to" her.

Doña María Meléndez ruled Nombre de Dios, a Timucua town just outside St. Augustine. The Spanish called her the *cacica*, their word for a woman chief. Her mother ruled the town before her and was one of the first Timucua people to become Christian. Doña María was a Christian too. She married a Spanish soldier named Clemente de Vernal, who lived with her and their children at Nombre de Dios.

The Franciscans set up their first *doctrina* at Nombre de Dios after 1587. Staff at the Florida Museum of Natural History write that the Franciscans' success there "was largely owing to" her. By 1606 she ruled the Timucua towns along the coast from St. Augustine north to about Cumberland Island, in what is now Georgia.

NOT KNOWN: how she gained that wider rule. The National Park Service's account says she may have had Spanish help, and the records do not settle it. Spaniards wrote down everything known about her, nothing survives in her own words, and no record says why she became a Christian.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 25 MINUTES

English Protestants founded several colonies in the 1600s so that they could worship in their own way, and in some of those colonies the leaders punished people who worshipped differently. The Puritans ran Massachusetts. They were English Protestants who wanted to clear the Church of England of what they saw as Catholic leftovers. Among the people they punished were Quakers, members of a Protestant group begun in England in 1652. Puritan leaders had the ears of some Quakers cut off, and they hanged four Quakers in Boston.

Other colonies set other rules. Maryland's lawmakers protected worship for Christians in a law of 1649, and the same law ordered death for anyone who denied the Trinity, the Christian belief that God is three persons in one. Roger Williams in Rhode Island let people of every faith live as neighbors with equal rights. William Penn wrote freedom of worship into Pennsylvania's Frame of Government, his plan for running the colony.

New Mexico. Spanish friars and governors punished Pueblo people for holding their own ceremonies. They whipped religious leaders and hanged some of them. In 1680 the Pueblo nations rose up and drove the Spanish out.

Why the Puritans left England

In England in the 1620s, the men who ran the church and the government ordered the Puritans to follow church practices they hated. Those leaders put Puritan ministers out of their jobs and threatened them with "extirpation from the earth," which means being wiped out.

Bishops, the high officials who ran the Church of England, had their critics punished hard. English judges had Protestants who wrote against the bishops whipped, cut and branded.

Alexander Leighton. The judges of a London court called the Star Chamber sentenced a Scottish minister named Alexander Leighton in June 1630, for writing a book against the bishops. They fined him 10,000 pounds and ordered him whipped. He was to stand in the pillory, a wooden frame that locked a person's head and hands in a public place. The judges ordered one of his ears cut off and one side of his nose slit open.

They also ordered his face branded with the letters S.S. Branding means pressing red-hot metal onto the skin to burn in a mark that never goes away. The letters stood for "sower of sedition," a person who stirs people up against the government.

Leighton escaped the night before the punishment, but he was caught again in Bedfordshire. The first part of the sentence was carried out on him on November 26, 1630.

> **NOT KNOWN:** which punishments that first part included, and who carried them out. The account of his life in the Dictionary of National Biography, a British book of life stories printed between 1885 and 1900, does not say.

He was held in the Fleet, a London prison, until 1640. That year the members of Parliament, England's lawmakers, set him free, canceled his fine and voted to pay him 6,000 pounds.

Crossing the ocean. As many as 20,000 Puritans left England for America from 1630 on, looking for the freedom to worship as they believed they should. John Winthrop's company of about 1,000 arrived in 1630 and settled on the Shawmut Peninsula, where Boston stands now.

It was the land of the Massachusett people, who called it Mushauwomuk, "the boat landing place." Between 1616 and 1618 an epidemic, a disease that spread fast from person to person, had killed all but about 25,000 of the roughly 100,000 people of that region. That is three people in every four. In 1633 the Massachusett leader Chickataubut signed a treaty with Winthrop that gave up his rights to the peninsula.

The Pilgrims who landed at Plymouth in 1620 were a different group. They had left the Church of England completely and formed churches of their own. The Puritans believed the Church of England was a true church that needed to be fixed.

Joining a church. In Massachusetts a person could not simply join a church. At first, only men and women who had gone through a conversion experience, a moment when a person feels that God has changed him or her, could become members. Each new member also had to convince the other members that it was real.

The Bay Psalm Book. Richard Mather and two other ministers turned the Psalms of the Bible into verses that church members could sing. Stephen Daye printed their book at Cambridge, Massachusetts, in 1640, and it was the first book printed in the English colonies of North America. Eleven copies still exist.

Reading for God

New England children were taught to read for a religious reason. In 1642 the men of the Massachusetts General Court, the colony's lawmakers, ordered parents and masters to teach their children and apprentices, young people learning a trade from a master, to read. The law required them to learn enough "to read the English tongue" and to know the colony's "Capital Lawes," the laws that set death as the punishment for a crime. The colonists pointed to the Bible as the authority for many of their criminal laws.

The Old Deluder Satan Act. A second law, in 1647, ordered towns of 50 households to hire someone to teach reading and writing. Towns of 100 families had to set up a grammar school, a school that prepared young people for "the university," which meant Harvard College. The law's opening words state that it is "one of the chief projects of that old deluder Satan to keep men from the knowledge of the Scriptures."

A deluder is someone who tricks people, and the law has been called the Old Deluder Satan Act ever since. The colonists wanted every child to be able to read the Bible without a minister in between.

Harvard College was founded in 1636 to train ministers. A pamphlet by its founders, *New Englands First Fruits*, printed in England in 1643, states the college's purpose. The founders were "dreading to leave an illiterate Ministery to the Churches, when our present Ministers shall lie in the Dust." They feared that when their ministers died, no trained ministers would be left to replace them.

New England children learned to read from *The New England Primer*, made by the printer Benjamin Harris. Staff at the American Antiquarian Society, a research library, date its first printing between 1687 and 1690, because the only evidence left is booksellers' advertisements. Harris placed his first advertisement for it in 1690. No copy printed before 1727 survives.

The picture of the fire. One page of the Primer showed John Rogers being burned alive at Smithfield, in England, on February 4, 1555. He was the first Protestant put to death under Mary, the Catholic queen of England. The

picture showed his wife and ten children beside the fire. A long poem next to it, written as his last advice to his children, told them to "Abhor the arrant Whore of Rome," which meant to hate the Catholic Church.

Library of Congress staff write that generations of New England children read these pages. In their words, the pages "doubtless helped to fuel" New Englanders' prejudice against Catholics, a dislike of a whole group of people without knowing them. That prejudice stayed strong there well into the 1800s.

BY 1650: Puritan leaders ran Massachusetts by their reading of the Bible, only people who had felt God change them could join its churches, and every town of 50 households had to hire someone to teach reading.

Praying towns and Deer Island

New England ministers also tried to bring the Native nations around them to Christianity. John Eliot, the minister at Roxbury, translated the whole Bible into the Algonquian language of the Massachusett people. It was printed at Cambridge in 1663, the first Bible printed in any language in North America. At one time Eliot ministered to about 1,100 Christian Native people, whom the English called Praying Indians. They lived in fourteen towns laid out in the English style.

Deer Island. The English were fighting what they called King Philip's War in 1675. That year the members of the Massachusetts General Court ordered the Christian Native people of Natick taken from their homes and put on Deer Island in Boston Harbor. Daniel Gookin, the Massachusetts official in charge of dealings with the Praying Indians, wrote that the order came from "false informations" given to the court to stir its members up against the people of Natick.

Samuel Shrimpton of Boston held the island, and he agreed to the plan as long as they did not cut down any trees. Captain Thomas Prentiss and a party of horsemen brought the Natick people to the Charles River.

About 200 people from Natick were put on three boats about midnight on October 30, 1675, under the court's order, and carried to Deer Island. Gookin does not name the men who took them. John Eliot met them at the landing and prayed with them. They were afraid they would never come home and would be shipped out of the country.

The winter. Massachusetts officials then sent the Punkapog people to the island too, "upon as little cause as the Naticks were," Gookin wrote. By late December about 500 people were held there. Gookin wrote that they lived mostly on clams and shellfish dug out of the sand at low tide. The island was cold, their wigwams, or houses, were poor, and they had few clothes, and thin ones.

Gookin wrote that the island kept them safe "from the fury of the people." His account of that winter gives no count of the dead. In 2025 the Episcopal bishops of Massachusetts and Western Massachusetts wrote that hundreds of Native people from the praying towns died on Deer Island from the cold and from starvation in the winter of 1675 to 1676.

Quakers punished in Massachusetts

The Quakers formed around a preacher named George Fox in England in 1652, and they call themselves the Religious Society of Friends. They taught that the Spirit, or "Light of Christ," lives inside every person. Quaker women spoke and preached at Quaker meetings for worship, and other colonists found this offensive. Many people at the time called Quaker teaching heresy, a belief that goes against the teaching of the church.

English officials had put 10,000 Quakers in prison by 1680. According to the Library of Congress, 243 of them died in the king's jails "of torture and mistreatment."

The Puritan lawmakers of Massachusetts passed law after law against Quakers, starting on October 14, 1656. They called Quakers "the cursed sect of heretics." They fined ship captains who brought Quakers into the colony and people who owned Quaker books.

Ears and tongues. A law of 1657 set the punishment for a Quaker who came back after being thrown out of the colony. A man would have one ear cut off the first time, and the other ear the second time. A woman would be whipped. A third return was to be punished by boring a hole through the person's tongue with a hot iron.

Christopher Holder, John Copeland and John Rous were Quakers who came back. The judges sentenced each of them to have his right ear cut off by the hangman. On September 16, 1658, the deputy of the colony's marshal, a law officer, came to the prison with a group of men and shut the door. Then he had the right ear of each of the three men cut off. Afterward the three men said: "They that do it Ignorantly, We desire the Lord from our hearts to forgive them."

Two children for sale. Massachusetts officials also punished the children of Quakers. Daniel and Provided Southwick were the son and daughter of Lawrence and Cassandra Southwick, Quakers of Salem. Judges at Salem and Ipswich fined them ten pounds for not going to the town's church meetings,

and they had no property to pay it with. The members of the General Court then gave the county treasurers the power to sell the two children to any English buyer, in Virginia or Barbados, to pay the fine.

The county treasurer, Edmond Batter, looked for a ship's captain to carry them to Barbados. The captain he found refused. He asked, "will ye offer to make Slaves of so harmless Creatures?" Batter sent the children home, and the order to sell them was never carried out.

Death for coming back. The lawmakers set the punishment of death on October 19, 1658, for any Quaker who came back after being banished, which means ordered out of a place and forbidden to return. Between 1659 and 1661 the colony's officials hanged four Quakers in Boston under that law: William Robinson and Marmaduke Stephenson in 1659, Mary Dyer in 1660 and William Leddra in 1661. The governor, John Endecott, sentenced three of them. The story of Mary Dyer, after this section, describes the first three hangings.

King Charles II ordered the Massachusetts lawmakers to stop punishing Quakers on September 9, 1661, after Quakers in England pressed him. By 1677 Quakers could hold their meetings openly in Massachusetts.

Virginia's lawmakers passed their own laws against Quakers from 1659, including death for Quakers who would not obey them. No one was put to death under them. Thomas Jefferson later wrote that the reason was not that the church held back or that the lawmakers went easy. In his words, it "was not owing to the moderation of the church, or the spirit of the legislature."

BY 1677: Quakers met openly in Massachusetts, and no Quaker had been hanged there since 1661.

Mary Dyer

Who: A Quaker woman hanged on Boston Common for returning to Massachusetts after she had been banished.

When and where: Massachusetts and Rhode Island. Hanged in Boston on June 1, 1660.

She stood on the ladder with the rope around her neck.

Her first trouble with the Massachusetts leaders came when she supported Anne Hutchinson, a Boston woman who was banished for her religious teaching. In 1638 the Massachusetts leaders forced Dyer and her family out, and they moved to Rhode Island. In the 1650s, in England, she became a Quaker.

She came back to New England to spread Quaker beliefs. The Massachusetts officials arrested her and banished her three times. In the spring of 1657 her husband, William Dyer, got her out of the Boston jail by confronting Governor John Endecott. Endecott let her go on the condition that she never return.

The first sentence. She returned anyway, and in October 1659 Endecott sentenced her to death along with two Quaker men, William Robinson, a merchant from London, and Marmaduke Stephenson, a farm worker from Yorkshire. He told them they would "be hanged until you are dead." On October 27, 1659, the officials led all three to the gallows and made Dyer watch the two men hanged. She stood on the ladder with the rope around her neck.

Then the officials announced a reprieve, which called off her hanging. Endecott had agreed to it the day before. The officials sent her out of the colony.

She came back once more in May 1660, and Endecott sentenced her to death a second time. Massachusetts officials hanged her on Boston Common on June 1, 1660.

Maryland, 1649: freedom for Christians, death for others

Most people in England hated the Catholic Church all through the 1600s, and they made life hard for Catholics and punished them. In 1632 George Calvert got a charter, a written grant of land and of the right to govern it, from King Charles I for the land that became Maryland. According to the Library of Congress, Calvert wanted a safe place for his fellow Catholics.

Two ships, the *Ark* and the *Dove*, carried about 200 settlers there. Two Catholic priests had to sneak on board to get away from England's laws against Catholics.

Maryland's colonists first landed on March 25, 1634. They later bought land from the Yaocomaco people and built St. Mary's City there. The records do not say what the Yaocomaco were paid or how they understood the sale. By 1649 the colony belonged to Cecil Calvert, the second Lord Baltimore.

The Toleration Act. Cecil Calvert sent the Maryland assembly, the colony's lawmakers, a set of proposals about religion in 1649. The members met at St. Mary's City from April 2. They rewrote part of what he sent, and on April 21 they voted it into law as An Act Concerning Religion. It has been called the Toleration Act ever since, toleration meaning letting people hold beliefs that others do not share.

The act protected only Christians. It stated that no person "professing to believe in Jesus Christ" should be "in any ways troubled, molested, or discountenanced" for his or her religion. Molested here meant bothered or attacked, and discountenanced meant treated with disapproval.

Death for other beliefs. The same act ordered death for anyone who cursed God, or denied that Jesus was the Son of God, or denied the Trinity of the Father, the Son and the Holy Ghost. Such a person "shall be punished with death," and the person's land and goods would be taken and given to Lord Baltimore. Under this law a Jew, or a Christian who did not believe in the Trinity, could be put to death.

The act also punished insults about religion. Calling someone a heretic, a Puritan, a Presbyterian, a "Popish Priest" or a similar name as an insult cost a fine of ten shillings (a shilling was an English coin). Half of the fine went to the

person insulted. A person who could not pay was "publickly whipt" and kept in jail until he or she asked forgiveness in public.

Puritans took control of Maryland in 1654 and threw out the act. They took legal rights away from Catholics, Jews, Quakers and others. Maryland's colonists canceled the Puritan laws in 1661, and the act was in force again. After 1689 the Church of England became Maryland's official church by law, and officials there enforced English laws that kept Catholics from voting, holding office or worshipping in public.

Two colonies that left worship alone

Roger Williams founded Providence in 1636, on land of the Narragansett people. He wanted it to be "a shelter for persons distressed for conscience." In Rhode Island, Anglicans (members of the Church of England), Puritans, Quakers, Catholics, Jews and people with no religion lived as neighbors with equal rights.

William Penn was a Quaker. In 1681 King Charles II owed a debt to Penn's father, and Penn turned it into a charter for Pennsylvania. The Lenape people, also called the Delaware, lived on both sides of the Delaware River. Beginning in 1682, the Lenape gave up land on the west bank of the river to Penn in exchange for cloth, guns, gunpowder, alcohol and other goods. By 1685 as many as 8,000 Quakers had moved to Pennsylvania.

Penn's plan. In 1682 Penn wrote a Frame of Government for the colony. It stated that everyone who believed in "One Almighty and eternal God" would "in no wayes be molested," meaning bothered, for their religion. It also stated that no one would "be compelled at any time to frequent or maintain any Religious Worship, Place or Ministry whatever." That meant no one in Pennsylvania had to attend a church, and no one had to pay for one.

The first group of Germans reached Philadelphia in 1683, from the town of Krefeld. They were Mennonites and possibly some Dutch Quakers. Mennonites, Dunkers, Schwenkfelders, Moravians and German Baptists followed. These were small Protestant groups from Germany and nearby lands,

and their members had been punished for their religion in Europe. People described Pennsylvania as "an asylum for banished sects," a safe place for religious groups that had been driven out.

Twenty-three Jewish refugees from Dutch Brazil arrived in New Amsterdam, the Dutch town that became New York, in 1654. Within a year they were holding religious services there. By 1658 Jews had also settled at Newport, Rhode Island, where they could worship freely.

Roger Williams

Who: A minister thrown out of Massachusetts who founded Rhode Island, where the government stayed out of religion.

When and where: Born in London about 1603. Massachusetts from 1631. Founded Providence in 1636. Died in Providence in 1683.

"Forced worship stinks in God's nostrils," he wrote in 1644.

Roger Williams taught that a government has no right to force religion on anyone. He called the freedom to follow one's own conscience "soul liberty," and he built a colony on that idea.

He had studied at Cambridge University and become a minister in the Church of England, and he disliked the church's ranks of officials, its wealth, its power in government and its decorations. He reached Massachusetts Bay in 1631, one year after the Puritans settled it. There, Puritan religious rules covered every part of daily life, and colony officials enforced them. Williams preached that the government should not enforce anyone's religion, and that a person's spiritual life should be kept separate from public life.

Thrown out. The judges of the colony's General Court ordered him out of Massachusetts for preaching these ideas. He fled in the winter, because he might have been put to death if he had been sent back to England. Wampanoag people took him in and kept him alive through that winter. He had traded with them and learned their language.

Providence. With help from Wampanoag and Narragansett people, he traveled by canoe to a freshwater spring at a place called Moshassuck. He settled there in 1636 with the permission of the Narragansett and named the place Providence. In 1638 the Narragansett leaders Canonicus and Miantonomi signed a deed with Williams for the settlement, a deed being a paper that hands over land. Under it, the Narragansett kept the right to trade inside the settlement.

In Rhode Island, people of every faith lived as neighbors with the same rights. Williams thought many of them were wrong about religion. He believed that correcting them was not the government's job.

His books. In 1643 he published *A Key into the Language of America*, a book about the Native languages of New England, mostly Narragansett, and about Narragansett daily life. In 1644 he published *The Bloudy Tenent of Persecution*, an argument against forced worship. In it he wrote: "God requireth not an uniformity of Religion to be inacted and inforced in any civill state." He meant that God does not require everyone to be made to follow the same religion.

Members of England's Parliament granted Rhode Island a patent, a paper that recognized the colony, in 1644. King Charles II granted it a new charter in 1663, at the request of John Clarke, a Rhode Island colonist who had stayed on in England to act for the colony. The charter stated that no one in the colony would be punished "for any differences in opinion in matters of religion."

Williams died in Providence in 1683. In 1965 members of Congress voted to set up the Roger Williams National Memorial in Providence.

Anne Hutchinson

Who: A Boston woman who held religious meetings in her home, taught her own reading of the Bible, and was banished and put out of her church for it.

When and where: Born in Alford, England, about 1591. Boston from 1634. Banished in November 1637. Killed in 1643.

Her church said she had "rather been a Husband than a Wife."

The leaders of Massachusetts put Anne Hutchinson out of the colony for what she taught about God, and then the members of her church put her out of the church. She had sailed to Boston in the autumn of 1634 with her husband, William, and their children, following their minister, John Cotton.

The meetings. From 1636 she held a meeting in her home once a week to talk over the Sunday sermon. At first only women came. Then men came too, including important men of the colony. In Massachusetts the minister's sermon in the meeting house was the official teaching, and a second meeting in a private house, led by a woman, was a rival to it.

What she taught. She taught that God's grace, God's free gift of love and forgiveness, alone decides who is saved, meaning who goes to heaven. She said good behavior played no part.

She also said she could tell which ministers and which neighbors had received God's grace and which had not. That second claim put her in judgment over the colony's ministers. Her opponents called her teaching antinomian, a word that means "against the law." They meant that her teaching let a saved person ignore the rules.

At her trial she said that God had spoken to her directly, by "an immediate revelation," and that God would curse the Puritans if they harmed her. The judges called that blasphemy, speaking against God.

Two trials. She faced two trials, five months apart. In November 1637 the members of the General Court tried her at Newtown, now Cambridge, and banished her. In March 1638 the members of the church at Boston

tried her separately and excommunicated her, which means they put her out of the church.

The church's charges said she had "rather been a Husband than a Wife and a preacher than a Hearer." The same charge went on: "and a Magistrate than a Subject," an official who governs or judges rather than a person who is governed.

Her death. In the spring of 1638 she moved to Aquidneck Island in what became Rhode Island and founded the town of Portsmouth. After her husband died, she moved in 1642 to land on Long Island Sound that the Dutch claimed for their colony.

A war broke out there the next year. In February 1643 soldiers of the Dutch governor, Willem Kieft, killed at least eighty, and perhaps as many as 120, Munsee people in one night, and that killing began the war. In August or September 1643, during that war, Siwanoy people attacked her home. They killed her and several of her children. One daughter survived.

Baptism and slavery in Virginia, 1667

Some colonists argued that a Christian could not be held as a slave. In September 1667 the members of Virginia's lawmaking assembly settled the question in favor of slavery. Their law said that being baptized "doth not alter the condition of the person as to his bondage or ffreedome," bondage meaning slavery. The law's own words state the lawmakers' reason. Once enslavers were sure that baptism would not free the people they held, they would be more willing to let them be baptized.

New Mexico: punished for their ceremonies

The Pueblo nations of New Mexico hold their religious ceremonies in kivas, rooms set aside for worship. Some of their ceremonies are masked dances. For about eighty years the Franciscan friars demanded that the kivas be destroyed and the masked dances ended. The friars called the dances idolatry, which means worshipping false gods.

Alonso de Posada, the head of the Franciscan missionaries from 1656 to 1665, forbade Pueblo ceremonies. He ordered the missionaries to go into the kivas, burn the masks, and take or destroy the sacred objects. In 1661 Spanish soldiers as well as friars raided kivas and burned sacred masks.

Whippings and hangings. Will Hansen and Analú María López, who work at the Newberry Library, a research library in Chicago, write that missionaries almost certainly whipped Native people who did not take part in Christian services. Spanish investigations in New Mexico between 1620 and 1680 recorded many cases of missionaries punishing Native people severely. According to the Architect of the Capitol's account of Po'pay, Pueblo people "found practicing their religion were tortured and sometimes executed."

Juan Francisco Treviño, the Spanish governor, ordered the arrest of 47 Pueblo medicine men in 1675. The charge was sorcery, using magic to cause harm, which is what Spanish officials called the medicine men's ceremonies. Spanish officials sentenced four of them to be hanged. Three were hanged, and the

fourth killed himself. Spanish officials whipped the others in public and put them in prison. The records do not name the men who did the hanging and the whipping.

Po'pay. One of the men whipped was Po'pay, of Ohkay Owingeh, also called San Juan Pueblo. He was born about 1630, and his name means "ripe squash" in the Tewa language. He was a farmer and a member of a medicine society, and his work was healing and the spiritual life of his people. Spanish officials had him flogged in public. Flogging is beating a person on the back with a whip. The whip leaves wounds that heal into scars.

When Pueblo leaders heard about the arrests, they sent armed men to Santa Fe, where the prisoners were held. Many Spanish soldiers were away at the time, so Treviño gave in and released the prisoners. Po'pay was one of them.

The rising. Po'pay led a rising of the Pueblo nations against the Spanish in 1680, and he planned it from Taos Pueblo. The leaders set it for August 13 and sent runners to the pueblos with knotted cords. Chapter 2, Native Nations, describes the rising in full, including what the knots meant.

On August 9 the Spanish captured two of the runners, Pedro Omtua and Nicolás Catua of Tesuque Pueblo. Spanish officials tortured them to make them explain the cords, and later hanged them. The records do not name the men who did it. With two of their runners in Spanish hands, the Pueblo leaders began the rising three days early, on August 10, 1680.

The Pueblo fighters killed about 400 people, including 21 of the 33 Spanish priests in New Mexico, and burned mission churches across the province. The Pueblo nations kept the Spanish out of New Mexico for twelve years.

The Pueblo nations run their own museum, the Indian Pueblo Cultural Center. Its account of those eighty years of Spanish rule states that "the Pueblo people suffered violence, cruelty, and hunger." It lists "exploitation of labor and resources," and "harassment and persecution for practicing long-held religious traditions that are interwoven in daily life." That means the Spanish forced Pueblo people to work for them, took what they had, and troubled and punished them again and again for keeping their own religion.

The statue. In 2005 a statue of Po'pay, given by the state of New Mexico, was placed in the National Statuary Hall Collection in the United States Capitol. Of all the people in that collection who were born on land that later became the United States, Po'pay was born the earliest.

No picture or written description of Po'pay survives, so the face is the sculptor's own work. The sculptor, Cliff Fragua of Jemez Pueblo, wrote: "On his back are the scars that remain from the whipping he received for his participation and faith in the Pueblo ceremonies and religion."

BY 1700

If you remember one thing: English colonists crossed the ocean to worship in their own way, and the leaders of Massachusetts, like the friars of New Mexico, punished people who worshipped any other way.

- 1636: Roger Williams founds Providence, where people of every faith live as neighbors.
- 1660: Massachusetts officials hang Mary Dyer on Boston Common.
- 1680: Po'pay leads the Pueblo rising that drives the Spanish out of New Mexico.

Next: In the 1700s traveling preachers held revivals across the colonies, and church members split over them.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 14 MINUTES

Traveling preachers held revivals, series of meetings meant to stir people to a stronger faith, across the colonies in the 1730s and 1740s. People in many colonies took part in the same religious movement at the same time, and it is called the Great Awakening.

Most colonists already went to church. By one estimate, 75 to 80 percent of colonists attended church between 1700 and 1740, and colonists were building new churches fast. The revival preachers wanted each person to go through the new birth, a conversion that a person could feel happening inside. Church members split into two sides over whether the new birth was real. Critics said many of the conversions were false, and even the work of the devil.

Killings over religion came in these years too. English colonists from Carolina and Creek fighters under James Moore destroyed the Spanish missions among the Apalachee of Florida in 1704. In 1741 New York officials hanged a man they called a Catholic priest.

The Apalachee missions destroyed, 1704

English colonists from Carolina and their Creek allies destroyed the Spanish missions among the Apalachee people in 1704. The Apalachee lived around what is now Tallahassee, Florida. Many of them were baptized Catholics who lived in fourteen Franciscan mission towns.

The members of the Carolina assembly approved the attack in September 1703 and asked James Moore, a former governor of Carolina, to lead it. Moore promised that the attack would cost the colony nothing, because he would pay for it "by the taking of loot and slaves." That meant stealing goods and capturing people to sell as slaves. He led fifty Englishmen and about one thousand Creek men, also called Muscogee.

Ayubale. Moore's men destroyed nearly all of the fourteen mission towns. At the church of Ayubale, on January 25, 1704, a friar named Angel de Miranda and 26 men held off the attackers for about nine hours. They surrendered when they ran out of arrows.

A magazine account based on the work of the historian John Hann describes what the attackers did there. They burned one friar alive and cut off his head, and the account does not give his name. They also tortured two Native people and burned them at the stake.

Patale. In a later raid that year, on June 23, 1704, attackers killed the friar Manuel de Mendoza at the Patale mission and burned the church. At Patale they tied prisoners to the Stations of the Cross, a set of markers used in Catholic prayer, and cut them with knives. Then they pushed burning sticks into the wounds and burned the prisoners to death. The records name James Moore as the commander of the January raid. They do not name the men who tortured and burned the prisoners.

How many were taken. The counts vary widely. Moore reported that his army killed more than 1,100 men, women and children and captured more than 4,300 people as slaves, most of them women and children. Moore had promised to pay for the raid with captives.

Other counts of the people captured run from hundreds to about 1,400. The historian John Hann counts about 2,000 Apalachee who went into exile after the raids, forced to live far from their homeland, and he writes that an unknown number were enslaved. In 1680 the fourteen mission towns had held about 8,000 people.

Many Apalachee were unhappy with the forced labor the Spanish demanded, which included carrying goods about 100 miles to St. Augustine. The Spanish had also forbidden mission Apalachee to own guns. Many Apalachee left with Moore. In his report, Moore said that most of the people of seven villages chose to join him.

New Lights and Old Lights

Church members argued over the revival, and many churches divided into two sides. People who supported the revival were called New Lights, and people who opposed it were called Old Lights. Those were the names among the Congregationalists, the members of the churches the Puritans had founded in New England. Among the Presbyterians, another family of Protestant churches, the two sides were called the New Side and the Old Side.

The revival had ended in New England by the middle of the 1740s. Preachers carried it south, where it started revivals that went on into the 1800s. The Presbyterians, Baptists and Methodists had backed the revival, and by the early 1800s they were the largest Protestant churches in the country. According to the Library of Congress, the Anglicans, the Quakers and the Congregationalists "were left behind."

A small number of colonists, among them Thomas Jefferson and John Adams, went a different way. They were deists, who stressed living a good life and rejected the Christian teaching that Jesus was God.

Drawings of New York City from 1690, 1730 and 1771 show the churches multiplying. The 1690 view shows an empty skyline. The 1771 view shows eighteen steeples, the tall towers on top of churches.

A college founded by revival supporters

Only graduates of Harvard or Yale could become Presbyterian ministers in the colonies in these years. Under that rule, the young men taught by William Tennent could not become Presbyterian ministers, so revival supporters set out to found new colleges.

Tennent was a Presbyterian minister, born in Ireland and educated in Edinburgh, Scotland. From 1726 until he died in 1745 he ran a school at Neshaminy, in Bucks County, Pennsylvania. People called it the Log College, because it was a log house about twenty feet long and almost as wide. He taught his students ancient languages and the Bible. Several of them became revival preachers anyway, among them his four sons.

Princeton. A group of New Lights planned a college in New Jersey in early 1746 to train young men for the ministry. On October 22, 1746, John Hamilton, the acting governor of New Jersey, granted a charter for the College of New Jersey. It is now Princeton University, and Princeton's own historians call its founding "a direct outgrowth" of the Great Awakening. Six months after the charter, three Log College graduates became trustees, the men in charge of running the college. They were Samuel Blair and two of Tennent's sons, Gilbert Tennent and William Tennent Jr.

Should enslaved people be taught Christianity?

Revival preachers pressed enslavers to let enslaved people be taught Christianity, and enslavers resisted. In 1740 George Whitefield printed an open letter in the *Pennsylvania Gazette* to the enslavers of Maryland, Virginia and the Carolinas. He wrote down their reason for keeping enslaved people from Christianity: that "teaching them Christianity would make them proud, and consequently unwilling to submit to Slavery."

Whitefield called that "a dreadful Reflection" on their own religion. He did not say in the letter that slavery was wrong, and he later held enslaved people himself.

Leah, who was enslaved in the household of the minister Jonathan Edwards, became a full member of the church at Northampton, Massachusetts, during the revival. Kenneth Minkema of the Jonathan Edwards Center at Yale records her membership.

George Whitefield

Who: An English preacher whose tours of the colonies were the high point of the Great Awakening, and who worked to make slavery legal in Georgia.

When and where: Born in Gloucester, England, December 16, 1714. Seven visits to America from 1738. Died in Newburyport, Massachusetts, September 30, 1770.

People in the crowd threw rotten eggs and pieces of dead cat.

George Whitefield drew crowds of up to 30,000 people in the Great Awakening, and he also pushed to make slavery legal in Georgia. He was a minister in the Church of England and was often in conflict with its leaders.

The tour. His preaching tour of the colonies from 1739 to 1741 was the high point of the Great Awakening in America. A sermon he gave in Boston drew as many as 30,000 people.

Many churches would not let him preach inside, so he preached outdoors from a folding pulpit that he carried with him. The first recorded use of it was at Moorfields, in England, on April 9, 1742, in front of a crowd estimated at twenty or thirty thousand. People in the crowd threw stones, rotten eggs and pieces of dead cat at him. He preached about two thousand sermons from that pulpit, by one estimate.

Why crowds came. He preached the new birth. In a printed sermon of 1739 he admitted that "how this glorious Change is wrought in the Soul cannot easily be explained." The Library of Congress lists the reasons crowds came to hear him: a strong voice, a way of speaking like an actor on a stage, the strong feelings he stirred up, simple messages, and clever use of the new ways of advertising.

Someone in London made an engraving, a printed picture, in 1763 that mocked his eye disease by calling him "Dr. Squintum." The picture's maker also accused him of sexual wrongdoing and of getting rich from

his preaching.

Bethesda. In 1740 he founded an orphanage called Bethesda near Savannah. Slavery was against the law in Georgia at that time. In 1741 he wrote that farming more land than his hired servants and older boys could manage was "impracticable without a few negroes." ("Negroes" was a word of his time for Black people.)

He wrote to a donor in 1747 that friends in South Carolina were giving money toward buying "a plantation and slaves" to support Bethesda. He added: "One negroe has been given me." He wrote that he planned to buy more that week.

Slavery in Georgia. From 1747 he pressed the trustees who ran Georgia to lift the ban on slavery. He wrote that "Georgia never can be a flourishing province, unless negroes are employed." He told them that unless the ban ended, he could not "cultivate the plantation in any manner." The trustees gave way.

Slavery became legal in Georgia in the early 1750s. The Penn and Slavery Project at the University of Pennsylvania dates it to 1752, when Georgia became a royal colony, run directly by the king's officials instead of the trustees. Other accounts give 1751.

Whitefield died at Newburyport, Massachusetts, on September 30, 1770. In his will he left 4,000 acres of land in Georgia and fifty enslaved people to the Countess of Huntingdon. Phillis Wheatley, a girl of about seventeen who was enslaved in Boston, wrote a poem about his death that was printed that year.

Jonathan Edwards

Who: The minister of the Northampton revival, whom the historian Perry Miller called "the first and greatest homegrown American philosopher." He bought and held enslaved people from 1731 until at least 1756.

When and where: Born in East Windsor, Connecticut, October 5, 1703. Northampton, Massachusetts, from 1726. Died in Princeton, New Jersey, March 22, 1758.

He compared his listeners to a spider held over a fire.

Jonathan Edwards led a revival in his own church at Northampton, Massachusetts, and wrote the books that other revival preachers learned from. He was the fifth of eleven children of a minister, and he finished first in Yale's class of 1720.

He joined his grandfather, Solomon Stoddard, as a minister of the church at Northampton in 1726. It was the largest church in Massachusetts outside Boston. He later took his grandfather's place. In 1727 he married Sarah Pierpont, and they had ten children.

A revival took place in his congregation, the people of his church, in 1734 and 1735. Edwards wrote a book about it, *A Faithful Narrative of the Surprising Work of God*. Revival preachers in America and Britain used it as a model, and it made him famous.

The sermon. On July 8, 1741, he preached a sermon at Enfield, Connecticut, called "Sinners in the Hands of an Angry God." He told the people that their own care and goodness had "no more influence to uphold you and keep you out of hell than a spider's web would have to stop a falling rock."

He also told them that God hated them: "The God that holds you over the pit of hell, much as one holds a spider or some loathsome insect over the fire, abhors you, and is dreadfully provoked."

He preached that God controls everything, that people are sinful by nature, that hell is real, and that a person has to be born again. Critics said many of the revival's conversions were false, or even the work of the devil. Edwards answered them in four books between 1741 and 1749, among them *A Treatise Concerning Religious Affections* in 1746. In them he tried to set out how a person could tell a real change of heart from a false one.

The people he enslaved. Edwards bought and held enslaved people from 1731 until at least 1756. Kenneth Minkema, of the Jonathan Edwards Center at Yale, lays out the record. The first was a teenage girl named Venus, whom Edwards bought at Newport in 1731. The last was a child named Titus, in 1756. Others whose names are known were Leah, Rose, and a married couple, Joseph and Sue.

His wife, Sarah, actively tried to get enslaved people from family members and from her husband's fellow ministers. Edwards defended slavery as allowed by the Bible. He argued that it was a way to bring enslaved people to Christianity. Late in his life he came to oppose the slave trade, writing that it was unjust to take away the freedom of people born free. After his death his son, Jonathan Edwards Jr., and his student Samuel Hopkins called for ending slavery completely.

Stockbridge. The members of his congregation dismissed him in 1750, after he tried to make the rules stricter for who could take communion, the Christian ceremony of sharing bread and wine in memory of Jesus. From 1751 to 1757 he worked at Stockbridge, a Mohican town on the western edge of Massachusetts. He preached to a small English congregation and was a missionary to about 150 Mohican and Mohawk families.

Stockbridge had 218 Mohican people living there in 1749. By 1774 the Mohicans had lost all but 1,200 of their 23,000 acres there, and in 1783 the Mohicans were forced to leave the rest. Their own history does not

name the people who took the land or forced them out. The Jonathan Edwards Center at Yale states that Edwards was "an agent in depriving Native Americans of their lands and culture."

His death. In late 1757 Edwards agreed to become president of the College of New Jersey. A few months after he arrived, he was inoculated against smallpox, given a small dose of live smallpox on purpose so that he would have a mild case and be safe afterward. A few of the people inoculated died from it, and Edwards was one of them. He died on March 22, 1758.

A man hanged as a Catholic priest, New York, 1741

Under a law the members of the New York assembly passed in 1700, any Catholic priest who came to New York faced prison for life.

The fires of 1741. After a string of fires in 1741, New York City officials claimed that enslaved people and poor white people had plotted to burn the city. By the end of August the judges had sentenced 30 people to death, by the count of the Historical Society of the New York Courts. They also sentenced 70 people to be sold as slaves in the Caribbean.

The historian Jill Lepore counts 34 people put to death. By her count, officials burned 13 Black men at the stake and hanged 17 Black men, two white men and two white women.

The hunt for priests. While those trials were going on, James Oglethorpe, the governor of Georgia, wrote to New York's lieutenant governor, the deputy governor. He warned that the Spanish had sent agents to burn English towns in North America, and that "many priests were employed" in disguise, pretending to be doctors or dancing masters. New York officials began hunting for hidden priests. They arrested John Ury, a schoolteacher who knew Latin. Ury said he was a minister of the Church of England.

Mary Burton, a servant, had been a witness in the earlier trials. A judge, Daniel Horsmanden, had promised her money and an end to her years of service. She had never mentioned Ury before. Now she said he was the leader of the plot

and had performed Catholic ceremonies for the plotters. Two more witnesses backed her. One was William Kane, a soldier Burton had accused, who went along to save himself. The other was Sarah Hughson, who was brought from the cell where she waited to be executed.

The trial. New York's attorney general, the colony's top lawyer, was a man named Bradley. At the trial he attacked the Catholic religion as "murderous" and read Oglethorpe's letter to the jury. Ury had no lawyer. His own witnesses said he was not a Catholic priest and that he had been elsewhere on the nights in question.

The jury took less than fifteen minutes to find him guilty. New York officials hanged John Ury on August 29, 1741. Before he died he said he was innocent and forgave the men who had prosecuted him. He was the last person executed for the supposed plot.

BY 1750

If you remember one thing: Traveling preachers set off the Great Awakening, and the argument over their new birth split churches into New Lights and Old Lights.

- In 1739: George Whitefield begins the preaching tour of the colonies that was the high point of the Great Awakening.
- In 1746: New Lights who backed the revival get a charter for a college of their own, now Princeton.

Next: After 1750 Americans argued over whether anyone should be taxed to pay for a church, and Black Americans born into slavery founded churches of their own.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 21 MINUTES

Americans spent these years arguing over whether the government should make people pay taxes to support a church. Before 1776, several colonies had an established church, one that the government supported with tax money. In Massachusetts it was the Congregational church. In Virginia it was the Church of England, also called the Anglican church, and people there had to attend its services as well as pay for its ministers.

Baptists and other people outside the tax-supported churches still had to pay those churches' ministers, and they organized against the church taxes. After the Revolution the lawmakers of each state decided for themselves whether to keep a church tax. The Baptists won in Virginia in 1786 and lost in Massachusetts, where the tax lasted until 1833. The First Amendment, the first of the changes added to the national Constitution in 1791, barred Congress, the national lawmakers, from setting up a national church.

New churches and new missions. Black Americans who had been born into slavery founded churches of their own in these years. On the Pacific coast, Spanish Franciscans built missions from 1769 on the lands of many Native nations. To keep people in the missions, missionaries and soldiers whipped them, locked them in the stocks, wooden frames that hold the feet or hands, and hunted down those who ran away.

Three more colleges with church backing

Three more colleges were founded with church backing in these years. Rhode Island College was founded in 1764 with the support of the Philadelphia Baptist Association, a group of Baptist churches, and it was renamed Brown University in 1804. Queen's College, tied to the Dutch Reformed Church, received its charter on November 10, 1766. It is now Rutgers University. Dartmouth College received its charter on December 13, 1769. Its founder was Eleazar Wheelock, a Congregational minister.

How Virginia ended its church tax

A parish was the area served by one church. In Virginia the men who ran each parish also cared for poor people and orphans, so they did the work of a local government as well as running a church.

The members of the Virginia assembly stopped the tax that paid Church of England ministers in 1779. Patrick Henry put forward a bill, a proposed law, for a new religious tax on December 24, 1784. Under his plan, each taxpayer could send his share to the church of his choice.

Some leading Virginians backed a religious tax. Richard Henry Lee wrote that "the experience of all times shows religion to be the guardian of morals." John Marshall, who later became Chief Justice of the United States, voted to move Henry's bill forward rather than put it off. George Washington wrote to George Mason that he did not object to "making people pay towards the support of that which they profess." He thought the bill would stir up trouble, though.

Madison's protest. James Madison led the other side. He got the vote put off until the autumn of 1785. On June 20, 1785, he published a long argument against religious taxes, without his name on it, called the *Memorial and Remonstrance*, a remonstrance being a formal protest.

Baptists and most Presbyterians were on his side. They argued that tax money spoils a church, and that a government has no right to control a person's religion. By the autumn of 1785 Madison had the votes, and the members of the assembly voted Henry's bill down.

Jefferson's law. In its place Madison brought back a bill that Thomas Jefferson had written in 1777. The members of the assembly passed it on January 16, 1786, and it became law three days later. It is called the Statute for Establishing Religious Freedom, a statute being a law. It states: "no man shall be compelled to frequent or support any religious worship, place, or ministry whatsoever."

Jefferson wrote that it was meant to protect "the Jew and the Gentile" (a person who is not Jewish), the "Mahometan" (an old word for a Muslim), the "Hindoo" (an old spelling of Hindu), and the "Infidel of every denomination," meaning people who believe in no religion at all.

Lawmakers in most states did not follow Virginia in these years. Those of Massachusetts, Connecticut and New Hampshire set up religious taxes that people had to pay. Maryland's and Georgia's lawmakers passed religious taxes but never collected them.

No religious test, and the First Amendment

The Constitution written in Philadelphia in 1787 says almost nothing about religion. Its one rule about religion is in Article VI. It states that "no religious Test shall ever be required" for a job in the national government. That meant no one could be made to swear an oath about religious beliefs before taking office. Charles Pinckney of South Carolina proposed the rule on August 20, 1787. The delegates at the convention, the meeting in Philadelphia that wrote the Constitution, adopted it on August 30.

Franklin's motion. Benjamin Franklin, who was eighty-one, asked the convention on June 28, 1787, to open each day with prayers. He told the delegates: "the longer I live, the more convincing proofs I see of this Truth." He went on: "that God governs in the Affairs of Men." The delegates did not approve his motion. According to reports of the time, the convention had no money to pay local ministers to lead the prayers.

Some people complained that the Constitution contained "no recognition of his mercies to us . . . or even of his existence," meaning God's. Many of the voters in Madison's part of Virginia were Baptists, and they wanted freedom of

religion protected in writing. On June 8, 1789, Madison introduced a set of amendments, or changes, to the Constitution. In his notes for that speech he wrote against a "national" religion.

The members of both houses of Congress voted on September 28, 1789, to send twelve amendments to the states. By December 1791 the lawmakers of three fourths of the states, the number needed, had approved ten of them. The first of the ten begins: "Congress shall make no law respecting an establishment of religion, or prohibiting the free exercise thereof." Those words mean that Congress may not set up an official church, and may not stop people from practicing their religion.

The last church tax

Massachusetts lawmakers kept their state's church tax for forty-two years after the First Amendment was approved. Article III of the state's 1780 constitution required every town and parish to pay for "public Protestant teachers of piety, religion and morality." It also allowed the lawmakers to order people to go and listen to those teachers.

If a man asked, his church tax went to the minister of his own church, as long as he went to hear that minister preach. If he had no such minister, his tax could go to the minister of the parish where the tax was collected.

The argument. Article III was the part of that constitution people argued about most. Phillips Payson, a Congregational minister at Chelsea, argued for it in a sermon on May 27, 1778. He warned that if "the restraints of religion" were "once broken down," nothing could "support and preserve order and government in the state." He meant that without religion, no one could keep order.

Isaac Backus, the leader of the Baptists of New England, answered him that year in a pamphlet. He argued that government money spoils churches, and that the best thing a government can do for religion is to leave it alone.

The Baptists lost. There was strong doubt that Article III had won the two thirds of the vote that it needed, and in 1780 the state's officials declared it adopted anyway.

Article III stood for fifty-three years. In 1833 a new amendment to the state constitution replaced it. Churches could still choose their ministers and agree to pay them, but they had to raise the money themselves. Any member could leave a church by handing its clerk a written notice. Massachusetts was the last state to end its church tax.

BY 1791: Virginia had ended its church tax and the First Amendment kept Congress from setting up a national church, but Massachusetts towns still had to pay Protestant ministers.

Black Baptists in Savannah

George Liele was born into slavery in Virginia in 1750. In 1775 he was ordained a Baptist minister, which means church leaders officially made him a minister, and he preached at plantations along the Savannah River. In 1778 he gathered a congregation of Black Baptists in Savannah, Georgia.

During the Revolution Liele told his followers to side with the British, who promised to treat African Americans as free people. When the British left their headquarters on Tybee Island, near Savannah, in 1782, many of Liele's followers went with them. They founded settlements and churches in Jamaica, Nova Scotia and Sierra Leone. Liele went to Jamaica.

Andrew Bryan kept the Savannah congregation going. He was born at Goose Creek, South Carolina, and was enslaved by a man named Jonathan Bryan. On January 20, 1788, in a barn at Brampton plantation, a white minister named Abraham Marshall and a Black minister named Jesse Peter baptized more than forty of Bryan's followers. They made the group a church and ordained Bryan as its minister. The church was named the Ethiopian Church of Jesus Christ.

Whipped for worshipping. Many white people in Savannah feared that any gathering of enslaved people could be the start of a revolt. A letter written from Savannah in 1790 by Jonathan Clarke describes what they did. White people of Savannah broke up the worship and took many of the worshippers before magistrates, local judges. The worshippers were jailed twice. About fifty

were severely whipped. Andrew Bryan was cut and "bled abundantly," which means he bled heavily. The letter does not say who used the whip or who gave the order.

Clarke wrote that Bryan said he "rejoiced not only to be whipped, but would freely suffer death for the cause of Jesus Christ." The chief justice, Henry Osborne, questioned the worshippers along with James Habersham and David Montague, and they released them. Another man, George Walton, said the treatment of the worshippers "would be condemned even among barbarians." Osborne then allowed them to worship between sunrise and sunset.

Bryan counted about 575 followers in 1790. Of those, 225 were full members. About 350 others were converts, and many of them did not have their enslavers' permission to be baptized. Bryan later bought freedom for himself and his family, and he bought land for the church. He died in 1812.

Richard Allen

Who: A preacher born into slavery who bought his freedom, walked out of a white-run church in Philadelphia, and founded the African Methodist Episcopal Church.

When and where: Born into slavery on February 14, 1760, held by a Philadelphia lawyer. Delaware, then Philadelphia. Died in Philadelphia, March 26, 1831.

A trustee pulled a praying man up off his knees.

Richard Allen founded Bethel Church in Philadelphia after a white trustee, one of the men who ran the church, pulled a Black member up off his knees during prayer. Allen was born enslaved by Benjamin Chew, a Philadelphia lawyer, who sold Allen and his family to Stokely Sturgis, a farmer near Dover, Delaware. Allen began preaching as a teenager, while he was still enslaved.

Buying freedom. In 1780 he bought his freedom for $2,000. He earned the money by working in his own time for about five years. The dollars were Continental money, paper money printed during the Revolution, which by 1780 had lost most of its value.

Philadelphia. He came to Philadelphia in February 1786. He wrote: "I frequently preached twice a day, at 5 o'clock in the morning and in the evening, and it was not uncommon for me to preach from four to five times a day." That year he gathered a group of forty-two members.

Absalom Jones, William White and Dorus Ginnings joined him in wanting a church for Black Philadelphians. The elder, the church leader in charge at St. George's Methodist Church, where they worshipped, forbade their prayer meetings. Allen wrote that the Black members "were considered as a nuisance."

The gallery. At St. George's, Allen wrote, the Black members were moved out of their usual seats and placed along the wall. He does not say who moved them. Then one Sunday the sexton, the church's caretaker, sent them up to the gallery, a balcony above the main floor. As they knelt there to pray, a trustee took hold of Absalom Jones and pulled him up off his knees.

Allen wrote down the words. The trustee said, "You must get up," and then, "you must not kneel here." Jones answered, "Wait until prayer is over, and I will get up and trouble you no more." The trustee said, "No, you must get up now, or I will call for aid and force you away." A second trustee went to pull up William White.

Allen wrote: "By this time, prayer was over, and we all went out of the church in a body, and they were no more plagued with us in the church." He added that the Black members had helped pay for the church, "in building the gallery and laying new floors."

When it happened. The date of the walkout is disputed, and Allen's own account gives none. The traditional date is 1787. The historian Richard S. Newman argues for 1792 or 1793, which would make it a planned protest rather than a sudden one. The Encyclopedia of Greater Philadelphia places it "in the early 1790s."

The group rented a storeroom for worship. Allen wrote that they were threatened with being "read publicly out of meeting," which meant expelled in front of the whole church, if they kept worshipping there. Benjamin Rush and Robert Ralston gave money for a new church. On their first day collecting money, Allen and Jones raised $360. Allen called it "the greatest day's collection that we met with."

Bethel. Allen bought a lot at Sixth and Lombard Streets in 1791. The first building was an old blacksmith's shop, and he opened Bethel Church there in July 1794. In 1799 he became the first Black man ordained a Methodist minister.

In Philadelphia in 1816 he brought together African Methodist churches from Pennsylvania, New Jersey, Delaware and Maryland. They formed a new church body, the African Methodist Episcopal Church, and elected Allen its first bishop, a leader over many churches. He died at his home on March 26, 1831, and was buried in the basement of Mother Bethel, as his church came to be called.

Absalom Jones

Who: A man born into slavery who taught himself to read and became the first priest of African descent in the Episcopal Church in the United States.

When and where: Born in Delaware, November 6, 1746. Philadelphia from 1762. Died 1818.

When he was sixteen, his mother, sister and five brothers were sold.

Absalom Jones was the first priest of African descent in the United States. He was born into slavery in Delaware and taught himself to read, and the New Testament was one of the books he learned from. When he was sixteen, his mother, his sister and his five brothers were sold. The records do not name who sold them. The man who enslaved Jones took him to Philadelphia in 1762, and there he went to a Quaker night school for Black people.

He bought his freedom in 1784. In 1787 he and Richard Allen started the Free African Society, which worked to end slavery and helped widows, orphans and poor people. In 1792, with help from local Quakers and Episcopalians, Jones and Allen set up the First African Church in Philadelphia.

St. Thomas. After the walkout from St. George's, Jones and Allen went separate ways. Jones led the congregation into the Episcopal Church, while Allen stayed a Methodist and opened Bethel Church in 1794. On October 17, 1794, Jones's congregation was formally received into the Episcopal Diocese of Pennsylvania, the group of churches under that state's bishop, as the African Episcopal Church of St. Thomas. Jones was ordained a deacon, a minister one rank below a priest, in 1795.

Bishop William White, a different man from the William White who walked out of St. George's, ordained Jones, then fifty-six, a priest. The Episcopal Church's own archives give the year as 1802, and Delaware's

> state historical marker gives 1804.
>
> Jones also started a day school, a Female Benevolent Society and an African Friendly Society. In 1800 he sent a petition, a written request, to Congress about slavery. He died in 1818.

Christian Lenape killed at Gnadenhutten, 1782

American militiamen, part-time soldiers from the local area, killed about 96 Christian Lenape and Mohican people at Gnadenhutten, in the Tuscarawas Valley of Ohio, on March 8, 1782. Chapter 22, War, tells the killing in full. The people killed were converts of the Moravian church and lived in Moravian mission villages. They had refused to fight for either side in the Revolution. Colonel David Williamson led between 100 and 200 militiamen.

The militiamen falsely accused the Christian converts of taking their goods in raids. They held them in houses and voted to kill them. Williamson asked the militiamen who wanted to spare the prisoners to stand apart, and only sixteen or eighteen did. The prisoners, knowing what was coming, prayed and sang hymns.

The militiamen tied them up and killed them by smashing their heads with a mallet, one after another, taking turns as each man got tired. Then they cut off their scalps, the skin and hair of the top of the head. They killed 62 adults and 34 children. Two boys escaped. One of them, named Thomas, lived after he was scalped.

The California missions

Spanish soldiers and Franciscan friars began building missions in Alta California, which is now the state of California, in 1769. Spain's officials wanted to hold the Pacific coast against Russia and England, and they had little money. They sent missionaries to live on Native homelands because it was a cheap way to claim the land. Soldiers and missionaries came together. The missionaries' job was to make Native people Catholics and to change their whole way of life.

Father Junípero Serra led the Franciscans. He was born in 1713 at Petra, on the island of Mallorca, and he had worked as a missionary in Mexico for seventeen years. He founded Mission San Carlos Borromeo at Monterey in 1770 and lived there until he died in 1784. By the end of 1800 the Franciscans had founded eighteen missions. By 1823 there were twenty-one.

Whose land. About 310,000 Native people lived in California in 1769. Each mission was built in the homeland of a people with its own government and language. By 1800 the missions stood on the lands of more than a dozen peoples, among them the Kumeyaay at San Diego, the Tongva at San Gabriel, the Chumash at Santa Bárbara and La Purísima, and different Ohlone peoples around Monterey Bay and San Francisco Bay.

Baptism and force. Serra's writings describe a plan to baptize young children first. He expected the children to learn Spanish and translate for their parents. Edward Ketchum, a tribal historian, has said that once a child was baptized, the child was the "property" of the church.

Researchers wrote a report on the missions for the state of California in 2023, with advisers from 19 California tribes. Drawing on the scholars Cook and Madley, it describes persuasion in the early years and more force after about 1790 to 1800. Native accounts make clear that force was used in many baptisms. When Native people kept resisting, the missionaries called in soldiers, who captured runaways and people who had not been baptized.

Missionaries punished baptized Native people to keep them from leaving the missions or to force them to come back. They punished people for running away, stealing, fighting and drinking, and for other acts they called disobedience. The missionaries sometimes gave the punishments themselves. They also had soldiers or Native officials deliver them. Soldiers patrolled the countryside around the missions and punished people there too.

Whipping. The 2023 report describes flogging, beating a person with a whip that cuts and bruises the skin, as "probably the most common" form of violence at the missions. It was done in the open, in front of others. Some Franciscans, Serra among them, whipped themselves as a religious practice, so whipping seemed to them a fair punishment for others.

Serra wrote to the military commander Fernando de Rivera y Moncada on July 31, 1775, about four men who had run from Mission Carmel. He asked to have them whipped "two or three" times. He offered to send shackles, iron rings locked around the legs or wrists. The journalist Elias Castillo, who has written against the missions, quotes the letter from Serra's collected writings.

Missionaries and soldiers used other punishments too. People were locked in the stocks, sometimes for days. People were hobbled, which means their legs were tied so they could only shuffle, and they were made to keep working. Others were beaten with a cudgel, a heavy club, or jailed and put to hard labor. For plotting, murder or armed rebellion, some people were executed. Soldiers raped Native women at missions including Soledad, San Luis Obispo, San Diego and San Gabriel.

Baptisms and deaths. The missionaries recorded 85,840 baptisms of Native people at the 21 missions from 1769 to 1834, and in the same years they recorded 59,538 Native deaths. That is about seven deaths for every ten baptisms. The death records for two missions are missing, so the real number of deaths is higher. These counts come from the Early California Population Project at the Huntington Library, and they run past 1800.

Other record collections cover different years and people, so their totals differ. One gives about 101,000 baptisms and 71,000 deaths. Another gives 95,493 baptisms and 56,776 deaths. The 2023 report lists the causes of death. They

include forced moves, hard labor, the banning of Native ways of life, poor living conditions, violence, and new diseases brought in from outside.

The Kumeyaay attack on Mission San Diego, 1775

Kumeyaay fighters burned Mission San Diego on the night of November 4 to 5, 1775. They came from fifteen of the twenty-five villages nearest the mission. Spanish soldiers estimated that 600 to 1,000 people took part.

Why they attacked. The attackers left no account of their own, but the Spanish records list what the Kumeyaay had been complaining about. From 1769 to the end of 1774 the friars had baptized only 100 people, and then they pushed much harder. On October 3, 1775, the friars baptized sixty people in a single day.

Spanish soldiers had raped Kumeyaay women, and at least one of those women was also murdered. Father Luis Jayme, a friar at the mission, had reported the rapes. Spanish cattle had destroyed village land. Writers at the Smithsonian's National Museum of the American Indian list the Spanish destruction of Native food sources and the Spanish taking of Native land among the causes. Disease was also spreading.

The attackers burned the mission and took goods from the church. They killed three people: Father Jayme, and a carpenter and a blacksmith from Mexico. They shot Jayme with an arrow and then beat him to death.

After the attack, Spanish soldiers took violent revenge. They went through the villages in force and singled out the people they believed had led it. The records give no number of the people they punished or killed, and the 2023 report does not say what the soldiers did to the people they singled out. The mission was rebuilt.

Toypurina

Who: A Tongva woman, often called a religious leader, who helped plan an attack on Mission San Gabriel.

When and where: Japchivit and Mission San Gabriel, near what is now Los Angeles, 1785.

The soldiers were warned and arrested twenty-one rebels before the attack.

Toypurina was a Tongva woman from the village of Japchivit who had never been baptized. She had influence and standing in her community, and writers often call her a shaman, a religious or spiritual leader. In October 1785 she helped plan an attack on Mission San Gabriel. Soldiers at the mission were warned, and they arrested twenty-one of the rebels before the attack began.

The plan. The plan started with Nicolás José, a baptized Tongva man. He had lost a son and two wives. He asked Toypurina to call together the leaders of the other villages. One of those village leaders carried her message into the mission. People from the mission and from up to eight Tongva villages joined the plan.

When they were questioned, Nicolás José and Toypurina gave their reasons. Nicolás José said that the ban on dancing kept him from holding the ceremonies of mourning for the dead. Toypurina said she was angry with the Spanish settlers. She was also angry with the Native people who had moved to the mission, which stood on her land.

All of the leaders were held in jail for two and a half years. Then Spanish officials freed two of the village leaders with a warning. They exiled Nicolás José and Toypurina, sending them away from their homes.

NOT KNOWN: the year of her birth and the rest of her life. The records of the plot do not give them.

BY 1800

If you remember one thing: Americans argued over whether anyone should be taxed to support a church, and in Virginia the side against church taxes won.

One number: 59,538 Native deaths recorded at the California missions from 1769 to 1834, against 85,840 baptisms. That is about seven deaths for every ten baptisms, and the real count of deaths is higher.

Next: After 1800 revivals spread across the whole country, and Protestant mobs turned on Catholics and Latter-day Saints.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 28 MINUTES

Protestant revivals were held in almost every corner of the country between 1800 and 1850, so many that one religious magazine stopped reporting them in 1827. Its editors wrote that revivals "are becoming too numerous in our country to admit of being generally mentioned in our Record." In 1833 Massachusetts lawmakers stopped using tax money to pay for churches, and no other state still did.

The kind of Protestant faith that spread in these years is called evangelicalism. Evangelical preachers called on each listener to be converted, to turn to God and become a true believer, and they called this change the "new birth." Converted people started societies to hand out Bibles and religious papers, teach children, and send out missionaries. Some of the societies worked to get people to stop drinking, because their members believed a sober person was readier to be converted.

A Library of Congress exhibit lists the six largest of these societies in 1826 and 1827, among them the American Sunday School Union and the American Tract Society.

Mobs and splits. In the same fifty years, Protestant mobs burned a school run by Catholic nuns near Boston and Catholic churches in Philadelphia, and a Missouri governor ordered the members of a new church driven from his state. Methodists and Baptists in the South broke away from their national churches over slavery and formed churches of their own. In California the whippings at the missions went on, and the Mexican officials who took the missions away from the friars often whipped people too.

Camp meetings

Big revivals began at both ends of the country in 1800, in New England and in Kentucky. A Library of Congress exhibit calls the New England revival "decorous," which means calm and proper, and the Kentucky revival "exuberant," which means full of wild feeling. In Kentucky, Presbyterian ministers started a new kind of gathering, the camp meeting.

The ministers copied the long outdoor "communion seasons" of the Church of Scotland, gatherings for communion, the church service in which believers share bread and wine in memory of Jesus. Families loaded wagons, drove to the meeting, pitched tents, and stayed for several days of preaching.

Cane Ridge. One camp meeting was held at Cane Ridge, near Lexington, Kentucky, in the summer of 1801. A Presbyterian minister named Barton Stone organized it. In 1832 Stone and Alexander Campbell left the Presbyterians and formed a new church, the Disciples of Christ.

Lorenzo Dow was a traveling Methodist preacher. While he preached, people in his crowds often jerked their bodies about, which was called the jerking exercise. The Presbyterians and Baptists who had started camp meetings soon turned against them because of the strong feelings they stirred in people. The Methodists kept them and carried them east, and for decades camp meetings were a Methodist practice.

A drawing from 1809. Benjamin Henry Latrobe drew a plan of a Methodist camp meeting in Fairfax County, Virginia, on August 8, 1809. His drawing shows the men's seats apart from the women's and the tents of Black worshippers apart from the tents of white worshippers. At one side of the

camp, the organizers had built a pen of boards filled with straw. Latrobe wrote that "the converted were thrown" into it "that they might kick about without injuring themselves."

A Library of Congress exhibit calls this drawing an example of separation by race. That separation led Black Methodists, a few years later, to leave and form a Methodist church of their own.

Mother Ann Lee, who founded the Shakers, came to America from England in 1774. The Shakers' full name was the United Society of Believers in Christ's Second Coming. They danced in worship, often in circles inside circles, and got their nickname from shaking and whirling during services. In the early 1800s Shakers from New York went to Kentucky to see the revivals, and they gained a small number of converts there.

Sunday schools taught reading

The first Sunday schools taught poor children to read. Sunday schools began in England in the 1780s. In 1791 a group of pastors and merchants in Philadelphia started the First Day Society. Its members paid schoolmasters to teach boys and girls, and grown men and women who could not read, to read and write. The Bible was the textbook. Sunday was the day poor working children were not at work, and there were no free public schools yet.

A group of Philadelphia businessmen started the Sunday and Adult School Union in 1817, with volunteer teachers instead of paid ones. Within a year they had opened forty-three Sunday schools in the city, with 5,970 pupils. The group grew into the American Sunday School Union. Its leaders hired missionaries who started Sunday schools in cities, in the countryside and on Native reservations, and it became a large publisher of children's books.

Tracts. The American Tract Society began in 1825 to print tracts, short papers about religion. A Library of Congress exhibit gives an estimate that its members handed out 35 million books and tracts in its first ten years. Its leaders printed papers that urged readers to be converted and to give up drinking and gambling, which the leaders believed kept people from being converted.

Rebecca Gratz started the first Hebrew Sunday School, a Sunday school for Jewish children, in Philadelphia in 1838.

The African Methodist Episcopal Church grows

The African Methodist Episcopal Church, which Richard Allen and other Black Methodists formed in 1816, grew fast. It began with 8 ministers and 5 churches. By 1846 it had 176 ministers, 296 churches and 17,375 members.

Women preached in the church from its early years. Many people objected to women preaching, in the A.M.E. Church as in the white churches. An engraving made in Philadelphia in 1844 by P. S. Duval, from a painting by Alfred Hoffy, shows one of these women: Juliann Jane Tillman, a preacher of the A.M.E. Church.

The convent burned at Charlestown, 1834

Some Protestants in these years turned to violence against Catholics. In Charlestown, Massachusetts, near Boston, Ursuline nuns, Catholic women who make religious promises and live together in a convent, ran a convent school from 1819. The Ursulines are an order of nuns who teach, and their students were girls.

The stories. Stories about the convent spread in Boston. Rebecca Reed, a Protestant woman, had entered the convent in 1832 and left after less than six months. Her stories about life inside went around before her book, *Six Months in a Convent*, came out in 1835.

A nun named Sister Mary John, whose name before she joined the order was Elizabeth Harrison, left the convent on July 28, 1834. Bishop Fenwick, the church leader in charge of all the Catholic churches of the area, talked her into going back. Rumors spread that she was being held against her will.

The Protestant minister Lyman Beecher preached against Catholic schools in three Boston churches on Sunday, August 10. The next day a group of Charlestown officials toured the convent, and Sister Mary John guided them. Historians disagree about whether Beecher's sermons led to what happened

that night. Ira Leonard argues that they did. Ray Billington points out that the workmen had met before the sermons. Beecher himself said that probably no rioter had heard him.

The fire. A mob of mostly Protestant workmen from Charlestown burned the convent school on the night of August 11 to 12, 1834. The nuns and 47 girl students were inside when the mob came, and only six of the girls were Catholic. The mob also broke open the convent's tomb, where several dead nuns were buried. Firemen stood by and watched. No deaths are recorded.

Police arrested 13 men. Juries found 12 of them not guilty, including John R. Buzzell, who admitted he had led the mob. The one man found guilty was 16 years old, and Governor John Davis pardoned him, which means he freed him from his punishment.

Whose Bible? Philadelphia, 1842 to 1844

Pennsylvania's lawmakers passed a law in 1838 that made the King James Bible, the Bible that Protestants used, a required textbook in the public schools. In Philadelphia, Protestant teachers also led their pupils in Protestant hymns, or church songs.

The bishop's letter. Bishop Francis Patrick Kenrick, the Catholic bishop of Philadelphia, wrote to the board that ran the city's public schools in 1842. He objected to the hymns and to the Bible readings. He asked that Catholic children be allowed to read the Douay Bible, the translation Catholics use, and that they not be made to join in Protestant worship.

Early in 1843 the board members agreed that children could be excused from the King James Bible and could use any Bible "without note or comment." The Douay Bible has notes and comments. Under the board's rule, the one Bible the Catholic bishop had asked for was still not allowed.

The rumor. Hugh Clark, a Catholic school director in the Kensington neighborhood, suggested in February 1844 stopping Bible reading until the board could write a rule both sides accepted. A rumor spread that Clark had ordered a teacher, Miss Bedford, to stop reading the Protestant Bible. Witnesses later showed that the rumor was false.

Nativists, people who opposed Catholic immigration and saw Catholics as a danger to the Protestant faith, spread it anyway. A newspaper editor named Lewis Levin had founded their group in Philadelphia, the American Republican Association, in December 1843.

The May riots. The nativists held a rally on May 3, 1844, and marched into Kensington, where Irish Catholic families lived. On May 6 a nativist named George Shiffler was shot dead. By the end of that day a second man, who seems to have been a bystander, had been shot dead. Two more nativists were shot that day and later died of their wounds. The Encyclopedia of Greater Philadelphia does not name who shot any of them.

Nativist crowds burned St. Michael's Church, St. Augustine's Church and Irish Catholic homes, including Hugh Clark's house, on May 8. They also burned a Catholic seminary, a school for training priests.

The July riots. Fighting started again in July, at the Church of St. Philip de Neri in the Southwark neighborhood. State militia, part-time soldiers called up by the state, fought the crowd with cannon under General George Cadwalader. By the morning of July 7, four militiamen and probably about a dozen rioters were dead, and many more were wounded.

Counts of the dead in the two riots differ. Counts for May run from at least 14 dead to 20, and one count gives at least 100 wounded. Counts for July run from at least 14 dead to about 16.

Who was blamed. A grand jury, a group of citizens who decide whether there is enough proof for a trial, charged 19 rioters with crimes, and only a few were found guilty. The grand jury blamed the May riots on "the efforts of a portion of the community to exclude the Bible from the public schools." The jurors meant the Catholics who had been attacked.

Catholics answered that the grand jury had called no Catholics to testify. After the riots, Catholics decided to build a school system of their own.

BY 1844: Protestant mobs had burned a convent school near Boston and two Catholic churches in Philadelphia, and Catholics had decided to build schools of their own.

The Latter-day Saints driven out of Missouri

Joseph Smith founded the Church of Jesus Christ of Latter-day Saints on April 6, 1830, at a farmhouse near Fayette, in western New York. Its members were often called Mormons. In 1831 Smith said that God had chosen western Missouri as the place where his people would gather. Church members moved there, and their Missouri neighbors drove them out twice.

Jackson County, 1833. The first time was in Jackson County in 1833. The older settlers feared that the newcomers would control the county's politics and trade, and they objected to the newcomers' religious beliefs. An article in the church newspaper about free Black converts coming to Missouri, a state that allowed slavery, also angered them.

A mob attacked the church's printing office on July 20, 1833, threw the press out of the window, and scattered the type in the street. They tarred and feathered Edward Partridge, the church's bishop in Missouri, and Charles Allen in the public square. In tarring and feathering, men tore off a person's clothes and smeared hot pine tar on the bare skin, which burned it. Then they covered the person with feathers, which stuck to the tar. The church's account does not describe the injuries the two men suffered.

Mobs drove the church members out of the county in November 1833. Three people were killed on November 4, one of them a church member, and the church's account does not say who killed them.

Caldwell County, 1836 to 1838. The Missouri lawmakers created Caldwell County in 1836 as a place for church members to live. More converts kept arriving and settled beyond its borders. In 1838 both sides used violence. The Missouri State Archives states that "Old Settler mobs and Mormon paramilitary units roamed the countryside," paramilitary units being armed groups organized like an army.

Some Latter-day Saint men formed the Danites, a group of vigilantes, people who act as their own police and punish others outside the law. In the middle of October, Mormon forces raided the towns of Gallatin and Millport and burned property there. Shortly before the end of October, at Crooked River, church

members attacked a unit of the state militia that they took for a mob. One Missourian and two church members were killed. One of the dead was David W. Patten, an apostle, one of the church's highest leaders.

The governor's order. Governor Lilburn Boggs issued an order on October 27, 1838. It said the Latter-day Saints must be driven from the state or "exterminated," killed every one, if necessary.

Haun's Mill. Three days later, on October 30, between about 200 and 240 armed men attacked the small church settlement at Haun's Mill at about 4 in the afternoon. Colonel Thomas Jennings of Livingston County commanded them, and Captain Nehemiah Comstock led one of the companies.

Boggs had not ordered this attack. The Missouri State Archives says it happened "even without official sanction," which means without the governor's approval. The historian Alma Blair writes that the governor's order "was now widely known."

Many of the Mormon men took cover in the blacksmith shop, and the attackers shot them through the cracks in its walls. The attackers then hacked some of the wounded men to death with corn knives, long blades used for cutting cornstalks. One of them shot Thomas McBride with McBride's own rifle as he surrendered, and then the attackers hacked him.

They killed seventeen or eighteen men and boys. The Missouri State Archives gives eighteen, and Blair and the church's history give seventeen. Among the dead was Sardius Smith, age 10. Charles Merrick, age 9, died of his wounds five weeks later. One woman was wounded. No record shows that anyone was ever tried for the killings at Haun's Mill.

General Samuel D. Lucas arrested Joseph Smith and other church leaders on October 31. Over the next year about 8,000 church members, driven out under the governor's order, went from Missouri to Illinois. The Missouri State Archives describes them as "often ragged and deprived of their property." Governor Boggs's order stayed in force until June 25, 1976, when Governor Bond canceled it.

Joseph Smith

Who: The founder of the Church of Jesus Christ of Latter-day Saints, whose followers were driven from state to state. A mob killed him in 1844.

When and where: Lived 1805 to 1844. Western New York, Ohio, Missouri, and Nauvoo, Illinois. Killed at Carthage, Illinois, on June 27, 1844.

About 100 men broke into the jail at Carthage, Illinois.

Joseph Smith founded a new American church in 1830, and a mob killed him fourteen years later. He grew up in a part of western New York that people called the Burned Over District, because so many revivals had passed through it. A Library of Congress exhibit states that his church did not split off from an older church, and it describes the church as the only one of its kind.

The Book of Mormon. E. B. Grandin of Palmyra, New York, published the Book of Mormon for Smith in 1830. Smith said the book came to him through revelations, messages he said came from God, on a set of gold plates written in a language he called reformed Egyptian. He said he had translated it. Eleven men signed statements saying that they had seen and handled the plates.

The book tells of Israelites, the people of ancient Israel in the Bible, who were led out of Jerusalem to the Americas about 600 years before the birth of Jesus. It also tells of Jesus appearing to their descendants in America. Smith's followers believed his church brought back the earliest form of Christianity. Church members accept the Book of Mormon as scripture, a holy book, alongside the Bible.

Nauvoo. The church moved from New York to Ohio, from Ohio to Missouri, and from Missouri to Illinois. In 1839 church leaders bought land on the Mississippi River in Illinois, at a nearly empty town called Commerce, and they renamed it Nauvoo in 1840. A large Meskwaki village had stood there, and accounts of how its people lost it differ.

The Illinois Department of Natural Resources runs the state park at Nauvoo. Its page about the park states that the site was signed away in a treaty in 1824, but it does not name who signed it. Another account states that a man named James White bought it from the Sauk leader Quashquame for liquor and corn, and that Quashquame's people moved across the river.

In Illinois, trouble grew again. Neighbors resented the church members' business dealings and the way they all voted together. Some accused them of theft. They were angry about Smith's plans in politics. News spread that some church leaders had more than one wife, and church members also argued among themselves about it.

Carthage. Men who opposed Smith started a newspaper in Nauvoo, and Smith ordered its printing press destroyed. He was jailed at Carthage, Illinois, for it. On June 27, 1844, about 100 men broke into the jail and killed Smith and his brother Hyrum. The men came from three units of the Illinois militia: the 59th Regiment, the Warsaw Independent Battalion and the Warsaw Rifle Company.

The militiamen guarding the jail barely resisted. Accounts of the killing differ. A Library of Congress exhibit states that Smith was "apparently" shot, and that one man in the mob came at him meaning to cut off his head and was stopped.

The trial. Nine men were charged with the murders in October 1844. Four ran away and were never arrested. Five were tried in May 1845: Thomas C. Sharp, the publisher of an anti-Mormon newspaper, and four militia officers, Levi Williams, Mark Aldrich, William N. Grover and Jacob C. Davis, who was also a state senator. The jury found all five not guilty.

The prosecutor, Josiah Lamborn, said of Davis: "I have no doubt in my own mind, not a particle, that Davis cooperated in the murder, but there is no legal evidence to convict him."

> After Smith's death, men attacked Mormon farms around Nauvoo. Brigham Young, who led the church after Smith, planned to move about 15,000 people west. On February 4, 1846, the first wagons crossed the Mississippi into Iowa.

The religion enslavers taught

In the South, enslavers did not let enslaved people attend the regular preaching with them on working days. Charles Colcock Jones was a Presbyterian minister in Liberty County, Georgia. In 1842 he wrote that enslaved people "were not permitted, on working days, to attend the public administration of the word in company with their masters." By "the public administration of the word" he meant the regular preaching. Where they did attend, part of the white church was set apart for them.

Jones wrote a book to urge planters and ministers to have Christianity taught to the people they held in slavery. In *The Religious Instruction of the Negroes in the United States*, finished on July 4, 1842, Jones told ministers which parts of the Bible to use. ("Negro" was the common word for a Black person in his time, and it is no longer used.)

He chose verses about obedience. One verse says "servants" should "count their masters worthy of all honor," and Jones used it to teach enslaved people to obey. He wrote that a minister should teach them "the duties of their station," meaning their duty to obey.

Frederick Douglass. Frederick Douglass, who had been enslaved in Maryland, wrote down what this religion looked like to the people it was used on. Thomas Auld, the man who held Douglass, was converted at a Methodist camp meeting in Talbot County, Maryland, in August 1832. Douglass hoped Auld would free him, or at least treat him better. "I was disappointed in both these respects," he wrote. "I believe him to have been a much worse man after his conversion than before."

Douglass wrote that after his conversion, Auld "found religious sanction and support for his slaveholding cruelty," sanction meaning approval. Auld became a leader in his church, and visiting preachers stayed at his house. Douglass

wrote: "while he starved us, he stuffed them."

At St. Michael's, Maryland, a young white man named Wilson started a Sunday school to teach enslaved people to read the New Testament. They met three times. Then Garrison West and Wright Fairbanks, two class leaders, each in charge of a small group of church members, came with many other men carrying sticks and other things to throw. The men drove the students away and forbade them to meet again.

Henny. Douglass saw Auld whip Henny, a young woman whose hands had been burned so badly in a fire when she was a child that she could not use them. Auld tied her up and whipped her bare shoulders with a heavy cowskin, a whip made of cowhide, "causing the warm red blood to drip." To justify it, he quoted the Bible: "He that knoweth his master's will, and doeth it not, shall be beaten with many stripes."

He left her tied up for four or five hours at a time. He whipped her before breakfast, went to his store, and came back and whipped her again at dinner "in the places already made raw." At last, in Auld's own words, he "set her adrift to take care of herself."

Douglass also named one minister, George Cookman, whom the enslaved people loved and believed was working to free them.

What Douglass was against. In 1845 Douglass added a note to the end of his book to explain that he was not against religion. He was against what he called "the slaveholding religion of this land." He wrote: "The man who wields the blood-clotted cowskin during the week fills the pulpit on Sunday." He wrote that the man who called it a religious duty to read the Bible denied him "the right of learning to read the name of the God who made me."

The churches split over slavery

Members of two of the largest Protestant churches in the country divided them over slavery more than fifteen years before the Southern states left the Union.

The Methodists. Bishop James Osgood Andrew of Georgia held enslaved people, whom he had gained through his marriage. Methodist leaders met for their national meeting in 1844. The delegates, the men chosen to speak and vote for their churches, voted to stop Andrew from working as a bishop for as long as he held enslaved people.

A few days later the delegates adopted a Plan of Separation. Under the plan, the Methodist conferences in the slave states, the regional groups of churches there, could form a church of their own. Delegates from those conferences met at Louisville, Kentucky, in 1845 and created the Methodist Episcopal Church, South.

The Baptists. The Baptists split the same year. Georgia Baptists put forward an enslaver, James E. Reeve, to be a missionary, and Northern Baptists refused to appoint him. From May 8 to 12, 1845, Baptists from the South met at Augusta, Georgia, and formed the Southern Baptist Convention. William Bullein Johnson was its first president.

BY 1845: Missouri's governor had ordered the Latter-day Saints driven out, a militia mob had killed their founder in Illinois, and Southern Methodists and Baptists had formed churches of their own over slavery.

Charles Grandison Finney

Who: A young lawyer who gave up the law after his conversion and became a revival preacher. His book on revivals became a handbook for other revival preachers.

When and where: Born in Warren, Connecticut, August 29, 1792. Upstate New York, then Oberlin, Ohio. Died 1875.

He told a client he now worked for "the Lord Jesus Christ."

Charles Grandison Finney taught that a person could choose to be converted, and he built new methods of preaching on that idea. He grew up in Oneida County, New York, which was then, in his words, "to a great extent, a wilderness." His parents did not belong to a church. He wrote that there were few religious books where he grew up and that he "seldom heard a sermon."

The retainer. He was training to be a lawyer at Adams, New York, when he was converted in 1821. The next morning a client came to ask whether he was ready to argue the client's case in court. The client was a deacon, an officer of his church. Finney answered with a lawyer's word, retainer, the money paid to a lawyer to take a case: "Deacon B, I have a retainer from the Lord Jesus Christ to plead His cause, and I cannot plead yours."

The anxious seat. He preached across upstate New York. From September 1830 he spent six months in Rochester. He preached every day of the week and three times on Sunday, and he spent most of each day praying with people one at a time. In Rochester he began using what was called the anxious seat, a set of seats at the front of the church where people who felt worried about their souls came and sat, in front of everyone.

Finney explained that people of the "higher classes" were "too proud to take any position that would reveal them to others as anxious for their souls." He wanted "something that would commit them publicly to the service of Christ."

His methods were called the New Measures. Finney taught that a person could decide to be converted, and his meetings stirred strong feelings in the people who came. Both things angered other ministers. Asahel Nettleton, a New England revival preacher, objected that Finney had changed the older Presbyterian teaching on how God saves people. Lyman Beecher feared that Finney's meetings let feelings run wild.

As early as July 1827, before Rochester, ministers had met at New Lebanon, New York, to look closely at the New Measures. Finney's *Lectures on Revivals*, published in 1835, became a handbook for revival preachers across the country.

Oberlin. Oberlin College in Ohio was close to shutting down in 1835. Arthur Tappan, a businessman in New York who worked to end slavery, agreed to rescue it on two conditions. Finney had to be made its teacher of theology, the study of religious belief. And the college had to admit Black students on the same terms as white students.

Finney took the job and also became minister of the Oberlin church. He was president of the college from 1851 to 1866 and stayed on as minister until 1872. He taught that the sign of God's grace in a person "was a person's benevolence toward others," meaning kindness and help given to other people. He died in 1875.

The California missions, 1800 to 1848

The California missions, run by Franciscan friars, went on through these years, and so did the whippings. Missionaries had Native people whipped for running away, stealing, fighting and drinking, for living with a partner without a church marriage, and for other acts that mission officials counted as disobedience. They often had soldiers or Native officials give the whippings, and some of them whipped people themselves. Father de la Peña did at Mission Santa Clara.

Santa Cruz, 1812. A 2023 report on the missions for the state of California says that at Mission Santa Cruz, Father Andrés Quintana used a whip tipped with metal for floggings, beatings on the bare back that cut and bruise the skin. In 1812 he had just beaten two men nearly to death with it when a group of Awaswas Ohlone people from the mission killed him. The records name five of them: Lacah, Yaquenonsat, Yachacxi, Ules and Lino.

For two years his death was taken to be a natural one. Then a guard overheard an argument, and the plan came out. Spanish officials tried sixteen men and found nine guilty. Seven of the men died while held at the presidio, the Spanish fort, four of them before they were sentenced. The story of the plan comes from Lorenzo Asisara, an Ohlone man whose father took part in it.

Punished for having no children. Diseases brought by the Spanish made Native women less able to have children, or made them lose their babies before birth. Some missionaries believed that women did this on purpose. Lorenzo Asisara told what Father Olbes did at Santa Cruz to a married couple who had no children. Olbes demanded to inspect the couple's genitals, the parts of the body used to make babies.

When the woman resisted, he had her flogged and made her stand in front of the church for nine days holding a wooden doll. Olbes also had her husband put in shackles, iron rings locked around the legs or wrists, and made him stand in front of the church wearing horns. At Mission San Gabriel, Father Zalvidea punished women whose babies were born dead. He had their heads shaved, made them carry a wooden doll, had them flogged, and kept them in iron ankle cuffs for three months.

The Chumash rising, 1824. Native people fought back. In February 1824 Chumash people rose against three missions, Santa Inés, La Purísima and Santa Bárbara, in the largest planned revolt in California in the mission years. It began at Santa Inés on February 21, a day earlier than planned, after soldiers whipped a Native man from La Purísima who had come to visit a relative in jail. The Chumash attacked the soldiers and burned mission buildings.

The next day troops from the Santa Bárbara fort burned the adobe houses, made of sun-dried mud bricks, where the rebels were camped. Up to nineteen Chumash people and one Mexican soldier were killed at Santa Inés. At Santa Bárbara the Chumash leader Andrés Sagimomatsee called people to fight. At least two Chumash were killed there, and more than 1,000 people fled the mission for the inland country of the Yokuts people.

At La Purísima, led by Pacomio, the rebels held the mission for almost a month. One Chumash man and four settlers were killed when the rebels took it. In March troops took it back and killed 16 Chumash people. Mexican officials then had seven Chumash men executed for killing the four settlers. They sentenced four leaders to ten years in prison followed by exile from their homeland, and eight more men to eight years.

Estanislao and Yozcolo. Hundreds of Yokuts people left Missions San José and Santa Clara in 1828 and 1829 under two mission leaders, Estanislao and Cipriano. They built a fort in the marshes of the San Joaquin River. Mexican officials sent soldiers, Native fighters and armed settlers against them, and these forces defeated them in a battle. The records give no count of the dead and do not name the commander. Estanislao later bargained for a pardon.

A man named Yozcolo freed the young women who were locked in the women's building at Mission Santa Clara in 1839, and fled to the hills. Soldiers defeated him. The records do not say who cut off his head. The Mexican officials there set it outside the mission church for people to see.

The end of the missions. Mexicans won their independence from Spain in 1821. Mexican officials then began to take the missions away from the Franciscans and turn them into ordinary parish churches, the local churches of

a town. This was called secularization, and Governor José Figueroa put it in place in 1833 and 1834.

The officials who took over the missions often used punishments like the friars' punishments, including whipping. At most missions, colonists got most of the land and supplies. Under Figueroa's rules, Native people had no right to their own homelands. In 1845 officials passed the final law ending the missions.

The historian Cook estimated that about 310,000 Native people lived in California before 1769, give or take about a tenth. By 1845, by his estimate, their number had dropped by about half, to no more than 150,000. That count includes all Native Californians, not only the people held at the missions.

BY 1850

If you remember one thing: Mobs burned Catholic churches and drove out the Latter-day Saints, and almost none of the attackers were ever found guilty.

A question for the next era: would any tax money help pay for the schools Catholics planned to build?

Next: Between 1850 and 1900 the numbers of Catholics and Jews grew fast, slavery ended, and federal officials made Native ceremonies a crime.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 19 MINUTES

Some Protestants met the fast growth of Catholics and Jews after 1850 with mobs, school laws and state constitutions. Catholics were about one person in twenty in the United States in 1850. By 1906 there were 14 million of them, about one person in six. Catholics answered by building their own schools. Jews started schools to train rabbis, the teachers and leaders of Jewish congregations.

The end of slavery. Census takers counted 3,953,760 enslaved people in the United States in 1860. Many were freed as the Union army advanced during the Civil War. In December 1865 enough state lawmakers had approved the Thirteenth Amendment, a change to the Constitution, to make it law. It bans slavery everywhere except as a punishment for a crime. Black members already ran some large churches of their own, and in 1865 their ministers spoke for the freed people of Savannah.

On the reservations. United States officials made Native religious ceremonies a crime in 1883, and that ban could be enforced until 1978. Churches also ran many of the government's boarding schools for Native children, with government money.

Bloody Monday in Louisville, 1855

A political party called the Know-Nothings, or the American Party, grew large in the 1850s. Its members opposed Catholic immigration, and they feared that Catholics were a danger to Protestant faith and to democracy.

On election day, Monday, August 6, 1855, in Louisville, Kentucky, mobs led by Know-Nothing members attacked the German immigrant neighborhoods east of downtown and the Irish neighborhoods to the west. The mobs set fires and robbed homes and businesses.

An armed crowd set out to attack St. Martin of Tours, a German Catholic church, and the mayor, John Barbee, who was himself a Know-Nothing, turned it back. The Kentucky Historical Society's marker counts at least 22 dead and does not say who they were. Five people were later charged with crimes, and none was found guilty.

Mountain Meadows, 1857

Members of the Church of Jesus Christ of Latter-day Saints killed about 120 people at Mountain Meadows, in Utah Territory, about 35 miles southwest of Cedar City, in September 1857. The victims belonged to the Baker-Fancher wagon train, a company of families mostly from Arkansas on their way to California. Only 17 children, all six years old or younger, were spared.

The killers were local militiamen, part-time soldiers of the territory, who were also church members, and Paiutes joined them. A statement from the church identifies five of the leaders: Isaac Haight, John D. Lee, William Dame, John Higbee and Philip Klingensmith. They held both church posts and militia or town posts. Haight was the regional church leader at Cedar City and second in command of the county militia. Dame was the colonel in command of the southern Utah militia. Higbee was the major in command at the meadows.

The white flag. The attack began on September 7, by Paiutes and by Mormon men dressed to look like Native Americans. On September 11, John D. Lee and William Bateman came to the wagon train under a white flag, the sign of a truce. They promised the travelers safe passage if they gave up their guns.

The militiamen walked the men of the wagon train out in a single line, each man beside an armed militiaman. Higbee called out, "Halt! Do your duty!" and each guard shot the man beside him. On the other side of the field, Nephi Johnson ordered the killing of the women and the older children.

Why. Douglas Linder of the University of Missouri-Kansas City gives three things behind the killings. Federal troops were marching on Utah that year, in what is called the Utah War. That spring, on May 13, 1857, Hector McLean had shot and stabbed Parley Pratt, a church leader, in Arkansas. Eleanor McLean had left her husband and become Pratt's twelfth plural wife, one of several wives he had at the same time. And in 1856 church leaders had held a preaching campaign called the Reformation.

DISPUTED: whether Brigham Young, the church's president, ordered the killings. In 2007 a church leader, Henry B. Eyring, said that Young's message telling local leaders not to interfere with the travelers arrived too late. Linder writes that Young reportedly said afterward, "It is all right. The only fear I have is from traitors."

The trials. Nine men were charged with crimes, and only John D. Lee was found guilty. A jury of church members found him guilty of murder on September 20, 1876. He was executed at Mountain Meadows on March 23, 1877, by a firing squad, a group of men who shoot a condemned person at the same time.

Speaking for the church on September 11, 2007, Eyring expressed "profound regret." He called the killings "a terrible and inexcusable departure from Christian teaching and conduct."

Synagogues, schools for rabbis, and an order to expel Jews

About 2,500 Jews lived in the United States in 1800. By 1860 there were between 125,000 and 200,000, and by 1880 between 230,000 and 300,000, about a hundred times as many as in 1800.

Czar Alexander II, the ruler of Russia, was murdered in 1881, and pogroms followed in Russia. A Library of Congress timeline defines a pogrom as a "massacre often with government collusion," which means a mass killing of Jews, often with the help of government officials. After that, Jews from Eastern Europe came to America in great numbers.

Schools for rabbis. American Jews divided in these years into a traditional wing and a Reform wing, which was less traditional, and both wings built schools to train rabbis. Isaac Leeser, a leader of the traditional wing, founded Maimonides College in Philadelphia in 1867, the first school in America to train rabbis.

In 1873 thirty-four congregations, groups of Jews who worship together, formed the Union of American Hebrew Congregations. Its founders hoped it would take in every synagogue, or Jewish house of worship, in the country, and it became the union of Reform congregations instead. Two years later Isaac Mayer Wise, a leader of the Reform wing, founded Hebrew Union College in Cincinnati, which became the Reform wing's school for rabbis.

The dinner. The food served in 1883 at a dinner for the first class of rabbis from Hebrew Union College was not kosher, which means it broke Jewish food laws. Traditional guests walked out. The argument that followed led to calls for a more traditional seminary, and in 1887 the Jewish Theological Seminary opened in New York.

General Orders No. 11. During the Civil War, on December 17, 1862, General Ulysses S. Grant issued General Orders No. 11. It said that "The Jews, as a class," were "hereby expelled from the department within twenty-four hours," and it accused them of "violating every regulation of trade." The order applied to Jewish civilians in Kentucky, Tennessee and Mississippi, the area under his command.

Eleven days later, Cesar Kaskel and all the other Jews known in Paducah, Kentucky, got papers from the army ordering them to leave within twenty-four hours. Kaskel sent a telegram to President Lincoln and went to Washington. He met Lincoln on January 3, 1863, and Lincoln had General Henry Halleck cancel the order. The historian Jonathan Sarna's account of the order gives no count of the people expelled.

That same year, 1862, Jacob Frankel became the first Jewish chaplain in the United States Army, a religious leader who serves soldiers.

Turned away at Saratoga. Henry Hilton, the manager of the Grand Union Hotel in Saratoga, New York, turned away Joseph Seligman, a New York banker, and his family in 1877 because they were Jewish. Hilton blamed a drop in the hotel's business on guests who did not want to stay where Jews were let in.

A Library of Congress timeline describes it as a sign of "the growth of social anti-Semitism in America," anti-Semitism being hatred of Jews. That same year New Hampshire's lawmakers gave Jews the same political rights as other citizens.

The ministers who spoke for the freedpeople

Secretary of War Edwin Stanton and General William Sherman met twenty Black men in Charles Green's house in Savannah, Georgia, on January 12, 1865, and asked them what freedom should look like. Thirteen of the twenty were ministers, ordained or licensed, which means their churches had made them ministers or given them permission to preach. The others included deacons, class leaders and other church officers.

The notes of the meeting record how large their churches already were. William J. Campbell, who had been freed by a will, was pastor of the First Baptist Church, with about 1,800 members. John Cox, who had bought his own freedom, led the Second African Baptist Church, with 1,222 members.

Another of the twenty, Jacob Godfrey, had been a class leader and a steward, a man who helped look after the church's business, in the Methodist church since 1836. Until the Union army came, he had been held as property by James E. Godfrey, a Methodist preacher who was serving in the Confederate army.

Garrison Frazier

Who: A Baptist minister who had bought his own freedom, chosen by the other Black church leaders at a meeting to tell the Secretary of War what freedom meant.

When and where: Born in Granville County, North Carolina, about 1798. Savannah, Georgia, January 12, 1865.

He said his people needed "land, and turn it and till it."

The twenty church leaders at the Savannah meeting chose Garrison Frazier to answer for all of them, and his answers were written down.

He was 67 years old. Eight years before the meeting he had bought his own freedom and his wife's for $1,000 in gold and silver. He had been an ordained Baptist minister for thirty-five years, but by 1865 his health was so poor that he no longer led a congregation of his own.

Slavery and freedom. Asked what slavery was, he said it was "receiving by irresistible power the work of another man, and not by his consent." Asked what freedom was, he said it was "taking us from under the yoke of bondage, and placing us where we could reap the fruit of our own labor."

Asked how his people could best take care of themselves, he answered: "land, and turn it and till it by our own labor." He added: "that is, by the labor of the women and children and old men."

Asked whether Black people would rather live among white people or by themselves, he said: "I would prefer to live by ourselves, for there is a prejudice against us in the South that will take years to get over." Each of the other men was then asked the same question, one at a time, and all of them agreed with him. At one point Sherman left the room, and the men were then asked what they thought of him.

Forty acres. Four days after the meeting, Sherman issued Special Field Orders No. 15. In it he set aside about 400,000 acres of land along the coast, from Charleston to northern Florida, for freed families in plots of

> forty acres, and about 40,000 people settled there. In the fall of 1865 President Andrew Johnson reversed the order and gave the land back to its former owners.

A Chinese temple in Weaverville, 1874

In Weaverville, in Trinity County, California, a temple of Taoism, a Chinese religion and way of life, burned down, and Chinese men there built a new one to replace it. Work began in February 1874, and the temple was dedicated that April. They named it the Temple of the Forest Beneath the Clouds.

It is the oldest Chinese temple in California that has been used without a break, and people still worship there. In 1956 Moon Lim Lee, a descendant of the men who built it, and his wife, Dorothy, gave it to California State Parks.

A second school system

In Philadelphia, Bishop Kenrick had asked the public school board to let Catholic children read their own Bible, and under the board's rule the Douay Bible was still not allowed. So Catholics built parish schools, schools run by a local Catholic church. In 1834 there had been fewer than a dozen parish schools in the whole country, for about half a million Catholics. In the 1850s Bishop John Neumann began building parish schools in Philadelphia.

Baltimore, 1884. The Catholic bishops of the United States met at the Third Plenary Council of Baltimore from November 9 to December 7, 1884. They declared that every parish that could afford a school should run one. Their letter to Catholics said that parish schools should be as good as "the best schools of the country." It also said: "A Catholic congregation without a Catholic school is like a family without a mother."

The Blaine amendment. Congressman James G. Blaine of Maine, at President Grant's suggestion, proposed an amendment to the Constitution in 1875. Its words said that no money raised by taxes for public schools "shall ever be under the control of any religious sect," a sect being a religious group.

Members of the House voted for it. In the Senate it fell four votes short. Lawmakers in thirty-seven states later put their own versions into their state constitutions.

Justice Clarence Thomas of the Supreme Court wrote about the amendment in 2000, in the case *Mitchell v. Helms*, for himself and three other justices. He said it came at "a time of pervasive hostility to the Catholic Church," a time when dislike of Catholics was everywhere. He wrote that "it was an open secret that 'sectarian' was code for 'Catholic,'" an open secret being something many people know but few say out loud.

Blaine ran for president in 1884. A pastor who supported him called the Democrats the party of "rum, Romanism and rebellion," Romanism being an insulting word for the Catholic Church. Blaine did not reject the remark. Lyle Denniston, writing for the National Constitution Center, says the remark and Blaine's amendment cost him many Irish Catholic votes. Blaine lost the election.

Belief and practice: the Reynolds case

George Reynolds, a Latter-day Saint in Utah Territory, married a second wife while his first wife was still living, which was the crime of bigamy. A court found him guilty and sentenced him to two years in prison at hard labor, forced heavy work, and a $500 fine. He argued that his religion required him to have more than one wife. In a case decided in the Supreme Court's 1878 term, *Reynolds v. United States*, the justices ruled against him.

Chief Justice Waite wrote that laws "cannot interfere with mere religious belief and opinions," but "they may with practices." He wrote that excusing a crime because of religious belief would "permit every citizen to become a law unto himself."

Hiram Price's rules against ceremonies, 1883

Hiram Price, the Commissioner of Indian Affairs, issued a set of rules on March 30, 1883, for the Court of Indian Offenses, which tried Native people for breaking them. He wrote them after the Secretary of the Interior, Henry M. Teller, sent him a letter on December 2, 1882. In it Teller objected to the work of medicine men, Native healers and religious leaders. He called Native dances "heathenish," his insult for the ways of people who were not Christian.

Price's rules named ceremonies as crimes: "The 'sun-dance,' the 'scalp-dance,' the 'war-dance,' and all other so-called feasts assimilating thereto, shall be considered 'Indian offenses.'" The words "assimilating thereto" mean "like those dances." For a first offense, officials took away the person's rations for up to ten days. For a second, they took away rations for fifteen to thirty days, or jailed the person for up to thirty days.

Food as punishment. Rations were the food that United States officials had agreed by treaty to give to Native nations whose hunting lands United States officials had taken. So the punishment for dancing was that a family went without food for as long as a month.

Another rule said that "the usual practices of so-called 'medicine-men' shall be considered 'Indian offenses.'" A medicine man could be jailed for at least ten days, and kept there until he showed that he had given up his practices. These rules could be enforced for ninety-five years, until 1978, when members of Congress passed the American Indian Religious Freedom Act.

Alaska. Missionaries worked against Native ceremonies in Alaska too. In 1885 missionaries of the Moravian Church, a Protestant church, arrived at Quinhagak, a Yup'ik village. They called Yup'ik masked dancing "heathen rites" and "idol worship," and they recruited Yup'ik helpers as missionaries. After the missionaries came, no traditional dancing took place in Quinhagak for more than a hundred years, until 2013. The 2015 *Archaeology* magazine report on the village does not say who stopped the dancing or how.

The churches and the boarding schools

Federal officials handed many of the Indian boarding schools, schools where children live as well as study, to churches to run, and they paid the churches to do it. Federal officials used these schools to assimilate Native children, which means to make them give up their own nations' ways of life. In a report published in 2024, investigators at the Department of the Interior counted 417 federal Indian boarding schools. A religious group ran 210 of them, about half.

At least 59 different religious groups received federal money to run or support the schools. That count leaves out the schools that churches ran without any federal money.

What staff did to the children. The same federal investigation recorded what staff at the schools did to the children. Staff cut the children's hair, took away their clothes for uniforms, gave them English names and forbade their own languages. Punishments included beatings, solitary confinement, which means being locked up alone, withheld food, and mouths washed out with soap.

At least 973 children died at the schools, and investigators found at least 74 marked and unmarked burial sites at 65 schools. The report states that the true number of deaths is higher.

Which churches. The investigators counted 132 schools tied to Protestant churches, 77 to the Catholic Church, and 5 to other groups. Some schools were tied to more than one. The Presbyterians had the most, with 31 schools. The Quakers, the American Board of Commissioners for Foreign Missions, the Methodists and the Episcopalians each had 12 or more.

On the Catholic side, the Jesuits, an order of priests, had 10. Many orders of Catholic nuns are on the list, among them the Sisters of Mercy, the Sisters of St. Joseph, the Ursulines and the Grey Nuns.

The money. The payments began long before. In the Civilization Fund Act of March 3, 1819, members of Congress set aside $10,000 a year to teach Native nations "the habits and arts of civilization." Officials often gave that money to religious groups, until members of Congress stopped the yearly payment in

1873. From 1871 to 1969, by the investigators' estimate, members of Congress set aside more than $23.3 billion, in 2023 dollars, for the boarding schools and related programs.

Government records show how much power officials handed to the churches. Each reservation was run from an office called an agency, by a government agent. In 1872 the Commissioner of Indian Affairs wrote that the agencies had been divided among the churches. Each church could put forward the person to be the agent in charge and could "assume charge of the intellectual and moral education of the Indians."

A Senate report of 1969 describes this as "an unprecedented delegation of power by the U.S. Government to church bodies," a handing over of power never done before. The report states that the "military was frequently called in to reinforce the missionaries' orders," which means soldiers were often brought in to back up what the missionaries ordered.

The Ghost Dance, 1889 and 1890

Wovoka's vision. A Northern Paiute man named Wovoka, whose English name was Jack Wilson, had a vision on January 1, 1889, during an eclipse of the sun. By one account, he saw a world made new, with the dead alive again, the buffalo back, the land restored, the settlers gone and everyone living in peace.

A young Arapaho man, Casper Edson, wrote down Wovoka's message for a group of visitors. In it Wovoka said: "You must not hurt anybody or do harm to anyone. You must not fight. Do right always." He said that "the dead are all alive again" and told people to dance every six weeks.

He also warned: "Do not refuse to work for the whites and do not make any trouble with them until you leave them." The dance and its teaching spread across the reservations of the Plains, and the religion became known as the Ghost Dance.

The teaching changed as it spread. At the end of October 1890, the Lakota leader Short Bull told a gathering near Pine Ridge that "the whites were interfering so much in the religious affairs of the Indians" that he would make

the great change, the new world of Wovoka's vision, come sooner. In that change, he said, "the white race itself would soon be annihilated," wiped out completely. He told the dancers to keep dancing even if soldiers surrounded them, because "the guns of the soldiers would be rendered harmless."

The agent's fear. Daniel F. Royer became the government agent at Pine Ridge in October 1890. On October 12 he reported that more than half of the 6,000 Lakota people at his agency were dancing, and he suggested calling in the army. General Miles told him the excitement would die down. On October 30 Royer wrote that 600 or 700 soldiers were needed.

Royer sent a telegram on November 15 saying "that the Indians were wild and crazy and that at least a thousand soldiers were needed." The Lakota called Royer "Young-man-afraid-of-Indians." Officials in the Indian Office passed his request to the War Department. In the middle of November 1890, President Benjamin Harrison ordered troops onto the reservations.

Wounded Knee. Soldiers killed between 250 and 300 Lakota people at Wounded Knee on December 29, 1890, most of them women and children. Chapter 2, Native Nations, tells what happened there.

BY 1900

If you remember one thing: As Catholics and Jews grew in number, some Protestants answered with mobs and with laws that kept tax money from Catholic schools.

- In 1850: about one American in twenty was Catholic.
- In 1906: about one in six was, 14 million people.

Next: After 1900 a new kind of worship began in Los Angeles, and a new Ku Klux Klan made opposing Catholics and Jews one of its central beliefs.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 23 MINUTES

New kinds of worship spread across the United States between 1900 and 1950. In the same years some Americans attacked other Americans because of their religion. A Black preacher, William J. Seymour, led a revival in Los Angeles in 1906, and Pentecostal worship began there. Pentecostals believe that God's Spirit fills a person, who then speaks in sounds he or she never learned. Aimee Semple McPherson preached from her Los Angeles church to listeners at home by radio.

Attacks. A new Ku Klux Klan, open only to white Protestant men born in America, made opposing Catholics and Jews one of its central beliefs. In 1915 a group of men in Georgia hanged a Jewish factory superintendent named Leo Frank. In 1940 mobs attacked Jehovah's Witnesses for refusing to salute the flag. Christians also argued with other Christians about how to read the Bible, and in 1925 both sides made their case at a trial in Tennessee.

On the reservations. Federal officials went on punishing Native people for holding their ceremonies. In the 1930s John Collier, the new Commissioner of Indian Affairs, the head of the federal office in charge of Native affairs, ordered the punishments stopped, but the rules of 1883 that made the ceremonies crimes stayed in the law.

Native ceremonies still punished, 1902 to 1940

Era 7 tells how Hiram Price's rules of 1883 made Native dances and the work of medicine men, Native healers and religious leaders, into crimes punished by the loss of food rations or by jail. Federal officials kept enforcing those rules after 1900. Commissioners of Indian Affairs kept issuing new orders against Native dances into the 1920s.

Hair and dances, 1902. Commissioner William A. Jones wrote to the superintendents, the officials in charge of each reservation, on January 11, 1902. He told them that "Indian dances and so-called Indian feasts should be prohibited," which means banned. He also wanted Native men to cut their long hair, and he suggested that superintendents hold back the men's rations and jobs until they did.

Charles Burton was the superintendent for the Hopi and the Navajo. The historian Tisa Wenger writes that he reportedly used "whips, guns, and sheep shears" to force men to obey the order. The surviving accounts do not say how many men were forced to cut their hair. People objected in print, including in the magazine *Harper's Weekly*, and officials then sent new directions advising "persuasion and example, rather than force."

The Kiowa Agency. At the Kiowa Agency, the federal office that ran the Kiowa reservation in Oklahoma, Superintendent C. V. Stinchecum held back people's own money to stop them from dancing. Kiowa families had trust money, money that belonged to them and that the government held for them. Stinchecum ruled that a man would not "receive his money until he has given his word of honor to the Superintendent that he will not again indulge in either a ghost dance or a give-away dance."

The Ghost Dance was the religion that spread across the Plains reservations in 1889 and 1890, and a give-away dance is a gathering at which people give gifts, such as horses, to others. By early December 1916 Stinchecum had held back the money of about ninety people.

Stinchecum sent the Indian Office, the federal office in charge of Native affairs, a list of 109 names in January 1917. Among them were Humming Bird, George Mopope and Silver Horn. To get their money, they had to sign a sworn paper promising not to dance.

Eleven horses were given away at a Fourth of July dance near Anadarko in 1921. Stinchecum accused nine Kiowa men of "the reckless giving away of property" and held their money until they signed a promise never to do it again. By early November only three of the nine had signed.

Orders from Washington. Charles Burke, the Commissioner of Indian Affairs, sent out Circular 1665, an order sent to every office at once, on April 26, 1921. It was titled "Indian Dancing." It allowed any superintendent to stop a dance he judged immoral or against the community's interest. Burke added a supplement on February 14, 1923, and a message "To All Indians" ten days later.

Collier's order. John Collier became Commissioner of Indian Affairs in April 1933. In 1934, with the approval of Harold Ickes, the Secretary of the Interior, Collier sent every agency a circular titled "Indian Religious Freedom and Indian Culture." It stated that "no interference with Indian religious life or ceremonial expression will hereafter be tolerated." The scholar Lee Irwin calls it the federal government's first policy statement protecting Native religious rights. At the Kiowa Agency, the punishments for dancing stopped soon after Collier took office.

Federal officials could still punish people under the 1883 rules until 1978. In 1940 federal rules under the Bald Eagle Protection Act also made it a federal crime to have eagle feathers. Under those rules, federal prosecutors took traditional practitioners, people who kept the old ceremonies, to court for holding eagle feathers.

Nicholas Black Elk

Who: An Oglala Lakota holy man who was also a Catholic catechist for decades.

When and where: Born December 1863 on the Little Powder River. Pine Ridge, South Dakota. Died August 1950.

"The only thing I really believe is the pipe religion," he said.

Nicholas Black Elk kept his standing as a Lakota holy man while he worked for decades as a Catholic catechist, teaching the faith and preparing people for baptism, the Christian ceremony with water in which a person joins the church.

The vision. He was born on the Little Powder River, in land that later became Wyoming, and in his ninth year he had a vision. In it, in the words of the National Park Service, he "met with a council of the Six Grandfathers representing the four cardinal directions as well as the earth and sky," the four directions being north, south, east and west. The vision gave him the power to heal and a pipe made of pipestone.

He was at the Battle of the Little Bighorn on June 25, 1876, when he was twelve. There he took his first scalp, cutting the skin and hair from a person's head. He also collected guns, ammunition and clothing from dead and dying soldiers.

Abroad. In 1886 he joined Buffalo Bill's Wild West, a traveling show, and performed at Madison Square Garden in New York. The next year the show sailed to England for Queen Victoria's Golden Jubilee, the celebration of her fifty years as queen. He was separated from the show, traveled around Europe on his own, found Buffalo Bill again and came home.

Wounded Knee. He was back at Pine Ridge in 1889 and took part in the Ghost Dance ceremonies. On December 29, 1890, soldiers of the 7th Cavalry, a regiment of soldiers on horseback, opened fire on a band of Lakota people at Wounded Knee Creek. They killed between 250 and 300

of them, most of them women and children. Black Elk heard the cannon and gunfire. He came to Wounded Knee, helped the survivors and wept for the dead.

He married Katie War Bonnet in 1892, and their children were baptized. Katie died in 1903. In 1904 he was baptized a Catholic and took the name Nicholas. In 1930 the poet John G. Neihardt interviewed him, and the book *Black Elk Speaks* came out in 1932.

He died in August 1950. Near the end of his life he told his daughter Lucy, "The only thing I really believe is the pipe religion." In 2017 Catholic leaders of the Diocese of Rapid City, the church's area centered on that city, opened his cause for sainthood, the first step toward the church naming him a saint.

Azusa Street, 1906

A new kind of Christian worship began in Los Angeles in April 1906, at meetings led by a Black preacher named William J. Seymour. It is called Pentecostal. The name comes from the Day of Pentecost, when, the Bible says, the first Christians were filled with the Holy Spirit, the Spirit of God. Pentecostals held that this still happens, and that speaking in tongues, speaking sounds that the speaker has not learned and does not understand, is the proof.

The first speaking in tongues came on April 9, 1906, at a house at 214 North Bonnie Brae Street, after a month of prayer and fasting.

312 Azusa Street. In mid-April the meetings moved to 312 Azusa Street, into a room about 40 feet by 60 feet. The building had been put up as a church of the African Methodist Episcopal Church, the church that Black Methodists formed in 1816. Biddy Mason, a woman who had been enslaved and won her freedom in a California court, had founded that church. By 1906 the building had been used as a tenement, a crowded apartment house, and as a livery stable, where horses were kept for hire.

The first people at the meetings were Black washerwomen, maids and laborers. Later most of the members and visitors were white. The University of Washington's architecture records call it "one of the first religious congregations in the U.S. that consisted of black and white parishioners together under a African-American pastor." That means Black and white worshippers met together under a Black pastor. The mission met there from 1906 to 1931.

William J. Seymour

Who: The Black preacher who led the Azusa Street revival in Los Angeles, where Pentecostal worship began.

When and where: Born May 2, 1870, at Centerville, Louisiana. Los Angeles from early 1906. Died September 28, 1922.

He learned the teaching sitting in a hallway outside the classroom.

William J. Seymour's parents, Simon and Phyllis Seymour, had both been enslaved.

Houston, 1905. In Houston in 1905 he asked to study at a Bible school run by a preacher named Charles Fox Parham. Under Texas law, Black and white students could not sit in the same classroom, so Parham had Seymour sit in the hallway and listen to the lectures through the doorway. There Seymour learned the teaching that speaking in tongues is the proof that the Holy Spirit has filled a person.

Locked out. Early in 1906 he came to Los Angeles. Julia Hutchins had invited him to help lead a Holiness church there, a church of one of the Protestant groups of the time. He preached what he had learned in Houston, and the congregation locked him out of the building. He kept preaching at a house on Bonnie Brae Street, where the speaking in tongues began on April 9, 1906. Within days the meetings moved to Azusa Street.

> The people of Seymour's mission began printing their own newspaper, *The Apostolic Faith*, in September 1906. It was sent out free, and at its height it went to about 50,000 subscribers around the world.
>
> Seymour ran the Apostolic Faith Mission at Azusa Street. He died on September 28, 1922, and the mission kept meeting there until 1931. The building is gone. A plaque on the spot reads "Site of the Azusa St. Revival from 1906 to 1931. Cradle of the Worldwide Pentecostal Movement."

Leo Frank, 1913 to 1915

A group of men in Georgia kidnapped and hanged Leo Frank, a Jewish factory superintendent, in 1915, after the governor spared his life. Frank ran the National Pencil Company in Atlanta. In 1913 a jury convicted him of murdering Mary Phagan, a thirteen-year-old girl who worked at the factory. The Georgia Historical Society's marker at the site describes it as "a highly controversial trial fueled by societal tensions and anti-Semitism," anti-Semitism being hatred of Jews.

Crowds outside the courthouse cheered the prosecutor. Much of the newspaper coverage was anti-Semitic in tone, according to The Temple, the Atlanta Jewish congregation that Frank belonged to. It is a Reform congregation, part of the less traditional branch of Judaism. Tom Watson, the publisher of a paper called *The Jeffersonian*, campaigned against Frank and against Jews in general, and more people bought his paper.

The lynching. Governor John M. Slaton commuted Frank's sentence from death to life in prison, changing it to a lighter one. On August 17, 1915, a group of men took Frank from the state prison at Milledgeville. They took him to Marietta, Mary Phagan's hometown, and hanged him before a local crowd. This was a lynching, a killing by a mob that had no legal right to kill.

The men called themselves the Knights of Mary Phagan. The New Georgia Encyclopedia describes them as twenty-five prominent citizens of Marietta. No one was ever tried for killing Frank.

The pardon. Georgia officials granted Frank a pardon, an official act that forgives a crime or cancels its punishment, in 1986, after his death. This pardon was given "without addressing guilt or innocence," so it did not say whether he had killed Mary Phagan. The marker says the pardon recognized the state's failure "to either protect Frank or bring his killers to justice." Georgia's officials had not protected him, and they had never put his killers on trial.

The Temple's own history says that no Jew ran for office for more than twenty years after the lynching. The New Georgia Encyclopedia says that after the case, Jewish southerners were afraid and watched their own behavior closely for the next fifty years. Some of the Knights of Mary Phagan later took part in the ceremony that started a new Ku Klux Klan.

BY 1915: Pentecostal worship had spread out from Azusa Street, federal officials were still punishing Native dances, and a Georgia mob had lynched Leo Frank.

A Klan that called itself Protestant

William J. Simmons and sixteen other men climbed Stone Mountain in Georgia on Thanksgiving evening in 1915. They set a cross on fire and announced the rebirth of the Knights of the Ku Klux Klan. Only white, American-born Protestant men could join. Opposing Catholics and Jews was one of its central beliefs, according to the New Georgia Encyclopedia. By 1924 it had an estimated five million members.

Robes and altars. Klan members borrowed the forms of Protestant worship. The historian Kelly J. Baker describes their white robes and masks, their "elaborate initiation ceremonies," which means ceremonies with many detailed steps for new members, their burning crosses, and their altars draped with the American flag.

In the middle of revival services, robed Klansmen gave money to the preachers. The traveling preacher Billy Sunday received $50 and a letter from Klansmen in Indiana. In Spokane, Washington, two robed Klansmen gave $50 to the traveling preacher George Wood, the *Idaho Daily Statesman* reported in March 1922. Other Protestants spoke against the Klan. In October 1922 the *Savannah Tribune* reported that the Federal Council of Churches, a group of Protestant churches, had condemned it.

The year Oregon voted to close the Catholic schools

Oregon's voters passed a law in 1922 that would have closed the state's Catholic schools, along with its other private schools, for children aged eight to sixteen. The law was an initiative, a law that citizens propose and then vote on themselves. It was proposed on June 26, 1922. Klan members put it on the ballot through the Scottish Rite Masons, a men's club.

The voters passed it on November 7, 1922, 115,506 to 103,685. It required children aged eight to sixteen to attend a public school, starting September 1, 1926. There were a few exceptions, such as a child who had finished the eighth grade, a child who lived far from a public school, or a child who was physically unable to attend school.

Fines and jail for parents. The penalties in the law were for parents, not for schools. A parent who kept a child out of public school could be fined $5 to $100, or jailed in the county jail for two to thirty days, or both. Each day a child stayed away counted as a separate crime. A child could be taught at home only with written permission from the county superintendent, renewed every year.

The Society of Sisters of the Holy Names of Jesus and Mary had run schools in Oregon since 1880. Their schools taught the Catholic religion along with the ordinary subjects. The Sisters sued. So did Hill Military Academy, a boys' school that was not religious at all. The National Catholic Welfare Conference, a Catholic organization, paid for the case to be carried to the Supreme Court.

The ruling. The justices of the Supreme Court struck the law down on June 1, 1925, in *Pierce v. Society of Sisters*, before it could take effect. They held that it "unreasonably interferes with the liberty of parents and guardians to direct the upbringing and education of children under their control."

Christians arguing with Christians at Dayton, 1925

The 1925 trial at Dayton, Tennessee, was an argument between two groups of Christians about how to read Genesis, the first book of the Bible. Governor Austin Peay of Tennessee signed a law, the Butler Act, on March 21, 1925. It made it a crime to teach "any theory that denies the story of Divine Creation of man as taught in the Bible," and to teach instead that people came "from a lower order of animals."

That was a ban on teaching evolution, the scientific theory that living things, people included, developed over very long ages from earlier forms of life. Tennessee prosecutors put John T. Scopes, a science teacher and football coach at Dayton's high school, on trial under the law. The trial opened on July 10, 1925.

Both sides of the argument were Christians, and both put their beliefs on the record. On one side were Christians often called fundamentalists, who read the Bible word for word. One of them was William Jennings Bryan, one of the lawyers against Scopes. On the other side, two scientists for the defense, both of them Christians, wrote that God works through evolution and that the Bible and evolution do not conflict.

Bryan on the stand. The prosecutor William Jennings Bryan took the witness stand himself on July 20, the seventh day of the trial. The defense lawyer Clarence Darrow questioned him about the Bible. Bryan said, "I accept the Bible absolutely." He said Darrow's purpose was "to cast ridicule on everybody who believes in the Bible."

Darrow answered that his purpose was "preventing bigots and ignoramuses from controlling the education of the United States." By bigots he meant people who hate or look down on those who believe differently, and by ignoramuses people who know very little.

Bryan had written a closing speech, but both sides gave up their closing speeches, so he never gave it. It was printed afterward with the trial record. In it he asked, "Can the fathers and mothers of Tennessee be blamed for trying to

protect their children from such a tragedy?" The tragedy he meant was a child losing "faith in God, trust in the Bible, and belief in the supernatural character of Christ."

The scientists' statements. The defense could not put its expert witnesses in front of the jury, so the defense lawyer Arthur Garfield Hays read their written statements into the record. Kirtley F. Mather led the geology department at Harvard and had graduated from Denison University, a Baptist college. He wrote: "Evolution is not a power, nor a force. It is a process, a method. God is a power, a force."

The zoologist Maynard M. Metcalf, a scientist who studied animals, wrote: "There is no conflict, no least degree of conflict, between the Bible and the fact of evolution." In the same sentence he went on to call the word-for-word reading of the Bible childish and "insulting, both to God and to human intelligence."

The verdict. The jury found Scopes guilty, and he was fined $100. In 1927 the judges who heard his appeal threw out the verdict on a technicality, a small point about how the case was handled. Chapter 20, Government, Law, and Politics, describes the case in full.

Laws against teaching evolution, 1921 to 1929

Lawmakers in twenty states introduced bills against teaching evolution between 1921 and 1929. Tennessee's passed in 1925. Oklahoma's lawmakers passed one and canceled it in 1926. Mississippi's law was still in force in 1968. Lawmakers in Florida and Texas passed resolutions against teaching evolution, which are statements of opinion rather than laws.

In Arkansas the voters passed the law themselves in 1928. It made it a crime "to teach the theory or doctrine that mankind ascended or descended from a lower order of animals," or to use a textbook that taught it. A teacher found guilty could be fined up to $500 and had to leave the job. In the forty years it stood, there is no record that anyone was prosecuted under it. Tennessee's lawmakers canceled the Butler Act in 1967.

BY 1929: Oregon's voters had passed a Klan-backed law to close the state's Catholic and other private schools, the Supreme Court had struck it down, and lawmakers in twenty states had tried to ban the teaching of evolution.

Aimee Semple McPherson

Who: A Pentecostal preacher who built the Angelus Temple in Los Angeles and preached from it by radio.

When and where: Born 1890 near Ingersoll, Canada. Los Angeles from 1918. Died 1944.

She rode a motorcycle onto the stage dressed as a policewoman.

Aimee Semple McPherson built a church in Los Angeles with seats for 5,300 people and preached from it by radio. She grew up in the country near Ingersoll, Canada. Her mother was a member of the Salvation Army, a Christian group, and her father led a Methodist choir. As a child she sang, prayed and recited Bible passages at Salvation Army services. She was the only child in her public school who belonged to the Salvation Army, and the other children taunted her for it.

In high school she was taught evolution and began to doubt her faith. After that she started going to a Pentecostal church. She married Robert Semple, a Pentecostal minister, and went with him to China as a missionary. He died there, and her first child was born a month after his death. Later she married Harold McPherson and lived in Providence, Rhode Island. In 1916 she left him and went preaching across the country.

The Angelus Temple. She settled in Los Angeles in 1918. The Angelus Temple on Glendale Boulevard opened in 1923, built and paid for with money from her preaching tours. The National Park Service writes that she "fell squarely into the fundamentalist camp." She called on believers to leave the old churches that she believed had been "tainted with Darwinian modernism," meaning they had accepted evolution. A newspaper in 1926 reported radio towers on top of the Temple carrying her words to listeners.

The National Park Service writes that she brought the entertainment of Hollywood films and vaudeville, stage shows of songs, comedy and acts, into her preaching. Once she dressed as a policewoman and rode a motorcycle onto the stage, and she preached on the subject "Stop! You are breaking God's law!" In 1926 she opened a Bible college. By 1944 more than 3,000 of its graduates, men and women, had become ministers or missionaries.

Each week, members brought an item to stock the Temple's shelves, and the staff gave the goods to anyone in need. During the Great Depression the Temple's workers gave out more than 1,500,000 meals.

In Florida she held special meetings for Black people because, in her words, "the dear colored people did not feel free to attend the white meetings." ("Colored" was a common word for Black people at the time, and it is no longer used.) In Oakland, California, she preached to the Ku Klux Klan.

Five weeks missing. She went into the ocean at Ocean Park, on the Santa Monica coast, on May 18, 1926, and did not come back. Police, lifeguards, lifeboats, motorboats and an airplane searched for her. Two days later her friends had given up hope that she was alive.

She walked out of the desert near Douglas, Arizona, on June 23 and was taken to a hospital. She said that two men and a woman had kidnapped her at the beach and held her in Mexico for half a million dollars in ransom. Armed search parties, including Mexican soldiers, searched the flats of Sonora for the shack she described.

California prosecutors brought a case against her afterward. On January 10, 1927, a judge dismissed the case at the request of District Attorney Asa Keyes, the county's chief prosecutor, who pointed to "the constantly conflicting stories" of a witness. No court ever decided what happened in those five weeks. She died in 1944.

Jehovah's Witnesses and the flag, 1940 to 1943

Jehovah's Witnesses, a Christian group, would not salute the American flag, and in 1940 people in towns across the country attacked them for it. They believed that saluting a flag was the same as bowing down to a graven image, an idol, which the Bible forbids.

The justices of the Supreme Court ruled 8 to 1 in 1940, in *Minersville School District v. Gobitis*, that school officials in Minersville, Pennsylvania, could expel two Witness children, Lillian and William Gobitas, for refusing to salute. A court clerk misspelled the family's name as Gobitis.

The attacks. After the ruling the attacks grew. The First Amendment Encyclopedia at Middle Tennessee State University lists beatings, the destruction of Kingdom Halls (the Witnesses' places of worship) and at least one castration. Castration means cutting off a man's testicles, the glands that make sperm. A man who loses both can never father a child. The encyclopedia does not name the man or say where or how it was done.

The encyclopedia states that the American Legion, a club of military veterans, was linked to more than 100 attacks by vigilantes, people who punish others with no legal right to do so.

Richwood, West Virginia. One attack was recorded in detail, at Richwood, West Virginia, on June 29, 1940. Deputy Sheriff Martin Catlette and Police Chief Bert Stewart held seven Witnesses whose loyalty local Legion members had questioned. According to West Virginia Public Broadcasting, members of the American Legion forced four of the seven to drink doses of castor oil.

Castor oil is a laxative, a medicine that makes the muscles of the bowels push out their contents. It can cause diarrhea, which is watery emptying of the bowels, along with stomach pain, sickness to the stomach and vomiting. The accounts do not give the size of the doses. Then the Legion members marched all seven through a jeering crowd to the post office, where the Witnesses refused to salute the flag.

Catlette and Stewart were found guilty in federal court at Charleston in 1942 of violating the Witnesses' civil rights, the rights the Constitution promises every citizen. Appeals judges upheld Catlette's conviction. West Virginia Public

Broadcasting calls it the only federal conviction for any of the hundreds of attacks on Witnesses in 1940.

Reversed in 1943. Three Witness families in Kanawha County, West Virginia, whose children had been expelled from school, took their case to the Supreme Court. In 1943, in *West Virginia State Board of Education v. Barnette*, the justices reversed their ruling of 1940 by a vote of 6 to 3.

Justice Robert Jackson wrote that "no official, high or petty, can prescribe what shall be orthodox in politics, nationalism, religion, or other matters of opinion." That means no official can order which beliefs are the correct ones. He called that rule the "fixed star in our constitutional constellation."

BY 1950

If you remember one thing: People who were attacked or punished for their faith began to win their cases in the Supreme Court.

- 1906: speaking in tongues begins at a house on Bonnie Brae Street in Los Angeles.
- 1925: the justices strike down Oregon's law to close its Catholic and other private schools.
- 1943: the justices rule that schools cannot force Witness children to salute the flag.

Next: After 1950 Black churches held the meetings of the civil rights movement, and men bombed them for it.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 30 MINUTES

Families took their fights over religion in the public schools to court between 1950 and 2000, and the justices of the Supreme Court drew the lines. Church membership held steady. About seven American adults in ten told the Gallup polling company that they belonged to a church, a synagogue or a mosque, the Jewish and Muslim houses of worship. Members of Congress added "under God" to the Pledge of Allegiance in 1954 and made "In God we trust" the national motto in 1956.

Churches and courts. In the South, Black churches held the meetings of the civil rights movement, and men bombed their buildings and killed their members. Families of several faiths, and some with none, brought the cases over prayer, Bible reading and the teaching of evolution. The justices of the Supreme Court ended official prayer and Bible reading in public schools. They also ruled for Amish fathers who kept their children out of high school for religious reasons.

Native religion allowed. Members of Congress declared in 1978 that Native Americans had a right to practice their own religions. Federal officials had been able to punish Native ceremonies under the 1883 rules for ninety-five years before that.

Deaths in two religious communities. Jim Jones, the leader of Peoples Temple, ordered the members at Jonestown, in Guyana, to take poison in 1978. Parents and nurses squirted it into the mouths of more than 200 children, and the adults drank it while armed guards surrounded them. In all, 909 people died there.

More than 70 Branch Davidians, 22 of them children, died in a fire near Waco, Texas, in 1993. A Special Counsel later found that some of the Davidians had set the fire, and that some had shot their own people.

God in the country's official words, 1954 and 1956

Members of Congress changed the Pledge of Allegiance, the promise of loyalty to the flag, by law on June 14, 1954. They added the words "under God." The pledge now reads: "I pledge allegiance to the flag of the United States of America and to the Republic for which it stands, one Nation under God, indivisible, with liberty and justice for all."

They passed a joint resolution, a kind of resolution that becomes a law, on July 30, 1956. It is one sentence long: "That the national motto of the United States is hereby declared to be 'In God we trust.'" A motto is a short saying chosen to stand for a group or a country.

Gallup is a private polling company, and its pollsters record what people say about themselves. Since 1937 they have asked Americans whether they belong to a church, synagogue or mosque. In 1937, 73 percent said they did. The figure stayed near 70 percent through the 1990s and was 70 percent in 1999.

A church that was bombed three times

Bethel Baptist Church in Birmingham, Alabama, served as the headquarters of a civil rights movement from 1956 to 1961, and it was bombed three times. The church stood in Collegeville, a Black working-class neighborhood six miles north of the center of the city. Fred L. Shuttlesworth had been its pastor since March 1953.

A movement in the churches. An Alabama judge banned the NAACP, a national civil rights organization, from working anywhere in the state on June 1, 1956. Four days later, on June 5, about a thousand people met at Sardis Baptist Church and started the Alabama Christian Movement for Human Rights. Shuttlesworth said that officials could outlaw an organization but not a movement.

He said the word "Christian" set the group apart from groups being called Communist, Communists being people who wanted the government to own the land and the businesses. He also said the word gave the group "a foundation and Christian orientation." Members met every Monday night in churches around the city. The National Park Service describes the meetings as

"emotional testimony, soul-lifting music, impassioned preaching, and fundraising." By 1958 weekly meetings were being held in 55 other working-class churches in Birmingham.

Christmas night, 1956. Someone set off twelve sticks of dynamite on Christmas night, December 25, 1956, in the space between the church and the parsonage, the pastor's house. The blast destroyed the parsonage. Shuttlesworth was inside talking with a deacon, a member who helps run the church, and his family was watching television. He had a minor head injury.

The bomb went off the night before the movement's members had said they would ride Birmingham's segregated buses, where Black riders had to sit apart from white riders. The next day 250 people rode the buses, and police arrested 22 of them.

A second bomb came on June 29, 1958, and a third in 1962, which broke the church's windows. The 1958 bombing was not solved until 1980, when J. B. Stoner was convicted of it. No one is recorded as convicted of the 1956 bomb or the 1962 bomb.

Other churches. Other Black churches were bombed and burned in these years. On September 15, 1963, a bomb at the 16th Street Baptist Church in Birmingham killed four girls: Addie Mae Collins, 14, Denise McNair, 11, Carole Robertson, 14, and Cynthia Wesley, 14. Robert Chambliss was convicted of the bombing in 1977, Thomas Blanton in 2001 and Bobby Frank Cherry in 2002.

Jason Wollschleger, a sociologist at Whitworth University who studies how groups of people live together, and his research team have mapped about 145 attacks on Black churches across the South from about 1954 to 1970. It is one team's count, not an official one.

In Albany, Georgia, in 1961 and 1962, four Black churches connected with the Student Nonviolent Coordinating Committee, a civil rights group of young people, were burned to the ground. Among them were Shady Grove Baptist and High Hope Baptist. Wollschleger's account does not name who set the fires.

The Revis family of Bethel Baptist Church

Who: A church deacon and his two daughters, whose front porch became the guard post for a church that kept being bombed.

When and where: Collegeville, Birmingham, Alabama, 1956 to the early 1960s.

Laverne Revis saw a bomb smoking beside the church.

The Revis family turned their own front porch into a guardhouse so that armed volunteers could watch Bethel Baptist Church day and night. James Revis was a deacon of the church and sat on the executive committee of the Alabama Christian Movement for Human Rights. His house stood across the street from the church.

The porch. After the bomb of Christmas night 1956 destroyed the pastor's house, members of the church and the movement volunteered to guard the church and the new parsonage. In 1957 the Revis family closed in their porch to make a room for the guards.

His daughter Robbie Revis Smith described it years later: "with our house sitting on the corner, it gave a very good view several ways that they could see because they had the glass out there and they put a little gas heater out there so they wouldn't be too cold sitting up. They did this year round."

Her sister Laverne Revis Martin said their parents made the room "just so the guards would be able to watch."

The guards. The volunteers were called the Civil Rights Guards. They worked in two shifts, from six in the evening to midnight and from midnight to six in the morning. They carried rifles, which they called their "nonviolent Winchesters." Women of the movement took a weekly shift, among them Desta Brooks, a florist, and Minnie Eaton, a schoolteacher.

The night of the bomb. At about half past one in the morning on Sunday, June 29, 1958, Laverne was coming home from work when she saw a bomb smoking beside the church. She told the guards. They carried it into the street, where it went off. It blew a hole in the road and broke

church windows, and the building stood. Afterward Birmingham police came to the Revis house and took four rifles from the guards on the porch.

Robbie Revis Smith said of coming home in those years: "So it was kind of a good feeling coming home because you never came home without a person out there watching you, so you didn't have to worry about that."

Billy Graham

Who: A Southern Baptist preacher who held large public revival meetings, which he called crusades, from 1949 to 2005.

When and where: Born in Charlotte, North Carolina, November 7, 1918. Died at Montreat, North Carolina, February 21, 2018.

Martin Luther King Jr. gave the opening prayer at his 1957 crusade.

Billy Graham held revival meetings from 1949 to 2005, and in 1957 Martin Luther King Jr. gave the opening prayer at one of them. Graham grew up on a dairy farm near Charlotte, North Carolina. In 1934, after hearing a traveling preacher in Charlotte, he decided to give his life to God. In 1943 he married Ruth Bell.

Crusades. From 1944 to 1948 he traveled the country speaking at rallies for a group called Youth for Christ. In 1949 he held a campaign in Los Angeles that was planned for three weeks and ran for seven. He then started the Billy Graham Evangelistic Association and a weekly radio program, *Hour of Decision*. He preached for twelve weeks in London in 1954, and every night for sixteen weeks at Madison Square Garden in New York in 1957.

King's prayer. On July 18, 1957, at the New York crusade, Martin Luther King Jr. gave the opening prayer. Graham and King had spoken by telephone, and on July 5 the evangelist Leighton Ford, a traveling

preacher, had sent King a telegram asking him to come. King asked for help "for a warless world and for a brotherhood that transcends race or color." Then he led the crowd in the Lord's Prayer.

King wrote to Graham on August 31, 1957: "You have courageously brought the Christian gospel to bear on the question of race." He wrote again on July 23, 1958, this time to object. Graham had shared a platform with Governor Price Daniel of Texas, and King called that an "endorsement of racial segregation and discrimination." At Easter in 1964, Graham held crusades in Birmingham, Alabama, that Black and white people attended together.

Honors. Members of Congress voted Billy and Ruth Graham a gold medal in 1996, and it was presented to them in the Capitol Rotunda on May 2. In the law, members of Congress wrote that his crusades had reached 100 million people in person and more than 2 billion on television. He stepped down from his association in 2000 and held a last crusade in Queens, New York, in 2005.

He died on February 21, 2018, aged 99. On February 28 his body lay in honor in the Rotunda, the round hall under the dome of the Capitol, where people could come to pay their respects. According to the House of Representatives, he was the fourth private citizen to be given that honor.

Dynamite at an Atlanta synagogue, 1958

Fifty sticks of dynamite exploded in the entrance of the Hebrew Benevolent Congregation in Atlanta, called The Temple, in the early hours of October 12, 1958. It was the congregation Leo Frank had belonged to. No one was hurt. Its rabbi, the congregation's religious leader, was Jacob Rothschild. He had preached against segregation in his sermons since 1947.

Within days police arrested five men, all members of anti-Semitic groups, groups that spread hatred of Jews. Only one of them, George Bright, was tried. His first trial ended in a mistrial, a trial stopped without a verdict, because the jury split 9 to 3. At his second trial the jury found him not guilty. The charges against the other four men were dropped. No one was ever convicted of the bombing.

BY 1958: Bombs had gone off at Bethel Baptist in Birmingham twice and at The Temple in Atlanta once, and no one had yet been convicted of any of them.

Prayer and the Bible in public schools, 1962 and 1963

The justices of the Supreme Court ended official prayer and Bible reading in public schools in 1962 and 1963. The families who brought the cases came from several faiths, and some had none.

Two rulings. On June 25, 1962, in *Engel v. Vitale*, the justices ruled that New York's public schools could not have pupils say a prayer written by the New York State Board of Regents, the state officials in charge of its schools. On June 17, 1963, in *School District of Abington Township v. Schempp*, they ruled that public schools could not open the day with Bible reading as a religious exercise.

The parents of ten pupils brought the New York case. They said the official prayer went against their own beliefs and their children's. Judges in a New York court had already found that the prayer did not fit the teachings of Jewish, Unitarian or Ethical Culture groups.

Unitarians belong to a church that does not teach the Trinity, the belief that God is three persons in one, and the Ethical Culture society is a group that meets to teach right conduct. One parent was an agnostic, a person who holds that no one can know whether God exists.

The Schempps. The Schempps, who brought the 1963 case, went to church. Edward and Sidney Schempp and their children Roger and Donna were members of the Unitarian church in Germantown, Philadelphia, and so was their son Ellory, who had started the case. Under Pennsylvania law, public schools had to read ten verses from the Bible each morning. A child could be excused if a parent asked, and Edward Schempp decided not to ask.

He testified that teachers and classmates would label his children "odd balls" every school day. He said classmates would lump every religious difference together as "atheism," the belief that there is no God, and in 1962 atheism was tied to communism and called un-American. He also said that an excused child had to stand in the hall outside the homeroom, which looked like a punishment for bad behavior.

Two scholars. Two scholars testified about the Bible itself. Solomon Grayzel, for the families, said the Jewish scriptures have no New Testament. He said that from the standpoint of Jewish faith, calling Jesus the Son of God is "practically blasphemous," which means close to an insult against God. Read to children without explanation, he said, parts of the New Testament had been "psychologically harmful to the child."

Luther A. Weigle, for the school district, said the Bible was "non-sectarian," meaning it belonged to no single religious group. He later said he meant non-sectarian among Christian churches.

A second case was decided with the Schempps'. In Baltimore, schools opened each day with a chapter from the Bible or the Lord's Prayer. Madalyn Murray and her son William J. Murray III were atheists. They argued that the rule threatened their religious liberty "by placing a premium on belief as against non-belief," which means it rewarded believers over people who do not believe.

The day after, and the years after

Calls to undo it. The day after the 1962 ruling, a member of Congress introduced an amendment, a change to the Constitution, to undo it, and others were drafting more. The *Evening Star* newspaper of Washington, D.C., reported on June 26, 1962, that Representative Taylor of North Carolina was first. The paper noted that Billy Graham was one of his constituents, the people he represented.

Senator Beall of Maryland had one ready for the Senate. Representative Chelf of Kentucky said that if he could not get a law passed, he would try to limit Supreme Court justices to a set number of years instead of life.

In Washington, D.C., the school board's rules required a Bible reading and the Lord's Prayer to open the school day. Superintendent Carl F. Hansen said that "on the surface it appears that it does invalidate the board's rule," meaning that the ruling seemed to cancel it. None of the proposed amendments became part of the Constitution.

Earlier limits. The justices had set some limits before 1962. In 1948, in a case brought by a mother named Vashti McCollum, the justices ruled against a program in Champaign, Illinois. There, Catholic, Protestant and Jewish teachers came into the public school buildings once a week to teach religion for thirty to forty-five minutes. Children whose parents asked had to attend.

The justices allowed a program in New York City in 1952 in which children left the school buildings to go to a church or synagogue for the lesson. So the justices ruled out religion lessons inside the school building, and they allowed lessons that children left the building to attend.

Justice Clark also wrote, in the 1963 ruling, that teaching *about* religion was allowed. He wrote that "it might well be said that one's education is not complete without a study of comparative religion or the history of religion and its relationship to the advancement of civilization." He wrote that the Bible "is worthy of study for its literary and historic qualities."

A moment of silence. After 1963 Alabama's lawmakers tried a moment of silence. They passed a law in 1978 for one minute of silence "for meditation." In 1981 they changed it to "for meditation or voluntary prayer," and in 1982 they let teachers lead "willing students" in a prayer to "Almighty God." In 1985, in *Wallace v. Jaffree*, the justices struck down the 1981 law because its purpose was to support religion. They left the 1978 law for plain silence in place.

The justices struck down a Kentucky law in 1980, in *Stone v. Graham*, that required the Ten Commandments to be posted in every public classroom.

A graduation prayer. The justices ruled in 1992 that a principal could not invite a member of the clergy, such as a minister or a rabbi, to pray at a school graduation. Principal Robert E. Lee of Nathan Bishop Middle School in Providence, Rhode Island, had invited Rabbi Leslie Gutterman of Temple Beth El to pray at Deborah Weisman's graduation. The principal gave the rabbi a pamphlet of guidelines and told him to keep the prayers non-sectarian. The rabbi began, "God of the Free, Hope of the Brave."

Deborah's father, Daniel Weisman, sued. In *Lee v. Weisman*, the justices held that by handing over the pamphlet and the instruction, the principal "directed and controlled the content of the prayers."

Evolution in court again, 1968 and 1987

Two state laws about teaching evolution reached the Supreme Court, in 1968 and 1987. The justices struck down both, and both times they named the religious belief behind the law.

Susan Epperson. Susan Epperson had gone to school in Arkansas and earned a master's degree in zoology. Little Rock school officials hired her in the autumn of 1964 to teach tenth-grade biology at Central High School. For the 1965 to 1966 school year, school officials adopted a new biology textbook with a chapter on evolution. Teaching that chapter was a crime under the Arkansas law of 1928. She sued, and H. H. Blanchard, a parent of children in the public schools, joined her case.

The justices struck the law down on November 12, 1968, in *Epperson v. Arkansas*. They wrote that "fundamentalist sectarian conviction was and is the law's reason for existence," meaning that the law existed because of one group's religious belief. They also wrote that the First Amendment requires the government to be neutral "between religion and religion, and between religion and nonreligion."

No justice voted to keep the law. They did not all agree on the reasons, and Justice Black wrote separately that he doubted the Court's reading of why Arkansas had passed it.

Creation science. Louisiana's lawmakers passed a different kind of law. It stated that a public school that taught evolution also had to teach "creation science," and a school that taught creation science also had to teach evolution. Creation science was the teaching that a supernatural being created humankind. The people who sued to stop the law included Louisiana parents, Louisiana teachers and religious leaders.

Religious groups stood on both sides. Groups may send the Court a friend-of-the-court brief, a written argument from people who are not in the case. Three groups, among them the Catholic League for Religious and Civil Rights and the Christian Legal Society, sent briefs asking the justices to keep the law. The New York Committee for Public Education and Religious Liberty sent one asking them to strike it down.

The justices struck the law down by 7 votes to 2 on June 19, 1987, in *Edwards v. Aguillard*. They held that "the preeminent purpose of the Louisiana Legislature was clearly to advance the religious viewpoint that a supernatural being created humankind." The justices meant that Louisiana's lawmakers passed the law mainly to promote one religious belief.

BY 1987: The justices had ended official prayer and Bible reading in public schools and struck down two state laws about teaching evolution, each time naming the religious belief behind it.

Jonas and Frieda Yoder

Who: An Old Order Amish father who would not send his daughter to high school, and the daughter, who told the court that the decision was her own religion too.

When and where: Green County, Wisconsin, 1968 to 1972.

Frieda Yoder, fifteen, was the only child who testified.

Frieda Yoder was the only child who testified in her family's case, which went to the Supreme Court and asked whether Amish parents could keep their children out of high school. Her father, Jonas Yoder, belonged to the Old Order Amish, a Christian church whose members keep apart from the outside world. They lived in Green County, Wisconsin.

Wisconsin law required children to stay in school until they turned sixteen. Frieda was fifteen and had finished the eighth grade at the public school. Her father did not send her on to high school.

One line from the Bible. His reason was one line from the Bible. In his letter to the Romans, Paul, an early leader of the Christian church, wrote, "be not conformed to this world." The Amish take that line as a rule for daily life. They did not object to the first eight grades, because a child needs to read the Bible, farm and deal with people outside the community. They saw the high school as the world they were told not to join.

The case. Jonas Yoder was charged, tried and convicted in the Green County Court along with two other fathers, Wallace Miller and Adin Yutzy. Miller was also Old Order Amish, and Yutzy belonged to the Conservative Amish Mennonite Church. On May 15, 1972, in *Wisconsin v. Yoder*, the justices of the Supreme Court decided for the three fathers.

Frieda testified that her own religious beliefs were the reason she had stopped going to school after the eighth grade. Neither side called the other children. Justice Douglas agreed with the ruling for her father. He disagreed with it for the other two fathers, because no one had asked their children, Barbara Miller and Vernon Yutzy, what they believed. He said the Court had never answered whose religion counts when a family's religion decides a child's schooling, the parents' or the child's.

Ministers on both sides of a textbook fight, 1974

Ministers spoke up on both sides of a fight over schoolbooks in Kanawha County, West Virginia, in 1974. During the protests people fired guns, and attackers firebombed and dynamited schools. The county had 46,000 pupils in 124 public schools. In April 1974 the five members of the county school board voted to adopt 325 language-arts textbooks and other books.

The next month a board member, Alice Moore, objected to what some of the books said and taught. She was the wife of a minister, and she had been elected in 1970 after campaigning against sex education. She and several ministers campaigned against the books. Twenty-seven ministers condemned them as immoral and indecent, and petitions carried 12,000 signatures. Another minister, Jim Lewis of Charleston, spoke out most strongly for the books. In June the board voted 3 to 2 to accept most of them.

The boycott. When school opened on September 3, 1974, a fifth of the pupils stayed home in a boycott, a refusal to take part. Picket lines, groups of protesters standing outside with signs, spread from the schools to businesses and coal mines, and 3,500 miners walked off the job in support.

The West Virginia Encyclopedia lists gunfire, firebombs and dynamite attacks on schools, and it does not name who carried them out. Police made 11 arrests. The schools were closed for four days starting September 12. The records of the protest give no count of the injured.

The protests went on into 1977, and outside groups joined them, among them the Ku Klux Klan and the John Birch Society, a political group. The Reverend Marvin Horan was tried for plotting to bomb schools, and the violence ended with his trial. Board members set up committees that gave parents a part in choosing textbooks.

The law that said Native religion was allowed, 1978 to 1997

Members of Congress passed the American Indian Religious Freedom Act on August 11, 1978. In it they declared that it was the policy of the United States to protect the right of Native people "to believe, express, and exercise the traditional religions of the American Indian, Eskimo, Aleut, and Native Hawaiians." That included the right to visit sacred sites, to have sacred objects, and to worship "through ceremonials and traditional rites."

The act came ninety-five years after the 1883 rules that made the Sun Dance, a sacred ceremony of the Plains nations, a crime.

What Congress admitted. The members of Congress wrote down what had been happening. Federal laws and policies "often deny American Indians access to sacred sites required in their religions, including cemeteries." They wrote that "traditional American Indian ceremonies have been intruded upon, interfered with, and in a few instances banned." Federal officials had done this, as Price's rules of 1883, Jones's order of 1902 and Burke's circular of 1921 show.

For example, a Native person had needed a federal permit since 1962 to hold an eagle feather for worship. Without one, the penalty was up to $500 or six months in prison.

Chimney Rock. The 1978 Act stated a policy. In 1988 the justices of the Supreme Court ruled that it created no right a court could enforce, in the case *Lyng v. Northwest Indian Cemetery Protective Association*.

Officials of the U.S. Forest Service planned to build a six-mile paved road through the Chimney Rock area of Six Rivers National Forest in California, and to allow logging there. The Yurok, Karuk and Tolowa had long used this high country for worship. Justice Brennan, who disagreed with the ruling, wrote that they had held it sacred "for at least 200 years and probably much longer." It is land of the Yurok, Karuk and Tolowa that the Forest Service holds as federal land.

The authors of a study that the Forest Service itself paid for found that the road "would cause serious and irreparable damage to the sacred areas" and advised against building it. In 1982 Forest Service officials chose a route through Chimney Rock anyway.

The justices ruled 5 to 3 that the Constitution did not stop the road. Justice O'Connor wrote that even if the road would "virtually destroy the Indians' ability to practice their religion, the Constitution simply does not provide a principle" for stopping it. The justices' opinion also records that the Native people of the area did not all oppose the road.

Peyote. Alfred Smith and Galen Black were members of the Native American Church. Their employer, a private drug treatment program, fired them because they ate peyote at a church ceremony. Peyote is a small cactus that causes hallucinations, which means it makes a person see or feel things that are not there. Members of the church take it as a sacrament, a sacred act of worship.

Oregon officials refused to pay the men unemployment money. The justices ruled 6 to 3 in 1990, in *Employment Division v. Smith*, that Oregon could ban peyote even for worship and deny the payments. Justice Scalia wrote that leaving such questions to lawmakers "will place at a relative disadvantage those religious practices that are not widely engaged in."

Congress answers. Members of Congress answered twice. In 1993 they passed the Religious Freedom Restoration Act. It stated that government officials could not put a heavy burden on a person's religion unless they had a reason of the highest importance, and that they had to choose the way that restricted religion least.

Members of Congress also amended the 1978 Act on October 6, 1994. They wrote that 28 states protected the peyote ceremony and 22 did not, and they made it lawful everywhere in the country for a Native person to use peyote in traditional religious ceremonies. In 1997, in a case brought by Archbishop Patrick Flores of San Antonio, the justices ruled that the 1993 act went beyond Congress's power over the states. That ruling did not touch the 1994 peyote law.

Jonestown, 1978

Jim Jones, the leader of an American congregation called Peoples Temple, ordered its members at its settlement in Guyana, in South America, to take poison on November 18, 1978. More than 900 people died. Jones had founded Peoples Temple in Indianapolis in the 1950s and moved it to San Francisco in 1971. Members then moved to a settlement they called Jonestown.

That day Temple members shot and killed Congressman Leo Ryan of California and four other people at the Port Kaituma airstrip, among them an NBC cameraman and a newspaper photographer.

How the children died. At Jonestown, at Jones's order, parents and nurses used syringes to squirt a mix of cyanide, sedatives and powdered fruit drink into the children's mouths. Cyanide is a poison that stops the cells of the body from using oxygen, so the cells cannot make the energy they need to live. Sedatives are drugs that make a person calm and sleepy.

More than 200 children were killed this way, according to the Association for Diplomatic Studies and Training, an organization of former American diplomats. The children could not choose, and the association calls them murdered. Then the adults drank the mix while armed guards surrounded the pavilion, the large roofed meeting place where they had gathered. In all, 909 members died at Jonestown, all but two of them from what appeared to be cyanide poisoning. One or two people there, among them Jim Jones, had bullet wounds instead.

Houses of worship attacked, 1991 to 2000

Gunmen shot nine people dead at Wat Promkunaram, a Thai Buddhist temple near Phoenix, Arizona, on August 10, 1991, during a robbery of the monks. Johnathan Doody and Alessandro Garcia were later convicted of the killings. The dead were six monks, a nun, a novice, who is a person training to become a monk, and a boy who helped at the temple. Each was shot in the head.

Doody was 17 at the time. His first conviction was thrown out because federal appeals judges found that his confession had been coerced, which means forced out of him. At his third trial he was convicted of nine counts of first-degree murder, a killing planned ahead of time, one for each person killed. He was sentenced to 249 years in prison on March 14, 2014. Alessandro Garcia pleaded guilty and was sentenced to life, NBC News reported.

Church fires. People set fire to churches across the country in the middle of the 1990s. Members of Congress passed the Church Arson Prevention Act on July 3, 1996, arson being the crime of setting a fire on purpose. They wrote that fires set at places of worship had increased, "especially in the context of places of religious worship that serve predominantly African-American congregations," which means mostly Black congregations.

Agents of a federal task force from the Justice and Treasury departments counted 297 arsons, bombings or attempted bombings at houses of worship in 1996. The count fell every year after that, to 140 in 1999. Between January 1995 and August 2000 the agents arrested 431 suspects, and courts found 305 people guilty.

Jay Scott Ballinger pleaded guilty on July 11, 2000, to setting fire to 26 churches in eight states between 1994 and 1999.

Mount Carmel, near Waco, 1993

A standoff between federal agents and a religious community near Waco, Texas, ended in 1993 when some of the Davidians set fire to their own building. More than 70 people died in it, 22 of them children, according to a report by members of Congress. A Special Counsel, an outside lawyer appointed to investigate, later found that some of the Davidians had spread fuel through their building and set it on fire.

The Branch Davidians lived at a property called Mount Carmel, outside Waco, and their leader was Vernon Howell, who called himself David Koresh.

The raid. Agents of the federal Bureau of Alcohol, Tobacco and Firearms, the ATF, stormed the building on February 28, 1993. There were 76 of them, and they had warrants, papers from a judge that allowed them to arrest people and search the building. The Davidians had learned of the raid in advance. The gun battle lasted almost 90 minutes.

The Special Counsel found that Davidians shot and killed four ATF agents, Conway C. LeBleu, Todd W. McKeehan, Robert J. Williams and Steven D. Willis, and wounded 20 others. Six Davidians were killed that day, by the Special Counsel's count. The House report says that ATF agents killed at least two of them.

DISPUTED: who fired first. In their 1996 report, members of two House subcommittees, small groups of members of the House of Representatives, found that the question could not be settled for certain, but that "It appears more likely, however, that the Davidians fired first." ATF agents testified that the Davidians did. The Davidians' lawyers said they believed an ATF gun went off by accident.

Fifty-one days. Agents of the FBI then took over and talked with the Davidians for 51 days. By March 23, 35 people had come out. The House majority report records 21 of them as children, and the minority report records 14. After March 23 no one else left.

A six-year-old girl, Melissa Morrison, told an FBI negotiator that she could not leave because "David won't let me." She died in the fire.

April 19. FBI agents in armored vehicles began punching holes in the walls and spraying tear gas inside at about 6 in the morning on April 19, 1993, and Davidians fired on the vehicles. At about 12:07 in the afternoon a fire broke out in the building. Two more began within two minutes, and within eight minutes the whole building was burning.

What the Special Counsel found. Former Senator John C. Danforth was the Special Counsel. The Attorney General, the head of the Justice Department, appointed him, and he studied the case for ten months. In 2000 he reported that government agents did not start or spread the fire and did not shoot at the Davidians.

He found that some of the Davidians spread fuel through the building and set it on fire in at least three places. Danforth placed the responsibility for the deaths on "certain of the Branch Davidians and their leader, David Koresh." He counted at least 80 dead in all, including the six Davidians killed on February 28.

Danforth found that some Davidians shot and killed their own people, among them at least five children. At least twenty Davidians were shot dead on April 19, many of them in the head or mouth at close range. A three-year-old, Dayland Gent, was stabbed in the chest and died. Danforth's report does not name who stabbed him.

Danforth also reported that an FBI agent had fired three tear-gas rounds of a kind that can start fires at 8:08 that morning, toward a pit about 75 feet from the living quarters. Government officials did not reveal this until August 1999. Danforth found that those rounds did not start the fire four hours later. A Texas jury later found eight Davidians guilty of gun crimes.

BY 2000

If you remember one thing: Families of many faiths, and some of none, won Supreme Court cases that ended official prayer and Bible reading in public schools.

One number: about 145, the attacks on Black churches across the South from about 1954 to 1970 that one research team has mapped.

Next: After 2000 a smaller share of Americans belonged to a congregation, and gunmen killed worshippers in a Sikh gurdwara, in churches and in a synagogue.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 29 MINUTES

Fewer Americans have belonged to a congregation since 2000. In a survey taken in 2023 and 2024, 29 percent of American adults said they had no religion, up from 16 percent in 2007. People attacked worshippers inside gurdwaras, the houses of worship of the Sikh religion, and inside churches and synagogues, and killed them where they prayed.

Abuse and courts. Records made public in these years showed that more than 10,000 people had told Catholic officials that priests sexually abused them as children. A grand jury in Pennsylvania, a group of citizens who hear evidence gathered by prosecutors, found that church leaders in six dioceses, the regions led by bishops, had hidden the abuse there.

A report on the Southern Baptist Convention found that its leaders had pushed back against people who reported abuse, and sometimes met them with outright hostility. Families and teachers kept going to court over what a public school may teach about religion and whether public money may pay for religious schools.

How many people say they have no religion

Twenty-nine percent of American adults, more than one in four, said they had no religion when researchers at the Pew Research Center, a private polling organization, asked 36,908 of them in 2023 and 2024. In Pew's study of 2007 the share was 16 percent. This group, often called the "nones," includes atheists, agnostics and people who answer "nothing in particular."

Sixty-two percent said they were Christian, down from 78 percent in 2007. Fewer than two adults in a hundred said they were Jewish, and about one in a hundred each said they were Muslim, Buddhist or Hindu. Pew's own report states that the fall in the Christian share had slowed and may have stopped, because it had stayed about the same for four or five years.

Gallup's question. Gallup's pollsters ask a different question: whether people belong to a church, synagogue or mosque. In 2020, for the first time since Gallup began asking in 1937, fewer than half said yes. The share was 47 percent, down from 70 percent in 1999.

Older Americans belonged far more often than younger ones. In the years 2018 to 2020, about two in three Americans born before 1946 belonged to a congregation, against a little more than one in three of those born between 1981 and 1996.

Killed where they were worshipping, 2001 to 2025

FBI officials count the crimes that police report as motivated by hatred of a group. Their report for 2001, the year of the September 11 attacks, records 481 incidents aimed at Muslims, with 554 victims. That was more than 1,600 percent higher than in 2000, or about seventeen times as many. Crimes aimed at Jews were still the largest group, at more than half of all religious hate crimes that year.

Oak Creek, 2012. A white supremacist, a person who believes white people are better than others and should rule over them, walked into the gurdwara at Oak Creek, Wisconsin, on Sunday morning, August 5, 2012. He carried a loaded gun, and he killed six people. Paramjit Kaur was sitting for morning prayers when he shot her. The others killed were Sita Singh, Ranjit Singh, Prakash Singh, Suvegh Singh Khattra and Satwant Singh Kaleka, the gurdwara's president, who was killed fighting the gunman.

Santokh Singh and Punjab Singh were shot and lived through the attack, but Punjab Singh later died of his wounds. In 2022 Senator Tammy Baldwin of Wisconsin introduced a resolution in the United States Senate that counts him as the seventh person killed. The resolution also names Lieutenant Brian Murphy, a police officer who was shot 15 times at close range.

Harpreet Singh Saini

Who: A young man whose mother was shot dead at morning prayers, and who asked the United States Senate to start counting crimes against Sikhs.

When and where: Oak Creek, Wisconsin, and Washington, D.C., August and September 2012.

He asked for his mother "the dignity of being a statistic."

Harpreet Singh Saini asked the United States Senate to count crimes against Sikhs, forty-five days after a gunman killed his mother at prayer. His parents brought him and his brother, Kamaljit, from India to the United States in 2004, when he was ten. "Like many other immigrants," he said, "they wanted us to have a better life, a better education."

His mother was Paramjit Kaur, the woman shot at morning prayers in the Oak Creek gurdwara. The five men killed with her all wore turbans, as Harpreet does.

The last rotis. Two days after the shooting, Harpreet and his brother ate the last rotis, a kind of flatbread, that their mother had made from scratch the night before she died. He said that with the last bite he understood it was the last food made by her hands that they would ever eat. He had just started college, and she was not there to send him off.

Before the Senate. On September 19, 2012, he testified before a subcommittee of the United States Senate. He said: "I want to tell the gunman who took her from me: You may have been full of hate, but my mother was full of love."

He told the senators what his mother's faith taught her. "My mother was a devout Sikh. Like all Sikhs, she was bound to live in Chardi Kala," which he described as "a state of high spirits and optimism." He said, "She was also taught as a Sikh to neither have fear of anyone nor strike fear in anyone."

> Then he asked for one thing. "I came here today to ask the government to give my mother the dignity of being a statistic. The FBI does not track hate crimes against Sikhs. My mother and those shot that day will not even count on a federal form. We cannot solve a problem we refuse to recognize."
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> The FBI's report for 2015 was the first to list crimes against Sikhs. It counted 6 offenses against Sikhs that year.

Charleston, Sutherland Springs and Pittsburgh, 2015 to 2018

Charleston, 2015. A 21-year-old man named Dylann Roof killed nine members of Mother Emanuel church in Charleston, South Carolina, at their weekly Bible study on Wednesday evening, June 17, 2015. Mother Emanuel is an African Methodist Episcopal church, part of the denomination, a group of churches sharing one set of teachings, that Richard Allen and other Black Methodists formed in 1816. Twelve members were meeting in the fellowship hall, and Roof came in at about 8:16.

Judges of a federal appeals court later wrote that "the parishioners welcomed Roof, handing him a Bible and a study sheet," and that for 45 minutes he "worshipped with the parishioners." Then: "They stood and shut their eyes for closing prayer. Roof then took out his gun and started shooting." He fired about seventy-four shots.

The nine killed were the Reverend Sharonda Coleman-Singleton, Cynthia Hurd, Susie Jackson, Ethel Lee Lance, the Reverend DePayne Middleton-Doctor, the Reverend Clementa Pinckney, Tywanza Sanders, the Reverend Daniel Simmons Sr. and the Reverend Myra Thompson. Roof told the FBI that he had looked up African American churches in Charleston, visited this one beforehand, and learned from a member that the Bible study met on Wednesday nights.

A federal jury convicted him on 33 counts, among them nine counts of obstructing religion resulting in death. That federal crime means using force to stop people from practicing their religion, in a way that kills someone. He

was sentenced to death, and appeals judges upheld the sentence on August 25, 2021.

Sutherland Springs, 2017. Devin Patrick Kelley walked into the First Baptist Church of Sutherland Springs, Texas, on Sunday, November 5, 2017, and opened fire. He killed twenty-six people and wounded twenty-two more, then died of his own gunshot after fleeing. Among the dead were Emily Hill, 11, and Megan Hill, 9. Kelley had been convicted of a crime of domestic violence while he was in the U.S. Air Force. In 2012 a military court found him guilty of attacking his wife at the time, Tessa Brennaman, and his stepson.

The missing record. Air Force agents and jail staff never sent his fingerprints or his conviction to the FBI. A gun store must run a background check, a search of national criminal records, before it sells a gun, and a reported conviction like his would have stopped the sale. Without the record, Kelley passed the check each time he bought a gun from a store.

Judge Xavier Rodriguez, a federal judge, found in 2021 that federal officials had failed "to exercise reasonable care" in sending his criminal record to the FBI. He held the United States government 60 percent responsible for the harm. He wrote that if the government had reported the record, it was "more likely than not" that Kelley would have been deterred, or put off, from the shooting.

Pittsburgh, 2018. Three Jewish congregations, Tree of Life, Dor Hadash and New Light, were at prayer in one building in Pittsburgh, Pennsylvania, on Saturday morning, October 27, 2018. Twelve Tree of Life members were in the chapel, partway through the opening prayers. Audrey Glickman had begun the service. David Rosenthal, who could not read but had memorized every prayer and song, was helping to lead it, and his brother Cecil was one of the greeters.

In Dor Hadash's room, three men were setting out prayer books and copies of the Torah, the Jewish scriptures: Jerry Rabinowitz, a doctor, Daniel Leger, a hospital chaplain, a minister who serves patients, and Martin Gaynor. Downstairs, Richard Gottfried and Daniel Stein of New Light were in the kitchen making a meal for a breakfast the next day.

Robert Bowers came in with three handguns and a rifle and killed eleven people: Jerry Rabinowitz, David Rosenthal, Cecil Rosenthal, Rose Mallinger, Bernice Simon, Sylvan Simon, Joyce Fienberg, Irving Younger, Richard Gottfried, Daniel Stein and Melvin Wax. The youngest was 54, and Rose Mallinger was 97.

Two more worshippers were badly wounded, and five police officers were injured going in after him. Federal prosecutors charged him with eleven counts of obstructing the free exercise of religious beliefs resulting in death, among other crimes. He was sentenced to death on August 3, 2023.

BY 2018: Gunmen had killed worshippers at morning prayers in a gurdwara in Oak Creek, at a Bible study in Charleston, at a Sunday service in Sutherland Springs and at Saturday prayers in Pittsburgh.

Two attacks on churches in 2025

Minneapolis. Children at the Church of the Annunciation in Minneapolis were at an all-school Mass, the Catholic worship service, on Wednesday, August 27, 2025, in the first week of school. A shooter fired through the church windows at them in the pews. Two children were killed. Twenty-six children and three adults were wounded by gunfire, and one more person was hurt in another way, according to the City of Minneapolis.

NPR reported that the shooter was Robin Westman, 23, who then fatally shot themself at the scene, according to police. NPR reported that FBI officials called it an act of domestic terrorism, an attack meant to frighten people inside the country. They said it was driven by a "hate-filled ideology," a set of hateful ideas.

Grand Blanc Township. Members of the Church of Jesus Christ of Latter-day Saints were at services in their meetinghouse in Grand Blanc Township, Michigan, on Sunday, September 28, 2025. Thomas Jacob Sanford, 40, of Burton, Michigan, drove his pickup truck through the front doors. He opened fire and set the building on fire. He killed four people: Craig Hayden, Pat Howard, John Bond and Thelma Armstrong.

Nine others, aged 6 to 78, were wounded. Police shot him dead within 3 minutes and 43 seconds of the first 911 call, the television station WNEM reported. The reports of the attack give no settled reason for it.

The count for 2025. FBI officials released their count of hate crimes for 2025 in August 2026. By the reading of the Anti-Defamation League, a group that speaks out against anti-Semitism, police reported 2,408 hate-crime incidents based on religion, and 1,528 of them, or 63 percent, were aimed at Jews. Staff of the Sikh Coalition, a group that speaks up for Sikhs, counted 177 incidents aimed at Sikhs in the FBI's figures for 2025, up from 142 in 2024.

What priests did to children, and what bishops did about it

More than 4,000 Catholic priests in the United States were accused of sexually abusing children between 1950 and 2002. Sexual abuse of a child means an adult using a child's body for sex. Church officials reported few of these priests to the police, and in Pennsylvania a grand jury found that church leaders in six dioceses had hidden the abuse.

The John Jay report. In 2002 the Catholic bishops of the United States asked John Jay College of Criminal Justice, in New York, to study how many of their priests had done it. Researchers at the college published their report in February 2004. It was based on surveys covering 98 percent of the country's diocesan priests, the priests who serve in a bishop's region.

The report records accusations against 4,392 priests for the years 1950 to 2002, about 4 percent of all the priests who served in those years, or one in twenty-five. These were accusations that had not been withdrawn and were not known to be false. In all, 10,667 people said they had been abused. Half of them were between 11 and 14 years old when it happened. Sixteen percent were between 8 and 10, and nearly 6 percent were younger than 7. Eighty-one percent were boys.

What was done to them. The acts reported most often were touching the child over or under the clothes. Next came the priest putting his mouth on the child's sex organs, the child being undressed, and rape, which means forcing sex on a person, or attempted rape. Rape or attempted rape was reported in 22.4 percent of the cases.

Most of the accused priests were said to have done more than one of these things to the same child. It happened most often in the priest's own home or the parish house, the house beside the church where priests live, and next most often inside the church.

Who was told. Only 384 of the 4,392 priests were ever charged with a crime. Victims reported fifteen percent of the priests to the police. The report says that dioceses and religious orders, the groups of priests, brothers or sisters who live under shared religious promises, reported "a much smaller number." Before 1993, church officials knew of only a third of the cases at all.

The dioceses and religious orders reported that they had already paid $472 million for victims, for treating priests and for lawyers, and 14 percent of them gave no figure.

Accusations after 2004. People kept coming forward after the report, and most of them told of abuse from decades before. Researchers at the Center for Applied Research in the Apostolate at Georgetown University sort the accusations made since 2004 by the years in which the abuse is said to have happened. They count 4,216 for the ten years of the 1960s, 2,813 for just the five years from 1970 to 1974, and 36 for the three years from 2020 to 2022.

These are accusations made years afterward. They are not a count of everything that happened in each period, and one priest can be accused by more than one person.

Pennsylvania, 2018. A grand jury in Pennsylvania released a report on six of the state's Catholic dioceses in August 2018. The jurors read half a million pages of the dioceses' own internal records. They found believable accusations against more than three hundred priests. They identified more than one thousand children from the church's own records, and they wrote that they believed the real number was in the thousands.

The jurors wrote that most of the victims were boys and some were girls, some were teenagers, and many were young children whose bodies had not yet begun to grow into adult bodies. Some were raped. "But all of them were brushed aside, in every part of the state, by church leaders who preferred to protect the abusers and their institution above all."

The playbook. The grand jury set out what it called a playbook for hiding the truth. Its first item was to use mild words instead of real ones in church documents: never to write "rape," but "inappropriate contact" or "boundary issues" instead. In the grand jury's account, church officials let fellow priests ask the questions rather than trained investigators. When they removed a priest, they told his parish he was on "sick leave," or said nothing.

They kept paying for his housing and living costs. If people found out, they moved him to a new place "where no one will know he is a child abuser." The last item was "Finally and above all, don't tell the police." Because of this hiding, the jurors wrote, "almost every instance of abuse we found is too old to be prosecuted."

The bishops released their own yearly audit, a check of their records, for July 2024 to June 2025 in May 2026. It records 1,070 new accusations made by 973 people, OSV News, a Catholic news service, reported. Only 24 of them came from people who were still children. Most of the rest concerned abuse that had happened decades earlier. Diocesan officials reported $389.9 million in costs connected with abuse that year.

The Southern Baptists. Members of the Executive Committee of the Southern Baptist Convention, a large group of Baptist churches, hired an outside firm, Guidepost Solutions, to look at how they had handled abuse. The firm published its report in May 2022. It states that people who reported abuse "were met, time and time again, with resistance, stonewalling, and even outright hostility," stonewalling meaning refusing to answer or act.

The firm's investigators found that two former leaders, D. August Boto and Roger Oldham, had kept a private list of ministers accused of abuse. On May 26, 2022, the committee released the list, 205 pages long, NBC News reported.

A sacred place traded to a mining company, 2014 to 2026

Members of Congress ordered in 2014 that Oak Flat, a place where Apache people pray, be traded to a mining company, and in March 2026 the trade was completed. Oak Flat, called Chí'chil Biłdagoteel in Western Apache, is a place in the Tonto National Forest in Arizona where Apache people hold ceremonies, among them the Sunrise Ceremony. It was listed on the National Register of Historic Places in 2016.

The company is Resolution Copper, owned by the mining companies Rio Tinto and BHP. Members of Congress put the order into the yearly bill that pays for the armed forces. With the exchange of about 2,400 acres in March 2026, the land became private property. Company officials expect the mine to leave a crater about two miles wide within about forty years, the radio station KJZZ reported.

A church divides, 2019 to 2024

More than a quarter of the United Methodist Church's American congregations left it between 2019 and the end of 2023. Its members had argued over gay members and gay ministers, gay meaning attracted to people of the same sex, since at least 1984. That year a ban on making gay people ministers went into the church's rule book.

The way out. At a special General Conference in 2019, the gathering where delegates write the church's rules, delegates added a rule. Under it, a congregation could leave the denomination, the whole group of United Methodist churches, and keep its buildings for "reasons of conscience" over the church's decisions about homosexuality, another word for being gay. The rule ran out on December 31, 2023. By then more than 7,600 congregations in the United States had left under it, about 26 percent of the church's American congregations.

The ban lifted. Delegates at the General Conference in Charlotte, North Carolina, voted 692 to 51 on May 1, 2024, to remove the ban that had been in the church's rule book since 1984. It had banned making gay people ministers, in the rule's words "self-avowed practicing homosexuals," meaning gay people

who said so openly and lived as gay people. Delegates at the same conference also removed the ban on ministers and churches holding weddings for same-sex couples.

BY 2024: Researchers working for the Catholic bishops had counted 10,667 people who said priests abused them as children, Southern Baptist leaders had released a private list of accused ministers, and about a quarter of United Methodist congregations had left their church.

A prisoner's dreadlocks, 2020 to 2026

Damon Landor was a prisoner at the Raymond Laborde Correctional Center in Louisiana. In 2020, on the warden's order, guards there handcuffed Landor to a chair and shaved off his dreadlocks. Landor had grown them for decades under a Rastafarian vow, a religious promise not to cut his hair. He showed the guards a court decision, and they threw it away. He had three weeks left to serve.

Landor sued under the Religious Land Use and Institutionalized Persons Act, a federal law that protects the religious practice of people held in prisons. The justices of the Supreme Court ruled 6 to 3 on June 23, 2026, that he could not sue the individual officers for money under that law. Justice Neil Gorsuch wrote for the majority. Justice Ketanji Brown Jackson disagreed, joined by Justices Sotomayor and Kagan. Published accounts of the case do not name the guards.

Worship on a screen

Twenty-seven percent of American adults were watching religious services on a screen by November 2022. Researchers at the Pew Research Center asked 11,377 adults how they took part in worship between November 16 and 27, 2022, and Pew published the answers in June 2023.

Sixteen percent were regularly going to services in person only, 10 percent were regularly watching only on a screen, and 17 percent were doing both. In all, 43 percent were attending or watching at least once a month, and 57 percent were doing neither. Among those who did both, 76 percent said they would rather be in the room. The reason people gave most often for watching on a screen was that it was convenient.

The churches answer for the boarding schools, 2022 to 2024

Catholic leaders apologized between 2022 and 2024 for the church's part in the boarding schools for Native children. Religious groups ran 210 of the 417 federal Indian boarding schools in the United States. At the 417 schools, staff cut children's hair, forbade their languages and beat them, and at least 973 children died, by the count of investigators at the Department of the Interior.

The pope. In Canada on July 25, 2022, Pope Francis asked forgiveness for the ways that church members "cooperated, not least through their indifference, in projects of cultural destruction and forced assimilation," forced assimilation meaning making people give up their own ways of life. He was speaking of Canada's schools.

The Doctrine of Discovery. Officials at the Vatican, the headquarters of the Catholic Church in Rome, rejected the Doctrine of Discovery on March 30, 2023. Catholic News Service described it as a set of old writings by popes that "appeared to bless the efforts of explorers to colonize and claim the lands of any people who were not Christian." The Vatican officials said it "is not part of the teaching of the Catholic Church."

The bishops' apology. The bishops of the United States Conference of Catholic Bishops voted 181 to 2 on June 14, 2024, to adopt a document called "Keeping Christ's Sacred Promise." It is a formal apology for the Catholic

Church's part in the schools in the United States. The Interior Department's 2024 report on the schools states that its authors know of no support from the bishops' conference for a Pope's apology for the schools in the United States.

Prayer at a football game, 2000

Families went to court again and again from 2000 on over religion in public schools and public money for religious schools. One case, decided on June 19, 2000, was about football.

At Santa Fe High School in Texas, before 1995, a student elected as chaplain of the student council said a prayer over the loudspeakers before every home game of the school's top football team. District officials then set up two student elections. One decided whether to have prayers at the games, and the other chose who would give them. Two families sued, one Mormon and one Catholic. In *Santa Fe Independent School District v. Doe*, the justices of the Supreme Court struck down the policy.

Public money and religious schools, 2022

The justices of the Supreme Court ruled in 2022 that Maine's officials could not refuse to pay for a child's schooling because the school taught religion. Under Maine law, the state pays the fees for children who live in towns with no high school of their own. Since 1981 it had paid only for schools that were nonsectarian, meaning not tied to any religious group.

David and Amy Carson of Glenburn, Maine, sent their daughter to Bangor Christian Schools, which Bangor Baptist Church had founded in 1970. They paid her fees themselves. Troy and Angela Nelson of Palermo, Maine, sent their son to Temple Academy, a school connected with Centerpoint Community Church. They could not afford its fees for their daughter too, so she went to a private school that was not religious. On June 21, 2022, in *Carson v. Makin*, the justices ruled 6 to 3 for the families.

What the schools taught. Justice Breyer, who disagreed, quoted the two schools' own descriptions of their teaching. One of Bangor Christian's goals is "lead[ing] each unsaved student to trust Christ as his/her personal savior," an

unsaved student being one who did not yet believe in Christ. School officials said there was no way to separate its religious teaching from its other teaching.

Temple Academy said the Bible is used in every subject it teaches. In mathematics that meant teaching that a creator made the universe so that "one plus one is always going to be two."

Joseph Kennedy

Who: A high school football coach who was put on paid leave over his prayers on the field after games, and who won his case in the Supreme Court.

When and where: Bremerton, Washington, 2008 to 2022.

He knelt at the 50-yard line after every game.

Joseph Kennedy was put on paid leave from coaching over prayers he said on the football field. In 2022 the justices of the Supreme Court ruled 6 to 3 that the school district's officials could not punish him for those prayers. He spent nearly twenty years in the Marine Corps. In 2008 he began coaching football at Bremerton High School in Washington State. After each game, once the players and coaches had shaken hands, he knelt at the 50-yard line and prayed for about thirty seconds.

At first he prayed alone. Then players asked whether they could join him, and he told them, "This is a free country. You can do what you want." After some games most of the team knelt with him. Sometimes players from the other team were invited to join. He began adding short speeches to encourage the players when others were there. No one complained for more than seven years.

The letters. In September 2015 an employee of another school praised the practice to Bremerton's principal, and the district's superintendent found out. District officials wrote to Kennedy on September 17, 2015. On October 16 they wrote again, forbidding "any overt actions" while he was on duty that could look like the school supporting prayer.

He prayed after the games of October 16, 23 and 26. On October 16 most of his players were singing the school fight song, and players from the other team and people from the crowd joined him before he finished. On October 23 no one joined him. On October 26 other adults gathered around him while he knelt. After that game, district officials put him on paid leave and barred him from the football program.

Two readings of one record. On June 27, 2022, in *Kennedy v. Bremerton School District*, the justices ruled for him, 6 to 3. Justice Gorsuch wrote for the majority that the First Amendment protects "an individual engaging in a personal religious observance from government reprisal," which means punishment by government officials.

Justice Sotomayor, who disagreed, described what happened differently. She wrote: "To the degree the Court portrays petitioner Joseph Kennedy's prayers as private and quiet, it misconstrues the facts," the petitioner being the person who brings a case to the Court. She wrote that the record showed that he "consistently invited others to join his prayers and for years led student athletes in prayer at the same time and location." The nine justices read the same record and described the events in two different ways.

The Ten Commandments in the classroom, 2024 to 2026

Louisiana's lawmakers passed a law requiring every public school classroom to display the Ten Commandments, and Governor Jeff Landry signed it on June 19, 2024. Families of several faiths sued to stop it. The justices of the Supreme Court had struck down a Kentucky law that required the same thing in 1980, in *Stone v. Graham*.

Louisiana's law requires a poster or framed copy at least 11 inches by 14 inches, with the Commandments as its "central focus" in large print. The law sets out the exact wording, which begins "I AM the LORD thy God."

The parents. The parents who sued include ministers and Jewish parents, and the case is named for one of them, the Reverend Darcy Roake. A Jewish parent swore that the law "misappropriates a Jewish text, ripping it from its Jewish context," which means it takes and misuses something that belongs to Jews.

The parent said it then requires the changed text to be shown to non-Jews, "in violation of core Jewish tenets that oppose proselytizing." Tenets are the main rules of a faith, and proselytizing means trying to convert people to a religion.

Another Jewish parent objected that Louisiana's version leaves out the freeing of the Jews from slavery in Egypt, "a key piece of the Ten Commandments and of the Jewish story." A Presbyterian minister who is also a parent objected to the state putting its "stamp of approval" on one version.

In the courts. A federal judge blocked the law, and on June 20, 2025, three judges of the Fifth Circuit Court of Appeals, the federal appeals court for Louisiana, Texas and Mississippi, agreed. Then all the judges of the Fifth Circuit reheard the case, and on February 20, 2026, they lifted the order blocking the law.

The law is called House Bill 71, or H.B. 71. The judges wrote that the question before them was "not whether H.B. 71 is constitutional, but whether that issue is fit for judicial resolution at this time." They held that it was too early to decide. Three judges disagreed in writing. As of September 2026, no court had decided whether Louisiana's law breaks the Constitution.

Texas. Texas's lawmakers passed a similar law. Rabbi Mara Nathan is one of the parents suing over it. She objects that the required text "erases this foundational Jewish tenet and historical context."

The full Fifth Circuit's judges ruled 9 to 8 on April 21, 2026, that the Texas law does not violate the First Amendment's religion clauses, the two parts of the amendment about religion. On August 17, 2026, the Texas families asked the Supreme Court to hear their case. As of late September 2026 the Court had not said whether it would.

Storybooks and parents' faith, 2025

Members of the school board of Montgomery County, Maryland, added five storybooks with LGBTQ+ characters, a short name for lesbian, gay, bisexual, transgender and queer people and others, for children from kindergarten to fifth grade in the 2022 to 2023 school year. At first board members let parents take their children out of those lessons. Then they ended that choice.

The parents. Parents of several faiths sued. Tamer Mahmoud and Enas Barakat are Muslims. Jeff Roman is Catholic and Svitlana Roman is Ukrainian Orthodox. Chris and Melissa Persak are Catholics.

The justices of the Supreme Court ruled 6 to 3 on June 27, 2025, in *Mahmoud v. Taylor*, that the parents were entitled to have their children excused while the case went on. Justice Alito wrote for the majority. He relied on the 1972 Amish case, *Wisconsin v. Yoder*, and its holding that parents have a right "to direct the religious upbringing of their children."

Justice Sotomayor wrote for the three who disagreed. She wrote that public schools "offer to children of all faiths and backgrounds an education and an opportunity to practice living in our multicultural society." She warned that this "will become a mere memory if children must be insulated from exposure to ideas and concepts that may conflict with their parents' religious beliefs."

Fewer sisters to teach in the Catholic schools

Members of religious orders built and staffed the Catholic school system of the 1800s. By 2025 there was about one Catholic sister, or nun, in the United States for every five there had been in 1965. The Center for Applied Research in the Apostolate at Georgetown University takes its figures from the Catholic Church's own yearly directory. Its researchers counted 178,740 religious sisters in the United States in 1965, 79,814 in 2000, and 33,135 on January 1, 2025.

Schools, priests and Mass. Over the same sixty years, Catholic elementary schools fell from 9,366 to 4,606, and Catholic high schools from 1,986 to 1,178. Priests fell from 59,426 to 33,462. In 1965, 530 parishes had no priest living there as pastor. In 2025, 3,674 did not. The same center estimates from surveys that 54.9 percent of Catholics went to Mass every week in 1970, and 17.6 percent in 2025.

Teaching religion at home

Some parents teach their children at home so that they can teach them religion. The National Center for Education Statistics, part of the U.S. Department of Education, surveyed parents of homeschooled children aged 5 to 17 about the 2022 to 2023 school year and published the answers in September 2024.

Fifty-three percent said that wanting to give religious instruction was an important reason for homeschooling, and 11 percent said it was the most important reason. More parents named a concern about the environment of other schools. That was important to 83 percent and the most important reason for 28 percent.

THE THREE QUESTIONS, ANSWERED

1. **Why did colonists who crossed the ocean to worship freely punish other people for their faith?** The Puritans wanted freedom for their own way of worship, and they ran Massachusetts

by it. Their leaders banished Roger Williams and Anne Hutchinson and hanged four Quakers in Boston.

2. **How did Americans stop paying taxes to support churches?** Baptists and James Madison defeated a new church tax in Virginia, where Jefferson's religious freedom law passed in 1786. Massachusetts ended the last state church tax in 1833.
3. **How long did US officials treat Native ceremonies as crimes?** For ninety-five years, from Hiram Price's rules of 1883 until members of Congress passed the American Indian Religious Freedom Act in 1978.

What changed: In the 1600s officials in New Mexico and Massachusetts punished people for how they worshipped, and later Americans fought for freedom of worship in court, sometimes winning and sometimes losing.

FIVE DATES TO KEEP

August 10, 1680: Po'pay leads the Pueblo rising in New Mexico.

January 16, 1786: Virginia's lawmakers pass Jefferson's religious freedom law.

December 1791: the First Amendment becomes part of the Constitution.

April 1906: Pentecostal worship begins at Azusa Street in Los Angeles.

August 11, 1978: the American Indian Religious Freedom Act passes.

NOT KNOWN: Nobody knows what the people of Cahokia believed as they watched the sun rise over Woodhenge.

Where to go: The statue of Po'pay, whipping scars carved on its back, stands in the US Capitol in Washington, D.C.

Next: Chapter 30, School and Education. It asks who got to go to school in America, and what they learned there.

Chapter 30: School and Education

CHAPTER 30 OF 37 · SCHOOL AND EDUCATION · 10 ERAS · ABOUT 225 MINUTES

School in America grew from a reading lesson for a few children into a right for every child. At every step, the adults in charge decided which children could come in and what could be done to them there.

On the grounds of Colonial Williamsburg, in Virginia, stands the oldest surviving schoolhouse for Black children in the United States. From 1760 to 1765, enslaved and free Black children as young as three learned to read in it. The men who ran the London charity that paid their teacher, Ann Wager, wanted them taught that slavery was God's will. In 1774 one of her former pupils, Isaac Bee, ran away. The man who claimed to own him warned in a newspaper that Isaac could read. The building was moved to Colonial Williamsburg on February 10, 2023.

This chapter follows American schooling through ten eras, from Native children learning beside their elders to the empty school buildings of 2020 and the fights that came after. Watch two things. First, watch who was let in. Lawmakers made it a crime to teach enslaved people to read. Officials sent Black, Chinese and Mexican American children to separate schools, took Native children far from home, and turned disabled children away, until families went to court or lawmakers changed the rules. Second, watch what adults were allowed to do to the children inside. You can follow the rules about hitting children from Ezekiel Cheever's birch rods in the 1600s to a Florida paddle in 1970.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. When did school become free, and required, for every child?
2. Which children were kept out of school, and how did they get in?

3. Is it still lawful for a teacher to hit a child?

Five minutes? Read "Punished for speaking Spanish" in 1950 to 2000.

Can skip: "The academies", "Catholic schools become a system", "The school year and the school day", "The last one-room schools". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 3 MINUTES

Native children in this land learned from their own families and communities, with no school buildings, no grades and no school year. Nobody taught as a full-time job. A child learned a skill by working beside an adult who had it. Children learned their nation's history by listening to the adults who kept it in memory.

Learning by doing and by listening

The Blackfeet are one nation whose way of teaching is on record. A National Park Service account states that "Blackfeet children learned by doing, by example, and through apprenticeship to their elders." That meant working beside the older people of the community until a child had their skills. On long winter evenings, the same account says, adults passed on the nation's history and ways of life around a fire the whole group shared.

This teaching by the spoken word, called oral tradition, never stopped. A teaching page from the Smithsonian's National Museum of the American Indian states that "the oral tradition continues to be an integral part of Indigenous teaching and learning," a necessary part of how the first peoples of this land teach. Native families and teachers still teach this way.

Memory in place of writing: wampum

No writing system was in use in this land before 1500. Without writing, people kept their nation's agreements and history in their memories, in speech, and in objects made to help them remember. Some people were trained to hold those memories and recite them.

Later, Spanish priests and English ministers wrote some Native languages down in the letters of their own alphabet. The first writing system invented for a language spoken in what is now the United States was the Cherokee syllabary of 1821, made by Sequoyah, a Cherokee man. It is a set of written signs in which each sign stands for one syllable.

The belts. The Haudenosaunee Confederacy is a union of five nations: the Seneca, Cayuga, Onondaga, Oneida and Mohawk. Its people record agreements and history in wampum, beads cut from the white and purple parts of clam shells, strung on strings or woven into belts. The Onondaga Nation says it was made the Wampum Keeper, the nation that looks after the belts, when the five nations joined under the Great Law of Peace.

How a belt holds a record. In the Onondaga Nation's own account, "The speaker puts the words of the agreement into the wampum as the strings or belts are woven together." After that, "each speaker thereafter uses the

wampum to remember the initial agreement and the history that has happened to date." The Onondaga call the belts "our living history." A speaker had to hold the agreement in memory and recite it, with the belt as the reminder.

DISPUTED: when the Great Law of Peace was agreed. Haudenosaunee oral tradition places it between about 1000 and 1400, and some scholars argue for the late 1100s, inside that range. Accounts based on written records put it at about 1450. No record settles it.

No names survive

NOT KNOWN: the name of any pupil or teacher in this land before 1500. With no writing, nobody made a list of who taught or who learned, and nobody counted them or the people who could recite a record.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 4 MINUTES

Spanish priests did the only European teaching in this land in the 1500s. They taught at their missions, settlements where they worked at conversion, the changing of a person's religion, to make Native people Catholic. At some missions they taught reading too.

The missions stood in Florida and, from 1598, in New Mexico. For a few months of 1570 and 1571, Spanish priests also ran one in Virginia. No schoolhouse stood anywhere in this land. Outside the missions, Native children went on learning from their families by the spoken word.

Mission teaching in Florida

Father Francisco Pareja taught Native people to read at a Spanish mission in Florida. He wrote of his own pupils: "many Indian men and women have learned to read in less than six months, and they write letters to one another in their own language." His mission, San Juan del Puerto, was one of the largest. It stood on Fort George Island among the Mocama, a coastal people who spoke Timucua.

At a mission the friars, Catholic men of the Franciscan order who had promised to live in poverty, taught Catholic doctrine, meaning the beliefs of the church. They also taught European customs, farming, and reading and writing, in Spanish and in Timucua, the local language.

How the missions began. Pedro Menéndez de Avilés and about 800 Spanish colonists, among them an unknown number of enslaved Africans, settled in the Timucua town of a chief named Seloy in September 1565, and the settlement became St. Augustine. Spanish priests began trying to convert Native people in Florida within a few years, and by the 1580s the Franciscans had missions across Florida. The National Park Service's history of San Juan del Puerto states only that it began "sometime in the 1570s."

New Mexico, 1598

Juan de Oñate led a Spanish expedition of colonists to the upper Rio Grande in 1598. Franciscan missions in New Mexico, and the religious teaching at them, began that year.

Paquiquineo (Don Luís de Velasco)

Who: A young man from the Chesapeake Bay country who spent years among Spanish priests and came home with a Jesuit mission.
When and where: 1561 to 1572. The Chesapeake, Spain, Mexico, and back.

He left home at about seventeen and came back nine years later.

Why this story is here: he is the one person of the 1500s whose schooling by Europeans is recorded by name and date. No pupil of a mission school in Florida or New Mexico is known by name.

He sailed from the Chesapeake on a Spanish ship in late June 1561. Whether he went willingly or was kidnapped is disputed, and the record does not settle it. Spanish captains of the time regularly captured boys and young men to serve as interpreters, people who translate between two languages. He came from the Powhatan chiefdoms, groups of towns each led by a chief.

Spain and Mexico. He reached Seville in September 1561, met King Philip II in Madrid that autumn, and sailed to Mexico in 1562. He fell so ill there that he was close to death, and he was baptized in Mexico City, the Christian ceremony of joining the church. He took the name of the viceroy of New Spain, the king's ruler in Mexico: Don Luís de Velasco.

The promise. He studied in Seville with the Jesuits, an order of Catholic priests, from 1566 to 1570. No surviving record says what they taught him. He promised Father Juan Bautista de Segura, the Jesuit leader in Florida, that many people in his homeland would become Christians, and Segura planned a mission on that promise.

Paquiquineo landed on the James River with Segura's party in September 1570, and his own people welcomed them. Encyclopedia Virginia states that soon after, he "appears to have left the missionaries" and gone back to his own family.

February 1571. Segura sent Father Quirós and two Brothers, Jesuits who were not priests, to meet him. The Brothers were Gabriel de Solís and Juan Baptista Méndez. On February 4, Paquiquineo killed the three Jesuits. Then he and his companions surrounded the mission house and killed Father Segura and the other Spaniards. The only one left alive was the altar boy, Alonso de Olmos, who helped the priests during church services.

What the soldiers did. Menéndez, the founder of St. Augustine and now the Spanish governor of Florida, sent soldiers to the James River in August 1572. They captured some of Paquiquineo's people in a fight and hanged several of them from the yardarms of their ship, the wooden beams that cross the masts.

One of the hanged men wore around his neck a paten, the metal dish the priests used in the Mass, their main church service. The Spanish records do not give the names of the men the soldiers hanged. The Powhatan people handed Alonso de Olmos back. The soldiers did not find Paquiquineo, and no European record mentions him again.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 11 MINUTES

English colonists in Massachusetts made teaching children to read a duty under law, and the men who wrote the laws gave a religious reason: every child had to be able to read the Bible. Schooling was not free. Parents or masters paid the teacher, and the Latin schools took only boys.

Outside New England, children learned at home, as apprentices, or at a few schools such as a Dutch one in Manhattan. New England colonists also opened schools and a college building for Native students, to make them Christian. Teachers beat children with rods, and one pupil wrote down what that was like.

Boston Latin School, 1635

Boston Latin School opened on April 23, 1635, and it is still open. It is the oldest public school in the country, which means a school that a town or a state runs for its children.

Whose land. Boston stood on the Shawmut peninsula, part of the land of the Massachusett nation, who called it Mushauwomuk, "the boat landing place." Between 1616 and 1618 an epidemic, a disease that spreads fast through a population, killed about three of every four people in the region, by the West End Museum's count. John Winthrop and about 1,000 English settlers arrived on the peninsula in 1630.

The first schoolmaster. Boston's townsmen met on April 13, 1635, and agreed to ask Philemon Pormort to teach. The town record states that he "shall be intreated (asked) to become scholemaster for the teaching and nourtering (bringing up) of children with us." It gives the date as "the second month," which was April, because the English year then began on March 25. The first classes met in Pormort's own house, and the boys learned Latin and Greek.

DISPUTED: which school is oldest. The leaders of Collegiate School in New York City, which Dutch colonists founded, say theirs is. In 1982 they moved its founding date back from 1638 to 1628, the year the minister Jonas Michaelius began teaching in Manhattan. The two schools' claims to be oldest of all have never been settled.

The two Massachusetts school laws

The 1642 law. The men of the Massachusetts General Court, the colony's lawmakers, passed the first law in this land that ordered children to be taught to read. Parents, and the masters that apprentices lived with and worked for, had to teach their children and apprentices to "read the English tongue." Children also had to know the "Capital Lawes," the colony's list of crimes punished by death.

The lawmakers set up no school and hired no teacher. Selectmen, the men elected to run a town's business, visited homes to check. A family that failed could be fined. The selectmen could also take a child or an apprentice away and place the child with another master who would do the teaching.

The Old Deluder Satan Act. Many families ignored that law, so the men of the General Court passed a second one in 1647. Its opening words give the reason: it is "one of the chief projects of that old deluder (trickster) Satan to keep men from the knowledge of the Scriptures," meaning the Bible. That line gave the law its nickname. The lawmakers wanted every child able to read the Bible without help.

Under the 1647 law, the men of every town of 50 households had to choose someone to teach children to read and write, and the parents or masters paid that teacher's wages. Every town of 100 families also had to set up a grammar school, a school that taught Latin, to prepare boys "for the university." That meant Harvard College, founded in 1636.

A town whose men did not obey had to pay 5 pounds, in English money, every year to the nearest school until they did. Lawmakers in the other New England colonies passed similar laws, except in Rhode Island.

The New England Primer

The English printer Benjamin Harris made *The New England Primer*, a first reading book, for buyers in the colonies by 1690. No copy printed before 1727 is known to survive, and the American Antiquarian Society holds a copy of that 1727 edition.

The book mixed lessons in reading, spelling and religious rules for behavior. Each letter of the alphabet had a small woodcut, a picture printed from a carved block of wood. After the alphabet, the 1727 edition tells the teacher, "let him learn these and such like Sentences by Heart." Children were expected to memorize the sentences and recite them.

Ezekiel Cheever

Who: A schoolmaster who taught Latin and Greek for 70 years, the last 38 of them as head of the Boston Latin School.

When and where: Born in London in 1615. Died in Boston on August 21, 1708. Taught at New Haven, Charlestown and Boston.

He taught Latin and Greek for 70 years without a break.

The Database of Classical Scholars, kept at Rutgers University, states that no one else in the history of this country has taught that long. Cheever ran the schools at New Haven from 1638 and taught at Charlestown from 1661. On December 29, 1670, he was invited to head the Boston Latin School. He was still teaching there when he died at 93.

His classroom. Accounts describe a smoky fireplace, poor light, and "a bundle of birch rods beside his desk." Birch rods were thin sticks, or bundles of twigs, cut from a birch tree and used to beat children.

Cotton Mather had been one of his pupils. He preached at Cheever's funeral and called him a teacher "than whom New England had known no better." Cheever's Latin textbook, *A Short Introduction to the Latin Tongue*, was printed in Boston in 1708, the year he died. Schools used it for nearly 200 years, longer than any other Latin textbook printed in this country.

Beaten at school: John Barnard

The rod was standard equipment in a schoolroom of the 1600s, and one of Cheever's pupils wrote down what it was like. John Barnard was born in Boston on November 6, 1681. He was seven when he was sent to Cheever's school, in the spring of 1689. He left after a few weeks, came back about a year and a half later, and stayed until he entered Harvard at fifteen.

Cheever told Barnard openly that he played too much and kept his classmates from their lessons. "If any of them cannot perform their duty," Cheever said, "I shall correct you for it." In the words of that time, to correct a child meant to beat him. Cheever was warning Barnard that he would be beaten whenever a classmate failed.

Beaten for a classmate. Then one classmate did not know his lesson, and "our master beat me," Barnard wrote. The classmate saw this and neglected his book on purpose, "for which I was still corrected, and that for several days." Barnard beat the classmate after school, and the classmate never came to school again.

Barnard wrote that he was "often beaten for my play." Once Cheever set the class to turn Aesop's fables, a set of old animal stories, into Latin verse. Barnard could not do it. Cheever beat him for it "two or three days going," until Barnard told him he had tried his hardest, and then Cheever stopped asking it of him.

NOT KNOWN: what Cheever hit him with, or how many times. Barnard does not say. He wrote his account as an old man, in the late 1760s, and it was printed in 1836.

Spare the rod. The saying "spare the rod and spoil the child" is not in the Bible, though Proverbs 13:24 comes close: it says that whoever spares the rod hates their children. The wording people use now comes from a comic poem of the 1660s by the English writer Samuel Butler.

Natick's school, and Deer Island

The English minister John Eliot and a group of Native leaders founded Natick, in Massachusetts, on a bend of the Charles River in 1651. Waban and Cutshamekin of the Massachusett nation and John Speen of the Nipmuc nation worked on it with him. Natick was the first of more than a dozen "praying towns," towns set up for Native people who had become Christians. Its meetinghouse had a school inside, and the schoolmaster was a Massachusett man named Monequassun.

Eliot printed schoolbooks in the Massachusett language: a catechism in 1653, which is a list of religious questions and answers that a pupil learns by heart, and an *Indian Primer* in 1669. In each of them the reading lessons were also lessons in Christian belief.

Deer Island. War began in 1675 between the English colonists of New England and a group of Native nations led by Metacom, a sachem, or leader, of the Wampanoag. The English called him King Philip, so they called it King Philip's War. On October 13, 1675, the men who governed the Massachusetts colony ordered the Christian Native people of the praying towns taken to Deer Island in Boston Harbor and held there as prisoners.

Colonists set up the prison camp, starting with the people of Natick. The surviving accounts do not name the men who voted for the order. The National Park Service's page on the island states that about 500 to 1,100 people were held there that winter, most of them women and children.

The prisoners starved and froze through the winter without shelter. The Park Service page states: "Due to starvation, freezing temperatures and lack of shelter, very few survived internment (imprisonment in a camp)." Members of a Boston history group, the Partnership of Historic Bostons, estimate that more than half of them died. Colonists released the survivors in May 1676.

An unknown number of people were smuggled off the island and sold into slavery in the West Indies or in Tangier, in North Africa. No record names the men who took them and sold them, and none says what became of Monequassun, the Natick schoolmaster. Every year, descendants of the Native nations of that war hold a remembrance of their ancestors on Deer Island.

Harvard, the Indian College, and the people Harvard men enslaved

Harvard's charter of 1650, the official paper from the colony's government that created the college, promised "the education of the English and Indian youth of this Country." A building called the Indian College stood in Harvard Yard from 1655. Accounts differ on when it came down: 1695 by Harvard Magazine's account, 1698 by that of Harvard's Native American Program. A printing press kept inside it printed the first Bible published in North America, in 1663, in the Algonquian language.

Only a handful of Native students came to the Indian College, and Caleb Cheeshahteaumuck, who graduated in 1665, was the only one of his time who finished. His classmate Joel Iacoomes drowned in a shipwreck just before the graduation ceremony. Two more students died of disease before they finished, and one left to become a sailor.

The people Harvard men enslaved. Harvard's leaders and staff enslaved people. In April 2022 a Harvard committee led by Tomiko Brown-Nagin reported records of "at least 79 people who were enslaved by Harvard presidents, overseers, and faculty and staff members" before slavery was outlawed in Massachusetts in 1783, the overseers being the men on one of Harvard's governing boards.

A woman named Jane was enslaved by the college steward, Andrew Bordman, the man in charge of food and supplies, who enslaved at least eight people. A woman named Cicely was enslaved by William Brattle, who taught students at the college and kept its money. Titus, Venus, Juba and Bilhah were enslaved at Wadsworth House, where they were made to live and work for two Harvard presidents.

Caleb Cheeshahteaumuck

Who: A Wampanoag student from Martha's Vineyard, the first Native American to graduate from Harvard.

When and where: Graduated in 1665 in Cambridge, Massachusetts. Died in 1666.

One letter in Latin is all he left in his own words.

Caleb Cheeshahteaumuck came from a Wampanoag community on Martha's Vineyard, an island off the coast of Massachusetts. His letter is addressed to the English donors who paid for the Indian College. It begins "Honoratissimi benefactores," Latin for "most honored benefactors," the people who gave the money. It is often described as the earliest surviving writing by a Native person on this continent.

He died in 1666, within a year of graduating, of a disease people then called consumption. One account says he died at Watertown, Massachusetts, at about twenty. Martha's Vineyard Magazine states that the disease was "presumably" tuberculosis, a disease of the lungs, which means the writer judged it likely. No record from his time names the disease that way.

Timucua in print

In Florida, Father Francisco Pareja wrote the Timucua language down in books printed in the early 1600s. He published three catechisms, a grammar explaining how the language works, and a book of questions for confession, dated 1613. In confession a person tells a priest their sins. Those books are now the only record of the Timucua language, which nobody speaks now.

About 200,000 people who spoke Timucua lived in what is now northern Florida and southern Georgia in the 1500s, by the estimate of the historian Matthew Holt Jennings. Nobody counted them. By the 1650s about 2,000 were left, one in every hundred. Most died in epidemics of diseases the Spanish had brought.

Raids, then exile. After the English founded Charles Town in Carolina in 1670, Carolina slave traders and Native groups allied with them raided the Spanish missions again and again to capture people and sell them. In 1763, at the end of a war, Spain's rulers gave Florida to Britain, and Spanish officials shipped the last Timucua people to Cuba. The last Timucua of full descent died there in 1767.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 12 MINUTES

Most children in these fifty years learned at home, at a small school in a neighbor's kitchen, or as apprentices. The schools that did exist mostly took boys whose families could pay. Church societies in London paid for charity schools and for schoolmasters to cross the Atlantic.

South Carolina's lawmakers made one kind of teaching a crime. In 1740 they banned teaching an enslaved person to write, the first law in these colonies that punished someone for teaching. In Virginia, two traders brought four captive Native boys to a college to be its first Native pupils.

The law against teaching, 1740

The law. South Carolina's lawmakers passed "An Act for the Better Ordering and Governing Negroes and Other Slaves in This Province" on May 10, 1740. In that title, "Negroes" is an old word for Black people.

The uprising before it. They passed it the year after the Stono Rebellion of 1739. In that uprising about 20 enslaved men near Charleston took up weapons and marched toward Spanish Florida, where they hoped to be free. They killed about 25 white colonists. William Bull, the lieutenant governor, or deputy governor, called out armed planters, who killed about 35 to 50 of the rebels. Colonists cut off the heads of some rebels and set them on stakes along the roads.

The act names the men who made it law: Bull, who was acting as the colony's head, and the King's Council and the Commons House of Assembly, the colony's two groups of lawmakers. It does not name the members who drafted it.

Section 45 of the act gives its reason: "the having of slaves taught to write, or suffering them to be employed in writing, may be attended with great inconveniences." Anyone who taught an enslaved person to write, had one

taught, or used one as a scribe, a person who writes documents for someone else, "shall, for every such offense, forfeit (lose as a punishment) the sum of one hundred pounds current money."

The teacher owed the fine, not the pupil. Under another section of the act, a person could collect the fine by suing the teacher in court. Half the money went to the colony and the other half to whoever reported or sued, which gave anyone a reason to watch for a teacher and turn the teacher in.

Why writing. Under the act, teaching writing was a crime and teaching reading was not, and the act gives only "great inconveniences" as its reason. Enslavers feared forged passes. A pass was the note from an enslaver that let an enslaved person travel, and a person who could write could forge, or fake, one and escape with it. One enslaver's newspaper advertisement of 1774, for a young man who had run away, warns that someone might forge a pass for him.

Harry and Andrew's school in Charleston

A school for Black pupils opened in Charleston, South Carolina, about three years after the law. The historian Carter G. Woodson wrote about it in 1915. In his account, leaders of the Society for the Propagation of the Gospel bought two young Black men named Harry and Andrew, and from then on the Society owned them as slaves.

The Society was a London group run by members of the Church of England, England's own Protestant church, one of the Christian churches that broke away from the Catholic Church. Woodson does not say who sold the two men.

The Society's leaders bought Harry and Andrew to train them as teachers "to their people." Alexander Garden, a Church of England official in Charleston, had a schoolhouse built for about 308 pounds, and Harry and Andrew taught in it.

Accounts give 1743 or 1744 for the opening. The school began with about sixty young pupils and ran in the evenings for people who worked in the day. Woodson judged it "highly probable" that after the 1740 law only free Black

people attended. The school closed in 1763, by Woodson's account in a later book, and some accounts give 1764. One of the two teachers had died, and the men who ran the Society judged that the other was not doing the job well.

Where a small child started: the dame school and the hornbook

In New England a small child usually learned to read first from a woman in her own home, at what was called a dame school, from an old word for a woman. The teacher took children from about three or four years old for a few hours a day, for a small fee.

Often she was a widow or an unmarried woman who needed the money. Families paid her in eggs, firewood, or washing and ironing instead of cash. She taught the letters, reading, the catechism and the Lord's Prayer, and sewing for girls.

The town schools of New England would not take a boy until he could "stand up and read words of two syllables and keep their places." Somebody had to teach him that first, and that somebody was usually a woman in her kitchen. Women were already teaching, for very little pay, before a woman could hold a teaching job in a town school.

The hornbook. A child's first reading tool was often a hornbook, a wooden paddle about the shape of a table-tennis bat with one printed sheet tacked to it. A thin, clear sheet of animal horn covered the paper so that it would not wear out. A leather cord through the handle let a child hang it from a belt.

The sheet showed a cross, then the alphabet in capital and small letters, then the vowels joined to consonants, then the Lord's Prayer. Because of the cross at the top, the alphabet row was called the "Christ's Cross row," later shortened to "crisscross row." The Smithsonian Institution holds a hornbook made in the 1700s.

Charity schools

Some schools in the colonies were paid for from London. King William III gave a charter to the Society for the Propagation of the Gospel in Foreign Parts on June 16, 1701. The minister Thomas Bray pushed for it, along with Bishop Henry Compton and others.

The Society's stated job was to send priests and schoolmasters to the colonies, and across the 1700s its members paid for more than 300 of them to cross the Atlantic. Members of a separate group, the Associates of Dr. Bray, founded in 1724, set up schools of their own.

These were charity schools. A society in London paid the teacher so that pupils could read the catechism and the Bible, and the town did not run them.

How many people could read and write

Nobody surveyed how many colonists could read. Historians such as Kenneth Lockridge count signatures instead, on wills, the papers saying who gets a person's belongings after death, and on deeds, the papers showing who owns land. A person who could write signed a name. A person who could not made a mark, such as an X. The method undercounts readers, because many children were taught to read and never taught to write, which was common for girls.

Men. Lockridge counted men's signatures in New England. Between 1650 and 1670 about 60 of every 100 white men could sign their names, and between 1787 and 1795 about 90 of every 100 could.

Women. Lockridge's counts, and later counts built on his method, cover women too. Of women born in the 1650s and 1660s, about 27 of every 100 could sign. Of women born in the 1730s and 1740s, about 81 of every 100 could. More people could sign in New England and the middle colonies than in the South, and more in towns than in the countryside.

Enslaved people. An article in Encyclopedia Virginia estimates that by 1775 as many as 5 percent of enslaved people in Virginia could read, one in twenty. The enslavers' own advertisements give one check on that number. Between

1736 and 1776 enslavers placed about 1,000 advertisements in the *Virginia Gazette*, a newspaper, to hunt for people who had run away. Fifty-five of them said the person could read or write.

Girls' schools: New Orleans and Pennsylvania

A group of Ursuline nuns, an order of Catholic women, came to New Orleans from France in 1727 and started a school for girls. New Orleans was then a French colony and became part of the United States in 1803. The school's own history states that it is the oldest Catholic school in the United States and the oldest school for girls that has run without stopping. It states that the nuns taught the first classes for enslaved Black women, free women of color and Native women.

Countess Benigna von Zinzendorf started classes for twenty-five girls at Germantown, near Philadelphia, in May 1742. She belonged to the Moravian Church, a Protestant church. The school moved to Bethlehem, Pennsylvania, in 1745 and settled there in 1748, taking in mainly the daughters of traveling Moravian church workers. The Moravian schools' own histories describe it as the first Protestant boarding school for girls in what became the United States, a school where pupils live as well as study.

Franklin's plan for an academy, 1749

Benjamin Franklin published *Proposals Relating to the Education of Youth in Pensilvania* in October 1749. He argued for a school that taught in English instead of Latin and Greek, with geography, natural history and modern languages. People called this kind of school an academy: a school for older pupils, a step above the town school and a step below a college.

What a school day was like, and what is not known

KNOWN:	the men of a town appointed one man to teach, the parents or masters paid him, and children of every age sat in the same room because there was only one teacher.
NOT KNOWN:	much more about the school day of these fifty years. The one-room school with a stove, slates to write on, and a bench where pupils stood to recite is described in records of the 1800s, and the records of 1700 to 1750 contain less.

The Indian school at William & Mary

The College of William & Mary in Virginia received its royal charter in 1693. The charter lists three purposes for the college, and the third was to teach Native students "in the Christian Religion." The money for this Indian school came from rents on a country estate in Yorkshire, England, bought in 1695 with 5,400 pounds from the estate of the scientist Robert Boyle.

Four captive boys. Governor Francis Nicholson tried to recruit Native pupils in 1700 and failed. William & Mary's own history states that about 1702 two traders, Robert Hicks and John Evans, "purchased or acquired four young Indian captives from the Catawba Nation." The traders got the boys along with deerskins and furs. The same history states that the boys were "likely destined to be sold by the Catawba as forced laborers."

The four boys became the school's first Native pupils, and the college's published records do not give their names. The Meherrin Nation's account states that the college "resorted to buying their pupils from local Native Americans who captured the boys from other tribes."

The college had twenty Native pupils besides the four captive boys by 1712, a third to a half of all its pupils. Enslaved workers built a building for the Indian school, the Brafferton, finished in 1723. More than 125 Native students from at least 26 nations lived and studied there between 1723 and 1776. They learned reading, writing, arithmetic and the prayers of the Church of England. The published records do not say how many of them died at the school.

The college's leaders bought people too. They paid 150 pounds for a tobacco plantation of 2,119 acres in Virginia in 1718, and 476 pounds more for seventeen enslaved people to work it.

Christopher Dock

Who: A Mennonite schoolmaster in Pennsylvania who wrote the first book about how to teach written in this country.

When and where: Skippack and Salford, Pennsylvania, from no later than 1718 until his death in 1771.

He rewarded good pupils with a penny or a sugar pretzel.

Christopher Dock wrote the first book about how to teach written in this country, and most of it is about how he punished children and how he rewarded them. He was a member of the Mennonite church, a Christian church.

He came to Pennsylvania between 1710 and 1714 and taught on Lenape land. The Lenape homeland covered southeastern Pennsylvania, central and southern New Jersey and Delaware. Between 1682 and 1684 William Penn bought land for his colony from the Lenape, paying at least 1,200 pounds in goods.

Dock opened a school at Skippack no later than 1718. He taught for ten years, then farmed for about ten more. In 1738 he went back to teaching and taught for the last 33 years of his life, running two schools, at Skippack and at Salford, on alternate days.

The book. He finished *Schul-Ordnung*, German for "school management," on August 8, 1750. He did not want it printed while he was alive, and the printer Christopher Saur printed it anyway. Accounts differ on the date. The historian Martin Brumbaugh, who wrote Dock's life in 1908, gives 1769 for the first printing and 1770 for the second, and other reference works give 1770 for the first. It was still being printed in English in the 1900s.

Penny, pretzel, rod. Dock sorted children by what worked on each one. He gave praise to children who wanted it, a penny for hard study to children who wanted money, and a sugar pretzel for good learning to children who wanted sweets. In Brumbaugh's words, "The ill-intentioned,

who were not affected by the promise of a penny or a cooky, he threatened with whipping," which means hitting a child again and again with a whip, a rod or a strap.

Brumbaugh names the rod: "With some he had to use the rod." When a child failed at a task, Dock would say "I no longer love you," and in Brumbaugh's words the children "wept until he comforted them."

He put a yoke, a wooden frame worn over the neck and shoulders, on a child who swore, and he made some children sit alone. Dock was milder than the schoolmasters around him, and he still used whipping and the threat of it.

One evening in the autumn of 1771 Dock did not come home from school at the usual time. In Brumbaugh's account, people went to look for him, and "he was found in his schoolroom on his knees," dead.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 12 MINUTES

The men who founded the United States argued that the new country needed schools, and they mostly did not pay for them. Thomas Jefferson wrote a plan for public schools in Virginia, and Virginia's lawmakers never passed it. The new schools of these fifty years included private academies, the first school for girls with a charter from a government, church charity schools, and a charity school and a college meant for Native students.

At a charity school in Williamsburg, Virginia, enslaved and free Black children learned to read, and their schoolhouse still stands.

Jefferson's plan, and who was left out

Thomas Jefferson wrote "A Bill for the More General Diffusion (spreading) of Knowledge." It went to the Virginia House of Delegates, the lower house of the state's lawmakers, on June 18, 1779. Jefferson described the plan in his book *Notes on the State of Virginia*.

The plan. Each county would be divided into districts five or six miles square, called hundreds, with a school in each one. Every free family could send its children for three years free, and longer if they paid. The district schools would teach reading, writing and arithmetic. Above them, grammar schools would teach Greek, Latin, geography and harder arithmetic.

Jefferson set out how poor boys would be chosen for more schooling. Each year, officials called visitors would "chuse (choose) the boy, of best genius (ability) in the school, of those whose parents are too poor to give them further education." In this way, he wrote, "twenty of the best geniusses will be raked from the rubbish annually (each year)." By the rubbish he meant the children of poor families.

Who was left out. The plan in *Notes on the State of Virginia* does not mention girls, and every reference in it is to boys. Enslaved children are not in it either, because it was a plan for free children only.

Virginia's lawmakers never passed the bill. James Madison brought it forward again while Jefferson was in Paris, and it failed again. In 1796 the lawmakers passed a much-changed version, an "Act to Establish Public Schools." Under that act the men who sat on each county's court chose whether to open schools, and in most counties they did nothing.

The Land Ordinance of 1785: school money from Native land

Members of Congress set aside land to pay for public schools in 1785. US commissioners had taken that land from Native nations by a treaty that the Shawnee and other nations refused to accept. The members passed the law, called the Land Ordinance, on May 20, 1785.

Under the Land Ordinance, surveyors were to cut the western land into townships six miles square, each split into 36 numbered sections of one square mile. The law states: "There shall be reserved the lot N 16, of every township, for the maintenance of public schools, within the said township."

Section 16 sat near the middle of each township. Under the law it was not to be sold to a settler, and the money it earned was to pay for a school. US officials sold the rest of the land to settlers.

Whose land. The first land surveyed under the Ordinance was in what is now eastern Ohio, where Wyandot, Lenape, Ojibwe, Odawa and Shawnee people lived. At Fort McIntosh on January 21, 1785, US commissioners signed a treaty with Wyandot, Lenape, Ojibwe and Odawa leaders. The commissioners' terms stated that most of the Ohio nations had to stay inside a reserve, an area set aside for them, between the Cuyahoga and Maumee rivers. Surveyors began measuring the land in September 1785, guarded by troops from Fort McIntosh.

George Knepper, an Ohio historian, writes that the Ohio nations, especially the Shawnee, refused to obey the treaty. In his words, it had been "negotiated under duress by minor tribal chiefs who lacked authority to speak for their people and for other tribes." He means that the treaty was worked out under threat or pressure, with lesser leaders who had no right to speak for everyone else.

The academies

Franklin's academy, the Academy and Charity School of Philadelphia, opened in 1751. Thomas and Richard Penn, the owners of the colony, granted it a charter in 1753, and the College of Philadelphia was chartered two years later.

Samuel Phillips founded Phillips Academy at Andover, Massachusetts, on April 25, 1778, and it opened five days later. Paul Revere cut its seal. John and Elizabeth Phillips signed the papers founding Phillips Exeter Academy in New Hampshire on April 3, 1781. That school opened on May 1, 1783, with one teacher and 56 students.

Girls' academies, and the first with a charter

John Poor founded the Young Ladies' Academy of Philadelphia on June 4, 1787. Officials of the state of Pennsylvania granted the school a charter on January 7, 1792, which made it the first school for the higher education of women in the country to be recognized by a government. Benjamin Rush was among the men who backed it.

Its pupils studied reading, writing, English grammar, mathematics, geography, composition, chemistry and natural philosophy, the old name for physics and the other sciences of how nature works.

Litchfield. Sarah Pierce, born in 1767, opened the Litchfield Female Academy in Connecticut in 1792 and ran it for 41 years. More than 2,000 girls attended over about 30 of those years, from 17 states and territories and from Canada and the West Indies. Researchers at the Litchfield Historical Society have found 1,848 of them by name.

More than 8 in 10 of the girls came from other towns and boarded with Litchfield families. They studied history, geography, arithmetic and composition, and later Latin, logic and natural philosophy. Girls who finished the whole course received diplomas.

Sarah Pierce took up an argument made after the Revolution: that mothers gave children their first moral training and their first lessons, and that the new country depended on it. In this argument, girls needed schooling because

of the children they would raise. Pierce herself believed girls and boys were equally able. She did not argue that women should enter the men's colleges or professions.

The Williamsburg Bray School, 1760 to 1774

Enslaved and free Black children in Williamsburg, Virginia, learned to read at a charity school from 1760 to 1774, and the men who paid for it meant to teach them to accept slavery as God's will. William & Mary's own account states that the school's "faith-based curriculum justified slavery," the curriculum being the set of lessons the school taught.

The Associates of Dr. Bray, a church charity in London, set up the school on Benjamin Franklin's advice. Thomas Dawson, the president of the College of William & Mary, backed it. One teacher, Ann Wager, taught there for all fourteen years.

The pupils were three to ten years old, and thirty or more might crowd into the room on an ordinary day. They learned to spell and to read aloud clearly. They worked up through *The Child's First Book*, the Psalms, which are the songs of the Bible, the New Testament, the Book of Common Prayer, which was the Church of England's prayer book, and the Bible.

Lessons began with the pupils reciting from memory the Apostles' Creed, a short statement of Christian belief, the Lord's Prayer and the catechism. Girls were also taught to sew.

How many pupils. The count is not settled. Colonial Williamsburg's history of the school states that "hundreds of students attended." William & Mary's Bray School Lab states 300 to 400, and an Encyclopedia Virginia article states about 200. The names of more than 80 pupils survive on three school lists, from 1762, 1765 and 1769. One of them is a child named Locust, sent by the Armistead family, who enslaved people near Williamsburg.

The school closed in 1774, when Ann Wager died. A second Bray school, at Fredericksburg, Virginia, ran from 1765 to about 1769 under the merchant Fielding Lewis, with 40 to 50 pupils. It closed because fewer pupils came and local enslavers opposed it.

The schoolhouse. The Williamsburg schoolhouse still stands. Researchers at Colonial Williamsburg and William & Mary confirmed in 2020 that a building on the William & Mary campus, later a private home and then student housing, was the one where the school began. The school met there from 1760 until 1765, when it moved to a larger building.

The house was moved to Colonial Williamsburg on February 10, 2023. William & Mary's Bray School Lab and *Smithsonian* magazine describe it as the oldest surviving schoolhouse for Black children in the United States.

Ann Wager

Who: The only teacher the Williamsburg Bray School ever had, and the teacher of a boy who later ran away from the man who claimed to own him.
When and where: Williamsburg, Virginia, 1760 to 1774.

She earned 20 pounds a year, about a skilled worker's pay.

Ann Wager's husband, William Wager, died in 1748, and she began teaching to support herself. She was born in about 1716, so she was about 44 when the Bray School opened.

Her rent was paid on top of her 20 pounds a year. Colonial Williamsburg's history of the school describes that pay as "roughly equivalent to a journeyman worker's yearly wage," the wage of a skilled worker who had finished an apprenticeship. She died on August 20, 1774.

Isaac Bee. One of her pupils on the 1765 list was Isaac Bee, the runaway named in the chapter opening. In the summer of 1774, at eighteen or nineteen, he ran away from Lewis Burwell, the man who claimed to own him. Burwell placed an advertisement in the *Virginia Gazette* that September. He wrote that Isaac Bee believed he had a right to be free, because his father was a free man.

Burwell offered a reward of 40 shillings, a shilling being one twentieth of a pound, for Isaac Bee's return. He warned that Isaac Bee could read, and that "he may easily get some One to forge a Pass for him." Bray School

researchers at William & Mary matched the runaway to the pupil on the 1765 list.

NOT KNOWN: whether Isaac Bee was caught, or what became of him. The Associated Press report on the researchers' work does not say.

Georgia copies South Carolina's ban

The members of Georgia's colonial House of Assembly passed the colony's first slave code, a set of laws for controlling enslaved people, in 1755. Following South Carolina's act of 1740, they made it a crime to teach an enslaved person to write, or to use one as a scribe. The members passed another slave law in 1770. The records disagree on whether the 1770 law also banned teaching enslaved people to read, or whether that ban came only in 1829.

Moor's school, Samson Occom and Dartmouth

Eleazar Wheelock, a Connecticut minister, opened Moor's Indian Charity School at Lebanon, Connecticut, for Native students in 1754, and from 1761 he took girls as well. Students past and present complained that Wheelock made the Native students do more manual labor, hard work with the hands, than was fair. He quarreled with many of his best students, including Samson Occom, a Mohegan man who had been one of his pupils.

Oneida families took all their children out of the school in 1769. The Dartmouth scholars who edit Wheelock's papers write that the Oneida acted "perhaps due to concerns about corporal punishment," which means hitting a child as a punishment.

The money Occom raised. Occom preached across England and Scotland from 1766 to 1768 to raise money for the school, and he raised 12,026 pounds. Wheelock used the money to start a college in New Hampshire instead. The royal governor of New Hampshire, John Wentworth, signed Dartmouth College's charter in 1769.

The charter stated its purpose as "the education and instruction of youth of the Indian tribes in this land," for "civilizing and christianizing children of pagans," meaning people who were not Christian. Only two Native students went with Wheelock to New Hampshire. Occom never set foot on the campus of the college his tour paid for. About 75 Native students enrolled at Dartmouth before 1972.

Wheelock enslaved people. Dartmouth's own researchers have found papers naming at least seventeen people he enslaved, among them Exeter, Chloe, Hercules, Peggy and Caesar. His papers record that between 1743 and 1769 he bought six men and two women.

Noah Webster

Who: A schoolteacher who wrote the spelling book American children learned from for more than a hundred years, and later a dictionary.
When and where: Born in West Hartford, Connecticut, on October 16, 1758. Died in New Haven on May 28, 1843.

By one count his spelling book sold 60 million copies by 1890.

Noah Webster wrote the spelling book that American children learned from for more than a hundred years. He grew up in the part of Hartford, Connecticut, that is now West Hartford, and finished Yale in 1778.

He taught school in Glastonbury, Hartford and West Hartford, and in 1782 in Goshen, New York. He believed the new country needed its own schoolbooks and one settled way of spelling to hold it together.

The Blue-Backed Speller. He published the first part of *A Grammatical Institute of the English Language* in 1783, and people called it the Blue-Backed Speller. In it Webster kept most spellings as they were. Earlier dictionaries had used some newer spellings only part of the time, and Webster used them every time.

Families and schools used the speller from 1783 into the early 1900s. By one estimate it had sold 15 million copies by 1837 and about 60 million by 1890. Other estimates run as high as 100 million copies in all. Nobody

> kept an exact count.

Teaching as a job

Some men taught for a few years on the way to other work. Noah Webster taught in four schools in about four years on his way to law and writing. Others made teaching their life's work: Ezekiel Cheever taught for 70 years and Christopher Dock for more than 40.

NOT KNOWN: which pattern was more common. No count of teachers from the 1700s is known.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 24 MINUTES

The free public school began as an argument in Massachusetts. Horace Mann made it there for eleven years: schools free to every family, paid for by taxes, with trained teachers. The first school paid for by a state to train teachers opened in 1839. School board members hired more and more women to teach, because they could pay a woman far less than a man. In Massachusetts in 1838 a woman teacher earned $6.49 a month and a man $23.10. Deaf children and blind children got schools of their own.

In the South, lawmakers made it a crime to teach an enslaved person to read. In North Carolina, Georgia and Virginia they wrote whipping into the punishment for Black people who broke these laws. In Boston, school officials kept a five-year-old Black girl out of the school nearest her home, and judges on the state's highest court ruled that the city could keep her out.

Horace Mann and the common school

Massachusetts lawmakers set up a state Board of Education in 1837 to look after the common schools. A common school was meant for all the children of a town and paid for by the town's taxes instead of fees from parents. Horace Mann, a lawyer and state senator, became the board's first secretary. Under the law the board's members could not make any town do anything. Mann's job was to gather facts, write reports and argue, and he did it for eleven years, until 1848.

Mann wrote twelve yearly reports, and in 1838 he started a magazine, the *Common School Journal*. He made the same three arguments again and again. Every child had a right to an education. Taxes on everyone should pay for schools, instead of fees charged to parents. Teachers should be trained before they taught.

Free for whom. The free school Mann argued for was in practice a school for white children. While he was secretary, Boston's school officials ran separate schools for Black children. Two years after he left the board, Chief Justice Lemuel Shaw of the state's highest court ruled that the city could keep them separate.

Horace Mann

Who: The first secretary of the Massachusetts Board of Education, who argued that schools should be free and paid for by taxes.

When and where: Born in Franklin, Massachusetts, on May 4, 1796. Died in 1859.

As a boy he got about ten weeks of school a year.

Horace Mann had very little school in his own boyhood. He grew up in a poor farming family in Franklin, Massachusetts. His father died when he was thirteen, and he braided straw for the hat-makers of Franklin to earn money.

Until he was about fifteen he spent no more than about ten weeks a year in school. A different retelling of his own memories puts it at no more than six weeks a year between the ages of ten and twenty.

A traveling teacher named Barrett taught him Latin and Greek, and he got into Brown University, where he graduated first in his class in 1819. He became a lawyer, then a member of the Massachusetts House of Representatives from 1827 and of the state Senate from 1835, where he became president of the Senate.

Europe. In 1843, as the board's secretary, he visited schools in Prussia and Saxony, two German states, and in Holland. He wrote about that trip in his seventh report, and thirty-one schoolmasters of Boston attacked it in print.

Voters elected him to the US House of Representatives in 1848, to the seat left empty when John Quincy Adams died, and there he spoke against slavery and against letting it spread. In 1853 he became the first president of Antioch College, in Yellow Springs, Ohio. He died in 1859.

The one-room school and the school day

A country school of these years had one room and one teacher. Pupils of every age, from small children to nearly grown young men and women, sat together. A wood stove stood in the middle of the room. Children wrote on slates, thin boards of dark stone, with slate pencils, and wiped them clean to use again.

At the front stood a recitation bench. A pupil came up to the bench and said a lesson out loud to the teacher while the others worked at their desks. Many children walked to school.

Two terms. Accounts of country schools from before about 1900 describe a school year of two terms. A summer term, roughly May to August, was for the younger children, and women taught it. A winter term, roughly November to April, was for the older boys, who could be spared from farm work then, and men were often hired for it to keep order. Children stayed home to plant in the spring and to harvest in the autumn.

Part of a teacher's pay was room and meals. The teacher lived in the house of each family in the district in turn, one family after another, through the term. This was called boarding round.

The first school for teachers, 1839

The first school paid for by a state to train teachers opened at Lexington, Massachusetts, on July 3, 1839, and three students came on the first day. Schools like it were called normal schools, from the French *école normale*, a school that taught the norms, or rules, of teaching. The principal was Cyrus Peirce, a minister whom Horace Mann had persuaded to come from Nantucket. By 1842 the school had 42 students. It moved twice and is now Framingham State University.

Members of the Massachusetts Board of Education set the entry rules for the state's normal schools. A man had to be at least 17 years old and a woman at least 16. Every student had to pass a test in spelling, geography and arithmetic. The course lasted one year and was free to a student who promised to teach in the state's schools.

Mary Elizabeth Miles graduated from the Lexington normal school in 1843, the first African American to graduate from any public teachers college in the United States. She married Henry Bibb, a man who had escaped slavery, and worked with him in Ontario, in Canada, in the movement to end slavery.

Mary Swift

Who: A seventeen-year-old in the first class of the first state school for teachers in America. She kept a journal of it.

When and where: Born in Nantucket, Massachusetts, on June 22, 1822. Died in Cambridge, Massachusetts, on March 2, 1909.

At seventeen she kept a journal of the first teachers' school.

Mary Swift was one of the first people in America trained at a state school for teachers, and her journal of it survives. She entered the new normal school at Lexington in 1839, three weeks after her seventeenth birthday. Her class graduated on April 4, 1840.

She wrote in her journal through the school's first term. The principal, Cyrus Peirce, kept a journal of the same days. Both survive, and in 1926 they were printed together as one book, *The First State Normal School in America: The Journals of Cyrus Peirce and Mary Swift*.

Laura Bridgman. She became the teacher of Laura Bridgman, a deaf and blind girl at the Perkins Institution in Boston, in 1841. Swift spoke to her by spelling words letter by letter into her hand, and she kept the job until she married Edwin Lamson in 1846.

Mary Swift's book about Laura Bridgman came out in 1878. She built it mostly from the journals that Bridgman's teachers had kept. It was called *Life and Education of Laura Dewey Bridgman, the Deaf, Dumb, and Blind Girl*, and in that title "dumb" meant unable to speak.

Why school boards hired women

Members of school boards hired women because they could pay them far less. School journals of the time printed the monthly wages, and a teacher usually got her room and meals on top of the wage.

The pay in 1838. In Massachusetts, where 3,826 women and 2,411 men taught, men were paid $23.10 a month and women $6.49, less than a third as much. In Connecticut women got less than half what men got. The gap was smallest in Pennsylvania, where many more men than women taught.

The officials who made the savings said so in writing. Samuel Lewis, the Superintendent of Common Schools, the official in charge of Ohio's common schools, wrote in his report of 1839 that counties that hired women "are able to do twice as much with the same money."

Lewis went on: "As the business of teaching is made more respectable, more females engage in it, and the wages are reduced." In his fourth report, in 1841, Horace Mann welcomed the "change . . . rapidly taking place" in the hiring of women teachers.

Catharine Beecher

Who: She ran a school for girls, then spent decades arguing that teaching was women's work.

When and where: Born in East Hampton, New York, on September 6, 1800. Died in Elmira, New York, in 1878.

By 1857 her board's own reports counted 452 teachers sent out.

Catharine Beecher spent decades arguing that teaching was women's work, and she helped send hundreds of women out to teach. She was the eldest child of the preacher Lyman Beecher and a sister of Harriet Beecher Stowe. In 1822 the man she was engaged to marry, a Yale mathematics professor named Alexander Metcalf Fisher, drowned when the ship *Albion* wrecked off the coast of Ireland. She never married, and she went into teaching.

Hartford. Accounts differ on when she opened the Hartford Female Seminary in Hartford, Connecticut. Two of them, one from the National Women's History Museum, give May 20, 1823, and say she opened it with her sister Mary. A third gives 1824 and her sister Isabella.

The girls there studied geography, world history, Latin, rhetoric, philosophy and the sciences, while most girls' schools of the time taught manners and fine arts. Within three years the school had 100 full-time pupils. In 1831 she moved west and opened the Western Female Institute in Cincinnati. It ran out of money after a money crisis called the Panic of 1837.

The salt. She published *An Essay on the Education of Female Teachers* in 1835. In it she argued that teaching was part of a mother's work and that women were the right people to raise the next generation. If the country's leading women became "refined, well educated, pious (devoted to religion) and active," she wrote, "the salt is scattered through the land to purify and save." By the salt she meant women like these, spread across the country as teachers.

She founded the Board of National Popular Education in 1847. Its members trained women in Hartford and sent them to teach in the West and the South. In 1852 she founded the American Woman's Educational Association, and by 1857 the board's own reports counted 452 teachers sent.

She opposed giving women the vote. She believed a woman's proper places were the home and the schoolroom.

Hitting children, and an argument about it

Horace Mann argued in his seventh report, in 1843, against beating children as the ordinary way to keep order in a classroom. He argued for teaching reading by whole words, instead of letter by letter, too.

The schoolmasters answer. Thirty-one Boston schoolmasters answered him in print in 1844, in a pamphlet called *Remarks on the Seventh Annual Report of the Hon. Horace Mann*. They attacked his plan to train teachers, and they defended beating. Mann answered them in a second pamphlet that same year, and in 1845 they answered him again. All three pamphlets survive, and the Library of Congress holds copies.

A novel of the time describes the tools of punishment in a village schoolroom. In *The Old Curiosity Shop*, which the English writer Charles Dickens published in 1840 and 1841, a cane and a ruler hang on hooks on the schoolroom wall, and on a shelf sits "the dunce's cap, made of old newspapers," a pointed paper hat that a teacher made a pupil wear to shame him. Nobody counted how often American teachers made a pupil wear one.

BY 1845: Massachusetts had a state Board of Education, a state school for teachers, and more women teachers than men, paid less than a third as much. Horace Mann and thirty-one Boston schoolmasters had argued in print over whether teachers should beat children.

The books children learned from

William Holmes McGuffey taught at Miami University in Ohio from 1826 to 1836. A Cincinnati publishing firm, Truman and Smith, paid him to make a set of reading books, the *Eclectic Readers*, eclectic meaning chosen from many sources. The First and Second came out in 1836, the Third and Fourth in 1837, the Fifth in 1844 and the Sixth in 1857.

The readers mixed proverbs, grammar and passages from Shakespeare with lessons about patriotism, religion, good behavior, games and sports. Librarians at Miami University keep more than 300 editions of the readers. They estimate that the readers sold more than 150 million copies over about a hundred years, most of them in the Midwest.

Children still learned spelling from Noah Webster's speller. In 1828 Webster published *An American Dictionary of the English Language*, with 70,000 words, their meanings and where they came from. A set cost $20.

Schools for deaf children and blind children

Dr. Mason Fitch Cogswell and other Hartford men asked Thomas Hopkins Gallaudet in 1815 to go to Europe and learn how teachers there taught deaf children. In Paris he studied at a school for the deaf, and he came back with Laurent Clerc, a deaf French teacher.

The school in Hartford. With Cogswell and ten other Hartford men behind them, Gallaudet and Clerc opened the American School for the Deaf in Hartford, Connecticut, on April 15, 1817. The first classes met in a hotel building, with seven full-time pupils, among them Cogswell's deaf daughter, Alice. Lessons were in sign language, a language made with the hands.

Connecticut's lawmakers voted the school a yearly grant of money in 1819, which the school's own history describes as the first state aid to a school in America. In 1820 members of Congress gave the school land in the Alabama Territory.

The school in Boston. Dr. John Dix Fisher founded the first school for blind children in the United States, in Boston, on March 2, 1829, after he saw a school for blind children in Paris. It took no pupils for three more years.

Samuel Gridley Howe became its first director in 1831, and in 1832 the school opened in the house of Howe's father, with six pupils. In 1839 Thomas Handasyd Perkins gave money for a larger building, and the school was named for him. It is Perkins School for the Blind now.

Laura Bridgman was born in Hanover, New Hampshire, on December 21, 1829. When she was two, scarlet fever, a disease that causes a high fever and a rash, left her deaf and blind.

She came to the Boston school on October 12, 1837, two months before her eighth birthday. Howe taught her by touch, giving her raised letters and objects together, again and again. After about two months she understood that the letters stood for the things. She was the first deaf and blind person in the United States to receive a formal education.

A Catholic school system begins

In the public schools of Philadelphia, the school day began with a reading aloud from the King James Bible, the Bible used by Protestants. Catholics used a different English Bible, the Douay-Rheims. In New York, Catholic families said the public schools were Protestant schools that mocked their religion.

The first free Catholic school in the country was older than this fight. In 1810 Elizabeth Ann Seton and her Sisters of Charity, a group of Catholic nuns, opened Saint Joseph's Academy and Free School at Emmitsburg, Maryland. They taught the poor girls of the parish, the area served by their church, for no charge.

New York. In New York City a private group called the Public School Society ran the public schools with state money. John Hughes challenged the Society in June 1840. He was a Catholic bishop in New York, a church leader in charge of the priests and churches of a whole area. Governor William Seward suggested that Catholics ask for a share of the school money, and New York Catholics asked for it, to run schools of their own.

State lawmakers shut down the Society in April 1842. They put elected school boards in its place and banned religious teaching in public schools, which was not what Hughes had asked for. During the fight someone threw stones at

Hughes's house. Protestants still controlled the elected boards, and Hughes set out to build a separate system of Catholic schools instead.

Philadelphia. In Philadelphia the fight ended in killings. Bishop Francis Patrick Kenrick asked the city's Board of Controllers of Public Schools in 1842 to let Catholic children read their church's Bible, and he objected that Catholic children had to sing Protestant hymns in school. In February 1844 a school director in Kensington, Hugh Clark, suggested stopping the Bible readings until the board could agree on a rule. A rumor spread that he had ordered the Bible out of the schools.

Nativists, people who opposed immigrants, held a meeting in Kensington on May 6, 1844, and a fight began. George Shiffler, a young nativist, was shot dead. The Encyclopedia of Greater Philadelphia's account of the riots does not say who shot him. By the end of the day a second man had been killed, and that account says he was apparently only watching. Several more people were wounded, and two of them later died.

Mobs burned houses on May 8, along with a Catholic seminary, which is a school for priests, and two Catholic churches, St. Michael's and St. Augustine's. Hugh Clark's house was among the houses burned. The Encyclopedia of Greater Philadelphia does not name the people in the mobs.

Southwark. A Catholic priest's brother stored guns in the basement of the Church of St. Philip Neri, in Southwark, in July. Crowds gathered and demanded the guns, and the sheriff called in the militia, the state's citizen soldiers.

The crowd and the militia fought each other with guns from July 5 to 7. At least two people in the crowd were killed at once. By the next morning four militiamen and probably about a dozen people in the crowd were dead. Counts that add up the dead from May and July put the total at about twenty or more.

Teaching an enslaved person to read becomes a crime

Lawmakers in six Southern states made it a crime to teach enslaved people to read between 1819 and 1834: Virginia, North Carolina, Georgia, Louisiana, Alabama and South Carolina. They wrote the punishments into the laws. South Carolina's law of 1740 had banned only the teaching of writing. The new laws banned reading too, and some reached free Black people as well.

Virginia. Virginia's lawmakers widened their laws in three steps. On January 24, 1804, they barred enslaved people from meeting at night at churches, meetinghouses or anywhere else. On March 2, 1819, they banned enslaved people from "any school or schools for teaching them reading or writing, either in the day or night."

A third law, passed on April 7, 1831, reached free Black people too: "All meetings of free negroes or mulattoes, at any school house…for teaching them reading or writing…shall be considered as an unlawful assembly." That meant a meeting the law banned, which officials could break up. Negro was a word used then for a Black person, and mulatto for a person with one Black parent and one white parent.

Under that law, any judge or justice of the peace, a local judge, could break up a school for free Black people and order up to twenty lashes for each person caught there. A whipping under these laws meant lashes, single blows, across the bare back with a leather whip or a switch, a thin stick cut from a tree, hard enough to break the skin. A white person found guilty of teaching Black people to read or write was fined $50 and could be jailed for two months. Teaching an enslaved person carried a fine of $10 to $100.

North Carolina. North Carolina's lawmakers passed their law in the session of 1830 and 1831. Its opening words give their reason: teaching enslaved people to read and write "has a tendency to excite dissatisfaction in their minds, and to produce insurrection and rebellion," meaning armed uprising. The law set the punishments by who the teacher was:

- A white person who taught an enslaved person to read or write was fined between $100 and $200, or jailed.

- A free Black person who did it could be whipped, with no fewer than 20 and no more than 39 lashes, or else fined or jailed, as the court chose.
- An enslaved person who taught another enslaved person got 39 lashes on the bare back.

Thirty-nine was the usual highest number of lashes in Southern law. Under the same North Carolina law, judges had to instruct grand juries, the groups of citizens who decide whether a person should be charged with a crime, to enforce it.

The other states. Lawmakers in the other four states wrote their own laws:

- Georgia, 1829: a Black person who taught another to read or write was fined and whipped, and a white person was fined up to $500 and could be jailed.
- Louisiana, 1830: anyone who taught an enslaved person to read or write could be sent to prison for one to twelve months.
- Alabama, 1832: anyone who taught any Black person, enslaved or free, was fined $250 to $500.
- South Carolina, 1834: lawmakers widened their law of 1740 from writing to reading as well, and barred Black people from working as clerks or salesmen in shops.

Only the North Carolina, Georgia and Virginia laws set the whip, and only for Black people. Louisiana's and Alabama's laws set the same prison term or fine for every teacher, whatever the teacher's race.

Walker and Turner. Two events fell in the middle of these years. David Walker, a free Black man from North Carolina who ran a used-clothing store in Boston, published his *Appeal to the Coloured Citizens of the World* in 1829. In it he urged Black people to resist slavery, and sailors carried copies into the South.

Nat Turner, an enslaved preacher, led a revolt of enslaved people in Southampton County, Virginia, in August 1831. They killed at least 55 white people. Afterward, militias and other armed white men killed more than 100

Black people, and some estimates run to 200. Many of them had no part in the revolt. A county court led by Judge Jeremiah Cobb sentenced Turner to death, and the sheriff hanged him in Jerusalem, Virginia, on November 11, 1831.

Virginia's laws of 1804 and 1819 came before both events. Every law after them came after Walker's *Appeal*, and Alabama's and South Carolina's came after Turner's revolt too.

Enslaved people learned to read anyway. About 5 percent of enslaved people in Virginia could read by 1775. In the decades before the Civil War, while these laws were in force, the share doubled to about 10 percent.

Federal money for schooling Native children, from 1819

Federal officials began paying for the schooling of Native children in 1819. Members of Congress passed the Civilization Fund Act on March 3 of that year. Under it they set aside $10,000 a year to hire people to teach Native nations farming and to teach their children "reading, writing, and arithmetic." The law states its purpose as "providing against the further decline and final extinction (dying out completely) of the Indian tribes" and "introducing among them the habits and arts of civilization."

Government officials gave much of the money to churches and other religious groups. Lawmakers in Congress kept paying into the fund until 1873. In a report published in 2022, investigators for the US Department of the Interior date the start of the federal Indian boarding school system to 1819.

The Choctaw Academy. The Choctaw Academy in Kentucky was one of these schools. It opened in November 1825 on the farm of Richard Mentor Johnson, a US senator who later became Vice President, and twenty-one Choctaw boys came first. Its first head was a Baptist minister, Thomas Henderson.

The Choctaw Nation's own money paid for it. Under a treaty of 1825, a written agreement between the United States and the Choctaw Nation, the US government owed the Choctaw $6,000 a year for schooling, and that money went to Johnson's school. Choctaw leaders tried to break with the academy and take back control of the money in 1829 and again in 1833. They failed.

More than 600 students from 17 Native nations went to the Choctaw Academy. Johnson enslaved at least sixty people. Among them were Julia Chinn and the two daughters she had with him, Adaline and Imogene. By law he owned Chinn, and she could not refuse him. Enslaved people did most of the farming, building and cleaning at the school.

Inspectors who visited in 1840 "confirmed the many complaints of poor living conditions, inadequate (not good enough) education, food, and clothing, as well as forced manual labor." In 1842 the Choctaw leader Peter Pitchlynn took every Choctaw student out of the school. The last thirteen Chickasaw students left in 1848.

A school open to Black and white students dragged away: Canaan, 1835

Men in Canaan, New Hampshire, drove Black students out of their school in 1835. Noyes Academy in Canaan had opened in March of that year with 28 white students and 14 Black students, after its trustees, the people in charge of the school, voted in 1834 to admit students of every race. The Black students included Henry Highland Garnet, Alexander Crummell, Thomas S. Sidney and Julia Williams.

The oxen. The voters at a Canaan town meeting declared the school a public nuisance, meaning a harm to the town, in July 1835. On August 10, men from Canaan and the towns around it hitched teams of oxen to the school building with chains. They pulled it off its foundation and dragged it down the road. The town's own history says it took two days, and the building ended as a heap near the meeting house.

One account counts about seventy-five men and nearly a hundred oxen. Another gives a hundred yoke of oxen, a yoke being a pair of oxen joined by a wooden frame, so a hundred yoke would be two hundred animals.

Garnet fired a shotgun from a window, and the crowd drew back. The townspeople drove the Black students out of Canaan, and no record shows that anyone was ever punished for it. Garnet, Crummell and Sidney went on to study at the Oneida Institute, near Utica, New York.

Sarah Roberts

Who: A five-year-old Black girl put out of the Boston school nearest her home. Her father sued the city.

When and where: Boston, Massachusetts, 1848 to 1850.

To reach her school she had to walk past five others.

Sarah Roberts was five years old when members of Boston's school committee kept her out of the school nearest her home, because she was Black. She lived on Andover Street. The school city officials let her attend was the Abiel Smith School, the school for Black children, and to get there she walked past five schools for white children.

Her father, Benjamin F. Roberts, asked the Boston Primary School Committee four times to let her into the nearest school, and each time the committee members refused. In 1848 he sent her to the nearest school anyway, and she was put out of it. The National Park Service's account of the case does not name the person who put her out.

The lawsuit. He sued the city under a Massachusetts law of 1845. That law let a child who had been wrongly shut out of a public school claim damages, a payment of money for the harm.

Her lawyers were Charles Sumner and Robert Morris, who was twenty-five and one of the first Black lawyers in the United States. The judges of the Massachusetts Supreme Judicial Court, the state's highest court, heard the case in 1849.

Chief Justice Lemuel Shaw ruled for the city in 1850. He wrote that the school committee "have come to the conclusion, that the good of both classes of school will be best promoted, by maintaining the separate primary schools for colored and for white children," colored being a word used then for Black people.

Black Bostonians kept up the fight. In 1855 Massachusetts lawmakers outlawed separate public schools by race.

BY 1850

If you remember one thing: Horace Mann made the case for free schools paid for by taxes in these fifty years, while Southern lawmakers made it a crime to teach enslaved people to read.

- 1830 to 1831: North Carolina's lawmakers set 39 lashes on the bare back for an enslaved person who teaches another to read.
- 1837: Massachusetts lawmakers set up a state Board of Education, and Horace Mann begins eleven years of arguing for free schools paid for by taxes.
- 1839: the first state school for teachers opens at Lexington, Massachusetts.

Next: After 1850, state lawmakers began to make children go to school, and after the Civil War freed people built schools of their own.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 31 MINUTES

State lawmakers began to order children to go to school, starting with Massachusetts in 1852, and school became a legal duty. Lawmakers in Congress gave states public land, which the United States had taken from Native nations, to found colleges.

Nearly four million people had been enslaved in 1860. Once freed, they built schools, and by 1869 they paid for all or part of 1,094 of them, by the count of a federal inspector. White men burned some of those schools, whipped their teachers and killed at least one teacher, and the same inspector wrote that down too. Southern lawmakers made separate schools for Black and white children the law, and the justices of the Supreme Court let it stand.

Federal officials began sending Native children to boarding schools far from home in 1879, where staff cut their hair, took their clothes and forbade their languages. More than 180 children died at Carlisle, the first federal boarding school built away from the reservations, the lands set aside for Native nations.

When school became the law

Massachusetts lawmakers passed the first compulsory attendance law, a law that requires children to go to school, in 1852. It covered children from 8 to 14 years old. Each child had to attend at least twelve weeks a year, six of them in a row. By 1900, 32 states had such a law, and by 1918 every state had one. Mississippi was the last. How strictly officials enforced the early laws is not known.

Colleges built with Native land: the Morrill Acts

The 1862 law. Under the Morrill Act, which became law on July 2, 1862, each state received 30,000 acres of public land for each of its senators and representatives in Congress. State officials sold the land and used the money to found colleges that taught agriculture and the mechanic arts, meaning engineering and machine work. These were called land-grant colleges, and at them the children of farmers and working people could go to college.

Whose land. The United States had taken the land from Native nations. Two journalists, Robert Lee and Tristan Ahtone, traced it through land records in 2020. By their count it came to about 10.7 million acres, taken from nearly 250 tribes, bands and communities through more than 160 land cessions, agreements in which a nation gave up land. Lee and Ahtone write that these were backed by violence.

US officials paid less than $400,000 for that land. Its sale raised $17.7 million for 52 colleges, more than forty times what the officials had paid.

Members of Congress passed a second Morrill Act in 1890, giving states money instead of land. The law required each state's lawmakers either to admit Black students to the land-grant college the state already had or to set up a separate one for them. Lawmakers in most Southern states set up separate colleges, and some of the historically Black colleges and universities, the colleges first founded for Black students, began that way. Alcorn State University in Mississippi, founded in 1871, is the oldest public historically Black land-grant college.

The schools freed people built

Members of Congress set up the Bureau of Refugees, Freedmen, and Abandoned Lands, called the Freedmen's Bureau, inside the War Department in March 1865. Its officials helped people who had just been freed from slavery. John W. Alvord, its inspector of schools, wrote a report on the schools every six months, from 1866 to 1870.

Freed people paid for many of these schools themselves. In the half year that ended on December 31, 1869, Alvord counted 1,094 schools that freed people "sustained wholly or in part," meaning they paid all or part of the cost. That half year his count reached 210,018 pupils, though a pupil who went to more than one kind of school was counted more than once.

Many children had to pick cotton in the autumn, so many schools did not open until December, and many others ran right through the summer.

Alvord estimated that in its five years the Bureau's schools had given "an elementary education to about one million of pupils." That did not count the parents and other adults who had learned to read at home, many of them from their own children. Bureau officials stopped paying for schools in 1870.

Attacks on the freed people's schools

White men burned freed people's schools, whipped their teachers and killed at least one of them, and Alvord recorded the attacks in his reports. In his report of January 1868 he wrote: "Teachers were proscribed (shut out) and ill-treated, school-houses burned, and threats so strong that many schools could not be opened." In the Oxford district of Mississippi, "A Mr. Borum started a school about twelve miles from this place and it was immediately burned." The report does not say who burned it.

Georgia, 1869. His report of January 1870 names places and people. Late in 1869, bands of "armed and disguised men," which the report calls "K. K. K.," attacked people in county after county in Georgia. K. K. K. stands for the Ku Klux Klan. Men who had fought for the Confederacy, the Southern states that broke away and fought the United States in the Civil War, started the Klan in Pulaski, Tennessee, in December 1865. Its members whipped, burned and murdered freed people and the people who helped them.

At Greensboro, Georgia, men drove the teacher R. H. Gladding out of town. The white man he boarded with was "taken out of his own house in the night and unmercifully whipped." Abram Colby, a Black man who had backed the school and had just been elected as one of the state's lawmakers, "was also taken out of his house and beaten nearly to death." The report does not name the men who did any of this. The mayor told Gladding that the town's officials could not protect him.

Tennessee and Delaware. In Dresden, Tennessee, two students from Fisk University ran a school of about fifty children. On the night of September 2, 1869, "men masked and armed" took them from their boarding house into the woods. The men "severely whipped" them and told them to stop teaching and leave or be killed. The report states that local officials were told about it and "nothing was done in the matter."

At Slaughter Neck, Delaware, drunken white men burned the school. The report states that the evidence was given to the state's attorney and the governor was told, "yet nothing was done to bring the parties to justice." It

does not say who gave the evidence. Black families there rebuilt the school with help from a Delaware charity and some Bureau money.

Memphis and Cross Plains. White mobs and police in Memphis, Tennessee, killed 46 Black people in May 1866. They burned every Black church and school in the city, 8 schools in all, by the count of a committee of Congress.

Members of the Ku Klux Klan lynched William Luke, killing him without any trial, at Cross Plains, Alabama, on July 11, 1870. Luke was a preacher from Flesherton, Ontario, in Canada, who taught freed people for Talladega College. The Klansmen killed Black men with him. One count gives six dead in all, and another gives eight.

BY 1870: Freed people had paid for all or part of more than a thousand schools, and the Freedmen's Bureau had stopped paying for any. White men had burned schools, whipped teachers and lynched one, and the Bureau's reports name none of the attackers.

Charlotte Forten

Who: The first Black graduate of the Salem Normal School and the first Black teacher in Beaufort County, South Carolina.

When and where: Born in Philadelphia in 1837. Died in Washington, D.C., on July 23, 1914.

In October 1862 she sailed south to teach freed people.

Charlotte Forten was the first Black teacher in Beaufort County, South Carolina. She grew up in Philadelphia in a well-off Black family that worked to end slavery. Philadelphia's school officials would not take her, so her family had her taught at home by tutors.

Salem. She moved to Salem, Massachusetts, in 1853 to live with the Remond family and go to school there. In 1856 she graduated from the Salem Normal School, its first African American graduate, having studied subjects from algebra and Latin to optics, the science of light. Her principal, Richard Edwards, found her a job at the Epes Grammar School. She became the first African American teacher in Salem's public schools, teaching white children.

In her journal she wrote that her time in Salem was "the happiest of my life." She also wrote: "I wonder that every colored person is not a misanthrope (a person who hates other people). Surely we have everything to make us hate mankind."

St. Helena Island. She sailed to St. Helena Island, off South Carolina, in October 1862, in the middle of the war, to teach at the Penn School with Laura Towne and Ellen Murray. The school had opened that year with 41 pupils, and by 1865 it had 436 pupils in four buildings. She taught there until the war ended and kept a journal the whole time. In May and June 1864 she published an account of it, "Life on the Sea Islands," in the *Atlantic Monthly*, a magazine.

> After the war she taught in Boston and in Charleston, South Carolina. By 1872 she was teaching in Washington, D.C., at the school that became Dunbar High School.

A teacher jailed in Norfolk, 1854

Officials in Norfolk, Virginia, jailed a white teacher in 1854 for teaching free Black children to read. Margaret Douglass, a white woman in Norfolk, and her daughter Rosa opened a school for free Black children in June 1852. A Mr. Robinson, whose two daughters Margaret Douglass had begun to teach, asked her to do it. The Douglasses charged three dollars for each quarter of a year. Twenty-five boys and girls came, and the school ran openly for about eleven months.

The arrest. The city constable, a town officer who kept order and made arrests, came to the house on the morning of May 9, 1853. His name was Cherry, and a second officer, Cox, came with him. Mayor Simon S. Stubbs had sent them. The officers wrote down the name of every child and every parent and marched the two teachers and the children to the mayor's office.

The mayor read out the Virginia law and told them its highest penalty was a fine of one hundred dollars and six months in jail. Those are his figures, as Margaret Douglass later reported them, and they are higher than the fine of $50 and two months in jail that the 1831 law set for a white teacher. A grand jury charged the two women with meeting with Black children "for the purpose of instructing them to read and to write." Rosa left Norfolk for New York before the trial.

The sentence. A jury found Margaret Douglass guilty in November 1853 and set her fine at one dollar. On January 10, 1854, Judge Baker of the Norfolk court added a month in the city jail. He said the jail term was "an example to all others in like cases disposed to offend." She served the month. Afterward she published her own account of the case, with the court papers printed in it.

Separate schools by law, upheld by the courts

After the Civil War, Southern lawmakers wrote the separation of Black and white pupils into law. Tennessee is one example. Its lawmakers passed a school law that separated the races just after the war. The men who wrote the state's new constitution of 1870 forbade mixed schools in it. In 1873 Tennessee's lawmakers wrote the rule out in full: "white and colored persons shall not be taught in the same school, but in separate schools."

North and West. In much of the North, members of city school boards kept Black children apart by their own rules. In San Francisco, school officials opened a separate school for Black children in 1854 and one for Chinese children in 1859. In 1870 they closed the Chinese school, and Chinese American children were left with no public school at all.

Plessy, 1896. The justices of the United States Supreme Court decided *Plessy v. Ferguson*, a case about train cars, in 1896. Louisiana's lawmakers had required railroads in 1890 to seat Black and white passengers in separate cars. Homer Plessy, a shoemaker in New Orleans, sat in a car for white passengers in 1892. He had arranged his arrest ahead of time, to test the law in court.

The justices ruled 7 to 1 that the law was allowed. In their ruling they called separate schools "the most common instance" of lawful separation by race. The first case they cited was Sarah Roberts's case from Boston. They cited state court rulings from Ohio, Missouri, California, Louisiana, New York, Indiana and Kentucky too, and they pointed out that laws passed by Congress required separate schools in Washington, D.C.

Augusta, 1899. The justices refused in 1899 to stop a Georgia school board that had closed a Black high school. The members of the school board of Richmond County, in Augusta, Georgia, had run the Ware High School for Black students since 1880. They closed it on July 10, 1897, saying they needed the money for Black primary schools, and they kept paying for high schools for white students. Three Black parents, J. W. Cumming, James S. Harper and John C. Ladeveze, sued the board and asked the courts to stop it from paying for the white high schools.

Justice John Marshall Harlan wrote the Court's ruling on December 18, 1899. Cutting off the white schools' money, he wrote, "would only be to take from white children educational privileges enjoyed by them, without giving to colored children additional opportunities." He wrote that school funding was "a matter belonging to the respective states." The board won, and the white high schools kept their money. Harlan did not rule on whether separate schools themselves were lawful, because the parents had not asked the Court to.

Women go to college

College leaders began to admit women in these years, and new colleges opened for women only. The University of Iowa's own archive states that it admitted women "on an equal basis with men" beginning in 1860. The University of Michigan and Cornell University took their first women students in 1870. At Cornell the first one, Jennie Spencer, left because there was nowhere near the campus for a woman to live.

Colleges for women. Matthew Vassar founded Vassar College in Poughkeepsie, New York, and it opened in 1865 with 353 students. Wellesley College and Smith College opened in 1875, and Bryn Mawr College in 1885.

Spelman. Sophia B. Packard and Harriet E. Giles, with the pastor Frank Quarles, started a school in the basement of Friendship Baptist Church in Atlanta on April 11, 1881. It had eleven pupils: ten women, some of whom had been enslaved, and one girl. It became Spelman College.

Sex in Education. Some men argued that study would harm women's bodies. Edward H. Clarke, a professor at Harvard Medical School, published *Sex in Education* in 1873. Smith College's own history states that he warned women against studying "in a boy's way." He warned that it could make a woman's womb and ovaries waste away, make her more like a man, leave her unable to have children, drive her insane, or even kill her. Clarke wrote that the results of such study were "monstrous brains and puny bodies."

Mary Putnam Jacobi answered him with evidence. She sent a thousand lists of questions to women and wrote up the answers. Her essay won Harvard's Boylston Prize in 1876, and she was the first woman to win it.

A one-room school, counted: the Freeman School

The Freeman School near Beatrice, Nebraska, was built of brick in 1872 and used as a school until 1967, for 95 years. It had one main room and a coat hall at the front. Eighteen desks stood in three rows that stepped up toward the back. A wood stove stood in the middle of the room, and a recitation bench stood at the front.

Pupils from 4 to 20 years old learned there together, seated by age. The most pupils in any one year was 33, in 1880. From 1882 to 1900 the teacher was paid $32 to $40 a month. Running the school cost $75 in 1872, and in 1873, $27 went for firewood. The building is now part of Homestead National Historical Park.

Beating children, by the rules

Teachers in this era still hit children, and school rules required them to write each beating down. Boston's school rules stated that each teacher had to send in a written report at the end of every month, a "report of all cases in which corporal punishment has been inflicted," corporal punishment meaning an adult hitting a child on purpose, to hurt the child, as a punishment.

The report had to "state the name of the pupil, the amount of punishment, and the reason for its infliction." The chairmen of the district committees then reported the totals every three months, along with the average for each teacher.

Under the same rules, a teacher had to explain the offense to the child first. The blows were to be only "blows on the hand with a rattan," a thin, bendable cane cut from a kind of palm. That limit did not hold when a pupil refused to take the punishment, and the rules did not say what was allowed then.

No teacher was to shut a child "in a closet or wardrobe." A girl in a grammar school, the school that came after primary school, could not be hit without the approval of the master, the head teacher.

New Jersey's lawmakers banned hitting children in the state's public schools in 1867, the first state lawmakers in the country to do it. No other state followed for 104 years, until Massachusetts in 1971.

BY 1899: Southern lawmakers had written separate schools into law, and the justices of the Supreme Court had called them lawful. Women studied at Vassar, Wellesley, Smith and Bryn Mawr. Only New Jersey's lawmakers had banned hitting children in school.

Josephine Foster

Who: A schoolgirl beaten for whispering in class. Her parents charged her teachers with cruelty, and the school committee took no action.

When and where: The Allston Grammar School, Cambridgeport, Massachusetts, June 12, 1866.

She was hit fifteen to twenty times for whispering in class.

A teacher hit Josephine Foster on the hand fifteen to twenty times for whispering and then refusing to be hit. The members of the Cambridge School Committee ordered four thousand copies of their own account of what happened to her printed.

Her trouble began in her writing lesson on the morning of June 12, 1866, when she refused to do what the master, B. W. Roberts, told her. He made her stand in front of the class.

That afternoon, while he was out of the room, an assistant teacher, Sarah M. Gray, caught her whispering and asked what she thought of whispering. Josephine answered "I don't know" in a way the teacher took as defiance. Miss Gray took her and the other girl to the recitation room. She hit the other girl's hand a few times with a rattan and sent her back. Then she told Josephine to hold out her hand. Josephine said, "I won't be whipped."

The ferule. Roberts came back. He would not hit her himself, and said it would be "less humiliating" for her to be hit by a woman teacher. He told Miss Gray to use his ferule instead of the cane. The committee's report describes a ferule as "a flexible instrument, made of two flat pieces of

leather, sewed together, and filled with some soft substance." It adds that the ferule was "a common implement of punishment in schools," in use in several other grammar schools in the city.

Josephine again said she would not be whipped, and she began to scream before anyone hit her. Another teacher, Miss Hodges, held her left hand down while Miss Gray struck her right hand. The screams carried into the schoolroom and set some of the boys laughing.

Roberts ordered the boys to march to piano music, a regular two-minute exercise, and used those two minutes to come into the recitation room. He held her hand himself and told her she would be hit until she stopped crying out. Miss Gray struck five blows while he was there. After he left, Josephine held out her hand herself and took three to five more light blows.

The report states: "In all, from fifteen to twenty blows were inflicted." For four days afterward, her left arm and thumb showed marks from being held. Red marks on her right hand were gone by eight o'clock that evening.

DISPUTED: her age. On one page the report says she was sixteen. On another it says her teachers thought she was "not over thirteen."

The hearing. Her parents, Charles and Elizabeth Foster, wrote to the school committee the next day. They charged all three teachers "for cruelty in punishment inflicted on our daughter," and asked the committee to "have justice done us." The committee members held a hearing on June 21, 1866, and voted, all of them in favor, that "the case does not require any action on the part of this Board."

James Walker and others sent the committee a separate petition, a written request signed by a group of people, asking its members to end the beating of girls. The committee members did not end it.

Teaching becomes mostly women's work

By the time federal officials first counted teachers across the whole country, most of them were already women. In the school year 1869 to 1870 there were about 201,000 public school teachers, 78,000 men and 123,000 women. By 1899 to 1900 there were about 423,000, of whom 127,000 were men and 296,000 women, seven women in every ten teachers.

The change came at different times in different regions. In 1850, 60 percent of teachers in New England were women. In the South the share was 24 percent, and in the Midwest it was 81 percent. The historians Mary Lou Breitborde and Kelly Kolodny write about school superintendents, the officials in charge of a district's schools. In their account, the superintendents were sure they were hiring young women skilled at managing children "at half the price of men."

After Massachusetts, normal schools for training teachers opened in New York and Philadelphia in the 1840s, and then in other states of the Northeast and the Midwest. By 1870 there were 39 normal schools in 18 states, and by 1900 there were 180.

Where the summer break came from

Many people say that schools close in summer so that children could help bring in the harvest. The historian Kenneth Gold, who wrote a history of summer schooling, says the record shows the opposite. A farming district's school year had a short winter term and a short summer term, because farm families needed their children most in the spring planting and the autumn harvest. Those are the months the school year covers now.

City schools stayed open far longer. New York City's schools were open 248 days in 1842, and since nobody had to attend, children came when they could. Across the whole country, the average school term was 132 days in 1869 to 1870, and the average pupil attended 78 of those days. By 1899 to 1900 the average term was 144 days, and the average pupil attended 99.

The middle ground. Late in the 1800s school reformers worked to make the city and country calendars match. They settled on a long summer break as the middle ground between the two. One reason for it was that teachers could be trained during the break.

Catholic schools become a system

The Catholic bishops of the United States met in Baltimore from November 9 to December 7, 1884. Fourteen archbishops came, an archbishop being a bishop who leads the main church area of a region. With them came sixty-one bishops or their representatives and six abbots, the heads of monasteries.

They declared that Catholic schools were an "absolute necessity" and required every pastor, the priest in charge of a local church, to set up a school, free where possible. They ruled: "Parents must send their children to such schools unless the bishop should judge the reason for sending them elsewhere to be sufficient." With that order the bishops set out to build a system of Catholic parish schools beside the public schools.

BY 1900: Seven of every ten public school teachers were women, and the Catholic bishops had required every pastor to set up a school beside the public schools.

Boarding schools for Native children: Carlisle

Federal officials opened a boarding school for Native children at Carlisle, Pennsylvania, far from the children's homes, in 1879. The man who ran it, Richard Henry Pratt, set out to wipe out the children's languages and ways of life. Speaking in Denver in 1892, he described the aim in his own words: "that all the Indian there is in the race should be dead. Kill the Indian in him and save the man."

Fort Marion. Pratt was an Army officer. In 1875 Army officers arrested 72 people from several Native nations, called them ringleaders, and sent them by train to Fort Marion, a fort in St. Augustine, Florida. War Department officials put Pratt in charge of them.

At Fort Marion, Pratt cut the prisoners' hair, put them in military uniforms, drilled them like soldiers, taught them English, and hired them out as laborers. He sold the things they made and drew. Ten of the prisoners died while he held them, between 1875 and 1878. At Carlisle, Pratt and his staff later did the same to the children: they cut their hair, put them in uniforms, drilled them and taught them English.

Setting up the school. Federal officials approved Pratt's request for a school in August 1879. The Commissioner of Indian Affairs, Ezra A. Hayt, allowed Pratt to choose 70 students from the Sioux nations and 30 from the Kiowa, Comanche, Cheyenne and Arapaho nations. The Secretary of War handed over the empty Carlisle Barracks, an old Army post, for the school.

The first children arrived on October 6, 1879: 83 of them, 24 girls and 59 boys, from the Rosebud and Pine Ridge agencies in Dakota Territory, the federal offices that ran those reservations. Some accounts give 82, and the National Park Service says about 140. The school formally opened a few weeks later.

Arrival. Staff cut the children's hair and took away their clothes, their moccasins and their family belongings. They dressed the boys in military uniforms and the girls in Victorian dresses, the long dresses white women

wore then, and gave the children English names. The published records name Pratt, who ran the school, but not the staff members who did the cutting and the punishing.

Luther Standing Bear, a Lakota boy brought to Carlisle in its first year, wrote later: "Now, after having my hair cut . . . I felt that I was no more Indian but would be an imitation of a white man."

An Army post. Pratt ran the school like one. The day began at six in the morning and ended with Taps, a bugle call, and room inspection at nine at night. The children marched in groups like soldiers in drills, and they changed movements when officials rang a bell. A fence six feet high ran around the grounds. Sunday school, chapel and Bible study were required.

The staff forbade the children to speak their own languages, and they often enforced that rule by hitting the children. Staff sent a child who broke a rule to the guardhouse, the post's old jail, or school officials sentenced the child to hard labor, heavy work given as a punishment. The published records do not name the staff members who did it.

Work without pay. Half of each day was lessons and half was forced work without pay, and the children's work kept the school running. The President's proclamation about Carlisle in 2024, an official statement, describes it as "compulsory labor." Boys worked at blacksmithing, tinsmithing, making wagons, carpentry, tailoring, making shoes and harnesses, baking, painting, printing and farming. Girls cooked, washed laundry and cleaned. The children built a printing office, a gymnasium, a hospital, a laundry and a warehouse with their own hands.

Under what Pratt called the outing system, staff sent children to live with white families and work for them for part of each year. The children's wages went into accounts held by school officials, not into their hands. School officials paid for the gymnasium with money they took from the children's own trust accounts, money federal officials kept for them, and called it donations.

The dead. More than 180 children died at Carlisle, by the federal government's count. The National Park Service counts at least 168 who died of three diseases: tuberculosis, which destroys the lungs, pneumonia, an infection

of the lungs, and influenza, the flu. Both counts are minimums, which means the real number may be higher. Many of the children are buried at the Carlisle Barracks cemetery.

Chief Spotted Tail and other Rosebud Sioux parents asked the government on May 23, 1881, to send their dead children home. Army officials returned the remains of nine of the children to the Rosebud Sioux Tribe on July 14, 2021, 140 years later.

Their names were Dennis Strikes First (Blue Tomahawk), Rose Long Face (Little Hawk), Lucy Take The Tail (Pretty Eagle), Warren Painter (Bear Paints Dirt), Ernest Knocks Off (White Thunder), Maud Little Girl (Swift Bear), Alvan, also called Roaster and Kills Seven Horses, Friend Hollow Horn Bear, and Dora Her Pipe (Brave Bull). The records printed with their names do not give their ages or what they died of.

By the time Carlisle closed in 1918, about 7,800 children from more than 140 Native nations had gone through the school, by the federal count. The National Park Service gives more than ten thousand from over 142 nations. Some children were five years old when they arrived. Carlisle was the first federal boarding school built away from the reservations, and officials opened many more.

Rations. Members of Congress passed a law about food rations, the set amounts of food handed out at the reservations, on March 3, 1893. Under it the Secretary of the Interior could stop the rations of any Native family whose children aged eight to twenty-one had not attended school the year before. That included rations a treaty had promised.

The count. Carlisle was one of 417 federal Indian boarding schools, and at least 973 children died in them, by the count federal investigators finished in 2024. That investigation is described under 2000 to Today.

The punishments. The US Department of the Interior's 2022 report lists the punishments staff used across the system. They locked children alone in a room, which is called solitary confinement, and held back their food. They flogged children, which means beating them hard again and again, and they whipped, slapped and cuffed them, cuffing being hitting with an open hand.

The report does not say how flogging differed from whipping, or what staff beat the children with. Staff sometimes made older children punish younger ones.

The President's 2024 proclamation about Carlisle also states that "many Native children were subjected to sexual abuse at the schools," which means forcing sexual acts on a child. The proclamation does not say who abused the children or how many.

Children kept their languages alive by speaking them in secret. At the schools they made friends with children from other nations, and those friendships were one root of the Native organizing of the next century.

Zitkala-Ša

Who: Quaker missionaries took her from her home at eight to a boarding school in Indiana. She wrote down what was done to her there and published it.

When and where: Born on the Yankton Indian Reservation, South Dakota, on February 22, 1876. Died in Washington, D.C., on January 26, 1938.

At eight she hid under a bed to save her braids.

Zitkala-Ša was eight years old when staff at a boarding school tied her to a chair and cut off her braids, and she later published her own account of it. Her Dakota name means Red Bird. Her English name was Gertrude Simmons.

Quaker missionaries, people sent out by the Quaker church to spread its faith, took her in 1884 from the Yankton Reservation to White's Indiana Manual Labor Institute, a boarding school at Wabash, Indiana. She stayed three years and went home. She went back at fifteen and graduated in 1895.

On her first day there, she learned that the staff meant to cut the girls' hair. She knew what that meant at home: "Our mothers had taught us that only unskilled warriors who were captured had their hair shingled (cut short) by the enemy. Among our people, short hair was worn by

mourners, and shingled hair by cowards!" Her friend Judewin said, "We have to submit, because they are strong." Zitkala-Ša answered, "No, I will not submit! I will struggle first!"

She hid under a bed in a dormitory. "I remember being dragged out, though I resisted by kicking and scratching wildly. In spite of myself, I was carried downstairs and tied fast in a chair."

Then, she wrote, "I cried aloud, shaking my head all the while until I felt the cold blades of the scissors against my neck, and heard them gnaw off one of my thick braids. Then I lost my spirit." She wrote that she "moaned for my mother, but no one came to comfort me," and that she was now "only one of many little animals driven by a herder."

She went on to Earlham College in Indiana and studied violin at the New England Conservatory in Boston. In 1899, at twenty-three, she went to Carlisle to teach music. In 1900 the *Atlantic Monthly* printed three of her essays about her school years.

Where it happened. Her braids were cut at White's Institute in Indiana, years before she worked at Carlisle. The federal proclamation of 2024 places her haircut at Carlisle, and that is wrong.

She spent the rest of her life working for Native people's rights. From 1916 she was secretary of the Society of American Indians, and she edited its *American Indian Magazine*. In 1926 she and her husband founded the National Council of American Indians, and she led it until she died. She is buried at Arlington National Cemetery. The year of death cut into her headstone, 1936, is two years wrong.

Booker T. Washington

Who: Born enslaved, he learned to read first from a number on a salt barrel, and he founded the school at Tuskegee.

When and where: Born at Hale's Ford, Franklin County, Virginia, in 1856. Died on November 14, 1915.

The first thing he learned to read was the number 18.

Booker T. Washington was born enslaved, and he founded a school for Black students at Tuskegee, Alabama, that he ran for 34 years. He grew up on the James Burroughs farm at Hale's Ford, in Franklin County, Virginia. He lived with his mother, Jane, who was the farm's cook, and two half-siblings in a one-room kitchen cabin. As a small boy he carried water to the field workers and carried other children's schoolbooks for them.

Slavery ended for him and his family in 1865. They moved to Malden, West Virginia, where he packed salt, mined coal and worked as a butler. The number 18 was on the salt barrels. It was the number of his stepfather's crew of salt packers.

At sixteen he traveled more than 500 miles to the Hampton Institute in Virginia. He arrived, in his own account, dirty, hungry and with fifty cents. He paid his way by working as the school's janitor, graduated at nineteen, and went home to Malden to teach. In 1879 he went back to Hampton as a teacher.

Tuskegee. He reached Tuskegee in June 1881. On July 4 he opened his school there with thirty students, in a shack lent by a Black church, Butler Chapel A.M.E. Zion. He was its first teacher and its principal, and he stayed principal until he died. The students learned trades and farming along with book subjects, and they made the bricks and put up the school's buildings themselves.

Washington argued that Black Americans should build skills and property first and press for their political rights later. He made that argument in a speech in Atlanta on September 18, 1895. W. E. B. Du Bois, another Black educator, named the speech the Atlanta Compromise and attacked it. In 1901 Washington became the first Black American invited to dine at the White House. He died in 1915, at 59.

BY 1900

If you remember one thing: School became something the law required in these fifty years, and the law also decided which school a child could go to.

In 1850: no state law made children go to school, and in six Southern states it was a crime to teach an enslaved person to read.

In 1900: 32 states made children go to school, Black and white children sat in separate schools by law across the South, and federal officials were sending Native children to boarding schools far from home.

Next: After 1900, finishing high school became ordinary, and school officials began to sort children with intelligence tests.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 42 MINUTES

Finishing high school went from rare to ordinary. In 1900 about 6 American children in every 100 finished high school, and by 1940 about half did. School board members closed one-teacher schools by the tens of thousands and sent buses to carry the children to bigger schools.

School officials also began to sort children. They gave written intelligence tests, sets of questions meant to measure how well a person can think, and put children into classes by the scores. Lewis Terman, a psychologist, a scientist who studies how people think, wrote the American version of the test. In his handbook for teachers he argued that the children with the lowest scores should not be allowed to have children of their own.

In Southern counties, officials paid for Black children's schools at a fraction of what they spent on white children's schools. A survey team reported to the Secretary of the Interior in 1928 that officials were trying to feed each child at the government's Indian boarding schools on eleven cents a day.

Finishing high school becomes ordinary

The high schools filled fast, by the federal government's own figures. In the school year 1899 to 1900, about 6 of every 100 young people aged seventeen graduated from high school. Forty years later, about half did.

The Depression years. The biggest jump came in the 1930s, during the Great Depression, when by 1933 about one worker in four had no job. Graduates rose from 29 for every 100 seventeen-year-olds in 1929 to 1930 to nearly 51 ten years later. Many teenagers stayed in school partly because there were no jobs to leave school for.

During the Second World War the share of teenagers in school fell back, because many older pupils left for wartime jobs. Girls finished high school more often than boys in every decade of this era.

Children also came to school on far more days. By 1949 to 1950 the average school term was 178 days, and the average pupil came on 158 of them, against 99 days in 1899 to 1900.

The one-room school closes

Members of local school boards closed one-room schools district by district in these years. They could close a small school only if its children had a way to reach a bigger one. School buses carried the children past the small school near their homes to a bigger school farther away.

Federal officials began counting one-teacher schools in 1929 to 1930, when there were about 149,000 of them. By 1949 to 1950 there were about 60,000, fewer than half as many. Local histories and popular accounts give a figure of about 200,000 one-room schools at the peak, but that is an estimate and not a count. The number of school districts fell too, by about 35,000 in the twelve years before 1950.

The buses carried more and more of the children. By 1950 tax money paid for nearly a third of the country's schoolchildren to ride to school, up from about 9 of every 100 in 1930.

Mamie Garvin Fields

Who: A Black teacher in South Carolina who started in a one-room school at twenty and taught for about thirty-five years.
When and where: Born in Charleston, South Carolina, in 1888. Died in 1987.

At twenty she took charge of a one-room school near Sumter.

Mamie Garvin Fields took her first teaching job in 1908, at Pine Wood, a one-room school near Sumter, South Carolina. She taught in one-teacher and two-teacher schools for Black children in rural South Carolina, and she kept teaching for about thirty-five years.

She grew up in Charleston and won a scholarship from her church to Claflin College in Orangeburg, first to its high school department and then to the college. In 1909 she came back to Charleston as one of the first Black teachers the county hired for its public schools. She was a principal for two years on Johns Island, and she retired in 1943.

Grown-ups too. She ran classes for adults on James Island and Johns Island in the early 1920s. During the Depression she started the first vacation Bible school for migrant workers in Charleston, farm workers who move from place to place to follow the harvests. From 1958 to 1964 she was president of the South Carolina Federation of Colored Women's Clubs, "colored" being a word then used for Black people.

She and her granddaughter Karen Fields began to record the story of her life in 1978, when she was ninety. Their book, *Lemon Swamp and Other Places*, came out in 1983. Her papers are kept at the Avery Research Center in Charleston. She died in 1987, at ninety-nine.

Children at work instead of at school

Many children in 1900 were at work, not in school. The census of 1900 counted about 2 million children working in mills, mines, fields, factories and stores and on city streets.

Members of Congress tried several times to stop it. They voted for the Keating-Owen Act in 1916, and President Woodrow Wilson signed it on September 1, 1916. Under it, a factory that employed children under 14 could not sell its goods across state lines. Neither could a mine that employed children under 16, or any workplace where children under 16 worked at night or more than eight hours a day.

The courts. The justices of the Supreme Court, the nine judges of the country's highest court, struck the law down in 1918, in *Hammer v. Dagenhart*. They ruled that the Constitution did not give Congress the power to pass it. Members of Congress then put a tax on goods made by children, and in 1922 the justices struck that law down too.

An amendment, a change to the Constitution, was proposed in the 1920s to allow a child labor law, but lawmakers in too few states approved it. Members of Congress finally set federal limits on child labor in the Fair Labor Standards Act of 1938, and in 1941 the justices ruled that this law was allowed.

Every state had a law by 1918 that required children to go to school. As fewer children went to work, more of them stayed in school, and the high schools filled up.

A school prayer case in Nebraska, 1902

A court fight over religious teaching in a public school began at the Freeman School in Nebraska, the one-room brick school near Beatrice. Daniel Freeman, the first person to file a claim for free land under the Homestead Act, sued in 1898 over the religious teaching there. On October 9, 1902, the justices of the Nebraska Supreme Court ruled in his favor. That was sixty years before the justices of the US Supreme Court ruled that public school staff anywhere in the country could not lead official prayers.

Who could hit a child, city by city

For the whole of this era, the law let teachers hit children in the public schools of every state except New Jersey. Members of city school boards wrote rules about who could do it and when, and some of their rule books survive. The rules call it corporal punishment, hitting a child's body as a punishment.

San Francisco. Members of the San Francisco Board of Education printed their rules in 1910. Section 63 said that corporal punishment "shall not be administered in the high schools, nor to girls in any of the schools of the department." Only principals could give it, only "in extreme cases when all other means fail to maintain discipline," and only in front of a witness.

The rule named the tools: "The use of either the rattan or the strap in administering punishment is optional," the rattan being a thin, bendy cane and the strap a long, flat strip of leather. Section 64 said the principal had to keep a book recording "the name and age of the pupil, the offense committed and the character of the punishment." At the end of every school month, the principal sent a copy to the Board and to the Superintendent, the official in charge of the city's schools.

New Haven and Omaha. The New Haven school rules of the same year began, "Teachers shall govern by kindness and by appeal to the better nature of pupils." The next lines set out when a teacher could hit a child anyway. The principal had to approve it first, and the teacher had to record it in the monthly report to the superintendent.

Another line read, "Corporal punishment shall not be inflicted in the presence of a school." In the English of the time, "a school" could mean the whole group of pupils, so a teacher could not hit a child in front of the others. In Omaha the rules of 1909 allowed it "as a last resort," under written instructions from the superintendent.

Each city's rules said who could hit a child and when. Only San Francisco's named the tools, and the rules in both San Francisco and New Haven required a written record of every case. Boston's rules had worked the same way in the 1860s.

BY 1920: Every state required children to go to school, and the high schools were filling as fewer children went to work. Teachers could still lawfully hit children everywhere except New Jersey.

Carlisle's last years: a beating, a petition and a hearing

A Pottawatomie student named Julia Hardin told a committee of Congress under oath in February 1914 how a staff member at the Carlisle boarding school had beaten her. The hearings began with a petition. Gus Welch, an Ojibwe student, gathered more than 200 students' signatures asking the Commissioner of Indian Affairs to look into wrongdoing by the school's leaders and its athletic department.

The hearings. Members of a joint committee of Congress came to the school without warning and held public hearings on the campus for a week. An inspector from the Indian Office, Edward B. Linnen, worked with them. They asked about everything from the laundry to the football team's money, and they heard testimony that the coach, Glenn "Pop" Warner, had used that money for his own gain.

Julia Hardin. She testified on February 7, 1914. She was 18, an orphan from Shawnee, Oklahoma, and a student in the business department, where she studied law, shorthand and typing. Every month she had received the mark "excellent" for her behavior.

She told the committee what had happened on the second day of one June. Under the school's outing program, staff sent students to live with white families and work for them. The outing agent had asked her if she wanted to go and told her, "Come and sign." Julia had answered, "Wait until I get a trunk and some clothes."

A note from the office of the matron, the woman on the staff in charge of the girls, told her she had to leave the next morning. The matron sent her to the superintendent, Moses Friedman, and Friedman sent her back to the matron.

The beating. Then Claude Stauffer, the school's music teacher and band leader, told her to sign a check for her train fare. She refused until her clothes were ready. He slapped her across the face with his hand and told her, "I am going to give you a sound thrashing, and I will stand the responsibility."

He pulled a board off a window sill, about three inches wide and two to two and a half feet long, and pushed her to the floor. Two matrons held her down, and one of them was Anna Ridenour. They pulled down the curtains so nobody could see in, and they locked the door.

Stauffer whipped her with the board "for at least ten minutes," she testified, "on the head, and every place." She lay face down with her hands over her head. Asked how many times he struck her, she said, "About sixty times." He threw her down five or six times, she said.

When Stauffer said again that he would take the responsibility, Ridenour said, "Yes, go ahead." Then he asked, "Shall I whip her some more?" He also said, "I will not let her get ahead of me."

Afterward. Stauffer and Ridenour took her to the school's lock-up and kept her there until after supper. Another staff member, Mr. Whitwell, spoke to her kindly. "Come over and sign this check and go to the country, and show them you are a lady," he said, and she signed the check.

The next morning she went to Mr. and Mrs. Crawford in Merchantville, New Jersey. For three months she did their washing, scrubbing, housecleaning and cooking, for $6 a month, and she paid her own train fare of $3.79. Someone at the school had sent the family a report that she had caused trouble, and she

was never told who sent it. When she arrived, the woman of the house told her, "Julia, we will have to go rather hard on you." The family treated her well after that, she said.

She told the committee that Stauffer had whipped several boys from his band in the guardhouse, and she named one of them, Robert Nash. She had seen Ridenour and another staff member, Miss Knight, whip a young woman named Rose Whipper and lock her up. After the hearings, officials suspended Stauffer. He was not fired, but his job was abolished. Warner left Carlisle.

The end of Carlisle. Officials handed Carlisle over to the War Department on September 1, 1918, and Army officers made it a hospital for the First World War. Federal officials moved the 279 children still at the school to other federal boarding schools.

John Dewey's school: children learn by doing

John Dewey, a professor at the University of Chicago, argued that children learn by doing things, not by sitting still and listening. To test the idea, he and his wife, Alice, ran a small school in Chicago for eight and a half years, from 1896 to 1904.

He described the classroom he was arguing against in his book *The School and Society*, published in 1899.

He had gone around the school supply stores in his city, looking for desks and chairs that children could work at. One dealer told him, "I am afraid we have not what you want. You want something at which the children may work. These are all for listening." Dewey described the ordinary schoolroom as having "rows of ugly desks placed in geometrical order, crowded together so that there shall be as little moving room as possible."

In his school, every child, boy or girl, had cooking, sewing and carpentry for an hour or two a week each. Each group of children cooked its own lunch once a week, set the table and served it to guests. Children also kept gardens and wove cloth.

The teachers planned for reading, writing and arithmetic to grow out of that work. In *The School and Society*, Dewey answered the question of what schools are for: "What the best and wisest parent wants for his own child, that must the community want for all of its children."

John Dewey

Who: A philosopher who started out as a high school teacher, argued that children learn by doing, and ran a school to test it.

When and where: Born in Burlington, Vermont, on October 20, 1859. Died in New York City on June 1, 1952.

On the first day, each child made a paper box for pencils.

John Dewey taught high school before he studied philosophy, and later he ran a school of his own to test his ideas about learning. He graduated from the University of Vermont in June 1879 and went straight to work as assistant principal of Oil City High School in Pennsylvania. There he taught the classics, the old Greek and Latin writers, along with science and algebra. He was nineteen, and he did that job for two years.

He earned a Ph.D., the highest degree a university gives, at Johns Hopkins University in 1884 and then taught at the University of Michigan. In March 1894 he accepted a job at the University of Chicago, and he was 34 when he arrived.

The school. He opened an elementary school in a house on 57th Street in January 1896, with sixteen pupils and two adults in charge. On the first day, the student newspaper reported, each child made a paper box for pencils. Two teachers who worked there later wrote the school's history. They called its first six months a time of trial and error, "chiefly indicative of what not to do."

The school had four homes in eight years. By October 1898 it had 82 children, a gymnasium, two science laboratories and rooms for manual training, which means work with tools. At its largest it had 140 children, 23 teachers and about ten graduate students helping.

The School and Society became a best seller in its first year. In 1916 he published *Democracy and Education*. In it he wrote that teaching which follows how children really grow "always sets out with situations which involve learning by doing."

His wife, Alice Chipman Dewey, resigned as principal of the school on April 5, 1904, and Dewey resigned from the University of Chicago on the same day. In 1905 he began teaching at Columbia University in New York. He retired in 1930 and died of pneumonia in 1952, at ninety-two.

Intelligence tests come to school

School officials began to give children written intelligence tests between 1916 and 1926. One of the men who made the tests wrote that the lowest scorers should be kept in separate classes and should not have children.

Terman's handbook. Lewis Terman published *The Measurement of Intelligence* in 1916. It was a handbook that told teachers how to give his American version of the test and what the scores meant. In it he wrote about two boys he had tested. He claimed that their level of intelligence was "very, very common" among Spanish-Indian and Mexican families of the Southwest and among Black people. He wrote that their low scores seemed "to be racial," meaning that he believed the cause was born into these families.

Then he wrote what he thought should be done, using the word "segregated," which means kept apart from other people. "Children of this group should be segregated in special classes and be given instruction which is concrete and practical," he wrote. "They cannot master abstractions (ideas), but they can often be made efficient workers, able to look out for themselves."

He went on: "There is no possibility at present of convincing society that they should not be allowed to reproduce." He added that "from a eugenic point of view they constitute a grave problem because of their unusually prolific breeding."

By "prolific breeding" Terman meant having many children, and eugenics was the idea that a country could improve its people by stopping those judged unfit from having children. The test makers gave names to ranges of low scores: "feeble-minded," "moron" and "imbecile." School officials used the labels to put children into separate classes.

The Army tests. Army officers gave tests like these to soldiers during the First World War. The psychologist Robert Yerkes gathered a committee at the Vineland Training School in New Jersey in May 1917, with Terman on it. They built tests that could be given to a whole room of soldiers at once. The Army Alpha was for men who could read English, and the Army Beta used pictures, for men who could not. About 1.75 million soldiers took them in 1917 and 1918.

The SAT. The first college entrance test of this kind came from the Army test. Officials of the College Entrance Examination Board gave the first Scholastic Aptitude Test, the SAT, on June 23, 1926, to about 8,000 students. Carl Brigham of Princeton University led the committee that built it. Brigham had already made a test for Princeton out of the Army Alpha, and five of the nine main sections of the first two SATs were copied word for word from that Princeton test.

Brigham had written a book in 1923 that used the Army scores to rank races by intelligence. By 1926 he had come to accept that a person's surroundings affect test scores, and later he took back the claims in his 1923 book. The first SAT was given in June, after most colleges had already chosen their students, so in most cases college officials made that year's choices without looking at the scores.

BY 1926: School officials gave children written intelligence tests and sorted them into classes by the scores, and the first SAT had been given.

Kept out of school: disabled children

No law before 1950 gave a disabled child the right to a public school classroom, and school board members could send a disabled child away. Merritt Beattie of Antigo, Wisconsin, had cerebral palsy, a condition that affects how a person controls the muscles. He could not control his drooling or the movements of his face, and his voice was rough.

He went through first grade to fifth grade in the Antigo public schools from 1914 to 1917 and kept up with the work. When he was thirteen, the members of the Antigo school board voted to put him out of school. The board members said that "his physical condition and ailment produces a depressing and nauseating effect upon the teachers and school children."

They offered him a place in a class for deaf children and children with speech problems. He and his parents refused it and went to court to get him back into his own school. On April 29, 1919, the justices of the Wisconsin Supreme Court ruled for the school board.

Only 161,000 children in the whole country were in special education, classes or programs set up for disabled children, in 1931 to 1932. That was 0.6 percent of public school pupils. In 1947 to 1948 there were 356,000, or 1.5 percent. For the whole of this era, about 1 child in 100 was in any special program. The other disabled children stayed at home, lived in institutions, or sat in ordinary classrooms with no extra help.

A teacher convicted for teaching in German

During and just after the First World War, lawmakers in Nebraska and several other states made it a crime to teach children in any language but English. The lawmakers aimed these laws at communities that spoke German.

Robert Meyer taught at Zion Parochial School, a school run by a church, in Hamilton County, Nebraska. On May 25, 1920, he taught a ten-year-old boy, Raymond Parpart, to read in German. Officials charged him with a crime, and a court found him guilty.

The state's law forbade teaching "any subject to any person in any language than the English language." A school could teach another language only to a pupil who had passed the eighth grade. The punishment was a fine of $25 to $100, or up to thirty days in the county jail, for each offense.

Meyer took his case to the Supreme Court of the United States and won. On June 4, 1923, in *Meyer v. Nebraska*, seven of the nine justices threw out his conviction. The other two, Justice Holmes and Justice Sutherland, would have let it stand.

Oregon voters try to close the private schools

A majority of Oregon's voters approved a law in 1922 to close every private school and church school in the state, and in 1925 the justices of the Supreme Court struck it down. Members of the Ku Klux Klan proposed the measure. After the Civil War the Klan's members had whipped and murdered freed Black people and the people who helped them, and they had burned Black schools and churches. Klan members put the measure on the ballot through another group, the Scottish Rite Masons.

The measure required every child aged 8 to 16 to attend the public school in the child's own district, and it made it a crime for a parent not to send them. Voters passed it on November 7, 1922, by 11,821 votes out of 219,191 cast. The count usually given is 115,506 for it and 103,685 against. It was to take effect in 1926.

Two schools went to court. One was the Society of the Sisters of the Holy Names of Jesus and Mary, a group of Catholic nuns who ran schools. The other was Hill Military Academy, which was not a religious school.

The nuns and the academy won. On June 1, 1925, in *Pierce v. Society of Sisters*, the justices ruled that the Oregon law broke the Constitution. They wrote: "The child is not the mere creature of the state." The people who raise a child, they wrote, "have the right, coupled with the high duty, to recognize and prepare him for additional obligations."

What the textbooks said

Members of the United Daughters of the Confederacy worked to control what school history books said about the Civil War. The group was founded in Nashville, Tennessee, in 1894, and its members were Southern women, all descended from people who had served the Confederacy or helped it.

Its members created an officer called the historian-general in 1908, partly to check school history books for anything the group called "unjust to the South." Mildred Lewis Rutherford held that job from 1911 to 1916, and in 1919 she published *A Measuring Rod to Test Text Books*, a checklist for judging them.

What they wanted the books to say. The group's members wanted the books to say that enslaved people had been faithful to their enslavers and that states' rights, not slavery, caused the war. A Texas chapter of the group wrote a list of questions and answers about the war in 1904, and a Virginia chapter printed it again. One question asked how enslaved people felt about the people who owned them. The answer given was, "They were faithful and devoted and were always ready and willing to serve them."

Mississippi's own leaders had said otherwise when the war began. On January 9, 1861, they voted to leave the United States and gave their reasons in writing: "Our position is thoroughly identified with the institution of slavery." Slavery, they were saying, was the center of everything they were defending.

By the end of the 1930s, officials in many states bought textbooks for every child. Every pupil in a state then read the same account of the Civil War, because all the books came from the state's approved list. Some of the science books stated that races could be ranked.

The Scopes trial. John Scopes, a Tennessee science teacher, was found guilty in 1925 and fined $100 for teaching evolution, the scientific account of how living things change over many generations. He taught from *A Civic Biology*, a high school book by George William Hunter, first printed in 1914 and approved by the state. The trial was about the book's pages on evolution, and no one at the trial was charged over its pages on race and on people it called feeble-minded.

The book listed five races and called "the civilized white inhabitants of Europe and America" "the highest type of all." About families it called feeble-minded and criminal, it states: "If such people were lower animals, we would probably kill them off to prevent them from spreading." It states that such people should be kept apart in asylums, places where people were locked away, and kept from marrying.

What the labels were used for: Carrie Buck

In the same years that school officials sorted children with labels like "feeble-minded," state officials used the same labels to choose people to sterilize. To sterilize a person is to operate on the body so that the person can never have a child. The justices of the Supreme Court said in 1927 that the states could do it.

Carrie Buck was born in Charlottesville, Virginia, on July 2, 1906. Her school records show that she made normal progress every year. Before she finished sixth grade, the foster family she lived with took her out of school to do their housework.

At seventeen she became pregnant. The National Archives states that the pregnancy was later reported to have been the result of rape, sex forced on her, allegedly by a relative of her foster parents. Her foster family then had her committed, which means sent by a court to live in an institution. A court hearing on January 23, 1924, declared her "epileptic and feebleminded," epileptic meaning that she had epilepsy, a disease of the brain that causes seizures.

Virginia's lawmakers passed a law on March 20, 1924, that let state officials sterilize people they found unfit "because of alcoholism, epilepsy, feeblemindedness, insanity, or other factors." Eight days later, on March 28, Carrie Buck's daughter, Vivian, was born. On June 4, 1924, under the court's order, Carrie Buck was locked up in the Virginia State Colony for Epileptics and Feeble-Minded.

The test case. Dr. Albert S. Priddy, who ran the Colony, picked her to be the first case under the new law, so that the courts would decide whether the law was allowed. Priddy died in 1925, and Dr. John H. Bell took over the Colony

and the case. On May 2, 1927, in *Buck v. Bell*, eight of the nine justices of the Supreme Court ruled for Virginia. Justice Butler voted against the ruling and wrote nothing.

Justice Oliver Wendell Holmes Jr. wrote for the others. He began, "It is better for all the world, if instead of waiting to execute degenerate offspring for crime, or to let them starve for their imbecility," and ended the sentence, "society can prevent those who are manifestly unfit from continuing their kind."

In plain words, he wrote that the state did not have to wait until the children of people it judged unfit committed crimes and were executed, or starved. It could stop such people from having children at all.

He went on: "The principle that sustains compulsory vaccination is broad enough to cover cutting the Fallopian tubes." Compulsory vaccination, a law that made people get a shot against a disease, was his comparison. If a state could force people to take a shot, he was saying, it could also force an operation that stopped a woman from ever having children.

Then he wrote, "Three generations of imbeciles are enough." He counted Carrie Buck as the second of the three generations and her daughter, Vivian, as the third.

The operation. Dr. John H. Bell, who ran the Colony and whose name is on the case, operated on her on October 19, 1927. The operation is called a salpingectomy. He cut a piece out of each of her two Fallopian tubes, the tubes that carry an egg from the ovary to the womb. After that, no egg could reach her womb, and she could never become pregnant again. The operation cannot be undone. She was twenty-one.

In April 1931 Vivian, the child Holmes had counted among the imbeciles, was on the honor roll at Venable Elementary School in Charlottesville. She died of an infection of the intestines in July 1932, at eight. Officials in Virginia sterilized about 8,300 people between 1927 and 1972. The law stayed in force until April 1974. The justices of the Supreme Court have never overruled *Buck v. Bell*.

BY 1927: The justices of the Supreme Court had let Virginia sterilize Carrie Buck, thrown out Robert Meyer's conviction for teaching in German, and struck down Oregon's law against private schools. In Tennessee a teacher had been convicted for teaching evolution.

Separate schools, and what separate meant

In the South, county officials spent far less on the schools for Black children than on the schools for white children, and the amounts are on record. In Lowndes County, Alabama, in 1909, white schools received $20 for each pupil and Black schools 67 cents. In Edgecombe County, North Carolina, county officials paid for janitors, electricity, water and transportation at the white schools and paid for none of them at the Black schools.

The 1917 survey. A federal survey published in 1917 measured the whole South. It divided what the Southern states paid their teachers by the number of children aged 6 to 14, which came to $10.32 for each white child and $2.89 for each Black child. The report identifies who made the split. State officials sent each county school money counted on every child, Black and white. "The county officers then divide these funds according to their interpretation of the needs of the white and black pupils."

The gap was widest where most people were Black. In counties where at least three people in four were Black, officers spent $22.22 on each white child and $1.78 on each Black child, more than twelve times as much. In Alabama, Florida, Louisiana, North Carolina and South Carolina, the Black public schools were open less than five months a year.

Rosenwald schools. Black families built schools themselves. Julius Rosenwald, a part-owner of the Sears, Roebuck company, met Booker T. Washington in 1912 and began paying part of the cost of school buildings for Black children. Between 1913 and 1932, more than 5,000 schoolhouses, teachers' homes and workshops went up in fifteen states, and more than 600,000 students went to these Rosenwald schools.

Rosenwald's money never paid the whole cost. Black families had to raise a matching share. They held fish fries and bake sales, and they gave building materials and their own work. The schools had no electricity, so the builders set large windows close together to let in daylight.

Martha Lum. Each state's officials decided which children had to go to the separate schools. Martha Lum was nine years old when she went to the first day of school at the Rosedale Consolidated High School in Bolivar County, Mississippi. At the noon recess the superintendent told her she could not come back.

The school's trustees, the board in charge of it, had ordered her out because she was Chinese, on instructions from the state superintendent of education. Under Mississippi's constitution, the white schools were for white children. Her father, Gong Lum, paid taxes that helped support the school, and he sued. No justice of the Supreme Court took his side. On November 21, 1927, in *Gong Lum v. Rice*, all of them ruled for Mississippi.

Lemon Grove and Bastrop. School officials sent Mexican American children to separate schools too. In Lemon Grove, California, on January 5, 1931, the principal, Jerome T. Green, stood in the school doorway. He let in every pupil except the Mexican American ones and told them to go to a separate two-room building. Their parents kept the children home and sued. On March 30, 1931, Judge Claude Chambers of the San Diego County court ordered the children let back in. Some accounts give March 31.

In Texas, Minerva Delgado and twenty other Mexican American parents sued the Bastrop school district and three others on June 15, 1948. Their lawyer was Gustavo C. Garcia. Judge Ben H. Rice ordered the separation ended by September 1949. His ruling still allowed a school to put first graders in a separate class if a test found that they spoke little English.

By the end of the era, more children of every group were in school. In 1900, 54 of every 100 white children aged 5 to 19 were enrolled, and 31 of every 100 children the government counted as Black and other races. By 1950 the

numbers were 79 and 75. The gap in who went to school had nearly closed, but in the South county officials still kept Black and white children in separate schools and still spent less on the Black ones.

The boarding schools at their largest

Tens of thousands of Native children lived at federal boarding schools in the 1920s. A report written for the Secretary of the Interior in 1928 describes what the children ate, where they slept and how they worked.

The numbers. Of 69,892 Native children enrolled in any school in 1926, 27,361 were in boarding schools, about two in five. Of the children in government schools, more than 4 in 5 were boarders. There were 19 boarding schools off the reservations and 59 on them, the reservations being the lands the United States government set aside for Native nations.

Officials could force parents to send their children. Under a law of February 14, 1920, the Secretary of the Interior could "make and enforce" rules to secure the "regular attendance" of Native children at federal schools or public schools. Under a law of 1929, state officers could go onto reservations to make Native children attend school under state laws.

The Meriam Report. Secretary of the Interior Hubert Work asked for the survey. Lewis Meriam led the team that wrote it, and he handed it to Work on February 21, 1928. "The survey staff finds itself obliged to say frankly and unequivocally (openly and with no doubt)," the report states, "that the provisions for the care of the Indian children in boarding schools are grossly inadequate (far short of enough)."

Food and sickness. Officials were trying to feed each child on eleven cents a day, plus what the school farm grew, the report states. It describes the food as "deficient in quantity, quality, and variety," meaning there was not enough of it, it was not good, and it was the same few things. The Indian Service's own rule was a quart of milk a day for each child, and staff at very few schools met it.

The report names the main diseases among Native people as tuberculosis, a lung disease that spreads by coughing and had no cure in the 1920s, and trachoma, an eye infection. Repeated trachoma infection scars the inside of the eyelid until the lashes turn inward and scrape the eye, and without treatment it blinds people. At the school at Fort Defiance, Arizona, which was set up for children with trachoma, "milk is not part of the normal diet."

The dormitories, the buildings where the children slept, held far more children than they were built for. The report states that they were "crowded materially beyond their capacities," and "the supply of soap and towels has been inadequate."

The children's work. The children's work paid for part of the cost of running the schools. "The boarding schools are frankly supported in part by the labor of the students," the report states. Pupils above the fourth grade worked half the day and went to class the other half. Its authors asked whether much of that work, "notably the work in the machine laundries," would be against the child labor laws of many states.

The authors also judged the idea behind the schools. Taking children, "even very young children, as completely as possible away from their home and family life," went against "modern views of education and social work," they wrote. By those views, children should not be pulled away from home and family. The report states: "Indian parents nearly everywhere ask to have their children during the early years, and they are right."

Uintah. A committee of senators led by Senator Lynn J. Frazier of North Dakota heard what those conditions meant at one school in November 1928, in Salt Lake City. It was the Uintah Boarding School at Whiterocks, Utah, on the Ute reservation.

In late October 1927, an eleven-year-old pupil, Tilford Denver, was climbing the rope of the school swing. The swing had been broken for a long time and had never been fastened down. The frame fell, a bar struck his head, and he died within the hour.

The agency doctor, George McClellan Hamilton, was asked whether officials were to blame. He answered, "Some of the officials of the reservation," and went on, "yes, sir ... Because this swing never had been secured." The record does not name the official in charge of the swing. Tilford's mother, Dulcia Denver, was never paid anything for his death, the doctor said.

Swanson Mowachean was an orphan who had come to the school at about ten. He touched a bare electric wire at the school, fell and hurt his head. His relatives wrote to the Commissioner of Indian Affairs, Charles Burke, on February 17, 1928. They wrote that the superintendent, H. M. Tidwell, "did not think this boy needed to go to the hospital," and "We fear that if we wait for the superintendent to act it will be too late and the boy will die."

Tidwell let him go back to the hospital later that month. The doctor testified that it was too late. Swanson died on March 6, 1928. The doctor also told the senators, "we have had Indians die there for a lack of attention and lack of proper food." It had happened, he said, "more than once."

The whippings. Fred Bruce, a Chippewa man hired in January 1928 to keep order among the boys, testified that the principal, George N. Shafer, wanted him to whip them. He quoted Shafer: "some of these Indians you have got to pound it into them." Shafer went on, "that is the only way they will learn anything." Bruce said he refused. About six months later he lost his job, and the record does not say who fired him.

A parent, Eugene Perank, told the senators through an interpreter what children had told him: "they have whipped them with a stick." He went on, "throw them down and whip them." He took his own children out of the school. The children who were whipped at Uintah are not named in the record.

John Collier, who led the American Indian Defense Association, charged "widespread brutality" at government boarding schools in May 1930. He persuaded senators to look into "flogging and other forms of excessive punishment," flogging meaning beating a person hard, again and again. His charges centered on the Phoenix Indian School in Arizona, whose leaders were accused of "whippings, beatings, and even death."

Officials read the Meriam Report in 1928, and the boarding schools stayed open. The federal count of the children who died at the schools, finished in 2024 and described under 2000 to Today, covers the whole life of the system and is not broken down by decade.

Teaching: colleges for teachers, and the marriage bar

In the first half of the century, many normal schools became teachers colleges. At a normal school, future teachers trained for one or two years after high school. At a teachers college, they studied for a four-year college degree. Between 1911 and 1930, 88 normal schools became teachers colleges, and by the 1950s the last of the normal schools were gone.

The marriage bar. Most teachers were women, and the members of most school boards made a woman choose between teaching and marriage, a rule called the marriage bar. The National Education Association, a teachers' organization, began asking school boards about it in 1928. That year 61 percent of the boards said they would not hire a married woman as a teacher, and 52 percent would not keep a teacher who married while she worked for them. By 1942 the figures had risen to 87 percent and 70 percent.

During the Second World War the number of men teaching in public schools fell from 195,000 in 1939 to 1940 to 127,000 in 1943 to 1944. Board members who had refused married women for years now needed them, and most boards dropped the rule. By 1951, 18 percent of boards still refused to hire a married woman, and 10 percent refused to keep one.

Teachers' pay rose over the fifty years. On average, a teacher earned $325 a year in 1899 to 1900, $1,420 in 1929 to 1930 and $3,010 in 1949 to 1950. Prices rose too, so what that pay could buy about tripled. The number of pupils for each teacher fell from about 37 to about 28.

A school explosion: New London, Texas, 1937

Members of the school board in New London, Texas, canceled the school's contract for natural gas early in 1937. To save money, they had plumbers connect the school to a gasoline company's pipe of leftover gas, which had no smell. On March 18, 1937, the gas exploded. About 540 people were in the building, 500 students and 40 teachers, and about 298 of them died.

No school official was found responsible, and more than seventy lawsuits failed. Texas lawmakers then passed a law requiring companies to add a smell to gas. Chapter 27, Disasters, describes the explosion in full.

School behind a fence: the Japanese American camps

During the Second World War, federal officials held Japanese American children in camps, and more than 30,000 of them went to school there. President Franklin D. Roosevelt signed Executive Order 9066, an order from the president, on February 19, 1942. Under it, Lieutenant General John L. DeWitt, the Army commander on the West Coast, ordered Japanese Americans there out of their homes.

More than 120,000 people were forced to leave under his orders, and two out of every three were American citizens. None of them was charged with a crime or tried. Officials of the War Relocation Authority, the federal agency that ran the camps, ran the camp schools as well.

At Manzanar, in California, more than 2,300 children arrived from over 200 schools, and no one had planned how to teach them. The first classes met in barracks and recreation halls with no furniture, no textbooks and no supplies.

Poston. Poston, in Arizona, held more than 18,000 people in three camps. It was built on the Colorado River Indian Reservation, and the Mohave were the largest of the nations who had lived on that land. John Collier, who was then the Commissioner of Indian Affairs, suggested the site. The reservation's tribal council opposed the camp, and officials built it anyway.

At Poston the people held there made about 750,000 adobe bricks, bricks of mud and straw dried in the sun, and built three school campuses. Elementary classes averaged 48 pupils to one teacher. Of the 100 teachers hired to come to Poston, 70 arrived, and by December 55 were still there.

Four Munemitsu children, Seiko, Saylo, Akiko and Kazuko, had gone to school in Westminster, California, until 1942. Then they were held at Poston. While they were there, the Mendez family leased and worked the Munemitsu family's farm in Westminster, and staff at a Westminster school turned the Mendez children away.

BY 1945: The Indian boarding schools were still open years after the Meriam Report, and Southern county officials still spent far less on each Black child than on each white child. Most school boards had dropped their rule against married women teachers.

Sylvia Mendez

Who: Turned away from a California school at eight, she was one of about 5,000 children in her father's lawsuit, and the families won.
When and where: Westminster, California, 1944 to 1947.

The staff enrolled her cousins and turned her away.

Sylvia Mendez was eight years old when staff at a school in Westminster, California, refused to enroll her, and her father went to court over it. Her family had moved from Santa Ana to a farm in Westminster in 1944. They leased it from the Munemitsus, a Japanese American family whom federal officials had forced into a camp.

Her aunt Soledad took Sylvia, her brothers Gonzalo Jr. and Jerome, and her own children, the Vidaurri cousins, to enroll at the Seventeenth Street School. The staff accepted the Vidaurri cousins, who had lighter skin and a last name that did not sound Spanish. They turned away the Mendez children. Her aunt answered: "My kids, they will not go to your school, if those of my brother cannot go, mine will not go!"

The staff sent the Mendez children to Hoover School, which people called the Mexican school. Teachers there taught the children embroidery, cooking or farm work instead of math and science.

The lawsuit. Her father, Gonzalo Mendez, and four other fathers, William Guzman, Frank Palomino, Thomas Estrada and Lorenzo Ramirez, went to federal court. Their lawyer, David Marcus, sued four Orange County school districts, Westminster, Santa Ana, Garden Grove and El Modena, for about 5,000 children.

After a six-day trial in Los Angeles, Judge Paul J. McCormick ruled for the families on February 18, 1946. On April 14, 1947, the judges of the Ninth Circuit Court of Appeals, a higher federal court, agreed with him.

California's governor, Earl Warren, signed a law in 1947 that made California the first state to ban separate public schools of every kind. In 1953 Warren became Chief Justice of the United States, and in 1954 he wrote the Supreme Court's ruling in *Brown v. Board of Education*.

Sylvia Mendez spoke about the case in public for decades. On February 15, 2011, she received the Presidential Medal of Freedom.

The G.I. Bill sends veterans to college

Members of Congress voted in 1944 to pay for veterans of the Second World War, the men and women who had served in the armed forces, to go to college or job training. President Roosevelt signed the Servicemen's Readjustment Act, called the G.I. Bill, on June 22, 1944.

Veterans made up 49 percent of the students admitted to colleges in 1947, the law's busiest year. By the time the first G.I. Bill ended on July 25, 1956, 7.8 million of the 16 million veterans of the war, about half, had used it for education or training.

Black veterans. The law did not mention race, but Black veterans in the South could not use it the same way. Local officials handled the benefits. A Black veteran in the South with money for tuition, the fee a college charges for classes, still had to find a college that would admit him.

The colleges open to him were separate Black colleges with a limited number of places. Many Black veterans had gone to the Southern schools for Black children, where officials spent far less on each child, and many had not been taught enough there to be ready for college. Many could not get into a college.

The number of Black college students still rose sharply. Medgar Evers was one of the Black veterans who got in. He had volunteered for the Army in 1942, and later he used the G.I. Bill to enroll at Alcorn A&M College in Lorman, Mississippi, a public college for Black students. In 1962 he helped James Meredith become the first Black student at the University of Mississippi.

BY 1950

If you remember one thing: Finishing high school became ordinary in these fifty years, while school officials sorted children by test scores, by race and by disability.

The number: 50.8 of every 100 seventeen-year-olds graduated from high school in 1939 to 1940, against 6.4 in 1899 to 1900, nearly eight times as many.

Next: After 1950 the justices of the Supreme Court ruled that separate schools broke the Constitution, and officials in the South spent years fighting the ruling.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 51 MINUTES

Judges and lawmakers opened the public schools to children who had been kept out or kept apart, and officials in many places spent years fighting them. The justices of the Supreme Court ruled in 1954 that separate public schools for Black and white children broke the Constitution. For the next ten years almost every Black child in the South still went to an all-Black school.

Officials in Arkansas and Virginia closed public schools rather than let Black children in, and two men were killed in a riot on the night federal marshals brought the first Black student onto the campus of the University of Mississippi.

Members of Congress began to pay for schooling on a large scale, for science in 1958 and for schools with poor children in 1965. Judges ruled on prayer in school, on disabled children shut out of school, on children who could not understand English, and on paddling. By 2000 college had become ordinary, and most people who finished a degree had borrowed money to pay for it.

Brown, and how slowly the classrooms changed

All nine justices of the Supreme Court ruled on May 17, 1954, in *Brown v. Board of Education of Topeka*, that separate public schools for Black and white children broke the Constitution. Chief Justice Earl Warren wrote the ruling, seven years after he had signed California's law against separate schools as its governor.

With all deliberate speed. The justices ordered the states on May 31, 1955, to desegregate, to end the separation of Black and white pupils, "with all deliberate speed." The justices set no date and asked only for "a prompt and reasonable start."

Ten years after *Brown*, one study estimated that 99 of every 100 Black children in the South were in schools with no white children. A team led by James Coleman made a national count in the school year 1965 to 1966, as the Civil Rights Act of 1964 required. Across the country, 65 of every 100 Black pupils were in schools that were more than 90 percent Black, and 80 of every 100 white pupils were in schools more than 90 percent white.

The city figures for elementary schools in the same survey record the South. In Mobile, Alabama, every Black child was in a school more than 90 percent Black, and in Charlotte, North Carolina, 96 of every 100 were.

The classrooms of the South changed later. By the mid-1980s, about 43 of every 100 Black students in the South went to schools where most of the students were white. That was the highest share in a count by researchers at the University of California, Los Angeles, that reaches back to 1968.

Little Rock: a year inside Central High, and a year with no high school

Nine Black students entered Central High School in Little Rock, Arkansas, in September 1957, only after President Dwight Eisenhower sent Army soldiers to take them in. The next year Governor Orval Faubus closed the city's high schools. The nine were Elizabeth Eckford, Minnijean Brown, Ernest Green, Thelma Mothershed, Gloria Ray, Jefferson Thomas, Carlotta Walls, Terrence Roberts and Melba Pattillo.

The Guard. Faubus said on television on September 2, 1957, that he had called out the Arkansas National Guard to "preserve order" at Central High. On September 4 the nine tried to go in. Elizabeth Eckford came alone, because she did not know that the others were being taken together.

Guard soldiers refused to let her in, and she walked to a bus stop through a crowd of more than 300 people. The other eight came with ministers that Daisy Bates had arranged, and Lieutenant Colonel Marion Johnson turned them away on Faubus's orders.

A federal judge, Ronald Davies, ruled on September 20 that Faubus had used the soldiers to stop integration, the mixing of Black and white students in one school, and not to prevent violence. Faubus pulled the Guard out that evening.

The mob. More than a thousand white people surrounded the school on September 23 while city police took the nine inside. People in the crowd attacked four Black reporters: Alex Wilson, James Hicks, Moses J. Newsom and Earl Davy. After three and a half hours, school staff took the nine out through a side door. The next day Mayor Woodrow Mann telegraphed the president that the "mob is armed" and the "situation is out of control."

Eisenhower sent 1,000 soldiers of the 101st Airborne Division. At 9:22 on the morning of September 25, the nine walked in through the front doors. More than twenty soldiers walked around them, more than 350 stood around the building, and an Army helicopter flew overhead.

Inside. White students kicked the nine, shoved them and called them names, again and again. Army officers then assigned soldiers to walk them to their classes. The soldiers could not go with them into the restrooms and locker

rooms, and the attacks went on there. On October 3 the vice principal, Elizabeth Huckaby, stopped white students who were harassing them, and about 150 white students walked out. She took 70 names, and all 70 were suspended.

Minnijean Brown dropped chili on two white boys who blocked her way in the cafeteria on December 17, and she was suspended until January 13. Two days after she came back, white students poured chili on her, and they were expelled. The National Park Service's timeline of that year does not say which school officials gave those punishments.

The school board members expelled Minnijean Brown for the rest of the year on February 17, 1958, after she called a white girl who hit her "white trash." They also suspended three white students, among them boys who wore badges that read "One Down and Eight to Go." She finished high school in New York. On May 27, 1958, Ernest Green became the first Black student to graduate from Central High.

The Lost Year. Judge Harry Lemley ruled in June 1958 that Little Rock could put off integration until January 1961, and in September the justices of the Supreme Court overturned his ruling. Faubus closed all four of Little Rock's public high schools on September 15, 1958, under laws that Arkansas lawmakers had passed for him. On September 27 the city's voters chose 19,470 to 7,561 against integration, and the high schools stayed shut for the whole school year. People in Little Rock call it the Lost Year.

White students went to new private schools, to schools in other towns and other states, or took courses by mail from the University of Arkansas. Faubus helped a private school company buy buildings with public money. Most Black students went to schools in other districts or other states. Some took a course by mail run by L. M. Christophe, the principal of the Black high school. State officials kept paying the teachers to sit in empty classrooms, and Central High's football team kept playing all year.

The segregationist members of the school board, people who wanted the races kept apart, voted in May 1959 not to renew the contracts of 44 teachers and school officials. On May 25, 1959, a group of women, the Women's Emergency

Committee to Open Our Schools, won a vote that removed three of those board members. The high schools opened again on August 12, 1959.

Science and languages: the 1958 law

Soviet scientists and engineers launched Sputnik, the first satellite to circle the Earth, on October 4, 1957. The launch shocked experts and ordinary people in the United States, who had hoped the United States would do it first. President Eisenhower signed the National Defense Education Act on September 2, 1958. Under it, federal officials paid for the teaching of science, mathematics and foreign languages, and for loans to college students.

Eisenhower called it "an emergency undertaking to be terminated after four years." Members of Congress did not end it after four years. Its student loans went on under that name until 1972 and then continued under other names.

Virginia closes its schools

Officials in Virginia made the same choice as Faubus. They closed public schools in five places rather than let Black children into white schools, and in Prince Edward County they kept the public schools shut for five years. In late February 1956, Senator Harry F. Byrd called for "massive resistance" against *Brown*, a campaign by Virginia's leaders to block it. Virginia's lawmakers, under Governor Thomas B. Stanley, then passed a law under which the governor would close any school that federal judges ordered to take Black pupils.

Governor J. Lindsay Almond Jr. used that law in September 1958. He closed nine schools in Front Royal, Charlottesville and Norfolk and locked out about 12,000 to 13,000 students. On January 19, 1959, judges of a state court and of a federal court both ruled that the closings broke the Constitution. The state's own scholarship program names five places that closed schools to avoid desegregation: Arlington, Charlottesville, Norfolk, Warren County and Prince Edward County. All but Prince Edward reopened within months.

Prince Edward County. Barbara Johns, who was 16, had led the Black students of Robert Russa Moton High School in Farmville out of their school on April 23, 1951, to protest its condition. Their case became one of the five

cases the justices decided together as *Brown*.

Local leaders spent years getting ready to keep the county's schools separate. From 1954, J. Barrye Wall, the editor of the *Farmville Herald*, and Robert Crawford, a laundry owner, built a group called the Defenders of State Sovereignty and Individual Liberties.

Hours after the second *Brown* ruling, on May 31, 1955, the county supervisors, the officials who ran the county, voted the schools only the legal minimum of $150,000, a fraction of what they needed. A week later about 1,250 people at a public meeting voted for a private fund to pay white teachers if the public schools closed.

Judges of two courts ordered the county to integrate its schools on May 1, 1959, and the supervisors then voted to stop paying for public schools at all. Accounts of that vote differ: two date it June 26, 1959, and a third gives June 2. The schools did not open that September. Roy Pearson of Farmville organized private schools for the white children, who went to Prince Edward Academy, a new private school, paid for partly by tuition grants from the state.

Five years without school. Between 1,700 and 2,000 Black children had no school. One account gives about 1,750, and another gives nearly 2,000. Some children went to live with relatives in other counties or states, and some were taught in church basements.

For one year, starting on September 16, 1963, private donors, backed by President John F. Kennedy's government, paid for free classes for about 1,500 children. On May 25, 1964, in *Griffin v. County School Board of Prince Edward County*, the justices of the Supreme Court ordered the county to reopen its schools. About 1,500 children went back on September 8, 1964.

Virginia's lawmakers voted in 2003 for a statement of "profound regret" for the closings. A news report in *Education Week* stated that some lawmakers would not vote for an apology because they thought an apology could lead to demands for payment. In 2004 the lawmakers set up a scholarship fund for people who had lost those years of school and put in $50,000. John Kluge, a rich donor, then gave $1 million on the condition that the lawmakers match it. They did, and the fund reached $2,050,000.

BY 1964: The justices of the Supreme Court had ordered Prince Edward County to reopen its schools, and Little Rock's high schools had been open again since 1959. Ten years after *Brown*, almost every Black child in the South still went to a school with no white children.

Ruby Bridges

Who: At six she walked into an all-white school in New Orleans, and for months she was the only pupil in it.

When and where: New Orleans, Louisiana, 1960 and 1961.

Every teacher but one refused to teach her.

Four federal marshals, officers who carry out the orders of federal courts, walked six-year-old Ruby Bridges into William Frantz Elementary School in New Orleans on November 14, 1960. For the rest of that school year, one teacher taught her in a classroom by herself. She was born on September 8, 1954, in Tylertown, Mississippi, the eldest child in her family. She grew up in New Orleans and went to a segregated kindergarten in 1959.

The test. A federal judge, J. Skelly Wright, ordered New Orleans in 1960 to desegregate its public schools one grade at a time, starting with first grade. Members of the Orleans Parish School Board made Black children take an entrance test and checked their families, and white children moving the other way had to do neither. Six children passed the test, and two of them chose to stay at their own school.

Ruby Bridges walked in with the marshals and her mother, Lucille. That same morning Leona Tate, Gail Etienne and Tessie Prevost, also six, went into McDonogh No. 19 school nearby with marshals of their own. The four girls were the first Black children in white public schools in Louisiana.

Louisiana's leaders fought the order. That summer the lawmakers had made it a crime, punished by fines and jail, to challenge the state's control over the schools. Under one of their laws, Governor Jimmie Davis took

over every decision about desegregation in New Orleans. Leaders of the White Citizens' Council, a group of white people who wanted the schools kept separate, held a rally of 5,000 people and told white parents to defy the court. On November 16 a riot broke out in front of the school board building.

A crowd stood outside Frantz and screamed at her. A group of young white women who called themselves the Cheerleaders came back day after day. They blocked the children and their parents, spat at the children, shouted racial insults and slashed tires. One protester threatened to poison her lunch. No record names the woman who made the threat.

White parents took their children out. By the end of November, fewer than ten white children attended the two schools together. From January to May 1961, Ruby Bridges was the only pupil at Frantz. Every teacher but one refused to teach her. Barbara Henry, a white teacher from Boston, taught her alone for the year. Ruby Bridges did not miss a day.

What it cost her family. Her father lost his job. Grocers refused to serve her mother. Her grandparents had sharecropped a farm for 25 years, farming land someone else owned and paying the owner with part of the crop, and they were thrown off it. The National Park Service's account does not name the owner who made them leave. Her parents later separated.

Robert Coles, a child psychiatrist, a doctor who treats problems of the mind, met with her through that year and had her draw and talk about what was happening. Norman Rockwell painted her walk to school in 1964 and called it *The Problem We All Live With*. On July 15, 2011, she saw the painting hanging in the White House.

Five years after she walked into Frantz, 96 of every 100 Black elementary pupils in New Orleans were still in schools more than 90 percent Black. She finished elementary school at Frantz and worked in travel for about

> fifteen years. In 1999 she started the Ruby Bridges Foundation. On November 14, 2015, fifty-five years to the day after she first went in, a bronze statue of her was unveiled at the school.

A university, 1962: two men killed

Two men were killed on the night federal marshals brought James Meredith, the first Black student at the University of Mississippi, onto its campus. Meredith was an Air Force veteran. He applied on January 20, 1961, and university officials turned him down after he wrote that he was Black.

Medgar Evers helped him, and Constance Baker Motley and Jack Greenberg, lawyers for the NAACP, a civil rights organization, took his case. On September 10, 1962, Justice Hugo Black ordered the university to admit him. Governor Ross Barnett worked for weeks to stop it.

The riot. About 500 federal marshals brought Meredith to the campus at Oxford on the evening of September 30, 1962, and put him in a dormitory, Baxter Hall. At about 7:25 that evening, State Senator George Yarbrough sent the Mississippi Highway Patrol away from the crowd, and at 7:30 a riot began.

The crowd grew to more than 2,500 people by one count and about 3,000 by another, students and people from off the campus. A former Army general, Edwin Walker, helped lead them. They threw rocks, bottles and firebombs at the marshals and fired guns.

Paul Guihard, 30, a reporter for a French news agency, was shot in the back within about ten minutes of reaching the campus. Ray Gunter, 23, a jukebox repairman from Oxford, was dead when he reached the hospital. Nobody was ever charged with killing Guihard, and no record names who killed Gunter.

The counts of the injured run from at least 75, in the first news report, to more than 300 in a later account, many of them marshals and soldiers. About 3,000 federal soldiers took control of the campus by the next morning, and 108 people were arrested.

Meredith registered as a student on October 1, 1962, and walked to his first class, a lesson in American colonial history. People shouted at him, "Was it worth two deaths?" Marshals went with him everywhere, and some students walked out of his classes. He graduated in August 1963.

Prayer and Bible reading in the school day

The justices of the Supreme Court ruled in 1962 and 1963 that public school staff could not lead pupils in official prayers or Bible readings. In New Hyde Park, New York, the members of the school board of Union Free School District No. 9 had told each class to say a prayer at the start of every day.

Members of the New York State Board of Regents had written it. The whole prayer is 22 words: "Almighty God, we acknowledge our dependence upon Thee, and we beg Thy blessings upon us, our parents, our teachers and our Country."

Engel, 1962. The justices struck it down, six votes to one, on June 25, 1962, in *Engel v. Vitale*. "It is no part of the business of government to compose official prayers for any group of the American people to recite," Justice Hugo Black wrote, "as a part of a religious program carried on by government."

Schempp, 1963. A Pennsylvania law stated that "at least ten verses from the Holy Bible shall be read, without comment, at the opening of each public school on each school day." At the high school in Abington Township, the verses and the Lord's Prayer went out over the school's loudspeakers. In Baltimore a rule from 1905 required a reading from the King James Bible and the Lord's Prayer.

The justices struck down both, eight votes to one, on June 17, 1963, in *School District of Abington Township v. Schempp*. Justice Tom Clark wrote that government must keep "strict neutrality, neither aiding nor opposing religion." It made no difference that a child could ask to be excused.

Yoder, 1972. The justices let Amish parents, members of a Christian church called the Old Order Amish, take their children out of school after the eighth grade.

Wisconsin's law required children to attend school until 16. Jonas Yoder, Wallace Miller and Adin Yutzy kept their children, Frieda Yoder and Barbara Miller, both 15, and Vernon Yutzy, 14, at home, and each father was fined $5. On May 15, 1972, in *Wisconsin v. Yoder*, Chief Justice Warren Burger wrote that the state could not force the parents to send the children to high school.

Federal money for poor children's schools, 1965

Members of Congress began paying in 1965 for schools where poor children studied, and for classes for children too young for school. President Lyndon Johnson signed the Elementary and Secondary Education Act on April 11, 1965, on the front lawn of the old Junction school in Johnson City, Texas, the one-room school he had started at when he was four.

His first teacher, Kate Deadrich Loney, sat beside him. He told her, "I started school when I was 4 years old," and "they tell me, Miss Kate, that I recited my first lessons while sitting on your lap." Members of the House had voted for the law 263 to 153, and senators had voted for it 73 to 18.

Johnson said the law would "bridge the gap between helplessness and hope for more than 5 million educationally deprived children." Under Title I, one part of the law, federal officials send money to schools with poor children. By the end of 1965, the first 100 Title I projects were reaching 347,047 children in 12 states.

Head Start. Head Start began the same year. Sargent Shriver, who ran the federal Office of Economic Opportunity, put together the group that designed it, and Johnson announced it on May 18, 1965. It was a summer program to prepare young children for school in September, and it gave them "medical and dental attention that they badly need."

On that day Johnson spoke of about 530,000 children in 11,000 centers, at a cost of $112 million. By August 31 he reported nearly 560,000 children in 13,400 centers in 2,500 communities.

Title IX. President Richard Nixon signed a new education law with a part called Title IX on June 23, 1972.

Title IX states that "no person in the United States shall, on the basis of sex, be excluded from participation in, be denied the benefits of," and it continues, "or be subjected to discrimination under any education program or activity receiving Federal financial assistance." That means a school or college that gets money from the federal government may not shut anyone out or treat anyone worse because the person is male or female.

BY 1972: Federal money paid for science teaching, student loans, schools with poor children and Head Start, and a new law barred schools that took federal money from treating anyone worse because of sex. Official prayers and Bible readings had been struck out of the public school day.

College for millions, and borrowing to pay for it

Over these fifty years, going to college changed from something a few Americans did to something millions did, and by the end most graduates had borrowed money to pay for it.

In the fall of 1950, about 2.3 million people were enrolled in American colleges and universities. By 1970 there were about 8.6 million, and by 2000 about 15.3 million, about seven times as many as in 1950.

The 1965 law. Federal money paid for much of the growth. Federal loans for students began under the 1958 defense education law. President Johnson signed the Higher Education Act on November 8, 1965, at Southwest Texas State College in San Marcos, Texas.

Under the law, students in need could get grants, money that does not have to be paid back, of up to $1,000 a year. They could get part-time jobs on campus worth about $400 a year, and loans backed by the government that a student did not have to start paying back until after graduating. Johnson said that 140,000 young people would be in college the next year who otherwise would not.

In the same speech he spoke of the year he had taught at the Welhausen Mexican School in Cotulla, Texas, for $125 a month. "I shall never forget the faces of the boys and the girls in that little Welhausen Mexican School," he

said. "And I remember even yet the pain of realizing and knowing then," he went on, "that college was closed to practically every one of those children because they were too poor."

The cost. College cost rose faster than other prices. Measured in dollars of the same value, a year at a public four-year college, with tuition, fees, a room and meals, cost about $7,700 in 1963 to 1964 and about $12,600 in 1999 to 2000. At private four-year colleges the cost rose from about $15,100 to about $31,500, more than double.

More students borrowed. Of the people who finished a four-year degree in 1992 to 1993, 49 in every 100 had borrowed, an average of $12,100. Of those who finished in 1999 to 2000, 65 in every 100 had borrowed, an average of $19,300. Among students from the richest families, the share who borrowed roughly doubled, from 24 to 46 in every 100.

What students could say and wear

Public school students have a right to free speech at school, the justices of the Supreme Court ruled in 1969. Four years later the justices let stand a school rule that punished Native boys for wearing braids.

The armbands. John Tinker, 15, and Christopher Eckhardt, 16, students at high schools in Des Moines, Iowa, and Mary Beth Tinker, 13, in junior high, planned in December 1965 to wear black armbands to protest the Vietnam War.

The Des Moines principals made a rule on December 14, 1965, that any student who wore an armband would be told to take it off and suspended for refusing. Mary Beth and Christopher wore armbands on December 16, and John wore his the next day. All three were suspended until after New Year's Day.

The justices ruled for the students, seven votes to two, on February 24, 1969, in *Tinker v. Des Moines*. Justice Abe Fortas wrote that students do not "shed their constitutional rights to freedom of speech or expression at the schoolhouse gate." He also wrote that the case did not concern rules about "the length of skirts or the type of clothing, to hair style, or deportment (behavior)."

The braids. At Pawnee Junior High School in Oklahoma, Pawnee boys wore their hair parted in the middle with a long braid on each side, to follow "the old traditional ways."

School officials suspended them with no end date on April 24, 1972, under a rule that a boy's hair could not reach his shirt collar or his ears. Federal appeals judges upheld the rule.

The justices of the Supreme Court refused to hear the case, *New Rider v. Board of Education*, on December 10, 1973. Justice William O. Douglas disagreed. He wrote that the boys "were in fact attempting to broadcast a clear and specific message to their fellow students and others, their pride in being Indian."

Black teachers lose their jobs, and the bus

When officials in Southern districts joined Black and white schools, school board members pushed Black teachers out of their jobs. The economist Owen Thompson counted teachers in 781 school districts in the eleven states that had formed the Confederacy. In 1964, 30.6 percent of their teachers were Black. By 1972 the share was 24.2 percent. The number of Black teachers fell by almost a third in eight years.

As districts joined schools together, the school board members let Black teachers go far more often than white ones, and they hired white teachers, many with less experience. Officials in the districts with the fewest Black pupils cut their Black teachers the hardest. Thompson estimates that about half of the teachers who lost their jobs went into work that needed less training, and the rest left the South to teach somewhere else.

The buses. The justices ruled unanimously on April 20, 1971, in *Swann v. Charlotte-Mecklenburg Board of Education*, that a federal judge could order children carried by bus to end separate schools. The Charlotte-Mecklenburg district in North Carolina had more than 84,000 pupils in 107 schools, and about 24,000 of them were Black.

The district's officials already bused children. The year before, its buses carried 26,600 pupils an average of 15 miles each way, a trip of more than an hour. Under the plan, elementary pupils would ride about seven miles, "not over 35 minutes at the most." The justices wrote that an objection to busing counted only when the trip was long enough to risk the children's health or to harm their schooling.

Boston. Fourteen families sued Boston's School Committee in 1972. On June 21, 1974, Judge W. Arthur Garrity ruled that "the entire school system of Boston was unconstitutionally segregated." Under his order, buses began carrying Black students from Roxbury to South Boston High School on September 9, 1974. Of the 800 Black students assigned there, 56 came on the first day. Protesters threw rocks and bottles at the buses.

Phyllis Ellison was one of the Black students bused to South Boston High, and she later described that year. People shouted racial insults at the bus and held up bananas at the students inside. In the school, she said, there were "anywhere between ten and fifteen fights" on an ordinary day. Black and white students sat on opposite sides of the classrooms, the lunchroom and the washrooms.

A white student, Michael Faith, was stabbed during a shoving match outside a classroom on December 11, 1974. Police arrested a Black student from Roxbury, James White. A crowd gathered at the school gates shouting racial abuse, and it threw bricks and bottles and turned over a police car. Boston police and 125 state troopers pushed the crowd back.

The Black students were locked in their classrooms while the white students were let out, Ellison said, and she did not say who gave that order.

Then four empty buses pulled up to the front door as decoys, while the Black students were put on buses at the back and driven home. *Time* magazine reported the decoy buses but did not say who planned them. South Boston High and seven other Boston schools were then closed for a time, and *Time*'s report does not say who closed them. Phyllis Ellison graduated in 1977.

The boarding schools: largest in the 1970s

Enrollment in the boarding schools for Native children reached its highest point about forty-five years after the Meriam Report, and children still died trying to get home from them. By the Heard Museum's count, enrollment peaked at about 60,000 children in 1973. The government's own counts were lower.

Officials of the Bureau of Indian Affairs ran 77 boarding schools in 1969, holding about 35,000 children, and nearly 9,000 of them were under nine years old. On the Navajo Reservation, 7,476 children aged nine and younger lived at 48 boarding schools because no day school was near their homes. One of them, at Tuba City, housed 1,200 elementary pupils.

Crownpoint, 1968. Three brothers ran away from the Crownpoint Boarding School in New Mexico in January 1968, just after the Christmas holiday. They were trying to walk about 50 miles home to Chaco Canyon, and they wore jeans, shorts and light jackets. That night the temperature fell to about 10 degrees below zero.

On January 13, searchers found Ronald Yazzie, who was 9, and Willie B. Yazzie, who was 13, frozen to death about 30 miles from the school. Their brother Ray, 12, lived, with frostbite, damage to skin and flesh from freezing. A Navajo police officer, Stewart Silentman, said, "They were lonesome for home and just took off."

A National Tragedy. A subcommittee of the Senate published a report in November 1969, *Indian Education: A National Tragedy, A National Challenge*. It describes the policy behind the schools as "coercive assimilation," which means forcing people to give up their own ways and take on another people's ways.

The report includes a letter from a teacher at Tuba City. At that school, parents could take a child home for a weekend only if staff judged the child's behavior good, and a child who had run away could be kept from going home for up to a month.

The report also includes the account of an anthropologist, a scientist who studies how groups of people live, of a Navajo boarding school where "Children are beaten" and where "teachers advocate the free labor of Navajo girls in their homes." The report does not name that school or the anthropologist. The senators who issued the report asked that Native nations run their own schools.

Navajo leaders opened Navajo Community College in 1968. The college's own history states that it was the first community college in the country run by a tribe, and it is now Diné College. Under the Indian Self-Determination and Education Assistance Act, which members of Congress passed in January 1975, Native nations could sign contracts to run schools themselves. Many federal boarding schools closed in the 1980s.

Taken from their families. Officials took Native children from their families in other ways too. Surveys by the Association on American Indian Affairs in 1969 and 1974 found that 25 to 35 of every 100 Native children had been separated from their families and put in foster homes, adoptive homes or institutions. State welfare workers, courts, private agencies and officials of the Bureau of Indian Affairs, through their boarding schools, removed them.

In 16 states surveyed in 1969, about 85 of every 100 Native children in foster care were living with non-Native families. The Bureau of Indian Affairs census of 1971 counted 34,538 children living in its boarding schools and dormitories instead of at home. The Indian Child Welfare Act became law on November 8, 1978. Lawmakers wrote it to keep Native children with Native families.

BY 1978: Native nations could contract to run their own schools, and lawmakers had passed a federal law to keep Native children with Native families. Federal judges had ordered buses to carry children to end separate schools in Charlotte and Boston.

Punished for speaking Spanish

In much of the Southwest, school staff punished Mexican American children for speaking Spanish, and in 1972 a federal commission printed how. At the U.S. Commission on Civil Rights' hearing in San Antonio in December 1968, a principal from El Paso testified that children caught speaking Spanish were sent to a Spanish detention class for an hour after school. The commission's report, *The Excluded Student*, includes a copy of the slip used to send a child there. No principal admitted hitting a child for speaking Spanish, but witnesses at the hearing said children were hit.

The children's own words. A class of Mexican American seventh graders in Texas wrote about their elementary schools in October 1964. A Texas teacher, Alonzo Perales, sent the essays to the commission, and the report does not name the children.

One wrote: "If we speak Spanish we had to pay 5 cents to the teacher or we had to stay after school." "In the first through the fourth grade," another wrote, "if the teacher caught us talking Spanish we would have to stand on the 'black square' for an hour or so."

A third had to "write three pages saying, 'I must not speak Spanish in school'." "In the sixth grade," a fourth wrote, "they kept a record of which if we spoke Spanish they would take it down and charge us a penny for every Spanish word." The same child went on: "If we spoke more than one thousand words our parents would have to come to school and talk with the principal."

How common the rule was. The commission surveyed about 5,800 schools with many Mexican American pupils in Arizona, California, Colorado, New Mexico and Texas, and about a third had rules that discouraged children from speaking Spanish in class. The rule was most common in Texas, where two-thirds of the schools surveyed had one, and in schools serving poor barrios, or Mexican American neighborhoods, where half did.

The law in Texas. Texas law had required teaching in English only since 1918. In October 1970 a Mexican American teacher in Crystal City, Texas, was charged with a crime for teaching a high school history class in Spanish, and

the charge was later dropped. On June 3, 1973, Governor Dolph Briscoe signed a law that ended the English-only rule, 55 years after it began.

Some districts kept Mexican American children apart and called it language teaching. In Driscoll, Texas, officials of the school district made Mexican American children spend four years getting through the first two grades. Linda Pérez, who spoke only English, was put in the "Mexican" first grade to learn English. In 1957 a federal judge called the system "arbitrary and unreasonable" and ordered it stopped.

Teachers at the Blackwell School in Marfa, Texas, taught Mexican American children from September 1909 until 1965. The school started in the old school building that district officials had left them in 1892, when they built a new school for white children.

The National Park Service states that schools like it ran in more than 120 Texas school districts in 59 counties, with a "no Spanish in the classroom" rule. Blackwell closed in 1965, when Marfa's school officials opened one elementary school for all children. In October 2022 it became a national historic site.

The walkouts. About 15,000 students walked out of seven schools in East Los Angeles from March 1 to 8, 1968. Mexican American students there faced a dropout rate of 60 percent in high school, bans on speaking Spanish, and being pushed into trade classes instead of classes that prepared them for college. They asked for bilingual teaching, classes taught in both Spanish and English, and for Mexican American school leaders. Members of the Board of Education turned them down and said there was no money.

Later that spring, Los Angeles police arrested thirteen of the organizers on felony conspiracy charges, which means they were accused of planning a serious crime together. One of those arrested was the teacher Sal Castro, and school officials took away his teaching job. After months of sit-ins at the Board of Education, he got his job back. About two years later, judges of a California appeals court threw out the charges against him and the others.

Lau, 1974. President Johnson signed the Bilingual Education Act on January 2, 1968, and it became Title VII of the 1965 school law. In 1974 the justices of the Supreme Court ruled on children placed in classes they could not

understand. In San Francisco about 2,856 pupils of Chinese descent did not speak English. About 1,000 were getting extra lessons in English, which left about 1,800 with none. One of the children was Kinney Kinmon Lau.

Justice William O. Douglas wrote in *Lau v. Nichols*, on January 21, 1974, that leaving those children without help "denies them a meaningful opportunity to participate in the public educational program." The ruling does not require schools to teach in Chinese. It states that a school may not put a child in a class taught only in a language the child did not understand and call that an equal education.

Hawaiian comes back into the schools

Hawaii's lawmakers required schools to teach in English in 1896, and in practice children were stopped from speaking Hawaiian at school. The history written by the Pūnana Leo teachers' group does not say who stopped them. A group of Hawaiian-language teachers met in January 1983 to plan how to keep the language alive, and in August 1984 they opened the first Pūnana Leo preschool, at Kekaha on the island of Kauaʻi. *Pūnana leo* means "nest of voices."

The teachers worked with the state's lawmakers in 1986 to end the 90-year ban on teaching in Hawaiian. In 1987 immersion classes, where children are taught in Hawaiian, opened at Keaukaha Elementary in Hilo and Waiau Elementary in Pearl City. The group's own history describes them as the first elementary classes in the United States taught through a Native language.

New kinds of school

Several new kinds of school began in these years, and two of them, the magnet school and the segregation academy, came out of the fight over desegregation.

Magnet schools. At a magnet school, officials offer special teaching to draw families from across a district. McCarver Elementary in Tacoma, Washington, opened as one in the fall of 1968. Before that, 91 of every 100 of its pupils were Black. Nobody was assigned to go there. School officials used a $200,000 federal grant, moved in the district's best teachers, and let children work through their lessons at their own speed. Counselors visited families at home to ask them to choose the school.

On the first day, about 200 white children enrolled where there had been about 50. By 1970 fewer than half of McCarver's pupils were Black, and there was a waiting list. The second magnet school opened in Boston in 1969. By 1981 there were about 1,000 in the country, and by 1991 more than 2,400.

Segregation academies. White families in the South also started private schools, called segregation academies, so that white children would not have to sit in class with Black children. Mississippi had only three private schools not run by a church before 1954. The state's lawmakers voted grants of $185 for each child to attend such a school in 1964, and in the school year 1964 to 1965 more than 500 children received more than $80,000.

One new school was Citizens' Council School No. 1 in Jackson. The Citizens' Council was a group founded in Indianola, Mississippi, in 1954, months after *Brown*. By the mid-1960s its leaders were promoting private schools as a way to avoid public schools with Black and white pupils together.

The justices of the Supreme Court ruled in 1969 that Mississippi's school districts had to desegregate by 1970. Between 1966 and 1970 the number of private schools in the state rose from 121 to 236, and the number of pupils in them tripled. In 1970 almost every white student in Canton left the public schools for a private academy, and more than 40 of every 100 white students in Jackson left.

The historian Michael Fuquay found that the academies got books, supplies, sports equipment, buildings and money straight from the public school systems, and his account does not name the officials who handed them over. In Tunica and Clay counties, private school teachers stayed on the public school payroll. In Forrest County, private school pupils rode public school buses.

William J. Simmons led the Mississippi Private School Association and took the minutes of its board meeting on January 23, 1971. They record the board's decision that its schools would stay members "until such a time as they accept a Negro student or faculty member," Negro being the usual word for Black people at the time. At that point, the minutes say, a school would be asked to leave the association.

Catholic schools. Catholic schools shrank in the same years. Their enrollment was highest in the school year 1964 to 1965, when 5,601,000 children attended 13,249 Catholic schools. By 1990 to 1991 there were about 2.5 million pupils, fewer than half as many, in 8,587 schools.

Charter schools. The charter school came last: a public school run by its own board under an agreement, called a charter, instead of by the school district. Minnesota's lawmakers passed the country's first charter school law in 1991, after the idea had failed three times in the state House, and it passed the House by one vote. The law allowed only eight charter schools. The first to open was City Academy in St. Paul, founded by Milo Cutter, in the fall of 1992.

Disabled children get a right to school

Until 1975 the officials of an American public school could refuse to take a disabled child. The federal Department of Education states that in 1970 American schools taught only one disabled child in five, and that more than 1 million children with disabilities were shut out of school altogether.

Seven children. Lawyers sued the school board of Washington, D.C., in 1971 for seven children who had been kept out of school. George Liddell was eight. Staff at Maury Elementary School had turned down his application because he needed a special class, and he had never been to school at all.

Janice King was thirteen, with weakness down one side of her body. She had been refused a place since she was old enough for school, because there was no program for her, and she had never been to school either.

Michael Williams, 16, had epilepsy, the brain disease that causes seizures, sudden bursts of activity in the brain that can make a person shake or pass out. He had been out of the Sharpe Health School since October 1969.

The other four had been put out of school or kept out, some of them for years, and two of them had been called "behavior problems." One of the four was Peter Mills, whose name is on the case.

The counts in the case did not agree. The families said perhaps 18,000 disabled children in the city were getting no special teaching, and the city's own report for 1971 estimated that 12,340 would not be served in the coming school year.

The 1975 law. President Gerald Ford signed the Education for All Handicapped Children Act on November 29, 1975. Under it, every disabled child in every state had the right to a free public education suited to that child.

Special education grew before the law as well as after it. The share of pupils in special education rose from 1.5 of every 100 in 1947 to 1948 to 5.9 of every 100 in 1969 to 1970, before the law, and to more than 11 of every 100, more than 1 child in 9, by 1989 to 1990. More than half of the rise in the number of children came before the law. After 1975, schooling was their right under federal law.

The groups children were sorted into changed too. The federal count had no group for children with learning disabilities until 1976 to 1977, and by 1989 to 1990 that group held nearly half of all children in special education. In the twenty years after 1969 to 1970, the number the government counted as "mentally retarded," its word then for children it judged to learn much more slowly than others, fell by about a third.

BY 1991: Every disabled child had a right under federal law to a free public education suited to that child, and more than 1 child in 9 was in special education. The country had more than 2,400 magnet schools, and Catholic schools had lost more than half their pupils since 1965.

Peter Mills

Who: A twelve-year-old put out of his Washington school and kept out, whose name is on the 1972 case in which a judge ruled that the city must teach every disabled child.
When and where: Washington, D.C., 1971 and 1972.

He was twelve, in fourth grade, and shut out of school.

Peter Mills was put out of Brent Elementary School in Washington, D.C., on March 23, 1971, and school officials did not place him in any public school after that. He was in the fourth grade. School staff called him a "behavior problem," in words the court record keeps.

He was a ward of the District of Columbia, which means the city government was legally responsible for him. In the court's words, he was "excluded from any publicly-supported education," and he stayed that way.

The case. The lawyers filed it on September 24, 1971, as *Mills v. Board of Education of the District of Columbia*. Part of the school board's answer was that it did not have the money to teach the seven children.

The ruling. Judge Joseph C. Waddy decided the case on August 1, 1972. He ruled that the District had to give a publicly supported education to every child of school age, whatever the child's disability, and that lack of money was no excuse. He wrote that if there was not enough money for everything, it had to be spent "in such a manner that no child is entirely excluded from a publicly supported education."

Before school officials could put a child out or move the child, they had to tell the family why and hold a hearing, a meeting where the family could answer.

Members of Congress applied the same rule to every state in a federal law in 1975. The federal Department of Education names two court cases as the ones that established that states had to teach disabled children, and *Mills* is one of them. The court record does not say what happened to Peter Mills afterward.

The paddle, in court and in the states

Staff at a public school could paddle a child without any hearing first, the justices of the Supreme Court ruled in 1977. After that ruling, state lawmakers banned school paddling one state at a time. Until 1971, hitting schoolchildren was lawful in every state except New Jersey. In American schools the usual method was a paddle, a wooden board that an adult swung against the child's buttocks.

James Ingraham. James Ingraham was an eighth grader at Charles R. Drew Junior High School in Dade County, Florida, in October 1970. "Because he was slow to respond to his teacher's instructions," the Supreme Court's opinion states, "Ingraham was subjected to more than 20 licks with a paddle while being held over a table in the principal's office." The beating left a hematoma, a pool of blood under the skin from a blow. He needed a doctor, and he missed several days of school.

Staff at the same school paddled Roosevelt Andrews, a ninth grader, several times for small things. He was also hit on the arms, and one blow left him unable to use his arm fully for a week.

The rules allowed paddling but set limits on it. Florida law allowed punishment that was not "degrading or unduly severe." Dade County's own rule described the paddle as "less than two feet long, three to four inches wide, and about one-half inch thick." It set the usual punishment at "one to five 'licks' or blows with the paddle." James Ingraham was hit more than twenty times.

Ingraham v. Wright. The families sued. On April 19, 1977, in *Ingraham v. Wright*, the justices ruled against the children, five votes to four. Justice Lewis Powell wrote that the Eighth Amendment, the part of the Constitution that forbids cruel and unusual punishment, "does not apply to the paddling of children as a means of maintaining discipline in public schools."

Powell also wrote that the Constitution did not require "notice and a hearing prior to the imposition of corporal punishment." He meant that school staff did not have to tell a child why, or hold a hearing, before they hit the child.

Justice Byron White wrote for the four who disagreed. The record, he wrote, showed "beatings so severe that if they were inflicted on a hardened criminal for the commission of a serious crime, they might not pass constitutional muster." He meant that the Constitution might forbid beating even a criminal that hard.

The states. State lawmakers ended school paddling instead, one state at a time. Massachusetts was the first state after New Jersey, in 1971. Hawaii followed in 1973, Maine in 1975, and Rhode Island and the District of Columbia in 1977. New Hampshire banned it in 1983, New York and Vermont in 1985 and California in 1986. Eight states banned it in 1989 and four in 1993.

Twenty-seven states and the District of Columbia had banned it by 1994, and then no state's lawmakers banned it again until 2003. When the century ended, school staff could still lawfully hit children in 23 states.

The districts' own reports to the federal government counted about 1.5 million children paddled in 1976. In 1978, 4 of every 100 American schoolchildren were hit at school. The count fell to 613,514 in 1990 and 342,038 in 2000.

District officials counted the children themselves, and nobody checked their numbers. In 2009 the law professor Deana Pollard Sacks wrote that some researchers had found the true numbers may be twice as high as reported.

Teaching at home

Some parents began to teach their children at home in the 1970s and 1980s, and in several states officials took them to court. Many state attendance laws said that a child had to be taught in a school, or by a teacher with a state certificate. The historian Milton Gaither, who wrote a history of American homeschooling, says the old laws in many states made it "either illegal or dubious (open to doubt)." Six states required parents to hold a teaching license.

Holt and the Moores. Two men who knew each other started the movement for different reasons. John Holt was a writer on school reform. He argued that ordinary schooling trained children to obey, and he told parents to take their children out of school. He started a newsletter, *Growing Without Schooling*, in August 1977, and he made up the word "unschooling." He died in 1985.

Raymond Moore was a researcher on education and a Seventh-day Adventist, a member of a Christian church. He and his wife, Dorothy Moore, argued that school before about age eight or nine harmed children and that they should be taught at home until then. Their book *Better Late Than Early* came out in 1975, and *Home Grown Kids* in 1981.

The courts. In Harrison County, West Virginia, Bobby and Esther Riddle taught their children, Tim and Jill, at home. They were convicted of breaking the state's school attendance law and fined ten dollars each. On December 11, 1981, the justices of the West Virginia Supreme Court upheld the convictions, although their ruling said, "By all accounts in the record below, the Riddles did an excellent job."

Larry Delconte moved to North Carolina in March 1981 with his wife, Michelle, and their four children and taught them at home. Judges of a state appeals court ruled that the attendance law forbade it. On May 7, 1985, the justices of the North Carolina Supreme Court reversed that ruling.

Lawmakers and judges in state after state changed the rules through the 1980s. Some states wrote detailed homeschooling laws, some added a sentence to an old law, and in some judges made the change. By 1989 most states had settled with homeschooling families, and a few held out into the early 1990s.

Mark and Chris DeJonge

Who: Parents in rural Michigan who were convicted of a crime for teaching their own children at home, and who appealed for about ten years.
When and where: Western Michigan, 1984 to 1993.

The trial judge praised their teaching and convicted them anyway.

Mark and Chris DeJonge were convicted in 1985 for teaching their own children at home, and the justices of the Michigan Supreme Court overturned the conviction in 1993. They began teaching their children in August 1984, in rural western Michigan. They used a course from the Church of Christian Liberty and Academy of Arlington Heights, Illinois. Neither parent held a Michigan teaching certificate, the state's license to teach.

Chris DeJonge was in her second week of teaching when officials from the state Department of Social Services came to the house and told the family that the children were truants, children missing school without permission. Under Michigan law, a child between 6 and 16 had to be taught in a public school, or in an approved private school by a certified teacher.

The trial. They were charged and convicted in the Ottawa County District Court in 1985. The trial judge had a "very, very favorable report on the education of the children." He convicted them anyway, because the law was about who did the teaching and not about how well it was done.

Each parent got two years of probation, which means living under a court's watch, and a fine of $200. The judge ordered them to have the children tested and to arrange for a certified teacher. The sentence was put on hold while they appealed.

Their reason was religious. They wanted what they called a "Christ centered education," because, as the court's opinion records, "the major purpose of education is to show a student how to face God, not just show him how to face the world."

Nine times in court. The judges of the Michigan Court of Appeals ruled against them on August 8, 1989. The justices of the Michigan Supreme Court sent the case back on October 17, 1990, and the appeals judges ruled against them again in 1991.

The justices of the Michigan Supreme Court ruled for them on May 25, 1993, four votes to three. Justice Riley wrote that the state had a strong interest in children being educated, but that Michigan had not shown that a certified teacher was necessary for it. The majority held that the rule could not be applied to a family with a sincere religious objection to it.

The case took them to court nine times over about ten years. In 2008 Mark DeJonge said, "We're glad we went through it. We grew as a family and we grew as a couple."

A Nation at Risk, 1983

Members of a federal commission published a report in April 1983 that stated American schools were failing. The modern argument over school standards began with this report. Secretary of Education T. H. Bell had set up the National Commission on Excellence in Education, led by David P. Gardner, on August 26, 1981. After 18 months its members published *A Nation at Risk*.

Its best-known sentences. The report compares weak schooling to an attack by an enemy country. "If an unfriendly foreign power had attempted to impose on America the mediocre educational performance that exists today," it states, "we might well have viewed it as an act of war. As it stands, we have allowed this to happen to ourselves." It states that the foundations of American schooling were "being eroded by a rising tide of mediocrity," meaning work that is only average or poor.

The report lists the commission's reasons. On 19 tests given in many countries, American students were never first or second, and compared with other industrial countries they were last seven times. It stated that some 23 million American adults could not read, write and understand well enough for everyday life. It stated that about 13 of every 100 seventeen-year-olds could not either, and as many as 40 of every 100 among minority youth.

Average SAT scores had fallen almost every year from 1963 to 1980, by more than 50 points in reading and nearly 40 in math. Of seventeen-year-olds, nearly 40 of every 100 could not draw conclusions from something they read, and only about one in three could solve a math problem that needed several steps.

These are the commission's own figures, and other researchers disputed several of them. In 1993 researchers at Sandia National Laboratories, a federal laboratory, published their own study of the same kinds of measures. On most of them, including test scores, dropout rates and college enrollment, they found steady or slight improvement.

Teachers strike, and their pay

Teachers in New York City and in Florida joined unions, groups of workers who join together to bargain with an employer over pay and working conditions, and went on strike in this era.

New York, 1960. Members of the new United Federation of Teachers held the first teachers' strike in New York City's history on November 7, 1960, Election Day. It lasted one day. The union's leaders said it had about a quarter of the city's 39,000 teachers, *Time* magazine reported. New York teachers voted in June 1961 to bargain as a group, and in December they chose the union to bargain for them. Later accounts call it the first big-city teachers' union in the country to win that right.

Florida, 1968. More than 27,000 Florida teachers and school officials handed in letters resigning their jobs and walked out on February 19, 1968. The Zinn Education Project, a group that makes history lessons for teachers, calls it the first strike by teachers across a whole state in the United States. The teachers had asked for better pay and more money for schools, and Governor Claude Kirk had refused.

The strike ended on March 8, when members of the state board of education agreed to more money for schools. Afterward, officials refused to rehire thousands of the teachers because they had gone on strike.

The pay. Measured in dollars of the same value, the average public school teacher earned about $58,700 a year in 1969 to 1970, about $52,900 in 1979 to 1980, and about $63,500 in 1989 to 1990. In 1999 to 2000 the figure was still about $63,500, so real teacher pay did not grow in the 1990s.

The economist Sylvia Allegretto compares teachers' weekly pay with that of other college graduates of the same age and schooling. By her count, teachers earned about 7 percent less in 1979, about 5 percent less in 1993 and about 6 percent less in 1996. After 1996 the gap grew.

The last one-room schools

The one-room school nearly disappeared in these years. The federal count of one-teacher public schools was 59,652 in 1949 to 1950, 20,213 in 1959 to 1960 and 1,815 in 1970 to 1971. By 1990 to 1991 there were 617, about one for every hundred there had been in 1950. The number of school districts kept falling, from 83,718 in 1949 to 1950 to 40,520 in 1959 to 1960 and 17,995 in 1970 to 1971.

BY 2000

If you remember one thing: Judges and lawmakers opened public schools to Black children, disabled children and children who did not speak English in these fifty years, and each change took years of fighting.

One question left open: Would the schools stay desegregated? The share of Southern Black students in mostly white schools was at its highest, about 43 percent, in the mid-1980s. The next era gives the answer.

Next: After 2000, members of Congress made every state test every child in reading and math each year.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 37 MINUTES

The arguments over American schooling after 2000 were about which school a child attended, what was taught there and how learning was measured. Under a federal law signed in 2002, every state had to test every child in reading and math each year from third grade to eighth, and the yearly tests began in the school year 2005 to 2006. More families chose charter schools, private schools paid for with state money, or teaching at home.

State and local officials closed the school buildings in the spring of 2020 because of the coronavirus pandemic, a disease spreading around the world, and children were taught over the internet for the rest of that school year.

Gunmen killed children and teachers inside schools. Federal officials counted the children who had died at the government's own Indian boarding schools, and a president apologized. In 2025 President Donald Trump ordered his Secretary of Education to begin closing the federal Department of Education. Federal officials publish their school counts years after the school year they describe.

Testing every child: No Child Left Behind

Under a federal law signed in 2002, every state had to test every child in reading and math every year from third grade through eighth grade. President George W. Bush signed the No Child Left Behind Act on January 8, 2002, at Hamilton High School in Hamilton, Ohio. The law is called the act "of 2001" because members of Congress passed it in 2001.

The law's first lines state its purpose: "to close the achievement gap with accountability, flexibility, and choice, so that no child is left behind," the achievement gap being the difference between the test scores of different groups of children, such as rich and poor children.

"The fundamental principle of this bill is that every child can learn, we expect every child to learn," Bush said at the signing, "and you must show us whether or not every child is learning."

What the tests covered. The yearly tests in grades 3 through 8 began in the school year 2005 to 2006. Science had to be tested at least once in grades 3 to 5, once in grades 6 to 9 and once in grades 10 to 12. States had to be able to report the scores separately for boys and girls, for each main racial and ethnic group, for children still learning English, for children of migrant farm workers, for disabled children and for children from poor families.

The law also set a deadline. By the end of the school year 2013 to 2014, "all students in each group" were to "meet or exceed the State's proficient level of academic achievement," proficient meaning good enough to meet the standard that officials in each state set. So every child in the country was to reach that level in reading and math by 2014.

The 2015 law. President Barack Obama signed a new law, the Every Student Succeeds Act, on December 10, 2015. The federal Department of Education's own description states that No Child Left Behind's "prescriptive (telling people exactly what to do) requirements became increasingly unworkable for schools and educators."

Under the new law the yearly tests in grades 3 through 8 stayed, along with one test in high school, and the single national deadline went away. Officials in each state now set their own "long-term goals" for all their pupils and for each group of them. Both laws are rewritten versions of the 1965 school law that President Lyndon Johnson signed at his old one-room school in Texas.

How many tests. Children in big cities took many tests. The Council of the Great City Schools, a group of big-city school districts, studied the tests in 66 of those districts in the school year 2014 to 2015. The districts had more than 7 million students in about three dozen states.

According to the Associated Press report of the study, a student took about 112 required standardized tests between pre-kindergarten and twelfth grade, a standardized test being the same test, given and scored the same way, for every child who takes it. Students spent 20 to 25 hours a year taking these tests, about 2.3 percent of class time for the average eighth grader. Eighth and tenth grade had the most required tests.

The researchers Thomas Dee and Brian Jacob called the law's effect on learning "a highly controversial but centrally important question." In 2011 they studied national test scores from before and after the law. They found that the law raised fourth graders' math scores, and eighth graders' math scores to some degree. They found no sign that it raised fourth graders' reading.

Choosing a school: charters, private schools and state money

More families sent their children to schools other than the local public school in these years, and state lawmakers paid for some of those choices with public money. The federal count of every public school records the growth of charter schools. In the school year 2012 to 2013 there were 6,079 charter schools, with 4.6 percent of all public school pupils. In 2022 to 2023 there were 7,998, with 3,717,857 pupils, or 7.6 percent.

Federal statisticians published that table in February 2024. As of September 2026, 2022 to 2023 was still the newest year in the federal count of charter schools.

Private schools. About 1 American child in 11 went to a private school. In the fall of 2021, 4.7 million children from kindergarten through twelfth grade were in private schools, 9 percent of all schoolchildren. Catholic schools had 1.7 million of them, other religious schools 2.0 million, and schools with no church behind them 1.1 million.

Ten years earlier, in the fall of 2011, the total was 4.5 million. Catholic schools had lost about 200,000 children over those ten years, and other religious schools had gained about 300,000. Catholic schools had 5,601,000 pupils in 1964 to 1965, their highest year, so the 1.7 million of 2021 was less than a third of that. The National Center for Education Statistics, the federal office that counts schools, published those figures in May 2024. As of September 2026, they were still the newest full federal count of private schools.

Public money for private school. State lawmakers created three main ways to pay for private schooling with public money.

One is the voucher, a payment from the state that a family can use toward a private school's fees instead of sending the child to the local public school. Another is the education savings account, or ESA, state money set aside for one child that the family can spend on private school fees, a tutor, books or an online course. The third, the tax-credit scholarship, works through taxes: a person or a company gives money to a scholarship fund, and the state takes the same amount off their tax bill.

EdChoice, a group that supports these programs, keeps the national count. It listed 69 programs for the school year 2024 to 2025. By its count, 488,736 children used education savings accounts, 349,923 used vouchers, 217,194 had tax-credit scholarships, and 106,442 in Florida had a tax-credit ESA, about 1,162,000 children in all.

Patrick Wolf, a researcher at the University of Arkansas, has said that studies of the programs open to all children find that about two of every three children using them had never gone to a public school. In Florida, which opened its programs to all children in 2023, *Politico* reported that 69 percent of the new students were already in private schools.

The argument over the money. EdChoice states that its aim is "advancing educational freedom and choice for all," and it quotes the economists Milton and Rose Friedman's idea that "every family could choose the learning environment that best fits their child." The National Education Association, a teachers' union, states that "vouchers take scarce funding from students in public schools and give those resources to unaccountable private schools."

Researchers who studied one program, Louisiana's vouchers, found different things about the children who used them and about the public schools nearby. The researchers Jonathan Mills and Patrick Wolf studied the program's first two years in 2016. Children who used a voucher to go to a private school scored lower in math than similar children who did not. The researcher Anna Egalite found that the public schools that stood to lose the most pupils to the program did the same or slightly better.

Teaching at home: the 2020 jump

The share of children taught at home rose in this era and about doubled after 2020. The federal survey of homeschooling counted 1.7 percent of children aged 5 to 17 taught at home in 1999 and 2.8 percent in 2019, about 1.5 million children.

In the 2019 survey, the reason parents gave most often was worry about the school environment, named for 80 of every 100 homeschooled children. Moral teaching and more family life together came next, at 75 each.

2020. Census Bureau staff asked households the same question twice in 2020, after the school buildings closed. In late April and early May, 5.4 percent of households with school-age children said they were homeschooling, and by early October 11.1 percent did. Among Black households it went from 3.3 percent to 16.1 percent. The staff worded the question so that children taking public school lessons over a screen were not counted.

Most of the rise stayed after the schools reopened. The researchers Genevieve Smith and Angela R. Watson studied the Census Bureau's surveys from September 2022 to August 2023 and found 5.82 percent of school-age children homeschooled, about twice the share before 2020. They wrote that education experts had predicted the number would fall once schools reopened, and it did not. It was highest in Alaska, at 12.6 percent, and lowest in Rhode Island, at 2.9 percent.

The rules. Lawmakers in each state make their own rules for homeschooling, and the rules range from almost none to many. The Coalition for Responsible Home Education, a group that argues for more checks on homeschooling, counted the rules in March 2026.

In 12 states a family does not have to tell anyone that it is homeschooling. In 29 states no one ever has to check a homeschooled child's progress, and only 3 states require a check for every homeschooled child. In 22 states a family does not have to keep any school records.

Texas law requires no notice, no check and no records. New York's requires all three. A parent there must send the local school superintendent a written notice by July 1 each year, and then a teaching plan for the child. Every three months the parent sends a report giving the hours taught and what was covered in each subject. Once a year the parent sends an assessment, a test or written review of how the child is doing.

BY 2023: More than 7 of every 100 public school pupils went to charter schools, about 9 of every 100 children went to private schools, and the share taught at home was about twice what it had been before 2020.

Special education now

About 1 public school student in 7 received special education in the 2020s. The 1975 law under which disabled children have a right to school is now called the Individuals with Disabilities Education Act, or IDEA. In the school year 2022 to 2023, 7.5 million students aged 3 to 21 received special education or related services under it, 15 percent of all public school students, according to the National Center for Education Statistics in May 2024.

The largest group, 32 in every 100, had specific learning disabilities, such as trouble learning to read or to do math. In 1989 to 1990 the count had been 4,641,000, or 11.4 percent.

How separate the schools are now

The schools of the South did not stay as desegregated as they had become by the 1980s. Researchers at the Civil Rights Project at the University of California, Los Angeles, count the Black students in the South who go to schools where most of the students are white. That share was highest, about 43 of every 100, in the mid-1980s. By 2021 it was about 16 of every 100, by their report of April 2024.

Across the whole country, more schools had almost no white students. The same researchers counted schools where 90 to 100 of every 100 students were not white. In 1988, 7.4 percent of American public schools were like that, and in 2021, 19.8 percent were. In 2021, 78 of every 100 students in those schools were poor.

The students themselves had changed. In 1968, about 81 of every 100 American public school students were white. In 2021, about 45 were white, and about 28 were Latino.

In New Orleans, the McDonogh No. 19 school building, where Leona Tate, Gail Etienne and Tessie Prevost walked in with federal marshals on November 14, 1960, reopened on May 4, 2022, as the Tate, Etienne and Prevost Center.

The school year and the school day

In most states that set a minimum, school must run at least 180 days a year, and a few districts have tried other calendars.

Researchers at the Pew Research Center checked state laws in 2023. Of the 38 states that set a minimum number of school days, the laws of 27 states and Washington, D.C., required 180. The laws of seven states set a lower minimum, and Colorado's 160 days was the lowest. The laws of four states set a higher one, and Kansas was highest, at 186 days for kindergarten through grade 11.

Year-round school spreads about the same number of days across the whole year, with shorter breaks. It became less common. The federal count found 4.4 percent of public schools with all their pupils on a year-round calendar in 2007 to 2008, 3.2 percent in 2011 to 2012 and 2.5 percent in 2017 to 2018.

The four-day week. The four-day school week spread instead. In September 2023 Paul Thompson, an economist at Oregon State University, counted about 2,100 schools in almost 900 districts in 26 states on a four-day week. He said that was about double the number ten years before and almost 40 percent more than in 2019.

In Missouri, by the count of Jon Turner of Missouri State University, 105 districts had a four-day week in 2020 and 128 in 2022, a quarter of the state's districts. His count and the state's list differ, so both are estimates.

Start times. In a 2014 policy statement, the American Academy of Pediatrics, a group of children's doctors, said that middle and high schools "should aim for a starting time of no earlier than 8:30 am," because teenagers who start earlier lose sleep. In the school year 2020 to 2021 the average American public high school started at 8:07 in the morning. More than 7 of every 10 started before 8:30.

California's lawmakers wrote a start time into law, and the governor approved it on October 13, 2019. Under Senate Bill 328, middle schools may begin no earlier than 8:00 a.m. and high schools no earlier than 8:30 a.m. Schools had

to change by July 1, 2022, or, where a teachers' contract in force on January 1, 2020, ended later, when that contract ended. The law does not apply to rural districts.

Florida's lawmakers passed a similar law in 2023, to begin in 2026. District officials said they did not have enough buses or drivers to do it. On March 27, 2025, Florida senators voted for Senate Bill 296 with no votes against. Under the bill, the requirement ended, and any district that kept early start times had to file a report on what stood in the way. Governor Ron DeSantis signed it on May 21, 2025.

The paddle and the suspension

The laws of 15 states still let an adult at school hit a child as a punishment in 2026, and in American schools the usual method is the paddle. At the end of the 1990s, 23 states had no ban on it. From 2003 to 2011 lawmakers in four more states banned it: Delaware in 2003, Pennsylvania in 2005, Ohio in 2009 and New Mexico in 2011.

The 2026 count. Reporters at NPR, the public radio network, counted 15 states in August 2026 whose laws allow it, and 30 states plus Washington, D.C., whose laws ban it in all public schools. The other five states neither allow it nor ban it outright. Some have other laws that have been interpreted as bans, and some have no law about it. Other groups count differently for that reason, and they say so.

Federal officials count the children who are hit, in a survey that every public school district must answer, called the Civil Rights Data Collection. In the school year 2021 to 2022, school staff hit about 24,500 public school students as a punishment. NPR counted almost 32,000 reported paddlings in the same data, because some children were hit more than once. Not every district reports it.

Almost all the reported cases were in eight states, most of them in the South. Mississippi had the highest rate in the country and Arkansas the second highest. Officials posted the next survey, for 2023 to 2024, in September 2026, about eight months late. They posted it as raw data files with no national totals, so as of September 2026, 2021 to 2022 was the newest national count.

The same survey records which children were hit: boys far out of proportion to their numbers, and girls of every group less often than their share of students. White boys were 23 of every 100 public school students and 47 of every 100 children hit. Black boys were 8 of every 100 students and 20 of every 100 children hit, and American Indian and Alaska Native boys, fewer than 1 of every 100 students, were 2 of every 100 children hit.

Suspensions. School staff suspend children, keeping them out of class for a set number of days in a separate room at school or at home, far more often than they paddle them. In 2021 to 2022, about 2.3 million children, 5 percent of public school students, were suspended at least once inside school, and about 2.4 million at least once out of school. Districts also reported about 220,000 children to the police. Even in public preschools, out of 1.4 million children, about 2,700 were suspended out of school and about 370 were expelled, which means put out of the school altogether.

School staff punished Black children and disabled children far out of proportion to their numbers. The federal report states that "Black boys were disciplined at higher rates, compared to boys of other races." The survey records each group's share of all students and its share of the children punished:

- Black boys were 8 of every 100 students and 22 of every 100 children suspended out of school.
- Children with disabilities were 17 of every 100 students and 29 of every 100 suspended out of school.
- Black children were 15 of every 100 students and 33 of every 100 children arrested at school.
- Black children were 18 of every 100 preschool children and 38 of every 100 preschool children suspended.
- Preschool children with disabilities were 23 of every 100 preschool children and 74 of every 100 of those expelled.

BY 2024: About 15 of every 100 public school students received special education. In the newest counts, from 2021 and 2022, about 16 of every 100 Black students in the South went to mostly white schools, and school staff hit

about 24,500 children in a year as a punishment.

Shootings at school

Gunmen shot and killed children and teachers inside American schools in these years. Statisticians at the National Center for Education Statistics count every time a gun is fired or shown at a school, or a bullet hits school property, whether anyone is hurt or not. In the school year 2000 to 2001 they counted 30 shootings, with 18 people killed and 29 hurt. In 2021 to 2022 they counted 327 shootings, with 81 people killed and 269 hurt, more than ten times as many shootings.

Both counts include the people who fired. The statisticians warned that 2021 to 2022 was far above every earlier year, and as of September 2026 it was the newest year the federal count covered.

A narrower count. Reporters at Education Week, a newspaper about schools, count only shootings on school grounds, during school or a school event, in which someone other than the shooter was hit. By their count, 7 people were killed in 2025, the fewest since 2020. The two counts use different rules and cannot be joined into one list.

Sandy Hook, 2012. Adam Lanza, who was 20, shot and killed his mother in her bed at their home in Newtown, Connecticut, on December 14, 2012. Then he went to Sandy Hook Elementary School with a rifle and shot his way into the building, killing the principal and the school psychologist in a hallway. In two first-grade classrooms he killed the two adults in each room, 15 children in one room and 5 in the other, and then he shot himself.

Stephen Sedensky investigated. He was the prosecutor for that part of Connecticut, the lawyer who brings criminal charges for the government there. Sedensky wrote that "in fewer than 11 minutes twenty first-grade pupils and six adults had lost their lives." Lanza also wounded two other adults. The report does not print the children's names. Sedensky found no one else criminally responsible.

Parkland, 2018. Nikolas Cruz, a 19-year-old former student, shot and killed 14 students and 3 staff members at Marjory Stoneman Douglas High School in Parkland, Florida, on February 14, 2018, and wounded 17 others. He fired inside one building for about six minutes, from 2:21 to 2:27 in the afternoon.

A state commission investigated. Its members found that Scot Peterson, the sheriff's deputy assigned to the school, "fled to a position of personal safety while Cruz shot and killed ... students and staff." Prosecutors charged Peterson with felony child neglect, a serious crime of failing to protect children, and with other crimes. On June 29, 2023, a jury found him not guilty.

Uvalde, 2022. An 18-year-old gunman went into Robb Elementary School in Uvalde, Texas, with a rifle at 11:33 in the morning on May 24, 2022. He went into two joined classrooms and killed 19 children and 2 teachers, and at least 17 others were wounded. Within 3 minutes, 11 police officers were inside the school. The gunman fired at them, and they pulled back.

The school district's police chief, Pete Arredondo, told officers several times to wait while they looked for keys, and he tried to talk to the gunman. Justice Department officials counted at least 380 officers from 24 agencies at the school, and a committee of the Texas House of Representatives counted 376. Children in the classrooms called 911, and within four minutes of the officers' arrival the dispatchers knew that class was in session.

At about 12:50 p.m. a team of Border Patrol agents and sheriff's deputies went in and killed the gunman, 77 minutes after the first officers entered the school.

Officials of the U.S. Justice Department published their review on January 18, 2024. The review states that the officers' "most significant failure" was how they saw the gunman. They treated the gunman as a "barricaded subject," a person shut in a room, instead of an "active shooter," a person who is still attacking people. The report does not name the gunman.

A grand jury, a group of citizens who decide whether there is enough proof for a trial, charged Arredondo and a former school police officer, Adrian Gonzales, with abandoning and endangering children in June 2024. They were the only officers charged. On January 21, 2026, a jury found Gonzales not guilty. Arredondo's trial on 10 counts was set for February 22, 2027.

Apalachee, 2024. Colt Gray shot and killed two 14-year-old students and two teachers at Apalachee High School in Winder, Georgia, on September 4, 2024. He pleaded guilty, and on July 27, 2026, a judge sentenced him to life in prison with no chance of ever being let out. His father, Colin Gray, had given him the rifle as a Christmas present.

A jury found Colin Gray guilty of two counts of second-degree murder on March 3, 2026. Under Georgia law, that crime means causing a death while, through criminal carelessness, causing a child under 18 "cruel or excessive physical or mental pain." It does not require meaning to kill, and it carries 10 to 30 years in prison.

The jury also found him guilty of two counts of involuntary manslaughter, which means causing a death without meaning to, and of reckless conduct and cruelty to children. On July 30, 2026, he was sentenced to 15 years.

The school buildings close, 2020

State and local officials closed nearly every school building in the country in the spring of 2020 because of COVID-19, an illness caused by a virus. Officials of the World Health Organization had declared it a pandemic on March 11, 2020. On March 17, Governor Laura Kelly of Kansas ordered every school building in her state closed for the rest of the school year, and Kansas was the first state to do that.

Reporters at Education Week tracked the closures from March 6 to May 15, 2020. By their count, at the peak at least 55.1 million students were shut out of 124,000 public and private school buildings. Nearly every state's officials ordered or advised that schools stay closed until the end of the school year.

Teachers taught over the internet, and not every child had a computer at home. In the spring of 2020, 45 percent of public school principals said their school gave pupils a computer or other device to use at home. In a Census Bureau survey in September 2020, 59 percent of adults with a child in school said the school or district had given the child a computer. By September 2021, staff at 96 percent of public schools said they gave devices such as tablets to pupils who needed them, and 70 percent supplied internet service at home.

Tabatha Rosproy

Who: A preschool teacher named National Teacher of the Year in the spring her classroom was shut by the pandemic.

When and where: Winfield, Kansas, 2020.

She put pinwheels in her pupils' yards.

Tabatha Rosproy kept teaching her preschool class from a distance after the governor of Kansas closed its school buildings in March 2020. In May 2020 she was named National Teacher of the Year, the first early childhood teacher to win the award.

She taught for the Winfield Early Learning Center in Winfield, Kansas. Her classroom was not in a school. It was inside Cumbernauld Village, a retirement community and nursing home in the town, and the program was called Little Vikes. Children with disabilities and children without them were in her class together, and residents of the home joined in with the children every day.

Teaching from home. When the Kansas school buildings closed, Rosproy was a co-chair of the state task force of teachers that wrote the guidance for teaching children at home. She recorded her lessons and put them on an app called Seesaw, and once a week she held a video call with each child, one at a time. When a family could not get online, she called them, texted them, or delivered the materials to their home.

For the families of her pupils with disabilities, she prepared materials that showed the parents how to use the methods she used in the classroom. She mailed each child a bin of sensory toys, the part of the classroom they liked best, and she put pinwheels in their yards.

"At first, I was nervous to be named National Teacher of the Year during this pandemic," she told Education Week, "because I wondered if I was able to make a difference." On May 22, 2020, she told the news site Chalkbeat, "Nothing will replace what we are able to do in person. But we are surviving through this and we are looking for the small victories when we have them."

Test scores after 2020

National test scores fell after the school closures and had not recovered by 2024. The National Assessment of Educational Progress, often called the Nation's Report Card, is the same federal test given to a sample of American children decade after decade. It was given from January to March 2022, the first time since the closures.

Compared with 2019, the average math score had fallen 5 points in fourth grade and 8 points in eighth grade. The report on the test called these the largest drops in math since those tests began in 1990.

It was given again from January to March 2024. The average fourth-grade reading score was 5 points below 2019, and 31 of every 100 fourth graders reached the level the test calls Proficient, 4 fewer than in 2019. Fourth-grade math was 3 points below 2019.

Twelfth grade. In twelfth grade, reading and math were each 3 points below 2019. Members of the National Assessment Governing Board, which oversees the test, released those twelfth-grade results on September 9, 2025. They called them "the lowest ever reported on each assessment," which means the lowest since each of these tests began, not the lowest in all of history. Forty-five of every 100 high school seniors scored below the lowest level the test names, called Basic, in math, and 32 of every 100 in reading.

The test does not measure why scores fell. Reading scores had already begun to drop between 2017 and 2019, before the closures.

BY 2025: Fourth-grade reading scores were still 5 points below 2019, and twelfth graders' scores in reading and math were the lowest ever reported on those tests.

Teachers: hard to hire, and paid less than other graduates

Officials at most public schools reported trouble filling teaching jobs in 2023 and 2024, and teachers earned less than other college graduates. Statisticians at the National Center for Education Statistics asked 1,392 public schools in every state in August 2024 how their hiring was going. Seventy-four of every 100 said they had trouble filling at least one teaching job for the school year 2024 to 2025.

Each school had six teaching jobs open on average, and it had filled 79 of every 100 of them with fully certified teachers. The reason schools gave most often was a lack of qualified people, at 64 of every 100 schools, and next was too few people applying, at 62 of every 100. A year earlier, 86 of every 100 schools had reported trouble hiring teachers.

The pay. The National Education Association, a teachers' union, keeps the pay figures. By its count the average public school teacher earned $74,495 in the school year 2024 to 2025, up from $71,985 the year before. A new teacher started at $48,112 on average. Once rising prices are counted, the union reports, teachers were earning about 5 percent less than teachers had ten years earlier.

The economist Sylvia Allegretto compares the weekly pay of public school teachers with that of other college graduates of about the same age, schooling and state. By her count, in 2024 teachers earned 26.9 percent less a week than those other graduates, the widest gap she had measured. In 1996 the gap had been about 6 percent.

Paying for college

A year at college cost tens of thousands of dollars in these years, and 42.3 million Americans owed money on federal student loans. The National Center for Education Statistics gives the full average cost of a year for a new full-time student living at a four-year college in the school year 2022 to 2023. At a public college it was $27,100. At a private college run for profit it was $33,600, and at a private college not run for profit it was $58,600.

Fewer new students borrowed than ten years before. In 2020 to 2021, 38 of every 100 new full-time students received a loan, down from 50 of every 100 in 2010 to 2011. Students who finished a four-year degree in 2015 to 2016 with federal loans still owed an average of $45,300 in 2020.

Canceling loans. Officials in President Joe Biden's Department of Education announced a plan to cancel some federal student loans in August 2022. A borrower who earned less than $125,000 a year could have up to $10,000 canceled, and some could have up to $20,000. About 43 million borrowers qualified. The Congressional Budget Office, which works out costs for Congress, estimated the plan would cancel about $430 billion.

The justices of the Supreme Court struck the plan down, six votes to three, on June 30, 2023, in *Biden v. Nebraska*. Chief Justice Roberts wrote that the law the officials relied on did not let them cancel "$430 billion of student loan principal," the money borrowed, not counting interest. Justice Kagan wrote for the three who disagreed: "Emergencies, after all, are emergencies, and more serious measures must be expected in response to more serious problems."

According to the federal office that runs student aid, 42.3 million people owed more than $1.7 trillion in federal student loans on June 30, 2026. More than 9.3 million of them were in default. Under federal law a loan is in default when the borrower has gone 270 days, about nine months, without making a payment on time.

The boarding schools investigated, and an apology

Federal officials investigated the government's own Indian boarding schools between 2021 and 2024, and in 2024 President Joe Biden apologized for them. Secretary of the Interior Deb Haaland ordered the investigation on June 22, 2021. She is a citizen of Laguna Pueblo and was the first Native person in a president's cabinet, the group of the president's top officials who each run a department.

Her order asked for a report on the government's records of the schools, "with an emphasis on cemeteries or potential burial sites." Her department's announcement said the order followed news from Canada. There, leaders of the Tk'emlúps te Secwepemc First Nation had reported 215 unmarked graves at the former Kamloops Indian Residential School.

The reports. Investigators published a first report in May 2022 and a final report in July 2024, under Bryan Newland, the Assistant Secretary for Indian Affairs. The final report counted 417 federal Indian boarding schools between 1819 and 1969 and at least 973 children who died at them, and it states that the real number of deaths is higher. Investigators found 74 burial sites at 65 of the schools. They named 18,624 children who had attended, and the report states that this list is incomplete. The punishments listed in the first report, from solitary confinement to flogging, are set out under 1850 to 1900.

The apology. Biden spoke at the Gila Crossing Community School in the Gila River Indian Community in Arizona on October 25, 2024, and Haaland introduced him. "I formally apologize today as President of the United States of America for what we did," he said. He also said, "After 150 years, the government eventually stopped the program but never formally apologized."

By a proclamation of December 9, 2024, a formal written order from the president, he made about 24.5 acres of the old Carlisle Indian Industrial School grounds in Pennsylvania into a national monument, a place the federal government keeps for its history. The proclamation states that "nearly 1,000 Native youths died in schools across the system, but the actual number of lives lost is likely much higher."

Members of Congress proposed a commission to study the boarding schools. Senator Lisa Murkowski introduced her bill on February 26, 2025, and Representative Tom Cole introduced his on February 3, 2026. By September 2026 neither the Senate nor the House had voted to pass either one.

Four federal boarding schools for Native students, away from reservations, were still open in 2026, according to the Bureau of Indian Education: Chemawa Indian School in Salem, Oregon, Flandreau Indian School in South Dakota, Riverside Indian School in Anadarko, Oklahoma, and Sherman Indian High School in Riverside, California.

Books taken off the shelves, and limits on teaching

School board members and state officials removed thousands of books from school libraries in the 2020s, and lawmakers or officials in 18 states limited what teachers could say about race and sex. Each side described the removals differently.

Counting bans. PEN America, a writers' group that defends free speech, calls a school book ban any removal of a book, or any limit on reading it, because of what is in it. The removal can follow a complaint from parents or community members, a decision by school officials, or an order or threat from government officials.

By PEN America's count there were just under 3,000 bans a year in the school years 2021 to 2022 and 2022 to 2023, 10,046 in 2023 to 2024, and 6,870 in 2024 to 2025. In 2024 to 2025 Florida had the most, 2,304, then Texas with 1,781 and Tennessee with 1,622.

Officials of the U.S. Department of Education sent out a memo in January 2025 calling book bans a "hoax." They also canceled earlier advice that removing books could break civil rights law. Defense Department officials removed nearly 600 titles in July 2025 from the schools the department runs for children on military bases. According to PEN America, the reason given was that the books did not fit the president's orders on diversity, race and what the orders called "gender ideology extremism."

The American Library Association, the national group for librarians, counts school, public and college libraries together. For 2025 it recorded 4,235 different titles that people tried to have removed. By its count, pressure groups, government officials and other decision makers started 92 of every 100 of those attempts.

Limits on teaching. Education Week reported in March 2024 that lawmakers or officials in 44 states had considered limits since 2021 on how race and gender could be taught, and that 18 states had put such limits in place.

In Florida, Governor Ron DeSantis signed the Parental Rights in Education law on March 28, 2022. Under it, classroom teaching about sexual orientation, which sex a person is attracted to, or gender identity, a person's own sense of being a boy or a girl, a man or a woman, was banned from kindergarten through third grade.

DeSantis said that parents "should be protected from schools using classroom instruction to sexualize their kids as young as 5 years old." Opponents called it the "Don't Say Gay" bill. U.S. Secretary of Education Miguel Cardona said the governor was targeting "some of Florida's most vulnerable students and families, all while under the guise of 'parents' rights,'" a guise being a false cover.

Mahmoud v. Taylor. Members of the Montgomery County school board in Maryland added five storybooks for kindergarten through fifth grade in the school year 2022 to 2023. The books had LGBTQ characters: lesbian, gay, bisexual, transgender and queer, words for people who are attracted to people of their own sex, or whose sense of being male or female differs from the body they were born with. At first board members let parents take their children out of those lessons, and then they ended that choice.

Parents of several religions sued, among them Tamer Mahmoud and Enas Barakat, who are Muslims, and Jeff and Svitlana Roman and Chris and Melissa Persak, who are Christians. On June 27, 2025, in *Mahmoud v. Taylor*, six justices of the Supreme Court ruled that the parents were entitled to have their children excused while their case went on.

Justice Alito, writing for the six, relied on *Wisconsin v. Yoder*, the 1972 Amish case, and wrote that parents have a right "to direct the religious upbringing of their children." Justice Sotomayor wrote for the three who disagreed that public schools "offer to children of all faiths and backgrounds an education and an opportunity to practice living in our multicultural society."

The federal Department of Education, 2025 and 2026

President Donald Trump ordered his Secretary of Education in 2025 to begin closing the federal Department of Education, and in 2026 his officials announced that they would move its special education and civil rights offices to other departments. The people who work in that department send federal money to schools and enforce the laws against discrimination in them. Staff in the department, or paid by it, also collect the civil rights survey, the national test and the other federal school counts in this era.

Secretary of Education Linda McMahon announced layoffs on March 11, 2025, that would cut the department's staff from 4,133 people to 2,183. On March 20, Trump signed an order telling her to "take all necessary steps to facilitate the closure of the Department of Education and return authority over education to the States and local communities."

The order gave his reasons. It cited the 2024 national test, by which "70 percent of 8th graders were below proficient in reading, and 72 percent were below proficient in math." Below proficient includes children who reached the Basic level.

The court fight. A federal judge in Massachusetts blocked the layoffs on May 22, 2025, in a lawsuit later called *McMahon v. New York*. On July 14, 2025, a majority of the justices of the Supreme Court, in an order that no justice signed, let the layoffs go ahead while the case continued.

Justice Sotomayor, joined by Justices Kagan and Jackson, disagreed. She wrote that the case "arises out of the President's unilateral efforts to eliminate a Cabinet-level agency established by Congress nearly half a century ago," unilateral meaning done by one person alone.

In June 2026 officials announced that the office for special education would move to the Department of Health and Human Services. The Office for Civil Rights, which investigates discrimination in schools, would move to the Justice Department. *Education Week* reported in June 2026 that "only Congress can formally close a cabinet agency." By then members of Congress had passed no law closing it.

BY TODAY

If you remember one thing: After 2000 more families left the local public school for charter schools, private schools and teaching at home, while the schools of the South grew more separate again.

- 2002: President Bush signs the law under which every child in grades 3 to 8 is tested each year.
- 2020: State and local officials close nearly every school building in the country.
- 2024: Federal investigators count at least 973 children who died at the government's Indian boarding schools, and President Biden apologizes.

Next: The chapter closing answers the three questions from the start of the chapter.

THE THREE QUESTIONS, ANSWERED

1. **When did school become free, and required, for every child?** It came in steps, not on one date. Horace Mann argued for free schools paid for by taxes from 1837, and Massachusetts lawmakers made school a legal duty in 1852. Every state had done the same by 1918, and disabled children won the right to a free public school only in 1975.
2. **Which children were kept out of school, and how did they get in?** Lawmakers barred enslaved people from learning to read,

and officials separated Black, Chinese and Mexican American children, turned disabled children away and took Native children far from home. Judges first backed keeping them out or apart, in Sarah Roberts's case in 1850, *Plessy* in 1896 and Martha Lum's case in 1927. Later judges let them in, from the Mendez case in 1946 and *Brown* in 1954 to the *Mills* case in 1972, and in 1975 members of Congress passed a law for disabled children and a law that let Native nations run their own schools.

3. **Is it still lawful for a teacher to hit a child?** Yes, in 15 states in 2026. The justices of the Supreme Court ruled in 1977 that the Constitution's ban on cruel and unusual punishment does not cover paddling at school, so state lawmakers have banned it one state at a time.

What changed: Native children learned from their elders before 1500, Massachusetts children had to be taught to read the Bible from 1642, and by 1975 every child in the country, disabled children included, had a right to a free public school.

FIVE DATES TO KEEP

April 23, 1635: Boston Latin School opens.

May 10, 1740: South Carolina's lawmakers make it a crime to teach an enslaved person to write.

1852: Massachusetts lawmakers pass the first law that makes children go to school.

October 6, 1879: The first children arrive at the Carlisle boarding school.

May 17, 1954: The justices of the Supreme Court rule in *Brown v. Board of Education*.

NOT KNOWN: How many children died at the federal Indian boarding schools is not known, because the investigators' own report states that its count of at least 973 is too low.

Where to go: The Williamsburg Bray School building, the oldest surviving schoolhouse for Black children in the United States, stands at Colonial Williamsburg in Virginia.

Next: Chapter 31, News and Communication. It asks how Americans learned what was happening, from messengers and the post to the press, radio, television and the feed.

Chapter 31: News and Communication

CHAPTER 31 OF 37 · NEWS AND COMMUNICATION · 10 ERAS · ABOUT 125 MINUTES

Again and again, from a runner with a knotted cord to a phone, a new way of carrying news in America has come with a fight over who may carry it and what it may say.

One copy of the first newspaper printed in the English colonies is known to survive, and it is in London. It is called *Publick Occurrences, Both Forreign and Domestick*. Benjamin Harris printed it in Boston on September 25, 1690. It has three printed pages and a blank fourth page, where readers could write more news before they passed the paper on. Four days later the governor and his council stopped the paper and ordered every copy handed back. There was never a second issue.

This chapter follows the news through ten eras. Watch two things as you go. The first is speed. In 1775 riders took five days and seven hours to carry the news of Lexington to Philadelphia. In 1860 the first Pony Express run took ten days to reach California. By the 1920s radio brought the news into homes as it happened. The second is who tried to stop the news, and who was kept from carrying it. Look for governors and judges, mobs that wrecked printing presses, a law that let only free white people carry the mail, and station managers who cut the news off the air.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Did the First Amendment stop the government from punishing people for what they printed?
2. Who was shut out of carrying and printing the news?

3. Why have so many local newspapers closed since 2005?

Five minutes? Read "The station that cut the news: WLBT, 1955 to 1971" in 1950 to 2000.

Can skip: "The telephone, 1876 to 1900", "Newspapers in Chinese, 1854 and 1855", "Pictures by wire, 1935", "Television begins, 1927 to 1941". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 4 MINUTES

News traveled on foot in this land before 1500. Runners carried messages from one nation to another, and nations that spoke different languages talked with their hands. What is known about how they did it comes from two places. One is the accounts that Native nations and Europeans gave later. The other is the old objects that archaeologists, scientists who study the past by digging, have found in the ground.

Runners and wampum

The Haudenosaunee have lived for a thousand years and more in what is now central New York. They are also called the Iroquois, or the Six Nations. Five nations joined together before Europeans came, and the Tuscarora became the sixth in 1722.

The runners. Haudenosaunee runners called the nations to council, a meeting of their leaders. They also carried news from nation to nation and warned of danger. A runner carried a string of wampum, small beads cut from the white and purple parts of a clam shell, to show that his message was official. His pouch held pounded, dried corn mixed with maple sugar.

The Onondaga Nation keeps the wampum records of the Haudenosaunee. Its people describe how a council is called. Each nation receives a string of wampum that stands for the topic the leaders will meet about. A wooden stick is tied to the end of the string. Its notches count the days until the meeting. As each day passes, someone cuts a notch off the stick.

Older beads. Shell beads as old as 4,500 years have been found in the Northeast. Early bead makers drilled them with stone drill bits. Drilling with stone was hard, so their beads were larger than later wampum, and there were fewer of them. After Europeans arrived, Native bead makers drilled with nails and with iron awls, thin pointed tools for making holes.

Hand signs and trade

People in this land spoke many different languages. Near the Gulf coast and in what is now Texas, nations that could not understand each other's speech talked with their hands.

The first and fullest written accounts of this sign language come from the survivors of Cabeza de Vaca's Spanish expedition, which sailed in 1527. They noted which groups spoke the same language and which groups could understand each other only through signs. Scholars who study these accounts, among them Jeffrey E. Davis, think the sign language was in use across that region well before the Spanish came.

Trade goods moved hundreds of miles. Copper from Lake Superior and shells from the Gulf of Mexico have been found far from where they came from. No record survives of what news the traders passed along with the goods.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 5 MINUTES

Newcomers from Spain, France and England sent their news home across the ocean in letters and printed books, and each side told its own version. On the plains, Native people made themselves understood to the Spanish in hand signs. On Roanoke Island, English colonists left a message carved in a post.

A letter to a king and a book by a survivor, 1565 and 1566

The news of the killings in Florida in 1565 reached Europe in two accounts. One came from the Spanish commander who ordered the killings, in a letter to his king. The other was a book by a French survivor.

The killings. The Spanish commander Pedro Menéndez de Avilés and his soldiers killed French Protestants, Christians who do not belong to the Catholic Church, at Fort Caroline in Florida in September 1565. The fort stood in the homeland of Timucua-speaking peoples. Menéndez's men then killed two more groups of Frenchmen at Matanzas Inlet, south of the fort. Both groups had given up before they were killed. Matanzas is the Spanish word for "slaughters." Menéndez gave his reason himself: the Frenchmen were Protestants.

The letter. Menéndez reported the killings to King Philip II of Spain in a letter dated October 15, 1565. By his own count, his men cut the throats of 132 men at the fort. The next day they cut the throats of 10 more, prisoners they had caught in the woods.

At the inlet, he wrote, the first group of Frenchmen gave up their weapons. "I had their hands tied behind them and put them to the knife." He meant that his men killed them with knives. He spared sixteen men of that group: twelve sailors, and four craftsmen he said he needed.

How many. The National Park Service counts about 245 Frenchmen killed at the inlet, among them their leader, Jean Ribault, on October 11 or 12. By these counts, Menéndez's men killed nearly 390 Frenchmen in about three weeks. Chapter 22, War and the Military, describes the killings in full.

The book. Nicolas Le Challeux, a carpenter who escaped from Fort Caroline, wrote the French account. A printer in Dieppe, France, published it in 1566. Its title describes the killings as the cruelty of the Spaniards against the French king's subjects. Le Challeux told how he fled from the fort into the woods. Two of the men with him climbed a tall tree and signaled to a small French ship, and a boat came to pick them up. The survivors then sailed home to France.

Hand signs on the plains, 1541

Francisco Vázquez de Coronado and his Spanish expedition met a Native nation on the plains of what is now the Texas Panhandle in 1541. The Spaniards called them Querechos. Pedro de Castañeda, a member of the expedition who later wrote its history, recorded that "although they conversed by means of signs they made themselves understood so well that there was no need of an interpreter," someone to put one language into another.

A message carved in a post, Roanoke, 1587 to 1590

John White left 117 English colonists on Roanoke Island, in what is now North Carolina, in 1587 and sailed to England for supplies. The island and the coast around it were the land of peoples who spoke Algonquian languages, a family of related languages. Among them were the Roanoke, the Secotan and the Croatoan.

The signal. Before White left, he and the colonists agreed on a signal. If they moved, they would carve the name of their new place on a tree or a doorpost. If they left because they were in danger, they would carve a cross above the name.

White could not sail back for three years. England's queen took every English ship for her navy in 1588, to fight the Spanish Armada, a great fleet of warships sent from Spain. When White landed on August 18, 1590, the houses were gone and a fence of posts stood around the empty site. The letters CROATOAN were carved on one post "in fayre Capitall letters," old spelling for neat capital letters. The letters CRO were cut into a tree. There was no cross.

White wrote that Croatoan was "the place where Manteo was borne," and he called the people of that island "our friends." Manteo was an Algonquian man who had learned English and interpreted for the English. A hurricane blew White's ship out to sea before he could search there. White never found the colonists.

Books for Europe. Readers in Europe learned about this coast from printed books. Thomas Harriot came to Roanoke Island with the English in 1585, and he and John White explored and mapped the area. Harriot published *A Briefe*

and True Report of the New Found Land of Virginia in London in 1588. Two years later the engraver Theodor de Bry, who cut pictures into metal plates for printing, printed it again in four languages. De Bry made his pictures from John White's watercolor paintings.

Manteo

Who: An Algonquian man, born on Croatoan by John White's account, who learned English and interpreted for the English.
When and where: The Outer Banks of North Carolina, and London, 1584 to 1590.

He crossed to London in 1584 and came home speaking English.

The English captains Philip Amadas and Arthur Barlowe spent six weeks on the Outer Banks in 1584. When they sailed home, Wingina, a leader of the Algonquian people there, chose two men to go with them. They were Manteo and Wanchese.

Both men began to learn English from the sailors during the voyage. The ship reached England in mid-September. In London they met Sir Walter Raleigh, who sent the voyages. By the end of 1584 Manteo could speak enough English to interpret. He sailed home on April 9, 1585, on the *Tyger*, the lead ship of Raleigh's fleet.

Back on Roanoke. Manteo helped the colonists of 1587 set up a village near the old English fort on Roanoke Island. On August 13 that year the English baptized him into their church, a Christian ceremony with water that made him a member. They also named him Lord of "Dasamongueponke." When White came back in 1590, he found neither the colonists nor Manteo.

NOT KNOWN: The National Park Service's account of his life says that what happened to Manteo is unknown.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 8 MINUTES

The men who governed the English colonies in the 1600s decided who could print news, and they stopped printers they had not approved. Virginia's governor, Thomas Culpeper, and his council, a small group of men who advised him, ordered the printer William Nuthead to stop printing in 1683. Seven years later, in Massachusetts, Governor Simon Bradstreet and his council shut down the first newspaper printed in the English colonies.

The colonies also got their first mail service, a rider who carried letters from New York to Boston. In New Mexico in 1680, Pueblo runners carried the plan for a revolt against Spanish rule from town to town.

Letters by ship, by tavern and by planter

Most letters in the early colonies crossed the ocean, between the colonies and the countries in Europe that ruled them. In 1639 the lawmakers of Massachusetts, who met as the General Court, named Richard Fairbanks's tavern in Boston as the place to leave and collect letters sent "beyond the seas." Inside the colonies, people sent letters with friends, with merchants and with Native messengers.

Virginia officials around the middle of the 1600s wrote "for public service" on the outside of government letters. Virginia's rulers required each planter to send such a letter on to the next plantation. A planter who failed to do it had to give up a hogshead, a large barrel, of tobacco as a fine. On the big plantations of the South, Postal Service historians write, the messengers were usually enslaved people.

The first post rider, 1673, and a mail service for all the colonies, 1692

The first post rider, a man on horseback who carried the mail along a set route, left New York for Boston on January 22, 1673. The ride was about 250 miles, through New Haven, Hartford, Springfield and Worcester. Governor Francis Lovelace of New York set up the route with John Winthrop Jr. of Connecticut. The rider was to make the trip once a month. It took him about two weeks.

The monthly service did not last long. The rider's trail became known as the Boston Post Road, and part of it is now US Route 1. In 1683 William Penn set up the first post office in Pennsylvania.

A mail service for all the colonies. A man named Thomas Neale got the right in 1692 to run a mail service for all the colonies for 21 years. The right came from the English Crown, the king or queen of England and the royal government. Neale never came to America. Andrew Hamilton, the governor of New Jersey, ran the service for him.

Virginia: no printing allowed, 1671 to 1730

William Berkeley was the governor of Virginia. In 1671 he wrote to the king's officials in London: "I thank God, there are no free schools nor printing, and I hope we shall not have these hundred years." He gave his reason. Printing, he wrote, had spread new religious groups, disobedience and attacks on the government.

Printing stopped. A merchant named John Buckner brought a printer, William Nuthead, to Virginia in 1682. By February 1683 Nuthead had printed the laws that Virginia's lawmakers had just passed. Governor Thomas Culpeper and his council ordered both men to stop printing, and Nuthead moved on to Maryland.

The next governor, Francis Howard, arrived in 1684 with orders from King Charles II that barred printing in Virginia. The first printing press that Virginia's rulers allowed arrived in 1730, when William Parks opened a print shop in Williamsburg.

The first newspaper, 1690, closed after one issue

Printers in Massachusetts had printed single sheets of news called broadsides since 1667, one at a time, with no plan for a next issue.

Numb. 1. Benjamin Harris printed *Publick Occurrences, Both Forreign and Domestick* in Boston on September 25, 1690. It was the first newspaper in the English colonies. Harris marked it "Numb. 1," number one, and planned to print it once a month.

Four days. Governor Simon Bradstreet and the men of his council ordered the paper "Suppressed and called in" on September 29. That meant it was stopped, and every copy had to be handed back. They said Harris had printed it without their permission.

They also objected to what he had printed. Harris had accused Native allies of the English of "barbarous" cruelty to French prisoners. He had complained that colonial officials told the Mohawks, one of the Six Nations of the Haudenosaunee, too much about the colonists' war plans. The paper also carried a rumor that King Louis XIV of France was having sex with his son's wife.

Bradstreet and his council then ordered that no one could print news in the colony without a license, written permission from the government.

Benjamin Harris

Who: The London printer who printed the first newspaper in the English colonies.

When and where: London, then Boston, 1679 to 1695.

Three arrests in London came before his one Boston newspaper.

In London in the 1670s and 1680s Harris printed pamphlets, short printed booklets, for Protestants who had broken away from the Church of England. He was a Baptist, a member of one of those Protestant groups, and he spoke out against King Charles II.

Harris printed a newspaper without a license in 1679, *The Domestick Intelligence*. English officials jailed him twice. After a third arrest in 1685 he fled to the Netherlands. A year later he sailed to Boston and opened a print shop.

The paper. His opening notice in *Publick Occurrences* said he wanted to stop the "many False Reports, maliciously made, and spread among us." The paper reported a fire that burned about 20 houses. It also reported an outbreak of smallpox, a disease that spreads from person to person and kills many of the people who catch it. The governor and his council forbade Harris to print any newspaper in the colony again.

Harris obeyed. Massachusetts officials named him the colony's official printer in 1692. He went back to England in 1695 and started a London paper, the *London Post*, in 1699. He died in 1720.

Knotted cords and the Pueblo Revolt, 1680

Pueblo leaders in New Mexico planned a revolt in 1680 to drive out the Spanish, and runners carried the plan. Spanish soldiers and settlers had come in 1598 under Juan de Oñate to take Pueblo land, labor and food.

The arrests of 1675. The Spanish governor Juan Francisco Treviño ordered the arrest of 47 Pueblo religious leaders in 1675 for "practicing sorcery," using magic. Spanish officials sentenced four of them to hang. They hanged three, and the fourth man killed himself. They whipped the others in public and put them in prison. Pueblo people went to Santa Fe and demanded that the prisoners be let go, and the Spanish released them. One of the men released was Po'pay, a Tewa leader from Ohkay Owingeh.

The knotted cords. Po'pay planned the revolt with towns that spoke at least five different languages. All of them had to rise on the same day. To set that day, he sent runners with knotted cords. Each town untied one knot a day, and when the last knot was untied, the revolt would begin. The accounts differ on the day planned. The National Museum of the American Indian gives August 11, and the Indian Pueblo Cultural Center gives August 13.

The revolt began on August 10, 1680, earlier than planned, because the Spanish caught two of the runners. Pueblo fighters killed about 400 people, including 21 of the 33 Spanish priests in New Mexico. Governor Antonio de Otermín led about 2,000 Spanish settlers out of Santa Fe on August 21 and south to El Paso. The Spanish did not return until 1692.

Nicolás Catua and Pedro Omtua

Who: Two young runners from Tesuque Pueblo who carried the knotted cords of the Pueblo Revolt.

When and where: New Mexico, August 8 and 9, 1680.

Their cord had two knots, and the Spanish hanged them for it.

Pedro Omtua and Nicolás Catua set out on the morning of August 8, 1680, with cords for the Tano towns. They came from Tesuque Pueblo, whose Tewa name means "village of the narrow place of the cottonwood trees." The first town they reached was Pecos.

After they left Pecos, Christian Pueblo people told Fernando de Velasco that two Tewa messengers had visited the war chief's house. Velasco was a leader of the Franciscan missionaries, Catholic priests sent to turn the Pueblo people Christian. Two Pueblo leaders from the Galisteo Basin who sided with the Spanish also sent word of a revolt to Governor Antonio de Otermín. Otermín ordered the messengers arrested, and the Spanish caught them on August 9.

Tortured, then hanged. Spanish officials tortured Catua and Omtua to make them explain the knotted cord. Torture means hurting a person's body on purpose to force him to talk. The surviving accounts do not say how the two young men were hurt.

The historian Charles W. Hackett worked from Otermín's own records. He wrote in 1911 that the two told Otermín they had been given a cord with two knots, to carry to the chiefs of the Tano towns, San Marcos and La Cienega. The two knots stood for the two days left before the revolt. Then the Spanish hanged them. The accounts do not name the men who tortured and hanged them.

When Po'pay heard that the runners had been caught, he ordered the revolt to start a day early.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 10 MINUTES

The first colonial newspaper that lasted began in Boston in 1704, and many of the men who printed the colonies' papers also ran the post office. Printers and postmasters started papers in Boston, Philadelphia, New York, Charleston and Williamsburg. Colonial governors and lawmakers still had printers jailed when the printers offended them. The owners made their money from advertisements, and some of those advertisements offered human beings for sale.

The Boston News-Letter, 1704

John Campbell, the postmaster of Boston, published the first issue of the *Boston News-Letter* on April 24, 1704. It was one small sheet printed on both sides, and it came out once a week. A printer named Bartholomew Green printed it for Campbell. The words "Published by Authority," meaning with the approval of the people in power, appeared in its title from 1704 to 1720, and again from 1723 to 1725.

Most of the first issue was copied from London newspapers of mid-December, more than four months old. Local news filled one column. The *News-Letter* came out every week until February 1776, the first newspaper in the colonies to keep going. Boston had no other newspaper until the *Boston Gazette* began on December 21, 1719. The *American Weekly Mercury* started in Philadelphia the next day.

The Courant: a paper the government did not approve, 1721 to 1726

James Franklin started the *New-England Courant* in Boston on August 7, 1721, during an epidemic of smallpox, an outbreak that spread to many people at once. He soon used the paper to take a side in a fight over how to stop the disease.

Inoculation. The minister Cotton Mather and the doctor Zabdiel Boylston urged people to try inoculation. A doctor put a little pus from a sick person into a cut in the skin of a healthy person. The healthy person usually got a mild case of smallpox and soon got well. Mather had learned of the method from Onesimus, an African man he had owned as a slave since 1706, and from other enslaved Africans.

Almost every doctor in Boston except Boylston opposed inoculation. James Franklin printed attacks in the *Courant* on the people who supported it. Someone threw a lit bomb through a window of Mather's house on November 14, 1721. A note with it read: "Cotton Mather, You Dog, Dam You: I'll inoculate you with this, with a Pox to you." The fuse came off, and the bomb did not explode. No record names the person who threw it.

Nearly 6,000 people caught smallpox in the epidemic, and 844 of them died. Boylston inoculated 248 people, and 98 out of every 100 of them lived. Among people who caught smallpox the ordinary way, 85 out of every 100 lived.

A month in jail. Pirates were raiding along the coast of New England in 1722. In June, James Franklin printed a short item in the *Courant* saying that Massachusetts officials were fitting out a ship to chase them. The ship, it said, would sail "some time this Month, if Wind and Weather permit." The item hinted that the officials were in no hurry to go after the pirates.

Massachusetts officials read it as a hidden attack on them. The members of the General Court, the colony's lawmakers, had James Franklin jailed for a month. His younger brother Benjamin, then 16, ran the paper while James was locked up. James was let out on July 2, 1722.

The members of the General Court then ordered censorship, checking and cutting what a printer may print before it comes out. Early in 1723 they forbade James Franklin to print the *Courant* unless officials checked each issue first. James then put the paper in Benjamin's name, and it came out that way until it closed in 1726.

The Weekly Journal and the Zenger trial, 1733 to 1736

For years New York had one newspaper, the *New-York Gazette*. Its printer, William Bradford, printed for the government and supported Governor William Cosby. In 1733 Lewis Morris, the chief judge of New York, wrote a dissent, a judge's written disagreement with the other judges, in a court case of Cosby's called Cosby v. Van Dam. Cosby removed Morris from office.

A second paper. Morris and two lawyers, James Alexander and William Smith, then started the *New-York Weekly Journal* on November 5, 1733. Alexander edited it and used it to accuse Cosby and his officials of tyranny, cruel and unjust rule. John Peter Zenger, a printer born in Germany, printed it. Cosby had copies of the *Journal* burned by the hangman.

Arrested for libel. Officials arrested Zenger on November 17, 1734, for seditious libel. Under the law of that time, seditious libel meant printing blame of any official, of the law, or of anything the law had set up, without a lawful excuse. Whether the words were true did not matter.

The court set Zenger's bail, the money paid to a court so that a prisoner can go free until his trial, at £400. Zenger stayed in jail for more than eight months, until his trial.

Not guilty. At the trial, the judge would not let Zenger's lawyer prove that the printed words were true. The jurors, the judge said, must decide only whether Zenger had printed the papers. Zenger's lawyer told them they had the right to judge the law as well. The jurors went further than the judge allowed. They found Zenger not guilty on August 4, 1735. Chapter 20, Government, Law, and Politics, describes the trial in full.

The verdict changed no law. In New York, a printer could not defend himself by showing that his words were true until 1805, when the state's lawmakers passed a law that allowed it. An account of the trial came out as a pamphlet in 1736, *A Brief Narrative of the Case and Tryal of John Peter Zenger*. Printers reprinted it many times.

Anna Catharina Zenger

Who: The printer's wife who kept the *New-York Weekly Journal* coming out while her husband was in jail, and who ran the print shop after he died.

When and where: New York City, 1734 to 1748.

Instructions came, the story goes, through a hole in the jail door.

Anna Catharina Zenger was born Anna Catharina Maulin, and she married John Peter Zenger on September 11, 1722. When officials jailed him, she took charge of the shop. She and his apprentices kept the *New-York Weekly Journal* coming out while he was held. They missed only one issue. The writer Ann T. Keene reports the story that Zenger gave her his instructions "through the hole of the door of the prison."

After the trial. Zenger became New York's official printer in 1737. He died on July 28, 1746. Anna Zenger then ran the family print shop for more than two years, until their eldest son, John, took it over in December 1748.

Printers who were postmasters, and papers in German

A printer who was also the postmaster got the letters and newspapers from other towns first, and so got the news first. He could also stop the post riders from carrying a rival's newspaper.

Bradford and Franklin. Andrew Bradford printed the *American Weekly Mercury* in Philadelphia and also kept the town's post office. He forbade the post riders to carry Benjamin Franklin's paper. Franklin wrote that he got his papers carried only by bribing the riders to take them in secret.

Alexander Spotswood, a former governor of Virginia, was in charge of the post office for the colonies. He was unhappy with how carelessly Bradford kept the post office's accounts, and in 1737 he took the job away from Bradford and gave it to Franklin, who was 31.

Franklin also set up partners as printers in other towns. He made Lewis Timothy the printer of the *South-Carolina Gazette* in Charleston in November 1733. In Virginia, William Parks printed the first issue of the *Virginia Gazette* in Williamsburg on August 6, 1736. Virginia's lawmakers allowed only one printing office in the colony, and they wanted it to print what the government needed first.

Papers in German. Franklin tried a German-language paper, the *Philadelphische Zeitung*, in May and June 1732, and it failed. Christopher Sauer, a German printer in Germantown, started a German-language paper on August 20, 1739. His family printed it, under changing names, until 1777. Each month sixteen newspapers from Europe reached Sauer, and he took his news of Europe from them.

Elizabeth Timothy

Who: A widow born in Holland who ran the *South-Carolina Gazette*. The *South Carolina Encyclopedia* names her the first woman in the American colonies to own and publish a newspaper.

When and where: Charleston, South Carolina, 1738 to 1746.

A widow with six children, she owned the paper within a year.

Elizabeth Timothy's husband, Lewis Timothy, printed the *South-Carolina Gazette* in Charleston under a six-year partnership with Franklin. He died in December 1738 after what was described at the time as "an unhappy Accident," meaning an unlucky one. He left her with six small children and a seventh about to be born. Franklin wrote that she was "born and bred in Holland."

Their eldest son, Peter, was too young to run the shop, so Elizabeth Timothy ran it herself. She printed the next issue on January 4, 1739, and promised to make the paper "as entertaining and correct as may be reasonably expected." By the end of 1739 she had bought out Franklin's share. Franklin praised how exactly she kept her accounts.

She handed the shop to Peter when he came of age in 1746 and opened a bookstore next to it. She died in 1757. In her will she left her children eight enslaved people, along with her other property.

Advertisements for people as property

Newspaper owners made much of their money from advertisements, and some of those advertisements were for human beings. Franklin printed advertisements in the *Pennsylvania Gazette* that slave owners and slave traders paid for. Some offered enslaved people for sale. Others were notices about enslaved people who had escaped, so that readers would help catch them. According to the International Printing Museum, by the 1750s these made up nearly a quarter of the paper's advertisements.

Franklin himself owned enslaved people. According to the Benjamin Franklin House museum in London, he and his household held enslaved people from about 1735 until 1790 and bought at least seven people. Their names were Joseph, Jemima, Peter, King, Othello, George and Bob. Peter and King went with Franklin to London.

Enslaved people had no say in how owners and traders described them in those advertisements. In South Carolina, lawmakers made it a crime in 1740 even to teach an enslaved person to write.

Benjamin Franklin

Who: A Boston printer's apprentice who became a Philadelphia newspaper owner and the town's postmaster.

When and where: Boston, 1718 to 1723, and Philadelphia, 1723 to 1753.

Movie: *Benjamin Franklin* (2022), a documentary by Ken Burns about his life, shown on PBS.

At 17 he ran away from his brother's print shop.

Benjamin Franklin grew rich as a printer, and much of his money came from the advertisements in his newspaper. In 1718 he became an apprentice, a young worker bound to a master for years to learn a trade, to his older brother James, a printer in Boston. Benjamin carried copies of the *New-England Courant* through the streets to the people who paid for it. In 1722, at 16, he wrote letters to the paper under the made-up name "Silence Dogood." He later wrote that his brother "had often beaten me."

Running away. Benjamin broke his apprenticeship and ran away in 1723, at 17. He landed at the Market Street wharf in Philadelphia on a Sunday morning.

The Gazette. Franklin bought the *Pennsylvania Gazette* from the printer Samuel Keimer on October 2, 1729. Advertising became the paper's main source of money, and Franklin grew the paper from two pages to four, partly to hold more advertisements.

He began *Poor Richard's Almanac*, a book that came out once a year with a calendar and other useful facts, in 1732. He wrote it under the made-up name Richard Saunders. He said that he sold "near ten thousand" copies a year, and that ordinary people "bought scarcely any other books."

The post office. As postmaster of Philadelphia, Franklin wrote that the post office "facilitated the correspondence that improv'd my newspaper." He meant that the job helped him get the letters and news that made his

> paper better, and with them more readers and more advertisements. British officials named him and William Hunter in 1753 to run the royal post office for all the colonies together.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 16 MINUTES

Printers, letter writers and post riders carried the news and the arguments that led to the American Revolution. Supporters of independence also wrecked the press of a New York printer who took Britain's side. Once the war was won, the new country's leaders fought over how free the press should be.

Lawmakers in three-fourths of the states approved the First Amendment to the Constitution in 1791, and it protects freedom of the press. Seven years later the Federalists, President John Adams's party, used their majority in Congress to pass the Sedition Act. Federal judges then jailed newspaper editors who had criticized Adams.

The Stamp Act and the printers, 1765

The members of Parliament in London, Britain's lawmakers, passed the Stamp Act in March 1765. It took effect on November 1, 1765. Under it, colonists had to pay a tax on newspapers, pamphlets, almanacs and legal papers. Printers had to print on stamped paper made in London, which showed that the tax was paid.

The tax. Colonists paid a halfpenny, a small British coin, on each copy of a newspaper printed on half a sheet of paper, and a penny on each copy printed on a whole sheet. They paid two shillings, or 24 pennies, on each advertisement in a newspaper.

Black borders. The Philadelphia printer William Bradford printed his *Pennsylvania Journal* on October 31, 1765, the day before the tax began, with thick black borders, like a tombstone. In the place for the official stamp he printed a skull and crossbones. Across the top of the front page he printed that the paper was "EXPIRING," or dying, "In Hopes of a Resurrection to LIFE again."

This William Bradford was born in 1722 and was a different man from the New York printer of the same name. He never actually stopped printing. He put out the next issue on November 7.

John Hughes had the job of handing out the stamps in Philadelphia. Opponents of the tax threatened him with violence and with destroying his property, and he promised not to enforce the act. The members of Parliament voted in 1766 to repeal, or cancel, the Stamp Act.

Committees of correspondence, 1772 to 1774

Samuel Adams proposed a committee of correspondence for Boston, a small group of men chosen to write letters to other towns and share news. Boston's town leaders voted on November 2, 1772, to set up a Committee of Correspondence of 21 men. They did it in answer to news that the colony's judges would now get their pay from the royal treasury, the king's own money.

The 21 men wrote a set of reports on the colonists' rights and on how Parliament had broken them. Printed as the Boston Pamphlet, the reports went to every town in Massachusetts.

Virginia's committee. Dabney Carr asked the members of Virginia's House of Burgesses, the colony's elected lawmakers, to set up an 11-man committee on March 12, 1773, and they did. Patrick Henry and Thomas Jefferson were two of its members.

The 11 men wrote to the lawmakers of the other colonies. They asked them to share news of laws from London and of what each colony's leaders planned to do about them. Within a year, lawmakers in eleven colonies had set up their own committees. North Carolina and Pennsylvania were the two that did not.

The news of Lexington, April 1775

British soldiers fired on the militia, ordinary men trained to fight when they were called, at Lexington, Massachusetts, on the morning of April 19, 1775. At about 10 o'clock that morning, in Watertown, Joseph Palmer wrote a letter "To all the Friends of American liberty." It said the soldiers had killed six men and wounded four.

Palmer was a member of the Massachusetts Committee of Safety. The colonists' own congress had set up this committee to gather supplies for a fight with the British army in Boston and to spread the alarm if the army marched.

Copy by copy. Palmer gave the letter to a rider, Isaac Bissell, to carry toward Connecticut. At Worcester the town clerk, Nathan Baldwin, copied the letter, wrote that the copy was true and signed it. Someone made another copy at Springfield. Other riders then carried copies on, and the alarm reached Philadelphia five days and seven hours after it left Watertown.

Palmer took care to make his letter believable. Seven months earlier, in September 1774, thousands of militiamen had turned out for nothing after false rumors of a British attack.

Isaac Bissell

Who: A 26-year-old blacksmith from Suffield, Connecticut, who carried a written report of the killings at Lexington from Massachusetts toward Connecticut.

When and where: From Watertown, Massachusetts, to Connecticut, April 19 and 20, 1775.

His name was Isaac, not Israel, and he stopped in Connecticut.

Isaac Bissell came from Suffield, a town on Connecticut's northern border. He was in Watertown, Massachusetts, on April 19, 1775, when Joseph Palmer handed him the letter. The letter said that "the Bearer Mr. Isaac Bissell" was "charged to alarm the country quite to Connecticutt." His job was to warn people all the way to Connecticut. The letter asked people along the way to give him the horses he needed.

Bissell reached Worcester in the early afternoon. On the morning of April 20 the letter reached Connecticut's leaders.

Too fast to believe. In Wethersfield, the home town of Governor Jonathan Trumbull, Silas Deane wrote in his journal that the news "was discredited," not believed, "from the suddeness of its arrival." After men from Massachusetts confirmed the news, Trumbull agreed to call Connecticut's lawmakers together.

The unpaid bill. Bissell then joined the siege of Boston, where colonial soldiers surrounded the British army in the town. He served as a private in Captain Elihu Kent's company from Suffield. In July 1775 the members of the Massachusetts committee of safety approved his bill as a post rider, two pounds and one shilling. The colony's treasurer then would not pay it without a signed approval from the colony's congress.

The story that grew. Many books call the rider "Israel Bissell" and say he rode all the way to Philadelphia. The historian J. L. Bell found that his name was Isaac, that he carried the letter only as far as Connecticut, and

that the long ride to Philadelphia is a story that grew up later. Many riders, most of them unnamed, carried the news the rest of the way.

Common Sense and the Declaration in print, 1776

The printer Robert Bell published *Common Sense* on January 10, 1776. It was a pamphlet by Thomas Paine that argued for independence from Britain. Printers printed it at least twenty-five times, sixteen of them in Philadelphia.

The claims. Paine claimed in April 1776 that 120,000 copies had sold, and later he claimed 150,000. Moncure Conway, who wrote a book about Paine's life, guessed half a million in 1892.

NOT KNOWN: No sales records survive, so nobody knows the real number. The historian Trish Loughran put the highest possible number at 75,000, and she thinks the real number was much lower.

The Declaration. Members of the Continental Congress, the meeting of men sent by each colony to act for all of them, declared independence in Philadelphia on July 4, 1776. Over that night the Philadelphia printer John Dunlap printed about 200 copies of the Declaration of Independence as broadsides.

The members of Congress ordered the copies sent to the new states and to the soldiers, read aloud, and put up in public places. George Washington's soldiers in New York City heard it read aloud on July 9. Twenty-six of Dunlap's copies are known to survive.

Thomas Paine

Who: An English-born writer whose pamphlet *Common Sense* argued that the colonies should declare independence.

When and where: Philadelphia, 1774 to 1776, and with the army, 1776.

He promised his share of the profits to buy mittens for soldiers.

Thomas Paine met Benjamin Franklin in London, and Franklin gave him letters of introduction to people in America. Paine arrived in Philadelphia on November 30, 1774. He helped edit the *Pennsylvania Magazine* and wrote articles under pen names, made-up names that hid who the writer was.

The slavery essay. For a long time Paine was named as the writer of "African Slavery in America," an essay printed in March 1775 under the name "Justice and Humanity." The essay calls slavery unjust. Gary Berton of the Thomas Paine National Historical Association later concluded that a Rhode Island minister, Samuel Hopkins, most likely wrote it, so the accounts differ.

Common Sense. The pamphlet came out on January 10, 1776. Paine agreed to pay for the printing whether it sold or not, and the printer, Robert Bell, was to get half of any profits. Paine promised his own half to buy mittens for the American soldiers outside the city of Quebec, in Canada, that winter. Bell handed over no money.

Then Bell printed a second edition that Paine had told him not to print. Paine took a new, longer edition to the printers of the *Pennsylvania Journal* instead. He never received any money from the sales. He later wrote that he gave the right to print it to every state.

Paine published *The American Crisis* on December 19, 1776. It began, "These are the times that try men's souls." Washington ordered it read to all his troops. Paine died in New York City on June 8, 1809.

A Loyalist printer's press wrecked, 1775

Colonists who stayed loyal to Britain were called Loyalists, and supporters of independence attacked the printers among them. The New York printer James Rivington printed the *New-York Gazetteer*, one of the most widely read newspapers in the colonies, and he used it to back the British government.

The press wrecked. Isaac Sears led a group of Sons of Liberty, colonists who organized against British taxes and rule, from New Haven, Connecticut, to Rivington's shop in New York City in November 1775. The men wrecked Rivington's press and his type, the set of metal letters that a printer puts together by hand to print a page. Type was a printer's most valuable possession. Rivington left for England a few weeks later.

After British soldiers captured New York, Rivington came back with the title of King's Printer, the official printer for the British government there. He printed his paper on the British side until the British left in November 1783.

A postal service for the new country, 1774 to 1792

British officials fired Benjamin Franklin from his job running the royal post office in the colonies in 1774. In the words of the Postal Service's own history, he was "too much of an American."

A rival post. The printer William Goddard set up a rival mail service, which he called the Constitutional Post. He warned that letters could be stopped and opened on the orders of the king's ministers, the men who ran Britain's government. What the letters said, he warned, could be twisted into proof of treason, the crime of betraying one's own government. By 1775 Goddard's service had nearly 50 post offices from Maine to Virginia.

The members of the Continental Congress made Franklin their postmaster general, the head of their post office, on July 26, 1775, at $1,000 a year.

Post roads. The Constitution, approved in 1788, lists among the powers of Congress the power "To establish Post Offices and post Roads." In 1789 the country had 75 post offices and about 2,400 miles of post roads, for almost four million people. The members of Congress passed a new law about the mail

on February 20, 1792. Under it, printers could send newspapers through the mail at very low rates, some for free. The law also forbade postal officials to open people's letters.

Mary Katharine Goddard

Who: A printer and newspaper publisher who ran Baltimore's post office for 14 years and printed the first copies of the Declaration that listed the signers.
When and where: Baltimore, Maryland, 1774 to 1789.

Readers could pay for her newspaper in beeswax or pork.

Mary Katharine Goddard learned printing in her brother William's shop in Providence, Rhode Island, in the 1760s. She moved to Baltimore in 1774. She ran his print shop and his newspaper, the *Maryland Journal*, during his many trips away, and she reprinted *Common Sense* in it.

Postmaster. She became the postmaster of Baltimore in 1775. Postal Service historians call her the first known woman postmaster in the colonies. Postmasters then earned a share of the postage instead of a salary. The members of Congress often failed to pay her, so she paid the post riders and the high rent of the office herself. In 1778 she let readers pay for the newspaper in beeswax, flour, lard, butter, beef or pork.

The Declaration with names. The Continental Congress was meeting in Baltimore in January 1777. Its members ordered new copies of the Declaration of Independence, and Goddard printed them on January 18. Hers were the first printed copies to show the signers' names. At the bottom she printed "Baltimore, in Maryland: Printed by Mary Katharine Goddard." Nine of her copies are known to survive.

Pushed out. Her brother William "apparently forced his sister out" of the *Maryland Journal* in January 1784, Smithsonian Magazine reports. The new postmaster general, Samuel Osgood, replaced her with a man in October 1789. Osgood's assistant, John Burrell, said the job might need "more travelling ... than a woman could undertake."

> More than 200 people in Baltimore signed a letter asking Osgood to give her the job back. She wrote to President George Washington and to the Senate, and neither helped her. She ran a bookshop in Baltimore until about 1810, and she died there on August 12, 1816.

The First Amendment and the party papers, 1789 to 1793

The First Amendment, an addition to the Constitution, took effect on December 15, 1791. It states that "Congress shall make no law ... abridging," or cutting back, "the freedom of speech, or of the press."

Party papers. Newspaper editors took sides in the quarrels inside Washington's government. John Fenno used his *Gazette of the United States* to push the plans of Alexander Hamilton, the head of the Treasury. Secretary of State Thomas Jefferson called Fenno's paper "a paper of pure Toryism," using the old name for colonists who had sided with Britain's king.

Jefferson hired Philip Freneau as a translating clerk at the State Department, paid $250 a year, even though Freneau knew only English and French. Freneau started the *National Gazette* on October 31, 1791. It came out twice a week and had about 1,700 subscribers, people who paid to get each issue. Freneau closed it on October 26, 1793.

Benjamin Franklin's grandson, Benjamin Franklin Bache, printed a paper in Philadelphia called the *Aurora*. In it he attacked Washington and John Adams.

The Sedition Act, 1798 to 1801

American and French warships began fighting each other in the Caribbean Sea in 1798. President John Adams's party, the Federalists, held a majority in Congress. The Federalist members voted through the Sedition Act, named for sedition, the crime of stirring people up against their government. Adams signed it on July 14, 1798.

What it banned. Under the act, it was a crime to write or print "any false, scandalous and malicious writing" against the federal government, Congress or the President. The punishment was a fine of up to $2,000 and up to two years in prison. Unlike the law in the Zenger trial, the act let a person charged try to prove that his words were true, and the jury could decide what the law meant as well as what the facts were.

Who was charged. Federal prosecutors and judges used the act mostly against editors of newspapers that supported Jefferson's party, the Democratic-Republicans. The counts of people found guilty differ. The lowest is 10, the number scholars long used. The highest, from the National Constitution Center, a museum about the Constitution, is more than two dozen. The legal historian Wendell Bird found cases against 16 people that the old count missed.

Some of those charged. They included Congressman Matthew Lyon of Vermont, William Duane of the *Aurora*, Ann Greenleaf of New York's *Argus*, and Anthony Haswell of the *Vermont Gazette*. Luther Baldwin and Brown Clark of Newark, New Jersey, were charged for words they spoke against President Adams in a tavern.

Bache. Federalist supporters physically attacked Benjamin Franklin Bache in Philadelphia. Jonathan Dayton of New Jersey, a Federalist, was Speaker of the House of Representatives, the member chosen to run its meetings. Dayton barred Bache from the floor of the House, the room where its members met, so that he could not report its debates.

A federal judge in Philadelphia, Judge Peters, signed an order for Bache's arrest on June 24, 1798, more than two weeks before the Sedition Act became law. The charge was seditious libel against President Adams, the same crime that officials had charged Zenger with in 1734. Peters let Bache out on bail, and his trial was set for October.

Bache kept printing the *Aurora* through an epidemic of yellow fever, a disease carried from person to person by mosquitoes. He died of the fever on September 10, 1798, at 29, a month before his trial date. His widow kept the paper going, with William Duane as editor. The text of the Sedition Act stated that it would stay in force only until March 3, 1801.

Matthew Lyon

Who: An Irish-born congressman and newspaper owner, the first person tried under the Sedition Act.
When and where: Vermont, 1798 to 1799.

He ran for Congress from a jail cell and won.

Matthew Lyon came to America from Ireland and fought in the Revolutionary War. He went to Congress for Vermont in 1797, and he also owned a newspaper. In 1798 he became the first person tried under the Sedition Act.

His words. In a letter that a newspaper printed, Lyon wrote of seeing the public good "swallowed up in a continual grasp for power, in an unbounded thirst for ridiculous pomp, foolish adulation, or selfish avarice." Adulation is an old word for too much praise, and avarice for greed. At campaign meetings he also read aloud a letter by the poet Joel Barlow, which said Adams should be sent to a "madhouse."

The trial. A federal grand jury in Vermont, a group of citizens who decide whether there is enough evidence to charge someone, charged Lyon with three crimes under the Sedition Act on October 5, 1798. A deputy marshal, an officer who made arrests for the federal courts,

arrested him the next day. At his trial in Rutland on October 9, Lyon defended himself. He pleaded not guilty, and in a second plea he stated that the Sedition Act broke the Constitution.

Justice William Paterson of the Supreme Court told the jury that its job had "nothing whatever to do with the constitutionality or unconstitutionality of the sedition law." He meant that the jurors must not decide whether the law broke the Constitution. The jurors came from towns that supported Adams's party. They found Lyon guilty within an hour. Paterson sentenced him to four months in jail and a $1,000 fine.

Lyon was held in the jail at Vergennes. He ran for Congress again from his cell, and the voters of his district re-elected him. He was let out on February 9, 1799.

BY 1800

If you remember one thing: Printers and riders carried the news of the Revolution, and seven years after the First Amendment took effect, federal judges were jailing editors under the Sedition Act.

- April 19, 1775: Joseph Palmer's letter about Lexington starts a chain of riders that reaches Philadelphia five days and seven hours later.
- December 15, 1791: The First Amendment takes effect.
- October 9, 1798: A jury finds Congressman Matthew Lyon guilty, the first person tried under the Sedition Act.

Next: By 1850 the country had 2,526 newspapers, and Black and Cherokee editors had started papers of their own.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 17 MINUTES

The fight over slavery reached into the print shops and the post office as newspapers spread across the country. The number of American newspapers grew from about 900 in the 1830s to 2,526 in 1850. That year they printed half a billion copies, for a population of about 23 million, or about 22 copies for every person.

A New York printer began selling a daily paper for one cent in 1833, when most daily papers cost six cents. From 1844 news could travel along a telegraph wire, faster than any horse or train.

Free Black New Yorkers started the first Black-owned newspaper in 1827, and Cherokee leaders started the first Native American paper a year later. Men who defended slavery attacked the editors who wrote against it. Men of the Georgia Guard, a band of armed men from the state of Georgia, seized the Cherokee press in 1835. That same year Amos Kendall, the Postmaster General, who ran the post office, let Southern postmasters hold back antislavery papers. In Illinois in 1837, a mob killed an editor who was defending his press.

The last Sedition Act trials, 1800

The Sedition Act was still the law in 1800, and federal judges used it that year against two more writers. Under the act it was a crime to print "false, scandalous and malicious" writing against the government, Congress or the President.

Callender. A grand jury, a group of citizens who decide whether there is enough proof for a trial, charged the writer James Callender in the federal court at Richmond, Virginia, on May 24, 1800. The charge was over his pamphlet *The Prospect Before Us*, in which Callender wrote words that harmed the good name of President John Adams. The trial began on June 3, and the jury found him guilty the same day.

Justice Samuel Chase sentenced Callender to nine months in prison and a fine. The Federal Judicial Center's history of the trials states the fine as $400 in one place and $200 in another.

Cooper. In Philadelphia the same year, Chase sentenced the writer Thomas Cooper to six months in prison. The same history gives Cooper's fine as $400 in one place and $500 plus court costs in another.

The act ended on March 3, 1801. After Thomas Jefferson became president that year, he pardoned, or officially forgave, Callender. Jefferson also paid $50 of his own money toward giving back Callender's fine.

Newspapers by mail, and who could carry it

Newspapers went to readers through the mail. Under the postal law of 1792, readers got newspapers by mail at very low rates, and editors mailed their papers to each other for free. A small-town editor could then reprint news from papers all over the country. Indiana had one newspaper in 1810 and 73 by 1840.

More than Europe. In 1840 the United States had about 17 million people, and Europe had about 230 million. Even so, more copies of American newspapers went out each week than of all the newspapers in Europe.

Most people who worked for the federal government worked for the post office. By 1831, 76 of every 100 federal workers outside the army and navy were postal workers. Postmasters outnumbered soldiers, 8,764 to 6,332.

Letters cost much more to mail than newspapers. From 1799 to 1815, one sheet of paper cost 8 cents to send 40 miles or less, and up to 25 cents to send more than 500 miles. A second sheet doubled the price. In 1845 postal officials began charging by weight instead, and in 1847 they began selling postage stamps.

"Free white persons." Some enslaved men carried the mail in the South until 1802. That March, Postmaster General Gideon Granger warned a senator that enslaved post riders, "hourly mixing with people," would "acquire information." "They will learn that a man's rights do not depend on his color," he wrote. "They will, in time, become teachers to their brethren," meaning their brothers.

Granger began his letter by pointing to "the scenes which St. Domingo has exhibited to the world." On Saint-Domingue, a French island colony in the Caribbean, the enslaved people had risen in a revolution.

Members of Congress then passed a law, and from 1802 to 1865 the postal laws said that mail carriers had to be "free white persons." A mail contractor, a person paid to carry the mail, who broke the rule had to pay a fine. The Postal Service's own history of Black postal workers names no one who was fined. For 63 years, no Black American, free or enslaved, could legally carry the mail.

Freedom's Journal, the first Black-owned newspaper, 1827

A group of free Black New Yorkers met in March 1827 to start a newspaper of their own. It was the first newspaper in the United States owned and run by Black Americans. Samuel Cornish and John Brown Russwurm edited it. Cornish was a Presbyterian minister, a preacher in one of the Protestant churches. Russwurm was one of the first Black Americans to graduate from a college in the United States.

Freedom's Journal came out on March 16, 1827, from 5 Varick Street in Manhattan. In the first issue the editors wrote: "We wish to plead our own cause. Too long have others spoken for us." Editors of other New York papers printed attacks on Black people, and Cornish and Russwurm answered them. They also wrote against slavery.

Readers in 11 states. A year of the paper cost $3. More than 800 copies went out to readers in 11 states and Washington, D.C., and copies also went to Canada, Europe and Haiti. David Walker was one of the agents who sold subscriptions to it.

Russwurm came to support a plan called colonization, which meant sending free Black Americans to live in Africa. Cornish opposed the plan and quit in September 1827. The last issue of *Freedom's Journal* came out on March 28, 1829. Russwurm moved to Liberia, in Africa, and started a newspaper there.

The Cherokee Phoenix, 1828 to 1835

The *Cherokee Phoenix* was the first newspaper printed by a Native American nation. The members of the Cherokee Nation's council set aside $1,500 in 1825 for a printing press and type. Samuel Worcester, a missionary, or Christian preacher sent to win people to his faith, helped the Cherokee set up a print shop at New Echota, their capital. New Echota was in land that the state of Georgia claimed. An iron press came by ship from Boston.

Type for Sequoyah's letters. Workers cast type for the Cherokee syllabary, the writing system that Sequoyah made, in which each of 85 symbols stands for a syllable. The first issue came out on February 21, 1828. Columns in Cherokee ran beside columns in English. The Cherokee leaders who ran the paper opposed every plan to move the Cherokee west, and they used it to argue their side.

Georgia and federal officials made the paper hard to keep going. Colonel Charles Nelson commanded the Georgia Guard, a unit that Georgia officials set up to police the Cherokee land their state claimed. He threatened to shut the paper down and to beat its editor.

The money cut off. President Andrew Jackson ordered the federal government's yearly payment to the Cherokee Nation paid to separate people instead of to the Nation. That cut off the paper's money. Georgia officials ordered the paper's white printer to leave. A local sheriff sued the editor, Elijah Hicks, for libel, printing false words that harm someone. The last issue came out in May 1834.

John Ross, the Principal Chief, the Cherokee Nation's top leader, planned in 1835 to start the paper again. Men of the Georgia Guard went to New Echota, took the press and threw out the type. Two accounts say they also burned the building, and that Stand Watie, a Cherokee who supported moving the Cherokee west, helped them. The records do not name the officer who led them.

A new Cherokee Nation newspaper, the *Cherokee Advocate*, began in Tahlequah in September 1844, with William Potter Ross as its editor.

Elias Boudinot

Who: A Cherokee writer who edited the *Cherokee Phoenix*, the first Native American newspaper, and later, with a small group of Cherokee, signed a treaty that gave up the Cherokee homeland.

When and where: New Echota, Cherokee Nation, in present-day Georgia, 1826 to 1835, and Park Hill, Indian Territory, 1837 to 1839.

He edited the Cherokee paper, then signed away the Cherokee homeland.

Elias Boudinot was born in the Cherokee Nation, in what is now northwest Georgia. His Cherokee name was Gallegina, which means "the Buck."

As a young man he studied at a school run by missionaries in Cornwall, Connecticut, and there he took the name of Elias Boudinot, the president of the American Bible Society. In Cornwall he became engaged to Harriet Gold, a white woman. Opponents of the marriage burned the couple in effigy, setting fire to dummies made to look like them. The two married in 1826.

The press. That year Boudinot went on a speaking tour that included Philadelphia. In a speech called "An Address to the Whites" he asked for money for a Cherokee press, and he collected it. When the *Cherokee Phoenix* began in 1828, he was its editor. The paper printed reports of the Georgia Guard's campaign against the missionaries in the Cherokee Nation, and the Guard's commander, Charles Nelson, threatened to beat Boudinot.

The treaty. Members of Congress passed the Indian Removal Act in 1830. Under it, the president could make treaties, written agreements between nations, with Native nations to move them west. In 1832 Boudinot's cousin John Ridge began to speak openly for moving west, and Boudinot began writing in the paper in favor of it. Principal Chief John Ross opposed moving west on any terms. Ross pressured Boudinot to resign, and he did on August 1, 1832.

In December 1835 Boudinot and a small group of Cherokee signed the Treaty of New Echota. In it they gave all Cherokee land east of the Mississippi River to the United States for five million dollars. They also agreed that the Cherokee would move west to Indian Territory, in what is now Oklahoma.

Killed under Cherokee law. Boudinot moved west to Park Hill, in Indian Territory, in 1837. A group of Cherokee who supported John Ross stabbed him to death there on June 22, 1839. On the same day, other men killed his uncle, Major Ridge, and his cousin John Ridge, who had also signed the treaty. The killers were never identified or tried.

A man named Allen Ross described a secret meeting before the killings. He told of it in 1890. The men at the meeting decided that the three should die under a Cherokee law. That law said anyone who agreed to sell the Nation's land "should forfeit their lives," meaning they should be put to death.

Newspapers against slavery, and the men who attacked them, 1831 to 1835

William Lloyd Garrison, a white abolitionist, someone who worked to end slavery, began *The Liberator* in Boston on January 1, 1831. Garrison demanded an immediate end to slavery, with no payment to the enslavers. About 3,000 people paid for his paper.

The drawing across the top of its front page showed whippings and the sale of children, set in Washington, D.C. Garrison believed that "slavery and freedom of the press cannot exist together." By 1835 members of the American Anti-Slavery Society were printing more than a million papers and pamphlets a year.

Mail burned in Charleston. Members of the society mailed their papers to people in the South in the summer of 1835. A small group of men broke into the post office in Charleston, South Carolina, on the night of July 29. They took the bags of antislavery papers, and a mob burned them. Opening or holding back the mail was against federal law. Even so, Postmaster General Amos Kendall, a close ally of President Andrew Jackson, gave Southern postmasters permission to refuse to deliver antislavery papers.

Garrison dragged. A crowd gathered outside *The Liberator*'s office in Boston on October 21, 1835, where the Boston Female Anti-Slavery Society was meeting. A newspaper of the time put the crowd at 5,000. A member of the society later described the men as "gentlemen of wealth and standing."

The men forced the women out, seized Garrison, tied a rope around his waist and dragged him through the streets. They nearly tarred and feathered him. Tarring and feathering means pouring warm pine tar over a person's body and then covering it with feathers, and the tar often blistered or burned the skin.

The mayor stopped them and had Garrison held in the city jail overnight for his own safety. Garrison later wrote that "every mail" brought letters telling him he had "only so many days to live."

The penny press, 1833

Benjamin Day, a New York printer, started *The Sun* in 1833, partly to bring more work to his print shop. Most daily papers cost six cents. Day sold *The Sun* for one cent a copy, a penny, so people with little money could buy it every day. Boys sold it on the street. Day printed crime stories, and he hired George Wisner, the first police reporter in the United States, to write short reports from the police office.

Life on the Moon. Day printed a made-up report in *The Sun* in 1835 that living creatures had been found on the Moon. Sales rose, and critics attacked Day when readers learned it was false. By 1839 *The Sun* sold 50,000 copies a day. Day paid for the paper mostly by selling advertisements. By 1842 New York City had nine daily papers that cost one or two cents.

Elijah Lovejoy

Who: A white Presbyterian minister and newspaper editor who wrote against slavery and was shot dead by a mob while defending his press.
When and where: St. Louis, Missouri, 1833 to 1836, and Alton, Illinois, 1836 to 1837.

Men who supported slavery broke four of his presses, then shot him.

Elijah Lovejoy wrote against slavery, as Garrison did, but in a state where slavery was legal. He became the editor of the *St. Louis Observer*, a Presbyterian weekly paper in St. Louis, Missouri, in 1833. A group of leading men in St. Louis signed a letter in 1835 asking him to tone down what he wrote. He refused.

Francis McIntosh. A white mob in St. Louis chained Francis McIntosh, a free Black steamboat worker, to a tree and burned him alive on April 28, 1836. Judge Luke Lawless refused to convict anyone and blamed abolitionists. Lovejoy attacked the judge in the *Observer*. In May 1836 men in St. Louis threatened him with mob violence and ran him out of town.

He moved his paper across the Mississippi River to Alton, in the free state of Illinois. Men in Alton who supported slavery destroyed his press three times. They threw one press from a second-floor window. A fourth press arrived by steamboat on November 6, 1837. Lovejoy and his supporters stored it in a stone warehouse owned by Benjamin Godfrey and Winthrop Gilman.

The warehouse. A mob surrounded the warehouse the next night. Its members said they would burn the building and kill every man inside. About twenty men inside had guns, and one of them shot and killed a man in the mob named Bishop. A man from the mob climbed a ladder to set fire to the roof. When Lovejoy stepped outside, men in the mob shot him five times. He was 34. The mob went into the warehouse, broke the press and threw it into the river.

The trials. No one was ever charged with killing Lovejoy. In January 1838 Alton's prosecutors, the lawyers who bring charges for the government, put twelve of the press's defenders on trial for riot. Illinois Attorney General Usher F. Linder, the state's top lawyer, joined the case against them. A jury found Gilman not guilty in ten minutes, and the city attorney dropped the charges against the other eleven.

Prosecutors then tried men from the mob, on January 19, 1838, for breaking into the warehouse and destroying the press. This time Linder was the lawyer for the mob. The jury found the men not guilty.

The men from the mob who were tried: Frederick Drucher, William Carr, James M. Hock, David Butler, Horace Beall, Levi Palmer, and two men named Nutter and Jennings.

The telegraph, 1844, and who owned it

Members of Congress voted Samuel Morse $30,000 in 1843 to build a test line from Washington to Baltimore for his telegraph, which sends messages along a wire as pulses of electricity. Morse sent the first message over the line from the US Capitol on May 24, 1844: "What hath God wrought," old words for "What has God done." Alfred Vail received it in Baltimore and sent it back.

Who owned the wires. Morse and his partners tried to sell the right to use his invention to the federal government and to groups of businessmen. No one bought it. Morse hired Amos Kendall, the former Postmaster General, to run the business. Kendall sold the right to use Morse's patent, the legal right to be the only one to make or use an invention, to separate companies, one region at a time.

Census Bureau officials, who count the people for the government, reported 75 telegraph companies with 21,147 miles of wire in 1851. Private companies owned the lines. From 1844 on, many newspapers printed a section headed "News by Telegraph."

The war between the United States and Mexico began in 1846, and fast news of the war cost much more than news had cost before. Moses Yale Beach, who owned *The Sun*, got the owners of the leading New York papers, six counting his own, to share the cost. That group grew into the New York Associated Press.

Frederick Douglass

Who: A man who escaped slavery in Maryland and started his own newspaper, the *North Star*.

When and where: Rochester, New York, from 1847.

Movie: *Becoming Frederick Douglass* (2022), a PBS documentary about his life, directed by Stanley Nelson.

His paper's motto began: "Right is of no sex."

Frederick Douglass started his own newspaper because he believed that Black Americans had to lead the fight for their own freedom. He was born enslaved in Talbot County, Maryland, in February 1818, and he escaped on September 3, 1838.

Speeches first. Douglass spoke about his life in slavery at a meeting of the Massachusetts Anti-Slavery Society on Nantucket island in 1841. The society's leaders then asked him to travel and give speeches for pay. He published the story of his life in 1845. He then gave speeches in Britain and Ireland, and the money from that tour paid to start his paper.

Douglass broke with Garrison in the late 1840s over whether politics could help end slavery, because Douglass came to believe that political action could. He published the first issue of the *North Star* in Rochester, New York, on December 3, 1847. Its motto was "Right is of no sex, Truth is of no color, God is the Father of us all, and we are brethren."

The *North Star* was four pages long and came out every week. It had more than 4,000 subscribers in the United States, Europe and the West Indies. On the front page Douglass wrote that the paper would "attack SLAVERY in all its forms and aspects." He also wrote that it would "promote the moral and intellectual improvement of the COLORED PEOPLE," using a word Americans of that time used for Black people. Its first printer and publisher was William Cooper Nell, a Black man from Boston.

BY 1850

If you remember one thing: Newspapers got cheaper and faster, and men who defended slavery burned antislavery mail, wrecked presses and killed an editor.

- In 1800: A federal judge sent James Callender to prison for his words about President Adams, and news traveled no faster than a horse or a ship could carry it.
- In 1850: The country had 2,526 newspapers, a daily paper could cost one cent, and news crossed the telegraph wires.

Next: A telegraph wire crossed the continent in 1861, and the Civil War sent reporters to the battlefields.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 19 MINUTES

One company, Western Union, came to carry nearly every telegram in the country, while army officers and white mobs shut down newspapers. A telegraph wire crossed the continent in 1861, and by 1866 Western Union had more than 90 percent of the country's telegraph business. During the Civil War, Northern editors sent dozens of reporters to the battlefields. Generals of the Union, as the Northern side in the war was called, and members of Lincoln's cabinet closed newspapers and had editors arrested without trials.

Boston's postmaster hired the first Black American known to work for the federal government in 1863. In New York, Joseph Pulitzer and William Randolph Hearst competed for readers, and some of the stories they printed were false. Postal workers began carrying mail to farm families at their homes in 1896. White men drove Black editors out of Memphis in 1892 and out of Wilmington, North Carolina, in 1898. In Wilmington a white mob burned a Black-owned newspaper and murdered Black men in the streets.

The Pony Express and the wire across the continent, 1860 to 1861

William H. Russell and his partners, Alexander Majors and William B. Waddell, started the Pony Express on April 3, 1860. Riders carried mail on horseback nearly 2,000 miles, from St. Joseph, Missouri, to Sacramento, California, across land where mostly Native Americans lived. The riders changed horses every 10 to 15 miles and covered 75 to 100 miles a day. The first run took ten days.

Lincoln's speech in under eight days. The riders carried President Abraham Lincoln's inaugural address, the speech a new president gives on taking office, across in 7 days and 17 hours in March 1861. The company charged as much as $5 for half an ounce of mail, when ordinary postage was ten cents or less. Workers finished the first telegraph line across the continent on October 24, 1861. Two days later the Pony Express shut down.

News of the Civil War, 1861 to 1865

The North had far more newspapers than the South when the Civil War began. In 1860 fewer than 10 of every 100 print shops in the country were in the South. The South had 70 of the country's 387 daily papers, and those 70 sold only about 10 of every 100 daily copies.

Soldiers of the Confederacy, the Southern states that had broken away from the United States, fired on Fort Sumter in April 1861. The next day the *New York Herald* printed 135,000 copies, about as many as all the South's daily papers together. Only about 20 Southern daily papers were still printing when the war ended.

Reporters with the armies. The editors of the *New York Herald* alone kept more than 40 reporters with the armies. At Gettysburg the telegraph lines had been cut. A. Homer Byington of the *New York Tribune* repaired the wires near Hanover, and in return he got the first use of them. George W. Smalley of the *Tribune* wrote his story of the Battle of Antietam, in September 1862, on a train to New York. The *Tribune* had thousands of copies on the streets the second morning after the battle.

Southern papers could no longer get the news of the Associated Press, the group of New York papers that shared news. So Southern publishers started their own Press Association in 1862.

Photographs of the time could not show action or reach many readers quickly. The editors of weekly papers such as *Frank Leslie's Illustrated Newspaper* and *Harper's Weekly* sent artists to draw the battles. Engravers carved the drawings into blocks of wood for printing. Readers saw the pictures within about a week.

Editors arrested and papers closed, 1861 to 1864

Union officers and members of Lincoln's cabinet shut down newspapers and arrested editors during the Civil War, most of the time without any court case. The First Amendment states that Congress shall make no law "abridging the freedom ... of the press." From April 1861, Lincoln's officials censored reporters' telegraph messages to and from Washington.

Soldiers arrested editors and closed their presses. Army officers sitting as military courts banished some editors to the Confederacy, forcing them to leave the North and not come back.

Seward and Stanton. William Seward, Lincoln's Secretary of State, the cabinet member in charge of dealings with other countries, ordered the arrest of an editor of the *Freeman's Journal*. Officials held the editor for 11 weeks and then let him go without a trial. Edwin Stanton, the Secretary of War, who ran the army for Lincoln, allowed a military governor, an army officer put in charge of a place, to destroy the office of the *Sunday Chronicle* in Washington.

McGregor. In Canton, Ohio, Archibald McGregor edited the *Stark County Democrat*. He printed words that his opponents called treason, and a mob then destroyed his paper's office. The record does not name the men in the mob. A year later soldiers arrested him without saying what the charge was.

General Ambrose Burnside closed the *Chicago Times*, a paper that backed the Democratic Party, in 1863. Lincoln canceled Burnside's order. He said he was "embarrassed" that his general had failed to weigh the needs of the Union against "the Liberty of the Press."

A forged announcement. Two New York reporters forged an announcement signed with Lincoln's name in May 1864, and the *New York World* printed it. It called for a day of fasting and a much larger draft, the government's order for men to join the army. The two men did it to change the price of gold. One of them, Joseph Howard Jr., had bought gold. A call for more soldiers meant a longer war, and he expected worried investors to buy gold and push its price up.

Lincoln ordered General John A. Dix to arrest the *World*'s editors and take over its offices. He canceled the order once it was clear the two reporters had faked the announcement.

The first Black postal workers, 1863 to 1869

Federal law had barred anyone but "free white persons" from carrying the mail since 1802. The law named the post riders and the drivers of the carriages that carried the mail along the roads. In early 1863, during the Civil War, John Palfrey, the postmaster of Boston, hired William Cooper Nell as a clerk inside the Boston Post Office. Nell was the first Black American known to work for the federal government.

Members of Congress ended the rule on March 3, 1865. Their new law said that "no person, by reason of color, shall be disqualified from employment in carrying the mails." That meant no one could be kept from the job because of the color of his skin.

James Christian began delivering mail in Richmond, Virginia, in June 1869. He was the first Black letter carrier known in the United States. Richmond's postmaster, Elizabeth Van Lew, hired him along with other Black clerks and letter carriers.

William Cooper Nell

Who: A Black writer and printer from Boston who printed Frederick Douglass's first newspaper and became the first Black American known to work for the federal government.

When and where: Boston, and Rochester, New York, 1840s to 1874.

His night shifts at the post office ran from 11 to 5:30.

William Cooper Nell was born in Boston. He went to Rochester, New York, in the 1840s and worked for nearly two years with Frederick Douglass, as the first printer and publisher of the *North Star*.

The schools. The members of the Boston School Committee, who ran the city's public schools, kept Black children apart from white children. Nell and other Black Bostonians sent the committee a petition in 1844, a written request signed by many people, asking it to open the schools to their children.

Eleven years later, in 1855, the governor of Massachusetts signed a law that made it illegal to turn a child away from a public school because of race. The same year, Nell published a 396-page book, *The Colored Patriots of the American Revolution*, about Black Americans who fought in the Revolution.

The post office. Nell wrote in a letter in 1873 that the post office was handling more and more letters, so he had been put on the night mail. His shifts began at 11 at night and usually ended at 5:30 in the morning.

He called himself "the first colored man employed about the United States Mail," using the word of his time for a Black man. He also wrote that he had "never lost a day from sickness or any cause." Nell worked at the post office until he died in 1874.

One company owns the telegraph, 1866 to 1900

Western Union bought its last two big rivals in 1866 and ended up with more than 90 percent of the country's telegraph business. Tomas Nonnenmacher of Allegheny College calls it the country's first nationwide industrial monopoly, a company with no real competition that can set its own prices.

Lobbyists. In almost every session of Congress, members brought bills to control the company or to put the government into the telegraph business itself. Western Union's lobbyists, people paid to persuade lawmakers, blocked nearly all of them.

The number of telegrams the company carried rose from 5.8 million a year in 1867 to 63.2 million in 1900. The average price of a telegram fell from $1.09 to 30 cents. Of every dollar customers paid, about 30 to 40 cents was profit for the company.

The New York Associated Press, run by the owners of a few New York papers, held what the American Antiquarian Society calls a news monopoly by the late 1800s. Chicago newspaper owners formed a rival group, the Western Associated Press, and they broke the New York group's hold. A new, national Associated Press began in 1892.

The telephone, 1876 to 1900

Alexander Graham Bell patented the telephone in 1876. He offered the patent to Western Union for $100,000, and the company's leaders turned it down. In 1900 the United States had about 1.36 million telephones, fewer than 2 for every 100 people. Of every 100 telephone calls, 97 were local calls within one town.

Newspapers in Chinese, 1854 and 1855

Before the Civil War there were already American newspapers in German, French, Spanish, Polish, Swedish, Norwegian, Hawaiian and other languages. Chinese was harder to print because it uses thousands of characters instead of an alphabet. The first Chinese-language paper in the country, *Golden Hill's News*, came out in San Francisco in 1854 and lasted a few months.

The second, *The Oriental*, began there on January 4, 1855. Its editor was William Speer, a Presbyterian minister. Its printers drew the characters on stone with a grease pencil and printed from the stone. The paper closed in 1857. No one tried another Chinese paper until the 1870s.

Nellie Bly

Who: A newspaper reporter who pretended to be mentally ill so that she could report from inside New York City's asylum for women.

When and where: Pittsburgh, 1885 to 1887, and New York City, 1887 to 1895.

Movie: *Escaping the Madhouse: The Nellie Bly Story* (2019), a Lifetime television film that its makers call a fictionalized account.

She pretended to be insane to get locked inside an asylum.

Nellie Bly was the pen name of Elizabeth Cochran, from Cochran's Mills, Pennsylvania. The *Pittsburgh Dispatch* printed an article called "What Girls Are Good For" in 1885. Cochran wrote an angry letter to the paper answering it, and its editor hired her. At the *Dispatch* she wrote about working girls and poor neighborhoods in Pittsburgh.

Ten days inside. Bly went to work for Joseph Pulitzer's *New York World* in 1887. For her first big story she pretended to be insane until she was locked in the city's asylum for women on Blackwell's Island, a hospital where people with mental illness were kept. She stayed ten days.

When she arrived, a nurse, Miss Grupe, took her to a cold, wet bathroom. The nurses pulled off her clothes while a group of patients watched from the door. A patient scrubbed her with soft soap. Then three buckets of ice-cold water came down over her head, and Bly did not say who poured them.

Bly saw 52 women tied together by wide leather belts locked around their waists and fastened to one long rope.

A nurse, Miss Grady, told a patient named Urena Little-Page that "the doctors say that you are thirty-three instead of eighteen," and the other nurses laughed. Then Grady and the other nurses slapped Little-Page, knocked her head and choked her, and dragged her into a closet. The marks of their fingers stayed on her throat all day. A patient told Bly that "the nurse would beat them if they told."

Bly's reports ran in the *World* and then as a book, *Ten Days in a Mad-House*. A grand jury looked into the asylum. Bly wrote that New York City officials had set aside $1,000,000 a year more than ever before for the care of people with mental illness.

Around the world. Bly then asked her editor to let her travel around the world faster than Phileas Fogg, the hero of Jules Verne's novel *Around the World in Eighty Days*. She set out in November 1889. In France she met Verne himself, who asked about her route. The whole trip took 72 days, 6 hours, 11 minutes and 14 seconds. The *World*'s contest to guess her time drew nearly a million entries. She died in New York on January 27, 1922.

Ida B. Wells

Who: A Black newspaper editor in Memphis who counted lynchings, printed what she found and was driven from her city for it.
When and where: Memphis, Tennessee, 1892, and New York City, 1892 to 1895.

She counted 728 lynchings in eight years and printed the count.

Ida B. Wells counted the lynchings in the South, killings by a crowd outside the law with no trial, and she printed what she found. White men in Memphis drove her out of the city for it. She was born into slavery in Holly Springs, Mississippi, on July 16, 1862. By 1892 she was an editor and part owner of the *Free Speech*, a Black newspaper in Memphis, Tennessee.

The People's Grocery. A white mob took three Black men out of their cells in the Memphis jail in the early morning of March 9, 1892, and shot them to death. They were Thomas Moss, Calvin McDowell, and a third man whom the records name as Will Stewart or Henry Steward. Moss was part owner of the People's Grocery, a store run by Black people that was taking business from a white-owned store nearby.

Wells wrote that the white store's owner, a man named Barrett, led twelve officers to the grocery at night. The grocery's defenders fired on them and wounded three. Officers then arrested 31 men, and the mob took the three from among them. Wells wrote that Black people in Memphis asked the authorities to bring the killers to justice, and "No attempt was made to do so."

Wells and the other editors of the *Free Speech* advised Black people to leave Memphis. Wells wrote that Black men "left the city by thousands." Those who stayed stopped riding the streetcars. The streetcar company lost so much money that its managers came to the *Free Speech* and asked the editors to tell people to ride again.

The count. Wells began to study lynchings, using the lists that the *Chicago Tribune* printed each year. She wrote in 1892 that mobs had killed 728 Black people in the past eight years, and that only one-third of the 728 had even been charged with rape. Rape means forcing someone into sex. Her list included "the boy Will Lewis," hanged in Tullahoma, Tennessee, "for being drunk and 'sassy' to white folks." It also included a fifteen-year-old girl hanged in Louisiana.

The editorial. Wells printed an editorial, an article that gives the paper's opinion, in the *Free Speech* on May 21, 1892. It listed eight Black men lynched since the last issue, and it called the claim that Black men raped white women "the old thread-bare lie." Wells was in New York at the time.

White men in Memphis held a meeting at the Cotton Exchange Building on May 24 and sent a group to find the paper's editors. Wells wrote that these men made "open threats of lynching" and did not carry them out only because they could not find the editors.

The business manager and half owner, J. L. Fleming, fled the city. Wells wrote that he was later "ordered not to return," and she did not say who gave the order. Letters and telegrams warned Wells that she would be hurt if she came home.

The office. Wells wrote in 1892 that people the paper owed money to took over the office and sold "the outfit," meaning the printing equipment. She described no fire. Later accounts by the National Park Service and the National Women's History Museum say a mob destroyed or burned the office.

Exiled. Wells did not go back. T. Thomas Fortune's *New York Age*, a Black weekly newspaper, printed most of her account on June 25, 1892, under the name "Exiled." It ran on the front page, and 10,000 copies went out across the country.

Black women in New York and Brooklyn held a meeting at Lyric Hall on October 5, 1892, and raised the money to print her account as a pamphlet. It was called *Southern Horrors: Lynch Law in All Its Phases*. In 1895 she published *A Red Record*, 100 pages of tables of lynchings.

Rural Free Delivery, 1896

People in cities had had their mail brought to their homes for free since 1863, but farm families still had to travel to a post office to get their letters and newspapers. In 1890 about 41 million Americans, 65 of every 100, lived in the country. One farmer asked, "Why should the cities have fancy mail service and the old colonial system still prevail in the country districts?"

Postmaster General John Wanamaker argued that rural people, those who lived in the country, needed the information in newspapers. Congressman Tom Watson of Georgia won $10,000 in 1893 to test home delivery in the country. The next Postmaster General, Wilson S. Bissell, did not use it. Members of Congress added $20,000 in 1895 and $10,000 in 1896.

The first routes. Postal officials began Rural Free Delivery on October 1, 1896, at Charles Town, Halltown and Uvilla, in West Virginia. Within a year 44 routes ran in 29 states. Rural families sent more than 10,000 petitions asking for routes. Rural delivery became permanent on July 1, 1902, with about 8,500 rural carriers.

Yellow journalism and the war with Spain, 1895 to 1898

Joseph Pulitzer owned the *World* in New York, and William Randolph Hearst owned the *Journal*. In 1896 Hearst hired away Pulitzer's cartoonist, Richard F. Outcault, whose comic strip had a character called the Yellow Kid. Pulitzer hired another cartoonist to keep drawing the Kid. People began to call the two men's kind of reporting "yellow journalism."

The State Department's historians define yellow journalism as reporting that put exciting stories ahead of facts. Pulitzer and Hearst printed stories about Cuba's fight for independence from Spain, and some of those stories turned out to be false.

The Maine. The US battleship *Maine* blew up and sank in the harbor of Havana, Cuba, on the night of February 15, 1898. Hearst and Pulitzer printed rumors that someone had planned to sink it, a plot that was never proved. Officials of Spain's government in Cuba wrote an early report that said the explosion started inside the ship.

A court of US Navy officers decided in March that a mine, a bomb placed in the water outside the ship, had set off the explosion. The court found no evidence of who was responsible, and no one has ever found out who placed a mine, or whether there was one. In 1976 a study led by Admiral Hyman Rickover found that a fire in a coal bin most likely set off the explosives stored next to it. People still argue over the cause.

By early May the war between the United States and Spain had begun. The State Department's historians say the two papers helped make Americans ready for a war, but that the papers alone did not cause it.

A Black newspaper burned in Wilmington, 1898

A mob of white men in Wilmington, North Carolina, burned the city's Black newspaper in 1898, murdered Black men in the streets and forced the elected city government out.

Alex Manly edited the paper, the *Daily Record*. In August 1898 he printed an editorial answering a speech by Rebecca Felton, a white woman who had urged the lynching of Black men accused of raping white women. Manly wrote that white newspapers blamed all Black people for the crimes of the few who might be guilty. He also wrote that in many cases called rape, the white woman had chosen to meet the Black man in secret. When people found out, he wrote, a mob lynched the man for rape.

Democratic Party leaders, with Furnifold Simmons at the head of their campaign, used the editorial to stir up white voters before the November election. Democratic newspapers reprinted it under headings such as "An Insult to the White Women of North Carolina."

The demands. The day after the Democrats won, a meeting of white men in Wilmington passed a set of demands. They ordered Manly to close his paper and leave the city, and they demanded that the mayor and the chief of police resign. Alfred Moore Waddell, a former Confederate officer and congressman, led a crowd of white men to the *Daily Record* office on November 10, 1898. About 500 men gathered at the start, and the crowd grew to as many as 2,000. They broke in and burned the building.

The killings. Later that morning, members of the white mob shot and killed Black men in the streets of Wilmington. Every person reported killed was Black. The shooting started where groups of Black and white men were arguing, and each side there said the other fired first. A white militia company, the Wilmington Light Infantry, had a machine gun mounted on a wagon, which white businessmen had bought before the election.

How many. The coroner, the official who looks into deaths, counted 14 men dead. A state commission, a group of people chosen to study the killings, found in 2006 that at least 14 and perhaps as many as 60 men were murdered. North

Carolina's state history office notes unconfirmed reports of many dozens or even hundreds of deaths. The records do not name the men who set the fire or the men who killed.

By late afternoon Waddell and his allies had forced the elected mayor and city council to resign, and the new board of aldermen, the city council, made Waddell mayor. They then banished people from the city. Manly came first, and he escaped. Black leaders, successful Black business owners and white Republicans came next. A federal investigation ended in 1900, and no one was charged.

BY 1900

If you remember one thing: By 1900 one company carried almost all the country's telegrams, and white mobs had driven Black editors out of Memphis and Wilmington.

One number: 728. That is how many Black people Ida B. Wells counted killed by mobs in the eight years before 1892, and only one-third of them had even been charged with rape.

Next: In the new century, magazine reporters dug into big companies and city governments, and radio brought the news into people's homes.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 17 MINUTES

Radio carried the news into American homes as it happened, and in two world wars federal officials worked to control what Americans printed and heard. Magazine reporters investigated companies and city governments from 1902 on and printed long reports about what they found. During World War I, federal officials ran the government's first large propaganda office. They also barred dozens of newspapers from the mail and filed more than 2,000 criminal cases against people for what they said or printed.

The Westinghouse company owned one of the first radio stations, KDKA, from 1920, and then networks paid for by advertisers took over. In World War II, government officials asked newspapers and radio stations to keep military secrets. Japanese Americans forced into camps printed their own newspapers, and camp officials checked what they printed.

The muckrakers, 1902 to 1912

Some reporters in the early 1900s dug into one subject and then wrote long reports about it, and editors at magazines such as *McClure's*, *Cosmopolitan* and *Collier's* printed them. The January 1903 issue of *McClure's* contained three such reports.

Three reports in one issue. One was part of Ida Tarbell's history of the Standard Oil Company. By 1880 that company controlled 90 to 95 percent of the oil refining in the United States, the work of turning crude oil from the ground into products people can use. Another was Lincoln Steffens's report on corruption, officials using their jobs to get money or favors illegally, in the city government of Minneapolis. The third was Ray Stannard Baker's report on coal miners.

The Jungle. Upton Sinclair spent about seven weeks undercover, hiding who he was, in the meatpacking plants of Chicago, the factories where animals were killed and the meat was cut up and packed. His novel *The Jungle* came out as a book in February 1906. He wanted readers to care about the workers, but readers were most upset about what went into their meat. "I aimed at the public's heart, and by accident I hit it in the stomach," Sinclair wrote.

Patent medicines. In a series that ran in *Collier's* in 1905 and 1906, Samuel Hopkins Adams reported on patent medicines, bottled cures sold with false claims. He wrote that Americans would "swallow huge quantities of alcohol, an appalling amount of opiates and narcotics" in them. By opiates he meant drugs made from the opium poppy.

President Theodore Roosevelt signed two new laws on June 30, 1906. Under the Meat Inspection Act, federal inspectors worked in the packing plants. The Pure Food and Drug Act banned food and medicine with false labels or harmful added ingredients from being sold across state lines.

Muckrakers. David Graham Phillips attacked members of the US Senate in a series called "The Treason of the Senate." Roosevelt answered with a speech on April 14, 1906. He compared these writers to the Man with the Muck Rake in

John Bunyan's religious book *The Pilgrim's Progress*, a man raking up muck, mud and filth, "who could look no way but downward."

Roosevelt meant it as an insult. The writers took his word "muckraker" as a name to be proud of. By 1912 this kind of reporting had mostly disappeared from the magazines, in part because large companies stopped buying advertisements in them.

Robert S. Abbott

Who: The son of formerly enslaved parents who founded the *Chicago Defender* and sent it into the South, where it urged Black families to move north.
When and where: Chicago, 1905 to 1940.

He printed "(white)" after white people's names.

Robert S. Abbott started the *Chicago Defender* with 300 copies in 1905, and by 1920 it outsold every other Black newspaper in the country. He was born on St. Simons Island, Georgia, in 1868. His parents, Thomas and Flora Abbott, had been enslaved.

Shut out. Abbott learned printing at Hampton Institute in Virginia. After he graduated in 1896 he moved to Chicago, where he was shut out of the printing business because he was Black. The record does not name the shops that turned him away. He earned a law degree in 1899, but because he was Black he could not get work as a lawyer in Indiana and Kansas.

Abbott printed the first issue of the *Chicago Defender* on May 5, 1905, and sold the 300 copies himself. In his paper's headlines and articles he attacked lynching and white supremacy, the belief that white people should rule over everyone else. One headline read, "When the Mob Comes and You Must Die Take at Least One with You."

White papers such as the *Chicago Tribune* printed "colored" after the names of Black people. Abbott printed "(white)" after the names of white people.

North. At that time 90 of every 100 Black Americans lived in the South, and Abbott sent the paper there. He urged readers to leave the South. In the *Defender* he printed job listings, train schedules, and photos of what he presented as the best schools, parks and housing in Chicago. For four years starting in 1916, more than 10,000 Black Southerners left the South each month. Between 1916 and 1919 about 500,000 left.

Smuggled south. Local police in Southern towns tried to stop the paper. They seized copies, and they arrested people who sold it under vagrancy laws, which made it a crime to have no job. Some police departments banned the sale of Black newspapers altogether. The accounts do not name the towns. Abbott arranged for Pullman porters, the Black workers who served passengers on railroad sleeping cars, to carry bundles of the paper south and sell them in secret.

The *Defender* sold more than 250,000 copies a week by 1929, and Abbott was one of the first Black millionaires. He died in Chicago on February 29, 1940.

World War I: the government's propaganda office, 1917 to 1919

The United States entered World War I on April 6, 1917. That month President Woodrow Wilson set up the Committee on Public Information and put George Creel, a former newspaper reporter, in charge. It was the federal government's first large office for propaganda, information made and spread to push people toward one opinion.

Creel's staff sent articles to newspapers, and Creel later estimated that their material filled 20,000 newspaper columns a week. The committee's staff printed their own paper, the *Official Bulletin*, and about 115,000 copies went out. Its artists made posters, and one showed a German soldier as a gorilla carrying a club.

Staff in one division of the committee watched the hundreds of newspapers printed in other languages in the United States. After the war, many Americans concluded that Creel and his staff had pushed the war too hard, and that they had made it harder for people to speak against it.

World War I: papers banned from the mail and people jailed for words, 1917 to 1921

Members of Congress passed the Espionage Act, named for espionage, or spying, on June 15, 1917. Under it, it was a crime to get in the way of the draft or to stir up disloyalty in the armed forces. Papers that urged treason could not go through the mail.

Banned from the mail. Postmaster General Albert S. Burleson told local postmasters to report anything suspicious. By 1918 his officials had barred 74 newspapers from the mail. The New York City postmaster, Thomas G. Patten, stopped *The Masses* magazine from going through the mail, and its November-December 1917 issue was its last.

Newspapers printed in German and other languages faced one more rule. Under the Trading with the Enemy Act of October 6, 1917, they had to file English translations of their war news with the local postmaster.

The Sedition Act of 1918. Members of Congress added the Sedition Act to the Espionage Act the next year, on May 16, 1918. Under it, it was a crime to speak or write anything "disloyal" about the government, the Constitution, the armed forces or the flag. The punishment was up to 20 years in prison and a $10,000 fine.

Federal prosecutors filed more than 2,000 cases under the two laws, and more than 1,000 ended in convictions. They used the Sedition Act mainly against pacifists, people who oppose all war. They also used it against anarchists, who want to do away with government, and socialists, who want the public rather than private owners to own big businesses.

The Socialist leader Eugene V. Debs gave a speech against the war to about 1,200 people in Canton, Ohio, on June 16, 1918. A judge sentenced him to 10 years in prison, and the justices of the Supreme Court upheld his conviction in 1919. President Warren G. Harding shortened his sentence, and Debs walked out of prison on Christmas Day 1921.

Victor Berger

Who: An immigrant newspaper owner in Milwaukee whose paper was barred from the mail for opposing World War I, and who was sentenced to 20 years in prison.

When and where: Milwaukee, Wisconsin, 1892 to 1929.

Voters kept electing him, and the House kept refusing to seat him.

Victor Berger was born to a Jewish family in Austria-Hungary. He came to the United States in 1878 and taught German in Milwaukee's public schools. He bought a German-language newspaper in 1892, and in 1898 he and Eugene V. Debs founded the Social Democratic Party. Milwaukee voters made him the first Socialist elected to the US House of Representatives in 1910. A year later he started a daily paper in English, the *Milwaukee Leader*.

The permit. Berger opposed the war and said so in his paper. In 1917 an official in Postmaster General Albert S. Burleson's department, the third assistant postmaster general, took away the *Leader*'s "second-class mailing privilege," the permit that let it go by mail as a newspaper.

Federal prosecutors charged Berger under the Espionage Act in 1918. He and four other Socialist leaders were convicted in Chicago, and in early 1919 Judge Kenesaw Mountain Landis sentenced each of them to 20 years in prison.

The empty seat. Milwaukee voters elected Berger to Congress again in 1918, while his case went on. In November 1919 the other members of the House refused to let him take his seat. Voters elected him again in a special election, a vote held to fill the empty seat, with 55 percent of the vote, and the members of the House refused him again.

The justices of the Supreme Court overturned his conviction in 1921. Voters elected him to Congress three more times, in 1922, 1924 and 1926. He died in 1929 after a streetcar accident.

Radio, and who owned it, 1920 to 1934

Radio station KDKA in Pittsburgh broadcast the results of the presidential election between Warren G. Harding and James M. Cox on the evening of November 2, 1920. KDKA belonged to the Westinghouse company. Its studio was a shed on the roof of a building at the Westinghouse works in East Pittsburgh.

Eight stations were on the air by the end of 1921, and by November 1, 1922, federal officials had licensed 564. At first, money for radio came from two places: selling radio sets and selling advertising. The Queensboro Corporation paid station WEAF in New York $50 on August 28, 1922, for about ten minutes of advertising for its apartments. Over the following years, advertising became the main way American radio was paid for.

Networks. The National Broadcasting Company, NBC, bought WEAF in 1926 and set up a permanent network, a group of stations in different cities that carry the same programs. Other networks followed, among them the Columbia Broadcasting System, CBS.

Rules for the air. By the middle of the 1920s, so many stations were on the air that their signals crossed and drowned each other out. Federal judges had ruled that the secretary of commerce, who gave out the licenses, could not stop giving them out even when the dial was full. The broadcasters themselves asked the federal government to restore order.

The biggest makers of radio equipment and the biggest broadcasters had also made business deals with each other. Those deals could have given a few companies a monopoly over radio, with no real competition from anyone else.

Members of Congress passed the Radio Act of 1927. Under it, the members of a federal commission decided which station could use which wavelength, the spot on the radio dial where a station is heard. They gave each of the 732 stations a set place on the dial, and they could turn down a new station when no room was left. In 1934 Congress's members passed the Communications Act and with it set up the Federal Communications Commission, the FCC, to oversee radio and messages sent by wire.

The president on the radio, 1933 to 1944

President Franklin D. Roosevelt gave his first fireside chat, a talk to the country by radio, on Sunday night, March 12, 1933. He had closed every bank in the country on March 6, two days after he took office.

In the talk Roosevelt explained in simple terms how banks worked to some 60 million listeners. He told them it was safer to keep money in a reopened bank than under the mattress. When the banks opened again, people put in more money than they took out. Harry Butcher of CBS gave these talks the name "fireside chats." According to Encyclopaedia Britannica, scholars count between 27 and 31 of them, from 1933 to 1944.

The War of the Worlds broadcast, 1938

The actor Orson Welles and his Mercury Theatre on the Air performed H. G. Wells's novel *The War of the Worlds* on CBS on the evening of October 30, 1938. Welles was 23. The writer Howard Koch, the producer John Houseman and Paul Stewart had rewritten the story as fake news bulletins about Martians landing in New Jersey. The program did not stop to say it was a play until almost 40 minutes in.

Some listeners believed it and called the police, newspapers and radio stations. The next morning, editors across the country printed stories about a nationwide panic.

The count. That night the C. E. Hooper company had asked 5,000 households what they were listening to, and 2 of every 100 named the Welles program. A. Brad Schwartz, who wrote a book about the broadcast, *Broadcast Hysteria*, found that only some listeners were fooled. Their worried phone calls convinced many journalists that the show had caused a panic across the country, and the journalists reported one.

Pictures by wire, 1935

Newspapers mostly got photographs by mail, train or airplane until 1935, and a picture could take up to 85 hours to arrive. The AT&T telephone company had run a service that sent photographs by wire from 1926, but it closed in 1933. Associated Press staff sent their first photograph over wires on January 1, 1935. It was a picture taken from the air of a plane crash near Morehouseville, New York. Forty-seven newspapers in 25 states received it.

Edward R. Murrow

Who: A CBS radio reporter whose eyewitness reports from Europe before and during World War II made him famous across the United States.

When and where: London, 1937 to 1945.

Movie: *Good Night, and Good Luck* (2005), directed by George Clooney, about Murrow's television reports on Senator Joseph McCarthy in 1953 and 1954.

From London he reported the war to Americans by radio.

Edward R. Murrow's radio reports from Europe for CBS, from 1938 to 1940, made him famous across the United States. He also helped bring to the United States German scholars whom the Nazis, Adolf Hitler's party, which ruled Germany, had driven out.

Murrow joined CBS in 1935. CBS executives sent him to London in 1937 to run the network's European office. On the radio he described Germany's moves across Europe as an eyewitness. In 1938 he reported Germany's takeover of Austria and the Munich Conference, a meeting of European leaders. He reported the takeover of Czechoslovakia in 1939, and the Battle of Britain, when German planes attacked Britain, in 1940.

McCarthy. After the war Murrow ran the news department at CBS. In 1954 he made a television program about Senator Joseph McCarthy. McCarthy had charged that Communists, people who wanted one party to run the government and the government to own the farms and factories, were working in the government. Murrow died on April 27, 1965.

World War II: secrets, propaganda and the Black press, 1941 to 1945

Japanese planes attacked the US Navy base at Pearl Harbor, Hawaii, on December 7, 1941. Twelve days later President Roosevelt set up the Office of Censorship. Its director, Byron Price, had been an editor at the Associated Press. The office had about 13,500 workers, and most of them read letters and cables, messages sent by wire under the ocean, going into and out of the country.

The code. Price and his staff gave newspapers and radio stations a code of rules, and editors agreed to follow it of their own free will. The code listed topics to keep out of the news: troop and ship movements, how war factories were doing, the president's travels, and even the weather. Price persuaded the reporters Drew Pearson and William L. Laurence not to write about the building of the atomic bomb.

Roosevelt set up the Office of War Information on June 13, 1942. Its staff ran programs, at home and in other countries, that told people how the government said the war was going and what its war plans were.

Federal prosecutors charged far fewer people for their words than in World War I. Philip A. Dynia, writing in the First Amendment Encyclopedia, counts dozens charged, against thousands in 1917 to 1919. Roosevelt kept pressing his attorney general, Francis Biddle, to "indict the seditionists," meaning to charge with crimes the people he saw as turning others against the government.

Double V. Writers for Black newspapers such as the *Chicago Defender* reported on the armed forces' separate units for Black troops, white officers placed over Black soldiers, blood supplies kept apart by race, and violence against Black servicemen. They ran a "Double V" campaign: victory over the enemy abroad and victory over prejudice at home.

Federal officials threatened to charge Black publishers with sedition. The *Defender*'s publisher, John H. Sengstacke, insisted on meeting Biddle. Biddle agreed to block the charges as long as Black editors did not make their criticism of the war any stronger.

Newspapers behind barbed wire, 1942 to 1945

President Roosevelt signed Executive Order 9066, a written command from the president, on February 19, 1942. Under it, Lieutenant General John L. DeWitt ordered Japanese Americans on the West Coast out of their homes, some with only 48 hours' notice. Soldiers and federal officials forced them into camps because of their Japanese ancestry.

How many. The National Archives counts about 122,000 people, and nearly 70,000 of them were American citizens. Another count gives 110,000 to 120,000. The First Amendment Encyclopedia describes the camps as concentration camps, places where people are held by force without a trial.

Inside the camps, Japanese Americans wrote, printed and handed out their own newspapers. The papers in the first camps, run by the army, were copied on mimeograph machines, which print copies from a cut paper stencil. Army officials required those papers to be written in English only. All ten camps run by the War Relocation Authority had papers in both English and Japanese, among them the *Manzanar Free Press* in California and the *Heart Mountain Sentinel* in Wyoming.

Officials checked what the papers printed. Army officials censored the papers in the first camps. In the later camps, officials of the War Relocation Authority interfered with what the papers printed, and in some cases they censored the papers outright.

Television begins, 1927 to 1941

Philo Farnsworth, who was 21, sent the first picture by all-electronic television, a straight line, in his laboratory in San Francisco on September 7, 1927. Station WNBT in New York showed the first television advertisement allowed by law on July 1, 1941, before a baseball game. It showed a Bulova watch face over a map of the United States. It lasted about ten seconds and cost $9.

BY 1950

If you remember one thing: Radio put the news in people's homes, and during World War I federal prosecutors won more than 1,000 convictions for what people said or printed.

The question this era leaves open: Television was only starting in 1950. Who would decide what its viewers were allowed to see? Era 9 answers it in Jackson, Mississippi.

Next: Television reached most American homes within ten years, and two newspapers printed what a president wanted kept secret.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 18 MINUTES

Television became the way most Americans got the news. About 9 of every 100 American homes had a television set in 1950, and by 1960, 87 of every 100 did. After the DuMont network closed in 1955, three national television networks were left: NBC, CBS and ABC.

Managers of one Mississippi station, WLBT, cut civil rights news off the air. A church leader and two Mississippi leaders fought the station before the government and in court, and its owners lost their license.

Reporters printed a secret history of the Vietnam War in 1971 and tied a burglary to the White House in 1972, and officials in Washington tried to stop them. By 1996 cable channels such as CNN and Fox News showed news all day and night, and the World Wide Web had been open to the public since 1991.

Television and the blacklist, 1950 to 1954

Senator Joseph McCarthy of Wisconsin gave a speech in Wheeling, West Virginia, in February 1950. He said that 205 communists, followers of a system in which one party runs the government and owns the farms and factories, worked in the State Department. When senators asked him, he could not name a single one.

The book. Three former FBI agents who ran a company called American Business Consultants published a book called *Red Channels* in June 1950. The book named 151 people who worked in radio and television and claimed they had ties to communism.

The company's owners then visited networks, advertisers and sponsors. For a fee, they offered to "clear" the people they had named. To be cleared, a person usually had to give the FBI or the House Un-American Activities Committee, a committee of Congress, the names of other people. Network executives and advertisers fired people on the list, which became a blacklist, a list of people whom employers refuse to hire or keep.

On camera. A Senate hearing on McCarthy's charges against Army officers and officials was shown on television across the country in 1954. It lasted 36 days. On camera, Joseph Welch, the Army's lawyer, asked McCarthy, "Have you no sense of decency, sir, at long last?"

Edward R. Murrow showed McCarthy's methods on his weekly CBS program *See It Now* that same year. Members of the Senate voted 67 to 22 on December 2, 1954, to condemn McCarthy.

Radio and television in Spanish, 1946 to 1961

Raoul A. Cortez was born in Xalapa, Mexico, in 1905. He worked as a reporter for *La Prensa*, a Spanish-language newspaper in San Antonio, Texas. In 1946 he opened KCOR in San Antonio, the first all-Spanish radio station owned and run by a Hispanic American.

Federal officials had limited broadcasts in languages other than English during World War II. To get his license, Cortez told officials that part of the station's purpose was to rally Mexican Americans behind the war effort.

Cortez started KCOR-TV in 1955, the first television station aimed only at Hispanic viewers. It broadcast on a kind of channel called UHF, and few television sets could pick it up, so advertisers were hard to find. He sold the station in 1961, and the new owners renamed it KWEX.

The station that cut the news: WLBT, 1955 to 1971

The managers of WLBT, a television station in Jackson, Mississippi, that carried NBC's programs, cut civil rights news off the air, and a group of citizens fought for years to take away the station's license. The Lamar Life Insurance Company started WLBT in 1953.

"Cable Trouble." The lawyer Thurgood Marshall appeared on an NBC program in the fall of 1955 to talk about the Supreme Court's decisions against segregation, keeping Black and white children in separate schools by law. WLBT staff cut him off and showed a sign that read "Sorry, Cable Trouble from New York." The general manager, Fred Beard, said he did it "because the TV networks were overloading the circuits with Negro propaganda," using a word of that time for Black people.

Station staff also cut away from NBC's evening news when it turned to civil rights. WLBT reporters used racial slurs on the air. The White Citizens' Council, a group that fought to keep segregation, ran a bookstore in the station's lobby. About half the people in WLBT's viewing area were Black, and the station's managers refused them time to answer.

Medgar Evers. Medgar Evers was the Mississippi organizer for the NAACP, the National Association for the Advancement of Colored People, a civil rights group. In May 1963 he spoke on WLBT to answer the mayor of Jackson, Allen Thompson. Three weeks later Byron De La Beckwith shot Evers in the back in the driveway of his home and killed him.

That year WLBT staff again ran the "Cable Trouble" sign, this time over NBC's report on white people attacking protesters at a lunch counter in Jackson. WLBT's own crew had filmed the attack.

The complaint. The Reverend Everett Parker of the United Church of Christ joined Aaron Henry, head of the Mississippi NAACP, and A. D. Beittel, the president of Tougaloo College, in 1963. They gathered complaints from viewers, knowing that they could be attacked for it. In 1964 they asked the Federal Communications Commission not to renew WLBT's license, the government permit a station needs to broadcast. The commissioners gave the station a one-year renewal without a hearing.

Judges on a federal appeals court ruled in 1966, in an opinion by Warren Burger, that viewers had the right to take part in these cases. In 1969 the judges ordered the commission to take away the owners' license. The owners appealed and lost in 1971. A board of Black and white citizens then ran the station, and the board hired some of the first Black television managers in the South.

Civil rights news on the page and on the screen, 1955 to 1965

Roy Bryant, his half-brother J. W. Milam and at least one other person took 14-year-old Emmett Till by force from his relatives' home in Mississippi in August 1955. Bryant and Milam later described beating him and shooting him in the head.

Three days later Till's body was found in the Tallahatchie River, weighed down with a 75-pound metal fan from a cotton gin, a machine that cleans cotton. The men had fastened the fan to his neck with barbed wire. An all-white jury found Bryant and Milam not guilty of murder. Afterward, both men told the writer William Bradford Huie, for *Look* magazine, that they had kidnapped and murdered Till.

The photographs. *Jet*, a weekly news magazine for Black readers published in Chicago by John H. Johnson's company, printed photographs of Till's body on September 15, 1955. *Jet* sold about 425,000 copies a week at the time. The Till issue sold out, and its publishers printed more.

The University of Mississippi. A crowd of more than 2,000 white segregationists rioted at the University of Mississippi on the night of September 30, 1962, to stop James Meredith from enrolling as its first Black student. Of the 212 people that marshals and soldiers held afterward, only 24 were students at the university, Justice Department lawyers wrote.

People in the crowd threw a pipe, rocks, bricks and bottles at US marshals. Hundreds of people were hurt, and two men died. One of them was Paul Guihard, a 30-year-old reporter for the French news agency Agence France-Presse, who was covering the story.

Someone shot Guihard in the back at close range, and the bullet hit his heart. No one has ever been charged with killing him. Justice Department lawyers closed the case in 2011 without naming anyone.

Selma. Alabama state troopers and other officers led by Major John Cloud attacked civil rights marchers with clubs and tear gas on the Edmund Pettus Bridge in Selma on March 7, 1965. At least 50 marchers went to the hospital. Television cameras filmed the attack, and the film was flown to New York.

The ABC newscaster Frank Reynolds interrupted the network's showing of *Judgment at Nuremberg*, a film about Nazi war crimes, at about 9:30 that night to show the film of the attack. Nearly 50 million people had tuned in to the film.

Walter Cronkite

Who: The CBS anchor who read the evening news to much of the country for 19 years, and who told viewers in 1968 that the Vietnam War was stuck.

When and where: New York, 1962 to 1981.

Movie: *Walter Cronkite: Witness to History* (2006), a PBS *American Masters* documentary about him.

"If I've lost Cronkite, I've lost Middle America."

Walter Cronkite was the anchor of the *CBS Evening News*, the person who reads the news and introduces the reports. A poll in 1972 named him "the most trusted man in America." He grew up in Houston, Texas, where he wrote for his school newspapers.

War reporter. In World War II Cronkite reported for the United Press news service. He flew on bombing raids over Germany and covered the D-Day landings in 1944. Edward R. Murrow hired him for CBS in 1950.

The anchor desk. Cronkite became the anchor of the *CBS Evening News* in 1962. Soon after, the program grew from 15 minutes to 30 minutes, the first half-hour nightly news program on network television. In 1963 Cronkite told viewers that President John F. Kennedy had been killed. He covered the Moon landing in 1969 and nearly every American spaceflight with a crew from 1961 to 1981.

Vietnam. Cronkite went to Vietnam in 1968, after the Tet Offensive, a large attack by North Vietnamese and Viet Cong forces. On February 27, 1968, he told viewers that the war was "mired in stalemate," stuck in a fight that neither side could win. President Lyndon Johnson told his staff, "If I've lost Cronkite, I've lost Middle America," meaning ordinary

American families. Encyclopaedia Britannica records that some people held this to be the reason Johnson decided not to run for president again that year.

Cronkite ended each program with "And that's the way it is." He left the anchor desk in 1981. He died in New York on July 17, 2009.

Vietnam and the reporters, 1965 to 1971

As late as 1964 fewer than two dozen American reporters were in Indochina, the region that includes Vietnam, Laos and Cambodia. By 1968 about 600 journalists from many countries were in Vietnam. More than 60 journalists were killed during the war, and the record of their deaths does not say who killed them.

US military officers did not censor the reports from Vietnam. The military's daily news briefings in Saigon were nicknamed "the five o'clock follies," after follies, silly shows.

Some Americans later blamed the press for turning people against the war, and Encyclopaedia Britannica records that the argument goes on. Experts who studied the news coverage, Britannica states, found that most of the reporting supported the war until 1968. They found that support at home fell most as more American soldiers were killed.

My Lai. American soldiers of Charlie Company killed unarmed villagers at My Lai in South Vietnam on March 16, 1968. The counts run from at least 175, found by the Army's own inquiry, to 504, the number of names on a memorial in the village. The soldiers also raped women and girls. Rape means forcing someone into sex by violence or threat.

Seymour Hersh, a freelance reporter, one who worked for himself rather than for one newspaper, followed a tip in 1969 and found Lieutenant William Calley. Hersh wrote the first public account of the killings, and he won a Pulitzer Prize, a top award for reporting, in 1970. Calley was found guilty in March 1971 of murdering 22 civilians. He was the only soldier convicted, and no one was convicted of rape.

The Pentagon Papers, 1971

Secretary of Defense Robert McNamara ordered a secret history in 1967 of what American leaders had done in Vietnam since World War II. It filled 47 volumes. Daniel Ellsberg, who had worked on it, gave copies to *The New York Times* without permission. The *Times* began printing stories about it on June 13, 1971.

The papers recorded that President Johnson had planned open war in 1964, a year before the public knew. They also recorded that he ordered the bombing of North Vietnam in 1965, even though intelligence officials had judged that it would not work.

Prior restraint. After three days, Justice Department lawyers got a prior restraint, a court order that stops something from being printed before it comes out, against the *Times*. *The Washington Post* also had the papers. The justices of the Supreme Court voted 6 to 3 on June 30, 1971, that the newspapers could keep printing.

Prosecutors charged Ellsberg under the Espionage Act, and he faced up to 115 years in prison. Men working for President Richard Nixon, led in this case by the former CIA officer E. Howard Hunt, broke into the office of Ellsberg's psychiatrist, a doctor who treats mental illness. They were looking for information to damage Ellsberg. In 1973 the judge dismissed all the charges because of this break-in and other wrongdoing by government officials.

The justices of the Supreme Court ruled in 1972, in a case called *Branzburg v. Hayes*, that the Constitution gives reporters no right to refuse to tell a grand jury who their sources are, the people who give them information. That same year two reporters at *The Washington Post* began getting information from a source whose name stayed hidden until 2005.

Bob Woodward and Carl Bernstein

Who: Two young *Washington Post* reporters who showed that a burglary in 1972 was tied to President Nixon's White House.

When and where: Washington, D.C., 1972 to 1974.

Movie: *All the President's Men* (1976), about their reporting, with Robert Redford as Woodward.

Five men were caught inside the Democrats' office at Watergate.

Bob Woodward and Carl Bernstein, two young reporters at *The Washington Post*, tied a 1972 burglary to President Richard Nixon's White House. Police arrested five men inside the office of the Democratic National Committee, the national leaders of the Democratic Party, in the Watergate building in Washington early on June 17, 1972. One of them, James W. McCord Jr., was the security chief of Nixon's re-election committee.

Woodward had covered minor crimes for the *Post* for nine months. Bernstein covered politics in Virginia. With a third reporter, Alfred E. Lewis, they wrote about the arrests for the next morning's paper.

Deep Throat. Nixon's press secretary, Ron Ziegler, called it a "third-rate burglary attempt." Most early news stories said the White House had nothing to do with it. Woodward and Bernstein kept reporting, and a secret source they called "Deep Throat" gave them information.

They reported on October 10, 1972, that the bugging of the Democrats' office, which means hiding microphones there to listen in, came from "a massive campaign of political spying and sabotage," or secret wrecking, run by White House officials for Nixon's re-election.

CBS cut short. CBS News aired the first part of a two-part report on the story on October 27, 1972. Nixon's aide Charles Colson then threatened William Paley, the head of CBS, and the second part was cut short. The records do not say who at CBS made the cut. That November Nixon

won re-election, carrying every state except Massachusetts. In 1973 the *Post* won the Pulitzer Prize for public service for Woodward and Bernstein's reporting.

Hearings on television. The reporting was one step among several that led to Nixon's resignation. Television stations showed a Senate committee's Watergate hearings live for two weeks in May 1973, and more than 150 public television stations showed them again in the evening. A month after the hearings began, 97 of every 100 Americans had heard of Watergate. A White House aide, Alexander Butterfield, told the committee that Nixon kept a voice-activated tape recorder in rooms of the White House.

The tapes. The justices of the Supreme Court ordered Nixon to hand over the tapes on July 24, 1974. One tape showed that on June 23, 1972, he had ordered the FBI to hold back its investigation. Members of the House Judiciary Committee approved three articles of impeachment, formal charges that could lead to his removal from office. Nixon resigned on August 9, 1974.

W. Mark Felt, who had been an associate director of the FBI, one of its top officials, said in 2005 that he was Deep Throat.

Reporters killed in the United States, 1976 to 1990

A car bomb in Phoenix. Don Bolles, a 47-year-old reporter for the *Arizona Republic*, went to a Phoenix hotel on June 2, 1976, to meet a man who had promised him information. It was to be about land fraud, cheating people in land deals, and organized crime, crime run by gangs. The man never came. When Bolles started his car, a bomb under it exploded.

Doctors had to cut off both his legs and one arm, and he died on June 13. In 1977 John Harvey Adamson pleaded guilty to building and planting the bomb. A jury later convicted Max Dunlap of murder and of conspiracy to murder, which means planning a murder with others. Dunlap died in prison in 2009.

Reporters from many other papers came to Arizona and spent five months finishing Bolles's reporting.

Five journalists for refugee papers. Killers murdered five Vietnamese American journalists between 1981 and 1990. They worked at small papers for refugees from the Vietnam War. Reporters at ProPublica who looked into the killings in 2015 tied them to what the papers printed about Vietnam and its government.

Duong Trong Lam, 27, was shot outside his home in San Francisco in 1981. A gunman chased Nguyen Dam Phong from his house in Houston in 1982 and shot him seven times.

Pham Van Tap was killed by the smoke from a fire that someone set in his office in Garden Grove, California, while he slept. A message sent to the Vietnamese press after his death said he was killed for printing advertisements for companies that did business with Vietnam.

The last two worked for the same magazine in Virginia, *Van Nghe Tien Phong*. A gunman shot and killed Do Trong Nhan, 56, who laid out the magazine's pages, in his car as he got ready to drive to work. Ten months later, on September 22, 1990, killers shot and killed the writer Le Triet and his wife, Dang-Tran Thi Tuyet, in their car in the driveway of their home.

FBI agents suspected a group led by former South Vietnamese military officers, but they never found enough evidence to charge anyone. No one was ever arrested or charged.

Public broadcasting, the fairness rule, cable news and talk radio, 1967 to 1996

Lawmakers, federal officials and owners changed who paid for the news on radio and television, and what it had to include, between 1967 and 1996.

Members of Congress set up the Corporation for Public Broadcasting in the Public Broadcasting Act of 1967, to hand out federal money to public television and radio stations. People at the corporation started PBS in 1969 and NPR, National Public Radio, in 1970.

Ted Turner's company started CNN, the first television channel to show news 24 hours a day, on June 1, 1980. It reached viewers through cable, wires that carried many channels into homes.

The fairness doctrine. The members of the Federal Communications Commission required stations from 1949 on to follow a rule called the fairness doctrine. Station owners had to cover important public questions fairly and air more than one side, and people attacked on the air had to get a chance to answer. The justices of the Supreme Court upheld the rule in 1969.

Members of Congress passed a bill in June 1987 to make the rule a law. President Ronald Reagan vetoed the bill, refusing to sign it. He wrote that the rule "requires Federal officials to supervise the editorial practices of broadcasters." That year the commissioners, led by their chairman, Dennis Patrick, voted 4 to 0 to end the rule.

Talk radio. Rush Limbaugh began a national radio talk show in 1988 on which he gave his conservative political opinions, views that favor keeping old ways and limiting government. Within five years an estimated 20 million people listened each week. His show began the year after the fairness doctrine ended. Whether the end of the rule helped talk shows like his grow is argued and not settled.

Rupert Murdoch's company started the Fox News Channel on October 7, 1996. Roger Ailes, a television producer and former Republican political adviser, ran it.

The internet, 1969 to 1996

Researchers at the University of California, Los Angeles, sent the first message over ARPANET, a computer network paid for by the Defense Department, to the Stanford Research Institute in late 1969.

Tim Berners-Lee, a British scientist working in Switzerland, opened the World Wide Web, the system of linked pages that people open over the internet, to the public in 1991. A team at the University of Illinois released Mosaic in 1993, the first widely used program for viewing web pages with pictures.

President Bill Clinton signed the Telecommunications Act in February 1996. Under it, one company could own more television stations, and many media companies merged after it passed. Under one part of it, sending "indecent" material to children online was a crime. The justices of the Supreme Court struck down that part in 1997. They ruled that the internet has the same protection for free speech as printed words.

BY 2000

If you remember one thing: As television brought the news into nearly every home, judges settled fights over keeping news from the public.

- 1969: Federal judges order the FCC to take WLBT's license from the owners who cut civil rights news off the air.
- June 30, 1971: The justices of the Supreme Court vote 6 to 3 that newspapers may keep printing the Pentagon Papers.
- July 24, 1974: The justices order President Nixon to hand over his tapes.

Next: After 2000 the news moved to phones and feeds, and thousands of newspapers closed.

2000 to Today

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Since 2000 the news has moved from paper to phones and feeds, and the men who own the biggest sites have changed what those feeds show. In a 2025 survey by the Pew Research Center, 86 of every 100 American adults said they got news at least sometimes from a phone, computer or tablet, and 64 of every 100 from television. Newspapers lost most of their advertising money. Researchers at Northwestern University counted close to 3,500 newspapers that closed between 2005 and 2025.

People angry at what reporters wrote killed journalists in their newsrooms and outside their homes. Police and federal agents arrested and held reporters at protests. In 2025 members of Congress took back the federal money for public radio and television.

The news on phones and feeds, 2004 to 2025

Mark Zuckerberg and three other Harvard students started Facebook in February 2004. Three former workers at the PayPal company started YouTube in 2005, and Google bought it in November 2006 for $1.65 billion. Twitter began in 2006.

The feed. Zuckerberg announced changes to Facebook's feed, the list of posts the site shows each user, in January 2018. Users would see fewer posts from news outlets and businesses and more posts from their friends and family.

Section 230. Under a law of 1996, known as Section 230, a website cannot be treated as the publisher of what its users post. That means the company that runs the site is not legally responsible, as a newspaper publisher would be, for what users write there.

Smartphones spread quickly. In 2011, according to the Pew Research Center, 35 of every 100 American adults owned one, and by 2025, 91 of every 100 did.

In a Pew survey in August 2025, about half of American adults said they got news at least sometimes from social media, most often on Facebook and YouTube. Only 7 of every 100 said they often read news in a printed newspaper

or magazine.

False news, measured, 2018

Three researchers at the Massachusetts Institute of Technology, Soroush Vosoughi, Deb Roy and Sinan Aral, published a study of news on Twitter in 2018. They followed about 126,000 chains of retweets, posts shared again on Twitter, from 2006 to 2017. Each chain began with one news story that people passed along. About 3 million people shared the stories more than 4.5 million times.

What they found. Six fact-checking groups had judged which stories were true and which were false. The researchers found that false stories were 70 percent more likely to be shared again than true ones. A true story took about six times as long as a false one to reach 1,500 people. False stories about politics spread fastest.

Some accounts on Twitter were run by computer programs, called bots. When the researchers left out the bots, the result stayed the same. So people, the researchers concluded, spread the false stories.

Who owns the feed, 2022 to 2026

Elon Musk bought Twitter for $44 billion in October 2022. He laid off about half its workers and cut back content moderation, the work of checking posts and removing those that break a site's rules. He began selling blue check marks for $8 a month.

Musk renamed the site X in 2023. In March 2025 he sold X to his artificial intelligence company, xAI, in a deal that valued X at $33 billion. In 2026 his rocket company, SpaceX, bought xAI.

On Facebook, Mark Zuckerberg made his own change. He announced in January 2025 that the company would end the fact-checking programs it had used for about ten years.

TikTok. TikTok belonged to ByteDance, a company in China. President Joe Biden signed a law in April 2024 that required ByteDance to sell TikTok within a year or see the app banned. The justices of the Supreme Court upheld the law in January 2025, and President Donald Trump then delayed the ban several times.

A new company took over TikTok in the United States in January 2026. The software company Oracle, the investment firm Silver Lake and a fund owned by the government of the United Arab Emirates each own 15 percent of it, and ByteDance kept 19.9 percent.

CBS and CNN. David Ellison bought Paramount, the company that owns CBS, in 2025, and he hired Bari Weiss to run CBS News. A *60 Minutes* report about a prison in El Salvador was pulled from the air in December 2025, three hours before it was to run. The correspondent, Sharyn Alfonsi, wrote that Weiss "spiked our story," which means stopped it, after CBS lawyers had cleared it.

Ellison's company also agreed to buy Warner Bros. Discovery, which owns CNN, for $111 billion. According to NPR, both deals depend on money from his father, Larry Ellison, a founder of Oracle. The top lawyers of 12 states, all of them Democrats, sued over the deal.

Paramount's owners settled the lawsuit in September 2026 by agreeing to set up a board meant to keep the owners out of news decisions at CBS and CNN. The purchase was not yet complete at the end of September 2026.

Newspapers close: the news deserts, 2005 to 2025

American newspapers took in less than a fifth as much money from advertising in 2022 as in 2005. Researchers at the Pew Research Center put the figures at about $49.4 billion in 2005 and an estimated $9.8 billion in 2022.

Where the money went. Advertisers spent more and more online, and most of that money went to big technology companies that do not report the news, such as Google and Facebook. Pew researchers wrote in 2014 that these companies could reach far more people than any news outlet, and aim their ads better than any one newspaper's website could. In 2014 five companies, including Google and Facebook, took in 61 of every 100 dollars spent on online ads in the United States.

Researchers at the Medill school of journalism at Northwestern University count newspapers every year. In their 2025 report they found that almost 2 of every 5 local newspapers in the country had closed since 2005, close to 3,500 papers in all. Papers were still closing in 2025, at more than two a week. Since 2005 the newspaper business had lost more than three of every four of its jobs.

News deserts. The Medill researchers call a county with no local newspaper, news website or station a news desert. The number of news deserts grew from just over 150 counties in 2005 to 212 in 2025, and most of them were in the countryside. Another 1,525 counties had only one local source left. In all, about 50 million people lived in a county with one local source or none.

Fewer owners. The number of companies and people who owned American newspapers fell by more than half, from 3,995 in 2005 to just under 1,900 in 2025.

The *Chicago Defender*, the paper that urged Black Southerners to move north in the 1910s, ended its print edition in 2019. Its owners now publish it only online.

Jeff German

Who: A Las Vegas newspaper reporter who was stabbed to death by an elected official he had written about.

When and where: Las Vegas, Nevada, 1980s to 2022.

The official he wrote about stabbed him outside his home.

Robert Telles, an elected official in Las Vegas, stabbed the reporter Jeff German to death in 2022 after German reported on Telles's office. German had reported in Las Vegas for four decades, and at the time of his death he worked for the *Las Vegas Review-Journal*. He wrote about organized crime and dishonest officials, and he hosted a *Review-Journal* podcast about the mob called *Mobbed Up*.

The stories. That year German wrote about the office of Robert Telles, who had been elected Clark County Public Administrator. When a person in Clark County dies and no family member can step in at once, workers in that office guard the dead person's property and money until they can hand it to the people who should get it.

German's stories described bullying in the office and the punishing of workers who complained. They also described an "inappropriate relationship" between Telles and a staff member. Telles denied it. In June 2022 he lost his election.

The evidence. Telles stabbed German to death outside German's home on September 2, 2022. Police found Telles's DNA, the chemical code inside every person's cells that can show who was at a place, on German's hands and fingernails. In Telles's home, police found a straw hat and sneakers, cut into pieces, like the ones the killer wore on a security camera video.

A jury found Telles guilty of murder on August 28, 2024. In October 2024 Judge Michelle Leavitt added more time because German was over 60 and Telles had used a deadly weapon. Telles was sentenced to 28 years to life in prison.

A newsroom attacked: Annapolis, 2018

Jarrod Ramos walked into the newsroom of the *Capital Gazette* in Annapolis, Maryland, on June 28, 2018, with a shotgun, smoke grenades and extra ammunition. He killed five people who worked there: Gerald Fischman, Rob Hiaasen, John McNamara, Rebecca Smith and Wendi Winters.

Ramos held a grudge because in 2011 the paper's reporters had written that he was convicted of harassing a woman he had gone to high school with. In 2021 a jury found that he was responsible for his crimes, and a judge sentenced him to five life terms in prison with no chance of release.

Darnella Frazier

Who: A 17-year-old who recorded a Minneapolis police officer killing George Floyd, and whose video became evidence at his trial.
When and where: Minneapolis, Minnesota, 2020 to 2021.

She was 17 and out walking with her nine-year-old cousin.

Darnella Frazier's phone video of a police officer killing George Floyd reached people around the world. She was walking with her cousin in Minneapolis on May 25, 2020, when she saw the police officer Derek Chauvin kneeling on the neck of George Floyd, a Black man, and she recorded it. Her video showed Chauvin kneeling on Floyd's neck for nine minutes and 29 seconds, and it recorded Floyd's murder.

The trial. According to NPR, the video played a major part in starting protests against police violence around the world. At Chauvin's trial in 2021, prosecutors played it again and again, and Frazier testified.

The jury found Chauvin guilty of three different legal grades of killing a person: second-degree murder, third-degree murder and second-degree manslaughter, which means killing a person without meaning to kill. The journalist Ann Marie Lipinski said, "There is no case without her."

The members of the Pulitzer Prize Board gave Frazier a special citation, an honor outside their usual prizes, on June 11, 2021. The citation identifies "the crucial role of citizens in journalists' quests for truth

> and justice." Frazier called recording the video a "traumatic life-changing experience."

A small-town paper raided: Marion, Kansas, 2023

Police Chief Gideon Cody of Marion, Kansas, led officers in a raid on the weekly *Marion County Record* and on the home of its publisher, Eric Meyer, on August 11, 2023. The officers took computers and reporters' phones. Cody claimed that Meyer and the reporter Phyllis Zorn had committed identity theft, pretending to be someone else by using that person's private information, or computer crimes. He said they had done it by checking a local business owner's driving record.

Joan Meyer. Meyer's mother, Joan Meyer, was 98 and owned part of the paper. A police camera recorded her telling officers, "Get out of my house!" She died of a heart attack the next afternoon.

Special prosecutors, lawyers brought in from outside to handle the case, later found that no one at the paper had committed a crime. They said Cody had reached a wrong conclusion through "a poor investigation." They said they could not charge anyone over Joan Meyer's death. Cody was charged in August 2024 with felony obstruction of justice, a serious crime of getting in the way of an investigation, for asking the business owner to delete text messages.

Reporters arrested, hit and deported, 2017 to 2026

Staff of the US Press Freedom Tracker, a project of the Freedom of the Press Foundation, have counted attacks on journalists in the United States since 2017. More than half of all the assaults they counted came in two years, 2020 and 2021: nearly 800 of the 1,487 they had counted by September 2026. They also counted 445 arrests or criminal charges.

Protests against deportation. The tracker counted at least 32 journalists arrested or held in 2025 and at least 170 assaulted. Most were covering protests against deportation, the sending of immigrants out of the country. Police in Los Angeles surrounded and held at least 10 journalists at one protest on June 9, 2025.

Minneapolis. In January 2026 the tracker counted at least nine assaults on journalists in the Minneapolis area. Most of those journalists were hit by pepper balls, small balls fired from a gun that burst into a stinging powder, or by pieces of stun grenades, both of them crowd-control weapons. The tracker's January report does not say who fired each one. Federal agents held the photographer John Abernathy on the ground on January 15, 2026.

Mario Guevara. Mario Guevara, a Spanish-language reporter who covered immigration near Atlanta, was arrested on June 14, 2025, while he livestreamed a protest, showing it live on the internet. The Committee to Protect Journalists' account of the case does not name the police who arrested him.

Federal officials held Guevara and argued that his filming of police was a danger to public safety. They deported him on October 3, 2025, sending him out of the country to El Salvador. The Committee to Protect Journalists said it was the first time it had recorded this kind of punishment for reporting.

WikiLeaks. Julian Assange published secret documents about the US wars in Afghanistan and Iraq on his WikiLeaks website starting in 2010. Federal prosecutors charged him in 2019 under the Espionage Act of 1917, the law once used against the editor Victor Berger and against Daniel Ellsberg. In June 2024 Assange pleaded guilty under that law.

The press pool. White House press secretary Karoline Leavitt told the Associated Press, the AP, on February 11, 2025, that its reporters could no longer come into the Oval Office with the press pool, the small group of reporters that covers the president up close. The ban was to last until the AP agreed to call the Gulf of Mexico the Gulf of America.

A federal judge, Trevor McFadden, ordered the AP's access restored in April. In October 2025 officials at the Pentagon made reporters agree to have their material checked in order to keep their passes.

Public broadcasting loses its federal money, 2025 to 2026

Republican majorities in both houses of Congress passed the Rescissions Act in July 2025, and President Trump signed it. With the act, members of Congress took back about $1.07 billion already promised to the Corporation for Public Broadcasting for 2026 and 2027, according to the research group USAFacts. That money had gone to NPR, PBS and local public radio and television stations.

Trump had signed an order in May 2025 calling for an end to government money for news outlets that did not meet standards of "fairness, accuracy, and nonpartisanship," which means not taking the side of any political party.

The members of the corporation's board voted in January 2026 to close it after 58 years. Public radio and television stations cut about 550 jobs between July 2025 and May 2026, according to a count reported by Nieman Lab.

Rural stations and stations run by Native nations had gotten a large share of their money from the federal government, and they were hit hardest. By mid-2026 at least one public television station had shut down, and more had announced plans to close. No full count had been published. Stations of every kind saw the number of people giving them money rise between 2024 and 2025.

Pressure on commercial stations. Officials also pressed commercial broadcasters, the stations paid for by advertising. The members of the Federal Communications Commission give out the licenses that those stations need to broadcast.

The talk-show host Jimmy Kimmel spoke on his show in 2025 about which political side the man accused of killing the activist Charlie Kirk was on. Brendan Carr, the chair of the FCC, then pressed ABC to take *Jimmy Kimmel Live!* off the air. In 2026 Carr opened an investigation of the talk show *The View* under the rule that stations must give political candidates equal time.

THE THREE QUESTIONS, ANSWERED

1. **Did the First Amendment stop the government from punishing people for what they printed?** No. Seven years after it took effect, federal judges jailed editors under the Sedition Act, and during World War I prosecutors won more than 1,000 convictions for words.
2. **Who was shut out of carrying and printing the news?** From 1802 to 1865 federal law let only free white persons carry the mail, and white mobs drove Black editors out of Memphis and Wilmington. Black and Cherokee Americans started papers of their own.
3. **Why have so many local newspapers closed since 2005?** Advertisers moved their money online, where most of it went to technology companies such as Google and Facebook, so newspapers' advertising money fell by more than four-fifths between 2005 and 2022. Readers moved to phones and feeds too, and by 2025, 212 counties had no local news source.

What changed: In 1673 a post rider took about two weeks to carry the mail from New York to Boston. By 1920 radio listeners heard election results on election night, and by 2025 most American adults got at least some news on a phone, computer or tablet.

FIVE DATES TO KEEP

September 25, 1690: *Publick Occurrences* appears in Boston and is stopped four days later.

December 15, 1791: The First Amendment takes effect.

March 3, 1865: Congress ends the whites-only rule for mail carriers.

November 2, 1920: KDKA broadcasts the election results.

June 30, 1971: The Supreme Court lets the Pentagon Papers be printed.

NOT KNOWN: No record survives of what news Native traders passed along with the copper and shells they carried before 1500.

Where to go: Part of the Boston Post Road, the first post rider's trail in 1673, is now US Route 1, and you can drive it.

Next: Chapter 32, Art. It asks who decided what counted as art, and who got to see it.

PART 9

Culture and Play

Chapter 32: Art

CHAPTER 32 OF 37 · ART · 10 ERAS · ABOUT 155 MINUTES

Who was allowed to make art in America, who got paid for it, and who ended up owning it has always depended on who held power.

An enslaved potter in South Carolina named Dave cut two lines into a big clay jar in 1857. They use lucre, an old word for money: "I made this Jar = for cash / Though its called Lucre trash." Dave was not paid for that jar. It was sold for the profit of the man who held him, Lewis J. Miles. On October 29, 2025, the Museum of Fine Arts in Boston gave Dave's descendants ownership of it and of a second jar he made that year.

This chapter follows the pictures, carvings, pots, photographs and books made in this land, from Ancestral Pueblo bowls to pictures made by computer programs. Keep an eye on the names. The carvers of the Ohio pipes left none, but the men who dug up, bought and sold such work often did. Then watch the money. Enslaved painters and potters earned it for other men, the mother in the 1936 photograph *Migrant Mother* never got a penny, and artists today are suing over computer programs built from their pictures. From the 1800s on, the fights are also about who gets into art schools, museums and shows.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. How did enslaved artists make art, and who got paid for it?
2. Who decided what counted as art, and who got to see it?
3. Why are museums giving back things they once took?

Five minutes? Read "Photographs of hunger, and photographs of the camps" in 1900 to 1950, and the Lange box after it.

Can skip: "Portrait painters in the colonies", "Writers", "Painters of the 1930s and 1940s", "A camera in every pocket". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 9 MINUTES

Every pot, carving and copper plate made in this land before 1500 was made by Native people, and no surviving record names the people who made them. Potters in the Southwest painted bowls and jars. Carvers in the Ohio valley shaped stone pipes into birds and animals. In the towns of the Mississippi valley and the Southeast, metalworkers hammered copper into sheets and pressed pictures into it, and on the Northwest Coast people carved bone, horn and stone.

Many of the objects that survive were buried with the dead in graves and mounds. In the 1800s and 1900s, diggers took them out of the ground, and many of those diggers sold what they found.

How did Ancestral Pueblo potters paint a pot?

Ancestral Pueblo potters painted black designs over the white of the clay, and the pottery is now called black-on-white. The Ancestral Pueblo people were the ancestors of today's Pueblo nations. A potter rolled clay into a long snake and coiled it around in a spiral to build up the pot. Then the potter scraped the coils together until the sides were smooth.

The black paint came from boiled plants, such as beeweed or tansy mustard, or from crushed rock with iron in it, such as hematite. Potters made their brushes from the fibers of the yucca plant. At Mesa Verde, people made this pottery from about the year 600 to about 1300.

Bowls with a hole. In the Mimbres valley of southwestern New Mexico, potters painted bowls from about 1000 to 1130 or 1150. Most of these bowls have been found in graves. Archaeologists, the scientists who dig up and study the things people left behind, think people set a bowl over the dead person's face or head. Before they did, someone punched a hole through the bottom of the bowl on purpose. Archaeologists call it a kill hole.

Pictures on the rock

Native artists cut or pecked some pictures into rock, called petroglyphs, and painted others, called pictographs. On the volcanic cliffs west of Albuquerque, New Mexico, there are about 24,000 to 25,000 petroglyphs. They show animals, people, crosses, the brand marks that ranchers burn into cattle, and other images.

Ancestral Pueblo people chipped most of them into the rock with chisels and other stones between 1300 and the 1680s. The oldest are much older and date from about 2000 BC. For some Native nations the petroglyphs still have religious meaning today, and a place in their ceremonies.

Pipes, copper and stone in the Ohio and Mississippi valleys

Carvers of the Hopewell culture made smoking pipes of stone in the shapes of animals. The Hopewell people lived in the Ohio valley from about 100 BC to about AD 400 or 500. Their pipes show birds of prey, the birds that hunt, and ducks and other water birds. They also show songbirds, frogs, toads, meat-eating animals and river otters.

Stone from far away. One Ohio mound, Tremper Mound in Scioto County, held a store of more than 500 objects, among them 136 pipes. Archaeologists thought most of the pipes were carved from Ohio pipestone. When scientists tested some of them, 16 out of every 100 turned out to be catlinite. That stone came from quarries, pits where people dug stone, in southwestern Minnesota.

Copper plates. Metalworkers at Etowah, near Cartersville, Georgia, hammered copper into thin sheets between about 1250 and 1325. They pushed the design up from the back of the sheet so it stood out on the front. The best-known plates show a dancing figure that archaeologists call the Birdman, or the falcon dancer.

The Birger figurine. Near Cahokia, in what is now Illinois, a carver made a figure of flint clay, a stone from near St. Louis, Missouri, between 1050 and 1200. It shows a kneeling woman with a pack on her back. She is using a hoe on the back of a snake that has the head of a cat. It was dug up in 1979.

At Spiro, in eastern Oklahoma, people built twelve mounds between 850 and 1450, along with grounds for ceremonies and a town. One of the mounds, Craig Mound, held objects of stone, copper, shell, basketry and cloth.

Carving on the Northwest Coast

On the Northwest Coast, people carved bone, stone and horn at least 5,000 years ago. The oldest Coast Salish basketry comes from an old fishing camp on the Hoko River, on the Olympic Peninsula. It is about 2,500 years old.

Most carving on the coast was done in cedar wood. Wood rots, so very little of it survives from before 1500. One piece that may be that old is a hat woven from twisted strands of cedar bark, found at the mouth of Wapato Creek. It dates from between 1450 and 1620. Carvers on the coast still work in this tradition today.

Who dug the objects up, and where did they go?

Two men named Ephraim Squier and Edwin Davis dug into the Mound City mounds, near Chillicothe, Ohio, in 1846 and took more than 200 stone pipes. Long before they came, the pipes had been broken and laid together under the floor of a building used for ceremonies.

Mound City to London. Davis sold his collection in 1864 for $10,000 to William Blackmore, who ran a museum in Salisbury, England. The collection was sold again in 1931, to the British Museum in London. Fewer than two dozen of the carved animal pipes have been on public display there.

Etowah. John Rogan dug at Etowah for the Smithsonian, the national museum in Washington, in 1885. He took the two best-known Birdman plates from a grave lined with stone. The Smithsonian's National Museum of Natural History still holds them.

Spiro. Craig Mound at Spiro was a burial mound. In 1933 six men who called themselves the Pocola Mining Company paid $50 each, $300 in all, to George Evans for the right to dig into it for two years. Evans was the grandfather of the children who owned the land. He needed the money to pay off a loan on the land and to treat his grandson's tuberculosis, a disease of the lungs.

The Pocola Mining Company hired coal miners, who dug tunnels into the mound. A business that sells copies of Spiro objects shows photographs of the dig from 1935 on its website, with notes drawn from a 1952 report on the mound. The notes name two of the diggers, John Hobbs and K. A. McKenzie, who pried up thirty copper axes and carried them out of one of the tunnels. Captions on the photographs name three more: the brothers Bill and Hayden Vandagriff, and W. Guinn Cooper.

What the diggers did. From 1933 to 1935 the diggers took things out of the graves to sell, and they destroyed about a third of the mound. They sold thousands of objects of stone, copper, shell, basketry and cloth to collectors around the world. One pile of objects, found in April 1935, went to a secondhand furniture dealer from Indianapolis, Indiana, for $100.

People who saw the dig described it. Beads, pearls and arrowheads lay spilled across the ground. The diggers trampled feather capes and woven cloth. They hauled out decorated shells by the wheelbarrow and dumped them near the entrance, where the shells were crushed under the diggers' feet. They burned ancient cedar poles as firewood. They piled human bones at the edge of their camp, where the bones crumbled to dust.

The blast. Forrest Clements, an archaeologist at the University of Oklahoma, tried to stop the dig. Oklahoma lawmakers passed a law in 1935 that required a license to dig at such sites, and Clements filed a complaint with the sheriff of LeFlore County. When he left for California, the diggers came back.

The historian David La Vere calls Craig Mound by its other name, the Great Temple Mound. In his words, "In a fit of spite, just to jab their finger in Clements' eye," the diggers "packed the central chamber of the Great Temple Mound with kegs of black powder and touched off a mighty explosion." The blast caved in part of the mound and left a large crack in it.

The Encyclopedia of Oklahoma History says the Pocola Mining Company was shut down for good in November 1935. It does not say who shut it down.

Then University of Oklahoma archaeologists led workers paid by a government jobs program, the Works Progress Administration, from 1936 to 1941. The workers took the remains of more than 600 people out of Craig Mound, some

whole skeletons and some only parts.

Mimbres. In the Mimbres valley, looters dug up graves to reach the painted bowls. In the 1960s they began using bulldozers and earth movers to get to the bowls faster, and they wiped out many village sites completely. Every Mimbres site has been looted. The records do not name the people who ran the bulldozers.

Nathan Jackson

Who: A Tlingit carver of totem poles and masks, still carving in the Northwest Coast tradition in his eighties.

When and where: Born August 29, 1938, in southeastern Alaska. He lives in Ketchikan, Alaska.

At 86, he carved an eight-foot pole about Raven and Fog Woman.

Carving on the Northwest Coast did not stop in 1500, and Nathan Jackson carves in that tradition today. He is Chilkoot Tlingit. Every Tlingit person belongs to one of two halves, Raven or Eagle, and each half is made up of clans, large groups of related families. Jackson's clan, the Sockeye Clan, is on the Raven side.

He carves totem poles, masks, canoes, carved doors and panels that show clan crests, the animal symbols that belong to a clan. He also works metal and dances. He learned his people's traditions from Jack David, his clan uncle and grandfather.

Totem poles in a hospital. Jackson served in the Army in Germany until 1959. Later he spent 54 days in a hospital with pneumonia, a serious sickness of the lungs, and while he was there he whittled small totem poles. "I told the nurses, 'Those totem poles there, they have a price on them,'" he said later.

NOT KNOWN: whether that hospital stay came soon after his Army service or in his mid-30s, because published accounts of his life disagree.

Learning the adze. He joined the Chilkat Dancers of Haines, Alaska, in the early 1960s. Then he studied for about three years at the Institute of American Indian Arts in Santa Fe, New Mexico. After that the carver Duane Pasco taught him to work with an adze, a carving tool with a curved blade set across the end of its handle. Through a program of the Alaska State Council on the Arts, he taught apprentices, younger carvers who learn by working beside a master.

Officials of the National Endowment for the Arts, a US government agency, gave him an honor called a National Heritage Fellowship in 1995. He has said that "the traditional way of life is something I identify with."

He began the Raven and Fog Woman pole in 2024 at the Edwin DeWill Carving Center in Saxman, Alaska. He finished it that autumn at the Arts Council of Princeton, in New Jersey, where it stands in the lobby. In the Tlingit story the pole shows, Raven marries Fog Woman, a woman who came out of the fog, for the salmon in the basket she weaves. When he mistreats her, she leaves him, and all the salmon follow her.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 4 MINUTES

The oldest European pictures of this land and its people were made by two artists who came with colonies that did not last. Jacques Le Moyne drew the French colony in Florida in 1564. John White painted the coast of what is now North Carolina in 1585. In Europe, Theodor de Bry cut copies of both men's pictures into metal plates so that he could print them in books, and he changed them.

How Le Moyne lost his Florida drawings

Jacques Le Moyne lost every drawing he made in Florida when Spanish soldiers took the French fort there in 1565, so the pictures that survive are ones he drew again from memory. French colonists had set up the colony, called Fort Caroline, in 1564, in a part of Florida that Spanish leaders claimed. They brought Le Moyne along as their official artist, to record it. He drew the fort, the settlers and the land, and the people who already lived there, the Timucua.

The attack. Pedro Menéndez de Avilés and 500 Spanish soldiers took the fort in an attack at dawn on September 20, 1565. They killed most of the French settlers they found. Menéndez spared about 50 women and children and shipped them to Havana. Le Moyne escaped.

Sold, and changed. Le Moyne died in France in 1588, and his widow sold his new drawings to de Bry, who printed them in 1591. De Bry changed the people and the landscapes to make his books sell better. He also made different versions of the pictures for different buyers.

One painting by Le Moyne, *Laudonnière and Athore, 1564*, shows the French commander René de Laudonnière with Athore, a Timucua leader. The year in the title is the year of the meeting the picture shows. Le Moyne made it after he got back to Europe. The New York Public Library holds it.

John White

Who: The English artist whose Roanoke watercolors are the oldest surviving pictures Europeans made on the spot in this land. Later he was governor of the 1587 colony there.

When and where: Roanoke Island and the coast of what is now North Carolina, 1585 and 1586, and again in 1587. Probably born in Cornwall, England. His years of birth and death are not known. He was working from 1577 to 1593.

More than seventy watercolors survive from his thirteen months at Roanoke.

John White made the first pictures of this land by an English artist.

The voyage. White sailed as the artist on Sir Richard Grenville's voyage to Roanoke Island in 1585. Walter Raleigh and other men who put money into the voyage paid for it. White worked beside the mathematician Thomas Harriot. The coast they recorded was the land of nations that spoke Algonquian languages, a family of related languages. Among them were the Roanoke, the Secotan and the Croatoan. In July 1585 White visited the town of Secotan and painted it.

What he painted. In about thirteen months he painted Algonquian towns, people, dances, clothing and fishing, and plants, birds and fish. He drew each outline in black lead, a kind of pencil. Then he filled it in with watercolor and added touches of gold and silver paint.

De Bry printed engravings, pictures cut into metal plates so that many copies could be printed, of White's pictures of people in 1590, in a book with Harriot's report on the colony. He changed some of them. In one of White's watercolors, a Secotan girl holds a small doll dressed in English clothes. De Bry's copy also puts a model of the globe and the sky in her hands.

The British Museum in London owns White's original watercolors, and its staff keep them stored away from light. In the middle of the 1800s, water badly damaged them during a fire at the Sotheby's auction house. Over

time, chemical changes have turned the silver paint black.

The governor. White returned to Roanoke in 1587 as the governor of a new colony. After the colonists landed, one of them, George Howe, was found dead. He had been shot with sixteen arrows and beaten. The Encyclopedia Virginia account of the colony does not name the men who killed him.

In August 1587 White led an attack on the town of Dasamonquepeuc to pay back Howe's killing. He believed its people were Roanoke, enemies of the English. They turned out to be Croatoan, who were friends of the English, and White's men killed a number of them. One account says the dead included a leader named Menatonon, and other accounts name the leader differently.

That same year the colonists talked White into sailing to England for more supplies. He could not return for three years. In 1588 Queen Elizabeth I took every English ship for her navy, to fight the Spanish Armada, a fleet of warships sent from Spain.

When White reached Roanoke again on August 18, 1590, the colonists were gone. The word "CROATOAN" was carved on a post. Chapter 31 tells what that word meant and how White tried to find the colonists.

NOT KNOWN: what became of the colonists.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 7 MINUTES

Art in the English colonies of the 1600s was made for families with money, for the graves of the dead and for the church. Painters made portraits for rich Boston families, carvers cut pictures into gravestones, and girls stitched needlework. A printer in Massachusetts produced the first full book printed in the English colonies in 1640, and a book of poems by a woman living in Massachusetts came out in London ten years later.

In New Mexico, a Spanish friar, a member of a Catholic religious order, had Acoma people build a church on top of the steep hill where their town stands, and they had to carry every load up to the top. Other friars, and Spanish soldiers, burned the masks that Pueblo people used in their ceremonies.

Portraits, gravestones and needlework in New England

English colonists built Boston on land of the Massachusett nation, on a peninsula the Massachusett called Mushauwomuk. Experts credit about ten portraits of people in Boston, all painted from 1670 to 1674, to one painter. No record gives the painter's name, so today the painter is called the Freake Painter, after the best-known portrait. An article in *The Magazine Antiques* calls the painter "probably Samuel Clement," but nobody has proved it.

A baby painted in later. That portrait shows Elizabeth Freake holding her baby daughter, Mary. The Freakes were one of the richest families in Boston. The painter first painted Elizabeth alone, in about 1671. Mary was born on May 6, 1674, and X-rays of the painting show that the baby was painted in afterward. The Worcester Art Museum, in Massachusetts, owns the painting.

Skulls with wings. Many settlers of New England were Puritans, English Protestants, Christians outside the Catholic Church, who wanted a plainer and stricter church. Puritans did not want religious symbols such as crosses and cherubs, carvings of baby angels, on their graves. Carvers cut other pictures

into Puritan gravestones: skulls with wings, crossed bones and hourglasses. Many stones carry the Latin words "Memento Mori," which mean "remember you will die."

Joseph Lamson of Charlestown, Massachusetts, who lived from 1658 to 1722, is one carver whose name is known. He carved winged skulls with eyebrows that hook at the ends, inside borders of fruit, vines and flowers. His sons Nathaniel and Caleb and his grandson Joseph carried on the work.

Girls learned needlework from their mothers. Many stitched samplers, strips of cloth that kept a record of stitches and patterns to copy later.

Loara Standish

Who: A daughter of Myles and Barbara Standish of Plymouth Colony, thought to be the maker of the oldest known sampler made in America.
When and where: Duxbury, in Plymouth Colony, in the 1640s or about 1653. She was born after 1627 and died before 1655.

"Loara Standish is my name," she stitched in silk on linen.

Loara Standish is thought to have stitched the oldest known sampler made in America, and the oldest known sampler with a verse. She made it in Duxbury, in Plymouth Colony. English settlers built Plymouth on the site of Patuxet, a Wampanoag town.

Loara was one of the seven children of Myles and Barbara Standish. According to the Pilgrim Hall Museum, she probably made the sampler in her teens. She stitched it in silk thread on linen cloth, with roses, carnations and oak leaves, and a letter S twisted around itself.

The verse. She stitched this into it: "Loara Standish is my name Lorde guide my hart that I may doe thy will also My hands with such Convenient skill as may Conduce to virtue void of Shame and I will give The glory to thy name." The verse asks God to guide her heart and to give her hands enough skill to do good. She promises to give God the glory.

> Lucius Alden gave the sampler to the Pilgrim Hall Museum in Plymouth in 1844. It has faded from being hung up for so many years. Experts at the museum date it to about 1653. Experts at the Textile Research Centre in Leiden, in the Netherlands, date it to the 1640s.

The first books printed and written here

The Reverend Jose Glover sailed for Massachusetts with a printing press in 1638 and died on the ship. His widow, Elizabeth Glover, set up the press in Cambridge, Massachusetts. She put it in the charge of Stephen Daye, a locksmith.

The Bay Psalm Book. Daye printed *The Whole Booke of Psalmes*, now called the *Bay Psalm Book*, in 1640. A group of thirty elders, men respected in the colony for their learning, had put the psalms, 150 poems in the Hebrew Bible, into English verse that could be sung. Among them were John Cotton, Richard Mather and John Eliot. Daye is thought to have printed about 1,700 copies, which sold for twenty pence, or pennies, each. Only eleven copies survive today.

Anne Bradstreet

Who: A poet in the Massachusetts Bay colony, whose book was the first book of poems by someone living in the English colonies.

When and where: Born in England in 1612. Came to Massachusetts in 1630. Died in 1672.

The title page called her only "a Gentlewoman of those parts."

Anne Bradstreet wrote the first book of poems by someone living in the English colonies. It was printed in London in 1650 as *The Tenth Muse Lately Sprung Up in America*. In old Greek stories, nine goddesses called the Muses watched over poetry and the other arts, so the title describes her as a tenth one. Her full name appears inside the book.

She was the daughter of Thomas and Dorothy Dudley. She married Simon Bradstreet in about 1628, sailed to Massachusetts with him and her parents in 1630, and had eight children.

How the poems got to London. Her brother-in-law, the Reverend John Woodbridge, took her poems to London. In the book's opening pages, Woodbridge wrote that she had not known he was doing it. Later, in a poem called "The Author to Her Book," Bradstreet called him a "friend more loving than true" for taking them.

What she knew. The biographer Charlotte Gordon studied the poems and the letters between Bradstreet and Woodbridge. She found that Bradstreet knew about the plan and helped with it from the start. One poem in the book is about events of 1649. Woodbridge had sailed in 1648, so someone had to send that poem after him.

Mission churches and burned masks in New Mexico

Acoma Pueblo is a town built on top of a mesa, a hill with steep sides and a flat top. Chapter 2 tells in full what happened there in 1598 and 1599. In the Acoma people's own account, the soldiers of the Spanish captain Juan de Zaldívar raped an Acoma woman, which means they forced her into sex. In December 1598 Acoma people killed Zaldívar and eleven to fourteen of his men.

The attack on Acoma. In January 1599 Spanish soldiers sent by Juan de Oñate, the leader of the Spanish colony in New Mexico, took the town and burned much of it. They killed hundreds of Acoma men, women and children, between about 300 and about 1,500 by the estimates.

Oñate sentenced the Acoma men older than twenty-five to have one foot cut off. He also sentenced them to twenty years of forced labor, work for others without pay and without the right to leave. Whether the feet were really cut off is disputed.

The church on the mesa. A friar named Juan Ramírez arrived at Acoma in 1629. He probably began soon after to have Acoma people build a church on the mesa, called San Esteban del Rey. The Acoma workers had to carry every load up the mesa. The loads included roof beams 40 feet long, cut in mountains 30 to 40 miles away. One account says the builders were enslaved. Another says only that they had to do the work.

NOT KNOWN: the exact years of the building, or whether the workers were paid, which the surviving records do not say.

San Esteban del Rey is the oldest European church in New Mexico that still stands.

Burned masks. Pueblo people made masks for their ceremonies. Katsina masks show spirit beings, and they were kept in kivas, rooms used for religious ceremonies. Alonso de Posada, the head of the Franciscan friars in New Mexico, banned Pueblo ceremonies from 1656 to 1665. He ordered friars to go into the kivas and burn the masks. They also burned prayer sticks and

carved figures, or took them away. In 1661 Spanish soldiers raided kivas and burned masks as well. No surviving record counts the masks that friars and soldiers burned.

Governor Juan Francisco Treviño had 47 Pueblo religious leaders arrested in 1675. Three of them were hanged, and one killed himself. The rest were whipped in public and jailed, until Pueblo men came to Santa Fe and forced their release. Treviño ordered the arrests. Dennis Zotigh's account of these events for the Smithsonian does not name the men who hanged and whipped the prisoners.

One of the freed men was Po'pay, of Ohkay Owingeh. He led the Pueblo Revolt on August 10, 1680, in which Pueblo people killed 21 friars and burned many of the mission churches across the province.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 7 MINUTES

The first trained painters in the colonies earned their living from families with money, and some of that money came from slavery and from Native land. A few painters trained in Europe settled here between 1700 and 1750. They painted merchants, planters and officials, because only families with money could pay for a portrait.

Two portraits from these years show slavery. One shows an enslaved boy in a metal collar beside the son of a rich Maryland family. The other shows the Royall family, whose money came from trading in sugar, rum and people. One painter made portraits of two Lenape leaders two years before William Penn's sons took a large part of Lenape land. A colonial governor took Cherokee baskets to London in 1725, and they are still kept there.

Portrait painters in the colonies

Henrietta Johnston drew portraits in Charleston, South Carolina, from 1708. John Smibert came from Scotland and painted in Boston from 1729 or 1730.

Justus Engelhardt Kühn came from Germany. He applied at Annapolis, Maryland, in 1708 to become an English subject, with the rights of a person born English. He painted the children of rich Maryland families until he died in 1717. He was the earliest artist known to work for pay in the Middle Atlantic colonies, the group of colonies that included Maryland.

Gustavus Hesselius was born in Sweden in about 1682 and studied art there. He painted portraits in Philadelphia. He also built church organs, among them one for the church of the Moravians, a Protestant group, in Bethlehem, Pennsylvania, in 1746. He died in Philadelphia on May 23, 1755. Robert Feke painted portraits in Newport, Rhode Island.

Henrietta Johnston

Who: A portrait artist who worked in pastels. She was the first woman recorded as working for pay as an artist in the English colonies.

When and where: Born in northern France in about 1674. Worked in Charleston, South Carolina, from 1708 until she died there on March 9, 1729.

Pastels came only from England, and pirates attacked her voyage home.

Henrietta Johnston was the first woman recorded as working for pay as an artist in the English colonies. She was also the first artist in the colonies to use pastels, crayons made from a paste of colored powder and resin, a sticky stuff that comes from trees.

Driven out of France. Her family were French Protestants. Two years before they fled to London in 1687, King Louis XIV of France had taken away Protestants' right to practice their religion. He sent soldiers to live in Protestant homes, and the king's officers let the soldiers loot the homes and ruin the families.

The soldiers tortured some people who would not become Catholic, which means they hurt them on purpose to make them give in. They stayed until the families became Catholic or left France.

Henrietta married in 1694 and moved to Ireland, where she learned to draw portraits in pastel. Her first husband died in 1704 and left her with two small children.

Charleston. Her second husband, Gideon Johnston, was a minister of the Church of England. In 1708 he moved the family to Charleston, where he served as the representative of the bishop of London. The Stono, Kiawah and Etiwan peoples had lived along the rivers at Charleston when English colonists settled there in 1670. The Johnstons were short of money, so Henrietta drew portraits of the town's richest families for pay.

Pastels could be bought only in England. She sailed to London for supplies in 1711, and on the voyage home pirates attacked her ship. She survived. After Gideon died in 1716, she earned the family's living alone. About three dozen of her portraits survive.

John Smibert

Who: A Scottish portrait painter in Boston, whose shop and studio showed younger painters copies of European art. He also drew the plans for Faneuil Hall.

When and where: Born in Edinburgh, Scotland, in 1688. Newport, Rhode Island, in 1729, then Boston until he died there in 1751.

Upstairs over his paint shop hung copies of Europe's famous paintings.

John Smibert was one of the first painters with formal art training to work in the thirteen colonies. In 1728 a church leader named George Berkeley asked him to come to America. Berkeley planned to start a college in Bermuda, where Smibert would teach art. The college was never started.

Smibert reached Newport, Rhode Island, with Berkeley's group in 1729 and painted Berkeley and his party in a picture now at Yale University. By 1729 or 1730 he had moved to Boston. There he painted merchants, such as Peter Faneuil, and Judge Samuel Sewall. He drew the plans for Faneuil Hall in Boston in 1742.

The shop and the studio. Portraits alone did not earn Smibert enough, so he kept a shop. In 1734 he began selling paint, other art supplies and prints. Upstairs, in his studio, he hung copies of famous European paintings that he had made in Europe. He also showed casts there, copies made by pouring material into molds.

The historian Richard Saunders calls it "America's first art gallery." Saunders writes that the collection gave Charles Willson Peale, Gilbert Stuart and John Trumbull much of their early training. A biography of Smibert on Encyclopedia.com names John Singleton Copley and Washington Allston as painters who learned from it. Stuart was born in 1755, four years after Smibert died, so he studied the copies after Smibert's death.

Two portraits that show slavery

The boy in the collar. Justus Engelhardt Kühn painted Henry Darnall III, the son of a rich Maryland family, in about 1710, when the boy was about eight. Henry wears fine clothes and holds a bow. Beside him stands a Black boy with a metal collar around his neck. The Black boy was enslaved.

His name was not recorded, and neither was the name of whoever put the collar on him. The Maryland Center for History and Culture owns the painting and calls him "Unnamed attendant." One art history website calls the picture the earliest known formal painting of an enslaved African in America.

The Royall family. Robert Feke painted Isaac Royall Jr. in 1741, when he was 22, with his wife Elizabeth, their baby daughter Elizabeth, his sister Penelope and his wife's sister-in-law Mary Palmer. Harvard Law School owns the painting.

Isaac's father, Isaac Royall Sr., had set up a sugar plantation on the island of Antigua. The father got rich by trading in sugar, in rum and in enslaved people. In 1737 the family moved to Medford, Massachusetts, and brought at least 27 enslaved Africans with them. When the father died in 1739, the son got his very large fortune, at the age of 20.

Two Lenape leaders painted, 1735

Gustavus Hesselius painted two Lenape leaders, Lapowinsa and Tishcohan, in 1735, for John Penn, a son of William Penn. English colonists led by William Penn had built Philadelphia on Lenape land. Lapowinsa is shown as a strong man of about 40. They are the only two Lenape chiefs whose portraits have survived. The Historical Society of Pennsylvania owns them.

The Walking Purchase. Two years later Lapowinsa and Tishcohan signed a land deal called the Walking Purchase with John Penn and his brother Thomas Penn. The Lenape leaders Nutimus and Menakihikon also signed it. James Logan, who worked for the Penns, showed the Lenape an unsigned copy of a deed from 1686, a paper selling land. He claimed it gave the Penns as much land as a man could walk in a day and a half.

The walk. The Penns' men cleared a path ahead of time. On September 19, 1737, three walkers set off at a fast pace, and in a day and a half one of them covered more than 60 miles. The Penns' surveyors, the men who measured and marked out the land, then bent the boundary line to take in even more. In all, the Penns took more than a thousand square miles of Lenape land, and they forced Nutimus and his people to leave their homes.

Cherokee baskets taken to London, 1725

Cherokee weavers made baskets from river cane, a tall plant with hollow stems that grows along rivers. They split the cane into thin strips and dyed some of the strips with walnut and pokeberry. Some baskets were double woven, two baskets woven one inside the other from the same strips, so both the inside and the outside show a finished pattern.

Francis Nicholson, the first governor of South Carolina chosen by the British king, left Charleston for London in 1725. He took a set of Cherokee double-woven baskets with him and passed them to Sir Hans Sloane, a collector in London. A note tied to them says, "They will keep anything in them from being wetted by rain." The British Museum in London holds them today.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 10 MINUTES

The men who led the Revolution had their portraits painted by Americans, several of them trained in London, and in the same cities enslaved Black artists painted and wrote poems, though only a few records name them. In Boston, Prince Demah painted portraits while he was enslaved, and later he served during the Revolution in the Massachusetts militia, a part-time army of local men.

Phillis Wheatley's *Poems*, the first book of poems by an African American, came out in London in 1773 while she was still enslaved in Boston. The painter Charles Willson Peale was paid for portraits with two enslaved people.

Painters who went to London

John Singleton Copley, a Boston painter born on July 3, 1738, sent a painting called *A Boy with a Flying Squirrel* to be shown in London in 1766. The London painters Joshua Reynolds and Benjamin West praised it. In 1769 Copley painted Isaac Royall Jr., the rich man from Medford, and his wife. Copley left America for England in 1774. He painted *Watson and the Shark* there in 1778, and he died in London on September 9, 1815.

The Pennsylvania painter Benjamin West settled in England in 1763 and became the king's official painter of history scenes. West was known for helping young American painters who came to London to learn.

Faces in wax. Patience Wright, born Patience Lovell in 1725, made portraits by shaping wax. Britannica gives her birthplace as Bordentown, New Jersey. The American Battlefield Trust says she was born at Oyster Bay, on Long Island in New York, and that her family moved to Bordentown when she was four. She took her wax figures around on tour to show them. She sailed to England in February 1772, and in London she showed wax figures of actors, politicians and nobles. She died in London on March 23, 1786.

Charles Willson Peale

Who: A painter of the leaders of the Revolution, who opened a museum in Philadelphia and held a young man in slavery to work in it.

When and where: Born in Chester, Maryland, in April 1741. Worked in Annapolis, then Philadelphia. Died in 1827.

A planter paid him for portraits with two enslaved people.

Charles Willson Peale painted the leaders of the Revolution, and George Washington sat for him seven times. Peale first trained to make saddles. Then he studied painting in England for three years. Back in Annapolis, Maryland, he joined the Sons of Liberty, groups of colonists who formed in 1765 to resist a British tax on printed paper, called the Stamp Act.

Mount Vernon. Peale came to Washington's home at Mount Vernon on May 18, 1772, and stayed two weeks. He painted small portraits of Martha Washington and her children. He also painted George Washington in oil, in the uniform of the Virginia militia.

The museum. Peale announced a show of 44 portraits of leaders of the Revolution in his home in Philadelphia in 1784. Visitors paid to get in, and Peale called it the Philadelphia Museum. One account dates the opening of his museum to 1786 instead.

Washington paid a yearly fee to support it and gave it two golden pheasants from China. In 1795 the museum was the site of the Columbianum, the first show in America put on by a society of artists. After Peale died, his family sold the museum's collection of things from nature in 1849 and its portraits at auction in 1854.

Paid in people. By 1776 a Maryland plantation owner had given Peale two enslaved people, Scarborough and Lucy, as payment for portraits. The account of this payment does not give the planter's name. Peale freed Scarborough and Lucy in 1786.

Their son, Moses Williams, was born in about 1777. Pennsylvania lawmakers passed a law in 1780 to end slavery slowly, over many years. Under that law, only children born after it was passed were to go free, and only when they turned 28. Moses had been born about three years before, so he was not one of them.

Peale kept him enslaved in his household until about 1802, when Moses was about 25 or 27. Peale trained him for work in the museum: cutting silhouettes, which are outline portraits cut from paper, preserving dead animals for display and building exhibits.

Prince Demah

Who: An enslaved portrait painter in Boston, who later lived free and served as a soldier in the Massachusetts militia.

When and where: Baptized in Boston in May 1745. Worked in Boston and London from 1769 to 1778. Died in Boston in March 1778.

He signed his portrait of a Boston merchant "Prince Demah Barnes."

Prince Demah painted portraits in Boston while he was enslaved. One of his signed paintings belongs today to the Metropolitan Museum of Art in New York. He and his mother, Daphney, were baptized at Trinity Church in Boston in May 1745, when Prince was a small child.

Bought for his talent. Henry Barnes was a merchant in Marlborough, Massachusetts, whose family already held Daphney in slavery. In 1769 Barnes bought her son, Prince, who was then in his late twenties. Barnes's wife, Christian, wrote to a friend that they planned on "improving his genius in painting," genius meaning talent. She also wrote that the Boston painter John Singleton Copley would not be willing to teach him.

London. Barnes took Prince to London in October 1770, and a painter named Pine gave him lessons there. He was probably Robert Edge Pine. Barnes was afraid that Prince would "attempt his freedom," which meant

try to escape. He wrote, "I do not let [Prince] converse with any of his own colour here." He meant that he kept Prince from talking with other Black people in London.

The advertisement. Back in Boston, an advertisement ran at least nine times in the *Boston News-Letter* in 1773. It offered the work of a "Negro Artist," using the word of that time for a Black person, who "takes Faces at the lowest Rates," which meant painted portraits cheaply. That February, Prince painted William Duguid, a Scottish cloth merchant living in Boston, and signed it "Prince Demah Barnes." He also painted, or copied from Copley, portraits of Henry and Christian Barnes.

Free. The Barnes family sided with the king. In late 1775 they left for England, and they left Daphney behind at their Marlborough home. The records do not say where Prince was when they sailed, or how he became free.

Prince joined a unit of cannon soldiers in the Massachusetts militia in April 1777. He signed up as Prince Demah, without the Barnes name. On March 11, 1778, he wrote his will as "Prince Demah, limner," a painter of portraits, and as a "free Negro." He left everything he had to his mother. He died one week later. The record does not give the cause.

Phillis Wheatley

Who: A poet, the first African American to publish a book of poems. She was enslaved in Boston when the book was printed.

When and where: Kidnapped in West Africa as a child and landed in Boston in 1761, at seven or eight years old. Her book was printed in London in 1773. She died in Boston in 1784, at about 31.

Eighteen men signed a statement at the front that she wrote it.

Phillis Wheatley wrote the first book of poems by an African American. It came out in London in September 1773, while she was enslaved in Boston. The title page describes her as "Negro Servant to Mr. John Wheatley, of Boston." Servant there stood for enslaved.

Named for a ship. She was kidnapped in West Africa as a child. In 1761, when she was seven or eight, she landed in Boston on a slave ship named the *Phillis*, and her first name came from the ship. John Wheatley's letter in her book says only that she "was brought from Africa to America," and it does not name the men who kidnapped her. In Boston, John Wheatley bought her. He wrote that she learned English in sixteen months, taught at home and not in any school.

When she was about seventeen, in 1770, a Boston printer named Ezekiel Russell printed her poem on the death of George Whitefield, a preacher who had toured the colonies.

The statement. The publisher of her book expected readers to doubt that an enslaved African girl could write poems. So he printed a statement at the front of the book, signed by 18 men of Boston. Among them were Governor Thomas Hutchinson, John Hancock, the Reverend John Moorhead and John Wheatley, listed as "her Master."

The men wrote that the poems were "written by PHILLIS, a young Negro Girl, who was but a few Years since, brought an uncultivated Barbarian from Africa, and has ever since been, and now is, under the Disadvantage of serving as a Slave" (uncultivated meant untaught, and barbarian was

> an insult for a person thought to be wild and without learning). They wrote that she "has been examined by some of the best Judges, and is thought qualified to write them."
>
> One poem in the book is called "To S. M. a young African Painter, on seeing his Works." The Wheatleys freed her by November 1773, soon after the book appeared. She married John Peters in 1778. She died poor in 1784, at about 31.

Two more Black painters, Scipio Moorhead and Joshua Johnson

Scipio Moorhead was an enslaved Black painter in Boston. The scholars Kwame Anthony Appiah and Henry Louis Gates Jr. write that Wheatley's poem to "S. M." may describe two of his paintings. He was held in slavery by the Reverend John Moorhead of Boston, one of the 18 men who signed the statement in Wheatley's book. Sarah Moorhead probably gave Scipio lessons.

Appiah and Gates also write that the engraved portrait of Wheatley at the front of her book was based on his painting of her. No signed painting by Scipio Moorhead survives.

Joshua Johnson. Joshua Johnson was born enslaved in Maryland in 1763. His father was a white man named George Johnson. His mother was a woman held in slavery by William Wheeler Sr., and her name was not written down. George Johnson bought his 19-year-old son for £25 in 1782. He sent him to learn the trade of a blacksmith and signed a manumission, a legal paper that freed him.

Joshua Johnson was listed in Baltimore as a portrait painter by 1796. In a 1798 newspaper advertisement he called himself a "self-taught genius." He wrote that he had faced "many insuperable obstacles," which means problems he could not get past. Amelia Peck of the Metropolitan Museum of Art and Paula M. Bagger write that before Prince Demah's work was found, Johnson was thought to be the earliest African American painter of portraits in oil paint.

Mercy Otis Warren's plays

Mercy Otis Warren wrote plays that made fun of the British governor of Massachusetts and attacked his policies. She was born in Barnstable, Massachusetts, on September 14, 1728. Her first play, *The Adulateur*, came out in a newspaper, the *Massachusetts Spy*, in 1772. At least four more of her plays followed by 1779.

All of them came out without her name. The National Women's History Museum says she likely did this to avoid punishment, and so that readers would take the work of a woman seriously. She died in 1814.

Gilbert Stuart

Who: A portrait painter whose unfinished picture of George Washington is the model for the face on the one-dollar bill.

When and where: Born in Kingstown, Rhode Island, on December 3, 1755. Worked in London and Dublin, then in the United States from 1793. Died in 1828.

He never delivered Washington's portrait, and he sold copies for $100.

Gilbert Stuart sailed to London in 1775. There Benjamin West hired him and gave him a studio and a place to live. In 1787 Stuart left London for Dublin to get away from the people he owed money. He arrived in New York in May 1793.

The unfinished head. Martha Washington ordered portraits of her husband and herself from Stuart in 1796. He painted only Washington's head and neck on a brown background, left the rest of the canvas unfinished and kept it. He never gave it to her, although she asked for it again and again after her husband died in 1799.

Stuart used the unfinished head as a model and painted more than a hundred copies of it, 130 by one count. He sold them for $100 each. An engraving of the head, flipped left to right, is on the one-dollar bill.

> The people in charge of the Boston Athenaeum, a library in Boston, bought the original for $1,500 in 1831, and the painting has been called the Athenaeum portrait ever since. Since 1980 the Museum of Fine Arts in Boston and the National Portrait Gallery in Washington have owned it together.

1800 to 1850

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City buyers paid for paintings of the Hudson valley and the Catskills in the same years that federal officials, the officials of the national government, forced Native nations off their land. From about 1825 a group of painters known as the Hudson River School made landscapes, pictures of the land and sky, of the Hudson River valley and the Catskill Mountains. The painter George Catlin travelled west and painted the people of about fifty nations.

Enslaved artists worked too. In South Carolina an enslaved potter named Dave began cutting his own verses into his jars in 1834, the year the state's lawmakers made it a crime for enslaved people to learn to read and write, and later he signed them "Dave." News of photography reached the United States in 1839. By the early 1850s, a small photograph in New York cost as little as 25 cents.

Black artists in Philadelphia and Baltimore

Moses Williams, the son of the two people a planter gave Charles Willson Peale as payment for portraits, was held by Peale until about 1802. Then he cut silhouettes at Peale's museum in Philadelphia, portraits that show only the outline of a face from the side. Williams used a machine called a physiognotrace, which traced the outline of a sitter's face onto paper through a set of connected rods.

Peale's own children learned painting from their father. Karl Cole's account of Williams's life for Davis Art says Williams was not given the chance to paint. The Philadelphia Museum of Art holds silhouettes he cut of the Peale family between 1803 and 1805.

Joshua Johnson, the Maryland painter whose white father bought him and freed him in 1782, was listed as a portrait painter in the Baltimore city directories, books that listed the town's people and their trades, from 1796 to 1824.

The Academy. Peale, the sculptor William Rush and other artists and business leaders founded the Pennsylvania Academy of the Fine Arts in Philadelphia in 1805. The academy's leaders very seldom let Black students in.

Robert Douglass Jr., a Black painter of Philadelphia who lived from 1809 to 1887, was one of the few. He trained at the academy and studied under the painter Thomas Sully. In 1834 he painted the portraits of William Lloyd Garrison and James Forten, two men who worked to end slavery. An article by the Library Company of Philadelphia describes him as Philadelphia's first African American photographer.

Women at the Academy. The academy's leaders let women in one step at a time. In 1844 they gave women artists the room of statues to draw in, on three mornings a week. In 1860 they let women take classes on the body's bones and muscles and draw from copies of ancient statues. They opened life classes, where students draw from a model without clothes, to women in 1868, with women as the models. Men became the models for these classes in 1874.

Painters on the road

Many portrait painters of these years had taught themselves. They travelled from town to town and charged somewhat less than painters with training. A story told later says these painters painted bodies ahead of time and added the customer's face when they arrived. The unfinished portraits that survive show the opposite. They have the face done and the body not yet painted, so the painters began with the head.

The "Border Limner." Ammi Phillips travelled and painted portraits in Connecticut, Massachusetts and New York from the middle of the 1810s until the early 1860s. He was born in Colebrook, Connecticut, on April 24, 1788. For years nobody knew who had painted many of his pictures, and collectors called the unknown painter the "Border Limner," limner being an old word for a painter of portraits. The collectors Barbara and Larry Holdridge matched those paintings to Phillips.

In the mid-1830s, in Dutchess County, New York, Phillips painted *Girl in Red Dress with Cat and Dog*. It was bought for the American Folk Art Museum in 1985 for one million dollars. Phillips died on July 11, 1865.

The Peaceable Kingdom. Edward Hicks, born in Pennsylvania on April 4, 1780, became an apprentice, a young worker learning a trade by working for its masters, to coach makers at 13. From 1812 he was a minister of the Quakers, a Christian group.

He painted 62 versions of *The Peaceable Kingdom*, based on a passage in the book of Isaiah in the Bible. Many versions show William Penn meeting Native people, who stand for the Lenape, on whose land Penn founded his colony. In his own lifetime, Hicks was better known as a minister than as a painter. He died on August 23, 1849.

Joseph Whiting Stock

Who: A travelling portrait painter who could not walk. He kept a journal of the paintings he made and what he was paid for them.

When and where: Born in Springfield, Massachusetts, on January 30, 1815. Painted in Massachusetts, Rhode Island and New York from 1832. Died in 1855, at the age of forty.

An oxcart fell on him at eleven, and he painted 900 pictures.

Joseph Whiting Stock painted more than 900 pictures, although he could not walk. When he was eleven, an oxcart turned over and fell on him. He was paralyzed from the waist down, which means he could not move or feel his legs. He could not leave his bed until he was nineteen.

A doctor's idea. A doctor suggested that he take up art. He began lessons at seventeen, with a painter named Franklin White, and tried a portrait of his sister Eliza. "I attempted my sister Eliza's portrait and made so good a likeness as to induce my friends to encourage me by their patronage," he wrote in his journal (patronage meant paying him for pictures).

A wheelchair. In 1834 a doctor named James Swan hired Stock to make drawings of the human body. Swan also built him a wheelchair. With it, Stock could paint large pictures and be lifted onto trains to travel for work.

His first painting trip was a month in Wilbraham, Massachusetts, in 1836. He later spent two years in Warren, Rhode Island, and New Bedford, Massachusetts, and several years in Orange County, New York. He lived with local families while he worked, and sometimes he paid for his room and meals with paintings.

His journal lists more than 900 paintings and what he was paid for them. About a hundred of those paintings are known today. He died of tuberculosis, a disease of the lungs, in 1855, at forty. The Springfield Museums hold his journal.

Writers

Some writers of these years turned to the American land and to nature, as the painters did. Washington Irving set his 1819 story "Rip Van Winkle" in the Catskill Mountains of New York, the same mountains the Hudson River School painted a few years later. Ralph Waldo Emerson's long essay *Nature* appeared in 1836.

James Fenimore Cooper's novel *The Last of the Mohicans* came out in Philadelphia on February 4, 1826. The Mohican people named in its title were living in New York and Wisconsin that year.

The Hudson River School, and whose land it painted

The Hudson River School was a group of American landscape painters who worked from about 1825 to 1870. They painted American land in careful, true-to-life detail, often with quiet farms next to the last of the wild forest. Many of them believed that God could be seen in the American landscape. Thomas Cole, Thomas Doughty and Asher B. Durand were the first leaders, and many of the group painted the Catskill Mountains, west of the Hudson River.

The Hudson valley was the homeland of the Mohican people. Cole made his home in the village of Catskill, New York, in 1826. That land is the homeland of the Muh-he-conneok, the People of the Waters That Are Never Still, who today are called the Stockbridge-Munsee Band of Mohican Indians. A marker at Cole's house states that they were forced from there.

Stockbridge. Through the 1600s and 1700s, before Cole was born, settlers pushed the Mohican out of the Hudson valley. Most moved to Stockbridge, Massachusetts, a mission town, which means a town built around a church where ministers taught Christianity. From 1764 white settlers poured into Stockbridge and took thousands of acres of Mohican land. They squatted on it, which means they settled on it without any right, and they bought it in ways the law did not allow. They also took land to pay off Mohican debts.

The Mohican had lost all their land in the town by 1783. In 1785 a group of them founded New Stockbridge in upstate New York, on land of the Oneida nation.

To Wisconsin. The Mohican were made to move again in the same years Cole began to paint the Catskills. Agents from New York and commissioners from the War Department arranged treaties, written agreements between governments. Under them, the Stockbridge, as the Mohican were by then called, and the Munsee had to leave New York for Wisconsin.

The moves began after an 1822 treaty, three years before Cole put his landscapes in a New York shop window. By 1831, five years after Cole settled in Catskill, 225 Stockbridge people and 100 Munsee people had gone. From 1832 to 1834, federal officials made them move once more, to the east shore of Lake Winnebago. Their reservation, land set aside for them by treaty, is in northeastern Wisconsin today.

Removal. While Cole was selling paintings of the Catskills, federal officials were moving Native nations west. President Andrew Jackson signed the Indian Removal Act on May 28, 1830. Under it, the president could make treaties in which Native nations gave up their lands east of the Mississippi River for lands in the west.

Federal officials and soldiers forced more than 16,000 Cherokee people from their homeland to Indian Territory, in what is now Oklahoma, in 1838 and 1839. At least 4,000 of them died of cold, hunger and disease along the way, by the usual count, and other counts are higher.

Thomas Cole

Who: A painter of landscapes and one of the first leaders of the Hudson River School.

When and where: Born February 1, 1801, in Bolton-le-Moors, Lancashire, England. Came to America in 1818. Lived in Catskill, New York, from 1826. Died there on February 11, 1848.

The cleared strips on the far hill seem to spell "Noah" in Hebrew.

Thomas Cole was one of the first leaders of the Hudson River School. He came to America from England with his family in 1818, and they lived first in Philadelphia and then in Steubenville, Ohio. A travelling portrait painter named Stein gave him lessons. Then he studied for two years at the Pennsylvania Academy of the Fine Arts.

A shop window. Cole put some landscapes in the window of a shop in New York City in 1825. The painters John Trumbull and Asher B. Durand saw them and bought them. The next year Cole moved to Catskill. He made pencil sketches outdoors, and in winter he painted in his studio. He also spent years painting in Europe, mostly in Italy, from 1829 to 1832 and again from 1841 to 1842.

The Oxbow. In 1836 he painted *The Oxbow*, a view from Mount Holyoke, near Northampton, Massachusetts. It shows the valley of the Connecticut River, not the Hudson, and Cole joined two views from the mountain into one picture. On the far hill, the bare strips where trees were cut seem to spell the Hebrew letters for "Noah." The art historian Matthew Baigell was the first to notice them.

Cole made the painting for a show in New York in April 1836. His patron, a person who pays an artist to make work, was Luman Reed, and Reed had suggested it. In March 1836 Cole wrote to Reed, "Fancy pictures seldom sell & they generally take more time than views so I have determined to paint one of the latter." By "fancy pictures" he meant pictures from his imagination, and by "views" he meant real places.

That same year he finished *The Course of Empire*, a set of five paintings. The Metropolitan Museum of Art in New York has owned *The Oxbow* since 1908.

Cole died in Catskill on February 11, 1848. Durand painted a picture called *Kindred Spirits* in his memory. Cole's house and studio in Catskill are now the Thomas Cole National Historic Site.

BY 1840: Painters were selling views of the Catskills in New York City, and federal officials and soldiers had forced the Mohican, the Munsee and the Cherokee out of their homelands.

George Catlin and the people he painted

George Catlin was a lawyer who became a painter. He lived from 1796 to 1872. In 1830 he went to St. Louis to meet William Clark, the federal official in charge of Indian affairs there. From St. Louis, Catlin made five trips west between 1830 and 1836, and in 1832 he travelled more than 2,000 miles up the Missouri River.

He wrote, "If my life be spared, nothing shall stop me from visiting every nation of Indians on the Continent of North America." By one count he visited 48 nations, and by another, fifty.

Four Bears. At a Mandan village in what is now North Dakota, Catlin painted Four Bears, a Mandan leader, in 1832. His Mandan name was Máh-to-tóh-pa. Catlin called him "the first and most popular man in the nation." Four Bears was a painter too, and he painted his own battles on buffalo robes. Catlin got one of those robes in 1832, and it is now at the Smithsonian.

A German prince, Maximilian of Wied, travelled up the Missouri with the painter Karl Bodmer in 1833 and 1834. He took another Four Bears robe to Germany, where it is in the Linden Museum in Stuttgart.

Catlin sometimes got objects from the people he painted, as proof that his pictures were accurate. One was a Sioux cradle. In 1836 he went to the place in southwestern Minnesota where people dug red stone for pipes. Sioux elders

protested strongly, and he went anyway.

Smallpox. Five years after Catlin painted the Mandan, smallpox killed almost all of them, Four Bears among them. Smallpox was a disease caused by a virus. It spread when people were close together, face to face. It caused a fever and a rash that turned into scabs, and about 3 of every 10 people who caught it died.

President Jackson signed the Indian Vaccination Act on May 5, 1832, a law to give Native people vaccinations, shots that protect a person from smallpox. Secretary of War Lewis Cass ran the program through federal Indian agents. The Mandan, the Hidatsa and the Arikara of the upper Missouri were kept out of it. The records do not name the officials who left them out.

The American Fur Company's steamboat *St. Peters* reached Fort Clark, next to the Mandan villages, on June 18, 1837. The fort's trader, Francis Chardon, learned from the captain, Bernard Pratte Jr., that people on board had smallpox. The crew unloaded goods and loaded furs in less than a day, and the boat went on up the river. It had brought smallpox to the Mandan villages. About 2,000 Mandan lived there that spring. By October, 138 were alive.

Four Bears caught smallpox and died on July 30, 1837. Chardon wrote down his last speech: "I have loved the Whites ... and how have they repaid it! With ingratitude!" (ingratitude means not being thankful).

Osceola. The Seminole leader Osceola sat for Catlin in January 1838 as a dying prisoner. General Thomas Jesup had ordered soldiers to seize Osceola near St. Augustine, Florida, on October 21, 1837, while he was under a white flag of truce, a sign that the two sides have stopped fighting to talk. Osceola was held at Fort Marion in St. Augustine and then at Fort Moultrie in South Carolina.

There he was very sick with malaria, a disease spread by mosquitoes that brings fever and chills, and with other illnesses. Three painters, Catlin, W. M. Laning and Robert John Curtis, talked him into letting them paint him anyway. Osceola died a prisoner on January 30, 1838. The army doctor, Frederick Weedon, cut off his head and then buried the body. Weedon kept the head and showed it in his drugstore.

The Indian Gallery. Catlin opened his paintings to the public at Clinton Hall in New York on September 23, 1837, at 50 cents a visit. He called the show his Indian Gallery. He tried to sell it to the United States government, and members of Congress did not buy it. In November 1839 he sailed for England with the gallery, a tepee of buffalo hide and two live bears. He showed the gallery in London, Brussels and Paris.

Groups of Native people who were touring Europe with promoters joined him there, 21 Ojibwe and 14 Iowa. Catlin talked them into acting out hunts, dances and even scalpings for the crowds. Scalping means cutting the skin and hair from the top of an enemy's head. In Paris in 1845, the Ojibwe group caught smallpox, and two of them died. The account that records their deaths does not name them.

In debt. Catlin ran out of money in 1852 and was put in a London prison for debt. Members of Congress still would not buy the gallery, even after he lowered his price from $65,000 to $25,000. Joseph Harrison, a Pennsylvania railroad owner, paid Catlin's debts and got the gallery for $20,000. He stored it in his boiler factory in Philadelphia. In 1879 Harrison's widow gave the paintings to the Smithsonian.

Audubon's birds, and the people he sold

John James Audubon's *The Birds of America* was printed in England from 1827 to 1838. It has 435 pictures of birds, each bird shown at its real size.

The people the Audubons sold. The historian Gregory Nobles records that the Audubons held nine enslaved people in their household in Henderson, Kentucky, in the 1810s. By the end of that decade, short of money, they sold them. In early 1819 Audubon took two enslaved men down the Mississippi River to New Orleans on a small boat, and there he offered the boat and the two men for sale.

The family bought more enslaved people in the 1820s and sold them in 1830, when Audubon moved to England to oversee the printing of the birds.

Audubon wrote to his wife in 1834 about the freeing of the enslaved people in Britain's colonies in the West Indies, the islands of the Caribbean. He wrote that the British government had "acted imprudently and too precipitously," which means without enough care and too fast. The surviving record counts the people Audubon held and sold. It does not give their names.

Dave (David Drake)

Who: An enslaved potter and poet who cut his name and his own verses into large jars of hard-fired clay.

When and where: Born about 1800 in South Carolina. Worked in the potteries of the Edgefield District from about 1818 to 1864. Listed in the 1870 census as "David Drake, Turner." Believed to have died in the 1870s.

He cut verses into clay when learning to read was a crime.

Dave was an enslaved potter who wrote his own verses on the jars he made. He is the only Black potter in Edgefield before the Civil War known to have signed his own work. He was born into slavery around 1800. In the Edgefield District of South Carolina, white factory owners made their potteries grow by using the unpaid work of enslaved men, women and children. At least 76 enslaved people worked at the district's 12 potteries.

Mortgaged. The first record of Dave is dated June 13, 1818. It describes "a boy about 17 years old country born," which meant born in America and not brought from Africa. His enslaver, Harvey Drake, mortgaged him to a man named Eldrid Simkins. That meant Drake used Dave to guarantee a loan: if Drake did not pay the money back, Simkins could take Dave.

Passed from man to man. At least five men held Dave in slavery during his life. Harvey Drake was the first, until he died in 1832. His brother Reuben Drake held Dave next, until 1835, and then the Reverend John Landrum. When Landrum died in 1846, all 18 of the people he enslaved were put up for sale. The records do not name who sold them. Franklin Landrum held Dave for a time. In 1849 Lewis Miles, who owned the Stony Bluff pottery, bought him.

Words on a jar. South Carolina's lawmakers made it a crime for enslaved people to learn to read and write in 1834. That same year, Dave cut words into a jar: "put every bit all between / surely this jar will hold

14." On the jar's other side he pressed fourteen dots. Arthur F. Goldberg and James Witkowski, two scholars who studied his pots, think such dots showed how many gallons a jar held, so the 14 means 14 gallons.

He wrote on another jar on July 31, 1840: "Dave belongs to Mr. Miles / wher the oven bakes and the pot biles." So by his own words, Lewis Miles held him in 1840, nine years before the 1849 sale. Accounts of his life do not agree on when Miles first held him or how.

NOT KNOWN: how he learned to read, though he may have learned while he worked at a newspaper, the *Edgefield Hive*, printed by Harvey Drake's business partner Abner Landrum.

Dave shaped the bottom of each jar on a potter's wheel. Then he built up the top by hand with thick coils of clay. Two of his jars at the Charleston Museum each hold more than 40 gallons.

What was taken from him. His family was sold away from him. The records do not say who sold them or where they were taken. One of his legs was amputated, which means it was cut off.

NOT KNOWN: how he lost the leg. One account says an enslaver beat him for writing, and another says he was hurt in a train accident. Neither has been proved.

He wrote three verses on his pots in 1857, eight in 1858 and seven in 1859. On August 16, 1857, he wrote: "I wonder where is all my relation / Friendship to all, and every nation."

His last known poem, from 1862, during the Civil War, reads: "I made this Jar all of cross, If you dont repent, you will be, lost." To repent means to be sorry for doing wrong and to change. Goldberg and Witkowski write that the poem carries political and religious meaning, because he wrote it in a time of slavery and civil war.

Free, and counted. In a survey of 2006, Goldberg and Witkowski counted 169 pots made by Dave or credited to him. More than a hundred, dated from 1849 to 1864, are signed "Dave." His verses and marks appear on 50 of them. Sometimes he added the names of other enslaved potters beside his own: Mark, Abram and Baddler. His last dated pot is from 1864.

Dave was free by the end of the Civil War in 1865. The Smithsonian American Art Museum credits his freedom to the Emancipation Proclamation, President Abraham Lincoln's order freeing enslaved people in the states at war with the United States. He took the last name Drake, probably from Harvey Drake, the first man who had held him.

The census of 1870, the government's count of everyone in the country, lists him as "David Drake," and his work as "turner," a person who shapes pots on a wheel. His name is not in the census of 1880. He is believed to have died in the 1870s.

Photography arrives, 1839

The French artist Louis Daguerre worked out a way of making pictures with light, and news of his process reached the United States in late August and September 1839. In October or November 1839, Robert Cornelius of Philadelphia took a picture of himself. It is the oldest American portrait photograph that survives.

These first photographs, called daguerreotypes, were each made directly on a copper plate coated with silver. Each was a single picture, and it could not be copied.

Studios on Broadway. A history of New York's photographers, published by the Bard Graduate Center, says the first American studio that sold portrait photographs opened in New York in 1840. By 1850 New York had 71 daguerreotype studios, and by 1853 it had more of them than all of England. Mathew Brady opened his first studio at 205 Broadway in 1844.

Cheaper studios on Broadway and on the Bowery made portraits for working people. In the early 1850s, a very small daguerreotype, about 1 5/8 inches by 1 3/8 inches, cost 25 cents. A cheap portrait from a "picture factory," a shop that turned out many portraits fast, cost 50 cents. A medium-sized portrait at Brady's studio cost $2.

Frederick Douglass

Who: A writer and speaker who escaped slavery. He wrote the story of his own life and sat for 160 known photographs.

When and where: Born in Talbot County, Maryland, in February 1818. Escaped slavery on September 3, 1838. Lived in Massachusetts, then Rochester, New York, then Washington. Died in 1895.

Movie: *Becoming Frederick Douglass* (2022), a documentary by Stanley Nelson, shown on PBS. It is about his life.

He sat for 160 photographs, more than Lincoln's 126.

By the count of the historians John Stauffer, Zoe Trodd and Celeste-Marie Bernier, Frederick Douglass sat for more photographs than any other American of the 1800s: 160 of them, against 126 of Abraham Lincoln. He was born into slavery in Maryland. He escaped on September 3, 1838, by train through Baltimore, dressed in a sailor's clothes. Anna Murray, a free Black woman, helped him, and he married her.

The book. His book came out from the Anti-Slavery Office in Boston on May 1, 1845. It was called *Narrative of the Life of Frederick Douglass, an American Slave. Written by Himself*. By one count it sold 4,500 copies in four months, and by another 5,000. By 1860 almost 30,000 copies had sold.

Pictures by white artists. In 1849 Douglass wrote about pictures of Black people drawn by white artists. He wrote that "negroes can never have impartial portraits at the hands of white artists" (negroes was a word of that time for Black people, and impartial means fair). "It seems to us next to impossible for white men to take likenesses of black men, without most grossly exaggerating their distinctive features," he wrote.

Douglass sat for photographs again and again. He said that "the great cheapness and universality of pictures must exert a powerful, though silent, influence upon the ideas and sentiment of present and future generations." By universality he meant that pictures were everywhere.

BY 1850

If you remember one thing: Painters in these years sold city buyers pictures of the Hudson valley and of Native leaders, while federal officials forced Native nations off their land.

- May 28, 1830: President Andrew Jackson signs the Indian Removal Act.
- 1834: Dave cuts words into a jar, the year South Carolina makes it a crime for enslaved people to learn to read and write.
- October or November 1839: Robert Cornelius of Philadelphia takes the oldest surviving American portrait photograph.

Next: In the next fifty years, photographers carry their cameras onto Civil War battlefields and into the valleys of the far West.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 23 MINUTES

Cameras and cheap printing put pictures and books in front of millions of Americans after 1850. Many Black and Native artists of the same years did their work far from home or as prisoners. Photographers showed Americans the bodies of soldiers killed in the Civil War, and the valleys and mountains of the far West. The first ten-cent novel, in 1860, sold more than 65,000 copies in a few months, and *Uncle Tom's Cabin* sold 300,000 in the United States in its first year.

The sculptor Edmonia Lewis said she was "practically driven to Rome" by prejudice against her color, and the painter Henry Ossawa Tanner made his career in Paris. At an army fort in Florida, Plains men held as prisoners without trial drew their lives in account books. In New York, the Metropolitan Museum of Art opened in 1872.

Writers and cheap books

Harriet Beecher Stowe's *Uncle Tom's Cabin*, a novel against slavery, sold more copies than any other book of the 1800s except the Bible. It first ran in parts in a newspaper, *The National Era*, from 1851, and came out as a book in 1852. In the United States it sold 10,000 copies in its first week and 300,000 in its first year.

Herman Melville's *Moby-Dick* sold far fewer. The novel, about a whaling captain who hunts the giant white whale that bit off his leg, came out in 1851, and its first printing in New York was 2,915 copies. It was out of print when Melville died in 1891. Readers began to call it a great American novel in the 1900s.

Walt Whitman paid to print his own book of poems, *Leaves of Grass*, on July 4, 1855. The first edition held twelve poems. Whitman kept adding to it and changing it until he died in 1892.

Dime novels. The brothers Erastus and Irwin Beadle began selling novels of about 100 pages for ten cents each in 1860. People called them dime novels, because a dime is ten cents. The first, Ann S. Stephens's *Malaeska, the Indian Wife of the White Hunter*, sold more than 65,000 copies in its first few months. The Beadles printed 321 of these books in the series.

Robert S. Duncanson in Cincinnati

Robert S. Duncanson was a free Black painter who had taught himself to paint. He was born in about 1821 in Fayette, New York, the son of a free Black tradesman. In 1840 he moved to Cincinnati, Ohio, where about 3,000 Black people lived.

Belmont. Nicholas Longworth hired Duncanson in about 1850 or 1851 to paint eight landscapes on the walls of the entrance hall of his house, Belmont. Each is about nine feet high. The house is now the Taft Museum of Art. In 1853 Duncanson painted *Uncle Tom and Little Eva*, from Stowe's novel, which is at the Detroit Institute of Arts.

During the Civil War he left the United States. He showed his work in Montreal, Canada, in 1864 and in Dublin, Ireland, in 1865. He died in Detroit on December 21, 1872.

Photographs of the Civil War dead

Alexander Gardner, born in Paisley, Scotland, in 1821, worked for the New York photographer Mathew Brady from 1856, and in 1858 Brady put him in charge of his studio in Washington. In September 1862 Gardner and his team photographed the dead soldiers on the battlefield at Antietam, in Maryland. The American Battlefield Trust estimates that 22,717 soldiers were killed, wounded or missing in that battle.

"The Dead of Antietam." Brady showed the photographs at his gallery in New York in October 1862, under a sign that read "The Dead of Antietam." A writer for *The New York Times* reported on October 20 that crowds kept climbing the stairs to see them. "If he has not brought bodies and laid them in our door-yards and along the streets, he has done something very like it," the writer said (door-yards are the yards in front of houses). The article gave Brady the credit.

Whose name went on the picture. Brady put the words "Photographed by Brady" on pictures his employees took. In about November 1862 Gardner stopped working for Brady, probably in part because of this. In May 1863

Gardner and his brother James opened their own studio in Washington and hired many of Brady's former workers.

Gardner published a book of 100 of the war's photographs in two volumes in 1866. Under each picture he printed the name of the man who took it, among them Timothy H. O'Sullivan and James F. Gibson. The book did not sell well.

A body moved for the camera. After the Battle of Gettysburg, on July 6, 1863, Gardner's team photographed a dead soldier among the rocks at a place called Devil's Den. Gardner later called him a Confederate, a soldier of the Southern states that fought the United States to keep slavery. He titled the picture *Home of a Rebel Sharpshooter*, a sharpshooter being a soldier who shoots at a distance with a rifle.

The historian William A. Frassanito showed in 1975 that the same body appears in six photographs. The photographers had carried the body from the south edge of Devil's Den to a stone wall, and the rifle beside it was probably put there for the picture. In 1998 the artist James Groves argued that the body was carried the other way, out of the wall and to the first spot.

KNOWN:	the photographers moved the body, which both men agree on.
NOT KNOWN:	the soldier's name.

Winslow Homer

Who: A painter who drew the Civil War for a magazine and later painted the sea.

When and where: Born in Boston on February 24, 1836. Drew the Civil War for *Harper's Weekly*. Lived in England from 1881 and in Prouts Neck, Maine, from 1883. Died there on September 29, 1910.

He sketched the Civil War at the front for *Harper's Weekly*.

Winslow Homer drew the Civil War at the front for *Harper's Weekly*, a magazine, and he drew everyday life in the soldiers' camps. He began as an apprentice at a lithography company, which printed pictures from drawings made on stone. Then he drew pictures for magazines, among them *Ballou's Pictorial* and *Harper's Weekly*.

The sea. He began painting in watercolor in 1873. In 1881 he went to live in Cullercoats, a town in England. In 1883 he moved to Prouts Neck, on the coast of Maine, and the sea became his main subject. He painted people at sea, in pictures such as *The Life Line* and *Fog Warning*. He died at Prouts Neck on September 29, 1910.

Edmonia Lewis

Who: A sculptor of Mississauga Ojibwe and African American descent, who carved in marble in Rome after she was beaten and falsely charged at college in Ohio.

When and where: Born about July 4, 1844, in Greenbush, New York. Oberlin, Ohio, from 1859. Boston. Rome from 1865 or 1866. Died in London on September 17, 1907.

She sold 100 plaster copies of a bust to pay her way to Rome.

Edmonia Lewis carved statues in marble in Rome. Before that, at college in Ohio, a group of white men beat her, officials charged her with poisoning two classmates, and college officials kept her from graduating.

She was born in Greenbush, New York, in about 1844. On official papers she gave her birth year as 1842, 1844 and 1854 at different times. Her mother, Catherine Mike Lewis, was of Mississauga Ojibwe and African American descent. Edmonia's name at birth was Wildfire. By the age of nine she had lost both parents. As a girl she and her aunts sold Ojibwe baskets, moccasins and embroidered blouses to tourists at Niagara Falls. Her brother Samuel paid for her schooling with money he made in the gold rush.

Oberlin. From 1859 she studied at Oberlin, in Ohio. In the winter of 1862, she gave spiced wine to two white students she lived with, Maria Miles and Christina Ennes, before a sleigh ride. They fell sick. Doctors said they had been poisoned with cantharides, a poison made from blister beetles. Both young women got well.

Beaten. One night soon after, Lewis was walking home alone. A group of white men dragged her into an open field, beat her badly and left her for dead. No one ever identified the men who did it. A few days later, officials arrested her for poisoning the two students. She was hurt so badly she could not walk, so her trial was put off.

John Mercer Langston, the first Black lawyer in Ohio, defended her. The court dropped the charges, because no one had tested what was in the young women's stomachs, so there was no proof of poison.

About a year later she was accused of stealing art supplies. The record does not name who accused her. That case was dropped too. But Oberlin officials did not let her sign up for her last term, so she could not graduate. In 2022, more than a hundred years after her death, the leaders of Oberlin College gave her a degree.

The bust. She went to Boston and made a bust, a sculpture of a person's head and shoulders, of Colonel Robert Gould Shaw. Shaw had led the 54th Massachusetts, a regiment of Black soldiers, in the Civil War. Lewis sold 100 plaster copies of the bust for $15 each, and the money paid her way to Rome.

Her passport of 1865 says she was "sent by subscription to Italy having displayed great talents as a sculptor" (sent by subscription means that a group of people paid for the trip). In Rome she carved her work in marble.

She told *The New York Times* in 1878: "I was practically driven to Rome in order to obtain the opportunities for art culture, and to find a social atmosphere where I was not constantly reminded of my color. The land of liberty had no room for a colored sculptor." (Colored was a word of that time for Black people.)

Cleopatra. Her marble statue *The Death of Cleopatra* was shown at the Centennial Exhibition in Philadelphia in 1876. In 1878 the critic William J. Clark called the way it showed death "ghastly" and "absolutely repellant," which means disgusting. Later a gambler named "Blind John" Condon bought it from a saloon in Chicago. He used it to mark the grave of a racehorse named Cleopatra.

The Smithsonian American Art Museum's record of the statue states that it was thought lost for almost a century. It was later moved to a construction yard in Cicero, Illinois. The scholars Marilyn Richardson

and Dorothy Porter Wesley told George Gurney, a curator at the Smithsonian, one of the museum workers who choose and care for its art, about it. In 1994 the Historical Society of Forest Park, Illinois, gave it to the Smithsonian American Art Museum.

BY 1866: Americans had seen photographs of the dead at Antietam and Gettysburg, and Gardner had printed the name of each man who took them under the war's photographs.

Pictures of the West, and whose land they showed

Carleton Watkins carried a very large camera into Yosemite Valley, in California, in July 1861. It took pictures on glass plates 18 inches by 22 inches. He came back with 30 of these large plates and 100 smaller negatives. His photographs were shown at a gallery in New York in 1862, and they helped persuade members of Congress to protect the valley. President Abraham Lincoln signed the Yosemite Grant on June 30, 1864.

The Ahwahneechee. Yosemite Valley was the home of the Ahwahneechee people. California's governor, John McDougal, raised a force of about 200 armed volunteers on horseback in 1851, called the Mariposa Battalion. Under Major James D. Savage, they invaded the valley in March 1851. According to the National Park Service, the battalion's men burned villages and food supplies, one after another, and forced men, women and children away from their homes.

They marched Tenaya, the leader of the Ahwahneechee, and his people to a reservation on the Fresno River. The National Park Service counts that by 1910, more than 90 out of every 100 of the Ahwahneechee who first lived there were dead or missing.

Yellowstone. The photographer William Henry Jackson and the painter Thomas Moran went to Yellowstone with the Hayden survey, a government expedition, in 1871. Jackson took the first known photographs of the Yellowstone falls. When President Ulysses S. Grant signed a law on March 1, 1872, Yellowstone became a national park. That year members of Congress

paid Moran $10,000 for his painting *The Grand Canyon of the Yellowstone* and hung it in the Capitol. It shows four small figures in front, and one of them is a Native man.

The Tukudika, a band of Mountain Shoshone, had lived in the Yellowstone area for thousands of years. According to the National Park Service, they were forced onto reservations. The Park Service account does not say who first forced them. In the fall of 1879, the park's superintendent, Philetus W. Norris, found one of their camps. He asked the government agent at Fort Washakie for help. The agent sent a party of Shoshone men to take the Tukudika to the Wind River Reservation.

Timothy O'Sullivan, who had photographed the war, later photographed for the Wheeler survey, an army survey of the West, from 1872 to 1879. The army's orders for that survey included a duty to "discover the numbers, habits, and disposition of Indians." Disposition here meant whether they were likely to be friendly or to fight.

Why was the Met closed on Sundays?

Many of the founders of the Metropolitan Museum of Art were Protestants, Christians outside the Catholic Church, who refused to open the museum on Sundays. Sunday was the one day off that most working New Yorkers had.

New York's lawmakers approved the founding of the museum on April 13, 1870. It opened on February 20, 1872, on Fifth Avenue, and moved to Central Park in 1880. City officials demanded Sunday opening. In 1885 officials of the city's parks department threatened to cut the museum's money if it stayed closed on Sundays. The trustees, the people in charge of the museum, agreed in the end, although they did not want to. The museum first opened on a Sunday on May 31, 1891, after nearly twenty years of argument.

Fort Marion: drawings by Plains prisoners

Plains artists had long painted their deeds on buffalo hides. About 30 million buffalo lived on the Plains in the early 1800s, and fewer than 1,000 were left by 1900. Hunters killed them for their hides, and army officers encouraged the killing, because the herds fed the Plains nations. Disease and dry years also cut the herds, and so did the railroads built across the Plains.

With the buffalo gone, Plains artists drew on paper instead, often in ledgers, lined account books meant for keeping records of money. This is now called ledger art. Prisoners held at an army fort in Florida made many ledger drawings.

Held without trial. US Army soldiers fought Comanche, Kiowa, Cheyenne and other Plains people on the southern Plains in 1874 and 1875, in what is called the Red River War. After it, army officers charged 72 Cheyenne, Arapaho, Kiowa, Comanche and Caddo people with war crimes, acts in war that break the rules of war, or with being "ringleaders," the leaders of the fighting. General Philip Sheridan ordered them sent "to an eastern fort to be held without trial for an undefined time."

They were 71 men and one woman, named Mochi. A girl of about nine, Black Horse's daughter, went with them. The trip from Fort Sill, in Indian Territory, took 26 days by wagon, train, steamship and oxcart. The poet Sidney Lanier saw them arrive "in leg-irons," chains locked around their ankles.

Fort Marion. They were held at Fort Marion, in St. Augustine, Florida, from 1875 to April 1878. Spanish builders had begun the fort in 1672 as the Castillo de San Marcos. Lieutenant Richard Henry Pratt, who was in charge of them, took off their chains. Then he cut the prisoners' hair, put them in army uniforms, made them march in drills, taught them English and sent them out to work for others as laborers.

Army officers kept them in an old fort without proper living quarters. In the Florida summer heat, many prisoners fell sick, and many were deeply sad and homesick. The writer Judith Lee Wilmot says the sickness, the sadness, the heat and the poor quarters together raised the number of prisoners who died.

"Only give us our women and children." Mah-mante, a Kiowa leader, asked for that in June 1875. President Grant agreed, and then changed his mind. In 1877 Making Medicine, a Cheyenne man, wrote for the young prisoners: "We have lived in this old place two years. It is old and we are young." He went on: "We want Washington to give us our wives and children, our fathers and mothers. We are willing to learn and want to work."

The drawings. Pratt gave the prisoners pencils, ink, crayons, watercolor paint and paper. Twenty-six of them made drawings, all younger Cheyenne, Arapaho and Kiowa men, and Pratt sold their drawings and crafts. The artists included Making Medicine, Howling Wolf, Bear's Heart and Tichkematse, who were Cheyenne, and Zotom, Wohaw, Etahdleuh Doanmoe and Koba, who were Kiowa. White Bear was Arapaho. More than a thousand of their drawings survive in collections around the world.

The dead. Ten of the prisoners died in Florida. Amy Larner Giroux, of the University of Central Florida's Center for Humanities and Digital Research, spent five years searching army records for their names. A January 2023 article in UCF News states that she found all ten.

Six of the dead were Cheyenne: Grey Beard, Lean Bear, Big Moccasin, Starving Wolf, Spotted Elk and Heap of Birds. Three were Kiowa: Sun, Mah-mante and Ih-pa-yah. The tenth, Nad-a-with-t, was Comanche.

Grey Beard died on the way to the fort. He jumped out of a window of the train that was taking the prisoners to St. Augustine. The guards had orders to catch him again, but he was shot instead, and the UCF article does not say who fired. He is buried in Piney Grove Cemetery in Baldwin, Florida.

For many years two graves in the St. Augustine National Cemetery, each holding six men, had stones that read only "Six Unknown Indians." That makes twelve men, three more than the nine prisoners who died at the fort. The UCF article does not say who the twelve men are.

Officials of the National Cemetery Administration, the federal office that runs national cemeteries, said in 2023 that they would replace the two markers with headstones that give the prisoners' names.

After their release, Bear's Heart, Koba and Buffalo Meat died of tuberculosis. Making Medicine became a deacon, a church minister, in the Episcopal Church in 1881. Pratt founded the Carlisle Indian School in Pennsylvania, a boarding school for Native children, in 1879. Staff there cut the children's hair, took away their clothes, gave them English names and forbade them to speak their own languages.

Howling Wolf

Who: A Southern Cheyenne man who lived through the Sand Creek Massacre at 15 and became a ledger artist while a prisoner at Fort Marion.

When and where: Born about 1849. The southern Plains, and St. Augustine, Florida, from 1875 to 1878. Died in 1927.

He drew the massacre at Sand Creek, which he lived through at 15.

Howling Wolf drew the attack he had lived through as a boy. His Cheyenne name was Ho-na-nist-to. He belonged to the band of the Cheyenne leader Black Kettle.

Sand Creek. Colonel John Chivington and the First Regiment of Colorado Volunteers attacked Black Kettle's camp at Sand Creek, in Colorado, on November 29, 1864. Howling Wolf was 15. He and his father, Eagle Head, were in the camp. The counts of Cheyenne and Arapaho people killed there run from about 70 to about 230. Most of the dead were women, children and old people. Among Howling Wolf's drawings is a picture of the massacre.

Army officers sent Howling Wolf to Fort Marion with the other prisoners in 1875. There he served as a sergeant in charge of a guard company made up of prisoners. He drew his people's lives and wars.

Home. He was released in 1878 and went home to the Cheyenne reservation in Indian Territory, now Oklahoma. His eyesight began to fail, and he sailed to Boston for an operation on his eyes.

> He spoke out against the Dawes Act of 1887. Under that law, federal officials broke up land that each nation held together into small plots for single owners and sold what they called the "surplus." Native nations held about 138 million acres in 1887 and about 48 million acres by 1934, a loss of about 90 million acres.
>
> Howling Wolf died in a car accident in 1927. He was on his way home to Oklahoma after performing in Houston, Texas, in a Wild West show, a traveling show of riding, shooting and scenes of frontier life.

Painters after the war

Mary Cassatt showed her paintings in Paris with the Impressionists, a group of painters in France. She was born in Allegheny City, now part of Pittsburgh, Pennsylvania, on May 22, 1844. She studied at the Pennsylvania Academy of the Fine Arts from 1861 to 1865, went to Europe in 1866 and settled in Paris in 1874.

The French painter Edgar Degas admired her drawing. At his request she showed her work with the Impressionists in 1879, and again in 1880, 1881 and 1886. Britannica's writers count her as the only American who showed with them in those years. She died near Paris on June 14, 1926.

Forced out. Thomas Eakins, born in Philadelphia on July 25, 1844, painted *The Gross Clinic* in 1875. He taught at the Pennsylvania Academy from 1876 and became head of its school in 1882.

During a class that women students attended, he removed the cloth that covered a male model's private parts. In 1886 he had to resign, and the record does not name the people who forced him out. Some of his students then formed their own school, the Art Students' League of Philadelphia, and he taught there. One of his students at the academy was Henry Ossawa Tanner. Eakins died on June 25, 1916.

Mark Twain

Who: A writer, born Samuel Langhorne Clemens, whose books were sold door to door.

When and where: Born November 30, 1835, in Florida, Missouri. Died in Redding, Connecticut, on April 21, 1910.

Movie: *Mark Twain* (2001), a documentary by Ken Burns, shown on PBS. It is about his whole life.

Salesmen going door to door sold 70,000 copies in one year.

Mark Twain sold his books through salesmen who went from door to door. His real name was Samuel Langhorne Clemens.

A subscription book. His book *The Innocents Abroad* came out on July 20, 1869. Elisha Bliss's American Publishing Company sold it as a subscription book. A salesman came to people's homes and showed them the book. They ordered a copy, and it was delivered later. *The Innocents Abroad* sold more than 70,000 copies in its first year. While Twain lived, it sold better than any of his other books.

The Adventures of Tom Sawyer came out in 1876. *Adventures of Huckleberry Finn* came out in Britain and Canada on December 10, 1884, and in the United States on February 18, 1885.

Kept off the shelf. In 1885 the members of the library committee in Concord, Massachusetts, decided to keep *Huckleberry Finn* out of the town's public library. According to the *Boston Transcript*, one member of the committee called it "the veriest trash," which meant the worst kind of trash.

Henry Ossawa Tanner

Who: A Black painter who studied under Thomas Eakins and made his career in France.

When and where: Born in Pittsburgh on June 21, 1859. Philadelphia. Paris from 1891. Died in Paris on May 25, 1937.

Students tied him to his easel in the middle of Broad Street.

Henry Ossawa Tanner was a Black painter who left the United States to work in France. His father, Benjamin Tucker Tanner, was a bishop, a high leader, of the African Methodist Episcopal Church.

Tanner wrote in 1909 about looking for a painting teacher when he was a boy. "The question was not, would the desired teacher have a boy who knew nothing and had little money, but would he have me, or would he keep me after he found out who I was," he wrote. He meant that a teacher might turn him away, or send him away later, because he was Black.

The academy. From 1879 to 1885 he studied at the Pennsylvania Academy of the Fine Arts, on and off, under Thomas Eakins. Joseph Pennell, a fellow student, later wrote about a night when other students attacked Tanner. "One night his easel was carried out into the middle of Broad Street and, though not painfully crucified, he was firmly tied to it and left there," Pennell wrote (crucified means fastened to a cross). Pennell did not name the students who did it.

Paris. Tanner went to Paris in 1891 to study at a school called the Académie Julian. A bishop from Atlanta, Joseph Crane Hartzell, and his wife paid for the trip by buying every painting in a show of Tanner's work. In Paris other American students mocked him. In 1893 he came home sick with typhoid fever, a disease spread by dirty water and food. In Philadelphia he painted *The Banjo Lesson*. Then he went back to Paris. Apart from short visits home, he lived there for the rest of his life.

> His painting *The Resurrection of Lazarus* won a third-place medal at the Paris Salon, a large yearly art show, in 1897, and French government officials then bought it. In 1923 he was made a member of the Legion of Honor, an order of people honored by France for their work.

The Shaw Memorial

A committee of 21 people in Boston hired the sculptor Augustus Saint-Gaudens in August 1882 to make a memorial to Colonel Robert Gould Shaw and the men of the 54th Massachusetts. The regiment had attacked Fort Wagner, in South Carolina, in July 1863. Joshua Bowen Smith, a Black man in Boston who had worked for the Shaw family, is credited with first proposing it. The writer Lydia Maria Child wrote that "the colored soldiers who fell ought to have a monument also."

Fourteen years. Saint-Gaudens worked on it for 14 years, longer than on anything else he made. He did not use photographs of the real soldiers or meet them. Instead he hired Black men he met on the street near his studio and made about 40 studies of their heads.

Whose names. Shaw's sister, Josephine Lowell, asked for the names of all the men of the 54th killed at Fort Wagner and afterward to be carved on the base. Her request failed, and the records do not say who turned it down. When the memorial went up, its base carried only the names of the white officers. The names of the Black soldiers were added in the early 1980s. The National Park Service account of the memorial does not say who added them.

The memorial was unveiled on Boston Common on May 31, 1897. Sixty-five veterans of the 54th led the parade, and Sergeant William Carney carried the flag.

Dickinson's poems, and Riis's flash photographs

Emily Dickinson of Amherst, Massachusetts, wrote nearly 1,800 poems. Only 10 were published while she was alive. She died on May 15, 1886, and her sister Lavinia found the poems after her death. Thomas Wentworth Higginson and Mabel Loomis Todd chose and prepared the poems for the first book of them, which came out in 1890.

That same year Jacob Riis published *How the Other Half Lives*, about the tenements of New York, the crowded buildings where poor families rented rooms. To photograph the inside of the dark tenements, he used magnesium flash powder, which burns with a sudden bright light.

BY 1900

If you remember one thing: By 1900 photographs and cheap books reached millions of Americans, and the Black and Native artists of the time had worked in Rome and Paris or drawn as prisoners in Florida.

In 1850: a photograph was a single picture on a silvered copper plate that could not be copied, and the Metropolitan Museum of Art did not yet exist.

In 1900: the Met had been open on Sundays since 1891, Gardner's war photographs were in print with each photographer's name, and Jacob Riis had photographed the inside of New York's tenements by flash.

Next: In the next fifty years, a show of modern European paintings draws jokes in New York, and federal officials begin paying artists a weekly wage.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 25 MINUTES

Americans argued over what counted as art from the first big show of modern paintings in 1913 on, and in the 1930s federal officials paid artists for the first time, though Black artists had to organize before they were given that work. About 75,000 people came to see modern European paintings at the Armory Show in New York in 1913, and its paintings drew jokes. In Harlem in the 1920s and 1930s, Black painters, sculptors, photographers and writers worked side by side.

Federal officials paid thousands of artists a weekly wage from 1935 to 1943. Government photographers recorded hungry farm families in the 1930s. In 1942 one of them photographed Japanese American families as soldiers and officials forced them into camps, and officials kept most of those pictures out of public view.

The Armory Show, 1913, and photographs as art

The International Exhibition of Modern Art, which people call the Armory Show, ran from February 17 to March 15, 1913, in the 69th Regiment's armory on Lexington Avenue in New York, a large building where the unit's soldiers trained and kept their weapons.

A group of American painters and sculptors organized it. Its catalog listed 1,090 works by 306 artists, and several hundred more were added during the show. About 75,000 people came to see it in New York. By the end of its tour to Chicago and Boston, nearly 300,000 people had seen it.

The jokes. Marcel Duchamp's painting *Nude Descending a Staircase* drew jokes. The critic Julian Street called it "an explosion in a shingle factory." Theodore Roosevelt said it looked like the Navajo rug in his bathroom. The sculptor Gutzon Borglum, one of the show's early organizers, called it "a staircase descending a nude."

A pretend trial. In Chicago, on April 16, 1913, students of the School of the Art Institute held a pretend trial of the painter Henri Matisse. They charged him with "pictorial arson," arson being the crime of setting fire to something on purpose, and "criminal misuse of line." Then they burned copies they had painted of three of his pictures, among them *Blue Nude*. School officials had told them not to. A reporter for the *Chicago Evening Post* counted two hundred students and more than a thousand people watching.

The photographer Alfred Stieglitz argued that a photograph could be fine art, as a painting was. He published *Camera Work*, a magazine of photographs, from 1903 to 1917. He also ran a gallery at 291 Fifth Avenue in New York from 1905 to 1917, and people called the gallery 291.

Georgia O'Keeffe

Who: A painter whose drawings Alfred Stieglitz showed in New York in 1916, and who later lived and painted in New Mexico.

When and where: Born November 15, 1887, on a dairy farm at Sun Prairie, Wisconsin. Texas, New York, and New Mexico from 1929. Died in Santa Fe on March 6, 1986.

Movie: *Georgia O'Keeffe* (about 1977), a documentary by Perry Miller Adato, made with the New York public television station WNET. It is about her life and work.

Stieglitz hung ten of her drawings at 291 in April 1916.

Georgia O'Keeffe first showed her work in Alfred Stieglitz's gallery, 291, in April 1916. That fall she became head of the art department at West Texas State Normal College, in Canyon, Texas, a college that trained teachers. She married Stieglitz in 1924.

New Mexico. From 1929 she spent her summers painting in New Mexico. In late 1945 she bought a house at Abiquiu, New Mexico, from the Catholic Church in Santa Fe. Stieglitz died in 1946, and in 1949 she moved to New Mexico for good.

An eye disease called macular degeneration had destroyed the center of her sight by 1972, and she could see only at the sides. She had stopped painting in oils without help. She died in Santa Fe on March 6, 1986, at the age of 98.

Native artists, the boarding schools, and the Studio

Angel De Cora was a Ho-Chunk painter from the Winnebago Agency in Nebraska, born on May 3, 1871. In her own account, a strange white man came to the reservation when she was a child and asked her, through an interpreter, if she would like to ride on a steam train. When the ride ended she was at the boarding school at Hampton, Virginia, a school where children live as well as study. "[It was] three years later when I returned to my mother," she said.

As an adult De Cora taught art at the Carlisle Indian School in Pennsylvania, the school Richard Henry Pratt had founded. She died on February 6, 1919.

Zitkala-Ša was a Yankton Dakota writer who lived from 1876 to 1938. In 1900 three of her essays ran in the *Atlantic Monthly* magazine. Two were about her childhood and her boarding school, and the third was about her work as a teacher. Her book *American Indian Stories* came out in 1921.

The Studio. A white teacher named Dorothy Dunn asked Chester Faris, the head of the Santa Fe Indian School, a federal boarding school in New Mexico, to let her open a painting studio there in 1932. The Studio opened on September 9, 1932, and by 1937 it had 170 students.

Dunn chose not to teach the students to draw real people and things as the eye sees them. She did not teach them to make objects look near or far, or to mix colors by rule. She taught one style: flat areas of color with heavy outlines, showing ceremonies, dances and old stories. She believed it "was the only authentic painting style for Native American artists to follow" (authentic means real and true to its roots).

What her students said. Her students disagreed about her. The Chiricahua Apache artist Allan Houser said, "she trained us all the same way... Her style lacked originality and creativity." Geronima Cruz Montoya, of Ohkay Owingeh Pueblo in New Mexico, said Dunn "made us realize how important our own Indian ways were, because we had been made to feel ashamed of them." She added, "She gave us something we could be proud of."

Dunn left in 1937. Montoya then ran the Studio until it closed in 1962. Other students included Pablita Velarde, the first girl in the art classes, and Harrison Begay, Quincy Tahoma, Oscar Howe, Pop Chalee and Andrew Tsinajinnie. Members of Congress created the Indian Arts and Crafts Board inside the Interior Department in 1935, while the Studio was running. They meant the board to widen the market for Native arts and crafts.

A basket worked out from a photograph. Lottie Queen Stamper, of the Eastern Band of Cherokee Indians in North Carolina, taught basket weaving for more than 30 years. In 1940 she found a photograph of the Cherokee double-woven baskets taken to London in 1725, two baskets woven one inside the other from the same strips.

She spent two and a half days working out the pattern from the photograph. Then she wove a copy from 500 splints, thin strips cut for weaving. "This, indeed, was the happiest day of my life," she wrote. "Now it was my duty to teach it to my brightest students."

People building along the rivers had destroyed many patches of cane by the middle of the 1900s. The weaver Eva Wolfe drove 80 miles to Hayesville, North Carolina, to gather river cane.

Maria Martinez

Who: A potter at San Ildefonso Pueblo, a Tewa town in New Mexico, known for black-on-black pottery.

When and where: Born Maria Poveka Montoya at San Ildefonso, probably in 1887. Died there on July 20, 1980.

Movie: *Maria Martinez: Indian Pottery of San Ildefonso* (1972), a documentary that shows her and her son Popovi Da making pottery at the pueblo.

The black on her pots comes from smoke and dried manure.

Maria Martinez and her husband, Julian Martinez, made black-on-black pottery at San Ildefonso Pueblo, a Tewa pueblo about 20 miles northwest of Santa Fe. Maria was born there, probably in 1887. Her aunt Nicolasa taught her to work clay.

Fairs and copies. In 1904 the couple showed how they made pottery at the World's Fair in St. Louis. Between 1907 and 1909, the archaeologist Edgar Lee Hewett asked Maria to make copies of ancient pots dug up at a site where he worked. Maria shaped the pots and Julian painted them. They showed their work again at fairs in San Diego in 1914 and Chicago in 1934.

The black. While the pots are fired, the potters smother them with dried manure, and the smoke turns the clay black. Potters at Santa Clara Pueblo had made polished black pots since the 1600s, and Maria learned the method from the Tafoya family there. In 1918 Julian finished the first pot with a dull black design on a shining black surface.

Maria signed the pots "Marie" from 1923 to 1925. From 1925 until Julian died in 1943, she signed them "Marie + Julian." Later she signed with her daughter-in-law, Santana, and then with her son, Popovi Da. Popovi Da built a shop at the pueblo and sold the family's work.

> Officials of the Santa Fe Indian School asked Maria to teach there in 1932. She said no. "I come and I work and they can watch ... nobody teaches," she said. She died at San Ildefonso on July 20, 1980.

BY 1932: Stieglitz had shown that a photograph could hang in a gallery, Maria and Julian Martinez had been making black-on-black pots since 1918, and the Studio in Santa Fe was teaching Native students a single style.

The Harlem Renaissance

Black painters, photographers and writers worked side by side in Harlem, in New York City, in the 1920s and 1930s. This time is called the Harlem Renaissance, renaissance being a French word that means rebirth.

Van Der Zee. James Van Der Zee was born in Lenox, Massachusetts, in 1886. He and Gaynella Greenlee opened the Guarantee Photo Studio on West 125th Street in Harlem in 1916, where he made portraits of his customers. In 1919 he photographed the victory parade of the 369th Infantry Regiment, an army unit. He also photographed the leader Marcus Garvey. His work became widely known in 1969, when it hung in a show called *Harlem on My Mind* at the Metropolitan Museum of Art.

Hughes. Langston Hughes was born in Joplin, Missouri, in 1901. His poem "The Negro Speaks of Rivers" came out in 1920 (Negro was a word people then used for Black people). Readers first noticed him through *The Crisis*, a magazine. His first book of poems, *The Weary Blues*, came out in 1926.

Douglas. Aaron Douglas was a painter from Topeka, Kansas. He meant to stop in Harlem in 1925 on his way to Paris, and he stayed. He designed the cover of the magazine *Fire!!* in 1926 and drew pictures for Alain Locke's book *The New Negro*.

Douglas painted a set of four murals, paintings made on a wall or for a wall, for the 135th Street branch of the New York Public Library in 1934, paid through a federal program. The murals are called *Aspects of Negro Life*. They show

Africa, slavery and the years after the Civil War, then lynching, the killing of a person by a mob without any trial, and segregation, keeping people apart by race, and then the move north.

Douglas started the art department at Fisk University, in Nashville, in 1944, and he taught there until 1966.

Augusta Savage

Who: A sculptor and art teacher in Harlem, who taught Jacob Lawrence and lost her largest sculpture for lack of money.

When and where: Born Augusta Christine Fells on February 29, 1892, in Green Cove Springs, Florida. Harlem from the 1920s. A farm at Saugerties, New York, from about 1940. Died in New York City on March 27, 1962.

Movie: *Searching for Augusta Savage* (2024), a short documentary by Charlotte Mangin and Sandy Rattley for the PBS series *American Masters*. It is about her life and her lost work.

Her 16-foot harp was bulldozed when the World's Fair closed.

Augusta Savage taught young artists in Harlem, among them Jacob Lawrence, and she lost her largest sculpture because she had no money to save it. Her father, Edward Fells, was a Methodist minister in Florida who said her clay figures were sinful. Her own words about him are printed by the public television station WMHT: "My father licked me four or five times a week," and he "almost whipped all the art out of me." (Licked here meant beat. Some other printings of her words say "kicked" instead.)

Turned away. In 1923 she was chosen for a summer art school at Fontainebleau, in France. French officials offered the school to 100 American women, and her place came with a $500 scholarship. When the American committee that chose the students learned that she was Black, its members took her place away.

Savage told the *New York World*: "My brother was good enough to be ... [in a] regiment that saw service in France during the war, but it seems his sister is not good enough to be a guest of the country for which he

fought." One committee member, the sculptor Hermon MacNeil, had supported her, and he invited her to study with him.

A teacher in Harlem. She opened the Savage Studio of Arts and Crafts in a basement on West 143rd Street in 1934. Jacob Lawrence, Norman Lewis and Gwendolyn Knight studied with her. In 1935 she helped start the Harlem Artists Guild. In 1937 she became the first director of the Harlem Community Art Center, which opened on December 27, 1937. By one count, 70,592 children and adults took part in its first sixteen months. Another count gives 1,500 people.

The Harp. The organizers of the 1939 New York World's Fair hired her to make a sculpture. She was the only African American woman given such a job. She made *Lift Every Voice and Sing*, named after a song by James Weldon Johnson. It was a harp 16 feet tall. Its strings were twelve Black singers, and the arm and hand of God formed its sounding board. The fair's organizers renamed it *The Harp*.

She made it of plaster painted to look like black stone. She needed money to cast it in bronze, which means pouring melted metal into a mold of the sculpture. She sold small metal copies as souvenirs, but she never raised enough. When the fair closed in 1940, its temporary buildings and plaster art were bulldozed, and *The Harp* with them. No record names who gave the order. Some of the small copies survive.

In about 1940 she moved to a farm at Saugerties, New York, and mostly left the art world.

Zora Neale Hurston

Who: A novelist and folklorist, someone who collects the stories, songs and sayings that people pass along by word of mouth.

When and where: Born January 7, 1891, in Notasulga, Alabama. Grew up in Eatonville, Florida. Harlem, Haiti and Jamaica. Died in Fort Pierce, Florida, on January 28, 1960.

Movie: *Zora Neale Hurston: Jump at the Sun* (2008), a documentary by Sam Pollard for the PBS series *American Masters*. It is about her life and writing.

She never earned more than $943.75 from any one of her books.

Zora Neale Hurston wrote novels and collected the stories of Black Southerners, and she died poor. She grew up in Eatonville, Florida. According to the official Hurston website, Eatonville was the nation's first incorporated Black township, which means the state recognized it as a town by law, with its own government.

Kossola. In 1927 she interviewed Kossola, also called Cudjo Lewis. He was one of the last survivors of the *Clotilda*, the last known ship to bring captives from Africa to the United States.

Soldiers of Ghezo, the king of Dahomey in West Africa, attacked Kossola's town in April 1860 and took him captive. Captain William Foster bought him and other captives there for Timothy Meaher, a businessman in Mobile, Alabama. In July 1860 Foster smuggled about 110 Africans into Mobile on the *Clotilda*, to be enslaved. Bringing enslaved people into the country had been against federal law for 52 years. In Mobile, Meaher's brother James enslaved Kossola.

The books. Hurston wrote a book from what Kossola told her and called it *Barracoon*, the word for a pen where captives were held until a ship took them. It was not published until 2018, 58 years after she died. She gathered Black folktales in Florida for her book *Mules and Men* (1935),

and stories in Haiti and Jamaica for *Tell My Horse* (1938). Her novel *Their Eyes Were Watching God* came out in 1937. In 1938 and 1939 she worked in Florida for the Federal Writers' Project.

Her royalties, the share of each book's price that goes to the writer, never came to more than $943.75 for any of her books. She had a stroke in early 1959. In October 1959 she had to enter the St. Lucie County Welfare Home, a home the county ran for people with no money, and she died there on January 28, 1960. Her neighbors in Fort Pierce collected money for her funeral, but not enough for a headstone.

Saved from the fire. After she died, a yardman cleaning out her house began burning her papers. Patrick DuVal, a law officer and her friend, put out the fire, and the papers went to the University of Florida. She lay in an unmarked grave in the Garden of Heavenly Rest, a cemetery for Black people only, until August 1973. That month the novelist Alice Walker found the spot and placed a marker there.

Painters of the 1930s and 1940s

Grant Wood painted *American Gothic* in 1930. His models were his sister, Nan Wood Graham, and the family dentist, Byron McKeeby. They stand in front of a house in Eldon, Iowa. The painting won a bronze medal and $300 at the Art Institute of Chicago. One judge called it a "comic valentine." Some Iowans who read about it in the *Cedar Rapids Gazette* were angry. They felt it showed them as "pinched, grim-faced, puritanical Bible-thumpers."

Edward Hopper painted *Nighthawks* in 1942, and on May 13 of that year the Art Institute of Chicago bought it for $3,000.

Norman Rockwell's four paintings called the *Four Freedoms* ran in the *Saturday Evening Post* in February and March 1943, one each week. The paintings then went on a tour of 16 cities, where people bought war bonds, loans to the government to pay for World War II. About 1.2 million people saw the paintings, and the tour raised about $132 million.

Who did the government pay to make art?

Federal officials paid thousands of artists a weekly wage through the Federal Art Project, but Black artists had to organize to be hired. The project ran from August 29, 1935, to June 30, 1943, as part of the Works Progress Administration, the federal jobs program of the Great Depression, the hard years of the 1930s when many people had no work. Holger Cahill directed it.

The pay. Most of its artists had to show they were poor enough to get government aid. They earned about $23.50 a week, and each had to turn in a finished work within a set number of weeks. According to Britannica, project officials employed more than 5,000 artists at its peak in 1936. Another count puts the total at about 10,000 artists and craft workers.

They made about 400,000 paintings, murals, prints and posters, and they opened more than 100 community art centers. When officials cut the budget and sent out notices that ended people's jobs, artists held sit-down strikes, refusing to leave their workplaces.

The Harlem Artists Guild. The project's rules said workers must be hired fairly, whatever their race, but Black artists were routinely refused work. The records do not name the people who turned them away. Augusta Savage, Charles Alston, Elba Lightfoot, Louise E. Jefferson, Arthur Schomburg and others started the Harlem Artists Guild in 1935, with Aaron Douglas as its president. Alston called it "a pressure group to get more black artists on the federal projects." According to *Black Art in America*, Alston became the first Black supervisor on the project.

Harlem Hospital. Alston, Vertis Hayes and Georgette Seabrooke designed murals for Harlem Hospital in 1936. According to the Harlem history project Harlem-is, it was the first mural job the WPA gave to African American artists anywhere in the country. The head of the hospital rejected the sketches because of their "Negro subject matter." Members of the Guild and the Artists Union wrote to Mayor Fiorello La Guardia and President Franklin Roosevelt, and the murals were then painted as the artists had planned.

Post office murals. Edward Bruce ran a second program, in the Treasury Department, from 1934 to 1943. Its artists painted about 1,400 murals for post offices in more than 1,300 towns and cities. For each new post office, about 1 percent of the building money was set aside to pay for art.

Bruce's officials pushed the artists toward American scenes and away from subjects that might start public quarrels. They also kept them away from abstract art, which shows shapes and colors instead of things. In Safford, Arizona, local people who were "still resentful about Apache raids" demanded a different subject for a mural, and the subject was changed. The account of the Safford mural does not say who changed it.

Federal officials also hired writers, for the Federal Writers' Project. From 1936 to 1938 its interviewers in 17 states recorded more than 2,300 first-person accounts by people who had been enslaved. The Library of Congress holds the accounts.

Jacob Lawrence

Who: The painter of the *Migration Series*, sixty panels about Black families moving north.

When and where: Born September 7, 1917, in Atlantic City, New Jersey. Philadelphia, Harlem from the age of 13, and Seattle. Died in Seattle on June 9, 2000.

He painted all 60 panels at once, one color at a time.

Jacob Lawrence painted all 60 panels of *The Migration Series* in 1940 and 1941, when he was 22 and 23. His parents split up in 1924. From 1927 to 1930 he and his younger brother and sister lived in foster care in Philadelphia, with families paid to raise children not their own, while their mother worked in New York City.

Harlem teachers. At 13 he moved with his mother to Harlem. He took art classes with Charles Alston at the Harlem Art Workshop. Augusta Savage got him a scholarship to the American Artists School and a paid place on the federal art project.

The series. Before *The Migration Series* he painted series of pictures about the lives of Toussaint L'Ouverture in 1938, Frederick Douglass in 1939 and Harriet Tubman in 1940. Toussaint L'Ouverture was born enslaved in the French colony that is now Haiti, and he became a leader of the revolution that ended slavery there. A grant, a gift of money, from the Julius Rosenwald Fund paid for *The Migration Series*. The panels show the Great Migration, when Black families moved from the South to cities in the North.

He wrote a caption for each panel first. Then he painted all 60 at the same time, one color at a time. Each panel is 12 inches by 18 inches. In November 1941 the editors of *Fortune* magazine printed panels from the series with an essay by Alain Locke. Today the Phillips Collection in Washington holds the odd-numbered panels, and the Museum of Modern Art in New York holds the even-numbered ones.

BY 1941: Federal officials had paid thousands of artists a wage, Black artists in Harlem had organized to get some of that work, and Jacob Lawrence had painted the Great Migration in 60 panels.

Photographs of hunger, and photographs of the camps

Officials of the federal Resettlement Administration, later called the Farm Security Administration, paid photographers in the 1930s to record poor farm families. In March 1936 Dorothea Lange stopped at a camp of pea pickers at Nipomo, California. Freezing rain had destroyed the crop, and as many as 2,500 to 3,500 people were camped there.

Migrant Mother. Lange made seven pictures of a mother and her children. One of them became known as *Migrant Mother*, and it is one of the most reprinted pictures of the Great Depression. The government caption reads: "Destitute pea pickers in California. Mother of seven children. Age thirty-two" (destitute means so poor that a person has nothing to live on).

Two accounts. Lange later wrote, "I did not ask her name or her history." She wrote that the mother said the family had been living on frozen vegetables from the fields and on birds the children killed. The mother had just sold the tires from her car to buy food, Lange wrote. The mother's son, Troy Owens, said later, "There's no way we sold our tires, because we didn't have any to sell." He thought Lange had mixed up two stories. The two accounts do not agree.

Florence Owens Thompson. The mother was Florence Owens Thompson, born on September 1, 1903, in Indian Territory, now Oklahoma. The records of her family's Cherokee ancestry disagree. Cherokee Nation records list her father as Cherokee, while her mother said she had Cherokee ancestors but told a federal commission that her own parents were both white. The government had paid for the photograph, so anyone could print it for free, and neither Lange nor Thompson earned money from it.

A reporter for the *Modesto Bee*, Emmett Corrigan, found her in 1978, about 42 years after the picture was taken. She was living in a mobile home park in Modesto, California, and he printed her name. "I wish she hadn't taken my

picture. I can't get a penny out of it. She didn't ask my name," she told him. "She said she wouldn't sell the pictures. She said she'd send me a copy. She never did."

Her children asked for help with her medical bills in 1983, when she was sick. More than 2,000 letters came with money. "For Mama and us, the photo had always been a bit of [a] curse," Troy Owens said. "After all those letters came in, I think it gave us a sense of pride." She died on September 16, 1983.

The camps. President Franklin Roosevelt signed Executive Order 9066, an order from the president that has the force of law, on February 19, 1942. Under it, Lieutenant General John L. DeWitt, who commanded the Army on the West Coast, ordered Japanese Americans out of their homes, some with only 48 hours' notice.

Soldiers and federal officials sent about 122,000 men, women and children to ten camps. They held them there behind wire fences watched by armed guards in towers. Nearly 70,000 of them were American citizens. That year officials of the War Relocation Authority, the agency that ran the camps, hired Lange to photograph the forced removal.

Dorothea Lange

Who: A photographer paid by the federal government to record poor families in the 1930s and the forced removal of Japanese Americans in 1942.

When and where: Born Dorothea Nutzhorn on May 26, 1895, in Hoboken, New Jersey. California and the West. Died October 11, 1965.

Movie: *Dorothea Lange: Grab a Hunk of Lightning* (2014), a documentary for the PBS series *American Masters*, directed by her granddaughter Dyanna Taylor. It is about her life and work.

At Manzanar, officials would not let her photograph the guard towers.

Dorothea Lange photographed Japanese American families as soldiers and officials forced them from their homes in 1942, and officials kept most of those photographs from the public. As a child she had polio, a disease that attacks the nerves, and it left her with a limp.

What she photographed. In 1942 she photographed families getting ready to leave their homes between Fresno and Sacramento. She photographed the Tanforan assembly center at San Bruno, a holding camp at a racetrack, where horse stalls had been turned into rooms for families. She also photographed the Manzanar camp. She made more than 800 photographs.

Officials discouraged her from talking with the prisoners. At Manzanar they did not allow her to photograph the wire fences, the watch towers, the armed guards or any sign that people were resisting. On July 30, 1942, officials of the War Relocation Authority let her go, with the note "completion of work."

Held back. Army officers judged her pictures critical of the camps and held them until the war ended, according to the National Coalition Against Censorship. The Densho Encyclopedia says officials of the War Relocation Authority kept most of them from public view. Neither account names the officers who gave the order.

> After the war, the coalition says, the photographs were "quietly placed" in the National Archives. It does not say who placed them there. In 2006 the historians Linda Gordon and Gary Y. Okihiro published a selection of them in a book called *Impounded*.
>
> Lange wrote to the photographer Ansel Adams on November 12, 1943: "I fear the intolerance and prejudice is constantly growing." She died on October 11, 1965.

A book banned and burned, 1939

John Steinbeck's novel *The Grapes of Wrath* came out on April 14, 1939. It was the best-selling book of 1939, and by February 1940 publishers had printed 430,000 copies. It won the National Book Award and the Pulitzer Prize.

The ban. The members of the Kern County Board of Supervisors in California voted 4 to 1 in August 1939 to take the book out of the county's libraries and schools. One supervisor called it a "libel and lie," a libel being a printed lie that harms someone's good name. Bill Camp, who led the Associated Farmers, a group of large landowners, pushed for the ban. He had one of his workers, Clell Pruett, burn a copy in a trash can for a newspaper photographer. Pruett had not read the book.

The librarian. The county librarian, Gretchen Knief, wrote to the supervisors: "Ideas don't die because a book is forbidden reading. If Steinbeck has written the truth, that truth will survive." She did not throw the books away. She offered them to other county libraries. The book went back on the shelves in January 1941.

Paintings sent abroad, and called back, 1946 and 1947

J. LeRoy Davidson, an art historian at the State Department, spent just under $50,000 in 1946 on 79 modern American paintings by 47 artists, among them Georgia O'Keeffe. He sent them to tour other countries in a show called *Advancing American Art*.

One congressman said, "I am just a dumb American who pays taxes for this kind of trash." President Harry Truman also mocked the paintings. He said that if that was art, then he was a member of an African people. The word he used for them was an insult that Europeans used for the Khoekhoe people of southern Africa.

Called back. Accounts differ on who stopped the tour, and when. The art historian Jennifer McComas writes that the tour was stopped in early April 1947. By her account, members of Congress had cancelled it by the end of that month, cut off the State Department's art program and done away with Davidson's job. The writer Matthew Algeo says instead that Secretary of State George Marshall ended the show in early May, and told a House committee that the paintings would be brought home and sold.

Officials sold the paintings, along with 38 watercolors, in June 1948 by sealed bid, in which buyers send in secret offers and the highest offer wins. Three universities in Alabama, Oklahoma and Georgia bought 81 of the 117 works.

BY 1950

If you remember one thing: For the first time, federal officials paid thousands of American artists to work, and other officials decided which pictures and books the public could see.

One number: About $23.50 a week was the pay of a Federal Art Project artist. In 1942 the Art Institute of Chicago paid $3,000 for one painting, Hopper's *Nighthawks*, more than two years of that wage.

Next: After 1950, painters in New York make pictures with no objects in them at all, and officers of the CIA secretly pay to send some of them abroad.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 21 MINUTES

After 1950 the fights over who made American art, who profited from it and who paid for it came into the open. Painters in New York after 1950 made pictures that showed no objects at all, and officers of the CIA secretly paid for shows that sent some of that painting to Europe. In the 1960s other artists painted soup cans and comic strips, and the prices of paintings climbed. A painting that Robert Rauschenberg sold for $900 went for $85,000 at an auction in 1973.

Black and Chicano artists painted murals on neighborhood walls and bridge pillars, and women and Black artists protested at museums that left them out. In 1987 volunteers laid 1,920 quilt panels on the National Mall for people who had died of AIDS, a disease that destroys the body's defenses against other illnesses.

Members of Congress fought in 1989 and 1990 over whether public money should pay for art that some of them called offensive. In 1990 they also passed laws to protect Native artists' work and to return Native remains and sacred objects to the nations.

Painting with no objects in it

Jackson Pollock laid his canvas on the floor from 1947 to 1950 and dripped and poured paint onto it. Other people called his way of working drip painting or action painting. His pictures showed no things that a viewer could name. This kind of painting, abstract because it shows no real objects, is called Abstract Expressionism. The historian Frances Stonor Saunders writes that the center of the art world had been moving from Paris to New York since the 1930s.

Jackson Pollock

Who: A painter who dripped and poured paint onto canvas laid on the floor.

When and where: Born January 28, 1912, in Cody, Wyoming. New York City, then Springs, on Long Island, New York. Died there on August 11, 1956.

Movie: *Pollock* (2000), a feature film directed by Ed Harris, who also plays Pollock. It is about his life.

"On the floor I am more at ease," he wrote in 1947.

Jackson Pollock explained his method in 1947, in a magazine called *Possibilities*: "I prefer to tack the unstretched canvas to the hard wall or the floor. I need the resistance of a hard surface. On the floor I am more at ease. I feel nearer, more a part of the painting, since this way I can walk around it, work from the four sides and literally be in the painting." He wrote that this was like the method of the Native sand painters of the West.

Pollock had married the painter Lee Krasner in 1945. On August 8, 1949, the editors of *Life* magazine put this question in a headline over a story about him: "Jackson Pollock: Is he the greatest living painter in the United States?"

The crash. On the night of August 11, 1956, in Springs, New York, Pollock drove his car while drunk. He crashed less than a mile from his home and was killed, and so was his passenger Edith Metzger. A second passenger, Ruth Kligman, survived.

The CIA and the painters

The historian Frances Stonor Saunders reported in *The Independent*, a British newspaper, in 1995 that officers of the CIA had helped promote Abstract Expressionism in other countries. The CIA, the Central Intelligence Agency, is the government's spy agency.

In those years the United States and the Soviet Union were rivals in the Cold War, a long contest for power and friends around the world that never became open war between the two. The Soviet government's official style of art was called Socialist Realism. Its paintings looked like real life, only better, and they showed simple stories in support of the government's ideas, such as workers winning their freedom.

What a CIA man said. Donald Jameson, a former CIA officer, told Saunders that CIA officers saw a chance in this new painting. He said "Abstract Expressionism was the kind of art that made Socialist Realism look even more stylised and more rigid." The support came "at two or three removes," he said, which means through other groups. The painters did not know.

Tom Braden, who led a CIA division from 1950, had worked at the Museum of Modern Art in New York in 1949. "We wanted to unite all the people who were writers, who were musicians, who were artists," Braden said, "to demonstrate that the West and the United States was devoted to freedom of expression."

The money's route. The money went through the Congress for Cultural Freedom, a group set up with CIA money in 1950 and run by a CIA agent. It had offices in 35 countries and paid for touring shows of American art. One, "The New American Painting," visited every large city in Europe in 1958 and 1959. Staff of the Museum of Modern Art organized most of these shows.

Accounts differ about how much this mattered to the painters or to their fame. The art historian Jennifer McComas writes that the claims hold "some truth." She also writes that some of the writers who made them got facts wrong and chose only the shows that fit their argument.

NOT KNOWN: the amount of money spent on the painting shows.

Photographs and books

The photographer Robert Frank won a Guggenheim Fellowship, a grant of money for artists and writers, in 1955. He used it to pay for trips across the United States. In the South, a sheriff told him he had "an hour to leave town." His book of photographs, *The Americans*, came out in Paris in 1958 and in New York in 1959. A writer in *Popular Photography* called the pictures "meaningless blur, grain, muddy exposures, drunken horizons and general sloppiness."

Black writers won national prizes and reached Broadway in the 1950s. Ralph Ellison's novel *Invisible Man*, about the problems Black Americans faced in the early 1900s, came out in 1952. It won the National Book Award in 1953, and Ellison was the first African American writer to win it.

Lorraine Hansberry's play *A Raisin in the Sun* opened on Broadway on March 11, 1959. It follows a Black family in Chicago that buys a house in a white neighborhood, where a man from the neighborhood offers to buy them out. Its director, Lloyd Richards, was the first Black director of a Broadway play, and the New York Drama Critics' Circle named it the best play of the year.

Gordon Parks

Who: A photographer, the first African American staff photographer at *Life* magazine.

When and where: Born November 30, 1912, in Fort Scott, Kansas. Washington, D.C., from 1942. New York. Died in New York on March 7, 2006.

Movie: *Half Past Autumn: The Life and Works of Gordon Parks* (2000), a documentary about his life.

He photographed a cleaning woman with her mop before the flag.

Gordon Parks was the first African American photographer on the staff of *Life* magazine. He grew up in Fort Scott, Kansas, the youngest of 15 children, and went to a segregated elementary school, a school for Black children only. He bought his first camera at a pawnshop, a store that lends money and sells things people leave there and do not come back for.

Washington, 1942. Parks won a Julius Rosenwald Fellowship in 1942 and went to Washington, D.C., to work for the Farm Security Administration. Workers in restaurants and shops in the segregated capital refused to serve him.

After that he photographed Ella Watson, a Black woman on the cleaning crew of the agency's building. She stands with a broom and a mop in front of an American flag. He titled the picture *American Gothic, Washington, D.C.*, after Grant Wood's 1930 painting *American Gothic*, which shows a farmer holding a pitchfork beside a woman in front of their house. His boss, Roy Stryker, said the picture was an attack on America and could get all his photographers fired.

Parks published a photo essay, a set of photographs that tell a story, about a young gang leader in Harlem in 1948. It won him a staff job at *Life*, and he stayed at the magazine until 1972.

Pop Art, and the price of a painting

Pop Art painters of the early 1960s made pictures of comic strips, soup cans and famous faces. Roy Lichtenstein painted scenes from comic strips in oil paint. His 1961 painting *Look Mickey* is now in the National Gallery of Art in Washington.

Thirty-two soup cans. Andy Warhol showed 32 paintings of Campbell's soup cans at the Ferus Gallery in Los Angeles in July 1962. Each canvas is 20 inches by 16 inches. The art dealer Irving Blum, who bought and sold art for a living, bought all 32 for $1,000, paid in ten monthly payments of $100.

The Museum of Modern Art got them from Blum in 1996. In 1970 one of Warhol's soup-can paintings sold at auction for $60,000. At that time it was the highest auction price for a living American artist.

$900, then $85,000. Robert Scull, who owned a fleet of taxis, sold 50 works from his art collection at an auction in New York on October 18, 1973, for about $2.2 million. Years earlier, Robert Rauschenberg had sold Scull his painting *Thaw* for $900. That night it sold for $85,000. Artists received nothing when their work was sold again. Rauschenberg shoved Scull and told him he had been working hard for Scull to make that profit.

Andy Warhol

Who: A painter who made pictures of soup cans, dollar bills and famous faces. His studio was called the Factory.

When and where: Born Andrew Warhola Jr. on August 6, 1928, in Pittsburgh. New York. Died in New York on February 22, 1987.

Movie: *Andy Warhol: A Documentary Film* (2006), a four-hour documentary by Ric Burns, shown on PBS. It is about his life.

Told to paint what he liked most, he answered, "money."

Andy Warhol started out drawing pictures for magazines before he made his paintings of soup cans. His parents were working-class immigrants, Rusyns from the village of Miková, in what is now northeastern Slovakia. His father worked in a coal mine.

Drawings for magazines. Warhol moved to New York in 1949. There Tina Fredericks, an editor at *Glamour* magazine, bought a drawing of an orchestra from him for $10. Through the early 1950s his drawings appeared regularly in fashion magazines and in *The New York Times*.

Silkscreens. In 1961 a printer, Floriano Vecchi, taught him silkscreen printing, in which the artist puts a design on a fine screen and pushes ink through it with a rubber blade, so the same picture can be printed again and again. His first silkscreens, made in early 1962, showed one-dollar and two-dollar bills.

Muriel Latow, who knew him from the art world, had told him to paint what he liked most, and he answered, "money." One of his best-known early silkscreen paintings is *Marilyn Diptych*, from 1962, a picture in two parts of the actress Marilyn Monroe.

His studio was called the Factory. On June 3, 1968, the writer Valerie Solanas shot Warhol and the critic Mario Amaya at the Factory. Warhol was in the hospital for nearly two months. He died in New York on February 22, 1987, after an operation on his gallbladder.

BY 1973: Paintings by New York artists had toured Europe in shows paid for in secret with CIA money, and a Rauschenberg painting bought for $900 had resold for $85,000.

Walls, bridges and museums: art made as protest

A walk to school. Ruby Bridges, who was six, walked into William Frantz Elementary School in New Orleans on November 14, 1960, between four federal officers called deputy US marshals. She and three other six-year-old girls at another school that day were the first Black children in white public schools in Louisiana. A crowd outside screamed at her.

Norman Rockwell painted her walk in 1963, the year he left the *Saturday Evening Post*, whose editors limited what political subjects he could paint. On the wall behind her he painted a racial slur, the letters "KKK" and a smashed tomato. KKK stands for the Ku Klux Klan, a white group that has threatened, beaten and killed Black people. The editors of *Look* magazine printed the painting, *The Problem We All Live With*, on January 14, 1964.

The Wall of Respect. Fourteen artists of the Organization of Black American Culture, among them William Walker, painted the *Wall of Respect* in 1967 on the side of a closed bar at 43rd Street and Langley Avenue in Chicago. It showed Nat Turner, Harriet Tubman, W. E. B. Du Bois, Marcus Garvey, Malcolm X, Gwendolyn Brooks, Muhammad Ali, Aretha Franklin and others. In the eight years after it, artists painted more than 1,500 murals across the country.

A fire damaged the building in 1971. The building was then torn down, and the mural with it. The records do not say who ordered the building torn down.

Chicano Park. Barrio Logan is a Mexican American neighborhood of San Diego, and many Mexican Americans call themselves Chicano. Highway builders cut the neighborhood in two with Interstate 5 in 1963, and in 1969 they cut it again with the ramps of the bridge to Coronado. City officials approved a park under the bridge that June and then did nothing.

A student named Mario Solis saw bulldozers there on April 22, 1970, getting the land ready for a parking lot for a California Highway Patrol station. Residents took over the land.

City officials set aside $21,814.96 for a park of 1.8 acres, called Chicano Park, on July 1, 1970. The artist Salvador Torres had proposed painting the bridge pillars. The artist Victor Ochoa brought 300 brushes on March 23, 1973, and nearly 300 people painted all weekend. The artists raised the money for the paint themselves.

At the museums. Faith Ringgold and other artists protested at the Whitney Museum of American Art in New York in 1970, because its show of American art had no Black artists in it. They also demanded that half the artists it showed be women.

Judy Chicago finished *The Dinner Party* in 1979. It is a table with three sides, set with 39 places for 39 women from history and legend, among them Sacajawea, Sojourner Truth, Susan B. Anthony, Virginia Woolf and Georgia O'Keeffe. Each place has a hand-painted plate. Most of the plates show a butterfly or flower shape that stands for a vulva, the outer part of a woman's sex organs.

The critic Hilton Kramer called it "very bad art ... failed art." Representative Robert K. Dornan called it "ceramic 3-D pornography" (pornography means pictures or objects made to show sex). The writer Alice Walker objected that Sojourner Truth, the one Black woman at the table, was shown in a different way from the white women. Truth's plate shows three faces instead.

The board of the University of the District of Columbia voted in 1990 to accept it as a gift. After the *Washington Times* printed stories against it, the plan was dropped. The records do not say who dropped it. *The Dinner Party* is now at the Brooklyn Museum.

Mining the Museum. The artist Fred Wilson rearranged the collection of the Maryland Historical Society in Baltimore in 1992 for a show called *Mining the Museum*. In a case of fine silver, he placed shackles that had been used to chain enslaved people. He also hung Justus Engelhardt Kühn's portrait of

Henry Darnall III, the Maryland boy painted in about 1710 with an enslaved Black boy beside him. Wilson lit the face of the enslaved boy and played a recorded voice asking, "Am I your brother?"

In another room he set old chairs facing a whipping post. Several writers said that the post had been used on enslaved people, but it came from the Baltimore City jail, where officials had used it to whip men convicted of beating their wives.

On the subway walls. Jean-Michel Basquiat and Al Diaz wrote graffiti, writing or drawing on walls without permission, signed "SAMO" on walls in Manhattan in the late 1970s. Basquiat later became a painter. He died of a heroin overdose on August 12, 1988, at 27.

Michael Stewart. New York City Transit Police officers arrested Michael Stewart, a 25-year-old Black artist, at the First Avenue subway station on September 15, 1983, for writing graffiti. While the officers held him, he went into a coma, a state like deep sleep that a person cannot wake from. He died on September 28. The city's medical examiner, the doctor who studies a body to find the cause of death, said his heart had stopped. Others disputed that finding.

Six officers were tried for causing his death by carelessness, for assault and for lying under oath. On November 24, 1985, an all-white jury found them not guilty. A lawsuit over his death was later settled for $1.7 million. Basquiat painted *Defacement (The Death of Michael Stewart)* about him.

Keith Haring drew in the early 1980s in white chalk on the black paper that covered empty advertising frames in New York subway stations. He learned he had AIDS in 1988, and in 1989 he started the Keith Haring Foundation. He died on February 16, 1990.

Faith Ringgold

Who: A painter and quilt maker who also wrote and illustrated children's books.

When and where: Born Faith Willi Jones on October 8, 1930, in Harlem. New York City. Died in Englewood, New Jersey, on April 13, 2024.

Her quilt *Tar Beach 2* shows a girl flying over Harlem.

Faith Ringgold painted stories on quilts, and police arrested her during the protest at the Whitney. For years she taught art in New York City public schools. In 1967 she painted *The Flag Is Bleeding* and *Die* for a series called *The American People*.

Arrested. With the critic Lucy Lippard and the artist Poppy Johnson, she started the Ad Hoc Women's Art Committee, which joined the protest at the Whitney. The museum owned work by Jacob Lawrence but left him out of its show. Police arrested Ringgold on November 13, 1970. That year she and her daughter, Michele Wallace, also started a group called Women Students and Artists for Black Art Liberation.

Quilts. From the 1980s she painted stories on quilts. *Tar Beach 2* (1990) shows Cassie Louise Lightfoot, an eight-year-old girl who dreams of flying over Harlem. She wrote and illustrated more than a dozen children's books.

Native artists and Native objects

Oscar Howe was a Yanktonai Dakota painter, born on the Crow Creek Sioux Reservation in South Dakota in 1915, who had studied at Dorothy Dunn's Studio in Santa Fe. In 1958 the judges of the Indian art show at the Philbrook Museum in Tulsa, Oklahoma, turned down his painting as "a fine painting, but not Indian."

Howe's answer. "Who ever said, that my paintings are not in the traditional Indian style, has poor knowledge of Indian art indeed," Howe wrote back. "There is much more to Indian Art, than pretty, stylized pictures." He added: "Are we to be herded like a bunch of sheep, with no right for individualism, dictated as the Indian has always been."

Scholder and Cannon. When the Studio closed in 1962, the Institute of American Indian Arts opened in Santa Fe. Fritz Scholder, a registered member of the La Jolla Band of Luiseño Indians, taught there from 1964. From 1967 he painted Native people with American flags, beer cans and cats. The paintings started an argument, among Native artists too, that still goes on.

The Cherokee painter Roy Boney Jr. said of *Indian with Beer Can* (1969): "While some might find the image offensive or even ugly," it "definitely was not a maiden on the river or a warrior on horseback, and I liked that."

T. C. Cannon, a Kiowa and Caddo painter, studied at the Institute. He served in Vietnam in 1967 and 1968 as a paratrooper, a soldier who jumps from planes, and he won two Bronze Stars, medals for brave or useful service in war. He died in 1978 at the age of 31.

Cannon was one of the Americans who served in that war. In 1981 a student, Maya Lin, 21, won the contest to design the memorial to them in Washington, the Vietnam Veterans Memorial. The judges knew the 1,421 entries only by number, not by name. Chapter 12 tells her story in full.

Two laws of 1990. Members of Congress passed the Indian Arts and Crafts Act in 1990. Under it, selling work as made by a Native artist when it was not is a crime. A person who does it for the first time can be fined up to $250,000 or sent to prison for five years, or both. A business can be fined up to $1,000,000.

President George H. W. Bush signed the Native American Graves Protection and Repatriation Act, called NAGPRA, on November 16, 1990. Representative Mo Udall had introduced it. Repatriation means sending something back home. Under the law, museums and agencies that take federal money must return Native human remains, objects buried with the dead, sacred objects and objects that belong to a whole nation, to descendants and to the Native nations.

Some of those remains and objects had been dug out of graves and mounds, among them the remains of more than 600 people that workers took out of Craig Mound at Spiro after the Pocola Mining Company's dig, which the first era of this chapter tells.

The AIDS Memorial Quilt, 1987

Cleve Jones, an activist in San Francisco, learned in November 1985 that more than 1,000 people in his city had died of AIDS. He made the first quilt panel, for his friend Marvin Feldman. In June 1987 he, Mike Smith, Gert McMullin and others started the NAMES Project Foundation.

Each panel is three feet by six feet, about the size of a grave, and each one is made in memory of people who died of AIDS. Many people who died of AIDS had no funeral. Some of their families were ashamed, and workers at many funeral homes and cemeteries refused to handle their bodies.

On the Mall. Volunteers laid 1,920 panels on the National Mall in Washington on October 11, 1987, over an area larger than a football field. They read all 1,920 names aloud. About half a million people came that weekend. By October 1989 the quilt had more than 12,000 panels. In October 1996 the whole quilt covered the Mall for the last time, and about 1.2 million people came to see it.

Fights over public money for art, 1989 and 1990

The National Endowment for the Arts, or NEA, is a federal agency that gives money to artists and museums. In 1987 the photographer Andres Serrano made a picture of a small plastic crucifix, a model of Jesus on the cross, in a glass of his own urine. It won an award of $15,000, paid for in part with NEA money.

Senators Al D'Amato and Jesse Helms attacked the award in 1989. Some members of Congress called art like this obscene, so shocking or indecent that some people want it banned, or blasphemous, insulting to God or religion. Serrano received death threats and hate mail, and he lost grants.

Mapplethorpe. Robert Mapplethorpe was a photographer who died of AIDS on March 9, 1989. A show of more than 150 of his photographs was then touring the country. It held portraits, nudes and pictures of flowers. It also held photographs that showed sex between men openly, some of it sex that involves pain, and two photographs of young children without clothes. At each museum, staff hung the sex pictures in a separate area that children could not enter.

Christina Orr-Cahall, director of the Corcoran Gallery of Art in Washington, cancelled the show in June 1989, because she feared for the NEA's money. Senator Helms then proposed that no NEA money could pay for "obscene" art. For a time, artists who took NEA grants had to sign a promise not to promote obscenity.

A museum on trial. The Mapplethorpe show opened at the Contemporary Arts Center in Cincinnati in 1990. After a campaign by a group called Citizens for Community Values, law officers ordered 400 visitors out of the museum and videotaped the photographs. Government lawyers charged the museum and its director, Dennis Barrie, with obscenity. It was the first time a museum in the United States was put on trial for art it showed. On October 5, 1990, a jury found them not guilty. More than 80,000 people saw the show in Cincinnati.

The NEA chairman, John Frohnmayer, turned down grants to four performance artists, who put on live shows, in June 1990 because of what their work was about. They were Karen Finley, Tim Miller, John Fleck and Holly Hughes. Panels of experts had already approved the grants. The four artists won in court in 1993. In 1998 the justices of the Supreme Court ruled in favor of the way the NEA chose its grants. After that, NEA officials stopped giving most grants to single artists.

Toni Morrison

Who: A novelist and book editor who won the Nobel Prize in Literature.

When and where: Born Chloe Ardelia Wofford on February 18, 1931, in Lorain, Ohio. Syracuse, then New York City. Died in the Bronx on August 5, 2019.

Movie: *Toni Morrison: The Pieces I Am* (2019), a documentary by Timothy Greenfield-Sanders about her life and work.

She found Margaret Garner's story while making *The Black Book*.

Toni Morrison published other Black writers as an editor before she became a famous novelist herself. In about 1965 she began work as an editor for a textbook company owned by Random House, in Syracuse, New York. About two years later she moved to Random House in New York City. There she was the first Black woman senior editor in the fiction department.

The editor. As an editor she published the novelist Gayl Jones, whose writing she had discovered, and the writer Toni Cade Bambara. She also published books by the activist Angela Davis and the boxer Muhammad Ali. She published the poet Henry Dumas, whom a transit officer had shot dead in a New York subway station in 1968. The records do not name the officer.

She and the collector Middleton A. Harris put together *The Black Book*, a collection of old papers, photographs, advertisements and pictures about Black life in America, in 1974.

Beloved. Her first novel, *The Bluest Eye*, came out in 1970. While working on *The Black Book*, she found the story of Margaret Garner. On January 28, 1856, Garner and her family escaped from slavery to a place near Cincinnati, Ohio. When slave catchers came for them, Garner killed her two-year-old daughter, Mary, with a knife rather than see the child returned to slavery. That story became the base for Morrison's novel *Beloved* (1987), which won the Pulitzer Prize in 1988. In 1993 she won the Nobel Prize in Literature.

BY 2000

If you remember one thing: Between 1950 and 2000 the fights over art moved into the open, in museums, in courts and in Congress, and in 1990 members of Congress ordered museums that take federal money to return Native remains and sacred objects.

Left open: how much the museums would really give back under the 1990 law, and how fast.

Next: After 2000 a new federal rule makes museums cover or close their Native displays, and computer programs begin making pictures in the style of living artists.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 24 MINUTES

Almost every adult now carries a camera in a phone, and the old arguments about art have come back over new things: computer programs that copy artists' styles, Native remains in museums, and what national museums may show. After 2022, computer programs could make pictures in the style of living artists, and some of those artists sued the companies that made the programs.

Under a rule that took effect in January 2024, museums must get a Native nation's consent before they show its ancestors' remains or sacred objects.

Two Black artists painted the official portraits of a president and a first lady in 2018. In 2025 one of them cancelled her show at a national museum after officials there questioned one of her paintings. That year officials of President Donald Trump's administration ended many federal arts grants and asked for a review of eight Smithsonian museums.

A camera in every pocket

A smartphone has a camera built in, so anyone who owns one can take a picture at any moment. In a survey taken from February to June 2025, researchers at the Pew Research Center found that 91 out of every 100 adults in the United States owned a smartphone. In Pew's first survey about smartphones, in 2011, only 35 out of every 100 did.

Who is in the picture

Officials of the National Portrait Gallery in Washington unveiled the official portraits of former President Barack Obama and former First Lady Michelle Obama on February 12, 2018. Kehinde Wiley painted the former president, and Amy Sherald painted the former first lady. According to the gallery's staff, Wiley and Sherald were the first African American artists chosen to paint the gallery's official portraits of a president or a first lady.

Wiley painted Obama sitting in front of a wall of leaves and flowers. The chrysanthemums stand for Chicago, the jasmine for Hawaii and the African blue lilies for Kenya, his father's country. Sherald titled her painting *Michelle LaVaughn Robinson Obama*.

The two portraits then toured museums, first the Art Institute of Chicago, and in 2022 Atlanta, Houston, San Francisco and Boston. The Los Angeles County Museum of Art was another stop. At the Art Institute of Chicago, the crowds broke the museum's records for daily attendance again and again, according to the Portrait Gallery.

Kara Walker. Kara Walker was born in Stockton, California, on November 26, 1969. She cuts figures out of black paper and sets them up in scenes, as big as a room, about slavery in the South. In 1997, when she was 28, she won a MacArthur Fellowship, a large prize of money given to people for creative work. That same year the artist Betye Saar sent letters to about 200 artists and politicians. Saar objected that Walker's work used racist pictures of Black people.

In a 1999 PBS documentary, Saar called the work "revolting and negative" and said it pandered to "the white art establishment," which means it gave those people what they wanted, to win their favor. Other writers call Walker one of the most prominent and praised Black American artists working today.

A sugar sphinx. Inside the old Domino Sugar Refinery in Brooklyn, from May to July 2014, before the building was torn down, Walker showed a huge white figure coated in sugar. It was a sphinx, a creature with a lion's body and a human head, and its face had the features of a Black woman. Fifteen smaller figures stood around it.

She called it *A Subtlety, or the Marvelous Sugar Baby*. Its full title calls it a tribute to "the unpaid and overworked Artisans who have refined our Sweet tastes from the cane fields to the Kitchens of the New World." The unpaid workers in the cane fields were enslaved people.

Amy Sherald

Who: The painter of Michelle Obama's official portrait and of Breonna Taylor, who cancelled her own show at the National Portrait Gallery in 2025.

When and where: Born August 30, 1973, in Columbus, Georgia. Baltimore, then New Jersey from 2018.

She cancelled her own show at the National Portrait Gallery.

Amy Sherald waited tables until she was 38 while she kept painting, and then she painted Michelle Obama's official portrait. She studied painting at Clark Atlanta University, where she finished in 1997, and at the Maryland Institute College of Art in Baltimore, where she finished in 2004. She taught art at the Baltimore City Detention Center.

A new heart. When she was 30, doctors found that she had congestive heart failure, which means her heart could not pump blood well enough. On December 18, 2012, when she was 39, she had a heart transplant, an operation that gave her the heart of a person who had died.

The portrait contest. In 2016 she won the National Portrait Gallery's portrait contest with a painting called *Miss Everything (Unsuppressed Deliverance)*. She was the first woman and the first African American to win it. Two years later, gallery officials unveiled her portrait of Michelle Obama. In 2018 she moved from Baltimore to New Jersey.

Breonna Taylor. Police officers in Louisville, Kentucky, forced their way into Breonna Taylor's apartment on March 13, 2020, with a warrant to search for drugs. Taylor was a Black woman. Her boyfriend fired one shot, the officers fired 32, and a shot by officer Myles Cosgrove killed her. No officer was charged with killing her. One officer, Brett Hankison, whose shots hit no one, was later convicted in federal court of using too much force.

For the September 2020 cover of *Vanity Fair*, Sherald painted Taylor's portrait. The Smithsonian's National Museum of African American History and Culture and the Speed Art Museum in Louisville bought the painting together.

The cancelled show. A large show of Sherald's work, *American Sublime*, hung at the San Francisco Museum of Modern Art and at the Whitney Museum of American Art in New York in 2024 and 2025. It was due at the National Portrait Gallery in September 2025.

Officials at the gallery, whom the reports do not name, raised "concerns" about one painting, *Trans Forming Liberty*. It shows a transgender woman, someone whose sense of being a woman is different from the sex written down at birth, in the pose of the Statue of Liberty. The officials talked about taking the painting out of the show.

Sherald cancelled the show on July 24, 2025. "I cannot in good conscience comply with a culture of censorship, especially when it targets vulnerable communities," she said. In a statement, Smithsonian officials said they had proposed adding a video to explain the painting. The Baltimore Museum of Art showed *American Sublime* instead, from November 2, 2025, to April 5, 2026.

Answering the Confederate statues

Delegates from seven Southern states that had left the United States met in Montgomery, Alabama, in February 1861 and set up the Confederate States of America, called the Confederacy. Its leaders then fought the United States in the Civil War to keep slavery.

Rumors of War. A show of the painter Kehinde Wiley's work opened at the Virginia Museum of Fine Arts in 2016. While he was there, he saw the Confederate statues on Monument Avenue in Richmond, Virginia. He then made *Rumors of War*, a bronze statue of a young Black man in jeans and sneakers riding a horse. The pose copies the statue of the Confederate general J. E. B. Stuart on Monument Avenue. It stands more than 27 feet tall.

It was first shown in Times Square in New York in September 2019. On December 10, 2019, it was unveiled outside the Virginia Museum of Fine Arts in Richmond, where it stays. Museum staff estimated the crowd at more than 5,000 people. Work by Wiley is also in the Freedom Monument Sculpture Park in Montgomery, Alabama, which opened on March 27, 2024.

Charlottesville. A bronze statue of the Confederate general Stonewall Jackson, made by Charles Keck in 1921, stood in Charlottesville, Virginia, for 100 years. At a rally in Charlottesville called Unite the Right, on August 12, 2017, James Alex Fields Jr. drove his car into people protesting against the rally. He killed Heather Heyer, who was 32. City workers took the Jackson statue down on July 10, 2021, along with a statue of Robert E. Lee.

The members of the City Council voted, all in favor, to give the Jackson statue to The Brick, an art gallery in Los Angeles. Kara Walker cut the bronze apart and joined the pieces into a new shape, which she finished in 2023. She called it *Unmanned Drone*. The Museum of Contemporary Art in Los Angeles bought it in February 2026. In October 2023 members of the Jefferson School African American Heritage Center melted down the Lee statue, in a project called Swords into Plowshares.

MONUMENTS. The Museum of Contemporary Art and The Brick showed ten statues that had been taken down from public places, most of them Confederate, in a show called *MONUMENTS*, from October 23, 2025, to May

3, 2026. New works by living artists stood beside them, and Kara Walker helped choose the works.

The oldest statue in the show was an 1887 figure of Roger B. Taney. As chief justice, the head judge of the Supreme Court, he wrote the Dred Scott decision of 1857, which held that Black Americans were not and could never be citizens. Hamza Walker, the director of The Brick, said the works "address the questions of who we want to be as a nation, and who and what is worth remembering, let alone celebrating."

Street murals after 2020

A Minneapolis police officer, Derek Chauvin, murdered George Floyd on May 25, 2020, by kneeling on his neck for more than nine minutes. In the weeks after the murder, people painted words and portraits on walls, on streets and on the plywood boards nailed over shop windows. In 2021 a jury convicted Chauvin of murder.

A team at the University of St. Thomas in St. Paul, Minnesota, led by the professors Heather Shirey, Todd Lawrence and Paul Lorah, began recording this art in early June 2020, from small stickers to large murals. Some of the painted boards were taken down, stored and later shown in exhibits.

Black Lives Matter Plaza. Washington's mayor, Muriel Bowser, had the words "Black Lives Matter" painted across two blocks of 16th Street in early June 2020, days after federal officers used tear gas on protesters in Lafayette Square near the White House. The yellow letters were 48 feet wide. She renamed the spot Black Lives Matter Plaza. Keyonna Jones, one of the artists who painted it, said, "We are saying it loud. We are here." According to NPR, the whole project cost more than $4 million.

Representative Andrew Clyde of Georgia introduced a bill in early March 2025 to cut federal money to the city unless it removed the words and renamed the spot Liberty Plaza. Bowser had earlier called the mural permanent. Now she announced it would go: "now we can't afford to be distracted by meaningless congressional interference." Workers of the District Department of Transportation began tearing it out on March 10, 2025.

BY 2025: Kehinde Wiley's *Rumors of War* stood in Richmond, Kara Walker had cut a Confederate general's statue apart, and workers were tearing out the painted words on 16th Street in Washington.

What museums hold, and what goes back

Officials of the Interior Department announced new rules for NAGPRA, the 1990 law on returning Native remains and sacred objects, on December 6, 2023. The rules took effect on January 12, 2024. A museum must now get the consent of the Native nation, or of the dead person's descendants, before it shows Native human remains or sacred objects. The same consent is needed before staff study them or let others get at them.

These remains and objects had been taken from Native graves and lands in earlier years, by men such as the diggers at Mound City in 1846 and at Spiro in the 1930s.

Halls closed. Museum leaders closed or covered Native displays in January 2024. Sean Decatur, president of the American Museum of Natural History in New York, closed its Eastern Woodlands and Great Plains halls on January 27, 2024, and covered seven more cases. He wrote to his staff that the halls came from "an era when museums such as ours did not respect the values, perspectives, and indeed shared humanity of Indigenous peoples."

About 500,000 children visit that museum each year on school or camp trips. In October 2023 museum officials had said that about 26 out of every 100 of the 12,000 human remains the museum held were Native American.

Staff at the Field Museum in Chicago, the Cleveland Museum of Art and the Metropolitan Museum of Art covered cases or took objects down. Harvard's Peabody Museum closed most of its floors on January 16, 2024, while its staff asked Native leaders which objects to remove, and they took down nearly 40. No one has published a count of every museum that did the same.

Fines. According to *The Art Newspaper*, from 1990 to early 2024 museums paid less than $60,000 in all in fines for breaking NAGPRA, and those fines came from 20 museums. Kelly Applegate, of the Mille Lacs Band of Ojibwe,

wrote to the Interior Department that museums had used the law's wording "to not give complete information about the entirety of their collections."

The count. The latest federal count comes from the National Park Service, in a report published on May 29, 2026. Since 1990, museums and federal agencies have reported holding the remains of 222,824 Native people. For 142,389 of them, officials have published a notice naming the nation or the family with the right to them. That notice is a legal step, and it does not mean the person's body has gone home.

As of September 30, 2025, the remains of 80,435 people had not yet gone through those legal steps. They were held by 381 museums and 104 federal offices. The report states that each year museums find and report remains they had never listed before. As of September 2026, the 2024 rule was still in force.

Dave's jars. Some returns fall outside NAGPRA. The Museum of Fine Arts in Boston gave the descendants of David Drake, the enslaved potter and poet, ownership of two jars he made in 1857, on October 29, 2025. The museum's officials said Drake "was not permitted to possess the jars and did not receive remuneration for them," which means he was not paid. The jars had been sold for the profit of his enslaver, Lewis J. Miles.

The family kept one jar and lent it back to the museum, and museum officials bought the other one from them. On that second jar, called the Poem Jar, Drake wrote: "I made this Jar = for cash / Though its called Lucre trash" (lucre is an old word for money).

Jeffrey Gibson

Who: A painter and sculptor, a member of the Mississippi Band of Choctaw Indians and of Cherokee descent, the first Native artist to represent the United States alone at the Venice Biennale.

When and where: Born March 31, 1972, in Colorado Springs, Colorado. Grew up in North Carolina, New Jersey, West Germany and South Korea. Lives near Hudson, New York.

In 2024 his work alone filled the United States pavilion in Venice.

Jeffrey Gibson was the only artist shown in the United States Pavilion at the Venice Biennale in 2024. The Biennale is an international art show held in Venice, Italy, where each country shows its artists in its own building, called a pavilion. According to SITE Santa Fe, one of the show's sponsors, he was the first Native artist to represent the United States there alone.

His parents' schools. Both of Gibson's parents had attended boarding schools as children. At the federal boarding schools for Native children, staff had for many years cut the children's hair, taken away their clothes and forbidden them to speak their own languages.

London. He studied painting at the School of the Art Institute of Chicago and at the Royal College of Art in London. The Mississippi Band of Choctaw Indians paid for his study in London. "My chief felt that me going there, being a strong artist, made him stronger," he said.

His paintings and sculptures use bright color, detailed patterns and written words. According to the Portland Art Museum, he mixes American, Native and queer history with music and popular culture, queer being a word many gay and lesbian people use for themselves. He won a MacArthur Fellowship in 2019.

> **Venice and after.** His Venice show ran from April 20 to November 24, 2024. He named it *the space in which to place me*, after a poem by the Oglala Lakota poet Layli Long Soldier. The same sponsor says it was also the first United States pavilion organized by a Native curator, a museum worker who chooses the art for a show and plans it. She was Kathleen Ash-Milby of the Portland Art Museum.
>
> Four large sculptures by Gibson, of a deer, a coyote, a squirrel and a hawk, stood on the front of the Metropolitan Museum of Art in New York from September 12, 2025, to June 9, 2026. On April 9, 2026, the Obama Foundation said it had asked him to make a wall of 17 round prints for the Obama Presidential Center.

Prices, and a picture that is only a file

The highest prices paid for paintings by American artists have reached hundreds of millions of dollars. A buyer paid $195 million at Christie's auction house in New York on May 9, 2022, for Andy Warhol's 1964 portrait of Marilyn Monroe, *Shot Sage Blue Marilyn*. The money went to a charity for children in Zurich, Switzerland.

The Associated Press reported that no work by an American artist had ever sold for more at auction, and as of September 2026 none had. Sixty years before that sale, an art dealer had bought all 32 of Warhol's soup-can paintings for $1,000.

Paintings by Basquiat and Pollock have also sold for more than $100 million. The Warhol beat the $110.5 million paid in 2017 for a 1982 painting of a skull by Jean-Michel Basquiat, the painter who had once written graffiti on Manhattan walls. In May 2026 a painting by Jackson Pollock, *Number 7A, 1948*, sold for $181.2 million, according to the *Robb Report*.

Paintings by women have sold for much less. Georgia O'Keeffe's *Jimson Weed/White Flower No. 1* sold for $44.4 million in 2014. According to the Associated Press, that was the most ever paid at auction for a work by a woman until November 20, 2025, when a painting by the Mexican painter Frida Kahlo sold for $54.7 million.

A picture that is only a file. A digital picture by Mike Winkelmann, who calls himself Beeple, sold at Christie's on March 11, 2021, for more than $69.3 million, according to Al Jazeera. It is a collage of 5,000 pictures he made, one each day, starting on May 1, 2007. The buyer, Vignesh Sundaresan, a programmer based in Singapore, paid in Ether, a kind of digital money.

What Sundaresan bought was a digital token, a record kept on a network of computers that names one person as the owner of a file. With it he got the right to show the picture. He did not get the copyright, the legal right to copy the work and sell copies.

Sundaresan and Beeple also both owned part of a different set of tokens, tied to other Beeple works. The price of those tokens rose to its highest during the sale and then fell sharply, and some people called the sale a publicity stunt.

Pictures made by computer programs

Anyone could type a few words into a website and get a picture back by the summer of 2022, as NPR reported.

The lawsuit. Kelly McKernan, Sarah Andersen, Karla Ortiz and other artists sued Stability AI, the company behind the program Stable Diffusion, along with Midjourney and the art website DeviantArt, in January 2023. Accounts give the date as January 12 or January 13.

In their lawsuit, the artists say the companies built these programs by feeding them huge collections of pictures gathered from the internet. They say the collection behind Stable Diffusion held more than five billion pictures, copied without asking the people who made them. These are the artists' claims, and no court has yet ruled on whether they are true.

Judge William Orrick in San Francisco let the artists' copyright claims go forward on August 12, 2024. As of September 2026, the case was still before him, and no trial had been held, according to the court's record of the case.

No human author. The computer scientist Stephen Thaler asked the US Copyright Office on November 3, 2018, to register a picture made by his computer program, with the program named as the only author. On August 12, 2019, officials refused, because the picture had no human author.

The officials relied in part on an 1884 Supreme Court case. In it the justices decided that a photograph of the writer Oscar Wilde counted as a work of art, because it came from the photographer's own ideas, his "intellectual invention." A picture with no person's ideas behind it, the officials reasoned, had no author who could own it.

Judges on a federal appeals court agreed with the refusal in 2025. On March 2, 2026, the justices of the Supreme Court declined to hear Thaler's appeal. As the law stood in 2026, according to the National Constitution Center, no one could hold the copyright to a picture made by a program with no human author.

Kelly McKernan

Who: An illustrator and painter, one of three artists who sued the makers of picture-making programs in January 2023.
When and where: Born 1986. Nashville, Tennessee.

People typed "in the style of Kelly McKernan" more than 12,000 times.

Kelly McKernan was one of three artists who sued the companies behind picture-making programs in January 2023. McKernan uses the words they and them. They live in Nashville, Tennessee, where they teach at the Nossi College of Art and Design. McKernan is a single parent.

Twenty years online. McKernan joined DeviantArt, a website where artists share their work, as a high school student in 2002. For about 20 years they posted paintings there, "not for money," and built a following. In January 2023 McKernan told NPR: "I almost barely make rent most months. I will spend, you know, 30 hours on a painting, and I won't see any money from that until it sells."

Their name on someone else's picture. People using picture programs had typed the words "in the style of Kelly McKernan" more than 12,000 times. Some people sold the results on websites such as

> Redbubble, with McKernan's name attached. "There's more and more images with my name attached to it that I can see my hand in, but it's not my work," McKernan said.
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> **In court.** Judge William Orrick dismissed most of the claims in the artists' case on October 30, 2023, and allowed them to rewrite them. Cleary Gottlieb, a law firm that represents Midjourney, wrote a report on the ruling. According to that report, he dismissed McKernan's copyright claims because McKernan had not registered those paintings with the Copyright Office. In 2023 *Time* magazine named McKernan to its list of the 100 most influential people in artificial intelligence. By September 2026 McKernan's name was no longer on the list of artists suing.

Books

More writers now publish their own books, without a publishing company. Every book for sale carries an identifying number called an ISBN, and in the United States a company called Bowker hands these numbers out. By Bowker's count, reported by the publishing writer Jane Friedman, writers who published their own books took out numbers for about one and a half times as many new books in 2021 as in 2018.

Spending on audiobooks grew too. Americans spent $2.43 billion on them in 2025, 9 percent more than in 2024, according to a survey for the Audio Publishers Association.

Books taken off the shelves. School and library officials removed or restricted thousands of books in the mid-2020s. Two groups count this, and their numbers differ because they count different things. PEN America, a group of writers that defends free expression, counts each time school officials remove or restrict a book. It counted 10,046 in the 2023-24 school year and 6,870 in 2024-25.

The American Library Association counts attempts it hears about to remove books from school, public and college libraries, listed by title. It counted 4,235 different titles in 2025, and it found that 92 out of every 100 of those attempts were started by "pressure groups, government officials and decision makers."

Which books, and why. The books PEN America found removed most often included *A Clockwork Orange*, a 1962 novel by Anthony Burgess, and *Sold*, a 2006 novel by Patricia McCormick about a girl sold into sexual slavery in India. PEN America says that books showing gay, lesbian or transgender people were often called "sexually explicit," among them *And Tango Makes Three*, a picture book about two male penguins at New York's Central Park Zoo who raise a chick together.

In July 2025 federal officials removed nearly 600 titles from the schools the Defense Department runs on military bases. According to PEN America, those books did not fit President Trump's orders on diversity, race and "gender ideology extremism."

Wilder's name. Laura Ingalls Wilder's Little House books, about her family's life as settlers on the frontier, came out from 1932 to 1943. The board of the Association for Library Service to Children took her name off its award for children's books on June 23, 2018. The award had carried her name since 1954.

The board members wrote that her books include "expressions of stereotypical attitudes inconsistent with ALSC's core values," a stereotype being an unfair, fixed idea about a whole group of people. They acted after critics pointed to how her books show Native and Black people, and after a group that included the American Indian Library Association reviewed the award.

Wilder described one place in her 1935 book *Little House on the Prairie* as somewhere "there were no people. Only Indians lived there." In that line, the Native people who lived there do not count as people. Later printings changed the line. Several characters in the Little House books say that "the only good Indian is a dead Indian." The prize is now called the Children's Literature Legacy Award.

Audubon's name. The members of the board of the National Audubon Society voted in March 2023 to keep the name of John James Audubon, who painted *The Birds of America* and who had enslaved people. "The name has come to represent not one person, but a broader love of birds and nature," said

the board chair, Susan Bell. Three board members resigned. The leaders of the local Audubon groups in Chicago, Seattle, Portland and Washington, D.C., had already decided to drop the name.

Public money and national museums, 2025 and 2026

White House officials sent Congress a budget for 2026 on May 2, 2025, that would have shut down the National Endowment for the Arts. The same day, NEA staff emailed arts groups across the country that their grants were withdrawn or ended because the projects fell outside the President's priorities. On May 5 about a dozen senior NEA staff members resigned.

Members of Congress did not close the agency. They gave it $207 million for 2026, the same amount as in 2024 and 2025, according to the Congressional Research Service. On April 3, 2026, White House officials proposed shutting the NEA down again, for 2027. As of September 2026, members of Congress had not decided.

The Smithsonian. President Donald Trump signed an order titled "Restoring Truth and Sanity to American History" on March 27, 2025. It stated that the Smithsonian Institution had "come under the influence of a divisive, race-centered ideology."

Trump announced on May 30 that he had fired Kim Sajet, who had directed the National Portrait Gallery for 12 years. He called her a "highly partisan person." He named no law under which he could fire her, and she kept coming to work. The members of the Smithsonian's Board of Regents, the group that governs it, then said that all decisions about staff belonged to the Smithsonian's head, Lonnie Bunch. Sajet resigned on June 13, 2025. She told her staff that her presence had become a distraction from the Smithsonian's mission.

Three White House officials, Lindsey Halligan, Vince Haley and Russell Vought, wrote to Bunch on August 12, 2025. They asked for a review of eight Smithsonian museums, among them the American Art Museum and the Portrait Gallery. They told the museums' leaders to begin changing their exhibits within 120 days. On July 24, 2026, Trump signed another order about the Smithsonian, aimed at the National Museum of American History.

BY TODAY

If you remember one thing: Since 2000 artists have sued over programs built from their pictures, museums have needed Native consent to show remains, and officials have pushed national museums to change what they show.

- January 12, 2024: a federal rule takes effect that requires a Native nation's consent before museums show its ancestors' remains or sacred objects.
- July 24, 2025: Amy Sherald cancels her show at the National Portrait Gallery.
- October 29, 2025: the Museum of Fine Arts in Boston gives Dave's descendants ownership of two of his jars.

THE THREE QUESTIONS, ANSWERED

1. **How did enslaved artists make art, and who got paid for it?** They painted, cut silhouettes, turned pots and wrote poems while other people held them as property, and the people who held them got the money. Dave's jars were sold for the profit of Lewis J. Miles, and Moses Williams learned to cut silhouettes and build exhibits in Peale's museum while Peale held him enslaved. Henry Barnes took Prince Demah to London for painting lessons, and Dave cut verses into his jars in years when it was a crime for him to learn to read.
2. **Who decided what counted as art, and who got to see it?** Mostly people with money or power: the Met's trustees, who kept it shut on Sundays until 1891, a teacher in Santa Fe who taught one "authentic" Native style, and a congressman who called a touring show of modern paintings trash. Artists pushed back, from Oscar Howe's letter in 1958 to Amy Sherald cancelling her show in 2025.
3. **Why are museums giving back things they once took?** Diggers took objects and human remains out of Native graves, at

Mound City in 1846 and at Spiro in the 1930s, and sold many of the objects. Under a 1990 law, NAGPRA, museums that take federal money must return Native remains and sacred objects, and since 2024 they need a nation's consent even to show them.

What changed: The chapter begins with potters and carvers whose names no record keeps and ends with artists suing over computer programs that copy their styles and attach their names.

FIVE DATES TO KEEP

1585: John White paints the town of Secotan, on the coast of what is now North Carolina.

1773: Phillis Wheatley's *Poems* comes out in London while she is enslaved in Boston.

1857: Dave cuts "I made this Jar = for cash" into the jar now called the Poem Jar.

1936: Dorothea Lange photographs Florence Owens Thompson at a pea pickers' camp in Nipomo, California.

November 16, 1990: President George H. W. Bush signs NAGPRA.

NOT KNOWN: No surviving record names the people who made the painted bowls, stone pipes and copper plates of this land before 1500.

Where to go: The Museum of Fine Arts in Boston holds the two jars Dave made in 1857, one bought from his descendants and one lent by them.

Next: Chapter 33, Music. It asks the same questions about songs: who made them, where they came from, and who got paid.

Chapter 33: Music

CHAPTER 33 OF 37 · MUSIC · 10 ERAS · ABOUT 135 MINUTES

For most of American history, the people who made the music were often not the people who got paid for it.

Willie Mae "Big Mama" Thornton, a blues singer from Alabama, recorded "Hound Dog" on August 13, 1952, and her record went to No. 1 on the rhythm and blues chart. Don Robey, who owned her record company, put his own name on the copyright form as one of the song's writers. Each writer named on that form has a right to part of the money the song earns.

Four years later Elvis Presley recorded the same song and sold about 10 million copies. "I got one check for $500 and never saw another," Thornton said. She died poor in a Los Angeles boarding house in 1984.

This chapter follows American music from wooden flutes made in the 600s to songs streamed over the internet. In every era, look first at who made the music: Native singers, enslaved fiddlers, New England singing teachers, blues singers in Mississippi, a teenage DJ in the Bronx. From the 1770s on, the records show who was paid, so look at that too.

Again and again the money and the credit went to someone else: a white stage performer who copied a Black man's song, a publisher who put his own name on a song, a company that paid a set fee and kept the rest. Watch, too, for the people who tried to stop music, from lawmakers who banned drums to officials who banned Native dances.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Where did the banjo and the blues come from?

2. Why did so many musicians make hit songs and still die poor?
3. Who tried to stop people from playing their own music, and why?

Five minutes? Read "Drums at Stono, and the 1740 law against drums and horns" in 1700 to 1750, then "Records for Black buyers, and who was paid, 1920 to 1938" in 1900 to 1950.

Can skip: "Jenny Lind's tour, 1850 to 1852", "Concerts for a ticket, and a fiddle for a prize", "The Kennedy Center, 2025 and 2026". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 6 MINUTES

Native nations in this land made music for nearly every part of life, and they passed their songs on by teaching them. Their songs were for ceremonies, healing, work, games, courting a sweetheart, storytelling, hunting, farming, war and social dances. None of the nations north of Mexico had a way of writing tunes down, so no singer's name and no song's name from these centuries survives in writing.

Some of the instruments survive. Archaeologists, scientists who dig up and study what people left behind, have found wooden flutes and metal-covered panpipes more than a thousand years old. Onondaga and Tlingit singers today still follow rules about where a song may be sung and who owns it.

How old are the oldest flutes?

The oldest known wooden flutes in North America were made between the years 620 and 670. Ancestral Pueblo people, the ancestors of today's Pueblo peoples, made them. The archaeologist Earl H. Morris found them in 1931, when he explored 15 caves in the Prayer Rock district of northeastern Arizona, in what is now the Navajo Nation. In the largest cave he found four wooden flutes. That cave is now called Broken Flute Cave.

The flutes. Each one is a tube of box elder wood between about 68 and 74 centimeters long, a little over two feet. The finger holes are almost the same on each flute, so the maker planned which notes the flute would play. Red, blue and black feathers were tied to one end. Two of the flutes lay broken in trash above a grave. The other two lay whole under a dirt floor, tied together. The Arizona State Museum in Tucson holds them today.

The panpipes. Farther east, people of the Hopewell culture lived in the woodlands of eastern North America between the years 1 and 400. "Hopewell" is a name archaeologists gave them, not the name of a nation. In southern Ohio they built earthworks, mounds and long banks of piled earth. A Hopewell panpipe is a row of three or four tubes of cane, reed or bone, held together by a metal jacket. A player blows across the tubes, and each tube makes a different note.

Almost every Hopewell panpipe that archaeologists have found was buried with the dead. They have found 105, from Wisconsin and Ontario down to the Gulf Coast, and very few came from places where people lived. Most of the metal jackets are copper. The rest are silver, copper covered with silver, or iron.

NOT KNOWN: Nobody knows what tunes anyone played on these flutes and panpipes.

Drums, rattles, and songs that belong to someone

Onondaga singers today play a water drum and a horn rattle. The Onondaga Nation, in what is now central New York, is one of the Haudenosaunee nations. The water drum is usually made of cedar, with deer hide stretched over the top. The horn rattle is a dried, hollowed animal horn with seeds inside. A rattle made of turtle shell is used only in a few ceremonies.

The Onondaga keep two kinds of songs. Social songs can be sung anywhere. Songs for ceremonies are sung only in the longhouse, the building where ceremonies are held, and the public may not watch them.

Songs a clan owns. Among the Tlingit of what is now southeast Alaska, a song can belong to a clan, a large family group that traces itself back to shared ancestors. A clan owns its songs, its stories, its crests and the clothing its members wear in ceremonies. The Tlingit word for these owned things is *at.óow*, "something owned or purchased."

Some songs and stories tell of a clan's losses in a certain place. In the Tlingit view, those songs are the clan's deed to that place, the proof that the place is theirs.

KNOWN:	These rules come from the nations' own accounts today.
NOT KNOWN:	No written record from before 1500 describes them.

R. Carlos Nakai

Who: A flute player from a Navajo (Diné) and Ute family, whose records were the first of the traditional Native flute to sell 500,000 copies.

When and where: Born April 16, 1946, in Flagstaff, Arizona. Still performing in 2026.

A car crash ruined his trumpet playing, and someone gave him a flute.

R. Carlos Nakai first studied the trumpet. A car accident ruined the way he shaped his lips against the trumpet's mouthpiece. Someone gave him a cedar flute as a gift and challenged him to master it. His family is Navajo and Ute, and his father, Raymond Nakai, was chairman of the Navajo Nation from 1963 to 1970.

The records. Nakai has recorded more than forty albums with Canyon Records in Phoenix, Arizona, starting with *Changes* in 1983. His albums *Canyon Trilogy* and *Earth Spirit* each sold 500,000 copies, and each earned a gold album award from the RIAA, the music business group that gives sales awards. They were the first recordings of the traditional Native flute to sell that many. *Canyon Trilogy* kept selling. In 2014 the RIAA named it a platinum album, its award for one million copies.

He was nominated for ten Grammy awards between 1994 and 2013. Nakai calls himself "an iconoclastic traditionalist," someone who breaks with old ways and keeps them at the same time. His concert calendar for 2026 lists shows in Utah and Colorado in September and more in October and November.

Why this story is here: Flutes survive from before 1500, but no flute player's name does, so this era's story belongs to a living player of the traditional cedar flute.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 4 MINUTES

The first European music on record in what is now the United States is church singing and soldiers' trumpets, and Native people watched and listened. It comes from three places: the coast of Florida, the coast of California, and New Mexico from 1598. Spanish priests sang Latin hymns, songs that praise God, when they came ashore and when they held services. French and English Protestants, Christians who had broken away from the Catholic Church, sang psalms, songs from the Bible.

Europeans wrote down a few of the Native listeners' reactions. No record from the 1500s names a mission choir or a Native singer trained in church music. The first ones on record come after 1598.

Hymns sung at the landings, Florida, 1564 and 1565

French Protestants called Huguenots built Fort Caroline in 1564, near the mouth of the St. Johns River and today's Jacksonville, Florida. They built it on the land of the Timucua people. Their leader, René de Laudonnière, wrote that on June 30 he ordered his men to sing aloud on the beach to thank God for their safe arrival. The historian Francis Parkman wrote in 1865 that on the morning they began the fort, the men sang a psalm before they started work.

St. Augustine. The Spanish commander Pedro Menéndez de Avilés landed at Seloy, a Timucua town, on September 8, 1565, and founded St. Augustine. His chaplain, the priest who served his soldiers, was Francisco López de Mendoza Grajales, and the chaplain wrote an account of that day. He wrote that Menéndez came ashore "to the sound of trumpets and salutes of artillery," meaning cannons fired as a greeting.

The chaplain walked to meet Menéndez carrying a cross and singing the *Te Deum*, a Latin hymn of praise to God. "A large number of Indians watched these proceedings and imitated all they saw done," the chaplain wrote.

The killings. Menéndez's soldiers captured Fort Caroline on September 20, 1565, and killed most of the Frenchmen there. In a letter of October 15, Menéndez wrote that 132 of them had "their throats cut." He gave his reason in

the same letter: "they spread the Lutheran sect," meaning the Protestant faith. When Menéndez came back to St. Augustine, the chaplain and his companions walked ahead of him in a line, "singing the Te Deum laudamus."

In the weeks after, shipwrecked Frenchmen surrendered to Menéndez at a place called Matanzas. His soldiers killed 111 of them. On October 12 they killed 134 more who had surrendered with their leader, Jean Ribault.

Psalms at Drake's landing, California, 1579

The English captain Francis Drake stopped on the California coast in June 1579 to repair his ship, probably at the place now called Drake's Bay. Coast Miwok people lived along that shore. Drake's men stayed for about five weeks. They held prayers, sang psalms and read from the Bible, and Miwok people came to watch.

"Gnaah." A book about the voyage, *The World Encompassed by Sir Francis Drake*, states that the Miwok "tooke such pleasure in our singing of Psalmes" that each time they came, "their first request was Gnaah." That word meant they wanted the Englishmen to sing. The anthropologist Robert Heizer, a scientist who studied people and the ways they live, wrote in 1947 that *gnaah* seems to be a Coast Miwok word.

Who wrote the book. It came out in London in 1628. Drake's nephew, also named Francis Drake, put it together. Its title page states that it was collected "out of the Notes of Master Francis Fletcher," the preacher on the voyage, and checked against other notes made on the same voyage. Heizer wrote that who wrote it is uncertain.

New Mexico, 1598

Juan de Oñate led Spanish colonists north into New Mexico in 1598 to start a colony on the lands of the Pueblo peoples. Pueblo people then began to hear Spanish military music and to sing Catholic church songs, some of them in parts, with different voices singing different notes at the same time. The music historian John Koegel dates this to "the Oñate expedition and its aftermath," meaning the journey and the years after it.

The earliest music teachers there that Koegel names are two friars, Catholic men who lived by the rules of a religious order and put the title Fray in front of their names. Fray Bernardo de Marta came in about 1605. Fray Cristóbal de Quiñones died in 1609.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 5 MINUTES

Colonists in the 1600s brought their church singing with them, and in New Mexico Spanish friars and officials punished Pueblo people for keeping their own ceremonies. In New England, congregations, the people who worship together at a church, sang psalms from a book printed in Massachusetts in 1640. For most of the century each printing of that book held only the words, with no musical notes.

In New Mexico, the friar Cristóbal de Quiñones put an organ in the church at San Felipe Pueblo and trained Pueblo singers. Later friars banned Pueblo religious ceremonies, and a Spanish governor had 47 Pueblo religious leaders arrested in 1675. Three were hanged. In Virginia, the crew of an English ship landed about twenty captured Africans in 1619. Enslaved Africans made music there through the century, but nobody who wrote in those years left a description of it that survives.

Singing without notes: the Bay Psalm Book and lining out

Stephen Daye printed *The Whole Booke of Psalmes Faithfully Translated into English Metre* at Cambridge, Massachusetts, in 1640. People today call it the *Bay Psalm Book*, and it was the first book printed in British North America. Richard Mather and other New England ministers turned the Psalms of the Bible into English verse with a steady beat, so that a congregation could sing them. Eleven copies of that first printing survive.

No notes. The 1640 book had no printed music. It lists tunes by name, tunes already printed in English psalm books. The ninth edition, printed in 1698, was the first to include music: 13 tunes taken from a music book by an Englishman, John Playford.

Lining out. A gathering of ministers at Westminster, in London, wrote a guide for church services in 1644. Where many people in a congregation could not read, the guide says, the minister or a person he chose should "read the psalm, line by line, before the singing thereof."

This way of singing is called lining out. A leader read or sang one line, and the congregation sang it back. Colonists from England and Scotland brought lining out to America, and members of Old Regular Baptist churches in the United States still sing this way today.

Fray Cristóbal de Quiñones

Who: A Franciscan friar who built the church at San Felipe Pueblo, put an organ in it, and trained Pueblo singers.

When and where: San Felipe Pueblo, a Keres-speaking town in New Mexico. Died in 1609.

He put an organ in a Pueblo church and trained its singers.

Fray Cristóbal de Quiñones belonged to the Franciscans, a Catholic religious order. He was the leader of the Franciscan friars in New Mexico. San Felipe is a Pueblo town whose people speak the Keres language. Quiñones learned Keres and baptized many Native people in the New Mexico missions, making them members of the Christian church in a ceremony with water. At San Felipe he built the church, a house for the friars, a pharmacy and a place to care for the sick.

What one writer recorded. Agustín de Vetancurt, a Franciscan writer in Mexico, described his work in a book of the lives of Franciscan friars, printed in the 1690s. Vetancurt wrote that Quiñones "installed an organ to enrich the Divine Service," meaning the church service, "and encouraged music-making by the skilled singers" he had trained.

Quiñones died in 1609. The music historian John Koegel calls him "perhaps the first trained teacher of European music" to live in what is now the US Southwest. Koegel gives neither his birthplace nor the year he came to New Mexico.

NOT KNOWN: Vetancurt tells only the friar's side, and his account does not record what the singers of San Felipe thought of the music they were taught.

Church music and banned ceremonies in New Mexico

Other friars taught music in more Pueblo towns. Fray Bernardo de Marta came to New Mexico in about 1605, taught Pueblo people in many towns to sing and to play European instruments, and died in 1635. By 1634 Fray Roque de Figueredo was teaching Zuni boys to sing in parts.

The ban. A Spanish church leader and a Spanish governor punished Pueblo people for keeping their own religion. The Franciscan leader Alonso de Posada forbade Pueblo ceremonies from 1656 to 1665. He ordered friars to invade the sacred kivas, the rooms where Pueblo people held their ceremonies. There the friars burned the masks and took or destroyed sacred objects.

The arrests. The governor, Juan Francisco Treviño, had 47 Pueblo religious leaders arrested in 1675 and charged with sorcery, using magic to do harm. Four were sentenced to be hanged. Three were hanged, and one killed himself. The rest were shamed and whipped in public, then put in prison. The records do not name the men who passed the sentences or carried out the hangings and the whippings. Pueblo people went to Santa Fe and demanded that the prisoners be freed, and Spanish officials let them go.

Po'pay. One of the whipped men was Po'pay, a leader from Ohkay Owingeh. On August 10, 1680, Po'pay led the Pueblo nations as they rose up together against the Spanish. The Pueblo fighters killed about 400 Spanish settlers, 21 of them Franciscan friars, and drove the Spanish out of New Mexico for 12 years.

What arrived with enslaved Africans

The crew of an English ship called the *White Lion* landed "20. and odd Negroes" at Point Comfort, Virginia, in late August 1619. Those are the words of the colonist John Rolfe, and "Negroes" was a word of that time for Black people. The crew had taken these Africans from a Portuguese slave ship. They came from the Ndongo and Kongo region of West Central Africa.

Enslaved Africans made music in the English colonies through the 1600s, but no one who wrote in those years left a description of it that survives. The first clear description of an early banjo comes from the island of Jamaica. An English doctor there, Hans Sloane, wrote in 1687 about an instrument he called a "strum strump."

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 9 MINUTES

Two fights over music ran through these fifty years. In New England, church members argued about singing from written notes, and in South Carolina, lawmakers took drums away from enslaved people. New England ministers began teaching their congregations to read notes around 1720. They started singing schools and printed the first books of music lessons in the colonies.

Boston had a concert for a paying audience by 1731, and a ticket cost five shillings. In Virginia, enslaved Black musicians played the fiddle. Three of them appear in notices their enslavers printed when the men ran away in 1745 and 1746. Enslaved men rose up near the Stono River in South Carolina in 1739, with drums beating. The next year South Carolina's lawmakers banned enslaved people from using drums and horns, and fined any master who let them.

Singing by note and the first singing schools

New England churches still sang by lining out at the start of the 1700s. "By the end of the seventeenth century singing had sunk to a low level in all the colonies," the historian Henry Wilder Foote wrote. In the first 25 years of the 1700s, ministers in New England turned against lining out. They called for "regular singing" instead, which meant singing from written notes.

The Reverend Thomas Symmes of Bradford, Massachusetts, made the case in a booklet of 1720, *The Reasonableness of Regular Singing, or Singing by Note*. He wrote that church singing had got worse because "Singing schools and singing books being laid aside, there was no way to learn." New England had had singing schools in its early days, Symmes wrote, but people had stopped holding them.

The first music books. The Reverend John Tufts of Newbury, Massachusetts, published *An Introduction to the Singing of Psalm Tunes*, the first book of music lessons printed in the English colonies.

No copy of the first printing is known, so its exact title and year are uncertain. The dates given for it run from about 1712 to 1721. Tufts traveled to give talks about singing and to set up singing schools, where a teacher taught a class to read and sing music.

Thomas Walter, a 25-year-old nephew of the Boston minister Cotton Mather, published *The Grounds and Rules of Musick Explained* in 1721, with 24 psalm tunes. James Franklin printed it in Boston. It was the first music in the colonies printed with bar lines, the lines that divide music into equal parts of time. It stayed a favorite for more than forty years and went through six editions.

New singing schools spread in the 1720s. The records that survive do not say where the first one met or what a pupil paid.

The young singers. Young people took up the new singing first. The Boston judge Samuel Sewall wrote in his diary on March 20, 1721, that "Dr. Mather" had preached that night "in the school house to the young musicians." The house was full, Sewall wrote, "and the singing extraordinarily excellent, such as hardly been heard before in Boston."

The fight. Other church members fought the change. In Symmes's town, he reported, "a great part of the town has for near half a year, been in a mere flame about it." He meant that people had been angry and arguing about it for almost six months.

The Reverend Nathaniel Chauncey of Connecticut wrote answers in 1728 to the objections people raised. One objection was that singing by note "leads to the Church of England and will bring in organs quickly." The Church of England was the official church back in England.

The churches in this fight had been started by Puritans, English Protestants who wanted to clear the Church of England of what they saw as leftovers of the Catholic Church. Puritan congregations sang with no instruments at all, because instruments looked to them like Catholic worship. Another objection was only about who liked the change: the new way must be wrong "because the young people fall in with it."

Concerts for a ticket, and a fiddle for a prize

A "Concert of Music" on several instruments was held in Mr. Pelham's great room in Boston on December 30, 1731. An advertisement in the *Boston News-Letter* that month listed it, and a ticket cost five shillings, a quarter of a British pound. It is the earliest advertised concert in Boston that the music historian Oscar Sonneck found. After a concert at Faneuil Hall in 1744, the town's selectmen, the men elected to run the town, received 205 pounds for the poor of Boston.

A fiddle for a prize. A notice in the *Virginia Gazette* of November 26, 1736, listed prizes for a holiday gathering at Page's Warehouse in Hanover County, Virginia, on November 30. One prize was "A fine Cremona Fiddle to be plaid for, by any number of Country Fiddlers." A Cremona fiddle was a violin made in the town of Cremona, in Italy. The races at John Pinkerton's Old Field in Virginia in November 1737 included a fiddling contest.

Fiddles, banjos, and the enslaved musicians who played them

The banjo has West African roots. A 2014 article on the Oxford University Press blog counts more than 60 plucked instruments in West Africa that are likely forerunners of it. The first known mention of a banjo in the mainland colonies is in John Peter Zenger's newspaper, the *New-York Weekly Journal*, in 1736.

Runaway notices. Virginia enslavers put notices in the *Virginia Gazette* when an enslaved person ran away. Each notice described the person's body, clothes and skills, so that strangers could recognize and catch the person. Playing the fiddle was one of the skills these notices listed. The University of Virginia's collection of Virginia runaway notices holds three from before 1751 that name a musician: two men named Tom, and a fiddler named Harry.

Robert Whitfield of Isle of Wight County offered four pistoles, Spanish gold coins used in the colonies, for a young man named Tom, who "plays very well on the Violin." His notice ran in the issue of November 28 to December 5, 1745. In April 1746 John Thornton of Fredericksburg advertised for another man named Tom, about 46 years old, who "plays on the Violin, and is a Sawyer," a man who saws logs into boards.

Harry, a fiddler who ran away, perhaps to his wife, 1746

Who: An enslaved man who played the fiddle. The man who held him put a notice in the newspaper for his capture.

When and where: Near Jamestown, Virginia, March 1746.

He wore a blue greatcoat. His enslaver wanted him whipped.

Harry was an enslaved fiddler who ran away in early March 1746 from the man who held him, near Jamestown, Virginia. Everything known about him comes from one notice in the *Virginia Gazette*, printed between March 20 and 27, 1746. The notice gives him no last name.

Harry was about 35 years old and about 5 feet 7 inches tall, with small eyes and "a very large Beard," and he "plays upon the Fiddle." He left wearing a double-breasted dark cloth coat, two dark cotton jackets, a blue greatcoat and buckskin breeches, which are pants that end at the knee. He had once belonged to Colonel Grymes of Richmond County, Virginia.

The notice. William Newgent signed it, though a later reprint spells his first name James. Newgent guessed that Harry had gone to Richmond County, "where he has a Wife." He offered one pistole for Harry's return. At the end he added: "As he ran away without any Cause, I desire he may be punish'd by Whipping, as the Law directs." Newgent was asking whoever caught Harry to have him beaten with a whip.

NOT KNOWN: The record does not say whether Harry was caught.

Drums at Stono, and the 1740 law against drums and horns

About 20 enslaved men seized weapons near the Stono River, south of Charleston, South Carolina, on September 9, 1739. A report written that year called them "Angola Negroes" and named their leader as a man called Jemmy. They marched south toward Spanish Florida.

More men joined them, "calling out Liberty," the report says, and they "marched on with Colours displayed," meaning flags, "and two Drums beating." When there were sixty to a hundred of them, they stopped in a field and "set to dancing, Singing and beating Drums, to draw more Negroes to them."

The killings. The same report says the rebels killed white men, women and children as they marched. Lieutenant Governor William Bull raised the militia, a force of armed planters, and the militiamen attacked the rebels in the field the same day. The report says the men captured in the field were questioned and then "shot on the Spot," and men who had run back to their plantations were caught there and shot. It does not name who fired.

The report counts about 40 Black people and 20 white people killed. Later counts give about 25 white colonists and 35 to 50 rebels killed. Colonists cut off the heads of rebels and set them on stakes along the main roads as a warning. They put most of the captured rebels to death and sold the survivors to the West Indies.

The law. The men of the South Carolina assembly passed the Negro Act on May 10, 1740. In section 36 they wrote that it was "absolutely necessary to the safety of this Province," meaning the colony, to stop enslaved people from "using or keeping of drums, horns, or other loud instruments."

They gave the reason in the same sentence: such instruments "may call together or give sign or notice to one another of their wicked designs and purposes." They meant that drums and horns could be used to call enslaved people together to rise up.

What the law allowed. A master or an overseer, the man who ran the work on a plantation, paid a fine of ten pounds each time he let enslaved people beat drums, blow horns or gather in public. The lawmakers wrote that enslaved people had to be kept from wandering and meeting at all times, and most of all on Saturday nights, Sundays and holidays. Any master, overseer or other person could seize an enslaved person found off the plantation without a written pass and with no white person along.

If an enslaved man found off the plantation carried a weapon, any master, overseer or other person could take the weapon away, seize him and whip him, even if he carried a pass.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 10 MINUTES

Americans began writing and printing their own music in these years. The names of the Black musicians who played beside them survive mainly in the notices enslavers printed when those musicians ran away. A Boston tanner named William Billings published a book of church songs he had written himself in 1770. By 1800 Americans had published about forty books of church music. During the Revolution, officers used drum beats to tell soldiers when to wake, march and fight, and American writers put new words to old tunes.

Black musicians played fiddles, horns, drums and banjos in Virginia, in Charleston and in New England. Slave traders took one of them, Newport Gardner, from Africa as a boy. He became a music teacher in Newport, Rhode Island, and bought his freedom in 1791. That same year, on the Pacific coast, a man sailing with a Spanish expedition wrote down Tlingit songs in musical notes.

William Billings and the New England singing masters

William Billings, a Boston tanner who turned animal skins into leather for a living, published *The New-England Psalm-Singer* in 1770. A Library of Congress research guide calls it "the first publication consisting of sacred music," meaning music for worship, "composed entirely by a native-born American." Paul Revere is credited with engraving the picture at the front of the book, cutting it into a metal plate so it could be printed.

Billings also taught singing schools. He opened one in Boston with John Barry in 1769, and he taught another in Stoughton, Massachusetts, in 1774. The historian Henry Wilder Foote counted about forty American books of church music published before 1800, most of them printed in Boston.

William Billings

Who: A tanner and singing teacher who wrote about 340 songs for choirs.

When and where: Boston. Born October 7, 1746. Died September 26, 1800.

A tanner with one blind eye wrote 340 songs and died poor.

William Billings wrote about 340 songs for choirs and published them in six books. He did it while working as a tanner and a singing teacher. His father died when Billings was 14, and he went to work learning to tan leather. He was mostly self-taught in music, though he may have had some lessons from John Barry of Boston's New South Church. A writer named Nathaniel Gould described him in 1853 as having "one eye blind, one leg shorter than the other, one arm somewhat withered."

"Chester." His 1770 book included his song "Chester." It opens with the words "Let tyrants shake their iron rod." The singers say that they do not fear tyrants or the chains of slavery, because they trust in God. Billings printed a new version with five verses in *The Singing Master's Assistant* in 1778. Soldiers "learned the words by heart," according to the writer David Stowe, and carried Billings's songs wherever the army sent them.

No copyright. Other printers copied Billings's tunes without paying him. Before the Revolution, the men who made Massachusetts's laws voted for a bill, a proposed law, to give Billings copyright over his first book: the legal right to decide who may copy and sell it. The governor, a Tory loyal to the British king, refused to sign it. Stowe writes that the governor may have refused because Billings had ties to patriots such as Samuel Adams.

Poor at the end. His friends paid to print his book *The Continental Harmony* in 1794. His wife, Lucy, died on March 26, 1795, and left him with six children under 18. In the 1790s he worked as a street cleaner. Billings died poor in Boston on September 26, 1800, and was buried in an unmarked grave on Boston Common.

Songs and signals of the Revolution

British officers first sang "Yankee Doodle" to make fun of the colonists. The man most often named as the writer of the words is Richard Shuckburgh, a British army surgeon, in about 1755 or 1758, during the French and Indian War. The date is disputed.

British soldiers arrived to occupy Boston in 1768, and their musicians played the song in the streets to mock the colonists. Early in the war, British troops and Loyalists, colonists who stayed on the British side, played it to make fun of the Continental soldiers. The Americans then made up their own verses, and by 1781 "Yankee Doodle" was a favorite American song.

"The Liberty Song." John Dickinson of Pennsylvania wrote its words to the tune of "Heart of Oak," the anthem, or official song, of the British Royal Navy. The *Boston Gazette* printed it in July 1768. Dickinson sent it to his Massachusetts friend James Otis with a note that "indifferent songs are very powerful on certain occasions." By indifferent he meant songs that were not very good.

Orders by drum. In the Continental Army, officers gave orders by drum beat. Under the army rules of Baron von Steuben, each beat had its own name and meaning. "The Reveille" woke the soldiers in the morning. "The General"

meant take down the tents and get ready to march, "The Assembly" meant line up at the flags, and "To Arms" meant pick up weapons and get ready to fight.

Drummers beat "The Tattoo" at 9 or 10 at night, and at that signal tavern keepers stopped serving the soldiers drinks. Boys under 14 were recommended as drummers, according to the American Battlefield Trust, and some boys who played the fife, a small, high-pitched flute, were 9 or 10 years old.

Black musicians, enslaved and free

Virginia enslavers kept naming enslaved musicians in their runaway notices. William Gregory advertised on May 4, 1769, for Peter, about 44, who had taken with him "a fiddle, which he is much delighted in when he gets any strong drink." David Walker of Dinwiddie County advertised on September 8, 1775, for Jemmy. He wrote that Jemmy "is very fond of singing hymns and preaching," and that he was living near Williamsburg under the name James Williams.

Mark. William Allason offered forty pounds in August 1780 for four men who had run away together from near Fauquier Court House, Virginia. One of them was Mark, about 25, born on the island of St. Jago, in the Cape Verde Islands off West Africa. He spoke Portuguese. The notice says Mark "blows the French horn, can play on the fiddle, whistles many tunes well," and that he loved marches and Catholic church music. Allason guessed that the four men planned to get aboard a ship.

At Nomini Hall. Philip Vickers Fithian was a tutor, a live-in teacher, to the children of Robert Carter III at Nomini Hall, a Virginia plantation, in 1774. His journal records that on a January evening, enslaved people gathered in the schoolroom "& began to play the Fiddle, & dance." Two of Carter's sons, Ben and Harry, joined them, and Fithian sent everyone away. A few days later, he wrote, "several Negroes & Ben, & Harry are playing on a Banjo & dancing."

In Charleston. A gentleman lost his pocketbook on the evening of December 2, 1766, as he passed "a croud of negroes assembled at the lower end of Elliott-street, with a banjer playing." A notice in the *South Carolina Gazette and General Advertiser* of December 12 reported the loss. A Charleston town rule of 1750 banned beating drums in town without an officer's permission.

The earliest banjo painting. A painting on paper called *The Old Plantation*, made between about 1785 and 1795, shows enslaved Black men and women dancing. One man plays an early banjo made from a gourd, a large fruit with a hard shell that can be dried and hollowed out. Another man plays a drum, and two women hold rattles. It is the earliest known painting of a banjo.

The librarian Susan Shames identified the painter in 2010 as John Rose, a South Carolina planter who held people in slavery. The Abby Aldrich Rockefeller Folk Art Museum in Williamsburg, Virginia, owns it.

Jefferson. Thomas Jefferson held people in slavery at Monticello, his plantation in Virginia. In the 1787 edition of his book *Notes on the State of Virginia*, he wrote about Black musicians: "In music they are more generally gifted than the whites." Then he doubted that they could write longer, more complicated music: that, he wrote, "is yet to be proved."

In a note he called the "Banjar" the instrument "which they brought hither," meaning here, "from Africa." In the same passage he wrote that Black people had "no poetry," and he dismissed the published poems of Phillis Wheatley, a Black woman.

Election days. Black communities in New England held their own election days from about 1750. In New Hampshire and Massachusetts they elected a "king," and in Rhode Island and Connecticut a "governor." The enslavers granted the day off and paid for the feast. The parades had "tambourines, banjos, fiddles and drums," and the feast had rum, gingerbread and "thick, fruit-studded election cakes."

Newport Gardner

Who: Born Occramer Marycoo in West Africa, enslaved in Newport, Rhode Island, where he became a music teacher and composer.

When and where: Born about 1746. Newport, from about 1760. Died in Liberia in 1826.

He won a lottery in 1791 and bought his family's freedom.

Newport Gardner was about 14 when slave traders took him from West Africa to Newport, Rhode Island. His name at birth was Occramer Marycoo. Caleb Gardner, a merchant and ship captain, bought him and renamed him Newport Gardner. The music writer Sophie Lowe writes that Caleb Gardner "helped enslave some 3,912 human beings" through the slave trade.

His teachers. The enslaved boy learned English and music in Newport. The best-known account names Andrew Law as his teacher, Lowe writes. But Law is first known to have been in Rhode Island in 1783, when Gardner was already in his late thirties. Lowe concludes that he "probably had a variety of musical teachers."

Freedom. Gardner and some friends won a lottery in late December 1791. With his share he bought freedom for himself and his family, then opened a school in Newport where he taught music. At Newport's First Congregational Church he was a deacon, a church member who helps lead it. In the 1820s he helped found a Black church, according to the historian Edward Andrews, and the historian John Millar writes that Gardner joined the new Colored Union Church in 1824.

His music. The writer Eileen Southern credited to him a tune called "Crooked Shanks," printed in 1803. Lowe reports newer research that found the same tune in print in London in 1768, so the credit is uncertain. The words of his anthem, a church song, survive. They begin "Hear the words of the LORD, O ye African race."

> **Liberia.** He led a group of Black New Englanders to Liberia, in West Africa. Millar counts about 30 of them, and Andrews counts about three dozen. Accounts also differ on when they sailed: late December 1825, or January 4, 1826. Gardner died of disease in Liberia within months, in 1826.

Tlingit songs written down, 1791

Two Spanish ships led by Malaspina and Bustamante anchored in Yakutat Bay, in what is now Alaska, in 1791. The bay was Tlingit land. Several times, Tlingit people sang "to and with" the Spanish sailors.

Tadeo Haenke was a botanist on the ships, a scientist who studies plants, and he had trained in music. He wrote the songs down in musical notes on four sheets of paper. The sheets stayed in the Museo de América, a museum in Madrid, Spain, for more than 200 years.

Hearing them again. The scholar Steve Langdon got pictures of the sheets and sent them to Maria Shaa Tláa Williams, a Tlingit ethnomusicologist, a scholar who studies the music of a people and its place in their lives. Williams found that the singers used harmony, two or more notes sung together, in part of their singing. She found one song, "Canta de la Paz," that was used to settle a quarrel over a stolen piece of clothing.

She then programmed a synthesizer, a machine that makes musical sounds electronically, to play the notes, so people could hear how the songs sounded. Judy Daxootsu Ramos, a Tlingit from Yakutat, said of the old songs, "we're still practicing that today."

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 15 MINUTES

Enslaved Black people in the South made songs that carried their own meanings, while white men in blackface copied Black music on stage and kept the money. The enslaved made their songs at work, at worship and on the days off their enslavers allowed, such as Christmas. Frederick Douglass later wrote down what the songs meant, and Harriet Tubman told a writer, Sarah Bradford, how she used hers.

The minstrel show. White performers darkened their faces with burnt cork and sang and danced as Black characters who were lazy, stupid, grinning fools. This blackface minstrel show became one of the most popular kinds of entertainment in the country. The money went to white performers and publishers. Stephen Foster, a white songwriter from Pittsburgh, wrote many of its best-known songs.

In Philadelphia a free Black bandleader, Francis Johnson, published more than 200 pieces of music. In the South, singers learned church music from books printed in shape notes, where the shape of each note shows the singer which sound to sing.

What the singers said the songs meant

Frederick Douglass was enslaved in Maryland, on Colonel Edward Lloyd's plantation in Talbot County and later on William Freeland's farm. He escaped in 1838 and published the story of his life in 1845. In it he wrote that enslaved people sang as they walked to the main farm to collect their monthly food, making up the words as they went.

Some people in the North took such singing as a sign that enslaved people were happy. Douglass wrote that this was a mistake: "Slaves sing most when they are most unhappy." He added, "I have often sung to drown my sorrow, but seldom to express my happiness."

"Make a noise." Enslavers wanted to hear singing, Douglass wrote in a second book, in 1855. "A silent slave is not liked by masters or overseers," he wrote. When the workers went quiet, the words they heard were "Make a noise." Teamsters, the men who drove the wagons, sang so that the overseer, the man who drove enslaved people at their work, would know where they were and that the work was going on.

A second meaning. Douglass and five other enslaved men planned to escape together in 1836. As they planned, they kept singing a hymn about being "bound for the land of Canaan." In the Bible, Canaan is the land God promised to the people of Israel after they escaped from slavery in Egypt.

Douglass wrote, "the north was our Canaan." Someone betrayed the plan, and it failed. Douglass never found out who. In his book he named the five men: Henry Harris, John Harris, Sandy Jenkins, Charles Roberts and Henry Bailey.

Juba. At Christmas, enslaved people on the Maryland farms played fiddles and beat out rhythms on their own bodies with their hands, which was called "juba" beating. The singers made up the words of the juba songs as they went. Douglass wrote down one verse that begins "We raise de wheat, Dey gib us de corn." It goes on to list the work the enslaved people did and the scraps they were given for it.

Tubman's signals. Harriet Tubman escaped from slavery in Maryland in September 1849 and later went back to lead others north. She told the writer Sarah Bradford how she used hymns as signals. People hiding in the woods came out only after she walked past singing a certain hymn twice. If she sang "Moses go down in Egypt" instead, they stayed hidden, "for dere's danger in de way." Bradford printed Tubman's account in 1869.

The "Drinking Gourd." Douglass and Tubman describe songs with hidden meanings and songs used as signals. A different claim is that the song "Follow the Drinking Gourd" was a map that led people from Alabama north to the Ohio River. That claim comes from a single article printed in 1928 by H. B. Parks, a Texas scientist who studied insects and collected songs.

A folk singer, Lee Hays, wrote the best-known words of the song in 1947. The researcher Joel Bresler found that no one ever collected the song in Alabama. He also found no older use of "drinking gourd" to mean the Big Dipper, the group of stars that points to the North Star. Bresler and the scholar James Kelley doubt that the song was ever a map.

Congo Square, New Orleans

Under a city law the mayor of New Orleans issued in 1817, enslaved people could gather in only one place: across Rampart Street at the back of town, on Sunday afternoons. The space was later called Congo Square. There enslaved people sold goods, sang, danced and played music. The architect Benjamin Latrobe wrote in his journal in 1819 of 500 to 600 enslaved people gathered there for dancing. He called the dancing "savage."

City officials stopped the Sunday music and dancing in 1835. It started again, and officials stopped it again in 1851. The records that survive do not give the names of the mayor who limited the gatherings or of the officials who stopped them.

What was blackface minstrelsy?

Blackface minstrelsy was a stage show in which white men darkened their faces with burnt cork and acted as Black people. White audiences liked seeing Black people shown as "singing, dancing, grinning fools," David Pilgrim of the Jim Crow Museum at Ferris State University writes. He writes that these shows helped spread the belief that Black people were lazy, stupid and "inherently less human," meaning less human by birth.

Jim Crow. Thomas Dartmouth Rice was a white actor, born in New York in 1808. He sang and danced a song called "Jump Jim Crow" in blackface. Pilgrim dates his first performance as Jim Crow to 1828. Other accounts say only that it had become his main act by 1832, when a Philadelphia newspaper notice announced that "Mr. Rice will appear and sing Jim Crow."

The man he copied. Rice said he copied a disabled Black man who cared for horses and sang as he worked, and that he bought the man's clothes to wear on stage. People still argue about when and where this happened, and whether it happened at all. No record names that man. No record shows that he or any other Black musician was paid for what the minstrels copied.

Rice played to sold-out theaters from about 1832 to 1844, and he took the act to London in 1836. By 1838 people were using "Jim Crow" as an insult for Black people. By the end of the 1800s, "Jim Crow" was the name for the

laws and customs that officials and lawmakers used against Black people. Under them, Black people were kept out of white schools, railroad cars and waiting rooms.

The Virginia Minstrels. Four white men performed in New York for a paying audience as the Virginia Minstrels on what was probably January 31, 1843. They were Dan Emmett, Billy Whitlock, Dick Pelham and Frank Brower. Earlier blackface acts had one singer or dancer. The Virginia Minstrels performed as a group, with fiddles, banjos, tambourines and the bones, a rhythm instrument.

A group led by Edwin P. Christy, the Christy Minstrels, started many of the features of the minstrel show from 1845. The players sat in a half circle. Two comic players called Mr. Tambo and Mr. Bones sat at the ends, and a leader called Mr. Interlocutor sat in the middle. The money went to the white performers and to the music publishers who printed their songs. Rice became rich, and he died poor in 1860.

The banjo goes on stage

Enslaved Black people played the banjo, which came from West African instruments, in America from at least 1736. That was about a hundred years before any white man played it on a stage.

Joel Walker Sweeney, a white man born in 1810 in Appomattox County, Virginia, learned to play it from enslaved people there. He played the banjo on a stage in New York in April 1839, and he is the earliest known person to have done so. Four years later, two of the Virginia Minstrels played banjos in their show.

One story says that Sweeney added the banjo's short fifth string. No proof of that exists. Older banjos, and pictures made before his time, already show the short fifth string.

Stephen Foster

Who: A white songwriter who wrote songs for blackface minstrel shows and for singing at home, and one of the first Americans to earn a living only by writing songs.

When and where: Born July 4, 1826, in Lawrenceville, Pennsylvania, now part of Pittsburgh. Worked in Pittsburgh, Cincinnati and New York. Died January 13, 1864, in New York.

When he died, his wallet held 37 cents and three pennies.

Stephen Foster wrote many of the best-known songs of the minstrel shows.

Kitty. Foster's family held a Black girl named Kitty as a bound servant, someone who had to work for a family without pay until a set number of years ran out. The writers of the University of Pittsburgh library's biography of Foster call this arrangement "enslavement in all but name."

Foster's father wrote in 1834 that a woman named Mrs. Collins had given the family the girl as "a present." The girl had more than three years left to serve. He wrote that it saved them the cost of hiring a girl.

Learning the style. At nine, Foster sang blackface songs such as "Jim Crow" in a boys' theater set up in a carriage house, and he was paid a small sum each week. He learned the style mostly from songs that white performers wrote to mock Black people. In 1846 and 1847 he kept the money records for his brother's shipping company in Cincinnati. The company made money from cotton picked by enslaved people in the South.

"Oh! Susanna." The song was first advertised in Pittsburgh in September 1847. Within a year the publisher William C. Peters printed it as one of the songs of the Sable Harmonists, a blackface group. At the time, no American song had sold more than 5,000 copies of its sheet music, the printed music people bought to play at home. "Oh! Susanna" sold more than 100,000. Its original second verse used a racist word for Black people and told, as comedy, of an explosion that killed 500 of them.

Christy's name on the song. The minstrel leader Edwin P. Christy paid Foster $10 a song from 1850 for the right to sing each new song first. For "Old Folks at Home" in 1851, Christy paid $15, and the sheet music named Christy as the writer. Foster asked in 1852 to have his own name put back. Christy refused.

That year Foster wrote to Christy about the words of his songs. He said he had made them suitable for "refined people," instead of the "trashy and really offensive words" of other minstrel songs. The University of Pittsburgh writers say he most likely did this to sell more sheet music.

Politics. His only openly political songs before the Civil War supported the Democrats who opposed ending slavery. One of them, "The Abolition Show" of 1856, attacked the people working to end slavery. Frederick Douglass, however, said in 1855 that songs such as "Old Kentucky Home" and "Uncle Ned" could make listeners feel for enslaved people.

Selling the future. In about 1857 Foster sold his rights to future payments on his songs, in return for money right away. He moved to New York in 1860 and kept writing, including more minstrel songs.

His death. In January 1864, while he was ill, Foster was badly cut on his neck and face in his hotel room. His writing partner, George Cooper, found him lying in a pool of blood, with what Cooper called "a cut in his throat." Foster died at Bellevue Hospital three days later, at 37. His wallet held 37 cents in paper money and three pennies.

DISPUTED: Biographers disagree about how he was cut. His brother Morrison wrote in 1896 that Stephen fainted and fell across the wash basin, which broke and cut him. Ken Emerson, writing in 1997, tells it as a fall onto a wash basin or a chamber pot, a bowl kept in the room for use as a toilet. JoAnne O'Connell argued in 2016, from Cooper's description of the wound, that Foster cut his own throat on purpose.

> **The words today.** Other people later changed the words that are sung today. Two Kentucky lawmakers, Representative Carl Hines and Senator Georgia Davis Powers, won votes in 1986 to sing "people" in place of a racist word in "My Old Kentucky Home," the state song since 1928.

Church songs: Richard Allen's hymn book and shape notes

Richard Allen, a Philadelphia minister who had been born enslaved, put together two hymn books in 1801, one of 54 hymns and one of 64. The books contain the words and no music. Allen formally organized the African Methodist Episcopal Church, a church run by Black Methodists, in 1816.

Shape notes. William Little and William Smith printed a music book called *The Easy Instructor* in Philadelphia in 1801, the same year as Allen's hymn books. Its music is written in shape notes: each note has one of four shapes, and each shape stands for one sound to sing. The book was made for singing schools.

The Sacred Harp. Two Georgia men, Benjamin Franklin White and Elisha J. King, published *The Sacred Harp* in 1844, a shape-note book of more than 250 songs. Some of its oldest songs are by William Billings, the Boston singing teacher of the late 1700s. By 1800 his tunes had mostly dropped out of use in Boston, but singers in the rural South kept them.

Sacred Harp singers sit in a square, facing inward, with an empty space in the middle, and they sing in four parts with no instruments. People still sing from the book, and a new edition was printed in 2025.

Francis Johnson

Who: A free Black bandleader, bugle player and composer in Philadelphia who published more than 200 pieces of music.

When and where: Born June 16, 1792, in Philadelphia. Died there April 6, 1844.

A congressman called him the "leader of the band at all balls."

Francis Johnson led the band at Philadelphia's dances and balls, and he published more than 200 pieces of music. He was a free Black man. A church record in Philadelphia gives his birth date as June 16, 1792. Some books say he was born on the island of Martinique, but they give no evidence.

He grew up in a free Black neighborhood of Philadelphia. By 15 he could play the flute, the violin, the bugle and the piano. He played in taverns, then as the fiddler at the Exchange Coffee House.

First in print. A Philadelphia publisher, George Willig, printed his dance music. The writer Richard Grant, in *Smithsonian* magazine, dates the first piece to 1810 and calls it the first music published by a Black American. Other accounts date his first wide notice to a collection Willig printed in 1818. That year Robert Waln, a former congressman, wrote a book about Philadelphia that called Johnson the "leader of the band at all balls." Johnson played the keyed bugle, a brass horn with keys, and wrote dances, marches and songs.

To England. Johnson took his four-man band to England in November 1837. No Black American musician had taken a band to Europe before, Grant writes. Some accounts say Johnson played for Queen Victoria, and others say that has not been proven.

Arrested in St. Louis. Johnson and nine Black musicians toured eight states in 1842 and 1843. In St. Louis in December 1842, the band members were served with warrants, papers from a court that order an

arrest. The charge was being "Free Negros within the State of Missouri without a license." Missouri law required free Black people to hold a license to be in the state.

The records do not name the officials who served the warrants. A firemen's club that had hired the band paid a lawyer. The judge closed the case without a ruling, and the band left.

The mob. In Allegheny City, Pennsylvania, a mob gathered outside the hall where the band played. The mob attacked the musicians in the street, threw stones, bricks and rotten eggs, and shouted racist insults. Several musicians were badly hurt. The band played the next night anyway. No record names anyone in the mob.

Johnson died on April 6, 1844, when a blood vessel near his heart burst. At his funeral his band played a funeral march he had written. One of the largest funeral crowds in Philadelphia's history came to it.

The Hutchinson Family sing against slavery

The Hutchinson Family Singers were brothers from Milford, New Hampshire: John, Asa, Jesse and Judson Hutchinson. They gave their first concert in Milford on November 6, 1840. Their sister Abby joined them when she was about 12. In 1842 they earned up to $130 a concert.

Songs against slavery. From 1843 the Hutchinsons sang against slavery. In 1844 they set "Get Off the Track!" to the tune of a minstrel song, "Old Dan Tucker." A year later they toured Britain with Frederick Douglass.

Philadelphia, 1847. The Hutchinsons sang in Philadelphia in the spring of 1847 to audiences of Black and white people together. The city's mayor, John Swift, said more concerts would cause a riot. He ruled that no antislavery speech could be given at the hall and that no Black person could sit in the audience. The Hutchinsons refused to sing for a white audience only.

BY 1850

If you remember one thing: In these fifty years enslaved people's songs held meanings of their own, and the white men who copied Black music on the minstrel stage kept the money.

- 1817: The mayor of New Orleans limits enslaved people's Sunday gatherings to the place later called Congo Square.
- 1843: Four white men perform in New York as the Virginia Minstrels, the first blackface act built as a group.
- 1845: Frederick Douglass writes, "Slaves sing most when they are most unhappy."

Next: After 1850 the songs of slavery were printed for the first time, and freed Black students sang them for a president and a queen.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 19 MINUTES

The songs of slavery reached print and the concert stage after the Civil War. Black singers carried them to audiences in America and Europe, while the officer who ran a boarding school for Native children set out to silence the children's drums. Three white Northerners printed the first book of spirituals, the religious songs that enslaved Black people had made, in 1867. Students at Fisk University, a school for freed Black people, sang those songs on tour to keep their school open.

That officer, at the Carlisle Indian School in Pennsylvania, gave the boys brass instruments to stop their drumming. The school's own paper said that as the band grew, the boys stopped singing their own songs. In the war itself, army bands played for soldiers on both sides, and bandsmen carried the wounded.

Singers recorded songs on wax cylinders from 1890. A wax cylinder was a tube of wax that held two or three minutes of sound, and a machine could play it back. Scott Joplin, a Black pianist in Missouri, published the "Maple Leaf Rag"

in 1899, a piece of ragtime, piano music with a bouncing rhythm. In the South, Black farm workers were making the music later called the blues.

Jenny Lind's tour, 1850 to 1852

The showman P. T. Barnum agreed in January 1850 to pay the Swedish singer Jenny Lind $1,000 a concert to tour the United States. She arrived in New York in September 1850, and Barnum sold tickets by auction. She sang 93 concerts for Barnum and earned about $350,000, while Barnum made at least $500,000. Lind gave her share to charities, most of it to free schools in Sweden.

War songs and army bands, 1859 to 1865

Bryant's Minstrels, a white blackface group, first sang "Dixie" in New York on April 4, 1859. The song is credited to Dan Emmett, one of the Virginia Minstrels.

Who wrote "Dixie"? The accounts differ. Since the 1910s or 1920s, people in Mount Vernon, Ohio, have said that Emmett took the song from the Snowdens, a Black family of musicians in Knox County, Ohio. The writers Howard and Judith Sacks suggest it came from the parents of the family, Thomas and Ellen Snowden. Other scholars, such as E. Lawrence Abel, reject the Snowden story outright.

A Confederate song. In Charleston, South Carolina, in December 1860, a band played "Dixie" after each vote to leave the Union, as people then called the United States. A band played it in February 1861 when Jefferson Davis became president of the Confederacy, the group of Southern states that broke away from the United States.

"Battle Hymn." Union soldiers from Massachusetts made up the song "John Brown's Body" in 1861, about John Brown, who led an 1859 raid on Harpers Ferry, Virginia. They sang it to the tune of a hymn from camp meetings, big outdoor church gatherings.

Julia Ward Howe heard soldiers singing it near Washington, D.C., and in November 1861 she wrote new words to the same tune. She sold them for $4 to *The Atlantic Monthly*, a magazine. Its editors printed her "Battle Hymn of the

Republic" on the front page in February 1862.

Banned from the camps. The Hutchinson Family sang a song against slavery to Union soldiers in Virginia in January 1862. Its words included "Let Slavery die!" Some of the soldiers booed. General George McClellan, who opposed using the war to end slavery, then banned the Hutchinsons from singing in the army camps.

Bandsmen. Under an army order of August 3, 1861, each Union regiment, one unit of soldiers, could have a band of 24 musicians. Members of Congress voted in July 1862 to send the regimental bands home. They kept one band for each brigade, a larger group of soldiers made up of several regiments.

Bandsmen also carried the wounded. Albert Mann, a bandsman from Massachusetts, wrote that the army surgeon taught the band to use "stretchers, bandages and tourniquets," bands tied tight around an arm or leg to stop heavy bleeding. The bandsmen were to go into battle with the regiment, "rescue the wounded, if possible, and carry them to the field hospital."

At the Battle of Gettysburg, fought from July 1 to 3, 1863, the bands of the 26th and 11th North Carolina regiments played while "heavy firing" went on, the 26th's band leader recalled. A British visitor heard a Confederate band play polkas and waltzes while cannon fired.

Spirituals written down

Thomas Wentworth Higginson, a white abolitionist who worked to end slavery, commanded the 1st South Carolina Volunteers in 1862. The regiment was a Union unit of men who had been enslaved. A drummer boy in the regiment told Higginson about a hymn with the words "We'll soon be free." He said that enslaved people in Georgetown, South Carolina, had been whipped for singing it when Abraham Lincoln was elected in 1860.

"De Lord." The enslavers thought "de Lord" in the song meant the Yankees, the Northern soldiers, the boy said. In a later version of the story, Higginson wrote that the singers had been put in jail. Neither account names the drummer boy, the singers, or the people who punished them.

The first book. William Francis Allen, Lucy McKim Garrison and Charles Pickard Ware, three white Northerners, published *Slave Songs of the United States* in 1867. It contains 136 songs and was the first book of spirituals ever printed. Most of its songs came from Gullah Geechee people, Black people of the South Carolina islands, who lived on St. Helena Island and had been freed during the war.

The Fisk Jubilee Singers, from 1871

Fisk University opened in Nashville, Tennessee, after the Civil War to teach freed Black people. By 1871 it was running out of money, and its classes met in old buildings of a Union army hospital. George White, a white church worker from the North, kept the school's money and led its music. He set out with nine students on a singing tour to raise money on October 6, 1871. Among them were Maggie Porter and Ella Sheppard.

At first they sang spirituals only as extra songs at the end of a concert. Audiences asked for more. George White named the group the Jubilee Singers, after the "year of jubilee" in the Bible, when slaves were set free.

Hard roads. After one early concert in Ohio, the singers gave their profit to the victims of the Chicago fire of October 1871. "We had thirty dollars and sent every penny to Chicago and didn't have anything for ourselves," Maggie Porter said. She also said, "There were many times, when we didn't have place to sleep or anything to eat."

Ella Sheppard wrote of "the ill treatment at hotels, on railroads, poorly attended concerts, and ridicule." George White and another man walked through sleet, snow and rain from hotel to hotel, looking for one that would take the singers in. The records do not name the hotels that turned them away.

What they raised. They sang for President Ulysses S. Grant at the White House in March 1872. By that spring they had raised the $20,000 that George White had promised. They sang for Queen Victoria in England in April 1873, and by May 1874 their total had reached nearly $50,000.

That money paid for Jubilee Hall, Fisk's first permanent building, which opened on January 1, 1876. It became a National Historic Landmark in 1975. Over seven years of touring, the singers raised about $150,000 in all.

Ella Sheppard

Who: Born into slavery, she became the pianist, arranger and assistant director of the Fisk Jubilee Singers.

When and where: Born February 4, 1851, at the Hermitage, Andrew Jackson's plantation near Nashville, Tennessee. Toured the United States and Europe. Died in Nashville in 1914 or 1915 (the records differ).

Movie: *Jubilee Singers: Sacrifice and Glory* (2000), a documentary about the group in the PBS series *American Experience*.

"She is mine and she shall die mine," said her mother's enslaver.

Ella Sheppard arranged most of the spirituals that the Fisk Jubilee Singers sang, writing out the parts for each voice. She was born into slavery at the Hermitage, Andrew Jackson's plantation near Nashville.

Her mother. Her father, Simon Sheppard, paid $1,800 to buy his own freedom, and then he bought Ella's. The woman who held Ella's mother, Sarah, refused to sell her. She said, "She is mine and she shall die mine. Let Simon get another wife." When Ella was three, her mother was sold to a plantation in Mississippi. Ella did not see her again until she was 15.

The records do not name the woman who held Sarah or the person who bought her. Her mother said, "My back was never struck, but my heart is like a checkerboard with its stripes of sorrow."

Lessons through the back door. After a riot in Nashville in 1856, white officials there made stricter rules for free Black people. Simon fell into debt and was afraid that his family would be seized and sold, so he took Ella to Cincinnati, Ohio. There a German woman taught her piano. A white teacher taught her in secret, at night, letting her in through the back door.

To Fisk. Her father died in 1866 of cholera, a gut infection that people catch from dirty food or water. It drains so much water from the body that a person can die. Ella taught music in Gallatin, Tennessee, and saved

about $6 in five months. She entered Fisk in 1868. The money lasted three weeks, and she stayed by giving music lessons and washing dishes.

With the singers. On the first tour, in 1871, she sang with eight other students: Isaac Dickerson, Maggie Porter, Minnie Tate, Jennie Jackson, Benjamin Holmes, Thomas Rutling, Eliza Walker and Green Evans. She played piano, organ and guitar, arranged the spirituals for the group to sing in four parts, and collected more than a hundred songs.

On stage she led the singers from her place among them. In the summer of 1875 she wrote down the music of eight spirituals for a printed book of the group's songs, and the book did not name her.

She later wrote that the old songs "were sacred to our parents, who used them in their religious worship," and that at first the students did not want to sing them for strangers. In 1890 she trained a new group of Jubilee Singers at Fisk.

Native songs, drums and the first recordings

The army officer Richard Henry Pratt opened the Carlisle Indian School in Pennsylvania, a boarding school for Native children, in 1879. Students came from 54 or more nations. Early in the school's history, a rule was made that students could not speak their own languages or practice "Indian customs." The school's paper does not say who made the rule.

"Even in song." An article in that paper, *The Red Man*, stated in 1896 that the school's aim was "the complete transformation of the Indians." It called any "display of savagery and barbarism, even in song and language," out of place in the school. Those were the paper's words for the students' own songs and languages. The article states that in the first year, the only music heard from the boys' rooms was the drum.

Brass for drums. Pratt told a Boston donor, Mrs. Walter Baker, that he wanted to stop the drumming. "If you will give me a set of brass band instruments I will give them to the 'tom tom' boys," he said, as the paper quotes him, and "this will stop the 'tom tom.'" The school band began in 1881.

The rule named languages and "Indian customs," but the same article states that "Indian singing was never prohibited" and that no order to stop the songs was ever given. As the band grew, the article says, the singers of those songs fell silent. Dennison Wheelock, an Oneida student who entered Carlisle in 1885, later led the school band.

Dances made offenses. Hiram Price, the federal Commissioner of Indian Affairs, issued rules on March 30, 1883, that made the Sun Dance and other religious dances "Indian offenses." His boss, Secretary of the Interior Henry Teller, had called Native dances "heathenish," an insulting word for not Christian. The rules named dances and feasts. They did not name songs. The punishment was the loss of rations, the food that federal officials handed out to Native people on reservations.

For a first offense, officials could hold back a person's rations for up to ten days, and for a later offense, 15 to 30 days. They could instead put the person in the agency prison, the jail at the government office on the reservation, for up to 30 days. Officials could enforce these rules until 1978.

The first recordings. The scientist Jesse Walter Fewkes recorded Passamaquoddy songs and stories on 36 wax cylinders in Calais, Maine, from March 15 to 18, 1890. The Passamaquoddy are a Native nation whose homeland is along the border between Maine and New Brunswick, in Canada. The singers were Peter Selmore, Newell Josephs and Peter Lacoute.

In a Library of Congress essay, Kelly Revak writes that these are "largely credited" as the first recordings of Native American music. Passamaquoddy speakers have used the recordings since the 1980s to bring old songs back. Wayne Newell said in 2009 that people were working on them at home "so we can sort of resurrect older songs."

Wounded Knee. Soldiers of the 7th Cavalry, an army unit on horseback, killed between 250 and 300 Lakota people at Wounded Knee on December 29, 1890, most of them women and children. The soldiers had been taking the guns of a Lakota band led by Spotted Elk. His people had taken up a religious movement called the Ghost Dance, whose followers danced in a circle, and the Ghost Dance had its own songs.

James Mooney, a scientist who studied Native peoples, recorded Ghost Dance songs in 1894. Library of Congress notes say the singer on those recordings is probably Mooney himself, not a Native singer.

John Philip Sousa

Who: Leader of the United States Marine Band and writer of more than 100 marches.

When and where: Born November 6, 1854, in Washington, D.C. Led the Marine Band from 1880 to 1892, then his own band. Died March 6, 1932, in Reading, Pennsylvania.

Movie: *Stars and Stripes Forever* (1952), a Hollywood film loosely based on his memoir.

His father signed him into the Marines at 13.

John Philip Sousa wrote more than 100 marches, including "The Stars and Stripes Forever," the national march of the United States. His father played in the Marine Band. When Sousa was 13, his father signed him into the Marine Corps as an apprentice musician, a boy learning the work. He did it to keep his son from joining a circus band. Sousa served seven years.

The Marine Band. Sousa became the 17th leader of the Marine Band on October 1, 1880. He led it for 12 years under five presidents, from Rutherford B. Hayes to Benjamin Harrison. In 1892 he formed a band of his own.

The march. Sousa was on a ship crossing the Atlantic on Christmas Day 1896. He had just learned that his band's manager, David Blakely, had died. On that voyage he began "The Stars and Stripes Forever," and he finished it in April 1897. Members of Congress voted in 1987 to make it the national march. Sousa also wrote 15 operettas, short comic operas, and he helped develop the sousaphone, a large brass horn made for marching bands.

Recording begins

Thomas Edison's workers built his first phonograph, a machine that recorded sound and played it back, in December 1877. Machines good enough to sell came on the market in 1888.

George W. Johnson. In 1890 two phonograph companies paid George W. Johnson 20 cents for each two-minute song he recorded on a wax cylinder. Johnson was a Black street whistler, born in Virginia in 1846. The cylinders played in coin-operated machines. No way yet existed to copy a cylinder in large numbers, so Johnson sang the same songs over and over, one cylinder at a time. Between 1890 and 1895 he made an estimated 25,000 to 50,000 cylinders.

His two best sellers were "The Laughing Song," which he wrote, and a whistling song by a white writer that mocked Black people. The title of that song used a racist insult for a Black person. "The Laughing Song" was the best-selling record in the country for ten weeks in 1891, and Johnson is counted as the first Black recording star.

What became of him. Johnson was put on trial in 1899 for the murder of Roskin Stuart, the woman he lived with as his wife, who had been found beaten in their apartment. He was found not guilty. By 1905 record makers could press many copies from one recording, so they no longer needed him to sing each copy. His friend Len Spencer hired him as a doorman for his office. Johnson died in 1914 and was buried in a grave without a marker.

Ragtime, and songs that mocked Black people

Ragtime is music with a "ragged" rhythm: the tune lands between the beats while a steady beat keeps time underneath. On the piano, the left hand keeps the steady beat and the right hand plays the tune. Ragtime began among African Americans in the late 1800s. Accounts differ about the first printed ragtime piece, and they date it to 1895 or 1897.

Ernest Hogan, a Black performer from Bowling Green, Kentucky, who had started out in minstrel shows, wrote a hit song in 1895 and 1896 whose title used a racist insult for Black people. The title said that all Black people look alike. Many other writers copied it. People at the time named these copies after the insult. They were songs that used racist words and pictures to mock Black people.

Scott Joplin

Who: A Black composer and pianist who wrote the "Maple Leaf Rag" and an opera, *Treemonisha*.

When and where: Born in northeastern Texas between mid-1867 and early 1868 (the date is not known for certain). Lived in Texarkana, Sedalia and St. Louis, Missouri, and New York. Died April 1, 1917, in New York.

His most famous rag earned him about $4 in its first year.

Scott Joplin wrote the "Maple Leaf Rag," and in his last years he paid to print his own opera. He was born in northeastern Texas, and the exact date is not known. The historian Edward A. Berlin places it between June 1867 and the middle of January 1868.

His father, Giles, had been enslaved in North Carolina. His mother, Florence, was born free in Kentucky. She cleaned houses, and Scott learned to play on her employer's piano. A teacher, Julius Weiss, gave him lessons.

Finding work. Churches and brothels, houses where men paid women for sex, were among the few places where a Black pianist could find steady work. In 1891 he was a member of the Texarkana Minstrels, a group raising money for a monument to Jefferson Davis, the former president of the Confederacy. He moved to Sedalia, Missouri, in 1894. There he played the cornet, a brass horn like a small trumpet, in a town band and studied music at George R. Smith College.

One cent a copy. Joplin signed a contract on August 10, 1899, with John Stark, a Sedalia music store owner and publisher, to publish the "Maple Leaf Rag." Joplin was to get one cent for each copy sold. His first biographer said it sold 75,000 copies in six months. Berlin later found that the first 400 copies took a year to sell, which paid Joplin about $4. Later sales gave him a small, steady income. Berlin found that by 1909 they would have paid him about $600 a year.

Treemonisha. He moved to St. Louis in 1901 and to New York in 1907. He wrote an opera, *Treemonisha*, set on a Texas plantation after the Civil War. It tells of a young, educated Black woman who helps free her people from ignorance and superstition. No one would pay to stage it. In 1911 he printed it with his own money. In 1915 he played it in a hall in Harlem, with only a piano.

His death. By 1916 Joplin had syphilis in its late stage. Syphilis is a disease passed from person to person during sex. Without treatment it can spread to the brain, where it causes confusion, changes in personality and dementia, the loss of memory and of the ability to think.

He was admitted on February 2, 1917, to Manhattan State Hospital, a hospital for mental illness. He died there of the disease on April 1, 1917. No record describes how he was treated there. He was buried in a grave with no marker.

Treemonisha was first fully staged in January 1972, in Atlanta. His grave was given a marker in 1974, 57 years after his burial. In 1976, 59 years after he died, Joplin won a Pulitzer Prize, a top American award.

The blues begins, and nobody wrote it down

The blues is a kind of Black music from the South. It grew out of the songs and calls that enslaved people sang as they worked in the fields, and out of the songs of Black workers on farms, docks and river boats after slavery ended. Writers on the blues place its beginning between 1870 and 1900.

Sharecropping. In those years, freed Black families in the South who had no land of their own farmed a landowner's fields in return for a share of the crop. This was called sharecropping. The families bought tools, seed and food on credit at the landowner's store prices. The debt rolled over from year to year, and families who owed it could not leave the land.

No written description of the blues from before 1900 is known. The first written accounts come from just after 1900.

NOT KNOWN: No one knows who first played the blues, or when.

BY 1900

If you remember one thing: The songs of slavery were written down and sung across America and Europe in these fifty years, and music could now be recorded and sold.

In 1850: No book of spirituals had ever been printed, and the people who made them were still enslaved.

In 1900: *Slave Songs of the United States* was in print, the Fisk Jubilee Singers had sung for a president and a queen, and a Black whistler, George W. Johnson, had made the best-selling record in the country.

Next: After 1900 record companies began to record the blues and jazz, and the fight over who was paid moved into the recording studio.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 29 MINUTES

Record companies began to record and sell the blues, jazz and country music in these years, and the companies and the men who held the copyrights kept much of the money. Black musicians in the South made the blues and jazz. Country music came from Southern singers and players, white ones like the Carter Family and Black ones like DeFord Bailey and Lesley Riddle. Bessie Smith's first record sold 780,000 copies, and Columbia paid her a set fee and no share of the sales.

Americans spent $106 million on records in 1921. From the 1920s, anyone with a radio could hear music at home for free. Owners and managers of clubs, hotels and concert halls kept Black people out, even Black stars. Billie Holiday recorded "Strange Fruit" in 1939, a song about Black people murdered by white mobs. That same year Marian Anderson sang outdoors in Washington to more than 75,000 people, after the leaders of two indoor halls in the city refused her.

The first written accounts of the blues, 1901 to 1903

The first written account of the kind of songs later called the blues comes from Mississippi in 1901 and 1902. Charles Peabody, an archaeologist, dug into a Native mound in Coahoma County in May and June of both years for Harvard University's Peabody Museum. He hired nine to fifteen Black men from the town of Clarksdale to do the digging, and the men sang while they dug. Peabody wrote in 1903 that bosses sometimes hired a strong singer just to keep workers going.

The men's songs were about hard luck, love and pay day. In one line the singer said he had been arrested for murder though he had never harmed anyone. Peabody did not use the word blues.

Tutwiler. The bandleader W. C. Handy was waiting for a train at Tutwiler, Mississippi, in about 1903. A man beside him played a guitar by sliding a knife along the strings, and sang. No record names the man. Handy later published the song as "Yellow Dog Blues."

"Father of the Blues." Handy sold "The Memphis Blues" for $100 in 1912, and in 1914 he published "Saint Louis Blues." With Harry Pace, a Black banker, he ran a company called Pace and Handy that published music. Handy called himself the "Father of the Blues." Jelly Roll Morton disagreed. In a letter to the magazine *Down Beat*, he wrote that Handy had not started the music.

"Mother of the Blues." The singer Gertrude "Ma" Rainey said that in 1902 a girl in Missouri sang her a song about a man who had left a woman. Rainey began singing it in her shows. Paramount Records later advertised her as the "Mother of the Blues." Between 1923 and 1928 she made more than 100 records.

Records for Black buyers, and who was paid, 1920 to 1938

Mamie Smith, a Black singer, recorded "Crazy Blues" for the Okeh record company on August 10, 1920. The songwriter Perry Bradford wrote it. Managers at Victor had turned Smith down after a try-out, so Bradford went to Fred Hager, who ran recording at Okeh. He told Hager that Smith could "do more with those songs than a monkey can do with a peanut."

Some white people had already warned Hager not to record Black musicians, and he recorded her anyway. Within two months, buyers had paid a dollar each for 75,000 copies. It was the first blues record to become a big hit.

"Race records." Record company managers then went looking for Black singers to sell records to Black buyers. By 1922 managers at Okeh were calling these records "race records." The Black businessman Harry Pace started Black Swan Records in 1921. Black Swan paid its singers fair wages and let them record under their own names. Paramount took over Black Swan in 1924.

A set fee. Columbia paid Bessie Smith a set fee for each side she recorded. A side was one song on one face of a record. She got no royalty, the small payment for every copy sold that lasts as long as the record sells. Columbia paid white singers more, and it paid them royalties. With a set fee, a singer earned the same whether the record sold a hundred copies or a million.

Twenty-nine songs. Robert Johnson, a Black guitarist from Mississippi, recorded 29 songs in 1936 and 1937, in a hotel room in San Antonio and in Dallas, both in Texas. His best seller, "Terraplane Blues," sold about 10,000 copies. His pay for the sessions is not on record. He died near Greenwood, Mississippi, on August 16, 1938, at 27. The cause of his death is not known. One story says he was poisoned, but no one has proved it.

Mississippi judges ruled in 1998 that his son, Claud Johnson, was his only heir, the one person with the right to what Johnson left. That right was worth more than $1 million in royalties.

The editors of *Billboard*, a music magazine, renamed its "race records" sales chart "rhythm and blues" in June 1949. A writer named Jerry Wexler suggested the new name.

Bessie Smith

Who: A Black blues singer whose first record sold 780,000 copies in six months.

When and where: Born in Chattanooga, Tennessee, on April 15, 1894 (the 1900 government count of the people gives July 1892). Recorded for Columbia Records from 1923. Died near Clarksdale, Mississippi, on September 26, 1937.

Movie: *Bessie* (2015), an HBO drama film, with Queen Latifah as Smith.

Her first record sold 780,000 copies. She got $125 a side.

Bessie Smith was born in Chattanooga, Tennessee. Her family kept April 15, 1894, as her birthday, but the 1900 census, the government's count of everyone in the country, gives July 1892. Both her parents died when she was a child. She and her brother Andrew sang and danced for coins on the streets of Chattanooga. In 1912 she joined a traveling show as a dancer, and Ma Rainey sang in the same show.

The pay. Frank Walker of Columbia Records signed her in 1923. Her first record, "Down Hearted Blues," sold 780,000 copies in six months. Her contract paid her $125 for each side that could be used. Later contracts paid $150 and then $200 a side, and none of them paid a royalty.

She recorded between 150 and 200 songs. Cary O'Dell's essay for the Library of Congress states 150 in one place and close to 200 in another. On tour she earned up to $2,000 a week, and she traveled in her own railroad car.

The crash. The car she rode in hit the back of a slow truck on Route 61 near Clarksdale, Mississippi, on September 26, 1937. Her partner, Richard Morgan, was driving. Her right arm was almost torn off at the elbow. A white doctor from Memphis, Hugh Smith, stopped to help. An ambulance took her to the G. T. Thomas Afro-American Hospital in Clarksdale. Doctors there amputated her arm, which means they cut it off, and she died that morning.

A story that was not true. The record producer John Hammond wrote in 1937 that she died because a white hospital refused to treat her. Dr. Smith said the ambulance would never have taken her to a white hospital. Her biographer, Chris Albertson, showed that Hammond's story was a myth.

More than 10,000 people came to her funeral in Philadelphia. Her grave had no stone for more than 30 years, until the singer Janis Joplin helped pay for one. Later Jack Gee Jr., the son of her husband, sued Columbia. The judges ruled against him because he had waited too long to sue. They did not decide whether she had been cheated.

Jazz on record, 1917 to 1931

Jazz began in the Black communities of New Orleans in the late 1800s and early 1900s, but white musicians made the first jazz record. Five white musicians from New Orleans recorded two songs in New York for the Victor company on February 26, 1917. They called themselves the Original Dixieland Jass Band. Victor's managers put the record on sale in May 1917, and by some counts it sold more than a million copies. The band's leader, Nick LaRocca, later claimed that he had invented jazz.

A legend. One story says that in 1915 Victor had asked a Black New Orleans player of the cornet, a brass horn like a small trumpet, to record first. The player was Freddie Keppard, and the story says he said no. The historian John Edward Hasse calls that story a legend.

The Cotton Club. Owney Madden, a gangster who sold illegal liquor, ran the Cotton Club in Harlem, New York. Madden's managers hired Black musicians and dancers, and they let in only white customers, apart from a few Black stars. The club's dancers had to be light-skinned, at least 5 feet 6 inches tall and under 21. Duke Ellington's band played there from December 4, 1927, to June 30, 1931. When Ellington asked, the people running the club let in a few more Black customers.

BY 1927: Blues records by Black singers sold in the hundreds of thousands, and the first jazz record, made by white musicians, had sold a million copies by some counts.

Louis Armstrong

Who: A Black trumpet player and singer from New Orleans who made jazz records with his Hot Five and Hot Seven.

When and where: Born in New Orleans. Most sources give August 4, 1901. He often said July 4, 1900. Lived in Chicago and New York. Died in New York on July 6, 1971.

Movie: *Louis Armstrong's Black & Blues* (2022), a documentary about him.

"The government can go to hell," he told a reporter in 1957.

Louis Armstrong learned to play the cornet in a home for Black boys in New Orleans, where a judge sent him after an arrest. On New Year's Eve 1912 he fired his stepfather's gun into the air. It was loaded with a blank, a shell with gunpowder but no bullet. Police arrested him, and the next day a judge sent him to the Colored Waif's Home, a reform home for Black boys. There a teacher, Peter Davis, taught him the cornet in the home's band.

Hot Five, Hot Seven. The bandleader King Oliver invited him to Chicago in 1922 to play second cornet in his band. From November 1925 Armstrong made records for Okeh with his own groups, the Hot Five and the Hot Seven.

Little Rock. A reporter in Grand Forks, North Dakota, asked Armstrong on September 17, 1957, about the fight over nine Black students entering Central High School in Little Rock, Arkansas. Armstrong said: "The way they are treating my people in the South, the government can go to hell." He called President Dwight Eisenhower "two faced" and said he had "no guts."

> Staff at WBKJ, a radio station in Hattiesburg, Mississippi, threw out his records. Officials at the University of Arkansas cancelled his invitation to play a dance. The congressman Adam Clayton Powell said Armstrong "didn't know what he was talking about." On September 24 Eisenhower sent soldiers to protect the students. Armstrong then sent him a telegram: "If you decide to walk into the schools with the little colored kids, take me along Daddy. God bless you."

Country music on record: Bristol, 1927

Ralph Peer of the Victor record company recorded 76 songs by 19 singers and groups in Bristol, Tennessee, from July 25 to August 5, 1927. He set up on the third floor of a hat and glove company's building. Each performer got $50 in cash for each side, plus 2½ cents for each record sold. The Carter Family recorded on August 1 and 2, and a singer named Jimmie Rodgers on August 4.

What Peer kept. Peer took no salary from Victor. Instead he took a royalty of one cent on each record side, which he split with the singer. He also kept the copyright of the songs he recorded, including old songs passed down from singer to singer, and with it the right to be paid whenever those songs were printed, recorded or played.

Peer started a company in 1928, Southern Music Publishing, to collect that money. He offered to publish the songs that A. P. Carter of the Carter Family had gathered, if Carter would claim that he wrote them.

Maybelle Carter

Who: Guitarist and singer in the Carter Family, who changed how the guitar is played in country music.

When and where: Born May 10, 1909, in Nickelsville, Virginia. Recorded at Bristol, Tennessee, in 1927, at 18. Died October 23, 1978, in Hendersonville, Tennessee.

Movie: *The Carter Family: Will the Circle Be Unbroken* (2005), a PBS *American Experience* documentary about the family.

Her thumb played the tune, and one finger kept the rhythm.

Maybelle Carter found a way to play the guitar that made it a lead instrument in country music. She picked out the tune with her thumb on the three low strings and strummed the three high strings with her index finger to keep the rhythm. It is called the thumb-lead style. She first recorded "Wildwood Flower," played this way, in 1928.

She sang and played with Sara Carter and A. P. Carter as the Carter Family. At Bristol in August 1927 she was 18. The Carter Family was paid Victor's rate of $50 a side and 2½ cents for each record sold.

Lesley Riddle. A. P. Carter went looking for old songs from December 1928 with Lesley Riddle, a Black guitarist from Burnsville, North Carolina. Riddle had lost his right leg at the knee in 1927 after an accident at a cement plant. On their trips, Carter wrote down the words while Riddle memorized the tunes. Riddle lived with the Carters for a time and taught Maybelle his finger-picking blues guitar style. Maybelle said she learned guitar runs from him, including the ones she played in "Cannonball Blues."

No record shows that Riddle was paid, or named as a writer, for any of the songs he helped gather.

Later Maybelle led a group with her daughters, called Mother Maybelle and the Carter Sisters. From 1968 she toured with her son-in-law, the singer Johnny Cash.

How did the Grand Ole Opry get its name?

George D. Hay named the Grand Ole Opry on the air on December 10, 1927, right after a Black harmonica player, DeFord Bailey, played his "Pan American Blues." Hay had started the show on November 28, 1925, on WSM, a Nashville radio station owned by the National Life and Accident Insurance Company. He first called it the *WSM Barn Dance*. The first player was a 77-year-old fiddler, Uncle Jimmy Thompson.

That December night the *Barn Dance* came on the air after a program of opera music. After Bailey played, Hay told listeners, "from now on, we will present 'The Grand Ole Opry.'" One hour of the show went out across the country on the NBC radio network from 1939 to 1956.

DeFord Bailey

Who: A Black harmonica player, for fifteen years one of the best-loved stars of the Grand Ole Opry.

When and where: Born December 14, 1899, in Smith County, Tennessee. Played on Nashville radio from 1926 to 1941. Died July 2, 1982.

Movie: *DeFord Bailey: A Legend Lost* (2002), a Nashville Public Television documentary about him.

White stars got him into hotels by the fire escape.

DeFord Bailey's family in Smith County, Tennessee, played what he called "Black hillbilly music." He caught polio at three. The virus that causes polio can damage the nerves that move the muscles, and it stunted Bailey's growth and left his back bent.

The pay. He played on the *WSM Barn Dance* from June 1926. Accounts of his pay differ. One says he got a fixed $5 at first, less than the white players he toured with. The Country Music Hall of Fame says he was one of the best-paid stars on the show.

On tour. He toured the South with white stars such as Roy Acuff, Bill Monroe and Uncle Dave Macon. Hotel and restaurant owners would not serve him because he was Black. The other players brought him food, or

got him into hotels by the fire escape.

Fired. The managers of most radio stations stopped playing songs licensed by ASCAP, a group that collected money for songwriters, during a fight over fees in 1941. Managers at WSM told Bailey not to play his ASCAP songs, which were some of his most popular ones. They fired him at the end of 1941.

"Lazy." Hay wrote a book about the show in 1945. In it he wrote, "Like some members of his race and other races, DeFord was lazy." Hay added that Bailey knew about a dozen tunes and "refused to learn more."

The historian Charles Wolfe reported in 1999 that Bailey remembered it the other way around. Bailey said he wanted to play new tunes, but Hay and the listeners kept asking for the same ones. Bailey also said that Hay had treated him kindly and had written those words to please his bosses.

Bailey ran a shoeshine stand and rented out rooms after 1941. He almost never played in public again until 1974. In 2005 he was named to the Country Music Hall of Fame.

Radio, the Depression and two fights over pay, 1906 to 1944

John Philip Sousa wrote in 1906 that records and player pianos were becoming a "substitute for human skill, intelligence and soul." He called records "canned music." Members of Congress passed a copyright law in 1909 that let anyone record a song without asking, as long as they paid the song's owner a set fee.

Sales up and down. Americans spent $106 million on records in 1921. A history of records from the University of Michigan Press says they did not spend that much again until 1945. IASA, a group of sound archivists, says sales did not reach their old peak until the 1950s. During the Great Depression, in the early 1930s, record sales fell to about one tenth of what they had been in the 1920s.

ASCAP and BMI. Radio station owners paid ASCAP a fee for the right to play songs by its songwriters, and ASCAP passed the money on to the writers. When ASCAP's leaders announced a big rise in fees in 1939, radio station owners started a group of their own, BMI, so they would have songs to play without paying ASCAP.

Managers at almost every station, and at all three national networks, stopped playing ASCAP songs in 1941. ASCAP's leaders had not wanted writers of blues, gospel, country, folk and Latin music, and BMI became the first such group to sign them.

The record ban. James Petrillo, the head of the musicians' union, told its members to stop making records on August 1, 1942. He wanted musicians to be paid when radio stations and jukeboxes played records instead of hiring live bands. Decca's managers signed an agreement in September 1943 to pay a fee on every record into a union fund. Victor and Columbia signed on November 11, 1944. Union rules still let singers record with voices alone, so for two years singers made records without bands.

Gospel, shape notes and prison recordings

Gospel is a kind of Black church music. Thomas A. Dorsey had played piano for Ma Rainey under the name "Georgia Tom." His wife, Nettie, and their newborn son died in August 1932, and he wrote "Take My Hand, Precious Lord." That year he started the first Black-owned company to publish gospel music, and in 1933 a national convention of gospel choirs. He led the music at Pilgrim Baptist Church in Chicago for fifty years. "I'm not ashamed of my blues," he said. "It's all the same talent."

Mahalia Jackson. She came to Chicago from New Orleans in 1927, and Dorsey trained her. She refused to sing the blues. Managers at Decca offered to keep recording her if she sang blues, and she refused. Her "Move On Up a Little Higher" of 1947 sold about two million copies.

The Colored Sacred Harp. Judge Jackson, a shape-note singer from Alabama, printed *The Colored Sacred Harp* in 1934. Judge was his first name, and "Colored" in the title was a word of that time for Black people. It was a book of 77 church songs printed in shape notes, notes whose shapes tell the singer which sound to sing. Eighteen of the songs were his own. He and Bishop J. D. Walker paid to print 1,000 copies, and Jackson sold them door to door.

Lead Belly. John Lomax and his son Alan recorded singers in Southern prisons from June 1933. John Lomax said he wanted music "not yet influenced by jazz and the radio." From 1934 he ran the Library of Congress's folk song collection, for a salary of one dollar. At Angola, the Louisiana state prison, they recorded Huddie Ledbetter, called Lead Belly. He was serving time for stabbing a man in a fight.

Years before, Lead Belly had been convicted in Texas of killing a relative, Will Stafford. He was in prison there from 1918 until the governor of Texas, Pat Neff, pardoned him in 1925, after Lead Belly sang him a song asking to be freed. He left Angola on August 1, 1934. A prison official later wrote that his singing had nothing to do with that release.

"Goodnight, Irene." John Lomax then managed Lead Belly and took a share of his pay. The two men split over money in 1935. Lomax published a book in 1936 that printed Lead Belly's song "Goodnight, Irene" with both men named as its writers. Lomax had recorded the song. He had not written it. A group called the Weavers sold 2 million copies of "Goodnight, Irene" in 1950, the year after Lead Belly died.

Native songs, Hawaiian guitar and a new songwriter

The band of the Carlisle Indian School, a boarding school for Native children in Pennsylvania, played the "Aboriginal Suite" at Carnegie Hall in New York on March 28, 1900. Its composer and conductor was Dennison Wheelock, an Oneida graduate of the school. Richard Henry Pratt had made him the Carlisle bandmaster in 1892, the first Native bandmaster there. Wheelock later became a lawyer and argued cases for Native nations before the Supreme Court.

Songs on wax. Scientists recorded thousands of Native songs in these years on wax cylinders, tubes of wax that each held two or three minutes of sound. The Library of Congress holds about 10,000 cylinders, and nearly 8,000 of them hold Native songs and speech. Many held sacred songs meant only for ceremonies.

One collector, Francis La Flesche, took singers from Oklahoma to Washington, D.C., so that neighbors who objected could not stop him from recording ceremonial songs. No record shows that the singers were paid. Frances Densmore recorded the songs of the Ojibwe, Sioux, Mandan, Hidatsa, Tohono O'odham, Seminole and other nations for the Smithsonian Institution from 1907.

The steel guitar. Joseph Kekuku, a Hawaiian student at the Kamehameha School for Boys, found in 1889 that he could change a guitar's pitch by sliding a steel bar along the strings. From 1904 he played his steel guitar in theaters across the mainland United States.

By then Hawaii belonged to the United States. A group of mostly American businessmen had overthrown Hawaii's queen, Lili‘uokalani, in 1893. Before they moved, John L. Stevens, the top US official in Hawaii, had 162 American sailors and Marines brought ashore. Native Hawaiians put 21,269 signatures

on petitions, written requests to the government, against joining the United States in 1897. Members of Congress voted to annex Hawaii anyway in 1898, making it part of the United States.

A new songwriter. Irving Berlin, born Israel Beilin, came to New York from the Russian Empire in 1893, at age five. He could not read music and could play the piano in only one key, and he wrote "Alexander's Ragtime Band" in 1911. Through his own publishing company he owned his own songs. He gave all the money from "God Bless America," which Kate Smith first sang in 1938, to the Boy Scouts and Girl Scouts.

BY 1939: Anyone with a radio could hear music at home for free, and record sales were still below their 1921 peak. That year one hour of the Grand Ole Opry began going out across the country on NBC.

Woody Guthrie

Who: A folk singer and songwriter who wrote about poor farm families, workers and migrants in the 1930s and 1940s.

When and where: Born July 14, 1912, in Okemah, Oklahoma. Lived in Oklahoma, California and New York. Died October 3, 1967, in New York.

Movie: *Bound for Glory* (1976), a Hollywood film based loosely on his own book about his life.

On his guitar he wrote "This machine kills fascists."

Woody Guthrie wrote "This Land Is Your Land" and many other songs about poor farm families and workers. His mother, Nora, had Huntington's disease and was put in a hospital. No account names who sent her there. Huntington's is a disease passed down in families that destroys parts of the brain. A person with it makes movements they cannot control, has changes in mood, and loses the ability to think.

The Dust Bowl. In the 1930s Guthrie joined the farm families who left the Great Plains for California after years of drought, the time called the Dust Bowl. He sang on a Los Angeles radio station, KFVD. In 1940 he made an album called *Dust Bowl Ballads* for Victor.

An answer to Berlin. He wrote "This Land Is Your Land" in a New York hotel on February 23, 1940, as an answer to Irving Berlin's "God Bless America," which Kate Smith sang on the radio. His first title was "God Blessed America for Me." His handwritten copy had two verses that were often left out later. One is about a sign that says "No Trespassing." The other is about hungry people at a relief office, a place that gave out food or money to the poor.

Twenty-six songs in a month. Paul Raver, the head of a federal agency called the Bonneville Power Administration, hired him for one month in May 1941 for $266. Guthrie's boss there, Stephen Kahn, wanted three pages of songs a day for a film about the agency's dams on the Columbia River. Guthrie wrote 26 songs in that month. One of them was "Roll On, Columbia." Kahn later said the government had never gotten "a better investment on its money."

His last years. The fascists of his guitar's slogan were the followers of the dictators who ruled Italy and Germany. In the Second World War he served in the Merchant Marine, the ships that carried supplies. Doctors found in 1952 that he had Huntington's disease. He spent his last years in hospitals in New Jersey and New York, and he died at Creedmoor hospital in New York in 1967, at 55.

Strange Fruit, 1939

"Strange Fruit" is a song about the lynching of Black Americans. Lynching means a mob killing a person without a trial, often by hanging. The song describes the bodies of Black people hanging from trees in the South: their bulging eyes and twisted mouths, and the smell of burning flesh.

Abel Meeropol, an English teacher in the Bronx, New York, wrote the song under the name Lewis Allan. He wrote it after he saw a photograph of the lynching of Thomas Shipp and Abram Smith in Marion, Indiana.

Marion, Indiana. A white mob of between 5,000 and 10,000 people broke into the jail in Marion on August 7, 1930. The mob wanted three young Black men held there for the murder of a white man, Claude Deeter, and the rape of a woman, Mary Ball. Rape means forcing a person to have sex. Men in the mob beat Shipp and Smith, who were both 19, and hanged them. People in the crowd took parts of their bodies as souvenirs.

James Cameron. The third prisoner, James Cameron, who was 16, was beaten but lived, after someone in the crowd said he was innocent. Cameron waited about a year in jail for his trial, and accounts of what Mary Ball said there differ. The historian James H. Madison writes that she said she had been raped but that the night was too dark to tell whether Cameron was one of the men. Fran Kaplan of America's Black Holocaust Museum writes that Ball testified she had not been raped at all.

Cameron was convicted on a lesser charge, of helping before the crime, and a judge sentenced him to 2 to 21 years. He served four years in prison. The governor of Indiana pardoned him in 1993.

The trials. Leaders of the NAACP, a group that fights for the rights of Black Americans, named 27 people they suspected of the lynching. All-white juries found the mob's leaders not guilty.

Café Society. Billie Holiday first sang "Strange Fruit" at Café Society, a club in New York, in 1939, and she closed her show with it. The waiters stopped serving, the room went dark except for a light on her face, and she sang no more songs after it. Managers at Columbia would not record it. John

Hammond of Columbia said record stores and radio stations in the South would object. Milt Gabler of Commodore Records recorded it on April 20, 1939. It became her best-selling record.

Billie Holiday

Who: A Black jazz singer who recorded "Strange Fruit," a song about lynching, in 1939.

When and where: Born Eleanora Fagan on April 7, 1915, in Philadelphia. Grew up in Baltimore. Sang in New York. Died July 17, 1959, in New York.

Movie: *Billie* (2019), a documentary about her.

Federal agents handcuffed her to her hospital bed as she was dying.

Billie Holiday was a Black jazz singer whom federal drug agents arrested three times, the last time in her hospital bed. She was born Eleanora Fagan in Philadelphia and grew up in Baltimore.

At nine she was brought before a juvenile court, a court for children, for missing school. The judge sent her to the House of the Good Shepherd, a Catholic reform school for girls, where courts sent children who were in trouble. As a young girl she ran errands in a brothel, a house where men paid women for sex, and she scrubbed steps and floors in the homes of her neighborhood.

The service elevator. She sang with Artie Shaw's band in 1938, and its other members were white. At the Lincoln Hotel in New York, the managers made her use the service elevator because white guests complained. She left the band soon after.

Anslinger's agents. Harry Anslinger ran the Federal Bureau of Narcotics, the government's drug police, from 1930 to 1962. Federal agents arrested Holiday on May 16, 1947, for having narcotics, illegal drugs, in her New York apartment. She pleaded guilty, and a judge sent

her to the Alderson Federal Prison Camp in West Virginia for a year. She said later: "It was called 'The United States of America versus Billie Holiday'. And that's just the way it felt."

Because of the conviction, she lost her New York City cabaret card, the police license a performer needed to work in any club that sold alcohol. After that she sang in concert halls and theaters. A federal agent named George White, a different man from the Fisk Jubilee Singers' leader, arrested her again in a San Francisco hotel in January 1949.

DISPUTED: No source before 2015 says that Anslinger's agents went after her because she sang "Strange Fruit," the jazz writer Lewis Porter points out. The writer Johann Hari wrote in 2015 that they did, from 1939.

Her last arrest. She was in Metropolitan Hospital in New York with liver and heart disease in 1959. Federal narcotics agents arrested her in her hospital bed on May 31 and handcuffed her to it. Writing in *TIME* in 2019, John Halpern and David Blistein said that Anslinger ordered the arrest. Police guarded her room until a judge ordered the handcuffs taken off. She died there on July 17, 1959, at 44, with 70 cents in the bank. Her liver was scarred, a disease called cirrhosis, and her heart failed.

Marian Anderson at the Lincoln Memorial, 1939

Officials of Howard University in Washington, D.C., asked in January 1939 to rent Constitution Hall for an Easter concert by the Black singer Marian Anderson. The hall belonged to the Daughters of the American Revolution, or DAR, a women's group. Its leaders refused. DAR leaders had let only white performers use the hall since 1932. Frank Ballou, the head of Washington's public schools, also refused to let her use the hall of Central High School, a white school, and the members of the school board backed him.

The Lincoln Memorial. First Lady Eleanor Roosevelt quit the DAR on February 26, 1939. She wrote that its leaders had "an opportunity to lead in an enlightened way" and had "failed." Harold Ickes, the Secretary of the Interior, then set up a concert on the steps of the Lincoln Memorial. Anderson sang there on Easter Sunday, April 9, 1939, to more than 75,000 people, and millions more heard her on the radio. She began with "My Country, 'Tis of Thee."

Marian Anderson

Who: A Black contralto, a singer with a low woman's voice, who sang at the Lincoln Memorial in 1939 and was the first Black American singer at the Metropolitan Opera.

When and where: Born February 27, 1897, in Philadelphia. Died April 8, 1993, in Portland, Oregon.

Movie: *Marian Anderson: The Lincoln Memorial Concert* (1939), a short film of the concert.

Toscanini told her she had "a voice heard once in a hundred years."

As a young woman, Marian Anderson applied to a music school in Philadelphia. The school's officials turned her away because they took only white students. In Salzburg, Austria, in 1935, the conductor Arturo Toscanini told her she had "a voice heard once in a hundred years."

Constitution Hall. The DAR's president in 1939, when its leaders kept Anderson out of the hall, was Sarah Corbin Robert.

The Met. She sang at the Metropolitan Opera on January 7, 1955, at 57, in the part of Ulrica, a fortune teller, in Verdi's opera *Un ballo in maschera*. She sang at the March on Washington in August 1963. That year she was given the Presidential Medal of Freedom, an award given by the president.

BY 1950

If you remember one thing: In these fifty years record companies and radio stations brought the blues, jazz, gospel and country music to listeners across the country, and the companies and copyright holders kept much of the money.

One number: $125. Columbia paid Bessie Smith that much for each side of "Down Hearted Blues," a record that sold 780,000 copies in six months, and it paid her no royalty on any of them.

Next: After 1950 Black musicians made rock and roll out of rhythm and blues and gospel, and white singers sold more copies of some of their songs than they did.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 22 MINUTES

Black musicians made rock and roll, soul and hip-hop. White singers then sold more copies of some of those songs than the Black singers who recorded them first. Record company owners, managers and publishers kept much of the money from songs such as "Hound Dog" and "That's All Right."

In Detroit, Berry Gordy built Motown, which became the largest Black-owned business on one magazine's list. Civil rights marchers sang old church songs with new words at their mass meetings.

A teenage DJ and his sister held a party in the Bronx, in New York City, in 1973 that people now name as the start of hip-hop. Senators held a hearing on the words of songs in 1985, and leaders of the RIAA, the record companies' own group, agreed to put warning labels on some albums.

Rock and roll, and who got paid for it

Sister Rosetta Tharpe, a Black gospel singer from Arkansas, played electric guitar in church music from the late 1930s. Her "Strange Things Happening Every Day," from 1944, was the first gospel record to reach the rhythm and blues sales chart. Some scholars call it the first rock and roll record. She was an influence on Elvis Presley, Little Richard, Chuck Berry and Johnny Cash.

Credit taken. Fats Domino's "The Fat Man," from 1949, had sold a million copies by 1951. Chuck Berry's "Maybellene," from 1955, sold more than a million. When Berry first saw his record, it listed Alan Freed, a disc jockey who played records on the radio, and another man as its co-writers. That gave them a right to part of the money the song earned its writers. Berry later won back full credit.

Cover versions. White singers made cover versions of songs by Black singers, new recordings of songs that someone else had already recorded. Georgia Gibbs, a white singer, copied LaVern Baker's "Tweedle Dee" note for note in 1955. Baker's record reached No. 14 on the national pop chart, and Gibbs's copy reached No. 1.

Baker tried to sue Gibbs for $250,000. She asked Congress to treat such copies as a breach of copyright. Representative Charles Diggs led an investigation by Congress into what he called "song theft." No law came of it.

Pat Boone, a white singer, covered Fats Domino's "Ain't That a Shame" in 1955. It was Boone's first No. 1 record. Domino and his co-writer, Dave Bartholomew, still earned writers' royalties from it. Boone also covered Little Richard's "Tutti Frutti," and Boone's record ranked higher on the national pop chart than Little Richard's own.

Little Richard's lawsuit. Little Richard later said that the people running his record company, Specialty Records, had cut his royalty rate without telling him. He sued the company and its owner, Art Rupe, in 1984 for $112 million in royalties he said he was owed. They settled out of court in 1986, which means they reached an agreement and the case never went to trial.

Payola. Members of a House committee, led by Representative Oren Harris, held hearings in 1959 and 1960 on payola, secret payments to a disc jockey or a radio station to play a record. Alan Freed refused to sign a statement that he had taken no bribes, and the managers of the radio station WABC fired him in November 1959.

Members of Congress made payola a crime in 1960, punished by up to a year in jail. Freed told a court in 1962 that he was guilty of taking bribes and paid a $300 fine. The older song publishers in ASCAP pushed for the hearings, partly because they opposed rock and roll.

Hound Dog: Big Mama Thornton and Elvis Presley

The songwriters Jerry Leiber and Mike Stoller wrote "Hound Dog." The chapter opening tells the start of this story: Big Mama Thornton's 1952 record, Don Robey's name on the copyright form, and her one check for $500. Robey's Peacock Records put her record out in 1953, and it stayed at No. 1 on the rhythm and blues chart for seven weeks. Robey guessed that it sold 500,000 to 750,000 copies. Thornton said, "That song sold over two million records."

Presley's record. Elvis Presley recorded "Hound Dog" for RCA on July 2, 1956. His record was No. 1 on the pop chart for 11 weeks. He followed a reworked version by Freddie Bell and the Bellboys, a band he had heard in Las Vegas that year. Leiber and Stoller earned large royalties only after Presley's record. Thornton died of heart and liver disease in 1984. She was buried in a grave shared with two other people.

Elvis Presley

Who: A white rock and roll singer from Mississippi and Tennessee whose records sold an estimated 500 million copies.

When and where: Born January 8, 1935, in Tupelo, Mississippi. Moved to Memphis, Tennessee, in 1948. Died August 16, 1977, at his Memphis home, Graceland, at 42.

Movie: *Elvis* (2022), a Hollywood drama film about him and his manager.

His manager took 25 percent of his pay, and later 50.

Elvis Presley recorded a Black singer's blues song at Sun Records in 1954, and his "Hound Dog" followed a song that had been a hit for a Black singer first. He first heard music at his family's church, an Assembly of God church. In Memphis he listened to Black blues and rhythm and blues singers on the radio.

Sun Records. Sam Phillips owned Sun Records in Memphis and recorded Black blues musicians such as Howlin' Wolf and B. B. King. His office manager, Marion Keisker, said he often told her: "If I could find a white man who had the Negro sound and the Negro feel, I could make a billion dollars." "Negro" was a word of that time for Black people. Presley sang "That's All Right," a blues song by the Black singer Arthur Crudup, at Sun on the night of July 5, 1954, and Phillips recorded it.

Arthur Crudup. Crudup had recorded the song for RCA in 1946. He said later: "I realised I was making everybody rich, and here I was poor." In 1968 a publisher backed out of a $60,000 settlement with him at the last moment. In 1971 a songwriters' group got him more than $10,000 in royalties he had not been paid. He died in 1974.

Colonel Parker's share. Phillips and Presley's manager, Colonel Tom Parker, sold Presley's contract to RCA for $40,000 in November 1955. Parker took 25 percent of Presley's pay. Presley agreed in 1967 to raise Parker's share to 50 percent of some of his income. Parker sold RCA the rights to all of Presley's old records in 1973 for $5.4 million. Because of that sale, Presley's family got no royalties on those records after he died.

> Blanchard Tual, a lawyer whom a judge had appointed, called the 50 percent share "extortionate" in 1981, meaning unfairly high. He called the 1973 sale "unethical and poorly handled." Presley served in the Army from 1958 to 1960. His record sales are estimated at about 500 million copies around the world. He died at Graceland on August 16, 1977, at 42.

Motown, soul and gospel

Berry Gordy Jr. of Detroit started a record company on January 12, 1959, with an $800 loan from his family's savings fund. It was later called Motown. From 1960 to 1969, 79 Motown songs reached the top ten of the national pop chart. The editors of *Black Enterprise* magazine began ranking the largest Black-owned businesses in 1973. Motown was first on that list, with $40 million in sales, and it stayed first until 1983. Gordy sold Motown in 1988 for $61 million.

The Funk Brothers. A group of Detroit studio musicians, later called the Funk Brothers, played on most Motown hits from 1959 to 1972. No Motown record listed their names until Marvin Gaye's album *What's Going On* in 1971. Gordy fined any of them he caught recording for other companies.

Ray Charles. He had been blind since he was about seven. In "I Got a Woman," from 1954, he mixed gospel with the blues, and some church people objected. He refused to play a dance in Augusta, Georgia, in 1961 when he learned that the audience would be split by race. The promoter sued him, and in 1962 a judge fined him $757. His 1959 contract with ABC-Paramount promised that he would own his master recordings, the original tapes that every copy is made from, in time.

Sam Cooke. A singer from Mississippi who had led the gospel group the Soul Stirrers, Cooke started his own record company and his own company to publish his songs. From 1963 his own company owned his master recordings, and he was among the first Black artists to control his own business.

A Los Angeles motel manager, Bertha Franklin, shot and killed him on December 11, 1964. A coroner's jury, a panel that decides how a person died, called it justifiable homicide, a killing the law excuses. His family has disputed

that account.

James Brown. Brown paid for the recording of his 1963 album *Live at the Apollo* himself. Martin Luther King Jr. was shot and killed on April 4, 1968, and James Earl Ray later pleaded guilty to his murder. The next day, Brown gave a free concert in Boston that was shown on television. His song "Say It Loud, I'm Black and I'm Proud" came out that August.

BY 1970: Motown had put 79 songs in the pop top ten, Sam Cooke's own company owned his master recordings, and Ray Charles had a contract that promised him his.

Aretha Franklin

Who: A Black singer who recorded "Respect" in 1967.

When and where: Born March 25, 1942, in Memphis, Tennessee. Grew up in Detroit, Michigan. Died there August 16, 2018.

Movie: *Amazing Grace* (2018), a documentary film of her 1972 gospel concert in a Los Angeles church.

She spelled it out: R-E-S-P-E-C-T.

Aretha Franklin took Otis Redding's song "Respect" and turned it into a woman's demand, and it went to No. 1 on the pop chart in 1967. Her father, the Reverend C. L. Franklin, led New Bethel Baptist Church in Detroit, and she sang there as a child. At 14 she recorded a gospel album, *Songs of Faith*. She had her first child at 12 and her second at 15.

She recorded for Columbia from 1960 to 1966 and had few hits. Jerry Wexler then brought her to Atlantic Records. In January 1967 she recorded for one day at FAME Studios in Muscle Shoals, Alabama, before a fight broke out in the studio.

"Respect." Redding had written and recorded "Respect" in 1965. In his song a man asks for respect when he comes home. Franklin recorded it in New York on February 14, 1967. She and her sister Carolyn made it a

woman's demand. They spelled out the word "R-E-S-P-E-C-T" and added new lines.

It was No. 1 on the pop chart for two weeks and on the rhythm and blues chart for eight. It became a song of the women's movement and of the civil rights movement. Redding called it the song "that a girl took away from me."

Angela Davis. Franklin spoke up in 1970 for Angela Davis, a Black activist whom FBI agents had arrested that October. Davis was charged with kidnapping and with the murder of Judge Harold Haley, who was killed in an attack on a California courthouse. Franklin said, "Angela Davis must go free." An all-white jury found Davis not guilty in June 1972.

Amazing Grace. Franklin recorded the gospel album *Amazing Grace* at New Temple Missionary Baptist Church in Los Angeles in January 1972. It sold more than two million copies. The film made at the same time was not shown until 2018, because the sound and the pictures could not be matched. In 1987 she became the first woman named to the Rock and Roll Hall of Fame. She died in Detroit on August 16, 2018.

Where did 'We Shall Overcome' come from?

"We Shall Overcome" started as a church hymn, "I'll Overcome Some Day," that Charles Albert Tindley wrote in 1901. Tobacco workers went on strike at a cigar factory in Charleston, South Carolina, in 1945 and 1946. One of them, Lucille Simmons, led the strikers in singing "We Will Overcome." Zilphia Horton of the Highlander Folk School learned it from her and printed it in 1947.

The folk singer Pete Seeger changed "will" to "shall," because, he said, "'We shall' opens the mouth wider." From 1959 Guy Carawan taught the song to young civil rights workers.

The March on Washington. Joan Baez led "We Shall Overcome" at the March on Washington on August 28, 1963. Mahalia Jackson, Marian Anderson, Odetta, Bob Dylan, and Peter, Paul and Mary sang there too. During Martin Luther King Jr.'s speech, Jackson called out to him, "Tell them about the dream, Martin!"

Selma. About 600 marchers set out from Selma, Alabama, on March 7, 1965. At the Edmund Pettus Bridge, Alabama state troopers and other officers, led by Major John Cloud, attacked them with nightsticks and tear gas. At least 50 marchers went to the hospital. President Lyndon Johnson spoke to Congress on March 15 and used the song's words: "And we shall overcome." In January 2018, after a court case, the owners of the song's copyright gave it up. Since then anyone may use it freely.

Bernice Johnson Reagon

Who: A Black student in Albany, Georgia, who led singing at civil rights meetings and sang with the SNCC Freedom Singers.

When and where: Born October 4, 1942, in Albany, Georgia. Died July 16, 2024.

"So instead I put in freedom," she said of an old song.

Bernice Johnson Reagon led the singing at civil rights meetings in Albany, Georgia, and changed the words of old spirituals to fit the protests. She joined the NAACP Youth Council in Albany. Officials at Albany State College suspended her for her part in the protests.

At mass meetings she took old spirituals and put new words in them. Of one song she said: "So instead I put in freedom and by the second line everyone was singing."

The Freedom Singers. She joined the Freedom Singers of SNCC, the Student Nonviolent Coordinating Committee, with Cordell Reagon, Rutha Mae Harris and Chuck Neblett. She married Cordell Reagon, who had started the group. The group traveled 50,000 miles in nine months to sing about the movement, without taking a single flight, and its first tour raised nearly $50,000 for SNCC. "There weren't soloists, there were song leaders," she said. She called the group "a singing newspaper."

Sweet Honey in the Rock. She started Sweet Honey in the Rock in 1973, a group of African American women who sing without instruments. She later worked for the Smithsonian Institution as a curator, a person in charge of a museum's collections. Through radio, film and concerts, she helped people study the songs and stories that Black Americans passed down by word of mouth. She died on July 16, 2024.

Bob Dylan

Who: A songwriter of the folk music revival of the early 1960s and after, and winner of the Nobel Prize in Literature.

When and where: Born Robert Allen Zimmerman on May 24, 1941, in Duluth, Minnesota. Grew up in Hibbing, Minnesota. Moved to New York in 1961.

Movie: *No Direction Home: Bob Dylan* (2005), a documentary by Martin Scorsese.

At the 1965 Newport Folk Festival he played an electric guitar.

Bob Dylan won the Nobel Prize in Literature, a world prize for writers, in 2016 for his songs. The prize was "for having created new poetic expressions within the great American song tradition."

He moved to New York in January 1961 and visited Woody Guthrie, who was then in a mental hospital in New Jersey with Huntington's disease. He released the album *The Freewheelin' Bob Dylan* in 1963. At the March on Washington that August he sang "Only a Pawn in Their Game."

Newport. He played an electric guitar at the Newport Folk Festival in 1965. The leaders of the folk music world met him with hostility.

Country music from Nashville and Bakersfield

Hank Williams learned guitar as a boy in Alabama from a Black street musician, Rufus "Tee-Tot" Payne, who taught him in return for money or meals. Williams had 12 No. 1 country records. He was born with a defect of the spine that caused him pain all his life, and he drank and took painkillers. Grand Ole Opry managers dropped him in August 1952 for drunkenness and missed shows. He died in the back seat of a car near Oak Hill, West Virginia, on January 1, 1953, at 29.

Patsy Cline recorded "Crazy," a song by Willie Nelson, in 1961. She died in a plane crash near Camden, Tennessee, on March 5, 1963, at 30. She became the first woman in the Country Music Hall of Fame in 1973.

Bakersfield. Buck Owens, Merle Haggard and other musicians in Bakersfield, California, played country music on electric guitars in the late 1950s, often two Fender Telecasters. Nashville records of those years added orchestras and a polished sound, and the Bakersfield musicians broke from it. Many of their listeners were the families of people who had come to California in the Dust Bowl years.

Johnny Cash recorded two concerts for prisoners at Folsom State Prison in California on January 13, 1968. The album ends with "Greystone Chapel," a song by a prisoner, Glen Sherley. It sold 500,000 copies by October 1968. Cash himself never served a prison sentence.

"The Pill." Loretta Lynn's song of 1975 is about a wife who is tired of being pregnant and starts taking birth-control pills. The people running a number of country radio stations refused to play it. It still reached No. 5 on the country chart.

Charley Pride. He became the first Black performer on the Grand Ole Opry in 1967 since DeFord Bailey last played there in 1941. Pride was one of eleven children of sharecroppers from Sledge, Mississippi, families who farmed another man's land for a share of the crop. His first records went to radio stations without his photograph. He had 30 No. 1 country records, and in 1971 he won the Country Music Association's award for Entertainer of the Year.

Hip-hop from the Bronx, 1973 onward

Clive Campbell, a teenage DJ called Kool Herc, played records for dancers at a party at 1520 Sedgwick Avenue in the Bronx, New York, on August 11, 1973. His sister Cindy Campbell planned the party to earn money for back-to-school clothes. City officials named the building the birthplace of hip-hop in 2007.

The fires. The first hip-hop DJs played in the Bronx in the same years that fires burned much of the South Bronx. Most of the fires were arson, fires set on purpose. Landlords paid people who lived nearby to set fires in the landlords' own buildings, so the landlords could collect insurance money. One landlord collected $125,000 from 21 fires.

After a study of the fire service by the RAND Corporation, a research company, city officials closed 12 fire companies in the South Bronx between 1974 and 1976, when the city was nearly out of money. Seven census tracts, small areas the census counts separately, lost more than 97 percent of their buildings to fire and abandonment. About 250,000 people lost their homes.

"Rapper's Delight." The Sugarhill Gang's "Rapper's Delight" came out on Sugar Hill Records on September 16, 1979. Sylvia Robinson produced it and ran the company's work. It reached No. 36 on the national pop chart, and many writers call it the first rap record that sold widely. It used the music of Chic's song "Good Times." Chic's writers, Nile Rodgers and Bernard Edwards, got credit only after they threatened to sue. One of its rappers, Big Bank Hank, used rhymes written by Grandmaster Caz and did not credit him.

"The Message." The 1982 song by Grandmaster Flash and the Furious Five described poverty in the inner city. Ed "Duke Bootee" Fletcher, Melle Mel, Clifton "Jiggs" Chase and Sylvia Robinson wrote it. In 2002 it became the first hip-hop record added to the National Recording Registry, the Library of Congress's list of recordings chosen to be kept for the future.

DJ Kool Herc

Who: Born Clive Campbell in Jamaica, the DJ at the 1973 Bronx party that people name as the start of hip-hop.

When and where: Born April 16, 1955, in Kingston, Jamaica. Moved to the Bronx, New York, in November 1967, at 13.

He played two copies of one record to stretch the drums.

The Merry-Go-Round. At parties he played funk records. The "break" of a funk record is its drum part. Herc used two copies of the same record. When the break ended on one, he started it again on the other, so the drum part lasted longer. He called this the "Merry-Go-Round." He called the dancers who danced to the breaks "b-boys" and "b-girls."

Later someone stabbed Herc at a club called the Executive Playhouse while he was trying to stop a fight. No record names who stabbed him. After that he pulled back from playing, and by 1980 he had stopped working as a DJ. No surviving document shows what he earned from hip-hop.

Woodstock, music television, compact discs and the lyrics hearing

Thirty-two acts played at the Woodstock music festival, on Max Yasgur's dairy farm in Bethel, New York, from August 15 to 18, 1969. The organizers expected about 50,000 people. Estimates of the crowd run from 400,000 to more than 460,000, and no record names who counted. The fences were not finished, so the festival became free. Three people died. Two died from taking too much of a drug, and a tractor ran over a 17-year-old as he slept. No record names the driver.

MTV. The MTV television channel began playing music videos on August 1, 1981. Its managers said they kept to a "rock format," and they played few Black artists. Walter Yetnikoff, president of CBS Records, threatened to pull CBS's videos unless MTV's managers played Black artists. They then put Michael Jackson's "Billie Jean" into heavy rotation from March 1983, which means they played it often. Compact disc players went on sale in Japan on October 1, 1982, and in the United States in March 1983.

The lyrics hearing. Tipper Gore, Susan Baker and two other women in Washington started a group called the Parents Music Resource Center in 1985. They made a list of 15 songs they objected to and called it the "Filthy Fifteen." It included Prince's "Darling Nikki." Members of the Senate Commerce Committee held a hearing on September 19, 1985. The musicians Frank Zappa, Dee Snider and John Denver spoke against labels. Zappa called the plan "an ill-conceived piece of nonsense," a badly thought-out idea.

Leaders of the RIAA, the record companies' own group, agreed on November 1, 1985, to put warning labels on some albums. Record companies began putting a standard "Parental Advisory" sticker on albums in 1990. Managers at Walmart and some other stores would not sell albums with the sticker.

The 1978 act. Many Native songs belong to religious ceremonies. Until 1978, federal officials could still enforce the rules against Native religious dances that Hiram Price, the head of the government's Indian office, had issued in 1883. That year members of Congress passed the American Indian Religious

Freedom Act. In it they wrote that Native "ceremonies have been intruded upon, interfered with, and in a few instances banned." The act does not name the people who did these things.

Selena Quintanilla

Who: A Mexican American singer of Tejano music, the Spanish-language music of Texas.

When and where: Born April 16, 1971, in Lake Jackson, Texas. Killed March 31, 1995, in Corpus Christi, Texas, at 23.

Movie: *Selena* (1997), a Hollywood drama film about her, with Jennifer Lopez.

The woman who ran her fan club shot her at 23.

Selena Quintanilla sang with her brother and sister in their father Abraham Quintanilla Jr.'s band, Selena y Los Dinos, and learned the Spanish words of her songs by their sound, with her father's help. She won the Tejano Music Award for Female Vocalist of the Year in 1987 and in each of the next nine years. In 1994 she won a Grammy Award for her album *Live!*

Yolanda Saldívar. Saldívar ran Selena's fan club and managed her clothing stores. She had secretly taken money from the business, more than $30,000 or more than $60,000 by different counts. Selena meant to fire her.

The shooting. Saldívar shot Selena with a gun at a Days Inn motel in Corpus Christi, Texas, on March 31, 1995, as Selena tried to leave the room. The bullet hit her lower right shoulder. Selena reached the lobby and named Saldívar. She died at the hospital that afternoon from loss of blood. Saldívar gave herself up to the police more than nine hours later.

The trial. A jury in Houston found Saldívar guilty of murder with a deadly weapon on October 23, 1995, and Judge Mike Westergren sentenced her to life in prison. Members of the Texas parole board, who decide whether a prisoner may leave prison early, refused to let her out on March 27, 2025. They set her next review for March 2030.

> Between 30,000 and 40,000 people came to see Selena's coffin. Her album *Dreaming of You* came out after her death. It made her the first Latin artist whose album entered the national album chart at No. 1.

BY 2000

If you remember one thing: Black musicians made rock and roll, soul and hip-hop in these fifty years, and the owners of the copyrights and the master recordings kept much of the money.

Left open: Who would own the recordings, the companies or the singers who made them? Sam Cooke's company owned his masters from 1963, and Ray Charles had a contract that promised him his. After 2000 one singer lost hers to a buyer and recorded four albums again.

Next: After 2000 Americans moved from buying records to streaming them, and a stream paid the owner of the recording a fraction of a cent.

2000 to Today

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Most Americans now stream their music, playing songs over the internet without buying copies, and a streaming service pays the owner of each recording a fraction of a cent a play. By 2025, 82 cents of every dollar that American record companies took in came from streaming.

Sales of recorded music in the United States had peaked in 1999, when most people bought CDs. Then a free program called Napster let people copy songs from each other's computers, and by the record companies' own count, sales fell in 15 of the 16 years from 2000 to 2015.

A streaming service pays the owner of the recording, usually a record company, and not the singer. In a 2018 survey, most working musicians said they earned money from playing live. Dolly Parton, a songwriter from the mountains of Tennessee, died on August 25, 2026, at 80. In September 2026 record companies and musicians were in court against companies whose computer programs make songs.

Napster, the iPod and streaming, 1999 to 2025

Shawn Fanning and Sean Parker started Napster in June 1999. It let people copy music files from each other's computers for free. Record companies sued in December 1999, and a federal judge ordered Napster to block copyrighted songs. Napster's owners could not do it, so they shut the service down in July 2001. At its peak, in February 2001, it had 26.4 million users by a checked count, though some estimates say 80 million.

The iPod. Apple's boss, Steve Jobs, showed the iPod, a pocket music player, on October 23, 2001. He said it held "1,000 songs in your pocket." Apple's iTunes Store opened on April 28, 2003, and sold most songs for 99 cents each. By 2014 it had sold 35 billion songs.

Streaming. Spotify, a company from Sweden, began streaming music in the United States in July 2011. By March 2026, the company said, Spotify had 300 million paying subscribers around the world.

The numbers. The Recording Industry Association of America, or RIAA, is the record companies' own group, and its staff count what the companies take in. Mark J. Perry of the American Enterprise Institute, a research group, used those figures in 2017. He found that sales peaked in 1999 at $20.9 billion, counted in 2016 dollars so that the rise in prices is taken out.

Sales then fell in 15 of the 16 years from 2000 to 2015. In 2016 they were $7.57 billion, a little more than a third of the 1999 figure. That year, for the first time, Americans paid more for streaming than for CDs, downloads and ringtones.

The RIAA reported in March 2026 that in 2025 the companies took in $11.5 billion at wholesale, the price the companies themselves are paid, which is lower than the price in stores. Counted that way, without taking out the rise in prices, the RIAA called 2025 the companies' best year. With the rise in prices taken out, the 1999 peak was still higher. Of the 2025 money, $9.5 billion came from streaming, and $1.04 billion came from vinyl records.

What does a stream pay?

Spotify pays the owners of the music less than half a cent for each play, by one company's estimate. It pays about 70 percent of the money it takes in to the owners of the music, by its own account.

About 58.5 percent goes to whoever owns the recording, which is usually a record company. About 12 percent goes to whoever owns the song itself: the songwriter and the publisher, a company that manages the rights to the song. A singer who does not own the recording gets a share from the record company, under the singer's contract.

The estimate. Spotify does not pay a set price for each play. It pays each owner by that owner's share of all the plays. iMusician, a company that sends music to streaming services, put the average in February 2025 at $0.003 to $0.005 a play, and about $0.004 for a listener in the United States.

The 1,000-play line. From April 1, 2024, Spotify stopped paying anything for a recording played fewer than 1,000 times in a year. In a November 2023 statement, Spotify's managers wrote that recordings under that line had earned an average of about 3 cents a month, and that 99.5 percent of all plays were of recordings above the line.

Spotify's March 2026 report states that it paid the music business more than $11 billion in 2025. By its count, more than 1,500 artists earned over $1 million from it that year. The artist ranked 100,000th earned about $7,300.

What musicians say. A musicians' group, the United Musicians and Allied Workers, held protests in 31 cities in March 2021 and asked Spotify to pay one cent a play. A group called the Music Industry Research Association asked 1,227 American musicians about their work in 2018. Most earned money from playing live, and most could not live on music alone.

Of the women who answered, 72 percent said they had been treated unfairly because they were women, and 67 percent said they had been sexually harassed, which means they faced unwanted sexual remarks or touching. Of the musicians who were not white, 63 percent said they had been treated unfairly because of their race. The musicians who answered did not name the people who did it.

Zoë Keating

Who: A cellist and composer who sells her own recordings and publishes what streaming pays her.

When and where: Born February 2, 1972, in Guelph, Ontario, Canada, to an English mother and an American father. Worked in San Francisco, California, and later Vermont.

"206,011 Spotify streams. $753."

Zoë Keating is a cellist who has published what streaming services paid her, so that other musicians could see real numbers. She posted those Spotify figures in September 2019. They came to a little more than a third of a cent for each play, and the company that sent her music to Spotify took no cut.

She played cello in the band Rasputina from 2002 to 2006. Alone, she uses a computer to record and repeat her own cello parts as she plays, so that one player sounds like many.

On her own. She released her album *One Cello x 16: Natoma* herself, without a record company. It reached No. 1 on the iTunes classical chart four times. She began publishing her income from sales and streaming in August 2012.

Her husband. Doctors found in May 2014 that her husband, Jeff Rusch, had stage 4 cancer, cancer that has spread through the body. Staff at their insurance company, Anthem, first refused to pay for his hospital stay. The records do not name them. Keating wrote about the refusal on her blog, as she publishes her pay. Anthem's staff reversed the refusal after local news stories about it. He died on February 19, 2015.

Who owns a recording: masters and songs

Every recorded song has two copyrights. One covers the song itself, its words and tune. The other covers the master, the first recording, which every copy is made from. Whoever owns the master is paid for every copy and every stream of that recording. Sam Cooke's own company owned his master recordings from 1963. Ray Charles's 1959 contract promised that he would own his in time.

The Universal fire. Jody Rosen, a writer for the New York Times Magazine, reported in 2019 that a fire in 2008 had destroyed 118,000 to 175,000 of Universal Music Group's master tapes. He wrote that they might include records by Chuck Berry, Muddy Waters and John Coltrane. A statement from Universal Music Group said the report held "numerous inaccuracies," meaning many mistakes. The company's leaders confirmed that the masters of one Soundgarden album had burned.

The fire began at Universal Studios Hollywood on June 1, 2008, when a worker used a blowtorch to heat roof shingles and left before they cooled. It spread across the studio's backlot, the outdoor area of sets and storage buildings.

Bob Dylan sold the rights to all of his songs to Universal Music Publishing Group in December 2020. Reporters at *The New York Times* estimated the price at more than $300 million, and other reports said about $400 million.

Taylor Swift

Who: A songwriter and singer who recorded four of her first six albums again after a music manager bought the master recordings.

When and where: Born December 13, 1989, in West Reading, Pennsylvania. Moved to Hendersonville, Tennessee, at 14.

Movie: *Miss Americana* (2020), a documentary about her.

She recorded four of her albums again, so she would own them.

Taylor Swift made new versions of four albums after a music manager bought the masters of her first six. Under her 2018 contract with Universal, she owns the masters of her new recordings.

At 14 her family moved to Hendersonville, Tennessee. That year a Nashville music publisher signed her as a songwriter, the youngest it had ever signed. After a show at Nashville's Bluebird Cafe on November 3, 2004, she signed with Scott Borchetta's new company, Big Machine Records. Big Machine owned the masters of her first six albums.

The sale. The music manager Scooter Braun bought Big Machine on June 30, 2019, for about $330 million, and with it her masters. Swift said she had tried for years to buy them. She said she was offered them only if she signed up to make six more albums for the company. Braun sold the masters to Shamrock Holdings in October 2020 for a reported $405 million. Shamrock is owned by the family of Roy E. Disney.

Taylor's Version. Swift recorded four of the albums again, and each came out as "Taylor's Version." *Fearless* came out on April 9, 2021, and *Red* on November 12, 2021. *Speak Now* followed on July 7, 2023, and *1989* on October 27, 2023.

Buying them back. Her Eras Tour ran from March 17, 2023, to December 8, 2024, with 149 shows. It took in $2.07 billion, more than any concert tour before it. On May 30, 2025, she said she had

> bought all six of the original masters from Shamrock. The price was not made public.

The Music Modernization Act, 2018

Members of Congress passed the Music Modernization Act in 2018. House members voted for it on April 25, and senators passed it on September 19 with no votes against. President Donald Trump signed it on October 11. Under the law, a nonprofit group, the Mechanical Licensing Collective, keeps a list of who owns each song and pays songwriters for streams. The group began paying out money in 2021 for songs whose owners nobody had identified.

Older records. Recordings made before February 15, 1972, had no federal copyright, so internet radio services did not have to pay for playing them. Many rhythm and blues records of the 1950s and 1960s were made before that date. Under the 2018 law, the singers and musicians on those older records are paid when digital radio plays them.

Hip-hop, Latin music and country radio

Hip-hop and R&B made up 24.5 percent of all the music Americans bought or streamed in 2017, more than any other kind. Rock came second, with just under 21 percent. Analysts at Nielsen Music, a company that tracks sales and streams, made the count, and CNBC's reporters wrote about it in January 2018.

Sales of Latin music came to $1 billion in 2025, 8.8 percent of what the record companies took in across the United States. The RIAA reported in 2026 that Latin music's sales had reached a new high for the tenth year in a row.

Country radio. Jada Watson, a researcher, counts the songs played on country radio. She found that songs by women got 11 percent of the plays by 2022. She reported in 2023 that no song by a Black, Indigenous or other woman of color was in regular daytime rotation, the list of songs a station plays over and over each day.

Beyoncé. The Black singer Beyoncé sang her country song "Daddy Lessons" at the Country Music Association Awards on November 2, 2016. Some country fans said she did not belong there, and the association's staff deleted its online posts about her performance.

Her country album *Cowboy Carter*, from 2024, made her the first Black woman with a No. 1 album on the country album chart. The album got no nominations at the Country Music Association Awards that year. It won the Grammy Awards for Album of the Year and Best Country Album on February 2, 2025.

BY 2025: Streaming made up 82 cents of every dollar that record companies took in, Latin music's sales reached a new high for the tenth year in a row, and Taylor Swift had bought back the masters of her first six albums.

Kendrick Lamar

Who: A Black rapper and songwriter, the first musician outside classical music and jazz to win the Pulitzer Prize for Music.

When and where: Born June 17, 1987, in Compton, California.

He won a Pulitzer Prize for an album of rap.

Kendrick Lamar became the first musician outside classical music and jazz to win the Pulitzer Prize for Music, in 2018. His parents had moved to Compton, California, from the South Side of Chicago in 1984. When he was five, he saw a teenage drug dealer killed in a drive-by shooting outside his apartment. No record names the shooter. In seventh grade an English teacher, Regis Inge, introduced him to poetry.

The albums. He signed with the record label Top Dawg Entertainment in 2005. His albums include *Good Kid, M.A.A.D City* in 2012 and *To Pimp a Butterfly* in 2015. Protesters in the Black Lives Matter movement used his songs as anthems.

The prize. His album *DAMN.* won the Pulitzer Prize for Music on April 16, 2018. The Pulitzer board called it "a virtuosic song collection unified by its vernacular authenticity and rhythmic dynamism," praising the skill of the songs, their everyday speech and their strong rhythm. The prize was $15,000.

He performed at the Super Bowl halftime show on February 9, 2025. About 133.5 million people watched, more than had watched any halftime show before. By 2026 he had won 27 Grammy Awards, more than any other rapper.

Deaths at music festivals, 2017 and 2021

More than 22,000 people were at the last night of the Route 91 Harvest country music festival in Las Vegas, Nevada, on October 1, 2017. At 10:05 p.m., while the country singer Jason Aldean was playing, Stephen Paddock, who was 64, began shooting into the crowd from his rooms on the 32nd floor of the Mandalay Bay hotel.

In ten minutes he fired more than 1,000 bullets. He killed 58 people that night and wounded at least 413 with gunfire, and about 867 people were hurt in all. Two wounded women died later, in 2019 and 2020, so the official count is 60 dead. Paddock shot himself. Investigators never found out why he did it.

Astroworld. About 50,000 people came to the Astroworld Festival in Houston, Texas, on November 5, 2021. When the rapper Travis Scott, who had started the festival, came on stage, the crowd pushed toward him. Ten people died of compressive asphyxiation: they were squeezed so hard they could not breathe. Thousands were hurt. The youngest to die was a boy of 9 or 10, and the reports differ on his age.

A Harris County grand jury, a group of citizens who decide whether someone should be charged with a crime, decided in June 2023 not to charge Scott or five others. Among the five was Brent Silberstein, a festival manager for the concert company Live Nation. Families of people who died sued Scott and Live Nation.

Machines that make music

Auto-Tune is a computer program that moves a singer's off-key notes onto the right pitch. The engineer Andy Hildebrand invented it, and the company Antares put it on sale on September 19, 1997. The singer Cher used it as a sound effect on her song "Believe" in 1998, the first record sold in stores to use it that way. The rapper and singer T-Pain used it that way on his 2005 album *Rappa Ternt Sanga* and after, and Kanye West used it on his 2008 album *808s & Heartbreak*.

Songs made by programs. Companies such as Suno and Udio sell computer programs that make whole songs. To train such a program, its makers feed recordings into it, and the program uses them to work out how to make new songs.

The lawsuits. Record companies, led by the RIAA, sued Suno and Udio in June 2024. They said the two companies had trained their programs on the record companies' recordings without permission.

The settlements. Some of the cases were settled in 2025 and 2026, which means the two sides ended them by agreement. Universal settled with Udio, and Udio's owners agreed to use only music they had permission to use. Warner and BMG settled with Suno and gave it permission to use their music.

Still in court. Universal and Sony kept suing Suno. In court papers, Suno's lawyers admitted that it had downloaded audio from YouTube. They argued that this was fair use, a use of someone else's work that the law allows in some cases.

Suno put out a new version of its program in September 2026, and its makers said they trained it on music from companies that had given permission. Universal and Sony sued again that month over 60,202 recordings, saying the new version grew out of the old ones. The songwriter Jason Isbell and three other musicians sued Suno that month too, and asked to speak for other artists as well. In October a music publisher sued Suno over 1,050 songs.

No final ruling. A separate case is about song lyrics. In March 2025 a federal judge, Eumi Lee, turned down music publishers' request to stop the company Anthropic from using lyrics to train its computer program while that case went on. As of October 26, 2026, no American court had given a final ruling on whether a company may train a song-making program on recordings without permission.

The Kennedy Center, 2025 and 2026

The Kennedy Center is a building for music, theater and dance in Washington, D.C., named for President John F. Kennedy. The National Symphony Orchestra plays there. Members of its board chose President Donald Trump as its chairman in February 2025. Trump then removed board members who had been appointed by Democrats and appointed his allies.

Board members voted on December 18, 2025, to add his name to the building, CBS News reporters wrote. Representative Joyce Beatty, a member of the board, said she had been muted on the call and that the vote had not been on the list of business for the meeting.

In court. Beatty went to court against the name change. A federal judge, Christopher Cooper, ruled on May 29, 2026, that the board could not rename the building. "Congress gave the Kennedy Center its name, and only Congress can change it," he wrote. He ordered the signs with Trump's name taken down within 14 days. Many performers canceled their shows at the center, saying it had become political, and ticket sales fell.

Closed. Board members voted to close the building on September 15, 2026, minutes after Judge Cooper again blocked the name change, Anastasia Tsioulcas of NPR reported. Trump wrote that repairs would not begin unless a higher court allowed his name. In a court filing, the center's executive director, Matt Floca, pointed to falling plaster and decay under the roof. He extended the closure to September 30, 2026, and said he would review it each week, reporters at WTOP, a Washington radio station, wrote.

The building did not reopen on September 30. Engineers had found rust and water damage on most of the panels under the building's overhanging roof, which could fall on people below.

NBC News and ABC News reported on October 1, 2026, an email in which Floca told the staff that the main building would "remain temporarily closed" while engineers checked whether people could safely use it and get out of it in an emergency. He expected that check to take one to two weeks, and he called that "an assessment timeline, not a reopening date." Judge Cooper asked for weekly reports on the repairs.

The Honors moved. The center's leaders moved the 2026 Kennedy Center Honors, a yearly award show, to Capital One Arena in Washington for December 7, Newsweek reported on October 2. It was the first time in nearly 50 years that the show would be held outside the Kennedy Center.

The texts. Beatty's lawyers asked Judge Cooper on October 25, 2026, to order the government's lawyers to hand over text messages they had kept back. They included 19 texts between Floca and Trump's assistant Natalie Harp and about 180 between Floca and Commerce Secretary Howard Lutnick, the Associated Press reported. The judge did not rule right away. As of October 26, 2026, the main building was still closed, with no date set for it to reopen.

Dolly Parton

Who: A singer and songwriter from Sevier County, in the mountains of Tennessee, who wrote more than 3,000 songs and mailed free books to children.

When and where: Born January 19, 1946, in a one-room cabin in Pittman Center, Sevier County, Tennessee. Moved to Nashville in 1964. Died August 25, 2026, in Nashville, at 80.

Movie: *Dolly Parton: Here I Am* (2019), a documentary about her.

She told Elvis Presley's manager no, and kept her song.

Dolly Parton wrote more than 3,000 songs, and in 1974 she refused to give Elvis Presley's manager half the rights to one of them. She started a program that mails free books to children, one a month from birth to age five.

The cabin. She was the fourth of 12 children, born in a one-room cabin in Pittman Center, in Sevier County, Tennessee. Her father, Robert Lee Parton, farmed tobacco and could not read or write. He paid the doctor who delivered her with a sack of cornmeal. Her mother was Avie Lee Parton. From age six Dolly sang in her grandfather's church.

By ten she sang on a radio and television show in Knoxville. At 13 she made her first record and sang at the Grand Ole Opry. The day after she finished high school in 1964, she moved to Nashville.

Porter Wagoner. The country singer Porter Wagoner hired her for his weekly television show in 1967. Their duets made the country Top 10 for six years in a row. From 1969 Wagoner owned nearly half of Owe-Par, the company that she and her uncle Bill Owens had started to publish her songs.

Her first No. 1 single as a solo singer was "Joshua," which she wrote by herself. It reached the top of the country chart in February 1971. That same year she recorded "Coat of Many Colors," about her family's

poverty. She wrote "Jolene" and "I Will Always Love You" within a few days of each other in 1973.

"I Will Always Love You" was her goodbye to Wagoner, and she left his show in 1974. The two later settled a lawsuit between them, and their 1980 album came out as part of that settlement.

Saying no. That year Elvis Presley wanted to record "I Will Always Love You." His manager, Colonel Tom Parker, told her that the songwriter always signed over half the publishing rights, the rights to the song itself, on any song Presley recorded. She said no. Whitney Houston recorded the song in 1992, and her record sold more than 24 million copies around the world. "I made enough money to buy Graceland," Parton said later, meaning Presley's home.

Hits. She wrote "9 to 5" for a 1980 film of the same name, in which she also acted. It reached No. 1 on the country and pop charts in February 1981. She sold more than 100 million records, had at least 25 No. 1 country songs, and won 11 Grammy Awards. She turned down the Presidential Medal of Freedom three times.

Who owned what. Ed Christman wrote in the magazine *Billboard* in August 2026 that Parton owned her own songs for her whole career. The records do not say how long Wagoner kept his share of Owe-Par, so they do not settle how that share fits with Christman's account. Her older recordings belong to Sony Music: the ones from her years with RCA, and probably from her Columbia years, about 23 albums in all. From 1995 she owned most of her new recordings, through her own record label.

Dollywood. She became a partner in a theme park in Pigeon Forge, Tennessee, which reopened as Dollywood on May 3, 1986. It has about 4,000 to 4,500 workers, more than any other employer in Sevier County. From 2022 Dollywood paid college costs for workers who chose to go.

The Imagination Library. She started the Imagination Library in Sevier County in 1995. The program's website states that she started it because her father could not read or write. Each child who signs up gets

one free book a month in the mail, from birth to age five.

Her foundation pays the program's running costs, and its staff bargain down the price of the books. Local partners raise the money for the books and the mailing, and families pay nothing. By 2026 more than 300 million books had gone out in five countries.

Gifts. After wildfires burned homes near Gatlinburg, Tennessee, in November 2016, she paid $1,000 a month for six months through her My People Fund to more than 900 families whose homes had burned, and $5,000 in the last month.

She gave $1 million to Vanderbilt University Medical Center in Nashville in the spring of 2020 for research on the coronavirus. Her longtime friend Dr. Naji Abumrad had told her that researchers there were making progress. A research paper on Moderna's COVID-19 vaccine listed her fund among its supporters. US government money for that vaccine came to about $2.5 billion. After Hurricane Helene in 2024, she and her businesses gave $2 million.

Her death. Her husband, Carl Thomas Dean, whom she married in 1966, died on March 3, 2025. She went into the Vanderbilt-Ingram Cancer Center in Nashville on August 21, 2026, and died four days later. Her family did not say what kind of cancer she had. Her nephew Bryan Seaver, her head of security for more than 20 years, announced her death, as she had asked him to.

That day Representative Tim Burchett of Tennessee said, "Tennessee lost one of our greatest ambassadors in Dolly Parton." President Trump ordered flags at federal buildings flown at half-staff for a week. Her family asked people who wanted to honor her to support the Imagination Library. They held a private funeral on August 28, and she was buried beside her husband in Nashville.

BY TODAY

If you remember one thing: Since 2000 most Americans have stopped buying their music and started streaming it, and the money from each play goes first to whoever owns the recording.

- 2001: Napster's owners shut it down after record companies sued.
- 2019: A music manager buys the masters of Taylor Swift's first six albums. She records four of them again and, in 2025, buys all six back.
- 2026: Record companies and musicians are in court against Suno over a computer program that makes songs.

Next: The chapter closing is below. Chapter 34, Storytelling Evolution, follows.

THE THREE QUESTIONS, ANSWERED

1. **Where did the banjo and the blues come from?** The banjo came from West African instruments, and enslaved Black people played it in the colonies from at least 1736, about a hundred years before a white man played it on a stage. The blues grew out of the work songs and field calls of enslaved people and the songs of Black workers in the South after slavery, between about 1870 and 1900.
2. **Why did so many musicians make hit songs and still die poor?** Many were paid a set fee and no royalty, as Bessie Smith was at $125 a side, or someone else put his name on their song, as Don Robey did on "Hound Dog." Others, like Stephen Foster, sold their rights to future payments for money right away.
3. **Who tried to stop people from playing their own music, and why?** South Carolina lawmakers banned drums and horns in 1740, saying the sound could call enslaved people together to rise up. Spanish friars burned Pueblo masks in the 1600s to punish Pueblo people for keeping their own religion. Richard Henry Pratt gave Carlisle students brass horns to stop their drumming, at a school

whose own paper gave its aim as "the complete transformation of the Indians." Federal rules from 1883 to 1978 punished Native religious dances, which the Secretary of the Interior had called "heathenish."

What changed: Music in this land began as songs one singer taught to the next, and by 2026 most of it reached listeners as streams that paid the owner of a recording less than half a cent a play.

FIVE DATES TO KEEP

May 10, 1740: South Carolina lawmakers ban enslaved people from using drums and horns.

1843: Four white men in blackface perform in New York as the Virginia Minstrels.

October 6, 1871: The Fisk Jubilee Singers set out on their first tour.

August 10, 1920: Mamie Smith records "Crazy Blues," the first blues record to become a big hit.

August 11, 1973: Kool Herc plays records at the Bronx party that people name as the start of hip-hop.

NOT KNOWN: No one knows who first played the blues, or when.

Where to go: *The Old Plantation*, the earliest known painting of a banjo, is at the Abby Aldrich Rockefeller Folk Art Museum in Williamsburg, Virginia.

Next: Chapter 34, Storytelling Evolution. It asks how Americans acted out stories, from Native performance and the minstrel stage to film, television and video games.

Chapter 34: Storytelling Evolution

CHAPTER 34 OF 37 · STORYTELLING EVOLUTION · 10 ERAS · ABOUT 125 MINUTES

People here have acted out stories for centuries, and in every era someone fought over who was allowed to act, whom an actor could play, and who owned the performance.

The court of Accomack County, Virginia, kept a record of three men who acted a play called *Ye Bare and Ye Cubb* on August 27, 1665. A man named Edward Martin reported them. On November 16 the judges ordered the three to come back to court "in those habilemts that they then acted in," which meant in their costumes, and to bring the script. In January 1666 the judges found them not guilty and made Martin pay all the costs. Those court orders are the only record of the play.

This chapter follows acting in this land from masked dancers before 1500 to computer copies of actors after 2000. Each new way of telling a story borrowed from the stage. The first films recorded stage acts, and the first television dramas were plays performed live.

Watch who is allowed to act. Priests burned Pueblo masks, lawmakers banned plays, and police arrested Black actors in the middle of a show. Watch, too, who plays whom. For more than a hundred years white actors blacked their faces to play Black people as fools, and film studios cast white actors as Asian and Native characters. After 2000 the fight moved to who owns an actor's face and voice.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who tried to stop Americans from acting, and why?

2. Why did white actors play Black, Asian and Native people for so long?
3. Who controls a computer copy of an actor's face and voice?

Five minutes? Read "The African Theatre, New York, 1821 to 1823" in 1800 to 1850, and the two stories after it.

Can skip: "Places built to hold a crowd", "Shakespeare on the rivers", "Big openings and home video", "YouTube and TikTok". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 7 MINUTES

In the Native nations here before 1500, acting was part of religion. A dancer in a mask took on the part of a spirit or an animal, in ceremonies and to ask the animals to give themselves to hunters. Among the Pueblo peoples, men carved dolls of these spirits so that girls would learn who each one was.

Some of what is known about these traditions comes from the nations themselves. Leaders of the Haudenosaunee, six nations whose lands stretch across what is now New York State and into Canada, published a policy on their masks in 1995. An Ojibwe teacher explained his people's storytelling rules in 2024. Older evidence survives in the ground. People built wide open squares called plazas and large round rooms for ceremonies, and they painted masked figures on walls and pots.

Who is inside the mask?

Pueblo men wear katsina masks and special clothing in dances, and the people believe that the spirit shown on the mask is present with or inside the dancer while the dance goes on. The Pueblo peoples, among them the Hopi, are Native nations of the Southwest. In their religion, katsinam are spirit beings, more than 500 of them. Men also carve katsina dolls and give them to girls, so the girls learn who each spirit is.

The katsina religion reached Homol'ovi around 1350. Homol'ovi is a group of towns near Winslow, in what is now Arizona, built by ancestors of today's Hopi.

Yup'ik masks. In what is now Alaska, the Yup'ik people held a winter ceremony in which dancers wore masks and took on the spirits of the animals the masks showed. Ann Fienup-Riordan, quoted in *Archaeology* magazine in 2015, said that the dancers asked the animals to give themselves to hunters in the coming year. After a dance, people usually burned the masks, broke them or left them out on the tundra, the flat treeless land of the far north.

People lived at Nunalleq, a village near Quinhagak, from 1300 to the 1650s. Archaeologists, scientists who dig up and study what people left behind long ago, found masks there that are the oldest complete Yup'ik masks anywhere. The masks were made before 1650, while people lived in the village.

Haudenosaunee masks. Among the Haudenosaunee, groups called medicine societies use masks of wood and of cornhusk. The Haudenosaunee have not published a date for when the societies began.

The Grand Council of the Haudenosaunee published its policy on these masks in 1995. The policy says only that a medicine society is made up of Haudenosaunee who have taken part in the medicine, and it does not explain the phrase further. It states that all the masks are sacred, whatever their size or age, and that only members of a medicine society should make masks. The masks must not be shown in public, and photographs and drawings of them should not be used in public displays.

Rules about who tells what, and when

Some nations set rules about which stories may be told, by whom, and at what time of year. An Ojibwe teacher described his people's rules in 2024. A Tlingit heritage group describes the Tlingit rules.

Ojibwe people tell their sacred stories, called aadizookaanag, only in winter. James Vukelich Kaagegaabaw, who is of Turtle Mountain Ojibwe descent, said in 2024 that tellers wait until the lakes freeze over and snow covers the ground. By then the frogs, snakes and other amphibians and reptiles are asleep for the winter, and so are the bears.

He said the spirits in the stories are thought to be asleep too, and "you wouldn't want to be thought of as gossiping about them." If someone tells the stories out of season, he said, "a frog or a toad may jump into your bed."

Tlingit crests. The Tlingit live in Southeast Alaska. They belong to clans, large family groups that trace back to the same ancestors. Each clan owns its crests, the animal and spirit figures that stand for the clan, and the stories, songs and names that go with each crest. The Tlingit word for this sacred clan property is at.óow.

The Sealaska Heritage Institute, led by Rosita Worl, published the first list of Tlingit clan crests in 2023. Its writers say that a clan got the right to a crest when an ancestor met a spirit being or what they call an "animal person." A clan "often acquired" the right "as payment for the life of an ancestor," they add, and the article does not say more about that payment. Clan members hold the crest and its stories through the generations.

Places built to hold a crowd

At Cahokia, in what is now Illinois across the Mississippi River from St. Louis, people built a Grand Plaza of about 47 acres. It was the largest of the city's five plazas. The builders cut down a ridge of sand and moved the sand to fill low places and make a level surface. Archaeologists see the plazas as the places where people from all around gathered for their most important ceremonies, meetings and social events.

Casa Rinconada. At Chaco Canyon, in what is now northwestern New Mexico, the people of Chaco built a great kiva called Casa Rinconada. A kiva is a round room, partly underground, used for ceremonies. Casa Rinconada is about 63 feet across inside, with a bench around the wall and 34 small openings, called niches, set into the wall.

A tunnel 39 feet long ran under the floor. Through it, a person, perhaps a leader of the ceremonies, could enter the kiva unseen and then come up suddenly in front of the people there. Building probably began in about the middle of the 1000s, and people stopped using the kiva in the early 1100s.

What the record does not have

No person from before 1500 in this land who acted, danced or told stories is named in any surviving record.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 7 MINUTES

The first European play on record in what is now the United States was acted on the day a Spanish leader claimed the land for his king, and his men made Native people act in it. Spanish soldiers and priests put it on beside the Rio Grande on April 30, 1598. They made Manso people, who lived near their camp, take part.

The Spanish belonged to an expedition, a large group of soldiers, priests and settlers, led by Juan de Oñate. They had come north from Mexico to take New Mexico for the king of Spain. Later that year Oñate's soldiers acted a second play at a pueblo, a Native town, farther north. No other European play acted in this land in the 1500s is on record.

The play at the river, 1598

Oñate's expedition reached the Rio Grande near present-day San Elizario, Texas, on April 20, 1598. Ten days later the colonists held a Mass of thanksgiving, a Catholic church service, in a chapel that Oñate had ordered set up in a grove of cottonwood trees.

"The taking." That day Oñate also read out a statement called "La Toma," which means "the taking." In it he claimed for King Philip II of Spain all the land that the river drained. Accounts differ on whether he read it before or after the play. Manso people lived near the camp.

After the Mass the expedition put on a play nearby. Captain Marcos Farfán de los Godos, one of Oñate's officers, wrote it. The stage was bare ground, and a canvas curtain hung between two tree trunks behind it. Soldiers and friars, Catholic priests who belong to a religious order, played the main parts.

The Manso actors. Members of the expedition made Manso people found near the camp take part in the play, the historian Marc Simmons writes. Simmons does not say who gave the order. In the play the Manso acted the parts of the Pueblo people who lived farther north.

An officer who watched described the story. Friars arrive in New Mexico, and the Native people there kneel and ask to be baptized, made members of the Catholic church in a ceremony with water. At the end, the people watching cheered and fired their guns into the air. No copy of the script survives, and its title is not known.

Marcos Farfán de los Godos

Who: A captain in Juan de Oñate's expedition who wrote the first European play on record in what is now the United States.

When and where: The Rio Grande near present-day San Elizario, Texas, April 30, 1598.

The author opened his play by calling it a poor one.

Marcos Farfán de los Godos wrote the play that Oñate's expedition acted on April 30, 1598, and he spoke first. In his opening speech he said he was unworthy to have written such a poor play. He offered it for the fun of his fellow soldiers and of Oñate.

Farfán came from the city of Seville, in Spain. When the expedition's officers checked each man and wrote down what he brought, Farfán had two servants, 16 horses, a tent, weapons and armor.

In November 1598 Farfán led a party west from the Hopi villages to look for mines. They reached the Verde Valley, in what is now Arizona. Near present-day Jerome they marked out 14 or 15 mining claims, pieces of ground where they said they had the right to dig for ore, rock that holds metal. What happened to Farfán after 1599, and when he died, is not known.

The same expedition at Acoma

About eight months after the play, men of the same expedition attacked Acoma Pueblo. The play had shown Pueblo characters kneeling and asking the friars to baptize them. At Acoma the expedition's soldiers killed real Pueblo people, and Oñate sentenced most of the people they captured to slavery. Acoma is a town built on top of a mesa, a steep hill with a flat top.

The demand for food. Oñate's nephew Juan de Zaldívar and his soldiers climbed the mesa in early December 1598 and demanded food. The All Pueblo Council of Governors is a council of the leaders of the Pueblo nations. In 2023 its leaders said that the soldiers had demanded food and supplies, and had forced the Acoma to promise loyalty to the king of Spain.

In the Acoma people's own account, as the Rio Grande Sun newspaper reported it in 2019, the soldiers also raped an Acoma woman, forcing her into sex. Acoma people killed Zaldívar and between 11 and 14 of his men. Accounts differ on the number.

War "by fire and blood." Oñate sent Zaldívar's brother, Vicente de Zaldívar, with about 70 soldiers in January 1599, and he declared a war "by fire and blood." The soldiers fought their way up the mesa. Over two or three days of fighting, they killed Acoma men, women and children and burned much of the pueblo. The soldiers took about 500 people prisoner.

No one counted the Acoma dead. Published estimates run from about 300 to about 1,500. The figure given most often is about 800 men, women and children, and it is the figure the Pueblos themselves use.

The sentences. Oñate put the prisoners on trial in February 1599 and sentenced them. Every Acoma man over 25 was to have one foot cut off and to be forced to work for Spanish masters for twenty years, as their slaves. Males from 12 to 25, and females over 12, got the same twenty years of forced work. Oñate also sentenced two Hopi men captured in the fighting to have their right hands cut off.

Oñate ordered the old people handed over as slaves to the Plains Apache, and children under 12 taken from their families and given to the friars. Under that order, Acoma girls were taken south to Mexico and placed in convents, houses where nuns live. The National Park Service's account gives about 70 girls, and the All Pueblo Council of Governors gives 60 children. The records do not name the men who took them. None is known to have come home.

The 24. Oñate's sentence is in the written record. Acoma people have passed down their own account since 1599: Oñate had the right feet of 24 captive men cut off. The Associated Press, NPR and the New York Times have reported that number. No one has traced it to a written record from the time. The historians Marc Simmons and John Kessell doubt that the cutting was carried out.

Oñate's own trial, years later. Spanish officials charged Oñate with 30 crimes in 1612 or 1613. In 1614 judges in Mexico found him guilty of 12 of them, including using too much force at Acoma. The judges ordered him out of New Mexico for life and barred him from Mexico City for four years. One account says five years. They also fined him 6,000 ducats, which were Spanish gold coins.

Los Moros y Cristianos

Oñate's soldiers had acted a second play months before the attack on Acoma. They put it on in the middle of 1598 to celebrate the expedition's safe arrival at a town of the Tewa, one of the Pueblo peoples, farther north. The Spanish renamed the town San Juan and made it their first capital. Today that pueblo is called Ohkay Owingeh.

The play was *Los Moros y Cristianos*, which means *The Moors and the Christians*. It shows the battles between the Moors, who were Muslims, and the Christians in Spain from the 700s to the 1400s. The leaders of each side make long speeches, and the battle scenes are full of clashing weapons.

New Mexicans have acted the play now and then in recent times, most of all a group of actors from Chimayó. Actors performed it at the University of New Mexico in 1998, for the 400th anniversary of Oñate's colony.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 6 MINUTES

People with power on both sides of the ocean tried to stop acting in the 1600s. Members of England's Parliament ordered stage plays stopped in 1642. In Virginia a man named Edward Martin reported three actors to a county court in 1665, and the court tried them. The first play in English on record in what is now the United States is known only from that court's orders.

In New Mexico, Spanish priests went into Pueblo ceremonial rooms and burned the masks used in the dances. In 1680 the Pueblos drove the Spanish out and brought back their own ceremonies.

Plays stopped in England, and a play at Harvard

England's lawmakers meet in Parliament, which has two parts, the House of Lords and the House of Commons. Members of both houses ordered on September 2, 1642, that "Public Stage Plays shall cease." Plays were to stop while "these sad causes and set Times of Humiliation do continue," meaning while the country was in trouble and kept times of sad, serious worship.

The order called plays "Spectacles of Pleasure, too commonly expressing lascivious Mirth and Levity." That meant shows put on for pleasure that too often made rude jokes about sex and joked about serious things.

Harvard. Students at Harvard College, in Cambridge, Massachusetts, acted *Gustavus Vasa* in 1690. It was a tragedy, a serious play with a sad ending, by Benjamin Colman. Local officials disapproved of the play. Arthur Hornblow, who wrote a history of the American theatre in 1919, called it the first play written by an American to be acted in America.

A play, a complaint and a court, 1665

Three men named Cornelius Watkinson, Philip Howard and William Darby acted a play called *Ye Bare and Ye Cubb* in Accomack County, Virginia, on August 27, 1665. The court record does not say where. A local history site, Secrets of the Eastern Shore, says they acted it at Fowkes Tavern near Pungoteague.

A man named Edward Martin reported the play to the court. The king's attorney for the county, the lawyer who brought cases to court in the king's name, took up the case. The record gives the charge only as "acting a play." It does not say what Martin objected to, or what law the attorney said the men had broken.

"In those habilemts." The king's attorney accused the three men in the county court on November 16, 1665. The judges ordered the men to come to the next court "in those habilemts that they then acted in." Habilemts was the court clerk's short spelling of habiliments, an old word for clothes, so the men had to come in their costumes. They also had to bring a written copy of the verses and speeches.

The judges ordered the sheriff to hold Watkinson and Howard until they gave security, a promise backed by money or property that they would come back. They ordered him to arrest Darby.

The verdict. The judges found the three men "not guilty of fault" at a court in January 1666. They ordered Edward Martin, who had reported the men, to pay all the costs of the case. The record does not say whether the men acted the play in front of the judges. The court orders are the only record of the play from its own time.

William Darby

Who: One of the three men who acted *Ye Bare and Ye Cubb* in Virginia in 1665, and the one the sheriff was ordered to arrest.

When and where: Accomack County, Virginia, August 27, 1665, and the county court, November 1665 to January 1666.

Later writers name him as the author of the lost play.

William Darby is one of the first people on record to act a play in English in what is now the United States. The Accomack County court record names him as an "actour," an old spelling of actor, in *Ye Bare and Ye Cubb*. He was the one actor of the three whom the judges ordered arrested, "to answere at his ma'ties suit." That meant he had to answer a case brought in the name of His Majesty, the king.

Later writers say he wrote the play or helped write it. The reference book *American Eras* lists it as "*Ye Bare and Ye Cubb*, by William Darby." Writers at the Oscar G. Brockett Center for Theatre History call him "one of the piece's authors."

NOT KNOWN: The court record is the only account of Darby from his own time, and it does not say what work he did, when he was born or when he died.

Pueblo ceremonies attacked, and brought back

In New Mexico, Spanish officials and priests tried to end the Pueblo ceremonies that the masked dancers performed. Franciscan friars, one group of Catholic friars, had set up churches in several Pueblo towns.

The Franciscan leader Alonso de Posada forbade the Pueblo people to hold their ceremonies from 1656 to 1665. He ordered the missionaries, priests sent to win people over to Christianity, to go into the kivas, burn the masks there, and take or destroy the sacred objects. Armed Spanish soldiers protected the missionaries while they did it.

The arrests. Governor Juan Francisco Treviño ordered the arrest of 47 Pueblo religious leaders for sorcery, using magic to do harm, in 1675. Spanish officials sentenced four of the men to death by hanging. They hanged three of them, and the fourth killed himself. They shamed the others, whipped them in public and put them in prison. Pueblo men then came to Santa Fe and demanded that the prisoners be freed, and the Spanish let them go. One of the men freed was Po'pay of Ohkay Owingeh, the pueblo the Spanish called San Juan.

The knotted ropes. Po'pay planned a revolt of all the Pueblo towns at once. He sent runners carrying knotted ropes to the towns, and the people in each town untied one knot a day. When the last knot was gone, the revolt would begin. On August 9, 1680, the Spanish caught two of the runners, Pedro Omtua and Nicolas Catua of Tesuque Pueblo, and tortured them. The Smithsonian's account does not say how.

When Po'pay learned the two had been caught, he started the revolt a day early, on August 10. Pueblo fighters killed about 400 people, including 21 of the 33 Spanish priests in New Mexico. About 2,000 settlers who survived fled south to El Paso. After the Spanish left, the Pueblo people destroyed the churches and went back to their own religious ceremonies. The Spanish stayed away for 12 years.

What the record does not show

English colonists in the 1600s enslaved Africans, owning them as property and forcing them to work without pay. Religion, storytelling, music and dancing were important parts of enslaved people's lives, and helped them pass African ways from parents to children. No record from the 1600s describes a performance by an enslaved person in the English colonies. The earliest known picture of a stick dance, a dance of enslaved people, is a watercolor called *The Old Plantation*, painted in the late 1700s.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 7 MINUTES

Two of the first actors in the English colonies were not free. They were servants, bound by contract to the man who built the first playhouse, in Williamsburg, Virginia, between 1716 and 1718. Builders in Charleston, South Carolina, put up a theatre made for plays in 1735 and 1736.

Some colonists wanted no plays at all. In Boston, a judge of the colony's high court wrote against acting one. The preacher George Whitefield, who had prepared to act plays as a schoolboy, preached in the fields. According to newspaper reports that Benjamin Franklin read, one crowd numbered 25,000. In New York, enslaved Black people gathered with music and dancing for a holiday called Pinkster.

The first actors and the first playhouses

The British actor Anthony Aston reached Charleston, South Carolina, in about 1702 and wrote a play about the colony there. Accounts give his arrival as 1701, 1702 or 1703. No copy of his play survives.

Williamsburg. A merchant named William Levingston signed a contract in 1716 with Charles Stagg and his wife, Mary Stagg. The Staggs were his indentured servants, people bound by a contract to work for a master for a set number of years. Under the contract, Levingston agreed to build a playhouse in Williamsburg, and the Staggs agreed to act and dance in it.

The first play on record there was acted in 1718. Lieutenant Governor Alexander Spotswood, the colony's leader at the time, sponsored it. No list of the plays acted in Levingston's playhouse survives.

Whose land. Williamsburg stands on land that the Paspahegh and Chiskiack peoples used before the English came. Both belonged to Tsenacomoco, a group of nations that Powhatan led until 1618. Its people spoke Algonquian languages, one family of related Native languages.

Charleston. A company of actors put on Thomas Otway's tragedy *The Orphan* in Charleston on January 24, 1735. They acted it in the "Court-room" of a tavern on Broad Street, a room that colony officials rented for trials. The company gave six shows through March.

A ticket cost forty shillings in South Carolina money, which was worth less than English money, so the ticket was worth a little less than six English shillings. Most of the actors' names are not recorded. A dancer named Henry Holt probably led them.

Builders then put up a theatre on Dock Street, made for plays. It opened on February 12, 1736, with George Farquhar's comedy *The Recruiting Officer*.

The Etiwan. The Etiwan people lived on the Charleston peninsula until 1672. That year the colony's Grand Council, the group of men who governed it, decided to build the town there. Colonists paid the Etiwan in trade goods to

move away. Maurice Mathews, one of the colonists who probably arranged it, wrote in 1680 that the settlers paid the Etiwan "to remove them from their old habitations," meaning their old homes.

Philadelphia. A company led by Walter Murray and Thomas Kean gave its first show in Philadelphia on August 22, 1749. It acted Joseph Addison's tragedy *Cato* that month.

Anthony Aston

Who: A British lawyer who became an actor and wrote a play about South Carolina while he was there.

When and where: Charleston, the Albemarle region of North Carolina, New York and Virginia, about 1701 to 1703.

He reached Charleston "full of lice, shame, poverty, nakedness and hunger."

Anthony Aston told his own story in the preface, the short opening piece, of his play *The Fool's Opera*, printed in 1730 or 1731. He listed himself as "a gentleman, lawyer, poet, actor, soldier, sailer, exciseman, publican," the last two meaning a collector of taxes on goods and a tavern keeper.

In Charleston, he wrote, he "turned player," or actor, "and poet and wrote one play on the subject of the country." That Charleston play was a different play from *The Fool's Opera*, and no copy of it survives.

He then spent about a month in the Albemarle region of North Carolina, where two settlers, Allen and Abraham Waights, supported him. On the way to New York, a storm blew his ship off course to the coast of Virginia. In New York he spent a winter "acting, writing, courting, fighting." Then he sailed to Virginia, and later he went back to England.

William Levingston

Who: A merchant and dancing teacher who built the first playhouse in the English colonies, with two indentured servants as his actors.

When and where: New Kent County and Williamsburg, Virginia, 1716 to 1723.

He built a theatre beside his bowling alley and stable.

William Levingston built the first theatre in the English colonies, in Williamsburg, Virginia. Before that, he ran a dancing school that traveled around New Kent County, with Charles and Mary Stagg as its star dancers.

Their contract, dated July 11, 1716, and recorded at Yorktown, set out how the theatre would work. Charles Stagg was an actor, a violin player and a dancer. Mary Stagg was an actor. The three agreed to share the costs of costumes, music and everything else needed to put on plays.

The theatre lot. Levingston bought land in Williamsburg on November 21, 1716. On it he built a house, a kitchen, a stable, a bowling alley and the theatre. He borrowed money from Dr. Archibald Blair with the property as security. In 1723 Blair foreclosed, which means he took the property because the debt was not paid.

Charles Stagg died in Williamsburg in 1735 or early 1736. After his death, Mary Stagg earned her living by holding dancing parties. College of William and Mary students acted plays in the theatre in 1735 and 1736. In 1745 a group of Virginia men bought the building and gave it to Williamsburg to use as a town hall.

Boston against plays

Samuel Sewall, a judge of the Massachusetts high court who later became its chief justice, wrote a letter on March 2, 1714, against a plan to act a play in the Council Chamber in Boston. The Council Chamber was the room where the colony's council met.

Sewall wrote that even the Romans had not been so set on plays as to "turn their Senate House into a Play-House." He went on: "Let not Christian Boston goe beyond Heathen Rome in the practice of Shamefull vanities." No record shows that the play was ever acted.

How far could one preacher's voice carry?

Benjamin Franklin worked out that the English preacher George Whitefield could be heard by more than 30,000 people at once. Franklin heard Whitefield preach from the courthouse steps on Market Street in Philadelphia, and he walked backward down the street until he could no longer hear clearly. Then he worked out how many people could stand in a half circle that size, giving each person two square feet. He "computed that he might well be heard by more than thirty thousand."

Whitefield first came to the colonies in 1738 and made a preaching tour from 1739 to 1741. According to newspaper reports that Franklin read, Whitefield once preached outdoors to 25,000 people. As a schoolboy of about twelve in Gloucester, England, he wrote later, he "was very fond of reading plays." He stayed away from school "for days together to prepare myself for acting them."

Practice on tour. Franklin noticed that Whitefield spoke each sermon better every time he repeated it on tour. By then "every accent, every emphasis, every modulation (rise and fall) of voice, was so perfectly well turn'd and well plac'd." Franklin wrote that a preacher who traveled could improve a sermon "by so many rehearsals," or practice runs.

Pinkster and Black election days

In New York, enslaved Black people gathered for a holiday called Pinkster, with music and dancing. In Connecticut, Black communities chose their own governor.

Dutch settlers brought Pinkster to New Netherland, later New York, in the 1620s. The oldest written account of Pinkster appeared in the *New-York Weekly Journal* in March 1737. It describes a gathering in fields outside New York City, with African-style musical instruments.

Enslavers gave the people they held time off for the holiday. For many enslaved Africans it was a chance to see loved ones and family members who lived some distance away. Africans and Europeans alike drank, played games, made music and danced at these gatherings, with dances called the jig, the breakdown and the double shuffle.

A Black governor. Black communities in Connecticut held their own elections for a governor by 1749, when a man named Hercules served as governor at New London. The voting took place the day after the white colonists' elections, and the celebration went on for a full week. Some of the clothes people wore were borrowed from the men and women who enslaved them. By custom, the enslaver of the man elected was expected to help pay for food, decorations and liquor.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 9 MINUTES

Lawmakers kept passing laws against plays even as paid actors arrived and found audiences. A family company of actors from London landed in Virginia in 1752 and toured the colonies. One reference work describes their *Merchant of Venice* in Williamsburg that year as the beginning of truly professional theatre in America.

Lawmakers in Massachusetts, Pennsylvania and Rhode Island passed laws against plays between 1750 and 1761. Even so, professional actors staged the first play by an American writer in 1767. In 1774, as the fight with Britain grew, the delegates to the First Continental Congress promised to discourage plays, and shortly before the war the leading company of actors left for the West Indies.

During the war, British soldiers acted a play in Boston, and American officers acted one at Valley Forge. After the war the actors came back. In 1787 an American comedy brought a new character to the stage, the plain-spoken Yankee.

Why lawmakers banned plays

Lawmakers in Massachusetts said that plays made people less moral and less religious. Members of the Massachusetts General Court, the colony's lawmakers, passed "An Act for preventing Stage-Plays, and other Theatrical Entertainments" in 1750. It stated that plays "tend to increase immorality, impiety, and a contempt of religion," impiety meaning a lack of respect for God. The act set fines for anyone who put on a play or took part in one.

Three more colonies. Members of the Pennsylvania House of Representatives voted through a law against acting plays on May 31, 1759, with a fine of five hundred pounds. Two years later the lawmakers of Rhode Island passed "an act to Prevent Stage Plays and other Theatrical Entertainments."

Members of the New Hampshire House of Representatives refused in 1762 to let a company of actors into the town of Portsmouth. They said that plays had a "peculiar influence on the minds of young people and greatly endanger their morals," peculiar meaning special, and morals meaning a person's ideas of right and wrong.

The crowd in Chapel Street. New Yorkers learned on May 5, 1766, that members of the British Parliament had voted to end the Stamp Act, a British tax that many colonists had protested. A crowd tore down the theatre in Chapel Street, where a comedy called *The Twin Rivals* was advertised for that night.

David Malinsky, writing in the *Journal of the American Revolution* in 2013, says that the New York Sons of Liberty did it. They were secret groups of colonists, sometimes violent, who fought the Stamp Act. Malinsky writes that the crowd severely beat an actor who was dressed as a woman and killed a small child. The older accounts record only that the theatre was destroyed. No record names the child, the actor, or who led the crowd.

The first touring company, 1752

The London actor Lewis Hallam led a company of actors to Virginia in 1752. It included members of his family, among them his son Lewis, about twelve years old. It opened in Williamsburg with Shakespeare's *The Merchant of Venice* on September 15, 1752. William Dunlap, an early historian of the American stage, gave the date as September 5.

The company acted tragedies by Shakespeare and Joseph Addison, and comedies by William Congreve and Richard Steele. *Hamlet*, *Othello* and *Richard III* were among its plays. It played Williamsburg, Annapolis, Philadelphia and New York. Hallam had trouble getting permission to act in New York and Philadelphia, and he did not even try in Boston. In New York, Hallam built a theatre on Nassau Street in 1754.

The American Company. Lewis Hallam then died in Jamaica, where the company had gone to perform. His widow, Sarah Hallam, married the actor and manager David Douglass. In 1758 they formed the American Company. Douglass built a theatre in Philadelphia in 1759. In Annapolis he opened a brick theatre on March 3, 1760, that held five to six hundred people.

Charles Daly, writing in 1896, says Douglass also built a theatre in Chapel Street, New York, in 1761. Gabriel Furman's notes name another man, Phil Miller, as its builder.

Lewis Hallam the Younger

Who: A boy in the first touring company of actors in the colonies, who grew up to be the leading actor and manager of the American Company.

When and where: Born about 1740 in England. Came to Virginia in 1752. Died in Philadelphia on November 1, 1808.

He landed in Virginia at about twelve and acted for fifty years.

Lewis Hallam the Younger spent about fifty years on the stage in America. He was about twelve years old when his father's company landed in Virginia in 1752. The historian William Dunlap got that story from Hallam himself.

At eighteen, in 1758, Hallam became the leading man of the American Company, which his mother and stepfather ran. He acted in *The Prince of Parthia* in 1767, the first play by an American writer that professional actors staged. He also acted in *The Contrast* in 1787.

His mother died in 1774, and his stepfather retired to Jamaica. Hallam then took over the family company. Shortly before the Revolution, he moved it to the West Indies. He brought the company back in 1784 or 1785, depending on the account, and managed it until 1796. He kept acting until shortly before he died. *Encyclopaedia Britannica* describes him as "known as a quarrelsome and financially inept manager," which means he argued with people and handled money badly.

Nancy Hallam

Who: An actress and singer in the American Company, and the subject of an early painting of an actress in a role in America.

When and where: Annapolis, Maryland, 1770 and 1771.

A newspaper letter praised "the Musick of her Tongue."

Nancy Hallam was an actress whom the painter Charles Willson Peale painted in the costume of one of her roles. She was the niece of Sarah Hallam Douglass. She trained as a singer in England from 1760 to 1765, and she was acting with the American Company by 1766. In 1769 she moved from the parts of young women to leading roles.

In Annapolis in 1770 she played Imogen in Shakespeare's *Cymbeline*, a princess who disguises herself as a young man named Fidele. On September 6, 1770, the *Maryland Gazette* printed a letter signed only with the initials "Y.Z." It praised her "Delicacy of Manner" and "the Musick of her Tongue."

Peale painted her dressed as Fidele, and the painting was on view by September 1771. Colonial Williamsburg owns it today. Nancy Hallam went back to Britain in 1773. What happened to her after that, and when she died, is not known.

The first American plays

Thomas Godfrey wrote a tragedy called *The Prince of Parthia*. The American Company acted it at the Southwark Theatre in Philadelphia on April 24, 1767. It was the first play by an American writer that professional actors performed. Godfrey had died of a disease called yellow fever in 1763, so he never saw it on the stage.

Mercy Otis Warren of Massachusetts wrote plays in which she attacked the policies of the colony's British governor. At the time, Massachusetts law banned acting plays.

The Yankee. Royall Tyler wrote a comedy called *The Contrast*. It opened at the John Street Theatre in New York in 1787, the first comedy by an American that professional actors performed. Tyler wrote it after he met Thomas Wignell, the American Company's leading comic actor.

Its leading character, Jonathan, has rough, plain-spoken manners. He is a Yankee, a word that can mean a person from New England. Playwrights after Tyler kept writing Yankee characters like him.

Mercy Otis Warren

Who: A Massachusetts writer who used plays to attack the royal government at a time when Massachusetts law banned acting them.

When and where: Born in Barnstable, Massachusetts, on September 14, 1728. Wrote plays from 1772 to 1779. Died in 1814.

She wrote five plays against the royal governor and signed none.

Mercy Otis Warren's first play, *The Adulateur*, came out in 1772 in a newspaper, the *Massachusetts Spy*. *The Defeat* followed in 1773, *The Group* in 1775, *The Blockheads* in 1776 and *The Motley Assembly* in 1779.

The National Women's History Museum's biography of her states that she likely left her name off the plays to avoid being punished, and also so that readers would take her work seriously even though a woman wrote it. In 1805 she published a history of the American Revolution in three volumes.

Congress against plays, and armies acting them

The First Continental Congress was a meeting of delegates, men chosen by each colony to speak for it. On October 20, 1774, they signed an agreement called the Continental Association. In its eighth article they promised to "discourage every species of extravagance and dissipation," meaning spending too much and wasting time and money on pleasure. The list that followed named horse racing, cockfighting, "shews, plays, and other expensive diversions and entertainments." Shortly before the war, the American Company left for the West Indies.

A play in Boston. British soldiers acted a play in Boston while American troops surrounded the city. On December 2, 1775, they staged *Zara*, a tragedy by the British writer Aaron Hill, in Faneuil Hall. The poster states that it was put on "by a Society of Ladies and Gentlemen." General John Burgoyne wrote a prologue for it, a speech given before the play begins. By one account, Francis Rawdon-Hastings, a British officer, read it. The money from tickets went to the widows and children of soldiers.

A play at Valley Forge. American officers at Valley Forge acted Joseph Addison's *Cato* in the spring of 1778. Lieutenant Colonel William Bradford Jr. wrote in a letter that "Cato was performed before a very numerous & splendid audience." He wrote that "His Excellency and Lady" were there, and that the performance was "admirable." His letter uses only that title and gives no names.

The historian Nancy K. Loane writes that General George Washington and his wife, Martha, attended the play. Accounts date the performance May 4 or May 11, 1778. The audience was officers and their guests. Shawn David McGhee, writing in the *Journal of the American Revolution* in 2024, argues that it may have been a reading of the play rather than a full staging.

Congress again. The delegates to Congress asked the states to stop "theatrical entertainments" in October 1778. On October 16 the delegates voted that any official who would "act, promote, encourage or attend such plays, shall be deemed unworthy to hold such office."

After the war

Lewis Hallam the Younger brought the American Company back from the West Indies in 1784 or 1785 and started it up again.

The Massachusetts law of 1750 still banned plays. In 1791 a group of men who wanted a theatre in Boston tried and failed to get it repealed, which means ended by a vote. In 1792 they opened the "New Exhibition Room" on Board Alley anyway. They sometimes advertised its plays as "moral lectures," talks meant to teach right from wrong. Governor John Hancock forced it to close in June 1793.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 15 MINUTES

White audiences paid to watch white actors play Black people as fools, while Black actors who played Shakespeare were attacked and arrested. The first known Black theatre company in the United States opened in New York in 1821, and police arrested its actors in the middle of a play. In 1824 a young actor from the company, Ira Aldridge, left for England. In the United States he faced discrimination, which means he was treated worse because he was Black.

Blackface. A white actor named Thomas Dartmouth Rice blacked his face with burnt cork in 1828 and played a Black man as a fool. White audiences paid to laugh at the act, and by 1843 four white men in the same makeup were performing together as a group.

Actors also played Shakespeare in city theatres and on boats on the rivers. In 1849 the fans of two rival actors fought outside a New York theatre. City officials gave soldiers the order to fire into the crowd, and the soldiers killed at least 22 people.

The African Theatre, New York, 1821 to 1823

The first known Black theatre company in the United States opened in New York in 1821. Its founder, William Brown, was a free Black man from the West Indies who had worked as a steward, a servant on a ship. He ran a tea garden behind his house at 38 Thomas Street, an outdoor place for refreshments and shows, with music on Sunday afternoons.

Brown opened a theatre in 1821 with about 300 seats on the second floor of a building at Mercer and Bleecker Streets. Its actors were Black. It opened with Shakespeare's *Richard III*. Accounts give the first night as September 17, 21 or 24, 1821. A ticket cost 25 cents, or 50 cents for a better seat.

A curtain for a robe. On the first night, an enslaved man played the king in a robe made from a window curtain. After that, an actor named James Hewlett took over the part. The company also played Shakespeare's *Othello* and other plays such as *The Poor Soldier*. White New Yorkers came too, and some of them made trouble.

A group of people that included a man named George Bellmont broke into the theatre. They attacked the actors and wrecked the costumes and scenery. Police arrested Bellmont and several of the others.

The arrests. Brown then moved his company next door to the Park Theatre, run by a white manager named Stephen Price. Brown's company played *Richard III* on the same nights as Price's company. White members of the audience shouted at Brown's actors and broke up the show. According to the Shakespeare Theatre Company, which has published Hewlett's history, Price started a riot because he hoped to destroy Brown's company.

During one performance, police arrested the actors and put them in jail. The police let them go only after they promised to stop acting Shakespeare. The surviving records do not give the date or name the officers.

The end. Mordecai Noah, the editor of a newspaper called the *National Advocate*, mocked the company in print. He warned that free Black New Yorkers might one day "outvote the whites." In October 1822 Noah's

newspaper included a notice that the theatre was closing because of an outbreak of a disease called yellow fever. The last known playbill of the company, a printed poster for a show, is from June 1823.

In London, the English comedian Charles Mathews mocked the company in his show *A Trip to America*, although he had never been to its theatre. In May 1824 James Hewlett answered him in a letter in the *National Advocate*: "Was this well for a brother actor?"

William Alexander Brown

Who: A Black ship's steward from the West Indies who started the first known Black theatre company in the United States, and wrote the first known play by a Black writer to be acted there.
When and where: New York City, 1816 to 1823.

His own play told of a 1796 war on St. Vincent.

William Brown wrote *The Drama of King Shotaway*, the first known play by a Black writer to be staged in the United States. Accounts give his full name two ways, as William Alexander Brown and as William Henry Brown.

Brown bought his house on Thomas Street in 1816. Neighbors and police complained about the shows in its tea garden again and again until the garden closed. After the attacks and arrests at his theatre, he moved his company several times.

Brown staged *King Shotaway* in 1823. It told the story of the war that the Black Caribs of the Caribbean island of St. Vincent fought against English and French settlers in 1796. The Black Caribs, now called the Garifuna, were a people whose ancestors were both free Africans and Native people of the Caribbean. No copy of the play survives. What happened to Brown after his company's last known playbill, in June 1823, is not known.

Ira Aldridge

Who: A Black actor from New York who left the United States in 1824 and spent more than forty years acting Shakespeare in Britain and Europe.

When and where: Born in New York about July 24, 1807. Went to England in 1824. Died on tour in Łódź, Poland, on August 7, 1867.

A London critic said his skin was not dark enough for Othello.

Ira Aldridge became a successful Shakespearean actor in Britain and across Europe. He is not known to have acted on a stage in the United States after he left. He went to the African Free School in New York, and as a teenager he acted with William Brown's company. Because he was Black, he faced discrimination in the United States, and in 1824 he sailed to England.

In October 1825 he played Oroonoko in a play called *The Revolt of Surinam, or A Slave's Revenge*, at a London theatre. A writer in *The Times* of London made fun of the shape of his lips. The same writer said his skin was not dark enough for him to play Othello.

Aldridge played Othello anyway, and King Lear and Macbeth. In 1828 he became the manager of the theatre in Coventry, England. In 1833 he took over the part of Othello at the Theatre Royal, Covent Garden, in London. *Encyclopaedia Britannica* describes that first appearance there as "highly successful."

He also played white characters, such as Shylock and Richard III, wearing greasepaint, a thick stage makeup, and a wig. On the last night of a run, he spoke to the audience against slavery. Rulers in Prussia, Austria and Russia gave him honors, and in 1863 he became a British citizen.

After the Civil War, Aldridge planned a tour of the United States and agreed to act in New York in September 1867. He died in Poland that August, before he could make the trip.

Blackface minstrelsy: the act and the makeup

In blackface minstrelsy, white actors darkened their faces with burnt cork and acted Black people as fools. They built caricatures of Black people, acts that made fun of them by blowing up some of their features, and played them as singing, dancing, grinning fools. White audiences paid to watch.

Writers at the Jim Crow Museum at Ferris State University state that Rice and the actors who copied him helped to spread the belief that Black people were lazy, stupid and less than human. They write that the shows also helped white people argue for keeping Black people apart from them.

"Jim Crow." Thomas Dartmouth Rice was a white actor who did short comic acts between the scenes of plays. In 1828 he first went on stage as "Jim Crow," a Black character acted as a fool, with his skin blacked with burnt cork. Rice said he copied a Black man he had seen. Some accounts say the man was old, enslaved and walked with difficulty. Others say he was a ragged stable boy. No record names him.

Rice took the act from Louisville to Cincinnati, Pittsburgh and Philadelphia, and he reached New York in 1832. The act made him rich and famous. He died poor in New York on September 19, 1860.

People were using "Jim Crow" as an insult for Black people by 1838. By the end of the 1800s, "Jim Crow" was the name for the segregation laws and customs, which kept Black Americans apart from white Americans by race. Those laws came after 1877 and lasted until the mid-1960s.

The first group. Four white men in New York performed together in blackface as the Virginia Minstrels on January 31, 1843, the first group act of its kind. Years later, Black performers took up minstrelsy too, and they also blacked their faces with burnt cork. They acted the same caricatures that white actors had invented. In the 1840s one young Black dancer was already working in blackface.

William Henry Lane (Master Juba)

Who: A young Black dancer, known on stage as Master Juba, who danced in blackface and who toured with a white minstrel company.
When and where: New York and London, 1840s. Born about 1825, died about 1854.

"A black man imitating a white man imitating a black man."

William Henry Lane was a Black dancer, known on stage as Master Juba. He learned to dance in the Five Points, a neighborhood in New York. By the early 1840s he was dancing for food and for coins people tossed him. Charles Dickens described a young Black dancer in the Five Points in a book in 1842, and later writers decided the dancer was Lane.

The reference book *Notable Black American Men* states that he was the only Black dancer of his time to get top billing, which means his name came first on the posters, in a minstrel company whose other members were all white. He toured with that company in the United States and in Britain.

The fake contests. A letter printed in a newspaper called the *Sunday Flash*, in 1841 or 1842, made a charge against the showman P. T. Barnum. The letter said that Barnum had made the Black boy up in blackface and advertised him as John Diamond, an Irish-American dancer who performed in blackface. It also said that Barnum staged fake dance contests with him: "Not one dollar is ever bet or staked."

The historian Eric Lott called the act "a black man imitating a white man imitating a black man." Lane later danced against the real John Diamond in public dance contests.

Lane went to London in 1848 with the Ethiopian Serenaders, a white minstrel company. A writer for *The Times* of London said that Lane "was compelled to repeat his efforts three times," which means the crowd made him dance again and again. He died in about 1854, when he was about 29.

NOT KNOWN: No record states the cause of his death. Historians think that overwork and too little food killed him.

Shakespeare on the rivers

Actors carried plays to river towns by boat. The actor Noah Ludlow bought a flatboat, a wide boat with a flat bottom, 25 feet long, at Olean, New York, in 1816. He named it Ludlow's Noah's Ark. He and eleven others floated down the Allegheny, Ohio and Mississippi rivers and stopped to perform wherever they could.

The Floating Theatre. William Chapman, a British-born actor, built a boat called the Floating Theatre at Pittsburgh in 1831. It was 14 feet wide and 100 feet long, the first boat built to be a theatre. Chapman and his family acted plays such as Shakespeare's *The Taming of the Shrew* at each landing, with songs and dances added.

When they reached New Orleans, they broke up the boat and went back to Pittsburgh on a steamboat to build another. From 1836 Chapman used a small steamboat to tow the theatre back up the river.

A night on the Floating Theatre began with a play. Then came a speech and a set of short scenes and songs. A ticket cost fifty cents, and people who had no cash could pay with fruit, potatoes or bacon. Minstrel shows in the same years acted comic versions of Shakespeare, with titles such as "Julius Sneezer."

The Astor Place riot, May 10, 1849

Soldiers in New York fired into a crowd outside the Astor Opera House, on Astor Place, on May 10, 1849, and at least 22 people died. City officials and the soldiers' own general had given the order. The crowd had come to stop the English actor William Charles Macready from playing Shakespeare's *Macbeth*.

The quarrel began between two actors. Edwin Forrest, the American star, was jealous of Macready. In 1846, in a theatre in Edinburgh, Scotland, Forrest hissed at Macready from a private box.

The first night. Forrest's supporters stopped Macready's *Macbeth* inside the opera house on May 7 by throwing rotten eggs, potatoes, shoes and pieces of ripped-up seats. Forty-seven New Yorkers, including the writers Herman Melville and Washington Irving, signed a letter asking Macready to play again.

Captain Isaiah Rynders and the writer Ned Buntline put up posters that spoke to working men. Rynders belonged to Tammany Hall, a political group in New York. The posters read "WORKINGMEN! SHALL AMERICANS OR ENGLISH RULE IN THIS CITY?" The two men also handed out free tickets.

The order to fire. A crowd of about 10,000 gathered outside the opera house on May 10. Mayor Caleb Woodhull had called out the militia, an army of part-time soldiers who have other jobs. General Charles Sandford brought about 350 soldiers of the Seventh Regiment, one unit of that army. People in the crowd threw stones at the building and at the soldiers.

Frederick Talmadge was the city's recorder, an elected city officer who also served as a judge. He told General William Hall, who commanded the soldiers on the spot, "You must order your men to fire." Sandford, the general who had brought the soldiers, said his men would not "stand and be pounded to death with stones." Then the sheriff, whose name no record gives, said, "you have permission to fire." Sandford gave the order. The soldiers fired first over the crowd's heads. Then they lowered their guns and fired into the crowd.

The dead. The counts differ. Betsy Golden Kellem, writing in *Smithsonian* magazine in 2017, counts 18 dead that night and five more who died of their wounds in the next few days, 23 in all. Miguel Hernandez, writing in the *New*

York Almanack, counts 25. The historian Nigel Cliff counts between 22 and 31.

Kellem writes that many of those shot were bystanders, people who were only watching. Seven of the dead were Irish immigrants. The dead included carpenters, clerks, printers and laborers, both men and women. About 48 people in the crowd were hurt, along with 50 to 70 police officers and 141 soldiers. The next night, in more fighting near Astor Place, a young boy was killed.

A jury later found that the events "justified the authorities in giving the order to fire." In September 1849 Buntline and nine others were found guilty of starting the riot, fined and jailed. Kellem does not name the judge. Kellem calls the shooting "the most significant loss of civilian life in New York since the Revolution."

Edwin Forrest

Who: An American-born star actor whose fans rioted against his English rival, William Charles Macready, at Astor Place in 1849.

When and where: Born in Philadelphia on March 9, 1806. Acted from about 1817 to 1872. Died December 12, 1872.

He paid prizes for new plays with a big part for himself.

Edwin Forrest was an American star actor. In 2006 the Theatre Alliance of Greater Philadelphia, a group of the city's theaters, called him "America's first superstar of the stage and the highest paid actor in the world in the mid-nineteenth century." He first acted at eleven, at the South Street Theatre in Philadelphia. He became a star playing Othello at New York's Bowery Theatre in 1826.

Metamora. From 1828 Forrest paid prizes for new American plays that had a leading part written for him. The first winner was *Metamora*, by John Augustus Stone. Its hero, Metamora, stands for Metacom, a leader of the Wampanoag nation whom the English called King Philip. Forrest, a white actor, played Metamora from December 15, 1829. At the end of the

play, Metamora kills his wife to save her from the settlers, and then he dies. The theatres made record profits from the play, and Stone got very little of the money.

An English visitor, Thomas Hamilton, called Forrest "a coarse and vulgar actor." Hamilton also wrote that his audience "were enraptured," meaning they loved him. No record shows that Forrest led the crowd at Astor Place in 1849, or that he was there that night.

Forrest beat the writer Nathaniel Parker Willis with a whip in Washington Square, in New York, in 1850. Willis took him to court and won $2,500. Forrest's wife, Catherine, won their divorce case on January 24, 1852, after a six-week trial in New York. Forrest had to pay her money every year, and he kept acting almost until he died, partly to pay it.

BY 1850

If you remember one thing: White actors in burnt-cork makeup made a paying business of mocking Black people, while the Black actors who played Shakespeare in New York were arrested and made to promise to stop.

- 1821: William Brown opened the African Theatre in New York.
- 1828: Thomas Dartmouth Rice first went on stage as "Jim Crow."
- May 10, 1849: soldiers fired into the crowd at Astor Place, and at least 22 people died.

Next: Harriet Beecher Stowe refused in 1852 to let her novel against slavery become a play, and theatre managers staged it anyway.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 16 MINUTES

The biggest touring shows after 1850 told the stories of Black and Native people, and white showmen decided how. From 1852 until well after 1900, white actors in blackface toured the country playing the enslaved people in stage versions of *Uncle Tom's Cabin*, a novel written against slavery. Buffalo Bill's Wild West opened in 1883, and Native performers in it acted out attacks on settlers. Black minstrels such as Billy Kersands worked in the same burnt-cork makeup that white minstrels wore.

Other stages. John Wilkes Booth, a working actor, shot President Abraham Lincoln in 1865 in a theatre where he often performed. From 1881 theatre managers in New York and Boston built a new kind of show called vaudeville. In 1894 the first paying customers in New York watched films, and most of those films showed stage performers doing their acts.

A novel against slavery, acted in blackface

Harriet Beecher Stowe published *Uncle Tom's Cabin*, a novel against slavery, in 1852. That year she refused to let it be made into a play, writing that "it would not be advisable." She never earned a cent from any stage version of it.

Theatre managers staged it anyway. The version that lasted opened in Troy, New York, on September 27, 1852. The manager, G. C. Howard, paid his 22-year-old cousin, George L. Aiken, $40 and a gold watch to write it in a week. When Aiken's play reached New York in July 1853, it ran for about 300 performances.

Blackface again. White actors in blackface played the enslaved characters. In Troy, the company's leading man, Green Germon, did not want to play a blacked-up character. He feared audiences would then expect minstrel dances and "Jim Crow" songs from him. The manager's wife, Mrs. Howard, played the enslaved girl Topsy and danced a minstrel dance called a breakdown. In January 1854 Thomas Dartmouth Rice, the actor who had created "Jim Crow," played Uncle Tom at New York's Bowery Theatre.

P. T. Barnum put on a version by Henry J. Conway at his American Museum in New York. His advertisement stated that this version "does not foolishly and unjustly elevate the Negro above the white man in intellect or morals," using a word of that time for a Black person. The advertisement promised white audiences that the play would not show Black people as being as smart or as good as white people.

Tom shows. People called these stage versions Tom shows. In the 1890s between 400 and 500 touring companies were playing the story. An agent's price list from 1893 offered actors to play Uncle Tom by the week, with "prime" ones at $60, "fair" ones at $50 and "culls," the leftovers, at $40.

One report counted a quarter of a million performances by 1902. A dozen companies were still touring the story in 1927.

Charlotte Cushman

Who: A Boston-born actress who played men's parts, including Romeo and Hamlet, as well as women's parts.

When and where: Born in Boston on July 23, 1816. Acted in the United States and Britain from 1835 to 1875. Lived in Rome from 1852. Died in Boston on February 18, 1876.

She played Romeo while her sister Susan played Juliet.

Encyclopaedia Britannica describes Charlotte Cushman as "the first native-born star on the American stage." A Philadelphia theater group has called Edwin Forrest, who was born in Philadelphia ten years before her, "America's first superstar of the stage," so the two descriptions do not agree on who came first. Cushman first trained to sing opera. Her voice failed after she sang parts that were too high for it, and she turned to acting. In 1836 she played Lady Macbeth.

Cushman played men's parts as well as women's. In London in 1845, and again in 1854 and 1855, she played Romeo while her sister Susan played Juliet. She also played Hamlet.

> Cushman lived with the writer Matilda Hays from 1848, and people in Europe knew the two women as a couple. In 1852 Cushman settled in Rome, among a group of American women who were artists. She later lived with the sculptor Emma Stebbins. Cushman helped the career of the sculptor Edmonia Lewis, whose mother was of Ojibwe and African American descent.
>
> During the Civil War, Cushman acted to raise money for the care of Union soldiers. In 1869 doctors began treating her for breast cancer. She kept acting and reading on stage. She announced her farewell at least seven times before her last performance in Boston on May 15, 1875.

The actor who shot Lincoln

The man who killed President Abraham Lincoln was an actor. John Wilkes Booth came from a family of actors, and he often performed at Ford's Theatre in Washington. On April 14, 1865, he learned that Lincoln would come that night to see a comedy called *Our American Cousin*.

That night Booth found the president's box almost unguarded. He went in, barred its outer door from the inside, and waited for a line that he knew would bring a big laugh. Then he burst into the box and shot Lincoln once in the back of the head with a small pistol called a derringer. Booth also slashed Major Henry Rathbone, one of Lincoln's guests, in the shoulder with a knife. The play stopped at the line "you sockdologizing old man-trap," a joking insult made of slang of the time.

Booth jumped from the box to the stage below. He broke his left leg in the fall, though some people believe the leg broke later. Accounts differ on what he shouted: "Sic semper tyrannis," Latin for "Thus always to tyrants," or "The South is avenged!", or both. Lincoln died the next morning.

The theatre after. The federal government bought the theatre in 1866. Edwin Stanton, the Secretary of War, ordered it turned into offices. Workers were digging in the basement on June 9, 1893, to put in an electric light plant,

when a brick pillar that held up the floors gave way. Part of the inside of the building fell in. Twenty-two clerks who worked there were killed, and at least 68 were hurt.

An inquest jury, a jury that looks into a death, met to find out why. It found three men guilty of criminal negligence, which means carelessness so serious it is a crime. The three were Colonel Fred C. Ainsworth, the head of the office whose clerks worked there, the engineer, and George Dant, the contractor in charge of the building. The jury blamed Dant most, because he had not propped up the brick pillars before the digging began. The charges against all of them were later dropped.

The restored theatre held its first public performance since Lincoln's death in 1968.

Edwin Booth

Who: An American actor of Shakespeare who played Hamlet a record hundred nights in a row, and the older brother of John Wilkes Booth.
When and where: Born in Bel Air, Maryland, on November 13, 1833. Acted mostly in New York and on tour. Died in New York on June 7, 1893.
Movie: *Prince of Players* (1955), a feature film about his life.

He pulled the president's son out from beside a moving train.

Edwin Booth once saved the life of Robert Lincoln, the president's son. Late in 1863 or early in 1864, on a train platform in Jersey City, New Jersey, Robert slipped into the gap between the platform and a moving train. Booth grabbed his coat collar and pulled him out. Robert Lincoln described the rescue in a letter in 1909.

Booth played Hamlet about a hundred nights in a row in New York in 1864 and 1865. That was a record, and no actor broke it until 1922. He was the son of the actor Junius Brutus Booth. Edwin supported the Union in the Civil War. His brother John Wilkes supported the Confederacy, the group of Southern states whose leaders broke away from the United States to keep slavery.

After the shooting. After his brother killed President Lincoln, Edwin Booth stayed off the stage for about nine months. He came back as Hamlet in January 1866. He opened his own theatre in New York in 1869. It never made a profit, and in 1874 it went bankrupt, which means Booth could not pay its debts.

A salesman named Mark Gray fired two shots at him on a stage in Chicago in 1879 and missed. In 1888 Booth bought a house in New York and founded The Players, a private club, in it. He kept an upper floor as his home, and he died there.

How vaudeville began

Vaudeville was a show made of many short, separate acts, one after another. Tony Pastor, a theatre manager, took over a theatre on 14th Street in New York in 1881. He sold no alcohol and kept the acts clean, so that women and families would come. *Encyclopaedia Britannica* credits Pastor with the first show of the kind later called vaudeville.

Boston. Benjamin Franklin Keith opened a museum of oddities on Washington Street in Boston in 1883. Its first exhibit was a very small baby, three months old, shown to paying customers as "Baby Alice." Keith added a theatre with 123 seats. He ran a "continuous" show there from 10 in the morning to 11 at night, so customers could walk in at any time.

From 1885 Keith and Edward F. Albee ran the Bijou Theatre in Boston together, and the two men went on to build a chain of vaudeville theatres.

Billy Kersands

Who: A Black minstrel comedian, the most popular Black comedian of his day, who performed in burnt-cork blackface.

When and where: Born about 1842 in Baton Rouge, Louisiana. Toured the United States and Europe. Died on June 30, 1915, in Artesia, New Mexico.

He filled his mouth with billiard balls to get laughs.

Billy Kersands was a Black comedian who blacked his face with burnt cork and played the character that white minstrels had invented. He worked in Black minstrel companies such as Sam Hague's Georgia Minstrels. On stage he played a slow, foolish Black man. He filled his wide mouth with billiard balls, or with a cup and saucer. Tom Fletcher, a Black performer who knew him, wrote that Kersands once told Queen Victoria that if his mouth were any bigger, his ears would have to be moved.

A tavern owner named Charles Callender bought the Black minstrel company that Kersands worked in. That was in 1872. When Kersands and other stars asked for more pay and better treatment, Callender fired them. Most of them later came back to him.

The pay. Kersands earned about $15 a week in 1879, and by 1882 he was reportedly earning $80 a week, only a little less than a white minstrel star. On tours of Europe he earned up to $250 a week. White and Black audiences in the South came to see him. Because he drew such crowds, some theatre owners relaxed their rules limiting Black customers to separate sections of the theatre.

People asked Kersands why he never moved to vaudeville. Fletcher recorded his answer: "All of my money came from the people of the South, the white and the colored (Black), while playing down there." Kersands died shortly after a show, at about 73.

The Wild West: Lakota performers

William F. Cody, called Buffalo Bill, started a show called Buffalo Bill's Wild West near North Platte, Nebraska, in 1883. In its arena, Native performers acted out attacks on stagecoaches, wagon trains and a settler's cabin. Then cowboys and soldiers drove them off, and in the last scene Cody rode to the rescue.

Sitting Bull's contract. The Hunkpapa Lakota leader Sitting Bull signed a contract on June 6, 1885, to tour with the show for four months. It paid him $50 a week and let him sell his own photographs and autographs. He signed at the Standing Rock Agency, the office of the Indian agent at Standing Rock. An Indian agent was the federal official who ran a reservation, land that US officials had set aside for a Native nation. James McLaughlin, the agent at Standing Rock, signed as a witness.

Deaths on tour. Several performers died on the show's tours of Europe. Surrounded by the Enemy, an Oglala Lakota man, died of a lung infection in London in 1887 and was buried there. Four months before him, a one-year-old boy named Red Penny, the son of Little Chief and Good Robe, died and was buried in the same London cemetery.

Chief Hawick died of typhoid fever on January 1, 1890, and Featherman died of smallpox, two diseases that often killed people then. Both men had been left in a hospital in Marseilles, France. Wounds One Another died after he fell from a train in Germany, with $108 in wages still owed to him. No full count of the deaths survives.

The pay. Most Lakota men in the show earned far less than Sitting Bull's $50 a week. The Lakota performer No Neck came home from Europe in July 1890 and told reporters that the men had been paid $25 a month. He said that no more Lakota would go for less than $50 a month. He also said that Cody had given him "a suit of clothes, paid my wages, and gave me $100 besides," and had told him to give "a good report of the treatment of the Indians."

The permits. To hire Lakota people from their reservations, Cody needed permits, written permission from federal officials. John Noble, the Secretary of the Interior, ordered an investigation because of the deaths on the 1889 to

1890 tour and the poor condition of the performers coming home. He also stopped giving new permits for Native performers.

Wounded Knee. Soldiers of the 7th Cavalry, a regiment of soldiers on horseback, were taking the guns of Spotted Elk's band of Lakota at Wounded Knee Creek, in South Dakota, on December 29, 1890. The soldiers opened fire with carbines, which are short rifles, and Hotchkiss guns, which are small cannon. They killed between 250 and 300 Lakota people, most of them women and children.

After the killings, army officers held Lakota Ghost Dancers, people who took part in a religious ceremony called the Ghost Dance, as prisoners at Fort Sheridan, Illinois.

In early 1891 Cody and Nate Salsbury, who had started the show with him, wrote to the Commissioner of Indian Affairs, the federal official over all the agents. They asked for 60 men and 15 women and children. They said the men would be paid $25 a month, that men they called "chiefs" would get $30 to $60 and that women would get $10. The officials in charge of Indian affairs agreed. In March 1891 federal officials released 23 of the Fort Sheridan prisoners to Cody, on the condition that they tour Europe with his show. General Nelson Miles, Cody's close friend, commanded the army in the region.

The prisoners included Short Bull and Kicking Bear. Short Bull said of Wounded Knee, "twenty-three of my own relations were killed in this fight, men, women and children, this is like butchery." In Cody's arena, the prisoners played "hostiles," the Indians who attacked.

The historian Sam Maddra studied these tours and concludes that the Lakota who joined the show got wages, travel and standing among their people. She adds that, perhaps more important, they got away from the control of government officials. They also earned far more than farming on the reservation paid.

Dances made crimes. United States officials and missionaries worked in the same years to stop Native ceremonies and dances. In 1883 Hiram Price, the Commissioner of Indian Affairs, issued rules that made the Sun Dance and

other dances "Indian offenses," or crimes, for the Courts of Indian Offenses on the reservations.

For a first offense, the rules specify stopping the person's rations, the food handed out to them on the reservation, for up to ten days. For a later offense, the rations could be stopped for 15 to 30 days, or the person could be locked in the agency jail for up to 30 days.

Missionaries from the Moravian Church arrived at Quinhagak, Alaska, in 1885. They called the masked dances of the Yup'ik people "heathen rites," meaning the religion of people who were not Christian. No traditional dancing took place in Quinhagak for more than a century, until 2013. *Archaeology* magazine's 2015 account of the village does not name the missionaries or say how the dancing was stopped.

What did the first films show?

Many of the first films were recordings of performers doing their stage acts. In December 1892 Thomas Edison's staff began building a film studio at West Orange, New Jersey. They called it the Black Maria. It was a studio lined with tar paper, with a roof on hinges that flipped open, and the whole building could turn on a track.

The first paying customers. The Holland brothers opened a Kinetoscope parlor at 1155 Broadway in New York on April 14, 1894. A Kinetoscope was a machine that showed one short film to one person looking into it. The parlor had ten machines and ten films, each 15 to 20 seconds long, all shot in the Black Maria.

They showed Sandow, a strongman, and Bertoldi, a contortionist who bent the body into unusual shapes. The other films showed wrestlers, a trapeze act, a Highland dancer, blacksmiths and a barber shop.

A stage kiss. A film projector called the Vitascope showed films on a screen to a seated audience in a theatre for the first time on April 23, 1896, at Koster and Bial's Music Hall, a vaudeville theatre in New York.

That same month, the actors May Irwin and John Rice made a film of a stage kiss. They had kissed at the end of a Broadway musical, *The Widow Jones*, in 1895. They acted the same kiss again in front of Edison's camera. The film, *The Kiss*, runs about 18 seconds.

BY 1900

If you remember one thing: Black and Native performers found paying work on the touring stage in these fifty years, mostly in parts that white managers had written to mock them or to act out their defeat.

In 1850: Theatres played Shakespeare and minstrel shows, and Cody's Wild West, vaudeville and films did not yet exist.

In 1900: Hundreds of companies had been touring *Uncle Tom's Cabin* in blackface, Keith and Albee ran a chain of vaudeville theatres, and New Yorkers had been paying to watch films for six years.

Next: The first film companies kept their actors' names secret, so fans called one star only "the Biograph Girl."

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 24 MINUTES

Acting moved onto film and into radio in these fifty years, and the people in charge kept the parts of the old minstrel stage, where white men had blacked their faces to mock Black people. Americans in 1903 saw films mostly in vaudeville theatres, which put on shows made of many short live acts. From 1905 small theatres charged five cents to see short films, and by 1908 there were about 8,000 of them.

The first film studio in Hollywood opened in 1911. In 1927 *The Jazz Singer* put recorded songs and speech into a full-length film, and by the 1930s eight companies made most American films.

Who played whom. White actors in blackface, dark makeup that made a white face look Black, played Black characters, and white actors made up to look Chinese played Chinese characters.

On the radio, two white men used put-on voices to play Black characters in one of the most popular shows of the time. Because of state laws against marriage between races, producers would not let the Chinese American star Anna May Wong kiss a white actor on screen. The studio heads' own rules banned showing romance between white and Black characters.

Radio plays and, from 1928, the first television plays brought actors into people's homes.

Story films, nickelodeons and actors with no names

Edwin S. Porter made *The Great Train Robbery* for Thomas Edison's company in November 1903. The film runs about 12 minutes. Porter cut between places to show things happening at the same time. He took the title from a stage play of 1896. Later legends called it the first film with a story. That claim is wrong: films with plots came before it.

The nickel theatres. Harry Davis and John P. Harris opened a small theatre in a Pittsburgh storefront on June 19, 1905, and charged a nickel to see films. They named it the Nickelodeon, and small theatres like it were called nickelodeons. By 1908 there were about 8,000 nickelodeons in the United States. Most of them seated fewer than 200 people. William Fox, Carl Laemmle and Louis B. Mayer, who later ran big film studios, all started out as nickelodeon owners.

Hollywood. The Nestor company opened the first permanent film studio in Hollywood on October 27, 1911, at Sunset Boulevard and Gower Street. Hollywood lies in the Los Angeles Basin, the homeland of the Tongva people, whom the Spanish called the Gabrielino. In 1994 California officials named the Gabrielino the first people of the Los Angeles Basin. Federal officials do not list any Tongva group as a tribe.

Actors with no names. The first film companies did not publish their actors' names. The owners of the Biograph company refused to name Florence Lawrence, one of their leading actors, so fans called her "the Biograph Girl." Her biographer writes that the owners feared famous actors would ask for more pay.

The producer Carl Laemmle hired her in 1910. He spread a false story that she had been killed by a streetcar. Then he named her in newspaper ads headed "We nail a lie." Lawrence is often called the first film star, and her headstone says "The First Movie Star." Evidence published in 2019 points to the French actor Max Linder as the first film actor named in public.

Films and free speech. All nine justices of the Supreme Court ruled in 1915 that showing films was "a business, pure and simple." That meant films had no protection as free speech. State and city censors, officials who decided which films could be shown, could cut any film or ban it. In 1952 the justices reversed that ruling.

The Birth of a Nation, 1915

D. W. Griffith's *The Birth of a Nation* showed members of the Ku Klux Klan hunting down and lynching a Black man, killing him with no lawful trial, and stopping Black men from voting. Many of its Black characters were played by white actors in blackface.

The Ku Klux Klan was a secret group that Confederate veterans founded in December 1865. Klansmen, armed and in disguise, whipped freed Black people and the white people who worked with them, burned their homes and murdered them.

The film opened in Los Angeles on February 8, 1915. It ran about three hours, longer than any film released before it. Griffith based it on a novel by Thomas Dixon called *The Clansman*.

What the film shows. The film shows its Black characters as unintelligent and as sexually aggressive toward white women, which means wanting to force sex on them. In one scene, Gus, a freed man played by a white actor in blackface, follows a white girl named Flora Cameron into the woods and says he wants to marry her. She runs from him to the edge of a cliff and jumps to her death.

Klansmen in the film hunt Gus down, hold their own trial, lynch him and leave his body at the house of a character named Silas Lynch.

At the end of the film, armed Klansmen on horseback stop Black men from voting. Griffith said that he had decided to have "no black blood among the principals," the main actors. Black extras, actors hired to fill out crowd scenes, appear in many shots. The filmmakers housed them apart from the white cast.

At the White House. President Woodrow Wilson watched the film in the East Room of the White House on February 18, 1915, with his family and members of his Cabinet, the officials who head the government's departments. Dixon and Wilson had been graduate students together. Wilson's secretary later wrote that Wilson had not known what the film contained.

In 1937, 22 years later, a magazine reported that Wilson had said of the film, "It is like writing history with lightning. And my only regret is that it is all so terribly true." No record from 1915 contains those words, and whether he said them is disputed. In May 1915 Dixon wrote to Wilson's secretary that "the real purpose of my film was to revolutionize Northern sentiments," meaning to change completely what Northerners thought.

The protests. Protests began at the Los Angeles opening in February 1915 and went on when the film opened in New York the next month. In Boston in April 1915, the managers of the Tremont Theatre refused to sell tickets to William Monroe Trotter, the editor of a newspaper called the *Guardian*, and to others with him. Police arrested Trotter and ten others.

Trotter and the Boston branch of the NAACP, the National Association for the Advancement of Colored People, then led rallies of thousands of people against the film.

The bans. Twelve mayors banned the film in 1915, and the mayor of Cedar Rapids, Iowa, was the first. In Chicago, Black voters pressed the mayor to act, and he banned showings on May 17, 1915. Officials in Denver, Pittsburgh, St. Louis and Minneapolis kept the film out too, and officials in the state of Ohio banned it on March 4, 1916. Most cities and states let it play.

The money. No one knows how much people paid to see the film. The company that made it reported $4.8 million by the end of 1917. The historian Richard Schickel estimates more than $60 million from its first run. No American film earned more until *Gone with the Wind* in 1939.

The Klan again. After the film opened in Atlanta, a preacher named William J. Simmons started the Klan again. He led a cross burning, setting a large cross on fire, on Stone Mountain, Georgia, in November or December 1915.

Encyclopaedia Britannica states that Simmons was "directly inspired" by the film. In the 1970s David Duke, a Klan leader, showed the film to bring in new members.

Lilian St. Cyr (Red Wing)

Who: A Ho-Chunk actress who played the leading woman in the first Western made in Hollywood.

When and where: Born on the Winnebago Reservation in Nebraska in February 1884. Acted in films from 1908 to 1925. Died in March 1974.

"He's working with a one hundred percent American."

Lilian St. Cyr played a leading part in the first Western made in what is now Hollywood, and she was paid far less than the man who starred with her, Dustin Farnum. She was Ho-Chunk, born on the Winnebago Reservation in Nebraska. By 1888 both of her parents had died.

The schools. She spent ten years in Indian boarding schools and graduated from the Carlisle Indian School in Pennsylvania in 1902. At these schools, run or paid for by the federal government, staff cut Native children's hair, gave them English names and forbade them to speak their own languages. Staff punished children with beatings, locked rooms and withheld food. A federal report in 2024 counted at least 973 children who died at the schools and said the true number is higher.

She married James Young Deer in 1906. The couple performed in Wild West shows, and from 1908 they acted in short Western films. She used the name Red Wing. In 1909 the director D. W. Griffith hired the couple as technical consultants, people who advise filmmakers on how to get details right.

The film. A film by Cecil B. DeMille opened on February 23, 1914, in which St. Cyr played a Ute woman who marries an Englishman and then kills herself. The film's title used an insult for Native women. St. Cyr made her own costumes. The male lead, Dustin Farnum, earned $1,000

a week. Someone told her she should be proud to act with such a star. She answered, "What about him? He's working with a one hundred percent American."

She acted until 1925. Then she made costumes in New York and worked for a national day to honor Native Americans. Accounts differ on her birth date, February 14 or February 23, 1884, and on the day she died, March 12 or March 13, 1974.

Wounded Knee acted again, 1913

Buffalo Bill Cody made a film in the fall of 1913 that acted out Wounded Knee. There, on December 29, 1890, soldiers of the 7th Cavalry had killed between 250 and 300 Lakota people, most of them women and children, with rifles and small cannon.

Cody filmed in the Badlands and the Black Hills of South Dakota. He got $50,000 for it from the Essanay film company and from the owners of two newspapers. Lindley Garrison, the Secretary of War, supplied soldiers of the 12th Cavalry to play the troops. Franklin K. Lane, the Secretary of the Interior, allowed more than 1,000 Lakota people to take part. General Nelson Miles advised on the film, although he had written that such a show was not "for the welfare of the Indians."

Short Bull. Short Bull acted in it. Twenty-three of his relatives had been killed at Wounded Knee. In 1891 federal officials had released him from prison at Fort Sheridan, an army post in Illinois, to tour Europe with Cody.

The film was called *The Indian Wars*. Only a few minutes of it survive. Ben Black Elk later said that "the government put a ban on it" and destroyed it. No surviving record names the official who did so.

BY 1915: Hollywood had its first film studio, the Supreme Court had ruled that films had no protection as free speech, and the longest film yet made showed Klansmen lynching a Black man.

Bert Williams

Who: A Black comedian who wore blackface on stage and starred in an otherwise all-white Broadway show.

When and where: Born in Nassau, in the Bahamas, on November 12, 1874. Performed in the United States and Britain from 1895. Died in New York City on March 4, 1922.

"I don't belong to either side. Nobody wants me."

Bert Williams was a Black comedian who blacked his face to work, even as a star on Broadway. He moved to the United States at ten. From 1895 he worked with another Black comedian, George Walker. The two men wore burnt-cork blackface, as white minstrels did.

Their stage name paired the word "Real" with an insult for Black people. It told audiences that these two were real Black men, unlike the many white performers in blackface. On stage Williams played the slow-witted, shuffling Black man that was then a standard part on the minstrel and vaudeville stage.

In August 1900, during a riot in New York, a white mob pulled Walker off a streetcar and beat him. In 1903 the two men starred in *In Dahomey*, a musical with an all-Black cast. It played on Broadway and then at Buckingham Palace in London.

One Black act. In 1909 a group of white vaudeville performers called the White Rats, who opposed Black performers and women gaining ground in vaudeville, made theatre managers cut Williams's billing, the size and place of his name on the posters. Managers of the big chains of vaudeville theatres allowed only one Black act on each program. Williams had to travel, eat and sleep apart from the white performers.

The Follies. Florenz Ziegfeld hired Williams for his Broadway revue, the *Follies*, a show of songs and comic scenes, in 1910. Several white cast members demanded that Ziegfeld fire him. Ziegfeld answered, "I can replace every one of you," except Williams. Williams starred in the *Follies* until 1919. On one three-year contract he was paid $62,400 a year.

In August 1919 the rest of the cast went on strike, and no one told him. He said, "I don't belong to either side. Nobody wants me." He collapsed on stage in Detroit in February 1922 and died in New York at 47. More than 5,000 people filed past his coffin.

Charlie Chaplin

Who: A London comedian who became a film star around the world by acting without speaking.

When and where: Born in London on April 16, 1889. Made films in Los Angeles from 1914 to 1952. Died in Switzerland on December 25, 1977.

Movie: *Chaplin* (1992), a feature film about his life, directed by Richard Attenborough.

When he came back, the audience stood and applauded for 12 minutes.

Charlie Chaplin became famous around the world for acting with his face and body, with no spoken words. He grew up poor in London. Before he was nine, he was sent twice to a workhouse, a place where poor people lived and worked in return for food and a bed.

Chaplin came to the United States with a touring comedy company. In September 1913 he signed with the Keystone studio for $150 a week. His character the Tramp first appeared on screen on February 7, 1914. In 1916 the Mutual company paid him $670,000 for one year of work.

His own company. Chaplin formed a company called United Artists on February 5, 1919, with the actors Mary Pickford and Douglas Fairbanks and the director D. W. Griffith. The four could then pay for their own films and control them. Chaplin kept making films with no spoken words after sound arrived, including *City Lights* (1931) and *Modern Times* (1936). In his first talking film, *The Great Dictator* (1940), Chaplin attacked Adolf Hitler and Benito Mussolini, the dictators, or rulers with total power, of Germany and Italy.

Shut out. FBI agents under J. Edgar Hoover investigated Chaplin. On September 18, 1952, he sailed for London. The next day Attorney General James McGranery canceled his permit to come back into the United States. Chaplin settled in Switzerland. He came back once, in April 1972, to accept an honorary Academy Award, and the audience stood and applauded for 12 minutes.

Sound, and the Production Code

The Jazz Singer opened in New York on October 6, 1927. It was the first full-length film with songs and speech recorded to match the actors' lips, in a few scenes. Its star, Al Jolson, was a Jewish performer born in Lithuania who worked in blackface. At the end of the film he sings in blackface.

The Code. The heads of the big studios agreed in 1930 to follow a set of rules called the Production Code. Martin Quigley, a Catholic publisher, and Daniel Lord, a Catholic priest, wrote it. The studio heads adopted the Code through their own trade group, led by Will H. Hays. From July 1, 1934, no film from those studios could open without a seal of approval from an office run by Joseph Breen.

One rule in the Code read: "Miscegenation (sex relationships between the white and black races) is forbidden," using an old word for marriage or romance between people of different races. The studio heads rewrote that rule in 1956. On November 1, 1968, they replaced the whole Code with movie ratings.

BY 1934: Films talked, eight companies made most of them, and no film from the big studios could open without the approval of Joseph Breen's office.

How a seven-year contract ran longer

Actors signed contracts with a studio for up to seven years, and studio heads did not count any time when an actor was not working. So a contract could run much longer than seven years. The actress Olivia de Havilland sued Warner Bros. over this in 1943. On December 8, 1944, California judges ruled that seven years meant seven years on the calendar.

Eight companies. Eight companies made most American films by the 1930s. Five of them, MGM, Paramount, Warner Bros., 20th Century-Fox and RKO, also owned chains of theatres. By 1939 there were 15,000 movie theatres in the country.

The justices of the Supreme Court ruled against the five studios that owned theatres on May 4, 1948, in a case called the Paramount case. The justices also outlawed block booking, the studios' practice of selling films to theatres only in large bundles. By 1954 the studios had sold their theatres.

Oscar Micheaux

Who: A Black homesteader and novelist who made more than 40 films for Black audiences outside the Hollywood studios.

When and where: Born in Metropolis, Illinois, on January 2, 1884. Made films from 1919 to 1948. Died in Charlotte, North Carolina, on March 25, 1951.

Movie: *Midnight Ramble: Oscar Micheaux and the Story of Race Movies* (1994), a PBS documentary about him and about films made for Black audiences.

The only known copy of his second film turned up in Spain.

Oscar Micheaux made more than 40 films for Black audiences outside the Hollywood studios. He worked as a Pullman porter, an attendant on railroad sleeping cars. Then he farmed a homestead in Gregory County, South Dakota, a farm on land that the federal government gave to settlers who lived on it and worked it. In 1906 he bought the claim from a settler who had given it up, and he lost the farm in 1911. He wrote novels about that life and sold them door to door.

George Johnson, the manager of the Lincoln Motion Picture Company, a Black-owned film company, offered in 1918 to film his novel *The Homesteader*. Micheaux wanted to make the film himself. He started his own company and sold shares in it for $75 to $100 each. The film came out in 1919.

Within Our Gates. His next film, *Within Our Gates* (1920), shows a white mob lynching a Black couple and burning their bodies. Micheaux gave it to Chicago's censor board in December 1919, and the board's members first rejected it. Five months earlier, gangs of white youths in Chicago had attacked Black residents during a riot in which 38 people were killed, 23 of them Black. Critics feared the film would set off more violence. Officials in Omaha and New Orleans also blocked it or demanded cuts.

> Micheaux's films played in about 700 theatres. He paid for each new film by touring with the last one and raising money from investors along the way, often carrying the copies of the film by hand. The only known copy of *Within Our Gates* was found in Spain in 1979.

Plays on the radio

Radio drama was acting for listeners who could not see the actors. Two white men, Freeman Gosden and Charles Correll, started *Amos 'n' Andy* on a Chicago radio station on March 19, 1928. They knew the traditions of the minstrel show. They voiced Amos, Andy and many other characters, more than 170 in the first ten years. One storyline in 1930 and 1931 drew about 40 million listeners. In 1930 Gosden and Correll played the parts in blackface in a film.

The petitions. Robert L. Vann, the publisher of the *Pittsburgh Courier*, a Black newspaper, collected petitions, written requests signed by many people, against the show from more than 700,000 Black Americans in 1931. Vann sent them to the Federal Radio Commission, the government office in charge of radio. Writers at the *Chicago Defender*, another Black newspaper, praised the show that same year.

The War of the Worlds. Orson Welles, who was 23, and his Mercury Theatre on the Air performed H. G. Wells's novel *The War of the Worlds* on the CBS radio network on October 30, 1938. They acted it as a string of fake news bulletins about Martians landing in New Jersey. The announcement that it was a play came almost 40 minutes in.

The next morning, newspaper stories across the country described a national panic. The *New York Times* ran an opinion piece headed "Terror by Radio." Later researchers found that the panic was far smaller than those stories said. Jefferson Pooley and Michael J. Socolow, writing in *Slate* in 2013, called it a myth, and they wrote that newspapers, which had been losing advertising money to radio, blew the story up.

That night workers for C. E. Hooper, a company that counted radio listeners, had telephoned 5,000 homes. Only 2 percent said they were listening to the Welles program, and none said they were hearing a news broadcast. Some

listeners were frightened and called police stations and newspapers. Six weeks later, Princeton researchers checked six New York City hospitals and found no one brought in because of the broadcast. No death has ever been traced to it.

The Federal Theatre Project, 1935 to 1939

Federal officials set up the Federal Theatre Project on August 27, 1935, to pay theatre workers to put on plays. It was part of the Works Progress Administration, a federal program that gave people jobs. Harry Hopkins, the head of that program, chose Hallie Flanagan, a theatre professor at Vassar College, to lead the project.

Within a year it employed 15,000 people at $23.86 a week, when the actors' union said an actor should get at least $40. About 30 million people saw its roughly 1,200 productions, and most of the shows were free.

Macbeth in Harlem. Federal Theatre officials set up 17 Negro Theatre Units, companies of Black actors, in cities from Seattle to Birmingham. On April 14, 1936, the New York unit opened *Macbeth* at the Lafayette Theatre in Harlem. It had an all-Black cast of about 150 and was set on an island like Haiti in the 1800s. Orson Welles, then 20, directed it. A free show on April 20 drew 3,000 more people than the theatre could seat.

The end. In 1938 Martin Dies Jr. and the other members of a House of Representatives committee he led held hearings and charged that the project spread communism, a system in which one party runs the government and the government owns the farms and factories. The committee's members argued that "racial equality forms a vital part of the Communist dictatorship." On December 6, 1938, Flanagan testified before the committee. Members of Congress then cut off the project's money in June 1939, and 8,000 people lost their jobs.

Anna May Wong

Who: A film actress often called the first Chinese American film star, who spoke out against the parts that film studios gave her.

When and where: Born Wong Liu Tsong in Los Angeles on January 3, 1905. Acted in the United States and Europe from 1919. Died in Santa Monica, California, on February 3, 1961.

"Why is it that the screen Chinese is always the villain?"

Anna May Wong became a film star while producers kept her out of leading parts written for Chinese women. Her parents were Chinese Americans. She got her first film job as an extra in 1919. She starred in *The Toll of the Sea* (1922), one of the first films made in color.

Studio producers would not let her kiss a white actor on screen, because state laws banned marriage between races. That held even when the white actor was playing an Asian man. Producers cast her as villains, as the sly and dishonest "Dragon Lady," or as the gentle "Butterfly" who gives up everything. In 1933 she wrote in a magazine article called "I Protest," "Why is it that the screen Chinese is always the villain?"

The Good Earth. The producers at MGM refused to consider her for O-Lan, the Chinese heroine of *The Good Earth* (1937). They gave the part to Luise Rainer, a white actress made up to play a Chinese woman. Rainer won the Academy Award for Best Actress. MGM's producers tested Wong for Lotus, a dishonest singer. She turned the part down. She told the producer Irving Thalberg that he was asking her, "with Chinese blood," to play "the only unsympathetic role" in a film where white actors played the Chinese characters.

She starred in *The Gallery of Madame Liu-Tsong* in 1951, the first American television series with an Asian American star. In 2022 her picture went on the quarter, which made her the first Asian American shown on United States coins.

Hattie McDaniel

Who: The first Black person to win an Academy Award.

When and where: Born in Wichita, Kansas, on June 10, 1893. Acted in Los Angeles from the 1930s. Won the award on February 29, 1940. Died on October 26, 1952.

She won the Academy Award and sat at a separate table.

Hattie McDaniel won the Academy Award in 1940 and had to sit apart from the white stars of her film to receive it. She was the youngest of 13 children, and her parents had been enslaved.

In *Gone with the Wind* she played Mammy, an enslaved house servant, for $450 a week. The film opened in Atlanta on December 15, 1939, at the Loew's Grand Theatre. The producer, David O. Selznick, asked that she attend. Officials at MGM advised against it because of Georgia's segregation laws, under which Black and white people had to stay apart. She did not go. Clark Gable, the film's star, threatened to stay away too, and she talked him into going.

The award. She won the Academy Award for Best Supporting Actress on February 29, 1940. The ceremony took place at the Cocoanut Grove in the Ambassador Hotel in Los Angeles. The hotel had a strict rule against Black guests, and Selznick had to ask a special favor to get her into the building. She sat at a small table with her escort and her white agent, away from the film's white stars. In her speech she said she hoped to "always be a credit to my race."

Accounts differ on where the table stood. *The Hollywood Reporter* places it against a far wall. A 2008 book places it at the back of the room. A 2019 article places it in a separate back room used to store the trophies.

At least 74 of her film parts were maids and other servants. Walter White of the NAACP and others criticized her for taking those parts. She often answered, "I'd rather play a maid than be a maid."

Her house and her grave. In 1945 white neighbors in West Adams Heights, her Los Angeles neighborhood, sued her and 30 other Black homeowners. They wanted to force them out under a rule that only white people could live there. Judge Thurmond Clarke threw the case out. McDaniel died of breast cancer in 1952. She had asked to be buried in Hollywood Cemetery, but the people who ran it refused, because the cemetery took only white people. She was buried at Rosedale Cemetery.

The first television plays

The radio station WGY in Schenectady, New York, broadcast a short spy play called *The Queen's Messenger* by television on September 11, 1928. It was the first television drama. Two actors, Izetta Jewell and Maurice Randall, performed it. Viewers watched on screens three inches wide.

Twenty years later, on June 8, 1948, the NBC network began broadcasting a comedy show called *Texaco Star Theater*. The comedian Milton Berle became its host that fall, and people called him "Mr. Television."

BY 1950

If you remember one thing: Films and radio carried the old minstrel parts to millions of people, and studio rules and state laws kept actors of color out of love scenes with white actors.

One number: about 40 million. That many people listened to one storyline of *Amos 'n' Andy*, in which two white men played Black characters, in 1930 and 1931. In 1931 more than 700,000 Black Americans signed petitions against the show.

Next: Ten Hollywood writers and directors refused in 1947 to tell a House committee whether they belonged to the Communist Party.

1950 to 2000

ERA 9 OF 10 · 1950 TO 2000 · ABOUT 17 MINUTES

The fight over who could act turned political after 1947. Members of a House committee questioned film workers about communism, a system in which one party runs the government and the government owns the farms and factories. Studio executives announced that they would not knowingly hire any Communist. Hundreds of writers, directors and actors lost work, some for more than ten years.

Live and on film. Actors in the early 1950s performed plays live on television, with no second takes. By 1961 about 80 percent of evening programs were filmed ahead of time.

The parts on offer. White actors still played Asian and Native characters. Actors of color such as Rita Moreno and Sidney Poitier won Academy Awards, and they were still offered a narrow set of parts.

Studio executives began opening big films in hundreds of theatres on the same day in 1975. Families could record shows at home from 1975, and in 1984 the Supreme Court justices ruled that taping a broadcast to watch later was legal. A physicist built one of the first video games in 1958, and by the 1990s game makers were hiring actors.

Who lost work in the blacklist?

Hundreds of people in film, radio and television lost their jobs in the late 1940s and 1950s because of their politics, or because they would not answer questions about it. Studio executives put them on a blacklist, a list of people they would not hire. The list was rarely written down. It grew out of many separate decisions by studio executives.

The hearings. Members of the House Un-American Activities Committee, a group of members of the House of Representatives, ordered 42 people from the film industry in late September 1947 to answer questions about communism. The hearings began in October. Walt Disney, the studio head Jack Warner, and the actors Gary Cooper and Ronald Reagan answered the committee's questions. Disney named former employees of his who, he said, were probably Communists.

The Hollywood Ten. Ten writers and directors refused to say whether they belonged to the Communist Party. They became known as the Hollywood Ten. On November 24, 1947, members of the House voted 346 to 17 to charge them with contempt of Congress, the crime of refusing to cooperate with Congress.

The next day, nearly 50 film executives met at the Waldorf-Astoria hotel in New York. Eric Johnston, who led the group that spoke for the film producers, then announced that the ten would be fired or suspended without pay. He said the studios would not "knowingly" hire any Communist. The promise is called the Waldorf Statement.

The ten were found guilty in 1948. They appealed, and the justices of the Supreme Court refused to hear their case again. They went to prison in 1950, for up to a year each. Others lost their jobs after witnesses named them, or after they refused to answer the committee's questions. The list grew to hundreds of names, and no one kept a complete count.

False names. Some writers kept working under false names. In 1957 the Academy Award for the story of *The Brave One* went to "Robert Rich." That was a false name for the blacklisted writer Dalton Trumbo. The real Robert

Rich was the nephew of the film's producer. In 1975 Academy officials reissued the award in Trumbo's name.

Some people named others to the committee, among them the director Elia Kazan and the writer Budd Schulberg. The actor Sterling Hayden said in 1963, "I was a rat, a stoolie, and the names I named of those close friends were blacklisted and deprived of their livelihood," using "stoolie" for a person who informs on others.

The blacklist began to break in 1960. The director Otto Preminger announced on January 20 that Trumbo had written *Exodus*, and the actor Kirk Douglas gave Trumbo credit on screen for *Spartacus*.

Paul Robeson. Paul Robeson was a Black actor who also gave concerts. In 1943 he became the first Black actor to play Othello on Broadway. In 1949 he gave a speech in Paris that was misquoted, and his concerts were canceled. Britannica's account does not name who misquoted him or who canceled the concerts.

That year near Peekskill, New York, people against communism started a riot and burned a straw figure of him. In 1950 officials of the State Department took away his passport, so he could not travel outside the country. His yearly income fell from $100,000 to $6,000 between 1947 and 1952.

He told the House committee in 1956, "I am not being tried for whether I am a communist." He said he was being tried "for fighting for the rights of my people, who are still second-class citizens in these United States of America."

Lee Grant

Who: An actress who was blacklisted for twelve years at the start of her film career, and who later won an Academy Award for acting. A documentary she directed won another Academy Award.

When and where: Born Lyova Haskell Rosenthal in New York City on October 31. The year is disputed, and every year from 1925 to 1931 has been given. Acted in New York and Los Angeles from 1948.

"Twelve working, acting years from my life."

Lee Grant lost twelve years of work in film and television to the blacklist. In 1949 she played a shoplifter in the play *Detective Story* on Broadway, and she played the part again in the 1951 film. It earned her an Academy Award nomination and the Best Actress prize at the Cannes Film Festival in France.

Two reasons. The actor J. Edward Bromberg died in 1951 soon after members of the House committee questioned him. At his memorial service, Grant gave a speech that suggested the committee's pressure had caused his death. In April 1951 the director Edward Dmytryk had named the writer Arnold Manoff to the House committee as a Communist. Grant married Manoff in 1952.

Accounts give two reasons why film and television employers put her on the blacklist, and they do not name those employers. One is her speech for Bromberg. The other, as the actor Kirk Douglas remembered it, is that she refused to testify about her husband.

For twelve years she could get little work in film or television. She taught acting and took small parts under other names. In her 2014 book about her own life she wrote of "my rage at those who had taken twelve working, acting years from my life."

She came back in the television series *Peyton Place* and won an Emmy award in 1966. She won the Academy Award for Best Supporting Actress for *Shampoo* (1975). Her documentary *Down and Out in America* (1986)

> won the Academy Award for Documentary Feature.

The Actors Studio, and plays on live television

Elia Kazan, Cheryl Crawford and Robert Lewis, all from a theatre company called the Group Theatre, opened the Actors Studio in New York in 1947, a place where working actors trained. About 700 actors tried out in the first year, and 50 were chosen, including Marlon Brando. Teachers at the studio taught "method acting," a way of acting built on the teaching of Konstantin Stanislavsky, a Russian director at the Moscow Art Theatre. Lee Strasberg ran the Actors Studio from 1952 until he died in 1982.

Brando played Stanley Kowalski in Tennessee Williams's play *A Streetcar Named Desire* on Broadway on December 3, 1947. Kazan directed it, and he directed the film version in 1951.

One chance. The television networks broadcast plays performed live from 1947, such as *Kraft Television Theatre*, from May 7, 1947, and *Studio One*, from November 7, 1948. The actors had one chance, with no second takes. The first practical videotape came in 1957. By 1961 about 80 percent of evening programs were filmed ahead of time.

I Love Lucy: a stage comedy on film

I Love Lucy was filmed like a stage play, in front of a live audience. It ran on CBS from October 15, 1951, to May 6, 1957. Lucille Ball and Desi Arnaz acted each episode in order, like a play, before an audience of about 300 people. Three film cameras recorded it at the same time.

Jerry Fairbanks had used the three-camera method on other shows before. Karl Freund, the man in charge of the cameras, filmed *I Love Lucy* with three 35 mm film cameras. It was the first scripted television show filmed on 35 mm film in front of a studio audience.

The deal. Executives at Philip Morris, the company that paid for the show, wanted it performed live in New York. Ball and Arnaz wanted to stay in Los Angeles, because Ball was pregnant, and to make the show on film. They offered to take a pay cut of $1,000 a week, and executives at CBS and Philip Morris agreed. In return Ball and Arnaz got 80 percent ownership of the films. Because they owned the films, they could sell the episodes again as reruns.

Lucille Ball

Who: A film actress who starred in the television comedy *I Love Lucy* and then became the first woman to run a major television studio.

When and where: Born in Jamestown, New York, on August 6, 1911. Worked in Los Angeles from the 1930s. Died on April 26, 1989.

Movie: *Being the Ricardos* (2021), a feature film about Ball and Arnaz, written and directed by Aaron Sorkin.

About 44 million people watched Lucy have a baby.

Lucille Ball became the first woman to run a major television studio in 1962. Through the 1930s she acted in films under contract to the RKO studio. People called her "Queen of the B's," after the low-budget films called B movies.

She starred in a radio comedy, *My Favorite Husband*, on CBS in 1948. Her real husband was Desi Arnaz, a Cuban American bandleader and actor. When CBS made *I Love Lucy*, the network's executives at first did not want Arnaz, a Cuban, cast as Ball's husband.

The episode of *I Love Lucy* in which Lucy has a baby aired on January 19, 1953, and about 44 million people watched it. In 1957 Desilu, the company Ball and Arnaz owned, bought the old RKO studios for more than $6 million. Ball had once worked there as a contract actress.

Running the studio. Ball and Arnaz divorced in 1960. In November 1962 she bought his share of Desilu and became its president. Desilu's staff developed the television series *Star Trek* and *Mission: Impossible*. She agreed in 1967 to sell Desilu for $17 million, and the sale closed in 1968.

Lorraine Hansberry

Who: The first Black woman to have a drama produced on Broadway.

When and where: Born in Chicago on May 19, 1930. *A Raisin in the Sun* opened in New York on March 11, 1959. Died in New York City on January 12, 1965, at 34.

Movie: *Lorraine Hansberry: Sighted Eyes/Feeling Heart* (2017), a documentary about her life and work.

Her family's fight over a house reached the US Supreme Court.

Lorraine Hansberry wrote *A Raisin in the Sun*, the first drama by a Black woman to be produced on Broadway. Her father, Carl Hansberry, bought and sold houses for other people in Chicago. When she was about seven, the family bought a house in Woodlawn, a white neighborhood. Owners there had signed a restrictive covenant, an agreement not to sell to Black people.

Anna M. Lee, a homeowner in the neighborhood, sued to cancel the sale. A violent mob harassed the family. The judges of the Illinois Supreme Court ruled against the Hansberrys, and the family had to leave. In 1940, in the case *Hansberry v. Lee*, the justices of the US Supreme Court reversed that ruling. Only 54 percent of the owners had signed the agreement, and it needed 95 percent.

The play. *A Raisin in the Sun* tells of the Younger family, a Black family in Chicago who buy a house in a white neighborhood. Karl Lindner, a white man sent by that neighborhood, offers to buy them out so they will stay away. The producer, Philip Rose, spent 18 months raising money for a play in which all but one character is Black.

Lloyd Richards directed it, the first Black director of a Broadway play. Sidney Poitier and Ruby Dee starred. The play ran for 530 performances. Richards said it was the first play on Broadway that large numbers of Black people came to see. Hansberry died of pancreatic cancer in 1965.

Who was allowed to play whom

White actors kept playing Asian characters. In *Breakfast at Tiffany's* (1961), Mickey Rooney, a white actor, played a Japanese landlord named I. Y. Yunioshi. In 2008 an Asian American group in Sacramento, California, told the city council that the part spread "hateful racial stereotypes," unfair ideas that every member of a group is the same, and the film was pulled from a free film series there. Rooney said that if he had known people would be so offended, "I wouldn't have done it."

Iron Eyes Cody. Espera de Corti was born in Kaplan, Louisiana, in 1904. Under the name Iron Eyes Cody, he played Native Americans in more than 200 films, and he said his parents were Cherokee and Cree. In 1971 he appeared in a television advertisement against litter, with a tear rolling down his face. In 1996 a New Orleans newspaper reported what his half-sister and his baptism record showed: his parents were Sicilian and Italian, and he was Italian American. He denied it.

Rita Moreno

Who: A Puerto Rican actress, dancer and singer who became the first Hispanic woman to win an Academy Award.

When and where: Born Rosa Dolores Alverío Marcano in Humacao, Puerto Rico, on December 11, 1931. Moved to New York in 1936. Acted in Hollywood from 1950. Won the award on April 9, 1962.

Movie: *Rita Moreno: Just a Girl Who Decided to Go for It* (2021), a documentary about her.

"I was always barefoot. It was humiliating, embarrassing stuff."

Rita Moreno became the first Hispanic woman to win an Academy Award on April 9, 1962, and film producers went on offering her the same narrow parts. Her mother brought her from Puerto Rico to New York City in 1936. Moreno was acting on Broadway at 13, and as a teenager she signed a contract with MGM.

Studio producers cast her again and again in the same kind of part. She later said, "I was always offered the stereotypical Latina roles. The Conchitas and Lolitas in westerns. I was always barefoot. It was humiliating, embarrassing stuff. But I did it because there was nothing else."

Anita. In the film *West Side Story* (1961) she played Anita, a Puerto Rican woman in New York. The part won her the Academy Award for Best Supporting Actress. She said the offers that followed were "pretty much the same thing. A lot of gang stories." She did not make another film for seven years. She went on to win a Grammy award for recorded music in 1972, a Tony award for theatre in 1975, and Emmy awards for television in 1977 and 1978.

Sidney Poitier

Who: The first Black man to win the Academy Award for Best Actor.

When and where: Born in Miami on February 20, 1927. Raised in the Bahamas. Acted in New York and Los Angeles from the 1940s. Won the award on April 13, 1964. Died on January 6, 2022.

Movie: *Sidney* (2022), a documentary about him, directed by Reginald Hudlin.

An older waiter helped him read the newspaper every night.

Sidney Poitier became the first Black man to win the Academy Award for Best Actor on April 13, 1964. His parents were tomato farmers on Cat Island in the Bahamas. He was born two months early in Miami, while they were there on business, so he was a US citizen. He grew up on Cat Island and in Nassau. He went to Miami at 15 and moved to New York at 16, where he worked washing dishes.

The audition. Poitier failed his first audition at the American Negro Theatre, a Black theatre company in New York, because he could not read the script well. An older waiter sat with him every night for several months and helped him read the newspaper. In November 1943, at 16, he joined the Army. Later he tried out again and won a part.

> He became the first Black actor nominated for Best Actor in 1958, for *The Defiant Ones*. In 1959 he starred on Broadway in *A Raisin in the Sun*. He won Best Actor for *Lilies of the Field* in 1964. In 1967 more people bought tickets to his films than to any other star's. Critics complained that his characters were too perfect. At the time, he was the only major Black actor being cast in leading parts.

Why Brando refused his Oscar, 1973

Marlon Brando won the Academy Award for Best Actor for *The Godfather* on March 27, 1973, and he refused it in protest. He stayed away from the ceremony and sent Sacheen Littlefeather, who said her father was White Mountain Apache and Yaqui, to refuse the award for him.

She spoke for about 60 seconds. She said Brando was protesting "the treatment of American Indians today by the film industry," and the armed standoff then going on at Wounded Knee, South Dakota. Some people in the audience booed, and some applauded. In 2022 she said that some people mocked her with the "tomahawk chop," a chopping arm motion that makes fun of Native people, as she was led off the stage.

Wounded Knee, 1973. About 200 Oglala Lakota people and members of the American Indian Movement held the village of Wounded Knee for 71 days, from February 27 to May 8, 1973. They were protesting broken treaties and the chairman of the Pine Ridge tribal government. Federal marshals, officers who work for the courts, and FBI agents surrounded them.

Two of the people holding the village were shot and killed. Frank Clearwater was shot in the head during a gun battle with federal forces, and a federal agent's sniper shot Lawrence "Buddy" Lamont through the heart. A US marshal, Lloyd Grimm, was shot and paralyzed from the waist down.

Afterward. Academy officials sent Littlefeather a letter of apology in June 2022. Littlefeather died on October 2, 2022. After her death, her two sisters told the Navajo writer Jacqueline Keeler that the family was not Native American. They said their father was of Spanish and Mexican descent.

Big openings and home video

Jaws opened on 409 screens in the United States at once on June 20, 1975, with advertisements on national television. Columbia Pictures executives had done something like it a month earlier with a film called *Breakout*. *Jaws* earned more than any film before it. *Star Wars* opened on May 25, 1977, in only a small number of theatres, because few theatre owners wanted it. It quickly became a surprise success and passed *Jaws* in earnings.

Taping at home. Sony's Betamax video recorder went on sale in the United States in November 1975. The first VHS recorder in the United States, made by RCA, followed on August 23, 1977. In 1976 Universal and Walt Disney Productions sued Sony to stop people from taping at home. On January 17, 1984, the justices of the Supreme Court ruled 5 to 4, in *Sony v. Universal*, that recording a broadcast at home to watch later was legal.

Games, and the first actors in them

The physicist William Higinbotham built *Tennis for Two* in 1958 for a public exhibition at Brookhaven National Laboratory in New York. Visitors lined up to play it on an oscilloscope, a screen that draws lines of light. In 1972 the Atari company sold *Pong* to arcades. It was the first video game to sell well.

Filmed actors. The early games had no actors. From 1983 some arcade games played filmed scenes stored on laser discs, shiny discs read by a beam of light. In the early 1990s, games on compact discs used filmed actors, often from low-budget films and television. *Night Trap* (1992) and *The 7th Guest* (1993) were two of them. In 1994 the makers of *Wing Commander III*, published by Electronic Arts, cast Hollywood actors in it, including Mark Hamill.

BY 2000

If you remember one thing: Actors lost work for their politics after 1947, and actors of color won the top awards while studios went on offering them narrow parts.

The question this era leaves open: Game makers had begun to hire actors. Would the people who acted in games be paid like film and television actors?

Next: The English actor Andy Serkis played the creature Gollum in a marker suit, and people argued over whether that was acting.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 26 MINUTES

The newest fight over acting is over computer copies of actors' faces, bodies and voices. Since 2000, actors have performed in suits covered with markers while computer artists turned their movements into creatures and game characters, and game makers now hire actors to perform whole stories. In 2023 film and television actors and writers went on strike together for the first time since 1960.

Video game performers struck from July 2024 to July 2025. Both strikes were partly about who controls a computer copy of a performer's face, body and voice.

Screens and seats. People could watch films and series streamed into their homes over the internet from 2007. Moviegoers in the United States and Canada spent 22 percent less on tickets in 2025 than in 2019.

Who is on screen. Michelle Yeoh and Ke Huy Quan won Academy Awards for acting in 2023, and in 2024 Lily Gladstone became the first Native American nominated for Best Actress.

Acting in a marker suit

In motion capture, cameras record the movements of an actor's body, and a computer uses those movements to move a character that artists have drawn. When the cameras also record the actor's face, the method is often called performance capture.

Gollum. In the *Lord of the Rings* films (2001 to 2003), the English actor Andy Serkis played the creature Gollum on set, beside the other actors. Artists at a company called Weta Digital used the recorded movements of his body and face. They painted Serkis out of each shot, one frame, or single picture, of the film at a time, and put the computer-made Gollum in his place.

For *The Two Towers*, the second film, the crew fed Serkis's movements into the Gollum model while he performed. It was the first feature film to use capture as the actor acted.

Other actors followed. Tom Hanks played several characters this way in *The Polar Express* (2004). The actors who played the Na'vi people in *Avatar* (2009) did too, and so did Mark Ruffalo as the Hulk in *The Avengers* (2012).

Andy Serkis

Who: An English actor whose performance-capture work started an argument. People asked whether a performance turned into a computer-made body still counts as acting.

When and where: Born April 20, 1964. Played Gollum in films released from 2001 to 2003. Played the ape Caesar in the *Planet of the Apes* films from 2011 to 2017.

He called the animators' work "painting digital makeup."

After Andy Serkis played Gollum, people argued in public over whether an actor whose body is replaced on screen by a computer image is still acting. Serkis also gave Gollum his voice, and for the second and third films the artists redesigned Gollum's face to look more like his.

The Oscar campaign. The films' producer, Barrie M. Osborne, argued that Serkis should be nominated for an Academy Award as an actor, because his voice, his body movements and his facial expressions were on screen. In 2011 staff at the studio 20th Century Fox campaigned for a nomination for his work as the ape Caesar in *Rise of the Planet of the Apes*.

The actor James Franco called Serkis "the undisputed master of the newest kind of acting called 'motion capture.'" Serkis said the apes were "infused with the heart and soul of an actor." He has never been nominated for an acting Oscar. *War for the Planet of the Apes* (2017) was nominated for its visual effects instead.

Serkis described the animators' work in 2014 as "painting digital makeup" onto an actor's performance. Animators objected that the phrase made their work sound small. The director Matt Reeves said the phrase had come from Weta's own artists. In 2011 Serkis opened a capture studio in London, called the Imaginarium.

Games as acted stories

The game company Naughty Dog released *The Last of Us* on June 14, 2013. Troy Baker played Joel, and Ashley Johnson played Ellie, with their voices and their bodies. About 85 percent of the movement in the game was recorded on a motion-capture stage. Some lines were recorded later in a sound studio. The two actors changed their characters as they worked.

The HBO television version began on January 15, 2023, with Pedro Pascal as Joel and Bella Ramsey as Ellie. Ramsey was told not to play the game, so as not to copy Johnson's performance.

The first game strike. Game actors went on strike for the first time from October 2016 to September 2017, for 340 days. They asked for residuals, payments that grow when a game sells well, and they did not get them. In 2016 the voice actor Jennifer Hale told NPR that game actors were paid $825 for a four-hour session, with nothing more however many copies sold.

Ashley Johnson

Who: A child television actor who later played Ellie in the video game *The Last of Us* with her voice and her body, and won the British Academy's game award for performance twice.

When and where: Born in Camarillo, California, on August 9, 1983. Acting in Los Angeles from 1990.

She talked the director into making Ellie tougher.

Ashley Johnson played Ellie in *The Last of Us*, and her acting changed who Ellie is. She began acting at six, as Chrissy Seaver on the comedy *Growing Pains*, from 1990 to 1992. By the time she was 12 she had been in the casts of eight television series.

She persuaded the game's director, Neil Druckmann, to rewrite Ellie as tougher and more independent. She said, "There were some days when we would shoot things that even at my age made me feel a little uncomfortable."

> **Her face.** Her own face shaped the character. The actor Elliot Page complained that the first design of Ellie looked like Page. The studio's artists then redrew Ellie to look younger and more like Johnson.
>
> Johnson won the British Academy Games Award for Performer for the 2013 game. She won it again on March 12, 2015, for the added chapter *Left Behind*. She is the only person to have won it twice. She played Ellie again in *The Last of Us Part II* (2020). In the 2023 television series, she played Ellie's mother, Anna.

Streaming, and fewer people in movie theatres

Since 2007 people have watched films and series streamed into their homes, and moviegoers now spend less at theatres than they did before 2020. Streaming means watching a film or show sent over the internet while it plays, instead of on a disc or at a set broadcast time. Netflix, a company that then mailed DVDs to renters, began its streaming service in January 2007, with 1,000 titles.

Every episode of *Lilyhammer*, a Norwegian series Netflix had bought, came out on the same day in February 2012. *House of Cards*, the first series Netflix had ordered made for itself, came out in February 2013. By 2026 Netflix had 325 million paid memberships in more than 190 countries.

The empty theatres. Theatres in North America had their lowest ticket sales since 1998 over one weekend in March 2020, as the COVID-19 disease spread. Days later, Universal executives announced that three new films would be sold for home viewing within the week. Disney put *Mulan* on its own streaming service instead of in American theatres.

In July 2020 executives at the AMC theatre chain and at Universal agreed that Universal could sell a film for home viewing after at least 17 days in AMC theatres.

Fewer tickets. Moviegoers in the United States and Canada bought about 780 million tickets in 2025, about 5 percent fewer than the more than 820 million of 2024, according to the research company Comscore. Because the

average ticket cost more, they still spent 1.5 percent more than in 2024, a total of $8.87 billion. That was 22 percent less than the $11.4 billion they spent in 2019, before the theatres closed.

Broadway dark for eighteen months

New York Governor Andrew Cuomo ordered on March 12, 2020, that gatherings of 500 people or more be canceled. Under his order, the Broadway theatres had to close at 5 p.m. that day, a day earlier than other large events. According to the theatre magazine *Playbill*, all 41 Broadway theatres stayed dark, which means closed, for 18 months.

Pass Over. The first play back on Broadway was *Pass Over*, by the playwright Antoinette Nwandu. It began previews, the shows before the official opening, at the August Wilson Theatre on August 4, 2021, and it opened on August 22. Full productions of other shows returned that fall.

Nwandu had started writing it in 2012, after Trayvon Martin, an unarmed Black 17-year-old, was shot and killed in Sanford, Florida. George Zimmerman, a man who volunteered to watch his neighborhood for crime, shot him. In 2013 a jury found Zimmerman not guilty.

The Broadway League reported in May 2026 that in the season ending May 24, 2026, audiences bought 14,577,322 tickets and paid $1.91 billion. That was the most money in any Broadway season on record.

YouTube and TikTok

Chad Hurley, Jawed Karim and Steve Chen, three former workers at the payment company PayPal, started YouTube in February 2005. The first video went up on April 23, 2005.

From May 2007, makers of videos in YouTube's Partner Program got a share of the advertising money, usually 55 percent, and YouTube kept 45 percent. From 2018 a channel needed at least 10,000 lifetime views to join the program. Since November 2020, YouTube's staff can put advertisements on any video in the United States. When the maker is not in the program, Google, which owns YouTube, keeps all of that money.

An app called Musical.ly was merged into TikTok, an app owned by the company ByteDance, on August 2, 2018. That October, TikTok was the most downloaded app in the United States.

BY MAY 2026: Actors in marker suits played creatures and game characters, Netflix had 325 million paid memberships, and Broadway had just finished the season that took in the most money on record.

Who is on screen, and who votes on the awards

All 20 actors nominated for Academy Awards on January 15, 2015, were white. That day April Reign started the Twitter hashtag #OscarsSoWhite, a tag that people added to their posts to join the protest. On January 14, 2016, all 20 were white again. The director Spike Lee and the actress Jada Pinkett Smith said they would stay away from the ceremony.

The voters. Members of the Academy of Motion Picture Arts and Sciences vote on the Oscars, and in 2015 most of them were white men. That year 92 percent of the members were white, and three of every four were men. In 2016 the Academy's president, Cheryl Boone Isaacs, promised to double the number of women and people of color among its members by 2020. The Academy then had about 6,000 members.

By 2019 women were 32 percent of the members, up from 25 percent, and people of color were 16 percent, twice the 8 percent of 2015. The Associated Press reported in June 2026 that if everyone invited that year joined, the Academy would have 11,319 members. Of them, 75 percent would be white, down from 92 percent in 2015, and 64 percent would be men.

The count. Hispanic and Native characters were rare in the most popular films of 2024. Researchers at the USC Annenberg Inclusion Initiative count the speaking characters in each year's 100 most popular films. In the films of 2024, they reported in August 2025, 63.6 percent of speaking characters were white, 13.2 percent Black and 13.5 percent Asian. Only 3.1 percent were Hispanic or Latino, and fewer than 1 percent were American Indian or Alaska Native. Ninety-eight of the 100 films had no Native American girls or women.

Asian people are about 6.3 percent of the US population, so Asian characters had about twice that share overall. They were bunched into some of the films, though: 29 of the 100 films had no Asian speaking characters at all.

Reservation Dogs. *Reservation Dogs* ran on the Hulu streaming service from August 9, 2021, to September 27, 2023. Sterlin Harjo, a filmmaker from Oklahoma, created it with Taika Waititi. It is about four Native teenagers in a

small town in the Muscogee Nation. It was the first American series with all Native writers and directors and an almost all Native cast.

Killers of the Flower Moon. *Killers of the Flower Moon* opened on October 20, 2023. It is about the murders of Osage people for their oil money in Oklahoma in the 1920s. Each Osage person held a share of the tribe's oil money, called a headright. When a holder died, the headright passed to the heirs, who could be people who were not Osage.

Under a 1921 law, local judges put many Osage adults' money under white "guardians," men who controlled it for them. Guardians and some of the men who married Osage women stole that money. Between about 1920 and 1925, more than 60 Osage headright holders died in murders and suspicious deaths, by the Osage Nation's count. The FBI counts at least two dozen.

The cattleman William Hale ordered the murders of the family of Mollie Kyle, an Osage woman, so that their headrights would pass to Hale's nephew, Ernest Burkhart, Mollie's husband. Mollie's sister Anna Brown was shot on May 22, 1921. The sisters' mother, Lizzie Q. Kyle, died in July 1921, and poison was suspected. Their cousin Henry Roan was shot in January 1923.

On March 10, 1923, the house of Mollie's sister Rita and Rita's husband, Bill Smith, was blown up, killing them and their servant, Nettie Brookshire. Ernest Burkhart pleaded guilty in 1926, and Hale was found guilty in 1929. Both were sentenced to life in prison.

Osage actors. The director, Martin Scorsese, met with the chief of the Osage Nation, Geoffrey Standing Bear. The film's makers held a meeting with about 200 Osage people. They cast Osage actors in at least 40 parts and hired Christopher Côté to teach the actors the Osage language.

At the premiere, Côté spoke about the film's telling of Mollie's marriage to Burkhart, the man who pleaded guilty in the murders of her family: "when somebody conspires to murder your entire family, that's not love. That's not love, that's just beyond abuse." In 2024 he said, "I still stand by what I said."

Lily Gladstone, who grew up on the Blackfeet Reservation in Montana, played Mollie Kyle, the Osage wife. In January 2024 she became the first Native American to win a Golden Globe for acting, and the first Native American

nominated for the Academy Award for Best Actress.

Everything Everywhere All at Once is about a Chinese immigrant woman who runs a laundromat. On March 12, 2023, it won seven Academy Awards. Michelle Yeoh won Best Actress, and Ke Huy Quan won Best Supporting Actor.

Ke Huy Quan

Who: A child actor from a refugee family who quit acting when he could not get parts as an adult Asian American actor, and came back decades later to win an Academy Award.

When and where: Born in Saigon, South Vietnam, in 1970. Came to the United States in 1979. Acted in Los Angeles from 1984. Won the award on March 12, 2023.

"My journey started on a boat."

Ke Huy Quan gave up acting because he could not get parts, and he did not act on screen again for 19 years, from 2002 to 2021. Then he came back to win an Academy Award.

His family's ancestors were Chinese, and the family lived in Vietnam. They left Vietnam three years after the war there ended in 1975. Quan, his father and five of his brothers and sisters went to a camp in Hong Kong for refugees, people who flee their country to be safe. His mother and three of the other children went to Malaysia. In 1979 the whole family came to the United States as refugees and settled in Los Angeles.

At 13 Quan played Short Round, Harrison Ford's young sidekick, in *Indiana Jones and the Temple of Doom* (1984). In 1985 he played Data in *The Goonies*. As he grew up, he said, "the opportunities got fewer, and the wait time longer. I was still working, but once a year, or once every two years. And the roles got smaller."

The wait. In 1993 he tried out for a part and never heard back. "I remember sitting at the edge of my bed, completely frozen," he said in 2023. He blamed himself: "I thought I wasn't tall enough, I wasn't good-

> looking enough, I was not a good enough actor." In time he quit acting and studied film at the University of Southern California. He arranged fights for the film *X-Men* (2000) and worked as an assistant director.
>
> He watched the film *Crazy Rich Asians* in 2018. "I was sitting in the theatre, fantasising, wishing I was up there with them," he said. He went back to acting and played Waymond Wang in *Everything Everywhere All at Once*. On March 12, 2023, he won the Academy Award for Best Supporting Actor, and he began his speech, "My journey started on a boat." He was the first actor born in Vietnam to win an Oscar.

#MeToo in Hollywood: Harvey Weinstein

Actresses were among the women who accused the film producer Harvey Weinstein of sexual attacks, and juries in New York and Los Angeles later found him guilty of rape and other sex crimes. Reporters at *The New York Times* and *The New Yorker* published the first accounts, from more than a dozen women, in October 2017. The women said Weinstein had sexually harassed them, meaning unwanted sexual remarks, demands or touching. Some said he had sexually assaulted them, meaning attacked them sexually, or raped them, meaning forced them into sex.

By the end of that month, more than 80 women had made such accusations. Leaders of his own company fired him, and the leaders of the Academy expelled him. Weinstein said he had never had sex with anyone against her will.

The spies. A month later a reporter for *The New Yorker* reported that Weinstein had hired a private spy company called Black Cube. Its investigators used false names to follow reporters and actresses. The people they followed included the actress Rose McGowan, who had said Weinstein raped her.

The first trials. A New York jury found Weinstein guilty in 2020 on two of the five charges against him: rape, and forcing oral sex on a woman. The jury found him not guilty on the other three, including the charge called predatory sexual assault. A judge sentenced him to 23 years in prison.

A Los Angeles jury found him guilty in 2022 on three of seven charges, for attacks in 2013: rape, forcing oral sex on a woman, and a third charge listed as sexual misconduct. The jury did not convict him on the other four. A judge sentenced him to 16 years.

The new trial. In 2024 the judges of New York's highest court threw out the two New York guilty decisions. They found that the trial judge had made "egregious errors," meaning very serious mistakes, and they ordered a new trial. Weinstein stayed in prison because of the California guilty decision.

At the new trial in 2025, a jury found him guilty of forcing oral sex on Miriam Haley, a former production assistant, a helper on a film crew, in his Manhattan apartment in 2006. The jury found him not guilty on the charge that involved the model Kaja Sokola. Two juries in a row could not agree on the charge that he raped the actress Jessica Mann, and in 2026 Manhattan's chief prosecutor announced that there would be no further trial on it.

The sentence. In September 2026 a judge, Curtis Farber, sentenced Weinstein to 15 years in prison for the attack on Haley. He told Weinstein, "You took what you wanted by force, and you have never accepted responsibility." Haley told the court that what he did to her was "a life sentence for me."

California judges had upheld his Los Angeles guilty decision earlier in 2026 but ordered a new sentence, and the California Supreme Court's justices declined to review the case. As of late September 2026, no date had been set for the new sentence.

BY SEPTEMBER 2026: Michelle Yeoh and Ke Huy Quan had won Academy Awards for acting, Lily Gladstone had become the first Native American nominated for Best Actress, and Harvey Weinstein was in prison.

The 2023 strikes, and scanning

The Writers Guild of America is the union of 11,500 screenwriters. Its members struck against the studios from May 2 to September 27, 2023, for 148 days. Their new contract forbids the studios to use writers' work to train AI, or artificial intelligence: computer programs that can make new text, images and voices by copying patterns from the work of real people. It also forbids the studios to use AI to cut writers or writers' pay.

Members of SAG-AFTRA, the actors' union, with about 160,000 members, struck from July 14 to November 9, 2023. It was the longest actors' strike against the film and television studios, and the first time since 1960 that actors and writers had struck at the same time.

The scans. The union's chief negotiator, Duncan Crabtree-Ireland, said on July 13, 2023, that the studios' negotiators had proposed scanning background actors, extras with no lines, for one day's pay, with the studio owning the scans. The studios' negotiators answered that a scan would be used only in the film the actor was hired for.

In August 2023 NPR interviewed five background actors who had been scanned. Most had signed agreements not to talk about their work. They said they feared that if they refused a scan, they would be called difficult and not hired again. No public count exists of how many background actors were scanned.

The contract. The union's members approved the contract that ended the strike on December 5, 2023. It requires a studio to get clear consent, meaning permission, from an actor before it makes or uses a digital replica, a computer copy of the actor's face, body or voice. A studio that wants to use a replica in a different production must ask again and bargain over pay. The consent lasts after the actor dies unless the contract says otherwise. Studios may not use replicas to avoid hiring background actors.

The writers approved a new four-year contract in April 2026. SAG-AFTRA's members approved theirs on June 4, 2026. Of the members who voted, 91 percent said yes, but only about 19 percent of members voted. According to

Los Angeles Magazine, it lets producers use a performer made entirely by computer only when that performer adds "significant additional value" over a live actor.

Alexandria Rubalcaba

Who: A background actor whom a television crew scanned into a digital copy of her face and body, with no word to her on how anyone would use it.

When and where: On the Disney+ series *WandaVision* during the COVID-19 pandemic. She told her story to NPR in 2023.

"Let us see your scared face. Let us see your surprised face."

Crew members on a television set scanned Alexandria Rubalcaba into a digital copy, and no one told her how or whether the copy would be used. She worked for four weeks as a background actor on the series *WandaVision*. Then the crew sent her to a truck trailer.

Inside, dozens of background actors stood one at a time in front of cameras mounted on metal frames. She remembered the instructions: "Have your hands out. Have your hands in. Look this way. Look that way. Let us see your scared face. Let us see your surprised face."

Each scan took about 15 minutes, and a digital copy of each actor was made from it. Rubalcaba earned the union rate for background actors, $187 a day. She would not be paid for any use of the scan, and she did not give permission for her copy to appear in any scene. "What if I don't want to be on MarioVision, or SarahVision?" she asked. "I fear that AI is eventually going to weed out background actors. They won't have any use for us anymore."

Disney did not answer NPR's questions, and no record shows whether her scan was ever used. Her account came out in August 2023, during the actors' strike that won the rules on consent for replicas.

The video game strike, 2024 to 2025

About 2,600 game performers who work in voice and motion capture struck from July 26, 2024, to July 9, 2025. They struck against companies including Activision, Electronic Arts, Epic Games, Take-Two and Warner Bros. Games. Their old contract had run out in November 2022. The one issue left unsettled was consent and pay for AI copies of the performers' voices and bodies.

Darth Vader's voice. Staff at Epic Games added Darth Vader to the game *Fortnite* on May 16, 2025. The character talked with players in a computer-made copy of the voice of James Earl Jones, who had died on September 9, 2024. In 2022 Jones had signed a deal that allowed his recorded voice to be used this way, and his family worked with Epic and Disney on it.

Union officials filed a charge with the National Labor Relations Board, the federal agency that hears complaints about unfair treatment of workers, on May 19, 2025. The charge was against Llama Productions, a company owned by Epic Games. It said the company's staff had replaced actors' work with AI without telling the union.

The contract. The strike ended when 95 percent of the members who voted approved a new contract on July 9, 2025. Before a company makes a digital replica of a performer, it must now get written consent that names what the replica will be used for. It must pay for the replica's use and tell the performer how it is used. During a future strike, performers can stop companies from using their replicas to make new material.

Faces and voices nobody performed

In *Rogue One* (December 16, 2016), the actor Guy Henry played the character Grand Moff Tarkin. Effects artists then placed a digital copy of the face of Peter Cushing, an actor who had died in 1994, over Henry's face. Lucasfilm staff got permission from Cushing's estate, the people in charge of his property after his death.

New laws. Tennessee Governor Bill Lee signed the ELVIS Act on March 21, 2024. Under it, copying a person's voice with AI without permission is against the law in Tennessee.

California Governor Gavin Newsom signed two laws at the actors' union headquarters on September 17, 2024. The first, AB 2602, requires a contract to describe how a digital replica of a living performer will be used. The second, AB 1836, forbids using a digital replica of a dead performer in films, games and recordings without the estate's consent. As of September 2026, members of Congress had not passed a national law, the NO FAKES Act.

Tilly Norwood. Eline Van der Velden, whose company made a computer-generated "actress" called Tilly Norwood, said in September 2025 that studios were interested in it and that a talent agency, a company that finds work for actors, would represent it. The leaders of SAG-AFTRA answered in a statement that Tilly Norwood "is not an actor." They said it was made by a computer program "trained on the work of countless professional performers," without their permission or pay.

Sora 2. The company OpenAI released its Sora 2 video app on September 30, 2025. Users soon made clips with the face and voice of the actor Bryan Cranston, without his consent. Cranston and the union complained.

OpenAI, the union, Cranston and several talent agencies put out a joint statement on October 20, 2025. It said OpenAI would require a person's permission before the app copies that person. Sam Altman, the head of OpenAI, said the company was "deeply committed to protecting performers

from misappropriation," meaning having their faces and voices taken. Cranston said he was "deeply concerned not just for myself, but for all performers whose work and identity can be misused in this way."

BY TODAY

If you remember one thing: Since 2000 actors' unions have won written rules that a studio or game company must get a performer's consent before it makes or uses a computer copy of that performer.

- June 14, 2013: *The Last of Us* came out, with Troy Baker and Ashley Johnson acting Joel and Ellie on a motion-capture stage.
- December 5, 2023: SAG-AFTRA's members approved the contract that ended their strike, with its rules on digital replicas.
- July 9, 2025: game performers approved a contract that requires written consent and pay for any replica.

Next: The chapter closing is below. Chapter 35, Styles, follows it.

THE THREE QUESTIONS, ANSWERED

1. **Who tried to stop Americans from acting, and why?** Early on, church leaders and lawmakers tried to stop it, for reasons of religion and morals. Spanish priests burned Pueblo masks to end the ceremonies, and Massachusetts lawmakers banned plays, saying they made people less moral and less religious. Later, police arrested Black actors after a rival white manager, by one account, started a riot to destroy their company, and studio heads blacklisted hundreds of film workers over communism.
2. **Why did white actors play Black, Asian and Native people for so long?** White audiences paid to laugh at blackface from 1828 on, and producers carried the old minstrel parts into film and radio. Because of the studio heads' own Code and state laws against

marriage between races, producers kept actors of color out of leading parts.

3. **Who controls a computer copy of an actor's face and voice?** Since the strikes of 2023 and 2024 to 2025, union contracts require a studio or game company to get the performer's consent before it makes or uses one, and the game contract also requires pay. Lawmakers in Tennessee and California passed laws of their own in 2024, and as of September 2026 Congress had passed no national law.

What changed: Acting moved from masked dancers in Native ceremonies to stages, screens and games, and the fight moved with it, from who was allowed to act at all to who owns a copy of an actor.

FIVE DATES TO KEEP

April 30, 1598: Spanish soldiers and priests acted the first European play on record here, beside the Rio Grande.

1821: William Brown opened the African Theatre in New York, the first known Black theatre company in the United States.

1828: Thomas Dartmouth Rice first went on stage as "Jim Crow," in blackface.

February 8, 1915: *The Birth of a Nation* opened in Los Angeles.

July 14, 2023: film and television actors began the longest actors' strike against the studios.

> **NOT KNOWN:** The title of that first European play of 1598 is not known, and no copy of its script survives.

Where to go: Ford's Theatre in Washington, D.C., where John Wilkes Booth shot President Lincoln in 1865, held its first public performance since that night in 1968.

Next: Chapter 35, Styles. It asks who made the rules about Americans' clothes and hair, and who made the cloth they wore.

Chapter 35: Styles

CHAPTER 35 OF 37 · STYLES · 10 ERAS · ABOUT 100 MINUTES

From the first colonial dress laws to the CROWN Acts, laws against punishing people for their natural hair, Americans with power made rules about who could wear what, and much of the cloth came from people who were enslaved, held captive or paid by the piece.

Hannah Lyman was about 16 when the judges of the Hampshire County court in Northampton, Massachusetts, fined her 10 shillings for wearing silk. The year was 1676. Massachusetts lawmakers had passed laws saying who was allowed to wear silk, lace and "great boots." The charge against Hannah said she wore her silk "in a flaunting manner," and that she wore it "not only before, but when she stood presented." She wore silk even while she stood charged before the court.

This chapter follows what Americans wore, and what it was made from, through all ten eras, from turkey-feather blankets to sneakers. It also looks at furniture in two eras, because styles changed in American homes too. As you go, watch who made the rules, such as the Massachusetts judges who fined people for wearing silk. Watch, too, who made the cloth: enslaved people in the cotton fields and indigo vats, and women sewing in Bangladesh and Los Angeles.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Why did lawmakers and officials make rules about other people's clothes and hair?
2. Who made the cloth Americans wore?
3. Which new materials changed what Americans wore?

Five minutes? Read "Cotton picked in the South, woven in the North" in 1800 to 1850, and the section after it.

Can skip: "Wigs in Williamsburg, 1769 to 1775", "The Federal style, 1780 to 1820", "Shirtwaists and the Gibson Girl, 1890s". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 7 MINUTES

The people of North America before 1500 dressed in what they could grow, hunt or get by trade. They wore deer hide, rabbit fur and turkey feathers, and cloth woven from cotton and from the fiber of the yucca, a desert plant. Their ornaments were sea shells from the Gulf of Mexico and copper from the country around Lake Superior. Their colors came from plants and from a mineral called red ochre.

No written record names anyone who wove this cloth or wore these beads. Archaeologists, scientists who dig up and study the things people left behind, have found the cloth, the feathers, the beads and the copper. What is known comes from those objects.

How many feathers went into one blanket?

A blanket made in the 1200s used about 11,550 turkey feathers, pulled from 4 to 10 turkeys. Its makers wrapped the feathers around more than 590 feet (180 meters) of cord twisted from yucca fiber. The blanket is kept at the Edge of the Cedars State Park Museum in southeastern Utah.

Its makers were Ancestral Puebloan people, the ancestors of today's Pueblo peoples. They lived in the Four Corners, the place in the Southwest where Utah, Colorado, Arizona and New Mexico meet. Most of their clothing was deer hide, rabbit skin and yucca fiber.

Bill Lipe and Shannon Tushingham, researchers at Washington State University, studied the blanket. In 2020 they reported that the makers probably pulled most of the feathers from live birds. In this region, feather blankets had replaced blankets made from strips of rabbit fur many centuries earlier.

Cotton. Cotton came north by trade from Mexico. The oldest farmed cotton found in the Tonto Basin of Arizona dates to about the year 250, and cotton reached the Four Corners around 600. By about 900, Ancestral Puebloan men were growing cotton, spinning it into thread and weaving large cloths and blankets on upright looms, frames that hold threads tight so a weaver can pass other threads over and under them.

The Salado people lived in the Tonto Basin from about 1250 to 1450, and by about 1300 their basin was one of the leading cotton-growing areas of the Southwest. They wove plain cotton cloth and openwork cloth, with small openings left in the weave as a pattern. They made blankets from strips of worn-out cotton cloth. One brightly dyed strap they wove is a tumpline, a band worn across the forehead to carry a load on the back. Their sandals were woven from agave and yucca, two desert plants.

Shell and copper along the Mississippi

Cahokia was a city of earth mounds in Illinois, a few miles east of the Mississippi River and today's St. Louis. Its people traded for copper from the Lake Superior region and for sea shells from the Gulf of Mexico. Archaeologists call the way of life of Cahokia and the towns like it the Mississippian culture.

Mississippian metalworkers heated copper, hammered it while the heat kept it soft, let it cool, and did the same again and again, a way of working metal called annealing. They made hair ornaments, pieces for headdresses and flat plates.

The bird-man. Some plates show a figure that is part bird and part man, pushed out by hammering from the back of the sheet. Artists at Cahokia began making pictures of this bird-man between 1100 and 1300. Mississippian copper plates were made from about 800 to 1600, and they have been found from Florida and Georgia to Oklahoma and Wisconsin.

At Cahokia, archaeologists found a copper workshop with raw copper nuggets, scraps of worked copper and stone tools for working it.

A bed of shell beads. People at Cahokia buried a man and a woman in the 1000s on a bed of shell disc beads laid out in the shape of a bird. The beads were made from sea shells. The magazine *Archaeology* gives the count as 20,000 beads, and another published count gives about 10,000.

Melvin Fowler dug up this burial in a mound called Mound 72 between 1967 and 1971, and he described both bodies at the center as men. In 2016 a team led by Thomas Emerson of the Illinois State Archaeological Survey studied the bones again. They found that the two were a man and a woman, with other pairs of a man and a woman buried around them.

Mound 72 holds about 272 people, buried at different times over about a century. Archaeologists judge that almost 62 percent of them were sacrifices, people whom others at Cahokia killed on purpose as part of the burial ceremonies. No written record names them or the people who killed them.

Colors from plants and earth

Ochre is a mineral made of iron and oxygen, and it can be brown, red or yellow. Red ochre is the red kind. Accounts written after Europeans arrived say that the Karankawa people of the Texas coast painted symbols on their faces with ochre, using their fingers or the stalks of plants. The Karankawa got their ochre by trade from inland. At a burial site called Mitchell Ridge, red ochre stains the bones of 18 people.

Native people of North America decorated clothing with porcupine quills, the stiff, hollow spines that cover a porcupine, before Europeans arrived. The Huron, Cheyenne, Arapaho, Cree and Gwich'in were among the nations that did quillwork. Men hunted the porcupines, and women did the work. They flattened the quills with an awl, a pointed tool, or with a piece of bone, then dyed them and sewed them onto moccasins, pouches, baby carriers and clothing.

The dyes. Red came from the bark of the tamarack tree, from spruce cones and from berries, and black from walnuts and wild grapes. For yellow, quillworkers used wild sunflower, pine bark or willow root. Blueberries and larkspur, a flower, made blue. Writers at the Penn Museum in Philadelphia list these dye plants. The museum's own quilled pieces date from the 1830s and later, and among the oldest are pieces that George Catlin collected between 1832 and 1839. Quillwork itself is much older, from before Europeans arrived.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 5 MINUTES

The first written descriptions of what Native people wore come from the Spanish, French and English explorers and soldiers who arrived in the 1500s. They wrote down deerskin, cloth of bark fiber and feathers, and ornaments of pearls. Some of them took what they found. Hernando de Soto's soldiers dug up Native graves at the town of Cofitachequi in 1540 to take the pearls buried there.

Spanish colonists drove churro sheep into New Mexico in 1598, onto the land of the Pueblo peoples. Wool from those sheep went into Native, Spanish and Mexican weaving for about 300 years.

Cloth and pearls at Cofitachequi, 1540

Hernando de Soto's army reached Cofitachequi, a Native town beside a river in the Southeast, in May 1540. A Portuguese man who marched with the army wrote an account of the march, printed in Portugal in 1557. His name is not known, so he is called the Gentleman of Elvas, after the Portuguese town of Elvas. He described what the people of the town wore and what they kept.

A woman ruled the town. No record gives her own name, and the Spanish called her the Lady of Cofitachequi. She crossed the river by canoe to meet de Soto and gave him mantles, or cloaks, and deerskins. Then she took a long string of pearls from her own neck and put it around his. He gave her a gold ring set with a ruby.

Elvas wrote that the town's storehouses held mantles woven from yarn made of tree bark. Other mantles were made of feathers colored white, green, red and yellow, and people wore them in winter. The storehouses also held deerskins with many panels drawn on them, and some of the deerskins had been made into leggings, stockings and shoes. Elvas wrote that everyone in the town wore shoes and clothes.

The graves. De Soto was looking for treasure. According to Elvas, the ruler told him to search the town's graves for pearls, because she had seen how much the Spaniards valued them. De Soto's men dug up the graves. They took the pearls, and small figures of babies and birds made from pearls.

The accounts disagree on how many pearls the men took. They give weights from about 200 pounds, about the weight of a large man, to 392 pounds.

In the same town, de Soto's men found a dagger and beads. They came from a colony that Lucas Vázquez de Ayllón had started in 1526 on the coast of today's South Carolina or Georgia. Ayllón died there, and the surviving colonists left within about two months.

De Soto then took the ruler with him as a prisoner, guarded by enslaved women. On the march she told her guards she needed to relieve herself. She walked into the woods and escaped, carrying off a cane box full of pearls that had not yet been drilled.

What an Algonquian leader wore, 1585

The English painter John White painted Native people who spoke Algonquian languages, a large family of related languages, in 1585. Thomas Hariot wrote the words that went with the pictures. One painting shows a leader of one of these peoples, dressed for a great hunt or a feast. He wears an apron of animal skin with a fringe, and a tail hangs down behind it. His quiver, the case that holds his arrows, is tied to the tail.

Feathers decorate his hair. One side of his head is shaved. Hariot wrote that the shaving kept the hair from tangling in the bowstring.

Spanish clothes in St. Augustine, 1565 to 1600

Spanish soldiers and settlers founded St. Augustine, in Florida, in 1565, on land where the Timucua people lived. Archaeologists from the Florida Museum have dug up many of the things the colonists wore. Rich residents wore costly cloth trimmed with metal lace, buttons, beads and metal ornaments. Buttons were worn mostly by soldiers and the rich.

Most people tied their plain clothes shut with laces. Each lace had a small metal tip, called an aglet, so it could pass through a hole. Many soldiers in the town married Native women.

A widow's bed. Juan de Cevadilla was the official in Mexico who collected the pay for the soldiers of St. Augustine. He died suddenly in 1590 and left his widow, Doña Petronilla (Doña is a Spanish title of respect for a woman), a carved bed covered with a thin layer of gold. Over the bed hung a cloth roof, called a canopy, made of taffeta, a smooth silk cloth, from China, with designs stitched into it. He also left her a large amount of costly cloth and clothing.

Churro sheep and red from Mexico

Juan de Oñate led Spanish colonists north from Mexico onto the land of the Pueblo peoples in New Mexico in 1598. They came on the road called El Camino Real de Tierra Adentro, the Royal Road of the Inland Country, and they brought churro sheep, a Spanish breed. By the 1700s the churros in New Mexico numbered in the hundreds of thousands. For the next 300 years their wool was the base of Native, Spanish and Mexican weaving there.

Before the sheep came, Pueblo weavers worked in cotton on upright looms. The Spanish brought a loom that lay flat on the floor. Weavers in New Mexico later used it to make Rio Grande blankets and serapes, blankets that can be worn as a cloak. After the Spanish brought churro wool and new dyes, wool became the main thread for weaving there, and the old Pueblo cotton weaving declined.

Cochineal. Far to the south, in Oaxaca, Puebla and Tlaxcala in Mexico, Native people had long bred a small insect called cochineal to make red dye. As early as the 100s BC they used it to dye cloth and feathers and to color books painted by hand. After Spanish soldiers conquered Mexico, traders carried cochineal around the world. Native Mexicans still did all the skilled work and all the labor of making it. Writers at Harvard's science museums note that they were never given credit for it.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 10 MINUTES

Lawmakers in the English colonies of the 1600s wanted clothes to show who ranked above whom, and in Virginia and Massachusetts they wrote laws to keep it that way. In Massachusetts, a person whose property was worth less than 200 pounds was not allowed to wear silk hoods, gold or silver lace, or boots called "great boots." Judges in the county courts fined people who did. Lists of belongings made after New England colonists died record clothes in red, blue, violet, grey and green, as well as black.

In New Mexico, Spanish soldiers and colonists demanded a woven cotton blanket from each Pueblo household every year. This was tribute, a payment that conquerors force a conquered people to make.

Virginia taxes clothing, 1619

The General Assembly of Virginia, the colony's lawmaking body, met for the first time on July 30, 1619. At that meeting its members passed a law "against excesse in apparel," meaning too much spending on clothes. They voted that each man's share of all public costs, collected through the church, would be set by his clothes. An unmarried man paid according to his own clothes. A married man paid according to his own clothes and his wife's.

A Virginia resolution of 1621 states the aim to "supress excess in cloaths." Under it, men who held no office in the colony could not wear gold in their clothes.

Mary Ring

Who: A widow of Plymouth Colony whose list of belongings shows the colonists there dressed in red, violet and blue.

When and where: Leiden in Holland, then Plymouth Colony, 1629 to 1631.

Her red petticoat was worth 16 shillings.

Mary Ring owned two petticoats, the skirts women wore, one red and one violet. She also owned two violet waistcoats, which were vests or short jackets, and three blue aprons. When she died, everything she owned was listed and given a value, in a list called a probate inventory. Hers gives values in pounds, shillings and pence, the English money of the time. Only one piece of her clothing was worth more than the red petticoat: a hooded cloak called a huke.

The inventory also lists a black kirtle, a kind of gown, and a waistcoat of "mingled coloured" cloth. It lists an apron in murrey, a dark purplish red, and pieces of blue cloth and red cloth. She owned blue stockings, white Irish stockings and a piece of grey kersey, a rough wool.

She also owned six ruffs, stiff pleated collars, and seven smocks, long linen shirts worn under the other clothes. She had three linen caps, two hats and two pairs of shoes. Her will left green cloth "to make a coat." Caleb Johnson wrote notes on her list for the Plymouth Colony Archive Project. He points out that it shows the English colonists of Plymouth wore violet, red, blue and grey as well as black and white.

Who she was. Mary Ring was the widow of William Ring, who died in Leiden, in Holland, in the 1620s. She came to Plymouth in 1629 or 1630 with her children Elizabeth, Susanna and Andrew. The English had built Plymouth in 1620 on the site of Patuxet, a town of the Wampanoag people. About 2,000 Wampanoag had lived in the area. Before 1620 a wave of sickness, now called the Great Dying, killed most of the people of Patuxet, and the few who lived left to join other Wampanoag towns.

> She died on July 15 or 19, 1631. She signed her own will, and she owned books. Governor William Bradford and Edward Winslow owed her money when she died.

Did the Puritans dress only in black?

The Puritan colonists of Massachusetts wore many colors, which they called "sad" colors, meaning serious rather than unhappy. A list from 1638 names some of them: liver color, tawny, russet, purple, French green, orange and "deer colour." The Puritans were English Protestants who wanted their leaders to strip the Church of England of the customs it still shared with the Catholic Church.

The historian David Hackett Fischer writes that black cloth was kept for the ruling elders, the men who led the church, and for the families who governed the colony. A portrait of Governor John Winthrop, painted in about 1629, shows him in black velvet with slashed sleeves, cut so that the cloth underneath showed through.

Lists of belongings and bills from the colony record "pinck hose," or pink stockings, "green sleeves," and "a Scarlet petticoat with Silver Lace." Fischer writes that Winthrop's own niece, Mary Downing, was scolded for wearing a little lace. His sentence about her does not say who scolded her.

Massachusetts dress laws, 1634 to 1682

The General Court of Massachusetts Bay was the colony's body of lawmakers and magistrates, its chief officials and judges. Its members passed their first sumptuary law, a law that limits what people may wear according to their rank, in 1634. It banned "new fashions, or long hair," silver and gold hatbands and belts, cloth woven with gold thread, and lace.

It also limited the slashes in a sleeve. The magazine *Atlas Obscura* gives the limit as two slashes in a sleeve, and the historian David Hackett Fischer gives it as one. A person caught wearing these things had to give them up. A law of 1639 banned more kinds of lace and banned short sleeves.

"Utter detestation." The members of the General Court wrote in 1651 that they felt "utter detestation," or hatred, when "men or weomen of meane condition," meaning people of low rank, dressed like gentlemen. They named gold or silver lace, gold or silver buttons, laces tied at the knees and "great boots." For women of that rank they named silk hoods and silk scarves. Anyone whose property was worth less than 200 pounds owed a fine of 10 shillings each time he or she wore them.

Magistrates, officers, soldiers and their families did not have to obey this law. Neither did people who had once been rich and had lost their money. The lawmakers gave the selectmen, the elected men who ran each town, the power to tax people who dressed above their rank.

In court. Judges in the county courts enforced the law. Judges in Essex County heard 18 cases in 1653, and they let 12 of the people go because they were worth more than 200 pounds. Ten years later, in 1663, the lawmakers made the law cover children and servants too, and barred tailors from making the forbidden clothes. A first offense brought a warning, a second a fine of 20 shillings, and later offenses up to 40 shillings.

The judges of the Hampshire County court charged 35 married women from five towns with wearing silk at one session in 1673. They found all but one of the seven from Northampton not guilty, and they warned, fined or let go the women from the other towns.

Whose land. Northampton stood on the land of the Nonotuck people. John Pynchon got a deed, a paper that hands over land, from seven Nonotuck men, including Chickwalloppe, on September 24, 1653. He paid 100 fathoms of wampum, strings of shell beads, and 10 wool coats. A fathom is 6 feet. The historian Margaret Bruchac writes that the Nonotuck signers probably understood the deed as an agreement to share the land, and that they probably expected to keep their own right to hunt, fish and plant there.

The war. King Philip's War was fought in 1675 and 1676. Metacom, a leader of the Wampanoag whom the English called King Philip, led several Native nations against the New England colonies. About 9,000 people died in the war, and by some counts about half of all the Native people of New England died. A Native man fighting for the colonists shot Metacom dead on August 12, 1676. The colonists sold hundreds of Native captives into slavery.

During the war the Northampton minister Solomon Stoddard asked the colony's leaders to act against "that intolerable pride in clothes and hair." In November 1675 the members of the General Court passed a law listing the sins they said were common in the colony. A law of 1675 also lowered the fine for a second or later offense to 10 shillings.

March 1676. The Hampshire County judges heard charges against 23 people of Northampton, women and young men, on March 27, 1676, "for wearing silk in a flaunting manner and for long hair and other extravagances" (flaunting means showing off, and extravagances meant spending too much on show). They warned them all and made each one pay court fees of 2 shillings 6 pence, except the wife of Samuel Holton. She had offended before, so they fined her 10 shillings. That March the same judges fined 45 more people from four other towns for silk and long hair.

The five towns of Northampton, Hadley, Hatfield, Westfield and Springfield were themselves charged at the county court between about 1680 and 1682. Their selectmen had failed to tax residents who wore silk "beyond their ranks." James Russell Trumbull, who wrote the history of Northampton, reports that nothing more was heard of these charges.

Hannah Lyman

Who: A Northampton girl of about 16 who was fined for wearing silk, and who wore it even while she stood charged before the court.

When and where: Northampton, Massachusetts, 1676.

Her charge said she wore silk even on days set apart for prayer.

Hannah Lyman of Northampton was about 16, and her father, Richard Lyman, had died. The charge against her said she wore silk "not only in ordinary times, but in extraordinary times, when the people of God were falling before the Lord in public humiliation." A day of public humiliation was a day when the people of the town prayed together and humbled themselves before God, because of "the heavy judgments and calamities that were threatening to come upon us."

Her case came to a later session of the county court than the one in March 1676. The judges fined her 10 shillings. The charge also said she wore silk "not only before, but when she stood presented," which means she kept wearing it even while she stood charged before the court.

Counts of the people charged in Northampton that year differ. The town's historian, James Russell Trumbull, who quoted the court records, gives 23 at the March session and Hannah at a later one. The magazine *American Heritage* and the historian David Hackett Fischer write that she was one of 36 young women of Northampton charged that year.

Blankets taken as tribute in New Mexico

Some Spanish colonists in New Mexico, called encomenderos, held the right to collect tribute from Pueblo households after 1598. Researchers for the Beyond Origins of New Mexico Families project report that by the 1660s each household owed one fanega of corn in the spring, a fanega being a Spanish measure of grain. In the fall each household owed one woven cotton blanket, called a manta, or one hide. The Spanish king allowed no more than 35 encomenderos at a time.

Ginés de Herrera Horta was a Spanish legal officer in Oñate's colony. In 1601 he testified that soldiers demanded a blanket each year from each household. When people said they had no blankets to give, the soldiers took the blankets off the women who were wearing them and left the women naked. The Spanish records of these seizures give no names for the women whose blankets were taken.

Most of the Pueblo peoples rose up against the Spanish colonists in 1680, in the Pueblo Revolt, led by Po'pay of San Juan Pueblo. They killed about 400 Spaniards and forced about 2,000 settlers to flee New Mexico for about 12 years. Among the Pueblo people's complaints was the tribute of labor, ground corn and cloth that the colonists had forced them to hand over.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 7 MINUTES

In the English colonies between 1700 and 1750, a wig showed that a man ranked near the top, and coarse cloth showed that a person was enslaved. The historian Thomas Ford calls the wig a badge of rank. A traveler in Maryland in the 1740s wrote that only a few men of the highest rank wore wigs, and that other men wore caps.

South Carolina's lawmakers wrote the coarse cloth into law in 1740. They named cheap fabrics and barred enslaved people from wearing anything finer. They ordered constables, local officers who kept order, and anyone else who found finer clothes on an enslaved person to take them away and keep them. In the 1740s South Carolina planters began growing indigo, a plant that gives a blue dye, in large amounts and shipping the dye to England. Enslaved people did the work of turning the leaves into dye.

Who wore a wig?

Only a few men of "the very elevated Sort" wore wigs, a traveler in Maryland wrote in the 1740s. Ordinary men wore caps of wool or yarn. Better-off men wore caps of white linen or cotton, even on journeys of 50 miles.

A wig of this time was called a peruke. A wigmaker built it from rows of hair sewn onto a cap of net and ribbon, using human hair and also hair from horses, cows and goats. The historian Thomas Ford writes that a wigmaker preferred to measure a customer whose head was shaved, so the wig would fit. He calls the wig "an important badge of social rank."

Some rich men, including Councillor Robert Carter, a member of the governor's council, and Dr. John Sequeira, had enslaved men trained to shave them and to dress their hair. Ford writes that rich planters may sometimes have dressed their enslaved coachmen and house servants in white wigs.

The 1740 clothing law in South Carolina

Enslaved people in South Carolina rose up against their enslavers in the Stono Rebellion in September 1739. On May 10, 1740, the lawmakers of the South Carolina assembly passed the Negro Act. Its full name was An Act for the Better Ordering and Governing Negroes and Other Slaves in this Province, and it used "Negroes," the lawmakers' word for Black people. Section 40 of the act was about clothes.

What Section 40 said. It began by claiming that "many of the slaves in this Province wear clothes much above the condition of slaves." The lawmakers said enslaved people got those clothes by "sinister and evil methods," meaning wrong and wicked ones. The section barred any owner from letting an enslaved person "have or wear any sort of apparel whatsoever, finer, other, or greater value" than the cloths it named. Servants in livery, a uniform the master chose, did not fall under this rule.

The list read: "Negro cloth, duffels, kerseys, osnabrigs, blue linen, check linen or coarse garlix, or calicoes, checked cottons, or Scotch plaids." All of them were cheap cloths, and many were coarse, which means rough and thick. Negro cloth was a coarse cloth made for enslaved people's clothes. Osnaburg,

which the law spells "osnabrigs," was a plain cloth first woven from flax in the German city of Osnabrück. An earlier South Carolina act, passed in 1735, had already named cheap cloth for enslaved people's clothing.

Who took the clothes. The law required "all and every constable and other persons" to seize finer clothes from any enslaved person wearing them. The person who seized the clothes could keep them "to his or their own use." An owner who thought the clothes had been taken wrongly could take the matter to a justice of the peace, a local judge. The surviving records do not show how often anyone seized clothes under this law.

Jack

Who: A man taken from Gambia and enslaved in Virginia, who ran away within five weeks of his sale.
When and where: Hanover County, Virginia, 1745.

Within five weeks of his sale, Jack ran away.

Slave traders brought Jack and another man from Gambia, in West Africa, on the ship *Ranger*. They sold the two men at Newcastle, Virginia, on September 5, 1745. Neither man spoke English, and both were nearly 6 feet tall.

Almost everything known about Jack comes from an advertisement that his owner, Margaret Arbuthnott of Hanover County, placed in the *Virginia Gazette* in October 1745 to get him back. The advertisement does not name the traders. Arbuthnott described Jack as having a scar over his right eyebrow. The other man, whose name she did not give, had what she described as "3 small Strokes on each Side of his Face."

The two men ran away in new jackets and in breeches, short pants that ended at the knee. Both were made of napped cotton, brushed so that its surface was soft and fuzzy, with no buttons and no buttonholes. The men also wore shirts of osnaburg and felt hats.

> Arbuthnott accused the two men of taking a damask tablecloth, its pattern woven into the cloth, along with fine Scottish linen, checked cloth, a white linen shirt and a silk handkerchief. She offered a reward to anyone who caught them and brought them back.
>
> The record does not say whether Jack was caught.

Indigo blue, 1739 to 1750

Indigo seeds came to South Carolina with the first English settlers in 1670. In the late 1730s and early 1740s, hundreds of South Carolina planters tried out new crops, and indigo was one of them.

Eliza Lucas. Eliza Lucas began trying to grow indigo in 1739 at Wappoo Plantation, on Wappoo Creek west of the Ashley River, near Charleston. She was 17 and managed the plantation and the 20 people enslaved on it. She worked first with an indigo maker from the island of Montserrat. She had the most success with an indigo maker of African descent whom her father hired from the French West Indies. In 1744 she used her crop to make seed and shared it with other planters.

The historian Nic Butler of the Charleston County Public Library writes that her part has been greatly exaggerated in recent times. Many of her neighbors were following the same steps. Her husband, Charles Pinckney, whom she married in 1744, published her methods in articles in 1744 and 1745 under the name "Agricola."

The bounty. South Carolina's lawmakers voted in 1744 to pay planters a bounty, a reward from the government, of 1 shilling for each pound of indigo. They canceled it in 1746. Members of the British Parliament, the lawmakers in London, voted a bounty of 6 pence a pound in 1748, starting in 1749.

Planters began shipping large amounts of indigo to England in 1747. Exports rose from about 5,000 pounds in 1745 and 1746 to about 130,000 pounds in 1748, about 26 times as much.

Who did the work. Enslaved people of African descent did almost all of the work. They planted and tended the indigo. They built the vats, big tubs for making dye, and filled them with water and cut leaves. After the leaves soaked, the workers beat the liquid with paddles or with buckets that had no bottoms. They added lime, made from burnt oyster shells, so that the blue settled out. Then workers scooped the blue mud into bags, dried it and cut it into cubes.

Charles Woodmason wrote in 1755 that 15 enslaved workers could plant and tend 50 acres of indigo, and that it took 25 skilled workers to turn that crop into dye. The accounts of indigo making from these years do not name the enslaved workers who made the dye.

One enslaved man who built the vats is named. He was Quash, a carpenter whom the Lucas family held in slavery at Wappoo. In 1744 George Lucas, Eliza's father, wrote that "Quashe" should build a wooden indigo works at the family's Garden Hill plantation. Eliza told her father that Quash had spent 33 days building indigo vats at other Lucas plantations. The minister Alexander Garden baptized him on April 16, 1746. The historian Tiffany Momon, who wrote about his life, calls him John "Quash" Williams.

South Carolina's indigo was shipped to England for British cloth makers. South Carolinians bought most of their own cloth from England, already dyed.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 8 MINUTES

Colonists who opposed British taxes turned their clothes into a protest. From the Stamp Act tax of 1765 until the Revolution, they agreed again and again to boycott, or refuse to buy, British goods, including cloth. Delegates at the First Continental Congress, a meeting of leaders from the colonies, voted in 1774 to ban British goods in every colony from December 1.

During the boycott of 1768 and 1769, a colonist who wore homespun, cloth spun and woven in America, often at home, showed that he or she backed the boycott. Women held public spinning meetings. Harvard's graduating class of 1768 voted not to wear imported British clothes at their graduation.

George Washington took the oath as the first president in 1789 in a suit of brown wool woven in Connecticut. Furniture makers in the new country copied a style of sharp edges, legs that grew thinner toward the floor, and inlaid eagles, now called the Federal style. In Spanish New Orleans in 1786, the governor ordered Black women and women of mixed race to stop wearing feathers and jewels in their hair and to wear it plain or covered with a cloth.

Homespun as a protest, 1765 to 1776

During the boycotts of British goods in the 1760s, Boston merchants opened spinning schools again. In Hartford, members of a society offered a prize of 12 pounds to whoever made the most yards of wool cloth in a year. Newspaper printers ran reports of spinning meetings in Massachusetts, Connecticut and Rhode Island. In Newport one spinner was a woman of 70 who had never spun a thread before. Only six of the newspaper stories called the spinners "Daughters of Liberty."

A day at the wheel. Women met at their minister's house in Brookfield, Massachusetts, at 5 in the morning one day in August 1769. A report of the meeting said the best young spinner at the linen wheel, a spinning wheel for linen thread, spun 70 knots of yarn, a knot being a measured length.

One married woman did her morning housework and made cheese first. Then she rode more than 2 miles carrying her own spinning wheel, spun 53 knots between 9 and 7, and went home to milk the cows.

The seniors at Harvard College voted in the autumn of 1767 not to drink tea or wear imported British clothes at their commencement, the ceremony at which students graduate. They did it to protest the Townshend taxes on goods from Britain. The printed sheets for the 1768 commencement, which listed the statements the graduates would argue for, were on paper made in New England. The next year's seniors did the same.

Charity Clarke

Who: A young New Yorker who wrote to her cousin in London that American women would spin, weave and dress in their own cloth rather than buy British clothes.

When and where: Her father's estate called Chelsea, on Manhattan, 1768 to 1774.

She promised "a fighting army of Amazons" armed with spinning wheels.

Charity Clarke was 21 or 22 when she wrote to her cousin in London in 1769. If the English did not give Americans the liberty they asked for, she told him, she would lead "a fighting army of Amazons," the women warriors of old Greek stories. She would gather "a number of ladies armed with spinning wheels." The historian Alan Singer dates the letter June 1769, and other accounts date it December 8, 1769.

Her ladies would have helpers "who shall all learn to weave and keep sheep," and they would all be "clothed with the work of our hands." They would be "no more slave to fashion and ceremony."

Her cousin was Joseph Jekyll, a lawyer in London, and she wrote to him between 1768 and 1774. Columbia University holds the letters today. She wrote to him again in 1774: "Though this body is not clad with silken garments, these limbs are armed with strength."

She was born in New York on June 28, 1747, and grew up on her father's estate, called Chelsea, on the island of Manhattan. She married the Rev. Benjamin Moore of Trinity Church in 1778. Their son, Clement Clarke Moore, wrote the poem "A Visit from St. Nicholas."

Wigs in Williamsburg, 1769 to 1775

Edward Charlton came to Williamsburg, Virginia, from London and became the town's leading barber and wigmaker. In 1769 eight barbers and wigmakers worked in the town. Charlton's account book for 1769 to about 1775 shows that in one year he sold sixty brown wigs of a style called the "dress bob." They cost 43 shillings each.

For shaving a man and dressing his wig for a year, Charlton usually charged 2 pounds 3 shillings. Peter Pelham paid 1 pound 5 shillings, and Speaker Randolph, the leader of the colony's lawmakers, paid 4 pounds.

Wigmakers needed hair. George Long of Petersburg, another Virginia town, advertised that he would pay one shilling an ounce for human hair.

Wigs were going out of fashion in England, and Charlton retired shortly before the Revolution.

Indigo's peak and end

South Carolina planters shipped more than one million pounds of indigo to England in 1775, more than seven times the 130,000 pounds of 1748. Enslaved workers did almost all of the work of making it. The British bounty on indigo ran out in 1777, during the war with Britain. Some planters grew indigo again after the war ended in 1783. In the 1790s most of them gave up indigo, and many turned to cotton.

Why did a governor ban feathers in women's hair?

Governor Esteban Rodríguez Miró of Spanish New Orleans ordered Black women and women of mixed race to stop wearing feathers and jewels in their hair in 1786. The historian Virginia Meacham Gould writes that he aimed the rule at women who dressed "too elegantly" or "competed too freely with white women for status."

The rule was part of a set of rules Miró issued on June 2, 1786, called the *bando de buen gobierno*, the order of good government. It said the women could "no longer have feathers nor jewelry in their hair." They had to wear their hair plain or cover it with a *pañuelo*, a head cloth. People later called such a cloth a tignon.

Gould found little written evidence that officials enforced the rule, and no record that anyone was punished under it. Portraits painted in New Orleans in the early 1800s show free Black women and women of mixed race wearing head wraps tied in many different styles. Gould reads the head wraps in these portraits as a mark of distinction.

George Washington

Who: The first president, who took the oath of office in a suit made from American wool.

When and where: Mount Vernon, Virginia, and New York City, January to April 1789.

He took the oath in brown wool woven in Connecticut.

George Washington wrote on April 8, 1789, three weeks before he took the oath as the first president of the United States: "I shall allways take a peculiar pleasure in giving every proper encouragement in my power to the manufactures of my Country" (manufactures meant goods made by hand or by machine). He took the oath on April 30, in New York.

Finding the cloth. In January 1789 a writer in *The Federal Gazette* had urged American leaders to wear clothing made in America. He signed himself "a Philadelphia mechanic," meaning a skilled worker who made things. The Hartford Woolen Manufactory in Connecticut had just begun making cloth. A merchant in Philadelphia could not get the cloth for Washington, so Henry Knox, then the secretary of war, dealt with the Hartford factory himself. The cloth reached Alexandria, Virginia, in early April. Washington left Mount Vernon for New York on April 16.

The staff at Mount Vernon believe that Cavan Boa made the coat, waistcoat and breeches. Boa was Washington's indentured tailor, a servant bound to work for him for a set number of years. Kathrin Breitt Brown makes copies of old clothes for Mount Vernon. She calls the coat "well-made but hastily finished," and she thinks Boa probably had help. Washington asked for six large engraved buttons.

In England the color of the cloth was called London brown. Americans called it Congress brown. The coat came to the collection at Mount Vernon in 1933.

Washington did not wear a wig. He powdered his own hair.

The Federal style, 1780 to 1820

Styles changed in American homes as well as in clothes. From about 1780 to 1820, American cabinetmakers, the skilled makers of fine furniture, worked in a style now called Federal. They copied many of its designs from English pattern books by George Hepplewhite and Thomas Sheraton. Federal pieces had legs that grew thinner toward the floor, and sharp edges. Makers covered them with veneers, thin sheets of fine wood laid over the surface. They inlaid pictures of eagles, urns and bellflowers, setting small pieces of different wood into the surface.

The main wood was mahogany, and makers also used satinwood, maple and birch. Makers of Federal furniture included John and Thomas Seymour in Boston and Duncan Phyfe in New York.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 10 MINUTES

Enslaved people in the South grew and picked the cotton that American mills spun between 1800 and 1850. Workers in New England mills spun and wove that cotton into cloth. In some mills in Rhode Island and in Lowell, Massachusetts, workers also wove a rough, cheap cloth that planters bought to dress the people they enslaved.

Fashionable women wore high-waisted white cotton dresses in the early 1800s. Later they wore corsets, stiff garments laced tight around the body, and wide skirts held out by layers of petticoats, the skirts worn under the outer skirt. Men stopped powdering their hair and cut it short. Some still wore knee breeches, pants that ended at the knee, and by the 1840s men wore narrow trousers.

Who made the clothes. A government report in 1811 stated that two-thirds of all the clothes Americans wore were made at home. People who could pay for it had a tailor or a seamstress, a woman who sewed for pay, make their clothes. The first shops that sold clothes already made stood near the docks, and their customers were sailors and other working men.

What fashionable people wore, 1800 to 1850

The fashionable dress for a woman in the first ten years of the 1800s had its waistline just under the breasts. It was made of muslin, a thin cotton cloth, and white was the most fashionable color. White dresses were hard to keep clean, and cleaning them cost money. For every day, women wore printed cottons and patterned silks, and wool in cold weather.

Men in the same years stopped powdering their hair and cut it short. They gave up embroidered silk and lace for plain wool coats in dark colors. Some still wore knee breeches. Others wore pantaloons, tight pants that reached the calf or the ankle. By the 1840s the frock coat was the standard coat for men. It was a fitted coat with a skirt that reached to just above the knee, worn with narrow trousers and vests of brightly patterned silk.

Women's sleeves puffed out wide from the shoulder in the early 1830s. This shape was called the gigot, or leg-of-mutton, sleeve, mutton being sheep meat. Skirts became very wide, and women wore more and more petticoats under them. Under the dress, a woman wore a corset laced tight from the chest to the hips.

The corset. A fashionable corset of the 1840s ran from a woman's breasts down over her hips. Heavy cords and strips of whalebone, a stiff, bendable material from the mouths of some whales, stiffened it. A long, flat piece of steel, wood or whalebone, called a busk, ran down the front. Sleeves were cut so tight that a woman often could not lift her arms higher than her shoulders. By the end of the decade some women wore six or seven petticoats.

Godey's Lady's Book, a magazine for women published in Philadelphia from 1830, showed its readers pictures of the latest dresses. Each issue opened with a fashion plate, a picture of the newest styles colored by hand. Sarah Josepha Hale edited the magazine from 1837 to 1877. The number of copies of each issue grew from about 10,000 in 1837 to about 150,000 by 1860.

Cotton picked in the South, woven in the North

The owners and workers of the cotton mills in Lowell, Massachusetts, depended on cotton grown in the South by enslaved people. US Senator Charles Sumner called the tie between Southern planters and New England mill owners an "unholy union." On one side, he said, stood "the cotton planters and fleshmongers of Louisiana and Mississippi." On the other stood "the cotton spinners and traffickers of New England" (a fleshmonger traded in human beings, and a trafficker was a trader).

In Lowell, workers at the Merrimack Company printed calico, cotton cloth printed with colored patterns. The company's owners hired skilled printers from England. The head printer they hired in 1825 was paid more than the company's treasurer, the officer in charge of its money.

Solomon Northup. Solomon Northup described how the cotton was picked. He was a free Black man, born in Minerva, New York. In 1841 two white men who called themselves Merrill Brown and Abram Hamilton offered him work with a circus. They took him to Washington and gave him a drink that drugged him.

He woke up alone in the dark, in chains, in a slave pen, a building where traders locked up enslaved people until they were sold. The pen stood near the US Capitol. The slave trader James H. Burch beat him for saying he was free. In New Orleans, Burch's partner Theophilus Freeman sold him, and enslavers in Louisiana held him until January 1853. For almost ten years he picked cotton on the plantation of Edwin Epps, on Bayou Boeuf.

The day's weight. Northup wrote that the pickers had to be in the field as soon as it was light. They stayed until it was too dark to see, with 10 or 15 minutes at noon to eat cold bacon. At night each picker's basket was weighed. An ordinary day's work was 200 pounds of cotton.

Epps and his drivers, the men who ran the work in the field, whipped anyone who brought in less. The more a picker came up short, the more lashes, or blows of the whip, he or she got. Epps set each picker's daily amount by what that person had shown he or she could pick.

Northup called a woman named Patsey "the most remarkable cotton picker on Bayou Boeuf." He wrote that 500 pounds a day was not unusual for her. In his words, she "would surely have been beaten if she failed to produce twice as much" as the 200 pounds of an ordinary day's work.

Cloth made for enslaved people's clothes

Planters called the rough fabric they bought for enslaved people's clothes "negro cloth" or "slave cloth." The names covered several coarse fabrics. Two of them were osnaburg, a coarse plain cloth, and kersey. Both were on the list in South Carolina's 1740 law, which barred enslaved people from wearing anything finer. Another was linsey-woolsey, a cheap cloth of wool mixed with cotton or linen. Jeans was a cotton and wool cloth woven in a twill, a weave that makes thin diagonal lines.

The Hazard family owned a woolen mill in Peace Dale, Rhode Island, where workers made this rough cloth for Southern planters to clothe the people they enslaved. In the 1820s Isaac Peace Hazard traveled through the South taking orders from plantation owners. By 1845 the owners of 17 of Rhode Island's 40 textile mills had made this cloth their main product. Workers in the Lowell mills made it too.

Southern buyers told the Hazards what they wanted. One wrote in 1824 that the cloth "would not suit servants unless it was blue." By "servants" he meant enslaved people. A complaint of the time described clothes "as thin as baize, threads not beat close together, others split all to pieces" (baize was a thin wool cloth).

Handing it out. Enslavers handed out clothing on set days. On Colonel Edward Lloyd's plantation in Maryland it came once a year. In Charleston, the people Mary Pringle enslaved as house servants got a summer share and a winter share.

Stephen Duncan owned the Carlisle plantation in Mississippi in the 1850s. There each man got enough cloth for three shirts, two pairs of summer pants and one pair of winter pants, along with a coat and two pairs of shoes. Each

child got four slips, short loose dresses. John Potter's rice plantation in South Carolina used 1,800 yards of cloth a year for the 400 people he enslaved, about 4.5 yards for each person.

Harriet Jacobs was enslaved in Edenton, North Carolina, in the household of Dr. James Norcom, whom she calls Dr. Flint in her book. She wrote: "I have a vivid recollection of the linsey-woolsey dress given me every winter by Mrs. Flint. How I hated it! It was one of the badges of slavery." Her grandmother's work, she wrote, "supplied my scanty wardrobe" (scanty means too little).

The slave trader Theophilus Freeman held Solomon Northup and other captives in his slave pen in New Orleans in 1841. Northup wrote that the traders made the captives wash and made the men shave. Then they gave each person "a new suit each, cheap, but clean." The men got a hat, a coat, a shirt, pants and shoes, and the women a calico dress and a handkerchief to tie around the head.

Frederick Douglass

Who: A man enslaved in Maryland who described the clothes he was given as a child, and who escaped to freedom dressed as a sailor.
When and where: Talbot County and Baltimore, Maryland, the 1820s to 1838.

As a child, his only clothing was one linen shirt.

As a child, Frederick Douglass had nothing to wear but one coarse linen shirt that reached to his knees. It came from the clothing that Colonel Edward Lloyd and his overseers, the men who ran the work for him, handed out each year. He had no shoes, stockings, jacket or trousers, "in hottest summer and coldest winter."

He had no bed. On the coldest nights he took a bag used for carrying corn to the mill, crawled into it and slept on the cold, damp clay floor. He wrote: "My feet have been so cracked with the frost, that the pen with which I am writing might be laid in the gashes."

Allowance-day. Douglass was born enslaved in Talbot County, Maryland, and as a child he lived on Lloyd's home plantation. Lloyd enslaved people on several farms, and they came to the home plantation for their food and their yearly clothing. Each enslaved adult got two coarse linen shirts, one pair of linen trousers, one jacket, one pair of winter trousers "made of coarse negro cloth," one pair of stockings and one pair of shoes a year. Douglass wrote that all of it "could not have cost more than seven dollars."

Children too young to work in the fields got only two coarse linen shirts a year. "When these failed them," he wrote, "they went naked until the next allowance-day."

When he was about seven or eight, his owners sent him to Baltimore. Before he left, Lucretia Auld, the daughter of his old master, told him he could have a pair of trousers if he scrubbed the dirt off his skin first. He wrote: "The thought of owning a pair of trousers was great indeed!"

The escape. Douglass escaped from Baltimore by train on September 3, 1838. He wrote: "I had on a red shirt and a tarpaulin hat, and a black cravat tied in sailor fashion carelessly and loosely about my neck" (a tarpaulin hat was a sailor's waterproof hat, and a cravat was a cloth tied around the neck). He carried a paper called a sailor's protection, borrowed from a friend who was a sailor. It described a free American sailor, and Douglass used it in place of papers showing he was free.

Anna Murray, a free Black woman who washed other people's clothes for pay, helped him. She married him in New York that same month. According to *The Frederick Douglass Encyclopedia*, her laundry work gave her access to sailors' clothing, and she got some of it for him. Other accounts say she sewed the sailor's outfit herself. She also gave him part of her savings, and she sold one of her feather beds to add to it.

Who bought the first ready-made clothes?

Sailors and other working men bought the first clothes sold ready to wear in America, in the early 1800s. They bought them at shops called slop shops, on the streets next to the docks of Boston, New York, Philadelphia and Baltimore, and of smaller whaling and fishing towns. The clothes came in one size, and a sailor heading out to sea made them fit as best he could.

More and more clothing makers sewed cheap clothes for planters to give to enslaved people between 1840 and 1860. Many New York firms made what they called "Negro clothing." A planter who bought these clothes did not have to use the time of enslaved workers to sew them.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 12 MINUTES

Sewing machines and chemists' new dyes changed how Americans' clothes were made and colored after 1850. Isaac Singer began making sewing machines in 1851, and from 1856 his partner, Edward Clark, let families pay for one a few dollars a month. A chemistry student in London made the first man-made dye to be sold in 1856, a purple, from chemicals in coal tar.

A tailor in Reno and a merchant in San Francisco got a patent in 1873 for work pants with metal rivets, small metal pins that hold cloth together. The patent, a government paper, gave them the right to be the only ones to make the pants that way for a set number of years. The riveted pants were later called blue jeans.

Mockery and rules. A few leaders of the fight for women's rights wore short skirts over trousers in 1851, and men mocked them for it. Members of San Francisco's Board of Supervisors, the men who made the city's laws, ordered in 1876 that every man in the county jail have his hair cut to one inch.

Sheriff Matthew Nunan used the order to cut off Chinese prisoners' long braids, and a federal judge later found that the order was meant only for the Chinese. From 1879, the staff at the Carlisle Indian School in Pennsylvania cut Native children's hair and took away their clothes.

The Bloomer costume, 1851 to 1854

Elizabeth Smith Miller visited Amelia Bloomer in Seneca Falls, New York, in February 1851, wearing a dress that reached the knee, over trousers. It was sometimes called the Turkish dress. Bloomer began wearing it too, and she printed articles and pictures of it in her newspaper, *The Lily*. Elizabeth Cady Stanton, a leader of the fight for women's rights, also wore it. People named the outfit bloomers, after Amelia Bloomer, the woman best known for wearing it and speaking up for it.

Bloomer wrote that "letters came pouring in upon me by the hundreds from women all over the country" asking about the dress. The number of copies of *The Lily* sent out each month rose from 500 to 4,000, by the count of the National Park Service. Another count gives 3,000.

Men mocked the women who wore it and called them "Amazons," after the women warriors of old Greek stories. Some young women who wore it were turned away when they asked to join a church. By about 1854, Stanton and other leaders had gone back to long skirts and corsets. They feared that wearing the dress took attention away from their fight for the right to vote.

Hoops, sewing machines and paper patterns

Women began to wear the cage crinoline, a frame of steel hoops that held a skirt out wide, in 1856. With it, women no longer needed the heavy stacks of petticoats, sometimes as many as seven, that they had worn to get the same shape.

The sewing machine. Isaac Singer began making sewing machines in 1851. He used the lockstitch, a stitch that locks two threads together, which Elias Howe had invented. Sewing a man's frock coat, a fitted coat with a skirt to the knee, by hand took almost 17 hours. By machine it took two and a half.

A Singer machine cost about $100 in 1856. That September, Edward Clark, Singer's partner, began to sell machines on installments, which means a buyer paid a little each month. The magazine *American Heritage* gives the terms as $5 a month. Earlier in 1856 Clark had offered $50 off a new Singer for any old sewing machine, even a worthless one.

Paper patterns. Ebenezer Butterick and his wife, Ellen Augusta Pollard Butterick, of Sterling, Massachusetts, began selling sewing patterns, paper shapes that a person who sews pins to cloth and cuts around, in 1863. They cut the patterns from tissue paper. By one report, the idea came from Ellen Butterick, who was tired of patterns that came in only one size.

The Buttericks' patterns came in a range of standard sizes. Members of the family cut and folded the first ones at home. They sold patterns for men's and boys' clothes first, and added women's dresses in 1866.

Elizabeth Keckley

Who: A dressmaker, born enslaved, who bought her freedom with money from her sewing and made Mary Todd Lincoln's dresses.
When and where: Virginia, St. Louis and Washington, D.C., 1818 to 1868.

"With my needle I kept bread in the mouths of seventeen persons."

Elizabeth Keckley wrote those words about St. Louis, where the Garland family who owned her lived on the money her sewing brought in. She kept them fed that way "for two years and five months." She wrote that "the best ladies in St. Louis" were her customers, and in the end those customers lent her the price of her freedom. Later she made the dresses of Mary Todd Lincoln, the wife of President Abraham Lincoln.

Growing up. Keckley was born enslaved in February 1818 at Dinwiddie Court House, Virginia. The Burwell family enslaved her and her mother, Agnes. Her father was Armistead Burwell, of the family that enslaved her. Agnes made clothes for 82 people: the 12 members of the Burwell family and the 70 people they enslaved. As Keckley grew up, she helped her mother make the clothes.

Her old master sent her to work for his eldest son, Robert Burwell, a minister, and his wife. When she was nearly 18, she moved with them to Hillsboro, North Carolina, where Robert Burwell led a church. A schoolmaster named Bingham had promised Mrs. Burwell that he would "subdue," or crush, what he called Keckley's "stubborn pride." He whipped her. When she asked him why, he hit her with a chair and knocked her to the floor.

While she lived in Hillsboro, a white man raped her again and again for four years, forcing her to have sex with him. Keckley wrote that the man "had base designs upon me" and "persecuted me for four years." She refused to name him. "I spare the world his name," she wrote. She gave birth to his son, George, the only child she ever had.

Freedom. Her owner agreed to free her and George for $1,200. One of her customers, Mrs. Le Bourgeois, raised the money among Keckley's other customers. Keckley and her son were freed on November 15, 1855. She took the money only as a loan, and she paid back every cent from her sewing.

The White House. She moved to Washington in 1860 and later employed 20 seamstresses. Abraham Lincoln took office on March 4, 1861. At eight o'clock the next morning, Tuesday, she went to the White House. Three other dressmakers were already waiting, and Mary Todd Lincoln chose Keckley. She wrote that she made "fifteen or sixteen dresses" for Mrs. Lincoln that spring and summer. Her customers also included Varina Davis, the wife of Jefferson Davis.

People freed from slavery were coming to Washington from Maryland and Virginia in the summer of 1862. In August 1862 Keckley and other Black residents of Washington started the Contraband Relief Association to help them. During the Civil War, Union soldiers used the word "contraband" for people who had escaped slavery. The association's members gave out food, shelter, clothing and medical care, and in the first year they handed out 5,150 pieces of clothing.

She published a book about her life in 1868, *Behind the Scenes, or, Thirty Years a Slave, and Four Years in the White House*. The Smithsonian holds a purple velvet gown she made for Mrs. Lincoln. She died in Washington in May 1907.

Where did the first man-made dye come from?

The first man-made dye came from coal tar, a thick black liquid made from coal. William Henry Perkin, an 18-year-old student at the Royal College of Chemistry in London, made it in 1856, while he was trying to make quinine, a drug for the disease malaria. He was on his Easter break.

Every dye before then came from plants, insects, shellfish or minerals: indigo blue from a plant, cochineal red from an insect and red ochre from a mineral. Perkin was working with chemicals that come from coal tar. Instead of quinine he made a purple dye, now called mauve or mauveine. It was the first man-made dye sold for use, and in 1857 he opened a factory near London.

Chemists were making other new colors from coal tar by 1859, such as fuchsine, a deep crimson red. These are called aniline dyes, after aniline, one of the coal-tar chemicals used to make them.

Levi Strauss and Jacob Davis

Who: A tailor in Reno who put metal rivets on work pants, and the San Francisco merchant who shared the patent with him.
When and where: Reno, Nevada, and San Francisco, California, 1870 to 1875.

A woodcutter's wife asked for strong work pants.

A woman asked Jacob Davis, a tailor in Reno, Nevada, to make strong work pants for her husband, a woodcutter. Davis had been using copper rivets on horse blankets. He put rivets at the weak points in the seams and pockets, so that the seams would not tear. In a court statement, Davis dated this to January 1871. Other accounts date it to December 1870.

The patent. Davis and Levi Strauss's company got a patent for the idea on May 20, 1873. The patent, number 139,121, covered "a metal rivet or eyelet at each edge of the pocket opening to prevent the ripping of the seam." The riveted pants were later called blue jeans.

Davis and Levi Strauss & Co. took A. B. Elfelt and others to court in January 1874, saying that they had used the patented rivets without permission. On February 10, 1875, the judge decided for Davis and Strauss. They had asked for $20,000 and were awarded $2,000.

The two men. Davis was born Jacob Youphes in the city of Riga on May 14, 1831. He came to New York in 1854, changed his name, and by 1869 ran a tailor shop in Reno. Levi Strauss was born in Buttenheim, Bavaria, on February 26, 1829. He came to the United States in 1847 and reached San Francisco in 1854, where he sold cloth and clothing to stores as Levi Strauss & Co.

Davis ran the San Francisco factory that made the riveted pants. Strauss died on September 26, 1902, and Davis died in 1908.

Cutting Chinese men's braids in San Francisco, 1876 to 1879

Chinese men in the 1800s shaved the front of the head and wore the rest of their hair in one long braid called a queue. Stephen Field, a federal judge, later wrote that Chinese men saw the loss of the queue as a disgrace that brought "misfortune and suffering after death."

Members of San Francisco's Board of Supervisors voted for an order to cut prisoners' hair in 1873. Mayor William Alvord vetoed, or rejected, the hair-cutting part.

The lodging law. California's lawmakers passed a law about lodging houses, buildings that rent out rooms, in April 1876. It made it a crime to sleep in a room with less than 500 cubic feet of air for each person. That is about the space of a room 8 feet long, 8 feet wide and 8 feet high. A history of the case by the Federal Judicial Center, the research office of the federal courts, states that the law was aimed at Chinese residents.

The hair order. Members of the Board of Supervisors voted on June 14, 1876, that every man jailed in the county have his hair cut to one inch from the scalp. Field later wrote: "The ordinance was intended only for the Chinese in San Francisco" (an ordinance is a city law).

Ho Ah Kow was found guilty under the lodging law in April 1878 and fined $10. Field's opinion in his case does not name the judge who sentenced him. His sentence was five days in the county jail if he did not pay. He did not pay, and he went to jail. Sheriff Matthew Nunan cut off his queue. Nunan and his deputies, the officers who worked under him, cut off other men's queues too. The records do not say how many.

Ho Ah Kow sued Nunan. On July 7, 1879, Justice Field ruled that the hair order went against the Fourteenth Amendment to the Constitution. That amendment states that no state may deny any person the equal protection of the laws, which means the same treatment under the law. Field wrote that the order was not meant to stop prisoners from escaping. Its purpose, he wrote, was "merely to aggravate the severity of his confinement," which means to make the prisoner's time in jail harder to bear.

Uniforms and short hair at Carlisle, 1879 on

Richard Henry Pratt, an Army officer, opened the Carlisle Indian Industrial School in Pennsylvania in 1879 for Native children. In 1892 Pratt said his aim was to "Kill the Indian in him and save the man." In a proclamation of December 2024, President Joe Biden stated that federal officials had tried to make Native children give up their own ways of life "by stripping them of their languages, religions, and cultures."

Hair and clothes. At Carlisle the staff cut the children's hair and took away their clothes. They dressed the boys in military uniforms and the girls in Victorian dresses, the long dresses of the time. Chapter 2, Native Nations, tells the full story of the boarding schools.

Carlisle's photographer, John N. Choate, took pictures of students when they arrived. He photographed the same students again after they had been at the school for a while. His "before and after" pictures show a student in the Native clothes he or she arrived in, then the same student in a school uniform. One pair has the caption "Tom Torlino, Navajo. As he entered the school in 1882. As he appeared three years later."

Shirtwaists and the Gibson Girl, 1890s

Women in the 1890s wore a separate skirt and a shirtwaist, a blouse cut like a man's shirt. Women who worked for pay wore it every day. Sleeves puffed out wide again and were biggest around 1895. Most working men wore the sack suit, a loose jacket with pants.

About 50,000 people rode bicycles in 1885, and about 10 million in 1896, by one estimate. Some women rode in bloomers, but most wore shorter skirts.

Charles Dana Gibson lived from 1867 to 1944. He drew for *Life*, *Collier's Weekly*, *Harper's Weekly* and other magazines. In the 1890s he drew a young woman whom people called the Gibson Girl. The Library of Congress showed his drawings in an exhibition. Its writers say she "pursued higher education, romance, marriage, physical well-being, and individuality" (pursued means went after) with more freedom than young women had had before. They also write that from the 1890s until World War I she "set the standard for beauty, fashion, and manners."

1900 to 1950

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Chemists made the new materials that Americans wore and handled every day in the first half of the 1900s. Leo Baekeland made Bakelite, a hard plastic, in 1907, and chemists at the DuPont company made nylon in 1935. Nylon went on sale as women's stockings in 1939. Many women in the 1920s cut their hair short and wore shorter, looser dresses. Black women built companies that sold hair products to Black customers, and Madam C. J. Walker and Annie Turnbo Malone each grew rich from that work.

Government officials also used clothes and hair to control people. The US commissioner of Indian affairs, the head of the federal office for Native affairs, told officials on the reservations, the lands the government had set aside for Native nations, to make Native men cut their hair in 1902.

Officials in Washington limited how much cloth a dress or a suit could use in 1942. In June 1943, servicemen in Los Angeles beat Mexican American young men and tore off their zoot suits, outfits with very wide pants and long coats. Police officers then arrested many of the bloodied young men.

The order to cut Native men's hair, 1902

William A. Jones was the US commissioner of Indian affairs. On January 11, 1902, he sent a letter to the superintendents, the officials who ran the federal reservations and the offices on them, telling them to make Native men cut their hair. He wrote that long hair on the men was "not in keeping with the advancement they are making, or will soon be expected to make, in civilization." He called short hair "a great step in advance."

Jones wrote that "Indian costume and blanket" should be discouraged, and that "Indian dances and so-called Indian feasts should be prohibited." He claimed that the paint Native people wore caused "the majority of blindness among the Indians of the United States." He told the superintendents they could force men to obey by holding back their jobs and their rations, the food and supplies that officials handed out on reservations. Each superintendent had to report his progress by June 30, 1902.

Charles Burton was the superintendent for the Hopi and the Navajo. The historian Tisa Wenger writes that he reportedly used "whips, guns, and sheep shears" to enforce the order. The surviving accounts do not say how many men were forced to cut their hair.

The letter drew an outcry, and writers at *Harper's Weekly* criticized it in February 1902. Jones backed down. Officials then received new instructions to use "persuasion and example, rather than force."

Shirtwaists, short hair and short skirts

The shirtwaist, the blouse cut like a man's shirt, stayed the everyday top for working women in the early 1900s. Young immigrant women sewed shirtwaists in New York factories.

The Triangle fire. The Triangle Waist Company made shirtwaists on the top floors of a building in New York. A fire there killed 146 workers on March 25, 1911. Most of them were young immigrant women, and some were only 14. A stairwell door on the ninth floor was locked, and dozens of workers jumped from the windows.

The owners, Max Blanck and Isaac Harris, were tried for manslaughter, a killing the law treats as less serious than murder. The jury found them not guilty on December 27, 1911, because the prosecutors could not prove that the owners knew the doors were locked.

The flapper look. Many women cut their hair into a short style called a bob in the 1920s. Dresses hung straight from the shoulders, with the waistline low on the hips. This slim, straight shape is often called the flapper look. Hemlines, the bottom edges of skirts, rose until 1926 and then began to fall.

Many women gave up stiff, formal layers for lighter, looser clothes. With new fabrics such as jersey, a soft knitted cloth, and "artificial silk," a man-made cloth that looked like silk, more women could dress in the new styles. Fashion historians at the Fashion Institute of Technology call this change a "democratization of fashion," which means that fashion spread to ordinary people.

Hair care built by Black women

Annie Turnbo, later known as Annie Turnbo Malone, made a hair product she called Wonderful Hair Grower and sold it door to door. She was born on August 9, 1877, in Metropolis, Illinois, and her parents had been enslaved in Kentucky. She moved to St. Louis in 1902 and opened a shop. She later named her company Poro.

She opened Poro College in St. Louis in 1918, a building that held a factory, a store, rooms where students slept, a gym, a chapel and an auditorium. Nearly 200 people worked there. About 75,000 women, in several parts of the world, trained in her Poro system. In 1924 she paid nearly $40,000 in taxes.

Madam C. J. Walker

Who: A Black businesswoman, born to parents who had been enslaved, who built a hair-care company and trained thousands of women to sell her products.

When and where: St. Louis, Denver and Indianapolis, 1889 to 1919.

Movie: *Two Dollars and a Dream* (1987), a documentary about her life by Stanley Nelson.

Her own hair falling out led her into the hair business.

Sarah Breedlove, the woman who later became Madam C. J. Walker, was washing clothes for pay and working as a cook in St. Louis when her hair began to fall out. A skin disease, harsh hair products, poor food and a home without running water all played a part. In 1904 she began using Annie Turnbo's hair product, and she became one of Turnbo's sales agents in St. Louis.

She was born on December 23, 1867, in Delta, Louisiana, and her parents had been enslaved. Her parents died when she was 7, and she married at 14. She moved to St. Louis in 1889.

Her own company. She moved to Denver in July 1905 and began selling her own product, "Madam Walker's Wonderful Hair Grower," almost the same name as Turnbo's Wonderful Hair Grower, which she had sold in St. Louis. She married Charles Joseph Walker in 1906 and took his name. In 1910 she moved her company to Indianapolis and built a factory there. The company made hair products for Black women, and she trained thousands of women to sell them.

Her company said that by 1917 nearly 20,000 women had trained with it. According to the National Women's History Museum, she employed 40,000 people. She gave money to the YWCA, the Young Women's Christian Association. She also gave $5,000 to the NAACP, the National Association for the Advancement of Colored People, for its work against lynching, the killing of a person by a mob without a trial.

She died on May 25, 1919. Accounts of her wealth differ. One account puts it between half a million and a million dollars. According to the National Women's History Museum, it was more than $1 million.

Plastics and nylon

The chemist Leo Baekeland made Bakelite in 1907 by heating two chemicals, phenol and formaldehyde, under pressure. Bakelite was the first plastic made entirely from chemicals, rather than from plants or animals. He got a patent for his method, the right to be the only one to use it for a set number of years, on December 7, 1909. Makers used Bakelite for jewelry, telephones, radios and billiard balls, and in cars.

Nylon stockings. Nylon went on sale as women's stockings. On October 24, 1939, the DuPont company sold its first 4,000 pairs to the wives of its employees in Wilmington, Delaware. They sold out in three hours. On May 16, 1940, called Nylon Day, four million pairs went on sale in stores across the country at about $1.15 a pair. They sold out in two days. Some sources give the date as May 15.

Japanese forces attacked Pearl Harbor, in Hawaii, on December 7, 1941. For the rest of World War II, DuPont's nylon went into parachutes, tire cords, ropes and other war goods instead of stockings. Stockings came back in 1945. In Pittsburgh, 40,000 people stood in a line more than a mile long for 13,000 pairs.

Wallace Carothers

Who: The chemist who led the DuPont laboratory where nylon was made.

When and where: DuPont's research station, called the Experimental Station, 1928 to 1937.

Half an ounce of nylon, made on February 28, 1935.

Gerard Berchet, a chemist working under Wallace Carothers at DuPont, made half an ounce of the nylon later used in stockings on February 28, 1935. Carothers led the team at DuPont that made two of the most widely used man-made materials of the 1900s, neoprene and nylon. In April 1930 his team made neoprene, a man-made rubber.

Carothers was born on April 27, 1896, in Burlington, Iowa. He started work at DuPont's Experimental Station on February 6, 1928.

His illness. Carothers had depression, an illness that brings long periods of deep sadness and makes it hard to think and to work. He wrote: "I suffer from neurotic spells of diminished capacity." He meant that at times his mind could not work as well as usual. In the summer of 1936 he was treated in a psychiatric hospital, a hospital for illnesses of the mind. His sister Isobel died in January 1937.

Carothers killed himself in Philadelphia on April 29, 1937, at age 41, by taking cyanide, a poison. His daughter, Jane, was born in November 1937, seven months after his death. DuPont officials announced nylon to the public on October 27, 1938.

Ann Lowe

Who: A Black dressmaker from Alabama who made gowns for rich families in Tampa and New York, while the wider public hardly knew her name.

When and where: Alabama, New York and Tampa, 1898 to 1950.

Her gown went to the Academy Awards under another name.

The actress Olivia de Havilland wore a gown Ann Lowe made when she accepted her Academy Award in 1946. The credit for that dress went to the name "Sonia Rosenberg," not to Lowe. The record does not say who gave that credit. Lowe also made gowns for Jacqueline Kennedy and the heiress Marjorie Merriweather Post, but she stayed almost unknown to the wider public.

A family of dressmakers. She was born about 1898 in Clayton, Alabama, into a family of Black dressmakers. Her great-grandmother was enslaved on an Alabama plantation, and the man who owned that plantation was Ann's great-grandfather. He held her as his property, and by law she could not refuse him. He was the father of her daughter, Georgia, who was Ann's grandmother.

Her mother, Jane, and her grandmother, Georgia, made dresses for well-known families in Montgomery, and Ann learned the trade from them. Her mother died in 1914, when Ann was 16, and Ann took over the business.

She went to the S. T. Taylor Design School in New York in 1917. She was taught in a classroom by herself, apart from the white students. The accounts of her life do not name who ordered it. She was ready to graduate after six months. She then made gowns in Tampa, Florida, and opened her own dress shop there in 1920.

She moved to New York in 1928 with $20,000. For years she made gowns to order for stores such as Henri Bendel, Neiman Marcus and Saks Fifth Avenue. The Winterthur Museum showed 40 of her gowns in 2023. The

> museum's description of the show describes her work as part of "a legacy of Black women's knowledge and skills that began as enslaved labor."

Cloth rules in World War II, 1942 to 1946

Officials of the War Production Board issued General Limitation Order L-85 on April 8, 1942, to save cloth and workers' time for the war. The War Production Board was the federal agency whose officials controlled materials during the war, and its chairman was Donald M. Nelson. The order set how long and how wide women's coats, dresses, suits and skirts could be.

What it banned. To save cloth, the order banned hoods and capes attached to coats. It banned belts and hems, the folded-up bottom edges of clothes, more than 2 inches wide, and it banned cuffs, turned-back bands of cloth, on slacks, or loose pants.

Wedding gowns, dresses for pregnant women, children's clothes, burial gowns and religious robes did not fall under the rules. The rules lasted until 1946.

Stanley Marcus, a co-owner of the Neiman-Marcus department stores, was asked on December 27, 1941, to head the board's section for women's and children's clothing. Makers and stores that broke the rules could be fined or jailed. In April 1942 the board's officials also limited the elastic used in bras and in girdles, tight underwear that shaped the waist and hips. Nylon and rubber went to the military.

The zoot suit attacks in Los Angeles, June 1943

A zoot suit had high-waisted, very wide pants that narrowed at the ankle and a long coat with wide lapels, the folded-back flaps at the front of a coat below the collar. It began among Black men in Harlem, in New York. In Los Angeles in the early 1940s young Mexican American men wore it. So did some Filipino American and Japanese American young men, and some Mexican American young women. The War Production Board's officials had stopped the making of zoot suits because they used so much cloth.

The attacks. Sailors, soldiers and Marines, joined by civilians, went through the Mexican American neighborhoods of Los Angeles from June 3 to June 8, 1943. They hunted young men in zoot suits, beat them, tore off their clothes and burned the suits in street fires. The National WWII Museum says the attacks began on June 3 with about 50 sailors. Another account says about 11 sailors started them, and about 200 joined the next day. By the later days, thousands of servicemen had joined.

More than 150 people were injured. Los Angeles police officers stood by while the attackers claimed they were defending themselves. The officers arrested bloodied and bruised Mexican American men for "disturbing the peace." They arrested nearly 600 Mexican Americans, by the National WWII Museum's count, or more than 500 by another count. The police arrested far fewer servicemen, and no soldiers were arrested for the beatings.

Navy and Marine Corps commanders ordered their men to stay out of Los Angeles on June 8, 1943, and the attacks stopped. Members of the Los Angeles City Council voted for a resolution, a statement of what they wanted, to make wearing a zoot suit in the city a crime. No such law was ever passed.

Governor Earl Warren of California chose a committee to study the attacks. Its members reported that most of the people attacked were Mexican Americans or Black people. On the cause of the attacks, they wrote that "the existence of race prejudice cannot be ignored," race prejudice meaning hating or looking down on people because of their race.

1950 to 2000

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After the mid-1950s, the people who designed American clothes began to take their ideas from teenagers, and plastics went into everyday clothes and furniture. In the 1950s young men copied the jeans, white T-shirts and leather jackets that film stars wore. The first suits made partly of polyester, a plastic made into thread, went on sale in 1951. From 1950 the furniture company Herman Miller sold chairs that the designers Charles and Ray Eames had shaped from plastic and glass fiber.

In court. People also went to court over clothes and hair. Seven of the nine justices of the Supreme Court, the highest court in the country, ruled in 1969 that students could wear black armbands to school to protest a war. Judges disagreed about boys' long hair, and in 1981 a federal judge allowed an airline to ban a Black woman's braids.

State and federal officers found 72 Thai workers held behind razor wire in El Monte, California, in 1995, sewing clothes that were sold in American chain stores.

Jeans, T-shirts and teenagers in the 1950s

The leading women's style of the early 1950s was still the "New Look" that Christian Dior showed in 1947. It had a tight waist and a full skirt. By the middle of the decade, dressmakers cut dresses straighter and slimmer, and designers began to take ideas from teenagers.

Young men took up a working-class look: jeans, a white T-shirt and a leather jacket, with the hair greased back. Marlon Brando wore it in the film *The Wild One* in 1953, and James Dean wore it in *Rebel Without a Cause* in 1955. Fashion historians at the Fashion Institute of Technology describe the look as young people turning away from their parents' generation. Young men who wore it got a reputation as rebels and troublemakers. By the end of the decade, boys wore jeans to most places.

Polyester and wash-and-wear

Two British chemists, John Rex Whinfield and James Tennant Dickson, patented polyester in 1941. The DuPont company bought the right to make it in the United States in 1946, and DuPont named its polyester fiber Dacron.

Hart, Schaffner & Marx of New York began selling men's suits made of cloth that was 55 percent Dacron and 45 percent wool on May 8, 1951. According to the Hagley Museum and Library, it was the first polyester fiber sold to the public anywhere in the world. The makers of the cloth liked it because it did not crease or wear through easily and was easy to clean. In 1953 DuPont opened a Dacron plant in Kinston, North Carolina.

Clothes made with Dacron could be washed and worn without ironing, so they were sold as "wash and wear." Most of the ironing time they saved was women's time.

The Polyester Decade. People in the 1970s could buy fashionable styles at any price, because of man-made cloth. Polyester and other man-made fabrics were so common that people called the decade the "Polyester Decade." Men wore leisure suits, a casual kind of suit often made of man-made cloth, as a cheap stand-in for a regular suit.

People dancing at disco clubs wore satin, sequins (tiny shiny disks sewn onto cloth) and Lycra, a stretchy man-made cloth. By the late 1970s people wore sweatshirt cloth and leggings, once clothes for the gym, every day.

Charles and Ray Eames

Who: A husband and wife who designed furniture of molded plywood and plastic, meant to be sold cheaply to families in small homes after World War II.

When and where: Los Angeles, 1941 to 1978.

Movie: *Eames: The Architect and the Painter* (2011), a documentary about them in the PBS series American Masters.

Their first molded plywood was leg splints for wounded servicemen.

During World War II, Charles and Ray Eames made leg splints of molded plywood for wounded servicemen, with money from the Navy. Plywood is made of thin sheets of wood glued together, and a splint holds a broken bone still. In 1945 and 1946 they used the same method to make plywood chairs. They wanted chairs that could be made from a few parts, so that good chairs could be sold at low prices.

The contest. The Museum of Modern Art in New York held a contest in 1948 for low-cost furniture for the small homes being built after the war. The Eameses entered a chair seat made in one piece of stamped steel, and they won second place. Managers at Herman Miller would not pay $80,000 for a machine to press the steel. So the Eameses used fiberglass instead, a plastic that hardens around threads of glass. The military had used it in the war.

Herman Miller sold their chairs from 1950, with seats molded in one piece from fiberglass. Workers at Zenith Plastics of Los Angeles shaped the seats in metal molds on hydraulic presses, machines that press with the force of liquid under pressure. The first colors were greige, a gray-beige, elephant-hide gray and parchment. Buyers could put the same

seat on metal legs, wooden legs or rockers. The Eameses described their aim as "getting the most of the best to the greatest number of people for the least."

Who got the credit. The Eameses designed a costly lounge chair of leather and molded rosewood plywood in 1956. Ray called it "comfortable and un-designy." Its first label said "designed by Charles Eames." While she was alive, Ray got less credit than writers, museums and filmmakers have given her since her death.

Charles Eames was born on June 17, 1907, in St. Louis, Missouri, and studied architecture. Ray Kaiser was born on December 15, 1912, in Sacramento, California, and trained as a painter. They married in June 1941 and moved to Los Angeles. Charles died on August 21, 1978. Ray died on August 21, 1988, ten years later to the day.

Who made Jacqueline Kennedy's wedding dress?

Ann Lowe, the Black dressmaker in New York, made it. Janet Auchincloss hired her in 1953 to make the wedding dress for her daughter, Jacqueline Bouvier, and Lowe also made the bridesmaids' dresses. Jacqueline Bouvier married Senator John F. Kennedy that September. The wedding dress used 50 yards of ivory silk taffeta, a smooth, crisp silk cloth, and cost $500.

Ten days before the wedding, water flooded Lowe's workroom. She and her workers made the ruined dresses again. Lowe paid the extra cost herself and did not tell the family. She got no public credit for the dress until after Lee Harvey Oswald shot and killed President Kennedy in Dallas on November 22, 1963.

Money troubles. Her rich customers often talked her out of charging hundreds of dollars for her gowns. After she paid her workers, she often lost money, and at the height of her career she was almost broke. She lost her New York shop in 1962 because she had not paid her taxes.

The same year doctors removed her right eye because of glaucoma, a disease in which pressure inside the eye destroys the nerve that carries sight. She declared bankruptcy, a step in court for a person who cannot pay what he or she owes, in 1963. She opened a new shop on Madison Avenue in 1968, and she retired in 1972. She died on February 25, 1981.

Mary Beth Tinker

Who: A 13-year-old junior high student who wore a black armband to school to protest the Vietnam War. She was suspended, and her case led to a Supreme Court ruling on students' rights.
When and where: Des Moines, Iowa, December 1965 to February 1969.

She was 13, and she wore a black armband to school.

Mary Beth Tinker wore a black armband to Warren Harding Junior High School in Des Moines, Iowa, on December 16, 1965. She wore it to protest the Vietnam War. School officials sent her home and suspended her.

That December a group of adults and students had met at the home of Christopher Eckhardt's family in Des Moines. They decided to wear black armbands through the holidays to object to the war and to call for a truce, an agreement to stop fighting. On December 14 the principals of the Des Moines schools agreed that any student who wore an armband would be told to take it off. A student who refused would be suspended.

Christopher Eckhardt, 16, wore an armband to his high school on the same day as Mary Beth. Her brother John, 15, wore his the next day. School officials suspended all three. They did not go back to school until after New Year's Day. Her younger brother Paul, 8, and her sister Hope, 11, also wore armbands, but the rule did not cover elementary schools.

The three students' fathers sued the school district for them, with help from the Iowa Civil Liberties Union. The Tinker family received hate mail and threats to kill them. No surviving record names who sent them. A federal judge ruled for the school district, and the judges of the appeals court, a higher court that checks the rulings of lower courts, split evenly.

> **The ruling.** Seven of the nine Supreme Court justices ruled for the students on February 24, 1969. Justice Abe Fortas wrote that students do not "shed their constitutional rights to freedom of speech or expression at the schoolhouse gate." He meant that students keep the right to free speech that the Constitution gives every American. He called the armbands "a silent, passive expression of opinion, unaccompanied by any disorder or disturbance," meaning that the students stated their opinion quietly and caused no trouble. Justices Hugo Black and John Harlan disagreed.

Long hair and braids at school

In the Tinker case, seven justices protected the armbands as a way of speaking. They did not decide what students could wear or how they could cut their hair. Justice Fortas wrote that the case did not concern "the length of skirts or the type of clothing, to hair style." Boys with long hair lost some court cases and won others.

Three Dallas boys tried to enroll at W. W. Samuell High School in September 1966. They were Phillip Ferrell, Stephen Webb and Paul Jarvis, and they played in a band called Sounds Unlimited. School officials turned them away because of their "Beatle" haircuts. In 1968 the judges of a federal appeals court sided with the school.

Chesley Karr of El Paso, Texas, challenged his school's rule in 1970 that a boy's hair could not hang over his ears or his collar. He lost. In 1972, in a case called Arnold v. Carpenter, judges on a different federal appeals court ruled for long-haired students.

The Supreme Court justices never agreed to hear a school hair case. In 1971 Justice Hugo Black wrote that he could not see why "anyone should think the Federal Constitution imposes on the United States courts the burden of supervising the length of hair that public school students should wear." Justice William O. Douglas disagreed when the other justices turned such cases away.

The Pawnee boys. One of those cases came from Pawnee boys in Oklahoma. The boys wore their hair parted in the middle with a long braid on each side, to follow "the old traditional ways." One of them said it was "one way of telling people that I am proud [to be an Indian]."

Officials at Pawnee Junior High School suspended the boys with no end date on April 24, 1972, under a rule that a boy's hair could not reach his collar or his ears. The first name on their case is Norman New Rider. Federal appeals judges ruled that the school could keep its rule. On December 10, 1973, the Supreme Court justices refused to hear the case. Justices Douglas and Thurgood Marshall disagreed. Douglas wrote that the boys were telling other students of "their pride in being Indian."

From Nebraska to Oklahoma. The Pawnee Nation once lived in Nebraska and northern Kansas, and the Sioux attacked the Pawnee there for four decades. In 1874 the Pawnee began a two-year removal, a move of the whole nation away from its homeland, to Indian Territory, which later became Oklahoma. Some accounts say the Pawnee were forced to go. The US government's agent for the Pawnee chose their new reservation, on land that had belonged to the Cherokee.

The Afro, braids and a court ruling

Many Black Americans straightened their hair before the 1960s. The creams they used to straighten it, called relaxers, often held lye, a chemical that could burn the scalp and the ears. In the 1960s many Black men and women in the "Black is beautiful" movement began wearing their hair natural. The Afro, hair combed out into a round shape, became a sign of Black pride. Angela Davis wore one in 1969. A person could keep up an Afro at home without paying a hairdresser.

Renee Rogers was an airport operations agent for American Airlines. The airline's managers had a rule against an all-braided hairstyle for workers in her job, and they told her to wear a hairpiece over her braids at work. She sued. On December 1, 1981, federal Judge Abraham Sofaer ruled for the airline.

Judge Sofaer ruled that a person can change a hairstyle but cannot change the texture of his or her hair. So, he ruled, the braid rule was not race or sex discrimination, treating people worse because of who they are, under the Civil Rights Act of 1964. The airline's lawyers said the style had been made popular by the actress Bo Derek, in the 1979 film *10*. Rogers said braids had a long history for Black women.

Sneakers and hip-hop in the 1980s

In New York in the 1980s, hip-hop groups such as Run DMC wore white Adidas sneakers, jeans, hooded sweatshirts, gold chains and baseball caps.

Nike began selling its Air Jordan basketball shoe, made for the player Michael Jordan, on April 1, 1985. NBA officials fined Jordan $5,000 a game for wearing shoes in colors that broke the league's rules. Reports of the shoe's sales in its first year range from $126 million to more than $150 million.

A killing for shoes. Costly sneakers could make the young person wearing them a target. James David Martin, 17, killed Michael Eugene Thomas, 15, on May 2, 1989, to take his Air Jordans. Thomas was a ninth grader at Meade Senior High School in Anne Arundel County, Maryland, and he had bought the shoes for $115.50. Martin led him into the woods near the school, raped him, forcing him into a sexual act, and strangled him to death. Then Martin took the shoes and ran.

Martin pleaded guilty to voluntary manslaughter, killing a person on purpose but without planning it ahead, a crime the law treats as less serious than murder. He was sentenced to seven years in prison.

In May 1990 the writer Rick Telander published a cover story in the magazine *Sports Illustrated*, "Your Sneakers or Your Life," about young people robbed for their sneakers and jackets, some of them killed. Police in a few cities counted such robberies. Atlanta police reported more than 50 muggings for sports clothes in four months, and a Chicago police sergeant counted about 50 jacket crimes and a dozen shoe crimes a month. Each count covered only one city. Richard Lapchick of Northeastern University called the news reports of these crimes "grossly exaggerated."

Slavery in a California sewing shop, 1995

State and federal officers and El Monte police raided a group of seven apartments in El Monte, California, near Los Angeles, on August 2, 1995. Inside they found 72 garment workers, people who sew clothes for a living, from Thailand. They had been held there and forced to sew.

Suni Manasurangkun, whom the workers called "Auntie," ran the shop with her sons. Recruiters in Thailand had promised the workers jobs, given them false passports and brought them to Los Angeles. The bosses took the passports and told each worker that he or she owed about $5,000 for the trip.

Behind the wire. A high fence topped with razor wire surrounded the apartments, and armed guards watched the workers. The guards showed them a picture of a man who had been beaten after he tried to escape. The records do not name who beat him. The workers sewed about 84 hours a week, from early morning to midnight, for about $1.60 an hour. Some of them had not been outside the fence for as long as seven years.

The officers learned of the shop after a woman escaped through a second-floor window. A letter written in pencil, with a map drawn by hand, reached the authorities. It said: "Very Dangerous. Please bring much manpower."

Who paid. The clothes carried "Made in the USA" labels and were sold in chain stores including Mervyn's, Montgomery Ward and Miller's Outpost. In February 1996 Suni Manasurangkun, three of her sons and three other people pleaded guilty to federal crimes. They were sentenced to up to seven years in prison. Two of her sons escaped the country and were never caught.

The workers sued the stores and clothing companies that had sold the clothes. In 1997 those companies agreed to pay the workers wages they were owed, and the lawsuits ended. The amount was $4 million by one account and more than $4.5 million by another.

Rotchana Cheunchujit

Who: A garment worker from Thailand who was held in the El Monte sewing shop until officers freed her in 1995.

When and where: El Monte, California, the early 1990s to 1995.

She was paid five to seven cents a piece.

Rotchana Cheunchujit sewed blouses and other clothes in the El Monte shop for five to seven cents a piece. That came to about $300 to $500 a month. The owners took half her pay first. Then they made the workers buy food and supplies from the owners' own store, at high prices. The bosses told her she owed about $5,000.

She came from Thailand, where she had heard that jobs in Los Angeles paid three times as much. She left her two young children with her mother there, hoping the job would help her family. The smugglers who brought her said that for about $4,800 they would get her to the United States and set her up with a job. She could pay them back from her wages. She arrived in 1994.

The raid. She was 24 and in bed when federal agents broke down the door in the early hours of August 2, 1995. Someone called out in Thai: "We're here to help you." "We were in shock," she said. "We didn't know what was going on." Immigration officers then held her and the other workers for nine days, until Asian American groups paid their bail, money paid to a court so that a person can go free while a case goes on.

Later she became a US citizen, learned English and was reunited with her children. She married Steve Sussman, a professor at the University of Southern California, and took his name. She spoke up for other garment workers and opened a vegetarian Thai restaurant in Arcadia, California.

BY 2000

If you remember one thing: Between 1950 and 2000, plastics such as polyester and fiberglass went into everyday clothes and chairs, and students and workers went to court over armbands, long hair and braids.

- 1951: Hart, Schaffner & Marx begins selling suits made partly of polyester.
- 1969: Seven Supreme Court justices rule that students may wear black armbands to school.
- 1995: Officers free 72 Thai garment workers held behind razor wire in El Monte, California.

Next: After 2000, almost all the clothes Americans bought were sewn in other countries, and in Bangladesh more than 1,100 people were killed when a building full of garment factories collapsed.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 14 MINUTES

Today almost all the clothes sold in the United States are sewn in other countries, and Americans throw away far more cloth than they did in 2000. They threw away 17 million tons of cloth in 2018, most of it clothing, by the count of the US Environmental Protection Agency. That was almost twice the 9.5 million tons they threw away in 2000.

Around the world, clothing sales more than doubled between 2000 and 2015, and people wore each piece fewer times, according to a 2017 report by the Ellen MacArthur Foundation.

Who sews them. Some of the clothes are sewn in Bangladesh, where a factory fire in 2012 and a building collapse in 2013 killed more than 1,200 people, most of them garment workers. Some of them had been sewing clothes for American stores.

In Los Angeles, garment workers were paid by the piece, often a few cents for each piece they sewed. Starting in 2022, a California law required pay by the hour, at least the minimum wage, the lowest hourly pay the law allows. After that, piece pay was allowed only as a bonus.

California lawmakers passed the first law against punishing workers or students for the natural texture of their hair in 2019. It also covers braids, twists and locs, long ropes of hair also called dreadlocks. Lawmakers in about 30 states have passed such laws since. Members of the CROWN Coalition, the group that worked for the first of those laws, counted the states.

Where do Americans' clothes come from?

About 97 percent of the clothes and shoes bought in the United States are made in other countries, most of them in Asia. The American Apparel & Footwear Association, a group of clothing and shoe companies, gave that figure in April 2025, from its latest data.

Workers in Bangladesh are among those who sew clothes for American stores. In the year from 2021 to 2022, the country's garment factories earned $42.6 billion, about 82 percent of the money Bangladesh earned from goods sold to other countries. About 4 million people worked in them. According to a 2023 report by the International Labour Organization, which studies work and workers around the world, the typical garment worker in Bangladesh was a woman of 23 who had moved to the city from a farming area.

The Tazreen fire and the Rana Plaza collapse

Tazreen. A fire started in a ground-floor warehouse of the Tazreen Fashions factory, on the edge of Dhaka, Bangladesh, on November 24, 2012. The building had nine floors. All three of its staircases led down through the ground floor, where the fire was. The fire killed at least 112 people, by the first count reported by the Associated Press. Later counts run from 117 to 124. Twelve people died jumping from windows.

Police arrested three supervisors and accused them of padlocking the exits. One survivor, Mohammad Ripu, said a manager told the workers: "The fire alarm had just gone out of order. Go back to work."

Workers at Tazreen had been sewing clothes sold by Walmart and other companies. In May 2011, inspectors checking the factory for Walmart had rated it high-risk. That same year Sridevi Kalavakolanu, Walmart's director of ethical sourcing, said that fire and electrical upgrades in factories were "not financially feasible for the brands." She meant they cost too much. Her job was checking how the companies that made goods for Walmart treated their workers.

After the fire, Walmart officials said a supplier, one of the companies that made clothes for Walmart, had sent work to the factory without permission, and they stopped doing business with that supplier.

Rana Plaza. Rana Plaza was a nine-story building in Savar, near Dhaka, with five garment factories on its upper floors. About 5,000 people worked in them. The building stood on a pond that had been filled in, and it was built in 2006 without proper permits, the government's written permission to build. Its owner, Sohel Rana, had the upper floors added without a permit. They were not built to hold the weight of factory machines and generators.

A Bangladeshi television channel showed cracks in the building on April 23, 2013. The shops and the bank on the lower floors closed. Sohel Rana told reporters that the building was safe. Factory managers ordered the garment workers back the next morning. Managers at the Ether Tex factory threatened to hold back a month's pay from any worker who stayed away.

The building fell at about 9 in the morning on April 24, 2013. It killed more than 1,100 people, 1,134 by the count of the Clean Clothes Campaign, a group that works for garment workers' rights. Counts of the injured run from about 1,500 to nearly 2,600. Some trapped workers could be freed only by amputating an arm or a leg, which means cutting it off.

Researchers for the Clean Clothes Campaign found at least 29 brands with recent orders at the factories in Rana Plaza. They included two American companies, Cato Fashions and The Children's Place. News reports also linked Walmart. Officials of J.C. Penney said that one factory there had made a small amount of clothing for another brand that Penney stores sold.

After Rana Plaza

After the collapse, officials of more than 200 clothing companies, most of them European, signed a safety agreement that bound them by law. Officials of Walmart and other North American companies did not sign it. In July 2013 leaders of 17 North American store chains, including Walmart, Gap, Target and Macy's, announced a safety plan of their own. It had no promise, binding by law, to pay for repairs to the factories.

Clothing brands paid into a fund for the families of the dead and for injured workers. The fund reached its target only after more than two years, when the last brands, such as Benetton, paid in. By October 2015, more than $34 million had been paid out, according to the International Labour Organization. According to the Clean Clothes Campaign, some of the brands linked to the building never paid.

In court. The husband of a worker who died and a worker who was injured sued J.C. Penney, The Children's Place and Walmart in a Delaware court. On May 4, 2016, the judge threw out the case, ending it without a trial, for two reasons. The law of Bangladesh allowed one year to bring such a case, and that year had passed.

The judge also ruled that the companies were not the workers' employers. So under Delaware law the companies were not required to keep those workers safe. No court has found an American company responsible for the deaths at Tazreen or Rana Plaza.

Sohel Rana was arrested near the border with India on April 28 or 29, 2013. He was still held in custody in April 2026. In 2017 a judge sentenced him to three years in prison for hiding his wealth. As of April 2026, judges in his murder trial had heard 145 of the 594 listed witnesses. The trial had not ended, *The Business Standard*, a newspaper in Dhaka, reported.

Paid by the piece in Los Angeles

Clothes are still sewn in Los Angeles, often in small shops that open and close quickly. In 2020 workers there were paid about 5 to 12 cents for each piece they sewed, according to the news site *CalMatters*. They worked about 10 hours a day, five and a half days a week, for about $300 a week in cash. That came to about $5.50 an hour, when the city's minimum wage was $15.

That year, at least 375 workers at the Los Angeles Apparel factory caught COVID-19, and four of them died.

The new law. Governor Gavin Newsom signed the Garment Worker Protection Act, which State Senator María Elena Durazo wrote, on September 27, 2021. From January 1, 2022, garment shops in California had to pay at least the minimum wage by the hour. They could pay by the piece only as a bonus on top. Workers could also claim unpaid wages from the brands and stores that sold the clothes.

Marissa Nuncio, director of the Garment Worker Center in Los Angeles, said in 2021 that the law applied to more than 45,000 garment workers in California. Most of them were immigrant women. In 2023 Remake, a group that works for garment workers, reported that a year after the law took effect some workers were earning the minimum wage, and others still reported wages they were owed and never paid.

Francisco Tzul

Who: A garment worker from Guatemala who sewed clothes in Los Angeles for decades, often for a few cents a piece, and told lawmakers about his work.

When and where: Los Angeles and Sacramento, the 1990s to 2021.

He sewed clothes that carried a TV star's name.

Francisco Tzul watched Kathie Lee Gifford's television talk show in the 1990s and recognized her name. It was on the labels of clothes he sewed in Los Angeles. He was paid a few cents for each piece of the sewing work, about a dollar for each finished item. He shared a one-bedroom apartment with three friends.

He came from Guatemala, and by 2020 he had sewn clothes in Los Angeles for decades. That year he was 56 and was living in the United States without legal papers. He has told his story in interviews with the news sites *CalMatters* in 2020 and Knock LA in 2021.

COVID-19. He sewed face masks at the Los Angeles Apparel factory in 2020, and the factory paid him by the hour: $15, the city's minimum wage, plus overtime. That was far more than the $5.50 an hour that piece pay came to in the small shops. He caught COVID-19 there. "My respiratory system was completely blocked and I wasn't able to breathe at all," he said. He spent 10 days in the hospital. When he came out, his roommates told him to leave. Staff at the Garment Worker Center found him a hotel room for two weeks.

He went to Sacramento in 2021 to tell California's lawmakers about his work. He called the day the Garment Worker Protection Act was signed the best day of his life.

What does the CROWN Act ban?

Under California's CROWN Act, employers and public schools may not treat people worse because of the texture of their hair, or because of styles such as braids, locs and twists. California became the first state with a CROWN Act on July 3, 2019. The name stands for "Creating a Respectful and Open World for Natural Hair." State Senator Holly Mitchell wrote the law with a group of supporters that included civil rights organizations and the company Dove.

New Jersey's governor, Phil Murphy, signed a CROWN Act on December 19, 2019. By July 2025, 27 states and Washington, D.C., had such laws, according to a list that GovDocs published. Pennsylvania's governor, Josh Shapiro, signed one on November 25, 2025. In September 2026 the CROWN Coalition, the group that worked for the first law, counted 30 states.

Members of the US House of Representatives passed a national CROWN Act in 2020 and again in 2022. Senators never voted on it. In September 2026 the United States had no national law against hair discrimination.

Damon Landor. Damon Landor's case over his hair went to court under a religious-freedom law, not a CROWN Act. Guards at a prison in Louisiana handcuffed him to a chair in 2020 and shaved off his dreadlocks. He had grown them for decades because of a promise he made as part of his Rastafarian religion, and he had three weeks left in his sentence. He showed the guards a court decision, and they threw it in the trash.

Six of the nine Supreme Court justices ruled on June 23, 2026, that he could not sue the guards for money under a federal law that protects the religious freedom of people in prison.

Andrew Johnson

Who: A high school wrestler in New Jersey whose locs were cut beside the mat so he could compete. His case led to a state law against hair discrimination.

When and where: Buena, New Jersey, December 2018 to December 2019.

The referee gave him 90 seconds: cut his hair or lose.

The team's athletic trainer cut off Andrew Johnson's locs beside the wrestling mat on December 19, 2018, while the crowd watched, so that Johnson could wrestle. The referee, Alan Maloney, had told Johnson that his hair broke the rules. Maloney rejected the cover Johnson wore over his locs and gave him 90 seconds to cut his hair or lose the match. Johnson was a junior at Buena Regional High School, and this was a home match.

The trainer who cut his hair was a white staff member. Johnson won the match in overtime. A local sports reporter, Mike Frankel, posted a video of the haircut. NBC News reported that the video had been viewed more than 14.5 million times.

Johnson's family said that Maloney had arrived late. The family said he had missed the weigh-ins, where referees check each wrestler before a match, and had raised no problem with the hair there. In 2016 Maloney had been accused of calling a Black referee an insulting name for Black people at a party. He was suspended for it, and then allowed back after an appeal.

What changed. In September 2019 officials of the New Jersey Division on Civil Rights and of the state's school sports association agreed that Maloney would be suspended for two seasons. They also agreed that all wrestling officials in the state would be trained that the rule is about the length of hair, not its style. On December 19, 2019, one year after the match, Governor Phil Murphy signed New Jersey's CROWN Act.

Where old clothes go

Americans threw away about 1.8 million tons of cloth in 1960. In 2018, the latest year the US Environmental Protection Agency has counted, they threw away 17 million tons, more than nine times as much. About 13 percent of old clothes and shoes were recycled that year. Another 3.2 million tons of cloth were burned to make power, and 11.3 million tons went to landfills.

The Ellen MacArthur Foundation reported in 2017 that, around the world, one garbage truck's worth of cloth is dumped or burned every second.

In April 2026 ThredUp, a company that sells used clothing online, published a report with the research firm GlobalData. It stated that in 2025, sales of used clothes in the United States grew nearly four times as fast as sales of clothes in general. ThredUp's researchers expect sales of used clothes to reach $78.8 billion a year by 2030. Nearly half of the shoppers they surveyed said they find used items through social media.

THE THREE QUESTIONS, ANSWERED

1. **Why did lawmakers and officials make rules about other people's clothes and hair?** To keep rank where everyone could see it, and to control the people below them: Massachusetts fined people worth less than 200 pounds for wearing silk, South Carolina held enslaved people to coarse cloth in 1740, and a federal official ordered Native men's hair cut in 1902. Since 2019, CROWN Acts in about 30 states have turned the rule around and forbidden punishing people for natural hair, braids, locs or twists.
2. **Who made the cloth Americans wore?** Native weavers first, then enslaved people who grew the cotton and made the indigo, and mill workers in New England. Today almost all of it is sewn in other countries, by workers in places like Bangladesh, and in small Los Angeles shops where workers were paid by the piece until a California law required hourly pay in 2022.

3. **Which new materials changed what Americans wore?** Churro wool, brought in 1598, became the main thread for weaving in New Mexico. Dyes made from coal tar, from 1856, gave clothes new colors such as mauve. In the 1920s "artificial silk" let more women dress in the new styles. Nylon made stockings from 1939, and polyester suits from 1951 could be washed and worn without ironing.

What changed: Before 1500 people here made blankets from turkey feathers and yucca cord, and today about 97 percent of the clothes and shoes Americans buy are made in other countries.

FIVE DATES TO KEEP

1651: Massachusetts lawmakers fine people worth less than 200 pounds for dressing like gentlemen.

1740: South Carolina's Negro Act holds enslaved people to coarse cloth.

1856: William Henry Perkin makes the first man-made dye, a purple, from coal tar.

1939: DuPont sells its first nylon stockings.

2013: Rana Plaza collapses in Bangladesh, and more than 1,100 people are killed.

NOT KNOWN: The surviving records do not show how often anyone took an enslaved person's clothes under South Carolina's 1740 law.

Where to go: George Washington's Mount Vernon, in Virginia, holds the brown wool coat he wore when he took the oath as president in 1789.

Next: Chapter 36, Sports and Play. It asks who was allowed to play, from chunkey at Cahokia to the leagues and screens of today.

Chapter 36: Sports and Play

CHAPTER 36 OF 37 · SPORTS AND PLAY · 10 ERAS · ABOUT 125 MINUTES

From chunkey at Cahokia to Fortnite, American games moved out of open fields and into leagues, arenas and screens run by adults and companies, and at almost every step someone in charge decided who was allowed to play.

On September 10, 1674, the judges of the York County court in Virginia punished a tailor named James Bullocke. He had bet 2,000 pounds of tobacco on a race between his mare and a gentleman's horse. Racing, the judges ruled, was "a sport only for Gentlemen," and they fined him. Then they found that the gentleman had fixed the race so that the tailor's mare would win. They sent the gentleman to sit in the stocks for an hour.

This chapter follows play through ten eras, from Native ball games on fields a mile long to games played on phones with strangers. Watch who runs the game. At first the players ran it themselves. Later club members wrote the rules, owners paid the players, adults ran leagues for children, and companies sold games played on a screen. And watch who gets kept out: a tailor in 1674, enslaved riders whose names the race cards left off, Black players after 1887, girls after 1951, Black children at city pools.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who invented baseball?
2. Who was kept out of American games, and how did they get in?
3. How has children's free time changed?

Five minutes? Read "Who wrote the rules of baseball?" in 1800 to 1850, then "Who could play Little League" and "How did a child's afternoon change?" in 1950 to 2000.

Can skip: "Games of chance", "Children, toys and the circus", "The Miracle on Ice", "Television and the Super Bowl". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 6 MINUTES

Native nations played games of skill and games of chance long before 1500, and their children made toys out of clay. Nobody wrote down the rules at the time. What survives are objects in the ground, found by archaeologists, the scientists who dig up and study the things people left behind. Stone disks for a game called chunkey, and little clay animals made by children, date from these centuries.

Archaeologists have also found split reeds used as dice and wooden pieces kicked along a race course, but no one has dated those. How the games were played is known from the nations' own traditions and from Europeans who watched them in later centuries.

Two kinds of games. Stewart Culin sorted Native games into two kinds in a long study printed in 1907 by the Bureau of American Ethnology, a government office in Washington that studied Native peoples.

In games of chance, players threw marked pieces and counted how they landed. In games of skill, players won by how well they used their hands and bodies. Culin found no game like chess, which a player wins by planning alone. Native children played many games with no objects at all, such as tag, he wrote, and he left those out.

Chunkey at Cahokia

Chunkey was a game played with a stone disk and long poles. One player rolled the stone along the ground. The players ran after it and threw their poles. The aim was to land a pole closest to the spot where the stone came to rest. The stones had rounded edges and hollowed sides, and some had a hole through the middle.

The game began about the year 600, in the area around present-day St. Louis. People built the city of Cahokia in that same area around 1050. By hand, they filled in and leveled a plaza, a huge open square, of about 50 acres. They used the plaza for ceremonies and for chunkey.

The stones got bigger. A chunkey stone made before Cahokia existed measures about 2.5 inches across. A stone from the 1100s measures about 3.5 inches. It is made of quartzite, a hard rock, from a single place on the Mississippi River. A shell collar found 90 miles south of Cahokia shows a chunkey player holding a pole and a stone.

Timothy Pauketat, an archaeologist at the University of Illinois, thinks messengers from Cahokia carried a chunkey stone in one hand and a war club in the other. In his account, they went out to the lands around the city to make peace or to make agreements with other peoples. He also writes that chunkey played a large part in how Cahokia's people lived together and how they were governed.

Chunkey stones from about 1300 have turned up as far away as Georgia, and in Maryland at sites in Frederick and Montgomery counties. Staff at the Jefferson Patterson Park and Museum in Maryland say Cahokia declined around 1500, and that people went on playing chunkey across North America after that.

Ball games with sticks

Nations in the East, among them the Choctaw, the Huron and the Haudenosaunee, played a ball game with sticks. The Choctaw call their game kapucha toli, which means "stick play." The Huron and the Haudenosaunee, also called the Iroquois, played the game known today as lacrosse. Writers for the Choctaw Nation, on the nation's website, say that stickball "has been the Choctaw national sport for unknown centuries."

No object or written record dates a stick-and-ball game before 1500. Jean de Brebeuf, a French missionary, a man who traveled to other peoples to spread the Christian religion, described Huron lacrosse in 1636. He belonged to the Jesuits, a group of Catholic priests. Another Jesuit missionary described a Choctaw match in 1729, the oldest written account of a Choctaw game.

How big a game could be. Fields ran from about 100 yards long to, perhaps, several miles, according to the same Choctaw writers. Some games had several hundred players on each side, and others had twenty or fewer. Among the Choctaw, women played alongside men. Betting was a large part of the game, especially for the women. The Choctaw writers also say that deaths were common in the old games.

People used the game to teach and to heal. The Choctaw have used it to show their young people how Choctaw families were arranged and what men and women each did, the Choctaw writers say. Among the Huron, Brebeuf wrote, a sick person or a healer could call for a lacrosse game to make the sick person well. Whole villages played against each other and bet beaver robes and what Brebeuf called "porcelain collars," his word for collars of wampum, beads made from shells.

Choctaw tradition says that towns sometimes played ball to settle an argument. The two games of this kind that someone wrote down took place long after 1500, around 1790 and 1800.

Games of chance

Pieces for games of chance survive in the Southwest. The archaeologist J. Walter Fewkes dug up split reeds used for a dice game at the Chevlon ruin, about 15 miles east of present-day Winslow, Arizona. Culin took the reeds as a sign that the Hopi and Zuni cane game was old. No one has dated them.

Pieces from a racing game survive too. Runners kicked a small wooden piece ahead of them along a race course. Pieces like these were found in cliff dwellings in Mancos Canyon, Colorado. One is 4 3/4 inches long and marked with lines cut into the wood.

What did children play with?

Most of the children's things that survive from before 1500 are made of clay. Toys made of wood, hide and grass rot away, so few of them last.

The archaeologist Kathryn Kamp studied clay objects about 800 years old from Sinagua village sites in northern Arizona. She measured the fingerprints left in the clay. The ridges of a child's fingerprint are narrower than an adult's. Kamp found that very young Sinagua children made little clay animals, and probably small dishes, to play with. While making them, the children learned how clay behaves.

People in the Southwest made animal figures out of split twigs about 2,000 years ago. Archaeologists had thought the figures were used in religious ceremonies for hunting. Then three of them, two deer and a duck, turned up in a child's grave. Archaeologists now think that some of the figures were toys.

Bacqueville de la Potherie, who wrote about the Huron in the early 1700s, said that Huron children played lacrosse. He wrote that they "never or rarely" played the dice games that Huron women played.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 2 MINUTES

Nearly everything written about games here in the 1500s describes Native games. The writers were Spanish and French soldiers, sailors and missionaries, and they set down what they saw Native people play. The Spanish soldiers at the forts of St. Augustine and Santa Elena on the Atlantic coast left guns, tools and pottery behind. No record from the 1500s describes what those soldiers played.

Timucua games in Florida, 1564 and 1565

René de Laudonnière led a French colony at Fort Caroline, near present-day Jacksonville, Florida, in 1564 and 1565. The Timucua lived there. Laudonnière wrote that they trained their young men to run. In one of their races, the runner who could keep going longest won. The young men also practiced often with bows and arrows.

Laudonnière described a Timucua ball game too. The players set up a tree trunk about 50 feet high in an open space. At the top they fixed a square mat woven from reeds. Whoever hit the mat with the ball won the game.

Jacques Le Moyne, the colony's artist, painted Timucua life. Theodor de Bry, an engraver in Europe, a man who cut pictures into metal plates so they could be printed many times, made prints from the paintings. He never went to Florida. One print shows young men running, shooting arrows and throwing a ball at a target on a tall pole.

Which record to trust. Most sources date the print to 1591, and a National Park Service web page gives 1587. The same page states that de Bry mixed in details from other parts of the Americas to sell more copies. Laudonnière's written account is the firmer record of the game.

Spanish games on horseback, 1598

Juan de Oñate led Spanish colonists north to the Rio Grande, in what is now New Mexico, in 1598. Gaspar Pérez de Villagrá wrote about that year in a book printed in 1610. Writers for the Museum of New Mexico's magazine, *El Palacio*, state that the colonists' celebrations in 1598 included "tilts with cane spears, running bulls and tilts at the ring."

In a tilt with cane spears, two lines of riders threw light spears made of cane at each other and blocked them with shields. In a tilt at the ring, a rider galloped at full speed and tried to push the point of a lance, a long spear, through a ring and carry it off.

Spanish friars, Catholic religious men who lived by strict rules, began building missions in Florida and Georgia in the 1590s. A mission was a church settlement where friars worked to make Native people Christians. About eighty years later, a friar in Florida set out to end a Native ball game.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 11 MINUTES

The men who governed the English and Dutch colonies in the 1600s wrote laws about when and where people could play. In Massachusetts and Plymouth, lawmakers and judges fined people for cards, dice and tavern games. In New Netherland, the Dutch director banned ball games, cards and bowling on Sunday before, between and during church services. Virginia planters, men who owned big farms, raced horses for thousands of pounds of tobacco, and one county court punished a tailor for racing, saying racing was "a sport only for Gentlemen."

Native nations kept playing their own games, and English and Spanish writers described some of them. In Spanish Florida, a friar and the governor worked to end the Apalachee ball game. Almost nothing written in these years describes what children did.

Bowling at Jamestown, 1611

Sir Thomas Dale arrived at Jamestown in May 1611 with about 300 soldiers, to govern the colony. The colonist Ralph Hamor later wrote that on his first day Dale found the colonists at "their daily and usuall works, bowling in the streetes."

The historians William Kelso and Karen Ordahl Kupperman argue that the colonists were sick and badly fed. In their view, hunger and disease made the men act in ways that could look like laziness.

Dale's code. Dale then put the colony under a code of military rules called the Lawes Divine, Morall and Martiall. Chapter 21 describes the punishments under those rules.

Massachusett football, 1634

William Wood, an Englishman, described the games of the Massachusett people on the coast in a book printed in 1634. He listed football, shooting, running and swimming. When one area played another, Wood wrote, "their goals be a mile long," set on sand as flat as a board. Players kicked a ball about the size of a handball with their bare feet. A game could last two days. The players marked the ground they had won and started again from that spot the next day.

The winners took the prizes hung on the goals: wampum, beaver skins and black otter skins. Players painted themselves before a game, as they did for war. Wood wrote that they did it so that no one could tell who had hurt him and take revenge later. While the men played, the boys played pipes and the women danced and sang. Wood also claimed that one Englishman could beat ten Native players at football.

Hubbub. Wood described a dice game called hubbub. A player put five small bones, black on one side and white on the other, in a small smooth tray. He thumped the tray on the ground so that the bones jumped, and scored by the colors that landed face up. The players shouted "Hub, Hub, Hub," and Wood wrote that they could be heard a quarter of a mile away.

Fines for games in New England

Lawmakers and judges in Massachusetts Bay and Plymouth fined people for playing games, although betting on the games was rare. The historian T. H. Breen found only a little betting in Massachusetts in the 1600s, most of it among poor people and servants, and none among the colony's leaders.

Tavern games. The lawmakers of Massachusetts Bay banned shuffleboard, a game of sliding pieces down a long board, in taverns in 1647. Three years later they banned bowling and every other game in taverns as well. They wrote that the games wasted "much precious time" and much wine and beer.

A tavern keeper who allowed the games paid 20 shillings, which made one pound in English money, and each player paid 5 shillings. Anyone who bet money had to pay three times the amount of the bet. A law of 1659 fined anyone caught playing cards or dice anywhere in the colony 5 shillings.

Fined by name. The judges of Plymouth Colony's court wrote down the name of each person they fined, wives among them. In June 1679 they fined John Holbrooke one pound for playing cards once in the woods, and "His man, Patricke," a servant who had played with him, 10 shillings. Mathew Gannet paid two pounds for playing cards several times and letting others play cards in his house.

New Netherland: Sunday bans, the goose and golf

New Netherland was the Dutch colony that is now New York. Peter Stuyvesant was its director, the man who governed it. He and the men of his council, who helped him govern, banned work on Sunday on October 26, 1656. They also banned "playing Ball, Cards, Tricktrack, Tennis, Cricket or Ninepins" (tricktrack was a board game like backgammon, and ninepins was bowling at nine wooden pins) before, between or during church services, along with boat and wagon trips. Anyone caught paid a double fine.

Pulling the goose. Dutch farmers brought a game from home called pulling the goose. They hung a goose by a rope between two poles and greased its neck and head with oil or soap. Riders galloped past and grabbed at the goose to pull it down. The winner was called king of the festival. The Dutch records printed in 1868 do not say whether the goose was alive.

Stuyvesant called the game "heathenish and Popish," which meant un-Christian and Catholic, and he banned it. Some farmers held it anyway. They were fined, and some were put in jail, even though the leaders of the town of New Amsterdam protested. Those printed Dutch records do not name the farmers or the officials who punished them.

The game first came to New Netherland in about 1654, and the records do not make clear in which year the farmers were fined. Farmers on Manhattan asked to hold the game in February 1658, and Stuyvesant refused. One later book dates his ban to that same month.

Golf in the streets. People in Fort Orange, now Albany, complained that men playing golf in the streets broke windows and put passers-by in danger of being hurt. The town's officers banned golf in the streets on December 10, 1659. The fine was 25 guilders, the Dutch unit of money.

Horse racing for tobacco

The English took New Netherland in 1664 and renamed it New York. A year later Colonel Richard Nicolls, its first English governor, laid out a two-mile racecourse on the Hempstead Plains of Long Island. He named it Newmarket, after a racecourse in England. According to the Encyclopaedia Britannica, he offered a silver cup to the best horses. Another account gives the prize as a silver crown or a bushel of wheat.

Native people had lived on that land. English settlers from Stamford, Connecticut, bought the land of the present towns of Hempstead and North Hempstead in 1643. They bought it from Tackapausha, a Marsapeague leader, and six other Native leaders. The town's history does not say what the leaders were paid or how they understood the sale.

Quarter races. Virginia planters ran quarter races, in which two horses sprinted a quarter mile along a dirt track. Men from leading families such as the Randolphs, Jeffersons and Washingtons appear in county court records of races. The largest bet in those records was made in Northumberland County in 1693. John Baker and John Haynie bet 4,000 pounds of tobacco and 40 shillings on their horses, Prince and Smoaker.

In early races a rider could shove, whip or knee the other rider and try to knock him off his horse. Later, riders agreed to "fair Rideing." They also chose a starter, judges and a stakeholder, the man who held the bets, before the race.

James Bullocke

Who: A tailor in York County, Virginia, punished for racing his horse against a gentleman's.

When and where: September 10, 1674, the York County court, Virginia.

The bet was 2,000 pounds of tobacco on one horse race.

James Bullocke bet a gentleman, Mathew Slader, that his mare would beat Slader's horse. On September 10, 1674, the justices of the York County court, the county's judges, punished Bullocke for it.

The justices ruled that it was "contrary to Law for a Labourer," a man who worked with his hands, "to make a race, being a sport only for Gentlemen." The court order, as it was printed in 1894, fined Bullocke 100 pounds of tobacco. The historian T. H. Breen read the same court book and gives the fine as 200 pounds.

The fixed race. The justices also found that the race had been fixed. Slader had signed a promise that his horse "should runn out of the way," which meant it would leave the track, so that Bullocke's mare would win. The justices called it "an apparent cheate." They ordered Slader, the gentleman, to sit for one hour in the stocks, a wooden frame that locked a person's ankles while he sat in a public place.

W. G. Stanard, a historian writing in 1894, called the court order the earliest notice of racing in Virginia's records.

NOT KNOWN: which law the justices meant. The order names none, and no such law has been found in the colony's printed laws.

The Apalachee ball game banned, 1676

The Apalachee and Yustaga nations of northern Florida played a ball game between towns. A town sent a messenger to challenge another town, naming the day and the number of players, about forty or fifty. Players kicked a small ball, a little bigger than a musket ball, at a single tall pole. Eleven hits won.

Fray Juan de Paiva was the Spanish friar at the mission of San Luis, where Tallahassee is now. He wrote a long paper against the game, dated September 23, 1676, with the royal interpreter Diego Salvador and Juan Mendoza, an Apalachee leader who interpreted for the church.

What the players did. Paiva described what players did to each other. They stepped on faces, heads and bellies. They pulled at arms and legs without caring whether the joints came apart. When a player swallowed the ball, others made him throw it up by squeezing his windpipe or kicking him in the stomach.

Paiva listed broken legs and ribs, people who had lost the use of one or both hands, and people blinded in one eye. He wrote of two players he knew who had been killed at San Luis. Five games in a row in 1676 ended in open fighting, he wrote, and soldiers were present at the games.

Paiva wanted the game banned. He also charged that people left their fields unplanted and their houses open to thieves while they watched. Governor Pablo de Hita Salazar backed him. So did Fray Francisco Perete, the head in Florida of the Franciscans, one group of Catholic friars.

The poles came down. Paiva wrote that the Apalachee chiefs pulled down the ball poles and put "the holy cross" in their place, and that Captain Juan Fernández de Florencia thanked the chiefs for it on the governor's behalf. Some people, he wrote, asked the governor in writing not to end the game. The historian Amy Bushnell dates the Spanish campaign against the game from 1675 to 1684. The Talimali Band of the Apalachee lives in Louisiana today.

Enslaved people and children

A law against gathering. Members of the Virginia General Assembly, the colony's lawmakers, passed a law against gatherings of enslaved people on June 8, 1680. They named it "An act for preventing Negroes Insurrections." "Negroes" was a word of the time for Black people. An insurrection is an uprising, so the lawmakers wrote the law to stop enslaved people from rising up against the men who held them.

The law's opening words stated that meetings of enslaved people "under pretence of feasts and burialls" were dangerous. The lawmakers meant that the feasts and funerals were only an excuse to meet.

An enslaved person who left the owner's land without a written pass was to get "twenty lashes on his bare back well layd on." That meant twenty hard blows with a whip on the bare skin. An enslaved person who raised a hand against "any Christian" was to get thirty lashes.

If an enslaved person hid out and fought back against the men sent by law to catch him, the same law said those men could kill him. Two years later the lawmakers barred enslaved people from gathering at another owner's plantation for more than four hours at a time.

Children. No one in the 1600s wrote down how a child spent the hours of a day. The only dated records of enslaved people's free time in these years are the laws of 1680 and 1682 that punished them for gathering.

The oldest marble that can be dated at Jamestown is made of brown stoneware, a hard kind of pottery, and dates from the middle of the 1600s. It may have been made in Germany, the Netherlands or England. Historic Jamestowne's description of the marble does not say whether a child or a grown person played with it.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 6 MINUTES

Horse races and cockfights, and the bets men made on them, fill most of the written record of play from 1700 to 1750. Virginia planters raced horses on straight quarter-mile paths. By the late 1730s they also held race meetings that lasted several days, with prizes for fiddlers, wrestlers and runners. Men in South Carolina and Massachusetts held cockfights, in which two roosters fought until one was dead.

No record shows adults organizing children's play into teams or clubs, and almost no one wrote down what children played. The few dated records are a boys' footrace at a Virginia race meeting in 1737 and an English children's book of 1744 with a rhyme called "Base-Ball."

Horses in Virginia

Robert Beverley wrote in 1705 that young Virginians loved to hunt wild horses. The horses were born in the woods of the upland country and had never been handled. An unmarked horse belonged to whoever caught it first.

The minister Hugh Jones wrote in 1724 that "almost every ordinary Person keeps a Horse." He had seen men spend a morning catching a horse in the woods, only to ride two or three miles to church, to court, "or to a Horse-Race."

Who could race? Historians disagree. In a quarter race, two horses sprinted along a straight, narrow path about a quarter mile long. Historians at Mount Vernon, George Washington's home in Virginia, write that owners rode their own farm horses, so ordinary farmers could race. The historian T. H. Breen found that gentlemen tried to keep racing to themselves, as the York County justices did in 1674.

A court record of 1678 shows a "boy" riding another man's horse, so not every rider owned his horse. The records of these races name the owners of the horses and seldom say who rode.

Horses from England. Planters began to bring racehorses from England. In 1730 a man named James Patton brought over a stallion, a male horse kept for breeding, named Bulle Rock. The horse was then 21 years old. Bulle Rock is thought to be the first English Thoroughbred, a breed raised in England for racing, brought to the colonies. Between 1740 and 1775 planters brought at least fifty stallions and thirty mares from England to Virginia.

Race meetings with prizes, 1736 to 1739

Prizes for fiddlers. A group of gentlemen in Hanover County, Virginia, offered prizes at Page's Warehouse on St. Andrew's Day, a church holiday, November 30, 1736. Their notice in the newspaper offered a hunting saddle to the fastest horse over a quarter mile. It offered a violin made in Cremona, Italy, to the best country fiddler. There were other prizes for dancing, singing, football, jumping and wrestling. A pair of silk stockings was to go to the young woman whom the crowd judged prettiest.

A later notice asked everyone who came to behave "with Decency and Sobriety," which meant politely and without getting drunk. Breen writes that poorer people apparently were welcome to come and watch.

A notice in the *Virginia Gazette* of July 1, 1737, stated that there would be horse races every Saturday until October near Williamsburg. That November, John Pinkerton held three days of races at the Old Field track in Virginia, with a three-mile race each day. Mount Vernon's historians call it the first documented long-distance race in Virginia.

Pinkerton's meeting also had a fiddling contest, a running race for boys, a contest for the prettiest unmarried woman, and a dinner with singing and dancing. Different sources spell the host's name Pinkerton or Brickerton, and the right spelling is not known.

A one-mile course. A Williamsburg fair in December 1739 had three days of races. On the first day, eight horses raced around a one-mile course for a saddle worth 40 shillings. Colonel Chiswell's horse Edgcomb won. The second horse won a bridle and the third won a whip.

Cricket

Colonists also played cricket, the English bat-and-ball game. William Byrd II owned the Westover plantation in Virginia. His diary for April 25, 1709, records that he played cricket and then "went to cricket again till dark." In Georgia in 1737, William Stephens wrote that townsmen, landowners and servants gathered in the main square to play cricket.

Cockfighting and gander pulling

In a cockfight, two roosters fought in a ring. Their owners strapped sharp spurs made of silver, steel or bone to the birds' legs, and the birds cut each other with the spurs. The fight went on until one bird was dead. Men bet on which bird would win. William Byrd II, Robert Carter and George Washington were among the men who followed cockfighting.

An advertisement in the *South Carolina Gazette* listed a cockfight at the house of a Mrs. Eldridge in 1732. Men in Salem, Massachusetts, held a cockfight in the Town House in 1744. Both Black and white people came to cockfights and bet on them. Some critics complained that "genteel people," meaning the well-off, mixed at the fights with people they called "the vulgar and debased," meaning common and low.

Gander pulling. Gander pulling was another game played on live birds. Men tied a live gander, a male goose, upside down by its feet to a tree branch and greased its neck. Riders galloped past and grabbed at the neck until one of them pulled the bird's head off.

Riders fell from their horses, and some lost fingers. In Virginia the game was often held on the Monday after Easter, and the crowd bet on the riders. Accounts describe gander pulling through the colonial years, but none gives a dated event before 1750.

Children

The London bookseller John Newbery printed *A Little Pretty Pocket-Book*, a book of rhymes and pictures for children, in 1744. He sold it with a ball for a boy or a pincushion for a girl. Its rhyme "Base-Ball" is the first known use of that word in print. It reads: "The Ball once struck off, / Away flies the Boy / To the next destin'd Post, / And then Home with Joy."

John Thorn, the historian of Major League Baseball, suggests that the game in the book may have been played without a bat. No copy of the 1744 printing survives. The rhyme is known only from later printings, such as one made in Worcester, Massachusetts, in 1787, and the words of the 1744 rhyme are taken from those later copies.

The records of these years do not show how much free time children had. Nor do they show how that time differed for free children, enslaved children and indentured children, who were bound to work for a master for a set number of years.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 10 MINUTES

The records of these fifty years start to say who owned the horses and who rode them. In Virginia the owners of racehorses came more and more from the top ranks of society, and enslaved men and boys often rode. The enslaved people at one Virginia plantation had a holiday on Easter Monday 1774 and went to cockfights. That June the planter's son Bob Carter pestered his tutor, his private teacher, for leave to go to a cockfight too.

The delegates to the First Continental Congress, men chosen in each colony to meet together, agreed in 1774 to stop buying British goods. They also promised to discourage horse racing, cockfighting and games.

People still played ball games. Soldiers at Valley Forge played a running game called base, and the voters of Pittsfield, Massachusetts, banned "baseball" near their new meeting house in 1791. No record from these years shows leagues, paid teams, or towns sharing written rules.

Native nations kept playing their own ball games. In Hawaii, British sailors wrote descriptions of surfing in 1779. Chapter 23 tells how those islands later became part of the United States.

Native games written down

James Adair described chunkey as the Choctaw played it, in a book printed in 1775. Only one or two players played on each side. The field was a square of ground, cleared and spread with fine sand, near the town house, the building where the town's leaders met.

Players ran after the rolling stone with 8-foot poles "most part of the day." They bet their silver ornaments, their nose, finger and ear rings, and even their clothes. Each town kept its chunkey stones from one generation to the next. Nobody buried a stone with the dead.

The beaver pond. The Creek and the Choctaw both claimed a large beaver pond on the Noxubee River, in present-day Mississippi, around 1790. Beaver furs sold well at the trading towns of Mobile and Pensacola. The two nations agreed to decide the claim with a game of stickball.

The account of that game comes from Stonie Hadjo, a Creek-born man whom the Choctaw adopted. He lived in Noxubee County, Mississippi, and often told the story to the white settlers there before he died in 1836. H. S. Halbert later wrote the story down, and H. B. Cushman printed it in 1899. Stonie Hadjo said that ten thousand people camped around the field. The Creek team won.

A fight broke out after the game, and the tellings differ on who started it. In Stonie Hadjo's telling, a Choctaw player insulted a Creek player, and the two began to fight. Choctaw accounts say the Creek player threw a petticoat, a woman's underskirt, on the Choctaw player first.

The fight spread to the warriors on both sides, who used sticks, knives, guns, tomahawks and bows and arrows. It lasted through the night until the chiefs arrived the next morning and stopped it. Stonie Hadjo said that the ball players "almost to a man perished," which means that nearly every ball player was killed. His account gives no count of the dead. The Choctaw kept the pond, even though the Creek team had won the game, and the account does not say why.

Writers for the Choctaw Nation describe two games around the 1790s played to settle land claims with the Creek, and both ended in battles. H. B. Cushman dates the second game, near the Black Warrior River, to about 1800.

Far to the north, the Mohawk and the Seneca played lacrosse at Grand River, in present-day Ontario, in 1797, with about 600 players on each side.

Surfing in Hawaii, 1779

No outside nation ruled Hawaii in 1779. Chiefs governed the islands, and Kalaniʻōpuʻu ruled the island of Hawaii. The islands were joined into one Hawaiian Kingdom starting in 1795.

British sailors with Captain James Cook watched Hawaiians surfing at Kealakekua Bay, on the island of Hawaii, in January 1779. The ship's surgeon's mate, David Samwell, the assistant to the ship's doctor, described surf riding there on January 22, 1779.

Lieutenant James King also wrote an account, which was printed in 1784. When storms made the waves highest, twenty or thirty people each took "a long narrow board, rounded at the ends," and swam out together. They dove under the breaking waves until they were past them. Then they lay on their boards and rode the largest wave back to shore.

Who owned the horses and who rode them

Almost every town in Virginia held yearly races by 1772, according to the English traveler J. F. D. Smyth. At Williamsburg the races ran for a week each spring and fall. The winner on the first day took a purse, a prize in money, of 100 pounds.

Jockey clubs. Horse owners in Fredericksburg, Petersburg, Richmond and other Virginia towns formed jockey clubs by the late 1700s. A jockey is a rider in horse races, but these clubs were groups of horse owners, and the members held races for one another. George Washington helped manage a race at Alexandria in May 1761.

The historian Gabrielle McCoy writes that the owners of racehorses came more and more from the top ranks of society. She writes that the riders "often came from the bottom of the social hierarchy," the ranking of people from the most powerful to the least. They were "enslaved men and boys."

Austin Curtis

Who: An enslaved jockey and horse trainer who became a free man.

When and where: Halifax, North Carolina, and the racecourses of Virginia and North Carolina, from the 1770s to 1807.

He kept his enslaver's racehorses safe from British raiders.

Willie Jones of Halifax, North Carolina, held Austin Curtis in slavery. In the 1770s Curtis rode Jones's horses in races in Virginia and North Carolina. During the Revolutionary War, British troops raided the Carolinas for horses to breed, and Curtis kept Jones's horses safe from them.

The petition. On December 5, 1791, Jones asked the lawmakers of North Carolina to free Curtis. He wrote that Curtis had shown "attachment to his Country during the War" and "fidelity," or loyalty, "to his Master," and so "has demonstrated that he deserves to be free." The lawmakers granted the request, and Curtis became free. His name was legally changed to Austin Curtis Jones.

Curtis kept working with horses as a trainer. He came to own more than 300 acres of land, and he bought the freedom of his son William. He and his wife, Nancy, had eleven children.

Curtis died at Halifax on December 10, 1807, at about 50 years old. A Raleigh newspaper, the *Minerva*, printed a notice of his death under the name Austin Jones on January 5, 1808. It called him "well known for many years past, as keeper of race horses," and said that he "particularly excelled" at managing them.

A plantation tutor's journal, 1773 and 1774

Philip Fithian taught the children of the planter Robert Carter at Nomini Hall, Virginia, from 1773 to 1774. He kept a journal. It records how the Carter children and the people Carter enslaved spent their free hours.

Easter Monday fell on April 4, 1774. Fithian wrote that the enslaved people had a holiday until Wednesday morning and were "at Cock Fights through the County." The next day his pupils complained that they had to study while the enslaved workers had the day off.

Sunday. Fithian saw a ring of enslaved people at the stable fighting cocks before breakfast on Sunday, April 10. Other enslaved people were digging the small garden plots that Carter allowed them to grow food on. Fithian explained that they did all their own work on Sundays because they worked for the Carters every other day.

On two winter evenings in 1774, enslaved people gathered in the schoolroom to play the fiddle and the banjo and to dance. Two of the Carter boys, Ben and Harry, joined them. The first time, Fithian sent them all away.

Another son, Bob, "teaz'd" Fithian, meaning he pestered him, on June 18 to let him go to a cockfight and a horse race two miles away. Fithian let him go after Bob promised to be home by sunset. Pupils at another school had "barred out" their teacher, Mr. Goodlet, in December 1773. They locked him out of the schoolhouse until he agreed to give them a Christmas holiday until January 6.

Fist fights with no rules

Fithian wrote on August 29, 1774, that four young men had arranged two fist fights near Mr. Lane's place. He listed what such fighters did, including kicking, scratching, biting, choking, "Gouging" and "Dismembring," which meant tearing or biting off parts of the other man's body. A crowd came to watch.

Gouging. Gouging was a kind of fighting in which only weapons were banned. A fighter tried to push his thumbs into the other man's eye socket and force the eye out. Jim L. Sumner, writing for NCpedia, says most historians think fighters seldom lost an eye. He writes that the fights still risked broken bones, scars, crippling injuries and death.

Gabriel Johnston, the governor of North Carolina, complained about this kind of fighting in 1746. North Carolina's lawmakers made it a crime to cut out tongues, pull out eyes, or bite off noses or fingers. Sumner's article does not give the year of that law.

The Revolution: a pause in racing and a game of base

A pause in racing. The delegates to the First Continental Congress agreed to the Continental Association, a pact among the colonies, on October 20, 1774. In it they agreed to stop buying goods from Britain, starting on December 1, 1774. They also promised to discourage "all horse-racing, and all kinds of games, cock fighting," plays and "other expensive diversions," meaning ways to have fun.

Members of the jockey clubs of Annapolis and New York called off their races that year. Four years later, on October 12, 1778, the members of Congress asked the lawmakers of each state to pass laws to stop "horse racing, gaming" and plays.

Base at Valley Forge. Soldiers in the Continental Army still played games. George Ewing was an ensign, a young officer, in the 3rd New Jersey Regiment, a unit of soldiers. At Valley Forge in April 1778 he wrote that the men drilled in the afternoon and "playd at base this evening."

General Washington dined with General Henry Knox on May 4, 1778. Afterward he played wicket, a bat-and-ball game like cricket, with Ewing and other officers. After the war, in 1793, Virginia's lawmakers banned bets of more than 7 dollars on horse races or "any other sport or pastime."

Toys and ball games

George Washington married Martha Custis in January 1759. Her children, Jacky and Patsy, were about six and four. In his first order of goods from London he asked for 10 shillings' worth of toys. He also ordered six little books for children learning to read, and a doll dressed in the latest fashion. When that doll was lost or broken, he ordered another that cost a guinea, which was 21 shillings. He also ordered a box of gingerbread toys and sugar figures.

Archaeologists at Mount Vernon found a small limestone marble at the House for Families. Most of the people George Washington enslaved at his Mansion House Farm lived in that building. Mount Vernon's historians think the marble may have belonged to one of the enslaved children who lived and worked there.

A reference book on leisure in America states that at fairs and on court days, the days when the county court met, poorer people ran footraces, wrestled, jumped, bowled and played football. They played for prizes such as a purse of money or a bottle of liquor, while wealthy people watched.

The printer Isaiah Thomas published *A Little Pretty Pocket-Book* in Worcester, Massachusetts, in 1787, with the English rhyme and a picture of "Base-Ball."

Baseball in Pittsfield. Voters at a town meeting in Pittsfield, Massachusetts, passed a rule on September 5, 1791, to protect the windows of their new meeting house, the town's main public building. They banned "any game of wicket, cricket, baseball, batball, football, cats, fives, or any other game played with ball" within 80 yards of the building. The rule is the oldest known record of people in an American town playing a game called baseball.

1800 to 1850

ERA 6 OF 10 · 1800 TO 1850 · ABOUT 15 MINUTES

Crowds of many thousands began to gather to watch sport in the first half of the 1800s, and the biggest crowds came to see horses race. By 1850 trotting races, in which a horse pulled a driver in a light cart, had begun to replace races with riders as the sport Americans most liked to watch. Prize fights, boxing matches with bare fists fought for money, had become about as popular to watch as trotting.

Members of clubs in Philadelphia and New York wrote down rules for their ball games, and in 1845 members of the Knickerbocker club of New York wrote the rules that grew into baseball.

Enslaved people played ball, wrestled and ran races on their few days off, and enslaved jockeys rode horses that other men owned. Children like Lucy Larcom in Massachusetts made their own toys from burrs and played outdoors. A Philadelphia firm began to make toys to sell in 1838.

Native ball games and the loss of Choctaw land

The Choctaw and the Creek both claimed the land between the Tombigbee and Black Warrior rivers, in present-day Alabama, about 1800. They agreed to settle the claim with a ball game about a mile below present-day Tuscaloosa. The Creek chief was Tuskeegee and the Choctaw chief was Luee. Both sides said they had won. A battle followed "in which many were killed and wounded on both sides," and more battles followed that fall. The account, which H. B. Cushman printed in 1899, gives no count of the dead.

The treaty. Choctaw leaders signed the Treaty of Dancing Rabbit Creek, a written agreement with the United States, on September 27, 1830. The men who bargained for the United States, Secretary of War John Eaton and John Coffee, gave the Choctaw two choices: move west of the Mississippi River, or stay and live under US and state law.

The chief Choctaw men at the talks were Greenwood LeFlore, Mosholatubbee and Nittucachee. Under the treaty, the Choctaw had to hand over about 11 million acres of their land to the United States, most of it in Mississippi. So staying meant living on land the Choctaw had handed over, under the laws of the state.

About 5,000 to 6,000 Choctaw stayed in Mississippi. US officials forced the rest, about 15,000 people, to leave their homeland and go west to Indian Territory, in present-day Oklahoma, in three groups between 1831 and 1833. A winter blizzard and an outbreak of cholera, a disease that kills by draining the body of water, struck them on the way. Writers for the Choctaw Nation say that more than a quarter of the Choctaw people forced out in those years died. Another count gives about 2,500 dead.

The graves. The ball players killed after the beaver-pond game of about 1790 had been buried at the ball ground with money and silver ornaments. A settler named Charles Dobbs took that ground in 1832, part of the land the Choctaw had to hand over under the 1830 treaty. Dobbs dug up their graves. He took about $500 in silver and about $250 in silver ornaments.

The game in Indian Territory. The Choctaw kept playing in their new land. The painter George Catlin watched a Choctaw ball game there in 1834. Several thousand people camped beside the field. Two leaders had chosen the players three or four months before and sent runners to each one with a decorated ball stick. The women placed the bets. They bet knives, dresses, blankets, pots and kettles, dogs, horses and guns.

Through the night before the game, the players danced by torchlight every half hour while the women sang. At 9 the next morning, at the firing of a gun, four old men who served as judges threw up the ball. Six or seven hundred players ran for it. Each player held a stick in each hand and could not touch the ball with his hands.

Catlin painted a Choctaw player named Tul-lock-chísh-ko, which means "Drinks the Juice of the Stone." Catlin called him "the most distinguished ball-player of the Choctaw nation." By the rules of the game he played barefoot, in a breechcloth, a strip of cloth worn around the hips, with a tail of white horsehair.

Surfing in Hawaii, 1847

Hawaii was still an independent kingdom. Hiram Bingham led the first American missionaries there from the Protestant churches, the Christian churches that broke away from the Catholic Church, and they arrived in 1820. He wrote in 1847 that surfing was "the favorite amusement of all classes." Surfers used boards from 3 feet to 14 feet long and paddled out as far as half a mile. People used to bet on it, he wrote.

Fewer Hawaiians surfed than before, Bingham wrote, because Hawaiians now wore cloth clothing in the missionaries' style and the people who made that clothing had less free time. He also gave the cause as more "modesty, industry or religion." In his time, modesty meant keeping the body covered, and industry meant hard work.

Was surfing banned? No law against surfing is on record. In the same passage, Bingham denied that the missionaries had caused any law against it. Peter T. Young, a historian of Hawaii, writes that only the king and the chiefs could make such a law, and that Hawaiians kept surfing through these years.

Games on the days off from slavery

Frederick Douglass was enslaved in Maryland. In his book of 1845 he wrote that the enslaved people he knew had the days from Christmas to New Year's Day off. Some of them made brooms, mats and baskets or hunted opossums, hares and raccoons. Most played ball, wrestled, ran footraces, fiddled, danced and drank whiskey. Douglass wrote that the drinking was what "our masters" liked best.

For a time Douglass taught other enslaved men to read on Sundays in St. Michael's, Maryland. Two class leaders, men who led small religious groups in the church, Wright Fairbanks and Garrison West, came in with other men carrying sticks and stones and broke up the school. Douglass wrote that the enslavers "had much rather see us engaged in those degrading sports," by which he meant wrestling, boxing and drinking whiskey.

In the South, planters set up boxing matches between enslaved men and bet on them. The reference book *American Eras*, which records this, gives the name of no fighter and no match.

Tom Molineaux

Who: A boxer from Virginia who fought twice for the championship of England.

When and where: Virginia, New York, England and Ireland, 1784 to 1818.

About 15,000 people watched the English champion break Molineaux's jaw.

Tom Molineaux fought the English champion, Tom Cribb, in Oxfordshire on December 18, 1810. In the 19th round the crowd pushed into the ring, and Molineaux hurt his left hand. Under the rules, a fighter had 30 seconds to come back to the line marked in the ring. Later in the fight, people argued over whether Cribb had taken longer than that. Cribb won. Accounts give the length of the fight as 35 rounds or about 39.

The English boxing writer Pierce Egan wrote that Molineaux surprised everyone with his power and his skill. About 15,000 people watched the second fight, on September 28, 1811. Cribb broke Molineaux's jaw and knocked him out in the 11th round.

Where he came from. Molineaux was born in Virginia on March 23, 1784. Writers at the Virginia Museum of History and Culture say he was born enslaved and fought other enslaved men while plantation owners bet on the fights. They say he won his freedom and $500 in one of those fights. Other accounts say the story comes from boxing writers of the 1800s and cannot be proved.

He had gone to New York and then to England by 1809. His trainer there was Bill Richmond, who had been born into slavery on Staten Island, New York, in 1763. The Virginia museum's writers say Richmond was born in Virginia.

Molineaux died in Ireland on August 4, 1818, at 34, with no money. He died in the band room of the 77th Regiment, a unit of soldiers, of tuberculosis, a lung disease, and of heavy drinking. The Galway City

> Museum in Ireland gives the place as an army barracks in Galway, and the Virginia museum's writers give Dublin.

Enslaved jockeys and trainers

Every state east of the Mississippi River had at least one racetrack by 1830, and Virginia had more than twelve. Enslaved men rode, trained and cared for horses at these tracks, and the records name a few of them.

An enslaved jockey named Simon raced against horses owned by Andrew Jackson, who later became president, from 1810 to 1815. The National Museum of Racing records nine races between them. Jackson's side lost eight times and gave up once without running.

An enslaver named John Henry Hammond bought an enslaved trainer and groom, a worker who cares for horses, named George in 1834. Hammond and other enslavers talked about splitting the ownership of George three ways.

Cato's win. An enslaved jockey named Cato rode the horse Wagner to victory at Louisville, Kentucky, on September 30, 1839, for a prize of $14,000. About 10,000 people watched. Some of them made bets in dollars and some bet enslaved people. Wagner's owner, John Campbell, also held Cato in slavery.

The Notable Kentucky African Americans Database states that Campbell freed Cato for the win and that Cato kept riding for him. Jessica Dallow, who has written about the painter Edward Troye's pictures of racehorses, writes that his freedom was only rumored at the time.

She points to the writer Edward Hotaling. He noted that each win made an enslaved jockey worth more to his owner, so freedom would have been an unusual reward. Enslaved jockeys could earn some money by riding, and Dallow writes that Cato may have been freed or may have bought his own freedom.

A race card, a printed list of the horses in a race, for the Washington Race Course at Charleston, South Carolina, in 1846 listed the owner of each horse, including Wade Hampton. It listed no rider. Each jockey appears only by the colors he wore.

Crowds for horses, runners and fighters

North against South. The northern horse American Eclipse beat the southern horse Sir Henry at the Union Course on Long Island, New York, on May 27, 1823. One history of the period puts the crowd at as many as 100,000 people. Horses from the North and the South raced each other in more matches of this kind until 1845.

People also came to watch men run. John Cox Stevens offered $1,000 in 1835 to any man who could run ten miles in under an hour. Henry Stannard did it in 59 minutes and 49 seconds. Trotting races began to take the place of races with riders on horseback about 1845.

Fights against the law. Prize fighting was against the law in most states, so organizers set up fights at the last minute. By 1850 prize fighting was about as popular to watch as trotting races. In the North, many fights set Irish Catholic immigrants against Protestants born in America.

Thomas McCoy, who was 21, died after a prize fight with Christopher Lilly at Hastings, New York, on September 13, 1842. A coroner's jury, a group of citizens who look into a sudden death, found that he died "by blows and injuries received in a fight with Christopher Lilly." The jury named eight men as the main people in the fight. None of them had been arrested when a Michigan newspaper printed the jury's finding on September 26.

A judge sentenced James "Yankee" Sullivan, who arranged the fight, to two years in prison. Sullivan served two months. He then paid a $200 bond, money a court keeps if a person breaks a promise, and promised not to fight for two years, and he was let out. The records do not say what happened to Lilly. Tom Hyer beat Sullivan in 17 minutes and 18 seconds in Maryland in 1849 and won a purse of $10,000.

Who wrote the rules of baseball?

No one person wrote them. Players in many towns played bat-and-ball games that went by local names, such as town ball, and clubs wrote down their own rules. Two groups of town ball players from Philadelphia met in Camden, New Jersey, on July 4, 1833, and joined to form one club, the Olympic. The club printed its rules and its list of members in 1838.

Twenty rules. The members of the Knickerbocker Base Ball Club of New York adopted twenty written rules on September 23, 1845. Two members, William Wheaton and William Tucker, wrote them down. A game ended when one side scored 21 runs. Three outs ended a side's turn at bat. A fielder put a runner out by throwing the ball to a baseman, not by hitting the runner with it. The members were merchants, lawyers and doctors. They played at the Elysian Fields in Hoboken, New Jersey.

The Knickerbockers played the New York club under the new rules on June 19, 1846, in one of the first recorded games between two clubs. One account says the New York club won 23 to 1. Another says the Knickerbockers won.

The Doubleday story. Albert Spalding and the members of a committee he asked for later spread a story about how baseball began that was not true. Spalding, a maker of sporting goods, asked for a study of how baseball began in 1905. Abner Graves, a 71-year-old mining engineer, wrote to an Ohio newspaper that Abner Doubleday had invented baseball in Cooperstown, New York, in 1839. Graves had been 5 years old in 1839. Doubleday was then a cadet, a student training to be an army officer, at West Point.

The historian Victor Salvatore found that baseball does not appear in any of Doubleday's 67 surviving diaries. Other writers say he mentioned it once in his letters or diaries. No member of Spalding's committee met Graves or wrote to him. Even so, the committee's chairman, Abraham Mills, wrote in a report of December 30, 1907, that Doubleday had invented the game.

Spalding wanted to show that baseball owed nothing to any English game. Salvatore titled his article about Doubleday "The Man Who Didn't Invent Baseball."

Children, toys and the circus

The toy firm Francis, Field and Francis opened on North Second Street in Philadelphia in 1838. It was Philadelphia's first toy maker and one of the first in the country. The owners sold toys brought from France and Germany and made toys of their own. According to the Encyclopedia of Greater Philadelphia, their toy fire engine pulled by horses is believed to be the first toy made in a factory in the United States.

The city leaders of Wilmington, Delaware, banned public shows in 1825. Joshua Purdy Brown put up a canvas tent for his circus just outside the city line. With his own tent, Brown could set up his circus in any town, and the circus became a traveling show.

Lucy Larcom

Who: A girl who went to work in a mill at 11 and later wrote about it.
When and where: Beverly and Lowell, Massachusetts, 1824 to the 1840s.

"The children will have to leave school and go into the mill."

Lucy Larcom grew up in Beverly, Massachusetts, in the 1820s and 1830s. She and her friends shaped burrs from burdock plants into chairs, cradles and sofas for their dolls. They played school on the doorsteps and climbed over the wall onto the hill. She wrote that the neighborhood children, from large families, enjoyed "a little wholesome neglect." She meant that the grown-ups mostly left them alone, and that this was good for them.

The move to Lowell. Her father, a retired sea captain, died in 1832. Three years later her mother, Lois Larcom, moved the family to Lowell, a mill town. There she ran a boarding house for women who worked in the mills, cooking and keeping house for up to forty of them. Each boarder paid her $1.25 a week. In her book about her childhood, Lucy wrote down the sentence that sent her to work: "The children will have to leave school and go into the mill." She was 11.

She went to work at the Lawrence Manufacturing Company as a doffer. A doffer took full bobbins, the spools that held thread, off the spinning machines and put on empty ones, about every 45 minutes. At first she told her family, "It is just like play." Between changes, she and the other little girls played games in a corner.

In winter the working day ran from first daylight to 7:30 at night. She earned one dollar a week plus her board, and her family needed the money, so she could not go on to high school. She wrote that mill work "was not, and could not be, the right sort of life for a child."

The mill rules banned books, so she pasted poems cut from newspapers all over the side of her window seat in the mill. She and her sister Emeline published poems in two magazines, the *Operatives' Magazine* and the *Offering*. Lucy Larcom grew up to teach school and to write poems. Her book about her childhood, *A New England Girlhood*, came out in 1889.

BY 1850

If you remember one thing: Sport turned into something huge crowds came to watch, while US officials forced the Choctaw west and enslaved jockeys rode horses that other men owned.

- 1823: as many as 100,000 people watch American Eclipse beat Sir Henry on Long Island.
- 1830: Choctaw leaders sign the Treaty of Dancing Rabbit Creek and must hand over about 11 million acres.
- 1845: the Knickerbocker club of New York writes down twenty rules for baseball.

Next: After 1850 a team paid its players openly, a teacher invented basketball, and the men who ran baseball shut Black players out.

1850 to 1900

ERA 7 OF 10 · 1850 TO 1900 · ABOUT 14 MINUTES

Sport turned into a business between 1850 and 1900, and the men who ran it decided who could play. Harry Wright put together the Cincinnati Red Stockings in 1869, the first known baseball team whose players were all openly paid. Students from Rutgers and Princeton played the first college football game that same year, and a teacher named James Naismith invented basketball in 1891. Black jockeys won most of the early Kentucky Derbies, and Isaac Murphy, who was born enslaved, won it three times.

The men who ran baseball's leagues shut Black players out, first in 1867 and again in 1887.

Adults also began to organize children's play. Members of a Boston health group put piles of sand in chapel yards for small children in 1885 or 1886, and people in other cities opened playgrounds in the 1890s. Many children had no time to play. Census takers in 1900, who counted everyone in the country, found more than 1.75 million children under 15 at work.

The first paid team and new games

Harry Wright hired ten players for the Cincinnati Red Stockings in 1869 and paid them openly. He and his brother George were two of the ten. They won all 57 of their games against clubs in the national association of that time. That September they traveled to San Francisco, and they became the first club to play on both coasts. The Brooklyn Atlantics beat them in 1870, in 11 innings, while 20,000 people watched.

Football. Rutgers beat Princeton 6 to 4 at New Brunswick, New Jersey, on November 6, 1869, in the first college football game. Each side had 25 players. They could kick the ball or hit it with their fists or hands, but they could not hold it or trip anyone. About 100 people watched. The game looked more like soccer than like football today.

Basketball. James Naismith was a young teacher from Canada at a YMCA training school in Springfield, Massachusetts, in the winter of 1891. Luther Gulick told him to take a class that needed a game to play indoors in winter: "Naismith, I want you to take that class and see what you can do with it."

Naismith wrote 13 rules. He asked the janitor for two boxes to use as goals, and the janitor brought two peach baskets instead. Nine players played on each side. Accounts date the first game to December 21 or 29, 1891.

James Connolly, a 27-year-old Harvard student from Boston, won the hop, skip and jump at Athens, Greece, on April 6, 1896. That event is now called the triple jump. It was the first final of the first modern Olympic Games, so Connolly was the first modern Olympic champion. A dean at Harvard had warned him that he might not be let back into the college if he left to compete, and he went anyway.

Games at home. Milton Bradley sold a board game called *The Checkered Game of Life* in 1860. Buyers took more than 45,000 copies by 1861. Six years later Bradley printed rules for croquet, a lawn game played by hitting wooden balls with mallets.

The color line in baseball

A color line was a rule, written or unwritten, that kept Black people out. The members of the National Association of Base Ball Players, the group that ran club baseball, put in place a formal ban on Black players and Black clubs in 1867.

Some Black men still played baseball for pay after that. Bud Fowler did in 1878. The Society for American Baseball Research calls him the first acknowledged Black professional baseball player. William Edward White, who had been born enslaved, played one game for Providence in 1879. He lived as a white man and gave his race as white in the census.

Moses Fleetwood Walker and his brother Weldy played for Toledo in the American Association, then a major league, in 1884. They were the first Black major-league players known to the public as Black men.

The vote in Buffalo. Cap Anson, the captain of the Chicago club, said in 1887 that his team would not play Newark if Newark's two Black players took the field. They were Walker and the pitcher George Stovey. On July 14, 1887, Newark played Chicago without them.

Newark played in the International League, a minor league, one of the lower leagues below the major leagues. That same day, at a meeting in Buffalo, New York, the league's owners, whom the surviving accounts do not name, voted 6 to 4 to tell the league's secretary, C. D. White, to approve no more contracts with Black players. The owners of teams with only white players voted for the ban, and those with a Black player voted against it.

Two Black players stayed in the league because their contracts had been signed before the vote: Frank Grant of Buffalo and Walker, who moved to Syracuse. Walker played in the league until 1889. After him, no Black player played in the International League until Jackie Robinson joined the Montreal Royals in 1946. The owners of the major-league teams, a step above, never wrote down a rule against Black players. They kept Black players out by an unwritten agreement among themselves.

Moses Fleetwood Walker

Who: A catcher, the first major-league player known to the public as a Black man.

When and where: Ohio, Toledo, Newark and Syracuse, 1856 to 1924.

A letter warned that a mob would attack if Walker played.

Moses Fleetwood Walker, born in Mount Pleasant, Ohio, in 1856, studied at Oberlin College and then studied law at the University of Michigan. He caught for Toledo in 1883. In a game that year that did not count in the standings, Cap Anson of Chicago used a racist slur about Walker and said his team would never again play a team with Walker on it.

Walker played his first major-league game for Toledo, at Louisville, on May 1, 1884. He played 42 games and batted .263, which means he got a hit in about 26 of every 100 times at bat. He often caught the ball barehanded, with little padding. Toledo's pitcher Tony Mullane later said Walker "was the best catcher I ever worked with, but I disliked a Negro." "Negro" was a word of the time for a Black person. Mullane said he threw his pitches without looking at the signals Walker gave him.

The letter. Toledo's manager got a letter in September 1884 warning that a mob would attack if Walker played in Richmond, Virginia. David Zang's account of Walker's life does not name the men who wrote it. The men who ran the Toledo team released Walker before the trip.

Syracuse, 1891. A group of white men attacked Walker outside a saloon in Syracuse on April 9, 1891. One of them, Patrick Murray, threw a stone that hit Walker in the head. Walker stabbed Murray with a pocketknife and killed him. On June 3, 1891, an all-white jury found Walker not guilty of second-degree murder, which means killing someone on purpose but without planning it ahead.

Walker was found guilty of robbing the mail in September 1898, and he spent a year in jail. He later ran an opera house, a theater, in Cadiz, Ohio. In 1908 he and his brother Weldy published *Our Home Colony*, a book

arguing that Black Americans should move to Africa. He died of pneumonia, an infection of the lungs, in Cleveland on May 11, 1924, at 67.

Black jockeys

Oliver Lewis rode the horse Aristides to win the first Kentucky Derby on May 17, 1875. The horse's trainer, Ansel Williamson, had been born enslaved in 1806. Black jockeys won 15 of the first 28 Kentucky Derbies.

Isaac Murphy

Who: A jockey born into slavery who won the Kentucky Derby three times.

When and where: Kentucky and the Eastern racetracks, 1861 to 1896.

By his own count he won 628 of his 1,412 races.

Isaac Burns Murphy was born into slavery in Clark County, Kentucky, on January 6, 1861. His father, Jerry Burns, joined the Union Army in 1864 and died that fall in the hospital at Camp Nelson.

Murphy won the Kentucky Derby in 1884, 1890 and 1891. He was the first jockey to win it three times and the first to win it two years in a row. By his own count, he won 628 of the 1,412 races he rode, about 44 of every 100.

The racing writer Terry Conway writes that Murphy was said to earn $15,000 to $20,000 a year, at a time when the average family of four earned about $1,200. He and his wife, Lucy, bought a 10-room house in Lexington for $10,000 in 1887.

The fall at Monmouth. Murphy rode the favorite, Firenze, in a race called the Monmouth Handicap in 1890. He swayed in the saddle and fell off after the finish. The race officials, called stewards, said he was drunk and barred him from riding for a time.

> Murphy said he had been drugged. One account says he blamed a sudden loss of weight. Someone had bet heavily against his horse, and that betting made some people suspect a crime. The accounts of the race do not say who placed those bets.
>
> Murphy died in Lexington in February 1896, at 35. Accounts give the cause as pneumonia or heart failure. How much money he left is not known. After he died, Lucy Murphy could not pay back a $5,500 loan from Gus Luigart, and she lost the house in a lawsuit. It was sold at auction in 1900.
>
> He was buried in African Cemetery No. 2 in Lexington. In later years the grave had no marker. He became the first jockey named to the Racing Hall of Fame in 1955. His body was moved in 1967 to a site near the grave of the racehorse Man o' War, and later to the Kentucky Horse Park.

Football deaths

Football players in the 1890s wore no helmets, no mouthpieces and no padding. Reporters at a New York newspaper, the *World*, counted at least eight deaths and more than 200 serious injuries in the 1897 season. Their list included broken bones, fractured skulls, twisted necks and crushed chests.

Richard Von Gammon, a 17-year-old University of Georgia player, dove into a pile of players to make a tackle against the University of Virginia in Atlanta on October 30, 1897. He lay still on the field and died early the next morning of a concussion, which is an injury to the brain from a hard blow to the head.

A mother's letter. Georgia's lawmakers voted to ban football at the state's colleges. Von Gammon's mother, Rosalind Burns Gammon, wrote to her representative. She asked that her son's death not be used as an argument against the sport. Governor William Atkinson vetoed the ban on December 7, 1897, which means he refused to sign it and it did not become law.

A Tennessee newspaper, the *Knoxville Tribune*, printed a full-page editorial, an article giving the paper's own opinion, calling for a ban on October 30, 1898. The editorial listed three deaths and gave no names. Two players' heads

struck together and one of them died. A player at the bottom of a pile had his neck broken. A player kicked in the head by a heavy shoe died three days later.

Women on bicycles

Frances Willard learned to ride a bicycle at 53. In 1895 she published a book about it. As a girl in the country she had almost no toys except the ones she made herself, such as a wooden plow. She practiced on the bicycle for three months. She wrote that women who rode "must, when riding, dress more rationally," which meant in clothes that made sense for the job.

Susan B. Anthony told the reporter Nellie Bly, in an article printed on February 2, 1896, that the bicycle "has done more to emancipate women," or set them free, "than anything else in the world." Anthony said that bloomers, which were loose pants gathered at the ankle, or very short skirts were the right clothes for riding.

The first playgrounds

Dr. Marie Zakrzewska wrote to a Boston health group, the Massachusetts Emergency and Hygiene Association. She told its leaders that in Berlin's public parks, children rich and poor dug and played in heaps of sand.

Which year? Accounts give two different years for the first sand in Boston. Joseph Lee, who wrote a book about helping the poor in 1902, gives 1886. That year, he wrote, members of the association put three piles of yellow sand in the yards of a children's mission, a church charity for children, and of two chapels, which are small churches.

A 1925 history by the Playground and Recreation Association of America says that a large sand pile was put in the yard of the Children's Mission on Parmenter Street in 1885.

The association had ten sand piles by 1887, and about 400 children came each day. The children's favorite games in the sand were "house" and "funeral." By 1900 members of the association ran 21 playgrounds, 20 of them in school yards.

Other cities. People in other cities opened playgrounds in the 1890s. Donors in Philadelphia started two summer playgrounds in 1893. A playground opened at Hull-House in Chicago in 1894, with swings, a may-pole, a tall pole that children danced around holding ribbons, a sand pile and benches. A kindergarten teacher and a policeman ran it, and the policeman usually captained one of the baseball teams.

The members of a New York City committee wrote in an 1897 report that New York "has as yet not a single municipal playground," meaning none run by the city. Jacob Riis was the committee's secretary.

People began to build amusement rides in these years too. LaMarcus Thompson opened a roller coaster, the Switchback Railway, at Coney Island in Brooklyn in 1884. A ride cost 5 cents, and the car went just over 6 miles an hour.

The children who had no time to play

Census takers counted working children for the first time in 1870. They found more than 750,000 workers under 15. That count left out children who worked on their own family's farm or in its shop. The number more than doubled by 1900, to more than 1.75 million children under 15 with jobs. After 1880 most working children aged 10 to 14 worked somewhere other than a farm.

One child in eight had a job in 1870, and more than one in five did by 1900. The census counts do not name the employers. About 25,000 of the nearly 100,000 workers in the cloth mills of the South in 1900 were children under 16. Children in the mills worked 12-hour days, six days a week.

Cherokee ball play, 1890

The researcher James Mooney described the Eastern Cherokee ball game in North Carolina in 1890. In the fall a game took place somewhere on the reservation, the land where the Eastern Cherokee lived, at least every other week. Each player carried two sticks just under 3 feet long, each with a net of twisted squirrel skin or hemp. For 7 to 28 days before a game, a player did not eat rabbit or frog or hot or salted food, and did not touch a woman.

The scratching. Before a game an old man named Standing Water scratched each player with a comb made of seven sharp splinters of turkey leg bone. He pressed the teeth into the skin from shoulder to elbow and from elbow to wrist, and down each leg. He also cut an X on each player's chest and back. The scratches added up to nearly three hundred cuts.

Mooney wrote that the cuts were shallow but painful, and that the players took them willingly as part of the ritual they believed would help them win. No order banning the Cherokee or the Choctaw ball game in these years is on record. The games drew large crowds, including local white people.

BY 1900

If you remember one thing: Paid teams, leagues and new games turned play into a business in these fifty years, and the men who ran baseball used that power to shut Black players out.

In 1850: no baseball team paid all its players, and nobody had played basketball or a college football game.

In 1900: paid baseball teams played in leagues closed to Black players, Isaac Murphy had won the Kentucky Derby three times, and more than 1.75 million children under 15 were at work.

Next: After 1900 adults filled children's free time with clubs, troops and leagues, and football deaths brought college coaches to the White House.

1900 to 1950

ERA 8 OF 10 · 1900 TO 1950 · ABOUT 23 MINUTES

Clubs, troops and leagues run by adults filled more and more of children's free time in the first half of the 1900s. Adults set up 4-H clubs in 1902, the Boy Scouts in 1910, the Girl Scouts in 1912 and Little League baseball in 1939. Children also spent many free hours watching and listening, at Saturday movies, beside the radio and over comic books. In Muncie, Indiana, in the 1930s, at least half of the children at Saturday movies stayed five or six hours.

At least 11 people died of football injuries in 1905, and President Theodore Roosevelt called coaches to the White House. The white owners of the major leagues, the top professional baseball teams, kept Black players out until Jackie Robinson played for Brooklyn in 1947. Black players had their own leagues from 1920. In the 1920s and 1930s, white officials and white swimmers in many Northern cities separated the public pools by race and kept Black children out of the pools where other children learned to swim.

Clubs and troops for children

A. B. Graham started a youth program in Clark County, Ohio, in 1902. Its first clubs were a Tomato Club and a Corn Growing Club, and they are counted among the beginnings of 4-H. William D. Boyce, a Chicago publisher, filed the legal papers that made the Boy Scouts of America an official organization on February 8, 1910.

Juliette Gordon Low signed up 18 girls in Savannah, Georgia, on March 12, 1912, as the first troop of what she called Girl Guides. The name changed to Girl Scouts in 1913. The Savannah girls started a basketball league and went camping.

Choctaw and Chickasaw ball play, 1903

Choctaw and Chickasaw people met near Allen, in Indian Territory, each spring and fall to race horses and play the stick-and-ball game. Allen is now in Oklahoma. In the spring of 1903 about 300 people came and bet on a game. A player hit a Choctaw player with a ball stick, and a fight spread through the crowd. The Choctaw Nation's mounted police, called Lighthorsemen, and US marshals, who were federal police officers, stopped it. After that fight the twice-yearly games between the two nations ended.

How many died playing football in 1905?

At least 11 people died of football injuries in the 1905 season, and counts of all the deaths tied to football that year run from 18 to 25. The firmest count comes from the writer Aaron Gordon, who later checked the newspaper reports. He found about 20 deaths tied to football, 11 of them directly from football injuries, and the cause of the others was not clear.

The NCAA's published history of its own beginnings states 18 deaths and 149 serious injuries. Reporters at the *Chicago Tribune* counted 19 deaths, and reporters at another newspaper, the *Cincinnati Commercial Tribune*, counted 25.

The dead included college players and high-school boys. Harold Moore, 19, a Union College player, was knocked out in a game against New York University. He died six hours later of bleeding inside his brain. In Rockville, Indiana, a broken rib pierced the heart of Carl Osborne, a high-school player, and killed him at once. Robert Brown, 16, of Sedalia, Missouri, was paralyzed by a neck injury and died without waking.

The White House meeting. President Theodore Roosevelt met coaches and graduates from Harvard, Yale and Princeton at the White House on October 9, 1905. They signed a statement against rough play. That December, Henry MacCracken, the head of New York University, called a meeting of college leaders for December 9.

At a second meeting, on December 28, the men from 62 colleges formed a group. Its members renamed it the National Collegiate Athletic Association, the NCAA, in 1910. The rules of the game changed for 1906. A team now had to gain 10 yards, not 5, to keep the ball. Jumping over the line was banned, and the forward pass was allowed.

Jack Johnson

Who: A Black heavyweight boxing champion. After he beat a white former champion in 1910, white crowds attacked Black people across the country.

When and where: Galveston, Texas, Sydney, Reno, Chicago and Leavenworth, 1878 to 1946.

His opponent said he fought to prove "a white man is better."

Jack Johnson, born in Galveston, Texas, in 1878, beat Tommy Burns in Sydney, Australia, on December 26, 1908. That made him the heavyweight boxing champion of the world, the champion of the boxers in the heaviest weight class.

He fought James Jeffries, a white former champion, in Reno, Nevada, on July 4, 1910. Before the fight Jeffries said he was fighting "for the sole purpose of proving that a white man is better than a Negro." "Negro" was a word of the time for a Black person. Jeffries's own helpers stopped the fight in the 15th round, and Johnson won.

That night. White crowds attacked Black people in more than 25 states. Between 11 and 26 people were killed, and hundreds were hurt. In Uvalda, Georgia, white gunmen fired into a work camp and killed three Black workers. In New York City a crowd clubbed a Black man to death. In Norfolk, Virginia, more than 300 white Navy sailors hunted Black people in the streets. The records do not name the people who were killed or the people who killed them.

Officials in many states and cities banned showing the film of the fight. The members of Congress banned carrying boxing films from one state to another in 1912. The ban lasted until 1940.

The arrest. Johnson opened a nightclub in Chicago, the Café de Champion, in July 1912, and Black and white customers mixed there. He was arrested on October 18, 1912, under the Mann Act, a federal law of 1910. It made it a crime to take "any woman or girl" across state lines "for

the purpose of prostitution or debauchery, or for any other immoral purpose." Prostitution means selling sex, and debauchery meant wild, immoral living.

Federal prosecutors, the lawyers who bring charges for the government, used the law against Johnson because he traveled with white women. An all-white jury, a group of citizens who decide whether a person is guilty, found him guilty in June 1913. Judge Kenesaw Mountain Landis ordered him to spend a year and a day in prison. Johnson fled to Canada and Europe and stayed away for seven years. He gave himself up on July 20, 1920, and served his sentence at Leavenworth prison in Kansas.

He died in a car crash near Franklinton, North Carolina, on June 10, 1946, after the staff of a diner that served only white people refused to serve him. President Donald Trump pardoned him on May 24, 2018. A pardon is an official act by a president or governor that forgives a crime and ends its punishment.

Black jockeys forced out

Jimmy Winkfield of Kentucky won the Kentucky Derby, a horse race held each spring in Louisville, in 1901 and 1902. No Black jockey has won it since. He left to ride in Russia in 1904, and later he rode in France. A Louisville newspaper, the *Courier-Journal*, printed the headline "Negro Jockeys Now Extinct" on January 21, 1912.

How it happened. Two economists, scholars who study money and business, Michael Leeds and Hugh Rockoff, studied why Black jockeys were driven out of American racing. They found that white jockeys attacked Black jockeys during races. The white riders boxed them in, ran them into the rail and hit them with their whips. Owners then stopped hiring Black jockeys, so that their horses would not be hurt. Henry King rode in the 1921 Kentucky Derby. No Black jockey rode in it again until 2000.

Jim Thorpe and his medals

Jim Thorpe was a Sac and Fox athlete, born in 1887 near Prague, in what is now Oklahoma. He played football at the Carlisle Indian Industrial School for the coach Glenn "Pop" Warner. At the 1912 Olympic Games in Stockholm, Sweden, he won the pentathlon, a contest of five track and field events, and the decathlon, a contest of ten. In the decathlon he beat the second-place man by 688 points.

The newspaper report. Roy Johnson, a reporter for the *Worcester Telegram* in Massachusetts, reported in January 1913 that Thorpe had been paid to play minor-league baseball, in the lower leagues below the major leagues, in North Carolina in 1909 and 1910. Accounts of his pay run from $2 a game to $35 a week. Olympic athletes then had to be amateurs, people who were never paid to play a sport.

Thorpe wrote to James E. Sullivan of the Amateur Athletic Union that he "was simply an Indian schoolboy and did not know all about such things." James Ring Adams wrote in the magazine of the National Museum of the American Indian that Warner and Moses Friedman, the head of the Carlisle school, may have written the letter for him.

Sullivan and the other officials of the Amateur Athletic Union took away Thorpe's amateur standing. The members of the International Olympic Committee struck out his wins. Their own rules allowed a complaint only within 30 days of the Games, and the newspaper report came about six months later. Warner took the medals from Thorpe's house and sent them back to Stockholm. Hugo Wieslander, who had come second in the decathlon, refused the gold medal.

Thorpe played major-league baseball for the New York Giants and other teams from 1913 to 1915 and from 1917 to 1919. He played pro football for the Canton Bulldogs for $250 a game. In 1920 he became the first president of the league that later took the name National Football League. In his last years he worked as a film extra, a laborer and a doorman. He died of heart failure in 1953, at 65.

The medals restored. A researcher, Florence Ridlon, found the 30-day rule in the 1912 rule book in 1982. On October 13 of that year, the leaders of the International Olympic Committee named Thorpe a co-champion, sharing each title with the second-place man: Hugo Wieslander in the decathlon and Ferdinand Bie in the pentathlon. On July 15, 2022, they named him the only champion of both events.

Baseball and the Negro Leagues

The Boston Americans beat the Pittsburgh Pirates 5 games to 3 in the first modern World Series, in October 1903. The owners of the two teams, Henry Killilea of Boston and Barney Dreyfuss of Pittsburgh, arranged it. About 100,000 people watched the eight games. There was no series in 1904, because the owners of the New York Giants refused to play.

BY 1920: Football had new rules after the deaths of 1905, Black jockeys were nearly gone from American racing, and the owners of Black baseball teams were about to start a league of their own.

A league of their own. Rube Foster, the owner and manager of the Chicago American Giants, met other owners of Black teams at a YMCA in Kansas City on February 13, 1920. Together they started the Negro National League. It had eight teams in its first year. The league closed in 1931, during the Great Depression, when millions of people lost their jobs. Teams in other Negro Leagues kept playing until 1948.

Major League Baseball's leaders began counting seven Negro Leagues of 1920 to 1948 as major leagues in December 2020. That added about 3,400 players to the major-league record. On May 28, 2024, they added the Negro League players' numbers to the official record books.

Babe Ruth

Who: A pitcher and home-run hitter for Boston and New York.
When and where: Baltimore, Boston and New York, 1895 to 1948.

His school's records called him "incorrigible," a child no one can control.

George Herman Ruth, born in Baltimore in 1895, was seven when he entered St. Mary's Industrial School for Boys on June 13, 1902. It was a reform school, a school meant to correct boys' behavior, and an orphanage, a home for children without parents. He spent most of the next 12 years there. A teacher there, Brother Matthias, taught him baseball.

Jacques Kelly, a writer for the *Baltimore Sun*, wrote in 1995 that Ruth's parents decided to place him at the school. The school's records listed the boy as "incorrigible," Kelly wrote. Writers have argued about the parents' reasons for many years.

The sale. Ruth pitched for the Boston Red Sox from 1914 to 1919. The owner of the Boston team, Harry Frazee, sold him to the New York Yankees for $100,000. The sale was made in December 1919 and announced in January 1920.

Ruth hit 54 home runs in 1920 and 60 in 1927, and he finished his career with 714. In 1936 he was one of the first five players chosen for the Baseball Hall of Fame. Cancer in the back of his nose and throat killed him on August 16, 1948, at 53.

Movies, radio and comic books

Children in the 1920s and 1930s spent many of their free hours at the movies. Walt Disney started Mickey Mouse Clubs that met at Saturday matinees, the afternoon showings, in 1929. Richard Butsch gathered studies of movie audiences from the 1930s in a study printed in 2001. In Muncie, Indiana, at least half of the children at Saturday matinees stayed five or six hours. In a working-class part of New Haven, Connecticut, more than half the children went twice a week or more.

Researchers of the time found that children were choosing movies over outdoor games. A child in the Bronx, in New York City, said that when the film broke, the children booed, stamped their feet and often got into fist fights.

Worried adults. The mayor of Chelsea, Massachusetts, closed a theater in 1930 because of how children in the audience behaved. Researchers paid by the Payne Fund, a private group that gave money to good causes, carried out 13 studies of how watching movies changed children's behavior from 1929 to 1933.

Little Orphan Annie, a radio show, began in Chicago in 1930 and was broadcast across the country from April 6, 1931. It had about 6 million listeners of all ages before it went off the air in 1942.

Comic books. Superman first appeared in a comic book, *Action Comics* number 1, on April 18, 1938. A comic book cost 10 cents. A survey for the publishers in 1944 found that 95 of every 100 boys and 91 of every 100 girls aged 6 to 11 read comic books, about 12 a month.

Sterling North, a newspaper book editor in Chicago, called comic books "badly drawn, badly written and badly printed" in 1940. He wrote that they strained children's eyes and nerves. Comic books were burned in the 1940s. The Comic Book Legal Defense Fund's history of these burnings does not name who burned them.

Lizzie Magie patented a board game called *The Landlord's Game* in 1904. Monopoly grew out of it, and Parker Brothers began selling Monopoly on November 5, 1935.

BY 1940: Children spent hours each week at the movies and with comic books, Babe Didrikson and Jesse Owens had won Olympic gold, and the owners of the major leagues still kept every Black player out.

Babe Didrikson Zaharias

Who: A track star, basketball player and golfer who won Olympic gold and helped start women's professional golf.

When and where: Texas, Los Angeles and golf courses in the United States and Britain, 1911 to 1956.

In three hours she won five events and tied for a sixth.

Mildred Didrikson was born in Port Arthur, Texas, in 1911, the sixth of seven children of Norwegian immigrants. She became known as Babe. Managers at the Employers Casualty Insurance Company in Dallas hired her in 1930 to do office work and to play on the company's women's basketball team, the Golden Cyclones.

At the national track championships of 1932 she was the company's whole team. In three hours she won five events and tied for first in a sixth. She scored more points than any other team.

The Olympics. At the 1932 Olympic Games in Los Angeles she won the javelin and the 80-meter hurdles. In the high jump she won second place, because the officials ruled that going over the bar head first was a foul.

Later that year the Amateur Athletic Union's officials banned her from amateur sport, the contests for unpaid athletes, because her name appeared in an advertisement for a car. Her family needed money, so she played for pay. She toured with a men's traveling baseball team, the House of David.

Golf. She took up golf. Officials of the US Golf Association barred her from amateur golf in 1935, after complaints from members of the Texas Women's Golf Association. She married the wrestler George Zaharias in 1938. She helped found the Ladies Professional Golf Association, in 1948

or 1950 by different accounts. She won the US Women's Open in 1948, 1950 and 1954. From 1950 she lived with the golfer Betty Dodd, and the two hid that they were a couple.

She had cancer of the bowel in April 1953. Surgeons removed the cancer and gave her a colostomy. In that operation, surgeons bring the end of the large bowel out through an opening cut in the belly, so that body waste collects in a bag outside the body. Fourteen weeks later she was playing golf again, and in 1954 she won the US Women's Open. The cancer came back in 1955, and she died of it on September 27, 1956, at 45.

Jesse Owens

Who: A runner and jumper who won four gold medals at the Olympic Games in Nazi Germany.
When and where: Alabama, Cleveland and Berlin, 1913 to 1980.
Movie: *Race* (2016), a drama film about Owens.

"I had four gold medals, but you can't eat four gold medals."

James Cleveland Owens was born in Oakville, Alabama, in 1913, the son of a sharecropper, a farmer who paid his rent with part of his crop. His family moved to Cleveland, Ohio, when he was nine. A teacher heard his initials, "J.C.," as "Jesse."

At a college track meet in Ann Arbor, Michigan, on May 25, 1935, he set five world records and tied a sixth in about 45 minutes.

Berlin. Owens won the 100 meters, the long jump, the 200 meters and the relay at the Olympic Games in Berlin in August 1936. Adolf Hitler's Nazi government ruled Germany then. The Nazis taught that white people they called "Aryans" were a superior race.

Back home, Owens had to ride the freight elevator, the one used for goods, to reach a party in his honor at the Waldorf Astoria hotel in New York. Accounts do not name who sent him there. President Franklin

> Roosevelt never invited him to the White House. Owens said later, "Hitler didn't snub me. It was our president who snubbed me. The president didn't even send me a telegram."
>
> When Owens refused to finish a tour of Europe, Avery Brundage and other officials of the Amateur Athletic Union barred him from amateur sport. He earned money by racing against horses. Later he went bankrupt, which means he could not pay his debts. He said, "I had four gold medals, but you can't eat four gold medals." He received the Presidential Medal of Freedom in 1976. Lung cancer killed him in Tucson, Arizona, on March 31, 1980.

Who was kept out of the pools

Jobless workers paid by a federal program, the Works Progress Administration, built about 6,000 athletic fields and playgrounds and 770 new swimming pools from 1935 to 1943. Each year tens of millions of Americans swam in city pools.

Jeff Wiltse, who wrote a book about public pools called *Contested Waters*, studied who was allowed in. Many cities in the North began letting men and women swim in the same public pools in the 1920s and 1930s. Wiltse found that white officials and white swimmers in those cities then separated the pools by race.

He wrote that "most northern whites did not want black men to have the opportunity to interact with white women" at the pools. Black children in those cities could not use the pools where other children learned to swim.

Fairground Park, 1949. John O'Toole, the head of public welfare in St. Louis, opened the Fairground Park pool to Black swimmers on June 21, 1949. He said, "They are taxpayers and citizens, too." About 30 to 40 Black people came.

Crowds of white young men gathered with baseball bats and bricks and beat Black people as they left the pool. The crowd grew to thousands. Twelve people were hurt, and police arrested seven, three white and four Black. The

published accounts of the attack do not name the attackers. Afterward, so few people came to the pool that it lost money, and it was closed.

Carl Stotz and the first Little League boys

Who: A Williamsport man who organized baseball for his nephews and the neighborhood boys, and two of the boys who played.
When and where: Williamsport, Pennsylvania, 1938 to 1955.

Three local businesses paid $30 each for the first uniforms.

Carl Stotz, 28, tried out rules and field sizes in the summer of 1938 with his nephews, Jimmy and Major Gehron, and other boys from his neighborhood in Williamsport, Pennsylvania. He had no children of his own.

The first season. Stotz, George Bebble and Bert Bebble each managed a team in 1939. Three local businesses paid $30 each for uniforms, and the teams took their names: Lycoming Dairy, Lundy Lumber and Jumbo Pretzel. Thirty boys played. The bases were 60 feet apart, and the pitcher threw from 40 feet. On June 6, 1939, Lundy Lumber beat Lycoming Dairy 23 to 8 on a vacant lot. It was the first Little League game.

Two boys on the Lycoming Dairy team were Bill Bair and Al "Sonny" Yearick. Yearick later became the first Little Leaguer to play baseball for pay, in the minor leagues. Both men died in January 2020. A volunteer remembered that they talked about Stotz and about how Little League had helped them stay out of trouble.

Only boys could play. Stotz said in 1973, "When we started the Little League, the idea of a girl playing baseball, even with other girls, was simply unthinkable." He left Little League's national company in November 1955 over "commercialization and central control," which meant that he objected to the league being run as a business from one head office.

Women's professional baseball, 1943 to 1954

More than 600 women played in the All-American Girls Professional Baseball League from 1943 to 1954. Philip Wrigley, the owner of the Chicago Cubs, started it. The league's rules required players to wear skirts and lipstick and banned pants. During spring training they went to evening classes at a charm school, a school that taught manners and how to look and dress. No Black women were recruited or hired. The accounts of the league do not name who made that choice.

Josh Gibson

Who: A catcher and home-run hitter whom the white major-league owners never signed.

When and where: Georgia and Pittsburgh, 1911 to 1947.

Movie: *Soul of the Game* (1996), a television drama about several Negro Leagues players, with Mykelti Williamson as Gibson.

His Hall of Fame plaque says he hit "almost 800" home runs.

The catcher for the Homestead Grays, a Black team near Pittsburgh, was hurt in a game on July 25, 1930. The Grays' owner, Cum Posey, called Josh Gibson, 18, out of the stands to catch. Gibson had been born in Buena Vista, Georgia, in 1911, and his family had moved to Pittsburgh in 1926. There he worked in an air-brake factory and at a steel company.

Gibson played for the Grays and for the Pittsburgh Crawfords. The Crawfords' owner, Gus Greenlee, sold him back to the Grays in 1937 for $2,500 and two players. People called him "the Black Babe Ruth."

Never in the majors. The white owners of the major-league teams kept Black players out, so Gibson never played in their leagues. Doctors told him in 1943 that he had a brain tumor. He died in Pittsburgh on January 20, 1947, at 35, less than three months before Jackie Robinson played his first game for Brooklyn.

Since 2024 the major-league record books have included the Negro Leagues. A batting average shows how often a player got a hit: .466 means about 47 hits in every 100 times at bat. Gibson's .466 for 1943 is the highest for a single season in those records, and his career average of .372 is also the highest.

The plaque. Gibson was named to the Baseball Hall of Fame in 1972. His plaque there says he hit "almost 800" home runs.

Jackie Robinson

Who: The first Black player in the white major leagues since the 1880s.

When and where: Georgia, California, Kansas City and Brooklyn, 1919 to 1972.

Movie: *42* (2013), a drama film with Chadwick Boseman as Robinson. An earlier film, *The Jackie Robinson Story* (1950), starred Robinson as himself.

"I need a Negro player with guts enough not to fight back."

Jackie Robinson was born in Cairo, Georgia, in 1919, the youngest of five children of sharecroppers. His father left, and in 1920 the family moved to Pasadena, California. At the University of California, Los Angeles, he was the first athlete to earn varsity letters, awards for playing on the college's top teams, in four sports.

As a soldier at Fort Hood, Texas, on July 6, 1944, he refused to move to the back of an Army bus. Officers charged him with disobeying orders. A panel of nine white officers found him not guilty.

The meeting with Rickey. Robinson played for the Kansas City Monarchs of the Negro Leagues in 1945. Branch Rickey, the head of the Brooklyn Dodgers, met him on August 28, 1945. Robinson asked, "Are you looking for a Negro who is afraid to fight back?" Rickey answered, "I need a Negro player with guts enough not to fight back." Rickey signed him for $600 a month. Robinson played the 1946 season for the Montreal Royals, a minor-league team.

He played his first game for Brooklyn on April 15, 1947, before 26,623 people. More than 14,000 of them were Black. During that first season, Ben Chapman, the manager of the Philadelphia team, told his players to "call Robinson everything and anything they wanted to." Chapman and his players shouted racial slurs, insulting names for Black people, at him. Chapman told him to "go back to the cotton fields." They kept it up until officials in the office of Happy Chandler, the commissioner who ran baseball, ordered them to stop.

Threats. Robinson got letters, signed by no one, telling him to leave baseball. Arthur Mann, who worked for Rickey, gave two of them to the police. A sportswriter, Stanley Woodward, reported that St. Louis players had threatened to strike rather than play against him. The St. Louis owners and managers denied it. The National League's president, Ford Frick, said he would suspend any player who went on strike. A St. Louis player, Enos Slaughter, cut Robinson's leg with the metal spikes on his shoe and left a seven-inch gash.

Robinson was named the first Rookie of the Year, the best first-year player, in 1947 and Most Valuable Player in 1949. He died on October 24, 1972, at 53. Officials of Major League Baseball retired his number, 42, for every team in 1997, so that new players could no longer be given it.

BY 1950

If you remember one thing: Adults filled children's free time with clubs, troops and leagues, and the white owners of the major leagues kept Black players out until 1947.

One number: 26,623 people watched Jackie Robinson's first game for Brooklyn, on April 15, 1947, and more than 14,000 of them were Black.

Next: After 1950 girls fought to get into Little League, television carried the Super Bowl into homes, and children began playing video games.

1950 to 2000

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Girls and Black children pushed their way into games that had kept them out in the second half of the 1900s, while more of children's afternoons went to lessons and organized sports. Fewer children walked or biked to school. Diaries of how children spent their days in 1981 and 1997 showed more time in organized sports and lessons, and less time in free play and watching television. Children could play video games for a quarter in arcades from 1972, and on game machines at home from 1975.

Little League officials kept girls out from 1951 until 1974. After Title IX became law in 1972, the number of girls playing high-school sports rose from about 294,000 to about 2.7 million by 1999.

Boxing officials took Muhammad Ali's title after he refused to be drafted in 1967. US Olympic officials suspended Tommie Smith and John Carlos and sent them out of the Olympic Village, where the athletes lived, after they protested on the medal stand in 1968. Two television networks, CBS and NBC, showed the first Super Bowl in 1967 to about 51 million viewers.

Kathryn "Tubby" Johnston

Who: A girl who played Little League baseball under a boy's name. Little League officials then banned girls.

When and where: Corning, New York, 1950.

"I ran into my brother's room, got a pair of his slacks."

Kathryn Johnston played baseball with boys on a sandlot near her home in Corning, New York, in the spring of 1950. She wanted to play on a Little League team, and only boys could play on Little League teams.

She described what she did next: "I ran into my brother's room, got a pair of his slacks, put on a baseball cap and signed up as Tubby Johnston." She took the name Tubby from a comic strip, *Little Lulu*. She tucked her hair under her cap and went to the tryouts with her brother.

She made the King's Dairy team and played first base. Then she told the coach she was a girl. He said she was good enough to make the team, so she could stay. More people came to the games once word got out.

The Tubby Rule. Little League officials made a rule in 1951 that no girls could play. Johnston said later, "It's known as the Tubby Rule because I was the reason why they put that rule in." She wrote to Little League's headquarters in 1974, after the ban ended. The officials there checked her story and wrote back that she was the first girl to play in Little League.

Who could play Little League

Little League officials also stopped a team of Black boys from playing on for a title. The Cannon Street YMCA in Charleston, South Carolina, entered its all-star team of 14 Black boys in the city Little League tournament in 1955. The adults who ran the 61 other teams, all of them white, pulled every team out rather than play them. Accounts do not name those adults. Cannon Street won the city and state titles because no one would play them.

At the regional tournament in Rome, Georgia, officials ruled that Cannon Street could not go on. The team had won without playing, and the officials said a team that had not won its games on the field could not advance. Little League's president agreed with the ruling. The managers of the white leagues left Little League and started their own league for white teams only.

Little League officials invited the Cannon Street boys to watch the Little League World Series in Williamsport. The crowd there chanted "Let them play." In 2023 one of the players, John Rivers, looked back on what had happened to the team: "So we didn't talk about it. So it tells me how painful it was."

Maria Pepe. Maria Pepe, 12, pitched three games for a Little League team in Hoboken, New Jersey, in 1972. Little League officials told the Hoboken league to drop her or lose its right to be part of Little League, and the Hoboken league's officials dropped her. Robert Stirrat, a Little League vice president, said, "as far as we're concerned, the incident is closed."

Members of the National Organization for Women, a women's rights group, took Pepe's case to New Jersey's civil rights office. Sylvia Pressler, the officer hearing the case, ruled in 1973 that keeping girls out broke the state's law. She called Little League "as American as the hot dog and apple pie," and wrote, "There is no reason why that part of Americana should be withheld from girls."

Little League's officials asked a higher court to overturn the ruling, and they lost in 1974. Little League held a charter from Congress, a paper that gave it official standing. President Gerald Ford signed a law in December 1974 that

changed the charter to let girls play. That year nearly 30,000 girls signed up for Little League's new softball program.

Who could swim

Some city officials closed their public pools rather than let Black residents swim in them. Officials in Jackson, Mississippi, closed four public pools in 1963. They gave up a fifth pool, and the YMCA then ran it for white people only. Five Supreme Court justices, the judges of the nation's highest court, ruled on June 14, 1971, in *Palmer v. Thompson*, that closing the pools to everyone was lawful. Four justices disagreed.

The Monson Motor Lodge. Black and white protesters jumped into the pool of the Monson Motor Lodge in St. Augustine, Florida, on June 18, 1964, in a protest against keeping Black and white swimmers apart. The motel's manager, James Brock, poured muriatic acid, a strong acid used to clean pools, into the water. A photograph of him doing it ran in newspapers across the country.

Jeff Wiltse, the historian of public pools, found that when Black swimmers won equal use of city pools, white swimmers "generally abandoned them for private pools," which means most of them left. He wrote that city officials then treated public pools as less important.

Television and the Super Bowl

The Green Bay Packers beat the Kansas City Chiefs 35 to 10 in Los Angeles on January 15, 1967. It was the first championship game between the two professional football leagues of the time. Tickets cost $6 to $12. Two television networks, CBS and NBC, both showed the game, to about 51 million viewers. A 30-second advertisement cost $42,000.

From the third game, in 1969, the official name was the Super Bowl. Lamar Hunt, who founded one of the two leagues, took the name from his children's toy, a Super Ball.

The 1993 game. NFL officials took the 1993 Super Bowl away from Arizona because its voters had said no to a holiday for Martin Luther King Jr. in November 1990. The league's officials moved the game to Pasadena, California. *PHOENIX* magazine says the game was to be played in Tempe, next to Phoenix, and the Pima County Public Library says Phoenix. People there had expected about $200 million from the game. Arizona hosted the Super Bowl in 1996. Chapter 37 tells the fight over the King holiday.

Muhammad Ali

Who: The heavyweight boxing champion whose title was taken from him when he refused to be drafted into the Army.

When and where: Louisville, Rome, Houston, Atlanta and Zaire, 1942 to 2016.

"I ain't got no quarrel with them Viet Cong."

Cassius Clay was born in Louisville, Kentucky, in 1942. When he was 12, someone stole his bicycle. A police officer, Joe Martin, taught boxing, and he got Clay started in the sport. Clay won an Olympic gold medal in Rome in 1960. On February 25, 1964, at 22, he beat Sonny Liston for the heavyweight title. He then joined the Nation of Islam, a Black Muslim religious group, and took the name Muhammad Ali.

Houston, 1967. Ali went to Houston to be drafted, which means ordered by law to join the Army, on April 28, 1967, during the Vietnam War. He had already said, "I ain't got no quarrel with them Viet Cong," the fighters the US Army was fighting in Vietnam. He refused three times to step forward when his name was called.

That same day the members of the New York State Athletic Commission suspended his boxing license, and officials of the World Boxing Association took away his title. Officials of other boxing commissions did the same. In a federal court in Houston on June 20, 1967, a jury of six white men and six white women found him guilty of draft evasion, which means refusing to serve when drafted. The judge sentenced him to five years in prison and a $10,000 fine.

Back in the ring. He could not box for more than three years. He fought again on October 26, 1970, in Atlanta. On June 28, 1971, all eight Supreme Court justices who heard his appeal overturned his conviction. On October 30, 1974, he beat George Foreman in Zaire, in Africa, and won the title back.

> Ali retired in 1981. In 1984 he said he had Parkinson's syndrome, a disease of the brain that makes the body shake and move slowly. He died on June 3, 2016.

The 1968 Olympic protest

Tommie Smith won the 200 meters at the Olympic Games in Mexico City on October 16, 1968, in a world-record 19.83 seconds. John Carlos came third. Peter Norman of Australia came second. During the US national anthem, Smith and Carlos each raised a fist in a black glove. They stood in black socks without shoes to stand for poor people. All three men wore the badge of a protest group, the Olympic Project for Human Rights.

Smith said they were protesting "the lack of black assistant coaches," the taking of Muhammad Ali's title, and the "lack of access to good housing."

Sent home. Avery Brundage, the head of the International Olympic Committee, called the protest "a domestic political statement unfit for the apolitical, international forum." He meant that the protest was about politics inside the United States, and that the Olympic Games were a world meeting place that should be kept free of politics.

The US Olympic Committee's officials refused at first to suspend the two men. Brundage then threatened to ban the whole US track team. The US officials suspended Smith and Carlos and sent them out of the Olympic Village. The sports commentator Brent Musburger called them "black-skinned storm troopers." Storm troopers were the street fighters of Hitler's Nazi party. Smith, Carlos and their families got death threats.

Both men later played pro football. Smith coached track at Oberlin College and taught at Santa Monica College. Norman was left off Australia's 1972 Olympic team, although his times qualified him, and accounts differ on why. Lawmakers in Australia's parliament apologized to him in 2012, six years after he died. Smith and Carlos carried his coffin at his funeral.

Players and owners

A team's owner could renew a player's contract year after year without his permission until 1975. The owner could also trade him to another team, and the player could not leave to sign with a team of his choice.

Curt Flood, a St. Louis Cardinals outfielder, refused to be traded to another team in December 1969. A television reporter, Howard Cosell, said to him, "You're a man who makes $90,000 a year, which isn't exactly slave wages." Flood answered, "A well-paid slave is nonetheless a slave." He sued baseball's commissioner. On June 19, 1972, five Supreme Court justices ruled against him and three sided with him.

About three years later, in 1975, an arbitrator named Peter Seitz ruled in a case brought by two pitchers, Andy Messersmith and Dave McNally. An arbitrator is a person both sides agree will settle a dispute. Seitz ruled that a team could renew a player's contract without his permission for only one season. After that the player was a free agent, free to sign with any team.

Girls' sports after Title IX

President Richard Nixon signed Title IX into law on June 23, 1972. Under it, no school or college program that gets money from the federal government may shut a person out because of sex.

The count. About nine times as many girls played high-school sports in 1999-2000 as in 1971-72, according to counts by the National Federation of State High School Associations. In the 1971-72 school year about 294,000 girls played, and girls were about 7 of every 100 players. By 1999-2000 about 2.7 million girls played, and girls were about 41 of every 100.

Yale University had 13 women's varsity teams, the college's top teams, in 1979. Writers for Yale's library later stated in an online exhibit that those teams got "only a grudging fraction of the resources" given to the men's teams.

Chris Ernst

Who: The captain of Yale's women's rowing team, who led a protest that won the team a locker room.

When and where: Yale University, New Haven, Connecticut, 1972 to 1977.

Movie: *A Hero for Daisy* (1999), a documentary about Ernst by Mary Mazzio.

Nineteen women wrote "Title IX" on their bodies and walked in.

Chris Ernst grew up in Scituate, Massachusetts. In high school she beat boys at arm wrestling to win time on the gymnastics equipment.

She began rowing at Yale in her second year and was captain of the women's team by her last year. The showers at the boathouse were for the men's team only. After practice, the women waited on a bus in wet clothes while the men showered. Then they rode back to campus to wash.

The protest. The 19 women on the team wrote "Title IX" on their bodies in early 1976 and went into the office of Joni Barnett, Yale's director of women's sports. They took off their clothes, and Ernst read a statement that began, "These are the bodies Yale is exploiting." To exploit someone means to use them for your own gain.

A *New York Times* reporter called the university that day. Yale officials set up a trailer with running water and planned a $250,000 addition to the boathouse. By 1977 the women had a locker room.

Ernst was a backup rower for the 1976 Olympic team, the first Olympic Games with women's rowing. She became the first woman in the plumbers' union in New Haven in 1980, and she won a world rowing title in 1986.

Billie Jean King

Who: A tennis champion who started the Women's Tennis Association and beat Bobby Riggs in a match called the "Battle of the Sexes."

When and where: Long Beach, California, and Houston, Texas, 1943 to 1973 and after.

She saved $8 to buy her first racket.

Billie Jean King was born in Long Beach, California, in 1943. At 11 she switched from softball to tennis after her parents suggested a more "ladylike" sport. She saved the $8 and took free lessons on the city's public courts. As a young player she was kept out of a group photograph because she wore shorts her mother had sewn instead of a tennis dress. The account does not name who left her out.

The Battle of the Sexes. King founded the Women's Tennis Association, a group that represents women players, in June 1973. Three months later she played Bobby Riggs in a match called the "Battle of the Sexes," because it set a man against a woman. Riggs was 55, and King was 29. She beat him 6-4, 6-3, 6-3 at the Houston Astrodome on September 20, 1973.

About 30,000 people watched in the stadium, about 50 million on television in the United States, and about 90 million more in 37 other countries. She won $100,000.

King won 39 Grand Slam titles, the four biggest tennis championships. She received the Presidential Medal of Freedom in 2009.

Arcades and game machines at home

Some computer games were made for school. Three student teachers from Carleton College, Don Rawitsch, Bill Heinemann and Paul Dillenberger, made *The Oregon Trail* for a junior high class in Minneapolis in 1971. It was first played on December 3, 1971. A Minnesota computer group for schools put out new versions in 1975 and 1985, and Oregon Trail games have sold more than 65 million copies.

Pong. Nolan Bushnell and Allan Alcorn of the company Atari put a test machine of their game *Pong* in a bar called Andy Capp's Tavern in Sunnyvale, California, in August 1972. *Pong* was a simple tennis game played on a screen. The bar's owner soon called them because the machine had stopped working. Its coin box was jammed full of quarters.

Atari sold *Pong* from November 29, 1972, at 25 cents a game, and sold more than 8,000 machines by the end of 1974. Bushnell said players sat "shoulder to shoulder" and could talk, laugh and challenge each other.

Sears sold a home version of *Pong* for $98.95 at Christmas 1975. Atari's own home game machine came out in 1977 for $199. By 1992 Atari had sold about 30 million of that one model.

The company Midway began selling the arcade game *Pac-Man* in North America in October 1980. Its designer, Toru Iwatani, wanted "a nonviolent, cheerful game" that women would play. By 1981 *Pac-Man* machines had taken in more than $1 billion in quarters.

The crash. About 30 million game machines of all makes sat in American homes by 1982, and about 35 million households had children aged 6 to 16. Then sales fell, from about $3.2 billion in 1983 to about $100 million in 1985. Workers for Atari buried about 728,000 unsold game cartridges in a landfill at Alamogordo, New Mexico, in September 1983.

Sales rose again after Nintendo began selling its home machine in the United States in 1985. Nintendo's Game Boy, a game machine small enough to hold in the hands, came out in 1989 for $89.95.

Senators Joe Lieberman and Herb Kohl led public hearings, meetings where senators question witnesses, on violent video games such as *Mortal Kombat* in December 1993 and March 1994. Leaders of the game companies then set up the Entertainment Software Rating Board. From September 13, 1994, its staff put age ratings on games.

The Miracle on Ice

The US Olympic hockey team beat the Soviet Union 4 to 3 at Lake Placid, New York, on February 22, 1980. The American players were college players with an average age of 21, coached by Herb Brooks. Two days later they beat Finland and won the gold medal.

The Iroquois Nationals

Haudenosaunee (Iroquois) people formed their own national lacrosse team in 1983, and the Grand Council of the Haudenosaunee approved it. Officials of the International Lacrosse Federation turned the team down for 1986 and accepted it as a member nation in 1988. In 1990 the team played in the world championship in Perth, Australia, and finished fifth of five.

How did a child's afternoon change?

Children spent less of it in free play and more of it in school, lessons and organized sports.

Walking or riding. Nearly half of children in kindergarten through eighth grade usually walked or biked to school in 1969, and about 12 of every 100 were driven, according to national travel surveys studied by Noreen McDonald and other researchers. A 1995 survey asked a slightly different question: how children got to school on the survey day. That day nearly half went by car, and about 14 of every 100 walked or biked.

Time diaries. The researchers Sandra Hofferth and John Sandberg compared diaries of how children spent their time in 1981 and in 1997. Children spent more time in school, day care, sports and art lessons. They spent less time in free play, housework and watching television. The drop in free play showed up among children whose mothers stayed home as well as those whose mothers had jobs.

More mothers did work for pay. In 1975, 47 of every 100 mothers with children under 18 were working or looking for work. By 2000 it was 73 of every 100, according to the US Bureau of Labor Statistics.

On their own. About 18 of every 100 children aged 5 to 14, some 6.9 million, regularly looked after themselves for part of the week in the fall of 1995. On average they did so for about six hours a week. More than twice as many children that age, about 15 million, took part in sports, lessons or clubs, according to the US Census Bureau.

Private teams. Organized sport also became more private. In a survey of adults published in 2025, about 4 of every 100 who were born in the 1950s said they had played on a private club or travel team as children. For those born in the 1990s it was 13 of every 100. The researchers, Chris Knoester and Chris Bjork, found that children whose parents had more schooling were more likely to play on these teams.

BY 2000

If you remember one thing: Girls won their way into Little League and school sports through a court case and a new law, while children spent less time in free play and more in lessons and organized sports.

One question this era leaves open: Senators held hearings on violent video games in 1993 and 1994. Do games and screens harm children? The next era gives what researchers later found about attention and mood, and where they disagree.

Next: After 2000 children's free time moved onto phones and into games played online with strangers.

2000 to Today

ERA 10 OF 10 · 2000 TO TODAY · ABOUT 22 MINUTES

Much of children's free time moved onto phones and into games played online, often with strangers, after 2000. Researchers at the Pew Research Center, which surveys American teenagers, found that 73 of every 100 teenagers aged 13 to 17 had or could use a smartphone in 2014-15, and 95 of every 100 by 2022. Fewer children walked to school or played outdoors. The average sports family spent $883 on one child's main sport in 2022. Researchers disagree about whether screens harm children's attention and mood.

Doctors found a brain disease in football players who had died. A team doctor who had sexually abused more than 500 girls and women pleaded guilty, and judges gave him prison sentences so long that he will die in prison. College athletes won the right to be paid, betting on games became legal in most states, and the men's soccer World Cup was played in the United States, Canada and Mexico in 2026.

Phones and social media

Pew researchers ask American teenagers aged 13 to 17 how they use phones and the internet. In 2014-15, 24 of every 100 said they were online "almost constantly." In 2022 it was 46 of every 100, and in 2025 it was 40. In Pew's 2025 survey, about 90 of every 100 teenagers used YouTube, and about 60 used TikTok. That year 64 of every 100 teenagers had used an AI chatbot, a computer program that writes answers in plain language.

Hours a day. Common Sense Media, a group that studies children and media, surveyed children in 2021. Children aged 8 to 12 spent an average of 5 hours 33 minutes a day on screens for fun, not counting schoolwork. Teenagers aged 13 to 18 spent 8 hours 39 minutes. In 2019 the figures had been 4 hours 44 minutes and 7 hours 22 minutes.

When a child used two screens at once, the researchers counted the time on both, so two hours of screen time can fit into one hour of the day. The survey's authors wrote that they could not tell how much of the rise came from the COVID-19 pandemic.

The warning. The US Surgeon General, Vivek Murthy, the country's top public health doctor, issued an advisory, a public warning, about social media on May 23, 2023. The advisory states that "we do not yet have enough evidence to determine if social media is sufficiently safe for children and adolescents."

It cites a study of 6,595 American children aged 12 to 15. Those who spent more than 3 hours a day on social media had double the risk of symptoms of depression and anxiety. In the advisory, Murthy asked lawmakers to set safety rules and limits for children's use.

State lawmakers began to keep phones out of classrooms. Florida's lawmakers passed a law banning students' phone use during class time in 2023.

By January 2026, lawmakers in 26 states had banned phones for the whole school day, from the first bell to the last, according to a count by the group Ballotpedia. By July 2026, lawmakers in at least 39 states and Washington,

DC, had passed rules requiring schools to ban or limit phones, according to a tally by *Education Week*. The two counts differ because they count different kinds of laws.

Games played online, and with strangers

Minecraft is a game of building and digging in a world of blocks, played alone or with others. It came out in 2011. By 2023 it had sold more than 300 million copies, more than any other video game.

Roblox opened in 2006. Its players build games and play each other's games. According to the company, half of all American children under 16 play it each month. *Fortnite* came out in 2017. In its best-known version, up to 100 players land on an island and fight until one player or team is left.

Fortnite is free to play. The companies behind *Fortnite* and *Roblox* sell game money, called V-Bucks in *Fortnite* and Robux in *Roblox*, which players buy with real money and spend on costumes and other items inside the game.

The Fortnite case. Officials of the Federal Trade Commission, the government agency that enforces laws protecting buyers, accused Epic Games, the maker of *Fortnite*, of breaking the law on December 19, 2022. They said the company had collected information on children under 13 without their parents' permission. They said the game's settings turned on live voice and text chat for children, and that the company matched "children and teens with strangers to play Fortnite together."

Children had been "bullied, threatened, harassed," some of them sexually, and "exposed to dangerous and psychologically traumatizing issues such as suicide," the officials said. Epic employees had urged the company to change those settings as early as 2017. In a settlement, an agreement that ended the case, Epic Games agreed to pay $520 million.

Roblox. People who brought four lawsuits in June 2025 accused the company that runs *Roblox* of letting adults use the game for the sexual exploitation of children, which means using children for sex acts or sexual pictures. Every

Roblox player had to prove their age, with an ID or a video scan of the face, before chatting from January 2026. Chat was also limited to players of similar ages.

Esports are video game contests played for prize money. The first *Fortnite* World Cup, held from July 26 to 28, 2019, in New York City, paid $30 million in prizes. About 40 million players tried to qualify.

Kyle "Bugha" Giersdorf

Who: A 16-year-old who won the first Fortnite World Cup and $3 million.

When and where: Arthur Ashe Stadium, New York City, July 28, 2019.

He scored 59 points. The second-place player scored 33.

Kyle Giersdorf was born in 2002. His father, who also played video games, got him started on *Fortnite*. Kyle played under the name "Bugha." He joined an esports team called Sentinels on March 25, 2019. That summer he qualified for the first *Fortnite* World Cup, out of about 40 million players who tried.

The final. At 16, he played six games against 99 other finalists in Arthur Ashe Stadium in New York City on July 28, 2019. He scored 59 points, and the second-place player scored 33. He won $3 million, more than any other *Fortnite* player has won in total.

Hours after he won, someone broke into his Twitter account and his account on Twitch, a website where players show their games live to viewers. Later he became a streamer on Twitch, playing his games live for viewers there.

What is measured

Researchers who surveyed children after 2000 found that they walked less and had less free time than children in earlier years. Team sports cost families hundreds of dollars a year, children from richer homes played more, and more children had obesity.

Fewer children walk or bike to school. Nearly half did in 1969. By 2009 it was about 13 of every 100, according to the national travel surveys studied by Noreen McDonald and other researchers. In 2017 about 11 of every 100 students aged 5 to 17 usually walked or biked, according to a later government travel survey.

Less free time. Sandra Hofferth's later time diaries of children aged 6 to 12 showed their free time falling from about 57 hours a week in 1981 to about 48 hours in 2003, the year her diaries end.

The share who played a sport on the diary days fell from 76 to 60 of every 100 between 1997 and 2003. The share who spent time in outdoor activities fell from 16 to 10 of every 100.

The National Survey of Children's Health is a government survey of parents. It found in 2023 that 55 of every 100 children aged 6 to 17 played on a sports team or took sports lessons. Children from richer homes played more.

The gap was already plain in 2012. About 35 of every 100 children in homes earning under $25,000 a year played a sport regularly, against about 49 of every 100 in homes earning $100,000 or more. By 2024 the difference between those two groups had grown to about 20 of every 100 children, according to the Sports and Fitness Industry Association.

What it costs. The average parent spent $883 in a year on one child's main sport, in a 2022 survey by the Aspen Institute and Utah State University. Travel to games and tournaments cost more than any other part. The richest families spent about four times as much as the poorest.

Fifteen of the 50 states had rules in 2012 that recommended or required a daily recess, according to a 2013 report by the research group Bridging the Gap.

Doctors measure obesity by comparing a child's weight with the child's height. From 1971 to 1974, about 5 of every 100 children aged 2 to 19 had obesity. From 2017 to March 2020 it was about 20 of every 100, nearly four times as many, or about 14.7 million children, according to the Centers for Disease Control and Prevention. In the poorest homes it was about 26 of every 100, and in the richest about 12.

What researchers disagree about

ADHD stands for attention-deficit/hyperactivity disorder. Doctors diagnose it when a child has trouble paying attention, or is overactive and acts without thinking, badly enough to cause problems in more than one place, such as at home and at school.

The count. In national surveys of parents, 9.5 of every 100 children aged 4 to 17 had been diagnosed with ADHD in 2007, and 11 of every 100 in 2011. The researchers who reported those counts wrote that the rise "could also reflect better detection" and more awareness, not only more children with ADHD.

The American Psychiatric Association changed its rules for diagnosing ADHD in 2013. Its manual of mental disorders now allowed a diagnosis when symptoms began by age 12 instead of age 7. That change came after the 2011 count, so it could explain only a rise that came later.

In 2022, 11.4 of every 100 children aged 3 to 17 had been diagnosed, about 7.1 million children, according to a 2024 study by M. L. Danielson and other researchers. They wrote that the survey changed its methods in 2022, which makes it harder to compare 2022 with earlier years.

A study of screens and attention. Researchers have studied whether screens are linked to attention problems. A team at the University of Southern California, led by Chaelin Ra and Adam Leventhal, followed 2,587 high-school students in Los Angeles County for two years, in a study printed in 2018. None had serious ADHD symptoms at the start. Students who used more kinds of digital media many times a day later reported more ADHD symptoms.

Among students who used no kind of media that often, 4.6 of every 100 reported the symptoms. Among students who used seven kinds, 9.5 of every 100 did. The researchers called the link "modest." They wrote that more research was needed "to determine whether this association is causal," which means whether the media use caused the symptoms. The students rated their own symptoms, and doctors did not diagnose them.

Other researchers studied something different: how well teenagers felt. The Oxford researchers Orben and Przybylski studied surveys of 355,358 teenagers in the United States and Britain in 2019. They found that technology use explained "at most 0.4%" of the differences in how well teenagers felt. That is less than half of one percent. They called the link "too small to warrant policy change," meaning too small to be a reason for new laws or rules.

BY 2022: Ninety-five of every 100 teenagers had or could use a smartphone, 46 of every 100 said they were online "almost constantly," and researchers still disagreed about what screens did to children.

The argument over a book. The psychologist Jonathan Haidt of New York University published a book called *The Anxious Generation* in 2024. He wrote that phone and social media companies had "rewired childhood" by taking the place of play and time with friends.

The psychologist Candice Odgers of the University of California, Irvine, answered in the science journal *Nature* on March 29, 2024. She wrote that the book's "repeated suggestion that digital technologies are rewiring our children's brains and causing an epidemic of mental illness is not supported by science." She wrote that studies had found "a mix of no, small and mixed associations." She agreed that the social media companies needed to make large changes.

Football and the brain

Mike Webster was a center for the Pittsburgh Steelers who played 17 seasons in the NFL, from 1974 to 1990, and won four Super Bowls. After he retired he lost his memory, suffered from depression and pain, and at times lived in his pickup truck or in train stations. He died of a heart attack on September 24, 2002, at 50.

What Omalu found. Bennet Omalu, a doctor in Pittsburgh who examines bodies to find out why people died, studied Webster's brain. He found clumps of protein unlike those of Alzheimer's disease. Omalu and other doctors described the disease in a medical journal in July 2005. It is called chronic traumatic encephalopathy, or CTE.

Doctors link CTE to repeated blows to the head. Its signs can include problems with behavior, mood and thinking. It often gets worse over time, and it can lead to dementia, the loss of the ability to think and remember. Doctors can find it only by examining the brain after death.

What the league said. Paul Tagliabue, the league's commissioner, had set up the NFL's own brain injury committee in 1994. He named Elliot Pellman to lead it, a doctor who treated joints and muscles, not the brain.

The committee's papers stated from 2003 on that concussions, brain injuries from a hard blow to the head, caused no lasting harm in NFL players. In May 2006 committee members wrote to the journal asking it to take back Omalu's paper. Years later Jeff Miller, a league official, told members of Congress that "there is a link between football-related head trauma and chronic traumatic encephalopathy."

More than 4,500 former players sued the NFL, saying the league had hidden the dangers. Judge Anita Brody approved a settlement in April 2015 to pay retired players with brain illness for 65 years, expected to cost more than $1 billion. Families of players who died before 2006 were not covered, so Webster's family got nothing from it. The NFL's own Retirement Board had already paid his family $1.8 million in 2005, after finding that football caused his brain injuries.

The scoring rule. Until 2021 the dementia tests used to judge claims were scored as if Black players started with lower thinking skills than other players. That made it harder for Black players to prove dementia and be paid. The news accounts of the lawsuit do not name who set that scoring rule. After a lawsuit, NFL officials said in June 2021 that they would stop.

Children were hurt in football too. About 23,000 football brain injuries a year that did not kill the player sent people to emergency rooms, and 90 of every 100 of those hurt were aged 5 to 18. Pop Warner, a youth football program, had 9.5 percent fewer players in 2012 than in 2010, according to Robert Boland of New York University.

Researchers at Boston University had found CTE in the brains of 345 of the 376 former NFL players they studied by 2023. They warned that families who suspected the disease were the ones who gave the brains. So the figure does not mean that 92 of every 100 NFL players have CTE.

The team doctor who abused gymnasts

Larry Nassar worked with the USA Gymnastics national team from 1986. From 1996 to 2014 he was its national medical coordinator, the doctor in charge of the gymnasts' care. From 1997 to 2016 he was also a doctor for athletes at Michigan State University.

What he did. During appointments he said were treatment for pain, he put his ungloved finger into girls' vaginas. Sometimes a parent was in the room and could not see what he was doing. More than 500 girls and women were abused. Some were as young as 6.

Who was told. Many adults were told and did not stop him. Athletes said they told Michigan State employees about him as early as 1997. Court papers say the Michigan State gymnastics coach, Kathie Klages, was told in 1997, and they accuse her of pressuring gymnasts to stay quiet. Michigan State officials looked into a complaint in 2014 and let him keep treating patients.

A coach, Sarah Jantzi, reported him to USA Gymnastics on June 17, 2015. Its president, Steve Penny, told agents of the FBI, the federal police, in Indianapolis. The agents did not open a formal investigation or tell officials in

Michigan. Reporters at *USA Today* wrote that for more than a year Nassar went on "treating, and raping, gymnasts" in Michigan. Rape means forcing a person into a sex act.

Michael Horowitz, the Justice Department's inspector general, the official who checks for wrongdoing inside the department, reported in July 2021 that the FBI agent W. Jay Abbott had failed to act and later lied about it. At least 70 more girls and women were abused after the FBI was told. The victims put the number at 120.

The sentences. Reporters at the *Indianapolis Star* published the story in September 2016. Nassar pleaded guilty to federal crimes of having sexual pictures of children, and to ten counts of first-degree criminal sexual conduct with girls in two Michigan counties. Under Michigan law, first-degree criminal sexual conduct means sexual penetration of a person in certain cases, such as when the victim is under 13, and it can be punished with life in prison.

Judges sentenced him to 60 years in federal prison, and to 40 to 175 years and 40 to 125 years in Michigan, so he will die in prison. Before he was sentenced, more than 200 women and girls spoke about what he did to them, over nine days in court.

What happened to the others. The heads of USA Gymnastics, Michigan State and the US Olympic Committee resigned. Nassar's dean at Michigan State, William Strampel, was convicted of neglect of duty and spent a year in jail. Klages was convicted in 2020 of lying to police, and that conviction was overturned in 2021.

John Geddert, an Olympic coach who had employed Nassar at his gym, was charged in 2021 with human trafficking, forced labor, sexual assault and other crimes. Human trafficking means using force, tricks or threats to make people work or perform sex acts. He killed himself shortly after the charges were announced.

Michigan State paid $500 million to 332 victims in 2018. USA Gymnastics and the US Olympic and Paralympic Committee together agreed to pay $380 million in 2021. Three years later the Justice Department paid $138.7 million to victims who sued over the FBI's failure.

The Olympic gymnasts Simone Biles, McKayla Maroney, Maggie Nichols and Aly Raisman told senators on September 15, 2021, how FBI agents had handled their reports. Raisman said it "was like serving innocent children up to a pedophile on a silver platter." A pedophile is an adult who is sexually drawn to children. The FBI's director, Christopher Wray, said he was "deeply and profoundly sorry."

Rachael Denhollander

Who: A former gymnast who was the first woman to accuse Larry Nassar in public.

When and where: Kalamazoo and Lansing, Michigan, 2000 to 2018.

Movie: *Athlete A* (2020), a Netflix documentary by Bonni Cohen and Jon Shenk about the *Indianapolis Star* reporters and the scandal.

"How much is a little girl worth?" she asked the judge.

Rachael Denhollander was a club gymnast from Kalamazoo, Michigan. She was 15 in 2000 when she saw Larry Nassar for pain in her lower back. He sexually abused her on five visits.

At least seven other girls and women had told someone about Nassar over twenty years before she spoke. By 2016 she was a lawyer. She reported Nassar to the Michigan State University police and filed a complaint with the university in August 2016. The *Indianapolis Star* printed her accusation on September 20, 2016. Michigan State officials fired Nassar that day. By January 2018 more than 265 women and girls had been identified as his victims, and the count later passed 500.

In court. She spoke last at both of his sentencing hearings, in January and February 2018. She asked the judge, "How much is a little girl worth?" She answered that "everything is what these survivors are worth." Judge Rosemarie Aquilina called her "the bravest person I have ever had in my courtroom."

Time magazine named her one of its 100 most influential people of 2018. In 2019 she published a book about the case, *What Is a Girl Worth?*

College athletes paid

NCAA rules let college athletes receive only scholarships and school costs until 2021. California's governor, Gavin Newsom, signed the first state law letting college athletes be paid for their name, image and likeness, meaning for ads and appearances that use who they are, in September 2019.

All nine Supreme Court justices ruled against NCAA limits on education benefits on June 21, 2021, in *NCAA v. Alston*. Justice Brett Kavanaugh wrote that antitrust law, which forbids businesses from agreeing to hold down pay, "should not be a cover for exploitation of the student athletes." From July 1, 2021, NCAA officials let athletes be paid for their name, image and likeness.

The settlement. Grant House, a swimmer at Arizona State, and Sedona Prince, a basketball player, sued the NCAA in 2020. Judge Claudia Wilken approved a settlement on June 6, 2025. Athletes who played from 2016 to 2024 will share about $2.8 billion over ten years.

Each college may pay its athletes directly from the 2025-26 school year, up to about $20.5 million that year, a limit that rises 4 percent a year. Eight women athletes asked a higher court to change the settlement. They argued that most of the money would go to men who played football and basketball.

BY 2025: Doctors had tied CTE to football, Larry Nassar was serving prison sentences so long that he will die in prison, and a judge had approved a settlement under which colleges may pay their athletes directly.

Betting on games

A federal law of 1992 barred states from allowing betting on sports, except in Nevada and three other states. On May 14, 2018, in *Murphy v. NCAA*, a case New Jersey brought, the Supreme Court justices struck the law down. By September 2023, 38 states and Washington, DC, allowed betting on sports. Americans bet $121 billion with licensed sports betting companies in 2023, according to the American Gaming Association, the casino industry's trade group.

Jontay Porter. Adam Silver, the NBA's commissioner, banned Jontay Porter, a Toronto Raptors player, for life on April 17, 2024. League investigators found that Porter told a bettor about his own health before a game in March 2024. He then left that game after three minutes, saying he felt ill. A person he knew had bet $80,000 that Porter would play poorly. Porter had also placed at least 13 bets on NBA games.

Transgender athletes

A transgender girl is a girl who was counted as a boy at birth. Idaho's governor, Brad Little, signed the first state law keeping transgender girls and women off girls' and women's teams in March 2020. By June 2026, 27 states had passed such laws, according to the Supreme Court.

Lia Thomas, a transgender swimmer at the University of Pennsylvania, won a women's college title in 2022. On February 5, 2025, President Donald Trump signed an order titled "Keeping Men Out of Women's Sports." The next day NCAA officials limited women's college events to athletes counted as female at birth. In July 2025 officials of the University of Pennsylvania agreed with the federal government to change Thomas's records.

The Supreme Court. Six Supreme Court justices ruled on June 30, 2026, in *West Virginia v. B.P.J.* and *Little v. Hecox*, that Title IX allows schools to keep separate teams "defined by biological sex." Three justices disagreed in part. B.P.J., who is named in court only by her initials, had wanted to run on the girls' cross-country and track teams at her school in West Virginia. She was 11 when she first asked to play, according to Justice Sotomayor's opinion.

Protest and equal pay

Colin Kaepernick, the quarterback for the San Francisco 49ers, knelt during the national anthem before games in 2016. He was protesting police brutality, which means violence by police officers, and unfair treatment of Black Americans. He chose to end his contract with the 49ers in March 2017, and after that no team signed him.

Kaepernick accused the NFL's owners in October 2017 of secretly agreeing among themselves to keep him out. He settled with the league in February 2019, and the terms were kept secret. The NFL's commissioner, Roger Goodell, apologized in June 2020 for not listening to Black players' concerns.

Equal pay. Five players on the US women's soccer team, among them Hope Solo, Carli Lloyd and Megan Rapinoe, filed a complaint in April 2016 that they were paid less than the men's team. The women won the World Cup in 2019. US Soccer officials settled for $24 million in February 2022. In May 2022 they signed contracts that pay the women and the men equally.

The Haudenosaunee Nationals

The Haudenosaunee national lacrosse team, formed in 1983, is the same team as the Iroquois Nationals. The Haudenosaunee have issued their own passports since 1927. British officials refused to accept those passports for the world lacrosse championship in England in 2010. They told the 23 players to travel on US or Canadian passports. The players refused, and they did not play.

The team won third place at the world championships of 2014, 2018 and 2023. Olympic officials added lacrosse to the 2028 Los Angeles Games in October 2023, and they said that only countries with a national Olympic committee could enter. President Joe Biden and Canada's prime minister, Justin Trudeau, asked them on January 17, 2025, to let the Haudenosaunee play under their own flag. No decision on that request is on record as of September 2026.

The 2026 World Cup

FIFA, the group that runs world soccer, held the men's World Cup in the United States, Canada and Mexico from June 11 to July 19, 2026. It was the first with 48 teams. Of the 104 games, 78 were played in 11 American cities. About 6.8 million people went to the games, more than at any World Cup before.

Spain beat Argentina 1 to 0 after extra time in the final at MetLife Stadium in New Jersey. The US team lost to Belgium 4 to 1 in the round of 16.

Tickets for the final started at $6,730. For the first time, FIFA officials raised ticket prices as more people wanted them.

THE THREE QUESTIONS, ANSWERED

1. **Who invented baseball?** No one person did. Clubs wrote the rules, and the Knickerbocker club of New York adopted twenty of them in 1845. The story that Abner Doubleday invented it in 1839 rests on one letter from a man who was 5 years old that year.
2. **Who was kept out of American games, and how did they get in?** The people who ran the games kept out laborers, Black players, girls and Black swimmers, and left enslaved riders' names off the race cards. Some got in through courts and a new law, like Maria Pepe and the girls of Title IX. Jackie Robinson got in when Branch Rickey signed him, and Kathryn Johnston played anyway, under a boy's name.
3. **How has children's free time changed?** In the 1830s Lucy Larcom made doll furniture from burrs while the grown-ups left the children alone. Since then more of children's time has gone to lessons and organized sports, and by 2022, 95 of every 100 teenagers had or could use a smartphone.

What changed: Play moved from fields the players ran themselves to leagues, arenas and screens run by adults and companies, and the fight over who could play moved from a county court in Virginia to the Supreme Court.

FIVE DATES TO KEEP

September 10, 1674: a Virginia court calls racing "a sport only for Gentlemen."

September 23, 1845: the Knickerbocker club adopts twenty rules for baseball.

April 15, 1947: Jackie Robinson plays his first game for Brooklyn.

June 23, 1972: President Nixon signs Title IX.

July 28, 2019: Kyle Giersdorf, 16, wins the first *Fortnite* World Cup.

NOT KNOWN: how old the Native ball games played with sticks are, because no object or written record dates one before 1500.

Where to go: The National Baseball Hall of Fame in Cooperstown, New York, holds Josh Gibson's plaque, which says he hit "almost 800" home runs.

Next: Chapter 37, Holidays. It asks how each day on the American calendar got there, and who had the power to name a day.

Chapter 37: Holidays

CHAPTER 37 OF 37 · HOLIDAYS · 10 ERAS · ABOUT 95 MINUTES

Many American holidays exist because someone asked for a day and kept asking, and the arguing over some of them is not over.

In May 1877 a woman of 88 in Philadelphia had a helper write a letter for her, because her eyes were too weak. Her name was Sarah Josepha Hale. For about thirty years she had been asking governors and presidents to make Thanksgiving one day for the whole country. President Lincoln had proclaimed it in 1863. She was still asking Congress for a law that made the last Thursday in November a legal holiday. She died two years later. Congress passed that law in 1941.

Most of the days in this chapter were made that way. Someone asked for a day, and someone else said no, often for years. The chapter starts with the ceremonies the Native nations keep for corn, salmon and the winter sun, and with the Spanish, French and English colonists who each held a "first Thanksgiving" of their own. It follows the days Black Americans made for themselves when white Americans' holidays left them out, from election days in Connecticut to Juneteenth in Texas. It ends with the fights still going on over Columbus Day, King's day and Juneteenth. Watch for two things as you read: who had the power to name a day, and who kept a day without anyone's permission.

THREE QUESTIONS THIS CHAPTER ANSWERS

1. Who decided when Americans would give thanks, and why did it take until 1941 to fix the date?
2. Which days did Black Americans make their own, from election days to Juneteenth?

3. Why did President Harrison proclaim the first Columbus Day in 1892, and why do some states no longer keep it?

Five minutes? Read "Juneteenth, 1865" in 1850 to 1900. The Jack Yates story follows it.

Can skip: "Places named for holy days", "Washington's birthday", "Pinkster banned in Albany, 1811", "Columbus Day, 1905 to 1937". The chapter still makes sense without them.

Before 1500

ERA 1 OF 10 · BEFORE 1500 · ABOUT 6 MINUTES

The Native nations marked their year with ceremonies, and each one came at its own time. Some came when a group of stars rose, or when the sun reached the end of its path along the horizon. Others came when the corn ripened or when the salmon swam back up the river. At many of them, people gave thanks for food as it came in: maple sap, strawberries, the first ripe corn, the first salmon.

The oldest written descriptions of the Green Corn Ceremony date from the 1700s, and outsiders wrote them. The nations have written their own since, and they still hold many of these ceremonies today. The Wampanoag writer Nancy Eldredge says that her people and other Native peoples of the Northeast have given thanks this way "from time immemorial," which means since before anyone can remember. The five ceremonies below come from five parts of the land, the Northeast first and Hawaii last.

Who sets the dates in the Northeast

The Faith Keepers. On the Onondaga Nation's land just south of Syracuse, New York, men and women called Faith Keepers decide when each ceremony takes place. The Onondaga are one of the Haudenosaunee nations. They list seven ceremonies in their year: Midwinter, Maple Sap, Planting, Bean, Strawberry, Green Corn and Harvest. The Faith Keepers go by the seasons and the moon, so the dates change from year to year. Midwinter lasts 21 days, and on each day people give thanks for different things.

Haudenosaunee people often start and end their meetings with the Thanksgiving Address. They also call it the Words That Come Before All Else. The speaker gives thanks to the people, the earth, the waters, the fish and the plants, one at a time. Then come the food plants and the medicine plants, and the thanks go on from there. Each part ends with the words "Now our minds are one." An English translation of the Mohawk version was published in 1993.

Farther east lived the Wampanoag. Nancy Eldredge, a Nauset Wampanoag writer, wrote about her people's thanksgiving ceremonies in 1998. They held one in early spring, for a good growing season, and one in the fall, for the harvest. They also gave thanks for strawberries and cranberries as each one ripened, through the thirteen moons of their year. The Wampanoag Tribe of Gay Head (Aquinnah), on the island of Martha's Vineyard, still holds Cranberry Day each October. The day opens with a greeting at sunrise. People pick cranberries, share a meal and set aside a share for the elders.

Why every fire in a Creek town went out

Once a year, every fire in a Muscogee (Creek) town went out. The Muscogee and other nations of the Southeast held the Green Corn Ceremony in midsummer, when the first corn was ripe. It lasted four or eight days. Men fasted, which means they went without food, and women, children and old people could eat certain foods. During the ceremony, people forgave every crime except murder.

What Bartram saw. William Bartram, a Philadelphia man who studied plants and animals, traveled through Creek towns in the 1770s. In a book printed in 1791, he wrote that the Creek had a feast for almost every month of the year. He called the Green Corn Ceremony the busk and said it was the most important one. Each town held its own busk in August, when its own corn was ripe.

Families made new clothes, pots and pans, and swept their houses and the whole town. Then everyone piled the worn-out clothes and the old grain in one heap and set it on fire. On the fourth morning, the high priest made a new fire by rubbing dry wood together, and every house took its fire from it. Then the women brought in the new corn, and people danced in the square at night.

Accounts differ on how long the men fasted. Bartram gives three days. Eric Bowne, writing for the Encyclopedia of Alabama, gives one and a half to two days. Three men who saw the ceremony in the 1700s wrote early accounts of it: James Adair in the middle of the century, Bartram in the 1770s, and Benjamin Hawkins, who wrote about the Creek country in 1798 and 1799. Creek and Yuchi towns in Oklahoma still hold the ceremony at their stomp grounds, the grounds each town keeps for it.

How a Sun-priest sets the calendar

In Pueblo towns today, a man called the Sun-priest watches where the sun comes up over the horizon. He usually watches at sunrise, from a quiet spot not far from the town. (*Pueblo* is the Spanish word for a town, and the Pueblo peoples of the Southwest get their name from the towns they live in.) The point where the sun rises moves along the horizon through the year, so from his sightings he can tell the time of year. From them he sets the planting calendar and the dates of the ceremonies.

The Hopi of northeastern Arizona begin their year of ceremonies with the Soyal ceremony at the winter solstice in December. The winter solstice is the shortest day of the year. After Soyal, each village holds a ceremony about once a month until the Niman, or Home Dance, in late July.

The First Salmon Feast on the Columbia River

The Yakama, Umatilla, Warm Springs and Nez Perce peoples say that when the Creator was about to bring people onto the earth, Salmon offered his body to feed them. These peoples fish for salmon in the Columbia River and its branches in the Northwest. Each spring, before the fishing season opens, they hold a First Salmon Feast.

At the feast, drummers sing, and a bell rings for everyone to stand for a prayer song. People bless water and drink it first. Then they eat the salmon and the other foods. At Celilo, on the Oregon bank of the Columbia, people hold the feast in the longhouse, a long building where the community gathers for ceremonies. There the feast usually falls on the second weekend of April.

Makahiki, the season when chiefs could not make war

Hawaiians kept their own calendar long before the islands became American land. Their new year began when the Pleiades, a small group of stars they call Makaliʻi, rose at sunset in late October or early November. The season that followed was called Makahiki. Kaʻanoʻi Walk of the Kamehameha Schools gives mid-November as the usual start of Makahiki and late January or February as its end. (The islands became American land in July 1898, when members of the US Congress voted to take them over and President William McKinley signed their resolution on July 7.)

The Hawaiian historian David Malo was born in about 1793. He wrote that Makahiki lasted four months. Priests carried an image of the god Lono around the island. At the edge of each district, the local chief handed over the taxes the people owed to the king. They paid in goods: pigs, taro, sweet potatoes, feathers, kapa cloth and mats.

Taro is a plant with a starchy root that Hawaiians cooked and ate, and kapa is cloth made by beating tree bark thin. Chiefs could not make war on each other during the season. People held races, boxing and wrestling matches, and games of skill.

The 1500s

ERA 2 OF 10 · THE 1500S · ABOUT 5 MINUTES

Spanish and French colonists brought the Christian church year to a few places on the coast of Florida and along the Rio Grande. The church year is a calendar of holy days: Sundays, Easter, Christmas and the feast days of saints, the days kept each year to honor holy people. The colonists named places after those days. When they landed, they held church services to give thanks.

Today people in St. Augustine, in Jacksonville and in Texas each call a different one of those services the "first Thanksgiving." The national holiday came from English days of thanksgiving instead, as the National Park Service's own page on St. Augustine says. Across the rest of the land, the Native nations kept their own ceremonies.

Places named for holy days

Juan Ponce de León claimed the coast of what is now Florida for Spain on April 2, 1513. It was the Easter season, which Spanish speakers called *Pascua Florida*. He named the land *La Florida* after it.

Pedro Menéndez de Avilés sighted the same coast on August 28, 1565, the feast day of Saint Augustine. Menéndez named the town he founded St. Augustine.

Two services of thanks on the Florida coast, and the killings that followed

French Protestants, Christians whose churches had broken away from the Catholic Church, built Fort Caroline in June 1564, near what is now Jacksonville. The French called these Protestants Huguenots. The land belonged to the Timucua people. The Huguenots' leader, René de Laudonnière, wrote that on June 30, 1564, he had his people gather on the beach. There he ordered them to sing aloud and give thanks to God for their safe arrival.

Menéndez and about 800 Spanish settlers came ashore at Seloy, a Timucua town, on September 8, 1565. A priest, Father Francisco López de Mendoza Grajales, said a Mass of thanksgiving, the main church service of the Catholic Church. Afterward, the priest wrote, Menéndez "had the Indians fed and dined himself."

The historian Michael Gannon called this Mass and meal "the first community act of religion and thanksgiving in the first permanent European settlement" in North America, and the National Park Service repeats his words. The Mass came 56 years before the English colonists' harvest meal at Plymouth, in New England. The same Park Service page states that the day "did not become the origin of a national annual tradition."

What the Spanish did next. Within five weeks of the Mass, Menéndez's soldiers killed hundreds of French men, many of them after they had surrendered. They captured Fort Caroline on September 20, 1565, twelve days after the Mass, and killed most of the men there. They spared the women and children and shipped them to Havana, in Cuba.

Shipwrecked French survivors then surrendered at an inlet south of St. Augustine, in two groups. Menéndez wrote that his men tied the hands of the first group behind their backs and "put them to the knife," which means they stabbed them to death. The second group came in with the French commander, Jean Ribault. By the Park Service's count, the soldiers killed 111 men from the first group, in late September, and 134 from the second, in October. The Spanish named the inlet Matanzas, which means "slaughters."

Two counts. Menéndez's own letter to King Philip II of Spain records different numbers, and Chapter 22 tells the killings in full. He wrote that his men cut the throats of 132 men at Fort Caroline, and of 10 more the next day. He wrote that Ribault came in with up to 70 men, and that he spared only five of them: two young men of about 18, and three musicians who played the fife (a small flute), the drum and the trumpet. He killed Ribault and all the rest.

In the same letter, dated October 15, 1565, Menéndez told the king why he had killed them. He wrote that they "spread the Lutheran sect," which meant the Protestant faith. He also wrote that they had built their fort "without permission of Your Majesty."

The thanksgiving Texans claim, 1598

Juan de Oñate led about 500 people north across the Chihuahuan Desert to start a Spanish colony in New Mexico. They were soldiers, colonists, wives and children. They brought about 7,000 farm animals. The crossing took about 50 days. They reached the Rio Grande near what is now San Elizario, Texas.

Oñate ordered a day of thanksgiving after ten days of rest, on April 30, 1598. Franciscan friars, men who had promised their lives to prayer and service in the Catholic Church, said Mass. The colonists ate game they had hunted and fish that Manso people, who lived nearby, brought them. One of the colonists wrote, "We built a great bonfire and roasted the meat and fish." Then Oñate read out a claim to the land for King Philip II. The claim is called La Toma, "the taking."

Some Texans call this day the first Thanksgiving. Others have made the same claim for a celebration held by Francisco Vázquez de Coronado's men in Palo Duro Canyon in 1541. The *Texas Almanac* states that this claim was later questioned.

What John White painted at Secotan, 1585

The English artist John White visited Secotan, a town in what is now North Carolina, in July 1585. Its people spoke an Algonquian language, one of a family of related Native languages. White painted a large ceremony there. Men and women sang and danced around a circle of wooden posts. Some shook gourd rattles, and some carried leafy branches.

Thomas Hariot, an Englishman on the same voyage, wrote a description of the scene. He wrote that the posts were carved with heads like the faces of nuns in their veils. He wrote that the dancers dressed in "the most strange fashion they can devise" and had marks on their backs to show which place they came from.

Encyclopedia Virginia describes the ceremony as "perhaps connected to the corn harvest." The records that survive do not say what the ceremony was called or what it meant to the people of Secotan.

The 1600s

ERA 3 OF 10 · THE 1600S · ABOUT 9 MINUTES

John Smith and his men kept Christmas in a Native town in Virginia in the winter of 1608 to 1609. Fifty years later, in 1659, the lawmakers of Massachusetts Bay made keeping Christmas a crime. English colonists did not agree about holidays. The Puritans, English Protestants whose leaders ran the New England colonies, kept Sundays as holy days but did not keep Christmas, because nothing in the Bible told Christians to keep it.

When a drought threatened the crops, Puritan leaders called a day of fasting and prayer. After rain, a good harvest, the safe arrival of ships or a victory, they called a day of thanksgiving, spent in church.

In the fall of 1621, the colonists at Plymouth held a harvest meal. About ninety Wampanoag men joined them, and the meal went on for three days. Americans now call it the first Thanksgiving. Nobody in print gave it that name until 1841.

Christmas and thanksgiving in Virginia

John Smith and his men were caught by bad weather at Kecoughtan, a Native town in Virginia, in the winter of 1608 to 1609. They stayed six or seven days, over Christmas. Smith wrote that they "were never more merry." He wrote that they had never eaten more good oysters, fish, meat, wild birds and bread. He also wrote that he had never had better fires in England than in the dry, smoky houses of Kecoughtan.

Berkeley Hundred. The ship *Margaret* landed 35 settlers and their captain, John Woodlief, at Berkeley Hundred on the James River on December 4, 1619. The land there was home to Native nations, among them the Chickahominy, the Paspahegh and the Weyanock.

Before the settlers left England, a group of men there called the Berkeley Company had given them ten written orders. The first order said the day of arrival "shall be yearly and perpetually kept holy as a day of Thanksgiving to Almighty God," and perpetually meant forever. The settlers held a service of thanks that day. The records do not show whether they held it again in the next two years.

Powhatan men attacked the English settlements along the James River on March 22, 1622. Their leader, Opechancanough, meant to stop the English from taking more Powhatan land, as Chapter 22 describes. Powhatan fighters killed eleven people at Berkeley. In 1958 Senator John Wicker, the Jamieson family and descendants of John Woodlief began holding the Berkeley thanksgiving again.

The harvest meal at Plymouth, 1621

The Plymouth colonists built their town in 1620 on the site of Patuxet, a Wampanoag town. An epidemic, a disease that spread through many people at once, had killed most of the people of Patuxet between 1616 and 1619. Of the 102 people who came on the *Mayflower*, 53 were still alive by the fall of 1621: 24 men, 5 women and 24 children and teenagers.

Only two people who were there wrote about that fall's harvest. One was Edward Winslow, in a letter of December 1621 that was printed in London in 1622. The book it was printed in is now called *Mourt's Relation*. The other was William Bradford, the governor, who later wrote a history of the colony that runs to the year 1650.

Winslow wrote that after the harvest, the governor sent four men to hunt birds "so we might after a special manner rejoice together." Massasoit, the leader the English called the Wampanoag's "greatest king," came "with some ninety men." The English "entertained and feasted" them for three days. The Wampanoag men went out, killed five deer, and gave them to the governor and other leaders.

Massasoit's own name was Ousamequin. The two sides had made an alliance earlier that year. To the Wampanoag, an alliance was a spoken agreement sealed by giving gifts. Historians at Plimoth Patuxet Museums think Ousamequin's visit was probably a show of strength, and that the two sides needed to renew such agreements, perhaps regularly, to keep the terms alive in memory.

Neither Winslow nor Bradford called the meal a thanksgiving. To the colonists, a thanksgiving was a day of prayer in church.

Later Americans called the Plymouth colonists the Pilgrims. By 1816 New Englanders already credited the Pilgrims with starting their yearly thanksgiving. That year a writer in the *Salem Gazette* called it the festival that "our Pilgrim forefathers" began. In 1841 a Boston minister, Alexander Young, printed Winslow's letter. His note read: "This was the first Thanksgiving,

the harvest festival of New England." The Plimoth Patuxet Museums page identifies that note as the first time anyone called the 1621 meal the first Thanksgiving.

William Bradford

Who: Governor of Plymouth Colony, who wrote the colony's history, *Of Plimoth Plantation*.

When and where: Plymouth, in what is now Massachusetts, 1621 to 1623.

Christmas Day 1621: the governor took away the bat and the ball.

On Christmas Day 1621, Bradford called the men of Plymouth out to work. Most of the men who had just arrived on the ship *Fortune* said that working on Christmas went against what they believed was right. Bradford let them stay behind. At noon he came back and found them playing games in the street. Some were "pitching the bar," a contest to see who could throw a heavy bar or pole the farthest. Others were playing stool-ball, a game played with a ball, a bat and a stool.

Bradford took their equipment away. He told them that if keeping the day was a matter of faith, they could keep it by praying in their houses. He said there would be no games or noisy fun in the streets.

In 1623 no rain fell on Plymouth from the third week of May until the middle of July. Bradford wrote that the corn began to wither. The colonists held what Bradford called "a solemn day of humiliation," a day set aside for "humble and fervent prayer." Rain came, and the harvest was good. Then they held a day of thanksgiving.

Fast days and thanksgiving days in New England

The Puritan leaders of New England called a fast or a thanksgiving when something happened. Each one was a day of church services, so the dates changed from year to year. In Massachusetts Bay, the members of the General Court, the colony's lawmakers, set these days together with the elders who led the churches.

One example. The lawmakers asked the elders in September 1638 to keep a thanksgiving on the last Thursday of the eighth month. The colony's lawmakers counted the months from March, so their records call August "the 6th Month." So the eighth month was October. The thanks were for the many ships that had arrived safely that year, for the good weather in the spring and for the ripening harvest.

A thanksgiving for a massacre, 1637

The Massachusetts Bay colonists held a day of thanksgiving in 1637 for a victory in which colonists killed hundreds of Pequot people. Before dawn on May 26, 1637, Connecticut colonists under Captain John Mason, with Mohegan and Narragansett allies, attacked the Pequot town at Mystic. The town had a wall of wooden posts around it. Mason's men set it on fire and shot the people who tried to escape. They killed between 400 and 700 Pequot, most of them old men, women and children, by the count in the Mashantucket Pequot Tribal Nation's own history.

The order. That August the lawmakers asked the magistrates, the colony's judges and officials, to talk with the church elders about a day of thanksgiving for the soldiers' return. They also asked each town to hold a feast for its soldiers. Later that year they ordered that October 12, 1637, be kept as a day of thanksgiving to God.

The order lists the reasons for the thanks: "subdewing the Pecoits," meaning defeating the Pequot, and "bringing the soldiers in safety." It also lists "the successe of the conference" and good news from Germany. "The conference"

was most likely the meeting of ministers at Newtown, now Cambridge, from August 30 to September 22, 1637. The ministers met to settle a quarrel over church teachings.

A claim that spread. William B. Newell, a Penobscot scholar, said that "the first official Thanksgiving Day commemorated (marked the memory of) the massacre of 700 Indian men, women and children during one of their religious ceremonies." A massacre is the killing of many people who cannot defend themselves. Newell was described as a former head of anthropology, the study of human peoples and their ways of life, at the University of Connecticut. Jeremy Bangs, writing for the History News Network in 2005, answered that Newell never headed that department and that Plymouth Colony took no part in the attack.

KNOWN:	The court records show that the lawmakers set October 12, 1637, as a day of thanksgiving for the victory over the Pequot, and set other days for other reasons, such as the 1638 day for the ships, the spring weather and the harvest.
NOT KNOWN:	No record of the attack mentions a ceremony at Mystic that morning.

Christmas made a crime, 1659 to 1681

The law. The lawmakers of Massachusetts Bay passed a law against Christmas on May 11, 1659. Anyone found "observing any such day as Christmas or the like, either by forbearing of labor, feasting, or any other way" had to pay a fine of five shillings. Forbearing of labor meant not working, and twenty shillings made an English pound.

The Puritans saw Christmas as a season of too much eating and drinking, with nothing in the Bible behind it. The minister Increase Mather called Christmas "a profane and superstitious custom," meaning one that was not holy and rested on false beliefs about magic and luck.

In 1681, under pressure from King Charles II's government in London, the lawmakers canceled the Christmas law. People in Massachusetts went back to keeping Christmas slowly. By the 1750s some hymn books in Massachusetts had Christmas songs in them. By 1760 most almanacs, small books printed each year with a calendar, marked December 25 as Christmas. Families there began to make it a home holiday in the early 1800s.

Ceremonies banned in New Mexico

In New Mexico, Spanish church leaders and governors tried to end the Pueblo people's own ceremonies. From 1656 to 1665, the head of the Franciscan friars there, Alonso de Posada, forbade those ceremonies. He ordered the friars to break into the kivas, the underground rooms where the Pueblo people worship. There the friars were to burn the masks and take or destroy the sacred objects.

Treviño's arrests. Governor Juan Francisco Treviño had 47 Pueblo religious leaders arrested in 1675 for practicing sorcery, which means using magic to do harm. Spanish officials sentenced four of them to be hanged. They hanged three, and the fourth killed himself. The officials shamed the rest, whipped them in public and sent them to prison.

Then a force of Pueblo men went to Santa Fe, and Treviño freed the prisoners. One of the men freed was Po'pay, who had been whipped in public. Cliff Fragua's statue of Po'pay shows the scars of the whipping on his back. In August 1680 he led the Pueblo Revolt, and the Pueblo people drove the Spanish out of New Mexico for twelve years.

1700 to 1750

ERA 4 OF 10 · 1700 TO 1750 · ABOUT 4 MINUTES

Boston boys paraded every November 5 with figures of the Pope, and New York's enslaved and free Black people gathered each spring for Pinkster, a holiday the Dutch had brought. Between 1700 and 1750, people in different colonies kept different days, and many of those days came from Britain. New England governors ordered days of thanksgiving and days of fasting by printed order. In 1746 a Massachusetts governor banned ordinary work on a day of thanksgiving. In New England, Black communities began choosing their own governors and kings.

Thanksgiving and fast days by the governor's printed order

In Connecticut and Massachusetts, the governor named the days of thanksgiving and the days of fasting. Printers published his order on a large single sheet of paper called a broadside.

Governor Gurdon Saltonstall of Connecticut set Wednesday, November 8, 1721, as a day of thanksgiving, in an order dated October 14. Smallpox, a disease that passes from person to person and can kill, was then spreading in Boston. In his order, Saltonstall gave thanks that the sickness had not reached Connecticut. He also gave thanks for the peace, for that year's harvest and for the birth of a prince in the royal family, Prince William Augustus.

Governor William Shirley of Massachusetts ordered a thanksgiving for Thursday, August 14, 1746. Soldiers under the Duke of Cumberland had defeated rebels in Scotland. Shirley called on people to go to worship that day. His order said, "all servile Labour is prohibited on said Day." Servile labor meant ordinary work. Massachusetts governors also ordered fast days. Jonathan Belcher ordered one in 1734, and Spencer Phips ordered another in 1745.

Why Boston boys burned the Pope every November 5

Pope's Night began with a plot in England. In the early hours of November 5, 1605, men loyal to King James I searched the cellars under Parliament, England's lawmaking body, in London. They found a man named Guy Fawkes there, in a cloak and hat, with boots and spurs, carrying fuses and matches. He and a group of other English Catholics had planned to kill the king when Parliament opened.

The members of Parliament then passed a law for a public thanksgiving every November 5. In New England the day became Pope's Night, and crowds burned figures of the Pope, the head of the Catholic Church.

In Boston, young men and older boys built wagons that carried moving figures of the Pope, devils and political enemies. One figure was the Stuart Pretender, a man who claimed he should be king of Britain. Some of the young men dressed as devils in tall pointed caps, and some dressed as women. They paraded through town and asked homeowners for money. The night ended with a meal outdoors and a bonfire.

Boston had two gangs, one from the North End and one from the South End. Each gang built its own Pope wagon, and on Pope's Night the two gangs fought each other. The winners burned the losing gang's figures on Copp's Hill or on the Common, a large open field in the middle of Boston.

The printer Isaiah Thomas, born in 1749, remembered the crowds as a boy. They blew on conch shells, beat on the sides of houses with clubs and shouted "North end forever" or "South end forever." Thomas was once knocked out by a brick. Another Boston boy, Henry Knox, once held up a corner of a Pope wagon. Knox later became a general in the Revolution.

Pinkster in New York

Pinkster is the Dutch name for Pentecost, a Christian feast held seven weeks after Easter. Dutch settlers brought it to their colony of New Netherland in the 1620s. That colony later became New York. The oldest known written account of Pinkster appeared in the *New-York Weekly Journal* in March 1737. It describes a Pinkster gathering in the fields outside New York City, with African-style musical instruments.

The celebration lasted several days. Enslavers gave enslaved men and women time off at Pinkster. Many traveled from farms into the city to see family members who lived some distance away. They sold berries, herbs, drinks, oysters and sassafras bark, which people boiled to make tea.

Black governors and kings in New England

Black communities in New England held elections of their own. Where white men elected their governor, as in Connecticut and Rhode Island, Black communities elected a Black governor. Where the king of England appointed the governor, as in Massachusetts and New Hampshire, they elected a king.

The earliest Black governor whose name survives is Hercules, of New London, Connecticut, from about 1749. He is known from the gravestone of his wife, Florio. The elections in Connecticut may have begun that year.

In Connecticut the Black election came the day after the white men's election, and the celebration went on for a week. Some people borrowed clothes from the men and women who enslaved them. The enslaver of the man who won was expected to help provide the food, the decorations and the liquor. People ate gingerbread and thick election cakes full of fruit, and drank rum. The parades had gunshots, tambourines, banjos, fiddles and drums, and some marchers spoke African languages.

1750 to 1800

ERA 5 OF 10 · 1750 TO 1800 · ABOUT 11 MINUTES

Americans held the first planned Fourth of July celebrations on July 4, 1777, the first anniversary of independence. Within a few years they also celebrated George Washington's birthday. Neither day was a national holiday yet. The Continental Congress was the meeting of men sent by each state to lead the Revolution. Its members asked Americans to keep days of thanksgiving on dates they chose. Later President Washington did the same. None of them set a yearly date.

Older customs changed during the Revolution. Boston's crowds turned Pope's Night into a protest against British taxes, and General Washington banned it in his army. At Washington's own plantation, the evidence points to four days off at Christmas for most enslaved workers. The Washingtons made the enslaved people who worked in the house do extra work, serving their guests.

Pope's Night turns into protest, 1764 to 1775

A Pope wagon in Boston ran over and killed a boy named Brown on Pope's Night, November 5, 1764. Accounts give his age as five to nine. The records that survive do not give his first name or say who was driving. The boy died early in the day. The parades and the fight between the gangs went on that night.

The Stamp Act crowds. Crowds in Boston protested the Stamp Act, a British tax on printed paper, in August 1765. They used the customs of Pope's Night: they hanged and burned figures, marched through town and lit bonfires. They broke into three or four large houses and smashed the furniture. Ebenezer Mackintosh, a shoemaker and the captain of the South End gang, led the crowd on August 14. On August 26 a crowd attacked the house of Lieutenant Governor Thomas Hutchinson. Afterward Sheriff Stephen Greenleaf arrested Mackintosh, then let him go.

That fall Mackintosh made a deal with Henry Swift, the captain of the North End gang. A Boston printer, John Boyle, wrote it down: "A Union established between the South and North End Popes." On Pope's Night the two gangs

joined as one, and they sat down to a dinner that wealthy merchants paid for.

Washington's order. General George Washington ordered his soldiers on November 5, 1775, not to keep "that ridiculous and childish Custom of burning the Effigy of the pope." An effigy is a figure made to look like a person. He wrote that the custom was improper "at a time when we are soliciting (asking for), and have really obtained the friendship and alliance of the people of Canada."

Pope's Night ended in Boston during the Revolution. By one account it ended in the mid-1770s. By another, Boston crowds kept it again in 1776 and 1777 and stopped when soldiers from France, a Catholic country, arrived to fight on the American side.

The first Fourth of July, 1777

The members of the Continental Congress voted for independence on July 2, 1776. The next day John Adams wrote to his wife, Abigail. He wrote that July 2 would be celebrated as "the great anniversary Festival" and that the day "ought to be solemnized with Pomp and Parade, with Shews, Games, Sports, Guns, Bells, Bonfires and Illuminations from one End of this Continent to the other." By solemnized he meant marked with ceremony, shews were shows, and illuminations were candles lit in the windows. The members of Congress formally approved the Declaration of Independence on July 4, and that is the date Americans celebrate.

Crowds greeted the public readings of the Declaration in July 1776 with parades, bonfires and gunfire. The first planned Fourth of July celebrations came a year later, in Philadelphia and in Boston.

The first planned Fourth. In Philadelphia on July 4, 1777, the armed ships in the Delaware River were hung with flags. At one o'clock each ship fired thirteen cannon, one for each state. That night bells rang and fireworks went up over the city's common, and people lit candles in their windows. A Philadelphia newspaper, the *Pennsylvania Evening Post* of July 5, 1777, describes the day. In 1781 Massachusetts lawmakers made the Fourth of July an official state holiday, the first state to do so.

Henry Wight

Who: A minister and Revolutionary War veteran who started the Fourth of July celebration in Bristol, which the town still holds.

When and where: Bristol, Rhode Island, from 1785.

The oldest Fourth of July that never stopped: Bristol, since 1785.

Henry Wight began the yearly Fourth of July ceremony in Bristol, Rhode Island, in 1785, when he led the town's first Patriotic Exercises. In a speech that day he honored the veterans of the war and celebrated the new country's freedoms. The town has held the ceremony every year since, and the town government calls it the oldest Fourth of July celebration in the United States that has never stopped.

Wight had left his studies at Harvard to serve in the Revolutionary War. One account says he served as a soldier. Another says he served as a surgeon, a doctor who operates on the body. In January 1785 the 36 members of the First Congregational Church of Bristol hired him as their minister.

Whose land Bristol stands on. Bristol stands on Mount Hope, called Montaup in the Pokanoket language, in the homeland of the Pokanoket Wampanoag. The Wampanoag leader Metacom, whom the English called King Philip, had his base there and led a war against the New England colonies in 1675 and 1676, called King Philip's War. On August 12, 1676, in an ambush at Mount Hope led by the English captain Benjamin Church, a Native man serving with the English shot Metacom dead.

The English put his head on display at Plymouth and sold his wife and nine-year-old son into slavery. In 1680 Plymouth Colony's leaders sold Mount Hope to four Boston merchants for 1,100 pounds, and the town of Bristol was founded there.

Before Wight. The people of Bristol had already marked the Fourth with cannon. In 1777, one account says, "they ushered in the morning by firing 13 cannons, one for each colony." What began in 1785 was the

> yearly ceremony. The town's Fourth of July committee gives the early 1800s as the likely start of Bristol's parade. Wight died in 1837.

Thanksgiving days for the new country

The British general John Burgoyne and his army surrendered to American soldiers at Saratoga, New York, on October 17, 1777. The members of the Continental Congress then set December 18, 1777, as a day of thanksgiving for the whole country. Over the war years they named at least one day of fasting or of thanksgiving in every year from 1775 to 1783.

Elias Boudinot of New Jersey was a member of the new House of Representatives. In 1789 he asked the other members of Congress to request a day of thanksgiving. The members of both houses agreed. They asked the president to name a day of thanks for the chance "peaceably to establish a form of government." On October 3, 1789, President George Washington named Thursday, November 26, 1789, as a day of thanksgiving.

Washington sent his order to the state governors. On the day, he went to church at St. Paul's Chapel in New York City. He sent beer and food to men held in prison because they owed money.

A second day, 1795. Thousands of farmers in western Pennsylvania took up arms in 1794 against a federal tax on whiskey. This uprising is called the Whiskey Rebellion. Washington led an army of militia, citizen soldiers, against it, and the rebel leaders fled. On January 1, 1795, Washington named February 19 as a day of thanksgiving. One of the reasons he gave was the end of the uprising. None of these days became a yearly holiday.

Washington's birthday

Soldiers at the army's winter camp at Valley Forge, Pennsylvania, gathered on George Washington's birthday in 1778 to wish him well. In 1779 people in Milton, Massachusetts, held a public celebration of the day.

Washington was not born on February 22 by the calendar used in his time. When he was born, the date was February 11, 1731. In 1751 the members of Parliament passed a law that moved Britain and its colonies to the calendar used today.

To catch up with that calendar, they took eleven days out of September 1752. In Britain and the colonies, the day after September 2, 1752, was September 14. The same law moved the start of the year from March 25 to January 1. Under the old count, a February birthday fell near the end of the year 1731. Under the new calendar, Washington's birth date became February 22, 1732.

Four days off at Christmas, and who got none

Mount Vernon was George Washington's plantation in Virginia. Mary V. Thompson, writing for Mount Vernon, says the evidence shows that most enslaved workers there had four days off at Christmas. They used the days to rest and to visit family, at Mount Vernon and on nearby plantations. For the enslaved cooks, housemaids, waiters and others who worked in the house, Christmas brought no rest. The Washingtons had extra guests over the holiday, and they made these workers do extra work.

Philip Vickers Fithian was a tutor from New Jersey, in his twenties, a teacher who lived with a family and taught its children. He spent Christmas 1773 at Nomini Hall, the Virginia plantation of Robert Carter. His journal records that the workers who served him kept working that Christmas morning, and he gave five of them small tips in bits, small silver coins. He wrote that guns fired all around the house woke him.

Nelson, a boy Fithian counted among the "Servants," came in early, dressed only in his shirt and breeches, short pants that ended at the knee. He built a big fire, blacked Fithian's shoes, tidied the room and wished him a joyful Christmas. Fithian gave him half a bit.

Before Fithian was dressed, the man who made the fire in the schoolroom came in. He was dressed neatly in green and was almost drunk, and he bowed three or four times. Fithian did not write his name.

Then an enslaved worker sent in Fithian's clean clothes with a message asking for a "Christmas Box," a small gift of money. Fithian gave a bit to each of these two men. He gave two bits to Tom the coachman, who was caring for his lame horse, and half a bit to Dennis, a boy who waited at table. In all he gave five bits, a little over three shillings.

Black governors in wartime Hartford, 1776

Black communities in Connecticut kept electing Black governors through these years. In Hartford, London was governor in 1755, while Thomas Seymour held him enslaved. Quaw, governor in 1760, was enslaved by George Wyllys, the secretary of the colony. Cuff was governor from 1766 to 1776.

These governors had little power of their own. A Black governor had no power to punish a white person who wronged a Black person.

John Anderson

Who: An enslaved man who became Hartford's Black governor in 1776, in a dispute that white officials investigated as a possible plot.
When and where: Hartford, Connecticut, 1776 to 1780.

A Black governor named without an election, two months before independence.

John Anderson became the Black governor of Hartford in May 1776, two months before the Declaration of Independence. White officials investigated his choice as a possible British plot.

That May, white men in Connecticut elected Jonathan Trumbull governor again. He had been governor since 1769. Soon after, Cuff, the Black governor of Hartford, resigned. He named John Anderson as governor in his place, without an election. Major French recorded it in his journal on May 11, 1776.

Anderson was enslaved by Philip Skene, who had been a major in the British army. The Americans held Skene as a prisoner, because they suspected he was loyal to Britain. White people in Hartford feared that Skene planned to use Anderson to bring Black people over to the British side. Members of a committee of Connecticut officials investigated.

Who paid. The committee found that money had been offered to win Black people's support for Anderson. Two accounts disagree about who offered it: Skene, or Anderson himself. Chris Menapace writes that Skene tried to pay Black people to back Anderson, and that many refused to back an enslaved man owned by a Loyalist, a colonist who stayed on the king's side. Skene told the committee he had offered the money "for amusement." Andy Piascik writes that Anderson himself paid some Black people to support him, and that the committee cleared Anderson of any plot. By Piascik's account, Connecticut officials approved a four-year term for Anderson. He served as governor until 1780.

1800 to 1850

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Wealthier families and newspaper editors in American towns worked, in the first half of the 1800s, to turn Christmas from a noisy street holiday into a day for families and children. In a poem printed in 1823, St. Nicholas comes down the chimney to fill the children's stockings. Thanksgiving was still mostly a New England day. Sarah Josepha Hale, a magazine editor, began asking governors and presidents to make it one day for the whole country.

Enslavers in the South gave enslaved people three to six days off at Christmas. In Harriet Jacobs's town in North Carolina, New Year's Day then brought the hiring out and selling of enslaved people. Free Black Americans chose days of their own to celebrate, such as July 5, 1827, in New York.

Christmas becomes a children's day

In the early 1800s, Christmas in American towns still meant heavy drinking, begging from door to door and rowdy parades. Wealthier families and newspaper editors feared that these crowds threatened public order. They worked to turn Christmas into a quiet day at home, centered on children.

The poem. The *Troy Sentinel*, a newspaper in Troy, New York, printed a poem called "Account of a Visit from St. Nicholas" on December 23, 1823. It carried no author's name. It begins "'Twas the night before Christmas," and it describes stockings hung by the chimney for St. Nicholas to fill. "Down the chimney St. Nicholas came with a bound," the poem says. Eight reindeer pull his sleigh, and in the 1823 printing two of them are named Dunder and Blixem.

Who wrote it. A book called the *New York Book of Poetry* listed Clement Clarke Moore, a professor at a school for ministers in New York City, as the author in 1837. Moore put the poem in his own book of poems in 1844. The descendants of Henry Livingston Jr., a New York farmer and poet, say he wrote it as early as 1808. Accounts still differ about who wrote it.

A few years after the poem appeared, Catharine Maria Sedgwick of Stockbridge, Massachusetts, published a story about children who found a Christmas tree with gifts around it. Shopkeepers began to advertise gifts for children.

By the 1840s lawmakers in many states had begun to make Christmas a legal holiday, a holiday set by law. Alabama is often called the first state, in 1836, but Alabama's state archivists searched and found no law from that year. In 1848 Alabama's lawmakers made Christmas a bank holiday, a day when payments on loans were not due.

Christmas under slavery, three days in the year

Enslavers in the South gave enslaved workers three to six days off at Christmas. Edwin Epps, who held Solomon Northup in the 1840s on a plantation on Bayou Boeuf, a slow swampy stream in Louisiana, gave three days. Northup wrote that other enslavers gave four, five or six.

Northup wrote that the planters on Bayou Boeuf took turns holding a "Christmas supper" for enslaved people from nearby plantations. Three to five hundred people came, and white people came to watch them eat. The supper and the days off were, Northup wrote, "southern life as it is, three days in the year." The other 362 days, he wrote, were "days of weariness, and fear, and suffering, and unremitting labor." Unremitting means without a break.

Frederick Douglass was enslaved in Maryland. In 1845 he wrote that enslavers gave the days from Christmas to New Year's as holidays. He wrote that they did it because they knew it "would be unsafe to deprive them of it," that is, to take it away. Douglass called the holidays "safety-valves." By that he meant that enslavers used the days off to drain away the anger that could lead enslaved people to rebel.

In Edenton, North Carolina, enslaved men from the plantations went from door to door in costume on Christmas Day, until noon. Harriet Jacobs, who was enslaved in Edenton, called these groups the "Johnkannaus." Each group had about a hundred men. Two of them wore calico, a cheap printed cotton cloth, covered with a net of bright stripes, with cows' tails and horns.

The men beat a drum made from a box covered in sheepskin, and others played triangles and jawbones. They sang songs they had made up during the month before. At every door they asked for a penny or a glass of rum, and a group could collect twenty or thirty dollars.

Hiring day. In Jacobs's town, January 1 was hiring day. On that day enslavers rented enslaved people to other white men for the coming year. The next day, each person had to go to the new master. Jacobs wrote that anyone who refused "is whipped, or locked up in jail, until he consents to go." She did not name who did the whipping. After the hiring, enslavers sold people.

The sale days. Jacobs wrote that on one of these sale days she watched a mother lead her seven children to the auction block, the platform where people were sold. The mother knew some of them would be taken from her, but the buyers took all seven. A slave trader bought the children. A man from the town bought the mother. Jacobs did not give the mother's name.

On another New Year's Day, Jacobs's friend Fanny was sold at auction to pay her enslaver's debts. Jacobs does not say who ordered the sale. One master bought Fanny, and another master, far away, bought her four daughters. Fanny later escaped.

Why Black Americans celebrated on January 1, July 5 and August 1

The members of Congress had voted to ban bringing enslaved people into the United States from other countries. The ban took effect on January 1, 1808. That day Absalom Jones preached a sermon of thanks at St. Thomas's African Episcopal Church in Philadelphia. He asked Black Americans to keep January 1 every year as a day of thanks. On that day they would teach their children "the history of the sufferings of our brethren (brothers and sisters), and of their deliverance (being set free)." Enslavers inside the country kept buying and selling people after the law.

July 5, not July 4. New York's lawmakers voted on March 31, 1817, to end slavery in the state on July 4, 1827. On that day about 4,600 people were freed. Some of them still had to work for their former enslavers for a set number of years under written contracts called indentures. Black New Yorkers held church services on July 4, 1827, and held their public celebration on July 5. Nathaniel Paul had led a meeting in Albany that voted to celebrate on the fifth, because the fourth was the day white citizens celebrated.

The people at the July 4 church services went home quietly. They feared that white men celebrating Independence Day would pick fights with them. In New York City on July 5, between 2,000 and 4,000 people gathered in St. John's Park. Samuel Hardenburgh led them as the marshal of the day, the man in charge of the celebration.

August 1. The members of Parliament, Britain's lawmakers, set August 1, 1834, as the day slavery would begin to end in the British colonies of the West Indies. The West Indies are the islands of the Caribbean Sea. On that day enslaved people in Antigua, Montserrat and Bermuda went free. In the other colonies they became "apprentices." Apprentices still had to work about 40 hours a week for their former enslavers, without pay. Their protests helped end that system in 1838.

Black Americans in New York City held a celebration on August 1, 1834. After freedom was complete in 1838, the editors of *The Emancipator*, Arthur Tappan's antislavery newspaper, called for a yearly holiday. In Oberlin, Ohio, Sabram Cox, who had escaped slavery, led the town's first August 1 celebration

in 1842. The historian Jeffrey Kerr-Ritchie writes that for about thirty years, August 1 was the most important yearly celebration of freedom among Black people in the northern United States.

Pinkster banned in Albany, 1811

By the late 1700s, large crowds of enslaved and free Black people gathered in Albany, New York, for Pinkster. City leaders worried about so many Black people gathering in one place. In 1811 the members of the Albany Common Council, the city's governing board, banned the celebration.

Thanksgiving, Washington's birthday and the Fourth

Thanksgiving in these years was a New England custom. People in other parts of the country did not keep it.

People in New York, Philadelphia and Richmond held balls, dinners and parades on Washington's birthday, February 22. The balls and dinners were for rich families, and the parades filled the streets for everyone else.

The fiftieth Fourth. The Fourth of July in 1826 was the fiftieth anniversary of the day the members of the Continental Congress approved the Declaration of Independence. John Adams and Thomas Jefferson both died that day.

Sarah Josepha Hale

Who: A magazine editor who asked governors and presidents for years to make Thanksgiving one national holiday.

When and where: New England and Philadelphia, 1827 to 1879.

Thirty years of letters, and the law came after her death.

Sarah Josepha Hale spent decades asking governors and presidents to make Thanksgiving one holiday for the whole country, on the same Thursday in every state. Her husband, David Hale, died after nine years of marriage and left her with five children. She wrote poems and books to support them. One of her poems for children was "Mary Had a Little Lamb."

In her 1827 novel *Northwood*, she described a New England Thanksgiving dinner. Roast turkey sat at the head of the table. Of all the pies, "the pumpkin pie occupied the most distinguished niche," which means the place of honor.

The magazine. She became the editor of *Godey's Lady's Book*, a women's magazine read across the country, in 1837, and held the job until 1877. She used the magazine and her letters to ask for a national Thanksgiving. She sent her articles to the governors of every state and every territory, a part of the country that was not yet a state.

Accounts disagree about when she began. Plimoth Patuxet Museums gives the 1830s. The National Women's History Museum gives 1846. In 1863 Hale herself wrote that she had asked "for the last fifteen years," which points to about 1848.

Some Southern leaders saw Thanksgiving as a Northern holiday. Governor Henry Wise of Virginia wrote back to Hale that he wanted nothing to do with "this theatrical national claptrap of Thanksgiving." Claptrap means silly talk meant to win cheers. Northern ministers preached against slavery on the day, and that was what Wise objected to.

Hale supported ending slavery, and later she backed a plan to send freed Black Americans to Africa. She did not support the movement to give women the vote.

Still asking. She got her national day in 1863. After that, each president proclaimed it each year. To proclaim a day is to name it in a public order. No law yet set Thanksgiving as a holiday for the whole country. From 1870, under a federal law, national government workers in Washington, D.C., had the day off on any thanksgiving day the president named.

In May 1877, when she was 88, her eyes were so weak that a helper wrote a letter for her to George Washington Warren. In it she was still asking for "a Resolution of Congress making the last Thursday in November a legal holiday." She died in 1879.

1850 to 1900

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President Abraham Lincoln proclaimed the first national Thanksgiving in 1863, and the first national holidays followed between 1870 and 1894. In 1870 the members of Congress made New Year's Day, the Fourth of July, Christmas and any thanksgiving day the president named holidays for federal workers, the people who work for the national government, in Washington, D.C. The members of Congress added Washington's birthday in 1879, Decoration Day in 1888 and Labor Day in 1894. Decoration Day was a day for putting flowers on soldiers' graves.

Freed Black Americans made days of their own after the Civil War. Freed people in Texas celebrated June 19, the day in 1865 when a Union general announced that slavery had ended there, and in Charleston, South Carolina, freed people reburied Union prisoners of war and decorated their graves. In Norfolk, Virginia, in 1866, white men attacked a march held to celebrate a new civil rights law.

The Commissioner of Indian Affairs, the official who ran the government's office for Native affairs, ordered the government's boarding schools for Native children in 1890 to celebrate the anniversary of the Dawes Act. Under that law, federal officials broke up the land set aside for Native nations.

A mob in New Orleans killed eleven Italian American men in 1891. The next year, President Benjamin Harrison proclaimed the first national Columbus Day, and some writers say he meant it partly to calm Italy and Italian Americans after the killings.

How Thanksgiving became national, 1863

Some Southern leaders in the 1850s called Thanksgiving a Northern holiday. In 1856 the Richmond *Whig*, a Virginia newspaper, printed an attack on the thanksgiving day in the District of Columbia. The editors wrote that the day cost working men a day's pay and led to drunkenness. They called New England a "crazy society."

Hale's letter. Sarah Josepha Hale wrote to President Abraham Lincoln on September 28, 1863, in the middle of the Civil War. She asked him to proclaim the last Thursday in November as a national day of thanksgiving. Getting every state to pass a law, she wrote, "would require years."

Lincoln issued the proclamation on October 3, 1863. Secretary of State William Seward, the head of the government's dealings with other countries, wrote it. In it Lincoln invited Americans everywhere, even those at sea and in other countries, to keep the last Thursday in November as a day of thanks. He also asked them to pray for "all those who have become widows, orphans, mourners or sufferers" in the war.

After 1863 every president proclaimed a Thanksgiving Day each year. Most chose the last Thursday in November. Two presidents chose other days: Andrew Johnson the first Thursday in December in 1865, and Ulysses S. Grant the third Thursday in November in 1869. In Texas, Governor Oran Roberts, who had been a Confederate officer, still refused to proclaim Thanksgiving in the 1880s.

The first federal holidays, 1870 to 1888

Santa in the army camp. During the Civil War, a magazine called *Harper's Weekly* printed a drawing by Thomas Nast on January 3, 1863. Nast had come to America from Bavaria, in Germany. The drawing showed Santa Claus visiting a Union army camp. Santa wore a coat covered in stars and striped trousers. In his hands was a toy of the Confederate president, Jefferson Davis, with a rope around its neck.

A writer in Smithsonian Magazine calls the drawing Union propaganda, a picture made to win people to the Union side. Seven years later, Christmas Day became one of the first federal holidays.

The 1870 law. The members of Congress passed the first law that gave federal workers in Washington, D.C., holidays on June 28, 1870. The days were New Year's Day, the Fourth of July, Christmas Day and any day the president named for fasting (going without food, as a way to pray) or thanksgiving. Researchers at the Congressional Research Service, the office that writes studies for Congress, write that the law was apparently an answer to a written request from bankers and businessmen in the city. Representative Burton Cook of Illinois said it would match the holidays of the states around the city.

Washington's birthday, 1879. The members of Congress added Washington's birthday, February 22, in January 1879. The main purpose was to make it a bank holiday. Lawmakers in Massachusetts had already made Christmas, Washington's birthday and the Fourth of July state holidays in 1856. Irish Catholic voters in the state helped pass that law. A law of January 6, 1885, gave paid holidays to federal workers paid by the day, all over the country.

Decoration Day, 1888. The members of Congress made Decoration Day, May 30, a holiday for federal workers in the city in 1888. Many of those workers belonged to the Grand Army of the Republic, the organization of Union veterans. Before the law, they lost a day's pay when they went to the Decoration Day ceremonies.

Juneteenth, 1865

The Union general Gordon Granger announced in Galveston, Texas, on June 19, 1865, that slavery had ended there. The Confederate general Robert E. Lee had surrendered his army in Virginia on April 9, 1865, ten weeks earlier. Granger arrived with about 1,800 soldiers, according to the historian Henry Louis Gates Jr. Other accounts give about 2,000.

Granger issued General Orders No. 3. It stated that "all slaves are free," and that former masters and the people they had enslaved now stood as "employer and hired labor." The order stated that the freed people were advised "to remain quietly at their present homes and work for wages." It also stated that they would not be allowed to gather at army posts, and that they would not be "supported in idleness," which meant fed without working.

About 250,000 people were enslaved in Texas. After 1862, enslavers from other Southern states who were fleeing the Union army had forced more than 150,000 of them to move there.

Many enslavers in Texas did not tell the people they held that they were free until after the harvest was in. One formerly enslaved Texan said an enslaver "didn't tell us we were free til a whole year after we were." In Galveston, the mayor, a former Confederate, forced freed people back to work.

Susan Merritt had been enslaved in Texas. She said that right after freedom, she saw many Black people hanging from trees in the Sabine River bottom, the low land along the river. She said white men caught them swimming across the river and shot them. She did not name the people who were killed, and no record names the killers. Katie Darling kept working for her former mistress for six more years. Darling said the woman whipped her after the war just as she had before.

The Freedmen's Bureau was the federal office set up to help freed people. Its officials did not reach Texas until September 1865. In 1868 Texans met to write a new state constitution, the state's basic set of laws. A committee of that meeting counted 939 murders in the state between 1865 and 1868. In 379 of them, white people killed Black people. The figures, as the Handbook of Texas

reports them, do not say who killed the other 560 people. Outlaw gangs and armed groups such as the Ku Klux Klan, a secret group of white men who used terror against Black people, attacked freed people and Union soldiers.

The first Juneteenth. Freed people in Texas celebrated the day a year later, on June 19, 1866. People began to call the day Juneteenth, a word made from June and nineteenth. Some early celebrations were political meetings, where freed people learned about their new right to vote. In some towns, city officials made Black people hold their celebrations outside the town. Black Texans raised money to buy land of their own for the day.

Jack Yates

Who: A Baptist minister, born enslaved, who led Black Houstonians in buying land for their Juneteenth celebrations.
When and where: Virginia, then Houston, Texas, 1828 to 1897.

Eight hundred dollars for ten acres, so Juneteenth had a home.

Jack Yates and three other men paid $800 in 1872 for ten acres in Houston, Texas, so that the city's Black people had a place of their own for Juneteenth. The other three men were Richard Allen, Richard Brock and Elias Dibble. The members of two churches bought the land for the Black people of Houston. One was Antioch Missionary Baptist Church, where Yates was the pastor. The other was Trinity Methodist Episcopal Church.

The land became Emancipation Park, named for the freeing of enslaved people. It is the oldest park in Houston. For parts of the years of legal segregation, the system of laws that kept Black people and white people apart, it was the only public park in the area that Black people were allowed to use. Juneteenth celebrations are still held there.

Yates was born enslaved in Gloucester County, Virginia, in 1828, as John Henry Yates. His enslaver's child taught him to read, which was against the law. He married Harriet Willis, who was enslaved on a nearby farm, and they had eleven children.

In about 1863, Harriet's enslaver moved to Matagorda County, Texas. Yates could not bear to be separated from his wife and children, so he begged to go along. The Emancipation Park Conservancy, a group that works to keep up the park, says he had already gained his freedom. By its account, he went back into slavery to stay with his family.

When freedom came in June 1865, the family moved to Houston. Yates hauled goods by day and preached at night. In January 1866 freed people organized Antioch Missionary Baptist Church, the first Black Baptist church in Houston. Yates became its first pastor, in 1868 by one account. He bought land for his own house in 1869. In 1885 he started Houston Academy, a school that taught formerly enslaved people reading, writing, arithmetic and trades. He died on December 22, 1897.

Freedom celebrations, and the attack at Norfolk, 1866

Black Americans marked freedom on days of their own, and in Norfolk, Virginia, white men attacked one of those celebrations. Before the war, on July 5, 1852, Frederick Douglass spoke in Rochester, New York. He asked, "What, to the American slave, is your 4th of July?" He answered that the day showed an enslaved person "the gross injustice and cruelty to which he is the constant victim."

Midnight, 1863. The Emancipation Proclamation, President Lincoln's order freeing the enslaved people in the states fighting against the Union, took effect on January 1, 1863. On the night it took effect, Douglass and other Black leaders in the North led large celebrations that began at midnight.

Norfolk. The members of Congress passed a Civil Rights Act in April 1866 over President Andrew Johnson's veto, his refusal to sign it. The members of Congress can still make a bill law if two-thirds of each house vote for it again. On April 16, Black residents of Norfolk, Virginia, marched to celebrate the law. Black Union soldiers marched with them in uniform, carrying their guns.

Someone fired a shot. In the chase that followed, a white Confederate veteran was shot and killed. The records do not say who fired either shot. White men then attacked Black people. They killed at least two Black people, badly

wounded six, and destroyed property that belonged to Black people. The attacks went on through the night until United States Army soldiers stopped them. The records do not give the names of the dead.

Joseph T. Wilson was a Black Union veteran who published a newspaper in Norfolk, the *True Southerner*. He printed his side of the story in it. After that issue came out, white men attacked his printing office and destroyed his press.

Who started Decoration Day?

The first national Decoration Day was May 30, 1868, and the man who ordered it was General John A. Logan, the head of the Grand Army of the Republic, the organization of Union veterans. On May 5, 1868, he ordered its members to decorate the graves of Union soldiers on that day. The first ceremony was held at Arlington National Cemetery in Virginia, on land that had been the home of the Confederate general Robert E. Lee. Americans now call the day Memorial Day.

The veterans' organization itself credited the idea to a member who had written to it about a German custom of decorating graves. Logan's wife, Mary Logan, said the idea came from her visit to a Confederate cemetery in Virginia.

People in several towns say their town did it first, and people in at least three places had decorated soldiers' graves before Logan's order. A page of the National Cemetery Administration, the federal office that runs the national military cemeteries, lists the claims.

Boalsburg, 1864. In October 1864, three women in Boalsburg, Pennsylvania, decorated the graves of soldiers who had died.

Charleston, 1865. In Charleston, South Carolina, Confederate soldiers had held Union prisoners of war at the Washington Race Course, a track for horse racing. The prisoners were kept outdoors in the middle of the track. According to the historian David Blight, at least 257 of them died of disease and of exposure, which means being left outside with no shelter from cold and weather. They were buried quickly in a mass grave, one pit for many bodies, behind the stands.

In April 1865 freed people and Black workmen dug up the bodies and buried each man in his own grave. They built a fence and an arch over the gate that read "Martyrs of the Race Course" (a martyr is a person who dies for a cause).

On May 1, 1865, freed people and white allies from the North held a ceremony there. Blight found the account of it in old papers at Harvard and in Charleston. An account from the College of Charleston says the parade may have included as many as 10,000 people. Among them were 3,000 Black schoolchildren, who sang the song "John Brown's Body."

Columbus, 1866. On April 25 and 26, 1866, women in Columbus, Mississippi, and Columbus, Georgia, held days to put flowers on the graves of Confederate soldiers. In Columbus, Mississippi, they put flowers on Union soldiers' graves too. Mary Ann Williams of the Ladies Memorial Association in Columbus, Georgia, had asked for the day in a letter to the newspaper. Supporters of the Confederacy in the South kept a separate day of their own for the Confederate dead.

KNOWN:	Logan's order of May 5, 1868, started the national day. The three earlier days are each documented, Charleston's most fully, in the newspapers Blight found.
DISPUTED:	Whether Logan copied any of them. The College of Charleston says historians debate whether the Charleston ceremony and the 1868 day are connected. The National Cemetery Administration says recent scholarship points to the women of Columbus, Georgia.

Labor Day, 1882 to 1894

Union workers in New York City held the first Labor Day parade on Tuesday, September 5, 1882. A union is a group of workers who join together to bargain for better pay and conditions. The city's Central Labor Union, a group of unions, planned the parade. People have credited two men with the idea: Peter J. McGuire of the carpenters' union and Matthew Maguire, a machinist, a worker who made and repaired machines. The US Department of Labor states that recent research favors Maguire.

Oregon's lawmakers made Labor Day a state holiday on February 21, 1887. By 1894 Labor Day was a holiday in 23 or 24 states, counting Oregon. A senator brought a bill for a national Labor Day to Congress in August 1893. Senators passed it in June 1894, and members of the House passed it on June 26. President Grover Cleveland signed it on June 28, 1894. It gave federal workers the first Monday in September off.

Cleveland signed it in the first days of a national railroad strike. On June 26, members of the American Railway Union, led by Eugene Debs, had begun refusing to run trains that pulled Pullman sleeping cars. They did it to support the workers at the Pullman factory, who were on strike. On July 3, five days after he signed the bill, Cleveland sent federal soldiers to Chicago to break the strike. Cleveland left no letter or statement that explains why he signed the Labor Day bill.

Columbus Day, 1892, and the New Orleans lynching

The first national Columbus Day came a year and a half after a mob in New Orleans lynched eleven Italian American men, that is, killed them without a trial. The writer Richard Gambino is among those who say President Benjamin Harrison meant the holiday partly to calm Italy and Italian Americans after the killings.

The police chief. Someone shot and killed David Hennessy, the police chief of New Orleans, in October 1890. Police arrested Italian Americans, and 19 were charged, among them a 14-year-old boy. Nine were tried. The jury found six of them not guilty and could not agree about the other three. No solid evidence had been shown against them.

The mob. Thousands of people gathered in the city on March 14, 1891, after the verdicts. A lawyer, William S. Parkerson, told the crowd to "set aside the verdict," the jury's decision. Parkerson led an armed group into the city jail with three other men: Walter Denegre, James D. Houston and John C. Wickliffe.

The armed men killed eleven Italian American men. They shot and clubbed nine of them to death inside the jail. Outside, they hanged Emmanuele Polizzi from a lamppost and Antonio Bagnetto from a tree, and shot both men. Among the mob were John M. Parker, who later became governor of Louisiana, and Walter C. Flower, who later became mayor of New Orleans.

The eleven men were Antonio Bagnetto, James Caruso, Loreto Comitis, Rocco Geraci, Joseph Macheca, Antonio Marchesi, Pietro Monasterio, Emmanuele Polizzi, Frank Romero, Antonio Scaffidi and Charles Traina. Three had been found not guilty. Three had had a mistrial, which means the jury could not agree. Five had never been tried.

No one charged. The grand jury, a group of citizens who decide whether someone should be charged with a crime, worked under Judge Robert H. Marr, a friend of several men in the mob. Its members said they could not identify the killers. They called the mob "several thousands of the first, best, and even the most law-abiding, of the citizens," and they charged no one.

Italy's leaders called home their ambassador, the official who spoke for Italy in the United States. President Harrison agreed to pay $25,000 to the dead men's families. Members of Congress objected, and federal officials paid $2,211.90 to each family.

The proclamation. Harrison proclaimed Friday, October 21, 1892, a general holiday for the people of the United States, in an order dated July 21. It was the 400th anniversary of Columbus's landing in the Americas. He asked schools to lead the celebrations and to fly the flag over every schoolhouse. The proclamation does not mention Italy or New Orleans.

Dances made crimes, boarding schools and a holiday for losing land

Dances made crimes. Commissioner of Indian Affairs Hiram Price issued rules in 1883 that made the Sun Dance and other Native ceremonies "Indian offenses" on the reservations. He wrote them after a letter from Henry Teller, the Secretary of the Interior, the head of the federal department in charge of Native affairs, who had called the ceremonies "old heathenish dances." Heathen was an insulting word for people who were not Christian.

Under the rules, government agents punished a person who held one of these dances by taking away the person's food rations for up to ten days. For later offenses they could take the rations for 15 to 30 days or put the person in the agency jail for up to thirty days. US officials had promised those rations in treaties, in return for the nations' land, which US officials had taken.

The boarding schools. Commissioner Thomas J. Morgan ordered a new holiday for the government's boarding schools for Native children in 1890. From 1879 on, federal officials took Native children from their families and put them in those schools. Federal officials meant the schools to make the children give up their nations' languages and ways. Richard Henry Pratt, the Army officer who opened the Carlisle school in Pennsylvania in 1879, described its aim in 1892: "Kill the Indian in him, and save the man."

In 1891 the members of Congress let officials make rules to force Native children to attend. In 1893 they let officials take away food rations, clothing and payments from parents who refused. Some government agents sent reservation police to seize children.

At the schools, staff cut the children's hair, took their clothes and gave them English names. Staff beat children for speaking their own languages, shut them alone in rooms, kept food from them and washed their mouths out with soap. Interior Department investigators later counted at least 973 children who died at the schools, and their 2024 report says the true number is higher.

A holiday for losing land. Morgan's holiday marked the anniversary of the Dawes Act of 1887. Under that law, federal officials broke up reservation land, the land set aside for Native nations, into farms for single families, and sold the land left over. Native nations held about 138 million acres in 1887. By 1934 they held about 48 million. They had lost about 90 million acres, two-thirds of their land.

BY 1900

If you remember one thing: The first national holidays came between 1863 and 1894, and three of them began as somebody's campaign: Hale's Thanksgiving, the Union veterans' Decoration Day and the New York unions' Labor Day. Juneteenth began in the same years, and no one in Washington made it a holiday for another 155 years.

- 1863: Lincoln proclaims a national Thanksgiving on the last Thursday in November.
- 1865: Granger's order at Galveston, June 19, and the first Juneteenth a year later.
- 1892: the first national Columbus Day, a year and a half after the New Orleans lynching.

Next: The 1900s add Mother's Day and Armistice Day, and a president moves Thanksgiving to help stores.

1900 to 1950

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Presidents and members of Congress added new national days between 1914 and 1941 and fixed the dates of old ones. After a campaign by Anna Jarvis, Mother's Day became a national day in 1914. She spent the rest of her life fighting the businesses that sold flowers and cards for it.

After the First World War, Americans marked November 11, the day the fighting stopped, as Armistice Day. In 1938 lawmakers in Congress made it a holiday set by law, dedicated to world peace. In 1934 President Franklin Roosevelt proclaimed, that is, named in a public order, October 12 of each year as Columbus Day.

Store owners used the holidays to sell. Workers at Macy's department store in New York City marched in the store's first Thanksgiving parade in 1924. In 1939 Roosevelt moved Thanksgiving a week earlier to give stores more shopping days before Christmas. Governors in sixteen states refused to follow him, by the National Archives' count, and in 1941 the members of Congress fixed the date by law.

Who thought of Mother's Day first?

Julia Ward Howe, who wrote "The Battle Hymn of the Republic," came first. In 1872 she started a Mothers' Peace Day, a day for peace held each June 2, and people kept it with services until about 1913. Two more proposals came before Anna Jarvis's. In 1887 Mary Towles Sasseen of Henderson, Kentucky, made a plan for schools to honor mothers. In February 1904 Frank Hering of the Fraternal Order of Eagles, a men's club, gave a speech about a day for mothers. The Eagles still call themselves the "true founders" of Mother's Day.

The law. President Woodrow Wilson approved a joint resolution, a measure that both houses of Congress pass together, on May 8, 1914. It named the second Sunday in May as Mother's Day. The next day, Wilson asked Americans to fly the flag on that day "as a public expression of our love and reverence for the mothers of our country." He asked for nothing more, and since Mother's Day fell on a Sunday, no one got a day off for it.

Anna Jarvis

Who: The woman who campaigned to create Mother's Day and then fought the businesses that made money from it.
When and where: West Virginia and Philadelphia, 1905 to 1948.

She made Mother's Day, then got arrested fighting it.

Anna Jarvis campaigned for years to create Mother's Day, and then spent her money and the rest of her life fighting the people who sold things for it. She was born in Webster, West Virginia, in 1864, during the Civil War. Her mother, Ann Reeves Jarvis, ran Mothers' Day Work Clubs, which cared for sick and wounded soldiers from both armies.

Her mother died in 1905. To honor her, Anna Jarvis began asking for a national day for mothers. She chose the second Sunday in May, because her mother had died near that date.

The first formal Mother's Day church service was held on May 10, 1908, at Andrews Methodist Episcopal Church in Grafton, West Virginia. Jarvis sent 500 white carnations to it. She chose white, she wrote, for the "truth, purity and broad-charity of mother love." That day she was in Philadelphia, at a Mother's Day gathering in the department store of John Wanamaker. National Geographic reports that 15,000 people came, and each one was given a carnation.

Jarvis wrote letters to business owners, politicians, ministers and newspaper editors across the country. In 1914 the members of Congress and President Wilson made the second Sunday in May a national Mother's Day.

The fight. Florists raised the price of carnations before Mother's Day, and card companies sold printed Mother's Day cards. Jarvis wrote, "A printed card means nothing except that you are too lazy to write to the woman who has done more for you than anyone in the world." In the 1920s she wrote that she did not want Mother's Day to become a "burdensome, wasteful, expensive gift day."

Arrested. Police in Philadelphia arrested her in 1925, when she was 61, for disturbing the peace, which means making a public disturbance. She had burst into a meeting of the American War Mothers, a group that sold carnations to raise money on Mother's Day. She brought many lawsuits against groups that used the name "Mother's Day," and she spent much of her money on lawyers.

She started a petition, a list of signatures, in 1943, asking to end the holiday. Later she was placed in the Marshall Square Sanitarium, a hospital in West Chester, Pennsylvania. She stayed there until she died. The National Park Service's account of her life does not say who had her placed there or why. The historian Katharine Antolini found that people connected with the flower and greeting-card businesses paid her bills there. Jarvis died on November 24, 1948, at 84, with no money left. She never had children.

Armistice Day, 1919 and 1938

The fighting in the First World War stopped at 11 o'clock in the morning on November 11, 1918. It stopped under an armistice, an agreement to stop fighting. In November 1919 President Wilson asked Americans to mark the first anniversary as Armistice Day. He asked for parades, public meetings and a short stop in work at 11 a.m.

By 1926 the lawmakers of 27 states had made November 11 a holiday. That year senators and representatives asked the president to call for flags and for ceremonies of "friendly relations with all other peoples."

The 1938 law. The members of Congress made Armistice Day a holiday set by law on May 13, 1938, "dedicated to the cause of world peace." By then all 48 states kept the day. Groups of First World War veterans backed the law.

Why Thanksgiving is the fourth Thursday, not the last

The date fight. Thanksgiving falls on the fourth Thursday of November because of a fight over the date that began in 1939. That year November had five Thursdays, and the last one fell on November 30. President Franklin Roosevelt was worried that a short shopping season before Christmas would slow the country's recovery from the Great Depression. The Great Depression was the long period of hard times in the 1930s, when many Americans had no work. Roosevelt moved Thanksgiving a week earlier, to November 23, to help stores.

The governors of 32 states followed him. The governors of 16 states kept the last Thursday, by the National Archives' count, and a history website called Ghosts of DC counts 23 states that refused. For two years, Americans kept two different Thanksgivings.

The members of Congress ended the confusion in 1941. They set Thanksgiving by law as the fourth Thursday in November. In a year when November has five Thursdays, that is a week before the last one. Roosevelt signed the law on December 26, 1941.

The parade. Workers at Macy's department store in New York City marched to the store in costume on November 27, 1924. They brought floats and animals from the Central Park Zoo. At the end of the parade, Santa Claus arrived at the store. The parade was held again in later years, and in 1933 more than a million people watched it.

New York children had their own Thanksgiving custom before the parade. From about 1870 into the 1920s, on Ragamuffin Day, they dressed in costumes and begged from door to door. Fewer children kept the custom over the years, and the last ragamuffin parade in New York City was held in 1956.

Columbus Day, 1905 to 1937

Angelo Noce, an Italian immigrant in Denver, asked Colorado's leaders to honor Columbus. In 1905 Colorado's governor, Jesse McDonald, proclaimed the first state Columbus Day.

The Knights of Columbus, a Catholic men's organization, and Generoso Pope, an Italian American leader in New York City, asked Congress for a national day. On April 30, 1934, the members of Congress asked the president to proclaim October 12 each year as Columbus Day. They asked for flags and ceremonies. President Roosevelt did so that September, and again in 1937.

Federal workers, the people who work for the national government, did not get the day off until a law passed in 1968 took effect in 1971.

Christmas, Labor Day, Juneteenth and Memorial Day, 1921 to 1937

The first National Christmas Tree. Grace Coolidge, the president's wife, gave the public schools of Washington, D.C., permission in November 1923 to put up a tree near the White House. On Christmas Eve 1923 President Calvin Coolidge lit it. It was the first National Christmas Tree, a fir 48 feet tall from Vermont. About 3,000 people watched.

Labor Day. The 1894 Labor Day law covered only federal workers. As late as the 1930s, unions encouraged workers to strike to make sure they got the day off.

Juneteenth grows smaller. Juneteenth, June 19, marks the day in 1865 when a Union general announced in Galveston that slavery had ended in Texas. A Boston Public Library history says Juneteenth celebrations began to grow smaller in the early 1900s. One reason, it says, is that teachers taught that the Emancipation Proclamation, President Lincoln's 1863 order freeing enslaved people in the states fighting the Union, had ended slavery. During the Great Depression, the same history says, many people were forced to work in cities, and employers there did not want to give workers the day off.

Memorial Day 1937. Decoration Day was by then also called Memorial Day. On that holiday, May 30, 1937, Chicago police shot and killed ten people who were marching in support of striking steelworkers at the Republic Steel plant. Chapter 14 describes the march and the killings.

The dances. Commissioner Hiram Price's 1883 rules, which made the Sun Dance and other ceremonies "Indian offenses" punished by taking away food rations, were still in force. Charles Burke, the Commissioner of Indian Affairs, the official who ran the government's office for Native affairs, sent out an order in 1921 that let the officials in charge of reservations shut down Native dances. He added to it in 1923.

1950 to 2000

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After 1950, Americans argued over whose days the country should mark. In 1968 the members of Congress moved three holidays to Mondays, so that workers got more three-day weekends, and made Columbus Day a federal holiday, a paid day off for the national government's workers. Maulana Karenga and his group, called Us, started Kwanzaa in 1966. Students and marchers in the civil rights movement brought Juneteenth back.

Coretta Scott King, the Southern Christian Leadership Conference, Stevie Wonder and millions of people who signed petitions campaigned for 15 years for a holiday for Martin Luther King Jr. President Ronald Reagan signed that law in 1983. Some state leaders refused the King holiday for years. The last state named a day for King in 1999.

Native people began to mark Thanksgiving as a National Day of Mourning in 1970. In 1992 the members of the city council of Berkeley, California, renamed Columbus Day for Indigenous peoples. Indigenous peoples are the first peoples of a place and their descendants.

Veterans Day and the Monday holidays, 1954 to 1975

On June 1, 1954, the members of Congress renamed Armistice Day as Veterans Day. By then Americans had also fought in the Second World War and the Korean War. The members of Congress chose the new name so that one day would honor the veterans of every war.

Memorial Day, the old Decoration Day, got an official birthplace in 1966. That year the members of Congress and President Lyndon Johnson named Waterloo, New York, the "birthplace" of Memorial Day. People in several other towns still say their town held the first one.

The Monday holidays. Under a 1968 bill called the Uniform Monday Holiday bill, several holidays would move to Mondays. Robert McClory, a member of the House of Representatives from Illinois, took it through the House Judiciary Committee. He was a Republican, a member of one of the country's two main political parties. That committee of House members worked on the bill before the full House voted.

Four business groups backed it: the US Chamber of Commerce, the National Association of Manufacturers, the National Association of Travel Organizations and the National Retail Federation. McClory said it would give families more time together and more chances to visit historic places. Representative H. R. Gross of Iowa asked whether store clerks would get the days off or spend them selling. McClory answered that labor unions supported the bill.

President Johnson signed it on June 28, 1968, and it took effect in 1971. Under the law, Washington's Birthday fell on the third Monday in February, Memorial Day on the last Monday in May and Veterans Day on the fourth Monday in October. In the same law, the members of Congress made Columbus Day a federal holiday, on the second Monday in October.

McClory had also wanted to rename Washington's Birthday "Presidents' Day." He dropped the idea to win the votes of Virginia's lawmakers. No act of Congress and no order of a president changed the federal name, although many people call it Presidents' Day.

Veterans' groups objected to moving Veterans Day, and 46 states kept November 11 or went back to it. In 1975 the members of Congress put Veterans Day back on November 11.

Kwanzaa, from 1966

Maulana Karenga, a Black activist who led a group called Us in the Los Angeles area, created Kwanzaa in 1966. The Root, a news site, says members of Us created it together. Kwanzaa runs for seven days, from December 26 to January 1. In 1965 Karenga wrote seven principles, the Nguzo Saba, and families talk about one of them on each night of Kwanzaa. The first is Umoja, the Swahili word for unity. Swahili is a language spoken in East Africa. The name Kwanzaa comes from a Swahili phrase meaning "first fruits." The last day ends with a feast called Karamu.

At first Karenga said that Kwanzaa gave Black Americans "an alternative to the existing holiday." In 1997 he wrote that it "was not created to give people an alternative to their own religion or religious holiday."

The crimes. A California court convicted Karenga in 1971 of two crimes, felony assault and false imprisonment. Felony assault is a serious crime of attacking a person. False imprisonment means holding people against their will so they cannot leave. The charges came from what was done to two women at his home in Inglewood, California. The Root describes the crimes as torturing the two women. The women, Deborah Jones and Gail Davis, were members of his group.

At the trial, Jones and Brenda Karenga testified that Karenga believed the two women were plotting to poison him. Jones testified that the two women were ordered to take off their clothes. Then they were whipped with an electrical cord and beaten with a karate baton. By one fuller version of the report, they were also hit on the head with toasters.

She testified that a hot soldering iron, a tool that heats metal to join wires, was put in Davis's mouth and pressed against Davis's face. One of Jones's own big toes was squeezed in a vise, a clamp that tightens with a screw. Jones said Karenga himself put detergent and running water hoses in their mouths. The news report of her testimony does not say who did each of the other acts.

Karenga was sentenced to one to ten years in prison and was released on parole in 1975. Parole is an early release from prison under rules. He has denied the charges and called them "trumped up," which means made up, and "politically motivated." He says he was one of many people the FBI went after for political reasons.

After the trial. People went on keeping Kwanzaa after the trial and Karenga's years in prison. The US Postal Service issued the first Kwanzaa stamp in 1997, and that year President Bill Clinton sent the first Kwanzaa message from a president. Counts of how many Americans celebrate Kwanzaa are only rough estimates. One estimate in 2009 put the number between 500,000 and 2 million. In a 2019 National Retail Federation poll, 2.6 percent of the people who celebrated a winter holiday said they planned to celebrate Kwanzaa.

Juneteenth comes back, 1968 to 1980

People in the civil rights movement brought Juneteenth back. In Atlanta, students in the movement wore Juneteenth Freedom buttons. The Poor People's March on Washington ended on June 19, 1968. The marchers took the celebration home to their own cities, among them Milwaukee and Minneapolis. People in those two cities still hold some of the largest Juneteenth events.

Al Edwards, a Texas state lawmaker from Houston and a Democrat, a member of the other main political party, brought a bill to make Juneteenth a Texas state holiday. Texas lawmakers passed it in 1979, and Governor Bill Clements signed it. The first Juneteenth celebration held by the state of Texas took place in 1980.

Thanksgiving as a day of mourning, from 1970

The Wampanoag leader Frank James and other Native people gathered on Cole's Hill in Plymouth, above Plymouth Rock, on Thanksgiving Day, November 26, 1970, for the first National Day of Mourning. About 500 Native people came.

The gathering has been held every Thanksgiving since then. United American Indians of New England organizes it. Many who come go without food from the evening before.

The arrests, 1997. Plymouth police stopped the marchers from walking through the center of town on Thanksgiving Day 1997. Earlier that day, people dressed as Pilgrims had walked the same road. The marchers' group and other accounts say police used pepper spray, a spray made from hot peppers that burns the eyes and skin. One witness, Chris Groden, told the Associated Press that police sprayed mace, a similar spray, straight into several people's eyes.

The police arrested 25 marchers. They charged them with disorderly conduct, which means disturbing public order, and with unlawful assembly, which means gathering where the law does not allow it. The records do not name the officers.

The settlement. The charges were dropped in October 1998, when the town and the marchers' group reached an agreement. Plymouth officials agreed that the marchers could walk on Thanksgiving without a permit, forever. The town paid $135,000 in all. By the marchers' group's account, $100,000 went to a fund for teaching about Metacom, $20,000 to the lawyers of the American Civil Liberties Union and $15,000 to two plaques.

One plaque stands on Cole's Hill. The other stands in Post Office Square in Plymouth and tells of Metacom, the Wampanoag leader. He was killed in 1676 in an ambush led by the English captain Benjamin Church. The English then put his head on a pole in that square and left it there for 25 years.

Frank James (Wamsutta)

Who: A Wampanoag leader whose speech was stopped in 1970 and who started the National Day of Mourning.

When and where: Massachusetts, 1970.

Invited to speak, then told his speech was out of place.

In September 1970, Massachusetts officials planning the celebration of the 350th anniversary of the Pilgrims' landing invited Frank James to speak. Governor Francis Sargent was among the organizers. James was Aquinnah Wampanoag, and he led the Wampanoag Tribe of Gay Head. His Wampanoag name was Wamsutta, the same name as Massasoit's eldest son, a Wampanoag leader of the 1600s.

James wrote his speech from the Pilgrims' own records, including William Bradford's history and *Mourt's Relation*, a book of Pilgrim writings printed in London in 1622. It began: "I speak to you as a man, a Wampanoag Man."

In the speech, James wrote that within four days of landing, the Pilgrims "had robbed the graves of my ancestors and stolen their corn and beans." He wrote that "before 50 years were to pass, the Wampanoag would no longer be a free people."

The organizers read the speech in advance. A state commerce official told him that "the theme of the anniversary celebration is brotherhood and anything inflammatory would have been out of place." Inflammatory means likely to make people angry. United American Indians of New England says he refused to give a speech written by a public relations person, someone paid to shape what the public thinks of an event. James did not speak at the celebration.

On Thanksgiving Day 1970, he spoke instead on Cole's Hill in Plymouth, beside the statue of the Wampanoag leader Massasoit. Accounts differ on whether he read the same speech or a changed one. That year he founded United American Indians of New England.

Fifteen years to get King's day, 1968 to 1983

It took 15 years of campaigning to make Martin Luther King Jr.'s birthday a national holiday. James Earl Ray, a white man, shot and killed King in Memphis on April 4, 1968, and pleaded guilty the next year. Chapter 25 tells the killing and the later dispute over whether others planned it.

The first bill. Four days after the killing, Representative John Conyers of Michigan brought a bill to Congress for a national holiday on King's birthday, January 15. The House members on the committee in charge of the bill took no action on it. Members of the Southern Christian Leadership Conference, the group King had led, gathered 3 million signatures on petitions in the early 1970s. In 1973 lawmakers in Illinois made the first state King holiday. State lawmaker Harold Washington was the bill's sponsor, the lawmaker who brought it forward.

Four or five votes short. Coretta Scott King, King's widow, spoke for the holiday before Congress in 1979. That November the members of the House voted 252 to 133 for it. Under the rules for that vote, the bill needed two-thirds of the votes, and it fell short. The Congressional Research Service, a group that studies laws for Congress, counts it as four votes short. The King Institute at Stanford University counts five.

Stevie Wonder's song. The singer Stevie Wonder released "Happy Birthday," a song asking for the holiday, in 1980. His money paid for a lobbying office, an office whose staff pressed lawmakers to vote for the bill. Petitions reached more than 6 million signatures. On January 15, 1983, more than 100,000 people rallied at the Washington Monument.

The Senate. The members of the House passed the bill 338 to 90 on August 2, 1983. In the Senate, Jesse Helms of North Carolina led the opposition. He called for the FBI's secret recordings of King to be made public. The King Institute says he wanted to hurt King's reputation. A court had sealed the recordings until 2027. He also gave senators a 300-page paper that claimed King had ties to communists, people who wanted the government to own all land and businesses. Senator Daniel Patrick Moynihan called it a "packet of filth" and stamped on it.

Helms's plan to open the FBI records lost, 3 votes to 90. Other senators said another paid day off would cost too much.

Senators passed the bill 78 to 22 on October 19, 1983. Reagan had wanted a day of remembrance, not a paid holiday. He signed the law on November 2, 1983. The first King holiday was kept on January 20, 1986.

The states that held out, 1986 to 1999

The 1983 law made King's birthday a federal holiday. Each state's leaders decided for themselves whether to keep the day. By January 1989, 44 states kept a King holiday.

Some state lawmakers joined King's day to a day for Confederate generals. The Confederacy was the group of Southern states that fought the Union in the Civil War to keep slavery. Alabama's holiday was named for Robert E. Lee and King, and Mississippi's for King and Lee. Arkansas lawmakers combined King and Lee from 1985. Virginia's lawmakers called their day Lee-Jackson-King Day, for Lee, the Confederate general Thomas "Stonewall" Jackson and King.

In Arizona, Governor Bruce Babbitt made King's day a paid state holiday by his own order in 1986. His successor, Evan Mecham, canceled it in January 1987. Mecham said Babbitt had no legal power to create it. Groups across the country moved their conventions out of Arizona. Stevie Wonder and other musicians refused to perform there.

The Super Bowl. Arizona voters voted no in November 1990 on two questions on the ballot about a King holiday. Officials of the National Football League then took the 1993 Super Bowl away from Arizona and moved it to Pasadena, California. *PHOENIX* magazine says the game was to be played in Tempe, next to Phoenix, and the Pima County Public Library says Phoenix. People there had expected about $200 million from the game. In November 1992, 61 percent of Arizona voters approved Martin Luther King Jr./Civil Rights Day. Arizona is the only state whose voters approved the holiday themselves.

New Hampshire's lawmakers made a "Civil Rights Day" in 1991. In June 1999 they renamed it for King. New Hampshire was the last state to name a day for King.

Liz Byrd

Who: A Cheyenne schoolteacher and lawmaker who brought the King holiday bill to Wyoming's legislature nine times.
When and where: Wyoming, 1981 to 1990.

Nine bills in nine years before Wyoming said yes.

Liz Byrd brought a bill for a Martin Luther King Jr. holiday before the Wyoming legislature, the state's lawmakers, nine times. In 1990 the lawmakers passed it. Her full name was Harriett Elizabeth Byrd. Her grandfather Charles Rhone had come to Wyoming Territory as a boy in 1876.

Officials at the University of Wyoming did not accept her application. Her son believes it was because she was Black. She earned her teaching degree at West Virginia State College in 1949. Cheyenne school officials turned her down at first too, WyoHistory says probably because of her race. They hired her in 1959. A reference book, *Black America: A State-by-State Historical Encyclopedia*, calls her the first fully certified full-time Black teacher in Wyoming. She taught second grade for most of her 37 years as a teacher.

She served in the Wyoming House of Representatives from 1981 to 1988. In 1989 she became the first Black member of the Wyoming Senate.

One Republican who voted against her King holiday bill was Loren Willford. He said a Wyoming holiday should honor someone who had been in Wyoming, such as Chief Washakie, a Shoshone leader, or Jim Bridger, a fur trapper and guide. In 1990 the lawmakers passed the holiday with a longer name, "Martin Luther King, Jr./Wyoming Equality Day." Her son said she never liked the added words but took them rather than lose the bill again. Another account gives the year as 1991.

> She died on January 27, 2015, at 88. Governor Matt Mead ordered flags lowered in her honor.

Pride marches, and Columbus Day renamed, 1970 to 1999

Several thousand people marched for gay and lesbian rights in New York City on June 28, 1970. They walked 51 blocks, from Sheridan Square in Greenwich Village to the Sheep Meadow in Central Park. Their permit came two hours before the start. The march came one year after police raided the Stonewall Inn, a gay bar in the city. The organizers' counts ranged from 3,000 to 20,000, and the *New York Times* put the march on its front page. Craig Rodwell and three others had proposed a yearly march in 1969.

Pride Month. The New York march has been held every June since 1970. People call these marches Pride marches. On June 11, 1999, President Bill Clinton proclaimed June "Gay and Lesbian Pride Month." It was the first time a president had proclaimed it.

Columbus Day renamed. Columbus's men enslaved Taino people, as Chapter 1 tells, and Native groups and others protested a national day in his honor. In 1977, people at a United Nations conference in Geneva, Switzerland, discussed replacing Columbus Day with a day for Indigenous peoples.

The 500th anniversary of Columbus's first voyage came in 1992. That year a group from the San Francisco Bay Area called Resistance 500 persuaded the city council of Berkeley, California, to declare October 12 a "Day of Solidarity with Indigenous People." Solidarity means standing together. The council members renamed the day Indigenous Peoples Day.

BY 2000

If you remember one thing: After 1950 the fights were over whose days counted, and the longest one, for Martin Luther King Jr.'s birthday, took from 1968 to 1983 in Congress and until 1999 in the last state.

In 1950: the federal holidays were New Year's Day, Washington's Birthday, Memorial Day, the Fourth of July, Labor Day, Armistice Day, Thanksgiving and Christmas, and not one was named for a Black American or for the Native nations.

In 2000: King's day was a federal holiday and every state had named a day for him, Juneteenth was a Texas state holiday, Berkeley kept Indigenous Peoples Day, and a president had proclaimed Pride Month.

Next: Juneteenth becomes the newest federal holiday in 2021, and two presidents disagree about Columbus Day.

2000 to Today

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Since 2000 the members of Congress have added one federal holiday, Juneteenth, in 2021. Lawmakers in Virginia and Arkansas stopped joining King's day to days for Confederate generals. Alabama and Mississippi still honor Robert E. Lee on the same Monday as King.

In 2021 President Joe Biden became the first president to proclaim Indigenous Peoples' Day. In 2025 President Donald Trump proclaimed Columbus Day only, and the Associated Press reported that he stayed silent on Juneteenth. Some states also added days for other communities, such as Lunar New Year in New York's schools and Diwali in California. As of September 2026, Juneteenth, King's day and Columbus Day are all still federal holidays.

The end of most combined days, 2000 to 2020

South Carolina's governor, Jim Hodges, signed a law in May 2000 that gave all state workers King's day off. South Carolina was the last state to do so. Before then, state workers there could choose King's day or one of three Confederate holidays.

Virginia. Virginians kept Lee-Jackson-King Day from 1984 to 2000. It honored the two Confederate generals and King on the same day. From 2000 to 2020 Virginia had two separate holidays. In April 2020 Virginia's governor, Ralph Northam, signed a law under which Lee-Jackson Day ended and Election Day became a state holiday. Northam said the old day "commemorates a lost cause. It's time to move on." A lost cause is a cause that has failed. CNN reported that both generals had enslaved people and had fought to keep slavery.

Arkansas lawmakers moved Lee's day off King's day in 2017. Alabama and Mississippi still honor Lee's birthday on the same Monday as King's day.

Juneteenth becomes a federal holiday, 2021

Juneteenth, the June 19 celebration of the end of slavery in Texas, became a federal holiday in 2021. On May 25, 2020, Derek Chauvin, a Minneapolis police officer, killed George Floyd. That June, many Juneteenth gatherings called for changes in the way police do their work.

The vote. The senators passed the Juneteenth National Independence Day Act in June 2021. On June 16, 2021, the members of the House passed it 415 votes to 14, according to the House's own record of the vote. All 14 votes against it came from Republicans. President Biden signed it on June 17, 2021. It was the first new federal holiday since King's day in 1983.

The leaders of each state still decide whether state workers get the day off. In 2025 the Congressional Research Service, which studies laws for Congress, counted 31 states and Washington, D.C., with a permanent paid Juneteenth holiday. PolitiFact, a fact-checking news site, reported that count in June 2026. Alabama was the most recent state to add it, in May 2025. PolitiFact

also reported that New Mexico names the day each year, and that the governors of Kansas and Kentucky have declared it. Private employers do not have to give the day off.

Juneteenth 2025. President Trump signed no proclamation for the day on Juneteenth 2025, the Associated Press reported. The White House had said earlier that he planned to sign one. "I'm not tracking his signature on a proclamation today," his press secretary, Karoline Leavitt, said. President Trump said that America has "too many non-working holidays," CBS News reported.

Park fees. In December 2025, CBS News reported a change in the National Park Service's list of days with free entry to the parks. The 2026 list no longer included Martin Luther King Jr. Day and Juneteenth, which President Biden's officials had made free-entry days. It added June 14, which is Flag Day and President Trump's birthday. The change affects only park fees. Both days are still federal holidays.

Opal Lee

Who: A retired Fort Worth teacher who organized Juneteenth celebrations for about 40 years and campaigned for it to become a national holiday.
When and where: Fort Worth, Texas, 1939 to today.

At 89 she walked toward Washington, 2.5 miles at a time.

Opal Lee was the best-known campaigner for making Juneteenth a national holiday. She was there when President Biden signed the law on June 17, 2021, and he asked the crowd to stand and cheer for her. She was born Opal Flake in Marshall, Texas, in 1926, and her family moved to Fort Worth when she was nine.

In June 1939, her parents bought a house at 940 East Annie Street in Fort Worth, in a white neighborhood. On June 19, 1939, when she was 12, a white mob of about 500 people drove the family out. The mob dragged

the family's furniture out and burned it, and burned the house to the ground.

Lee remembered that police officers were there and did not stop the mob. She said her father came home from work with a gun. The police told him that if he fired, they would let the mob have the family. No record names anyone in the mob, and no record shows that anyone was arrested.

She became a teacher after she graduated from Wiley College in 1953. When she retired in 1977, she helped found the Tarrant County Black Historical and Genealogical Society. She began planning Fort Worth's Juneteenth celebrations, and she led them for about 40 years.

The walk. In 2016, when she was 89, she organized a walk of 1,400 miles from her home in Texas to Washington, D.C. The walk drew attention across the country to Juneteenth. Her family worried about her health. So she walked about 300 miles herself, in pieces of 2.5 miles each. The 2.5 miles stand for the two and a half years it took for the news of freedom to reach Texas. Her petition for the holiday gathered 1.6 million signatures, according to the National Women's History Museum.

President Biden gave her the Presidential Medal of Freedom, the country's highest honor for a civilian, a person who is not in the military, on May 3, 2024. That year builders put up a new house for her on the lot where her family's house had burned. She paid $10 for the land, KERA News reported in 2024.

On Juneteenth 2026, nearly 600 people walked Opal's Walk for Freedom in Fort Worth, the Fort Worth Report counted. Lee did not attend. Her family led the walk while she rested, a few months before her 100th birthday.

Two presidents, two Octobers: Indigenous Peoples' Day and Columbus Day, 2021 to 2026

President Biden proclaimed Indigenous Peoples' Day on October 8, 2021. He wrote that for generations "Federal policies systematically sought to assimilate and displace Native people and eradicate Native cultures." To assimilate people means to make them give up their own ways and take on someone else's, to displace them means to push them off their land, and to eradicate means to wipe out. Biden also proclaimed Columbus Day each year from 2021 to 2024. In 2020 Colorado's lawmakers replaced Columbus Day in their state with Cabrini Day, named for an Italian American nun.

October 2025. President Trump proclaimed Columbus Day in October 2025 and did not mention Indigenous Peoples' Day. His proclamation said of Columbus, "Left-wing radicals toppled his statues, vandalized his monuments, tarnished his character." NPR reported a count by the Pew Research Center, a group that studies public life. In 2025 Pew researchers counted 30 states and three territories that recognize Columbus Day in some way. They also counted 17 states and Washington, D.C., that honor Indigenous people on that day. Some states do both.

The National Day of Mourning is still held at noon on Cole's Hill in Plymouth every Thanksgiving. The most recent one was on November 27, 2025. Only Native people speak, and many of those who come go without food from the evening before. Frank James's granddaughter, Kisha James, helps organize it.

Pride Month, and new days for more communities

President Barack Obama proclaimed Lesbian, Gay, Bisexual and Transgender Pride Month in 2009. President Biden added more groups to the name in 2022.

Halloween, on October 31, is not a federal holiday, but Americans spend billions of dollars on it. The National Retail Federation, a trade group for stores, has surveyed shoppers each year since 2005. In September 2026 it reported that Halloween spending had reached a record $13.1 billion in 2025. Its survey forecast $13.5 billion for 2026. These are estimates from a survey, not counts of actual sales.

Americans also began to keep more communities' days. Cinco de Mayo, the fifth of May, marks the Mexican army's defeat of French troops at Puebla, Mexico, on May 5, 1862. Latinos in California began celebrating it during the Civil War, according to David Hayes-Bautista, a scholar at the University of California, Los Angeles. Hayes-Bautista says it is scarcely observed in Mexico. It is not Mexico's Independence Day, which is September 16.

New York's governor, Kathy Hochul, signed a law in September 2023 that closes all the state's public schools for Lunar New Year, the start of the year in calendars based on the moon. California's governor, Gavin Newsom, signed a law in October 2025 making Diwali a state holiday from 2026. California was the third state to do so, after Pennsylvania and Connecticut. Diwali is the festival of lights, kept mostly by Hindus, Sikhs, Jains and Buddhists.

THE THREE QUESTIONS, ANSWERED

1. **Who decided when Americans would give thanks, and why did it take until 1941 to fix the date?** Governors and church elders set the days one at a time, until Lincoln proclaimed a national Thursday in 1863 at Sarah Josepha Hale's request. Every president after him named the day by proclamation, so when Franklin Roosevelt moved it in 1939 to help stores, sixteen states refused by

the National Archives' count, and Congress ended the confusion with the 1941 law.

2. **Which days did Black Americans make their own, from election days to Juneteenth?** Black governors' election days in New England from about 1749, January 1 from 1808, July 5 in New York in 1827, August 1 after 1834, and Juneteenth from 1866, which Congress made a federal holiday in 2021.
3. **Why did President Harrison proclaim the first Columbus Day in 1892, and why do some states no longer keep it?** He proclaimed it for the 400th anniversary of Columbus's landing, a year and a half after a New Orleans mob lynched eleven Italian American men, and some writers say the holiday was partly meant to calm Italy. Native groups and others object to honoring a man whose men enslaved Taino people. Since Berkeley in 1992, cities and 17 states have used the day to honor Indigenous peoples, and some of those states keep Columbus Day too.

What changed: Ten eras ago the Faith Keepers and the Sun-priests set the days. By 2021 a majority in Congress did, and the days they chose included one, Juneteenth, that freed people in Texas had kept for 155 years without asking anyone.

FIVE DATES TO KEEP

Fall 1621: the harvest meal at Plymouth, called the first Thanksgiving only from 1841.

October 3, 1863: Lincoln proclaims a national Thanksgiving on the last Thursday in November.

June 19, 1865: General Granger announces in Galveston that slavery has ended in Texas.

October 21, 1892: the first national Columbus Day.

June 17, 2021: President Biden signs the law making Juneteenth a federal holiday.

NOT KNOWN: No surviving record gives the first name of the boy Brown, killed by a Pope wagon in Boston on November 5, 1764, or says who was driving it.

Where to go: Cole's Hill in Plymouth, Massachusetts, where the statue of Massasoit stands and the National Day of Mourning is held at noon every Thanksgiving.

Next: This is the last of the 37 chapters. Chapter 1, Exploration and Discovery, is where the grid begins.

How We Know What Happened

Nobody alive today saw the Pueblo Revolt, the founding of Jamestown or the first Thanksgiving. Nobody alive saw Cahokia when it was the largest city north of Mexico. Everything known about those times comes from what people left behind: papers, objects, bones, buildings, photographs, and the stories that families and nations told their children.

Historians are people who study the past and write about it. They gather that evidence and check it. Each kind of evidence keeps some facts and loses others, so a historian gathers as many kinds as possible and checks them against each other. Then the historian states three things: what is known, what is not known, and where the evidence disagrees. Those three show up all through this book. A fact the records show is stated as a fact. A fact no record kept is marked "not known." Where records conflict, the better-supported version comes first, and the other is given with its source.

Papers written at the time

A record made when something happened, by someone who was there, is called a primary source. Letters, diaries, court papers and newspaper reports are primary sources. A book written later by someone who studied those papers is called a secondary source. Historians trust a primary source most, but they still check it, because the person who wrote it saw only part of what happened.

A diary is a book in which a person writes down each day as it happens. Amelia Stewart Knight kept one in 1853, when she traveled by wagon from Iowa to Oregon with her husband and seven children. She wrote about rain, mud, dust, sick children and hard river crossings. She was pregnant the whole way. She did not mention it in the diary until her son Adam was born, on September 18, 1853. The family had reached Oregon the day before. A diary records what its writer chose to write down, and a careful reader notices what was left out.

Court records contain the judges' own words about how they punished people. On July 9, 1640, a court in Virginia judged three servants who had run away together from their master, Hugh Gwyn, and been caught. The judges gave the two European men four more years of work as punishment. They sentenced the third man, John Punch, an African, to serve "for the time of his natural Life." Historians count that ruling as the first known sentence of lifetime slavery in Virginia, because Punch had to serve until he died and his master could sell him or give him away. It is also the first record in which Virginia judges punished one man more harshly than the others because of his race. Nobody wrote down why.

Bones, ruins and things in the ground

Archaeology is the study of the past through the things people left in the ground: tools, pots, buildings, trash and graves. Archaeologists dig slowly and record exactly where each object lay, because where something was found is part of what it shows. For most of the thousands of years that people lived in North America, nobody wrote anything on paper. Most of what is known about those years comes from archaeology.

Sometimes bones confirm a written account. At Jamestown in Virginia, the English colonists ran out of food in the winter of 1609 to 1610. Powhatan, the leader of the Native nations around them, had cut off trade and surrounded their fort. The colonists ate their horses, their dogs, rats and leather. Written accounts from the time say starving colonists also ate the bodies of the dead. In 2013, researchers at the Smithsonian announced a find from that winter. The bones of a girl showed that starving colonists had eaten from her body. Researchers call her Jane. No record states her real name. For that winter, the written accounts and the bones agree.

Stories passed down

Oral tradition is history that people keep by telling it, from parents to children, across many lifetimes. Native nations kept their laws, treaties and histories this way for thousands of years before Europeans arrived, and many still do. Records written by colonists describe Native nations from the outside. A nation's own tradition records what its people remember and how they understood events. Where this book gives a Native nation's history, it gives the nation's own account beside the written one and says which is which.

Chapter 2, Native Nations, and Chapter 5, Migration Across America, describe a 2023 study in which scientists tested old horse bones to find their age. The bones showed that many Plains nations had horses by the early 1600s, as the nations' own histories say.

Counting

Numbers look exact, but every count depends on who did the counting, what they counted, and why. When two counts of the same thing differ, historians look for the reason.

A count changes when someone decides what to leave out. At Hoover Dam, built in the 1930s, the official count is 96 workers killed on the job. That count comes from the Bureau of Reclamation, the federal office that runs the dam, and it leaves out deaths from heat and sickness. The records list 42 more workers as dying of pneumonia, a sickness of the lungs. Workers blamed those deaths on carbon monoxide, a gas from engine exhaust, in the tunnels. The true number of workers who died is higher than 96, and nobody knows exactly how many.

The words "at least" in a count mean the real number may be higher. Where this book gives a number, it gives the range when the sources give a range, and it names who counted.

Who wrote the records, and who is missing

For most of American history, the people who wrote things down and kept the papers were the people with power: officials, officers, owners and ministers. Their records describe other people from the outside, and many people appear in them without names. In late August 1619, an English ship called the *White Lion*, which had sailed from the Netherlands, landed captive Africans at Point Comfort, Virginia. The colonist John Rolfe wrote that it brought "20. and odd Negroes." His record lists none of their names. The first record of Africans in English Virginia records them only as a number.

When an account names a person, this book names the person. When the only record names nobody, the book states the event without a name, because an invented name would be a false fact.

When sources disagree

When two records tell a story differently, historians check who wrote each one and whether that person was there. They check how long after the event it was written, whether the writer checked it with the people involved, and what the writer gained from telling it one way. A record made soon after, by a witness who checked it, is the better source. When neither is clearly better, a careful historian reports both. Chapter 25, Rights and Movements, contains the clearest case in this book: the two versions of Sojourner Truth's 1851 speech, one printed three weeks later by a man who knew her, the other written twelve years later by a woman who never checked it with her.

Questions with no answer

Some questions cannot be answered, because no record of the answer was ever made or none survived. When that happens, a careful historian states that the record is missing and leaves the gap empty instead of guessing. No record made in 1619 names any of the Africans who landed that year, and the records do not settle whether the Virginia colonists treated them as servants, who could earn their freedom after some years, or as slaves for life. Virginia's laws did not yet use the word "slave." This book leaves that gap open, and every gap like it.

A quotation keeps the words the person actually used, even words people today find hurtful, because changing them would change the record. Rolfe's "Negroes" was the common word of his time for Black people. Words made to insult a whole people are not repeated. Instead, the account says plainly what the word called people.

The past keeps changing

What is known about the past changes as new evidence is found. New tests find new facts in old bones. Lost letters turn up in attics and in archives, the places where old papers are kept. People who were left out of the old records are found again when someone looks for them. Each new piece of evidence can confirm an old account, correct it, or show that it was wrong.

Glossary

Every hard word the chapters explain, from A to Z, with the chapter where it first appears.

0 to 9

10 a.m. policy: the 1935 Forest Service rule that crews bring each fire under control by 10 o'clock the morning after it was spotted (Chapter 3) 215

3D printer: a machine that builds a solid object out of many thin layers, one on top of another (Chapter 8) 570

55th parallel: an east-west line across North America. North of it, hunters went on using the atlatl beside the bow for more than 1,000 years (Chapter 8) 512

A

AAA: the Agricultural Adjustment Administration, the federal office that paid farmers to grow less in the 1930s (Chapter 16) 1084

AAA purge: the firing, in February 1935, of the lawyers at the Agricultural Adjustment Administration who ruled that landlords had to keep the same tenants (Chapter 16) 1084

aadizookaanag: the Ojibwe sacred stories, told only in winter (Chapter 34) 2814

abbots: the heads of monasteries, the houses where monks live and pray together (Chapter 30) 2392

abolished: ended by law (Chapter 2) 126

abolition society: a group that worked to end slavery (Chapter 9) 591

abolitionist: someone who worked to end slavery (Chapter 5) 352

abortion: a medical procedure that ends a pregnancy (Chapter 25) 1909

about going: about to die (Chapter 19) 1256

abridged: cut back (Chapter 20) 1341

absentee ballots: ballots mailed to voters who vote from home (Chapter 20) 1390

abstract art: art that shows shapes and colors instead of things (Chapter 32) 2660

Abstract Expressionism: painting that shows no real objects, as Pollock's did (Chapter 32) 2679

abstractions: ideas (Chapter 30) 2414

abuse of power: using the powers of an office for wrong purposes (Chapter 20) 1372

academy: a school for older pupils, a step above the town school and a step below a college (Chapter 30) 2359

Acadia: a French colony on the Atlantic coast of what is now Canada (Chapter 5) 346

Acadians: French-speaking Catholic farm families of Nova Scotia (Chapter 5) 346

accessory: a person who helps a crime happen without doing the act (Chapter 21) 1442

accountability: judging schools by how their students scored on tests (Chapter 20) 1390

achievement gap: the difference between the test scores of different groups of children, such as rich and poor children (Chapter 30) 2474

Achilles tendon: the cord just above the heel (Chapter 24) 1676

Acoma Pueblo: the Acoma people's town, with their fields around it (Chapter 2) 102

acre: a piece of land a little smaller than a football field (Chapter 1) 39

acre-foot: enough water to cover one acre of land one foot deep (Chapter 3) 240

act of God: a natural event that no person could be blamed for or could have prevented (Chapter 27) 2068

act of oblivion: a Cherokee act of 1810 in which the clans forgave every life one clan still owed another. Oblivion means forgetting (Chapter 21) 1434

acting governor: the official running a colony for the king while the governor was away (Chapter 18) 1182

active duty: serving full-time in the military (Chapter 22) 1585

active shooter: a person who is still attacking people (Chapter 30) 2474

active volcano: a volcano that is erupting, throwing out gas and melted rock (Chapter 9) 633

activists: people who campaign for a cause (Chapter 6) 452

actour: an old spelling of actor, used in the 1665 court record (Chapter 34) 2823

addiction: a need for a drug so strong that stopping is very hard (Chapter 15) 1039

adelantado: a Spanish office that a man received by signing a contract with the king (Chapter 20) 1308

ADHD: attention-deficit/hyperactivity disorder, a condition doctors find in a child who has trouble paying attention, or is overactive and acts without thinking, badly enough to cause problems in more than one place (Chapter 36) 3058

admirable: in Roger Williams's English, worth wondering at (Chapter 11) 726

adobe: sun-dried mud brick (Chapter 6) 395

adulation: too much praise (Chapter 20) 1322

adulterated: made impure or mixed with something that should not be in it (Chapter 15) 1015

adultery: sex between a married person and someone outside the marriage (Chapter 21) 1412

advance: money paid before the work is done (Chapter 18) 1204

advertising agency: a business that plans advertisements for sellers and buys space for them in newspapers and magazines (Chapter 18) 1193

advertising exchange: the online market where websites sell ad space to advertisers (Chapter 19) 1291

advisory: a public warning (Chapter 36) 3058

advocacy groups: groups that campaign for a cause (Chapter 7) 495

adze: a carving tool with a curved blade set across the end of its handle (Chapter 32) 2598

Aesop's fables: a set of old animal stories (Chapter 30) 2351

affirmative action: letting colleges consider an applicant's race, in a limited way, to gain the benefits of having students of many races (Chapter 25) 1909

affluence: wealth (Chapter 20) 1372

African Methodist Episcopal Church: a church run by Black Methodists (Chapter 12) 832

Afro: hair combed out into a round shape (Chapter 35) 2959

aftershocks: smaller quakes that follow a large one (Chapter 27) 2041

agency: the office of the US official who ran a reservation (Chapter 2) 134

agency prison: the jail at the government office on a reservation (Chapter 33) 2746

agent: an organizer for a society (Chapter 25) 1778

Agent Orange: a plant-killing chemical that American forces sprayed over South Vietnam, tainted with a poison called dioxin (Chapter 22) 1575

aglet: the small metal tip of a lace, so it can pass through a hole (Chapter 35) 2912

agnostic: a person who holds that no one can know whether God exists (Chapter 29) 2304

agricultural experiment station: a place where scientists run tests to improve farming (Chapter 9) 604

AI (artificial intelligence): computer programs that do tasks such as recognizing a face or writing a paragraph (Chapter 8) 570

AI chatbot: a program that answers typed questions in sentences (Chapter 8) 570

aide: a helper to a general (Chapter 22) 1522

AIDS: a disease caused by a virus that destroys the body's defenses against other illnesses (Chapter 4) 310

air brake: a brake worked by air pressure (Chapter 11) 745

air shafts: narrow gaps between buildings meant to let in air and light (Chapter 7) 480

air traffic controllers: the workers who guide airplanes in and out of airports (Chapter 14) 963

airship: a giant aircraft held up by a gas lighter than air, with engines to push it and a cabin for passengers (Chapter 13) 868

alcalde: an official who acted as the judge in a Spanish town (Chapter 21) 1409

alchemy: the older practice that came before chemistry, in which alchemists tried to turn ordinary metals such as lead into gold (Chapter 13) 847

alcoholic: a person who cannot stop drinking (Chapter 28) 2127

alcoholism: the illness of being addicted to alcohol (Chapter 28) 2165

aldermen: the members of a city council (Chapter 18) 1178

Algonquian: a large group of related Native languages (Chapter 9) 584

alibi: proof that a person was somewhere else when a crime happened (Chapter 4) 275

aliens: people from other countries who were not citizens (Chapter 25) 1770

alimony: the money a person pays a former husband or wife after a divorce (Chapter 18) 1214

All Pueblo Council of Governors: the governors who lead the 20 Pueblo nations of New Mexico and Texas (Chapter 2) 102

alliance: a promise to stand together against enemies (Chapter 2) 108

Allies: the countries fighting Germany in the two world wars (Chapter 10) 691

allotment: cutting land that a whole nation owned together into private pieces (Chapter 2) 99

almanac: a book printed once a year with a calendar and tables of the sun, moon and stars (Chapter 6) 407

almshouse: a public home for poor people (Chapter 9) 596

alpha particles: the tiny particles americium gives off (Chapter 8) 561

alpha radiation: the kind of radiation radium gives off (Chapter 13) 868

alphabet houses: Oak Ridge houses built in a few standard types named by letters, such as the A house and the B house (Chapter 6) 431

ALS: a disease that destroys the nerve cells that carry orders from the brain to the muscles (Chapter 25) 1909

altar boy: a boy who helped the priests during church services (Chapter 2) 102

alter, modify, or repeal: change or cancel (Chapter 19) 1250

alternating current (AC): electricity that flows back and forth many times a second (Chapter 8) 541

aluminum oxide: a compound of aluminum and oxygen (Chapter 13) 863

amaranth: a plant whose tiny seeds are eaten like grain (Chapter 15) 1039

amateur sport: the contests for unpaid athletes (Chapter 36) 3028

amateurs: people who were never paid to play a sport (Chapter 2) 149

Amazons: the women warriors of old Greek stories (Chapter 35) 2928

ambassador: an official who speaks for one country in another (Chapter 21) 1442

ambrotype: a small photograph on glass (Chapter 22) 1543

amendment: a change to the Constitution (Chapter 3) 240

American national: a person who can live in the United States and carry an American passport but is not a citizen (Chapter 23) 1634

americium: a radioactive element, used in ionization smoke detectors (Chapter 8) 561

ammonium nitrate: a chemical used as fertilizer that can explode (Chapter 27) 2078

amplifier: a device to make a signal stronger (Chapter 9) 614

amputate: to cut off an arm or leg (Chapter 32) 2627

amputation: an operation that cuts off an arm or leg (Chapter 22) 1534

anarchist: a person who believes there should be no government (Chapter 14) 929

ancestral lands: the lands a people's ancestors lived on (Chapter 3) 215

Ancestral Puebloan people: the ancestors of today's Pueblo peoples (Chapter 2) 99

anchor: the person who reads the news and introduces the reports on a news program (Chapter 31) 2572

anchor store: a big store that draws shoppers to the smaller shops around it (Chapter 18) 1222

Anderson Bill: the 1947 California law that repealed the state's law on segregated schools (Chapter 20) 1356

Andes: the high mountains of South America (Chapter 1) 74

anemia: having too few red blood cells, which leaves a person weak and tired (Chapter 13) 868

anemic: pale and weak because the blood is not carrying enough of what the body needs (Chapter 14) 941

anesthesia: the use of a drug so that a patient does not feel pain during an operation (Chapter 9) 633

anesthetic: a drug that puts a patient to sleep so the patient feels no pain (Chapter 22) 1534

Anglican: a member of the Church of England (Chapter 29) 2213

aniline dyes: dyes named after aniline, one of the coal-tar chemicals used to make them (Chapter 35) 2940

Anishinaabe: a family of related nations that includes the Ojibwe, the Odawa and the Potawatomi (Chapter 15) 986

annealing: heating copper, hammering it while the heat keeps it soft, letting it cool, and doing the same again and again (Chapter 35) 2908

annex: to take over as part of the United States (Chapter 1) 64

annexation: moving a city's legal boundary out to take in neighboring land (Chapter 6) 443

annihilate: wipe out (Chapter 20) 1341

annual percentage rate: the cost of a loan over a full year as a share of the amount borrowed (Chapter 17) 1160

annually: each year (Chapter 30) 2367

annuities: the food and goods US officials promised a tribe every year in a treaty, as payment for land (Chapter 3) 215

anode: a battery's negative end (Chapter 8) 570

Anschluss: the takeover of Austria by Nazi Germany in March 1938 (Chapter 18) 1214

anteroom: a small room next to a classroom (Chapter 25) 1871

anthem: an official song, or a church song (Chapter 25) 1909

anthracite: a hard kind of coal from northeastern Pennsylvania, which people then called stone coal (Chapter 10) 677

anthropologist: a scientist who studies how groups of people live (Chapter 2) 102

anthropology: the study of human peoples and their ways of life (Chapter 9) 614

antibiotic: a medicine that kills bacteria (Chapter 26) 1990

Anti-Masonic Party: the party formed by men who opposed the Freemasons (Chapter 20) 1333

antinomian: a word that means "against the law," used by Anne Hutchinson's opponents to say her teaching let a saved person ignore the rules (Chapter 29) 2213

antiproton: a tiny particle with the opposite electric charge of the proton, a particle in the center of every atom (Chapter 9) 614

antisemitic: hateful toward Jewish people (Chapter 9) 614

anti-Semitism: hatred of Jews (Chapter 29) 2274

antitrust laws: laws against trusts and other combinations of companies (Chapter 19) 1256

anxious seat: seats at the front of a church where people worried about their souls sat in front of everyone (Chapter 29) 2256

aplastic anemia: radiation damage to the bone marrow until the marrow stops making blood cells (Chapter 13) 868

apocryphal: probably made up (Chapter 3) 200

Apollo: the Moon program (Chapter 1) 82

apoplexy: an old word for a sudden stroke (Chapter 21) 1434

apostate: a person who has abandoned their faith (Chapter 25) 1767

apostle: one of the highest leaders of the Church of Jesus Christ of Latter-day Saints (Chapter 29) 2256

Apostles' Creed: a short statement of Christian belief (Chapter 30) 2367

Appalachian backcountry: the land in and near the Appalachian Mountains, far from the coast (Chapter 4) 267

apparent suicides: deaths in which people appeared to have killed themselves (Chapter 4) 320

appeal: a request for a higher court to change a ruling (Chapter 2) 170

appeals court: a higher court that checks the rulings of lower courts (Chapter 4) 320

appendicitis: an infection of a small part of the gut (Chapter 1) 44

appointed: provided free, said of a lawyer for a person who cannot pay for one (Chapter 20) 1372

apportionment population: the count used to divide the seats in the House among the states (Chapter 20) 1390

appraiser: a person whose job is to judge what a property is worth (Chapter 6) 431

apprehended: caught (Chapter 4) 320

apprentice: a young worker who learns a trade from a master (Chapter 4) 267

apprentice musician: a boy learning the work of a band musician (Chapter 33) 2746

apprentices (British West Indies): freed people who, after 1834, still had to work about 40 hours a week for their former enslavers, without pay (Chapter 37) 3101

apprenticeship: working beside the older people of the community until a child has their skills (Chapter 30) 2345

aqua vitae: a strong liquor like brandy (Chapter 28) 2127

aquanauts: people who live and work underwater (Chapter 1) 82

aqueduct: a channel built to carry water a long way, as the Croton Aqueduct (Chapter 6) 413

arbitrary power: power used without limits (Chapter 20) 1317

arbitration: settling a dispute by letting a neutral panel decide (Chapter 23) 1605

arbitrator: a person both sides agree will settle a dispute (Chapter 36) 3045

arc lamp: a lamp that makes light from a spark between two rods (Chapter 8) 541

arcade: a covered walkway (Chapter 18) 1214

archaeologist: a scientist who digs up and studies the things people left behind (Chapter 1) 17

archbishop: a bishop who leads the main church area of a region (Chapter 4) 260

Archbishop of Canterbury: the head of the Church of England (Chapter 4) 260

archive: a building that stores old papers (Chapter 1) 21

Archivist of the United States: the federal official who certifies and publishes amendments (Chapter 25) 1909

ardent spirits: strong liquor such as rum, gin and whiskey (Chapter 28) 2138

arks: the boats that carried anthracite down the Lehigh River to Philadelphia (Chapter 10) 677

armistice: an agreement to stop fighting (Chapter 22) 1559

Armistice Day: November 11, the day the fighting in the First World War stopped (Chapter 37) 3119

armorer: a man who makes and repairs weapons (Chapter 17) 1123

armory: a building where weapons are made or stored (Chapter 8) 528

armory and arsenal: places where the government made and stored guns (Chapter 22) 1543

Army Corps of Engineers: the part of the Army that builds dams and waterways (Chapter 2) 170

Army National Guard: a part-time state army force that can be sent to war (Chapter 22) 1585

ARPANET: a computer network paid for by the Department of Defense to link computers at universities, the network the internet began with (Chapter 8) 561

arrange: to write out the parts of a song for each voice (Chapter 33) 2746

arranged marriage: a marriage planned by other people rather than by the couple (Chapter 7) 480

arsenic: an element and a strong poison (Chapter 13) 868

arson: the crime of setting a fire on purpose (Chapter 27) 2104

arsonist: a person who sets fires on purpose (Chapter 21) 1456

art dealer: a person who buys and sells art for a living (Chapter 32) 2679

artery: a large tube that carries blood from the heart (Chapter 26) 1983

Article 5: the part of the NATO treaty that states that an attack on one member is an attack on all of them (Chapter 23) 1656

articles: the written rules of a group (Chapter 6) 404

Articles of Confederation: the country's plan of government before the Constitution (Chapter 20) 1322

articles of impeachment: formal charges that could lead to a president's removal from office (Chapter 20) 1372

artificial silk: a man-made cloth that looked like silk (Chapter 35) 2949

artillery: soldiers who fire the big guns (Chapter 8) 553

artillery firing tables: charts that told soldiers how to aim their big guns (Chapter 8) 553

Aryans: the name the Nazis gave the white people they taught were a superior race (Chapter 36) 3028

asbestos: a mineral that separates into thin, strong fibers, used to fireproof and insulate buildings (Chapter 6) 431

asbestosis: a lung disease from breathing asbestos fibers, which scar the lungs. It brings shortness of breath, coughing and lasting lung damage (Chapter 19) 1280

ASCAP: a group that collected money for songwriters (Chapter 33) 2759

Ashley's Hundred: the men who answered William Ashley and Andrew Henry's 1822 newspaper advertisement for one hundred "enterprising young men" (Chapter 1) 44

asiento: the contract that gave British traders the only right, for thirty years, to carry enslaved Black people into Spain's colonies in the Americas and sell them there (Chapter 23) 1602

ask no quarter and give none: no one would be spared, even people who tried to surrender (Chapter 1) 44

assaulted: attacked (Chapter 20) 1390

assembly: in colonial Virginia, the colony's lawmakers (Chapter 15) 991

assembly center: a short-term holding camp, such as the Tanforan racetrack, where horse stalls were turned into rooms for families (Chapter 20) 1356

assembly line: a line of workers where each worker did one job over and over (Chapter 8) 553

assessment: a test or written review of how a child is doing (Chapter 30) 2474

assigns: anyone a master sold or handed a servant to (Chapter 24) 1670

assimilate: to make people give up their own ways and take on someone else's (Chapter 29) 2274

assumption: the national government taking over the states' war debts (Chapter 16) 1062

astronomy: the science of the stars and planets (Chapter 6) 407

asylum: a hospital where people thought to be insane were locked up (Chapter 21) 1442

at two or three removes: through other groups (Chapter 32) 2679

Athabascan migration: the long move of the ancestors of the Navajo and Apache south from western Canada and eastern Alaska (Chapter 5) 333

atheism: the belief that there is no God (Chapter 29) 2304

atlatl: a spear-thrower stick that holds the back end of the spear during the throw (Chapter 8) 512

atoll: a ring of small coral islands (Chapter 13) 886

atom: one of the tiny pieces, far too small to see, that everything is made of (Chapter 9) 614

atomic bomb: a bomb that gets its huge power from the tiny centers of atoms (Chapter 6) 431

Atomic Energy Commission (AEC): the federal agency in charge of atomic energy and nuclear weapons (Chapter 10) 702

at.óow: the Tlingit word for sacred clan property: crests and the stories, songs and names that go with them (Chapter 33) 2713

atrophy: organs shrinking and stopping working (Chapter 25) 1795

attained: reached (Chapter 25) 1871

attendance zones: the areas whose children go to each school (Chapter 20) 1390

Attorney General: the top lawyer of the nation, or of a state (Chapter 4) 310

auction block: the platform where people were sold (Chapter 37) 3101

audit: a check of records (Chapter 29) 2324

authentic: real and true to its roots (Chapter 32) 2660

authenticated: proven by records (Chapter 14) 920

autobahns: Germany's highways (Chapter 11) 764

autobiography: a book a person writes about their own life (Chapter 2) 134

automated teller machine: a machine that pays out cash without a bank clerk (Chapter 17) 1153

autopsy: an examination of a body to find the cause of death (Chapter 10) 711

Auto-Tune: a computer program that moves a singer's off-key notes onto the right pitch (Chapter 33) 2795

avarice: greed (Chapter 20) 1322

awl: a thin pointed tool for punching holes (Chapter 13) 844

aye: yes (Chapter 20) 1372

B

B movies: low-budget films (Chapter 34) 2877

Baby Bells: the seven regional companies made from AT&T's local telephone companies in 1984 (Chapter 19) 1280

baby boom: the years 1946 to 1964, when about 76 million babies were born in the United States (Chapter 7) 495

bachelor: a man who has never married (Chapter 4) 298

back a loan or a bond: to promise to pay the lender or the buyer if the borrower cannot (Chapter 16) 1067

back pay: pay that workers were owed and had not been paid (Chapter 14) 963

backcountry: the valleys and hills behind the older coastal settlements (Chapter 4) 267

background actors: extras with no lines (Chapter 34) 2889

background check: a search of national criminal records that a gun store must run before it sells a gun (Chapter 29) 2324

backlot: the outdoor area of a film studio's sets and storage buildings (Chapter 33) 2795

Bacon's Rebellion: a revolt in Virginia in 1676 (Chapter 16) 1052

bacteria: tiny living things, one kind of germ (Chapter 13) 899

bad faith: dishonestly (Chapter 8) 570

badge of inferiority: a mark that one group is lower than another (Chapter 20) 1341

badly vented stove: a stove that did not let its smoke out properly (Chapter 1) 74

baffle: a hollow iron wall at the back of the Franklin stove that warmed room air (Chapter 8) 524

bail: money paid to a court so that a person can leave jail until the trial (Chapter 4) 267

bailout: government money given to businesses in trouble (Chapter 17) 1160

baize: a thin wool cloth (Chapter 35) 2934

Bakelite: the first plastic made entirely from chemicals, rather than from plants or animals (Chapter 8) 553

bale: a large bundle of cotton pressed tight and tied (Chapter 8) 528

ballad: a song that tells a story (Chapter 12) 795

ballast: heavy dirt carried low in a ship to keep it steady (Chapter 3) 182

ballot question: a question put on the ballot for voters to decide, such as California's Proposition 22 (Chapter 11) 773

band: a group of families who lived and traveled together (Chapter 2) 134

bando de buen gobierno: the order of good government, a set of rules Governor Miró issued in New Orleans (Chapter 35) 2928

baneful: harmful (Chapter 20) 1322

banish: to order a person to leave and never return (Chapter 2) 102

bank examiner: a person who checks that banks follow the rules (Chapter 27) 2093

bank holiday: a day when payments on loans were not due (Chapter 37) 3101

bank holiday (1933): Roosevelt's closing of every bank in the country on March 6, 1933 (Chapter 17) 1143

bank note reporters, counterfeit detectors: printed guides that listed each bank's discount and described the fake notes going around (Chapter 17) 1128

bank run: a rush by many people to take their money out of banks (Chapter 16) 1074

banknote: paper money printed by a bank (Chapter 17) 1123

bankrupt: unable to pay your debts, so that your lenders can take what you own (Chapter 7) 502

bankruptcy: the court process for a person or a company that cannot pay its debts (Chapter 10) 691

bankruptcy law: the rules for what happens when a person or a company cannot pay its debts (Chapter 10) 691

banter: to haggle over the price (Chapter 18) 1188

baptism: the Christian ceremony with water that makes a person a member of a church (Chapter 7) 463

baptize: to make a person a member of the Christian church in a ceremony with water (Chapter 2) 102

bar: the body of lawyers allowed to practice in a court (Chapter 25) 1795

bar iron: iron hammered into bars (Chapter 13) 852

bar lines: the lines that divide music into equal parts of time (Chapter 33) 2723

barbarian: an insult for a person thought to be wild and without learning (Chapter 32) 2619

barbarity: cruelty (Chapter 25) 1767

Barbary states: Algiers, Tunis, Tripoli and Morocco, states on the coast of North Africa (Chapter 23) 1605

baristas: the workers who make the coffee drinks (Chapter 14) 974

barracoon: a pen for captives (Chapter 24) 1712

barred out: locked a teacher out of the schoolhouse until he agreed to a holiday (Chapter 36) 2999

barricaded subject: a person shut in a room (Chapter 30) 2474

barrios: Mexican American neighborhoods (Chapter 30) 2442

barter: swapping one good for another (Chapter 16) 1049

base: a running game with bases (Chapter 36) 2999

baseless allegations: claims with nothing true behind them (Chapter 14) 941

Basque Country: a mountain region on the border of Spain and France (Chapter 18) 1173

bastinado: a punishment in which men threw a person down, tied him and beat the soles of his bare feet with sticks. James Leander Cathcart took 28 blows and lost four toenails (Chapter 23) 1605

bastions: towers that stuck out from a wall, where archers could stand and shoot at attackers (Chapter 22) 1499

bathyscaphe: a deep-diving craft (Chapter 1) 82

battalion: an army unit of a few hundred soldiers (Chapter 3) 200

batteries: gun positions (Chapter 24) 1712

battery electric car: a car that runs on electricity alone (Chapter 10) 711

batting average: how often a player got a hit, so .263 means about 26 hits in every 100 times at bat (Chapter 36) 3019

bayonet: a blade fixed to the end of a rifle (Chapter 2) 134

bayou: a slow-moving stream through low, wet land (Chapter 5) 346

BC: years counted back before the year 1, so 1700 BC to 1100 BC is about 3,700 to 3,100 years ago (Chapter 12) 783

Bell Labs: the research laboratory of the telephone company AT&T (Chapter 8) 553

bellows: giant bags that squeeze air into a fire (Chapter 8) 517

benefactors: the people who gave the money (Chapter 30) 2351

benefit of clergy: an old English rule that let a first-time offender who could read escape a death sentence once (Chapter 21) 1427

benevolence: kindness and help given to other people (Chapter 29) 2256

benign offices: kind duties (Chapter 25) 1795

Bering Strait: the narrow strip of sea between Asia and North America (Chapter 1) 36

besieged: surrounded a town or fort and cut it off (Chapter 2) 108

bibles: the chunks of blubber whalers cut up for boiling (Chapter 10) 677

Biennale: an international art show held in Venice, Italy (Chapter 32) 2695

Big Bang: the hot beginning of the universe (Chapter 9) 633

Big Dipper: the group of stars that points to the North Star (Chapter 33) 2736

bigamy: the crime of marrying a second wife while the first is still living (Chapter 29) 2274

bigot: a person who hates or looks down on people who believe differently (Chapter 29) 2287

bigotry: hatred of people for who they are or what they believe (Chapter 22) 1534

bilboes: iron rings locked around the ankles and joined to a bar, so the person could not walk away (Chapter 21) 1412

bilingual teaching: classes taught in both Spanish and English (Chapter 30) 2442

bill: a proposed law (Chapter 3) 196

Bill of Rights: the first ten amendments to the Constitution (Chapter 20) 1322

billing: the size and place of a performer's name on the posters (Chapter 34) 2861

bills of credit: the paper money Massachusetts lawmakers printed from 1690 (Chapter 17) 1115

biochemist: a scientist who studies the chemistry inside living things (Chapter 9) 648

birch rods: thin sticks, or bundles of twigs, cut from a birch tree and used to beat children (Chapter 30) 2351

birth control: ways to keep from becoming pregnant (Chapter 23) 1634

birth defect: a problem in a baby's body that is there when the baby is born (Chapter 3) 229

bishop: a church leader in charge of the priests and churches of a whole area (Chapter 4) 275

bit: a small silver coin, one of the eight parts of a piece of eight, so two bits made a quarter of a dollar (Chapter 17) 1113

bitcoin: a digital coin that no government and no bank puts out (Chapter 17) 1160

Black Codes: laws Southern lawmakers passed from November 1865 that let officials arrest, hire out and force to work Black people who could not prove they had a job (Chapter 14) 929

Black Drink: a tea made from the leaves of yaupon holly or dahoon holly (Chapter 28) 2120

Black governor or king: a leader Black communities in New England elected for themselves, a governor where white men elected the colony's governor and a king where the king of England appointed it (Chapter 37) 3091

black hole: a place in space where gravity is so strong that not even light can get out (Chapter 9) 648

black lead: a kind of pencil (Chapter 32) 2605

black lung: a disease caused by breathing coal dust, which scars the lungs. In the worst form, masses of scar tissue leave the miner "gasping for every breath" (Chapter 10) 702

black powder: an early kind of gunpowder (Chapter 4) 283

Black Sunday: the dust storm of April 14, 1935 (Chapter 3) 215

black vomit: blood from the stomach, mixed with stomach acid and vomited up black, in the late stage of yellow fever (Chapter 6) 407

blackface: dark makeup that made a white face look Black (Chapter 33) 2736

blackface minstrelsy: shows in which white actors darkened their faces with burnt cork and acted Black people as fools (Chapter 33) 2736

blackjack: a short heavy club (Chapter 22) 1559

blacklist: a list of workers that employers agree never to hire (Chapter 19) 1256

blackmail: to make people pay money by threatening to tell their secrets (Chapter 20) 1356

bladder: where urine is held (Chapter 22) 1575

blank: a gun shell with gunpowder but no bullet (Chapter 1) 64

blasphemous: insulting to God (Chapter 29) 2304

blasphemy: insulting God (Chapter 21) 1412

blast furnace: a furnace where air blown into the fire makes it hot enough, about 3,000 degrees Fahrenheit, to melt iron out of rock (Chapter 8) 517

blatant: done openly (Chapter 22) 1585

blight: run-down property (Chapter 21) 1484

blighted: a legal label for an area that officials judged run-down enough to tear down (Chapter 6) 443

block booking: the studios' practice of selling films to theatres only in large bundles (Chapter 34) 2861

blockade: closing off a harbor with a line of warships (Chapter 20) 1341

blockbusting: agents and speculators pushing white families to sell and move out by telling them Black families were moving in, then selling the same houses to Black families for much more (Chapter 6) 443

bloodletting, letting blood: cutting open a vein, most often in the arm, and letting blood run out into a bowl (Chapter 26) 1954

bloomers: a dress that reached the knee, worn over loose pants gathered at the ankle, named after Amelia Bloomer (Chapter 35) 2940

blower and striker: the blacksmith's helper who pumps the bellows and swings the heavy hammer (Chapter 8) 535

blowout: an uncontrolled burst of oil and gas up a well (Chapter 3) 229

blubber: the thick layer of fat under a whale's skin (Chapter 10) 677

blue-collar jobs: jobs working with one's hands in factories, mines and building trades (Chapter 14) 963

bluff: a high, steep bank (Chapter 6) 407

BMI: the group radio station owners started in 1939 to collect fees for songwriters (Chapter 33) 2759

Board of Fire Wards: a group of town officers for fires (Chapter 6) 404

Board of Regents (New York): the New York state board that oversees the state's schools and licenses doctors (Chapter 20) 1372

Board of Regents (Smithsonian): the group that governs the Smithsonian (Chapter 9) 596

Board of Supervisors: the officials who made San Francisco's laws (Chapter 25) 1795

Board of Trade: the group of British officials in London who oversaw the colonies and, later, British ships (Chapter 20) 1317

boarders: people who paid to sleep in a family's rooms (Chapter 7) 480

boarding around: a teacher living with the students' families, staying with one family and then the next, as part of the pay (Chapter 7) 474

boarding school: a school where children live as well as study. Federal officials put Native children taken from their families in government boarding schools to make them give up their nations' languages and ways (Chapter 2) 134

boardinghouse: a house where people paid to sleep and eat (Chapter 4) 275

boat people: Vietnamese who escaped in small boats (Chapter 4) 310

bob: a short hairstyle of the 1920s (Chapter 35) 2949

bobbin: a spool that holds thread (Chapter 16) 1074

bobbin boy: a boy who changed the spools that held the thread in a cotton mill (Chapter 16) 1074

body camera: a camera worn on an officer's uniform (Chapter 21) 1484

bog iron: iron ore that workers collected from swamps and ponds (Chapter 13) 848

boiler: a closed tank of water heated by a fire to make steam (Chapter 8) 528

boll weevil: a beetle that eats cotton plants (Chapter 5) 376

bolts: iron shackles (Chapter 28) 2127

bona fide: real (Chapter 18) 1193

bond: a paper promise to repay a loan with interest (Chapter 11) 745

bond (in court): money paid to a court so a person can go home. The court keeps it if the person breaks a promise (Chapter 25) 1909

bond (in slavery): held in slavery (Chapter 20) 1309

bondage: slavery (Chapter 24) 1670

bondholders: people who had lent money by buying bonds (Chapter 17) 1136

bone marrow: the soft center of the bones that makes blood cells (Chapter 13) 868

bonus: an extra payment for war service (Chapter 16) 1084

Book of Common Prayer: the Church of England's prayer book (Chapter 30) 2367

boomers: settlers who entered the Unassigned Lands against the law before 1889 (Chapter 5) 363

booming out: the Mohawk ironworkers' name for traveling to bridge and skyscraper jobs wherever the steel work was (Chapter 6) 418

boomtown: a town whose number of people grows very fast, in a few years (Chapter 6) 413

booster: a person who promoted a city to bring in money and business (Chapter 6) 418

bootleggers: people who made or sold liquor against the law (Chapter 28) 2165

borax: a white powder made from a mineral (Chapter 26) 1961

Border Patrol: the federal agents who guard the border (Chapter 4) 310

bosun chairs: seats hung from a cable (Chapter 12) 815

botanic garden: a garden where plants are grown so they can be studied (Chapter 1) 44

botanist: a plant scientist (Chapter 9) 581

bots: accounts run by computer programs (Chapter 31) 2585

bottomlands: the low, flat lands beside a river (Chapter 10) 702

bound laborer: a person who had to work for whoever bought them until their term ended (Chapter 4) 267

bound out: made to work for a farmer until turning 21 (Chapter 18) 1188

bound servant: someone who had to work for a family without pay until a set number of years ran out (Chapter 33) 2736

bound workers: workers who could not leave their work for years or for life (Chapter 4) 267

bounty: a reward paid for each dead animal, or for each pound of a crop (Chapter 3) 215

bow priests: the Zuni priests who handled dealings with outsiders (Chapter 20) 1304

bowels: the long tube in the belly that food passes through after the stomach (Chapter 26) 1949

boycott: refusing to buy, use or handle something as a protest (Chapter 11) 756

Bozeman Trail: a road opened in 1863 for white settlers to reach the Montana gold fields, through Lakota country (Chapter 2) 134

bracero: a Mexican farm worker hired under contract. The Spanish word means a man who works with his arms (Chapter 4) 298

brakeman: the crew member who set the brakes and joined the cars (Chapter 11) 745

branding: burning the skin with a hot iron to leave a mark that never goes away (Chapter 4) 260

brass: a yellow metal made by melting copper and zinc together (Chapter 8) 516

breadbasket colonies: Pennsylvania, New York and New Jersey, where farmers grew more wheat than their families needed (Chapter 15) 995

breadline: a line for free food (Chapter 15) 1015

break ties: choose between students who ranked the same (Chapter 20) 1390

breakdown: a minstrel dance (Chapter 34) 2851

breaker: the wooden building where coal was broken and sorted (Chapter 10) 681

breaker boys: boys who sat on benches in the breakers and picked slate, a kind of rock, out of the coal (Chapter 14) 929

breaking on the wheel: a way of putting a person to death in which the executioner broke the person's arms and legs with a heavy bar or club and tied the body to a large wooden wheel (Chapter 21) 1412

breechcloth: a strip of cloth worn around the hips (Chapter 36) 3008

breeches: pants that end at the knee (Chapter 33) 2723

breeding: each year taking seed only from the plants farmers chose and planting it again, so the crop changed a little at a time (Chapter 15) 986

brethren: brothers, or brothers and sisters (Chapter 2) 116

Bretton Woods system: the agreement the governments of many countries made after World War II, under which foreign governments could trade dollars for gold at $35 an ounce (Chapter 17) 1153

brew-house: a building for making beer (Chapter 28) 2127

brewing: the making of beer (Chapter 7) 471

brief: a written argument filed with a court (Chapter 19) 1291

brigade: a larger group of soldiers made up of several regiments (Chapter 33) 2746

brine: very salty water, used to keep meat (Chapter 11) 734

Brittany: a part of France (Chapter 22) 1502

Broad Arrow: the mark on the king's trees, three slashes cut with a hatchet (Chapter 3) 189

broadband: a fast internet connection that is always on (Chapter 7) 502

broadside: a single large printed sheet (Chapter 26) 1954

broiler: a young chicken raised for its meat (Chapter 15) 1015

broke the buck: a money market fund's shares fell below $1 (Chapter 17) 1160

broker: a person who arranges sales for others (Chapter 5) 363

Bronze Star: a medal for brave or useful service in war (Chapter 32) 2679

brothel: a house where people paid for sex (Chapter 21) 1442

Brother: a Jesuit who was not a priest (Chapter 30) 2348

brushes: the sliding electrical contacts that older motors used to carry current to their turning part (Chapter 8) 541

bubble: a time when prices climb far above what things are worth and then fall (Chapter 16) 1102

budget surplus: a year when federal tax money coming in is more than federal spending (Chapter 16) 1096

buffalo runners: what the hide hunters called themselves (Chapter 3) 200

bugging: hiding microphones in a place to listen in (Chapter 31) 2572

building code: a set of rules for how strong buildings must be (Chapter 27) 2093

building trades unions: groups that carpenters, plumbers, ironworkers and other builders join to bargain together for jobs and pay (Chapter 12) 825

bull boat: a boat made of buffalo hide (Chapter 1) 44

bulletin: a small printed booklet (Chapter 15) 1015

bulwark: a strong wall that protects (Chapter 25) 1778

bum blockade: the Los Angeles police chief's 1936 use of his officers at the state line to turn back poor migrants (Chapter 5) 376

bumpers: glasses filled to the brim and drunk off at once (Chapter 28) 2133

bundle: to build one program into another (Chapter 19) 1280

bungalow: a house of one main floor with a low roof and a front porch, and sometimes rooms tucked under the roof (Chapter 7) 485

Bureau of American Ethnology: a government office in Washington whose staff studied Native peoples (Chapter 36) 2981

Bureau of Indian Affairs: the federal office in charge of the government's dealings with Native nations (Chapter 2) 158

Bureau of Labor Statistics: a federal office whose staff gather facts about work (Chapter 8) 553

Bureau of Reclamation: the federal office whose staff run Hoover Dam (Chapter 3) 240

burgess: an elected member of the assembly (Chapter 20) 1309

burial mound: a mound people built over the graves of their dead (Chapter 13) 844

burned in effigy: setting fire to dummies made to look like people (Chapter 31) 2533

burning at the stake: chaining a person to an upright post with wood piled around it and setting the wood on fire, so the person burned to death (Chapter 21) 1422

burning lens: a glass lens that bends sunlight to one hot point (Chapter 13) 855

bushel: a measure of grain or other dry goods that fills a large basket (Chapter 3) 200

busing: carrying children by bus to schools outside their neighborhoods (Chapter 25) 1871

busk: a long, flat piece of steel, wood or whalebone down the front of a corset (Chapter 35) 2934

busk (ceremony): William Bartram's name for the Green Corn Ceremony (Chapter 21) 1406

bust: a sculpture of a person's head and shoulders (Chapter 32) 2643

Butler Act: the 1925 Tennessee law that made it unlawful to teach evolution in the state's public schools (Chapter 20) 1356

butte: a steep hill with a flat top (Chapter 9) 581

buy out an indenture: pay to free a servant from the rest of his years of work (Chapter 4) 260

buying on contract: buying a house while the seller keeps the deed until the last payment, so that one missed payment loses the down payment, every monthly payment and the house (Chapter 7) 495

buying on credit: taking the goods home and paying for them later (Chapter 18) 1174

bystanders: people who were only watching (Chapter 24) 1712

C

cabaret card: the New York City police license a performer needed to work in any club that sold alcohol (Chapter 33) 2759

Cabinet: the group of the president's top officials who each run a department (Chapter 14) 941

cabinet office: the job of heading one of the main federal departments (Chapter 2) 170

cabinetmakers: the skilled makers of fine furniture (Chapter 35) 2928

cable: a message sent by wire under the ocean (Chapter 31) 2559

cable television: wires that carried many channels into homes (Chapter 18) 1214

Cabrini Day: the day Colorado's lawmakers put in place of Columbus Day in 2020, named for an Italian American nun (Chapter 37) 3135

cacao: the bean that chocolate is made from (Chapter 15) 989

cache: a hidden store of food (Chapter 3) 200

cacos: Haitian fighters who rose against the American occupation (Chapter 23) 1634

cadaver: a dead body that doctors and medical students cut open to study the human body (Chapter 21) 1442

cadet: a student training to be an army officer (Chapter 36) 3008

cage crinoline: a frame of steel hoops that held a skirt out wide (Chapter 35) 2940

Cahokia: a city on the Mississippi River near present-day St. Louis (Chapter 1) 17

caisson disease, or the bends: the illness that struck men who came up out of the air under pressure too quickly, when bubbles of gas formed in their blood, causing great pain in the joints and sometimes paralysis or death (Chapter 12) 807

caissons: huge boxes, open underneath, sunk to the bottom of a river, where men dug out the riverbed in air pumped in under pressure (Chapter 12) 807

Cajun: a descendant of the Acadians in Louisiana, from the word Acadian (Chapter 5) 346

calico: cotton cloth printed with colored patterns (Chapter 18) 1182

California cuisine: cooking built on fresh food from local farms (Chapter 15) 1028

California Gold Rush: when crowds of people went to California to look for gold (Chapter 4) 283

California Labor Commissioner: the state office that handles pay disputes (Chapter 11) 773

call in loans: ask borrowers to pay back money right away (Chapter 16) 1067

camas: a root that people dug up and ate (Chapter 18) 1170

Camino Real: the Royal Road, which ran about 1,600 miles north from Mexico City (Chapter 5) 336

camp meeting: an outdoor revival where families pitched tents and stayed for several days of preaching (Chapter 29) 2256

camphene: a cheap lamp fuel of one part turpentine, four parts alcohol and a little camphor, whose fumes caught fire easily (Chapter 10) 677

Canaan: in the Bible, the land God promised to the people of Israel after they escaped from slavery in Egypt (Chapter 33) 2736

canal: a waterway that people dug so boats could travel where no river ran (Chapter 1) 74

cannabis: the plant marijuana comes from (Chapter 28) 2165

cannibalism: eating human flesh (Chapter 4) 310

canning: sealing food in a container and then heating it, so it keeps without spoiling (Chapter 14) 941

canopy: a cloth roof hung over a bed (Chapter 35) 2912

cantharides: a poison made from blister beetles (Chapter 32) 2643

Cantino map: a map made in 1502 that already shows a shape like Florida (Chapter 1) 21

cap-and-trade: a state program that makes polluters pay for permits to pollute (Chapter 11) 773

capital: money or property that an owner uses to make more money (Chapter 16) 1067

capital laws: the laws whose penalty was death (Chapter 20) 1309

capitol: the building where a state's lawmakers meet (Chapter 4) 275

Capitol: the building where Congress meets (Chapter 2) 108

captains: the party workers under a district leader (Chapter 20) 1356

captives: people seized by force and traded away with the other goods (Chapter 4) 260

carbines: short rifles (Chapter 2) 134

carbon dioxide: the gas people breathe out (Chapter 1) 82

carbon monoxide: a gas with no smell that stops the blood from carrying oxygen (Chapter 1) 74

carbonated water: water with bubbles of gas in it (Chapter 13) 855

carcasses: the dead bodies of animals (Chapter 3) 200

carcinogen: a substance that causes cancer (Chapter 19) 1280

carcinogenic: causing cancer (Chapter 3) 240

cardiac arrest: the heart stopped (Chapter 21) 1468

cardinal directions: north, south, east and west (Chapter 29) 2287

carding machine: a machine that combed raw cotton so the fibers lay straight (Chapter 8) 528

carhops: the servers who had carried food out to people's cars (Chapter 15) 1028

caricature: an act that makes fun of people by blowing up some of their features (Chapter 34) 2841

carroting: soaking fur in a liquid that contained mercury nitrate so it matted into felt (Chapter 13) 863

carte blanche: full freedom (Chapter 19) 1256

cartels: Mexican crime groups that make fentanyl outside the law (Chapter 28) 2190

cash crop: a crop grown to sell rather than to eat (Chapter 16) 1052

cash tenant: a farmer who rented land and paid in money (Chapter 16) 1084

cast: a copy made by pouring material into a mold (Chapter 32) 2613

cast iron: iron poured into a mold while it is melted (Chapter 10) 669

caste system: a system that sorts people at birth into ranks they can never leave (Chapter 16) 1052

casting in bronze: pouring melted metal into a mold of a sculpture (Chapter 32) 2660

castor oil: a laxative, a medicine that makes the muscles of the bowels push out their contents. It can cause diarrhea, stomach pain, sickness to the stomach and vomiting (Chapter 29) 2287

castration: cutting off a man's testicles, which left him unable to father children (Chapter 24) 1670

casualties: soldiers killed, wounded or missing (Chapter 2) 120

catalog: a printed list of goods that customers could order by mail (Chapter 7) 485

catechism: a set of questions and answers about Christian beliefs that people learn by heart (Chapter 29) 2207

catechist: a person who teaches the faith and prepares people for baptism (Chapter 29) 2287

cathode: a battery's positive end (Chapter 8) 570

Catholic: a Christian who follows the pope in Rome (Chapter 1) 21

catlinite: a pipe stone from quarries in southwestern Minnesota (Chapter 32) 2598

cat-o'-nine-tails: a whip of nine knotted cords, made to tear the skin and cause great pain (Chapter 1) 44

Caucasian: white (Chapter 6) 443

causal: whether one thing caused another (Chapter 19) 1280

cause for sainthood: the first step toward the Catholic Church naming a person a saint (Chapter 29) 2287

cautery: burning the skin at a painful spot on purpose, as a treatment (Chapter 26) 1949

cavalry: soldiers on horseback (Chapter 2) 134

caveat: a notice that an inventor was working on an invention and planned to apply for a patent (Chapter 8) 541

CDC: the federal disease agency (Chapter 13) 899

ceiling price: the highest price government officials allowed (Chapter 18) 1204

cemesto: ready-made panels of cement and asbestos (Chapter 6) 431

censors: army officials who read soldiers' mail (Chapter 22) 1559

censors: officials who decided which films could be shown (Chapter 34) 2861

censorship: checking and cutting what a printer may print before it comes out (Chapter 31) 2512

censure: a vote to blame someone officially (Chapter 17) 1128

census: the government's count of everyone in the country, taken every ten years (Chapter 1) 64

Census Bureau: the federal office that counts the population (Chapter 4) 320

Census of Agriculture: the federal government's count of the country's farms (Chapter 8) 553

census takers: the government workers who count every person in the country (Chapter 2) 102

census tract: a small area the census counts separately (Chapter 33) 2779

Centennial Exhibition: a big fair in Philadelphia in 1876 that marked 100 years since 1776 (Chapter 12) 807

central bank: a bank that holds other banks' cash and lends to them in a crisis (Chapter 6) 431

Central Labor Union: the group of unions in New York City that planned the first Labor Day parade (Chapter 37) 3107

central power station: a plant that makes electricity in one place and sends it through wires to many buildings (Chapter 10) 681

Cepheid: a variable star that grows brighter and dimmer over and over, in a steady rhythm (Chapter 9) 614

cerebral palsy: a condition from birth that affects movement (Chapter 25) 1829

ceremonial: built for religious gatherings and ceremonies (Chapter 29) 2203

certificate of disability: a paper saying a soldier was not fit to serve (Chapter 22) 1543

certified: officially declared part of the Constitution (Chapter 20) 1356

certified teachers: teachers licensed by the state (Chapter 20) 1372

cervix: the narrow lower end of the womb (Chapter 9) 633

chain gang: a crew of prisoners chained together and forced to work (Chapter 24) 1712

chain reaction: each atom that splits setting off more atoms splitting (Chapter 9) 614

chain store: one of many stores that belong to the same owners and sell the same kinds of goods (Chapter 18) 1193

Challenger Deep: the deepest known place in the ocean, in the Mariana Trench near Guam (Chapter 1) 82

Chamber of Commerce: a group of a city's business owners (Chapter 5) 376

chamber pot: a bowl kept in the room for use as a toilet (Chapter 33) 2736

Chancellor: the British government's top money official (Chapter 17) 1160

chapel: a small church (Chapter 12) 789

chaplain: a minister who serves people far from home (Chapter 9) 585

charcoal: wood burned slowly with little air until only a black fuel is left (Chapter 1) 74

charge: an accusation, not proof (Chapter 2) 102

charge card: a card whose holder had to pay the whole bill each month (Chapter 17) 1153

charity schools: schools where a society in London paid the teacher so that pupils could read the catechism and the Bible (Chapter 30) 2359

charm school: a school that taught manners and how to look and dress (Chapter 36) 3028

charter: the legal paper that gives a colony, bank or company the right to exist and states what it may do (Chapter 3) 189

charter from Congress: a paper that gave Little League official standing (Chapter 36) 3045

charter school: a public school run by its own board under an agreement, called a charter, instead of by the school district (Chapter 20) 1372

chassis: the frame of a car, with its wheels (Chapter 8) 553

chattel: movable property (Chapter 17) 1119

chattel slavery: holding a person as property for life and passing that status on to the person's children, from chattel, an old word for property (Chapter 24) 1666

chemic process: a kind of chemistry, Henry David Thoreau's words for cooking on a stove (Chapter 7) 474

chemical thermodynamics: the science of energy in chemical changes, whose rules let chemists work out from the energy in a change whether it will happen on its own (Chapter 9) 604

chemotherapy: strong drugs that kill cells, here the cells left in the bone marrow (Chapter 26) 2022

chert: a hard stone that holds a sharp edge (Chapter 1) 17

cherubs: carvings of baby angels (Chapter 32) 2608

Chesapeake: the country around the Chesapeake Bay, in Virginia and Maryland (Chapter 2) 102

Chesapeake colonies: Virginia and Maryland, the two colonies around Chesapeake Bay (Chapter 16) 1052

Chevalier: a French title, which means knight (Chapter 1) 36

Chief Justice: the head judge of the Supreme Court (Chapter 2) 126

chiefdom: a group of towns under a single ruler (Chapter 2) 102

child psychiatrist: a doctor who treats problems of the mind (Chapter 30) 2442

children's mission: a church charity for children (Chapter 36) 3019

Children's Online Privacy Protection Act: a law that requires a company to get a parent's permission before it collects personal information from a child under 13 (Chapter 18) 1222

chimney bay: the part of the house that held the chimney, the front door and the stairs (Chapter 7) 465

Chinese Exclusion Act: the 1882 law under which no Chinese laborer could come into the country for ten years (Chapter 4) 283

chloride and lime: chemicals that kill germs (Chapter 26) 1983

chlorinated hydrocarbons: a kind of chemical that contains chlorine (Chapter 3) 229

chlorine: the chemical that kills germs in drinking water (Chapter 3) 229

cholera: a disease people catch from dirty water or food, which causes heavy diarrhea and can kill within days (Chapter 4) 275

chondrosarcoma: a slow-growing bone cancer (Chapter 9) 614

Christmas Box: a small gift of money (Chapter 37) 3094

chromosome: a thread inside a cell that carries genes (Chapter 3) 229

chronic bronchitis: a long-lasting swelling of the air passages (Chapter 26) 2005

chronic traumatic encephalopathy (CTE): a brain disease doctors link to repeated blows to the head, which can bring problems with behavior, mood and thinking and lead to dementia, and can be found only by examining the brain after death (Chapter 36) 3058

chuckwagon: the kitchen on wheels that fed cowhands on cattle drives (Chapter 15) 1009

chunkey: a game in which one player rolled a stone disk across the ground and the players threw spears to see whose landed closest to where the stone stopped (Chapter 6) 395

Church of England: the official church of England (Chapter 4) 260

church year: a calendar of holy days, among them Sundays, Easter, Christmas and the feast days of saints (Chapter 37) 3080

churchwardens: officers of the local church (Chapter 16) 1052

churro sheep: a Spanish breed of sheep (Chapter 35) 2912

chuse: choose (Chapter 30) 2367

CIA: the Central Intelligence Agency, the American government's spy service (Chapter 19) 1280

Cibao: the part of Hispaniola with the gold mines (Chapter 1) 21

Cibola: seven rich cities the Spanish went looking for (Chapter 1) 21

cilia: the tiny hairs that sweep the airways clean (Chapter 15) 1015

Cinco de Mayo: the fifth of May, which marks the Mexican army's defeat of French troops at Puebla, Mexico, on May 5, 1862 (Chapter 37) 3135

circuit: the path electricity flows through and the parts along it that do jobs such as switching it on and off (Chapter 8) 561

circular: an order sent to every office at once (Chapter 9) 604

cirrhosis: damage to the liver from heavy drinking (Chapter 28) 2165

cistern: a tank for catching rain (Chapter 6) 413

citation: an honor outside the Pulitzer Board's usual prizes (Chapter 25) 1909

citations: the official notices for breaking safety rules (Chapter 10) 711

cite (a court decision): to point to an earlier decision to show a judge how courts have ruled in similar cases (Chapter 8) 570

citizen science: research done by ordinary volunteers (Chapter 9) 614

citizen's arrest: an arrest by someone who is not a police officer (Chapter 20) 1390

citrus fruit: the family of fruit that includes oranges and lemons (Chapter 5) 376

city charter: a city's basic set of laws (Chapter 4) 283

city directory: a book that listed a town's people and their trades (Chapter 32) 2627

civil case: a lawsuit rather than a criminal trial (Chapter 25) 1871

civil government: one run by elected or appointed officials outside the armed forces (Chapter 23) 1623

civil judgment: a court's order in a lawsuit to pay money (Chapter 20) 1390

Civil Liberties Act of 1988: the law in which members of Congress apologized for the wartime imprisonment and voted to pay $20,000 to each person who had been held (Chapter 20) 1356

civil list: a set list of paid government posts (Chapter 20) 1317

civil rights: the rights every citizen has to equal treatment under the law (Chapter 2) 149

Civil Rights Act of 1964: the law that banned segregation in places that serve the public and discrimination in jobs and in programs paid for with federal money (Chapter 12) 825

civil service exam: a test a person had to pass to get a government job (Chapter 21) 1442

civil war: a war inside one country (Chapter 4) 320

civilian: a person who is not in the military (Chapter 9) 604

Civilian Conservation Corps: a government jobs program of the 1930s (Chapter 3) 215

civilian noninstitutionalized population: everyone except people who live in institutions (Chapter 25) 1909

civilian use: use that is not for the military (Chapter 10) 702

Civill body politick: a government the signers of the Mayflower Compact promised to form together (Chapter 20) 1309

clan: a large group of families who come down from the same ancestor (Chapter 7) 460

clan crest: an animal symbol that belongs to a clan (Chapter 32) 2598

clan mother: the eldest or most respected woman of the clan, the head of a longhouse (Chapter 7) 460

claptrap: silly talk meant to win cheers (Chapter 37) 3101

class leader: a church member in charge of a small group of members (Chapter 29) 2256

Classical Revival: a style of building that copies the temples of ancient Greece and Rome (Chapter 12) 799

classics: the old Greek and Latin writers (Chapter 30) 2414

classified: officially secret (Chapter 2) 149

clause: one part of a written law or constitution (Chapter 7) 485

clear estate: an old legal phrase meaning owning property outright (Chapter 20) 1322

clemency: a governor's mercy that cuts a punishment short (Chapter 21) 1468

clerical work: office work such as typing, filing and answering phones (Chapter 14) 963

clerk: a worker who keeps written records for a business (Chapter 1) 44

clockwork motor: a motor that runs on gears like the inside of a wind-up clock (Chapter 8) 535

close-in neighborhood: a neighborhood near downtown (Chapter 6) 452

cloture: the motion that ends debate (Chapter 20) 1372

cloud computing: renting out computer storage and computing power over the internet (Chapter 19) 1291

coal heaver: a man who moved coal on and off boats (Chapter 14) 922

coal tar: a thick black liquid made from coal (Chapter 35) 2940

coalition: a group that agrees to act together (Chapter 2) 108

coarse: rough and thick (Chapter 15) 991

cocaine: a drug made from coca leaves (Chapter 20) 1390

cochineal: a small insect bred in Mexico to make red dye (Chapter 35) 2912

cochlear implants: devices placed in the ear that help deaf people hear (Chapter 25) 1909

cockfight: a fight in a ring between two roosters with sharp spurs strapped to their legs, until one was dead (Chapter 36) 2995

code talkers: Native soldiers who sent military messages in their own languages, which the enemy could not understand (Chapter 2) 149

codes of fair competition: rules for whole industries (Chapter 20) 1356

coerced: forced out of a person (Chapter 29) 2304

coerced emigration: being forced to leave the country (Chapter 25) 1829

coercive assimilation: forcing a group of people to give up their own language and ways of life (Chapter 2) 158

coffin ships: what people called the crowded ships that carried the Irish in the famine years (Chapter 4) 275

coffle: a chained line of enslaved people that traders marched overland (Chapter 5) 352

coke: a fuel baked from coal that steel makers burn to melt iron ore (Chapter 19) 1256

cold front: the edge of a large mass of cold air (Chapter 27) 2068

Cold War: the long rivalry between the United States and the Soviet Union after World War II (Chapter 9) 633

colic: sharp pains in the belly (Chapter 13) 868

collateral: property a borrower promises to hand over if the loan is not repaid (Chapter 16) 1067

collective bargaining: bargaining as a union with employers (Chapter 14) 963

collier: a man who made charcoal (Chapter 8) 517

colonization: a plan to send free Black Americans to live in Africa (Chapter 31) 2533

colony: a settlement ruled from a faraway country (Chapter 1) 21

color line: a rule, written or unwritten, that kept Black people out (Chapter 25) 1829

colorblind: taking no notice of race at all (Chapter 25) 1909

Colored: a common word for Black people in the 1800s and 1900s, no longer used (Chapter 1) 82

colors: a regiment's flag (Chapter 22) 1543

colostomy: an operation in which surgeons bring the end of the large bowel out through an opening cut in the belly, so that body waste collects in a bag outside the body (Chapter 36) 3028

Columbian Exchange: the carrying of crops and farm animals across the Atlantic in both directions, named by the historian Alfred W. Crosby in 1972 (Chapter 15) 989

column: the long line of people, wagons and animals on the march (Chapter 5) 336

coma: a state like deep sleep that a person cannot wake from (Chapter 21) 1468

co-managers of the fishery: tribal governments and the state running the salmon catch and its waters together (Chapter 2) 158

Comanchería: Comanche country, the land the Comanche held on the southern Plains (Chapter 2) 116

comfit-maker: a maker of sugar-coated sweets (Chapter 28) 2127

command module: the part of the spacecraft that would carry all three men home (Chapter 1) 82

commemorates: honors (Chapter 12) 832

commencement: the ceremony at which students graduate (Chapter 35) 2928

commercial bank: a bank that keeps people's money safe and lends money to merchants and businesses (Chapter 19) 1245

commercial broadcasters: stations paid for by advertising (Chapter 31) 2585

commercial herds: herds raised for sale (Chapter 3) 240

commercialization and central control: running Little League as a business from one head office (Chapter 36) 3028

commission: a group of people chosen to study a question and recommend what to do (Chapter 1) 82

commission: the paper that gave a person the job (Chapter 20) 1333

commissioner: the head of the INS (Chapter 4) 310

Commissioner of Indian Affairs: the federal official in charge of the government's dealings with Native nations (Chapter 2) 134

commissioner of the Patent Office: the head of the government office that grants patents (Chapter 8) 535

commissioners: officials sent to do a certain job (Chapter 3) 192

committed: sent to live in a state institution by a court's order (Chapter 20) 1356

committee of correspondence: a small group of men chosen to write letters to other towns and share news (Chapter 31) 2520

commodity: a thing to be bought and sold (Chapter 17) 1115

commodity money: money that is a useful thing in itself (Chapter 17) 1115

commodore: a navy officer who commands a group of ships (Chapter 11) 745

Common: a large open field in the middle of Boston (Chapter 22) 1499

Common Council: Albany's governing board (Chapter 37) 3101

common school: a free public school open to the children of the town (Chapter 20) 1333

Commons House: South Carolina's lawmaking body (Chapter 26) 1967

Commonwealth (Estado Libre Asociado): the name Puerto Rico took in 1952, under which it stayed a territory of the United States (Chapter 23) 1647

communion: the Christian ceremony of sharing bread and wine in memory of Jesus (Chapter 4) 267

communism: a system in which one party runs the government and the government owns the farms and factories (Chapter 29) 2304

Communist: a person who wanted the government to own the land and the businesses (Chapter 29) 2304

communist country: a country where the government owned the businesses and only one party was allowed (Chapter 4) 310

communities of color: places where most people are not white (Chapter 3) 229

commute: to change a sentence to a lighter one (Chapter 29) 2287

compact: a formal agreement between a nation and a state about gaming (Chapter 2) 158

compactor: the part of a garbage truck that crushes the garbage (Chapter 14) 963

company town: a town where one company owned the houses and its managers collected the rent (Chapter 14) 929

compensated: paid (Chapter 12) 807

compensation: money paid to make up for a loss (Chapter 2) 158

compensation claims: payments owed when a worker is hurt or killed on the job (Chapter 12) 815

compiler: a program that takes instructions written closer to human language and translates them into instructions the computer can carry out (Chapter 8) 561

complexion: skin color (Chapter 20) 1317

compliance: obeying the ruling (Chapter 20) 1372

compost: rotted plant and animal matter (Chapter 15) 1028

compound: two or more elements joined together, such as iron and oxygen in iron oxide (Chapter 13) 844

comprehensive plan: a plan setting out what a whole city should become, with its streets, parks and waterfront planned together (Chapter 6) 431

compress: to squeeze together (Chapter 16) 1096

compressive asphyxiation: being squeezed so hard that a person cannot breathe (Chapter 33) 2795

compulsory: required (Chapter 20) 1356

compulsory attendance law: a law that requires children to go to school (Chapter 30) 2392

compulsory vaccination: a law that made people get a shot against a disease (Chapter 30) 2414

computer: a person whose job was to do calculations and measurements (Chapter 1) 82

computer terminal: a keyboard and a printer or screen connected to a computer somewhere else (Chapter 8) 561

concentration camp: a place where people are held by force without a trial (Chapter 4) 298

concept artist: an artist who draws the first designs for characters and scenes (Chapter 8) 570

concrete deterioration: concrete breaking down (Chapter 27) 2104

concubine: an old word for a woman a man kept for sex without marrying her (Chapter 24) 1684

concussion: an injury in which a blow shakes the brain (Chapter 22) 1585

condemn: to take land by law without the owner's agreement, at a price the government sets (Chapter 6) 443

condolence: comfort for people in grief (Chapter 21) 1406

condominium: an apartment that each owner buys (Chapter 6) 452

conduct to the prejudice of good order and military discipline: behaving in a way that harmed order and discipline in the armed forces (Chapter 23) 1634

confederacy: a group of nations that keep their own governments but act together on matters they share (Chapter 2) 99

Confederacy: the group of Southern states whose leaders broke away from the United States to keep slavery (Chapter 5) 363

Confederate army: the army of the Southern states that fought to keep slavery in the Civil War (Chapter 12) 815

Confederate states: the Southern states whose leaders broke away from the United States to keep slavery, and which then fought the United States in the Civil War (Chapter 8) 541

Confederates: the Southerners who fought against the United States in the Civil War to keep slavery (Chapter 5) 363

Confederation Congress: the national government of the United States before the Constitution (Chapter 3) 192

conference: a regional group of Methodist churches (Chapter 29) 2256

confession: telling a priest your sins (Chapter 30) 2351

confirm: to approve (Chapter 20) 1372

congestive heart failure: a heart that cannot pump blood well enough (Chapter 32) 2695

conglomerate: one company that owns many businesses that have nothing to do with each other (Chapter 19) 1280

congregation: the people of a church (Chapter 5) 339

Congregationalist: a member of the churches the Puritans founded in New England (Chapter 29) 2231

Congressional Budget Office: the office that estimates the cost of laws for Congress (Chapter 15) 1039

Congressional Research Service: the office that does research for members of Congress (Chapter 15) 1028

Connecticut Compromise: the deal that gave every state two senators and gave each state seats in the House by its population (Chapter 20) 1322

conscientious objector: a person who refuses to fight because their religion forbids it (Chapter 22) 1559

consensual: agreed to by both adults (Chapter 25) 1871

consensus: a way of deciding in which every member must agree, so no law passes while even one member is against it (Chapter 20) 1304

consent: agreeing to something after being told what it is (Chapter 4) 283

consent decree: an agreement that a judge enforces (Chapter 25) 1909

conservation: protecting land, water, plants and animals so that people do not use them up (Chapter 3) 185

conservation of charge: the idea that electric charge is never made or destroyed, only moved from one object to another (Chapter 9) 588

conservationist: a person who works to protect wild land (Chapter 6) 431

conservative: political views that favor keeping old ways and limiting government (Chapter 24) 1744

conspiracy: a secret plan by two or more people to break the law (Chapter 14) 922

conspiracy to murder: planning a murder with others (Chapter 31) 2572

conspiring: plotting with others (Chapter 10) 711

constable: a town officer who kept order and made arrests (Chapter 21) 1427

constituent: one of the people a member of Congress represents (Chapter 29) 2304

constitution: a written plan for a government and its basic laws (Chapter 2) 126

constitutional convention: a meeting of men chosen to rewrite the state constitution (Chapter 20) 1333

constitutionality: whether a law is allowed by the Constitution (Chapter 20) 1322

construction: stretching the meaning of the words (Chapter 20) 1333

consul: an official who looks after his country's trade and citizens in another country (Chapter 23) 1605

consumer: a person who buys goods to use them (Chapter 16) 1096

Consumer Financial Protection Bureau: a government office that watches over lenders (Chapter 17) 1153

consumer prices: the prices of the things people buy for daily life (Chapter 16) 1096

consumer revolution: historians' name for the big change in buying in the 1700s, when colonists began buying the clothes, tools and kitchen things they had once made (Chapter 18) 1178

consumption: the name people then used for the disease Caleb Cheeshahteaumuck died of, which Martha's Vineyard Magazine states was "presumably" tuberculosis (Chapter 30) 2351

container: a large shipping box (Chapter 11) 764

containment: American leaders' plan, from 1947, to block the spread of Soviet power (Chapter 23) 1647

contempt of Congress: the crime of refusing to obey Congress or one of its committees (Chapter 20) 1356

contempt of court: disobeying a court's order (Chapter 14) 929

content moderation: checking posts and removing those that break a site's rules (Chapter 31) 2585

Continental Association: a pact among the colonies to stop buying goods from Britain (Chapter 34) 2833

Continental bills: the paper money members of the Continental Congress printed from June 1775 to pay for the war (Chapter 17) 1123

Continental Congress: the meeting of leaders from the colonies (Chapter 5) 346

Continental Divide: the line of high ground that separates rivers flowing east from rivers flowing west (Chapter 1) 44

Continental dollars: the paper money the Continental Congress printed to pay for the Revolution (Chapter 16) 1062

Continental money: paper money printed during the Revolution, which by 1780 had lost most of its value (Chapter 29) 2242

contortionist: a performer who bends the body into unusual shapes (Chapter 34) 2851

contour plowing: plowing along the curves of a slope instead of up and down it (Chapter 3) 215

contraband: the word Union soldiers used for people who had escaped slavery (Chapter 22) 1543

contraband camps: camps run by Union army officials, and later the Freedmen's Bureau, for people who had escaped slavery (Chapter 24) 1712

contraband of war: property of the enemy that an army may seize (Chapter 24) 1712

contraception: ways to prevent pregnancy (Chapter 25) 1795

contract: a written deal that a court can make people keep (Chapter 1) 21

contract laborer: a worker brought over without free consent, already bound to work for someone for a set time (Chapter 4) 283

contract sale: a sale under which a buyer who missed a single payment could be put out and lose everything already paid (Chapter 6) 443

contract workers (Hawaii): workers the sugar planters brought in from overseas, bound to work for a set number of years (Chapter 15) 1015

contractor: a man who ran a section of the work (Chapter 11) 734

contralto: a singer with a low woman's voice (Chapter 33) 2759

contrived: arranged (Chapter 22) 1522

controlled fires: fires started on purpose and kept in bounds, to shape the land (Chapter 8) 512

convent: a house of nuns, women who have promised their lives to the Catholic church (Chapter 2) 102

convent school: a school that nuns ran in their own home (Chapter 4) 275

convention: a large organized meeting (Chapter 11) 732

convention: a written agreement between governments, such as the Convention of Kanagawa (Chapter 23) 1623

conversion: the changing of a person's religion (Chapter 30) 2348

conversion experience: a moment when a person feels that God has changed him or her (Chapter 29) 2213

conversion therapy: any practice that tries to change a person's sexual orientation or gender identity (Chapter 25) 1909

converted: turned to God and became a true believer (Chapter 29) 2256

convict leasing: state officials renting out prisoners to mines, railroads and planters, who forced them to work (Chapter 14) 929

convicted: found guilty (Chapter 4) 275

convicts: people found guilty of crimes (Chapter 4) 267

convoy: a group of vehicles traveling together (Chapter 11) 764

convulsions: fits of violent shaking a person cannot control (Chapter 13) 886

cooper: a person who makes barrels (Chapter 4) 275

cooperative: a business owned by the people who use it (Chapter 10) 691

coppersmith: a person who makes things out of copper (Chapter 13) 844

copyright: the legal right to control who copies and sells a piece of writing (Chapter 2) 134

copyright law: the law that protects an artist's right to decide who copies their work (Chapter 8) 570

coquina: a soft stone made of packed seashells (Chapter 12) 789

cord (of wood): a stack 8 feet long, 4 feet wide and 4 feet high (Chapter 8) 517

corduroy road: logs laid side by side across swampy ground, named after a cloth with raised ribs (Chapter 11) 728

cordwainers: shoemakers (Chapter 14) 920

core: the center of a reactor, where the fuel sits (Chapter 10) 702

corn knife: a long blade used for cutting cornstalks (Chapter 29) 2256

cornea: the clear front of the eye (Chapter 25) 1829

corner a stock: to buy up so much of it that the buyers can set its price (Chapter 17) 1143

cornerstone: the first stone set at a corner of a new building (Chapter 12) 795

cornet: a brass horn like a small trumpet (Chapter 33) 2746

Coromantee: Akan people from the Gold Coast of West Africa, which is Ghana today (Chapter 24) 1676

corona: the bronze-colored shape in three levels on the outside of the National Museum of African American History and Culture (Chapter 12) 832

coroner: the official who looks into deaths and finds out how people died (Chapter 2) 149

coroner's jury: a group of citizens who decide how people died (Chapter 27) 2078

corporal punishment: an adult hitting a child on purpose, to hurt the child, as a punishment (Chapter 7) 485

corporate culture: a company's habits and ways of working (Chapter 10) 711

corporation: a company that goes on after its first owners are gone, one the law treats as one body (Chapter 10) 691

correct: in the words of that time, to punish (Chapter 8) 541

correction: a beating (Chapter 21) 1412

corridor: a long, narrow strip of land (Chapter 3) 240

corrosion control: a treatment that keeps water from eating into pipes (Chapter 13) 899

corruption: officials using their jobs to get money or favors illegally (Chapter 2) 158

corruptly: by dishonest deals (Chapter 20) 1356

corset: a stiff garment laced tight around the body (Chapter 35) 2934

corvée: a system of forced labor (Chapter 23) 1634

cosmic microwave background: the cooled light left over from the early universe (Chapter 9) 633

cosmonaut: the Russian word for an astronaut (Chapter 1) 82

cotton belt: the part of the South where cotton was grown (Chapter 16) 1074

cotton gin: a machine that pulls the seeds out of cotton (Chapter 8) 528

cotton press: a machine that squeezed cotton into bales (Chapter 8) 528

cotton ridge: a raised row of soil where the cotton grows (the middles are the low strips between the rows) (Chapter 8) 541

council: a meeting of leaders (Chapter 17) 1112

council house: the building where a town's leaders met (Chapter 1) 21

Council of Economic Advisers: a group of economists who advise the president (Chapter 16) 1096

Council of State: the leaders who ran England's government in 1650 (Chapter 8) 517

Council of the Indies: the king's council in Spain for the American colonies (Chapter 1) 21

counsel: a lawyer (Chapter 25) 1829

counterfeit bill: a fake banknote (Chapter 25) 1909

counterfeiters: people who make fake money (Chapter 17) 1119

counterfeiting: making fake money (Chapter 17) 1123

country born: born in America and not brought from Africa (Chapter 32) 2627

country pay: payment in goods (Chapter 17) 1115

counts: charges (Chapter 20) 1390

county supervisors: the officials who ran the county (Chapter 25) 1829

county unit system: Georgia's system for statewide elections, under which the candidate who won a county got all of that county's unit votes (Chapter 20) 1372

coup: a sudden takeover of a government by a small group, often the army (Chapter 23) 1647

coupling: the job of joining railroad cars together (Chapter 11) 745

court days: the days when the county court met (Chapter 36) 2999

Court of Appeals: a court that hears appeals from trial courts (Chapter 12) 832

Court of Assistants: the Massachusetts colony's ruling council (Chapter 11) 726

court of inquiry: a panel of officers looking into what Marines had done in Haiti (Chapter 23) 1634

Court of Oyer and Terminer: a special court, the words mean "to hear and to decide" (Chapter 21) 1412

court-martial: a trial held by the military (Chapter 1) 44

covenant (in a deed): a section written into the deed of a house, such as one that forbade the owner to sell or rent it to a Black buyer. It is also called a restrictive covenant (Chapter 6) 431

covenant (promise): a serious promise (Chapter 2) 108

cover version: a new recording of a song that someone else has already recorded (Chapter 33) 2779

coverture: the old English rule that a married woman was "covered" by her husband and had no legal standing of her own (Chapter 25) 1760

COVID-19: an illness caused by a coronavirus that spreads in the droplets and tiny particles an infected person breathes out (Chapter 14) 974

cowpeas: a kind of bean (Chapter 15) 1015

cowpox: a mild disease of cows (Chapter 26) 1975

co-writer: a person named as one of a song's writers, with a right to part of the money the song earns its writers (Chapter 33) 2779

cowskin: a whip made of dried, twisted cow's hide, also called the cowhide (Chapter 24) 1693

crack: a form of cocaine that is smoked (Chapter 21) 1468

craft union: a union of the skilled workers of one trade, such as cigar makers or carpenters (Chapter 14) 929

Cranberry Day: the day each October when the Wampanoag Tribe of Gay Head (Aquinnah) greets the sunrise, picks cranberries, shares a meal and sets aside a share for the elders (Chapter 37) 3075

craniometry: measuring skulls, part of a false "racial science" used to rank the races (Chapter 9) 604

crash: a sudden fall in prices, of stocks or of homes (Chapter 7) 480

cravat: a cloth tied around the neck (Chapter 35) 2934

crawlspace: the low space under a floor (Chapter 27) 2078

craze: something that very many people took up at the same time (Chapter 11) 745

creation science: the teaching that a supernatural being created humankind (Chapter 29) 2304

credentials: the letters that proved an official's position (Chapter 23) 1605

credentials committee: the group that decided which delegates could take their seats (Chapter 25) 1871

credit: a person's record of paying back debts (Chapter 7) 502

credit: a loan, or the right to buy now and pay later (Chapter 15) 1009

credit card: a card whose holder could pay part of the bill and owe the rest later (Chapter 17) 1153

creditor: a person who is owed money (Chapter 16) 1067

credulity: a willingness to believe things (Chapter 9) 614

creed: a religion (Chapter 14) 941

Cremona fiddle: a violin made in the town of Cremona, in Italy (Chapter 33) 2723

crest: an animal or spirit figure that stands for a clan (Chapter 34) 2814

cricket: the English bat-and-ball game (Chapter 36) 2995

Crime of '73: the silver side's name for the Coinage Act of 1873 (Chapter 17) 1136

criminal negligence: carelessness so serious it is a crime (Chapter 26) 1975

criminal sexual act: under New York law, forcing a person into sex acts such as oral sex (Chapter 25) 1909

criminalizing: making something a crime (Chapter 28) 2179

criminally negligent homicide: causing a death through carelessness (Chapter 21) 1468

cripple: an old, unkind word for a person who cannot use part of the body (Chapter 26) 2005

CRISPR: a tool for changing genes (Chapter 8) 570

crisscross row: the alphabet row on a hornbook, from "Christ's Cross row" (Chapter 30) 2359

crop rotation: changing the crop planted in a field from one year to the next (Chapter 15) 1015

cropping: cutting off part or all of each ear (Chapter 17) 1123

croquet: a lawn game played by hitting wooden balls with mallets (Chapter 36) 3019

cross burning: setting a large cross on fire (Chapter 34) 2861

CROWN Act: a law, first passed in California, that bars employers and public schools from treating people worse because of their hair's texture or styles such as braids, locs and twists. The name stands for "Creating a Respectful and Open World for Natural Hair" (Chapter 25) 1909

crowning utility: the most useful thing a person ever did (Chapter 20) 1333

crucifix: a model of Jesus on the cross (Chapter 32) 2679

crucify: to put someone to death by nailing them to a cross (Chapter 17) 1136

crude oil: the oil that comes straight out of the ground (Chapter 3) 240

crypt: a burial room beneath a church (Chapter 4) 271

cryptocurrency: digital money such as bitcoin (Chapter 17) 1160

cudgel: a heavy club (Chapter 29) 2242

cuffing: hitting with an open hand (Chapter 30) 2392

cuffs: turned-back bands of cloth (Chapter 35) 2949

culls: the leftovers (Chapter 34) 2851

cult of domesticity: the strongly held belief that the home was a place set apart from paid work and that it was the mother's to run, named by the historian Aileen Kraditor in 1968 (Chapter 7) 474

cultivate: to break up the soil and clear weeds between rows of growing crops (Chapter 8) 553

cultural burning: lighting small fires on purpose and keeping them under control to help the plants and animals people use for food, clothing and ceremonies (Chapter 3) 240

cultural fire: burning done by Native people to care for the land (Chapter 3) 200

Cumberland Gap: a low pass through the Appalachians (Chapter 1) 39

curator: the person in charge of a museum's collection (Chapter 9) 604

curfew: an order to stay off the streets at night (Chapter 6) 452

curie: a unit that measures how much radioactive material is let loose (Chapter 10) 702

current: the flow of electricity (Chapter 8) 553

current (ocean): a stream of moving water inside the ocean (Chapter 9) 596

curriculum: the set of lessons a school teaches (Chapter 30) 2367

custody: control over a child and where the child lives (Chapter 3) 200

custody: being held by officers (Chapter 4) 320

customs informer: a person paid by customs officials, who collected the taxes on goods brought in by ship, to report smugglers (Chapter 18) 1182

customs officers: officers who collected the taxes on goods coming in (Chapter 23) 1600

customs officials: officials who check what comes into a country at its border (Chapter 13) 899

cut: a channel dug through high ground so railroad tracks can run level (Chapter 14) 922

cutlass: a short curved sword (Chapter 1) 39

cyanide: a poison that stops the cells of the body from using oxygen, so the cells cannot make the energy they need to live (Chapter 29) 2304

cyclotron: a machine that speeds tiny particles around in a circle and smashes them into atoms (Chapter 8) 553

D

DACA: Deferred Action for Childhood Arrivals, the 2012 policy under which some people brought to the country as children can apply for protection from being deported and for permission to work (Chapter 4) 320

Dacron: DuPont's name for its polyester fiber (Chapter 35) 2959

daguerreotype: an early kind of photograph made on a polished metal plate (Chapter 8) 535

damages: a payment of money for a harm (Chapter 9) 604

damask: cloth with its pattern woven into it (Chapter 35) 2923

dame school: a school a woman ran in her own home for small children, from an old word for a woman (Chapter 30) 2359

Danbury shakes: the mercury poisoning of the hatters of Danbury, Connecticut (Chapter 13) 863

Darién: a jungle between Colombia and Panama (Chapter 4) 320

dark: closed, said of a theatre (Chapter 34) 2889

dark money: election spending by groups that do not have to name their donors (Chapter 19) 1291

dark patterns: confusing buttons and screens that led players into charges they did not mean to make (Chapter 18) 1222

dastardly: an old word for cowardly and cruel (Chapter 2) 134

data brokers: companies that collect records of what people look at online, combine them with records from outside the internet and sell what they gather (Chapter 18) 1222

data elements: separate facts (Chapter 18) 1222

datura: a flowering plant that holds two drugs that make people see and hear things that are not there (Chapter 28) 2120

Daughters of Liberty: a name for women on the side of the protest against British taxes (Chapter 18) 1182

Dawes Act: the 1887 law under which federal officials broke up the land set aside for Native nations into farms for single families and sold the land left over (Chapter 2) 134

day laborer: a worker paid by the day (Chapter 15) 1015

day of fasting: a day when people went without food and prayed (Chapter 27) 2047

DDT: a white powder for killing insects, which thinned the eggshells of birds that hunt (Chapter 3) 229

deacon: a church officer who helps the minister (Chapter 11) 726

deadline: at Andersonville, a line inside the wall that guards shot any prisoner for crossing (Chapter 22) 1543

dean: the head of a law school (Chapter 25) 1829

death row: where people sentenced to death wait to be executed (Chapter 21) 1468

debauchery: wild, immoral living (Chapter 36) 3028

debt bondage: being forced to keep working to pay off a debt (Chapter 18) 1204

debtors' prison: a jail for people who could not pay what they owed (Chapter 15) 1004

decathlon: a contest of ten track and field events (Chapter 2) 149

Decency and Sobriety: politely and without getting drunk (Chapter 36) 2995

deceptive lending: lending by misleading borrowers (Chapter 7) 502

decimal ratio: counting up by tens (Chapter 17) 1123

declaration of taking: a paper under which the government could take land right away, whether or not the owners wanted to sell, and settle the price later (Chapter 6) 431

declared legally dead: counted as dead under the law (Chapter 1) 74

Decoration Day: the first name of Memorial Day, the day for putting flowers on soldiers' graves (Chapter 37) 3107

decorous: calm and proper (Chapter 29) 2256

dedicated: opened with a public ceremony (Chapter 12) 804

deed: a paper that says who owns a piece of land (Chapter 1) 39

deemed: counted (Chapter 8) 528

"deemed a mere brush": thought of as a small punishment (Chapter 8) 528

deep-learning era: the time after 2012 when neural networks came into wide use (Chapter 8) 570

Deep South: the cotton lands from Georgia west to Texas (Chapter 5) 352

deer tallow: the fat of a deer (Chapter 8) 512

defamation: harming a person's good name with false statements (Chapter 20) 1390

default: to be unable to repay a loan (Chapter 15) 1028

defendants: the people and companies being sued (Chapter 27) 2104

defense: a reason that can keep a person from being found guilty (Chapter 20) 1317

deferred: put off (Chapter 4) 320

defrauded: cheated (Chapter 19) 1291

degenerate: Buffon's word for animals and people growing smaller and weaker in the Americas (Chapter 9) 591

dehumanizing: treating people as if they were not human (Chapter 25) 1757

DEI: diversity, equity and inclusion (Chapter 3) 240

deist: a person who stressed living a good life and rejected the Christian teaching that Jesus was God (Chapter 29) 2231

delegate: a person chosen to speak and vote for others at a meeting (Chapter 2) 116

delegation: a group sent to speak for its people (Chapter 3) 200

deliberate: careful and unhurried (Chapter 20) 1372

delirious: confused and out of one's mind (Chapter 27) 2061

delirium tremens: an illness of heavy drinkers who suddenly stop, in which the body shakes, the mind is confused and the person sees things that are not there. It can kill (Chapter 28) 2147

deliverance: being set free (Chapter 37) 3101

Delmarva Peninsula: the land where Delaware, Maryland and Virginia meet (Chapter 15) 1015

delta: the low, wet land where a river splits into many channels near the sea (Chapter 1) 64

deluder: someone who tricks people (Chapter 20) 1309

dementia: the loss of the ability to think and remember (Chapter 15) 1015

democracy: a government in which the people have a say (Chapter 2) 99

Democrat: a member of the Democratic Party, one of the two main political parties (Chapter 20) 1322

Democratic-Republicans: the party led by Thomas Jefferson and James Madison, whose members wanted more of the government's power left with the states (Chapter 20) 1322

democratization of fashion: fashion spreading to ordinary people (Chapter 35) 2949

denatured alcohol: factory alcohol with chemicals added so that no one could drink it (Chapter 28) 2165

denomination: a group of churches that share the same beliefs and leaders (Chapter 24) 1693

dentalium shells: the shells of a small sea animal, each shaped like a tiny elephant tusk, which traders used to measure what other goods were worth (Chapter 18) 1170

department store: a large store with many kinds of goods, each kind in its own section, called a department (Chapter 18) 1188

deport: to force a person to leave the country (Chapter 4) 298

deportment: behavior (Chapter 30) 2442

deposit insurance: a promise backed by the government to repay people's savings if their bank fails (Chapter 17) 1136

deposition: sworn testimony taken outside a courtroom and written down (Chapter 19) 1268

deposits: money kept in a bank's care (Chapter 17) 1128

depot for immigrants: a building where officials received newcomers, checked them and recorded them (Chapter 4) 283

depression: a long time when many businesses fail and many people lose their jobs, as in the Depression of the 1930s (Chapter 4) 298

depression (illness): an illness that brings long periods of deep sadness and makes it hard to think and to work (Chapter 35) 2949

deprive: to take away (Chapter 37) 3101

deputies: the officers who worked under a sheriff (Chapter 4) 283

deputize: to give men the power to act as officers of the law (Chapter 14) 929

deputy marshal: a federal officer who makes arrests (Chapter 20) 1322

derringer: a small pistol (Chapter 34) 2851

desegregation: an end to keeping children of different races apart (Chapter 25) 1829

deserter: a soldier who leaves his army without permission (Chapter 22) 1534

despotism: rule by a tyrant (Chapter 25) 1770

destitute: so poor that a person has nothing to live on (Chapter 32) 2660

detainees: people being held (Chapter 4) 298

detention: holding people in custody while officials decide whether to deport them (Chapter 4) 320

deterred: put off from doing something (Chapter 29) 2324

Dey: the ruler of Algiers (Chapter 23) 1605

dialect spelling: spelling meant to show how a person's speech sounded (Chapter 8) 535

dialysis: a treatment that cleans the blood of people whose kidneys have failed (Chapter 27) 2104

dictator: a ruler with total power (Chapter 34) 2861

dictatorial tyrannies: countries ruled by dictators (Chapter 25) 1829

dies (for coins): the stamps that make coins (Chapter 8) 517

diethylene glycol: a chemical used in antifreeze, and a poison (Chapter 26) 1990

diffusion: spreading (Chapter 9) 596

digital replica: a computer copy of an actor's face, body or voice (Chapter 34) 2889

digital token: a record kept on a network of computers that names one person as the owner of a file (Chapter 32) 2695

dike: a bank of earth built to keep water back (Chapter 27) 2078

dilapidated: falling apart (Chapter 25) 1871

dime novel: a novel of about 100 pages sold for ten cents (Chapter 32) 2643

Diné: the Navajo people's own name for themselves (Chapter 2) 99

Dinka: an ethnic group of southern Sudan (Chapter 4) 320

diocesan priest: a priest who serves in a bishop's region (Chapter 29) 2324

diocese: the group of churches under one bishop (Chapter 29) 2242

dioxin: a group of chemicals that cause cancer and harm babies before they are born (Chapter 3) 229

diphtheria: an infection by bacteria that kills the tissue of the throat and nose until the coating of dead tissue stops a child breathing (Chapter 26) 1961

diplomacy: the work of dealing with other nations through talks and agreements (Chapter 2) 116

diplomat: a person sent to speak and bargain for a nation (Chapter 1) 39

diptych: a picture in two parts (Chapter 32) 2679

direct current (DC): electric current that flows in one direction only, like the current from a battery (Chapter 10) 681

director general: the man who governed the Dutch colony of New Netherland for the Dutch West India Company (Chapter 6) 399

disaffected: unfriendly (Chapter 22) 1522

disappeared: taken by the government's forces and never seen again (Chapter 23) 1647

disc jockey: a person who plays records on the radio or for dancers, called a DJ for short (Chapter 33) 2779

disclaimer: a statement that refuses blame (Chapter 3) 229

discount: the difference between the amount printed on a bank's note and the smaller amount people far from the bank would take it for (Chapter 17) 1128

discount chain: stores selling goods at prices below what other stores charged (Chapter 18) 1214

discountenanced: treated with disapproval (Chapter 29) 2213

discretion: the power to decide (Chapter 11) 756

discrimination: being treated worse because of who you are, such as your race (Chapter 12) 825

disfranchise: to take away a person's vote (Chapter 20) 1333

dishonest graft: blackmailing gamblers and squeezing money out of illegal businesses (Chapter 20) 1356

dismembring: tearing or biting off parts of another man's body (Chapter 36) 2999

disorderly conduct: disturbing public order (Chapter 37) 3125

disowning: a Quaker meeting's formal statement that a member no longer shared an essential part of the faith (Chapter 25) 1767

dispatcher: a worker who directs trains, drivers or police officers and keeps track of them (Chapter 8) 541

dispatches: official letters for a government (Chapter 23) 1605

displace: to push people off their land (Chapter 37) 3135

displaced persons: Europeans who had lost their homes in World War II (Chapter 4) 298

Displaced Persons Act: the 1948 law, the first American law for refugees (Chapter 4) 298

disposition: whether people were likely to be friendly or to fight (Chapter 1) 64

disproportionate: more than their share (Chapter 11) 764

dissection: the cutting open of a dead body to look at its insides (Chapter 18) 1188

dissent: a judge's written disagreement with the other judges (Chapter 20) 1317

dissident beliefs: beliefs that go against those of the people in power (Chapter 25) 1871

dissolution: the breakup of a company (Chapter 19) 1268

distemper: an old word for a sickness (Chapter 26) 1961

distill: to heat a drink and catch the steam, which cools into a liquor stronger than beer or wine (Chapter 28) 2120

distillery: a place where liquor is made (Chapter 6) 418

Distinguished Flying Cross: a medal for bravery or great skill in flying (Chapter 1) 74

Distinguished Service Cross: an army medal for bravery in battle (Chapter 22) 1559

distributing reservoir: a large basin that held the water and fed it out to the city's pipes (Chapter 6) 413

district: the land set aside for the capital (Chapter 6) 407

district leader: the party's boss in one part of the city (Chapter 20) 1356

District of Columbia: the capital city (Chapter 12) 799

district schools: country schools, which usually held a summer term and a winter term (Chapter 7) 474

disturbing the peace: making a public disturbance (Chapter 35) 2949

diversion tunnels: tunnels dug to carry a river around a building site (Chapter 12) 815

diversions: ways to have fun (Chapter 34) 2833

dividend: a share of a company's profits paid to its owners (Chapter 19) 1245

Divine Service: the church service (Chapter 33) 2719

Diwali: the festival of lights, kept mostly by Hindus, Sikhs, Jains and Buddhists (Chapter 37) 3135

Dixiecrats: the States' Rights Democratic Party, formed by Southern delegates in 1948 (Chapter 20) 1356

DNA: the set of chemical instructions inside every living cell (Chapter 9) 633

doctorate: the highest degree a university gives (Chapter 9) 604

doctrina: a mission town with a friar living in it (Chapter 29) 2207

doctrine: the beliefs of the church (Chapter 30) 2348

doctrine of discovery: a rule in American law under which a land that no European had known before belonged to the European government whose explorers found it first, and no other European government could claim it. No one asked the Native nations who already lived there (Chapter 1) 44

doffer: a mill worker who took full bobbins off the spinning machines and put on empty ones (Chapter 14) 922

doffing: taking the full bobbins, the spools that held the spun thread, off a spinning machine and putting empty ones on (Chapter 14) 922

dolly: a wooden stick for stirring clothes as they boiled (Chapter 7) 485

domestic servants: people paid to cook, clean and care for other families' homes (Chapter 7) 485

domestic service: paid work in another family's home (Chapter 7) 480

domestic terrorism: an attack meant to frighten people inside the country (Chapter 29) 2324

domestic workers: people who cleaned, cooked and cared for children in other people's homes (Chapter 16) 1084

Doña: a Spanish title of respect for a woman (Chapter 29) 2207

door-yards: the yards in front of houses (Chapter 32) 2643

dormitory: a large room full of beds (Chapter 4) 298

Dorset: the name archaeologists give a Native people who lived at L'Anse aux Meadows about 300 years before the Norse (Chapter 1) 17

dot-com bubble: the boom and crash in technology stocks from about 1995 to 2002, named after companies that did business on the internet (Chapter 16) 1102

Douay Bible: the Bible translation Catholics use (Chapter 29) 2256

double helix: two long strands wound around each other in a spiral (Chapter 9) 633

double jeopardy: the rule that a person found not guilty cannot be tried a second time for the same crime (Chapter 21) 1468

double-woven basket: two baskets woven one inside the other from the same strips, so both the inside and the outside show a finished pattern (Chapter 32) 2613

Dow Jones average: a number built from the stock prices of a group of large companies (Chapter 16) 1084

down payment: part of the price paid up front (Chapter 7) 485

downwinders: people who lived where the wind carried fallout from nuclear bomb tests (Chapter 9) 648

dowry: the money or goods a bride's family gives when she marries (Chapter 17) 1115

draft: the system that called men into the military, or ordered them to come and work (Chapter 7) 474

draft board: the local officials who chose which men in an area were drafted (Chapter 22) 1575

draft evasion: refusing to serve when drafted (Chapter 36) 3045

draftsman: someone who makes exact drawings of machines and buildings (Chapter 8) 541

dram: a small drink of liquor (Chapter 28) 2147

drawback: money a railroad would pay a shipper on every barrel his rivals shipped (Chapter 19) 1256

drawing wire: pulling iron through small holes to make it thin (Chapter 8) 517

Dred Scott decision: the 1857 Supreme Court ruling that African Americans were not citizens (Chapter 8) 541

drilling: practicing marching and handling weapons with a company of soldiers (Chapter 22) 1522

drive pipe: a pipe driven down to the rock so the drill could work inside it, which kept water and loose ground out (Chapter 10) 681

driver: an enslaved man put in charge of other enslaved workers (Chapter 9) 604

drought: a long time with too little rain (Chapter 2) 99

dry goods: cloth, clothing and sewing supplies (Chapter 18) 1188

dubious: open to doubt (Chapter 30) 2442

ducat: a Spanish gold coin (Chapter 2) 102

due process: the rules a court must follow to give an accused person a fair trial (Chapter 21) 1442

duff: a thick, soft layer of fallen leaves on a forest floor (Chapter 3) 182

duffel: a heavy wool used for blankets and coats (Chapter 35) 2923

dugout: a home dug into the side of a hill or a creek bank (Chapter 7) 480

dugout (canoe): a canoe made from one log that builders hollowed out with fire and tools (Chapter 11) 721

dumb (in an old book title): unable to speak (Chapter 30) 2375

dumbbell tenement: a tenement pinched in at the middle around a narrow air shaft, wide at both ends and thin in the middle, like an exercise weight with a ball at each end of a short bar (Chapter 6) 418

dunce's cap: a pointed paper hat that a teacher made a pupil wear to shame him (Chapter 30) 2375

Dunmore's War: the war Virginians fought against the Shawnee in 1774 (Chapter 1) 39

Dust Bowl: the parts of Oklahoma, Texas, Kansas, Colorado and New Mexico where drought and windstorms tore the topsoil off farms in the 1930s (Chapter 3) 215

dust pneumonia: an illness that came when dust settled deep in a person's lungs and kept the tiny hairs that sweep the airways clean from moving. The person had fever, chest pain, coughing and trouble breathing (Chapter 15) 1015

duster: a dust storm (Chapter 27) 2078

Dutch West India Company: the trading company whose officials ran New Netherland and gave its governor his orders (Chapter 4) 260

duty: a tax on goods brought in (Chapter 16) 1057

dynamo: a machine that makes electricity when it is turned (Chapter 8) 541

dysentery: an infection of the gut that causes severe, bloody diarrhea (Chapter 1) 31

E

eagle: a ten-dollar gold coin (Chapter 13) 855

earth lodge: a dome made of timber and packed over with earth (Chapter 7) 460

earthworks: mounds and long banks of piled earth (Chapter 12) 783

easement: legal permission to cross land (Chapter 2) 170

Eastern Agricultural Complex: the group of crops Native people in eastern North America bred on their own, a squash and six seed plants (Chapter 15) 986

Eastern Woodlands: the forests of the eastern part of the continent (Chapter 7) 460

eclectic: chosen from many sources (Chapter 30) 2375

ecology: the ways living things depend on each other and on the land (Chapter 9) 581

economist: an expert on money, prices and trade (Chapter 11) 764

economy: the ways people have of making, sharing and trading the things they need (Chapter 16) 1049

ecumenical: bringing together people of different churches (Chapter 16) 1096

Edict of Nantes: a French law of 1598 under which French Protestants could worship in the towns they held (Chapter 4) 260

editorial: an article that gives a newspaper's opinion (Chapter 5) 376

education savings account (ESA): state money set aside for one child that the family can spend on private school fees, a tutor, books or an online course (Chapter 30) 2474

effigy: a figure made to look like a person (Chapter 31) 2533

egregious errors: very serious mistakes (Chapter 25) 1909

Eighteenth Amendment: the change to the Constitution that banned alcoholic drinks from January 1920 (Chapter 20) 1356

Eighth Amendment: the part of the Constitution that forbids cruel and unusual punishment (Chapter 20) 1372

El Camino Real de Tierra Adentro: Spanish for "the royal road of the interior," the route from Mexico City to northern New Mexico (Chapter 5) 336

elder: a leader in charge of a church, or a man respected for his learning (Chapter 29) 2242

election cakes: thick cakes full of fruit, eaten at the Black election celebrations (Chapter 33) 2729

elective franchise: the right to vote (Chapter 25) 1778

Electoral Commission: the fifteen representatives, senators and Supreme Court justices who settled the disputed electoral votes of 1876 (Chapter 20) 1341

Electoral Count Reform and Presidential Transition Improvement Act: the 2022 law that changed the rules for counting the electoral votes (Chapter 20) 1390

electoral votes: the votes cast by electors, the people each state chooses to pick the president (Chapter 20) 1322

electors: the people who pick the president (Chapter 20) 1322

electric chair: a chair that kills a person by sending a strong electric current through the body (Chapter 10) 681

electrocution: killing a person by sending a strong electric current through the body, which stops the heart and burns the skin where the metal touches it (Chapter 21) 1442

electromagnet: a magnet that works only while electricity flows through it (Chapter 8) 535

electromagnetic induction: the making of an electric current with a magnet (Chapter 9) 596

electron: one of the tiny particles inside every atom, which carries electricity (Chapter 8) 553

electron volt: a very small unit of energy used for particles (Chapter 8) 553

element: a substance made of only one kind of atom (Chapter 8) 553

elementary and secondary schools: the schools for children from first grade through high school (Chapter 20) 1372

elevator (on an airplane): a movable part at the front that controls pitch (Chapter 8) 553

elimination: a disease no longer spreading nonstop inside the country for at least a year (Chapter 26) 2022

emaciated: so thin from hunger that the bones show (Chapter 22) 1522

emancipate: to set enslaved people free (Chapter 11) 745

Emancipation Park: the ten acres Black Houstonians bought in 1872, named for the freeing of enslaved people (Chapter 37) 3107

Emancipation Proclamation: President Abraham Lincoln's 1863 order declaring the enslaved people in the Confederate states free (Chapter 8) 541

embargo: an order that stops trade (Chapter 10) 702

embezzlement: stealing money that others trusted a person to handle (Chapter 10) 691

embroidery: sewing pictures and patterns with colored thread (Chapter 18) 1182

emergency manager: an official the governor appointed to run a city in place of its elected leaders (Chapter 13) 899

Emergency Quota Act: the 1921 law, the first with a set limit on how many immigrants could come (Chapter 4) 298

emigrant: a person leaving one home to settle in another (Chapter 5) 352

emigrate: to leave your own country to live in another (Chapter 4) 267

eminent: famous and respected (Chapter 27) 2078

eminent domain: the power of a government to take private property for public use, paying the owner whether the owner wants to sell or not (Chapter 6) 418

emissions: the pollution a plant sends into the air (Chapter 3) 240

Emmys: the top awards for television (Chapter 24) 1741

emotional distress: harm to a person's mind and feelings (Chapter 9) 604

emotionally disturbed: having serious trouble controlling feelings and behavior (Chapter 20) 1372

emphysema: a lung disease in which the thin walls between the lungs' tiny air sacs break down, so the person cannot get enough air (Chapter 26) 2005

employer sanctions: punishments for employers who knowingly hire workers without legal permission (Chapter 4) 310

encomenderos: Spanish colonists in New Mexico who held the right to collect tribute from Pueblo households (Chapter 35) 2916

encomienda: a system set up by the king of Spain under which Spanish settlers could force Native people to work for them and hand over tribute (Chapter 14) 912

encroachments: steps that go beyond what is allowed (Chapter 20) 1322

encumber: to crowd or block (Chapter 16) 1067

endangered species: a kind of plant or animal in danger of dying out (Chapter 3) 229

endowment: a sum of money kept and invested so that its earnings pay for a school's work (Chapter 25) 1778

energy independence: a country producing more energy in total than it uses, while still buying some fuel from other countries (Chapter 10) 711

engine knock: a pinging sound in the engine (Chapter 19) 1268

engineered road: a road planned and built to a design, with a prepared surface (Chapter 11) 732

English Separatists: people who wanted to worship apart from the Church of England, among them the church that moved to Leiden and founded Plymouth (Chapter 4) 260

English subject: a person with the rights of someone born English (Chapter 32) 2613

engraver: a man who cuts designs into metal plates for printing (Chapter 6) 407

engraving: a picture printed from lines cut into metal or wood (Chapter 6) 418

enlisted soldier: a soldier who is not an officer (Chapter 22) 1543

enrolled citizens: people officially listed as members of a nation (Chapter 2) 170

ensign: a young officer (Chapter 36) 2999

enslaved person: a person owned by another person as property and forced to work for no pay (Chapter 4) 275

entering wedge: a first step that opens the way to more (Chapter 16) 1084

enterprising young men: bold young men ready for hard work (Chapter 1) 44

entradas: the Spanish name for armed expeditions into the north (Chapter 2) 102

entrails: the organs inside the body (Chapter 11) 734

enumerated goods: listed goods (Chapter 16) 1052

environmental justice: the campaign to stop companies and government officials from putting polluting sites next to poor, Black, brown and Native communities (Chapter 3) 229

Environmental Protection Agency (EPA): the federal office whose staff work to protect the country's air, water and land (Chapter 3) 229

environmental racism: Benjamin Chavis's name for choosing communities of color for waste sites and polluting factories (Chapter 3) 229

envoys: men sent to bargain for their government (Chapter 23) 1616

epidemic: a disease that spreads quickly to many people (Chapter 1) 64

epilepsy: a disease of the brain that causes seizures (Chapter 25) 1829

Equal Employment Opportunity Commission: the federal office that handles complaints of unfair treatment at work (Chapter 11) 764

equal protection: the rule that a state has to apply its laws the same way to everyone (Chapter 20) 1341

Equal Protection Clause: the part of the Fourteenth Amendment that requires states to treat people equally under the law (Chapter 20) 1356

equinox: one of the two days, in spring and fall, when day and night are the same length (Chapter 6) 395

eradicate: to wipe out (Chapter 37) 3135

erosion: the wearing away of soil by wind or water (Chapter 3) 215

erroneous: wrong (Chapter 25) 1829

espionage: spying (Chapter 31) 2559

esports: video game contests played for prize money (Chapter 36) 3058

essential workers: workers in grocery stores, warehouses, delivery and health care who kept going to work in person during the pandemic (Chapter 14) 974

established church: a church that the government supports with tax money (Chapter 29) 2242

establishing a religion: setting up an official religion (Chapter 20) 1372

estate: the money and property a person leaves when he dies (Chapter 7) 480

estate (land): a large holding of land (Chapter 1) 64

estate taxes: a tax on what a person leaves at death (Chapter 25) 1909

ether: an unseen substance that scientists once thought light traveled through as waves (Chapter 9) 604

ether (drug): a drug breathed as a vapor that kept a patient from feeling an operation (Chapter 22) 1543

Ether (money): a kind of digital money (Chapter 32) 2695

Ethical Culture: a group that meets to teach right conduct (Chapter 29) 2304

ethnicity: the group a person is born into, with its own language and customs (Chapter 4) 283

ethnomusicologist: a scholar who studies the music of a people and its place in their lives (Chapter 33) 2729

eugenics: the false idea that a country could improve its people by stopping some people from having children (Chapter 4) 298

evacuee: a person who leaves a place to escape danger (Chapter 5) 387

evangelicalism: the kind of Protestant faith whose preachers called on each listener to be converted (Chapter 29) 2256

evasive and nonresponsive: dodging the questions (Chapter 19) 1280

Everglades: the swamps of southern Florida (Chapter 2) 126

Every Student Succeeds Act: the 2015 law that replaced No Child Left Behind (Chapter 20) 1390

evict: to force people out of their homes (Chapter 3) 229

evolution: the scientific idea that living things, people included, developed over a very long time from earlier forms (Chapter 20) 1356

excess in apparel: too much spending on clothes (Chapter 35) 2916

excessive force: far more force than was needed (Chapter 14) 941

excise: a tax on goods made and sold inside the country (Chapter 16) 1062

exciseman: a collector of taxes on goods (Chapter 34) 2827

exclusion orders: the military orders that kept people of Japanese descent away from the west coast (Chapter 20) 1356

excommunication: putting a person out of the church (Chapter 21) 1412

execrable: hateful (Chapter 26) 1967

executive order: a written order from the president (Chapter 3) 200

exemption: a rule that lets some people off a requirement (Chapter 20) 1372

exile: being forced to live far from one's homeland (Chapter 25) 1778

Exodusters: Black Southerners who moved to Kansas around 1879, named after the Exodus, the Bible story of the Israelites' escape from slavery in Egypt (Chapter 5) 363

exoneration: a judge or a governor clearing a person of a crime after a conviction, because new evidence shows the person did not do it (Chapter 21) 1484

exoplanet: a planet that circles a star other than our Sun (Chapter 9) 648

expansion: a stretch of growth without a downturn (Chapter 16) 1096

expedited: quick (Chapter 25) 1909

expedition: a group of men sent out to explore (Chapter 1) 21

expelled: put out of the school altogether (Chapter 30) 2474

experiment station: a station run by a land-grant college where scientists studied soil, seed and farm animals and sent their results to farmers (Chapter 9) 604

experimental physicist: a scientist who runs experiments and measurements (Chapter 9) 648

exploit: to use someone for your own gain (Chapter 36) 3045

Explorers Club: a society of explorers (Chapter 1) 74

export trade: the business of sending goods away to be sold in another place (Chapter 16) 1052

exposition: a large public show of new machines and goods (Chapter 6) 418

exposure: dying from cold or other harsh weather without shelter (Chapter 2) 149

expressway: a wide road for fast traffic (Chapter 6) 443

expropriation: taking property away from its owners (Chapter 3) 200

expunge: to strike from the record (Chapter 20) 1333

exterminate: to kill off entirely (Chapter 3) 200

extinction: dying out completely (Chapter 30) 2375

extinguish a title: to end someone's ownership of land (Chapter 11) 745

extirpate: to wipe out completely (Chapter 22) 1522

extort: to force people by threats to pay (Chapter 21) 1442

extras: actors hired to fill out crowd scenes (Chapter 34) 2861

extravagance and dissipation: spending too much, and wasting time and money on pleasure (Chapter 34) 2833

extravagant: spending too much on show, or living too freely (Chapter 4) 310

exuberant: full of wild feeling (Chapter 29) 2256

F

faʻa Sāmoa: the Samoan way of life, which includes land held in common by families and the matai system of chiefs (Chapter 23) 1656

face recognition: a computer comparison of a photo of a face with a collection of photos that names possible matches (Chapter 8) 570

fair use: a use of someone else's work that the law allows in some cases (Chapter 33) 2795

fairness doctrine: the rule of the Federal Communications Commission that stations cover important public questions fairly, air more than one side, and give people attacked on the air a chance to answer (Chapter 31) 2572

Faith Keepers: the Onondaga men and women who decide when each ceremony takes place (Chapter 37) 3075

fallacy: a mistake in reasoning (Chapter 20) 1341

fallopian tubes: the tubes that carry eggs to the womb (Chapter 20) 1356

fallout: radioactive dust that falls back to the ground after a nuclear explosion (Chapter 9) 614

falls (of a river): where the river drops over rocks (Chapter 8) 517

false imprisonment: holding people against their will so they cannot leave (Chapter 37) 3125

famine: a time when food runs so short that many people die of hunger (Chapter 2) 102

famine relief: the food and work given to people who were starving (Chapter 4) 275

"fancy" trade: the sale of enslaved women and teenage girls, at much higher prices, to men who bought them in order to rape them (Chapter 24) 1693

fanega: a Spanish measure of grain (Chapter 35) 2916

fanner baskets: coiled baskets in which enslaved workers tossed rice to separate the grain from its husks (Chapter 15) 995

Farewell Address: Washington's written message to the country in 1796 saying he would not run again (Chapter 20) 1322

farm-to-table: buying food straight from local farms (Chapter 15) 1028

fascists: the followers of the dictators who ruled Italy and Germany (Chapter 33) 2759

fashion plate: a picture of the latest dresses colored by hand (Chapter 35) 2934

fasted: went without food (Chapter 21) 1406

fathom: six feet (Chapter 11) 745

fathom of wampum: a string of beads six feet long (Chapter 18) 1174

fault: a crack in the earth's crust where the rock on each side can move (Chapter 27) 2032

FBI: the Federal Bureau of Investigation, a national police agency (Chapter 2) 149

FCC: the Federal Communications Commission, the government office that licenses radio and television stations (Chapter 31) 2559

FDA: the Food and Drug Administration, the name given in 1930 to the Bureau of Chemistry, the office that enforced the food law (Chapter 15) 1015

FDIC: the Federal Deposit Insurance Corporation, the government agency that insures bank deposits (Chapter 17) 1143

feast day: a day the church sets aside to honor a saint (Chapter 4) 258

federal: part of the national government in Washington (Chapter 1) 82

Federal Bureau of Narcotics: the government's drug police (Chapter 28) 2165

Federal Communications Commission: the government office that licenses stations (Chapter 18) 1204

federal corporation: a company owned by the national government (Chapter 10) 691

federal crime: a crime the national government can charge (Chapter 20) 1356

federal district judges: the judges of the lowest federal courts (Chapter 20) 1372

federal funds rate: the rate banks charge one another for overnight loans (Chapter 17) 1153

federal grants: money the national government had promised (Chapter 11) 773

federal holiday: a paid day off for the national government's workers (Chapter 14) 929

federal law: a law made by Congress for the whole country (Chapter 3) 200

federal marshals: officers who work for the federal courts (Chapter 2) 158

federal prosecutors: the government lawyers who bring criminal charges for the United States (Chapter 4) 320

federal regulation: rules made and enforced by the national government (Chapter 19) 1256

Federal Reserve: a bank for the nation's banks, the central bank of the United States (Chapter 6) 431

Federal Reserve Bank of Chicago: a branch of the country's central bank (Chapter 6) 431

federal territories: the lands that were not yet states (Chapter 20) 1341

Federal Trade Commission: the federal office whose officials act against companies that cheat or mislead buyers (Chapter 13) 868

Federal Writers' Project: a government program that gave jobs to writers (Chapter 24) 1735

Federalists: the party led by Alexander Hamilton and John Adams, whose members wanted a strong national government (Chapter 20) 1322

feeble-minded: a label officials put on people they judged to be slow to learn (Chapter 4) 298

feeble-minded, moron, imbecile: names test makers gave to ranges of low scores (Chapter 30) 2414

feeblemindedness, insanity, criminality, dependency: old words for low intelligence, mental illness, crime and needing public charity (Chapter 4) 298

feed: the list of posts a site shows each user (Chapter 31) 2585

felling saws: saws for cutting down trees (Chapter 8) 517

fellow (of the Royal Society): a full member (Chapter 9) 585

felony: one of the most serious kinds of crime (Chapter 13) 899

felony assault: a serious crime of attacking a person (Chapter 37) 3125

felony child neglect: a serious crime of failing to protect children (Chapter 21) 1484

felony conspiracy: planning together to commit a serious crime (Chapter 25) 1871

felony obstruction of justice: a serious crime of getting in the way of an investigation (Chapter 31) 2585

felt: a thick cloth made by pressing fur together (Chapter 3) 185

FEMA: the Federal Emergency Management Agency, the national government's office for disaster help (Chapter 5) 387

female complaint: a health problem of women, in the words of patent-medicine advertisements (Chapter 18) 1193

female seminary: a school for girls and young women (Chapter 7) 474

feme covert: the legal name for a married woman, a covered woman (Chapter 25) 1760

feme sole: an adult woman who had never married or whose husband had died, a woman on her own in law (Chapter 25) 1760

fencerow to fencerow: planting every acre up to the fences (Chapter 15) 1028

fentanyl: an opioid made in a lab, 50 to 100 times stronger than morphine (Chapter 26) 2022

feral hogs: pigs from farm stock that live in the wild (Chapter 3) 182

ferule: in the Cambridge report's words, "a flexible instrument, made of two flat pieces of leather, sewed together, and filled with some soft substance" (Chapter 30) 2392

fetus: the medical word for the unborn offspring growing in the womb after the first weeks of pregnancy (Chapter 25) 1909

feud: a long run of killings in which each side took revenge for the last death (Chapter 21) 1406

feudal: like the old system in Europe in which farm workers were bound to a lord's land (Chapter 25) 1757

Feynman diagrams: simple drawings that show how particles act on each other (Chapter 9) 633

fiat money: money backed by no metal and worth what it can buy (Chapter 17) 1153

fiberglass: plastic that hardens around threads of glass (Chapter 35) 2959

fidelity: loyalty (Chapter 25) 1909

fife: a small, high-pitched flute (Chapter 33) 2729

Fifteenth Amendment: the 1870 change to the Constitution that barred the states from denying a man the vote because of his race (Chapter 20) 1333

filament: a thin thread inside an electric lamp that glows when an electric current heats it (Chapter 8) 541

filibuster: a way of blocking a bill by talking on and on so that no vote can happen (Chapter 20) 1372

fill: earth dumped on top, used in Chicago to raise the streets over the sewer pipes (Chapter 6) 418

finished goods: things already made (Chapter 13) 852

finished ironware: iron goods ready for use (Chapter 13) 852

fire engine: the name people in the 1700s used for a steam engine (Chapter 10) 671

fire insurance: many owners each pay a small amount into a shared fund, and the company that keeps the fund pays any owner whose house burns (Chapter 7) 495

fire limits: an area with strict building rules meant to stop fires spreading (Chapter 25) 1829

fire tower: a shaft that ran up through a building (Chapter 6) 431

firemaster: an officer chosen by a vote whose orders at a fire everyone had to obey as if they were the law (Chapter 6) 404

fireside chat: a radio talk by President Franklin Roosevelt to the country (Chapter 17) 1143

firestorm: many fires joined into one huge fire that uses up the oxygen in the air (Chapter 22) 1559

firing squad: a group of shooters who fire rifles at the prisoner's heart (Chapter 21) 1468

First Continental Congress: the meeting of leaders from the colonies in 1774 (Chapter 34) 2833

first-degree criminal sexual conduct: under Michigan law, a sex act that goes inside another person's body, in certain cases, such as when the victim is under 13. It can be punished with life in prison (Chapter 36) 3058

first-degree murder: the most serious kind of murder, planned ahead of time (Chapter 12) 832

First International Polar Year: the time when scientists from several countries studied the far north and far south at the same time (Chapter 1) 64

First Salmon Feast: the feast the Yakama, Umatilla, Warm Springs and Nez Perce peoples hold each spring before the fishing season opens (Chapter 37) 3075

fiscal year: the government's budget year (Chapter 14) 974

fish-in: people fishing in the open in front of state officers and letting themselves be arrested, to force the treaties into a courtroom (Chapter 2) 158

fish ladder: rising pools that let salmon swim upstream past a dam (Chapter 10) 691

fish weir: a fence of wooden stakes set across the water, with bundles of brush woven between the stakes, that traps fish when the tide goes out (Chapter 8) 512

fission: the splitting of the center of an atom, called the nucleus, into two pieces, which gives off energy (Chapter 9) 614

five-and-dime stores: stores that sold small goods for 5 and 10 cents (Chapter 18) 1214

Five Tribes: the Cherokee, Choctaw, Muscogee (Creek), Chickasaw and Seminole nations (Chapter 5) 363

fixed rate: an interest charge that never changes (Chapter 7) 485

flag of truce: a white flag, a sign that both sides promise not to attack or seize anyone while they talk (Chapter 2) 126

flagrant disregard: openly ignoring something (Chapter 27) 2093

flagship: the ship that carried a fleet's leader (Chapter 4) 260

flapper look: the slim, straight shape of 1920s dresses (Chapter 35) 2949

flash-bang: an explosive device that makes a loud noise and a bright flash of light (Chapter 2) 170

flash flood emergency: a more urgent alert than a flash flood warning (Chapter 27) 2104

flat-top: a flat-roofed house (Chapter 6) 431

flatboat: a large, flat-bottomed boat that carried farm goods and people downriver (Chapter 27) 2061

flatirons: solid metal irons heated over the coals and held with a rag around the handle (Chapter 7) 485

flaunting: showing off (Chapter 35) 2916

flax: a plant whose stems give the threads for linen (Chapter 15) 1000

fleshmonger: a person who traded in human beings (Chapter 35) 2934

flint and chert: two hard stones that people made into tools (Chapter 13) 844

flint clay: a stone from near St. Louis, Missouri, used for the Birger figurine (Chapter 32) 2598

flogging: beating a person's bare back with a whip, which cuts the skin and leaves welts and scars (Chapter 1) 44

flogging round the fleet: a sentence under which a sailor was whipped aboard each ship of the squadron in turn (Chapter 1) 44

flood plain: the low land beside a river that the river covers when it floods (Chapter 27) 2093

florins: Dutch coins (Chapter 26) 1954

flue: a channel that draws hot air and smoke upward, as a chimney does (Chapter 6) 418

flux, bloody flux: the old word for diarrhea, and diarrhea with blood in it (Chapter 26) 1954

flying boat: a plane that takes off from and lands on water (Chapter 11) 756

flywheel magneto: an engine part that makes electricity (Chapter 8) 553

folklorist: someone who collects the stories, songs and sayings people pass along by word of mouth (Chapter 32) 2660

follies: silly shows (Chapter 31) 2572

folly: a foolish project (Chapter 1) 64

foment a rebellion: to stir one up (Chapter 25) 1770

food deserts: low-income areas far from a supermarket, more than 1 mile away in a city or more than 10 miles in the country (Chapter 15) 1039

food sovereignty: Native nations' work to control their own food (Chapter 15) 1039

forbearing of labor: not working (Chapter 37) 3083

forced assimilation: making people give up their own ways of life (Chapter 29) 2324

forced labor: work people could not refuse and could not leave (Chapter 3) 200

forcible feeding: holding a person down, pushing a tube through the nose or mouth down the throat into the stomach, and pouring liquid food through it (Chapter 25) 1829

forcibly removed: made to go whether they wanted to or not (Chapter 1) 64

ford: a shallow place where people and horses can wade across a river (Chapter 11) 728

foreclose: to take a house back from a borrower who cannot pay (Chapter 7) 485

foreclosure filing: the paper a lender files to start taking back a home (Chapter 7) 502

foreign agent: someone acting on behalf of another country's government (Chapter 20) 1390

foreman: the head juror (Chapter 21) 1434

forerunner: an earlier event of the same kind (Chapter 3) 192

Forest Finns: settlers of New Sweden who were skilled at building with logs (Chapter 7) 471

forest reserve: public forest land that a president could set aside under the 1891 Forest Reserve Act, the start of the national forests (Chapter 3) 200

forester: a person trained to study and manage forests (Chapter 3) 215

forfeit: to lose something as a punishment, or to lose a match without playing it (Chapter 25) 1909

forge: a workshop where workers heat iron and hammer it (Chapter 8) 517

forge (papers): to fake (Chapter 24) 1693

"formal racial barrier": a bar based on race written into the rules (Chapter 8) 541

fortify: to make stronger (Chapter 20) 1322

Forty-Eighters: German revolutionaries of 1848 who came to the United States (Chapter 4) 275

forty-niners: the people who rushed to California for gold in 1849 (Chapter 5) 352

forum: a website where users wrote to one another (Chapter 11) 745

foster care: living with families paid to raise children not their own (Chapter 30) 2442

foundry: a workshop where metal is melted and poured into molds (Chapter 7) 474

Four Corners: the place in the Southwest where Utah, Colorado, Arizona and New Mexico meet (Chapter 6) 395

Fourteenth Amendment: the 1868 change to the Constitution that made all persons born or naturalized in the United States citizens and promised them the equal protection of the laws (Chapter 4) 283

fracking: short for hydraulic fracturing, a way of getting gas and oil out of rock deep underground by pumping water, sand and chemicals down a well at high pressure to crack the rock (Chapter 10) 711

frame: a single picture of a film (Chapter 34) 2889

Frame of Government: William Penn's plan for running Pennsylvania (Chapter 29) 2213

franchise: a deal that lets other people open stores under an owner's name and system in return for fees (Chapter 15) 1028

Franciscans: one group of Catholic friars (Chapter 29) 2207

frankly and unequivocally: openly and with no doubt (Chapter 30) 2414

Fraternal Order of Eagles: a men's club (Chapter 37) 3119

fraud: cheating people by lying (Chapter 10) 691

fraud, deceit and unprofessional conduct: legal words for cheating, lying and breaking the rules of one's profession (Chapter 26) 2005

free agent: a player free to sign with any team (Chapter 36) 3045

free association: being a separate country tied to the United States by a formal agreement, as the Marshall Islands are (Chapter 23) 1656

free-banking laws: state laws with few rules for opening a bank (Chapter 17) 1128

free labor: workers who were not enslaved (Chapter 11) 734

free people of color: people of African or mixed descent who were not enslaved (Chapter 4) 271

free, prior and informed consent: a yes given freely and beforehand, by people who have been told everything about the request (Chapter 9) 648

Free Public School Act: the 1834 Pennsylvania law for a statewide system of common schools paid for by taxes (Chapter 20) 1333

free silver: the government coining all the silver brought to it, at 16 ounces of silver to 1 ounce of gold (Chapter 17) 1136

Free Speech Movement: a 1964 protest by Berkeley students for the right to speak and organize about politics on campus (Chapter 8) 561

Freedmen: the people the Five Tribes had enslaved and freed after 1866 (Chapter 24) 1712

Freedmen's Bureau: a federal office set up to help people freed from slavery (Chapter 24) 1712

freedom dues: a payment of clothes, tools or money at the end of an indenture (Chapter 4) 260

freedpeople: the men, women and children who had been freed from slavery (Chapter 15) 1009

freelance reporter: a reporter who works for himself rather than for one newspaper (Chapter 31) 2572

freeman: a man who was neither enslaved nor bound as a servant (Chapter 20) 1309

Freemasons: a secret society of men (Chapter 20) 1333

freight: goods being shipped (Chapter 5) 352

freight elevator: the elevator used for goods (Chapter 36) 3028

freight station: a building where trains loaded and unloaded goods (Chapter 18) 1193

friar: a member of a Catholic religious order (Chapter 1) 21

friend-of-the-court brief: a written argument sent to a court by people who are not in the case (Chapter 25) 1829

frock coat: a fitted coat with a skirt that reached to just above the knee (Chapter 35) 2934

from time immemorial: since before anyone can remember (Chapter 37) 3075

frontier: the edge of the colonists' settlements, on land where Native nations lived (Chapter 1) 64

frontier line: the edge of thinly settled land in the West, where two to six people lived on each square mile (Chapter 1) 64

frostbite: damage to skin and flesh from freezing (Chapter 27) 2068

fuchsine: a deep crimson red made from coal tar (Chapter 35) 2940

fuel (for fires): the dead wood and brush that burns (Chapter 3) 240

fuel assembly: a tall bundle of fuel rods (Chapter 10) 711

fuel-economy standards: rules for how far a car must go on a gallon of gas (Chapter 10) 702

fugitive: a person who has escaped (Chapter 24) 1712

fugitive clause: the part of the Constitution that said a person "held to Service or Labour" who escaped to another state had to be handed back (Chapter 24) 1684

fully synthetic: made entirely from chemicals rather than from anything that grew (Chapter 8) 553

fundamental right: a right the Constitution itself protects (Chapter 20) 1372

fundamentalist: a Christian who reads the Bible word for word (Chapter 29) 2287

funeral procession: the line of people walking behind the coffin (Chapter 18) 1182

fur trade: the buying and selling of animal skins between European traders and Native trappers (Chapter 2) 108

furnace: the tall oven where iron is melted out of rock (Chapter 8) 517

furnish: the food, seed, tools and cabin a landowner gave a sharecropping family on credit, to be paid for later (Chapter 15) 1009

G

gag rule: the House rule, from 1836 to 1844, that set aside every antislavery petition without reading it (Chapter 25) 1778

galena: a lead ore (Chapter 13) 859

gallery: a balcony above a church's main floor (Chapter 29) 2242

gallows: a frame for hanging people (Chapter 4) 283

game: the wild animals people hunted (Chapter 3) 180

Game Boy: a game machine small enough to hold in the hands (Chapter 36) 3045

gamed: tricked an app (Chapter 14) 974

gaming: gambling businesses such as casinos (Chapter 2) 158

gander: a male goose (Chapter 36) 2995

gander pulling: a game in which riders grabbed at the greased neck of a live gander hung upside down until one pulled its head off (Chapter 36) 2995

gang system: overseers driving enslaved people in gangs, row by row across the fields (Chapter 24) 1693

gangrene: the rotting of dead flesh in a living body (Chapter 1) 44

garlix: a coarse linen (Chapter 35) 2923

garment workers: people who sew clothes for a living (Chapter 14) 941

garroted: strangled to death with a cord or an iron collar that was tightened around the neck (Chapter 1) 21

gas chamber: a sealed room filled with poison gas (Chapter 21) 1484

gas works: plants that made gas (Chapter 10) 677

gash: a deep cut (Chapter 12) 825

gasometers: large tanks for storing gas made from coal (Chapter 10) 677

gavilleros: guerrillas in the east of the Dominican Republic (Chapter 23) 1634

gay: a man who loves men or a woman who loves women (Chapter 1) 82

GDP: the total value of all the goods and services made in the country in a year (Chapter 16) 1084

Gendarmerie: the Haitian police force that Marines commanded (Chapter 23) 1634

gender dysphoria: the medical term for the distress a person feels when the sex recorded at birth does not match their own sense of their gender (Chapter 25) 1909

gender identity: a person's own sense of being a boy or a girl, a man or a woman (Chapter 25) 1909

genealogist: a person who traces family histories through old records (Chapter 4) 283

General Accounting Office: the federal office that checks how federal money is spent (Chapter 2) 158

General Assembly of Virginia: the colony's lawmaking body (Chapter 35) 2916

General Conference: the gathering where United Methodist delegates write the church's rules (Chapter 29) 2324

General Court (Massachusetts): the Massachusetts colony's lawmakers (Chapter 4) 260

General Court (Virginia): the governor of Virginia and his council, sitting together as the colony's highest court (Chapter 14) 914

general incorporation law: a law that let any group form a corporation by following set steps, without a special vote (Chapter 19) 1250

General Orders No. 3: General Gordon Granger's order at Galveston on June 19, 1865, which stated that "all slaves are free" (Chapter 37) 3107

general strike: workers in many different trades stopping work at the same time (Chapter 14) 922

general-purpose computer: one that could be set up to solve many kinds of problems (Chapter 8) 553

generative AI: AI that makes new pictures, text or sound (Chapter 8) 570

generator: a machine that turns motion into electric current (Chapter 10) 681

genes: the tiny instructions inside every living thing that parents pass on to their young (Chapter 3) 229

genetically engineered seed: seed whose DNA scientists changed in a laboratory (Chapter 15) 1039

genetics: the study of how living things pass on their traits (Chapter 9) 614

genitals: the parts of the body used to make babies (Chapter 4) 298

genius: unusual ability (Chapter 25) 1829

genome: the complete set of DNA in a living thing (Chapter 9) 633

genteel: well-off (Chapter 36) 2995

Gentile: a person who is not Jewish (Chapter 29) 2242

gentrification: what happens when buyers and lenders return to a neighborhood they had avoided for years: new owners fix up the houses, rents and prices rise, and the people who lived there can no longer afford to stay (Chapter 6) 452

genus: a group of closely related kinds of animals or plants (Chapter 9) 591

geologist: a scientist who studies rocks and the earth (Chapter 1) 64

Georgian house: a style of house with both sides built to match and a balanced look taken from ancient Greek and Roman buildings (Chapter 7) 468

germ: a living thing too small to see that can cause disease (Chapter 7) 485

German measles: an illness that spreads from person to person (Chapter 1) 82

germanium: a chemical element (Chapter 8) 553

gerrymandering: drawing the lines of voting districts so that one side wins more seats (Chapter 20) 1390

Ghost Dance: the religion that spread across the Plains reservations in 1889 and 1890 (Chapter 2) 134

ghost forests: the dead trunks of cedar trees killed by salt water when the coast dropped in 1700 (Chapter 27) 2047

G.I. Bill: a 1944 law under which federal officials paid for veterans' schooling and backed their home loans (Chapter 16) 1096

giardia and amoebas: tiny living things that cause stomach infections (Chapter 26) 1946

gig platform: an app or website that hands out short jobs one at a time (Chapter 14) 974

gigafire: a fire that burned more than a million acres (Chapter 3) 240

gigot or leg-of-mutton sleeve: a sleeve that puffed out wide from the shoulder (Chapter 35) 2934

gilded: covered with a thin layer of gold (Chapter 16) 1074

Gilded Age: the name for the years 1850 to 1900, from an 1873 novel by Mark Twain and Charles Dudley Warner (Chapter 16) 1074

gin-house: the building where the cotton gin stood (Chapter 8) 528

girdle: tight underwear that shaped the waist and hips (Chapter 14) 963

give-away dance: a gathering at which people give gifts, such as horses, to others (Chapter 29) 2287

glaciers: thick sheets of ice that once covered the north (Chapter 3) 182

glaucoma: a disease in which pressure inside the eye destroys the nerve that carries sight (Chapter 35) 2959

glider: an airplane with no engine (Chapter 8) 553

Godey's Lady's Book: a women's magazine read across the country (Chapter 35) 2934

gold album: the RIAA's award for 500,000 copies sold (Chapter 33) 2713

gold certificates: paper notes that could be traded for gold (Chapter 17) 1143

gold leaf: thin sheets of gold (Chapter 12) 825

gold standard: a dollar backed by gold alone (Chapter 17) 1136

Golden Rule: the teaching that people should treat others as they want to be treated (Chapter 25) 1760

gonorrhea and chancroid: diseases passed during sex (Chapter 26) 1990

Good Friday: the Christian holy day before Easter (Chapter 22) 1515

gospel: a kind of Black church music (Chapter 33) 2759

gouging: fighting in which only weapons were banned and a fighter tried to push his thumbs into the other man's eye socket and force the eye out (Chapter 36) 2999

gourd: a large fruit with a hard shell that can be dried and hollowed out (Chapter 33) 2729

gout: painful, swollen joints (Chapter 13) 868

Government Accountability Office: the office that checks federal agencies for Congress (Chapter 4) 320

government agent: an official sent to deal with a Native nation (Chapter 21) 1406

government bonds: papers the government promises to pay back with interest (Chapter 11) 745

governor's council: the six men who advised the governor (Chapter 20) 1309

GPS: the Global Positioning System, which uses signals from satellites to show a GPS device where it is (Chapter 8) 561

grace: in Puritan teaching, God's gift of being saved and going to heaven (Chapter 4) 260

grade: to level the ground for railroad tracks (Chapter 14) 922

gradual abolition: ending slavery step by step over many years (Chapter 24) 1684

graft: money a person gets by using a public office (Chapter 20) 1356

grain elevator: a tall building where grain is stored before it is shipped (Chapter 16) 1074

grammar (book): a book that explains how a language works (Chapter 30) 2351

grammar school: a school to prepare boys for the university (Chapter 20) 1309

grammar school (1800s): the school that came after primary school (Chapter 25) 1795

Grand Army of the Republic: the organization of Union veterans (Chapter 37) 3107

Grand Council (Carolina): the group of men who governed the Carolina colony (Chapter 34) 2827

Grand Council (Haudenosaunee): the Haudenosaunee league's lawmaking body, with fifty seats (Chapter 2) 99

grand jury: a group of citizens who decide whether there is enough proof for a trial (Chapter 4) 275

Grand Sachem: the head of Tammany Hall (Chapter 20) 1341

Grand Slam titles: the four biggest tennis championships (Chapter 36) 3045

grandfather clause: a rule that let a man who failed the voting tests still register if his father or grandfather had voted in 1867 (Chapter 20) 1341

Grange: the Patrons of Husbandry, the first large farmers' group with members across the whole country (Chapter 15) 1009

grant: a gift of money (Chapter 20) 1390

graphical browser: a program that shows web pages with pictures (Chapter 8) 561

grate: a rack that holds coal up off the floor of a fireplace so air can flow in underneath (Chapter 10) 677

graven image: an idol (Chapter 29) 2287

gravitational waves: ripples in space and time (Chapter 9) 648

greasepaint: a thick stage makeup (Chapter 34) 2841

Great American Desert: Stephen Long's name for the high plains (Chapter 1) 44

Great Awakening: the revivals that swept the colonies in the 1730s and 1740s (Chapter 29) 2231

Great Basin: a huge dry region between the Rocky Mountains and the Sierra Nevada (Chapter 1) 44

Great Charter: the paper the leaders of the Virginia Company put out in 1618 that let the governor call a General Assembly to make laws (Chapter 20) 1309

Great Compression: the sharp narrowing of the gap between high and low wages in the 1940s that lasted into the 1970s (Chapter 16) 1096

Great Depression: the hard years of the 1930s when many people had no work (Chapter 4) 298

Great Famine: the Irish famine of 1845 to 1852 (Chapter 4) 275

great house: one of the large stone buildings of many stories at Chaco Canyon (Chapter 6) 395

Great Law of Peace: the laws the five Haudenosaunee nations agreed to live by when they joined together (Chapter 2) 99

Great Migration: the coming of about 20,000 Puritans to New England in the 1630s (Chapter 4) 260

Great Migration: the move of about 6 million Black Americans out of the South between 1910 and 1970 (Chapter 5) 376

Green Corn Ceremony: the midsummer ceremony of the Muscogee (Creek) and other nations of the Southeast, held when the first corn was ripe (Chapter 21) 1406

Green Revolution: the rise in grain harvests in poor countries that followed Borlaug's wheat (Chapter 15) 1028

green wood: freshly cut wood (Chapter 10) 671

greenbacks: the national paper money of 1862, called United States Notes, printed with green ink on the back (Chapter 17) 1136

greenhouse gas: a gas that takes in heat and gives it off again (Chapter 9) 633

greige: a gray-beige (Chapter 35) 2959

grid: a pattern of straight streets crossing at right angles (Chapter 6) 398

grid: the network of power plants and wires that serves an area (Chapter 10) 711

grid (in a vacuum tube): a bent wire between the filament and the plate (Chapter 8) 553

grievance: a wrong worth complaining about (Chapter 25) 1770

griot: a keeper of history who tells it from memory (Chapter 24) 1741

grist mill: a mill that grinds grain into flour (Chapter 8) 517

grog shop: a saloon (Chapter 28) 2154

groom: a worker who cares for horses (Chapter 36) 3008

gross negligence: a serious failure to take the care a job requires (Chapter 27) 2093

grossly inadequate: far short of enough (Chapter 13) 868

GU272: the 272 enslaved people whom the Maryland Jesuits, priests of a Catholic religious order, sold to plantations in Louisiana in 1838, in part to keep Georgetown College from running out of money (Chapter 24) 1744

guano: the droppings of seabirds, which make crops grow (Chapter 23) 1623

guardhouse: the old jail of an Army post (Chapter 30) 2392

guardian: a person given legal control of someone else's money (Chapter 2) 149

guerrillas: fighters in small bands who hide among the people and attack by surprise (Chapter 23) 1623

Guggenheim Fellowship: a grant of money for artists and writers (Chapter 32) 2679

guilder: a Dutch coin (Chapter 6) 399

guinea: a coin worth 21 shillings (Chapter 36) 2999

guise: a false cover (Chapter 30) 2474

gurdwara: a Sikh house of worship (Chapter 29) 2324

gynecologist: a doctor who treats the parts of a woman's body that carry and deliver babies (Chapter 9) 633

H

H-2A visa: a paper that lets a farmworker from another country come for one season of farm work (Chapter 14) 974

habeas corpus: a person's right to be brought before a judge, who decides whether the government may go on holding that person (Chapter 4) 283

haberdashery: small sewing goods such as thread, ribbons and buttons (Chapter 18) 1178

habiliments: an old word for clothes ("habilemts" in the 1665 court record) (Chapter 34) 2823

half-freedom: the Dutch status under which a man could hold land of his own but still owed the Dutch West India Company work and a yearly payment, and his children stayed the Company's property (Chapter 6) 399

halfpenny: a small British coin (Chapter 31) 2520

hall: the single room of the basic English colonial house (Chapter 7) 465

hall-and-parlor house: a house with a hall for cooking and work and a second room, the parlor, the best room (Chapter 7) 465

hallucination: seeing or hearing things that are not there (Chapter 13) 863

handbill: a short printed notice (Chapter 5) 363

handcart: a small two-wheeled cart pulled by hand (Chapter 5) 352

handicapped: an older word for disabled (Chapter 20) 1372

haole: the Hawaiian word for white people who are not Native Hawaiian (Chapter 23) 1623

harassment: a steady stream of acts meant to frighten people (Chapter 7) 495

hard labor: heavy work done as a punishment (Chapter 2) 126

hardrock mine: a mine dug into solid rock (Chapter 13) 863

harmony: two or more notes sung together (Chapter 33) 2729

harpoon: a heavy iron spear thrown into a whale from a rowed boat, which whalers called an iron (Chapter 10) 677

hate crimes: crimes against people chosen because of their race, religion or a similar trait (Chapter 12) 832

Haudenosaunee: a league of Native nations in what is now New York State (Chapter 1) 31

hawk's bell: a small round bell (Chapter 1) 21

hazardous waste: waste that can harm people's health (Chapter 3) 229

head: the word for counting cattle, used in an 1824 loan to count enslaved people (Chapter 17) 1128

headman: a leader (Chapter 3) 200

headright: an equal share of the Osage nation's oil money (Chapter 2) 149

headright: Virginia's rule that whoever paid for a person's passage received 50 acres of land for that person (Chapter 16) 1052

health insurance: a plan that pays doctor and hospital bills (Chapter 26) 2005

hearing: a meeting where each side tells its story to an official who decides (Chapter 2) 102

hearing officer: the official who runs a hearing (Chapter 11) 773

heart transplant: an operation that gives a person the heart of someone who has died (Chapter 32) 2695

hearth: the open fireplace that gave a room its heat and light (Chapter 7) 465

heat illness: what happens when the body gets too hot and cannot cool itself down (Chapter 18) 1222

heat shield: the part that protects a spacecraft from the heat of coming back through the atmosphere (Chapter 1) 92

heater's helper: one of the workers who heated the steel (Chapter 19) 1256

heathen: an insulting word for people who were not Christian (Chapter 29) 2274

heathen rites: missionaries' words for the religion of people who were not Christian (Chapter 29) 2274

heathenish: an insult for the ways of people who were not Christian (Chapter 29) 2274

heathenish and Popish: un-Christian and Catholic (Chapter 36) 2987

heavy rotation: playing a song often on television or radio (Chapter 33) 2779

heavyweights: the boxers in the heaviest weight class (Chapter 36) 3028

hectare: a square of land 100 meters on each side (Chapter 3) 182

heirs: the family members who inherit a person's things (Chapter 8) 524

heirs' property: land that all of a landowner's heirs owned together after the owner died without a will, each owning a share of the whole farm (Chapter 15) 1028

HeLa: the name of Henrietta Lacks's cells, from the first two letters of her first and last names (Chapter 9) 633

heliotrope: a small purple flower (Chapter 18) 1193

helium: an element, a gas that does not burn (Chapter 6) 431

hematoma: a pocket of blood under the skin (Chapter 20) 1372

hemlines: the bottom edges of skirts (Chapter 35) 2949

hemophilia: a condition in which the blood does not clot (Chapter 26) 2005

hemp and flax: plants whose fibers are made into rope and cloth (Chapter 15) 1000

hems: the folded-up bottom edges of clothes (Chapter 35) 2949

henceforth: from now on (Chapter 23) 1616

hepatitis: an infection of the liver, a disease that inflames the liver (Chapter 26) 1946

hereditary diseases: illnesses passed from parents to children (Chapter 25) 1829

heredity: the passing of traits from parents to children (Chapter 9) 614

heresy: teaching against the church's beliefs (Chapter 4) 260

heroin: a drug made from morphine (Chapter 19) 1291

herpes: a virus that causes sores (Chapter 26) 1946

hides: the skins of animals such as bison (Chapter 3) 200

Higgs boson: a particle tied to how other particles get their mass, the amount of matter in an object (Chapter 9) 648

high cuffing: chaining a prisoner standing to the cell bars with the hands crossed and pulled up high, so only the toes touch the floor, which strained the arms and shoulders until they swelled (Chapter 22) 1559

high steel: joining the steel frames of bridges and buildings high above the ground (Chapter 6) 418

hired out: rented by an owner to another employer, who paid wages that the owner kept (Chapter 1) 44

hiring day: January 1, when enslavers rented enslaved people to other white men for the coming year (Chapter 37) 3101

Hispaniola: an island in the Caribbean (Chapter 1) 21

historically Black colleges and universities: the colleges first founded for Black students (Chapter 30) 2392

HIV: a virus that destroys the white blood cells that fight infection, passed through blood, semen and other body fluids (Chapter 26) 2005

Hmong: an ethnic group from the mountains of Laos (Chapter 4) 310

hoax: a lie made up on purpose (Chapter 28) 2154

hobbled: having the legs tied so a person can only shuffle (Chapter 29) 2242

hogan: a traditional Navajo house (Chapter 13) 899

hoggee: a boy who drove the mules that pulled canal boats (Chapter 11) 734

hogshead: a large barrel (Chapter 31) 2506

hogtied: ankles bound and hands tied to the feet (Chapter 21) 1468

holding company: a company that owns shares, or pieces of ownership, of other companies (Chapter 10) 691

Holocaust: the Nazis' murder of Jews and other people during World War II (Chapter 4) 298

Homebrew Computer Club: a group of people who built their own computers (Chapter 8) 561

homespun: cloth made at home (Chapter 18) 1182

homestead: a farm on land that the federal government gave to settlers who lived on it and worked it (Chapter 3) 200

Homestead Act: the 1862 law under which a person could claim up to 160 acres of public land and keep it after five years of living on it or farming it (Chapter 3) 200

homesteader: a settler who claimed land from the US government to live on and farm (Chapter 3) 200

homesteading: claiming public land and farming it for years in order to own it (Chapter 3) 215

homicide: on a death record, a death that another person caused, which is not a finding that anyone broke the law (Chapter 4) 320

hominy: dried corn cooked in water (Chapter 15) 991

homogeneity: sameness (Chapter 4) 298

homogeneous population: a group of people who are all alike (Chapter 13) 886

homosexuality: another word for being gay (Chapter 29) 2324

honest graft: learning where city officials planned to build something, buying the land first and then selling it to the city at a profit (Chapter 20) 1356

honeycombing: drilling rows of holes close together (Chapter 12) 815

honorable discharge: a record that a soldier served well and left in good standing (Chapter 22) 1522

honorary degree: a degree given as an honor instead of for study (Chapter 4) 298

Hooverville: the name people gave the shantytowns of the 1930s (Chapter 16) 1084

Hopewell culture: a name archaeologists gave the Native people who built the Ohio earthworks, not the name of any nation (Chapter 12) 783

horizontal drilling: drilling wells that turn and run sideways through the rock for thousands of feet (Chapter 10) 711

hormone treatment: giving a person the hormone of the sex they identify as, which changes the body in that direction (Chapter 25) 1909

hornbook: a wooden paddle about the shape of a table-tennis bat with one printed sheet tacked to it, covered by a thin, clear sheet of animal horn (Chapter 30) 2359

horse litter: a stretcher carried by horses (Chapter 5) 339

Horse People, Bow People and Little Cherry People: the French names for Plains nations the La Verendrye brothers traveled with (Chapter 1) 36

horsepower: a measure of power based on the work one horse can do (Chapter 8) 541

hose: stockings (Chapter 35) 2916

hostage: a person held prisoner so that his or her people do what the captor wants, such as letting an army pass safely (Chapter 1) 21

Hotchkiss guns: small cannon (Chapter 2) 134

hotplate: a small electric burner for cooking (Chapter 6) 431

house arrest: being kept under guard in one's own quarters (Chapter 22) 1575

House of Burgesses: colonial Virginia's elected lawmaking body (Chapter 21) 1422

House of Commons: one of the two groups of lawmakers in England's Parliament (Chapter 19) 1235

house of correction: a kind of jail for people found guilty of small crimes (Chapter 25) 1778

hoyaneh: a leader who holds a seat on the Grand Council (Chapter 20) 1304

Hudson River School: a group of American landscape painters, about 1825 to 1870, who painted American land in careful, true-to-life detail (Chapter 32) 2627

Huguenots: French Protestants (Chapter 4) 258

huke: a hooded cloak (Chapter 35) 2916

hulk: the hull of an old ship that no longer sailed (Chapter 6) 413

hull: the body of a boat (Chapter 8) 512

human trafficking: using force, tricks or threats to make people work or perform sex acts (Chapter 14) 963

humbug: a trick or a false claim made to fool people (Chapter 18) 1188

humor: an old word for a fluid of the body (Chapter 26) 1961

hundreds: districts five or six miles square in Jefferson's plan (Chapter 30) 2367

hunger strike: refusing to eat as a protest (Chapter 25) 1829

Huntington's disease: a disease passed down in families that destroys parts of the brain, causing movements a person cannot control, changes in mood, and the loss of the ability to think (Chapter 33) 2759

husbandry: an old word for farming (Chapter 15) 1009

hush harbors: brush arbors, shelters built of branches, where enslaved people worshipped out of their owners' sight (Chapter 24) 1693

hush money: money paid to keep someone quiet (Chapter 21) 1484

hybrid: a car with both a gasoline engine and an electric motor (Chapter 10) 711

hybrid seed: seed that comes from crossing two different parent plants (Chapter 15) 1015

hydraulic: worked by liquid under pressure (Chapter 1) 92

hydroelectric dam: a dam that holds back a river, so that water rushing down through it turns machines that make electricity (Chapter 10) 691

hydrogen: a gas that burns (Chapter 8) 553

hydrogen bomb: a bomb far more powerful than the first atomic bombs (Chapter 13) 886

hydrogen ions: hydrogen atoms with an electric charge (Chapter 8) 553

hydropower: electricity made from flowing water (Chapter 10) 711

hymn: a church song (Chapter 29) 2256

hypodermic syringe: a hollow needle for pushing a drug under the skin (Chapter 28) 2154

hypothermia: a body temperature fallen too low to keep a person alive (Chapter 10) 711

hysterectomy: an operation in which a surgeon cuts out the womb, the organ in which a baby grows, after which a woman can never become pregnant (Chapter 2) 158

I

ICE: a federal immigration agency, whose agents arrest and deport people (Chapter 4) 256

icebox: a cabinet cooled by a block of ice (Chapter 7) 485

iconoclastic traditionalist: someone who breaks with old ways and keeps them at the same time (Chapter 33) 2713

identity theft: pretending to be someone else by using that person's private information (Chapter 31) 2585

ideological deviation: holding ideas that Cuba's rulers banned (Chapter 4) 310

ideology: a set of ideas (Chapter 25) 1909

idolatry: worshipping false gods (Chapter 29) 2213

ignoramus: a person who knows very little (Chapter 29) 2287

illuminations: candles lit in the windows (Chapter 37) 3094

image dissector: Philo Farnsworth's camera tube (Chapter 8) 553

ImageNet: a contest that scored computer programs on naming what is in photographs (Chapter 8) 570

Imbangala: African fighters paid to fight for others (Chapter 4) 260

imbecile: an insulting label for a person judged slow to learn (Chapter 20) 1356

immersion school: a school where teachers give every lesson in the language the children are learning (Chapter 2) 170

Immigration and Nationality Act: the 1965 law that ended the national quotas and admitted immigrants on the basis of family ties and skills (Chapter 4) 310

Immigration and Naturalization Service (INS): the federal office in charge of immigration (Chapter 4) 310

immune: able to fight off a disease, so the person will not catch it again (Chapter 26) 1946

immunities: the protections the law gives every citizen (Chapter 23) 1616

impairing: weakening (Chapter 19) 1250

impartial: fair (Chapter 21) 1427

impeach: to charge an official with wrongdoing in office, so that the Senate puts him on trial (Chapter 20) 1322

imperial wizard: the top leader of a Ku Klux Klan group (Chapter 25) 1871

impiety: a lack of respect for God (Chapter 34) 2833

importation: bringing people or goods in from abroad (Chapter 4) 275

impress: to seize men and force them to serve on a navy ship (Chapter 14) 918

Impressionists: a group of painters in France (Chapter 32) 2643

impressment: British navy officers taking sailors off American ships and forcing them to serve on British warships (Chapter 22) 1534

impunity: no fear of punishment (Chapter 21) 1442

imputation: an accusation (Chapter 25) 1795

in blast: lit and making iron (Chapter 10) 671

in custody: held by police and not free to leave (Chapter 4) 320

in restraint of trade: blocking free buying and selling between the states (Chapter 19) 1256

inadequate: not good enough (Chapter 13) 868

inalienable: a right that nobody can take away (Chapter 25) 1778

inaugural address: a new president's first speech as president (Chapter 20) 1322

Inca: the people who ruled an empire in the Andes Mountains before the Spanish arrived (Chapter 1) 74

incitement of insurrection: stirring people up to rise against the government (Chapter 20) 1390

inciting to riot: urging a crowd to violence (Chapter 6) 443

inconsequential: mostly unimportant (Chapter 19) 1242

incorporate: to form a corporation (Chapter 19) 1250

incorporated: recognized by the state as a town by law, with its own government (Chapter 32) 2660

incorporated in the Union: made part of the country (Chapter 23) 1616

incorrigible: a child no one can control (Chapter 36) 3028

indenture: a written contract that required a person to work for a master for a set number of years (Chapter 4) 260

indentured servant: a person who worked for a master for a set number of years, often four to seven, to pay for the trip to America (Chapter 3) 189

independent contractor: a person who works for themselves and sells their work to companies (Chapter 11) 773

Indian agent: a US official who dealt with Native nations (Chapter 1) 44

Indian Citizenship Act: the 1924 law under which American Indians became citizens (Chapter 2) 149

Indian country: the legal name for Native land (Chapter 9) 604

Indian offenses: the crimes that Hiram Price's 1883 rules made of the Sun Dance and other Native ceremonies on the reservations (Chapter 29) 2274

Indian police: Native men who worked as police for the agency that ran the reservation (Chapter 2) 134

Indian removal: the plan to force the Cherokee and other Native nations off their land (Chapter 5) 352

Indian Removal Act: the 1830 law under which United States soldiers forced the Native nations of the Southeast out of their homes and moved them west (Chapter 2) 126

Indian Territory: the land in present-day Oklahoma where United States officials sent Native nations they forced from their homes (Chapter 2) 126

indicted: charged with a crime (Chapter 21) 1456

indictment: the criminal charge against a person (Chapter 25) 1829

Indies: Spain's islands and lands across the Atlantic (Chapter 1) 21

Indigenous: belonging to the people who lived in a place first (Chapter 2) 170

Indigenous Peoples Day: the name Berkeley's city council gave Columbus Day in 1992 (Chapter 37) 3125

indigent: poor (Chapter 26) 1975

indigo: a plant used to make blue dye (Chapter 15) 995

Indochina: the region that includes Vietnam, Laos and Cambodia (Chapter 31) 2572

induction: a changing electric current in one wire making a current flow in another wire nearby, without the two touching (Chapter 8) 541

induction center: the place where new soldiers were sworn in (Chapter 22) 1575

industrial country: a country where much of the work is done in mines, mills and factories (Chapter 16) 1074

industrial union: a union that takes in every kind of worker in an industry, instead of one craft at a time (Chapter 14) 929

industry: in the 1800s, hard work (Chapter 36) 3008

ineligible for citizenship: unable to become a citizen under US law (Chapter 4) 298

inestimable: too precious to measure (Chapter 25) 1770

inevitable: certain to happen (Chapter 19) 1268

infamous: known for something bad (Chapter 21) 1442

infantry: soldiers on foot (Chapter 16) 1084

inferior: lower or less able (Chapter 9) 591

infidel: a person who believes in no religion (Chapter 29) 2242

infiltration: a word for an enemy sneaking in, which appraisers used for Black families moving in (Chapter 6) 431

inflammatory: likely to make people angry (Chapter 37) 3125

inflation: a general rise in prices, so that each dollar buys less than before (Chapter 16) 1096

influenza: the flu, a disease caused by a virus that infects the nose, throat and lungs (Chapter 3) 182

informant: a person who secretly passes information to the FBI (Chapter 25) 1871

informed consent: the rule that every person in a study must first be told what will be done and what the risks are, and must agree (Chapter 26) 2005

informer: a person who secretly passes information to the police or the FBI (Chapter 18) 1182

infringing: breaking (Chapter 25) 1829

ingratitude: giving no thanks in return for kindness (Chapter 26) 1975

inhaler: a breathing device (Chapter 26) 1975

inherently: by their very nature (Chapter 9) 633

initialed: wrote one's initials on (Chapter 20) 1372

initiative: a law that citizens propose and then vote on themselves (Chapter 29) 2287

injunction: a court order to stop doing something (Chapter 6) 431

inlaid: made by setting small pieces of different wood into the surface (Chapter 35) 2928

inlet: a narrow arm of the sea that reaches into the land (Chapter 22) 1502

inoculation: giving a healthy person a small case of smallpox on purpose, by putting dried pus from a smallpox sore into a cut, so the person would not catch it again. To inoculate is to do this (Chapter 7) 471

inquest jury: a jury that looks into a death (Chapter 34) 2851

inquiry: an official hearing to find out what had happened (Chapter 2) 102

insane: the word of the time for a person whose mind worked in ways that their family or their town could not manage (Chapter 13) 868

insinuation: a hint (Chapter 12) 815

insolent: rude to someone in charge (Chapter 25) 1770

inspector general: an official who checks the work of a government department (Chapter 4) 320

installments: paying for goods in small amounts over time, after taking them home (Chapter 18) 1193

institution: a company or other group (Chapter 19) 1235

insubordination: refusing to obey the people in charge of you (Chapter 6) 407

Insular Cases: the Supreme Court decisions of 1901 to 1922 that the islands taken in 1898 belonged to the United States but were not fully part of it, insular meaning having to do with islands (Chapter 23) 1634

insuperable obstacles: problems a person cannot get past (Chapter 32) 2619

insurance company: a company that takes a regular fee from people and pays them back after a loss such as a fire (Chapter 19) 1245

insurance marketplaces: websites where people compare health plans and sign up (Chapter 26) 2022

insurance policy: a written deal in which the buyer pays a fee and the company pays out money when something happens, such as a death (Chapter 16) 1067

insure a loan: to promise to repay the lender if the buyer cannot (Chapter 6) 431

insurrection: armed uprising (Chapter 20) 1390

intake towers: towers that take in water (Chapter 12) 815

integral: a necessary part (Chapter 30) 2345

integrate: to let Black and white students learn in the same schools (Chapter 25) 1829

integrated circuit (chip): a whole electronic circuit built on one small piece of material (Chapter 8) 561

integration: the mixing of Black and white students in one school (Chapter 30) 2442

intellectual disability: a condition that makes learning and thinking much harder (Chapter 20) 1372

intelligence test: a set of questions meant to measure how well a person can think (Chapter 25) 1829

interchangeable parts: gun parts made by machine so that any part would fit any gun of the same model (Chapter 8) 528

interest: the fee for borrowing money (Chapter 2) 158

International Astronomical Union: the scientists' group that names places in space (Chapter 1) 74

International Hydrographic Organization: a group of countries that work together on sea charts (Chapter 1) 92

International Labour Organization: a world body that sets rules for workplaces (Chapter 13) 868

International Space Station: a laboratory that circles the Earth (Chapter 1) 92

internment: imprisonment in a camp (Chapter 30) 2351

interpret: to put each side's words into the other's language (Chapter 15) 991

interpreter: a person who translates between two languages (Chapter 1) 44

interrogation room: a room for questioning suspects (Chapter 20) 1372

interstate commerce: trade between the states (Chapter 19) 1256

Interstate Commerce Commission: the federal officials in charge of trade and travel between states (Chapter 11) 764

Interstate Highway System: a network of big highways across the country (Chapter 5) 383

interstellar space: the space between the stars (Chapter 1) 92

intervention: a meeting at which a family asks a person to go into treatment (Chapter 28) 2179

intoxicating liquors: alcoholic drinks (Chapter 20) 1356

intreated: asked (Chapter 30) 2351

Inuit: the Native people of the far north, from Alaska to Greenland (Chapter 1) 64

Inuk: one of the Inuit (Chapter 1) 64

invalid: a person too sick to live a normal life (Chapter 25) 1795

inventory: a list of a dead person's belongings, written down one by one (Chapter 7) 465

inverted siphon: the U-shaped passage behind the stove's baffle that made the smoke go up, down and up again (Chapter 8) 524

investment bank: a firm that sells bonds and shares to raise money for businesses such as railroads (Chapter 16) 1074

investor: a person who puts in money hoping for a profit (Chapter 7) 502

involuntary land loss: land lost when the owners did not choose to sell, as when a land buyer bought one heir's share of a farm and then had a court sell the whole farm at auction (Chapter 15) 1028

involuntary manslaughter: causing a person's death through carelessness without meaning to kill (Chapter 13) 899

iron furnace: a tall oven where fire melts iron out of rock (Chapter 8) 524

iron gag: an iron piece clamped over a prisoner's tongue and chained to his hands, held behind his back (Chapter 21) 1434

iron lung: a metal tube about seven feet long in which a pump changed the air pressure to pull air into a paralyzed patient's lungs (Chapter 26) 2005

iron ore: the rock that iron comes from (Chapter 8) 524

iron oxide: a compound of iron and oxygen, which gives ochre its color (Chapter 13) 844

ironmonger: a seller of iron goods (Chapter 8) 524

ironworkers: workers who put up the steel frames of buildings and bridges (Chapter 2) 149

ironworks: a place where workers make iron from rock dug out of the ground (Chapter 4) 267

Iroquois: a league of six nations, also called the Six Nations or the Haudenosaunee (Chapter 1) 31

irradiate: expose to radiation (Chapter 22) 1575

irrigation: water brought to the fields by pipes and ditches (Chapter 15) 1028

ISBN: the identifying number books for sale carry (Chapter 32) 2695

ISIS-K: a terror group in Afghanistan that was also fighting the Taliban (Chapter 22) 1585

island of stability: a group of superheavy elements that physicists predict would last longer before breaking apart (Chapter 9) 648

isolate: to collect a gas apart from everything else (Chapter 13) 855

Israelites: the people of ancient Israel in the Bible (Chapter 5) 363

IUPAC (International Union of Pure and Applied Chemistry): the world organization of chemists whose officials approve element names (Chapter 9) 648

J

jackscrew: a screw inside a case that a man turned by hand with a lever, used by the hundreds to lift buildings (Chapter 6) 418

jake leg: the illness people got in 1930 from Jamaica ginger, a medicine that was mostly alcohol, after Harry Gross began making his with a poisonous chemical called TOCP. The poison destroyed nerve cells in the spinal cord and took away the use of the hands and feet (Chapter 28) 2165

Japanese ancestry: having a family that came from Japan (Chapter 5) 376

jaundice: an illness that turns the skin and eyes yellow (Chapter 1) 44

jeans (cloth): a cotton and wool cloth woven in a twill (Chapter 35) 2934

jerking exercise: the jerking of people's bodies in the crowds at Lorenzo Dow's preaching (Chapter 29) 2256

jersey: a soft knitted cloth (Chapter 35) 2949

Jesuit mission: a group of priests from the Jesuit order of the Catholic church, with helpers, sent to turn people to Christianity (Chapter 2) 102

Jesuits: a group of priests within the Catholic Church (Chapter 1) 31

Jim Crow: the laws and customs in the South under which white lawmakers and officials kept Black people apart from white people and took away their rights. White mobs enforced it with murder (Chapter 5) 376

Jim Crow car: the railroad car set aside for Black passengers (Chapter 5) 376

jockey: a rider in horse races (Chapter 36) 2999

jockey club: a group of horse owners who held races for one another (Chapter 36) 2999

Johnkannaus: Harriet Jacobs's name for the groups of enslaved men who went from door to door in costume on Christmas Day (Chapter 37) 3101

johnnycakes: flat cakes baked from cornmeal (Chapter 15) 991

joiner: a woodworker who makes the fine wooden parts inside a house, such as doors, trim and furniture (Chapter 12) 799

joint resolution: a measure passed by members of both the House and the Senate, which passes when more than half of each vote for it (Chapter 23) 1616

joint-stock company: a company with many owners at once, each holding shares (Chapter 19) 1234

journal (a shopkeeper's): the book that listed every sale and every purchase in the order it happened (Chapter 18) 1174

journeyman: the middle rank of a trade, a skilled worker paid wages by the day (Chapter 14) 918

juba: beating out rhythms on one's own body with the hands (Chapter 33) 2736

Judge Advocate General: the Army's top lawyer (Chapter 27) 2068

judicial department: the courts (Chapter 20) 1333

judicial review: the power of judges to refuse to follow a law that goes against the Constitution (Chapter 20) 1333

Juneteenth: the holiday that marks June 19, 1865, the day Major General Gordon Granger reached Galveston, Texas, with the news that enslaved African Americans were free (Chapter 12) 832

Juneteenth National Independence Day Act: the 2021 law that made Juneteenth a federal holiday (Chapter 24) 1744

jurisdiction: the area a government rules (Chapter 4) 260

jury: a group of citizens who decide whether a person is guilty (Chapter 2) 108

jury commission: the board that chose jurors (Chapter 25) 1871

just war: a war that is right and fair (Chapter 21) 1409

justice of the peace: a local judge (Chapter 20) 1333

justices: the judges of the Supreme Court (Chapter 2) 158

justices: a county's judges (Chapter 36) 2987

justifiable homicide: a killing the law excuses (Chapter 33) 2779

juvenile court: a separate court for children accused of crimes, run to help the child instead of only punishing (Chapter 21) 1442

K

Kanien'kehaka: the Mohawk people's own name for themselves (Chapter 22) 1515

kapa: cloth made by beating tree bark thin (Chapter 37) 3075

kapucha toli: the Choctaw stick-and-ball game, "stick play" (Chapter 36) 2981

Karamu: the feast that ends Kwanzaa (Chapter 37) 3125

Katipunan: a secret society fighting Spanish rule in the Philippines (Chapter 23) 1623

Katsina masks: Pueblo ceremonial masks that show spirit beings (Chapter 32) 2608

katsinam: spirit beings in Pueblo religion, more than 500 of them (Chapter 34) 2814

Keeling Curve: the line that the carbon dioxide measurements on Mauna Loa make on a graph (Chapter 9) 633

Keepers of the Central Fire, of the Eastern Door and of the Western Door: the Onondaga, the Mohawk and the Seneca (Chapter 20) 1304

keloid scars: thick raised scars that grow over a healed wound (Chapter 24) 1712

Kentucky Derby: a horse race held each spring in Louisville (Chapter 36) 3028

kerosene: a cheap lamp fuel that refiners made from oil (Chapter 7) 485

kersey: a rough, thick, coarse wool (Chapter 35) 2916

keyed bugle: a brass horn with keys (Chapter 33) 2736

kickback: a secret payment for a favor (Chapter 27) 2068

kill hole: a hole punched on purpose through the bottom of a Mimbres bowl before it was set over a dead person's face or head (Chapter 32) 2598

kilns: very hot ovens (Chapter 12) 789

kin: a person's relatives by birth or by marriage (Chapter 7) 460

Kinetoscope: a machine that showed one short film to one person looking into it (Chapter 34) 2851

King Cotton: the name for cotton, the most valuable American export, grown by enslaved people (Chapter 16) 1067

King James Bible: the Bible that Protestants used (Chapter 29) 2256

King Philip's War: the war Metacom led against the New England colonies in 1675 and 1676 (Chapter 14) 922

King Tides: the name used in Miami Beach for the highest tides of the year (Chapter 6) 452

king's attorney: the lawyer who brought cases to court in the king's name (Chapter 34) 2823

kirtle: a kind of gown (Chapter 35) 2916

kitchenette: a one-room unit split from an apartment and rented separately, with a hotplate and a bathroom shared with other families (Chapter 6) 431

kiva: a round room used for ceremonies and meetings, usually sunk into the ground (Chapter 6) 395

KKK: the Ku Klux Klan, a white group that has threatened, beaten and killed Black people (Chapter 32) 2679

Knights of Columbus: a Catholic men's organization (Chapter 37) 3119

knock: a pinging sound in a gasoline engine (Chapter 13) 868

knot: a measured length of yarn (Chapter 18) 1182

Know-Nothings: the American Party of the 1850s, against immigrants and Catholics, named for members' answer "I know nothing" (Chapter 4) 283

kosher: following Jewish food laws (Chapter 29) 2274

Kristallnacht: the Night of Broken Glass, November 9 and 10, 1938, when Nazi leaders set off attacks on Jews across Germany (Chapter 4) 298

Ku Klux Klan: a group started in December 1865 by men who had fought for the Confederacy, whose members whipped, burned and murdered freed people and the people who helped them (Chapter 1) 74

Ku Klux Klan Act: the 1871 law that made it a federal crime to deny anyone the rights the Constitution protects (Chapter 20) 1341

Kwanzaa: the seven-day holiday from December 26 to January 1, created in 1966, whose name comes from a Swahili phrase meaning "first fruits" (Chapter 37) 3125

L

la junta: Puerto Ricans' name for the seven-member board that has power over Puerto Rico's budget. Its members are appointed in Washington under PROMESA, a law Congress passed in 2016 (Chapter 23) 1656

la operación: the Puerto Rican name for tubal ligation, an operation in which a doctor cut or tied the two tubes that carry eggs from the ovaries to the womb, so the woman could never become pregnant again (Chapter 23) 1634

La Toma: "the taking," the ceremony in which Oñate claimed New Mexico for Spain (Chapter 5) 336

lab-grown meat: meat, such as chicken, grown from animal cells instead of from a live animal (Chapter 15) 1039

labor force: everyone who has a paid job or is looking for one (Chapter 7) 495

labor recruiter: a person hired to find workers (Chapter 5) 376

labourer: a man who worked with his hands (Chapter 36) 2987

Lake Itasca: Schoolcraft's name for Elk Lake, made up from pieces of the Latin words for "truth" and "head" (Chapter 1) 44

lance: a long spear with a sharp blade (Chapter 10) 677

lancet: a small blade (Chapter 7) 471

land bridge: a stretch of dry ground that joined Asia to North America during the ice age (Chapter 4) 256

land cession: an agreement in which a Native nation gave up land to the United States. Two journalists who traced many of these agreements write that they were backed by violence (Chapter 30) 2392

land ethic: Aldo Leopold's idea that people count themselves as members of the land community and stop acting as its conquerors (Chapter 3) 215

land fraud: cheating people in land deals (Chapter 31) 2572

land grant: a gift of land from a government (Chapter 19) 1256

land held in trust: land the United States holds on behalf of a Native nation or person (Chapter 2) 149

land office: an office where settlers filed their claims (Chapter 6) 418

Land Ordinance: the 1785 law that stated how the public land would be sold (Chapter 3) 192

land patent: the government paper that made homestead land the owner's (Chapter 15) 1009

land reform: a plan to share out farmland (Chapter 23) 1647

land run: a race in which settlers rushed in at a set signal to claim pieces of land (Chapter 6) 418

land-grant colleges: colleges started with money from selling public land under the Morrill Act, teaching farming and the skills of working with machines (Chapter 9) 604

landscape: a picture of the land and sky (Chapter 32) 2627

landscape architect: a person who plans outdoor spaces such as parks (Chapter 12) 832

landscape architecture: designing the land itself, its paths, lawns, trees, water and roads (Chapter 6) 418

landslide: a win by a huge margin (Chapter 20) 1372

lapels: the folded-back flaps at the front of a coat below the collar (Chapter 35) 2949

lard oil: oil made from pig fat (Chapter 10) 677

Large Hadron Collider: a machine that smashes tiny particles together at very high speed (Chapter 9) 648

large language models: programs that write text by predicting the next word, over and over (Chapter 8) 570

largest private employer: the company with more workers than any other company (Chapter 18) 1214

larkspur: a flower (Chapter 35) 2908

laser discs: shiny discs read by a beam of light (Chapter 34) 2877

laser interferometer: a detector that uses laser beams to measure very small changes in distance (Chapter 9) 648

lash: one stroke of a whip on a person's bare body, hard enough to cut the skin (Chapter 8) 528

last hired, first fired: the pattern of employers hiring Black workers last and firing them first (Chapter 16) 1084

latchkey children: children who went home after school to an empty house and carried their own key (Chapter 7) 495

late blight: a potato disease caused by a water mold, a tiny living thing that destroys potato plants (Chapter 15) 1004

Latino or Hispanic: Americans whose families came from Spanish-speaking countries (Chapter 16) 1102

latrine trenches: the ditches soldiers used as toilets (Chapter 26) 1983

laudanum: a painkiller made of opium dissolved in alcohol (Chapter 28) 2147

launder money: to hide where money from crime came from (Chapter 21) 1468

Laws of the Indies: the written orders of 1573 for Spain's lands in the Americas, which the Spanish called the Indies. They stated how the founders of Spanish towns had to lay them out (Chapter 6) 398

Le Grand Dérangement: "the great upheaval," the French name for the forced removal of the Acadians that British officials ordered in 1755 (Chapter 5) 346

lead encephalopathy: swelling and damage of the brain caused by lead (Chapter 13) 886

leadsman: the crew member who measured the depth of the water (Chapter 11) 745

league: a measure of distance of about 3 miles, so 370 leagues is about 1,185 miles (Chapter 23) 1597

League of Nations: a group of countries that would settle quarrels by talking instead of fighting (Chapter 23) 1634

learned society: a group of people who meet to share and discuss what they have found out (Chapter 9) 588

lease: an agreement to use land or a building for a set time in return for payment. To lease is to rent or to rent out (Chapter 3) 215

ledger: the shopkeeper's book that gave each customer a page, purchases on the left and payments on the right (Chapter 18) 1174

ledger art: Plains drawings made on paper in ledgers, lined account books (Chapter 32) 2643

Lee-Jackson-King Day: Virginia's day for Robert E. Lee, Thomas "Stonewall" Jackson and Martin Luther King Jr. (Chapter 37) 3125

leg irons: metal rings locked around the ankles and chained together (Chapter 2) 134

legal guardian: a person the law makes responsible for children (Chapter 1) 44

legal holiday: a holiday set by law (Chapter 37) 3101

legal segregation: the system of laws that required Black people and white people to be kept apart (Chapter 37) 3107

legal status: the right to stay in the country (Chapter 4) 310

legal tender: money the law says people have to accept to pay a debt (Chapter 17) 1113

legion: an army unit, such as Anthony Wayne's legion of 3,300 men (Chapter 2) 120

Legion of Honor: an order of people honored by France for their work (Chapter 32) 2643

Legionnaires' disease: a lung infection and a kind of pneumonia, caused by bacteria that grow in poorly treated water (Chapter 13) 899

legislate: make laws (Chapter 20) 1309

legislative body: a group of lawmakers (Chapter 25) 1778

legislative districts: the areas that elect a state's lawmakers (Chapter 20) 1372

legislature: a state's lawmakers (Chapter 25) 1770

leisure suit: a casual kind of suit often made of man-made cloth (Chapter 35) 2959

Lenapehoking: the Lenape name for their homeland (Chapter 4) 260

leprosy: a disease caused by bacteria that damages the skin and nerves (Chapter 26) 1990

leptospirosis: an infection by bacteria that rats carry in their urine (Chapter 26) 1954

lesbian: a woman who loves women (Chapter 1) 82

Lesser Antilles: the chain of small islands at the eastern edge of the Caribbean (Chapter 27) 2054

less-lethal bullet: a bullet made to hurt rather than kill (Chapter 25) 1909

lest: for fear that (Chapter 20) 1341

lethal injection: killing a person with drugs put into a vein through a tube in the arm. In the usual method, the first drug puts the person to sleep, the second stops the muscles and the breathing, and the third stops the heart (Chapter 21) 1468

letter of reprimand: a written scolding (Chapter 4) 310

leukemia: a cancer of the blood (Chapter 9) 633

levee: a wall of earth or concrete built to hold back water (Chapter 5) 387

LGBT: short for lesbian, gay, bisexual and transgender (Chapter 25) 1909

LGBTQ: short for lesbian, gay, bisexual, transgender and queer, words for people who are attracted to people of their own sex, or whose sense of being male or female differs from the body they were born with (Chapter 1) 82

LGBTQ+: a short name for lesbian, gay, bisexual, transgender and queer people and others (Chapter 29) 2324

liable: able to be made to pay a debt (Chapter 19) 1242

libel: printing something false that harms a person's good name (Chapter 18) 1193

license: written permission (Chapter 2) 126

licks: blows on the buttocks with a flat wooden paddle (Chapter 20) 1372

lien: a legal claim on property, such as a crop, until a debt is paid (Chapter 15) 1009

lieutenant: a junior officer (Chapter 1) 44

lieutenant governor: the official just below the governor (Chapter 5) 346

life class: a class where students draw from a model without clothes (Chapter 32) 2627

life expectancy: the average age a newborn would reach if that year's death rates held (Chapter 26) 2022

light infantry: lightly armed foot soldiers (Chapter 22) 1522

light trucks: pickups, sport utility vehicles (SUVs) and minivans (Chapter 11) 773

Lighthorsemen: the Choctaw Nation's mounted police (Chapter 36) 3028

lightning rod: a metal rod on a building that carries lightning into the ground (Chapter 8) 528

light-year: the distance light travels in a year (Chapter 9) 614

likely: a sellers' word of praise for an enslaved person, meaning healthy, good-looking and fit for work (Chapter 19) 1242

lime: a white powder, made by burning oyster shells, that went into the mortar (Chapter 12) 789

limited liability: the rule that a company's owners cannot also be made to pay its debts from their own property (Chapter 19) 1245

limner: an old word for a painter of portraits (Chapter 32) 2619

Lincoln Memorial Commission: the group in charge of building the Lincoln Memorial (Chapter 12) 815

lineal descendant: a child or grandchild (Chapter 20) 1356

linen wheel: a spinning wheel for linen thread (Chapter 18) 1182

linguistics: the study of how languages work (Chapter 2) 170

lining out: a way of singing in which a leader read or sang one line and the congregation sang it back (Chapter 33) 2719

linsey-woolsey: a cheap cloth of wool mixed with cotton or linen (Chapter 35) 2934

lip-pointing: drawing a paintbrush between the lips to shape its tip, which made the dial painters swallow radium (Chapter 13) 868

liquid transmitter: a sending part that used a liquid (Chapter 8) 541

liquidity problems: banks not having enough cash on hand to pay the people who had put money in them (Chapter 17) 1143

literacy test: a reading and writing test that officials used to keep Black citizens from voting (Chapter 20) 1372

lithium-ion battery: a kind of battery that can be recharged many times (Chapter 8) 570

lithography: printing pictures from drawings made on stone (Chapter 32) 2643

livery: a uniform the master chose (Chapter 35) 2923

livery stable: a place where horses were kept for hire (Chapter 29) 2287

livestream: to show something live on the internet (Chapter 31) 2585

loan guarantee: the government's promise to repay a loan if the borrowers cannot (Chapter 16) 1096

lobby: to try to talk lawmakers into acting (Chapter 11) 764

lobbying office: an office whose staff pressed lawmakers to vote for the bill (Chapter 37) 3125

lobbyist: a person hired to persuade lawmakers (Chapter 19) 1245

loblollie: a thick porridge (Chapter 4) 260

local: a branch of a union (Chapter 16) 1096

lockout: an employer shutting workers out of their jobs, as Henry Clay Frick did at Homestead in 1892 (Chapter 14) 929

lockstep: marching in step with arms locked together (Chapter 21) 1434

lockstitch: a stitch in which two threads lock together through the cloth (Chapter 8) 535

locomotive: a train engine (Chapter 11) 734

locomotive fireman: the worker who shoveled coal into the engine's fire (Chapter 14) 929

locs: long ropes of hair, also called dreadlocks (Chapter 25) 1909

locusts of Egypt: the swarms of insects sent on Egypt in the Bible (Chapter 25) 1795

lode: a long, rich band of ore inside the rock (Chapter 13) 863

lodges: the tent homes of a Plains band (Chapter 22) 1543

lodging houses: buildings that rent out rooms (Chapter 35) 2940

loft: the open upper floor of a building (Chapter 14) 929

logwood: a wood that gives a dye (Chapter 19) 1242

longhouse: a long building with a frame of wooden posts, covered with bark, that held many related families (Chapter 7) 460

loom: a machine that weaves thread into cloth (Chapter 16) 1067

looters: people who steal from damaged buildings (Chapter 27) 2078

looting: stealing during a riot (Chapter 21) 1456

Lords Proprietors: the English noblemen who owned the Carolina colony (Chapter 24) 1670

lost cause: a cause that has failed. Virginia's governor used the words in 2020 for a holiday that honored two Confederate generals who had fought to keep slavery (Chapter 37) 3135

lottery: a drawing of names by chance (Chapter 13) 859

Louisiana: La Salle's name for the whole Mississippi River basin, after King Louis XIV (Chapter 1) 31

Lowcountry: the low coastal land of South Carolina (Chapter 4) 267

lower 48 states: every state except Alaska and Hawaii (Chapter 3) 229

low-to-moderate income: earning well below the typical family in the area (Chapter 6) 431

Loyalists: colonists who sided with the British king against the Revolution (Chapter 4) 271

LSD: a drug that makes people see and hear things that are not there (Chapter 28) 2179

lucre: an old word for money (Chapter 32) 2695

lunar module: a small landing craft (Chapter 1) 82

Lunar New Year: the start of the year in calendars based on the moon (Chapter 37) 3135

lung fibrosis: scarring that makes the lungs stiff, so that breathing gets harder (Chapter 13) 886

Lutheran sect: Menéndez's words for the Protestant faith (Chapter 22) 1502

Lycra: a stretchy man-made cloth (Chapter 35) 2959

lye: a chemical in some hair straighteners that could burn the scalp and the ears (Chapter 35) 2959

lynch mob: a crowd that kills a person without any trial (Chapter 5) 376

lynching: the murder of a person by a mob, without any trial (Chapter 12) 832

M

MacArthur Fellowship: a large prize of money given to people for creative work (Chapter 2) 170

Mace: a stinging chemical spray (Chapter 14) 963

machinist: a worker who made and repaired machines (Chapter 37) 3107

macular degeneration: an eye disease that destroys the center of sight (Chapter 32) 2660

magistrates: officials with the power to judge cases (Chapter 8) 517

magnesium flash powder: a powder that burns with a sudden bright light (Chapter 32) 2643

magnet school: a school where officials offer special teaching to draw families from across a district (Chapter 30) 2442

magnitude: the number scientists use for an earthquake's size (Chapter 10) 711

Mahicannittuk: the Mohican name for the Hudson River (Chapter 1) 31

Mahometan: an old word for a Muslim (Chapter 29) 2242

maids: unmarried women (Chapter 25) 1770

mail contractor: a person paid to carry the mail (Chapter 31) 2533

main stockholder: the person who owns the most shares in a company (Chapter 19) 1268

maize: corn (Chapter 9) 614

major leagues: the top professional baseball teams (Chapter 36) 3028

majority leader: the leader of the larger party (Chapter 20) 1372

majority-minority: where most of the people living there are not white (Chapter 6) 431

Makahiki: the Hawaiian season after the new year began, when chiefs could not make war on each other (Chapter 37) 3075

malaria: a disease spread by mosquitoes that causes high fevers and chills again and again (Chapter 11) 734

Malcontents: Georgia colonists who complained that the rum ban hurt trade (Chapter 28) 2133

malignant: cancerous (Chapter 9) 633

malnutrition: not getting enough of the right food (Chapter 6) 418

"mammy": a made-up picture of an enslaved Black woman who cared for a white family's children and seemed happy in slavery, which hid the truth that enslavers separated these women from their own children (Chapter 12) 832

mandatory evacuation: an order for everyone to leave (Chapter 27) 2104

mandatory minimum: the shortest sentence a law allows for a crime. A judge cannot give less (Chapter 21) 1468

Manhattan Project: the Army's secret program to build an atomic bomb (Chapter 6) 431

Manifest Destiny: the idea, held by many Americans, that the United States was meant to spread across the continent to the Pacific Ocean. People used it to argue for going to war with Mexico (Chapter 5) 352

Mann Act: a federal law of 1910 that made it a crime to take "any woman or girl" across state lines for prostitution or any "immoral purpose" (Chapter 36) 3028

manoomin: a wild rice that grows in shallow water. In Ojibwe the word means "the good berry" (Chapter 15) 986

manslaughter: the crime of killing someone without planning to (Chapter 4) 283

manta: a woven cotton blanket (Chapter 35) 2916

mantles: cloaks (Chapter 35) 2912

manual labor: hard work with the hands (Chapter 30) 2367

manual training: work with tools (Chapter 30) 2414

manufactures: goods made by hand or by machine (Chapter 16) 1062

manufacturing: the making of goods in workshops, mills and factories (Chapter 3) 240

manumission: freeing an enslaved person by a legal paper (Chapter 27) 2054

manure (to manure ground): to bury something in the soil to feed the crops, as the Plymouth colonists buried fish with their corn seed (Chapter 15) 991

manuscript: written by hand (Chapter 9) 591

maple sugaring: boiling down the sap of maple trees to make sugar (Chapter 14) 911

marasmus: severe malnutrition, a form of starvation in which the body gets too little food of every kind, uses up its own fat and then its own muscle, and the child wastes away (Chapter 6) 418

margin: the gap between the two candidates' totals (Chapter 20) 1390

marijuana: the dried leaves and flowers of the cannabis plant (Chapter 21) 1468

marine biologist: a scientist who studies life in the sea (Chapter 1) 82

marine hospital: a hospital for sailors (Chapter 26) 1967

marines: soldiers who served on ships (Chapter 1) 39

mark: an English unit of money (Chapter 27) 2041

market day: a set day when farmers brought meat, milk, butter, cheese, fruit and vegetables into town to sell (Chapter 18) 1174

market dominance: a lead in a market (Chapter 19) 1268

market revolution: the change from families growing and making most of what they used to growing crops and making goods to sell, and buying what they needed (Chapter 16) 1067

market value: the price of one share times the number of shares (Chapter 19) 1291

marriage bar: the rule that made a woman teacher choose between teaching and marriage (Chapter 30) 2414

marrow: the soft center of the bone, the tissue that makes blood (Chapter 13) 868

marshal: an officer who carried out a court's orders (Chapter 2) 126

marshal of the day: the man in charge of the celebration (Chapter 37) 3101

mart: an old word for a market (Chapter 18) 1170

martial law: rule by army officers instead of by ordinary courts (Chapter 21) 1412

martyr: a person who dies for a cause (Chapter 37) 3107

masculinization: a girl's body taking on a man's traits (Chapter 25) 1795

mason: a builder in stone (Chapter 12) 789

masonry building: a building of brick or stone (Chapter 6) 418

masonry fort: a fort built of stone or brick (Chapter 12) 789

Mass: the main Catholic church service (Chapter 29) 2207

mass casualty: an event in which many people are hurt or killed at once (Chapter 27) 2093

mass execution: putting many people to death at one time (Chapter 2) 134

mass grave: one pit for many bodies (Chapter 27) 2078

Massa: master, as Jo Anderson's words were written down in 1885 (Chapter 8) 535

Massachusetts General Court: the colony's lawmakers (Chapter 8) 517

massacre: the killing of many people who could not defend themselves (Chapter 2) 102

massive resistance: a campaign by Virginia's leaders to block the Supreme Court's *Brown* ruling and keep Black children out of white schools (Chapter 30) 2442

mast: the tall pole that holds up a ship's sails (Chapter 3) 189

master (of a school): the head teacher of a school (Chapter 30) 2392

master (of a servant or an enslaved person): the person a servant or an enslaved person had to work for (Chapter 1) 44

master (of an apprentice): the adult an apprentice lived with and worked for (Chapter 4) 267

master collier: a collier in charge of the pits and of the helpers who worked them (Chapter 10) 681

master recording: the original recording that every copy is made from (Chapter 33) 2779

Matanzas: the Spanish word for "slaughters," the name of the Florida inlet where Spanish soldiers killed about 245 French prisoners who had surrendered (Chapter 1) 21

mate: the officer just below the captain (Chapter 1) 31

matinee: an afternoon showing of a movie (Chapter 36) 3028

matron: the woman in charge of housekeeping and care in a hospital or a boarding school (Chapter 25) 1778

mauve or mauveine: the purple dye Perkin made, the first man-made dye sold for use (Chapter 35) 2940

Mayflower Compact: the agreement forty-one men signed off Cape Cod on November 11, 1620, to keep order (Chapter 20) 1309

may-pole: a tall pole that children danced around holding ribbons (Chapter 36) 3019

mean condition: low rank (Chapter 35) 2916

Meat Inspection Act: the 1906 law that required federal inspectors in meatpacking plants, set rules for cleanliness and banned false labels (Chapter 15) 1015

meatpacker: a businessman who ran a plant where animals were killed and cut up into meat for sale (Chapter 15) 1009

meatpacking plant: a plant where animals are killed and cut up for meat (Chapter 14) 974

mechanic arts: the skills of building things and working with machines (Chapter 3) 200

mechanical refrigerator: a refrigerator cooled by a motor instead of a block of ice (Chapter 8) 553

mechanics: people who build and repair machines. In 1789 a mechanic was a skilled worker who made things (Chapter 8) 528

Medal of Honor: the top award the United States military gives for bravery in battle (Chapter 1) 74

median: the one in the middle when all are ranked by size, with half above and half below (Chapter 7) 502

mediator: a go-between who helps both sides reach a deal (Chapter 14) 941

Medicaid: health insurance for people with little money (Chapter 26) 2005

medical examiners: the doctors who find out why a person died (Chapter 4) 320

medical sociology: the study of health and illness in society (Chapter 25) 1909

Medicare: health insurance for people 65 and older (Chapter 26) 2005

medicine men: Native healers and religious leaders (Chapter 5) 339

medicine society: a group made up of Haudenosaunee who have taken part in the medicine, as the Haudenosaunee Grand Council's 1995 policy describes it (Chapter 34) 2814

mediocrity: work that is only average or poor (Chapter 30) 2442

meetinghouse: a building used as a church and as a town's main public building (Chapter 12) 789

megawatt: a million watts (Chapter 10) 711

meltdown: an accident in which a reactor's fuel gets so hot that it melts (Chapter 10) 702

melting pot: a pot in which different metals melt together, used to mean newcomers blending into one nation (Chapter 4) 298

membrane: a thin sheet that shakes in time with sound (Chapter 8) 541

Memento Mori: Latin words on gravestones that mean "remember you will die" (Chapter 32) 2608

memoir: a book about the writer's own life (Chapter 2) 149

memorial: a formal written statement (Chapter 25) 1778

Memorial Day: the day for putting flowers on soldiers' graves, first called Decoration Day (Chapter 19) 1268

menace: a danger. To menace someone is to threaten them (Chapter 3) 215

mental age: a test score given as the age of an ordinary child who would score the same (Chapter 25) 1871

mercantilism: the belief that colonies existed to make the home country richer (Chapter 16) 1052

merchant fleet: all the ships that carry a country's goods for trade (Chapter 16) 1057

Merchant Marine: the crews of American cargo ships (Chapter 14) 963

mercuric oxide: a compound of the element mercury and oxygen (Chapter 13) 855

mercury (quicksilver): a metal and element that stays liquid at room temperature (Chapter 13) 863

merger: two or more companies joining into one (Chapter 19) 1268

meridian: a north-south line on a map (Chapter 1) 64

Merino sheep: a kind of sheep raised for its wool (Chapter 3) 196

mesa: a hill with a flat top and steep sides (Chapter 2) 102

Mesoamerica: the region that is now Mexico and Central America (Chapter 11) 721

mesothelioma: a cancer of the thin linings of the chest and belly that asbestos fibers can cause (Chapter 19) 1280

mess attendant: a sailor who cooked and served food (Chapter 22) 1559

meteorite: a piece of rock or metal from space that has fallen to Earth (Chapter 1) 64

methane: the main gas in natural gas (Chapter 10) 711

method acting: a way of acting built on the teaching of Konstantin Stanislavsky, a Russian director (Chapter 34) 2877

methyl alcohol: wood alcohol, which blinds and kills people who drink it (Chapter 28) 2165

methyl isocyanate: the poison gas that leaked at Bhopal. It damaged people's eyes and filled their lungs with fluid (Chapter 19) 1280

methylmercury: the form of mercury that builds up most easily in living bodies and is most poisonous to people (Chapter 13) 863

metric ton: about 2,205 pounds (Chapter 16) 1096

Metro: Washington's subway (Chapter 6) 452

metro area: short for metropolitan area, a city together with its suburbs (Chapter 5) 387

Mexican Cession: the land Mexican officials gave up to the United States after losing the Mexican-American War (Chapter 5) 352

Mexican Repatriation: officials pushing people of Mexican descent out of the United States from 1929 to 1936. The name comes from a word for sending people back to their own country, but by the larger counts most of them had been born in the United States (Chapter 4) 298

mica: a shiny rock that splits into thin sheets (Chapter 17) 1119

microgram: one millionth of a gram (Chapter 3) 240

microplastic: bits of plastic too small to see easily (Chapter 3) 240

midazolam: a drug that makes people calm and sleepy and in large doses slows and then stops a person's breathing (Chapter 27) 2104

Middle America: ordinary American families (Chapter 31) 2572

Middle Atlantic colonies: the group of colonies that included Maryland (Chapter 32) 2613

Middle Passage: the crossing of the Atlantic Ocean on which ship crews carried captive Africans to be sold (Chapter 5) 352

middle-class families: the families in between the rich and the poor (Chapter 7) 474

middlemen: the sellers between the maker and the buyer (Chapter 18) 1193

midwife: a woman who helps other women give birth (Chapter 7) 471

migrant farm workers: people who moved from farm to farm to pick crops (Chapter 27) 2078

migrant workers: farm workers who move from place to place to follow the harvests (Chapter 30) 2414

migration path: the route a people traveled as they moved from one home to another (Chapter 29) 2203

militant: ready to fight hard for a cause (Chapter 25) 1829

military attaché: an officer posted to an embassy (Chapter 23) 1647

military commission: a court run by the army (Chapter 2) 134

military demonstration: a show of armed force (Chapter 23) 1623

military governor: an army officer put in charge of a place (Chapter 31) 2545

military installations: Army forts (Chapter 1) 64

military-industrial complex: the ties between the armed forces and the companies that made their weapons (Chapter 19) 1280

militia: a force of ordinary citizens called up to serve as soldiers, or any armed group outside the regular army (Chapter 2) 126

militiamen: part-time citizen soldiers (Chapter 5) 352

mill: one tenth of a cent (Chapter 17) 1123

Mill Creek chert: a hard stone, found only in southern Illinois, that could be shaped into tools (Chapter 1) 17

milligram: a thousandth of a gram (Chapter 13) 899

millisievert: a unit that measures how much radiation a body takes in (Chapter 10) 702

mimeograph: a machine that prints copies from a cut paper stencil (Chapter 31) 2559

mine: a bomb placed in the water (Chapter 31) 2545

mineral leasing: renting out the right to take minerals from the ground (Chapter 3) 240

mineralogist: a person who studies rocks (Chapter 1) 44

minié ball: a soft lead bullet shaped like a cone that often flattened when it hit a body (Chapter 22) 1543

minimum wage: the lowest hourly pay the law allows (Chapter 11) 773

mining: using a computer to create new bitcoins (Chapter 17) 1160

mining claim: a legal claim to the right to mine minerals on public land (Chapter 3) 240

miniseries: a story told on television over several nights (Chapter 7) 471

minister: a man who led church services, or a government's representative in another country (Chapter 2) 108

Minister Plenipotentiary: a title meaning full power to act for one's country in another country (Chapter 23) 1605

minister to France: the man who represented the United States to the French government (Chapter 12) 799

minor leagues: the lower leagues below the major leagues (Chapter 36) 3019

minority leader: the leader of the smaller party (Chapter 20) 1372

minors: young people who are not yet adults (Chapter 4) 320

mint: the place where workers make the country's coins (Chapter 9) 591

mintmaster: the man in charge of the mint (Chapter 17) 1115

minutes: the written record of what was said at a meeting (Chapter 2) 116

mirliton: a pale green squash (Chapter 15) 1015

misanthrope: a person who hates other people (Chapter 30) 2392

misappropriate: to take and misuse something that belongs to others (Chapter 29) 2324

misappropriation: taking and misusing something that belongs to someone else, such as a performer's face and voice (Chapter 34) 2889

misbranding: selling a drug with false or misleading claims about it (Chapter 19) 1291

miscarriage: the death of an unborn baby before it can live outside the mother's body (Chapter 3) 229

miscegenation: an old word for marriage or romance between people of different races (Chapter 34) 2861

misconduct and unprofessional conduct: breaking the rules of one's profession (Chapter 27) 2093

misdemeanor: a less serious kind of crime, such as being drunk in public (Chapter 10) 711

misdemeanor manslaughter: a minor charge of killing a person without planning to (Chapter 21) 1468

misprision of treason: knowing about a plot against the government and not reporting it (Chapter 23) 1623

mission: a settlement run by Catholic priests, where Native people lived and were taught the Catholic religion (Chapter 1) 31

mission town: a town built around a church where ministers taught Christianity (Chapter 29) 2207

missionaries: people sent to bring others into their religion (Chapter 2) 102

Mississippi appendectomy: Fannie Lou Hamer's name for operations like hers, in which a doctor cut out a woman's womb without asking her, so she could never become pregnant. She named them after the ordinary operation that takes out the appendix, a small part of the gut (Chapter 25) 1871

Mississippian culture: the name archaeologists give the way of life of Cahokia and the towns like it (Chapter 35) 2908

Mississippians: the town-building peoples of the Mississippi Valley and the Southeast (Chapter 2) 102

Missouri Compromise: the 1820 law that banned slavery in the northern part of the Louisiana Purchase lands. Supreme Court justices struck it down in *Dred Scott* (Chapter 20) 1341

mistrial: a trial that ends with no verdict, as when the jury cannot agree (Chapter 21) 1442

mita: the system under which Spanish officials forced Native men to work in the mines in turns (Chapter 17) 1113

mixed schools: schools for Black and white children together (Chapter 20) 1341

model law: a sample law for lawmakers to copy (Chapter 4) 298

modesty: in the 1800s, keeping the body covered (Chapter 36) 3008

modulation: the rise and fall of the voice (Chapter 34) 2827

molasses: a thick, dark syrup left over when sugar is made (Chapter 7) 468

moldboard: the curved blade of a plow that turns the soil over (Chapter 8) 535

molecule: a group of atoms joined together (Chapter 9) 633

molested: in the old laws, bothered or attacked (Chapter 29) 2213

Molotov cocktails: bottles filled with fuel and set on fire (Chapter 7) 495

monetary compensation: payment in money (Chapter 12) 815

money laundering: hiding where money from a crime came from (Chapter 17) 1160

money market fund: a fund that pools many people's money to buy short-term loans to the government and to large companies, a way for small savers to earn more than the legal limit on bank interest (Chapter 17) 1153

Mongolian: a word white Americans once used for people from East Asia (Chapter 25) 1795

monopolize: to become the only seller of something (Chapter 19) 1256

monopoly: the right to be the only company doing a kind of business somewhere, or a company with no real rivals (Chapter 11) 734

Montaup: the Pokanoket name for Mount Hope (Chapter 37) 3094

Moors: Muslims who fought Christians in Spain from the 700s to the 1400s (Chapter 34) 2818

moral lectures: talks meant to teach right from wrong (Chapter 34) 2833

morals: a person's ideas of right and wrong (Chapter 29) 2242

Moravians: members of a Protestant church from central Europe (Chapter 5) 343

Mormons: members of the Church of Jesus Christ of Latter-day Saints (Chapter 1) 64

morphine: a strong painkilling drug made from opium that slows a person's breathing, so that too much can kill, especially a child (Chapter 18) 1193

Morrill Act: the 1862 law under which the federal government gave each state 30,000 acres of federal land for each of its members of Congress, to pay for colleges (Chapter 3) 200

Morse code: the pattern of dots and dashes for every letter and number (Chapter 8) 535

mortar: a paste that hardens (Chapter 12) 789

mortgage: a loan in which the borrower promises property, such as a house or enslaved people, to the lender, who can take that property if the loan is not repaid (Chapter 6) 431

mortgage lenders: lenders who lend money to buy houses (Chapter 18) 1214

mosque: a Muslim house of worship (Chapter 29) 2304

most distinguished niche: the place of honor (Chapter 37) 3101

most integrated setting: at home or in the community rather than in an institution (Chapter 25) 1909

most rigid scrutiny: the closest possible look (Chapter 25) 1829

motion capture: cameras record an actor's movements, and a computer uses them to move a character artists have drawn (Chapter 34) 2889

motions (in a court case): formal requests to the judge (Chapter 8) 570

mountain man: a trapper who lived in the western mountains (Chapter 1) 44

mounted: stuffed and set up for display (Chapter 3) 215

mourning war: a Haudenosaunee custom in which a family that had lost someone in war adopted a captive to take the dead person's place (Chapter 24) 1666

Mourt's Relation: the book of Pilgrim writings printed in London in 1622 that contains Edward Winslow's letter about the 1621 harvest (Chapter 37) 3083

moving assembly line: a line on which a motor pulls the car or part past the workers, and each worker adds parts or does one task as it goes by (Chapter 8) 553

MRI machine: a machine that takes pictures of the inside of the body (Chapter 13) 899

muckraker: a reporter of the early 1900s who dug into one subject and wrote long reports about it. Theodore Roosevelt meant the word as an insult, after a man raking up muck (mud and filth) in *The Pilgrim's Progress*. The reporters took it as a name to be proud of (Chapter 31) 2559

Muhheakantuck: the Lenape name for the Hudson River (Chapter 1) 31

mulatto: an old word for a person with both Black and white parents or ancestors (Chapter 20) 1317

mulch: a covering that keeps soil from drying out (Chapter 27) 2032

municipal playground: a playground run by the city (Chapter 36) 3019

muriatic acid: a strong acid used to clean pools (Chapter 36) 3045

murrey: a dark purplish red (Chapter 35) 2916

Muses: in old Greek stories, nine goddesses who watched over poetry and the other arts (Chapter 32) 2608

music publisher: a company that owns or manages the rights to songs and collects money when people use them (Chapter 33) 2795

musket: a long gun (Chapter 14) 914

muslin: a thin cotton cloth (Chapter 35) 2934

muster list: a count of the living and the dead at each settlement (Chapter 14) 914

mutable loci: places on a chromosome that can change (Chapter 9) 614

mutiny: a rising up against the people in charge, such as a ship's captain or an army's officers (Chapter 1) 31

mutual fire society: a group of neighbors who promised to help save one another's houses, and only those (Chapter 6) 404

My Lai: a village in South Vietnam where American soldiers killed the unarmed people on March 16, 1968, at least 175 of them by the Army's own count (Chapter 22) 1575

myriads: huge numbers (Chapter 3) 200

N

NAACP: the National Association for the Advancement of Colored People, a group that works for the rights of Black Americans (Chapter 6) 431

NAACP Legal Defense and Educational Fund: a law office whose lawyers took civil rights cases to court (Chapter 20) 1372

NACA: the government agency that came before NASA (Chapter 1) 82

NAGPRA: the Native American Graves Protection and Repatriation Act of 1990, under which museums and agencies that take federal money must return Native remains and sacred objects (Chapter 2) 158

naloxone: a drug that can undo an opioid overdose if someone gives it fast enough (Chapter 26) 2022

name, image and likeness: payment for ads and appearances that use who an athlete is (Chapter 36) 3058

napalm: gasoline thickened into a jelly that sticks to whatever it lands on, including skin, and keeps burning (Chapter 22) 1559

napped cotton: cotton brushed so its surface is soft and fuzzy (Chapter 35) 2923

narcotics: in the 1914 law, drugs made from opium or from coca (Chapter 28) 2165

NASA: the US space agency, the National Aeronautics and Space Administration (Chapter 1) 82

NASDAQ: an index, a single number built from the stock prices of many companies, among them many technology companies (Chapter 16) 1096

Nation of Islam: a Black Muslim religious group (Chapter 22) 1575

national bank: a bank that got its permission to do business from the national government, such as the bank members of Congress set up in 1791 to hold the government's money (Chapter 17) 1128

National Bureau of Economic Research: the group of economists that sets the dates when hard times begin and end (Chapter 16) 1074

National Cemetery Administration: the federal office that runs the national military cemeteries (Chapter 32) 2643

national convention: a large meeting of a party, where delegates choose its candidate for president and write its platform (Chapter 20) 1356

National Day of Mourning: the gathering Native people have held on Cole's Hill in Plymouth every Thanksgiving since 1970 (Chapter 37) 3125

national debt: the total the federal government owes on money it has borrowed (Chapter 16) 1096

National Endowment for the Arts: a US government agency that gives money and honors to artists (Chapter 8) 512

national forest: public land where federal foresters look after the trees (Chapter 3) 215

National Guard: a state's own force of part-time soldiers (Chapter 10) 691

National Heritage Fellowship: the highest honor in the United States for folk and traditional arts, the crafts and music that families and communities hand down (Chapter 8) 512

National Historic Person: a title Canadian officials give to people who were important in Canada's history (Chapter 5) 346

National Institute of Standards and Technology: a federal science office (Chapter 3) 240

National Labor Relations Board: the government office that hears complaints about unfair treatment of workers (Chapter 14) 974

national laboratories: the government's big science labs (Chapter 9) 614

national medical coordinator: the doctor in charge of the care of the national team's gymnasts (Chapter 36) 3058

national monument: land a president protects by proclamation under the Antiquities Act (Chapter 3) 215

national origin: the country a person or their family came from (Chapter 14) 941

National Priorities List: the list of the country's most dangerous waste sites, chosen for cleanup under the federal Superfund program (Chapter 13) 886

National Recording Registry: the Library of Congress's list of recordings chosen to be kept for the future (Chapter 33) 2779

National Retail Federation: a trade group for stores (Chapter 37) 3135

National Transportation Safety Board: the federal board that investigates crashes (Chapter 11) 773

nationality: the country a person comes from (Chapter 4) 283

national-origins quotas: the rules that limited immigration country by country from the 1920s to 1965 (Chapter 4) 310

nationals: people who belong to the United States without being full citizens (Chapter 20) 1390

native copper: copper found pure in the ground (Chapter 11) 721

nativism: hatred of immigrants (Chapter 14) 922

nativists: people who opposed immigrants and wanted power kept for native-born Protestants (Chapter 4) 275

NATO: the North Atlantic Treaty Organization, a military alliance (Chapter 23) 1634

natural history: the study of plants, animals and rocks (Chapter 9) 596

natural philosophy: the old name for the study of nature, so a philosophical society was a group for studying nature (Chapter 9) 585

natural selection: Darwin's name for the way species change, when living things with a helpful trait have more young and the trait spreads (Chapter 9) 604

naturalist: a scientist who studies plants and animals (Chapter 1) 36

naturalization: the legal process by which a person born in another country becomes a citizen (Chapter 4) 271

naturalized: made a citizen by a legal process (Chapter 20) 1341

naturallists: Bradford's spelling of naturalists, people who study nature (Chapter 27) 2041

nature printing: printing the shapes of real leaves on bills to stop counterfeiters (Chapter 17) 1119

naval magazine: a Navy base for storing explosives (Chapter 27) 2078

Navigation Acts: the laws by which members of Parliament decided what English colonists could buy and sell (Chapter 16) 1052

navigation engineer: a person who works out where a spacecraft is and where it is heading (Chapter 9) 633

navigator: the person who worked out where they were and which way to go (Chapter 1) 74

Navy Cross: a navy medal for bravery in battle (Chapter 22) 1559

Nazis: Adolf Hitler's party, which ruled Germany (Chapter 4) 298

NBA draft: the yearly event in which basketball teams choose new players (Chapter 4) 320

NEA: the National Endowment for the Arts, a federal agency that gives money to artists and museums (Chapter 32) 2679

nebula: a cloud of gas and dust in space (Chapter 9) 604

necrosis of the jaw: the jawbone dying and breaking down, which the Essex County medical examiner named "radium jaw" (Chapter 13) 868

neglect of duty: failing to do one's job (Chapter 13) 899

negligence: carelessness, in the law's word (Chapter 17) 1160

negotiated under duress: worked out under threat or pressure (Chapter 30) 2367

Negro: an old word for a Black person, in common use until the 1960s (Chapter 1) 74

negro cloth: a cheap, rough cloth Northern mills made and sold to planters to clothe the people they enslaved, named with the word white Americans of that time used for Black people (Chapter 16) 1067

"Negro hire": the commissioners' name for renting enslaved men from their owners by the year, using "Negro," a word of that time for Black people (Chapter 12) 799

neonatal tetanus: an infection in newborn babies that locks the jaw and other muscles in spasms (Chapter 26) 1975

neoprene: a kind of man-made rubber (Chapter 3) 240

net domestic migration: the number of people who move into a state from other states, minus the number who move out to other states (Chapter 5) 387

network: a group of stations in different cities that carry the same programs (Chapter 31) 2559

neural network: a kind of AI program that learns from examples: a computer compares each of its answers with the right answer and adjusts its settings so the next answer comes closer (Chapter 8) 570

neutral: taking neither side (Chapter 2) 116

neutrally: favoring no religion and no lack of one (Chapter 20) 1390

neutron: a tiny particle in the center of an atom (Chapter 13) 886

never know want: never be poor (Chapter 19) 1256

new birth: a conversion that a person could feel happening inside (Chapter 29) 2231

New Deal: the set of programs President Franklin Roosevelt started in the 1930s (Chapter 14) 941

New France: the French colony in North America (Chapter 2) 116

New Great Migration: the return of Black Americans to the South since the 1970s (Chapter 5) 383

New Laws: laws King Charles I of Spain issued in 1542 stating that no one could enslave Native people and that an encomienda ended when its holder died (Chapter 24) 1667

New Lights and Old Lights: the Congregationalists who supported and opposed the revival, called the New Side and the Old Side among Presbyterians (Chapter 29) 2231

New Netherland: the Dutch colony that later became New York (Chapter 4) 260

New Spain: the Spanish colony that included Mexico (Chapter 5) 336

New York State Supreme Court: the state's main trial court (Chapter 6) 418

New York Stock Exchange: the market in New York City where people buy and sell shares of companies (Chapter 16) 1074

news desert: a county with no local newspaper, news website or station (Chapter 31) 2585

newsreels: short news films shown in movie theaters (Chapter 27) 2078

Nguzo Saba: the seven principles Maulana Karenga wrote in 1965, one talked about on each night of Kwanzaa (Chapter 37) 3125

niacin: a vitamin, also called vitamin B3 (Chapter 15) 1015

niches: small openings set into a wall (Chapter 34) 2814

nickelodeon: a small theatre that charged a nickel to see films (Chapter 34) 2861

nicotine: the drug in tobacco, which makes people addicted (Chapter 19) 1280

Niman: the Home Dance, the Hopi ceremony in late July (Chapter 37) 3075

ninepins: bowling at nine wooden pins (Chapter 36) 2987

Nineteenth Amendment: the 1920 change to the Constitution stating that no state and no federal official may keep a citizen from voting because of whether the citizen is a man or a woman (Chapter 18) 1193

NIOSH: the federal agency that studies workers' health (Chapter 10) 711

nitrogen: a gas in the air that plants need to grow (Chapter 3) 180

nitroglycerin: a liquid explosive that can go off from a jolt (Chapter 4) 283

No Child Left Behind Act: the 2002 law that members of Congress wrote as a new version of the 1965 school law. With it they began to focus on judging schools by how their students scored on tests (Chapter 20) 1390

NOAA: the US government's ocean and weather agency (Chapter 1) 82

Nobel Prizes: awards first given in 1901 under the will of Alfred Nobel, a Swedish inventor (Chapter 9) 614

nominated: named (Chapter 4) 320

nominating convention: a meeting where a party's members choose its candidates (Chapter 20) 1333

non compos mentis: Latin for "not of sound mind" (Chapter 25) 1829

noncommissioned officer: a soldier who had risen through the ranks to lead other soldiers (Chapter 24) 1712

nonconsensual sex: sex the other person did not agree to (Chapter 25) 1909

nones: people who say they have no religion (Chapter 29) 2324

nonpartisanship: not taking the side of any political party (Chapter 31) 2585

nonprofit: a group run for a cause rather than for profit (Chapter 8) 561

nonprofit corporation: a group organized like a company that does not pay profits to owners (Chapter 20) 1390

nonsectarian: not tied to any religious group (Chapter 29) 2324

non-venereal syphilis: another name for bejel, syphilis not passed by sex (Chapter 26) 1946

normal college: a college that trained teachers (Chapter 32) 2660

normal school: a school for training teachers, from the French école normale, a school that taught the norms, or rules, of teaching (Chapter 30) 2375

Norse: the people often called Vikings (Chapter 1) 17

Northwest Territory: the US name for the land north and west of the Ohio River, which later became Ohio, Indiana, Illinois, Michigan and Wisconsin (Chapter 3) 192

nose-slitting: cutting a person's nose open (Chapter 24) 1670

nostrums and quackery: fake cures and the selling of fake medicine (Chapter 18) 1193

not worth a Continental: worth nothing at all (Chapter 16) 1062

notched-corner log cabin: a cabin whose builders cut notches near the ends of the logs so that the logs locked together at the corners (Chapter 7) 471

notes: written promises to pay (Chapter 18) 1182

nourtering: bringing up (Chapter 30) 2351

nova: a star that suddenly grows much brighter for a time (Chapter 9) 604

novice: a person training to become a monk, a man who gives his life to his religion and lives by its rules (Chapter 29) 2304

nuclear chemist: a chemist who studies atoms that break apart and give off radiation (Chapter 9) 633

nuclear reactor: a machine that makes heat from uranium fuel in its core, with water flowing around the core to cool it (Chapter 10) 711

Nuclear Regulatory Commission: the federal agency that oversees nuclear plants (Chapter 10) 702

nuclear weapons: atomic bombs and the stronger bombs that came after them, all of which get their huge power from the tiny centers of atoms (Chapter 22) 1575

null result: a result where the test finds nothing (Chapter 9) 604

nullity: something with no force (Chapter 20) 1333

nun: a Catholic woman who makes religious promises and lives with others in a convent (Chapter 29) 2256

nylon: a strong thread-like material made from chemicals (Chapter 1) 82

O

oakum: old rope fiber picked apart and used to seal the seams of ships (Chapter 27) 2047

oath: a solemn promise before the law (Chapter 4) 298

oath of office: the promise each new president makes (Chapter 20) 1322

obelisk: a tall stone pillar with four sides that narrows toward a pointed top (Chapter 12) 804

obesity: a body weight high enough to raise the risk of illness (Chapter 15) 1039

obituary: the published notice of a person's death (Chapter 1) 82

obscene: too sexual for the law to allow (Chapter 21) 1468

observatory: a building for studying the sky (Chapter 9) 604

obsidian: a black natural glass formed by volcanoes, which chips to a sharp edge (Chapter 1) 17

obstructing religion resulting in death: the federal crime of using force to stop people from practicing their religion, in a way that kills someone (Chapter 29) 2324

obstructing traffic: blocking the road (Chapter 25) 1829

obstruction of Congress: blocking Congress's investigation (Chapter 20) 1390

obstruction of justice: trying to block an investigation (Chapter 20) 1372

obtrusive and ostentatious: pushing forward to be seen (Chapter 25) 1778

occupy: to hold a country with troops and control its government (Chapter 23) 1634

oceanographer: a scientist who studies the ocean (Chapter 1) 82

ochre: a colored earth that went into paint (Chapter 13) 844

Office of the Comptroller of the Currency: the government office that oversees national banks (Chapter 17) 1136

official misconduct: police, prosecutors or other officials breaking the rules (Chapter 21) 1484

offspring: children (Chapter 20) 1356

oil reserves: land kept for the Navy's future oil (Chapter 20) 1356

Okies: what Californians called the newcomers from the Dust Bowl (Chapter 5) 376

Old Order Amish: a Christian church whose members keep apart from the outside world (Chapter 20) 1372

old-age pension: regular payments to people too old to work (Chapter 16) 1084

olive jars: big Spanish storage jars that carried wine and olive oil by ship (Chapter 28) 2124

Olympic Village: where the athletes lived during the Games (Chapter 36) 3045

omnibus: a horse-drawn coach (Chapter 11) 734

on credit: getting something now and paying for it later (Chapter 15) 1009

on relief: getting government help (Chapter 15) 1015

one-judge grand jury: a single judge who decides alone whether to charge someone with a crime (Chapter 13) 899

only for cause: for a real reason such as wrongdoing (Chapter 23) 1656

Onondaga: one of the Haudenosaunee nations (Chapter 2) 99

OPEC: a group of countries that sell oil (Chapter 10) 702

open range: the unfenced grassland where cattle could graze and travel freely (Chapter 5) 363

open secret: something many people know but few say out loud (Chapter 29) 2274

open source: free for anyone to copy and change (Chapter 8) 570

openwork cloth: cloth with small openings left in the weave as a pattern (Chapter 35) 2908

opera house: a theater (Chapter 34) 2841

operating license: permission to run a reactor (Chapter 10) 711

operating system: the main program that runs a computer and the other programs on it (Chapter 19) 1280

operetta: a short comic opera (Chapter 33) 2746

opiates: drugs made from the opium poppy (Chapter 31) 2559

opinion: a judge's written explanation of a court's decision (Chapter 2) 170

opioid: a drug from the same family of chemicals as heroin, such as the painkillers oxycodone and fentanyl. Too much slows a person's breathing and can kill (Chapter 19) 1291

opium: a drug made from the opium poppy that dulls pain and brings sleep (Chapter 26) 1975

opium den: a room where people paid to lie down and smoke opium (Chapter 28) 2154

optics: the science of light (Chapter 30) 2392

oral histories: accounts passed down by word of mouth, or recorded interviews about people's lives (Chapter 2) 108

oral tradition: teaching by the spoken word (Chapter 30) 2345

ordained: officially made a minister by church leaders (Chapter 29) 2242

order: a group of Catholic men or women who promise to live by one set of rules, such as the Jesuits or the Franciscans (Chapter 1) 21

Orderly Departure Program: a legal way to leave Vietnam without the boat trip (Chapter 4) 310

ordinance: a city law (Chapter 25) 1829

ordinance of secession: a law declaring that a state was leaving the United States (Chapter 20) 1341

ordinary: the old name for a tavern (Chapter 28) 2127

ore: rock that holds a metal (Chapter 8) 524

organic farming: growing food without man-made chemical fertilizers and pesticides (Chapter 15) 1028

organic matter: the carbon-based material that living things are made of (Chapter 9) 633

organized crime: crime run by gangs (Chapter 31) 2572

Oriental: an old word for Asian (Chapter 25) 1795

Original Proprietors: the owners of the land when Washington, D.C., was laid out (Chapter 6) 407

O-rings: rubber rings that seal a joint, such as a joint of a booster rocket, so that hot gas cannot escape (Chapter 1) 82

ornithologist: a scientist who studies birds (Chapter 9) 614

orrery: a machine that shows the planets moving around the sun (Chapter 9) 591

Orteig Prize: a $25,000 prize offered for the first nonstop flight between New York and Paris (Chapter 1) 74

orthophosphate: a chemical that keeps water from eating into pipes (Chapter 13) 899

oscilloscope: a screen that draws lines of light (Chapter 34) 2877

OSHA: the government's workplace safety office (Chapter 18) 1222

osnaburg: a coarse, plain cloth first woven from flax in the German city of Osnabrück (Chapter 35) 2923

Ousamequin: Massasoit's own name (Chapter 37) 3083

out of order: against a meeting's rules (Chapter 25) 1871

outing system: Pratt's name for sending children to live with white families and work for them for part of each year (Chapter 2) 134

outpost: a small fort or settlement (Chapter 13) 847

overdose: too much of a drug at once, enough to make a person very sick or kill them (Chapter 19) 1291

overhang: a place where the rock of a cliff sticks out above like a roof (Chapter 12) 783

override a veto: to make a bill law without the president's signature (Chapter 20) 1372

overrule: to cancel a ruling with a new ruling (Chapter 20) 1356

overseer: the man in charge of the workers, such as the man who drove enslaved people at their work on a plantation. At Harvard, the overseers were the men on one of the college's governing boards (Chapter 8) 524

overseer of the highways: a man put in charge of the roads (Chapter 11) 728

overshot wheel: a water wheel turned by the weight of water poured onto its top (Chapter 10) 665

overtime: extra hours of work. Overtime pay is extra pay for hours past 40 in a week (Chapter 14) 941

Oviraptor: a meat-eating dinosaur (Chapter 1) 74

P

pacifist: a person who refuses to take part in war or violence (Chapter 25) 1871

pack line: the rope a person used to carry loads (Chapter 27) 2047

packing house: a building where fruit was packed for sale (Chapter 5) 376

paddle: a wooden board that an adult swung against a child's buttocks (Chapter 20) 1372

pagans: in Dartmouth's charter, people who were not Christian (Chapter 30) 2367

Page Act: the 1875 law, the first federal law to limit immigration, which barred Chinese laborers brought over under contract (Chapter 4) 283

Palatines: people from the Palatinate, a region along the Rhine River (Chapter 4) 267

Pale of Settlement: the region of the Russian Empire where Jews were allowed to live (Chapter 4) 283

paleontology: the study of ancient life through fossils (Chapter 9) 591

palisade: a wall of tall upright logs (Chapter 1) 21

pamphlet: a short printed booklet (Chapter 8) 524

pandemic: a disease that spreads around the world (Chapter 6) 452

pandered: gave people what they wanted, to win their favor (Chapter 32) 2695

panic: a time when fear spreads, many people rush to take their money out of the banks at once, and businesses close (Chapter 7) 480

Panic of 1873: a money crash in which many people lost their jobs (Chapter 7) 480

panpipe: a row of tubes held together, which a player blows across, each tube making a different note (Chapter 33) 2713

pantaloons: tight pants that reached the calf or the ankle (Chapter 35) 2934

pañuelo: a head cloth (Chapter 35) 2928

paper sons and paper daughters: Chinese arrivals who claimed a family tie that existed only on paper (Chapter 4) 298

paradox: something that seems to contradict itself (Chapter 19) 1291

paralegal: a worker who helps lawyers (Chapter 12) 832

parallax: looking at an object from two different places, so that a near object seems to shift against the stars far more than a distant one (Chapter 9) 585

parallel: an east-west line on a map (Chapter 1) 64

paralysis: being unable to move part of the body (Chapter 13) 868

paralyzed: unable to move parts of the body (Chapter 12) 807

paramedic: a medical worker who rides an ambulance (Chapter 5) 387

paramilitary unit: an armed group organized like an army (Chapter 29) 2256

paramount: most important (Chapter 2) 108

paramount chief (mamanatowick): a chief whom other chiefs answer to (Chapter 2) 108

paramount chiefdom: a group of many towns and chiefs under one head chief (Chapter 27) 2041

paratrooper: a soldier trained to jump from planes by parachute (Chapter 2) 149

parcel: a group (Chapter 19) 1242

parch (corn): to swirl corn kernels in a hot pan until they toast (Chapter 15) 1039

pardon: an official act that forgives a crime and ends the punishment (Chapter 2) 102

Paris climate agreement: an agreement in which the governments that join promise to cut the gases that are warming the earth (Chapter 23) 1656

Paris Salon: a large yearly art show (Chapter 32) 2643

Paris Universal Exposition: a world's fair in Paris (Chapter 12) 807

parish: the area served by one church. In Louisiana, a parish is what other states call a county (Chapter 3) 240

parish house: the house beside a church where priests live (Chapter 29) 2324

parish school: a school run by a local Catholic church (Chapter 29) 2274

Parkinson's syndrome: a disease of the brain that makes the body shake and move slowly (Chapter 36) 3045

Parliament: the lawmakers in London (Chapter 4) 275

parochial school: a school run by a church (Chapter 30) 2414

parole: letting a prisoner out early, watched by officials (Chapter 2) 149

parole board: the officials who decide whether a prisoner may leave prison early (Chapter 33) 2779

parole camp: a camp where freed prisoners waited to go home (Chapter 27) 2068

parsonage: the pastor's house (Chapter 29) 2304

partisan: favoring one political party (Chapter 20) 1390

partnership: a business in which two or more men put their money together and shared the profits (Chapter 18) 1178

parts per million: molecules of a gas in every million molecules of air (Chapter 9) 633

partus sequitur ventrem: a Latin phrase, "the child follows the womb", the rule that a child born to an enslaved mother was enslaved for life (Chapter 24) 1670

Pascua Florida: what Spanish speakers called the Easter season (Chapter 37) 3080

pasha: the ruler of Tripoli (Chapter 23) 1616

pass: a note from an owner that allowed an enslaved person to travel (Chapter 24) 1693

passage: the cost of the ocean trip (Chapter 16) 1052

passbooks: the small books that recorded depositors' savings (Chapter 17) 1136

passenger pigeon: a wild pigeon that flew in huge flocks (Chapter 3) 196

pastel: a crayon made from a paste of colored powder and resin (Chapter 32) 2613

pastor: the priest in charge of a local church (Chapter 5) 339

paten: the metal dish the priests used in the Mass, their main church service (Chapter 30) 2348

patent: a government paper that gives only the inventor the right to make and sell an invention for a set number of years. In the 1600s, a patent was also a paper giving permission to settle a colony (Chapter 8) 512

patent examiner: a person who checks patent applications for the government (Chapter 8) 541

patent medicines: medicines sold under a brand name whose makers did not have to say what was in them (Chapter 18) 1188

Patriots: the colonists fighting the British in the Revolution (Chapter 24) 1684

patron: a person who pays an artist to make work (Chapter 32) 2627

patronage: paying an artist for pictures (Chapter 32) 2627

pauper: a person too poor to feed themselves without help (Chapter 16) 1067

pavilion: a large roofed meeting place, or the building where a country shows its artists at the Biennale (Chapter 29) 2304

pawnshop: a store that lends money and sells things people leave there and do not come back for (Chapter 32) 2679

payday loan: a small loan meant to be paid back from the borrower's next paycheck (Chapter 17) 1160

paymaster: the man who pays out an army's or a navy's money (Chapter 19) 1242

payola: secret payments to a disc jockey or a radio station to play a record (Chapter 33) 2779

payroll: a company's list of the people it paid (Chapter 11) 745

PCBs: a group of chemicals made by people, used in electrical, heating and hydraulic equipment (Chapter 3) 229

Peace Corps: a US government program that sends volunteers to work in other countries (Chapter 1) 82

pearlash: a salt made from wood ashes, used to make dough rise before baking powder existed (Chapter 15) 1000

peasants: poor farm workers (Chapter 4) 283

peck: a measure of about 8 quarts, as in the weekly peck of cornmeal Jefferson gave each enslaved worker at Monticello (Chapter 15) 1000

peculiar: special (Chapter 20) 1333

peddler: a seller who traveled a route through the countryside and sold goods at people's doors (Chapter 18) 1188

pedestal: the stone base a statue stands on (Chapter 4) 283

pedophile: an adult who is sexually drawn to children (Chapter 21) 1484

Peking: the city in China now called Beijing (Chapter 1) 74

pellagra: a disease caused by a diet missing a vitamin called niacin. It brought skin rashes, diarrhea and dementia and could end in death (Chapter 15) 1015

pen name: a made-up name for signing one's writing (Chapter 11) 745

pence: two or more of the small English coins called pennies (Chapter 11) 726

pence, shillings and pounds: English money, with twelve pence to a shilling and twenty shillings to a pound (Chapter 17) 1115

pendant: an ornament that hangs from a cord (Chapter 13) 844

pending: undecided (Chapter 3) 240

Pendleton Act: the 1883 law under which some federal jobs went to people who passed a competitive exam (Chapter 20) 1341

penicillin: a drug made from a mold that kills many kinds of bacteria, the first antibiotic made in large amounts (Chapter 26) 1990

peninsula: a piece of land with water on almost every side (Chapter 1) 21

penitentiary: a prison. The word comes from penitence, being sorry for wrongdoing (Chapter 21) 1427

Pennsylvania Dutch: German settlers of Pennsylvania, from Deutsch, the German word for German (Chapter 4) 267

pension: money paid to someone every year, such as a former soldier or a worker who has retired (Chapter 10) 681

Pentagon: the headquarters of the US military, near Washington (Chapter 4) 320

Pentagon Papers: the secret history of what American leaders had done in Vietnam since World War II. *The New York Times* began printing stories about it in 1971 (Chapter 31) 2572

pentathlon: a contest of five track and field events (Chapter 2) 149

Pentecostal: the kind of Christian worship in which believers hold that God's Spirit fills a person, and that speaking in tongues is the proof (Chapter 29) 2287

peonage: forcing a person to work to pay off a debt (Chapter 16) 1074

pepper balls: small balls fired from a gun that burst into a stinging powder (Chapter 31) 2585

pepper spray: a spray made from hot peppers that burns the eyes and skin (Chapter 21) 1484

performance artist: an artist who puts on live shows (Chapter 32) 2679

performance capture: motion capture that also records the actor's face (Chapter 34) 2889

periauger: a shallow sailboat with two masts (Chapter 11) 745

periodic table: the chart of all the chemical elements (Chapter 9) 614

period-luminosity law: Henrietta Leavitt's rule that the time of a Cepheid's blink tells its true brightness (Chapter 9) 614

perjury: lying under oath (Chapter 20) 1372

permanent memory: a part of a computer that keeps what is stored in it even when the power is off (Chapter 8) 561

permits: written permission from government officials (Chapter 4) 320

pernicious sect: a harmful religious group (Chapter 25) 1760

perpetually: forever (Chapter 37) 3083

persecuted: punished again and again for who a person is and what he says or writes (Chapter 4) 310

persistent pesticide: a pesticide that stays in soil, water and living things for a long time (Chapter 3) 229

peruke: a wig of the 1700s (Chapter 35) 2923

perverse: stubbornly wrong (Chapter 4) 275

pesos: Spanish coins (Chapter 12) 789

pest house: a building where people who might carry disease were held and checked (Chapter 4) 267

pesticide: a chemical made to kill insects or weeds (Chapter 3) 229

pesticide plant: a factory that makes poisons to kill insects (Chapter 19) 1280

pestilence: a deadly disease that spreads (Chapter 26) 1983

pet banks: the state banks Jackson chose for the government's money, as his opponents called them (Chapter 17) 1128

petition: a paper that many people sign to ask for something (Chapter 7) 495

petition for habeas corpus: a request that a judge order a person held prisoner brought to court, and freed if no one can show a lawful reason to hold the prisoner (Chapter 25) 1829

petitioner: the person who brings a case to the Supreme Court (Chapter 25) 1778

petroglyph: a picture cut or pecked into rock (Chapter 32) 2598

petticoat: a woman's underskirt (Chapter 25) 1770

Pew Research Center: a group that studies public life (Chapter 4) 320

pewter: a gray metal made mostly of tin and used for plates, bowls, mugs and spoons (Chapter 13) 852

peyote: a small cactus whose top holds mescaline, a drug that causes visions (Chapter 28) 2120

pharmacy degree: a degree in preparing medicines (Chapter 25) 1795

PhD: the highest degree a university gives (Chapter 9) 581

Philadelphia lawyer: a very clever lawyer, skilled at the fine points of the law (Chapter 20) 1317

phlegm: the thick slime that comes up from the lungs (Chapter 10) 702

phone contract: a promise to pay a phone company for service every month (Chapter 8) 570

phonograph: a machine that records sound and plays it back (Chapter 8) 541

photo essay: a set of photographs that tell a story (Chapter 32) 2679

photo lineup: a set of photos shown to someone who might pick out the thief (Chapter 8) 570

photographic plate: a glass sheet coated with chemicals that change where light hits them (Chapter 6) 418

physic: medicine (Chapter 26) 1967

physical anthropology: the study of human bodies and bones (Chapter 9) 614

physicist: a scientist who studies matter and energy (Chapter 1) 82

physiognotrace: a machine that traced the outline of a sitter's face onto paper through a set of connected rods (Chapter 32) 2627

picket line: a group of protesters standing outside with signs (Chapter 25) 1829

picketing: standing outside a place with signs as a public protest (Chapter 25) 1829

pictograph: a picture painted on rock (Chapter 32) 2598

picture factory: a shop that turned out many portraits fast (Chapter 32) 2627

piece of eight: a Spanish silver coin worth 8 reales, also called the Spanish dollar (Chapter 17) 1113

Piedmont: the hill country of the Carolinas (Chapter 5) 343

pig iron: the iron that came out of a blast furnace (Chapter 13) 848

pikes: long spears (Chapter 25) 1760

Pilgrims: what later Americans called the Plymouth colonists (Chapter 4) 260

pillory: a wooden frame on a post, on a platform in a public place, that locked a person's neck and wrists while the person stood (Chapter 21) 1412

pilot: the man who steered a ship (Chapter 11) 745

Pinkerton detective agency: a private company that hired out guards and detectives (Chapter 14) 929

Pinkster: the Dutch name for Pentecost, a Christian feast held seven weeks after Easter (Chapter 34) 2827

pinnace: a small sailing ship (Chapter 27) 2041

pinole: flour made from ground seeds (Chapter 1) 39

pinta: a slow skin disease in the syphilis family (Chapter 26) 1946

pious: devoted to religion (Chapter 30) 2375

pistole: a Spanish gold coin used in the colonies (Chapter 33) 2723

piston: a tight-fitting plug pushed into the cylinder (Chapter 8) 528

pit: the pile of stacked wood, covered with leaves and dust, in which a collier made charcoal (Chapter 10) 671

pitch: a sticky tree sap (Chapter 8) 512

pitching the bar: a contest to see who could throw a heavy bar or pole the farthest (Chapter 37) 3083

piteous: full of pain (Chapter 25) 1778

place of public accommodation: a place that serves the public (Chapter 20) 1372

placebo: a shot with no medicine in it (Chapter 26) 2005

placer miner: a miner who washes gold out of sand and gravel in streams (Chapter 13) 863

plague: a disease that spread quickly and killed many people (Chapter 7) 465

Plains: the wide grasslands in the middle of the continent (Chapter 1) 17

plaintiff: a person who brings a case to court (Chapter 25) 1909

plank: one position in a platform (Chapter 20) 1356

planned city: a city whose builders decide ahead of time where each open square, wall and mound will go (Chapter 6) 395

plantation: a large farm, owned by a planter. Until slavery ended, planters forced enslaved people to do the work (Chapter 3) 189

planter: the owner of a plantation, a large farm (Chapter 1) 64

plaque: a metal sign with words on it (Chapter 4) 298

plate tectonics: the idea that the Earth's outer shell is broken into huge pieces, called plates, that slowly shift (Chapter 9) 633

plates: the huge slabs the outer layer of the earth is broken into (Chapter 27) 2032

platform: a party's list of positions (Chapter 17) 1136

platinum album: the RIAA's award for one million copies sold (Chapter 33) 2713

playbill: a printed poster for a show (Chapter 34) 2841

player: an actor (Chapter 34) 2827

plaza: a public square of open ground with the town built around it (Chapter 6) 398

plea agreement: a statement in which a person accused of a crime admits to it under a deal with prosecutors (Chapter 25) 1909

plea deal: an agreement to admit a crime in return for a lighter punishment (Chapter 22) 1585

plead guilty: to tell a court one has broken the law (Chapter 14) 963

pledge: a written promise not to drink (Chapter 28) 2147

Pledge of Allegiance: the promise of loyalty to the flag (Chapter 29) 2304

Pleiades: a small group of stars Hawaiians call Makali‘i (Chapter 37) 3075

plow under: to turn growing plants back into the soil (Chapter 16) 1084

ploy: a trick (Chapter 22) 1534

plural wife: one of several wives a man had at the same time (Chapter 29) 2274

plutonium: a radioactive metal that gives off heat (Chapter 1) 92

plywood: thin sheets of wood glued together (Chapter 2) 170

pneumonia: a sickness of the lungs (Chapter 12) 815

pocket veto: what happens when the president does not sign a bill and Congress ends its meeting, so the bill never becomes law (Chapter 3) 200

pogrom: a violent riot in which a crowd attacks Jewish people and their homes, shops and synagogues (Chapter 4) 283

police brutality: violence by police officers (Chapter 25) 1909

polio: a disease that can paralyze muscles (Chapter 9) 633

political action committee (PAC): a group that collects money and gives it to candidates (Chapter 19) 1280

political machine: a party organization that wins votes by handing out jobs and favors (Chapter 20) 1341

political party: a group of people who work together to win elections and run the government their way (Chapter 4) 283

poll tax: a tax a man had to pay before he could vote (Chapter 20) 1341

polyester: a plastic made into thread (Chapter 35) 2959

polygenism: the false and racist claim that the races of people were created separately (Chapter 9) 604

Polynesians: the people of the many islands of the central and southern Pacific (Chapter 1) 17

pomp: showy display (Chapter 20) 1322

Pong: a simple tennis game played on a screen (Chapter 34) 2877

pools: agreements among owners to hold prices up (Chapter 19) 1268

poorhouse: a building where a town kept people who had no money (Chapter 25) 1778

Pope's Night: November 5 in New England, when crowds burned figures of the Pope (Chapter 37) 3091

porcelain collars: Brebeuf's word for collars of wampum (Chapter 36) 2981

porcupine quills: the stiff, hollow spines that cover a porcupine (Chapter 35) 2908

pornography: pictures or objects made to show sex (Chapter 32) 2679

port: a small opening in a ship's side (Chapter 1) 31

portage: a crossing over land, carrying the canoes, from one river or lake to the next (Chapter 11) 721

porter: a worker who carried passengers' bags on the trains (Chapter 9) 614

positive and negative (electricity): Franklin's words for an object holding more or less of the electric fluid than normal (Chapter 9) 588

posse: a group of men gathered to chase someone (Chapter 21) 1442

post rider: a man on horseback who carried letters (Chapter 11) 726

postdoc: a scientist doing research for a few years after a PhD before taking a permanent job (Chapter 9) 648

posterity: the people who live later (Chapter 18) 1182

Postmaster General: the official in charge of the national mail service (Chapter 18) 1193

post-polio syndrome: new muscle pain and weakness 15 to 40 years after polio (Chapter 26) 2005

post-traumatic stress disorder: nightmares and memories of combat that keep coming back (Chapter 22) 1559

potash: a salt made from wood ashes (Chapter 19) 1242

potato blight: a plant disease, a water mold, that rotted Ireland's potatoes (Chapter 4) 275

potter's field: a burial ground for poor people (Chapter 9) 596

pound (£): the main unit of English and British money (Chapter 16) 1057

power loom: a weaving machine driven by water power (Chapter 19) 1250

Powhatan: a union of Native towns and peoples under one head chief. The English also called that head chief, Wahunsenacawh, the Powhatan (Chapter 2) 102

Praying Indians: the English name for the Christian Native people of New England's praying towns (Chapter 29) 2213

praying towns: villages of Native people who had become Christians (Chapter 2) 108

precarious: easily lost (Chapter 3) 200

precedent: the Court's own earlier rulings (Chapter 25) 1909

precinct: one small voting area (Chapter 20) 1356

preclearance: the federal permission listed places needed before changing a voting rule (Chapter 25) 1909

predator: an animal that hunts other animals for food (Chapter 3) 215

preface: the short opening piece of a book (Chapter 34) 2827

prehistoric: built before anyone here kept written records (Chapter 6) 395

prejudice: a dislike of a whole group of people without knowing them (Chapter 5) 376

Presbyterians: members of a Protestant church (Chapter 4) 267

prescribed burns: fires set on purpose by fire managers to burn off fuel (Chapter 3) 240

prescription: a doctor's written order for a medicine (Chapter 19) 1291

prescriptive: telling people exactly what to do (Chapter 30) 2474

preservation: keeping something safe (Chapter 3) 196

President of the Senate: the vice president (Chapter 20) 1390

Presidential Medal of Freedom: the country's highest honor for a civilian (Chapter 1) 82

Presidents' Day: the name many people use for Washington's Birthday, though no act of Congress and no order of a president changed the federal name (Chapter 37) 3125

presides: runs the meeting (Chapter 20) 1390

presidio: a Spanish fort (Chapter 29) 2256

press gangs: the groups of men sent to seize sailors and dock workers for the navy (Chapter 14) 918

press pool: the small group of reporters that covers the president up close (Chapter 31) 2585

pressing: forcing a plea by laying a prisoner under a wooden board and piling heavier and heavier stones on it until the prisoner answered or died (Chapter 21) 1412

pretext: a false excuse (Chapter 4) 283

prevalent: very common (Chapter 26) 1990

previews: the shows before a play's official opening (Chapter 34) 2889

previous condition of servitude: having once been enslaved (Chapter 20) 1341

price discrimination: charging different buyers different prices for the same goods (Chapter 19) 1268

Pride marches: the name for marches like the one held in New York City every June since 1970 (Chapter 37) 3125

primary: the election in which a party chooses its candidates (Chapter 20) 1356

prime contract: a contract made directly between the government and a company (Chapter 19) 1268

prime minister: the head of the British government (Chapter 4) 275

primer: a first reading book (Chapter 29) 2213

Principal Chief: the Cherokee Nation's top leader (Chapter 2) 158

principal (of a loan): the money borrowed, not counting interest (Chapter 30) 2474

printing press: a machine that presses paper hard onto metal letters covered in ink (Chapter 8) 517

prior restraint: a court order that stops something from being printed before it comes out (Chapter 31) 2572

prisoners of war: people an army holds after capturing them in a war (Chapter 2) 134

pristine: untouched and pure (Chapter 26) 1946

privateer: a privately owned armed ship with a government's permission to attack enemy ships (Chapter 4) 260

privy: an outdoor toilet built over a pit in the ground (Chapter 6) 413

Privy Council: the English king's inner council (Chapter 19) 1235

prize fight: a boxing match with bare fists fought for money (Chapter 36) 3008

probable cause or reasonable suspicion: the legal words for a good reason to believe a person has done something wrong (Chapter 25) 1829

probate inventory: a list of everything a person owned, each thing given a value, made after the person died (Chapter 35) 2916

probation: a time when a person found guilty lives under a court's watch instead of in prison, or any trial period under watch (Chapter 20) 1356

probes: robot spacecraft sent to study faraway places (Chapter 1) 92

process: a series of steps for making something, such as Hall's way of making aluminum (Chapter 8) 535

processed foods: foods that were canned, dried or frozen (Chapter 15) 1015

proclaim: to name a day in a public order (Chapter 37) 3101

proclamation: an official public announcement of an order (Chapter 2) 120

Proclamation of Neutrality: Washington's 1793 statement that the United States would take neither side (Chapter 23) 1605

production assistant: a helper on a film crew (Chapter 25) 1909

Production Code: the set of rules for what films could show, which the heads of the big studios agreed in 1930 to follow (Chapter 34) 2861

profane: not holy (Chapter 37) 3083

profession: a skilled job that people study for and that schools teach (Chapter 7) 474

professionals: people trained for skilled work such as law or medicine (Chapter 4) 275

proficient: good enough to meet the standard that officials in each state set (Chapter 30) 2474

profit-sharing: a share of a company's profits (Chapter 14) 941

programming language: a set of words and rules for writing instructions a computer can follow (Chapter 8) 561

Prohibition: the years from 1920 to 1933 when making and selling alcohol was against the law (Chapter 21) 1456

prologue: a speech given before a play begins (Chapter 34) 2833

promiscuity: sex with many partners (Chapter 20) 1356

promissory notes: written promises to pay money later (Chapter 17) 1128

prone: lying face down (Chapter 25) 1909

propaganda: messages meant to push people to one side (Chapter 12) 815

propaganda campaign: a huge effort to push one side of the story (Chapter 12) 832

propagandist: a person who spreads one side's story (Chapter 9) 614

property tax: a tax on the value of land and buildings (Chapter 20) 1372

prophecy: a teaching about what was to come (Chapter 15) 986

Proposition 22: a law put to California voters directly (Chapter 11) 773

proscribed: shut out (Chapter 30) 2392

prosecutor: the lawyer who brings criminal charges for the government (Chapter 8) 570

proselytizing: trying to convert people to a religion (Chapter 29) 2324

prospector: a person who searches for gold or other minerals (Chapter 1) 64

prostitution: selling sex for money (Chapter 4) 283

protein: a molecule that does work inside living cells, and the part of food the body uses to build muscle (Chapter 9) 648

Protestant: a Christian whose church broke away from the Catholic Church (Chapter 4) 258

Protestant church: one of the Christian churches that broke away from the Catholic Church (Chapter 4) 267

Protoceratops: a dinosaur that scientists first thought laid the Gobi eggs (Chapter 1) 74

proton: a tiny particle with a positive electric charge in the center of every atom, whose count gives each element its number (Chapter 9) 614

proved (a will): agreed by court officials to be real, so it could be carried out (Chapter 7) 465

province: proper job, or a colony (Chapter 20) 1333

provisional government: a government set up for a short time until a lasting one is formed (Chapter 23) 1623

provost marshal: an army officer who ran the draft (Chapter 22) 1543

Prussia: a kingdom in what is now Germany (Chapter 4) 275

psalm: a religious song from the Bible (Chapter 7) 465

psalm book: a book with the songs from the Bible's Book of Psalms (Chapter 7) 465

psilocybin: a drug from certain mushrooms that causes visions (Chapter 28) 2179

psychiatric hospital: a hospital for illnesses of the mind (Chapter 35) 2949

psychiatric unit: the ward for mental illness (Chapter 25) 1909

psychiatrist: a doctor who treats illnesses of the mind (Chapter 25) 1829

psychological warfare: using fear, rumors and false stories to turn people against their leaders (Chapter 23) 1647

psychologist: a scientist who studies how people think (Chapter 30) 2414

psychology: the study of the mind and of how people behave (Chapter 7) 485

psychopathic ward: the part of a hospital for people thought to be insane (Chapter 25) 1829

puberty blocker: a drug that pauses the hormones that start puberty (Chapter 25) 1909

public assistance: money from a government aid program (Chapter 17) 1153

public conveyances: trains, boats and coaches for hire (Chapter 25) 1795

public execution: a hanging in front of a crowd (Chapter 21) 1434

public hearings: meetings where senators question witnesses (Chapter 30) 2414

public housing: apartments owned by a government and rented to families with low incomes (Chapter 6) 443

public humiliation: a day when the people of a town prayed together and humbled themselves before God (Chapter 35) 2916

public land: land that US officials claimed for the nation, much of it lived on by Native nations (Chapter 3) 192

public land bank: a public office through which colonial officials lent paper money to people who pledged their land as security (Chapter 17) 1119

public nuisance: a place that harms the people around it (Chapter 25) 1795

Public Proclamation: an official public order (Chapter 20) 1356

public relations firm: a company paid to shape what the public thinks of its clients (Chapter 19) 1280

public relations person: someone paid to shape what the public thinks of an event (Chapter 37) 3125

public school: a school that a town or a state runs for its children (Chapter 7) 495

public works: projects that the government pays for and the public uses (Chapter 11) 764

publican: a tavern keeper (Chapter 34) 2827

"Published by Authority": printed with the approval of the people in power (Chapter 31) 2512

publishing rights: the rights to a song itself (Chapter 33) 2795

Publius: the made-up name Hamilton, Madison and Jay signed their essays with (Chapter 20) 1322

pueblo: a Native town of connected houses built of stone or dried mud, named from the Spanish word for town (Chapter 1) 21

Pueblo Revolt: the 1680 uprising, led by Po'pay, in which the Pueblo people drove the Spanish out of New Mexico for twelve years (Chapter 2) 108

puffing: praising someone too much (Chapter 9) 591

Pulitzer Prize: a top yearly American prize for writing, reporting or music (Chapter 6) 443

pulley and tackle: a set of ropes and wheels for lifting heavy loads (Chapter 12) 807

pulling the goose: a Dutch game in which riders galloped past a goose hung by a rope and grabbed at it to pull it down (Chapter 36) 2987

Pullman porter: a Black worker who served passengers on railroad sleeping cars (Chapter 31) 2559

pulpit: preachers in churches (Chapter 19) 1280

Pūnana leo: the name of the Hawaiian-language preschools, meaning "nest of voices" (Chapter 30) 2442

punch: a mixed drink made with liquor (Chapter 21) 1422

purchasing power: how much people can buy with their money (Chapter 16) 1096

Pure Food and Drug Act: the 1906 law that banned selling mislabeled or impure food, drink or drugs across state lines (Chapter 15) 1015

purge: a medicine that emptied the bowels (Chapter 26) 1967

purify: to remove everything else from a substance (Chapter 13) 899

Puritan: an English Protestant who wanted to "purify" the Church of England. Puritans started the Massachusetts colony (Chapter 4) 260

Purple Heart: the medal for being wounded or killed in battle (Chapter 22) 1559

purse: a prize in money (Chapter 36) 2999

pursued: went after (Chapter 35) 2940

put them to the knife: stabbed them to death (Chapter 22) 1502

Q

quahogs: hard-shelled clams of the Atlantic coast (Chapter 17) 1112

Quaker: a member of the Religious Society of Friends, a Christian group that began in England in the 1600s (Chapter 4) 260

Quaker meeting: a group of Quakers who worship and make decisions together. A monthly meeting is local, a quarterly meeting joins several local groups, and a yearly meeting joins a whole region (Chapter 25) 1760

Quaker missionaries: people sent out by the Quaker church to spread its faith (Chapter 30) 2392

qualified immunity: a rule that shields government officials from being sued unless the right they broke was already clearly established, meaning already made clear in the law (Chapter 25) 1909

quantum mechanics: the branch of physics that describes very tiny particles (Chapter 9) 614

quarantine: keeping people or animals that may carry a disease apart from others so it does not spread. President Kennedy also used the word for stopping ships carrying weapons to Cuba (Chapter 4) 267

quarantine law: a law keeping animals apart to stop the spread of disease (Chapter 5) 363

quarantine line: the border Texas cattle could not cross into Kansas under a Kansas law of 1885 and 1886 (Chapter 15) 1009

quarantine station: a place where new arrivals were kept apart so that any disease they carried would not spread (Chapter 4) 267

quarks: the smaller particles that protons and neutrons are built from (Chapter 9) 633

quarried: cut from the ground (Chapter 12) 799

quarry: a pit where stone is cut out of the ground (Chapter 12) 789

quarter: one of the three-month periods for which companies report their results (Chapter 19) 1291

quarter race: a race in which two horses sprinted a quarter mile along a straight track (Chapter 36) 2987

quartermaster: an army supply officer (Chapter 27) 2068

quartzite: a hard rock (Chapter 36) 2981

Quasi-War: the undeclared war at sea with France, 1798 to 1800, from a word meaning "almost" (Chapter 23) 1605

queer: a word many gay and lesbian people use for themselves (Chapter 1) 82

Query: an old word for a question, Jefferson's name for each chapter of his book (Chapter 9) 591

queue: one long braid, worn by Chinese men who shaved the front of the head (Chapter 35) 2940

quinine: a drug for the disease malaria (Chapter 35) 2940

quitrent: a small yearly rent on land (Chapter 19) 1235

quiver: the case that holds arrows (Chapter 35) 2912

Quivira: a rich land the Turk told of, where Coronado found Wichita towns and no gold (Chapter 1) 21

quota: a fixed share (Chapter 4) 298

quota laws: laws that fixed how many people could come from each country (Chapter 4) 298

QWERTY: the keyboard arrangement that begins Q, W, E, R, T, Y on the top row of letters (Chapter 8) 541

R

rabbi: a teacher and leader of a Jewish congregation (Chapter 29) 2274

race card: a printed list of the horses in a race (Chapter 36) 3008

race prejudice: hating or looking down on people because of their race (Chapter 5) 376

race records: record company managers' name in the 1920s for records made for Black buyers (Chapter 33) 2759

racial bias: unfair treatment because of race (Chapter 16) 1084

racial gerrymander: a set of districts drawn by race in a way the Constitution forbids (Chapter 25) 1909

racial slurs: insulting names for Black people (Chapter 22) 1559

racially nondiscriminatory basis: no child could be turned away because of race (Chapter 20) 1372

rack: an old device for torturing a person by stretching the body (Chapter 21) 1456

racketeering law: the law against running a business through a pattern of fraud (Chapter 19) 1291

radiation: energy that substances such as a reactor's fuel give off, which can damage the cells of the body (Chapter 10) 702

radiation sickness: fever within two days, then hair falling out and bleeding from the gums, nose and under the skin, caused by radiation (Chapter 22) 1559

radio direction finder: a radio that locks onto a signal to show which way it came from (Chapter 1) 74

radioactive: describes a material whose atoms break down and give off energy and tiny particles (Chapter 1) 92

radiocarbon date: an age measured from something that was once alive, such as wood or bone (Chapter 13) 844

radiocarbon dating: a way to measure the age of anything that was once alive, by how much of one kind of carbon is left in it (Chapter 2) 108

radium: a radioactive metal (Chapter 9) 633

radium jaw: a jaw bone wasting away and breaking apart from radium (Chapter 13) 868

radon: a radioactive gas from uranium that caused lung cancer in miners who breathed it (Chapter 10) 702

radon daughters: radioactive specks that radon atoms break apart into, which miners breathed in (Chapter 13) 886

Ragamuffin Day: a Thanksgiving custom, from about 1870 into the 1920s, when New York children dressed in costumes and begged from door to door (Chapter 37) 3119

ragtime: piano music with a "ragged" rhythm, in which the tune lands between the beats while a steady beat keeps time (Chapter 33) 2746

raid: a quick attack to seize goods, horses or people (Chapter 2) 116

railhead: a town at the end of a railroad line (Chapter 5) 363

ranch house: a long, low house of one story (Chapter 7) 485

Ranger: a soldier of a specially trained Army unit (Chapter 22) 1585

ransacked: tore through a house and wrecked it (Chapter 28) 2138

rape: forcing a person to have sex (Chapter 1) 21

raptors: birds that hunt other animals (Chapter 3) 229

Rastafarian vow: a religious promise not to cut one's hair (Chapter 29) 2324

ratchet: a bar with teeth (Chapter 6) 418

ratify: to approve formally (Chapter 2) 120

ration: a set amount of food given out (Chapter 3) 200

ration points: numbers printed on ration stamps and used, along with money, to pay for rationed foods (Chapter 18) 1204

rational: having a sensible reason (Chapter 20) 1372

rationing: limiting how much of a food each person could buy (Chapter 15) 1015

rations, clothing and annuities: the food, goods and yearly payments promised to a nation in its treaties (Chapter 2) 134

rattan: a thin, bendable cane cut from a kind of palm (Chapter 30) 2392

Raven and Eagle sides: the two halves of the Tlingit people, each made up of clans (Chapter 32) 2598

raw iron: iron not yet made into finished goods (Chapter 13) 852

razed: torn down completely (Chapter 3) 229

reactor: a machine where atoms are split in a chain reaction (Chapter 9) 614

read publicly out of meeting: expelled in front of the whole church (Chapter 29) 2242

read the Constitution strictly: hold that the government can do only what the document says it can (Chapter 20) 1333

real: a Spanish silver coin (Chapter 12) 789

real GDP: GDP counted with the change in prices taken out, to show how much the country actually made (Chapter 16) 1084

reaper: a machine that cuts ripe grain (Chapter 8) 535

rear admiral: one of the highest ranks in the Navy (Chapter 1) 74

reasonable diligence: reasonable care (Chapter 14) 974

rebate: money a railroad would pay back to a shipper on every barrel he shipped (Chapter 19) 1256

receiver: an official who takes over a failed bank and pays its debts from what is left (Chapter 17) 1128

recess appointment: a president filling a seat while the Senate is away (Chapter 20) 1372

recession: a time when business shrinks and people lose jobs (Chapter 16) 1049

recitation bench: the bench at the front where a pupil said a lesson out loud to the teacher (Chapter 30) 2375

reckless endangerment: putting people in danger through carelessness (Chapter 21) 1468

reclamation: taking something back (Chapter 2) 170

recognition: United States officials dealing with a nation as a government (Chapter 2) 158

recognize (a country): to accept officially that it exists (Chapter 2) 120

recognize a union: to accept it and bargain with it (Chapter 14) 941

recognized (federally): accepted by federal officials as a nation with its own government (Chapter 2) 126

reconcentration: forcing civilians out of their homes into camps or zones that soldiers guarded (Chapter 23) 1623

reconciliation: making peace after a wrong (Chapter 24) 1744

Reconstruction: the rebuilding of the Southern states' governments after the Civil War, in the years when members of Congress wrote Black Americans' rights into three amendments to the Constitution and into new federal laws (Chapter 20) 1341

Reconstruction Acts: the laws members of Congress passed for Reconstruction (Chapter 20) 1341

recorded Quaker minister: a speaker whom a Quaker meeting officially recognized (Chapter 25) 1767

recorder: a city official who also served as a judge (Chapter 25) 1795

recording secretary: the keeper of the written record of a union's meetings (Chapter 16) 1096

recount: a second count of the votes (Chapter 20) 1390

rectum: the end of the bowel (Chapter 26) 1975

redlining: drawing a line around a neighborhood on a map and then refusing to lend money for houses inside it (Chapter 6) 431

refinery: a plant that turns crude oil into fuels (Chapter 3) 240

refining: the work of turning crude oil from the ground into fuels people can use (Chapter 16) 1074

reform school: a school where courts sent children who were in trouble (Chapter 33) 2759

refuge: a safe place, or land where wild animals are kept safe (Chapter 2) 102

refuge of last resort: a place to go when there is nowhere else (Chapter 27) 2104

refugee: a person who flees their home to be safe (Chapter 2) 102

refugee (Refugee Act of 1980): a person outside their own country who cannot go back because of a real fear of being harmed for their race, religion, nationality, group or political beliefs (Chapter 4) 310

regiment: a unit of the army (Chapter 7) 463

Register of Deeds: the county office that keeps records of land (Chapter 17) 1119

registrar: the official who signed up voters or students (Chapter 20) 1341

regular singing: singing from written notes (Chapter 33) 2723

regulating parties: companies of men who made sure people obeyed the Cherokee Nation's laws (Chapter 21) 1434

Regulation Q: a federal rule that forbade banks to pay more than a set rate of interest on a savings account (Chapter 17) 1153

regulator clock: a very exact clock (Chapter 6) 407

rejection: the body's defenses attacking an organ from almost anyone else (Chapter 26) 2005

rekindle: to start a fire burning again (Chapter 27) 2104

relaxers: creams used to straighten hair (Chapter 35) 2959

relay: a device that boosts a weakening signal at points along a telegraph line (Chapter 8) 535

relays: one party hauled loads part of the way and left them, and the next party carried them on (Chapter 1) 74

relief: public money for people without work (Chapter 4) 298

relief office: a place that gave out food or money to the poor (Chapter 25) 1829

relief valve: a valve that opens to let out pressure (Chapter 10) 702

religion clauses: the two parts of the First Amendment about religion (Chapter 29) 2324

religious order: a group of priests, brothers or sisters who live under shared religious promises (Chapter 1) 21

religious test: an oath about religious beliefs that a person had to swear before taking office (Chapter 29) 2242

relocation: the program under which officials of the Bureau of Indian Affairs moved Native people off reservations and into cities, giving those who signed up a one-way ticket and short-term help (Chapter 2) 158

remonstrance: a formal protest (Chapter 29) 2242

removal (from the country): the government's word for deportation, forcing a person to leave the country (Chapter 4) 320

removal (of a Native nation): forcing a whole Native nation out of its homeland and moving it away, as United States soldiers did to the nations of the Southeast (Chapter 2) 126

renaissance: a French word that means rebirth (Chapter 32) 2660

rendezvous: the gathering where the trappers met every summer, from the French word for a meeting (Chapter 1) 44

renewable energy: wind, solar, flowing water and other sources that do not run out (Chapter 10) 711

repair: payments or other steps to make up for what slavery did (Chapter 24) 1666

reparations: payments made to make up for a wrong (Chapter 16) 1067

repartimiento: a Spanish system of forced labor under which officials sent Native workers in groups, by turns, to work on Spanish projects, and were supposed to pay them (Chapter 14) 912

repatriation: sending human remains and sacred objects home to the nations and families they came from (Chapter 2) 158

repeal: to cancel a law (Chapter 19) 1250

repellant: disgusting (Chapter 32) 2643

repent: to be sorry for doing wrong and to change (Chapter 32) 2627

repoussé: shaping thin copper sheets by hammering (Chapter 12) 807

reprehensible: deserving strong blame (Chapter 9) 633

reprieve: an order that calls off or puts off an execution (Chapter 2) 134

reprimand: scold officially (Chapter 4) 310

reprisal: punishment by government officials (Chapter 29) 2324

republic: a country governed by elected representatives (Chapter 20) 1322

Republican: a member of one of the country's two main political parties (Chapter 1) 44

research and development: the work of trying out new ideas (Chapter 18) 1214

reservation: a piece of land set aside for a Native nation in a treaty or by an order. United States officials and soldiers forced the nations onto reservations, and officials and settlers took the rest of their land (Chapter 1) 44

reserve: an area set aside for a nation to live on (Chapter 3) 192

reserves: the money a bank keeps on hand instead of lending it out (Chapter 17) 1143

reservoir: a lake made by a dam to store water (Chapter 3) 215

Resident Commissioner: Puerto Rico's one member in the House, who cannot vote on the House floor (Chapter 20) 1390

residuals: payments that grow when a game sells well (Chapter 34) 2889

resin: a sticky stuff that comes from trees (Chapter 32) 2613

resolution: a written statement that a meeting or a group of lawmakers votes on (Chapter 1) 64

restrictive covenant: a section written into the deed of a house that forbade the owner to sell or rent it to a Black buyer, and sometimes to Jewish or Asian buyers, and bound every later owner (Chapter 6) 431

retailer: a business that sells goods to the people who use them (Chapter 18) 1204

retainer: the money paid to a lawyer to take a case (Chapter 29) 2256

retarded: the word then used for children and people with intellectual disabilities (Chapter 25) 1829

retweet: a post shared again on Twitter (Chapter 31) 2585

revelation: a message a person says came from God (Chapter 25) 1760

revenue: the money a company takes in (Chapter 19) 1291

revival: a series of meetings meant to stir people to a stronger faith (Chapter 29) 2231

revue: a show of songs and comic scenes (Chapter 34) 2861

rhetoric: the art of speaking and writing well (Chapter 25) 1778

RIAA: the Recording Industry Association of America, the record companies' own group, which gives sales awards and counts what the companies take in (Chapter 33) 2713

Rice Coast: the part of West Africa that included Senegambia and Sierra Leone (Chapter 15) 995

ridden out on a rail: carried out of town on a wooden pole (Chapter 21) 1442

rider: a short passage attached to a spending bill (Chapter 2) 134

riding bosses: the men on horseback who watched over the field workers (Chapter 15) 1015

rift valley: a valley that forms where the Earth's crust, its rocky outer layer, is pulling apart (Chapter 9) 633

right of Representation in the Legislature: a say in the body that makes the laws (Chapter 25) 1770

rights movement: a group of people who organize together to change a rule that denies them something (Chapter 12) 832

ringleaders: the leaders of the fighting (Chapter 30) 2392

Rio Grande: the river along the border between Texas and Mexico (Chapter 1) 21

river basin: all the land whose rain drains into one river (Chapter 1) 31

river bottom: the low land along a river (Chapter 37) 3107

river cane: a tall plant with hollow stems that grows along rivers (Chapter 32) 2613

riveting gang: a crew that joined steel beams together with hot metal pins called rivets (Chapter 2) 149

rivets: hot metal pins that lock steel beams together, or small metal pins that hold cloth together (Chapter 2) 149

rocket-propelled grenade: a small rocket fired from a tube carried on a fighter's shoulder (Chapter 22) 1585

rod: a stick or switch used for punishment (Chapter 7) 474

roentgen: a unit for measuring radiation (Chapter 9) 614

roll, pitch and yaw: an airplane's wing tips moving up and down, its nose moving up and down, and its nose swinging from side to side (Chapter 8) 553

rolling and slitting mill: a mill that pressed iron into flat sheets and cut it into thin rods (Chapter 8) 517

rolling outages: planned power cuts that move from area to area (Chapter 10) 711

Romanism: an insulting word for the Catholic Church (Chapter 29) 2274

Rookie of the Year: the best first-year player (Chapter 36) 3028

Ross Ice Shelf: a huge sheet of ice floating on the sea (Chapter 1) 74

rostrum: a speaker's stand (Chapter 20) 1356

rotation: the list of songs a station plays over and over each day (Chapter 33) 2795

rotation in office: Jackson's name for filling government jobs with his own supporters (Chapter 20) 1333

rote: learned by repeating things over and over (Chapter 7) 495

Rotunda: the round central hall of the US Capitol (Chapter 4) 320

roundhouse: a round building for gatherings (Chapter 3) 240

rovers: robot vehicles (Chapter 1) 92

royal colony: a colony with a governor and council chosen by the king (Chapter 19) 1235

Royal Commission: a group set up to find the cause of the Quebec Bridge collapse (Chapter 12) 815

Royal Proclamation of 1763: the British king's order of 1763 that set aside much of the western land for Native nations and banned private buyers from buying Native land (Chapter 5) 346

royal prosecutor: the lawyer who brought a case for the king (Chapter 22) 1502

royal treasury: the king's own money (Chapter 31) 2520

royalties: a share of the money from every sale (Chapter 2) 149

rudder: a movable upright part at the back of an airplane that controls yaw, the nose swinging from side to side (Chapter 8) 553

ruffs: stiff pleated collars (Chapter 35) 2916

rule of reason: the court's test that the Sherman Act banned only "unreasonable" limits on trade (Chapter 19) 1268

ruling elders: the men who led the Puritan church (Chapter 35) 2916

rum: a strong liquor made from molasses (Chapter 15) 1000

run: the yearly trip of fish up a river, or a rush on a single bank, when people crowd in to trade its notes or their deposits for coins (Chapter 3) 196

runoff: a second election between the top two candidates in a party's primary (Chapter 20) 1372

rural free delivery: a service in which postal workers carried mail to farm families' homes (Chapter 18) 1193

Rust Belt: a nickname for the older factory states of the North (Chapter 5) 383

RV: a recreational vehicle, a home on wheels (Chapter 16) 1102

S

Sabbath: the day of rest and worship (Chapter 21) 1412

sabotage: secretly wrecking someone else's work (Chapter 31) 2572

sachem: a leader of a Native nation (Chapter 2) 108

sack suit: a loose jacket with pants (Chapter 35) 2940

sacrament: a sacred act of worship (Chapter 29) 2304

sacred music: music for worship (Chapter 33) 2729

sacrifices: people whom others at Cahokia killed on purpose as part of the burial ceremonies (Chapter 35) 2908

sad colors: the colors Puritans wore, "sad" then meaning serious rather than unhappy (Chapter 35) 2916

sadistic: taking pleasure in hurting people (Chapter 22) 1585

"safe level" theory: the lead industry's claim that lead below a certain amount did no harm (Chapter 13) 868

safety valve: a part that lets steam out before the pressure gets too high (Chapter 27) 2068

safety-valves: Frederick Douglass's word for the Christmas holidays, which enslavers used to drain away the anger that could lead enslaved people to rebel (Chapter 37) 3101

sagas: old stories written down in Iceland (Chapter 1) 17

sailing master: the officer in charge of sailing a ship (Chapter 27) 2061

sailor's protection: a paper describing a free American sailor (Chapter 35) 2934

saloon: a bar, mostly for men, that sold drinks by the glass (Chapter 17) 1136

salpingectomy: an operation that cuts out a section of each of the two tubes that carry an egg from an ovary to the womb, so that a woman can never become pregnant (Chapter 20) 1356

salt domes: large masses of salt buried deep underground (Chapter 10) 691

salt lick: a place where animals lick salt from the ground (Chapter 11) 721

salutes of artillery: cannons fired as a greeting (Chapter 33) 2717

salvage: bringing up goods from a wreck (Chapter 27) 2047

same-sex couple: two men or two women (Chapter 25) 1909

samp: corn pounded into coarse meal and boiled into a mush. The name comes from the Narragansett word nasàump (Chapter 15) 991

sampler: a strip of cloth that kept a record of stitches and patterns to copy later (Chapter 32) 2608

sanction: approval (Chapter 29) 2256

sanitized depictions: pictures with the cruelty cleaned out (Chapter 12) 832

saplings: thin young trees (Chapter 7) 460

sappers and miners: soldiers who dug the trenches that let an army surround a fort (Chapter 22) 1522

sarcophagus: a stone coffin (Chapter 12) 825

sassafras bark: a bark people boiled to make tea (Chapter 37) 3091

satisfied: said of a judgment whose terms have been met (Chapter 20) 1390

savings and loans, thrifts: businesses that took in people's savings and made long-term home loans at fixed interest rates (Chapter 17) 1153

sawmill: a mill that cuts logs into boards (Chapter 3) 192

sawyer: a man who saws logs into boards (Chapter 12) 799

scalp: the skin and hair cut away from the top of a person's head (Chapter 2) 120

scalp hunters: men who hunted Native people for their scalps (Chapter 22) 1515

scalpel: a surgeon's knife (Chapter 22) 1543

scandal: wrongdoing that becomes publicly known (Chapter 20) 1372

scanty: too little (Chapter 35) 2934

scapegoating: putting the blame on people who were not at fault (Chapter 25) 1770

scarlet fever: a disease that causes a high fever and a rash (Chapter 30) 2375

Schedule I: the federal list for drugs with a high risk of abuse and no accepted use as medicine (Chapter 28) 2179

Schedule III: a federal list of drugs that doctors may order as medicine (Chapter 28) 2190

school book ban: in PEN America's use, any removal of a book, or any limit on reading it, because of what is in it (Chapter 30) 2474

schooner: a sailing ship with two masts or more (Chapter 27) 2061

science journal: a magazine where scientists publish their research (Chapter 5) 387

scientific racism: the use of the look of science to claim that some races are better than others (Chapter 9) 596

scintilla of proof: almost no evidence (Chapter 27) 2078

Scots-Irish: families of Scottish background who had lived in the north of Ireland (Chapter 4) 267

scour: for a plow, to clean itself as it moves (Chapter 8) 535

scourged: whipped (Chapter 24) 1712

scribe: a person who writes documents for someone else (Chapter 30) 2359

scrip: paper that stood in for money (Chapter 15) 1015

scripture: a holy book. "The Scriptures" means the Bible (Chapter 20) 1309

scrub: brush and small trees (Chapter 3) 189

scrutiny: close study (Chapter 16) 1084

scurvy: a disease caused by a severe lack of vitamin C, which comes from fresh food, that makes people weak, damages their gums and makes them bleed under the skin (Chapter 1) 21

scythe: a long curved blade for cutting grass (Chapter 3) 185

Sea of Tranquility: a flat plain on the Moon (Chapter 1) 82

Seabed 2030: an international project in which people from many countries are working to map the whole floor of the ocean (Chapter 1) 92

seafloor spreading: the idea that new ocean floor forms along the undersea mountain ranges and spreads outward (Chapter 9) 633

sealed bid: a sale where buyers send in secret offers and the highest offer wins (Chapter 32) 2660

seamstress: a woman who sewed for pay (Chapter 14) 929

search index: the huge list of web pages a search engine keeps (Chapter 19) 1291

search warrant: a paper from a judge that allows a search (Chapter 21) 1468

seasonal round: a yearly circuit from one camp to the next (Chapter 5) 333

seasoning: the sickness that killed between 50 and 90 percent of early newcomers to Virginia (Chapter 8) 517

Second Bank of the United States: a national bank (Chapter 16) 1067

Second Continental Congress: the meeting of delegates each colony chose to speak for it (Chapter 23) 1605

Second Middle Passage: the forced move of enslaved people inside the country, from the Upper South to the Deep South (Chapter 5) 352

second-class mailing privilege: the permit that let a paper go by mail as a newspaper (Chapter 31) 2559

second-degree murder: killing someone on purpose but without planning it ahead (Chapter 21) 1468

second-degree murder (Georgia): causing a death while, through criminal carelessness, causing a child under 18 "cruel or excessive physical or mental pain" (Chapter 30) 2474

secondhand smoke: the smoke other people breathe in (Chapter 28) 2190

Secretary of State: the official who runs the country's dealings with other countries (Chapter 1) 64

Secretary of the Interior: one of the heads of government departments who advise the president (Chapter 2) 134

Secretary of the Treasury: the head of the government department in charge of money (Chapter 17) 1123

sect: a religious group (Chapter 22) 1502

section: one square mile, which is 640 acres (Chapter 3) 192

Section 230: the 1996 law under which a website is not treated as the publisher of what its users post (Chapter 31) 2585

secularization: Mexican officials turning the missions into ordinary parish churches (Chapter 29) 2256

security: something a lender can take if a loan is not paid back. Security is also a promise backed by money or property that a person will come back to court (Chapter 11) 734

sedative: a drug that makes a person calm and sleepy (Chapter 29) 2304

sedition: words or acts that stir people up against the government (Chapter 20) 1322

Sedition Act: the 1798 law that made it a crime to write or publish "any false, scandalous and malicious writing" against the government (Chapter 20) 1322

seditious libel: the crime of printing words that blame a public official or the government, whether or not the words were true (Chapter 4) 267

seed-stock: the animals or plants that farmers keep for breeding, a word Harry Laughlin used for people (Chapter 25) 1829

segregated: kept apart by race (Chapter 12) 815

segregated schools: schools that kept children of different races apart (Chapter 20) 1356

segregation: keeping people apart by race (Chapter 1) 82

segregation academy: a private school started so that white children would not have to sit in class with Black children (Chapter 30) 2442

segregationist: a person who wants the races kept apart (Chapter 12) 832

seismologist: a scientist who studies earthquakes (Chapter 27) 2061

seized: taken by force (Chapter 1) 21

seizure: under the Constitution, a way of stopping or holding a person by force. A seizure is also a sudden burst of activity in the brain that can make a person shake or pass out (Chapter 25) 1909

Selective Availability: the policy under which US government officials made public GPS signals less exact on purpose until 2000 (Chapter 8) 561

selectmen: a town's elected leaders (Chapter 20) 1309

self-deportation: leaving the country on one's own, without being deported by officers (Chapter 4) 320

self-determination: a nation deciding its own affairs (Chapter 2) 158

self-induction: a coil of wire making its own push of electricity against a change in its current (Chapter 9) 596

self-service: customers picking goods off the shelves themselves instead of asking a clerk to fetch them (Chapter 18) 1204

selling short: a bet that a stock's price will fall, made by selling borrowed shares that must later be bought back and returned, for less if the price has fallen (Chapter 18) 1204

semi-dwarf wheat: wheat with short stems (Chapter 15) 1028

seminary: a school that trains people for church work. A seminary was also a school for older girls or young women (Chapter 4) 275

Senegambia: the land along the Senegal and Gambia rivers (Chapter 15) 995

sensible: aware (Chapter 20) 1317

sent by subscription: paid for by a group of people (Chapter 32) 2643

separate but equal: the rule of Plessy v. Ferguson (1896) that let states keep Black and white people apart by law (Chapter 24) 1712

Separatists: English Pilgrims who wanted to worship apart from the Church of England (Chapter 4) 260

Sephardic Jews: Jews whose families had come from Spain and Portugal (Chapter 4) 267

sequins: tiny shiny disks sewn onto cloth (Chapter 35) 2959

serape: a blanket that can be worn as a cloak (Chapter 35) 2912

sergeant major: a regiment's top sergeant (Chapter 22) 1543

sermon: a talk given in church (Chapter 12) 795

service line: the pipe that carries water from the main line under the street into a house (Chapter 13) 899

servile labor: ordinary work (Chapter 37) 3091

servitude: slavery (Chapter 20) 1341

setback: a place where the upper floors of a building step back from the street (Chapter 6) 431

settle: to agree to take money and end a lawsuit (Chapter 18) 1222

settlement: an agreement that ends a lawsuit (Chapter 4) 320

Seventeenth Amendment: the 1913 change to the Constitution under which voters, not state lawmakers, choose senators (Chapter 20) 1356

Seventh-day Adventist: a member of a Christian church (Chapter 30) 2442

Seward's Folly, Seward's Icebox: what some opponents called the purchase of Alaska (Chapter 1) 64

sewing patterns: paper shapes that a person who sews pins to cloth and cuts around (Chapter 35) 2940

sexton: the caretaker of a church (Chapter 12) 795

sexual abuse of a child: an adult using a child's body for sex (Chapter 29) 2324

sexual assault: any sexual touching or attack against a person's will, including rape (Chapter 15) 1039

sexual exploitation of children: using children for sex acts or sexual pictures (Chapter 36) 3058

sexual harassment: unwanted sexual words or touching (Chapter 25) 1909

sexual orientation: which sex a person is attracted to (Chapter 25) 1909

sexually aggressive: wanting to force sex on someone (Chapter 34) 2861

shackles: metal rings locked around a person's wrists or ankles (Chapter 14) 914

shaft: a deep hole that leads down into a mine (Chapter 10) 681

shale: a soft rock made of hardened mud (Chapter 10) 681

shallop: a small open boat (Chapter 11) 726

shaman: a religious or spiritual leader (Chapter 29) 2242

shanty: a small, roughly built house (Chapter 7) 480

shantytown: a group of shacks (Chapter 16) 1084

shape notes: printed notes whose shape tells the singer which sound to sing (Chapter 33) 2736

share tenant: a tenant who owned mules or tools and kept two-thirds to three-quarters of the crop (Chapter 16) 1074

sharecropper: a person who farms someone else's land and pays the owner with a share of the crop (Chapter 1) 74

sharecropping: farming a landowner's fields in return for a share of the crop, buying tools, seed and food on credit at the landowner's prices (Chapter 15) 1009

shares: small pieces of ownership in a company (Chapter 4) 260

sharpshooter: a soldier who shoots at a distance with a rifle (Chapter 32) 2643

sheet music: the printed music people bought to play at home (Chapter 33) 2736

shelterbelts: rows of trees planted to slow the wind (Chapter 3) 215

shews: shows (Chapter 34) 2833

shilling: a coin of English money, 20 of which made one pound (Chapter 3) 192

shingled: cut short (Chapter 30) 2392

shipping container: a metal box 35 feet long, the size of a truck trailer (Chapter 11) 764

ship's log: the daily record a crew keeps of the ship's position, the wind and the sea (Chapter 9) 596

shirtwaist: a woman's blouse (Chapter 14) 941

shock: what happens when too little blood is left in the body, so the person becomes confused, the heart beats fast and the blood pressure drops (Chapter 14) 922

short rations: less food than people need (Chapter 27) 2036

shortgrass: the short native grass of the southern Plains (Chapter 3) 215

shuffleboard: a game of sliding pieces down a long board (Chapter 36) 2987

sickle: a curved blade for cutting wheat by hand (Chapter 15) 1004

sickle cell disease: a disease that bends red blood cells into a curved shape, so they slow the blood and carry less oxygen, causing attacks of severe pain (Chapter 26) 2022

side: one song on one face of a record (Chapter 33) 2759

side arms: the small weapons officers carried (Chapter 22) 1543

siege: surrounding a place and cutting it off until the people inside give up (Chapter 2) 120

Siete Partidas: an old Spanish code of laws that listed rights for enslaved people (Chapter 24) 1667

sieve: a strainer full of holes (Chapter 16) 1084

Signal Corps: the Army's communications branch (Chapter 8) 553

silhouette: an outline portrait cut from paper (Chapter 32) 2619

silica: the hard mineral that sand is made of (Chapter 14) 941

silicon: a chemical element, used for chips (Chapter 8) 561

Silicon Valley: the area south of San Francisco where many computer companies started (Chapter 8) 561

silicosis: a disease caused by breathing silica dust. The dust forms scar tissue in the lungs, the scarred lungs cannot take in enough oxygen, it cannot be cured, and in bad cases it kills (Chapter 14) 941

silkscreen printing: pushing ink through a design on a fine screen with a rubber blade, so the same picture can be printed again and again (Chapter 32) 2679

Silver Star: a medal for bravery in battle (Chapter 22) 1585

silversmith: a craftsman who makes things out of silver (Chapter 2) 126

singing in parts: different voices singing different notes at the same time (Chapter 33) 2717

singing school: a class where a teacher taught people to read and sing music (Chapter 33) 2723

sinister and evil methods: wrong and wicked means (Chapter 35) 2923

sister: a nun, a woman who belongs to a Catholic religious order (Chapter 29) 2324

sit-down strike: a strike in which workers stop working but stay inside the plant, so the company cannot bring in other workers (Chapter 14) 941

site plan: the layout of the grounds (Chapter 6) 418

sit-in: a protest in which people sit inside a building and refuse to leave (Chapter 8) 561

Six Nations: six Native nations joined in one league, the Iroquois (Chapter 2) 116

sixpence: six British pennies (Chapter 16) 1057

Sixteenth Amendment: the 1913 change to the Constitution that lets Congress tax incomes (Chapter 20) 1356

skepel: a Dutch measure for dry goods (Chapter 6) 399

skid: one of the long metal bars a helicopter lands on (Chapter 27) 2093

Skraelings: the Norse name for the people they met in Vinland (Chapter 1) 17

slacks: loose pants (Chapter 35) 2949

slashed sleeves: sleeves cut so the cloth underneath showed through (Chapter 35) 2916

slates: thin boards of dark stone that children wrote on and wiped clean (Chapter 30) 2375

slave broker: a dealer in enslaved people (Chapter 17) 1119

slave code: a set of laws listing what enslaved people could not do, how they could be punished and what their owners could do to them (Chapter 16) 1052

slave patrol: a group of white men who rode out to watch and control enslaved people (Chapter 21) 1422

slave pen: a building where traders locked up enslaved people until they sold them (Chapter 24) 1693

slave-trade clause: the part of the Constitution that barred Congress from ending the import of enslaved people before 1808 (Chapter 24) 1684

slaw: a salad of shredded cabbage (Chapter 15) 1000

sleeping car: a railroad car with beds (Chapter 11) 745

slickwater fracturing: a cheaper way of fracking that an engineer, Nick Steinsberger, used on the Barnett Shale in the late 1990s (Chapter 10) 711

slighted: treated without respect (Chapter 25) 1770

slips: short loose dresses (Chapter 35) 2934

sloop: a small sailing ship (Chapter 24) 1684

slop shops: the first shops that sold ready-made clothes, next to the docks (Chapter 35) 2934

SLS: Space Launch System, the rocket for NASA's Orion spacecraft (Chapter 1) 92

slump: a time when business falls off (Chapter 16) 1096

smallpox: a disease that spreads from person to person and can kill (Chapter 2) 120

smelter: a plant where workers melt ore to get the metal out (Chapter 13) 868

smelting: heating rock until the metal inside it melts out (Chapter 13) 844

smiths: people who shape metal into tools (Chapter 8) 516

smocks: long linen shirts worn under the other clothes (Chapter 35) 2916

smoke easement: a legal right to send a company's smoke over a farm (Chapter 13) 868

smolder: to burn slowly with very little air (Chapter 10) 671

smuggling: bringing goods in secretly (Chapter 12) 789

Snake: the expedition members' name for the Shoshone (Chapter 1) 44

SNAP: the Supplemental Nutrition Assistance Program, once called food stamps (Chapter 15) 1039

SNCC: the Student Nonviolent Coordinating Committee (Chapter 33) 2779

sniper: a hidden marksman (Chapter 2) 158

social hierarchy: the ranking of people from the most powerful to the least (Chapter 36) 2999

Social Security: the government's program of payments to people who are old or cannot work (Chapter 14) 941

socialist: a person who wants the public, rather than private owners, to own big businesses (Chapter 14) 929

Socialist Realism: the Soviet government's official style of art, paintings that looked like real life, only better, and showed simple stories in support of the government's ideas (Chapter 32) 2679

sociologist: a scholar who studies how groups of people live together (Chapter 29) 2304

sod: the top layer of soil, held together by the tangled roots of grass (Chapter 3) 215

sod corn: corn set straight into newly plowed prairie sod (Chapter 15) 1009

sojourner: a traveler who stays in a place for a while (Chapter 25) 1778

soldering iron: a tool that heats metal to join wires (Chapter 37) 3125

Soldier's Medal: the Army's highest award for bravery outside combat with an enemy (Chapter 22) 1575

solemn day of humiliation: a day set aside for "humble and fervent prayer" (Chapter 37) 3083

solemnized: marked with ceremony (Chapter 37) 3094

soliciting: asking for (Chapter 37) 3094

Solicitor General: the officer who argues the government's own cases in the Supreme Court (Chapter 20) 1372

solidarity: standing together (Chapter 37) 3125

solitary confinement: being locked up alone (Chapter 2) 134

solstices: the longest day and the shortest day of the year (Chapter 6) 395

Sons of Liberty: colonists who organized against British taxes and rule (Chapter 19) 1245

Sooners: people who sneaked in before the start of the 1889 Land Run to grab land (Chapter 5) 363

soot: the black powder that smoke leaves behind (Chapter 3) 192

sorcery: using magic to do harm (Chapter 2) 108

soul liberty: Roger Williams's name for the freedom to follow one's own conscience (Chapter 29) 2213

sounding: a measurement of how deep the water is (Chapter 9) 633

sources: the people who give a reporter information (Chapter 31) 2572

sousaphone: a large brass horn made for marching bands (Chapter 33) 2746

South Pass: a crossing of the Rocky Mountains in present-day Wyoming (Chapter 1) 44

Southern Christian Leadership Conference: the group Martin Luther King Jr. had led (Chapter 37) 3125

Southern Poverty Law Center: a group that campaigns for civil rights (Chapter 12) 832

sovereignty: the right of a nation to govern itself (Chapter 2) 120

Soviet Union: a large country ruled from Moscow, in Russia, from 1922 to 1991 (Chapter 1) 82

sower of sedition: a person who stirs people up against the government (Chapter 29) 2213

Soyal: the Hopi ceremony at the winter solstice that begins their year of ceremonies (Chapter 37) 3075

space shuttle: a spacecraft that rode rockets up into space and landed on a runway when it came back (Chapter 1) 82

Spanish Armada: a fleet of Spanish warships (Chapter 4) 258

spawn: to lay eggs, for fish (Chapter 3) 196

speakeasy: a hidden bar that sold liquor against the law (Chapter 28) 2165

Speaker: the officer who runs an assembly's meetings (Chapter 20) 1309

Speaker of the House: the member chosen to run the House's meetings (Chapter 28) 2179

speaking in tongues: speaking sounds that the speaker has not learned and does not understand (Chapter 29) 2287

special counsel: the title of the NAACP's chief lawyer. A Special Counsel is also an outside lawyer appointed to investigate (Chapter 25) 1829

special deputies: men given a police officer's powers for a time (Chapter 21) 1456

special education: classes or programs set up for disabled children (Chapter 25) 1871

special election: a vote held to fill an empty seat (Chapter 20) 1341

special prosecutor: a lawyer brought in from outside to handle a case (Chapter 20) 1372

special session: an extra meeting of lawmakers (Chapter 27) 2104

specialist: a person who spends most of their time on one kind of work, such as making tools, instead of growing food (Chapter 16) 1049

specie: coins of gold and silver (Chapter 17) 1128

spectral evidence: an accuser's word that the spirit of the accused person had come to her in a dream or a vision and hurt her (Chapter 21) 1412

spectroscope: an instrument that splits light into its colors (Chapter 13) 863

speculate: to buy something in the hope of selling it later at a higher price (Chapter 16) 1067

speculation: buying something only to sell it later at a higher price (Chapter 15) 1004

speculator: a person who buys property to sell it again at a profit (Chapter 6) 443

spermaceti: a waxy oil found in the head of the sperm whale, made into the brightest and costliest candles (Chapter 10) 669

spherical trigonometry: the mathematics of angles and distances on a sphere, such as the sky (Chapter 9) 591

sphinx: a creature with a lion's body and a human head (Chapter 32) 2695

spiked: stopped a story (Chapter 31) 2585

spillway: the channel that lets extra water out of a dam (Chapter 27) 2068

spinal cord: the bundle of nerves inside the backbone that carries messages between the brain and the body (Chapter 22) 1575

spinal tap: pushing a needle between the bones of the lower back to draw out fluid from around the spinal cord (Chapter 26) 1990

spindle: the rod that twists cotton fibers into thread (Chapter 19) 1250

spire: a tall pointed top (Chapter 6) 431

spirits: a strong liquor (Chapter 16) 1062

spiritual: a religious song made by enslaved Black people (Chapter 24) 1693

spittoon: a bowl for spitting into (Chapter 28) 2147

spleen: an organ that cleans the blood (Chapter 9) 614

splint: something that holds a broken bone still (Chapter 35) 2959

splints: thin strips cut for weaving (Chapter 32) 2660

spoils: the goods an army takes from the side it beats (Chapter 20) 1333

spoils system: the practice of presidents handing out government jobs to their supporters (Chapter 20) 1333

sponsor: the lawmaker who brings a bill forward (Chapter 37) 3125

sprawl: the spreading of cities outward in subdivisions of houses (Chapter 6) 443

spreadsheet program: a program for working with tables of numbers (Chapter 8) 561

Sputnik: the first satellite to circle the Earth (Chapter 1) 82

squadron: a group of warships (Chapter 1) 44

squatters: people living on land without the owner's permission (Chapter 11) 732

squatting: settling on land without any right (Chapter 32) 2627

stabilize: make firm (Chapter 12) 832

stablecoins: digital coins whose makers promise to trade each one for a fixed amount, such as one dollar (Chapter 17) 1160

staff: a mix of plaster, cement and plant fiber that workers poured into molds, used for the fronts of the 1893 fair buildings (Chapter 6) 418

stage: the raised platform where a Hidatsa family dried its corn (Chapter 7) 474

stage 4 cancer: cancer that has spread through the body (Chapter 33) 2795

stage line: a wagon or coach carrying passengers along a fixed route between towns (Chapter 11) 728

stagflation: a mix of inflation and a stagnant economy, one that is not growing (Chapter 16) 1096

stakeholder: the man who held the bets (Chapter 36) 2987

stalemate: a fight that neither side can win (Chapter 31) 2572

stallion: a male horse kept for breeding (Chapter 36) 2995

Stamp Act: a British tax on printed paper (Chapter 31) 2520

stand presented: to stand charged before the court (Chapter 35) 2916

standard wage: the same pay for the same work across a trade (Chapter 14) 920

standardized test: the same test, given and scored the same way, for every child who takes it (Chapter 30) 2474

standards: lists of what students should learn (Chapter 20) 1390

starboard: a ship's right side (Chapter 27) 2061

Starvation Winter: the winter of 1883 and 1884, when about 600 Blackfeet died of starvation. Hide hunters had killed nearly all the bison, and members of Congress had cut the money for the food promised to the tribe in a treaty (Chapter 3) 215

Starving Time: the winter of 1609 to 1610 at Jamestown, when about 500 colonists fell to about 61 (Chapter 15) 991

state assembly: one of the two groups of lawmakers in New York State (Chapter 20) 1356

state constitution: the state's basic set of laws (Chapter 10) 681

state geologist: a scientist the state paid to study its rocks (Chapter 13) 859

State Law Librarian: the librarian in charge of the state's law library (Chapter 6) 443

state of emergency: an official order saying that people are in danger and need help fast (Chapter 3) 229

State Security: the political police of Fidel Castro's government (Chapter 4) 310

state troopers: the state police (Chapter 7) 495

stateless: counted by no country as its citizen (Chapter 4) 320

States General: the lawmaking assembly of the Netherlands (Chapter 19) 1235

station license: the government permit a station needs to broadcast (Chapter 31) 2572

Stations of the Cross: a set of markers used in Catholic prayer (Chapter 29) 2231

statute: a law (Chapter 25) 1795

statute of limitations: a time limit after which a person can no longer sue (Chapter 25) 1829

stay: an order that stops a hanging until a court can hear the case (Chapter 21) 1456

steamboat pilot: the person who steers the boat (Chapter 11) 745

steelhead: a trout that swims to the sea and back (Chapter 10) 691

steelyard: a kind of weighing scale (Chapter 8) 517

steeple: the tall pointed tower on top of a church (Chapter 12) 795

steer: one of the male cattle raised for beef (Chapter 15) 1009

steerage: the cheapest space on a ship, below the decks (Chapter 4) 298

stenographer: a clerk who wrote down every word (Chapter 4) 298

stereolithography: a method of 3D printing in which the machine hardens each thin layer in turn (Chapter 8) 570

stereotype: an unfair, fixed idea about a whole group of people (Chapter 32) 2695

sterility: being unable to have children (Chapter 25) 1795

sterilization: an operation after which a person can never have a baby. For a woman a doctor cut or tied the tubes that carry eggs to the womb, cut out a piece of each, or removed the womb. For a man a doctor cut the tubes that carry sperm. It could not be undone. To sterilize a person is to do this operation (Chapter 2) 158

steward: a ship's servant. A steward was also the man in charge of supplies at a hospital or a college, or a man who helped look after a church's business (Chapter 20) 1341

stewardess: a woman who served passengers on an airplane, now called a flight attendant (Chapter 11) 764

stewards: race officials (Chapter 36) 3019

stick dance: a dance of enslaved people (Chapter 34) 2823

still: the equipment used to distill liquor (Chapter 28) 2138

stillbirth: a baby born dead (Chapter 9) 614

stillborn: born dead (Chapter 20) 1333

stimulus bill: a plan to spend federal money to bring back jobs (Chapter 16) 1102

stock: shares in owning a company (Chapter 8) 541

stock exchange: a place where people buy and sell shares of companies (Chapter 16) 1074

stock yards: pens where cattle, hogs and sheep were held until they were sold (Chapter 15) 1009

stockade: a pen fenced with upright logs. A stockade is also a military jail (Chapter 2) 126

stockholders: the people who own a company's shares (Chapter 19) 1250

stocks: a wooden board with holes that locked a person's ankles while the person sat in a public place (Chapter 21) 1412

stomp grounds: the grounds each Creek and Yuchi town keeps for the ceremony (Chapter 37) 3075

Stonewall Inn: a gay bar in New York City that police raided in 1969 (Chapter 25) 1871

stonewalling: refusing to answer or act (Chapter 29) 2324

stoneware: a hard kind of pottery (Chapter 36) 2987

stool: bowel movements (Chapter 4) 298

stool-ball: a game played with a ball, a bat and a stool (Chapter 37) 3083

stoolie: a person who informs on others (Chapter 34) 2877

stop and frisk: police stopping a person they suspect is armed and patting the outside of the clothes for weapons (Chapter 21) 1468

storm and sewer lines: pipes that carry away rainwater and waste (Chapter 6) 431

storm surge: the wall of seawater that a hurricane's winds push onto land (Chapter 27) 2093

storm troopers: the street fighters of Hitler's Nazi party (Chapter 36) 3045

straitjacket: a coat that pins a person's arms to the body (Chapter 13) 868

strap: a long, flat strip of leather used to hit children (Chapter 30) 2414

Strategic Petroleum Reserve: a store of crude oil kept for emergencies (Chapter 10) 702

streetcar: a passenger car that ran on rails laid in the streets (Chapter 6) 418

streetcar suburb: a neighborhood of homes at the edge of a city, along a streetcar line, where people lived and from which they traveled in to work (Chapter 6) 418

strike: a time when workers stop working until their employer agrees to better pay or conditions (Chapter 4) 283

strike fund: a fund of money the strikers set up to support one another during a strike (Chapter 14) 920

stripe: one stroke of the whip (Chapter 24) 1670

structural iron: the metal frames that hold up bridges and tall buildings (Chapter 2) 149

strychnine: a poison park managers put inside dead animals for wolves to eat (Chapter 3) 215

Stuart Pretender: a man who claimed he should be king of Britain (Chapter 37) 3091

student deferment: putting off a full-time college student's draft call while he stayed in school (Chapter 22) 1575

stylus: a pen-like point on the end of an arm (Chapter 8) 535

sub-captain: one of the leaders under a company's captain (Chapter 5) 363

subcommittee: a small group of senators or House members assigned to one subject (Chapter 2) 158

subdewing the Pecoits: defeating the Pequot (Chapter 37) 3083

subdivision: a tract of land divided into lots for single-family houses (Chapter 6) 443

subduction: one plate sliding down under another (Chapter 27) 2032

subprime loan: a loan with a higher interest rate, meant for borrowers that lenders think are more likely to fall behind (Chapter 7) 502

subscribers: the people who paid to get every copy of a newspaper (Chapter 18) 1178

subsidy: money paid to help a business grow (Chapter 16) 1062

subsist: to stay alive (Chapter 3) 200

substitute: a man hired to go into the army in a drafted man's place (Chapter 22) 1543

suburb: a town at the edge of a city (Chapter 4) 320

suffocate: to die because one cannot get air (Chapter 14) 941

suffrage: the right to vote (Chapter 20) 1341

sumptuary law: a law that limits what people may wear according to their rank (Chapter 35) 2916

Sun Belt: the warm states of the South and the West (Chapter 5) 383

Sun Dance: a sacred ceremony of the Plains nations (Chapter 29) 2304

Sun-priest: the man in a Pueblo town who watches where the sun comes up over the horizon and sets the planting calendar and the dates of the ceremonies (Chapter 37) 3075

sunstroke: an illness in which the body gets so hot that it can no longer cool itself (Chapter 4) 310

super PAC: a committee that may raise and spend unlimited sums on elections (Chapter 19) 1291

Superfund: the 1980 law under which federal officials could make companies that polluted a site pay for cleaning it up (Chapter 3) 229

superintendent: the official in charge of a national park, a reservation or a district's schools (Chapter 1) 64

Superintendent of Finance: the official in charge of the Continental Congress's money (Chapter 19) 1245

supermarket: a store that meets the Smithsonian's five tests: separate departments, self-service, low prices, many stores under one name and buying goods in huge amounts (Chapter 14) 963

superstitious: resting on false beliefs about magic and luck (Chapter 3) 200

supervised release: a time after prison when a court officer checks on a person (Chapter 10) 711

Supplemental Security Income: a federal payment for people with very low incomes who are elderly, blind or disabled (Chapter 23) 1656

supplier: one of the companies that made clothes for a store chain (Chapter 35) 2970

supply chains: the network of factories, ships and trucks that moves goods (Chapter 16) 1102

supported in idleness: fed without working (Chapter 37) 3107

suppress: stop (Chapter 19) 1291

Supreme Court: the highest court in the country (Chapter 2) 126

Supreme Court justices: the nine judges of the country's highest court (Chapter 2) 158

Surgeon General: the country's top public health doctor (Chapter 13) 868

surplus: extra, the name officials gave to land left over after allotment (Chapter 2) 134

surreptitiously: secretly (Chapter 20) 1390

surveillance: a close watch (Chapter 19) 1280

survey: to measure land and mark its borders (Chapter 3) 192

survey teams: groups sent to measure and map the western lands in detail (Chapter 1) 64

surveyor: a worker who measures land (Chapter 3) 192

surveyor-general: the chief surveyor of a colony (Chapter 6) 399

suspend: to shut a bank's doors or stop paying out cash (Chapter 17) 1136

suspension: keeping a child out of class for a set number of days, in a separate room at school or at home (Chapter 30) 2474

suspension bridge: a bridge whose roadway hangs from cables strung between tall towers (Chapter 12) 807

swab pole: a long pole with a mop on the end for putting out sparks on roofs (Chapter 27) 2041

Swahili: a language spoken in East Africa (Chapter 37) 3125

sweatshop: a crowded workshop where people sewed clothing for long hours and low pay (Chapter 14) 929

sweetgrass: a plant used to make baskets (Chapter 9) 581

swill milk: thin, dirty milk from cows kept in stables beside distilleries and fed the leftover grain mush from making liquor (Chapter 6) 418

swindles: tricks to cheat people out of their money (Chapter 18) 1188

switch: a thin stick cut from a tree (Chapter 7) 474

switchboard: the board where an operator connected each telephone caller to the person being called (Chapter 14) 929

sworn statement: a statement made with a promise under the law to tell the truth (Chapter 10) 665

syllabary: a set of written characters in which each character stands for a whole syllable, not a single sound (Chapter 2) 126

synagogue: a building where Jews worship (Chapter 4) 283

synthesizer: a machine that makes musical sounds electronically (Chapter 33) 2729

syphilis: a disease passed during sex, or from a mother to her unborn baby, caused by bacteria (Chapter 21) 1456

systemic: built into the way a company worked (Chapter 14) 974

T

taffeta: a smooth silk cloth (Chapter 35) 2912

tailings: the ground-up rock left behind after the uranium is taken out (Chapter 13) 886

"taken indiscriminately": taken without anyone choosing which ones (Chapter 8) 570

taking faces: painting portraits (Chapter 32) 2619

talent agency: a company that finds work for actors (Chapter 34) 2889

Taliban: the armed group American forces fought in Afghanistan from 2001 (Chapter 22) 1585

tallow candles: candles made of animal fat (Chapter 10) 677

Tammany Hall: a political machine of the Democratic Party in New York (Chapter 20) 1341

tanner: a worker who turns animal skins into leather (Chapter 33) 2729

tannery: a place where animal skins are made into leather (Chapter 3) 200

tanning (hides): treating animal skins so they stay soft and do not rot (Chapter 8) 512

Taoism: a Chinese religion and way of life (Chapter 29) 2274

Taps: a bugle call (Chapter 30) 2392

tariff: a tax on goods brought in from other countries (Chapter 16) 1062

taro: a plant with a starchy root that Hawaiians cooked and ate (Chapter 37) 3075

TARP: the Troubled Asset Relief Program, the rescue plan for financial firms that members of Congress passed in October 2008 (Chapter 16) 1102

tarpaulin hat: a sailor's waterproof hat (Chapter 35) 2934

tarred and feathered: stripped of clothes, coated with hot pine tar that burned the skin, and covered with feathers (Chapter 16) 1062

Taser: a weapon that fires an electric shock (Chapter 21) 1468

task: the set amount of work a planter gave each enslaved rice worker every day, usually a quarter of an acre (Chapter 15) 995

task force: a group a state sets up to study a question (Chapter 13) 899

task system: the way overseers on the rice plantations gave each enslaved person a set task for the day. When it was done, the rest of the day was the person's own (Chapter 24) 1693

Tava-kaavi: the Ute name for Pikes Peak (Chapter 1) 44

taverns and inns: places where people drank, ate and stayed the night (Chapter 18) 1188

tax credit: money taken off a buyer's tax bill (Chapter 10) 711

tax evasion: not paying taxes that are owed (Chapter 21) 1456

tax-credit scholarship: a person or a company gives money to a scholarship fund, and the state takes the same amount off their tax bill (Chapter 30) 2474

TCP/IP: the internet's shared set of rules for moving data between computers (Chapter 8) 561

Te Deum: a Latin hymn of praise to God (Chapter 33) 2717

tea garden: an outdoor place for refreshments and shows (Chapter 34) 2841

teachers college: a college where future teachers studied for a four-year degree (Chapter 30) 2414

teaching certificate: the state's license to teach (Chapter 30) 2442

teamster: a man who drove a wagon (Chapter 10) 671

Teapot Dome: the case in which a member of the president's cabinet leased Navy oil land to two oil men who gave him money (Chapter 20) 1356

tear gas: a gas that stings the eyes, nose and throat, so people cough and their eyes stream with tears (Chapter 2) 170

teaz'd: pestered (Chapter 36) 2999

technetium: element 43, the first element made by people, named from a Greek word meaning "artificial" (Chapter 9) 614

technetium-99m: a radioactive chemical that gives off rays a camera outside the body picks up, to make pictures of organs (Chapter 26) 2005

technical consultants: people who advise filmmakers on how to get details right (Chapter 34) 2861

technicality: a small point about how a court case was handled (Chapter 29) 2287

teething: growing first teeth (Chapter 18) 1193

Tejano music: the Spanish-language music of Texas (Chapter 33) 2779

Tektite project: a project that let scientists live in a laboratory under the sea (Chapter 1) 82

telegram: a short message sent over electric wires (Chapter 11) 764

telegraph: a machine that sent messages over a wire as pulses of electricity (Chapter 3) 200

telegraph line: the wire that carried messages by electric signals (Chapter 8) 535

telephone operator: the person who connected each caller to the person being called (Chapter 8) 541

tellers: the members of Congress chosen to count the electoral votes (Chapter 20) 1390

temperance: drinking little or no alcohol (Chapter 28) 2147

tenant families: families who rent the land they farm (Chapter 4) 275

tenant farmer: a farmer who works land that someone else owns (Chapter 15) 1015

tenement: a cheap apartment building packed with families (Chapter 4) 283

tenets: the main rules of a faith (Chapter 29) 2324

teosinte: the wild grass Native farmers made corn out of. Its seed head holds only a few hard kernels (Chapter 15) 986

termination: members of Congress and federal officials no longer treating a tribe as a nation with its own government, and the tribe losing its federal services (Chapter 2) 158

terrace: a wide step, here on the side of a mound (Chapter 6) 395

territory: land that belongs to the United States but is not a state (Chapter 1) 64

Test Act: a 1704 Irish law under which only people who took communion in the Church of Ireland could hold a public job (Chapter 4) 267

test case: a trial arranged on purpose so that a court would rule on a law (Chapter 20) 1356

testicles: the organs that make sperm (Chapter 9) 633

testify: to speak as a witness in court or at a hearing (Chapter 4) 298

testimony: a statement to lawmakers (Chapter 8) 570

tetanus: an infection from a wound that makes the muscles tighten and lock (Chapter 12) 807

tetraethyl lead: a liquid made of lead joined with other chemicals, added to gasoline to stop engine knock (workers called it "loony gas") (Chapter 13) 868

Tewa: one of the Pueblo peoples (Chapter 2) 108

Texas fever: a cattle disease that a Kansas law of 1885 and 1886 was passed to keep out (Chapter 15) 1009

Texas Revolution: the war Texans fought to break away from Mexico (Chapter 12) 795

thanksgiving: to the colonists, a day of prayer in church, called after rain, a good harvest, the safe arrival of ships or a victory (Chapter 22) 1508

Thanksgiving Address: the words of thanks, to the people, the earth, the waters, the fish, the plants and more, one at a time, that Haudenosaunee people often use to start and end their meetings, also called the Words That Come Before All Else (Chapter 37) 3075

thatch: dried straw or reeds used as roofing (Chapter 6) 398

the bones: a rhythm instrument (Chapter 33) 2736

the Crown: the king or queen and the royal government (Chapter 5) 346

The Federalist: the 85 essays Hamilton, Madison and Jay wrote to persuade New Yorkers to vote for the Constitution (Chapter 20) 1322

the Union: the side of the United States government in the Civil War (Chapter 4) 283

theology: the study of religious belief (Chapter 29) 2256

theory of relativity: Einstein's account of space, time and light, and of how they are tied to gravity (Chapter 9) 604

theses: a printed sheet of the ideas graduates were ready to argue for in public at the ceremony (Chapter 18) 1182

thesis: a long paper written at the end of college (Chapter 8) 553

third degree: police beating, threatening or questioning a prisoner for hours or days to force a confession (Chapter 21) 1456

third-degree burns: burns that destroy the skin all the way through (Chapter 1) 82

Thirteenth Amendment: the part of the Constitution that bans slavery, except as a punishment for crime (Chapter 16) 1074

Thoroughbred: a breed of horse raised in England for racing (Chapter 36) 2995

Three Sisters: corn, beans and squash grown together in the same field (Chapter 3) 180

three-fifths: the rule in the Constitution that counted each enslaved person as three-fifths of a free person when working out a state's seats in the House (Chapter 20) 1322

throwing a person to the dogs: setting trained attack dogs on a captive to bite and tear the person, often to death (Chapter 22) 1502

thumb-lead style: Maybelle Carter's way of picking the tune on the low strings with the thumb while strumming the high strings with one finger (Chapter 33) 2759

thyroid: a gland in the neck (Chapter 22) 1575

tignon: the name people later gave the head cloth Black women in New Orleans were ordered to wear (Chapter 35) 2928

tilt at the ring: a game in which a galloping rider tried to push a lance through a ring and carry it off (Chapter 36) 2985

tilt with cane spears: a game in which two lines of riders threw light cane spears at each other and blocked them with shields (Chapter 36) 2985

timber: wood for building (Chapter 2) 120

time capsule: a box of papers sealed to be opened in the future (Chapter 9) 604

time zone: a region where all clocks show the same time (Chapter 11) 745

time-sharing computer: a computer that many people could use at once from different terminals (Chapter 8) 561

timing chain: an engine part a car cannot run without (Chapter 5) 376

tinware: pots, pans, cups and other things made of thin sheets of metal coated with tin (Chapter 18) 1188

tipi: a house covered with animal hides that a family could take down and pack up to move with the hunt (Chapter 7) 460

tiswin: the Apache corn beer (Chapter 28) 2120

tithables: the people counted for a tax on each person, all free men and all enslaved people 16 and older (Chapter 11) 728

Title I: the part of the 1965 school law under which federal officials send money to school districts according to how many of their children come from families with low incomes (Chapter 20) 1372

Title IX: the part of the Education Amendments of 1972 under which a school or college that gets federal money may not treat a student worse because the student is a girl or a boy (Chapter 20) 1372

Title VII: the part of the Civil Rights Act of 1964 that bans job discrimination because of sex (Chapter 20) 1372

titles: the parts of an act (Chapter 20) 1372

tobacco inspection house: a building where tobacco was checked before it was shipped (Chapter 6) 407

tobacco note: a receipt for inspected tobacco, passed from hand to hand as money (Chapter 17) 1119

toleration: letting people hold beliefs that others do not share (Chapter 29) 2213

toll road: a road that people paid a fee to use (Chapter 11) 732

Tom shows: stage versions of *Uncle Tom's Cabin* (Chapter 34) 2851

tomahawk chop: a chopping arm motion that makes fun of Native people (Chapter 34) 2877

tonic: a drink sellers claimed would make people healthy (Chapter 13) 868

top billing: having your name first on the posters (Chapter 34) 2841

top income-tax rate: the tax on each dollar of income above a high level (Chapter 16) 1096

topsoil: the top layer of soil, where crops grow (Chapter 3) 215

Torah: the Jewish scriptures (Chapter 29) 2324

torso: the middle of the body (Chapter 4) 320

torture: hurting a person's body on purpose to force them to talk (Chapter 1) 36

Tory: the old name for colonists who sided with Britain's king (Chapter 31) 2520

total abstinence: no alcohol at all, not even beer or wine (Chapter 28) 2147

total solar eclipse: the moon blocking all of the sun's light for a short time (Chapter 9) 604

tour: a stretch of service in a war zone (Chapter 22) 1575

tourniquet: a tight band tied around an arm or leg to slow bleeding (Chapter 22) 1543

town ball: a bat-and-ball game played under local rules (Chapter 36) 3008

town house: the building where a Native town's leaders met (Chapter 36) 2999

township: a square of land six miles on each side, cut into 36 sections (Chapter 3) 192

tracheostomy: a hole cut through the front of the neck into the windpipe, with a short tube held in it (Chapter 25) 1909

tracheotomy: an operation that cuts an opening in the front of the neck into the windpipe so that air can get in (Chapter 26) 2005

trachoma: an eye infection that can cause blindness (Chapter 4) 298

tract: a short printed paper about religion (Chapter 29) 2256

trade cards: small printed cards given out to customers (Chapter 18) 1193

trade union: a group of workers in one trade who join together to bargain, or make a deal, with their employers over pay and hours (Chapter 14) 920

trading post: a place where traders swapped goods for furs (Chapter 1) 44

trading stamps: small stamps that shoppers got with purchases, pasted into books and traded for goods (Chapter 18) 1214

Traditional Ecological Knowledge (TEK): the knowledge of ecology that Native nations built up by watching closely over many generations (Chapter 9) 581

traditional practitioners: people who kept the old ceremonies (Chapter 29) 2287

traditional tobacco: tobacco grown or gathered for ceremony or medicine and offered in prayer, set apart from commercial tobacco, the kind sold in stores (Chapter 28) 2120

trafficker: a trader (Chapter 35) 2934

tragedy: a serious play with a sad ending (Chapter 34) 2823

trail boss: the man in charge of the whole cattle drive (Chapter 15) 1009

Trail of Tears: the Cherokee removal of 1838 and 1839, when soldiers forced the Cherokee people west to Indian Territory (Chapter 2) 126

trainmen: the workers who ran the trains (Chapter 11) 745

trait: a feature of a living thing, such as color or size (Chapter 9) 604

trance: a state like a dream while awake (Chapter 28) 2120

transatlantic: across the Atlantic (Chapter 19) 1235

transfermium: the elements after fermium, element 100 (Chapter 13) 886

transformer: a machine that can easily raise or lower the voltage of alternating current (Chapter 10) 681

transgender: a person whose own sense of being male or female differs from the sex recorded at birth (Chapter 1) 82

transistor: a small part that controls the flow of electricity and can make a weak signal stronger (Chapter 8) 553

transit of Venus: the planet Venus passing across the face of the sun, as seen from Earth (Chapter 9) 591

transportation: sending convicts away across the ocean as a punishment (Chapter 4) 267

transported convict: a person found guilty of a crime in Britain and sent to the colonies to be sold into 7 to 14 years of forced work (Chapter 10) 669

transposition: genes moving from one place to another on the chromosomes (Chapter 9) 614

Transylvania: Richard Henderson's name for the colony in Kentucky he hired Boone to help start (Chapter 1) 39

traumatic brain injury: damage to the brain from a blow or a blast (Chapter 22) 1585

travois: a frame of two poles that a dog dragged along the ground (Chapter 7) 474

treason: the crime of making war against your own country (Chapter 16) 1062

treasurer: the official in charge of a colony's money (Chapter 2) 108

treasury: the office that handles a government's money (Chapter 4) 267

Treasury bills: short loans to the government (Chapter 17) 1153

treating: giving voters food and drink before and after an election (Chapter 20) 1309

treatise on domestic economy: a long book about running a home (Chapter 7) 474

treaty: a written agreement between nations (Chapter 1) 39

treaty council: a meeting where Native leaders and colonial officials bargained and made agreements (Chapter 2) 116

Treaty of Amity and Commerce: a treaty of friendship and trade (Chapter 23) 1605

Treaty of Paris (1763): a peace agreement that ended the war between Britain and France (Chapter 5) 346

Treaty of Paris (1783): the peace agreement in which British diplomats accepted that the United States was independent and handed it Britain's claim to land that Native nations lived on (Chapter 22) 1522

Treaty of Paris (1898): the peace agreement in which Spanish diplomats gave up Cuba, signed Puerto Rico and Guam over to the United States and sold it the Philippines (Chapter 23) 1623

tremor: shaking a person cannot stop (Chapter 13) 863

trespass: going onto private land without permission (Chapter 25) 1909

trestle: a raised track (Chapter 10) 691

tribunals: special courts in which planters sat as judges (Chapter 24) 1693

tribute: payments of money, goods or work that rulers force people to give (Chapter 1) 21

tricktrack: a board game like backgammon (Chapter 36) 2987

Trinity: the Christian belief that God is three persons in one, the Father, the Son and the Holy Ghost (Chapter 29) 2213

tripartite: three-sided (Chapter 23) 1623

triple jump: the event once called the hop, skip and jump (Chapter 36) 3019

troopers: soldiers on horseback (Chapter 18) 1174

Tropic of Cancer: a line on maps north of the equator (Chapter 19) 1235

trotting race: a race in which a horse pulled a driver in a light cart (Chapter 36) 3008

truants: children missing school without permission (Chapter 30) 2442

truce: an agreement to stop fighting (Chapter 2) 126

trumped up: made up (Chapter 37) 3125

trust: a way to run many companies as one business, with one small group in charge (Chapter 19) 1256

trust accounts: money federal officials kept for the Native children at Carlisle (Chapter 30) 2392

trust company: a kind of bank that took people's deposits and managed their money (Chapter 16) 1074

trust money: money that belonged to Native families and that the government held for them (Chapter 29) 2287

trust territory: islands one country governs for the United Nations, with a duty to look after the people (Chapter 23) 1647

trustee: a person chosen to look after property or a project, such as a school or a museum, for other people (Chapter 12) 807

try-works: a brick furnace built on a whaling ship's deck for boiling blubber into oil (Chapter 10) 677

Tsenacomoco: the land of the Powhatan chiefdom (Chapter 4) 260

tub mill: a small water mill whose wheel lies flat in the stream and turns the millstone directly (Chapter 10) 663

tubal ligation: an operation in which a surgeon cuts or ties the two tubes that carry eggs to the womb, so that a girl or woman can never become pregnant (Chapter 2) 158

tuberculosis: a lung disease that spreads through the air when a sick person coughs or talks (Chapter 1) 64

tuition: the cost of attending a school (Chapter 20) 1390

Tukudika: a band of Mountain Shoshone who had lived in the Yellowstone country for thousands of years (Chapter 1) 64

tularemia: a fever people catch from animals (Chapter 26) 1946

tumor: a lump of tissue that grows where it should not (Chapter 9) 614

tumpline: a band worn across the forehead to carry a load on the back (Chapter 35) 2908

tundra: the flat treeless land of the far north (Chapter 34) 2814

turbine: a wheel with many blades, spun by steam (Chapter 10) 702

turbulent: unruly (Chapter 4) 275

turner: a person who shapes pots on a wheel (Chapter 32) 2627

turn-out: the shoemakers' and mill workers' word for a strike (Chapter 14) 920

turnover: how often workers quit and must be replaced. At Ford in 1913 it meant hiring nearly four men a year for every job (Chapter 14) 941

turnpike: a road whose owners charge travelers a fee, called a toll, at gates along the way (Chapter 11) 732

turnstile: a gate with turning arms that lets one person through at a time (Chapter 18) 1204

turpentine: a liquid made from pine trees (Chapter 10) 677

turret lathe: a metal-cutting machine for making many copies of the same part (Chapter 14) 941

tutor: a teacher who lived with a family and taught its children (Chapter 8) 528

Tweed Ring: William M. Tweed and the men around him, who stole money from New York City (Chapter 20) 1341

Twenty-fifth Amendment: the 1967 change to the Constitution that states what happens when a president dies, resigns or is removed, and how to fill an empty vice presidency (Chapter 20) 1372

Twenty-first Amendment: the 1933 change to the Constitution that ended the ban on alcohol (Chapter 20) 1356

Twenty-fourth Amendment: the 1964 change to the Constitution that bans poll taxes in elections for president, vice president, senator and representative (Chapter 20) 1372

Twenty-second Amendment: the 1951 change to the Constitution under which no person may be elected president more than twice (Chapter 20) 1372

Twenty-sixth Amendment: the 1971 change to the Constitution that gave eighteen-year-olds the vote (Chapter 20) 1372

Twenty-third Amendment: the 1961 change to the Constitution that gave the District of Columbia electors for president (Chapter 20) 1372

twill: a weave that makes thin diagonal lines (Chapter 35) 2934

Twitch: a website where players show their games live to viewers (Chapter 36) 3058

type: the set of metal letters a printer puts together by hand to print a page (Chapter 8) 535

type 2 diabetes: an illness in which the body cannot keep the sugar in the blood at a healthy level (Chapter 15) 1039

typhoid: a disease that causes a high fever, caught from food or water dirtied by human waste (Chapter 5) 352

typhus: a fever spread by body lice (Chapter 22) 1522

tyranny: cruel and unfair rule (Chapter 20) 1317

tyrants: rulers who use their power cruelly (Chapter 25) 1770

U

ulcer: a sore in the stomach (Chapter 9) 614

Umoja: the Swahili word for unity (Chapter 37) 3125

unanimous: with every justice agreeing (Chapter 20) 1372

Unassigned Lands: the last part of Indian Territory federal officials had not given to any Native nation, opened in the 1889 Land Run (Chapter 5) 363

unauthorized charges: charges that players had not agreed to (Chapter 18) 1222

unceded: never given up by treaty or agreement (Chapter 25) 1778

unconditional discharge: a sentence of no jail, no fine and no probation (Chapter 21) 1484

unconstitutional: against the Constitution (Chapter 5) 376

uncouth: rude and rough (Chapter 25) 1829

uncultivated: untaught (Chapter 32) 2619

under guardianship: placed in someone else's care by law (Chapter 20) 1356

underage: younger than an adult (Chapter 4) 298

undercover: working in secret, hiding who you are (Chapter 15) 1015

Underground Railroad: a network of people, with guides and safe houses, who helped enslaved people escape to the free states and to Canada (Chapter 24) 1693

understanding clause: a rule that a man had to read any part of the state constitution, or understand it when read to him, or give "a reasonable interpretation" of it, before he could vote (Chapter 24) 1735

understory: the layer of brush and small plants under the tall trees (Chapter 3) 180

undertaker: a person who prepares the dead for burial (Chapter 5) 363

undertakers (old sense): people who take on a big job (Chapter 8) 517

uneducable: judged unable to be taught (Chapter 25) 1871

unemployment: the share of people who want work and cannot find it (Chapter 16) 1067

unemployment insurance: payments to workers who have lost their jobs (Chapter 16) 1084

UNESCO: the part of the United Nations that keeps the list of World Heritage Sites (Chapter 6) 395

unethical: wrong (Chapter 9) 596

unexecuted: not carried out (Chapter 20) 1341

Uniform Monday Holiday bill: the 1968 bill under which several holidays moved to Mondays (Chapter 37) 3125

uniform practice: the same rules everywhere (Chapter 20) 1333

uniformity of religion: everyone being made to follow the same religion (Chapter 29) 2213

unilateral: done by one person alone (Chapter 30) 2474

unincorporated territory: a place the United States owns and rules, where the people do not get the whole Constitution unless Congress chooses to give it to them (Chapter 23) 1634

union: a group of workers who join together to bargain with an employer over pay and working conditions (Chapter 10) 691

Union army: the army of the United States government in the Civil War (Chapter 4) 283

Unitarian: a member of a church that does not teach the Trinity (Chapter 29) 2304

United States Exploring Expedition: a Navy exploring voyage paid for by the US government (Chapter 1) 44

Universal Product Code (UPC): the barcode printed on packages in stores, read by a scanner at the checkout (Chapter 18) 1214

unlawful assembly: a meeting the law banned, which officials could break up (Chapter 30) 2375

unofficial spokesman: a man who speaks for a club without holding any title (Chapter 12) 815

unprecedented: never done before (Chapter 29) 2274

unremitting: without a break (Chapter 37) 3101

unsaved: not yet believing in Christ (Chapter 29) 2324

unseasonable night meetings: meetings at an improper hour of the night (Chapter 18) 1174

unsigned opinion: an opinion that no single justice signed as its author (Chapter 20) 1390

unsubstantiated: not backed by proof (Chapter 9) 633

unwarranted: without good reason (Chapter 19) 1280

upland cotton: cotton that grows away from the coast (Chapter 8) 528

Upper South: Virginia, Maryland, the Carolinas, and later Kentucky and Tennessee (Chapter 5) 352

uppowoc: the Algonquian name for tobacco (Chapter 26) 1949

uprising: an armed revolt (Chapter 2) 134

uranium: a heavy metal element whose atoms break apart slowly and give off radiation (Chapter 3) 240

uranium mill: a plant that takes uranium out of crushed rock (Chapter 13) 886

urban renewal: a federal program, run from 1949 to 1974, under which city officials forced families out of whole neighborhoods, had the buildings torn down and sold the cleared land to developers (Chapter 6) 443

Ursuline nuns: an order of Catholic women (Chapter 29) 2256

US attorney: a federal prosecutor (Chapter 21) 1468

US Commission of Fine Arts: a federal board that reviews designs in Washington (Chapter 12) 825

US marshals: federal police officers (Chapter 21) 1468

US Sanitary Commission: a group that checked on the health of soldiers and their camps (Chapter 22) 1543

USB ports: the sockets that computer memory sticks plug into (Chapter 20) 1390

USDA organic seal: a label for food that meets the national organic rules (Chapter 15) 1028

utilities: the companies that supplied things like gas and water (Chapter 14) 929

utter detestation: hatred (Chapter 35) 2916

V

vacate: to cancel a conviction (Chapter 20) 1356

vaccination: scratching fluid from a cowpox sore into the skin to protect a person against smallpox (Chapter 26) 1975

vaccine: a medicine that trains the body to fight off a germ before the germ can make a person sick (Chapter 9) 633

vacuum: a space with no air in it, like outer space (Chapter 8) 553

vacuum tube: a glass tube with the air taken out, like a light bulb (Chapter 8) 553

vagina: the passage a baby is born through (Chapter 9) 633

vagrancy: the crime of having no home or job (Chapter 5) 376

valet: a personal servant (Chapter 20) 1322

vanguard company: the group that goes ahead of the rest (Chapter 5) 352

vaping: using an e-cigarette, which heats a liquid, usually with nicotine, into a mist the user breathes in (Chapter 28) 2190

vapor: a gas rising off a liquid or a solid (Chapter 10) 681

variable star: a star whose brightness changes (Chapter 9) 604

variety stores: stores that sold many kinds of small, cheap goods (Chapter 18) 1214

varsity letters: awards for playing on a college's top teams (Chapter 36) 3028

vasectomy: an operation that cuts the tubes that carry sperm so a man can no longer father children (Chapter 9) 633

vassal: a person bound to obey a ruler (Chapter 2) 102

vat: a large tub (Chapter 16) 1074

Vatican: the headquarters of the Catholic Church in Rome (Chapter 29) 2324

vaudeville: a show made of many short, separate acts, one after another (Chapter 29) 2287

V-Bucks: game money that Fortnite players buy with real money to get costumes and dance moves for their characters (Chapter 18) 1222

vein: a long thin band of ore or metal running through the rock (Chapter 10) 671

veneers: thin sheets of fine wood laid over a surface (Chapter 35) 2928

Venerable: a title the pope gives to a person whom church leaders are studying as a possible saint (Chapter 4) 271

ventilated: with fresh air pumped in (Chapter 26) 2005

ventilating furnace: a small fire kept burning at the bottom of a mine shaft to make a draft that moved fresh air through the mine (Chapter 10) 681

ventilation: the fans and shafts that blow fresh air into a mine and push radon out (Chapter 13) 886

ventilator: a machine that pushes air through a tube into the lungs (Chapter 25) 1909

verdict: the jury's decision (Chapter 14) 922

vertical farms: farms where workers grow vegetables indoors on shelves stacked high (Chapter 15) 1039

vertical integration: one company owning each stage of making a product, from the raw material to the finished goods (Chapter 19) 1256

vesicovaginal fistula: a hole torn through the wall between the bladder and the vagina by a long, blocked childbirth, so that urine runs out of the woman's body all the time (Chapter 26) 1975

vestry: the governing board of a church, here St. Philip's in Charleston (Chapter 27) 2047

Veterans Day: the 1954 name for Armistice Day, chosen so that one day would honor the veterans of every war (Chapter 37) 3125

veto: a refusal by a president, a governor or a mayor to sign a bill into law. Members of Congress can pass a law over a president's veto when two-thirds of each house vote for it (Chapter 3) 200

viaduct: here, a two-level section of a highway raised above the ground (Chapter 27) 2093

vice versa: the other way around (Chapter 19) 1280

viceroy: the king's governor of a colony (Chapter 1) 21

Victorian dresses: the long dresses of the late 1800s (Chapter 30) 2392

victory gardens: home vegetable gardens grown to help feed the country during World War II (Chapter 15) 1015

Viet Cong: the communist fighters in South Vietnam who fought on North Vietnam's side (Chapter 22) 1575

vigilance committee: a group of private citizens who arrested, tried and punished people themselves, outside the courts (Chapter 21) 1442

vigilance committees: groups, run by free Black communities in the North, that hid people who had escaped slavery and helped them move on (Chapter 24) 1693

vigilante: a person who uses force to punish others without any legal right to (Chapter 5) 352

vilest miscreants: the worst villains (Chapter 1) 44

virgin soil epidemic: an epidemic among people never exposed to the disease, in which everyone in a town can be infected at the same time (Chapter 26) 1946

Virginia Company: the English company that ran the Virginia colony (Chapter 4) 260

Virginia General Assembly: Virginia's lawmakers (Chapter 12) 799

Virginia State Colony for Epileptics and Feebleminded: a place where Virginia's officials locked up people who had seizures or were judged slow to learn. Dr. John H. Bell, who ran it, sterilized Carrie Buck against her will in 1927 (Chapter 20) 1356

vise: a clamp that tightens with a screw (Chapter 37) 3125

Vitascope: a film projector (Chapter 34) 2851

void: having no force (Chapter 20) 1333

volt: a measure of how strongly electricity pushes through a wire or a body (Chapter 21) 1442

voltage: the force that pushes electric current through a wire (Chapter 8) 553

voluntary manslaughter: killing a person on purpose but without planning it ahead, a charge less serious than murder (Chapter 21) 1468

voter rolls: the lists of people allowed to vote (Chapter 25) 1829

Voting Rights Act: the 1965 law under which federal examiners could register voters and some states and counties had to get permission before changing any voting rule (Chapter 20) 1372

voucher: a payment from the state that a family can use toward a private school's fees instead of sending the child to the local public school (Chapter 30) 2474

vulcanized rubber: rubber treated with sulfur so that heat does not melt it (Chapter 8) 535

vulgar and debased: common and low (Chapter 36) 2995

vulva: the outer part of a woman's sex organs (Chapter 32) 2679

W

wage theft: an employer not paying workers all the money the law says they have earned (Chapter 14) 974

waggoner: a man who drove wagons (Chapter 13) 855

waistcoat: a vest or short jacket (Chapter 35) 2916

Wall Street: a street in New York City where people buy and sell shares in companies (Chapter 6) 399

wall-bearing building: a building whose outside walls hold up everything (Chapter 6) 418

wampum: beads cut from the white and purple parts of clam shells, strung on strings or woven into belts (Chapter 2) 99

wampum belt: shell beads woven into patterns in which each pattern stands for the terms of an agreement (Chapter 2) 99

Wampum Keeper: the nation that looks after the wampum belts (Chapter 30) 2345

wanton: done for no reason (Chapter 22) 1585

wapato: a root that people dug up and ate (Chapter 18) 1170

war bonds: loans people made to the government to pay for the war (Chapter 2) 149

war bride: a woman engaged or married to a man who served in the war (Chapter 4) 298

war crimes: acts that break the laws of war (Chapter 22) 1543

War Department: the part of the government that ran the army (Chapter 1) 44

War Powers Resolution: the 1973 law that limits how long a president may keep troops fighting without the approval of Congress (Chapter 20) 1372

War Production Board: the federal agency whose officials controlled materials during World War II (Chapter 35) 2949

warden: the official in charge of a prison or of prisoners (Chapter 16) 1074

wards: people a government is legally responsible for (Chapter 20) 1356

warrant: a paper from a court that allows an arrest or a search (Chapter 3) 192

wash and wear: clothes that could be washed and worn without ironing (Chapter 35) 2959

washboard: a ridged board set in a tub of warm water and soap, for scrubbing clothes (Chapter 7) 485

water cure: a torture in which soldiers poured water into a prisoner's mouth and nose until his stomach swelled, then pressed or stood on his stomach to force it out, and started again (Chapter 22) 1559

water engine: a pump that sprays water (Chapter 27) 2041

water frame: Richard Arkwright's spinning machine, named for the waterwheel that drove it (Chapter 8) 528

water main: a large pipe that carries water under the streets (Chapter 6) 431

water mold: a tiny living thing that grows on a plant and makes it rot (Chapter 4) 275

water table: the level underground where the soil is full of water (Chapter 3) 229

watercourts: Calusa pens with walls of shell where live fish were kept (Chapter 2) 99

watered stock: stock issued for more than the property behind it (Chapter 19) 1268

Watergate: the 1972 break-in at the Democratic National Committee's offices in the Watergate building, and the scandal that followed (Chapter 20) 1372

watershed: all the land that drains into one stream or river (Chapter 3) 180

wavelength: the spot on the radio dial where a station is heard (Chapter 31) 2559

WAVES: the women's branch of the Navy in World War II (Chapter 8) 561

wax cylinder: a tube of wax that held two or three minutes of sound, which a machine could play back (Chapter 33) 2746

wealth or net worth: what a household owns minus what it owes (Chapter 16) 1102

weapons of mass destruction: nuclear, chemical or biological weapons (Chapter 23) 1656

web browser: the program people use to look at websites (Chapter 18) 1222

weevils: small beetles (Chapter 22) 1522

weld: a place where two pieces of metal have been melted together (Chapter 6) 431

welfare: government help for people without enough money (Chapter 4) 298

welfare officers: the officials who handed out relief money (Chapter 4) 298

welts: raised and swollen stripes on the skin (Chapter 8) 528

Wendat: the people also called the Huron (Chapter 1) 31

weroance: the leader, or chief, of a tribe. Also spelled werowance and wiroans (Chapter 2) 108

West Indies: the islands of the Caribbean Sea (Chapter 2) 108

Western Hemisphere: the half of the world that includes North and South America (Chapter 4) 310

Westinghoused: a word Edison's supporters used to mean killed by electricity (Chapter 10) 681

whalebone: a stiff, bendable material from the mouths of some whales (Chapter 18) 1188

"What hath God wrought": words from the Bible, Numbers 23:23, meaning "What has God done" (Chapter 8) 535

whelk: a large sea snail (Chapter 16) 1049

wherefore: why (Chapter 4) 260

whip: the senator who counts votes for a party and keeps its members in line (Chapter 20) 1372

whipping: being struck on the bare back with a whip, often at a post in a public place. The whip cuts the skin and leaves welts and scars (Chapter 11) 734

Whiskey Rebellion: the 1794 uprising of thousands of farmers in western Pennsylvania against a federal tax on whiskey (Chapter 16) 1062

White Citizens' Council: a group founded in Mississippi in 1954, after the Supreme Court's *Brown v. Board of Education* ruling, to keep segregation in place (Chapter 25) 1871

white flag: the sign that both sides agree not to fight while they talk (Chapter 2) 126

white supremacist: a person who believes white people are better than others and should rule over them (Chapter 20) 1341

white supremacy: the belief that white people should rule over everyone else (Chapter 11) 764

white-collar jobs: jobs in offices, stores and schools (Chapter 14) 963

wholesale: selling goods in bulk to other sellers (Chapter 18) 1188

wholesome: healthy to live in (Chapter 6) 399

wicket: a bat-and-ball game like cricket (Chapter 36) 2999

wigwam: a house with a frame of saplings, covered with bark or with woven mats (Chapter 7) 460

Wild West show: a traveling show of riding, shooting and scenes of frontier life (Chapter 32) 2643

wildcat banks: banks set up under free-banking laws. The story went that some stood so far out in the woods that only wildcats lived nearby (Chapter 17) 1128

wilderness area: land kept free of roads and buildings (Chapter 3) 215

Wilderness Road: the trail Boone's work party cut through the Cumberland Gap into central Kentucky, first marked as Boone's Trace (Chapter 1) 39

wildlife refuge: land set aside to protect wild animals (Chapter 2) 170

will: the paper that says who gets a person's belongings after death (Chapter 7) 465

willful neglect of duty: knowingly failing to do one's job (Chapter 13) 899

willfully: knowingly and on purpose (Chapter 14) 974

winch: a machine that winds a cable (Chapter 12) 815

Wind River Reservation: land in Wyoming the US government had set aside for the Shoshone. A US Indian agent had the Tukudika moved there from their homeland by force (Chapter 1) 44

wind tunnel: a tunnel in which a fan blows a steady stream of air past a model (Chapter 8) 553

windlass: the winch that pulled up a ship's anchor chain (Chapter 24) 1670

Winnebago: the name long used in English for the Ho-Chunk people (Chapter 8) 535

winter solstice: the shortest day of the year (Chapter 37) 3075

wire fraud: cheating people by phone, email or the internet (Chapter 17) 1160

witchcraft: using the Devil's power to hurt others, something many colonists believed people could do. After the Salem trials of 1692 and 1693, Massachusetts lawmakers cancelled many of the convictions and, in 1957, apologized (Chapter 21) 1412

withdrawal: the sickness that comes when the body loses a drug it depends on (Chapter 28) 2154

within its jurisdiction: inside the area where its laws apply (Chapter 20) 1341

without apportionment: without dividing the money among the states by population (Chapter 20) 1356

woodcut: a picture printed from a carved block of wood (Chapter 30) 2351

woodlot: a piece of land kept for growing trees (Chapter 3) 196

workhouse: a building where poor people, or people who broke small laws, were held and made to work (Chapter 2) 120

working copper cold: shaping copper without melting it (Chapter 8) 512

working level months: a unit that measures how much radon a miner breathed over time (Chapter 13) 886

workmen's compensation: money employers pay to workers who are hurt or made sick by their jobs, or to their families when they are killed (Chapter 13) 868

Works Progress Administration (WPA): a federal program that hired people to build public projects such as roads, pipes and schools (Chapter 6) 431

World Heritage List: a list of places chosen because they matter to all people (Chapter 12) 783

World Trade Organization: a group of countries that agree on rules for trade between them (Chapter 23) 1656

World Wide Web: the system of linked pages that people open over the internet (Chapter 8) 561

writ: a written order from a court (Chapter 4) 260

writ of habeas corpus: a court order that the people holding someone bring that person before a judge and give a lawful reason for holding them (Chapter 4) 283

wrought iron: a tougher iron that bends instead of breaking (Chapter 13) 848

Y

Yankee: someone from New England (Chapter 10) 669

yardarm: a wooden beam that crosses a ship's mast (Chapter 2) 102

yarn: the thread a spinning machine twists from cotton fibers (Chapter 8) 528

yaws and bejel: diseases in the same family as syphilis that spread mainly by touching skin and mouths, not by sex (Chapter 26) 1946

yearly rate: how far output would fall in a full year at the speed it was falling (Chapter 16) 1102

year-round school: about the same number of school days spread across the whole year, with shorter breaks (Chapter 30) 2474

yellow fever: a deadly disease spread by mosquitoes (Chapter 4) 271

yellow journalism: reporting that put exciting stories ahead of facts (Chapter 31) 2545

Yiddish: a language spoken by many Jews in Eastern Europe (Chapter 7) 480

yoke (of oxen): a pair of oxen joined by a wooden frame (Chapter 30) 2375

yoke (on a child): a wooden frame worn over the neck and shoulders (Chapter 30) 2359

Yoruba: an African people (Chapter 12) 832

yucca: a desert plant (Chapter 32) 2598

Z

zeal: great eagerness (Chapter 19) 1291

zoning: dividing a city into districts and setting rules for what may be built in each one: how tall, how close to the street, and for what use (Chapter 6) 431

zoologist: a scientist who studies animals (Chapter 29) 2287

zoot suit: an outfit with high-waisted, very wide pants that narrowed at the ankle and a long coat with wide lapels (Chapter 35) 2949

Timeline

The key dates from every chapter and the country's big turning points, oldest first, each with the chapter where you can read the whole story. When a line gives only a year, it comes first in that year.

Before 1500

About 9,000 years ago: Farmers in southern Mexico begin breeding corn from a wild grass called teosinte (Chapter 15) 986

1021: Norse sailors cut trees at L'Anse aux Meadows, in Newfoundland (Chapter 1) 17

About 1100: Cahokia, the largest city north of Mexico, reaches its peak of roughly 10,000 to 20,000 people (Chapter 6) 395

October 12, 1492: Rodrigo de Triana, a lookout on Columbus's ship *Pinta*, sees Guanahani, an island in the Bahamas (Chapter 1) 17

June 7, 1494: The rulers of Spain and Portugal split the lands across the Atlantic at Tordesillas (Chapter 23) 1597

The 1500s

1540: Hopi guides lead Spanish soldiers to the rim of the Grand Canyon (Chapter 1) 21

September 8, 1565: At Seloy, a Timucua town, Pedro Menéndez de Avilés founds St. Augustine, the oldest European town on the US mainland (Chapter 4) 258

1573: In the Laws of the Indies, Spain's king orders a plaza and a grid for every new Spanish town (Chapter 6) 398

1585: John White paints the town of Secotan, on the coast of what is now North Carolina (Chapter 32) 2605

April 30, 1598: Juan de Oñate claims New Mexico for Spain in La Toma, "the taking" (Chapter 5) 336

April 30, 1598: On the day of La Toma, Spanish soldiers and priests act the first European play on record in what is now the United States, beside the Rio Grande (Chapter 34) 2818

The 1600s

May 13, 1607: The Virginia Company's settlers found Jamestown (Chapter 19) 1235

1614: John Rolfe sends his first tobacco to London (Chapter 16) 1052

July 30 to August 4, 1619: Twenty-two burgesses, two elected men from each of Virginia's settlements, meet at Jamestown as the first elected assembly in English North America (Chapter 20) 1309

August 1619: The English ship *White Lion* carries about twenty captive Africans to Point Comfort, Virginia (Chapter 24) 1670

November 1620: The *Mayflower* men dig up and take the seed corn that Nauset farmers had buried at Cornhill, on Cape Cod (Chapter 15) 991

Fall 1621: The harvest meal at Plymouth, called the first Thanksgiving only from 1841 (Chapter 37) 3083

1631: Massachusetts judges order settlers to keep a night watch (Chapter 21) 1412

1634: Massachusetts judges order Robert Coles to wear a red D for a year as punishment for drunkenness (Chapter 28) 2127

April 23, 1635: Boston Latin School opens (Chapter 30) 2351

May 26, 1637: English soldiers burn the Pequot town at Mystic and shoot the people who try to get out (Chapter 22) 1508

July 9, 1640: Virginia judges sentence John Punch, the only African among three runaways, to serve for life (Chapter 24) 1670

1642: The Narragansett leader Miantonomi describes what English settlers are doing to the land (Chapter 3) 185

1646: Joseph Jenks gets a patent for his water-driven mills (Chapter 8) 517

1651: Massachusetts lawmakers fine people worth less than 200 pounds for dressing like gentlemen (Chapter 35) 2916

December 29, 1653: John Pynchon's account book records that John Stewart, a Scottish prisoner of war sold into forced labor, owes him 30 pounds for his freedom (Chapter 18) 1174

1662: Virginia's lawmakers pass a law that a child born to an enslaved mother is enslaved for life (Chapter 24) 1670

October 2, 1672: Workers break ground for the Castillo de San Marcos, the stone fort at St. Augustine (Chapter 12) 789

January 1673: The first post rider leaves New York for Boston (Chapter 11) 726

September 10, 1674: Virginia judges punish a tailor for racing and call racing "a sport only for Gentlemen" (Chapter 36) 2987

August 10, 1680: Po'pay leads the Pueblo Revolt, and Pueblo fighters drive the Spanish out of New Mexico (Chapter 29) 2213

April 18, 1688: Four men at Germantown sign a protest against slavery (Chapter 25) 1760

September 25, 1690: Benjamin Harris prints the newspaper *Publick Occurrences* in Boston, and the governor and his council stop it four days later (Chapter 31) 2506

December 1690: Massachusetts lawmakers vote for paper money (Chapter 17) 1115

1700 to 1750

January 26, 1700: The Cascadia fault breaks along the Pacific coast, and its tsunami reaches Japan the next day (Chapter 27) 2047

November 30, 1711: New York's aldermen, the men who run the city, set up a market at Wall Street for selling enslaved people (Chapter 18) 1178

1721: The Boston doctor Zabdiel Boylston inoculates people, giving them a small case of smallpox on purpose so they will not catch it again. Cotton Mather first heard of the method from Onesimus, a man he enslaved (Chapter 26) 1961

September 9, 1739: About 20 enslaved men rise up at the Stono River, south of Charleston, South Carolina (Chapter 24) 1676

May 10, 1740: South Carolina's lawmakers pass the Negro Act and ban enslaved people from using drums and horns (Chapter 33) 2723

May 10, 1740: In the same law, South Carolina's lawmakers also make it a crime to teach an enslaved person to write (Chapter 30) 2359

May 10, 1740: In the same law, South Carolina's lawmakers also bar enslaved people from wearing anything finer than coarse cloth (Chapter 35) 2923

1750 to 1800

June 3, 1769: Four men at Norriton time the planet Venus crossing the sun (Chapter 9) 591

1773: Phillis Wheatley's *Poems* comes out in London while she is enslaved in Boston (Chapter 32) 2619

December 16, 1773: Colonists dump the East India Company's tea into Boston Harbor (Chapter 19) 1245

April 19, 1775: The shooting of the Revolution starts at Lexington, Massachusetts, where British soldiers kill eight militiamen, ordinary men called up to fight (Chapter 22) 1522

July 4, 1776: Delegates to the Continental Congress adopt the Declaration of Independence, which states that "all men are created equal" (Chapter 25) 1770

September 3, 1783: In the Treaty of Paris, British diplomats accept American independence (Chapter 23) 1605

May 20, 1785: Members of Congress pass the Land Ordinance and order the land northwest of the Ohio River measured into a grid (Chapter 3) 192

July 6, 1785: Members of Congress vote for a dollar counted in tens (Chapter 17) 1123

January 16, 1786: Virginia's lawmakers pass Jefferson's Statute for Establishing Religious Freedom (Chapter 29) 2242

June 2, 1786: Philadelphia's printers begin the earliest proven strike of workers in one trade in the United States (Chapter 14) 920

September 17, 1787: Thirty-nine delegates sign the Constitution in Philadelphia (Chapter 20) 1322

1790: Members of Congress let only free white people born in other countries become citizens (Chapter 4) 271

1790: Pennsylvania lawmakers turn the Walnut Street Jail in Philadelphia into the country's first state prison (Chapter 21) 1427

December 15, 1791: The First Amendment, on freedom of speech and of the press, takes effect (Chapter 31) 2520

1793: Eli Whitney builds the cotton gin (Chapter 8) 528

September 18, 1793: George Washington lays the cornerstone of the Capitol (Chapter 12) 799

1799: Conrad Reed, 12, finds a 17-pound rock in a North Carolina creek, and his family uses it as a doorstop until a jeweler identifies it as gold in 1802 (Chapter 13) 855

1800 to 1850

February 1803: In *Marbury v. Madison*, the justices of the Supreme Court first strike down part of a law passed by Congress (Chapter 20) 1333

April 30, 1803: American diplomats agree to pay France $15 million for the right to take Louisiana from Native nations (Chapter 1) 44

1806: Lewis and Clark reach St. Louis with about 140 maps (Chapter 1) 44

January 1, 1808: A federal law against bringing captives from abroad to be enslaved takes effect (Chapter 24) 1693

February 11, 1808: Jesse Fell burns hard coal in his tavern fireplace (Chapter 10) 677

1821: William Brown opens the African Theatre in New York, the first known Black theatre company in the United States (Chapter 34) 2841

October 26, 1825: The Erie Canal opens (Chapter 11) 734

February 13, 1826: Reformers in Boston found the American Temperance Society (Chapter 28) 2147

1828: Thomas Dartmouth Rice first goes on stage as "Jim Crow," in blackface (Chapter 34) 2841

May 28, 1830: President Andrew Jackson signs the Indian Removal Act (Chapter 2) 126

May 1838: US soldiers begin rounding up about 16,000 Cherokee for the Trail of Tears, and more than 4,000 die by the usual count, or as many as twice that by the scholar Russell Thornton's estimate (Chapter 2) 126

1839: In *Crania Americana*, Samuel Morton ranks the races by skull size (Chapter 9) 596

1841: Catharine Beecher's *A Treatise on Domestic Economy*, a book on running a home, comes out (Chapter 7) 474

June 22, 1842: Water from the Croton Aqueduct reaches New York City (Chapter 6) 413

October 4, 1842: Ojibwe bands give up the Keweenaw copper country in the Treaty of La Pointe (Chapter 13) 859

1843: Four white men in blackface perform in New York as the Virginia Minstrels (Chapter 33) 2736

May 24, 1844: Samuel Morse sends the first public telegraph message (Chapter 8) 535

September 23, 1845: The Knickerbocker club adopts twenty rules for baseball (Chapter 36) 3008

February 2, 1848: In the Treaty of Guadalupe Hidalgo, Mexican officials sign over the Southwest (Chapter 23) 1616

July 19 and 20, 1848: More than 300 people meet at Seneca Falls, New York, and vote to demand the vote for women (Chapter 25) 1778

1850 to 1900

1852: Massachusetts lawmakers pass the first law that makes children go to school (Chapter 30) 2392

1856: William Henry Perkin makes the first man-made dye, a purple, from coal tar (Chapter 35) 2940

1857: Dave, an enslaved potter, cuts "I made this Jar = for cash" into the Poem Jar (Chapter 32) 2643

March 6, 1857: The justices of the Supreme Court rule against Dred and Harriet Scott, saying no Black American can be a citizen (Chapter 24) 1712

August 27, 1859: Drake's crew strikes oil in Pennsylvania (Chapter 10) 681

1860: Census takers count 3,953,760 enslaved people (Chapter 16) 1074

April 12, 1861: Confederate gunners fire on Fort Sumter, in South Carolina, and the Civil War begins (Chapter 22) 1543

May 20, 1862: President Lincoln signs the Homestead Act (Chapter 5) 363

January 1, 1863: Lincoln's Emancipation Proclamation takes effect, and Union soldiers begin freeing enslaved people as they advance (Chapter 24) 1712

July 1 to 3, 1863: The Union and Confederate armies fight the battle of Gettysburg, in Pennsylvania, and about 7,000 soldiers are killed (Chapter 22) 1543

October 3, 1863: Lincoln proclaims a national Thanksgiving on the last Thursday in November (Chapter 37) 3107

March 3, 1865: Members of Congress end the whites-only rule for mail carriers (Chapter 31) 2545

April 14, 1865: The actor John Wilkes Booth shoots President Lincoln at Ford's Theatre, and Lincoln dies the next morning (Chapter 34) 2851

June 19, 1865: General Granger announces in Galveston that slavery has ended in Texas, the day later called Juneteenth (Chapter 37) 3107

December 6, 1865: The Thirteenth Amendment, which forbids slavery except as a punishment for crime, becomes part of the Constitution (Chapter 24) 1712

May 10, 1869: Chinese workers lay the last track of the first railroad across the country, at Promontory Summit (Chapter 11) 745

1870: The Fifteenth Amendment bars every state from denying a man the vote because of race (Chapter 20) 1341

October 6, 1871: The Fisk Jubilee Singers set out on their first tour (Chapter 33) 2746

March 1, 1872: Yellowstone becomes the first national park (Chapter 3) 200

August 18, 1872: Montgomery Ward mails his first catalog from Chicago (Chapter 18) 1193

June 25, 1876: Lakota, Cheyenne and Arapaho fighters defeat George Custer's soldiers at the Little Bighorn (Chapter 2) 134

February 28, 1877: Members of Congress take the Black Hills from the Sioux (Chapter 12) 807

April 24, 1877: The last federal troops leave the Louisiana statehouse, and Reconstruction, the rebuilding after the war, ends (Chapter 24) 1712

October 6, 1879: The first Native children arrive at the federal boarding school at Carlisle, Pennsylvania, far from their homes, a few weeks before it formally opens (Chapter 30) 2392

November 1, 1879: Richard Henry Pratt opens the Carlisle Indian Industrial School in Pennsylvania (Chapter 2) 134

May 6, 1882: President Arthur signs the Chinese Exclusion Act (Chapter 4) 283

September 4, 1882: Thomas Edison's workers start the country's first permanent central power station, on Pearl Street in New York City (Chapter 10) 681

May 24, 1883: The Brooklyn Bridge opens, after at least 21 workers died building it (Chapter 12) 807

1887: Under the Dawes Act, federal officials begin cutting Native nations' land into private pieces (Chapter 2) 134

May 31, 1889: The South Fork Dam breaks, and 2,208 people die in Johnstown (Chapter 27) 2068

July 2, 1890: The Sherman Antitrust Act becomes law (Chapter 19) 1256

December 29, 1890: Soldiers of the 7th Cavalry kill between 250 and 300 Lakota people at Wounded Knee (Chapter 2) 134

October 21, 1892: The first national Columbus Day (Chapter 37) 3107

February 1, 1893: A white mob burns Henry Smith to death in Paris, Texas (Chapter 21) 1442

May 18, 1896: In *Plessy v. Ferguson*, the justices of the Supreme Court rule against Homer Plessy and allow states to keep separating Black and white passengers by law (Chapter 11) 745

December 10, 1898: In the Treaty of Paris, Spanish diplomats sign over Puerto Rico and Guam and sell the Philippines (Chapter 23) 1623

1900 to 1950

1900: Army doctors near Havana prove that mosquitoes spread yellow fever (Chapter 26) 1990

September 8, 1900: A hurricane kills between 6,000 and 12,000 people in Galveston, Texas, the deadliest natural disaster in the country's history (Chapter 27) 2078

May 27, 1901: In *Downes v. Bidwell*, the justices rule that Puerto Rico "belongs to, but is not a part of, the United States" (Chapter 23) 1634

December 17, 1903: The Wright brothers make the first powered flights (Chapter 8) 553

April 1906: Pentecostal worship begins at the Azusa Street meetings in Los Angeles (Chapter 29) 2287

June 30, 1906: President Theodore Roosevelt signs the Pure Food and Drug Act and the Meat Inspection Act (Chapter 15) 1015

December 6, 1907: The Monongah mine explosions kill at least 362 miners (Chapter 10) 691

1909: Peary says his party has reached the North Pole (Chapter 1) 74

March 25, 1911: The Triangle fire kills 146 workers in a New York factory where a stairway door was locked (Chapter 14) 941

May 15, 1911: The justices of the Supreme Court rule that Standard Oil has broken the Sherman Antitrust Act and give its owners six months to split it up (Chapter 19) 1268

1912: Henrietta Leavitt publishes her rule for measuring how far away stars are (Chapter 9) 614

December 23, 1913: President Wilson signs the Federal Reserve Act (Chapter 17) 1143

September 1, 1914: Martha, the last passenger pigeon, dies at the Cincinnati Zoo (Chapter 3) 215

February 8, 1915: *The Birth of a Nation* opens in Los Angeles (Chapter 34) 2861

July 25, 1916: New York City officials adopt the first zoning rules for a whole city, setting what may be built where (Chapter 6) 431

September 6, 1916: Clarence Saunders opens the first Piggly Wiggly, where shoppers serve themselves (Chapter 18) 1204

January 17, 1920: National Prohibition, the federal ban on making and selling alcoholic drinks, begins (Chapter 28) 2165

August 10, 1920: Mamie Smith records "Crazy Blues," the first blues record to become a big hit (Chapter 33) 2759

August 26, 1920: The Nineteenth Amendment, on votes for women, becomes part of the Constitution (Chapter 25) 1829

November 2, 1920: KDKA, one of the first radio stations, broadcasts the presidential election results (Chapter 31) 2559

May 31 to June 1, 1921: White Tulsans burn Greenwood, the Black district of Tulsa, and kill as many as 300 people (Chapter 25) 1829

May 26, 1924: President Coolidge signs the quota law, which fixes how many immigrants may come from each country (Chapter 4) 298

June 8, 1928: The New Jersey dial painters, the Radium Girls, sign their settlement with the U.S. Radium Corporation (Chapter 13) 868

October 29, 1929: Stock prices fall hard on Black Tuesday (Chapter 16) 1084

1932: Public Health Service doctors begin the Tuskegee study, keeping 399 Black men with syphilis from treatment for 40 years (Chapter 26) 1990

January 1, 1934: Deposit insurance, the government's promise to pay savers back if their bank fails, begins (Chapter 17) 1143

August 14, 1935: President Roosevelt signs the Social Security Act (Chapter 16) 1084

March 1936: Dorothea Lange photographs Florence Owens Thompson at Nipomo, California, in the picture known as *Migrant Mother* (Chapter 32) 2660

June 25, 1938: President Roosevelt signs the law that sets the first federal minimum wage, 25 cents an hour (Chapter 14) 941

1939: DuPont sells its first nylon stockings (Chapter 35) 2949

December 14, 1940: Glenn Seaborg's team makes plutonium at Berkeley (Chapter 13) 868

December 7, 1941: Japanese Navy pilots attack Pearl Harbor, Hawaii, and kill 2,403 Americans (Chapter 22) 1559

February 19, 1942: President Roosevelt signs Executive Order 9066, and soldiers then force Japanese Americans on the West Coast into camps (Chapter 5) 376

August 6, 1945: An American bomber crew drops an atomic bomb on Hiroshima, and 90,000 to 166,000 people die within four months (Chapter 22) 1559

April 15, 1947: Jackie Robinson plays his first game for Brooklyn (Chapter 36) 3028

October 1, 1947: The first families move into Levittown, New York (Chapter 7) 485

1948: In *Shelley v. Kraemer*, the justices of the Supreme Court rule that courts can no longer enforce restrictive covenants, sections of house deeds that forbid selling to Black buyers (Chapter 6) 431

1950 to 2000

April 25, 1953: Watson and Crick publish the shape of DNA, which they worked out from Rosalind Franklin's X-ray pictures without her knowledge (Chapter 9) 633

May 17, 1954: In *Brown v. Board of Education*, all nine justices rule that separating children by race in public schools breaks the Constitution (Chapter 25) 1871

1955: Jonas Salk's polio vaccine is shown to work and approved for use (Chapter 26) 2005

December 1, 1955: Police in Montgomery, Alabama, arrest Rosa Parks for refusing to give up her bus seat (Chapter 25) 1871

June 29, 1956: President Eisenhower signs the Interstate highway law (Chapter 11) 764

March 18, 1963: The justices of the Supreme Court rule that a poor person charged with a serious crime must get a lawyer (Chapter 21) 1468

July 2, 1964: President Lyndon Johnson signs the Civil Rights Act, which makes segregation in businesses and public places illegal (Chapter 25) 1871

1965: Members of Congress end the national-origins quotas, which limited immigration country by country (Chapter 4) 310

August 6, 1965: President Lyndon Johnson signs the Voting Rights Act, which bans literacy tests for voting (Chapter 25) 1871

1968: Members of Congress pass the Fair Housing Act (Chapter 7) 495

1969: Armstrong and Aldrin walk on the Moon (Chapter 1) 82

April 22, 1970: About 20 million Americans take part in Earth Day events (Chapter 3) 229

June 17, 1971: President Nixon calls drug abuse "public enemy number one" (Chapter 28) 2179

June 30, 1971: The justices of the Supreme Court vote 6 to 3 that newspapers may keep printing the Pentagon Papers (Chapter 31) 2572

August 15, 1971: President Nixon ends the dollar's last tie to gold (Chapter 17) 1153

June 23, 1972: President Nixon signs Title IX, the law against shutting people out of school programs because of sex (Chapter 36) 3045

June 30, 1973: The law allowing the draft runs out (Chapter 22) 1575

August 11, 1973: Kool Herc plays records at the Bronx party that people name as the start of hip-hop (Chapter 33) 2779

February 12, 1974: Judge George Boldt upholds the Northwest fishing treaties (Chapter 2) 158

August 9, 1974: Richard Nixon resigns the presidency rather than face a House vote on three charges (Chapter 20) 1372

August 11, 1978: The American Indian Religious Freedom Act becomes law (Chapter 29) 2304

November 16, 1990: President George H. W. Bush signs NAGPRA, under which museums must return Native remains and sacred objects (Chapter 32) 2679

July 16, 1995: Amazon.com opens for orders (Chapter 18) 1214

1996: Purdue Pharma begins selling the painkiller OxyContin (Chapter 28) 2179

April 14, 1999: Judge Paul L. Friedman approves the *Pigford v. Glickman* settlement for Black farmers (Chapter 15) 1028

2000 to today

September 11, 2001: Nineteen al-Qaeda hijackers take over four passenger jets and kill 2,977 people. They fly three into the World Trade Center and the Pentagon, and the fourth crashes in Pennsylvania after passengers and crew fight back (Chapter 27) 2104

August 29, 2005: Hurricane Katrina comes ashore, and New Orleans floods when its floodwalls break (Chapter 27) 2104

September 15, 2008: The financial firm Lehman Brothers fails during the crash in home prices, owing about 619 billion dollars (Chapter 16) 1102

2009: The number of people in prison reaches its highest point, 1,617,970 (Chapter 21) 1484

January 21, 2010: The justices of the Supreme Court decide *Citizens United*, and from then on corporations and unions may spend without limit on elections if they do not work directly with a candidate (Chapter 19) 1291

2013: Rana Plaza, a building in Bangladesh that holds five clothing factories, collapses and kills more than 1,100 people (Chapter 35) 2970

April 2014: Emergency managers switch the water of Flint, Michigan, to the Flint River, and lead from old pipes gets into the city's tap water (Chapter 13) 899

June 17, 2015: Dylann Roof kills nine Black worshippers at Mother Emanuel church in Charleston (Chapter 29) 2324

2016: For the first time, more American electricity comes from natural gas than from coal (Chapter 10) 711

July 28, 2019: Kyle Giersdorf, 16, wins the first *Fortnite* World Cup (Chapter 36) 3058

2020: COVID-19 reaches the United States (Chapter 26) 2022

July 9, 2020: Five Supreme Court justices rule in *McGirt v. Oklahoma* that the Muscogee (Creek) reservation never ended (Chapter 2) 170

June 17, 2021: President Biden signs the law making Juneteenth a federal holiday (Chapter 37) 3135

April 1, 2022: Workers at an Amazon warehouse on Staten Island win the first union at an Amazon warehouse in the country (Chapter 14) 974

November 30, 2022: OpenAI releases ChatGPT (Chapter 8) 570

July 14, 2023: Film and television actors begin the longest actors' strike against the studios (Chapter 34) 2889

People

Everyone with a story box in this book is here, by last name or by their only name, with the chapter and era where you will find them.

A

Robert S. Abbott: founder of the *Chicago Defender*, which urged Black Southerners to move north (Chapter 31, 1900 to 1950) 2561

Abigail Adams: ran her family's farm, its money and her children's schooling through the Revolutionary War, and had her four children inoculated against smallpox in 1776 (Chapter 7, 1750 to 1800) 472

Abigail Adams: wrote to her husband to "Remember the Ladies" in the new country's laws (Chapter 25, 1750 to 1800) 1772

John Adams: the first American minister to Great Britain, presented to King George III in 1785 (Chapter 23, 1750 to 1800) 1610

John Quincy Adams: Secretary of State who got Florida from Spain and wrote the warning now called the Monroe Doctrine (Chapter 23, 1800 to 1850) 1619

Emilio Aguinaldo: Filipino leader who declared independence from Spain and led the war against the United States (Chapter 23, 1850 to 1900) 1632

Persis Foster Eames Albee: the first woman to sell door to door for the company that became Avon (Chapter 18, 1850 to 1900) 1202

Pedro Albizu Campos: Army officer in the First World War who became the leader of the movement for Puerto Rican independence (Chapter 23, 1900 to 1950) 1644

Ira Aldridge: a Black actor from New York who left in 1824 and acted Shakespeare in Britain and Europe for more than forty years (Chapter 34, 1800 to 1850) 2844

Buzz Aldrin: Apollo 11 astronaut who landed on the Moon with Neil Armstrong and followed him out 19 minutes later (Chapter 1, 1950 to 2000) 86

Paul Alexander: Dallas boy paralyzed by polio who lived in an iron lung for more than 70 years and became a lawyer (Chapter 26, 1950 to 2000) 2009

Muhammad Ali: the heavyweight champion who refused the draft and won his case in the Supreme Court (Chapter 22, 1950 to 2000) 1580

Muhammad Ali: the heavyweight champion whose title was taken when he refused to be drafted in 1967 (Chapter 36, 1950 to 2000) 3050

Virginia Ali: a co-founder of Ben's Chili Bowl on U Street in Washington, D.C., which she has helped run since 1958 (Chapter 6, 2000 to Today) 454

Richard Allen: Black Philadelphian who, with Absalom Jones, led volunteers to nurse the sick and bury the dead in the yellow fever of 1793 (Chapter 26, 1750 to 1800) 1974

Richard Allen: preacher born into slavery who walked out of St. George's and founded the African Methodist Episcopal Church (Chapter 29, 1750 to 1800) 2247

Kate Alterman: Triangle fire survivor who told the jury how Margaret Schwartz died at the locked door (Chapter 27, 1900 to 1950) 2084

Anarcha: enslaved young woman on whom J. Marion Sims operated about thirty times with nothing to stop the pain (Chapter 26, 1800 to 1850) 1981

John Anderson: enslaved man who became Hartford's Black governor in 1776, in a dispute that white officials investigated as a possible plot (Chapter 37, 1750 to 1800) 3100

Marian Anderson: Black contralto who sang at the Lincoln Memorial in 1939 and was the first Black American singer at the Metropolitan Opera (Chapter 33, 1900 to 1950) 2778

Marvin Roger Anderson: a Rondo resident whose family lost its townhomes to Interstate 94 and who now leads ReConnect Rondo (Chapter 6, 1950 to 2000) 446

Roy Chapman Andrews: naturalist who started as a janitor at the American Museum of Natural History and whose team found the first known nests of dinosaur eggs (Chapter 1, 1900 to 1950) 78

Giannis Antetokounmpo: basketball player born stateless in Greece to Nigerian parents, who came to play for the Milwaukee Bucks (Chapter 4, 2000 to Today) 328

Susan B. Anthony: voted in Rochester in 1872 and was the only one of fifteen women put on trial for it (Chapter 25, 1850 to 1900) 1803

Kathleen McNulty Mauchly Antonelli: one of the six women who programmed ENIAC (Chapter 8, 1900 to 1950) 559

Michael Arad: the architect whose design for the National September 11 Memorial was chosen from 5,201 entries (Chapter 12, 2000 to Today) 833

Reinaldo Arenas: Cuban novelist jailed for his writing and for being gay, who left in the Mariel boatlift (Chapter 4, 1950 to 2000) 315

Louis Armstrong: Black trumpet player and singer who told a reporter in 1957 that "the government can go to hell" over Little Rock (Chapter 33, 1900 to 1950) 2764

Neil Armstrong: Apollo 11 astronaut, the first person to step onto the Moon (Chapter 1, 1950 to 2000) 86

Anthony Aston: a British lawyer turned actor who wrote a play about South Carolina (Chapter 34, 1700 to 1750) 2829

John James Audubon: painter of *The Birds of America*, who put one 1813 flock of passenger pigeons at more than 1 billion birds (Chapter 3, 1800 to 1850) 198

B

Doukenie Bacos: Greek girl held three days at Ellis Island at 16 (Chapter 4, 1900 to 1950) 300

DeFord Bailey: Black harmonica player whose playing came just before the Grand Ole Opry got its name (Chapter 33, 1900 to 1950) 2767

Bill Bair: boy who played for Lycoming Dairy in the first Little League season, in 1939 (Chapter 36, 1900 to 1950) 3041

Jessie Little Doe Baird: Mashpee Wampanoag linguist who brought the Wôpanâak language back into use (Chapter 2, 2000 to Today) 177

The Baldizzi family: Sicilian immigrant family at 97 Orchard Street through the Depression, forced out in 1935, whose apartment museum staff rebuilt from the daughter's memories (Chapter 7, 1900 to 1950) 492

Charles Ball: enslaved man marched in a chained coffle from Maryland to South Carolina, who escaped and told his story (Chapter 5, 1800 to 1850) 354

Lucille Ball: star of *I Love Lucy* and the first woman to run a major television studio (Chapter 34, 1950 to 2000) 2883

Martha Ballard: Maine midwife whose diary of 27 years records 816 births and the daily work of her household (Chapter 7, 1750 to 1800) 473

Robert Ballard: Navy oceanographer who helped find life around hot springs on the sea floor and found the wreck of the *Titanic* (Chapter 1, 1950 to 2000) 89

Benjamin Banneker: a free Black tobacco farmer and self-taught astronomer whose observations fixed the starting corner of the federal district (Chapter 6, 1750 to 1800) 410

Benjamin Banneker: free Black farmer and self-taught astronomer who sent Thomas Jefferson his almanac as proof against Jefferson's claim that Black people were less able than white people (Chapter 9, 1750 to 1800) 594

P.T. Barnum: a showman whose first paid show was Joice Heth, an enslaved woman he advertised as 161 years old (Chapter 18, 1800 to 1850) 1190

Frédéric Auguste Bartholdi: the French sculptor who designed the Statue of Liberty (Chapter 12, 1850 to 1900) 813

Jean Jennings Bartik: one of the six women who programmed ENIAC (Chapter 8, 1900 to 1950) 559

Clara Barton: took supplies to Civil War battlefields and founded the American Red Cross (Chapter 26, 1850 to 1900) 1984

Clara Barton: founder of the American Red Cross, who led its relief after fires, floods and hurricanes (Chapter 27, 1850 to 1900) 2077

John Bartram: Quaker farmer and botanist who built the oldest surviving botanic garden in America (Chapter 9, 1700 to 1750) 590

Jim Beckwourth: Black fur trapper, born into slavery, who lived with the Crow and found the lowest pass through the Sierra Nevada in California (Chapter 1, 1850 to 1900) 65

Catharine Beecher: wrote *A Treatise on Domestic Economy* (1841) and argued that keeping house was a profession schools should teach (Chapter 7, 1800 to 1850) 476

Catharine Beecher: ran a school for girls and spent decades arguing that teaching was women's work (Chapter 30, 1800 to 1850) 2382

Alexander Graham Bell: Boston inventor who received the first telephone patent in 1876 (Chapter 8, 1850 to 1900) 545

Lucas Benitez: tomato picker who helped start the Coalition of Immokalee Workers and its Fair Food Program, and who investigated slavery in the orange groves around Lake Placid, Florida (Chapter 15, 2000 to Today) 1042

Adelaide Wisdom Benjamin: girl whose family grew a victory garden and shopped with ration stamps during World War II (Chapter 15, 1900 to 1950) 1025

Bruce Bent: set up the first money market fund, the Reserve Fund, with Henry Brown (Chapter 17, 1950 to 2000) 1157

Victor Berger: Milwaukee Socialist editor sentenced to 20 years under the Espionage Act and twice refused his House seat (Chapter 31, 1900 to 1950) 2565

Vitus Bering: Danish-born officer in the Russian navy who led the voyage that reached Alaska in 1741 (Chapter 1, 1700 to 1750) 36

Irving Berlin: Jewish boy from the Russian Empire who came through Ellis Island and wrote "God Bless America" (Chapter 4, 1850 to 1900) 296

Carl Bernstein: *Washington Post* reporter who, with Bob Woodward, tied the Watergate burglary to President Nixon's White House (Chapter 31, 1950 to 2000) 2580

Barbara Ann Berwick: San Francisco Uber driver whom a California hearing officer ruled to be Uber's employee in 2015 (Chapter 11, 2000 to Today) 774

Jeff Bezos: the founder of Amazon (Chapter 18, 2000 to Today) 1223

Nicholas Biddle: head of the Second Bank of the United States who cut off its loans in his fight with President Andrew Jackson (Chapter 19, 1800 to 1850) 1254

William Billings: Boston tanner who wrote about 340 choir songs, including "Chester," and died poor (Chapter 33, 1750 to 1800) 2730

Hiram Bingham: Yale history teacher, not a trained archaeologist, who made the Inca site of Machu Picchu famous (Chapter 1, 1900 to 1950) 77

Isaac Bissell: Suffield blacksmith who carried Joseph Palmer's Lexington letter toward Connecticut (Chapter 31, 1750 to 1800) 2524

Colleen Black: a teenager from Nashville who worked as a leak detector at Oak Ridge's K-25 plant and lived in Oak Ridge the rest of her life (Chapter 6, 1900 to 1950) 441

Nicholas Black Elk: Oglala Lakota holy man who also taught the Catholic faith for decades as a catechist (Chapter 29, 1900 to 1950) 2290

Elizabeth Blackwell: first woman in the United States to earn a medical degree (Chapter 26, 1800 to 1850) 1982

Theodore and Patricia Bladykas: the first family to move into Levittown, New York, on October 1, 1947 (Chapter 7, 1900 to 1950) 493

Nellie Bly: reporter who got herself locked in New York's asylum for women and reported what she saw (Chapter 31, 1850 to 1900) 2552

Daniel Boone: hunter who marked the road through the Cumberland Gap into Kentucky (Chapter 1, 1750 to 1800) 40

Edwin Booth: a Shakespeare actor who played Hamlet a record hundred nights in a row, brother of John Wilkes Booth (Chapter 34, 1850 to 1900) 2854

Gutzon Borglum: the sculptor who designed Mount Rushmore, after starting a Confederate carving at Stone Mountain that the Ku Klux Klan helped pay for (Chapter 12, 1900 to 1950) 819

Norman Borlaug: plant scientist whose wheat helped poor countries grow far more grain, in the Green Revolution (Chapter 15, 1950 to 2000) 1032

Elias Boudinot: editor of the *Cherokee Phoenix* who later signed the Treaty of New Echota. A group of Cherokee who supported John Ross stabbed him to death in 1839 (Chapter 31, 1800 to 1850) 2538

Zabdiel Boylston: the one Boston doctor who tried inoculation in 1721 (Chapter 26, 1700 to 1750) 1964

William Bradford: governor of Plymouth Colony, who wrote the colony's history, *Of Plimoth Plantation* (Chapter 37, The 1600s) 3086

Anne Bradstreet: author of the first book of poems by someone living in the English colonies (Chapter 32, The 1600s) 2611

Jim Bridger: trapper and guide who never learned to read or write and reached the Great Salt Lake in a buffalo-hide boat (Chapter 1, 1800 to 1850) 58

Ruby Bridges: at six she walked into an all-white school in New Orleans, and for months she was its only pupil (Chapter 30, 1950 to 2000) 2448

Sergey Brin: Jewish boy from Moscow who came at six and started Google (Chapter 4, 1950 to 2000) 318

Joseph Broussard (Beausoleil): Acadian who fought the British removal, then led about 200 Acadians to Louisiana (Chapter 5, 1750 to 1800) 349

Henry Brown: set up the first money market fund, the Reserve Fund, with Bruce Bent (Chapter 17, 1950 to 2000) 1157

Josie Mabel Brown: Navy nurse on the 1918 influenza wards at Great Lakes, Illinois (Chapter 26, 1900 to 1950) 1996

William Alexander Brown: the founder of the African Theatre, the first known Black theatre company in the United States, and author of *King Shotaway* (Chapter 34, 1800 to 1850) 2843

Blanche K. Bruce: man born enslaved who became a United States senator for Mississippi and the first Black senator to serve a full six-year term (Chapter 24, 1850 to 1900) 1730

Jette Bruns: German settler in Missouri who wrote letters home to her brother for more than fifty years (Chapter 4, 1800 to 1850) 277

William Jennings Bryan: former congressman from Nebraska who became the Democratic candidate for president in 1896 with his "cross of gold" speech for silver (Chapter 17, 1850 to 1900) 1142

Sharon Buchanan: the cashier who made the first sale ever scanned with a barcode (Chapter 18, 1950 to 2000) 1220

Carrie Buck: young Virginia woman picked for a rigged test case in which the Supreme Court's justices allowed Virginia doctors to sterilize her. Dr. John H. Bell then operated on her so that she could never have a child (Chapter 25, 1900 to 1950) 1853

Buffalo Bird Woman (Waheenee): Hidatsa woman who grew up in her mothers' earth lodge and told the story of her life to Gilbert Wilson (Chapter 7, 1800 to 1850) 477

James Bullocke: a York County, Virginia, tailor punished in 1674 for racing his mare against a gentleman's horse (Chapter 36, The 1600s) 2992

Tarana Burke: youth organizer who used the words "me too" with survivors years before the hashtag (Chapter 25, 2000 to Today) 1926

Daniel Burnham: the Chicago architect who directed the building of the 1893 world's fair and wrote the 1909 Plan of Chicago with Edward Bennett (Chapter 6, 1900 to 1950) 434

Liz Byrd: Cheyenne schoolteacher and lawmaker who brought the King holiday bill before her fellow Wyoming lawmakers nine times (Chapter 37, 1950 to 2000) 3133

Richard Byrd: Navy pilot who led the first flight to the South Pole, whose claim to have flown over the North Pole is disputed (Chapter 1, 1900 to 1950) 79

C

Álvar Núñez Cabeza de Vaca: Spanish member of the Narváez expedition, one of the four survivors who walked from Texas to Mexico City (Chapter 1, The 1500s) 25

Caesar: enslaved healer whose cure for poison led South Carolina's lawmakers to buy his freedom (Chapter 26, 1750 to 1800) 1967

Rachel Calof: young woman from Russia whose memoir records a 12-by-14-foot homestead shanty in North Dakota, shared in winter with her in-laws and the chickens (Chapter 7, 1850 to 1900) 484

Teresa and Francisco Camacho: a Spanish soldier and fisherman and his Native wife, who shared their St. Augustine household with her sister and her sister's son (Chapter 7, The 1500s) 464

James Cameron: Canadian film director, the first person to dive alone to the Challenger Deep (Chapter 1, 2000 to Today) 94

Canasatego: Onondaga speaker who told the colonies to unite at Lancaster in 1744 (Chapter 2, 1700 to 1750) 119

Al Capone: the Chicago gang boss who sold illegal beer and liquor and went to prison for not paying the taxes he owed (Chapter 21, 1900 to 1950) 1465

Frank Capra: Sicilian boy who came through Ellis Island and won three Academy Awards as a director (Chapter 4, 1900 to 1950) 299

Addie Card: child spinner in a Vermont cotton mill whose Lewis Hine photograph appeared on a 1998 postage stamp (Chapter 14, 1900 to 1950) 944

Andrew Carnegie: a Scottish immigrant who started as a mill boy, sold his steel company for 480 million dollars and gave away more than 350 million (Chapter 16, 1850 to 1900) 1082

Andrew Carnegie: Pittsburgh steelmaker whose company owned every step of making steel (Chapter 19, 1850 to 1900) 1261

William Harvey Carney: a man born enslaved who carried the flag at Fort Wagner, the earliest act of Black American bravery recognized with the Medal of Honor (Chapter 24, 1850 to 1900) 1722

Wallace Carothers: chemist who led the DuPont team that made neoprene and nylon (Chapter 35, 1900 to 1950) 2955

Rachel Carson: marine biologist who wrote *Silent Spring* about persistent pesticides (Chapter 3, 1950 to 2000) 230

Lillian Carter: nurse from Plains, Georgia, who joined the Peace Corps at 68 and served in India (Chapter 23, 1950 to 2000) 1649

Maybelle Carter: guitarist of the Carter Family who made the guitar a lead instrument in country music (Chapter 33, 1900 to 1950) 2766

George Washington Carver: scientist who taught poor Southern farmers how to rebuild worn-out cotton land and grow their own food (Chapter 15, 1900 to 1950) 1017

James Leander Cathcart: American sailor enslaved in Algiers for eleven years who became chief Christian clerk to the Dey, the ruler of Algiers, and put the 1795 treaty into English (Chapter 23, 1750 to 1800) 1612

Nicolás Catua: young Tesuque runner who, with Pedro Omtua, carried the knotted cords of the Pueblo Revolt and was hanged by the Spanish (Chapter 31, The 1600s) 2511

William Chadbourne: an English carpenter who helped build one of the first sawmills run by water in New England (Chapter 10, The 1600s) 666

Charlie Chaplin: a London comedian who became a film star around the world by acting without speaking (Chapter 34, 1900 to 1950) 2868

Frank Chapman: bird scientist who started the Christmas Bird Count in 1900 (Chapter 9, 1900 to 1950) 616

Cesar Chavez: organizer of the farmworkers' union in the California grape fields. In March 2026 *The New York Times* reported that two women, Debra Rojas and Ana Murguia, said he sexually abused them as girls. Dolores Huerta then said he had raped her. He died in 1993, and no court heard the accusations (Chapter 14, 1950 to 2000) 965

Caleb Cheeshahteaumuck: a Wampanoag student from Martha's Vineyard, the first Native American to graduate from Harvard (Chapter 30, The 1600s) 2357

Ezekiel Cheever: a schoolmaster who taught Latin and Greek for 70 years, the last 38 as head of the Boston Latin School (Chapter 30, The 1600s) 2353

Rotchana Cheunchujit: Thai garment worker held in the El Monte sewing shop until officers freed her in 1995 (Chapter 35, 1950 to 2000) 2969

Santino Chuor: Dinka refugee from Sudan who, like Peter Dut, was one of the "Lost Boys" resettled in the United States (Chapter 4, 2000 to Today) 329

William Clark: Army officer who led the expedition with Meriwether Lewis and held York as a slave (Chapter 1, 1800 to 1850) 45

Charity Clarke: New Yorker who promised her cousin "a fighting army of Amazons" armed with spinning wheels (Chapter 35, 1750 to 1800) 2930

Nick Clifford: the youngest worker on Mount Rushmore and the last of its carvers to die (Chapter 12, 1900 to 1950) 820

Isaac Cline: Galveston's chief weather forecaster, who had called a hurricane harming the city "a crazy idea" and whose wife died in the 1900 storm (Chapter 27, 1900 to 1950) 2080

DeWitt Clinton: New York governor who pushed the Erie Canal through while people mocked it (Chapter 11, 1800 to 1850) 740

Samuel Cole: sweet-maker who kept the earliest licensed tavern known in Boston (Chapter 28, The 1600s) 2129

Thomas Cole: landscape painter and a first leader of the Hudson River School (Chapter 32, 1800 to 1850) 2633

Robert Coles: Massachusetts colonist made to wear a red letter D for drunkenness (Chapter 28, The 1600s) 2130

David and Micki Colfax and their sons: California ranch family who taught their four sons at home, three of whom went to Harvard (Chapter 7, 1950 to 2000) 501

John Collins: an English mason the Spanish held prisoner, who made the lime and ran the quarry for the Castillo de San Marcos (Chapter 12, The 1600s) 793

Michael Collins: Apollo 11 astronaut who stayed in orbit around the Moon in the command module (Chapter 1, 1950 to 2000) 86

John Colter: Lewis and Clark soldier who stayed to trap, the first person of European descent known to see the Yellowstone country (Chapter 1, 1800 to 1850) 51

Christopher Columbus: sea captain from Genoa who sailed west for Spain looking for a sea route to Asia and never set foot in what is now the United States (Chapter 1, The 1500s) 22

Edward Converse: Puritan settler and deacon who ran the first ferry between Charlestown and Boston (Chapter 11, The 1600s) 727

Martin Cooper: Motorola engineer who made the first call on a handheld cell phone in 1973 (Chapter 8, 1950 to 2000) 564

Thornton Copeland: a freed man who advertised in 1865 for his mother, from whom he had been sold away in 1844 (Chapter 24, 1850 to 1900) 1725

Nephi Craig: White Mountain Apache and Diné (Navajo) chef who opened a café built around Apache foods (Chapter 15, 2000 to Today) 1045

Prudence Crandall: Connecticut teacher arrested, tried and attacked for teaching Black girls (Chapter 25, 1800 to 1850) 1783

Walter Cronkite: CBS anchor who told viewers in 1968 the Vietnam War was "mired in stalemate" (Chapter 31, 1950 to 2000) 2577

Cuffee: enslaved man in New York City, sentenced by Judge Horsmanden in the trials of 1741 and burned at the stake with Quack (Chapter 21, 1700 to 1750) 1426

Ame and Elizabeth Cuming: Boston shopkeeper sisters who refused to join the boycott and were named in the newspaper and the town record for it (Chapter 18, 1750 to 1800) 1185

Austin Curtis: an enslaved jockey and trainer who kept his enslaver's horses from British raiders and was freed in 1791 (Chapter 36, 1750 to 1800) 3003

Jennie Curtis: Pullman seamstress whose speech helped start the Pullman boycott (Chapter 14, 1850 to 1900) 938

Charlotte Cushman: a Boston-born actress who played men's parts, including Romeo and Hamlet (Chapter 34, 1850 to 1900) 2852

D

William Darby: one of three men tried for acting *Ye Bare and Ye Cubb* in Virginia, the one the sheriff was ordered to arrest (Chapter 34, The 1600s) 2825

Dave (David Drake): enslaved potter and poet who cut his name and verses into his jars (Chapter 32, 1800 to 1850) 2638

Benjamin O. Davis Jr.: commander of the Black fighter pilots trained at Tuskegee and the first Black general in the Air Force (Chapter 22, 1900 to 1950) 1569

Jacob Davis: Reno tailor who put copper rivets on work pants (Chapter 35, 1850 to 1900) 2945

Joycelyn Davis: a descendant of *Clotilda* survivors who organizes gatherings in their honor in Africatown (Chapter 24, 2000 to Today) 1751

Eugene V. Debs: locomotive fireman who founded the American Railway Union and went to jail for leading the Pullman boycott (Chapter 14, 1850 to 1900) 939

John Deere: plow maker at Grand Detour, Illinois, whose polished steel plow cleaned itself in sticky prairie soil (Chapter 8, 1800 to 1850) 539

Mark and Chris DeJonge: Michigan parents convicted of a crime for teaching their children at home, who won in the state's highest court in 1993 (Chapter 30, 1950 to 2000) 2469

C. L. Dellums: retired Pullman porter, fired by Pullman managers in 1927 for organizing porters into a union, who described the porters' work on film in 1982 (Chapter 11, 1900 to 1950) 761

Prince Demah: enslaved portrait painter in Boston who later served free in the Massachusetts militia (Chapter 32, 1750 to 1800) 2622

Rachael Denhollander: a former gymnast who was the first woman to accuse Larry Nassar in public (Chapter 36, 2000 to Today) 3068

John Dewey: a philosopher who argued that children learn by doing and ran a school in Chicago to test it (Chapter 30, 1900 to 1950) 2423

Gerald Dickey: a Youngstown steelworker who proposed that the workers and the town buy their closed mill and reopen it (Chapter 16, 1950 to 2000) 1099

Dorothea Dix: visited every Massachusetts jail and poorhouse and reported how people with mental illness were held (Chapter 25, 1800 to 1850) 1789

Christopher Dock: a Mennonite schoolmaster in Pennsylvania who wrote the first book about how to teach written in this country (Chapter 30, 1700 to 1750) 2366

Jennifer Doudna: Berkeley biochemist who with Emmanuelle Charpentier made CRISPR-Cas9 a tool for cutting DNA (Chapter 9, 2000 to Today) 652

Frederick Douglass: a man born enslaved who learned to read in secret, escaped, and became the most heard abolitionist of his century (Chapter 24, 1800 to 1850) 1700

Frederick Douglass: escaped from slavery and started the *North Star* (Chapter 31, 1800 to 1850) 2544

Frederick Douglass: writer who escaped slavery and sat for 160 known photographs (Chapter 32, 1800 to 1850) 2642

Frederick Douglass: man enslaved in Maryland who described his childhood shirt and escaped dressed as a sailor (Chapter 35, 1800 to 1850) 2938

Edwin Drake: the man whose 1859 well near Titusville, Pennsylvania, started the American oil business, though he lost his money afterward (Chapter 10, 1850 to 1900) 682

Tammy Duckworth: a helicopter pilot who lost both legs in Iraq and later became a US senator (Chapter 22, 2000 to Today) 1591

Monica Dunford: physicist on the ATLAS experiment followed in *Particle Fever* (Chapter 9, 2000 to Today) 651

Peter Dut: Dinka refugee from Sudan who, like Santino Chuor, was one of the "Lost Boys" resettled in the United States (Chapter 4, 2000 to Today) 329

Mary Dyer: Quaker hanged on Boston Common for returning to Massachusetts after she was banished (Chapter 29, The 1600s) 2221

Bob Dylan: songwriter who won the Nobel Prize in Literature (Chapter 33, 1950 to 2000) 2788

E

Charles Eames: designer who, with Ray Eames, made molded plywood and fiberglass chairs (Chapter 35, 1950 to 2000) 2961

Ray Eames: designer and painter who, with Charles Eames, made molded plywood and fiberglass chairs (Chapter 35, 1950 to 2000) 2961

Amelia Earhart: the first woman to fly alone and nonstop across the Atlantic, lost over the Pacific in 1937 (Chapter 1, 1900 to 1950) 81

Sylvia Earle: marine biologist who holds the women's record for the deepest dive to the sea floor and led the first all-woman team to live underwater (Chapter 1, 1950 to 2000) 88

Thomas Edison: inventor whose team at Menlo Park made a long-lasting electric lamp, the phonograph and a system to sell electric light (Chapter 8, 1850 to 1900) 548

Thomas Edison: the inventor who built his Pearl Street electric system on direct current, which could not travel far (Chapter 10, 1850 to 1900) 684

Jonathan Edwards: minister of the Northampton revival, preacher of "Sinners in the Hands of an Angry God," and an enslaver (Chapter 29, 1700 to 1750) 2238

Albert Einstein: physicist who came to Princeton in 1933, signed the 1939 bomb letter and called racism "a disease of white people" (Chapter 9, 1900 to 1950) 623

Izzy Einstein: federal Prohibition agent who made arrests in disguise (Chapter 28, 1900 to 1950) 2172

Dwight Eisenhower: president who signed the Interstate highway law 37 years after crossing the country in an Army convoy (Chapter 11, 1950 to 2000) 765

Chris Ernst: the Yale rowing captain whose 1976 protest won the women's team a locker room (Chapter 36, 1950 to 2000) 3053

Estevanico: enslaved African man from Morocco, one of the four survivors, whom people of the Zuni town of Hawikuh killed in 1539 (Chapter 1, The 1500s) 25

Estevanico: an enslaved man from Morocco who walked across the south of the continent after a shipwreck. The people of the Zuni town of Hawikuh killed him in 1539, while he was guiding a Spanish expedition (Chapter 24, The 1500s) 1669

F

David Falconer: a government photographer who recorded Oregon's gasoline shortage of 1973 and 1974 in pictures and captions (Chapter 10, 1950 to 2000) 707

Katty Familia: immigrant mother from the Dominican Republic who, with Pierre Solon, lost the family's Boston house to foreclosure in 2009 (Chapter 7, 2000 to Today) 506

Marcos Farfán de los Godos: a captain with Juan de Oñate who wrote the first European play on record in what is now the United States (Chapter 34, The 1500s) 2820

Alpheus Felch: Michigan bank commissioner, later the state's governor and a US senator, who wrote down the frauds he found in 1838 (Chapter 17, 1800 to 1850) 1134

Lee Felsenstein: engineer who set up Community Memory, ran the Homebrew Computer Club's meetings and designed the Sol-20 and the Osborne 1 (Chapter 8, 1950 to 2000) 565

Richard Feynman: physicist who won a Nobel Prize for his work on how light and charged matter act on each other, and who showed the *Challenger* O-ring failure in ice water (Chapter 9, 1950 to 2000) 642

Mamie Garvin Fields: a Black teacher in South Carolina who started in a one-room school at twenty and taught for about thirty-five years (Chapter 30, 1900 to 1950) 2417

Charles Grandison Finney: lawyer turned revival preacher whose New Measures and lectures guided other revival preachers (Chapter 29, 1800 to 1850) 2269

John Fitch: builder of the first steamboat service anywhere to carry paying passengers on a schedule (Chapter 11, 1750 to 1800) 733

John Fitisemanu: American national born in American Samoa, living in Utah, who sued to be counted a citizen from birth (Chapter 23, 2000 to Today) 1661

Christian Fleetwood: a free Black sergeant major who carried his regiment's flag at New Market Heights and received the Medal of Honor (Chapter 22, 1850 to 1900) 1550

Williamina Fleming: Scottish immigrant hired as a maid who sorted more than 10,000 stars at Harvard's observatory (Chapter 9, 1850 to 1900) 613

Viola Fletcher: Greenwood survivor who at 107 asked Congress for justice (Chapter 25, 1900 to 1950) 1846

Betty Ford: wife of President Gerald Ford, who told the country she was addicted to pills and alcohol and founded a treatment center (Chapter 28, 1950 to 2000) 2185

Henry Ford: carmaker whose workers at Highland Park, Michigan, built the Model T on a moving assembly line (Chapter 8, 1900 to 1950) 555

Henry Ford: the carmaker who roughly doubled his workers' pay in 1914 (Chapter 16, 1900 to 1950) 1085

Edwin Forrest: an American star actor whose fans rioted against his English rival at Astor Place (Chapter 34, 1800 to 1850) 2849

Charlotte Forten: the first Black graduate of the Salem Normal School and the first Black teacher in Beaufort County, South Carolina (Chapter 30, 1850 to 1900) 2397

Adam Fortunate Eagle: Ojibwe writer, Pipestone boarding-school student and a lead organizer of the Alcatraz occupation (Chapter 2, 1950 to 2000) 169

Josephine Foster: a schoolgirl beaten for whispering in class, whose parents charged her teachers with cruelty (Chapter 30, 1850 to 1900) 2402

Robert Joseph Pershing Foster: Black surgeon who drove from Louisiana to Los Angeles in 1953 to practice medicine (Chapter 5, 1950 to 2000) 384

Stephen Foster: white songwriter of minstrel songs such as "Oh! Susanna," who died with 37 cents and three pennies (Chapter 33, 1800 to 1850) 2741

Naomi Parker Fraley: war worker who drilled, patched airplane wings and riveted in the machine shop of the Naval Air Station at Alameda, California, and the woman in a 1942 photograph linked to the "We Can Do It!" poster (Chapter 14, 1900 to 1950) 961

Billy Frank Jr.: Nisqually fisherman arrested more than fifty times, a leader of the fish-ins (Chapter 2, 1950 to 2000) 168

Aretha Franklin: Black singer who turned "Respect" into a woman's demand (Chapter 33, 1950 to 2000) 2784

Benjamin Franklin: a Philadelphia printer who helped start the Union Fire Company, the first volunteer fire company in the colonies to protect a whole town (Chapter 6, 1700 to 1750) 406

Benjamin Franklin: Philadelphia printer who designed the Franklin stove and the lightning rod and refused to patent his inventions (Chapter 8, 1700 to 1750) 527

Benjamin Franklin: Philadelphia experimenter whose electricity letters made him the first American scientist known abroad (Chapter 9, 1700 to 1750) 589

Benjamin Franklin: printer's apprentice who ran away from his brother's Boston shop at 17 (Chapter 14, 1700 to 1750) 919

Benjamin Franklin: Pennsylvania printer who argued for paper money, then printed it with the veins of real leaves to stop counterfeiters (Chapter 17, 1700 to 1750) 1121

Benjamin Franklin: the Philadelphia printer who made advertising pay for his newspaper and ran advertisements offering enslaved people for sale (Chapter 18, 1700 to 1750) 1181

Benjamin Franklin: American commissioner and then minister in France, and one of the three Americans who made the peace with Britain (Chapter 23, 1750 to 1800) 1608

Benjamin Franklin: printer, newspaper owner and Philadelphia postmaster whose paper earned money from advertisements, including ones selling enslaved people (Chapter 31, 1700 to 1750) 2519

Darnella Frazier: 17-year-old whose phone video recorded the police murder of George Floyd and became trial evidence (Chapter 31, 2000 to Today) 2590

Garrison Frazier: Baptist minister who told the Secretary of War in 1865 what freedom meant (Chapter 29, 1850 to 1900) 2280

Rose Rosenfeld Freedman: Triangle fire survivor who told the truth about the locked doors for ninety years (Chapter 14, 1900 to 1950) 946

Elizabeth Freeman (Mum Bett): an enslaved woman who sued for her freedom under the Massachusetts constitution and won (Chapter 24, 1750 to 1800) 1688

Herbert Freeman Jr.: New Orleans man who rescued his mother by boat and stayed by her body after she died at the Convention Center (Chapter 27, 2000 to Today) 2108

John C. Frémont: Army mapmaker whose maps guided thousands west and whose men killed at least 120 Wintu people in California (Chapter 1, 1800 to 1850) 61

Richard Frethorne: English indentured servant whose letters home from Virginia survive (Chapter 4, The 1600s) 262

Richard Frethorne: indentured servant at Martin's Hundred, Virginia, whose letters home begging for food survive (Chapter 14, The 1600s) 916

Limor Fried: engineer who founded Adafruit Industries, which makes electronics kits (Chapter 8, 2000 to Today) 573

Grace Fryer: radium dial painter who sued the company while she was dying (Chapter 14, 1900 to 1950) 952

Grace Fryer: radium watch-dial painter who sued the company that poisoned her (Chapter 25, 1900 to 1950) 1848

Samuel Fuller: *Mayflower* wool weaver with no known medical training who became the doctor for Plymouth and its neighbors (Chapter 26, The 1600s) 1956

Robert Fulton: the man who made the steamboat a business that lasted (Chapter 11, 1800 to 1850) 736

G

Lloyd Gaines: won the NAACP's first school case at the Supreme Court and then disappeared (Chapter 25, 1900 to 1950) 1861

Antonio García: Mexican farmworker who joined the Bracero Program at seventeen (Chapter 4, 1950 to 2000) 313

Newport Gardner: born Occramer Marycoo in West Africa, enslaved in Newport, Rhode Island, music teacher who bought his freedom with a lottery prize (Chapter 33, 1750 to 1800) 2734

James Garfield: 16-year-old canal boat hand in 1848 who later became president (Chapter 11, 1800 to 1850) 741

Bill Gates: head of Microsoft when government lawyers sued it for keeping an illegal monopoly (Chapter 19, 1950 to 2000) 1287

Jeff German: Las Vegas reporter stabbed to death by an elected official he had written about (Chapter 31, 2000 to Today) 2589

Albert Ghiorso: the builder of instruments who helped discover 12 elements without ever earning a doctorate (Chapter 13, 1950 to 2000) 890

Josiah Willard Gibbs: Yale physicist who founded chemical thermodynamics and worked nine years without pay (Chapter 9, 1850 to 1900) 611

Lois Gibbs: Love Canal mother who organized the Love Canal Homeowners Association in 1978 (Chapter 3, 1950 to 2000) 236

Jeffrey Gibson: Choctaw and Cherokee artist, the first Native artist to represent the United States alone at the Venice Biennale (Chapter 32, 2000 to Today) 2703

Josh Gibson: a Negro Leagues catcher and home-run hitter the white major-league owners never signed (Chapter 36, 1900 to 1950) 3042

Clarence Earl Gideon: a poor man in Florida whose pencil letter to the US Supreme Court won the right to a lawyer for every person charged with a serious crime (Chapter 21, 1950 to 2000) 1471

Kyle "Bugha" Giersdorf: a 16-year-old who won the first Fortnite World Cup and $3 million in 2019 (Chapter 36, 2000 to Today) 3061

Ida Mae Brandon Gladney: sharecropper's wife who left Mississippi for Chicago in 1937 after white men nearly beat a cousin to death (Chapter 5, 1900 to 1950) 378

Hugh Glass: fur trapper who crawled and walked about 200 miles after a grizzly attack, abandoned by the men left to watch over him (Chapter 1, 1800 to 1850) 54

Victor Glover: NASA pilot of Artemis II, the first Black man to travel 252,756 miles from Earth (Chapter 1, 2000 to Today) 95

Mary Katharine Goddard: Baltimore printer and postmaster who printed the first Declaration copies with the signers' names (Chapter 31, 1750 to 1800) 2528

Robert Goddard: Clark University physicist who flew the first liquid-fuel rocket in 1926 (Chapter 9, 1900 to 1950) 617

Nan Goldin: photographer who became addicted to OxyContin and led protests against the Sackler family (Chapter 28, 2000 to Today) 2195

Samuel Gompers: cigar maker who led the American Federation of Labor for almost forty years (Chapter 14, 1850 to 1900) 935

Charles Goodnight: Texas cattleman who invented the chuckwagon (Chapter 15, 1850 to 1900) 1013

John Tackett Goolrick: Confederate soldier whose family's letters record his 30 years of morphine addiction (Chapter 28, 1850 to 1900) 2157

John B. Gough: bookbinder whose drinking cost him his work, who then spent his life asking others to sign the temperance pledge (Chapter 28, 1800 to 1850) 2151

Billy Graham: Southern Baptist preacher who held crusades from 1949 to 2005 (Chapter 29, 1950 to 2000) 2308

Moses Grandy: enslaved boatman on the Dismal Swamp canals who bought his freedom three times and described the men who dug the canals (Chapter 11, 1800 to 1850) 742

Milla Granson (Lilly Ann Granderson): an enslaved woman who ran a secret night school in Natchez and taught hundreds to read and write (Chapter 24, 1800 to 1850) 1702

Lee Grant: an actress blacklisted for twelve years who later won an Academy Award (Chapter 34, 1950 to 2000) 2880

Ulysses S. Grant: commander of all the Union armies from 1864, who took Lee's surrender at Appomattox (Chapter 22, 1850 to 1900) 1551

Angelina Grimké: antislavery speaker, the first woman in the United States to speak before a body of lawmakers (Chapter 25, 1800 to 1850) 1787

Sarah Grimké: South Carolina-born antislavery speaker who answered the ministers with letters on the equality of the sexes (Chapter 25, 1800 to 1850) 1787

Victor Gruen: the architect who designed Southdale, the first enclosed American mall, and later rejected what developers built from his idea (Chapter 18, 1950 to 2000) 1215

James Guild: a Vermont farm boy who peddled goods from a trunk and wrote down what happened to him (Chapter 18, 1800 to 1850) 1189

Nathalie Gumpertz: tenement mother of four at 97 Orchard Street who supported her children as a dressmaker after her husband disappeared (Chapter 7, 1850 to 1900) 483

Henry Gunther: a drafted bank clerk listed as the last American soldier killed in World War I (Chapter 22, 1900 to 1950) 1564

Woody Guthrie: folk singer who wrote "This Land Is Your Land" (Chapter 33, 1900 to 1950) 2772

H

Deb Haaland: Laguna Pueblo, one of the first two Native American women in Congress and the first Native American cabinet secretary (Chapter 2, 2000 to Today) 176

Fred Haise: Apollo 13 astronaut whose crew survived an explosion on the way to the Moon (Chapter 1, 1950 to 2000) 87

Sarah Josepha Hale: magazine editor who asked governors and presidents for years to make Thanksgiving one national holiday (Chapter 37, 1800 to 1850) 3106

Alex Haley: the writer of *Roots*, which led many Black Americans to search for their families. Experts who trace family histories through records have challenged parts of his research (Chapter 24, 1950 to 2000) 1743

Nelson Cole Haley: a New Bedford harpooner on the whaling ship *Charles W. Morgan* whose book tells how a crew turned a whale into lamp oil (Chapter 10, 1800 to 1850) 680

Charles Martin Hall: the 22-year-old who found the cheap way to make aluminum, in his family's woodshed (Chapter 13, 1850 to 1900) 867

Julia Brainerd Hall: Charles Martin Hall's older sister, a chemist whose notes helped him win the fight over his patent (Chapter 13, 1850 to 1900) 867

Lewis Hallam the Younger: a boy in the first touring company who became the leading actor and manager of the American Company (Chapter 34, 1750 to 1800) 2836

Nancy Hallam: an actress in the American Company whom Charles Willson Peale painted in costume (Chapter 34, 1750 to 1800) 2837

Fannie Lou Hamer: Mississippi organizer who told the country on television how state highway patrolmen made two prisoners beat her in jail (Chapter 25, 1950 to 2000) 1884

Alexander Hamilton: orphaned Caribbean clerk sent to school in New York after his letter about a hurricane (Chapter 4, 1750 to 1800) 271

Alexander Hamilton: wrote the plan to pay the Revolution's debts in full (Chapter 16, 1750 to 1800) 1064

Alexander Hamilton: first Secretary of the Treasury, who planned the national bank and the United States Mint (Chapter 17, 1750 to 1800) 1127

Dr. Alexander Hamilton: Annapolis doctor whose 1744 diary records colonial taverns and drinking clubs, a different man from the Alexander Hamilton who ran the Treasury (Chapter 28, 1700 to 1750) 2137

Andrew Hamilton: Philadelphia lawyer who defended John Peter Zenger without pay and told the jury it could judge the law as well as the facts (Chapter 20, 1700 to 1750) 1321

Thomas Hancock: Boston merchant whose firm rented out the ships that carried the Acadians away in 1755 (Chapter 19, 1700 to 1750) 1244

Lorraine Hansberry: the first Black woman to have a drama produced on Broadway (Chapter 34, 1950 to 2000) 2884

Jeremy Hansen: Canadian Space Agency astronaut on Artemis II, the first Canadian to travel 252,756 miles from Earth (Chapter 1, 2000 to Today) 95

Laszlo Hanyecz: programmer who bought two pizzas for 10,000 bitcoins, the first recorded time anyone paid with bitcoin (Chapter 17, 2000 to Today) 1162

Thomas Hariot (also spelled Harriot): English mathematician who smoked tobacco the Algonquian way at Roanoke and wrote down how the Algonquian people used it (Chapter 28, The 1500s) 2125

Thomas Harriot (also spelled Hariot): English mathematician and scientist who wrote down, town by town, the deaths among the Carolina Algonquians (Chapter 26, The 1500s) 1953

Benjamin Harris: London printer who printed the first newspaper in the English colonies, *Publick Occurrences* (Chapter 31, The 1600s) 2509

Sarah Harris: young Black woman whose wish to be taught began Prudence Crandall's school for Black girls (Chapter 25, 1800 to 1850) 1783

Harry: enslaved fiddler who ran away near Jamestown, Virginia, in 1746, known from one runaway notice (Chapter 33, 1700 to 1750) 2727

Ira Hayes: Akimel O'odham Marine who helped raise the second flag on Iwo Jima (Chapter 2, 1900 to 1950) 156

Victor Heiser: 16-year-old who rode the Johnstown Flood on a barn roof, lost both parents, and became a doctor (Chapter 27, 1850 to 1900) 2075

John Hemmings: an enslaved joiner who made the fine woodwork of Monticello's rebuilt rooms, one of five people Jefferson freed in his will (Chapter 12, 1800 to 1850) 806

Caroline Henderson: Oklahoma Panhandle homesteader who stayed through the dust years and wrote "Letters from the Dust Bowl" (Chapter 3, 1900 to 1950) 225

Joseph Henry: discoverer of self-induction and first head of the Smithsonian, who drew the first national weather maps (Chapter 9, 1800 to 1850) 601

Matthew Henson: Black sailor and explorer, Robert Peary's navigator and dog-sled driver on seven Arctic voyages (Chapter 1, 1900 to 1950) 75

Judy Heumann: refused a school place and a teaching license, then helped lead the 1977 Section 504 sit-in (Chapter 25, 1950 to 2000) 1905

Darleane Hoffman: nuclear chemist who found plutonium in nature (Chapter 9, 1950 to 2000) 639

Darleane Hoffman: the nuclear chemist who found plutonium in ancient rock, proving it occurs in nature (Chapter 13, 1950 to 2000) 891

Betty Snyder Holberton: one of the six women who programmed ENIAC (Chapter 8, 1900 to 1950) 559

Billie Holiday: Black jazz singer who recorded "Strange Fruit" and was arrested in her hospital bed (Chapter 33, 1900 to 1950) 2775

John Campanius Holm: New Sweden chaplain who kept the first known weather record in what is now the United States (Chapter 9, The 1600s) 587

John W. Holway: medical student hired as a Pinkerton guard, beaten by a crowd at Homestead in 1892 (Chapter 19, 1850 to 1900) 1265

Winslow Homer: drew the Civil War for *Harper's Weekly* and later painted the sea (Chapter 32, 1850 to 1900) 2647

Philip Hone: a New York merchant and former mayor whose diary recorded the Panic of 1837 as it happened (Chapter 16, 1800 to 1850) 1074

Thomas Hooker: minister who led about 100 members of his church from Massachusetts to found Hartford (Chapter 5, The 1600s) 342

Grace Hopper: mathematician and Navy officer who built the A-0 compiler (Chapter 8, 1950 to 2000) 562

Josiah Hornblower: the young English engine builder who came over with the first steam engine in America and later ran it (Chapter 10, 1750 to 1800) 673

Lafayette Houck: one of the last colliers at Hopewell Furnace, who taught the National Park Service how charcoal was made (Chapter 10, 1850 to 1900) 689

Howling Wolf: Southern Cheyenne survivor of Sand Creek who became a ledger artist as a prisoner at Fort Marion (Chapter 32, 1850 to 1900) 2654

Edwin Hubble: astronomer who showed Andromeda is a separate galaxy and the universe is expanding (Chapter 9, 1900 to 1950) 619

Henry Hudson: English sea captain who sailed the *Half Moon* up the Hudson River in 1609, whose crew later put him into an open boat and sailed away (Chapter 1, The 1600s) 32

Dolores Huerta: farmworkers' union organizer who negotiated the first grape contracts and made up the slogan "Sí, se puede" (Chapter 14, 1950 to 2000) 965

Fountain Hughes: a man born into slavery whose own voice survives on a 1949 Library of Congress recording (Chapter 24, 1900 to 1950) 1740

John Hull: Boston silversmith who ran the Massachusetts mint against English law and struck the pine-tree shilling (Chapter 17, The 1600s) 1118

Amos Humiston: a sergeant killed at Gettysburg, identified by the photograph of his three children (Chapter 22, 1850 to 1900) 1547

Hung Wah: the leading Chinese headman on the Central Pacific (Chapter 11, 1850 to 1900) 749

Zora Neale Hurston: novelist and folklorist who never earned more than $943.75 in royalties from any book (Chapter 32, 1900 to 1950) 2669

Anne Hutchinson: English woman who followed her minister to Boston and was banished by the Massachusetts court (Chapter 4, The 1600s) 264

Anne Hutchinson: Boston woman tried and banished for what she taught at meetings in her house (Chapter 25, The 1600s) 1762

Anne Hutchinson: Boston woman banished and put out of her church for her teaching (Chapter 29, The 1600s) 2226

I

Daniel Inouye: a Japanese American soldier who lost his right arm in Italy and later served Hawaii in the Senate for 50 years (Chapter 22, 1900 to 1950) 1570

Samuel Insull: Thomas Edison's former secretary, who tied more than 85 electric companies into one chain that collapsed in 1932 (Chapter 10, 1900 to 1950) 695

J

Jack: man taken from Gambia and enslaved in Virginia, who ran away within five weeks of his sale (Chapter 35, 1700 to 1750) 2925

Andrew Jackson: president who vetoed the Second Bank's new charter and moved the government's money out of it (Chapter 17, 1800 to 1850) 1131

Andrew Jackson: president who signed the law to remove Native nations from the East, used the veto to fight the national bank and gave government jobs to his supporters (Chapter 20, 1800 to 1850) 1337

Nathan Jackson: Tlingit carver of totem poles and masks, still carving in his eighties (Chapter 32, Before 1500) 2603

Jane Jacobs: the writer who helped defeat three New York projects, among them the Lower Manhattan Expressway (Chapter 6, 1950 to 2000) 448

Frank James (Wamsutta): Wampanoag leader who started the National Day of Mourning in 1970, after the Massachusetts officials planning the Pilgrims' 350th anniversary stopped his speech (Chapter 37, 1950 to 2000) 3130

Anna Jarvis: woman who campaigned to create Mother's Day and then fought the businesses that made money from it (Chapter 37, 1900 to 1950) 3120

Thomas Jefferson: the owner of Monticello and an enslaver, who designed his own house and worked on a design for Virginia's capitol modeled on a Roman temple (Chapter 12, 1750 to 1800) 803

Mae Jemison: doctor, the first African American woman in space (Chapter 1, 1950 to 2000) 90

Joseph Jenks: London sword maker who ran the Saugus blacksmith shop and won the Massachusetts colony's protection for his water-driven mills in 1646, which US Patent Office officials credit as the first patent for a machine in what is now the United States (Chapter 8, The 1600s) 522

Steve Jobs: cofounder of Apple Computer who made its first sale (Chapter 8, 1950 to 2000) 567

Barbara Rose Johns: sixteen-year-old who led her school out on strike, which led to one of the Brown cases (Chapter 25, 1950 to 2000) 1874

Andrew Johnson: New Jersey high school wrestler whose locs were cut beside the mat (Chapter 35, 2000 to Today) 2977

Ashley Johnson: the actor who played Ellie in *The Last of Us* with her voice and body (Chapter 34, 2000 to Today) 2892

Ed Johnson: a Black man wrongly convicted of rape and lynched after the US Supreme Court ordered him kept safe (Chapter 21, 1900 to 1950) 1457

Francis Johnson: free Black bandleader in Philadelphia who published more than 200 pieces of music (Chapter 33, 1800 to 1850) 2744

Jack Johnson: the Black heavyweight champion whose 1910 win was followed by white attacks on Black people in more than 25 states (Chapter 36, 1900 to 1950) 3031

Katherine Johnson: Black mathematician who calculated the flight paths of the first American spaceflights (Chapter 1, 1950 to 2000) 85

Lyndon B. Johnson: Senate leader who kept count of how each senator would vote, then a president who used the same skill on civil rights laws (Chapter 20, 1950 to 2000) 1379

Henrietta Johnston: first woman recorded as working for pay as an artist in the English colonies, a pastel portraitist in Charleston (Chapter 32, 1700 to 1750) 2614

Kathryn "Tubby" Johnston: a girl who played Little League under a boy's name in 1950, after which officials banned girls (Chapter 36, 1950 to 2000) 3046

Louis Jolliet: fur trader born near Quebec who paddled down the Mississippi with Jacques Marquette in 1673 (Chapter 1, The 1600s) 34

Tryntje Jonas: official midwife of New Amsterdam (Chapter 26, The 1600s) 1959

Absalom Jones: Black Philadelphian who, with Richard Allen, led volunteers to nurse the sick and bury the dead in the yellow fever of 1793 (Chapter 26, 1750 to 1800) 1974

Absalom Jones: man born into slavery who became the first priest of African descent in the Episcopal Church (Chapter 29, 1750 to 1800) 2250

Mother Jones (Mary Harris Jones): union organizer for coal miners who led the March of the Mill Children in 1903 (Chapter 14, 1900 to 1950) 942

Scott Joplin: Black composer of the "Maple Leaf Rag" and the opera *Treemonisha* (Chapter 33, 1850 to 1900) 2756

Russel and Mary Jane Jordan: Iowa farm couple who sold their machinery, livestock and household goods at auction to pay their bank (Chapter 15, 1950 to 2000) 1038

Chief Joseph: Nez Perce leader in the flight of 1877 who surrendered forty miles from Canada (Chapter 2, 1850 to 1900) 141

Ona Judge: an enslaved woman who walked out of President George Washington's household and was never caught (Chapter 24, 1750 to 1800) 1692

K

Zoë Keating: cellist who publishes what streaming services pay her (Chapter 33, 2000 to Today) 2798

Elizabeth Keckley: dressmaker born enslaved who bought her freedom by sewing and made Mary Todd Lincoln's dresses (Chapter 35, 1850 to 1900) 2943

Joseph Kennedy: football coach put on leave over his prayers on the field, who won his case in the Supreme Court (Chapter 29, 2000 to Today) 2338

Patrick Kennedy: barrel maker who left County Wexford, Ireland, during the Great Famine and married Bridget Murphy in Boston in 1849, great-grandfather of John F. Kennedy (Chapter 4, 1800 to 1850) 282

Billy Kersands: the most popular Black minstrel comedian of his day, who performed in burnt-cork blackface (Chapter 34, 1850 to 1900) 2856

Lina Khan: law student whose 2017 article on Amazon's power came before she led the Federal Trade Commission (Chapter 19, 2000 to Today) 1297

Robin Wall Kimmerer: Citizen Potawatomi botanist who uses Native knowledge and laboratory science together (Chapter 9, Before 1500) 583

Billie Jean King: a tennis champion who founded the Women's Tennis Association and beat Bobby Riggs in 1973 (Chapter 36, 1950 to 2000) 3054

Martin Luther King Jr.: minister chosen to lead the Montgomery boycott. He was shot in Memphis in 1968, and James Earl Ray pleaded guilty to his murder (Chapter 25, 1950 to 2000) 1876

Amelia Stewart Knight: Oregon Trail traveler whose 1853 diary does not mention her pregnancy until the birth (Chapter 5, 1850 to 1900) 364

Jack Knight: airmail pilot who flew through a night snowstorm in 1921 to finish the test that kept the airmail going (Chapter 11, 1900 to 1950) 759

Sarah Kemble Knight: Boston businesswoman whose 1704 journal records the households where she stayed on the road to New York (Chapter 7, 1700 to 1750) 470

Sarah Kemble Knight: Boston businesswoman whose journal of her 1704 ride is an eyewitness record of the road (Chapter 11, 1700 to 1750) 731

Christina Koch: NASA astronaut on Artemis II, the first woman to travel 252,756 miles from Earth (Chapter 1, 2000 to Today) 95

Kondiaronk: Huron-Wendat chief who argued for the Great Peace of Montréal while dying of a fever (Chapter 23, 1700 to 1750) 1603

DJ Kool Herc: Clive Campbell, the DJ at the 1973 Bronx party that people name as the start of hip-hop (Chapter 33, 1950 to 2000) 2791

Fred Korematsu: refused to leave his home under the removal orders and had his conviction thrown out in 1983 (Chapter 25, 1900 to 1950) 1868

Oluale Kossola (Cudjo Lewis): a man carried to Alabama on the *Clotilda*, the last known slave ship, who helped found Africatown (Chapter 24, 1850 to 1900) 1717

Ron Kovic: a Marine paralyzed in Vietnam who came home and spoke against the war (Chapter 22, 1950 to 2000) 1579

Ray Kroc: milkshake-machine salesman who became the McDonald brothers' franchise agent and then bought them out (Chapter 15, 1950 to 2000) 1031

L

Henrietta Lacks: Black woman whose cancer cells doctors took without asking, the first human cells to keep growing in a laboratory (Chapter 26, 1950 to 2000) 2006

Kendrick Lamar: Black rapper, the first musician outside classical music and jazz to win the Pulitzer Prize for Music (Chapter 33, 2000 to Today) 2803

Rebecca Lamar: survivor of the *Pulaski* explosion who lived five days on a piece of the wreck and wrote it down (Chapter 27, 1800 to 1850) 2064

William Henry Lane (Master Juba): a Black dancer who danced in blackface and toured with a white minstrel company (Chapter 34, 1800 to 1850) 2846

Dorothea Lange: government photographer of 1930s farm families and the 1942 forced removal of Japanese Americans (Chapter 32, 1900 to 1950) 2676

Lucy Larcom: a girl who went to work in a Lowell mill at 11 and later wrote about her childhood (Chapter 36, 1800 to 1850) 3017

Robert de La Salle: French fur trader and explorer who followed the Mississippi to the sea and claimed its whole valley for France (Chapter 1, The 1600s) 35

Lewis Latimer: self-taught draftsman who drew the telephone patent and patented a better way to make lamp filaments (Chapter 8, 1850 to 1900) 549

Ernest Lawrence: scientist at Berkeley who built the cyclotron (Chapter 8, 1900 to 1950) 560

Ernest Lawrence: the physicist who invented the cyclotron, the machine behind most of the elements discovered in the United States (Chapter 13, 1900 to 1950) 881

Jacob Lawrence: painter of the 60-panel *Migration Series* (Chapter 32, 1900 to 1950) 2673

Raef Lawson: Grubhub delivery driver whose court case asked whether an app driver is an employee (Chapter 14, 2000 to Today) 976

Opal Lee: retired Fort Worth teacher who organized Juneteenth celebrations for about 40 years and campaigned for it to become a national holiday (Chapter 37, 2000 to Today) 3137

Lee Puey You: Chinese "paper daughter" held about 20 months on Angel Island and deported (Chapter 4, 1900 to 1950) 302

Robert E. Lee: commander of the Confederate Army of Northern Virginia, who surrendered at Appomattox (Chapter 22, 1850 to 1900) 1551

Leif Erikson: Norse sailor from Greenland, son of Erik the Red, who led one of the first European voyages to North America (Chapter 1, Before 1500) 20

Pierre L'Enfant: the French-born engineer who drew the plan of Washington, D.C., and was dismissed in 1792 before it was built (Chapter 6, 1750 to 1800) 409

Aldo Leopold: forester who got the Gila Wilderness set aside in 1924 and wrote *A Sand County Almanac* (Chapter 3, 1900 to 1950) 227

William Levingston: a merchant and dancing teacher who built the first playhouse in the English colonies, with two indentured servants as his actors (Chapter 34, 1700 to 1750) 2830

Edmonia Lewis: Mississauga Ojibwe and African American sculptor who carved in marble in Rome after being beaten and falsely charged at Oberlin (Chapter 32, 1850 to 1900) 2648

Meriwether Lewis: Army officer sent by Jefferson with William Clark across the Louisiana Purchase to the Pacific (Chapter 1, 1800 to 1850) 45

Hugh Kwong Liang: 15-year-old Chinatown survivor of the 1906 San Francisco earthquake and fire (Chapter 27, 1900 to 1950) 2082

Queen Lili‘uokalani: last queen of Hawaii, overthrown with American help in 1893 and tried in her own palace in 1895 (Chapter 23, 1850 to 1900) 1628

Maya Lin: the 21-year-old college student whose design won the contest for the Vietnam Veterans Memorial (Chapter 12, 1950 to 2000) 830

Abraham Lincoln: president who suspended habeas corpus during the Civil War and kept doing it after the Chief Justice told him he had no power to (Chapter 20, 1850 to 1900) 1343

Charles Lindbergh: airmail pilot who made the first solo nonstop flight across the Atlantic Ocean (Chapter 1, 1900 to 1950) 80

Sandra Lindsay: Queens nurse who got the first COVID-19 vaccine given in the United States outside a trial (Chapter 26, 2000 to Today) 2026

Little Turtle (Mishikinaakwa): Miami war leader who helped destroy St. Clair's army in 1791 and signed the Treaty of Greenville (Chapter 2, 1750 to 1800) 125

Lydia Lobsiger: the first depositor paid back by federal deposit insurance, with a $1,250 check, her life savings, dated July 3, 1934 (Chapter 17, 1900 to 1950) 1151

Elijah Lovejoy: antislavery editor shot dead by a mob in Alton, Illinois, while defending his press (Chapter 31, 1800 to 1850) 2541

Jim Lovell: commander of Apollo 13, whose crew lived in the lunar module after an explosion and came home safely (Chapter 1, 1950 to 2000) 87

Ann Lowe: Black dressmaker who made gowns for rich families, Olivia de Havilland's 1946 Oscar gown and Jacqueline Kennedy's wedding dress (Chapter 35, 1900 to 1950) 2956

Frank Lumpkin: Chicago steelworker who fought for 17 years to win back the pay and pensions lost when Wisconsin Steel closed (Chapter 14, 1950 to 2000) 970

Tristán de Luna y Arellano: Spanish noble who led the colony at Pensacola Bay when a hurricane sank its ships and food (Chapter 27, The 1500s) 2038

Emilia Ruth Palmquist Lundberg: a widowed mother of six in Jersey City who shopped for her family with ration books in the Second World War (Chapter 18, 1900 to 1950) 1212

Hannah Lyman: Northampton girl of about 16 fined 10 shillings for wearing silk, who wore it even while charged (Chapter 35, The 1600s) 2921

Matthew Lyon: Vermont congressman, the first person tried under the Sedition Act, re-elected from jail (Chapter 31, 1750 to 1800) 2531

Patrick Lyon: a Philadelphia blacksmith held three months in the Walnut Street Prison for a bank robbery he did not commit (Chapter 21, 1750 to 1800) 1433

M

Clara Maass: nurse who died after letting infected mosquitoes bite her in a yellow fever experiment (Chapter 26, 1900 to 1950) 1992

James Madison: Constitutional Convention delegate who first argued against a bill of rights, then wrote one and steered it through Congress (Chapter 20, 1750 to 1800) 1328

Wilma Mankiller: the first woman elected Principal Chief of the Cherokee Nation (Chapter 2, 1950 to 2000) 167

Horace Mann: the first secretary of the Massachusetts Board of Education, who argued for free schools paid for by taxes (Chapter 30, 1800 to 1850) 2377

Manteo: Algonquian man from Croatoan who learned English in London and interpreted for the English (Chapter 31, The 1500s) 2505

Jacques Marquette: French Jesuit priest who paddled down the Mississippi with Louis Jolliet in 1673 (Chapter 1, The 1600s) 34

John Marshall: Chief Justice who wrote that judges may refuse to follow a law that breaks the Constitution (Chapter 20, 1800 to 1850) 1334

Thurgood Marshall: lawyer who won 29 of the 32 cases he argued in the Supreme Court and later became its first Black justice (Chapter 20, 1950 to 2000) 1375

Joseph Plumb Martin: a Connecticut teenager who joined at 15 and wrote down what the war was like (Chapter 22, 1750 to 1800) 1530

Maria Martinez: San Ildefonso potter known for black-on-black pottery (Chapter 32, 1900 to 1950) 2665

Frank Mayer: buffalo runner from 1872 to 1878 who said army posts gave hunters free ammunition (Chapter 3, 1850 to 1900) 210

Barbara McClintock: scientist who found that genes can move, in corn (Chapter 9, 1900 to 1950) 632

Harold McComber: a Mohawk ironworker from Kahnawake who left home to work on Manhattan skyscrapers, the man the film *High Steel* follows (Chapter 12, 1950 to 2000) 827

Cyrus McCormick: held the patent on the McCormick reaper and built the business that sold it (Chapter 8, 1800 to 1850) 539

Antron McCray: one of the Exonerated Five, New York teenagers convicted in 1990 of a Central Park rape and cleared in 2002 (Chapter 21, 2000 to Today) 1486

Hattie McDaniel: the first Black person to win an Academy Award (Chapter 34, 1900 to 1950) 2875

Kelly McKernan: illustrator who sued the makers of picture-making programs in 2023 (Chapter 32, 2000 to Today) 2706

Malcom McLean: North Carolina trucking owner who started shipping truck-sized containers by sea in 1956 (Chapter 11, 1950 to 2000) 771

Aimee Semple McPherson: Pentecostal preacher who built the Angelus Temple and preached from it by radio (Chapter 29, 1900 to 1950) 2300

Doña María Meléndez: Timucua chief of Nombre de Dios, beside St. Augustine, and a Catholic (Chapter 29, The 1500s) 2212

Marlyn Wescoff Meltzer: one of the six women who programmed ENIAC (Chapter 8, 1900 to 1950) 559

Sylvia Mendez: turned away from a California school at eight, one of about 5,000 children in her father's lawsuit, which the families won (Chapter 30, 1900 to 1950) 2439

Francisco Menéndez: a Mandinga man who escaped slavery in Carolina and led the free Black militia at Fort Mose (Chapter 24, 1700 to 1750) 1681

Metacom (King Philip): Wampanoag sachem who led the coalition war of 1675 and 1676 against the New England colonies (Chapter 2, The 1600s) 112

Miantonomi: Narragansett sachem who described what English settlers were doing to the land in 1642, sentenced to death by English officials and killed in 1643 (Chapter 3, The 1600s) 187

Oscar Micheaux: a Black homesteader and novelist who made more than 40 films for Black audiences (Chapter 34, 1900 to 1950) 2871

Albert Michelson: Prussian-born physicist who measured the speed of light to within about 0.05 percent (Chapter 9, 1850 to 1900) 612

Peter Mills: a twelve-year-old shut out of his Washington school, whose name is on the 1972 case that gave disabled children a right to public school (Chapter 30, 1950 to 2000) 2465

Maria Mitchell: Nantucket librarian and astronomer who found "Miss Mitchell's Comet" in 1847 (Chapter 9, 1800 to 1850) 603

Tom Molineaux: a boxer from Virginia who fought the English champion Tom Cribb twice (Chapter 36, 1800 to 1850) 3013

Annie Moore: Irish girl of 17, the first immigrant processed at Ellis Island (Chapter 4, 1850 to 1900) 295

Bridget and Joseph Moore: an Irish immigrant couple at 97 Orchard Street whose daughter Agnes Mary died of marasmus, a form of starvation, at five months in 1869 (Chapter 6, 1850 to 1900) 424

Linda Morabito: navigation engineer who found the first active volcano seen beyond Earth, on Io (Chapter 9, 1950 to 2000) 641

Rita Moreno: the first Hispanic woman to win an Academy Award (Chapter 34, 1950 to 2000) 2885

Irene Morgan: Baltimore war worker who refused to give up her bus seat in 1944 and won her case in the Supreme Court in 1946 (Chapter 11, 1900 to 1950) 763

J.P. Morgan: New York banker who ran the rescue in the Panic of 1907 from his own library (Chapter 17, 1900 to 1950) 1145

J.P. Morgan: Wall Street banker who joined rival companies into giant ones, among them US Steel (Chapter 19, 1900 to 1950) 1270

Robert Morris: ran the country's finances near the end of the Revolution, bought most of the Seneca Nation's land in New York, then went to a debtors' prison (Chapter 19, 1750 to 1800) 1248

Toni Morrison: editor and novelist who won the Nobel Prize in Literature (Chapter 32, 1950 to 2000) 2693

Samuel Morse: inventor who held the patents on the electric telegraph and sent the first public message in 1844 (Chapter 8, 1800 to 1850) 537

Robert Moses: the New York official who built bridges, parks and expressways for decades without winning an election (Chapter 6, 1950 to 2000) 448

Wandrea ArShaye "Shaye" Moss: Fulton County election worker falsely accused of cheating in the 2020 count, who told a committee of Congress under oath what followed (Chapter 20, 2000 to Today) 1397

Lucretia Mott: Quaker antislavery organizer who called the Seneca Falls convention with Elizabeth Cady Stanton (Chapter 25, 1800 to 1850) 1792

Ebenezer Mudgett: Weare, New Hampshire, sawmill owner fined for cutting the king's pines, who led the Pine Tree Riot of 1772 (Chapter 3, 1750 to 1800) 193

John Muir: Scottish-born writer whose articles made Yosemite known, first president of the Sierra Club (Chapter 3, 1850 to 1900) 213

The Munoz family: Las Vegas family whose four school-age children did school online at the dining room table in August 2020 (Chapter 7, 2000 to Today) 507

Bridget Murphy: woman from County Wexford, Ireland, who married Patrick Kennedy in Boston in the famine years, great-grandmother of John F. Kennedy (Chapter 4, 1800 to 1850) 282

Isaac Murphy: a jockey born into slavery who won the Kentucky Derby three times (Chapter 36, 1850 to 1900) 3023

Edward R. Murrow: CBS radio reporter in London before and during World War II (Chapter 31, 1900 to 1950) 2568

Daisy and William Myers: the first Black family in Levittown, Pennsylvania, who stayed in their house through weeks of mob attacks in 1957 (Chapter 7, 1950 to 2000) 498

N

R. Carlos Nakai: Navajo and Ute flute player whose records were the first of the traditional Native flute to sell 500,000 copies (Chapter 33, Before 1500) 2716

Carry Nation: Kansas woman who smashed saloons to force officials to enforce the state's liquor ban (Chapter 28, 1900 to 1950) 2166

William Cooper Nell: printer of the *North Star* and the first Black American known to hold a federal job (Chapter 31, 1850 to 1900) 2549

Eliot Ness: the federal agent whose team raided Al Capone's breweries (Chapter 21, 1900 to 1950) 1465

Ed Neufeldt: an Elkhart, Indiana, worker who built RV cabinets for 32 years, lost his job in 2008 and introduced President Obama in 2009 (Chapter 16, 2000 to Today) 1106

Rebecca Neugin: Cherokee woman taken west on the Trail of Tears at about three, interviewed in 1932 (Chapter 2, 1800 to 1850) 132

Rebecca Neugin: Cherokee woman forced west on the Trail of Tears at about three, interviewed at 97 or 98 (Chapter 5, 1800 to 1850) 357

Chester Nez: Navajo Marine, one of the twenty-nine who built the Navajo code (Chapter 2, 1900 to 1950) 157

Chester Nez: one of the first 29 Navajo code talkers, who fought on four Pacific islands (Chapter 22, 1900 to 1950) 1571

Emanuel Ninger: sign painter who drew fake money by hand and was caught when his ink ran on a wet bar in 1896 (Chapter 17, 1850 to 1900) 1138

E. D. Nixon: retired Pullman porter who called the meeting that organized the 1955 Montgomery bus boycott and described the porters' work on film in 1982 (Chapter 11, 1900 to 1950) 761

Solomon Northup: a free New Yorker kidnapped and held in slavery in Louisiana for twelve years, who named the people who did it (Chapter 24, 1800 to 1850) 1707

Rebecca Nurse: a member of the Salem church found not guilty of witchcraft, then convicted and hanged (Chapter 21, The 1600s) 1420

Emma Nutt: the first woman telephone operator, hired in Boston in 1878 (Chapter 8, 1850 to 1900) 546

O

Jim Obergefell: widower whose case made same-sex marriage legal in every state (Chapter 25, 2000 to Today) 1916

Uriah and Mattie Oblinger: Nebraska homesteaders whose letters describe the food and work of a farm claim (Chapter 15, 1850 to 1900) 1010

Georgia O'Keeffe: painter shown by Alfred Stieglitz in 1916 who later lived and painted in New Mexico (Chapter 32, 1900 to 1950) 2662

Frederick Law Olmsted: the designer, with Calvert Vaux, of Central Park (Chapter 6, 1850 to 1900) 427

Pedro Omtua: young Tesuque runner who, with Nicolás Catua, carried the knotted cords of the Pueblo Revolt and was hanged by the Spanish (Chapter 31, The 1600s) 2511

Juan de Oñate: Spanish leader of the 1598 colony that came over land into New Mexico, who ordered the attack on Acoma Pueblo (Chapter 5, The 1500s) 338

Onesimus: enslaved African man who described inoculation to Cotton Mather (Chapter 26, 1700 to 1750) 1963

Karla Ortiz: concept artist who told US senators how AI image programs affected artists and sued AI companies (Chapter 8, 2000 to Today) 575

Osceola: Seminole war leader seized under a flag of truce in 1837 (Chapter 2, 1800 to 1850) 134

Jesse Owens: a runner and jumper who won four gold medals at the 1936 Olympic Games in Berlin (Chapter 36, 1900 to 1950) 3039

P

Thomas Paine: writer of *Common Sense* and *The American Crisis* (Chapter 31, 1750 to 1800) 2526

Paquiquineo (Don Luís): Chesapeake man who met the king of Spain, came home with a Spanish mission in 1570 and led an attack on it (Chapter 2, The 1500s) 104

Paquiquineo (Don Luís de Velasco): a young man from the Chesapeake who spent years among Spanish priests and came home with a Jesuit mission (Chapter 30, The 1500s) 2349

Gordon Parks: first African American staff photographer at *Life* magazine (Chapter 32, 1950 to 2000) 2683

Rosa Parks: NAACP secretary whose arrest set off the Montgomery bus boycott (Chapter 25, 1950 to 2000) 1876

Dolly Parton: Tennessee songwriter who refused to give Elvis Presley's manager half the rights to "I Will Always Love You" (Chapter 33, 2000 to Today) 2808

Matthew Patten: New Hampshire farmer, carpenter and judge whose diary records what his family grew, caught and ate (Chapter 15, 1750 to 1800) 1001

Alice Paul: organizer of the White House picket line, jailed and fed by force, author of the Equal Rights Amendment (Chapter 25, 1900 to 1950) 1833

Cristian Pavón Pineda: an 11-year-old sixth grader who died in his family's mobile home in Conroe, Texas, during the 2021 blackout. Early reports blamed the cold, but police later said an autopsy found carbon monoxide poisoning (Chapter 10, 2000 to Today) 716

Charles Willson Peale: painter of the Revolution's leaders, whom a Maryland planter paid for portraits with two enslaved people, Scarborough and Lucy (Chapter 32, 1750 to 1800) 2621

Robert Peary: Navy officer who said his party reached the North Pole in 1909, a claim still disputed (Chapter 1, 1900 to 1950) 75

John Stith Pemberton: Georgia druggist, addicted to morphine, who invented Coca-Cola (Chapter 28, 1850 to 1900) 2159

William Penn: the founder of Pennsylvania, a Quaker, who had Philadelphia planned on paper before it existed on the ground (Chapter 6, The 1600s) 403

Frances Perkins: witness to the Triangle fire who became the first woman in a US Cabinet and pushed for the first federal minimum wage (Chapter 14, 1900 to 1950) 947

Frances Perkins: the Secretary of Labor who chaired the committee that planned Social Security, and the first woman in a United States Cabinet (Chapter 16, 1900 to 1950) 1091

Philippe Petit: a French high-wire walker who crossed between the Twin Towers on a cable eight times (Chapter 12, 1950 to 2000) 828

Clarice Phelps: the Oak Ridge nuclear chemist and Navy veteran who helped purify the berkelium used to confirm element 117 (Chapter 13, 2000 to Today) 901

Elizabeth Phillips: licensed midwife whose gravestone says she delivered more than 3,000 babies in and around Boston (Chapter 26, 1700 to 1750) 1965

Susan La Flesche Picotte: first Native American woman to earn a medical degree, doctor for the Omaha Nation (Chapter 26, 1850 to 1900) 1989

Sengbe Pieh (Joseph Cinqué): a Mende man who led the *Amistad* captives in taking the ship and won their freedom in the Supreme Court (Chapter 24, 1800 to 1850) 1706

Zebulon Pike: Army lieutenant sent to explore the edges of the Louisiana Purchase, arrested by Spanish soldiers in 1807 (Chapter 1, 1800 to 1850) 50

Gifford Pinchot: first chief of the U.S. Forest Service, fired by President Taft in 1910 (Chapter 3, 1900 to 1950) 221

Eliza Lucas Pinckney: planter's daughter who ran her family's South Carolina plantations as a teenager and made the colony's first successful indigo dye. She took charge of the 20 enslaved people who worked Wappoo plantation (Chapter 15, 1700 to 1750) 999

Eliza Lucas Pinckney: ran her father's three South Carolina plantations from the age of 16 and directed the colony's first successful indigo dye crop, which enslaved workers grew and processed (Chapter 16, 1700 to 1750) 1059

Allan Pinkerton: a Scottish barrel maker who started a private detective agency in Chicago (Chapter 21, 1850 to 1900) 1448

George Washington Plunkitt: New York machine politician who explained to a reporter, from a shoeshine stand, how he made his money and kept his votes (Chapter 20, 1900 to 1950) 1357

Sidney Poitier: the first Black man to win the Academy Award for Best Actor (Chapter 34, 1950 to 2000) 2886

Charles Pollard: farmer kept in the Tuskegee study for 40 years who brought the men's lawsuit (Chapter 26, 1900 to 1950) 2000

Jackson Pollock: painter who dripped and poured paint onto canvas laid on the floor (Chapter 32, 1950 to 2000) 2680

Po'pay (Popé): Tewa religious leader who organized the Pueblo rising of 1680 (Chapter 2, The 1600s) 115

John Pory: secretary of the Virginia colony, who served as Speaker of the first elected assembly in English North America and wrote the record of what it did (Chapter 20, The 1600s) 1313

John Wesley Powell: one-armed Civil War officer who led the first recorded boat journey by white men through the whole Grand Canyon (Chapter 1, 1850 to 1900) 69

Elvis Presley: white rock and roll singer whose manager took 25 and then 50 percent of his income (Chapter 33, 1950 to 2000) 2782

Joseph Priestley: the English chemist credited with discovering oxygen, who spent his last ten years in Pennsylvania (Chapter 13, 1750 to 1800) 857

John Punch: the first African known to have been sentenced by a Virginia court to serve for life (Chapter 24, The 1600s) 1673

Q

Quack: enslaved man in New York City, sentenced by Judge Horsmanden in the trials of 1741 and burned at the stake with Cuffee (Chapter 21, 1700 to 1750) 1426

Ke Huy Quan: a child actor from a refugee family who quit acting and came back to win an Academy Award (Chapter 34, 2000 to Today) 2898

Fray Cristóbal de Quiñones: Franciscan friar who put an organ in the church at San Felipe Pueblo and trained its singers (Chapter 33, The 1600s) 2721

Selena Quintanilla: Mexican American Tejano singer killed at 23 by the woman who ran her fan club (Chapter 33, 1950 to 2000) 2793

R

The Radium Girls: young women, among them Grace Fryer and Catherine Wolfe Donohue, poisoned by the radium paint they used on watch dials. Their supervisors told them to shape their brushes with their lips and said the paint was safe (Chapter 13, 1900 to 1950) 871

Joseph Hayne Rainey: man born to enslaved parents and forced to work on Confederate defenses, who became the first Black member of the US House and the first to preside over it (Chapter 20, 1850 to 1900) 1350

A. Philip Randolph: union leader who won the first contract between a major American company and a union led by Black workers (Chapter 14, 1900 to 1950) 958

A. Philip Randolph: union leader whose threatened marches won two presidents' orders against discrimination (Chapter 25, 1900 to 1950) 1863

Philip Ratcliffe: a servant whipped, his ears cut off, fined and banished for speaking against the Massachusetts government and the Salem church (Chapter 21, The 1600s) 1416

Bernice Johnson Reagon: Albany, Georgia, song leader of the civil rights movement and a SNCC Freedom Singer (Chapter 33, 1950 to 2000) 2787

Conrad Reed: the 12-year-old who found the first gold in the United States that an expert confirmed was real (Chapter 13, 1750 to 1800) 858

Philip Reid: an enslaved foundry worker who worked out how to take apart the model of the Statue of Freedom, free by the day it was raised (Chapter 12, 1850 to 1900) 808

Minnie Lee and Mary Alice Relf: Black sisters in Alabama, aged 12 and 14, whom surgeons sterilized through a federally paid clinic, operating on them so that they could never have children (Chapter 26, 1950 to 2000) 2019

Hiram Revels: Mississippi minister who in 1870 became the first Black United States senator (Chapter 24, 1850 to 1900) 1730

The Revis family: Bethel Baptist deacon James Revis and his daughters, whose porch became the guard post for a bombed church (Chapter 29, 1950 to 2000) 2307

Margie Richard: Norco, Louisiana, schoolteacher who led the Old Diamond campaign that made Shell pay to move the families (Chapter 3, 2000 to Today) 244

Kevin Richardson: one of the Exonerated Five, New York teenagers convicted in 1990 of a Central Park rape and cleared in 2002 (Chapter 21, 2000 to Today) 1486

Sally Ride: physicist, the first American woman in space (Chapter 1, 1950 to 2000) 90

Jacob Riis: a Danish immigrant and police reporter whose flash photographs showed how New York's poor lived (Chapter 6, 1850 to 1900) 423

John Riley: an Irish-born sergeant who led the San Patricios on the Mexican side and was whipped and branded (Chapter 22, 1800 to 1850) 1541

Mary Ring: Plymouth widow whose will and inventory record what one household owned (Chapter 7, The 1600s) 467

Mary Ring: Plymouth Colony widow whose list of belongings shows the colonists wore red, violet and blue (Chapter 35, The 1600s) 2917

Faith Ringgold: painter and quilt maker arrested for protesting at the Whitney Museum (Chapter 32, 1950 to 2000) 2689

David Rittenhouse: self-taught instrument maker and astronomer who timed Venus at Norriton in 1769 (Chapter 9, 1750 to 1800) 592

Barbara "Dusty" Roads: American Airlines stewardess who fought the rule that fired women at 32 or when they married (Chapter 11, 1950 to 2000) 768

Margo Robbins: Yurok woman who helped found the Cultural Fire Management Council in 2013 (Chapter 3, 2000 to Today) 242

Kimberly Rivers Roberts: New Orleans woman who filmed the Katrina flood from inside it (Chapter 5, 2000 to Today) 390

Sarah Roberts: a five-year-old Black girl put out of the Boston school nearest her home, whose father sued the city (Chapter 30, 1800 to 1850) 2391

Harriet Hanson Robinson: Lowell mill girl who led her room out on strike at 11 and later wrote a book about the mill girls (Chapter 14, 1800 to 1850) 927

Jackie Robinson: the first Black player in the white major leagues since the 1880s, from April 15, 1947 (Chapter 36, 1900 to 1950) 3043

John D. Rockefeller: founder of Standard Oil, which by 1880 controlled about 90 percent of American oil refining (Chapter 16, 1850 to 1900) 1083

John D. Rockefeller: built Standard Oil and set up the first large trust (Chapter 19, 1850 to 1900) 1257

John, Washington and Emily Roebling: the man who designed the Brooklyn Bridge, the son who took charge of building it, and the son's wife, who carried his instructions to the engineers for 11 years (Chapter 12, 1850 to 1900) 810

John Rolfe: the Jamestown colonist whose tobacco shipments to London began Virginia's export trade (Chapter 16, The 1600s) 1053

Franklin D. Roosevelt: president who lost key cases in the Supreme Court, tried to add justices to it and was elected four times (Chapter 20, 1900 to 1950) 1367

Theodore Roosevelt: president who put about 230 million acres under federal protection and camped in Yosemite with John Muir in 1903 (Chapter 3, 1900 to 1950) 217

Theodore Roosevelt: the president known as the trust-buster, whose lawyers won the Northern Securities case (Chapter 19, 1900 to 1950) 1272

Hannah Ropes: Civil War head nurse who reported a thieving steward to the Secretary of War and died of typhoid (Chapter 26, 1850 to 1900) 1985

Tabatha Rosproy: a preschool teacher named National Teacher of the Year in the spring of 2020, when her classroom was shut by the pandemic (Chapter 30, 2000 to Today) 2488

Clyde Ross: veteran refused a mortgage who bought his Chicago house on contract and became vice president of the Contract Buyers League (Chapter 7, 1950 to 2000) 500

Benjamin Roth: a Youngstown, Ohio, lawyer who kept a diary of the Great Depression for more than ten years (Chapter 16, 1900 to 1950) 1088

Alexandria Rubalcaba: a background actor scanned into a digital copy with no word on how it would be used (Chapter 34, 2000 to Today) 2902

Benjamin Rush: Philadelphia doctor who bled and purged patients in the 1793 yellow fever epidemic and wrongly said Black people could not catch it (Chapter 26, 1750 to 1800) 1973

Benjamin Rush: Philadelphia doctor and signer of the Declaration of Independence who wrote that heavy drinking was a disease (Chapter 28, 1750 to 1800) 2142

Bayard Rustin: planned the March on Washington, convicted in 1953 for sex with men, pardoned in 2020 (Chapter 25, 1950 to 2000) 1891

Babe Ruth: a pitcher and home-run hitter for Boston and New York who grew up in a reform school (Chapter 36, 1900 to 1950) 3036

S

Sacagawea: Lemhi Shoshone woman who interpreted for Lewis and Clark while carrying her baby son (Chapter 1, 1800 to 1850) 47

Harpreet Singh Saini: son of a woman killed at the Oak Creek gurdwara, who asked the Senate to count crimes against Sikhs (Chapter 29, 2000 to Today) 2327

Yusef Salaam: one of the Exonerated Five, New York teenagers convicted in 1990 of a Central Park rape and cleared in 2002 (Chapter 21, 2000 to Today) 1486

Jonas Salk: doctor who made the first polio vaccine and did not patent it (Chapter 26, 1950 to 2000) 2008

Deborah Sampson: a woman who served 17 months as Robert Shurtleff and was wounded in combat (Chapter 22, 1750 to 1800) 1531

Sir Edwin Sandys: ran the Virginia Company while its officers shipped thousands of settlers to a colony where most of them died (Chapter 19, The 1600s) 1238

Raymond Santana: one of the Exonerated Five, New York teenagers convicted in 1990 of a Central Park rape and cleared in 2002 (Chapter 21, 2000 to Today) 1486

Clarence Saunders: the Memphis grocer who started self-service shopping with Piggly Wiggly and lost the company in a fight over its stock (Chapter 18, 1900 to 1950) 1207

Augusta Savage: Harlem sculptor and teacher whose 16-foot sculpture *The Harp* was bulldozed after the 1939 World's Fair (Chapter 32, 1900 to 1950) 2667

Levi Savage: a sub-captain of the Willie handcart company, one of the leaders under the captain, who warned against leaving late in 1856 and kept a journal (Chapter 5, 1850 to 1900) 366

Carl Schurz: German Forty-Eighter who became a Union general, the first German-born senator and Secretary of the Interior (Chapter 4, 1850 to 1900) 285

Dred and Harriet Scott: an enslaved couple whose freedom suit ended in the Supreme Court ruling that no Black American could be a citizen (Chapter 24, 1850 to 1900) 1715

Glenn Seaborg: the chemist whose teams found plutonium and nine more elements, and the first person to see an element named for him while he was alive (Chapter 13, 1900 to 1950) 885

Richard Sears: the railroad station agent who sold watches and built the Sears catalog (Chapter 18, 1850 to 1900) 1201

Emilio Segrè: Italian physicist who lost his job under Mussolini's anti-Jewish laws and stayed at Berkeley (Chapter 9, 1900 to 1950) 624

Emilio Segrè: the Italian physicist who stayed in the United States after Mussolini's officials barred Jews from university jobs, and who helped find technetium and astatine (Chapter 13, 1900 to 1950) 884

Sequoyah: Cherokee silversmith who could not read any language and made a way to write Cherokee (Chapter 2, 1800 to 1850) 129

Andy Serkis: the English actor whose performance-capture work as Gollum started an argument about what counts as acting (Chapter 34, 2000 to Today) 2891

William J. Seymour: Black preacher who led the Azusa Street revival where Pentecostal worship began (Chapter 29, 1900 to 1950) 2292

Ella Sheppard: born into slavery, pianist and arranger of the Fisk Jubilee Singers (Chapter 33, 1850 to 1900) 2751

Amy Sherald: painter of Michelle Obama's official portrait and of Breonna Taylor, who cancelled her National Portrait Gallery show in 2025 (Chapter 32, 2000 to Today) 2697

Mélisande Short-Colomb: a descendant of two of the GU272, the 272 enslaved people the Maryland Jesuits sold in 1838, in part to keep Georgetown College going. She enrolled at Georgetown University at 63 (Chapter 24, 2000 to Today) 1747

Amelia Simmons: author of *American Cookery*, the first cookbook written by an American for American kitchens (Chapter 15, 1750 to 1800) 1003

Upton Sinclair: novelist whose book about the Chicago meatpacking plants came out months before the 1906 food-safety laws (Chapter 15, 1900 to 1950) 1016

Benjamin "Pap" Singleton: man born into slavery who organized Black Southerners to move to Kansas, the "Father of the Exodus" (Chapter 5, 1850 to 1900) 373

Sitting Bull: Hunkpapa Lakota leader and holy man killed by Indian police at Standing Rock in 1890 (Chapter 2, 1850 to 1900) 140

Samuel Slater: English mill worker who rebuilt England's spinning machines from memory at Pawtucket and hired children to run them (Chapter 8, 1750 to 1800) 534

Chris Smalls: fired Amazon warehouse worker who organized the first union at an Amazon warehouse in the United States (Chapter 14, 2000 to Today) 978

Robert Smalls: an enslaved harbor pilot who sailed a Confederate ship out of Charleston to freedom and later served five terms in Congress (Chapter 24, 1850 to 1900) 1719

John Smibert: Scottish portrait painter in Boston whose studio of copies trained younger painters (Chapter 32, 1700 to 1750) 2616

Bessie Smith: Black blues singer paid $125 a side, with no royalty, for a record that sold 780,000 copies (Chapter 33, 1900 to 1950) 2762

Henry Smith: a Black man tortured and burned to death by a white mob in Paris, Texas, with no trial (Chapter 21, 1850 to 1900) 1452

Jedediah Smith: fur trapper who led the first party of US citizens known to reach California over land (Chapter 1, 1800 to 1850) 56

Joseph Smith: founder of the Church of Jesus Christ of Latter-day Saints, killed by a militia mob at Carthage in 1844 (Chapter 29, 1800 to 1850) 2264

Kemba Smith: a college student sentenced to 24 and a half years in federal prison in a crack cocaine case, later freed and pardoned (Chapter 21, 1950 to 2000) 1479

James Smithson: English chemist who left his fortune to found the Smithsonian without ever visiting the country (Chapter 9, 1800 to 1850) 600

Pierre Solon: immigrant father from Haiti who, with Katty Familia, lost the family's Boston house to foreclosure in 2009 (Chapter 7, 2000 to Today) 506

John Philip Sousa: leader of the Marine Band and writer of "The Stars and Stripes Forever" (Chapter 33, 1850 to 1900) 2754

Press Clay Southworth: 14-year-old who shot the last confirmed wild passenger pigeon in 1900 (Chapter 3, 1900 to 1950) 222

Frances Bilas Spence: one of the six women who programmed ENIAC (Chapter 8, 1900 to 1950) 559

Frank Sprague: engineer whose 1888 Richmond streetcar system showed that electric streetcars could work in a city (Chapter 11, 1850 to 1900) 755

Squanto (Tisquantum): Patuxet man who taught the Plymouth colonists to plant corn after an English captain kidnapped him (Chapter 15, The 1600s) 993

Loara Standish: Plymouth Colony girl thought to have stitched the oldest known American sampler (Chapter 32, The 1600s) 2609

Elizabeth Cady Stanton: called the Seneca Falls convention and wrote the Declaration of Sentiments (Chapter 25, 1800 to 1850) 1792

George Swanson Starling: Florida fruit picker who fled to Harlem in 1945 when grove owners plotted to hang him (Chapter 5, 1900 to 1950) 382

Lilian St. Cyr (Red Wing): a Ho-Chunk actress who played the leading woman in the first Western made in Hollywood (Chapter 34, 1900 to 1950) 2865

Cecile Steele: Delaware farmer who began the business of raising chickens for meat after a hatchery sent her 500 chicks by mistake (Chapter 15, 1900 to 1950) 1021

John Stewart: a Scottish prisoner of war, sold into forced labor, who bought his freedom on credit and became Springfield's blacksmith (Chapter 18, The 1600s) 1176

Joseph Whiting Stock: travelling portrait painter who could not walk and kept a journal of 900 paintings (Chapter 32, 1800 to 1850) 2630

Carl Stokes: the first Black mayor of a major American city, elected in Cleveland in 1967 (Chapter 6, 1950 to 2000) 449

Carl Stotz: the Williamsport man who organized the first Little League games in 1939 (Chapter 36, 1900 to 1950) 3041

Joseph Strauss: the chief engineer of the Golden Gate Bridge, who hung a safety net under it that caught 19 men (Chapter 12, 1900 to 1950) 823

Levi Strauss: San Francisco merchant who shared the 1873 riveted-pants patent with Jacob Davis (Chapter 35, 1850 to 1900) 2945

George Templeton Strong: a young New York lawyer whose diary, kept for forty years, records the coming of Croton water (Chapter 6, 1800 to 1850) 417

Gilbert Stuart: painter of the unfinished Washington portrait on the one-dollar bill (Chapter 32, 1750 to 1800) 2626

Mary Swift: a seventeen-year-old in the first class of the first state school for teachers, who kept a journal of it (Chapter 30, 1800 to 1850) 2380

Taylor Swift: songwriter who recorded four albums again after a music manager bought her masters (Chapter 33, 2000 to Today) 2800

Jack Swigert: Apollo 13 astronaut who took the place of Ken Mattingly (Chapter 1, 1950 to 2000) 87

T

Henry Ossawa Tanner: Black painter who studied under Thomas Eakins and made his career in France (Chapter 32, 1850 to 1900) 2657

Joseph Tape: Chinese American father who sued the San Francisco Board of Education (Chapter 25, 1850 to 1900) 1819

Mary Tape: Chinese American mother who sued for her daughter's place in school and wrote to the school board (Chapter 25, 1850 to 1900) 1819

Franklin Tarbell: oil producer who fought Standard Oil from 1872 into the 1890s, father of Ida Tarbell (Chapter 19, 1850 to 1900) 1260

Ida Tarbell: reporter whose history of Standard Oil showed how John D. Rockefeller took over the oil business (Chapter 19, 1900 to 1950) 1271

Susie King Taylor: a girl who learned to read in secret schools and at 14 became the first Black teacher of freed children in a school working openly in Georgia (Chapter 24, 1850 to 1900) 1727

Tecumseh: Shawnee leader who argued that no single nation could sell land that belonged to all of them (Chapter 2, 1800 to 1850) 128

Ruth Lichterman Teitelbaum: one of the six women who programmed ENIAC (Chapter 8, 1900 to 1950) 559

Nikola Tesla: Serbian immigrant whose induction motor ran on alternating current (Chapter 8, 1850 to 1900) 551

Nikola Tesla: the inventor whose AC motors George Westinghouse bought the right to use in 1888 (Chapter 10, 1850 to 1900) 687

Anthony Thacher: colonist who, with his wife, alone survived a boat wreck in the hurricane of 1635 and wrote it down (Chapter 27, The 1600s) 2043

Marie Tharp: geologist and mapmaker who found the rift valley in the Mid-Atlantic Ridge (Chapter 9, 1950 to 2000) 637

Florence Owens Thompson: farmworker of Cherokee descent, the woman in Dorothea Lange's "Migrant Mother" (Chapter 5, 1900 to 1950) 380

Hugh Thompson Jr.: the Army helicopter pilot who landed between American soldiers and Vietnamese villagers at My Lai to stop the soldiers from killing unarmed villagers (Chapter 22, 1950 to 2000) 1582

Henry David Thoreau: writer and land surveyor who lived at Walden Pond and wrote "in Wildness is the preservation of the world" (Chapter 3, 1800 to 1850) 199

Jim Thorpe: Sac and Fox athlete and Carlisle student, the first Native American to win an Olympic gold medal (Chapter 2, 1900 to 1950) 155

Emmett Till: a 14-year-old Black boy from Chicago kidnapped and murdered in Mississippi, whose killers an all-white jury set free (Chapter 21, 1950 to 2000) 1469

Pat Tillman: a football player who joined the Army Rangers and was killed by soldiers of his own platoon (Chapter 22, 2000 to Today) 1589

Elizabeth Timothy: widow who ran and then owned the *South-Carolina Gazette* (Chapter 31, 1700 to 1750) 2517

Mary Beth Tinker: 13-year-old whose black armband led to the 1969 Supreme Court ruling on students' rights (Chapter 35, 1950 to 2000) 2963

Tom, Peter, Ben, Harry and Daniel: five enslaved carpenters who helped build the White House, known only by their first names on the carpenters' payroll for May 1795 (Chapter 12, 1750 to 1800) 801

Pierre Toussaint: man brought enslaved from Saint-Domingue who became a leading New York hairdresser and was freed in 1807 (Chapter 4, 1750 to 1800) 273

Toypurina: Tongva woman who helped plan an attack on Mission San Gabriel in 1785 (Chapter 29, 1750 to 1800) 2255

Tung Trinh: survivor of a Vietnamese refugee boat that drifted 37 days in 1988 (Chapter 4, 1950 to 2000) 317

Nicholas Trist: American diplomat who made peace with Mexico after President Polk ordered him home (Chapter 23, 1800 to 1850) 1623

Sojourner Truth: a woman who walked away from slavery in New York and sued to get back her son, sold south against the law (Chapter 24, 1800 to 1850) 1709

Sojourner Truth: woman born into slavery in New York who spoke for Black people's and women's rights for over thirty years (Chapter 25, 1850 to 1900) 1796

Harriet Tubman: a woman who escaped slavery in Maryland and went back about 13 times to lead about 70 people out (Chapter 24, 1800 to 1850) 1710

Rosina Tucker: porter's wife who worked to build the porters' union and tells their story in the 1982 film *Miles of Smiles, Years of Struggle* (Chapter 11, 1900 to 1950) 761

Frederic Tudor: Boston merchant called the Ice King, who shipped New England pond ice to warm places around the world (Chapter 15, 1800 to 1850) 1005

Nat Turner: an enslaved preacher who led the deadliest slave revolt in United States history (Chapter 24, 1800 to 1850) 1705

Renica Turner: an Amazon delivery driver in the California desert who joined a union to push for protection from the heat (Chapter 18, 2000 to Today) 1229

Mark Twain (Samuel Clemens): Mississippi River steamboat pilot before he wrote books (Chapter 11, 1850 to 1900) 751

Mark Twain: writer whose books were sold door to door (Chapter 32, 1850 to 1900) 2656

Francisco Tzul: Guatemalan garment worker in Los Angeles who told California lawmakers about piece pay (Chapter 35, 2000 to Today) 2975

V

José Luis Vaello Madero: man who moved from New York to Puerto Rico, where members of Congress have not made his federal payment available. Lawyers for the United States government sued him to get back $28,081 in payments that kept coming after the move (Chapter 23, 2000 to Today) 1662

Wayne Valliere: Ojibwe builder of birchbark canoes who teaches the craft and the Ojibwe language at Lac du Flambeau, Wisconsin (Chapter 8, Before 1500) 514

Cornelius Vanderbilt: ferryman who moved his money into railroads and left the largest American fortune of his time (Chapter 11, 1850 to 1900) 750

Jesse Vazquez: retired veteran in Puerto Rico who lived 96 days without electricity after Hurricane Maria with his mother (Chapter 27, 2000 to Today) 2110

Philip Vigol: one of the first two Americans found guilty of treason, for attacking whiskey-tax officers (Chapter 28, 1750 to 1800) 2145

Paul Volcker: chairman of the Federal Reserve who stopped the Great Inflation by letting interest rates rise past 20 percent (Chapter 17, 1950 to 2000) 1158

W

Ann Wager: the only teacher the Williamsburg Bray School ever had (Chapter 30, 1750 to 1800) 2372

Madam C. J. Walker: Black businesswoman who built a hair-care company and trained thousands of women to sell her products (Chapter 35, 1900 to 1950) 2952

Moses Fleetwood Walker: a catcher, the first major-league player known to the public as a Black man (Chapter 36, 1850 to 1900) 3022

Glenna Wallace: the first woman to lead the Eastern Shawnee Tribe of Oklahoma, chief from 2006 to 2026, who fought to get a golf course off the Octagon Earthworks (Chapter 12, Before 1500) 787

Don Walsh: Navy lieutenant who, with Jacques Piccard, made the first dive to the deepest point in the ocean (Chapter 1, 1950 to 2000) 83

Sam Walton: the Arkansas storekeeper who put discount stores in small towns and built Walmart (Chapter 18, 1950 to 2000) 1217

John Wanamaker: the Philadelphia merchant credited with spreading the price tag and the money-back promise (Chapter 18, 1850 to 1900) 1195

Aaron Montgomery Ward: the traveling salesman who started selling to farm families by mail from a printed catalog (Chapter 18, 1850 to 1900) 1200

Andy Warhol: painter of soup cans, dollar bills and famous faces (Chapter 32, 1950 to 2000) 2685

Mercy Otis Warren: a Massachusetts writer who attacked the royal government in five unsigned plays (Chapter 34, 1750 to 1800) 2839

Booker T. Washington: born enslaved, he founded the school at Tuskegee and ran it for 34 years (Chapter 30, 1850 to 1900) 2412

George Washington: first president, elected twice with every electoral vote, who chose to leave office after two terms. He was an enslaver who owned 123 people at Mount Vernon (Chapter 20, 1750 to 1800) 1326

George Washington: Virginia officer in the French and Indian War and commander of the Continental Army through the Revolution (Chapter 22, 1750 to 1800) 1529

George Washington: first president, who took the oath in brown wool woven in Connecticut (Chapter 35, 1750 to 1800) 2933

Alice Waters: cook who built her restaurant around food from nearby farms and started a school garden (Chapter 15, 1950 to 2000) 1036

Noah Webster: a schoolteacher who wrote the spelling book American children learned from for more than a hundred years (Chapter 30, 1750 to 1800) 2374

Rainer Weiss: MIT physicist and child refugee from Nazi Germany who worked out the idea for the LIGO detectors (Chapter 9, 2000 to Today) 653

Ida B. Wells: Memphis editor who counted lynchings and was driven out of the South for it (Chapter 25, 1850 to 1900) 1824

Ida B. Wells: Memphis editor who counted lynchings, printed the count and was driven from the city (Chapter 31, 1850 to 1900) 2553

George Westinghouse: the businessman who sold alternating current and won the jobs of lighting the 1893 Chicago world's fair and building the Niagara Falls generators (Chapter 10, 1850 to 1900) 687

Phillis Wheatley: a girl kidnapped from West Africa who became the first African American to publish a book of poetry (Chapter 24, 1750 to 1800) 1685

Phillis Wheatley: first African American to publish a book of poems, enslaved in Boston when it came out (Chapter 32, 1750 to 1800) 2624

John White: English artist whose 1585 Roanoke watercolors are the oldest surviving European pictures made here, later governor of the 1587 colony (Chapter 32, The 1500s) 2606

Ryan White: Indiana teenager with HIV from his hemophilia medicine who was kept out of school and spoke across the country about AIDS (Chapter 26, 1950 to 2000) 2021

George Whitefield: English preacher of the Great Awakening who pushed to make slavery legal in Georgia (Chapter 29, 1700 to 1750) 2236

Eli Whitney: builder of the cotton gin, patented in 1794, and the subject of a myth about interchangeable gun parts (Chapter 8, 1750 to 1800) 532

Jeffrey Wigand: tobacco company research chief who told the public his company changed its tobacco to strengthen the effect of nicotine (Chapter 28, 1950 to 2000) 2188

Henry Wight: minister and Revolutionary War veteran who started the Fourth of July celebration in Bristol, Rhode Island, which the town still holds (Chapter 37, 1750 to 1800) 3096

Laura Ingalls Wilder: author of the Little House books, whose family moved by covered wagon again and again (Chapter 5, 1850 to 1900) 370

Charles Wilkes: Navy lieutenant who led the United States Exploring Expedition, whose men burned villages in Fiji (Chapter 1, 1800 to 1850) 59

Will: enslaved man who put out the fire on St. Philip's Church roof and whom the church's leaders bought and freed (Chapter 27, 1750 to 1800) 2060

Bert Williams: a Black comedian who wore blackface and starred in an otherwise all-white Broadway show (Chapter 34, 1900 to 1950) 2867

Cathay Williams: a woman born enslaved who served two years as William Cathay and was refused a pension (Chapter 22, 1850 to 1900) 1556

Charles Williams: the first prisoner at Eastern State Penitentiary (Chapter 21, 1800 to 1850) 1438

Eunice Kanenstenhawi Williams: John Williams's daughter, taken at seven, who became Kanien'kehaka (Mohawk) at Kahnawake and refused to come back (Chapter 22, 1700 to 1750) 1518

John Williams: Deerfield's minister, captured in the 1704 raid, who wrote the best-known captivity story of colonial America (Chapter 22, 1700 to 1750) 1517

Roger Williams: minister thrown out of Massachusetts who founded Providence, where people of every faith lived as neighbors (Chapter 29, The 1600s) 2224

Woodrow Wilson: president who proposed the League of Nations and sent the Marines into Haiti (Chapter 23, 1900 to 1950) 1642

Melvin Windsor: Park Police officer who stood on a helicopter skid to pull the last Air Florida survivor from the Potomac (Chapter 27, 1950 to 2000) 2100

Sarah Winnemucca: Northern Paiute interpreter, lecturer and writer of the first known autobiography by a Native American woman (Chapter 2, 1850 to 1900) 145

Sarah Winnemucca: Northern Paiute interpreter who campaigned against her people's forced removal (Chapter 25, 1850 to 1900) 1813

Korey Wise: one of the Exonerated Five, tried as an adult at 16 and locked up about 13 years until another man's confession and DNA cleared all five in 2002 (Chapter 21, 2000 to Today) 1486

Reid Wiseman: NASA commander of Artemis II, the first flight to take people to the Moon since 1972 (Chapter 1, 2000 to Today) 95

Alice Wong: disabled organizer who founded the Disability Visibility Project (Chapter 25, 2000 to Today) 1933

Anna May Wong: often called the first Chinese American film star, who protested the parts studios gave her (Chapter 34, 1900 to 1950) 2874

Wong Kim Ark: San Francisco-born cook whose Supreme Court case made everyone born in the United States a citizen (Chapter 4, 1850 to 1900) 297

Granville T. Woods: Black inventor who patented a telegraph for messages between moving trains and stations (Chapter 8, 1850 to 1900) 550

Bob Woodward: *Washington Post* reporter who, with Carl Bernstein, tied the Watergate burglary to President Nixon's White House (Chapter 31, 1950 to 2000) 2580

F. W. Woolworth: the storekeeper who built a national chain of five-cent and ten-cent stores and paid cash for the tallest building in the world (Chapter 18, 1900 to 1950) 1205

George Worthylake: the first keeper of Boston Light, who drowned with his family in 1718 (Chapter 12, 1700 to 1750) 796

Steve Wozniak: cofounder of Apple Computer who designed the Apple I and the Apple II (Chapter 8, 1950 to 2000) 567

Orville Wright: bicycle builder from Dayton who made the first powered flight in 1903 and survived the 1908 crash that killed Thomas Selfridge (Chapter 8, 1900 to 1950) 554

Orville and Wilbur Wright: the brothers who flew their powered airplane four times on December 17, 1903 (Chapter 11, 1900 to 1950) 757

Wilbur Wright: bicycle builder from Dayton who, with his brother, flew the first powered airplane in 1903 (Chapter 8, 1900 to 1950) 554

Y

Jack Yates: Baptist minister, born enslaved, who led Black Houstonians in buying land for their Juneteenth celebrations (Chapter 37, 1850 to 1900) 3111

Pearl Yates: a country schoolteacher in western North Carolina who remembered when the lights first came on in her home in 1939 (Chapter 10, 1900 to 1950) 700

Al "Sonny" Yearick: boy who played for Lycoming Dairy in the first Little League season and later became the first Little Leaguer to play baseball for pay (Chapter 36, 1900 to 1950) 3041

Jonas and Frieda Yoder: Amish father and daughter whose case won the right to keep Amish children out of high school (Chapter 29, 1950 to 2000) 2315

York: Black man held as a slave by William Clark, who crossed the continent with the expedition and came home still a slave (Chapter 1, 1800 to 1850) 49

Brigham Young: Latter-day Saint leader of the first company to the Salt Lake Valley in 1847 (Chapter 5, 1800 to 1850) 360

Steve Young: Vlasic pickle executive who described selling to Walmart (Chapter 19, 1950 to 2000) 1289

Z

Babe Didrikson Zaharias: a track star, basketball player and golfer who won Olympic gold and helped found women's professional golf (Chapter 36, 1900 to 1950) 3038

Al Zampa: an ironworker who fell into the Golden Gate Bridge's net and lived, and who later had a bridge named for him (Chapter 12, 1900 to 1950) 823

Anna Catharina Zenger: kept the *New-York Weekly Journal* coming out while her husband was jailed, then ran the shop after he died (Chapter 31, 1700 to 1750) 2515

John Peter Zenger: Palatine boy who became a New York printer, jailed and found not guilty of seditious libel (Chapter 4, 1700 to 1750) 268

John Peter Zenger: New York printer jailed for more than eight months for printing attacks on the governor, then found not guilty by a jury (Chapter 20, 1700 to 1750) 1320

Zitkala-Ša: Yankton Dakota writer taken to a boarding school as a child, who later wrote against the schools (Chapter 2, 1850 to 1900) 147

Zitkala-Ša: a Yankton Dakota girl whose braids were cut at a boarding school in Indiana, and who wrote down what was done to her (Chapter 30, 1850 to 1900) 2410

Places to See

Museums, sites and archives the chapters name, grouped by state so you can find the ones near you. Then come places in more than one state, places in other countries, online collections, and last, other places whose state the chapters do not give. Places close and exhibits change, so check before you go.

Alabama

16th Street Baptist Church, Birmingham: where the children's march began, and where a bomb set by Ku Klux Klan members killed four girls in 1963 (Chapter 25) 1871

Africatown, Mobile: the town founded by survivors of the *Clotilda*, the last known slave ship to reach the United States. It has the church and school they started, and Lewis Quarters, one of the first pieces of land they bought (Chapter 24) 1712

Edmund Pettus Bridge, Selma: where state troopers attacked civil rights marchers with clubs and tear gas on Bloody Sunday, March 7, 1965. Television cameras filmed the attack (Chapter 25) 1871

Freedom Monument Sculpture Park, Montgomery: the National Monument to Freedom, with more than 122,000 last names from the 1870 census (Chapter 12) 832

Freedom Monument Sculpture Park, Montgomery: work by Kehinde Wiley (Chapter 32) 2695

National Memorial for Peace and Justice and the Legacy Museum, Montgomery: 805 steel blocks, one for each county where a lynching was recorded (Chapter 12) 832

Alaska

Castle Hill, Sitka: site of the 1867 handover of Alaska, on Tlingit land the Russians took from the Kiksadi clan in 1804 (Chapter 23) 1623

Arizona

Arizona State Museum, Tucson: the four Ancestral Pueblo flutes from Broken Flute Cave, made between 620 and 670 (Chapter 33) 2713

Café Gozhóó, Whiteriver, on the White Mountain Apache Reservation: Nephi Craig's café built around Apache foods, opened in 2021 (Chapter 15) 1039

Arkansas

Central High School, Little Rock: where nine Black students entered in 1957 (Chapter 25) 1871

Daisy Bates home, Little Rock: where the nine students met every school day (Chapter 25) 1871

California

50 UN Plaza, San Francisco: the federal offices disabled protesters occupied in 1977 (Chapter 25) 1871

88-Inch Cyclotron, Berkeley: a later cyclotron at Berkeley that still runs (Chapter 8) 553

Alcatraz Island, San Francisco Bay: held by Native activists from 1969 to 1971 (Chapter 25) 1871

Alfred Zampa Memorial Bridge, Carquinez Strait: the bridge named for the ironworker in 2003 (Chapter 12) 815

Angel Island Immigration Station, San Francisco Bay: the barracks whose wooden walls carry more than 200 poems carved by Chinese people held there (Chapter 4) 298

Azusa Street plaque, 312 Azusa Street, Los Angeles: marks the site of the 1906 to 1931 revival (Chapter 29) 2287

Beckwourth Riverfront Park, Marysville: the town's largest park, named in 1996 to recognize the debt Marysville still owed Jim Beckwourth (Chapter 1) 64

Chez Panisse, Berkeley: Alice Waters's restaurant, opened in 1971 (Chapter 15) 1028

Chicano Park, San Diego: murals on the bridge pillars, first painted in 1973 (Chapter 32) 2679

The Edible Schoolyard, Martin Luther King Jr. Middle School, Berkeley: a school garden where students grow food, cook it and eat it (Chapter 15) 1028

Golden Gate Bridge, San Francisco: the bridge opened in 1937 (Chapter 12) 815

Hetch Hetchy Valley, Yosemite National Park: under water behind the O'Shaughnessy Dam since 1923 (Chapter 3) 215

LA Plaza, Los Angeles: the 2012 plaque apologizing for the Mexican Repatriation (Chapter 4) 298

Mount Wilson Observatory: the Hooker telescope Hubble used (Chapter 9) 614

Museum of Contemporary Art, Los Angeles: Kara Walker's Unmanned Drone, made from a Stonewall Jackson statue (Chapter 32) 2695

Presidio of San Francisco, San Francisco: the Army post where officers kept Chinese survivors apart in 1906 (Chapter 27) 2078

Room 307, Gilman Hall, Berkeley: the room where plutonium was proved a new element, now a National Historic Landmark (Chapter 13) 868

Temple of the Forest Beneath the Clouds, Weaverville: Chinese Taoist temple of 1874, still in use, held by California State Parks (Chapter 29) 2274

Tule Lake: the prison camp where Japanese Americans harvested almost 30 crops and most field workers were paid $12 a month (Chapter 15) 1015

Yosemite Valley, Yosemite National Park: the Ahwahneechee homeland, and Wahhoga, the site of the last Native village, where a roundhouse was finished in 2025 (Chapter 3) 200

Colorado

Cliff Palace, Mesa Verde: the largest cliff dwelling in North America, built between about 1190 and 1280, with about 150 rooms and 23 kivas, round rooms for ceremonies (Chapter 12) 783

Mesa Verde: where Ancestral Pueblo potters made black-on-white pottery from about 600 to 1300 (Chapter 32) 2598

Sand Creek Massacre National Historic Site: the place of the 1864 massacre, looked after by National Park Service staff (Chapter 2) 134

Connecticut

Mashantucket Pequot Museum, Mashantucket: the Pequot Nation's museum and history of the Pequot War (Chapter 22) 1508

Mystic Seaport Museum, Mystic: the Charles W. Morgan, the last wooden whaling ship left in the world, on which Nelson Cole Haley sailed as a harpooner (Chapter 10) 677

Florida

Anhaica, Tallahassee: the Apalachee town de Soto's army seized in 1539. Archaeologists have found burned corncobs at the site (Chapter 15) 989

Castillo de San Marcos National Monument (Fort Marion), St. Augustine: the fort of coquina, a stone made of packed seashells, that Spanish builders began in 1672. Army officers held Native prisoners from the Plains there from 1875 to 1878 (Chapter 12) 789

Fort Mose Historic State Park, near St. Augustine: the site of Fort Mose, the first free Black town in what is now the United States to be recognized by law (Chapter 24) 1676

Fountain of Youth Park, St. Augustine: the remains of Seloy's Timucua town lie under it (Chapter 7) 463

Mass grave site and state historical marker, West Palm Beach: where Black victims of the 1928 hurricane were buried, marked in 2003 (Chapter 27) 2078

Matanzas Inlet, south of St. Augustine: where Menéndez's soldiers killed French prisoners who had given up, in 1565 (Chapter 23) 1599

Mound Key: the shell mounds of Calos, the Calusa capital (Chapter 2) 99

Pelican Island: the first federal bird reservation, named in 1903 (Chapter 3) 215

Plaza de la Constitucion, St. Augustine: the oldest public square in the country (Chapter 6) 398

St. Augustine: the oldest town Europeans founded on the US mainland, begun by the Spanish in September 1565 at Seloy, a Timucua village (Chapter 4) 258

St. Augustine National Cemetery, St. Augustine: the graves of Fort Marion prisoners (Chapter 32) 2643

Georgia

Andersonville: the Confederate prison camp where almost 13,000 Union soldiers died (Chapter 22) 1543

Fort Pulaski National Monument, near Savannah: a national park site whose exhibit showed a copy of the 1863 photograph of Peter's back, covered in scars from whipping. Peter had escaped slavery in Louisiana. In 2025 newspapers reported that officials ordered the photograph removed, and an Interior Department spokesperson said park sites were not asked to remove it (Chapter 24) 1744

Jekyll Island Club, Jekyll Island: where six men secretly wrote a plan for a central bank in November 1910 (Chapter 17) 1143

Hawaii

Bellows Beach site, Waimānalo, Oʻahu: the one early home site in Hawaii dated with confidence (Chapter 23) 1597

ʻIolani Palace, Honolulu: where officials of the Republic of Hawaii tried Queen Liliʻuokalani in her throne room in 1895 and held her prisoner in an upstairs bedroom (Chapter 23) 1623

Kealakekua Bay, island of Hawaii: where Captain Cook's sailors watched Hawaiians surf in 1779 (Chapter 36) 2999

Mauna Loa: where the Keeling Curve measurements have run since 1958 (Chapter 9) 633

USS Arizona, Pearl Harbor: the sunken ship, with more than 900 of its crew still inside (Chapter 22) 1559

Illinois

Cahokia, across the Mississippi from St. Louis: the largest town north of Mexico before Europeans came. Its remains are now Cahokia Mounds State Historic Site, on the World Heritage List. The site has Monks Mound, the Grand Plaza, the burial mound called Mound 72, and the places where the log wall, the Woodhenge circles of sunrise posts and a copper workshop stood (Chapter 6) 395

Chicago History Museum, Chicago: its Haymarket history counts eight police officers dead and an unknown number in the crowd killed or wounded (Chapter 14) 929

Chicago History Museum, Chicago: its count of the 163 items on Montgomery Ward's first catalog, a single sheet of paper he mailed from Chicago in 1872 (Chapter 18) 1193

Chicago's bungalows, Chicago: more than 80,000 brick bungalows built between 1910 and the mid-1930s (Chapter 7) 485

Field Museum, Chicago: covered display cases in two halls under the 2024 NAGPRA rule (Chapter 9) 648

Haymarket Square, Chicago: the square where the bomb was thrown at the police on May 4, 1886 (Chapter 14) 929

Henry Gerber House, Chicago: a National Historic Landmark, home of the founder of the Society for Human Rights (Chapter 25) 1829

Monks Mound, Cahokia: the largest structure made of earth in North and South America (Chapter 12) 783

Nauvoo: the town on the Mississippi River where the Latter-day Saints settled in 1839, on ground where a large Meskwaki village had stood. After a mob killed their founder, Joseph Smith, in 1844, they began leaving for the West in 1846. A state park is there today (Chapter 29) 2256

Newberry Library, Chicago: a research library whose staff write about the Spanish missions in New Mexico (Chapter 29) 2213

Kentucky

Kentucky Horse Park: Isaac Murphy's grave (Chapter 36) 3019

Louisiana

Congo Square, New Orleans: the place where enslaved people gathered to sing, dance and play music on Sunday afternoons (Chapter 33) 2736

LIGO detector, Livingston: one of the two detectors that first picked up gravitational waves, on September 14, 2015 (Chapter 9) 648

Poverty Point: five mounds and six curved ridges built 3,100 to 3,700 years ago, on the World Heritage List (Chapter 12) 783

Tate, Etienne and Prevost Center, New Orleans: the old McDonogh No. 19 school. On November 14, 1960, Leona Tate, Gail Etienne and Tessie Prevost, all six years old, walked in with federal marshals, among the first Black children in white public schools in Louisiana. The building reopened as the center on May 4, 2022 (Chapter 30) 2442

William Frantz Elementary School, New Orleans: a bronze statue of Ruby Bridges, unveiled on November 14, 2015 (Chapter 30) 2442

Maine

Great Works bridge, Brattle Street, South Berwick: the rocky gorge below the bridge where the 1634 sawmill stood (Chapter 10) 665

Unity College: 16 of the Carter White House solar panels, put back up to heat water for the cafeteria (Chapter 10) 702

Maryland

Catoctin Furnace: the iron furnace the Johnson brothers built, where at least 271 enslaved African Americans worked, and the cemetery of its African American workers, with burials going back to 1774 (Chapter 13) 855

Clara Barton National Historic Site, Glen Echo: Clara Barton's house, built partly from the lumber of the Red Cross buildings at Johnstown (Chapter 27) 2117

Fort McHenry, Baltimore: the fort the British shelled for 25 hours in 1814 (Chapter 22) 1534

Jefferson Patterson Park and Museum: its staff say people went on playing chunkey across North America after Cahokia declined around 1500 (Chapter 36) 2981

Maryland Center for History and Culture, Baltimore: Kühn's portrait of Henry Darnall III with the enslaved boy in a collar (Chapter 32) 2613

Most Holy Redeemer Cemetery, Baltimore: Henry Gunther's grave (Chapter 22) 1559

Mount Clare Museum House, Baltimore: the museum whose staff record the history of the Baltimore Iron Works and the enslaved people who worked there (Chapter 13) 852

Massachusetts

Boston Light, Little Brewster Island, Boston Harbor: the tower rebuilt in 1783 on the site of the first lighthouse (Chapter 12) 795

Bunker Hill Monument, Charlestown: the 221-foot granite obelisk on Breed's Hill (Chapter 12) 804

Cole's Hill, Plymouth: the hill above Plymouth Rock where the statue of Massasoit stands and the National Day of Mourning is held at noon every Thanksgiving. Its plaque comes from the 1998 agreement between the town and the Day of Mourning marchers, after Plymouth police arrested 25 of them in 1997 (Chapter 37) 3125

Cornhill, Cape Cod: where the *Mayflower* men dug up Nauset seed corn in November 1620 (Chapter 15) 991

Deer Island, Boston Harbor: where colonists held the Christian Native people of the praying towns as prisoners in the winter of 1675 to 1676, and many of them died of hunger and cold. Descendants hold a remembrance there every year (Chapter 30) 2351

Faneuil Hall, Boston: the hall where a 1744 concert raised 205 pounds for the poor of Boston, and where British officers staged the play *Zara* in 1775 (Chapter 33) 2723

Harvard Law School, Cambridge: Feke's portrait of the Royall family (Chapter 32) 2613

Lowell: the mill city the Boston Associates built at Pawtucket Falls on the Merrimack River, on the land of the Pennacook people. Lowell National Historical Park there is about its mills and the slave-grown cotton they spun (Chapter 14) 922

Massachusetts Archives: Joseph Jenks's patent petition, dated May 10, 1646 (Chapter 8) 517

Massachusetts Historical Society, Boston: keeps the letters of Abigail and John Adams (Chapter 25) 1767

Museum of Fine Arts, Boston: the two jars Dave made in 1857, one bought from his descendants and one lent by them (Chapter 32) 2619

Old North Church, Boston: the steeple where two lanterns were hung on April 18, 1775 (Chapter 12) 795

Old Ship Meetinghouse, Hingham: the only Puritan meetinghouse from the 1600s still standing (Chapter 12) 789

Old State House, Boston: the brick Town House rebuilt after the fire of 1711 (Chapter 27) 2047

Phipps Street Burying Ground, Charlestown: Elizabeth Phillips's gravestone, which says she "brought into this world above 3000 Children" (Chapter 26) 1961

Pilgrim Hall Museum, Plymouth: Loara Standish's sampler (Chapter 32) 2608

Pocumtuck Valley Memorial Association, Deerfield: Deerfield's history museum, which keeps the town's old records of the 1704 raid (Chapter 22) 1515

Post Office Square, Plymouth: a plaque that tells of Metacom, whose head the English put on a pole in that square and left there for 25 years (Chapter 37) 3125

Rebecca Nurse's house, Danvers: her house, which still stands (Chapter 21) 1412

Saugus Iron Works National Historic Site, Saugus: the Hammersmith ironworks, which ran from the 1640s until about 1670, rebuilt on its old site (Chapter 8) 517

Shaw Memorial, Boston Common, Boston: Saint-Gaudens's memorial to Shaw and the 54th Massachusetts (Chapter 32) 2643

Thacher Island, off Cape Ann: the island where Anthony Thacher and his wife came ashore in 1635 (Chapter 27) 2041

Walden Pond, near Concord: where Thoreau built his cabin and lived from 1845 to 1847. Workers also cut ice from the pond in winter (Chapter 3) 196

West End Museum, Boston: a history page on Edward Converse's ferry, which it calls the first on that crossing (Chapter 11) 726

West End Museum, Boston: a history of the Shawmut Peninsula and the Massachusett people, with the count of the 1616 to 1618 epidemic (Chapter 19) 1235

Worcester Art Museum, Worcester: the Freake portrait of Elizabeth and Mary Freake (Chapter 32) 2608

Minnesota

Ingalls dugout site, Plum Creek near Walnut Grove: the hollow in the creek bank where the Ingalls family's dugout stood, on the former Gordon farm (Chapter 7) 480

University of Minnesota: holds the Eugenics Record Office records (Chapter 9) 614

University of Minnesota St. Paul campus, St. Paul: four of the Rangh Court townhomes moved for Interstate 94 (Chapter 6) 443

Mississippi

Natchez City Cemetery, Natchez: Milla Granson's grave (Chapter 24) 1693

Missouri

Gateway Arch National Park, St. Louis: the 630-foot stainless steel arch. On July 14, 1964, Percy Green and Richard Daly climbed 125 feet up its unfinished north leg to protest that Black workers were kept out of the jobs building it (Chapter 12) 832

Missouri State Archives: records of the 1838 fighting between older Missouri settlers and armed Mormon groups (Chapter 5) 352

Sugarloaf Mound, St. Louis: the last Native mound in the city, now owned by the Osage Nation (Chapter 12) 807

Truman Library, Independence: where President Johnson signed Medicare and Medicaid in 1965 (Chapter 26) 2005

Montana

Glacier National Park: on land federal officials say the Blackfeet sold in 1895. Many Blackfeet say they only leased it (Chapter 3) 215

National Bison Range, Flathead Reservation: the bison refuge that members of Congress gave back to the Confederated Salish and Kootenai Tribes in 2020 (Chapter 3) 240

Nebraska

Homestead National Historical Park, near Beatrice: the Freeman School, a one-room brick school used from 1872 to 1967 (Chapter 30) 2392

Nebraska State Historical Society: keeps the letters of Uriah and Mattie Oblinger, a homestead family in Nebraska in the 1870s (Chapter 15) 1009

Nevada

Hoover Dam, near Boulder City: the dam dedicated in 1935 (Chapter 12) 815

University of Nevada, Las Vegas library, Las Vegas: keeps a history of Hoover Dam that tells how company doctors called tunnel workers' illness pneumonia (Chapter 14) 941

New Hampshire

Amoskeag Falls, on the Merrimack River: Pennacook fishing place where Matthew Patten fished in 1767 (Chapter 15) 1000

New Jersey

Arts Council of Princeton, Princeton: Nathan Jackson's Raven and Fog Woman pole, in the lobby (Chapter 32) 2598

Falls of the Passaic River, Paterson: where Hamilton's Society for Establishing Useful Manufactures started a factory town (Chapter 16) 1062

Nothnagle Log House, Greenwich Township: a log house built in about 1638 to 1643 that still stands (Chapter 7) 471

Rutgers Oral History Archives: Donald Lundberg's 2008 interview about shopping under rationing (Chapter 18) 1204

New Mexico

Acoma Pueblo: the Acoma people's town on its mesa. Spanish soldiers under Vicente de Zaldívar attacked it in January 1599 and killed men, women and children. Juan de Oñate then sentenced the survivors over 12 to twenty years of forced work, the men over 25 to have a foot cut off as well, and the old people to slavery (Chapter 2) 102

Chaco Canyon, northwestern New Mexico: the stone great houses of the Ancestral Puebloans, begun in the 800s and lined up with the sun and moon, and their planned roads. Casa Rinconada there is a great kiva, a round room for ceremonies, about 63 feet across (Chapter 29) 2203

El Camino Real de Tierra Adentro National Historic Trail: the road from Mexico City to northern New Mexico that Oñate's colonists traveled north in 1598 (Chapter 5) 336

Fajada Butte, Chaco Canyon: the Sun Dagger slabs and carved spirals (Chapter 9) 581

Gila Wilderness: the first area in the world officially named a wilderness area (Chapter 3) 215

Palace of the Governors, Santa Fe: on the plaza, used as a public building without a break for longer than any other building in the United States (Chapter 6) 399

Pueblo Bonito, Chaco Canyon: the largest great house, four stories high, where jars from about 1000 to 1125 held traces of a drink made from cacao, the bean chocolate comes from (Chapter 12) 783

San Esteban del Rey, Acoma Pueblo: the oldest European church in New Mexico that still stands, built by Acoma workers (Chapter 32) 2608

San Miguel Chapel, Santa Fe: an adobe chapel on foundations from about 1620 (Chapter 12) 789

Taos Pueblo, Taos: the North House and the South House, five stories of adobe built between about 1000 and 1450, where people have lived for more than 1,000 years (Chapter 7) 460

Volcanic cliffs west of Albuquerque, Petroglyph National Monument: about 24,000 to 25,000 petroglyphs, pictures that Native artists cut or pecked into the rock (Chapter 32) 2598

White Sands National Park: human footprints at least 21,000 years old (Chapter 1) 17

New York

1520 Sedgwick Avenue, the Bronx: the building city officials named the birthplace of hip-hop (Chapter 33) 2779

Arthur Ashe Stadium, New York City: where the first Fortnite World Cup final was played in 2019 (Chapter 36) 3058

Brooklyn Bridge, New York: the suspension bridge opened in 1883 (Chapter 12) 807

Brooklyn Museum, Brooklyn: Judy Chicago's The Dinner Party (Chapter 32) 2679

Castle Clinton, Battery Park, New York: the site of Castle Garden, the first official immigrant depot, 1855 to 1890 (Chapter 4) 283

Central Park, New York: the site of Seneca Village, where about 225 people lived by 1855, most of them African American. City officials forced them out by 1857 to make the park (Chapter 6) 413

Cold Spring Harbor Laboratory: its archive website publishes the Eugenics Record Office history (Chapter 9) 614

Columbia University: Charity Clarke's letters to her cousin Joseph Jekyll (Chapter 35) 2928

Cooper Union, New York: the hall where Clara Lemlich called the shirtwaist makers out on strike in 1909 (Chapter 14) 941

East Hampton Library: Robert Townsend's account books of 1773 to 1785, copied out and published by its staff (Chapter 18) 1182

Ellis Island, New York Harbor: the immigration station where about 12 million immigrants came through from 1892 to 1954. Its main building has been a museum of immigration since 1990, and a statue of Annie Moore, the first immigrant to come through it, stands there (Chapter 4) 267

Empire State Building, New York: the 102-story building finished in 1931 (Chapter 12) 815

Lewis Latimer House Museum, East Flushing, Queens: the house Lewis Latimer lived in from 1903 (Chapter 8) 541

Metropolitan Museum of Art, New York: the Temple of Dendur, in the wing where Nan Goldin's group protested in 2018 and that no longer carries the Sackler name (Chapter 28) 2190

Metropolitan Museum of Art, New York: a signed painting by Prince Demah, and Cole's The Oxbow (Chapter 32) 2619

Morgan Library & Museum, New York City: an exhibition on the Panic of 1907 that describes Morgan's locked-door meeting (Chapter 17) 1143

Museum of Modern Art, New York: the even-numbered panels of Jacob Lawrence's The Migration Series (Chapter 32) 2660

National Baseball Hall of Fame, Cooperstown: Josh Gibson's plaque, which says he hit "almost 800" home runs (Chapter 36) 3072

National September 11 Memorial and Museum, New York: two pools with the names of 2,983 people, the people killed on September 11, 2001, and six killed in a 1993 bombing of the towers. The museum has an exhibit about Mohawk ironworkers called "Skywalkers" (Chapter 12) 832

New York Public Library, New York: Le Moyne's painting Laudonnière and Athore, 1564 (Chapter 32) 2605

One World Trade Center, New York: the tower 1,776 feet to the tip of its spire (Chapter 12) 832

Onondaga Nation, central New York: keeps the wampum records of the Haudenosaunee (Chapter 31) 2500

St. Patrick's Cathedral, New York: the crypt where Pierre Toussaint's remains lie (Chapter 4) 271

Statue of Liberty, New York Harbor: the statue, with Emma Lazarus's poem mounted on a bronze plaque inside its base in 1903, and the original 1886 torch in the statue's museum (Chapter 12) 807

Stonewall National Monument, New York: Christopher Park, in memory of the 1969 uprising at the Stonewall Inn (Chapter 25) 1909

Sun Building, 280 Broadway, New York City: A. T. Stewart's Marble Palace store building of 1846, which still stands (Chapter 18) 1188

Tenement Museum, New York: the Moores' apartment at 97 Orchard Street, recreated as it was in 1869 (Chapter 6) 418

Tenement Museum, 97 Orchard Street, New York: the tenement where about 7,000 people lived, with the Baldizzi family's apartment rebuilt from Josephine Baldizzi's memories (Chapter 7) 480

Thomas Cole National Historic Site, Catskill: Cole's house and studio (Chapter 32) 2627

Tompkins Square Park, New York City: where police beat people with clubs and drove out a crowd of about 7,000 jobless people on January 13, 1874 (Chapter 16) 1074

Wall Street and Water Street, New York City: a sign marking where the city's market for enslaved people stood from 1711 to 1762 (Chapter 18) 1178

Wesleyan Chapel, Seneca Falls: where the 1848 women's rights convention met (Chapter 25) 1778

Woolworth Building, New York: the tallest building in the world when it was finished in 1913 (Chapter 12) 815

North Carolina

Berry site, Burke County: the remains of Fort San Juan, built in 1567 (Chapter 22) 1502

Biltmore House, Asheville: the largest privately owned house in the United States (Chapter 12) 807

Roanoke Island: where John White left 117 colonists in 1587 and found CROATOAN carved on a post in 1590 (Chapter 31) 2502

Southern Oral History Program, University of North Carolina: the 1984 interview with Pearl and Oral Yates about electricity reaching Haywood County (Chapter 10) 691

Ohio

Hopewell Ceremonial Earthworks: earthen walls and mounds built between about the years 1 and 400, a World Heritage Site (Chapter 29) 2203

Miami University: its librarians keep more than 300 editions of the McGuffey Readers (Chapter 30) 2375

Octagon Earthworks, Newark: earth walls about 2,000 years old that line up with the moonrise, a golf course from 1910 to 2024 and open to the public since January 1, 2025 (Chapter 12) 783

Ohio History Connection, Columbus: keeps "Buttons," the last confirmed wild passenger pigeon, shot in 1900 (Chapter 3) 215

Serpent Mound: a snake of earth about a quarter of a mile long (Chapter 12) 783

Taft Museum of Art, Cincinnati: Duncanson's eight landscape murals in the entrance hall (Chapter 32) 2643

Oklahoma

Greenwood, Tulsa: the Black district a white mob burned in 1921, which its residents rebuilt (Chapter 17) 1143

Greenwood Cultural Center, Tulsa: where President Biden met the three Greenwood survivors in 2021 (Chapter 25) 1829

Oaklawn Cemetery, Tulsa: where searchers have dug up the remains of 55 people in the search for the massacre's graves (Chapter 25) 1829

Oregon

Crater Lake, southern Oregon: the lake that filled the collapsed top of Mount Mazama, 1,943 feet deep (Chapter 27) 2032

Hells Canyon, Snake River: where a gang of white horse thieves killed as many as 34 Chinese miners in 1887, with a 2012 stone marker in English, Chinese and Nez Perce (Chapter 4) 283

Pennsylvania

Amos Humiston monument, Gettysburg: on North Stratton Street, the only monument on the battlefield to a single enlisted soldier (Chapter 22) 1593

Carlisle Indian Industrial School grounds, Carlisle: a national monument since December 9, 2024, with the Carlisle Barracks cemetery, where many of the children who died at the school are buried (Chapter 30) 2474

Eastern State Penitentiary, Philadelphia: the prison where Charles Williams became Prisoner Number 1 in 1829, now a historic site (Chapter 21) 1434

Eden Cemetery, Collingdale: burial place of 19 Black Philadelphians from the Morton collection (Chapter 9) 596

Franklin Institute, Philadelphia: a science museum with a history of Franklin's inventions (Chapter 8) 524

Haverford College, Haverford: keeps the 1688 Germantown protest against slavery (Chapter 25) 1942

Historical Society of Pennsylvania, Philadelphia: Hesselius's portraits of Lapowinsa and Tishcohan (Chapter 32) 2613

Hopewell Furnace National Historic Site, French Creek, Berks County: the charcoal iron furnace that ran from 1771 to 1883, on land federal officials bought in the 1930s. More than 100 woodcutters and about 20 colliers, the men who made the charcoal, worked there in the 1830s (Chapter 10) 671

Independence Hall, Philadelphia: the Pennsylvania State House, whose planning Andrew Hamilton led, where the delegates signed the Constitution on September 17, 1787, open to visitors as part of a national park (Chapter 20) 1317

John Bartram's botanic garden, Kingsessing, on the Schuylkill River: the oldest botanic garden in America that survives today (Chapter 9) 588

Logan, Franklin, Washington and Rittenhouse squares and City Hall, Philadelphia: the five squares of Holme's 1682 plan (Chapter 6) 399

Mother Bethel, Sixth and Lombard Streets, Philadelphia: Richard Allen's church, where he is buried in the basement (Chapter 29) 2242

Mütter Museum, Philadelphia: a museum of medical history (Chapter 26) 1967

Penn Museum, University of Pennsylvania, Philadelphia: holds Samuel Morton's skull collection (Chapter 9) 596

Penn Museum, Philadelphia: quilled pieces from the 1830s and later, including pieces George Catlin collected between 1832 and 1839 (Chapter 35) 2908

Philadelphia Mint, Philadelphia: where the last pennies made for everyday use were struck on November 12, 2025 (Chapter 17) 1160

Philadelphia Museum of Art, Philadelphia: Augustin de Saint-Aubin's 1777 engraving of Benjamin Franklin in his fur cap (Chapter 23) 1605

Philadelphia Museum of Art, Philadelphia: Moses Williams's silhouettes of the Peale family (Chapter 32) 2627

President's House site, Philadelphia: where George Washington lived while he was president, and where Ona Judge, an enslaved woman, walked out to freedom in 1796. Its exhibit panels describe nine people Washington held in slavery there. Officials of the Trump administration replaced the old panels with their own in July 2026 (Chapter 12) 832

Priestley's house, Northumberland: Joseph Priestley's house, finished in 1798, where American chemists met in 1874, now a National Historic Landmark (Chapter 13) 855

Soldiers' National Cemetery, Gettysburg: the graves of the Gettysburg dead, 979 of them marked "Unknown" (Chapter 22) 1543

Valley Forge: the winter camp of 1777 to 1778 (Chapter 22) 1522

Rhode Island

Bristol: the Fourth of July celebration held every year since 1785, which the town calls the oldest in the United States that has never stopped (Chapter 37) 3094

Roger Williams National Memorial, Providence: the memorial Congress voted to set up in 1965 (Chapter 29) 2213

South Carolina

Avery Research Center, Charleston: the papers of Mamie Garvin Fields (Chapter 30) 2414

Charleston County Register of Deeds, Charleston County: hundreds of colonial land mortgages (Chapter 17) 1119

Charleston Museum, Charleston: two Dave jars that each hold more than 40 gallons (Chapter 32) 2627

International African American Museum, Charleston: the 1850 Zealy daguerreotypes of Renty, Delia and five others, on public view since March 11, 2026 (Chapter 9) 604

International African American Museum, Charleston: a museum at Gadsden's Wharf, one of the main places where slave ships landed their captives (Chapter 24) 1744

South Carolina Department of Archives and History: thousands of early chattel mortgages, in which enslavers and lenders listed enslaved people as property (Chapter 17) 1119

St. Philip's Church, Charleston: the church whose burning roof Will saved in 1796 (Chapter 27) 2047

Sullivan's Island, Charleston: the harbor island where enslaved Africans were held in pest houses (Chapter 4) 267

South Dakota

Mount Rushmore, Black Hills: the carving on the mountain the Lakota call Tunkasila Sakpe, the Six Grandfathers (Chapter 12) 815

South Dakota Cultural Heritage Center: the La Verendrye lead plate from 1743, found at Pierre in 1913 (Chapter 1) 36

Wounded Knee, Pine Ridge Reservation: the site of the 1890 massacre and the 1973 occupation (Chapter 2) 134

Tennessee

Clayborn Temple, Memphis: the church where police fired tear gas and clubbed marchers on March 28, 1968 (Chapter 14) 963

Dollywood, Pigeon Forge: the theme park Dolly Parton became a partner in (Chapter 33) 2795

Ed Johnson memorial, Chattanooga: a memorial to Ed Johnson, the Black man a mob lynched in 1906, dedicated on September 19, 2021 (Chapter 21) 1456

Graceland, Memphis: Elvis Presley's home, where he died (Chapter 33) 2779

Jubilee Hall, Fisk University, Nashville: the building paid for by the Fisk Jubilee Singers' tours, a National Historic Landmark (Chapter 33) 2746

Lorraine Motel, Memphis: where James Earl Ray shot and killed Martin Luther King Jr. on April 4, 1968, during the sanitation strike (Chapter 14) 963

Reelfoot Lake, northwest Tennessee: the lake formed when the ground sank in the New Madrid earthquakes (Chapter 27) 2061

Texas

The Alamo, San Antonio: the unfinished mission church begun in 1758 (Chapter 12) 795

Anchor monument, Texas City: lists 576 people known to have died in the 1947 explosions (Chapter 27) 2078

Blackwell School, Marfa: a school for Mexican American children from 1909 to 1965, a national historic site since October 2022 (Chapter 30) 2442

Emancipation Park, Houston: the ten acres Black Houstonians bought for Juneteenth in 1872, the oldest park in Houston, where Juneteenth celebrations are still held (Chapter 37) 3107

Galveston seawall, Galveston: the seawall built after the 1900 hurricane, 17 feet above the low-tide line (Chapter 27) 2068

Witte Museum, San Antonio: two dried peyote tops about 5,700 years old (Chapter 28) 2120

Utah

Bears Ears National Monument: the monument five tribes proposed and President Donald Trump cut by about 91 percent in July 2026 (Chapter 3) 240

Edge of the Cedars State Park Museum, southeastern Utah: a blanket of about 11,550 turkey feathers made in the 1200s (Chapter 35) 2908

Golden Spike National Historical Park, Promontory Summit: marks where the last track of the first railroad across the country was laid in 1869. Working copies of the engines Jupiter and No. 119 run there from May to October (Chapter 11) 779

Topaz camp site, near Delta: one of the ten camps where officials held Japanese Americans (Chapter 25) 1829

Vermont

Vermont Historical Society: printed James Guild's journal in 1937 (Chapter 18) 1188

Virginia

Abby Aldrich Rockefeller Folk Art Museum, Williamsburg: *The Old Plantation*, the earliest known painting of a banjo (Chapter 33) 2729

Arlington National Cemetery: where Matthew Henson and his wife were buried again in 1988 (Chapter 1) 74

Arlington National Cemetery: L'Enfant's grave, on a hillside above Washington (Chapter 6) 407

Arlington National Cemetery: the grave of William Hushka, a veteran police shot and killed during the Bonus Army protest of 1932 (Chapter 16) 1084

Arlington National Cemetery: Zitkala-Ša's grave, whose headstone gives the wrong year of death (Chapter 30) 2392

Arlington National Cemetery: where the first national Decoration Day ceremony was held in 1868, on land that had been the home of Robert E. Lee (Chapter 37) 3107

Colonial Williamsburg, Williamsburg: owns Charles Willson Peale's painting of Nancy Hallam dressed as Fidele (Chapter 34) 2833

Historic Jamestowne, Jamestown: the oldest playing marble that can be dated at Jamestown, made of brown stoneware, a hard kind of pottery (Chapter 36) 2987

Jamestown Island: the foundations of rows of houses at New Towne, dug up by archaeologists (Chapter 6) 399

Monticello: Thomas Jefferson's home and plantation, where enslaved people did the farm work. The house has woodwork by enslaved craftsmen such as John Hemmings (Chapter 12) 799

Mount Vernon: George Washington's home and plantation, built and enlarged by enslaved workers. His whiskey distillery there made almost 11,000 gallons in 1799. Mount Vernon holds the brown wool coat he wore when he took the oath as president in 1789. Archaeologists found a limestone marble there that may have belonged to an enslaved child (Chapter 12) 799

Salem Witch Trials Documentary Archive, University of Virginia: the papers of the 1692 cases, among them Rebecca Nurse's case file, number 94 (Chapter 21) 1412

Virginia Museum of Fine Arts, Richmond: Kehinde Wiley's Rumors of War (Chapter 32) 2695

Virginia State Capitol, Richmond: the capitol modeled on the Maison Carrée, an ancient Roman temple in France (Chapter 12) 799

Williamsburg Bray School building, Colonial Williamsburg, Williamsburg: the oldest surviving schoolhouse for Black children in the United States, moved there on February 10, 2023 (Chapter 30) 2496

Washington

LIGO detector, Hanford: one of the two detectors that first picked up gravitational waves, on September 14, 2015 (Chapter 9) 648

Space Needle, Seattle: the 605-foot tower built for the 1962 world's fair (Chapter 12) 825

Washington, D.C.

Ben's Chili Bowl, 1213 U Street NW: the diner, still open (Chapter 6) 452

Blair House: where President Truman was living when two Puerto Rican Nationalists attacked it in 1950 (Chapter 23) 1647

Emancipation Hall, Capitol Visitor Center: the block of Aquia Creek sandstone that honors the enslaved laborers who quarried the Capitol's stone (Chapter 12) 832

Ford's Theatre: the theatre where John Wilkes Booth shot President Lincoln, restored, with its first public performance since that night in 1968 (Chapter 34) 2851

Kennedy Center: the building for music, theater and dance, closed as of October 26, 2026, with no date set to reopen (Chapter 33) 2795

Library of Congress: Robert Cornelius's 1839 self-portrait, the oldest surviving portrait photograph made in America (Chapter 8) 535

Library of Congress: John Adams's letter of June 3, 1785, to Thomas Jefferson about meeting King George III (Chapter 23) 1605

Library of Congress: "Born in Slavery," more than 2,300 life stories of formerly enslaved people recorded from 1936 to 1938, with photographs, and the "Voices Remembering Slavery" recordings, including Fountain Hughes (Chapter 24) 1735

Library of Congress: holds copies of the three pamphlets of the 1844 and 1845 argument between Horace Mann and the Boston schoolmasters (Chapter 30) 2375

Library of Congress: about 10,000 wax cylinders, nearly 8,000 of them holding Native songs and speech (Chapter 33) 2729

Library of Congress: Charles Dana Gibson's Gibson Girl drawings, shown in an exhibition (Chapter 35) 2940

Lincoln Memorial: Daniel Chester French's seated Lincoln. Marian Anderson sang on its steps on April 9, 1939, and the March on Washington ended there in 1963 (Chapter 12) 815

Martin Luther King, Jr. Memorial: the 30-foot Stone of Hope (Chapter 12) 832

National Archives: all 374 ratified treaties between the United States and Native nations, every one online (Chapter 2) 120

National Archives: records of the 1942 forced removal of Japanese Americans, which count about 120,000 people removed (Chapter 5) 376

National Archives: David Falconer's photographs and captions of Oregon's 1973 and 1974 gasoline shortage (Chapter 10) 702

National Archives: the Freedman's Savings Bank's signature books and deposit records from 29 branches, which list depositors' names, ages, jobs and family members (Chapter 17) 1136

National Archives: the Freedmen's Bureau marriage registers (Chapter 24) 1712

National Museum of African American History and Culture: stands on part of Notley Young's plantation (Chapter 6) 407

National Museum of African American History and Culture: the building whose metal screen copies ironwork by enslaved and free Black craftsmen (Chapter 12) 832

National Museum of African American History and Culture: history galleries that begin with the slave trade, and a cabin where enslaved people lived (Chapter 24) 1744

National Museum of the American Indian: the national museum of Native peoples on the National Mall, opened in 2004 (Chapter 2) 108

National Portrait Gallery: the Obama portraits by Kehinde Wiley and Amy Sherald, and (with the Museum of Fine Arts, Boston) Stuart's Athenaeum portrait of Washington (Chapter 32) 2619

Phillips Collection: the odd-numbered panels of Jacob Lawrence's The Migration Series (Chapter 32) 2660

Smithsonian: the Ontonagon Boulder, a 3,708-pound lump of pure copper sacred to the Ojibwe, held there since 1860 (Chapter 13) 844

Smithsonian: a purple velvet gown Elizabeth Keckley made for Mary Todd Lincoln (Chapter 35) 2940

Smithsonian American Art Museum: Edmonia Lewis's The Death of Cleopatra (Chapter 32) 2627

Smithsonian Institution: collections that the Wilkes expedition's more than 60,000 plant and bird specimens helped start (Chapter 1) 44

Smithsonian Institution: has Martha, the last passenger pigeon (Chapter 3) 200

Smithsonian Institution: half of the cylinder of the first steam engine in America (Chapter 10) 671

Smithsonian Institution: a hornbook, a child's wooden reading paddle, made in the 1700s, in its National Museum of American History (Chapter 30) 2359

Smithsonian Institution Building: the red sandstone building called the Castle (Chapter 12) 804

Smithsonian's National Air and Space Museum: the collection whose curator calls the DC-3 the first airliner able to make money just by carrying passengers (Chapter 11) 756

Smithsonian's National Museum of American History: the paper tape of Morse's first public telegraph message, May 24, 1844 (Chapter 8) 535

Smithsonian's National Museum of American History: one of President Carter's 1979 White House solar panels (Chapter 10) 702

Smithsonian's National Museum of American History: one of the first ten barcode scanners, a Spectra-Physics Model A (Chapter 18) 1214

Smithsonian's National Museum of American History: Sandra Lindsay's hospital scrubs and ID badge from the first COVID-19 vaccine outside a trial (Chapter 26) 2022

Smithsonian's National Museum of Natural History: in 2023 reporters found that the museum still held at least 255 human brains, most of them collected by Aleš Hrdlička or people working for him (Chapter 9) 614

Smithsonian's National Museum of Natural History: the two best-known Etowah Birdman copper plates (Chapter 32) 2598

Statue of Freedom, US Capitol dome: the bronze statue Philip Reid helped cast (Chapter 12) 807

Statue of Po'pay, National Statuary Hall Collection, US Capitol: the 2005 statue of Po'pay, the Pueblo man who led the rising against the Spanish in 1680, with whipping scars carved on its back (Chapter 29) 2213

United States Capitol: Sarah Winnemucca's statue for Nevada (Chapter 25) 1795

Vietnam Veterans Memorial, National Mall: Maya Lin's black granite walls, with the names of 58,281 Americans who died in the Vietnam War (Chapter 12) 825

Washington Monument: the change in the marble's color about 150 feet up, where work stopped for 25 years, and the aluminum pyramid set on its top on December 6, 1884 (Chapter 12) 804

West Virginia

West Virginia Mine Wars Museum: it puts the Monongah mine dead of 1907 at more than 500, higher than the official count of 362 (Chapter 14) 941

West Virginia Mine Wars Museum: a page describing coal-company scrip as debt bondage (Chapter 18) 1204

Wyoming

Obsidian Cliff: a source of obsidian that traders carried as far as western Canada (Chapter 16) 1049

Yellowstone National Park: the first national park, made a public park in 1872, where wolves returned in 1995. John Colter walked through its hot springs country in the winter of 1807 to 1808 (Chapter 3) 229

More than one state

Boston Post Road (part of US Route 1), between New York and Boston: the trail of the first post rider in 1673 (Chapter 31) 2506

The Dalles Dam, on the Columbia River between Oregon and Washington: the dam whose lake covers Celilo Falls and the old trading place (Chapter 18) 1170

Emmett Till and Mamie Till-Mobley National Monument: places connected to Emmett Till and his mother, made a national monument by President Joe Biden on July 25, 2023 (Chapter 21) 1468

Ghost forests, rivers and bays of Washington and Oregon: dead cedar trunks from the coast's drop in the 1700 earthquake (Chapter 27) 2047

Trail of Tears National Historic Trail, nine states: more than 2,200 miles of the Cherokee removal routes, run by the National Park Service (Chapter 5) 392

Outside the United States

Benjamin Franklin House museum, London, England: a museum about Franklin that names seven people he and his household bought (Chapter 31) 2512

British Museum, London, England: John White's watercolors, the Mound City pipes and the Cherokee double-woven baskets of 1725 (Chapter 32) 2598

Cobh, Ireland: a statue of Annie Moore (Chapter 4) 283

Kahnawake, Canada: crosses made from the Quebec Bridge's steel in the village cemetery, and the 2007 monument to the men killed (Chapter 12) 815

L'Anse aux Meadows, Newfoundland, Canada: the Norse camp where trees were cut in 1021 (Chapter 1) 17

Lomanikoro, Rewa, Fiji: the chiefs' burial ground where Vendovi's remains were buried in December 2025. Vendovi was a Fijian chief the Wilkes expedition took prisoner, and he died in New York in 1842 (Chapter 1) 44

Machu Picchu, Andes Mountains, Peru: the Inca site Hiram Bingham made famous, where Agustin Lizarraga had written "A. Lizarraga 1902" on a wall (Chapter 1) 74

Museo de América, Madrid, Spain: Tadeo Haenke's 1791 sheets of Tlingit songs (Chapter 33) 2729

National Archives of the United Kingdom: the one surviving copy of John Pory's report on the first General Assembly at Jamestown in 1619 (Chapter 20) 1309

Qaanaaq, Greenland: where the remains of Qisuk and three other Inuit were buried after their return in 1993. Robert Peary had tricked them into sailing to New York in 1897, and they died there in 1898 (Chapter 1) 64

Solar museum, Dezhou, China: some of the Carter White House solar panels (Chapter 10) 702

Online

Bracero History Archive: more than 700 recorded interviews with braceros, including Antonio García's (Chapter 4) 310

discoverfreedmen.org: about 1.8 million searchable names from the Freedmen's Bureau records (Chapter 24) 1744

DoHistory.org: the whole of Martha Ballard's diary (Chapter 7) 471

K-25 Virtual Museum: Colleen Black's account (Chapter 6) 431

"Last Seen: Finding Family After Slavery": more than 5,000 "Information Wanted" advertisements, including Thornton Copeland's, item 918 (Chapter 24) 1712

Library of Congress, Prairie Settlement collection: 318 of the Oblinger family's letters, written from 1862 to 1911 (Chapter 15) 1009

Mapping Inequality project, University of Richmond: the 1930s redlining maps and the appraisers' notes (Chapter 6) 431

Voices of the Manhattan Project, Atomic Heritage Foundation: Colleen Black's 2005 oral history (Chapter 6) 431

Other places

American Antiquarian Society: holds a copy of the 1727 edition of The New England Primer (Chapter 30) 2351

Baltimore and Ohio Railroad Museum: its page on the Tom Thumb describes the 1830 race with a horse as a legend (Chapter 11) 734

Carter Library: some of the Carter White House solar panels (Chapter 10) 702

Computer History Museum: its profile of Lee Felsenstein, who received its Fellow Award in 2016 (Chapter 8) 561

Devils Tower National Monument: the first national monument, sacred to dozens of Native nations. The Lakota call it Mato Tipila, or Bear Lodge (Chapter 3) 252

Eli Whitney Museum: its history of the cotton gin (Chapter 8) 528

Jim Crow Museum at Ferris State University: its writers' account of blackface minstrelsy (Chapter 34) 2841

NASA's Johnson Space Center: keeps the 2,200 Apollo Moon samples (Chapter 9) 633

National WWII Museum: Adelaide Wisdom Benjamin's 2013 oral history (Chapter 15) 1015

Smithsonian's National Museum of the American Indian: the exhibition Booming Out: Mohawk Ironworkers Build New York (Chapter 6) 418

United States Holocaust Memorial Museum: its count of 254 *St. Louis* passengers killed in the Holocaust (Chapter 23) 1634

US Botanic Garden: the garden that grew from the 254 live plants the Wilkes expedition brought home (Chapter 1) 44

Voices of Rondo oral-history collection: Marvin Roger Anderson's account (Chapter 6) 443

Walter P. Reuther Library, Wayne State University: its account that children in several San Joaquin Valley towns had cancer at rates "800% above the norm" in the 1980s (Chapter 15) 1028

Winterthur Museum: showed 40 of Ann Lowe's gowns in 2023 (Chapter 35) 2949

Movies

Every movie and TV show listed in the story boxes, from A to Z, with what each box says about it. A title that begins with "The" or "A" is sorted by its next word, so *The Revenant* is under R. Some of these were made for adults, so check each one's rating and ask a grown-up before you watch it.

10,000 Black Men Named George (2002): a television drama with Andre Braugher as A. Philip Randolph (Chapter 14) 958

12 Years a Slave (2013): a film directed by Steve McQueen that won the Academy Award for Best Picture (Chapter 24) 1707

17 Miracles (2011): a drama about the Willie handcart company that draws on Levi Savage's journal (Chapter 5) 366

42 (2013): a drama with Chadwick Boseman as Jackie Robinson (Chapter 36) 3043

A. Philip Randolph: For Jobs and Freedom (1996): a documentary shown on PBS (Chapter 14) 958

Alexander Hamilton (2007): a two-hour documentary in the PBS series *American Experience*, with Brían F. O'Byrne as Hamilton (Chapter 16) 1064

Alice Waters and Her Delicious Revolution (2003): a documentary in the PBS series *American Masters* (Chapter 15) 1036

All the Beauty and the Bloodshed (2022): a documentary by Laura Poitras about Nan Goldin's life and her protests (Chapter 28) 2195

All the President's Men (1976): a drama about Bob Woodward and Carl Bernstein's reporting, with Robert Redford as Woodward (Chapter 31) 2580

Amazing Grace (2018): a documentary film of Aretha Franklin's 1972 gospel concert in a Los Angeles church (Chapter 33) 2784

Amazon Empire: The Rise and Reign of Jeff Bezos (2020): a documentary by James Jacoby and Anya Bourg in the PBS series *Frontline*, an investigation that argues a case against Amazon's power (Chapter 18) 1223

American Socialist: The Life and Times of Eugene Victor Debs (2017): a documentary about Debs's life, directed by Yale Strom (Chapter 14) 939

Amistad (1997): a drama about the *Amistad* case (Chapter 24) 1706

Andrew Jackson: Good, Evil and the Presidency (2007): a two-hour PBS documentary narrated by Martin Sheen. It covers Jackson's whole presidency, not only the Bank War (Chapter 17) 1131

Andy Warhol: A Documentary Film (2006): a four-hour documentary by Ric Burns about Warhol's life, shown on PBS (Chapter 32) 2685

Apollo 13 (1995): a film based on Jim Lovell's book about the flight, *Lost Moon* (Chapter 1) 87

Athlete A (2020): a Netflix documentary by Bonni Cohen and Jon Shenk about the *Indianapolis Star* reporters and Larry Nassar's sexual abuse of gymnasts (Chapter 36) 3068

Becoming Frederick Douglass (2022): a documentary by Stanley Nelson about Douglass's life, shown on PBS (Chapter 24) 1700

Before Night Falls (2000): a drama based on Reinaldo Arenas's life story (Chapter 4) 315

Being the Ricardos (2021): a drama about Lucille Ball and Desi Arnaz, written and directed by Aaron Sorkin (Chapter 34) 2883

Benjamin Franklin (2022): a two-part documentary by Ken Burns about Franklin's whole life, shown on PBS in April 2022 (Chapter 8) 527

Bessie (2015): an HBO drama with Queen Latifah as Bessie Smith (Chapter 33) 2762

Billie (2019): a documentary about Billie Holiday (Chapter 33) 2775

The Birth of a Nation (1915): a drama by D. W. Griffith that showed Ku Klux Klan members lynching a Black man and stopping Black men from voting. After it opened in Atlanta, a preacher named William J. Simmons started the Klan again (Chapter 34) 2861

The Birth of a Nation (2016): a drama about Nat Turner's revolt. Its makers took the title on purpose from a 1915 film by D. W. Griffith that inspired a new Ku Klux Klan (Chapter 24) 1705

Bolinao 52 (2008): a documentary directed by Duc Nguyen about a boat of refugees from Vietnam that drifted at sea for 37 days in 1988. Tung Trinh was one of its 52 survivors (Chapter 4) 317

Born in Freedom: The Story of Colonel Drake (1954): a 30-minute drama paid for by the oil industry's American Petroleum Institute, with Vincent Price as Drake (Chapter 10) 682

Born on the Fourth of July (1989): a drama directed by Oliver Stone and based on Ron Kovic's memoir. Kovic helped write the script (Chapter 22) 1579

Bound for Glory (1976): a Hollywood drama based loosely on Woody Guthrie's own book about his life (Chapter 33) 2772

Brooklyn Bridge (1981): Ken Burns' first documentary, about the bridge and the Roeblings, narrated by David McCullough. It was nominated for the Academy Award for Best Documentary Feature (Chapter 12) 810

The Carter Family: Will the Circle Be Unbroken (2005): a documentary about the Carter family in the PBS series *American Experience* (Chapter 33) 2766

The Central Park Five (2012): a documentary by Ken Burns, Sarah Burns and David McMahon (Chapter 21) 1486

Cesar Chavez (2014): a drama directed by Diego Luna, with Michael Peña as Chavez. It came out before March 2026, when the New York Times reported that Chavez had sexually abused two girls in the 1970s and Dolores Huerta said he had raped her (Chapter 14) 965

Chaplin (1992): a drama about Charlie Chaplin's life, directed by Richard Attenborough (Chapter 34) 2868

The Circus (2018): a two-part documentary about the American circus in the PBS series *American Experience*. It tells a large part of P. T. Barnum's story (Chapter 18) 1190

Citizen Jane: Battle for the City (2016): a 92-minute documentary directed by Matt Tyrnauer about Jane Jacobs's fights with Robert Moses over his plans for New York City (Chapter 6) 448

The Computers (2014): a short documentary about the six ENIAC programmers (Chapter 8) 559

Crisis in Levittown (1957): a documentary by Lee Bobker and Lester Becker, filmed while the events were happening (Chapter 7) 498

The Current War (2017): a Hollywood drama, not a documentary, with Michael Shannon as George Westinghouse and Nicholas Hoult as Nikola Tesla (Chapter 10) 687

DeFord Bailey: A Legend Lost (2002): a documentary about Bailey made by Nashville Public Television (Chapter 33) 2767

Descendant (2022): a documentary by Margaret Brown about the *Clotilda*, Africatown and the families of the survivors. Joycelyn Davis appears in it (Chapter 24) 1717

Dolly Parton: Here I Am (2019): a documentary about Dolly Parton (Chapter 33) 2808

Dolores (2017): a documentary about Dolores Huerta directed by Peter Bratt. It came out before March 2026, when Huerta said Cesar Chavez had raped her in 1966 (Chapter 14) 965

Dorothea Lange: Grab a Hunk of Lightning (2014): a documentary about Lange's life and work in the PBS series *American Masters*, directed by her granddaughter Dyanna Taylor (Chapter 32) 2676

Eames: The Architect and the Painter (2011): a documentary about Charles and Ray Eames in the PBS series *American Masters* (Chapter 35) 2961

Edison (2015): a documentary about Thomas Edison's life and work in the PBS series *American Experience* (Chapter 8) 548

Elvis (2022): a Hollywood drama about Elvis Presley and his manager (Chapter 33) 2782

Escaping the Madhouse: The Nellie Bly Story (2019): a Lifetime television drama that its makers call a fictionalized account, meaning parts of it are made up (Chapter 31) 2552

The Fantastic Mr Feynman (2013): a BBC documentary with interviews with Richard Feynman, his family and his colleagues (Chapter 9) 642

The Far Horizons (1955): a drama that invents a romance between Sacagawea and William Clark, and a rivalry between Clark and Meriwether Lewis (Chapter 1) 45

The Farthest (2017): a documentary about the Voyager missions, in which Linda Morabito tells the story of her discovery (Chapter 9) 641

Flags of Our Fathers (2006): a drama in which Adam Beach plays Ira Hayes (Chapter 2) 156

Fly With Me (2024): a documentary about flight attendants in the PBS series *American Experience*, directed by Sarah Colt and Helen Dobrowski. Dusty Roads appears in it (Chapter 11) 768

Food Chains (2014): a documentary about the Coalition of Immokalee Workers. Lucas Benitez appears in it (Chapter 15) 1042

The Founder (2016): a drama based on Ray Kroc's story and the McDonald brothers' (Chapter 15) 1031

Frank Capra's American Dream (1997): a documentary made for television (Chapter 4) 299

Franklin (2024): an eight-part drama series on Apple TV+, not a documentary, about Benjamin Franklin's time in France, with Michael Douglas as Franklin. It is based on Stacy Schiff's book *A Great Improvisation* (Chapter 23) 1608

Frederick Law Olmsted: Designing America (2014): a PBS documentary first shown on June 20, 2014 (Chapter 6) 427

Gather (2020): a documentary about Native food sovereignty that features Nephi Craig (Chapter 15) 1045

George Washington Carver: An Uncommon Life (2018): a documentary made by Iowa PBS (Chapter 15) 1017

Georgia O'Keeffe (about 1977): a documentary about O'Keeffe's life and work by Perry Miller Adato, made with the New York public television station WNET (Chapter 32) 2662

Gideon's Trumpet (1980): a television drama with Henry Fonda as Clarence Earl Gideon, based on Anthony Lewis's 1964 book (Chapter 21) 1471

Good Night, and Good Luck (2005): a film directed by George Clooney about Edward R. Murrow's television reports on Senator Joseph McCarthy in 1953 and 1954 (Chapter 31) 2568

Grant (2020): a History channel documentary in three parts about Ulysses S. Grant, with acted scenes (Chapter 22) 1551

The Greatest Showman (2017): a musical movie about P. T. Barnum. Its story is mostly made up, and Joice Heth does not appear in it (Chapter 18) 1190

Green Fire: Aldo Leopold and a Land Ethic for Our Time (2011): a documentary about Aldo Leopold (Chapter 3) 227

The Gruen Effect: Victor Gruen and the Shopping Mall (2009): a documentary by Anette Baldauf and Katharina Weingartner about Gruen and the malls he started (Chapter 18) 1215

Half Past Autumn: The Life and Works of Gordon Parks (2000): a documentary about Gordon Parks's life (Chapter 32) 2683

Hamilton (2020): the Broadway stage musical, filmed with its original cast, and not a documentary (Chapter 4) 271

Harriet (2019): a drama about Harriet Tubman's life, with Cynthia Erivo as Tubman (Chapter 24) 1710

Henry Ford (2013): a documentary about Ford's life and his company in the PBS series *American Experience*, directed by Sarah Colt (Chapter 8) 555

A Hero for Daisy (1999): a documentary about Chris Ernst by Mary Mazzio (Chapter 36) 3053

Hidden Figures (2016): a drama in which Taraji P. Henson plays Katherine Johnson (Chapter 1) 85

High Steel (1965): a documentary about Harold McComber's working life by the National Film Board of Canada, directed by Don Owen (Chapter 12) 827

Human Nature (2019): a documentary about CRISPR in which Jennifer Doudna is one of the main people interviewed (Chapter 9) 652

I Will Fight No More Forever (1975): a television drama whose title quotes Chief Joseph's words at his surrender in 1877 (Chapter 2) 141

The Immortal Life of Henrietta Lacks (2017): a drama made for HBO about Henrietta Lacks and her daughter Deborah (Chapter 26) 2006

Inside Bill's Brain: Decoding Bill Gates (2019): a Netflix series directed by Davis Guggenheim (Chapter 19) 1287

The Insider (1999): a drama about Jeffrey Wigand, directed by Michael Mann, with Russell Crowe as Wigand (Chapter 28) 2188

Is Wal-Mart Good for America? (2004): a documentary about Walmart in the PBS series *Frontline* (Chapter 19) 1289

Izzy and Moe (1985): a television film about two Prohibition agents, with Jackie Gleason as Izzy Einstein and Art Carney as Moe Smith (Chapter 28) 2172

The Jackie Robinson Story (1950): a drama in which Jackie Robinson played himself (Chapter 36) 3043

Jim Thorpe – All-American (1951): a drama about Jim Thorpe (Chapter 2) 155

John Adams (2008): a drama miniseries made for HBO about John Adams and his family. Abigail Adams is one of its main characters (Chapter 7) 472

John James Audubon: Drawn from Nature (2007): a documentary in the PBS series *American Masters* (Chapter 3) 198

John Muir in the New World (2011): a documentary in the PBS series *American Masters* (Chapter 3) 213

Jubilee Singers: Sacrifice and Glory (2000): a documentary about the Fisk Jubilee Singers in the PBS series *American Experience* (Chapter 33) 2751

Kemba (2024): a film about Kemba Smith's life, shown on BET+ (Chapter 21) 1479

Lincoln (2012): a film about the fight in January 1865 to get the House of Representatives to pass the Thirteenth Amendment. It is about the amendment, not about habeas corpus (Chapter 20) 1343

Little House on the Prairie (NBC series, 1974 to 1983): a television drama series with Michael Landon as Charles Ingalls, loosely based on Laura Ingalls Wilder's books (Chapter 5) 370

Lois Gibbs and the Love Canal (1982): a CBS television drama with Marsha Mason as Lois Gibbs (Chapter 3) 236

Lorraine Hansberry: Sighted Eyes/Feeling Heart (2017): a documentary about Hansberry's life and work (Chapter 34) 2884

Lost Boys of Sudan (2003): a documentary that records the lives of Peter Dut and Santino Chuor (Chapter 4) 329

Louis Armstrong's Black & Blues (2022): a documentary about Louis Armstrong (Chapter 33) 2764

Man in the Wilderness (1971): a film based on Hugh Glass's story, like *The Revenant* (Chapter 1) 54

Man on Wire (2008): a documentary about Philippe Petit's walk, directed by James Marsh. It won the Academy Award for Best Documentary Feature (Chapter 12) 828

The Man Who Tried to Feed the World (2020): a documentary about Norman Borlaug in the PBS series *American Experience* (Chapter 15) 1032

Mankiller (2017): a documentary about Wilma Mankiller (Chapter 2) 167

Maria Martinez: Indian Pottery of San Ildefonso (1972): a documentary that shows Maria Martinez and her son Popovi Da making pottery at the pueblo (Chapter 32) 2665

Marian Anderson: The Lincoln Memorial Concert (1939): a short film of Marian Anderson's concert at the Lincoln Memorial (Chapter 33) 2778

Mark Twain (2001): a documentary by Ken Burns about Mark Twain's whole life, shown on PBS in January 2002 (Chapter 11) 751

Maya Lin: A Strong Clear Vision (1994): a documentary about Maya Lin that won the Academy Award for Best Documentary Feature (Chapter 12) 830

The Men Who Built America (2012): a History Channel series about John D. Rockefeller, Andrew Carnegie, J. P. Morgan and others. It is a docudrama, in which actors play the real people in scripted scenes (Chapter 19) 1257

Midnight Ramble: Oscar Micheaux and the Story of Race Movies (1994): a PBS documentary about Oscar Micheaux and about films made for Black audiences (Chapter 34) 2871

A Midwife's Tale (1997): a documentary about Martha Ballard, made for the PBS series *American Experience* (Chapter 7) 473

Miles of Smiles, Years of Struggle (1982): a documentary by Jack Santino and Paul Wagner, made from the Pullman porters' own accounts (Chapter 11) 761

Miss Americana (2020): a documentary about Taylor Swift (Chapter 33) 2800

Miss Lillian: More Than a President's Mother (2021): a documentary about Lillian Carter directed by Vivian Winther (Chapter 23) 1649

Mr. Sears' Catalogue (1989): a documentary about Richard Sears and his catalog in the PBS series *American Experience* (Chapter 18) 1201

No Direction Home: Bob Dylan (2005): a documentary by Martin Scorsese (Chapter 33) 2788

One Man's Hero (1999): a feature film about John Riley, with Tom Berenger as Riley. It is a drama, not a documentary (Chapter 22) 1541

The Outsider (1961): a drama about Ira Hayes (Chapter 2) 156

Particle Fever (2013): a documentary that follows six physicists, Monica Dunford among them, from the start-up of the Large Hadron Collider to the announcement of the Higgs boson in 2012 (Chapter 9) 651

Penn of Pennsylvania (1941): a drama released in the United States as *The Courageous Mr. Penn* (1942). It is about William Penn's life in general, not the making of the plan for Philadelphia (Chapter 6) 403

Pollock (2000): a drama about Jackson Pollock's life, directed by Ed Harris, who also plays Pollock (Chapter 32) 2680

Prince of Players (1955): a drama about Edwin Booth's life (Chapter 34) 2854

The Queen of Code (2015): a short documentary about Grace Hopper's life and work, directed by Gillian Jacobs (Chapter 8) 562

Race (2016): a drama about Jesse Owens (Chapter 36) 3039

Rachel Carson (2017): a documentary in the PBS series *American Experience* (Chapter 3) 230

Radium Girls (2018): a drama inspired by the dial painters' case. Its main characters are two invented sisters, not Grace Fryer (Chapter 14) 952

The Revenant (2015): a film based on Hugh Glass's story (Chapter 1) 54

The Richest Man in the World: Andrew Carnegie (1997): a documentary about Andrew Carnegie in the PBS series *American Experience*, first shown on January 20, 1997 (Chapter 16) 1082

Rise (2022): a Disney+ drama, not a documentary, about the Antetokounmpo family, made with the family (Chapter 4) 328

Rita Moreno: Just a Girl Who Decided to Go for It (2021): a documentary about Rita Moreno (Chapter 34) 2885

The River Ran Red (1993): a documentary about the Homestead lockout that tells the story from the workers' side (Chapter 19) 1265

The Rockefellers (2000): a documentary in the PBS series *American Experience* about four generations of John D. Rockefeller's family, not about him alone (Chapter 16) 1083

The Roosevelts: An Intimate History (2014): a Ken Burns documentary on PBS about the whole Roosevelt family (Chapter 3) 217

Roots (1977): a television miniseries on ABC based on Alex Haley's book. It tells his family's story as a drama and is not a documentary about Haley (Chapter 24) 1743

The Ryan White Story (1989): a drama made for ABC television, in which Ryan White had a small part (Chapter 26) 2021

Searching for Augusta Savage (2024): a short documentary about Augusta Savage's life and her lost work, made for the PBS series *American Masters* (Chapter 32) 2667

Seeking the Greatest Good (2013): a documentary about Gifford Pinchot, from WVIA and PBS (Chapter 3) 221

Selena (1997): a Hollywood drama about Selena Quintanilla, with Jennifer Lopez (Chapter 33) 2793

Sidney (2022): a documentary about Sidney Poitier, directed by Reginald Hudlin (Chapter 34) 2886

Sitting Bull: A Stone in My Heart (2006): a documentary built from Sitting Bull's own recorded words (Chapter 2) 140

Soul of the Game (1996): a television drama about several Negro Leagues players, with Mykelti Williamson as Josh Gibson (Chapter 36) 3042

Squanto: A Warrior's Tale (1994): a drama that mixes Squanto's real story with invented events (Chapter 15) 993

Stars and Stripes Forever (1952): a Hollywood drama loosely based on John Philip Sousa's memoir (Chapter 33) 2754

Steve Jobs (2015): a drama, not a documentary, directed by Danny Boyle and written by Aaron Sorkin. Actors play the real people, and it is based on Walter Isaacson's biography of Jobs (Chapter 8) 567

Steve Jobs: The Man in the Machine (2015): a documentary about Jobs directed by Alex Gibney (Chapter 8) 567

Summoned: Frances Perkins and the General Welfare (2020): a documentary about Perkins's life by Mick Caouette, shown on PBS (Chapter 14) 947

Ten Who Dared (1960): a Disney drama about John Wesley Powell's expedition (Chapter 1) 69

Till (2022): a drama about Emmett Till's mother, Mamie Till-Mobley, and her fight after his murder (Chapter 21) 1469

Toni Morrison: The Pieces I Am (2019): a documentary by Timothy Greenfield-Sanders about Morrison's life and work (Chapter 32) 2693

TR: The Story of Theodore Roosevelt (1996): a documentary in the PBS series *American Experience*, first shown in October 1996 (Chapter 19) 1272

Trouble the Water (2008): a documentary built largely on Kimberly Rivers Roberts's own video (Chapter 5) 390

Troublesome Creek: A Midwestern (1995): a documentary about the Jordan family. It won the Grand Jury Prize at the Sundance Film Festival and was nominated for an Academy Award (Chapter 15) 1038

Two Dollars and a Dream (1987): a documentary about Madam C. J. Walker's life by Stanley Nelson (Chapter 35) 2952

Union (2024): a documentary by Brett Story and Stephen Maing about the union campaign at Amazon's JFK8 warehouse. It won a Special Jury Award at the Sundance Film Festival in 2024 (Chapter 14) 978

The Untouchables (1987): a drama, not a documentary, and it changes many of the facts (Chapter 21) 1465

Wal-Mart: The High Cost of Low Price (2005): a documentary by Robert Greenwald that argues against the company and is not a life story of Sam Walton (Chapter 18) 1217

Walter Cronkite: Witness to History (2006): a documentary about Cronkite in the PBS series *American Masters* (Chapter 31) 2577

Washington (2020): a History channel documentary series in three parts, with acted scenes (Chapter 22) 1529

We Shall Remain: Tecumseh's Vision (2009): a PBS documentary about Tecumseh and his brother (Chapter 2) 128

We Still Live Here: Âs Nutayuneân (2011): a documentary about the work of Jessie Little Doe Baird, who brought back the Wôpanâak language (Chapter 2) 177

When They See Us (2019): a Netflix drama series directed by Ava DuVernay (Chapter 21) 1486

With Byrd at the South Pole (1930): a documentary film of the trip (Chapter 1) 79

Wright Brothers' Flying Machine (2003): a documentary about the brothers' flying machine in the PBS series *NOVA*, first shown on November 11, 2003 (Chapter 8) 554

Zora Neale Hurston: Jump at the Sun (2008): a documentary by Sam Pollard about Hurston's life and writing, made for the PBS series *American Masters* (Chapter 32) 2669

Answers to the Chapter Questions

Each chapter opens with three questions, and here they are with their answers, chapter by chapter.

Chapter 1: Exploration and Discovery

How do we know people were living here at least 21,000 years ago? Human footprints at White Sands, in New Mexico, lie between layers of seeds, pollen and quartz sand, and three different tests date those layers to about 21,000 to 23,000 years ago. 96

What did the explorers do to the people whose land they crossed? Some traded with them and were fed by them. Others took captives, from the 500 Taino Columbus's men shipped to Spain to the six Inuit Peary brought to New York, and some killed: de Soto's men at Mabila, Coronado's at Arenal, Frémont's on the Sacramento River. 96

Which famous "firsts" are still in doubt? Peary's North Pole in 1909 and Byrd's flight over it in 1926. Wally Herbert found Peary's records unreliable, and most researchers think Byrd's plane could not carry enough fuel. Older firsts are in doubt too: Spanish slave raiders may have reached Florida before Ponce de León, Ute tradition says Ute ancestors climbed Pikes Peak before Long's men, and nobody knows whether the Norse camp in Newfoundland was Leif's own. 96

Chapter 2: Native Nations

How did Native nations govern themselves and deal with other nations before the United States existed? They ran their own councils and laws, as the Haudenosaunee did under the Great Law of Peace. Their leaders made alliances and treaties and recorded them on wampum belts. 177

What did United States officials and soldiers do to take the nations' land and break the nations up? Soldiers forced whole nations west and onto reservations, and killed hundreds of people at a time at Sand Creek, Wounded Knee and other places. Officials cut the nations' land from about 138 million acres to 48 million and took children to boarding schools, where at least 973 died. 177

How did Native nations hold on to their governments, their languages and some of their land? They went to court, from *Worcester v. Georgia* in 1832 to *McGirt v. Oklahoma* in 2020. They organized until members of Congress restored the Menominee, and they kept their languages, first in secret and now in schools of their own. 177

Chapter 3: Land and Environment

How did Native nations shape the land before Europeans came? They burned the eastern forests on purpose, again and again, to keep them open and grow food for game, and they planted corn, beans and squash together to keep the soil good. 252

What happened to the 30 million bison and the billions of passenger pigeons? Hide hunters, carried by the railroads and helped by army officers who wanted the Plains nations' food gone, left fewer than 1,000 bison by 1900. Netters and shooters killed the pigeons at their nestings, and the last one died in a zoo in 1914. 252

Who was pushed out when parks were made, and who lived next to the waste? Native people were pushed out: the Ahwahneechee in Yosemite, the Tukudika in Yellowstone, the Blackfeet at Glacier. Studies from 1983 on found waste sites far more often beside Black and poor neighborhoods. 252

Chapter 4: Immigration to America

Why did people leave home to come here? Many came for land and work, like the Germans who farmed the Midwest and the Chinese who built the western railroad, or to worship in their own way, like the Puritans. Others ran from danger: the Irish famine, the pogroms against Jews in the Russian Empire, the Nazis, and wars in Vietnam and Sudan. 330

Who did Americans try to keep out, and how? Lawmakers shut out Chinese workers in 1882 and almost everyone from Asia in 1924, and they cut the number of southern and eastern Europeans with quotas. Mobs burned Catholic churches in Philadelphia and drove Chinese people out of Eureka and Tacoma. Officials pushed hundreds of thousands of people of Mexican descent, many of them citizens, across the border in the 1930s and again in 1954. In 2018 agents took thousands of children from their parents at the border, and in January 2026 officers held more than 71,000 people in detention, the most ever. 330

Who could become a citizen, and how did that change? In 1790 only "a free white person" could become a citizen. In 1870 members of Congress added people of African birth or descent. In 1898 the Supreme Court ruled in Wong Kim Ark's case that anyone born here is a citizen, and in 1952 members of Congress removed the limit to white people. 330

Chapter 5: Migration Across America

How did horses reach the Native nations of the Plains? In the older account, horses spread north through trade after the Pueblo Revolt of 1680, when the Spanish left perhaps more than 1,500 of them behind. A 2023 study of horse bones found that many nations had horses by the early 1600s, decades earlier, as the nations' own histories say. 392

How many Cherokee died when soldiers forced them west? Nobody knows exactly. The count most often given is about 4,000, and estimates run from more than 1,000 on the journey itself to about 8,000 or more. 392

Why did about 6 million Black Americans leave the South between 1910 and 1970? They left lynch mobs, Jim Crow laws and cotton crops ruined by the boll weevil and floods, and they went toward jobs in Northern and Western cities. In Georgia, one example, lynch mobs killed more than 450 people between 1880 and 1930, and nearly all Black Georgians could not vote. 392

Chapter 6: City Building

Why do so many American cities have straight streets that cross in a grid? Town founders chose the grid on purpose, from Spain's town rules of 1573 to Penn's Philadelphia in 1682. The three men who laid a grid over Manhattan in 1811 wrote that square-cornered houses were "the most cheap to build and the most convenient to live in." 456

How did people get clean water into crowded cities? City leaders paid for big waterworks after epidemics and fires: steam pumps in Philadelphia in 1801, the Croton Aqueduct in New York in 1842, and a canal that turned Chicago's river around in 1900. 456

Who decided which families could live in which neighborhoods? White owners wrote rules called covenants into their deeds to bar Black buyers, and judges enforced them until 1948. Federal officials refused to back home loans where Black families lived, and after 1949 city officials cleared about 2,000 neighborhoods. 456

Chapter 7: Home and Family Life

Who lived together under one roof, from the longhouse to today? A longhouse held 30 to 60 relatives, and colonial households held servants, apprentices and enslaved people too. By 2025 the average household held 2.50 people. 507

Who did the work of keeping house, and did stoves, pipes and wires make it lighter? Women and children did most of it. In Ruth Schwartz Cowan's argument, the machines did not cut a housewife's hours, because families expected more. 507

How did lenders and builders keep Black families out of new neighborhoods, and what did those families do? FHA officials usually refused to back loans where Black families lived, and William Levitt sold only to white families. The Myers family stayed through weeks of attacks, and about 500 families in the league Clyde Ross helped lead stopped paying together. 507

Chapter 8: Technology

Why did planters enslave far more people after the cotton gin? One gin cleaned about 50 pounds of cotton a day, work that took several hundred hours by hand. Planters grew far more cotton and enslaved far more people to grow and pick it, nearly 4 million by 1860. 578

Why could an enslaved inventor not get a patent? Only the inventor could get a patent, and only by swearing an oath. In 1858 Attorney General Jeremiah Black ruled that an enslaved man was not a citizen and could not swear it, and that the enslaver could not take the patent instead. 578

What went wrong when people trusted a computer's answer? In 2023 two lawyers cited court decisions that ChatGPT had made up, and a judge ordered them and their law firm to pay a $5,000 penalty. In 2020 Detroit police held Robert Williams for 30 hours after a face-recognition search matched him by mistake. 578

Chapter 9: Science

How did American scientists learn how far away the stars are? Henrietta Leavitt showed in 1912 that the blink of a Cepheid star tells how bright it truly is, and so how far away it is. Edwin Hubble used her rule in 1923 to place Andromeda far outside our own galaxy. 658

Why did scientists collect human skulls and brains, and where are those remains now? Samuel Morton, the Army's museum scientists and Aleš Hrdlička measured them to rank the races, an idea that is false. Morton's skulls are at the Penn Museum, whose staff buried 19 of them in 2024, and the Smithsonian still held at least 255 of the brains in 2023. Many Native American remains have been returned since the laws of 1989 and 1990, but in January 2025 museums still had 90,831 not yet made available for return. 658

Who did the work but was left out of the credit? Lise Meitner explained fission, and Rosalind Franklin's X-ray pictures showed the shape of DNA. The Nobel Prizes went to Otto Hahn alone in 1944 and to Watson, Crick and Wilkins in 1962. 658

Chapter 10: Energy

Did kerosene really save the whales? No. Whaling was shrinking before kerosene came, and in most homes kerosene replaced camphene, after a Civil War tax on alcohol raised camphene's price. 718

How did electricity reach almost every American home? Edison's Pearl Street station began selling current in 1882, and alternating current, which a transformer could raise or lower, carried power from Niagara Falls to Buffalo by 1896. After 1935 federal loans paid for farmers' own power lines, and by 1950, 94 households in 100 had electricity. 718

Who paid for America's energy with their freedom, their land or their lives? Enslaved people fed the iron furnaces, at least 271 at Catoctin alone, and about 105,000 coal miners died in accidents from 1900 through 2025. Federal officials made 82,000 people in the Tennessee Valley leave their homes for dams, other dams flooded Kettle Falls and the Fort Berthold farmlands, and husbands, guardians and others murdered Osage people for their oil money. 718

Chapter 11: Transportation

How much faster and cheaper did travel get? On the Erie Canal a ton of goods went from Buffalo to New York City for under $10 instead of about $100 by wagon, and on the first railroad across the country a coast-to-coast trip took a week instead of four months. By 1956, loading a ton of cargo onto a ship cost about 16 cents in containers, against about $5.86 by hand. 779

Who built the new routes, and whose land and homes did they take? Irish and American-born laborers dug the Erie Canal, enslaved people dug the Dismal Swamp Canal and built most of the Southern railroads, and Chinese workers built the western part of the first railroad across the country. Builders and US officials took Native nations' land, and highway officials building federal roads took the homes of more than 475,000 households, many in Black neighborhoods. 779

How did Black travelers fight the laws that made them sit apart? They refused to move, went to court and stayed off the buses: Ida B. Wells sued a railroad in 1884, Homer Plessy tested a Louisiana law and lost in 1896, Irene Morgan won in the Supreme Court in 1946, and Black riders in Montgomery, Alabama, stayed off the city buses for more than a year in 1955 and 1956. After white mobs and Klansmen beat the Freedom Riders in Alabama in 1961, federal officials banned separating passengers in bus and train stations. 779

Chapter 12: Landmarks

Who built the White House and the Capitol, and who was paid for their work? Enslaved men did much of the work, from quarrying the stone to sawing boards and hauling loads. The officials in charge rented them from their owners by the year, and the owners collected the pay. 840

How many workers died building Hoover Dam? By the official count, 96. The records list 42 more as dying of pneumonia, deaths that workers blamed on carbon monoxide in the tunnels, so the exact number is not known. 840

What has happened to the Confederate monuments since 2015? By April 2025 officials and work crews had taken down, moved or renamed 415 Confederate memorials, from statues to place names. Hundreds of them came down or were renamed after Dylann Roof killed nine Black worshippers in Charleston in 2015, James Alex Fields Jr. killed Heather Heyer in Charlottesville in 2017 and the police officer Derek Chauvin murdered George Floyd in Minneapolis in 2020. In 2025 federal officials began putting some back. 840

Chapter 13: The Elements

Who lost their land when Americans found gold and copper on it? The Cherokee lost theirs after gold turned up in Georgia in 1829, when Georgia lawmakers gave it away by lottery and soldiers forced them west in 1838. The Ojibwe bands of Lake Superior gave up the copper country in the Treaty of La Pointe in 1842, and members of Congress took the Black Hills from the Sioux in 1877, after Custer's men found gold there. 906

What did the people in charge say about the radium, lead and uranium that poisoned workers? Supervisors told the dial painters the paint was safe, the Bayway supervisor said his poisoned men had worked too hard, and Thomas Midgley breathed the vapor of tetraethyl lead in front of reporters to show it was safe. Public Health Service researchers agreed with mine owners not to tell the uranium miners what they found. 906

When did someone in the United States first discover a new element? In 1940, at Berkeley, California, when teams using Ernest Lawrence's cyclotron made astatine and neptunium, and then plutonium. Every element before that had been found abroad, and Fred Allison's 1931 claims were false. 906

Chapter 14: Work and Workers

Which workers were not free to leave their work? Native people forced to work under Spanish rule, indentured servants bound for years, enslaved people, prisoners leased to Southern mines and planters, and the Thai garment workers held behind a fence in El Monte until 1995. Virginia judges sentenced John Punch in 1640 to serve for the rest of his life. 983

What happened when workers went on strike? Some won, like the Philadelphia printers in 1786 and the Flint autoworkers in 1937. Many times police, hired guards and soldiers broke the strike, and at Lattimer in 1897, Ludlow in 1914 and Chicago in 1937 they killed strikers, their families and people marching with them. 983

What made work less deadly? Laws written after workers died: more than one way out of every Pennsylvania mine after the Avondale fire, safer couplers after thousands of brakemen died, new factory laws in New York after the Triangle fire, and the 1969 coal mine law after Farmington. Deaths on the job fell from about 38 a day in 1970 to about 14 a day in 2024. 983

Chapter 15: Food and Farming

How did Native farmers turn a wild grass into corn? For thousands of years, starting in southern Mexico about 9,000 years ago, they saved seed only from the teosinte plants they chose, until the grass became corn. 1046

Who did the hardest farm work, and what did they get for it? Enslaved people worked the rice and cotton fields for no pay, and sharecroppers got half the crop's value or less and stayed in debt. From 1992 to 2006, crop workers died of heat at nearly 20 times the rate of all American workers. 1046

How did Americans come to eat food grown far away by people they never met? With canning, pond ice and railroad cars cooled with ice, food could be sold far from the farm, and the farms grew fewer, down to 1.9 million in 2022. 1046

Chapter 16: Economy

How did slavery make money for people far from any plantation? Bankers in London, Amsterdam and New York sold bonds backed by mortgages on enslaved people, Northern companies insured their lives, and Boston mill owners spun the cotton that enslaved people grew. Rhode Island merchants sent out ships that carried more than 100,000 Africans into slavery. 1109

What happened to ordinary families when a boom turned into a crash? Judges jailed debtors in 1819, creditors sold enslaved families apart in 1837, one worker in four had no job by 1933, and lenders took 7.7 million to 10 million homes in the crisis that began in 2007. In the 1930s and after 2007, Black families lost jobs and homes at higher rates than white families. 1109

When did the national government start helping people who lost their jobs? In the 1930s. Before 1933 the federal government paid no unemployment insurance or old-age pensions, and members of Congress set up both in the Social Security Act of 1935, though farm and domestic workers had to wait until 1950 and 1954. 1109

Chapter 17: Money

What has stood behind the dollar, and what stands behind it now? Silver and gold stood behind it, then gold alone. Since August 15, 1971, only the law and people's trust in the government that prints it have stood behind it. 1167

Why did people rush to pull their money out of banks, and who lost when a bank failed? A bank keeps only part of its money in cash, so the first in line got paid. Until deposit insurance began in 1934, the rest took the loss, about $1.34 billion from 1930 to 1933. 1167

How did enslavers and bankers use enslaved people to raise money? Enslavers pledged them for loans, and in the 1830s Louisiana banks lent out their own paper money against them. When the loans went unpaid, the people were sold. 1167

Chapter 18: Marketplace and Consumer Life

How did colonists pay for goods when they had almost no coins? They took goods home on credit and paid later in crops, other goods, work and wampum, while the shopkeeper wrote each debt in a ledger. 1230

Where were people bought and sold like goods? At the market New York's aldermen set up on Wall Street in 1711, in newspaper advertisements such as the 277 or more in Franklin's *Pennsylvania Gazette*, and at a racetrack near Savannah, where Pierce Butler sold between 429 and 436 people in 1859. 1230

How much of what Americans buy today is bought online? About one dollar in six: online sales were 17.1 percent of all retail sales from April to June 2026, by the US Census Bureau's count. 1230

Chapter 19: Big Business and Its Power

How did a few companies come to control whole industries? Their owners made secret deals with railroads, bought out rivals and joined companies into trusts and mergers. By 1880 Standard Oil refined 90 to 95 percent of the country's oil. 1300

What did owners do when their workers went on strike? They locked workers out and hired armed guards, and officials sent soldiers and court orders. At Homestead in 1892 Frick's Pinkerton men fought the steelworkers, and at Ludlow in 1914 soldiers paid by the mine owners burned the strikers' tent camp. 1300

Did breaking up a giant company ever work? Not clearly. Judges split Standard Oil into 34 companies in 1911, yet Rockefeller was three times as rich by 1913. AT&T gave up its local telephone companies in 1984 in a deal most newspapers called a win for AT&T, the order to split Microsoft was thrown out in 2001, and judges in 2025 and 2026 would not make Google sell off parts of its business. 1300

Chapter 20: Government, Law, and Politics

Who decided who could vote, and how did the answer change? Each colony's and then each state's lawmakers decided, and the men who wrote the Constitution left it that way. Virginia's lawmakers took the vote from free Black men in 1723, New Jersey's lawmakers took it from women and free Black men in 1807, and Southern delegates took it from Black men again after 1890. Members of Congress took most of that power from the states with four amendments (1870, 1920, 1964, 1971), which state lawmakers approved, and with the Voting Rights Act of 1965. 1402

When did judges first cancel a law, and what have they done with that power since? In 1803, in *Marbury v. Madison*. The justices have used it to allow forced sterilization, separate railroad cars and the wartime camps, and to strike down school segregation, official prayers, poll taxes and unequal districts. 1402

What happens when people in power break the law? A jury, a court or the Senate decides. In 1614 royal judges found Juan de Oñate, a former governor of New Mexico, guilty of 12 charges and ordered him out of New Mexico for life. A jury found a cabinet secretary guilty of taking a bribe in 1929, and a judge sentenced him to a year in prison. A president resigned in 1974 rather than face a vote. Bill Clinton and Donald Trump were impeached, Trump twice, and the senators found neither guilty. A jury found a senator guilty of bribery in 2024, and he went to prison. 1402

Chapter 21: Crime, Police, and Justice

How did punishment move from the whipping post to the prison cell? Colonial judges had people whipped, branded and hanged in public, and had their ears cut off. Pennsylvania's lawmakers turned the Walnut Street Jail into the country's first state prison in 1790, and by 2009 American prisons held about 1.62 million people. 1496

Who kept order before there were police? Among the Native nations, families and clans settled killings, and in the colonies settlers took turns on a night watch while white men in the South rode on slave patrols. Full-time police began work in Boston in 1838 and in New York in 1845, and the US attorney general put 34 federal agents to work in 1908. 1496

What happened to the people in mobs who killed without a trial? In almost every case the chapter describes, nobody punished them. No one was charged for killing twenty Conestoga people in 1763, for burning Francis McIntosh in 1836 or for burning Henry Smith in 1893, and no white person went to prison for the attack on Greenwood, the Black district of Tulsa, in 1921. 1496

Chapter 22: War and the Military

What killed most American soldiers before World War II? Disease. In the war with Mexico seven of every eight American dead died of something other than enemy fire, and in World War I flu and pneumonia killed more soldiers than the war's deadliest battle did. 1593

Who was made to fight, and who could get out of it? Drafts filled the armies of the Civil War, both world wars and Vietnam. In 1863 a drafted man could pay $300 and go home, and in the Vietnam years college students could put off the draft. Since 1973 everyone who serves has volunteered. 1593

Were soldiers who killed people who could not fight back ever punished? Rarely. No one was charged for killing 96 Christian Lenape and Mohican people at Gnadenhutten in 1782, and no one was tried for Sand Creek. Lieutenant William Calley, the only man convicted for My Lai, went free on parole in 1974. 1593

Chapter 23: America and the World

Who got to choose when this land and its islands changed hands? Almost never the people who lived there: no Native nation was at Paris in 1783, and no one from Puerto Rico, Guam or the Philippines was at Paris in 1898. Of today's five territories, only the Northern Mariana Islands voted to join, and Hawaii's voters chose statehood in 1959 from a ballot that left out independence. 1662

How did a weak new country come to rule islands and overthrow governments? First it grew: by 1848 it reached the Pacific, and by 1900 its mills made more steel than Britain's and Germany's together. Then its leaders used force. Marines helped overthrow Hawaii's queen in 1893, American leaders took Puerto Rico, Guam and the Philippines after a war with Spain in 1898, and after 1950 CIA officers secretly helped overthrow governments in Iran, Guatemala and Chile. 1662

Are the people of the territories American citizens? Most are, but people born in American Samoa are nationals, not citizens, and no one living in a territory can vote for president. 1662

Chapter 24: Slavery and Freedom

How did a servant in Virginia become a slave for life, and a child a slave from birth? Virginia's judges and lawmakers did it one ruling at a time. Judges sentenced John Punch to serve for life in 1640, and in 1662 lawmakers ruled that a child born to an enslaved mother was enslaved for life. 1752

How did enslaved people fight back? They worked slowly, ran away, sued, learned to read in secret and rose up with weapons. During the Civil War about half a million freed themselves by reaching the Union army. 1752

What happened to the freedom people won in 1865? Freed people voted, built schools and sent 16 men to Congress. But within twelve years Southern lawmakers passed Black Codes, Klansmen whipped and murdered freed people, and federal officials stopped enforcing their rights in the South until members of Congress passed the laws of 1964 and 1965. 1752

Chapter 25: Rights and Movements

How did people without power get the people in power to listen? They signed petitions, printed pamphlets and spoke to crowds, and later they planned test cases, boycotted, marched, sat in and filmed the police, from the Germantown protest of 1688 to Darnella Frazier's video in 2020. 1942

What did organizing cost the people who did it? Their jobs, their homes and their freedom, and sometimes their bodies and their lives. A prison doctor fed Lucy Burns by force through a tube, state highway patrolmen had Fannie Lou Hamer beaten in a Mississippi jail, Byron De La Beckwith shot Medgar Evers in his driveway, and Klansmen killed four girls with a bomb in a Birmingham church. 1942

Can a right that has been won be taken away again? Yes. New Jersey's lawmakers took the vote from women and free Black men in 1807, members of Congress took Utah women's vote in 1887, and the justices of the Supreme Court cut back the Voting Rights Act in 2013 and 2026 and ended the right to an abortion in 2022. 1942

Chapter 26: Health, Disease, and Medicine

Why did new diseases kill so many Native people after 1492? The crowd diseases, such as smallpox and measles, had never reached the Americas, so nobody here was immune and a whole town could fall sick at once. Smallpox killed about half of the Cherokee in 1738 and 1739. 2029

How did Americans learn to stop smallpox, yellow fever and polio? Boston doctors tried inoculation in 1721, using a method Onesimus described, and safer vaccination followed in 1800. Army doctors proved in 1900 that mosquitoes spread yellow fever, and after the Salk and Sabin vaccines the last polio case caught inside the country came in 1979. 2029

Who was used in medical experiments without agreeing, and what changed because of it? Enslaved women operated on by J. Marion Sims, the Black men of the Tuskegee study, prisoners and patients in Guatemala, and children at Willowbrook. After the Tuskegee study became public, members of Congress passed the National Research Act in 1974, and researchers in a federally paid study on people must now get each person's informed consent. 2029

Chapter 27: Disasters and Rescue

What rules did Americans make after disasters, and did they work? Some worked, some failed and some are still being tested. Lawmakers and officials wrote new rules after many disasters, from Boston's fire wardens in 1711 to one building code for Florida after Hurricane Andrew. The rule to put a smell in natural gas after the New London school explosion was in use within weeks, but the 1838 steamboat law had no one to enforce it. In late September 2026 members of Congress were still debating FEMA's future. 2117

Was anyone punished when people's choices killed others? Rarely. No one was punished for the *Sultana*, the Johnstown club won in court, a jury found the Triangle owners not guilty, and no Pittston official was charged after Buffalo Creek. Barnett Welansky went to prison for the Cocoanut Grove fire, and PG&E pleaded guilty to 84 counts of involuntary manslaughter after the Camp Fire. 2117

Did everyone get the same help after a disaster? No. Officials paid out London's money for Charleston in 1740 by the size of each loss, so most of it went to rich merchants. Armed men forced Black workers to handle Galveston's dead in 1900, and camp leaders forced Black people to work in the 1927 flood. In 2005 police turned back people walking out of New Orleans at gunpoint, and in 2025 orders to leave reached west Altadena hours late. 2117

Chapter 28: Drugs and Alcohol

How long have people here used tobacco, and what did they use it for? For at least 12,300 years, since the fire at the Wishbone site. Native people offered it in prayer, burning it or smoking it in pipes, and the Algonquian people at Roanoke smoked it to keep the body healthy. 2198

What happened when the whole country banned alcohol from 1920 to 1933? Drinking fell at first to about 30 percent of what it had been, then rose to about 60 to 70 percent. Treasury officials had more poison added to factory alcohol, and the poison killed thousands of drinkers before delegates in Utah ended the ban on December 5, 1933. 2198

Who has been punished for drugs, and who has not? Robert Coles with his red D, the laborer Samuel Caldwell with four years of hard labor, Billie Holiday handcuffed to her hospital bed, and the people sentenced for crack, 91 of every 100 of them Black in 1992, were all punished. Harry Gross never served his jail sentence, and no member of the Sackler family has been charged with a crime. 2198

Chapter 29: Religion

Why did colonists who crossed the ocean to worship freely punish other people for their faith? The Puritans wanted freedom for their own way of worship, and they ran Massachusetts by it. Their leaders banished Roger Williams and Anne Hutchinson and hanged four Quakers in Boston. 2342

How did Americans stop paying taxes to support churches? Baptists and James Madison defeated a new church tax in Virginia, where Jefferson's religious freedom law passed in 1786. Massachusetts ended the last state church tax in 1833. 2342

How long did US officials treat Native ceremonies as crimes? For ninety-five years, from Hiram Price's rules of 1883 until members of Congress passed the American Indian Religious Freedom Act in 1978. 2342

Chapter 30: School and Education

When did school become free, and required, for every child? It came in steps, not on one date. Horace Mann argued for free schools paid for by taxes from 1837, and Massachusetts lawmakers made school a legal duty in 1852. Every state had done the same by 1918, and disabled children won the right to a free public school only in 1975. 2496

Which children were kept out of school, and how did they get in? Lawmakers barred enslaved people from learning to read, and officials separated Black, Chinese and Mexican American children, turned disabled children away and took Native children far from home. Judges first backed keeping them out or apart, in Sarah Roberts's case in 1850, *Plessy* in 1896 and Martha Lum's case in 1927. Later judges let them in, from the Mendez case in 1946 and *Brown* in 1954 to the *Mills* case in 1972, and in 1975 members of Congress passed a law for disabled children and a law that let Native nations run their own schools. 2496

Is it still lawful for a teacher to hit a child? Yes, in 15 states in 2026. The justices of the Supreme Court ruled in 1977 that the Constitution's ban on cruel and unusual punishment does not cover paddling at school, so state lawmakers have banned it one state at a time. 2496

Chapter 31: News and Communication

Did the First Amendment stop the government from punishing people for what they printed? No. Seven years after it took effect, federal judges jailed editors under the Sedition Act, and during World War I prosecutors won more than 1,000 convictions for words. 2594

Who was shut out of carrying and printing the news? From 1802 to 1865 federal law let only free white persons carry the mail, and white mobs drove Black editors out of Memphis and Wilmington. Black and Cherokee Americans started papers of their own. 2594

Why have so many local newspapers closed since 2005? Advertisers moved their money online, where most of it went to technology companies such as Google and Facebook, so newspapers' advertising money fell by more than four-fifths between 2005 and 2022. Readers moved to phones and feeds too, and by 2025, 212 counties had no local news source. 2594

Chapter 32: Art

How did enslaved artists make art, and who got paid for it? They painted, cut silhouettes, turned pots and wrote poems while other people held them as property, and the people who held them got the money. Dave's jars were sold for the profit of Lewis J. Miles, and Moses Williams learned to cut silhouettes and build exhibits in Peale's museum while Peale held him enslaved. Henry Barnes took Prince Demah to London for painting lessons, and Dave cut verses into his jars in years when it was a crime for him to learn to read. 2710

Who decided what counted as art, and who got to see it? Mostly people with money or power: the Met's trustees, who kept it shut on Sundays until 1891, a teacher in Santa Fe who taught one "authentic" Native style, and a congressman who called a touring show of modern paintings trash. Artists pushed back, from Oscar Howe's letter in 1958 to Amy Sherald cancelling her show in 2025. 2710

Why are museums giving back things they once took? Diggers took objects and human remains out of Native graves, at Mound City in 1846 and at Spiro in the 1930s, and sold many of the objects. Under a 1990 law, NAGPRA, museums that take federal money must return Native remains and sacred objects, and since 2024 they need a nation's consent even to show them. 2710

Chapter 33: Music

Where did the banjo and the blues come from? The banjo came from West African instruments, and enslaved Black people played it in the colonies from at least 1736, about a hundred years before a white man played it on a stage. The blues grew out of the work songs and field calls of enslaved people and the songs of Black workers in the South after slavery, between about 1870 and 1900. 2811

Why did so many musicians make hit songs and still die poor? Many were paid a set fee and no royalty, as Bessie Smith was at $125 a side, or someone else put his name on their song, as Don Robey did on "Hound Dog." Others, like Stephen Foster, sold their rights to future payments for money right away. 2811

Who tried to stop people from playing their own music, and why? South Carolina lawmakers banned drums and horns in 1740, saying the sound could call enslaved people together to rise up. Spanish friars burned Pueblo masks in the 1600s to punish Pueblo people for keeping their own religion. Richard Henry Pratt gave Carlisle students brass horns to stop their drumming, at a school whose own paper gave its aim as "the complete transformation of the Indians." Federal rules from 1883 to 1978 punished Native religious dances, which the Secretary of the Interior had called "heathenish." 2811

Chapter 34: Storytelling Evolution

Who tried to stop Americans from acting, and why? Early on, church leaders and lawmakers tried to stop it, for reasons of religion and morals. Spanish priests burned Pueblo masks to end the ceremonies, and Massachusetts lawmakers banned plays, saying they made people less moral and less religious. Later, police arrested Black actors after a rival white manager, by one account, started a riot to destroy their company, and studio heads blacklisted hundreds of film workers over communism. 2905

Why did white actors play Black, Asian and Native people for so long? White audiences paid to laugh at blackface from 1828 on, and producers carried the old minstrel parts into film and radio. Because of the studio heads' own Code and state laws against marriage between races, producers kept actors of color out of leading parts. 2905

Who controls a computer copy of an actor's face and voice? Since the strikes of 2023 and 2024 to 2025, union contracts require a studio or game company to get the performer's consent before it makes or uses one, and the game contract also requires pay. Lawmakers in Tennessee and California passed laws of their own in 2024, and as of September 2026 Congress had passed no national law. 2905

Chapter 35: Styles

Why did lawmakers and officials make rules about other people's clothes and hair? To keep rank where everyone could see it, and to control the people below them: Massachusetts fined people worth less than 200 pounds for wearing silk, South Carolina held enslaved people to coarse cloth in 1740, and a federal official ordered Native men's hair cut in 1902. Since 2019, CROWN Acts in about 30 states have turned the rule around and forbidden punishing people for natural hair, braids, locs or twists. 2978

Who made the cloth Americans wore? Native weavers first, then enslaved people who grew the cotton and made the indigo, and mill workers in New England. Today almost all of it is sewn in other countries, by workers in places like Bangladesh, and in small Los Angeles shops where workers were paid by the piece until a California law required hourly pay in 2022. 2978

Which new materials changed what Americans wore? Churro wool, brought in 1598, became the main thread for weaving in New Mexico. Dyes made from coal tar, from 1856, gave clothes new colors such as mauve. In the 1920s "artificial silk" let more women dress in the new styles. Nylon made stockings from 1939, and polyester suits from 1951 could be washed and worn without ironing. 2978

Chapter 36: Sports and Play

Who invented baseball? No one person did. Clubs wrote the rules, and the Knickerbocker club of New York adopted twenty of them in 1845. The story that Abner Doubleday invented it in 1839 rests on one letter from a man who was 5 years old that year. 3072

Who was kept out of American games, and how did they get in? The people who ran the games kept out laborers, Black players, girls and Black swimmers, and left enslaved riders' names off the race cards. Some got in through courts and a new law, like Maria Pepe and the girls of Title IX. Jackie Robinson got in when Branch Rickey signed him, and Kathryn Johnston played anyway, under a boy's name. 3072

How has children's free time changed? In the 1830s Lucy Larcom made doll furniture from burrs while the grown-ups left the children alone. Since then more of children's time has gone to lessons and organized sports, and by 2022, 95 of every 100 teenagers had or could use a smartphone. 3072

Chapter 37: Holidays

Who decided when Americans would give thanks, and why did it take until 1941 to fix the date? Governors and church elders set the days one at a time, until Lincoln proclaimed a national Thursday in 1863 at Sarah Josepha Hale's request. Every president after him named the day by proclamation, so when Franklin Roosevelt moved it in 1939 to help stores, sixteen states refused by the National Archives' count, and Congress ended the confusion with the 1941 law. 3140

Which days did Black Americans make their own, from election days to Juneteenth? Black governors' election days in New England from about 1749, January 1 from 1808, July 5 in New York in 1827, August 1 after 1834, and Juneteenth from 1866, which Congress made a federal holiday in 2021. 3140

Why did President Harrison proclaim the first Columbus Day in 1892, and why do some states no longer keep it? He proclaimed it for the 400th anniversary of Columbus's landing, a year and a half after a New Orleans mob lynched eleven Italian American men, and some writers say the holiday was partly meant to calm Italy. Native groups and others object to honoring a man whose men enslaved Taino people. Since Berkeley in 1992, cities and 17 states have used the day to honor Indigenous peoples, and some of those states keep Columbus Day too. 3140